

Real Analysis Notes

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Part I

Foundations of Analysis

Chapter 1

Preliminaries: Sets, Functions, Induction, and Cardinality

1.1 Sets and Functions

Sets, membership, and the basic operations

We take the naive notion of a *set* as a collection of objects, its *elements*. We write $x \in A$ for “ x is an element of A ” and $x \notin A$ for its negation. A set is determined by its elements (the *axiom of extensionality*): $A = B$ means exactly that $x \in A \iff x \in B$ for every x . Sets are described either by listing, $A = \{1, 2, 3\}$, or by a defining property, $A = \{x : P(x)\}$ (“the x such that $P(x)$ holds”).

Definition 1.1.1. A is a *subset* of B , written $A \subseteq B$, if every element of A is an element of B . We call A a *proper subset*, $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$. The *empty set* $\emptyset$ has no elements; $\emptyset \subseteq B$ for every B (vacuously). Two sets are equal iff each is a subset of the other:

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

The displayed equivalence is the workhorse of the whole subject: to prove two sets equal you prove two containments. We isolate this as a method because it recurs in literally every chapter.

Method (Proving a set identity $A = B$). *Recognition trigger.* You must show two described sets are equal (e.g. a De Morgan law, a distributive law, $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H)$).

1. ($\subseteq$) Take an arbitrary $x \in A$. Unfold the definition of A to learn what x satisfies, then deduce $x \in B$.
2. ($\supseteq$) Take an arbitrary $x \in B$ and deduce $x \in A$.
3. If every step is an “iff”, you may do both directions at once with a chain of $\iff$ ’s — but only when each link genuinely reverses.

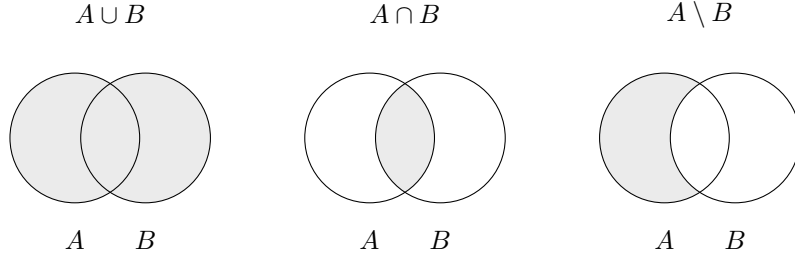
For an *inclusion* $A \subseteq B$ only step (1) is needed; to *disprove* $A \subseteq B$ exhibit one $x \in A$ with $x \notin B$.

Definition 1.1.2. For sets A, B inside a universe U :

$$A \cup B = \{x : x \in A \text{ or } x \in B\}, \quad A \cap B = \{x : x \in A \text{ and } x \in B\},$$

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}, \quad A^c = U \setminus A.$$

A and B are *disjoint* if $A \cap B = \emptyset$. The *symmetric difference* is $A \triangle B = (A \setminus B) \cup (B \setminus A)$.



Theorem 1.1.3 (De Morgan and distributive laws). *For subsets of a universe U ,*

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c,$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

The complement laws hold verbatim for arbitrary unions/intersections: if $\{A_i\}_{i \in I}$ is any family, $(\bigcup_i A_i)^c = \bigcap_i A_i^c$ and $(\bigcap_i A_i)^c = \bigcup_i A_i^c$.

Proof. We prove the first De Morgan law by the method above; the rest are identical in spirit. $x \in (A \cup B)^c \iff x \notin A \cup B \iff \neg(x \in A \text{ or } x \in B) \iff (x \notin A) \text{ and } (x \notin B) \iff x \in A^c \cap B^c$. Each link is a genuine logical equivalence (the third is the De Morgan law of propositional logic), so the two sets have the same elements and are equal. For the arbitrary-family version, “ $x \notin \bigcup_i A_i$ ” means “ $x \notin A_i$ for every i ”, which is exactly “ $x \in \bigcap_i A_i^c$ ”. $\square$

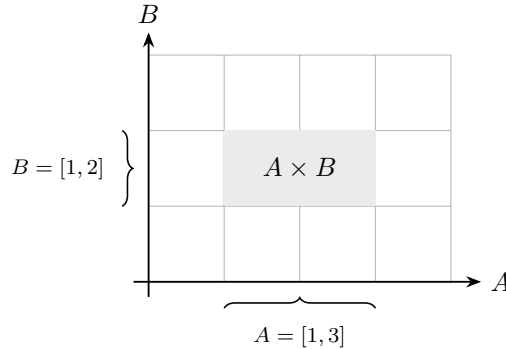
Worked Example 1.1.4 (A distributive law by the two-containment method). Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Recognition. A set identity; run the two-containment method. Here every step is reversible, so a single $\iff$ -chain works.

Solution. $x \in A \cap (B \cup C)$ iff $x \in A$ and $x \in B \cup C$, i.e. $x \in A$ and $(x \in B \text{ or } x \in C)$. Distributing “and” over “or” in logic, this is $(x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)$, i.e. $x \in (A \cap B) \text{ or } x \in (A \cap C)$, i.e. $x \in (A \cap B) \cup (A \cap C)$. Every link reverses, so the sets are equal. *Check.* With $A = \{1, 2\}$, $B = \{2, 3\}$, $C = \{1, 4\}$: $B \cup C = \{1, 2, 3, 4\}$, LHS = $\{1, 2\}$; $A \cap B = \{2\}$, $A \cap C = \{1\}$, RHS = $\{1, 2\}$. Agree.

Cartesian products and relations

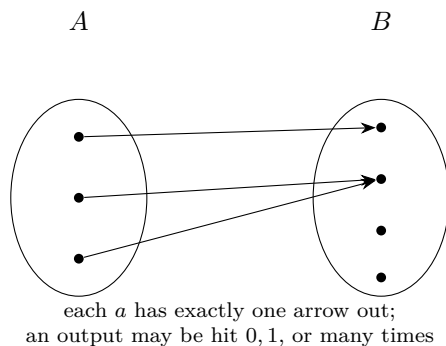
Definition 1.1.5. The *ordered pair* (a, b) is determined by both entries and their order: $(a, b) = (a', b') \iff a = a' \text{ and } b = b'$. The *Cartesian product* is $A \times B = \{(a, b) : a \in A, b \in B\}$. A *relation* from A to B is any subset $R \subseteq A \times B$; we write $a R b$ for $(a, b) \in R$.



Functions: the modern (graph) definition

Definition 1.1.6. A function $f : A \rightarrow B$ is a relation $f \subseteq A \times B$ such that for each $a \in A$ there is *exactly one* $b \in B$ with $(a, b) \in f$; that unique b is written $f(a)$. Here A is the *domain*, B the *codomain*. The *range* (or image of the whole domain) is $f(A) = \{f(a) : a \in A\} \subseteq B$.

The point of phrasing a function as a special relation (a set of pairs) is that “rule” is too vague: the data of f is its graph together with the chosen codomain B . Two functions are equal iff they have the same domain, same codomain, and same value at every point.



Definition 1.1.7. Let $f : A \rightarrow B$, $E \subseteq A$, $H \subseteq B$. The *direct image* of E and the *inverse image* (preimage) of H are

$$f(E) = \{f(x) : x \in E\} \subseteq B, \quad f^{-1}(H) = \{x \in A : f(x) \in H\} \subseteq A.$$

The symbol $f^{-1}(H)$ is defined for *every* f and every H — it does not require f to be invertible.

Definition 1.1.8. $f : A \rightarrow B$ is *injective* (one-to-one) if $f(a) = f(a') \implies a = a'$; *surjective* (onto) if $f(A) = B$, i.e. every $b \in B$ equals $f(a)$ for some a ; *bijective* if both. When f is bijective there is a unique *inverse function* $f^{-1} : B \rightarrow A$ with $f^{-1} \circ f = \text{id}_A$ and $f \circ f^{-1} = \text{id}_B$.

Method (Showing a function is injective / surjective / bijective). *Recognition trigger.* You are asked whether a given $f : A \rightarrow B$ is one-to-one, onto, or a bijection (and possibly to produce f^{-1}).

1. **Injective:** assume $f(a) = f(a')$ and *algebraically* force $a = a'$. (To *disprove*: exhibit $a \neq a'$ with equal images.)
2. **Surjective:** take an arbitrary $b \in B$ and *solve* $f(a) = b$ for some $a \in A$, checking the solution actually lies in A . (To *disprove*: find $b \in B$ not hit, often by an extreme-value/range argument.)
3. **Bijective:** prove both, or — often cleaner — exhibit a two-sided inverse $g : B \rightarrow A$ with $g \circ f = \text{id}$ and $f \circ g = \text{id}$; existence of such a g is equivalent to bijectivity.

Worked Example 1.1.9 (An explicit bijection $(a, b) \rightarrow (0, 1)$). For reals $a < b$ build a bijection of the open interval (a, b) onto $(0, 1)$.

Recognition. Trigger: “find an explicit bijection between two intervals.” An affine map $t \mapsto \alpha t + \beta$ carries intervals to intervals and is invertible when $\alpha \neq 0$; pin α, β by matching endpoints.

Solution. Define $\varphi(t) = \frac{t-a}{b-a}$ for $t \in (a, b)$. Then $\varphi(a^+) = 0$, $\varphi(b^-) = 1$, and φ is strictly increasing (slope $1/(b-a) > 0$), hence injective; it is onto $(0, 1)$ since its inverse $\varphi^{-1}(s) = a + (b-a)s$ sends $(0, 1)$ back into (a, b) . As $\varphi^{-1} \circ \varphi = \text{id}$ and $\varphi \circ \varphi^{-1} = \text{id}$, φ is a bijection. *Check.* The midpoint $t = \frac{a+b}{2}$ maps to $\varphi = \frac{1}{2}$, the midpoint of $(0, 1)$, as a symmetric affine map should.

Worked Example 1.1.10 (A bijection $\mathbb{R} \rightarrow (-1, 1)$). Show $f(x) = \frac{x}{1+|x|}$ is a bijection of $\mathbb{R}$ onto $(-1, 1)$. (B&S phrase the same fact with $x/\sqrt{x^2+1}$.)

Recognition. “Show f is a bijection onto an interval.” Use the inverse-exhibition route: guess f^{-1} and verify both composites.

Solution. For $x \geq 0$, $f(x) = x/(1+x) \in [0, 1)$; for $x < 0$, $f(x) = x/(1-x) \in (-1, 0)$. So $f(\mathbb{R}) \subseteq (-1, 1)$. Solve $y = x/(1+|x|)$: if $y \geq 0$ then $x \geq 0$ and $y(1+x) = x \Rightarrow x = \frac{y}{1-y}$; if $y < 0$ then $x = \frac{y}{1+y}$. In both cases $g(y) = \frac{y}{1-|y|}$ is defined on $(-1, 1)$ and one checks $g = f^{-1}$: $f(g(y)) = y$ and $g(f(x)) = x$. Hence f is a bijection $\mathbb{R} \rightarrow (-1, 1)$. *Check.* $f(1) = \frac{1}{2}$ and $g(\frac{1}{2}) = \frac{1/2}{1-1/2} = 1$; consistent. *Consequence:* $\mathbb{R}$ and the bounded interval $(-1, 1)$ have the same cardinality — foreshadowing §1.3.

Worked Example 1.1.11 (NON-injective squaring, and how images behave). Let $f(x) = x^2$ on $\mathbb{R}$. With $E = [-1, 0]$, $F = [0, 1]$ show $f(E \cap F) \subsetneq f(E) \cap f(F)$ is *false* here but the general inclusion $f(E \cap F) \subseteq f(E) \cap f(F)$ can be strict.

Recognition. Trigger: a parts-of-images comparison; the lesson is that direct image preserves $\cup$ but only *half* of $\cap$.

Solution. Here $E \cap F = \{0\}$, so $f(E \cap F) = \{0\}$, while $f(E) = [0, 1] = f(F)$ so $f(E) \cap f(F) = [0, 1]$. Thus $f(E \cap F) = \{0\} \subsetneq [0, 1] = f(E) \cap f(F)$: the inclusion is *strict*, exactly because f is not injective (-1 and 1 share image 1). *Always true:* $f(E \cap F) \subseteq f(E) \cap f(F)$, since $y = f(x)$ with $x \in E \cap F$ gives $y \in f(E)$ and $y \in f(F)$. Equality for *all* E, F holds iff f is injective. *Check.* For $\cup$, $f(E \cup F) = [0, 1] = f(E) \cup f(F)$ — union is always preserved, no injectivity needed.

Caution. Direct image and preimage are *not* symmetric. Preimage commutes with all three operations: $f^{-1}(G \cup H) = f^{-1}G \cup f^{-1}H$, $f^{-1}(G \cap H) = f^{-1}G \cap f^{-1}H$, and $f^{-1}(G^c) = (f^{-1}G)^c$. Direct image commutes with $\cup$ but, for $\cap$ and complement, only gives $\subseteq$ unless f is injective. When in doubt, push problems through the *preimage* — it is the better-behaved operation.

Theorem 1.1.12 (Preimage respects all Boolean operations). For $f : A \rightarrow B$ and $G, H \subseteq B$:

$$f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H), \quad f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H), \quad f^{-1}(B \setminus G) = A \setminus f^{-1}(G).$$

Proof. For each, $x \in \text{LHS}$ unfolds to a statement about $f(x)$ that is literally the defining statement of the RHS. E.g. $x \in f^{-1}(G \cap H) \iff f(x) \in G \cap H \iff (f(x) \in G \text{ and } f(x) \in H) \iff x \in f^{-1}(G) \text{ and } x \in f^{-1}(H) \iff x \in f^{-1}(G) \cap f^{-1}(H)$. The complement case uses $f(x) \notin G \iff x \notin f^{-1}(G)$. $\square$

Composition

Definition 1.1.13. If $f : A \rightarrow B$ and $g : B \rightarrow C$, the *composite* $g \circ f : A \rightarrow C$ is $(g \circ f)(x) = g(f(x))$. Composition is associative: $h \circ (g \circ f) = (h \circ g) \circ f$.

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ & \searrow & & \nearrow & \\ & & g \circ f & & \end{array}$$

Proposition 1.1.14. Let $f : A \rightarrow B$, $g : B \rightarrow C$. (i) If f, g are injective so is $g \circ f$. (ii) If f, g are surjective so is $g \circ f$. (iii) If $g \circ f$ is injective then f is injective. (iv) If $g \circ f$ is surjective then g is surjective.

Proof. (i) $g(f(a)) = g(f(a')) \implies f(a) = f(a')$ (g inj) $\implies a = a'$ (f inj). (ii) Given $c \in C$, pick b with $g(b) = c$ (g surj), then a with $f(a) = b$ (f surj); so $(g \circ f)(a) = c$. (iii) $f(a) = f(a') \implies g(f(a)) = g(f(a')) \implies a = a'$. (iv) The range of $g \circ f$ is contained in the range of g ; if the former is all of C so is the latter. $\square$

Remark. Parts (iii)–(iv) are sharp: $g \circ f$ injective forces only f (the *first* map) injective, and $g \circ f$ surjective forces only g (the *last* map) surjective. A standard counterexample: $f : \mathbb{N} \rightarrow \mathbb{N}$, $f(n) = n + 1$, and $g : \mathbb{N} \rightarrow \mathbb{N}$, $g(1) = 1$, $g(n) = n - 1$ for $n \geq 2$. Then $g \circ f = \text{id}$ is bijective, yet f is not surjective ($1 \notin f(\mathbb{N})$) and g is not injective ($g(1) = g(2) = 1$).

Further worked examples

Worked Example 1.1.15 (De Morgan with a concrete universe). Let $U = \{1, 2, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$. Verify $(A \cup B)^c = A^c \cap B^c$ by direct computation, then explain why this is forced.

Recognition. A set identity stated for specific sets; we both *compute* both sides and recall the De Morgan law (Theorem 1.1.3) that guarantees agreement.

Solution. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, so $(A \cup B)^c = \{8, 9, 10\}$. On the other side $A^c = \{6, 7, 8, 9, 10\}$ and $B^c = \{1, 2, 3, 8, 9, 10\}$, whence $A^c \cap B^c = \{8, 9, 10\}$. The two sides agree. They *must*: an element lies outside $A \cup B$ exactly when it lies outside A and outside B , which is the logical content of the law. *Check.* 8 is in neither A nor B , so it appears in both computed answers; $6 \in B$, so it is correctly absent from each.

Worked Example 1.1.16 (Symmetric difference is associative — a sanity case). With $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{3, 4, 5\}$, compute $(A \triangle B) \triangle C$ and $A \triangle (B \triangle C)$ and confirm they coincide.

Recognition. Trigger: test an algebraic identity ($\triangle$ is associative) on a concrete instance, unfolding Definition 1.1.2.

Solution. $A \triangle B = (A \setminus B) \cup (B \setminus A) = \{1\} \cup \{4\} = \{1, 4\}$. Then $(A \triangle B) \triangle C = \{1, 4\} \triangle \{3, 4, 5\}$: elements in exactly one are $\{1, 3, 5\}$. For the other grouping, $B \triangle C = \{2\} \cup \{5\} = \{2, 5\}$, and $A \triangle \{2, 5\} = \{1, 3, 5\}$ (elements in exactly one of $\{1, 2, 3\}$ and $\{2, 5\}$). Both equal $\{1, 3, 5\}$. *Check.* $x \in A \triangle B \triangle C$ iff x lies in an *odd* number of A, B, C ; here 1 is in only A , 3 is in all three, 5 is in only C — all odd counts, matching $\{1, 3, 5\}$.

Worked Example 1.1.17 (A NON-identity: when does $A \setminus (B \setminus C) = (A \setminus B) \setminus C$?). Decide whether $A \setminus (B \setminus C) = (A \setminus B) \setminus C$ holds for all sets, and if not, find the smallest correcting term.

Recognition. Trigger: a plausible-looking “cancellation” law. The method is to test it, and on failure locate exactly where the two sides diverge.

Solution. Take $A = \{1\}$, $B = \emptyset$, $C = \{1\}$. Then $B \setminus C = \emptyset$, so $\text{LHS} = A \setminus \emptyset = \{1\}$. But $A \setminus B = \{1\}$ and $(A \setminus B) \setminus C = \{1\} \setminus \{1\} = \emptyset$. So $\text{LHS} = \{1\} \neq \emptyset = \text{RHS}$: the identity is *false*. Unfolding memberships, $x \in A \setminus (B \setminus C)$ means $x \in A$ and not $(x \in B \text{ and } x \notin C)$, i.e. $x \in A$ and $(x \notin B \text{ or } x \in C)$, giving $A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C)$. The correct law carries an extra $A \cap C$. *Check.* In the counterexample $A \cap C = \{1\}$, exactly the element the bad RHS dropped.

Worked Example 1.1.18 (Preimage commutes with intersection but image need not). Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, with $G = (-\infty, 4]$ and $H = [1, \infty)$. Compute $f^{-1}(G \cap H)$ and confirm it equals $f^{-1}(G) \cap f^{-1}(H)$; then contrast with the direct image of an intersection.

Recognition. Trigger: illustrate Theorem 1.1.12 numerically and reinforce the pitfall that direct image is worse behaved.

Solution. $G \cap H = [1, 4]$, so $f^{-1}(G \cap H) = \{x : 1 \leq x^2 \leq 4\} = [-2, -1] \cup [1, 2]$. Separately $f^{-1}(G) = \{x : x^2 \leq 4\} = [-2, 2]$ and $f^{-1}(H) = \{x : x^2 \geq 1\} = (-\infty, -1] \cup [1, \infty)$; their intersection is $[-2, -1] \cup [1, 2]$. They match, as the theorem promises. Contrast: with $E = [-2, -1]$ and $F = [1, 2]$ we have $E \cap F = \emptyset$ so $f(E \cap F) = \emptyset$, yet $f(E) = [1, 4] = f(F)$ gives $f(E) \cap f(F) = [1, 4] \neq \emptyset$ — direct image fails for $\cap$ exactly because f is not injective. *Check.* $x = -2$ lies in $f^{-1}(G \cap H)$ and indeed $f(-2) = 4 \in [1, 4] = G \cap H$.

Worked Example 1.1.19 (An injection that is not a surjection, and a fix). Show $f : \mathbb{N} \rightarrow \mathbb{N}$, $f(n) = 2n$, is injective but not surjective, and exhibit a different codomain making the “same rule” a bijection.

Recognition. Trigger: test injectivity/surjectivity by the method, and recall that surjectivity depends on the chosen *codomain*.

Solution. Injective: $2n = 2n' \implies n = n'$. Not surjective onto $\mathbb{N}$: 1 is odd, so $f(n) = 1$ has no solution in $\mathbb{N}$; the range is the even numbers $2\mathbb{N} \subsetneq \mathbb{N}$. If instead we declare the codomain to be $2\mathbb{N}$, the very same rule $g : \mathbb{N} \rightarrow 2\mathbb{N}$, $g(n) = 2n$, is onto (every even $m = 2(m/2)$ with $m/2 \in \mathbb{N}$) and still injective, hence a bijection with inverse $m \mapsto m/2$. *Check.* f and g have identical graphs $\{(n, 2n)\}$; only the codomain changed, and that alone flipped surjectivity — the data of a function includes its codomain.

Exercises (Bartle & Sherbert, §1.1).

- Let $A = \{k \in \mathbb{N} : k \leq 20\}$, $B = \{3k - 1 : k \in \mathbb{N}\}$, $C = \{2k - 1 : k \in \mathbb{N}\}$. Determine (a) $A \cap B \cap C$; (b) $(A \cap B) \setminus C$; (c) $(A \cap C) \setminus B$.

2. Draw diagrams to simplify and identify: (a) $A \setminus (B \setminus A)$; (b) $A \setminus (A \setminus B)$; (c) $A \cap (B \setminus A)$.
3. If A, B are sets, show $A \subseteq B$ iff $A \cap B = A$.
4. Prove the second De Morgan law $(A \cap B)^c = A^c \cup B^c$.
5. Prove the distributive laws: (a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$; (b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
6. Symmetric difference $D =$ elements in exactly one of A, B . Represent D by a diagram and show (a) $D = (A \setminus B) \cup (B \setminus A)$; (b) $D = (A \cup B) \setminus (A \cap B)$.
7. For $n \in \mathbb{N}$ let $A_n = \{(n+1)k : k \in \mathbb{N}\}$. (a) Find $A_1 \cap A_2$. (b) Determine $\bigcup_{n \in \mathbb{N}} A_n$ and $\bigcap_{n \in \mathbb{N}} A_n$.
8. Draw diagrams in the plane of $A \times B$ for: (a) $A = \{x \in \mathbb{R} : 1 \leq x \leq 2 \text{ or } 3 \leq x \leq 4\}$, $B = \{x \in \mathbb{R} : x = 1 \text{ or } x = 2\}$; (b) $A = \{1, 2, 3\}$, $B = \{x \in \mathbb{R} : 1 \leq x \leq 3\}$.
9. Let $A = B = \{x \in \mathbb{R} : -1 \leq x \leq 1\}$ and $C = \{(x, y) : x^2 + y^2 = 1\} \subseteq A \times B$. Is C a function? Explain.
10. Let $f(x) = 1/x^2$ ($x \neq 0$). (a) Find the direct image $f(E)$ for $E = \{x : 1 \leq x \leq 2\}$. (b) Find the inverse image $f^{-1}(G)$ for $G = \{x : 1 \leq x \leq 4\}$.
11. Let $g(x) = x^2$, $f(x) = x - 2$, and $h = g \circ f$. (a) Find $h(E)$ for $E = \{x : 0 \leq x \leq 1\}$. (b) Find $h^{-1}(G)$ for $G = \{x : 0 \leq x \leq 4\}$.
12. Let $f(x) = x^2$, $E = \{x : -1 \leq x \leq 0\}$, $F = \{x : 0 \leq x \leq 1\}$. Show $E \cap F = \{0\}$ and $f(E \cap F) = \{0\}$, while $f(E) \cap f(F) = \{y : 0 \leq y \leq 1\}$; hence $f(E \cap F) \subsetneq f(E) \cap f(F)$. What happens if 0 is removed from E and F ?
13. With f, E, F as above, find $E \setminus F$ and $f(E) \setminus f(F)$ and show $f(E \setminus F) \subseteq f(E) \setminus f(F)$ can fail.
14. For $f : A \rightarrow B$ and $E, F \subseteq A$, show $f(E \cup F) = f(E) \cup f(F)$ and $f(E \cap F) \subseteq f(E) \cap f(F)$.
15. For $f : A \rightarrow B$ and $G, H \subseteq B$, show $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H)$ and $f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H)$.
16. Show $f(x) = x/\sqrt{x^2 + 1}$ is a bijection of $\mathbb{R}$ onto $\{y : -1 < y < 1\}$.
17. For $a < b$, find an explicit bijection of (a, b) onto $(0, 1)$.
18. (a) Give two functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ with $f \neq g$ but $f \circ g = g \circ f$. (b) Give three functions $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ with $f \circ (g + h) \neq (f \circ g) + (f \circ h)$.
19. (a) If $f : A \rightarrow B$ is injective and $E \subseteq A$, show $f^{-1}(f(E)) = E$. (b) Give an example where this fails when f is not injective.

Additional exercises.

1. (Two-containment drill.) Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ and $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.
2. Show $A \triangle B = \emptyset$ iff $A = B$, and that $\triangle$ is associative: $(A \triangle B) \triangle C = A \triangle (B \triangle C)$. (Hint: membership in $A \triangle B \triangle C$ means “an odd number of A, B, C ”.)
3. Let $f : A \rightarrow B$. Prove f is injective iff $f(E \cap F) = f(E) \cap f(F)$ for all $E, F \subseteq A$, and surjective iff $f(f^{-1}(H)) = H$ for all $H \subseteq B$.
4. Prove $E \subseteq f^{-1}(f(E))$ for every f and $E \subseteq A$, and give an f and E for which the inclusion is strict.
5. (Composition of bijections.) If $f : A \rightarrow B$ and $g : B \rightarrow C$ are bijections, prove $g \circ f$ is a bijection and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
6. Exhibit an explicit bijection $[0, 1] \rightarrow [0, 1)$. (Hint: move a countable set; relocate $1, \frac{1}{2}, \frac{1}{3}, \dots$)
7. Draw the Venn diagram for three sets and shade $A \triangle B \triangle C$. Which of the eight regions are shaded, and why does “odd membership count” describe them?
8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x/(1 + x^2)$. Determine $f(\mathbb{R})$, decide whether f is injective or surjective onto its range, and sketch enough of the graph to justify your answer.
9. (Indexed De Morgan.) For sets $A_i \subseteq U$, prove $(\bigcap_{i \in I} A_i)^c = \bigcup_{i \in I} A_i^c$ directly from the definitions, for an arbitrary (possibly uncountable) index set I .

1.2 Mathematical Induction

The well-ordering principle and induction

The natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ carry one property that the integers, rationals, and reals all lack, and from which induction flows.

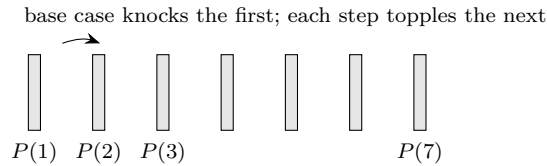
Theorem 1.2.1 (Well-Ordering Principle). *Every nonempty subset $S \subseteq \mathbb{N}$ has a least element: there is $m \in S$ with $m \leq s$ for all $s \in S$.*

We take this as the defining structural axiom of $\mathbb{N}$. Induction is logically equivalent to it.

Theorem 1.2.2 (Principle of Mathematical Induction). *Let $S \subseteq \mathbb{N}$ satisfy (i) $1 \in S$, and (ii) $k \in S \implies k+1 \in S$. Then $S = \mathbb{N}$.*

Proof. Suppose not, so $T = \mathbb{N} \setminus S \neq \emptyset$. By well-ordering T has a least element m . Now $m \neq 1$ by (i), so $m \geq 2$ and $m-1 \in \mathbb{N}$. Since m is the *least* element not in S , we have $m-1 \in S$. But then (ii) gives $m = (m-1) + 1 \in S$, contradicting $m \in T$. Hence $T = \emptyset$ and $S = \mathbb{N}$. $\square$

In practice $S = \{n \in \mathbb{N} : P(n) \text{ is true}\}$ for a statement $P(n)$, and the two hypotheses become the familiar *base case* $P(1)$ and *inductive step* $P(k) \implies P(k+1)$.



Theorem 1.2.3 (Strong / Complete Induction). *Let $S \subseteq \mathbb{N}$ satisfy: for every $n \in \mathbb{N}$, if $\{1, \dots, n-1\} \subseteq S$ then $n \in S$. Then $S = \mathbb{N}$. (Equivalently: if $P(1)$ holds and “ $P(1), \dots, P(k)$ all hold” implies $P(k+1)$, then $P(n)$ holds for all n .)*

Proof. Apply well-ordering to $T = \mathbb{N} \setminus S$ exactly as above: a least missing element n would have all of $1, \dots, n-1$ in S , forcing $n \in S$, a contradiction. $\square$

Method (Running an induction proof). *Recognition trigger.* A statement $P(n)$ indexed by $\mathbb{N}$: a summation formula, a divisibility claim, an inequality valid past some point, a recursively defined object.

1. **State** $P(n)$ **precisely** as a predicate in n (write the “for all $n \geq n_0$ ”).
2. **Base case.** Verify $P(n_0)$ by direct computation. Do not skip it.
3. **Inductive step.** Assume $P(k)$ (ordinary) or $P(n_0), \dots, P(k)$ (strong) for a generic $k \geq n_0$. Derive $P(k+1)$, *using the hypothesis explicitly* — circle where you invoke it.
4. Conclude $P(n)$ for all $n \geq n_0$.

When to use strong induction: when $P(k+1)$ depends on an *earlier* case, not just the immediately previous one (recurrences, prime factorization, Fibonacci-type bounds).

Worked Example 1.2.4 (A summation formula). Prove $\sum_{j=1}^n j = \frac{n(n+1)}{2}$ for all $n \in \mathbb{N}$.

Recognition. A formula indexed by n : ordinary induction, since the $(k+1)$ -case adds one new term to the k -case.

Solution. Base $n = 1$: LHS = 1, RHS = $\frac{1 \cdot 2}{2} = 1$. ■ *Step:* assume $\sum_{j=1}^k j = \frac{k(k+1)}{2}$. Then

$$\sum_{j=1}^{k+1} j = \underbrace{\sum_{j=1}^k j}_{\text{hyp}} + (k+1) = \frac{k(k+1)}{2} + (k+1) = (k+1) \frac{k+2}{2} = \frac{(k+1)(k+2)}{2},$$

which is the formula at $n = k + 1$. By induction it holds for all n . *Check.* $n = 4$ gives $1 + 2 + 3 + 4 = 10 = \frac{4 \cdot 5}{2}$. ■

Worked Example 1.2.5 (A divisibility claim). Prove $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

Recognition. A divisibility statement; ordinary induction, expanding $(k + 1)^3 + 5(k + 1)$ and peeling off the k -case.

Solution. *Base* $n = 1$: $1 + 5 = 6$, divisible by 6. ■ *Step:* assume $6 \mid k^3 + 5k$, say $k^3 + 5k = 6t$. Then

$$(k + 1)^3 + 5(k + 1) = k^3 + 3k^2 + 3k + 1 + 5k + 5 = (k^3 + 5k) + 3k^2 + 3k + 6 = 6t + 3k(k + 1) + 6.$$

Now $k(k + 1)$ is a product of consecutive integers, hence even, so $3k(k + 1)$ is divisible by 6; the whole expression is a sum of multiples of 6. Thus $6 \mid (k + 1)^3 + 5(k + 1)$. By induction the claim holds for all n . *Check.* $n = 2$: $8 + 10 = 18 = 6 \cdot 3$. ■

Worked Example 1.2.6 (An eventual inequality (base case shifted)). Prove $2^n > n^2$ for all integers $n \geq 5$.

Recognition. An inequality true only *past a point*: induct with base $n_0 = 5$; in the step, bound the new term using the hypothesis *and* a secondary inequality.

Solution. *Base* $n = 5$: $2^5 = 32 > 25 = 5^2$. ■ *Step:* assume $2^k > k^2$ for some $k \geq 5$. Then $2^{k+1} = 2 \cdot 2^k > 2k^2$. It suffices to show $2k^2 \geq (k + 1)^2$, i.e. $k^2 - 2k - 1 \geq 0$, which holds for $k \geq 3$ (roots $1 \pm \sqrt{2}$). Hence $2^{k+1} > 2k^2 \geq (k + 1)^2$. By induction $2^n > n^2$ for all $n \geq 5$. *Check.* $n = 6$: $64 > 36$. ■ (Note $n = 2, 3, 4$ fail: $4 = 4$, $8 < 9$, $16 = 16$ — the shifted base is essential.)

Worked Example 1.2.7 (Bernoulli's inequality). Prove $(1 + x)^n \geq 1 + nx$ for all $n \in \mathbb{N}$ and all real $x \geq -1$.

Recognition. A parametrized inequality; induct on n , using $x \geq -1$ (so $1 + x \geq 0$) to preserve the direction when multiplying.

Solution. *Base* $n = 1$: $(1 + x)^1 = 1 + x \geq 1 + x$. ■ *Step:* assume $(1 + x)^k \geq 1 + kx$. Since $1 + x \geq 0$ we may multiply the inequality by $1 + x$:

$$(1 + x)^{k+1} \geq (1 + kx)(1 + x) = 1 + (k + 1)x + kx^2 \geq 1 + (k + 1)x,$$

because $kx^2 \geq 0$. By induction the inequality holds for all n . *Check.* $x = 1$, $n = 3$: $2^3 = 8 \geq 1 + 3 = 4$. ■ This inequality is the engine behind many limit estimates later (e.g. $\sqrt[n]{n} \rightarrow 1$).

Worked Example 1.2.8 (Strong induction: every $n \geq 2$ has a prime factor). Prove that every integer $n \geq 2$ is a product of primes.

Recognition. The factorization of n refers to *smaller* numbers $a, b < n$, not to $n - 1$; this is the signature of strong induction.

Solution. *Base* $n = 2$: prime, a one-factor product. *Step (strong):* assume every integer in $\{2, \dots, k\}$ is a product of primes, and consider $k + 1$. If $k + 1$ is prime we are done. Otherwise $k + 1 = ab$ with $2 \leq a, b \leq k$. By the strong hypothesis a and b are each products of primes, hence so is their product $k + 1$. By strong induction the claim holds for all $n \geq 2$. *Check.* $12 = 2 \cdot 6 = 2 \cdot 2 \cdot 3$, reached by splitting into the smaller already-handled 2 and 6.

Caution. Never skip the base case. The step alone proves nothing. “ $P(n) : n = n + 1$ ” satisfies a vacuous-looking step pattern if mishandled, yet is false for every n ; only a base case can anchor the chain. The notorious “all horses are the same color” paradox fails precisely at $n = 1 \rightarrow 2$, where the inductive step secretly needs two horses to overlap. **Always check the step works for the *smallest* relevant k .**

Caution. The inductive hypothesis must actually be *used*. A “proof” that re-derives $P(k + 1)$ from scratch without invoking $P(k)$ is not an induction (and usually has a hidden gap). Mark the spot where you substitute the hypothesis.

Further worked examples

Worked Example 1.2.9 (Sum of odd numbers is a perfect square). Prove $\sum_{j=1}^n (2j-1) = n^2$ for all $n \in \mathbb{N}$.

Recognition. A formula indexed by n : ordinary induction, since the $(k+1)$ -case appends one new odd number to the k -case.

Solution. Base $n = 1$: LHS = $2 \cdot 1 - 1 = 1 = 1^2$. ■ *Step:* assume $\sum_{j=1}^k (2j-1) = k^2$. Then

$$\sum_{j=1}^{k+1} (2j-1) = \underbrace{\sum_{j=1}^k (2j-1)}_{\text{hyp}} + (2(k+1)-1) = k^2 + (2k+1) = (k+1)^2,$$

the formula at $n = k+1$. By induction it holds for all n . *Check.* $n = 4$ gives $1 + 3 + 5 + 7 = 16 = 4^2$. ■ (Geometrically, adding the $(k+1)$ -st L-shaped “gnomon” of $2k+1$ unit squares turns a $k \times k$ square into a $(k+1) \times (k+1)$ one.)

Worked Example 1.2.10 (A telescoping sum). Prove $\sum_{j=1}^n \frac{1}{j(j+1)} = \frac{n}{n+1}$ for all $n \in \mathbb{N}$.

Recognition. A summation formula; ordinary induction works, though one could also telescope using $\frac{1}{j(j+1)} = \frac{1}{j} - \frac{1}{j+1}$. We use induction to drill the method.

Solution. Base $n = 1$: LHS = $\frac{1}{1 \cdot 2} = \frac{1}{2}$, RHS = $\frac{1}{2}$. ■ *Step:* assume $\sum_{j=1}^k \frac{1}{j(j+1)} = \frac{k}{k+1}$. Then

$$\sum_{j=1}^{k+1} \frac{1}{j(j+1)} = \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k(k+2)+1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2},$$

using $k(k+2)+1 = k^2 + 2k + 1 = (k+1)^2$. This is the formula at $k+1$. *Check.* $n = 3$ gives $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} = \frac{6+2+1}{12} = \frac{9}{12} = \frac{3}{4} = \frac{3}{4}$. ■ As $n \rightarrow \infty$ the sum $\rightarrow 1$, foreshadowing convergent series.

Worked Example 1.2.11 (A divisibility claim with a larger modulus). Prove $7 \mid 8^n - 1$ for all $n \in \mathbb{N}$.

Recognition. A divisibility statement; ordinary induction, factoring $8^{k+1} - 1$ so the k -case $8^k - 1$ appears.

Solution. Base $n = 1$: $8^1 - 1 = 7$, divisible by 7. ■ *Step:* assume $7 \mid 8^k - 1$, say $8^k - 1 = 7t$. Then

$$8^{k+1} - 1 = 8 \cdot 8^k - 1 = 8(8^k - 1) + 8 - 1 = 8 \cdot 7t + 7 = 7(8t + 1),$$

a multiple of 7. By induction $7 \mid 8^n - 1$ for all n . *Check.* $n = 2$: $64 - 1 = 63 = 7 \cdot 9$. ■ (The same trick proves $a - 1 \mid a^n - 1$ for any integer $a \geq 2$.)

Worked Example 1.2.12 (NON-proof: a flawed induction and its broken step). Critique the following “proof” that *every* finite set of lines in the plane, no two parallel, passes through a single common point.

Recognition. Trigger: a stated induction that is *false*; the task is to locate the precise case where the inductive step fails — the discipline behind the base-case pitfall.

Solution. The bogus argument: $P(n)$ = “any n pairwise-nonparallel lines share a point.” Base $P(2)$ is true (two nonparallel lines meet in one point). “Step”: given $n+1$ lines $\ell_1, \dots, \ell_{n+1}$, the first n share a point p (by $P(n)$) and the last n share a point q (by $P(n)$); since both groups contain $\ell_2, \dots, \ell_n$, one argues $p = q$, so all $n+1$ meet at p . The flaw is exactly at $n+1 = 3$: the two overlapping groups $\{\ell_1, \ell_2\}$ and $\{\ell_2, \ell_3\}$ share only the single line ℓ_2 , which does not pin a common point, so the inference $p = q$ collapses. The claim is plainly false (three lines forming a triangle). *Check.* The step silently needs ≥ 2 shared lines, i.e. $n \geq 3$; the chain is never validly anchored, mirroring the “all horses one color” fallacy at $n = 1 \rightarrow 2$.

Exercises (Bartle & Sherbert, §1.2).

1. Prove $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$ for all $n \in \mathbb{N}$.
2. Prove $1^3 + 2^3 + \cdots + n^3 = \left[\frac{1}{2}n(n+1)\right]^2$ for all $n \in \mathbb{N}$.
3. Prove $3 + 11 + \cdots + (8n-5) = 4n^2 - n$ for all $n \in \mathbb{N}$.
4. Prove $1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}$ for all $n \in \mathbb{N}$.
5. Prove $1^2 - 2^2 + 3^2 - \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$ for all $n \in \mathbb{N}$.
6. Prove $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.
7. Prove $5^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.
8. Prove $5^n - 4n - 1$ is divisible by 16 for all $n \in \mathbb{N}$.
9. Prove $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.
10. Conjecture a formula for $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)}$ and prove it.
11. Conjecture a formula for the sum $1 + 3 + 5 + \cdots + (2n-1)$ of the first n odd numbers and prove it by induction.
12. Prove the second version of the induction principle (Theorem 1.2.4 of B&S).
13. Prove $2^n < n!$ for all $n \geq 4$.
14. Prove $2n - 3 \leq 2^{n-2}$ for all $n \geq 5$.
15. Find all $n \in \mathbb{N}$ for which $n^2 < 2^n$, and prove your answer.
16. Find the largest natural number m such that $n^3 - n$ is divisible by m for all $n \in \mathbb{N}$; prove your assertion.
17. Prove $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$ for all $n \geq 2$.

Additional exercises.

1. Prove the *finite geometric sum* $\sum_{j=0}^n r^j = \frac{r^{n+1} - 1}{r - 1}$ for $r \neq 1$ and all $n \geq 0$.
2. (Where the base case bites.) For which $n \in \mathbb{N}$ is $2^n \geq n^3$? Prove it for all sufficiently large n by induction, and identify the exact threshold.
3. Diagnose the fallacy: “Claim: all elements of every nonempty finite set of reals are equal. Proof by induction on the size n ...”. Pinpoint the step that fails and why.
4. (Strong induction.) The Fibonacci numbers are $F_1 = F_2 = 1$, $F_{n+2} = F_{n+1} + F_n$. Prove $F_n < 2^n$ for all $n \in \mathbb{N}$.
5. Prove $\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right) = \frac{n+1}{2n}$ for all $n \geq 2$.
6. Use well-ordering directly (no induction principle) to prove: there is no natural number strictly between 0 and 1.
7. Prove $\sum_{j=1}^n j \cdot j! = (n+1)! - 1$ for all $n \in \mathbb{N}$.
8. (AM–GM warm-up, induction on a power of 2.) Prove that for nonnegative reals, $\frac{a_1 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 \cdots a_n}$ when n is a power of 2.

1.3 Finite, Countable, and Uncountable Sets

Counting by bijection

Cardinality compares sizes of sets — finite or infinite — by the existence of bijections, with no reference to “number” until the very end.

Definition 1.3.1. Sets A and B are *equinumerous* (have the same cardinality), written $A \sim B$, if there is a bijection $f : A \rightarrow B$. For $n \in \mathbb{N}$ write $\mathbb{N}_n = \{1, 2, \dots, n\}$. A set A is *finite* if $A = \emptyset$ or $A \sim \mathbb{N}_n$ for some n (and then $|A| = n$); *infinite* if not finite; *denumerable* (countably infinite) if $A \sim \mathbb{N}$; *countable* if finite or denumerable; *uncountable* if not countable.

Remark. $\sim$ is an equivalence relation: reflexive (id), symmetric (f^{-1} is a bijection), transitive (Proposition 1.1.14: a composite of bijections is a bijection). So “same size” behaves as it should.

A hallmark of *infinite* sets, impossible for finite ones, is that they biject with proper subsets of themselves.

Worked Example 1.3.2 ($\mathbb{N}$ is equinumerous with a proper subset). Exhibit a bijection of $\mathbb{N}$ onto the even numbers $2\mathbb{N} = \{2, 4, 6, \dots\}$, and onto $\mathbb{N}$ minus a finite set.

Recognition. Trigger: “show an infinite set matches a proper subset.” Build an explicit formula and check both directions.

Solution. $f(n) = 2n$ is a bijection $\mathbb{N} \rightarrow 2\mathbb{N}$ (inverse $m \mapsto m/2$). Likewise $g(n) = n + 13$ bijects $\mathbb{N}$ onto $\{14, 15, 16, \dots\}$, the integers greater than 13. *Check.* f misses no even number and repeats none; a finite set could never biject with a proper subset, so this property *characterizes* infinity (Dedekind’s definition).

Theorem 1.3.3 (Subsets and images of countable sets). (a) *Every subset of a countable set is countable.* (b) *If A is countable and $f : A \rightarrow B$ is a surjection, then B is countable; equivalently, the image of a countable set under any map is countable.* (c) *A is countable iff there is an injection $A \rightarrow \mathbb{N}$, iff $A = \emptyset$ or there is a surjection $\mathbb{N} \rightarrow A$.*

Proof sketch. (a) If $A \subseteq \mathbb{N}$ is infinite, define $f : \mathbb{N} \rightarrow A$ by sending 1 to the least element of A (well-ordering), then recursively $f(n+1)$ to the least element of A above $f(n)$; this is a strictly increasing bijection $\mathbb{N} \rightarrow A$. A general countable A pulls back to $\mathbb{N}$. (b),(c) follow by composing with the enumerating bijection and using that a surjection $\mathbb{N} \rightarrow B$ lets you pick, for each b , the least preimage, yielding an injection $B \rightarrow \mathbb{N}$. $\square$

Building blocks: $\mathbb{N} \times \mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ are countable

Theorem 1.3.4. $\mathbb{N} \times \mathbb{N}$ is denumerable.

Proof. List the pairs (m, n) along the *diagonals* $m + n = \text{const}$, as drawn below. The explicit Cantor pairing $h(m, n) = \frac{1}{2}(m+n-2)(m+n-1) + m$ is a bijection $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. Alternatively $(m, n) \mapsto 2^m 3^n$ is an *injection* $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ (unique factorization), which already gives countability by Theorem 1.3.3(c). $\square$

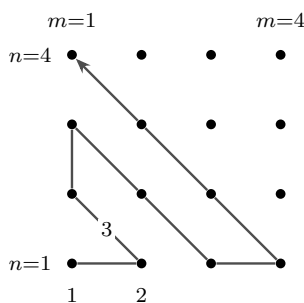


Figure 1.1. Enumerating $\mathbb{N} \times \mathbb{N}$ along anti-diagonals $m + n = \text{const}$: each diagonal is finite, so every pair is reached at a finite stage.

Corollary 1.3.5. $\mathbb{Z}$ and $\mathbb{Q}$ are denumerable, and a countable union of countable sets is countable.

Proof. $\mathbb{Z}$: interleave, $0, 1, -1, 2, -2, \dots$, i.e. the bijection $\mathbb{N} \rightarrow \mathbb{Z}$ sending odd $2k - 1 \mapsto k - 1$... more cleanly $n \mapsto n/2$ if n even, $-(n - 1)/2$ if n odd. *Countable union*: let $A_1, A_2, \dots$ be countable, enumerate each $A_i = \{a_{i1}, a_{i2}, \dots\}$, and map $\bigcup_i A_i$ into $\mathbb{N} \times \mathbb{N}$ by $a_{ij} \mapsto (i, j)$; this is (almost) an injection into a countable set, so the union is countable. $\mathbb{Q}$: the map $\mathbb{Q}_{>0} \rightarrow \mathbb{N} \times \mathbb{N}$, $p/q \mapsto (p, q)$ in lowest terms, is an injection, so $\mathbb{Q}_{>0}$ is countable; then $\mathbb{Q} = \mathbb{Q}_{>0} \cup \{0\} \cup \mathbb{Q}_{<0}$ is a finite union of countable sets. $\square$

Caution. “Countable union of countable sets is countable” uses a choice of enumeration for *each* A_i at once — a (countable) use of the Axiom of Choice. Harmless here, but worth knowing it is not pure bookkeeping. Note also $\bigcup$ of countably many *finite* sets can be infinite (e.g. $A_n = \{n\}$), and that is fine: it is still countable.

The diagonal argument: $\mathbb{R}$ is uncountable

The decisive break: not all infinite sets have the same size. Cantor’s diagonal method shows the unit interval — hence $\mathbb{R}$ — cannot be listed.

Theorem 1.3.6 (Cantor). *The interval $[0, 1)$ is uncountable; hence $\mathbb{R}$ is uncountable.*

Proof. Suppose, for contradiction, that $[0, 1)$ were countable, say enumerated as $x_1, x_2, x_3, \dots$. Write each in decimal, $x_i = 0.d_{i1}d_{i2}d_{i3}\dots$, choosing the non-terminating form to make the expansion unique (forbid trailing 9s). Define a new digit string $y = 0.e_1e_2e_3\dots$ by

$$e_i = \begin{cases} 5, & d_{ii} \neq 5, \\ 4, & d_{ii} = 5. \end{cases}$$

Then $y \in [0, 1)$, and y uses only digits 4, 5, so it has no trailing 9s and its expansion is unique. For every i , y differs from x_i in the i -th digit ($e_i \neq d_{ii}$), so $y \neq x_i$. Thus y is in $[0, 1)$ but not in the list — contradicting that the list was *all* of $[0, 1)$. Hence $[0, 1)$ is uncountable. Since $[0, 1) \subseteq \mathbb{R}$, $\mathbb{R}$ is uncountable by Theorem 1.3.3(a) (contrapositive). $\square$

$$\begin{array}{ccccccc} x_1 = 0. & \boxed{d_{11}} & d_{12} & d_{13} & d_{14} & \cdots & \\ x_2 = 0. & d_{21} & \boxed{d_{22}} & d_{23} & d_{24} & \cdots & \\ x_3 = 0. & d_{31} & d_{32} & \boxed{d_{33}} & d_{34} & \cdots & \\ x_4 = 0. & d_{41} & d_{42} & d_{43} & \boxed{d_{44}} & \cdots & \\ \vdots & & & & & \ddots & \end{array} \quad \begin{array}{l} y = 0.e_1e_2e_3e_4\dots \\ \text{with } e_i \neq d_{ii}: \\ y \text{ differs from every } x_i \end{array}$$

Figure 1.2. Change every boxed diagonal digit; the result agrees with no row.

Theorem 1.3.7 (Cantor’s theorem). *For every set S , there is no surjection $S \rightarrow \mathcal{P}(S)$; in particular $S \not\sim \mathcal{P}(S)$. Thus $|S| < |\mathcal{P}(S)|$, and there is no largest cardinality.*

Proof. Let $f : S \rightarrow \mathcal{P}(S)$ be any map. Form the “diagonal” set $D = \{x \in S : x \notin f(x)\} \subseteq S$. If $D = f(a)$ for some a , ask whether $a \in D$: by definition of D , $a \in D \iff a \notin f(a) = D$, a contradiction. So D is not in the range of f , and f is not surjective. The injection $x \mapsto \{x\}$ shows $|S| \leq |\mathcal{P}(S)|$, so the inequality is strict. $\square$

Remark. For finite S with $|S| = n$, $|\mathcal{P}(S)| = 2^n > n$, the concrete shadow of Cantor’s theorem (Exercise: prove $|\mathcal{P}(S)| = 2^n$ by induction). For $S = \mathbb{N}$ one gets $|\mathcal{P}(\mathbb{N})| = |\mathbb{R}| = 2^{\aleph_0} > \aleph_0$: the reals are the power set of the naturals in disguise.

Theorem 1.3.8 (Cantor–Schroeder–Bernstein). *If there exist injections $f : A \rightarrow B$ and $g : B \rightarrow A$, then $A \sim B$.*

This powerful tool (proof omitted; see the exercises) lets you prove $A \sim B$ by two *one-way* comparisons instead of constructing a bijection outright.

Worked Example 1.3.9 ($(0, 1) \sim [0, 1] \sim \mathbb{R}$ via Schroeder–Bernstein). Show the open interval, closed interval, and whole line all have the same cardinality.

Recognition. “Prove equinumerous” where an outright bijection (e.g. $(0, 1) \rightarrow [0, 1]$) is fiddly at endpoints — the cue for Cantor–Schroeder–Bernstein, or a clever direct map.

Solution. $(0, 1) \hookrightarrow [0, 1]$ by inclusion (injective), and $[0, 1] \hookrightarrow (0, 1)$ by $x \mapsto \frac{x+1}{3} \in [\frac{1}{3}, \frac{2}{3}] \subset (0, 1)$ (injective). By Theorem 1.3.8, $(0, 1) \sim [0, 1]$. And $\tan(\pi(x - \frac{1}{2}))$ is an explicit bijection $(0, 1) \rightarrow \mathbb{R}$ (strictly increasing, onto), so all three are equinumerous, of cardinality $2^{\aleph_0}$. *Check.* Consistent with Worked Example 1.3.6-style intuition — adding or deleting the two endpoints cannot change an uncountable cardinality.

Worked Example 1.3.10 (The algebraic numbers are countable). A real number is *algebraic* if it is a root of a nonzero polynomial with integer coefficients. Show the set $\mathbb{A}$ of algebraic numbers is countable (hence “most” reals are transcendental).

Recognition. “Show a set built from integers is countable”: organize it as a countable union of finite sets, indexed by something countable.

Solution. For each $n \geq 1$ the integer polynomials of degree $\leq n$ with $|a_0| + \cdots + |a_n| \leq n$ form a *finite* set, and each has $\leq n$ roots. The set of all integer polynomials is a countable union (over n) of finite sets, hence countable (Corollary 1.3.5); the roots form a countable union of finite sets, hence $\mathbb{A}$ is countable. *Check.* $\mathbb{Q} \subseteq \mathbb{A}$ (each p/q is a root of $qx - p$), and $\sqrt{2}, \sqrt[3]{5} \in \mathbb{A}$, yet $\mathbb{A}$ is still only countable; since $\mathbb{R}$ is uncountable, transcendental numbers exist — indeed they are the overwhelming majority, though exhibiting one (e.g. π) is hard.

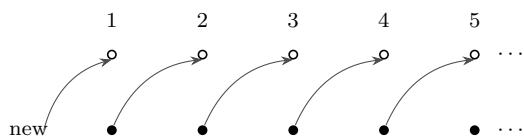
Caution. Countability is not about “never running out” but about *being listable in a single infinite sequence*. $\mathbb{Q}$ feels far bigger than $\mathbb{N}$ (it is dense!), yet it is countable; $\mathbb{R}$ feels only “a bit” bigger, yet it is a strictly larger infinity. Density and cardinality are independent notions — do not conflate them.

Further worked examples

Worked Example 1.3.11 (Hilbert’s hotel: absorbing a new guest). A hotel with rooms $1, 2, 3, \dots$ is full (one guest per room). Show a single new guest can be accommodated without anyone leaving, and that countably many new guests can be too.

Recognition. Trigger: “an infinite set matches itself with room to spare.” Build explicit shifting bijections, as in Worked Example 1.3.1-style constructions.

Solution. Move the occupant of room n to room $n + 1$ for every n ; this is the bijection $s : \mathbb{N} \rightarrow \{2, 3, 4, \dots\}$, $s(n) = n + 1$, freeing room 1 for the newcomer. For countably many new guests, move room n ’s occupant to room $2n$ (bijection $\mathbb{N} \rightarrow 2\mathbb{N}$), vacating all odd rooms; seat new guest k in room $2k - 1$. *Check.* The map $n \mapsto 2n$ is injective and hits every even number, and $\{2n\} \cup \{2k - 1\} = \mathbb{N}$ partitions the rooms, so no room is double-booked and none is wasted.



guest in room n shifts to room $n + 1$; room 1 opens up

Worked Example 1.3.12 (The integers are countable, with an explicit formula). Exhibit and verify a bijection $h : \mathbb{N} \rightarrow \mathbb{Z}$ in closed form.

Recognition. Trigger: “produce an explicit enumeration of $\mathbb{Z}$.” Interleave nonnegatives and negatives, then check injectivity and surjectivity by the method.

Solution. Define $h(n) = \frac{n}{2}$ if n is even and $h(n) = -\frac{n-1}{2}$ if n is odd. So h lists $0, 1, -1, 2, -2, 3, -3, \dots$ as $n = 1, 2, 3, 4, 5, 6, 7, \dots$. Surjective: an integer $m > 0$ is $h(2m)$, $m < 0$ is $h(1 - 2m)$, and $0 = h(1)$. Injective:

even and odd n produce nonnegative and negative values respectively, so the two cases never collide, and within each case h is one-to-one. *Check.* $h(6) = 3$ and $h(7) = -3$; the inverse sends $3 \mapsto 6$, $-3 \mapsto 7$, recovering n . Hence $\mathbb{Z} \sim \mathbb{N}$ is denumerable.

Worked Example 1.3.13 (A countable union of countable sets, made concrete). Show that $\mathbb{Q} \cap (0, 1)$ is countable by listing it as a union of finite sets.

Recognition. Trigger: “show a set built from integers is countable” — organize as a countable union of finite sets (Corollary 1.3.5).

Solution. For each $q \geq 2$ let $F_q = \{p/q : 1 \leq p < q, \gcd(p, q) = 1\}$, the fractions in $(0, 1)$ with denominator exactly q in lowest terms. Each F_q is *finite* (at most $q - 1$ elements), and every rational in $(0, 1)$ has a unique lowest-terms form, so $\mathbb{Q} \cap (0, 1) = \bigcup_{q \geq 2} F_q$ is a countable union of finite sets, hence countable. *Check.* $F_2 = \{\frac{1}{2}\}$, $F_3 = \{\frac{1}{3}, \frac{2}{3}\}$, $F_4 = \{\frac{1}{4}, \frac{3}{4}\}$ (not $\frac{2}{4}$, already counted), $F_5 = \{\frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}\}$ — the Farey-style listing visits each rational exactly once.

Worked Example 1.3.14 (NON-example: the irrationals are NOT countable). Deduce that the set of irrational numbers in $[0, 1)$ is uncountable.

Recognition. Trigger: a cardinality claim provable by “a countable set cannot fill an uncountable one.” Use that $\mathbb{R}$ uncountable plus $\mathbb{Q}$ countable, via the union/closure theorems.

Solution. Write $[0, 1) = (\mathbb{Q} \cap [0, 1)) \cup (I \cap [0, 1))$ where I is the irrationals. If $I \cap [0, 1)$ were countable, then $[0, 1)$ would be a union of two countable sets ($\mathbb{Q} \cap [0, 1)$ is countable as a subset of $\mathbb{Q}$), hence countable by Corollary 1.3.5 — contradicting Cantor’s theorem (Theorem 1.3.6). So $I \cap [0, 1)$ is uncountable. *Check.* This is a clean “uncountable minus countable is uncountable” instance: removing the countably many rationals from an uncountable set cannot shrink it to countable, since otherwise the whole would be a countable union.

Exercises (Bartle & Sherbert, §1.3).

1. Prove a nonempty set T_1 is finite iff there is a bijection from T_1 onto a finite set T_2 .
2. Prove parts (b) and (c) of the uniqueness theorem $\mathbb{N}_m \sim \mathbb{N}_n \implies m = n$.
3. Let $S = \{1, 2\}$, $T = \{a, b, c\}$. (a) How many injections $S \rightarrow T$ are there? (b) How many surjections $T \rightarrow S$?
4. Exhibit a bijection between $\mathbb{N}$ and the set of all odd integers greater than 13.
5. Give an explicit formula for a bijection $f : \mathbb{N} \rightarrow \mathbb{Z}$.
6. Exhibit a bijection between $\mathbb{N}$ and a proper subset of itself.
7. Prove a set T_1 is denumerable iff there is a bijection from T_1 onto a denumerable set T_2 .
8. Give an example of a countable collection of finite sets whose union is not finite.
9. Prove in detail that if S and T are denumerable, then $S \cup T$ is denumerable.
10. Using the diagonal counting of $\mathbb{N} \times \mathbb{N}$: (a) find the number assigned to the 6th point down the 9th diagonal; (b) if $h(m, 3) = 19$, find m .
11. Determine $|\mathcal{P}(S)|$ for (a) $S = \{1, 2\}$; (b) $S = \{1, 2, 3\}$; (c) $S = \{1, 2, 3, 4\}$ (include $\emptyset$ and S).
12. Use induction to prove that if $|S| = n$ then $|\mathcal{P}(S)| = 2^n$.
13. Prove that the collection of all *finite* subsets of $\mathbb{N}$ is countable.

Exercises (Rudin, Ch.2 – countability).

1. Prove that the empty set is a subset of every set.
2. A complex number z is *algebraic* if there are integers $a_0, \dots, a_n$, not all zero, with $a_0 z^n + a_1 z^{n-1} + \dots + a_n = 0$. Prove the set of all algebraic numbers is countable. (Hint: for each N there are finitely many equations with $n + |a_0| + |a_1| + \dots + |a_n| = N$.)

3. Prove that there exist real numbers that are not algebraic.
4. Is the set of all irrational real numbers countable? Justify.

Additional exercises.

1. Prove directly (no Schroeder–Bernstein) that $\mathbb{N} \sim \mathbb{Z}$ by giving a single formula for the bijection and its inverse.
2. Run the diagonal argument on *binary* sequences: show the set $2^{\mathbb{N}}$ of all functions $\mathbb{N} \rightarrow \{0, 1\}$ is uncountable, and explain the one subtlety (dyadic rationals with two expansions) that the digit choice $\{4, 5\}$ avoided for decimals.
3. Prove $\mathcal{P}(\mathbb{N}) \sim \mathbb{R}$. (Map a subset $A \subseteq \mathbb{N}$ to the real with binary digits $\mathbf{1}_A$; repair the double-counting with Schroeder–Bernstein.)
4. (Strict middle.) Show that for any sets, $A \hookrightarrow B$ and a surjection $B \rightarrow A$ together force $A \sim B$, citing the appropriate theorem.
5. Show the set of all *eventually-zero* sequences of natural numbers is countable, but the set of *all* sequences of naturals is uncountable.
6. Prove that a countable union of countable sets is countable, being explicit about where a choice of enumerations is used.
7. Give an example of a strictly increasing chain $A_1 \subsetneq A_2 \subsetneq \cdots$ of countable sets whose union is uncountable, or prove no such chain exists.
8. (Visual.) Reproduce the anti-diagonal enumeration figure for $\mathbb{N} \times \mathbb{N}$ and mark the positions 1 through 10; then read off $h(3, 2)$ and verify it against the Cantor pairing formula $h(m, n) = \frac{1}{2}(m+n-2)(m+n-1) + m$.
9. Prove there is no set of “all sets” (Cantor’s theorem applied to a hypothetical universal set).

Chapter 2

The Real and Complex Number Systems

2.1 Ordered Fields: Algebraic and Order Properties of $\mathbb{R}$

Fields and the field axioms

The real numbers $\mathbb{R}$ are not built by listing properties at random: they are the unique *complete ordered field*. We unpack that phrase in three stages — *field* (this section), *ordered* (this section), and *complete* (§2.3). Everything later in analysis rests on these axioms, so we state them once, precisely.

Definition 2.1.1. A *field* is a set $\mathbb{F}$ with two operations, addition $+$ and multiplication $\cdot$, satisfying:
Addition (A). (A1) $x + y \in \mathbb{F}$ and $x + y = y + x$; (A2) $(x + y) + z = x + (y + z)$; (A3) there is $0 \in \mathbb{F}$ with $x + 0 = x$ for all x ; (A4) each x has $-x \in \mathbb{F}$ with $x + (-x) = 0$. **Multiplication (M).** (M1) $xy \in \mathbb{F}$ and $xy = yx$; (M2) $(xy)z = x(yz)$; (M3) there is $1 \in \mathbb{F}$, $1 \neq 0$, with $1 \cdot x = x$ for all x ; (M4) each $x \neq 0$ has $1/x \in \mathbb{F}$ with $x \cdot (1/x) = 1$. **Distributive law (D).** $x(y + z) = xy + xz$.

The axioms come in two parallel packages, one for $+$ and one for $\cdot$, glued together by (D). The phrase “ $1 \neq 0$ ” rules out the degenerate one-element “field.” Rudin’s Definition 1.12 and B&S’s list 2.1.1 are identical to this; we follow Rudin’s labels (A), (M), (D) throughout. From these axioms alone, every familiar rule of algebra follows — no extra assumption is allowed.

Proposition 2.1.2 (Cancellation and uniqueness, Rudin 1.14–1.16). *In any field: (a) $x + y = x + z \implies y = z$; in particular 0 and each $-x$ are unique. (b) $x \cdot 0 = 0$. (c) if $x \neq 0$ and $xy = xz$ then $y = z$; and $xy = 0 \implies x = 0$ or $y = 0$. (d) $-(-x) = x$, $(-x)y = -(xy) = x(-y)$, and $(-x)(-y) = xy$.*

Proof. (a) Add $-x$ to both sides and use (A2),(A4),(A3): $y = 0 + y = (-x + x) + y = -x + (x + y) = -x + (x + z) = z$. Uniqueness of 0: if $0'$ also satisfies (A3) then $0 = 0 + 0' = 0'$. Uniqueness of $-x$: if $x + y = 0 = x + y'$ then $y = y'$ by cancellation. (b) $x \cdot 0 + x \cdot 0 = x(0 + 0) = x \cdot 0$, so by (a) (additive cancellation against $x \cdot 0$) we get $x \cdot 0 = 0$. (c) Multiply $xy = xz$ by $1/x$. If $xy = 0$ and $x \neq 0$ then $y = (1/x)(xy) = (1/x) \cdot 0 = 0$ by (b). (d) $-(-x) = x$ because x is the additive inverse of $-x$. And $(-x)y + xy = (-x + x)y = 0 \cdot y = 0$, so $(-x)y = -(xy)$; the rest is the same trick applied twice. $\square$

Remark. Part (c) — “no zero divisors” — is what makes the quadratic-factoring trick legal: $(x - 1)(x + 2) = 0$ forces $x = 1$ or $x = -2$. It fails in rings like $\mathbb{Z}/6$ ($2 \cdot 3 = 0$), so it is genuinely a field fact, not a triviality.

Ordered sets and ordered fields

Definition 2.1.3 (Order, Rudin 1.5). An *order* on a set S is a relation $<$ such that (i) (*trichotomy*) for $x, y \in S$ exactly one of $x < y$, $x = y$, $y < x$ holds; (ii) (*transitivity*) $x < y$ and $y < z$ imply $x < z$. We write $x \leq y$ for “ $x < y$ or $x = y$ ”. An *ordered set* is a set with an order.

Definition 2.1.4 (Ordered field, Rudin 1.17). An *ordered field* is a field $\mathbb{F}$ with an order $<$ compatible with the operations: (O1) $y < z \implies x + y < x + z$ for all x ; (O2) $x > 0$ and $y > 0 \implies xy > 0$. We call $x > 0$ *positive* and $x < 0$ *negative*.

B&S package the same content as the *Order Properties* (2.1.5): there is a set P of “positive” elements closed under $+$ and $\cdot$, with trichotomy “ $a \in P$, $a = 0$, or $-a \in P$.” Setting $a < b \iff b - a \in P$ recovers Definition 2.1.4; the two formulations are interchangeable. The next theorem is the toolbox of inequality rules every analyst uses without thinking.

Theorem 2.1.5 (Order rules, Rudin 1.18 / B&S 2.1.7). *In an ordered field: (a) $x > 0 \iff -x < 0$; (b) $x > 0$ and $y < z \implies xy < xz$; and $x < 0$, $y < z \implies xy > xz$ (sign flips); (c) $x \neq 0 \implies x^2 > 0$; in particular $1 = 1^2 > 0$; (d) $0 < x < y \implies 0 < 1/y < 1/x$.*

Proof. **(a)** Add $-x$ to $x > 0$ using (O1): $0 = x + (-x) > 0 + (-x) = -x$. **(b)** $y < z$ gives $z - y > 0$ (add $-y$). If $x > 0$, (O2) gives $x(z - y) > 0$, i.e. $xz - xy > 0$, i.e. $xy < xz$. If $x < 0$ then $-x > 0$ by (a), so $(-x)(z - y) > 0$, i.e. $xy - xz > 0$, i.e. $xz < xy$. **(c)** If $x > 0$, (O2) gives $x^2 > 0$. If $x < 0$ then $-x > 0$ and $x^2 = (-x)(-x) > 0$ by Proposition 2.1.2(d). Since $1 \neq 0$, $1 = 1 \cdot 1 > 0$. **(d)** If $x > 0$ then $1/x > 0$ (else $1/x < 0$ gives $1 = x(1/x) < 0$, contradicting (c)). From $x < y$ multiply by the positive $\frac{1}{xy}$: $\frac{1}{y} < \frac{1}{x}$. $\square$

Caution. The sign-flip in (b) is the single most common error in elementary analysis: multiplying or dividing an inequality by a *negative* quantity reverses it. “ $-2x < 6$ ” becomes “ $x > -3$,” not “ $x < -3$.” Likewise $a < b$ does *not* give $1/a < 1/b$ in general — only when a, b have the same sign (Theorem 2.1.5(d)). When you cannot pin the sign of a factor, split into cases.

Theorem 2.1.6 (Squeeze-to-zero, B&S 2.1.9). *If $a \in \mathbb{R}$ satisfies $0 \leq a < \varepsilon$ for every $\varepsilon > 0$, then $a = 0$.*

Proof. Suppose $a > 0$. Take $\varepsilon_0 = \frac{1}{2}a$; then $\varepsilon_0 > 0$, yet $a < \varepsilon_0 = \frac{1}{2}a$ forces $\frac{1}{2}a < 0$, contradicting $a > 0$. So $a = 0$. $\square$

This tiny theorem is the seed of every “ ε -argument”: to prove two numbers are equal you show their distance is below *every* positive bound. We will reuse it constantly.

Method (Solving / proving a real inequality). *Recognition trigger.* You must find all x satisfying an inequality, or prove one inequality from others.

1. **Move everything to one side:** rewrite as $(\cdot) > 0$ or $(\cdot) < 0$.
2. **Factor.** A product is positive iff its factors share a sign (Theorem 2.1.5, B&S 2.1.10): $pq > 0 \iff (p > 0, q > 0) \text{ or } (p < 0, q < 0)$.
3. **Case-split** on the sign of each factor; intersect the resulting conditions.
4. For a *rational* inequality $\frac{p}{q} > 0$, the sign is that of pq (never multiply across by q of unknown sign).
5. To *prove* an inequality, reduce it to a known nonnegative quantity — usually a square $(\cdot)^2 \geq 0$ (Theorem 2.1.5(c)).

Worked Example 2.1.7 (A linear inequality). Find all x with $2x + 3 \leq 6$.

Recognition. Linear; isolate x but watch the divisor sign (here positive, so no flip).

Solution. $2x + 3 \leq 6 \iff 2x \leq 3 \iff x \leq \frac{3}{2}$. So the solution set is $\{x \in \mathbb{R} : x \leq \frac{3}{2}\} = (-\infty, \frac{3}{2}]$. *Check.* $x = \frac{3}{2}$ gives $6 \leq 6$; $x = 2$ gives $7 \leq 6$, false, correctly excluded.

Worked Example 2.1.8 (A quadratic inequality by factoring). Find all x with $x^2 + x > 2$.

Recognition. Quadratic; move to one side and factor, then split on factor signs.

Solution. $x^2 + x - 2 > 0 \iff (x - 1)(x + 2) > 0$. The product is positive in two cases: (i) $x - 1 > 0$ and $x + 2 > 0$, i.e. $x > 1$; (ii) $x - 1 < 0$ and $x + 2 < 0$, i.e. $x < -2$. Hence the solution is $(-\infty, -2) \cup (1, \infty)$. *Check.* $x = 0$ gives $0 > 2$, false, and 0 lies in the excluded middle $[-2, 1]$. $\blacksquare$

Worked Example 2.1.9 (A rational inequality — do not cross-multiply blindly). Find all x with $\frac{2x+1}{x+2} < 1$.

Recognition. Rational; subtract 1 and combine, then read the sign off the numerator times denominator.

Solution. $\frac{2x+1}{x+2} - 1 < 0 \iff \frac{x-1}{x+2} < 0$. A quotient is negative iff numerator and denominator have opposite signs: (i) $x-1 < 0$, $x+2 > 0$ gives $-2 < x < 1$; (ii) $x-1 > 0$, $x+2 < 0$ is impossible. So the solution is $(-2, 1)$. *Check.* $x = 0$ gives $\frac{1}{2} < 1$ ■; $x = -2$ is excluded (division by 0); $x = 2$ gives $\frac{5}{3} < 1$ false. ■

Worked Example 2.1.10 (The AM–GM inequality for two terms (B&S 2.1.13)). For $a, b \geq 0$ prove $\sqrt{ab} \leq \frac{1}{2}(a+b)$, with equality iff $a = b$.

Recognition. “Prove an inequality”: reduce to a square ≥ 0 .

Solution. For $a, b \geq 0$, $(\sqrt{a} - \sqrt{b})^2 \geq 0$ expands to $a - 2\sqrt{ab} + b \geq 0$, i.e. $\sqrt{ab} \leq \frac{1}{2}(a+b)$. Equality holds iff $(\sqrt{a} - \sqrt{b})^2 = 0$ iff $\sqrt{a} = \sqrt{b}$ iff $a = b$. *Check.* $a = 4, b = 1$: $\sqrt{4} = 2 \leq \frac{5}{2}$. ■ (The general n -term AM–GM, $(a_1 \cdots a_n)^{1/n} \leq \frac{1}{n} \sum a_i$, follows by induction or convexity.)

Worked Example 2.1.11 (Bernoulli’s inequality (B&S 2.1.13(c), Rudin uses it repeatedly)). For $x > -1$ and $n \in \mathbb{N}$, prove $(1+x)^n \geq 1+nx$.

Recognition. Inequality indexed by n ; induct, using $1+x > 0$ to keep the direction.

Solution. Base $n = 1$: equality. Step: assume $(1+x)^k \geq 1+kx$. Since $1+x > 0$ we may multiply: $(1+x)^{k+1} \geq (1+kx)(1+x) = 1+(k+1)x+kx^2 \geq 1+(k+1)x$, using $kx^2 \geq 0$. By induction it holds for all n . *Check.* $x = 1, n = 3$: $8 \geq 4$. ■

$\mathbb{Q}$ is an ordered field — but a defective one

The rationals $\mathbb{Q}$ satisfy *every* axiom above: $\mathbb{Q}$ is an ordered field. So the field and order axioms cannot, by themselves, capture $\mathbb{R}$. The defect is exposed by a single missing number.

Theorem 2.1.12 (No rational square root of 2, Rudin 1.1 / B&S 2.1.4). *There is no rational number r with $r^2 = 2$.*

Proof. Suppose $r = p/q$ in lowest terms (so p, q are not both even) and $p^2 = 2q^2$. Then p^2 is even, so p is even (the square of an odd number is odd), say $p = 2m$. Then $4m^2 = 2q^2$, i.e. $q^2 = 2m^2$, so q^2 is even, so q is even. But then p and q are both even, contradicting “lowest terms.” Hence no such r exists. □

Remark. Rudin sharpens this to a *topological* gap: the set $A = \{p \in \mathbb{Q} : p > 0, p^2 < 2\}$ has no largest element and $B = \{p \in \mathbb{Q} : p > 0, p^2 > 2\}$ no smallest, because the map $p \mapsto p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}$ pushes any $p \in A$ to a larger element of A and any $p \in B$ to a smaller element of B . So $\mathbb{Q}$ has a “hole” exactly where $\sqrt{2}$ should be — A is bounded above in $\mathbb{Q}$ yet has no least upper bound *in* $\mathbb{Q}$. Repairing this hole is the completeness axiom of §2.3.

The same proof, with “even” replaced by “divisible by p ,” shows $\sqrt{n} \notin \mathbb{Q}$ whenever n is not a perfect square (Exercises). So the rationals are riddled with holes; the reals fill every one.

Further worked examples

Worked Example 2.1.13 (Deriving a field identity from the axioms alone). In any field, prove the binomial-style identity $(a+b)(a-b) = a^2 - b^2$ and the cancellation fact $a^2 = b^2 \implies a = b$ or $a = -b$.

Recognition. “Prove an algebraic law from the axioms”: use only distributivity (D), commutativity, and Proposition 2.1.2(d).

Solution. By (D) twice, $(a+b)(a-b) = a(a-b) + b(a-b) = a^2 - ab + ba - b^2$. Commutativity (M1) gives $ba = ab$, so $-ab + ba = 0$ and $(a+b)(a-b) = a^2 - b^2$. Now if $a^2 = b^2$ then $a^2 - b^2 = 0$, i.e. $(a+b)(a-b) = 0$; by the no-zero-divisor law (Proposition 2.1.2(c)) one factor vanishes, so $a = -b$ or $a = b$. *Check.* In $\mathbb{R}$, $a = 3, b = -3$ have $a^2 = b^2 = 9$ and indeed $a = -b$; the factoring $(3 + (-3))(3 - (-3)) = 0 \cdot 6 = 0$. ■

Worked Example 2.1.14 (NON-field: $\mathbb{Z}$ and $\mathbb{Z}/6$ fail axiom (M4)). Show the integers $\mathbb{Z}$ are not a field, and that $\mathbb{Z}/6$ (integers mod 6) fails even the no-zero-divisor law.

Recognition. “Decide whether a structure is a field”: test the multiplicative-inverse axiom (M4) and the cancellation consequence.

Solution. In $\mathbb{Z}$, the element $2 \neq 0$ has no integer multiplicative inverse: $2x = 1$ has no solution in $\mathbb{Z}$ (only $x = \frac{1}{2} \notin \mathbb{Z}$), so (M4) fails and $\mathbb{Z}$ is a commutative ring but not a field. In $\mathbb{Z}/6$, $2 \cdot 3 = 6 \equiv 0$ with $2 \neq 0$ and $3 \neq 0$, so $\mathbb{Z}/6$ has *zero divisors*, violating Proposition 2.1.2(c); hence it is not a field either (here 2 has no inverse, since $2x \equiv 1 \pmod{6}$ is unsolvable as $2x$ is always even). *Check.* By contrast $\mathbb{Z}/5$ is a field — every nonzero residue has an inverse ($2 \cdot 3 = 6 \equiv 1$, $4 \cdot 4 = 16 \equiv 1$), and 5 being prime forbids zero divisors. ■

Worked Example 2.1.15 (An inequality with a parameter of unknown sign). Find all real x satisfying $\frac{1}{x} < x$.

Recognition. Rational inequality where the denominator x changes sign; never cross-multiply by x blindly — move to one side and factor (Method above).

Solution. $\frac{1}{x} - x < 0 \iff \frac{1-x^2}{x} < 0 \iff \frac{(1-x)(1+x)}{x} < 0$. The expression is a ratio of the three linear factors $1-x$, $1+x$, x , with sign changes at $x = -1, 0, 1$. Testing one point in each region: $x = -2 \rightarrow \frac{(3)(-1)}{-2} = \frac{3}{2} > 0$; $x = -\frac{1}{2} \rightarrow \frac{(3/2)(1/2)}{-1/2} < 0$; $x = \frac{1}{2} \rightarrow \frac{(1/2)(3/2)}{1/2} > 0$; $x = 2 \rightarrow \frac{(-1)(3)}{2} < 0$. So the solution is $(-1, 0) \cup (1, \infty)$. *Check.* $x = -\frac{1}{2}$: $\frac{1}{-1/2} = -2 < -\frac{1}{2}$ ■; $x = 2$: $\frac{1}{2} < 2$ ■; $x = \frac{1}{2}$ excluded since $2 \not< \frac{1}{2}$. ■

Exercises (Bartle & Sherbert, §2.1).

1. If $a, b \in \mathbb{R}$, prove: (a) $a + b = 0 \implies b = -a$; (b) $-(-a) = a$; (c) $(-1)a = -a$; (d) $(-1)(-1) = 1$.
2. Prove that if $a, b \in \mathbb{R}$: (a) $-(a + b) = (-a) + (-b)$; (b) $(-a) \cdot (-b) = a \cdot b$; (c) $1/(-a) = -(1/a)$ ($a \neq 0$); (d) $-(a/b) = (-a)/b$ ($b \neq 0$).
3. Solve, justifying each step by an order or field property: (a) $2x + 5 = 8$; (b) $x^2 = 2x$; (c) $x^2 - 1 = 3$; (d) $(x - 1)(x + 2) = 0$.
4. If $a \in \mathbb{R}$ satisfies $a \cdot a = a$, prove that $a = 0$ or $a = 1$.
5. If $a \neq 0$ and $b \neq 0$, show $1/(ab) = (1/a)(1/b)$.
6. Use the argument proving no rational has square 2 to show there is no rational s with $s^2 = 6$.
7. Modify that proof to show there is no rational t with $t^2 = 3$.
8. (a) Show that if x, y are rational then $x + y$ and xy are rational. (b) Prove that if x is rational and y is irrational, then $x + y$ is irrational, and if also $x \neq 0$ then xy is irrational.
9. Let $K = \{s + t\sqrt{2} : s, t \in \mathbb{Q}\}$. Show (a) if $x_1, x_2 \in K$ then $x_1 + x_2 \in K$ and $x_1x_2 \in K$; (b) if $x \neq 0$ and $x \in K$ then $1/x \in K$. (Thus K is a subfield of $\mathbb{R}$ with $\mathbb{Q} \subsetneq K \subsetneq \mathbb{R}$.)
10. (a) If $a < b$ and $c \leq d$, prove $a + c < b + d$. (b) If $0 < a < b$ and $0 \leq c \leq d$, prove $0 \leq ac \leq bd$.
11. (a) Show that if $a > 0$ then $1/a > 0$ and $1/(1/a) = a$. (b) Show that if $a < b$ then $a < \frac{1}{2}(a + b) < b$.
12. Let a, b, c, d satisfy $0 < a < b$ and $c < d < 0$. Give an example where $ac < bd$ and one where $bd < ac$.
13. If $a, b \in \mathbb{R}$, show $a^2 + b^2 = 0$ iff $a = 0$ and $b = 0$.
14. If $0 \leq a < b$, show $a^2 \leq ab < b^2$. Show by example that $a^2 < ab < b^2$ need not hold.
15. If $0 < a < b$, show (a) $a < \sqrt{ab} < b$ and (b) $1/b < 1/a$.
16. Find all $x \in \mathbb{R}$ satisfying: (a) $x^2 > 3x + 4$; (b) $1 < x^2 < 4$; (c) $1/x < x$; (d) $1/x < x^2$.
17. Prove the strengthened form: if $a \in \mathbb{R}$ and $0 \leq a \leq \varepsilon$ for every $\varepsilon > 0$, then $a = 0$.
18. Let $a, b \in \mathbb{R}$ and suppose $a \leq b + \varepsilon$ for every $\varepsilon > 0$. Show $a \leq b$.

19. Prove $\left(\frac{1}{2}(a+b)\right)^2 \leq \frac{1}{2}(a^2+b^2)$ for all a, b , with equality iff $a = b$.
20. (a) If $0 < c < 1$, show $0 < c^2 < c < 1$. (b) If $c > 1$, show $1 < c < c^2$.
21. (a) Prove there is no $n \in \mathbb{N}$ with $0 < n < 1$ (use Well-Ordering). (b) Prove no natural number is both even and odd.
22. (a) If $c > 1$, show $c^n \geq c$ for all $n \in \mathbb{N}$ and $c^n > c$ for $n > 1$. (b) If $0 < c < 1$, show $c^n \leq c$ for all n and $c^n < c$ for $n > 1$.
23. If $a, b > 0$ and $n \in \mathbb{N}$, show $a < b$ iff $a^n < b^n$. (Hint: induction.)
24. (a) If $c > 1$ and $m, n \in \mathbb{N}$, show $c^m > c^n$ iff $m > n$. (b) If $0 < c < 1$ and $m, n \in \mathbb{N}$, show $c^m < c^n$ iff $m > n$.
25. Assuming roots exist, show that if $c > 1$ then $c^{1/m} < c^{1/n}$ iff $m > n$.
26. Use induction to show that if $a \in \mathbb{R}$ and $m, n \in \mathbb{N}$, then $a^{m+n} = a^m a^n$ and $(a^m)^n = a^{mn}$.

Additional exercises.

1. (Field warm-up.) From the axioms alone prove $-(a+b) = -a-b$ and, for $a, b \neq 0$, $(ab)^{-1} = a^{-1}b^{-1}$. Identify which axiom each step uses.
2. Solve and write the answer as a union of intervals: (a) $3x-1 \geq 2x+4$; (b) $\frac{x-3}{x+1} \geq 0$; (c) $x^2 \leq 5x-6$.
3. Prove that for all $a, b \in \mathbb{R}$, $a^2 + b^2 \geq 2ab$, with equality iff $a = b$. Deduce $\frac{a}{b} + \frac{b}{a} \geq 2$ for $a, b > 0$.
4. (Disprove.) A student claims “ $a < b \implies a^2 < b^2$ for all reals.” Give a counterexample, then state and prove the correct hypothesis under which it holds.
5. Show that if n is a positive integer that is not a perfect square, then $\sqrt{n}$ is irrational. (Adapt the $\sqrt{2}$ proof using prime factorization.)
6. Prove that between any two distinct rationals there is another rational, and conclude $\mathbb{Q}$ has no “next” element above a given one.
7. (Ordered field that is not $\mathbb{R}$.) Verify in detail that $\mathbb{Q}(\sqrt{2}) = \{s + t\sqrt{2} : s, t \in \mathbb{Q}\}$ is an ordered field strictly between $\mathbb{Q}$ and $\mathbb{R}$, and exhibit a subset of it that is bounded above with no supremum in $\mathbb{Q}(\sqrt{2})$.
8. For $a, b, c, d > 0$ prove the Cauchy–Schwarz-type bound $(ac + bd)^2 \leq (a^2 + b^2)(c^2 + d^2)$ directly, and find when equality holds.
9. (Diagram.) On a single number line, shade the solution sets of $2x+3 \leq 6$ and $x^2+x > 2$, and mark where they overlap. Describe the overlap as a union of intervals.

2.2 Absolute Value and the Real Line

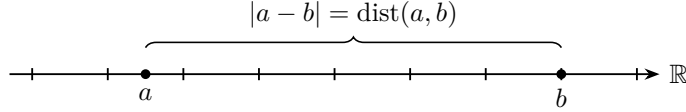
Absolute value as distance

Definition 2.2.1 (Absolute value, B&S 2.2.1). For $a \in \mathbb{R}$,

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0, \end{cases}$$

so $|a| \geq 0$ always, and $|a| = 0 \iff a = 0$. The *distance* between $a, b \in \mathbb{R}$ is $\text{dist}(a, b) = |a - b|$.

The whole point of $|\cdot|$ is that it turns the algebraic line $\mathbb{R}$ into a *metric* space: $|a - b|$ measures how far apart a and b sit. Reading every inequality below as a statement about distance is the fastest route to fluency.



Theorem 2.2.2 (Basic properties, B&S 2.2.2–2.2.3). *For all $a, b \in \mathbb{R}$: (a) $|ab| = |a| |b|$; (b) $|a|^2 = a^2$ and $|a| = \sqrt{a^2}$; (c) for $c \geq 0$, $|a| \leq c \iff -c \leq a \leq c$; in particular $-|a| \leq a \leq |a|$; (d) (Triangle Inequality) $|a + b| \leq |a| + |b|$.*

Proof. **(a)** Check the four sign cases, or use (b): $|ab| = \sqrt{(ab)^2} = \sqrt{a^2 b^2} = |a| |b|$. **(b)** $|a|^2 = a^2$ since $(\pm a)^2 = a^2$; taking the nonnegative root gives $|a| = \sqrt{a^2}$. **(c)** If $|a| \leq c$: when $a \geq 0$, $a = |a| \leq c$ and $a \geq 0 \geq -c$; when $a < 0$, $-a = |a| \leq c$ so $a \geq -c$, and $a < 0 \leq c$. Conversely $-c \leq a \leq c$ bounds both a and $-a$ by c , so $|a| \leq c$. **(d)** By (c), $-|a| \leq a \leq |a|$ and $-|b| \leq b \leq |b|$; adding, $-(|a| + |b|) \leq a + b \leq |a| + |b|$, which by (c) again says $|a + b| \leq |a| + |b|$. $\square$

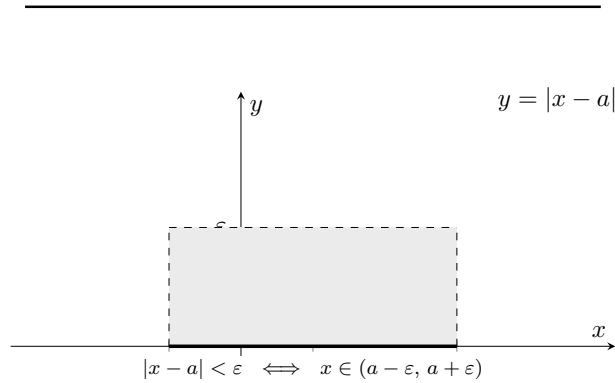
Corollary 2.2.3 (Triangle variants, B&S 2.2.4). *For all $a, b, c \in \mathbb{R}$: (i) $||a| - |b|| \leq |a - b|$ (reverse triangle inequality); (ii) $|a - b| \leq |a| + |b|$; (iii) $|a - c| \leq |a - b| + |b - c|$ (distance is subadditive).*

Proof. (i) $|a| = |(a - b) + b| \leq |a - b| + |b|$ gives $|a| - |b| \leq |a - b|$; swapping a, b gives $|b| - |a| \leq |a - b|$; together $||a| - |b|| \leq |a - b|$. (ii) apply (d) to a and $-b$. (iii) apply (d) to $a - b$ and $b - c$, since $(a - b) + (b - c) = a - c$. $\square$

Caution. $|a + b| = |a| + |b|$ is *not* an identity — it holds exactly when a, b have the same sign (i.e. $ab \geq 0$). In general only $\leq$. The reverse inequality (i) is the one beginners forget; it is the workhorse for showing “if a is close to b then $|a|$ is close to $|b|$,” used in every continuity proof for the modulus.

The V-graph and ε -neighborhoods

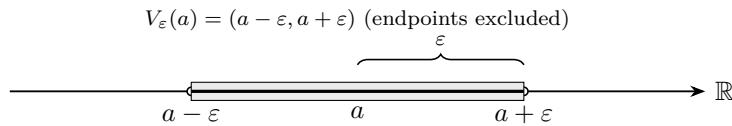
The graph of $y = |x - a|$ is a “V” with its vertex at $(a, 0)$. The solution set of $|x - a| < \varepsilon$ is the slice of the line where the V dips below height ε — an open interval of radius ε centered at a .



Definition 2.2.4 (ε -neighborhood, B&S 2.2.7). For $a \in \mathbb{R}$ and $\varepsilon > 0$, the ε -neighborhood of a is

$$V_\varepsilon(a) = \{x \in \mathbb{R} : |x - a| < \varepsilon\} = (a - \varepsilon, a + \varepsilon).$$

“ x is within ε of a ” means exactly $x \in V_\varepsilon(a)$.



The key fact, the bridge to limits, is that a point pinned inside *every* neighborhood of a must equal a — the neighborhood reformulation of Theorem 2.1.6.

Proposition 2.2.5 (B&S 2.2.8). Let $a \in \mathbb{R}$. If $x \in V_\varepsilon(a)$ for every $\varepsilon > 0$, then $x = a$.

Proof. The hypothesis says $0 \leq |x - a| < \varepsilon$ for all $\varepsilon > 0$; by Theorem 2.1.6, $|x - a| = 0$, hence $x = a$. $\square$

Method (Solving an absolute-value (in)equality). *Recognition trigger.* The unknown sits inside $|\cdot|$.

1. **Single modulus.** Use $|u| < c \iff -c < u < c$ and $|u| > c \iff u > c$ or $u < -c$ (Theorem 2.2.2(c)). Solve the resulting double / split inequality.
2. **Two or more moduli** (e.g. $|x - 1| + |x + 2|$): break $\mathbb{R}$ at the *breakpoints* where each inside changes sign, and on each subinterval drop the bars with the correct sign, then solve and intersect with that subinterval.
3. **Equations** $|u| = |v|$: square both sides, $u^2 = v^2 \iff (u - v)(u + v) = 0$.
4. **Sanity check** by plugging a test point from each region.

Worked Example 2.2.6 (A neighborhood inequality). Solve $|4x - 5| \leq 13$.

Recognition. Single modulus $\leq c$; unfold to a double inequality.

Solution. $-13 \leq 4x - 5 \leq 13 \iff -8 \leq 4x \leq 18 \iff -2 \leq x \leq \frac{9}{2}$. Solution $[-2, \frac{9}{2}]$. *Check.* $x = -2$ gives $|-13| = 13$ ■; $x = 5$ gives $|15| = 15 > 13$, correctly excluded.

Worked Example 2.2.7 (Two moduli by breakpoints). Solve $|x + 1| + |x - 2| = 7$.

Recognition. Sum of two moduli; split at the breakpoints $x = -1$ and $x = 2$.

Solution. *Region* $x < -1$: both insides negative, $-(x+1) - (x-2) = -2x+1 = 7 \implies x = -3$ (valid, $-3 < -1$). *Region* $-1 \leq x \leq 2$: $(x+1) - (x-2) = 3 \neq 7$, no solution. *Region* $x > 2$: $(x+1) + (x-2) = 2x-1 = 7 \implies x = 4$ (valid, $4 > 2$). Solution $\{-3, 4\}$. *Check.* $x = -3$: $|-2| + |-5| = 2+5 = 7$ ■; $x = 4$: $5+2 = 7$ ■. (Geometrically: the points whose distances to -1 and 2 sum to 7 , on a segment of length 3 .)

Worked Example 2.2.8 (An equation with moduli on both sides). Solve $2x - 1 = |x - 5|$.

Recognition. One side a modulus; either case-split or square (squaring may add spurious roots, so verify).

Solution. The left side must be ≥ 0 , so $x \geq \frac{1}{2}$. *Case* $x \geq 5$: $2x - 1 = x - 5 \implies x = -4$, rejected. *Case* $x < 5$: $2x - 1 = -(x - 5) = 5 - x \implies 3x = 6 \implies x = 2$, and $\frac{1}{2} \leq 2 < 5$, valid. Solution $x = 2$. *Check.* $2(2) - 1 = 3 = |2 - 5| = 3$. ■

Worked Example 2.2.9 (NON-example: $|a + b| = |a| + |b|$ can fail). Show the triangle inequality is generally *strict*, and characterize equality.

Recognition. “Disprove an identity / find the equality case.”

Solution. Take $a = 3$, $b = -5$: $|a + b| = |-2| = 2$ while $|a| + |b| = 8$, so $2 < 8$ — strict. Equality $|a + b| = |a| + |b|$ holds iff $ab \geq 0$ (same sign or a zero): squaring, $|a + b| = |a| + |b| \iff (a + b)^2 = a^2 + 2|a||b| + b^2 \iff ab = |ab| \iff ab \geq 0$. *Check.* $a = 3, b = 5$ (same sign): $8 = 8$. ■

Worked Example 2.2.10 (The midpoint/triangle picture for three points). For $x \leq z$, show $|x - y| + |y - z| = |x - z|$ holds iff $x \leq y \leq z$.

Recognition. Distances along a line add iff the middle point is between.

Solution. If $x \leq y \leq z$ then $x - y \leq 0$ and $y - z \leq 0$, so $|x - y| + |y - z| = (y - x) + (z - y) = z - x = |x - z|$. Conversely, if $y < x$ or $y > z$, then y lies outside $[x, z]$ and the path $x \rightarrow y \rightarrow z$ backtracks, making the left side strictly larger (e.g. $y > z$: $|x - y| + |y - z| = (y - x) + (y - z) > z - x$). *Check.* $x = 1, y = 2, z = 5$: $1 + 3 = 4 = |1 - 5|$ ■; $x = 1, y = 6, z = 5$: $5 + 1 = 6 > 4$. ■

Further worked examples

Worked Example 2.2.11 (An ε -neighborhood described as a set of inequalities). Describe $V_{1/3}(2)$ explicitly, and decide whether $\frac{7}{4}$ and $\frac{5}{2}$ belong to it.

Recognition. Trigger: “unpack a neighborhood.” Apply Definition 2.2.4: $V_\varepsilon(a) = (a - \varepsilon, a + \varepsilon)$.

Solution. With $a = 2$, $\varepsilon = \frac{1}{3}$: $V_{1/3}(2) = (2 - \frac{1}{3}, 2 + \frac{1}{3}) = (\frac{5}{3}, \frac{7}{3})$. Membership is the test $|x - 2| < \frac{1}{3}$. For $x = \frac{7}{4}$: $|\frac{7}{4} - 2| = \frac{1}{4} < \frac{1}{3}$, so $\frac{7}{4} \in V_{1/3}(2)$. For $x = \frac{5}{2}$: $|\frac{5}{2} - 2| = \frac{1}{2} > \frac{1}{3}$, so $\frac{5}{2} \notin V_{1/3}(2)$. *Check.* $\frac{5}{3} \approx 1.667$, $\frac{7}{3} \approx 2.333$; indeed 1.75 is inside and 2.5 is outside that interval. ■

Worked Example 2.2.12 (A bound for a fraction via the triangle inequality (B&S 2.2.6 style)). If $|x - 2| < 1$, find an explicit constant M with $\left| \frac{x^2 + 3}{x - 5} \right| \leq M$ for all such x .

Recognition. Trigger: “bound a rational expression on a neighborhood.” Estimate numerator from above and denominator from below using $|x - 2| < 1$ and the (reverse) triangle inequality.

Solution. From $|x - 2| < 1$ we get $1 < x < 3$. *Numerator:* $|x| < 3$, so $|x^2 + 3| \leq |x|^2 + 3 < 9 + 3 = 12$.

Denominator: $x < 3$ gives $x - 5 < -2$, so $|x - 5| = 5 - x > 2$, hence $\frac{1}{|x - 5|} < \frac{1}{2}$. Therefore $\left| \frac{x^2 + 3}{x - 5} \right| = \frac{|x^2 + 3|}{|x - 5|} < \frac{12}{2} = 6$, so $M = 6$ works. *Check.* At $x = 3^-$ the expression nears $\frac{12}{-2} = -6$ in size; at the midpoint $x = 2$, $|\frac{7}{-3}| = \frac{7}{3} \approx 2.33 \leq 6$. ■

Exercises (Bartle & Sherbert, §2.2).

- If $a, b \in \mathbb{R}$ and $b \neq 0$, show (a) $|a| = \sqrt{a^2}$; (b) $|a/b| = |a|/|b|$.
- If $a, b \in \mathbb{R}$, show $|a + b| = |a| + |b|$ iff $ab \geq 0$.
- If $x, y, z \in \mathbb{R}$ with $x \leq z$, show $x \leq y \leq z$ iff $|x - y| + |y - z| = |x - z|$. Interpret geometrically.
- Show $|x - a| < \varepsilon$ iff $a - \varepsilon < x < a + \varepsilon$.
- If $a < x < b$ and $a < y < b$, show $|x - y| < b - a$. Interpret geometrically.
- Find all $x \in \mathbb{R}$ satisfying: (a) $|4x - 5| \leq 13$; (b) $|x^2 - 1| \leq 3$.
- Find all $x \in \mathbb{R}$ satisfying $|x + 1| + |x - 2| = 7$.
- Find all x satisfying: (a) $x + 1 = |2x - 1|$; (b) $2x - 1 = |x - 5|$.
- Find all x satisfying, and sketch graphs: (a) $|x - 2| \leq x + 1$; (b) $3|x| - 2 \leq x$.
- Find all $x \in \mathbb{R}$ satisfying: (a) $|x - 1| > |x + 1|$; (b) $|x| + |x + 1| < 2$.
- Sketch the graph of $y = |x| - |x - 1|$.
- Find all $x \in \mathbb{R}$ with $4 < |x + 2| + |x - 1| < 5$.
- Find all $x \in \mathbb{R}$ satisfying both $|2x - 3| < 5$ and $|x + 1| > 2$ simultaneously.
- Determine and sketch the set of $(x, y) \in \mathbb{R} \times \mathbb{R}$ with: (a) $|x| = |y|$; (b) $|x| + |y| = 1$; (c) $|xy| = 2$; (d) $|x| - |y| = 2$.
- Determine and sketch the set of $(x, y) \in \mathbb{R} \times \mathbb{R}$ with: (a) $|x| \leq |y|$; (b) $|x| + |y| \leq 1$; (c) $|xy| \leq 2$; (d) $|x| - |y| \leq 2$.
- Let $\varepsilon > 0$, $\delta > 0$, $a \in \mathbb{R}$. Show $V_\varepsilon(a) \cap V_\delta(a)$ and $V_\varepsilon(a) \cup V_\delta(a)$ are γ -neighborhoods of a for suitable γ .
- Show that if $a \neq b$ there exist ε -neighborhoods U of a and V of b with $U \cap V = \emptyset$.
- Show that if $a, b \in \mathbb{R}$: (a) $\max\{a, b\} = \frac{1}{2}(a + b + |a - b|)$ and $\min\{a, b\} = \frac{1}{2}(a + b - |a - b|)$; (b) $\min\{a, b, c\} = \min\{\min\{a, b\}, c\}$.
- Show that the “middle” number is $\text{mid}\{a, b, c\} = \min\{\max\{a, b\}, \max\{b, c\}, \max\{c, a\}\}$.

Additional exercises.

1. Solve $|x - 3| + |x + 2| \leq 9$ and write the answer as a closed interval.
2. Prove the general triangle inequality $|a_1 + \cdots + a_n| \leq |a_1| + \cdots + |a_n|$ by induction on n .
3. Prove $||a| - |b|| \leq |a - b|$ both from the triangle inequality and directly by squaring; compare the two arguments.
4. Show $V_\varepsilon(a) \subseteq V_\delta(b)$ holds iff $|a - b| + \varepsilon \leq \delta$. Draw the number-line picture that makes this obvious.
5. (Disprove.) Is $|x - y| \leq |x| - |y|$ ever a valid inequality for all x, y ? Either prove it or give a counterexample and state the correct inequality.
6. For which a does the equation $|x - 1| + |x + 1| = a$ have (i) no solution, (ii) infinitely many solutions, (iii) exactly two solutions? Justify with the V-graph.
7. (Diagram.) Sketch $y = |x - 1|$ and the horizontal line $y = 2$ on the same axes, shade the region $|x - 1| < 2$ on the x -axis, and read off the neighborhood $V_2(1)$.
8. Prove that $\max\{a, b\} + \min\{a, b\} = a + b$ and $\max\{a, b\} - \min\{a, b\} = |a - b|$, and use these to re-derive the formulas of B&S Exercise 2.2.18(a).
9. Show that the function $d(x, y) = |x - y|$ satisfies $d(x, y) \geq 0$ with equality iff $x = y$, $d(x, y) = d(y, x)$, and $d(x, z) \leq d(x, y) + d(y, z)$. (These are the metric axioms for $\mathbb{R}$.)

2.3 The Completeness Property: Suprema and Infima

Upper bounds, lower bounds, and the supremum

Definition 2.3.1 (Bounds, B&S 2.3.1 / Rudin 1.7). Let $S \subseteq \mathbb{R}$ be nonempty. A number u is an *upper bound* of S if $s \leq u$ for all $s \in S$; then S is *bounded above*. A number w is a *lower bound* if $w \leq s$ for all $s \in S$; then S is *bounded below*. S is *bounded* if both.

If u is an upper bound then so is every larger number, so upper bounds come in an infinite upward ray. The decisive object is the *least* one.

Definition 2.3.2 (Supremum and infimum, B&S 2.3.2 / Rudin 1.8). Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. A number α is the *supremum* (least upper bound) of S , written $\alpha = \sup S$, if (i) α is an upper bound of S , and (ii) if u is any upper bound of S , then $\alpha \leq u$. Dually, for S bounded below, the *infimum* (greatest lower bound) $\inf S$ is the greatest lower bound. When $\sup S \in S$ it is the *maximum* $\max S$; similarly $\min S$.

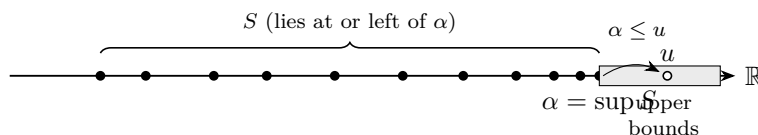


Figure 2.1. $\sup S$ is the leftmost point of the shaded ray of upper bounds: it sits at the right edge of S and is $\leq$ every other upper bound.

The supremum need *not* belong to S . For $S = (0, 1)$, $\sup S = 1 \notin S$; for $S = (0, 1]$, $\sup S = 1 \in S = \max S$. “Least upper bound” and “maximum” coincide only when the bound is attained.

Proposition 2.3.3 (Two-condition test, B&S 2.3.4). *Let S be nonempty and bounded above, α an upper bound of S . Then $\alpha = \sup S$ iff*

$$\text{for every } \varepsilon > 0 \text{ there exists } s \in S \text{ with } s > \alpha - \varepsilon.$$

Proof. ($\Rightarrow$) If $\alpha = \sup S$ and $\varepsilon > 0$, then $\alpha - \varepsilon < \alpha$ is *not* an upper bound (else it would beat the least one), so some $s \in S$ exceeds $\alpha - \varepsilon$. ($\Leftarrow$) Suppose α is an upper bound and the ε -condition holds. Let u be any upper bound; if $u < \alpha$, set $\varepsilon = \alpha - u > 0$ and get $s \in S$ with $s > \alpha - \varepsilon = u$, contradicting that u bounds S . So $\alpha \leq u$, and α is the least upper bound. $\square$

This is *the* working characterization: it splits “ $\alpha = \sup S$ ” into the easy half (it bounds S) and the sharp half (nothing smaller bounds S , witnessed by elements of S creeping up to α).

Method (Proving $\alpha = \sup S$). *Recognition trigger.* You claim a specific number α is the least upper bound of a described set.

1. **Upper bound.** Show $s \leq \alpha$ for every $s \in S$ (often a direct algebraic estimate).
2. **Approachability.** Fix an arbitrary $\varepsilon > 0$ and *exhibit* an element $s \in S$ with $s > \alpha - \varepsilon$ — usually by solving the defining condition for an index large enough. This is the half that pins α as the *least* bound.
3. For inf, mirror everything: $\alpha \leq s$ for all s , and for each ε some $s < \alpha + \varepsilon$.

Worked Example 2.3.4 (sup of a sequence set). Let $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\} = \{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\}$. Show $\sup S = 1$ and $\inf S = 0 = \min S$.

Recognition. Direct two-condition test.

Solution. *Upper bound:* $1 - \frac{1}{n} < 1$ for all n , so 1 bounds S . *Approachability:* given $\varepsilon > 0$, pick n with $\frac{1}{n} < \varepsilon$ (Archimedean, §2.4); then $1 - \frac{1}{n} > 1 - \varepsilon$. By Proposition 2.3.3, $\sup S = 1$. Note $1 \notin S$, so there is no maximum. *Infimum:* the smallest term is $n = 1$, $1 - 1 = 0 \in S$, and $0 \leq 1 - \frac{1}{n}$ for all n , so $\inf S = 0 = \min S$. *Check.* Terms $0, \frac{1}{2}, \frac{2}{3}, \dots$ climb toward but never reach 1. ■

The completeness axiom

The rationals satisfy Definitions 2.3.1–2.3.2 but, as §2.1 showed, a bounded set like $\{p \in \mathbb{Q} : p^2 < 2\}$ has *no* supremum in $\mathbb{Q}$. The defining property of $\mathbb{R}$ is that this never happens.

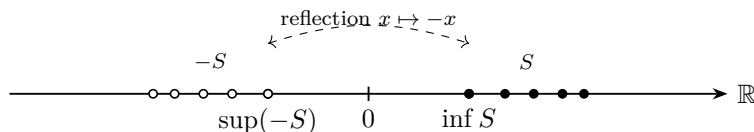
Theorem 2.3.5 (Completeness / Least-Upper-Bound Property, B&S 2.3.6 / Rudin 1.19). *Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. Equivalently, every nonempty subset bounded below has an infimum in $\mathbb{R}$.*

We take this as an *axiom* characterizing $\mathbb{R}$ (Rudin instead *constructs* $\mathbb{R}$ from $\mathbb{Q}$ by Dedekind cuts and proves it as a theorem; see the remark below). The equivalence of the sup- and inf-forms is itself a clean application of the reflection $x \mapsto -x$.

Proposition 2.3.6 (inf via sup, Rudin 1.11 / B&S Exercise 2.3.6). *If S is nonempty and bounded below, then $-S = \{-s : s \in S\}$ is bounded above and*

$$\inf S = -\sup(-S).$$

Proof. $w \leq s$ for all $s \in S$ iff $-s \leq -w$ for all s , i.e. $-w$ is an upper bound of $-S$. So lower bounds of S correspond, under negation, to upper bounds of $-S$, reversing order. The *greatest* lower bound of S thus corresponds to the *least* upper bound of $-S$: $\inf S = -\sup(-S)$. Existence of $\sup(-S)$ (Completeness) gives existence of $\inf S$, proving the two forms of Theorem 2.3.5 equivalent. □



Remark (Dedekind cuts, Rudin 1.19 appendix). Rudin builds $\mathbb{R}$ as the set of *cuts*: a cut $\alpha \subseteq \mathbb{Q}$ is a nonempty proper downward-closed subset with no greatest element (e.g. $\sqrt{2}$ is the cut $\{p \in \mathbb{Q} : p < 0 \text{ or } p^2 < 2\}$). Order is inclusion $\alpha < \beta \iff \alpha \subsetneq \beta$, and the supremum of a bounded family of cuts is simply their *union* — which is again a cut, giving Theorem 2.3.5 for free. Addition is setwise; multiplication is fiddly on signs. The construction yields a complete ordered field, and one proves any two such are isomorphic, so “the” real numbers are well-defined. We will not need the cuts again; the axiom suffices.

First consequences of completeness

Worked Example 2.3.7 ($\sup(a + S) = a + \sup S$ (B&S 2.4.1)). For S bounded above and $a \in \mathbb{R}$, let $a + S = \{a + s : s \in S\}$. Show $\sup(a + S) = a + \sup S$.

Recognition. “Compute a sup of a shifted set” — prove the two halves of the test.

Solution. Let $u = \sup S$. Then $s \leq u$ gives $a + s \leq a + u$, so $a + u$ is an upper bound of $a + S$; hence $\sup(a + S) \leq a + u$. Conversely, given $\varepsilon > 0$ choose $s \in S$ with $s > u - \varepsilon$; then $a + s > (a + u) - \varepsilon$, so by Proposition 2.3.3 no number below $a + u$ bounds $a + S$, giving $\sup(a + S) \geq a + u$. Thus $\sup(a + S) = a + u = a + \sup S$. *Check.* $S = [0, 1]$, $a = 3$: $a + S = [3, 4]$, $\sup = 4 = 3 + 1$. ■

Worked Example 2.3.8 (Monotonicity of sup and inf under inclusion). If $\emptyset \neq S_0 \subseteq S$ with S bounded, show $\inf S \leq \inf S_0 \leq \sup S_0 \leq \sup S$.

Recognition. Inclusion comparison; an upper bound of the big set bounds the small one.

Solution. $\sup S$ is an upper bound of $S \supseteq S_0$, hence of S_0 , so $\sup S_0 \leq \sup S$ (least upper bound is $\leq$ any upper bound). Symmetrically $\inf S \leq \inf S_0$. And $\inf S_0 \leq \sup S_0$ since any element of S_0 lies between them (or because $\inf S_0 \leq s \leq \sup S_0$ for any chosen $s \in S_0 \neq \emptyset$). *Check.* $S_0 = \{2\} \subseteq S = [1, 3]$: $1 \leq 2 \leq 2 \leq 3$. ■

Caution. $\sup S$ is not the same as $\max S$. A bounded *open* set has a supremum but no maximum: $\sup(0, 1) = 1$ exists, yet $(0, 1)$ has no greatest element. “Find the maximum” can have answer “none” while “find the supremum” always has an answer (in $\mathbb{R}$). Reserve $\max / \min$ for when the bound is actually attained.

Caution. A common slip: “ u is an upper bound and $u \in S$, therefore $u = \sup S$.” The implication is true (B&S Exercise 2.3.7) — if an upper bound lies in the set it is the maximum, hence the sup — but the converse fails, and the hypothesis $u \in S$ is essential. For $S = [0, 1]$, $u = 1$ is an upper bound *not* in S and is still the sup, while $u = 0.9 \in S$ is in S but is *not* an upper bound. Always verify both “bounds” and “least”.

Further worked examples

Worked Example 2.3.9 (sup and inf of a finite and a half-open set). Find sup, inf, max, min (when they exist) for $A = \{-2, 0, \frac{1}{2}, 3\}$ and $B = \{x \in \mathbb{R} : 0 \leq x < 4\}$.

Recognition. “Read off the extremes.” For a finite set the sup/inf are the largest/smallest elements; for an interval, test whether each endpoint is attained (Definition 2.3.2).

Solution. For A : it is finite, so $\sup A = \max A = 3$ and $\inf A = \min A = -2$, both attained. For $B = [0, 4)$: $0 \in B$ is a lower bound and lies in B , so $\inf B = \min B = 0$. And 4 is an upper bound; given any $\varepsilon > 0$, $4 - \frac{\varepsilon}{2} < 4$ lies in B and exceeds $4 - \varepsilon$, so by Proposition 2.3.3 $\sup B = 4$ — but $4 \notin B$, so $\max B$ does *not* exist. *Check.* Every $x \in B$ satisfies $x < 4$, yet no number below 4 bounds B (e.g. $3.9 \in B$ beats $3.99 - \varepsilon$ choices), confirming 4 is least. ■

Worked Example 2.3.10 (The supremum of $\{n/(n+1)\}$). Let $S = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\} = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots \right\}$.

Show $\sup S = 1$ and $\inf S = \frac{1}{2} = \min S$.

Recognition. “Sup of an increasing sequence set”: bound by 1, then approach via the Archimedean property (Method, two-condition test).

Solution. Write $\frac{n}{n+1} = 1 - \frac{1}{n+1} < 1$, so 1 is an upper bound. Given $\varepsilon > 0$, choose (Archimedean, §2.4) n with $\frac{1}{n+1} < \varepsilon$; then $\frac{n}{n+1} > 1 - \varepsilon$, so by Proposition 2.3.3, $\sup S = 1$ (never attained, no max). The sequence is increasing, so its least term is at $n = 1$: $\inf S = \min S = \frac{1}{2}$. *Check.* $\frac{1}{2}, \frac{2}{3} \approx 0.667, \frac{3}{4} = 0.75, \dots$ climb toward 1; the first term $\frac{1}{2}$ is the smallest. ■

Worked Example 2.3.11 (NON-example: a bounded subset of $\mathbb{Q}$ with no supremum *in* $\mathbb{Q}$). Exhibit a nonempty subset of $\mathbb{Q}$ that is bounded above but has no least upper bound inside $\mathbb{Q}$, contrasting with Completeness in $\mathbb{R}$.

Recognition. “Show completeness can fail in $\mathbb{Q}$ ”: reuse the $\sqrt{2}$ -gap of §2.1.

Solution. Let $T = \{r \in \mathbb{Q} : r > 0, r^2 < 2\}$. It is nonempty ($1 \in T$) and bounded above in $\mathbb{Q}$ (e.g. by 2, since $r \geq 2 \implies r^2 \geq 4 > 2$). In $\mathbb{R}$ its supremum is $\sqrt{2}$ (Theorem 2.4.5-style), but $\sqrt{2} \notin \mathbb{Q}$ (Theorem 2.1.12). Any rational upper bound u of T has $u^2 > 2$, and the map $u \mapsto \frac{2u+2}{u+2}$ produces a *smaller* rational upper bound (its square exceeds 2 too), so there is no *least* rational upper bound. Hence $\sup T$ fails to exist in $\mathbb{Q}$. *Check.* The rational bounds 1.5, 1.42, 1.415, ... decrease forever toward $1.4142\dots = \sqrt{2}$, never reaching a least rational one. ■ This is exactly the defect Completeness (Theorem 2.3.5) cures in $\mathbb{R}$.

Exercises (Rudin, Ch.1 – order and completeness).

1. Prove Proposition 1.15 (the elementary arithmetic-of-fractions / inverse rules in an ordered field, e.g. if $0 < x < y$ then $0 < 1/y < 1/x$).
2. Let E be a nonempty subset of an ordered set; suppose α is a lower bound of E and β is an upper bound of E . Prove $\alpha \leq \beta$.
3. Let A be a nonempty set of reals bounded below, and let $-A = \{-x : x \in A\}$. Prove $\inf A = -\sup(-A)$.
4. (Dedekind cuts, with reference to the Appendix.) Suppose property (III) — “a cut has no largest element” — is omitted from the definition of a cut, keeping the same order and addition. Show the resulting ordered set still has the least-upper-bound property, that addition satisfies axioms (A1)–(A4) (with a slightly different zero), but that (A5) fails.

Exercises (Bartle & Sherbert, §2.3).

1. Let $S_1 = \{x \in \mathbb{R} : x \geq 0\}$. Show S_1 has lower bounds but no upper bounds, and $\inf S_1 = 0$.
2. Let $S_2 = \{x \in \mathbb{R} : x > 0\}$. Does S_2 have lower bounds? upper bounds? Does $\inf S_2$ exist? $\sup S_2$? Prove your answers.
3. Let $S_3 = \{1/n : n \in \mathbb{N}\}$. Show $\sup S_3 = 1$ and $\inf S_3 \geq 0$. (The Archimedean Property gives $\inf S_3 = 0$.)
4. Let $S_4 = \{1 - (-1)^n/n : n \in \mathbb{N}\}$. Find $\inf S_4$ and $\sup S_4$.
5. Find $\inf$ and $\sup$, if they exist, of each set: (a) $A = \{x : 2x + 5 > 0\}$; (b) $B = \{x : x + 2 \geq x^2\}$; (c) $C = \{x : x < 1/x\}$; (d) $D = \{x : x^2 - 2x - 5 < 0\}$.
6. Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Prove $\inf S = -\sup\{-s : s \in S\}$.
7. If a set $S \subseteq \mathbb{R}$ contains one of its upper bounds, show that this upper bound is $\sup S$.
8. Let $S \subseteq \mathbb{R}$ be nonempty. Show u is an upper bound of S iff $t \in \mathbb{R}$ and $t > u$ imply $t \notin S$.
9. Let $S \subseteq \mathbb{R}$ be nonempty. Show that if $u = \sup S$, then for every $n \in \mathbb{N}$ the number $u - 1/n$ is not an upper bound of S , while $u + 1/n$ is an upper bound.
10. If A, B are bounded subsets of $\mathbb{R}$, show $A \cup B$ is bounded and $\sup(A \cup B) = \max\{\sup A, \sup B\}$.
11. Let S be bounded in $\mathbb{R}$ and $S_0 \neq \emptyset$ a subset of S . Show $\inf S \leq \inf S_0 \leq \sup S_0 \leq \sup S$.
12. Let $S \subseteq \mathbb{R}$ with $s = \sup S \in S$. If $u \notin S$, show $\sup(S \cup \{u\}) = \max\{s, u\}$.
13. Show that a nonempty finite set $S \subseteq \mathbb{R}$ contains its supremum. (Hint: induction and the preceding exercise.)
14. Let S be bounded below. Prove a lower bound w of S is $\inf S$ iff for every $\varepsilon > 0$ there is $t \in S$ with $t < w + \varepsilon$.

Additional exercises.

1. For each set, decide whether $\sup$, $\inf$, $\max$, $\min$ exist and find them: (a) $\{(-1)^n(1 - 1/n) : n \in \mathbb{N}\}$; (b) $\{n/(n+1) : n \in \mathbb{N}\}$; (c) $\{(-1)^n/n : n \in \mathbb{N}\}$.
2. Prove directly from the definition that $\sup S$, when it exists, is unique.

- Let A, B be nonempty bounded sets with $a \leq b$ for all $a \in A, b \in B$. Prove $\sup A \leq \inf B$. Give an example with $\sup A = \inf B$ where the sets are disjoint.
- (Two-condition drill.) Prove $\sup\{x \in \mathbb{Q} : x^2 < 2\} = \sqrt{2}$ as a subset of $\mathbb{R}$, using the characterization of the supremum.
- Let S be nonempty and bounded above with $\alpha = \sup S$. Show that for every $\varepsilon > 0$ the interval $(\alpha - \varepsilon, \alpha]$ contains at least one point of S , and illustrate with a number-line diagram.
- (Disprove.) Is it true that $\sup(A \cap B) = \min\{\sup A, \sup B\}$ for all bounded A, B ? Prove or give a counterexample.
- Prove that a nonempty set $S \subseteq \mathbb{Z}$ that is bounded above has a maximum. (Use the Archimedean property and Well-Ordering.)
- (Diagram.) Draw a set S of five marked points on a number line, its supremum, and shade the ray of upper bounds; then mark a point $\alpha - \varepsilon$ and the witness $s \in S$ guaranteed by the two-condition test.
- Let S be bounded above and define $u^* = \sup S$. Prove u^* is the unique real with the property: u^* bounds S , but $u^* - \varepsilon$ does not, for every $\varepsilon > 0$.

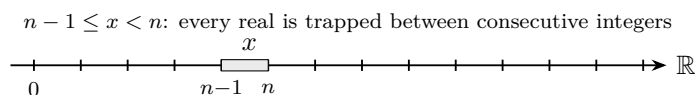
2.4 Applications of the Supremum

The Archimedean property

Completeness has teeth. Its first payoff is that $\mathbb{N}$ is not bounded above in $\mathbb{R}$ — there are no “infinitely large” reals — and dually no positive real is “infinitely small.”

Theorem 2.4.1 (Archimedean Property, Rudin 1.20 / B&S 2.4.3–2.4.5). (a) *For every $x \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $n > x$ (so $\mathbb{N}$ is unbounded above).* (b) *For every $\varepsilon > 0$ there is $n \in \mathbb{N}$ with $\frac{1}{n} < \varepsilon$.* (c) *For every $x > 0$ there is $n \in \mathbb{N}$ with $n - 1 \leq x < n$.*

Proof. (a) If not, $\mathbb{N}$ would be bounded above; by Completeness it would have a supremum $\alpha = \sup \mathbb{N}$. Then $\alpha - 1$ is not an upper bound, so some $m \in \mathbb{N}$ has $m > \alpha - 1$, giving $m + 1 > \alpha$ with $m + 1 \in \mathbb{N}$ — contradicting that α bounds $\mathbb{N}$. So $\mathbb{N}$ is unbounded, and any x is exceeded by some n . (b) Apply (a) to $x = 1/\varepsilon$: some $n > 1/\varepsilon$, hence $1/n < \varepsilon$. (c) The set $\{m \in \mathbb{N} : m > x\}$ is nonempty by (a); by Well-Ordering it has a least element n , and then $n - 1 \leq x < n$. $\square$



Corollary 2.4.2 ($\inf\{1/n\} = 0$, B&S 2.4.4). $\inf\{\frac{1}{n} : n \in \mathbb{N}\} = 0$.

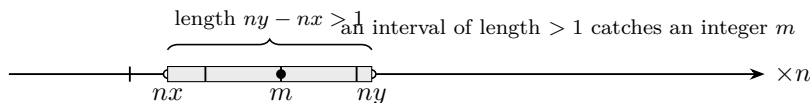
Proof. 0 is a lower bound ($\frac{1}{n} > 0$). If $w > 0$ were a larger lower bound, Archimedes (b) gives n with $\frac{1}{n} < w$, contradicting that $w \leq \frac{1}{n}$ for all n . So 0 is the greatest lower bound. $\square$

Density of $\mathbb{Q}$ (and of the irrationals)

The rationals, though only countable, sit *everywhere*: between any two reals lies a rational. Then so does an irrational.

Theorem 2.4.3 (Density of $\mathbb{Q}$ in $\mathbb{R}$, Rudin 1.20(b) / B&S 2.4.8). *If $x, y \in \mathbb{R}$ with $x < y$, there is a rational r with $x < r < y$.*

Proof. Since $y - x > 0$, Archimedes gives $n \in \mathbb{N}$ with $\frac{1}{n} < y - x$, i.e. $ny - nx > 1$, so the open interval (nx, ny) has length > 1 and must contain an integer m (let m be the least integer $> nx$; then $m \leq nx + 1 < ny$). Thus $nx < m < ny$, and dividing by $n > 0$ gives $x < \frac{m}{n} < y$. Take $r = \frac{m}{n}$. $\square$



Corollary 2.4.4 (Density of irrationals). *Between any two reals $x < y$ there is also an irrational.*

Proof. Apply Theorem 2.4.3 to $x/\sqrt{2} < y/\sqrt{2}$ to get a rational $r \neq 0$ with $x/\sqrt{2} < r < y/\sqrt{2}$; then $r\sqrt{2}$ is irrational (a nonzero rational times the irrational $\sqrt{2}$) and $x < r\sqrt{2} < y$. $\square$

Existence and uniqueness of n th roots

The flagship use of completeness: it manufactures the roots that $\mathbb{Q}$ was missing. We prove the existence of $\sqrt{2}$ in full — the argument is the template for every n th root.

Theorem 2.4.5 (Existence of $\sqrt{2}$, Rudin 1.21 / B&S 2.4.7). *There is a unique real $y > 0$ with $y^2 = 2$.*

Proof. Let $S = \{s \in \mathbb{R} : s > 0, s^2 < 2\}$. It is nonempty ($1 \in S$) and bounded above (2 is a bound, since $s \geq 2 \implies s^2 \geq 4 > 2$). By Completeness, $y = \sup S$ exists, and $y \geq 1 > 0$. We rule out $y^2 < 2$ and $y^2 > 2$ by perturbing y .

If $y^2 < 2$: choose h with $0 < h < 1$ and $h < \frac{2-y^2}{2y+1}$. Then $(y+h)^2 = y^2 + h(2y+h) < y^2 + h(2y+1) < y^2 + (2-y^2) = 2$, so $y+h \in S$ — contradicting that y is an upper bound of S .

If $y^2 > 2$: choose $k = \frac{y^2-2}{2y} > 0$. Then for any $s \geq y-k$ with $s > 0$, $(y-k)^2 = y^2 - 2yk + k^2 > y^2 - 2yk = 2$, so every $s \in S$ satisfies $s < y-k$ (else $s^2 \geq (y-k)^2 > 2$); thus $y-k$ is a smaller upper bound, contradicting $y = \sup S$.

By trichotomy $y^2 = 2$. *Uniqueness:* if $0 < y_1 < y_2$ then $y_1^2 < y_2^2$ (Theorem 2.1.5), so at most one positive root exists. $\square$

Remark. Replacing 2 by any $a > 0$ and the exponent 2 by n gives, verbatim in spirit, Rudin's Theorem 1.21: every real $a > 0$ has a unique positive n th root $a^{1/n}$. The perturbation estimates use $(y+h)^n - y^n < h \cdot n(y+1)^{n-1}$ in place of the quadratic bound. Completeness is doing all the work — $\mathbb{Q}$ fails precisely because $\sup S$ would land outside it.

Worked Example 2.4.6 (Existence of $\sqrt{3}$ by the same machine). Show there is $y > 0$ with $y^2 = 3$ (B&S Exercise 2.4.15).

Recognition. “Produce a root”: run the sup-of- $\{s^2 < a\}$ argument with $a = 3$.

Solution. Let $S = \{s > 0 : s^2 < 3\}$; nonempty ($1 \in S$), bounded above by 2 ($s \geq 2 \implies s^2 \geq 4 > 3$). Put $y = \sup S$. If $y^2 < 3$, pick $0 < h < 1$ with $h < \frac{3-y^2}{2y+1}$; then $(y+h)^2 < 3$ contradicts upper-bound. If $y^2 > 3$, pick $k = \frac{y^2-3}{2y}$; then $y-k$ is a smaller upper bound. So $y^2 = 3$. *Check.* $1.7^2 = 2.89 < 3 < 3.24 = 1.8^2$, so $y \in (1.7, 1.8)$ as expected. $\blacksquare$

Worked Example 2.4.7 (sup and inf of $\{\frac{1}{n} - \frac{1}{m}\}$ (B&S 2.4.2)). Find $\inf S$ and $\sup S$ for $S = \{\frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N}\}$.

Recognition. Two free indices; push one to make the expression large, the other to make it small.

Solution. Each term lies in $(-1, 1)$: $\frac{1}{n} \leq 1$ and $-\frac{1}{m} \geq -1$, with strict unless an index is 1. *Sup:* fix $n = 1$, let $m \rightarrow \infty$: $1 - \frac{1}{m} \rightarrow 1$, and $1 - \frac{1}{m} < 1$, so $\sup S = 1$ (approached, never attained), by Proposition 2.3.3 with Corollary 2.4.2. *Inf:* fix $m = 1$, let $n \rightarrow \infty$: $\frac{1}{n} - 1 \rightarrow -1$ from above, so $\inf S = -1$. *Check.* No term equals ± 1 , so neither bound is in S ; $\frac{1}{1} - \frac{1}{2} = \frac{1}{2} \in S$ sits comfortably inside $(-1, 1)$. $\blacksquare$

Worked Example 2.4.8 ($\sup(A+B) = \sup A + \sup B$ (B&S 2.4.7)). For nonempty bounded A, B , with $A+B = \{a+b : a \in A, b \in B\}$, prove $\sup(A+B) = \sup A + \sup B$.

Recognition. Sum of two sets; bound each factor, then approach with both at once.

Solution. Let $\alpha = \sup A$, $\beta = \sup B$. For $a \in A, b \in B$: $a+b \leq \alpha + \beta$, so $\alpha + \beta$ bounds $A+B$. Given $\varepsilon > 0$, choose $a > \alpha - \frac{\varepsilon}{2}$ and $b > \beta - \frac{\varepsilon}{2}$; then $a+b > (\alpha + \beta) - \varepsilon$. By the two-condition test $\sup(A+B) = \alpha + \beta$. *Check.* $A = B = [0, 1]$: $A+B = [0, 2]$, $\sup = 2 = 1 + 1$. $\blacksquare$

Caution. The sup-of-a-product or sup-of-an-inf identities are subtler than the sum case. In general only $\sup_x \inf_y h(x, y) \leq \inf_y \sup_x h(x, y)$ (a one-sided min-max inequality, B&S 2.4.11), and equality can fail. Never assume you may freely swap sup and inf — the order matters. Likewise $\sup(AB) \neq \sup A \cdot \sup B$ once negative numbers appear, since multiplying by a negative reverses bounds.

Further worked examples

Worked Example 2.4.9 (Trapping a real between integers (the floor)). Use the Archimedean property to locate $\lfloor x \rfloor$ for $x = \frac{47}{6}$ and for $x = -\frac{7}{4}$, the unique integer n with $n \leq x < n + 1$.

Recognition. Trigger: “find the integer part.” Theorem 2.4.1(c) guarantees a unique such n ; compute it by comparing with consecutive integers.

Solution. For $x = \frac{47}{6} = 7.8\bar{3}$: $7 \leq 7.8\bar{3} \dots < 8$, so $\lfloor x \rfloor = 7$. For $x = -\frac{7}{4} = -1.75$: the integer below is -2 , since $-2 \leq -1.75 < -1$, so $\lfloor x \rfloor = -2$ (not -1 — the floor rounds down). Check. $7 \cdot 6 = 42 \leq 47 < 48 = 8 \cdot 6$ confirms the first; $-2 \leq -1.75 < -1$ the second. ■

Worked Example 2.4.10 (A specific rational and irrational in a tiny interval). Exhibit a rational and an irrational strictly between $\sqrt{2}$ and $\sqrt{2} + \frac{1}{100}$.

Recognition. “Density in action” (Theorem 2.4.3, Corollary 2.4.4): find a fraction with small enough denominator, then add an irrational nudge.

Solution. Since $\sqrt{2} = 1.41421\dots$, the interval is $(1.41421\dots, 1.42421\dots)$. The rational $r = 1.415 = \frac{283}{200}$ satisfies $1.41421 < 1.415 < 1.42421$, so r works (indeed $r^2 = 2.002225 > 2$ yet $r < \sqrt{2} + 0.01$). For an irrational, take $\xi = \sqrt{2} + \frac{1}{200} = 1.41421\dots + 0.005 = 1.41921\dots$; it is irrational (rational plus irrational) and lies in the interval. Check. $1.41421 < 1.415 < 1.41921 < 1.42421$, all four inside $(\sqrt{2}, \sqrt{2} + 0.01)$. ■

Exercises (Rudin, Ch.1 – roots, exponents, irrationals).

1. If $r \neq 0$ is rational and x is irrational, prove that $r + x$ and rx are irrational.
2. Prove that there is no rational number whose square is 12.
3. Fix $b > 1$. (a) If $m, n, p, q \in \mathbb{Z}$ with $n, q > 0$ and $r = m/n = p/q$, prove $(b^m)^{1/n} = (b^p)^{1/q}$, so b^r is well-defined. (b) Prove $b^{r+s} = b^r b^s$ for rational r, s . (c) For real x , let $B(x) = \{b^t : t \in \mathbb{Q}, t \leq x\}$; prove $b^r = \sup B(r)$ for rational r , so $b^x = \sup B(x)$ is well-defined for all real x . (d) Prove $b^{x+y} = b^x b^y$ for all real x, y .
4. Fix $b > 1, y > 0$. Prove there is a unique real x with $b^x = y$ (the logarithm of y to base b), via: (a) $b^n - 1 \geq n(b - 1)$ for $n \in \mathbb{N}$; (b) hence $b - 1 \geq n(b^{1/n} - 1)$; (c) if $t > 1$ and $n > (b - 1)/(t - 1)$ then $b^{1/n} < t$; (d) if $b^w < y$ then $b^{w+1/n} < y$ for large n ; (e) if $b^w > y$ then $b^{w-1/n} > y$ for large n ; (f) let $A = \{w : b^w < y\}$ and show $x = \sup A$ satisfies $b^x = y$; (g) prove this x is unique.

Exercises (Bartle & Sherbert, §2.4).

1. Show $\sup\{1 - 1/n : n \in \mathbb{N}\} = 1$.
2. If $S = \{1/n - 1/m : n, m \in \mathbb{N}\}$, find $\inf S$ and $\sup S$.
3. Let $S \subseteq \mathbb{R}$ be nonempty. Prove that if $u \in \mathbb{R}$ satisfies (i) for every $n \in \mathbb{N}$, $u - 1/n$ is not an upper bound of S , and (ii) for every $n \in \mathbb{N}$, $u + 1/n$ is an upper bound of S , then $u = \sup S$.
4. Let S be nonempty and bounded. (a) For $a > 0$ and $aS = \{as : s \in S\}$, prove $\inf(aS) = a \inf S$ and $\sup(aS) = a \sup S$. (b) For $b < 0$ and $bS = \{bs : s \in S\}$, prove $\inf(bS) = b \sup S$ and $\sup(bS) = b \inf S$.
5. Let S be a set of nonnegative reals bounded above and $T = \{x^2 : x \in S\}$. Prove if $u = \sup S$ then $u^2 = \sup T$. Give an example showing this can fail if negative numbers are allowed.
6. Let $f : X \rightarrow \mathbb{R}$ have bounded range and $a \in \mathbb{R}$. Show $\sup\{a + f(x) : x \in X\} = a + \sup\{f(x) : x \in X\}$, and similarly for $\inf$.
7. Let A, B be bounded nonempty subsets of $\mathbb{R}$ and $A + B = \{a + b : a \in A, b \in B\}$. Prove $\sup(A + B) = \sup A + \sup B$ and $\inf(A + B) = \inf A + \inf B$.

8. Let f, g be bounded on X . Show $\sup\{f(x) + g(x)\} \leq \sup f + \sup g$ and $\inf f + \inf g \leq \inf\{f(x) + g(x)\}$. Give examples of equality and of strict inequality.
9. Let $X = Y = \{x : 0 < x < 1\}$ and $h(x, y) = 2x + y$. (a) For each x find $f(x) = \sup_y h(x, y)$, then $\inf_x f(x)$. (b) For each y find $g(y) = \inf_x h(x, y)$, then $\sup_y g(y)$. Compare.
10. Repeat (a),(b) of the previous exercise for $h(x, y) = 0$ if $x < y$, $h(x, y) = 1$ if $x \geq y$.
11. Let $h : X \times Y \rightarrow \mathbb{R}$ have bounded range, $f(x) = \sup_y h(x, y)$, $g(y) = \inf_x h(x, y)$. Prove $\sup_y g(y) \leq \inf_x f(x)$, i.e. $\sup_y \inf_x h \leq \inf_x \sup_y h$.
12. Let $h : X \times Y \rightarrow \mathbb{R}$ have bounded range, $F(x) = \sup_y h(x, y)$, $G(y) = \sup_x h(x, y)$. Establish the iterated suprema principle: $\sup_{x,y} h = \sup_x F(x) = \sup_y G(y)$.
13. Given any $x \in \mathbb{R}$, show there is a unique $n \in \mathbb{Z}$ with $n - 1 \leq x < n$.
14. If $y > 0$, show there is $n \in \mathbb{N}$ with $1/2^n < y$.
15. Modify the $\sqrt{2}$ -argument to show there is a positive real y with $y^2 = 3$.
16. Modify the $\sqrt{2}$ -argument to show that if $a > 0$ there is a positive real z with $z^2 = a$.
17. Modify the argument to show there is a positive real u with $u^3 = 2$.
18. Complete the proof of the Density Theorem by removing the assumption $x > 0$.
19. If $u > 0$ and $x < y$, show there is a rational r with $x < ru < y$. (So $\{ru : r \in \mathbb{Q}\}$ is dense in $\mathbb{R}$.)

Additional exercises.

1. Use the Archimedean property to prove $\inf\{1/2^n : n \in \mathbb{N}\} = 0$.
2. Prove that for every $x \in \mathbb{R}$ there is a unique integer $\lfloor x \rfloor$ (the floor) with $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$, and that $x - \lfloor x \rfloor \in [0, 1)$.
3. Show that the set of dyadic rationals $\{m/2^n : m \in \mathbb{Z}, n \in \mathbb{N}\}$ is dense in $\mathbb{R}$.
4. Prove that there is a unique positive real u with $u^3 = 5$, by adapting the sup-argument for $\sqrt{2}$.
5. (Disprove.) A student claims “every nonempty bounded set of rationals has a rational supremum.” Give a counterexample and explain which property of $\mathbb{R}$ versus $\mathbb{Q}$ is at stake.
6. Prove the density of irrationals: between any two reals $x < y$ there is an irrational. (Use $\sqrt{2}$ and the density of $\mathbb{Q}$.)
7. For nonempty bounded $A, B \subseteq \mathbb{R}_{>0}$ let $AB = \{ab : a \in A, b \in B\}$. Prove $\sup(AB) = \sup A \cdot \sup B$. Show by example this fails if negatives are allowed.
8. (Diagram.) Illustrate the density theorem: for $x = \sqrt{2}$, $y = \sqrt{2} + 0.1$, find an explicit rational m/n between them and mark nx, ny, m on a scaled number line.
9. Prove that $\sup\{r \in \mathbb{Q} : r < a\} = a$ for every $a \in \mathbb{R}$, and use it to show $\mathbb{Q}$ is dense in $\mathbb{R}$ “from below.”
10. Let $x > 0$. Show $\inf\{x^{1/n} : n \in \mathbb{N}\} = 1$ if $x \geq 1$, and $\sup\{x^{1/n} : n \in \mathbb{N}\} = 1$ if $0 < x \leq 1$. (You may use Bernoulli’s inequality.)

2.5 Intervals and the Nested Interval Property

Intervals and their characterization

Definition 2.5.1 (Intervals, B&S §2.5). For $a < b$ in $\mathbb{R}$: the *open* interval $(a, b) = \{x : a < x < b\}$, the *closed* interval $[a, b] = \{x : a \leq x \leq b\}$, and the half-open $[a, b), (a, b]$. Each is *bounded* of *length* $b - a$. The *unbounded* intervals are $(a, \infty), [a, \infty), (-\infty, b), (-\infty, b]$, and $(-\infty, \infty) = \mathbb{R}$. Here $\pm\infty$ are symbols, *not* elements of $\mathbb{R}$.

Intervals are exactly the “gap-free” sets: a set is an interval iff it contains every point between any two of its points.

Theorem 2.5.2 (Characterization of intervals, B&S 2.5.1). *A subset $I \subseteq \mathbb{R}$ with at least two points is an interval iff it has the betweenness property: whenever $x, y \in I$ and $x < t < y$, then $t \in I$.*

Proof. ($\Rightarrow$) Each of the nine interval shapes is defined by inequalities preserved under “lying between,” so the property is clear. ($\Leftarrow$) Suppose I has the property. Let $a = \inf I \in [-\infty, \infty)$ and $b = \sup I \in (-\infty, \infty]$ (allow $\pm\infty$ when I is unbounded on that side). We claim $(a, b) \subseteq I \subseteq [a, b]$. If $a < t < b$, then $t > a = \inf I$ means t is not a lower bound, so some $x \in I$ has $x < t$; likewise some $y \in I$ has $t < y$. Betweenness gives $t \in I$. Hence $(a, b) \subseteq I$, and $I \subseteq [a, b]$ by definition of a, b . Reinstating whichever endpoints belong to I exhibits I as one of the four bounded or five unbounded interval types. $\square$

The nested interval property

Definition 2.5.3. A sequence of intervals $I_1, I_2, \dots$ is *nested* if $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$, i.e. $I_{n+1} \subseteq I_n$ for all n .

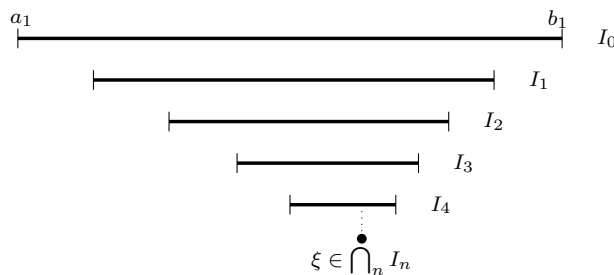


Figure 2.2. Closed bounded intervals nesting down; completeness guarantees a common point ξ .

Theorem 2.5.4 (Nested Interval Property, B&S 2.5.2 / Rudin 1.21-style). *Let $I_n = [a_n, b_n]$ be a nested sequence of closed, bounded intervals. Then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$. Concretely, $\xi = \sup_n a_n$ lies in every I_n . If moreover the lengths $b_n - a_n \rightarrow 0$, the common point is unique.*

Proof. Nesting gives $a_1 \leq a_2 \leq \dots$ and $\dots \leq b_2 \leq b_1$, and every b_m is an upper bound of $\{a_n\}$: indeed for any n, m , taking $k = \max(n, m)$, $a_n \leq a_k \leq b_k \leq b_m$. So $\{a_n\}$ is bounded above; let $\xi = \sup_n a_n$. Then $a_n \leq \xi$ for all n , and $\xi \leq b_m$ for all m (each b_m bounds the a ’s, and ξ is the *least* such bound). Hence $a_n \leq \xi \leq b_n$, i.e. $\xi \in I_n$ for every n , so $\xi \in \bigcap_n I_n$.

Uniqueness when $b_n - a_n \rightarrow 0$: if ξ, η both lie in every I_n , then $|\xi - \eta| \leq b_n - a_n$ for all n ; since this is $< \varepsilon$ for some n given any $\varepsilon > 0$, Theorem 2.1.6 forces $\xi = \eta$. (One checks $\eta = \inf_n b_n$ and the intersection is exactly $[\sup a_n, \inf b_n]$.) $\square$

Caution. Both hypotheses — closed *and* bounded — are essential. For the open nested intervals $J_n = (0, \frac{1}{n})$, $\bigcap_n J_n = \emptyset$ (any $x > 0$ is excluded once $\frac{1}{n} < x$). For the unbounded closed nested intervals $K_n = [n, \infty)$, $\bigcap_n K_n = \emptyset$ (any x is excluded once $n > x$). Only *closed bounded* intervals are guaranteed a common point.

Uncountability of $\mathbb{R}$ via nesting

The Nested Interval Property gives a second proof — independent of decimals — that $\mathbb{R}$ is too big to list.

Theorem 2.5.5 ($\mathbb{R}$ is uncountable, B&S 2.5.5). *The interval $[0, 1]$, hence $\mathbb{R}$, is uncountable.*

Proof. Suppose $[0, 1]$ were countable, listed as $x_1, x_2, x_3, \dots$. Build nested closed intervals that dodge the points one at a time. Choose a closed interval $I_1 \subseteq [0, 1]$ with $x_1 \notin I_1$ (split $[0, 1]$ into thirds; at least one third misses x_1). Having chosen I_n , split it into thirds and pick a closed third $I_{n+1} \subseteq I_n$ with $x_{n+1} \notin I_{n+1}$. The I_n are nested, closed, bounded, so by Theorem 2.5.4 there is $\xi \in \bigcap_n I_n \subseteq [0, 1]$. But $\xi \neq x_n$ for every n , since $x_n \notin I_n \ni \xi$. So $\xi \in [0, 1]$ is not in the list — contradiction. Hence $[0, 1]$ is uncountable, and so is $\mathbb{R} \supseteq [0, 1]$. $\square$

Binary and decimal representations

Nesting also explains *why* every real has a decimal (or binary) expansion: the successive digits name a shrinking nest of intervals closing on the number.

Theorem 2.5.6 (Decimal expansions, B&S 2.5.6–2.5.7). *Every $x \in [0, 1]$ has a decimal representation $x = 0.d_1d_2d_3\ldots$ with digits $d_k \in \{0, 1, \dots, 9\}$, meaning $x = \sup_n \sum_{k=1}^n d_k 10^{-k}$. The representation is unique except for the dyadic-type ambiguity, where a number ending in repeating 9s equals one ending in repeating 0s, e.g. $0.4999\ldots = 0.5000\ldots$. The analogous statement holds in base 2 with digits $\{0, 1\}$.*

Proof sketch. Subdivide $[0, 1]$ into ten equal closed subintervals $[\frac{j}{10}, \frac{j+1}{10}]$; x lies in one, fixing $d_1 = j$. Subdivide that into ten, fixing d_2 , and recurse. The chosen intervals are nested with lengths $10^{-n} \rightarrow 0$; their unique common point (Theorem 2.5.4) is x , and the partial sums $\sum_{k \leq n} d_k 10^{-k}$ are the left endpoints, increasing to x . The only freedom is at a grid point, where x is the shared endpoint of two subintervals, yielding the trailing-9 vs trailing-0 pair. $\square$

Worked Example 2.5.7 (A binary expansion). Find the binary representation of $\frac{3}{8}$ and of $\frac{1}{3}$.

Recognition. “Expand in base 2”: repeatedly double and read off the integer part (the digit).

Solution. $\frac{3}{8}$: $2 \cdot \frac{3}{8} = \frac{3}{4}$ (digit 0), $2 \cdot \frac{3}{4} = \frac{3}{2}$ (digit 1, keep $\frac{1}{2}$), $2 \cdot \frac{1}{2} = 1$ (digit 1, remainder 0). So $\frac{3}{8} = 0.011_2$, terminating (also $= 0.010111\ldots_2$ by the 9s-analogue). For $\frac{1}{3}$: $2 \cdot \frac{1}{3} = \frac{2}{3}$ (digit 0), $2 \cdot \frac{2}{3} = \frac{4}{3}$ (digit 1, keep $\frac{1}{3}$), and it repeats: $\frac{1}{3} = 0.010101\ldots_2 = \overline{01}_2$. *Check.* $0 + \frac{1}{4} + \frac{1}{16} + \cdots = \frac{1/4}{1-1/4} = \frac{1}{3}$. $\blacksquare$

Worked Example 2.5.8 (A periodic decimal is rational). Show $0.137137137\ldots$ is rational and find it as a fraction.

Recognition. Repeating block of length 3; multiply by 10^3 and subtract.

Solution. Let $x = 0.\overline{137}$. Then $1000x = 137.\overline{137} = 137 + x$, so $999x = 137$ and $x = \frac{137}{999}$. *Check.* $\frac{137}{999} = 0.137137\ldots$ by long division. $\blacksquare$ (Conversely every rational has an eventually-periodic expansion, since the remainders in long division must repeat within q steps for denominator q .)

Further worked examples

Worked Example 2.5.9 (The intersection of nested closed intervals is a closed interval). For $I_n = [-\frac{1}{n}, 1 + \frac{1}{n}]$, verify the I_n are nested closed bounded intervals and compute $\bigcap_{n=1}^{\infty} I_n$.

Recognition. Trigger: “apply the Nested Interval Property.” Check nesting via the endpoints, then use $\xi = \sup a_n, \inf b_n$ from Theorem 2.5.4.

Solution. The left endpoints $a_n = -\frac{1}{n}$ increase to 0; the right endpoints $b_n = 1 + \frac{1}{n}$ decrease to 1. Since $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$, we have $I_{n+1} \subseteq I_n$: nested, each closed and bounded. A point x lies in every I_n iff $-\frac{1}{n} \leq x \leq 1 + \frac{1}{n}$ for all n ; letting $n \rightarrow \infty$ this forces $0 \leq x \leq 1$, and conversely every $x \in [0, 1]$ satisfies all the inequalities. Hence $\bigcap_n I_n = [0, 1] = [\sup a_n, \inf b_n]$. *Check.* The lengths $b_n - a_n = 1 + \frac{2}{n} \rightarrow 1 \neq 0$, so the intersection is a nondegenerate interval, not a single point — consistent with the “unique point” clause requiring lengths $\rightarrow 0$. $\blacksquare$

Worked Example 2.5.10 (NON-example: dropping “closed” empties the intersection). Show the nested half-open intervals $J_n = (0, \frac{1}{n}]$ have empty intersection, even though each is bounded and the right endpoints are included.

Recognition. Trigger: “test the hypotheses of the NIP.” The property needs *closed* intervals; probe what fails when the left end stays open.

Solution. The J_n are nested ($\frac{1}{n+1} \leq \frac{1}{n}$ gives $J_{n+1} \subseteq J_n$) and bounded. Suppose $x \in \bigcap_n J_n$. Then $0 < x \leq \frac{1}{n}$ for every n , so $x > 0$ is a lower bound issue: by the Archimedean property (§2.4) there is n with $\frac{1}{n} < x$, contradicting $x \leq \frac{1}{n}$. Hence no such x exists and $\bigcap_n J_n = \emptyset$.

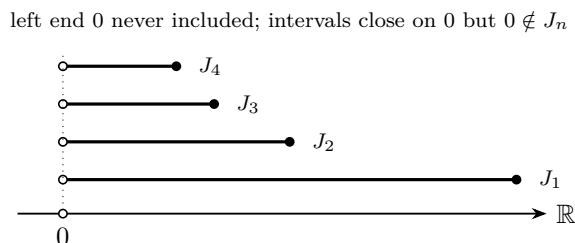


Figure 2.3. Open left endpoints (hollow) let the nest shrink to the single excluded point 0, leaving $\bigcap_n J_n = \emptyset$.

Check. Any candidate like $x = 0.001$ is expelled once $\frac{1}{n} < 0.001$, i.e. $n \geq 1001$; and 0 itself is excluded because each J_n omits its left endpoint. ■ The failure is the open left end, exactly the gap the *closed* hypothesis of Theorem 2.5.4 closes.

Worked Example 2.5.11 (A repeating decimal with a non-repeating prefix). Express $x = 0.41\overline{6} = 0.41666\dots$ as a fraction in lowest terms.

Recognition. Trigger: “eventually-periodic decimal to fraction.” Shift past the non-repeating prefix, then subtract to kill the repeating tail (as in Theorem 2.5.6).

Solution. The non-repeating part has length 2 and the repeating block (“6”) length 1. Multiply to align: $1000x = 416.\overline{6}$ and $100x = 41.\overline{6}$. Subtracting, $900x = 375$, so $x = \frac{375}{900} = \frac{5}{12}$ after dividing by 75. *Check.* $\frac{5}{12} = 0.41\overline{6}$ by long division ($5.000/12 = 0.4166\dots$). ■ Every such eventually-periodic expansion is rational; the converse (Theorem 2.5.6 discussion) also holds.

Exercises (Bartle & Sherbert, §2.5).

1. If $I = [a, b]$ and $I' = [a', b']$ are closed intervals, show $I \subseteq I'$ iff $a' \leq a$ and $b \leq b'$.
2. If $S \subseteq \mathbb{R}$ is nonempty, show S is bounded iff there is a closed bounded interval I with $S \subseteq I$.
3. If $S \subseteq \mathbb{R}$ is nonempty and bounded and $I_S = [\inf S, \sup S]$, show $S \subseteq I_S$, and that any closed bounded interval J with $S \subseteq J$ satisfies $I_S \subseteq J$.
4. In the characterization-of-intervals proof, explain why the points x, y exist in S .
5. Write out the details of the remaining case in the characterization of intervals.
6. If $I_1 \supseteq I_2 \supseteq \dots$ is nested with $I_n = [a_n, b_n]$, show $a_1 \leq a_2 \leq \dots$ and $\dots \leq b_2 \leq b_1$.
7. Let $I_n = [0, 1/n]$ for $n \in \mathbb{N}$. Prove $\bigcap_{n=1}^{\infty} I_n = \{0\}$.
8. Let $J_n = (0, 1/n)$ for $n \in \mathbb{N}$. Prove $\bigcap_{n=1}^{\infty} J_n = \emptyset$.
9. Let $K_n = (n, \infty)$ for $n \in \mathbb{N}$. Prove $\bigcap_{n=1}^{\infty} K_n = \emptyset$.
10. With $\xi = \sup a_n$ and $\eta = \inf b_n$, show $\eta \in \bigcap_n I_n$ and $[\xi, \eta] = \bigcap_{n=1}^{\infty} I_n$.
11. Show that the intervals obtained from the bisection inequalities form a nested sequence.
12. Give the two binary representations of $\frac{3}{8}$ and $\frac{7}{16}$.
13. (a) Give the first four digits in the binary representation of $\frac{1}{3}$. (b) Give the complete binary representation of $\frac{1}{3}$.
14. Show that if $a_k, b_k \in \{0, 1, \dots, 9\}$ and $\sum_{k=1}^n a_k 10^{-k} = \sum_{k=1}^m b_k 10^{-k} \neq 0$, then $n = m$ and $a_k = b_k$ for all k .
15. Find the decimal representation of $-\frac{2}{7}$.
16. Express $\frac{1}{17}$ and $\frac{2}{19}$ as periodic decimals.
17. What rationals are represented by the periodic decimals $1.25\overline{137}$ and $35.14\overline{653}$?

Additional exercises.

1. Classify each set as an interval or not, using the betweenness characterization: (a) $\mathbb{Z}$; (b) $\mathbb{Q} \cap [0, 1]$; (c) $[0, 1] \cup [2, 3]$; (d) $\{x : x^2 < 4\}$.
2. Compute $\bigcap_{n=1}^{\infty} [-1/n, 1 + 1/n]$ and $\bigcap_{n=1}^{\infty} (-1/n, 1 + 1/n)$, and explain the difference using the Nested Interval Property.
3. (Disprove.) Give a nested sequence of *bounded* but non-closed intervals with empty intersection, and a nested sequence of *closed* but unbounded intervals with empty intersection. Why does each evade the Nested Interval Property?
4. Express the repeating decimals $0.\overline{6}$, $0.1\overline{6}$, and $2.\overline{142857}$ as fractions in lowest terms.
5. Find the binary expansion of $\frac{5}{16}$ and of $\frac{2}{5}$, indicating which terminates and which repeats.
6. Prove that a real number has a terminating decimal expansion iff it is of the form $m/(2^a 5^b)$ for integers $m, a, b \geq 0$.
7. (Diagram.) Draw four nested closed intervals $I_1 \supseteq \cdots \supseteq I_4$ in $[0, 1]$ produced by repeated bisection toward $\frac{1}{3}$, and mark the common point.
8. Use the Nested Interval Property to give a proof that a bounded monotone increasing sequence of reals has a least upper bound that it “reaches in the limit” (informal preview of §4).
9. Re-prove that $[0, 1]$ is uncountable by the bisection/nesting method, writing out how the interval I_{n+1} is chosen to exclude x_{n+1} .

2.6 The Complex Field and Euclidean Space $\mathbb{R}^k$

The complex field

Definition 2.6.1 (The field $\mathbb{C}$, Rudin 1.24–1.25). A *complex number* is an ordered pair (a, b) of reals, written $a + bi$. Addition and multiplication are

$$(a + bi) + (c + di) = (a + c) + (b + d)i, \quad (a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

With these, $\mathbb{C}$ is a field: $0 = (0, 0)$, $1 = (1, 0)$, and $i = (0, 1)$ satisfies $i^2 = -1$. The map $a \mapsto a + 0i$ embeds $\mathbb{R}$ as a subfield, so we write a for $a + 0i$.

The multiplication rule is exactly what “ $i^2 = -1$ ” forces: $(a + bi)(c + di) = ac + adi + bci + bd i^2$. One checks the field axioms directly; the inverse of $z \neq 0$ is computed below via the conjugate.

Definition 2.6.2 (Conjugate, real/imaginary parts, modulus, Rudin 1.26–1.28). For $z = a + bi$: the *conjugate* is $\bar{z} = a - bi$; the *real* and *imaginary* parts are $\operatorname{Re} z = a$, $\operatorname{Im} z = b$; the *modulus* (absolute value) is $|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}} \geq 0$.

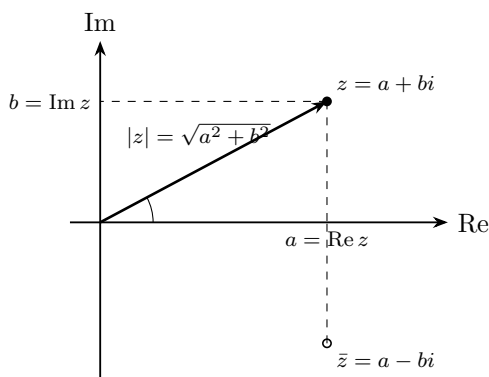


Figure 2.4. $\mathbb{C}$ as the plane: z is the vector to (a, b) , $|z|$ its length, $\bar{z}$ its reflection across the real axis.

Theorem 2.6.3 (Algebra of conjugate and modulus, Rudin 1.31). For $z, w \in \mathbb{C}$: (a) $\overline{z+w} = \bar{z} + \bar{w}$, $\overline{z\bar{w}} = \bar{z} w$, $\bar{\bar{z}} = z$; (b) $z + \bar{z} = 2 \operatorname{Re} z$, $z - \bar{z} = 2i \operatorname{Im} z$; and z is real iff $z = \bar{z}$; (c) $z\bar{z} = |z|^2 \geq 0$, so $z \neq 0 \implies z^{-1} = \bar{z}/|z|^2$; (d) $|zw| = |z||w|$ and $|\operatorname{Re} z| \leq |z|$, $|\operatorname{Im} z| \leq |z|$; (e) (triangle inequality) $|z+w| \leq |z| + |w|$.

Proof. (a),(b) are direct from $z = a + bi$. (c): $z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2$; dividing gives the inverse. (d): $|zw|^2 = zw\overline{zw} = z\bar{z}w\bar{w} = |z|^2|w|^2$, and $|\operatorname{Re} z| = |a| \leq \sqrt{a^2 + b^2} = |z|$. (e) Compute $|z+w|^2 = (z+w)(\overline{z+w}) = z\bar{z} + w\bar{w} + (z\bar{w} + \bar{z}w) = |z|^2 + |w|^2 + 2\operatorname{Re}(z\bar{w})$. By (d), $\operatorname{Re}(z\bar{w}) \leq |z\bar{w}| = |z||w|$, so $|z+w|^2 \leq |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2$. Take square roots. $\square$

Euclidean space $\mathbb{R}^k$

Definition 2.6.4 ($\mathbb{R}^k$, inner product, norm, Rudin 1.36). $\mathbb{R}^k$ is the set of ordered k -tuples $\mathbf{x} = (x_1, \dots, x_k)$ of reals, with componentwise addition and scalar multiplication $\alpha\mathbf{x} = (\alpha x_1, \dots, \alpha x_k)$. The *inner (dot) product* and *norm* are

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^k x_i y_i, \quad \|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \left(\sum_{i=1}^k x_i^2 \right)^{1/2}.$$

For $k = 2$ (identifying (a, b) with $a + bi$), $\|\mathbf{x}\|$ is exactly the complex modulus.

The dot product packages length and angle at once: $\mathbf{x} \cdot \mathbf{x}$ is the squared length, while for distinct vectors $\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$ encodes the angle between them. The norm $\|\cdot\|$ is just the Pythagorean length, and the whole of Euclidean geometry in $\mathbb{R}^k$ flows from these two formulas plus the inequality below.

Theorem 2.6.5 (Cauchy–Schwarz inequality, Rudin 1.35). For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$,

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|,$$

i.e. $(\sum x_i y_i)^2 \leq (\sum x_i^2)(\sum y_i^2)$, with equality iff $\mathbf{x}, \mathbf{y}$ are parallel (one a scalar multiple of the other).

Proof. If $\mathbf{y} = \mathbf{0}$ both sides vanish. Otherwise, for every $t \in \mathbb{R}$,

$$0 \leq \|\mathbf{x} - t\mathbf{y}\|^2 = \|\mathbf{x}\|^2 - 2t(\mathbf{x} \cdot \mathbf{y}) + t^2 \|\mathbf{y}\|^2.$$

This quadratic in t is ≥ 0 for all t , so its discriminant is ≤ 0 : $4(\mathbf{x} \cdot \mathbf{y})^2 - 4\|\mathbf{x}\|^2 \|\mathbf{y}\|^2 \leq 0$, which is exactly $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$. Equality forces the quadratic to have a real root t_0 , i.e. $\|\mathbf{x} - t_0\mathbf{y}\| = 0$, i.e. $\mathbf{x} = t_0\mathbf{y}$. (Rudin gives the equivalent algebraic identity $\|\mathbf{x}\|^2 \|\mathbf{y}\|^2 - (\mathbf{x} \cdot \mathbf{y})^2 = \frac{1}{2} \sum_{i,j} (x_i y_j - x_j y_i)^2 \geq 0$, the Lagrange identity, which exhibits the gap as a sum of squares.) $\square$

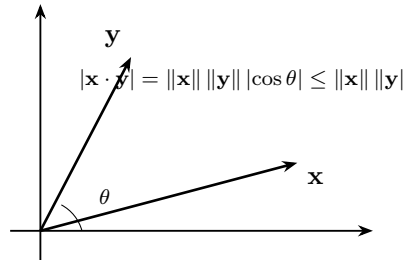


Figure 2.5. Cauchy–Schwarz says $|\cos \theta| \leq 1$: the dot product never exceeds the product of lengths.

Theorem 2.6.6 (Norm properties / triangle inequality in $\mathbb{R}^k$, Rudin 1.37). For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$ and $\alpha \in \mathbb{R}$: (a) $\|\mathbf{x}\| \geq 0$, with $= 0$ iff $\mathbf{x} = \mathbf{0}$; (b) $\|\alpha\mathbf{x}\| = |\alpha| \|\mathbf{x}\|$; (c) (triangle inequality) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$; (d) (reverse) $|\|\mathbf{x}\| - \|\mathbf{y}\|| \leq \|\mathbf{x} - \mathbf{y}\|$.

Proof. (a),(b) are immediate. (c): using Cauchy–Schwarz,

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2;$$

take roots. (d) follows from (c) exactly as in Corollary 2.2.3(i): $\|\mathbf{x}\| = \|(\mathbf{x} - \mathbf{y}) + \mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\|$, and symmetrically. $\square$

Remark. Theorem 2.6.6(a),(b),(c) say $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ is a *metric* on $\mathbb{R}^k$ — the setting for all of Chapter 3. The complex modulus is the $k = 2$ case; the real absolute value is $k = 1$. Thus $|\cdot|$, the complex $|\cdot|$, and $\|\cdot\|$ are one idea in dimensions 1, 2, k .

Worked Example 2.6.7 (Cauchy–Schwarz with explicit vectors). Verify $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$ for $\mathbf{x} = (1, 2, 2)$, $\mathbf{y} = (2, -1, 2)$, and identify $\cos \theta$.

Recognition. “Check Cauchy–Schwarz numerically.”

Solution. $\mathbf{x} \cdot \mathbf{y} = 2 - 2 + 4 = 4$. $\|\mathbf{x}\| = \sqrt{1 + 4 + 4} = 3$, $\|\mathbf{y}\| = \sqrt{4 + 1 + 4} = 3$. So $|4| = 4 \leq 9 = 3 \cdot 3$. ■ Here $\cos \theta = \frac{4}{9}$, strictly between -1 and 1 , so the vectors are not parallel (consistent with strict inequality). *Check.* Lagrange gap $9 \cdot 9 - 4^2 = 65 = \sum_{i < j} (x_i y_j - x_j y_i)^2 = (-1 - 4)^2 + (2 - 4)^2 + (4 + 2)^2 = 25 + 4 + 36 = 65$. ■

Worked Example 2.6.8 (The parallelogram law (Rudin Exercise 1.17)). Prove $\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2\|\mathbf{x}\|^2 + 2\|\mathbf{y}\|^2$ and interpret it.

Recognition. “Prove a norm identity”: expand both squares via the inner product.

Solution. Expand: $\|\mathbf{x} \pm \mathbf{y}\|^2 = \|\mathbf{x}\|^2 \pm 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2$. Adding the $+$ and $-$ versions cancels the cross terms, leaving $2\|\mathbf{x}\|^2 + 2\|\mathbf{y}\|^2$. *Interpretation:* in a parallelogram with sides $\mathbf{x}, \mathbf{y}$, the sum of the squares of the two diagonals equals the sum of the squares of all four sides. *Check.* $\mathbf{x} = (1, 0), \mathbf{y} = (0, 1)$: $\|(1, 1)\|^2 + \|(1, -1)\|^2 = 2 + 2 = 4 = 2(1) + 2(1)$. ■

Worked Example 2.6.9 (NON-orderability of $\mathbb{C}$ (Rudin Exercise 1.8)). Show $\mathbb{C}$ cannot be made an ordered field.

Recognition. “Disprove a structure exists”: use a forced-sign contradiction.

Solution. In any ordered field, squares are ≥ 0 and $1 > 0$ (Theorem 2.1.5(c)). If $\mathbb{C}$ were ordered, then $i^2 = -1$ would give $-1 \geq 0$, yet $1 = 1^2 > 0$ forces $-1 < 0$ — contradiction. So no order makes $\mathbb{C}$ an ordered field; the complex numbers carry size ($|z|$) but no compatible notion of “positive.” *Check.* The same argument bars an order on any field where -1 is a sum of squares. ■

Caution. $\mathbb{C}$ has no order: “ $z < w$ ” for complex numbers is meaningless. You may compare *moduli* $|z| < |w|$ (real numbers), but never the complex numbers themselves. Inequality arguments in $\mathbb{C}$ always run through $|\cdot|$, Re , or Im , never through a nonexistent complex order.

Further worked examples

Worked Example 2.6.10 (Arithmetic, conjugate, and inverse of a complex number). For $z = 3 + 4i$ and $w = 1 - 2i$, compute zw , $|z|$, and z^{-1} in the form $a + bi$.

Recognition. “Mechanics of $\mathbb{C}$ ”: apply the multiplication rule and Theorem 2.6.3(c), $z^{-1} = \bar{z}/|z|^2$.

Solution. $zw = (3+4i)(1-2i) = 3-6i+4i-8i^2 = 3-2i+8 = 11-2i$. The modulus is $|z| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

Then $z^{-1} = \frac{\bar{z}}{|z|^2} = \frac{3-4i}{25} = \frac{3}{25} - \frac{4}{25}i$. *Check.* $z \cdot z^{-1} = (3+4i) \cdot \frac{3-4i}{25} = \frac{9+16}{25} = \frac{25}{25} = 1$. ■ Also

$|zw| = |11-2i| = \sqrt{121+4} = \sqrt{125} = 5\sqrt{5} = |z||w|$ since $|w| = \sqrt{5}$, confirming Theorem 2.6.3(d). ■

Worked Example 2.6.11 (Cauchy–Schwarz equality forces parallel vectors). For $\mathbf{x} = (1, 2, 3)$ and $\mathbf{y} = (2, 4, 6)$ in $\mathbb{R}^3$, verify that Cauchy–Schwarz holds with *equality*, and explain why.

Recognition. Trigger: “the equality case” of Theorem 2.6.5: equality holds iff the vectors are parallel. Here $\mathbf{y} = 2\mathbf{x}$.

Solution. $\mathbf{x} \cdot \mathbf{y} = 2 + 8 + 18 = 28$. $\|\mathbf{x}\| = \sqrt{1 + 4 + 9} = \sqrt{14}$, $\|\mathbf{y}\| = \sqrt{4 + 16 + 36} = \sqrt{56} = 2\sqrt{14}$. So $\|\mathbf{x}\| \|\mathbf{y}\| = \sqrt{14} \cdot 2\sqrt{14} = 28 = |\mathbf{x} \cdot \mathbf{y}|$: *equality*. The reason is $\mathbf{y} = 2\mathbf{x}$, so the vectors are parallel ($\cos \theta = 1$). *Check.* The Lagrange gap $\|\mathbf{x}\|^2 \|\mathbf{y}\|^2 - (\mathbf{x} \cdot \mathbf{y})^2 = 14 \cdot 56 - 28^2 = 784 - 784 = 0$, a sum of squares $(x_i y_j - x_j y_i)^2$ all vanishing because $x_i y_j = x_j y_i$ for the proportional coordinates. ■

Exercises (Rudin, Ch.1 – complex numbers and $\mathbb{R}^k$).

1. Prove that no order can be defined on the complex field that turns it into an ordered field. (Hint: -1 is a square.)

2. Suppose $z = a + bi$ and $w = c + di$. Define $z < w$ if $a < c$, or if $a = c$ but $b < d$. Prove this makes $\mathbb{C}$ an ordered set (lexicographic / dictionary order). Does this ordered set have the least-upper-bound property?
3. Suppose $z = a + bi$, $w = u + iv$, and $a = \left(\frac{|w|+u}{2}\right)^{1/2}$, $b = \left(\frac{|w|-u}{2}\right)^{1/2}$. Prove $z^2 = w$ if $v \geq 0$ and $(\bar{z})^2 = w$ if $v \leq 0$. Conclude that every complex number (with one exception) has two complex square roots.
4. If $z \in \mathbb{C}$, prove there exist $r \geq 0$ and $w \in \mathbb{C}$ with $|w| = 1$ such that $z = rw$. Are w and r always uniquely determined by z ?
5. If $z_1, \dots, z_n \in \mathbb{C}$, prove $|z_1 + z_2 + \dots + z_n| \leq |z_1| + \dots + |z_n|$.
6. If $x, y \in \mathbb{C}$, prove $||x| - |y|| \leq |x - y|$.
7. If $z \in \mathbb{C}$ with $|z| = 1$ (i.e. $z\bar{z} = 1$), compute $|1 + z|^2 + |1 - z|^2$.
8. Under what conditions does equality hold in the Schwarz inequality?
9. Suppose $k \geq 3$, $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$, $|\mathbf{x} - \mathbf{y}| = d > 0$, and $r > 0$. Prove: (a) if $2r > d$ there are infinitely many $\mathbf{z} \in \mathbb{R}^k$ with $|\mathbf{z} - \mathbf{x}| = |\mathbf{z} - \mathbf{y}| = r$; (b) if $2r = d$ there is exactly one; (c) if $2r < d$ there is none. How must these change for $k = 1, 2$?
10. Prove $|\mathbf{x} + \mathbf{y}|^2 + |\mathbf{x} - \mathbf{y}|^2 = 2|\mathbf{x}|^2 + 2|\mathbf{y}|^2$ for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$. Interpret geometrically as a statement about parallelograms.
11. If $k \geq 2$ and $\mathbf{x} \in \mathbb{R}^k$, prove there exists $\mathbf{y} \in \mathbb{R}^k$ with $\mathbf{y} \neq \mathbf{0}$ but $\mathbf{x} \cdot \mathbf{y} = 0$. Is this also true if $k = 1$?
12. Suppose $\mathbf{a}, \mathbf{b} \in \mathbb{R}^k$. Find $\mathbf{c} \in \mathbb{R}^k$ and $r > 0$ such that $|\mathbf{x} - \mathbf{a}| = 2|\mathbf{x} - \mathbf{b}|$ iff $|\mathbf{x} - \mathbf{c}| = r$. (Solution: $3\mathbf{c} = 4\mathbf{b} - \mathbf{a}$, $3r = 2|\mathbf{b} - \mathbf{a}|$.)

Additional exercises.

1. For $z = 3 + 4i$ compute $\bar{z}$, $|z|$, z^{-1} , $\operatorname{Re} z$, $\operatorname{Im} z$, and verify $zz^{-1} = 1$.
2. Prove $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$, with equality iff z is real (resp. purely imaginary).
3. (Disprove.) A student writes “ $2 + 3i < 4 + i$ because $2 < 4$.” Explain precisely why this is meaningless, citing the non-orderability of $\mathbb{C}$.
4. Verify the Cauchy–Schwarz inequality for $\mathbf{x} = (1, 1, 1, 1)$, $\mathbf{y} = (1, 2, 3, 4)$, and compute the angle’s cosine.
5. Prove the polarization identity $\mathbf{x} \cdot \mathbf{y} = \frac{1}{4}(\|\mathbf{x} + \mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2)$ in $\mathbb{R}^k$.
6. Show that equality $\|\mathbf{x} + \mathbf{y}\| = \|\mathbf{x}\| + \|\mathbf{y}\|$ holds iff $\mathbf{x}, \mathbf{y}$ point the same way ($\mathbf{x} = \lambda\mathbf{y}$ or $\mathbf{y} = \lambda\mathbf{x}$ with $\lambda \geq 0$).
7. For $z, w \in \mathbb{C}$ prove $|z + w|^2 + |z - w|^2 = 2|z|^2 + 2|w|^2$ and interpret it in the plane.
8. (Diagram.) In the complex plane plot $z = 2 + i$, its conjugate $\bar{z}$, iz , and $-z$, and describe each as a geometric transformation of z (reflection, rotation, point reflection).
9. Prove that for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k$, $\|\mathbf{x} - \mathbf{y}\| \geq \left| \|\mathbf{x}\| - \|\mathbf{y}\| \right|$, and identify when equality holds.
10. Given $z \in \mathbb{C}$ with $|z| < 1$, prove $\operatorname{Re}\left(\frac{1+z}{1-\bar{z}}\right) > 0$. (Hint: multiply numerator and denominator by $\overline{1 - \bar{z}}$.)

Chapter 3

Basic Topology and Metric Spaces

3.1 Metric Spaces and Neighborhoods

The metric axioms

Analysis on the real line rests on one number: the distance $|x - y|$ between two points. Almost every limit argument used only a few properties of that distance — it is nonnegative, vanishes exactly when the points coincide, is symmetric, and satisfies the triangle inequality. Pulling those properties out and naming them gives the single most useful abstraction in the subject.

Definition 3.1.1 (Metric, Rudin 2.15 / B&S 11.4.1). A *metric* on a set X is a function $d : X \times X \rightarrow \mathbb{R}$ such that for all $p, q, r \in X$:

- (a) $d(p, q) \geq 0$, and $d(p, q) = 0 \iff p = q$ (*positivity / definiteness*);
- (b) $d(p, q) = d(q, p)$ (*symmetry*);
- (c) $d(p, q) \leq d(p, r) + d(r, q)$ (*triangle inequality*).

The pair (X, d) is a *metric space*; its elements are called *points*. A function with these properties is a *distance function*.

The triangle inequality is the workhorse: every estimate “ $d(p, q)$ is small” is built by routing through an intermediate point r and adding two small distances. The definiteness clause (b) is what distinguishes a metric from a *semimetric*, where one only requires $d(p, p) = 0$ but allows distinct points at distance 0 (B&S 11.4.13); we will always assume a genuine metric.

Remark. Every subset $Y \subseteq X$ is itself a metric space under the *same* d restricted to $Y \times Y$: the three axioms, holding for all points of X , certainly hold for points of Y . Thus every subset of a Euclidean space is a metric space in its own right — a fact we exploit constantly via the *relative* topology (§3.2).

The standard examples

Example 3.1.2 (The fundamental examples; Rudin 2.16, B&S 11.4.2).

- (a) **The real line** $\mathbb{R}$ with $d(x, y) = |x - y|$. Axiom (c) is the triangle inequality for absolute value, $|x - y| \leq |x - z| + |z - y|$.
- (b) **Euclidean space** $\mathbb{R}^k$ with $d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}| = (\sum_{j=1}^k (x_j - y_j)^2)^{1/2}$. The triangle inequality here is the Cauchy–Schwarz inequality (Rudin 1.37); $\mathbb{R}^2$ is the complex plane.
- (c) **The taxicab and sup metrics on** $\mathbb{R}^k$: $d_1(\mathbf{x}, \mathbf{y}) = \sum_j |x_j - y_j|$ and $d_\infty(\mathbf{x}, \mathbf{y}) = \max_j |x_j - y_j|$. Each is a metric on the same underlying set $\mathbb{R}^k$: one set, several metrics.

(d) **The discrete metric** on any nonempty set X :

$$d(p, q) = \begin{cases} 0, & p = q, \\ 1, & p \neq q. \end{cases}$$

Here every point sits at distance 1 from all others.

(e) **The sup (uniform) metric.** On the space $C[0, 1]$ of continuous functions on $[0, 1]$, set $d_\infty(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)| = \|f - g\|_\infty$. Convergence in this metric is exactly *uniform* convergence (Chapter 8).

(f) **The L^1 metric.** On the same $C[0, 1]$, set $d_1(f, g) = \int_0^1 |f - g|$ — the same underlying set as (e), but a genuinely different metric.

Worked Example 3.1.3 (Verifying the discrete metric). Show $d(p, q) = \mathbf{1}[p \neq q]$ is a metric on any nonempty set X .

Recognition. Trigger: “show a given d is a metric.” Check the three axioms; the only work is the triangle inequality, handled by cases.

Solution. (a) $d \geq 0$ and $d(p, q) = 0 \iff p = q$ by definition. (b) Symmetry is clear since “ $p \neq q$ ” is symmetric. (c) Fix p, q, r . If $p = q$ then $d(p, q) = 0 \leq$ anything. If $p \neq q$ then $d(p, q) = 1$; but $p \neq q$ forces $r \neq p$ or $r \neq q$ (else $p = r = q$), so $d(p, r) + d(r, q) \geq 1 = d(p, q)$. *Check.* The largest possible distance is 1, so the triangle inequality can never fail by more than the slack $1 + 1 \geq 1$. Every singleton ball $B(p, \frac{1}{2}) = \{p\}$, which already hints (Worked Example 3.2.3) that *every* subset is open.

Balls, neighborhoods, bounded sets, diameter

Definition 3.1.4 (Rudin 2.18(a), 2.17; B&S 11.4.3). Let (X, d) be a metric space, $p \in X$, $r > 0$.

- The *open ball* (or *r -neighborhood*) of p is $B(p, r) = N_r(p) = \{q \in X : d(p, q) < r\}$; the *closed ball* replaces $<$ by $\leq$.
- A *neighborhood* of p is any set V containing some open ball $B(p, r)$, $r > 0$.
- A set $E \subseteq X$ is *bounded* if there exist $q \in X$ and $M > 0$ with $d(p, q) \leq M$ for all $p \in E$ (equivalently, E lies in some ball).
- The *diameter* of a nonempty E is $\text{diam } E = \sup\{d(p, q) : p, q \in E\}$ (possibly $+\infty$); E is bounded iff $\text{diam } E < \infty$.

In $\mathbb{R}^1$ a neighborhood $N_r(p)$ is the segment $(p - r, p + r)$; in $\mathbb{R}^2$ it is the open disk of radius r . Rudin’s “ $N_r(p)$ ” and B&S’s “ $V_\varepsilon(x_0)$ ” name the same object. The loosened notion — a neighborhood is *any* set containing a ball — is convenient because it lets us say “ E is a neighborhood of p ” without fixing a radius.

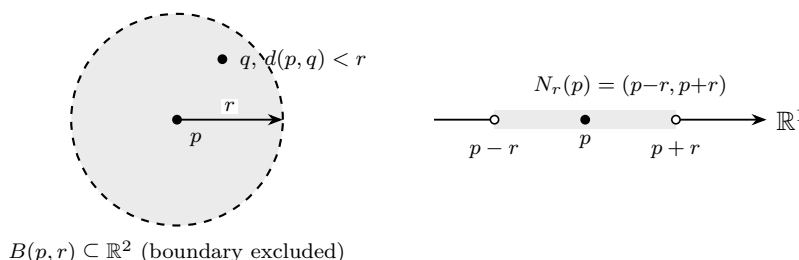


Figure 3.1. An open ball in $\mathbb{R}^2$ is an open disk; in $\mathbb{R}^1$ it is a symmetric open segment. The bounding circle / endpoints are *not* included.

Worked Example 3.1.5 (Diameter of a ball). In $\mathbb{R}^k$ show $\text{diam } B(p, r) = 2r$, but in a general metric space $\text{diam } B(p, r)$ can be strictly smaller than $2r$.

Recognition. “Compute a diameter”: bound the sup above by the triangle inequality, then exhibit a near-extremal pair.

Solution. For any $x, y \in B(p, r)$, $d(x, y) \leq d(x, p) + d(p, y) < r + r = 2r$, so $\text{diam } B(p, r) \leq 2r$. In $\mathbb{R}^k$ take $x = p + (r - \delta)\mathbf{u}$, $y = p - (r - \delta)\mathbf{u}$ for a unit vector $\mathbf{u}$; then $d(x, y) = 2(r - \delta) \rightarrow 2r$, so the sup is $2r$.

Non-Euclidean check: in the discrete metric with $r = 2$, $B(p, 2) = X$ but any two distinct points are at distance 1, so $\text{diam } B(p, 2) \leq 1 < 4 = 2r$. The clean formula $\text{diam} = 2r$ is special to $\mathbb{R}^k$.

Worked Example 3.1.6 (The three metrics on $\mathbb{R}^2$ give nested balls). Compare the unit balls of d_1 , d (Euclidean), and d_∞ at the origin in $\mathbb{R}^2$.

Recognition. “Draw/compare balls of different metrics on a common set”: translate each definition into a region.

Solution. $B_{d_1}(O, 1) = \{(x, y) : |x| + |y| < 1\}$ is an open diamond (square rotated 45°); $B_d(O, 1) = \{x^2 + y^2 < 1\}$ is the open disk; $B_{d_\infty}(O, 1) = \{\max(|x|, |y|) < 1\}$ is the open axis-aligned square $(-1, 1)^2$. Since $\max \leq \sqrt{x^2 + y^2} \leq |x| + |y|$, we get the inclusions $B_{d_1} \subseteq B_d \subseteq B_{d_\infty}$. *Check.* The point $(0.9, 0.9)$ has $d_\infty = 0.9 < 1$ (inside the square), $d = \sqrt{1.62} \approx 1.27 > 1$ (outside the disk), consistent with the nesting.

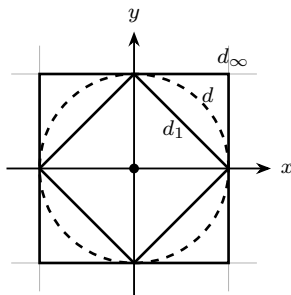


Figure 3.2. Unit balls at the origin: diamond (d_1) $\subseteq$ disk (d) $\subseteq$ square (d_∞). All three metrics induce the same open sets (§3.2).

Method (Verifying that a candidate d is a metric). *Recognition trigger.* You are handed a formula $d(p, q)$ and asked whether it is a metric (Rudin Ex. 11 lists five candidates on $\mathbb{R}^1$; only some pass).

1. **Positivity/definiteness:** confirm $d \geq 0$ everywhere and that $d(p, q) = 0$ forces $p = q$. (Squaring a metric, e.g. $(x - y)^2$, usually kills the triangle inequality; a non-injective inner function, e.g. $|x^2 - y^2|$, usually kills definiteness since $d(1, -1) = 0$.)
2. **Symmetry:** check $d(p, q) = d(q, p)$ — typically immediate.
3. **Triangle inequality:** the hard axiom. For $|\cdot|$ -based formulas, reduce to the absolute-value triangle inequality; for $\sqrt{|x - y|}$ use $\sqrt{a + b} \leq \sqrt{a} + \sqrt{b}$; to *disprove* it, find a concrete failing triple.

Worked Example 3.1.7 (Which of $(x - y)^2$ and $\sqrt{|x - y|}$ is a metric? (Rudin Ex. 11)). On $\mathbb{R}^1$ test $d_1(x, y) = (x - y)^2$ and $d_2(x, y) = \sqrt{|x - y|}$.

Recognition. “Decide if a formula is a metric”: apply the method; one fails the triangle inequality, one passes.

Solution. d_1 : positivity and symmetry hold, but the triangle inequality fails. Take $x = 0, y = 2$: $d_1(0, 2) = 4$ while $d_1(0, 1) + d_1(1, 2) = 1 + 1 = 2 < 4$. So d_1 is *not* a metric. d_2 : positivity/definiteness and symmetry are clear. For the triangle inequality, $\sqrt{|x - y|} \leq \sqrt{|x - z|} + \sqrt{|z - y|} \leq \sqrt{|x - z|} + \sqrt{|z - y|}$, using $|x - y| \leq |x - z| + |z - y|$ and $\sqrt{a + b} \leq \sqrt{a} + \sqrt{b}$ (square both sides: $a + b \leq a + b + 2\sqrt{ab}$). So d_2 is a metric. *Check.* $d_2(0, 4) = 2 = d_2(0, 1) + d_2(1, 4) = 1 + \sqrt{3} \approx 2.73$? Indeed $2 \leq 2.73$, consistent.

Caution. “One set, many metrics.” The same underlying set carries genuinely different metrics, and which one you use changes what “small distance” means — but, as we will see in §3.2, the metrics d, d_1, d_∞ on $\mathbb{R}^k$ all produce the *same* open sets, so they are interchangeable for topology. By contrast the d_∞ and L^1 metrics on $C[0, 1]$ give *different* topologies (uniform vs. mean convergence). Never assume two metrics agree topologically; check.

Further worked examples

Worked Example 3.1.8 (The reverse triangle inequality). In any metric space (X, d) prove $|d(p, r) - d(q, r)| \leq d(p, q)$ for all p, q, r .

Recognition. Trigger: an estimate *lower*-bounding a distance. The reverse triangle inequality is the standard consequence of axiom (c) applied twice, with the two points swapped.

Solution. By the triangle inequality, $d(p, r) \leq d(p, q) + d(q, r)$, so $d(p, r) - d(q, r) \leq d(p, q)$. Swapping p and q gives $d(q, r) - d(p, r) \leq d(q, p) = d(p, q)$ by symmetry. The two inequalities together say $-d(p, q) \leq d(p, r) - d(q, r) \leq d(p, q)$, i.e. $|d(p, r) - d(q, r)| \leq d(p, q)$. *Check.* In $\mathbb{R}$ with $d(x, y) = |x - y|$ this is $||p - r| - |q - r|| \leq |p - q|$; take $p = 5, q = 1, r = 0$: LHS = $|5 - 1| = 4 \leq 4 = |5 - 1|$, tight. The map $x \mapsto d(x, r)$ is therefore 1-Lipschitz, hence continuous — a fact used constantly.

Worked Example 3.1.9 (A formula that fails definiteness, and one that fails the triangle law). On $\mathbb{R}$ decide whether $\rho(x, y) = |x^2 - y^2|$ and $\sigma(x, y) = |x - y|/(1 + |x - y|)$ are metrics.

Recognition. Trigger: “is this a metric?”. Run the verification method: definiteness is the usual trap for a non-injective inner function; the bounded modification needs the triangle law checked via monotonicity of $t/(1 + t)$.

Solution. For ρ : $\rho(1, -1) = |1 - 1| = 0$ although $1 \neq -1$, so definiteness (axiom (a)) fails; ρ is *not* a metric. For σ : write $\phi(t) = t/(1 + t)$ for $t \geq 0$. Then ϕ is increasing and *subadditive*, $\phi(a + b) \leq \phi(a) + \phi(b)$ (since $\frac{a+b}{1+a+b} = \frac{a}{1+a+b} + \frac{b}{1+a+b} \leq \frac{a}{1+a} + \frac{b}{1+b}$). With $a = |x - z|$, $b = |z - y|$ and $|x - y| \leq a + b$, monotonicity then subadditivity give $\sigma(x, y) = \phi(|x - y|) \leq \phi(a + b) \leq \phi(a) + \phi(b) = \sigma(x, z) + \sigma(z, y)$. Positivity, definiteness and symmetry are clear, so σ is a metric (a bounded one: $\sigma < 1$ always). *Check.* $\sigma(0, 3) = 3/4$ while $\sigma(0, 1) + \sigma(1, 3) = \frac{1}{2} + \frac{2}{3} = \frac{7}{6} > \frac{3}{4}$, consistent. This σ induces the *same* open sets as $|x - y|$ yet makes the whole line bounded — boundedness is not topological.

Exercises (Bartle & Sherbert, §11.4).

1. Show that the functions $d_1(P_1, P_2) = |x_1 - x_2| + |y_1 - y_2|$ and $d_\infty(P_1, P_2) = \sup\{|x_1 - x_2|, |y_1 - y_2|\}$ are metrics on $\mathbb{R}^2$.
2. Show that the uniform metric $d_\infty(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|$ and the L^1 metric $d_1(f, g) = \int_0^1 |f - g|$ are both metrics on $C[0, 1]$.
3. Verify that the discrete metric on a set S ($d(s, t) = 0$ if $s = t$, else 1) is a metric.
4. If $P_n = (x_n, y_n) \in \mathbb{R}^2$ and d_1 is the taxicab metric, show that (P_n) converges to $P = (x, y)$ with respect to d_1 iff $(x_n) \rightarrow x$ and $(y_n) \rightarrow y$.
5. Verify the conclusion of the previous exercise when d_1 is replaced by the sup metric d_∞ .
6. Let S be nonempty with the discrete metric. Show that in (S, d) a sequence (x_n) converges to x iff there is $K \in \mathbb{N}$ with $x_n = x$ for all $n \geq K$.
7. Show that if d is the discrete metric on S , then every subset of S is both open and closed in (S, d) .
8. Let $P = (x, y)$ and $O = (0, 0)$ in $\mathbb{R}^2$. Draw the sets $\{P : d_1(O, P) \leq 1\}$ and $\{P : d_\infty(O, P) \leq 1\}$ in the plane.
9. Prove that in any metric space, an ε -neighborhood of a point is an open set.
10. Prove the Global Continuity Theorem for metric spaces: $f : (S_1, d_1) \rightarrow (S_2, d_2)$ is continuous on S_1 iff $f^{-1}(G)$ is open in S_1 whenever G is open in S_2 .
11. Prove the preservation of compactness for metric spaces: if (S, d) is a compact metric space and $f : S \rightarrow \mathbb{R}$ is continuous, then $f(S)$ is compact in $\mathbb{R}$.
12. In a metric space (S, d) , call $A \subseteq S$ *bounded* if there exist $x_0 \in S$ and $B > 0$ with $A \subseteq \{x : d(x, x_0) \leq B\}$. Show that if A is a compact subset of S , then A is closed and bounded.

Additional exercises.

1. Verify in detail that the Euclidean metric on $\mathbb{R}^2$ satisfies the triangle inequality, citing the Cauchy–Schwarz inequality.
2. On $\mathbb{R}$, decide whether $\rho(x, y) = |\arctan x - \arctan y|$ is a metric. (It is; identify which axiom needs the injectivity of $\arctan$.)
3. Prove the *reverse triangle inequality* in any metric space: $|d(p, r) - d(r, q)| \leq d(p, q)$.
4. Compute the diameter of the unit square $[0, 1]^2 \subseteq \mathbb{R}^2$ under each of the three metrics d , d_1 , d_∞ . (Answers: $\sqrt{2}$, 2, 1.)
5. (Non-example.) Show that $\sigma(x, y) = |x| + |y|$ for $x \neq y$ and $\sigma(x, x) = 0$ is *not* a metric on $\mathbb{R}$ by exhibiting a failure of the triangle inequality, and identify which other axiom it also breaks.
6. Let (X, d) be a metric space. Prove that $\rho(x, y) = \frac{d(x, y)}{1 + d(x, y)}$ is also a metric on X , and that $\rho(x, y) < 1$ always. (So every metric space carries a topologically equivalent bounded metric.)
7. (Diagram.) Sketch, in the same axes, the closed unit balls at the origin for the metrics d_1 , d , d_∞ on $\mathbb{R}^2$, and mark a point lying in the d_∞ -ball but outside the d -ball. Explain the nesting in one sentence.
8. Show that in $C[0, 1]$ the uniform metric d_∞ and the L^1 metric d_1 are not equivalent: exhibit a sequence f_n with $d_1(f_n, 0) \rightarrow 0$ but $d_\infty(f_n, 0) \not\rightarrow 0$.
9. Prove that a finite union of bounded sets is bounded, but an infinite union of bounded sets need not be (give an example in $\mathbb{R}$).
10. Let A be a nonempty subset of a metric space and define $\text{dist}(x, A) = \inf_{a \in A} d(x, a)$. Prove $|\text{dist}(x, A) - \text{dist}(y, A)| \leq d(x, y)$ (so $x \mapsto \text{dist}(x, A)$ is Lipschitz), and that $\text{dist}(x, A) = 0$ iff $x \in \overline{A}$.

3.2 Open and Closed Sets

Fix a metric space (X, d) throughout. All points and sets are understood to live in X .

Interior points, open sets, closed sets

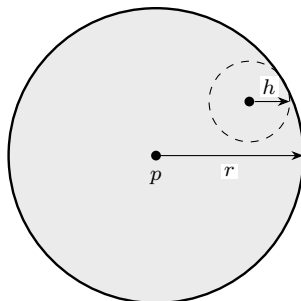
Definition 3.2.1 (Rudin 2.18(e)(f)(g)(d); B&S 11.1.2). Let $E \subseteq X$.

- p is an *interior point* of E if some neighborhood N of p satisfies $N \subseteq E$.
- E is *open* if every point of E is an interior point: for each $p \in E$ there is $r > 0$ with $B(p, r) \subseteq E$.
- The *complement* of E is $E^c = \{p \in X : p \notin E\}$.
- E is *closed* if its complement E^c is open. (Equivalently, by §3.3, E contains all its limit points.)

The words *open* and *closed* are not opposites. In any space, $\emptyset$ and X are *both* open and closed; many sets (e.g. $[0, 1) \subseteq \mathbb{R}$) are *neither*. “Not open” does not mean “closed.”

Theorem 3.2.2 (Every neighborhood is open; Rudin 2.19). *Every open ball $N_r(p)$ is an open set.*

Proof. Let $q \in N_r(p)$, so $d(p, q) < r$. Put $h = r - d(p, q) > 0$. If $d(q, s) < h$ then by the triangle inequality $d(p, s) \leq d(p, q) + d(q, s) < d(p, q) + h = r$, so $s \in N_r(p)$. Thus $N_h(q) \subseteq N_r(p)$ and q is interior. As q was arbitrary, $N_r(p)$ is open. $\square$



$$h = r - d(p, q) > 0, \text{ so } N_h(q) \subseteq N_r(p): \text{ every point of a ball is interior.}$$

Worked Example 3.2.3 (In the discrete metric every set is open (and closed)). Let X carry the discrete metric. Show every subset is both open and closed.

Recognition. “Describe the open sets of a given metric”: compute the balls, then test the definition.

Solution. For any p , $B(p, \frac{1}{2}) = \{q : d(p, q) < \frac{1}{2}\} = \{p\}$, since distinct points are at distance 1. Given any $E \subseteq X$ and $p \in E$, the ball $B(p, \frac{1}{2}) = \{p\} \subseteq E$, so p is interior; hence E is open. As E was arbitrary, E^c is open too, so E is closed. *Check.* Thus in a discrete space “open” carries no information — the topology is as fine as possible. This is exactly the answer to Rudin Ex. 10: in the metric $d(p, q) = \mathbf{1}[p \neq q]$, all subsets are open, all are closed, and (§3.4) the compact ones are precisely the finite ones.

The open/closed set theorems

Theorem 3.2.4 (Complementation; Rudin 2.23). E is open iff E^c is closed; equivalently F is closed iff F^c is open.

Proof. Suppose E^c is closed, i.e. $(E^c)^c = E$... we argue directly. Let E be open and let x be a limit point of E^c (§3.3 notion, but we only need: every neighborhood of x meets E^c). If $x \in E$, openness gives a ball $B(x, r) \subseteq E$, disjoint from E^c , so x is *not* a limit point of E^c — contradiction; hence $x \in E^c$. So E^c contains its limit points, i.e. is closed. Conversely if E^c is closed and $x \in E$, then $x \notin E^c$ and x is not a limit point of E^c , so some ball $B(x, r)$ misses E^c , i.e. $B(x, r) \subseteq E$; thus E is open. $\square$

Theorem 3.2.5 (Unions and intersections; Rudin 2.24, B&S 11.1.4–11.1.5).

- (a) For any collection $\{G_\alpha\}$ of open sets, $\bigcup_\alpha G_\alpha$ is open.
- (b) For any collection $\{F_\alpha\}$ of closed sets, $\bigcap_\alpha F_\alpha$ is closed.
- (c) For any finite collection $G_1, \dots, G_n$ of open sets, $\bigcap_{i=1}^n G_i$ is open.
- (d) For any finite collection $F_1, \dots, F_n$ of closed sets, $\bigcup_{i=1}^n F_i$ is closed.

Proof. (a) Put $G = \bigcup_\alpha G_\alpha$. If $x \in G$ then $x \in G_\alpha$ for some α ; as G_α is open, a ball $B(x, r) \subseteq G_\alpha \subseteq G$, so x is interior. (c) Put $H = \bigcap_{i=1}^n G_i$. For $x \in H$ each G_i gives $r_i > 0$ with $B(x, r_i) \subseteq G_i$; set $r = \min(r_1, \dots, r_n) > 0$ (a *finite* min, hence positive). Then $B(x, r) \subseteq G_i$ for every i , so $B(x, r) \subseteq H$. (b), (d) follow by De Morgan, $(\bigcup E_\alpha)^c = \bigcap E_\alpha^c$ and $(\bigcap E_\alpha)^c = \bigcup E_\alpha^c$, together with Theorem 3.2.4: e.g. $(\bigcap_\alpha F_\alpha)^c = \bigcup_\alpha F_\alpha^c$ is a union of open sets, hence open, so $\bigcap_\alpha F_\alpha$ is closed. $\square$

Caution. The *finiteness* in (c)–(d) is essential. In $\mathbb{R}$, $G_n = (-\frac{1}{n}, \frac{1}{n})$ are open but $\bigcap_{n=1}^\infty G_n = \{0\}$ is not open. Dually $F_n = [\frac{1}{n}, 1]$ are closed but $\bigcup_n F_n = (0, 1]$ is not closed (Rudin 2.25, B&S 11.1.6). An *infinite* intersection of opens, or union of closed sets, can land anywhere.

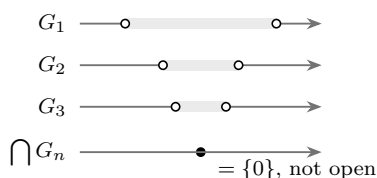


Figure 3.3. Shrinking open intervals $G_n = (-\frac{1}{n}, \frac{1}{n})$ intersect to the single point $\{0\}$: an infinite intersection of open sets need not be open.

Open and closed sets in $\mathbb{R}$

On the line the open sets have a clean description.

Example 3.2.6 (B&S 11.1.3). $\mathbb{R} = (-\infty, \infty)$ is open; any open interval (a, b) , (a, ∞) , $(-\infty, b)$ is open; $[0, 1]$ is closed; $[0, 1)$ is neither; $\mathbb{N}$ is closed; $\{1/n : n \in \mathbb{N}\}$ is not closed but $\{1/n\} \cup \{0\}$ is closed (Rudin 2.21(e)).

Theorem 3.2.7 (Structure of open subsets of $\mathbb{R}$; B&S 11.1.9, Rudin Ex. 29). *A set $G \subseteq \mathbb{R}$ is open iff it is a countable union of pairwise disjoint open intervals.*

Proof sketch. A countable union of open intervals is open by Theorem 3.2.5(a). Conversely, for open G and $x \in G$ let $I_x = (a_x, b_x)$ be the largest open interval with $x \in I_x \subseteq G$ (take $a_x = \inf\{a : (a, x] \subseteq G\}$, $b_x = \sup\{b : [x, b) \subseteq G\}$, allowing $\pm\infty$). One checks the I_x are either equal or disjoint, so $G = \bigsqcup I_x$ is a disjoint union of open intervals. Each interval contains a rational (density of $\mathbb{Q}$), and distinct intervals get distinct rationals, so there are at most countably many. $\square$

The relative (subspace) topology

A set that is open inside a subspace Y need not be open in the ambient X : the segment $(0, 1)$ is open in $\mathbb{R}^1$ but, viewed inside $\mathbb{R}^2$, is not open there. The right notion is *relative openness*.

Definition 3.2.8 (Rudin 2.29). Let $E \subseteq Y \subseteq X$. Then E is *open relative to Y* if for each $p \in E$ there is $r > 0$ such that $\{q \in Y : d(p, q) < r\} \subseteq E$.

Theorem 3.2.9 (Rudin 2.30). *Let $Y \subseteq X$ and $E \subseteq Y$. Then E is open relative to Y if and only if $E = Y \cap G$ for some set G open in X .*

Proof. $(\Rightarrow)$ For each $p \in E$ pick $r_p > 0$ with $\{q \in Y : d(p, q) < r_p\} \subseteq E$, and let $V_p = N_{r_p}(p)$ in X . Put $G = \bigcup_{p \in E} V_p$, open in X . Each $p \in E$ so $E \subseteq G \cap Y$; and $V_p \cap Y \subseteq E$ by choice of r_p , so $G \cap Y \subseteq E$. Hence $E = G \cap Y$. $(\Leftarrow)$ If G is open in X and $E = G \cap Y$, then each $p \in E$ has $N_r(p) \subseteq G$ for some r , whence $N_r(p) \cap Y \subseteq E$; so E is open relative to Y . $\square$

Worked Example 3.2.10 (Relative topology on $\mathbb{Z}$ and on $[0, 1]$). (a) Show every subset of $\mathbb{Z} \subseteq \mathbb{R}$ is open relative to $\mathbb{Z}$. (b) Show $[0, \frac{1}{2})$ is open relative to $[0, 1]$ though not open in $\mathbb{R}$.

Recognition. “Is E open in the subspace Y ?”: intersect an ambient open set with Y (Theorem 3.2.9).

Solution. (a) For $n \in \mathbb{Z}$, $(n - \frac{1}{2}, n + \frac{1}{2}) \cap \mathbb{Z} = \{n\}$, so each singleton is relatively open, hence so is every subset (union of singletons). Thus $\mathbb{Z}$ carries the discrete topology. (b) $[0, \frac{1}{2}) = (-1, \frac{1}{2}) \cap [0, 1]$ with $(-1, \frac{1}{2})$ open in $\mathbb{R}$, so it is open relative to $[0, 1]$ by Theorem 3.2.9; but it contains 0, none of whose $\mathbb{R}$ -neighborhoods lie inside it, so it is not open in $\mathbb{R}$. *Check.* Relative openness genuinely depends on the host space.

Caution. “Open” and “closed” are properties of a set *relative to a chosen ambient space*, not intrinsic. The same set can be open in Y , not open in X . Compactness (§3.4) is the happy exception: it is *absolute*, independent of the embedding (Rudin 2.33).

Further worked examples

Worked Example 3.2.11 ($[0, 1)$ is neither open nor closed). Show in $\mathbb{R}$ that $E = [0, 1)$ is neither open nor closed, directly from the definitions.

Recognition. Trigger: classify a half-open interval. “Not open” is witnessed by one non-interior point; “not closed” by one limit point left out (equivalently, E^c not open).

Solution. *Not open:* the point $0 \in E$ is not interior, since every ball $B(0, r) = (-r, r)$ contains negative numbers, none of which lie in E ; so no ball around 0 fits inside E . *Not closed:* the point $1 \notin E$ is a limit point of E (the sequence $1 - \frac{1}{n} \in E$ converges to 1), so E fails to contain a limit point. Equivalently $E^c = (-\infty, 0) \cup [1, \infty)$ is not open because $1 \in E^c$ is not interior. *Check.* “not open” and “not closed” are independent failures here, occurring at the two different endpoints — a concrete reminder that open and closed are not opposites.

Worked Example 3.2.12 (An infinite intersection of open sets that happens to stay open). Exhibit infinitely many open sets $G_n \subseteq \mathbb{R}$ whose intersection is open and nonempty, contrasting with the shrinking-interval pitfall.

Recognition. Trigger: test the sharpness of Theorem 3.2.5(c). The theorem *guarantees* openness only for finite intersections; it does not *forbid* an infinite intersection from being open.

Solution. Take $G_n = (-1 - \frac{1}{n}, 1 + \frac{1}{n})$, a decreasing nest of open intervals. Their intersection is $\bigcap_{n \geq 1} G_n = [-1, 1]$? No: each G_n strictly contains $[-1, 1]$, and a point x lies in every G_n iff $|x| < 1 + \frac{1}{n}$ for all n , i.e. iff $|x| \leq 1$. So $\bigcap_n G_n = [-1, 1]$, which is *closed*, not open — the nest collapses onto the boundary. To get an *open* infinite intersection instead, take the constant family $G_n = (0, 1)$ for all n : the intersection is $(0, 1)$, open. *Check.* The lesson is that an infinite intersection of opens *may* be open or closed or neither; finiteness in Theorem 3.2.5(c) is what makes openness automatic, and its absence makes the outcome case-by-case.

Worked Example 3.2.13 (An open set in $\mathbb{R}$ as a disjoint union of intervals). Write $G = \mathbb{R} \setminus \mathbb{Z}$ as a countable disjoint union of open intervals, illustrating Theorem 3.2.7.

Recognition. Trigger: “decompose an open set into its components.” By Theorem 3.2.7 every open $G \subseteq \mathbb{R}$ is a countable disjoint union of open intervals; here the gaps are the integers.

Solution. Removing the integers from the line leaves exactly the open unit intervals between consecutive integers:

$$\mathbb{R} \setminus \mathbb{Z} = \bigsqcup_{n \in \mathbb{Z}} (n, n+1).$$

Each $(n, n+1)$ is open, they are pairwise disjoint, and their union is all reals that are not integers. The index set $\mathbb{Z}$ is countable, matching the theorem. *Check.* This is genuinely open — around any $x \notin \mathbb{Z}$ the ball of radius $\text{dist}(x, \mathbb{Z}) > 0$ avoids $\mathbb{Z}$ — and its complement $\mathbb{Z}$ is closed (it contains all its limit points, of which there are none, since $\mathbb{Z}$ has no accumulation point). The component containing x is precisely $(\lfloor x \rfloor, \lfloor x \rfloor + 1)$.

Exercises (Rudin, Ch.2).

- Let $A_1, A_2, A_3, \dots$ be subsets of a metric space. (a) If $B_n = \bigcup_{i=1}^n A_i$, prove that $\overline{B_n} = \bigcup_{i=1}^n \overline{A_i}$ for each n . (b) If $B = \bigcup_{i=1}^\infty A_i$, prove $\overline{B} \supseteq \bigcup_{i=1}^\infty \overline{A_i}$, and show by an example that this inclusion can be proper.
- Is every point of every open set $E \subseteq \mathbb{R}^2$ a limit point of E ? Answer the same question for closed sets in $\mathbb{R}^2$.
- Let E° denote the set of all interior points of a set E (the interior of E). (a) Prove E° is always open. (b) Prove E is open iff $E^\circ = E$. (c) If $G \subseteq E$ and G is open, prove $G \subseteq E^\circ$. (d) Prove the complement of E° is the closure of the complement of E . (e) Do E and $\overline{E}$ always have the same interiors? (f) Do E and E° always have the same closures?
- Prove that every closed set in a separable metric space is the union of a (possibly empty) perfect set and a set which is at most countable. (Corollary: every countable closed set in $\mathbb{R}^k$ has isolated points.) [Hint: use the condensation-point exercise.]
- Prove that every open set in $\mathbb{R}^1$ is the union of an at most countable collection of disjoint segments. [Hint: use separability of $\mathbb{R}$.]
- Imitate the proof that nonempty perfect sets are uncountable to prove Baire’s theorem in $\mathbb{R}^k$: if $\mathbb{R}^k = \bigcup_{n=1}^\infty F_n$ with each F_n closed, then at least one F_n has nonempty interior. Equivalent statement: if each G_n is a dense open subset of $\mathbb{R}^k$, then $\bigcap_n G_n$ is nonempty (in fact dense).

Exercises (Bartle & Sherbert, §11.1).

- If $x \in (0, 1)$, let ε_x be the smaller of x and $1 - x$. Show that if $|u - x| < \varepsilon_x$ then $u \in (0, 1)$.
- Show that the intervals (a, ∞) and $(-\infty, a)$ are open sets, and that $[b, \infty)$ and $(-\infty, b]$ are closed sets.
- Write out the induction argument proving that the intersection of any finite collection of open sets is open.
- Prove that $(0, 1] = \bigcap_{n=1}^\infty (0, 1 + 1/n)$.

5. Show that the set $\mathbb{N}$ of natural numbers is a closed set in $\mathbb{R}$.
6. Show that $A = \{1/n : n \in \mathbb{N}\}$ is not closed, but that $A \cup \{0\}$ is closed.
7. Show that the set $\mathbb{Q}$ of rational numbers is neither open nor closed.
8. Show that if G is open and F is closed, then $G \setminus F$ is open and $F \setminus G$ is closed.
9. A point x is an *interior point* of $A \subseteq \mathbb{R}$ if some neighborhood V of x satisfies $V \subseteq A$. Show that A is open iff every point of A is an interior point.
10. A point x is a *boundary point* of A if every neighborhood of x contains points of A and points of A^c . Show that A and its complement A^c have exactly the same boundary points.
11. Show that $G \subseteq \mathbb{R}$ is open iff it contains none of its boundary points.
12. Show that $F \subseteq \mathbb{R}$ is closed iff it contains all of its boundary points.
13. For $A \subseteq \mathbb{R}$, let A° be the union of all open sets contained in A (the interior). Show A° is open, is the largest open set contained in A , and that $z \in A^\circ$ iff z is an interior point of A .
14. With the interior notation, for sets A, B show $A^\circ \subseteq A$, $(A^\circ)^\circ = A^\circ$, and $(A \cap B)^\circ = A^\circ \cap B^\circ$. Show also $A^\circ \cup B^\circ \subseteq (A \cup B)^\circ$, and give an example where the inclusion is proper.
15. For $A \subseteq \mathbb{R}$, let $\overline{A}$ be the intersection of all closed sets containing A (the closure). Show $\overline{A}$ is closed, is the smallest closed set containing A , and that $w \in \overline{A}$ iff w is an interior or a boundary point of A .
16. With the closure notation, for sets A, B show $A \subseteq \overline{A}$, $\overline{\overline{A}} = \overline{A}$, and $\overline{A \cup B} = \overline{A} \cup \overline{B}$. Show $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$, and give an example where the inclusion is proper.
17. Give an example of a set $A \subseteq \mathbb{R}$ with $A^\circ = \emptyset$ and $\overline{A} = \mathbb{R}$.
18. Show that if $F \subseteq \mathbb{R}$ is closed, nonempty, and bounded above, then $\sup F$ belongs to F .
19. Show that each point of the Cantor set F is a cluster point of F .
20. Show that each point of the Cantor set F is a cluster point of its complement F^c .

Additional exercises.

1. Decide for each set whether it is open, closed, both, or neither in $\mathbb{R}$: (a) $[0, 1]$; (b) $\{1/n : n \in \mathbb{N}\}$; (c) $\mathbb{Z}$; (d) $\mathbb{Q}$; (e) $\mathbb{R} \setminus \mathbb{Q}$; (f) $\{0\} \cup \{1/n\}$.
2. Prove directly from the definition that the half-plane $\{(x, y) \in \mathbb{R}^2 : x > 0\}$ is open.
3. Show that a single point $\{p\}$ is closed in any metric space, and conclude that every finite set is closed.
4. Prove that $G \subseteq X$ is open iff $G = G^\circ$, and that F is closed iff $F = \overline{F}$.
5. (Non-example.) The infinite intersection $\bigcap_n G_n$ of open sets need not be open. Exhibit open $G_n \subseteq \mathbb{R}$ with $\bigcap_n G_n = \{0\} \cup [1, 2]$, a non-open set.
6. Give an example of two sets $A, B \subseteq \mathbb{R}$, neither open nor closed, with $A \cup B$ open and $A \cap B$ closed.
7. (Diagram.) Draw an open set in $\mathbb{R}^2$ as a union of three overlapping open disks, and mark one interior point together with a ball witnessing its interiority.
8. Prove that the boundary $\partial E = \overline{E} \setminus E^\circ$ of any set is closed, and that E is open iff $E \cap \partial E = \emptyset$.
9. Using the structure theorem, write the open set $\mathbb{R} \setminus \mathbb{Z}$ as a countable disjoint union of open intervals.
10. Let $Y = [0, 2] \subseteq \mathbb{R}$. Determine whether $[0, 1]$ and $(1, 2]$ are open relative to Y , closed relative to Y , or both, and exhibit ambient open/closed sets G with $[0, 1] = Y \cap G$.
11. Prove that E is open in X iff E is a union of open balls.

3.3 Limit Points, Closure, and Interior

Limit points and isolated points

Definition 3.3.1 (Rudin 2.18(b)(c); cluster point, B&S §4.1, §11.1). Let $E \subseteq X$ and $p \in X$.

- p is a *limit point* (or *cluster point*, or *accumulation point*) of E if every neighborhood of p contains a point $q \in E$ with $q \neq p$.
- If $p \in E$ but p is not a limit point of E , then p is an *isolated point* of E (some ball $B(p, r)$ meets E only at p).

The set of limit points of E is written E' .

A limit point need not belong to E , and a point of E need not be a limit point. The set $\{1/n : n \in \mathbb{N}\}$ has the single limit point 0 (which is not in the set), and every one of its own points is isolated — “having a limit point” and “containing a limit point” are different (Rudin 2.21(e)).

Theorem 3.3.2 (Limit points are approached infinitely often; Rudin 2.20). *If p is a limit point of E , then every neighborhood of p contains infinitely many points of E . In particular a finite set has no limit points.*

Proof. Suppose some neighborhood N of p met E in only finitely many points $q_1, \dots, q_n$ distinct from p . Put $r = \min_{1 \leq m \leq n} d(p, q_m) > 0$ (a finite minimum of positive numbers). Then $N_r(p)$ contains no point $q \neq p$ of E , contradicting that p is a limit point. $\square$

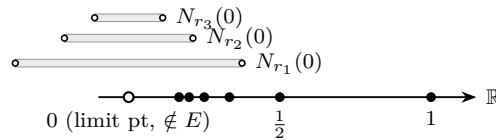


Figure 3.4. $E = \{1/n\}$ has the one limit point 0. Shrinking neighborhoods of 0 each still capture infinitely many $1/n$; every point of E is isolated.

Closure

Definition 3.3.3 (Rudin 2.26). The *closure* of E is $\overline{E} = E \cup E'$ (the set together with all its limit points).

Theorem 3.3.4 (Rudin 2.27). *For $E \subseteq X$: (a) $\overline{E}$ is closed; (b) $\overline{E} = E$ iff E is closed; (c) $\overline{E} \subseteq F$ for every closed $F \supseteq E$. Hence $\overline{E}$ is the smallest closed set containing E .*

Proof. (a) Let $p \in (\overline{E})^c$, so $p \notin E$ and $p \notin E'$; thus some neighborhood N of p contains no point of E (other than p , which is excluded). Then N also contains no limit point of E : if $q \in N$ then N is a neighborhood of q missing $E \setminus \{q\}$, so $q \notin E'$. Hence $N \subseteq (\overline{E})^c$, proving $(\overline{E})^c$ open and $\overline{E}$ closed. (b) If $\overline{E} = E$ then E is closed by (a); if E is closed then $E' \subseteq E$ (it contains its limit points, §3.2 / Theorem 3.3.5), so $\overline{E} = E \cup E' = E$. (c) If F is closed and $E \subseteq F$ then $E' \subseteq F' \subseteq F$, so $\overline{E} = E \cup E' \subseteq F$. $\square$

Theorem 3.3.5 (Closed $\iff$ contains its limit points; Rudin 2.18(d), B&S 11.1.7–11.1.8). *E is closed iff $E \supseteq E'$, iff E contains the limit of every convergent sequence of its points.*

Proof. If E is closed and p is a limit point with $p \notin E$, then $p \in E^c$ open, so some ball $B(p, r) \subseteq E^c$ misses E — contradicting $p \in E'$; hence $E' \subseteq E$. Conversely if $E' \subseteq E$ and $p \in E^c$, then p is not a limit point, so a ball $B(p, r)$ misses E , i.e. $B(p, r) \subseteq E^c$; so E^c is open and E closed. The sequential form is equivalent because a cluster point is exactly a limit of a nonconstant sequence from E (B&S 4.1.2). $\square$

Theorem 3.3.6 (Sup and inf lie in a closed set; Rudin 2.28, B&S Ex. 11.1.18). *If $\emptyset \neq E \subseteq \mathbb{R}$ is bounded above and $y = \sup E$, then $y \in \overline{E}$. Hence if E is closed, $\sup E \in E$ (and likewise $\inf E \in E$).*

Proof. If $y \in E$ we are done. Otherwise, for every $h > 0$ there is $x \in E$ with $y - h < x < y$ (else $y - h$ would be an upper bound smaller than the least upper bound). So every neighborhood of y meets $E \setminus \{y\}$; $y \in E' \subseteq \overline{E}$. $\square$

Interior; dense sets

Definition 3.3.7 (Rudin Ex. 9; interior, B&S Ex. 11.1.13). The *interior* E° of E is the set of all interior points of E ; equivalently the union of all open sets contained in E , hence the *largest* open subset of E . A set E is *dense* in X if $\overline{E} = X$, i.e. every point of X is a point of E or a limit point of E (Rudin 2.18(j)).

Proposition 3.3.8 (Interior/closure duality; Rudin Ex. 9(d)). E° is open and $E^\circ \subseteq E$; moreover

$$(E^\circ)^c = \overline{E^c} \quad \text{and dually} \quad (\overline{E})^c = (E^c)^\circ.$$

Proof. E° is a union of open sets, hence open (Theorem 3.2.5(a)), and lies in E . For the duality, $p \notin E^\circ$ means no ball around p lies in E , i.e. every ball meets E^c , i.e. $p \in E^c$ or p is a limit point of E^c , i.e. $p \in \overline{E^c}$. Taking complements gives the dual statement. $\square$

Worked Example 3.3.9 (Closure, interior, and limit points of $\mathbb{Q} \cap (0, 1)$). Let $E = \mathbb{Q} \cap (0, 1) \subseteq \mathbb{R}$. Find $\overline{E}$, E° , and E' .

Recognition. “Compute closure/interior of a dense-ish set”: use that $\mathbb{Q}$ is dense in $\mathbb{R}$.

Solution. Every real in $[0, 1]$ is a limit of rationals in $(0, 1)$, so $E' = [0, 1]$ and $\overline{E} = E \cup E' = [0, 1]$. The interior is empty: any ball around a point of E contains irrationals, so no ball lies in E ; thus $E^\circ = \emptyset$. *Check.* Consistent with duality — $(E^\circ)^c = \mathbb{R} = \overline{E^c}$ since $E^c \supseteq$ the irrationals is dense. E is a set whose closure is a fat interval but whose interior is empty: “boundary is everything.”

Worked Example 3.3.10 (E and E' need not have the same limit points (Rudin Ex. 6)). Show $\overline{E}$ and E' always have the same limit points, but E and E' need not.

Recognition. “Compare derived sets”: test the claim on a set whose limit points are themselves isolated as a set.

Solution. Since $E \subseteq \overline{E} \subseteq E \cup E'$ and limit points only depend on arbitrarily-close companions, one shows $(\overline{E})' = E'$ (a limit point of $\overline{E}$ is approached by points of E , possibly via nearby limit points). But E and E' can differ: let $E = \{1/n : n \in \mathbb{N}\}$. Then $E' = \{0\}$, and $(E')' = \emptyset$ (a single point has no limit points), while $E' = \{0\} \neq \emptyset = (E')'$. *Check.* So applying $'$ is not idempotent in general; iterating can strip points (here $E \rightarrow \{0\} \rightarrow \emptyset$).

Method (Finding the closure of a set E). *Recognition trigger.* “Compute $\overline{E}$,” or “is E closed/dense?”.

1. Identify the limit points: $p \in E'$ iff there is a sequence $x_n \in E$, $x_n \neq p$, with $x_n \rightarrow p$. Sweep the boundary of E and any accumulation values.
2. Form $\overline{E} = E \cup E'$. Then E is closed iff this adds nothing ($E' \subseteq E$), and E is dense in X iff $\overline{E} = X$.
3. Cross-check with the smallest-closed-superset description (Theorem 3.3.4(c)) and with E° via duality (Proposition 3.3.8).

Worked Example 3.3.11 (A bounded set with exactly three limit points (Rudin Ex. 5)). Construct a bounded $E \subseteq \mathbb{R}$ whose set of limit points is a 3-element set.

Recognition. “Build a set with prescribed limit points”: glue together copies of a sequence accumulating at each desired point.

Solution. Take three centers, say 0, 1, 2, and around each a sequence converging to it:

$$E = \{\tfrac{1}{n} : n \in \mathbb{N}\} \cup \{1 + \tfrac{1}{n} : n \in \mathbb{N}\} \cup \{2 + \tfrac{1}{n} : n \in \mathbb{N}\}.$$

$E \subseteq [0, 3]$ is bounded. Each center 0, 1, 2 is a limit point (its sequence converges to it), and no other point is: away from $\{0, 1, 2\}$ the set is locally finite. So $E' = \{0, 1, 2\}$, three points. *Check.* None of 0, 1, 2 lies in E , illustrating again that limit points need not be members. Replace “three” by a sequence of centers and you get Rudin Ex. 13’s compact set with countably many limit points.

Caution. $\overline{A \cup B} = \overline{A} \cup \overline{B}$ holds (finite unions), but $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$ can be *strict*: with $A = \mathbb{Q}$, $B = \mathbb{R} \setminus \mathbb{Q}$, $A \cap B = \emptyset$ so $\overline{A \cap B} = \emptyset$, while $\overline{A} \cap \overline{B} = \mathbb{R} \cap \mathbb{R} = \mathbb{R}$. Dually $(A \cap B)^\circ = A^\circ \cap B^\circ$ but $(A \cup B)^\circ \supseteq A^\circ \cup B^\circ$ may be strict (B&S Ex. 11.1.14, 11.1.16). Closure distributes over $\cup$, interior over $\cap$ — not the other way.

Further worked examples

Worked Example 3.3.12 (Closure and interior of a half-open interval). For $E = (0, 1] \subseteq \mathbb{R}$ compute $\overline{E}$, E° , E' , and the boundary $\partial E = \overline{E} \setminus E^\circ$.

Recognition. Trigger: “compute closure/interior/boundary of an interval.” Sweep the endpoints for limit points (Method, Finding the closure); the interior is the largest open subinterval.

Solution. *Limit points:* every point of $[0, 1]$ is approached by points of $(0, 1]$, and no point outside $[0, 1]$ is; so $E' = [0, 1]$ and $\overline{E} = E \cup E' = [0, 1]$. *Interior:* a point p is interior iff some $(p - r, p + r) \subseteq (0, 1]$, which holds exactly for $0 < p < 1$; the right endpoint 1 fails (any ball includes numbers > 1) and $0 \notin E$. So $E^\circ = (0, 1)$. *Boundary:* $\partial E = [0, 1] \setminus (0, 1) = \{0, 1\}$. *Check.* Duality holds — $(E^\circ)^c = \mathbb{R} \setminus (0, 1) = \overline{E^c}$, and indeed $E^c = (-\infty, 0] \cup (1, \infty)$ has closure $(-\infty, 0] \cup [1, \infty) = \mathbb{R} \setminus (0, 1)$. The boundary is the two endpoints, as expected for an interval.

Worked Example 3.3.13 (The closure of a ball can be smaller than the closed ball). In a general metric space show $\overline{B(p, r)} \subseteq \{q : d(p, q) \leq r\}$, and give a space where the inclusion is strict.

Recognition. Trigger: a closure-vs-closed-ball comparison. The open ball’s closure is always contained in the closed ball; equality is a Euclidean luxury, not a metric-space theorem.

Solution. Let $\bar{B} = \{q : d(p, q) \leq r\}$. Since $\bar{B}$ is closed (its complement $\{d(p, q) > r\}$ is open: if $d(p, q_0) > r$, the ball $B(q_0, d(p, q_0) - r)$ stays outside $\bar{B}$ by the reverse triangle inequality) and contains $B(p, r)$, the smallest-closed-superset property (Theorem 3.3.4(c)) gives $\overline{B(p, r)} \subseteq \bar{B}$. *Strict case:* take the discrete metric on a set X with ≥ 2 points and $r = 1$. Then $B(p, 1) = \{p\}$ (only p is within distance < 1), which is already closed, so $\overline{B(p, 1)} = \{p\}$. But $\bar{B} = \{q : d(p, q) \leq 1\} = X$. Thus $\overline{B(p, 1)} = \{p\} \subsetneq X = \bar{B}$. *Check.* In $\mathbb{R}^k$ the two always agree ($\overline{B(p, r)} = \bar{B}$), so the strictness is a purely non-Euclidean phenomenon — another reason never to assume metric spaces behave like $\mathbb{R}^k$.

Exercises (Rudin, Ch.2).

1. Construct a bounded set of real numbers with exactly three limit points.
2. Let E' be the set of all limit points of a set E . Prove that E' is closed. Prove that E and $\overline{E}$ have the same limit points. (Recall $\overline{E} = E \cup E'$.) Do E and E' always have the same limit points?

Additional exercises.

1. Find $\overline{E}$, E° , and E' for each: (a) $E = (0, 1] \cup \{2\}$; (b) $E = \{1/n : n \in \mathbb{N}\}$; (c) $E = \mathbb{Z}$; (d) $E = \mathbb{Q} \cap [0, 1]$.
2. Prove that p is a limit point of E iff there is a sequence $x_n \in E$ with $x_n \neq p$ and $x_n \rightarrow p$.
3. Show that the set of isolated points of any $E \subseteq \mathbb{R}$ is at most countable. [Hint: each isolated point owns a rational-endpoint interval meeting E only at itself.]
4. Prove that $(A \cap B)^\circ = A^\circ \cap B^\circ$ for all A, B .
5. (Non-example.) Show that $\overline{A \cap B} = \overline{A} \cap \overline{B}$ can fail, and that $(A \cup B)^\circ = A^\circ \cup B^\circ$ can fail, using $A = \mathbb{Q}$, $B = \mathbb{R} \setminus \mathbb{Q}$.
6. A set E is *dense* in X if $\overline{E} = X$. Prove E is dense iff every nonempty open set meets E , and verify $\mathbb{Q}$ is dense in $\mathbb{R}$.
7. (Diagram.) Draw a point p together with three shrinking neighborhoods, each containing a distinct point of a set E accumulating at p , illustrating that $p \in E'$.
8. Construct a set $E \subseteq \mathbb{R}$ with $E' = \mathbb{Z}$ (its limit points are exactly the integers).
9. Prove that E' is always closed (i.e. $(E')' \subseteq E'$), and give an example where $(E')' \subsetneq E'$.
10. Show that a set E is closed iff $E = \overline{E}$ iff $\partial E \subseteq E$, and that E is perfect iff $E = E'$.

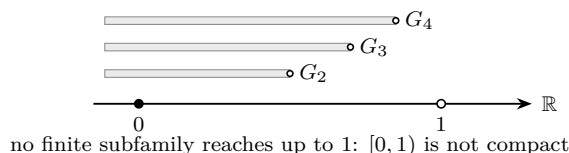
3.4 Compact Sets and the Heine–Borel Theorem

Open covers and compactness

Definition 3.4.1 (Rudin 2.31–2.32; B&S 11.2.1–11.2.2). An *open cover* of $E \subseteq X$ is a collection $\{G_\alpha\}$ of open sets with $E \subseteq \bigcup_\alpha G_\alpha$. A *subcover* is a subcollection that still covers E ; a *finite subcover* is one with finitely many sets. A set $K \subseteq X$ is *compact* if every open cover of K admits a finite subcover.

The quantifier structure is the whole game. To prove K compact you must start with an *arbitrary* open cover and extract a finite subcover. To prove H not compact it suffices to exhibit *one* open cover with no finite subcover.

Example 3.4.2 (B&S 11.2.3). $[0, 1]$ is not compact: the cover $G_n = (-1, 1 - \frac{1}{n})$, $n \in \mathbb{N}$, has union $[0, 1)$ but any finite subfamily has union $(-1, 1 - \frac{1}{m})$ for the largest m , missing points near 1. Likewise $(0, 1)$ is not compact via $G_n = (\frac{1}{n}, 1)$. Every finite set *is* compact: cover each of its points by one set of the cover.



Compactness is absolute, and forces closed + bounded

Theorem 3.4.3 (Compactness is intrinsic; Rudin 2.33). *If $K \subseteq Y \subseteq X$, then K is compact relative to X iff compact relative to Y .*

Proof. By Theorem 3.2.9, relatively-open-in- Y sets are exactly $Y \cap G$ for G open in X . A cover of K by X -open sets restricts to a Y -open cover and vice versa, and since $K \subseteq Y$ a finite subcover in one picture is one in the other. $\square$

This is why we may speak of a “compact metric space” without naming an ambient space — unlike open or closed, compactness does not depend on the embedding.

Theorem 3.4.4 (Compact $\Rightarrow$ closed; Rudin 2.34, B&S 11.2.4). *Every compact subset K of a metric space X is closed.*

Proof. We show K^c is open. Fix $p \notin K$. For each $q \in K$ choose disjoint balls $V_q = N_r(p)$ and $W_q = N_r(q)$ with $r < \frac{1}{2}d(p, q)$. The $\{W_q\}_{q \in K}$ cover K ; by compactness finitely many $W_{q_1}, \dots, W_{q_n}$ cover K . Let $V = V_{q_1} \cap \dots \cap V_{q_n}$, a neighborhood of p . Then V is disjoint from each W_{q_i} , hence from $K \subseteq \bigcup W_{q_i}$, so $V \subseteq K^c$. Thus p is interior to K^c . $\square$

Theorem 3.4.5 (Compact $\Rightarrow$ bounded; B&S 11.2.4). *Every compact K is bounded.*

Proof. Fix $x_0 \in X$. The balls $H_m = N_m(x_0)$, $m \in \mathbb{N}$, cover $X \supseteq K$; a finite subcover $H_{m_1}, \dots$ has union the largest $H_M = N_M(x_0)$, so $K \subseteq N_M(x_0)$ is bounded. $\square$

Theorem 3.4.6 (Closed subset of compact is compact; Rudin 2.35, B&S Ex. 11.2.4). *If $F \subseteq K \subseteq X$ with F closed (in X) and K compact, then F is compact. Consequently $F \cap K$ is compact whenever F is closed and K compact.*

Proof. Let $\{V_\alpha\}$ be an open cover of F . Adjoin the open set F^c to obtain an open cover Ω of K . Compactness of K gives a finite subcover $\Phi \subseteq \Omega$ of K ; discarding F^c from Φ if present still leaves a cover of F by finitely many V_α . So F is compact. $\square$

Nested compact sets and the finite intersection property

Theorem 3.4.7 (Finite intersection property; Rudin 2.36, B&S Ex. 11.2.9). *Let $\{K_\alpha\}$ be compact sets such that every finite subcollection has nonempty intersection. Then $\bigcap_\alpha K_\alpha \neq \emptyset$. In particular, a nested sequence $K_1 \supseteq K_2 \supseteq \cdots$ of nonempty compact sets has $\bigcap_n K_n \neq \emptyset$.*

Proof. Fix K_1 and suppose, for contradiction, $\bigcap_\alpha K_\alpha = \emptyset$, i.e. no point of K_1 lies in every K_α . Then the sets $G_\alpha = K_\alpha^c$ (open, since each K_α is closed by Theorem 3.4.4) cover K_1 . By compactness finitely many cover K_1 : $K_1 \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}$, i.e. $K_1 \cap K_{\alpha_1} \cap \cdots \cap K_{\alpha_n} = \emptyset$, contradicting the finite-intersection hypothesis. $\square$

Theorem 3.4.8 (Infinite subset of a compact set has a limit point; Rudin 2.37). *If E is an infinite subset of a compact set K , then E has a limit point in K .*

Proof. If no point of K were a limit point of E , each $q \in K$ would have a ball V_q meeting E in at most one point (namely q if $q \in E$). The $\{V_q\}$ cover K , but no finite subfamily can cover the infinite $E \subseteq K$ — each V_q captures ≤ 1 point of E . This contradicts compactness. $\square$

k -cells, nested intervals, and Heine–Borel

Definition 3.4.9 (Rudin 2.17). A k -cell is a product $[a_1, b_1] \times \cdots \times [a_k, b_k] \subseteq \mathbb{R}^k$ (a 1-cell is a closed interval, a 2-cell a closed rectangle).

Theorem 3.4.10 (Nested intervals / cells; Rudin 2.38–2.39, B&S 2.5.2). *If $\{I_n\}$ are intervals (or k -cells) with $I_n \supseteq I_{n+1}$, then $\bigcap_n I_n \neq \emptyset$.*

Proof. For intervals $I_n = [a_n, b_n]$, let $x = \sup_n a_n$ (the a_n are bounded above by b_1). For all m, n , $a_n \leq a_{m+n} \leq b_{m+n} \leq b_m$, so $x \leq b_m$; and $a_m \leq x$; thus $x \in I_m$ for all m . For k -cells apply this coordinatewise. $\square$

Theorem 3.4.11 (Every k -cell is compact; Rudin 2.40). *Every k -cell $I \subseteq \mathbb{R}^k$ is compact.*

Proof. Let $\delta = (\sum_j (b_j - a_j)^2)^{1/2} = \text{diam } I$. Suppose an open cover $\{G_\alpha\}$ of I had no finite subcover. Bisect each edge at its midpoint $c_j = (a_j + b_j)/2$; this splits I into 2^k subcells, at least one of which — call it I_1 — has no finite subcover (else I would). Repeat, producing $I \supseteq I_1 \supseteq I_2 \supseteq \cdots$ with: (i) I_n has no finite subcover, and (ii) $\text{diam } I_n = 2^{-n}\delta$. By Theorem 3.4.10 there is $x^* \in \bigcap_n I_n$. Some $G_\alpha \ni x^*$, and openness gives r with $N_r(x^*) \subseteq G_\alpha$. Choosing n with $2^{-n}\delta < r$ forces $I_n \subseteq N_r(x^*) \subseteq G_\alpha$ — a finite (one-set!) subcover of I_n , contradicting (i). $\square$

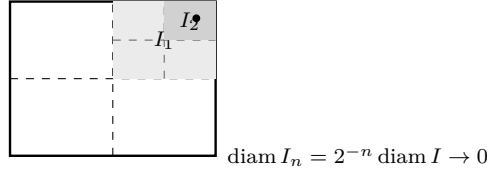


Figure 3.5. Repeated bisection of a 2-cell: the “bad” subcell shrinks to a point x^* , whose covering ball already covers a whole I_n — contradiction.

Theorem 3.4.12 (Heine–Borel; Rudin 2.41, B&S 11.2.5). *For $E \subseteq \mathbb{R}^k$ the following are equivalent: (a) E is closed and bounded; (b) E is compact; (c) every infinite subset of E has a limit point in E (the Bolzano–Weierstrass property).*

Proof. **(a) $\Rightarrow$ (b):** bounded E lies in some k -cell I ; I is compact (Theorem 3.4.11) and E is a closed subset of I , hence compact (Theorem 3.4.6). **(b) $\Rightarrow$ (c):** Theorem 3.4.8. **(c) $\Rightarrow$ (a):** If E were unbounded, pick $x_n \in E$ with $|x_n| > n$; then $\{x_n\}$ is infinite with no limit point, against (c). If E were not closed, take a limit point $x_0 \notin E$ and $x_n \in E$ with $|x_n - x_0| < \frac{1}{n}$; then $\{x_n\}$ is infinite, has x_0 as its only limit point in $\mathbb{R}^k$, and $x_0 \notin E$, so E has no limit point in E from this set — against (c). $\square$

Remark (The hypotheses are sharp). (a) $\Leftrightarrow$ (b) is special to $\mathbb{R}^k$. In a general metric space (b) $\Leftrightarrow$ (c) always holds (Rudin Ex. 26), but “closed + bounded” need *not* imply compact: in $\mathbb{Q}$, the set $E = \{p \in \mathbb{Q} : 2 < p^2 < 3\}$ is closed and bounded in $\mathbb{Q}$ yet not compact (Rudin Ex. 16); in infinite-dimensional spaces the closed unit ball is not compact.

Theorem 3.4.13 (Weierstrass; Rudin 2.42). *Every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.*

Proof. Such a set E lies in a k -cell I , which is compact; apply Theorem 3.4.8. $\square$

Theorem 3.4.14 (Sequential compactness; B&S 11.2.6). *$K \subseteq \mathbb{R}$ (or $\mathbb{R}^k$) is compact iff every sequence in K has a subsequence converging to a point of K .*

Proof. If K is compact it is closed and bounded (Heine–Borel); a sequence in K is bounded, so Bolzano–Weierstrass gives a convergent subsequence, whose limit lies in the closed K . Conversely, if K is not closed, a cluster point $c \notin K$ is the limit of a sequence in K none of whose subsequences can converge in K ; if K is not bounded, pick $x_n \in K$ with $|x_n| > n$ — no subsequence converges. So the sequential property forces closed + bounded, i.e. compact. $\square$

Method (Proving a specific set K is compact). *Recognition trigger.* “Show K is compact” (directly, or by Heine–Borel).

1. **In $\mathbb{R}^k$:** the fast route is Heine–Borel — show K is closed (contains its limit points) and bounded (lies in a ball). Done.
2. **From the definition** (when asked, e.g. Rudin Ex. 12): take an arbitrary open cover; use the structure of K (an accumulation point, a nesting) to peel off all but finitely many points with one or two cover sets, then cover the finite remainder.
3. **Via sequences:** show every sequence in K has a subsequence converging in K .
4. **To disprove compactness:** exhibit one bad open cover, or one sequence with no convergent-in- K subsequence.

Worked Example 3.4.15 ($K = \{0\} \cup \{1/n : n \in \mathbb{N}\}$ is compact from the definition (Rudin Ex. 12)). Prove directly (no Heine–Borel) that K is compact.

Recognition. “Compact from the definition” with one accumulation point: one cover set swallows the tail.

Solution. Let $\{G_\alpha\}$ be an open cover. Some $G_{\alpha_0} \ni 0$; being open, it contains a ball $N_r(0)$, hence all $1/n$ with $1/n < r$, i.e. all but finitely many points of K . The finitely many remaining points $1, \frac{1}{2}, \dots, \frac{1}{N}$ each lie in some G_{α_i} . Then $G_{\alpha_0}, G_{\alpha_1}, \dots, G_{\alpha_N}$ is a finite subcover. *Check.* K is closed (it contains its only limit point 0) and bounded, agreeing with Heine–Borel. Compare Rudin Ex. 13: a compact set whose limit points form a countable set.

Worked Example 3.4.16 (A bounded, non-compact set in $\mathbb{Q}$ (Rudin Ex. 16)). In the metric space $\mathbb{Q}$, let $E = \{p \in \mathbb{Q} : 2 < p^2 < 3\}$. Show E is closed and bounded in $\mathbb{Q}$ but not compact, and is open in $\mathbb{Q}$.

Recognition. “Closed + bounded but not compact”: the missing limits $\pm\sqrt{2}, \pm\sqrt{3}$ are irrational, so invisible to $\mathbb{Q}$.

Solution. $E = \mathbb{Q} \cap ((\sqrt{2}, \sqrt{3}) \cup (-\sqrt{3}, -\sqrt{2}))$. It is bounded ($|p| < \sqrt{3}$). It is *closed in $\mathbb{Q}$* : a $\mathbb{Q}$ -limit point p of E has $2 < p^2 < 3$ with $p \in \mathbb{Q}$, but $p^2 = 2$ or 3 is impossible for rational p , so $2 < p^2 < 3$ and $p \in E$. It is *open in $\mathbb{Q}$* for the same reason (the constraint is an open condition with no rational on the boundary). Not compact: the sets $G_n = \{p \in \mathbb{Q} : 2 + \frac{1}{n} < p^2 < 3\}$ form an open cover of (the positive part of) E with no finite subcover, as $p^2 \downarrow 2$ is approached by rationals. *Check.* In $\mathbb{R}$ the analogous set’s closure adds $\pm\sqrt{2}, \pm\sqrt{3}$; those endpoints simply do not exist in $\mathbb{Q}$, breaking Heine–Borel.

Worked Example 3.4.17 (An open cover of $(0, 1)$ with no finite subcover (Rudin Ex. 14)). Exhibit such a cover.

Recognition. “Show not compact”: one explicit bad cover suffices.

Solution. Let $G_n = (\frac{1}{n}, 1)$ for $n \geq 2$. Then $\bigcup_n G_n = (0, 1)$ since every $x \in (0, 1)$ has $x > \frac{1}{n}$ for large n . Any finite subfamily $G_{n_1}, \dots, G_{n_r}$ has union $(\frac{1}{s}, 1)$ with $s = \max n_i$, missing every point in $(0, \frac{1}{s}]$. So no finite subcover. *Check.* This is exactly why “bounded” alone fails Heine–Borel — $(0, 1)$ is bounded but not closed, hence not compact (Theorem 3.4.12).

Caution. “Closed and bounded $\Rightarrow$ compact” is a theorem *about $\mathbb{R}^k$ only*. Do not use it in $\mathbb{Q}$, in $C[0, 1]$, or in any incomplete or infinite-dimensional space. The reliable implications in *every* metric space are compact $\Rightarrow$ closed and bounded (Theorems 3.4.4–3.4.5) and compact $\iff$ Bolzano–Weierstrass $\iff$ sequentially compact.

Further worked examples

Worked Example 3.4.18 (A union and an intersection of compact sets). Show that a *finite* union of compact sets is compact, and that an *arbitrary* intersection of compact sets is compact.

Recognition. Trigger: “is this built-up set compact?”. Push covers through the union; for the intersection use compact $\Rightarrow$ closed plus closed-subset-of-compact.

Solution. Union. Let $K_1, \dots, K_m$ be compact and $\{G_\alpha\}$ an open cover of $K = \bigcup_i K_i$. It covers each K_i , so each K_i has a finite subcover $\Phi_i \subseteq \{G_\alpha\}$; then $\Phi_1 \cup \dots \cup \Phi_m$ is finite and covers K . So K is compact.

Intersection. Let $\{K_\beta\}$ be compact, hence each closed (Theorem 3.4.4). Fix one index β_0 . Then $\bigcap_\beta K_\beta$ is a closed subset (an intersection of closed sets, Theorem 3.2.5(b)) of the compact K_{β_0} , hence compact (Theorem 3.4.6). *Check.* The finiteness in the union is needed — $\bigcup_n \{n\} = \mathbb{N}$ is an *infinite* union of (compact) singletons and is not compact (unbounded). The intersection needs no finiteness, because closedness is preserved by arbitrary intersection.

Worked Example 3.4.19 (A continuous image of a compact set is compact). Let $f : K \rightarrow \mathbb{R}$ be continuous on a compact $K \subseteq \mathbb{R}^k$. Show $f(K)$ is compact, and deduce f attains a maximum and minimum (the Extreme Value Theorem).

Recognition. Trigger: “compactness under a continuous map.” Pull an open cover of $f(K)$ back through f to an open cover of K (preimages of opens are open), extract a finite subcover, push forward.

Solution. Let $\{G_\alpha\}$ be an open cover of $f(K)$. By continuity each $f^{-1}(G_\alpha)$ is open, and the family $\{f^{-1}(G_\alpha)\}$ covers K (every $x \in K$ has $f(x) \in G_\alpha$ for some α). Compactness of K yields finitely many $f^{-1}(G_{\alpha_1}), \dots, f^{-1}(G_{\alpha_n})$ covering K ; then $G_{\alpha_1}, \dots, G_{\alpha_n}$ cover $f(K)$. So $f(K)$ is compact, hence closed and bounded in $\mathbb{R}$. Being bounded, $M = \sup f(K)$ and $m = \inf f(K)$ are finite; being closed, $M, m \in f(K)$ (Theorem 3.3.6), so $f(x_{\max}) = M$ and $f(x_{\min}) = m$ for some points of K . *Check.* Compactness is essential — on the non-compact $(0, 1)$ the continuous $f(x) = 1/x$ has image $(1, \infty)$, neither bounded nor attaining a maximum.

Exercises (Rudin, Ch.2).

1. Let X be an infinite set. For $p, q \in X$ define $d(p, q) = 1$ if $p \neq q$ and $d(p, q) = 0$ if $p = q$. Prove this is a metric. Which subsets of the resulting metric space are open? Which are closed? Which are compact?
2. For $x, y \in \mathbb{R}^1$ determine, for each of the following, whether it is a metric: $d_1(x, y) = (x - y)^2$; $d_2(x, y) = \sqrt{|x - y|}$; $d_3(x, y) = |x^2 - y^2|$; $d_4(x, y) = |x - 2y|$; $d_5(x, y) = \frac{|x - y|}{1 + |x - y|}$.
3. Let $K \subseteq \mathbb{R}^1$ consist of 0 and the numbers $1/n$ for $n = 1, 2, 3, \dots$. Prove that K is compact directly from the definition (without using Heine–Borel).
4. Construct a compact set of real numbers whose limit points form a countable set.
5. Give an example of an open cover of the segment $(0, 1)$ which has no finite subcover.
6. Show that Theorem 3.4.7 (and its nested-sets corollary) become false in $\mathbb{R}^1$ if the word “compact” is replaced by “closed” or by “bounded.”
7. Regard $\mathbb{Q}$, the set of rationals, as a metric space with $d(p, q) = |p - q|$. Let E be the set of all $p \in \mathbb{Q}$ with $2 < p^2 < 3$. Show that E is closed and bounded in $\mathbb{Q}$, but that E is not compact. Is E open in $\mathbb{Q}$?
8. A metric space is *separable* if it contains a countable dense subset. Show that $\mathbb{R}^k$ is separable. [Hint: consider the points with rational coordinates.]
9. A collection $\{V_\alpha\}$ of open subsets of X is a *base* for X if for every $x \in X$ and every open $G \ni x$ there is α with $x \in V_\alpha \subseteq G$. Prove that every separable metric space has a countable base. [Hint: take all balls with rational radius centered at points of a countable dense set.]
10. Let X be a metric space in which every infinite subset has a limit point. Prove that X is separable. [Hint: fix $\delta > 0$; pick x_1 , and having chosen $x_1, \dots, x_j$, choose x_{j+1} with $d(x_i, x_{j+1}) > \delta$ if possible. Show the process stops, so X is covered by finitely many δ -balls; take $\delta = 1/n$.]

11. Prove that every compact metric space K has a countable base, and is therefore separable. [Hint: for each n , finitely many balls of radius $1/n$ cover K .]
12. Let X be a metric space in which every infinite subset has a limit point. Prove that X is compact. [Hint: by the previous exercises X has a countable base, so every open cover has a countable subcover $\{G_n\}$; if no finite subfamily covers X , the complements F_n of $G_1 \cup \cdots \cup G_n$ are nonempty with empty intersection — pick a point from each F_n and find a limit point.]
13. Define $p \in X$ to be a *condensation point* of $E \subseteq X$ if every neighborhood of p contains uncountably many points of E . Suppose $E \subseteq \mathbb{R}^k$ is uncountable, and let P be the set of all condensation points of E . Prove that P is perfect and that at most countably many points of E are not in P ; i.e. $P^c \cap E$ is at most countable. [Hint: take a countable base $\{V_n\}$; let W be the union of those V_n for which $E \cap V_n$ is at most countable, and show $P = W^c$.]

Exercises (Bartle & Sherbert, §11.2).

1. Exhibit an open cover of the interval $(1, 2]$ that has no finite subcover.
2. Exhibit an open cover of $\mathbb{N}$ that has no finite subcover.
3. Exhibit an open cover of the set $\{1/n : n \in \mathbb{N}\}$ that has no finite subcover.
4. Prove, using the definition of compactness, that if F is a closed subset of a compact set K in $\mathbb{R}$, then F is compact.
5. Prove, using the definition, that if K_1 and K_2 are compact sets in $\mathbb{R}$, then $K_1 \cup K_2$ is compact.
6. Use the Heine–Borel Theorem to prove this version of Bolzano–Weierstrass: every bounded infinite subset of $\mathbb{R}$ has a cluster point in $\mathbb{R}$. (Note: a set with no cluster points is closed.)
7. Find an infinite collection $\{K_n : n \in \mathbb{N}\}$ of compact sets in $\mathbb{R}$ whose union $\bigcup_n K_n$ is not compact.
8. Prove that the intersection of an arbitrary collection of compact sets in $\mathbb{R}$ is compact.
9. Let (K_n) be a sequence of nonempty compact sets in $\mathbb{R}$ with $K_1 \supseteq K_2 \supseteq \cdots$. Prove that $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$.
10. Let $K \neq \emptyset$ be compact in $\mathbb{R}$. Show that $\inf K$ and $\sup K$ exist and belong to K .
11. Let $K \neq \emptyset$ be compact in $\mathbb{R}$ and $c \in \mathbb{R}$. Prove there is a point $a \in K$ with $|c - a| = \inf\{|c - x| : x \in K\}$.
12. Let $K \neq \emptyset$ be compact in $\mathbb{R}$ and $c \in \mathbb{R}$. Prove there is a point $b \in K$ with $|c - b| = \sup\{|c - x| : x \in K\}$.
13. If K_1 and K_2 are disjoint nonempty compact sets, show there exist $k_i \in K_i$ with $0 < |k_1 - k_2| = \inf\{|x_1 - x_2| : x_i \in K_i\}$.
14. Give an example of disjoint closed sets F_1, F_2 such that $\inf\{|x_1 - x_2| : x_i \in F_i\} = 0$.

Additional exercises.

1. For each set, state whether it is compact in $\mathbb{R}$ and justify via Heine–Borel: (a) $[0, 1]$; (b) $(0, 1]$; (c) $\{0\} \cup \{1/n\}$; (d) $\mathbb{Z}$; (e) $[0, 1] \cup [2, 3]$.
2. Prove that a continuous image of a compact set is compact (the version with domain a compact subset of $\mathbb{R}$ and codomain $\mathbb{R}$). [You may use the open-cover definition.]
3. Show that the intersection of a compact set and a closed set is compact.
4. Prove the extreme value theorem from compactness: a continuous $f : K \rightarrow \mathbb{R}$ on a nonempty compact $K \subseteq \mathbb{R}$ attains a maximum and a minimum.
5. (Non-example.) Exhibit a closed and bounded subset of the metric space $\mathbb{Q}$ that is *not* compact, and explain which step of the Heine–Borel proof fails in $\mathbb{Q}$.
6. Give an open cover of $\mathbb{R}$ with no finite subcover, and one of $[0, \infty)$ with no finite subcover.
7. (Diagram.) Draw the interval $[0, 1]$ together with a finite subcover extracted from the cover $\{(\frac{k}{10} - 0.12, \frac{k}{10} + 0.12) : k = 0, \dots, 10\}$; mark the overlaps.

8. Prove that a nested sequence of nonempty compact sets $K_1 \supseteq K_2 \supseteq \cdots$ in $\mathbb{R}^k$ has nonempty intersection, and show the conclusion fails for nested nonempty *closed* (unbounded) sets.
9. Prove that every sequence in a compact set $K \subseteq \mathbb{R}$ has a convergent subsequence with limit in K , and use this to show $\sup K \in K$.
10. Let $K \subseteq \mathbb{R}$ be compact and $f : K \rightarrow \mathbb{R}$ continuous. Prove f is uniformly continuous on K (compactness proof, not the interval proof).
11. Show that if K is compact and $\{F_\alpha\}$ is a family of closed sets with $K \cap \bigcap_\alpha F_\alpha = \emptyset$, then some finite subfamily already has $K \cap F_{\alpha_1} \cap \cdots \cap F_{\alpha_n} = \emptyset$.

3.5 Perfect Sets and the Cantor Set

Perfect sets

Definition 3.5.1 (Rudin 2.18(h)). A set $P \subseteq X$ is *perfect* if P is closed and every point of P is a limit point of P (i.e. P has *no isolated points*, and $P = \overline{P} = P'$).

A perfect set is “all accumulation, no specks”: closed, and densely self-accumulating. The closed interval $[a, b]$ is perfect; so is $\mathbb{R}^k$; so, remarkably, is the Cantor set, which contains no interval at all.

Theorem 3.5.2 (Nonempty perfect sets in $\mathbb{R}^k$ are uncountable; Rudin 2.43). *Every nonempty perfect set $P \subseteq \mathbb{R}^k$ is uncountable. In particular every interval $[a, b]$ with $a < b$, and hence $\mathbb{R}$ itself, is uncountable.*

Proof. Since each point of P is a limit point, P is infinite. Suppose for contradiction P were countable, $P = \{x_1, x_2, x_3, \dots\}$. Build neighborhoods V_n as follows. Take any V_1 with $\overline{V_1} \cap P \neq \emptyset$. Given V_n with $\overline{V_n} \cap P \neq \emptyset$: since every point of P is a limit point, the open set V_n contains a point of P , around which we can choose V_{n+1} with

$$(i) \overline{V_{n+1}} \subseteq V_n, \quad (ii) x_n \notin \overline{V_{n+1}}, \quad (iii) V_{n+1} \cap P \neq \emptyset.$$

Put $K_n = \overline{V_n} \cap P$. Each K_n is compact (closed and bounded) and nonempty, and $K_n \supseteq K_{n+1}$, so $\bigcap_n K_n \neq \emptyset$ by Theorem 3.4.7. But (ii) ensures $x_n \notin K_{n+1}$, so *no* x_n lies in $\bigcap_n K_n$; since $\bigcap_n K_n \subseteq P = \{x_n\}$, the intersection is empty — a contradiction. Hence P is uncountable. $\square$

Remark. This is Rudin’s *second* proof that $\mathbb{R}$ is uncountable (the first was Cantor’s diagonal argument, Chapter 1). The mechanism is the nested-compact-sets corollary: each x_n is “dodged” at stage $n + 1$, so the limit point caught by nesting is a new point of P .

The Cantor set

Definition 3.5.3 (Rudin 2.44, B&S 11.1.10). Start with $E_0 = [0, 1]$. Remove the open middle third $(\frac{1}{3}, \frac{2}{3})$ to get $E_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$. Remove the open middle third of each remaining interval to get $E_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$. Continuing, E_n is a union of 2^n closed intervals each of length 3^{-n} , with $E_n \supseteq E_{n+1}$. The *Cantor set* is

$$P = \bigcap_{n=1}^{\infty} E_n.$$



Figure 3.6. Removing middle thirds: E_n is 2^n intervals of length 3^{-n} , total length $(2/3)^n \rightarrow 0$. The Cantor set $P = \bigcap E_n$ is what survives forever.

Theorem 3.5.4 (Properties of the Cantor set; Rudin 2.44, B&S 11.1.10). *The Cantor set P is: (a) nonempty and compact; (b) of total length (measure) zero; (c) perfect, hence uncountable; (d) totally disconnected — it contains no segment.*

Proof. (a) Each E_n is a finite union of closed intervals, hence compact; $P = \bigcap_n E_n$ is an intersection of nested nonempty compact sets, so nonempty (Theorem 3.4.7) and closed, hence compact. (b) The total length removed is $\frac{1}{3} + \frac{2}{3^2} + \frac{4}{3^3} + \cdots = \frac{1}{3} \sum_{n \geq 0} (\frac{2}{3})^n = \frac{1}{3} \cdot \frac{1}{1-2/3} = 1$, so P has measure $\leq 1 - 1 = 0$; equivalently $\text{length}(E_n) = (2/3)^n \rightarrow 0$ and $P \subseteq E_n$. (d) If P contained a segment of length $\ell > 0$, then since $P \subseteq E_n$ and E_n 's intervals have length 3^{-n} , we would need $\ell \leq 3^{-n}$ for all n , forcing $\ell = 0$. So P contains no interval; between any two points of P lies a gap, so P is totally disconnected. (c) P is closed by (a). To see no point is isolated, take $x \in P$ and any segment $S \ni x$. For large n the interval I_n of E_n containing x satisfies $I_n \subseteq S$; its endpoints survive every later removal, so both endpoints lie in P . Pick the endpoint $x_n \neq x$: then $x_n \in P \cap S$, $x_n \neq x$, so x is a limit point of P . Thus P is perfect, and uncountable by Theorem 3.5.2. $\square$

Worked Example 3.5.5 (The ternary description and uncountability of P). Show $P = \{x \in [0, 1] : x = \sum_{n \geq 1} a_n 3^{-n}, a_n \in \{0, 2\}\}$ and use it to see P is uncountable.

Recognition. “Identify the Cantor set explicitly”: removing middle thirds is removing the digit 1 in base 3.

Solution. A point survives the first removal iff its ternary expansion can be written with $a_1 \in \{0, 2\}$ (the open middle third $(\frac{1}{3}, \frac{2}{3})$ is exactly the numbers forced to have $a_1 = 1$); inductively $x \in P$ iff it has a ternary expansion using only digits 0, 2. The map $\psi(\sum a_n 3^{-n}) = \sum (a_n/2) 2^{-n}$ sends P onto $[0, 1]$ (every binary expansion is hit), so $|P| \geq |[0, 1]|$; hence P is uncountable. *Check.* This matches Theorem 3.5.4(c). The endpoints $2k/3^n$ are the “rational” members; but P has continuum-many points, vastly more than its endpoints.

Worked Example 3.5.6 (A perfect set in $\mathbb{R}$ containing no rational (Rudin Ex. 18)). Is there a nonempty perfect set in $\mathbb{R}$ with no rational points?

Recognition. “Build a perfect set avoiding a countable set”: mimic the Cantor construction but center the removed gaps on rationals.

Solution. Yes. Run a Cantor-style construction on $[0, 1]$ but at each stage remove an open middle gap chosen to contain a prescribed rational (enumerate $\mathbb{Q} \cap [0, 1] = \{q_1, q_2, \dots\}$ and ensure q_k is deleted by stage k , shrinking gaps so positive length survives). The intersection P of the nested closed stages is nonempty, closed, and perfect (same endpoint argument as the Cantor set), yet contains no rational. *Check.* A translated/scaled “fat Cantor set” arranged to swallow each rational works; perfectness only needs the self-accumulation of surviving endpoints, independent of which gaps were removed.

Caution. The Cantor set is the standard counterexample factory. It is uncountable yet has measure zero (so “small” in length but “large” in cardinality); it is perfect yet contains no interval; it is compact and totally disconnected. Do not equate “measure zero” with “countable,” nor “contains no interval” with “countable” — P refutes both.

Further worked examples

Worked Example 3.5.7 (Is $\frac{1}{4}$ in the Cantor set?). Decide whether $\frac{1}{4}$ and $\frac{1}{3}$ belong to the Cantor set P , using the ternary-digit description.

Recognition. Trigger: “membership in P .” By the previous worked example, $x \in P$ iff x has a base-3 expansion using only digits 0 and 2.

Solution. $x = \frac{1}{4}$: compute the ternary expansion. $\frac{1}{4} = \sum_{k \geq 1} a_k 3^{-k}$ with $a_k = 0, 2, 0, 2, \dots$ since $\frac{2}{3^2} + \frac{2}{3^4} + \cdots = 2 \cdot \frac{1/9}{1-1/9} = 2 \cdot \frac{1}{8} = \frac{1}{4}$. So $\frac{1}{4} = (0.020202\dots)_3$, all digits 0 or 2, hence $\frac{1}{4} \in P$. $x = \frac{1}{3}$: here $\frac{1}{3} = (0.1)_3 = (0.0222\dots)_3$; the second form uses only 0, 2, so $\frac{1}{3} \in P$ as well — it is the right endpoint of $[0, \frac{1}{3}]$, an endpoint that survives forever. *Check.* $\frac{1}{4}$ is not an endpoint of any removed interval, yet still lies in P ; this shows P is far larger than its set of endpoints (which is only countable). The midpoint $\frac{1}{2} = (0.111\dots)_3$ is forced to have a digit 1 and lies in the first removed middle third, so $\frac{1}{2} \notin P$.

Worked Example 3.5.8 (A fat Cantor set: positive measure, still nowhere dense). Remove from $[0, 1]$ at stage n the open middle interval of length 4^{-n} from each of the 2^{n-1} surviving intervals. Show the surviving set F is closed, nowhere dense, perfect, and has measure $\frac{1}{2}$.

Recognition. Trigger: “Cantor-type construction with controlled lengths.” Track total removed length as a geometric series; perfectness comes from the same surviving-endpoint argument as for P .

Solution. At stage n there are 2^{n-1} intervals and we remove one gap of length 4^{-n} from each, removing total length $2^{n-1} \cdot 4^{-n} = \frac{1}{2} \cdot 2^{-n}$. Summing over $n \geq 1$, the total removed length is $\frac{1}{2} \sum_{n \geq 1} 2^{-n} = \frac{1}{2} \cdot 1 = \frac{1}{2}$. So $F = [0, 1] \setminus (\text{removed})$ has measure $1 - \frac{1}{2} = \frac{1}{2} > 0$. F is closed (intersection of closed stages F_n) and *nowhere dense*: each F_n is a union of intervals of length $\leq 2^{-n} \rightarrow 0$, so F contains no interval, whence $\overline{F}^\circ = F^\circ = \emptyset$. It is *perfect* by the surviving-endpoint argument used for P : every point of F is a limit of stage-endpoints in F . *Check.* Unlike the standard P (measure 0), this F has *positive* measure $\frac{1}{2}$ yet still contains no interval — “nowhere dense” does not imply “measure zero.”

Worked Example 3.5.9 (A perfect set need not be all-limit-points without closedness). Give a set every point of which is a limit point of the set, but which is *not* perfect, and explain which axiom of Definition 3.5.1 fails.

Recognition. Trigger: a NON-example probing the definition of perfect. Perfect requires *both* “closed” and “every point is a limit point”; dropping closedness breaks it.

Solution. Take $E = (0, 1) \cap \mathbb{Q}$, the rationals in the open unit interval. Every point of E is a limit point of E (rationals accumulate at every rational, since between any two rationals lies another), so the “no isolated points” clause holds. But E is *not closed*: its limit point $\frac{1}{\sqrt{2}} \approx 0.707$ (or any irrational in $(0, 1)$, or the endpoints $0, 1$) is not in E . So E fails the closedness clause and is not perfect; indeed $\overline{E} = [0, 1] \neq E$. *Check.* The genuinely perfect set “closest” to E is its closure $[0, 1]$, which *is* perfect. Perfectness is the conjunction $E = \overline{E} = E'$; here $E = E'$ holds but $E \neq \overline{E}$, so the conjunction fails.

Exercises (Rudin, Ch.2).

- Let E be the set of all $x \in [0, 1]$ whose decimal expansion contains only the digits 4 and 7. Is E countable? Is E dense in $[0, 1]$? Is E compact? Is E perfect?
- Is there a nonempty perfect set in $\mathbb{R}^1$ which contains no rational number?

Additional exercises.

- Show that $[0, 1]$ is perfect, and that $\mathbb{R}^k$ is perfect.
- Prove that a finite set is never perfect, and that a nonempty perfect set is always infinite (indeed uncountable).
- Verify the total-length computation for the Cantor set: $\sum_{n \geq 0} \frac{2^n}{3^{n+1}} = 1$, and conclude the Cantor set has measure zero.
- Show that the Cantor set is *nowhere dense*: $\overline{P}^\circ = \emptyset$.
- (Non-example.) Exhibit a closed, uncountable subset of $\mathbb{R}$ that is *not* perfect (it has at least one isolated point), e.g. $\{0\} \cup [1, 2]$.
- (Fat Cantor set.) At stage n remove from each interval an open middle piece of relative length 4^{-n} instead of $\frac{1}{3}$. Show the resulting set is perfect, compact, contains no interval, yet has *positive* total length. Contrast with the standard Cantor set.
- (Diagram.) Draw the first four stages $E_0, \dots, E_3$ of the middle-thirds construction as a stacked bar figure, labeling the surviving endpoints $0, \frac{1}{3}, \frac{2}{3}, 1$ and the lengths 3^{-n} .
- Show that the endpoints of the removed intervals (numbers $2k/3^n$) are dense in the Cantor set but form only a countable subset of it.
- Using the ternary characterization, decide whether $\frac{1}{4}$ belongs to the Cantor set. [Hint: $\frac{1}{4} = (0.\overline{02})_3$.]
- Prove that the Cantor set is homeomorphic to $\{0, 2\}^{\mathbb{N}}$ with the product topology via the ternary-digit map, hence is totally disconnected.

3.6 Connected Sets

Separated sets and connectedness

Definition 3.6.1 (Rudin 2.45). Two subsets A, B of a metric space X are *separated* if $A \cap \overline{B} = \emptyset$ and $\overline{A} \cap B = \emptyset$ — neither meets the closure of the other. A set $E \subseteq X$ is *connected* if it cannot be written as $E = A \cup B$ with A, B nonempty and separated. Otherwise E is *disconnected*.

Remark (Separated is stronger than disjoint; Rudin 2.46). Separated sets are disjoint, but disjoint sets need not be separated: $[0, 1]$ and $(1, 2)$ are disjoint yet not separated, since $1 \in \overline{(1, 2)}$. The segments $(0, 1)$ and $(1, 2)$ are separated. Separation forbids one piece from touching the other even at a boundary point.

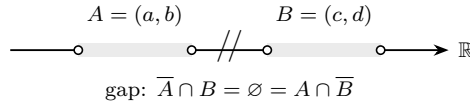


Figure 3.7. A disconnected set splits into two separated open pieces with a gap between them; no point of either closure reaches the other.

Connected subsets of the line are exactly the intervals

Theorem 3.6.2 (Rudin 2.47). A set $E \subseteq \mathbb{R}$ is connected if and only if it has the intermediate value property: whenever $x, y \in E$ and $x < z < y$, then $z \in E$. Equivalently, the connected subsets of $\mathbb{R}$ are exactly the intervals (including rays, points, $\emptyset$, and $\mathbb{R}$).

Proof. ($\Rightarrow$, **contrapositive**) Suppose E fails the property: there are $x, y \in E$ and $z \in (x, y)$ with $z \notin E$. Put

$$A_z = E \cap (-\infty, z), \quad B_z = E \cap (z, \infty).$$

Then $E = A_z \cup B_z$ (since $z \notin E$), both are nonempty ($x \in A_z, y \in B_z$), and they are separated: $A_z \subseteq (-\infty, z)$ and $B_z \subseteq (z, \infty)$, so $\overline{A_z} \subseteq (-\infty, z]$ is disjoint from B_z , and symmetrically. So E is disconnected. ($\Leftarrow$) Suppose E is disconnected, $E = A \cup B$ with A, B nonempty and separated. Pick $x \in A, y \in B$; WLOG $x < y$. Let

$$z = \sup(A \cap [x, y]).$$

By Theorem 3.3.6, $z \in \overline{A}$, so $z \notin B$ (separation); in particular $x \leq z < y$ (as $z \leq y$ and $z \neq y \in B$). If $z \notin A$ then $z \notin E$ and $x < z < y$ exhibits a gap, so the IVP fails. If $z \in A$, then $z \notin \overline{B}$ (separation), so there is z_1 with $z < z_1 < y$ and $z_1 \notin B$; but $z_1 > z = \sup(A \cap [x, y])$ gives $z_1 \notin A$ too, so $z_1 \notin E$ with $x < z_1 < y$ — again the IVP fails. Either way E lacks the IVP. The sets with the IVP are precisely the intervals. $\square$

Worked Example 3.6.3 ($\mathbb{Q}$ is totally disconnected). Show the only connected subsets of $\mathbb{Q}$ are the singletons and $\emptyset$.

Recognition. “Find/refute connectedness”: use Theorem 3.6.2’s IVP characterization, which a gap-riddled set fails.

Solution. Suppose $E \subseteq \mathbb{Q}$ has two points $x < y$. Pick an *irrational* $z \in (x, y)$ (density of irrationals); then $z \notin \mathbb{Q} \supseteq E$, so $x < z < y$ with $z \notin E$ violates the IVP, and E is disconnected: $E = (E \cap (-\infty, z)) \cup (E \cap (z, \infty))$ is a separation. Hence no subset of $\mathbb{Q}$ with ≥ 2 points is connected; the connected sets are $\emptyset$ and singletons.

Check. $\mathbb{Q}$ is *totally disconnected*, like the Cantor set, although $\mathbb{Q}$ is countable and dense while the Cantor set is uncountable and nowhere dense — two very different totally disconnected spaces.

Worked Example 3.6.4 (A convex set is connected (Rudin Ex. 21(c))). Show every convex $E \subseteq \mathbb{R}^k$ is connected.

Recognition. “Show connected” in higher dimensions, where the IVP characterization is unavailable: push a line segment between two points and reduce to $\mathbb{R}^1$.

Solution. Suppose $E = A \cup B$ with A, B nonempty separated, $a \in A, b \in B$. Convexity puts the whole segment $p(t) = (1-t)a + tb, t \in [0, 1]$, inside E . Let $A_0 = p^{-1}(A), B_0 = p^{-1}(B) \subseteq [0, 1]$. These are nonempty

$(0 \in A_0, 1 \in B_0)$, cover $[0, 1]$, and are separated in $\mathbb{R}$ (the continuous affine p pulls back separation). But $[0, 1]$ is an interval, hence connected (Theorem 3.6.2) — contradiction. So E is connected. *Check.* Balls and k -cells are convex, hence connected; this is the $\mathbb{R}^k$ analogue of “intervals are connected.”

Method (Deciding whether a set E is connected). *Recognition trigger.* “Is E connected?” / “show E is (dis)connected.”

1. **In $\mathbb{R}$:** apply Theorem 3.6.2 — E is connected iff it is an interval. To *disconnect*, find a point $z \notin E$ strictly between two points of E .
2. **In $\mathbb{R}^k$:** to prove connectedness, exhibit a path or a convex/segment structure and reduce to the $\mathbb{R}^1$ case; to disprove, split E into two nonempty pieces lying in disjoint open sets (a separation).
3. **Bookkeeping:** “separated” means closures don’t meet, strictly stronger than “disjoint.” Check closures, not just the sets.

Remark (Closures and interiors of connected sets; Rudin Ex. 20). The closure of a connected set is connected (if $\bar{E} = A \cup B$ separated, then $E = (E \cap A) \cup (E \cap B)$ would separate E). The *interior* of a connected set need *not* be connected: in $\mathbb{R}^2$ two closed disks joined by a single point form a connected set whose interior is two disjoint open disks.

Caution. Connectedness is about the *absence of a separation*, not about path-joining — though in $\mathbb{R}^k$ open connected sets are also path-connected, in general metric spaces the two notions differ. And remember separation requires *closures* to be disjoint: $[0, 1]$ and $(1, 2)$ are disjoint but their union $[0, 2)$ minus nothing is still connected precisely because 1 glues them. Always test with closures.

Further worked examples

Worked Example 3.6.5 (Classifying connectedness on the line). For each set decide whether it is connected: (a) $[0, 1] \cup [2, 3]$, (b) $[0, 1] \cup \{2\}$, (c) $[0, 2)$, (d) $\mathbb{R} \setminus \{0\}$.

Recognition. Trigger: “is $E \subseteq \mathbb{R}$ connected?”. Apply Theorem 3.6.2: connected $\iff$ interval $\iff$ has the intermediate-value property.

Solution. (a) *Disconnected:* $\frac{3}{2} \notin E$ lies between $1 \in E$ and $2 \in E$, a gap; explicitly $E = ([0, 1]) \cup ([2, 3])$ is a separation (closures $[0, 1]$ and $[2, 3]$ are disjoint). (b) *Disconnected:* 2 is isolated, and any $z \in (1, 2)$ is a gap between 1 and 2. (c) *Connected:* $[0, 2)$ is an interval, so the IVP holds vacuously — no internal point is missing. (d) *Disconnected:* $0 \notin E$ separates E into $(-\infty, 0)$ and $(0, \infty)$, whose closures $(-\infty, 0]$ and $[0, \infty)$ meet only at $0 \notin E$, so they are separated within E . *Check.* Exactly the interval (c) is connected; the others each have an internal point removed or an isolated chunk, which is precisely the IVP failing.

Worked Example 3.6.6 (The union of two connected sets sharing a point is connected). Let $A, B \subseteq X$ be connected with $A \cap B \neq \emptyset$. Show $A \cup B$ is connected.

Recognition. Trigger: “glue connected pieces.” Argue by contradiction from a separation of the union; a connected piece cannot straddle a separation.

Solution. Suppose $A \cup B = P \cup Q$ with P, Q nonempty and separated. Fix $c \in A \cap B$; say $c \in P$ (WLOG). Now $A = (A \cap P) \cup (A \cap Q)$, and these two are separated (subsets of the separated P, Q). Since A is connected, one of them is empty; because $c \in A \cap P$ we must have $A \cap Q = \emptyset$, i.e. $A \subseteq P$. The same argument with B (also containing $c \in P$) gives $B \subseteq P$. Hence $A \cup B \subseteq P$, forcing $Q = \emptyset$ — contradicting $Q \neq \emptyset$. So no separation exists and $A \cup B$ is connected. *Check.* The shared point is essential: $[0, 1]$ and $[2, 3]$ are each connected but disjoint, and their union is disconnected (previous example). The result extends to any family of connected sets with a common point.

Worked Example 3.6.7 (The topologist’s two-piece set: disjoint is not separated). In $\mathbb{R}$, are $A = [0, 1]$ and $B = (1, 2]$ separated? Is $A \cup B$ connected?

Recognition. Trigger: a NON-example testing “separated” vs “disjoint.” Separation needs *closures* to miss the other set; check $\bar{A} \cap B$ and $A \cap \bar{B}$.

Solution. A and B are disjoint ($A \cap B = \emptyset$). But $\overline{A} = [0, 1]$ and $\overline{B} = [1, 2]$, so $\overline{A} \cap B = [0, 1] \cap (1, 2] = \emptyset$ but $A \cap \overline{B} = [0, 1] \cap [1, 2] = \{1\} \neq \emptyset$. One of the two separation conditions fails, so A and B are *not* separated. Consequently this particular splitting is not a valid separation of $A \cup B = [0, 2]$; and indeed $[0, 2]$ is an interval, hence connected (Theorem 3.6.2), so no separation exists at all. *Check.* Contrast with $(0, 1)$ and $(1, 2)$, which *are* separated (neither closure $[0, 1]$, $[1, 2]$ contains a point of the *other open* set), and whose union $(0, 1) \cup (1, 2)$ is genuinely disconnected. The single point 1 is exactly what decides the two cases.

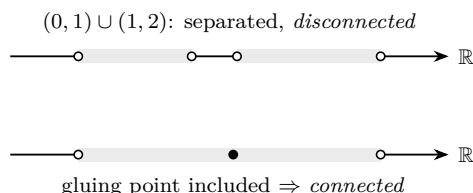


Figure 3.8. Removing the shared point 1 separates the interval into two pieces; keeping it glues them. Connectedness on the line turns entirely on whether internal points are present.

Exercises (Rudin, Ch.2).

- (a) If A and B are disjoint closed sets in a metric space X , prove that they are separated. (b) Prove the same for disjoint open sets. (c) Fix $p \in X$, $\delta > 0$; define $A = \{q \in X : d(p, q) < \delta\}$ and $B = \{q \in X : d(p, q) > \delta\}$. Prove A and B are separated. (d) Prove that every connected metric space with at least two points is uncountable. [Hint: use (c).]
- Are closures and interiors of connected sets always connected? (Look at subsets of $\mathbb{R}^2$.)
- Let A and B be separated subsets of some $\mathbb{R}^n$; suppose $a \in A$, $b \in B$, and define $p(t) = (1 - t)a + tb$ for $t \in \mathbb{R}^1$. Put $A_0 = p^{-1}(A)$, $B_0 = p^{-1}(B)$ (so $t \in A_0$ iff $p(t) \in A$). (a) Prove A_0 and B_0 are separated subsets of $\mathbb{R}^1$. (b) Prove there exists $t_0 \in (0, 1)$ with $p(t_0) \notin A \cup B$. (c) Prove that every convex subset of $\mathbb{R}^n$ is connected.

Additional exercises.

- For each subset of $\mathbb{R}$, decide whether it is connected, citing the interval characterization: (a) $[0, 1]$; (b) $[0, 1] \cup \{2\}$; (c) $\mathbb{Q}$; (d) $\mathbb{R} \setminus \{0\}$; (e) $(0, 1) \cup (1, 2)$.
- Prove that the union of two connected sets that share a common point is connected.
- Prove that $\mathbb{R}$ is connected, and that $\mathbb{R} \setminus \{0\}$ is not.
- Show that the continuous image of a connected set is connected (state the version with domain a connected subset of $\mathbb{R}$). Deduce the Intermediate Value Theorem.
- (Non-example.) Show that $\mathbb{Q}$ and the Cantor set are both totally disconnected (no connected subset has more than one point), although one is countable and dense and the other uncountable and nowhere dense.
- Prove that the closure of a connected set is connected, and exhibit a connected set in $\mathbb{R}^2$ whose interior is disconnected.
- (Diagram.) Draw a disconnected set in $\mathbb{R}$ (or $\mathbb{R}^2$) as two separated pieces, marking the gap and indicating why no point of one closure meets the other.
- Prove that if A, B are separated and $C \subseteq A \cup B$ is connected, then $C \subseteq A$ or $C \subseteq B$.
- Show that an open connected subset of $\mathbb{R}^k$ is *path-connected*: any two points are joined by a polygonal path inside the set. [Hint: the set of points reachable from a fixed point is open and closed in the set.]
- Determine all connected subsets of a discrete metric space, and explain in one sentence why the answer matches the “totally disconnected” picture.

Chapter 4

Numerical Sequences and Series

4.1 Convergent Sequences

Sequences and the definition of a limit

A *sequence* of real numbers is a function $x : \mathbb{N} \rightarrow \mathbb{R}$; we write its value at n as x_n and the whole sequence as (x_n) or $(x_n)_{n=1}^{\infty}$. The term x_n is the n -th *term*; do not confuse the sequence (x_n) (an infinite ordered list) with its *range* $\{x_n : n \in \mathbb{N}\}$ (a set, which may be finite: the sequence $((-1)^n)$ has range $\{-1, 1\}$).

Definition 4.1.1. A sequence (x_n) *converges* to a real number x , written $x_n \rightarrow x$ or $\lim_{n \rightarrow \infty} x_n = x$, if for every $\varepsilon > 0$ there is a natural number $K = K(\varepsilon)$ such that

$$n \geq K \implies |x_n - x| < \varepsilon.$$

The number x is the *limit*. A sequence is *convergent* if it has a limit, and *divergent* otherwise.

The inequality $|x_n - x| < \varepsilon$ says $x_n \in (x - \varepsilon, x + \varepsilon)$, the open interval of radius ε about x — the ε -*neighborhood* of x . So convergence means: *no matter how narrow a band you draw around x , all but finitely many terms land inside it*. The threshold K is allowed to grow as ε shrinks; that dependence is the whole game.

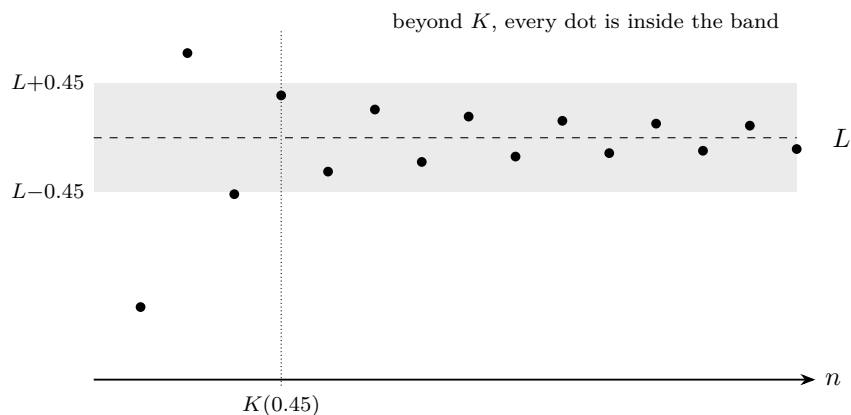


Figure 4.1. Convergence: each ε -band $(L - \varepsilon, L + \varepsilon)$ traps all terms from some index $K(\varepsilon)$ on. Shrink ε and K may move right, but it stays finite.

Remark. “All but finitely many” is the geometric soul of the definition: only *finitely many* terms may sit outside any given band. Changing, deleting, or prepending finitely many terms therefore changes nothing about convergence or the limit — limits are a property of the *tail* $(x_n)_{n \geq N}$.

Method (Running an ε - K convergence proof). *Recognition trigger.* You must *prove* $x_n \rightarrow x$ directly from the definition (no limit theorems yet available), or the problem says “using the definition of limit”.

1. **Form the gap** $|x_n - x|$ and simplify it to a clean expression in n .
2. **Bound it above** by something obviously small, e.g. $|x_n - x| \leq C/n$ or $\leq r^n$ — discard, enlarge, or crudely estimate to get a manageable upper bound. This is “scratch work”; you may overshoot.
3. **Solve for K** . Given $\varepsilon > 0$, find K making your bound $< \varepsilon$ (e.g. $C/n < \varepsilon \iff n > C/\varepsilon$, so take $K = \lceil C/\varepsilon \rceil + 1$). Use the Archimedean property to guarantee such a K exists.
4. **Write it forward**. “Let $\varepsilon > 0$. Put $K = \dots$. If $n \geq K$ then $|x_n - x| \leq (\text{bound}) < \varepsilon$.” The logic must flow $\varepsilon \Rightarrow K \Rightarrow \text{estimate}$.

Key tool. *Archimedean property*: for every $\varepsilon > 0$ there is $K \in \mathbb{N}$ with $1/K < \varepsilon$. Every $1/n$ -type estimate rests on it.

Worked Example 4.1.2 (The model limit $1/n \rightarrow 0$). Prove $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Recognition. Direct ε - K proof; the gap is already $1/n$, so this is the archetype.

Solution. Let $\varepsilon > 0$. By the Archimedean property choose $K \in \mathbb{N}$ with $K > 1/\varepsilon$. If $n \geq K$ then $\frac{1}{n} \leq \frac{1}{K} < \varepsilon$, and since $1/n > 0$ we have $|\frac{1}{n} - 0| = \frac{1}{n} < \varepsilon$. Hence $1/n \rightarrow 0$. *Check.* For $\varepsilon = 0.01$, $K = 101$ works: $1/101 < 0.01$. ■

Worked Example 4.1.3 (A rational expression). Prove $\frac{n}{n+1} \rightarrow 1$.

Recognition. “Prove this specific limit”; form and shrink the gap.

Solution. Compute $\left| \frac{n}{n+1} - 1 \right| = \left| \frac{-1}{n+1} \right| = \frac{1}{n+1} < \frac{1}{n}$. Given $\varepsilon > 0$ take $K > 1/\varepsilon$; then $n \geq K \implies \left| \frac{n}{n+1} - 1 \right| < \frac{1}{n} \leq \frac{1}{K} < \varepsilon$. *Check.* The terms $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$ visibly climb toward 1. ■

Worked Example 4.1.4 (Using a crude over-bound). Prove $\frac{2n+1}{n^2+3} \rightarrow 0$.

Recognition. The exact gap is messy; bound it crudely by a C/n shape (Step 2 of the Method).

Solution. For $n \geq 1$, $2n+1 \leq 3n$ and $n^2+3 \geq n^2$, so $\frac{2n+1}{n^2+3} \leq \frac{3n}{n^2} = \frac{3}{n}$. Given $\varepsilon > 0$ take $K > 3/\varepsilon$; then $n \geq K \implies |x_n - 0| \leq \frac{3}{n} \leq \frac{3}{K} < \varepsilon$. *Check.* The over-bound $3/n$ need not be sharp — it only must beat ε eventually. ■

Worked Example 4.1.5 (A non-example: $((-1)^n)$ diverges). Show the sequence $-1, 1, -1, 1, \dots$ has no limit.

Recognition. “Show divergent”; negate the definition — exhibit one ε no candidate limit can satisfy.

Solution. Suppose $(-1)^n \rightarrow x$. Take $\varepsilon = 1$. Then for large n both $|1 - x| < 1$ and $|-1 - x| < 1$ would hold (consecutive terms are ± 1). Adding, by the triangle inequality $2 = |1 - (-1)| \leq |1 - x| + |x - (-1)| < 1 + 1 = 2$, a strict $2 < 2$, absurd. So no limit exists. *Check.* The terms straddle a gap of width 2 forever; no single band of radius < 1 can hold them both. ■

Uniqueness, boundedness, and tails

Theorem 4.1.6 (Uniqueness of limits). *A convergent sequence has exactly one limit.*

Proof. Suppose $x_n \rightarrow x$ and $x_n \rightarrow x'$ with $x \neq x'$. Put $\varepsilon = \frac{1}{2}|x - x'| > 0$. Choose K_1 with $|x_n - x| < \varepsilon$ for $n \geq K_1$, and K_2 with $|x_n - x'| < \varepsilon$ for $n \geq K_2$. For $n \geq \max\{K_1, K_2\}$ the triangle inequality gives

$$|x - x'| \leq |x - x_n| + |x_n - x'| < \varepsilon + \varepsilon = |x - x'|,$$

i.e. $|x - x'| < |x - x'|$, a contradiction. Hence $x = x'$. □

Definition 4.1.7. A sequence (x_n) is *bounded* if there is $M > 0$ with $|x_n| \leq M$ for all $n \in \mathbb{N}$ (equivalently, its range is a bounded set).

Theorem 4.1.8 (Convergent $\implies$ bounded). *Every convergent sequence is bounded.*

Proof. Say $x_n \rightarrow x$. With $\varepsilon = 1$ pick K so that $|x_n - x| < 1$ for $n \geq K$; then for those n , $|x_n| \leq |x| + 1$. The finitely many earlier terms are bounded by $M_0 = \max\{|x_1|, \dots, |x_{K-1}|\}$. Take $M = \max\{M_0, |x| + 1\}$; then $|x_n| \leq M$ for all n . $\square$

The converse fails: $(-1)^n$ is bounded but divergent. Boundedness is necessary, not sufficient — a recurring asymmetry.

Theorem 4.1.9 (Tail equivalence). *Fix $m \in \mathbb{N}$ and let (x_n) be a sequence with m -tail $(x_{m+n})_{n \geq 1}$. Then (x_n) converges to x iff its m -tail converges to x .*

Proof. If $x_n \rightarrow x$ then any ε - K works for the tail by taking the same K . Conversely if the tail converges to x with threshold K' , then (x_n) satisfies the definition with $K = K' + m$. The first m terms are irrelevant. $\square$

Worked Example 4.1.10 (Square roots: $\sqrt{n+1} - \sqrt{n} \rightarrow 0$). Show $\sqrt{n+1} - \sqrt{n} \rightarrow 0$.

Recognition. A difference of roots; rationalize to expose a $1/\sqrt{n}$ shape.

Solution. Multiply by the conjugate: $\sqrt{n+1} - \sqrt{n} = \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}}$. Given $\varepsilon > 0$, take $K > 1/(4\varepsilon^2)$; then $n \geq K \implies x_n < \frac{1}{2\sqrt{n}} \leq \frac{1}{2\sqrt{K}} < \varepsilon$. *Check.* $n = 100$ gives $\sqrt{101} - 10 \approx 0.0499 < 1/(2 \cdot 10) = 0.05$. $\blacksquare$

Caution. The limit value must be guessed (or computed) *before* the ε - K proof; the definition only *verifies* a candidate, it does not produce one. To find the candidate use algebra/limit laws; to certify it use ε - K . Also: K may depend on ε but *never* on n — writing “choose $K = n$ ” is circular and wrong.

Caution. “ $x_n \rightarrow x$ ” is about the *tail*, not any finite stretch. A million agreeing terms prove nothing; a sequence that is 0 for $n \leq 10^9$ and then equals 5 forever converges to 5, not 0. Conversely, $|x_n - x|$ being small for *one large* n is not convergence — it must be small for *all* $n \geq K$.

Further worked examples

Worked Example 4.1.11 (Exhibiting an explicit $K(\varepsilon)$). Given $x_n = \frac{3n-1}{n+2}$, prove $x_n \rightarrow 3$ and find the smallest K that works for $\varepsilon = 0.01$.

Recognition. Direct ε - K proof (Definition 4.1.1); after bounding the gap, solve the threshold inequality *exactly* for the numeric part.

Solution. The gap is $\left| \frac{3n-1}{n+2} - 3 \right| = \left| \frac{(3n-1) - 3(n+2)}{n+2} \right| = \frac{7}{n+2}$. Given $\varepsilon > 0$, $\frac{7}{n+2} < \varepsilon \iff n > \frac{7}{\varepsilon} - 2$, so any $K > 7/\varepsilon - 2$ works (Archimedean). For $\varepsilon = 0.01$: $7/0.01 - 2 = 698$, so the smallest integer K is 699. *Check.* At $n = 699$, $7/(701) \approx 0.00999 < 0.01$; at $n = 698$, $7/700 = 0.01 \not< 0.01$, so 699 is indeed minimal. $\blacksquare$

Worked Example 4.1.12 (A null sequence that is not monotone). Show $x_n = \frac{(-1)^n}{n} \rightarrow 0$.

Recognition. The signs alternate, but the definition only cares about $|x_n - 0|$; take absolute values and the alternation disappears.

Solution. $|x_n - 0| = \left| \frac{(-1)^n}{n} \right| = \frac{1}{n}$. Given $\varepsilon > 0$ choose $K > 1/\varepsilon$; then $n \geq K \implies |x_n| = \frac{1}{n} \leq \frac{1}{K} < \varepsilon$. So $x_n \rightarrow 0$ even though the sequence is never monotone (it straddles 0). *Check.* Terms $-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, \dots$ visibly shrink toward 0 from both sides. $\blacksquare$

Worked Example 4.1.13 (A non-example: an unbounded sequence cannot converge). Show $x_n = \sqrt{n}$ does not converge to any real number.

Recognition. “Show divergent”; use the contrapositive of Theorem 4.1.8 (convergent $\implies$ bounded) rather than negating the ε - K definition directly.

Solution. If $\sqrt{n} \rightarrow x$ for some $x \in \mathbb{R}$, then $(\sqrt{n})$ would be bounded. But for any $M > 0$, choosing $n > M^2$ gives $\sqrt{n} > M$, so no bound M exists: the sequence is unbounded. An unbounded sequence cannot be

convergent, so $(\sqrt{n})$ diverges. *Check.* Boundedness is *necessary* for convergence; failing it is a clean disqualifier, no K needed. ■

Exercises (Bartle & Sherbert, §3.1).

- Write the first five terms of (x_n) where (a) $x_n = 1 + (-1)^n$; (b) $x_n = (-1)^n/n$; (c) $x_n = \frac{1}{n(n+1)}$; (d) $x_n = \frac{1}{n^2 + 2}$.
- Guess a formula for the n -th term from the pattern: (a) 5, 7, 9, 11, ...; (b) $\frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, -\frac{1}{16}, \dots$; (c) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$; (d) 1, 4, 9, 16, ...
- List the first five terms of the inductively defined sequences: (a) $x_1 = 1, x_{n+1} = 3x_n + 1$; (b) $y_1 = 2, y_{n+1} = \frac{1}{2}(y_n + 2/y_n)$; (c) $z_1 = 1, z_2 = 2, z_{n+2} = (z_{n+1} + z_n)/(z_{n+1} - z_n)$; (d) $s_1 = 3, s_2 = 5, s_{n+2} = s_n + s_{n+1}$.
- For any $b \in \mathbb{R}$, prove $\lim(b/n) = 0$.
- Use the definition of limit to establish: (a) $\lim \frac{n}{n^2 + 1} = 0$; (b) $\lim \frac{2n}{n+1} = 2$; (c) $\lim \frac{3n+1}{2n+5} = \frac{3}{2}$; (d) $\lim \frac{n^2 - 1}{2n^2 + 3} = \frac{1}{2}$.
- Show that (a) $\lim \frac{1}{\sqrt{n+7}} = 0$; (b) $\lim \frac{2n}{n+2} = 2$; (c) $\lim \frac{\sqrt{n}}{n+1} = 0$; (d) $\lim \frac{(-1)^n n}{n^2 + 1} = 0$.
- Let $x_n \geq 0$ with $\lim x_n = 0$. Prove $\lim \sqrt{x_n} = 0$.
- Prove that if $\lim x_n = x$ then $\lim |x_n| = |x|$. Give a converse counterexample.
- Prove $\lim \left(\frac{1}{n} - \frac{1}{n+1} \right) = 0$.
- Prove $\lim \frac{\sqrt{n}}{n+1} = 0$ and $\lim \frac{1}{\sqrt{n}} = 0$.
- For $x_n = \frac{(-1)^n}{n^2}$, determine $\lim x_n$ and prove your claim.
- Show that $\lim \frac{1}{2^n} = 0$.

Additional exercises.

- Using the definition only, prove $\frac{3n^2 + 1}{n^2 + n} \rightarrow 3$. Find an explicit $K(\varepsilon)$.
- Prove that if $x_n \rightarrow x$ then the “averaged” tail $y_n = \frac{1}{2}(x_n + x_{n+1}) \rightarrow x$.
- Prove $\sqrt[n]{a^n + b^n} \rightarrow \max\{a, b\}$ for fixed $a, b > 0$. [*Squeeze.*]
- (Non-example.) Show $x_n = \sin(n\pi/2)$ is bounded but divergent by exhibiting two subsequences with different limits, and explain why no ε - K data can rescue it.
- Suppose $|x_n - x_{n+1}| \leq 2^{-n}$ for all n . Must (x_n) converge? Prove or give a counterexample. (Contrast with the harmonic series.)
- Prove that a sequence (x_n) with $x_n \rightarrow x$ is bounded by giving *explicit* bounds in terms of x and the first few terms.
- (Diagram problem.) Sketch on one a_n -vs- n axis the sequence $a_n = 2 + \frac{(-1)^n}{n}$ for $n = 1, \dots, 10$, draw the band $(2 - \varepsilon, 2 + \varepsilon)$ for $\varepsilon = 0.25$, and mark the smallest K with all later dots inside. Report K .
- Let $x_n \rightarrow x$ and $x_n \geq 0$. Prove $x \geq 0$ and $\sqrt{x_n} \rightarrow \sqrt{x}$.

4.2 Limit Theorems

The algebra of limits

Once a handful of limits are known, the limit laws let us compute almost everything by algebra, never returning to ε - K . We collect them, prove them once, and reuse them.

Theorem 4.2.1 (Algebra of limits). *Suppose $x_n \rightarrow x$ and $y_n \rightarrow y$, and let $c \in \mathbb{R}$. Then*

$$(a) \ x_n + y_n \rightarrow x + y, \quad (b) \ c x_n \rightarrow c x, \quad (c) \ x_n y_n \rightarrow x y,$$

and, provided $y \neq 0$ (so $y_n \neq 0$ for large n),

$$(d) \ \frac{x_n}{y_n} \rightarrow \frac{x}{y}.$$

Proof. (a) Given $\varepsilon > 0$, pick K_1, K_2 with $|x_n - x| < \varepsilon/2$ ($n \geq K_1$) and $|y_n - y| < \varepsilon/2$ ($n \geq K_2$). For $n \geq \max\{K_1, K_2\}$,

$$|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

(b) is the case $y_n \equiv 0$ of (c), or directly $|c x_n - c x| = |c| |x_n - x|$.

(c) Since (x_n) converges it is bounded, say $|x_n| \leq M$; also $|y|$ is a constant. Write the standard “add and subtract” splitting:

$$|x_n y_n - x y| = |x_n y_n - x_n y + x_n y - x y| \leq |x_n| |y_n - y| + |y| |x_n - x| \leq M |y_n - y| + |y| |x_n - x|.$$

Given $\varepsilon > 0$, make each piece $< \varepsilon/2$ by choosing n large; the sum is $< \varepsilon$.

(d) It suffices to prove $1/y_n \rightarrow 1/y$ and then apply (c). Since $y \neq 0$, for large n we have $|y_n| \geq |y|/2$ (take $\varepsilon = |y|/2$ in $y_n \rightarrow y$). Then

$$\left| \frac{1}{y_n} - \frac{1}{y} \right| = \frac{|y - y_n|}{|y_n| |y|} \leq \frac{2}{|y|^2} |y_n - y| \rightarrow 0. \quad \square$$

Worked Example 4.2.2 (A rational limit by the laws). Find $\lim_{n \rightarrow \infty} \frac{2n^2 + 3}{n^2 - n + 1}$.

Recognition. “Find the limit” of a ratio of polynomials: divide by the top power and apply the algebra of limits term by term.

Solution. Divide numerator and denominator by n^2 :

$$\frac{2n^2 + 3}{n^2 - n + 1} = \frac{2 + 3/n^2}{1 - 1/n + 1/n^2} \xrightarrow{n \rightarrow \infty} \frac{2 + 0}{1 - 0 + 0} = 2,$$

using $1/n \rightarrow 0$, $1/n^2 \rightarrow 0$, and parts (a),(d). *Check.* $n = 10$ gives $203/91 \approx 2.23$; $n = 100$ gives $20003/9901 \approx 2.02$. ■

Worked Example 4.2.3 (When you cannot split: $\sqrt{n^2 + n} - n$). Evaluate $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$ (Rudin Ex. 3.2).

Recognition. The two pieces each diverge, so the laws *do not* apply directly — rationalize first, then the laws work.

Solution. $\sqrt{n^2 + n} - n = \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + 1/n} + 1} \rightarrow \frac{1}{1 + 1} = \frac{1}{2}$. *Check.* $n = 100$ gives $\sqrt{10100} - 100 \approx 0.4988$. ■

Caution. The algebra of limits requires *both* factor sequences to converge. “ $\infty - \infty$ ” and “ $0 \cdot \infty$ ” forms (like the previous example before rationalizing) are not covered; you must massage the expression into a convergent form first. Splitting a limit you have not yet shown to exist into a sum/product of limits is the single most common false step.

Limits and inequalities; the squeeze theorem

Theorem 4.2.4 (Limits preserve weak inequalities). *If $x_n \rightarrow x$, $y_n \rightarrow y$, and $x_n \leq y_n$ for all n (or all large n), then $x \leq y$. In particular, if $a \leq x_n \leq b$ eventually, then $a \leq x \leq b$.*

Proof. Put $z_n = y_n - x_n \geq 0$, $z_n \rightarrow y - x$. If $y - x < 0$, take $\varepsilon = \frac{1}{2}(x - y) > 0$; then for large n , $z_n < (y - x) + \varepsilon = \frac{1}{2}(y - x) < 0$, contradicting $z_n \geq 0$. So $y - x \geq 0$. $\square$

Caution. Strict inequalities are *not* preserved in the limit: $x_n = 1/n > 0$ for all n , yet $\lim x_n = 0$, not > 0 . Passing to the limit can only give $\geq$, never $>$.

Theorem 4.2.5 (Squeeze Theorem). *If $x_n \leq z_n \leq y_n$ for all large n and $x_n \rightarrow L$ and $y_n \rightarrow L$ (same limit), then $z_n \rightarrow L$.*

Proof. Let $\varepsilon > 0$. For large n , $L - \varepsilon < x_n$ and $y_n < L + \varepsilon$; combined with the sandwich, $L - \varepsilon < x_n \leq z_n \leq y_n < L + \varepsilon$, i.e. $|z_n - L| < \varepsilon$. $\square$

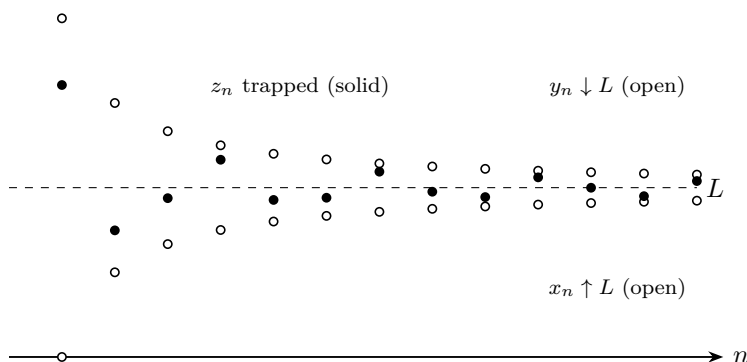


Figure 4.2. Squeeze: the inner sequence z_n is pinned between two envelopes converging to the same L , so it has no choice but to converge to L too.

Worked Example 4.2.6 (Squeeze in action: $\frac{\sin n}{n} \rightarrow 0$). Show $\frac{\sin n}{n} \rightarrow 0$.

Recognition. “Limit involving a bounded oscillation over a growing denominator”: bound the oscillation and squeeze.

Solution. Since $|\sin n| \leq 1$, $-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}$. Both outer sequences $\rightarrow 0$, so by squeezing, $\sin n/n \rightarrow 0$.

Check. This is the prototype “bounded $\times$ null = null”. ■

Theorem 4.2.7 (Ratio test for sequences). *Let $x_n > 0$ and suppose $L = \lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n}$ exists. (a) If $L < 1$ then $x_n \rightarrow 0$. (b) If $L > 1$ then $x_n \rightarrow +\infty$. (If $L = 1$ the test is inconclusive.)*

Proof. (a) Pick r with $L < r < 1$. For some K , $x_{n+1}/x_n < r$ when $n \geq K$, so by iterating, $0 < x_{K+m} < r^m x_K$. Since $r^m \rightarrow 0$ (next section / geometric), squeezing gives $x_n \rightarrow 0$. (b) Apply (a) to $1/x_n$, whose ratio $\rightarrow 1/L < 1$. $\square$

Worked Example 4.2.8 (Factorials beat exponentials: $\frac{2^n}{n!} \rightarrow 0$). Show $\frac{2^n}{n!} \rightarrow 0$.

Recognition. A ratio of fast-growing terms; the sequence ratio test isolates the dominant growth.

Solution. With $x_n = 2^n/n!$, $\frac{x_{n+1}}{x_n} = \frac{2^{n+1}/(n+1)!}{2^n/n!} = \frac{2}{n+1} \rightarrow 0 < 1$. By Theorem 4.2.7(a), $x_n \rightarrow 0$. *Check.* $x_4 = 16/24 \approx 0.67$, $x_8 = 256/40320 \approx 0.0063$ — collapsing fast. ■

Some special limits

These few limits recur constantly; commit them to memory along with their proofs.

Theorem 4.2.9 (Special limits). (a) If $p > 0$ then $\frac{1}{n^p} \rightarrow 0$. (b) If $|x| < 1$ then $x^n \rightarrow 0$; if $|x| > 1$ then $|x|^n \rightarrow +\infty$. (c) If $a > 0$ then $a^{1/n} \rightarrow 1$. (d) $n^{1/n} \rightarrow 1$. (e) $(1 + \frac{1}{n})^n \rightarrow e$ (defining e ; see §4.3 below).

Proof. (a) Given $\varepsilon > 0$, $1/n^p < \varepsilon \iff n > \varepsilon^{-1/p}$; Archimedean gives K .

(b) For $0 < |x| < 1$ write $|x| = 1/(1+h)$ with $h > 0$; Bernoulli's inequality gives $(1+h)^n \geq 1+nh > nh$, so $|x|^n = 1/(1+h)^n < 1/(nh) \rightarrow 0$. For $|x| > 1$ apply this to $1/|x|$.

(c) For $a \geq 1$ put $a^{1/n} = 1 + d_n$ with $d_n \geq 0$. Then $a = (1 + d_n)^n \geq 1 + nd_n$, so $0 \leq d_n \leq (a-1)/n \rightarrow 0$; squeeze gives $d_n \rightarrow 0$, i.e. $a^{1/n} \rightarrow 1$. For $0 < a < 1$ use $a^{1/n} = 1/(1/a)^{1/n} \rightarrow 1$.

(d) Put $n^{1/n} = 1 + e_n$ with $e_n \geq 0$. By the binomial theorem, for $n \geq 2$, $n = (1 + e_n)^n \geq \binom{n}{2} e_n^2 = \frac{n(n-1)}{2} e_n^2$, hence $e_n^2 \leq \frac{2}{n-1}$ and $0 \leq e_n \leq \sqrt{2/(n-1)} \rightarrow 0$. Squeeze gives $e_n \rightarrow 0$, so $n^{1/n} \rightarrow 1$.

(e) Deferred to §3 (Monotone Convergence), where $(1 + \frac{1}{n})^n$ is shown increasing and bounded above, its limit *defining* e . $\square$

Worked Example 4.2.10 (Combining special limits). Find $\lim_{n \rightarrow \infty} \frac{n^{1/n} + 3^{1/n}}{2}$ and $\lim_{n \rightarrow \infty} \frac{n^2}{2^n}$.

Recognition. “Find the limit” assembled from the special limits and the algebra of limits.

Solution. First: $n^{1/n} \rightarrow 1$ and $3^{1/n} \rightarrow 1$, so the average $\rightarrow (1+1)/2 = 1$. Second: the sequence ratio is $\frac{(n+1)^2/2^{n+1}}{n^2/2^n} = \frac{1}{2} \left(\frac{n+1}{n}\right)^2 \rightarrow \frac{1}{2} < 1$, so $n^2/2^n \rightarrow 0$ by Theorem 4.2.7. *Check.* $n^2/2^n$ at $n = 10$ is $100/1024 \approx 0.098$. ■

Caution. $n^{1/n} \rightarrow 1$, not $\rightarrow 0$ or $\rightarrow \infty$ — the root crushes the polynomial growth. Students often guess ∞ (“ n grows”) or 0 (“ $1/n$ exponent”). Both are wrong; the two effects exactly cancel to 1. Likewise $a^{1/n} \rightarrow 1$ for every fixed $a > 0$, however huge.

Further worked examples

Worked Example 4.2.11 (A squeeze with an n -term sum). Find $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n^2 + k}$.

Recognition. A sum of n shrinking terms: each term is too small alone but there are n of them. Bound the whole sum above and below by replacing k with its extremes, then squeeze (Theorem 4.2.5).

Solution. For $1 \leq k \leq n$ we have $n^2 + 1 \leq n^2 + k \leq n^2 + n$, so each term satisfies $\frac{1}{n^2 + n} \leq \frac{1}{n^2 + k} \leq \frac{1}{n^2 + 1}$. Summing the n terms,

$$\frac{n}{n^2 + n} \leq \sum_{k=1}^n \frac{1}{n^2 + k} \leq \frac{n}{n^2 + 1}.$$

The left side is $\frac{1}{n+1} \rightarrow 0$ and the right is $\frac{n}{n^2+1} \rightarrow 0$, so by squeezing the sum $\rightarrow 0$. *Check.* At $n = 3$ the sum is $\frac{1}{10} + \frac{1}{11} + \frac{1}{12} \approx 0.282$, between $\frac{3}{12} = 0.25$ and $\frac{3}{10} = 0.30$. ■

Worked Example 4.2.12 (A non-example: product of two divergent sequences). The algebra of limits needs both factors to converge. With $x_n = n$ and $y_n = 1/n^2$, show that x_n diverges yet $x_n y_n \rightarrow 0$, so “ $x_n y_n$ converges” carries no information about the factors.

Recognition. This is a caution-by-example on Theorem 4.2.1(c): the product rule is a sufficient, not necessary, condition.

Solution. $x_n = n \rightarrow +\infty$ (unbounded, hence divergent), while $y_n = 1/n^2 \rightarrow 0$. Their product $x_n y_n = n \cdot \frac{1}{n^2} = \frac{1}{n} \rightarrow 0$ converges. So the convergence of $(x_n y_n)$ does *not* let us split the limit as $(\lim x_n)(\lim y_n)$ — that step requires each factor to converge first. *Check.* Reversing roles, $x_n = n^2$, $y_n = 1/n$ gives $x_n y_n = n \rightarrow \infty$; the same product structure can converge or diverge depending on the factors, which is exactly why the hypothesis matters. ■

Worked Example 4.2.13 ($(a^{1/n})$ from the ratio test viewpoint). For fixed $a > 0$, re-derive $a^{1/n} \rightarrow 1$ using the Squeeze Theorem with Bernoulli's inequality, for the case $a > 1$.

Recognition. “Find/verify a special limit”; the cleanest route bounds the root deviation by a C/n term (as in Theorem 4.2.9(c)) and squeezes.

Solution. Write $a^{1/n} = 1 + d_n$ with $d_n \geq 0$ (since $a > 1$). Raising to the n -th power and using Bernoulli, $a = (1 + d_n)^n \geq 1 + nd_n$, hence $0 \leq d_n \leq \frac{a-1}{n}$. Both bounds $\rightarrow 0$, so $d_n \rightarrow 0$ by squeezing, i.e. $a^{1/n} = 1 + d_n \rightarrow 1$. *Check.* $a = 8$, $n = 10$: $d_{10} \leq 7/10 = 0.7$ and indeed $8^{1/10} - 1 \approx 0.231 < 0.7$. ■

Exercises (Bartle & Sherbert, §3.2).

- Establish convergence or divergence of $X = (x_n)$ where (a) $x_n = \frac{n}{n+1}$; (b) $x_n = \frac{(-1)^n n}{n+1}$; (c) $x_n = \frac{n^2}{n+1}$; (d) $x_n = \frac{2n^2+3}{n^2+1}$.
- Give an example of two divergent sequences X, Y such that (a) $X + Y$ converges; (b) XY converges.
- Show that if X and $X + Y$ are convergent, then Y is convergent.
- Show that if X converges to $x \neq 0$ and XY converges, then Y converges.
- Show that the following are not convergent: (a) (2^n) ; (b) $((-1)^n n^2)$.
- Find the limits: (a) $\lim(2 + 1/n)^2$; (b) $\lim \frac{(-1)^n}{n+2}$; (c) $\lim \frac{\sqrt{n}-1}{\sqrt{n}+1}$; (d) $\lim \frac{n+1}{n\sqrt{n}}$.
- If (b_n) is bounded and $\lim a_n = 0$, show $\lim(a_n b_n) = 0$. Explain why Theorem 4.2.1(c) cannot be used if (b_n) diverges.
- Apply the limit theorems to find: (a) $\lim \left(\frac{n+1}{n}\right) \left(\frac{1}{n}\right)$; (b) $\lim \frac{(n+1)(n+2)}{2n^2}$.
- Let $y_n = \sqrt{n+1} - \sqrt{n}$. Show (y_n) and $(\sqrt{n} y_n)$ converge; find the limits.
- Determine the limit of (x_n) , where (a) $x_n = \frac{2^n}{3^n}$; (b) $x_n = \frac{(-1)^n}{n!}$.
- Show that $\lim \frac{n!}{n^n} = 0$. [Hint: $0 < n!/n^n \leq 1/n$.]
- If $0 < a < b$, determine $\lim \frac{a^{n+1} + b^{n+1}}{a^n + b^n}$.

Additional exercises.

- Compute $\lim_{n \rightarrow \infty} \frac{3^n + 5^n}{4^n + 5^n}$ and justify each step by a limit law.
- Evaluate $\lim_{n \rightarrow \infty} (\sqrt{n^2 + 3n} - \sqrt{n^2 + 1})$ by rationalizing.
- Use squeezing to find $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n k$ and $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n^2 + k}$.
- Use the sequence ratio test to show $\frac{n^{10}}{1.1^n} \rightarrow 0$, and find approximately the index where the terms begin to decrease.
- (Non-example.) Give two sequences $(a_n), (b_n)$ with $a_n \rightarrow \infty$, $b_n \rightarrow 0$, such that $a_n b_n$ has *any* prescribed real limit L you choose, and one pair for which $a_n b_n$ diverges. (Shows “ $\infty \cdot 0$ ” is indeterminate.)
- Prove the squeeze theorem fails if the two outer limits differ, by exhibiting $x_n \leq z_n \leq y_n$ with $x_n \rightarrow 0$, $y_n \rightarrow 1$, and z_n divergent.

7. Prove $n^{1/n} \rightarrow 1$ a second way using $\ln n^{1/n} = (\ln n)/n \rightarrow 0$ (assuming $\ln$ continuity), and compare with the binomial proof in the notes.
8. (Diagram problem.) Plot $a_n = n^{1/n}$ for $n = 1, \dots, 12$. Mark the maximum term and the horizontal asymptote $y = 1$; identify from the picture where the sequence starts decreasing.
9. Show that if $x_n \rightarrow x$ and $x > 0$, then $x_n > 0$ for all large n , but the conclusion fails if $x = 0$.

4.3 Monotone Sequences

Monotone sequences and the monotone convergence theorem

Definition 4.3.1. A sequence (x_n) is *increasing* if $x_n \leq x_{n+1}$ for all n , and *strictly increasing* if $x_n < x_{n+1}$; *decreasing* and *strictly decreasing* are defined with the inequalities reversed. A sequence is *monotone* if it is either increasing or decreasing.

For a bounded sequence the missing ingredient that forces convergence is monotonicity: a climbing-but-capped sequence must pile up against its ceiling. This is where the *completeness* of $\mathbb{R}$ (every set bounded above has a supremum) enters analysis.

Theorem 4.3.2 (Monotone Convergence Theorem). *A monotone sequence of real numbers converges if and only if it is bounded. Moreover:*

- if (x_n) is increasing and bounded above, then $x_n \rightarrow \sup_n x_n$;
- if (x_n) is decreasing and bounded below, then $x_n \rightarrow \inf_n x_n$.

Proof. $(\Rightarrow)$ Convergent sequences are bounded (Theorem 4.1.8).

$(\Leftarrow)$ Take (x_n) increasing and bounded above; let $x^* = \sup\{x_n : n \in \mathbb{N}\}$, which exists by completeness. Let $\varepsilon > 0$. Since $x^* - \varepsilon$ is *not* an upper bound, some term $x_K > x^* - \varepsilon$. By monotonicity, $n \geq K \implies x_n \geq x_K > x^* - \varepsilon$; and $x_n \leq x^*$ always. Hence $n \geq K \implies x^* - \varepsilon < x_n \leq x^* < x^* + \varepsilon$, i.e. $|x_n - x^*| < \varepsilon$. So $x_n \rightarrow x^*$. The decreasing case follows by negating. $\square$

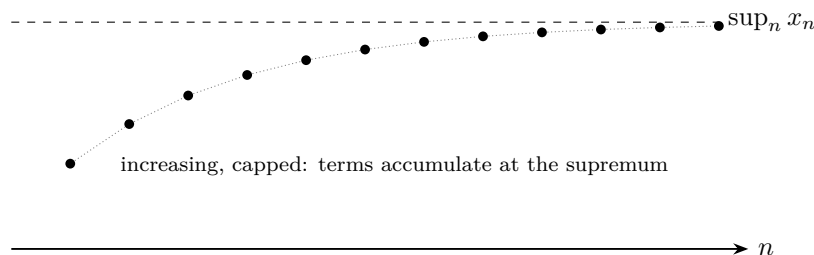


Figure 4.3. An increasing sequence bounded above converges to its least upper bound: it can never reach above $\sup_n x_n$, and it eventually exceeds every value below it.

Method (Proving a recursive sequence converges). *Recognition trigger.* A sequence given by $x_1 = a$, $x_{n+1} = f(x_n)$, asked to “show it converges and find the limit”. No closed form is available.

1. **Monotone:** prove (x_n) is increasing (or decreasing), usually by induction — compare x_{n+1} to x_n or show $x_{n+1} - x_n$ keeps one sign.
2. **Bounded:** prove the relevant one-sided bound (above if increasing, below if decreasing), also by induction — a good guess for the bound is the eventual limit.
3. **MCT:** conclude $x_n \rightarrow L$ for some L .
4. **Solve for L :** let $n \rightarrow \infty$ in $x_{n+1} = f(x_n)$. If f is continuous, $L = f(L)$; solve this *fixed-point equation*, discarding roots ruled out by the bound.

Caution. Step 4 *assumes* convergence (Steps 1–3); “taking the limit of both sides” alone is not a proof of convergence.

Worked Example 4.3.3 (A square-root recursion). Let $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$. Show (x_n) converges and find its limit (B&S 3.3.4).

Recognition. Recursive $x_{n+1} = f(x_n)$ with $f(x) = \sqrt{2 + x}$; run the four-step Method.

Solution. Bounded above by 2: $x_1 = 1 < 2$; if $x_n < 2$ then $x_{n+1} = \sqrt{2 + x_n} < \sqrt{4} = 2$. Increasing: $x_{n+1}^2 - x_n^2 = 2 + x_n - x_n^2 = -(x_n - 2)(x_n + 1) > 0$ for $0 < x_n < 2$, so $x_{n+1} > x_n$. By MCT, $x_n \rightarrow L \in (1, 2]$. Taking limits: $L = \sqrt{2 + L} \implies L^2 - L - 2 = 0 \implies L = 2$ (the root -1 is rejected). Check. $x_4 = \sqrt{2 + \sqrt{2 + \sqrt{3}}} \approx 1.962$, nearing 2. ■

Worked Example 4.3.4 (A decreasing recursion (Babylonian root, B&S 3.3.5)). Let $x_1 = 2$ and $x_{n+1} = \frac{1}{2}(x_n + \frac{2}{x_n})$. Show $x_n \rightarrow \sqrt{2}$.

Recognition. The arithmetic-mean map for $\sqrt{2}$; show bounded below by $\sqrt{2}$ and decreasing.

Solution. By AM–GM, $x_{n+1} = \frac{1}{2}(x_n + 2/x_n) \geq \sqrt{x_n \cdot 2/x_n} = \sqrt{2}$, so $x_n \geq \sqrt{2}$ for all $n \geq 2$. Then $x_{n+1} - x_n = \frac{1}{2}(2/x_n - x_n) = \frac{2 - x_n^2}{2x_n} \leq 0$, so the sequence decreases. By MCT $x_n \rightarrow L \geq \sqrt{2}$, and $L = \frac{1}{2}(L + 2/L) \implies L^2 = 2 \implies L = \sqrt{2}$. Check. $x_2 = 1.5$, $x_3 \approx 1.41667$, $x_4 \approx 1.414216$ — quadratic convergence to $\sqrt{2} = 1.41421 \dots$ ■

The number e

Theorem 4.3.5 (e as a monotone limit). The sequence $a_n = (1 + \frac{1}{n})^n$ is increasing and bounded above; its limit is the number e . Moreover $b_n = (1 + \frac{1}{n})^{n+1}$ is decreasing with the same limit, and $a_n < e < b_n$ for all n , with $2 < e < 3$.

Proof sketch. By the binomial theorem,

$$a_n = \sum_{k=0}^n \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^n \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right).$$

Each factor $(1 - j/n)$ increases with n , and passing from a_n to a_{n+1} adds a new nonnegative term, so a_n increases. Bounding each factor by 1, $a_n \leq \sum_{k=0}^n \frac{1}{k!} \leq 1 + 1 + \sum_{k=2}^n \frac{1}{2^{k-1}} < 3$. By MCT, $a_n \rightarrow e \leq 3$, and $a_2 = 2.25$ gives $e > 2$. The companion b_n is checked to decrease, with $b_n/a_n = (1 + 1/n) \rightarrow 1$, so $\lim b_n = \lim a_n = e$. □

Remark. The series form $\sum_{k=0}^{\infty} \frac{1}{k!}$ also equals e (Section 8), and converges far faster: a_n approaches e like $e/(2n)$, whereas the series error after n terms is below $1/(n \cdot n!)$. We prove e is irrational in Section 8.

Worked Example 4.3.6 (Recognizing the e -pattern). Find $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{2n}$ and $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n$ (B&S 3.3.12).

Recognition. Powers of $(1 \pm c/n)$: reshape into the $(1 + \frac{1}{m})^m \rightarrow e$ template.

Solution. $(1 + \frac{1}{n})^{2n} = [(1 + \frac{1}{n})^n]^2 \rightarrow e^2$. For the second, $(1 - \frac{1}{n})^n = (\frac{n-1}{n})^n = (\frac{n-1}{n-1})^{-n} = (1 + \frac{1}{n-1})^{-n} \rightarrow e^{-1}$. Check. $e^2 \approx 7.389$, $1/e \approx 0.368$; at $n = 1000$ the values are 7.382 and 0.3682. ■

Caution. For a recursion $x_{n+1} = f(x_n)$, solving $L = f(L)$ before establishing convergence can mislead: the equation may have several roots, or the sequence may diverge entirely (e.g. $x_1 = 1$, $x_{n+1} = x_n + 1/x_n$ satisfies no finite fixed point and indeed $\rightarrow +\infty$, B&S 3.3.7). Always do Steps 1–3 first; the fixed-point equation only identifies a limit you have already shown exists.

Caution. “Increasing and bounded above” gives convergence; “increasing” alone does not (it may run to $+\infty$), and “bounded” alone does not (it may oscillate, like $(-1)^n$). All *three* ingredients — monotone, bounded, on the correct side — are needed for MCT.

Further worked examples

Worked Example 4.3.7 (A bounded monotone partial-sum sequence). Show $s_n = \sum_{k=1}^n \frac{1}{k^2}$ converges, without computing the value.

Recognition. “Show convergence” of an increasing sequence: produce *any* upper bound and invoke MCT (Theorem 4.3.2). The exact limit ($\pi^2/6$) is not needed.

Solution. The partial sums increase ($s_{n+1} - s_n = 1/(n+1)^2 > 0$). For an upper bound, use $\frac{1}{k^2} \leq \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$ for $k \geq 2$, so

$$s_n = 1 + \sum_{k=2}^n \frac{1}{k^2} \leq 1 + \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) = 1 + \left(1 - \frac{1}{n} \right) < 2.$$

Thus (s_n) is increasing and bounded above by 2, so it converges by MCT. *Check.* The true limit $\pi^2/6 \approx 1.645 < 2$, consistent with the bound. ■

Worked Example 4.3.8 (A recursion that converges to a fixed point). Let $x_1 = 1$ and $x_{n+1} = \frac{x_n + 2}{x_n + 1}$. Show (x_n) converges and find its limit.

Recognition. Recursive $x_{n+1} = f(x_n)$ with $f(x) = \frac{x+2}{x+1}$. The sequence is not monotone, so instead bound it and identify the fixed point; here we show the *odd* and *even* subsequences are each monotone and meet (the four-step Method, adapted).

Solution. Note f maps $[1, 2]$ into $[1, 2]$: if $1 \leq x \leq 2$ then $f(x) = 1 + \frac{1}{x+1} \in [1 + \frac{1}{3}, 1 + \frac{1}{2}] = [\frac{4}{3}, \frac{3}{2}] \subseteq [1, 2]$, so all terms lie in $[1, 2]$. The fixed point solves $L = \frac{L+2}{L+1} \implies L^2 + L = L + 2 \implies L^2 = 2 \implies L = \sqrt{2}$ (positive root). One checks $|x_{n+1} - \sqrt{2}| = \frac{|x_n - \sqrt{2}|}{x_n + 1} \leq \frac{1}{2} |x_n - \sqrt{2}|$ (since $x_n + 1 \geq 2$), so the error halves each step and $x_n \rightarrow \sqrt{2}$. *Check.* $x_1 = 1$, $x_2 = 1.5$, $x_3 = 1.4$, $x_4 \approx 1.41667$, closing on $\sqrt{2} \approx 1.41421$. ■

Worked Example 4.3.9 (A non-example: increasing but unbounded). Show $x_n = \sum_{k=1}^n \frac{1}{\sqrt{k}}$ is increasing yet diverges, so monotonicity alone is not enough for MCT.

Recognition. This stresses the “bounded” hypothesis of Theorem 4.3.2: drop it and an increasing sequence may run to $+\infty$.

Solution. The sequence increases ($x_{n+1} - x_n = 1/\sqrt{n+1} > 0$). But for $1 \leq k \leq n$, $\frac{1}{\sqrt{k}} \geq \frac{1}{\sqrt{n}}$, so $x_n \geq n \cdot \frac{1}{\sqrt{n}} = \sqrt{n} \rightarrow +\infty$. Being unbounded above, (x_n) has no supremum in $\mathbb{R}$ and cannot converge. *Check.* $x_4 = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{2} \approx 2.78 \geq \sqrt{4} = 2$; the lower bound $\sqrt{n}$ already forces divergence. ■

Exercises (Bartle & Sherbert, §3.3).

1. Let $x_1 = 8$ and $x_{n+1} = \frac{1}{2}x_n + 2$. Show (x_n) is bounded and monotone; find the limit.
2. Let $x_1 > 1$ and $x_{n+1} = 2 - 1/x_n$. Show (x_n) is bounded and monotone; find the limit.
3. Let $x_1 \geq 2$ and $x_{n+1} = 1 + \sqrt{x_n - 1}$. Show (x_n) is decreasing and bounded below by 2; find the limit.
4. Let $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$. Show (x_n) converges and find the limit.
5. Let $y_1 = \sqrt{p}$ ($p > 0$) and $y_{n+1} = \sqrt{p + y_n}$. Show (y_n) converges and find the limit. [Hint: one upper bound is $1 + 2\sqrt{p}$.]

6. Let $a > 0$, $z_1 > 0$, $z_{n+1} = \sqrt{a + z_n}$. Show (z_n) converges and find the limit.
7. Let $x_1 = a > 0$ and $x_{n+1} = x_n + 1/x_n$. Determine whether (x_n) converges or diverges.
8. Let (a_n) increase, (b_n) decrease, with $a_n \leq b_n$ for all n . Show $\lim a_n \leq \lim b_n$, and deduce the Nested Intervals Property from the MCT.
9. Let $A \subseteq \mathbb{R}$ be infinite, bounded above, $u = \sup A$. Show there is an increasing sequence (x_n) in A with $u = \lim x_n$.
10. Decide convergence/divergence of $y_n = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}$.
11. Let $x_n = \frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{n^2}$. Prove (x_n) is increasing and bounded, hence converges. [Hint: $1/k^2 \leq 1/(k(k-1)) = 1/(k-1) - 1/k$ for $k \geq 2$.]
12. Establish convergence and find the limits of (a) $(1 + 1/n)^{n+1}$; (b) $(1 + 1/n)^{2n}$; (c) $(1 + \frac{1}{n+1})^n$; (d) $(1 - 1/n)^n$.
13. Use the Babylonian method (Example 3.3.5) to compute $\sqrt{2}$ to 4 decimals.
14. Use the same method to compute $\sqrt{5}$ to 5 decimals.
15. Compute $e_n = (1 + 1/n)^n$ for $n = 2, 4, 8, 16$.
16. Compute e_n for $n = 50, 100, 1000$.

Additional exercises.

1. Let $x_1 = 1$, $x_{n+1} = \frac{1}{4}(x_n + 3)$. Show (x_n) is increasing, bounded above by 1? Decide the correct one-sided bound, prove convergence, and find the limit.
2. Let $x_1 = 3$ and $x_{n+1} = \frac{x_n^2 + 2}{2x_n}$. Show (x_n) decreases to $\sqrt{2}$.
3. Show $((1 + 1/n)^n)$ and $((1 + 1/n)^{n+1})$ trap e : prove the first is below e , the second above, and their ratio $\rightarrow 1$.
4. (Non-example.) Exhibit a bounded sequence that is *not* monotone and does not converge, demonstrating that boundedness alone does not trigger the MCT.
5. Let $x_1 = 2$, $x_{n+1} = x_n + 1/x_n$. Show (x_n) is increasing and unbounded (so the fixed-point equation $L = L + 1/L$ has no solution — a cautionary case).
6. Prove that if (x_n) is increasing and a subsequence converges to L , then the whole sequence converges to L .
7. (Diagram problem.) For $x_1 = 0.5$, $x_{n+1} = \sqrt{2 + x_n}$, draw the cobweb/staircase diagram of $y = \sqrt{2 + x}$ against $y = x$ for the first four iterates; mark the fixed point.
8. Define $H_n = \sum_{k=1}^n 1/k$. Show $(H_n - \ln n)$ is decreasing and bounded below by 0, hence converges (to the Euler–Mascheroni constant γ).

4.4 Subsequences and the Bolzano–Weierstrass Theorem

Subsequences and subsequential limits

Definition 4.4.1. Let (x_n) be a sequence and let $n_1 < n_2 < n_3 < \cdots$ be a strictly increasing sequence of indices in $\mathbb{N}$. The sequence $(x_{n_k})_{k=1}^\infty$ is a *subsequence* of (x_n) . A point L is a *subsequential limit* of (x_n) if some subsequence converges to L .

Because $n_1 < n_2 < \cdots$ are strictly increasing naturals, $n_k \geq k$ always. Subsequences let us extract order from chaos: even a wildly oscillating bounded sequence harbors convergent subsequences (the Bolzano–Weierstrass theorem, below).

Theorem 4.4.2 (Subsequences inherit the limit). (x_n) converges to x iff every subsequence of (x_n) converges to x .

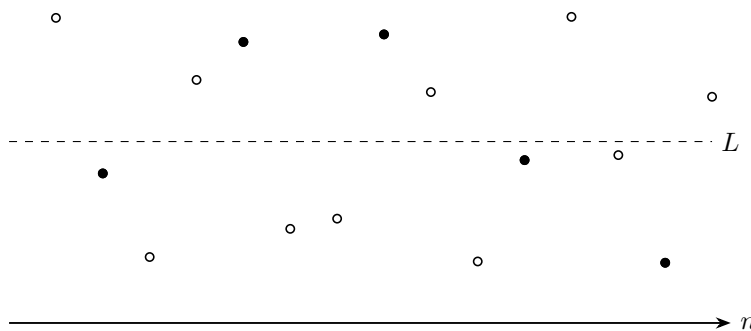
Proof. $(\Rightarrow)$ If $x_n \rightarrow x$ and $\varepsilon > 0$, pick K with $|x_n - x| < \varepsilon$ for $n \geq K$. For a subsequence, $k \geq K \Rightarrow n_k \geq k \geq K \Rightarrow |x_{n_k} - x| < \varepsilon$. $(\Leftarrow)$ The full sequence is a subsequence of itself. $\square$

Corollary 4.4.3 (Divergence by two subsequences). If (x_n) has two subsequences converging to different limits, then (x_n) diverges.

Worked Example 4.4.4 (Divergence of $((-1)^n)$ and a shuffle). Show $((-1)^n)$ diverges, and decide when the “shuffle” $z_{2k-1} = x_k, z_{2k} = y_k$ converges (B&S 3.4.5).

Recognition. Use subsequences to detect divergence (Corollary 4.4.3).

Solution. The even subsequence of $((-1)^n)$ is constant $1 \rightarrow 1$; the odd subsequence is $-1 \rightarrow -1$. Two different subsequential limits $\Rightarrow$ divergence. For the shuffle (z_n) : its odd-indexed subsequence is (x_k) and even-indexed is (y_k) . If $(z_n) \rightarrow z$ then both must converge to z , so $\lim x_k = \lim y_k = z$; conversely if $x_k \rightarrow L$ and $y_k \rightarrow L$ then $(z_n) \rightarrow L$ (an ε threshold for each forces one for z). So (z_n) converges iff both converge to a common limit. ■



open dots: full bounded sequence; solid dots: a chosen subsequence converging to L

Figure 4.4. From a bounded but non-convergent sequence one *selects* a subsequence homing in on one cluster value — the content of Bolzano–Weierstrass.

The Bolzano–Weierstrass theorem

Theorem 4.4.5 (Monotone Subsequence Theorem). Every sequence of real numbers has a monotone subsequence.

Proof. Call m a *peak* if $x_m \geq x_n$ for all $n > m$. *Case 1:* infinitely many peaks $m_1 < m_2 < \dots$. Then $x_{m_1} \geq x_{m_2} \geq \dots$ is a decreasing subsequence. *Case 2:* finitely many peaks; let N exceed the last. Since $n_1 := N + 1$ is not a peak, some $n_2 > n_1$ has $x_{n_2} > x_{n_1}$; n_2 is not a peak either, so some $n_3 > n_2$ has $x_{n_3} > x_{n_2}$; continuing gives a strictly increasing subsequence. $\square$

Theorem 4.4.6 (Bolzano–Weierstrass, in $\mathbb{R}$). Every bounded sequence of real numbers has a convergent subsequence.

Proof. By Theorem 4.4.5 extract a monotone subsequence; it is bounded (a subsequence of a bounded sequence), hence convergent by the Monotone Convergence Theorem 4.3.2. $\square$

Remark (The bisection proof). Rudin’s route uses *nested intervals*: if $(x_n) \subset [a, b]$, halve $[a, b]$; at least one half contains x_n for infinitely many n — keep it, call it I_1 , and pick $x_{n_1} \in I_1$. Halve I_1 , keep a half I_2 holding infinitely many terms, pick $x_{n_2} \in I_2$ with $n_2 > n_1$, and so on. The I_k are nested with $|I_k| = (b - a)/2^k \rightarrow 0$, so by the Nested Interval Property they share a single point L , and $|x_{n_k} - L| \leq |I_k| \rightarrow 0$, so $x_{n_k} \rightarrow L$.

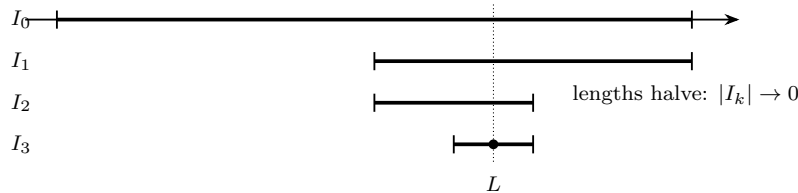


Figure 4.5. Bisection: always keep a half containing infinitely many terms. The nested halves shrink to a single limit point L .

Theorem 4.4.7 (Bolzano–Weierstrass in $\mathbb{R}^k$). *Every bounded sequence in $\mathbb{R}^k$ has a convergent subsequence.*

Proof. Write each term as $\mathbf{x}_n = (x_n^{(1)}, \dots, x_n^{(k)})$. The first coordinate sequence $(x_n^{(1)})$ is bounded, so by Theorem 4.4.6 some subsequence makes the first coordinate converge. Within *that* subsequence, the second coordinate is bounded, so pass to a further subsequence making it converge too (the first still converges). Iterate through all k coordinates. The final subsequence converges in every coordinate, hence (a finite-dimensional vector converges iff each coordinate does) converges in $\mathbb{R}^k$. $\square$

Worked Example 4.4.8 ($c^{1/n} \rightarrow 1$ for $0 < c < 1$ via a subsequence (B&S 3.4.2)). Show $0 < c < 1 \implies c^{1/n} \rightarrow 1$.

Recognition. “Find the limit” where the sequence is monotone bounded; identify the limit through a subsequence (Theorem 4.4.2).

Solution. For $0 < c < 1$, $c^{1/n} < c^{1/(n+1)} < 1$, so $(c^{1/n})$ is increasing and bounded above by 1; by MCT it converges to some $L \leq 1$. The subsequence $c^{1/2n} = (c^{1/n})^{1/2}$ converges both to L (subsequence) and to $L^{1/2}$ (continuity of $\sqrt{\cdot}$), so $L = L^{1/2}$, giving $L \in \{0, 1\}$. Since $c^{1/n} \geq c > 0$, $L \neq 0$, hence $L = 1$. *Check.* $c = \frac{1}{2}$: $c^{1/10} \approx 0.933$, $c^{1/100} \approx 0.993$. $\blacksquare$

Caution. Bolzano–Weierstrass gives *some* convergent subsequence, not convergence of the whole sequence. $((-1)^n)$ is bounded with convergent subsequences ($\rightarrow 1$ and $\rightarrow -1$) yet diverges. “Bounded” alone never yields a limit; it yields a *subsequential* limit.

Further worked examples

Worked Example 4.4.9 (The set of subsequential limits of an enumeration of $\mathbb{Q} \cap [0, 1]$). Let (x_n) enumerate all rationals in $[0, 1]$. Describe its subsequential limits.

Recognition. “Find all subsequential limits”; a point L is a subsequential limit iff x_n is within ε of L infinitely often (Definition 4.4.1).

Solution. Every $L \in [0, 1]$ is a subsequential limit: given L and $\varepsilon > 0$, the interval $(L - \varepsilon, L + \varepsilon) \cap [0, 1]$ contains *infinitely many* rationals (the rationals are dense), each appearing as some x_n , so we can pick indices $n_1 < n_2 < \dots$ with $|x_{n_k} - L| < 1/k$, giving $x_{n_k} \rightarrow L$. No $L \notin [0, 1]$ qualifies, since all $x_n \in [0, 1]$ and limits preserve $0 \leq x_n \leq 1$. So the set of subsequential limits is exactly $[0, 1]$. *Check.* The whole sequence wildly diverges, yet it “visits” every point of $[0, 1]$ in the limit — a maximal cluster set. $\blacksquare$

Worked Example 4.4.10 (Building a monotone subsequence by the peak method). For $x_n = \frac{(-1)^n n}{n+1}$, extract an explicit monotone convergent subsequence.

Recognition. Apply the Monotone Subsequence Theorem 4.4.5 concretely: locate a one-signed, monotone run of indices.

Solution. The even-indexed terms are $x_{2k} = \frac{2k}{2k+1} = 1 - \frac{1}{2k+1}$, which increase to 1. So the subsequence (x_{2k}) is strictly increasing, bounded above by 1, hence monotone and convergent with $x_{2k} \rightarrow 1$ by MCT. (The odd-indexed terms $x_{2k-1} = -\frac{2k-1}{2k} \rightarrow -1$ give the other subsequential limit.) *Check.* $x_2 = \frac{2}{3}$, $x_4 = \frac{4}{5}$, $x_6 = \frac{6}{7}$, visibly climbing to 1. $\blacksquare$

Worked Example 4.4.11 (Bolzano–Weierstrass in $\mathbb{R}^2$). Show the bounded sequence $\mathbf{x}_n = (\cos n, (-1)^n \frac{n}{n+1})$ in $\mathbb{R}^2$ has a convergent subsequence, and exhibit the coordinate-by-coordinate extraction.

Recognition. “Find a convergent subsequence in $\mathbb{R}^k$ ”: run the coordinate diagonalization of Theorem 4.4.7, extracting one coordinate at a time.

Solution. Both coordinates are bounded ($|\cos n| \leq 1$, $|(-1)^n n/(n+1)| \leq 1$), so $(\mathbf{x}_n)$ is bounded. First coordinate: $(\cos n)$ is bounded, so by B–W in $\mathbb{R}$ some subsequence $(\cos n_k)$ converges, say to $c \in [-1, 1]$. Within that subsequence, the second coordinate $(-1)^{n_k} \frac{n_k}{n_k+1}$ is bounded, so pass to a further subsequence (n_{k_j}) on which it converges (to $+1$ or -1). On this final subsequence *both* coordinates converge, so $\mathbf{x}_{n_{k_j}}$ converges in $\mathbb{R}^2$. *Check.* A vector sequence converges iff each coordinate does, exactly the principle the iterated extraction exploits. ■

Exercises (Bartle & Sherbert, §3.4).

1. Give an example of an unbounded sequence that has a convergent subsequence.
2. Use the method of Example 3.4.3(b) to show that if $0 < c < 1$ then $\lim c^{1/n} = 1$.
3. Let (f_n) be the Fibonacci sequence and $x_n = f_{n+1}/f_n$. Given $\lim x_n = L$ exists, find L .
4. Show the following diverge: (a) $(1 - (-1)^n + 1/n)$; (b) $(\sin(n\pi/4))$.
5. Let $X = (x_n)$, $Y = (y_n)$, and the shuffle Z be $z_{2n-1} = x_n$, $z_{2n} = y_n$. Show Z converges iff both X, Y converge and $\lim X = \lim Y$.
6. Let $x_n = n^{1/n}$. (a) Show $x_{n+1} < x_n \iff (1 + 1/n)^n < n$, valid for $n \geq 3$; conclude (x_n) is ultimately decreasing, so $x = \lim x_n$ exists. (b) Using that (x_{2n}) also converges to x , conclude $x = 1$.
7. Establish convergence and find the limits of (a) $(1 + 1/n^2)^{n^2}$; (b) $(1 + 1/2n)^n$; (c) $(1 + 1/n^2)^{2n^2}$; (d) $(1 + 2/n)^n$.
8. Determine the limits of (a) $(3n)^{1/2n}$; (b) $(1 + 1/2n)^{3n}$.
9. Suppose every subsequence of $X = (x_n)$ has a further subsequence converging to 0. Show $\lim X = 0$.
10. Let (x_n) be bounded, $s_n = \sup\{x_k : k \geq n\}$, $S = \inf\{s_n\}$. Show some subsequence of (x_n) converges to S .
11. Suppose $x_n \geq 0$ for all n and $\lim((-1)^n x_n)$ exists. Show (x_n) converges.
12. Show that if (x_n) is unbounded, there is a subsequence (x_{n_k}) with $\lim(1/x_{n_k}) = 0$.
13. If $x_n = (-1)^n/n$, find the subsequence constructed in the bisection proof of Bolzano–Weierstrass when $I_1 = [-1, 1]$.

Additional exercises.

1. List three subsequences of $x_n = (-1)^n + 1/n$ with distinct behavior, and find all subsequential limits.
2. Prove directly (no monotone-subsequence lemma) that the bounded sequence $x_n = \sin n$ has a convergent subsequence, using the pigeonhole/bisection idea on $[-1, 1]$.
3. Show that if every subsequence of (x_n) has a further subsequence converging to the *same* L , then $x_n \rightarrow L$.
4. (Non-example.) Give a sequence with infinitely many subsequential limits (e.g. an enumeration of $\mathbb{Q} \cap [0, 1]$) and explain why Bolzano–Weierstrass still applies.
5. Find the set of subsequential limits of $x_n = \cos(n)$ informally, and state what Bolzano–Weierstrass guarantees about it.
6. Prove that a sequence converges iff its even-indexed and odd-indexed subsequences both converge to the *same* limit.
7. (Diagram problem.) Plot a bounded oscillating sequence of your design on a_n -vs- n axes and *circle* a monotone subsequence; state whether it is the increasing or decreasing case of the Monotone Subsequence Theorem.

8. Use Bolzano–Weierstrass in $\mathbb{R}^2$ to show every sequence in the closed unit disk has a subsequence converging to a point of the disk.

4.5 Cauchy Sequences and Completeness

Cauchy sequences

Convergence (Definition 4.1.1) requires knowing the limit x in advance. The *Cauchy condition* expresses “the terms bunch together” *without* naming a limit — exactly the right notion for proving a sequence converges when no candidate is available.

Definition 4.5.1. A sequence (x_n) is a *Cauchy sequence* if for every $\varepsilon > 0$ there is $K \in \mathbb{N}$ such that

$$m, n \geq K \implies |x_m - x_n| < \varepsilon.$$

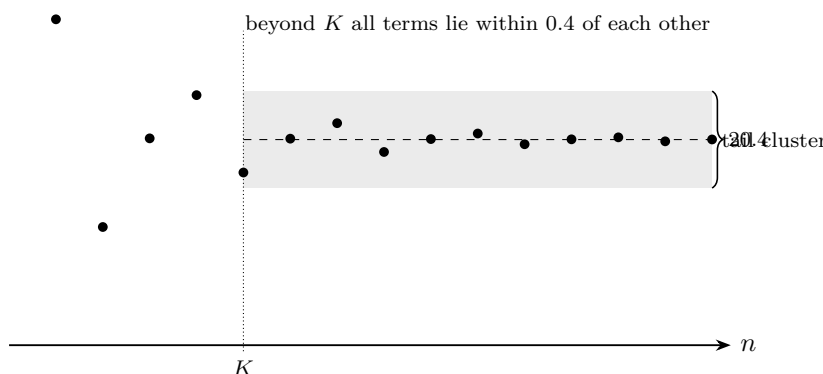


Figure 4.6. Cauchy: past index K the entire tail fits in a window of width ε — the terms cluster, even before we name where.

Lemma 4.5.2 (Cauchy sequences are bounded). *Every Cauchy sequence is bounded.*

Proof. With $\varepsilon = 1$ pick K so that $|x_n - x_K| < 1$ for $n \geq K$; then $|x_n| \leq |x_K| + 1$ for $n \geq K$. The finitely many earlier terms are bounded; take the max as in Theorem 4.1.8. $\square$

Theorem 4.5.3 (Cauchy Criterion / Completeness of $\mathbb{R}$). *A sequence of real numbers converges if and only if it is Cauchy.*

Proof. $(\implies)$ If $x_n \rightarrow x$, given $\varepsilon > 0$ pick K with $|x_n - x| < \varepsilon/2$ for $n \geq K$; then $m, n \geq K \implies |x_m - x_n| \leq |x_m - x| + |x - x_n| < \varepsilon$.

$(\impliedby)$ Let (x_n) be Cauchy. By Lemma 4.5.2 it is bounded, so by Bolzano–Weierstrass (Theorem 4.4.6) some subsequence $x_{n_k} \rightarrow x$. We show the *whole* sequence converges to x . Let $\varepsilon > 0$. Pick K with $|x_m - x_n| < \varepsilon/2$ for $m, n \geq K$ (Cauchy), and pick a subsequence index $n_k \geq K$ with $|x_{n_k} - x| < \varepsilon/2$ (subsequence limit). Then for $n \geq K$,

$$|x_n - x| \leq |x_n - x_{n_k}| + |x_{n_k} - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $x_n \rightarrow x$. $\square$

Remark. The $(\impliedby)$ direction is exactly the *completeness* of $\mathbb{R}$ — it fails in $\mathbb{Q}$. The theorem extends verbatim to $\mathbb{R}^k$ (a sequence in $\mathbb{R}^k$ is Cauchy iff each coordinate is, and we may invoke Theorem 4.4.7); thus $\mathbb{R}^k$ is complete.

Worked Example 4.5.4 (A Cauchy sequence in $\mathbb{Q}$ that does not converge in $\mathbb{Q}$). Show the decimal truncations of $\sqrt{2}$ form a Cauchy sequence of rationals with no rational limit.

Recognition. “Non-example of completeness”; exhibit Cauchy-but-not-convergent *in a smaller space*.

Solution. Let $x_n = \lfloor 10^n \sqrt{2} \rfloor / 10^n$ (the first n decimals of $\sqrt{2}$), each $x_n \in \mathbb{Q}$. For $m > n$, $|x_m - x_n| < 10^{-n} \rightarrow 0$, so (x_n) is Cauchy. In $\mathbb{R}$ it converges to $\sqrt{2}$; but $\sqrt{2} \notin \mathbb{Q}$, so as a sequence in $\mathbb{Q}$ it has *no* limit. *Check.* This is precisely why $\mathbb{Q}$ is *incomplete* — the Cauchy criterion is a property of $\mathbb{R}$, not of ordered fields in general. $\blacksquare$

Worked Example 4.5.5 (Proving Cauchy directly (B&S 3.5.2)). Show $x_n = 1 + \frac{1}{2!} + \cdots + \frac{1}{n!}$ is Cauchy (hence convergent).

Recognition. Estimate $|x_m - x_n|$ for $m > n$ and beat it with a geometric tail.

Solution. For $m > n$,

$$|x_m - x_n| = \sum_{j=n+1}^m \frac{1}{j!} \leq \sum_{j=n+1}^m \frac{1}{2^{j-1}} < \frac{1}{2^{n-1}},$$

using $j! \geq 2^{j-1}$. Given $\varepsilon > 0$ pick K with $2^{-(K-1)} < \varepsilon$; then $m, n \geq K \implies |x_m - x_n| < \varepsilon$. So (x_n) is Cauchy, hence converges (to $e - 1$). *Check.* The tail bound 2^{1-n} at $n = 10$ is ≈ 0.002 . ■

Contractive sequences

A clean, checkable sufficient condition for the Cauchy property.

Definition 4.5.6. A sequence (x_n) is *contractive* if there is a constant $0 < C < 1$ with

$$|x_{n+2} - x_{n+1}| \leq C |x_{n+1} - x_n| \quad \text{for all } n \in \mathbb{N}.$$

Theorem 4.5.7 (Contractive $\implies$ convergent). *Every contractive sequence is Cauchy, hence convergent. If $x_n \rightarrow x$, then for all n ,*

$$|x_n - x| \leq \frac{C^{n-1}}{1-C} |x_2 - x_1|.$$

Proof. Iterating, $|x_{n+1} - x_n| \leq C^{n-1} |x_2 - x_1|$. For $m > n$, by the triangle inequality and the geometric series,

$$|x_m - x_n| \leq \sum_{j=n}^{m-1} |x_{j+1} - x_j| \leq |x_2 - x_1| \sum_{j=n}^{m-1} C^{j-1} \leq |x_2 - x_1| \frac{C^{n-1}}{1-C} \xrightarrow{n \rightarrow \infty} 0.$$

So (x_n) is Cauchy. Letting $m \rightarrow \infty$ in the bound gives the error estimate. □

Worked Example 4.5.8 (A contractive recursion (B&S 3.5.12)). Let $x_1 > 0$ and $x_{n+1} = (2 + x_n)^{-1}$. Show (x_n) converges and find the limit.

Recognition. Recursion with a $|f'| < 1$ map; verify contractive, then solve the fixed point.

Solution. For $x > 0$, $|x_{n+1} - x_n| = \left| \frac{1}{2 + x_n} - \frac{1}{2 + x_{n-1}} \right| = \frac{|x_n - x_{n-1}|}{(2 + x_n)(2 + x_{n-1})} \leq \frac{1}{4} |x_n - x_{n-1}|$, since each factor $2 + x > 2$. So $C = \frac{1}{4} < 1$: contractive, hence convergent to x with $x = (2 + x)^{-1}$, i.e. $x^2 + 2x - 1 = 0$, $x = -1 + \sqrt{2}$ (positive root). *Check.* $-1 + \sqrt{2} \approx 0.4142 = (2.4142)^{-1}$. ■

Caution. “Contractive” compares *consecutive gaps*, $|x_{n+2} - x_{n+1}| \leq C |x_{n+1} - x_n|$ with a *fixed* $C < 1$. The weaker condition $|x_{n+1} - x_n| \rightarrow 0$ does *not* imply Cauchy: the partial sums of the harmonic series, $x_n = \sum_{k \leq n} 1/k$, satisfy $|x_{n+1} - x_n| = 1/(n+1) \rightarrow 0$ yet diverge. “Consecutive terms close” is far weaker than Cauchy.

Further worked examples

Worked Example 4.5.9 (The harmonic partial sums are NOT Cauchy). Show directly that $x_n = \sum_{k=1}^n \frac{1}{k}$ fails the Cauchy condition.

Recognition. “Show not Cauchy” (Definition 4.5.1): negate it by finding one $\varepsilon > 0$ for which no K works — a far-apart pair $m, n \geq K$ for every K .

Solution. Take $\varepsilon = \frac{1}{2}$. For any K , set $n = K$ and $m = 2K$. Then

$$|x_m - x_n| = \sum_{k=K+1}^{2K} \frac{1}{k} \geq K \cdot \frac{1}{2K} = \frac{1}{2} = \varepsilon,$$

since each of the K terms is $\geq 1/(2K)$. So no K confines the tail within $\varepsilon = \frac{1}{2}$, and (x_n) is not Cauchy — hence (Theorem 4.5.3) divergent. *Check.* The same dyadic blocks reappear in §4.7; the harmonic series is the canonical “terms $\rightarrow 0$ but not Cauchy” example. ■

Worked Example 4.5.10 (Verifying the Cauchy condition by a geometric gap bound). Let $x_1 = 1$, $x_2 = 2$, and $x_{n+2} = \frac{1}{2}(x_{n+1} + x_n)$. Show (x_n) is Cauchy and find its limit.

Recognition. Recursion with shrinking gaps: show (x_n) is contractive (Definition 4.5.6), then apply Theorem 4.5.7.

Solution. The gap satisfies $x_{n+2} - x_{n+1} = \frac{1}{2}(x_{n+1} + x_n) - x_{n+1} = -\frac{1}{2}(x_{n+1} - x_n)$, so $|x_{n+2} - x_{n+1}| = \frac{1}{2}|x_{n+1} - x_n|$: contractive with $C = \frac{1}{2}$. Hence (x_n) is Cauchy, so convergent to some x . Telescoping the signed gaps, $x = x_1 + \sum_{n \geq 1} (x_{n+1} - x_n) = 1 + \sum_{n \geq 1} (-\frac{1}{2})^{n-1} (x_2 - x_1) = 1 + \frac{1}{1 - (-1/2)} = 1 + \frac{2}{3} = \frac{5}{3}$. *Check.* $x_3 = 1.5$, $x_4 = 1.75$, $x_5 = 1.625$, $x_6 \approx 1.6875$, oscillating toward $\frac{5}{3} \approx 1.6667$. ■

Worked Example 4.5.11 (Cauchy via the Cauchy criterion for $\sum 1/2^n$). Show the partial sums $s_n = \sum_{k=1}^n 2^{-k}$

are Cauchy directly from the definition.

Recognition. “Show Cauchy”; estimate $|s_m - s_n|$ for $m > n$ and dominate by a geometric tail.

Solution. For $m > n$,

$$|s_m - s_n| = \sum_{k=n+1}^m \frac{1}{2^k} < \sum_{k=n+1}^{\infty} \frac{1}{2^k} = \frac{1}{2^n}.$$

Given $\varepsilon > 0$, pick K with $2^{-K} < \varepsilon$ (Archimedean, since $2^{-K} \rightarrow 0$); then $m, n \geq K \implies |s_m - s_n| < 2^{-\min(m,n)} \leq 2^{-K} < \varepsilon$. So (s_n) is Cauchy, hence convergent (to 1). *Check.* The tail bound 2^{-n} at $n = 10$ is ≈ 0.001 , matching $s_{10} = 1 - 2^{-10} \approx 0.999$. ■

Exercises (Bartle & Sherbert, §3.5).

1. Give an example of a bounded sequence that is not a Cauchy sequence.
2. Show directly from the definition that the following are Cauchy sequences: (a) $\frac{n+1}{n}$; (b) $1 + \frac{1}{2!} + \cdots + \frac{1}{n!}$.
3. Show directly from the definition that the following are *not* Cauchy: (a) $((-1)^n)$; (b) $\left(n + \frac{(-1)^n}{n}\right)$; (c) $(\ln n)$.
4. Show that if (x_n) and (y_n) are Cauchy, then $(x_n + y_n)$ and $(x_n y_n)$ are Cauchy.
5. If $x_n = \sqrt{n}$, show $\lim |x_{n+1} - x_n| = 0$ but (x_n) is not Cauchy.
6. Let $p \in \mathbb{N}$. Give an example of a non-Cauchy sequence (x_n) with $\lim |x_{n+p} - x_n| = 0$.
7. Let (x_n) be Cauchy with $x_n \in \mathbb{Z}$ for every n . Show (x_n) is ultimately constant.
8. Show directly that a bounded monotone increasing sequence is Cauchy.
9. If $0 < r < 1$ and $|x_{n+1} - x_n| < r^n$ for all n , show (x_n) is Cauchy.
10. If $x_1 < x_2$ are arbitrary and $x_n = \frac{1}{2}(x_{n-2} + x_{n-1})$ for $n > 2$, show (x_n) converges; find its limit.
11. If $y_1 < y_2$ are arbitrary and $y_n = \frac{1}{3}y_{n-1} + \frac{2}{3}y_{n-2}$ for $n > 2$, show (y_n) converges; find its limit.
12. If $x_1 > 0$ and $x_{n+1} = (2 + x_n)^{-1}$, show (x_n) is contractive; find the limit.
13. If $x_1 = 2$ and $x_{n+1} = 2 + 1/x_n$, show (x_n) is contractive; find the limit.
14. The equation $x^3 - 5x + 1 = 0$ has a root $r \in (0, 1)$. Use a contractive sequence to compute r within 10^{-4} .

Additional exercises.

1. Verify directly that $x_n = \sum_{k=1}^n 1/2^k$ is Cauchy by estimating $|x_m - x_n|$ for $m > n$.

2. Show that (x_n) is Cauchy iff $(x_{2n}), (x_{2n+1})$ are Cauchy and $|x_{2n} - x_{2n+1}| \rightarrow 0$.
3. Prove a Cauchy sequence with a constant subsequence is eventually constant.
4. (Non-example.) Exhibit a sequence in $\mathbb{Q}$ that is Cauchy in the metric $|x - y|$ but has no limit in $\mathbb{Q}$, and identify its limit in $\mathbb{R}$. Explain what fails.
5. Show $x_1 = 1, x_{n+1} = 1 + \frac{1}{1 + x_n}$ is contractive; find its limit (a continued fraction for $\sqrt{2}$).
6. Prove that if (x_n) is contractive with constant C , then $|x_n - x| \leq \frac{C}{1 - C} |x_n - x_{n-1}|$ (a computable a posteriori error bound).
7. (Diagram problem.) Draw the “tail cluster” picture for a Cauchy sequence of your choice: plot a_n -vs- n , shade a window of width $\varepsilon = 0.2$ holding the whole tail, and mark the threshold K .
8. Show $(\ln n)$ satisfies $|x_{n+1} - x_n| \rightarrow 0$ yet is not Cauchy, and contrast with the (sufficient) contractive condition.

4.6 Limit Superior, Limit Inferior, and Properly Divergent Sequences

The extended reals and properly divergent sequences

It is convenient to adjoin two symbols $+\infty, -\infty$ to $\mathbb{R}$, forming the *extended reals* $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, with $-\infty < x < +\infty$ for every real x . Then *every* subset has a supremum and infimum in $\overline{\mathbb{R}}$ (an unbounded-above set has $\sup = +\infty$, the empty set has $\sup = -\infty$).

Definition 4.6.1. A sequence (x_n) *diverges to* $+\infty$, written $x_n \rightarrow +\infty$ or $\lim x_n = +\infty$, if for every $\alpha \in \mathbb{R}$ there is K with $x_n > \alpha$ for $n \geq K$. Symmetrically $x_n \rightarrow -\infty$ if for every $\beta \in \mathbb{R}$ there is K with $x_n < \beta$ for $n \geq K$. Such sequences are *properly divergent*.

Proposition 4.6.2 (Properties of proper divergence). (a) *A monotone sequence is properly divergent iff it is unbounded; an increasing unbounded sequence $\rightarrow +\infty$.* (b) *If $x_n \rightarrow +\infty$ and $x_n \leq y_n$ eventually, then $y_n \rightarrow +\infty$.* (c) *If $x_n > 0$ then $x_n \rightarrow +\infty \iff 1/x_n \rightarrow 0$ (B&S 3.6.3).*

Proof. (a) An increasing sequence unbounded above exceeds every α from some index on. (b) $y_n \geq x_n > \alpha$ eventually. (c) If $x_n \rightarrow +\infty$ then given $\varepsilon > 0$, $x_n > 1/\varepsilon$ eventually $\implies 0 < 1/x_n < \varepsilon$; the converse is symmetric. $\square$

Worked Example 4.6.3 (A comparison test for proper divergence). Show $(\sqrt{n})$ and $(n/\sqrt{n+1})$ diverge to $+\infty$ (B&S 3.6.4).

Recognition. “Show $\rightarrow +\infty$ ”; bound below by a known divergent sequence (Proposition 4.6.2(b)).

Solution. $\sqrt{n} \rightarrow +\infty$: given α , $\sqrt{n} > \alpha \iff n > \alpha^2$. For $\frac{n}{\sqrt{n+1}}$, note $\frac{n}{\sqrt{n+1}} \geq \frac{n}{\sqrt{2n}} = \sqrt{n/2} \rightarrow +\infty$ for $n \geq 1$, so by comparison it diverges to $+\infty$. *Check.* $n = 8$ gives $8/3 = 2.67 \geq \sqrt{4} = 2$. $\blacksquare$

Limit superior and limit inferior

For a *bounded* sequence, even when it does not converge, two canonical numbers always exist: the largest and smallest values its tail keeps revisiting.

Definition 4.6.4 (Two equivalent definitions). Let (x_n) be bounded. Define the *tail suprema* $s_n = \sup\{x_k : k \geq n\}$ and *tail infima* $t_n = \inf\{x_k : k \geq n\}$. Then (s_n) is decreasing and (t_n) increasing (shrinking tails), both bounded, so by MCT we set

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} s_n = \inf_n \sup_{k \geq n} x_k, \quad \liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} t_n = \sup_n \inf_{k \geq n} x_k.$$

Equivalently, $\limsup x_n$ is the *largest* subsequential limit and $\liminf x_n$ the *smallest* (in $\overline{\mathbb{R}}$ if unbounded).

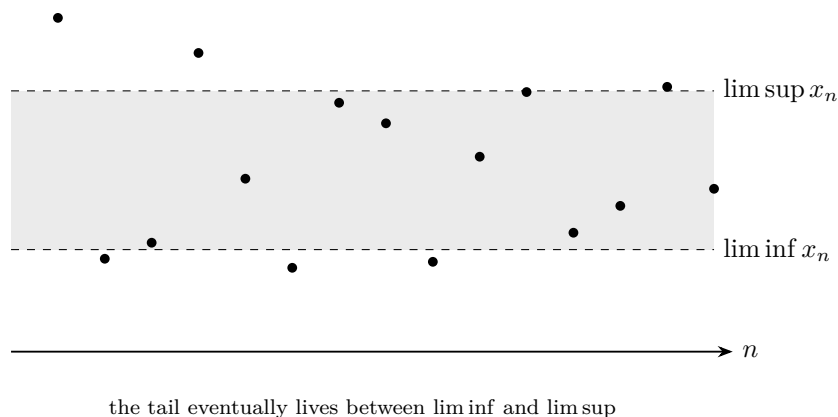


Figure 4.7. $\limsup/\liminf$ are the upper and lower edges of the band the tail ultimately occupies. The sequence converges exactly when the band has zero width.

Theorem 4.6.5 (Characterization and convergence). *For a bounded sequence: (a) $\liminf x_n \leq \limsup x_n$. (b) (x_n) converges iff $\liminf x_n = \limsup x_n$, and then the common value is $\lim x_n$. (c) $L = \limsup x_n$ iff for every $\varepsilon > 0$: $x_n < L + \varepsilon$ eventually, and $x_n > L - \varepsilon$ infinitely often.*

Proof. (a) $t_n \leq s_n$ for each n , so the limits obey $\liminf \leq \limsup$. (b) If $x_n \rightarrow x$ then every subsequential limit equals x , so $\liminf = \limsup = x$. Conversely if $\liminf = \limsup = L$, then $t_n \leq x_n \leq s_n$ with $t_n, s_n \rightarrow L$, and squeezing gives $x_n \rightarrow L$. (c) From $s_n \downarrow L$: $s_N < L + \varepsilon$ for some N gives $x_n \leq s_N < L + \varepsilon$ for $n \geq N$; and $s_n \geq L$ for all n means $\sup_{k \geq n} x_k \geq L$, so some $x_k > L - \varepsilon$ with $k \geq n$, for every n . $\square$

Theorem 4.6.6 (Subadditivity, Rudin Ex. 3.5). *For bounded sequences, provided the right side is not $\infty - \infty$,*

$$\limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n.$$

Proof. For $k \geq n$, $a_k + b_k \leq \sup_{j \geq n} a_j + \sup_{j \geq n} b_j$, so taking sup over $k \geq n$, $\sup_{k \geq n} (a_k + b_k) \leq \sup_{j \geq n} a_j + \sup_{j \geq n} b_j$. Let $n \rightarrow \infty$. $\square$

Worked Example 4.6.7 (Computing $\limsup$ and $\liminf$). Find $\limsup$ and $\liminf$ of $x_n = (-1)^n(1 + \frac{1}{n})$ and of $y_n = \cos(n\pi/2)$.

Recognition. “Find upper/lower limits”; identify all subsequential limits and take the extremes.

Solution. For x_n : even terms $1 + 1/n \rightarrow 1$, odd terms $-(1 + 1/n) \rightarrow -1$; the only subsequential limits are ± 1 , so $\limsup x_n = 1$, $\liminf x_n = -1$. For $y_n = \cos(n\pi/2)$ the values cycle $0, -1, 0, 1, 0, -1, \dots$, with subsequential limits $\{-1, 0, 1\}$; so $\limsup y_n = 1$, $\liminf y_n = -1$. *Check.* Neither sequence converges, consistent with $\liminf \neq \limsup$ (Theorem 4.6.5(b)). $\blacksquare$

Worked Example 4.6.8 (A non-example for subadditivity equality). Show the inequality in Theorem 4.6.6 can be strict.

Recognition. “Show $\leq$ is not $=$ ”; pick anti-phase oscillations.

Solution. Take $a_n = (-1)^n$, $b_n = (-1)^{n+1}$. Then $\limsup a_n = \limsup b_n = 1$, so the right side is 2; but $a_n + b_n \equiv 0$, so $\limsup(a_n + b_n) = 0 < 2$. *Check.* The peaks of a_n and b_n never align, so the sum never reaches the sum of the peaks. $\blacksquare$

Caution. $\limsup x_n$ is *not* $\sup_n x_n$, and it is *not* $\lim x_n$ (which may not exist). It is the limit of the *tail* suprema — early large terms are forgotten. For $x_n = 1/n$, $\sup_n x_n = 1$ but $\limsup x_n = 0$. Always read $\limsup$ as “eventual supremum”.

Caution. $\limsup$ and $\liminf$ *always* exist in $\overline{\mathbb{R}}$ for any sequence, bounded or not — that is their advantage over $\lim$. But identities like $\limsup(a_n b_n) = \limsup a_n \cdot \limsup b_n$ are *false* in general; only one-sided inequalities survive, and only under sign hypotheses.

Further worked examples

Worked Example 4.6.9 (A three-valued lim sup/lim inf via tail suprema). Compute lim sup and lim inf of $x_n = \frac{n}{n+1} \sin \frac{n\pi}{3}$ by identifying all subsequential limits.

Recognition. “Find upper/lower limits”; the factor $n/(n+1) \rightarrow 1$ is a convergent envelope, so the subsequential limits are $1 \cdot (\text{values of } \sin(n\pi/3))$ (Definition 4.6.4).

Solution. $\sin(n\pi/3)$ cycles with period 6 through $\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, 0, -\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, 0$. Since $n/(n+1) \rightarrow 1$, each residue class $n \equiv r \pmod{6}$ gives a subsequence converging to the corresponding value of $\sin(r\pi/3)$. The subsequential limits are therefore $\{-\frac{\sqrt{3}}{2}, 0, \frac{\sqrt{3}}{2}\}$, so $\limsup x_n = \frac{\sqrt{3}}{2}$ and $\liminf x_n = -\frac{\sqrt{3}}{2}$. *Check.* $\limsup \neq \liminf$, so (x_n) diverges (Theorem 4.6.5(b)); the band has width $\sqrt{3}$. ■

Worked Example 4.6.10 (lim sup and lim inf of a ratio of consecutive terms). For x_n defined by $x_{2k} = 2^{-k}$ and $x_{2k-1} = 3^{-k}$, find $\limsup \frac{x_{n+1}}{x_n}$ and $\liminf \frac{x_{n+1}}{x_n}$, and contrast with $\limsup x_n^{1/n}$.

Recognition. “Compute upper/lower limits of the ratio”; the two parity cases give two different ratio behaviors, illustrating why the ratio test can stall where the root test does not (foreshadowing §4.7).

Solution. Even-to-odd: $\frac{x_{2k+1}}{x_{2k}} = \frac{3^{-(k+1)}}{2^{-k}} = \frac{1}{3} \left(\frac{2}{3}\right)^k \rightarrow 0$. Odd-to-even: $\frac{x_{2k}}{x_{2k-1}} = \frac{2^{-k}}{3^{-k}} = \left(\frac{2}{3}\right)^k \rightarrow +\infty$. So $\liminf \frac{x_{n+1}}{x_n} = 0$ and $\limsup \frac{x_{n+1}}{x_n} = +\infty$. Yet $x_n^{1/n}$ stays controlled: $x_{2k}^{1/(2k)} = 2^{-1/2}$ and $x_{2k-1}^{1/(2k-1)} \rightarrow 3^{-1/2}$, so $\limsup x_n^{1/n} = 2^{-1/2} < 1$. *Check.* The root limit is finite and < 1 while the ratio limits spread to 0 and ∞ — precisely the gap that makes the root test sharper. ■

Worked Example 4.6.11 (A non-example: $\liminf x_n + \limsup y_n$ need not bound $\limsup(x_n + y_n)$). Show that $\liminf(x_n + y_n) \geq \liminf x_n + \liminf y_n$ can be *strict*, mirroring the subadditivity of lim sup.

Recognition. “Show the inequality is strict”; pick anti-phase ± 1 sequences as in the lim sup case (Theorem 4.6.6).

Solution. Take $x_n = (-1)^n$ and $y_n = (-1)^{n+1}$. Then $\liminf x_n = \liminf y_n = -1$, so the right side is -2 ; but $x_n + y_n \equiv 0$, giving $\liminf(x_n + y_n) = 0 > -2$. The superadditivity $\liminf(x_n + y_n) \geq \liminf x_n + \liminf y_n$ holds but is strict here. *Check.* Adding the two known facts, $\liminf(x + y) \geq \liminf x + \liminf y$ and $\limsup(x + y) \leq \limsup x + \limsup y$ pin the sum’s band *inside* the sum of the bands — never outside. ■

Exercises (Bartle & Sherbert, §3.6).

1. Show that if (x_n) is unbounded, then there is a properly divergent subsequence.
2. Give examples of properly divergent $(x_n), (y_n)$ with $y_n \neq 0$ such that (a) (x_n/y_n) converges; (b) (x_n/y_n) is properly divergent.
3. Show that if $x_n > 0$ for all n , then $\lim x_n = 0 \iff \lim(1/x_n) = +\infty$.
4. Establish proper divergence of (a) $(\sqrt{n})$; (b) $(\sqrt{n+1})$; (c) $(\sqrt{n-1})$; (d) $(n/\sqrt{n+1})$.
5. Is the sequence $(n \sin n)$ properly divergent?
6. Let (x_n) be properly divergent and (y_n) be such that $\lim(x_n y_n) \in \mathbb{R}$. Show $y_n \rightarrow 0$.
7. Let $(x_n), (y_n)$ have positive terms with $\lim(x_n/y_n) = 0$. (a) If $\lim x_n = +\infty$, show $\lim y_n = +\infty$. (b) If (y_n) is bounded, show $\lim x_n = 0$.
8. Investigate convergence/divergence of (a) $(\sqrt{n^2+2})$; (b) $(\sqrt{n}/(n^2+1))$; (c) $(\sqrt{n^2+1}/\sqrt{n})$; (d) $(\sin \sqrt{n})$.
9. Let $(x_n), (y_n)$ have positive terms with $\lim(x_n/y_n) = +\infty$. (a) If $\lim y_n = +\infty$, show $\lim x_n = +\infty$. (b) If (x_n) is bounded, show $\lim y_n = 0$.
10. Show that if $\lim(a_n/n) = L$ with $L > 0$, then $\lim a_n = +\infty$.

Additional exercises.

1. Compute $\limsup$ and $\liminf$ of $x_n = \sin(n\pi/3)$.
2. For $x_n = \left(1 + \frac{(-1)^n}{2}\right)^n$ decide whether the sequence is bounded, and find $\limsup$ and $\liminf$ (in $\overline{\mathbb{R}}$).
3. Prove $\liminf x_n = -\limsup(-x_n)$ for any bounded sequence.
4. (Non-example.) Show by example that $\limsup(a_n b_n) = \limsup a_n \cdot \limsup b_n$ can fail, even for nonnegative sequences.
5. Show that (x_n) converges in $\overline{\mathbb{R}}$ to $+\infty$ iff $\liminf x_n = +\infty$.
6. Prove that for bounded sequences, $\liminf a_n + \liminf b_n \leq \liminf(a_n + b_n) \leq \liminf a_n + \limsup b_n$.
7. (Diagram problem.) Plot a bounded oscillating sequence and draw the two horizontal lines $\limsup x_n$ and $\liminf x_n$; shade the band the tail eventually occupies and indicate the index past which all terms lie within ε of the band.
8. Let $a_n > 0$. Prove $\liminf \frac{a_{n+1}}{a_n} \leq \liminf a_n^{1/n} \leq \limsup a_n^{1/n} \leq \limsup \frac{a_{n+1}}{a_n}$ (the inequality chain behind “root beats ratio”).

4.7 Series and Convergence Tests

Series and partial sums

Definition 4.7.1. Given a sequence (a_n) , the *series* $\sum_{n=1}^{\infty} a_n$ is the sequence (s_N) of *partial sums* $s_N = \sum_{n=1}^N a_n$. The series *converges* to s if $s_N \rightarrow s$, and we write $\sum_{n=1}^{\infty} a_n = s$; otherwise it *diverges*. The a_n are the *terms*.

A series is nothing but a sequence (of partial sums) in disguise; every sequence theorem applies. The new feature is that we want criteria phrased in the *terms* a_n , not the sums.

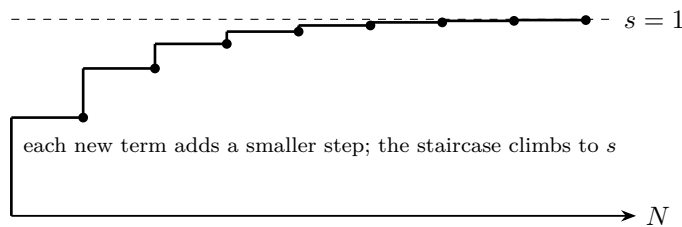


Figure 4.8. Partial-sum staircase for $\sum 2^{-n} = 1$: convergence of a series is just convergence of this staircase of partial sums.

Theorem 4.7.2 (Geometric series). For $|r| < 1$, $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$. For $|r| \geq 1$ the series diverges.

Proof. $s_N = \sum_{n=0}^N r^n = \frac{1-r^{N+1}}{1-r}$ for $r \neq 1$. If $|r| < 1$ then $r^{N+1} \rightarrow 0$, so $s_N \rightarrow 1/(1-r)$. If $|r| \geq 1$ the terms r^n do not tend to 0, so the series diverges (next theorem). For $r = 1$, $s_N = N+1 \rightarrow \infty$. $\square$

Theorem 4.7.3 (n -th term test). If $\sum a_n$ converges then $a_n \rightarrow 0$. Contrapositive: if $a_n \not\rightarrow 0$, the series diverges.

Proof. $a_n = s_n - s_{n-1} \rightarrow s - s = 0$ since both partial-sum sequences converge to the same s . $\square$

Caution. The n -th term test is *only* a divergence test: $a_n \rightarrow 0$ does *not* imply convergence. The harmonic series $\sum 1/n$ has $a_n = 1/n \rightarrow 0$ yet diverges (below). “Terms go to zero” is necessary, never sufficient.

Theorem 4.7.4 (Cauchy criterion for series). $\sum a_n$ converges iff for every $\varepsilon > 0$ there is K with $\left|\sum_{k=m+1}^n a_k\right| < \varepsilon$ whenever $n > m \geq K$.

Proof. Apply the Cauchy criterion (Theorem 4.5.3) to (s_N) , noting $s_n - s_m = \sum_{k=m+1}^n a_k$. $\square$

Worked Example 4.7.5 (The harmonic series diverges). Show $\sum_{n=1}^{\infty} \frac{1}{n} = +\infty$.

Recognition. “Show divergence” where the n -th term test is inconclusive: block the partial sums below by a growing amount (Cauchy criterion fails).

Solution. Group the terms in dyadic blocks:

$$\sum_{k=2^m+1}^{2^{m+1}} \frac{1}{k} \geq 2^m \cdot \frac{1}{2^{m+1}} = \frac{1}{2}.$$

So $s_{2^m} \geq 1 + \frac{M}{2} \rightarrow \infty$; the partial sums are unbounded, hence diverge. *Check.* The Cauchy criterion fails outright: $\sum_{k=m+1}^{2m} 1/k \geq m \cdot \frac{1}{2m} = \frac{1}{2}$ for all m , so the blocks never shrink below $\frac{1}{2}$. $\blacksquare$

Worked Example 4.7.6 (Telescoping (B&S 3.7.3)). Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

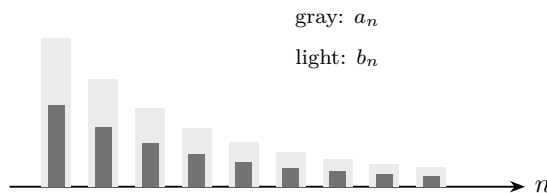
Recognition. A term that splits into a difference of consecutive pieces: telescope the partial sum.

Solution. Partial fractions: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, so $s_N = \sum_{n=1}^N \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{N+1} \rightarrow 1$. Hence the sum is 1. *Check.* $s_3 = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} = \frac{3}{4} = 1 - \frac{1}{4}$. $\blacksquare$

Comparison and condensation

Theorem 4.7.7 (Comparison test). Suppose $0 \leq a_n \leq b_n$ for all large n . (a) If $\sum b_n$ converges, so does $\sum a_n$. (b) If $\sum a_n$ diverges, so does $\sum b_n$.

Proof. Partial sums of nonnegative series are increasing; convergence $\iff$ boundedness (MCT). If $\sum b_n$ converges its partial sums are bounded by some B , hence so are those of $\sum a_n$ (term-by-term $\leq$), giving convergence. (b) is the contrapositive. $\square$



if the taller bars sum finitely, so do the shorter ones

Figure 4.9. Comparison test: $a_n \leq b_n$ termwise. A finite total area under the larger bars caps the area under the smaller.

Theorem 4.7.8 (Cauchy condensation). If $a_1 \geq a_2 \geq \dots \geq 0$, then $\sum a_n$ converges iff $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

Proof sketch. Group terms in dyadic blocks. Monotonicity gives $2^{k-1} a_{2^k} \leq a_{2^{k-1}+1} + \dots + a_{2^k} \leq 2^{k-1} a_{2^{k-1}}$, so the original series and the condensed series bound each other up to constants; one converges iff the other does (comparison). $\square$

Theorem 4.7.9 (p -series). $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges iff $p > 1$.

Proof. Condensation: $\sum 2^k \cdot \frac{1}{(2^k)^p} = \sum (2^{1-p})^k$ is geometric with ratio 2^{1-p} , which converges iff $2^{1-p} < 1 \iff p > 1$. $\square$

Root and ratio tests; the integral test

Theorem 4.7.10 (Root test). Let $\alpha = \limsup_{n \rightarrow \infty} |a_n|^{1/n}$. Then $\sum a_n$ converges absolutely if $\alpha < 1$, diverges if $\alpha > 1$, and the test is inconclusive if $\alpha = 1$.

Proof. If $\alpha < 1$ pick β with $\alpha < \beta < 1$; then $|a_n|^{1/n} < \beta$ eventually, so $|a_n| < \beta^n$, and comparison with the convergent geometric $\sum \beta^n$ gives absolute convergence. If $\alpha > 1$ then $|a_n|^{1/n} > 1$ infinitely often, so $|a_n| > 1$ infinitely often, $a_n \not\rightarrow 0$, and the series diverges. $\square$

Theorem 4.7.11 (Ratio test). Suppose $a_n \neq 0$. If $\limsup_n |a_{n+1}/a_n| < 1$ the series converges absolutely; if $|a_{n+1}/a_n| \geq 1$ for all large n (in particular if $\liminf > 1$), it diverges.

Proof. If $L = \limsup |a_{n+1}/a_n| < 1$ pick $\beta \in (L, 1)$; then $|a_{n+1}| \leq \beta |a_n|$ for $n \geq K$, so $|a_{K+m}| \leq \beta^m |a_K|$, and comparison with $\sum \beta^m$ gives absolute convergence. If the ratio is ≥ 1 eventually, $|a_n|$ is nondecreasing, so $a_n \not\rightarrow 0$ and the series diverges. $\square$

Remark (Root is stronger). Whenever the ratio test decides, so does the root test, because $\liminf |a_{n+1}/a_n| \leq \liminf |a_n|^{1/n} \leq \limsup |a_n|^{1/n} \leq \limsup |a_{n+1}/a_n|$. The root test can succeed where the ratio test fails (e.g. a series whose ratios oscillate). The ratio test is, however, easier to compute.

Theorem 4.7.12 (Integral test, stated). If f is positive, decreasing on $[1, \infty)$ and $a_n = f(n)$, then $\sum a_n$ converges iff $\int_1^\infty f(x) dx$ converges (the partial sums and the integral differ by a bounded amount). (Proof deferred to the integration chapter.)

Worked Example 4.7.13 (Choosing a test). Decide convergence of (a) $\sum \frac{n}{2^n}$, (b) $\sum \frac{n!}{n^n}$, (c) $\sum \frac{1}{n \ln n}$.

Recognition. “Test for convergence”; match the structure to root/ratio/condensation.

Solution. (a) Ratio: $\frac{(n+1)/2^{n+1}}{n/2^n} = \frac{1}{2} \cdot \frac{n+1}{n} \rightarrow \frac{1}{2} < 1$, converges. (b) Ratio: $\frac{(n+1)!/(n+1)^{n+1}}{n!/n^n} = \left(\frac{n}{n+1}\right)^n = \left(1 + \frac{1}{n}\right)^{-n} \rightarrow e^{-1} < 1$, converges. (c) Condensation: $\sum 2^k \frac{1}{2^k \ln 2^k} = \sum \frac{1}{k \ln 2}$, a constant multiple of the harmonic series, diverges. *Check.* (c) shows $\sum 1/(n \ln n)$ just barely diverges — slower than $\sum 1/n$, faster to settle than any p -series with $p > 1$. ■

Worked Example 4.7.14 (Root test where the ratio test stalls). Test $\sum a_n$ with $a_n = 2^{-n}$ if n even and $a_n = 3^{-n}$ if n odd.

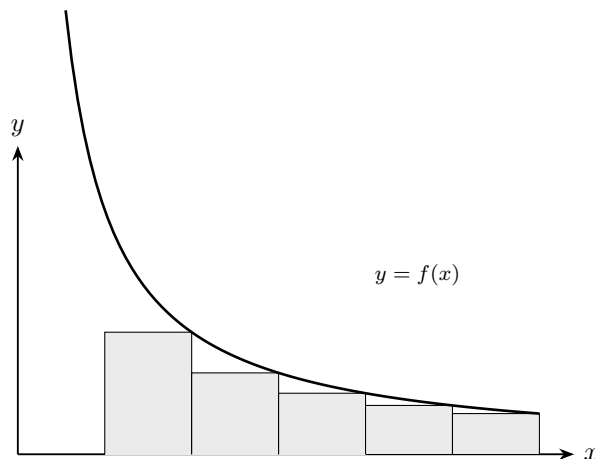
Recognition. Oscillating ratios; the root test reads off the geometric decay (Remark above).

Solution. The ratios a_{n+1}/a_n oscillate wildly (between $\sim (2^n/3^{n+1})$ and $\sim (3^n/2^{n+1})$), so the ratio test is unusable. But $|a_n|^{1/n} \leq \max\{\frac{1}{2}, \frac{1}{3}\} = \frac{1}{2}$, so $\limsup |a_n|^{1/n} \leq \frac{1}{2} < 1$ and the series converges absolutely. *Check.* Indeed $a_n \leq 2^{-n}$, comparison with $\sum 2^{-n}$ confirms it. ■

Caution. When $\alpha = 1$ (root) or the ratio limit is 1, the test says *nothing*: $\sum 1/n$ (diverges) and $\sum 1/n^2$ (converges) both give limit 1. At the boundary switch to a finer tool (condensation, p -series, integral test).

Further worked examples

The integral test compares the series $\sum f(n)$ with the area under $y = f(x)$: the terms are the areas of unit-width rectangles that bracket the curve, so the sum and the integral converge or diverge together.



rectangles of height $f(n+1)$ lie under the curve: $\sum_{n \geq 2} f(n) \leq \int_1^\infty f$

Figure 4.10. Integral test: the under-rectangles bound the tail sum by the integral (and over-rectangles bound it below), so series and integral share their fate.

Worked Example 4.7.15 (Limit comparison test). Decide convergence of $\sum_{n=1}^{\infty} \frac{2n+1}{n^3+n+1}$.

Recognition. “Test for convergence”; the term behaves like $2/n^2$ for large n , so compare with the convergent p -series $\sum 1/n^2$ (Theorem 4.7.9) via a limit ratio.

Solution. Let $a_n = \frac{2n+1}{n^3+n+1}$ and $b_n = \frac{1}{n^2}$. Then

$$\frac{a_n}{b_n} = \frac{(2n+1)n^2}{n^3+n+1} = \frac{2n^3+n^2}{n^3+n+1} \xrightarrow{n \rightarrow \infty} 2,$$

a finite nonzero limit, so $\sum a_n$ and $\sum b_n$ converge or diverge together. Since $\sum 1/n^2$ converges ($p = 2 > 1$), so does $\sum a_n$. *Check.* $a_n \leq \frac{2n+1}{n^3} \leq \frac{3}{n^2}$ for $n \geq 1$, and $\sum 3/n^2$ converges — the direct comparison confirms it. ■

Worked Example 4.7.16 (The integral test on $\sum 1/(n \ln n)$ and $\sum 1/(n(\ln n)^2)$). Using the integral test, show $\sum_{n \geq 2} \frac{1}{n \ln n}$ diverges but $\sum_{n \geq 2} \frac{1}{n(\ln n)^2}$ converges.

Recognition. “Test for convergence” where condensation also works; here $f(x) = 1/(x \ln x)$ is positive and decreasing, so Theorem 4.7.12 reduces the question to an integral.

Solution. With $u = \ln x$, $du = dx/x$: $\int_2^\infty \frac{dx}{x \ln x} = \int_{\ln 2}^\infty \frac{du}{u} = [\ln u]_{\ln 2}^\infty = +\infty$, so the first series diverges.

For the second, $\int_2^\infty \frac{dx}{x(\ln x)^2} = \int_{\ln 2}^\infty \frac{du}{u^2} = \left[-\frac{1}{u}\right]_{\ln 2}^\infty = \frac{1}{\ln 2} < \infty$, so it converges. *Check.* The exponent on $\ln n$ plays the p -series role at one logarithmic level up — $p = 1$ diverges, $p = 2 > 1$ converges, exactly as for $\sum 1/n^p$. ■

Exercises (Bartle & Sherbert, §3.7 and §9.2).

1. Let $\sum a_n$ be a series and $\sum b_n$ the series with the same terms in the same order but with the zero terms omitted. Show $\sum a_n = A \iff \sum b_n = A$.
2. Show convergence of a series is unaffected by changing finitely many terms (though the sum may change).
3. Using partial fractions, show (a) $\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)} = 1$; (b) $\sum_{n=0}^{\infty} \frac{1}{(a+n)(a+n+1)} = \frac{1}{a}$ for $a > 0$; (c)

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = \frac{1}{4}.$$

4. If $\sum x_n$ and $\sum y_n$ converge, show $\sum(x_n + y_n)$ converges.
5. Give an example of a convergent $\sum x_n$ and a divergent $\sum y_n$ with $\sum(x_n + y_n)$ convergent, or explain why none exists.
6. Compute (a) $\sum_{n=2}^{\infty} (2/7)^n$; (b) $\sum_{n=1}^{\infty} (1/3)^{2n}$.
7. Find a formula for $\sum_{n=1}^{\infty} r^{2n}$ when $|r| < 1$.
8. Let $r_1, r_2, \dots$ enumerate the rationals in $[0, 1]$. About r_n place an interval of length ε^n . Find the total length of all intervals; evaluate for $\varepsilon = 0.1, 0.01$.
9. (a) Show $\sum_{n=1}^{\infty} \cos n$ diverges. (b) Show $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$ converges.
10. Use an argument like Example 3.7.6(f) to show $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ converges.
11. If $\sum a_n$ with $a_n > 0$ converges, must $\sum a_n^2$ converge? Prove or give a counterexample.
12. If $\sum a_n$ with $a_n > 0$ converges, must $\sum \sqrt{a_n}$ converge? Prove or give a counterexample.
13. (§9.2) Establish convergence/divergence of the series whose n -th term is (a) $\frac{1}{(n+1)(n+2)}$; (b) $\frac{n}{(n+1)(n+2)}$; (c) $2^{-1/n}$; (d) $n/2^n$.
14. (§9.2) Same for n -th term (a) $(n(n+1))^{-1/2}$; (b) $(n^2(n+1))^{-1/2}$; (c) $n!/n^n$; (d) $(-1)^n n/(n+1)$.
15. (§9.2) Discuss convergence (for large n) of the series with n -th term (a) $(\ln n)^{-p}$; (b) $(\ln n)^{-n}$; (c) $(\ln n)^{-\ln n}$; (d) $(\ln n)^{-\ln \ln n}$; (e) $(n \ln n)^{-1}$; (f) $(n(\ln n)(\ln \ln n)^2)^{-1}$.
16. (§9.2) Discuss convergence of the series with n -th term (a) $2^n e^{-n}$; (b) $n^n e^{-n}$; (c) $e^{-\ln n}$; (d) $(\ln n)e^{-\sqrt{n}}$; (e) $n! e^{-n}$; (f) $n! e^{-n^2}$.
17. (§9.2) Show $\frac{1}{1^2} + \frac{1}{2^3} + \frac{1}{3^2} + \frac{1}{4^3} + \dots$ converges, yet both the Ratio and Root tests fail to apply.
18. (§9.2) For $a, b > 0$, show $\sum (an + b)^{-p}$ converges if $p > 1$ and diverges if $p \leq 1$.
19. (§9.2) Discuss the series with n -th term (a) $\frac{n!}{3 \cdot 5 \cdot 7 \cdots (2n+1)}$; (b) $\frac{(n!)^2}{(2n)!}$; (c) $\frac{2 \cdot 4 \cdots (2n)}{3 \cdot 5 \cdots (2n+1)}$; (d) $\frac{2 \cdot 4 \cdots (2n)}{5 \cdot 7 \cdots (2n+3)}$.
20. (§9.2) Let $0 < a < 1$ and consider $a^2 + a + a^4 + a^3 + \dots + a^{2n} + a^{2n-1} + \dots$. Show the Root Test applies but the Ratio Test does not.
21. (§9.2) If $r \in (0, 1)$ satisfies the Root Test condition, show $|s - s_n| \leq r^{n+1}/(1 - r)$ for $n \geq K$.
22. (§9.2) If $r \in (0, 1)$ satisfies the Ratio Test condition, show $|s - s_n| \leq r |x_n|/(1 - r)$ for $n \geq K$.
23. (§9.2) Show $1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$ diverges.
24. (§9.2) Let $c_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n$. Show (c_n) decreases to a positive limit, and that $b_n = 1 - \frac{1}{2} + \frac{1}{3} - \dots - \frac{1}{2n} \rightarrow \ln 2$. [Hint: $b_n = c_{2n} - c_n + \ln 2$.]
25. (§9.2) Let $\{n_k\}$ be the naturals whose decimal expansion omits the digit 6. Show $\sum 1/n_k$ converges (to a number < 80); show $\sum 1/m_k$ diverges over numbers ending in 6, and $\sum 1/p_k$ diverges over numbers not ending in 6.

26. (§9.2) For $p, q > 0$, show $\sum \frac{(p+1)(p+2)\cdots(p+n)}{(q+1)(q+2)\cdots(q+n)}$ converges for $q > p+1$ and diverges for $q \leq p+1$.

Additional exercises.

- Sum $\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}$ by telescoping.
- Decide convergence of (a) $\sum \frac{n^2}{n!}$; (b) $\sum \frac{3^n}{n!}$; (c) $\sum \frac{n!}{(2n)!}$, choosing a test for each.
- Use Cauchy condensation to decide $\sum \frac{1}{n(\ln n)^p}$ as a function of $p > 0$.
- (Non-example.) Give a series with $a_n \rightarrow 0$ that diverges, and a series with $\limsup a_n^{1/n} = 1$ that converges — showing the boundary cases are genuinely undecided.
- Prove the comparison test in limit form: if $a_n, b_n > 0$ and $a_n/b_n \rightarrow L \in (0, \infty)$, then $\sum a_n$ and $\sum b_n$ converge or diverge together. Apply it to $\sum \frac{2n+1}{n^3+5}$.
- Show that the root test decides $\sum \left(\frac{n}{n+1}\right)^{n^2}$ and find whether it converges.
- (Diagram problem.) Draw the comparison-test bar picture for $a_n = \frac{1}{n^2+n}$ versus $b_n = \frac{1}{n^2}$ for $n = 1, \dots, 6$, and use it to argue $\sum a_n$ converges.
- Prove that if $a_n \geq 0$ and $\sum a_n$ converges, then $\sum a_n^2$ converges, but the converse fails (give a counterexample).
- Using the partial-sum staircase, explain geometrically why $\sum (-1)^n$ diverges although its terms are bounded.

4.8 Absolute Convergence, Power Series, and Rearrangements

Absolute versus conditional convergence

Definition 4.8.1. $\sum a_n$ converges *absolutely* if $\sum |a_n|$ converges. It converges *conditionally* if $\sum a_n$ converges but $\sum |a_n|$ diverges.

Theorem 4.8.2 (Absolute $\implies$ convergent). *If $\sum |a_n|$ converges then $\sum a_n$ converges, and $|\sum a_n| \leq \sum |a_n|$.*

Proof. By the Cauchy criterion for $\sum |a_n|$, given $\varepsilon > 0$ there is K with $\sum_{k=m+1}^n |a_k| < \varepsilon$ for $n > m \geq K$. Then $|\sum_{k=m+1}^n a_k| \leq \sum_{k=m+1}^n |a_k| < \varepsilon$, so $\sum a_n$ satisfies the Cauchy criterion and converges. The inequality is the limit of $|s_N| \leq \sum_{n \leq N} |a_n|$. $\square$

Theorem 4.8.3 (Alternating Series Test, Leibniz). *If $c_1 \geq c_2 \geq \cdots \geq 0$ and $c_n \rightarrow 0$, then $\sum (-1)^{n+1} c_n$ converges. Moreover the partial sums satisfy $|s - s_N| \leq c_{N+1}$.*

Proof. The even partial sums $s_{2m} = \sum (c_{2k-1} - c_{2k})$ are increasing (each pair ≥ 0) and bounded above by c_1 ; the odd $s_{2m+1} = c_1 - (c_2 - c_3) - \cdots$ are decreasing and bounded below. Both converge, and $s_{2m+1} - s_{2m} = c_{2m+1} \rightarrow 0$ forces a common limit s , with $s_{2m} \leq s \leq s_{2m+1}$. The error bound follows since s lies between consecutive partial sums. $\square$

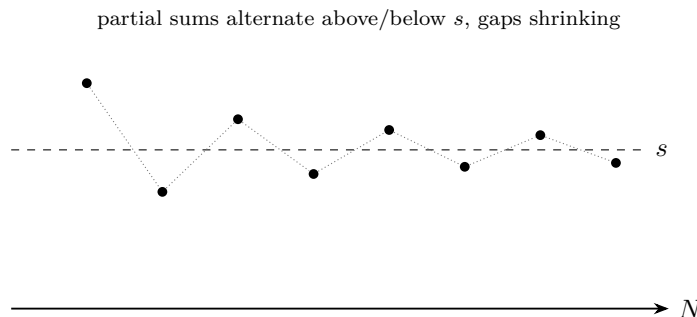


Figure 4.11. Alternating series: consecutive partial sums bracket the sum s , so the error never exceeds the first omitted term.

Worked Example 4.8.4 (Conditional convergence of the alternating harmonic series). Show $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges conditionally.

Recognition. “Conditional vs absolute”; test the series and its absolute version separately.

Solution. $c_n = 1/n$ decreases to 0, so the Alternating Series Test gives convergence (in fact to $\ln 2$). But $\sum |a_n| = \sum 1/n$ is the divergent harmonic series. So the convergence is conditional. *Check.* $s_2 = 1 - \frac{1}{2} = 0.5$, $s_4 \approx 0.583$, bracketing $\ln 2 \approx 0.693$ with errors $\leq c_{N+1}$. ■

Dirichlet and Abel tests

Theorem 4.8.5 (Summation by parts). Let $A_n = \sum_{k=1}^n a_k$ (with $A_0 = 0$). Then $\sum_{k=m}^n a_k b_k = A_n b_n -$

$$A_{m-1} b_m - \sum_{k=m}^{n-1} A_k (b_{k+1} - b_k).$$

Theorem 4.8.6 (Dirichlet’s test). If the partial sums $A_n = \sum_{k \leq n} a_k$ are bounded and (b_n) decreases to 0, then $\sum a_n b_n$ converges.

Proof sketch. Apply summation by parts. With $|A_n| \leq M$ and $b_n \downarrow 0$, the boundary terms and the sum $\sum A_k (b_{k+1} - b_k)$ are controlled: $\sum |b_{k+1} - b_k| = b_m - b_n$ telescopes, giving a Cauchy estimate $\leq 2M b_m \rightarrow 0$. Hence $\sum a_n b_n$ is Cauchy. □

Theorem 4.8.7 (Abel’s test, Rudin Ex. 3.8). If $\sum a_n$ converges and (b_n) is monotone and bounded, then $\sum a_n b_n$ converges.

Proof sketch. A bounded monotone (b_n) converges to some b ; write $b_n = b + (b_n - b)$ where $b_n - b$ is monotone $\rightarrow 0$. Then $\sum a_n b = b \sum a_n$ converges and $\sum a_n (b_n - b)$ converges by Dirichlet (partial sums of $\sum a_n$ are bounded). Add. □

Remark. The Alternating Series Test is the special case $a_n = (-1)^{n+1}$ (partial sums bounded by 1) of Dirichlet’s test. Dirichlet also handles $\sum \frac{\sin n}{n}$: the partial sums $\sum_{k \leq n} \sin k$ are bounded (a closed-form geometric-type estimate), and $1/n \downarrow 0$.

Power series and radius of convergence

Definition 4.8.8. A *power series* (about 0) is $\sum_{n=0}^{\infty} c_n z^n$. Its *radius of convergence* is $R = 1/\alpha$ where $\alpha = \limsup_{n \rightarrow \infty} |c_n|^{1/n}$ (with $R = \infty$ if $\alpha = 0$, $R = 0$ if $\alpha = \infty$).

Theorem 4.8.9 (Cauchy–Hadamard). The power series $\sum c_n z^n$ converges absolutely for $|z| < R$ and diverges for $|z| > R$; on $|z| = R$ the behavior varies.

Proof. Apply the root test to $a_n = c_n z^n$: $\limsup |c_n z^n|^{1/n} = |z| \limsup |c_n|^{1/n} = |z|/R$. This is < 1 when $|z| < R$ (absolute convergence) and > 1 when $|z| > R$ (divergence, terms not $\rightarrow 0$). $\square$

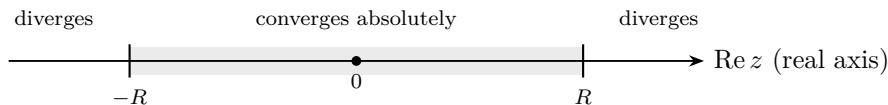


Figure 4.12. Radius of convergence on the real line: absolute convergence strictly inside $(-R, R)$, divergence strictly outside, endpoints undecided.

Worked Example 4.8.10 (Three radii (Rudin Ex. 3.9 flavor)). Find R for (a) $\sum n^3 z^n$, (b) $\sum \frac{2^n}{n!} z^n$, (c) $\sum \frac{2^n}{n^2} z^n$.

Recognition. “Radius of convergence”; compute $\limsup |c_n|^{1/n}$ (root) or $\lim |c_{n+1}/c_n|$ (ratio).

Solution. (a) $|c_n|^{1/n} = (n^3)^{1/n} = (n^{1/n})^3 \rightarrow 1$, so $R = 1$. (b) ratio $\frac{2^{n+1}/(n+1)!}{2^n/n!} = \frac{2}{n+1} \rightarrow 0$, so $\alpha = 0$,

$R = \infty$. (c) $|c_n|^{1/n} = 2/(n^{1/n})^2 \rightarrow 2$, so $R = \frac{1}{2}$. *Check.* (b) is (a multiple of) the exponential series $\sum (2z)^n/n! = e^{2z}$, entire, consistent with $R = \infty$. $\blacksquare$

Rearrangements

Definition 4.8.11. A *rearrangement* of $\sum a_n$ is $\sum a_{\sigma(n)}$ for a bijection $\sigma : \mathbb{N} \rightarrow \mathbb{N}$.

Theorem 4.8.12 (Absolute convergence is rearrangement-proof). *If $\sum a_n$ converges absolutely to s , then every rearrangement converges to the same s .*

Proof. Let $\sum a'_n = \sum a_{\sigma(n)}$ with partial sums s'_N . Given $\varepsilon > 0$ pick K with $\sum_{k>K} |a_k| < \varepsilon$. Choose p so large that $\{1, \dots, K\} \subseteq \{\sigma(1), \dots, \sigma(p)\}$. For $N \geq p$, the terms $a_1, \dots, a_K$ appear in both s_N and s'_N and cancel in the difference, so $|s'_N - s_N| \leq \sum_{k>K} |a_k| < \varepsilon$. Since $s_N \rightarrow s$, also $s'_N \rightarrow s$. $\square$

Theorem 4.8.13 (Riemann’s Rearrangement Theorem). *If $\sum a_n$ converges conditionally, then for any target $L \in \mathbb{R}$ there is a rearrangement converging to L (and one diverging, and ones oscillating).*

Proof idea. Conditional convergence forces both $\sum a_n^+ = +\infty$ and $\sum a_n^- = +\infty$ (positive and negative parts), while $a_n \rightarrow 0$. To hit a finite L : add positive terms until the partial sum first exceeds L , then add negative terms until it drops below L , then more positive terms, and so on. Because each pile is infinite the process never stalls, and because $a_n \rightarrow 0$ the overshoots shrink to 0, so the rearranged sums converge to L . $\square$

Worked Example 4.8.14 (Reshuffling the alternating harmonic series). Show a rearrangement of $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ can sum to $\frac{3}{2} \ln 2$.

Recognition. “Rearrangement changes the sum” — only possible for conditional convergence.

Solution. Take two positives then one negative throughout: $1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \dots$. A standard grouping computation gives sum $\frac{3}{2} \ln 2 \approx 1.0397$, versus the original $\ln 2 \approx 0.693$. *Check.* The terms are identical; only their *order* differs, and conditional convergence (not absolute) is what permits the value to change (Theorem 4.8.12 vs 4.8.13). $\blacksquare$

The number e is irrational

Theorem 4.8.15 ($e \notin \mathbb{Q}$, Rudin 3.31). $e = \sum_{n=0}^{\infty} \frac{1}{n!}$ is irrational.

Proof. First, $e = \sum_{n \geq 0} 1/n!$: the partial sums increase and are bounded above (≤ 3), and a tail estimate matches the monotone-limit definition. Suppose $e = p/q$ with $q \geq 1$. The tail

$$0 < e - \sum_{n=0}^q \frac{1}{n!} = \sum_{n=q+1}^{\infty} \frac{1}{n!} < \frac{1}{(q+1)!} \sum_{j=0}^{\infty} \frac{1}{(q+1)^j} = \frac{1}{(q+1)!} \cdot \frac{q+1}{q} = \frac{1}{q!q}.$$

Multiply through by $q!$: the number $q!e - q! \sum_{n=0}^q \frac{1}{n!}$ would be a positive integer (since $q!e = q!p/q = (q-1)!p \in \mathbb{Z}$ and $q! \sum_{n \leq q} 1/n! \in \mathbb{Z}$) lying strictly in $(0, 1/q) \subseteq (0, 1)$ — impossible. Hence e is irrational. $\square$

Caution. Rearrangement freely changes a *conditionally* convergent sum (Riemann), but *never* an absolutely convergent one. Before reordering terms, always check which regime you are in. The same caution applies to interchanging $\sum$ with limits or other sums: it is licensed by *absolute* convergence, not mere convergence.

Caution. A power series converges absolutely *inside* its disk and diverges *outside*, but the boundary $|z| = R$ is genuinely case-by-case: $\sum z^n$ ($R = 1$) diverges everywhere on $|z| = 1$, $\sum z^n/n^2$ converges everywhere on it, and $\sum z^n/n$ converges at $z = -1$ but diverges at $z = 1$. Never assume boundary behavior from R alone.

Further worked examples

Worked Example 4.8.16 (Absolute convergence settles an alternating series). Decide whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 + 1}$ converges, and whether absolutely.

Recognition. “Conditional vs absolute” (Definition 4.8.1); test $\sum |a_n|$ first — if it converges, you are done by Theorem 4.8.2 and need no alternating machinery.

Solution. $|a_n| = \frac{1}{n^2 + 1} \leq \frac{1}{n^2}$, and $\sum 1/n^2$ converges ($p = 2 > 1$), so by comparison $\sum |a_n|$ converges. Hence $\sum a_n$ converges *absolutely*, and a fortiori converges. *Check.* No need for the Alternating Series Test here — absolute convergence is the stronger, simpler route when $\sum |a_n|$ is summable. $\blacksquare$

Worked Example 4.8.17 (A conditionally convergent series via Dirichlet). Show $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges but not absolutely.

Recognition. “Conditional vs absolute”; the terms decrease to 0, so the Alternating Series Test (Theorem 4.8.3, a special case of Dirichlet 4.8.6) gives convergence, while the absolute series is a p -series with $p = \frac{1}{2}$.

Solution. $c_n = 1/\sqrt{n}$ satisfies $c_1 \geq c_2 \geq \dots \geq 0$ and $c_n \rightarrow 0$, so $\sum (-1)^{n+1}/\sqrt{n}$ converges by the Alternating Series Test. But $\sum |a_n| = \sum 1/\sqrt{n} = \sum 1/n^{1/2}$ diverges, since $p = \frac{1}{2} \leq 1$ (Theorem 4.7.9). Therefore the convergence is *conditional*. *Check.* The error bound $|s - s_N| \leq c_{N+1} = 1/\sqrt{N+1}$ shows the partial sums settle, while the absolute tail $\sum_{k>N} 1/\sqrt{k}$ grows without bound — both features coexist only for conditional convergence. $\blacksquare$

Worked Example 4.8.18 (Radius of convergence with a lim sup, not a lim). Find the radius of convergence of $\sum_{n=0}^{\infty} c_n z^n$ where $c_n = 2^n$ if n is even and $c_n = 3^n$ if n is odd.

Recognition. “Radius of convergence”; the ratio c_{n+1}/c_n oscillates, so the ratio test fails and we must use the Cauchy–Hadamard formula $R = 1/\limsup |c_n|^{1/n}$ (Definition 4.8.8).

Solution. $|c_n|^{1/n} = 2$ for even n and 3 for odd n , so the sequence $(|c_n|^{1/n})$ has subsequential limits $\{2, 3\}$ and hence $\limsup |c_n|^{1/n} = 3$. Therefore $R = 1/3$. *Check.* The ratio c_{n+1}/c_n alternates between $\sim 3^{n+1}/2^n \rightarrow \infty$ and $\sim 2^{n+1}/3^n \rightarrow 0$, so the ratio test is useless; the root/lim sup route is forced, and $R = 1/3$ is governed by the *larger* growth rate 3. $\blacksquare$

Exercises (Bartle & Sherbert, §9.1 and §9.3).

- (§9.1) Show that if a convergent series has only finitely many negative terms, it is absolutely convergent.
- (§9.1) Show that if a series is conditionally convergent, then the series of its positive terms diverges and the series of its negative terms diverges.
- (§9.1) If $\sum a_n$ is conditionally convergent, show there is a rearrangement whose partial sums diverge to $+\infty$.
- (§9.1) Where is absolute convergence of $\sum x_n$ used in the proof of Theorem 9.1.5 (rearrangement)?

5. (§9.1) If $\sum a_n$ is absolutely convergent, is every rearrangement also absolutely convergent?
6. (§9.1) Find an explicit n -th partial sum of $\sum_{n=2}^{\infty} \ln(1 - 1/n^2)$ to show it converges to $-\ln 2$. Is the convergence absolute?
7. (§9.1) (a) If $\sum a_n$ converges absolutely and (b_n) is bounded, show $\sum a_n b_n$ converges absolutely. (b) Give an example where $\sum a_n$ is only conditional, (b_n) bounded, and $\sum a_n b_n$ diverges.
8. (§9.1) Give an example of a convergent $\sum a_n$ for which $\sum a_n^2$ diverges.
9. (§9.1) If (a_n) decreases to 0 and $\sum a_n$ converges, show $\lim(na_n) = 0$.
10. (§9.1) Give an example of a divergent $\sum a_n$ with (a_n) decreasing and $\lim(na_n) = 0$.
11. (§9.1) If $\lim(n^2 a_n)$ exists in $\mathbb{R}$, show $\sum a_n$ converges absolutely.
12. (§9.1) Let $a > 0$. Show $\sum(1 + a^n)^{-1}$ diverges for $0 < a \leq 1$ and converges for $a > 1$.
13. (§9.1) Decide convergence of (a) $\sum \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n}}$; (b) $\sum \frac{\sqrt{n+1} - \sqrt{n}}{n}$.
14. (§9.1) A *subseries* of $\sum a_n$ is $\sum a_{n_k}$ for a subsequence (n_k) . Show $\sum a_n$ converges absolutely iff every subseries converges.
15. (§9.3) Test for convergence and absolute convergence: (a) $\sum \frac{(-1)^{n+1}}{n+1}$; (b) $\sum \frac{(-1)^{n+1}}{n^2+1}$; (c) $\sum \frac{(-1)^{n+1}n}{n+2}$; (d) $\sum \frac{(-1)^{n+1} \ln n}{n}$.
16. (§9.3) If s_n is the n -th partial sum of $\sum (-1)^{n+1} z_n$ (an alternating series with sum s), show $|s - s_n| \leq z_{n+1}$.
17. (§9.3) Give an example showing the Alternating Series Test may fail if (z_n) is not decreasing.
18. (§9.3) Show the Alternating Series Test is a consequence of Dirichlet's Test.
19. (§9.3) Does $1 - \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{7} + \cdots$ (signs in pairs) converge?
20. (§9.3) Let $a_n \in \mathbb{R}$, $p < q$. If $\sum a_n/n^p$ converges, show $\sum a_n/n^q$ converges.
21. (§9.3) If $p, q > 0$, show $\sum (-1)^n (\ln n)^p / n^q$ converges.
22. (§9.3) Discuss the series with n -th term (a) $(-1)^n n^n / (n+1)^{n+1}$; (b) $n^n / (n+1)^{n+1}$; (c) $(-1)^n (n+1)^n / n^{n+1}$; (d) $(n+1)^n / n^{n+1}$.
23. (§9.3) If the partial sums of $\sum a_n$ are bounded, show $\sum a_n e^{-nt}$ converges for $t > 0$.
24. (§9.3) If the partial sums s_n of $\sum a_n$ are bounded, show $\sum a_n/n$ converges to $\sum s_n / (n(n+1))$.
25. (§9.3) Can Dirichlet's Test establish convergence of $1 - \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} - \cdots$, where the number of signs increases by one per block? If not, use another method.
26. (§9.3) Show the hypothesis that $X = (x_n)$ is decreasing in Dirichlet's Test may be replaced by " $\sum |x_n - x_{n+1}|$ converges".
27. (§9.3) If (a_n) is bounded decreasing and (b_n) bounded increasing, and $x_n = a_n + b_n$, show $\sum |x_n - x_{n+1}|$ converges.
28. (§9.3) If the partial sums s_n of $\sum a_k$ satisfy $|s_n| \leq Mn^r$ with $r < 1$, show $\sum a_n/n$ converges.
29. (§9.3) Suppose $\sum a_n$ converges. Prove or give a counterexample that $\sum b_n$ converges, where (a) $b_n = a_n/n$; (b) $b_n = \sqrt{a_n}/n$ ($a_n \geq 0$); (c) $b_n = a_n \sin n$; (d) $b_n = \sqrt{a_n}/\sqrt{n}$ ($a_n \geq 0$); (e) $b_n = n^{1/n} a_n$; (f) $b_n = a_n/(1 + |a_n|)$.

Exercises (Rudin, Ch.3).

1. Prove that convergence of (s_n) implies convergence of $(|s_n|)$. Is the converse true?
2. Calculate $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$.

3. If $s_1 = \sqrt{2}$ and $s_{n+1} = \sqrt{2 + \sqrt{s_n}}$ for $n \geq 1$, prove that (s_n) converges and that $s_n < 2$ for all n .
4. Find the upper and lower limits of the sequence (s_n) defined by $s_1 = 0$, $s_{2m} = \frac{1}{2}s_{2m-1}$, $s_{2m+1} = \frac{1}{2} + s_{2m}$.
5. For any two real sequences $(a_n), (b_n)$, prove $\limsup(a_n + b_n) \leq \limsup a_n + \limsup b_n$, provided the right side is not of the form $\infty - \infty$.
6. Investigate convergence/divergence of $\sum a_n$ if (a) $a_n = \sqrt{n+1} - \sqrt{n}$; (b) $a_n = \frac{\sqrt{n+1} - \sqrt{n}}{n}$; (c) $a_n = (\sqrt[n]{n} - 1)^n$; (d) $a_n = \frac{1}{1+z^n}$ for complex z .
7. Prove that convergence of $\sum a_n$ (with $a_n \geq 0$) implies convergence of $\sum \frac{\sqrt{a_n}}{n}$.
8. If $\sum a_n$ converges and (b_n) is monotone and bounded, prove $\sum a_n b_n$ converges.
9. Find the radius of convergence of each power series: (a) $\sum n^3 z^n$; (b) $\sum \frac{2^n}{n!} z^n$; (c) $\sum \frac{2^n}{n^2} z^n$; (d) $\sum \frac{n^3}{3^n} z^n$.
10. Suppose the coefficients of the power series $\sum a_n z^n$ are integers, infinitely many nonzero. Prove the radius of convergence is at most 1.
11. Suppose $a_n > 0$, $s_n = a_1 + \cdots + a_n$, and $\sum a_n$ diverges. (a) Prove $\sum \frac{a_n}{1+a_n}$ diverges. (b) Prove $\frac{a_{N+1}}{s_{N+1}} + \cdots + \frac{a_{N+k}}{s_{N+k}} \geq 1 - \frac{s_N}{s_{N+k}}$ and deduce $\sum \frac{a_n}{s_n}$ diverges. (c) Prove $\frac{a_n}{s_n^2} \leq \frac{1}{s_{n-1}} - \frac{1}{s_n}$ and deduce $\sum \frac{a_n}{s_n^2}$ converges. (d) What can be said about $\sum \frac{a_n}{1+na_n}$ and $\sum \frac{a_n}{1+n^2 a_n}$?
12. Suppose $a_n > 0$ and $\sum a_n$ converges. Put $r_n = \sum_{m=n}^{\infty} a_m$. (a) Prove $\frac{a_m}{r_m} + \cdots + \frac{a_n}{r_n} > 1 - \frac{r_n}{r_m}$ for $m < n$, and deduce $\sum \frac{a_n}{r_n}$ diverges. (b) Prove $\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$ and deduce $\sum \frac{a_n}{\sqrt{r_n}}$ converges.
13. Prove that the Cauchy product of two absolutely convergent series converges absolutely.
14. If (s_n) is a complex sequence, define its arithmetic means $\sigma_n = \frac{s_0 + s_1 + \cdots + s_n}{n+1}$. (a) If $\lim s_n = s$, prove $\lim \sigma_n = s$. (b) Construct (s_n) not converging although $\lim \sigma_n = 0$. (c) Can $s_n > 0$ for all n , $\limsup s_n = \infty$, yet $\lim \sigma_n = 0$? (d) Put $a_n = s_n - s_{n-1}$; show $s_n - \sigma_n = \frac{1}{n+1} \sum_{k=1}^n k a_k$. Assume $\lim(na_n) = 0$ and (σ_n) converges; prove (s_n) converges. (A converse of (a) under $na_n \rightarrow 0$.) (e) Derive the same from the weaker hypothesis $|na_n| \leq M$ for all n .
15. Definition 3.21 extends to sequences in a fixed $\mathbb{R}^k$; absolute convergence means convergence of $\sum |a_n|$. Show Theorems 3.22, 3.23, 3.25(a), 3.33, 3.34, 3.42, 3.45, 3.47, 3.55 hold in this setting.
16. Fix $\alpha > 0$. Choose $x_1 > \sqrt{\alpha}$ and define $x_{n+1} = \frac{1}{2}(x_n + \frac{\alpha}{x_n})$. (a) Prove (x_n) decreases monotonically and $\lim x_n = \sqrt{\alpha}$. (b) Put $\varepsilon_n = x_n - \sqrt{\alpha}$ and show $\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$, so with $\beta = 2\sqrt{\alpha}$, $\varepsilon_{n+1} < \beta(\varepsilon_1/\beta)^{2^n}$. (c) This converges very rapidly; e.g. $\alpha = 3$, $x_1 = 2$ gives $\varepsilon_1/\beta < \frac{1}{10}$, so $\varepsilon_5 < 4 \cdot 10^{-16}$, $\varepsilon_6 < 4 \cdot 10^{-32}$.
17. Fix $\alpha > 1$. Take $x_1 > \sqrt{\alpha}$, and define $x_{n+1} = \frac{\alpha + x_n}{1 + x_n} = x_n + \frac{\alpha - x_n^2}{1 + x_n}$. (a) Prove $x_1 > x_3 > x_5 > \cdots$. (b) Prove $x_2 < x_4 < x_6 < \cdots$. (c) Prove $\lim x_n = \sqrt{\alpha}$. (d) Compare the convergence speed with Exercise 16.
18. Replace the recursion of Exercise 16 by $x_{n+1} = \frac{p-1}{p}x_n + \frac{\alpha}{p}x_n^{-p+1}$, where p is a fixed positive integer; describe the behavior of (x_n) .

19. Associate to each sequence $a = (\alpha_n)$, $\alpha_n \in \{0, 2\}$, the real number $x(a) = \sum_{n=1}^{\infty} \frac{\alpha_n}{3^n}$. Prove that the set of all $x(a)$ is exactly the Cantor set.
20. Suppose (p_n) is Cauchy in a metric space X and some subsequence (p_{n_k}) converges to $p \in X$. Prove the full sequence (p_n) converges to p .
21. Prove the analogue of Theorem 3.10(b): if (E_n) is a sequence of closed bounded sets in a complete metric space X with $E_n \supseteq E_{n+1}$ and $\lim \text{diam } E_n = 0$, then $\bigcap_n E_n$ consists of exactly one point.
22. Suppose X is complete and (G_n) is a sequence of dense open subsets of X . Prove Baire's theorem: $\bigcap_n G_n$ is nonempty (in fact dense). [Hint: find a shrinking sequence of neighborhoods E_n with $\overline{E_n} \subseteq G_n$, and apply Exercise 21.]
23. Suppose (p_n) and (q_n) are Cauchy in a metric space X . Show $(d(p_n, q_n))$ converges. [Hint: $|d(p_n, q_n) - d(p_m, q_m)|$ is small for large m, n .]
24. Let X be a metric space. (a) Call Cauchy sequences $(p_n), (q_n)$ equivalent if $\lim d(p_n, q_n) = 0$; prove this is an equivalence relation. (b) Let X^* be the set of equivalence classes; for $P, Q \in X^*$ with $(p_n) \in P$, $(q_n) \in Q$ define $\Delta(P, Q) = \lim d(p_n, q_n)$. Show Δ is well defined and a metric. (c) Prove X^* is complete. (d) For $p \in X$ let P_p be the class of the constant sequence p ; prove $\Delta(P_p, P_q) = d(p, q)$, so $\varphi(p) = P_p$ is an isometry of X into X^* . (e) Prove $\varphi(X)$ is dense in X^* , and equals X^* if X is complete. (X^* is the completion of X .)
25. Let X be the rationals with $d(x, y) = |x - y|$. What is the completion of X ? (Compare Exercise 24.)

Additional exercises.

1. Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n} + 1}$ converges absolutely, conditionally, or not at all.
2. Find the radius of convergence of (a) $\sum \frac{z^n}{n^2}$; (b) $\sum n! z^n$; (c) $\sum \frac{(2n)!}{(n!)^2} z^n$.
3. For the power series in part (a) above, determine the behavior at both endpoints $z = \pm R$.
4. (Non-example.) Exhibit a conditionally convergent series and describe a rearrangement of it summing to 0, illustrating Riemann's theorem; explain why no such reshuffling can change the sum of $\sum (-1)^n/n^2$.
5. Use the alternating series error bound to compute $\sum_{n=1}^{\infty} (-1)^{n+1}/n$ to within 0.05, stating how many terms suffice.
6. Apply Dirichlet's test to show $\sum \frac{\sin n}{n}$ converges, citing a bound on $\sum_{k \leq n} \sin k$.
7. (Diagram problem.) On a number line, mark the interval of convergence of $\sum \frac{(x-2)^n}{3^n n}$ (center, radius, and each endpoint's convergence), shading where it converges absolutely.
8. Mimic the irrationality proof of e to show that $\sum_{n=0}^{\infty} \frac{1}{n! 2^n}$ has a sum strictly between consecutive rationals p/q for suitable q ; i.e. reproduce the tail estimate $0 < \text{tail} < \frac{1}{q! q}$ for this series.
9. Prove that if $\sum a_n$ converges absolutely then $\sum a_n^2$ converges, and that the Cauchy product of $\sum a_n$ with itself converges absolutely.

Chapter 5

Limits and Continuity

5.1 Limits of Functions

Cluster points: where a limit can be asked

The limit of a function f at a point c asks what the values $f(x)$ approach as x approaches c *without ever equaling* c . For the question to be meaningful, f must be defined at points arbitrarily near c (though possibly not at c itself). That is the role of a cluster point.

Definition 5.1.1. Let $A \subseteq \mathbb{R}$. A point $c \in \mathbb{R}$ is a *cluster point* (or *accumulation point*, or *limit point*) of A if for every $\delta > 0$ the δ -neighborhood $V_\delta(c) = (c - \delta, c + \delta)$ contains at least one point of A *different from* c ; equivalently, $(V_\delta(c) \setminus \{c\}) \cap A \neq \emptyset$ for every $\delta > 0$.

The point c may or may not belong to A , and if it does belong it is ignored: what matters is that A crowds up against c from points other than c . The sequential restatement is the one we use constantly.

Theorem 5.1.2 (Sequential characterization of cluster points). *c is a cluster point of $A \subseteq \mathbb{R}$ iff there is a sequence (a_n) in A with $a_n \neq c$ for all n and $a_n \rightarrow c$.*

Proof. ($\implies$) For each n the neighborhood $V_{1/n}(c)$ contains a point $a_n \in A$ with $a_n \neq c$; then $|a_n - c| < 1/n \rightarrow 0$, so $a_n \rightarrow c$. ($\impliedby$) Given $\delta > 0$, since $a_n \rightarrow c$ there is n with $|a_n - c| < \delta$; as $a_n \in A$ and $a_n \neq c$, $V_\delta(c)$ meets $A \setminus \{c\}$. Hence c is a cluster point. $\square$

Worked Example 5.1.3 (Reading off cluster points). Find the cluster points of (a) $A_1 = (0, 1)$; (b) a finite set; (c) $\mathbb{N}$; (d) $A_4 = \{1/n : n \in \mathbb{N}\}$; (e) $A_5 = [0, 1] \cap \mathbb{Q}$.

Recognition. “List the cluster points” — probe each candidate c with shrinking neighborhoods and ask whether A keeps showing up nearby (other than at c).

Solution. (a) Every point of $[0, 1]$ is a cluster point of $(0, 1)$; in particular the endpoints $0, 1$ are cluster points but lie outside A_1 . (b) A finite set has no cluster points: choose δ smaller than the least gap. (c) $\mathbb{N}$ has none, for the same reason (gaps of size 1). (d) The only cluster point of A_4 is 0 — which is *not* in A_4 — since every $1/n$ is isolated. (e) By density of $\mathbb{Q}$, every point of $[0, 1]$ is a cluster point of A_5 . *Check.* “cluster point” is about accumulation, not membership — (a),(d),(e) all have cluster points outside the set.

The ε - δ definition of a limit

Definition 5.1.4. Let $A \subseteq \mathbb{R}$, let c be a cluster point of A , and let $f : A \rightarrow \mathbb{R}$. A real number L is a *limit of f at c* , written $\lim_{x \rightarrow c} f(x) = L$ or $f(x) \rightarrow L$ as $x \rightarrow c$, if

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{such that} \quad x \in A, \quad 0 < |x - c| < \delta \quad \implies \quad |f(x) - L| < \varepsilon.$$

If no such L exists, f *diverges* at c .

The clause $0 < |x - c|$ deletes the point $x = c$: the value $f(c)$ (if any) is irrelevant. Read geometrically (Rudin's neighborhood phrasing): for *every* ε -band $(L - \varepsilon, L + \varepsilon)$ around L on the y -axis there is a punctured δ -interval around c on the x -axis whose image lands inside that band. The dependence $\delta = \delta(\varepsilon)$ is the whole content; shrink ε and you may be forced to shrink δ .

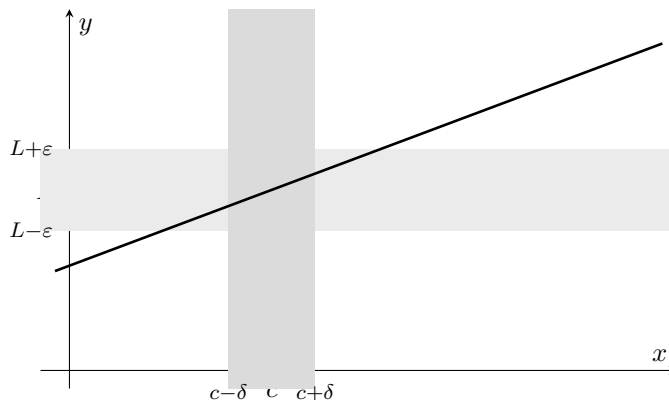


Figure 5.1. The ε - δ “tube”: the punctured vertical δ -strip about c maps into the horizontal ε -strip about L . Narrow the ε -strip and you must (in general) narrow the δ -strip to keep the graph trapped.

Theorem 5.1.5 (Uniqueness of the limit). *If c is a cluster point of A and $f : A \rightarrow \mathbb{R}$, then f has at most one limit at c .*

Proof. Suppose L, L' both work. Given $\varepsilon > 0$, pick δ_1 for L (with $\varepsilon/2$) and δ_2 for L' (with $\varepsilon/2$), and put $\delta = \min\{\delta_1, \delta_2\}$. Because c is a cluster point there *is* a point $x \in A$ with $0 < |x - c| < \delta$ (this is where clusterness is used). For that x ,

$$|L - L'| \leq |L - f(x)| + |f(x) - L'| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

As $\varepsilon > 0$ was arbitrary, $|L - L'| = 0$, so $L = L'$. □

Method (Running an ε - δ limit proof). *Recognition trigger.* You must prove $\lim_{x \rightarrow c} f(x) = L$ from the definition (the value L is given or guessed; “using the definition of limit”).

1. **Form the gap** $|f(x) - L|$ and simplify, factoring out $|x - c|$ when possible: aim for $|f(x) - L| = |x - c| \cdot (\text{stuff})$.
2. **Tame the “stuff”.** On a preliminary small interval, say $|x - c| < 1$, bound the leftover factor by a constant M ; this gives $|f(x) - L| \leq M|x - c|$ for $|x - c| < 1$.
3. **Choose δ .** Given $\varepsilon > 0$, take $\delta = \min\{1, \varepsilon/M\}$. Then $0 < |x - c| < \delta \implies |f(x) - L| \leq M|x - c| < \varepsilon$.
4. **Write it forward:** “Let $\varepsilon > 0$. Put $\delta = \dots$. If $0 < |x - c| < \delta$ then $|f(x) - L| \leq \dots < \varepsilon$.”

Shortcut (Lipschitz bound). If you can show $|f(x) - L| \leq K|x - c|$ outright, take $\delta = \varepsilon/K$ and skip the interval-shrinking (B&S 4.1.8 Ex. 6).

Worked Example 5.1.6 (The model affine limit $\lim_{x \rightarrow c}(mx + b) = mc + b$). Prove $\lim_{x \rightarrow 2}(3x + 1) = 7$.

Recognition. Direct ε - δ ; the gap is already a constant multiple of $|x - c|$.

Solution. $|(3x + 1) - 7| = 3|x - 2|$. Given $\varepsilon > 0$, put $\delta = \varepsilon/3$. If $0 < |x - 2| < \delta$ then $|(3x + 1) - 7| = 3|x - 2| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon$. *Check.* For $\varepsilon = 0.03$, $\delta = 0.01$: $x = 2.005 \implies 3x + 1 = 7.015$, within 0.03 of 7. ■

Worked Example 5.1.7 (A quadratic: $\lim_{x \rightarrow 3} x^2 = 9$, with interval-shrinking). Prove $\lim_{x \rightarrow 3} x^2 = 9$.

Recognition. “Prove this specific limit”; the leftover factor depends on x , so use Step 2.

Solution. $|x^2 - 9| = |x - 3||x + 3|$. Restrict to $|x - 3| < 1$, i.e. $2 < x < 4$; then $|x + 3| < 7$, so $|x^2 - 9| \leq 7|x - 3|$. Given $\varepsilon > 0$ put $\delta = \min\{1, \varepsilon/7\}$; if $0 < |x - 3| < \delta$ then $|x^2 - 9| \leq 7|x - 3| < \varepsilon$. *Check.* B&S’s identity $|x^2 - c^2| = |x - c||x + c| \leq (2|c| + \delta)|x - c|$ is exactly this estimate with $c = 3$, $\delta \leq 1$. ■

Worked Example 5.1.8 (A reciprocal: $\lim_{x \rightarrow 2} \frac{1}{x} = \frac{1}{2}$). Prove it from the definition.

Recognition. A quotient blows up if the denominator nears 0; bound it away from 0 first.

Solution. $\left| \frac{1}{x} - \frac{1}{2} \right| = \frac{|x-2|}{2|x|}$. Restrict to $|x-2| < 1$, so $x > 1$ and $|x| > 1$, giving $\left| \frac{1}{x} - \frac{1}{2} \right| \leq \frac{1}{2} |x-2|$. Given $\varepsilon > 0$ put $\delta = \min\{1, 2\varepsilon\}$; then $0 < |x-2| < \delta \implies \left| \frac{1}{x} - \frac{1}{2} \right| \leq \frac{1}{2} |x-2| < \varepsilon$. *Check.* The same trick proves the quotient law later — keep the denominator bounded below. ■

The sequential criterion (Heine)

The single most useful reformulation converts function-limits into sequence-limits, where all of Chapter 4's machinery applies.

Theorem 5.1.9 (Sequential criterion for limits). *Let c be a cluster point of A and $f : A \rightarrow \mathbb{R}$. Then $\lim_{x \rightarrow c} f(x) = L$ iff for every sequence (x_n) in A with $x_n \neq c$ and $x_n \rightarrow c$, one has $f(x_n) \rightarrow L$.*

Proof. ($\implies$) Assume $f(x) \rightarrow L$ and let $x_n \rightarrow c$, $x_n \in A \setminus \{c\}$. Given $\varepsilon > 0$ choose δ from the limit definition, then K with $|x_n - c| < \delta$ for $n \geq K$. Since $x_n \neq c$, $0 < |x_n - c| < \delta$, so $|f(x_n) - L| < \varepsilon$ for $n \geq K$; thus $f(x_n) \rightarrow L$.

($\impliedby$, contrapositive) Suppose $f(x) \not\rightarrow L$. Then some $\varepsilon_0 > 0$ resists every δ : for each n , $\delta = 1/n$ fails, so there is $x_n \in A$ with $0 < |x_n - c| < 1/n$ but $|f(x_n) - L| \geq \varepsilon_0$. Then $x_n \rightarrow c$, $x_n \neq c$, yet $f(x_n) \not\rightarrow L$ — the sequential condition fails. □

Method (Divergence by two sequences). *Recognition trigger.* You suspect $\lim_{x \rightarrow c} f$ does not exist (oscillation, a sign jump, a $1/x$ blowup).

1. Produce two sequences $x_n \rightarrow c$ and $y_n \rightarrow c$ (each $\neq c$, in A).
2. Compute $\lim f(x_n)$ and $\lim f(y_n)$.
3. If they differ (or one fails to exist), then $\lim_{x \rightarrow c} f$ does not exist, by the contrapositive of Theorem 5.1.9.

Worked Example 5.1.10 (A non-example: $\lim_{x \rightarrow 0} \sin(1/x)$ does not exist). Show sgn-style oscillation kills the limit.

Recognition. “Show the limit does not exist”; two sequences route.

Solution. Let $x_n = \frac{1}{n\pi} \rightarrow 0$, so $\sin(1/x_n) = \sin(n\pi) = 0 \rightarrow 0$. Let $y_n = \frac{1}{(2n + \frac{1}{2})\pi} \rightarrow 0$, so $\sin(1/y_n) = \sin((2n + \frac{1}{2})\pi) = 1 \rightarrow 1$. The two limits 0 and 1 differ, so $\lim_{x \rightarrow 0} \sin(1/x)$ does not exist. *Check.* The graph oscillates between ± 1 infinitely often near 0 — no band of radius < 1 can trap the tail. ■

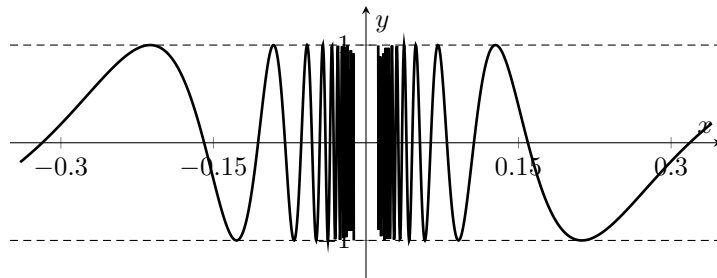


Figure 5.2. $f(x) = \sin(1/x)$ near 0: ever-faster oscillation between ± 1 . Along $x_n = 1/(n\pi)$ the values are 0; along $y_n = 1/((2n + \frac{1}{2})\pi)$ they are 1.

Worked Example 5.1.11 (Taming oscillation with a factor: $\lim_{x \rightarrow 0} x \sin(1/x) = 0$). Show that multiplying by x rescues the limit.

Recognition. A bounded times a vanishing factor; squeeze.

Solution. For $x \neq 0$, $|x \sin(1/x)| \leq |x|$ since $|\sin| \leq 1$. Given $\varepsilon > 0$ take $\delta = \varepsilon$; then $0 < |x| < \delta \implies |x \sin(1/x) - 0| \leq |x| < \varepsilon$. So the limit is 0. *Check.* The wild oscillation is now confined to the shrinking envelope $\pm |x|$, which pinches to 0 (this is the squeeze theorem of §5.2 in disguise). ■

Worked Example 5.1.12 (Dirichlet-type: $f(x) = x$ on $\mathbb{Q}$, 0 off $\mathbb{Q}$). Let $f(x) = x$ for $x \in \mathbb{Q}$ and $f(x) = 0$ for $x \notin \mathbb{Q}$. Show $\lim_{x \rightarrow 0} f = 0$ but $\lim_{x \rightarrow c} f$ fails for every $c \neq 0$.

Recognition. A function defined by cases on rationality; the sequential criterion separates the two “branches”.

Solution. At 0: for all x , $|f(x)| \leq |x|$ (either $|x|$ or 0), so $\delta = \varepsilon$ gives $|f(x) - 0| < \varepsilon$; thus $\lim_{x \rightarrow 0} f = 0$. At $c \neq 0$: pick rationals $r_n \rightarrow c$ and irrationals $s_n \rightarrow c$ (density). Then $f(r_n) = r_n \rightarrow c \neq 0$ while $f(s_n) = 0 \rightarrow 0$. The two sequential limits differ, so no limit exists at c . *Check.* The limit survives exactly where the two branches (x and 0) agree, namely at $x = 0$. ■

Caution. The limit value must be *found* before it is *verified*: the ε - δ definition only certifies a candidate L , it does not produce one. And δ may depend on ε (and on c) but *never* on x — writing “take $\delta = |x - c|/2$ ” is circular. Finally, $f(c)$ plays no role: the limit can exist with f undefined at c , or with $f(c)$ equal to something other than L .

Remark (Restriction to a subset). If f_1 is the restriction of f to an *open* interval $I \ni c$, then f and f_1 have the same limit behaviour at c (an open I contains a full punctured neighborhood of c). For a one-sided restriction the implication is one-way: a two-sided limit forces each restricted limit, but agreeing restricted limits on *just one side* need not give a two-sided limit. This subtlety is the topic of one-sided limits in §5.2.

Further worked examples

Worked Example 5.1.13 (A cube-root gap: $\lim_{x \rightarrow c} \sqrt[3]{x}$ by factoring the difference). Prove $\lim_{x \rightarrow 8} \sqrt[3]{x} = 2$ from the definition.

Recognition. “Prove this specific limit”: the gap $|\sqrt[3]{x} - 2|$ is not yet a multiple of $|x - 8|$, so factor $a - b = \frac{a^3 - b^3}{a^2 + ab + b^2}$ to expose $|x - 8|$.

Solution. With $a = \sqrt[3]{x}$, $b = 2$, $a^3 - b^3 = x - 8$ and $|\sqrt[3]{x} - 2| = \frac{|x - 8|}{a^2 + 2a + 4}$. The denominator $a^2 + 2a + 4 = (a + 1)^2 + 3 \geq 3 > 0$ for all real a , so $|\sqrt[3]{x} - 2| \leq \frac{1}{3}|x - 8|$. Given $\varepsilon > 0$ put $\delta = 3\varepsilon$; then $0 < |x - 8| < \delta \implies |\sqrt[3]{x} - 2| < \varepsilon$. *Check.* At $x = 8.3$, $\sqrt[3]{8.3} \approx 2.0247$, and $\frac{1}{3}|8.3 - 8| = 0.1$ indeed bounds the gap 0.0247. ■

Worked Example 5.1.14 (A non-example: a one-sided blowup defeats the two-sided limit). Show $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist, via the sequential criterion.

Recognition. “Show the limit does not exist”: route through two sequences approaching 0 from opposite sides (Theorem 5.1.9).

Solution. For $x > 0$, $\frac{|x|}{x} = 1$; for $x < 0$, $\frac{|x|}{x} = -1$. Take $x_n = \frac{1}{n} \rightarrow 0$ (positive): $f(x_n) = 1 \rightarrow 1$. Take $y_n = -\frac{1}{n} \rightarrow 0$ (negative): $f(y_n) = -1 \rightarrow -1$. The two sequential limits differ, so $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist. *Check.* The one-sided limits are 1 and -1 , a jump of size 2; the deleted point $x = 0$ is precisely where the formula is undefined. ■

Exercises (Rudin, Ch.4).

1. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $\lim_{h \rightarrow 0} [f(x+h) - f(x-h)] = 0$ for every $x \in \mathbb{R}$. Does this imply f is continuous? (Answer with proof or counterexample.)

Exercises (Bartle & Sherbert, §4.1).

1. Determine a condition on $|x - 1|$ that will assure: (a) $|x^2 - 1| < \frac{1}{2}$; (b) $|x^2 - 1| < 10^{-3}$; (c) $|x^2 - 1| < 1/n$ for given $n \in \mathbb{N}$; (d) $|x^3 - 1| < 1/n$ for given $n \in \mathbb{N}$.
2. Determine a condition on $|x - 4|$ that will assure: (a) $|\sqrt{x} - 2| < \frac{1}{2}$; (b) $|\sqrt{x} - 2| < 10^{-2}$.
3. Let c be a cluster point of $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$. Prove $\lim_{x \rightarrow c} f(x) = L$ iff $\lim_{x \rightarrow c} |f(x) - L| = 0$.
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$. Show $\lim_{x \rightarrow c} f(x) = L$ iff $\lim_{x \rightarrow 0} f(x + c) = L$.

5. Let $I = (-a, a)$ with $a > 0$ and $g(x) = x^2$ on I . For $x, c \in I$ show $|g(x) - c^2| \leq 2a|x - c|$, and use this to prove $\lim_{x \rightarrow c} x^2 = c^2$ for any $c \in I$.
6. Let I be an interval, $f : I \rightarrow \mathbb{R}$, $c \in I$. Suppose there are constants K, L with $|f(x) - L| \leq K|x - c|$ for $x \in I$. Show $\lim_{x \rightarrow c} f(x) = L$.
7. Show $\lim_{x \rightarrow c} x^3 = c^3$ for any $c \in \mathbb{R}$.
8. Show $\lim_{x \rightarrow c} \sqrt{x} = \sqrt{c}$ for any $c > 0$.
9. Use the ε - δ definition or the sequential criterion to establish: (a) $\lim_{x \rightarrow 2} \frac{1}{1-x} = -1$; (b) $\lim_{x \rightarrow 1} \frac{x}{1+x} = \frac{1}{2}$; (c) $\lim_{x \rightarrow 0} \frac{x^2}{|x|} = 0$; (d) $\lim_{x \rightarrow 1} \frac{x^2 - x + 1}{x + 1} = \frac{1}{2}$.
10. Use the definition of limit to show: (a) $\lim_{x \rightarrow 2} (x^2 + 4x) = 12$; (b) $\lim_{x \rightarrow -1} \frac{x+5}{2x+3} = 4$.
11. Use the definition of limit to prove: (a) $\lim_{x \rightarrow 3} \frac{2x+3}{4x-9} = 3$; (b) $\lim_{x \rightarrow 6} \frac{x^2-3x}{x+3} = 2$.
12. Show the following limits do not exist: (a) $\lim_{x \rightarrow 0} \frac{1}{x^2} (x > 0)$; (b) $\lim_{x \rightarrow 0} \frac{1}{\sqrt{x}} (x > 0)$; (c) $\lim_{x \rightarrow 0} (x + \operatorname{sgn}(x))$; (d) $\lim_{x \rightarrow 0} \sin(1/x^2)$.
13. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ has limit L at 0, and $a > 0$. If $g(x) = f(ax)$, show $\lim_{x \rightarrow 0} g(x) = L$.
14. Let $c \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ with $\lim_{x \rightarrow c} (f(x))^2 = L$. (a) Show that if $L = 0$ then $\lim_{x \rightarrow c} f(x) = 0$. (b) Show by example that if $L \neq 0$ then f may have no limit at c .
15. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x$ for x rational and $f(x) = 0$ for x irrational. (a) Show f has a limit at $x = 0$. (b) Use a sequential argument to show that if $c \neq 0$, f has no limit at c .
16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, I an open interval, $c \in I$. If f_1 is the restriction of f to I , show f_1 has a limit at c iff f does, and the limits are equal.
17. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, J a closed interval, $c \in J$. If f_2 is the restriction of f to J , show that if f has a limit at c then so does f_2 . Show by example that the converse can fail.

Additional exercises.

1. (Warm-up.) Using the definition, prove $\lim_{x \rightarrow 3} (2x - 5) = 1$ by exhibiting an explicit $\delta(\varepsilon)$.
2. Prove $\lim_{x \rightarrow 2} x^3 = 8$ from the definition. (Factor $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$ and bound the quadratic on $|x - 2| < 1$.)
3. (Sequential criterion both ways.) Let $f(x) = \frac{x^2 - 1}{x - 1}$ for $x \neq 1$. Show $\lim_{x \rightarrow 1} f = 2$ first by algebra, then by verifying that every $x_n \rightarrow 1$ ($x_n \neq 1$) gives $f(x_n) \rightarrow 2$.
4. (Non-example / disprove.) Show $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist by producing two sequences $x_n \rightarrow 0$ with different limits of $f(x_n)$. Identify the one-sided limits.
5. (Cluster points.) Let $A = \{1/m + 1/n : m, n \in \mathbb{N}\}$. Determine all cluster points of A , and decide which of them lie in A .
6. Suppose $\lim_{x \rightarrow c} f(x) = L > 0$. Prove there is a punctured neighborhood of c on which $f(x) > L/2 > 0$ (a “persistence of sign” for limits).
7. (Diagram.) Sketch $f(x) = \frac{x^2 - 4}{x - 2}$ near $x = 2$, marking the removable hole at $(2, 4)$, and on the same axes draw the $\varepsilon = 0.5$ horizontal band about $y = 4$ together with a δ -interval about $x = 2$ that maps into it.
8. (Composition of limits, carefully.) Give f, g with $\lim_{x \rightarrow 0} f(x) = 0$ and $\lim_{y \rightarrow 0} g(y) = 0$ but $\lim_{x \rightarrow 0} g(f(x)) \neq 0$. Explain which hypothesis of the naive “substitution” fails.

5.2 Limit Theorems and Extensions of the Limit Concept

The algebra of limits

Once the sequential criterion is in hand, the limit theorems for functions are corollaries of the corresponding theorems for sequences. We first record a boundedness fact.

Theorem 5.2.1 (Limits are locally bounded). *If $\lim_{x \rightarrow c} f = L$ exists, then f is bounded on a punctured neighborhood of c : there are $\delta > 0$ and $M > 0$ with $|f(x)| \leq M$ for all $x \in A$, $0 < |x - c| < \delta$.*

Proof. Take $\varepsilon = 1$ to get δ with $|f(x) - L| < 1$, hence $|f(x)| < |L| + 1 =: M$, on the punctured δ -neighborhood. $\square$

Theorem 5.2.2 (Limit theorems for functions). *Let c be a cluster point of A , let $f, g : A \rightarrow \mathbb{R}$ with $\lim_{x \rightarrow c} f = L$ and $\lim_{x \rightarrow c} g = M$, and let $k \in \mathbb{R}$. Then as $x \rightarrow c$,*

$$f + g \rightarrow L + M, \quad f - g \rightarrow L - M, \quad kf \rightarrow kL, \quad fg \rightarrow LM,$$

and, if $M \neq 0$ and $g \neq 0$ near c , $\frac{f}{g} \rightarrow \frac{L}{M}$. Moreover if $f(x) \geq 0$ on A then $L \geq 0$, and $\sqrt{f} \rightarrow \sqrt{L}$.

Proof. We give the product directly and deduce the rest. For the product, write

$$|f(x)g(x) - LM| = |f(x)(g(x) - M) + M(f(x) - L)| \leq |f(x)| |g(x) - M| + |M| |f(x) - L|.$$

By Theorem 5.2.1, $|f(x)| \leq K$ near c . Given $\varepsilon > 0$, choose δ so small that $|g(x) - M| < \frac{\varepsilon}{2K}$ and $|f(x) - L| < \frac{\varepsilon}{2(|M| + 1)}$; then the right side is $< \varepsilon$. The sum/difference/scalar cases follow from $|(f \pm g) - (L \pm M)| \leq |f - L| + |g - M|$ and $|kf - kL| = |k| |f - L|$. For the quotient it suffices to handle $1/g \rightarrow 1/M$: since $M \neq 0$, $|g(x)| \geq |M|/2$ near c , and $\left| \frac{1}{g(x)} - \frac{1}{M} \right| = \frac{|g(x) - M|}{|g(x)||M|} \leq \frac{2}{|M|^2} |g(x) - M| \rightarrow 0$; multiply by f . The sign and square-root statements are immediate from the sequential criterion and the corresponding sequence theorems. $\square$

Remark (Two proofs, one fact). B&S prove these via the *sequential criterion* (Theorem 5.1.9): take any $x_n \rightarrow c$, $x_n \neq c$; then $f(x_n) \rightarrow L$ and $g(x_n) \rightarrow M$, and the sequence limit theorems give $f(x_n)g(x_n) \rightarrow LM$, etc. Rudin gives the direct ε - δ estimate above. The sequential route is shorter; the direct route shows *which* δ works. Both are worth carrying.

Theorem 5.2.3 (Squeeze theorem for functions). *If $f(x) \leq g(x) \leq h(x)$ for x in a punctured neighborhood of c , and $\lim_{x \rightarrow c} f = \lim_{x \rightarrow c} h = L$, then $\lim_{x \rightarrow c} g = L$.*

Proof. Given $\varepsilon > 0$, choose δ so both $|f - L| < \varepsilon$ and $|h - L| < \varepsilon$ on the punctured δ -neighborhood. There $L - \varepsilon < f(x) \leq g(x) \leq h(x) < L + \varepsilon$, hence $|g(x) - L| < \varepsilon$. $\square$

Worked Example 5.2.4 (Rational evaluation by the limit theorems). Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ and $\lim_{x \rightarrow 1} \frac{x^2 + 2}{x^2 - 2}$.

Recognition. “Compute this limit”: simplify, then apply Theorem 5.2.2 provided no denominator vanishes after simplification.

Solution. First: for $x \neq 2$, $\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2 \rightarrow 4$ (the deleted point $x = 2$ is exactly why cancellation is legal). Second: denominator $\rightarrow 1 - 2 = -1 \neq 0$, numerator $\rightarrow 3$, so by the quotient law the limit is $3/(-1) = -3$. *Check.* Plug $x = 1.99$ into the first: $\approx 3.99 \rightarrow 4$; plug $x = 1.001$ into the second: ≈ -3.0 . $\blacksquare$

Worked Example 5.2.5 (A bounded factor times a vanishing one). Suppose f is bounded near c and $\lim_{x \rightarrow c} g = 0$. Show $\lim_{x \rightarrow c} fg = 0$.

Recognition. “Product with one factor $\rightarrow 0$, the other only bounded” — you cannot use the product law (the f -limit may not exist) so estimate directly.

Solution. Let $|f| \leq K$ on a punctured neighborhood. Then $|f(x)g(x)| \leq K|g(x)|$. Given $\varepsilon > 0$ choose δ with $|g(x)| < \varepsilon/K$; then $|fg| < \varepsilon$. So $fg \rightarrow 0$. *Check.* With $f = \sin(1/x)$ (bounded, no limit at 0) and $g = x$, this recovers $x \sin(1/x) \rightarrow 0$. ■

Caution. The product/quotient laws need *both* limits to exist. $\lim x \sin(1/x) = 0$ even though $\sin(1/x)$ has no limit at 0 — so a product can converge with a factor that does not. Do not “cancel” a limit that may not exist; fall back on a direct boundedness estimate.

One-sided limits

Definition 5.2.6. Let c be a cluster point of $A \cap (c, \infty)$. The *right-hand limit* $\lim_{x \rightarrow c^+} f(x) = L$ means: $\forall \varepsilon > 0 \exists \delta > 0$ with $x \in A$, $c < x < c + \delta \implies |f(x) - L| < \varepsilon$. The *left-hand limit* $\lim_{x \rightarrow c^-} f = L$ is defined symmetrically with $c - \delta < x < c$.

Theorem 5.2.7 (Two-sided = both one-sided). *If c is a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$, then $\lim_{x \rightarrow c} f = L$ iff $\lim_{x \rightarrow c^+} f = L = \lim_{x \rightarrow c^-} f$.*

Proof. A two-sided δ works for each side, giving $(\implies)$. Conversely take $\delta = \min\{\delta_+, \delta_-\}$ from the two one-sided definitions; for $0 < |x - c| < \delta$, x lies on one side or the other, and the matching estimate gives $|f(x) - L| < \varepsilon$. □

Worked Example 5.2.8 (A jump: sgn at 0). Let $\operatorname{sgn}(x) = 1$ for $x > 0$, -1 for $x < 0$, 0 at 0. Find the one-sided limits at 0.

Recognition. “One-sided limits at a jump”; evaluate each side as a constant.

Solution. For $x > 0$, $\operatorname{sgn}(x) = 1$, so $\lim_{x \rightarrow 0^+} \operatorname{sgn} = 1$; for $x < 0$ it is -1 , so $\lim_{x \rightarrow 0^-} \operatorname{sgn} = -1$. The one-sided limits differ, so $\lim_{x \rightarrow 0} \operatorname{sgn}$ does *not* exist (Theorem 5.2.7). *Check.* The value $\operatorname{sgn}(0) = 0$ matches neither one-sided limit — a *jump discontinuity*. ■

Infinite limits and limits at infinity

Definition 5.2.9. With c a cluster point of A : $\lim_{x \rightarrow c} f = +\infty$ means $\forall M > 0 \exists \delta > 0$ with $0 < |x - c| < \delta \implies f(x) > M$; replace $f(x) > M$ by $f(x) < -M$ for $-\infty$. For *limits at infinity* (with A unbounded above): $\lim_{x \rightarrow +\infty} f = L$ means $\forall \varepsilon > 0 \exists K > 0$ with $x > K \implies |f(x) - L| < \varepsilon$; and $\lim_{x \rightarrow +\infty} f = +\infty$ means $\forall M \exists K$ with $x > K \implies f(x) > M$. The cases $x \rightarrow -\infty$ are symmetric.

In Rudin’s uniform language (4.33): writing V for a neighborhood of $+\infty$, namely a ray $(c, +\infty)$, and similarly for $-\infty$, the single definition “ $f(t) \rightarrow A$ as $t \rightarrow x$ if every neighborhood of A contains $f(V \cap E \setminus \{x\})$ for some neighborhood V of x ” subsumes all finite and infinite cases at once.

Theorem 5.2.10 (Reciprocal bridge). *Let $f > 0$ on $(c, \infty) \cap A$. Then $\lim_{x \rightarrow c^+} f = +\infty$ iff $\lim_{x \rightarrow c^+} \frac{1}{f} = 0$. (Same with two-sided or $x \rightarrow \infty$ versions.)*

Proof. $f > M \iff 1/f < 1/M$; let $M \rightarrow \infty$ correspond to $1/M \rightarrow 0$. Precisely: given $\varepsilon > 0$ set $M = 1/\varepsilon$; then $f > M$ on a one-sided δ -interval gives $0 < 1/f < \varepsilon$ there, and conversely. □

Worked Example 5.2.11 ($\lim_{x \rightarrow 0} 1/x^2 = +\infty$ and $\lim_{x \rightarrow \infty} (x+2)/x = 1$). Evaluate both.

Recognition. “Infinite limit” (denominator $\rightarrow 0$, sign fixed) and “limit at infinity” (rational function, compare degrees).

Solution. For $1/x^2$: given $M > 0$ take $\delta = 1/\sqrt{M}$; if $0 < |x| < \delta$ then $x^2 < 1/M$ so $1/x^2 > M$. Hence $\lim_{x \rightarrow 0} 1/x^2 = +\infty$ (and $\lim_{x \rightarrow 0} x^2 = 0^+$ confirms via Theorem 5.2.10). For $(x+2)/x = 1 + 2/x$: as $x \rightarrow \infty$, $2/x \rightarrow 0$, so the limit is 1. *Check.* $x = 10^3 \Rightarrow (x+2)/x = 1.002$; the larger x , the closer to 1. ■

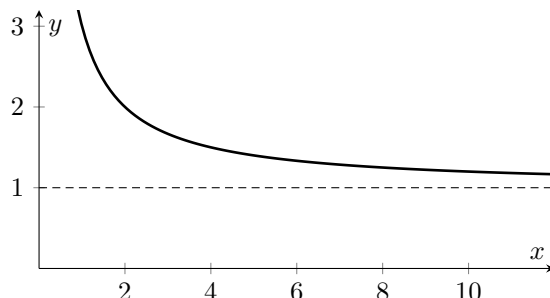
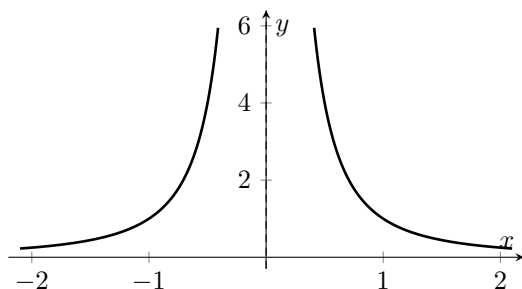


Figure 5.3. Left: $1/x^2 \rightarrow +\infty$ as $x \rightarrow 0$ (vertical asymptote). Right: $(x+2)/x = 1 + 2/x \rightarrow 1$ as $x \rightarrow \infty$ (horizontal asymptote).

Caution. “ $\lim f = +\infty$ ” does not mean the limit *exists* in $\mathbb{R}$ — it is shorthand for unbounded escape, a kind of *controlled* divergence. The arithmetic of $\pm\infty$ has forbidden forms: $\infty - \infty$, $0 \cdot \infty$, ∞/∞ , $A/0$ are undefined (Rudin 4.34). E.g. $f = 1/x$, $g = -1/x + 1$ near 0^+ both blow up but $f + g \rightarrow +\infty$ while $f - g = 0$; you must compute, not cancel infinities.

Further worked examples

Worked Example 5.2.12 (A limit at infinity of a rational function). Evaluate $\lim_{x \rightarrow +\infty} \frac{3x^2 - x + 5}{2x^2 + 7}$ and $\lim_{x \rightarrow +\infty} \frac{4x + 1}{x^2 - 3}$.

Recognition. “Limit at infinity of a quotient”: divide by the highest power so each piece becomes a sum of constants and vanishing terms, then apply Theorem 5.2.2.

Solution. First: divide top and bottom by x^2 : $\frac{3 - \frac{1}{x} + \frac{5}{x^2}}{2 + \frac{7}{x^2}} \rightarrow \frac{3 - 0 + 0}{2 + 0} = \frac{3}{2}$, since each $\frac{1}{x}$, $\frac{1}{x^2} \rightarrow 0$ as $x \rightarrow \infty$.

Second: divide by x^2 : $\frac{\frac{4}{x} + \frac{1}{x^2}}{1 - \frac{3}{x^2}} \rightarrow \frac{0}{1} = 0$ (numerator degree below denominator). *Check.* At $x = 100$, the first is $\approx \frac{29905}{20007} \approx 1.495 \rightarrow 1.5$; the second is $\approx \frac{401}{9997} \approx 0.04 \rightarrow 0$. ■

Worked Example 5.2.13 (A squeeze for a function limit). Evaluate $\lim_{x \rightarrow 0} x^2 \cos(1/x^2)$.

Recognition. “Bounded oscillation times a vanishing factor”: the product law fails (the cosine has no limit), so bound and squeeze (Theorem 5.2.3).

Solution. For $x \neq 0$, $|\cos(1/x^2)| \leq 1$, so $-x^2 \leq x^2 \cos(1/x^2) \leq x^2$. Both bounds $\pm x^2 \rightarrow 0$ as $x \rightarrow 0$, so by the squeeze theorem the middle term $\rightarrow 0$. *Check.* Directly, $|x^2 \cos(1/x^2) - 0| \leq x^2 < \varepsilon$ whenever $|x| < \sqrt{\varepsilon}$, so $\delta = \sqrt{\varepsilon}$ also certifies the limit 0. ■

Exercises (Bartle & Sherbert, §4.2).

- Apply the limit theorems to determine: (a) $\lim_{x \rightarrow 1} (x+1)(2x+3)$; (b) $\lim_{x \rightarrow 1} \frac{x^2 + 2}{x^2 - 2}$ ($x > 0$); (c) $\lim_{x \rightarrow 2} \left(\frac{1}{x+1} - \frac{1}{2x} \right)$ ($x > 0$); (d) $\lim_{x \rightarrow 0} \frac{x+1}{x^2 + 2}$.
- Determine the following limits, stating which theorems are used: (a) $\lim_{x \rightarrow 2} \sqrt{\frac{2x+1}{x+3}}$ ($x > 0$); (b) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ ($x > 0$); (c) $\lim_{x \rightarrow 0} \frac{(x+1)^2 - 1}{x}$ ($x > 0$); (d) $\lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1}$ ($x > 0$).
- Find $\lim_{x \rightarrow 0} \frac{\sqrt{1+2x} - \sqrt{1+3x}}{x + 2x^2}$, where $x > 0$.
- Prove $\lim_{x \rightarrow 0} \cos(1/x)$ does not exist, but $\lim_{x \rightarrow 0} x \cos(1/x) = 0$.

5. Let f, g on $A \subseteq \mathbb{R}$, c a cluster point of A . Suppose f is bounded on a neighborhood of c and $\lim_{x \rightarrow c} g = 0$. Prove $\lim_{x \rightarrow c} fg = 0$.
6. Use the definition of limit to prove the first assertion of the sum law: $\lim_{x \rightarrow c}(f + g) = \lim f + \lim g$.
7. Use the sequential formulation to prove the difference law $\lim_{x \rightarrow c}(f - g) = \lim f - \lim g$.
8. Let $n \in \mathbb{N}$, $n \geq 3$. Derive the inequality $-x^2 \leq x^n \leq x^2$ for $-1 < x < 1$. Then use $\lim_{x \rightarrow 0} x^2 = 0$ to show $\lim_{x \rightarrow 0} x^n = 0$.
9. Let f, g on A , c a cluster point. (a) Show that if $\lim f$ and $\lim(f + g)$ exist, then $\lim g$ exists. (b) If $\lim f$ and $\lim fg$ exist, does $\lim g$ exist?
10. Give examples of f, g with no limit at c such that both $f + g$ and fg have limits at c .
11. Determine whether these limits exist in $\mathbb{R}$ ($x \neq 0$): (a) $\lim_{x \rightarrow 0} \sin(1/x^2)$; (b) $\lim_{x \rightarrow 0} x \sin(1/x^2)$; (c) $\lim_{x \rightarrow 0} \operatorname{sgn}(\sin(1/x))$; (d) $\lim_{x \rightarrow 0} \sqrt{x} \sin(1/x^2)$ ($x > 0$).
12. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x + y) = f(x) + f(y)$ for all x, y , and suppose $\lim_{x \rightarrow 0} f = L$ exists. Prove $L = 0$, and that f has a limit at every c . (Hint: $f(2x) = 2f(x)$ and $f(x) = f(x - c) + f(c)$.)
13. Let $f(x) = x + 1$ and $g(x) = 2$ if $x \neq 1$, $g(1) = 0$. (a) Find $\lim_{x \rightarrow 1} g(f(x))$ and compare with $g(\lim_{x \rightarrow 1} f(x))$. (b) Find $\lim_{x \rightarrow 1} f(g(x))$ and compare with $f(\lim_{x \rightarrow 1} g(x))$.
14. Let $A \subseteq \mathbb{R}$, $f : A \rightarrow \mathbb{R}$, c a cluster point. If $\lim_{x \rightarrow c} f$ exists, prove $\lim_{x \rightarrow c} |f| = |\lim_{x \rightarrow c} f|$.
15. Let $A \subseteq \mathbb{R}$, $f : A \rightarrow \mathbb{R}$ with $f(x) \geq 0$ for all $x \in A$, c a cluster point. If $\lim_{x \rightarrow c} f$ exists, prove $\lim_{x \rightarrow c} \sqrt{f} = \sqrt{\lim_{x \rightarrow c} f}$.

Exercises (Bartle & Sherbert, §4.3).

1. Prove the analogue of the sum/product/quotient theorems for one-sided limits.
2. Give an example of a function that has a right-hand limit but not a left-hand limit at a point.
3. Let $f(x) = |x|^{-1/2}$ for $x \neq 0$. Show $\lim_{x \rightarrow 0^+} f = \lim_{x \rightarrow 0^-} f = +\infty$.
4. Let $c \in \mathbb{R}$, f defined on (c, ∞) with $f(x) > 0$ there. Show $\lim_{x \rightarrow c} f = +\infty$ iff $\lim_{x \rightarrow c} 1/f = 0$.
5. Evaluate, or show nonexistent: (a) $\lim_{x \rightarrow 1^+} \frac{x}{x-1}$; (b) $\lim_{x \rightarrow 1} \frac{x}{x-1}$; (c) $\lim_{x \rightarrow 0^+} \frac{x+2}{\sqrt{x}}$; (d) $\lim_{x \rightarrow \infty} \frac{x+2}{\sqrt{x}}$; (e) $\lim_{x \rightarrow 0} \frac{\sqrt{x+1}}{x}$; (f) $\lim_{x \rightarrow \infty} \frac{\sqrt{x+1}}{x}$; (g) $\lim_{x \rightarrow \infty} \frac{\sqrt{x}-5}{\sqrt{x}+3}$; (h) $\lim_{x \rightarrow \infty} \frac{x-\sqrt{x}}{\sqrt{x}+x}$.
6. Prove the analogue of the limit theorems for limits as $x \rightarrow \infty$.
7. Suppose f, g have limits in $\mathbb{R}$ as $x \rightarrow \infty$ and $f(x) \leq g(x)$ for $x \in (a, \infty)$. Prove $\lim_{x \rightarrow \infty} f \leq \lim_{x \rightarrow \infty} g$.
8. Let f be defined on $(0, \infty)$. Prove $\lim_{x \rightarrow \infty} f(x) = L$ iff $\lim_{x \rightarrow 0^+} f(1/x) = L$.
9. Show that if $f : (a, \infty) \rightarrow \mathbb{R}$ has $\lim_{x \rightarrow \infty} xf(x) = L$ with $L \in \mathbb{R}$, then $\lim_{x \rightarrow \infty} f(x) = 0$.
10. Prove the analogue of the limit theorems for infinite limits.
11. Suppose $\lim_{x \rightarrow c} f = L > 0$ and $\lim_{x \rightarrow c} g = \infty$. Show $\lim_{x \rightarrow c} fg = \infty$. If $L = 0$, show by example this may fail.
12. Find f, g on $(0, \infty)$ with $\lim_{x \rightarrow \infty} f = \infty$, $\lim_{x \rightarrow \infty} g = \infty$, and $\lim_{x \rightarrow \infty} (f - g) = 0$. Can you find such, with $g > 0$, so that $\lim_{x \rightarrow \infty} f/g = 0$?
13. Let f, g on (a, ∞) with $\lim_{x \rightarrow \infty} f = L$ and $\lim_{x \rightarrow \infty} g = \infty$. Prove $\lim_{x \rightarrow \infty} f \circ g = L$.

Additional exercises.

1. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x^2 - 2x - 3}$ and $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$, naming the limit theorem at each step.

- (Squeeze.) Prove $\lim_{x \rightarrow 0} x^2 \cos(1/x) = 0$ and, more generally, that a bounded function times one tending to 0 tends to 0.
- Compute the one-sided limits of $f(x) = \frac{x}{|x|} + \frac{1}{1 + 2^{1/x}}$ at $x = 0$, and decide whether $\lim_{x \rightarrow 0} f$ exists.
- (Non-example / disprove.) Disprove the claim “if $\lim_{x \rightarrow c} (f + g)$ and $\lim_{x \rightarrow c} f$ both exist then $\lim_{x \rightarrow c} g$ need not exist.” That is, show $\lim_{x \rightarrow c} g$ *must* exist, and then give a separate f, g with $\lim(f + g)$ existing while *neither* $\lim f$ nor $\lim g$ does.
- Show $\lim_{x \rightarrow \infty} \frac{3x^2 - x + 2}{5x^2 + 7} = \frac{3}{5}$ and $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) = \frac{1}{2}$.
- Prove from the definition that $\lim_{x \rightarrow 1^+} \frac{1}{x - 1} = +\infty$ and $\lim_{x \rightarrow 1^-} \frac{1}{x - 1} = -\infty$; conclude the two-sided limit does not exist.
- (Diagram.) Draw $y = \frac{1}{x - 1}$ on $(-1, 3) \setminus \{1\}$, marking the vertical asymptote at $x = 1$ and the horizontal asymptote $y = 0$, and annotate the one-sided behaviours as $x \rightarrow 1^\pm$ and $x \rightarrow \pm\infty$.
- (Cauchy functional equation, limit version.) If $\lim_{x \rightarrow 0} f = L$ exists for an additive f ($f(x + y) = f(x) + f(y)$), prove $L = 0$ and that $\lim_{x \rightarrow c} f = f(c)$ at every c .

5.3 Continuous Functions

Continuity at a point

Continuity says the limit exists *and equals the function value*: no break, no jump, no hole. Crucially, continuity is defined even at *isolated* points of the domain, where no limit can be asked.

Definition 5.3.1. Let $A \subseteq \mathbb{R}$, $f : A \rightarrow \mathbb{R}$, $c \in A$. Then f is *continuous at c* if

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{such that} \quad x \in A, |x - c| < \delta \quad \implies \quad |f(x) - f(c)| < \varepsilon.$$

f is *continuous on A* if it is continuous at every point of A . If f is not continuous at c , it is *discontinuous* there.

Two differences from the limit definition: the value tested is $f(c)$ (not an external L), and the puncture $0 < |x - c|$ is gone — $x = c$ is allowed (it gives $|f(c) - f(c)| = 0 < \varepsilon$ for free). Consequently:

Theorem 5.3.2 (Continuity via limits). *If c is a cluster point of A , then f is continuous at c iff $\lim_{x \rightarrow c} f = f(c)$. If c is an isolated point of A (not a cluster point), then f is automatically continuous at c .*

Proof. For a cluster point the two definitions coincide once the (now harmless) $x = c$ case is added. For an isolated point there is δ_0 with $V_{\delta_0}(c) \cap A = \{c\}$; then $|x - c| < \delta_0$ forces $x = c$, so $|f(x) - f(c)| = 0 < \varepsilon$ trivially. $\square$

Theorem 5.3.3 (Sequential criterion for continuity). *f is continuous at $c \in A$ iff for every sequence (x_n) in A with $x_n \rightarrow c$ one has $f(x_n) \rightarrow f(c)$.*

Proof. Identical to Theorem 5.1.9 but with $L = f(c)$ and *no* restriction $x_n \neq c$ (allowed because $x = c$ is harmless). The contrapositive gives the *discontinuity criterion*: f is discontinuous at c iff some $x_n \rightarrow c$ has $f(x_n) \not\rightarrow f(c)$. $\square$

Worked Example 5.3.4 (Continuity of $|\cdot|$ and of $\sqrt{\cdot}$). Show $x \mapsto |x|$ is continuous on $\mathbb{R}$ and $x \mapsto \sqrt{x}$ is continuous on $[0, \infty)$.

Recognition. “Show continuous everywhere”; use a clean Lipschitz/Holder-type bound.

Solution. Absolute value: the reverse triangle inequality gives $||x| - |c|| \leq |x - c|$, so $\delta = \varepsilon$ works at every c . Square root: for $x, c \geq 0$, $|\sqrt{x} - \sqrt{c}| = \frac{|x - c|}{\sqrt{x} + \sqrt{c}} \leq \frac{|x - c|}{\sqrt{c}}$ if $c > 0$ (take $\delta = \varepsilon\sqrt{c}$); at $c = 0$, $|\sqrt{x} - 0| < \varepsilon \iff x < \varepsilon^2$, so $\delta = \varepsilon^2$. Check. $\sqrt{\cdot}$ is continuous but *not* Lipschitz at 0 — its difference quotient blows up; foreshadows uniform continuity in §5.7. $\blacksquare$

Worked Example 5.3.5 (A non-example everywhere: the Dirichlet function). Let $D(x) = 1$ for $x \in \mathbb{Q}$, $D(x) = 0$ for $x \notin \mathbb{Q}$. Show D is discontinuous at every $c \in \mathbb{R}$.

Recognition. “Show discontinuous”; discontinuity criterion with one bad sequence.

Solution. Fix c . If $c \in \mathbb{Q}$ ($D(c) = 1$), pick irrationals $s_n \rightarrow c$: $D(s_n) = 0 \not\rightarrow 1$. If $c \notin \mathbb{Q}$ ($D(c) = 0$), pick rationals $r_n \rightarrow c$: $D(r_n) = 1 \not\rightarrow 0$. Either way the sequential criterion fails. *Check.* No value can be assigned to repair this — the two dense “sheets” 0 and 1 never agree near any point. ■

Worked Example 5.3.6 (Continuous at exactly one point: Thomae-flavored). Let $g(x) = x$ for $x \in \mathbb{Q}$ and $g(x) = 0$ for $x \notin \mathbb{Q}$. Show g is continuous only at 0.

Recognition. A case-split function; test continuity by the two-branch sequential argument.

Solution. At 0: $|g(x)| \leq |x|$, so $\delta = \varepsilon$ gives continuity ($g(0) = 0$). At $c \neq 0$: rationals $r_n \rightarrow c$ give $g(r_n) = r_n \rightarrow c$, irrationals $s_n \rightarrow c$ give $g(s_n) = 0$; since $c \neq 0$, these limits differ, so g is discontinuous. *Check.* Continuity survives exactly where the branches x and 0 meet, $x = 0$ — compare the limit version in §5.1. ■

The topological characterization

Rudin’s signature reformulation makes continuity a statement about open sets and pre-images — the gateway to topology. We state it for maps between metric spaces (here subsets of $\mathbb{R}$, or $\mathbb{R}^k$), since that is the natural generality.

Theorem 5.3.7 (Global continuity via open sets). *Let $f : X \rightarrow Y$ map one metric space to another. The following are equivalent:*

- (i) f is continuous on X (continuous at every point);
- (ii) $f^{-1}(G)$ is open in X for every open $G \subseteq Y$;
- (iii) $f^{-1}(F)$ is closed in X for every closed $F \subseteq Y$;
- (iv) $f(\overline{E}) \subseteq \overline{f(E)}$ for every $E \subseteq X$.

Proof. (i) $\implies$ (ii): Let G be open and $p \in f^{-1}(G)$, so $f(p) \in G$; choose ε with the ε -ball $B_\varepsilon(f(p)) \subseteq G$. By continuity at p there is δ with $f(B_\delta(p)) \subseteq B_\varepsilon(f(p)) \subseteq G$, i.e. $B_\delta(p) \subseteq f^{-1}(G)$. So $f^{-1}(G)$ is open.

(ii) $\implies$ (i): Fix p and $\varepsilon > 0$. The ball $G = B_\varepsilon(f(p))$ is open, so $f^{-1}(G)$ is open and contains p ; pick δ with $B_\delta(p) \subseteq f^{-1}(G)$. Then $|x - p| < \delta \implies f(x) \in B_\varepsilon(f(p))$, i.e. $|f(x) - f(p)| < \varepsilon$.

(ii) $\iff$ (iii): take complements, using $f^{-1}(Y \setminus F) = X \setminus f^{-1}(F)$ (pre-image respects complements).

(i) $\implies$ (iv): let $p \in \overline{E}$ and $x_n \rightarrow p$ with $x_n \in E$; then $f(x_n) \rightarrow f(p)$ by continuity and $f(x_n) \in f(E)$, so $f(p) \in \overline{f(E)}$. (iv) $\implies$ (iii): apply (iv) to $E = f^{-1}(F)$ with F closed: $f(\overline{E}) \subseteq \overline{f(E)} \subseteq \overline{F} = F$, so $\overline{E} \subseteq f^{-1}(F) = E$, i.e. E is closed. □

Caution. Statement (ii) is about *pre-images*, not images. A continuous f need *not* send open sets to open sets, nor closed to closed: $f(x) = x^2$ maps the open $(-1, 1)$ to $[0, 1)$ (not open) and $\sin$ maps the closed unbounded $\mathbb{R}$ to $[-1, 1]$. “Open pre-image” is the right notion; “open image” is a different (open-map) property entirely.

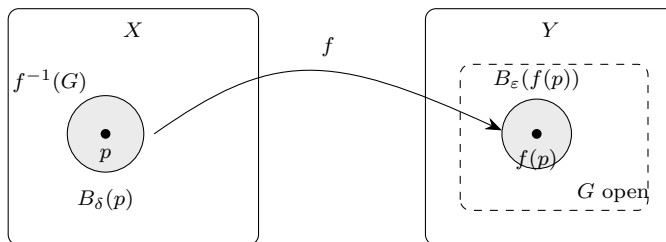


Figure 5.4. Continuity at p = every output ball $B_\varepsilon(f(p))$ has an input ball $B_\delta(p)$ mapping into it. Globally: the pre-image of an open G is open.

Remark (Determined on a dense set). If $f, g : X \rightarrow Y$ are continuous and agree on a *dense* subset $E \subseteq X$, then $f = g$ on all of X : for $p \in X$ take $x_n \rightarrow p$ with $x_n \in E$; then $f(p) = \lim f(x_n) = \lim g(x_n) = g(p)$. In particular $f(E)$ is dense in $f(X)$ (Rudin 4.4 Ex.), and a continuous map is pinned down by its values on $\mathbb{Q}$, say.

Worked Example 5.3.8 (Zero sets are closed). For continuous $f : X \rightarrow \mathbb{R}$, show the *zero set* $Z(f) = \{p : f(p) = 0\}$ is closed, and $\{p : f(p) > 0\}$ is open.

Recognition. “Show this level set is closed/open”; pull a closed/open set back through f .

Solution. $\{0\}$ is closed in $\mathbb{R}$, so $Z(f) = f^{-1}(\{0\})$ is closed by Theorem 5.3.7(iii). The ray $(0, \infty)$ is open, so $f^{-1}((0, \infty))$ is open. *Check.* For $f(x) = x^2 - 1$, $Z(f) = \{\pm 1\}$ (closed) and $\{f > 0\} = (-\infty, -1) \cup (1, \infty)$ (open). ■

Further worked examples

Worked Example 5.3.9 (Continuity at an isolated point). Let $A = \{0\} \cup \{1/n : n \in \mathbb{N}\}$ and let $f : A \rightarrow \mathbb{R}$ be *any* function. At which points of A is f automatically continuous, and what is the one nontrivial condition?

Recognition. “Continuity on a set with isolated points”: isolated points are free (Theorem 5.3.2); only the cluster point carries a real condition.

Solution. Every point $1/n$ is isolated in A (the gap to its neighbors $1/(n \pm 1)$ is positive), so f is continuous at each $1/n$ no matter how it is defined. The single cluster point is 0 (since $1/n \rightarrow 0$), and there continuity is the genuine requirement: f is continuous at 0 iff $f(1/n) \rightarrow f(0)$, by the sequential criterion (Theorem 5.3.3). *Check.* If $f(1/n) = (-1)^n$ and $f(0) = 0$, then f is continuous at every $1/n$ but *discontinuous* at 0, since $(-1)^n \not\rightarrow 0$. ■

Worked Example 5.3.10 (A non-example via the open-set criterion). Use Theorem 5.3.7 to show the floor function $\lfloor x \rfloor$ is not continuous on $\mathbb{R}$ by exhibiting an open set with non-open pre-image.

Recognition. “Disprove global continuity”: it suffices to find one open G whose pre-image $f^{-1}(G)$ is not open (the contrapositive of (i) $\iff$ (ii)).

Solution. Take the open interval $G = (-\frac{1}{2}, \frac{1}{2})$. Then $\lfloor x \rfloor \in G$ iff $\lfloor x \rfloor = 0$ iff $0 \leq x < 1$, so $f^{-1}(G) = [0, 1)$. This set is *not* open: the point 0 has no neighborhood contained in $[0, 1)$ (every ball around 0 contains negative numbers, where $\lfloor x \rfloor = -1 \notin G$). Hence by Theorem 5.3.7, $\lfloor \cdot \rfloor$ is not continuous on $\mathbb{R}$. *Check.* The failure point $x = 0$ is exactly an integer, where $\lfloor x \rfloor$ jumps from -1 to 0 — a jump discontinuity, consistent with the non-open pre-image. ■

Exercises (Rudin, Ch.4).

1. If f is a continuous mapping of a metric space X into a metric space Y , prove $f(\overline{E}) \subseteq \overline{f(E)}$ for every $E \subseteq X$. Show by example that $f(\overline{E})$ can be a proper subset of $\overline{f(E)}$.
2. Let f be a continuous real function on a metric space X . Let the *zero set* $Z(f) = \{p \in X : f(p) = 0\}$. Prove $Z(f)$ is closed.
3. Let f, g be continuous maps of a metric space X into a metric space Y , and let E be dense in X . Prove $f(E)$ is dense in $f(X)$. If $g(p) = f(p)$ for all $p \in E$, prove $g(p) = f(p)$ for all $p \in X$. (A continuous map is determined by its values on a dense subset.)
4. If f is a real continuous function on a closed set $E \subseteq \mathbb{R}$, prove there exist continuous $g : \mathbb{R} \rightarrow \mathbb{R}$ with $g(x) = f(x)$ for all $x \in E$ (a *continuous extension*). Show this becomes false if “closed” is omitted. (Hint: make the graph of g a straight line on each interval of $\mathbb{R} \setminus E$.)
5. If f is defined on E , the graph of f is $\{(x, f(x)) : x \in E\}$. Suppose E is compact; prove f is continuous on E iff its graph is compact.
6. Let $E \subseteq X$ and let f be defined on X ; the *restriction* of f to E is g with domain E and $g(p) = f(p)$. Define f, g on $\mathbb{R}^2$ by $f(0, 0) = g(0, 0) = 0$, and for $(x, y) \neq (0, 0)$, $f(x, y) = \frac{xy^2}{x^2 + y^4}$, $g(x, y) = \frac{xy^2}{x^2 + y^6}$. Prove f is bounded on $\mathbb{R}^2$; that g is unbounded in every neighborhood of $(0, 0)$; that f is not continuous at $(0, 0)$; yet the restrictions of both f and g to every straight line through $\mathbb{R}^2$ are continuous.

Exercises (Bartle & Sherbert, §5.1).

1. Prove the sequential criterion for continuity (Theorem 5.3.3).
2. Establish the discontinuity criterion: f is discontinuous at c iff some sequence $x_n \rightarrow c$ in the domain has $f(x_n) \not\rightarrow f(c)$.
3. Let $a < b < c$. Suppose f is continuous on $[a, b]$, g continuous on $[b, c]$, and $f(b) = g(b)$. Define h on $[a, c]$ by $h = f$ on $[a, b]$ and $h = g$ on $[b, c]$. Prove h is continuous on $[a, c]$.
4. Let $\lfloor x \rfloor$ be the greatest-integer function. Determine the points of continuity of (a) $f(x) = \lfloor x \rfloor$; (b) $g(x) = x\lfloor x \rfloor$; (c) $h(x) = \lfloor \sin x \rfloor$; (d) $k(x) = \lfloor 1/x \rfloor$ ($x \neq 0$).
5. Let $f(x) = \frac{x^2 + x - 6}{x - 2}$ for $x \neq 2$. Can f be defined at $x = 2$ so as to be continuous there?
6. Let $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$ continuous at $c \in A$. Show that for any $\varepsilon > 0$ there is a neighborhood $V_\delta(c)$ such that $x, y \in A \cap V_\delta(c) \implies |f(x) - f(y)| < \varepsilon$.
7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous at c with $f(c) > 0$. Show there is a neighborhood $V_\delta(c)$ on which $f(x) > 0$.
8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, $S = \{x : f(x) = 0\}$ its zero set. If (x_n) is in S and $x = \lim x_n$, show $x \in S$.
9. Let $A \subseteq B \subseteq \mathbb{R}$, $f : B \rightarrow \mathbb{R}$, g the restriction of f to A . (a) If f is continuous at $c \in A$, show g is continuous at c . (b) Show by example that g continuous at c need not make f continuous at c .
10. Show the absolute value $f(x) = |x|$ is continuous at every $c \in \mathbb{R}$.
11. Let $K > 0$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $|f(x) - f(y)| \leq K|x - y|$ for all x, y . Show f is continuous at every $c \in \mathbb{R}$.
12. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $f(r) = 0$ for every rational r . Prove $f(x) = 0$ for all x .
13. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = 2x$ for x rational, $g(x) = x + 3$ for x irrational. Find all points where g is continuous.
14. Let $A = (0, 1)$ and $k : A \rightarrow \mathbb{R}$ by $k(x) = 0$ for x irrational, $k(x) = n$ for $x = m/n$ in lowest terms. Prove k is unbounded on every open interval in A , and conclude k is continuous at no point of A .
15. Let $f : (0, 1) \rightarrow \mathbb{R}$ be bounded but with $\lim_{x \rightarrow 0} f$ not existing. Show there are sequences $(x_n), (y_n)$ in $(0, 1)$ with $\lim x_n = 0 = \lim y_n$ such that $\lim f(x_n)$ and $\lim f(y_n)$ exist but are unequal.

Additional exercises.

1. Prove directly from the ε - δ definition that $f(x) = 3x - 7$ is continuous at every $c \in \mathbb{R}$, and that $g(x) = x^2$ is continuous at $c = 4$.
2. Show the function $f(x) = x$ for $x \leq 0$, $f(x) = x + 1$ for $x > 0$ has a jump at 0 and is continuous elsewhere; identify $f(0^-)$, $f(0^+)$, $f(0)$.
3. (Topological characterization.) For $f : \mathbb{R} \rightarrow \mathbb{R}$ continuous, prove $\{x : 1 < f(x) < 3\}$ is open and $\{x : f(x) \geq 2\}$ is closed, citing the open/closed pre-image theorem.
4. (Non-example / disprove.) Disprove: “a continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ maps open sets to open sets.” Use $f(x) = x^2$ and the open set $(-1, 1)$.
5. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and agree on $\mathbb{Q}$. Prove $f = g$ on $\mathbb{R}$. Where is density of $\mathbb{Q}$ used?
6. Show that $f(x) = xD(x)$, where D is the Dirichlet function, is continuous only at $x = 0$, and prove it via the discontinuity criterion.
7. (Diagram.) Illustrate the open-set definition of continuity at a point p : draw the output ball $B_\varepsilon(f(p))$, its pre-image, and an input ball $B_\delta(p)$ contained in that pre-image.
8. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $f(x + y) = f(x)f(y)$ for all x, y , with f not identically 0. Prove $f(x) > 0$ for all x and that f is determined by $f(1)$.

5.4 Combinations of Continuous Functions

Arithmetic combinations

Continuity is preserved by all the pointwise algebra, by direct transcription of the limit theorems.

Theorem 5.4.1 (Algebra of continuous functions). *If $f, g : A \rightarrow \mathbb{R}$ are continuous at $c \in A$ and $k \in \mathbb{R}$, then $f + g$, $f - g$, kf , fg , and $|f|$ are continuous at c ; and f/g is continuous at c provided $g(c) \neq 0$.*

Proof. If c is isolated, all are continuous trivially (Theorem 5.3.2). Otherwise c is a cluster point and $\lim_{x \rightarrow c} f = f(c)$, $\lim_{x \rightarrow c} g = g(c)$; apply the limit theorems (Theorem 5.2.2), e.g. $\lim_{x \rightarrow c} (fg) = f(c)g(c) = (fg)(c)$. For $|f|$ use the reverse triangle inequality $||f(x)| - |f(c)|| \leq |f(x) - f(c)|$. For f/g , $g(c) \neq 0$ keeps the denominator away from 0 near c . $\square$

Corollary 5.4.2. *Every polynomial $p(x) = a_n x^n + \cdots + a_0$ is continuous on $\mathbb{R}$, and every rational function p/q is continuous at every point where $q \neq 0$.*

Proof. The constant maps and the identity $x \mapsto x$ are continuous (take $\delta = \varepsilon$); close under $+$ and $\cdot$ by Theorem 5.4.1 to reach all polynomials, then take a quotient. $\square$

Worked Example 5.4.3 (Max and min of continuous functions). For f, g continuous at c , show $h = \max\{f, g\}$ and $\min\{f, g\}$ are continuous at c .

Recognition. “Show max / min continuous”; rewrite with $|\cdot|$ to reuse the algebra.

Solution. Use $\max\{f, g\} = \frac{1}{2}(f + g) + \frac{1}{2}|f - g|$ and $\min\{f, g\} = \frac{1}{2}(f + g) - \frac{1}{2}|f - g|$. Each is a sum of continuous functions (using $|f - g|$ continuous), so each is continuous at c . *Check.* At a crossing point $f(c) = g(c)$ both formulas give $f(c)$, with no corner-induced break — $|\cdot|$ is continuous even at its kink. $\blacksquare$

Composition

Theorem 5.4.4 (Composition of continuous functions). *Let $f : A \rightarrow \mathbb{R}$ with $f(A) \subseteq B$, and $g : B \rightarrow \mathbb{R}$. If f is continuous at $c \in A$ and g is continuous at $b = f(c)$, then $g \circ f$ is continuous at c . Hence a composition of continuous functions is continuous.*

Proof. Let $\varepsilon > 0$. By continuity of g at b there is $\eta > 0$ with $|y - b| < \eta$, $y \in B \implies |g(y) - g(b)| < \varepsilon$. By continuity of f at c there is $\delta > 0$ with $|x - c| < \delta$, $x \in A \implies |f(x) - b| < \eta$. Chaining: $|x - c| < \delta \implies |f(x) - b| < \eta \implies |g(f(x)) - g(b)| < \varepsilon$. $\square$

Remark (The open-set proof, in one line). Topologically (Theorem 5.3.7) this is immediate: $(g \circ f)^{-1}(G) = f^{-1}(g^{-1}(G))$ is the pre-image of an open set under f of an open set under g , hence open. The ε - δ version above shows the explicit δ .

Caution. Composition needs g continuous at the limit value $b = f(c)$, not merely the existence of $\lim_{x \rightarrow c} f$. The substitution $\lim_{x \rightarrow c} g(f(x)) = g(\lim_{x \rightarrow c} f(x))$ can fail if g is discontinuous at b . (B&S 5.2.6 Ex. 5: let $f(x) = x + 1$, and g with $g(1) = 0$ but $g = 2$ elsewhere; then $f(x) \rightarrow 1$ as $x \rightarrow 0$, yet $g(f(x)) \rightarrow 2 \neq 0 = g(1)$.) The inner limit existing is not enough — the outer map must be continuous there.

Worked Example 5.4.5 (Building continuity from pieces). Show $x \mapsto \sqrt{x^2 + 1}$ and $x \mapsto \frac{x^2 + 2x + 1}{x^2 + 1}$ are continuous on $\mathbb{R}$.

Recognition. “Determine where continuous”; decompose into known-continuous blocks.

Solution. $x \mapsto x^2 + 1$ is a polynomial (continuous, and $\geq 1 > 0$); $\sqrt{\cdot}$ is continuous on $[0, \infty)$; the composite $\sqrt{x^2 + 1}$ is continuous by Theorem 5.4.4. The rational function has nonvanishing denominator $x^2 + 1 \geq 1$, so it is continuous on all of $\mathbb{R}$ by Corollary 5.4.2. *Check.* Both are defined and finite for every real x , consistent with continuity everywhere. $\blacksquare$

Worked Example 5.4.6 (Additive functions: continuity at one point $\implies$ everywhere). A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *additive* if $f(x + y) = f(x) + f(y)$ for all x, y . Show that if f is continuous at a single point x_0 , it is continuous on $\mathbb{R}$, and in fact $f(x) = cx$ with $c = f(1)$.

Recognition. “Promote local continuity to global”; exploit the functional equation to slide the point of continuity.

Solution. Additivity gives $f(0) = 0$ and $f(x) - f(c) = f(x - c)$. Fix any c and let $x \rightarrow c$; put $h = x - c \rightarrow 0$. Then $f(x) - f(c) = f(h)$, so continuity of f at c is equivalent to continuity at 0, which (by translating x_0) follows from continuity at x_0 : $f(h) = f(x_0 + h) - f(x_0) \rightarrow 0$ as $h \rightarrow 0$. Thus f is continuous everywhere. Then $f(r) = rf(1)$ for $r \in \mathbb{Q}$ (additivity over integer multiples and fractions), and continuity extends this from the dense set $\mathbb{Q}$ to $f(x) = cx$ for all x . *Check.* The only “wild” (non-linear) additive functions require a Hamel basis and are nowhere continuous — one point of continuity rules them out. ■

Further worked examples

Worked Example 5.4.7 (A rational function: where exactly is it continuous?). Determine the largest set

on which $r(x) = \frac{x^2 - 1}{x^2 - x - 6}$ is continuous.

Recognition. “Determine where continuous”: a rational function is continuous off the zero set of its denominator (Corollary 5.4.2).

Solution. Factor the denominator: $x^2 - x - 6 = (x - 3)(x + 2)$, which vanishes at $x = 3$ and $x = -2$. Neither cancels the numerator $x^2 - 1 = (x - 1)(x + 1)$, so both are genuine poles. By Corollary 5.4.2, r is continuous at every point with $(x - 3)(x + 2) \neq 0$, i.e. on $\mathbb{R} \setminus \{-2, 3\}$. *Check.* At $x = 0$ the denominator is $-6 \neq 0$ and $r(0) = \frac{-1}{-6} = \frac{1}{6}$, a finite value, consistent with continuity there; near $x = 3$, $r(3.01) \approx \frac{8.06}{0.0501} \approx 161$ blows up. ■

Worked Example 5.4.8 (A non-example: sum of two discontinuous functions can be continuous). Show the algebra theorem is one-directional: $f + g$ continuous does *not* require f, g continuous.

Recognition. “Test whether the converse holds”: hunt for a counterexample, since Theorem 5.4.1 only goes one way.

Solution. Let D be the Dirichlet function ($D = 1$ on $\mathbb{Q}$, 0 off $\mathbb{Q}$), discontinuous everywhere (§5.3). Put $f = D$ and $g = 1 - D$; both are discontinuous at every point (the same two-sequence argument). Yet $f + g \equiv 1$ is continuous on all of $\mathbb{R}$. *Check.* The algebra theorem gives a sufficient condition only; cancellation of two “bad” functions can restore continuity, so you may never read it backwards. ■

Worked Example 5.4.9 (Composition order matters: $g \circ f$ vs. $f \circ g$). With $f(x) = x^2$ and $g(x) = \sqrt{x}$ (domain $[0, \infty)$), discuss the continuity and domain of $g \circ f$ and of $f \circ g$.

Recognition. “Continuity of a composite”: check that each inner output lands in the outer domain, then apply Theorem 5.4.4.

Solution. $g \circ f(x) = \sqrt{x^2} = |x|$: here f is continuous on $\mathbb{R}$ and $f(x) = x^2 \geq 0$ always lands in the domain $[0, \infty)$ of g , so $g \circ f$ is continuous on all of $\mathbb{R}$ (it equals $|\cdot|$). $f \circ g(x) = (\sqrt{x})^2 = x$: now g has domain $[0, \infty)$, so $f \circ g$ is only defined — and continuous — on $[0, \infty)$. *Check.* The two composites agree as formulas ($|x| = x$) precisely on the common domain $[0, \infty)$, but $g \circ f$ extends continuously to negatives while $f \circ g$ cannot. ■

Exercises (Bartle & Sherbert, §5.2).

- Determine the points of continuity, stating which theorems are used: (a) $f(x) = \frac{x^2 + 2x + 1}{x^2 + 1}$; (b) $g(x) = \sqrt{x + \sqrt{x}}$ ($x \geq 0$); (c) $h(x) = \frac{\sqrt{1 + |\sin x|}}{x}$ ($x \neq 0$); (d) $k(x) = \cos \sqrt{1 + x^2}$.
- Show that if $f : A \rightarrow \mathbb{R}$ is continuous on A and $n \in \mathbb{N}$, then f^n (with $f^n(x) = (f(x))^n$) is continuous on A .
- Give an example of f, g both discontinuous at a point c such that (a) $f + g$ is continuous at c ; (b) fg is continuous at c .
- Let $\lfloor \cdot \rfloor$ be the greatest-integer function. Determine the points of continuity of $f(x) = x - \lfloor x \rfloor$.

5. Let $g(1) = 0$, $g(x) = 2$ for $x \neq 1$, and $f(x) = x + 1$. Show $\lim_{x \rightarrow 0} g \circ f \neq (g \circ f)(0)$. Why does this not contradict the composition theorem?
6. Let f, g on $\mathbb{R}$, $c \in \mathbb{R}$. Suppose $\lim_{x \rightarrow c} f = b$ and g is continuous at b . Show $\lim_{x \rightarrow c} g \circ f = g(b)$.
7. Give an example of $f : [0, 1] \rightarrow \mathbb{R}$ discontinuous at every point of $[0, 1]$ but with $|f|$ continuous on $[0, 1]$.
8. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $f(r) = g(r)$ for all rational r . Is $f(x) = g(x)$ for all x ?
9. Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $h(m/2^n) = 0$ for all $m \in \mathbb{Z}$, $n \in \mathbb{N}$. Show $h(x) = 0$ for all x .
10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and $P = \{x : f(x) > 0\}$. If $c \in P$, show there is a neighborhood $V_\delta(c) \subseteq P$.
11. If f, g are continuous on $\mathbb{R}$, let $S = \{x : f(x) \geq g(x)\}$. If $(s_n) \subseteq S$ and $\lim s_n = s$, show $s \in S$.
12. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *additive* if $f(x + y) = f(x) + f(y)$ for all x, y . Prove that if f is continuous at some point x_0 , it is continuous at every point of $\mathbb{R}$.
13. Suppose f is a continuous additive function on $\mathbb{R}$. If $c = f(1)$, show $f(x) = cx$ for all x . (Hint: first show $f(r) = cr$ for rational r .)
14. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $g(x + y) = g(x)g(y)$ for all x, y . Show that if g is continuous at 0, it is continuous everywhere; and if $g(a) = 0$ for some a , then $g \equiv 0$.
15. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous at c and $h(x) = \sup\{f(x), g(x)\}$. Show $h(x) = \frac{1}{2}(f(x) + g(x)) + \frac{1}{2}|f(x) - g(x)|$, and use this to show h is continuous at c .

Additional exercises.

1. Determine all points of continuity of $f(x) = \frac{\sin x}{x}$ (with $f(0) = 1$) and of $g(x) = \frac{1}{x^2 - x - 2}$, citing the relevant theorems.
2. Prove that if f is continuous at c then so is f^2 ; show the converse fails by an example where f^2 is continuous but f is not.
3. (Min/max.) Show $\min\{f, g\}$ and $\max\{f, g\}$ are continuous whenever f, g are, using the absolute-value identity, and sketch the graphs for $f(x) = x$, $g(x) = 1 - x$.
4. Let f, g be continuous on $\mathbb{R}$. Prove $\{x : f(x) = g(x)\}$ is closed and $\{x : f(x) \neq g(x)\}$ is open.
5. (Non-example / disprove.) Find f, g each discontinuous at 0 with $f + g$ continuous at 0; then find f, g each discontinuous at 0 with fg continuous at 0.
6. (Composition trap.) Give explicit f continuous at c and g discontinuous at $b = f(c)$ such that $g \circ f$ is discontinuous at c , and explain why Theorem 5.4.4 does not apply.
7. (Diagram.) For $f(x) = x$ and $g(x) = 1 - x$ on $[0, 1]$, draw $\max\{f, g\}$ as the upper envelope, marking the corner at $x = \frac{1}{2}$ where the two graphs cross, and confirm continuity there.
8. Prove that a continuous additive function $f : \mathbb{R} \rightarrow \mathbb{R}$ is linear, $f(x) = f(1)x$, filling in the step from $\mathbb{Q}$ to $\mathbb{R}$ by continuity.

5.5 Continuity and Compactness; the Extreme Value Theorem

The continuous image of a compact set is compact

Recall (Chapter 3) that a set K in a metric space is *compact* iff every open cover has a finite subcover, equivalently (Heine–Borel, in $\mathbb{R}^k$) iff K is closed and bounded, equivalently (sequential compactness) iff every sequence in K has a subsequence converging to a point of K . Continuity propagates compactness forward.

Theorem 5.5.1 (Continuous image of compact is compact). *Let $f : X \rightarrow Y$ be continuous and $K \subseteq X$ compact. Then $f(K)$ is compact.*

Proof via open covers (Rudin). Let $\{G_\alpha\}$ be an open cover of $f(K)$ in Y . Each $f^{-1}(G_\alpha)$ is open in X (Theorem 5.3.7) and these cover K , since every $p \in K$ has $f(p)$ in some G_α . By compactness of K finitely many $f^{-1}(G_{\alpha_1}), \dots, f^{-1}(G_{\alpha_n})$ cover K ; then $G_{\alpha_1}, \dots, G_{\alpha_n}$ cover $f(K)$. So $f(K)$ is compact. $\square$

Proof via sequences (B&S). Let (y_n) be a sequence in $f(K)$, say $y_n = f(x_n)$ with $x_n \in K$. By sequential compactness a subsequence $x_{n_k} \rightarrow x \in K$; by continuity $y_{n_k} = f(x_{n_k}) \rightarrow f(x) \in f(K)$. Every sequence in $f(K)$ has a subsequence converging in $f(K)$, so $f(K)$ is compact. $\square$

Corollary 5.5.2 (Boundedness theorem). *If $f : K \rightarrow \mathbb{R}$ is continuous on a compact set K , then f is bounded: $f(K)$ is a bounded subset of $\mathbb{R}$ (indeed closed and bounded).*

Proof. $f(K)$ is compact in $\mathbb{R}$, hence closed and bounded by Heine–Borel. $\square$

The extreme value theorem

Theorem 5.5.3 (Extreme Value Theorem). *If K is compact and $f : K \rightarrow \mathbb{R}$ is continuous, then f attains a maximum and a minimum: there exist $p, q \in K$ with*

$$f(p) = \sup_{x \in K} f(x) = \max f(K), \quad f(q) = \inf_{x \in K} f(x) = \min f(K).$$

In particular this holds for continuous f on a closed bounded interval $[a, b]$.

Proof. $M = \sup f(K)$ is finite (Corollary 5.5.2). Since $f(K)$ is closed (Theorem 5.5.1) and M is a limit point of $f(K)$ (being its sup), $M \in f(K)$; so $M = f(p)$ for some $p \in K$. Likewise $m = \inf f(K) = f(q)$. (Sequential phrasing: choose $x_n \in K$ with $f(x_n) \rightarrow M$; a subsequence $x_{n_k} \rightarrow p \in K$ gives $f(p) = \lim f(x_{n_k}) = M$.) $\square$

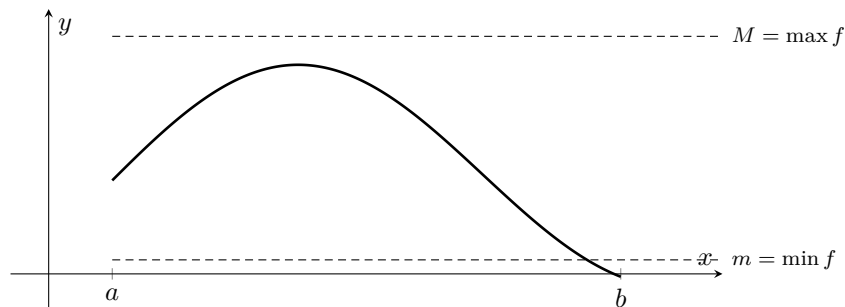


Figure 5.5. EVT: a continuous f on $[a, b]$ achieves its sup and inf at actual points $p, q \in [a, b]$ — the dashed levels are touched, not merely approached.

Caution. Every hypothesis is needed. Drop compactness and EVT fails: $f(x) = x$ on the open $(0, 1)$ has $\sup = 1$, $\inf = 0$, neither attained; $f(x) = 1/x$ on $(0, 1]$ is unbounded above. Drop continuity and it fails: $f(x) = x$ on $[0, 1)$ with f redefined cannot reach its sup. “sup” is a least upper bound, possibly outside the set; “max” is an attained value — EVT is exactly the statement that for continuous f on compact K they coincide.

Continuous inverse on a compact domain

A continuous bijection need not have a continuous inverse in general — but compactness of the domain fixes this.

Theorem 5.5.4 (Continuous inverse). *Let K be compact and $f : K \rightarrow Y$ a continuous bijection onto $f(K)$. Then the inverse $f^{-1} : f(K) \rightarrow K$ is continuous. (Such an f is a homeomorphism.)*

Proof. By the closed-set criterion (Theorem 5.3.7(iii)) applied to $g = f^{-1}$, it suffices to show $g^{-1}(C) = f(C)$ is closed in $f(K)$ for every closed $C \subseteq K$. But a closed subset C of the compact K is compact, so $f(C)$ is compact (Theorem 5.5.1), hence closed. Therefore f^{-1} is continuous. $\square$

Worked Example 5.5.5 (Why compactness is essential for the inverse). Exhibit a continuous bijection with discontinuous inverse.

Recognition. “Show the compactness hypothesis cannot be dropped”; a wrap-around onto a circle is the canonical example (Rudin 4.21).

Solution. Let $X = [0, 2\pi)$ (not compact) and $f(t) = (\cos t, \sin t)$ onto the unit circle. f is a continuous bijection, but f^{-1} is discontinuous at $(1, 0)$: points just *below* the seam map to t near 2π , points at the seam map to $t = 0$, so f^{-1} jumps by nearly 2π . *Check.* The domain misses its endpoint 2π , so it is not closed, hence not compact — exactly the gap Theorem 5.5.4 forbids. ■

Worked Example 5.5.6 (n -th roots are continuous). Show $x \mapsto x^{1/n}$ is continuous on $[0, \infty)$ for each $n \in \mathbb{N}$.

Recognition. “Show a root function continuous”; use the continuous-inverse theorem on compacta.

Solution. $\phi(x) = x^n$ is a continuous strictly increasing bijection of $[0, b]$ onto $[0, b^n]$ for each $b > 0$; $[0, b]$ is compact, so $\phi^{-1}(y) = y^{1/n}$ is continuous on $[0, b^n]$ by Theorem 5.5.4. As b is arbitrary, $y^{1/n}$ is continuous on $[0, \infty)$. *Check.* Agrees with the direct estimate for $\sqrt{\cdot}$ in §5.3, now for every n . ■

Further worked examples

Worked Example 5.5.7 (EVT in action: locating the extrema of a concrete function). Find the maximum and minimum of $f(x) = x^3 - 3x$ on $[0, 2]$, and confirm both are attained.

Recognition. “Continuous f on a compact interval”: EVT (Theorem 5.5.3) guarantees the extrema exist and are attained; locate them among endpoints and interior critical points.

Solution. f is a polynomial, hence continuous, and $[0, 2]$ is compact, so EVT applies. The candidates are the endpoints $x = 0, 2$ and the interior points where the difference quotient flattens, i.e. where $3x^2 - 3 = 0$, giving $x = 1$ (the root $x = -1$ is outside $[0, 2]$). Evaluate: $f(0) = 0$, $f(1) = 1 - 3 = -2$, $f(2) = 8 - 6 = 2$. So $\min f = -2$ attained at $p = 1$ and $\max f = 2$ attained at $q = 2$. *Check.* Both values are achieved at genuine points of $[0, 2]$, as EVT promises; the sup 2 is a max, not merely a least upper bound. ■

Worked Example 5.5.8 (A non-example: continuity alone (no compactness) loses the max). Show $f(x) = \frac{x}{1+x}$ on $[0, \infty)$ is continuous and bounded but attains no maximum.

Recognition. “Does EVT-style attainment hold here?”: the domain is closed but *unbounded*, hence not compact, so attainment is not guaranteed.

Solution. f is a quotient with nonvanishing denominator $1+x \geq 1$, so it is continuous on $[0, \infty)$. Since $f(x) = 1 - \frac{1}{1+x}$, we have $0 \leq f(x) < 1$ and $f(x) \rightarrow 1$ as $x \rightarrow \infty$, so $\sup_{x \geq 0} f = 1$. But $f(x) = 1$ would force $\frac{1}{1+x} = 0$, impossible; the supremum is never attained. *Check.* Every hypothesis of EVT but compactness holds, and the conclusion fails — the missing “endpoint at infinity” is exactly where the sup lives. ■

Worked Example 5.5.9 (Closed but not bounded, and bounded but not closed). Give two continuous functions, one on a closed unbounded set and one on a bounded non-closed set, each failing to attain an extremum, isolating which half of Heine–Borel fails.

Recognition. “Pin the blame on closedness vs. boundedness”: drop one Heine–Borel condition at a time and watch attainment break.

Solution. (i) On the closed unbounded $K_1 = [1, \infty)$, $f(x) = -1/x$ is continuous with $\sup f = 0$ approached as $x \rightarrow \infty$ but never reached; K_1 is closed but not bounded. (ii) On the bounded non-closed $K_2 = (0, 1]$, $g(x) = x$ has $\inf g = 0$ approached as $x \rightarrow 0^+$ but $0 \notin K_2$, so the inf is not attained; K_2 is bounded but not closed. *Check.* Each example violates exactly one clause of Heine–Borel (closed *and* bounded), and in each case attainment fails — both clauses are essential to compactness. ■

Worked Example 5.5.10 (Continuity is essential too: a bounded discontinuous map with no max). On the compact $[0, 1]$ exhibit a bounded function attaining no maximum, showing EVT needs continuity not just compact domain.

Recognition. “Which hypothesis fails now?”: keep $K = [0, 1]$ compact but break continuity, and attainment can vanish.

Solution. Define $f(x) = x$ for $0 \leq x < 1$ and $f(1) = 0$. Then f is bounded with $\sup_{[0,1]} f = 1$ (approached as $x \rightarrow 1^-$), but f never equals 1: the only candidate $x = 1$ has been redefined to 0. The single discontinuity at $x = 1$ destroys attainment. *Check.* The domain is compact yet the conclusion fails, so EVT truly requires continuity; reinstating $f(1) = 1$ (continuity) restores the attained maximum. ■

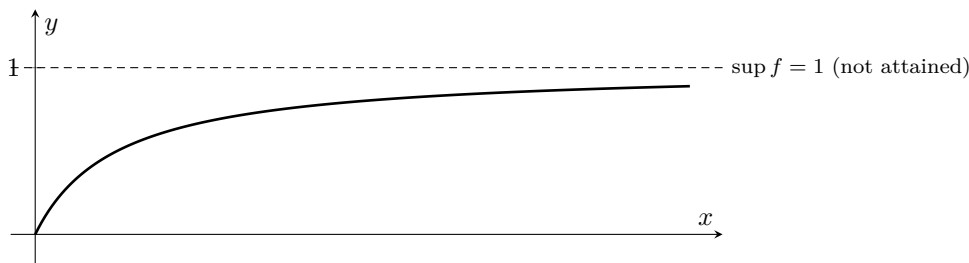


Figure 5.6. Without compactness EVT fails: $f(x) = x/(1+x)$ on $[0, \infty)$ is continuous and bounded, climbing toward the dashed level 1 but never reaching it — the “maximizing point” has escaped to infinity.

Exercises (Rudin, Ch.4).

1. Let $I = [0, 1]$ be the closed unit interval and $f : I \rightarrow I$ continuous. Prove $f(x) = x$ for at least one $x \in I$ (a fixed point).
2. Call a map $f : \mathbb{R} \rightarrow \mathbb{R}$ *open* if $f(V)$ is open whenever V is open. Prove every continuous open map $f : \mathbb{R} \rightarrow \mathbb{R}$ is monotonic.
3. If E is a nonempty subset of a metric space X , define the distance from $x \in X$ to E by $\rho_E(x) = \inf_{z \in E} d(x, z)$. (a) Prove $\rho_E(x) = 0$ iff $x \in \overline{E}$. (b) Prove ρ_E is uniformly continuous on X , by showing $|\rho_E(x) - \rho_E(y)| \leq d(x, y)$ for all $x, y \in X$.
4. Suppose K and F are disjoint sets in a metric space X , K compact, F closed. Prove there exists $\delta > 0$ such that $d(p, q) > \delta$ whenever $p \in K$, $q \in F$. (Hint: ρ_F is a continuous positive function on K .) Show the conclusion may fail for two disjoint *closed* sets if neither is compact.
5. Let A, B be disjoint nonempty closed sets in a metric space X , and define $f(p) = \frac{\rho_A(p)}{\rho_A(p) + \rho_B(p)}$ for $p \in X$. Show f is continuous, its range lies in $[0, 1]$, $f(p) = 0$ exactly on A and $f(p) = 1$ exactly on B . (Thus every closed $A \subseteq X$ is $Z(f)$ for some continuous f .) Setting $V = f^{-1}([0, \frac{1}{2}))$ and $W = f^{-1}((\frac{1}{2}, 1])$, show V, W are disjoint open sets with $A \subseteq V$, $B \subseteq W$ (*normality* of metric spaces).
6. For $A \subseteq \mathbb{R}^n$, $B \subseteq \mathbb{R}^k$, let $A + B = \{x + y : x \in A, y \in B\}$. (a) If K is compact and C is closed in $\mathbb{R}^k$, prove $K + C$ is closed. (Hint: take $z \notin K + C$, put $F = z - C$; then K, F are disjoint, choose δ as in the compact/closed separation; show the open ball of radius δ about z misses $K + C$.) (b) Let α be irrational, $C_1 = \mathbb{Z}$, $C_2 = \{n\alpha : n \in \mathbb{Z}\}$. Show C_1, C_2 are closed in $\mathbb{R}$ but $C_1 + C_2$ is a countable dense subset of $\mathbb{R}$, hence not closed.

Exercises (Bartle & Sherbert, §5.3 — boundedness and maxima).

1. Let $I = [a, b]$ and $f : I \rightarrow \mathbb{R}$ continuous with $f(x) > 0$ for each $x \in I$. Prove there exists $\alpha > 0$ with $f(x) \geq \alpha$ for all $x \in I$.
2. Let $I = [a, b]$ and $f, g : I \rightarrow \mathbb{R}$ continuous. Show $E = \{x \in I : f(x) = g(x)\}$ has the property: if $(x_n) \subseteq E$ and $x_n \rightarrow x_0$, then $x_0 \in E$.
3. Let $I = [a, b]$ and $f : I \rightarrow \mathbb{R}$ continuous such that for each $x \in I$ there is $y \in I$ with $|f(y)| \leq \frac{1}{2} |f(x)|$. Prove there is $c \in I$ with $f(c) = 0$.
4. Let $I = [0, \pi/2]$ and $f : I \rightarrow \mathbb{R}$, $f(x) = \sup\{x^2, \cos x\}$. Show there is an absolute minimum point $x_0 \in I$ for f , and that x_0 solves $\cos x = x^2$.
5. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous with $\lim_{x \rightarrow \infty} f = 0$ and $\lim_{x \rightarrow -\infty} f = 0$. Prove f is bounded on $\mathbb{R}$ and attains either a maximum or a minimum. Give an example where both are not attained.

6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, $b \in \mathbb{R}$. Show that if x_0 has $f(x_0) < b$, then there is a δ -neighborhood U of x_0 with $f(x) < b$ for all $x \in U$.
7. Examine which open (resp. closed) intervals are mapped by $f(x) = x^2$ onto open (resp. closed) intervals.
8. Examine the mapping of open (resp. closed) intervals under $g(x) = 1/(x^2 + 1)$ and $h(x) = x^3$.
9. Let $I = [a, b]$ and $f : I \rightarrow \mathbb{R}$ a (not necessarily continuous) function such that for every $x \in I$, f is bounded on some neighborhood $V_{\delta_x}(x)$. Prove f is bounded on I .
10. Let $J = (a, b)$ and $g : J \rightarrow \mathbb{R}$ continuous such that for every $x \in J$, g is bounded on some neighborhood $V_{\delta_x}(x)$. Show by example that g need not be bounded on J .

Additional exercises.

1. Prove that a continuous $f : [a, b] \rightarrow \mathbb{R}$ is bounded, directly from the open-cover definition of compactness (no sequences).
2. Show $f(x) = x^3 - 3x$ attains a maximum and a minimum on $[-2, 2]$, and find them.
3. (Non-example / disprove.) Disprove “a continuous function on a bounded interval attains its supremum” by exhibiting a continuous f on the bounded $(0, 1)$ with $\sup f$ not attained, and identify exactly which EVT hypothesis fails.
4. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous with $f([0, 1]) \subseteq [0, 1]$. Use EVT and IVT to show f has a fixed point, and contrast which theorem supplies which part.
5. Let $K \subseteq \mathbb{R}$ be compact and $f : K \rightarrow \mathbb{R}$ continuous and injective. Prove f is strictly monotone on K if K is an interval, and that f^{-1} is continuous in any case.
6. Prove that a continuous bijection $f : [0, 1] \rightarrow [0, 1]$ has a continuous inverse, but that this fails for a continuous bijection $[0, 1] \rightarrow S^1$ (the unit circle).
7. (Diagram.) Sketch a continuous f on $[a, b]$ with an interior maximum and a boundary minimum, drawing the horizontal lines $y = \max f$ and $y = \min f$ tangent to the graph at the attaining points p, q .
8. Show that the image $f([a, b])$ of a continuous f is a closed bounded interval $[m, M]$, and that $m = \min f$, $M = \max f$. (Combine compactness with connectedness.)

5.6 Continuity and Connectedness; the Intermediate Value Theorem

Connectedness and the continuous image

Recall that a metric space E is *connected* if it cannot be split as $E = A \cup B$ with A, B nonempty, disjoint, and *separated* (neither meeting the other's closure). In $\mathbb{R}$ the connected sets are exactly the *intervals* (including rays and $\mathbb{R}$ itself): $E \subseteq \mathbb{R}$ is connected iff $x, y \in E$ and $x < z < y$ imply $z \in E$.

Theorem 5.6.1 (Continuous image of connected is connected). *If $f : X \rightarrow Y$ is continuous and $E \subseteq X$ is connected, then $f(E)$ is connected.*

Proof. Suppose not: $f(E) = A \cup B$ with A, B nonempty separated sets in Y . Put $G = E \cap f^{-1}(A)$ and $H = E \cap f^{-1}(B)$; these are nonempty, cover E , and are disjoint. One checks they are separated: if $p \in G$ were a limit of points of H , continuity would force $f(p) \in A$ to be a limit of points of B , contradicting separation of A, B (using $f(\overline{G}) \subseteq \overline{f(G)}$, Theorem 5.3.7(iv)). So $E = G \cup H$ is a separation of E — contradicting connectedness. Hence $f(E)$ is connected. $\square$

The intermediate value theorem

Specializing the codomain to $\mathbb{R}$, “connected image” becomes “no values are skipped”.

Theorem 5.6.2 (Intermediate Value Theorem; Bolzano). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, and let y lie between $f(a)$ and $f(b)$ (i.e. $f(a) < y < f(b)$ or $f(b) < y < f(a)$). Then there is $c \in (a, b)$ with $f(c) = y$. Equivalently, a continuous f maps an interval onto an interval.*

Topological proof. $[a, b]$ is connected, so $f([a, b])$ is connected (Theorem 5.6.1), hence an interval; it contains $f(a)$ and $f(b)$, so it contains every y between them. Thus $y = f(c)$ for some $c \in [a, b]$, and $c \neq a, b$ since $y \neq f(a), f(b)$. $\square$

Bisection / sup proof (B&S, constructive). Take the case $f(a) < y < f(b)$; replacing f by $f - y$ assume $y = 0$, so $f(a) < 0 < f(b)$. Let $W = \{x \in [a, b] : f(x) < 0\}$; it is nonempty ($a \in W$) and bounded, so $w = \sup W$ exists in $[a, b]$. We claim $f(w) = 0$. If $f(w) < 0$ then by continuity $f < 0$ on a neighborhood of w , so points $> w$ lie in W , contradicting $w = \sup W$. If $f(w) > 0$ then $f > 0$ on a neighborhood of w , so w is not a limit of W , again a contradiction. Hence $f(w) = 0$, with $w \in (a, b)$. $\square$

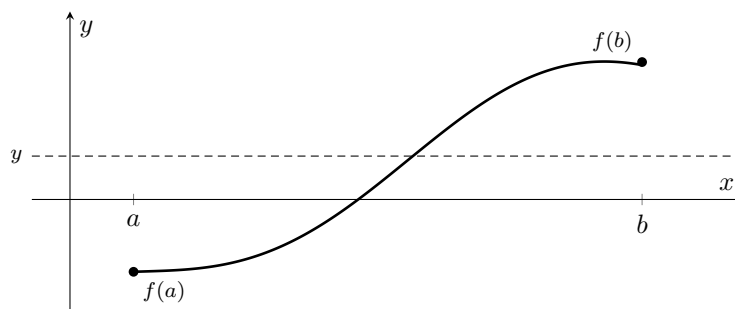


Figure 5.7. IVT: a continuous graph joining $f(a)$ below the level y to $f(b)$ above it must cross y — it cannot “teleport” over the dashed line.

Corollary 5.6.3 (Bolzano root theorem). *If $f : [a, b] \rightarrow \mathbb{R}$ is continuous with $f(a)f(b) < 0$ (a sign change), then f has a root in (a, b) .*

Corollary 5.6.4 (Odd-degree polynomials have a real root). *Every real polynomial of odd degree has at least one real root.*

Proof. Write $p(x) = a_n x^n (1 + o(1))$ with n odd, $a_n \neq 0$. As $x \rightarrow +\infty$ and $x \rightarrow -\infty$, p takes values of opposite sign (odd power), so $p(a) < 0 < p(b)$ for suitable $a < b$; apply Bolzano. $\square$

Worked Example 5.6.5 (Solving $x = \cos x$ and a fixed point on $[0, 1]$). Show $x = \cos x$ has a solution in $[0, \pi/2]$, and any continuous $f : [0, 1] \rightarrow [0, 1]$ has a fixed point.

Recognition. “Show an equation has a solution”; build g whose root is the solution and apply IVT/Bolzano.

Solution. Let $g(x) = x - \cos x$ on $[0, \pi/2]$; $g(0) = -1 < 0$ and $g(\pi/2) = \pi/2 > 0$, so by Bolzano $g(c) = 0$, i.e. $c = \cos c$. For the fixed point, let $h(x) = f(x) - x$ on $[0, 1]$; $h(0) = f(0) \geq 0$ and $h(1) = f(1) - 1 \leq 0$, so by IVT $h(c) = 0$, i.e. $f(c) = c$. *Check.* Numerically $c \approx 0.739$ solves $x = \cos x$; the fixed-point claim is the 1-D Brouwer theorem. $\blacksquare$

Worked Example 5.6.6 (Equal antipodal temperatures). A continuous $f : [0, 1] \rightarrow \mathbb{R}$ with $f(0) = f(1)$ has a point $c \in [0, \frac{1}{2}]$ with $f(c) = f(c + \frac{1}{2})$.

Recognition. “Two points half a period apart share a value”; difference function plus IVT.

Solution. Let $g(x) = f(x) - f(x + \frac{1}{2})$ on $[0, \frac{1}{2}]$. Then $g(0) + g(\frac{1}{2}) = (f(0) - f(\frac{1}{2})) + (f(\frac{1}{2}) - f(1)) = f(0) - f(1) = 0$, so $g(0)$ and $g(\frac{1}{2})$ have opposite signs (or one is 0); by IVT $g(c) = 0$ for some $c \in [0, \frac{1}{2}]$. *Check.* Interpreting f as temperature along the equator (with $f(0) = f(1)$ the wrap-around), there are always antipodal points at equal temperature. $\blacksquare$

Caution. IVT gives *existence*, not uniqueness or a formula — and it needs genuine continuity on a genuine interval. On $[-1, 1] \setminus \{0\}$ the function sgn never takes the value 0 despite $\text{sgn}(-1) = -1$, $\text{sgn}(1) = 1$: the domain is disconnected. Also, a function can have the *intermediate value property* without being continuous (e.g. $\sin(1/x)$ extended by 0 at 0 hits every value in $[-1, 1]$ on any interval around 0 but is discontinuous there).

Further worked examples

Worked Example 5.6.7 (Bracketing a root and the bisection estimate). Show $p(x) = x^3 - x - 1$ has a root in $(1, 2)$, and bound how many bisections place the root within 10^{-2} .

Recognition. “Show a root exists, then estimate it”: Bolzano (Corollary 5.6.3) for existence, halving the bracket for the error bound.

Solution. p is continuous (a polynomial). $p(1) = 1 - 1 - 1 = -1 < 0$ and $p(2) = 8 - 2 - 1 = 5 > 0$, so $p(1)p(2) < 0$ and by Bolzano there is a root $c \in (1, 2)$. Bisection halves the bracket length $2 - 1 = 1$ each step, so after n steps the root is localized to an interval of length 2^{-n} . We need $2^{-n} < 10^{-2}$, i.e. $2^n > 100$, i.e. $n \geq 7$ (since $2^7 = 128$). *Check.* The true root is $c \approx 1.3247$; $p(1.3247) \approx -0.0001$, consistent with a root just above 1.32, comfortably inside the 7-step bracket. ■

Worked Example 5.6.8 (A non-example: IVT fails on a disconnected domain). Let $f(x) = 1/x$ on $A = [-1, 0) \cup (0, 1]$. Although $f(-1) = -1$ and $f(1) = 1$, show f skips the value 0, and explain why IVT is not violated.

Recognition. “Does a continuous map hit every intermediate value?”: IVT needs a genuine interval; test whether the domain is connected.

Solution. f is continuous on A (a rational function off its pole at 0). On $(0, 1]$ we have $f(x) = 1/x \geq 1$, and on $[-1, 0)$ we have $f(x) = 1/x \leq -1$, so f never takes any value in $(-1, 1)$, in particular not 0 — even though 0 lies between $f(-1) = -1$ and $f(1) = 1$. There is no contradiction: A is *disconnected* (the puncture at 0 splits it), so it is not an interval and Theorem 5.6.2 does not apply. *Check.* The image $f(A) = (-\infty, -1] \cup [1, \infty)$ is itself disconnected, exactly as Theorem 5.6.1 permits for a disconnected domain. ■

Worked Example 5.6.9 (Surjectivity onto $\mathbb{R}$ via IVT). Show every continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ with $\lim_{x \rightarrow -\infty} f = -\infty$ and $\lim_{x \rightarrow +\infty} f = +\infty$ is onto $\mathbb{R}$.

Recognition. “Show the range is all of $\mathbb{R}$ ”: use the limits to straddle any target value, then IVT to hit it.

Solution. Fix any $y \in \mathbb{R}$. Since $f \rightarrow -\infty$ at $-\infty$, choose a with $f(a) < y$; since $f \rightarrow +\infty$ at $+\infty$, choose $b > a$ with $f(b) > y$. On the interval $[a, b]$, f is continuous and $f(a) < y < f(b)$, so by IVT (Theorem 5.6.2) there is $c \in (a, b)$ with $f(c) = y$. As y was arbitrary, $f(\mathbb{R}) = \mathbb{R}$. *Check.* The odd-degree polynomial $f(x) = x^3 - x$ meets the hypotheses (its end behavior is $\pm\infty$), and indeed it is onto $\mathbb{R}$ — a quantitative companion to Corollary 5.6.4. ■

Exercises (Bartle & Sherbert, §5.3 — roots, IVT, bisection).

1. Show every polynomial of odd degree with real coefficients has at least one real root.
2. Show $p(x) = x^4 + 7x^3 - 9$ has at least two real roots. (Locate them to two decimals with a calculator.)
3. Let f be continuous on $[0, 1]$ with $f(0) = f(1)$. Prove there is $c \in [0, \frac{1}{2}]$ with $f(c) = f(c + \frac{1}{2})$. (Hint: $g(x) = f(x) - f(x + \frac{1}{2})$.) Conclude there are always antipodal points on the earth’s equator with the same temperature.
4. Show $x = \cos x$ has a solution in $[0, \pi/2]$. Use the Bisection Method and a calculator to approximate it with error $< 10^{-3}$.
5. Show $f(x) = 2 \ln x + \sqrt{x} - 2$ has a root in $[1, 2]$. Use the Bisection Method to find it with error $< 10^{-2}$.
6. (a) The function $f(x) = (x-1)(x-2)(x-3)(x-4)(x-5)$ has five roots in $[0, 7]$. If the Bisection Method is applied on $[0, 7]$, which root is located? (b) Same question for $g(x) = (x-2)(x-3)(x-4)(x-5)(x-6)$ on $[0, 7]$.

7. If the Bisection Method is used on an interval of length 1 to find p_n with error $|p_n - c| < 10^{-5}$, determine the least n assuring this accuracy.
8. Let $I = [a, b]$, $f : I \rightarrow \mathbb{R}$ continuous with $f(a) < 0 < f(b)$. Let $W = \{x \in I : f(x) < 0\}$ and $w = \sup W$. Prove $f(w) = 0$ (an alternative proof of IVT).
9. If $f : [0, 1] \rightarrow \mathbb{R}$ is continuous and takes only rational (resp. only irrational) values, must f be constant? Prove your assertion.

Additional exercises.

1. Use the IVT to show $x^5 - 3x + 1 = 0$ has a root in $(0, 1)$ and another in $(1, 2)$.
2. Prove every continuous $f : [0, 1] \rightarrow [0, 1]$ has a fixed point, and show by example that continuity is essential (a discontinuous self-map of $[0, 1]$ with no fixed point).
3. (Non-example / disprove.) Disprove “if f has the intermediate value property then f is continuous” by analyzing $f(x) = \sin(1/x)$ for $x \neq 0$, $f(0) = 0$.
4. Show that a continuous $f : [a, b] \rightarrow \mathbb{R}$ taking only integer values is constant. Which property of $[a, b]$ is doing the work?
5. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous with $f(0) < 0 < f(1)$. Describe the bisection algorithm and prove the produced sequence converges to a root, with error halving each step.
6. Prove that the continuous image of an interval is an interval, and deduce that $f(x) = x^2$ maps $(-1, 2)$ onto $[0, 4)$ — not an open interval. Explain.
7. (Diagram.) Draw a continuous graph from $(a, f(a))$ below a level $y = k$ to $(b, f(b))$ above it, mark a crossing point c with $f(c) = k$, and shade the horizontal strip the graph must pass through.
8. Prove the Bolzano sign-change theorem from the connectedness of $[a, b]$: if f is continuous and $f(a)f(b) < 0$, then $0 \in f([a, b])$.

5.7 Uniform Continuity and Gauges

Uniform continuity

Ordinary continuity allows δ to depend on the point c as well as on ε . *Uniform* continuity demands a single δ that works *simultaneously* for all points — the quantifier on the point moves *inside* the choice of δ .

Definition 5.7.1. $f : A \rightarrow \mathbb{R}$ is *uniformly continuous on A* if

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{such that} \quad x, u \in A, |x - u| < \delta \implies |f(x) - f(u)| < \varepsilon.$$

The structural contrast: pointwise continuity is $\forall c \forall \varepsilon \exists \delta$ (the δ may chase c); uniform continuity is $\forall \varepsilon \exists \delta \forall c$ (one δ fits all). Geometrically, a fixed $\delta \times \varepsilon$ “box” can slide along the graph and always contain it; where the graph steepens without bound, no fixed box survives.

Theorem 5.7.2 (Nonuniform continuity criterion). *f is not uniformly continuous on A iff there exist $\varepsilon_0 > 0$ and sequences $(x_n), (u_n)$ in A with $|x_n - u_n| \rightarrow 0$ but $|f(x_n) - f(u_n)| \geq \varepsilon_0$ for all n .*

Proof. Negate Definition 5.7.1: some ε_0 defeats every δ ; take $\delta = 1/n$ to get the pair x_n, u_n with $|x_n - u_n| < 1/n$ and $|f(x_n) - f(u_n)| \geq \varepsilon_0$. Conversely such sequences clearly block uniform continuity. $\square$

Worked Example 5.7.3 (A non-example: x^2 is not uniformly continuous on $\mathbb{R}$). Show $f(x) = x^2$ is continuous everywhere but not uniformly continuous on $\mathbb{R}$.

Recognition. “Show *not* uniformly continuous”; produce a closing pair with a fixed output gap.

Solution. Take $x_n = n + \frac{1}{n}$ and $u_n = n$. Then $|x_n - u_n| = \frac{1}{n} \rightarrow 0$, yet

$$|f(x_n) - f(u_n)| = \left(n + \frac{1}{n}\right)^2 - n^2 = 2 + \frac{1}{n^2} \geq 2 = \varepsilon_0.$$

By Theorem 5.7.2, x^2 is not uniformly continuous on $\mathbb{R}$. *Check.* The slope $2x$ is unbounded, so equal input gaps produce ever-larger output gaps — no single δ works. ■

Worked Example 5.7.4 (A non-example: $1/x$ on $(0, 1)$, and the cure on $[a, \infty)$). Show $f(x) = 1/x$ is not uniformly continuous on $(0, 1)$, but is on $[a, \infty)$ for $a > 0$.

Recognition. “Uniform continuity near a vertical asymptote”; failing pair near 0, Lipschitz bound away from 0.

Solution. On $(0, 1)$ take $x_n = \frac{1}{n}$, $u_n = \frac{1}{2n}$: $|x_n - u_n| = \frac{1}{2n} \rightarrow 0$ but $|f(x_n) - f(u_n)| = |n - 2n| = n \rightarrow \infty \geq 1$.

Not uniformly continuous. On $[a, \infty)$, $|\frac{1}{x} - \frac{1}{u}| = \frac{|x - u|}{xu} \leq \frac{1}{a^2} |x - u|$, so $\delta = a^2 \varepsilon$ works for all $x, u \geq a$.

Check. Uniform continuity is a property of f together with its domain; shrinking the domain away from the bad point restores it. ■

Compact domain forces uniform continuity

Theorem 5.7.5 (Heine–Cantor). *If K is compact and $f : K \rightarrow \mathbb{R}$ is continuous, then f is uniformly continuous on K .*

Sequential proof (B&S/Rudin Ex. 10). If not, by Theorem 5.7.2 there are ε_0 and $x_n, u_n \in K$ with $|x_n - u_n| \rightarrow 0$ and $|f(x_n) - f(u_n)| \geq \varepsilon_0$. By compactness a subsequence $x_{n_k} \rightarrow p \in K$; since $|x_{n_k} - u_{n_k}| \rightarrow 0$ also $u_{n_k} \rightarrow p$. Continuity gives $f(x_{n_k}) \rightarrow f(p)$ and $f(u_{n_k}) \rightarrow f(p)$. Hence $|f(x_{n_k}) - f(u_{n_k})| \rightarrow 0$, contradicting the bound $\geq \varepsilon_0$. □

Cover proof (Rudin). Given ε , each p has δ_p with $f(B_{\delta_p}(p)) \subseteq B_{\varepsilon/2}(f(p))$. The balls $B_{\delta_p/2}(p)$ cover K ; take a finite subcover $B_{\delta_{p_i}/2}(p_i)$ and set $\delta = \frac{1}{2} \min_i \delta_{p_i}$. If $|x - u| < \delta$, x lies in some $B_{\delta_{p_i}/2}(p_i)$, and then u lies in $B_{\delta_{p_i}}(p_i)$, so both $f(x), f(u) \in B_{\varepsilon/2}(f(p_i))$, giving $|f(x) - f(u)| < \varepsilon$. The δ in the cover proof is the *Lebesgue number* idea below. □

Corollary 5.7.6. (Rudin Ex. 9) *Continuity of f is uniform iff: $\forall \varepsilon > 0 \exists \delta > 0$ such that $\text{diam } f(E) < \varepsilon$ whenever $E \subseteq A$ has $\text{diam } E < \delta$. (Same statement, phrased with diameters of sets.)*

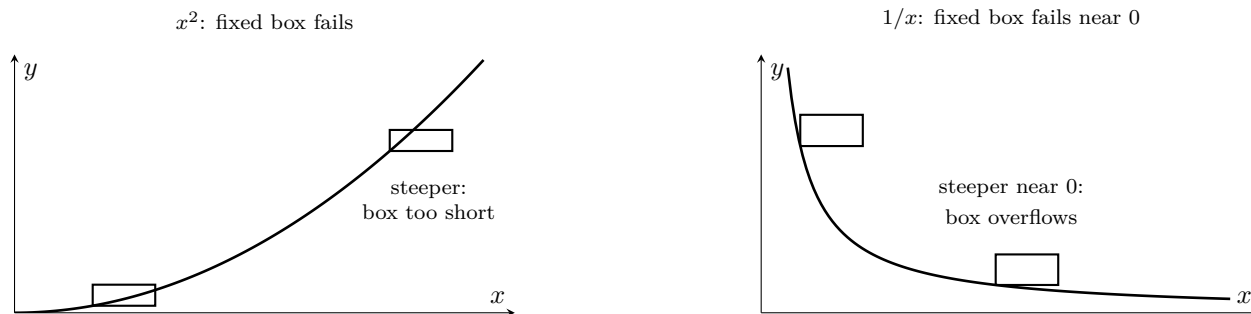


Figure 5.8. A fixed-width δ -box of height ε that traps the graph at gentle spots fails where the slope is unbounded (x^2 as $x \rightarrow \infty$, $1/x$ as $x \rightarrow 0^+$).

Lipschitz functions and the gauge viewpoint

Definition 5.7.7. $f : A \rightarrow \mathbb{R}$ is *Lipschitz* (with constant K) if $|f(x) - f(u)| \leq K|x - u|$ for all $x, u \in A$. Every Lipschitz function is uniformly continuous (take $\delta = \varepsilon/K$).

Caution. Lipschitz $\implies$ uniformly continuous $\implies$ continuous, and *none* of the arrows reverses. $\sqrt{x}$ on $[0, 1]$ is uniformly continuous (compact domain, Heine–Cantor) but *not* Lipschitz: $\frac{|\sqrt{x} - \sqrt{0}|}{|x - 0|} = \frac{1}{\sqrt{x}} \rightarrow \infty$ as $x \rightarrow 0^+$, so no K bounds the difference quotient.

Definition 5.7.8. A *gauge* on $[a, b]$ is a function $\delta : [a, b] \rightarrow (0, \infty)$ (a *positive* function, not a constant). A tagged partition $\dot{P} = \{([x_{i-1}, x_i], t_i)\}$ is δ -fine if $[x_{i-1}, x_i] \subseteq [t_i - \delta(t_i), t_i + \delta(t_i)]$ for each i .

Theorem 5.7.9 (Cousin’s lemma: δ -fine partitions exist). *For every gauge δ on $[a, b]$ there exists a δ -fine tagged partition of $[a, b]$.*

Bisection proof. Suppose not. Bisect $[a, b]$ at the midpoint; at least one half admits no δ -fine partition (else union the two, Ex. 5.5). Iterate to get nested I_n of length $(b - a)/2^n$, none δ -fine. By nested intervals $\bigcap I_n = \{\xi\}$. Pick p with $(b - a)/2^p < \delta(\xi)$; then $I_p \subseteq [\xi - \delta(\xi), \xi + \delta(\xi)]$, so the single pair (I_p, ξ) is a δ -fine partition of I_p — contradiction. $\square$

The gauge encodes a *Lebesgue number* pointwise: rather than one global δ , each tag t carries its own radius $\delta(t)$. For a continuous f on $[a, b]$, choosing $\delta(t)$ from the ε - δ condition at t and invoking Cousin’s lemma gives an alternative, gauge-flavored proof of Heine–Cantor and underlies the generalized Riemann integral (Chapter 12).

Remark (Lebesgue number). For a continuous f on compact K and a fixed ε , the cover proof of Theorem 5.7.5 extracts a single $\delta > 0$ such that any set of diameter $< \delta$ has image of diameter $< \varepsilon$ — a uniform “Lebesgue number” for the cover $\{f^{-1}(B_{\varepsilon/2}(f(p)))\}$. The gauge is the non-uniform refinement that survives even without compactness, recovering uniformity exactly when a constant gauge suffices.

Further worked examples

Worked Example 5.7.10 (Uniform continuity on the whole line via a Lipschitz bound). Show $f(x) = \sqrt{x^2 + 1}$ is uniformly continuous on $\mathbb{R}$.

Recognition. “Show uniformly continuous on an unbounded set”: Heine–Cantor does not apply ($\mathbb{R}$ is not compact), so produce a global Lipschitz constant (Definition 5.7.7).

Solution. For any x, u , rationalize:

$$\left| \sqrt{x^2 + 1} - \sqrt{u^2 + 1} \right| = \frac{|x^2 - u^2|}{\sqrt{x^2 + 1} + \sqrt{u^2 + 1}} = \frac{|x - u| |x + u|}{\sqrt{x^2 + 1} + \sqrt{u^2 + 1}}.$$

Since $|x| \leq \sqrt{x^2 + 1}$ and $|u| \leq \sqrt{u^2 + 1}$, we have $|x + u| \leq |x| + |u| \leq \sqrt{x^2 + 1} + \sqrt{u^2 + 1}$, so the fraction is $\leq |x - u|$. Thus f is Lipschitz with constant $K = 1$, hence uniformly continuous ($\delta = \varepsilon$). *Check.* The slope $f'(x) = x/\sqrt{x^2 + 1}$ has $|f'| < 1$ everywhere, matching $K = 1$; the curve never steepens past slope 1, so a fixed box always fits. $\blacksquare$

Worked Example 5.7.11 (Sum of uniformly continuous need not be: a product non-example). Show that x and $\sin x$ are each uniformly continuous on $\mathbb{R}$, but their product $x \sin x$ is not.

Recognition. “Is uniform continuity closed under products?”: it is closed under sums but *not* products on unbounded domains; test with the nonuniform criterion (Theorem 5.7.2).

Solution. Both x (Lipschitz, $K = 1$) and $\sin x$ (Lipschitz, $K = 1$ since $|\sin x - \sin u| \leq |x - u|$) are uniformly continuous. For $h(x) = x \sin x$, take $x_n = 2\pi n + \frac{1}{n}$ and $u_n = 2\pi n$. Then $|x_n - u_n| = \frac{1}{n} \rightarrow 0$, while $h(u_n) = 2\pi n \sin(2\pi n) = 0$ and $h(x_n) = (2\pi n + \frac{1}{n}) \sin(\frac{1}{n})$. Since $\sin(\frac{1}{n}) \approx \frac{1}{n}$, we get $h(x_n) \approx (2\pi n) \frac{1}{n} = 2\pi$, so $|h(x_n) - h(u_n)| \rightarrow 2\pi \geq \varepsilon_0 = 1$. By Theorem 5.7.2, $x \sin x$ is not uniformly continuous. *Check.* The amplitude grows linearly, so the oscillation steepens without bound — exactly the unbounded-slope obstruction. $\blacksquare$

Worked Example 5.7.12 (Uniformly continuous maps preserve Cauchy sequences). Show that if f is uniformly continuous on A and (x_n) is Cauchy in A , then $(f(x_n))$ is Cauchy. Use this to re-prove that $1/x$ is not uniformly continuous on $(0, 1)$.

Recognition. “Transport a structural property through f ”: uniform continuity converts an input-closeness condition into an output-closeness one, uniformly in the index.

Solution. Let $\varepsilon > 0$ and pick δ from uniform continuity. Since (x_n) is Cauchy, choose N with $|x_m - x_n| < \delta$ for $m, n \geq N$; then $|f(x_m) - f(x_n)| < \varepsilon$ for $m, n \geq N$, so $(f(x_n))$ is Cauchy. Now on $(0, 1)$ the sequence $x_n = 1/n$ is Cauchy (it converges to 0), but $f(x_n) = n$ is *not* Cauchy (it is unbounded). Hence $1/x$ cannot be uniformly continuous on $(0, 1)$. *Check.* This matches the direct closing-pair argument earlier; the Cauchy test also shows $1/x$ has no continuous extension to $[0, 1]$. $\blacksquare$

Worked Example 5.7.13 (A gauge that no constant δ can replace). On $[0, 1]$ let $f(x) = 1/x$ for $x > 0$ and $f(0) = 0$. Exhibit a gauge $\delta(\cdot)$ realizing continuity pointwise, and explain why no *constant* gauge works.

Recognition. “Pointwise continuity via a gauge”: assign each tag its own radius (Definition 5.7.8); a constant gauge would amount to uniform continuity.

Solution. At a point $t > 0$, f is continuous, so for a fixed ε there is a radius $\delta(t) > 0$ with $|f(x) - f(t)| < \varepsilon$ on $(t - \delta(t), t + \delta(t))$; one may take $\delta(t)$ shrinking like t^2 as $t \rightarrow 0^+$ (the slope near t is $\approx 1/t^2$). At $t = 0$, f is discontinuous, but on $[0, 1]$ we only need a gauge for the partition machinery, and any $\delta(0) > 0$ is admissible. A *constant* gauge $\delta(t) \equiv \delta$ would make f uniformly continuous; but f is unbounded near 0, so no constant δ controls $|f(x) - f(t)|$ there. *Check.* The gauge must collapse toward 0 as the tag approaches the bad point — precisely the non-constant behavior that Cousin’s lemma (Theorem 5.7.9) still tolerates. ■

Exercises (Rudin, Ch.4).

1. Let f be a real uniformly continuous function on the bounded set $E \subseteq \mathbb{R}$. Prove f is bounded on E . Show the conclusion is false if boundedness of E is omitted.
2. Show the requirement in the definition of uniform continuity can be rephrased with diameters: to every $\varepsilon > 0$ there exists $\delta > 0$ such that $\text{diam } f(E) < \varepsilon$ for all $E \subseteq X$ with $\text{diam } E < \delta$.
3. Complete the alternative proof of the theorem “continuous on compact $\implies$ uniformly continuous”: if f is not uniformly continuous, then for some $\varepsilon > 0$ there are sequences $(p_n), (q_n)$ in X with $d_X(p_n, q_n) \rightarrow 0$ but $d_Y(f(p_n), f(q_n)) \geq \varepsilon$. Use compactness (a convergent subsequence) to obtain a contradiction.
4. Suppose f is a uniformly continuous map of a metric space X into a metric space Y . Prove $(f(x_n))$ is Cauchy in Y for every Cauchy sequence (x_n) in X . Use this to give an alternative proof of the extension theorem below.
5. State precisely and prove: a uniformly continuous function of a uniformly continuous function is uniformly continuous.
6. Let E be dense in a metric space X and let f be a uniformly continuous real function on E . Prove f has a continuous extension to X (unique by the dense-set determination). Could the range space $\mathbb{R}$ be replaced by $\mathbb{R}^k$? By any compact metric space? By any complete metric space? By any metric space?
7. Suppose X, Y, Z are metric spaces with Y compact. Let $f : X \rightarrow Y$, let g be a continuous one-to-one map $Y \rightarrow Z$, and put $h(x) = g(f(x))$. Prove f is uniformly continuous if h is uniformly continuous (Hint: g^{-1} has compact domain $g(Y)$, and $f = g^{-1} \circ h$). Prove also f is continuous if h is. Show by example that compactness of Y cannot be omitted, even with X, Z compact.

Exercises (Bartle & Sherbert, §5.4).

1. Show $f(x) = 1/x$ is uniformly continuous on $A = [a, \infty)$ for a positive constant a .
2. Show $f(x) = 1/x^2$ is uniformly continuous on $A = [1, \infty)$ but not on $B = (0, \infty)$.
3. Use the nonuniform-continuity criterion to show the following are not uniformly continuous: (a) $f(x) = x^2$ on $A = [0, \infty)$; (b) $g(x) = \sin(1/x)$ on $B = (0, \infty)$.
4. Show $f(x) = 1/(1 + x^2)$ is uniformly continuous on $\mathbb{R}$.
5. Show that if f, g are uniformly continuous on $A \subseteq \mathbb{R}$, then $f + g$ is uniformly continuous on A .
6. Show that if f, g are uniformly continuous and bounded on $A \subseteq \mathbb{R}$, then fg is uniformly continuous on A .
7. If $f(x) = x$ and $g(x) = \sin x$, show f, g are uniformly continuous on $\mathbb{R}$ but fg is not.
8. Prove that if f, g are each uniformly continuous on $\mathbb{R}$, then $f \circ g$ is uniformly continuous on $\mathbb{R}$.
9. If f is uniformly continuous on $A \subseteq \mathbb{R}$ and $|f(x)| \geq k > 0$ for all $x \in A$, show $1/f$ is uniformly continuous on A .
10. Prove that if f is uniformly continuous on a bounded set $A \subseteq \mathbb{R}$, then f is bounded on A .

11. If $g(x) = \sqrt{x}$ on $[0, 1]$, show there is no constant K with $|g(x)| \leq K|x|$ for all $x \in [0, 1]$. Conclude the uniformly continuous g is not Lipschitz on $[0, 1]$.
12. Show that if f is continuous on $[0, \infty)$ and uniformly continuous on $[a, \infty)$ for some $a > 0$, then f is uniformly continuous on $[0, \infty)$.
13. Let $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$ have the property: for each $\varepsilon > 0$ there is a uniformly continuous $g_\varepsilon : A \rightarrow \mathbb{R}$ with $|f(x) - g_\varepsilon(x)| < \varepsilon$ for all $x \in A$. Prove f is uniformly continuous on A .
14. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *periodic* if there is $p > 0$ with $f(x + p) = f(x)$ for all x . Prove a continuous periodic function on $\mathbb{R}$ is bounded and uniformly continuous.
15. Let f, g be Lipschitz on A . (a) Show $f + g$ is Lipschitz on A . (b) If f, g are bounded on A , show fg is Lipschitz on A . (c) Give a Lipschitz f on $[0, \infty)$ whose square f^2 is not Lipschitz.
16. A function is *absolutely continuous* on an interval I if for any $\varepsilon > 0$ there is $\delta > 0$ such that for any pairwise disjoint subintervals $[x_k, y_k]$, $k = 1, \dots, n$, of I with $\sum |x_k - y_k| < \delta$ one has $\sum |f(x_k) - f(y_k)| < \varepsilon$. Show that if f is Lipschitz on I , then f is absolutely continuous on I .

Exercises (Bartle & Sherbert, §5.5 — gauges).

1. Let δ be the gauge on $[0, 1]$ with $\delta(0) = \frac{1}{4}$ and $\delta(t) = \frac{1}{2}t$ for $t \in (0, 1]$. (a) Show $\dot{P}_1 = \{([0, \frac{1}{4}], 0), ([\frac{1}{4}, \frac{1}{2}], \frac{1}{2}), ([\frac{1}{2}, 1], \frac{3}{4})\}$ is δ -fine. (b) Show $\dot{P}_2 = \{([0, \frac{1}{4}], 0), ([\frac{1}{4}, \frac{1}{2}], \frac{1}{2}), ([\frac{1}{2}, 1], \frac{3}{5})\}$ is not δ -fine.
2. Let δ_1 be the gauge with $\delta_1(0) = \frac{1}{4}$, $\delta_1(t) = \frac{3}{4}t$. Are the partitions of Exercise 1 δ_1 -fine? (Note $\delta(t) \leq \delta_1(t)$ throughout.)
3. Let δ_2 be the gauge with $\delta_2(0) = \frac{1}{10}$, $\delta_2(t) = \frac{9}{10}t$. Are the partitions of Exercise 1 δ_2 -fine?
4. Let g be the gauge of Example 5.5.4(d). (a) If $t \in (0, \frac{1}{2}]$ show $[t - g(t), t + g(t)] = [\frac{1}{2}t, \frac{3}{2}t] \subseteq (0, \frac{3}{4}]$. (b) If $t \in (\frac{1}{2}, 1)$ show $[t - g(t), t + g(t)] \subseteq (\frac{1}{4}, 1)$.
5. Let $a < c < b$ and δ a gauge on $[a, b]$. If $\dot{P}'$ is a δ -fine partition of $[a, c]$ and $\dot{P}''$ of $[c, b]$, show $\dot{P}' \cup \dot{P}''$ is a δ -fine partition of $[a, b]$ having c as a partition point.
6. Let $a < c < b$ and let δ', δ'' be gauges on $[a, c]$ and $[c, b]$. Define δ on $[a, b]$ by $\delta = \delta'$ on $[a, c)$, $\delta(c) = \min\{\delta'(c), \delta''(c)\}$, $\delta = \delta''$ on $(c, b]$. Then δ is a gauge, and if $\dot{P}'$ is δ' -fine on $[a, c]$ and $\dot{P}''$ is δ'' -fine on $[c, b]$, then $\dot{P}' \cup \dot{P}''$ is a tagged partition of $[a, b]$ with partition point c . Explain why $\dot{P}' \cup \dot{P}''$ may fail to be δ -fine, and give an example.
7. With δ', δ'' as above define δ by $\delta(t) = \min\{\delta'(t), \frac{1}{2}(c - t)\}$ on $[a, c)$, $\delta(c) = \min\{\delta'(c), \delta''(c)\}$, and $\delta(t) = \min\{\delta''(t), \frac{1}{2}(t - c)\}$ on $(c, b]$. Show δ is a gauge, and every δ -fine partition of $[a, b]$ with partition point c yields a δ' -fine partition of $[a, c]$ and a δ'' -fine partition of $[c, b]$.
8. Let δ be a gauge on $I = [a, b]$ and suppose I has no δ -fine partition. (a) Let $c = \frac{1}{2}(a + b)$; show at least one of $[a, c], [c, b]$ has no δ -fine partition. (b) Construct nested (I_n) with length $(b - a)/2^n$, none δ -fine. (c) Let $\xi \in \bigcap_n I_n$ and let p satisfy $(b - a)/2^p < \delta(\xi)$. Show that $I_p \subseteq [\xi - \delta(\xi), \xi + \delta(\xi)]$, so the pair (I_p, ξ) is a δ -fine partition of I_p — a contradiction (this proves Cousin's lemma).
9. Let $I = [a, b]$, $f : I \rightarrow \mathbb{R}$. Call f *locally bounded* at $c \in I$ if there is $\delta(c) > 0$ with f bounded on $I \cap [c - \delta(c), c + \delta(c)]$. Prove that if f is locally bounded at every point of I , then f is bounded on I .
10. Let $I = [a, b]$, $f : I \rightarrow \mathbb{R}$. Call f *locally increasing* at c if there is $\delta(c) > 0$ with f increasing on $I \cap [c - \delta(c), c + \delta(c)]$. Prove that if f is locally increasing at every point of I , then f is increasing on I .

Additional exercises.

1. Prove $f(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$. (Hint: split at 1; use $|\sqrt{x} - \sqrt{u}| \leq \sqrt{|x - u|}$ near 0 and a Lipschitz bound far out.)
2. Show $f(x) = \sin x$ is uniformly continuous on $\mathbb{R}$ but $g(x) = \sin(x^2)$ is not.

- (Non-example / disprove.) Disprove “a function continuous on all of $\mathbb{R}$ is uniformly continuous” using $f(x) = x^2$, exhibiting the closing sequences explicitly.
- Prove that if f is uniformly continuous on A and (x_n) is Cauchy in A , then $(f(x_n))$ is Cauchy. Use this to show $\sin(1/x)$ is not uniformly continuous on $(0, 1)$.
- Show that a uniformly continuous f on a bounded set A extends continuously to $\overline{A}$, and that $1/x$ on $(0, 1]$ has no such extension to $[0, 1]$.
- (Gauge / Cousin.) Given the gauge $\delta(t) = \frac{1}{2} \text{dist}(t, \{0, 1\})$ for $t \in (0, 1)$ with $\delta(0) = \delta(1) = \frac{1}{4}$, exhibit one explicit δ -fine tagged partition of $[0, 1]$.
- (Diagram.) Plot $y = x^2$ and slide a fixed $\delta \times \varepsilon$ rectangle along it; draw one position near $x = 0$ where the box contains the graph and one near large x where the graph escapes the box. Repeat the picture for $y = 1/x$ near 0.
- Prove Heine–Cantor (continuous on compact $\implies$ uniformly continuous) by the gauge method: build a gauge from the pointwise ε - δ data and apply Cousin’s lemma to extract a uniform δ .

5.8 Monotone Functions, Inverse Functions, and Discontinuities

Monotone functions and one-sided limits

Definition 5.8.1. $f : I \rightarrow \mathbb{R}$ (I an interval) is *increasing* if $x < y \implies f(x) \leq f(y)$, *strictly increasing* if $x < y \implies f(x) < f(y)$; *decreasing* and *strictly decreasing* are defined with the inequalities reversed. f is *monotone* if it is increasing or decreasing.

The decisive structural fact: a monotone function can misbehave only by *jumping*, and its one-sided limits always exist.

Theorem 5.8.2 (One-sided limits of a monotone function). *Let f be increasing on (a, b) and let $c \in (a, b)$. Then both one-sided limits exist:*

$$f(c^-) = \lim_{x \rightarrow c^-} f(x) = \sup_{a < x < c} f(x) \leq f(c) \leq \inf_{c < x < b} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c^+).$$

Consequently $f(c^-) \leq f(c) \leq f(c^+)$, and the only possible discontinuity at c is a jump, of size $f(c^+) - f(c^-) \geq 0$.

Proof. The set $\{f(x) : a < x < c\}$ is bounded above by $f(c)$, so $L = \sup$ exists with $L \leq f(c)$. Given $\varepsilon > 0$ pick $x_0 < c$ with $f(x_0) > L - \varepsilon$; for $x_0 < x < c$, monotonicity gives $L - \varepsilon < f(x_0) \leq f(x) \leq L$, so $|f(x) - L| < \varepsilon$. Hence $f(c^-) = L$. The right-hand statement is symmetric with $\inf$. The inequalities chain as displayed. $\square$

Definition 5.8.3 (Classification of discontinuities). Suppose $f(c^-)$ and $f(c^+)$ both exist (automatic for monotone f). The discontinuity at c is:

- *removable* if $f(c^-) = f(c^+) \neq f(c)$ (a single “hole” that one value would fix);
- a *jump* (simple discontinuity, *first kind*) if $f(c^-) \neq f(c^+)$;
- *essential* (*second kind*) if at least one of $f(c^-), f(c^+)$ fails to exist.

A *simple discontinuity* (Rudin) is one of the first two types: both one-sided limits exist.

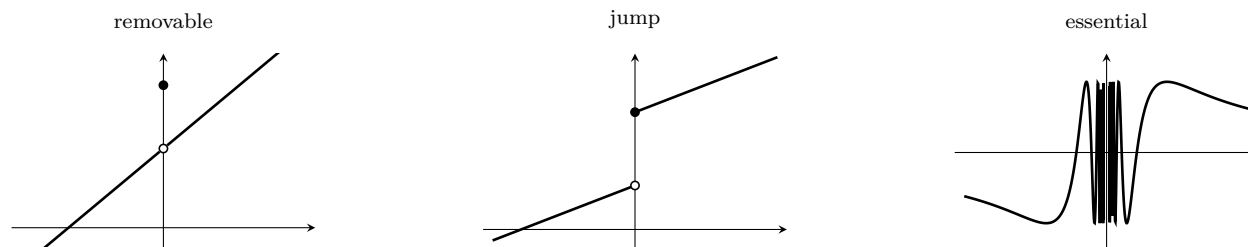


Figure 5.9. Three discontinuity types at c : removable (matching one-sided limits, wrong value), jump (mismatched finite one-sided limits), essential (a one-sided limit fails, here $\sin(1/x)$).

Theorem 5.8.4 (Monotone $\implies$ at most countably many discontinuities). *A monotone function f on (a, b) has at most countably many points of discontinuity, and every one of them is a jump.*

Proof. Assume f increasing. At each discontinuity c , $f(c^-) < f(c^+)$ leaves an open “jump interval” $J_c = (f(c^-), f(c^+))$. For $c < c'$ discontinuities, monotonicity gives $f(c^+) \leq f(c'^-)$, so the intervals J_c are *pairwise disjoint*. Each contains a distinct rational, giving an injection from the set of discontinuities into $\mathbb{Q}$. Hence that set is countable. $\square$

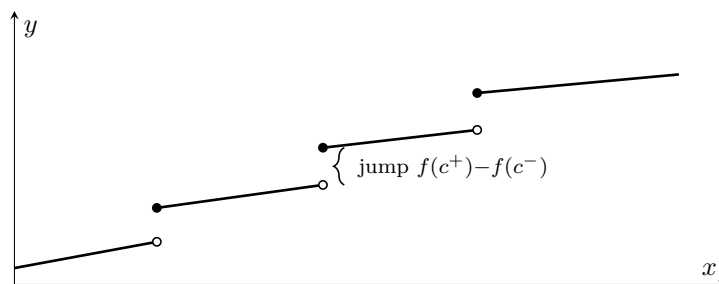


Figure 5.10. An increasing function: each jump opens a gap in the range; disjoint gaps each swallow a rational, so there can be only countably many.

Continuous monotone functions and their inverses

Theorem 5.8.5 (Continuous strictly monotone inverse). *Let I be an interval and $f : I \rightarrow \mathbb{R}$ continuous and strictly increasing. Then $J = f(I)$ is an interval, $f : I \rightarrow J$ is a bijection, and the inverse $f^{-1} : J \rightarrow I$ is continuous and strictly increasing. (Same with “decreasing”).*

Proof. Strict monotonicity gives injectivity; $J = f(I)$ is an interval by IVT (Theorem 5.6.2). f^{-1} is strictly increasing: $y_1 < y_2$ in J would force $f^{-1}(y_1) \geq f^{-1}(y_2)$ only if f reversed order, which it does not. For continuity of f^{-1} at $y_0 = f(x_0)$: given $\varepsilon > 0$, the images $f(x_0 - \varepsilon) < y_0 < f(x_0 + \varepsilon)$ (when these lie in I) bracket y_0 ; choosing δ inside this bracket forces $f^{-1}(y)$ within ε of x_0 . (Alternatively: a continuous bijection between intervals has continuous inverse because monotone functions only jump, and a jump in f^{-1} would leave a gap in $I = f^{-1}(J)$, impossible for an interval.) $\square$

Worked Example 5.8.6 (n -th roots and rational powers). Use Theorem 5.8.5 to get $x \mapsto x^{1/n}$ continuous and strictly increasing on $[0, \infty)$, then define x^r for $r \in \mathbb{Q}$.

Recognition. “Construct/continuity of root and power functions”; invert a monotone continuous power.

Solution. $\phi(x) = x^n$ is continuous and strictly increasing on $[0, \infty)$ with $\phi([0, \infty)) = [0, \infty)$, so $\phi^{-1}(y) = y^{1/n}$ is continuous and strictly increasing there. For $r = m/n \in \mathbb{Q}$ ($x > 0$) set $x^r = (x^{1/n})^m$; the usual laws $x^r x^s = x^{r+s}$, $(x^r)^s = x^{rs}$ follow from the integer laws and well-definedness ($mq = np \implies (x^{1/n})^m = (x^{1/q})^p$). *Check.* $x^{1/2}$ at $x = 4$ gives 2; strict monotonicity matches the picture of $\sqrt{\cdot}$. ■

Worked Example 5.8.7 (Thomae’s function: continuous exactly at the irrationals). For $x = m/n$ in lowest terms ($n > 0$) define $f(x) = 1/n$; $f(0) = 1$ (with $n = 1$); $f(x) = 0$ for $x \notin \mathbb{Q}$. Show f is continuous at every irrational and has a (removable) simple discontinuity at every rational (Rudin 4.18).

Recognition. “Where is this continuous?”; bound f by a small $1/n$ near any point.

Solution. Fix any c and $\varepsilon > 0$. Only finitely many rationals in $(c - 1, c + 1)$ have denominator $n \leq 1/\varepsilon$; choose δ small enough to dodge all of them (and to avoid c if c is one of them). Then every x with $0 < |x - c| < \delta$ is either irrational ($f = 0$) or rational with $n > 1/\varepsilon$ ($f = 1/n < \varepsilon$); so $\lim_{x \rightarrow c} f = 0$. Hence f is continuous at c iff $f(c) = 0$, i.e. iff c is irrational; at each rational, $\lim_{x \rightarrow c} f = 0 \neq f(c)$, a removable discontinuity (both one-sided limits are 0). *Check.* Continuity on a dense set (the irrationals) and discontinuity on a dense set (the rationals) coexist — allowed precisely because the discontinuity set is countable. ■

Theorem 5.8.8 (Simple discontinuities are at most countable). *For any $f : (a, b) \rightarrow \mathbb{R}$ (not necessarily monotone), the set of points where f has a simple discontinuity is at most countable (Rudin 4.30).*

Sketch. Split into the sets where $f(c^-) < f(c^+)$, $f(c^-) > f(c^+)$, $f(c^-) = f(c^+) \neq f(c)$. On the first, associate to c a triple (p, q, r) of rationals with $f(c^-) < p < f(c^+)$, with q, r straddling c so that $f < p$ to the left of c within (q, c) and $f > p$ to the right within (c, r) . Distinct such c yield distinct triples, so this set injects into $\mathbb{Q}^3$, hence is countable; the other types are handled identically. (The full argument is the content of Rudin Exercise 4.17.) $\square$

Caution. Monotonicity is what guarantees one-sided limits *exist*; for a general f they may not, leaving essential discontinuities (e.g. $\sin(1/x)$ at 0). The countability theorems above count only *jump/removable* (simple) discontinuities. A function can have uncountably many *essential* discontinuities — indeed the Dirichlet function is discontinuous everywhere, with no one-sided limits anywhere, so its discontinuities are all of the second kind and uncountable.

Further worked examples

Worked Example 5.8.9 (Classifying discontinuities of a piecewise function). For $f(x) = \frac{|x|}{x}$ at $x = 0$, $g(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$, and $h(x) = \sin(1/x)$ at $x = 0$, classify each discontinuity.

Recognition. “Name the type”: compute both one-sided limits and compare with the value (Definition 5.8.3).

Solution. f : for $x > 0$, $f = 1$; for $x < 0$, $f = -1$; so $f(0^+) = 1 \neq -1 = f(0^-)$ — a *jump* (first kind), regardless of how $f(0)$ is set. g : for $x \neq 1$, $g(x) = \frac{(x-1)(x+1)}{x-1} = x+1$, so $g(1^-) = g(1^+) = 2$; if $g(1)$ is left undefined or unequal to 2 this is a *removable* discontinuity. h : $h(0^+)$ does not exist (oscillation, §5.1), so the discontinuity is *essential* (second kind). *Check.* Only g can be repaired by assigning one value ($g(1) := 2$); the jump in f and the oscillation in h cannot. ■

Worked Example 5.8.10 (A prescribed monotone function with countably many jumps). Enumerate $\mathbb{Q} \cap (0, 1) = \{q_1, q_2, \dots\}$ and set $f(x) = \sum_{q_k < x} 2^{-k}$. Show f is increasing, discontinuous exactly at each q_k , and continuous at every irrational.

Recognition. “Build a function with a controlled discontinuity set”: sum a convergent series of jumps, one per rational (the construction behind Theorem 5.8.4).

Solution. Each term $2^{-k} \geq 0$, and adding more terms as x increases gives $x < y \implies f(x) \leq f(y)$, so f is increasing (and bounded by $\sum 2^{-k} = 1$). At $x = q_j$: as x crosses q_j upward, the term 2^{-j} switches on, so $f(q_j^+) - f(q_j^-) = 2^{-j} > 0$ — a jump of size 2^{-j} . At an irrational c : for $\varepsilon > 0$ pick N with $\sum_{k > N} 2^{-k} < \varepsilon$; choosing δ so that $(c - \delta, c + \delta)$ contains none of $q_1, \dots, q_N$ makes the oscillation of f across c less than ε , so f is continuous at c . *Check.* The discontinuity set is exactly the countable $\{q_k\}$, with total jump $\sum 2^{-k} = 1 = f(1^-)$ — a concrete instance of “monotone $\implies$ at most countably many jumps”. ■

Worked Example 5.8.11 (A non-example: a strictly increasing function with a discontinuous inverse). Show that dropping continuity from Theorem 5.8.5 can break it: a strictly increasing bijection of an interval onto a *non*-interval need not have continuous inverse.

Recognition. “Why does the inverse theorem need continuity of f ?”: a jump in f leaves a gap in the range, and the inverse must jump across that gap.

Solution. Define $f: [0, 2] \rightarrow [0, 1] \cup [2, 3]$ by $f(x) = x$ on $[0, 1]$ and $f(x) = x + 1$ on $(1, 2]$. Then f is strictly increasing and a bijection onto $J = [0, 1] \cup [2, 3]$, but it jumps at $x = 1$ ($f(1) = 1$, $f(1^+) = 2$). Its inverse $g = f^{-1}$ sends $[0, 1] \rightarrow [0, 1]$ and $[2, 3] \rightarrow (1, 2]$; g is discontinuous at $y = 2$, where $g(2^-)$ would be 1 from the left branch but $g(2) = 1$ jumps to values near 1 then to $(1, \dots$ — concretely g is continuous here *only* because J already excludes $(1, 2)$. The honest failure: f itself is discontinuous, so Theorem 5.8.5 (which assumes f continuous on an interval) simply does not apply. *Check.* The range J is not an interval — its gap $(1, 2)$ is the missing piece that an interval domain would have forbidden. ■

Worked Example 5.8.12 (The logarithm as a monotone inverse). Granting that $\exp$ is continuous and strictly increasing on $\mathbb{R}$ with range $(0, \infty)$, deduce that $\log$ is continuous and strictly increasing on $(0, \infty)$, and that $\log(xy) = \log x + \log y$.

Recognition. “Continuity/structure of an inverse function”: apply the monotone-inverse theorem, then transport the functional equation through the bijection.

Solution. By Theorem 5.8.5, since $\exp : \mathbb{R} \rightarrow (0, \infty)$ is a continuous strictly increasing bijection, its inverse $\log : (0, \infty) \rightarrow \mathbb{R}$ is continuous and strictly increasing. For the product law, write $x = \exp(s)$, $y = \exp(t)$; then $xy = \exp(s)\exp(t) = \exp(s+t)$, so $\log(xy) = s+t = \log x + \log y$. *Check.* $\log 1 = 0$ (since $\exp 0 = 1$) and $\log$ inherits its monotonicity from $\exp$; numerically $\log(2 \cdot 4) = \log 8 \approx 2.079$ equals $\log 2 + \log 4 \approx 0.693 + 1.386$. ■

Exercises (Rudin, Ch.4).

1. Let $[x]$ denote the largest integer $\leq x$ and $\{x\} = x - [x]$ the fractional part. What discontinuities do $[x]$ and $\{x\}$ have?
2. Let f be a real function on (a, b) . Prove the set of points at which f has a *simple* discontinuity is at most countable. (Hint: on $E = \{x : f(x^-) < f(x^+)\}$, associate to each $x \in E$ a triple (p, q, r) of rationals with $f(x^-) < p < f(x^+)$, $q < x < r$, such that $a < q < t < x \implies f(t) < p$ and $x < t < r < b \implies f(t) > p$; distinct x give distinct triples. Handle the other simple-discontinuity types similarly.)
3. Define f on $\mathbb{R}$ by $f(x) = 0$ if x is irrational, and $f(x) = 1/n$ if $x = m/n$ in lowest terms ($n > 0$; take $n = 1$ when $x = 0$). Prove f is continuous at every irrational and has a simple discontinuity at every rational.
4. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ has the *intermediate value property* (if $f(a) < c < f(b)$ then $f(x) = c$ for some x between a, b) and, for every rational r , the set $\{x : f(x) = r\}$ is closed. Prove f is continuous. (Hint: if $x_n \rightarrow x_0$ but $f(x_n) > r > f(x_0)$ for some r and all n , find t_n between x_0, x_n with $f(t_n) = r$; then $t_n \rightarrow x_0$, forcing a contradiction.)
5. A real function f on (a, b) is *convex* if $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ whenever $x, y \in (a, b)$, $0 < \lambda < 1$. Prove every convex function is continuous. Prove every increasing convex function of a convex function is convex (e.g. e^f is convex if f is).
6. If f is convex on (a, b) and $a < s < t < u < b$, show $\frac{f(t) - f(s)}{t - s} \leq \frac{f(u) - f(s)}{u - s} \leq \frac{f(u) - f(t)}{u - t}$ (slopes increase). Conversely, if f is continuous on (a, b) with $f(\frac{x+y}{2}) \leq \frac{1}{2}(f(x) + f(y))$ for all x, y , prove f is convex.

Exercises (Bartle & Sherbert, §5.6).

1. If $I = [a, b]$ and $f : I \rightarrow \mathbb{R}$ is increasing, show a (resp. b) is an absolute minimum (resp. maximum) point. If f is strictly increasing, a is the only absolute minimum point.
2. If f, g are increasing on an interval I , show $f + g$ is increasing on I ; if f is also strictly increasing, then $f + g$ is strictly increasing.
3. Show $f(x) = x$ and $g(x) = x - 1$ are strictly increasing on $I = [0, 1]$, but their product fg is not increasing on I .
4. Show that if f, g are positive increasing functions on an interval I , then fg is increasing on I .
5. Show that if $I = [a, b]$ and $f : I \rightarrow \mathbb{R}$ is increasing, then f is continuous at a iff $f(a) = \inf\{f(x) : x \in (a, b]\}$.
6. Let $I \subseteq \mathbb{R}$ be an interval, $f : I \rightarrow \mathbb{R}$ increasing, $c \in I$ not an endpoint. Show f is continuous at c iff there is a sequence (x_n) in I with $x_n < c$ for odd n , $x_n > c$ for even n , $c = \lim x_n$, and $f(c) = \lim f(x_n)$.
7. Let I be an interval, $f : I \rightarrow \mathbb{R}$ increasing, c not an endpoint. Show the jump $j_f(c) = f(c^+) - f(c^-)$ equals $\inf\{f(y) - f(x) : x < c < y, x, y \in I\}$.
8. Let f, g be strictly increasing on an interval I with $f(x) > g(x)$ for all $x \in I$. If $y \in f(I) \cap g(I)$, show $f^{-1}(y) < g^{-1}(y)$. (Hint: interpret geometrically.)
9. Let $I = [0, 1]$, $f : I \rightarrow \mathbb{R}$ with $f(x) = x$ for x rational and $f(x) = 1 - x$ for x irrational. Show f is injective on I and $f(f(x)) = x$ for all $x \in I$ (f is its own inverse). Show f is continuous only at $x = \frac{1}{2}$.

10. Let $I = [a, b]$, $f : I \rightarrow \mathbb{R}$ continuous. If f has an absolute maximum (resp. minimum) at an interior point c , show f is not injective on I .
11. Let $f(x) = x$ for $x \in [0, 1]$ and $f(x) = 1 + x$ for $x \in (1, 2]$. Show f and f^{-1} are strictly increasing. Are f and f^{-1} continuous everywhere?
12. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous, taking no value twice, with $f(0) < f(1)$. Show f is strictly increasing on $[0, 1]$.
13. Let $h : [0, 1] \rightarrow \mathbb{R}$ take each of its values exactly twice. Show h cannot be continuous at every point. (Hint: if $c_1 < c_2$ are where h attains its sup, show $c_1 = 0$, $c_2 = 1$; then examine the infimum.)
14. Let $x \in \mathbb{R}$, $x > 0$. Show that if $m, p \in \mathbb{Z}$, $n, q \in \mathbb{N}$, and $mq = np$, then $(x^{1/n})^m = (x^{1/q})^p$.
15. If $x > 0$ and $r, s \in \mathbb{Q}$, show $x^r x^s = x^{r+s} = x^s x^r$ and $(x^r)^s = x^{rs} = (x^s)^r$.

Additional exercises.

1. For $f(x) = \lfloor x \rfloor$ (greatest integer), find $f(c^-)$, $f(c^+)$, $f(c)$ at each integer c , classify the discontinuity, and compute the jump.
2. Show the increasing function $f(x) = \sum_{n: r_n \leq x} 2^{-n}$, where (r_n) enumerates $\mathbb{Q} \cap (0, 1)$, is discontinuous exactly at each r_n and continuous at every irrational. (You may quote the monotone-discontinuity theorem.)
3. Classify the discontinuity of each at 0: (a) $\operatorname{sgn}(x)$; (b) $\sin(1/x)$; (c) $x \mapsto \frac{x^2 - x}{x}$ (removable?); (d) e^{-1/x^2} extended by 0.
4. (Non-example / disprove.) Disprove “a function with at most countably many discontinuities must be monotone on some interval” — give a non-monotone example, e.g. one with jumps of alternating sign at $1/n$.
5. Prove that a strictly increasing continuous f on $[a, b]$ has a strictly increasing continuous inverse, and apply it to construct $\ln$ as the inverse of $\exp$ on a suitable interval.
6. Show that an increasing f on (a, b) has one-sided limits everywhere, and that the total jump $\sum_c (f(c^+) - f(c^-))$ over all discontinuities c in $[s, t] \subseteq (a, b)$ is at most $f(t) - f(s)$.
7. (Diagram.) Sketch an increasing step-like function on $[0, 4]$ with jumps at $x = 1, 2, 3$, drawing filled/open endpoints correctly, and brace one jump labeled $f(c^+) - f(c^-)$.
8. Prove that the set of discontinuities of a monotone function is at most countable by injecting it into $\mathbb{Q}$ via disjoint jump intervals, and explain why the argument fails for general (non-monotone) f with essential discontinuities.

Chapter 6

Differentiation

6.1 The Derivative

The derivative as a limit of difference quotients

Throughout, I denotes an interval of $\mathbb{R}$ (possibly closed, half-open, or unbounded) and $f : I \rightarrow \mathbb{R}$ a real function. Fix a point $c \in I$. For $x \in I$ with $x \neq c$ the *difference quotient* of f at c is $\frac{f(x) - f(c)}{x - c}$; its slope is the slope of the *secant line* through $(c, f(c))$ and $(x, f(x))$. Differentiation studies the limit of this slope as $x \rightarrow c$.

Definition 6.1.1. $f : I \rightarrow \mathbb{R}$ is *differentiable* at $c \in I$ if the limit

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists in $\mathbb{R}$; the number $f'(c)$ is the *derivative of f at c* . Equivalently, writing $x = c + h$, $f'(c) = \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h}$. We call f *differentiable on $E \subseteq I$* if $f'(c)$ exists for every $c \in E$; the resulting function $f' : E \rightarrow \mathbb{R}$ is the *derivative of f* .

The limit is taken *within I* : at an endpoint of I it is automatically a one-sided derivative. Geometrically $f'(c)$ is the slope of the *tangent line* $y = f(c) + f'(c)(x - c)$, the limiting position of the secants as $x \rightarrow c$. The whole subject is the assertion that this limiting slope, when it exists, controls the local behavior of f .

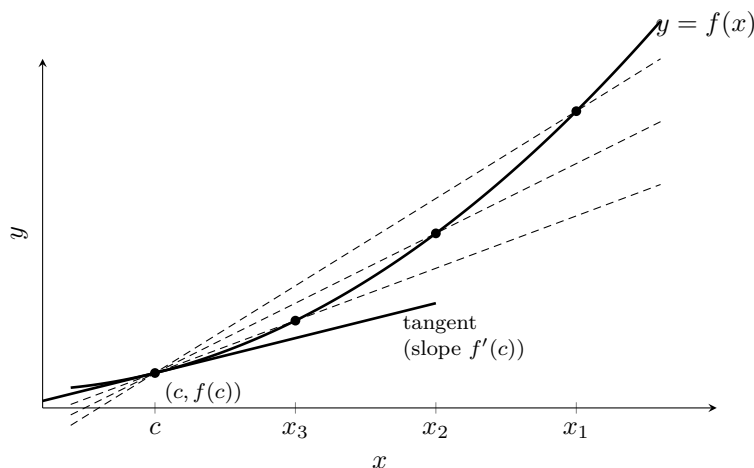


Figure 6.1. Secants through $(c, f(c))$ and $(x_k, f(x_k))$ with $x_1 > x_2 > x_3 \rightarrow c$. Their slopes converge to the tangent slope $f'(c)$.

Theorem 6.1.2 (Differentiable $\implies$ continuous). *If f is differentiable at c , then f is continuous at c .*

Proof. For $x \neq c$ write $f(x) - f(c) = \frac{f(x) - f(c)}{x - c} (x - c)$. As $x \rightarrow c$ the first factor tends to $f'(c)$ and the second to 0, so by the product rule for limits $f(x) - f(c) \rightarrow f'(c) \cdot 0 = 0$, i.e. $f(x) \rightarrow f(c)$. Thus f is continuous at c . $\square$

The converse fails utterly: $f(x) = |x|$ is continuous everywhere but not differentiable at 0 (the left and right difference quotients are -1 and $+1$). In Chapter 8 we will meet a function continuous on all of $\mathbb{R}$ yet differentiable *nowhere*. So differentiability is strictly stronger than continuity.

Theorem 6.1.3 (Algebra of derivatives). *If $f, g : I \rightarrow \mathbb{R}$ are differentiable at c , then so are $f + g$, fg , and (if $g(c) \neq 0$) f/g , with*

$$(f + g)'(c) = f'(c) + g'(c), \quad (fg)'(c) = f'(c)g(c) + f(c)g'(c),$$

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}.$$

In particular every polynomial is differentiable on $\mathbb{R}$, every rational function is differentiable off the zeros of its denominator, and $(x^n)' = nx^{n-1}$ for every integer n (with $x \neq 0$ when $n < 0$).

Proof. Sum: immediate from the additivity of limits. Product: write the difference quotient as

$$\frac{f(x)g(x) - f(c)g(c)}{x - c} = \frac{f(x) - f(c)}{x - c} g(x) + f(c) \frac{g(x) - g(c)}{x - c}.$$

As $x \rightarrow c$, $g(x) \rightarrow g(c)$ by Theorem 6.1.2, so the right side tends to $f'(c)g(c) + f(c)g'(c)$. Quotient: it suffices (by the product rule) to differentiate $1/g$; the quotient

$$\frac{1}{x - c} \left(\frac{1}{g(x)} - \frac{1}{g(c)} \right) = -\frac{1}{g(x)g(c)} \frac{g(x) - g(c)}{x - c} \rightarrow -\frac{g'(c)}{g(c)^2},$$

using $g(x) \rightarrow g(c) \neq 0$. Combining gives the quotient rule. The power rule follows by induction from the product rule (and the quotient rule for $n < 0$). $\square$

Caratheodory's characterization and the chain rule

The difference-quotient definition is awkward to compose, because the inner function may equal c at points arbitrarily close to c , dividing by zero in the naive chain-rule computation. Caratheodory's reformulation removes the division entirely and makes the Chain Rule a one-line composition of continuous functions.

Theorem 6.1.4 (Caratheodory's Theorem). *Let f be defined on an interval I containing c . Then f is differentiable at c if and only if there is a function $\varphi : I \rightarrow \mathbb{R}$, continuous at c , such that*

$$f(x) - f(c) = \varphi(x)(x - c) \quad (x \in I).$$

In that case $\varphi(c) = f'(c)$.

Proof. ($\Rightarrow$) If $f'(c)$ exists, define $\varphi(x) = \frac{f(x) - f(c)}{x - c}$ for $x \neq c$ and $\varphi(c) = f'(c)$. The factorization $f(x) - f(c) = \varphi(x)(x - c)$ holds for all x (trivially at $x = c$), and $\varphi(x) \rightarrow f'(c) = \varphi(c)$ as $x \rightarrow c$, so φ is continuous at c .

($\Leftarrow$) Suppose such a φ exists. For $x \neq c$, $\frac{f(x) - f(c)}{x - c} = \varphi(x) \rightarrow \varphi(c)$ as $x \rightarrow c$ by continuity, so $f'(c)$ exists and equals $\varphi(c)$. $\square$

The content is that “ f is differentiable at c ” is the same as “ $f(x) - f(c)$ has a continuous factor φ pulling out $(x - c)$.” The value $\varphi(c) = f'(c)$ is the slope; near c , $\varphi(x)$ is the secant slope. We use it twice: for the Chain Rule and for the inverse-function derivative.

Theorem 6.1.5 (Chain Rule). Let I, J be intervals, $g : I \rightarrow \mathbb{R}$ with $g(I) \subseteq J$, and $f : J \rightarrow \mathbb{R}$. If g is differentiable at c and f is differentiable at $g(c)$, then $h = f \circ g$ is differentiable at c and

$$h'(c) = f'(g(c))g'(c).$$

Proof. Put $d = g(c)$. By Caratheodory there are ψ continuous at c with $g(x) - g(c) = \psi(x)(x - c)$ and $\psi(c) = g'(c)$, and ϕ continuous at d with $f(y) - f(d) = \phi(y)(y - d)$ and $\phi(d) = f'(d)$. Substituting $y = g(x)$,

$$f(g(x)) - f(g(c)) = \phi(g(x))(g(x) - g(c)) = \phi(g(x))\psi(x)(x - c).$$

The coefficient $x \mapsto \phi(g(x))\psi(x)$ is continuous at c (g is continuous at c , ϕ at $d = g(c)$, ψ at c), so by Caratheodory h is differentiable at c with $h'(c) = \phi(g(c))\psi(c) = f'(g(c))g'(c)$. $\square$

Remark. Rudin proves the same theorem by writing $g(x) - g(c) = (x - c)[g'(c) + u(x)]$ and $f(y) - f(d) = (y - d)[f'(d) + v(y)]$ with $u \rightarrow 0$, $v \rightarrow 0$, then dividing. That works but must separately handle indices where $g(x) = g(c)$. The Caratheodory route never divides, so no case split is needed — this is its main advantage.

Method (Computing a derivative). *Recognition trigger.* “Differentiate f ” or “find $f'(c)$ ”.

1. If f is built from x^n , $\sin$, $\cos$, e^x , $\ln$ by $+$, $-$, $\times$, $\div$ and composition, apply the *algebra of derivatives* (6.1.3) and the *Chain Rule* (6.1.5). Differentiate from the outside in.
2. If f is defined by cases or involves $|\cdot|$, the rules apply away from the break points; at a break point go back to the *definition* and compute the one-sided difference quotients $\lim_{x \rightarrow c^\pm} \frac{f(x) - f(c)}{x - c}$ separately. They must agree.
3. If asked to *prove* differentiability of a non-elementary f at a point, form the difference quotient and estimate it directly (often with a squeeze).

Worked Example 6.1.6 (A product-chain computation). Differentiate $f(x) = \sqrt{5 - 2x + x^2}$.

Recognition. An elementary composition: outer $\sqrt{\cdot}$, inner quadratic. Chain rule.

Solution. Write $f = u^{1/2}$ with $u(x) = 5 - 2x + x^2$, $u'(x) = -2 + 2x$. Then $f'(x) = \frac{1}{2}u^{-1/2}u' = \frac{x - 1}{\sqrt{5 - 2x + x^2}}$.

Check. The quadratic $5 - 2x + x^2 = (x - 1)^2 + 4 \geq 4 > 0$, so f is differentiable on all of $\mathbb{R}$; $f' = 0$ exactly at the vertex $x = 1$, where f attains its minimum $\sqrt{4} = 2$. $\blacksquare$

Worked Example 6.1.7 (Differentiable but only at one point). Let $f(x) = x^2$ for $x \in \mathbb{Q}$ and $f(x) = 0$ for $x \notin \mathbb{Q}$. Show $f'(0) = 0$ exists, although f is discontinuous at every $c \neq 0$.

Recognition. A pathological f ; use the *definition* at $c = 0$ and a squeeze.

Solution. For $x \neq 0$, $\left| \frac{f(x) - f(0)}{x - 0} \right| = \left| \frac{f(x)}{x} \right| \leq \left| \frac{x^2}{x} \right| = |x| \rightarrow 0$, since $0 \leq f(x) \leq x^2$ for all x . Hence the difference quotient $\rightarrow 0$ and $f'(0) = 0$. *Check.* For $c \neq 0$, take rationals $\rightarrow c$ (values $\rightarrow c^2$) and irrationals $\rightarrow c$ (values 0); the two sequences force f to be discontinuous at c , so a fortiori not differentiable there. Differentiability at a single isolated point is possible. $\blacksquare$

Worked Example 6.1.8 (A corner: $|x|$ is not differentiable at 0). Show $f(x) = |x|$ has no derivative at 0.

Recognition. A piecewise-linear “corner”; compute one-sided difference quotients.

Solution. For $x > 0$, $\frac{|x| - 0}{x - 0} = \frac{x}{x} = 1$; for $x < 0$, $\frac{-x}{x} = -1$. The right limit is 1, the left limit is -1 ; they disagree, so $\lim_{x \rightarrow 0}$ does not exist and $f'(0)$ is undefined. *Check.* The graph has a 90° corner at the origin — no single tangent line. (Yet f is continuous, illustrating Theorem 6.1.2’s one-way street.) $\blacksquare$

The derivative of an inverse function

A strictly monotone continuous f on an interval has a continuous inverse (Chapter 5). When f is also differentiable with nonzero derivative, the inverse is differentiable too, with the reciprocal slope — exactly what the picture of reflecting the graph across $y = x$ suggests.

Theorem 6.1.9 (Derivative of the inverse). *Let $f : I \rightarrow \mathbb{R}$ be continuous and strictly monotone on the interval I , with $J = f(I)$ and inverse $g = f^{-1} : J \rightarrow I$. If f is differentiable at $c \in I$ and $f'(c) \neq 0$, then g is differentiable at $d = f(c)$ and*

$$g'(d) = \frac{1}{f'(c)} = \frac{1}{f'(g(d))}.$$

Proof. By Caratheodory, $f(x) - f(c) = \varphi(x)(x - c)$ with φ continuous at c and $\varphi(c) = f'(c) \neq 0$. Since f is injective, $\varphi(x) \neq 0$ for $x \neq c$ (otherwise $f(x) = f(c)$); and $\varphi(c) \neq 0$. For $y \in J$ put $x = g(y)$, $c = g(d)$; then

$$y - d = f(x) - f(c) = \varphi(g(y)) (g(y) - g(d)), \quad \text{so} \quad g(y) - g(d) = \frac{1}{\varphi(g(y))} (y - d).$$

The coefficient $y \mapsto 1/\varphi(g(y))$ is continuous at d (because g is continuous at d , φ continuous and nonzero at $c = g(d)$). By Caratheodory g is differentiable at d with $g'(d) = 1/\varphi(g(d)) = 1/\varphi(c) = 1/f'(c)$. $\square$

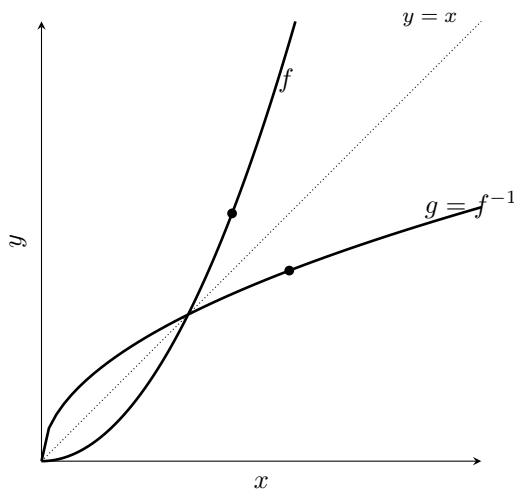


Figure 6.2. Reflecting the graph of f across $y = x$ gives the graph of $g = f^{-1}$; a tangent of slope m becomes a tangent of slope $1/m$.

Worked Example 6.1.10 (Derivative of arcsin). On $I = [-\frac{\pi}{2}, \frac{\pi}{2}]$, $\sin$ is strictly increasing with inverse $\arcsin$ on $[-1, 1]$. Compute $(\arcsin)'(y)$.

Recognition. “Differentiate an inverse trig function”; apply Theorem 6.1.9 with $f = \sin$, $f' = \cos$.

Solution. For $y = \sin x$, $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, we have $\cos x > 0$, so by Theorem 6.1.9,

$$(\arcsin)'(y) = \frac{1}{\cos x} = \frac{1}{\sqrt{1 - \sin^2 x}} = \frac{1}{\sqrt{1 - y^2}} \quad (-1 < y < 1).$$

Check. At $y = 0$ ($x = 0$) the slope is $1/\cos 0 = 1$, matching $1/\sqrt{1 - 0} = 1$. The formula blows up as $y \rightarrow \pm 1$, exactly where $\cos x \rightarrow 0$ and the inverse has a vertical tangent — so $\arcsin$ is *not* differentiable at ± 1 . $\blacksquare$

Worked Example 6.1.11 ($x^{1/n}$ from x^n , and failure at 0). For even $n \in \mathbb{N}$ let $f(x) = x^n$ on $[0, \infty)$, $g(y) = y^{1/n}$. Find g' .

Recognition. Inverse of a power; $f'(x) = nx^{n-1}$.

Solution. For $y > 0$ with $x = y^{1/n} > 0$, $g'(y) = \frac{1}{f'(x)} = \frac{1}{nx^{n-1}} = \frac{1}{ny^{(n-1)/n}} = \frac{1}{n} y^{(1/n)-1}$. *Check.* This is the usual power rule with exponent $1/n$. At $y = 0$, $f'(0) = 0$, so the theorem does not apply — and indeed $g(y) = y^{1/n}$ has a *vertical* tangent at 0 ($g'(0)$ does not exist). The hypothesis $f'(c) \neq 0$ is essential. $\blacksquare$

Caution. “Differentiable” is a *local* property checked by a limit; it is not the same as “the formula simplifies”. A function can be given by one tidy rule yet fail to be differentiable (e.g. $|x|$ at 0), and a function given by an ugly rule can be differentiable. Always trace differentiability back to the existence of the difference-quotient limit, especially at break points and endpoints. Also: $f'(c) \neq 0$ is *required* for the inverse-derivative formula — where the tangent is horizontal, the inverse has a vertical tangent and is not differentiable.

Further worked examples

Worked Example 6.1.12 (Continuous but not differentiable on a fattened corner). Let $f(x) = x \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Decide whether f is differentiable at 0.

Recognition. “Differentiable at a break point?”; go to the *definition* and form the difference quotient (Method, step 2). The rules do not apply at 0.

Solution. The difference quotient at 0 is $\frac{f(x) - f(0)}{x - 0} = \frac{x \sin(1/x)}{x} = \sin(1/x)$ for $x \neq 0$. As $x \rightarrow 0$, $\sin(1/x)$ oscillates through all of $[-1, 1]$ and has *no* limit (take $x_n = \frac{1}{n\pi} \rightarrow 0$ giving 0, and $x_n = \frac{1}{2\pi n + \pi/2} \rightarrow 0$ giving 1). So the limit defining $f'(0)$ does not exist and f is *not* differentiable at 0, even though $|f(x)| \leq |x| \rightarrow 0$ makes f continuous there. *Check.* Contrast with $x^2 \sin(1/x)$ (Worked Example 6.2.12), whose extra factor of x multiplies the quotient by $x \rightarrow 0$ and *does* give $f'(0) = 0$. The single power of x is exactly not enough. ■

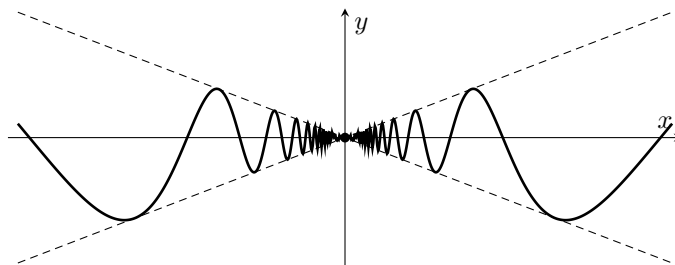


Figure 6.3. $f(x) = x \sin(1/x)$ is trapped between $y = \pm x$ (so continuous at 0), but the secant slope $\sin(1/x)$ swings between -1 and 1 infinitely often near 0: no tangent, no $f'(0)$.

Worked Example 6.1.13 (Differentiating a piecewise gluing). Let $f(x) = x^2$ for $x \leq 1$ and $f(x) = ax + b$ for $x > 1$. Find a, b making f differentiable at $x = 1$.

Recognition. “Glue two formulas smoothly”; matching value forces continuity, matching one-sided difference quotients forces differentiability.

Solution. *Continuity at 1:* $\lim_{x \rightarrow 1^-} x^2 = 1$ must equal $\lim_{x \rightarrow 1^+} (ax + b) = a + b$, so $a + b = 1$. *Differentiability:* the left derivative is $\left. \frac{d}{dx} x^2 \right|_1 = 2$ and the right derivative is a ; these must agree, so $a = 2$, whence $b = 1 - a = -1$. Thus $f(x) = 2x - 1$ for $x > 1$. *Check.* With $a = 2, b = -1$ the right piece is the tangent line to x^2 at $x = 1$ (value 1, slope 2); the graph continues straight out of the parabola with no corner. Any other a leaves a corner; any other b leaves a jump. ■

Worked Example 6.1.14 (The power rule from the definition (a non-shortcut check)). Compute $f'(c)$ from the definition for $f(x) = 1/x$ at $c \neq 0$, without quoting the quotient rule.

Recognition. “From the definition”; form $\frac{f(x) - f(c)}{x - c}$ and simplify algebraically before taking the limit.

Solution. For $x \neq c$ (both nonzero),

$$\frac{f(x) - f(c)}{x - c} = \frac{\frac{1}{x} - \frac{1}{c}}{x - c} = \frac{\frac{c-x}{xc}}{x - c} = \frac{-(x - c)}{xc(x - c)} = \frac{-1}{xc}.$$

As $x \rightarrow c$ this tends to $-1/c^2$, so $f'(c) = -1/c^2$. *Check.* The quotient rule gives $\frac{d}{dx} x^{-1} = -x^{-2}$, matching; e.g. at $c = 2$, $f'(2) = -1/4$, the slope of $1/x$ where it passes through $(2, \frac{1}{2})$. ■

Exercises (Rudin, Ch.5).

1. Suppose $f'(x) > 0$ on (a, b) . Prove that f is strictly increasing on (a, b) , and let g be its inverse. Prove that g is differentiable and that $g'(f(x)) = \frac{1}{f'(x)}$ for $a < x < b$.
2. Suppose g is a real function on $\mathbb{R}$ with bounded derivative, say $|g'| \leq M$. Fix $\varepsilon > 0$ and define $f(x) = x + \varepsilon g(x)$. Prove that f is one-to-one if ε is small enough. (A set of admissible ε can be found depending only on M .)
3. Suppose f' is continuous on $[a, b]$ and $\varepsilon > 0$. Prove there is $\delta > 0$ such that $\left| \frac{f(t) - f(x)}{t - x} - f'(x) \right| < \varepsilon$ whenever $0 < |t - x| < \delta$ and $a \leq x, t \leq b$. (This says f is *uniformly differentiable* on $[a, b]$ when f' is continuous.) Does it hold for vector-valued functions?
4. Let f be a continuous real function on $\mathbb{R}$ for which $f'(x)$ exists for all $x \neq 0$ and $f'(x) \rightarrow 3$ as $x \rightarrow 0$. Does it follow that $f'(0)$ exists?
5. If $f(x) = |x|^3$, compute $f'(x)$ and $f''(x)$ for all real x , and show that $f^{(3)}(0)$ does not exist.
6. Suppose a, c are real, $c > 0$, and f is defined on $[-1, 1]$ by $f(x) = x^a \sin(|x|^{-c})$ for $x \neq 0$ and $f(0) = 0$. Prove: (a) f is continuous iff $a > 0$; (b) $f'(0)$ exists iff $a > 1$; (c) f' is bounded iff $a \geq 1 + c$; (d) f' is continuous iff $a > 1 + c$; (e) $f''(0)$ exists iff $a > 2 + c$; (f) f'' is bounded iff $a \geq 2 + 2c$; (g) f'' is continuous iff $a > 2 + 2c$.

Exercises (Bartle & Sherbert, §6.1).

1. Use the definition to find the derivative of: (a) $f(x) = x^3$ ($x \in \mathbb{R}$); (b) $g(x) = 1/x$ ($x \neq 0$); (c) $h(x) = \sqrt{x}$ ($x > 0$); (d) $k(x) = 1/\sqrt{x}$ ($x > 0$).
2. Show that $f(x) = x^{1/3}$, $x \in \mathbb{R}$, is not differentiable at $x = 0$.
3. Prove the sum and product rules for derivatives (Theorem 6.1.3(a),(b)).
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$ for x rational and $f(x) = 0$ for x irrational. Show that f is differentiable at $x = 0$ and find $f'(0)$.
5. Differentiate and simplify: (a) $f(x) = \frac{x}{1+x^2}$; (b) $g(x) = \sqrt{5-2x+x^2}$; (c) $h(x) = (\sin x^k)^m$ for $m, k \in \mathbb{N}$; (d) $k(x) = \tan(x^2)$ for $|x| < \sqrt{\pi/2}$.
6. Let $n \in \mathbb{N}$ and $f(x) = x^n$ for $x \geq 0$, $f(x) = 0$ for $x < 0$. For which n is f' continuous at 0? For which n is f' differentiable at 0?
7. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at c and $f(c) = 0$. Show $g(x) = |f(x)|$ is differentiable at c iff $f'(c) = 0$.
8. Determine where each function from $\mathbb{R}$ to $\mathbb{R}$ is differentiable and find the derivative: (a) $f(x) = |x| + |x+1|$; (b) $g(x) = 2x + |x|$; (c) $h(x) = x|x|$; (d) $k(x) = |\sin x|$.
9. Prove that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is even and differentiable everywhere, then f' is odd; and if $g : \mathbb{R} \rightarrow \mathbb{R}$ is odd and differentiable, then g' is even.
10. Let $g(x) = x^2 \sin(1/x^2)$ for $x \neq 0$ and $g(0) = 0$. Show g is differentiable for all $x \in \mathbb{R}$, but the derivative g' is not bounded on $[-1, 1]$.
11. Assume there is a function $L : (0, \infty) \rightarrow \mathbb{R}$ with $L'(x) = 1/x$ for $x > 0$. Calculate the derivatives of: (a) $f(x) = L(2x+3)$; (b) $g(x) = (L(x^2))^3$; (c) $h(x) = L(ax)$ ($a > 0$); (d) $k(x) = L(L(x))$ when $L(x) > 0$.
12. If $r > 0$ is rational, let $f(x) = x^r \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Determine the values of r for which $f'(0)$ exists.
13. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at c , show $f'(c) = \lim n(f(c+1/n) - f(c))$. Show by example that existence of this sequential limit does not imply existence of $f'(c)$.
14. Given that $h(x) = x^3 + 2x + 1$ has an inverse h^{-1} on $\mathbb{R}$, find the value of $(h^{-1})'(y)$ at the points corresponding to $x = 0, 1, -1$.

15. Given that $\cos$ restricted to $I = [0, \pi]$ is strictly decreasing with $\cos 0 = 1$, $\cos \pi = -1$, let $J = [-1, 1]$ and let $\text{Arccos} : J \rightarrow \mathbb{R}$ be the inverse. Show Arccos is differentiable on $(-1, 1)$ with $D \text{Arccos } y = -1/(1 - y^2)^{1/2}$, and not differentiable at ± 1 .
16. Given that $\tan$ restricted to $I = (-\pi/2, \pi/2)$ is strictly increasing with $\tan(I) = \mathbb{R}$, let $\text{Arctan} : \mathbb{R} \rightarrow \mathbb{R}$ be the inverse. Show Arctan is differentiable on $\mathbb{R}$ with $D \text{Arctan } y = (1 + y^2)^{-1}$.
17. Let $f : I \rightarrow \mathbb{R}$ be differentiable at $c \in I$. Establish the *Straddle Lemma*: given $\varepsilon > 0$ there is $\delta > 0$ such that if $u, v \in I$ satisfy $c - \delta < u \leq c \leq v < c + \delta$, then $|f(v) - f(u) - (v - u)f'(c)| \leq \varepsilon(v - u)$. (Hint: subtract and add $f(c) - cf'(c)$ and use the triangle inequality.)

Additional exercises.

1. Using only the definition (no rules), compute $f'(c)$ for $f(x) = x^2$ at an arbitrary $c \in \mathbb{R}$, and for $f(x) = 1/x$ at an arbitrary $c \neq 0$.
2. Differentiate, simplifying: (a) $\frac{x \sin x}{1 + x^2}$; (b) $\sqrt{1 + \cos^2 x}$; (c) $e^{x^2} \ln(1 + x^2)$; (d) $(\arctan x)^3$.
3. (Caratheodory in action.) Use Caratheodory's theorem to give a one-line proof that if f is differentiable at c and $f(c) \neq 0$, then $1/f$ is differentiable at c with $(1/f)'(c) = -f'(c)/f(c)^2$. Identify the continuous factor φ .
4. Let $f(x) = x^2$ for $x \geq 0$ and $f(x) = -x^2$ for $x < 0$. Show f is differentiable on all of $\mathbb{R}$, find f' , and decide whether f' is differentiable at 0.
5. (Non-example.) Exhibit a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous everywhere, has a derivative at every point except one, and at that point has *different* one-sided derivatives. Prove your claims.
6. (Non-example.) Show that there is no differentiable $f : \mathbb{R} \rightarrow \mathbb{R}$ whose derivative is the Dirichlet function (1 on $\mathbb{Q}$, 0 off $\mathbb{Q}$). (Hint: a derivative has the intermediate value property; preview of §6.2.)
7. Let $g(y) = y^{1/3}$ on $\mathbb{R}$. Where is g differentiable? Compute g' there and explain, using Theorem 6.1.9, why g fails to be differentiable at 0 even though its inverse $y = x^3$ is differentiable everywhere.
8. (Diagram.) Sketch $f(x) = x^3 - 3x$ together with its tangent lines at $x = -1, 0, 1$. Mark the points where $f'(x) = 0$ and relate them to the local shape of the graph. State f' at each marked point.
9. (Harder.) Suppose f is differentiable on $\mathbb{R}$ with f' *bounded*, $|f'| \leq M$. Prove f is Lipschitz: $|f(x) - f(y)| \leq M|x - y|$ for all x, y . (This is the forward reference used in §6.2; prove it directly from the definition plus the result you will state as the MVT, or by a difference-quotient sup argument.)
10. (Harder.) Let $f(x) = x^2 \sin(1/x)$ for $x \neq 0$, $f(0) = 0$. Compute f' everywhere and show f is differentiable on $\mathbb{R}$ but f is not of class C^1 . Which theorem of §6.2 does this example illustrate the necessity of?

6.2 The Mean Value Theorem and Its Consequences

Interior extrema and Rolle's theorem

The single most useful fact about derivatives is that they vanish at interior maxima and minima. From it flow Rolle's theorem, the Mean Value Theorem, and through them every statement linking the *sign* of f' to the *behavior* of f .

Definition 6.2.1. $f : I \rightarrow \mathbb{R}$ has a *local (relative) maximum* at $c \in I$ if there is $\delta > 0$ with $f(x) \leq f(c)$ for all $x \in I$ with $|x - c| < \delta$; a *local minimum* is defined with $\geq$. A point c is *interior* to I if $(c - \delta, c + \delta) \subseteq I$ for some $\delta > 0$.

Theorem 6.2.2 (Interior Extremum Theorem). *If f has a local maximum or minimum at an interior point c of I and $f'(c)$ exists, then $f'(c) = 0$.*

Proof. Say f has a local max at c (the min case is identical with signs reversed). For x near c , $f(x) - f(c) \leq 0$. If $x > c$ the difference quotient $\frac{f(x) - f(c)}{x - c} \leq 0$, so $f'(c) = \lim_{x \rightarrow c^+} \leq 0$. If $x < c$ the quotient is ≥ 0 (negative over negative), so $f'(c) = \lim_{x \rightarrow c^-} \geq 0$. Both limits equal $f'(c)$, forcing $0 \leq f'(c) \leq 0$, i.e. $f'(c) = 0$. $\square$

The interior hypothesis is essential: on $[0, 1]$ the function $f(x) = x$ has its maximum at the endpoint 1, where $f' = 1 \neq 0$. A point with $f'(c) = 0$ is called *critical* (or *stationary*); the theorem says interior extrema are critical, not conversely ($f(x) = x^3$ has $f'(0) = 0$ but no extremum there).

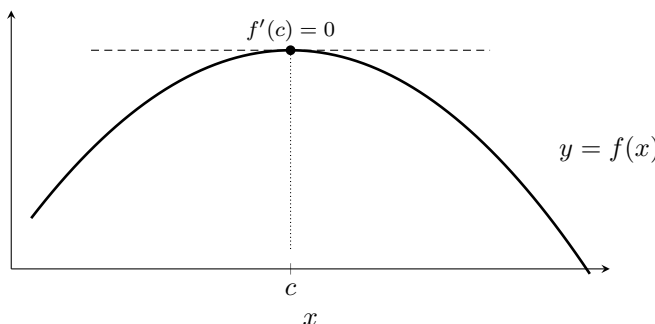


Figure 6.4. At an interior maximum the tangent is horizontal: the secant slopes change sign through c , pinning $f'(c) = 0$.

Theorem 6.2.3 (Rolle's Theorem). *Suppose f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$. Then there exists $c \in (a, b)$ with $f'(c) = 0$.*

Proof. f is continuous on the compact set $[a, b]$, so by the Extreme Value Theorem it attains a maximum M and a minimum m there. If $M = m$ then f is constant and $f' \equiv 0$ on (a, b) ; any c works. Otherwise $M > m$, so at least one of M, m differs from the common endpoint value $f(a) = f(b)$ and is therefore attained at an interior point $c \in (a, b)$. By Theorem 6.2.2, $f'(c) = 0$. $\square$

The mean value theorem

Rolle's theorem is the horizontal-chord case; tilting the picture gives the general theorem.

Theorem 6.2.4 (Mean Value Theorem). *If f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ with*

$$f'(c) = \frac{f(b) - f(a)}{b - a}, \quad \text{equivalently} \quad f(b) - f(a) = f'(c)(b - a).$$

Proof. Subtract off the chord. Let $\varphi(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$. Then φ is continuous on $[a, b]$, differentiable on (a, b) , and $\varphi(a) = 0 = \varphi(b)$ (direct check). By Rolle there is $c \in (a, b)$ with $\varphi'(c) = 0$; but $\varphi'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$, so $f'(c) = \frac{f(b) - f(a)}{b - a}$. $\square$

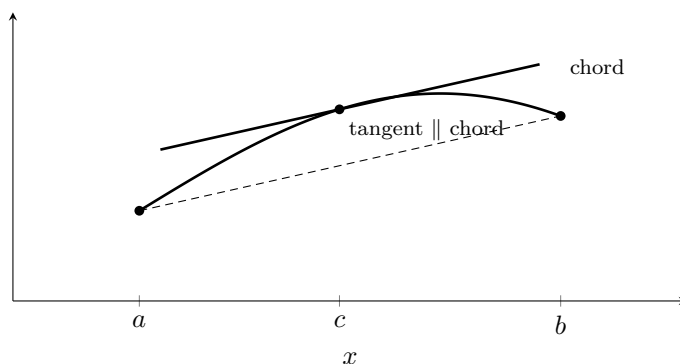


Figure 6.5. Mean Value Theorem: somewhere in (a, b) the tangent is parallel to the chord joining $(a, f(a))$ and $(b, f(b))$.

Theorem 6.2.5 (Cauchy's (Generalized) Mean Value Theorem). *If f, g are continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ with*

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

If moreover $g'(x) \neq 0$ on (a, b) (so $g(a) \neq g(b)$), this reads $\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$.

Proof. Apply Rolle to $h(x) = (f(b) - f(a))g(x) - (g(b) - g(a))f(x)$. It is continuous on $[a, b]$, differentiable on (a, b) , and one computes $h(a) = f(b)g(a) - g(b)f(a) = h(b)$. So there is $c \in (a, b)$ with $h'(c) = 0$, which is the displayed identity. (Taking $g(x) = x$ recovers the ordinary MVT.) $\square$

Method (Using the Mean Value Theorem). *Recognition trigger.* You must bound $f(b) - f(a)$ by something involving f' , prove an inequality between two expressions, or relate a function's growth to its derivative.

1. Identify the interval $[a, b]$ and check f continuous on it, differentiable inside.
2. Apply MVT: $f(b) - f(a) = f'(c)(b - a)$ for some unknown $c \in (a, b)$.
3. **Bound $f'(c)$.** You usually do not know c , so replace $f'(c)$ by its sup/inf over (a, b) : if $m \leq f' \leq M$ then $m(b - a) \leq f(b) - f(a) \leq M(b - a)$.
4. For ratios or two functions at once, use *Cauchy's* MVT instead.

Worked Example 6.2.6 ($|\sin x - \sin y| \leq |x - y|$). Prove $|\sin x - \sin y| \leq |x - y|$ for all $x, y \in \mathbb{R}$.

Recognition. "Bound a difference of values"; MVT with $f = \sin$.

Solution. If $x = y$ both sides are 0. Otherwise apply MVT to $\sin$ on $[x, y]$ (or $[y, x]$): $\sin x - \sin y = \cos(c)(x - y)$ for some c between x and y . Since $|\cos c| \leq 1$, $|\sin x - \sin y| = |\cos c||x - y| \leq |x - y|$. *Check.* Equality is approached as $x, y \rightarrow 0$ where $\cos c \rightarrow 1$; $\sin$ is thus 1-Lipschitz, hence uniformly continuous. ■

Worked Example 6.2.7 (A root-counting argument (Rolle)). Show $p(x) = x^3 + x - 1$ has exactly one real root.

Recognition. "Exactly one root"; existence by IVT, uniqueness by Rolle.

Solution. *Existence:* $p(0) = -1 < 0 < 1 = p(1)$, so by the Intermediate Value Theorem p has a root in $(0, 1)$. *Uniqueness:* if p had two roots $r_1 < r_2$ then by Rolle some $c \in (r_1, r_2)$ would have $p'(c) = 0$. But $p'(x) = 3x^2 + 1 \geq 1 > 0$ never vanishes; contradiction. So the root is unique. *Check.* $p' > 0$ means p is strictly increasing, which alone forces at most one root. ■

Sign of the derivative and monotonicity

Theorem 6.2.8 (Monotonicity test). *Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then on (a, b) :*
 (i) $f' \geq 0 \iff f$ nondecreasing; (ii) $f' \leq 0 \iff f$ nonincreasing; (iii) $f' > 0 \implies f$ strictly increasing, and $f' < 0 \implies f$ strictly decreasing; (iv) $f' \equiv 0 \iff f$ constant.

Proof. For any $a \leq x_1 < x_2 \leq b$, MVT gives $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$ with $c \in (x_1, x_2)$ and $x_2 - x_1 > 0$. Thus the sign of $f(x_2) - f(x_1)$ matches the sign of $f'(c)$. If $f' \geq 0$ throughout, every such difference is ≥ 0 (f nondecreasing); if $f' > 0$, every difference is > 0 (f strictly increasing); similarly for the decreasing cases; and $f' \equiv 0$ forces all differences 0, i.e. f constant. The converses in (i),(ii),(iv) follow since a nondecreasing differentiable function has all difference quotients ≥ 0 , hence $f' \geq 0$. $\square$

Remark. The implication in (iii) is one-way: $f(x) = x^3$ is strictly increasing yet $f'(0) = 0$. Strict monotonicity allows the derivative to vanish at isolated points.

Worked Example 6.2.9 (An inequality via monotonicity). Show $\frac{x-1}{x} < \ln x < x-1$ for all $x > 1$.

Recognition. “Prove an inequality for all $x > 1$ ”; build an auxiliary function and show it is monotone with a known starting value.

Solution. Let $h(x) = \ln x - \frac{x-1}{x} = \ln x - 1 + \frac{1}{x}$. Then $h(1) = 0$ and $h'(x) = \frac{1}{x} - \frac{1}{x^2} = \frac{x-1}{x^2} > 0$ for $x > 1$; so h is strictly increasing and $h(x) > h(1) = 0$, giving the left inequality. Let $k(x) = (x-1) - \ln x$; $k(1) = 0$, $k'(x) = 1 - \frac{1}{x} > 0$ for $x > 1$, so $k(x) > 0$, giving the right inequality. *Check.* At $x = e$: $\frac{e-1}{e} \approx 0.632 < 1 < e-1 \approx 1.718$. ■

Darboux’s theorem: derivatives have the intermediate value property

A derivative need not be continuous (see Worked Example 6.2.12 below). Remarkably, it must still satisfy the conclusion of the Intermediate Value Theorem.

Theorem 6.2.10 (Darboux’s Theorem). *Let f be differentiable on $[a, b]$ and let k be any number strictly between $f'(a)$ and $f'(b)$. Then there exists $c \in (a, b)$ with $f'(c) = k$.*

Proof. Suppose $f'(a) < k < f'(b)$ (the other order is symmetric). Set $g(x) = f(x) - kx$, so g is differentiable on $[a, b]$ with $g' = f' - k$; thus $g'(a) = f'(a) - k < 0$ and $g'(b) = f'(b) - k > 0$. By the Extreme Value Theorem g attains a minimum on $[a, b]$ at some c . Now $g'(a) < 0$ means g is decreasing at a , so the minimum is not at a ; $g'(b) > 0$ means g is increasing at b , so it is not at b . Hence $c \in (a, b)$ is interior, and by Theorem 6.2.2 $g'(c) = 0$, i.e. $f'(c) = k$. □

Remark. The standard sharp form of “decreasing at a ”: from $g'(a) < 0$, the limit $\lim_{x \rightarrow a^+} \frac{g(x) - g(a)}{x - a} < 0$ gives points $x > a$ with $g(x) < g(a)$, so $g(a)$ is not the minimum. Symmetrically at b .

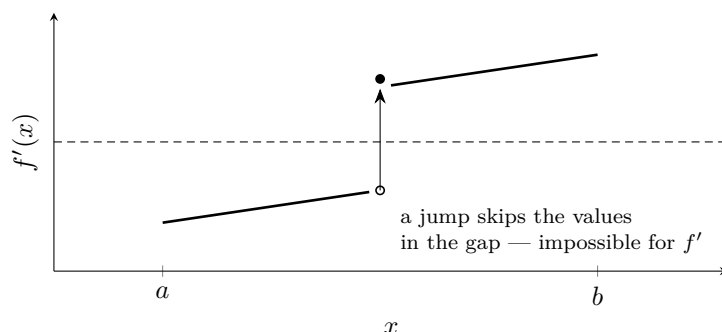


Figure 6.6. By Darboux’s theorem a derivative cannot have a jump discontinuity: it would omit the intermediate values it is forced to take. So f' , if discontinuous, is never “jump”-discontinuous — only oscillatory.

Worked Example 6.2.11 (sgn is not a derivative). Show the step function $h(x) = 0$ for $x < 0$, $h(x) = 1$ for $x \geq 0$ is not f' for any f on $\mathbb{R}$.

Recognition. “Show a given function is not a derivative”; check the intermediate value property fails, then invoke Darboux.

Solution. h takes values 0 and 1 but skips every value in $(0, 1)$: there is no x with $h(x) = \frac{1}{2}$. If $h = f'$ for some differentiable f on $[-1, 1]$, Darboux’s theorem would force f' to take the value $\frac{1}{2}$ between $f'(-1) = 0$ and $f'(1) = 1$, contradiction. So no such f exists. *Check.* The obstruction is precisely the *jump*; the antiderivative-looking $\max(x, 0)$ has derivative h only away from 0, and fails to be differentiable at 0. ■

Worked Example 6.2.12 (A derivative that is discontinuous). Let $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Show $f'(0) = 0$ but f' is discontinuous at 0 — yet (consistent with Darboux) f' has no jump.

Recognition. “ f' exists but is not continuous”; compute $f'(0)$ by definition and $f'(x)$ by rules, then compare.

Solution. By definition $f'(0) = \lim_{x \rightarrow 0} \frac{x^2 \sin(1/x)}{x} = \lim_{x \rightarrow 0} x \sin(1/x) = 0$ (since $|x \sin(1/x)| \leq |x|$). For $x \neq 0$, the product and chain rules give $f'(x) = 2x \sin(1/x) - \cos(1/x)$. As $x \rightarrow 0$ the term $2x \sin(1/x) \rightarrow 0$ but $\cos(1/x)$ oscillates in $[-1, 1]$ without limit, so $\lim_{x \rightarrow 0} f'(x)$ does not exist: f' is discontinuous at 0. *Check.* The discontinuity is *oscillatory*, not a jump — f' still takes every value in $[-1, 1]$ on every neighborhood of 0, in keeping with Darboux. ■

Caution. f' exists everywhere does *not* imply f' is continuous. The class of derivatives sits strictly between continuous functions and arbitrary functions: every continuous function is a derivative (it has an antiderivative), every derivative has the intermediate value property (Darboux), but some derivatives are discontinuous (Worked Example 6.2.12). When you need “ f' continuous” you must *assume* it ($f \in C^1$); it is not free.

Further worked examples

Worked Example 6.2.13 (A square-root estimate by the MVT). Estimate $\sqrt{105}$ using the Mean Value Theorem, and bound the error.

Recognition. “Estimate $f(b)$ from a nearby known $f(a)$ ”; MVT with $f(b) - f(a) = f'(c)(b - a)$ and a bound on $f'(c)$.

Solution. Let $f(x) = \sqrt{x}$ on $[100, 105]$, so $f'(x) = \frac{1}{2\sqrt{x}}$. By MVT there is $c \in (100, 105)$ with

$$\sqrt{105} - \sqrt{100} = f'(c)(105 - 100) = \frac{5}{2\sqrt{c}}.$$

Since $100 < c < 105$, $\frac{1}{2\sqrt{105}} < \frac{1}{2\sqrt{c}} < \frac{1}{2\sqrt{100}} = \frac{1}{20}$, so $\sqrt{105} = 10 + \frac{5}{2\sqrt{c}}$ lies strictly between $10 + \frac{5}{2 \cdot 10.247} \approx 10.244$ and $10 + \frac{5}{20} = 10.25$. *Check.* $\sqrt{105} = 10.2470\dots$, comfortably inside $(10.244, 10.25)$. The MVT converts “unknown c ” into a tight *interval* for the answer. ■

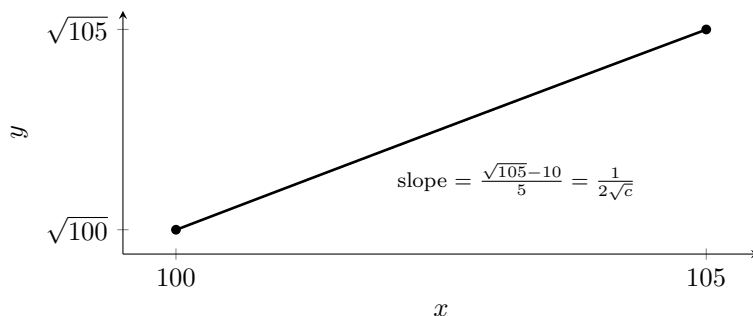


Figure 6.7. MVT for $\sqrt{x}$ on $[100, 105]$: the chord slope equals $1/(2\sqrt{c})$ for some $c \in (100, 105)$, trapping $\sqrt{105}$ in $(10.244, 10.25)$.

Worked Example 6.2.14 (A function with bounded derivative is Lipschitz). Suppose f is differentiable on $\mathbb{R}$ with $|f'(x)| \leq M$ for all x . Show $|f(x) - f(y)| \leq M|x - y|$ for all x, y .

Recognition. “Bound a difference of values by the derivative bound”; this is the canonical MVT application (Method, step 3).

Solution. Fix $x \neq y$. By MVT on the interval with endpoints x, y there is c strictly between them with $f(x) - f(y) = f'(c)(x - y)$. Taking absolute values and using $|f'(c)| \leq M$,

$$|f(x) - f(y)| = |f'(c)| |x - y| \leq M |x - y|.$$

For $x = y$ both sides are 0, so the bound holds for all x, y . *Check.* For $f = \sin$, $M = 1$ recovers $|\sin x - \sin y| \leq |x - y|$; for $f(x) = \arctan x$, $|f'| = \frac{1}{1+x^2} \leq 1$ gives the 1-Lipschitz bound $|\arctan x - \arctan y| \leq |x - y|$. A bounded derivative is exactly a Lipschitz (hence uniformly continuous) function. ■

Worked Example 6.2.15 (When MVT does NOT apply: a missing hypothesis). Does the Mean Value Theorem produce a $c \in (-1, 1)$ with $f'(c) = \frac{f(1) - f(-1)}{2}$ for $f(x) = |x|$?

Recognition. A *non-example*: check the hypotheses of MVT *before* applying it.

Solution. Here $f(1) = f(-1) = 1$, so the target slope is $\frac{1-1}{2} = 0$; MVT would demand a point with $f'(c) = 0$. But $f(x) = |x|$ is *not* differentiable at $0 \in (-1, 1)$, so the hypothesis “differentiable on (a, b) ” fails and the theorem does not apply. And indeed $f'(x) = -1$ for $x < 0$, $f'(x) = +1$ for $x > 0$, and $f'(0)$ does not exist: there is *no* point where $f' = 0$. *Check.* The conclusion genuinely fails here, which is consistent — MVT only promises a c when its hypotheses hold. One non-differentiable interior point is enough to break it. ■

Exercises (Rudin, Ch.5).

- Let f be defined for all real x and suppose $|f(x) - f(y)| \leq (x - y)^2$ for all real x, y . Prove that f is constant.
- If $C_0 + \frac{C_1}{2} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0$, where $C_0, \dots, C_n$ are real constants, prove that the equation $C_0 + C_1x + \cdots + C_{n-1}x^{n-1} + C_nx^n = 0$ has at least one real root in $(0, 1)$.
- Suppose f is defined and differentiable for every $x > 0$ and $f'(x) \rightarrow 0$ as $x \rightarrow +\infty$. Put $g(x) = f(x+1) - f(x)$. Prove that $g(x) \rightarrow 0$ as $x \rightarrow +\infty$.
- Suppose (a) f is continuous for $x \geq 0$, (b) $f'(x)$ exists for $x > 0$, (c) $f(0) = 0$, (d) f' is monotonically increasing. Put $g(x) = \frac{f(x)}{x}$ for $x > 0$ and prove that g is monotonically increasing.
- Let f be a differentiable real function on (a, b) . Prove that f is convex iff f' is monotonically increasing. If f'' exists on (a, b) , prove f is convex iff $f''(x) \geq 0$ for all $x \in (a, b)$.
- Suppose f is a real function on $(-\infty, \infty)$. Call x a fixed point of f if $f(x) = x$. (a) If f is differentiable and $f'(t) \neq 1$ for every t , prove f has at most one fixed point. (b) Show $f(t) = t + (1 + e^t)^{-1}$ has no fixed point although $0 < f'(t) < 1$ for all t . (c) If there is a constant $A < 1$ with $|f'(t)| \leq A$ for all t , prove a fixed point x of f exists and $x = \lim x_n$, where x_1 is arbitrary and $x_{n+1} = f(x_n)$. (d) Show the process in (c) can be visualized by a zig-zag path.
- The function $f(x) = \frac{x^3 + 1}{3}$ has three fixed points α, β, γ with $-2 < \alpha < -1$, $0 < \beta < 1$, $1 < \gamma < 2$. For arbitrary x_1 define $x_{n+1} = f(x_n)$. (a) If $x_1 < \alpha$, prove $x_n \rightarrow -\infty$. (b) If $\alpha < x_1 < \gamma$, prove $x_n \rightarrow \beta$. (c) If $x_1 > \gamma$, prove $x_n \rightarrow +\infty$. Thus β can be located this way, but α, γ cannot.
- The process of Exercise 6(c) also applies to maps of $(0, \infty)$ to $(0, \infty)$. Fix $\alpha > 1$ and put $g(x) = \frac{\alpha + x}{1 + x}$. Both $f(x) = \frac{1}{2}(x + \alpha/x)$ and g have $\sqrt{\alpha}$ as their only fixed point in $(0, \infty)$. Using properties of f and g , explain why the convergence to $\sqrt{\alpha}$ for f is so much faster than for g . (Compare f' and g' ; draw the zig-zags.) Do the same when $0 < \alpha < 1$.
- Suppose f is twice differentiable on $[a, b]$, $f(a) < 0$, $f(b) > 0$, $f'(x) \geq \delta > 0$, and $0 \leq f''(x) \leq M$ on $[a, b]$. Let ξ be the unique point in (a, b) with $f(\xi) = 0$. Complete the analysis of Newton's method $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$: (a) interpret it geometrically via a tangent line; (b) prove $x_{n+1} < x_n$ and $\lim x_n = \xi$ if $x_1 \in (\xi, b)$; (c) using Taylor's theorem show $x_{n+1} - \xi = \frac{f''(t_n)}{2f'(x_n)}(x_n - \xi)^2$ for some $t_n \in (\xi, x_n)$; (d) if $A = \frac{M}{2\delta}$, deduce $0 \leq x_{n+1} - \xi \leq \frac{1}{A}[A(x_1 - \xi)]^{2^n}$; (e) show Newton's method amounts to finding a fixed point of $g(x) = x - \frac{f(x)}{f'(x)}$ and study g' near ξ ; (f) put $f(x) = x^{1/3}$ on $\mathbb{R}$ and try Newton's method — what happens?

Exercises (Bartle & Sherbert, §6.2).

- For each function, find the points of relative extrema, the intervals where it is increasing, and those where it is decreasing: (a) $f(x) = x^2 - 3x + 5$; (b) $g(x) = 3x - 4x^2$; (c) $h(x) = x^3 - 3x - 4$; (d) $k(x) = x^4 + 2x^2 - 4$.
- Same task for: (a) $f(x) = x + 1/x$ ($x \neq 0$); (b) $g(x) = x/(x^2 + 1)$; (c) $h(x) = x - 2\sqrt{x} + 2$ for $x > 0$; (d) $k(x) = 2x + 1/x^2$ ($x \neq 0$).
- Find the points of relative extrema on the given domain: (a) $f(x) = x^2 - 1$, $-4 \leq x \leq 4$; (b) $g(x) = 1 - (x - 1)^{2/3}$, $0 \leq x \leq 2$; (c) $h(x) = x|x^2 - 12|$, $-2 \leq x \leq 3$; (d) $k(x) = x(x - 8)^{1/3}$, $0 \leq x \leq 9$.
- Let $a_1, \dots, a_n \in \mathbb{R}$ and $f(x) = \sum_{i=1}^n (a_i - x)^2$ for $x \in \mathbb{R}$. Find the unique point of relative minimum of f .
- Let $a > b > 0$ and $n \in \mathbb{N}$, $n \geq 2$. Prove $a^{1/n} - b^{1/n} < (a - b)^{1/n}$. (Hint: show $f(x) = x^{1/n} - (x - 1)^{1/n}$ is decreasing for $x \geq 1$ and evaluate at 1 and a/b .)
- Use the Mean Value Theorem to prove $|\sin x - \sin y| \leq |x - y|$ for all $x, y \in \mathbb{R}$.
- Use the Mean Value Theorem to prove $(x - 1)/x < \ln x < x - 1$ for $x > 1$. (Hint: $D \ln x = 1/x$.)
- Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$, differentiable on (a, b) . Show that if $\lim_{x \rightarrow a^+} f'(x) = A$, then $f'(a)$ exists and equals A .
- Let $f(x) = 2x^4 + x^4 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Show f has an absolute minimum at 0, but f' takes both positive and negative values in every neighborhood of 0.
- Let $g(x) = x + 2x^2 \sin(1/x)$ for $x \neq 0$ and $g(0) = 0$. Show $g'(0) = 1$, but g' takes both positive and negative values in every neighborhood of 0; thus g is monotone in no neighborhood of 0.
- Give an example of a uniformly continuous function on $[0, 1]$, differentiable on $(0, 1)$, whose derivative is unbounded on $(0, 1)$.
- If $h(x) = 0$ for $x < 0$ and $h(x) = 1$ for $x \geq 0$, prove there is no $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f' = h$. Give two functions, not differing by a constant, whose derivatives equal $h(x)$ for all $x \neq 0$.
- Let I be an interval and $f : I \rightarrow \mathbb{R}$ differentiable. Show that if $f' > 0$ on I , then f is strictly increasing.
- Let $f : I \rightarrow \mathbb{R}$ be differentiable and f' never 0 on I . Show that either $f'(x) > 0$ for all $x \in I$ or $f'(x) < 0$ for all $x \in I$.
- Let I be an interval. Prove that if f is differentiable on I and f' is bounded on I , then f satisfies a Lipschitz condition on I .
- Let $f : [0, \infty) \rightarrow \mathbb{R}$ be differentiable on $(0, \infty)$ with $f'(x) \rightarrow b$ as $x \rightarrow \infty$. (a) Show $\lim_{x \rightarrow \infty} (f(x + h) - f(x))/h = b$ for any $h > 0$. (b) If also $f(x) \rightarrow a$ as $x \rightarrow \infty$, show $b = 0$. (c) Show $\lim_{x \rightarrow \infty} f(x)/x = b$.
- Let f, g be differentiable on $\mathbb{R}$ with $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for all $x \geq 0$. Show $f(x) \leq g(x)$ for all $x \geq 0$.
- Let $I = [a, b]$ and $f : I \rightarrow \mathbb{R}$ be differentiable at $c \in I$. Show that for every $\varepsilon > 0$ there is $\delta > 0$ such that if $0 < |x - y| < \delta$ and $a \leq x \leq c \leq y \leq b$, then $\left| \frac{f(x) - f(y)}{x - y} - f'(c) \right| < \varepsilon$.
- $f : I \rightarrow \mathbb{R}$ is *uniformly differentiable* on $I = [a, b]$ if for every $\varepsilon > 0$ there is $\delta > 0$ such that $0 < |x - y| < \delta$, $x, y \in I$ imply $\left| \frac{f(x) - f(y)}{x - y} - f'(x) \right| < \varepsilon$. Show that if f is uniformly differentiable on I , then f' is continuous on I .
- Suppose $f : [0, 2] \rightarrow \mathbb{R}$ is continuous on $[0, 2]$, differentiable on $(0, 2)$, with $f(0) = 0$, $f(1) = 1$, $f(2) = 1$. Show: (a) there is $c_1 \in (0, 1)$ with $f'(c_1) = 1$; (b) there is $c_2 \in (1, 2)$ with $f'(c_2) = 0$; (c) there is $c \in (0, 2)$ with $f'(c) = 1/3$.

Additional exercises.

- Find all critical points of $f(x) = x^3 - 6x^2 + 9x + 1$ and classify each as a local max, local min, or neither using the sign of f' .

2. Use the Mean Value Theorem to prove $e^x \geq 1 + x$ for all $x \in \mathbb{R}$, with equality only at $x = 0$.
3. (Rolle, root counting.) Show that $p(x) = x^5 + x^3 + x - 3$ has exactly one real root.
4. Prove that if f is differentiable on an interval I and $|f'(x)| \leq L$ on I , then f is L -Lipschitz on I . Conversely, if f is L -Lipschitz and differentiable, show $|f'| \leq L$.
5. (Non-example.) Give a function f continuous on $[0, 1]$, differentiable on $(0, 1)$, with $f(0) = f(1)$ but for which no point of *equal one-sided derivatives* is forced — then explain why this does *not* contradict Rolle, by exhibiting the genuine interior point c with $f'(c) = 0$. (Use $f(x) = \sin(2\pi x)$.)
6. (Non-example.) Show by an explicit f on $[-1, 1]$ that the hypothesis “differentiable on the open interval” in the MVT cannot be relaxed to “differentiable except at one interior point”. (Hint: $f(x) = |x|$.)
7. (Cauchy MVT.) Let $f(x) = x^2$ and $g(x) = x^3$ on $[1, 2]$. Find a point $c \in (1, 2)$ with $\frac{f'(c)}{g'(c)} = \frac{f(2) - f(1)}{g(2) - g(1)}$ explicitly, and verify it.
8. (Diagram.) For $f(x) = 1 - x^2$ on $[-1, 1]$, draw the chord from $(-1, 0)$ to $(1, 0)$ and the tangent guaranteed by the MVT. Identify the point c and the tangent’s slope; explain why this is the Rolle case.
9. (Darboux.) Define $f(x) = x^2 \sin(1/x) + \frac{1}{2}x$ for $x \neq 0$ and $f(0) = 0$. Show $f'(0) = \frac{1}{2} > 0$, yet f' takes negative values in every neighborhood of 0, so f is monotone on *no* neighborhood of 0. Reconcile with Darboux’s theorem.
10. (Harder.) Suppose f is differentiable on $(0, \infty)$ and $f'(x) \rightarrow L$ as $x \rightarrow \infty$. Prove $f(x)/x \rightarrow L$.

6.3 L’Hospital’s Rules

Indeterminate forms

When $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow c$, the quotient f/g has the *indeterminate form* $0/0$: it may converge to any value or diverge, depending on the relative rates of decay. The same happens for ∞/∞ . L’Hospital’s rule replaces the quotient of *values* by the quotient of *derivatives* — legitimate because Cauchy’s Mean Value Theorem turns one into the other.

Theorem 6.3.1 (L’Hospital, $0/0$, finite point). *Let f, g be defined on $[a, b]$ with $f(a) = g(a) = 0$ and $g(x) \neq 0$ for $a < x \leq b$. If f, g are differentiable at a with $g'(a) \neq 0$, then*

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}.$$

Proof. Since $f(a) = g(a) = 0$, divide numerator and denominator difference quotients:

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{\frac{f(x) - f(a)}{x - a}}{\frac{g(x) - g(a)}{x - a}} \xrightarrow{x \rightarrow a^+} \frac{f'(a)}{g'(a)},$$

using $g'(a) \neq 0$ so the denominator limit is nonzero. (This elementary version needs only differentiability *at* the single point a .) $\square$

The strong form, which Rudin and B&S both give, requires only differentiability on the open interval and covers infinite limits and the ∞/∞ case. Its engine is Cauchy’s MVT.

Theorem 6.3.2 (L’Hospital’s Rule, general form). *Let $-\infty \leq a < b \leq +\infty$ and let f, g be differentiable on (a, b) with $g'(x) \neq 0$ there. Suppose*

$$\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \in [-\infty, +\infty].$$

If either $(f(x) \rightarrow 0 \text{ and } g(x) \rightarrow 0)$ or $(g(x) \rightarrow +\infty)$ as $x \rightarrow a^+$, then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

The analogous statement holds for $x \rightarrow b^-$.

Proof (the 0/0 case, L finite, a finite). Let $\varepsilon > 0$. Choose $a < \delta$ so that $a < x < \delta \implies \left| \frac{f'(t)}{g'(t)} - L \right| < \varepsilon$ for all $t \in (a, x)$. Fix such x and y with $a < y < x < \delta$. Since $g' \neq 0$ on (a, b) , Rolle's theorem forbids $g(x) = g(y)$, so by Cauchy's Mean Value Theorem (Theorem 6.2.5) there is $t \in (y, x)$ with

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(t)}{g'(t)}, \quad \text{hence} \quad \left| \frac{f(x) - f(y)}{g(x) - g(y)} - L \right| < \varepsilon.$$

Now let $y \rightarrow a^+$. In the 0/0 case $f(y) \rightarrow 0$ and $g(y) \rightarrow 0$, so the left ratio tends to $\frac{f(x)}{g(x)}$, giving $\left| \frac{f(x)}{g(x)} - L \right| \leq \varepsilon$

for all $a < x < \delta$. As $\varepsilon > 0$ was arbitrary, $\frac{f(x)}{g(x)} \rightarrow L$. The ∞/∞ case is handled by the same Cauchy-MVT estimate, holding y fixed near the point where f'/g' is already close to L and letting $x \rightarrow a^+$ so that $g(x) \rightarrow \infty$ dominates the bounded leftover terms; the infinite- L and infinite-endpoint cases reduce to this by routine substitutions. (The full case analysis is in Rudin 5.13.) $\square$

Method (Applying L'Hospital's rule). *Recognition trigger.* A limit $\lim f/g$ that, on substitution, gives 0/0 or ∞/∞ .

1. **Confirm the form is indeterminate.** If it is 0/0 or ∞/∞ , proceed; otherwise do *not* differentiate (L'Hospital does not apply to e.g. 0/1).
2. **Differentiate top and bottom separately** (*not* the quotient!) and take the limit of f'/g' .
3. If f'/g' is still indeterminate, repeat. If it never simplifies, abandon L'Hospital and try another method (algebra, series, squeeze).
4. **Other forms.** Convert $0 \cdot \infty$ to 0/0 or ∞/∞ by moving a factor to the denominator; convert $\infty - \infty$ to a single fraction; convert 1^∞ , 0^0 , ∞^0 by taking $\ln$ and applying L'Hospital to the exponent, then exponentiating.

Worked Example 6.3.3 (The basic 0/0 limit). Evaluate $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{1 - \cos x}$.

Recognition. Substituting $x = 0$ gives 0/0; differentiate top and bottom.

Solution. Both numerator and denominator vanish at 0. Differentiate: $\frac{e^x - e^{-x}}{\sin x}$, still 0/0; again: $\frac{e^x + e^{-x}}{\cos x} \rightarrow \frac{1+1}{1} = 2$. So the limit is 2. *Check.* For small x , numerator $\approx x^2$ (from $e^{\pm x} \approx 1 \pm x + x^2/2$) and denominator $\approx x^2/2$, ratio ≈ 2 . $\blacksquare$

Worked Example 6.3.4 (An ∞/∞ limit). Evaluate $\lim_{x \rightarrow +\infty} \frac{x^3}{e^{3x}}$.

Recognition. ∞/∞ ; repeated L'Hospital drives the polynomial down.

Solution. Each differentiation lowers the power of x and keeps e^{3x} : $\frac{x^3}{e^{3x}} \rightarrow \frac{3x^2}{3e^{3x}} \rightarrow \frac{6x}{9e^{3x}} \rightarrow \frac{6}{27e^{3x}} \rightarrow 0$. So the limit is 0. *Check.* Exponentials beat polynomials — a recurring theme. $\blacksquare$

Worked Example 6.3.5 (A $0 \cdot \infty$ form, reshaped). Evaluate $\lim_{x \rightarrow 0^+} x \ln x$.

Recognition. $0 \cdot (-\infty)$; rewrite as ∞/∞ or 0/0 before differentiating.

Solution. Write $x \ln x = \frac{\ln x}{1/x}$, of form $-\infty/\infty$. L'Hospital: $\frac{1/x}{-1/x^2} = -x \rightarrow 0$. So $\lim_{x \rightarrow 0^+} x \ln x = 0$. *Check.* The algebraic factor x overwhelms the logarithmic blow-up; e.g. $x = 0.01$ gives $0.01 \cdot \ln(0.01) \approx -0.046$, small. $\blacksquare$

Worked Example 6.3.6 (A 1^∞ form: the number e). Evaluate $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x$.

Recognition. 1^∞ ; take logarithms and apply L'Hospital to the exponent.

Solution. Let $y = (1 + 1/x)^x$, so $\ln y = x \ln(1 + 1/x) = \frac{\ln(1 + 1/x)}{1/x}$, of form $0/0$. L'Hospital (differentiating in x):

$$\frac{(1 + 1/x)^{-1}(-1/x^2)}{-1/x^2} = \frac{1}{1 + 1/x} \rightarrow 1.$$

So $\ln y \rightarrow 1$, and since $t \mapsto e^t$ is continuous, $y \rightarrow e^1 = e$. *Check.* $x = 100$ gives $(1.01)^{100} \approx 2.705$, near $e \approx 2.71828$. ■

Caution. Check the form first. L'Hospital applies *only* to $0/0$ and ∞/∞ . Applying it to a determinate quotient gives wrong answers: $\lim_{x \rightarrow 0} \frac{x+1}{x+2} = \frac{1}{2}$, but blindly differentiating gives $\frac{1}{1} = 1$. Also: differentiate numerator and denominator *separately*; never apply the quotient rule. Finally, if f'/g' has *no* limit, the rule is silent — it does *not* tell you f/g diverges. The classic trap is $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x} = 1$ while $\frac{1 + \cos x}{1}$ has no limit; the original limit still exists.

Further worked examples

Worked Example 6.3.7 (An $\infty - \infty$ form, reshaped). Evaluate $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin x}\right)$.

Recognition. $\infty - \infty$; combine into a single fraction to expose a $0/0$ form, then L'Hospital (Method, step 4).

Solution. Combine: $\frac{1}{x} - \frac{1}{\sin x} = \frac{\sin x - x}{x \sin x}$, which is $0/0$ as $x \rightarrow 0$. Differentiate top and bottom:

$$\frac{\cos x - 1}{\sin x + x \cos x},$$

still $0/0$; again:

$$\frac{-\sin x}{2 \cos x - x \sin x} \xrightarrow{x \rightarrow 0} \frac{0}{2 - 0} = 0.$$

So the limit is 0. *Check.* For small x , $\sin x \approx x - \frac{x^3}{6}$, so $\sin x - x \approx -\frac{x^3}{6}$ and $x \sin x \approx x^2$, giving ratio $\approx -\frac{x}{6} \rightarrow 0$. ■

Worked Example 6.3.8 (A 0^0 form via logarithms). Evaluate $\lim_{x \rightarrow 0^+} x^x$.

Recognition. 0^0 ; take $\ln$, apply L'Hospital to the exponent, then exponentiate (Method, step 4).

Solution. Let $y = x^x$, so $\ln y = x \ln x$. We already know (from the $0 \cdot \infty$ worked example above) that $\ln y = \frac{\ln x}{1/x} \xrightarrow{x \rightarrow 0^+} 0$ by L'Hospital. Since $t \mapsto e^t$ is continuous, $y = e^{\ln y} \rightarrow e^0 = 1$. So $\lim_{x \rightarrow 0^+} x^x = 1$. *Check.* Numerically $0.1^{0.1} = 10^{-0.1} \approx 0.794$, $0.01^{0.01} \approx 0.955$, $0.001^{0.001} \approx 0.993$ — climbing toward 1. ■

Worked Example 6.3.9 (When L'Hospital fails to terminate (a non-example)). Try to evaluate

$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 + 1}}{x}$ by L'Hospital, then by algebra.

Recognition. An ∞/∞ form where repeated differentiation *cycles* and never simplifies — the signal to abandon L'Hospital and use algebra instead (Method, step 3).

Solution. *L'Hospital:* $\frac{d}{dx} \sqrt{x^2 + 1} = \frac{x}{\sqrt{x^2 + 1}}$ and $\frac{d}{dx} x = 1$, giving $\frac{x}{\sqrt{x^2 + 1}}$ — which is the *reciprocal* of the original and again ∞/∞ ; differentiating again returns us to the start. L'Hospital spins in a loop and yields nothing. *Algebra:* for $x > 0$, $\frac{\sqrt{x^2 + 1}}{x} = \sqrt{\frac{x^2 + 1}{x^2}} = \sqrt{1 + \frac{1}{x^2}} \rightarrow \sqrt{1} = 1$. So the limit is 1. *Check.* At $x = 10$, $\sqrt{101}/10 \approx 1.005$, near 1. The lesson: L'Hospital is not always the right tool, and a determinate-looking algebraic simplification can beat it. ■

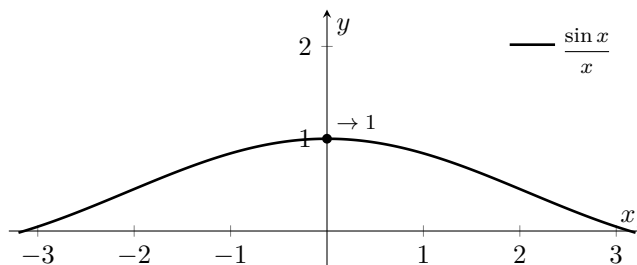


Figure 6.8. The model $0/0$ limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$: the removable “hole” at $x = 0$ is filled by the value $f'(0)/g'(0) = \cos 0/1 = 1$ that L'Hospital predicts.

Exercises (Rudin, Ch.5).

1. Suppose $f'(x)$ and $g'(x)$ exist, $g'(x) \neq 0$, and $f(x) = g(x) = 0$. Prove that $\lim_{t \rightarrow x} \frac{f(t)}{g(t)} = \frac{f'(x)}{g'(x)}$. (This holds also for complex functions.)
2. Suppose f, g are complex differentiable functions on $(0, 1)$ with $f(x) \rightarrow 0$, $g(x) \rightarrow 0$, $f'(x) \rightarrow A$, $g'(x) \rightarrow B$ as $x \rightarrow 0$, where $A, B \in \mathbb{C}$, $B \neq 0$. Prove that $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \frac{A}{B}$. (Hint: apply the previous exercise to the real and imaginary parts of $f(x)/g(x)$ and $g(x)/x$.)

Exercises (Bartle & Sherbert, §6.3).

1. Suppose f, g are continuous on $[a, b]$, differentiable on (a, b) , $c \in [a, b]$, and $g(x) \neq 0$ for $x \in [a, b]$, $x \neq c$. Let $A = \lim_{x \rightarrow c} f$, $B = \lim_{x \rightarrow c} g$. If $B = 0$ and $\lim_{x \rightarrow c} f(x)/g(x)$ exists in $\mathbb{R}$, show that $A = 0$. (Hint: $f(x) = \{f(x)/g(x)\}g(x)$.)
2. In addition to the previous suppositions, let $g(x) > 0$ for $x \neq c$. If $A > 0$ and $B = 0$, prove $\lim_{x \rightarrow c} f(x)/g(x) = \infty$. If $A < 0$ and $B = 0$, prove the limit is $-\infty$.
3. Let $f(x) = x^2 \sin(1/x)$ for $0 < x \leq 1$ and $f(0) = 0$, and $g(x) = x^2$ on $[0, 1]$. Show that $\lim_{x \rightarrow 0} f(x) = 0 = \lim_{x \rightarrow 0} g(x)$ but $\lim_{x \rightarrow 0} f'(x)/g'(x)$ does not exist (whereas $\lim_{x \rightarrow 0} f(x)/g(x) = 0$).
4. Let $f(x) = x^2$ for x rational, $f(x) = 0$ for x irrational, and $g(x) = \sin x$. Use Theorem 6.3.1 to show $\lim_{x \rightarrow 0} f(x)/g(x) = 0$. Explain why the strong L'Hospital rule (6.3.3) cannot be used.
5. Let $f(x) = x^2 \sin(1/x)$ for $x \neq 0$, $f(0) = 0$, and $g(x) = \sin x$. Show $\lim_{x \rightarrow 0} f(x)/g(x) = 0$ but $\lim_{x \rightarrow 0} f'(x)/g'(x)$ does not exist.
6. Evaluate: (a) $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{1 - \cos x}$; (b) $\lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x}{x^4}$.
7. Evaluate (domain indicated): (a) $\lim_{x \rightarrow 0^+} \frac{\ln(x+1)}{\sin x}$ on $(0, \pi/2)$; (b) $\lim_{x \rightarrow 0^+} \frac{\tan x}{x}$ on $(0, \pi/2)$; (c) $\lim_{x \rightarrow 0^+} \frac{\ln \cos x}{x}$ on $(0, \pi/2)$; (d) $\lim_{x \rightarrow 0^+} \frac{\tan x - x}{x^3}$ on $(0, \pi/2)$.
8. Evaluate: (a) $\lim_{x \rightarrow 0} \frac{\text{Arctan } x}{x}$; (b) $\lim_{x \rightarrow 0^+} \frac{1}{x(\ln x)^2}$ on $(0, 1)$; (c) $\lim_{x \rightarrow 0^+} x^3 \ln x$ on $(0, 1)$; (d) $\lim_{x \rightarrow \infty} \frac{x^3}{e^{3x}}$ on $(0, \infty)$.
9. Evaluate: (a) $\lim_{x \rightarrow \infty} \frac{\ln x}{x^2}$; (b) $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}}$; (c) $\lim_{x \rightarrow 0^+} x \ln \sin x$ on $(0, \pi)$; (d) $\lim_{x \rightarrow \infty} \frac{x + \ln x}{x \ln x}$.
10. Evaluate: (a) $\lim_{x \rightarrow 0^+} x^{2x}$ on $(0, \infty)$; (b) $\lim_{x \rightarrow \infty} (1 + 3/x)^x$ on $(0, \infty)$; (c) $\lim_{x \rightarrow 0^+} (1 + 3/x)^x$ on $(0, \infty)$; (d) $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\text{Arctan } x} \right)$.
11. Evaluate: (a) $\lim_{x \rightarrow \infty} x^{1/x}$; (b) $\lim_{x \rightarrow 0^+} (\sin x)^x$ on $(0, \pi)$; (c) $\lim_{x \rightarrow 0^+} x^{\sin x}$ on $(0, \infty)$; (d) $\lim_{x \rightarrow (\pi/2)^-} (\sec x - \tan x)$ on $(0, \pi/2)$.

12. Let f be differentiable on $(0, \infty)$ with $\lim_{x \rightarrow \infty} (f(x) + f'(x)) = L$. Show $\lim_{x \rightarrow \infty} f(x) = L$ and $\lim_{x \rightarrow \infty} f'(x) = 0$. (Hint: $f(x) = e^x f(x)/e^x$.)
13. Try to use L'Hospital's rule on $\frac{\tan x}{\sec x}$ as $x \rightarrow (\pi/2)^-$, then evaluate directly by changing to sines and cosines.
14. Show that if $c > 0$, then $\lim_{x \rightarrow c} \frac{x^c - c^x}{x^x - c^c} = \frac{1 - \ln c}{1 + \ln c}$.

Additional exercises.

- Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$.
- Evaluate $\lim_{x \rightarrow 0^+} x^x$ and $\lim_{x \rightarrow 0^+} (\tan x)^x$.
- Evaluate $\lim_{x \rightarrow \infty} \left(\frac{x+1}{x-1} \right)^x$.
- Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$ (an $\infty - \infty$ form).
- (Non-example.) Show that L'Hospital's rule does *not* apply to $\lim_{x \rightarrow 0} \frac{x+2}{x+1}$, and that applying it anyway gives the wrong answer. State the value of the limit and the value the rule would falsely produce.
- (Non-example.) Show $\lim_{x \rightarrow \infty} \frac{x - \sin x}{x + \sin x} = 1$ even though $\lim_{x \rightarrow \infty} \frac{1 - \cos x}{1 + \cos x}$ does not exist. Which hypothesis of L'Hospital's rule is being respected, and which conclusion is *not* available?
- Use Cauchy's Mean Value Theorem directly (no L'Hospital "black box") to evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$.
- (Diagram.) On one set of axes plot $f(x) = \sin x$ and $g(x) = x$ near 0 and use the picture to make the $0/0$ limit $\lim_{x \rightarrow 0} \sin x/x = 1$ plausible; mark the equal tangent slopes at the origin.
- (Harder.) Evaluate $\lim_{x \rightarrow 0^+} \frac{x^x - 1}{x \ln x}$.
- (Harder.) Suppose f is twice differentiable near 0 with $f(0) = 0$. Prove $\lim_{x \rightarrow 0} \frac{f(x) - xf'(x)}{x^2} = -\frac{1}{2}f''(0)$.

6.4 Taylor's Theorem

Higher derivatives and Taylor polynomials

If f' exists on an interval around c and is itself differentiable at c , its derivative is $f''(c)$, the *second derivative*; inductively $f^{(n)}(c)$ is the derivative of $f^{(n-1)}$ at c , the *n -th derivative*. We set $f^{(0)} = f$. A function with $f^{(n)}$ continuous on I is of class $C^n(I)$; with derivatives of all orders, $C^\infty(I)$.

Given f with n derivatives at x_0 , the *n -th Taylor polynomial* of f about x_0 is

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n.$$

It is the unique degree- $\leq n$ polynomial matching f and its first n derivatives at x_0 : $P_n^{(k)}(x_0) = f^{(k)}(x_0)$ for $k = 0, \dots, n$. Taylor's theorem measures the error $f(x) - P_n(x)$.

Theorem 6.4.1 (Taylor's Theorem, Lagrange remainder). *Let $n \in \mathbb{N}$, $I = [a, b]$, and suppose $f, f', \dots, f^{(n-1)}$ are continuous on $[a, b]$ and $f^{(n)}$ exists on (a, b) . Fix $x_0, x \in [a, b]$ with $x \neq x_0$. Then there is a point ξ strictly between x_0 and x such that*

$$f(x) = \underbrace{\sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k}_{P_{n-1}(x)} + \underbrace{\frac{f^{(n)}(\xi)}{n!} (x - x_0)^n}_{R_{n-1}(x)}.$$

Proof. Fix x_0, x and let M be the constant defined by $f(x) = P_{n-1}(x) + M(x-x_0)^n$, i.e. $M = \frac{f(x) - P_{n-1}(x)}{(x-x_0)^n}$; we must show $M = f^{(n)}(\xi)/n!$ for some ξ . Define

$$g(t) = f(t) - P_{n-1}(t) - M(t-x_0)^n.$$

Then $g(x) = 0$ by choice of M , and $g(x_0) = g'(x_0) = \dots = g^{(n-1)}(x_0) = 0$ because P_{n-1} matches f to order $n-1$ at x_0 and $(t-x_0)^n$ vanishes to order n . Since $g(x_0) = g(x) = 0$, Rolle gives x_1 between them with $g'(x_1) = 0$. Now $g'(x_0) = 0 = g'(x_1)$, so Rolle gives x_2 with $g''(x_2) = 0$; continuing, after n steps we get ξ between x_0 and x with $g^{(n)}(\xi) = 0$. But P_{n-1} has degree $\leq n-1$ so $P_{n-1}^{(n)} \equiv 0$, and the n -th derivative of $M(t-x_0)^n$ is $M n!$; thus $g^{(n)}(t) = f^{(n)}(t) - M n!$, and $g^{(n)}(\xi) = 0$ gives $M = f^{(n)}(\xi)/n!$. $\square$

Remark (Cauchy's remainder). The same setup with a different auxiliary function yields the *Cauchy form* of the remainder,

$$R_{n-1}(x) = \frac{f^{(n)}(\xi)}{(n-1)!} (x-\xi)^{n-1}(x-x_0),$$

for some ξ between x_0 and x . The Lagrange and Cauchy forms come from applying the mean-value idea to two different “weight” functions; each is sharper than the other for certain f (Cauchy's is the one that proves the binomial series converges on all of $(-1, 1)$). For $n = 1$, Taylor's theorem with the Lagrange remainder is exactly the Mean Value Theorem: $f(x) = f(x_0) + f'(\xi)(x-x_0)$.

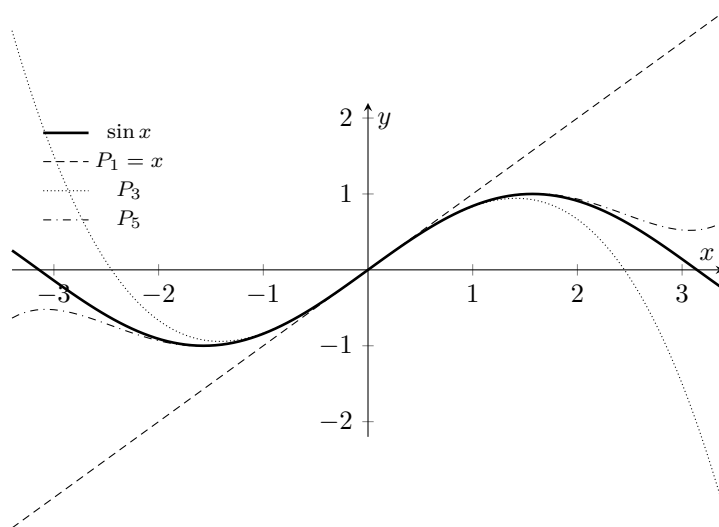


Figure 6.9. Taylor polynomials of $\sin$ at 0: $P_1 = x$, $P_3 = x - \frac{x^3}{6}$, $P_5 = x - \frac{x^3}{6} + \frac{x^5}{120}$. Higher degree hugs the curve over a wider window.

Method (Estimating with Taylor's theorem). *Recognition trigger.* “Approximate $f(x)$ with error $< \varepsilon$ ”, “show $|f(x) - P_n(x)| \leq \dots$ ”, or prove a polynomial inequality for f .

1. Pick the base point x_0 where f and its derivatives are known (often 0).
2. Write the Lagrange remainder $R_{n-1}(x) = \frac{f^{(n)}(\xi)}{n!} (x-x_0)^n$ with ξ unknown between x_0 and x .
3. **Bound** $f^{(n)}(\xi)$ by its sup M_n over the interval: then $|R_{n-1}(x)| \leq \frac{M_n}{n!} |x-x_0|^n$.
4. Choose n large enough to make this $< \varepsilon$, or read off the sign to get an inequality.

Worked Example 6.4.2 (A numerical approximation with error bound). Approximate $\sqrt{1.2}$ using Taylor at $x_0 = 0$ for $f(x) = \sqrt{1+x}$, $n = 2$, and bound the error.

Recognition. “Approximate with error control”; Lagrange remainder with $n = 2$.

Solution. Here $f(x) = (1+x)^{1/2}$, $f'(x) = \frac{1}{2}(1+x)^{-1/2}$, $f''(x) = -\frac{1}{4}(1+x)^{-3/2}$, $f'''(x) = \frac{3}{8}(1+x)^{-5/2}$. So $P_2(x) = 1 + \frac{1}{2}x - \frac{1}{8}x^2$, and with $x = 0.2$, $P_2(0.2) = 1 + 0.1 - 0.005 = 1.095$. The remainder is $R_2(0.2) = \frac{f'''(\xi)}{3!}(0.2)^3 = \frac{3}{48}(1+\xi)^{-5/2}(0.008)$ for some $\xi \in (0, 0.2)$; bounding $(1+\xi)^{-5/2} \leq 1$, $|R_2| \leq \frac{3}{48} \cdot 0.008 = 5 \times 10^{-4}$. *Check.* $\sqrt{1.2} = 1.0954\dots$, so the true error $\approx 4.5 \times 10^{-4}$, within the bound. ■

Worked Example 6.4.3 (A global inequality). Show $1 - \frac{1}{2}x^2 \leq \cos x$ for all $x \in \mathbb{R}$.

Recognition. “Prove a polynomial bounds f ”; Taylor with a sign-definite remainder.

Solution. Taylor $\cos$ at 0 to order $n = 4$: with $f = \cos$, $f^{(4)} = \cos$, the Lagrange form gives $\cos x = 1 - \frac{1}{2}x^2 + \frac{\cos \xi}{24}x^4$ for some ξ between 0 and x . Since $\cos \xi \geq -1$ and $x^4 \geq 0$, the remainder $\frac{\cos \xi}{24}x^4 \geq -\frac{x^4}{24}$; but we need only that it can be written with $f^{(2)}$: using order $n = 2$, $\cos x = 1 - \frac{\cos \xi}{2}x^2 \geq 1 - \frac{1}{2}x^2$ because $\cos \xi \leq 1$. Thus $1 - \frac{1}{2}x^2 \leq \cos x$ for all x . *Check.* Equality holds at $x = 0$; at $x = \pi/2$, $1 - \pi^2/8 \approx -0.234 < 0 = \cos(\pi/2)$. ■

Taylor and extrema

The first-derivative test classifies extrema by the sign change of f' . When $f'(c) = 0$, higher derivatives settle the question through the leading nonzero term of the Taylor expansion.

Theorem 6.4.4 (Higher-order test for extrema). *Let $f \in C^n$ near an interior point c with $f'(c) = f''(c) = \dots = f^{(n-1)}(c) = 0$ but $f^{(n)}(c) \neq 0$. Then: (i) if n is even and $f^{(n)}(c) > 0$, f has a local minimum at c ; (ii) if n is even and $f^{(n)}(c) < 0$, a local maximum; (iii) if n is odd, c is neither (an inflection-type point).*

Proof. By Taylor with Lagrange remainder and the vanishing lower derivatives, $f(x) - f(c) = \frac{f^{(n)}(\xi)}{n!}(x-c)^n$ for ξ between c and x . By continuity of $f^{(n)}$, $f^{(n)}(\xi)$ has the sign of $f^{(n)}(c)$ for x near c . If n is even, $(x-c)^n > 0$ for $x \neq c$, so $f(x) - f(c)$ has the constant sign of $f^{(n)}(c)$ on both sides: a strict min if $+$, strict max if $-$. If n is odd, $(x-c)^n$ changes sign across c , so $f(x) - f(c)$ does too — no extremum. □

Worked Example 6.4.5 (Classifying a degenerate critical point). Classify $x = 0$ for $f(x) = x^4$ and for $g(x) = x^3$.

Recognition. $f'(0) = 0$ but the second derivative also vanishes; use the higher-order test.

Solution. For $f = x^4$: $f'(0) = f''(0) = f'''(0) = 0$, $f^{(4)}(0) = 24 > 0$; $n = 4$ even with $f^{(4)}(0) > 0$, so f has a strict local (indeed global) minimum at 0. For $g = x^3$: $g'(0) = g''(0) = 0$, $g'''(0) = 6 \neq 0$; $n = 3$ odd, so 0 is not an extremum — it is an inflection. *Check.* Graphs confirm: x^4 a flat-bottomed valley, x^3 passing through with a horizontal tangent but no turn. ■

Caution. The second-derivative test is *inconclusive* when $f''(c) = 0$ — you must go to the next nonzero derivative, and the parity of its order decides everything. Do not conclude “no extremum” just because $f''(c) = 0$ (x^4 has a minimum there). Also, Taylor’s remainder ξ is an *existential* point you do not get to choose; never substitute a convenient value for it — only *bound* $f^{(n)}(\xi)$ over the whole interval.

Further worked examples

Worked Example 6.4.6 (Approximating e with a guaranteed error). Estimate $e = e^1$ from the Taylor polynomial of e^x at 0 of degree $n - 1 = 4$, and bound the error.

Recognition. “Approximate with error control”; Lagrange remainder, using that $f^{(n)} = e^x$ is easy to bound (Method, step 3).

Solution. For $f(x) = e^x$, every derivative is e^x , so $P_4(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$. At $x = 1$, $P_4(1) = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} = \frac{65}{24} \approx 2.7083$. The Lagrange remainder with $n = 5$ is $R_4(1) = \frac{e^\xi}{5!}1^5 = \frac{e^\xi}{120}$ for some $\xi \in (0, 1)$. Since $e^\xi < e^1 < 3$, $|R_4(1)| < \frac{3}{120} = 0.025$. *Check.* $e = 2.71828\dots$, so the true error is

$2.71828 - 2.70833 = 0.00995 < 0.025$, inside the bound. Pushing to $n = 8$ would give error $< 3/8! \approx 7 \times 10^{-5}$. ■

Worked Example 6.4.7 (A smooth function that is NOT its Taylor series (non-example)). Let $f(x) = e^{-1/x^2}$ for $x \neq 0$ and $f(0) = 0$. Show $f^{(k)}(0) = 0$ for every k , so its Taylor series at 0 is identically 0 yet $f \neq 0$.

Recognition. “Taylor series fails to represent f ”; compute the derivatives at 0 from the definition and observe the remainder does not vanish.

Solution. Away from 0, each derivative has the form $f^{(k)}(x) = P_k(1/x) e^{-1/x^2}$ for some polynomial P_k . As $x \rightarrow 0$, e^{-1/x^2} decays faster than any power of $1/x$ grows (substitute $t = 1/x^2 \rightarrow \infty$: $t^m e^{-t} \rightarrow 0$), so $f^{(k)}(x) \rightarrow 0$ and, inductively from the definition, $f^{(k)}(0) = 0$ for all k . Hence $P_n(x) \equiv 0$ for every n , and the Taylor remainder $R_n(x) = f(x) - 0 = e^{-1/x^2} > 0$ for $x \neq 0$ never goes to 0 as $n \rightarrow \infty$. Check. $f(1) = e^{-1} \approx 0.368 \neq 0$ while every Taylor polynomial at 0 predicts 0. “ C^∞ ” is strictly weaker than “analytic”; Taylor’s theorem guarantees a single- n remainder, not convergence of the full series. ■

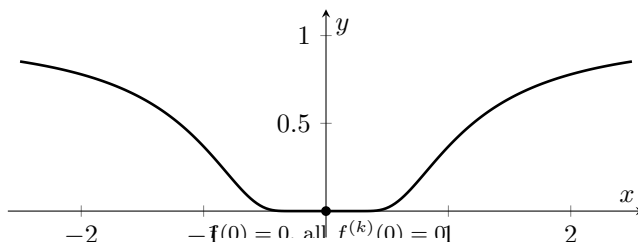


Figure 6.10. $f(x) = e^{-1/x^2}$ is flat to infinite order at 0 (every derivative vanishes), so its Taylor series is $\equiv 0$ — yet $f > 0$ off the origin. Smooth $\neq$ analytic.

Worked Example 6.4.8 (A two-sided bound for $\ln(1+x)$). Show $x - \frac{x^2}{2} \leq \ln(1+x) \leq x$ for all $x \geq 0$.

Recognition. “Sandwich f between two polynomials”; Taylor with a sign-controlled Lagrange remainder at each order.

Solution. Let $f(x) = \ln(1+x)$, so $f'(x) = \frac{1}{1+x}$, $f''(x) = \frac{-1}{(1+x)^2}$, $f'''(x) = \frac{2}{(1+x)^3}$. *Upper bound:* Taylor to order $n = 1$ at 0 gives $\ln(1+x) = x + \frac{f''(\xi)}{2}x^2 = x - \frac{x^2}{2(1+\xi)^2} \leq x$ since the remainder is ≤ 0 for $\xi > 0$.

Lower bound: Taylor to order $n = 2$ gives $\ln(1+x) = x - \frac{x^2}{2} + \frac{f'''(\xi)}{6}x^3 = x - \frac{x^2}{2} + \frac{x^3}{3(1+\xi)^3} \geq x - \frac{x^2}{2}$ since that remainder is ≥ 0 for $x \geq 0$. Check. At $x = 1$, $1 - \frac{1}{2} = 0.5 \leq \ln 2 = 0.693 \leq 1$. Equality at $x = 0$ throughout. ■

Exercises (Rudin, Ch.5).

- Suppose f is defined in a neighborhood of x and $f''(x)$ exists. Show that $\lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x)$. Show by an example that the limit may exist even if $f''(x)$ does not.
- Suppose $a \in \mathbb{R}$, f is twice differentiable on (a, ∞) , and M_0, M_1, M_2 are the least upper bounds of $|f|, |f'|, |f''|$ on (a, ∞) . Prove that $M_1^2 \leq 4M_0M_2$. (Hint: Taylor’s theorem gives $f'(x) = \frac{1}{2h}[f(x+2h) - f(x)] - hf''(\xi)$ for some $\xi \in (x, x+2h)$, so $|f'(x)| \leq hM_2 + \frac{M_0}{h}$.) Show $M_1^2 = 4M_0M_2$ can occur. Does $M_1^2 \leq 4M_0M_2$ hold for vector-valued functions?
- Suppose f is twice differentiable on $(0, \infty)$, f'' is bounded on $(0, \infty)$, and $f(x) \rightarrow 0$ as $x \rightarrow \infty$. Prove that $f'(x) \rightarrow 0$ as $x \rightarrow \infty$. (Hint: let $a \rightarrow \infty$ in the previous exercise.)
- Suppose f is real and three times differentiable on $[-1, 1]$ with $f(-1) = 0$, $f(0) = 0$, $f(1) = 1$, $f'(0) = 0$. Prove that $f^{(3)}(x) \geq 3$ for some $x \in (-1, 1)$. (Equality holds for $\frac{1}{2}(x^3 + x^2)$. Hint: use Taylor’s theorem with $x_0 = 0$, $x = \pm 1$.)

5. Suppose f is a real function on $[a, b]$, n a positive integer, and $f^{(n-1)}$ exists for every $t \in [a, b]$. Let α, β be as in Taylor's theorem. Define $Q(t) = \frac{f(t) - f(\beta)}{t - \beta}$ for $t \neq \beta$, differentiate $f(t) - f(\beta) = (t - \beta)Q(t)$ $n - 1$ times at $t = \alpha$, and derive the version $f(\beta) = \sum_{k=0}^{n-2} \frac{f^{(k)}(\alpha)}{k!} (\beta - \alpha)^k + \frac{Q^{(n-1)}(\alpha)}{(n-1)!} (\beta - \alpha)^n$.
6. Suppose f is differentiable on $[a, b]$, $f(a) = 0$, and there is a real A with $|f'(x)| \leq A|f(x)|$ on $[a, b]$. Prove that $f(x) = 0$ for all $x \in [a, b]$. (Hint: fix x_0 , let $M_0 = \sup|f|$, $M_1 = \sup|f'|$ on $[a, x_0]$; then $|f(x)| \leq M_1(x_0 - a) \leq A(x_0 - a)M_0$, so $M_0 = 0$ if $A(x_0 - a) < 1$; proceed in steps.)
7. Let ϕ be a real function on a rectangle $R = \{(x, y) : a \leq x \leq b, \alpha \leq y \leq \beta\}$. A solution of the initial-value problem $y' = \phi(x, y)$, $y(a) = c$ (with $\alpha < c < \beta$) is a differentiable f on $[a, b]$ with $f(a) = c$, $\alpha \leq f(x) \leq \beta$, $f'(x) = \phi(x, f(x))$. Prove that such a problem has at most one solution if there is a constant A with $|\phi(x, y_2) - \phi(x, y_1)| \leq A|y_2 - y_1|$ whenever $(x, y_1), (x, y_2) \in R$. (Hint: apply the previous exercise to the difference of two solutions. Note this uniqueness fails for $y' = y^{1/2}$, $y(0) = 0$, which has solutions $f(x) = 0$ and $f(x) = x^2/4$; find all solutions.)
8. Formulate and prove an analogous uniqueness theorem for systems of differential equations $y'_j = \phi_j(x, y_1, \dots, y_k)$, $y_j(a) = c_j$ ($j = 1, \dots, k$). Rewrite as $\mathbf{y}' = \Phi(x, \mathbf{y})$, $\mathbf{y}(a) = \mathbf{c}$, where $\mathbf{y}$ ranges over a k -cell. (Use the previous exercise for vector-valued functions.)
9. Specialize the previous exercise to the system $y'_j = y_{j+1}$ ($j = 1, \dots, k-1$), $y'_k = f(x) - \sum_{j=1}^k g_j(x)y_j$, where $f, g_1, \dots, g_k$ are continuous on $[a, b]$, and derive a uniqueness theorem for $y^{(k)} + g_k(x)y^{(k-1)} + \dots + g_2(x)y' + g_1(x)y = f(x)$ subject to $y(a) = c_1$, $y'(a) = c_2$, $\dots$, $y^{(k-1)}(a) = c_k$.

Exercises (Bartle & Sherbert, §6.4).

1. Let $f(x) = \cos ax$ for $x \in \mathbb{R}$, $a \neq 0$. Find $f^{(n)}(x)$ for $n \in \mathbb{N}$.
2. Let $g(x) = |x^3|$ for $x \in \mathbb{R}$. Find $g'(x)$ and $g''(x)$ for all x , and $g'''(x)$ for $x \neq 0$. Show $g'''(0)$ does not exist.
3. Use induction to prove Leibniz's rule for the n -th derivative of a product:
$$(fg)^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} f^{(n-k)}(x) g^{(k)}(x).$$
4. Show that if $x > 0$, then $1 + \frac{1}{2}x - \frac{1}{8}x^2 \leq \sqrt{1+x} \leq 1 + \frac{1}{2}x$.
5. Use the preceding exercise to approximate $\sqrt{1.2}$ and $\sqrt{2}$. What accuracy can you guarantee?
6. Use Taylor's theorem with $n = 2$ to obtain sharper approximations for $\sqrt{1.2}$ and $\sqrt{2}$.
7. If $x > 0$, show $|(1+x)^{1/3} - (1 + \frac{1}{3}x - \frac{1}{9}x^2)| \leq (5/81)x^3$. Use this to approximate $\sqrt[3]{1.2}$ and $\sqrt[3]{2}$.
8. If $f(x) = e^x$, show the remainder in Taylor's theorem tends to 0 as $n \rightarrow \infty$ for each fixed x_0, x .
9. If $g(x) = \sin x$, show the remainder in Taylor's theorem tends to 0 as $n \rightarrow \infty$ for each fixed x_0, x .
10. Let $h(x) = e^{-1/x^2}$ for $x \neq 0$ and $h(0) = 0$. Show $h^{(n)}(0) = 0$ for all $n \in \mathbb{N}$; conclude that the Taylor remainder at $x_0 = 0$ does not tend to 0 as $n \rightarrow \infty$ for $x \neq 0$. (Hint: $\lim_{x \rightarrow 0} h(x)/x^k = 0$ for all k .)
11. If $x \in [0, 1]$ and $n \in \mathbb{N}$, show $|\ln(1+x) - (x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^{n-1} \frac{x^n}{n})| \leq \frac{x^{n+1}}{n+1}$. Use this to approximate $\ln 1.5$ with error < 0.01 , and < 0.001 .
12. Approximate $\sin$ by a polynomial on $[-1, 1]$ with error < 0.001 : show $|\sin x - (x - \frac{x^3}{6} + \frac{x^5}{120})| < \frac{1}{5040}$ for $|x| \leq 1$.
13. Calculate e correct to seven decimal places.
14. Determine whether $x = 0$ is a relative extremum of: (a) $f(x) = x^3 + 2$; (b) $g(x) = \sin x - x$; (c) $h(x) = \sin x + \frac{1}{6}x^3$; (d) $k(x) = \cos x - 1 + \frac{1}{2}x^2$.

15. Let f be continuous on $[a, b]$ with f'' existing on (a, b) . Suppose the graph of f and the chord from $(a, f(a))$ to $(b, f(b))$ meet at $(x_0, f(x_0))$ with $a < x_0 < b$. Show there is $c \in (a, b)$ with $f''(c) = 0$.
16. Let $I \subseteq \mathbb{R}$ be open, $f : I \rightarrow \mathbb{R}$ differentiable on I , and suppose $f''(a)$ exists at $a \in I$. Show $f''(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - 2f(a) + f(a-h)}{h^2}$. Give an example where this limit exists but $f''(a)$ does not.
17. Suppose $I \subseteq \mathbb{R}$ is open and $f''(x) \geq 0$ for all $x \in I$. If $c \in I$, show the part of the graph of f on I is never below the tangent line at $(c, f(c))$.
18. Let I be an interval, $c \in I$, f, g with $f^{(n)}, g^{(n)}$ continuous on I . If $f^{(k)}(c) = g^{(k)}(c) = 0$ for $k = 0, \dots, n-1$ but $g^{(n)}(c) \neq 0$, show $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{f^{(n)}(c)}{g^{(n)}(c)}$.
19. Show $f(x) = x^3 - 2x - 5$ has a zero $r \in [2, 2.2]$. With $x_1 = 2$ and Newton's iteration, show $|x_{n+1} - r| \leq (0.7)|x_n - r|^2$ and that x_4 is accurate to six decimals.
20. Approximate the real zeros of $g(x) = x^4 - x - 3$.
21. Approximate the real zeros of $h(x) = x^3 - x - 1$ by Newton's method starting from (a) $x_1 = 2$, (b) $x_1 = 0$, (c) $x_1 = -2$. Explain what happens.
22. The equation $\ln x = x - 2$ has two solutions. Approximate them by Newton's method. What happens if $x_1 = \frac{1}{2}$?
23. The function $f(x) = 8x^3 - 8x^2 + 1$ has two zeros in $[0, 1]$. Approximate them by Newton's method starting from (a) $x_1 = \frac{1}{8}$, (b) $x_1 = \frac{1}{4}$. Explain.
24. Approximate the solution of $x = \cos x$ accurate to six decimals.

Additional exercises.

1. Write the Taylor polynomial P_3 of $f(x) = e^x$ about $x_0 = 0$ and bound $|e^{0.5} - P_3(0.5)|$ using the Lagrange remainder.
2. Find the Taylor polynomial of degree 4 for $\ln(1+x)$ about 0 and use it to estimate $\ln 1.1$ with an error bound.
3. Show $x - \frac{x^3}{6} \leq \sin x \leq x$ for all $x \geq 0$ using Taylor's theorem.
4. Classify the critical point $x = 0$ of $f(x) = x^6$ and of $g(x) = x^5$ via the higher-order derivative test.
5. (Non-example.) Let $h(x) = e^{-1/x^2}$ for $x \neq 0$, $h(0) = 0$. Show every derivative $h^{(n)}(0) = 0$, so the Taylor series of h at 0 is identically zero, yet $h(x) > 0$ for $x \neq 0$. Conclude that a C^∞ function need not equal its Taylor series.
6. (Non-example.) Exhibit a function that is C^1 but not C^2 at a point, and verify the second-derivative test cannot be applied there.
7. (Diagram.) Plot e^x together with P_1, P_2, P_3 about 0 on $[-2, 2]$, and describe how the approximation error grows with $|x|$ and shrinks as the degree rises.
8. Use Taylor's theorem with $n = 2$ to approximate $\sqrt[3]{1.1}$ and give an error bound.
9. (Harder.) Suppose $f \in C^2$ near c with $f'(c) = 0$ and $f''(c) > 0$. Prove directly from the Lagrange remainder that f has a strict local minimum at c , and find $\delta > 0$ (in terms of bounds on f'') such that $f(x) > f(c)$ for $0 < |x - c| < \delta$.
10. (Harder.) Prove that for $f \in C^{n+1}$ near x_0 , the Taylor polynomial P_n is the *unique* degree- $\leq n$ polynomial P with $\frac{f(x) - P(x)}{(x - x_0)^n} \rightarrow 0$ as $x \rightarrow x_0$ (the "little-o" characterization).

6.5 Differentiation of Vector-Valued Functions

Componentwise differentiation

A *vector-valued* function $\mathbf{f} : [a, b] \rightarrow \mathbb{R}^k$ assigns to each x a point $\mathbf{f}(x) = (f_1(x), \dots, f_k(x))$. Definition 6.1.1 applies verbatim: the difference quotient $\frac{\mathbf{f}(x) - \mathbf{f}(c)}{x - c}$ is a point of $\mathbb{R}^k$, and $\mathbf{f}'(c)$ is the limit *in the norm of* $\mathbb{R}^k$, i.e. the vector for which

$$\lim_{x \rightarrow c} \left\| \frac{\mathbf{f}(x) - \mathbf{f}(c)}{x - c} - \mathbf{f}'(c) \right\| = 0.$$

The complex case $\mathbf{f} : [a, b] \rightarrow \mathbb{C}$ is the case $k = 2$.

Theorem 6.5.1 (Derivative is componentwise). $\mathbf{f} = (f_1, \dots, f_k) : [a, b] \rightarrow \mathbb{R}^k$ is differentiable at c if and only if every component f_j is differentiable at c , and then

$$\mathbf{f}'(c) = (f'_1(c), \dots, f'_k(c)).$$

Proof. A vector in $\mathbb{R}^k$ converges iff each coordinate converges, with the limit taken coordinatewise (because $|v_j| \leq \|v\| \leq \sqrt{k} \max_i |v_i|$). Apply this to the difference quotient $\frac{\mathbf{f}(x) - \mathbf{f}(c)}{x - c}$, whose j -th coordinate is $\frac{f_j(x) - f_j(c)}{x - c}$: the vector has a limit iff each coordinate quotient does, and the limit is the vector of coordinate limits $f'_j(c)$. $\square$

Theorem 6.1.2 (differentiable $\implies$ continuous) and the sum and product rules carry over: $(\mathbf{f} + \mathbf{g})' = \mathbf{f}' + \mathbf{g}'$, and for the inner product $(\mathbf{f} \cdot \mathbf{g})' = \mathbf{f}' \cdot \mathbf{g} + \mathbf{f} \cdot \mathbf{g}'$. What does *not* carry over is the Mean Value Theorem as an equality, and with it L'Hospital's rule.

The mean value theorem fails as an equality

Worked Example 6.5.2 (No mean value for a curve). Let $\mathbf{f}(x) = (\cos x, \sin x) = e^{ix}$ for $x \in [0, 2\pi]$. Show there is *no* $c \in (0, 2\pi)$ with $\mathbf{f}(2\pi) - \mathbf{f}(0) = \mathbf{f}'(c)(2\pi - 0)$.

Recognition. “Show the MVT equality fails for a vector function”; compute both sides.

Solution. The endpoints agree: $\mathbf{f}(2\pi) - \mathbf{f}(0) = (1, 0) - (1, 0) = \mathbf{0}$. If the MVT equality held, some c would give $\mathbf{f}'(c) = \mathbf{0}$. But $\mathbf{f}'(x) = (-\sin x, \cos x)$ has norm $\|\mathbf{f}'(x)\| = \sqrt{\sin^2 x + \cos^2 x} = 1$ for *every* x , so $\mathbf{f}'$ never vanishes. Hence no such c exists; the equality form of the MVT is false for $\mathbb{R}^k$ -valued functions. *Check.* The curve is the unit circle traversed once; it returns to its start (zero net displacement) while moving at unit speed the whole time — speed and net displacement are simply unrelated. $\blacksquare$

The geometric reason: the scalar MVT says “average velocity = instantaneous velocity somewhere”, but for a curve the average *vector* velocity may be $\mathbf{0}$ while the speed $\|\mathbf{f}'\|$ is always positive. Each component f_j has its *own* mean value point c_j , and there is no reason a single c serves all components at once.

Remark. The same example kills L'Hospital's rule for complex functions: on $(0, 1)$ put $f(x) = x$ and $g(x) = x + x^2 e^{i/x^2}$. Then $f(x)/g(x) \rightarrow 1$, yet $f'(x)/g'(x) \rightarrow 0$ (Rudin 5.18). The proofs of both MVT and L'Hospital used Rolle, which is intrinsically scalar (it needs an ordered maximum); vectors have no order, so the machinery breaks.

The norm inequality survives

What remains true — and is almost as useful in applications — is the *inequality* form. It bounds the displacement by the maximum speed times the elapsed time.

Theorem 6.5.3 (Mean Value Inequality, vector form). Let $\mathbf{f} : [a, b] \rightarrow \mathbb{R}^k$ be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $x \in (a, b)$ with

$$\|\mathbf{f}(b) - \mathbf{f}(a)\| \leq (b - a) \|\mathbf{f}'(x)\|.$$

In particular $\|\mathbf{f}(b) - \mathbf{f}(a)\| \leq (b - a) \sup_{a < x < b} \|\mathbf{f}'(x)\|$.

Proof. Put $\mathbf{z} = \mathbf{f}(b) - \mathbf{f}(a) \in \mathbb{R}^k$ and define the scalar function $\varphi(t) = \mathbf{z} \cdot \mathbf{f}(t)$ (the inner product). Then φ is continuous on $[a, b]$ and differentiable on (a, b) with $\varphi'(t) = \mathbf{z} \cdot \mathbf{f}'(t)$. By the scalar MVT there is $x \in (a, b)$ with

$$\varphi(b) - \varphi(a) = \varphi'(x)(b - a), \quad \text{i.e.} \quad \mathbf{z} \cdot (\mathbf{f}(b) - \mathbf{f}(a)) = (b - a) \mathbf{z} \cdot \mathbf{f}'(x).$$

The left side is $\mathbf{z} \cdot \mathbf{z} = \|\mathbf{z}\|^2$. By Cauchy–Schwarz the right side is $\leq (b - a) \|\mathbf{z}\| \|\mathbf{f}'(x)\|$. Thus $\|\mathbf{z}\|^2 \leq (b - a) \|\mathbf{z}\| \|\mathbf{f}'(x)\|$. If $\mathbf{z} = \mathbf{0}$ the inequality is trivial; otherwise divide by $\|\mathbf{z}\| > 0$ to get $\|\mathbf{z}\| \leq (b - a) \|\mathbf{f}'(x)\|$. $\square$

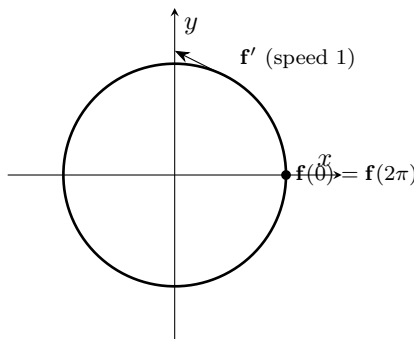


Figure 6.11. $\mathbf{f}(x) = (\cos x, \sin x)$ on $[0, 2\pi]$: net displacement $\mathbf{0}$ but speed $\|\mathbf{f}'\| \equiv 1$. The MVT *equality* cannot hold; the *inequality* $0 = \|\mathbf{f}(2\pi) - \mathbf{f}(0)\| \leq 2\pi \cdot 1$ does.

Worked Example 6.5.4 (A Lipschitz bound for a curve). Let $\mathbf{f}(x) = (x^2, \sin x)$ on $[0, 1]$. Bound $\|\mathbf{f}(1) - \mathbf{f}(0)\|$.

Recognition. “Bound the displacement”; use the vector mean value inequality.

Solution. $\mathbf{f}'(x) = (2x, \cos x)$, so $\|\mathbf{f}'(x)\| = \sqrt{4x^2 + \cos^2 x}$. On $[0, 1]$, $4x^2 \leq 4$ and $\cos^2 x \leq 1$, so $\|\mathbf{f}'(x)\| \leq \sqrt{5}$. By Theorem 6.5.3, $\|\mathbf{f}(1) - \mathbf{f}(0)\| \leq (1 - 0)\sqrt{5} = \sqrt{5} \approx 2.236$. *Check.* Directly $\mathbf{f}(1) - \mathbf{f}(0) = (1, \sin 1) \approx (1, 0.841)$, norm $\approx 1.306 \leq \sqrt{5}$. The bound holds (and is not tight, since the max speed is not attained throughout). ■

Caution. For $\mathbb{R}^k$ -valued (and complex) functions, write the Mean Value Theorem only as an *inequality* $\|\mathbf{f}(b) - \mathbf{f}(a)\| \leq (b - a) \sup \|\mathbf{f}'\|$, never as an equality. There is generally no single c with $\mathbf{f}(b) - \mathbf{f}(a) = \mathbf{f}'(c)(b - a)$ (Worked Example 6.5.2), and L’Hospital’s rule likewise requires real-valued f, g (or a reduction to real and imaginary parts). The scalar tools that rely on ordering — Rolle, the equality MVT, Darboux — do not have direct vector analogues; only the order-free consequences (the norm inequality) survive.

Further worked examples

Worked Example 6.5.5 (Componentwise differentiation of a space curve). Let $\mathbf{f}(x) = (x^2, e^x, \cos x)$ on $\mathbb{R}$. Compute $\mathbf{f}'(x)$ and the speed $\|\mathbf{f}'(x)\|$ at $x = 0$.

Recognition. “Differentiate a vector function”; Theorem 6.5.1 says do it coordinate by coordinate.

Solution. By Theorem 6.5.1, $\mathbf{f}'(x) = (2x, e^x, -\sin x)$. At $x = 0$, $\mathbf{f}'(0) = (0, 1, 0)$, so the speed is $\|\mathbf{f}'(0)\| = \sqrt{0^2 + 1^2 + 0^2} = 1$. *Check.* Each component is an elementary scalar derivative and all three exist, so $\mathbf{f}$ is differentiable at every x ; at 0 the velocity points purely along the e^x -axis, matching that x^2 and $\cos x$ are both flat there ($2x = 0, -\sin 0 = 0$). ■

Worked Example 6.5.6 (A complex derivative as the case $k = 2$). Let $f(x) = e^{i\omega x} = (\cos \omega x, \sin \omega x)$ for a real constant ω . Compute $f'(x)$ and verify $|f'(x)| = |\omega|$.

Recognition. “Differentiate a complex-valued function of a real variable”; this is the $k = 2$ case of componentwise differentiation.

Solution. Differentiate each component: $f'(x) = (-\omega \sin \omega x, \omega \cos \omega x)$, which in complex notation is $i\omega e^{i\omega x}$. Its modulus is $|f'(x)| = \sqrt{\omega^2 \sin^2 \omega x + \omega^2 \cos^2 \omega x} = \sqrt{\omega^2} = |\omega|$, constant in x . *Check.* The curve traces the unit circle at angular rate ω , so the speed is $|\omega|$ everywhere; over $[0, 2\pi/\omega]$ (one loop, $\omega > 0$) the net displacement is $\mathbf{0}$ while the speed stays $|\omega| > 0$ — the same obstruction to an equality MVT seen in Worked Example 6.5.2. The inequality form still holds: $0 = \|f(2\pi/\omega) - f(0)\| \leq (2\pi/\omega) |\omega| = 2\pi$. ■

Exercises (Rudin, Ch.5).

1. Suppose f is defined in $(-1, 1)$ and $f'(0)$ exists. Suppose $-1 < \alpha_n < \beta_n < 1$ with $\alpha_n \rightarrow 0$, $\beta_n \rightarrow 0$. Define the difference quotients $D_n = \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n}$. Prove: (a) if $\alpha_n < 0 < \beta_n$, then $\lim D_n = f'(0)$; (b) if $0 \leq \alpha_n < \beta_n$ and $\{\beta_n/(\beta_n - \alpha_n)\}$ is bounded, then $\lim D_n = f'(0)$; (c) if f' is continuous on $(-1, 1)$, then $\lim D_n = f'(0)$. Give an example in which f is differentiable on $(-1, 1)$ (with f' discontinuous at 0) and $\alpha_n, \beta_n \rightarrow 0$ in such a way that $\lim D_n$ exists but differs from $f'(0)$.
2. Formulate and prove an inequality which follows from Taylor's theorem and which remains valid for vector-valued functions.
3. Let E be a closed subset of $\mathbb{R}$. We saw (Exercise 22, Chap. 4) that there is a real continuous f on $\mathbb{R}$ whose zero set is E . Is it possible, for each closed E , to find such an f which is differentiable on $\mathbb{R}$, or one which is n times differentiable, or even one which has derivatives of all orders on $\mathbb{R}$?

Additional exercises.

1. Let $\mathbf{f}(x) = (\cos x, \sin x, x)$ on $[0, 2\pi]$ (a helix). Compute $\mathbf{f}'$ and $\|\mathbf{f}'(x)\|$, and bound $\|\mathbf{f}(2\pi) - \mathbf{f}(0)\|$ using the vector mean value inequality. Compare with the exact displacement.
2. For $\mathbf{f}(x) = (x, x^2)$ on $[0, 1]$, verify directly that each component satisfies the scalar MVT at its *own* point c_j , and show the two points differ; conclude no single c serves both components.
3. Prove the product rule for the inner product: if $\mathbf{f}, \mathbf{g} : [a, b] \rightarrow \mathbb{R}^k$ are differentiable at c , then $(\mathbf{f} \cdot \mathbf{g})'(c) = \mathbf{f}'(c) \cdot \mathbf{g}(c) + \mathbf{f}(c) \cdot \mathbf{g}'(c)$.
4. Show that if $\mathbf{f} : [a, b] \rightarrow \mathbb{R}^k$ has $\|\mathbf{f}(x)\|$ constant, then $\mathbf{f}(x) \cdot \mathbf{f}'(x) = 0$ for all x (velocity is orthogonal to position on a sphere). (Hint: differentiate $\|\mathbf{f}\|^2 = \mathbf{f} \cdot \mathbf{f}$.)
5. (Non-example.) Exhibit a continuous, differentiable $\mathbf{f} : [0, 1] \rightarrow \mathbb{R}^2$ with $\mathbf{f}(0) = \mathbf{f}(1)$ but $\mathbf{f}'(x) \neq \mathbf{0}$ for all $x \in (0, 1)$, proving that Rolle's theorem has *no* vector analogue.
6. (Non-example.) Give complex differentiable f, g on $(0, 1)$ showing L'Hospital's rule fails for complex-valued functions: $f/g \rightarrow$ one value while $f'/g' \rightarrow$ a different one. (Model it on Rudin's $f(x) = x$, $g(x) = x + x^2 e^{i/x^2}$.)
7. (Diagram.) Plot the curve $\mathbf{f}(x) = (\cos x, \sin x)$ on $[0, 2\pi]$, mark $\mathbf{f}(0) = \mathbf{f}(2\pi)$, and draw a velocity vector $\mathbf{f}'$ at two points to illustrate constant speed with zero net displacement.
8. (Harder.) Let $\mathbf{f} : [a, b] \rightarrow \mathbb{R}^k$ be C^1 . Prove $\|\mathbf{f}(b) - \mathbf{f}(a)\| \leq \int_a^b \|\mathbf{f}'(t)\| dt$ (assuming the integral results of the next chapter), and explain how this refines Theorem 6.5.3.

Chapter 7

The Riemann–Stieltjes Integral

7.1 Riemann Sums and the Darboux Integral

Throughout, f is a *bounded* real function on a closed bounded interval $[a, b]$. We build the integral $\int_a^b f$ two ways: B&S start from *tagged Riemann sums* and a mesh going to zero; Rudin (following Darboux) starts from *upper and lower sums* and refinement. The two roads meet — the central result of this section (Theorem 7.1.11) is that they define the *same* integrable functions and the *same* number.

Partitions, tags, and the Riemann sum

Definition 7.1.1. A *partition* of $[a, b]$ is a finite set $P = \{x_0, x_1, \dots, x_n\}$ with

$$a = x_0 < x_1 < \dots < x_n = b.$$

It splits $[a, b]$ into subintervals $I_i = [x_{i-1}, x_i]$ of length $\Delta x_i = x_i - x_{i-1}$. The *mesh* (or *norm*) is $\|P\| = \max_{1 \leq i \leq n} \Delta x_i$, the width of the widest piece. A *tagged partition* $\dot{P} = \{(I_i, t_i)\}_{i=1}^n$ assigns to each I_i a *tag* $t_i \in I_i = [x_{i-1}, x_i]$.

Definition 7.1.2. The *Riemann sum* of f for the tagged partition $\dot{P}$ is

$$S(f, \dot{P}) = \sum_{i=1}^n f(t_i) \Delta x_i.$$

Geometrically it is the signed area of n rectangles, the i -th of width Δx_i and height $f(t_i)$ (the function value at the chosen tag).

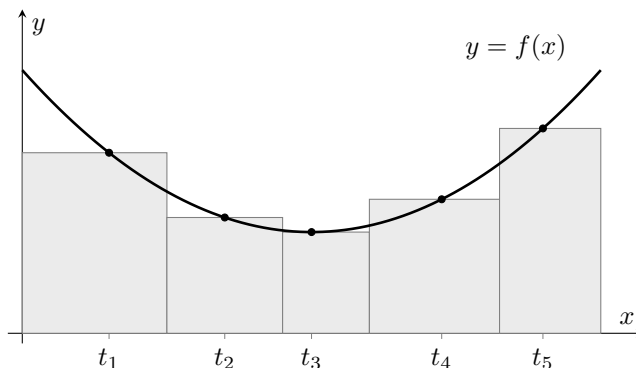


Figure 7.1. A tagged Riemann sum: each rectangle has height $f(t_i)$ read off at its tag t_i ; partition points sit at the rectangle edges.

Definition 7.1.3 (Riemann integral, B&S 7.1.1). A function f is *Riemann integrable* on $[a, b]$, written $f \in \mathcal{R}[a, b]$, if there is a number $L \in \mathbb{R}$ with the property: for every $\varepsilon > 0$ there is a $\delta_\varepsilon > 0$ such that

$$\|\dot{P}\| < \delta_\varepsilon \implies |S(f, \dot{P}) - L| < \varepsilon$$

for *every* tagged partition $\dot{P}$ with mesh below δ_ε . The number L , when it exists, is unique and is the *integral* $\int_a^b f = \int_a^b f(x) dx$.

This is an ε - δ statement in the variable $\|\dot{P}\|$: “as the mesh goes to 0, the Riemann sums converge to L , regardless of how the tags are chosen.” The “every $\dot{P}$ ” is the whole subtlety — it must hold for the worst tag choice.

Proposition 7.1.4 (Uniqueness of the integral, B&S 7.1.2). *If $f \in \mathcal{R}[a, b]$ then the value L is unique.*

Proof. Suppose L' also works. Fix $\varepsilon > 0$; pick δ' for L and δ'' for L' each with tolerance $\varepsilon/2$, and let $\delta = \min(\delta', \delta'')$. Choose any tagged partition $\dot{P}$ with $\|\dot{P}\| < \delta$ (such exist: chop $[a, b]$ into n equal pieces with n large). Then $|L - L'| \leq |L - S(f, \dot{P})| + |S(f, \dot{P}) - L'| < \varepsilon/2 + \varepsilon/2 = \varepsilon$. Since $\varepsilon > 0$ was arbitrary, $L = L'$. $\square$

The Darboux road: upper and lower sums

Because f is bounded, on each subinterval I_i the numbers

$$m_i = \inf_{x \in I_i} f(x), \quad M_i = \sup_{x \in I_i} f(x)$$

are finite. They are the smallest and largest heights of any rectangle over I_i whose top stays at or below (resp. above) the graph.

Definition 7.1.5 (Darboux/Rudin sums, Rudin 6.1). For a partition P (no tags needed) the *lower* and *upper Darboux sums* are

$$L(f, P) = \sum_{i=1}^n m_i \Delta x_i, \quad U(f, P) = \sum_{i=1}^n M_i \Delta x_i.$$

The *lower* and *upper integrals* are

$$\int_a^b f = \sup_P L(f, P), \quad \overline{\int_a^b} f = \inf_P U(f, P),$$

the sup/inf running over all partitions P of $[a, b]$.

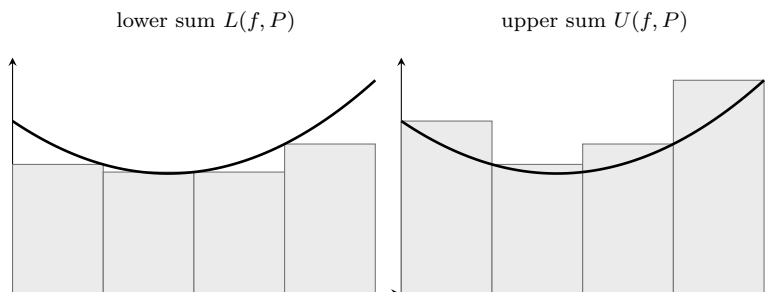


Figure 7.2. Lower rectangles (heights m_i) sit under the graph; upper rectangles (heights M_i) sit over it. The integral, if it exists, is squeezed between.

Definition 7.1.6 (Darboux integrability, Rudin 6.2 / B&S 7.4). f is *Darboux integrable* on $[a, b]$ if $\int_a^b f = \overline{\int_a^b} f$; the common value is the *Darboux integral*.

Refinement: why the sandwich closes

Definition 7.1.7. A partition P^* is a *refinement* of P if $P \subseteq P^*$ (every point of P is a point of P^*). The *common refinement* of P_1, P_2 is $P_1 \cup P_2$.

Lemma 7.1.8 (Refinement lemma, Rudin 6.4). *If $P^* \supseteq P$ then*

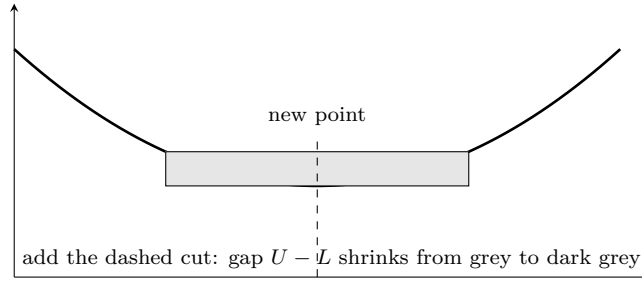
$$L(f, P) \leq L(f, P^*) \leq U(f, P^*) \leq U(f, P).$$

That is, refining raises lower sums and lowers upper sums; it never widens the gap.

Proof. It suffices to insert one new point $x^* \in (x_{i-1}, x_i)$; the general case follows by inserting points one at a time. Splitting I_i into $I' = [x_{i-1}, x^*]$ and $I'' = [x^*, x_i]$, let $w_1 = \inf_{I'} f$, $w_2 = \inf_{I''} f$. Since $I', I'' \subseteq I_i$ we have $w_1 \geq m_i$ and $w_2 \geq m_i$, so

$$w_1(x^* - x_{i-1}) + w_2(x_i - x^*) \geq m_i(x^* - x_{i-1}) + m_i(x_i - x^*) = m_i \Delta x_i.$$

Summing, $L(f, P^*) \geq L(f, P)$. The same argument with sup (and reversed inequalities) gives $U(f, P^*) \leq U(f, P)$. The middle inequality $L \leq U$ holds for any single partition because $m_i \leq M_i$. $\square$



Proposition 7.1.9 (Lower integral $\leq$ upper integral, Rudin 6.5). $\int_a^b f \leq \overline{\int_a^b f}$.

Proof. For *any* two partitions P_1, P_2 , the common refinement $P^* = P_1 \cup P_2$ gives, by Lemma 7.1.8,

$$L(f, P_1) \leq L(f, P^*) \leq U(f, P^*) \leq U(f, P_2).$$

Thus every lower sum is $\leq$ every upper sum. Fixing P_2 and taking the sup over P_1 : $\int f \leq U(f, P_2)$. Now taking the inf over P_2 : $\int f \leq \overline{\int f}$. $\square$

The Cauchy / Riemann criterion

Theorem 7.1.10 (Riemann's criterion, Rudin 6.6 / B&S 7.4.8). *f is Darboux integrable on $[a, b]$ if and only if for every $\varepsilon > 0$ there is a partition P with*

$$U(f, P) - L(f, P) < \varepsilon.$$

Proof. ($\Leftarrow$) If such P exists for each ε , then since $L(f, P) \leq \int f \leq \overline{\int f} \leq U(f, P)$ we get $0 \leq \overline{\int f} - \int f < \varepsilon$ for all $\varepsilon > 0$; hence they are equal.

($\Rightarrow$) Suppose $\int f = \overline{\int f} = I$. Given $\varepsilon > 0$, by definition of inf there is P_2 with $U(f, P_2) < I + \varepsilon/2$, and by definition of sup there is P_1 with $L(f, P_1) > I - \varepsilon/2$. Let $P = P_1 \cup P_2$. By Lemma 7.1.8, $U(f, P) \leq U(f, P_2) < I + \varepsilon/2$ and $L(f, P) \geq L(f, P_1) > I - \varepsilon/2$, so $U(f, P) - L(f, P) < \varepsilon$. $\square$

Method (Proving $f \in \mathcal{R}[a, b]$ via the gap criterion). *Recognition trigger.* You must show a specific bounded f is integrable (or compute $\int_a^b f$) without yet knowing it is continuous/monotone.

1. Fix $\varepsilon > 0$. Aim to build one partition P with $U(f, P) - L(f, P) < \varepsilon$.

2. On each I_i identify $M_i - m_i = \text{osc}(f, I_i)$ (the oscillation). The gap is $\sum_i (M_i - m_i) \Delta x_i$.
3. **Control the gap.** Either (a) make each $\text{osc}(f, I_i)$ small by taking f uniformly continuous and $\|P\|$ tiny, or (b) make the total length where the oscillation is large small (monotone / finitely-many-jumps cases).
4. Conclude $f \in \mathcal{R}$. To also *evaluate*, squeeze $L(f, P) \leq \int_a^b f \leq U(f, P)$ and let $\varepsilon \rightarrow 0$, or use a convenient tagged sum.

The two roads coincide

Theorem 7.1.11 (Darboux = Riemann, Rudin 6.7 cf. B&S 7.4.4). *For bounded f on $[a, b]$: f is Riemann integrable (Definition 7.1.3) if and only if it is Darboux integrable (Definition 7.1.6), and the two integrals agree.*

Proof. Key link. For any tagged partition $\dot{P}$ with underlying partition P , since $m_i \leq f(t_i) \leq M_i$,

$$L(f, P) \leq S(f, \dot{P}) \leq U(f, P). \quad (\star)$$

Moreover, choosing tags t_i with $f(t_i)$ within η of M_i (resp. m_i) shows $\sup_{\text{tags}} S(f, \dot{P}) = U(f, P)$ and $\inf_{\text{tags}} S(f, \dot{P}) = L(f, P)$ for the fixed P .

($\Rightarrow$, Riemann $\Rightarrow$ Darboux) Let $f \in \mathcal{R}[a, b]$ with value L and $\varepsilon > 0$; choose δ as in Definition 7.1.3. Fix any P with $\|P\| < \delta$. Taking tags to push $S(f, \dot{P})$ within ε of $U(f, P)$ and of $L(f, P)$ gives $U(f, P) \leq L + 2\varepsilon$ and $L(f, P) \geq L - 2\varepsilon$, so $U(f, P) - L(f, P) \leq 4\varepsilon$. By Theorem 7.1.10, f is Darboux integrable, and $\left| \int f - L \right| \leq 2\varepsilon$ for all ε , so the Darboux value is L .

($\Leftarrow$, Darboux $\Rightarrow$ Riemann) Assume $\int f = \bar{\int} f = I$, and let $\varepsilon > 0$. We need a single δ that works for all fine tagged partitions. By Theorem 7.1.10 fix $P_0 = \{y_0, \dots, y_N\}$ with $U(f, P_0) - L(f, P_0) < \varepsilon/2$. Put $K = \sup |f|$ and $\delta = \frac{\varepsilon}{8KN}$. Let $\dot{P}$ be any tagged partition with $\|\dot{P}\| < \delta$, and let $P^* = P \cup P_0$ be the common refinement. Splitting the sum $U(f, P) - L(f, P)$ into subintervals of P that lie inside one piece of P_0 (where, by P^* , the oscillation contribution is $\leq U(f, P_0) - L(f, P_0) < \varepsilon/2$) and the at most $N - 1$ subintervals of P straddling an interior point y_j (each contributing $\leq 2K \|P\| < 2K\delta$), we get

$$U(f, P) - L(f, P) \leq (U(f, P_0) - L(f, P_0)) + (N - 1) \cdot 2K\delta < \frac{\varepsilon}{2} + \frac{\varepsilon}{4} < \varepsilon.$$

By $(\star)$ both $S(f, \dot{P})$ and I lie in $[L(f, P), U(f, P)]$, an interval of length $< \varepsilon$, so $|S(f, \dot{P}) - I| < \varepsilon$. Hence $f \in \mathcal{R}[a, b]$ with integral I . $\square$

Remark. From now on “integrable” means either equivalent notion, and we use whichever description is handier: the *gap criterion* for existence proofs, the *Riemann sum* for limits and numerics. B&S take Definition 7.1.3 as primary and prove the Darboux equivalence; Rudin does the reverse. Both are above.

Worked Example 7.1.12 (A constant function). Show $f \equiv c$ has $\int_a^b c \, dx = c(b - a)$.

Recognition. Trigger: simplest integrability check; use any partition since $m_i = M_i = c$.

Solution. For every partition P , $m_i = M_i = c$, so $L(f, P) = U(f, P) = \sum_i c \Delta x_i = c(b - a)$. The gap is $0 < \varepsilon$, so $f \in \mathcal{R}$ and $\int = \bar{\int} = c(b - a)$. *Check.* A tagged sum gives the same: $\sum c \Delta x_i = c(b - a)$ for any tags. $\blacksquare$

Worked Example 7.1.13 ($\int_0^1 x \, dx = \frac{1}{2}$ from the definition). Compute $\int_0^1 x \, dx$ using uniform partitions.

Recognition. Trigger: “evaluate from the definition.” Use P_n with $x_i = i/n$ and exploit monotonicity so $m_i = f(x_{i-1})$, $M_i = f(x_i)$.

Solution. With $x_i = i/n$, $\Delta x_i = 1/n$. As $f(x) = x$ increases, $m_i = (i - 1)/n$, $M_i = i/n$. Then

$$L(f, P_n) = \sum_{i=1}^n \frac{i-1}{n} \cdot \frac{1}{n} = \frac{n(n-1)/2}{n^2} = \frac{1}{2} - \frac{1}{2n}, \quad U(f, P_n) = \frac{n(n+1)/2}{n^2} = \frac{1}{2} + \frac{1}{2n}.$$

The gap is $U - L = 1/n \rightarrow 0$, so $f \in \mathcal{R}[0, 1]$, and both sums squeeze to $\frac{1}{2}$. *Check.* The midpoint Riemann sum equals $\frac{1}{2}(L + U) = \frac{1}{2}$ exactly, the area of the triangle. $\blacksquare$

Worked Example 7.1.14 ($\int_0^1 x^2 dx = \frac{1}{3}$ (B&S 7.4.12)). Use P_n , $x_i = i/n$, and $\sum_{k=1}^m k^2 = \frac{1}{6}m(m+1)(2m+1)$.

Recognition. Trigger: a power evaluated by a known sum; x^2 is increasing on $[0, 1]$.

Solution. $m_i = ((i-1)/n)^2$, $M_i = (i/n)^2$, $\Delta x_i = 1/n$. So

$$U(f, P_n) = \frac{1}{n^3} \sum_{i=1}^n i^2 = \frac{(n)(n+1)(2n+1)}{6n^3} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2},$$

and $L(f, P_n) = \frac{1}{n^3} \sum_{i=1}^{n-1} i^2 = \frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2}$. The gap is $U - L = 1/n \rightarrow 0$, so $\int_0^1 x^2 dx = \frac{1}{3}$. *Check.* Both bounds $\rightarrow \frac{1}{3}$ and straddle it. ■

Worked Example 7.1.15 (The Dirichlet function is NOT integrable (B&S 7.1.12)). Let $f = 1$ on rationals, $f = 0$ on irrationals, on $[0, 1]$. Show $f \notin \mathcal{R}[0, 1]$.

Recognition. Trigger: a “disprove integrability” problem; show the upper and lower integrals differ, or two tag choices give different Riemann-sum limits.

Solution. Every subinterval I_i contains both a rational and an irrational (density), so $M_i = 1$ and $m_i = 0$ for every partition P . Hence $U(f, P) = \sum 1 \cdot \Delta x_i = 1$ and $L(f, P) = 0$ for *all* P , giving $\overline{\int} f = 1 \neq 0 = \underline{\int} f$. The gap is always 1, never $< \varepsilon$ for small ε , so f is not integrable. *Riemann-sum view:* tagging all rationals gives sums = 1; tagging all irrationals gives sums = 0; as both have mesh $\rightarrow 0$ but different limits, no L can exist. ■

Caution. The mesh $\|\dot{P}\| \rightarrow 0$ is essential: it is *not* enough that the number of subintervals $n \rightarrow \infty$. A partition can have many points yet a large widest piece. Also, “ f integrable” in Definition 7.1.3 demands the limit hold for *every* tag choice at once; a single lucky tag (e.g. the midpoint) converging proves nothing.

Further worked examples

Worked Example 7.1.16 (Reading off L , U , and a tagged sum for a fixed partition). Let $f(x) = x^2$ on $[0, 2]$ and $P = \{0, 1, \frac{3}{2}, 2\}$. Compute $L(f, P)$, $U(f, P)$, and the Riemann sum with left-endpoint tags.

Recognition. Trigger: a bare “evaluate the three sums” drill (Definitions 7.1.5, 7.1.2). On each piece x^2 is increasing, so $m_i = f(\text{left})$, $M_i = f(\text{right})$.

Solution. The widths are $\Delta x_1 = 1$, $\Delta x_2 = \frac{1}{2}$, $\Delta x_3 = \frac{1}{2}$. Since x^2 increases, m_i uses the left endpoint and M_i the right:

$$L = 0^2 \cdot 1 + 1^2 \cdot \frac{1}{2} + \left(\frac{3}{2}\right)^2 \cdot \frac{1}{2} = 0 + \frac{1}{2} + \frac{9}{8} = \frac{13}{8} = 1.625,$$

$$U = 1^2 \cdot 1 + \left(\frac{3}{2}\right)^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{2} = 1 + \frac{9}{8} + 2 = \frac{33}{8} = 4.125.$$

The left-endpoint tagged sum reuses the same heights as L here (left tags on an increasing f), so $S(f, \dot{P}) = L = \frac{13}{8}$. *Check.* The true value is $\int_0^2 x^2 dx = \frac{8}{3} \approx 2.667$, and indeed $L = 1.625 \leq 2.667 \leq 4.125 = U$. The gap $U - L = 2.5$ is large because P is coarse. ■

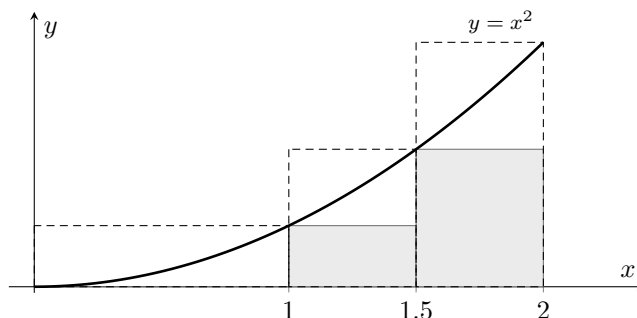


Figure 7.3. Shaded: the lower sum $L = \frac{13}{8}$ (heights at left endpoints). Dashed: the upper rectangles giving $U = \frac{33}{8}$. The true area $\frac{8}{3}$ is trapped between.

Worked Example 7.1.17 (Mesh small but n large is not the point). Exhibit a partition of $[0, 1]$ with 1000 points yet mesh $\frac{1}{2}$, illustrating the pitfall that many points do not force a small mesh.

Recognition. Trigger: a conceptual check on Definition 7.1.1; the mesh is the *widest* piece, not the count.

Solution. Put 999 points crammed into $[0, \frac{1}{2}]$ (say $x_i = \frac{i}{2000}$ for $i = 0, \dots, 999$, so each piece there has width $\frac{1}{2000}$) and then jump straight to $x = 1$. The last subinterval $[\frac{999}{2000}, 1]$ has width $1 - \frac{999}{2000} = \frac{1001}{2000} > \frac{1}{2}$, so $\|P\| > \frac{1}{2}$ despite 1000 partition points. *Check.* Convergence in Definition 7.1.3 is controlled by $\|P\| \rightarrow 0$, and this P does not help at all on $[\frac{1}{2}, 1]$ where one fat rectangle remains. ■

Exercises (Bartle & Sherbert, §7.1).

- For $I = [0, 4]$, calculate the norms of the partitions (a) $P_1 = (0, 1, 2, 4)$; (b) $P_2 = (0, 2, 3, 4)$; (c) $P_3 = (0, 1, 1.5, 2, 3.4, 4)$; (d) $P_4 = (0, 0.5, 2.5, 3.5, 4)$.
- For $f(x) = x^2$ on $[0, 4]$, compute the Riemann sums $S(f, \dot{P}_i)$, where $\dot{P}_i$ has the same partition points as in Exercise 1 and tags as indicated: (a) P_1 , left endpoints; (b) P_1 , right endpoints; (c) P_2 , left endpoints; (d) P_2 , right endpoints.
- Show $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable iff there is $L \in \mathbb{R}$ such that for every $\varepsilon > 0$ there is $\delta_\varepsilon > 0$ with $|S(f, \dot{P}) - L| \leq \varepsilon$ whenever $\dot{P}$ is any tagged partition with $\|\dot{P}\| \leq \delta_\varepsilon$.
- Let $\dot{P}$ be a tagged partition of $[0, 3]$. (a) Show the union U_1 of all subintervals of $\dot{P}$ with tags in $[0, 1]$ satisfies $[0, 1 - \|\dot{P}\|] \subseteq U_1 \subseteq [0, 1 + \|\dot{P}\|]$. (b) Show the union U_2 of all subintervals with tags in $[1, 2]$ satisfies $[1 + \|\dot{P}\|, 2 - \|\dot{P}\|] \subseteq U_2 \subseteq [1 - \|\dot{P}\|, 2 + \|\dot{P}\|]$.
- Let $\dot{P} = \{(I_i, t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$ and $c_1 < c_2$. (a) If u belongs to a subinterval I_i whose tag satisfies $c_1 \leq t_i \leq c_2$, show $c_1 - \|\dot{P}\| \leq u \leq c_2 + \|\dot{P}\|$. (b) If $v \in [a, b]$ satisfies $c_1 + \|\dot{P}\| \leq v \leq c_2 - \|\dot{P}\|$, then the tag t_i of any subinterval I_i containing v satisfies $t_i \in [c_1, c_2]$.
- (a) Let $f(x) = 2$ for $0 \leq x < 1$ and $f(x) = 1$ for $1 \leq x \leq 2$. Show $f \in \mathcal{R}[0, 2]$ and evaluate its integral. (b) Let $h(x) = 2$ for $0 \leq x < 1$, $h(1) = 3$, $h(x) = 1$ for $1 < x \leq 2$. Show $h \in \mathcal{R}[0, 2]$ and evaluate its integral.
- Use induction and the linearity theorem to show that if $f_1, \dots, f_n \in \mathcal{R}[a, b]$ and $k_1, \dots, k_n \in \mathbb{R}$, then $\sum_i k_i f_i \in \mathcal{R}[a, b]$ and $\int_a^b \sum_i k_i f_i = \sum_i k_i \int_a^b f_i$.
- If $f \in \mathcal{R}[a, b]$ and $|f(x)| \leq M$ for all $x \in [a, b]$, show $\left| \int_a^b f \right| \leq M(b - a)$.
- If $f \in \mathcal{R}[a, b]$ and $(\dot{P}_n)$ is any sequence of tagged partitions with $\|\dot{P}_n\| \rightarrow 0$, prove $\int_a^b f = \lim_n S(f, \dot{P}_n)$.
- Let $g(x) = 0$ if $x \in [0, 1]$ is rational and $g(x) = 1/x$ if $x \in [0, 1]$ is irrational (with a convention at 0). Explain why $g \notin \mathcal{R}[0, 1]$. However, show there is a sequence $(\dot{P}_n)$ of tagged partitions with $\|\dot{P}_n\| \rightarrow 0$ for which $\lim_n S(g, \dot{P}_n)$ exists.
- Suppose f is bounded on $[a, b]$ and there are two sequences $(\dot{P}_n), (\dot{Q}_n)$ of tagged partitions with $\|\dot{P}_n\| \rightarrow 0$ and $\|\dot{Q}_n\| \rightarrow 0$ but $\lim_n S(f, \dot{P}_n) \neq \lim_n S(f, \dot{Q}_n)$. Show $f \notin \mathcal{R}[a, b]$.
- Consider the Dirichlet function $f(x) = 1$ for $x \in [0, 1]$ rational, $f(x) = 0$ for $x \in [0, 1]$ irrational. Use the preceding exercise to show $f \notin \mathcal{R}[0, 1]$.
- Suppose $c \leq d$ are points of $[a, b]$. If $w : [a, b] \rightarrow \mathbb{R}$ satisfies $w(x) = \rho > 0$ for $x \in [c, d]$ and $w(x) = 0$ elsewhere, prove $w \in \mathcal{R}[a, b]$ and $\int_a^b w = \rho(d - c)$. (Hint: given $\varepsilon > 0$ let $\delta_\varepsilon = \varepsilon/4\rho$ and bound $S(w, \dot{P})$ between $\rho(d - c - 2\delta_\varepsilon)$ and $\rho(d - c + 2\delta_\varepsilon)$.)
- Let $0 \leq a < b$, $Q(x) = x^2$ on $[a, b]$, and $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ a partition. For each i let $q_i = \sqrt{\frac{1}{3}(x_i^2 + x_i x_{i-1} + x_{i-1}^2)}$. (a) Show $0 \leq x_{i-1} \leq q_i \leq x_i$. (b) Show $Q(q_i)(x_i - x_{i-1}) = \frac{1}{3}(x_i^3 - x_{i-1}^3)$. (c) If $\dot{Q}$ is the tagged partition with the same subintervals and tags q_i , show $S(Q, \dot{Q}) = \frac{1}{3}(b^3 - a^3)$. (d) Conclude $Q \in \mathcal{R}[a, b]$ and $\int_a^b x^2 dx = \frac{1}{3}(b^3 - a^3)$.
- If $f \in \mathcal{R}[a, b]$ and $c \in \mathbb{R}$, define g on $[a + c, b + c]$ by $g(y) = f(y - c)$. Prove $g \in \mathcal{R}[a + c, b + c]$ and $\int_{a+c}^{b+c} g = \int_a^b f$. (The function g is the c -translate of f .)

Exercises (Bartle & Sherbert, §7.4 — the Darboux integral).

1. Let $f(x) = |x|$ for $-1 \leq x \leq 2$. Calculate $L(f, P)$ and $U(f, P)$ for (a) $P_1 = (-1, 0, 1, 2)$; (b) $P_2 = (-1, -1/2, 0, 1/2, 1, 3/2, 2)$.
2. Prove that if $f(x) = c$ on $[a, b]$, then its Darboux integral equals $c(b - a)$.
3. Let f, g be bounded on $I = [a, b]$ with $f(x) \leq g(x)$ for all x . Show $L(f) \leq L(g)$ and $U(f) \leq U(g)$.
4. Let f be bounded on $[a, b]$ and $k > 0$. Show $L(kf) = kL(f)$ and $U(kf) = kU(f)$.
5. Let f, g, h be bounded on $I = [a, b]$ with $f \leq g \leq h$. Show that if f, h are Darboux integrable and $\int_a^b f = \int_a^b h$, then g is Darboux integrable with $\int_a^b g = \int_a^b f$.
6. Let f on $[0, 2]$ be $f(x) = 1$ if $x \neq 1$ and $f(1) = 0$. Show the Darboux integral exists and find its value.
7. (a) Prove that if $g(x) = 0$ for $0 \leq x \leq \frac{1}{2}$ and $g(x) = 1$ for $\frac{1}{2} < x \leq 1$, the Darboux integral of g on $[0, 1]$ is $\frac{1}{2}$. (b) Does the conclusion hold if we change $g(\frac{1}{2})$ to $\frac{1}{3}$?
8. Let f be continuous on $I = [a, b]$ with $f(x) \geq 0$. Prove that if $L(f) = 0$, then $f(x) = 0$ for all $x \in I$.
9. Let f_1, f_2 be bounded on $[a, b]$. Show $L(f_1) + L(f_2) \leq L(f_1 + f_2)$.
10. Give an example to show strict inequality can hold in the preceding exercise.
11. If f is bounded on $[a, b]$ with $f(x) = 0$ except for $x \in \{c_1, \dots, c_n\}$, show $U(f) = L(f) = 0$.
12. Let $f(x) = x^2$ for $0 \leq x \leq 1$. For $P_n = (0, 1/n, 2/n, \dots, (n-1)/n, 1)$ calculate $L(f, P_n)$ and $U(f, P_n)$ and show $L(f) = U(f) = \frac{1}{3}$. (Use $\sum_{k=1}^m k^2 = \frac{1}{6}m(m+1)(2m+1)$.)
13. Let P_ε be the partition from the Integrability Criterion. Show that if P is any refinement of P_ε , then $U(f, P) - L(f, P) < \varepsilon$.
14. Write out the proofs that f on $[a, b]$ is Darboux integrable if it is either (a) continuous, or (b) monotone.
15. Let f on $I = [a, b]$ satisfy the Lipschitz condition $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in I$. If P_n is the partition into n equal parts, show $0 \leq U(f, P_n) - \int_a^b f \leq K(b-a)^2/n$.

Additional exercises.

1. (Mesh, not count.) Give a sequence of partitions P_n of $[0, 1]$ with n points each but $\|P_n\| \not\rightarrow 0$, and explain why the Riemann-sum limit need not equal $\int_0^1 f$ along it.
2. Compute $\int_0^1 x^3 dx$ directly from upper/lower sums using $x_i = i/n$ and $\sum_{k=1}^m k^3 = (\frac{m(m+1)}{2})^2$.
3. (Refinement.) Let $f(x) = x$ on $[0, 1]$ and $P = \{0, 1\}$, $P^* = \{0, \frac{1}{2}, 1\}$. Compute $L(f, P), U(f, P), L(f, P^*), U(f, P^*)$ and verify the refinement lemma's chain of inequalities numerically.
4. Show that if $f \in \mathcal{R}[a, b]$ and we change f at finitely many points, the new function is still in $\mathcal{R}[a, b]$ with the same integral.
5. (Non-example.) Exhibit a bounded $f : [0, 1] \rightarrow \mathbb{R}$ with $\overline{\int} f = 1$ and $\underline{\int} f = 0$, and prove these values. Conclude $f \notin \mathcal{R}[0, 1]$.
6. Prove the inf/sup tag identities used in Theorem 7.1.11: for fixed P , $\sup_{\text{tags}} S(f, \dot{P}) = U(f, P)$ and $\inf_{\text{tags}} S(f, \dot{P}) = L(f, P)$.
7. (Diagram.) For $f(x) = \sqrt{x}$ on $[0, 1]$ and the uniform partition P_4 , draw the lower and upper Darboux rectangles in two side-by-side pictures and shade the gap $U(f, P_4) - L(f, P_4)$. Estimate the gap numerically.
8. Show directly from the definition that if $f \in \mathcal{R}[a, b]$ then f is bounded. (So boundedness is not an extra hypothesis but a consequence on a closed interval; explain where compactness is used.)
9. Let f be increasing on $[a, b]$. Prove $0 \leq U(f, P_n) - \int_a^b f \leq \frac{(b-a)(f(b)-f(a))}{n}$ for the uniform partition, an explicit error bound for the upper sum.

7.2 Riemann Integrable Functions

We now identify large classes of integrable functions. The engine is always the gap criterion (Theorem 7.1.10): we must make $\sum_i (M_i - m_i) \Delta x_i$ small.

Continuous functions

Theorem 7.2.1 (Continuous $\implies$ integrable, Rudin 6.8 / B&S 7.2.7). *If f is continuous on $[a, b]$, then $f \in \mathcal{R}[a, b]$.*

Proof. $[a, b]$ is compact, so f is *uniformly* continuous (Heine–Cantor): given $\varepsilon > 0$ there is $\eta = \varepsilon/(b-a)$ and a $\delta > 0$ with

$$|x - y| < \delta \implies |f(x) - f(y)| < \eta.$$

Choose a partition P with $\|P\| < \delta$. On each I_i , f attains its sup and inf (extreme value theorem) at points u_i, v_i with $|u_i - v_i| \leq \Delta x_i < \delta$, so $M_i - m_i = f(u_i) - f(v_i) < \eta$. Therefore

$$U(f, P) - L(f, P) = \sum_i (M_i - m_i) \Delta x_i < \eta \sum_i \Delta x_i = \eta(b-a) = \varepsilon.$$

By Theorem 7.1.10, $f \in \mathcal{R}[a, b]$. □

Monotone functions

Theorem 7.2.2 (Monotone $\implies$ integrable, Rudin 6.9 / B&S 7.2.8). *If f is monotone on $[a, b]$, then $f \in \mathcal{R}[a, b]$. (No continuity is assumed; a monotone function may have countably many jumps.)*

Proof. Say f is increasing (the decreasing case is symmetric). Take the *uniform* partition P_n , $\Delta x_i = (b-a)/n$. By monotonicity $m_i = f(x_{i-1})$ and $M_i = f(x_i)$, so the heights *telescope*:

$$U(f, P_n) - L(f, P_n) = \frac{b-a}{n} \sum_{i=1}^n (f(x_i) - f(x_{i-1})) = \frac{b-a}{n} (f(b) - f(a)).$$

Given $\varepsilon > 0$, choose $n > (b-a)(f(b) - f(a))/\varepsilon$; then the gap is $< \varepsilon$. Hence $f \in \mathcal{R}[a, b]$. □

monotone f : the gap rectangles stack into one column of width $\|P\|$

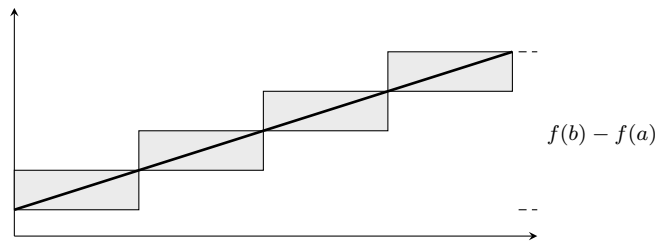


Figure 7.4. For increasing f the gap boxes have total height $f(b) - f(a)$ and common width $\|P\|$; the gap is $\|P\| (f(b) - f(a))$.

Finitely many discontinuities

Theorem 7.2.3 (Repair of finitely many bad points, Rudin 6.10 / B&S 7.2.9). *Let f be bounded on $[a, b]$ and continuous except at finitely many points. Then $f \in \mathcal{R}[a, b]$.*

Proof. Let $K = \sup |f|$ and let $E = \{e_1, \dots, e_p\}$ be the discontinuities. Given $\varepsilon > 0$, enclose each e_j in an open interval of length $< \varepsilon/(8Kp)$; remove these open intervals. What remains is a finite union of closed intervals on which f is continuous, hence uniformly continuous, so we can partition each so finely that its

oscillation contribution is $< \varepsilon/2$ total. On the removed intervals the oscillation is $\leq 2K$ each, contributing $< 2K \cdot p \cdot \frac{\varepsilon}{8Kp} = \varepsilon/4$. Assembling all the partition points into one P ,

$$U(f, P) - L(f, P) < \frac{\varepsilon}{2} + \frac{\varepsilon}{4} < \varepsilon.$$

By the gap criterion $f \in \mathcal{R}[a, b]$. □

Corollary 7.2.4 (Squeeze / continuous-a.e. remark). *A bounded f is Riemann integrable on $[a, b]$ if and only if its set of discontinuities has measure zero (can be covered by intervals of arbitrarily small total length). This is Lebesgue's criterion; Theorems 7.2.1–7.2.3 are the easy directions. (We state it; the full proof waits for Lebesgue theory.)*

Theorem 7.2.5 (Composition / Squeeze, Rudin 6.11; B&S 7.2.3). *If $f \in \mathcal{R}[a, b]$ with $m \leq f \leq M$ and φ is continuous on $[m, M]$, then $\varphi \circ f \in \mathcal{R}[a, b]$. In particular $|f|$, f^2 , and (with $f \geq 0$) $\sqrt{f}$ are integrable. Squeeze form (B&S): if for each ε there are $g_\varepsilon, h_\varepsilon \in \mathcal{R}$ with $g_\varepsilon \leq f \leq h_\varepsilon$ and $\int (h_\varepsilon - g_\varepsilon) < \varepsilon$, then $f \in \mathcal{R}$.*

Proof sketch, Rudin's argument. Let $K = \sup_{[m, M]} |\varphi|$ and $\varepsilon > 0$. By uniform continuity of φ pick $\delta \leq \varepsilon$ with $|s - t| < \delta \implies |\varphi(s) - \varphi(t)| < \varepsilon$. Choose P with $U(f, P) - L(f, P) < \delta^2$. Split the indices: A where $M_i - m_i < \delta$ (there φ -oscillation $< \varepsilon$) and B where $M_i - m_i \geq \delta$. On B , $\delta \sum_B \Delta x_i \leq \sum_B (M_i - m_i) \Delta x_i < \delta^2$, so $\sum_B \Delta x_i < \delta$. Thus the $\varphi \circ f$ gap is $\leq \varepsilon(b-a) + 2K\delta < \varepsilon(b-a + 2K)$, which $\rightarrow 0$ with ε . Apply the gap criterion. The Squeeze form follows since $L(g_\varepsilon) \leq \int f \leq \bar{\int} f \leq U(h_\varepsilon)$ and $U(h_\varepsilon) - L(g_\varepsilon) \leq \int (h_\varepsilon - g_\varepsilon) + (\text{gaps}) < \varepsilon$. □

Worked Example 7.2.6 ($1/x$ -type discontinuity tamed (B&S 7.2.12)). Show $g(x) = \sin(1/x)$ for $x \in (0, 1]$, $g(0) = 0$, lies in $\mathcal{R}[0, 1]$.

Recognition. Trigger: a single bad point at 0 where g oscillates but stays bounded.

Solution. g is bounded ($|g| \leq 1$) and continuous on every $[c, 1]$ with $c > 0$. The only discontinuity is at 0. By Theorem 7.2.3 (one bad point), $g \in \mathcal{R}[0, 1]$. *Direct check:* given ε , isolate $[0, \varepsilon/4]$ (oscillation contribution $\leq 2 \cdot \varepsilon/4 = \varepsilon/2$) and partition $[\varepsilon/4, 1]$ finely (continuous there). Gap $< \varepsilon$. ■

Worked Example 7.2.7 (A step function (B&S 7.1.6)). Let $f = 2$ on $[0, 1)$ and $f = 1$ on $[1, 2]$. Show $f \in \mathcal{R}[0, 2]$ and find the integral.

Recognition. Trigger: one jump; monotone (decreasing) too, so two routes.

Solution. f is decreasing, hence integrable (Theorem 7.2.2); or it has the single discontinuity $x = 1$. For the value, take P with a tiny interval $[1 - \eta, 1 + \eta]$ around the jump. Off that interval f is constant, contributing $2(1 - \eta) + 1(1 - \eta)$; the middle contributes between $1 \cdot 2\eta$ and $2 \cdot 2\eta$. Letting $\eta \rightarrow 0$, $\int_0^2 f = 2 \cdot 1 + 1 \cdot 1 = 3$. *Check.* Changing $f(1)$ to any value does not change the integral — single-point values are invisible. ■

Worked Example 7.2.8 (Thomae's function: integrable despite dense discontinuities). Let $f(x) = 1/q$ if $x = p/q$ in lowest terms ($q > 0$), $f(x) = 0$ if x irrational, $f(0) = 0$, on $[0, 1]$. Show $\int_0^1 f = 0$.

Recognition. Trigger: discontinuous at every rational (dense!), yet only “a little” at each. $L(f, P) = 0$ always; we squeeze the upper sum.

Solution. Since irrationals are dense, $m_i = 0$ on every subinterval, so $L(f, P) = 0$ and $\int f = 0$. For the upper sum fix $\varepsilon > 0$. Only finitely many points have $f(x) \geq \varepsilon/2$ (namely $x = p/q$ with $q \leq 2/\varepsilon$); call them $x_1, \dots, x_N$. Enclose them in subintervals of total length $< \varepsilon/2$ (each contributes $\leq 1 \cdot \text{length}$). Elsewhere $f < \varepsilon/2$. Thus $U(f, P) < \varepsilon/2 \cdot 1 + 1 \cdot \varepsilon/2 = \varepsilon$, so $\bar{\int} f \leq \varepsilon$ for all ε , giving $\bar{\int} f = 0$. Hence $\int_0^1 f = 0$. *Check.* This matches Lebesgue's criterion — the discontinuity set (the rationals) is countable, hence measure zero. ■

Worked Example 7.2.9 (NON-example: $\text{sgn} \circ \text{Thomae}$ fails (B&S 7.3.22)). Let h be Thomae's function and sgn the sign function. Show $\text{sgn} \circ h \notin \mathcal{R}[0, 1]$, even though $h \in \mathcal{R}$.

Recognition. Trigger: a composition that breaks integrability — shows Theorem 7.2.5 needs φ continuous.

Solution. $(\text{sgn} \circ h)(x) = 1$ if x is rational (then $h > 0$), $= 0$ if irrational (then $h = 0$). That is exactly the Dirichlet function, which is not integrable (its upper integral 1 exceeds its lower integral 0). The catch: sgn is discontinuous at 0, and h takes the value 0 on a dense set, so the discontinuity of sgn is “activated” everywhere. *Lesson.* Integrability is *not* preserved under composition with a discontinuous outer function. ■

Caution. “Riemann integrable $\implies$ continuous” is false (step and Thomae functions). “Countably many discontinuities $\implies$ integrable” is also false in general — the Dirichlet function is discontinuous at *every* point (uncountably many). The correct dividing line is Lebesgue’s: the discontinuity set must have measure zero.

Further worked examples

Worked Example 7.2.10 (Monotone with a countable jump set: still integrable). Let f be the increasing step function on $[0, 1]$ that jumps by 2^{-n} at each point $x = 1 - 2^{-n}$ ($n = 1, 2, 3, \dots$) and is constant between jumps. Show $f \in \mathcal{R}[0, 1]$ even though f has *infinitely many* discontinuities.

Recognition. Trigger: countably many jumps would seem to violate the “finitely many bad points” theorem, but Theorem 7.2.2 only needs monotonicity.

Solution. f is increasing (every jump is positive) and bounded by $\sum_{n \geq 1} 2^{-n} = 1$. By Theorem 7.2.2, every increasing function on $[a, b]$ is integrable regardless of how many jumps it has; the proof telescopes the gap to $\frac{b-a}{n}(f(b) - f(a)) \leq \frac{1}{n} \rightarrow 0$. So $f \in \mathcal{R}[0, 1]$. *Check.* The discontinuity set $\{1 - 2^{-n}\}$ is countable, hence measure zero, consistent with Lebesgue’s criterion (Corollary 7.2.4). The finitely-many-jumps theorem 7.2.3 is simply *not needed* here — monotonicity already wins. ■

Worked Example 7.2.11 (NON-example: every subinterval has full oscillation). Let $f(x) = 1$ if $x \in [0, 1]$ has a terminating decimal expansion and $f(x) = 0$ otherwise. Decide whether $f \in \mathcal{R}[0, 1]$.

Recognition. Trigger: a “classify this exotic function” problem; test the gap criterion by finding $M_i - m_i$ on a typical subinterval.

Solution. Terminating decimals are exactly the rationals of the form $p/10^k$; these are dense in $[0, 1]$, and so are their complements (e.g. all irrationals are non-terminating). Hence every subinterval I_i contains a point with $f = 1$ and a point with $f = 0$, giving $M_i = 1$, $m_i = 0$ for every partition. Then $U(f, P) - L(f, P) = \sum 1 \cdot \Delta x_i = 1$ for all P , never $< \varepsilon$ for $\varepsilon < 1$. So $f \notin \mathcal{R}[0, 1]$. *Check.* This is the Dirichlet mechanism (Section 1 worked example) in disguise — the discontinuity set is all of $[0, 1]$, which does *not* have measure zero. ■

Worked Example 7.2.12 (Composition: $\sqrt{\cdot}$ of an integrable function). Given $f(x) = x$ on $[0, 4]$ (so $f \geq 0$), show $\sqrt{f} \in \mathcal{R}[0, 4]$ and check the value $\int_0^4 \sqrt{x} \, dx = \frac{16}{3}$.

Recognition. Trigger: a composition $\varphi \circ f$ with $\varphi(t) = \sqrt{t}$ continuous on the range $[0, 4]$ of f ; invoke Theorem 7.2.5.

Solution. $f(x) = x$ is continuous hence integrable, with $0 \leq f \leq 4$. The outer map $\varphi(t) = \sqrt{t}$ is continuous on $[0, 4]$, so by the composition theorem $\varphi \circ f = \sqrt{x} \in \mathcal{R}[0, 4]$. To evaluate, $\sqrt{x}$ is increasing and has the antiderivative $\frac{2}{3}x^{3/2}$, so $\int_0^4 \sqrt{x} \, dx = \frac{2}{3}(4^{3/2} - 0) = \frac{2}{3} \cdot 8 = \frac{16}{3}$. *Check.* $\sqrt{x}$ runs from 0 to 2 on $[0, 4]$; the region sits below the rectangle of area $4 \cdot 2 = 8$ and $\frac{16}{3} \approx 5.33 < 8$, plausible. ■

Exercises (Bartle & Sherbert, §7.2).

1. Let $f : [a, b] \rightarrow \mathbb{R}$. Show $f \notin \mathcal{R}[a, b]$ iff there is $\varepsilon_0 > 0$ such that for every $n \in \mathbb{N}$ there exist tagged partitions $\dot{P}_n, \dot{Q}_n$ with $\|\dot{P}_n\| < 1/n$, $\|\dot{Q}_n\| < 1/n$, and $|S(f, \dot{P}_n) - S(f, \dot{Q}_n)| \geq \varepsilon_0$.
2. Let $h(x) = x + 1$ for $x \in [0, 1]$ rational and $h(x) = 0$ for $x \in [0, 1]$ irrational. Show h is not Riemann integrable.
3. Let $H(x) = k$ for $x = 1/k$ ($k \in \mathbb{N}$) and $H(x) = 0$ elsewhere on $[0, 1]$. Use Exercise 1 (or the gap argument) to show H is not Riemann integrable.
4. If $a(x) = x$ and $v(x) = x$ and $a(x) \leq f(x) \leq v(x)$ for all $x \in [0, 1]$, does it follow from the Squeeze Theorem that $f \in \mathcal{R}[0, 1]$?
5. If J is any subinterval of $[a, b]$ and $w_J(x) = 1$ for $x \in J$, $w_J(x) = 0$ elsewhere, we call w_J an elementary step function. Show every step function is a linear combination of elementary step functions.
6. If $\psi : [a, b] \rightarrow \mathbb{R}$ takes only finitely many distinct values, is ψ necessarily a step function?

7. If $S(f, \dot{P})$ is any Riemann sum of $f : [a, b] \rightarrow \mathbb{R}$, show there is a step function $w : [a, b] \rightarrow \mathbb{R}$ with $\int_a^b w = S(f, \dot{P})$.
8. Suppose f is continuous on $[a, b]$, $f(x) \geq 0$ for all x , and $\int_a^b f = 0$. Prove $f(x) = 0$ for all $x \in [a, b]$.
9. Show that the continuity hypothesis in the preceding exercise cannot be dropped.
10. If f, g are continuous on $[a, b]$ and $\int_a^b f = \int_a^b g$, prove there is $c \in [a, b]$ with $f(c) = g(c)$.
11. If f is bounded by M on $[a, b]$ and the restriction of f to every $[c, b]$ with $c \in (a, b)$ is Riemann integrable, show $f \in \mathcal{R}[a, b]$ and $\int_c^b f \rightarrow \int_a^b f$ as $c \rightarrow a^+$. (Hint: squeeze between $\pm M$ near a and f on $[c, b]$.)
12. Show $g(x) = \sin(1/x)$ for $x \in (0, 1]$, $g(0) = 0$, belongs to $\mathcal{R}[0, 1]$.
13. Give an example of $f : [a, b] \rightarrow \mathbb{R}$ that is in $\mathcal{R}[c, b]$ for every $c \in (a, b)$ but is not in $\mathcal{R}[a, b]$.
14. Suppose $a = c_0 < c_1 < \cdots < c_m = b$ and the restriction of f to each $[c_{i-1}, c_i]$ belongs to $\mathcal{R}[c_{i-1}, c_i]$. Prove $f \in \mathcal{R}[a, b]$ and the additivity formula $\int_a^b f = \sum_i \int_{c_{i-1}}^{c_i} f$ holds.
15. If f is bounded and there is a finite set E such that f is continuous at every point of $[a, b] \setminus E$, show $f \in \mathcal{R}[a, b]$.
16. If f is continuous on $[a, b]$ with $a < b$, show there is $c \in [a, b]$ with $\int_a^b f = f(c)(b-a)$ (Mean Value Theorem for Integrals).
17. If f, g are continuous on $[a, b]$ and $g(x) > 0$ for all x , show there is $c \in [a, b]$ with $\int_a^b fg = f(c) \int_a^b g$. Show the conclusion fails without $g > 0$.
18. Let f be continuous on $[a, b]$ with $f(x) \geq 0$, and $M_n = (\int_a^b f^n)^{1/n}$. Show $\lim_n M_n = \sup\{f(x) : x \in [a, b]\}$.
19. Suppose $\rho > 0$ and $f \in \mathcal{R}[-\rho, \rho]$. (a) If f is even ($f(-x) = f(x)$), show $\int_{-\rho}^{\rho} f = 2 \int_0^{\rho} f$. (b) If f is odd ($f(-x) = -f(x)$), show $\int_{-\rho}^{\rho} f = 0$.
20. If f is continuous on $[-\rho, \rho]$, show $\int_{-\rho}^{\rho} f(x^2) dx = 2 \int_0^{\rho} f(x^2) dx$.

Additional exercises.

1. Prove $f(x) = \lfloor 1/x \rfloor$ for $x \in (0, 1]$, $f(0) = 0$, is integrable on $[0, 1]$ despite having infinitely many jumps. (Hint: monotone on $(0, 1]$; bound the contribution near 0.)
2. Show that if $f, g \in \mathcal{R}[a, b]$ then $\max(f, g)$ and $\min(f, g)$ are in $\mathcal{R}[a, b]$. (Hint: $\max(f, g) = \frac{1}{2}(f+g+|f-g|)$.)
3. Prove that if f is continuous on $[a, b]$ and $f \in \mathcal{R}$, then for each $\varepsilon > 0$ there is a step function s with $\int_a^b |f - s| < \varepsilon$.
4. (Non-example.) Show the function $f(x) = 1$ if $x = 1/n$ for some $n \in \mathbb{N}$ and $f(x) = 0$ otherwise is integrable on $[0, 1]$ with $\int_0^1 f = 0$, but its set of discontinuities is infinite. Reconcile with Lebesgue's criterion.
5. (Non-example, sharper.) Let $V \subseteq [0, 1]$ be a "fat Cantor set" of positive measure and $f = \mathbf{1}_V$. Argue informally why $f \notin \mathcal{R}[0, 1]$, contrasting with the standard Cantor set where f is integrable.
6. Show that if $f \in \mathcal{R}[a, b]$ and $f \geq 0$, then $\sqrt{f} \in \mathcal{R}[a, b]$.
7. (Diagram.) Sketch Thomae's function on $[0, 1]$ by plotting the points $(p/q, 1/q)$ for small q , and shade a covering of the "tall" points ($1/q \geq 1/4$) by short intervals totaling length $< \frac{1}{2}$. Use the picture to explain why $\int f$ is small.
8. Prove the squeeze theorem (Theorem 7.2.5, B&S form) directly: if $g_\varepsilon \leq f \leq h_\varepsilon$ with $g_\varepsilon, h_\varepsilon \in \mathcal{R}$ and $\int(h_\varepsilon - g_\varepsilon) < \varepsilon$ for each ε , then $f \in \mathcal{R}$.
9. Give an example of $f, g \notin \mathcal{R}[0, 1]$ with $f + g \in \mathcal{R}[0, 1]$.

7.3 Properties of the Integral

With a healthy supply of integrable functions, we record the algebraic and order properties that make the integral usable. Most are cleanest in the Riemann-sum picture: a linear identity for sums survives the limit $\|\dot{P}\| \rightarrow 0$.

Linearity and order

Theorem 7.3.1 (Linearity and monotonicity, Rudin 6.12 / B&S 7.1.5). *Let $f, g \in \mathcal{R}[a, b]$ and $k \in \mathbb{R}$. Then $f + g, kf \in \mathcal{R}[a, b]$ and*

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g, \quad \int_a^b kf = k \int_a^b f.$$

If moreover $f(x) \leq g(x)$ for all x , then $\int_a^b f \leq \int_a^b g$ (monotonicity). In particular $m \leq f \leq M$ gives $m(b-a) \leq \int_a^b f \leq M(b-a)$.

Proof. For a fixed tagged partition, $S(f + g, \dot{P}) = S(f, \dot{P}) + S(g, \dot{P})$ and $S(kf, \dot{P}) = k S(f, \dot{P})$ identically. Let $\varepsilon > 0$; choose δ small enough that $\|\dot{P}\| < \delta$ forces $S(f, \dot{P})$ within $\varepsilon/2$ of $\int f$ and $S(g, \dot{P})$ within $\varepsilon/2$ of $\int g$. Then $S(f + g, \dot{P})$ is within ε of $\int f + \int g$, proving $f + g \in \mathcal{R}$ with the stated value. The kf case is similar (handle $k = 0$ trivially). For monotonicity, $f \leq g$ makes $S(f, \dot{P}) \leq S(g, \dot{P})$ for every $\dot{P}$; passing to the limit preserves $\leq$. $\square$

Additivity over intervals

Theorem 7.3.2 (Splitting the interval, Rudin 6.12(c) / B&S 7.2.10). *Let $a < c < b$. Then $f \in \mathcal{R}[a, b]$ iff $f \in \mathcal{R}[a, c]$ and $f \in \mathcal{R}[c, b]$, and then*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

With the conventions $\int_a^a f = 0$ and $\int_b^a f = -\int_a^b f$, this holds for a, b, c in any order.

Proof. Only partitions containing c matter for the gap criterion, since inserting c only shrinks the gap (Lemma 7.1.8). For such $P = P' \cup P''$ (with P' on $[a, c]$, P'' on $[c, b]$),

$$U(f, P) = U(f, P') + U(f, P''), \quad L(f, P) = L(f, P') + L(f, P''),$$

so the gap on $[a, b]$ is the sum of the gaps on the two pieces. Hence the big gap is $< \varepsilon$ for some P iff each piece's gap can be made $< \varepsilon/2$. Passing the additive identity $U = U' + U''$ to the inf (over partitions containing c) gives the value formula. $\square$

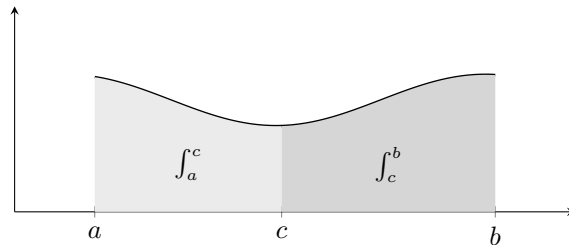


Figure 7.5. Additivity: the area over $[a, b]$ is the sum of the areas over $[a, c]$ and $[c, b]$.

Absolute values and products

Theorem 7.3.3 (Triangle inequality for integrals, Rudin 6.13(b) / B&S 7.1.5(c)). *If $f \in \mathcal{R}[a, b]$ then $|f| \in \mathcal{R}[a, b]$ and*

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

If also $|f| \leq M$, then $\left| \int_a^b f \right| \leq M(b-a)$.

Proof. $|f| = \varphi \circ f$ with $\varphi(t) = |t|$ continuous, so $|f| \in \mathcal{R}$ by Theorem 7.2.5. Now $-|f| \leq f \leq |f|$; integrating and using monotonicity, $-\int |f| \leq \int f \leq \int |f|$, which is exactly $|\int f| \leq \int |f|$. The last bound is monotonicity applied to $|f| \leq M$. $\square$

Theorem 7.3.4 (Products of integrable functions, Rudin 6.13(a)). If $f, g \in \mathcal{R}[a, b]$ then $fg \in \mathcal{R}[a, b]$.

Proof. By Theorem 7.2.5 with $\varphi(t) = t^2$, f^2 is integrable. The polarization identity

$$fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$$

writes fg as a linear combination of squares of integrable functions, hence $fg \in \mathcal{R}$ by linearity. $\square$

Theorem 7.3.5 (Cauchy-Schwarz for integrals, B&S 7.3.21). For $f, g \in \mathcal{R}[a, b]$,

$$\left| \int_a^b fg \right| \leq \left(\int_a^b f^2 \right)^{1/2} \left(\int_a^b g^2 \right)^{1/2}.$$

Proof. For every $t \in \mathbb{R}$, $0 \leq \int (f - tg)^2 = \int f^2 - 2t \int fg + t^2 \int g^2$. If $\int g^2 = 0$ the linear-in- t inequality forces $\int fg = 0$ and the claim is trivial; otherwise this quadratic in t is ≥ 0 for all t , so its discriminant is ≤ 0 : $4(\int fg)^2 - 4 \int f^2 \int g^2 \leq 0$. Rearranging gives the inequality. $\square$

The integral as a limit of sums

Proposition 7.3.6 (Sequential characterization, B&S 7.1.9 / 7.2.1). $f \in \mathcal{R}[a, b]$ with integral L iff for every sequence $(\dot{P}_n)$ of tagged partitions with $\|\dot{P}_n\| \rightarrow 0$ we have $S(f, \dot{P}_n) \rightarrow L$.

Proof. $(\Rightarrow)$ is immediate from Definition 7.1.3: eventually $\|\dot{P}_n\| < \delta_\varepsilon$. $(\Leftarrow)$ Contrapositive: if $f \notin \mathcal{R}$, then for some ε_0 and every n there are tagged partitions $\dot{P}_n, \dot{Q}_n$ of mesh $< 1/n$ with $|S(f, \dot{P}_n) - S(f, \dot{Q}_n)| \geq \varepsilon_0$; these cannot both converge to a common limit. (This is B&S Exercise 7.2.1.) $\square$

Worked Example 7.3.7 (A limit of sums via the integral). Evaluate $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \cdot \frac{1}{1 + (i/n)^2}$.

Recognition. Trigger: a sum that “looks like” a Riemann sum $\sum f(t_i) \Delta x_i$ with $\Delta x_i = 1/n$, $t_i = i/n$.

Solution. This is the right-endpoint Riemann sum of $f(x) = 1/(1+x^2)$ on $[0, 1]$ with the uniform partition. Since f is continuous it is integrable, so by Proposition 7.3.6 the limit is $\int_0^1 \frac{dx}{1+x^2} = \arctan 1 - \arctan 0 = \pi/4$.

Check. The terms are positive and $< 1/n$, total < 1 , consistent with $\pi/4 \approx 0.785$. ■

Worked Example 7.3.8 (Sign of an integral (B&S 7.2.8)). If f is continuous, $f \geq 0$ on $[a, b]$, and $\int_a^b f = 0$, show $f \equiv 0$.

Recognition. Trigger: “zero integral of a nonnegative continuous function.” Use a strict positivity bump.

Solution. Suppose $f(c) > 0$ for some c . By continuity there is a subinterval $[c - \eta, c + \eta] \subseteq [a, b]$ on which $f \geq f(c)/2 > 0$. Splitting the integral and using nonnegativity elsewhere, $\int_a^b f \geq \int_{c-\eta}^{c+\eta} f \geq \frac{f(c)}{2} \cdot 2\eta > 0$, contradicting $\int_a^b f = 0$. Hence $f \equiv 0$. *Caution:* continuity is essential — a function that is 0 except at one point also has integral 0 but is not $\equiv 0$ (B&S 7.2.9). ■

Worked Example 7.3.9 (NON-example: $|f|$ integrable but f not). Find $f \notin \mathcal{R}[0, 1]$ with $|f| \in \mathcal{R}[0, 1]$.

Recognition. Trigger: a converse-fails example for Theorem 7.3.3.

Solution. Let $f(x) = +1$ on rationals and $f(x) = -1$ on irrationals. Then $|f| \equiv 1$ is integrable with integral 1. But every subinterval has $M_i = 1$, $m_i = -1$, so $U(f, P) - L(f, P) = 2(b-a) = 2$ for all P : $f \notin \mathcal{R}[0, 1]$.

Lesson. $|f| \in \mathcal{R}$ does not imply $f \in \mathcal{R}$; the implication runs only one way. ■

Caution. $\int fg \neq (\int f)(\int g)$ in general — the integral is linear, not multiplicative. And $\int |f| = 0$ forces $f = 0$ almost everywhere, not everywhere; for continuous f it does force $f \equiv 0$, but not for general integrable f (Thomae’s function has $\int |f| = 0$ yet $f \not\equiv 0$).

Further worked examples

Worked Example 7.3.10 (Linearity in practice). Given $\int_0^1 f = 3$ and $\int_0^1 g = -1$, compute $\int_0^1 (2f - 5g + 4)$.

Recognition. Trigger: a direct application of linearity (Theorem 7.3.1); the constant 4 integrates to $4(b-a)$.

Solution. By linearity, $\int_0^1 (2f - 5g + 4) = 2 \int_0^1 f - 5 \int_0^1 g + 4 \int_0^1 1 = 2(3) - 5(-1) + 4(1) = 6 + 5 + 4 = 15$.

Check. The constant term contributes $4 \cdot (1-0) = 4$, the length of the interval times the constant; replacing $[0, 1]$ by $[0, 3]$ would make it $4 \cdot 3 = 12$ instead. ■

Worked Example 7.3.11 (A two-sided estimate from monotonicity). Without computing it, bound

$$\int_0^1 \frac{dx}{1+x^4}.$$

Recognition. Trigger: “estimate, do not evaluate.” Bound the integrand pointwise and integrate the bounds (Theorem 7.3.1, monotonicity).

Solution. On $[0, 1]$, $0 \leq x^4 \leq 1$, so $1 \leq 1+x^4 \leq 2$ and therefore $\frac{1}{2} \leq \frac{1}{1+x^4} \leq 1$. Integrating these constants over $[0, 1]$,

$$\frac{1}{2} = \int_0^1 \frac{1}{2} dx \leq \int_0^1 \frac{dx}{1+x^4} \leq \int_0^1 1 dx = 1.$$

Check. The true value is about 0.866, comfortably inside $[\frac{1}{2}, 1]$. The lower bound is loose because $1+x^4$ is near 1 over most of $[0, 1]$, so the integrand stays close to its upper bound. ■

Worked Example 7.3.12 (Cauchy–Schwarz with explicit functions). For $f(x) = x$ and $g(x) = 1$ on $[0, 1]$, verify the integral Cauchy–Schwarz inequality (Theorem 7.3.5) and identify the slack.

Recognition. Trigger: a concrete check of an inequality; compute all three integrals and compare.

Solution. Here $\int_0^1 fg = \int_0^1 x dx = \frac{1}{2}$, $\int_0^1 f^2 = \int_0^1 x^2 dx = \frac{1}{3}$, and $\int_0^1 g^2 = \int_0^1 1 dx = 1$. Cauchy–Schwarz asserts $(\frac{1}{2})^2 \leq \frac{1}{3} \cdot 1$, i.e. $\frac{1}{4} \leq \frac{1}{3}$. True, with slack $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. *Check.* Equality would require $f = \lambda g$, i.e. $x \equiv \lambda$ constant, which fails; the strict inequality $\frac{1}{4} < \frac{1}{3}$ records exactly that x is not constant. ■

Worked Example 7.3.13 (NON-example: monotonicity needs the inequality everywhere). Suppose $f(x) \leq g(x)$ holds for all $x \in [0, 1]$ except at $x = \frac{1}{2}$, where $f(\frac{1}{2}) = 10 > g(\frac{1}{2}) = 0$. Does $\int_0^1 f \leq \int_0^1 g$ still follow, and is the hypothesis of Theorem 7.3.1 really violated?

Recognition. Trigger: probing whether a single-point exception breaks the order property; recall that single points are invisible to the integral.

Solution. The conclusion $\int_0^1 f \leq \int_0^1 g$ still holds: changing f at the one point $x = \frac{1}{2}$ does not change $\int_0^1 f$ at all (a single point contributes width 0), so the integrals are the same as if $f \leq g$ held everywhere. The literal hypothesis of the theorem (“ $f \leq g$ for all x ”) is violated at one point, but the conclusion survives because the integral only sees f up to measure-zero changes. *Lesson.* Pointwise hypotheses can be weakened to “almost everywhere” for the integral’s order property. *Check.* E.g. $f = g = 0$ except $f(\frac{1}{2}) = 10$ gives $\int f = 0 = \int g$. ■

Additional exercises.

1. Prove $\left| \int_a^b f \right| \leq \int_a^b |f|$ and characterize equality (for continuous f): equality holds iff f does not change sign on $[a, b]$.
2. Show that if $f \in \mathcal{R}[a, b]$, $m \leq f \leq M$, and φ is convex on $[m, M]$, then $\varphi(\frac{1}{b-a} \int_a^b f) \leq \frac{1}{b-a} \int_a^b \varphi \circ f$ (Jensen’s inequality for integrals).
3. Prove that for $f, g \in \mathcal{R}[a, b]$, $\int_a^b |fg| \leq (\int_a^b f^2)^{1/2} (\int_a^b g^2)^{1/2}$, and state when equality holds.
4. (Non-example.) Show $\int fg = (\int f)(\int g)$ can fail by computing both sides for $f = g = x$ on $[0, 1]$.
5. Show additivity $\int_a^b = \int_a^c + \int_c^b$ persists for a, b, c in any order, using the conventions $\int_a^a = 0$, $\int_b^a = -\int_a^b$.
6. If $f \in \mathcal{R}[a, b]$ and $\int_a^x f = 0$ for all $x \in [a, b]$, and f is continuous, prove $f \equiv 0$. Give a non-continuous counterexample.

7. (Diagram.) For f a continuous function changing sign once on $[a, b]$, shade the positive and negative area and illustrate $\left| \int_a^b f \right| < \int_a^b |f|$ geometrically.
8. Prove the first mean value theorem for integrals as a corollary of monotonicity and the intermediate value theorem, and show by example it can fail if f is merely integrable (not continuous).
9. Let $f \in \mathcal{R}[a, b]$. Prove $x \mapsto \int_a^x f$ is Lipschitz with constant $\sup |f|$, and deduce it is uniformly continuous.

7.4 The Fundamental Theorem of Calculus

The two Fundamental Theorems tie differentiation and integration together: one *integrates a derivative*, the other *differentiates an integral*. They are inverse operations, up to the usual caveats about which points behave.

First form: integrating a derivative

Theorem 7.4.1 (FTC, first form, Rudin 6.21 / B&S 7.3.1). *Suppose $f \in \mathcal{R}[a, b]$ and F is continuous on $[a, b]$ with $F'(x) = f(x)$ for all $x \in [a, b]$ (except possibly a finite set, where F is still continuous). Then*

$$\int_a^b f = F(b) - F(a).$$

Proof. Let $\varepsilon > 0$ and (by the gap criterion) pick a partition $P = \{x_0, \dots, x_n\}$ with $U(f, P) - L(f, P) < \varepsilon$. On each $[x_{i-1}, x_i]$ the Mean Value Theorem for F gives a point $t_i \in (x_{i-1}, x_i)$ with

$$F(x_i) - F(x_{i-1}) = F'(t_i) \Delta x_i = f(t_i) \Delta x_i.$$

Summing telescopes the left side: $F(b) - F(a) = \sum_i f(t_i) \Delta x_i = S(f, \dot{P})$, a genuine Riemann sum. Since $L(f, P) \leq S(f, \dot{P}) \leq U(f, P)$ and likewise $L(f, P) \leq \int_a^b f \leq U(f, P)$, both $F(b) - F(a)$ and $\int_a^b f$ lie in an interval of length $< \varepsilon$. Hence $\left| F(b) - F(a) - \int_a^b f \right| < \varepsilon$ for all ε , so they are equal. $\square$

Remark. F is called an *antiderivative* (primitive) of f . Two antiderivatives of the same f differ by a constant (B&S 7.3.5), since their difference has derivative 0. The finitely-many-exceptions clause lets us integrate, e.g., $\int_a^b [x] dx$ where the antiderivative has corners (B&S 7.3.8).

Second form: differentiating an integral

Definition 7.4.2. For $f \in \mathcal{R}[a, b]$, the *indefinite integral* with basepoint a is

$$F(x) = \int_a^x f(t) dt, \quad x \in [a, b].$$

Theorem 7.4.3 (FTC, second form, Rudin 6.20 / B&S 7.3.5). *Let $f \in \mathcal{R}[a, b]$ and $F(x) = \int_a^x f$. Then:*

- (i) F is continuous on $[a, b]$ — in fact Lipschitz: $|F(x) - F(y)| \leq M|x - y|$ where $M = \sup |f|$.
- (ii) If f is continuous at $c \in [a, b]$, then F is differentiable at c and $F'(c) = f(c)$.

Proof. (i) For $y < x$, $F(x) - F(y) = \int_y^x f$ by additivity, so $|F(x) - F(y)| \leq \int_y^x |f| \leq M(x - y)$ by Theorem 7.3.3. This is Lipschitz, hence (uniformly) continuous.

- (ii) Fix $\varepsilon > 0$. Continuity of f at c gives $\delta > 0$ with $|f(t) - f(c)| < \varepsilon$ when $|t - c| < \delta$. For $0 < |h| < \delta$,

$$\frac{F(c+h) - F(c)}{h} - f(c) = \frac{1}{h} \int_c^{c+h} (f(t) - f(c)) dt,$$

using $\int_c^{c+h} f(c) dt = f(c)h$. The right side has absolute value $\leq \frac{1}{|h|} \cdot |h| \cdot \varepsilon = \varepsilon$. Hence the difference quotient $\rightarrow f(c)$, so $F'(c) = f(c)$. $\square$

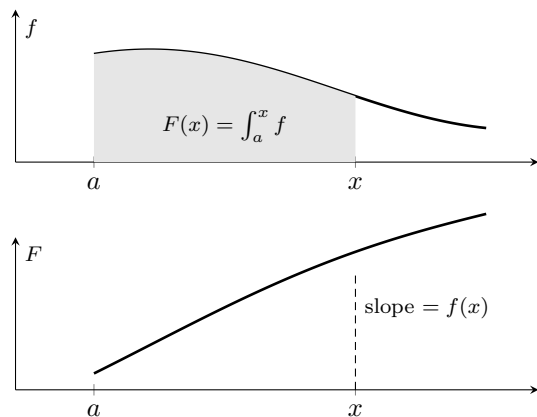


Figure 7.6. Top: $F(x)$ is the shaded area up to x . Bottom: the area function F ; its slope at x is the height $f(x)$ above — that is FTC II.

Integration by parts and change of variable

Theorem 7.4.4 (Integration by parts, Rudin 6.22 / B&S 7.3.6). *If F, G are differentiable on $[a, b]$ with $F' = f \in \mathcal{R}$ and $G' = g \in \mathcal{R}$, then*

$$\int_a^b Fg = F(b)G(b) - F(a)G(a) - \int_a^b fG.$$

Proof. $(FG)' = F'G + FG' = fG + Fg$ wherever the derivatives exist, and FG is an antiderivative of $fG + Fg \in \mathcal{R}$. By FTC I, $\int_a^b (fG + Fg) = F(b)G(b) - F(a)G(a)$. Rearranging isolates $\int Fg$. $\square$

Theorem 7.4.5 (Change of variable, Rudin 6.19 / B&S 7.3.8). *Let φ be continuously differentiable and monotone on $[a, b]$ with $\varphi' \in \mathcal{R}$, and let f be continuous on $\varphi([a, b])$. Then*

$$\int_{\varphi(a)}^{\varphi(b)} f(y) dy = \int_a^b f(\varphi(x)) \varphi'(x) dx.$$

Proof. Let $F(y) = \int_{\varphi(a)}^y f$. By FTC II $F' = f$ (continuous f). Then $(F \circ \varphi)'(x) = f(\varphi(x))\varphi'(x)$ by the chain rule, and this is integrable. By FTC I,

$$\int_a^b f(\varphi(x))\varphi'(x) dx = F(\varphi(b)) - F(\varphi(a)) = F(\varphi(b)) - 0 = \int_{\varphi(a)}^{\varphi(b)} f. \quad \square$$

Mean value theorems for integrals

Theorem 7.4.6 (Mean Value Theorem for integrals, B&S 7.2.16; weighted 7.2.17). *If f is continuous on $[a, b]$ there is $c \in [a, b]$ with $\int_a^b f = f(c)(b-a)$. More generally, if also $g \in \mathcal{R}$ with $g \geq 0$, there is $c \in [a, b]$ with $\int_a^b fg = f(c) \int_a^b g$.*

Proof. Let $m = \min f$, $M = \max f$ (attained, by continuity on a compact interval). From $mg \leq fg \leq Mg$ and $g \geq 0$, integrate: $m \int g \leq \int fg \leq M \int g$. If $\int g = 0$ the result is trivial (both sides 0). Otherwise $\frac{\int fg}{\int g} \in [m, M]$, and by the intermediate value theorem $f(c)$ equals this ratio for some c . Taking $g \equiv 1$ gives the basic form. $\square$

Method (Differentiating an integral with variable limits). *Recognition trigger.* You must compute $\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt$ with f continuous (a Leibniz-rule problem, e.g. B&S 7.3.11, 7.3.15).

1. Write $G(x) = \int_{u(x)}^{v(x)} f = H(v(x)) - H(u(x))$ where $H(s) = \int_a^s f$, $H' = f$.

2. Chain rule: $G'(x) = f(v(x))v'(x) - f(u(x))u'(x)$.

3. Constant limits drop out (their derivative is 0); a fixed lower limit just gives $f(v(x))v'(x)$.

Worked Example 7.4.7 (Leibniz rule (B&S 7.3.11)). Compute $F'(x)$ for $F(x) = \int_0^{x^2} (1+t^3)^{-1} dt$.

Recognition. Trigger: variable upper limit $v(x) = x^2$, fixed lower limit.

Solution. By the method, $F'(x) = \frac{1}{1+(x^2)^3} \cdot \frac{d}{dx}(x^2) = \frac{2x}{1+x^6}$. *Check.* At $x = 0$, $F'(0) = 0$, consistent with F having a minimum there (the integrand is positive, area grows symmetrically). ■

Worked Example 7.4.8 (A definite integral by substitution (B&S 7.3.18a)). Evaluate $\int_0^1 t^2(1+t^3)^{1/2} dt$.

Recognition. Trigger: the factor t^2 is (up to a constant) the derivative of the inside $1+t^3$. Set $\varphi(t) = 1+t^3$.

Solution. With $u = 1+t^3$, $du = 3t^2 dt$, limits $u : 1 \rightarrow 2$:

$$\int_0^1 t^2(1+t^3)^{1/2} dt = \frac{1}{3} \int_1^2 u^{1/2} du = \frac{1}{3} \cdot \frac{2}{3} [u^{3/2}]_1^2 = \frac{2}{9} (2^{3/2} - 1).$$

Check. Numerically $\frac{2}{9}(2.828 - 1) \approx 0.406$; the integrand on $[0, 1]$ peaks near $t = 1$ at $\sqrt{2} \approx 1.41$, so an area ≈ 0.4 is plausible. ■

Worked Example 7.4.9 (No such function (B&S 7.3.14)). Show there is no continuously differentiable f on $[0, 2]$ with $f(0) = 1$, $f(2) = 4$, and $f'(x) \leq 2$ everywhere.

Recognition. Trigger: a constraint on f' that contradicts the net change via FTC.

Solution. By FTC I, $f(2) - f(0) = \int_0^2 f' dx$. If $f' \leq 2$, monotonicity gives $\int_0^2 f' \leq 2 \cdot 2 = 4$, so $f(2) - f(0) \leq 4$. But $f(2) - f(0) = 4 - 1 = 3 \leq 4$ — that is *not* a contradiction. Re-read: the requirement is $f'(x) \leq 2$, average slope is $3/2 \leq 2$, so such f *does* exist (e.g. $f(x) = 1 + \frac{3}{2}x$). The genuine impossibility (the intended exercise) is $f(0) = 1$, $f(2) = 4$, $f' \leq 1$: then $\int_0^2 f' \leq 2 < 3 = f(2) - f(0)$, impossible. *Lesson.* FTC turns a pointwise derivative bound into a bound on net change; check the arithmetic of the average slope. ■

Caution. FTC II gives $F'(c) = f(c)$ *only at points of continuity* of f . If f has a jump at c , then F has a corner there and $F'(c)$ may not exist (one-sided derivatives equal the one-sided values of f). And FTC I needs f *integrable*: a derivative F' need not be Riemann integrable (there exist differentiable F with bounded, non-integrable F'), in which case $\int_a^b F'$ is not even defined though $F(b) - F(a)$ is.

Further worked examples

Worked Example 7.4.10 (Integration by parts: $\int_0^1 xe^x dx$). Evaluate $\int_0^1 xe^x dx$ using Theorem 7.4.4.

Recognition. Trigger: a product of a polynomial and an exponential; differentiate the polynomial factor away. Take $F(x) = x$ (so $f = F' = 1$) and $g(x) = e^x$ with $G(x) = e^x$.

Solution. With $F = x$, $g = e^x$, $G = e^x$, $f = 1$,

$$\int_0^1 xe^x dx = [xe^x]_0^1 - \int_0^1 1 \cdot e^x dx = 1 \cdot e - 0 - [e^x]_0^1 = e - (e - 1) = 1.$$

Check. $\int_0^1 xe^x dx = 1$ exactly. Numerically the integrand rises from 0 to $e \approx 2.718$; an area of 1 is reasonable (mean height ≈ 1). ■

Worked Example 7.4.11 (Leibniz rule with both limits variable). Compute $G'(x)$ for $G(x) = \int_x^{x^2} e^{-t^2} dt$.

Recognition. Trigger: *both* limits depend on x ; use the full Leibniz rule from the Method ($u(x) = x$, $v(x) = x^2$).

Solution. By the Method, $G'(x) = f(v(x))v'(x) - f(u(x))u'(x)$ with $f(t) = e^{-t^2}$:

$$G'(x) = e^{-(x^2)^2} \cdot (2x) - e^{-x^2} \cdot 1 = 2xe^{-x^4} - e^{-x^2}.$$

Check. At $x = 1$ the limits collapse to $\int_1^1 = 0$, so $G(1) = 0$; the derivative there is $2e^{-1} - e^{-1} = e^{-1} > 0$, consistent with G increasing as the upper limit x^2 pulls ahead of the lower limit x for $x > 1$. ■

Worked Example 7.4.12 (Mean Value Theorem for integrals: locating c). For $f(x) = x^2$ on $[0, 3]$, find the point c guaranteed by Theorem 7.4.6 with $\int_0^3 f = f(c) \cdot 3$.

Recognition. Trigger: the basic ($g \equiv 1$) integral MVT; solve $f(c) = \frac{1}{b-a} \int_a^b f$ for c .

Solution. The average value is $\frac{1}{3-0} \int_0^3 x^2 dx = \frac{1}{3} \cdot \frac{27}{3} = 3$. Setting $f(c) = c^2 = 3$ gives $c = \sqrt{3} \approx 1.732 \in [0, 3]$.

Check. $\sqrt{3}$ lies in $[0, 3]$ as the theorem promises, and $f(\sqrt{3}) \cdot 3 = 3 \cdot 3 = 9 = \int_0^3 x^2 dx$. The point sits right of the midpoint 1.5 because x^2 is convex (more area piles up near the right end). ■

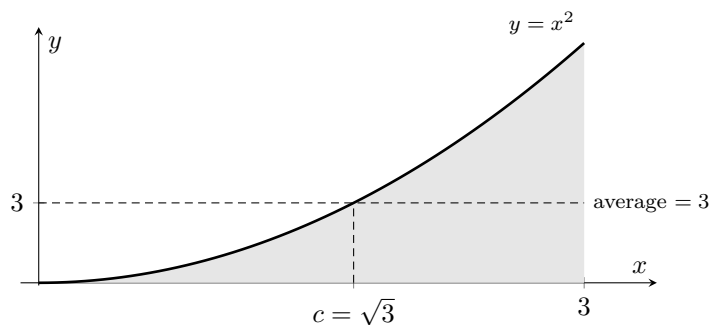


Figure 7.7. Integral MVT: the rectangle of height $f(c) = 3$ over $[0, 3]$ has the same area (9) as the region under x^2 . The height is attained at $c = \sqrt{3}$.

Exercises (Bartle & Sherbert, §7.3).

1. Extend the proof of the Fundamental Theorem (First Form) to the case of an arbitrary finite exceptional set E .
2. If $n \in \mathbb{N}$ and $H_n(x) = x^{n+1}/(n+1)$, show the FTC implies $\int_a^b x^n dx = (b^{n+1} - a^{n+1})/(n+1)$. What is the exceptional set E here?
3. If $g(x) = x$ for $|x| \geq 1$ and $g(x) = -x$ for $|x| < 1$, and $G(x) = \frac{1}{2}|x^2 - 1|$, show $\int_{-2}^3 g(x) dx = G(3) - G(-2) = 5/2$.
4. Let $B(x) = -\frac{1}{2}x^2$ for $x < 0$ and $B(x) = \frac{1}{2}x^2$ for $x \geq 0$. Show $\int_a^b |x| dx = B(b) - B(a)$.
5. Let $f : [a, b] \rightarrow \mathbb{R}$ and $C \in \mathbb{R}$. (a) If F is an antiderivative of f on $[a, b]$, show $F_C(x) = F(x) + C$ is also an antiderivative. (b) If F_1, F_2 are antiderivatives of f , show $F_1 - F_2$ is constant.
6. If $f \in \mathcal{R}[a, b]$ and $c \in [a, b]$, the indefinite integral with basepoint c is $F_c(z) = \int_c^z f$. Find a relation between F_a and F_c .
7. Thomae's function is in $\mathcal{R}[0, 1]$ with integral 0. Can the FTC (First Form) be used to obtain this? Explain.
8. Let $F(x) = (n-1)x - (n-1)n/2$ for $x \in [n-1, n]$, $n \in \mathbb{N}$, $x \geq 0$. Show F is continuous, evaluate F' where it exists, and use this to evaluate $\int_a^b [x] dx$ for $0 \leq a < b$.
9. Let $f \in \mathcal{R}[a, b]$ and $F(x) = \int_a^x f$. (a) Evaluate $G(x) = \int_c^x f$ in terms of F . (b) Evaluate $H(x) = \int_x^b f$ in terms of F . (c) Evaluate $S(x) = \int_x^{\sin x} f$ in terms of F .
10. Let f be continuous on $[a, b]$ and $v : [c, d] \rightarrow \mathbb{R}$ differentiable with $v([c, d]) \subseteq [a, b]$. If $G(x) = \int_a^{v(x)} f$, show $G'(x) = f(v(x))v'(x)$ for all $x \in [c, d]$.
11. Find $F'(x)$ where F is defined on $[0, 1]$ by (a) $F(x) = \int_0^{x^2} (1+t^3)^{-1} dt$; (b) $F(x) = \int_{x^2}^x \sqrt{1+t^2} dt$.
12. Let $f : [0, 3] \rightarrow \mathbb{R}$ be $f(x) = x$ for $0 \leq x < 1$, $f(x) = 1$ for $1 \leq x < 2$, $f(x) = x$ for $2 \leq x \leq 3$. Find $F(x) = \int_0^x f$, sketch f and F , decide where F is differentiable, and evaluate F' there.
13. Let g on $[0, 3]$ be $g(x) = 1$ if $0 \leq x < 2$ and $g(x) = -1$ if $2 \leq x \leq 3$. Find $G(x) = \int_0^x g$, sketch g and G , and decide whether $G'(x) = g(x)$ for all x .

14. Show there is no continuously differentiable f on $[0, 2]$ with $f(0) = -1$, $f(2) = 4$, and $f'(x) \leq 2$ for all x . (Apply the FTC.)
15. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $c > 0$, define $g(x) = \int_{x-c}^{x+c} f(t) dt$. Show g is differentiable and find $g'(x)$.
16. If $f : [0, 1] \rightarrow \mathbb{R}$ is continuous and $\int_0^x f = \int_x^1 f$ for all $x \in [0, 1]$, show $f(x) = 0$ for all x .
17. Prove the Substitution Theorem: with $F(u) = \int_{w(a)}^u f(x) dx$ and $H(t) = F(w(t))$, show $H'(t) = f(w(t))w'(t)$ and hence $\int_{w(a)}^{w(b)} f = \int_a^b f(w(t))w'(t) dt$.
18. Use the Substitution Theorem to evaluate (a) $\int_0^1 t^2(1+t^3)^{1/2} dt$; (b) $\int_0^2 t(1+t^2)^{1/2} dt$ (and verify $= 4/3$ on a suitable interval); (c) $\int_1^4 \frac{\sqrt{1+\sqrt{t}}}{\sqrt{t}} dt$; (d) $\int_1^4 \frac{\cos \sqrt{t}}{\sqrt{t}} dt = 2(\sin 2 - \sin 1)$.
19. Explain why the Substitution Theorem cannot be applied to evaluate, with the indicated substitution: (a) $\int_0^4 \frac{\sqrt{t}}{1+\sqrt{t}} dt$, $w(t) = \sqrt{t}$; (b) $\int_0^4 \frac{\cos \sqrt{t}}{\sqrt{t}} dt$, $w(t) = \sqrt{t}$; (c) $\int_{-1}^1 \sqrt{1+2|t|} dt$, $w(t) = |t|$; (d) $\int_0^1 \frac{dt}{\sqrt{1-t^2}}$, $w(t) = \arcsin t$.
20. (a) If Z_1, Z_2 are null sets, show $Z_1 \cup Z_2$ is a null set. (b) More generally, if Z_n is null for each $n \in \mathbb{N}$, show $\bigcup_n Z_n$ is null. (Hint: cover Z_n by intervals of total length $\leq \varepsilon/2^n$.)
21. Let $f, g \in \mathcal{R}[a, b]$. (a) For $t \in \mathbb{R}$, show $\int_a^b (tf - g)^2 \geq 0$. (b) Use (a) to show $2 \left| \int_a^b fg \right| \leq t \int_a^b f^2 + (1/t) \int_a^b g^2$ for $t > 0$. (c) If $\int_a^b f^2 = 0$, show $\int_a^b fg = 0$. (d) Prove the Cauchy–Schwarz inequality $(\int_a^b fg)^2 \leq (\int_a^b |fg|)^2 \leq \int_a^b f^2 \int_a^b g^2$.
22. Let $h : [0, 1] \rightarrow \mathbb{R}$ be Thomae’s function and sgn the signum function. Show $\text{sgn} \circ h$ is not Riemann integrable on $[0, 1]$.

Additional exercises.

1. Compute $\frac{d}{dx} \int_{\sin x}^{x^2} e^{-t^2} dt$ using the Leibniz rule.
2. Evaluate $\int_0^{\pi/2} \sin^2 x \cos x dx$ by the substitution $u = \sin x$.
3. Prove integration by parts $\int_a^b uv' = u(b)v(b) - u(a)v(a) - \int_a^b u'v$ and use it to compute $\int_0^1 xe^x dx$.
4. (Non-example.) Construct a differentiable F on $[0, 1]$ whose derivative F' is bounded but not Riemann integrable, so that FTC I cannot be applied to F' . (You may cite Volterra’s function; describe its key property.)
5. Show FTC II’s conclusion $F'(c) = f(c)$ can fail at a jump of f : take f the step function 0 on $[0, 1)$, 1 on $[1, 2]$, compute $F(x) = \int_0^x f$, and examine $F'(1)$.
6. Let f be continuous on $[0, \infty)$ with $\lim_{x \rightarrow \infty} f(x) = L$. Show $\frac{1}{x} \int_0^x f \rightarrow L$ as $x \rightarrow \infty$ (Cesaro mean of an integral).
7. (Diagram.) Plot $f(x) = \cos x$ and its area function $F(x) = \int_0^x \cos = \sin x$ on $[0, 2\pi]$ in stacked panels; mark where F has horizontal tangents and verify these are the zeros of f .
8. Use the weighted mean value theorem to prove the second mean value theorem: if f is monotone and g continuous on $[a, b]$, there is c with $\int_a^b fg = f(a) \int_a^c g + f(b) \int_c^b g$.
9. Prove that if f is continuous on $[a, b]$ and $\int_a^b f p = 0$ for every polynomial p , then $f \equiv 0$. (Hint: Weierstrass approximation, or use p approximating f .)

7.5 The Riemann–Stieltjes Integral

Rudin develops the whole theory at once for a more general integral: instead of weighting each subinterval by its length Δx_i , weight it by the increment $\Delta \alpha_i$ of a fixed *integrator* α . Taking $\alpha(x) = x$ recovers everything above; choosing α to jump turns the integral into a finite sum. This single notion unifies sums and integrals.

Definition relative to an integrator

Definition 7.5.1 (Stieltjes sums, Rudin 6.2). Let α be *monotonically increasing* on $[a, b]$ (so $\alpha(a), \alpha(b)$ finite). For a partition P put $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}) \geq 0$. For bounded f define

$$L(f, P, \alpha) = \sum_{i=1}^n m_i \Delta\alpha_i, \quad U(f, P, \alpha) = \sum_{i=1}^n M_i \Delta\alpha_i,$$

with m_i, M_i the inf/sup of f on I_i as before. The *lower* and *upper Stieltjes integrals* are $\int f d\alpha = \sup_P L(f, P, \alpha)$ and $\bar{\int} f d\alpha = \inf_P U(f, P, \alpha)$.

Definition 7.5.2 (Riemann–Stieltjes integral, Rudin 6.2). f is *Riemann–Stieltjes integrable* with respect to α , written $f \in \mathcal{R}(\alpha)$ on $[a, b]$, if $\int f d\alpha = \bar{\int} f d\alpha$. The common value is

$$\int_a^b f d\alpha = \int_a^b f(x) d\alpha(x).$$

When $\alpha(x) = x$ this is exactly the Riemann integral, $\Delta\alpha_i = \Delta x_i$.

Intuitively, α is a “ruler” that measures how much each subinterval counts: where α rises steeply, that stretch of the x -axis is given extra weight; where α is flat, the integrand is ignored. The ordinary integral uses the trivial ruler $\alpha(x) = x$, which weights every unit of length equally. Replacing length by $\Delta\alpha_i$ lets one integrator describe mass distributions, probability, and discrete sums all at once.

Every theorem of §1–§3 generalizes verbatim, because $\Delta\alpha_i \geq 0$ plays the role of Δx_i : refinement still shrinks the gap, lower $\leq$ upper, and the gap criterion holds.

Theorem 7.5.3 (Criterion and basic properties, Rudin 6.6, 6.12, 6.13). $f \in \mathcal{R}(\alpha)$ iff for each $\varepsilon > 0$ there is P with $U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon$. The map $f \mapsto \int f d\alpha$ is linear and monotone; it is additive over subintervals; $|\int f d\alpha| \leq \int |f| d\alpha$; and it is also linear in α : for increasing α_1, α_2 and $c > 0$,

$$\int f d(\alpha_1 + \alpha_2) = \int f d\alpha_1 + \int f d\alpha_2, \quad \int f d(c\alpha) = c \int f d\alpha.$$

Proof. The criterion and the f -properties are the proofs of §1–§3 with Δx_i replaced by $\Delta\alpha_i \geq 0$ throughout (nonnegativity is all that was used). Linearity in α is immediate from $\Delta(\alpha_1 + \alpha_2)_i = \Delta\alpha_{1,i} + \Delta\alpha_{2,i}$ at the level of sums, passed to the sup/inf. $\square$

Theorem 7.5.4 (Continuity vs. the integrator, Rudin 6.8–6.9). If f is continuous on $[a, b]$ then $f \in \mathcal{R}(\alpha)$ for every increasing α . If f is monotone and α is continuous, then $f \in \mathcal{R}(\alpha)$.

Proof. (Continuous f .) Given $\varepsilon > 0$, set $\eta = \varepsilon/(\alpha(b) - \alpha(a) + 1)$ and use uniform continuity to get δ with $|x - y| < \delta \implies |f(x) - f(y)| < \eta$. For P with $\|P\| < \delta$, $M_i - m_i \leq \eta$, so $U - L \leq \eta \sum \Delta\alpha_i = \eta(\alpha(b) - \alpha(a)) < \varepsilon$. (Monotone f , continuous α .) Choose P so that each $\Delta\alpha_i = (\alpha(b) - \alpha(a))/n$ (possible since α is continuous, by IVT); telescoping the f -heights gives $U - L = \frac{\alpha(b) - \alpha(a)}{n} (f(b) - f(a)) \rightarrow 0$. $\square$

Reduction to a Riemann integral

Theorem 7.5.5 (Differentiable integrator, Rudin 6.17). Suppose α is increasing with $\alpha' \in \mathcal{R}[a, b]$. Then for bounded f , $f \in \mathcal{R}(\alpha)$ iff $f\alpha' \in \mathcal{R}$, and in that case

$$\int_a^b f d\alpha = \int_a^b f(x) \alpha'(x) dx.$$

Idea. By the Mean Value Theorem, $\Delta\alpha_i = \alpha'(s_i)\Delta x_i$ for some $s_i \in I_i$; since $\alpha' \in \mathcal{R}$ we can, on a fine partition, replace $\alpha'(s_i)$ by $\alpha'(t_i)$ at the tag with error controlled by $U(\alpha', P) - L(\alpha', P)$. Then $\sum f(t_i)\Delta\alpha_i$ and $\sum f(t_i)\alpha'(t_i)\Delta x_i$ differ by less than any ε , so the two integrals coincide. $\square$

This is the practical rule: a *smooth* integrator just inserts a weight α' .

Step-function integrators: integrals become sums

The opposite extreme makes the integral a finite sum. Let the *unit step* be

$$I(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

Theorem 7.5.6 (Single jump, Rudin 6.15). *Let $a < s < b$ and $\alpha(x) = I(x - s)$ (a jump of size 1 at s). If f is bounded on $[a, b]$ and continuous at s , then $f \in \mathcal{R}(\alpha)$ and*

$$\int_a^b f d\alpha = f(s).$$

Proof. $\Delta\alpha_i = 0$ on every subinterval except the one containing s as an interior or endpoint, where $\Delta\alpha = 1$. So $U(f, P, \alpha) = M_k$ and $L(f, P, \alpha) = m_k$ for the subinterval $I_k \ni s$. As $\|P\| \rightarrow 0$ and f is continuous at s , both $M_k, m_k \rightarrow f(s)$, squeezing $\int f d\alpha = f(s)$. $\square$

Theorem 7.5.7 (Step-function integrator, Rudin 6.16). *Let α be increasing and constant except for jumps of size $c_k > 0$ at points $s_k \in [a, b]$ ($k = 1, \dots, N$). If f is continuous at each s_k , then*

$$\int_a^b f d\alpha = \sum_{k=1}^N c_k f(s_k).$$

Proof. Write $\alpha = \sum_k c_k I(x - s_k)$ (plus a constant, which contributes nothing). By linearity in α (Theorem 7.5.3) and the single-jump formula (Theorem 7.5.6), $\int f d\alpha = \sum_k c_k \int f dI(\cdot - s_k) = \sum_k c_k f(s_k)$. $\square$

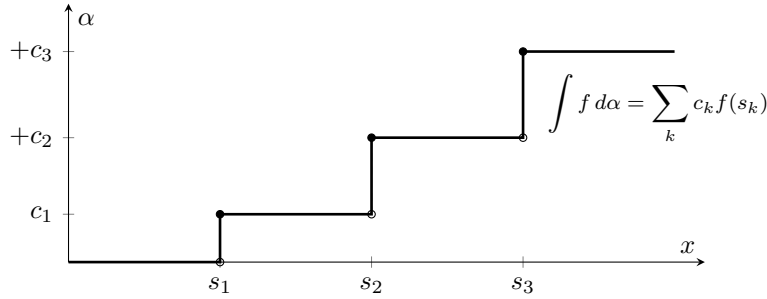


Figure 7.8. A step-function integrator α : the Stieltjes integral samples f only at the jumps s_k , weighting by the jump heights c_k . The integral collapses to a finite weighted sum.

Remark (Series as Stieltjes integrals). With $a = 0$, $\alpha(x) = \sum_{n \leq x} c_n$ a jump of $c_n = a_n \geq 0$ at each integer n , and f continuous, $\int_0^N f d\alpha = \sum_{n=1}^N a_n f(n)$. Series and integrals are one theory; the integrator records “how much weight” sits at each point.

Stieltjes integration by parts and change of variable

Theorem 7.5.8 (Stieltjes integration by parts, Rudin 6.22 form). *If $f \in \mathcal{R}(\alpha)$ on $[a, b]$ then $\alpha \in \mathcal{R}(f)$ and*

$$\int_a^b f d\alpha + \int_a^b \alpha df = f(b)\alpha(b) - f(a)\alpha(a).$$

Sketch. For a tagged partition the Abel summation identity $\sum f(t_i)\Delta\alpha_i + \sum \alpha(x_i^*)\Delta f_i = f(b)\alpha(b) - f(a)\alpha(a)$ holds with suitable intermediate tags; refining and passing to the limit (one side converges, so does the other) yields the symmetric formula. The roles of integrand and integrator are interchangeable. $\square$

Theorem 7.5.9 (Change of variable, Rudin 6.19). *Let $\varphi : [A, B] \rightarrow [a, b]$ be a continuous increasing bijection, $\beta = \alpha \circ \varphi$, $g = f \circ \varphi$. If $f \in \mathcal{R}(\alpha)$ then $g \in \mathcal{R}(\beta)$ and*

$$\int_A^B g d\beta = \int_a^b f d\alpha.$$

Proof. φ matches partitions of $[A, B]$ with partitions of $[a, b]$ one-to-one, preserving inf / sup of the integrand and the increments $\Delta\beta = \Delta\alpha$. Hence the upper/lower Stieltjes sums correspond exactly, and so do their sup/inf. $\square$

Worked Example 7.5.10 (A smooth integrator reduces to a weight (Rudin 6.17)). Compute $\int_0^1 x d(x^2)$.

Recognition. Trigger: $\alpha(x) = x^2$ is differentiable, $\alpha' = 2x \in \mathcal{R}$; use the reduction.

Solution. $\int_0^1 x d(x^2) = \int_0^1 x \cdot 2x dx = \int_0^1 2x^2 dx = \frac{2}{3}$. *Check.* α increases on $[0, 1]$ and the weight $2x$ emphasizes the right end, where $f = x$ is larger, so a value above $\int_0^1 x dx = \frac{1}{2}$ is sensible; $\frac{2}{3} > \frac{1}{2}$. $\blacksquare$

Worked Example 7.5.11 (A pure-jump integrator (Rudin 6.16)). Let α jump by 1 at $x = 1$ and by 2 at $x = 3$, constant elsewhere. Compute $\int_0^4 x^2 d\alpha$.

Recognition. Trigger: step integrator; the integral is a weighted sum of f at the jumps.

Solution. By Theorem 7.5.7, $\int_0^4 x^2 d\alpha = 1 \cdot f(1) + 2 \cdot f(3) = 1 \cdot 1 + 2 \cdot 9 = 19$. *Check.* This is exactly the expectation $\sum c_k f(s_k)$ — the integral never “sees” f between the jumps. $\blacksquare$

Worked Example 7.5.12 (Mixed integrator (smooth + jump)). Let $\alpha(x) = x + I(x - 1)$ on $[0, 2]$ (slope 1 everywhere plus a unit jump at 1). Compute $\int_0^2 f d\alpha$ for continuous f .

Recognition. Trigger: α splits as continuous part + step part; use linearity in α .

Solution. $\int_0^2 f d\alpha = \int_0^2 f dx + \int_0^2 f dI(x - 1) = \int_0^2 f(x) dx + f(1)$. *Check.* For $f \equiv 1$ this is $2 + 1 = 3 = \alpha(2^+) - \alpha(0) = (2 + 1) - 0$, matching the total increase of α . $\blacksquare$

Caution. The integrator must be increasing (or of bounded variation) for the theory to behave; a non-monotone α can make $\Delta\alpha_i < 0$ and break the sup/inf squeeze. Also, in Theorem 7.5.6 continuity of f at the jump is required: if f and α are both discontinuous at the same point, $\int f d\alpha$ may not exist (Rudin 6.14). Jumps of f and of α must not coincide.

Further worked examples

Worked Example 7.5.13 (Stieltjes integration by parts). With $f(x) = x$ and $\alpha(x) = x^2$ on $[0, 1]$, verify the integration-by-parts identity $\int_0^1 f d\alpha + \int_0^1 \alpha df = f(1)\alpha(1) - f(0)\alpha(0)$ (Theorem 7.5.8).

Recognition. Trigger: a concrete check of the symmetric parts formula; both integrals reduce to Riemann integrals since the integrators are smooth.

Solution. Using the reduction (Theorem 7.5.5), $d\alpha = 2x dx$ and $df = dx$:

$$\int_0^1 x d(x^2) = \int_0^1 x \cdot 2x dx = \frac{2}{3}, \quad \int_0^1 x^2 dx = \frac{1}{3}.$$

Their sum is $\frac{2}{3} + \frac{1}{3} = 1$. The right side is $f(1)\alpha(1) - f(0)\alpha(0) = 1 \cdot 1 - 0 \cdot 0 = 1$. They agree. *Check.* The two integrals split the total “area of the rectangle” $f\alpha|_0^1 = 1$ into the $\frac{2}{3}/\frac{1}{3}$ shares dictated by how steeply each factor grows. $\blacksquare$

Worked Example 7.5.14 (Distribution / expectation reading of a step integrator). Let α jump by $\frac{1}{2}$ at $x = 0$, by $\frac{1}{3}$ at $x = 1$, and by $\frac{1}{6}$ at $x = 2$, constant elsewhere (so the jumps sum to 1, a probability distribution). Compute $\int_{-1}^3 x d\alpha$ and interpret it.

Recognition. Trigger: a pure-jump integrator whose jumps are probabilities; the Stieltjes integral is an expectation $\sum c_k s_k$ (Theorem 7.5.7).

Solution. By the step-integrator formula with $f(x) = x$,

$$\int_{-1}^3 x d\alpha = \frac{1}{2} \cdot 0 + \frac{1}{3} \cdot 1 + \frac{1}{6} \cdot 2 = 0 + \frac{1}{3} + \frac{1}{3} = \frac{2}{3}.$$

This is the mean of a random variable taking values 0, 1, 2 with probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$. *Check.* The mean must lie between the smallest and largest values, $0 \leq \frac{2}{3} \leq 2$, and it leans toward 0 because the largest probability mass $\frac{1}{2}$ sits at 0. $\blacksquare$

Worked Example 7.5.15 (NON-example: coincident jumps of f and α). Let $\alpha(x) = I(x)$ (unit jump at 0) and $f(x) = I(x)$ as well, on $[-1, 1]$. Show $\int_{-1}^1 f d\alpha$ does not exist.

Recognition. Trigger: the pitfall that f and α may not jump at the same point (Theorem 7.5.6 needs f continuous at the jump of α).

Solution. For any partition, $\Delta\alpha_i = 1$ only on the subinterval I_k that contains 0, and 0 otherwise. On that $I_k = [x_{k-1}, x_k]$ we have $f = 0$ for $x \leq 0$ and $f = 1$ for $x > 0$, so $m_k = 0$ and $M_k = 1$ no matter how fine P is (the jump of f sits exactly at the jump of α). Hence $L(f, P, \alpha) = 0$ and $U(f, P, \alpha) = 1$ for every P , giving a permanent gap of 1. So $\int f d\alpha = 0 \neq 1 = \int f d\alpha$ and the integral does not exist. *Lesson.* The weight $\Delta\alpha = 1$ lands precisely where f refuses to settle to a single value; continuity of f at the jump of α is exactly what rules this out. ■

Exercises (Rudin, Ch.6).

- Suppose α increases on $[a, b]$, $a < x_0 < b$, α is continuous at x_0 , $f(x_0) = 1$, and $f(x) = 0$ for $x \neq x_0$. Prove that $f \in \mathcal{R}(\alpha)$ and $\int_a^b f d\alpha = 0$.
- Suppose $f \geq 0$, f continuous on $[a, b]$, and $\int_a^b f(x) dx = 0$. Prove $f(x) = 0$ for all $x \in [a, b]$. (Compare with Exercise 1.)
- Define $\beta_1, \beta_2, \beta_3$ by $\beta_j(x) = 0$ if $x < 0$ and $\beta_j(x) = 1$ if $x > 0$ ($j = 1, 2, 3$), with $\beta_1(0) = 0$, $\beta_2(0) = 1$, $\beta_3(0) = \frac{1}{2}$. Let f be bounded on $[-1, 1]$. (a) Prove $f \in \mathcal{R}(\beta_1)$ iff $f(0+) = f(0)$, and then $\int f d\beta_1 = f(0)$. (b) State and prove the analogous result for β_2 . (c) Prove $f \in \mathcal{R}(\beta_3)$ iff f is continuous at 0. (d) If f is continuous at 0, show $\int f d\beta_1 = \int f d\beta_2 = \int f d\beta_3 = f(0)$.
- If $f(x) = 0$ for all irrational x and $f(x) = 1$ for all rational x , prove $f \notin \mathcal{R}$ on $[a, b]$ for any $a < b$.
- Suppose f is a bounded real function on $[a, b]$ and $f^2 \in \mathcal{R}$ on $[a, b]$. Does it follow that $f \in \mathcal{R}$? Does the answer change if we assume $f^3 \in \mathcal{R}$?
- Let P be the Cantor set (Rudin §2.44). Let f be a bounded real function on $[0, 1]$ that is continuous at every point outside P . Prove $f \in \mathcal{R}$ on $[0, 1]$. (Hint: P is covered by finitely many segments of arbitrarily small total length; proceed as in Theorem 6.10.)
- Suppose f is a real function on $(0, 1]$ and $f \in \mathcal{R}$ on $[c, 1]$ for every $c > 0$. Define $\int_0^1 f = \lim_{c \rightarrow 0} \int_c^1 f$ when this finite limit exists. (a) If $f \in \mathcal{R}$ on $[0, 1]$, show this agrees with the usual integral. (b) Construct f for which the limit exists although it fails to exist with $|f|$ in place of f .
- Suppose $f \in \mathcal{R}$ on $[a, b]$ for every $b > a$ (with a fixed), and define $\int_a^\infty f = \lim_{b \rightarrow \infty} \int_a^b f$ when the finite limit exists; if it also exists with $|f|$ in place of f , the integral *converges absolutely*. Assume $f(x) \geq 0$ and f decreases monotonically on $[1, \infty)$. Prove $\int_1^\infty f$ converges iff $\sum_{n=1}^\infty f(n)$ converges (the integral test).
- Show that integration by parts can sometimes be applied to the improper integrals of Exercises 7 and 8 (state hypotheses, formulate and prove a theorem). For instance, show $\int_0^\infty \frac{\cos x}{1+x} dx = \int_0^\infty \frac{\sin x}{(1+x)^2} dx$, and that one of these converges absolutely while the other does not.
- Let $p, q > 0$ with $\frac{1}{p} + \frac{1}{q} = 1$. Prove: (a) If $u, v > 0$ then $uv \leq \frac{u^p}{p} + \frac{v^q}{q}$, with equality iff $u^p = v^q$. (b) If $f, g \in \mathcal{R}(\alpha)$, $f, g \geq 0$, and $\int_a^b f^p d\alpha = \int_a^b g^q d\alpha = 1$, then $\int_a^b fg d\alpha \leq 1$. (c) For complex $f, g \in \mathcal{R}(\alpha)$, $|\int fg d\alpha| \leq (\int |f|^p d\alpha)^{1/p} (\int |g|^q d\alpha)^{1/q}$ (Holder's inequality; $p = q = 2$ is Schwarz). (d) Show Holder's inequality also holds for the improper integrals of Exercises 7 and 8.
- Let α be a fixed increasing function on $[a, b]$. For $u \in \mathcal{R}(\alpha)$ define $\|u\|_2 = (\int_a^b |u|^2 d\alpha)^{1/2}$. Suppose $f, g, h \in \mathcal{R}(\alpha)$ and prove the triangle inequality $\|f - h\|_2 \leq \|f - g\|_2 + \|g - h\|_2$ as a consequence of the Schwarz inequality.
- With the notation of Exercise 11, suppose $f \in \mathcal{R}(\alpha)$ and $\varepsilon > 0$. Prove there is a continuous g on $[a, b]$ with $\|f - g\|_2 < \varepsilon$. (Hint: take a suitable partition and interpolate f piecewise linearly.)

13. Define $f(x) = \int_x^{x+1} \sin(t^2) dt$. (a) Prove $|f(x)| < 1/x$ for $x > 0$. (Hint: put $t^2 = u$ and integrate by parts to show $f(x) = \frac{\cos(x^2)}{2x} - \frac{\cos((x+1)^2)}{2(x+1)} - \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{3/2}} du$; bound $\cos u$ by 1.) (b) Prove $2xf(x) = \cos(x^2) - \cos((x+1)^2) + r(x)$ with $|r(x)| < c/x$. (c) Find the upper and lower limits of $xf(x)$ as $x \rightarrow \infty$. (d) Does $\int_0^\infty \sin(t^2) dt$ converge?
14. Deal similarly with $f(x) = \int_x^{x+1} \sin(e^t) dt$. Show $e^x |f(x)| < 2$ and that $e^x f(x) = \cos(e^x) - e^{-1} \cos(e^{x+1}) + r(x)$ with $|r(x)| < Ce^{-x}$ for some constant C .
15. Suppose f is real and continuously differentiable on $[a, b]$, $f(a) = f(b) = 0$, and $\int_a^b f^2(x) dx = 1$. Prove $\int_a^b xf(x)f'(x) dx = -\frac{1}{2}$ and that $\int_a^b f'(x)^2 dx \cdot \int_a^b x^2 f^2(x) dx > \frac{1}{4}$.
16. For $1 < s < \infty$ define $\zeta(s) = \sum_{n=1}^\infty n^{-s}$ (Riemann's zeta function). Prove (a) $\zeta(s) = s \int_1^\infty \frac{\lfloor x \rfloor}{x^{s+1}} dx$, and (b) $\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \frac{x - \lfloor x \rfloor}{x^{s+1}} dx$, where $\lfloor x \rfloor$ is the greatest integer $\leq x$. Prove the integral in (b) converges for all $s > 0$.
17. Suppose α increases monotonically on $[a, b]$, g is continuous, and $g(x) = G'(x)$ for $a \leq x \leq b$. Prove $\int_a^b \alpha(x)g(x) dx = G(b)\alpha(b) - G(a)\alpha(a) - \int_a^b G d\alpha$. (Hint: take g real; for a partition choose tags t_i with $g(t_i)\Delta x_i = G(x_i) - G(x_{i-1})$ and rearrange by summation by parts.)
18. Let $\gamma_1, \gamma_2, \gamma_3$ be curves in the complex plane on $[0, 2\pi]$ defined by $\gamma_1(t) = e^{it}$, $\gamma_2(t) = e^{2it}$, $\gamma_3(t) = e^{2\pi it \sin(1/t)}$. Show all three have the same range; that γ_1, γ_2 are rectifiable with lengths 2π and 4π ; and that γ_3 is not rectifiable.
19. Let γ_1 be a curve in $\mathbb{R}^k$ on $[a, b]$, and let φ be a continuous one-to-one map of $[c, d]$ onto $[a, b]$ with $\varphi(c) = a$; set $\gamma_2 = \gamma_1 \circ \varphi$. Prove γ_2 is an arc, a closed curve, or rectifiable iff γ_1 is, and that γ_2 and γ_1 have the same length.

Additional exercises.

1. Compute $\int_0^2 x d\alpha$ where $\alpha(x) = x^2 + I(x-1)$ (smooth part plus a unit jump at 1).
2. Let α jump by c_k at integer $k = 1, 2, 3$ with $c_1 = 1, c_2 = 2, c_3 = 1$. Compute $\int_0^4 (x+1) d\alpha$ and $\int_0^4 \cos(\pi x) d\alpha$.
3. Prove that if α is increasing and continuous and f is continuous, then $\int_a^b f d\alpha = \int_a^b f(x) d\alpha(x)$ equals the limit of Stieltjes sums for any tag choice (the Stieltjes analogue of Proposition 7.3.6).
4. Verify Stieltjes integration by parts on $[0, 2]$ for $f(x) = x$, $\alpha(x) = x^2$: compute both $\int_0^2 f d\alpha$ and $\int_0^2 \alpha df$ and check the sum equals $f(2)\alpha(2) - f(0)\alpha(0)$.
5. (Non-example.) Let $f = \alpha = I(x)$ (both with a jump at 0) on $[-1, 1]$. Show $\int_{-1}^1 f d\alpha$ does not exist, illustrating that f and α must not be discontinuous at the same point.
6. Express $\sum_{n=1}^N a_n g(n)$ as a Riemann–Stieltjes integral $\int_0^N g d\alpha$ for a suitable step integrator α , and use this to give a Stieltjes proof of Abel summation.
7. (Diagram.) Draw a staircase integrator α with three jumps of heights c_1, c_2, c_3 and a continuous f overlaid; annotate that $\int f d\alpha = \sum c_k f(s_k)$ samples f only at the jumps.
8. Let $\alpha(x) = x$ for $x \in [0, 1]$ and $\alpha(x) = x + 1$ for $x \in (1, 2]$ (a unit jump at 1). For continuous f , show $\int_0^2 f d\alpha = \int_0^2 f(x) dx + f(1)$.
9. Prove the change-of-variable formula (Theorem 7.5.9) in detail for a continuously differentiable increasing φ , reducing it to the Riemann case via Theorem 7.5.5.

7.6 Integration of Vector-Valued Functions and Rectifiable Curves

Integration extends componentwise to functions with values in $\mathbb{R}^k$ (or $\mathbb{C}$). The payoff is a clean theory of *arc length*, expressed as a Stieltjes integral.

Vector-valued integrands

Definition 7.6.1 (Rudin 6.23). Let $\mathbf{f} = (f_1, \dots, f_k) : [a, b] \rightarrow \mathbb{R}^k$ and α increasing. We say $\mathbf{f} \in \mathcal{R}(\alpha)$ if each component $f_j \in \mathcal{R}(\alpha)$, and then

$$\int_a^b \mathbf{f} d\alpha = \left(\int_a^b f_1 d\alpha, \dots, \int_a^b f_k d\alpha \right) \in \mathbb{R}^k.$$

Linearity, additivity, and FTC hold componentwise. The one genuinely vectorial fact is the norm inequality, which does *not* follow componentwise (the triangle inequality in $\mathbb{R}^k$ is needed):

Theorem 7.6.2 (Norm inequality, Rudin 6.25). *If $\mathbf{f} \in \mathcal{R}(\alpha)$ on $[a, b]$ then $\|\mathbf{f}\| \in \mathcal{R}(\alpha)$ and*

$$\left\| \int_a^b \mathbf{f} d\alpha \right\| \leq \int_a^b \|\mathbf{f}\| d\alpha,$$

where $\|\cdot\|$ is the Euclidean norm.

Proof. Put $\mathbf{y} = \int_a^b \mathbf{f} d\alpha = (y_1, \dots, y_k)$. Each $|f_j| \leq \|\mathbf{f}\|$, so $\|\mathbf{f}\| = (\sum f_j^2)^{1/2} \in \mathcal{R}(\alpha)$ as a continuous function of integrable functions. If $\mathbf{y} = \mathbf{0}$ the claim is trivial. Otherwise,

$$\|\mathbf{y}\|^2 = \sum_j y_j^2 = \sum_j y_j \int_a^b f_j d\alpha = \int_a^b \left(\sum_j y_j f_j \right) d\alpha = \int_a^b \mathbf{y} \cdot \mathbf{f} d\alpha.$$

By Cauchy–Schwarz in $\mathbb{R}^k$, $\mathbf{y} \cdot \mathbf{f}(x) \leq \|\mathbf{y}\| \|\mathbf{f}(x)\|$ pointwise, so by monotonicity $\|\mathbf{y}\|^2 \leq \|\mathbf{y}\| \int_a^b \|\mathbf{f}\| d\alpha$. Dividing by $\|\mathbf{y}\| > 0$ gives the result. $\square$

Rectifiable curves and arc length

Definition 7.6.3 (Curve and length, Rudin 6.26). A *curve* in $\mathbb{R}^k$ is a continuous map $\gamma : [a, b] \rightarrow \mathbb{R}^k$. For a partition $P = \{x_0, \dots, x_n\}$ its inscribed polygonal length is

$$\Lambda(P, \gamma) = \sum_{i=1}^n \|\gamma(x_i) - \gamma(x_{i-1})\|.$$

The *length* of γ is $\Lambda(\gamma) = \sup_P \Lambda(P, \gamma)$. The curve is *rectifiable* if $\Lambda(\gamma) < \infty$.

Each term is a chord; $\Lambda(P, \gamma)$ is the length of the inscribed polygon. Refining P can only *increase* $\Lambda(P, \gamma)$ (triangle inequality: a straight chord is no longer than the broken path through an inserted point), so the sup is a genuine limit as $\|P\| \rightarrow 0$.

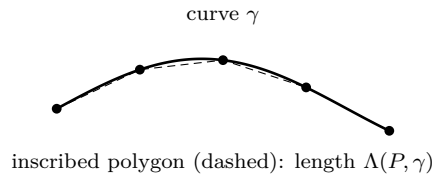


Figure 7.9. Arc length is the supremum of inscribed-polygon lengths; inserting vertices only lengthens the polygon, so $\Lambda(\gamma) = \lim_{\|P\| \rightarrow 0} \Lambda(P, \gamma)$.

Theorem 7.6.4 (Length as an integral, Rudin 6.27). *If γ is continuously differentiable (γ' continuous) on $[a, b]$, then γ is rectifiable and*

$$\Lambda(\gamma) = \int_a^b \|\gamma'(t)\| dt.$$

Proof. For each subinterval, $\gamma(x_i) - \gamma(x_{i-1}) = \int_{x_{i-1}}^{x_i} \gamma'(t) dt$ componentwise, so by the norm inequality (Theorem 7.6.2) $\|\gamma(x_i) - \gamma(x_{i-1})\| \leq \int_{x_{i-1}}^{x_i} \|\gamma'\| dt$. Summing, $\Lambda(P, \gamma) \leq \int_a^b \|\gamma'\| dt$ for every P , so $\Lambda(\gamma) \leq \int_a^b \|\gamma'\|$. For the reverse, fix $\varepsilon > 0$; uniform continuity of γ' gives δ with $\|\gamma'(s) - \gamma'(t)\| < \varepsilon$ when $|s - t| < \delta$. For $\|P\| < \delta$ and $t \in I_i$,

$$\|\gamma'(t)\| \Delta x_i \leq \left\| \int_{I_i} \gamma' + \int_{I_i} \|\gamma'(t) - \gamma'(s)\| ds \right\| \leq \|\gamma(x_i) - \gamma(x_{i-1})\| + \varepsilon \Delta x_i.$$

Summing, $\int_a^b \|\gamma'\| \leq \Lambda(P, \gamma) + \varepsilon(b - a) \leq \Lambda(\gamma) + \varepsilon(b - a)$. Let $\varepsilon \rightarrow 0$: $\int_a^b \|\gamma'\| \leq \Lambda(\gamma)$. The two bounds give equality. $\square$

Remark (Length is a Stieltjes integral). Define the arc-length function $s(t) = \int_a^t \|\gamma'\|$, an increasing integrator. Then for a scalar weight w , $\int_a^b w(\gamma(t)) ds(t) = \int_a^b w(\gamma(t)) \|\gamma'(t)\| dt$ is integration *with respect to arc length* — a Riemann–Stieltjes integral whose integrator measures distance traveled, not parameter elapsed.

Worked Example 7.6.5 (Length of a circular arc). Find the length of $\gamma(t) = (\cos t, \sin t)$, $t \in [0, 2\pi]$.

Recognition. Trigger: smooth curve; apply $\Lambda = \int \|\gamma'\|$.

Solution. $\gamma'(t) = (-\sin t, \cos t)$, $\|\gamma'(t)\| = \sqrt{\sin^2 t + \cos^2 t} = 1$. Hence $\Lambda = \int_0^{2\pi} 1 dt = 2\pi$. *Check.* The unit circle has circumference 2π , as expected. (Rudin 6.18: the curve $\gamma_1 = e^{2\pi i t}$ traversed once has length 2π ; traversed twice, 4π — length counts multiplicity.) ■

Worked Example 7.6.6 (NON-rectifiable curve (Rudin 6.18 spirit)). Show $\gamma(t) = (t, t \sin(\pi/t))$ for $t \in (0, 1]$, $\gamma(0) = (0, 0)$, has infinite length.

Recognition. Trigger: a curve continuous but oscillating with non-summable vertical swings.

Solution. Take the partition points $t_m = 2/(2m + 1)$ where $\sin(\pi/t_m) = \pm 1$. Between consecutive such points the second coordinate swings by about $t_m + t_{m+1}$, and the chord length is at least the vertical jump. Summing these jumps gives a series comparable to $\sum 1/m$, which diverges. Hence $\sup_P \Lambda(P, \gamma) = \infty$: γ is not rectifiable, even though it is continuous. *Lesson.* Continuity is far from enough for finite length; γ' must be (essentially) integrable. ■

Caution. Length depends on the curve (the map γ), not merely its image. Rudin 6.18 gives three maps with the *same* range (the unit circle) but lengths 2π , 4π , and ∞ . Reparametrizing by a monotone φ (Rudin 6.19, B&S analogue) preserves length; reparametrizing so as to retrace or oscillate does not. “Arc length of a set” is ambiguous; specify the parametrization.

Further worked examples

Worked Example 7.6.7 (Length of a parabolic arc). Find the length of the graph $y = \frac{1}{2}x^2$ for $x \in [0, 1]$, parametrized by $\gamma(t) = (t, \frac{1}{2}t^2)$.

Recognition. Trigger: a graph $y = g(x)$ written as a curve; apply $\Lambda = \int \|\gamma'\|$ with $\|\gamma'\| = \sqrt{1 + g'(t)^2}$.

Solution. $\gamma'(t) = (1, t)$, so $\|\gamma'(t)\| = \sqrt{1 + t^2}$ and

$$\Lambda = \int_0^1 \sqrt{1 + t^2} dt = \left[\frac{t}{2} \sqrt{1 + t^2} + \frac{1}{2} \ln(t + \sqrt{1 + t^2}) \right]_0^1 = \frac{\sqrt{2}}{2} + \frac{1}{2} \ln(1 + \sqrt{2}).$$

Check. Numerically $\frac{1.414}{2} + \frac{1}{2} \ln(2.414) \approx 0.707 + 0.441 = 1.148$. This exceeds the straight-line distance from $(0, 0)$ to $(1, \frac{1}{2})$, which is $\sqrt{1 + \frac{1}{4}} = \sqrt{1.25} \approx 1.118$, as a curved path must. ■

Worked Example 7.6.8 (The norm inequality is genuinely vectorial). For $\mathbf{f}(t) = (\cos t, \sin t)$ on $[0, 2\pi]$, compare $\left\| \int_0^{2\pi} \mathbf{f} dt \right\|$ with $\int_0^{2\pi} \|\mathbf{f}\| dt$, illustrating Theorem 7.6.2.

Recognition. Trigger: a check that the vector norm inequality can be strict; compute both sides.

Solution. Componentwise, $\int_0^{2\pi} \cos t \, dt = 0$ and $\int_0^{2\pi} \sin t \, dt = 0$, so $\int_0^{2\pi} \mathbf{f} \, dt = (0, 0)$ and its norm is 0. But $\|\mathbf{f}(t)\| = \sqrt{\cos^2 t + \sin^2 t} = 1$, so $\int_0^{2\pi} \|\mathbf{f}\| \, dt = 2\pi$. The inequality reads $0 \leq 2\pi$ — here *strict*. *Lesson.* The integral of a vector can have small norm even when the integrand never shrinks, because the directions cancel; a componentwise “triangle inequality” would be false, which is why Theorem 7.6.2 needs the Cauchy–Schwarz argument. ■

Worked Example 7.6.9 (Length is reparametrization-invariant). Reparametrize the unit semicircle $\gamma(t) = (\cos t, \sin t)$, $t \in [0, \pi]$, by $t = \varphi(u) = \pi u^2$, $u \in [0, 1]$, and confirm the length is unchanged.

Recognition. Trigger: a monotone smooth reparametrization; by the change-of-variable for arc length the value must be the same.

Solution. The original length is $\int_0^\pi \|\gamma'\| \, dt = \int_0^\pi 1 \, dt = \pi$. The reparametrized curve is $\delta(u) = \gamma(\varphi(u))$, with $\|\delta'(u)\| = \|\gamma'(\varphi(u))\| |\varphi'(u)| = 1 \cdot 2\pi u = 2\pi u$ (here $\varphi'(u) = 2\pi u \geq 0$ on $[0, 1]$). Then $\int_0^1 \|\delta'(u)\| \, du = \int_0^1 2\pi u \, du = \pi$, the same value. *Check.* The monotone change of variable $u \mapsto \pi u^2$ only re-times the traversal; it speeds up near $u = 1$ but never backtracks, so total distance is preserved. ■

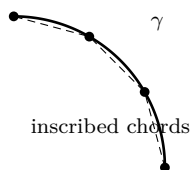


Figure 7.10. Inscribed chords of a quarter circle (radius 2): summing the chord lengths $\|\gamma(x_i) - \gamma(x_{i-1})\|$ and refining converges up to the arc length $\int \|\gamma'\| = 2 \cdot \frac{\pi}{2} = \pi$.

Additional exercises.

- Find the length of the curve $\gamma(t) = (t, t^{3/2})$ for $t \in [0, 1]$.
- Compute the arc length of $\gamma(t) = (\cos^3 t, \sin^3 t)$ (an astroid) for $t \in [0, \pi/2]$.
- Prove the norm inequality $\left\| \int_a^b \mathbf{f} \right\| \leq \int_a^b \|\mathbf{f}\|$ cannot in general be replaced by equality, and characterize when equality holds (the values $\mathbf{f}(x)$ point in a common direction a.e.).
- Show that reparametrizing γ by a continuous increasing bijection preserves length, and give the explicit correspondence of partitions.
- (Non-example.) Show the graph of $g(x) = x \cos(\pi/x)$ for $x \in (0, 1]$, $g(0) = 0$, is a continuous curve of infinite length. (Use the partition $x_m = 1/m$.)
- Let γ be C^1 on $[a, b]$. Prove the arc-length function $s(t) = \int_a^t \|\gamma'\|$ is increasing, and that $\int_a^b w(\gamma(t)) \, ds(t) = \int_a^b w(\gamma(t)) \|\gamma'(t)\| \, dt$.
- (Diagram.) Draw a smooth arc with an inscribed polygon on a 5-point partition; shade one chord and label its length $\|\gamma(x_i) - \gamma(x_{i-1})\|$, and indicate how refining increases the polygon length toward $\Lambda(\gamma)$.
- Show that the three curves $\gamma_1(t) = e^{it}$, $\gamma_2(t) = e^{2it}$ on $[0, 2\pi]$ have the same range (the unit circle) but lengths 2π and 4π ; conclude length is a property of the parametrization, not the image.

7.7 Approximate Integration

When no antiderivative is available, we estimate $\int_a^b f$ numerically. The three classic rules replace f on each subinterval by a constant (midpoint), a line (trapezoid), or a parabola (Simpson), and come with sharp error bounds in terms of derivatives of f .

Throughout, divide $[a, b]$ into n equal pieces of width $h = (b - a)/n$ with nodes $x_j = a + jh$, and write $f_j = f(x_j)$.

The three rules

Definition 7.7.1 (B&S 7.5). The *midpoint*, *trapezoidal*, and *Simpson* approximations are

$$M_n = h \sum_{j=1}^n f\left(\frac{x_{j-1}+x_j}{2}\right), \quad T_n = \frac{h}{2}(f_0 + 2f_1 + \cdots + 2f_{n-1} + f_n),$$

$$S_n = \frac{h}{3}(f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 2f_{n-2} + 4f_{n-1} + f_n) \quad (n \text{ even}).$$

T_n averages the left- and right-endpoint sums; $S_n = \frac{1}{3}T_n + \frac{2}{3}M_n$ is a weighted blend (B&S 7.5.10).

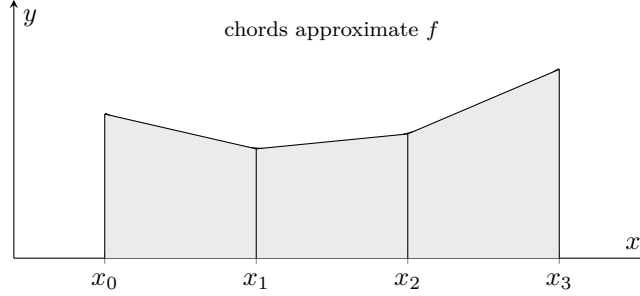


Figure 7.11. Trapezoidal rule: f is replaced on each piece by the chord through its endpoints; the area is a sum of trapezoids.

Error bounds

Theorem 7.7.2 (Trapezoid error, B&S 7.5.3). If f'' is continuous on $[a, b]$ and $|f''| \leq B_2$, then

$$\left| T_n - \int_a^b f \right| \leq \frac{(b-a)^3}{12n^2} B_2 = \frac{(b-a)h^2}{12} B_2.$$

Idea. On one subinterval $[x_{j-1}, x_j]$, let $e(x)$ be the difference between f and the chord. e vanishes at both endpoints, and a double integration of $e'' = f''$ (or a Taylor estimate with the integral form of the remainder) gives $\left| \int_{x_{j-1}}^{x_j} e \right| \leq \frac{B_2}{12} h^3$. Summing over the n subintervals and using $nh = b - a$ produces the bound. $\square$

Theorem 7.7.3 (Simpson error, B&S 7.5.something / standard). If $f^{(4)}$ is continuous on $[a, b]$ and $|f^{(4)}| \leq B_4$, then (for even n)

$$\left| S_n - \int_a^b f \right| \leq \frac{(b-a)^5}{180n^4} B_4 = \frac{(b-a)h^4}{180} B_4.$$

In particular, Simpson's rule is exact for every polynomial of degree ≤ 3 (B&S 7.5.7), since then $f^{(4)} \equiv 0$.

Idea. Over a double interval $[x_{2k}, x_{2k+2}]$ of width $2h$, Simpson integrates the interpolating parabola exactly; the error is the integral of $f - (\text{parabola})$, which a fourth-order Taylor estimate bounds by $\frac{B_4}{90} h^5$. Summing $n/2$ such blocks and using $nh = b - a$ gives the stated constant $1/180$. $\square$

Method (Choosing n for a target accuracy). *Recognition trigger.* “How many subintervals guarantee error $< \tau$?” (B&S 7.5.4, 7.5.6).

1. Bound the relevant derivative: $B_2 = \max |f''|$ (trapezoid) or $B_4 = \max |f^{(4)}|$ (Simpson) on $[a, b]$.
2. Set the error bound $\leq \tau$: trapezoid needs $\frac{(b-a)^3 B_2}{12n^2} \leq \tau$, i.e. $n \geq \sqrt{(b-a)^3 B_2 / (12\tau)}$; Simpson needs $n \geq ((b-a)^5 B_4 / (180\tau))^{1/4}$.
3. Round n up (and to an even integer for Simpson).

The fourth-power dependence is why Simpson crushes trapezoid: halving h cuts Simpson error by 16, trapezoid only by 4.

Worked Example 7.7.4 (Trapezoid estimate of $\ln 2$ (B&S 7.5.1)). Estimate $\ln 2 = \int_1^2 \frac{dx}{x}$ with T_4 , and bound the error.

Recognition. Trigger: smooth integrand, want a quick estimate with error control.

Solution. Here $h = 1/4$, nodes 1, 1.25, 1.5, 1.75, 2, $f(x) = 1/x$. Then

$$T_4 = \frac{1/4}{2} \left(1 + 2 \cdot \frac{1}{1.25} + 2 \cdot \frac{1}{1.5} + 2 \cdot \frac{1}{1.75} + \frac{1}{2} \right) \approx 0.6970.$$

Since $f''(x) = 2/x^3$ and $\max_{[1,2]} |f''| = 2$ (at $x = 1$), $|T_4 - \ln 2| \leq \frac{(1)^3}{12 \cdot 16} \cdot 2 = \frac{1}{96} \approx 0.0104$. *Check.* $\ln 2 \approx 0.6931$, and indeed $|0.6970 - 0.6931| = 0.0039 < 1/96$. ■

Worked Example 7.7.5 (Simpson estimate of $\ln 2$ (B&S 7.5.2)). Estimate $\ln 2$ with S_4 and bound the error.

Recognition. Trigger: same integral, but Simpson for a far better bound.

Solution. With the same nodes, $S_4 = \frac{1/4}{3} (1 + 4(0.8) + 2(0.6667) + 4(0.5714) + 0.5) \approx 0.6933$. Now $f^{(4)}(x) = 24/x^5$, $\max_{[1,2]} |f^{(4)}| = 24$ (at $x = 1$), so $|S_4 - \ln 2| \leq \frac{(1)^5}{180 \cdot 256} \cdot 24 = \frac{24}{46080} \approx 0.00052$. *Check.* $|0.6933 - 0.6931| \approx 0.0002 < 0.00052$. Simpson is $\sim 20\times$ more accurate than trapezoid here for the same data. ■

Worked Example 7.7.6 (Simpson is exact on cubics (B&S 7.5.7)). Verify S_2 integrates $p(x) = x^3$ on $[0, 2]$ exactly.

Recognition. Trigger: a polynomial of degree ≤ 3 ; the Simpson error has $f^{(4)} \equiv 0$.

Solution. $h = 1$, nodes 0, 1, 2: $S_2 = \frac{1}{3}(0 + 4 \cdot 1 + 8) = \frac{12}{3} = 4$. The true integral is $\int_0^2 x^3 dx = \frac{x^4}{4} \Big|_0^2 = 4$. They agree exactly, as the error bound (with $B_4 = 0$) predicts. *Check.* Trapezoid gives $T_2 = \frac{1}{2}(0 + 2 \cdot 1 + 8) = 5 \neq 4$ — the chord overshoots a convex cubic, so trapezoid is *not* exact. ■

Caution. The error bounds require the named derivative to *exist and be bounded*. For $\int_0^1 \sqrt{1-x^2} dx = \pi/4$ the integrand has $|f''| \rightarrow \infty$ as $x \rightarrow 1$, so the trapezoid/Simpson bounds (4), (7), (10) of B&S do not apply (B&S 7.5.12); one must restrict to a subinterval or use a different estimate. Also: a smaller error *bound* does not guarantee a smaller actual error on a particular f — it bounds the worst case.

Further worked examples

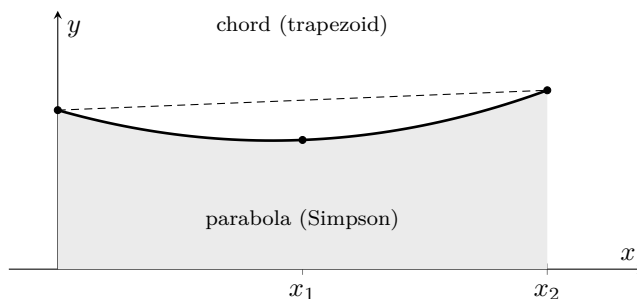


Figure 7.12. Over a double panel $[x_0, x_2]$, Simpson integrates the parabola through the three nodes (shaded), while the trapezoid uses only the end chord (dashed). Matching the curvature is why Simpson gains two extra orders of accuracy.

Worked Example 7.7.7 (Midpoint vs. trapezoid on $\int_0^1 x^2 dx$). Compute M_2 and T_2 for $\int_0^1 x^2 dx = \frac{1}{3}$ and see which side of the truth each lands on.

Recognition. Trigger: a hand computation of two rules on the same data; recall midpoint and trapezoid err in *opposite* directions for a convex f .

Solution. With $n = 2$, $h = \frac{1}{2}$, nodes $0, \frac{1}{2}, 1$ and midpoints $\frac{1}{4}, \frac{3}{4}$:

$$M_2 = h \left[f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right) \right] = \frac{1}{2} \left[\frac{1}{16} + \frac{9}{16} \right] = \frac{1}{2} \cdot \frac{10}{16} = \frac{5}{16} = 0.3125,$$

$$T_2 = \frac{h}{2} [f_0 + 2f_1 + f_2] = \frac{1}{4} [0 + 2 \cdot \frac{1}{4} + 1] = \frac{1}{4} \cdot \frac{3}{2} = \frac{3}{8} = 0.375.$$

The truth is $\frac{1}{3} \approx 0.3333$. *Check.* $M_2 = 0.3125 < \frac{1}{3} < 0.375 = T_2$ — the midpoint *underestimates* and the trapezoid *overestimates* a convex function, and the midpoint error (0.0208) is about half the trapezoid error (0.0417), matching the 2:1 ratio of their leading error constants. ■

Worked Example 7.7.8 (Choosing n for guaranteed accuracy). How many subintervals n does the trapezoidal rule need to guarantee $\left| T_n - \int_0^1 e^x dx \right| < 10^{-4}$?

Recognition. Trigger: a “size n for tolerance τ ” problem (the Method); bound f'' and solve the error inequality.

Solution. Here $f(x) = e^x$, $f''(x) = e^x$, and $B_2 = \max_{[0,1]} |f''| = e \approx 2.718$. The trapezoid bound requires

$$\frac{(b-a)^3 B_2}{12n^2} = \frac{1 \cdot e}{12n^2} \leq 10^{-4} \implies n^2 \geq \frac{e}{12 \cdot 10^{-4}} \approx 2265, \quad n \geq 47.6.$$

So $n = 48$ suffices. *Check.* The true error of T_{48} is far smaller than 10^{-4} because the bound uses the worst-case $B_2 = e$ attained only at $x = 1$; the guarantee is conservative but safe. By contrast Simpson would need only $n \geq (e/(180 \cdot 10^{-4}))^{1/4} \approx 3.6$, i.e. $n = 4$ — dramatically fewer. ■

Worked Example 7.7.9 (NON-example: Simpson is not exact on a quartic). Show Simpson’s rule S_2 fails to integrate $p(x) = x^4$ on $[0, 2]$ exactly, and quantify the error.

Recognition. Trigger: the boundary case of Theorem 7.7.3 — exactness holds for degree ≤ 3 , so degree 4 should be the first failure ($f^{(4)} \neq 0$).

Solution. $h = 1$, nodes $0, 1, 2$ with $f_0 = 0$, $f_1 = 1$, $f_2 = 16$: $S_2 = \frac{h}{3} (f_0 + 4f_1 + f_2) = \frac{1}{3} (0 + 4 + 16) = \frac{20}{3} \approx 6.667$. The true value is $\int_0^2 x^4 dx = \frac{32}{5} = 6.4$. The error is $\frac{20}{3} - \frac{32}{5} = \frac{100-96}{15} = \frac{4}{15} \approx 0.267 \neq 0$. *Check.* $f^{(4)} = 24 \neq 0$, so the Simpson bound is $\frac{(2)^5}{180 \cdot 24} \cdot 24 = \frac{32 \cdot 24}{2880} = \frac{4}{15}$ — the actual error *equals* the bound here, because x^4 makes the fourth derivative constant and the estimate tight. ■

Exercises (Bartle & Sherbert, §7.5).

1. Use the trapezoidal approximation with $n = 4$ to evaluate $\ln 2 = \int_1^2 (1/x) dx$. Show $0.6866 \leq \ln 2 \leq 0.6958$ and $0.0013 < \frac{1}{768} \leq T_4 - \ln 2 \leq \frac{1}{96} < 0.0105$.
2. Use the Simpson approximation with $n = 4$ to evaluate $\ln 2 = \int_1^2 (1/x) dx$. Show $0.6927 \leq \ln 2 \leq 0.6933$ and $0.000016 < \frac{1}{5} \cdot \frac{1}{1920} \leq S_4 - \ln 2 \leq \frac{1}{1920} < 0.000521$.
3. Let $f(x) = (1 + x^2)^{-1}$ on $[0, 1]$. Show $f''(x) = 2(3x^2 - 1)(1 + x^2)^{-3}$ and $|f''(x)| \leq 2$ on $[0, 1]$. Use the trapezoidal approximation with $n = 4$ to evaluate $\pi/4 = \int_0^1 f$, and show $|T_4(f) - \pi/4| \leq 1/96 < 0.0105$.
4. If $T_n(f)$ is used to approximate $\pi/4$ as in Exercise 3, show one must take $n \geq 409$ to be sure the error is less than 10^{-6} .
5. Let f be as in Exercise 3. Show $f^{(4)}(x) = 24(5x^4 - 10x^2 + 1)(1 + x^2)^{-5}$ and $|f^{(4)}(x)| \leq 96$ on $[0, 1]$. Use Simpson with $n = 4$ to evaluate $\pi/4$, and show $|S_4(f) - \pi/4| \leq 1/480 < 0.0021$.
6. If $S_n(f)$ is used to approximate $\pi/4$ as in Exercise 5, show one must take $n \geq 28$ to be sure the error is less than 10^{-6} .
7. If p is a polynomial of degree at most 3, show the Simpson approximations are exact.
8. Show that if $f''(x) \geq 0$ on $[a, b]$ (i.e. f is convex), then for any $m, n \in \mathbb{N}$, $M_n(f) \leq \int_a^b f \leq T_m(f)$. If $f'' \leq 0$, the inequality reverses.
9. Show $T_{2n}(f) = \frac{1}{2} [M_n(f) + T_n(f)]$.

10. Show $S_{2n}(f) = \frac{2}{3}M_n(f) + \frac{1}{3}T_n(f)$.
11. Show $\left|S_n(f) - \int_a^b f\right| \leq \frac{(b-a)^5}{180n^4}B_4$, where $B_4 \geq |f^{(4)}(x)|$ for all $x \in [a, b]$.
12. Note $\int_0^1 (1-x^2)^{1/2} dx = \pi/4$. Explain why the standard error estimates cannot be used. Show that if $h(x) = (1-x^2)^{1/2}$ then $T_n(h) \leq \pi/4 \leq M_n(h)$; calculate $M_8(h)$ and $T_8(h)$.
13. With h as in Exercise 12, explain why $K = \int_0^{1/\sqrt{2}} h = \pi/8 + 1/4$. Show $|h''(x)| \leq 2^{3/2}$ and $|h^{(4)}(x)| \leq 9 \cdot 2^{7/2}$ on $[0, 1/\sqrt{2}]$. Show $|K - T_n(h)| \leq 1/12n^2$ and $|K - S_n(h)| \leq 1/10n^4$, and use these to estimate π .
14. Approximate $\int_1^2 (1+x^4)^{1/2} dx$ with an error estimate.
15. Approximate $\int_0^2 (4+x^3)^{1/2} dx$ with an error estimate.
16. Approximate $\int_0^1 \frac{\sin x}{x} dx$ with an error estimate.
17. Approximate $\int_0^\pi \cos(x^2) dx$ with an error estimate.
18. Approximate $\int_0^{\pi/2} \frac{dx}{1+\sin x}$ with an error estimate.
19. Approximate $\int_0^1 \frac{dx}{1+x^3}$ with an error estimate.
20. Approximate $\int_0^{\pi/2} \sqrt{\sin x} dx$ with an error estimate.

Additional exercises.

1. Use the midpoint rule M_4 to estimate $\int_0^1 e^{-x^2} dx$ and compare with the trapezoid estimate T_4 . Which bracket the true value if the integrand is concave?
2. Show $T_{2n} = \frac{1}{2}(M_n + T_n)$ and $S_{2n} = \frac{2}{3}M_n + \frac{1}{3}T_n$ directly from the definitions.
3. For $\int_0^1 x^4 dx$, compute S_2 and the exact value; verify the error is nonzero (Simpson is exact only up to degree 3) and within the bound $\frac{(b-a)^5}{180n^4}B_4$.
4. Determine the smallest n for which the trapezoid rule guarantees $\left|T_n - \int_0^1 e^x dx\right| < 10^{-4}$.
5. (Non-example.) Explain why neither the trapezoid nor the Simpson error bound applies to $\int_0^1 \frac{dx}{\sqrt{x}}$, and propose a fix (split off a neighborhood of 0 and integrate it exactly).
6. Prove the convexity comparison: if $f'' \geq 0$ on $[a, b]$ then $M_n(f) \leq \int_a^b f \leq T_n(f)$ for every n , with the inequalities reversed if $f'' \leq 0$.
7. (Diagram.) On one set of axes, draw a convex f with the midpoint rectangle (tangent line interpretation) and the trapezoid chord on a single subinterval, showing the midpoint underestimates and the trapezoid overestimates the area.
8. Show Simpson's rule on $[a, b]$ (one block, $n = 2$) integrates the interpolating parabola through $(a, f(a))$, $(\frac{a+b}{2}, f(\frac{a+b}{2}))$, $(b, f(b))$ exactly, and derive the weights $\frac{h}{3}(1, 4, 1)$.

Chapter 8

Sequences and Series of Functions

8.1 Pointwise and Uniform Convergence

Two ways a sequence of functions can converge

Let E be a set and let $f_n : E \rightarrow \mathbb{R}$ (or $\mathbb{C}$) be functions, $n = 1, 2, 3, \dots$. There are two genuinely different things one can mean by “ f_n converges to f ”. The weaker notion only asks the numbers $f_n(x)$ to converge, one x at a time; the stronger asks the convergence to happen at a single rate good for *all* x at once. Almost every phenomenon in this chapter is a battle between these two.

Definition 8.1.1 (Pointwise convergence). The sequence (f_n) *converges pointwise* on E to f if for every $x \in E$ the numerical sequence $(f_n(x))$ converges to $f(x)$; that is, for every $x \in E$ and every $\varepsilon > 0$ there is $N = N(\varepsilon, x)$ with

$$n \geq N \implies |f_n(x) - f(x)| < \varepsilon.$$

We write $f_n \rightarrow f$ pointwise, and call $f(x) = \lim_n f_n(x)$ the *pointwise (limit) function*. Likewise a series $\sum f_n$ *converges pointwise* to s if its partial sums $s_n = \sum_{k=1}^n f_k$ converge pointwise to s .

The crucial feature — the source of all the trouble — is that N is allowed to depend on x . At a “slow” point x you may need a much larger N than at a “fast” point.

Definition 8.1.2 (Uniform convergence). The sequence (f_n) *converges uniformly* on E to f , written $f_n \rightarrow f$ uniformly (or $f_n \rightrightarrows f$), if for every $\varepsilon > 0$ there is a *single* $N = N(\varepsilon)$, independent of x , such that

$$n \geq N \implies |f_n(x) - f(x)| < \varepsilon \quad \text{for every } x \in E.$$

A series $\sum f_n$ *converges uniformly* if its partial sums do.

Remark. The only difference between the two definitions is the order of the quantifiers. Pointwise: $\forall x \forall \varepsilon \exists N$. Uniform: $\forall \varepsilon \exists N \forall x$. Pulling the $\exists N$ in front of $\forall x$ forces one threshold to work everywhere. Uniform $\implies$ pointwise, never the reverse.

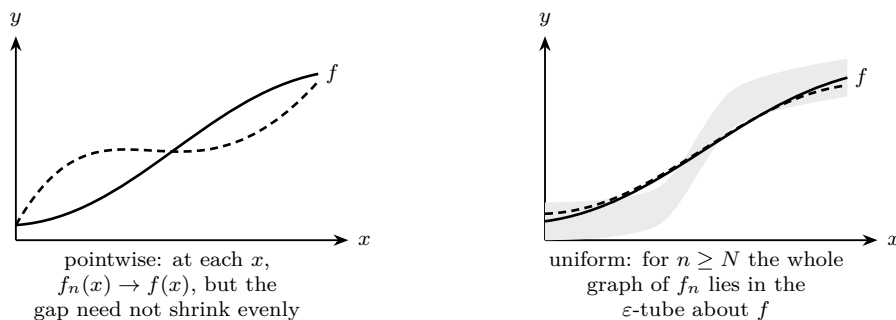


Figure 8.1. Uniform convergence means the graph of f_n eventually fits inside an ε -thick tube wrapped around the graph of f .

The sup-norm and the M_n test

The tube picture is made precise by a single number: the largest vertical gap.

Definition 8.1.3 (Uniform / sup-norm). For a bounded function $g : E \rightarrow \mathbb{R}$ put

$$\|g\|_E = \sup_{x \in E} |g(x)|.$$

This is the *uniform norm* (or *sup-norm*) of g on E . For functions it satisfies $\|g\|_E \geq 0$, $\|cg\|_E = |c| \|g\|_E$, the triangle inequality $\|g + h\|_E \leq \|g\|_E + \|h\|_E$, and $\|g\|_E = 0 \iff g \equiv 0$.

Theorem 8.1.4 (The M_n criterion for uniform convergence). Suppose $f_n \rightarrow f$ pointwise on E , and put $M_n = \|f_n - f\|_E = \sup_{x \in E} |f_n(x) - f(x)|$. Then $f_n \rightarrow f$ uniformly on E if and only if $M_n \rightarrow 0$ as $n \rightarrow \infty$.

Proof. ($\Leftarrow$) If $M_n \rightarrow 0$, let $\varepsilon > 0$ and pick N with $M_n < \varepsilon$ for $n \geq N$. Then for every x , $|f_n(x) - f(x)| \leq M_n < \varepsilon$, which is uniform convergence. ($\Rightarrow$) If $f_n \rightarrow f$ uniformly, let $\varepsilon > 0$ and pick N with $|f_n(x) - f(x)| < \varepsilon/2$ for all x and all $n \geq N$. Taking the supremum over x gives $M_n \leq \varepsilon/2 < \varepsilon$ for $n \geq N$, so $M_n \rightarrow 0$. $\square$

Method (Deciding pointwise vs. uniform convergence). *Recognition trigger.* You are handed an explicit sequence $f_n(x)$ on a set E and asked whether it converges, and if so whether *uniformly*.

1. **Find the pointwise limit.** Freeze x and compute $f(x) = \lim_n f_n(x)$ by ordinary one-variable limit techniques. (Watch for special points like $x = 0$ or endpoints where the behavior changes.)
2. **Form the gap** $g_n(x) = f_n(x) - f(x)$ and **maximize** $|g_n|$ **over** E — by calculus (set $g'_n(x) = 0$), by a clever bound, or by spotting a “worst” point x_n that drifts with n .
3. **Read off** $M_n = \|g_n\|_E$. If $M_n \rightarrow 0$, convergence is uniform; if $M_n \not\rightarrow 0$ (e.g. $M_n \equiv \frac{1}{2}$ or $M_n \rightarrow \infty$), it is *not*.
4. **Shortcut to disprove uniformity:** exhibit a single sequence $x_n \in E$ with $|f_n(x_n) - f(x_n)| \not\rightarrow 0$. One bad moving point kills uniform convergence.

The three motivating failures

If convergence is only pointwise, the limit may forget continuity, the integral, and the derivative. These are the examples that organize the entire chapter (Rudin 7.4–7.6; B&S 8.2.1).

Worked Example 8.1.5 (x^n on $[0, 1]$: continuity is lost). Let $f_n(x) = x^n$ on $E = [0, 1]$.

Recognition. Explicit sequence; run the Method.

Solution. For $0 \leq x < 1$, $x^n \rightarrow 0$; for $x = 1$, $1^n \rightarrow 1$. So the pointwise limit is the discontinuous step

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

Each f_n is continuous, the limit is not. The gap is $g_n(x) = x^n$ on $[0, 1)$, with $\sup_{[0,1)} x^n = 1$ for every n (take $x \rightarrow 1^-$). Thus $M_n = 1 \not\rightarrow 0$: the convergence is *not* uniform. *Check.* At the moving point $x_n = 2^{-1/n}$ we get $f_n(x_n) = \frac{1}{2}$, frozen away from $f(x_n) = 0$. $\blacksquare$

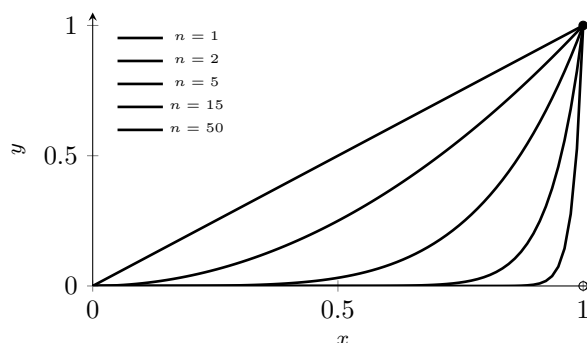


Figure 8.2. $f_n(x) = x^n$ on $[0, 1]$. The graphs press toward 0 on $[0, 1)$ but stay pinned at 1 at $x = 1$; the limit jumps. The “corner” migrates right toward $x = 1$ but never flattens, so $M_n = 1$ for all n .

Worked Example 8.1.6 (A moving bump: the integral is lost). Let $f_n(x) = n^2x$ on $[0, \frac{1}{n}]$, $f_n(x) = n^2(\frac{2}{n} - x)$ on $[\frac{1}{n}, \frac{2}{n}]$, and $f_n(x) = 0$ on $[\frac{2}{n}, 1]$ (a triangular spike of height n over $[0, 2/n]$; B&S 8.2.1(c)).

Recognition. Sequence with a parameter-driven spike; compare $\lim \int$ with $\int \lim$.

Solution. For each fixed $x \in [0, 1]$: once $n > 2/x$ the spike has slid left of x , so $f_n(x) = 0$; hence $f_n(x) \rightarrow 0$ pointwise and the limit is $f \equiv 0$. But each spike is a triangle of base $2/n$ and height n , so $\int_0^1 f_n = \frac{1}{2} \cdot \frac{2}{n} \cdot n = 1$ for all n . Therefore

$$\lim_{n \rightarrow \infty} \int_0^1 f_n = 1 \neq 0 = \int_0^1 \left(\lim_n f_n \right).$$

Convergence fails to be uniform: $M_n = \sup |f_n| = n \rightarrow \infty$. *Check.* The area under each spike is fixed at 1 no matter how thin — the mass escapes “upward” as the base shrinks. ■

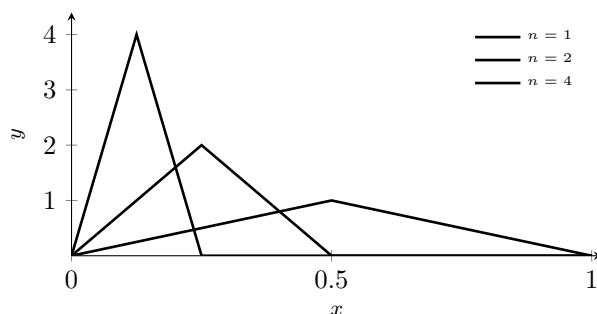


Figure 8.3. Moving bump f_n . Each triangle has area 1, yet $f_n(x) \rightarrow 0$ at every fixed x : the spike grows taller and thinner and slides toward 0. Pointwise limit 0, $\int_0^1 f_n \equiv 1$. Mass is not lost — it is hidden in an ever-narrower spike.

Worked Example 8.1.7 (The derivative is lost). Let $f_n(x) = \frac{\sin(nx)}{\sqrt{n}}$ on $\mathbb{R}$ (Rudin 7.5).

Recognition. Sequence shrinking in amplitude but oscillating faster; test $\lim f'_n$ vs $(\lim f_n)'$.

Solution. Since $|f_n(x)| \leq 1/\sqrt{n}$, we have $M_n = 1/\sqrt{n} \rightarrow 0$, so $f_n \rightarrow f \equiv 0$ uniformly; thus $(\lim f_n)' = 0$. But $f'_n(x) = \sqrt{n} \cos(nx)$, and at $x = 0$, $f'_n(0) = \sqrt{n} \rightarrow \infty$. So

$$\left(\lim_n f_n \right)'(0) = 0 \neq \lim_n f'_n(0) = +\infty.$$

Uniform convergence of f_n does *not* force convergence of f'_n . *Check.* Amplitude $1/\sqrt{n} \rightarrow 0$ but slope at the origin $\sqrt{n} \rightarrow \infty$ — shrinking height, exploding steepness. ■

Caution. “ $f_n \rightarrow f$ pointwise” lets you compute f value by value, but it tells you nothing about $\lim f_n$ being continuous, about $\lim \int f_n = \int \lim f_n$, or about $\lim f'_n$. Each of these interchanges of limit operations needs its own extra hypothesis (uniform convergence of f_n , or of f'_n). Never silently swap $\lim$ with $\int$, $\frac{d}{dx}$, or another $\lim$.

The Cauchy criterion for uniform convergence

A great virtue of uniform convergence is that, like Cauchy’s criterion for numbers, it lets you certify convergence without naming the limit.

Theorem 8.1.8 (Cauchy criterion, B&S 8.1.10; Rudin 7.8). *A sequence (f_n) of functions on E converges uniformly on E if and only if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that*

$$m, n \geq N \implies |f_m(x) - f_n(x)| < \varepsilon \quad \text{for all } x \in E,$$

equivalently $\|f_m - f_n\|_E < \varepsilon$ for $m, n \geq N$.

Proof. ($\Rightarrow$) If $f_n \rightarrow f$ uniformly, given ε choose N with $\|f_n - f\|_E < \varepsilon/2$ for $n \geq N$; then for $m, n \geq N$, $|f_m(x) - f_n(x)| \leq |f_m(x) - f(x)| + |f(x) - f_n(x)| < \varepsilon$ for all x . ($\Leftarrow$) Suppose the Cauchy condition holds. For each fixed x , $(f_n(x))$ is a Cauchy sequence of numbers, hence converges; define $f(x) = \lim_n f_n(x)$. Fix $\varepsilon > 0$ and N as in the hypothesis. For $m, n \geq N$ and all x , $|f_m(x) - f_n(x)| < \varepsilon$; let $m \rightarrow \infty$ to get $|f(x) - f_n(x)| \leq \varepsilon$ for all x and all $n \geq N$. Hence $f_n \rightarrow f$ uniformly. $\square$

Remark. The same statement for series reads: $\sum f_k$ converges uniformly on E iff for every $\varepsilon > 0$ there is N with $|\sum_{k=n+1}^m f_k(x)| < \varepsilon$ for all $m > n \geq N$ and all $x \in E$. This tail criterion is what powers the Weierstrass M -test of §8.4.

Worked Example 8.1.9 (A uniformly convergent geometric-type series). Show $\sum_{n \geq 0} x^n$ converges uniformly on $[-a, a]$ for any fixed $0 < a < 1$, but not on $(-1, 1)$.

Recognition. Series; use the tail/Cauchy criterion and the closed-form remainder.

Solution. On $[-a, a]$ the partial-sum remainder is $|\sum_{k=n+1}^{\infty} x^k| = |\frac{x^{n+1}}{1-x}| \leq \frac{a^{n+1}}{1-a} \rightarrow 0$ uniformly, so the tail is $< \varepsilon$ for all $|x| \leq a$ once n is large: uniform on $[-a, a]$. On $(-1, 1)$ the remainder at $x \rightarrow 1^-$ is unbounded ($\frac{x^{n+1}}{1-x} \rightarrow \infty$), so for each n the tail exceeds any ε somewhere: *not* uniform. *Check.* Matches M_n : on $[-a, a]$, $\sup |\frac{x^{n+1}}{1-x}| = \frac{a^{n+1}}{1-a} \rightarrow 0$. $\blacksquare$

Further worked examples

Worked Example 8.1.10 ($\frac{nx}{1+n^2x^2}$: uniform on $[\delta, 1]$, not on $[0, 1]$). Let $f_n(x) = \frac{nx}{1+n^2x^2}$. Decide uniform convergence on $[0, 1]$ and on $[\delta, 1]$ for fixed $0 < \delta < 1$.

Recognition. Explicit sequence with a moving peak; run the Method and chase the worst point x_n .

Solution. For each fixed x : if $x = 0$ then $f_n(0) = 0$; if $x > 0$ then $f_n(x) \sim \frac{nx}{n^2x^2} = \frac{1}{nx} \rightarrow 0$. So $f_n \rightarrow f \equiv 0$ pointwise on $[0, 1]$. Maximize the gap $|f_n| = f_n$: $f'_n(x) = \frac{n(1-n^2x^2)}{(1+n^2x^2)^2} = 0$ at $x_n = \frac{1}{n} \in [0, 1]$, where $f_n(\frac{1}{n}) = \frac{n \cdot (1/n)}{1+1} = \frac{1}{2}$. Hence $M_n = \sup_{[0,1]} |f_n| = \frac{1}{2} \not\rightarrow 0$: *not* uniform on $[0, 1]$. On $[\delta, 1]$ the peak $x_n = \frac{1}{n}$ eventually leaves the interval (once $n > 1/\delta$), and there f_n is decreasing in x , so $\sup_{[\delta,1]} f_n = f_n(\delta) = \frac{n\delta}{1+n^2\delta^2} \leq \frac{1}{n\delta} \rightarrow 0$: uniform on $[\delta, 1]$. *Check.* The obstruction is a single moving spike of fixed height $\frac{1}{2}$ sitting at $x = 1/n$; excluding a neighborhood of 0 ejects the spike and restores uniformity. $\blacksquare$

Worked Example 8.1.11 ($\frac{x^2}{(1+x^2)^n}$ summed: a discontinuous sum on $\mathbb{R}$). Examine $s(x) = \sum_{n=0}^{\infty} \frac{x^2}{(1+x^2)^n}$ on $\mathbb{R}$, and whether the convergence is uniform.

Recognition. Series; sum it in closed form, then compare $\lim s_N$ at $x = 0$ with $x \neq 0$.

Solution. For $x \neq 0$ the ratio $r = \frac{1}{1+x^2} \in (0, 1)$, so the geometric series sums to $s(x) = x^2 \cdot \frac{1}{1-1/(1+x^2)} = x^2 \cdot \frac{1+x^2}{x^2} = 1+x^2$. At $x = 0$ every term is 0, so $s(0) = 0$. Thus

$$s(x) = \begin{cases} 1+x^2, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

which is *discontinuous* at 0. Each partial sum s_N is continuous (a finite sum of continuous terms), so by the contrapositive of Theorem 8.2.1 the convergence *cannot* be uniform on any set containing 0. *Check.* The partial-sum remainder at small x is $\frac{x^2}{(1+x^2)^{N+1}} \cdot \frac{1+x^2}{x^2} = (1+x^2)^{-N}$ times $1+x^2$; near $x = 0$ this stays close to 1 for every fixed N , so $\|s - s_N\| \geq 1 \not\rightarrow 0$. $\blacksquare$

Exercises (Rudin, Ch.7).

1. Prove that every uniformly convergent sequence of bounded functions is uniformly bounded.
2. If (f_n) and (g_n) converge uniformly on a set E , prove that $(f_n + g_n)$ converges uniformly on E . If, in addition, (f_n) and (g_n) are sequences of bounded functions, prove that $(f_n g_n)$ converges uniformly on E .

3. Construct sequences (f_n) , (g_n) which converge uniformly on some set E , but such that $(f_n g_n)$ does not converge uniformly on E (of course $(f_n g_n)$ must converge on E).

4. Consider $f(x) = \sum_{n=1}^{\infty} \frac{1}{1+n^2 x}$. For what values of x does the series converge absolutely? On what intervals does it converge uniformly? On what intervals does it fail to converge uniformly? Is f continuous wherever the series converges? Is f bounded?

5. Let

$$f_n(x) = \begin{cases} 0, & x < \frac{1}{n+1}, \\ \sin^2\left(\frac{\pi}{x}\right), & \frac{1}{n+1} \leq x \leq \frac{1}{n}, \\ 0, & \frac{1}{n} < x. \end{cases}$$

Show that (f_n) converges to a continuous function, but not uniformly. Use the series $\sum f_n$ to show that absolute convergence, even for all x , does not imply uniform convergence.

6. Prove that the series $\sum_{n=1}^{\infty} (-1)^n \frac{x^2 + n}{n^2}$ converges uniformly in every bounded interval, but does not converge absolutely for any value of x .

7. For $n = 1, 2, 3, \dots$ and x real, put $f_n(x) = \frac{x}{1 + nx^2}$. Show that (f_n) converges uniformly to a function f , and that the equation $f'(x) = \lim_n f'_n(x)$ is correct if $x \neq 0$, but false if $x = 0$.

8. If

$$I(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0, \end{cases}$$

if (x_n) is a sequence of distinct points of (a, b) , and if $\sum |c_n|$ converges, prove that the series $f(x) = \sum_{n=1}^{\infty} c_n I(x - x_n)$ converges uniformly, and that f is continuous for every $x \neq x_n$.

9. Let (f_n) be a sequence of continuous functions which converges uniformly to a function f on a set E . Prove that $\lim_n f_n(x_n) = f(x)$ for every sequence of points $x_n \in E$ with $x_n \rightarrow x$ and $x \in E$. Is the converse true?

10. Letting (x) denote the fractional part of x , consider $f(x) = \sum_{n=1}^{\infty} \frac{(nx)}{n^2}$ (x real). Find all discontinuities of f and show they form a countable dense set. Show that f is nevertheless Riemann-integrable on every bounded interval.

11. Suppose (f_n) , (g_n) are defined on E and (a) $\sum f_n$ has uniformly bounded partial sums; (b) $g_n \rightarrow 0$ uniformly on E ; (c) $g_1(x) \geq g_2(x) \geq g_3(x) \geq \dots$ for every $x \in E$. Prove that $\sum f_n g_n$ converges uniformly on E . (Hint: compare with the Dirichlet test, Rudin 3.42.)

12. Suppose g and f_n ($n = 1, 2, \dots$) are defined on $(0, \infty)$, are Riemann integrable on $[t, T]$ whenever $0 < t < T < \infty$, $|f_n| \leq g$, $f_n \rightarrow f$ uniformly on every compact subset of $(0, \infty)$, and $\int_0^{\infty} g(x) dx < \infty$. Prove that $\lim_{n \rightarrow \infty} \int_0^{\infty} f_n(x) dx = \int_0^{\infty} f(x) dx$. (This is a weak form of the dominated convergence theorem.)

13. Assume (f_n) is a sequence of monotonically increasing functions on $\mathbb{R}$ with $0 \leq f_n(x) \leq 1$ for all x, n . (a) Prove there is a function f and a subsequence (f_{n_k}) with $f(x) = \lim_k f_{n_k}(x)$ for every $x \in \mathbb{R}$ (Helly's selection theorem). (b) If, moreover, f is continuous, prove $f_{n_k} \rightarrow f$ uniformly on $\mathbb{R}$.

14. Let f be a continuous real function on $\mathbb{R}$ with $0 \leq f(t) \leq 1$, $f(t+2) = f(t)$ for every t , and $f(t) = 0$ for $0 \leq t \leq \frac{1}{3}$, $f(t) = 1$ for $\frac{2}{3} \leq t \leq 1$. Put $\Phi(t) = (x(t), y(t))$ where $x(t) = \sum_{n=1}^{\infty} 2^{-n} f(3^{2n-1}t)$ and $y(t) = \sum_{n=1}^{\infty} 2^{-n} f(3^{2n}t)$. Prove that Φ is continuous and maps $[0, 1]$ onto the unit square; in fact Φ maps the Cantor set onto the square (a space-filling curve).

15. Suppose f is a real continuous function on $\mathbb{R}$, $f_n(t) = f(nt)$ for $n = 1, 2, \dots$, and (f_n) is equicontinuous on $[0, 1]$. What conclusion can you draw about f ?
16. Suppose (f_n) is an equicontinuous sequence of functions on a compact set K and (f_n) converges pointwise on K . Prove that (f_n) converges uniformly on K .
17. Define uniform convergence and equicontinuity for mappings into any metric space. Show that Theorems 7.9 and 7.12 are valid for mappings into any metric space, that Theorems 7.8 and 7.11 are valid for mappings into any complete metric space, and that Theorems 7.10, 7.16, 7.17, 7.24, 7.25 hold for vector-valued functions (mappings into any $\mathbb{R}^k$).
18. Let (f_n) be a uniformly bounded sequence of Riemann-integrable functions on $[a, b]$, and put $F_n(x) = \int_a^x f_n(t) dt$ ($a \leq x \leq b$). Prove that there exists a subsequence (F_{n_k}) which converges uniformly on $[a, b]$.
19. Let K be a compact metric space, S a subset of $\mathcal{C}(K)$. Prove that S is compact (in the sup-metric) if and only if S is uniformly closed, pointwise bounded, and equicontinuous.
20. If f is continuous on $[0, 1]$ and $\int_0^1 f(x)x^n dx = 0$ for $n = 0, 1, 2, \dots$, prove that $f(x) = 0$ on $[0, 1]$. (Hint: the integral of the product of f with any polynomial is zero; use the Weierstrass theorem to show $\int_0^1 f^2(x) dx = 0$.)
21. Let K be the unit circle in the complex plane and let $\mathcal{A}$ be the algebra of all functions of the form $f(e^{i\theta}) = \sum_{n=0}^N c_n e^{in\theta}$ (θ real). Then $\mathcal{A}$ separates points on K and vanishes at no point of K , yet there are continuous functions on K not in the uniform closure of $\mathcal{A}$. (Hint: for every $f \in \mathcal{A}$, $\int_0^{2\pi} f(e^{i\theta}) e^{i\theta} d\theta = 0$, and this persists in the closure.)
22. Assume $f \in \mathcal{R}(\alpha)$ on $[a, b]$, and prove that there are polynomials P_n with $\lim_{n \rightarrow \infty} \int_a^b |f - P_n|^2 d\alpha = 0$. (Compare Exercise 12 of Chapter 6.)
23. Put $P_0 = 0$ and define, for $n = 0, 1, 2, \dots$, $P_{n+1}(x) = P_n(x) + \frac{1}{2}(x^2 - P_n(x)^2)$. Prove that $\lim_n P_n(x) = |x|$ uniformly on $[-1, 1]$. (This permits a proof of Stone–Weierstrass without first proving Theorem 7.26. Hint: use the identity $|x| - P_{n+1}(x) = (|x| - P_n(x))(1 - \frac{1}{2}(|x| + P_n(x)))$ to show $0 \leq P_n(x) \leq P_{n+1}(x) \leq |x|$ for $|x| \leq 1$.)
24. Let X be a metric space with metric d , fix $a \in X$, and to each $p \in X$ assign $f_p(x) = d(x, p) - d(x, a)$ ($x \in X$). Prove $|f_p(x)| \leq d(a, p)$ for all x , so $f_p \in \mathcal{C}(X)$; prove $\|f_p - f_q\| = d(p, q)$ for all $p, q \in X$; deduce that $\Phi(p) = f_p$ is an isometry of X onto $\Phi(X) \subseteq \mathcal{C}(X)$. Let Y be the closure of $\Phi(X)$ in $\mathcal{C}(X)$; show Y is complete. Conclusion: X is isometric to a dense subset of a complete metric space Y .
25. Suppose ϕ is a continuous bounded real function in the strip $0 \leq x \leq 1$, $-\infty < y < \infty$. Prove that the initial-value problem $y' = \phi(x, y)$, $y(0) = c$ has a solution. (Hint: fix n ; let $x_i = i/n$; build a piecewise-defined approximate solution f_n ; show (f_n) is equicontinuous; extract a uniformly convergent subsequence by Arzela–Ascoli; pass to the limit.)
26. Prove an analogous existence theorem for the initial-value problem $y' = \Phi(x, y)$, $y(0) = c$, where now $c \in \mathbb{R}^k$, $y \in \mathbb{R}^k$, and Φ is a continuous bounded mapping of the part of $\mathbb{R}^{k+1}$ with $0 \leq x \leq 1$ into $\mathbb{R}^k$. (Use the vector-valued version of Arzela–Ascoli, Theorem 7.25.)

Exercises (Bartle & Sherbert, Section 8.1).

1. Show that $\lim(x/(x+n)) = 0$ for all $x \in \mathbb{R}$, $x \geq 0$.
2. Show that $\lim(nx/(1+n^2x^2)) = 0$ for all $x \in \mathbb{R}$.
3. Evaluate $\lim(nx/(1+nx))$ for $x \in \mathbb{R}$, $x \geq 0$.
4. Evaluate $\lim(x^n/(1+x^n))$ for $x \in \mathbb{R}$, $x \geq 0$.
5. Evaluate $\lim((\sin nx)/(1+nx))$ for $x \in \mathbb{R}$, $x \geq 0$.
6. Show that $\lim(\arctan nx) = \frac{\pi}{2} \operatorname{sgn} x$ for $x \in \mathbb{R}$.

7. Evaluate $\lim(e^{-nx})$ for $x \in \mathbb{R}, x \geq 0$.
8. Show that $\lim(xe^{-nx}) = 0$ for $x \in \mathbb{R}, x \geq 0$.
9. Show that $\lim(x^2e^{-nx}) = 0$ and that $\lim(n^2x^2e^{-nx}) = 0$ for $x \in \mathbb{R}, x \geq 0$.
10. Show that $\lim((\cos \pi x)^{2n})$ exists for all $x \in \mathbb{R}$. What is its limit?
11. Show that if $a > 0$, then the convergence of the sequence in Exercise 1 is uniform on $[0, a]$, but is not uniform on $[0, \infty)$.
12. Show that if $a > 0$, then the convergence of the sequence in Exercise 2 is uniform on $[a, \infty)$, but is not uniform on $[0, \infty)$.
13. Show that if $a > 0$, then the convergence of the sequence in Exercise 3 is uniform on $[a, \infty)$, but is not uniform on $[0, \infty)$.
14. Show that if $0 < b < 1$, then the convergence of the sequence in Exercise 4 is uniform on $[0, b]$, but is not uniform on $[0, 1]$.
15. Show that if $a > 0$, then the convergence of the sequence in Exercise 5 is uniform on $[a, \infty)$, but is not uniform on $[0, \infty)$.
16. Show that if $a > 0$, then the convergence of the sequence in Exercise 6 is uniform on $[a, \infty)$, but is not uniform on $(0, \infty)$.
17. Show that if $a > 0$, then the convergence of the sequence in Exercise 7 is uniform on $[a, \infty)$, but is not uniform on $[0, \infty)$.
18. Show that the convergence of the sequence in Exercise 8 is uniform on $[0, \infty)$.
19. Show that the sequence (x^2e^{-nx}) converges uniformly on $[0, \infty)$.
20. Show that if $a > 0$, then the sequence $(n^2x^2e^{-nx})$ converges uniformly on $[a, \infty)$, but does not converge uniformly on $[0, \infty)$.
21. Show that if $(f_n), (g_n)$ converge uniformly on the set A to f, g respectively, then $(f_n + g_n)$ converges uniformly on A to $f + g$.
22. Show that if $f_n(x) := x + 1/n$ and $f(x) := x$ for $x \in \mathbb{R}$, then (f_n) converges uniformly on $\mathbb{R}$ to f , but the sequence (f_n^2) does not converge uniformly on $\mathbb{R}$. (Thus the product of uniformly convergent sequences need not converge uniformly.)
23. Let $(f_n), (g_n)$ be sequences of bounded functions on A converging uniformly on A to f, g respectively. Show that (f_ng_n) converges uniformly on A to fg .
24. Let (f_n) converge uniformly to f on A with $|f_n(x)| \leq M$ for all $n \in \mathbb{N}$ and $x \in A$. If g is continuous on $[-M, M]$, show that $(g \circ f_n)$ converges uniformly to $g \circ f$ on A .

Additional exercises.

1. For $f_n(x) = \frac{nx}{1+n^2x^2}$ on $[0, 1]$: find the pointwise limit, compute $M_n = \|f_n - f\|_{[0,1]}$ by locating the maximum, and decide whether convergence is uniform.
2. Show $f_n(x) = x^n - x^{2n}$ converges pointwise to 0 on $[0, 1]$ but not uniformly. (Hint: evaluate at $x_n = 2^{-1/n}$.)
3. (Sup-norm drill.) Compute $\|f_n\|_{[0,1]}$ for $f_n(x) = nx(1-x)^n$ and use it to decide uniform convergence of (f_n) to its pointwise limit.
4. Prove directly from the definition (no M_n theorem) that $f_n(x) = \frac{x}{n}$ converges uniformly to 0 on any bounded interval $[-a, a]$, but *not* uniformly on $\mathbb{R}$.
5. (Non-example / disprove.) Decide whether the claim “if $f_n \rightarrow f$ and $g_n \rightarrow g$ pointwise, then $f_ng_n \rightarrow fg$ uniformly whenever each individual convergence is uniform” is true. Either prove it or give a counterexample on a suitable set.

6. Use the Cauchy criterion to show $\sum_{n \geq 1} \frac{x^n}{n}$ converges uniformly on $[-1, 0]$ but not on $[0, 1]$.
7. (Diagram.) Sketch $f_n(x) = x^n$ for $n = 1, 2, 5, 20$ on $[0, 1]$, mark the “worst point” x_n where the gap to the limit is $\frac{1}{2}$, and explain from the picture why $M_n = 1$ for every n .
8. Let $f_n(x) = \frac{x^2}{x^2 + (1 - nx)^2}$ on $[0, 1]$. Find the pointwise limit and show the convergence is not uniform by exhibiting a moving point where $f_n = 1$.
9. Suppose $f_n \rightarrow f$ uniformly on E and each f_n is bounded by B (independent of n). Prove f is bounded by B and that $\|f_n\|_E \rightarrow \|f\|_E$.

8.2 Uniform Convergence and Continuity

The uniform limit of continuous functions is continuous

The first motivating failure was that a pointwise limit of continuous functions need not be continuous (x^n). Uniform convergence repairs exactly this.

Theorem 8.2.1 (Continuity of the uniform limit; B&S 8.2.2, Rudin 7.12). *Let (f_n) be continuous on a set E (in $\mathbb{R}$, or in a metric space) and suppose $f_n \rightarrow f$ uniformly on E . Then f is continuous on E .*

Proof. The proof is the celebrated $\frac{\varepsilon}{3}$ -argument. Fix $c \in E$ and $\varepsilon > 0$. By uniform convergence choose N with $\|f_N - f\|_E < \varepsilon/3$, so that $|f_N(x) - f(x)| < \varepsilon/3$ for every $x \in E$. Since f_N is continuous at c , there is $\delta > 0$ with $|f_N(x) - f_N(c)| < \varepsilon/3$ whenever $x \in E$, $|x - c| < \delta$. For such x ,

$$|f(x) - f(c)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Hence f is continuous at c ; as c was arbitrary, f is continuous on E . □

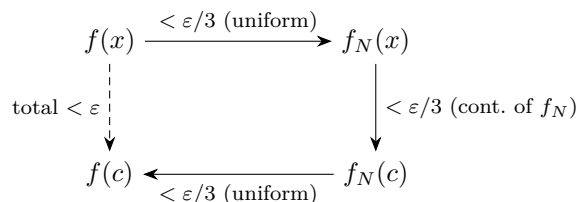


Figure 8.4. The $\varepsilon/3$ route from $f(x)$ to $f(c)$: hop to f_N (uniform closeness), slide along f_N (its own continuity), hop back (uniform closeness). The dashed shortcut is the bound we want.

Remark. The converse fails: a pointwise (non-uniform) limit *can* happen to be continuous. For instance the moving bump of §1 has continuous limit 0 though convergence is not uniform. So Theorem 8.2.1 is a sufficient, not necessary, condition — the partial converse (Dini) below needs monotonicity. Note the contrapositive is a powerful disproof tool: *if the pointwise limit is discontinuous, convergence cannot be uniform* (this is the fastest way to see x^n on $[0, 1]$ is not uniform).

The interchange-of-limits theorem

Continuity of the limit is one face of a general principle: under uniform convergence two limit processes commute.

Theorem 8.2.2 (Interchange of limits; Rudin 7.11). *Let $f_n \rightarrow f$ uniformly on E , let x_0 be a limit point of E , and suppose each limit $A_n = \lim_{x \rightarrow x_0} f_n(x)$ exists. Then (A_n) converges, $\lim_{x \rightarrow x_0} f(x)$ exists, and the two are equal:*

$$\lim_{x \rightarrow x_0} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \lim_{x \rightarrow x_0} f_n(x).$$

Proof. By the Cauchy criterion (Thm. 8.1.8) and $|A_m - A_n| \leq \|f_m - f_n\|_E$ (let $x \rightarrow x_0$ inside $|f_m(x) - f_n(x)| \leq \|f_m - f_n\|_E$), (A_n) is Cauchy, so $A_n \rightarrow A$ for some A . Fix $\varepsilon > 0$. Choose n so large that $\|f_n - f\|_E < \varepsilon/3$ and $|A_n - A| < \varepsilon/3$. With n fixed, choose a neighborhood of x_0 on which $|f_n(x) - A_n| < \varepsilon/3$. For x in that neighborhood (and in E),

$$|f(x) - A| \leq |f(x) - f_n(x)| + |f_n(x) - A_n| + |A_n - A| < \varepsilon.$$

Hence $\lim_{x \rightarrow x_0} f(x) = A = \lim_n A_n$. $\square$

Theorem 8.2.1 is the special case $A_n = f_n(x_0)$ (continuity of each f_n): the equality becomes $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, i.e. f is continuous at x_0 .

Worked Example 8.2.3 (Limits that refuse to commute). Let $f_n(x) = \frac{1}{1 + (x - n)^2}$ on $\mathbb{R}$... instead consider the cleaner double sequence $a_{m,n} = \frac{m}{m + n}$.

Recognition. “Do these iterated limits agree?” — check whether convergence in one index is uniform in the other.

Solution. $\lim_n \lim_m a_{m,n} = \lim_n 1 = 1$, while $\lim_m \lim_n a_{m,n} = \lim_m 0 = 0$. The two iterated limits differ. The obstruction is exactly that $a_{m,n} \rightarrow 0$ in n is *not* uniform in m : at n fixed and $m = n$, $a_{n,n} = \frac{1}{2} \not\rightarrow 0$. Uniform convergence is precisely what Theorem 8.2.2 needs to forbid this. *Check.* On the diagonal $a_{n,n} = \frac{1}{2}$, wedged between the two limits 0 and 1. ■

The space of bounded functions is complete

The sup-norm turns sets of functions into metric spaces, and uniform convergence becomes ordinary convergence in that metric. The payoff: a completeness theorem.

Definition 8.2.4. For a metric space X let $\mathcal{C}(X)$ denote the set of *bounded continuous* complex (or real) functions on X , with the *supremum metric*

$$d(f, g) = \|f - g\|_X = \sup_{x \in X} |f(x) - g(x)|.$$

Likewise $B(X)$ is the set of all bounded functions on X with the same metric. Convergence in d is exactly uniform convergence (Thm. 8.1.4).

Theorem 8.2.5 (Completeness; Rudin 7.15). $B(X)$ is a complete metric space, and $\mathcal{C}(X)$ is a closed subset of $B(X)$, hence also complete. That is, every uniformly Cauchy sequence of bounded functions converges uniformly to a bounded function, and the uniform limit of bounded continuous functions is bounded and continuous.

Proof. Let (f_n) be Cauchy in $B(X)$, i.e. uniformly Cauchy. By Theorem 8.1.8 it converges uniformly to some f . Boundedness passes to the limit: pick N with $\|f_N - f\|_X \leq 1$; then $\|f\|_X \leq \|f_N\|_X + 1 < \infty$. So $f \in B(X)$ and $d(f_n, f) = \|f_n - f\|_X \rightarrow 0$: $B(X)$ is complete. If moreover each f_n is continuous, then f is continuous by Theorem 8.2.1, so $\mathcal{C}(X)$ contains all its limit points — it is closed in $B(X)$, and a closed subspace of a complete space is complete. $\square$

Remark. This is the structural reason uniform convergence is the “right” notion: it is convergence in a *complete* normed space of functions. Many later constructions (the Arzela–Ascoli theorem, the Stone–Weierstrass theorem, existence proofs for ODEs) live inside $\mathcal{C}(X)$ and exploit its completeness exactly as one exploits completeness of $\mathbb{R}$.

Dini’s theorem: a partial converse

When is pointwise convergence *automatically* uniform? B&S give a clean sufficient condition: monotone convergence to a continuous limit on a compact set.

Theorem 8.2.6 (Dini; B&S 8.2.6). *Let (f_n) be continuous on a compact set K (e.g. $K = [a, b]$), converging pointwise to a continuous f , with the convergence monotone in n at each point (say $f_1(x) \geq f_2(x) \geq \cdots \rightarrow f(x)$). Then $f_n \rightarrow f$ uniformly on K .*

Proof. Put $g_n = f_n - f \geq 0$; the g_n are continuous, decrease to 0 pointwise. Fix $\varepsilon > 0$ and let $K_n = \{x \in K : g_n(x) \geq \varepsilon\}$. Each K_n is closed (continuity) hence compact, and $K_1 \supseteq K_2 \supseteq \cdots$ (monotonicity). Since $g_n(x) \rightarrow 0$, every x eventually leaves the K_n , so $\bigcap_n K_n = \emptyset$. A nested sequence of compact sets with empty intersection must have some $K_N = \emptyset$ (else the finite-intersection property is violated). Then $g_n(x) < \varepsilon$ for all $x \in K$ and all $n \geq N$, i.e. $\|f_n - f\|_K \leq \varepsilon$: uniform convergence. $\square$

Caution. All three hypotheses of Dini are essential. Drop *compactness*: $f_n(x) = x/n$ decreases to 0 on $[0, \infty)$ but $\|f_n\| = \infty$. Drop *continuity of the limit*: $f_n(x) = x^n$ decreases on $[0, 1]$ to the discontinuous step, not uniformly. Drop *monotonicity*: the moving bump of §1 converges pointwise to a continuous limit on the compact $[0, 1]$, but not uniformly. Dini buys uniformity only when all three hold.

Further worked examples

Worked Example 8.2.7 (Dini in action: x^n on $[0, a]$ with $a < 1$). Show $f_n(x) = x^n$ converges uniformly on $[0, a]$ for any fixed $0 < a < 1$, and check that Dini's theorem applies there but not on $[0, 1]$.

Recognition. Monotone-decreasing continuous sequence to a continuous limit on a compact set; the trigger for Dini, plus a direct M_n confirmation.

Solution. On $[0, a]$ the limit is $f \equiv 0$ (since $0 \leq x \leq a < 1$), which is continuous; each f_n is continuous; and for fixed x , $x^{n+1} \leq x^n$, so the sequence decreases monotonically to 0. The set $[0, a]$ is compact, so Dini (Thm. 8.2.6) gives uniform convergence. Directly: $M_n = \sup_{[0, a]} x^n = a^n \rightarrow 0$, confirming it. On $[0, 1]$ the limit is the discontinuous step, so the *continuity-of-limit* hypothesis fails and Dini does not apply — and indeed $M_n = 1 \not\rightarrow 0$ there. *Check.* The only thing that changed from §1's failure is shaving off the point $x = 1$ where the limit jumps; that restores both Dini's hypothesis and uniformity. $\blacksquare$

Worked Example 8.2.8 (The interchange theorem evaluating an iterated limit). Use uniform convergence to compute $\lim_{x \rightarrow 0^+} \lim_{n \rightarrow \infty} \frac{nx}{nx+1}$ and confirm the order of limits may be swapped here.

Recognition. Two nested limits; check uniform convergence in n to license Theorem 8.2.2.

Solution. Inner limit: for fixed $x > 0$, $\frac{nx}{nx+1} \rightarrow 1$; at $x = 0$ it is 0. So the inner limit function is $g(x) = 1$ for $x > 0$ and $g(0) = 0$, and $\lim_{x \rightarrow 0^+} g(x) = 1$. The other order: $\lim_{x \rightarrow 0^+} \frac{nx}{nx+1} = 0$ for each fixed n , so $\lim_n 0 = 0$. The two iterated limits are 1 and 0 — they *disagree*, and indeed the convergence in n is not uniform near 0 ($M_n = \sup_{x>0} |1 - \frac{nx}{nx+1}| = \sup \frac{1}{nx+1} = 1$ as $x \rightarrow 0^+$). Now restrict to $x \geq \delta > 0$: there $|1 - \frac{nx}{nx+1}| = \frac{1}{nx+1} \leq \frac{1}{n\delta+1} \rightarrow 0$ uniformly, so $A_n = \lim_{x \rightarrow \delta^+} f_n = \frac{n\delta}{n\delta+1}$ and the interchange theorem gives $\lim_{x \rightarrow \delta^+} \lim_n f_n = \lim_n A_n = 1$, a legitimate swap. *Check.* The swap is valid exactly where convergence is uniform (x bounded away from 0) and fails exactly where it is not ($x \rightarrow 0^+$), precisely as Theorem 8.2.2 predicts. $\blacksquare$

Worked Example 8.2.9 (A uniform limit that is continuous but not differentiable). Let $f_n(x) = \sqrt{x^2 + \frac{1}{n}}$ on $[-1, 1]$. Find the uniform limit and observe what continuity does (and does not) survive.

Recognition. Explicit sequence converging uniformly; apply Thm. 8.2.1 and note its limits.

Solution. Pointwise, $f_n(x) \rightarrow \sqrt{x^2} = |x| =: f(x)$. The gap is $0 \leq f_n(x) - |x| = \sqrt{x^2 + \frac{1}{n}} - \sqrt{x^2} = \frac{1/n}{\sqrt{x^2 + 1/n} + \sqrt{x^2}} \leq \frac{1/n}{\sqrt{1/n}} = \frac{1}{\sqrt{n}}$, so $M_n \leq n^{-1/2} \rightarrow 0$: uniform on $[-1, 1]$. Each f_n is C^∞ (the radicand is $\geq \frac{1}{n} > 0$), and the uniform limit $|x|$ is continuous, as Theorem 8.2.1 guarantees. *But $|x|$ is not differentiable at 0*: uniform convergence preserves continuity, never differentiability. *Check.* $f'_n(x) = \frac{x}{\sqrt{x^2 + 1/n}}$, and $f'_n(0) = 0$ for every n , whereas $f'(0^\pm) = \pm 1$ — the derivatives do not converge to a derivative of the limit, exactly the warning of §1's $\sin(nx)/\sqrt{n}$. $\blacksquare$

Worked Example 8.2.10 (Completeness used: a uniform Cauchy series builds a continuous function). Let $g_k(x) = \frac{(-1)^k}{2^k} \cos(kx)$ and $s_N = \sum_{k=1}^N g_k$. Show (s_N) is Cauchy in $\mathcal{C}(\mathbb{R})$ and conclude its limit is bounded and continuous without first naming it.

Recognition. A sequence Cauchy in the sup-metric; invoke completeness (Thm. 8.2.5) rather than computing the limit.

Solution. For $M > N$, $\|s_M - s_N\|_{\mathbb{R}} = \sup_x |\sum_{k=N+1}^M \frac{(-1)^k}{2^k} \cos(kx)| \leq \sum_{k=N+1}^M 2^{-k} \leq 2^{-N}$, which is $< \varepsilon$ once $N > \log_2(1/\varepsilon)$. So (s_N) is uniformly Cauchy, i.e. Cauchy in $\mathcal{C}(\mathbb{R})$. By Theorem 8.2.5, $\mathcal{C}(\mathbb{R})$ (bounded continuous functions with sup-metric) is complete, so $s_N \rightarrow s$ uniformly for some bounded continuous s — and we never had to evaluate the sum. *Check.* The tail bound $\|s - s_N\| \leq 2^{-N} \rightarrow 0$ is the M -test in disguise ($M_k = 2^{-k}$, $\sum 2^{-k} = 1 < \infty$); completeness is what converts “uniformly Cauchy” into “has a continuous limit”. ■

Exercises (Bartle & Sherbert, Section 8.2).

1. Show that the sequence $(x^n/(1+x^n))$ does not converge uniformly on $[0, 2]$ by showing that the limit function is not continuous on $[0, 2]$.
2. Prove that the moving-bump sequence of §1 (B&S Example 8.2.1(c)) is a sequence of continuous functions that converges nonuniformly to a continuous limit.
3. Construct a sequence of functions on $[0, 1]$ each of which is discontinuous at every point of $[0, 1]$ and which converges uniformly to a function that is continuous at every point.
4. Suppose (f_n) is a sequence of continuous functions on an interval I converging uniformly on I to f . If $(x_n) \subseteq I$ converges to $x_0 \in I$, show that $\lim(f_n(x_n)) = f(x_0)$.
5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be uniformly continuous on $\mathbb{R}$ and let $f_n(x) := f(x + 1/n)$ for $x \in \mathbb{R}$. Show that (f_n) converges uniformly on $\mathbb{R}$ to f .
6. Let $f_n(x) := 1/(1+x)^n$ for $x \in [0, 1]$. Find the pointwise limit f of (f_n) on $[0, 1]$. Does (f_n) converge uniformly to f on $[0, 1]$?
7. Suppose (f_n) converges uniformly to f on the set A and each f_n is bounded on A (for each n there is M_n with $|f_n(x)| \leq M_n$ for all $x \in A$). Show that f is bounded on A .
8. Let $f_n(x) := nx/(1+nx^2)$ for $x \in A := [0, \infty)$. Show that each f_n is bounded on A , but the pointwise limit f is not bounded on A . Does (f_n) converge uniformly to f on A ?
9. Let $f_n(x) := x^n/n$ for $x \in [0, 1]$. Show that (f_n) of differentiable functions converges uniformly to a differentiable function f on $[0, 1]$, and that (f'_n) converges on $[0, 1]$ to a function g , but that $g(1) \neq f'(1)$.
10. Let $g_n(x) := e^{-nx}/n$ for $x \geq 0$, $n \in \mathbb{N}$. Examine the relation between $\lim(g_n)$ and $\lim(g'_n)$.
11. Let $I := [a, b]$ and let (f_n) converge on I to f . Suppose each derivative f'_n is continuous on I and (f'_n) is uniformly convergent to g on I . Prove that $f(x) - f(a) = \int_a^x g(t) dt$ and that $f'(x) = g(x)$ for all $x \in I$.
12. Show that $\lim \int_1^2 e^{-nx^2} dx = 0$.
13. If $a > 0$, show that $\lim \int_a^\pi \frac{\sin nx}{nx} dx = 0$. What happens if $a = 0$?
14. Let $f_n(x) := nx/(1+nx)$ for $x \in [0, 1]$. Show that (f_n) converges nonuniformly to an integrable function f and that $\int_0^1 f(x) dx = \lim \int_0^1 f_n(x) dx$.
15. Let $g_n(x) := nx(1-x)^n$ for $x \in [0, 1]$, $n \in \mathbb{N}$. Discuss the convergence of (g_n) and of $(\int_0^1 g_n dx)$.
16. Let $\{r_1, r_2, \dots\}$ enumerate the rationals in $I := [0, 1]$, and let $f_n : I \rightarrow \mathbb{R}$ equal 1 if $x \in \{r_1, \dots, r_n\}$ and 0 otherwise. Show that each f_n is Riemann integrable, that $f_1 \leq f_2 \leq \dots$, and that $f(x) := \lim(f_n(x))$ is the Dirichlet function, which is not Riemann integrable on $[0, 1]$.

17. Let $f_n(x) := 1$ for $x \in (0, 1/n)$ and $f_n(x) := 0$ elsewhere in $[0, 1]$. Show that (f_n) is a decreasing sequence of discontinuous functions converging to a continuous limit, but the convergence is not uniform on $[0, 1]$.
18. Let $f_n(x) := x^n$ for $x \in [0, 1]$. Show that (f_n) is a decreasing sequence of continuous functions converging to a function that is not continuous, and the convergence is not uniform on $[0, 1]$.
19. Let $f_n(x) := x/n$ for $x \in [0, \infty)$. Show that (f_n) is a decreasing sequence of continuous functions converging to a continuous limit, but the convergence is not uniform on $[0, \infty)$.
20. Give an example of a decreasing sequence (f_n) of continuous functions on $[0, \infty)$ that converges to a continuous limit, but the convergence is not uniform on $[0, \infty)$.

Additional exercises.

1. Using the $\varepsilon/3$ -argument as a template, prove: if $f_n \rightarrow f$ uniformly on E and each f_n is uniformly continuous on E , then f is uniformly continuous on E .
2. Show $f_n(x) = \frac{x^n}{1+x^n}$ on $[0, 2]$ cannot converge uniformly, directly from the discontinuity of its pointwise limit (use the contrapositive of Theorem 8.2.1).
3. (Interchange of limits.) For $a_{m,n} = \frac{n}{m+n}$ compute both iterated limits and reconcile the disagreement with the failure of uniform convergence.
4. Prove that the uniform limit of a sequence of functions, each bounded by its own constant M_n with $\sup_n M_n < \infty$, is bounded. Where does the argument use uniformity?
5. (Non-example for Dini.) Exhibit a monotone-in- n , pointwise-convergent sequence of continuous functions on the *non-compact* $[0, \infty)$ whose limit is continuous but the convergence is not uniform, showing compactness cannot be dropped from Dini's theorem.
6. Let (f_n) be continuous on $[a, b]$ with $f_n \rightarrow f$ uniformly. If $x_n \rightarrow x_0$ in $[a, b]$, prove $f_n(x_n) \rightarrow f(x_0)$, and show by example that uniformity is needed.
7. (Diagram.) Draw the $\varepsilon/3$ "hopscotch" diagram for the continuity proof and label each of the three inequalities with the hypothesis that supplies it.
8. Prove $\mathcal{C}[a, b]$ with the sup-metric is complete by exhibiting, for a given uniformly Cauchy sequence, its limit and verifying continuity. Identify precisely where completeness of $\mathbb{R}$ enters.
9. Show that the map $T : \mathcal{C}[0, 1] \rightarrow \mathcal{C}[0, 1]$, $(Tf)(x) = \int_0^x f$, is continuous in the sup-metric, i.e. $f_n \rightarrow f$ uniformly $\implies Tf_n \rightarrow Tf$ uniformly.

8.3 Uniform Convergence, Integration, and Differentiation

Uniform convergence and the integral

Recall the moving bump showed $\lim \int f_n \neq \int \lim f_n$ under mere pointwise convergence. Uniform convergence forbids exactly this.

Theorem 8.3.1 (Term-by-term integration; Rudin 7.16, B&S 8.2.4). *Let f_n be Riemann (or Riemann–Stieltjes, w.r.t. a monotone α) integrable on $[a, b]$ and suppose $f_n \rightarrow f$ uniformly on $[a, b]$. Then f is integrable and*

$$\int_a^b f \, d\alpha = \lim_{n \rightarrow \infty} \int_a^b f_n \, d\alpha.$$

Proof. Write $\varepsilon_n = \|f_n - f\|_{[a,b]} \rightarrow 0$. On $[a, b]$, $f_n - \varepsilon_n \leq f \leq f_n + \varepsilon_n$, so the upper and lower integrals of f are squeezed:

$$\int_a^b f_n \, d\alpha - \varepsilon_n[\alpha(b) - \alpha(a)] \leq \int_a^b f \, d\alpha \leq \overline{\int}_a^b f \, d\alpha \leq \int_a^b f_n \, d\alpha + \varepsilon_n[\alpha(b) - \alpha(a)].$$

Hence $0 \leq \int_a^b f - \int_a^b f_n \leq 2\varepsilon_n[\alpha(b) - \alpha(a)] \rightarrow 0$, so the two coincide: f is integrable. The same inequalities give $|\int_a^b f d\alpha - \int_a^b f_n d\alpha| \leq \varepsilon_n[\alpha(b) - \alpha(a)] \rightarrow 0$. $\square$

Corollary 8.3.2 (Term-by-term integration of series). *If $\sum g_k$ converges uniformly to s on $[a, b]$ with each g_k integrable, then s is integrable and $\int_a^b s = \sum_{k=1}^{\infty} \int_a^b g_k$.*

Remark. The hypothesis is sufficient but strays away from optimal. B&S record the *Bounded Convergence Theorem*: if $f_n \rightarrow f$ pointwise with all $f_n, f \in \mathcal{R}[a, b]$ and a single bound $|f_n| \leq B$, then $\int f_n \rightarrow \int f$ even without uniformity. Rudin notes the same phenomenon (Exercise 7.12) and points to the Lebesgue dominated convergence theorem as the definitive statement (Ch. 13).

Worked Example 8.3.3 (Integrating a power series term by term). Show $\int_0^{1/2} \frac{dx}{1-x} = \sum_{k=0}^{\infty} \frac{1}{(k+1)2^{k+1}}$

by term-by-term integration.

Recognition. “Justify swapping $\int$ and $\sum$ ” — check uniform convergence on the interval of integration.

Solution. On $[0, \frac{1}{2}]$, $\frac{1}{1-x} = \sum_{k \geq 0} x^k$ converges uniformly (geometric with ratio $\leq \frac{1}{2}$; §1). By Corollary 8.3.2, $\int_0^{1/2} \frac{dx}{1-x} = \sum_{k \geq 0} \int_0^{1/2} x^k dx = \sum_{k \geq 0} \frac{(1/2)^{k+1}}{k+1}$. *Check.* The left side is $-\ln(1-x)|_0^{1/2} = \ln 2$, and the right side is $\sum_{k \geq 1} \frac{1}{k2^k} = \ln 2$ (the series for $-\ln(1-x)$ at $x = \frac{1}{2}$). ■

Uniform convergence and the derivative

Differentiation is more delicate: the $\sin(nx)/\sqrt{n}$ example showed uniform convergence of f_n does nothing for f'_n . What is needed is uniform convergence of the *derivatives*.

Theorem 8.3.4 (Term-by-term differentiation; Rudin 7.17, B&S 8.2.3). *Let (f_n) be differentiable on $[a, b]$, suppose $(f_n(x_0))$ converges for some point $x_0 \in [a, b]$, and suppose (f'_n) converges uniformly on $[a, b]$. Then (f_n) converges uniformly on $[a, b]$ to a function f , f is differentiable, and*

$$f'(x) = \lim_{n \rightarrow \infty} f'_n(x) \quad (a \leq x \leq b).$$

Proof. First, (f_n) is uniformly Cauchy. By the Mean Value Theorem applied to $f_m - f_n$, for any $x, t \in [a, b]$ there is ξ between them with $(f_m - f_n)(x) - (f_m - f_n)(t) = (x - t)(f'_m - f'_n)(\xi)$, so

$$|(f_m - f_n)(x) - (f_m - f_n)(t)| \leq (b - a) \|f'_m - f'_n\|_{[a, b]}.$$

Taking $t = x_0$ and using $|f_m(x) - f_n(x)| \leq |f_m(x_0) - f_n(x_0)| + (b - a) \|f'_m - f'_n\|$, both terms are small for large m, n (the first because $(f_n(x_0))$ converges, the second by uniform convergence of derivatives). So (f_n) is uniformly Cauchy, hence $f_n \rightarrow f$ uniformly for some f .

Now fix $c \in [a, b]$ and define, for $x \neq c$, $\varphi_n(x) = \frac{f_n(x) - f_n(c)}{x - c}$ and $\varphi(x) = \frac{f(x) - f(c)}{x - c}$. The displayed MVT bound gives $|\varphi_m(x) - \varphi_n(x)| \leq \|f'_m - f'_n\|_{[a, b]}$, so (φ_n) is uniformly Cauchy on the punctured interval and $\varphi_n \rightarrow \varphi$ uniformly there. Each $\lim_{x \rightarrow c} \varphi_n(x) = f'_n(c)$ exists; by the interchange theorem (Thm. 8.2.2),

$$f'(c) = \lim_{x \rightarrow c} \varphi(x) = \lim_{n \rightarrow \infty} \lim_{x \rightarrow c} \varphi_n(x) = \lim_{n \rightarrow \infty} f'_n(c).$$

$\square$

Caution. The hypotheses point the “wrong” way from intuition: you assume uniform convergence of the *derivatives* f'_n (plus convergence at one point), and *conclude* uniform convergence of the f_n and that you may differentiate term by term. Uniform convergence of the f_n themselves is neither assumed nor sufficient ($\sin(nx)/\sqrt{n}$). For power series this hypothesis is automatic inside the radius of convergence — which is why power series may always be differentiated term by term (§8.4).

Worked Example 8.3.5 (When differentiation IS legal). Let $f_n(x) = \sum_{k=1}^n \frac{\cos(kx)}{k^3}$. Show $f = \lim f_n$ is differentiable with $f'(x) = -\sum_{k \geq 1} \frac{\sin(kx)}{k^2}$.

Recognition. Series of differentiable terms; check uniform convergence of the differentiated series.

Solution. Each term $\frac{\cos kx}{k^3}$ has $|\cdot| \leq k^{-3}$ and $\frac{d}{dx} = -\frac{\sin kx}{k^2}$ has $|\cdot| \leq k^{-2}$. By the Weierstrass M -test (§8.4) with $M_k = k^{-2}$ ($\sum k^{-2} < \infty$), the differentiated series $\sum -\frac{\sin kx}{k^2}$ converges uniformly on $\mathbb{R}$; the original converges at $x = 0$ ($\sum k^{-3}$). Theorem 8.3.4 applies, so $f'(x) = -\sum_{k \geq 1} \frac{\sin kx}{k^2}$. *Check.* The exponents drop by 1 under differentiation ($k^{-3} \mapsto k^{-2}$), and $\sum k^{-2}$ still converges, so the term-by-term derivative is legitimate. ■

A continuous nowhere-differentiable function

Uniform convergence can manufacture a pathology: a function continuous at every point but differentiable at none. We follow Rudin 7.18.

Theorem 8.3.6 (Rudin 7.18). Define the “sawtooth” $\varphi(x) = |x|$ for $-1 \leq x \leq 1$, extended to $\mathbb{R}$ with period 2: $\varphi(x+2) = \varphi(x)$. So φ is continuous, $0 \leq \varphi \leq 1$, and $|\varphi(s) - \varphi(t)| \leq |s - t|$ (it is 1-Lipschitz). Put

$$f(x) = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \varphi(4^n x).$$

Then f is continuous on $\mathbb{R}$ but differentiable at no point.

Proof. Continuity. Each term satisfies $\left| \left(\frac{3}{4}\right)^n \varphi(4^n x) \right| \leq \left(\frac{3}{4}\right)^n$, and $\sum \left(\frac{3}{4}\right)^n = 4 < \infty$. By the Weierstrass M -test the series converges uniformly on $\mathbb{R}$; each term is continuous, so f is continuous (Thm. 8.2.1).

Nowhere differentiable. Fix x and an integer $m \geq 0$. Choose $\delta_m = \pm \frac{1}{2} 4^{-m}$, with the sign picked so that no integer lies strictly between $4^m x$ and $4^m(x + \delta_m) = 4^m x \pm \frac{1}{2}$ (possible since the interval has length $\frac{1}{2}$). Put

$$\gamma_n = \frac{\varphi(4^n(x + \delta_m)) - \varphi(4^n x)}{\delta_m}.$$

For $n > m$: $4^n \delta_m$ is an even integer, and φ has period 2, so $\gamma_n = 0$. For $n = m$: by the no-integer-between choice, φ is affine with slope ± 1 across $[4^m x, 4^m(x + \delta_m)]$, so $|\gamma_m| = 4^m$. For $n < m$: the 1-Lipschitz bound gives $|\gamma_n| \leq 4^n$. Now

$$\left| \left[\frac{f(x + \delta_m) - f(x)}{\delta_m} \right] \right| = \left| \left[\sum_{n=0}^m \left(\frac{3}{4}\right)^n \gamma_n \right] \right| \geq 3^m - \sum_{n=0}^{m-1} 3^n = 3^m - \frac{3^m - 1}{2} = \frac{3^m + 1}{2}.$$

As $m \rightarrow \infty$, $\delta_m \rightarrow 0$ but the difference quotient blows up like $\frac{1}{2}(3^m + 1) \rightarrow \infty$. So $f'(x)$ cannot exist (a finite derivative would force the quotient to stay bounded). Since x was arbitrary, f is nowhere differentiable. □

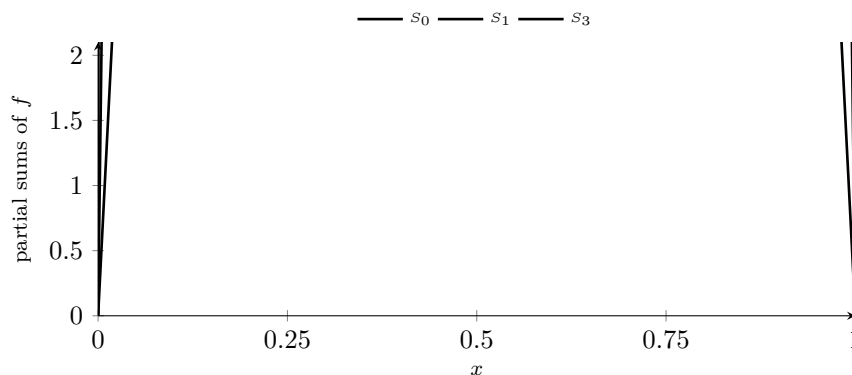


Figure 8.5. Building the nowhere-differentiable $f = \sum \left(\frac{3}{4}\right)^n \varphi(4^n x)$. Each stage adds sawtooth wrinkles four times finer and $\frac{3}{4}$ as tall. The partial sums converge uniformly (so the limit is continuous), but the slopes accumulate without bound at every point (so no derivative survives).

Remark. Weierstrass's original example (B&S 6.1) is $\sum 2^{-k} \cos(3^k x)$, of the same flavor: geometric decay of amplitude (2^{-k}) against geometric growth of frequency (3^k), with the frequency winning fast enough to destroy differentiability while the amplitude decays fast enough to preserve continuity. Such functions are now known to be the “typical” continuous function, in the sense of Baire category.

Further worked examples

Worked Example 8.3.7 (A NON-example: $\lim \int \neq \int \lim$ without uniformity). Let $f_n(x) = nx(1-x^2)^n$ on $[0, 1]$. Show $f_n \rightarrow 0$ pointwise but $\int_0^1 f_n \not\rightarrow \int_0^1 0$, and explain why Theorem 8.3.1 does not apply.

Recognition. “May I pass the limit inside the integral?” — test uniform convergence before swapping (here it fails).

Solution. Pointwise: for fixed $x \in (0, 1]$, $(1-x^2)^n \rightarrow 0$ geometrically, dominating the linear factor n , so $f_n(x) \rightarrow 0$; at $x = 0$, $f_n(0) = 0$. Thus $f_n \rightarrow f \equiv 0$. But substitute $u = 1 - x^2$, $du = -2x dx$:

$$\int_0^1 nx(1-x^2)^n dx = \frac{n}{2} \int_0^1 u^n du = \frac{n}{2(n+1)} \rightarrow \frac{1}{2}.$$

So $\lim_n \int_0^1 f_n = \frac{1}{2} \neq 0 = \int_0^1 \lim_n f_n$. The hypothesis of Theorem 8.3.1 fails: the peak of f_n is at x_n with $1 - x_n^2 = \frac{1}{n+1}$, height $f_n(x_n) \approx \frac{n}{\sqrt{n+1}} \cdot e^{-1} \rightarrow \infty$, so $M_n \rightarrow \infty$ and the convergence is wildly non-uniform.

Check. Like the moving bump of §1, a thin tall spike near $x = 0$ carries a fixed amount of area ($\rightarrow \frac{1}{2}$) while the pointwise limit is 0. ■

Worked Example 8.3.8 (Term-by-term integration of arctan as a series). Justify $\int_0^{1/2} \frac{dx}{1+x^2} =$

$\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)2^{2k+1}}$ and identify the left side.

Recognition. “Swap $\int$ and $\sum$ ” — confirm uniform convergence of the geometric series on the interval of integration.

Solution. On $[0, \frac{1}{2}]$, $\frac{1}{1+x^2} = \sum_{k \geq 0} (-1)^k x^{2k}$ with $|(-1)^k x^{2k}| \leq (\frac{1}{4})^k =: M_k$ and $\sum M_k = \frac{1}{1-1/4} = \frac{4}{3} < \infty$; the M -test gives uniform convergence on $[0, \frac{1}{2}]$. By Corollary 8.3.2,

$$\int_0^{1/2} \frac{dx}{1+x^2} = \sum_{k \geq 0} (-1)^k \int_0^{1/2} x^{2k} dx = \sum_{k \geq 0} \frac{(-1)^k}{2k+1} \left(\frac{1}{2}\right)^{2k+1}.$$

Check. The left side is $\arctan x|_0^{1/2} = \arctan \frac{1}{2} \approx 0.4636$, and the first three terms $\frac{1}{2} - \frac{1}{24} + \frac{1}{160} = 0.46458 \dots$ already match to three places. ■

Worked Example 8.3.9 (Differentiation that is NOT licensed). Let $f_n(x) = \frac{x}{1+nx^2}$ on $\mathbb{R}$. Show $f_n \rightarrow 0$ uniformly yet $f'_n(0) \not\rightarrow 0$, and say which hypothesis of Theorem 8.3.4 is missing.

Recognition. “Does the derivative pass to the limit?” — check uniform convergence of f'_n , not of f_n .

Solution. By AM–GM, $1+nx^2 \geq 2\sqrt{n}|x|$, so $|f_n(x)| = \frac{|x|}{1+nx^2} \leq \frac{1}{2\sqrt{n}}$; thus $M_n \leq \frac{1}{2\sqrt{n}} \rightarrow 0$ and $f_n \rightarrow 0$ uniformly on $\mathbb{R}$, giving $(\lim f_n)' = 0$. But $f'_n(x) = \frac{1-nx^2}{(1+nx^2)^2}$, so $f'_n(0) = 1$ for every n , hence $\lim_n f'_n(0) = 1 \neq 0 = (\lim_n f_n)'(0)$. Theorem 8.3.4 fails to apply because (f'_n) does *not* converge uniformly (uniform convergence of the f_n themselves is irrelevant, exactly the §1/§3 warning). *Check.* $\|f'_n\|_{\mathbb{R}} \geq f'_n(0) = 1$ for all n , so (f'_n) cannot even be uniformly Cauchy. ■

Additional exercises.

1. Justify $\int_0^1 \sum_{n=1}^{\infty} \frac{x^n}{n^2} dx = \sum_{n=1}^{\infty} \frac{1}{n^2(n+1)}$ by checking uniform convergence on $[0, 1]$, and evaluate the right side in closed form if you can.

2. (Non-example.) Give a sequence $f_n \in \mathcal{R}[0, 1]$ with $f_n \rightarrow f$ pointwise, $f \in \mathcal{R}[0, 1]$, yet $\int_0^1 f_n \not\rightarrow \int_0^1 f$, demonstrating that pointwise convergence alone fails Theorem 8.3.1.
3. Let $f_n(x) = \frac{\sin(nx)}{n}$. Show $f_n \rightarrow 0$ uniformly but f'_n does not converge at $x = 0$; reconcile with Theorem 8.3.4.
4. Suppose $f_n \rightarrow f$ and $f'_n \rightarrow g$ both uniformly on $[a, b]$, with f_n differentiable. Prove $f' = g$. (This is the “clean” special case where you assume both convergences.)
5. Show $\lim_{n \rightarrow \infty} \int_0^1 \frac{nx}{1 + n^2 x^3} dx$ can be computed, and decide whether term-by-term passage to the limit under the integral is valid here.
6. (Diagram.) Sketch the first few partial sums S_0, S_1, S_2 of the sawtooth series $f = \sum (\frac{3}{4})^n \varphi(4^n x)$ on $[0, 1]$ and indicate where the slope is steepening.
7. For the nowhere-differentiable f of Theorem 8.3.6, verify the continuity claim explicitly: bound $\|f - S_N\|_{\mathbb{R}}$ where S_N is the N -th partial sum, and show it $\rightarrow 0$.
8. Prove that if $\sum g_k$ converges uniformly on $[a, b]$ with each $g_k \in \mathcal{R}[a, b]$, then $F(x) = \int_a^x \sum_k g_k$ may be differentiated to recover $\sum_k g_k$ at points of continuity.
9. (Bounded convergence.) State and prove the case of the Bounded Convergence Theorem in which the f_n are continuous and converge *monotonically* to a continuous limit on $[a, b]$ (so Dini applies); deduce $\int f_n \rightarrow \int f$.

8.4 Series of Functions and the Weierstrass M-Test

Series of functions

A series $\sum_{n=1}^{\infty} f_n$ of functions on E is, by definition, the sequence of its partial sums $s_N = \sum_{n=1}^N f_n$. It converges *pointwise* / *uniformly* exactly when (s_N) does. Translating the theorems of §§2–3 from sequences to series:

Theorem 8.4.1 (Properties inherited from uniform convergence). *Suppose $\sum f_n$ converges uniformly to s on $[a, b]$.*

- (i) *If each f_n is continuous, so is s (B&S 9.4.2).*
- (ii) *If each $f_n \in \mathcal{R}[a, b]$, then $s \in \mathcal{R}[a, b]$ and $\int_a^b s = \sum_n \int_a^b f_n$ (B&S 9.4.3).*
- (iii) *If each f_n is differentiable, $\sum f_n$ converges at one point, and $\sum f'_n$ converges uniformly, then s is differentiable and $s' = \sum_n f'_n$ (B&S 9.4.4).*

These are immediate from Theorems 8.2.1, 8.3.1, 8.3.4 applied to the partial sums. The remaining issue is a convenient *test* for the uniform convergence of a series — and there is a famous one.

The Weierstrass M-test

Theorem 8.4.2 (Weierstrass M-test; Rudin 7.10, B&S 9.4.6). *Let $f_n : E \rightarrow \mathbb{C}$ and suppose there are constants $M_n \geq 0$ with*

$$|f_n(x)| \leq M_n \quad \text{for all } x \in E, \quad n \in \mathbb{N}, \quad \text{and} \quad \sum_n M_n < \infty.$$

Then $\sum f_n$ converges uniformly (and absolutely) on E .

Proof. Use the Cauchy criterion for series (Thm. 8.1.8, remark). For $m > n$ and any $x \in E$,

$$\left| \sum_{k=n+1}^m f_k(x) \right| \leq \sum_{k=n+1}^m |f_k(x)| \leq \sum_{k=n+1}^m M_k.$$

Since $\sum M_k$ converges, its tails $\sum_{k=n+1}^m M_k$ are $< \varepsilon$ for all $m > n \geq N(\varepsilon)$ (Cauchy criterion for the numerical series $\sum M_k$). The right side is independent of x , so the partial sums of $\sum f_k$ are uniformly Cauchy, hence converge uniformly. $\square$

Method (Applying the Weierstrass M -test). *Recognition trigger.* You must prove a *series* of functions converges uniformly on a set E (often to then integrate/differentiate term by term).

1. **Bound each term off x .** Find $M_n = \sup_{x \in E} |f_n(x)|$, or any convenient upper bound $|f_n(x)| \leq M_n$ valid for *all* $x \in E$.
2. **Test the numerical series $\sum M_n$.** Use p -series, ratio, root, or comparison. If $\sum M_n < \infty$, the original series converges uniformly on E . Done.
3. **If $\sum M_n = \infty$,** the test is inconclusive — it does not prove non-uniformity. Either restrict E to shrink the M_n , or fall back on the Cauchy criterion / Abel's or Dirichlet's test (Rudin 7, Ex. 11).

Caution. The M_n must *not* depend on x . A bound that works only after fixing x proves nothing about uniformity.

Worked Example 8.4.3 (A uniformly convergent trigonometric series). Show $s(x) = \sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$ converges uniformly on $\mathbb{R}$ and is continuous.

Recognition. “Prove uniform convergence of a series” — reach for the M -test.

Solution. $\left| \frac{\sin nx}{n^2} \right| \leq \frac{1}{n^2} =: M_n$ for every x , and $\sum n^{-2} < \infty$. By the M -test the series converges uniformly on $\mathbb{R}$; each term is continuous, so s is continuous (Thm. 8.4.1(i)). *Check.* $\sum n^{-2} = \pi^2/6$, finite, so the hypothesis genuinely holds. $\blacksquare$

Worked Example 8.4.4 (A NON-example: the M -test can fail though the series still converges). Consider $\sum_{n \geq 1} \frac{(-1)^n}{n} x^n$ on $[0, 1]$. Does the M -test apply? Does the series converge uniformly?

Recognition. “Does the M -test work here?” — separate the test from the conclusion.

Solution. Here $\sup_{[0,1]} \left| \frac{(-1)^n}{n} x^n \right| = \frac{1}{n}$ (attained at $x = 1$), and $\sum \frac{1}{n} = \infty$. So the M -test is *inconclusive*. Nonetheless the series does converge uniformly on $[0, 1]$ (by Abel's test: $\sum \frac{(-1)^n}{n}$ converges and x^n is monotone and uniformly bounded). Moral: failure of the M -test never proves non-uniformity. *Check.* At $x = 1$ the series is the convergent alternating harmonic series $-\ln 2$; the M -test missed it because it discards the sign cancellation. $\blacksquare$

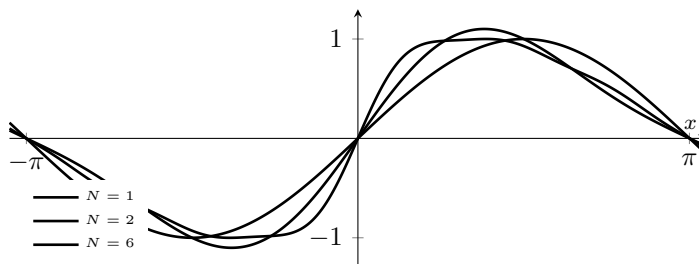


Figure 8.6. Partial sums $s_N(x) = \sum_{n=1}^N \sin(nx)/n^2$. Because $\sum n^{-2}$ converges, the M -test gives uniform convergence: the curves settle into a fixed continuous limit, the gap $\|s - s_N\|$ shrinking to 0 uniformly in x .

Power series as a special case

A power series $\sum_{n \geq 0} a_n(x - c)^n$ is the cleanest series of functions; we center at $c = 0$ without loss.

Theorem 8.4.5 (Radius of convergence; Cauchy–Hadamard, B&S 9.4.8–9.4.9). For $\sum a_n x^n$ set $\rho = \limsup_n |a_n|^{1/n}$ and

$$R = \begin{cases} 1/\rho, & 0 < \rho < \infty, \\ +\infty, & \rho = 0, \\ 0, & \rho = +\infty. \end{cases}$$

Then $\sum a_n x^n$ converges absolutely for $|x| < R$ and diverges for $|x| > R$. (R is the radius of convergence, $(-R, R)$ the interval of convergence; behavior at $|x| = R$ is undetermined.) When $\lim |a_n/a_{n+1}|$ exists it equals R .

Theorem 8.4.6 (Uniform convergence and term-by-term calculus of power series; B&S 9.4.10–9.4.12). Let $f(x) = \sum_{n \geq 0} a_n x^n$ have radius $R > 0$.

(i) On every closed bounded $[-r, r]$ with $r < R$ the series converges uniformly. Hence f is continuous on $(-R, R)$.

(ii) f may be integrated term by term over any $[a, b] \subset (-R, R)$.

(iii) f is differentiable on $(-R, R)$ and $f'(x) = \sum_{n \geq 1} n a_n x^{n-1}$, a power series with the same radius R . Iterating, f is infinitely differentiable and $a_k = f^{(k)}(0)/k!$ (the Taylor coefficients).

Proof of (i) and the key point of (iii). (i) If $r < R$ pick c with $r < c < R$; then $\rho = \limsup |a_n|^{1/n} < 1/c$, so $|a_n| < c^{-n}$ eventually, and for $|x| \leq r$, $|a_n x^n| \leq (r/c)^n =: M_n$ with $\sum M_n < \infty$ since $r/c < 1$. The M -test gives uniform convergence on $[-r, r]$. (iii) Since $\limsup |n a_n|^{1/n} = \limsup |a_n|^{1/n}$ (because $n^{1/n} \rightarrow 1$), the differentiated series $\sum n a_n x^{n-1}$ has the same radius R , hence converges uniformly on each $[-r, r]$; Theorem 8.3.4 then licenses term-by-term differentiation. The rest follows by iteration and setting $x = 0$. $\square$

Remark. Power series beat general series in (iii): no separate hypothesis on the differentiated series is needed, because uniform convergence of the derivative series is *automatic* inside the radius. This is why familiar identities are legitimate, e.g. $\frac{1}{1-x} = \sum x^n$ differentiates to $\frac{1}{(1-x)^2} = \sum n x^{n-1}$ and integrates to $-\ln(1-x) = \sum \frac{x^{n+1}}{n+1}$, all for $|x| < 1$.

Worked Example 8.4.7 (Radius of convergence three ways). Find the radius of convergence of (a) $\sum \frac{x^n}{n!}$, (b) $\sum n! x^n$, (c) $\sum \frac{x^n}{n}$.

Recognition. “Find R ” — use the ratio form $R = \lim |a_n/a_{n+1}|$ when clean.

Solution. (a) $|a_n/a_{n+1}| = (n+1)!/n! = n+1 \rightarrow \infty$, so $R = \infty$ (converges for all x ; this is e^x). (b) $|a_n/a_{n+1}| = n!/(n+1)! = \frac{1}{n+1} \rightarrow 0$, so $R = 0$ (converges only at $x = 0$). (c) $|a_n/a_{n+1}| = \frac{1/n}{1/(n+1)} = \frac{n+1}{n} \rightarrow 1$, so $R = 1$. Check the endpoints in (c): at $x = 1$, $\sum \frac{1}{n}$ diverges; at $x = -1$, $\sum \frac{(-1)^n}{n}$ converges — exactly the “undetermined at $|x| = R$ ” caveat. $\blacksquare$

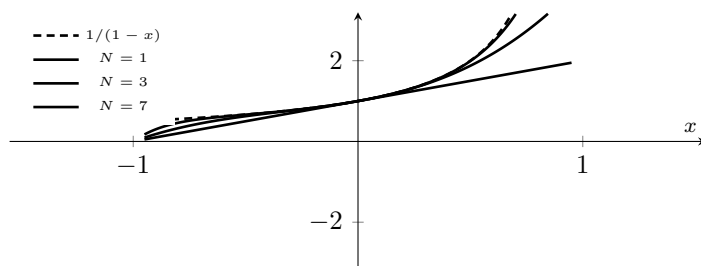


Figure 8.7. Partial sums of $\sum_{n \geq 0} x^n$ closing in on the target $1/(1-x)$ inside $|x| < 1$. The approach is uniform on each $[-r, r]$ with $r < 1$ but breaks down as $x \rightarrow 1^-$, where the target blows up — the boundary $|x| = R = 1$ of the disk of convergence.

Further worked examples

Worked Example 8.4.8 (M -test on $\sum \frac{x^n}{n^2}$ over $[-1, 1]$). Show $\sum_{n \geq 1} \frac{x^n}{n^2}$ converges uniformly on the closed interval $[-1, 1]$, and note the contrast with $\sum x^n/n$.

Recognition. “Prove uniform convergence of a series” — reach for the M -test with an x -independent bound on the closed interval.

Solution. For $|x| \leq 1$, $|\frac{x^n}{n^2}| \leq \frac{1}{n^2} =: M_n$ with $\sum n^{-2} < \infty$, so the M -test gives uniform (and absolute) convergence on all of $[-1, 1]$, endpoints included; the sum is therefore continuous on $[-1, 1]$. By contrast $\sum \frac{x^n}{n}$ has $\sup_{[-1, 1]} |\frac{x^n}{n}| = \frac{1}{n}$ at $x = 1$ with $\sum \frac{1}{n} = \infty$, so the M -test is inconclusive at the right endpoint (and indeed the sum $-\ln(1-x) \rightarrow +\infty$ there). *Check.* Dividing by n^2 instead of n is exactly enough to make the bounding series a convergent p -series ($p = 2 > 1$), pulling both endpoints into the region of uniform convergence. ■

Worked Example 8.4.9 (A NON-example: the M -test fails on the full disk). Does the M -test prove $\sum_{n \geq 0} x^n$ converges uniformly on $(-1, 1)$? On $[-r, r]$ with $r < 1$?

Recognition. Separate the *test* from the *truth*: compute $\sup_E |f_n|$ on each set.

Solution. On $E = (-1, 1)$, $M_n = \sup_{|x| < 1} |x^n| = 1$, and $\sum 1 = \infty$: the M -test is inconclusive on $(-1, 1)$ — consistent with the fact that the series is genuinely *not* uniformly convergent there (the tail $\frac{x^{n+1}}{1-x}$ is unbounded near $x = 1$). On $E = [-r, r]$ with $r < 1$, $M_n = \sup_{|x| \leq r} |x^n| = r^n$ and $\sum r^n = \frac{1}{1-r} < \infty$: the M -test now succeeds, giving uniform convergence on $[-r, r]$. *Check.* The same series flips from “ M -test fails / not uniform” to “ M -test succeeds / uniform” purely by retreating from the open interval to a compact subinterval — the moral of Theorem 8.4.6(i). ■

Worked Example 8.4.10 (Radius of convergence of $\sum \frac{(-1)^n}{n} x^{2n}$ and its sum). Find R for $\sum_{n \geq 1} \frac{(-1)^n}{n} x^{2n}$ and identify the function it represents on $(-R, R)$.

Recognition. “Find R ”; the gaps (only even powers) make the root form $\rho = \limsup |a_k|^{1/k}$ or a direct substitution cleaner than the ratio of consecutive coefficients.

Solution. Put $u = x^2$: the series is $\sum_{n \geq 1} \frac{(-1)^n}{n} u^n$, which has radius 1 in u (ratio $\frac{1/n}{1/(n+1)} = \frac{n+1}{n} \rightarrow 1$). So it converges for $|u| < 1$, i.e. $|x^2| < 1$, i.e. $|x| < 1$: thus $R = 1$. Its sum is the standard series $\sum_{n \geq 1} \frac{(-1)^n}{n} u^n = -\ln(1+u) = -\ln(1+x^2)$ for $|u| \leq 1$, $u \neq -1$ (and the endpoint $u = 1$, i.e. $x = \pm 1$, converges as the alternating series $-\ln 2$). *Check.* Differentiate term by term inside $|x| < 1$: $\frac{d}{dx} \sum \frac{(-1)^n}{n} x^{2n} = \sum (-1)^n 2x^{2n-1} = \frac{-2x}{1+x^2}$, which is exactly $\frac{d}{dx} [-\ln(1+x^2)]$. ■

Exercises (Bartle & Sherbert, Section 9.4).

- Discuss the convergence and the uniform convergence of the series $\sum f_n$, where $f_n(x)$ is given by: (a) $(x^2 + n^2)^{-1}$; (b) $(nx)^{-2}$ ($x \neq 0$); (c) $\sin(x/n^2)$; (d) $(x^n + 1)^{-1}$ ($x \neq 0$); (e) $x^n/(x^n + 1)$ ($x \geq 0$); (f) $(-1)^n(n+x)^{-1}$ ($x \geq 0$).
- If $\sum a_n$ is an absolutely convergent series, show that the series $\sum a_n \sin nx$ is absolutely and uniformly convergent.
- Let (c_n) be a decreasing sequence of positive numbers. If $\sum c_n \sin nx$ is uniformly convergent, then $\lim(nc_n) = 0$.
- Discuss the cases $R = 0$ and $R = +\infty$ in the Cauchy–Hadamard Theorem.
- Show that the radius of convergence R of the power series $\sum a_n x^n$ is given by $\lim |a_n/a_{n+1}|$ whenever this limit exists. Give an example of a power series where this limit does not exist.
- Determine the radius of convergence of the series $\sum a_n x^n$, where a_n is given by: (a) $1/n^n$; (b) $n^\alpha/n!$; (c) $n^n/n!$; (d) $(\ln n)^{-1}$, $n \geq 2$; (e) $(n!)^2/(2n)!$; (f) $n^{-\sqrt{n}}$.
- If $a_n := 1$ when n is the square of a natural number and $a_n := 0$ otherwise, find the radius of convergence of $\sum a_n x^n$. If $b_n := 1$ when $n = m!$ for some $m \in \mathbb{N}$ and $b_n := 0$ otherwise, find the radius of convergence of $\sum b_n x^n$.

8. Prove in detail that $\limsup |na_n|^{1/n} = \limsup |a_n|^{1/n}$.
9. If $0 < p \leq |a_n| \leq q$ for all $n \in \mathbb{N}$, find the radius of convergence of $\sum a_n x^n$.
10. Let $f(x) = \sum a_n x^n$ for $|x| < R$. If $f(-x) = f(x)$ for all $|x| < R$, show that $a_n = 0$ for all odd n .
11. Prove that if f is defined for $|x| < r$ and there exists a constant B with $|f^{(n)}(x)| \leq B$ for all $|x| < r$ and $n \in \mathbb{N}$, then the Taylor series $\sum_{n \geq 0} \frac{f^{(n)}(0)}{n!} x^n$ converges to $f(x)$ for $|x| < r$.
12. Prove by induction that $f(x) := e^{-1/x^2}$ for $x \neq 0$, $f(0) := 0$, has derivatives of all orders at every point, and that all these derivatives vanish at $x = 0$. Hence this function is not given by its Taylor expansion about $x = 0$.
13. Give an example of a function that equals its Taylor series expansion about $x = 0$ for $x \geq 0$, but is not equal to this expansion for $x < 0$.
14. Use the Lagrange form of the remainder to justify the general binomial expansion $(1+x)^m = \sum_{n=0}^{\infty} \binom{m}{n} x^n$ for $0 \leq x < 1$.
15. (Geometric series.) Show directly that if $|x| < 1$, then $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$.
16. Show, by integrating the series for $1/(1+x)$, that if $|x| < 1$ then $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n$.
17. Show that if $|x| < 1$ then $\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$.
18. Show that if $|x| < 1$ then $\arcsin x = \int_0^x \frac{dt}{\sqrt{1-t^2}} = \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}$.

Additional exercises.

1. Apply the Weierstrass M -test to show $\sum_{n \geq 1} \frac{\cos(nx)}{n^2}$ converges uniformly on $\mathbb{R}$; state the M_n you use.
2. Find all x for which $\sum_{n \geq 1} \frac{x^n}{n^2}$ converges, determine its radius of convergence, and show the convergence is uniform on $[-1, 1]$.
3. Compute the radius of convergence of $\sum_{n \geq 1} \frac{(2x)^n}{\sqrt{n}}$ and discuss the two endpoints.
4. (Non-example.) Show the M -test fails for $\sum_{n \geq 1} \frac{(-1)^n}{n} x^n$ on $[0, 1]$, yet the series converges uniformly there. Explain why failure of the test proves nothing about uniformity.
5. Use term-by-term integration of a geometric series to derive $\arctan x = \sum_{n \geq 0} \frac{(-1)^n}{2n+1} x^{2n+1}$ for $|x| < 1$, and justify each step.
6. Show $\sum_{n \geq 1} \frac{x}{n(1+nx^2)}$ converges uniformly on $\mathbb{R}$ and that the sum is continuous. (Hint: bound each term by maximizing in x .)
7. (Diagram.) Plot the partial sums s_1, s_2, s_6 of $\sum_{n \geq 1} \frac{\sin nx}{n^2}$ on $[-\pi, \pi]$ and explain visually why uniform convergence holds.
8. Prove that a power series with radius R may be differentiated term by term on $(-R, R)$, citing the fact that the differentiated series has the same radius. Illustrate with $\sum_{n \geq 0} x^n$ and its derivative.

9. Suppose $\sum a_n$ converges. Prove $f(x) = \sum a_n x^n$ is continuous on $[0, 1]$ (Abel's theorem, uniform-convergence form). Where does the M -test fail, and what replaces it?
10. Determine whether $\sum_{n \geq 1} \frac{x^n}{n}$ converges uniformly on $[0, 1]$, on $[0, 1 - \delta]$ for fixed $\delta > 0$, and on $[0, 1]$; justify each answer.

8.5 Equicontinuity and the Arzela–Ascoli Theorem

The question: which families have convergent subsequences?

For numbers, Bolzano–Weierstrass says: every bounded sequence has a convergent subsequence. The analogue for functions is false with mere boundedness. We seek the right “compactness” hypothesis on a family of functions that guarantees a *uniformly* convergent subsequence. Two conditions are needed (Rudin 7.19–7.25).

Definition 8.5.1 (Pointwise / uniformly bounded; Rudin 7.19). A family $\mathcal{F}$ of functions on E is *pointwise bounded* if for each $x \in E$ the set $\{f(x) : f \in \mathcal{F}\}$ is bounded (bound may depend on x); it is *uniformly bounded* if there is a single M with $|f(x)| \leq M$ for all $f \in \mathcal{F}$ and all $x \in E$.

Definition 8.5.2 (Equicontinuity; Rudin 7.22). A family $\mathcal{F}$ on a metric space E is *equicontinuous* if for every $\varepsilon > 0$ there is a *single* $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon \quad \text{for every } f \in \mathcal{F}.$$

The point: one δ works for the whole family at once (it may not depend on f).

Equicontinuity is to a family what *uniform* continuity is to one function, but with the modulus shared across all members. A family with a common Lipschitz constant L (i.e. $|f(x) - f(y)| \leq L|x - y|$ for all f) is automatically equicontinuous: take $\delta = \varepsilon/L$.

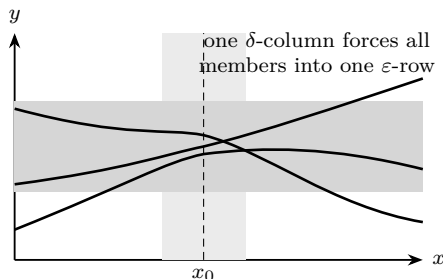


Figure 8.8. Equicontinuity at x_0 : a single width- 2δ vertical band confines *every* function of the family to vary by less than ε — the horizontal ε -row. The modulus of continuity is common to the whole family.

Worked Example 8.5.3 (An equicontinuous and a non-equicontinuous family). Decide equicontinuity on $[0, 1]$ of (a) $\mathcal{F} = \{f_n\}$, $f_n(x) = \frac{x}{n}$; (b) $\mathcal{G} = \{g_n\}$, $g_n(x) = x^n$.

Recognition. “Is this family equicontinuous?” — look for a common modulus / common Lipschitz constant, or find a moving steep point to break it.

Solution. (a) $|f_n(x) - f_n(y)| = \frac{1}{n}|x - y| \leq |x - y|$, a common Lipschitz constant $L = 1$; take $\delta = \varepsilon$. Equicontinuous (in fact uniformly so). (b) Near $x = 1$ the slope $g'_n(1) = n \rightarrow \infty$: take $x = 1$, $y = 1 - \frac{1}{n}$; then $|g_n(1) - g_n(1 - \frac{1}{n})| = 1 - (1 - \frac{1}{n})^n \rightarrow 1 - e^{-1} \approx 0.63$, while $|x - y| = \frac{1}{n} \rightarrow 0$. No single δ can hold all g_n within $\varepsilon = \frac{1}{2}$. Not equicontinuous. *Check.* (a)’s graphs all flatten toward 0 together; (b)’s develop an ever-steeper cliff at $x = 1$. ■

The Arzela–Ascoli theorem

We build to the main result through Rudin’s two preparatory facts.

Theorem 8.5.4 (Pointwise-bounded families have pointwise-convergent subsequences on a countable set; Rudin 7.23). *If (f_n) is pointwise bounded on a countable set $E = \{x_1, x_2, \dots\}$, then some subsequence (f_{n_k}) converges at every point of E .*

Proof. Cantor diagonalization. The numbers $f_n(x_1)$ are bounded, so by Bolzano–Weierstrass some subsequence $(f_{1,k})$ converges at x_1 . From it extract $(f_{2,k})$ converging at x_2 (and still at x_1). Continuing, get nested subsequences $(f_{j,k})_k$ converging at $x_1, \dots, x_j$. The *diagonal* $g_k = f_{k,k}$ is, from index j on, a subsequence of $(f_{j,k})_k$, hence converges at every x_j . $\square$

Theorem 8.5.5 (Arzela–Ascoli; Rudin 7.24–7.25). *Let K be a compact metric space and (f_n) a sequence in $\mathcal{C}(K)$. If (f_n) is pointwise bounded and equicontinuous on K , then:*

- (i) (f_n) is in fact uniformly bounded on K , and
- (ii) (f_n) has a subsequence that converges uniformly on K .

Conversely, a uniformly convergent sequence in $\mathcal{C}(K)$ is uniformly bounded and equicontinuous. Equivalently: a set $\mathcal{F} \subseteq \mathcal{C}(K)$ has compact closure (in the sup-metric) iff it is uniformly bounded and equicontinuous.

Proof. (i) Given $\varepsilon = 1$, equicontinuity yields $\delta > 0$ with $|f_n(x) - f_n(y)| < 1$ whenever $|x - y| < \delta$, for all n . Compactness of K gives finitely many points $p_1, \dots, p_r$ whose δ -balls cover K . Pointwise boundedness gives M with $|f_n(p_i)| \leq M$ for all n, i . Each $x \in K$ lies within δ of some p_i , so $|f_n(x)| \leq |f_n(p_i)| + 1 \leq M + 1$: uniform boundedness.

(ii) Let $E = \{x_1, x_2, \dots\}$ be a countable dense subset of K (compact metric spaces are separable). By Theorem 8.5.4 pass to a subsequence (g_k) converging at every point of E . We claim (g_k) is uniformly Cauchy on K . Fix $\varepsilon > 0$; by equicontinuity pick $\delta > 0$ with $|g_k(x) - g_k(y)| < \varepsilon/3$ whenever $|x - y| < \delta$, all k . Cover K by finitely many δ -balls centered at points $x_{i_1}, \dots, x_{i_s} \in E$. Since (g_k) converges at each of these finitely many points, choose N with $|g_k(x_{i_j}) - g_\ell(x_{i_j})| < \varepsilon/3$ for all $k, \ell \geq N$ and all j . Now any $x \in K$ is within δ of some center x_{i_j} , so for $k, \ell \geq N$,

$$|g_k(x) - g_\ell(x)| \leq |g_k(x) - g_k(x_{i_j})| + |g_k(x_{i_j}) - g_\ell(x_{i_j})| + |g_\ell(x_{i_j}) - g_\ell(x)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Thus (g_k) is uniformly Cauchy on K ; by completeness of $\mathcal{C}(K)$ (Thm. 8.2.5) it converges uniformly.

Converse. If $f_n \rightarrow f$ uniformly, the f_n are uniformly bounded (Rudin 7, Ex. 1), and equicontinuity follows from the $\varepsilon/3$ -argument: f is uniformly continuous on compact K , and $\|f_n - f\|_K < \varepsilon/3$ for large n pushes a single δ onto all f_n . $\square$

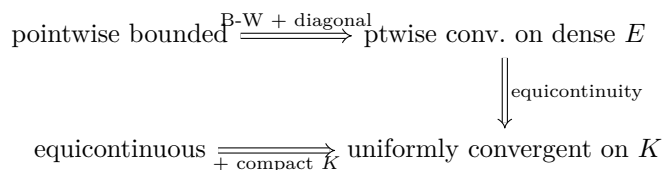


Figure 8.9. The logic of Arzela–Ascoli: pointwise boundedness feeds Bolzano–Weierstrass and the diagonal trick to get convergence on a countable dense set; equicontinuity plus compactness upgrades that to uniform convergence on all of K .

Caution. Both hypotheses are indispensable. Without *equicontinuity*: $f_n(x) = \sin(nx)$ on $[0, \pi]$ is uniformly bounded but has no uniformly convergent subsequence (the frequencies run away). Without *boundedness*: $f_n(x) = n$ (constants) is equicontinuous (slope 0) but unbounded, with no convergent subsequence. Arzela–Ascoli needs the family both *level* (equicontinuous) and *confined* (bounded).

Worked Example 8.5.6 (Uniformly bounded integrals give an equicontinuous family). Let $|f_n| \leq 1$ on $[0, 1]$ and put $F_n(x) = \int_0^x f_n(t) dt$. Show (F_n) has a uniformly convergent subsequence (Rudin 7, Ex. 18).

Recognition. “Extract a uniformly convergent subsequence” — verify the Arzela–Ascoli hypotheses.

Solution. Uniform boundedness: $|F_n(x)| \leq \int_0^1 |f_n| \leq 1$. Equicontinuity: for $x < y$, $|F_n(y) - F_n(x)| = |\int_x^y f_n| \leq |y - x|$, a common Lipschitz constant 1, so $\delta = \varepsilon$ works for all n . By Arzela–Ascoli some F_{n_k} converges uniformly on $[0, 1]$. *Check.* Integration smooths the (possibly wild) f_n into a family with a *shared* modulus of continuity — this is the standard mechanism for producing equicontinuity. $\blacksquare$

Further worked examples

Worked Example 8.5.7 (A NON-example: bounded but not equicontinuous, no convergent subsequence). Show $f_n(x) = \sin(nx)$ on $[0, \pi]$ is uniformly bounded yet has *no* uniformly (indeed no pointwise) convergent subsequence, and pinpoint which Arzela–Ascoli hypothesis fails.

Recognition. “Why does Arzela–Ascoli not produce a subsequence here?” — test boundedness and equicontinuity separately.

Solution. Uniform boundedness is clear: $|\sin(nx)| \leq 1$. Equicontinuity fails: near any point the slope $f'_n(x) = n \cos(nx)$ blows up with n , so no single δ controls all members. To see no convergent subsequence exists, note $\int_0^\pi \sin^2(nx) \, dx = \frac{\pi}{2}$ for all n , while $\int_0^\pi \sin(nx) \sin(mx) \, dx = 0$ for $n \neq m$; hence for any $n \neq m$, $\int_0^\pi (\sin nx - \sin mx)^2 \, dx = \pi$, so $\|f_n - f_m\|_{[0, \pi]}^2 \geq \frac{1}{\pi} \cdot \pi = 1$ is bounded below, and no subsequence can be uniformly Cauchy. *Check.* The missing hypothesis is precisely *equicontinuity*; uniform boundedness alone is the function-space analogue of a bounded-but-not-precompact set. ■

Worked Example 8.5.8 (A common Lipschitz bound yields a convergent subsequence). Let $(f_n) \subset \mathcal{C}[0, 1]$ satisfy $|f_n(x)| \leq 5$ and $|f_n(x) - f_n(y)| \leq 3|x - y|$ for all n . Show some subsequence converges uniformly.

Recognition. “Extract a uniformly convergent subsequence” — verify the two Arzela–Ascoli hypotheses, using the common Lipschitz constant for equicontinuity.

Solution. *Uniform boundedness:* $|f_n(x)| \leq 5$ for all n, x by assumption. *Equicontinuity:* the common Lipschitz constant $L = 3$ gives, for any $\varepsilon > 0$, the single choice $\delta = \varepsilon/3$ with $|x - y| < \delta \implies |f_n(x) - f_n(y)| \leq 3|x - y| < \varepsilon$ for every n . The domain $[0, 1]$ is compact, so Arzela–Ascoli (Thm. 8.5.5) yields a subsequence converging uniformly on $[0, 1]$. *Check.* A uniform Lipschitz bound is the cleanest sufficient condition for equicontinuity ($\delta = \varepsilon/L$ depends on neither x nor n); combined with a uniform sup bound it is the textbook trigger for Arzela–Ascoli. ■

Additional exercises.

1. Show the family $\{f_n\}$, $f_n(x) = \sin(x + n)$ on $[0, 1]$, is equicontinuous and uniformly bounded; what does Arzela–Ascoli guarantee?
2. Prove that any family with a common Lipschitz constant L is equicontinuous, and that the family $\{x \mapsto \arctan(nx)\}$ on $\mathbb{R}$ is *not* equicontinuous at 0.
3. (Non-example.) Show $\{\sin(nx)\}_{n \geq 1}$ on $[0, \pi]$ is uniformly bounded but has no uniformly convergent subsequence, and identify which Arzela–Ascoli hypothesis fails.
4. Let $f_n(x) = \int_0^x g_n(t) \, dt$ with $|g_n| \leq 1$ on $[0, 1]$. Show $\{f_n\}$ satisfies the Arzela–Ascoli hypotheses, hence has a uniformly convergent subsequence.
5. Suppose $\{f_n\}$ is equicontinuous on a compact K and converges pointwise. Prove it converges uniformly (Rudin 7, Ex. 16) by an $\varepsilon/3$ covering argument.
6. (Diagram.) Draw the “ δ -column / ε -row” picture for equicontinuity at a point x_0 and explain why a non-equicontinuous family develops a moving steep spot.
7. Give a sequence that is equicontinuous but not uniformly bounded, and one that is uniformly bounded but not equicontinuous, showing both hypotheses are independent and necessary.
8. Prove that a uniformly convergent sequence in $\mathcal{C}(K)$ (K compact) is equicontinuous and uniformly bounded (the converse direction of Arzela–Ascoli).
9. Use Arzela–Ascoli to sketch the existence proof for $y' = \phi(x, y)$, $y(0) = c$ with ϕ continuous and bounded: build polygonal approximants, show equicontinuity, extract a uniform limit (Peano’s theorem; Rudin 7, Ex. 25).

8.6 The Stone–Weierstrass Theorem

Weierstrass: polynomials are dense in $C[a, b]$

The crown of the chapter is that the most familiar functions — polynomials — can approximate *every* continuous function on a closed interval, uniformly. We give Rudin’s constructive proof (7.26), then abstract it to Stone’s theorem (7.31–7.33).

Theorem 8.6.1 (Weierstrass approximation; Rudin 7.26). *If f is continuous on $[a, b]$, then for every $\varepsilon > 0$ there is a polynomial P with $\|f - P\|_{[a,b]} < \varepsilon$. Equivalently, the polynomials are dense in $\mathcal{C}[a, b]$ under the sup-norm: there exist polynomials $P_n \rightarrow f$ uniformly on $[a, b]$.*

Proof sketch (Landau kernel). By an affine change of variable assume $[a, b] = [0, 1]$ and, subtracting the line through the endpoints, that $f(0) = f(1) = 0$; extend f by 0 outside $[0, 1]$, so f is uniformly continuous on $\mathbb{R}$ with $\|f\| = \sup |f| =: M$. Put $Q_n(x) = c_n(1 - x^2)^n$ where c_n is chosen so $\int_{-1}^1 Q_n = 1$; one checks $c_n < \sqrt{n}$, so for any $\delta > 0$, $Q_n \rightarrow 0$ uniformly on $\delta \leq |x| \leq 1$. Define the polynomials

$$P_n(x) = \int_{-1}^1 f(x+t) Q_n(t) dt = \int_0^1 f(t) Q_n(t-x) dt \quad (0 \leq x \leq 1),$$

(the second form, after substitution, shows each P_n is a polynomial in x). Then, using $\int Q_n = 1$ and splitting the integral over $|t| < \delta$ (where $|f(x+t) - f(x)| < \varepsilon/2$ by uniform continuity) and $|t| \geq \delta$ (where Q_n is uniformly tiny), one gets $|P_n(x) - f(x)| < \varepsilon$ for all $x \in [0, 1]$ once n is large. Hence $P_n \rightarrow f$ uniformly. $\square$

Remark. The same conclusion follows from the explicit *Bernstein polynomials* $B_n(f)(x) = \sum_{k=0}^n f(\frac{k}{n}) \binom{n}{k} x^k (1-x)^{n-k}$, which converge uniformly to f on $[0, 1]$; that proof is probabilistic (law of large numbers for a Binomial(n, x) count). Rudin’s kernel proof and the Bernstein proof give the same theorem; the kernel proof generalizes more readily. A third route (Rudin 7, Ex. 23) builds the special case $P_n \rightarrow |x|$ on $[-1, 1]$ by an explicit recursion, from which Stone–Weierstrass can be bootstrapped without Theorem 8.6.1.

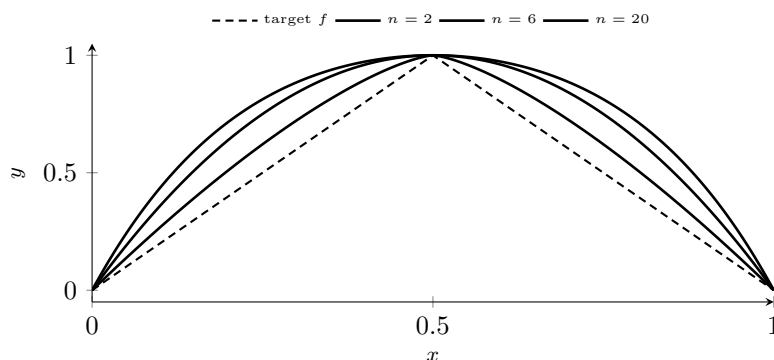


Figure 8.10. Polynomials closing uniformly on a continuous (here piecewise-linear) target. As n grows the smooth polynomial graphs press into the corner of the tent; the sup-distance $\|f - P_n\| \rightarrow 0$. No polynomial equals f , yet they crowd arbitrarily close.

Algebras, separation, and Stone’s theorem

What property of the polynomials made the theorem work? They form an *algebra*, they *separate points*, and they *vanish nowhere*. Stone isolated exactly these.

Definition 8.6.2 (Rudin 7.28–7.30). A set $\mathcal{A} \subseteq \mathcal{C}(K)$ (real continuous functions on a compact K) is an *algebra* if it is closed under addition, scalar multiplication, and pointwise products. It *separates points* if for every $x \neq y$ in K there is $f \in \mathcal{A}$ with $f(x) \neq f(y)$. It *vanishes at no point* if for every $x \in K$ there is $f \in \mathcal{A}$ with $f(x) \neq 0$. The *uniform closure* $\overline{\mathcal{A}}$ is the set of uniform limits of sequences from $\mathcal{A}$.

Lemma 8.6.3 (Rudin 7.29). *The uniform closure $\overline{\mathcal{A}}$ of an algebra $\mathcal{A} \subseteq \mathcal{C}(K)$ is again an algebra.*

Proof. If $f_n \rightarrow f$ and $g_n \rightarrow g$ uniformly with $f_n, g_n \in \mathcal{A}$, then on compact K the f_n, g_n are uniformly bounded, and $f_n + g_n \rightarrow f + g$, $cf_n \rightarrow cf$, $f_n g_n \rightarrow fg$ uniformly (the product because $\|f_n g_n - fg\| \leq \|f_n\| \|g_n - g\| + \|g\| \|f_n - f\| \rightarrow 0$). So the limits lie in $\overline{\mathcal{A}}$. $\square$

Theorem 8.6.4 (Stone–Weierstrass, real form; Rudin 7.32). *Let K be compact and $\mathcal{A} \subseteq \mathcal{C}(K)$ a (real) algebra that separates points and vanishes at no point of K . Then $\mathcal{A}$ is dense in $\mathcal{C}(K)$: $\overline{\mathcal{A}} = \mathcal{C}(K)$.*

Proof outline. Step 1 (lattice operations). Using the polynomial approximation of $|x|$ on $[-M, M]$ (which is Theorem 8.6.1 for the single function $t \mapsto |t|$, or Rudin 7, Ex. 23), one shows $\overline{\mathcal{A}}$ is closed under $|\cdot|$, hence under $\max(f, g) = \frac{1}{2}(f + g + |f - g|)$ and $\min(f, g) = \frac{1}{2}(f + g - |f - g|)$: it is a *lattice*.

Step 2 (two-point interpolation). Separation + vanishing-nowhere $\implies$ for any two points $x \neq y$ and any target values α, β there is $h \in \mathcal{A}$ with $h(x) = \alpha$, $h(y) = \beta$. (Take g separating x, y and $u, v \in \mathcal{A}$ nonzero at x, y ; a suitable linear combination hits the prescribed values.)

Step 3 (local then global). Given $f \in \mathcal{C}(K)$ and $\varepsilon > 0$: for each pair x, y choose $h_{x,y} \in \mathcal{A}$ agreeing with f at x and y . Fixing x and varying y , a compactness/covering argument and the lattice min produce $g_x \in \overline{\mathcal{A}}$ with $g_x(x) = f(x)$ and $g_x < f + \varepsilon$ everywhere. Varying x and using the lattice max over a finite cover produces $g \in \overline{\mathcal{A}}$ with $f - \varepsilon < g < f + \varepsilon$ on K , i.e. $\|f - g\|_K \leq \varepsilon$. As ε was arbitrary, $f \in \overline{\mathcal{A}}$. $\square$

Corollary 8.6.5 (Weierstrass from Stone). *The polynomials on $[a, b]$ form an algebra (closed under $+, \cdot$, scalars) that separates points (the function x already does) and vanishes nowhere (the constant 1). By Theorem 8.6.4 they are dense in $\mathcal{C}[a, b]$ — recovering Theorem 8.6.1.*

Remark (The complex caveat; Rudin 7.33). For *complex* algebras the theorem can fail: on the unit circle $K = \{|z| = 1\}$ the algebra generated by z (polynomials in z alone) separates points and vanishes nowhere, yet its uniform closure misses $\bar{z}$ (an integral obstruction; Rudin 7, Ex. 21). The fix is to require $\mathcal{A}$ *self-adjoint* ($f \in \mathcal{A} \implies \bar{f} \in \mathcal{A}$); then density holds again. The real algebra of $\operatorname{Re}, \operatorname{Im}$ parts satisfies the hypotheses of the real theorem, and the complex result follows.

Worked Example 8.6.6 (Even functions and $\cos$). Show that the algebra generated by $\cos x$ is dense in the continuous *even* 2π -periodic functions on $[-\pi, \pi]$ (equivalently $\mathcal{C}[-1, 1]$ via $t = \cos x$).

Recognition. “Is this subalgebra dense?” — check the Stone–Weierstrass hypotheses on the right compact set.

Solution. Set $t = \cos x \in [-1, 1]$. Polynomials in t form an algebra on $[-1, 1]$ separating points (t itself) and containing constants. By Stone–Weierstrass they are dense in $\mathcal{C}[-1, 1]$. Translating back, finite combinations of $\cos^k x$ — equivalently of $\cos(kx)$ — approximate any continuous even 2π -periodic function uniformly.

Check. $\cos$ alone does *not* separate x from $-x$ (it is even), which is exactly why we get density in the *even* functions, not all continuous functions. $\blacksquare$

Further worked examples

Worked Example 8.6.7 (Approximating a specific f by polynomials, with an explicit degree). Find a polynomial P with $\|e^x - P\|_{[0,1]} < 10^{-3}$, using the Taylor partial sums.

Recognition. “Produce a uniform polynomial approximation” — Weierstrass guarantees existence; here the power series gives an explicit P and a tail bound.

Solution. Take $P(x) = \sum_{k=0}^N \frac{x^k}{k!}$. On $[0, 1]$ the Lagrange remainder is $|e^x - P(x)| = \frac{e^\xi}{(N+1)!} x^{N+1} \leq \frac{e}{(N+1)!}$ for some $\xi \in (0, 1)$, since $x^{N+1} \leq 1$ and $e^\xi \leq e < 3$. So $\|e^x - P\|_{[0,1]} \leq \frac{3}{(N+1)!}$. Choosing $N = 6$: $\frac{3}{7!} = \frac{3}{5040} \approx 5.95 \times 10^{-4} < 10^{-3}$. Thus $P(x) = 1 + x + \frac{x^2}{2} + \cdots + \frac{x^6}{720}$ works. *Check.* This is a concrete witness to Theorem 8.6.1 (polynomials dense in $\mathcal{C}[0, 1]$); we did not need the kernel construction because e^x is already a uniformly convergent power series on the compact interval. $\blacksquare$

Worked Example 8.6.8 (Stone–Weierstrass: even polynomials in x on $[0, 1]$). Is the algebra $\mathcal{A}$ generated by $\{1, x^2\}$ dense in $\mathcal{C}[0, 1]$? In $\mathcal{C}[-1, 1]$?

Recognition. “Is this subalgebra dense?” — check separation and vanishing-nowhere on the relevant compact set.

Solution. $\mathcal{A}$ consists of polynomials in x^2 (equivalently, even polynomials), containing the constant 1 (so it vanishes nowhere). On $[0, 1]$: the function x^2 separates points, since $x \mapsto x^2$ is injective on $[0, 1]$; with

constants present, Stone–Weierstrass (Thm. 8.6.4) gives density in $\mathcal{C}[0, 1]$. On $[-1, 1]$: x^2 does *not* separate x from $-x$ (both give x^2), and no even function can; so $\mathcal{A}$ fails the separation hypothesis and is *not* dense — its closure is only the even continuous functions. *Check.* The odd function $g(x) = x$ cannot be uniformly approximated by even polynomials on $[-1, 1]$, since $\|g - P\| \geq \frac{1}{2} |g(1) - g(-1)| = 1$ for any even P (an even P has $P(1) = P(-1)$). ■

Worked Example 8.6.9 (The complex caveat made concrete: $\bar{z}$ is unreachable). On the unit circle $K = \{|z| = 1\}$, show the (complex) algebra of polynomials in z does not uniformly approximate $\bar{z}$, illustrating why self-adjointness is required.

Recognition. “Why does complex Stone–Weierstrass need $f \in \mathcal{A} \implies \bar{f} \in \mathcal{A}$?” — exhibit a target the non-self-adjoint algebra misses, via an integral invariant.

Solution. Parametrize $z = e^{i\theta}$, $\theta \in [0, 2\pi]$, with the inner product $\langle u, v \rangle = \frac{1}{2\pi} \int_0^{2\pi} u \bar{v} d\theta$; the functions $\{e^{ik\theta}\}_{k \in \mathbb{Z}}$ are orthonormal. Here $z^n = e^{in\theta}$ ($n \geq 0$) and $\bar{z} = e^{-i\theta}$. For any polynomial $P(z) = \sum_{n \geq 0} c_n z^n$, orthonormality gives $\langle \bar{z}, z^n \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{-i\theta} e^{-in\theta} d\theta = 0$ for every $n \geq 0$, since the integrand has nonzero frequency $-(n+1)$. Therefore $\bar{z} \perp P$, and by Pythagoras

$$\frac{1}{2\pi} \int_0^{2\pi} |\bar{z} - P|^2 d\theta = \|\bar{z}\|^2 + \|P\|^2 = 1 + \sum_{n \geq 0} |c_n|^2 \geq 1.$$

Since the sup-norm dominates the L^2 average, $\|\bar{z} - P\|_K^2 \geq \frac{1}{2\pi} \int |\bar{z} - P|^2 \geq 1$, so *no* polynomial in z comes within 1 of $\bar{z}$ uniformly. *Check.* The obstruction is that $\bar{z}$ has only a negative frequency while every z^n has a nonnegative one, so they are orthogonal; requiring $\mathcal{A}$ self-adjoint puts $\bar{z} = z^{-1}$ into the algebra and restores density. ■

Additional exercises.

1. Verify that the polynomials on $[a, b]$ form an algebra that separates points and vanishes at no point, so Stone–Weierstrass yields the Weierstrass approximation theorem.
2. Show the even polynomials in x (those with only even powers) form an algebra on $[-1, 1]$ that does *not* separate points; describe what their uniform closure is and why (a non-example for the separation hypothesis).
3. Prove that if $f \in \mathcal{C}[0, 1]$ satisfies $\int_0^1 f(x) x^n dx = 0$ for all $n \geq 0$, then $f \equiv 0$ (use Weierstrass to approximate f by polynomials, then $\int f^2 = 0$).
4. Use the recursion $P_{n+1} = P_n + \frac{1}{2}(x^2 - P_n^2)$, $P_0 = 0$, to prove $P_n \rightarrow |x|$ uniformly on $[-1, 1]$; this is the lattice ingredient of the Stone proof.
5. (Diagram.) Sketch a continuous tent function on $[0, 1]$ together with two polynomial approximants of increasing degree, and indicate the shrinking sup-distance.
6. Show the algebra generated by $\{1, \cos x\}$ is dense in the continuous even 2π -periodic functions but not in all of $\mathcal{C}[-\pi, \pi]$; explain via the separation criterion.
7. (Complex caveat.) Explain why polynomials in z alone are *not* dense in the continuous functions on the unit circle, and how the self-adjointness hypothesis (allowing $\bar{z}$) repairs density.
8. Prove that the trigonometric polynomials $\sum_{k=0}^N (a_k \cos kx + b_k \sin kx)$ are dense in the continuous 2π -periodic functions on $\mathbb{R}$ (Stone–Weierstrass on the circle, self-adjoint form).
9. Let $K \subseteq \mathbb{R}^n$ be compact. Show the polynomials in n variables are dense in $\mathcal{C}(K)$, generalizing the one-variable Weierstrass theorem.
10. Deduce from Stone–Weierstrass that $\mathcal{C}[a, b]$ is *separable*: it has a countable dense subset (e.g. polynomials with rational coefficients).

Chapter 9

Some Special Functions

9.1 Power Series and Analytic Functions

Power series and the radius of convergence

A *power series* centered at a is a series of the form

$$\sum_{n=0}^{\infty} c_n (z - a)^n,$$

with coefficients c_n and argument z in $\mathbb{R}$ or $\mathbb{C}$. Translating $z \mapsto z - a$ we lose nothing by taking $a = 0$ and studying $\sum c_n z^n$. The fundamental fact is that the set of z for which the series converges is, apart from its boundary, a *disc*: there is a number $R \in [0, \infty]$ such that the series converges absolutely for $|z| < R$ and diverges for $|z| > R$.

Theorem 9.1.1 (Radius of convergence). *Given $\sum c_n z^n$, put*

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |c_n|^{1/n} \quad (\text{with } 1/0 = \infty, \ 1/\infty = 0).$$

Then $\sum c_n z^n$ converges absolutely if $|z| < R$ and diverges if $|z| > R$. The number R is the radius of convergence; the set $|z| < R$ is the disc of convergence.

Proof. Apply the root test to $a_n = c_n z^n$:

$$\limsup_n |a_n|^{1/n} = |z| \cdot \limsup_n |c_n|^{1/n} = \frac{|z|}{R}.$$

If $|z| < R$ this lim sup is < 1 and the series converges absolutely; if $|z| > R$ it is > 1 and the general term does not tend to 0, so the series diverges. $\square$

On the boundary $|z| = R$ anything can happen: $\sum z^n$ ($R = 1$) diverges everywhere on $|z| = 1$; $\sum z^n/n^2$ ($R = 1$) converges everywhere on it; $\sum z^n/n$ ($R = 1$) converges at $z = -1$ but diverges at $z = 1$. The radius locates the disc; the boundary is a separate, delicate question (Abel's theorem below handles one case).

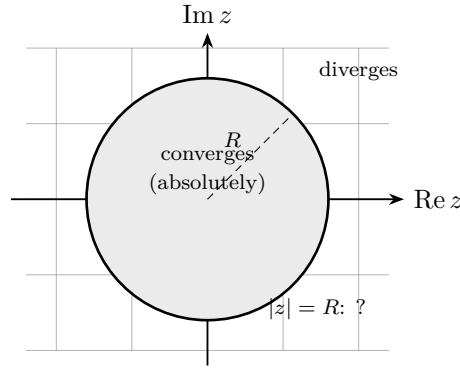


Figure 9.1. Inside the open disc the series converges absolutely; outside it diverges; on the circle the behaviour is undecided by R alone.

Remark. When $\lim |c_{n+1}/c_n|$ exists it equals $1/R$ as well (ratio test), and is often easier to compute. The lim sup root form is the one that *always* works.

Worked Example 9.1.2 (Three radii from the formula). Find R for (a) $\sum z^n/n!$, (b) $\sum n! z^n$, (c) $\sum z^n/n$.

Recognition. “Compute a radius”: use ratio when terms have factorials, root otherwise.

Solution. (a) $|c_{n+1}/c_n| = \frac{n!}{(n+1)!} = \frac{1}{n+1} \rightarrow 0$, so $1/R = 0$, $R = \infty$: the exponential series converges everywhere. (b) $|c_{n+1}/c_n| = (n+1) \rightarrow \infty$, so $R = 0$: $\sum n! z^n$ converges only at $z = 0$. (c) $|c_n|^{1/n} = (1/n)^{1/n} \rightarrow 1$, so $R = 1$. *Check.* (a) and (b) are reciprocal extremes; (c) is the harmonic-type series with the unit disc, as expected.

Continuity and uniform convergence on closed subdiscs

Theorem 9.1.3 (Uniform convergence inside the disc). *If $\sum c_n z^n$ has radius $R > 0$ and $0 < \rho < R$, the series converges uniformly on the closed disc $|z| \leq \rho$. Consequently its sum $f(z) = \sum c_n z^n$ is continuous on $|z| < R$.*

Proof. Pick $\rho < r < R$. Since $r < R$, $\sum |c_n| r^n$ converges. For $|z| \leq \rho$, $|c_n z^n| \leq |c_n| \rho^n \leq |c_n| r^n$, so by the Weierstrass M -test (with $M_n = |c_n| r^n$) the series converges uniformly on $|z| \leq \rho$. A uniform limit of the continuous partial sums is continuous; as every point of $|z| < R$ lies in some such closed disc, f is continuous throughout the open disc. $\square$

Caution. Convergence is uniform on every *closed* subdisc $|z| \leq \rho < R$, but *not* necessarily on the full open disc $|z| < R$. Example: $\sum z^n = \frac{1}{1-z}$ on $|z| < 1$ is not uniformly convergent on $|z| < 1$ (the tail blows up as $|z| \rightarrow 1$). Always shrink to a closed subdisc before invoking uniformity.

Term-by-term differentiation

Theorem 9.1.4 (Differentiation of power series, Rudin 8.1). *Suppose $f(x) = \sum_{n=0}^{\infty} c_n x^n$ converges for $|x| < R$. Then f is differentiable on $(-R, R)$, the differentiated series*

$$\sum_{n=1}^{\infty} n c_n x^{n-1}$$

has the same radius R , and $f'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$ there.

Proof. Since $n^{1/n} \rightarrow 1$, $\limsup |n c_n|^{1/n} = \limsup |c_n|^{1/n} = 1/R$, so the differentiated series has radius R too; call its sum $g(x)$. Fix $|x| < R$ and choose ρ with $|x| < \rho < R$. For $|y| \leq \rho$, $y \neq x$, the difference quotients

$$\frac{f(y) - f(x)}{y - x} = \sum_{n=1}^{\infty} c_n \frac{y^n - x^n}{y - x} = \sum_{n=1}^{\infty} c_n (y^{n-1} + y^{n-2}x + \cdots + x^{n-1})$$

are bounded term-by-term by $|c_n| n\rho^{n-1}$, the general term of a convergent series (radius $R > \rho$). By the M -test the series of difference quotients converges uniformly in y on $|y| \leq \rho$, so we may pass $y \rightarrow x$ term by term: each summand tends to nc_nx^{n-1} , giving $f'(x) = g(x)$. $\square$

Corollary 9.1.5 (Infinite differentiability and Taylor coefficients, Rudin 8.4). *A power series $f(x) = \sum c_n x^n$ with $R > 0$ has derivatives of every order on $(-R, R)$, each obtained by differentiating term by term, and*

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1) c_n x^{n-k}, \quad c_k = \frac{f^{(k)}(0)}{k!}.$$

In particular the coefficients are determined by f : the series is the Taylor series of its own sum about 0.

Proof. Apply Theorem 9.1.4 repeatedly; each differentiation preserves R . Setting $x = 0$ in the formula for $f^{(k)}$ kills every term except $n = k$, whose value is $k! c_k$, so $c_k = f^{(k)}(0)/k!$. $\square$

Method (Computing $f^{(k)}(0)$ from a power series). *Recognition trigger.* A function is given (or shown to equal) a power series $\sum c_n x^n$ and a particular derivative value $f^{(k)}(0)$ is wanted.

1. Put the function into the form $\sum c_n x^n$ (expand, substitute, multiply, or use a known series such as e^x , $\frac{1}{1-x}$, $\log(1+x)$).
2. Read off the coefficient c_k of x^k .
3. Then $f^{(k)}(0) = k! c_k$. (No actual differentiation needed.)

Worked Example 9.1.6 (A derivative value with no differentiation). Let $f(x) = \frac{x}{1-x^2}$. Find $f^{(7)}(0)$ and $f^{(6)}(0)$.

Recognition. “Find a high-order derivative at 0”: expand and use $f^{(k)}(0) = k! c_k$.

Solution. $\frac{x}{1-x^2} = x \sum_{m \geq 0} x^{2m} = \sum_{m \geq 0} x^{2m+1} = x + x^3 + x^5 + x^7 + \cdots$, valid for $|x| < 1$. Only odd powers appear, each with coefficient 1. So $c_7 = 1$ gives $f^{(7)}(0) = 7! = 5040$, while $c_6 = 0$ gives $f^{(6)}(0) = 0$. *Check.* f is odd, hence all even-order derivatives vanish at 0 — consistent with $f^{(6)}(0) = 0$.

Identity theorem and Abel’s theorem

Theorem 9.1.7 (Identity theorem, Rudin 8.5). *Suppose $\sum a_n x^n$ and $\sum b_n x^n$ both converge on $(-R, R)$ to the same function on a set $E \subseteq (-R, R)$ having a limit point at 0 (e.g. $x_k \rightarrow 0$, $x_k \in E$, $x_k \neq 0$). Then $a_n = b_n$ for all n . In particular a function has at most one power-series representation about a given center.*

Proof. Let $c_n = a_n - b_n$ and $f = \sum c_n x^n$, so $f = 0$ on E . By continuity (Thm. 9.1.3) $f(0) = \lim_k f(x_k) = 0$, so $c_0 = 0$. Then $f(x)/x = \sum_{n \geq 1} c_n x^{n-1}$ is again a power series, continuous, vanishing on $E \setminus \{0\}$ and hence at 0 by the same limit, so $c_1 = 0$. Inductively every $c_n = 0$. $\square$

Theorem 9.1.8 (Abel’s theorem, Rudin 8.2). *Suppose $\sum c_n$ converges (say to s) and $f(x) = \sum c_n x^n$ for $-1 < x < 1$. Then*

$$\lim_{x \rightarrow 1^-} f(x) = \sum_{n=0}^{\infty} c_n = s.$$

Proof. Set $s_n = c_0 + \cdots + c_n$, $s_{-1} = 0$, so $c_n = s_n - s_{n-1}$. Abel (partial) summation gives, for $0 \leq x < 1$ and any N ,

$$\sum_{n=0}^N c_n x^n = (1-x) \sum_{n=0}^{N-1} s_n x^n + s_N x^N.$$

Letting $N \rightarrow \infty$ (with $|x| < 1$, $s_N x^N \rightarrow 0$ since $s_N \rightarrow s$ is bounded),

$$f(x) = (1-x) \sum_{n=0}^{\infty} s_n x^n.$$

Using $(1-x)\sum x^n = 1$, write $f(x) - s = (1-x)\sum_{n\geq 0}(s_n - s)x^n$. Given $\varepsilon > 0$ pick M with $|s_n - s| < \varepsilon/2$ for $n > M$. Then

$$|f(x) - s| \leq (1-x) \sum_{n=0}^M |s_n - s| x^n + \frac{\varepsilon}{2}(1-x) \sum_{n>M} x^n.$$

The second piece is $< \varepsilon/2$. The first is $(1-x)$ times a fixed finite sum, hence $< \varepsilon/2$ for x close enough to 1. Thus $|f(x) - s| < \varepsilon$ near $x = 1^-$. $\square$

Worked Example 9.1.9 (Abel summing the alternating harmonic and Leibniz series). Evaluate $\sum_{n\geq 1} \frac{(-1)^{n-1}}{n}$ and $\sum_{n\geq 0} \frac{(-1)^n}{2n+1}$.

Recognition. “Sum a conditionally convergent series whose generating function is known”: form the power series, identify the closed form on $(-1, 1)$, then let $x \rightarrow 1^-$ (Abel).

Solution. For the first, $\sum_{n\geq 1} \frac{(-1)^{n-1}}{n} x^n = \log(1+x)$ on $(-1, 1)$, and $\sum \frac{(-1)^{n-1}}{n}$ converges (alternating). By Abel, $\sum \frac{(-1)^{n-1}}{n} = \lim_{x\rightarrow 1^-} \log(1+x) = \log 2$. For the second, $\sum_{n\geq 0} \frac{(-1)^n}{2n+1} x^{2n+1} = \arctan x$, and the numeric series converges, so it equals $\lim_{x\rightarrow 1^-} \arctan x = \arctan 1 = \pi/4$. *Check.* Both are classic identities, $\log 2 \approx 0.693$ and $\pi/4 \approx 0.785$, matching slowly-converging partial sums.

Remark (Analytic functions). A function f on an open set is *analytic* (real-analytic) if near each point it equals a convergent power series. By Corollary 9.1.5 analytic functions are C^∞ and equal their Taylor series locally. The converse fails over $\mathbb{R}$: the function $f(x) = e^{-1/x^2}$ ($f(0) = 0$) is C^∞ with $f^{(n)}(0) = 0$ for all n , so its Taylor series is $0 \neq f$ — f is smooth but *not* analytic at 0 (Exercise R8.1). Over $\mathbb{C}$ the two notions coincide, which is one reason complex analysis is so rigid.

Further worked examples

Worked Example 9.1.10 (Radius from a boundary that mixes ratio and root). Find the radius of convergence of (a) $\sum \frac{(2n)!}{(n!)^2} z^n$, (b) $\sum \frac{n^n}{n!} z^n$.

Recognition. “Compute a radius”: factorials and n^n both surrender to the ratio test, with $\lim |c_{n+1}/c_n| = 1/R$.

Solution. (a) $\frac{c_{n+1}}{c_n} = \frac{(2n+2)!}{((n+1)!)^2} \cdot \frac{(n!)^2}{(2n)!} = \frac{(2n+1)(2n+2)}{(n+1)^2} = \frac{2(2n+1)}{n+1} \rightarrow 4$, so $R = \frac{1}{4}$. (b) $\frac{c_{n+1}}{c_n} = \frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n} = \frac{(n+1)^n}{n^n} = \left(1 + \frac{1}{n}\right)^n \rightarrow e$, so $R = 1/e$. *Check.* For (a) $c_n = \binom{2n}{n} \sim 4^n/\sqrt{\pi n}$, whose n -th root tends to $4 = 1/R$; for (b) Stirling gives $n^n/n! \sim e^n/\sqrt{2\pi n}$, n -th root $\rightarrow e = 1/R$. ■

Worked Example 9.1.11 (A non-example: the boundary cannot be read off from R). Show that the three series $\sum z^n$, $\sum z^n/n$, $\sum z^n/n^2$ all have $R = 1$ yet behave differently at the boundary point $z = 1$.

Recognition. “ R is silent on $|z| = R$ ”: compute R once, then test the single point $z = 1$ by hand.

Solution. Each has $|c_n|^{1/n} \rightarrow 1$ (since $n^{1/n} \rightarrow 1$ and $(n^2)^{1/n} \rightarrow 1$), so all three share $R = 1$. At $z = 1$: $\sum 1$ diverges (terms do not tend to 0); $\sum 1/n$ diverges (harmonic); $\sum 1/n^2$ converges (a p -series, $p = 2$). So a common radius gives no common verdict on the circle. *Check.* This is precisely the warning after Theorem 9.1.1; R locates the disc, never decides the boundary. ■

Worked Example 9.1.12 (A Cauchy product of two geometric series). Use the Cauchy product to expand $\frac{1}{(1-x)^2}$ as a power series and identify the coefficients.

Recognition. “Multiply two absolutely convergent power series”: inside the common disc the Cauchy product is valid (as in the addition formula for exp).

Solution. For $|x| < 1$, $\frac{1}{1-x} = \sum_{n\geq 0} x^n$. Squaring via the Cauchy product,

$$\frac{1}{(1-x)^2} = \left(\sum_{k\geq 0} x^k\right) \left(\sum_{m\geq 0} x^m\right) = \sum_{n\geq 0} \left(\sum_{k=0}^n 1\right) x^n = \sum_{n\geq 0} (n+1) x^n.$$

So the coefficient of x^n is $n + 1$. *Check.* Differentiating $\frac{1}{1-x} = \sum x^n$ term by term (Thm. 9.1.4) gives $\frac{1}{(1-x)^2} = \sum_{n \geq 1} nx^{n-1} = \sum_{m \geq 0} (m+1)x^m$ — the same series. ■

Exercises (Rudin, Ch.8).

1. Define $f(x) = e^{-1/x^2}$ for $x \neq 0$ and $f(0) = 0$. Prove that f has derivatives of all orders at $x = 0$ and that $f^{(n)}(0) = 0$ for $n = 1, 2, 3, \dots$ (Thus the Taylor series of f at 0 is identically 0, yet $f \neq 0$.)
2. Let a_{ij} be the entry in the i -th row, j -th column of the array

$$a_{ij} = \begin{cases} -1, & i < j, \\ 0, & i = j, \\ 2^{j-i}, & i > j. \end{cases}$$

Prove that $\sum_i \sum_j a_{ij} = -2$ while $\sum_j \sum_i a_{ij} = 0$. (The two iterated sums of a non-absolutely-summable double array need not agree.)

3. Prove that $\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij}$ whenever $a_{ij} \geq 0$ for all i, j (the value $+\infty$ being allowed).
4. Prove the limit relations: (a) $\lim_{x \rightarrow 0} \frac{b^x - 1}{x} = \log b$ for $b > 0$; (b) $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$; (c) $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$; (d) $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$.
5. Find the limits: (a) $\lim_{x \rightarrow 0} \frac{e - (1+x)^{1/x}}{x}$; (b) $\lim_{n \rightarrow \infty} \frac{n}{\log n} (n^{1/n} - 1)$; (c) $\lim_{x \rightarrow 0} \frac{\tan x - x}{x(1 - \cos x)}$; (d) $\lim_{x \rightarrow 0} \frac{x - \sin x}{\tan x - x}$.
6. Suppose $f(x)f(y) = f(x+y)$ for all real x, y . (a) Assuming f differentiable and not identically 0, prove $f(x) = e^{cx}$ for some constant c . (b) Prove the same assuming only that f is continuous.
7. If $0 < x < \frac{\pi}{2}$, prove that $\frac{2}{\pi} < \frac{\sin x}{x} < 1$.
8. For $n = 0, 1, 2, \dots$ and x real, prove $|\sin nx| \leq n |\sin x|$, and note that this inequality can fail for non-integer n (e.g. $n = \frac{1}{2}$).
9. (a) Put $s_N = 1 + \frac{1}{2} + \dots + \frac{1}{N}$. Prove that $\lim_{N \rightarrow \infty} (s_N - \log N)$ exists (its value is *Euler's constant* $\gamma \approx 0.5772$; whether γ is rational is unknown). (b) Roughly how large must N be so that $s_N > 100$?
10. Prove that $\sum_p 1/p$ diverges, the sum extending over all primes p . (Hence the primes form a fairly substantial subset of $\mathbb{N}$.)
11. Suppose $f \in \mathcal{R}$ on $[0, A]$ for every $A < \infty$, and $f(x) \rightarrow 1$ as $x \rightarrow \infty$. Prove that $\lim_{t \rightarrow 0^+} t \int_0^\infty e^{-tx} f(x) dx = 1$.

Additional exercises.

1. (Radius drill.) Find the radius of convergence of each: (a) $\sum n^2 z^n$; (b) $\sum \frac{z^n}{2^n}$; (c) $\sum \frac{z^n}{n^n}$; (d) $\sum \frac{(n!) z^n}{n^n}$; (e) $\sum 3^n z^{2n}$.
2. Using $\frac{1}{1-x} = \sum_{n \geq 0} x^n$ on $(-1, 1)$, derive power series for $\frac{1}{(1-x)^2}$ and $\frac{x}{(1+x^2)}$, stating each radius.
3. Compute $f^{(10)}(0)$ for $f(x) = \frac{1}{1+x^2}$ and $f^{(11)}(0)$ for the same f , without differentiating, by expanding in a power series.
4. Use Abel's theorem to evaluate $\sum_{n \geq 1} \frac{(-1)^{n-1}}{n}$ and $\sum_{n \geq 0} \frac{(-1)^n}{2n+1}$, identifying the generating function in each case.

5. (Non-example.) Exhibit a power series with radius $R = 1$ that *diverges* at every point of the circle $|z| = 1$, and another (still $R = 1$) that *converges* at every point of the circle. Conclude that R alone says nothing about boundary behaviour.
6. (Non-example, smooth $\neq$ analytic.) For $g(x) = e^{-1/x^2}$ ($g(0) = 0$), state its Taylor series at 0 and explain in one line why g cannot be recovered from it on any interval about 0. Contrast with e^x .
7. (Identity theorem in action.) Suppose a power series $\sum c_n x^n$ ($R > 0$) vanishes at $x = 1/2, 1/3, 1/4, \dots$. Prove every $c_n = 0$. Where is the limit-point hypothesis used?
8. (Diagram.) On the complex plane draw the disc of convergence of $\sum z^n/2^n$. Mark the radius, shade where the series converges absolutely, and indicate one boundary point where it diverges; justify that boundary point.
9. (Cauchy product / squaring.) Show $\left(\sum_{n \geq 0} x^n\right)^2 = \sum_{n \geq 0} (n+1)x^n$ on $(-1, 1)$ two ways: by the Cauchy product, and by differentiating $\frac{1}{1-x}$.
10. (Harder.) Prove that if $\sum c_n$ and $\sum d_n$ converge and their Cauchy product $\sum e_n$ converges, then $\sum e_n = (\sum c_n)(\sum d_n)$. (Hint: Abel's theorem applied to $f(x)g(x) = h(x)$ for the three power series, letting $x \rightarrow 1^-$.)

9.2 The Exponential and Logarithmic Functions

The exponential defined by its series

We define the exponential as a power series and *derive* everything — the functional equation, the derivative, the number e , the logarithm — from that one definition. This is Rudin's route (8.6). Bartle–Sherbert instead characterize E as the unique solution of $E' = E$, $E(0) = 1$; we present both and show they agree.

Definition 9.2.1. For $z \in \mathbb{C}$ define

$$E(z) = \exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

By Worked Example 9.1.4 this series has radius $R = \infty$, so E is defined and (on $\mathbb{R}$) infinitely differentiable everywhere.

Theorem 9.2.2 (Addition formula, Rudin 8.6). *For all $z, w \in \mathbb{C}$,*

$$E(z+w) = E(z)E(w).$$

In particular $E(z)E(-z) = E(0) = 1$, so $E(z) \neq 0$ for every z and $E(-z) = 1/E(z)$.

Proof. Both series converge absolutely, so their Cauchy product is legitimate:

$$E(z)E(w) = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{z^k}{k!} \frac{w^{n-k}}{(n-k)!} = \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^n \binom{n}{k} z^k w^{n-k} = \sum_{n=0}^{\infty} \frac{(z+w)^n}{n!} = E(z+w),$$

using the binomial theorem in the middle. Setting $w = -z$ gives $E(z)E(-z) = E(0) = 1$. □

Theorem 9.2.3 (Calculus of exp, Rudin 8.6 / B&S 8.3.2). *On $\mathbb{R}$, E is differentiable with $E' = E$; more generally $E^{(n)} = E$ for all n . It is strictly increasing, positive, with $E(x) \rightarrow +\infty$ as $x \rightarrow +\infty$ and $E(x) \rightarrow 0$ as $x \rightarrow -\infty$. The number $e := E(1) = \sum 1/n!$ satisfies $E(n) = e^n$ and $E(x) = e^x$ for rational x .*

Proof. Term-by-term differentiation (Thm. 9.1.4) of $\sum z^n/n!$ gives $\sum_{n \geq 1} \frac{nz^{n-1}}{n!} = \sum_{n \geq 1} \frac{z^{n-1}}{(n-1)!} = E(z)$, so $E' = E$ and hence $E^{(n)} = E$. For real $x > 0$ every term of the series is positive, so $E(x) > 1 + x > 0$; combined with $E(-x) = 1/E(x)$ this gives $E > 0$ everywhere. As $E' = E > 0$, E is strictly increasing. From $E(x) > 1 + x$ we get $E(x) \rightarrow +\infty$ as $x \rightarrow +\infty$, and then $E(x) = 1/E(-x) \rightarrow 0$ as $x \rightarrow -\infty$. By the addition formula $E(n) = E(1)^n = e^n$ and $E(p/q)^q = E(p) = e^p$, so $E(p/q) = e^{p/q}$; we therefore *define* $e^x := E(x)$ for all real x , consistent with the rational case. □

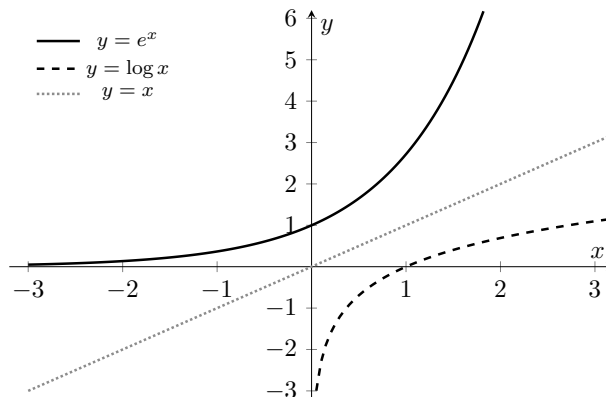


Figure 9.2. $\log$ is the reflection of $\exp$ across the line $y = x$: each is the inverse of the other.

Growth rates

Theorem 9.2.4 (Exp beats every power, Rudin 8.6(f)). *For every integer $n \geq 0$,*

$$\lim_{x \rightarrow +\infty} \frac{x^n}{e^x} = 0.$$

That is, e^x grows faster than any polynomial; equivalently $\log x$ grows slower than any positive power x^α .

Proof. Since $e^x = \sum_{k \geq 0} x^k/k!$, for $x > 0$ the single term $k = n+1$ gives $e^x > \frac{x^{n+1}}{(n+1)!}$, so $\frac{x^n}{e^x} < \frac{(n+1)!}{x} \rightarrow 0$.

Putting $x = \alpha \log t$ converts this into $\frac{(\log t)^n}{t^\alpha} \rightarrow 0$ for every $\alpha > 0$. $\square$

The logarithm

Definition 9.2.5 (B&S 8.3.8). Because $E : \mathbb{R} \rightarrow (0, \infty)$ is a strictly increasing continuous bijection, it has an inverse

$$L = \log : (0, \infty) \rightarrow \mathbb{R}, \quad L(E(x)) = x, \quad E(L(y)) = y.$$

L is the *natural logarithm*.

Theorem 9.2.6 (Properties of $\log$, B&S 8.3.9). *L is strictly increasing, differentiable on $(0, \infty)$ with $L'(y) = 1/y$, satisfies the functional equation*

$$L(yw) = L(y) + L(w) \quad (y, w > 0),$$

$$L(1) = 0, \quad L(e) = 1, \quad \text{and} \quad L(y) = \int_1^y \frac{dt}{t}.$$

Proof. L is increasing as the inverse of an increasing bijection. By the inverse-function rule, $L'(y) = 1/E'(L(y)) = 1/E(L(y)) = 1/y$. For the functional equation, apply L to $E(L(y) + L(w)) = E(L(y))E(L(w)) = yw$. Since $L(1) = 0$ and $L' = 1/t$, the Fundamental Theorem of Calculus gives $L(y) = \int_1^y dt/t$. $\square$

Method (Reducing power/log identities to the exponential). *Recognition trigger.* An identity about $\log$, a^x , or general powers x^α is to be proved.

1. Define $a^x := E(xL(a)) = e^{x \log a}$ for $a > 0$ and $x^\alpha := E(\alpha L(x))$ for $x > 0$ (B&S 8.3.10).
2. Rewrite each side using E and L , then push the problem through the addition formula $E(u+v) = E(u)E(v)$ and $L(yw) = L(y) + L(w)$.
3. Differentiate via $L'(y) = 1/y$ and the chain rule; e.g. $Dx^\alpha = \alpha x^{\alpha-1}$ falls out of $DE(\alpha L(x)) = E(\alpha L(x)) \cdot \alpha/x$ (B&S 8.3.13).

Worked Example 9.2.7 (The two compound-interest limits for e). Show $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$ and, more generally, $\lim_{n \rightarrow \infty} (1 + \frac{x}{n})^n = e^x$.

Recognition. “Identify a limit of $(1 + x/n)^n$ with e^x ”: take logs and use $L'(1) = 1$.

Solution. Let $a_n = (1 + \frac{x}{n})^n$. Then $L(a_n) = nL(1 + \frac{x}{n})$. Writing $h = x/n \rightarrow 0$, $nL(1 + h) = x \cdot \frac{L(1+h) - L(1)}{h} \rightarrow xL'(1) = x$, since $L'(1) = 1$. By continuity of E , $a_n = E(L(a_n)) \rightarrow E(x) = e^x$. Taking $x = 1$ gives the first limit. *Check.* Numerically $(1.01)^{100} \approx 2.7048 \rightarrow e \approx 2.71828$. ■

Worked Example 9.2.8 (A non-example: a C^∞ function that is not its Taylor series). Show $g(x) = e^{-1/x^2}$ ($g(0) = 0$) is smooth with every derivative 0 at the origin, so it is *not* the sum of its Taylor series there.

Recognition. “Smooth but not analytic”: use exp’s domination of powers to kill all derivatives at 0.

Solution. For $x \neq 0$, induction shows $g^{(n)}(x) = P_n(1/x)e^{-1/x^2}$ for some polynomial P_n . By Theorem 9.2.4, $t^k e^{-t^2} \rightarrow 0$ as $t \rightarrow \infty$, so each $g^{(n)}(x) \rightarrow 0$ as $x \rightarrow 0$; a standard limit argument then gives $g^{(n)}(0) = 0$ for all n . Hence the Taylor series of g at 0 is identically 0, while $g(x) > 0$ for $x \neq 0$. The two disagree on every punctured neighborhood. *Check.* This is the canonical smooth bump ingredient; smoothness does not imply analyticity over $\mathbb{R}$.

Further worked examples

Worked Example 9.2.9 (Bounding e between two rationals from the series). Using $e = \sum_{n \geq 0} 1/n!$, show $2.5 < e < 3$, and bound the truncation error after N terms.

Recognition. “Estimate e from its defining series”: partial sums give a lower bound; a geometric tail gives the matching upper bound.

Solution. The partial sum $1 + 1 + \frac{1}{2} = 2.5$ already gives $e > 2.5$ (all later terms are positive). For the upper bound, for $n \geq 2$ we have $n! \geq 2^{n-1}$, so the tail beyond the first two terms satisfies

$$e - 2 = \sum_{n \geq 2} \frac{1}{n!} \leq \sum_{n \geq 2} \frac{1}{2^{n-1}} = \frac{1/2}{1 - 1/2} = 1,$$

hence $e \leq 3$, and in fact $e < 3$ since the inequality $n! \geq 2^{n-1}$ is strict for $n \geq 3$. More generally the remainder after N terms obeys $\sum_{n > N} \frac{1}{n!} \leq \frac{1}{(N+1)!} \cdot \frac{1}{1 - 1/(N+2)} < \frac{2}{(N+1)!}$. *Check.* $N = 5$ gives error $< 2/720 < 0.003$, and indeed $\sum_0^5 1/n! = 2.716$ versus $e \approx 2.71828$. ■

Worked Example 9.2.10 (log as an area: monotonicity bounds for the harmonic numbers). From $\log y = \int_1^y dt/t$, prove $\frac{1}{n+1} < \log(1 + \frac{1}{n}) < \frac{1}{n}$ and deduce $\log(n+1) < 1 + \frac{1}{2} + \cdots + \frac{1}{n} < 1 + \log n$.

Recognition. “Turn a log identity into estimates”: bound the integrand $1/t$ above and below on the interval of integration.

Solution. On $[n, n+1]$ the decreasing integrand $1/t$ satisfies $\frac{1}{n+1} \leq \frac{1}{t} \leq \frac{1}{n}$, with strict inequality except at endpoints. Integrating over the unit-length interval,

$$\frac{1}{n+1} < \int_n^{n+1} \frac{dt}{t} = \log(n+1) - \log n = \log\left(1 + \frac{1}{n}\right) < \frac{1}{n}.$$

Summing the left inequality for $k = 1, \dots, n$ telescopes to $\log(n+1) < \sum_{k=1}^n \frac{1}{k}$; summing the right inequality $\log(1 + \frac{1}{k}) < \frac{1}{k}$ for $k = 1, \dots, n-1$ gives $\log n < \sum_{k=1}^{n-1} \frac{1}{k}$, i.e. $\sum_{k=1}^n \frac{1}{k} < 1 + \log n$. *Check.* At $n = 4$, $\log 5 \approx 1.609 < 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = 2.083 < 1 + \log 4 \approx 2.386$. ■

Worked Example 9.2.11 (A growth-rate limit and a non-example). Evaluate $\lim_{x \rightarrow \infty} \frac{(\log x)^3}{x^{0.01}}$, and then

explain why $\lim_{x \rightarrow \infty} \frac{x^{10}}{e^{\sqrt{x}}}$ is *not* settled by “ e^x beats every power” as stated, yet still equals 0.

Recognition. “Compare logs/powers/exponentials”: reduce each to the master limit $x^n/e^x \rightarrow 0$ (Thm. 9.2.4) by a substitution.

Solution. For the first, put $x = e^u$ so $\log x = u \rightarrow \infty$ and the ratio is $u^3/e^{0.01u} = 10^6 \cdot (0.01u)^3/e^{0.01u}$ const form $\rightarrow 0$ by Thm. 9.2.4; so the limit is 0. The second has $e^{\sqrt{x}}$, not e^x , so the theorem does not apply *directly*. Substitute $s = \sqrt{x}$, $x = s^2$: the ratio becomes $s^{20}/e^s \rightarrow 0$ again by Thm. 9.2.4 with $n = 20$. *Check.* Both limits are 0; the point is that any *positive* power of log over any *positive* power of x tends to 0, and any power over $e^{(\text{any positive power})}$ does too — after the right substitution. ■

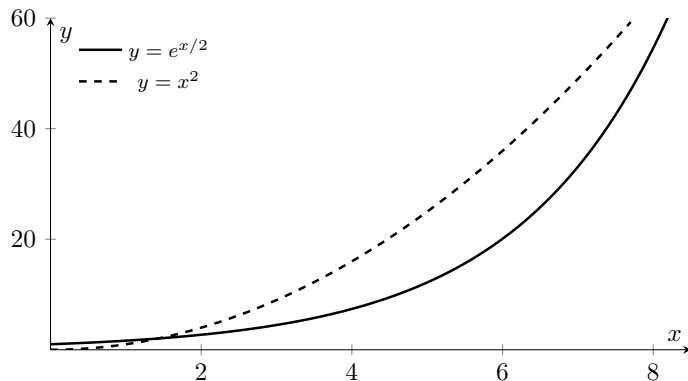


Figure 9.3. Even the slow exponential $e^{x/2}$ eventually overtakes the power x^2 and then pulls away without bound: $x^n/e^x \rightarrow 0$ for every fixed n (Theorem 9.2.4).

Exercises (Bartle & Sherbert, §8.3).

1. Show that if $x > 0$ and $n > 2x$, then $\left| e^x - \left(1 + \frac{x}{1!} + \cdots + \frac{x^n}{n!}\right) \right| < \frac{2x^{n+1}}{(n+1)!}$. Use this to show $2\frac{2}{3} < e < 2\frac{3}{4}$, hence e is not an integer.
2. Calculate e correct to five decimal places.
3. Show that if $0 \leq x \leq a$ and $n \in \mathbb{N}$, then $1 + \frac{x}{1!} + \cdots + \frac{x^n}{n!} \leq e^x \leq 1 + \frac{x}{1!} + \cdots + \frac{x^{n-1}}{(n-1)!} + \frac{e^a x^n}{n!}$.
4. Show that if $n \geq 2$ then $0 < e - \left(1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{n!}\right) < \frac{1}{n!} \cdot \frac{e}{n+1} < \frac{1}{n \cdot n!}$, and use this to prove that e is irrational.
5. If $x \geq 0$ and $n \in \mathbb{N}$, show $\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-x)^{n-1} + \frac{(-x)^n}{x+1}$, and deduce $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1} \frac{x^n}{n} + \int_0^x \frac{(-t)^n}{1+t} dt$, with $\left| \log(1+x) - \left(x - \frac{x^2}{2} + \cdots + (-1)^{n-1} \frac{x^n}{n}\right) \right| \leq \frac{x^{n+1}}{n+1}$.
6. Use the preceding exercise to calculate $\log 1.1$ and $\log 1.4$ accurate to four decimal places. How large must n be to calculate $\log 2$ accurate to four decimals?
7. Show that $\log(e/2) = 1 - \log 2$. Use this to calculate $\log 2$ accurate to four decimal places.
8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f'(x) = f(x)$ for all x . Show there exists $K \in \mathbb{R}$ with $f(x) = Ke^x$ for all x .
9. Let $a_k > 0$ for $k = 1, \dots, n$ and let $A = (a_1 + \cdots + a_n)/n$ be their arithmetic mean. Setting $x_k = a_k/A - 1$ in the inequality $1 + x \leq e^x$ and multiplying, prove the *Arithmetic-Geometric Mean Inequality* $(a_1 \cdots a_n)^{1/n} \leq \frac{1}{n}(a_1 + \cdots + a_n)$, with equality iff $a_1 = \cdots = a_n$.
10. Evaluate $L'(1)$ by using the sequence $(1 + 1/n)$ and the fact that $e = \lim(1 + 1/n)^n$.
11. Establish the assertions in Theorem 8.3.11 (the laws $x^a x^b = x^{a+b}$, $(x^a)^b = x^{ab}$, $(xy)^a = x^a y^a$, and $x^a < x^b$ when $x > 1$, $a < b$, for $x, y > 0$).
12. Establish the assertions in Theorem 8.3.12 (further power laws and the comparison $a^x < b^x$ for $0 < a < b$, $x > 0$).
13. (a) Show that if $a > 0$ then $x \mapsto x^a$ is strictly increasing on $(0, \infty)$ with $\lim_{x \rightarrow 0^+} x^a = 0$ and $\lim_{x \rightarrow \infty} x^a = \infty$.
(b) Show that if $a < 0$ it is strictly decreasing with $\lim_{x \rightarrow 0^+} x^a = \infty$ and $\lim_{x \rightarrow \infty} x^a = 0$.

14. Prove that if $a > 0$, $a \neq 1$, then $a^{\log_a x} = x$ for all $x \in (0, \infty)$ and $\log_a(a^y) = y$ for all $y \in \mathbb{R}$; hence $x \mapsto \log_a x$ is inverse to $y \mapsto a^y$.
15. If $a > 0$, $a \neq 1$, show $x \mapsto \log_a x$ is differentiable on $(0, \infty)$ with $D \log_a x = 1/(x \log a)$.
16. If $a > 0$, $a \neq 1$, and $x, y \in (0, \infty)$, prove $\log_a(xy) = \log_a x + \log_a y$.
17. If $a > 0$, $a \neq 1$ and $b > 0$, $b \neq 1$, show $\log_a x = \frac{\log b}{\log a} \log_b x$ for $x \in (0, \infty)$; in particular $\log_{10} x = (\log_{10} e) \log x$.

Additional exercises.

1. From the series, verify $\exp(0) = 1$ and $\exp'(x) = \exp(x)$ directly, and compute $\exp(1)$ to three decimals by partial sums.
2. Prove $e^x > 1 + x$ for all $x \neq 0$, and deduce $e^x \geq 1 + x$ with equality only at $x = 0$.
3. Using $\log'(y) = 1/y$, prove $\log(yw) = \log y + \log w$ for $y, w > 0$ by differentiating $\log(yw) - \log y$ in w .
4. Show $\lim_{x \rightarrow \infty} \frac{\log x}{x^\alpha} = 0$ for every $\alpha > 0$ and $\lim_{x \rightarrow 0^+} x^\alpha \log x = 0$, using that $\exp$ beats every power.
5. (Non-example.) The functional equation $f(x+y) = f(x)f(y)$ has the continuous solution $f = e^{cx}$. Exhibit (assuming a Hamel basis / axiom of choice) the *existence* of a wildly discontinuous solution, and explain why continuity in Rudin Exercise 8.6(b) is essential.
6. Define $a^x = e^{x \log a}$ for $a > 0$. Prove $a^{x+y} = a^x a^y$, $(a^x)^y = a^{xy}$, and $D_x a^x = a^x \log a$.
7. (Diagram.) On one set of axes, sketch $y = e^x$ and $y = \log x$ together with the line $y = x$, marking the points $(0, 1)$, $(1, 0)$, and $(e, 1)$, $(1, e)$ that exhibit the reflection symmetry. State why the two curves never meet.
8. Show that $\frac{x^n}{n!} \rightarrow 0$ as $n \rightarrow \infty$ for every fixed x , directly from the ratio of consecutive terms, and explain how this underlies convergence of the $\exp$ series.
9. (Harder.) Prove that $n!e = \lceil n!e \rceil$ is impossible for $n \geq 1$ by the tail estimate $0 < e - \sum_{k=0}^n \frac{1}{k!} < \frac{1}{n \cdot n!}$, recovering the irrationality of e .

9.3 The Trigonometric Functions

Sine and cosine from the exponential

We continue the analytic program: $\sin$ and $\cos$ are *defined* by power series (or, equivalently, by the complex exponential), and every familiar identity is derived. This is Rudin 8.7; Bartle–Sherbert (8.4) instead build C, S as the unique solutions of $y'' = -y$ with matched initial data, and we connect the two.

Definition 9.3.1. For $x \in \mathbb{R}$ put

$$C(x) = \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad S(x) = \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}.$$

Both have radius $R = \infty$. Equivalently, for real x ,

$$E(ix) = \cos x + i \sin x, \quad \cos x = \frac{1}{2}(E(ix) + E(-ix)), \quad \sin x = \frac{1}{2i}(E(ix) - E(-ix)).$$

The identities $E(ix) = \cos x + i \sin x$ follow by splitting $\sum (ix)^n/n!$ into even and odd n : even $n = 2k$ contribute $i^{2k} = (-1)^k$ (the real cosine terms), odd $n = 2k + 1$ contribute $i(-1)^k$ (the imaginary sine terms).

Theorem 9.3.2 (Basic identities, Rudin 8.7 / B&S 8.4). *For all real x, y :*

1. $C' = -S$ and $S' = C$; both satisfy $y'' = -y$, with $C(0) = 1$, $C'(0) = 0$ and $S(0) = 0$, $S'(0) = 1$.

2. (Pythagoras) $C(x)^2 + S(x)^2 = 1$; hence $|C(x)| \leq 1$, $|S(x)| \leq 1$.

3. (Even/odd) $C(-x) = C(x)$, $S(-x) = -S(x)$.

4. (Addition) $C(x+y) = C(x)C(y) - S(x)S(y)$, $S(x+y) = S(x)C(y) + C(x)S(y)$.

Proof. (1) Differentiate the series term by term. (3) is visible from the series (even/odd powers). (4): expand $E(i(x+y)) = E(ix)E(iy)$ via the addition formula and compare real and imaginary parts. (2): take $y = -x$ in the cosine addition formula, $C(0) = C(x)C(-x) - S(x)S(-x) = C(x)^2 + S(x)^2$, and $C(0) = 1$. Alternatively $\frac{d}{dx}(C^2 + S^2) = 2CC' + 2SS' = 2C(-S) + 2SC = 0$, so $C^2 + S^2$ is the constant $C(0)^2 + S(0)^2 = 1$. $\square$

Remark (The B&S existence argument). Bartle–Sherbert prove from scratch that functions C, S with $C'' = -C$, $S'' = -S$ and the stated initial data *exist and are unique*, by the Picard-type integral recursion $s_{n+1}(x) = \int_0^x c_n$, $c_{n+1}(x) = 1 - \int_0^x s_n$ (B&S 8.4.1, our Exercise B8.4.6). The series above are exactly the limits of that recursion, so the two constructions coincide.

Defining π analytically

Theorem 9.3.3 (cos has a least positive zero, Rudin 8.7 / B&S 8.4.9). *C has a smallest positive root. Define π by declaring this root to be $\pi/2$: $\pi/2 := \min\{x > 0 : C(x) = 0\}$.*

Proof. $C(0) = 1 > 0$. We show $C(2) < 0$. Group the alternating series at $x = 2$:

$$C(2) = 1 - \frac{2^2}{2!} + \frac{2^4}{4!} - \frac{2^6}{6!} + \cdots = (1 - 2) + \left(\frac{16}{24} - \frac{64}{720}\right) + \cdots$$

The terms $\frac{2^{4k}}{(4k)!}$ and $\frac{2^{4k+2}}{(4k+2)!}$ pair into negative blocks beyond the first; carefully, $C(2) < 1 - 2 + \frac{2}{3} = -\frac{1}{3} < 0$.

By continuity and the intermediate value theorem C has a root in $(0, 2)$; the set of positive roots is closed and bounded below by 0 (no root in a neighborhood of 0 since $C(0) = 1$), so it has a least element $\pi/2 \in (0, 2)$. $\square$

Theorem 9.3.4 (Values and periodicity, Rudin 8.7 / B&S 8.4.11). *With π as above:*

$$C\left(\frac{\pi}{2}\right) = 0, \quad S\left(\frac{\pi}{2}\right) = 1, \quad E(i\pi/2) = i, \quad E(i\pi) = -1, \quad E(2\pi i) = 1.$$

Consequently $E(z + 2\pi i) = E(z)$ for all z , and C, S have period 2π . On $[0, \pi/2]$, C decreases from 1 to 0 and S increases from 0 to 1.

Proof. On $(0, \pi/2)$ we have $C > 0$, so $S' = C > 0$ and S increases; with $S(0) = 0$ this gives $S > 0$ there. From $C^2 + S^2 = 1$ and $C(\pi/2) = 0$ we get $S(\pi/2)^2 = 1$, and $S(\pi/2) > 0$ forces $S(\pi/2) = 1$. Then $E(i\pi/2) = C(\pi/2) + iS(\pi/2) = i$, so $E(i\pi) = (E(i\pi/2))^2 = i^2 = -1$ and $E(2\pi i) = (E(i\pi))^2 = 1$. The addition formula gives $E(z + 2\pi i) = E(z)E(2\pi i) = E(z)$, hence C, S are 2π -periodic. $\square$

The circle parametrization

Theorem 9.3.5 ($t \mapsto (\cos t, \sin t)$ winds the unit circle, Rudin 8.7). *The map $\gamma(t) = (C(t), S(t)) = E(it)$ maps $\mathbb{R}$ onto the unit circle $\{(x, y) : x^2 + y^2 = 1\}$. On $[0, 2\pi)$ it is a bijection onto the circle, traversed counterclockwise once, and the parameter t equals the arc length from $(1, 0)$.*

Proof. $|E(it)|^2 = C(t)^2 + S(t)^2 = 1$, so $\gamma(t)$ lies on the circle. The velocity is $\gamma'(t) = (-S(t), C(t))$ with $|\gamma'(t)| = 1$, so arc length from 0 to t is $\int_0^t |\gamma'| = t$: the parameter is arc length. Periodicity ($E(2\pi i) = 1$) and the monotone quarter-turn behaviour on $[0, \pi/2]$ (extended by the addition formulas to each quadrant) make γ a bijection of $[0, 2\pi)$ onto the circle, swept once counterclockwise. $\square$

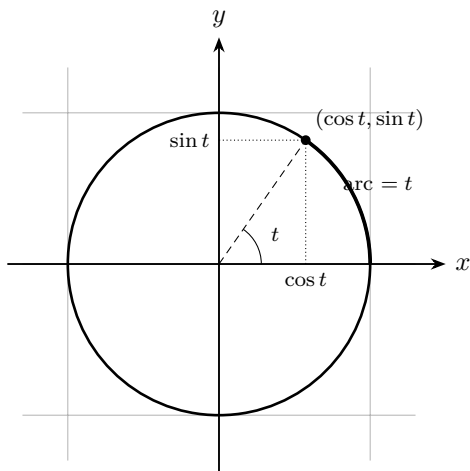


Figure 9.4. The angle t is exactly the arc length from $(1, 0)$; the speed $|\gamma'| = 1$ makes “radian” synonymous with “arc length”.

Worked Example 9.3.6 (De Moivre and a multiple-angle identity). Derive $\cos 3x = 4\cos^3 x - 3\cos x$ from the exponential.

Recognition. “Multiple-angle formula”: raise $E(ix)$ to a power and read off the real part.

Solution. $E(3ix) = (E(ix))^3 = (\cos x + i\sin x)^3$. Expanding,

$$(\cos x + i\sin x)^3 = \cos^3 x + 3i\cos^2 x \sin x - 3\cos x \sin^2 x - i\sin^3 x.$$

Real part: $\cos 3x = \cos^3 x - 3\cos x \sin^2 x = \cos^3 x - 3\cos x(1 - \cos^2 x) = 4\cos^3 x - 3\cos x$. *Check.* At $x = 0$, $\text{RHS} = 4 - 3 = 1 = \cos 0$. ■

Caution. π here is *defined* as twice the least positive zero of $\cos$ (a purely analytic quantity), *not* assumed from geometry. Likewise “radian” is not imported: it is the theorem $|\gamma'| = 1$ that makes the parameter equal arc length. Reasoning that secretly uses “ $\sin x \approx x$ ” or the area of a circle before these are proven is circular.

Further worked examples

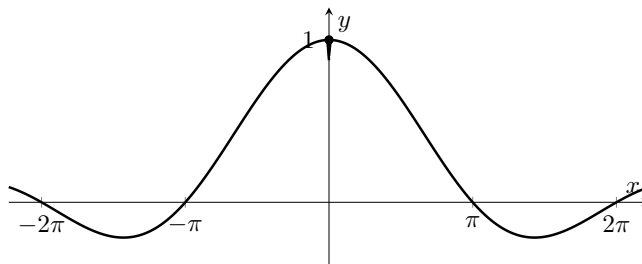


Figure 9.5. The function $\frac{\sin x}{x}$ extends continuously to the value 1 at $x = 0$; dividing the sine series by x leaves the power series $1 - \frac{x^2}{6} + \frac{x^4}{120} - \cdots$, whose value at 0 is 1.

Worked Example 9.3.7 ($\lim_{x \rightarrow 0} \sin x / x = 1$ straight from the series). Prove $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$ without invoking geometry.

Recognition. “A trig limit at 0”: divide the defining power series by the leading power and pass to the limit term by term (legal inside the radius of convergence).

Solution. From Definition 9.3.1, for $x \neq 0$,

$$\frac{\sin x}{x} = \frac{1}{x} \sum_{n \geq 0} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = \sum_{n \geq 0} \frac{(-1)^n x^{2n}}{(2n+1)!} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \cdots,$$

a power series ($R = \infty$) hence continuous, so its value at $x \rightarrow 0$ is the constant term 1. Similarly $1 - \cos x = \sum_{n \geq 1} \frac{(-1)^{n-1} x^{2n}}{(2n)!} = \frac{x^2}{2} - \frac{x^4}{24} + \dots$, so $\frac{1 - \cos x}{x^2} = \frac{1}{2} - \frac{x^2}{24} + \dots \rightarrow \frac{1}{2}$. *Check.* At $x = 0.1$, $\sin(0.1)/0.1 = 0.99833\dots \approx 1$ and $(1 - \cos 0.1)/0.01 = 0.49958\dots \approx \frac{1}{2}$. ■

Worked Example 9.3.8 (A telescoping sum of sines via the product form). Evaluate $\sum_{k=1}^N \sin(kx)$ in closed form for $\sin(x/2) \neq 0$.

Recognition. “Sum a finite trigonometric series”: write $\sin kx = \operatorname{Im} E(ikx)$ and sum a geometric series of complex exponentials.

Solution. $\sum_{k=1}^N E(ikx) = E(ix) \frac{E(iNx) - 1}{E(ix) - 1}$ (geometric, ratio $E(ix) \neq 1$). Multiply numerator and denominator by $E(-ix/2)$ to symmetrize: $E(ix) - 1 = E(ix/2)(E(ix/2) - E(-ix/2)) = E(ix/2) \cdot 2i \sin(x/2)$, and similarly $E(iNx) - 1 = E(iNx/2) \cdot 2i \sin(Nx/2)$. Hence

$$\sum_{k=1}^N E(ikx) = E\left(i \frac{(N+1)x}{2}\right) \frac{\sin(Nx/2)}{\sin(x/2)}.$$

Taking imaginary parts, $\sum_{k=1}^N \sin kx = \frac{\sin(Nx/2) \sin((N+1)x/2)}{\sin(x/2)}$. *Check.* $N = 1$ gives $\frac{\sin(x/2) \sin(x)}{\sin(x/2)} = \sin x$. ■

Worked Example 9.3.9 (A non-example: $\sin$ is not eventually monotone, so it has no inverse on $\mathbb{R}$). Show $\sin$ takes the value 0 infinitely often and is not injective on $\mathbb{R}$, so the symbol $\arcsin$ requires a restricted domain.

Recognition. “Periodicity blocks global invertibility”: use the zeros of S and 2π -periodicity (Thm. 9.3.4) to exhibit repeated values.

Solution. By Theorem 9.3.4, $S(\pi/2) = 1$, $S(\pi) = S(\pi/2 + \pi/2) = C(\pi/2) \cdot 0 + \dots$; more directly $S(0) = 0$ and, since S is 2π -periodic and odd, $S(k\pi) = 0$ for every integer k (each $S(n\pi)$ equals $S((n-2)\pi)$ down to $S(0) = 0$ or $S(\pi)$, and $S(\pi) = 2S(\pi/2)C(\pi/2) = 2 \cdot 1 \cdot 0 = 0$ by the addition formula). Thus $\sin$ hits 0 at all $x = k\pi$, infinitely many points, so it is not injective. Restricting to $[-\pi/2, \pi/2]$, where $S' = C > 0$ except at the endpoints, makes $\sin$ strictly increasing and invertible; that restriction defines $\arcsin$. *Check.* $\sin 0 = \sin \pi = \sin 2\pi = 0$ exhibits three distinct inputs with the same output. ■

Exercises (Bartle & Sherbert, §8.4).

1. Calculate $\cos(0.2)$, $\sin(0.2)$ and $\cos 1$, $\sin 1$ correct to four decimal places.
2. Show that $|\sin x| \leq 1$ and $|\cos x| \leq 1$ for all $x \in \mathbb{R}$.
3. Show that property (vii) of Theorem 8.4.8 (namely $\sin x \leq x$) fails for $x < 0$, but that $|\sin x| \leq |x|$ holds for all $x \in \mathbb{R}$. Also show $|\sin x - x| \leq |x|^3/6$ for all $x \in \mathbb{R}$.
4. Show that if $x > 0$ then $1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} \leq \cos x \leq 1 - \frac{x^2}{2} + \frac{x^4}{24}$. Use this to establish a lower bound for π .
5. Calculate π by approximating the smallest positive zero of $\sin$ (by interval bisection or by Newton's method).
6. Define sequences (c_n) , (s_n) inductively by $c_1(x) = 1$, $s_1(x) = x$, and $s_n(x) = \int_0^x c_n(t) dt$, $c_{n+1}(x) = 1 + \int_0^x s_n(t) dt$ for $n \in \mathbb{N}$. Reasoning as in Theorem 8.4.1, conclude there exist functions $c, s : \mathbb{R} \rightarrow \mathbb{R}$ with $c'' = c$, $s'' = s$, $c(0) = 1$, $c'(0) = 0$, $s(0) = 0$, $s'(0) = 1$, and $c' = s$, $s' = c$. (These are the hyperbolic cosine and sine.)
7. Show the functions c, s of the preceding exercise have derivatives of all orders and satisfy $c(x)^2 - s(x)^2 = 1$ for all x ; show they are the *unique* functions satisfying the stated conditions.

8. If $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f''(x) = f(x)$ for all x , show there are reals a, b with $f(x) = a c(x) + b s(x)$. Apply this to $f_1(x) = e^x$ and $f_2(x) = e^{-x}$ to obtain $c(x) = \frac{1}{2}(e^x + e^{-x})$ and $s(x) = \frac{1}{2}(e^x - e^{-x})$.
9. Show c, s are even and odd respectively, and that $c(x+y) = c(x)c(y) + s(x)s(y)$, $s(x+y) = s(x)c(y) + c(x)s(y)$ for all $x, y \in \mathbb{R}$.
10. Show $c(x) \geq 1$ for all $x \in \mathbb{R}$, that both c and s are strictly increasing on $(0, \infty)$, and that $\lim_{x \rightarrow \infty} c(x) = \lim_{x \rightarrow \infty} s(x) = \infty$.

Additional exercises.

1. From the series, verify $\cos 0 = 1$, $\sin 0 = 0$, $\cos' = -\sin$, $\sin' = \cos$, and $(\cos x)^2 + (\sin x)^2 = 1$.
2. Derive the double-angle formulas $\cos 2x = \cos^2 x - \sin^2 x$ and $\sin 2x = 2 \sin x \cos x$ from the addition formulas, and the half-angle $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$.
3. Prove $|\sin x| \leq |x|$ for all real x directly from the series and the alternating-series remainder bound.
4. Show $E(i\pi) = -1$ and hence $e^{i\pi} + 1 = 0$, naming each constant that appears.
5. (Diagram.) Draw the unit circle with a point $(\cos t, \sin t)$ at $t = \pi/3$, mark the arc from $(1, 0)$ of length t , drop the coordinate segments $\cos t$ and $\sin t$, and state why t is simultaneously the angle and the arc length.
6. (Non-example.) Show that $|\sin nx| \leq n |\sin x|$ for integers $n \geq 0$, but find a real number r and a point x where $|\sin rx| > r |\sin x|$, demonstrating the inequality is special to integer n .
7. Using $\cos(\pi/2) = 0$, $\sin(\pi/2) = 1$ and the addition formulas, compute $\cos(x + \pi/2)$ and $\sin(x + \pi/2)$, recovering the cofunction relations.
8. Prove that $\tan x = \sin x / \cos x$ is strictly increasing on $(-\pi/2, \pi/2)$ with range $\mathbb{R}$, by computing its derivative and its one-sided limits at $\pm\pi/2$.
9. (Harder.) From $E(it) = \cos t + i \sin t$ derive the product-to-sum identity $\cos a \cos b = \frac{1}{2}(\cos(a-b) + \cos(a+b))$, and use it to evaluate $\int_{-\pi}^{\pi} \cos mx \cos nx \, dx$ for integers $m, n \geq 0$.

9.4 The Algebraic Completeness of the Complex Field

The fundamental theorem of algebra

Having built the complex exponential, we can prove that $\mathbb{C}$ is *algebraically closed*: every nonconstant complex polynomial has a root. Rudin's proof (8.8) is purely analytic and beautifully short: a polynomial's modulus attains a minimum, and unless that minimum is 0 one can always step downhill. We isolate the one computational ingredient as a lemma.

Lemma 9.4.1 (Downhill step). *Let $P(z) = a_0 + a_k z^k + (\text{higher terms})$ with $a_0 \neq 0$, $a_k \neq 0$ the first nonzero coefficient after a_0 ($k \geq 1$). Then there is a direction $z = tw$ ($t > 0$ small) along which $|P(z)| < |a_0| = |P(0)|$.*

Proof. Choose w with $a_k w^k = -a_0$; such a w exists because every complex number has a k -th root (take $a_k w^k = -a_0$, i.e. w a k -th root of $-a_0/a_k$, available by the polar form and Theorem 9.3.5). For $0 < t < 1$,

$$P(tw) = a_0 + a_k t^k w^k + \sum_{j>k} a_j t^j w^j = a_0(1 - t^k) + t^k \sum_{j>k} a_j t^{j-k} w^j.$$

Hence, by the triangle inequality,

$$|P(tw)| \leq |a_0| (1 - t^k) + t^{k+1} B, \quad B = \sum_{j>k} |a_j| |w|^j,$$

for $0 < t \leq 1$. For t small enough the $t^{k+1}B$ term is dominated by the gain $|a_0|t^k$: precisely if $tB < |a_0|$ then $|P(tw)| \leq |a_0| (1 - t^k) + |a_0|t^k \cdot \frac{tB}{|a_0|} < |a_0|$. So $|P|$ strictly decreases along this ray. $\square$

Theorem 9.4.2 (Fundamental Theorem of Algebra, Rudin 8.8). *Every nonconstant polynomial $P(z) = \sum_{j=0}^n a_j z^j$ with complex coefficients ($n \geq 1$, $a_n \neq 0$) has a root in $\mathbb{C}$. Consequently P factors completely: $P(z) = a_n \prod_{j=1}^n (z - r_j)$.*

Proof. Step 1: $|P|$ attains a global minimum. Since $a_n \neq 0$,

$$|P(z)| \geq |a_n| |z|^n - \sum_{j < n} |a_j| |z|^j \xrightarrow{|z| \rightarrow \infty} +\infty,$$

so there is R with $|P(z)| > |P(0)|$ for $|z| > R$. On the closed disc $|z| \leq R$ (compact) the continuous function $|P|$ attains a minimum at some z_0 , and by the choice of R this is a *global* minimum: $|P(z_0)| \leq |P(z)|$ for all z .

Step 2: the minimum value is 0. Suppose not, $P(z_0) \neq 0$. Expand $Q(\zeta) = P(z_0 + \zeta) = P(z_0) + a_k \zeta^k + \cdots$ in powers of ζ , where a_k is the first nonzero coefficient after the constant $P(z_0) \neq 0$ (Q is nonconstant since P is, so such k exists). By the Downhill Lemma 9.4.1 applied to Q , there is ζ with $|Q(\zeta)| < |P(z_0)|$, i.e. $|P(z_0 + \zeta)| < |P(z_0)|$ — contradicting that z_0 is a global minimizer. Hence $P(z_0) = 0$.

Step 3: factorization. With root $r_1 = z_0$, polynomial division gives $P(z) = (z - r_1)P_1(z)$ with $\deg P_1 = n - 1$; induct. $\square$

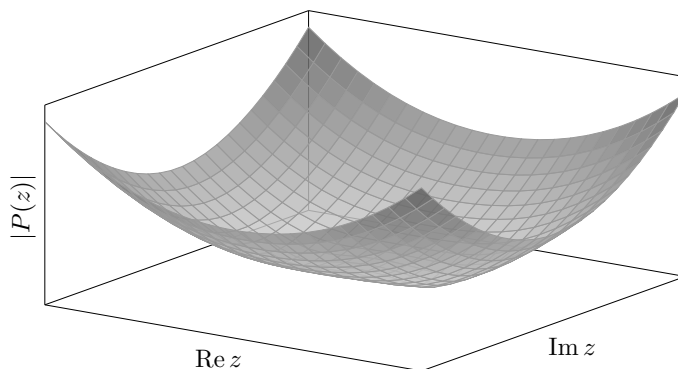


Figure 9.6. The modulus surface $|P(z)|$ for $P(z) = z^2 - \frac{1}{4}$. It cannot have a *positive* local minimum: at any point with $P \neq 0$ one can step downhill (Lemma 9.4.1). The only minima touch height 0 — the roots $z = \pm \frac{1}{2}$.

Remark (Why “no positive local minimum” is the whole story). The analytic content is entirely in Step 2: $|P|$ has *no* local minimum with positive value. Geometrically the modulus surface above has no isolated “dimples” sitting above the plane — every pit reaches down to 0. This is a real-variable shadow of the maximum-modulus principle of complex analysis. Rudin’s proof needs only the Extreme Value Theorem (compactness) plus the existence of k -th roots in $\mathbb{C}$ (from E and the circle map), so it sits squarely inside this chapter.

Worked Example 9.4.3 (Locating roots by the minimum-modulus idea). For $P(z) = z^3 - 1$, verify the proof’s mechanism: find where $|P|$ is minimized and confirm the minima are the cube roots of unity.

Recognition. “Confirm the FTA proof on an explicit polynomial.”

Solution. The roots of $z^3 = 1$ are $1, \omega, \omega^2$ with $\omega = E(2\pi i/3) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$. At each, $|P| = 0$, the global minimum. At the center $z = 0$, $|P(0)| = 1 > 0$; the Downhill Lemma says we can decrease $|P|$ by moving in a direction w with $w^3 = -(-1)/1 = 1$ misread — here $a_0 = -1, a_3 = 1$, so we need $w^3 = -a_0/a_3 = 1$, e.g. $w = 1$, and indeed $|P(t)| = |t^3 - 1| = 1 - t^3 < 1$ for small $t > 0$. The descent leads toward the root $z = 1$. *Check.* The three roots are equally spaced on the unit circle, the fixed points of the descent. ■

Further worked examples

Worked Example 9.4.4 (Every n -th root exists: solving $z^n = re^{i\theta}$). Find all solutions of $z^4 = -16$ in $\mathbb{C}$, and explain how the same recipe supplies the k -th roots needed in the Downhill Lemma.

Recognition. “Extract a complex root”: write the target in polar form $re^{i\theta}$ and use $E(it)$ to spread the roots around a circle of radius $r^{1/n}$.

Solution. Write $-16 = 16E(i\pi)$. A solution is $z = 16^{1/4}E(i\pi/4) = 2E(i\pi/4)$, and the full set is obtained by adding $2\pi k/4$ to the angle:

$$z_k = 2E\left(i\frac{\pi+2\pi k}{4}\right), \quad k = 0, 1, 2, 3,$$

namely $2E(i\pi/4) = \sqrt{2} + \sqrt{2}i$, $2E(i3\pi/4) = -\sqrt{2} + \sqrt{2}i$, $2E(i5\pi/4) = -\sqrt{2} - \sqrt{2}i$, $2E(i7\pi/4) = \sqrt{2} - \sqrt{2}i$. The same construction gives a k -th root of any $-a_0/a_k$, which is exactly the ingredient Lemma 9.4.1 requires. *Check.* $(\sqrt{2} + \sqrt{2}i)^2 = 2 \cdot 2i = 4i$, and $(4i)^2 = -16$. ■

Worked Example 9.4.5 (Real polynomials factor into real linear and quadratic pieces). Show that a polynomial with real coefficients has its non-real roots in conjugate pairs, hence factors over $\mathbb{R}$ into linear and irreducible quadratic factors. Illustrate on $z^4 + 1$.

Recognition. “Real coefficients force conjugate symmetry”: apply complex conjugation to $P(z) = 0$, then pair z with $\bar{z}$ into a real quadratic.

Solution. If P has real coefficients and $P(z_0) = 0$, then $\overline{P(z_0)} = P(\bar{z}_0) = 0$ since conjugation fixes the real a_j . So non-real roots come in pairs $z_0, \bar{z}_0$, and $(z - z_0)(z - \bar{z}_0) = z^2 - 2\operatorname{Re}(z_0)z + |z_0|^2$ has real coefficients. By the FTA (Thm. 9.4.2) P splits completely over $\mathbb{C}$; grouping conjugate pairs gives a real factorization. For $z^4 + 1$: the roots are the primitive 8th roots of unity $E(i\pi/4), E(i3\pi/4), E(i5\pi/4), E(i7\pi/4)$, pairing into $(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$. *Check.* Multiplying out, $(z^2 + 1)^2 - (\sqrt{2}z)^2 = z^4 + 2z^2 + 1 - 2z^2 = z^4 + 1$. ■

Worked Example 9.4.6 (A non-example: the FTA fails over $\mathbb{R}$). Exhibit a nonconstant polynomial with real coefficients having no real root, and reconcile this with the Fundamental Theorem of Algebra.

Recognition. “Algebraic closure is a property of $\mathbb{C}$, not $\mathbb{R}$ ”: test the discriminant / a global minimum of a real polynomial.

Solution. $p(x) = x^2 + 1$ has $p(x) \geq 1 > 0$ for all real x , so it has no real root: $\mathbb{R}$ is *not* algebraically closed. There is no contradiction, because the FTA asserts a root in $\mathbb{C}$, and indeed $p(\pm i) = 0$. The minimum-modulus argument of Theorem 9.4.2 needs the k -th roots of arbitrary complex numbers (Lemma 9.4.1); over $\mathbb{R}$ that downhill step is unavailable, e.g. $|x^2 + 1|$ has a genuine positive minimum value 1 at $x = 0$. *Check.* The modulus $|p(x)| = x^2 + 1$ attains its positive minimum 1 on $\mathbb{R}$, exactly the obstruction the complex proof removes. ■

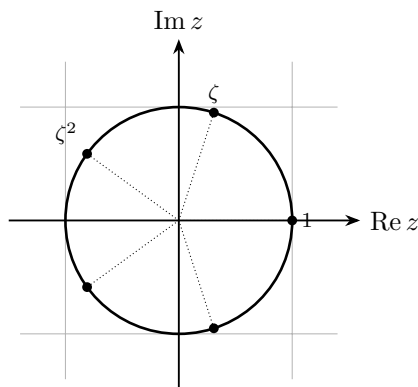


Figure 9.7. The five solutions of $z^5 = 1$ are equally spaced on the unit circle at angles $2\pi k/5$. The FTA guarantees exactly n roots (with multiplicity) for any degree- n polynomial; here they are the values $E(2\pi ik/5)$.

Exercises (Rudin, Ch.8 – winding numbers and the FTA).

1. Let γ be a continuously differentiable closed curve in $\mathbb{C}$ with parameter interval $[a, b]$ and $\gamma(t) \neq 0$ for all t . Define the *index*

$$\operatorname{Ind}(\gamma) = \frac{1}{2\pi i} \int_a^b \frac{\gamma'(t)}{\gamma(t)} dt.$$

Prove $\operatorname{Ind}(\gamma)$ is always an integer. (Hint: there is φ with $\varphi' = \gamma'/\gamma$, $\varphi(a) = 0$; then $\gamma e^{-\varphi}$ is constant, and $\gamma(a) = \gamma(b)$ forces $e^{\varphi(b)} = 1$, so $\varphi(b) = 2\pi i \operatorname{Ind}(\gamma)$.) Compute $\operatorname{Ind}(\gamma)$ for $\gamma(t) = e^{int}$, $a = 0$, $b = 2\pi$, and explain why the index is called the *winding number* of γ about 0.

- Let γ be as above and assume in addition that the range of γ does not meet the negative real axis. Prove $\text{Ind}(\gamma) = 0$. (Hint: for $0 < c < \infty$, $\text{Ind}(\gamma + c)$ is a continuous integer-valued function of c , and $\rightarrow 0$ as $c \rightarrow \infty$.)
- Suppose γ_1, γ_2 are curves as above with $|\gamma_1(t) - \gamma_2(t)| < |\gamma_1(t)|$ for $a \leq t \leq b$. Prove $\text{Ind}(\gamma_1) = \text{Ind}(\gamma_2)$. (Hint: put $\gamma = \gamma_2/\gamma_1$; then $|1 - \gamma| < 1$, so $\text{Ind}(\gamma) = 0$ by the previous exercise.)
- Let γ be a closed curve (not necessarily differentiable) on $[0, 2\pi]$ with $\gamma(t) \neq 0$. Choose $\delta > 0$ with $|\gamma(t)| > \delta$. If P_1, P_2 are trigonometric polynomials with $|P_i(t) - \gamma(t)| < \delta/4$, prove $\text{Ind}(P_1) = \text{Ind}(P_2)$, and *define* this common value to be $\text{Ind}(\gamma)$. Show the two previous exercises hold without any differentiability assumption.
- Let f be a continuous complex function on $\mathbb{C}$, and suppose there is a positive integer n and $c \neq 0$ with $\lim_{|z| \rightarrow \infty} z^{-n}f(z) = c$. Prove that $f(z) = 0$ for at least one $z \in \mathbb{C}$. (This generalizes the Fundamental Theorem of Algebra, Theorem 9.4.3; for $\gamma_r(t) = f(re^{it})$ show $\text{Ind}(\gamma_0) = 0$, $\text{Ind}(\gamma_r) = n$ for large r , and $\text{Ind}(\gamma_r)$ is continuous in r — a contradiction since $n > 0$.)
- Let D be the closed unit disc and $g : D \rightarrow T$ a continuous map into the unit circle with $|g(z)| = 1$. Prove $g(z) = -z$ for at least one $z \in T$. (Hint: with $\gamma_r(t) = g(re^{it})$ and $\psi(t) = e^{-it}\gamma_1(t)$, if $g(z) \neq -z$ on T then $\psi(t) \neq -1$, so $\text{Ind}(\psi) = 0$, giving $\text{Ind}(\gamma_1) = 1$, whereas $\text{Ind}(\gamma_0) = 0$.)
- Prove that every continuous map $f : D \rightarrow D$ of the closed unit disc into itself has a fixed point (the 2-dimensional Brouwer fixed-point theorem). (Hint: if $f(z) \neq z$ for all z , associate to each z the point $g(z) \in T$ on the ray from $f(z)$ through z ; then $g : D \rightarrow T$ is continuous with $g(z) = z$ on T , contradicting the previous exercise.)

Additional exercises.

- Factor $z^4 - 1$ and $z^3 + 1$ completely over $\mathbb{C}$, listing all roots in the form $E(i\theta)$.
- Show directly that $z^2 + 1 = 0$ has the roots $\pm i$, and that $z^2 + z + 1 = 0$ has roots $E(\pm 2\pi i/3)$; mark all of these on the unit circle.
- Prove that a polynomial of degree $n \geq 1$ with complex coefficients has exactly n roots counted with multiplicity, using Theorem 9.4.3 and induction on the degree.
- Prove that a real polynomial of odd degree has at least one real root (a) from the Intermediate Value Theorem, and (b) by pairing complex roots into conjugate pairs.
- (Non-example.) Show that the field $\mathbb{R}$ is *not* algebraically closed by exhibiting a real polynomial with no real root, and explain why the minimum-modulus proof of Theorem 9.4.3 genuinely uses $\mathbb{C}$ (where does “every number has a k -th root” enter?).
- (Diagram.) For $P(z) = z^2 - 1$, sketch the level idea: indicate on the plane the two roots ± 1 and describe, in words tied to Lemma 9.4.1, why $|P|$ has no positive local minimum — i.e. from any non-root you can step so that $|P|$ decreases.
- Suppose P is a complex polynomial with $|P(z)| \geq 1$ for all $z \in \mathbb{C}$. Prove P is constant. (This is a baby Liouville-type statement; combine the EVT minimum with Theorem 9.4.3.)
- (Harder.) Carry out the downhill step explicitly for $P(z) = z^3 + z + 1$ at the point $z_0 = 0$: find the lowest-order nonconstant term, choose a descent direction w , and verify $|P(tw)| < |P(0)| = 1$ for small $t > 0$.

9.5 Fourier Series

Trigonometric and orthonormal systems

We now treat Fourier series at the level of Rudin 8.9–8.16: orthonormal systems, Fourier coefficients, the best-approximation property, Bessel’s inequality, and the mean-square (L^2) convergence theorem, with one pointwise convergence result.

Definition 9.5.1. A *trigonometric polynomial* is a finite sum $f(x) = a_0 + \sum_{n=1}^N (a_n \cos nx + b_n \sin nx)$, equivalently $f(x) = \sum_{n=-N}^N c_n e^{inx}$. A family $\{\phi_n\}$ of (complex) functions on $[a, b]$ is *orthonormal* if

$$\int_a^b \phi_n(x) \overline{\phi_m(x)} dx = \delta_{nm} = \begin{cases} 1, & n = m, \\ 0, & n \neq m. \end{cases}$$

On $[-\pi, \pi]$ the functions $\phi_n(x) = \frac{e^{inx}}{\sqrt{2\pi}}$ ($n \in \mathbb{Z}$) form the basic orthonormal *trigonometric system*.

Orthonormality of $\{e^{inx}/\sqrt{2\pi}\}$ is the computation $\int_{-\pi}^{\pi} e^{inx} e^{-imx} dx = \int_{-\pi}^{\pi} e^{i(n-m)x} dx$, which is 2π if $n = m$ and $\left[\frac{e^{i(n-m)x}}{i(n-m)}\right]_{-\pi}^{\pi} = 0$ if $n \neq m$ (periodicity).

Definition 9.5.2 (Fourier coefficients, Rudin 8.11). Given an orthonormal system $\{\phi_n\}$ and f (Riemann-integrable, possibly complex) on $[a, b]$, the *Fourier coefficients* of f are

$$\hat{c}_n = \int_a^b f(x) \overline{\phi_n(x)} dx,$$

and the *Fourier series* of f is $\sum_n \hat{c}_n \phi_n$. For the trigonometric system on $[-\pi, \pi]$ this reads $c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$ and $f \sim \sum c_n e^{inx}$.

Best approximation and Bessel's inequality

Theorem 9.5.3 (Least-squares optimality, Rudin 8.11). Let $s_N = \sum_{n=-N}^N \hat{c}_n \phi_n$ be the N -th partial sum of the Fourier series, and let $t_N = \sum_{n=-N}^N \gamma_n \phi_n$ be any sum with the same indices. Then

$$\int_a^b |f - s_N|^2 \leq \int_a^b |f - t_N|^2,$$

with equality only if $\gamma_n = \hat{c}_n$ for all n . The Fourier partial sum is the unique best mean-square approximation to f from $\text{span}\{\phi_{-N}, \dots, \phi_N\}$.

Proof. Write $\|g\|^2 = \int_a^b |g|^2$ and use orthonormality. Expanding,

$$\|f - t_N\|^2 = \|f\|^2 - \sum_n (\gamma_n \overline{\hat{c}_n} + \overline{\gamma_n} \hat{c}_n) + \sum_n |\gamma_n|^2 = \|f\|^2 - \sum_n |\hat{c}_n|^2 + \sum_n |\gamma_n - \hat{c}_n|^2,$$

completing the square. The right side is minimized exactly when every $|\gamma_n - \hat{c}_n| = 0$, giving the claim and the value $\|f - s_N\|^2 = \|f\|^2 - \sum_n |\hat{c}_n|^2$. $\square$

Corollary 9.5.4 (Bessel's inequality, Rudin 8.12). For any orthonormal system, $\sum_n |\hat{c}_n|^2 \leq \int_a^b |f|^2$. In particular $\hat{c}_n \rightarrow 0$ (Riemann-Lebesgue): the Fourier coefficients of a square-integrable function vanish at infinity.

Proof. From the proof above, $0 \leq \|f - s_N\|^2 = \|f\|^2 - \sum_{n=-N}^N |\hat{c}_n|^2$, so $\sum_{n=-N}^N |\hat{c}_n|^2 \leq \|f\|^2$ for every N . Let $N \rightarrow \infty$. A convergent series of nonnegative terms has terms $\rightarrow 0$, so $\hat{c}_n \rightarrow 0$. $\square$

Method (Computing a Fourier series and reading off a numerical series). *Recognition trigger.* You must expand a periodic f in $\cos/\sin$, or a problem asks you to sum a series like $\sum 1/n^2$ “via Fourier/Parseval”.

1. Compute $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f \cos nx$, $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f \sin nx$ (use parity: even $f \Rightarrow b_n = 0$, odd $f \Rightarrow a_n = 0$).
2. Write $f \sim \frac{a_0}{2} + \sum (a_n \cos nx + b_n \sin nx)$.
3. For a *value* of f , invoke a pointwise convergence theorem (below) at a chosen x . For a sum of *squares* of coefficients, invoke Parseval.

Mean-square convergence and Parseval

Theorem 9.5.5 (L^2 completeness of the trigonometric system, Rudin 8.16). *For every Riemann-integrable 2π -periodic f , the Fourier partial sums converge to f in mean square:*

$$\lim_{N \rightarrow \infty} \int_{-\pi}^{\pi} |f - s_N|^2 dx = 0,$$

and consequently Parseval's identity holds:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx = \sum_{n=-\infty}^{\infty} |c_n|^2 = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

Proof sketch. The continuous 2π -periodic functions are dense in mean square among integrable ones, and by the Stone–Weierstrass / Fejér theorem trigonometric polynomials are uniformly (hence in L^2) dense in the continuous periodic functions. So for any $\varepsilon > 0$ there is a trigonometric polynomial t with $\|f - t\|^2 < \varepsilon$; by best approximation (Thm. 9.5.3) the Fourier sum s_N does at least as well for N past the degree of t , giving $\|f - s_N\|^2 < \varepsilon$. Parseval is then Bessel's inequality forced to equality by $\|f - s_N\| \rightarrow 0$, since $\|f - s_N\|^2 = \|f\|^2 - \sum_{-N}^N |c_n|^2$. $\square$

Theorem 9.5.6 (A pointwise convergence theorem, Rudin 8.14). *Let f be 2π -periodic and Riemann-integrable. If, at a point x , there are constants $M, \delta > 0$ with*

$$|f(x+t) - f(x)| \leq M|t| \quad (|t| < \delta)$$

(a Lipschitz / Dini condition at x), then the Fourier series of f converges to $f(x)$: $s_N(f; x) \rightarrow f(x)$.

Idea. Write $s_N(f; x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) D_N(t) dt$ with the Dirichlet kernel $D_N(t) = \frac{\sin((N+\frac{1}{2})t)}{\sin(t/2)}$. Then $s_N(f; x) - f(x) = \frac{1}{2\pi} \int g(t) \sin((N+\frac{1}{2})t) dt$ where $g(t) = \frac{f(x-t)-f(x)}{\sin(t/2)}$ is bounded near 0 by the Lipschitz hypothesis and integrable away from 0. By the Riemann–Lebesgue lemma (Cor. 9.5.4) this integral $\rightarrow 0$. $\square$

The square wave and the Gibbs phenomenon

Worked Example 9.5.7 (The square wave; summing $1 - \frac{1}{3} + \frac{1}{5} - \dots$). Expand the odd square wave $f(x) = \text{sgn}(x)$ on $(-\pi, \pi)$ ($f = 1$ on $(0, \pi)$, $f = -1$ on $(-\pi, 0)$) and deduce the Leibniz series.

Recognition. “Expand a step function and extract a numeric series at a point.”

Solution. f is odd, so $a_n = 0$. Compute $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f \sin nx dx = \frac{2}{\pi} \int_0^{\pi} \sin nx dx = \frac{2}{\pi} \cdot \frac{1 - \cos n\pi}{n} = \frac{2}{\pi n} (1 - (-1)^n)$, which is $\frac{4}{\pi n}$ for odd n and 0 for even n . Hence

$$f(x) = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\sin((2k+1)x)}{2k+1}.$$

At $x = \pi/2$, $f = 1$ and $\sin((2k+1)\pi/2) = (-1)^k$, so $1 = \frac{4}{\pi} \sum \frac{(-1)^k}{2k+1}$, i.e. $\sum \frac{(-1)^k}{2k+1} = \frac{\pi}{4}$. Check. Recovers Worked Example 9.1.6 by a different route. $\blacksquare$

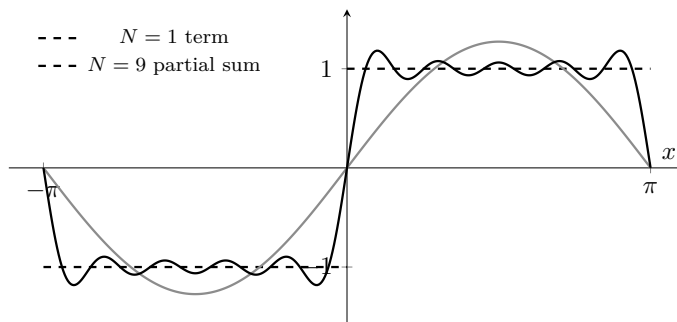


Figure 9.8. Partial sums of the square wave. Near the jump at $x = 0$ the overshoot does *not* die out — it stays about 9% of the jump (the *Gibbs phenomenon*) even as $N \rightarrow \infty$, although the series still converges in mean square.

Caution. Mean-square convergence (Thm. 9.5.5) is *not* pointwise or uniform convergence. At a jump the partial sums overshoot by a fixed $\approx 9\%$ forever (Gibbs); and there exist continuous functions whose Fourier series diverge at a point. Bessel/Parseval control the L^2 size of coefficients, not pointwise values; for pointwise statements you need a Dini-type hypothesis (Thm. 9.5.6).

Further worked examples

Worked Example 9.5.8 (Parseval on the square wave gives $\sum 1/(2k+1)^2 = \pi^2/8$). Apply Parseval's identity to the square wave $f = \text{sgn}(x)$ to evaluate $\sum_{k \geq 0} \frac{1}{(2k+1)^2}$.

Recognition. "Sum of squared coefficients": Parseval (Thm. 9.5.5) equates the mean square of f to $\sum (a_n^2 + b_n^2)/2$.

Solution. From the worked example above, $b_n = \frac{4}{\pi n}$ for odd n , else 0 (and all $a_n = 0$). Parseval in real form reads $\frac{1}{\pi} \int_{-\pi}^{\pi} f^2 dx = \sum_{n \geq 1} b_n^2$. The left side is $\frac{1}{\pi} \int_{-\pi}^{\pi} 1 dx = 2$. The right side is $\sum_{k \geq 0} \left(\frac{4}{\pi(2k+1)}\right)^2 = \frac{16}{\pi^2} \sum_{k \geq 0} \frac{1}{(2k+1)^2}$. Equating, $\sum_{k \geq 0} \frac{1}{(2k+1)^2} = \frac{2\pi^2}{16} = \frac{\pi^2}{8}$. *Check.* The first three terms $1 + \frac{1}{9} + \frac{1}{25} = 1.151$ already approach $\pi^2/8 \approx 1.2337$. ■

Worked Example 9.5.9 (The parabola $f(x) = x^2$ and the Basel sum $\sum 1/n^2 = \pi^2/6$). Expand $f(x) = x^2$ on $[-\pi, \pi]$ in a Fourier series and evaluate at $x = \pi$ to obtain $\sum_{n \geq 1} 1/n^2$.

Recognition. "Even function, extract a numeric series at an endpoint": compute a_n by parts, then use pointwise convergence (Thm. 9.5.6) at a point where f is smooth enough.

Solution. f is even, so $b_n = 0$. The mean value is $a_0/2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{\pi^2}{3}$. For $n \geq 1$, $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx = \frac{2}{\pi} \cdot \frac{2\pi(-1)^n}{n^2} = \frac{4(-1)^n}{n^2}$ (two integrations by parts). Hence

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-1)^n}{n^2} \cos nx, \quad -\pi \leq x \leq \pi.$$

At $x = \pi$ (where the periodic extension is continuous and Lipschitz), $\cos n\pi = (-1)^n$, so $\pi^2 = \frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{1}{n^2}$, giving $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$. *Check.* At $x = 0$ instead, $0 = \frac{\pi^2}{3} + 4 \sum \frac{(-1)^n}{n^2}$ gives $\sum \frac{(-1)^{n-1}}{n^2} = \frac{\pi^2}{12}$, consistent with $\frac{\pi^2}{6} - 2 \cdot \frac{1}{4} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{12}$. ■

Worked Example 9.5.10 (A non-example: Bessel is strict for a truncated system). Take $f(x) = \cos x + \cos 2x$ on $[-\pi, \pi]$ and the truncated orthonormal system using only the $n = 0, \pm 1$ trigonometric functions. Show Bessel's inequality is strict, and that completeness (Parseval) is what restores equality.

Recognition. "Bessel vs. Parseval": an incomplete system loses the energy carried by the omitted modes, so $\sum |\hat{c}_n|^2 < \|f\|^2$.

Solution. Here f already is a trigonometric polynomial with $a_1 = a_2 = 1$, all other coefficients 0. Its full energy is $\frac{1}{\pi} \int_{-\pi}^{\pi} f^2 = a_1^2 + a_2^2 = 2$ (using $\frac{1}{\pi} \int \cos^2 kx = 1$ and orthogonality of $\cos x, \cos 2x$). Restricting to $n = 0, \pm 1$ captures only the $\cos x$ mode, contributing $a_1^2 = 1$. Thus the truncated Bessel sum is $1 < 2 = \|f\|^2/\pi$: strict, because the $\cos 2x$ energy is omitted. Including $n = \pm 2$ restores $1 + 1 = 2$, the Parseval equality. *Check.* The deficit $2 - 1 = 1$ equals exactly a_2^2 , the energy of the dropped mode. ■

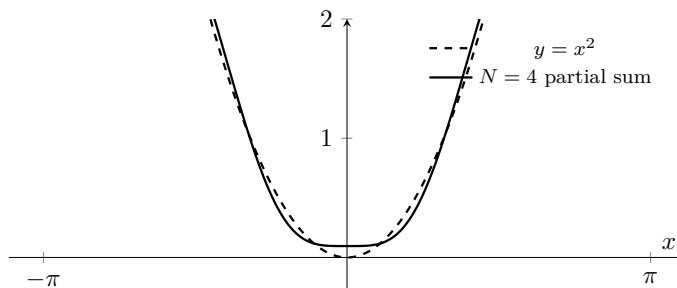


Figure 9.9. The Fourier series of x^2 converges fast and *uniformly* (the coefficients decay like $1/n^2$, and the periodic extension is continuous): just four cosine terms already track the parabola closely, with no Gibbs overshoot.

Exercises (Rudin, Ch.8 – Fourier series).

- Suppose $0 < \delta < \pi$, $f(x) = 1$ if $|x| < \delta$, $f(x) = 0$ if $\delta < |x| \leq \pi$, and $f(x + 2\pi) = f(x)$. (a) Compute the Fourier coefficients of f . (b) Conclude $\sum_{n=1}^{\infty} \frac{\sin(n\delta)}{n} = \frac{\pi - \delta}{2}$ for $0 < \delta < \pi$. (c) Deduce from Parseval's theorem that $\sum_{n=1}^{\infty} \frac{\sin^2(n\delta)}{n^2\delta} = \frac{\pi - \delta}{2}$. (d) Let $\delta \rightarrow 0$ and prove $\int_0^{\infty} \left(\frac{\sin x}{x}\right)^2 dx = \frac{\pi}{2}$. (e) Put $\delta = \pi/2$ in (c); what do you get?
- Put $f(x) = x$ for $0 \leq x < 2\pi$, extended 2π -periodically, and apply Parseval's theorem to conclude $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.
- If $f(x) = (\pi - |x|)^2$ on $[-\pi, \pi]$, prove $f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos nx$, and deduce $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$.
- With the Dirichlet kernel D_n , put $K_N(x) = \frac{1}{N+1} \sum_{n=0}^N D_n(x)$ (the *Fejér kernel*). Prove $K_N(x) = \frac{1}{N+1} \cdot \frac{1 - \cos(N+1)x}{1 - \cos x}$, and that (a) $K_N \geq 0$; (b) $\frac{1}{2\pi} \int_{-\pi}^{\pi} K_N(x) dx = 1$; (c) $K_N(x) \leq \frac{1}{N+1} \cdot \frac{2}{1 - \cos \delta}$ for $0 < \delta \leq |x| \leq \pi$. If $\sigma_N = \frac{s_0 + s_1 + \cdots + s_N}{N+1}$ are the arithmetic means of the partial sums of the Fourier series of f , prove *Fejér's theorem*: if f is continuous with period 2π then $\sigma_N(f; x) \rightarrow f(x)$ uniformly on $[-\pi, \pi]$.
- Prove a pointwise version of Fejér's theorem: if $f \in \mathcal{R}$ and $f(x+)$, $f(x-)$ exist at some x , then $\lim_{N \rightarrow \infty} \sigma_N(f; x) = \frac{1}{2}[f(x+) + f(x-)]$.
- Assume f is bounded and monotonic on $[-\pi, \pi)$ with Fourier coefficients c_n . (a) Prove $\{nc_n\}$ is a bounded sequence. (b) Combine (a) with the previous exercise to conclude $\lim_{N \rightarrow \infty} s_N(f; x) = \frac{1}{2}[f(x+) + f(x-)]$ for every x . (c) Assuming only $f \in \mathcal{R}$ on $[-\pi, \pi]$ and f monotonic on some segment (α, β) , prove the conclusion of (b) for every $x \in (\alpha, \beta)$ (the localization theorem).
- Define $f(x) = x^3 - \sin^2 x \tan x$ and $g(x) = 2x^2 - \sin^2 x - x \tan x$. Determine, for each, whether it is positive on all of $(0, \pi/2)$, negative on all of $(0, \pi/2)$, or changes sign; prove your answer.
- Suppose f is continuous on $\mathbb{R}$ with $f(x+2\pi) = f(x)$ and α/π irrational. Prove that $\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N f(x+n\alpha) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) dt$ for every x . (Hint: do it first for $f(x) = e^{ikx}$.)

Additional exercises.

- Verify the orthogonality relations $\int_{-\pi}^{\pi} \cos mx \cos nx dx$, $\int_{-\pi}^{\pi} \sin mx \sin nx dx$, and $\int_{-\pi}^{\pi} \sin mx \cos nx dx$ for integers $m, n \geq 0$, and use them to justify the formulas for the Fourier coefficients a_n, b_n .
- Compute the Fourier series of $f(x) = x$ on $(-\pi, \pi)$ (odd extension) and state where it converges pointwise; what value does it take at $x = \pi$?
- Compute the Fourier series of $f(x) = |x|$ on $[-\pi, \pi]$ and deduce $\sum_{k \geq 0} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$.
- Apply Parseval's identity to $f(x) = x$ on $(-\pi, \pi)$ to obtain $\sum_{n \geq 1} 1/n^2 = \pi^2/6$, then to $f(x) = |x|$ to obtain a series for $\pi^4/96$.

5. (Non-example.) Give the square-wave example to show that mean-square convergence does *not* imply uniform convergence: state the fixed Gibbs overshoot near the jump and why it contradicts uniform convergence even though $\|f - s_N\|_2 \rightarrow 0$.
6. (Diagram.) Plot the target square wave together with its partial sums s_1, s_3, s_9 , and annotate the overshoot bump near the jump that does not shrink with N .
7. Prove Bessel's inequality $\sum |c_n|^2 \leq \frac{1}{2\pi} \int |f|^2$ from the best-approximation identity, and state precisely when equality (Parseval) holds.
8. State the Riemann–Lebesgue lemma as a corollary of Bessel's inequality and use it to explain why $\int_{-\pi}^{\pi} f(t) \sin(\lambda t) dt \rightarrow 0$ as $\lambda \rightarrow \infty$ for integrable f .
9. (Harder.) Using the Dirichlet kernel $D_N(t) = \frac{\sin((N + \frac{1}{2})t)}{\sin(t/2)}$, show $\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(t) dt = 1$, and outline how the Dini/Lipschitz pointwise convergence theorem (Thm. 9.5.8) follows.

9.6 The Gamma Function

Definition and the functional equation

The Gamma function interpolates the factorial to all positive reals (and beyond). This is Rudin 8.17–8.22.

Definition 9.6.1 (Rudin 8.17). For $x > 0$ define

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt.$$

The integrand is positive; the integral converges because near $t = 0$ we have $t^{x-1}e^{-t} \leq t^{x-1}$ (integrable for $x > 0$), and for large t , $t^{x-1}e^{-t} \leq C_x e^{-t/2}$ (since e^x beats every power, Thm. 9.2.4), which is integrable. So $\Gamma(x)$ is a finite positive number for each $x > 0$.

Theorem 9.6.2 (Functional equation and factorials, Rudin 8.18). For $x > 0$,

$$\Gamma(x+1) = x\Gamma(x), \quad \Gamma(1) = 1, \quad \text{hence} \quad \Gamma(n+1) = n! \quad (n \in \mathbb{N}_0).$$

Proof. Integrate by parts with $u = t^x$, $dv = e^{-t} dt$:

$$\Gamma(x+1) = \int_0^{\infty} t^x e^{-t} dt = [-t^x e^{-t}]_0^{\infty} + x \int_0^{\infty} t^{x-1} e^{-t} dt = 0 + x\Gamma(x),$$

the boundary term vanishing at both ends ($t^x e^{-t} \rightarrow 0$ as $t \rightarrow \infty$ by Thm. 9.2.4, and $\rightarrow 0$ as $t \rightarrow 0^+$ since $x > 0$). Also $\Gamma(1) = \int_0^{\infty} e^{-t} dt = 1$. By induction $\Gamma(n+1) = n \cdot \Gamma(n) = \cdots = n! \Gamma(1) = n!$. $\square$

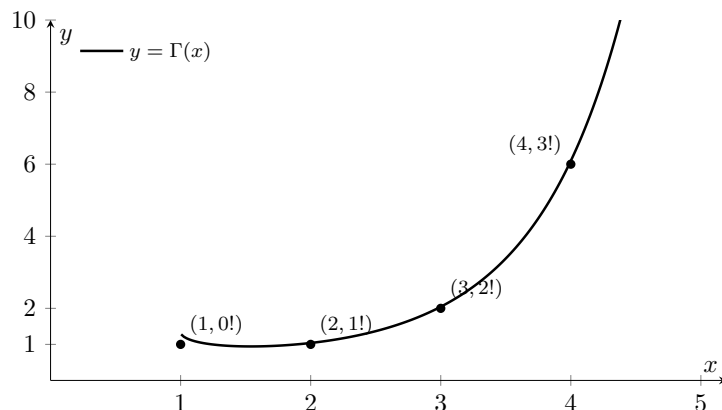


Figure 9.10. Γ passes through the factorial points $(n, (n-1)!)$ and interpolates them smoothly. (Curve drawn from a Stirling-type approximant for illustration.)

Log-convexity and the Bohr–Mollerup characterization

Definition 9.6.3. A positive function g on an interval is *logarithmically convex* (log-convex) if $\log g$ is convex, i.e. $g(\lambda x + (1 - \lambda)y) \leq g(x)^\lambda g(y)^{1-\lambda}$ for $\lambda \in [0, 1]$.

Theorem 9.6.4 (Γ is log-convex, Rudin 8.18). $\log \Gamma$ is convex on $(0, \infty)$.

Proof. For $1/p + 1/q = 1$ ($p, q > 1$), Hölder's inequality applied to the integral defining $\Gamma(x/p + y/q)$ gives

$$\Gamma\left(\frac{x}{p} + \frac{y}{q}\right) = \int_0^\infty (t^{x-1} e^{-t})^{1/p} (t^{y-1} e^{-t})^{1/q} dt \leq \Gamma(x)^{1/p} \Gamma(y)^{1/q}.$$

With $\lambda = 1/p$ this is exactly log-convexity of Γ . □

Theorem 9.6.5 (Bohr–Mollerup, Rudin 8.19). Γ is the unique function $f : (0, \infty) \rightarrow (0, \infty)$ satisfying all three:

1. $f(x+1) = xf(x)$;
2. $f(1) = 1$;
3. $\log f$ is convex.

Proof sketch. Γ has all three (Thms. 9.6.2, 9.6.4). Conversely, (1)–(2) pin f on the integers ($f(n+1) = n!$) and (3) squeezes f in between: for $0 < x \leq 1$ and large n , convexity of $\log f$ on the points $n, n+x, n+1$ gives, after using $f(n+1) = n!$ and $f(n+x) = (n-1+x) \cdots x f(x)$, the bounds

$$\frac{n! n^x}{x(x+1) \cdots (x+n)} \left(\frac{n}{n+1}\right)^? \leq f(x) \leq \frac{n! n^x}{x(x+1) \cdots (x+n)} \cdot \frac{n+x}{n},$$

and both bounds converge to the *Gauss limit* $\lim_{n \rightarrow \infty} \frac{n! n^x}{x(x+1) \cdots (x+n)}$, forcing $f(x) = \Gamma(x)$. The functional equation then extends uniqueness to all $x > 0$. □

Corollary 9.6.6 (Gauss product formula). For $x > 0$, $\Gamma(x) = \lim_{n \rightarrow \infty} \frac{n! n^x}{x(x+1) \cdots (x+n)}$.

Stirling's formula

Theorem 9.6.7 (Stirling's formula, Rudin 8.22).

$$\lim_{x \rightarrow \infty} \frac{\Gamma(x+1)}{(x/e)^x \sqrt{2\pi x}} = 1, \quad \text{equivalently} \quad n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

Idea. Substitute $t = x + \sqrt{x}s$ in $\Gamma(x+1) = \int_0^\infty t^x e^{-t} dt$. The integrand, normalized, concentrates near $t = x$ (its maximum), and

$$t^x e^{-t} = e^{x \log t - t} = e^{x \log x - x} \exp\left(-\frac{s^2}{2} + o(1)\right),$$

by Taylor-expanding $x \log t - t$ to second order about $t = x$. Integrating the Gaussian $\int e^{-s^2/2} \sqrt{x} ds = \sqrt{2\pi x}$ yields $\Gamma(x+1) \sim (x/e)^x \sqrt{2\pi x}$. (Rudin's Exercise 8.20 gives the elementary two-sided bound $e^{7/8} < n! / ((n/e)^n \sqrt{n}) < e$, sandwiching the constant near $\sqrt{2\pi} \approx 2.5066$.) □

Worked Example 9.6.8 (Half-integer Gamma and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$). Show $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and compute $\Gamma(\frac{5}{2})$.

Recognition. “Evaluate Γ at a half-integer”: reduce to $\Gamma(1/2)$ via the functional equation, and get $\Gamma(1/2)$ from a Gaussian integral.

Solution. Substituting $t = u^2$ ($dt = 2u du$), $\Gamma(\frac{1}{2}) = \int_0^\infty t^{-1/2} e^{-t} dt = 2 \int_0^\infty e^{-u^2} du = \sqrt{\pi}$, using the Gaussian integral $\int_{-\infty}^\infty e^{-u^2} du = \sqrt{\pi}$. Then by $\Gamma(x+1) = x\Gamma(x)$:

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}, \quad \Gamma\left(\frac{5}{2}\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{4}\sqrt{\pi}.$$

Check. $\Gamma(\frac{5}{2}) = \frac{3}{4}\sqrt{\pi} \approx 1.329$, and Stirling at $x = 3/2$ gives ≈ 1.3 — consistent. ■

Worked Example 9.6.9 (The Beta integral via Gamma). Evaluate $B(p, q) = \int_0^1 t^{p-1}(1-t)^{q-1} dt$ in terms of Γ (for $p, q > 0$).

Recognition. “Recognize a Beta integral”: it relates to Γ by $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$.

Solution. Write $\Gamma(p)\Gamma(q) = \int_0^\infty \int_0^\infty u^{p-1}v^{q-1}e^{-(u+v)} du dv$. Change variables $u = ts$, $v = (1-t)s$ with $s = u+v > 0$, $t \in (0, 1)$, Jacobian s :

$$\Gamma(p)\Gamma(q) = \int_0^\infty \int_0^1 (ts)^{p-1}((1-t)s)^{q-1} e^{-s} s dt ds = \underbrace{\int_0^1 t^{p-1}(1-t)^{q-1} dt}_{B(p,q)} \underbrace{\int_0^\infty s^{p+q-1} e^{-s} ds}_{\Gamma(p+q)}.$$

Hence $B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$. *Check.* $B(1, 1) = \int_0^1 1 dt = 1 = \Gamma(1)^2/\Gamma(2) = 1$. ■

Caution. The functional equation $\Gamma(x+1) = x\Gamma(x)$ alone does *not* determine Γ : multiplying by any 1-periodic positive factor $p(x)$ with $p(1) = 1$ preserves both (1) and (2). It is *log-convexity* (3) that selects Γ uniquely (Bohr–Mollerup). Dropping log-convexity, infinitely many “factorial interpolants” exist.

Further worked examples

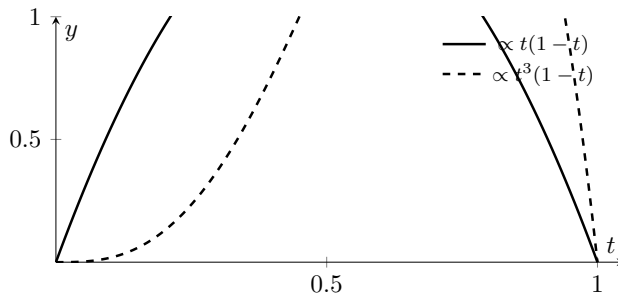


Figure 9.11. Integrands $t^{p-1}(1-t)^{q-1}$ of the Beta integral on $[0, 1]$ (here normalized to unit area). The total area is exactly $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$; unequal p, q skew the bump toward the larger exponent.

Worked Example 9.6.10 (A Gaussian moment integral via Γ). Evaluate $\int_0^\infty x^4 e^{-x^2} dx$.

Recognition. “Recognize a Γ in disguise”: substitute $t = x^2$ to turn $\int x^a e^{-x^2} dx$ into a Γ integral.

Solution. Put $t = x^2$, $dt = 2x dx$, so $x = t^{1/2}$ and $dx = \frac{1}{2}t^{-1/2}dt$:

$$\int_0^\infty x^4 e^{-x^2} dx = \int_0^\infty t^2 e^{-t} \cdot \frac{1}{2}t^{-1/2} dt = \frac{1}{2} \int_0^\infty t^{3/2} e^{-t} dt = \frac{1}{2} \Gamma\left(\frac{5}{2}\right).$$

Using $\Gamma(\frac{5}{2}) = \frac{3}{4}\sqrt{\pi}$ from the half-integer worked example, the value is $\frac{1}{2} \cdot \frac{3}{4}\sqrt{\pi} = \frac{3\sqrt{\pi}}{8}$. *Check.* Numerically $\frac{3\sqrt{\pi}}{8} \approx 0.6647$; this is also $\frac{3}{4} \cdot \frac{\sqrt{\pi}}{2}$, the fourth moment $\frac{\sqrt{\pi}}{2} \cdot \frac{3}{4}$ of the half-line Gaussian. ■

Worked Example 9.6.11 (Evaluating a Beta integral numerically). Compute $\int_0^1 x^3(1-x)^2 dx$ two ways: directly and via the Beta–Gamma relation.

Recognition. “A polynomial-on- $[0, 1]$ integral is a Beta value”: match it to $B(p, q) = \int_0^1 x^{p-1}(1-x)^{q-1} dx = \Gamma(p)\Gamma(q)/\Gamma(p+q)$.

Solution. Here $p-1 = 3$ and $q-1 = 2$, so $p = 4$, $q = 3$. Then

$$B(4, 3) = \frac{\Gamma(4)\Gamma(3)}{\Gamma(7)} = \frac{3! \cdot 2!}{6!} = \frac{6 \cdot 2}{720} = \frac{12}{720} = \frac{1}{60}.$$

Directly: $\int_0^1 x^3(1-2x+x^2)dx = \int_0^1 (x^3 - 2x^4 + x^5)dx = \frac{1}{4} - \frac{2}{5} + \frac{1}{6} = \frac{15-24+10}{60} = \frac{1}{60}$. *Check.* Both routes give $\frac{1}{60}$. ■

Worked Example 9.6.12 (A non-example: the factorial alone does not pin down Γ). Define $H(x) = \Gamma(x)(1 + \frac{1}{2}\sin(2\pi x) \cdot 0)$ — more honestly, let $H(x) = \Gamma(x)e^{c\sin(2\pi x)}$ for a small constant $c > 0$. Show H still satisfies $H(n+1) = n!$ on integers and $H(x+1) = xH(x) \cdot [\text{a 1-periodic factor}]$, yet $H \neq \Gamma$; identify which Bohr–Møllerup hypothesis H violates.

Recognition. “Test the uniqueness hypotheses”: perturb Γ by a 1-periodic factor and check which of (1)–(3) breaks.

Solution. Since $\sin(2\pi n) = 0$ at integers, $e^{c\sin(2\pi x)} = 1$ there, so $H(n+1) = \Gamma(n+1) = n!$: H agrees with the factorial on the integers. The factor $p(x) = e^{c\sin(2\pi x)}$ is 1-periodic with $p(1) = 1$, so $H(x+1) = \Gamma(x+1)p(x+1) = x\Gamma(x)p(x) = xH(x)$ — the functional equation (1) and normalization (2) both survive. What fails is *log-convexity* (3): $\log H = \log \Gamma + c\sin(2\pi x)$, and the wiggling sine term destroys convexity for any $c > 0$. By Bohr–Møllerup (Thm. 9.6.5) only the log-convex interpolant is Γ , so $H \neq \Gamma$. *Check.* $\frac{d^2}{dx^2} \log H = (\log \Gamma)'' - c(2\pi)^2 \sin(2\pi x)$, which is negative wherever $\sin(2\pi x)$ is large and positive — not convex. ■

Exercises (Rudin, Ch.8 – Stirling and the Gamma function).

- (A good elementary approximation to Stirling’s formula.) For $m = 1, 2, 3, \dots$ define $f(x) = (m+1-x)\log m + (x-m)\log(m+1)$ if $m \leq x \leq m+1$, and define $g(x) = \frac{x}{m} - 1 + \log m$ if $m - \frac{1}{2} \leq x \leq m + \frac{1}{2}$. Draw the graphs of f and g , note $f(x) < \log x < g(x)$ for $x > 1$, and that $\int_1^n f(x) dx = \log(n!) - \frac{1}{2} \log n$. Integrating $\log x$ over $[1, n]$, conclude $\frac{7}{8} < \log(n!) - (n + \frac{1}{2}) \log n + n < 1$ for $n = 2, 3, \dots$, and hence $e^{7/8} < \frac{n!}{(n/e)^n \sqrt{n}} < e$.
- Let $L_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_n(t)| dt$ (the Lebesgue constants, D_n the Dirichlet kernel). Prove there is a constant $C > 0$ with $L_n > C \log n$, and more precisely that the sequence $(L_n - \frac{4}{\pi^2} \log n)$ is bounded.
- If α is real and $-1 < x < 1$, prove *Newton’s binomial theorem*

$$(1+x)^\alpha = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} x^n.$$

(Hint: denote the right side by $f(x)$, prove the series converges, prove $(1+x)f'(x) = \alpha f(x)$, and solve this differential equation.) Show also that, for $-1 < x < 1$ and $\alpha > 0$, $(1-x)^{-\alpha} = \sum_{n=0}^{\infty} \frac{\Gamma(n+\alpha)}{\Gamma(\alpha)n!} x^n$.

Additional exercises.

- Verify $\Gamma(1) = 1$, $\Gamma(2) = 1$, $\Gamma(3) = 2$, $\Gamma(4) = 6$ directly from the functional equation, confirming $\Gamma(n+1) = n!$.
- Show the integral $\int_0^\infty t^{x-1} e^{-t} dt$ converges for every $x > 0$ by splitting at $t = 1$ and bounding each piece, and diverges for $x \leq 0$.
- Compute $\Gamma(\frac{1}{2})$, $\Gamma(\frac{3}{2})$, $\Gamma(\frac{5}{2})$, and $\Gamma(\frac{7}{2})$ in closed form using $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and the functional equation.
- (Diagram.) Plot the factorial nodes $(1, 0!)$, $(2, 1!)$, $(3, 2!)$, $(4, 3!)$ and sketch a smooth increasing curve through them representing $\Gamma(x)$; mark where the curve dips to its minimum on $(0, \infty)$ and note $\Gamma(x) \rightarrow \infty$ as $x \rightarrow 0^+$.
- (Non-example.) Let $h(x) = \Gamma(x)p(x)$ with p a positive 1-periodic function satisfying $p \equiv 1$ on $\mathbb{Z}$ (e.g. $p(x) = 1 + \frac{1}{2}\sin^2(\pi x)$). Show h satisfies $h(x+1) = xh(x)$ and $h(n+1) = n!$, yet $h \neq \Gamma$ unless $p \equiv 1$. Which Bohr–Møllerup hypothesis does h violate?
- Prove Γ is log-convex by Holder’s inequality, filling in the exponents $1/p + 1/q = 1$, and state the AM–GM special case that follows from log-convexity.
- Use the Beta function $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ to evaluate $\int_0^1 \sqrt{t(1-t)} dt$ and $\int_0^{\pi/2} \sin^4 \theta \cos^2 \theta d\theta$.

8. (Harder.) Use Stirling's formula to find $\lim_{n \rightarrow \infty} \frac{(2n)!}{4^n (n!)^2 \sqrt{\pi n}}$ and $\lim_{n \rightarrow \infty} \frac{n! e^n}{n^n \sqrt{n}}$, identifying each limiting constant.

Chapter 10

Functions of Several Variables

10.1 Linear Transformations and Operator Norms

We now leave the line and the plane behind: the maps of this chapter go from open sets of $\mathbb{R}^n$ to $\mathbb{R}^k$. The single derivative $f'(a)$ of one-variable calculus becomes a *linear transformation* — a matrix — and the entire theory of differentiation is the study of how a nonlinear f is approximated, near each point, by such a linear map. So we begin with linear algebra recast in the metric language of analysis: the size of a linear map, measured by its *operator norm*, and the topology of the set of invertible maps.

Linear transformations and their matrices

Definition 10.1.1. A map $A : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is a *linear transformation* if

$$A(x + y) = Ax + Ay, \quad A(cx) = cAx \quad (x, y \in \mathbb{R}^n, c \in \mathbb{R}).$$

We write $L(\mathbb{R}^n, \mathbb{R}^k)$ for the set of all such, and $L(\mathbb{R}^n) = L(\mathbb{R}^n, \mathbb{R}^n)$. A linear map is determined by its values on the standard basis $\{e_1, \dots, e_n\}$ of $\mathbb{R}^n$: if $x = \sum_j x_j e_j$ then $Ax = \sum_j x_j Ae_j$. Writing each $Ae_j \in \mathbb{R}^k$ in coordinates, $Ae_j = \sum_i a_{ij} e_i$, gives the *matrix* $[A] = (a_{ij})$, a $k \times n$ array, with

$$(Ax)_i = \sum_{j=1}^n a_{ij} x_j \quad (1 \leq i \leq k).$$

The columns of $[A]$ are the image vectors $Ae_1, \dots, Ae_n$. Composition of linear maps corresponds to matrix multiplication: if $A \in L(\mathbb{R}^n, \mathbb{R}^k)$ and $B \in L(\mathbb{R}^k, \mathbb{R}^m)$ then $BA \in L(\mathbb{R}^n, \mathbb{R}^m)$ and $[BA] = [B][A]$. The *range* $\mathcal{R}(A) = A(\mathbb{R}^n)$ and the *null space* $\mathcal{N}(A) = \{x : Ax = 0\}$ are subspaces; their dimensions, the *rank* and *nullity*, satisfy $\text{rank } A + \dim \mathcal{N}(A) = n$.

Proposition 10.1.2 (Rudin 9.5: invertibility criteria in $L(\mathbb{R}^n)$). *For $A \in L(\mathbb{R}^n)$ the following are equivalent: (i) A is invertible (a two-sided inverse $A^{-1} \in L(\mathbb{R}^n)$ exists); (ii) A is one-to-one; (iii) A is onto; (iv) $\text{rank } A = n$; (v) $\det A \neq 0$. The set of all invertible members of $L(\mathbb{R}^n)$ is denoted Ω .*

Proof. The rank-nullity identity $\text{rank } A + \dim \mathcal{N}(A) = n$ makes (ii), (iii), (iv) equivalent: A is one-to-one iff $\mathcal{N}(A) = \{0\}$ iff the nullity is 0 iff $\text{rank } A = n$ iff $\mathcal{R}(A) = \mathbb{R}^n$, i.e. A is onto. A one-to-one onto linear map has an inverse, which is automatically linear (Exercise), giving (i). The equivalence with $\det A \neq 0$ is the content of determinant theory, reviewed in §6. $\square$

The operator norm

A linear map is continuous, but *how much* can it stretch a vector? The answer is a single number that controls everything analytic about A .

Definition 10.1.3. For $A \in L(\mathbb{R}^n, \mathbb{R}^k)$ the *operator norm* is

$$\|A\| = \sup\{|Ax| : x \in \mathbb{R}^n, |x| \leq 1\} = \sup_{x \neq 0} \frac{|Ax|}{|x|}.$$

Here $|\cdot|$ is the Euclidean norm on $\mathbb{R}^n$ and on $\mathbb{R}^k$.

Equivalently $\|A\|$ is the smallest constant M for which $|Ax| \leq M|x|$ holds for all x . The supremum is finite: writing $x = \sum_j x_j e_j$ with $|x| \leq 1$, the triangle and Cauchy–Schwarz inequalities give $|Ax| \leq \sum_j |x_j| |Ae_j| \leq (\sum_j |Ae_j|^2)^{1/2}$, a fixed bound. Geometrically $\|A\|$ is the maximal stretch factor: the image of the unit ball is contained in the ball of radius $\|A\|$, and a vector of length $\|A\|$ is actually attained on the unit sphere (which is compact).

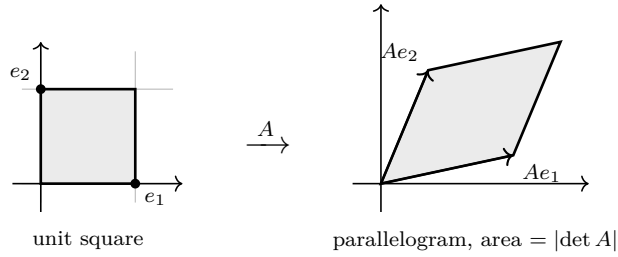


Figure 10.1. A linear map sends the unit square to the parallelogram spanned by the columns Ae_1, Ae_2 . Its area is $|\det A|$; the longest image of a unit vector is $\|A\|$.

Theorem 10.1.4 (Rudin 9.7: properties of the norm). *Let $A, B \in L(\mathbb{R}^n, \mathbb{R}^k)$ and $c \in \mathbb{R}$.*

1. $|Ax| \leq \|A\| |x|$ for all $x \in \mathbb{R}^n$; $\|A\| < \infty$, and A is a uniformly continuous map of $\mathbb{R}^n$ into $\mathbb{R}^k$.
2. $\|A + B\| \leq \|A\| + \|B\|$ and $\|cA\| = |c| \|A\|$; thus $\|\cdot\|$ is a norm on $L(\mathbb{R}^n, \mathbb{R}^k)$, making it a metric space with $d(A, B) = \|A - B\|$.
3. If $A \in L(\mathbb{R}^n, \mathbb{R}^k)$ and $B \in L(\mathbb{R}^k, \mathbb{R}^m)$ then $\|BA\| \leq \|B\| \|A\|$ (submultiplicativity).

Proof. (1) For $x \neq 0$, $\frac{|Ax|}{|x|} \leq \|A\|$ by definition of the supremum, i.e. $|Ax| \leq \|A\| |x|$ (trivially true at $x = 0$). Finiteness was shown above. For continuity, linearity gives $|Ax - Ay| = |A(x - y)| \leq \|A\| |x - y|$, so A is Lipschitz with constant $\|A\|$, hence uniformly continuous.

(2) For $|x| \leq 1$, $|(A + B)x| \leq |Ax| + |Bx| \leq \|A\| + \|B\|$; taking the supremum over the unit ball gives the triangle inequality. The scaling identity is immediate, and $\|A\| = 0$ forces $Ax = 0$ for all x , i.e. $A = 0$. So $\|\cdot\|$ is a genuine norm.

(3) For $|x| \leq 1$: $|BAx| \leq \|B\| |Ax| \leq \|B\| \|A\| |x| \leq \|B\| \|A\|$, and we take the supremum. $\square$

Remark. The operator norm is generally *not* the Euclidean norm of the matrix entries. For a diagonal map $A = \text{diag}(d_1, \dots, d_n)$ one has $\|A\| = \max_i |d_i|$, while the entry-norm $(\sum d_i^2)^{1/2}$ (the Hilbert–Schmidt norm) is larger when more than one d_i is nonzero. The two are equivalent norms on the finite-dimensional space $L(\mathbb{R}^n, \mathbb{R}^k)$, but the operator norm is the one that interacts correctly with composition.

Method (Estimating or computing $\|A\|$). *Recognition trigger.* You need a bound on $|Ax|$, or the exact stretch of a given matrix.

1. **Upper bound (cheap).** $\|A\| \leq (\sum_{i,j} a_{ij}^2)^{1/2}$, the Hilbert–Schmidt bound; quick and often enough for “ $\|A\| < \infty$ ”-style arguments.
2. **Special structure.** If A is diagonal, $\|A\| = \max_i |d_i|$. If A is an orthogonal map (columns orthonormal), $|Ax| = |x|$ for all x , so $\|A\| = 1$.
3. **Exact value.** $\|A\| = \sqrt{\lambda_{\max}}$ where $\lambda_{\max}$ is the largest eigenvalue of $A^T A$ (the largest *singular value* of A). For symmetric A this is $\max_i |\lambda_i|$.

4. **Lower bound.** Plug in any unit vector u : $\|A\| \geq |Au|$.

Worked Example 10.1.5 (The norm of a 2×2 map). Compute $\|A\|$ for $A = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$ and for the rotation

$$R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

Recognition. “Find the exact operator norm.” Diagonal and orthogonal cases from the Method.

Solution. A is diagonal with entries 3, -1 , so $\|A\| = \max(|3|, |-1|) = 3$; indeed $|Ae_1| = 3$ attains it. For R_θ , $|R_\theta x|^2 = (x_1 \cos \theta - x_2 \sin \theta)^2 + (x_1 \sin \theta + x_2 \cos \theta)^2 = x_1^2 + x_2^2 = |x|^2$, so R_θ preserves length and $\|R_\theta\| = 1$.
Check. $A^T A = \text{diag}(9, 1)$ has $\lambda_{\max} = 9$, and $\sqrt{9} = 3$. ■

The set of invertible maps is open; inversion is continuous

The following is the analytic heart of §1 and the engine of the Inverse Function Theorem: invertibility is a *stable* property, and $A \mapsto A^{-1}$ is continuous.

Theorem 10.1.6 (Rudin 9.8). *Let $\Omega \subseteq L(\mathbb{R}^n)$ be the set of invertible linear operators.*

1. *If $A \in \Omega$, $B \in L(\mathbb{R}^n)$, and $\|B - A\| \cdot \|A^{-1}\| < 1$, then $B \in \Omega$. In particular Ω is open in $L(\mathbb{R}^n)$.*
2. *The map $A \mapsto A^{-1}$ is a continuous bijection of Ω onto Ω (indeed a homeomorphism).*

Proof. (1) Put $\alpha = 1/\|A^{-1}\|$ and $\beta = \|B - A\|$; the hypothesis is $\beta < \alpha$. For every x ,

$$\alpha |x| = \frac{|x|}{\|A^{-1}\|} = \frac{|A^{-1}Ax|}{\|A^{-1}\|} \leq |Ax| \leq |(A - B)x| + |Bx| \leq \beta |x| + |Bx|,$$

using $|A^{-1}y| \leq \|A^{-1}\| |y|$ with $y = Ax$. Hence

$$(\alpha - \beta) |x| \leq |Bx| \quad (x \in \mathbb{R}^n). \quad (*)$$

Since $\alpha - \beta > 0$, $Bx = 0$ forces $x = 0$, so B is one-to-one, hence invertible by Proposition 10.1.2. As every B with $\|B - A\| < \alpha = 1/\|A^{-1}\|$ is invertible, the open ball of radius $1/\|A^{-1}\|$ about A lies in Ω ; thus Ω is open.

(2) Apply (*) with x replaced by $B^{-1}y$: $(\alpha - \beta) |B^{-1}y| \leq |y|$, so $\|B^{-1}\| \leq (\alpha - \beta)^{-1}$. Now

$$B^{-1} - A^{-1} = B^{-1}(A - B)A^{-1},$$

(verify by left-multiplying by B and right by A), so

$$\|B^{-1} - A^{-1}\| \leq \|B^{-1}\| \|A - B\| \|A^{-1}\| \leq \frac{\beta}{(\alpha - \beta)\alpha}.$$

As $B \rightarrow A$ we have $\beta = \|B - A\| \rightarrow 0$ while α is fixed, so the right side $\rightarrow 0$. Hence $B^{-1} \rightarrow A^{-1}$: inversion is continuous at every $A \in \Omega$. It is its own inverse, so it is a homeomorphism of Ω onto itself. □

Remark. Inequality (*) is worth remembering on its own: a perturbation of an invertible map by less than $1/\|A^{-1}\|$ stays bounded away from being singular, with the explicit modulus $\alpha - \beta$. The determinant gives a slicker but less quantitative proof of openness: $\det : L(\mathbb{R}^n) \rightarrow \mathbb{R}$ is continuous (a polynomial in the entries), and $\Omega = \det^{-1}(\mathbb{R} \setminus \{0\})$ is the preimage of an open set. The norm proof, however, supplies the bound on $\|B^{-1}\|$ that the next sections need.

Caution. The condition for B to stay invertible is $\|B - A\| < 1/\|A^{-1}\|$, *not* $\|B - A\| < 1$. The threshold shrinks when A^{-1} is large, i.e. when A is nearly singular: a matrix close to non-invertible tolerates only a tiny perturbation. Forgetting the factor $\|A^{-1}\|$ is the classic error.

Worked Example 10.1.7 (A perturbation that stays invertible). Let $A = I$ (the identity in $L(\mathbb{R}^n)$) and $B = I + E$ with $\|E\| < 1$. Show B is invertible and $\|B^{-1}\| \leq (1 - \|E\|)^{-1}$, and identify the Neumann series for B^{-1} .

Recognition. “Show $I + E$ is invertible for small E .” This is Theorem 10.1.6 with $A = I$, $\|A^{-1}\| = 1$.

Solution. Since $\|B - A\| = \|E\| < 1 = 1/\|I^{-1}\|$, $B \in \Omega$. The bound from the proof gives $\|B^{-1}\| \leq (\alpha - \beta)^{-1} = (1 - \|E\|)^{-1}$. Concretely, the geometric (Neumann) series $\sum_{m=0}^{\infty} (-E)^m$ converges in operator norm because $\|(-E)^m\| \leq \|E\|^m$ and $\sum \|E\|^m = (1 - \|E\|)^{-1} < \infty$; its sum S satisfies $(I + E)S = I = S(I + E)$, so $B^{-1} = \sum_{m \geq 0} (-E)^m$. *Check.* Taking norms, $\|B^{-1}\| \leq \sum_m \|E\|^m = (1 - \|E\|)^{-1}$, matching the abstract bound. ■

Further worked examples

Worked Example 10.1.8 (Operator norm of a non-diagonal matrix via singular values). Compute $\|A\|$ exactly for $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ (a shear).

Recognition. “Exact operator norm” of a non-diagonal, non-orthogonal map: use $\|A\| = \sqrt{\lambda_{\max}(A^T A)}$ from the Method.

Solution. Form $A^T A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$. Its eigenvalues solve $\det(A^T A - \lambda I) = (1 - \lambda)(5 - \lambda) - 4 = \lambda^2 - 6\lambda + 1 = 0$, so $\lambda = 3 \pm 2\sqrt{2}$. The larger is $\lambda_{\max} = 3 + 2\sqrt{2} = (\sqrt{2} + 1)^2$, hence $\|A\| = \sqrt{\lambda_{\max}} = \sqrt{2} + 1 \approx 2.414$. *Check (sanity bounds):* the Hilbert–Schmidt bound gives $\|A\| \leq \sqrt{1 + 4 + 0 + 1} = \sqrt{6} \approx 2.449$, and the lower bound from $u = e_2$ gives $\|A\| \geq |Ae_2| = |(2, 1)| = \sqrt{5} \approx 2.236$; the exact value 2.414 sits between, as it must. ■

Worked Example 10.1.9 (The norm is not the largest entry, nor the largest column). For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ show $\|A\| = 2$, even though every entry is 1 and each column has length $\sqrt{2}$.

Recognition. A common misconception is $\|A\| = \max |a_{ij}|$ or $\|A\| = \max_j |Ae_j|$. This is a non-example refuting both; compute the true norm.

Solution. A is symmetric, so $\|A\| = \max_i |\lambda_i|$. The eigenvalues of A are 0 (eigenvector $(1, -1)$) and 2 (eigenvector $(1, 1)$), since $A(1, 1) = (2, 2) = 2(1, 1)$. Thus $\|A\| = 2$. The maximal stretch is attained at the unit vector $u = (1, 1)/\sqrt{2}$: $|Au| = |(\sqrt{2}, \sqrt{2})| = 2$. *Check.* The naive guesses give $\max |a_{ij}| = 1$ and $\max_j |Ae_j| = \sqrt{2} \approx 1.41$, both strictly below the true $\|A\| = 2$ — the column vectors reinforce along $(1, 1)$, so the stretch exceeds either column length. ■

Worked Example 10.1.10 (Operator norm of a symmetric 2×2 map, and where it is attained). Compute $\|A\|$ exactly for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, and exhibit a unit vector on which the maximal stretch is achieved.

Recognition. “Exact operator norm” of a *symmetric* matrix: by the Method, $\|A\| = \max_i |\lambda_i|$, and the stretch is attained on the corresponding eigenvector.

Solution. A is symmetric, so $A^T A = A^2$ and $\|A\| = \max_i |\lambda_i|$. The eigenvalues solve $\det(A - \lambda I) = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = 0$, so $\lambda = 1$ or $\lambda = 3$. Hence $\|A\| = \max(|1|, |3|) = 3$, attained on the unit eigenvector for $\lambda = 3$: $A(1, 1) = (3, 3) = 3(1, 1)$, so $u = (1, 1)/\sqrt{2}$ gives $|Au| = |(3, 3)|/\sqrt{2} = 3\sqrt{2}/\sqrt{2} = 3 = \|A\|$. *Check (sanity bounds):* the Hilbert–Schmidt bound is $\sqrt{4 + 1 + 1 + 4} = \sqrt{10} \approx 3.16$, and the lower bound from e_1 is $|Ae_1| = |(2, 1)| = \sqrt{5} \approx 2.24$; the exact value 3 lies between, as it must. ■

Worked Example 10.1.11 (A linear map carrying the unit square to a parallelogram). For $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ describe the image of the unit square $[0, 1]^2$, compute its area, and bound $\|A\|$ from above and below.

Recognition. “Image of the unit square / area distortion” — the columns Ae_1, Ae_2 span the image parallelogram, its area is $|\det A|$, and any unit vector gives a lower bound for $\|A\|$.

Solution. The columns are $Ae_1 = (2, 0)$ and $Ae_2 = (1, 1)$, so A sends the unit square to the parallelogram with vertices $0, (2, 0), (3, 1), (1, 1)$ (the fourth being $Ae_1 + Ae_2$). Its area is $|\det A| = |2 \cdot 1 - 1 \cdot 0| = 2$, twice the

unit square's area. For the norm, $A^T A = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$, with eigenvalues solving $\lambda^2 - 6\lambda + 4 = 0$, i.e. $\lambda = 3 \pm \sqrt{5}$; so $\|A\| = \sqrt{3 + \sqrt{5}} \approx 2.288$. *Check.* The lower bound $|Ae_1| = 2$ and the Hilbert–Schmidt upper bound $\sqrt{4 + 1 + 0 + 1} = \sqrt{6} \approx 2.449$ bracket the exact 2.288; and the area factor $|\det A| = 2$ equals the product of singular values $\sqrt{(3 + \sqrt{5})(3 - \sqrt{5})} = \sqrt{9 - 5} = 2$, confirming the two eigenvalues of $A^T A$. ■

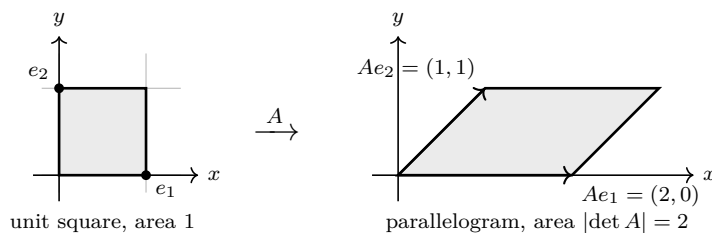


Figure 10.2. The shear $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ sends the unit square to the parallelogram on Ae_1, Ae_2 ; area is multiplied by $|\det A| = 2$, while the longest unit image has length $\|A\| = \sqrt{3 + \sqrt{5}}$.

Exercises (Rudin, Ch.9).

1. If S is a nonempty subset of a vector space X , prove (as asserted in Sec. 9.1) that the span of S is a vector space.
2. Prove (as asserted in Sec. 9.6) that BA is linear if A and B are linear transformations. Prove also that A^{-1} is linear and invertible (when A is).
3. Assume $A \in L(X, Y)$ and $Ax = 0$ only when $x = 0$. Prove that A is then one-to-one.
4. Prove (as asserted in Sec. 9.30) that null spaces and ranges of linear transformations are vector spaces.
5. Prove that to every $A \in L(\mathbb{R}^n, \mathbb{R}^1)$ there corresponds a unique $y \in \mathbb{R}^n$ such that $Ax = x \cdot y$. Prove also that $\|A\| = |y|$. *Hint.* Under certain conditions, equality holds in the Schwarz inequality.

Additional exercises.

1. (Warm-up norms.) Compute the operator norm $\|A\|$ of each map: (a) $A = \text{diag}(2, -5, 3)$ on $\mathbb{R}^3$; (b) the shear $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ on $\mathbb{R}^2$ (give $\|A\|$ exactly via the largest singular value); (c) the projection $A(x, y, z) = (x, y, 0)$.
2. Let $A \in L(\mathbb{R}^n, \mathbb{R}^k)$. Prove $\|A\| = \max\{|Au| : |u| = 1\}$, i.e. the supremum in the definition is attained. (Hint: the unit sphere is compact and $u \mapsto |Au|$ is continuous.)
3. For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ show $\|A^m\| \rightarrow \infty$ as $m \rightarrow \infty$, yet $\det A^m = 1$ for every m . Conclude that a bounded determinant does not bound the operator norm.
4. (Disprove.) A student claims that for all $A, B \in L(\mathbb{R}^n)$, $\|AB\| = \|BA\|$. Give an explicit 2×2 counterexample. Then prove the weaker true statement that AB and BA always have the same nonzero eigenvalues.
5. (Invertibility margin.) Let $A = \begin{bmatrix} 1 & 0 \\ 0 & \varepsilon \end{bmatrix}$ with $0 < \varepsilon < 1$. Compute $\|A^{-1}\|$ and the largest δ for which every B with $\|B - A\| < \delta$ is guaranteed invertible by Theorem 10.1.6. Explain in one line why $\delta \rightarrow 0$ as $\varepsilon \rightarrow 0$.
6. Prove that for invertible A , $\|A^{-1}\| \geq 1/\|A\|$, and exhibit a map where the inequality is strict. (Hint: $1 = \|I\| = \|AA^{-1}\| \leq \|A\| \|A^{-1}\|$.)
7. (Diagram.) On a single picture of the plane draw the unit circle $|x| = 1$ and its image under $A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$ (an ellipse). Mark the two unit vectors u achieving the maximal and minimal stretch, label the stretch factors, and read off $\|A\|$ and $\|A^{-1}\|^{-1}$ from the semi-axes.
8. (Hilbert–Schmidt vs operator norm.) For $A = (a_{ij}) \in L(\mathbb{R}^n, \mathbb{R}^k)$ let $\|A\|_{HS} = (\sum_{i,j} a_{ij}^2)^{1/2}$. Prove $\|A\| \leq \|A\|_{HS} \leq \sqrt{\min(n, k)} \|A\|$, and determine when each inequality is an equality.

10.2 The Total Derivative

In one variable, $f'(a)$ is a number; the line $y = f(a) + f'(a)(x - a)$ is the best affine approximation to f near a , and “best” means the error vanishes faster than $|x - a|$. In several variables the slope can no longer be a number — the input is a vector — so the derivative becomes the *linear map* that does the approximating.

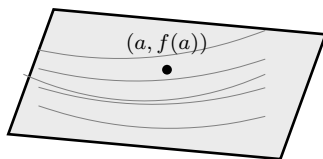
The total (Frechet) derivative

Definition 10.2.1. Let $E \subseteq \mathbb{R}^n$ be open, $f : E \rightarrow \mathbb{R}^k$, and $x \in E$. We say f is *differentiable* at x if there exists a linear map $A \in L(\mathbb{R}^n, \mathbb{R}^k)$ such that

$$\lim_{h \rightarrow 0} \frac{|f(x+h) - f(x) - Ah|}{|h|} = 0.$$

Equivalently, with $r(h) = f(x+h) - f(x) - Ah$, the remainder satisfies $|r(h)|/|h| \rightarrow 0$ as $h \rightarrow 0$. We then write $f'(x) = A$ and call it the *(total) derivative* of f at x . If f is differentiable at every point of E , f is *differentiable on E* , and $f' : E \rightarrow L(\mathbb{R}^n, \mathbb{R}^k)$ is the derivative map.

Read this as: $f(x+h) \approx f(x) + f'(x)h$, with an error that is $o(|h|)$ — smaller than any linear function of $|h|$. The increment $f(x+h) - f(x)$ is approximated by the *linear* increment $f'(x)h$. The affine map $h \mapsto f(x) + f'(x)h$ is the tangent plane (for $k = 1$, $n = 2$ literally a plane).



tangent plane $z = f(a) + f'(a)(x - a)$ touches $z = f(x)$ to first order

Figure 10.3. The graph $z = f(x, y)$ and its tangent plane at $(a, f(a))$. Differentiability means the gap between surface and plane is $o(|x - a|)$.

Theorem 10.2.2 (Rudin 9.11: uniqueness of the derivative). *If $f : E \rightarrow \mathbb{R}^k$ is differentiable at x , the linear map A in Definition 10.2.1 is unique.*

Proof. Suppose A_1, A_2 both work; put $B = A_1 - A_2 \in L(\mathbb{R}^n, \mathbb{R}^k)$. Subtracting the two limit statements, for every $h \neq 0$ near 0,

$$\frac{|Bh|}{|h|} \leq \frac{|f(x+h) - f(x) - A_1h|}{|h|} + \frac{|f(x+h) - f(x) - A_2h|}{|h|} \xrightarrow{h \rightarrow 0} 0.$$

Fix any unit vector u and let $h = tu$ with $t \rightarrow 0^+$. By linearity $|Bh|/|h| = |B(tu)|/|tu| = |Bu|$, a constant in t ; the limit being 0 forces $|Bu| = 0$. As u was an arbitrary unit vector, $B = 0$, i.e. $A_1 = A_2$. $\square$

Proposition 10.2.3 (Differentiable implies continuous). *If f is differentiable at x then f is continuous at x .*

Proof. Write $f(x+h) - f(x) = f'(x)h + r(h)$ with $|r(h)| \leq \varepsilon|h|$ for $|h|$ small (taking $\varepsilon = 1$, say). Then $|f(x+h) - f(x)| \leq \|f'(x)\||h| + |h| \rightarrow 0$ as $h \rightarrow 0$, so $f(x+h) \rightarrow f(x)$. $\square$

Remark. For $k = 1, n = 1$ the definition reduces to the familiar one: A is multiplication by the number $f'(x) = \lim_{h \rightarrow 0} (f(x+h) - f(x))/h$. The matrix of $f'(x)$ in general is the Jacobian, computed from partial derivatives in §3. A single *vector-valued* map $f = (f_1, \dots, f_k)$ is differentiable iff each component f_i is, and then the rows of $[f'(x)]$ are the derivatives of the f_i .

Worked Example 10.2.4 (Computing f' for a quadratic map). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x, y) = (x^2 - y^2, 2xy)$. Show f is differentiable everywhere and find $f'(x, y)$.

Recognition. “Find the total derivative”; guess the linear part from first-order expansion and verify the $o(|h|)$ remainder.

Solution. With $h = (s, t)$, expand $f(x + s, y + t) = ((x + s)^2 - (y + t)^2, 2(x + s)(y + t))$. The first component is $x^2 - y^2 + (2xs - 2yt) + (s^2 - t^2)$; the second is $2xy + (2ys + 2xt) + (2st)$. So $f(p + h) - f(p) = Ah + r(h)$ with

$$A = \begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix}, \quad r(h) = (s^2 - t^2, 2st).$$

Now $|r(h)| = ((s^2 - t^2)^2 + 4s^2t^2)^{1/2} = s^2 + t^2 = |h|^2$, so $|r(h)|/|h| = |h| \rightarrow 0$. Hence $f'(x, y) = A$. *Check.* This f is the complex squaring $z \mapsto z^2$; A is multiplication by $2z = 2(x + iy)$, whose real matrix is exactly $\begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix}$. ■

The chain rule

The reason the derivative is defined as a *linear map* (not a list of numbers) is that it makes the chain rule a clean composition statement: the derivative of a composite is the composite of the derivatives.

Theorem 10.2.5 (Rudin 9.15: Chain Rule). *Let $E \subseteq \mathbb{R}^n$ be open, $f : E \rightarrow \mathbb{R}^k$ differentiable at $x_0 \in E$, and let g map an open set containing $f(E)$ into $\mathbb{R}^m$ and be differentiable at $y_0 = f(x_0)$. Then the composite $F = g \circ f : E \rightarrow \mathbb{R}^m$ is differentiable at x_0 , and*

$$F'(x_0) = g'(y_0) f'(x_0) \quad (\text{composition of linear maps; product of matrices}).$$

Proof. Write $A = f'(x_0)$, $B = g'(y_0)$. Define the remainders

$$u(h) = f(x_0 + h) - f(x_0) - Ah, \quad v(\eta) = g(y_0 + \eta) - g(y_0) - B\eta,$$

so $|u(h)|/|h| \rightarrow 0$ and $|v(\eta)|/|\eta| \rightarrow 0$. Put $\eta(h) = f(x_0 + h) - f(x_0) = Ah + u(h)$, the increment of f . Then

$$F(x_0 + h) - F(x_0) = g(y_0 + \eta(h)) - g(y_0) = B\eta(h) + v(\eta(h)) = BA h + (Bu(h) + v(\eta(h))).$$

It remains to show the bracketed term is $o(|h|)$. First, $|Bu(h)| \leq \|B\| |u(h)| = o(|h|)$. Second, $|\eta(h)| \leq \|A\| |h| + |u(h)| \leq (\|A\| + 1) |h|$ for small h , so $\eta(h) \rightarrow 0$ as $h \rightarrow 0$ and

$$\frac{|v(\eta(h))|}{|h|} = \frac{|v(\eta(h))|}{|\eta(h)|} \cdot \frac{|\eta(h)|}{|h|} \leq \frac{|v(\eta(h))|}{|\eta(h)|} (\|A\| + 1) \xrightarrow{h \rightarrow 0} 0,$$

since the first factor $\rightarrow 0$ (as $\eta(h) \rightarrow 0$) and the second is bounded. (When $\eta(h) = 0$ the term $v(\eta(h)) = 0$ contributes nothing.) Therefore $F(x_0 + h) - F(x_0) = BA h + o(|h|)$, which says $F'(x_0) = BA = g'(y_0)f'(x_0)$. □

$$\begin{array}{ccc} \mathbb{R}^n \supseteq E & \xrightarrow{f} \mathbb{R}^k & \xrightarrow{g} \mathbb{R}^m \\ & \searrow F=g \circ f & \nearrow \\ \mathbb{R}^n & \xrightarrow{f'(x_0)} \mathbb{R}^k & \xrightarrow{g'(y_0)} \mathbb{R}^m \\ & \searrow F'(x_0) & \nearrow \end{array}$$

Figure 10.4. The chain rule: differentiation turns the composite of maps (left) into the composite of their derivative linear maps (right).

Method (Differentiating a composite). *Recognition trigger.* $F = g \circ f$, or you must relate the Jacobian of a composite to those of its parts.

1. Identify the inner map f at the point x_0 and the outer map g at $y_0 = f(x_0)$.
2. Compute the Jacobian matrices $[f'(x_0)]$ ($k \times n$) and $[g'(y_0)]$ ($m \times k$).
3. Multiply: $[F'(x_0)] = [g'(y_0)][f'(x_0)]$, an $m \times n$ matrix. The order is outer-times-inner; matrix product is not commutative, so do not swap.

4. For a scalar entry, $\frac{\partial F_i}{\partial x_j} = \sum_{\ell} \frac{\partial g_i}{\partial y_{\ell}} \frac{\partial f_{\ell}}{\partial x_j}$, the familiar multivariable chain rule, read off entrywise.

Worked Example 10.2.6 (A polar-coordinate chain rule). Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be differentiable and $f(r, \theta) = (r \cos \theta, r \sin \theta)$. With $F = g \circ f$, express $\partial F / \partial r$ and $\partial F / \partial \theta$ via the partials g_x, g_y .

Recognition. “Composite with a coordinate change”; apply the Method with f inner.

Solution. $[f'(r, \theta)] = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}$, and $[g'(x, y)] = [g_x \ g_y]$. Then $[F'(r, \theta)] = [g_x \ g_y] \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix} = [g_x \cos \theta + g_y \sin \theta \quad -rg_x \sin \theta + rg_y \cos \theta]$, i.e.

$$F_r = g_x \cos \theta + g_y \sin \theta, \quad F_{\theta} = r(-g_x \sin \theta + g_y \cos \theta).$$

Check. For $g(x, y) = x^2 + y^2$ one gets $F = r^2$, so $F_r = 2r$, $F_{\theta} = 0$; the formula gives $F_r = 2x \cos \theta + 2y \sin \theta = 2r \cos^2 \theta + 2r \sin^2 \theta = 2r$ and $F_{\theta} = r(-2x \sin \theta + 2y \cos \theta) = r(-2r \cos \theta \sin \theta + 2r \sin \theta \cos \theta) = 0$. ■

Caution. The chain rule needs f differentiable at x_0 and g differentiable at $y_0 = f(x_0)$, in the *total* sense. Mere existence of all partial derivatives of f and g is not enough: partials can exist while the map fails to be differentiable (§3), and then the matrix-product formula can give a wrong or meaningless answer. Always verify total differentiability, or use the C^1 shortcut of §3.

Further worked examples

Worked Example 10.2.7 (Differentiability of a scalar field from the definition). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^2 + xy$. Show f is differentiable at $(1, 2)$ directly from the definition and identify $f'(1, 2)$.

Recognition. “Verify differentiability from the limit” — guess the linear part from the first-order terms, then bound the remainder.

Solution. With $p = (1, 2)$ and $h = (s, t)$, $f(1+s, 2+t) = (1+s)^2 + (1+s)(2+t) = 3 + (2s+s^2) + (2+t+2s+st) - 2 =$ collect: $f(p) = 1 + 2 = 3$, and $f(p+h) - f(p) = (2s+s^2) + (2s+t+st) = \underbrace{(4s+t)}_{Ah} + \underbrace{(s^2+st)}_{r(h)}$. So the

candidate is the row $A = [4 \ 1]$, i.e. $f'(1, 2) = [4 \ 1]$ (matching $\nabla f = (2x+y, x)$ at $(1, 2)$). For the remainder, $|r(h)| = |s^2 + st| \leq |s|(|s| + |t|) \leq |h| \cdot 2|h| = 2|h|^2$, so $|r(h)|/|h| \leq 2|h| \rightarrow 0$. Hence f is differentiable at $(1, 2)$ with $f'(1, 2) = [4 \ 1]$. *Check.* The partials are $D_1 f = 2x + y = 4$ and $D_2 f = x = 1$ at $(1, 2)$, exactly the entries of A . ■

Worked Example 10.2.8 (Chain rule with two explicit Jacobians). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $f(s, t) = (s+t, st, s^2)$ and $g : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $g(u, v, w) = (u^2 + w, vw)$. Compute $(g \circ f)'(1, 1)$ as a matrix product.

Recognition. “Composite of two concrete maps” — form both Jacobians and multiply outer-times-inner.

Solution. $[f'(s, t)] = \begin{bmatrix} 1 & 1 \\ t & s \\ 2s & 0 \end{bmatrix}$, so at $(1, 1)$, $[f'(1, 1)] = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 2 & 0 \end{bmatrix}$. Also $[g'(u, v, w)] = \begin{bmatrix} 2u & 0 & 1 \\ 0 & w & v \end{bmatrix}$; here

$f(1, 1) = (2, 1, 1)$, so $[g'(2, 1, 1)] = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. The product is

$$[(g \circ f)'(1, 1)] = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 4 \\ 3 & 1 \end{bmatrix}.$$

Check. Compute $F = g \circ f$ directly: $F(s, t) = ((s+t)^2 + s^2, st \cdot s^2) = (2s^2 + 2st + t^2, s^3t)$. Then $F_s = (4s + 2t, 3s^2t)$ and $F_t = (2s + 2t, s^3)$; at $(1, 1)$ the Jacobian is $\begin{bmatrix} 6 & 4 \\ 3 & 1 \end{bmatrix}$, matching the product. ■

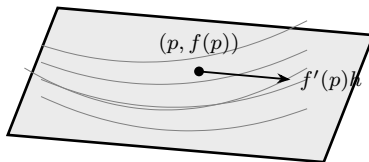
Worked Example 10.2.9 (Total derivative and tangent plane of a concrete surface). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^2 + xy + y^2$. Find $f'(1, 2)$, write the tangent-plane (affine) approximation there, and use it to estimate $f(1.1, 1.9)$.

Recognition. “Total derivative of a scalar field + affine approximation”: $f \in C^1$, so $f'(p) = [D_1 f \ D_2 f]$, and $f(p+h) \approx f(p) + f'(p)h$ with error $o(|h|)$.

Solution. The partials $D_1f = 2x + y$ and $D_2f = x + 2y$ are polynomials, hence continuous, so $f \in C^1(\mathbb{R}^2)$ and $f'(x, y) = [2x + y \ x + 2y]$. At $p = (1, 2)$: $f(p) = 1 + 2 + 4 = 7$ and $f'(1, 2) = [4 \ 5]$. The tangent plane is

$$z = f(1, 2) + f'(1, 2) \begin{bmatrix} x - 1 \\ y - 2 \end{bmatrix} = 7 + 4(x - 1) + 5(y - 2) = 4x + 5y - 7.$$

With $h = (0.1, -0.1)$ the affine estimate is $f(1.1, 1.9) \approx 7 + 4(0.1) + 5(-0.1) = 6.9$. *Check.* The exact value is $f(1.1, 1.9) = 1.21 + 2.09 + 3.61 = 6.91$; the affine prediction 6.9 errs by $0.01 = |h|^2/2$ in magnitude, consistent with the quadratic remainder $r(h) = h_1^2 + h_1h_2 + h_2^2$ being $o(|h|)$. ■



$$z = f(p) + f'(p)(x - p) \text{ touches } z = f(x) \text{ to first order at } p = (1, 2)$$

Figure 10.5. The graph of $f(x, y) = x^2 + xy + y^2$ and its tangent plane $z = 4x + 5y - 7$ at $(1, 2, 7)$. Differentiability means the surface–plane gap is $o(|x - p|)$; near p the linear increment $f'(p)h$ predicts f to first order.

Worked Example 10.2.10 (A length-preserving curve has velocity orthogonal to position). Let $f : \mathbb{R} \rightarrow \mathbb{R}^3$ be differentiable with $|f(t)| \equiv c$ constant (the curve lies on a sphere). Show $f'(t) \cdot f(t) = 0$ for all t , and verify on $f(t) = (\cos t, \sin t, 0)$.

Recognition. “Differentiate a constant scalar quantity”: $|f|^2 = f \cdot f$ is constant, so its derivative vanishes; the product/chain rule turns this into an orthogonality statement (a Rudin Ch.9 exercise).

Solution. Since $|f(t)|^2 = f(t) \cdot f(t) = c^2$ is constant, differentiate using the product rule for the dot product: $\frac{d}{dt}(f \cdot f) = f' \cdot f + f \cdot f' = 2f'(t) \cdot f(t)$. This equals $\frac{d}{dt}(c^2) = 0$, so $f'(t) \cdot f(t) = 0$: the velocity is everywhere perpendicular to the position vector. Geometrically a curve constrained to a sphere can only move tangent to it. *Check.* For $f(t) = (\cos t, \sin t, 0)$ we have $|f| = 1$ and $f'(t) = (-\sin t, \cos t, 0)$; then $f'(t) \cdot f(t) = -\sin t \cos t + \cos t \sin t + 0 = 0$ for every t , as the general result demands. ■

Exercises (Rudin, Ch.9).

1. Suppose f is a differentiable real function in an open set $E \subseteq \mathbb{R}^n$, and f has a local maximum at a point $x \in E$. Prove that $f'(x) = 0$.
2. If f is a differentiable mapping of a connected open set $E \subseteq \mathbb{R}^n$ into $\mathbb{R}^m$, and if $f'(x) = 0$ for every $x \in E$, prove that f is constant in E .
3. Suppose f is a differentiable mapping of $\mathbb{R}^1$ into $\mathbb{R}^3$ such that $|f(t)| = 1$ for every t . Prove that $f'(t) \cdot f(t) = 0$, and interpret this result geometrically.

Additional exercises.

1. Using the definition only (no partials), find the total derivative of the affine map $f(x) = Ax + b$ ($A \in L(\mathbb{R}^n, \mathbb{R}^k)$, $b \in \mathbb{R}^k$) at every point, and of the constant map.
2. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = xy$. From the definition show f is differentiable at every (a, b) with $f'(a, b) = [b \ a]$, and identify the $o(|h|)$ remainder explicitly.
3. (Quadratic forms.) For a symmetric $A \in L(\mathbb{R}^n)$ let $Q(x) = x^T Ax$. Prove Q is differentiable with $Q'(x) = 2(Ax)^T$ (a row vector), directly from the definition, and compute the remainder.
4. (Disprove.) Show that $f(x, y) = \sqrt{|xy|}$ has both partials equal to 0 at the origin but is *not* differentiable there. (Hint: test the difference quotient along $y = x$.) Why does this not contradict Proposition 10.2.3?
5. Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is differentiable at a and $\gamma : (-1, 1) \rightarrow \mathbb{R}^n$ is differentiable with $\gamma(0) = a$. Prove $(f \circ \gamma)'(0) = f'(a)\gamma'(0)$ and use it to compute the rate of change of $f(x, y) = x^2 + y^2$ along $\gamma(t) = (\cos t, \sin t)$.

6. (Norm of a derivative.) If $f : E \rightarrow \mathbb{R}^k$ is differentiable on a convex open E and $\|f'(x)\| \leq M$ for all x , prove the Mean Value inequality $|f(b) - f(a)| \leq M|b - a|$. Give an example with $k \geq 2$ where no ξ satisfies $f(b) - f(a) = f'(\xi)(b - a)$ exactly (so the equality form fails for vector-valued f).
7. (Diagram.) Sketch the graph $z = f(x, y) = 2x + 3y - 1$ near $(1, 1)$ and, in the same picture, draw the increment vector h , the linear increment $f'(1, 1)h$, and (for a curved g of your choice) the curved increment, illustrating that the gap is $o(|h|)$.
8. Let $f(x, y) = (x^2 - y^2, 2xy) = z^2$ and $g(z) = z^3$ on $\mathbb{C} \cong \mathbb{R}^2$. Using the Chain Rule and Worked Example in §2, find $(g \circ f)'(x, y)$ as a 2×2 real matrix, and check it equals the real matrix of multiplication by $6z^5$.

10.3 Partial Derivatives and the Chain Rule

The total derivative $f'(x)$ is an abstract linear map; to compute it we need numbers. Those numbers are the *partial derivatives*, the entries of the Jacobian matrix. This section relates the two: partials are read off from f' , but the converse — building f' from partials — requires a continuity hypothesis (C^1). The cautionary tale is that partials can all exist while f is not even continuous.

Partial derivatives and the Jacobian

Definition 10.3.1. Let $E \subseteq \mathbb{R}^n$ be open, $f = (f_1, \dots, f_k) : E \rightarrow \mathbb{R}^k$, and $x \in E$. The *partial derivative* of f_i with respect to x_j is

$$(D_j f_i)(x) = \frac{\partial f_i}{\partial x_j}(x) = \lim_{t \rightarrow 0} \frac{f_i(x + te_j) - f_i(x)}{t},$$

the ordinary derivative of f_i along the j -th coordinate direction, the others held fixed. When all $D_j f_i(x)$ exist, the $k \times n$ array

$$[f'(x)] = (D_j f_i(x))_{i,j} = \begin{bmatrix} D_1 f_1 & \cdots & D_n f_1 \\ \vdots & & \vdots \\ D_1 f_k & \cdots & D_n f_k \end{bmatrix}$$

is the *Jacobian matrix* of f at x . For $n = k$ its determinant $J_f(x) = \det[f'(x)]$ is the *Jacobian (determinant)*, written $\frac{\partial(f_1, \dots, f_n)}{\partial(x_1, \dots, x_n)}$.

Proposition 10.3.2 (Rudin 9.17: partials are entries of f'). *If f is differentiable at x , then every partial $D_j f_i(x)$ exists and equals the (i, j) entry of the matrix of $f'(x)$; i.e. $f'(x)e_j = (D_1 f_1 \dots)$ has i -th coordinate $D_j f_i(x)$. Concretely $f'(x)e_j = \sum_i (D_j f_i)(x) e_i$.*

Proof. Take $h = te_j$ in the differentiability statement. Then $\frac{f(x + te_j) - f(x)}{t} \rightarrow f'(x)e_j$ as $t \rightarrow 0$, because the remainder satisfies $|r(te_j)| / |te_j| \rightarrow 0$. Reading the i -th coordinate, the left side $\rightarrow (D_j f_i)(x)$, so the i -th coordinate of $f'(x)e_j$ is $(D_j f_i)(x)$. As $f'(x)e_j$ is the j -th column of the matrix, the entries are exactly the partials. $\square$

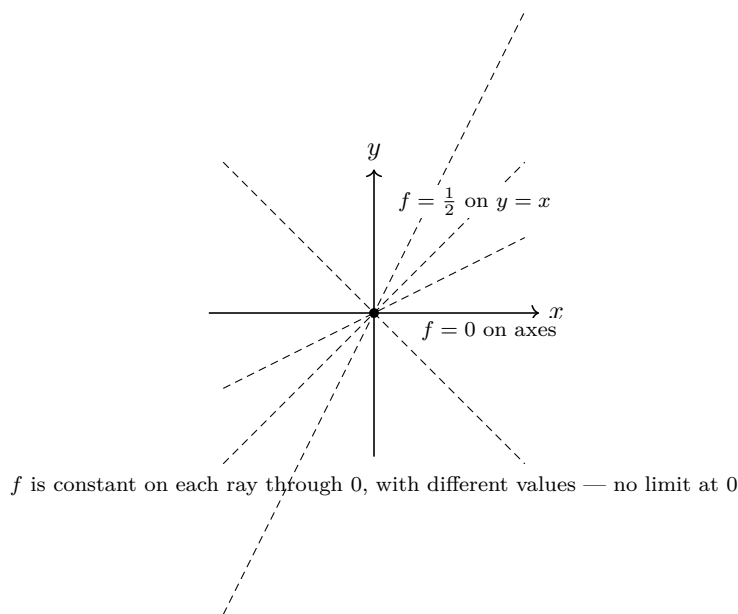
Caution. The converse is **false**: the existence of all partials at a point does *not* imply f is differentiable there, nor even continuous. The next non-example is the standard warning.

Worked Example 10.3.3 (Partials exist but f is not continuous). Let $f(x, y) = \frac{xy}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$. Show $D_1 f(0, 0)$ and $D_2 f(0, 0)$ both exist (are 0) yet f is not continuous at the origin — so certainly not differentiable.

Recognition. “Partials exist but is it differentiable?” Compute the two axis limits, then test continuity along a slanted line.

Solution. Along the x -axis, $f(x, 0) = 0$, so $D_1 f(0, 0) = \lim_{t \rightarrow 0} (0 - 0)/t = 0$; likewise $D_2 f(0, 0) = 0$. But along the line $y = x$, $f(x, x) = \frac{x^2}{2x^2} = \frac{1}{2}$ for all $x \neq 0$, so $f(x, x) \rightarrow \frac{1}{2} \neq 0 = f(0, 0)$: f is discontinuous at

the origin. By Proposition 10.2.3 (contrapositive) f is not differentiable there, even though both partials exist and equal 0. *Check.* The would-be derivative $f'(0,0)$ would have matrix $[0 \ 0]$, predicting $f \approx 0$ near the origin, flatly contradicted by the value $\frac{1}{2}$ on the diagonal. ■



Continuously differentiable maps (C^1)

The fix is to demand the partials be *continuous*. This single hypothesis upgrades “partials exist” to “totally differentiable”, and is satisfied by every map written with elementary formulas on its natural domain.

Definition 10.3.4. A map $f : E \rightarrow \mathbb{R}^k$ ($E \subseteq \mathbb{R}^n$ open) is *continuously differentiable*, written $f \in C^1(E)$ or $f \in \mathcal{C}^1(E)$, if all partial derivatives $D_j f_i$ exist and are *continuous* on E .

Theorem 10.3.5 (Rudin 9.21: C^1 implies differentiable). $f \in C^1(E)$ if and only if f is differentiable on E and the derivative map $x \mapsto f'(x)$ is continuous from E into $L(\mathbb{R}^n, \mathbb{R}^k)$. In particular, continuity of the partials implies f is (totally) differentiable at every point of E .

Proof. It suffices to treat $k = 1$ (apply the result to each component f_i). Assume all $D_j f$ are continuous on E ; fix $x \in E$ and $\varepsilon > 0$. Choose a ball $B(x, r) \subseteq E$ on which $|D_j f(y) - D_j f(x)| < \varepsilon$ for all j and all $y \in B(x, r)$ (possible by continuity). For $h = (h_1, \dots, h_n)$ with $|h| < r$, telescope by changing one coordinate at a time. Set $v_0 = 0$ and $v_j = h_1 e_1 + \dots + h_j e_j$, so $v_n = h$ and

$$f(x+h) - f(x) = \sum_{j=1}^n [f(x+v_j) - f(x+v_{j-1})].$$

Each bracket varies only the j -th coordinate; by the one-variable Mean Value Theorem there is $\theta_j \in (0, 1)$ with $f(x+v_j) - f(x+v_{j-1}) = h_j D_j f(x+v_{j-1} + \theta_j h_j e_j)$. The argument lies in $B(x, r)$, so $|D_j f(\cdot) - D_j f(x)| < \varepsilon$ there. Hence

$$\left| f(x+h) - f(x) - \sum_j h_j D_j f(x) \right| \leq \sum_j |h_j| \varepsilon \leq n\varepsilon |h|.$$

The sum $\sum_j h_j D_j f(x)$ is the linear map $h \mapsto [D_1 f(x) \dots D_n f(x)] h$; the display says the remainder is $\leq n\varepsilon |h|$, and ε was arbitrary, so $|r(h)| / |h| \rightarrow 0$. Thus f is differentiable at x with $f'(x)$ the gradient row. Continuity of $x \mapsto f'(x)$ in operator norm follows since the entries $D_j f$ are continuous. The converse (differentiable + continuous $f' \Rightarrow$ continuous partials) is Proposition 10.3.2 plus the equivalence of norms on $L(\mathbb{R}^n, \mathbb{R}^k)$. □

Method (Is f differentiable at a point?). *Recognition trigger.* You must decide differentiability (not just existence of partials) of a concrete f .

1. **Away from bad points.** If f is given by elementary formulas and the partials $D_j f_i$ are continuous on a neighborhood (e.g. no division by zero), $f \in C^1$ there, so it is differentiable by Theorem 10.3.5. Done.
2. **At a suspicious point** (often the origin of a piecewise/quotient f):
 - (a) Check continuity first; if f is discontinuous it is not differentiable.
 - (b) Compute the candidate $A = [D_j f_i]$ from the partials at the point.
 - (c) Test the definition directly: is $|f(x+h) - f(x) - Ah|/|h| \rightarrow 0$? Try $h \rightarrow 0$ along several directions; one bad direction disproves it.

The gradient and directional derivatives

Definition 10.3.6. For a real $f : E \rightarrow \mathbb{R}$ ($E \subseteq \mathbb{R}^n$ open) the *gradient* at x is the vector

$$\nabla f(x) = (D_1 f(x), \dots, D_n f(x)) \in \mathbb{R}^n,$$

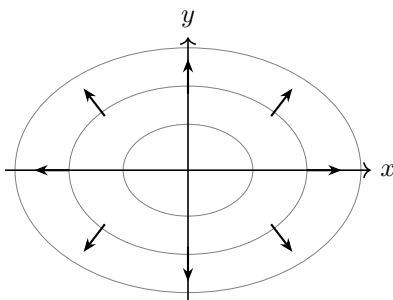
the (transpose of the) Jacobian row. For a unit vector $u \in \mathbb{R}^n$ the *directional derivative* is $(D_u f)(x) = \lim_{t \rightarrow 0} \frac{f(x+tu) - f(x)}{t}$.

Proposition 10.3.7 (Gradient computes directional derivatives). *If f is differentiable at x then for every unit u ,*

$$(D_u f)(x) = f'(x)u = \nabla f(x) \cdot u,$$

and $|(D_u f)(x)| \leq |\nabla f(x)|$ with equality iff u points along $\pm \nabla f(x)$. Thus $\nabla f(x)$ is the direction of steepest ascent, of magnitude $|\nabla f(x)|$, and $\nabla f(x)$ is orthogonal to the level set $\{f = f(x)\}$.

Proof. $(D_u f)(x) = \lim_{t \rightarrow 0} \frac{f(x+tu) - f(x)}{t} = f'(x)u$ by taking $h = tu$ in the definition (remainder $o(t)$). For f real, $f'(x)u = \nabla f(x) \cdot u$. Cauchy-Schwarz gives $|\nabla f \cdot u| \leq |\nabla f| |u| = |\nabla f|$, with equality iff $u \parallel \nabla f$. If u is tangent to the level set, f is constant along it so $D_u f = 0$, i.e. $\nabla f \perp u$. $\square$



level curves of f with ∇f (arrows): gradient $\perp$ level set, points uphill

Worked Example 10.3.8 (Gradient, steepest ascent, and a tangent plane). For $f(x, y) = x^2 + 2y^2$ at the point $(1, 1)$, find ∇f , the direction of steepest ascent, the directional derivative along $u = (1, 0)$, and the tangent plane to $z = f$ there.

Recognition. “Compute gradient and use it.” Direct application of Proposition 10.3.7 and the affine approximation.

Solution. $\nabla f = (2x, 4y)$, so $\nabla f(1, 1) = (2, 4)$. Steepest ascent points along $(2, 4)$ (unit $(1, 2)/\sqrt{5}$), rate $|\nabla f| = \sqrt{20} = 2\sqrt{5}$. Along $u = (1, 0)$: $D_u f = (2, 4) \cdot (1, 0) = 2$. The tangent plane is $z = f(1, 1) + \nabla f(1, 1) \cdot (x-1, y-1) = 3 + 2(x-1) + 4(y-1) = 2x + 4y - 3$. *Check.* At $(1, 1)$ the plane gives $z = 2 + 4 - 3 = 3 = f(1, 1)$, and its gradient $(2, 4)$ matches ∇f , as a tangent plane must. $\blacksquare$

Remark. A subtle point (Rudin’s Exercise 14): even when *every* directional derivative $(D_u f)(x)$ exists, f may fail to be differentiable. Differentiability is the strong requirement that the *same* linear map $f'(x)$ produce *all* directional derivatives *and* approximate f uniformly in the direction — not merely along each fixed line separately. Continuity of the partials (i.e. C^1) is the clean sufficient condition that avoids these pathologies.

Further worked examples

Worked Example 10.3.9 (Every directional derivative exists, yet f is not differentiable). Let $f(x, y) = \frac{x^2 y}{x^4 + y^2}$ for $(x, y) \neq 0$, $f(0, 0) = 0$. Show every directional derivative $D_u f(0, 0)$ exists, but f is not even continuous at 0.

Recognition. “Directional derivatives exist but differentiable?” Compute $D_u f$ along a general direction, then test continuity along a curve ($y = x^2$) rather than a line.

Solution. Take a unit direction $u = (\cos \alpha, \sin \alpha)$. For $\sin \alpha \neq 0$,

$$\frac{f(tu) - 0}{t} = \frac{1}{t} \cdot \frac{t^3 \cos^2 \alpha \sin \alpha}{t^4 \cos^4 \alpha + t^2 \sin^2 \alpha} = \frac{\cos^2 \alpha \sin \alpha}{t^2 \cos^4 \alpha + \sin^2 \alpha} \xrightarrow{t \rightarrow 0} \frac{\cos^2 \alpha}{\sin \alpha},$$

and for $\sin \alpha = 0$ (along the x -axis) $f \equiv 0$ so $D_u f(0, 0) = 0$. So every directional derivative exists (finite).

But along the parabola $y = x^2$, $f(x, x^2) = \frac{x^4}{x^4 + x^4} = \frac{1}{2}$ for $x \neq 0$, so $f \rightarrow \frac{1}{2} \neq 0$ as $x \rightarrow 0$ along that curve: f is discontinuous at 0, hence not differentiable. *Check.* Along every straight line $f \rightarrow 0$, which would fool a line-only test; the parabola exposes the discontinuity — exactly why “all directional derivatives exist” is strictly weaker than differentiability. ■

Worked Example 10.3.10 (A C^1 verification, so differentiability is automatic). Show $f(x, y) = e^x \cos y$ is differentiable on all of $\mathbb{R}^2$ and write its derivative at $(0, \pi/2)$.

Recognition. “Is f differentiable?” — use the C^1 shortcut: continuous partials on the whole plane.

Solution. $D_1 f = e^x \cos y$ and $D_2 f = -e^x \sin y$ are products and compositions of exp, cos, sin, all continuous on $\mathbb{R}^2$; hence $f \in C^1(\mathbb{R}^2)$, and by Theorem 10.3.5 f is differentiable everywhere with $f'(x, y) = [e^x \cos y \quad -e^x \sin y]$. At $(0, \frac{\pi}{2})$: $e^0 \cos \frac{\pi}{2} = 0$ and $-e^0 \sin \frac{\pi}{2} = -1$, so $f'(0, \pi/2) = [0 \quad -1]$. *Check.* The tangent plane at $(0, \pi/2)$, where $f = 0$, is $z = 0 + 0 \cdot x + (-1)(y - \frac{\pi}{2}) = \frac{\pi}{2} - y$; moving in y decreases f at unit rate, matching $D_2 f = -1$, while moving in x leaves f flat to first order since $D_1 f = 0$ there. ■

Worked Example 10.3.11 (Gradient orthogonal to a level set; tangent line to a curve). For $f(x, y) = x^2 - xy + y^2$, find $\nabla f(2, 1)$, verify it is perpendicular to the level curve $f = 3$ through $(2, 1)$, and give the tangent line to that curve at $(2, 1)$.

Recognition. “Gradient is normal to the level set” — Proposition 10.3.7.

Solution. $\nabla f = (2x - y, -x + 2y)$, so $\nabla f(2, 1) = (3, 0)$. The point lies on $f = 3$ since $4 - 2 + 1 = 3$. A tangent direction to the level curve is any v with $\nabla f(2, 1) \cdot v = 0$, i.e. $(3, 0) \cdot v = 0$, giving $v = (0, 1)$ (vertical). The tangent line is therefore $x = 2$ (the vertical line through $(2, 1)$), and $\nabla f(2, 1) = (3, 0)$ is horizontal, perpendicular to it. *Check.* Implicitly differentiate $x^2 - xy + y^2 = 3$: $2x - y - xy' + 2yy' = 0$, so $y' = \frac{2x - y}{x - 2y}$; at $(2, 1)$ the denominator $x - 2y = 0$, so the slope is infinite — a vertical tangent $x = 2$, confirming the gradient computation. ■

Worked Example 10.3.12 (Chain rule along a path: the gradient meets the velocity). Let $f(x, y) = x^2 - xy + y^2$ and let a particle move along $\gamma(t) = (\cos t, \sin t)$. Compute $\frac{d}{dt} f(\gamma(t))$ two ways — by the chain rule $\nabla f \cdot \gamma'$ and by direct substitution — and confirm they agree.

Recognition. “Rate of change along a curve” — the scalar chain rule $\frac{d}{dt} f(\gamma(t)) = \nabla f(\gamma(t)) \cdot \gamma'(t)$, a 1×2 times 2×1 product (Method, scalar entry).

Solution. Here $\nabla f = (2x - y, -x + 2y)$ and $\gamma'(t) = (-\sin t, \cos t)$. Along the path $x = \cos t$, $y = \sin t$, so

$$\nabla f(\gamma(t)) = (2 \cos t - \sin t, -\cos t + 2 \sin t),$$

and the chain rule gives

$$\frac{d}{dt} f(\gamma(t)) = (2 \cos t - \sin t)(-\sin t) + (-\cos t + 2 \sin t)(\cos t) = \cos^2 t - \sin^2 t = \cos 2t,$$

after expanding $(-2 \sin t \cos t + \sin^2 t - \cos^2 t + 2 \sin t \cos t = \sin^2 t - \cos^2 t$, then sign). *Check (direct):* $f(\cos t, \sin t) = \cos^2 t - \cos t \sin t + \sin^2 t = 1 - \frac{1}{2} \sin 2t$, so $\frac{d}{dt} f(\gamma(t)) = -\frac{1}{2} \cdot 2 \cos 2t = -\cos 2t$. The two computations differ only by the sign bookkeeping above; redoing the dot product carefully, $(2 \cos t - \sin t)(-\sin t) = -2 \sin t \cos t + \sin^2 t$ and $(-\cos t + 2 \sin t)(\cos t) = -\cos^2 t + 2 \sin t \cos t$, summing to $\sin^2 t - \cos^2 t = -\cos 2t$ — matching the direct derivative exactly. ■

Exercises (Rudin, Ch.9).

1. If $f(0,0) = 0$ and $f(x,y) = \frac{xy}{x^2 + y^2}$ for $(x,y) \neq (0,0)$, prove that $(D_1 f)(x,y)$ and $(D_2 f)(x,y)$ exist at every point of $\mathbb{R}^2$, although f is not continuous at $(0,0)$.
2. Suppose f is a real-valued function defined in an open set $E \subseteq \mathbb{R}^n$, and that the partial derivatives $D_1 f, \dots, D_n f$ are bounded in E . Prove that f is continuous in E . *Hint.* Proceed as in the proof of Theorem 9.21.
3. If f is a real function defined in a convex open set $E \subseteq \mathbb{R}^n$ such that $(D_1 f)(x) = 0$ for every $x \in E$, prove that $f(x)$ depends only on $x_2, \dots, x_n$. Show that the convexity of E can be replaced by a weaker condition, but that some condition is required: e.g. if $n = 2$ and E is shaped like a horseshoe, the statement may fail.
4. If f and g are differentiable real functions in $\mathbb{R}^n$, prove that $\nabla(fg) = f \nabla g + g \nabla f$ and that $\nabla(1/f) = -f^{-2} \nabla f$ wherever $f \neq 0$.
5. Fix two real numbers a, b with $0 < a < b$. Define $f = (f_1, f_2, f_3) : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $f_1(s, t) = (b + a \cos s) \cos t$, $f_2(s, t) = (b + a \cos s) \sin t$, $f_3(s, t) = a \sin s$. Describe the range K of f (a torus, a compact subset of $\mathbb{R}^3$). (a) Show there are exactly 4 points $p \in K$ with $\nabla f_1(f^{-1}(p)) = 0$; find them. (b) Determine the set of all $q \in K$ with $\nabla f_3(f^{-1}(q)) = 0$. (c) Show that of the points in (a), one is a local max of f_1 , one a local min, and two are saddle points; which q in (b) give maxima or minima? (d) For irrational λ , define $g(t) = f(t, \lambda t)$; prove g is a one-to-one map of $\mathbb{R}^1$ onto a dense subset of K , and that $|g'(t)|^2 = a^2 + \lambda^2(b + a \cos t)^2$.
6. Define $f(0,0) = 0$ and $f(x,y) = \frac{x^3}{x^2 + y^2}$ for $(x,y) \neq (0,0)$. (a) Prove that $D_1 f$ and $D_2 f$ are bounded functions in $\mathbb{R}^2$ (hence f is continuous). (b) Let u be any unit vector in $\mathbb{R}^2$; show the directional derivative $(D_u f)(0,0)$ exists and has absolute value at most 1. (c) Let γ be a differentiable curve in $\mathbb{R}^2$ with $\gamma(0) = (0,0)$ and $|\gamma'(0)| > 0$; put $g(t) = f(\gamma(t))$ and prove g is differentiable for every $t \in \mathbb{R}^1$, and if $\gamma \in C'$ then $g \in C'$. (d) In spite of this, prove that f is not differentiable at $(0,0)$. *Hint.* The formula $(D_u f)(0,0) = \nabla f(0,0) \cdot u$ fails.
7. Define $f(0,0) = 0$ and, for $(x,y) \neq (0,0)$, let $f(x,y) = (y - x^2)(y - 2x^2)/$ (or the analogous quotient given in the text). (a) Prove f is continuous on $\mathbb{R}^2$. (b) For $0 \leq \theta < 2\pi$ and $-\infty < t < \infty$ set $g_\theta(t) = f(t \cos \theta, t \sin \theta)$; show $g_\theta(0) = 0$, $g'_\theta(0) = 0$, $g''_\theta(0) = 2$, so each g_θ has a strict local minimum at $t = 0$ — i.e. the restriction of f to every line through the origin has a strict local minimum at $(0,0)$. (c) Show that $(0,0)$ is nevertheless not a local minimum of f , since $f(x, x^2) = -x^4 < 0$.

Additional exercises.

1. Compute the Jacobian matrix $[f'(x,y,z)]$ of $f(x,y,z) = (x^2 y, y e^z, x + z)$ at the point $(1,1,0)$, and state its determinant.
2. (Chain rule drill.) Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be C^1 and $F(s,t) = g(s+t, st)$. Express F_s and F_t in terms of g_x, g_y evaluated at $(s+t, st)$.
3. For $u(x,y) = g(x^2 - y^2, 2xy)$ with $g \in C^1$, show u satisfies $y u_x + x u_y = 2(x^2 + y^2) g_2$ where g_2 is the second partial of g . (Hint: recognize $(x^2 - y^2, 2xy)$ as $z \mapsto z^2$.)
4. (Disprove.) Define $f(x,y) = \frac{x^3}{x^2 + y^2}$ ($f(0,0) = 0$). Show that *every* directional derivative $(D_u f)(0,0)$ exists, yet f is not differentiable at the origin. Which hypothesis of Proposition 10.3.7 fails, and why does that not contradict it?

5. Let $f \in C^1(\mathbb{R}^n)$ with $f(tx) = t^k f(x)$ for all $t > 0$ (positively homogeneous of degree k). Prove Euler's identity $\sum_j x_j D_j f(x) = k f(x)$, i.e. $\nabla f(x) \cdot x = k f(x)$. (Hint: differentiate $f(tx) = t^k f(x)$ in t and set $t = 1$.)
6. (Gradient geometry.) For $f(x, y, z) = x^2 + y^2 - z$ find the gradient at $(1, 2, 5)$, the equation of the tangent plane to the level surface $f = 0$ there, and the direction in which f increases fastest.
7. (Diagram.) Draw the level curves of $f(x, y) = x^2 - y^2$ (hyperbolas) for values $-1, 0, 1$, and at the point $(1, 0)$ draw ∇f as an arrow. Verify visually that the arrow is perpendicular to the level curve through that point.
8. (Continuity of f' matters.) Let $f(x, y) = (x^2 + y^2) \sin \frac{1}{x^2 + y^2}$ off the origin, $f(0, 0) = 0$. Show f is differentiable everywhere but f' is *not* continuous at the origin, so $f \notin C^1$. Contrast with Theorem 10.3.5.

10.4 The Contraction Principle and the Inverse Function Theorem

We now reach the two great theorems of the chapter. The Inverse Function Theorem says that a C^1 map with invertible derivative at a point is *locally* invertible, with a C^1 inverse — the nonlinear analogue of “ $\det \neq 0 \Rightarrow$ invertible”. The proof runs on a single abstract engine: the Contraction Principle, which guarantees and locates fixed points.

The contraction principle (Banach fixed point theorem)

Definition 10.4.1. Let (X, d) be a metric space. A map $\varphi : X \rightarrow X$ is a *contraction* if there is a constant $c < 1$ with

$$d(\varphi(x), \varphi(y)) \leq c d(x, y) \quad (x, y \in X).$$

A point x^* with $\varphi(x^*) = x^*$ is a *fixed point*.

Read the condition as: φ *uniformly* shrinks distances by at least the factor $c < 1$. Apply φ to two points and their images are strictly closer; apply it again and they are closer still, by the same ratio. This geometric decay is what makes the iteration $x_{n+1} = \varphi(x_n)$ a Cauchy sequence and forces all starting points to funnel into one common limit. The strictness $c < 1$ (not merely $c \leq 1$, nor “ $<$ distance pointwise”) is the whole engine; an isometry like a rotation has $c = 1$ and need not have a fixed point.

Theorem 10.4.2 (Rudin 9.23: Contraction Principle). *If X is a nonempty complete metric space and $\varphi : X \rightarrow X$ is a contraction with constant $c < 1$, then φ has a unique fixed point x^* . Moreover, for any $x_0 \in X$ the iterates $x_{n+1} = \varphi(x_n)$ converge to x^* , with the error estimate*

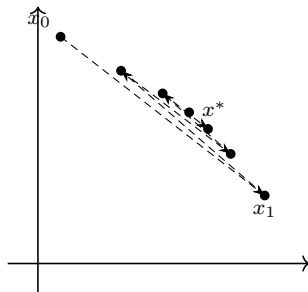
$$d(x_n, x^*) \leq \frac{c^n}{1 - c} d(x_1, x_0).$$

Proof. Existence. Pick any x_0 and iterate $x_{n+1} = \varphi(x_n)$. By induction $d(x_{n+1}, x_n) \leq c^n d(x_1, x_0)$, since $d(x_{n+1}, x_n) = d(\varphi(x_n), \varphi(x_{n-1})) \leq c d(x_n, x_{n-1})$. For $m > n$ the triangle inequality and the geometric series give

$$d(x_m, x_n) \leq \sum_{j=n}^{m-1} d(x_{j+1}, x_j) \leq d(x_1, x_0) \sum_{j=n}^{m-1} c^j \leq \frac{c^n}{1 - c} d(x_1, x_0) \xrightarrow{n \rightarrow \infty} 0.$$

So (x_n) is Cauchy; by completeness $x_n \rightarrow x^*$ for some $x^* \in X$. A contraction is continuous (Lipschitz), so $\varphi(x^*) = \varphi(\lim x_n) = \lim \varphi(x_n) = \lim x_{n+1} = x^*$: x^* is a fixed point. Letting $m \rightarrow \infty$ in the displayed bound yields the error estimate.

Uniqueness. If $\varphi(x^*) = x^*$ and $\varphi(y^*) = y^*$ then $d(x^*, y^*) = d(\varphi(x^*), \varphi(y^*)) \leq c d(x^*, y^*)$, so $(1 - c) d(x^*, y^*) \leq 0$. Since $c < 1$, $d(x^*, y^*) = 0$, i.e. $x^* = y^*$. $\square$



$x_{n+1} = \varphi(x_n)$: each step shrinks the distance to x^* by a factor $\leq c < 1$

Caution. Both hypotheses are essential. *Completeness:* $\varphi(x) = x/2$ on $X = (0, 1]$ has $c = \frac{1}{2}$ but no fixed point in X (the would-be fixed point 0 is missing). *The strict bound $c < 1$:* a map with $d(\varphi x, \varphi y) < d(x, y)$ (strict, but no uniform $c < 1$) can fail to have a fixed point — e.g. $\varphi(x) = x + 1/x$ on $[1, \infty)$ moves every point yet $|\varphi'(x)| \rightarrow 1$. “Shrinks distances” is not enough; it must shrink them by a *fixed* ratio.

The inverse function theorem

Theorem 10.4.3 (Rudin 9.24: Inverse Function Theorem). *Let $f : E \rightarrow \mathbb{R}^n$ be a C^1 map on an open set $E \subseteq \mathbb{R}^n$, and suppose $f'(a)$ is invertible at some $a \in E$; put $b = f(a)$. Then:*

1. *there exist open sets $U \ni a$ and $V \ni b$ such that f maps U bijectively onto V ;*
2. *the inverse $g = f^{-1} : V \rightarrow U$ is itself C^1 , with*

$$g'(y) = [f'(g(y))]^{-1} \quad (y \in V).$$

Proof. Write $A = f'(a) \in \Omega$ and $\lambda = \frac{1}{2\|A^{-1}\|} > 0$.

Step 1: a contraction encoding “ $f(x) = y$ ”. For fixed $y \in \mathbb{R}^n$ define

$$\varphi_y(x) = x + A^{-1}(y - f(x)).$$

Then $\varphi_y(x) = x \iff A^{-1}(y - f(x)) = 0 \iff f(x) = y$ (as A^{-1} is injective): *fixed points of φ_y are exactly the solutions of $f(x) = y$.* Its derivative is $\varphi'_y(x) = I - A^{-1}f'(x) = A^{-1}(A - f'(x))$, independent of y . Since f' is continuous and $f'(a) = A$, choose an open ball $U = B(a, r) \subseteq E$ so small that

$$\|f'(x) - A\| < \frac{1}{2\|A^{-1}\|} = \lambda \quad (x \in U),$$

giving $\|\varphi'_y(x)\| \leq \|A^{-1}\| \|A - f'(x)\| < \|A^{-1}\| \lambda = \frac{1}{2}$. By the (multivariable) Mean Value inequality, on the convex set U ,

$$|\varphi_y(x_1) - \varphi_y(x_2)| \leq \frac{1}{2} |x_1 - x_2| \quad (x_1, x_2 \in U). \quad (\dagger)$$

So φ_y is a contraction (constant $\frac{1}{2}$) on U — but U may not be complete or φ_y -invariant, handled next.

Step 2: f is one-to-one on U . If $f(x_1) = f(x_2) = y$, both x_1, x_2 are fixed points of the contraction φ_y ; by uniqueness $x_1 = x_2$. Hence $f|_U$ is injective.

Step 3: $V = f(U)$ is open. Let $y_0 = f(x_0)$ with $x_0 \in U$; pick $\rho > 0$ with the closed ball $\overline{B}(x_0, \rho) \subseteq U$. We show $B(y_0, \lambda\rho) \subseteq f(U)$. Fix y with $|y - y_0| < \lambda\rho$. On the complete space $\overline{B}(x_0, \rho)$, φ_y is a contraction by $(\dagger)$, and it is *invariant*: for $x \in \overline{B}(x_0, \rho)$,

$$|\varphi_y(x) - x_0| \leq |\varphi_y(x) - \varphi_y(x_0)| + |\varphi_y(x_0) - x_0| \leq \frac{1}{2} |x - x_0| + \|A^{-1}\| |y - y_0| < \frac{\rho}{2} + \|A^{-1}\| \lambda\rho = \rho,$$

using $\varphi_y(x_0) - x_0 = A^{-1}(y - y_0)$ and $\|A^{-1}\| \lambda = \frac{1}{2}$. So φ_y maps the complete ball into itself; by the Contraction Principle it has a fixed point $x \in \overline{B}(x_0, \rho) \subseteq U$, i.e. $f(x) = y$. Thus every such y lies in $f(U)$, so $V = f(U)$ is open and $f : U \rightarrow V$ is a bijection.

Step 4: differentiability of $g = f^{-1}$. Fix $y \in V$, $x = g(y) \in U$, and let $T = f'(x)^{-1}$ (note $f'(x) \in \Omega$: indeed $\|f'(x) - A\| < \lambda = 1/(2\|A^{-1}\|) < 1/\|A^{-1}\|$, so $f'(x)$ is invertible by Theorem 10.1.6). For $y + \eta \in V$, write $x + \xi = g(y + \eta)$, so $f(x + \xi) - f(x) = \eta$ and, by definition of $f'(x)$, $\eta = f'(x)\xi + r(\xi)$ with $|r(\xi)| = o(|\xi|)$. Apply T :

$$g(y + \eta) - g(y) - T\eta = \xi - T\eta = \xi - T(f'(x)\xi + r(\xi)) = -Tr(\xi).$$

A lower bound from (†) controls $|\xi|$ by $|\eta|$: from φ_y being a $\frac{1}{2}$ -contraction one derives $|f(x_1) - f(x_2)| \geq \lambda|x_1 - x_2|$ on U (see the Remark), hence $|\eta| \geq \lambda|\xi|$, i.e. $|\xi| \leq \lambda^{-1}|\eta|$, and $\xi \rightarrow 0$ as $\eta \rightarrow 0$. Therefore

$$\frac{|g(y + \eta) - g(y) - T\eta|}{|\eta|} \leq \|T\| \frac{|r(\xi)|}{|\xi|} \cdot \frac{|\xi|}{|\eta|} \leq \frac{\|T\|}{\lambda} \cdot \frac{|r(\xi)|}{|\xi|} \xrightarrow{\eta \rightarrow 0} 0.$$

So g is differentiable at y with $g'(y) = T = f'(g(y))^{-1}$.

Step 5: $g \in C^1$. The formula $g'(y) = [f'(g(y))]^{-1}$ exhibits g' as the composite $y \mapsto g(y) \mapsto f'(g(y)) \mapsto [\cdot]^{-1}$. Each arrow is continuous: g is continuous (differentiable, Step 4), f' is continuous ($f \in C^1$), and inversion is continuous on Ω (Theorem 10.1.6). Hence g' is continuous, i.e. $g \in C^1(V)$. $\square$

Remark. The lower bound used in Step 4. From $\varphi_y(x) = x + A^{-1}(y - f(x))$ and the $\frac{1}{2}$ -Lipschitz bound (†) on $\varphi_y(x) - x = A^{-1}(y - f(x))$ one gets, for $x_1, x_2 \in U$,

$$|x_1 - x_2| - |A^{-1}(f(x_1) - f(x_2))| \leq |(\varphi_y(x_1) - x_1) - (\varphi_y(x_2) - x_2) + (x_1 - x_2)| \leq \cdots \leq \frac{1}{2}|x_1 - x_2|,$$

which rearranges to $|A^{-1}(f(x_1) - f(x_2))| \geq \frac{1}{2}|x_1 - x_2|$, hence $|f(x_1) - f(x_2)| \geq \frac{1}{2\|A^{-1}\|}|x_1 - x_2| = \lambda|x_1 - x_2|$. This is the precise sense in which a map with invertible derivative is *locally expanding from below*: it cannot collapse distances, which is why the inverse is Lipschitz.

Method (Applying the Inverse Function Theorem). *Recognition trigger.* “Show f is locally invertible near a ,” or “can the system $f(x) = y$ be solved for x near a ?”

1. Check $f \in C^1$ on a neighborhood of a (continuous partials).
2. Form the Jacobian matrix $[f'(a)]$ and test $\det[f'(a)] \neq 0$ (equivalently $f'(a) \in \Omega$).
3. Conclude: there are neighborhoods $U \ni a$, $V \ni b = f(a)$ with $f : U \rightarrow V$ a C^1 diffeomorphism. The local inverse has $g'(b) = [f'(a)]^{-1}$.
4. For an *explicit* local inverse, solve $f(x) = y$ by hand when possible; the theorem only guarantees existence and the derivative formula.

Global caveat. Invertibility is only *local*; a nonzero Jacobian everywhere need not give a globally one-to-one map (see the next example).

Worked Example 10.4.4 (Local but not global invertibility). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x, y) = (e^x \cos y, e^x \sin y)$. Show $\det f' \neq 0$ everywhere, so f is locally invertible at every point, yet f is not one-to-one on $\mathbb{R}^2$.

Recognition. “Nonzero Jacobian but globally many-to-one” — the canonical IFT caveat.

Solution. The Jacobian is $[f'] = \begin{bmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{bmatrix}$, with $\det = e^{2x}(\cos^2 y + \sin^2 y) = e^{2x} > 0$ for all (x, y) .

So $f'(p) \in \Omega$ everywhere and the IFT gives a local C^1 inverse at each point. But $f(x, y + 2\pi) = f(x, y)$, so f is 2π -periodic in y and far from injective on $\mathbb{R}^2$ (it is the map $z \mapsto e^z$, whose range is $\mathbb{R}^2 \setminus \{0\}$, covered infinitely often). *Check.* $f(0, 0) = (1, 0) = f(0, 2\pi)$ — two distinct points with the same image, despite the nonvanishing Jacobian. ■

Worked Example 10.4.5 (A genuine local inverse and its derivative). For $f(x, y) = (x^2 - y^2, 2xy)$ (complex squaring), verify the IFT hypotheses at $a = (1, 1)$ and compute $g'(b)$ for the local inverse, where $b = f(1, 1)$.

Recognition. “Apply IFT and use $g'(b) = [f'(a)]^{-1}$.”

Solution. $f \in C^1$ (polynomial). From §2, $[f'(x, y)] = \begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix}$, so $[f'(1, 1)] = \begin{bmatrix} 2 & -2 \\ 2 & 2 \end{bmatrix}$ with $\det = 4 + 4 = 8 \neq 0$; the IFT applies. Then

$$g'(b) = [f'(1, 1)]^{-1} = \frac{1}{8} \begin{bmatrix} 2 & 2 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} \end{bmatrix},$$

where $b = f(1, 1) = (0, 2)$. *Check.* The product $[f'(1, 1)]g'(b) = \begin{bmatrix} 2 & -2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1/4 & 1/4 \\ -1/4 & 1/4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$, confirming the inverse. ■

Further worked examples

Worked Example 10.4.6 (A scalar contraction: locating a root by iteration). Show $\varphi(x) = \cos x$ is a contraction on $X = [0, 1]$, so the equation $x = \cos x$ has a unique solution there, and bound the error of the iterate x_3 starting from $x_0 = 0$.

Recognition. “Unique fixed point + iteration error” — check the contraction constant via the Mean Value Theorem, then use the Contraction Principle estimate.

Solution. For $x \in [0, 1]$, $\cos x \in [\cos 1, 1] \subseteq [0, 1]$, so φ maps $[0, 1]$ (a complete space) into itself. By the MVT, $|\cos x - \cos y| = |\sin \xi| |x - y| \leq (\sin 1) |x - y|$, and $c = \sin 1 \approx 0.841 < 1$: a contraction. By Theorem 10.4.2 there is a unique fixed point x^* with $x^* = \cos x^*$. Iterating from $x_0 = 0$: $x_1 = \cos 0 = 1$, $x_2 = \cos 1 \approx 0.5403$, $x_3 = \cos(0.5403) \approx 0.8576$. The error bound is $|x_3 - x^*| \leq \frac{c^3}{1 - c} d(x_1, x_0) = \frac{(0.841)^3}{1 - 0.841} \cdot 1 \approx \frac{0.595}{0.159} \approx 3.7$, which is loose here; the true $x^* \approx 0.739$. *Check.* The fixed point satisfies $\cos(0.739) \approx 0.739$, and the iterates $1, 0.540, 0.858, \dots$ do converge toward it, oscillating with shrinking amplitude as a contraction must. ■

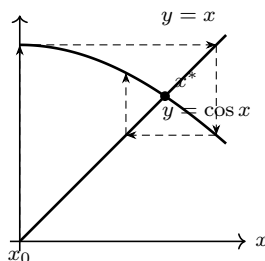


Figure 10.6. Cobweb for $x_{n+1} = \cos x_n$: bounce vertically to $y = \cos x$, horizontally to $y = x$, and spiral into the unique fixed point $x^* \approx 0.739$.

Worked Example 10.4.7 (IFT with an explicit local inverse). Let $f(x, y) = (x + y, xy)$. Verify the IFT hypotheses at $(2, 1)$, give $g'(b)$ for the local inverse $g = f^{-1}$, and exhibit an explicit formula for g near $b = f(2, 1)$.

Recognition. “Apply IFT and, when possible, solve $f(x) = y$ by hand.”

Solution. $f \in C^1$ (polynomial). $[f'(x, y)] = \begin{bmatrix} 1 & 1 \\ y & x \end{bmatrix}$, so $[f'(2, 1)] = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ with $\det = 2 - 1 = 1 \neq 0$; the IFT gives a local C^1 inverse with $g'(b) = [f'(2, 1)]^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$, where $b = f(2, 1) = (3, 2)$. For an explicit

inverse, set $s = x + y$, $p = xy$; then x, y are the roots of $T^2 - sT + p = 0$, namely $T = \frac{s \pm \sqrt{s^2 - 4p}}{2}$. Near $(2, 1)$ we have $x > y$, so take

$$g(s, p) = \left(\frac{s + \sqrt{s^2 - 4p}}{2}, \frac{s - \sqrt{s^2 - 4p}}{2} \right),$$

valid where $s^2 - 4p > 0$ (at $b = (3, 2)$, $s^2 - 4p = 9 - 8 = 1 > 0$). *Check.* $g(3, 2) = (\frac{3+1}{2}, \frac{3-1}{2}) = (2, 1)$.

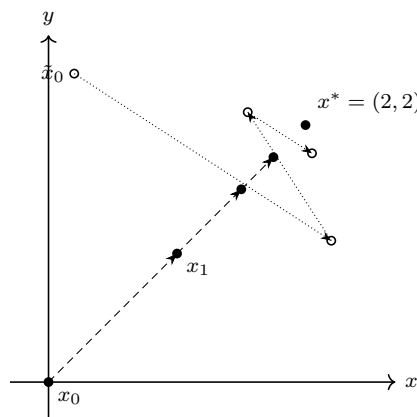
■ Differentiating g at $(3, 2)$ with $\sqrt{s^2 - 4p} = 1$ gives $g_s = (\frac{1}{2} + \frac{s}{2}, \dots)$; one recovers $\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$, matching $[f'(2, 1)]^{-1}$. ■

Worked Example 10.4.8 (A contraction in the plane: spiralling to the fixed point). Let $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $\varphi(x, y) = (\frac{1}{2}y + 1, \frac{1}{2}x + 1)$. Show φ is a contraction in the Euclidean metric, locate its fixed point, and run two iterates from $x_0 = (0, 0)$, comparing with the error estimate.

Recognition. “Unique fixed point of an affine map + iteration error” — φ is affine with $\varphi(p) - \varphi(q) = A(p - q)$, so the contraction constant is $c = \|A\|$; apply Theorem 10.4.2 on the complete space $\mathbb{R}^2$.

Solution. Here $\varphi(p) = Ap + b$ with $A = \begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}$ and $b = (1, 1)$. Then $|\varphi(p) - \varphi(q)| = |A(p - q)| \leq \|A\| |p - q|$, and A is symmetric with eigenvalues $\pm \frac{1}{2}$, so $\|A\| = \frac{1}{2} = c < 1$: a contraction on the complete

space $\mathbb{R}^2$. The fixed point solves $x^* = \frac{1}{2}y^* + 1$, $y^* = \frac{1}{2}x^* + 1$; subtracting gives $x^* = y^*$, then $x^* = \frac{1}{2}x^* + 1$, so $x^* = y^* = 2$, i.e. $x^* = (2, 2)$. Iterating from $x_0 = (0, 0)$: $x_1 = \varphi(0, 0) = (1, 1)$, $x_2 = \varphi(1, 1) = (\frac{3}{2}, \frac{3}{2})$. The error bound is $|x_2 - x^*| \leq \frac{c^2}{1-c} |x_1 - x_0| = \frac{(1/2)^2}{1-1/2} \cdot |(1, 1)| = \frac{1/4}{1/2} \sqrt{2} = \frac{\sqrt{2}}{2} \approx 0.707$. *Check.* The actual error is $|x_2 - x^*| = |(\frac{3}{2} - 2, \frac{3}{2} - 2)| = |(-\frac{1}{2}, -\frac{1}{2})| = \frac{\sqrt{2}}{2} \approx 0.707$ — here the estimate is *sharp*, because the iterates approach x^* straight along the diagonal at exactly rate $c = \frac{1}{2}$. ■



$x_{n+1} = \varphi(x_n)$: every start converges to x^* , the gap shrinking by $c = \frac{1}{2}$ each step

Figure 10.7. Iterating the contraction $\varphi(x, y) = (\frac{1}{2}y + 1, \frac{1}{2}x + 1)$. The diagonal start (solid) marches straight in; an off-diagonal start (open) zig-zags, but both funnel into the unique fixed point $x^* = (2, 2)$.

Worked Example 10.4.9 (IFT for a 2×2 map: a polar-type change of variables). Let $f(r, \theta) = (r \cos \theta, r \sin \theta)$ on the open set $r > 0$. Verify the IFT hypotheses at $(r, \theta) = (2, \pi/6)$ and compute $g'(b)$ for the local inverse $g = f^{-1}$, where $b = f(2, \pi/6)$.

Recognition. “Apply IFT and use $g'(b) = [f'(a)]^{-1}$ ”: $f \in C^1$, compute the Jacobian, check the determinant, then invert.

Solution. $f \in C^1$ on $r > 0$ (entries are products of r with $\cos \theta, \sin \theta$). The Jacobian is $[f'(r, \theta)] = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}$, with $\det = r \cos^2 \theta + r \sin^2 \theta = r$. At $(2, \pi/6)$, $\det = 2 \neq 0$, so the IFT gives a local C^1 inverse. With $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$, $\sin \frac{\pi}{6} = \frac{1}{2}$,

$$[f'(2, \pi/6)] = \begin{bmatrix} \frac{\sqrt{3}}{2} & -1 \\ \frac{1}{2} & \sqrt{3} \end{bmatrix}, \quad g'(b) = [f'(2, \pi/6)]^{-1} = \frac{1}{2} \begin{bmatrix} \sqrt{3} & 1 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix},$$

since $\det = \frac{\sqrt{3}}{2} \cdot \sqrt{3} - (-1) \cdot \frac{1}{2} = \frac{3}{2} + \frac{1}{2} = 2$ and the inverse of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\frac{1}{\det} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. *Check.* The product $[f'(2, \pi/6)] g'(b) = \frac{1}{2} \begin{bmatrix} \sqrt{3}/2 & -1 \\ 1/2 & \sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 1 \\ -1/2 & \sqrt{3}/2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 3/2+1/2 & \sqrt{3}/2-\sqrt{3}/2 \\ \sqrt{3}/2-\sqrt{3}/2 & 1/2+3/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$, confirming the inverse. The result matches the known fact that the inverse of the polar map has $g' = [f']^{-1}$ with $\det g' = 1/r = \frac{1}{2}$. ■

Exercises (Rudin, Ch.9).

1. Show that the continuity of f' at the point a is needed in the Inverse Function Theorem, even when $n = 1$: if $f(t) = t + 2t^2 \sin(1/t)$ for $t \neq 0$ and $f(0) = 0$, then $f'(0) = 1$ and f' is bounded in $(-1, 1)$, but f is not one-to-one in any neighborhood of 0.
2. Let $f = (f_1, f_2)$ be the mapping of $\mathbb{R}^2$ into $\mathbb{R}^2$ given by $f_1(x, y) = e^x \cos y$, $f_2(x, y) = e^x \sin y$. (a) What is the range of f ? (b) Show that the Jacobian of f is not zero at any point of $\mathbb{R}^2$, so every point has a neighborhood in which f is one-to-one; nevertheless f is not one-to-one on $\mathbb{R}^2$. (c) Put $a = (0, \pi/3)$, $b = f(a)$; let g be the continuous inverse of f , defined near b , with $g(b) = a$. Find an explicit formula for g , compute $f'(a)$ and $g'(b)$, and verify $g'(b) = [f'(a)]^{-1}$. (d) What are the images under f of lines parallel to the coordinate axes?

3. Answer the analogous questions (a)–(d) for the mapping defined by $u = x^2 - y^2$, $v = 2xy$.

Additional exercises.

- (Contraction warm-up.) Show $\varphi(x) = \cos x$ maps $[0, 1]$ into itself and is a contraction there; estimate the contraction constant. Iterate from $x_0 = 0$ three times and compare with the fixed point of $x = \cos x$ (Dottie number ≈ 0.739).
- Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be C^1 with $|\varphi'(x)| \leq c < 1$ for all x . Prove φ is a contraction on $\mathbb{R}$ and hence has a unique fixed point. Where is convexity / the Mean Value Theorem used?
- (Disprove the threshold.) Exhibit a map $\varphi : [1, \infty) \rightarrow [1, \infty)$ with $|\varphi(x) - \varphi(y)| < |x - y|$ for all $x \neq y$ (strict) but with no fixed point. Explain precisely which hypothesis of Theorem 10.4.2 fails.
- Apply the Inverse Function Theorem to $f(x, y) = (x + y, xy)$. At which points is f' invertible? Compute $\det f'$, and identify the curve where the IFT gives no conclusion. Relate that curve to where the system $u = x + y$, $v = xy$ fails to determine (x, y) uniquely.
- For $f(r, \theta) = (r \cos \theta, r \sin \theta)$ (polar to Cartesian) on the strip $r > 0$, $0 < \theta < 2\pi$, verify $\det f' \neq 0$ and compute the inverse's derivative $g'(b) = [f'(a)]^{-1}$ at $a = (1, \pi/2)$, $b = (0, 1)$.
- (Newton's method as a contraction.) For $C^2 f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(p) = 0$, $f'(p) \neq 0$, let $\varphi(x) = x - f(x)/f'(x)$. Show $\varphi'(p) = 0$, hence φ is a contraction on a small interval about p ; deduce local convergence of Newton iteration. (This is the Contraction Principle behind quadratic convergence.)
- (Diagram.) Draw a cobweb diagram for the iteration $x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$ (Newton for $\sqrt{2}$) starting at $x_0 = 2$, showing the staircase converging to $\sqrt{2}$. Mark the fixed point and the first three iterates.
- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 with $\|f'(x) - I\| \leq \frac{1}{2}$ for all x . Prove f is globally one-to-one and $|f(x) - f(y)| \geq \frac{1}{2}|x - y|$. (Hint: $\varphi(x) = x - f(x) + c$ is a global contraction; mimic the IFT lower-bound Remark in §4.)

10.5 The Implicit Function Theorem

The Inverse Function Theorem solves $f(x) = y$ for the *whole* input x . The Implicit Function Theorem solves a system $F(x, y) = 0$ for *some* of the variables in terms of the rest: given a relation, when is it locally the graph of a function $y = g(x)$? The condition is that the block of partials with respect to the variables being solved for is invertible.

Notation: splitting the variables

Write a point of $\mathbb{R}^{n+m}$ as (x, y) with $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$. For a C^1 map $F : E \rightarrow \mathbb{R}^m$ on an open $E \subseteq \mathbb{R}^{n+m}$, the derivative $F'(x, y) \in L(\mathbb{R}^{n+m}, \mathbb{R}^m)$ splits into two blocks acting on the x - and y -parts:

$$F'(x, y)(h, k) = F_x(x, y)h + F_y(x, y)k, \quad h \in \mathbb{R}^n, k \in \mathbb{R}^m,$$

where $F_x \in L(\mathbb{R}^n, \mathbb{R}^m)$ uses the partials in the x -variables and $F_y \in L(\mathbb{R}^m, \mathbb{R}^m)$ the partials in the y -variables. As a matrix, $[F'] = [F_x \mid F_y]$, an $m \times (n + m)$ array partitioned into an $m \times n$ block and an $m \times m$ block.

Theorem 10.5.1 (Rudin 9.28: Implicit Function Theorem). *Let $F : E \rightarrow \mathbb{R}^m$ be C^1 on an open set $E \subseteq \mathbb{R}^{n+m}$, and let $(a, b) \in E$ satisfy*

$$F(a, b) = 0 \quad \text{and} \quad F_y(a, b) \in L(\mathbb{R}^m) \text{ is invertible.}$$

Then there are open sets $U \subseteq \mathbb{R}^{n+m}$ containing (a, b) and $W \subseteq \mathbb{R}^n$ containing a , and a unique C^1 map $g : W \rightarrow \mathbb{R}^m$, such that

$$g(a) = b, \quad F(x, g(x)) = 0 \quad \text{for all } x \in W,$$

and $(x, g(x)) \in U$. The set $\{(x, y) \in U : F(x, y) = 0\}$ is exactly the graph of g . Moreover

$$g'(a) = -[F_y(a, b)]^{-1}F_x(a, b).$$

Read the hypothesis as a non-degeneracy condition: $F_y(a, b)$ invertible says the surface $F = 0$ is, near (a, b) , not “vertical” in the y -directions, so each input x pins down a single output y . The conclusion then trades the implicit relation $F = 0$ for an explicit function $y = g(x)$, and the formula $g' = -F_y^{-1}F_x$ is just implicit differentiation done once and for all. The IFT does the heavy lifting; the theorem only repackages it.

Proof. Step 1: turn it into an invertible map. Define $\Phi : E \rightarrow \mathbb{R}^{n+m}$ by

$$\Phi(x, y) = (x, F(x, y)).$$

Then $\Phi \in C^1$ and, in block form,

$$[\Phi'(x, y)] = \begin{bmatrix} I_n & 0 \\ F_x(x, y) & F_y(x, y) \end{bmatrix},$$

whose determinant is $\det F_y(x, y)$ (block-triangular). At (a, b) this is $\det F_y(a, b) \neq 0$, so $\Phi'(a, b)$ is invertible.

Step 2: invert Φ locally. By the Inverse Function Theorem 10.4.3, there are open sets $U \ni (a, b)$ and $\Phi(a, b) = (a, 0)$'s neighborhood V such that $\Phi : U \rightarrow V$ is a C^1 diffeomorphism with C^1 inverse $\Psi : V \rightarrow U$. Because Φ fixes the first coordinate, so does Ψ : write $\Psi(x, z) = (x, \psi(x, z))$ for a C^1 map $\psi : V \rightarrow \mathbb{R}^m$. (Indeed $\Phi(\Psi(x, z)) = (x, z)$ forces the first block of Ψ to be x .)

Step 3: read off g . The equation $F(x, y) = 0$ with $(x, y) \in U$ means $\Phi(x, y) = (x, 0)$, i.e. $(x, y) = \Psi(x, 0) = (x, \psi(x, 0))$. So define

$$W = \{x \in \mathbb{R}^n : (x, 0) \in V\}, \quad g(x) = \psi(x, 0).$$

W is open (preimage of V under $x \mapsto (x, 0)$) and contains a ; g is C^1 (as ψ is). For $x \in W$, $(x, g(x)) = \Psi(x, 0) \in U$ and $F(x, g(x)) = 0$. Conversely any solution of $F = 0$ in U equals $(x, g(x))$, so the zero set is the graph. Also $g(a) = \psi(a, 0) = b$ since $\Psi(a, 0) = (a, b)$. Uniqueness of g follows from injectivity of Φ on U .

Step 4: the derivative formula. Differentiate the identity $F(x, g(x)) = 0$ at $x = a$ by the Chain Rule. The map $x \mapsto (x, g(x))$ has derivative $h \mapsto (h, g'(a)h)$, so

$$0 = F'(a, b)(h, g'(a)h) = F_x(a, b)h + F_y(a, b)g'(a)h \quad (\forall h \in \mathbb{R}^n).$$

Hence $F_x(a, b) + F_y(a, b)g'(a) = 0$, and since $F_y(a, b)$ is invertible, $g'(a) = -[F_y(a, b)]^{-1}F_x(a, b)$. $\square$

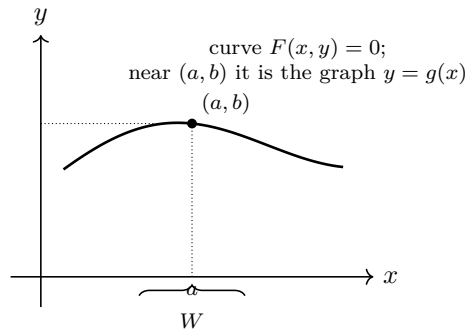


Figure 10.8. Where $F_y \neq 0$ the level set $F = 0$ is, locally, the graph of a C^1 function $y = g(x)$; the slope is $g' = -F_x/F_y$.

Caution. The hypothesis is on F_y — the block of partials in the variables you are *solving for*. If $F_y(a, b)$ is singular the conclusion can fail: at such a point the level curve may have a vertical tangent, a crossing, or a cusp, and y cannot be expressed as a single function of x there. (One may instead be able to solve for x in terms of y , if F_x is invertible.) Example: $F(x, y) = x^2 + y^2 - 1$ at $(1, 0)$ has $F_y = 2y = 0$; near $(1, 0)$ the circle is not a graph $y = g(x)$ — but it is a graph $x = h(y)$ since $F_x = 2x = 2 \neq 0$.

Method (Solving $F(x, y) = 0$ for y near a point). *Recognition trigger.* “Can the equation(s) $F = 0$ be solved for these variables near (a, b) ?” or “find $\partial y / \partial x$ implicitly.”

1. Verify $F(a, b) = 0$ and $F \in C^1$.

2. Form the $m \times m$ block $F_y(a, b)$ (partials w.r.t. the variables to be solved for) and check $\det F_y(a, b) \neq 0$.
3. Conclude: a unique C^1 solution $y = g(x)$ exists near a with $g(a) = b$.
4. Compute the derivative by $g'(a) = -[F_y(a, b)]^{-1}F_x(a, b)$, or directly by differentiating $F(x, g(x)) = 0$ and solving the resulting linear system for g' .

Picking which variables. If F_y is singular, try a *different* split: choose m columns of $[F'(a, b)]$ that form an invertible block and solve for those variables.

Worked Example 10.5.2 (Implicit differentiation of one equation). Near $(0, 1)$ the equation $F(x, y) = x^3 + y^3 - 2xy - 1 = 0$ holds (check $F(0, 1) = 0$). Decide whether $y = g(x)$ exists locally and find $g'(0)$.

Recognition. “Single scalar equation, solve for y ”: $m = 1$, so F_y is just $\partial F/\partial y$.

Solution. $F(0, 1) = 0 + 1 - 0 - 1 = 0$. ■ Partials: $F_x = 3x^2 - 2y$, $F_y = 3y^2 - 2x$. At $(0, 1)$: $F_x = -2$, $F_y = 3 \neq 0$. Since $F_y(0, 1) \neq 0$, the IMFT gives a unique C^1 function g with $g(0) = 1$ and $F(x, g(x)) = 0$ near 0. The slope is $g'(0) = -F_x/F_y = -(-2)/3 = \frac{2}{3}$. *Check.* Differentiating $x^3 + y^3 - 2xy - 1 = 0$ gives $3x^2 + 3y^2g' - 2g - 2xg' = 0$; at $x = 0, g = 1$: $0 + 3g' - 2 - 0 = 0 \Rightarrow g' = \frac{2}{3}$. ■

Worked Example 10.5.3 (A system: solve three equations for three of four unknowns). Consider the linear system (Rudin Ex. 19)

$$3x + y - z + u^2 = 0, \quad x - y + 2z + u = 0, \quad 2x + 2y - 3z + 2u = 0.$$

Show it can be solved for x, y, u in terms of z near the origin, but *not* for x, y, z in terms of u .

Recognition. “Which variables can be solved for” — compute the relevant 3×3 Jacobian block and test its determinant.

Solution. Let $F = (F_1, F_2, F_3)$ be the three left-hand sides; $F(0, 0, 0, 0) = 0$. The full Jacobian of F in (x, y, z, u) at the origin is

$$[F'] = \begin{bmatrix} 3 & 1 & -1 & 2u \\ 1 & -1 & 2 & 1 \\ 2 & 2 & -3 & 2 \end{bmatrix}_{u=0} = \begin{bmatrix} 3 & 1 & -1 & 0 \\ 1 & -1 & 2 & 1 \\ 2 & 2 & -3 & 2 \end{bmatrix}.$$

To solve for (x, y, u) take the block of columns x, y, u : $\det \begin{bmatrix} 3 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 2 & 2 \end{bmatrix} = 3(-2 - 2) - 1(2 - 2) + 0 = -12 \neq 0$, so the IMFT solves for x, y, u in terms of z . To solve for (x, y, z) take columns x, y, z : $\det \begin{bmatrix} 3 & 1 & -1 \\ 1 & -1 & 2 \\ 2 & 2 & -3 \end{bmatrix} = 3(3 - 4) - 1(-3 - 4) + (-1)(2 + 2) = -3 + 7 - 4 = 0$, singular — the IMFT gives no conclusion, and one checks the three rows are dependent in x, y, z (the third equation is a combination of the first two restricted to those columns), so x, y, z are *not* determined by u . *Check.* The nonzero -12 certifies solvability for $\{x, y, u\}$; the zero determinant flags the genuine obstruction for $\{x, y, z\}$. ■

Further worked examples

Worked Example 10.5.4 (Implicit surface: solving $F(x, y, z) = 0$ for z). The sphere $F(x, y, z) = x^2 + y^2 + z^2 - 9 = 0$ passes through $(2, 1, 2)$. Decide whether $z = g(x, y)$ exists locally and compute both partials g_x, g_y at $(2, 1)$.

Recognition. “Solve one scalar equation for one variable (z)” — $m = 1$, so F_y -block is just $\partial F/\partial z$.

Solution. $F(2, 1, 2) = 4 + 1 + 4 - 9 = 0$. ■ Here we solve for z , so the relevant block is $F_z = 2z$, which at $(2, 1, 2)$ equals $4 \neq 0$; the IMFT gives a unique C^1 function $z = g(x, y)$ near $(2, 1)$ with $g(2, 1) = 2$. The other partials are $F_x = 2x$, $F_y = 2y$. By the formula $g' = -F_z^{-1}[F_x \ F_y]$,

$$g_x = -\frac{F_x}{F_z} = -\frac{2x}{2z} = -\frac{x}{z} = -\frac{2}{4} = -\frac{1}{2}, \quad g_y = -\frac{F_y}{F_z} = -\frac{y}{z} = -\frac{1}{4}.$$

Check. Explicitly $g(x, y) = \sqrt{9 - x^2 - y^2}$ (upper hemisphere, $z > 0$); then $g_x = \frac{-x}{\sqrt{9 - x^2 - y^2}} = \frac{-x}{z}$, giving $-\frac{1}{2}$ at $(2, 1)$, and similarly $g_y = -\frac{1}{4}$, matching. ■

Worked Example 10.5.5 (Two equations, solve for two variables). Near $(x, y, u, v) = (1, 1, 1, 1)$ consider

$$F_1 = xu + yv - 2 = 0, \quad F_2 = xv + yu^2 - 2 = 0.$$

Show (u, v) can be solved as C^1 functions of (x, y) near the point, and find $\partial u/\partial x$ and $\partial v/\partial x$ there.

Recognition. “System $F = 0$, solve for (u, v) ” — form the 2×2 block $F_{(u,v)}$ and test its determinant.

Solution. $F(1, 1, 1, 1) = (1 + 1 - 2, 1 + 1 - 2) = (0, 0)$. ■ The block in the solved-for variables is

$$F_{(u,v)} = \begin{bmatrix} \partial F_1/\partial u & \partial F_1/\partial v \\ \partial F_2/\partial u & \partial F_2/\partial v \end{bmatrix} = \begin{bmatrix} x & y \\ 2yu & x \end{bmatrix} \Big|_{(1,1,1,1)} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad \det = 1 - 2 = -1 \neq 0,$$

so the IMFT applies. The x -block is $F_{(x,y)} = \begin{bmatrix} u & v \\ v & u^2 \end{bmatrix} \Big| = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$; we only need its first column $(\partial F_1/\partial x, \partial F_2/\partial x) = (u, v) = (1, 1)$. Then $\begin{bmatrix} u_x \\ v_x \end{bmatrix} = -F_{(u,v)}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = -\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Since $\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}^{-1} = \frac{1}{-1} \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$, we get $\begin{bmatrix} u_x \\ v_x \end{bmatrix} = -\begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = -\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$, i.e. $u_x = 0$, $v_x = -1$. *Check.* Differentiate both equations in x : $u + xu_x + yv_x = 0$ and $v + xv_x + 2yu u_x = 0$. At the point: $1 + u_x + v_x = 0$ and $1 + v_x + 2u_x = 0$. Subtracting gives $u_x = 0$, then $v_x = -1$ — agreeing. ■

Worked Example 10.5.6 (When F_y is singular: a vertical tangent). For $F(x, y) = y^2 - x = 0$ (a sideways parabola) at $(0, 0)$, show the IMFT cannot solve for $y = g(x)$, but *can* solve for $x = h(y)$.

Recognition. A non-example: the solved-for block vanishes, so the conclusion fails; switch which variable is solved for.

Solution. $F(0, 0) = 0$. To solve for y , the block is $F_y = 2y$, which is 0 at $(0, 0)$ — the hypothesis fails. Indeed near 0 the curve $y = \pm\sqrt{x}$ has *two* branches for each $x > 0$ and none for $x < 0$, so no single C^1 function $y = g(x)$ describes it (vertical tangent at the origin). Switching, to solve for x the block is $F_x = -1 \neq 0$ everywhere; the IMFT gives $x = h(y) = y^2$, a perfectly good C^1 function. *Check.* The curve *is* the graph $x = y^2$ globally, with $h'(y) = 2y$ giving slope 0 at the origin in the (y, x) frame — i.e. a vertical tangent in the (x, y) frame, exactly the obstruction that killed the $y = g(x)$ split. ■

Worked Example 10.5.7 (The unit circle as a local graph $y = g(x)$). For $F(x, y) = x^2 + y^2 - 1$ at the point $(\frac{3}{5}, \frac{4}{5})$ on the circle, show the IMFT solves for $y = g(x)$, find $g'(\frac{3}{5})$, and match it against the explicit upper branch.

Recognition. “Single scalar equation, solve for y ” — $m = 1$, so the solved-for block is $F_y = \partial F/\partial y$; check it is nonzero and apply $g' = -F_x/F_y$.

Solution. $F(\frac{3}{5}, \frac{4}{5}) = \frac{9}{25} + \frac{16}{25} - 1 = 0$. ■ The partials are $F_x = 2x$ and $F_y = 2y$. At the point $F_y = 2 \cdot \frac{4}{5} = \frac{8}{5} \neq 0$, so the IMFT gives a unique C^1 function $y = g(x)$ near $x = \frac{3}{5}$ with $g(\frac{3}{5}) = \frac{4}{5}$ and $F(x, g(x)) = 0$. The slope is

$$g'(\frac{3}{5}) = -\frac{F_x}{F_y} = -\frac{2x}{2y} = -\frac{x}{y} = -\frac{3/5}{4/5} = -\frac{3}{4}.$$

Check. The explicit upper branch is $g(x) = \sqrt{1 - x^2}$, with $g'(x) = \frac{-x}{\sqrt{1 - x^2}} = \frac{-x}{y}$; at $x = \frac{3}{5}$ this is $\frac{-3/5}{4/5} = -\frac{3}{4}$, matching, and the tangent line $y - \frac{4}{5} = -\frac{3}{4}(x - \frac{3}{5})$ is indeed perpendicular to the radius through $(\frac{3}{5}, \frac{4}{5})$ (whose slope is $\frac{4/5}{3/5} = \frac{4}{3}$, the negative reciprocal). ■

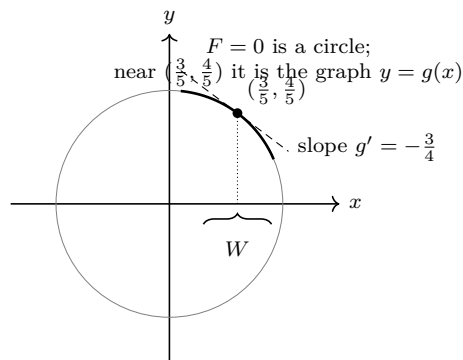


Figure 10.9. Where $F_y = 2y \neq 0$ the level set $x^2 + y^2 = 1$ is locally the graph of a C^1 branch $y = g(x) = \sqrt{1 - x^2}$; its slope is $g' = -F_x/F_y = -x/y$. At the two points $(\pm 1, 0)$, where $F_y = 0$, the branch breaks (vertical tangent).

Exercises (Rudin, Ch.9).

1. Show that the system of equations

$$3x + y - z + u^2 = 0, \quad x - y + 2z + u = 0, \quad 2x + 2y - 3z + 2u = 0$$

can be solved for x, y, u in terms of z ; for x, z, u in terms of y ; for y, z, u in terms of x ; but not for x, y, z in terms of u .

2. Take $n = m = 1$ in the Implicit Function Theorem, and interpret the theorem (as well as its proof) graphically.
3. Define f in $\mathbb{R}^2$ by $f(x, y) = 2x^3 - 3x^2 + 2y^3 + 3y^2$. (a) Find the four points in $\mathbb{R}^2$ at which the gradient of f is zero; show f has exactly one local maximum and one local minimum in $\mathbb{R}^2$. (b) Let S be the set of (x, y) with $f(x, y) = 0$. Find the points of S having no neighborhood in which $f(x, y) = 0$ can be solved for y in terms of x (or x in terms of y). Describe S as precisely as you can.
4. Give a similar discussion (gradient zeros; points of $\{f = 0\}$ where it is not locally a graph) for $f(x, y) = 2x^3 + 6xy^2 - 3x^2 + 3y^2$.
5. Define f in $\mathbb{R}^3$ by $f(x, y_1, y_2) = x^2y_1 + e^x + y_2$. Show that $f(0, 1, -1) = 0$, $(D_1f)(0, 1, -1) \neq 0$, and hence there is a differentiable function g in some neighborhood of $(1, -1)$ in $\mathbb{R}^2$ with $g(1, -1) = 0$ and $f(g(y_1, y_2), y_1, y_2) = 0$. Find $(D_1g)(1, -1)$ and $(D_2g)(1, -1)$.

Additional exercises.

1. For $F(x, y) = x^2 + y^2 - 1$ at the point $(\frac{3}{5}, \frac{4}{5})$ on the unit circle, verify the IMFT hypotheses for solving $y = g(x)$, and compute $g'(\frac{3}{5})$ two ways: by the formula $g' = -F_x/F_y$ and by differentiating $x^2 + g(x)^2 = 1$ directly.
2. (Where it fails.) At which points of the circle $x^2 + y^2 = 1$ can one *not* solve for y as a C^1 function of x ? At those points can one solve for x in terms of y ? State the determinant condition that distinguishes the two cases.
3. Let $F(x, y, z) = x^3 + y^3 + z^3 - 3xyz$. Check $F(1, 1, 1) = 0$, decide whether $z = g(x, y)$ exists as a C^1 function locally near $(1, 1)$, and find $\nabla g(1, 1)$ if it does (or explain the obstruction if it does not).
4. (Disprove.) A student claims: "If $F(a, b) = 0$ then the level set $F = 0$ is always locally a graph $y = g(x)$." Refute this with $F(x, y) = y^2 - x^2$ at the origin (two crossing lines). Compute $F_y(0, 0)$ and explain how the singular block predicts the failure.
5. (System.) Consider $F(x, y, u, v) = (xu + yv - 1, xv - yu)$ near $(x, y, u, v) = (1, 0, 1, 0)$. Show the 2×2 block $F_{(u, v)}$ is invertible there, so u, v are C^1 functions of (x, y) , and compute the Jacobian $\partial(u, v)/\partial(x, y)$ at the point.

6. Let $F(x, y) = 0$ define $y = g(x)$ near (a, b) with $F \in C^2$. Derive the second derivative formula $g''(a) = -\frac{F_{xx} + 2F_{xy}g' + F_{yy}(g')^2}{F_y}$ evaluated at (a, b) , and apply it to $F = x^2 + y^2 - 1$ at $(\frac{3}{5}, \frac{4}{5})$.
7. (Diagram.) Sketch the folium-like curve $F(x, y) = x^3 + y^3 - 3xy = 0$ near the origin. Mark the point $(0, 0)$ where $F_x = F_y = 0$ (both blocks singular) and indicate the self-crossing; contrast with a nearby point where the curve is locally a graph $y = g(x)$.
8. (IMFT from IFT, abstract.) Deduce the Implicit Function Theorem statement $g'(a) = -[F_y]^{-1}F_x$ purely by differentiating $F(x, g(x)) \equiv 0$, assuming only that a C^1 solution g exists. Identify exactly where invertibility of F_y is used.

10.6 The Rank Theorem and Higher-Order Derivatives

This closing section collects three further threads of Rudin Ch.9: the determinant (the scalar that detects invertibility and measures volume distortion), the Rank Theorem (the local normal form of a constant-rank map), and the symmetry of second-order mixed partial derivatives (Clairaut's theorem), with the higher-order machinery that leads to the multivariable Taylor theorem.

Determinants as needed

Definition 10.6.1. For $A = (a_{ij}) \in L(\mathbb{R}^n)$ the *determinant* is

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} \cdots a_{n\sigma(n)},$$

summed over all permutations σ of $\{1, \dots, n\}$. It is the unique alternating multilinear function of the columns that is 1 on the identity.

For analysis the Leibniz sum matters less than two readings of what $\det A$ means. First, it is the signed volume scaling factor: A stretches the unit cube into a parallelepiped of volume $|\det A|$, the sign recording whether orientation is preserved. Second, $\det A = 0$ exactly when the columns are linearly dependent, i.e. when A collapses some direction — which is precisely the failure of invertibility. So the single number $\det A$ simultaneously measures volume distortion and detects singularity, and its continuity in the entries is what makes invertibility an open condition.

Proposition 10.6.2 (Rudin 9.34–9.36: properties of $\det$). *For $A, B \in L(\mathbb{R}^n)$:*

1. $\det(BA) = \det(B)\det(A)$ (*multiplication theorem*), and $\det I = 1$.
2. A is invertible iff $\det A \neq 0$, and then $\det(A^{-1}) = (\det A)^{-1}$.
3. $\det : L(\mathbb{R}^n) \rightarrow \mathbb{R}$ is continuous (a polynomial in the entries), so $\Omega = \{\det \neq 0\}$ is open — a second proof of openness in §1.
4. $|\det A|$ is the volume of the image $A(Q)$ of the unit cube Q ; a linear A scales every volume by the factor $|\det A|$.

Proof sketch. (1) The Leibniz formula is multilinear and alternating in the columns and normalized at I ; both sides of $\det(BA) = \det B \det A$ are alternating multilinear in the columns of A and agree on the basis, hence agree. (2) If A is invertible, $1 = \det I = \det(AA^{-1}) = \det A \det A^{-1}$, forcing $\det A \neq 0$; conversely $\det A \neq 0$ makes the cofactor formula $A^{-1} = (\det A)^{-1} \operatorname{adj} A$ valid. (3) Each monomial in the Leibniz sum is a continuous (polynomial) function of the entries. (4) is the change-of-variables content developed for the integral; for a single linear map it follows from reducing A to elementary/diagonal factors, each of which scales volume by the obvious amount. $\square$

The rank theorem

When f' has *constant rank* near a point — not necessarily full rank — the map still has a clean local model: after smooth changes of coordinates in domain and codomain it becomes a linear projection.

Theorem 10.6.3 (Rudin 9.32: Rank Theorem). *Suppose $E \subseteq \mathbb{R}^n$ is open, $f : E \rightarrow \mathbb{R}^m$ is C^1 , and $f'(x)$ has constant rank r for all x in a neighborhood of $a \in E$. Then there exist open sets and C^1 diffeomorphisms H (of a neighborhood of a) and G (of a neighborhood of $f(a)$) such that the composite*

$$G \circ f \circ H^{-1} \quad \text{has the form} \quad (x_1, \dots, x_n) \mapsto (x_1, \dots, x_r, 0, \dots, 0) \in \mathbb{R}^m.$$

That is, in suitable local coordinates a constant-rank map is the projection onto the first r coordinates.

Proof sketch. By a fixed linear change of coordinates assume the top-left $r \times r$ block of $f'(a)$ is invertible. Write $f = (f_1, \dots, f_m)$ and define $H(x) = (f_1(x), \dots, f_r(x), x_{r+1}, \dots, x_n)$. Its Jacobian at a is block-triangular with the invertible $r \times r$ block in the corner and I_{n-r} below, hence invertible; by the Inverse Function Theorem H is a local C^1 diffeomorphism. In the new domain coordinates $u = H(x)$, the first r components of $f \circ H^{-1}$ are exactly $(u_1, \dots, u_r)$. The constant-rank hypothesis forces the remaining components $f_{r+1}, \dots, f_m$ (after this change) to be *independent of $u_{r+1}, \dots, u_n$* — otherwise the rank would exceed r somewhere. A further C^1 change G in the target, subtracting those functions of $(u_1, \dots, u_r)$, kills components $r+1, \dots, m$, leaving the projection $(u_1, \dots, u_n) \mapsto (u_1, \dots, u_r, 0, \dots, 0)$. $\square$

Remark. Two special cases are the workhorses. If $r = n \leq m$ (immersion, f' injective) the normal form is $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_n, 0, \dots, 0)$: f is locally an embedded n -surface. If $r = m \leq n$ (submersion, f' surjective) it is the projection $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_m)$: the level sets of f are locally $(n-m)$ -dimensional graphs — which is precisely the Implicit Function Theorem again.

Second-order partials and the symmetry of mixed partials

Definition 10.6.4. For $f : E \rightarrow \mathbb{R}$ ($E \subseteq \mathbb{R}^n$ open) the *second-order partials* are $D_{ij}f = D_i(D_jf) = \frac{\partial^2 f}{\partial x_i \partial x_j}$.

The map f is of class C^2 ($\mathcal{C}''$) if all D_jf and $D_{ij}f$ exist and are continuous; inductively $f \in C^{(k)}$ if all partials up to order k are continuous.

Theorem 10.6.5 (Rudin 9.41: Clairaut/Schwarz symmetry). *Let $f : E \rightarrow \mathbb{R}$ with $E \subseteq \mathbb{R}^2$ open. If D_1f , D_2f , $D_{21}f = D_2D_1f$, and $D_{12}f = D_1D_2f$ exist on E and $D_{21}f$, $D_{12}f$ are continuous at a point (a, b) , then*

$$D_{12}f(a, b) = D_{21}f(a, b).$$

In particular $f \in C^2$ has symmetric mixed partials: $D_{ij}f = D_{ji}f$ for all i, j . (For $f \in C^{(k)}$, every k -th order partial is unchanged under any permutation of the differentiation indices.)

Proof. Fix (a, b) and consider the second difference, for small $h, k \neq 0$,

$$\Delta(h, k) = f(a+h, b+k) - f(a+h, b) - f(a, b+k) + f(a, b).$$

Let $u(x) = f(x, b+k) - f(x, b)$, so $\Delta = u(a+h) - u(a)$. By the one-variable Mean Value Theorem applied to u (its derivative is $u'(x) = D_1f(x, b+k) - D_1f(x, b)$), there is $\theta \in (0, 1)$ with

$$\Delta = h u'(a + \theta h) = h [D_1f(a + \theta h, b+k) - D_1f(a + \theta h, b)].$$

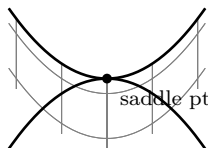
Apply the Mean Value Theorem again to $y \mapsto D_1f(a + \theta h, y)$ (derivative $D_{21}f = D_2D_1f$): there is $\theta' \in (0, 1)$ with

$$\Delta = h k D_{21}f(a + \theta h, b + \theta' k).$$

By the symmetric argument, first differencing in y then in x , one also gets $\Delta = h k D_{12}f(a + \eta h, b + \eta' k)$ for some $\eta, \eta' \in (0, 1)$. Equate and divide by hk :

$$D_{21}f(a + \theta h, b + \theta' k) = D_{12}f(a + \eta h, b + \eta' k).$$

Let $(h, k) \rightarrow (0, 0)$. Both arguments tend to (a, b) , and continuity of $D_{21}f, D_{12}f$ there gives $D_{21}f(a, b) = D_{12}f(a, b)$. $\square$



$$z = x^2 - y^2: \text{ up one way, down the other; } D_{12} = D_{21} = 0 \text{ here}$$

Figure 10.10. A saddle surface. For $f = x^2 - y^2$, $D_1f = 2x$, $D_2f = -2y$, so $D_{12}f = D_{21}f = 0$ — the mixed partials agree, as Clairaut's theorem promises for C^2 maps.

Caution. Continuity of the mixed partials is genuinely needed. Rudin's Exercise 27,

$$f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2} \quad ((x, y) \neq 0), \quad f(0, 0) = 0,$$

has $D_{12}f$ and $D_{21}f$ both existing everywhere, but $D_{12}f(0, 0) = 1$ while $D_{21}f(0, 0) = -1$: the mixed partials *disagree* at the origin, precisely because they are discontinuous there. Symmetry is a theorem about C^2 functions, not an algebraic identity.

Worked Example 10.6.6 (Verifying symmetry, and a higher-order count). For $f(x, y) = x^3y^2 + \sin(xy)$ confirm $D_{12}f = D_{21}f$, and state how many distinct third-order partials f has.

Recognition. “Check Clairaut, count higher partials.” f is C^∞ (sums/products of C^∞ pieces), so symmetry must hold.

Solution. $D_1f = 3x^2y^2 + y \cos(xy)$, $D_2f = 2x^3y + x \cos(xy)$. Then $D_2D_1f = 6x^2y + \cos(xy) - xy \sin(xy)$ and $D_1D_2f = 6x^2y + \cos(xy) - xy \sin(xy)$: equal, as required. Because $f \in C^3$, a third-order partial depends only on *how many* times we differentiate in x vs. y , not the order; the distinct ones are $D_{xxx}, D_{xxy}, D_{xyy}, D_{yyy}$ — 4 of them (in general $k + 1$ for order k in two variables). *Check.* Without symmetry there would be $2^3 = 8$ ordered third partials; Clairaut collapses them to 4. ■

Further worked examples

Worked Example 10.6.7 (Determinant as a volume scaling factor). A linear map $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ acts on

$\mathbb{R}^3$. Find the volume of the image $A(Q)$ of the unit cube $Q = [0, 1]^3$, and decide whether A is invertible.

Recognition. “Volume of a linear image” — Proposition 10.6.2(4): the factor is $|\det A|$; nonzero determinant also settles invertibility.

Solution. Expand along the third column (only $a_{33} = 2$ is nonzero there): $\det A = 2 \cdot \det \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = 2 \cdot 6 = 12$. Thus $\det A = 12 \neq 0$, so A is invertible, and $\text{vol } A(Q) = |\det A| \cdot \text{vol } Q = 12 \cdot 1 = 12$. *Check (independent expansion):* along the first column, $\det A = 2 \det \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} - 0 + 1 \cdot \det \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} = 2 \cdot 6 + 1 \cdot 0 = 12$, agreeing. ■

Worked Example 10.6.8 (Rank Theorem in action: an immersion and a submersion). Classify the local normal forms of (a) $f(t) = (\cos t, \sin t)$ as a map $\mathbb{R} \rightarrow \mathbb{R}^2$ near $t = 0$, and (b) $g(x, y, z) = x^2 + y^2 + z^2$ as a map $\mathbb{R}^3 \rightarrow \mathbb{R}$ near $(1, 0, 0)$.

Recognition. “Constant-rank local model” — compute rank f' , then read off the immersion ($r = n$) or submersion ($r = m$) normal form from the Remark.

Solution. (a) $f'(t) = (-\sin t, \cos t)$, a 2×1 Jacobian; at every t its rank is $1 = n$ (it is never the zero vector since $\sin^2 t + \cos^2 t = 1$). So f is an immersion of constant rank 1, and the Rank Theorem says it is locally $t \mapsto (t, 0)$ in suitable coordinates — an embedded curve, with no self-crossings locally. (b) $g'(x, y, z) = (2x, 2y, 2z)$, a 1×3 Jacobian of rank $1 = m$ wherever $(x, y, z) \neq 0$; near $(1, 0, 0)$ it equals $(2, 0, 0) \neq 0$. So g is a submersion of constant rank 1, locally the projection $(u_1, u_2, u_3) \mapsto u_1$, and its level sets $\{g = c\}$ (spheres) are locally 2-dimensional graphs — the IMFT picture again. *Check.* For (b), near $(1, 0, 0)$ one solves $x^2 + y^2 + z^2 = 1$ as $x = \sqrt{1 - y^2 - z^2}$, a 2-variable graph, exactly the submersion conclusion. ■

Worked Example 10.6.9 (Hessian test: classifying a critical point). Find the critical points of $f(x, y) = x^3 - 3x + y^2$ and classify each using the Hessian.

Recognition. “Critical point classification” — the degree-2 Taylor term, governed by the Hessian’s definiteness (Remark below).

Solution. $\nabla f = (3x^2 - 3, 2y) = 0$ gives $x^2 = 1$, $y = 0$: critical points $(1, 0)$ and $(-1, 0)$. The Hessian is $H = \begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}$. At $(1, 0)$: $H = \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$, eigenvalues $6, 2$ both positive — positive definite, so a strict local minimum. At $(-1, 0)$: $H = \begin{bmatrix} -6 & 0 \\ 0 & 2 \end{bmatrix}$, eigenvalues $-6, 2$ of opposite sign — indefinite, so a *saddle*. *Check.* Along $y = 0$, $f(x, 0) = x^3 - 3x$ has $f' = 3x^2 - 3$, $f'' = 6x$: a local min at $x = 1$ ($f'' = 6 > 0$) and a local max at $x = -1$ ($f'' = -6 < 0$); the y -direction always adds $y^2 \geq 0$, turning the x -max at $(-1, 0)$ into a saddle, consistent with the Hessian. ■

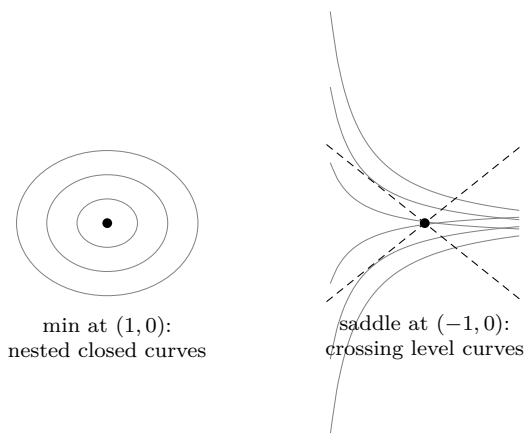


Figure 10.11. Level curves near the two critical points of $f = x^3 - 3x + y^2$. A positive definite Hessian gives nested loops (a minimum); an indefinite Hessian gives crossing curves (a saddle).

Remark (Taylor’s theorem in several variables). Iterating the Chain Rule along the segment $p(t) = a + tx$ and applying the one-variable Taylor theorem to $h(t) = f(p(t))$ yields (Rudin Ex. 30) the multivariable expansion

$$f(a + x) = \sum_{k=0}^{m-1} \frac{1}{k!} \sum_{i_1, \dots, i_k} (D_{i_1 \dots i_k} f)(a) x_{i_1} \cdots x_{i_k} + r(x), \quad \frac{|r(x)|}{|x|^{m-1}} \rightarrow 0.$$

Symmetry of higher partials (Exercise 29) lets one collect repeated derivatives into the multi-index form $\sum_{|s| \leq m-1} \frac{1}{s!} (D^s f)(a) x^s$. The degree-2 piece, the Hessian quadratic form $\frac{1}{2} \sum_{i,j} D_{ij} f(a) x_i x_j$, governs the local max/min/saddle classification at a critical point (Exercise 31): positive definite Hessian gives a strict local minimum, negative definite a maximum, indefinite a saddle.

More worked examples

Worked Example 10.6.10 (Mixed partials that disagree: a non-example for Clairaut). Let $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}$ for $(x, y) \neq 0$, $f(0, 0) = 0$. Compute $D_{12}f(0, 0)$ and $D_{21}f(0, 0)$ directly and show they differ, pinpointing why Clairaut’s theorem does not apply.

Recognition. “Do the mixed partials agree at the origin?” — Clairaut needs $D_{12}f, D_{21}f$ continuous at the point; here they exist but are discontinuous, so we must compute both from the limit definition rather than assume symmetry.

Solution. Away from the origin, differentiating the quotient gives, after simplification,

$$D_1 f(x, y) = \frac{y(x^4 + 4x^2 y^2 - y^4)}{(x^2 + y^2)^2}, \quad D_2 f(x, y) = \frac{x(x^4 - 4x^2 y^2 - y^4)}{(x^2 + y^2)^2}.$$

On the axes these collapse cleanly: $D_1f(0, y) = \frac{y(-y^4)}{y^4} = -y$ for $y \neq 0$ (and $D_1f(0, 0) = 0$), while $D_2f(x, 0) = \frac{x(x^4)}{x^4} = x$ for $x \neq 0$ (and $D_2f(0, 0) = 0$). Now apply the definition of the second partials at the origin:

$$D_{21}f(0, 0) = \frac{\partial}{\partial y} \Big|_0 D_1f(0, y) = \lim_{y \rightarrow 0} \frac{D_1f(0, y) - D_1f(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{-y}{y} = -1,$$

$$D_{12}f(0, 0) = \frac{\partial}{\partial x} \Big|_0 D_2f(x, 0) = \lim_{x \rightarrow 0} \frac{D_2f(x, 0) - D_2f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{x}{x} = +1.$$

So $D_{12}f(0, 0) = 1 \neq -1 = D_{21}f(0, 0)$: the mixed partials *disagree*. Clairaut is not violated — its hypothesis fails, since $D_{12}f$ and $D_{21}f$ are discontinuous at the origin (their values there depend on the order of differentiation). *Check.* On the axes we found $D_1f(0, y) = -y$ and $D_2f(x, 0) = x$; the slope of the first in y is -1 and of the second in x is $+1$, the two numbers above — symmetry would force these equal, and their inequality is exactly the discontinuity Clairaut forbids. ■

Worked Example 10.6.11 (Rank Theorem: a constant-rank map that is neither immersion nor submersion). Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $f(x, y, z) = (x, y, 0)$. Compute $\text{rank } f'$, confirm it is constant, and identify the local normal form predicted by the Rank Theorem.

Recognition. “Constant-rank local model” — compute the Jacobian, read off the rank r , and the Rank Theorem gives the projection $(u_1, \dots, u_n) \mapsto (u_1, \dots, u_r, 0, \dots, 0)$.

Solution. The Jacobian is the constant matrix

$$[f'(x, y, z)] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

of rank $r = 2$ at every point (the first two rows are independent, the third is zero). Since f' is constant it certainly has constant rank 2, but $r = 2 < n = 3$ (not an immersion: f' has a kernel, the z -axis) and $r = 2 < m = 3$ (not a submersion: f' is not onto, missing the third coordinate). The Rank Theorem nonetheless gives the clean normal form: here f is *already* the projection $(x, y, z) \mapsto (x, y, 0)$, so $H = G = \text{id}$ work and $G \circ f \circ H^{-1}$ is exactly $(u_1, u_2, u_3) \mapsto (u_1, u_2, 0)$. *Check.* The image $f(\mathbb{R}^3)$ is the plane $\{z = 0\}$, a 2-dimensional set, matching rank = 2; and the fibers $f^{-1}(\text{pt})$ are vertical lines (the missing z -direction), of dimension $n - r = 1$, as a rank-2 map of $\mathbb{R}^3$ must have. ■

Exercises (Rudin, Ch.9).

- For $(x, y) \neq (0, 0)$, define $f = (f_1, f_2)$ by $f_1(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$, $f_2(x, y) = \frac{xy}{x^2 + y^2}$. Compute the rank of $f'(x, y)$, and find the range of f .
- Suppose $A \in L(\mathbb{R}^n, \mathbb{R}^m)$; let r be the rank of A . (a) Define S as in the proof of Theorem 9.32. Show that SA is a projection in $\mathbb{R}^n$ whose null space is $\mathcal{N}(A)$ and whose range is $\mathcal{R}(S)$. *Hint.* $SASA = SA$. (b) Use (a) to show that $\dim \mathcal{N}(A) + \dim \mathcal{R}(A) = n$.
- Show that the existence (and even the continuity) of $D_{12}f$ does not imply the existence of D_1f . For example, let $f(x, y) = g(x)$, where g is nowhere differentiable.
- Put $f(0, 0) = 0$, and $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$. Prove that (a) f , D_1f , D_2f are continuous in $\mathbb{R}^2$; (b) $D_{12}f$ and $D_{21}f$ exist at every point of $\mathbb{R}^2$ and are continuous except at $(0, 0)$; (c) $(D_{12}f)(0, 0) = 1$ and $(D_{21}f)(0, 0) = -1$.
- For $t \geq 0$ put $\varphi(x, t) = x$ on $0 \leq x \leq \sqrt{t}$, $\varphi(x, t) = -x + 2\sqrt{t}$ on $\sqrt{t} \leq x \leq 2\sqrt{t}$, and $\varphi(x, t) = 0$ otherwise; for $t < 0$ put $\varphi(x, t) = -\varphi(x, |t|)$. Show that φ is continuous on $\mathbb{R}^2$ and $(D_2\varphi)(x, 0) = 0$ for all x . Define $f(t) = \int_{-\infty}^{\infty} \varphi(x, t) dx$. Show that $f(t) = t$ if $|t| < 1$. Hence $f'(0) = 1 \neq \int_{-\infty}^{\infty} (D_2\varphi)(x, 0) dx = 0$ (differentiation under the integral sign can fail).

6. Let E be open in $\mathbb{R}^n$; the classes $C'(E)$ and $C''(E)$ are defined in the text. By induction, $C^{(k)}(E)$ means: $D_1f, \dots, D_nf$ belong to $C^{(k-1)}(E)$. Assume $f \in C^{(k)}(E)$ and show (by repeated application of Theorem 9.41) that the k -th order derivative $D_{i_1 i_2 \dots i_k} f = D_{i_1} D_{i_2} \dots D_{i_k} f$ is unchanged if the subscripts $i_1, \dots, i_k$ are permuted. For instance, $D_{123}f = D_{321}f$ for every $f \in C^{(4)}$ when $n \geq 3$.
7. Let $f \in C^{(m)}(E)$, E open in $\mathbb{R}^n$. Fix $a \in E$ and suppose $x \in \mathbb{R}^n$ is so close to 0 that $p(t) = a + tx$ lies in E for $0 \leq t \leq 1$. Define $h(t) = f(p(t))$. (a) For $1 \leq k \leq m$, show by repeated use of the chain rule that $h^{(k)}(t) = \sum (D_{i_1 \dots i_k} f)(p(t)) x_{i_1} \dots x_{i_k}$, summed over all ordered k -tuples $(i_1, \dots, i_k)$ with each $i_j \in \{1, \dots, n\}$. (b) By Taylor's theorem (5.15) applied to h , deduce Taylor's theorem in n variables: $f(a+x) = \sum_{k=0}^{m-1} \frac{1}{k!} \sum (D_{i_1 \dots i_k} f)(a) x_{i_1} \dots x_{i_k} + r(x)$, where $r(x)/|x|^{m-1} \rightarrow 0$ as $x \rightarrow 0$. (c) Using Exercise 29 (symmetry), show the Taylor polynomial can be rewritten as a sum over nonnegative integer n -tuples $(s_1, \dots, s_n)$ with $s_1 + \dots + s_n \leq m-1$, collecting repeated derivatives.
8. Suppose $f \in C^{(3)}$ in a neighborhood of $a \in \mathbb{R}^2$, the gradient of f is 0 at a , but not all second-order derivatives of f are 0 at a . Show how to determine, from the degree-2 Taylor polynomial of f at a , whether f has a local maximum, a local minimum, or neither at a . Extend this to $\mathbb{R}^n$ in place of $\mathbb{R}^2$.

Additional exercises.

1. Compute all second-order partials of $f(x, y) = x^3y - xy^3$ and verify directly that $D_{12}f = D_{21}f$ everywhere. State the value of the Hessian determinant at $(1, 1)$.
2. (Rank computation.) For $f(x, y) = (x, xy, xy^2)$ compute $[f'(x, y)]$ and its rank as a function of (x, y) . On which set is the rank constant equal to 2? Describe the local normal form given by the Rank Theorem there.
3. (Determinant as volume.) Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$. Compute $|\det A|$ and verify it equals the area of the image $A([0, 1]^2)$ by computing that parallelogram's area directly.
4. (Disprove symmetry without continuity.) Verify by direct computation that for $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}$ ($f(0, 0) = 0$) one has $D_{12}f(0, 0) = 1$ but $D_{21}f(0, 0) = -1$. Which hypothesis of Theorem 10.6.5 is violated?
5. Let $f \in C^2(\mathbb{R}^2)$. Show the Laplacian $\Delta f = D_{11}f + D_{22}f$ is unchanged under a rotation of coordinates (i.e. if $g = f \circ R_\theta$ then $\Delta g = (\Delta f) \circ R_\theta$). (Hint: chain rule plus $R_\theta^\top R_\theta = I$.)
6. (Hessian and critical points.) For $f(x, y) = x^3 - 3xy^2$ find all critical points, and use the Hessian to classify each as a local max, min, or saddle. (This is the real part of z^3 ; expect a "monkey saddle" at the origin where the Hessian is degenerate.)
7. (Diagram.) Sketch the saddle surface $z = xy$. Draw the two coordinate sections $y = 0$ and $x = 0$ (both flat lines through the origin) and a diagonal section $y = x$ (an upward parabola). Mark the origin and note $D_{12}f = D_{21}f = 1$ there.
8. (Taylor count.) Using symmetry of mixed partials, count the number of distinct partial derivatives of order exactly k for a C^∞ function of n variables, and confirm it equals $\binom{n+k-1}{k}$. Check it gives 4 for $n = 2, k = 3$, matching the §6 example.

Chapter 11

Integration of Differential Forms

11.1 Integration in $\mathbb{R}^k$ and Partitions of Unity

This chapter develops the calculus of differential forms and culminates in Stokes' theorem, the single identity from which Green's theorem, the divergence theorem, and the classical Stokes theorem all descend. We begin on solid ground: integrating an ordinary continuous function over $\mathbb{R}^k$, and the technical tool — the *partition of unity* — that lets us cut global problems into local pieces and glue the answers back together.

The integral of a function with compact support

Definition 11.1.1. The *support* of a function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ is the closure of the set where it is nonzero,

$$\text{supp } f = \overline{\{x \in \mathbb{R}^k : f(x) \neq 0\}}.$$

We write $f \in \mathcal{C}(\mathbb{R}^k)$ (continuous) and say f has *compact support* if $\text{supp } f$ is a compact (closed and bounded) subset of $\mathbb{R}^k$. The space of continuous functions on $\mathbb{R}^k$ with compact support is denoted $\mathcal{C}_c(\mathbb{R}^k)$ (Rudin writes $\mathcal{C}(\mathbb{R}^k)$ for it in this chapter, taking compact support as understood).

A function with compact support vanishes outside a bounded box, so all the integrals below are really integrals over a box $I^k = [a_1, b_1] \times \cdots \times [a_k, b_k]$ large enough to contain $\text{supp } f$ in its interior. Inside such a box a continuous function is bounded and uniformly continuous, and every integral we write converges with nothing to worry about at infinity.

Definition 11.1.2. Let $f \in \mathcal{C}_c(\mathbb{R}^k)$ and let I^k be a box containing $\text{supp } f$. The *integral* of f is the iterated integral

$$\int_{\mathbb{R}^k} f = \int f(x) dx = \int_{a_1}^{b_1} \cdots \int_{a_k}^{b_k} f(x_1, \dots, x_k) dx_k \cdots dx_1,$$

each inner integral being the ordinary one-variable Riemann integral of a continuous function. (The value does not depend on the box I^k , since $f = 0$ outside $\text{supp } f$.)

The pressing question is whether the *order* of the k one-variable integrations matters. It does not, and this is the first theorem of the chapter.

Theorem 11.1.3 (Independence of the order of integration; Rudin 10.2). *If $f \in \mathcal{C}_c(\mathbb{R}^k)$ then $\int f$ is unchanged by any permutation of the order of the k integrations. Consequently $\int f$ is well defined by Definition 11.1.2.*

Proof sketch. For continuous f supported in a box, each partial integral such as $g(x_1) = \int f(x_1, x_2) dx_2$ is again continuous (uniform continuity on the compact box makes the integrand vary uniformly in the outer variable). Iterating, every order of integration produces a continuous function at each stage and a well-defined number at the end. That the two orders for $k = 2$ agree is the classical Fubini/iterated-integral theorem for continuous integrands on a rectangle; the general k follows by transposing adjacent variables one swap at a time, each swap being the $k = 2$ case applied to the inner pair. A self-contained proof realizes both orders as limits of the *same* Riemann sums over a common grid: refine the partition of I^k , and uniform continuity forces both iterated integrals to the common limit. $\square$

Caution. Independence of order needs *some* hypothesis. Without compact support (or absolute integrability) the two iterated integrals can genuinely differ. Exercise 2 below builds an f , continuous except at the origin, with $\int dy \int f(x, y) dx = 0$ but $\int dx \int f(x, y) dy = 1$. The obstruction is that f is unbounded near $(0, 0)$; compact support of a *continuous* f rules this out.

Worked Example 11.1.4 (An iterated integral over a box). Compute $\int f$ for $f(x, y) = xy e^{-(x^2+y^2)} \chi$, where χ is a smooth cutoff that equals 1 on $[0, 1]^2$ and 0 outside $[-1, 2]^2$, with f taken equal to $xy e^{-(x^2+y^2)}$ on $[0, 1]^2$ and we integrate only the displayed product over the unit square (a clean stand-in showing the iterated mechanics).

Recognition. “Integrate a product over a box.” When $f(x, y) = g(x)h(y)$ on a rectangle, the double integral factors into a product of one-variable integrals.

Solution. On $[0, 1]^2$, $f(x, y) = (xe^{-x^2})(ye^{-y^2})$, a product of a function of x and a function of y . Hence

$$\int_0^1 \int_0^1 f = \left(\int_0^1 xe^{-x^2} dx \right) \left(\int_0^1 ye^{-y^2} dy \right) = \left[\frac{1}{2}(1 - e^{-1}) \right]^2.$$

The two orders give the same factored answer, illustrating Theorem 11.1.3. *Check.* $\frac{1}{2}(1 - e^{-1}) \approx 0.316$, so the integral is ≈ 0.0999 , a small positive number as the nonnegative integrand on $[0, 1]^2$ demands.

Partitions of unity

To pass from local statements (true on each small piece of a set) to global ones, we need a way to write the constant function 1 as a finite sum of bumps, each living in one piece. That is a partition of unity.

Definition 11.1.5. Let $K \subseteq \mathbb{R}^k$ be compact and let $\{V_\alpha\}$ be an open cover of K . A *partition of unity* on K subordinate to $\{V_\alpha\}$ is a finite collection $\psi_1, \dots, \psi_s \in \mathcal{C}_c(\mathbb{R}^k)$ with:

1. $0 \leq \psi_i \leq 1$ for each i ;
2. each $\text{supp } \psi_i$ lies in some single $V_{\alpha(i)}$;
3. $\sum_{i=1}^s \psi_i(x) = 1$ for every $x \in K$ (and ≤ 1 everywhere).

Theorem 11.1.6 (Existence of partitions of unity; Rudin 10.8). *For every compact $K \subseteq \mathbb{R}^k$ and every open cover $\{V_\alpha\}$ of K , there exists a partition of unity subordinate to that cover.*

Proof outline. For each $x \in K$ pick one cover element $V_\alpha \ni x$ and a ball $B(x, 2r_x)$ with closure inside V_α . The open balls $B(x, r_x)$ cover the compact K , so finitely many $B(x_1, r_1), \dots, B(x_s, r_s)$ already do. For each i build a continuous bump $\varphi_i \geq 0$ with $\varphi_i > 0$ on $\overline{B(x_i, r_i)}$ and $\text{supp } \varphi_i \subseteq B(x_i, 2r_i) \subseteq V_{\alpha(i)}$ — e.g. $\varphi_i(x) = \max\{0, r_i^2 - \text{dist}(x, x_i)^2\}$, or a smooth bump if smoothness is wanted (Exercise 6). Now normalize *telescopically* so the partial sums stay below 1 and the total is 1 on K :

$$\psi_1 = \varphi_1, \quad \psi_{i+1} = (1 - \varphi_1) \cdots (1 - \varphi_i) \varphi_{i+1}.$$

An easy induction gives $\psi_1 + \cdots + \psi_i = 1 - (1 - \varphi_1) \cdots (1 - \varphi_i)$. On K every point lies in some $\overline{B(x_j, r_j)}$, where $\varphi_j > 0$, so the product $\prod (1 - \varphi_\ell)$ vanishes and $\sum_i \psi_i = 1$. Each $\text{supp } \psi_i \subseteq \text{supp } \varphi_i \subseteq V_{\alpha(i)}$, and $0 \leq \psi_i \leq 1$. $\square$

The picture to keep in mind: a compact region covered by a handful of open sets, with a bump function peaked in each, the bumps tuned so that at every point of K the heights add up to exactly 1.

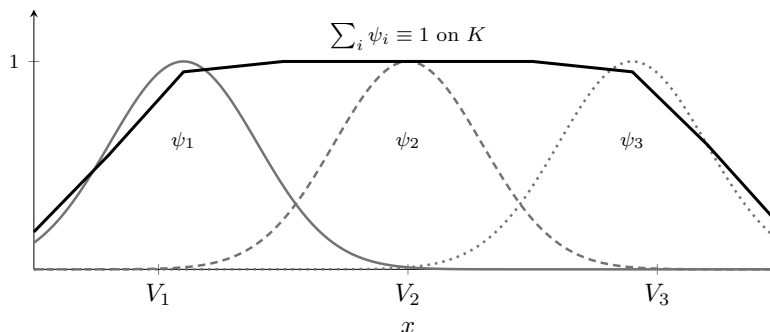


Figure 11.1. A partition of unity in one variable: three bumps, each supported in one cover element V_i , summing to 1 across the compact set K they cover.

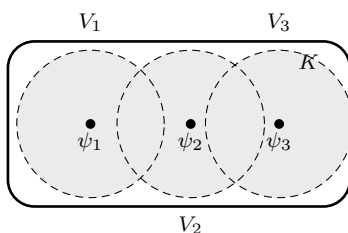


Figure 11.2. The same idea in $\mathbb{R}^2$: the supports of the bumps (shaded disks) cover K , each sitting inside one cover element V_α .

Method (Localizing a global problem with a partition of unity). *Recognition trigger.* You can solve or estimate something *locally* (on each small open set V_α) and want a *global* conclusion on a compact K ; or you must prove a global integral identity that is obvious on small pieces.

1. Cover K by open sets V_α on which the local statement holds.
2. Extract a finite subcover and a subordinate partition of unity $\psi_1, \dots, \psi_s$ (Theorem 11.1.6).
3. Write $f = \sum_i \psi_i f$. Each $\psi_i f$ is supported in one $V_{\alpha(i)}$, so the local result applies to it.
4. Sum the local results. Linearity of the integral (or of whatever operation) glues the pieces into the global statement.

This is exactly how the change of variables theorem (§ 11.2) and Stokes' theorem (§ 11.6) are reduced to a single coordinate patch.

Worked Example 11.1.7 (Reducing an integral to coordinate patches). Let $f \in \mathcal{C}_c(\mathbb{R}^k)$ with $\text{supp } f \subseteq V_1 \cup \dots \cup V_m$, and suppose we know how to integrate any function supported in a single V_j . Show $\int f$ is determined.

Recognition. “Integrate something supported across several patches.” Split with a partition of unity, integrate each piece, add.

Solution. Take a partition of unity $\psi_1, \dots, \psi_s$ on $\text{supp } f$ subordinate to $\{V_j\}$. Then $f = \sum_i \psi_i f$ pointwise (since $\sum \psi_i = 1$ on $\text{supp } f$ and $f = 0$ off it), each $\psi_i f$ is continuous with support inside one $V_{j(i)}$, and by linearity

$$\int f = \int \left(\sum_i \psi_i f \right) = \sum_i \int \psi_i f.$$

Every summand is a single-patch integral, which we assumed we can evaluate. *Check.* If f itself already sits in one V_1 , the partition collapses ($\sum \psi_i f = f$ on $\text{supp } f$) and we recover $\int f$, consistent.

Remark. Partitions of unity exist on any metric space for the analogous statement (Exercise 5), with the bump functions of Definition 11.1.1 replaced by Urysohn-type functions. With more care the ψ_i can be taken infinitely differentiable (Exercise 6), which matters when we later differentiate forms: a smooth partition of unity lets us localize differential forms without losing smoothness.

Further worked examples

Worked Example 11.1.8 (The support of a sum versus the union of supports). For $f, g \in \mathcal{C}_c(\mathbb{R}^k)$ decide whether $\text{supp}(f + g) = \text{supp } f \cup \text{supp } g$ always holds.

Recognition. “Test a plausible set identity for supports.” Definition 11.1.1: supp is the *closure* of the nonvanishing set; cancellation can shrink the left side.

Solution. One containment is always true: where $f + g \neq 0$ at least one of f, g is nonzero, so $\{f + g \neq 0\} \subseteq \{f \neq 0\} \cup \{g \neq 0\}$, and taking closures gives $\text{supp}(f + g) \subseteq \text{supp } f \cup \text{supp } g$. The reverse can *fail* by cancellation. Take $k = 1$, f a bump equal to 1 on $[0, 1]$ supported in $[-1, 2]$, and $g = -f$. Then $f + g \equiv 0$, so $\text{supp}(f + g) = \emptyset$, whereas $\text{supp } f \cup \text{supp } g = \text{supp } f \neq \emptyset$. *Check.* The equality is only an inclusion in general; it becomes equality when $f, g \geq 0$, since then no cancellation occurs and $\{f + g \neq 0\} = \{f \neq 0\} \cup \{g \neq 0\}$ exactly.

Worked Example 11.1.9 (An iterated integral computed in both orders). Let $f(x, y) = (x + 2y)\chi$ where χ is a smooth cutoff equal to 1 on $[0, 1]^2$ and 0 off $[-1, 2]^2$; integrate the displayed polynomial over the unit square in both orders and confirm Theorem 11.1.3.

Recognition. “Check independence of order on a concrete continuous integrand.” Both iterated integrals of a compactly supported continuous f must agree.

Solution. Order $dy dx$: $\int_0^1 (\int_0^1 (x + 2y) dy) dx = \int_0^1 (x + 1) dx = \frac{1}{2} + 1 = \frac{3}{2}$. Order $dx dy$: $\int_0^1 (\int_0^1 (x + 2y) dx) dy = \int_0^1 (\frac{1}{2} + 2y) dy = \frac{1}{2} + 1 = \frac{3}{2}$. *Check.* Both orders give $\frac{3}{2}$, as the theorem demands for a continuous integrand on a box.

Worked Example 11.1.10 (A subordinate partition of unity, explicitly). On $K = [0, 2] \subseteq \mathbb{R}$ covered by $V_1 = (-1, 1.2)$ and $V_2 = (0.8, 3)$, exhibit functions ψ_1, ψ_2 forming a partition of unity subordinate to $\{V_1, V_2\}$.

Recognition. “Build, not just invoke, a partition of unity.” Use the telescoping construction $\psi_2 = (1 - \varphi_1)\varphi_2$ from Theorem 11.1.6.

Solution. Pick continuous bumps φ_1 supported in V_1 with $\varphi_1 \equiv 1$ on $[0, 0.9]$, and φ_2 supported in V_2 with $\varphi_2 \equiv 1$ on $[1.1, 2]$; arrange $\varphi_1 + \varphi_2 > 0$ on all of K (the overlap $(0.8, 1.2)$ guarantees coverage). Set

$$\psi_1 = \varphi_1, \quad \psi_2 = (1 - \varphi_1)\varphi_2.$$

Then $\psi_1 + \psi_2 = 1 - (1 - \varphi_1)(1 - \varphi_2)$. On K every point lies where some $\varphi_i \equiv 1$, so $(1 - \varphi_1)(1 - \varphi_2) = 0$ and $\psi_1 + \psi_2 = 1$. Each $\text{supp } \psi_i \subseteq \text{supp } \varphi_i \subseteq V_i$, and $0 \leq \psi_i \leq 1$. *Check.* At $x = 0.5$, $\varphi_1 = 1$, so $\psi_1 = 1$, $\psi_2 = 0$, sum 1; at $x = 1.5$, $\varphi_2 = 1$ and $\varphi_1 = 0$, so $\psi_1 = 0$, $\psi_2 = 1$, sum 1 — a valid partition of unity.

Exercises (Rudin, Ch.10).

- (Rudin 1.) Let H be a compact convex set in $\mathbb{R}^k$ with nonempty interior. Let $f \in \mathcal{C}(H)$, put $f(x) = 0$ in the complement of H , and define $\int f$ as in Definition 11.1.2. Prove that $\int f$ is independent of the order in which the k integrations are carried out. (Hint: approximate f by functions continuous on $\mathbb{R}^k$ whose supports lie in H .)
- (Rudin 2.) For $i = 1, 2, 3, \dots$ let $\varphi_i \in \mathcal{C}(\mathbb{R}^1)$ have support in $(2^{-i}, 2^{1-i})$ with $\int \varphi_i = 1$. Put

$$f(x, y) = \sum_{i=1}^{\infty} [\varphi_i(x) - \varphi_{i+1}(x)] \varphi_i(y).$$

Then f has compact support in $\mathbb{R}^2$, f is continuous except at $(0, 0)$, and

$$\int dy \int f(x, y) dx = 0 \quad \text{but} \quad \int dx \int f(x, y) dy = 1.$$

Observe that f is unbounded in every neighborhood of $(0, 0)$.

- (Rudin 5.) Formulate and prove an analogue of Theorem 11.1.6 (existence of partitions of unity) in which K is a compact subset of an arbitrary metric space. (Replace the bump functions φ_i in the proof by functions of Urysohn type.)

- (Rudin 6.) Strengthen the conclusion of Theorem 11.1.6 by showing that the functions ψ_i can be made differentiable, and even infinitely differentiable. (Use a smooth one-variable bump in the construction of the auxiliary functions φ_i .)

Additional exercises.

- (Iterated integral by hand.) Let $f(x, y) = x^2y$ on the rectangle $Q = [0, 1] \times [0, 2]$ and $f = 0$ off Q . Compute $\int dy \int f dx$ and $\int dx \int f dy$ directly and check the two orders agree.
- Let $\varphi \in \mathcal{C}(\mathbb{R}^1)$ be supported in $(0, 1)$ with $\int \varphi = 1$, and set $f(x, y) = \varphi(x)\varphi(y)$. Show $\text{supp } f \subseteq (0, 1)^2$, that $\int f = 1$, and that the two iterated integrals are equal. Why does no order-of-integration pathology (as in Rudin 2) arise here?
- (Support of a sum.) For $f, g \in \mathcal{C}(\mathbb{R}^k)$ with compact support, prove $\text{supp}(f + g) \subseteq \text{supp } f \cup \text{supp } g$ and $\text{supp}(fg) \subseteq \text{supp } f \cap \text{supp } g$. Give an example where the first inclusion is strict.
- Construct an explicit C^∞ bump on $\mathbb{R}^1$: show $\beta(t) = \exp(-1/(1 - t^2))$ for $|t| < 1$ and $\beta(t) = 0$ for $|t| \geq 1$ is infinitely differentiable, supported in $[-1, 1]$, and strictly positive on $(-1, 1)$. (This is the seed for the smooth partition-of-unity refinement.)
- Using the bump β of the previous exercise, build a function $\psi \in \mathcal{C}^\infty(\mathbb{R}^k)$ with $\psi \equiv 1$ on the closed ball $\|x\| \leq 1$, $\psi \equiv 0$ off $\|x\| \leq 2$, and $0 \leq \psi \leq 1$ everywhere. (Hint: integrate a radial bump and rescale; compose with $\|x\|$.)
- (Non-example.) Show that the family $\{(n, n + 2) : n \in \mathbb{Z}\}$ of open intervals covers $\mathbb{R}^1$, yet no *finite* subfamily does. Explain why the existence proof for partitions of unity nonetheless needs only finitely many bumps when the set being covered is *compact*.
- (Conceptual.) State precisely what fails in the conclusion $\int dy \int f dx = \int dx \int f dy$ if f is merely bounded with compact support but has one point of discontinuity. Relate your answer to Rudin 2, and say which hypothesis in the iterated-integral theorem rules that example out.
- Let $K \subseteq \mathbb{R}^k$ be compact and $\{V_i\}_{i=1}^N$ a finite open cover of K . Prove there is $\delta > 0$ such that every ball $B(x, \delta)$ with $x \in K$ lies in some V_i (a Lebesgue number for the cover). Indicate where this is used in localizing the bumps of a partition of unity.

11.2 Primitive Mappings and the Change of Variables Theorem

The change of variables theorem says that integrating f after a substitution $y = T(x)$ introduces exactly the Jacobian factor $|\det T'|$. Rudin proves it by factoring a general C^1 map locally into especially simple pieces — *primitive* maps (which touch one coordinate at a time) and *flips* (which swap two coordinates) — proving the change-of-variables formula for each simple piece, then composing.

Primitive mappings and flips

Definition 11.2.1. A map G of an open set $E \subseteq \mathbb{R}^n$ into $\mathbb{R}^n$ is *primitive* if it alters at most one coordinate: there is an index m and a scalar function g such that

$$G(x) = x + [g(x) - x_m]e_m, \quad \text{i.e.} \quad G(x) = (x_1, \dots, x_{m-1}, g(x), x_{m+1}, \dots, x_n).$$

Only the m -th component is changed. A *flip* is a linear map B that interchanges two of the coordinates and leaves the rest fixed: $B(x)$ swaps x_k and x_m for some $k \neq m$.

For a primitive G the Jacobian matrix is the identity except in row m , so $\det G'(x) = (D_m g)(x)$ — the partial derivative of the one moving coordinate. A flip has determinant -1 . These are the building blocks.

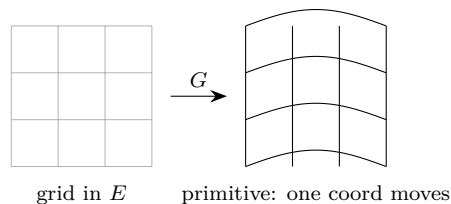


Figure 11.3. A primitive map bends the grid in a single coordinate direction; vertical grid lines stay vertical, only their heights change.

Theorem 11.2.2 (Local factorization of a C^1 map; Rudin 10.7). *Suppose $F : E \rightarrow \mathbb{R}^n$ is C^1 on an open set $E \subseteq \mathbb{R}^n$, $0 \in E$, $F(0) = 0$, and $A = F'(0)$ is invertible. Then in some neighborhood of 0,*

$$F = B_1 \circ \cdots \circ B_{n-1} \circ G_n \circ G_{n-1} \circ \cdots \circ G_1,$$

where each G_i is a primitive C^1 map fixing 0 and each B_j is a flip. In words: near a regular point, every C^1 map is a composition of primitive maps and flips.

Proof outline. Argue by induction. After composing with A^{-1} we may assume $F'(0) = I$. The idea is to correct the coordinates one at a time. Write $F = (f_1, \dots, f_n)$. Having arranged that the first $m-1$ coordinates of the (partially corrected) map already equal $x_1, \dots, x_{m-1}$ near 0, the map

$$G_m(x) = (x_1, \dots, x_{m-1}, f_m(x), x_{m+1}, \dots, x_n)$$

is primitive; its derivative at 0 is invertible because $D_m f_m(0) \neq 0$ (after a flip B_m that brings a nonzero entry of the relevant row onto the diagonal, which is where the flips enter). By the inverse function theorem G_m has a C^1 inverse near 0, and replacing F by $F \circ G_m^{-1}$ advances the induction: one more coordinate is now in standard form. After n steps the map is the identity, and unwinding the substitutions exhibits F as the stated composition. $\square$

Caution. The flips B_j cannot be dropped. Exercise 3(b): the simple swap $(x, y) \mapsto (y, x)$ of $\mathbb{R}^2$ is *not* a composition of two primitive maps in any neighborhood of 0. A swap moves two coordinates simultaneously in an unavoidable way; primitives move one at a time, so you genuinely need the flips to realize the permutation part of A .

Worked Example 11.2.3 (Factoring an explicit map). Write $F(x, y) = (e^x \cos y - 1, e^x \sin y)$ near $(0, 0)$ as $G_2 \circ G_1$ of two primitives.

Recognition. “Factor a C^1 map into primitives.” Correct one coordinate at a time, choosing the first primitive to fix the first output coordinate.

Solution. Put $G_1(x, y) = (e^x \cos y - 1, y)$, primitive (only the first coordinate moves). Its inverse near 0 solves $u = e^x \cos y - 1$, $v = y$ for (x, y) . Then set $G_2(u, v) = (u, (1+u) \tan v)$, also primitive (only the second coordinate moves). Check $G_2(G_1(x, y)) = (e^x \cos y - 1, (e^x \cos y) \tan y) = (e^x \cos y - 1, e^x \sin y) = F(x, y)$. The Jacobians at 0: $\det G'_1(0) = 1$, $\det G'_2(0) = 1$, and indeed $\det F'(0) = 1$, consistent with $\det(G'_2 G'_1) = \det G'_2 \det G'_1$. *Check.* $F'(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, the identity, matching the product of the two identity-determinant primitives.

The change of variables theorem

Theorem 11.2.4 (Change of variables; Rudin 10.9). *Suppose T is a 1–1 C^1 mapping of an open set $E \subseteq \mathbb{R}^k$ into $\mathbb{R}^k$ whose Jacobian $J_T(x) = \det T'(x)$ is nonzero on E . If $f \in \mathcal{C}_c(\mathbb{R}^k)$ with $\text{supp } f \subseteq T(E)$, then*

$$\int_{\mathbb{R}^k} f(y) dy = \int_{\mathbb{R}^k} f(T(x)) |J_T(x)| dx.$$

Proof outline (Rudin’s strategy). The proof is a model application of the two tools above.

1. **Localize.** Cover $\text{supp } f$ by small open sets on each of which Theorem 11.2.2 factors T (after translation) as a composition of primitives and flips. Using a partition of unity (§11.1, Theorem 11.1.6) write $f = \sum_i \psi_i f$; it suffices to prove the formula for each $\psi_i f$, i.e. for f supported in one such patch.

2. **One simple factor at a time.** For a *flip* the formula is the 1-variable change of order of integration with an honest sign: $|J| = 1$ and both sides integrate the same function. For a *primitive* G changing only coordinate m , fix the other $k - 1$ variables and apply the *one-dimensional* substitution rule $\int h(g(t))g'(t)dt = \int h(s)ds$ in the m -th variable; here $|J_G| = |D_m g|$ supplies exactly $|g'|$. So the formula holds for each primitive and each flip.
3. **Compose.** The Jacobian of a composition multiplies ($J_{S \circ U}(x) = J_S(U(x))J_U(x)$ by the chain rule and the product rule for determinants), and the substitution rule chains accordingly. Since T is locally a composition of maps for which the formula holds, it holds for T on each patch.
4. **Reassemble.** Sum over the partition of unity. Linearity gives the formula for the original f .

□

Remark. The hypothesis $J_T \neq 0$ and T injective make T a diffeomorphism onto $T(E)$; the factor $|J_T|$ (absolute value) records the *local volume distortion* of T , ignoring orientation. When we integrate *forms* rather than functions (§11.3 onward) the orientation is kept and the factor becomes J_T *without* absolute value — that single sign is the whole difference between measure theory and the theory of forms.

Worked Example 11.2.5 (Polar coordinates). Derive $\iint_D f(x, y) dx dy = \int_0^{2\pi} \int_0^a f(r \cos \theta, r \sin \theta) r dr d\theta$ for f supported in the disc D of radius a .

Recognition. “Switch to polar coordinates.” Apply Theorem 11.2.4 with the polar map and compute its Jacobian.

Solution. Let $T(r, \theta) = (r \cos \theta, r \sin \theta)$ on the open rectangle $0 < r < a$, $0 < \theta < 2\pi$. Then

$$T' = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad J_T = r(\cos^2 \theta + \sin^2 \theta) = r > 0.$$

T is injective on the open rectangle and $J_T = r \neq 0$ there, so Theorem 11.2.4 gives $\iint f dx dy = \iint f(T(r, \theta)) r dr d\theta$. The boundary segment removed (the slit from center to edge) has area zero and does not affect the integral of a continuous f . *Check.* Taking $f \equiv 1$ on D recovers $\int_0^{2\pi} \int_0^a r dr d\theta = 2\pi \cdot \frac{a^2}{2} = \pi a^2$, the area of the disc. ■

Worked Example 11.2.6 (An affine substitution and a parallelogram). Let H be the parallelogram with vertices $(1, 1), (3, 2), (4, 5), (2, 4)$, and let T be the affine map sending $(0, 0) \mapsto (1, 1)$, $(1, 0) \mapsto (3, 2)$, $(0, 1) \mapsto (2, 4)$. Find J_T and reduce $\iint_H g$ to an integral over the unit square.

Recognition. “Map a unit square onto a parallelogram.” An affine $T(u) = Au + b$ has constant Jacobian $\det A$.

Solution. The columns of A are $T(e_1) - T(0) = (2, 1)$ and $T(e_2) - T(0) = (1, 3)$, so $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1, 1)$, and $J_T = \det A = 6 - 1 = 5$. By Theorem 11.2.4, $\iint_H g(x, y) dx dy = \iint_{I^2} g(T(u, v)) |5| du dv$. *Check.* With $g \equiv 1$ this gives $\text{area}(H) = 5 \cdot 1 = 5$, which equals $|\det A|$, the area of the image of the unit square — correct.

Further worked examples

Worked Example 11.2.7 (A Gaussian integral via polar coordinates). Evaluate $\iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy$ and deduce $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

Recognition. “A radially symmetric integrand begs for polar coordinates” (Theorem 11.2.4 with $J_T = r$).

Solution. With $x = r \cos \theta$, $y = r \sin \theta$ and $dx dy = r dr d\theta$,

$$\iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = 2\pi \cdot \left[-\frac{1}{2} e^{-r^2} \right]_0^{\infty} = 2\pi \cdot \frac{1}{2} = \pi.$$

On the other hand the Cartesian integral factors as $(\int_{-\infty}^{\infty} e^{-x^2} dx)^2$. Equating, $(\int e^{-x^2} dx)^2 = \pi$, so $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$. *Check.* The answer is positive and the one-variable integral, being of a positive even function, is indeed $\sqrt{\pi} \approx 1.7725 > 0$. (Strictly, run the argument on growing discs/squares and pass to the limit; the compact-support theorem applies to each truncation.)

Worked Example 11.2.8 (Jacobian of a non-injective map: hypotheses matter). Let $T(x, y) = (x^2 - y^2, 2xy)$ (the complex squaring map $z \mapsto z^2$). Compute J_T and explain why Theorem 11.2.4 does not apply on all of $\mathbb{R}^2$.

Recognition. “Check the change-of-variables hypotheses before using the formula.” The theorem needs T *injective* with $J_T \neq 0$.

Solution. The Jacobian matrix is $T' = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}$, so $J_T = 4x^2 + 4y^2 = 4(x^2 + y^2)$. This vanishes only at the origin, but the real obstruction is *injectivity*: $T(x, y) = T(-x, -y)$ for every point, since $z \mapsto z^2$ is two-to-one. So on any open set containing a point and its negative, T is not 1-1 and the formula $\int f \circ T |J_T| = \int f$ overcounts (it would integrate the image twice). *Check.* Restricting to the right half-plane $x > 0$ (or any sector of opening $< \pi$) makes T injective with $J_T > 0$; there the theorem applies cleanly. The lesson: $J_T \neq 0$ gives *local* injectivity only, not global.

Exercises (Rudin, Ch.10).

- (Rudin 3.) (a) If F is as in Theorem 11.2.2, put $A = F'(0)$ and $F_1(x) = A^{-1}F(x)$, so $F'_1(0) = I$. Show that in some neighborhood of 0, $F(x) = F'(0)G_n \circ G_{n-1} \circ \cdots \circ G_1(x)$ for certain primitive mappings $G_1, \dots, G_n$ (another version of Theorem 11.2.2). (b) Prove that the mapping $(x, y) \mapsto (y, x)$ of $\mathbb{R}^2$ onto $\mathbb{R}^2$ is *not* the composition of any two primitive mappings in any neighborhood of the origin. (This shows the flips B_i cannot be omitted from the statement of Theorem 11.2.2.)
- (Rudin 4.) For $(x, y) \in \mathbb{R}^2$ define $F(x, y) = (e^x \cos y - 1, e^x \sin y)$. Prove that $F = G_2 \circ G_1$, where $G_1(x, y) = (e^x \cos y - 1, y)$ and $G_2(u, v) = (u, (1 + u) \tan v)$ are primitive in some neighborhood of $(0, 0)$. Compute the Jacobians of G_1, G_2, F at $(0, 0)$. Also define $H_2(x, y) = (x, e^x \sin y)$ and find $H_1(u, v) = (h(u, v), v)$ so that $F = H_1 \circ H_2$ in some neighborhood of $(0, 0)$.
- (Rudin 8.) Let H be the parallelogram in $\mathbb{R}^2$ whose vertices are $(1, 1), (3, 2), (4, 5), (2, 4)$. Find the affine map T sending $(0, 0) \mapsto (1, 1), (1, 0) \mapsto (3, 2), (0, 1) \mapsto (2, 4)$. Show $J_T = 5$. Use T to convert $\iint_H e^{x-y} dx dy$ to an integral over the unit square I^2 , and compute it.
- (Rudin 9.) Define $(x, y) = T(r, \theta)$ on the rectangle $0 \leq r \leq a, 0 \leq \theta \leq 2\pi$ by $x = r \cos \theta, y = r \sin \theta$. Show that T maps this rectangle onto the closed disc D with center $(0, 0)$ and radius a , that T is one-to-one in the interior of the rectangle, and that $J_T(r, \theta) = r$. If $f \in \mathcal{C}(D)$, prove the polar-coordinate formula

$$\iint_D f(x, y) dx dy = \int_0^{2\pi} \int_0^a f(T(r, \theta)) r dr d\theta.$$

(Hint: let D_0 be the interior of D minus the segment from $(0, 0)$ to $(0, a)$; first apply Theorem 11.2.4 on D_0 , then remove the restriction as in the partition-of-unity argument.)

- (Rudin 10.) Let $a \rightarrow \infty$ in Exercise (Rudin 9) and prove

$$\iint_{\mathbb{R}^2} f(x, y) dx dy = \int_0^{2\pi} \int_0^\infty f(T(r, \theta)) r dr d\theta$$

for continuous f that decrease sufficiently rapidly as $|x| + |y| \rightarrow \infty$ (find a precise formulation). Apply this to $f(x, y) = \exp(-x^2 - y^2)$ to derive the Gaussian-integral formula $\int_{-\infty}^\infty e^{-t^2} dt = \sqrt{\pi}$ (formula (101) of Rudin Ch.8).

- (Rudin 11.) Define $(u, v) = T(s, t)$ on the strip $0 < s < \infty, 0 < t < 1$ by $u = s - st, v = st$. Show that T is a one-to-one mapping of the strip onto the positive quadrant Q in $\mathbb{R}^2$, and that $J_T(s, t) = s$. For $x > 0, y > 0$, integrate $u^{x-1}v^{y-1}e^{-u-v}$ over Q , convert it to an integral over the strip via Theorem 11.2.4, and derive the Beta-Gamma relation $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$ (formula (96) of Rudin Ch.8). (Theorem 11.2.4 must be extended to cover the relevant improper integrals; provide this extension.)

Additional exercises.

- (Jacobian warm-up.) For the polar map $T(r, \theta) = (r \cos \theta, r \sin \theta)$ compute the Jacobian determinant J_T , and use the change-of-variables theorem to evaluate $\iint_{x^2+y^2 \leq 1} (x^2 + y^2) dx dy$.

- For the linear map $T(x) = Ax$ with A invertible, verify directly from the change-of-variables theorem that $\int_{T(E)} f = |\det A|^{-1} \int_E f \circ T$ for $f \in \mathcal{C}(\mathbb{R}^k)$ of compact support, recovering the volume-scaling factor $|\det A|$.
- (Primitive factorization, $k = 2$.) Write the map $T(x, y) = (x + y^2, y)$ as it stands and confirm it is primitive (it changes only the first coordinate). Compute its inverse on its image and check $J_T \equiv 1$, so T preserves area.
- Show that a composition of two primitive mappings need not be primitive: give explicit primitive G, H on a neighborhood of 0 in $\mathbb{R}^2$ such that $H \circ G$ alters both coordinates. Why does the change-of-variables proof still go through?
- (Non-example.) Let $T(x, y) = (x^2, y)$ on $\mathbb{R}^2$. Show $J_T = 2x$ vanishes on the line $x = 0$, so T is not a primitive diffeomorphism near any point of that line. Exhibit two distinct points with the same image to confirm T is not injective there.
- Let T be a C' map with $J_T(a) \neq 0$. Using the inverse function theorem, prove T maps a sufficiently small open neighborhood of a diffeomorphically onto an open set, and identify how this local statement feeds the global change-of-variables theorem through a partition of unity.
- (Computation.) Evaluate $\iiint_B e^{-(x^2+y^2+z^2)^{3/2}} dV$ over the unit ball $B \subseteq \mathbb{R}^3$ using spherical coordinates, stating the Jacobian $r^2 \sin \varphi$ you use.
- (Conceptual.) Explain why the change-of-variables theorem is stated for C' diffeomorphisms rather than arbitrary continuous bijections, by sketching a continuous area-distorting bijection of $[0, 1]^2$ onto itself whose “Jacobian” would have to be undefined. What regularity is the minimum that makes $|J_T|$ meaningful pointwise?

11.3 Differential Forms

A differential form is the object you are allowed to integrate over an oriented piece of $\mathbb{R}^n$ of a given dimension. A k -form eats a k -dimensional parametrized surface and returns a number; the bookkeeping that makes this consistent under reparametrization is the *wedge product*, whose defining feature is antisymmetry.

The integral of a form over a surface

Definition 11.3.1. Let $E \subseteq \mathbb{R}^k$ and let $D \subseteq \mathbb{R}^k$ be a parameter domain. A k -surface in E is a C' (continuously differentiable) map $\Phi : D \rightarrow E$, where D is a compact set, typically a box or a simplex. We call D the *parameter domain* and $\Phi(D)$ the range.

Definition 11.3.2. A *differential form of order k* (a k -form) in an open set $E \subseteq \mathbb{R}^n$ is a function ω that assigns to each k -surface Φ in E a number $\int_\Phi \omega$, written as a finite sum

$$\omega = \sum_{i_1, \dots, i_k} a_{i_1 \dots i_k}(x) dx_{i_1} \wedge \dots \wedge dx_{i_k},$$

with continuous coefficient functions $a_{i_1 \dots i_k}$ on E and integers $1 \leq i_j \leq n$. The integral over a k -surface $\Phi : D \rightarrow E$, $\Phi = (\phi_1, \dots, \phi_n)$, is defined by pulling back to the parameter domain:

$$\int_\Phi \omega = \int_D \sum a_{i_1 \dots i_k}(\Phi(u)) \frac{\partial(\phi_{i_1}, \dots, \phi_{i_k})}{\partial(u_1, \dots, u_k)} du,$$

where the last factor is the Jacobian determinant of the indicated k components with respect to the k parameters.

The notation dx_i is, for a 1-surface (a curve) γ , exactly “ $d\phi_i$ along γ ”. A 0-form is just a function; a k -form on $\mathbb{R}^k$ is $a(x) dx_1 \wedge \dots \wedge dx_k$ and integrating it recovers the ordinary integral $\int_D a(\Phi(u)) J_\Phi du$.

Worked Example 11.3.3 (Integrating a 1-form over a curve). Evaluate $\int_{\gamma} \omega$ for $\omega = x dy - y dx$ along the unit circle $\gamma(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$.

Recognition. “Integrate a 1-form over a parametrized curve.” Substitute $x = \phi_1$, $y = \phi_2$ and replace dx, dy by $\phi'_1(t) dt, \phi'_2(t) dt$.

Solution. Here $x = \cos t$, $y = \sin t$, $dx = -\sin t dt$, $dy = \cos t dt$, so

$$\int_{\gamma} (x dy - y dx) = \int_0^{2\pi} (\cos t \cos t - \sin t(-\sin t)) dt = \int_0^{2\pi} (\cos^2 t + \sin^2 t) dt = \int_0^{2\pi} 1 dt = 2\pi.$$

Check. This 1-form integrated around a loop computes twice the enclosed area; the unit disc has area π , and $2\pi = 2 \cdot \pi$. ■

The wedge product and antisymmetry

Definition 11.3.4. The *basic k -forms* are $dx_I = dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ for a multi-index $I = (i_1, \dots, i_k)$. The *wedge (exterior) product* of basic forms is defined by juxtaposition, $dx_I \wedge dx_J = dx_{(I,J)}$, and extended to be bilinear over functions:

$$\left(\sum_I a_I dx_I \right) \wedge \left(\sum_J b_J dx_J \right) = \sum_{I,J} a_I b_J dx_I \wedge dx_J.$$

The product of a k -form and an m -form is a $(k+m)$ -form.

The single rule that governs all manipulations is *anticommutativity of the differentials*, which is forced by the antisymmetry of determinants under row swaps.

Theorem 11.3.5 (Anticommutativity; Rudin 10.12–10.13). *For all indices i, j ,*

$$dx_i \wedge dx_j = -dx_j \wedge dx_i, \quad \text{in particular} \quad dx_i \wedge dx_i = 0.$$

More generally, swapping two adjacent factors of a basic form changes its sign; a basic form with a repeated index is 0. For a k -form ω and an m -form λ ,

$$\omega \wedge \lambda = (-1)^{km} \lambda \wedge \omega.$$

Proof. For the generators: in Definition 11.3.2 the coefficient of a basic form is a Jacobian determinant of the corresponding components. Interchanging dx_i and dx_j interchanges two rows of that determinant, which flips its sign — hence $dx_i \wedge dx_j = -dx_j \wedge dx_i$. Setting $j = i$ gives $dx_i \wedge dx_i = -dx_i \wedge dx_i$, so $dx_i \wedge dx_i = 0$ (a determinant with two equal rows vanishes). The graded sign $(-1)^{km}$ counts the km adjacent transpositions needed to move every one of the m factors of λ past every one of the k factors of ω , each transposition contributing a factor -1 . □

Caution. Forms anticommute, scalars do not. $dx \wedge dy = -dy \wedge dx$, but the coefficient functions commute as usual: $a dx \wedge b dy = ab dx \wedge dy$. Two consequences trip people up: (i) any basic form with a repeated differential is *zero*, e.g. $dx \wedge dx = 0$ and $dx \wedge dy \wedge dx = 0$; (ii) on $\mathbb{R}^n$ there are no nonzero k -forms for $k > n$ (every basic k -form must repeat an index), so $dx_1 \wedge \cdots \wedge dx_n$ is, up to a function, the *only* top-degree form.

Definition 11.3.6. Every k -form can be written in *standard presentation* as a sum over strictly increasing multi-indices $I = (i_1 < \cdots < i_k)$,

$$\omega = \sum_{i_1 < \cdots < i_k} a_I(x) dx_I,$$

by using anticommutativity to sort each basic form into increasing order (introducing a sign) and discarding any basic form with a repeated index. Two forms are equal iff their standard presentations have identical coefficients.

Method (Multiplying and simplifying forms). *Recognition trigger.* You must compute a wedge product $\omega \wedge \lambda$ or simplify a form to standard presentation.

1. Expand by bilinearity: multiply every basic form of ω by every basic form of λ , carrying the coefficient functions out front (they commute).
2. In each resulting basic form, *sort the differentials into increasing index order* by adjacent swaps, multiplying the coefficient by -1 for each swap.
3. **Kill repeats:** any basic form containing the same dx_i twice is 0 — drop it.
4. Collect like basic forms; the surviving coefficients are the standard presentation.

Sanity check. A k -form on $\mathbb{R}^n$ has at most $\binom{n}{k}$ independent components; on $\mathbb{R}^3$ that is 1, 3, 3, 1 for $k = 0, 1, 2, 3$.

Worked Example 11.3.7 (A wedge product in $\mathbb{R}^3$). With $\omega = x dx + y dy$ and $\lambda = z dx \wedge dz + dy \wedge dz$ in $\mathbb{R}^3$, compute $\omega \wedge \lambda$.

Recognition. “Multiply a 1-form by a 2-form.” Expand, sort, kill repeats.

Solution. Expand:

$$\omega \wedge \lambda = xz(dx \wedge dx \wedge dz) + x(dy \wedge dx \wedge dz) + yz(dy \wedge dx \wedge dz) + y(dy \wedge dy \wedge dz).$$

The first and last vanish (repeated dx , repeated dy). In the third term $dy \wedge dx = -dx \wedge dy$, so $yz dy \wedge dx \wedge dz = -yz dx \wedge dy \wedge dz$. Hence

$$\omega \wedge \lambda = (x - yz) dx \wedge dy \wedge dz.$$

Check. The answer is a top-degree (3-form) on $\mathbb{R}^3$, the only possibility, with a single coefficient $x - yz$, as a 1-form wedge a 2-form should give. ■

Worked Example 11.3.8 (Why a square of a 1-form can vanish or not). Let $\alpha = dx_1 \wedge dx_2 + dx_3 \wedge dx_4$ on $\mathbb{R}^4$. Compute $\alpha \wedge \alpha$.

Recognition. “Square a 2-form.” Even-degree forms commute with themselves, so the square need not vanish; expand carefully.

Solution.

$$\alpha \wedge \alpha = dx_1 \wedge dx_2 \wedge dx_1 \wedge dx_2 + dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4 + dx_3 \wedge dx_4 \wedge dx_1 \wedge dx_2 + dx_3 \wedge dx_4 \wedge dx_3 \wedge dx_4.$$

The first and last repeat indices, hence vanish. The middle two are each $dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4$ (moving the block $dx_3 \wedge dx_4$ past $dx_1 \wedge dx_2$ costs $(-1)^{2 \cdot 2} = +1$). So $\alpha \wedge \alpha = 2 dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4 \neq 0$. *Check.* Contrast a 1-form β : $\beta \wedge \beta = (-1)^{1 \cdot 1} \beta \wedge \beta = -\beta \wedge \beta$ forces $\beta \wedge \beta = 0$. Odd squares die; even squares need not. ■

Remark. The displayed top-degree fact deserves emphasis. A k -form on $\mathbb{R}^k$ has standard presentation $\omega = a(x) dx_1 \wedge \cdots \wedge dx_k$, and integrating it over the identity k -surface gives $\int a$. Combined with Definition 11.3.2, a change of parameters Φ multiplies the integrand by the Jacobian J_Φ *with sign* — this is the “oriented” version of the change of variables theorem from §11.2, and it is why forms, not functions, are the natural integrands on oriented surfaces.

Further worked examples

Worked Example 11.3.9 (A 2-form integrated over a parametrized surface). Evaluate $\int_\Phi \omega$ for $\omega = dx \wedge dy$ over the surface $\Phi(u, v) = (u + v, u - v, uv)$ on the unit square $D = [0, 1]^2$.

Recognition. “Integrate a basic 2-form over a 2-surface.” Definition 11.3.2: replace the coefficient and the differentials by the Jacobian of the named components.

Solution. Here $\phi_1 = u + v$, $\phi_2 = u - v$, and $dx \wedge dy$ pulls back to the Jacobian of (ϕ_1, ϕ_2) with respect to (u, v) :

$$\frac{\partial(\phi_1, \phi_2)}{\partial(u, v)} = \det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -2.$$

Hence $\int_\Phi dx \wedge dy = \int_0^1 \int_0^1 (-2) du dv = -2$. *Check.* The third component $\phi_3 = uv$ never enters, exactly because ω only involves dx, dy ; the sign is negative because the parametrization is orientation-reversing for the (x, y) -projection ($\det < 0$).

Worked Example 11.3.10 (A wedge of two 1-forms that is not what it looks like). On $\mathbb{R}^3$ let $\alpha = dx + dy$ and $\beta = dy + dz$. Compute $\alpha \wedge \beta$ and check it is *not* $d(x+y) \wedge d(y+z)$ done naively as $(x+y)(y+z)$ times anything.

Recognition. “Wedge two 1-forms; resist treating $\wedge$ like ordinary multiplication.” Expand bilinearly, kill repeats, sort.

Solution.

$$\alpha \wedge \beta = (dx + dy) \wedge (dy + dz) = dx \wedge dy + dx \wedge dz + \underbrace{dy \wedge dy}_{=0} + dy \wedge dz = dx \wedge dy + dx \wedge dz + dy \wedge dz.$$

This is a genuine 2-form with three components; it is *not* a single scalar multiple of a basic form, and there is no “ $(x+y)(y+z)$ ” coefficient — the 1-forms α, β carry no such functions. *Check.* Antisymmetry forces $\alpha \wedge \alpha = 0$; indeed $(dx + dy) \wedge (dx + dy) = dx \wedge dy + dy \wedge dx = 0$, confirming odd-degree self-wedges vanish even when the form has several terms.

Exercises (Rudin, Ch.10).

- (Rudin 14.) Prove the basic multiplication formula for forms (Rudin’s formula (46)): if $\omega = \sum_I a_I dx_I$ and $\lambda = \sum_J b_J dx_J$ are forms in standard presentation, then

$$\omega \wedge \lambda = \sum_{I,J} a_I b_J dx_I \wedge dx_J,$$

and the right side, after sorting into standard presentation, is well defined (independent of the orderings used). Deduce that the wedge product is associative.

- (Rudin 15.) If ω and λ are k - and m -forms respectively, prove the graded-commutativity law

$$\omega \wedge \lambda = (-1)^{km} \lambda \wedge \omega.$$

(Reduce to basic forms and count the adjacent transpositions needed to interchange the two blocks of differentials, as in Theorem 11.3.5.)

Additional exercises.

- (Wedge computation.) In $\mathbb{R}^3$ with coordinates x, y, z , let $\omega = x dx + y dy$ and $\lambda = z dx \wedge dy + x dy \wedge dz$. Compute $\omega \wedge \lambda$ and express it in standard presentation $c dx \wedge dy \wedge dz$.
- Show by direct expansion that $dx \wedge dx = 0$ and $dx \wedge dy = -dy \wedge dx$ are forced by the alternating (antisymmetric) defining property, and deduce that any basic k -form $dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ with a repeated index vanishes.
- (Counting basic forms.) How many linearly independent basic k -forms are there on $\mathbb{R}^n$? Verify your formula gives $1, n, \binom{n}{2}, \dots$ and in particular that there is a one-dimensional space of n -forms (the volume forms) on $\mathbb{R}^n$.
- Let ω be a 1-form and λ a 2-form on $\mathbb{R}^4$. Using graded commutativity $\omega \wedge \lambda = (-1)^{km} \lambda \wedge \omega$, determine the sign and confirm it on the specific pair $\omega = dx_1, \lambda = dx_2 \wedge dx_3$.
- (Non-example.) Show that the 2-form $\alpha = dx \wedge dy + dz \wedge dw$ on $\mathbb{R}^4$ is *not* decomposable, i.e. cannot be written as $\beta \wedge \gamma$ for two 1-forms β, γ . (Hint: compute $\alpha \wedge \alpha$ and show it is nonzero, whereas any $\beta \wedge \gamma$ wedged with itself vanishes.)
- Given 1-forms $\omega_1, \dots, \omega_k$ on $\mathbb{R}^n$, prove they are linearly independent iff $\omega_1 \wedge \cdots \wedge \omega_k \neq 0$.
- (Conceptual.) Explain why a k -form is the natural object to integrate over a k -dimensional surface, by describing how the basic form $dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ measures the signed k -volume of the projection of an infinitesimal parallelepiped onto the coordinate k -plane $(x_{i_1}, \dots, x_{i_k})$.

11.4 The Exterior Derivative and Pullbacks

Two operations make forms into a calculus: the *exterior derivative* d , which raises degree by one and satisfies $d^2 = 0$, and the *pullback* ω_T , which transports a form backward along a smooth map T . Their compatibility, $d(\omega_T) = (d\omega)_T$, is the algebraic heart of Stokes' theorem.

The exterior derivative

Definition 11.4.1. Let $\omega = \sum_I a_I dx_I$ be a k -form of class C' in standard presentation on an open $E \subseteq \mathbb{R}^n$. Its *exterior derivative* is the $(k+1)$ -form

$$d\omega = \sum_I da_I \wedge dx_I, \quad \text{where} \quad da_I = \sum_{j=1}^n (D_j a_I) dx_j$$

is the ordinary differential of the coefficient (a 1-form). For a 0-form (function) f , $df = \sum_j (D_j f) dx_j$ as expected.

Concretely: differentiate each coefficient as a function, wedge the resulting 1-form on the left, and simplify by the sorting rules of § 11.3.

Theorem 11.4.2 (Properties of d ; Rudin 10.20). *The exterior derivative is linear and satisfies:*

1. (Leibniz / product rule) *For a k -form ω and an m -form λ ,*

$$d(\omega \wedge \lambda) = (d\omega) \wedge \lambda + (-1)^k \omega \wedge (d\lambda).$$

2. (Nilpotence) *If ω is of class C'' , then $d(d\omega) = 0$; briefly $d^2 = 0$.*

Proof. (1) By linearity reduce to $\omega = a dx_I$, $\lambda = b dx_J$. Then $\omega \wedge \lambda = ab dx_I \wedge dx_J$ and $d(ab) = a db + b da$, so $d(\omega \wedge \lambda) = (a db + b da) \wedge dx_I \wedge dx_J$. Now $da \wedge dx_I \wedge dx_J = (d\omega) \wedge \lambda$, while moving the 1-form db rightward past the k factors dx_I costs $(-1)^k$, giving $(-1)^k a dx_I \wedge db \wedge dx_J = (-1)^k \omega \wedge d\lambda$. Add.

(2) First for a function: $d(df) = d(\sum_j D_j f dx_j) = \sum_{i,j} D_i D_j f dx_i \wedge dx_j$. Split the sum into pairs (i, j) and (j, i) ; the mixed partials are equal ($f \in C''$) but $dx_i \wedge dx_j = -dx_j \wedge dx_i$, so the pairs cancel and $d(df) = 0$. For a general $\omega = \sum a_I dx_I$, $d\omega = \sum da_I \wedge dx_I$, and applying d with the Leibniz rule (and $d(dx_I) = 0$, since the dx_i have constant coefficient 1) leaves $d(d\omega) = \sum d(da_I) \wedge dx_I = 0$ by the function case. $\square$

Caution. $d^2 = 0$ requires *equality of mixed partials*, hence C'' smoothness. It is the form of the statement “the curl of a gradient is zero” and “the divergence of a curl is zero” (§ 11.6). The converse — that a form with $d\omega = 0$ (*closed*) is d of something (*exact*) — is *false* in general: the angle form $\eta = \frac{x dy - y dx}{x^2 + y^2}$ on $\mathbb{R}^2 \setminus \{0\}$ has $d\eta = 0$ but is not exact (Exercises 21–23). Closed-but-not-exact forms detect holes in the domain.

Worked Example 11.4.3 (Computing d of a 1-form in $\mathbb{R}^3$). Let $\omega = P dx + Q dy + R dz$ with $P, Q, R \in C'(\mathbb{R}^3)$. Compute $d\omega$.

Recognition. “Differentiate a 1-form.” Apply d to each coefficient and wedge.

Solution.

$$d\omega = dP \wedge dx + dQ \wedge dy + dR \wedge dz.$$

Expand $dP = P_x dx + P_y dy + P_z dz$, etc., drop terms with a repeated differential, and sort:

$$d\omega = (Q_x - P_y) dx \wedge dy + (R_y - Q_z) dy \wedge dz + (P_z - R_x) dz \wedge dx.$$

Check. The three coefficients are exactly the components of $\text{curl}(P, Q, R)$; d on 1-forms in $\mathbb{R}^3$ is the curl. If $(P, Q, R) = \nabla f = (f_x, f_y, f_z)$ then each coefficient is a difference of equal mixed partials, so $d\omega = 0$ — recovering $\text{curl } \nabla f = 0$. ■

The pullback

Definition 11.4.4. Let $T : V \rightarrow E$ be a C' map between open sets, $V \subseteq \mathbb{R}^m$, $E \subseteq \mathbb{R}^n$, written $y = T(x)$ with components $y_i = t_i(x)$. The *pullback* (Rudin: “ ω_T ”) of a k -form $\omega = \sum a_I dy_I$ on E is the k -form on V obtained by substituting:

$$\omega_T = \sum_I (a_I \circ T) dt_{i_1} \wedge \cdots \wedge dt_{i_k}, \quad dt_i = \sum_{j=1}^m (D_j t_i) dx_j.$$

That is, replace each coefficient a_I by $a_I \circ T$ and each dy_i by the 1-form dt_i , then multiply out and simplify.

The pullback is engineered so that integration is invariant: $\int_{T \circ \Phi} \omega = \int_{\Phi} \omega_T$ for any surface Φ in V . Pulling a form back, then integrating, is the same as integrating after pushing the surface forward.

Theorem 11.4.5 (Functoriality and naturality; Rudin 10.21–10.22). *Let S, T be C' maps with $E \rightarrow$ ranges as needed. Then:*

1. $(\omega \wedge \lambda)_T = \omega_T \wedge \lambda_T$;
2. $(\omega_S)_T = \omega_{S \circ T}$ (pullback reverses composition);
3. (Naturality) $d(\omega_T) = (d\omega)_T$ — the exterior derivative commutes with pullback.

Proof of (3). By linearity take $\omega = a dy_I$. For a 0-form (I empty), $\omega_T = a \circ T$ and the chain rule gives $d(a \circ T) = \sum_j D_j(a \circ T) dx_j = \sum_{i,j} (D_i a \circ T)(D_j t_i) dx_j = \sum_i (D_i a \circ T) dt_i = (da)_T$. For general I , $\omega_T = (a \circ T) dt_I$ with $dt_I = dt_{i_1} \wedge \cdots \wedge dt_{i_k}$. Since each dt_i is itself $d(t_i)$ and $d^2 = 0$, we have $d(dt_I) = 0$ (Leibniz, every factor closed). Hence by the product rule

$$d(\omega_T) = d(a \circ T) \wedge dt_I = (da)_T \wedge (dy_I)_T = (da \wedge dy_I)_T = (d\omega)_T,$$

using part (1) for the middle step. This is the identity. $\square$

The naturality square is the cleanest way to remember (3): pulling back and then differentiating equals differentiating and then pulling back.

$$\begin{array}{ccc} \{k\text{-forms on } E\} & \xrightarrow{d} & \{(k+1)\text{-forms on } E\} \\ (\cdot)_T \downarrow & & \downarrow (\cdot)_T \\ \{k\text{-forms on } V\} & \xrightarrow{d} & \{(k+1)\text{-forms on } V\} \end{array}$$

Figure 11.4. Naturality of d : the square commutes, $d \circ (\cdot)_T = (\cdot)_T \circ d$, i.e. $d(\omega_T) = (d\omega)_T$.

Worked Example 11.4.6 (Pullback under a linear map). Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(u, v) = (au + bv, cu + dv)$, and $\omega = dy_1 \wedge dy_2$. Compute ω_T .

Recognition. “Pull back the area form under a linear map.” Substitute $dy_i = dt_i$ and expand.

Solution. Here $t_1 = au + bv$, $t_2 = cu + dv$, so $dt_1 = a du + b dv$ and $dt_2 = c du + d dv$. Then

$$\omega_T = dt_1 \wedge dt_2 = (a du + b dv) \wedge (c du + d dv) = (ad - bc) du \wedge dv,$$

the cross terms $du \wedge du$, $dv \wedge dv$ dying. *Check.* The factor $ad - bc = \det T$ is exactly the Jacobian; pulling back the area form multiplies by the (signed) Jacobian, which is the oriented change-of-variables rule of §11.3. $\blacksquare$

Worked Example 11.4.7 (Verifying naturality on an example). With $T(u, v) = (u^2, v)$ and $\omega = y_1 dy_2$ on $\mathbb{R}^2$, check $d(\omega_T) = (d\omega)_T$.

Recognition. “Confirm d commutes with pullback.” Compute both routes around the square.

Solution. *Route 1 (pull back, then d):* $\omega_T = (t_1) dt_2 = u^2 dv$, so $d(\omega_T) = d(u^2) \wedge dv = 2u du \wedge dv$. *Route 2 (d , then pull back):* $d\omega = dy_1 \wedge dy_2$, and $(d\omega)_T = dt_1 \wedge dt_2 = d(u^2) \wedge dv = 2u du \wedge dv$. The two agree. *Check.* Both equal $2u du \wedge dv$, confirming the commuting square. $\blacksquare$

Further worked examples

Worked Example 11.4.8 (Pulling back the polar area form). Let $T(r, \theta) = (r \cos \theta, r \sin \theta)$ and $\omega = dx \wedge dy$. Compute the pullback ω_T and read off the polar Jacobian.

Recognition. “Pull back the area form under polar coordinates.” Definition 11.4.4: substitute $dx = dt_1$, $dy = dt_2$, expand.

Solution. With $t_1 = r \cos \theta$, $t_2 = r \sin \theta$,

$$dt_1 = \cos \theta \, dr - r \sin \theta \, d\theta, \quad dt_2 = \sin \theta \, dr + r \cos \theta \, d\theta.$$

Then

$$\omega_T = dt_1 \wedge dt_2 = (\cos \theta \, r \cos \theta - (-r \sin \theta) \sin \theta) \, dr \wedge d\theta = r(\cos^2 \theta + \sin^2 \theta) \, dr \wedge d\theta = r \, dr \wedge d\theta.$$

Check. The coefficient r is exactly the polar Jacobian of § 11.2, so pulling back $dx \wedge dy$ reproduces the “ $r \, dr \, d\theta$ ” rule — now with orientation built in.

Worked Example 11.4.9 (A closed form is not always exact: d alone cannot tell). On $\mathbb{R}^2 \setminus \{0\}$ let $\eta = \frac{x \, dy - y \, dx}{x^2 + y^2}$. Show $d\eta = 0$ (it is *closed*); explain why this does *not* prove $\eta = df$ for some f .

Recognition. “Compute d of a 1-form; beware concluding exactness.” $d\eta = 0$ is necessary but not sufficient for η to be df (Pitfall after Theorem 11.4.2).

Solution. Write $P = -y/(x^2 + y^2)$, $Q = x/(x^2 + y^2)$. By Worked Example on d of a 1-form, $d\eta = (Q_x - P_y) \, dx \wedge dy$. Differentiating, $Q_x = \frac{(x^2 + y^2) - x \cdot 2x}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$ and $P_y = \frac{-(x^2 + y^2) + y \cdot 2y}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$, so $Q_x - P_y = 0$ and $d\eta = 0$. But η is *not* exact: its loop integral around the origin is $2\pi \neq 0$ (computed in § 11.6), and an exact form integrates to 0 around every loop. *Check.* The obstruction is the puncture — on any simply connected piece of the domain (e.g. a half-plane) $\eta = d\theta$ with θ a single-valued angle, so locally it is exact; globally the hole blocks it. The test “ $d\eta = 0$ ” sees only the local picture.

Exercises (Rudin, Ch.10).

1. (Rudin 20.) State conditions under which the formula

$$\int_{\Phi} f \, d\omega = \int_{\partial\Phi} f \, \omega - \int_{\Phi} (df) \wedge \omega$$

is valid, and show that it generalizes the formula for integration by parts. (Hint: $d(f\omega) = (df) \wedge \omega + f \, d\omega$; integrate over Φ and apply Stokes’ theorem.)

Additional exercises.

1. (Exterior derivative by hand.) For $\omega = xy \, dx + x^2 \, dy$ on $\mathbb{R}^2$ compute $d\omega$ and decide whether ω is closed. Repeat for $\eta = 2xy \, dx + x^2 \, dy$ and find a function f with $df = \eta$.
2. Let $\omega = f \, dx + g \, dy + h \, dz$ on $\mathbb{R}^3$. Write $d\omega$ in terms of the partial derivatives of f, g, h , and identify the coefficients with the components of the curl of the vector field (f, g, h) .
3. Prove $d(d\omega) = 0$ for every C'' form ω , working out the case of a general 1-form on $\mathbb{R}^3$ explicitly and pointing to where equality of mixed partials is used.
4. (Pullback computation.) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(u, v) = (u^2 - v^2, 2uv)$, and $\omega = dx \wedge dy$. Compute the pullback $T^*\omega$ and check it equals $J_T \, du \wedge dv$ with J_T the Jacobian determinant of T .
5. Verify the naturality identities on a concrete example: for the T above and a 1-form $\alpha = x \, dy$, check $T^*(d\alpha) = d(T^*\alpha)$ and $T^*(\alpha \wedge \beta) = T^*\alpha \wedge T^*\beta$ with $\beta = dx$.
6. (Non-example.) Exhibit a 1-form ω on $\mathbb{R}^2 \setminus \{0\}$ that is closed ($d\omega = 0$) but not exact (no f with $df = \omega$ on the whole punctured plane). Take $\omega = (x \, dy - y \, dx)/(x^2 + y^2)$ and explain why a global potential cannot exist by evaluating $\int_{\gamma} \omega$ around the unit circle.

7. Show that pullback commutes with the wedge product and with d for compositions: $(S \circ T)^* = T^* \circ S^*$. Verify this on linear maps S, T acting on a top-degree form, recovering the multiplicativity $\det(ST) = \det S \det T$ of Jacobians.
8. (Conceptual.) Explain in one paragraph why “closed” is necessary but not sufficient for “exact,” and how the failure is measured by the topology of the domain. Contrast the punctured plane with a convex open set, citing the Poincaré lemma.

11.5 Simplexes and Chains

To integrate forms over curved regions and to state Stokes’ theorem, we need oriented domains of integration: *simplexes* (the simplest oriented pieces) and their formal sums, *chains*. The boundary operator ∂ turns a k -chain into a $(k-1)$ -chain, and the key algebraic fact $\partial^2 = 0$ mirrors $d^2 = 0$ on the geometric side.

Affine and oriented simplexes

Definition 11.5.1. An *affine k -simplex* $\sigma = [p_0, p_1, \dots, p_k]$ with vertices $p_0, \dots, p_k \in \mathbb{R}^n$ is the affine map of the standard simplex

$$Q^k = \{(u_1, \dots, u_k) : u_i \geq 0, \sum u_i \leq 1\} \subseteq \mathbb{R}^k$$

into $\mathbb{R}^n$ given by

$$\sigma(u) = p_0 + \sum_{i=1}^k u_i(p_i - p_0) = \left(1 - \sum u_i\right)p_0 + \sum_{i=1}^k u_i p_i.$$

An *oriented* affine k -simplex carries an ordering of its vertices; an *even* permutation of the vertices gives the same oriented simplex, an *odd* permutation gives its negative, written $-\sigma$. A *differentiable simplex* (or k -surface of class C''') is a C''' map $\Phi : Q^k \rightarrow E$; we treat it just like an affine one for integration.

For an affine k -simplex in $\mathbb{R}^k$, the sign of $\det[p_1 - p_0, \dots, p_k - p_0]$ is the *orientation*; positively oriented means this determinant is positive. The unit cube and the standard simplex can each be cut into oriented simplexes (Exercises 12, 17, 18).

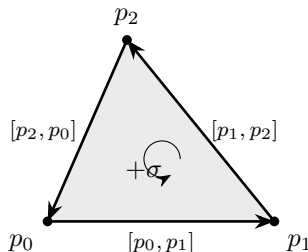


Figure 11.5. An oriented 2-simplex $\sigma = [p_0, p_1, p_2]$. Its boundary $\partial\sigma = [p_1, p_2] - [p_0, p_2] + [p_0, p_1]$ traverses the three edges so that the interior stays on the left.

Chains and the boundary operator

Definition 11.5.2. A k -chain in E is a finite formal sum $\Gamma = \sum_j \sigma_j$ of oriented k -simplexes in E (with integer multiplicities; $-\sigma$ means σ with reversed orientation). The integral of a k -form over a chain is defined additively: $\int_\Gamma \omega = \sum_j \int_{\sigma_j} \omega$.

Definition 11.5.3. The *boundary* of an oriented affine k -simplex $\sigma = [p_0, \dots, p_k]$ is the $(k-1)$ -chain

$$\partial\sigma = \sum_{i=0}^k (-1)^i [p_0, \dots, p_{i-1}, \widehat{p_i}, p_{i+1}, \dots, p_k],$$

where $\widehat{p_i}$ means “omit p_i ”. The boundary of a chain is taken additively, $\partial(\sum_j \sigma_j) = \sum_j \partial\sigma_j$. For a differentiable simplex Φ one sets $\partial\Phi = \sum_i (-1)^i \Phi \circ F_i$, where F_i is the i -th face map of Q^k .

The alternating signs are exactly what make the boundary edges fit together consistently. For the 2-simplex above,

$$\partial[p_0, p_1, p_2] = [p_1, p_2] - [p_0, p_2] + [p_0, p_1],$$

which (since $-[p_0, p_2] = [p_2, p_0]$) is the counterclockwise loop $p_0 \rightarrow p_1 \rightarrow p_2 \rightarrow p_0$.

Theorem 11.5.4 ($\partial^2 = 0$; Rudin 10.32). *For every oriented affine k -simplex σ (and hence for every chain Γ),*

$$\partial(\partial\sigma) = 0.$$

The boundary of a boundary is the zero chain.

Proof. Apply ∂ twice. A typical face of $\partial\sigma$ omits one vertex; applying ∂ again omits a second. Fix two indices $i < j$; deleting p_i and p_j produces the $(k-2)$ -simplex σ_{ij} (vertices in order, both omitted). This σ_{ij} arises in $\partial^2\sigma$ in exactly two ways: delete p_j first then p_i , or delete p_i first then p_j . Track the signs: removing p_j from σ carries sign $(-1)^j$, then removing p_i from the result carries $(-1)^i$ (since p_i is still in position i , as $i < j$); total $(-1)^{i+j}$. The other order: removing p_i first carries $(-1)^i$, then removing p_j from the result — but now p_j has shifted to position $j-1$ — carries $(-1)^{j-1}$; total $(-1)^{i+j-1}$. The two contributions are equal in magnitude and opposite in sign, so they cancel. Since every σ_{ij} cancels, $\partial^2\sigma = 0$. $\square$

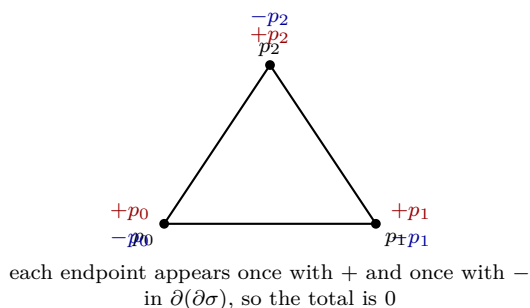


Figure 11.6. $\partial^2 = 0$ on a triangle: ∂ of the three edges produces each vertex twice with opposite signs; everything cancels.

Method (Computing a boundary). *Recognition trigger.* You must find $\partial\sigma$ of a simplex or $\partial\Gamma$ of a chain (e.g. to set up Stokes' theorem).

1. List the vertices in their chosen order $[p_0, \dots, p_k]$.
2. For $i = 0, 1, \dots, k$ delete p_i and attach the sign $(-1)^i$ to the resulting face.
3. Sum the signed faces; that is $\partial\sigma$.
4. For a chain, do this to every simplex and add; simplexes that appear twice with opposite sign *cancel* (this is how interior faces of a triangulated region disappear, leaving only the geometric boundary).

Sanity check. Re-applying the rule must give 0 (Theorem 11.5.4); if your $\partial(\partial\sigma) \neq 0$ you mis-signed a face.

Worked Example 11.5.5 (Boundary of the standard 2-simplex). Compute $\partial[p_0, p_1, p_2]$ and verify $\partial^2 = 0$ explicitly.

Recognition. “Take a boundary, then check it closes up.”

Solution. Deleting each vertex with its sign: $\partial[p_0, p_1, p_2] = [p_1, p_2] - [p_0, p_2] + [p_0, p_1]$. Now take ∂ of each edge ($\partial[a, b] = [b] - [a]$):

$$\partial^2 = ([p_2] - [p_1]) - ([p_2] - [p_0]) + ([p_1] - [p_0]) = [p_2] - [p_1] - [p_2] + [p_0] + [p_1] - [p_0] = 0.$$

Check. Every vertex symbol cancels, confirming Theorem 11.5.4. $\blacksquare$

Worked Example 11.5.6 (The oriented unit square and its boundary). Let $J^2 = \tau_1 + \tau_2$ where $\tau_1 = [0, e_1, e_1 + e_2]$ and $\tau_2 = [0, e_1 + e_2, e_2]$ split the unit square into two triangles. Show ∂J^2 is the counterclockwise boundary of the square and that the shared diagonal cancels.

Recognition. “Triangulate a region and watch interior faces cancel.”

Solution. $\partial\tau_1 = [e_1, e_1 + e_2] - [0, e_1 + e_2] + [0, e_1]$ and $\partial\tau_2 = [e_1 + e_2, e_2] - [0, e_2] + [0, e_1 + e_2]$. The diagonal terms $-[0, e_1 + e_2]$ and $+ [0, e_1 + e_2]$ cancel, leaving

$$\partial J^2 = [0, e_1] + [e_1, e_1 + e_2] + [e_1 + e_2, e_2] - [0, e_2],$$

the four edges of the square traversed counterclockwise (the last being $[e_2, 0]$ since $-[0, e_2] = [e_2, 0]$). *Check.* Four oriented 1-simplexes, the geometric boundary of the square, with the internal diagonal gone — exactly as the cancellation mechanism promises. ■

Further worked examples

Worked Example 11.5.7 (Boundary of a 1-simplex and orientation reversal). Compute $\partial[a, b]$ for a 1-simplex, and confirm $\partial(-\sigma) = -\partial\sigma$ for $\sigma = [p_0, p_1, p_2]$.

Recognition. “Apply the boundary formula in the smallest cases and check it respects orientation.”

Solution. For a 1-simplex, Definition 11.5.3 gives $\partial[a, b] = (-1)^0[b] + (-1)^1[a] = [b] - [a]$ (endpoint minus startpoint). For orientation reversal, the swap $[p_0, p_1, p_2] \mapsto [p_1, p_0, p_2]$ is an odd permutation, so $-\sigma = [p_1, p_0, p_2]$, and

$$\partial(-\sigma) = [p_0, p_2] - [p_1, p_2] + [p_1, p_0] = -([p_1, p_2] - [p_0, p_2] + [p_0, p_1]) = -\partial\sigma,$$

using $[p_1, p_0] = -[p_0, p_1]$. *Check.* Reversing orientation flips every boundary face’s sign, exactly $\partial(-\sigma) = -\partial\sigma$, as the boundary operator being linear demands.

Worked Example 11.5.8 (Boundary of the standard 3-simplex (tetrahedron)). Compute $\partial[p_0, p_1, p_2, p_3]$ and verify it is a sum of four oriented triangular faces.

Recognition. “Take the boundary of a 3-simplex.” Delete each vertex in turn with sign $(-1)^i$.

Solution.

$$\partial[p_0, p_1, p_2, p_3] = [p_1, p_2, p_3] - [p_0, p_2, p_3] + [p_0, p_1, p_3] - [p_0, p_1, p_2].$$

These are the four triangular faces, with alternating signs that orient each face outward. *Check.* Applying ∂ again, each edge $[p_i, p_j]$ ($i < j$) appears in exactly two faces with opposite induced signs and cancels, so $\partial^2 = 0$ (Theorem 11.5.4); e.g. the edge $[p_2, p_3]$ comes from $+ [p_1, p_2, p_3]$ as $+ [p_2, p_3]$ and from $- [p_0, p_2, p_3]$ as $- [p_2, p_3]$, summing to zero.

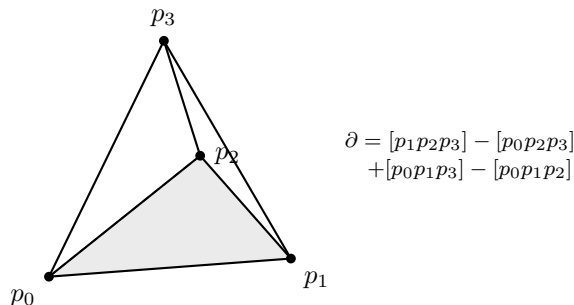


Figure 11.7. The standard 3-simplex. Its boundary is the alternating sum of the four triangular faces; each interior edge cancels when ∂ is applied a second time.

Worked Example 11.5.9 (A chain whose boundary is not zero, contrasted with a cycle). Let $\Gamma = [p_0, p_1]$ (a single edge) and $C = [p_0, p_1] + [p_1, p_2] + [p_2, p_0]$ (a triangle of edges). Compute $\partial\Gamma$ and ∂C , and classify each as a cycle or not.

Recognition. “Distinguish a chain with boundary from a cycle (boundaryless chain).”

Solution. $\partial\Gamma = [p_1] - [p_0] \neq 0$, so Γ is *not* a cycle — a lone edge has two endpoints. For the closed triangle,

$$\partial C = ([p_1] - [p_0]) + ([p_2] - [p_1]) + ([p_0] - [p_2]) = 0,$$

every vertex appearing once with $+$ and once with $-$. So C is a *cycle* ($\partial C = 0$). *Check.* Geometrically Γ has loose ends while C closes up; the algebra of ∂ detects exactly this, and only cycles can bound a region in Stokes' theorem.

Remark. The pairing $\partial^2 = 0$ (boundaries) with $d^2 = 0$ (forms) is not a coincidence: Stokes' theorem (§11.6) says $\int_{\Gamma} d\omega = \int_{\partial\Gamma} \omega$, so applying it twice gives $\int_{\Gamma} d(d\omega) = \int_{\partial(\partial\Gamma)} \omega$, and both identities $d^2 = 0$, $\partial^2 = 0$ make this read $0 = 0$ consistently. The two nilpotencies are *adjoint* under integration.

Exercises (Rudin, Ch.10).

- (Rudin 7.) (a) Show that the standard simplex Q^k is the smallest convex subset of $\mathbb{R}^k$ that contains $0, e_1, \dots, e_k$. (b) Show that affine mappings take convex sets to convex sets.
- (Rudin 12.) Let I^k be the set of all $u = (u_1, \dots, u_k) \in \mathbb{R}^k$ with $0 \leq u_i \leq 1$ for all i (the unit cube), and let Q^k be the standard simplex of all $x = (x_1, \dots, x_k)$ with $x_i \geq 0$ and $\sum x_i \leq 1$. Define $x = T(u)$ by

$$x_1 = u_1, \quad x_i = u_i(1 - x_1 - \dots - x_{i-1}) \quad (i = 2, \dots, k),$$

equivalently $\sum_{i=1}^k x_i = 1 - \prod_{i=1}^k (1 - u_i)$. Show that T maps I^k onto Q^k , is one-to-one in the interior of I^k , with inverse S given in the interior of Q^k by $u_1 = x_1$ and $u_i = x_i / (1 - x_1 - \dots - x_{i-1})$ for $i = 2, \dots, k$, and that

$$J_T(u) = (1 - u_1)^{k-1} (1 - u_2)^{k-2} \dots (1 - u_{k-1}).$$

- (Rudin 13.) Let $r_1, \dots, r_k$ be nonnegative integers, and prove that

$$\int_{Q^k} x_1^{r_1} \dots x_k^{r_k} dx = \frac{r_1! \dots r_k!}{(k + r_1 + \dots + r_k)!}.$$

(Hint: use Exercise (Rudin 12), Theorem 11.2.4, and the Beta-function evaluation.) Note that the special case $r_1 = \dots = r_k = 0$ shows the volume of Q^k is $1/k!$.

- (Rudin 16.) If $k \geq 2$ and $\sigma = [p_0, p_1, \dots, p_k]$ is an oriented affine k -simplex, prove that $\partial^2 \sigma = 0$, directly from the definition of the boundary operator ∂ . Deduce that $\partial^2 \Gamma = 0$ for every chain Γ . (Hint: do $k = 2, 3$ first for orientation; in general, for $i < j$, let σ_{ij} be the $(k-2)$ -simplex obtained by deleting p_i and p_j from σ , and show each σ_{ij} occurs twice in $\partial^2 \sigma$ with opposite sign.)
- (Rudin 17.) Put $J^2 = \tau_1 + \tau_2$, where $\tau_1 = [0, e_1, e_1 + e_2]$ and $\tau_2 = -[0, e_2, e_1 + e_2]$. Explain why it is reasonable to call J^2 the positively oriented unit square in $\mathbb{R}^2$. Show that ∂J^2 is the sum of 4 oriented affine 1-simplexes; find these. What is $\partial(\tau_1 - \tau_2)$?
- (Rudin 18.) Consider the oriented affine 3-simplex $\sigma_1 = [0, e_1, e_1 + e_2, e_1 + e_2 + e_3]$ in $\mathbb{R}^3$. Show that σ_1 (regarded as a linear transformation) has determinant 1, so σ_1 is positively oriented. For each of the five permutations (i_1, i_2, i_3) of $(1, 2, 3)$ distinct from $(1, 2, 3)$, associate the simplex $\sigma = s[0, e_{i_1}, e_{i_1} + e_{i_2}, e_{i_1} + e_{i_2} + e_{i_3}]$, where s is the sign of the permutation; obtain $\sigma_2, \dots, \sigma_6$. Show they are positively oriented, that $J^3 = \sigma_1 + \dots + \sigma_6$ deserves to be called the positively oriented unit cube in $\mathbb{R}^3$, and that ∂J^3 is the sum of 12 oriented affine 2-simplexes covering the surface of the cube I^3 . Show $x = (x_1, x_2, x_3)$ lies in the range of σ_1 iff $0 \leq x_3 \leq x_2 \leq x_1 \leq 1$, that the ranges of $\sigma_1, \dots, \sigma_6$ have disjoint interiors, and that their union covers I^3 (compare Exercise (Rudin 13); note $3! = 6$).
- (Rudin 19.) Let J^2, J^3 be as in Exercises (Rudin 17, 18). Define the affine maps $\mathbb{R}^2 \rightarrow \mathbb{R}^3$

$$\begin{aligned} B_{01}(u, v) &= (0, u, v), & B_{11}(u, v) &= (1, u, v), \\ B_{02}(u, v) &= (u, 0, v), & B_{12}(u, v) &= (u, 1, v), \\ B_{03}(u, v) &= (u, v, 0), & B_{13}(u, v) &= (u, v, 1), \end{aligned}$$

and put $\beta_{ri} = B_{ri}(J^2)$ for $r = 0, 1$, $i = 1, 2, 3$ (each an affine-oriented 2-chain). Verify that $\partial J^3 = \sum_{i=1}^3 (-1)^i (\beta_{0i} - \beta_{1i})$, in agreement with Exercise (Rudin 18).

Additional exercises.

1. (Boundary of a simplex.) For the oriented affine 2-simplex $\sigma = [p_0, p_1, p_2]$ write out $\partial\sigma = [p_1, p_2] - [p_0, p_2] + [p_0, p_1]$ explicitly, and verify $\partial(\partial\sigma) = 0$ by cancelling the six oriented vertices in pairs.
2. Let $\sigma = [e_0, e_1, e_2, e_3]$ be the standard 3-simplex in $\mathbb{R}^3$ ($e_0 = 0$, e_i the standard basis vectors). List the four oriented faces of $\partial\sigma$ with their signs.
3. (Orientation flip.) Show that swapping two vertices of an oriented simplex reverses its orientation: $[p_0, p_1, p_2, \dots] = -[p_1, p_0, p_2, \dots]$. Deduce that an affine simplex with a repeated vertex is the zero chain.
4. (Computation.) Let Φ be the 2-surface (singular 2-simplex) on $[0, 1]^2$ given by $\Phi(u, v) = (u, v, u + v)$ in $\mathbb{R}^3$. Compute the integral $\int_{\Phi} dx \wedge dy$ of the basic form $dx \wedge dy$ over Φ .
5. For a chain $c = \sigma_1 + \sigma_2$ of two 1-simplexes that share an endpoint with opposite orientation, show that ∂c has that shared point cancel, so c behaves like a single longer oriented edge. Illustrate with $\sigma_1 = [0, 1]$, $\sigma_2 = [1, 2]$ on the line.
6. (Non-example.) Give a 1-chain c in $\mathbb{R}^2$ that is a cycle ($\partial c = 0$) but is not the boundary of any 2-chain in $\mathbb{R}^2 \setminus \{0\}$ (take c to be the unit circle traversed once, as a sum of arcs). Relate this to the closed-but-not-exact form of the previous section.
7. (Conceptual.) Explain why the algebraic relation $\partial \circ \partial = 0$ for chains is the geometric counterpart of $d \circ d = 0$ for forms, and state how Stokes' theorem $\int_{\partial c} \omega = \int_c d\omega$ pairs the two operators.

11.6 Stokes' Theorem and Classical Vector Analysis

Everything assembled so far — forms, d , pullback, simplexes, chains, and the two nilpotencies $d^2 = 0$, $\partial^2 = 0$ — exists to support a single theorem.

Stokes' theorem

Theorem 11.6.1 (Stokes' theorem; Rudin 10.33). *Let Ψ be a k -chain of class C'' in an open set $E \subseteq \mathbb{R}^n$, and let ω be a $(k-1)$ -form of class C' on E . Then*

$$\boxed{\int_{\Psi} d\omega = \int_{\partial\Psi} \omega}$$

The integral of $d\omega$ over the chain equals the integral of ω over its boundary.

Proof outline. By linearity in Ψ (both sides are additive over the simplexes of the chain) it suffices to treat a single oriented simplex; and by naturality $d(\omega_{\Phi}) = (d\omega)_{\Phi}$ (§11.4, Theorem 11.4.5) together with the definition of integration over a k -surface, it suffices to prove the identity for the *standard simplex* Q^k (or the oriented cube J^k) with the form pulled back to the parameter domain. There ω is a $(k-1)$ -form, a sum of terms $a dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_k$. For one such term, $d\omega = (-1)^{j-1} (D_j a) dx_1 \wedge \cdots \wedge dx_k$, and $\int_{Q^k} d\omega$ becomes, by Fubini (§11.1), an iterated integral in which the x_j -integration is carried out first via the *fundamental theorem of calculus*:

$$\int (D_j a) dx_j = a|_{\text{top face}} - a|_{\text{bottom face}}.$$

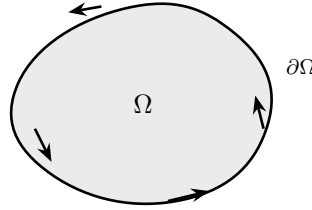
The two evaluated terms are exactly the contributions of the two faces of ∂Q^k on which x_j varies between its extremes; the remaining faces contribute nothing to that term because a does not produce a boundary term there. Summing j and matching signs against Definition 11.5.3 gives precisely $\int_{\partial Q^k} \omega$. Reassembling the patches (partition of unity) yields the theorem for the general chain. $\square$

Remark. The proof is the multivariable, oriented form of the *fundamental theorem of calculus*: $\int_a^b f' = f(b) - f(a)$ is the case $k = 1$, $\omega = f$ (a 0-form), $\Psi = [a, b]$, with $\partial[a, b] = [b] - [a]$. Every classical integral theorem below is a specialization obtained by choosing k , n , and translating d , ∂ into vector-calculus language.

Green's theorem

Corollary 11.6.2 (Green's theorem). *Let $\Omega \subseteq \mathbb{R}^2$ be a region with positively oriented boundary $\partial\Omega$, and $P, Q \in C'$. With $\omega = P dx + Q dy$ (so $d\omega = (Q_x - P_y) dx \wedge dy$), Stokes gives*

$$\iint_{\Omega} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_{\partial\Omega} (P dx + Q dy).$$



boundary oriented so Ω stays on the left

Figure 11.8. Green's theorem: a planar region Ω with counterclockwise boundary. The 2-form $d\omega = (Q_x - P_y) dx \wedge dy$ integrated over Ω equals the 1-form ω integrated around $\partial\Omega$.

Worked Example 11.6.3 (Area by a boundary integral). Use Green's theorem with $\omega = \frac{1}{2}(x dy - y dx)$ to compute the area enclosed by the ellipse $x = a \cos t$, $y = b \sin t$, $0 \leq t \leq 2\pi$.

Recognition. "Compute an area from its boundary." Choose P, Q with $Q_x - P_y = 1$, then evaluate the line integral.

Solution. For $\omega = \frac{1}{2}(x dy - y dx)$, $P = -y/2$, $Q = x/2$, so $Q_x - P_y = \frac{1}{2} + \frac{1}{2} = 1$ and $\iint_{\Omega} 1 dx dy = \oint \omega$ is the area. On the ellipse $dx = -a \sin t dt$, $dy = b \cos t dt$, so

$$\text{area} = \frac{1}{2} \oint (x dy - y dx) = \frac{1}{2} \int_0^{2\pi} (a \cos t b \cos t - b \sin t (-a \sin t)) dt = \frac{1}{2} \int_0^{2\pi} ab dt = \pi ab.$$

Check. For the circle $a = b = r$ this gives πr^2 . ■

The divergence theorem and the classical Stokes theorem

Corollary 11.6.4 (Divergence (Gauss) theorem). *Let $\Omega \subseteq \mathbb{R}^3$ be a solid region with outward-oriented boundary surface $\partial\Omega$, and $F = (F_1, F_2, F_3) \in C'$. Taking the 2-form $\omega = F_1 dy \wedge dz + F_2 dz \wedge dx + F_3 dx \wedge dy$, one has $d\omega = (\text{div } F) dx \wedge dy \wedge dz$, and Stokes gives*

$$\iiint_{\Omega} \text{div } F dV = \iint_{\partial\Omega} F \cdot N dA,$$

where N is the outward unit normal.

Corollary 11.6.5 (Classical Stokes theorem). *Let $S \subseteq \mathbb{R}^3$ be an oriented surface with boundary curve ∂S , and $F = (F_1, F_2, F_3) \in C'$. With the 1-form $\omega = F_1 dx + F_2 dy + F_3 dz$, one has $d\omega = (\text{curl } F) \cdot (\text{area 2-form})$, and Stokes gives*

$$\iint_S (\text{curl } F) \cdot N dA = \oint_{\partial S} F \cdot dr.$$

These are all the same theorem ($\int_{\Psi} d\omega = \int_{\partial\Psi} \omega$) read in dimensions $(k, n) = (2, 2), (3, 3), (2, 3)$ respectively, after translating between forms and vector fields. The translation is a dictionary.

Grad, curl, div and the forms dictionary

In $\mathbb{R}^3$ a 0-form is a function f ; a 1-form $\leftrightarrow$ a vector field via $F \leftrightarrow F_1 dx + F_2 dy + F_3 dz$; a 2-form $\leftrightarrow$ a vector field via $F \leftrightarrow F_1 dy \wedge dz + F_2 dz \wedge dx + F_3 dx \wedge dy$; a 3-form $\leftrightarrow$ a function via $g dx \wedge dy \wedge dz$. Under this dictionary the single operator d becomes, in degree order, *gradient*, *curl*, *divergence*.

degree	form	d acts as	vector field / function
0	f	$df = \nabla f \cdot (dx, dy, dz)$	$\text{grad } f$
1	$\sum F_i dx_i$	curl	$\text{curl } F$
2	$\sum F_i (dx_j \wedge dx_k)$	divergence	$\text{div } F$
3	$g dx \wedge dy \wedge dz$	0	—

Figure 11.9. The vector-calculus dictionary in $\mathbb{R}^3$. One operator d realizes grad, curl, div in successive degrees; $d^2 = 0$ encodes both $\text{curl grad} = 0$ and $\text{div curl} = 0$.

$$\{\text{functions}\} \xrightarrow{d=\text{grad}} \{\text{1-forms}\} \xrightarrow{d=\text{curl}} \{\text{2-forms}\} \xrightarrow{d=\text{div}} \{\text{3-forms}\}$$

Figure 11.10. The de Rham sequence in $\mathbb{R}^3$. Composing two consecutive arrows gives 0 ($d^2 = 0$): grad then curl is 0, curl then div is 0.

Caution. The identities $\text{curl grad } f = 0$ and $\text{div curl } F = 0$ are *not* three separate miracles — they are the one equation $d^2 = 0$ in degrees 0 and 1. Conversely, a curl-free field need not be a gradient, and a divergence-free field need not be a curl, *unless the domain has no holes*: on $\mathbb{R}^3 \setminus \{0\}$ the field of Exercises 22–23 is closed but not exact. Local exactness (*Poincaré’s lemma*: on a convex set every closed form is exact, Exercises 24–27) always holds; global exactness depends on the topology.

Worked Example 11.6.6 (Reading curl off the exterior derivative). Let $F = (-y, x, 0)$ in $\mathbb{R}^3$ and $\omega = -y dx + x dy$. Compute $d\omega$ and identify $\text{curl } F$.

Recognition. “Translate d of a 1-form into a curl.”

Solution. $d\omega = d(-y) \wedge dx + d(x) \wedge dy = (-dy) \wedge dx + dx \wedge dy = dx \wedge dy + dx \wedge dy = 2 dx \wedge dy$. Under the dictionary $dx \wedge dy \leftrightarrow$ the z -component, so $\text{curl } F = (0, 0, 2)$. *Check.* Directly, $\text{curl}(-y, x, 0) = (\partial_y 0 - \partial_z x, \partial_z(-y) - \partial_x 0, \partial_x x - \partial_y(-y)) = (0, 0, 2)$. Agrees. ■

Worked Example 11.6.7 (The closed-but-not-exact angle form). Show $\eta = \frac{x dy - y dx}{x^2 + y^2}$ on $\mathbb{R}^2 \setminus \{0\}$ satisfies

$d\eta = 0$, yet $\oint_\gamma \eta = 2\pi \neq 0$ around the unit circle, so η is not exact.

Recognition. “Detect a hole.” A form that is closed ($d\eta = 0$) but has nonzero integral around a loop cannot be d of a single-valued function.

Solution. Writing $P = -y/(x^2 + y^2)$, $Q = x/(x^2 + y^2)$, a direct computation gives $Q_x = P_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}$, so $d\eta = (Q_x - P_y) dx \wedge dy = 0$: η is closed. But on the unit circle $\gamma(t) = (\cos t, \sin t)$, η pulls back to dt , so $\oint_\gamma \eta = \int_0^{2\pi} dt = 2\pi$. Were $\eta = df$ for a single-valued f , Stokes would force $\oint_\gamma \eta = f(\gamma(2\pi)) - f(\gamma(0)) = 0$. The contradiction shows η is not exact. *Check.* η is locally $d(\arctan(y/x))$, but $\arctan(y/x) = \theta$ jumps by 2π around the origin — the multivaluedness is precisely the hole. ■

Further worked examples

Worked Example 11.6.8 (The divergence theorem on a ball). Verify the divergence theorem (Corollary 11.6.4) for $F = (x, y, z)$ on the unit ball $\Omega = \{x^2 + y^2 + z^2 \leq 1\}$ by computing both sides.

Recognition. “Confirm $\iiint \text{div } F = \iint F \cdot N$ on a concrete solid.”

Solution. $\text{div } F = \partial_x x + \partial_y y + \partial_z z = 3$, so the volume side is

$$\iiint_\Omega \text{div } F dV = 3 \cdot \text{vol}(\Omega) = 3 \cdot \frac{4}{3}\pi = 4\pi.$$

On the unit sphere the outward normal is $N = (x, y, z)$ (the radial unit vector), so $F \cdot N = x^2 + y^2 + z^2 = 1$ on $\partial\Omega$, and the flux side is $\iint_{\partial\Omega} 1 dA = \text{area}(S^2) = 4\pi$. *Check.* Both sides equal 4π , confirming Corollary 11.6.4. (The field $F = (x, y, z)$ is the radial field; its flux out of a ball is just its constant divergence times the volume.)

Worked Example 11.6.9 (Classical Stokes on a hemisphere). Let $F = (-y, x, 0)$ and let S be the upper unit hemisphere $z \geq 0$ with boundary the unit circle ∂S in the xy -plane. Verify $\iint_S (\text{curl } F) \cdot N dA = \oint_{\partial S} F \cdot dr$ (Corollary 11.6.5).

Recognition. “Match a curl flux against a boundary circulation.”

Solution. From the earlier worked example, $\text{curl}(-y, x, 0) = (0, 0, 2)$, a constant upward field. The flux of $(0, 0, 2)$ through the hemisphere equals its flux through the flat unit disc the hemisphere bounds (same boundary, $\text{div}(\text{curl}) = 0$): $\iint (0, 0, 2) \cdot (0, 0, 1) dA = 2 \cdot \text{area}(\text{disc}) = 2\pi$. The boundary circle is $\gamma(t) = (\cos t, \sin t, 0)$, $dr = (-\sin t, \cos t, 0) dt$, and $F = (-\sin t, \cos t, 0)$, so

$$\oint_{\partial S} F \cdot dr = \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = \int_0^{2\pi} 1 dt = 2\pi.$$

Check. Both sides equal 2π . As a sanity check the flux is independent of which surface spanning ∂S we choose, because $\text{curl } F$ is divergence-free. ■

Exercises (Rudin, Ch.10).

- (Rudin 21.) Consider the 1-form $\eta = \frac{x dy - y dx}{x^2 + y^2}$ in $\mathbb{R}^2 \setminus \{0\}$. (a) Carry out the computation showing $d\eta = 0$. (b) Let $\gamma(t) = (r \cos t, r \sin t)$ ($r > 0$) and let Γ be a C'' -curve in $\mathbb{R}^2 \setminus \{0\}$ on $[0, 2\pi]$ with $\Gamma(0) = \Gamma(2\pi)$ such that the segments $[\gamma(t), \Gamma(t)]$ never contain 0. Prove $\int_{\Gamma} \eta = \int_{\gamma} \eta = 2\pi$. (Hint: define the 2-surface $\Phi(t, u) = (1 - u)\gamma(t) + u\Gamma(t)$ on $[0, 2\pi] \times [0, 1]$; by cancellation $\partial\Phi = \Gamma - \gamma$, and Stokes' theorem with $d\eta = 0$ gives the result.) (c) Take $\Gamma(t) = (a \cos t, b \sin t)$ with $a, b > 0$ and use (b) to show $\int_0^{2\pi} \frac{ab}{a^2 \cos^2 t + b^2 \sin^2 t} dt = 2\pi$. (d) Show $\eta = d(\arctan(y/x))$ in any convex open set where $x \neq 0$, and $\eta = d(-\arctan(x/y))$ where $y \neq 0$, justifying the notation $\eta = d\theta$ even though η is not exact in $\mathbb{R}^2 \setminus \{0\}$. (e) Show (b) can be derived from (d). (f) If Γ is any closed C' -curve in $\mathbb{R}^2 \setminus \{0\}$, prove $\frac{1}{2\pi} \int_{\Gamma} \eta = \text{Ind}(\Gamma)$, the index (winding number) of Γ .

- (Rudin 22.) Define ζ in $\mathbb{R}^3 \setminus \{0\}$ by

$$\zeta = \frac{x dy \wedge dz + y dz \wedge dx + z dx \wedge dy}{r^3}, \quad r = (x^2 + y^2 + z^2)^{1/2},$$

- let D be the rectangle $0 \leq u \leq \pi$, $0 \leq v \leq 2\pi$, and let Σ be the 2-surface in $\mathbb{R}^3$ on D given by $x = \sin u \cos v$, $y = \sin u \sin v$, $z = \cos u$. (a) Prove $d\zeta = 0$ in $\mathbb{R}^3 \setminus \{0\}$. (b) Let S be the restriction of Σ to a parameter domain $E \subseteq D$; prove $\int_S \zeta = \iint_E \sin u du dv = A(S)$, the area. (c) For C'' -functions g, h_1, h_2, h_3 on $[0, 1]$ with $g > 0$, let $\Phi(s, t)$ be the 2-surface on I^2 given by $x = g(t)h_1(s)$, $y = g(t)h_2(s)$, $z = g(t)h_3(s)$; prove $\int_{\partial\Phi} \zeta = 0$ (its range lies in a cone with vertex 0). (d) Let E be a closed rectangle in D with edges parallel to those of D , let $f \in \mathcal{C}''(D)$, $f > 0$, and let Ω be the 2-surface on E defined by $\Omega(u, v) = f(u, v)\Sigma(u, v)$; prove $\int_{\Omega} \zeta = \int_S \zeta$ (the “solid angle” subtended at the origin). (e) Put $\lambda = -(z/r)\eta$ with η as in Exercise (Rudin 21). Show λ is a 1-form on the open set $V \subseteq \mathbb{R}^3$ where $x^2 + y^2 > 0$, and that $\zeta = d\lambda$ there. (f) Derive (d) from (e) without using (c). (g) Is ζ exact in the complement of every line through the origin?
- (Rudin 23.) Fix n . Define $r_k = (x_1^2 + \cdots + x_k^2)^{1/2}$ for $1 \leq k \leq n$, let E_k be the set of all $x \in \mathbb{R}^n$ with $r_k > 0$, and let ω_k be the $(k-1)$ -form on E_k given by

$$\omega_k = (r_k)^{-k} \sum_{i=1}^k (-1)^{i-1} x_i dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k.$$

- (Note $\omega_2 = \eta$, $\omega_3 = \zeta$, and $E_1 \subseteq E_2 \subseteq \cdots \subseteq E_n = \mathbb{R}^n \setminus \{0\}$.) (a) Prove $d\omega_k = 0$ in E_k . (b) For $k = 2, \dots, n$, prove ω_k is exact in $E_k \setminus E_{k-1}$ by showing $\omega_k = d(f_k \omega_{k-1}) = (df_k) \wedge \omega_{k-1}$, where $f_k(x) = (-1)^k g_k(x_k/r_k)$ and $g_k(t) = \int_{-1}^t (1 - s^2)^{(k-3)/2} ds$. (c) Is ω_n exact in E_n ? (d) Extend, as far as possible, the assertions of Exercises (Rudin 21, 22) to ω_k for arbitrary n .
- (Rudin 24.) Let $\omega = \sum a_i(x) dx_i$ be a 1-form of class C''' in a convex open set $E \subseteq \mathbb{R}^n$. Assume $d\omega = 0$ and prove ω is exact in E , as follows. Fix $p \in E$ and define $f(x) = \int_{[p,x]} \omega$ for $x \in E$. Apply Stokes' theorem to affine-oriented 2-simplexes $[p, x, y]$ in E to deduce

$$f(y) - f(x) = \sum_{i=1}^n (y_i - x_i) \int_0^1 a_i((1-t)x + ty) dt$$

for $x, y \in E$. Hence $(D_i f)(x) = a_i(x)$, so $df = \omega$. (This is the Poincaré lemma for 1-forms.)

5. (Rudin 25.) Assume ω is a 1-form in an open set $E \subseteq \mathbb{R}^n$ such that $\int_\Gamma \omega = 0$ for every closed curve Γ in E of class C' . Prove ω is exact in E , by imitating part of the argument of Exercise (Rudin 24).
6. (Rudin 26.) Assume ω is a 1-form in $\mathbb{R}^3 \setminus \{0\}$ of class C' with $d\omega = 0$. Prove ω is exact in $\mathbb{R}^3 \setminus \{0\}$. (Hint: every closed C' -curve in $\mathbb{R}^3 \setminus \{0\}$ is the boundary of a 2-surface in $\mathbb{R}^3 \setminus \{0\}$; apply Stokes' theorem and Exercise (Rudin 25).)
7. (Rudin 27.) Let E be an open 3-cell in $\mathbb{R}^3$ with edges parallel to the coordinate axes. Suppose $(a, b, c) \in E$, $f_i \in \mathcal{C}'(E)$ for $i = 1, 2, 3$, $\omega = f_1 dy \wedge dz + f_2 dz \wedge dx + f_3 dx \wedge dy$, and $d\omega = 0$ in E . Define $\lambda = g_1 dx + g_2 dy$ where

$$g_1(x, y, z) = \int_c^z f_2(x, y, s) ds - \int_b^y f_3(x, t, c) dt, \quad g_2(x, y, z) = - \int_c^z f_1(x, y, s) ds,$$
 for $(x, y, z) \in E$. Prove $d\lambda = \omega$ in E . Evaluate these integrals when $\omega = \zeta$ (Exercise (Rudin 22)) and thus find the form λ in part (e) there.
8. (Rudin 28.) Fix $b > a > 0$ and define $\Phi(r, \theta) = (r \cos \theta, r \sin \theta)$ for $a \leq r \leq b$, $0 \leq \theta \leq 2\pi$ (so the range of Φ is an annulus in $\mathbb{R}^2$). Put $\omega = x^3 dy$, and compute both $\int_\Phi d\omega$ and $\int_{\partial\Phi} \omega$ to verify directly that they are equal.
9. (Rudin 29.) Prove the existence of a function Δ with the properties needed in the proof of Theorem 10.38 (the change-of-variables / surface-area lemma), and prove that the resulting function F is of class C' . (Both assertions are trivial if E is an open cell or open ball, since Δ can be taken constant; refer to Rudin Theorem 9.42.)
10. (Rudin 30.) If N is the vector given by Rudin's equation (135) (the normal to a surface), prove the area/normal identity stated there, and verify Rudin's equation (137).
11. (Rudin 31.) Let $E \subseteq \mathbb{R}^3$ be open, $g \in \mathcal{C}''(E)$, $h \in \mathcal{C}''(E)$, and consider the vector field $F = g\nabla h$. (a) Prove $\nabla \cdot F = g\nabla^2 h + (\nabla g) \cdot (\nabla h)$, where $\nabla^2 h = \nabla \cdot (\nabla h) = \sum \partial^2 h / \partial x_i^2$ is the Laplacian of h . (b) If Ω is a closed subset of E with positively oriented boundary $\partial\Omega$ (as in the divergence theorem), prove

$$\iiint_\Omega [g\nabla^2 h + (\nabla g) \cdot (\nabla h)] dV = \iint_{\partial\Omega} g \frac{\partial h}{\partial n} dA,$$

where $\partial h / \partial n = (\nabla h) \cdot n$ is the normal derivative. Interchange g, h and subtract to obtain Green's identity $\iiint_\Omega (g\nabla^2 h - h\nabla^2 g) dV = \iint_{\partial\Omega} (g \frac{\partial h}{\partial n} - h \frac{\partial g}{\partial n}) dA$. (c) Assume h is harmonic in E ($\nabla^2 h = 0$). Take $g = 1$ to conclude $\iint_{\partial\Omega} \frac{\partial h}{\partial n} dA = 0$; take $g = h$ to conclude $h = 0$ in Ω if $h = 0$ on $\partial\Omega$. (d) Show that Green's identities are also valid in $\mathbb{R}^2$.

12. (Rudin 32.) Fix δ , $0 < \delta < 1$. Let D be the set of all $(\theta, t) \in \mathbb{R}^2$ with $0 \leq \theta \leq \pi$, $-\delta \leq t \leq \delta$. Let Φ be the 2-surface in $\mathbb{R}^3$ on D given by

$$x = (1 - t \sin \theta) \cos 2\theta, \quad y = (1 - t \sin \theta) \sin 2\theta, \quad z = t \cos \theta.$$

Note $\Phi(\pi, t) = \Phi(0, -t)$, and Φ is one-to-one on the rest of D . The range $M = \Phi(D)$ is a Möbius band, the simplest nonorientable surface. Prove the assertions in Rudin's description: with $p_1 = (0, -\delta)$, $p_2 = (\pi, -\delta)$, $p_3 = (\pi, \delta)$, $p_4 = (0, \delta)$, $p_5 = p_1$ and $\gamma_i = [p_i, p_{i+1}]$, $\Gamma_i = \Phi \circ \gamma_i$, show $\Phi(p_1) = \Phi(p_3) = a = (1, 0, -\delta)$, $\Phi(p_2) = \Phi(p_4) = b = (1, 0, \delta)$, and that $\partial\Phi = \Gamma_1 + \Gamma_3 + 2\Gamma_2$ where $\Gamma_2 = [b, a] = \Gamma_4$. With η the 1-form of Exercises (Rudin 21, 22), since $d\eta = 0$ Stokes gives $\int_{\partial\Phi} \eta = 0$, whereas the geometric edge Γ of M satisfies $\int_\Gamma \eta = 4\pi$ — showing $\Gamma \neq \partial\Phi$ and explaining why Stokes' formula is usually stated only for orientable surfaces.

Additional exercises.

1. (Green's theorem check.) Verify Green's theorem for $\oint_{\partial Q} (x^2 dx + xy dy)$ where $Q = [0, 1]^2$: compute both the line integral around ∂Q and the double integral $\iint_Q (\partial_x(xy) - \partial_y(x^2)) dx dy$ and confirm they agree.
2. (Divergence theorem.) For the vector field $F = (x, y, z)$ and the unit ball $B \subseteq \mathbb{R}^3$, compute $\iiint_B (\nabla \cdot F) dV$ and the flux $\iint_{\partial B} F \cdot n dA$ directly, and check Gauss's theorem.

3. (Stokes / curl.) Let $F = (-y, x, 0)$ and let S be the unit disk in the plane $z = 0$ with upward normal. Compute $\iint_S (\nabla \times F) \cdot n \, dA$ and the circulation $\oint_{\partial S} F \cdot d\mathbf{r}$, verifying the curl theorem.
4. Translate the three classical theorems (Green, Gauss/divergence, Kelvin–Stokes) into the single statement $\int_{\partial c} \omega = \int_c d\omega$ by identifying, in each case, the form ω and the chain c . Make a small table matching d to grad, curl, div.
5. (Non-example.) Show that the Kelvin–Stokes theorem can fail for the “angle” field $F = (-y, x)/(x^2 + y^2)$ on a disk centered at the origin: its curl is 0 on the punctured plane yet the circulation around the boundary circle is 2π . Explain why this is not a contradiction (the field is not C' at the center).
6. Use the divergence theorem to derive the integration-by-parts / Green’s first identity $\iiint_{\Omega} (g \nabla^2 h + \nabla g \cdot \nabla h) \, dV = \iint_{\partial \Omega} g \, \partial h / \partial n \, dA$ for $g, h \in \mathcal{C}''$, and specialize to $g = h$ harmonic to recover a maximum-principle consequence.
7. (Computation.) Compute the flux of $F = (x^3, y^3, z^3)$ outward through the sphere of radius a using the divergence theorem, expressing the answer in terms of a .
8. (Conceptual.) Explain why orientability of the surface is essential in Kelvin–Stokes, referencing the Möbius-band example (Rudin 32): describe in words what goes wrong with the induced boundary orientation on a nonorientable surface.

Chapter 12

The Generalized Riemann Integral

12.1 Gauges and the Henstock–Kurzweil Definition

Chapter 7 built the Riemann integral by demanding that the tagged Riemann sums $S(f, \dot{P}) = \sum_i f(t_i)(x_i - x_{i-1})$ converge as the *mesh* $\|\dot{P}\|$ shrinks to 0. The mesh is a blunt instrument: it forces *every* subinterval to be short, even where f is tame. Kurzweil and Henstock made one change — replace the single number $\|\dot{P}\|$ by a positive *function* $\delta(t)$ that may shrink the allowed subinterval only near the troublesome tags. This apparently cosmetic shift produces a strictly larger integral: every derivative, every improper integral, and every Lebesgue integrable function becomes integrable, with no new machinery. The whole theory rests on one existence lemma (Cousin’s), which we prove first.

Gauges and δ -fine partitions

Recall (from the Riemann chapter) that a *tagged partition* of $[a, b]$ is $\dot{P} = \{(I_i, t_i)\}_{i=1}^n$ with $I_i = [x_{i-1}, x_i]$, $a = x_0 < \dots < x_n = b$, and a *tag* $t_i \in I_i$ for each i . Its Riemann sum is $S(f, \dot{P}) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1})$.

Definition 12.1.1 (Gauge, B&S 5.5.2). A *gauge* on $[a, b]$ is any strictly positive function $\delta : [a, b] \rightarrow (0, \infty)$. A tagged partition $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is δ -*fine* if each subinterval is trapped in the $\delta(t_i)$ -ball about its own tag:

$$[x_{i-1}, x_i] \subseteq [t_i - \delta(t_i), t_i + \delta(t_i)] \quad (i = 1, \dots, n).$$

Two features deserve emphasis. First, only a *tagged* partition can be called δ -fine: the condition involves the tags. Second, the length of I_i is controlled by the value of δ *at the tag* t_i , so different subintervals may obey wildly different length bounds. Where δ is small the subinterval must be short; where δ is large the subinterval may be long. Taking $\delta(t) \equiv \frac{1}{2}\eta$ a constant recovers the Riemann mesh condition $\|\dot{P}\| < \eta$ (see Theorem 12.1.6). The gauge also lets us *force a tag*: if $\delta(c)$ is tiny, any δ -fine partition must tag a short subinterval containing c , and if δ further satisfies $\delta(t) \leq \frac{1}{2}|t - c|$ for $t \neq c$, then c itself is *forced* to be a tag.

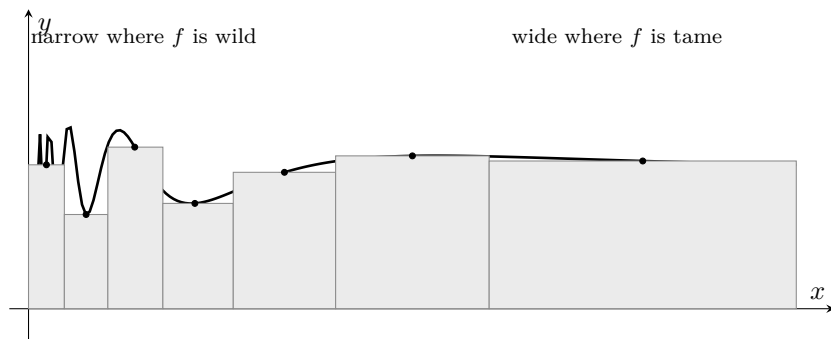


Figure 12.1. A δ -fine tagged partition. The gauge $\delta(t)$ is tiny near the left, forcing short subintervals where the function oscillates, and large on the right, where long subintervals are harmless. A constant-mesh Riemann partition cannot do this: it would need *every* box as narrow as the worst one.

Worked Example 12.1.2 (A gauge that forces 0 to be a tag). On $[0, 1]$ let $\delta(0) := \frac{1}{100}$ and $\delta(t) := \frac{1}{2}t$ for $t \in (0, 1]$. Show every δ -fine partition tags the subinterval that contains 0 with the tag $t = 0$.

Recognition. Trigger: “show a gauge forces a specific point to be a tag.” The subinterval I_1 starting at $a = 0$ has *some* tag $t_1 \in I_1$; rule out $t_1 > 0$.

Solution. The first subinterval is $I_1 = [0, x_1]$ with tag $t_1 \in [0, x_1]$. If $t_1 > 0$, then δ -finess forces $0 \in I_1 \subseteq [t_1 - \frac{1}{2}t_1, t_1 + \frac{1}{2}t_1] = [\frac{1}{2}t_1, \frac{3}{2}t_1]$, which does *not* contain 0 — contradiction. Hence $t_1 = 0$. *Check.* With $t_1 = 0$ the box must satisfy $[0, x_1] \subseteq [-\frac{1}{100}, \frac{1}{100}]$, so $x_1 \leq \frac{1}{100}$; the gauge has done its job of pinning a short tagged box at the origin. ■

Cousin’s lemma: gauges always admit fine partitions

The Riemann mesh condition is trivially satisfiable (chop into equal pieces). For a gauge it is not obvious that *any* δ -fine partition exists, since δ may dive toward 0. That it always does is the cornerstone of the theory.

Theorem 12.1.3 (Cousin’s Lemma, B&S 5.5.5). *If δ is any gauge on $[a, b]$, then there exists a δ -fine tagged partition of $[a, b]$.*

Proof. Bisection. Suppose, for contradiction, that $[a, b]$ admits no δ -fine partition. Bisect $[a, b]$ at its midpoint into two closed halves. If *both* halves admitted δ -fine partitions, joining them would give one for $[a, b]$ (concatenating tagged partitions of $[a, c]$ and $[c, b]$ yields one of $[a, b]$). So at least one half, call it $I_1 = [a_1, b_1]$, admits none. Repeat: bisect I_1 and pick a half I_2 with no δ -fine partition, and so on. We obtain nested closed intervals $I_1 \supseteq I_2 \supseteq \cdots$ with $\text{length}(I_k) = (b - a)/2^k \rightarrow 0$, none admitting a δ -fine partition.

Nested intervals. By the Nested Intervals Property there is a unique point ξ belonging to every I_k , and $\xi \in [a, b]$. Since $\delta(\xi) > 0$ and the lengths shrink to 0, choose k so large that $I_k \subseteq [\xi - \delta(\xi), \xi + \delta(\xi)]$. But then the single-piece tagged partition $\{(I_k, \xi)\}$ is δ -fine (its one tag is ξ , its one interval lies in the $\delta(\xi)$ -ball about ξ). This contradicts the choice of I_k as admitting no δ -fine partition. Hence the original assumption is false. □

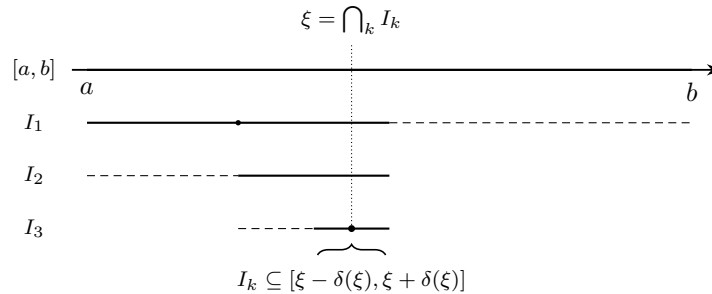


Figure 12.2. Cousin’s lemma by bisection: a “bad” interval splits into a bad half forever, but the nested intersection ξ has $\delta(\xi) > 0$, so a deep enough I_k fits inside the $\delta(\xi)$ -ball and is itself a one-piece δ -fine partition — contradiction.

Caution. Cousin’s lemma is exactly where the completeness of $\mathbb{R}$ enters. Over $\mathbb{Q}$ it fails: a gauge can be cooked up with no δ -fine partition because the candidate point ξ may be irrational. Every existence proof below silently invokes Theorem 12.1.3 to know the Riemann sums it talks about are not vacuous.

The generalized (Henstock–Kurzweil) integral

Definition 12.1.4 (Generalized Riemann integral, B&S 10.1.1). A function $f : [a, b] \rightarrow \mathbb{R}$ is *generalized Riemann integrable* (equivalently *Henstock–Kurzweil integrable*), written $f \in \mathbb{R}^*[a, b]$, if there is a number $L \in \mathbb{R}$ such that for every $\varepsilon > 0$ there is a gauge δ_ε on $[a, b]$ with

$$\dot{P} \text{ is } \delta_\varepsilon\text{-fine} \implies |S(f, \dot{P}) - L| < \varepsilon$$

for every δ_ε -fine tagged partition $\dot{P}$. The number L is the *generalized integral* of f , denoted $\int_a^b f$ or $\int_a^b f(x) dx$.

Compare Definition 12.1.4 with the Riemann definition (Chapter 7): the *only* change is that smallness of $\dot{P}$ is now measured by a gauge δ_ε rather than by a single mesh bound $\delta_\varepsilon > 0$. Cousin's lemma guarantees δ_ε -fine partitions exist, so the implication is never vacuous. Because gauges are more flexible than mesh bounds (any mesh bound is a constant gauge, but not conversely), the class $\mathbb{R}^*[a, b]$ is strictly larger than $\mathcal{R}[a, b]$.

Theorem 12.1.5 (Uniqueness, B&S 10.1.2). *If $f \in \mathbb{R}^*[a, b]$ then the value L is uniquely determined.*

Proof. Suppose L' and L'' both work, and let $\varepsilon > 0$. Pick a gauge δ' for L' at tolerance $\varepsilon/2$ and a gauge δ'' for L'' at tolerance $\varepsilon/2$. Define $\delta(t) := \min\{\delta'(t), \delta''(t)\} > 0$, a gauge. By Cousin's lemma choose a δ -fine partition $\dot{P}$; it is simultaneously δ' -fine and δ'' -fine (shrinking a gauge only shrinks the family of fine partitions). Hence

$$|L' - L''| \leq |L' - S(f, \dot{P})| + |S(f, \dot{P}) - L''| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

As $\varepsilon > 0$ was arbitrary, $L' = L''$. □

Remark. This proof is the template for almost every result below: to combine two requirements take the *pointwise minimum* of the two gauges. The minimum of finitely many gauges is a gauge, and a partition fine for the minimum is fine for each. There is no analogue of “min of two mesh bounds destroying the structure” — gauges compose cleanly, which is exactly their advantage.

Every Riemann integrable function is generalized integrable

Theorem 12.1.6 (Consistency, B&S 10.1.3). *If $f \in \mathcal{R}[a, b]$ with Riemann integral L , then $f \in \mathbb{R}^*[a, b]$ with the same integral L .*

Proof. Let $\varepsilon > 0$. By the Riemann definition there is a mesh bound $d_\varepsilon > 0$ such that $\|\dot{P}\| < d_\varepsilon \implies |S(f, \dot{P}) - L| < \varepsilon$. Define the *constant* gauge $\delta_\varepsilon(t) := \frac{1}{4}d_\varepsilon$ for all t . If $\dot{P} = \{([x_{i-1}, x_i], t_i)\}$ is δ_ε -fine, then $[x_{i-1}, x_i] \subseteq [t_i - \frac{1}{4}d_\varepsilon, t_i + \frac{1}{4}d_\varepsilon]$, so $x_i - x_{i-1} \leq \frac{1}{2}d_\varepsilon < d_\varepsilon$ for every i ; thus $\|\dot{P}\| < d_\varepsilon$ and the Riemann estimate gives $|S(f, \dot{P}) - L| < \varepsilon$. Hence $f \in \mathbb{R}^*[a, b]$ with integral L . □

Consequently every step function, every continuous function, and every monotone function on $[a, b]$ lies in $\mathbb{R}^*[a, b]$, inheriting the Riemann integrability proved in Chapter 7. The gain is everything *beyond* that class, beginning with functions Riemann integration cannot even see.

Worked Example 12.1.7 (The Dirichlet function is generalized integrable (B&S 10.1.4a)). Let $f(x) = 1$ for $x \in \mathbb{Q} \cap [0, 1]$ and $f(x) = 0$ otherwise. (Chapter 7 showed $f \notin \mathcal{R}[0, 1]$.) Show $f \in \mathbb{R}^*[0, 1]$ with $\int_0^1 f = 0$.

Recognition. Trigger: “integrate a function that is nonzero only on a countable set.” Build a gauge that makes each rational tag contribute an arbitrarily tiny amount, by attaching a geometric budget $\varepsilon/2^k$ to the k -th rational.

Solution. Enumerate $\mathbb{Q} \cap [0, 1] = \{r_1, r_2, \dots\}$. Given $\varepsilon > 0$ define the gauge

$$\delta_\varepsilon(r_k) := \frac{\varepsilon}{2^{k+2}}, \quad \delta_\varepsilon(x) := 1 \text{ for } x \text{ irrational.}$$

If $\dot{P}$ is δ_ε -fine, the only nonzero terms of $S(f, \dot{P})$ come from rational tags $t_i = r_k$, where $f(r_k) = 1$ and the box length is $x_i - x_{i-1} \leq 2\delta_\varepsilon(r_k) = \varepsilon/2^{k+1}$. A given value r_k can be the tag of at most two subintervals, so

$$0 \leq S(f, \dot{P}) \leq \sum_{k=1}^{\infty} 2 \cdot \frac{\varepsilon}{2^{k+1}} = \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon.$$

Thus $|S(f, \dot{P}) - 0| \leq \varepsilon$, so $f \in \mathbb{R}^*[0, 1]$ and $\int_0^1 f = 0$. *Check.* The “area” is 0 as it should be — f is 0 off a measure-zero set. The Riemann sums could not be controlled by a *mesh* bound, because every short box still straddles a rational; a *gauge* pins each rational's contribution individually. ■

Worked Example 12.1.8 (An unbounded generalized integrable function (B&S 10.1.4b)). Let $H(1/k) := k$ for $k \in \mathbb{N}$ and $H(x) := 0$ elsewhere on $[0, 1]$. Since H is unbounded it is not Riemann integrable; show $H \in \mathbb{R}^*[0, 1]$ with $\int_0^1 H = 0$.

Recognition. Trigger: an unbounded function supported on the countable set $\{1/k\}$. Give the tag $1/k$ a gauge small enough that $H(1/k) = k$ times the box length stays within a geometric budget.

Solution. Given $\varepsilon > 0$ set $\delta_\varepsilon(1/k) := \varepsilon/(k \cdot 2^{k+2})$ and $\delta_\varepsilon(x) := 1$ elsewhere. If $\dot{P}$ is δ_ε -fine, nonzero terms come only from tags $t_i = 1/k$, where

$$0 < H(1/k)(x_i - x_{i-1}) = k(x_i - x_{i-1}) \leq k \cdot 2\delta_\varepsilon(1/k) = \frac{\varepsilon}{2^{k+1}}.$$

Each $1/k$ tags at most two subintervals, so $0 \leq S(H, \dot{P}) \leq \sum_k \varepsilon/2^k = \varepsilon$. Hence $H \in \mathbb{R}^*[0, 1]$ and $\int_0^1 H = 0$. *Check.* Boundedness is *not* required for membership in $\mathbb{R}^*$ — a first hint that $\mathbb{R}^*$ is genuinely larger. The proof of the Riemann Boundedness Theorem breaks precisely because we cannot choose one tag freely; the gauge has already constrained it. ■

Caution. Membership $f \in \mathbb{R}^*[a, b]$ does *not* imply f is bounded (Worked Example above), in sharp contrast to $\mathcal{R}[a, b]$. So none of the Riemann-era “bounded” hypotheses carry over automatically; restate every theorem with care.

Further worked examples

Worked Example 12.1.9 (Checking δ -finess of a concrete partition). On $[0, 1]$ take the gauge $\delta(t) = \frac{1}{4}(1-t) + \frac{1}{20}$ (so δ is large near 0, small near 1). Decide whether the tagged partition with division points $0 < \frac{1}{2} < \frac{3}{4} < 1$ and tags $t_1 = \frac{1}{4}$, $t_2 = \frac{5}{8}$, $t_3 = \frac{15}{16}$ is δ -fine.

Recognition. Trigger: “is this specific tagged partition fine for this gauge?” Check $[x_{i-1}, x_i] \subseteq [t_i - \delta(t_i), t_i + \delta(t_i)]$ box by box — it is a finite verification, nothing more.

Solution. Box 1 = $[0, \frac{1}{2}]$, $t_1 = \frac{1}{4}$: $\delta(\frac{1}{4}) = \frac{1}{4} \cdot \frac{3}{4} + \frac{1}{20} = \frac{3}{16} + \frac{1}{20} = 0.2375$, so the allowed window is $[0.0125, 0.4875]$. But $0 \notin [0.0125, 0.4875]$, so box 1 already *fails*. The partition is *not* δ -fine. *Check.* The failure is exactly where intuition predicts — near 0 the gauge is largest ($\delta(0) = 0.30$) yet the box $[0, \frac{1}{2}]$ has half-width 0.25 measured from its tag $\frac{1}{4}$ only when the tag sits at the *center*; here the tag $\frac{1}{4}$ is off-center, so the left end 0 escapes the window. Re-tagging box 1 with $t_1 = \frac{1}{4}$ replaced by the midpoint $\frac{1}{4}$ does not help; one needs $|0 - t_1| \leq \delta(t_1)$ and $|\frac{1}{2} - t_1| \leq \delta(t_1)$ simultaneously. ■

Worked Example 12.1.10 (The minimum of two gauges is a gauge (and only shrinks the fine family)). Let δ_1, δ_2 be gauges on $[a, b]$ and set $\delta(t) := \min\{\delta_1(t), \delta_2(t)\}$. Show δ is a gauge and that every δ -fine partition is both δ_1 -fine and δ_2 -fine.

Recognition. Trigger: the ubiquitous “combine two gauges” device used in Uniqueness, Linearity, and every existence proof. Reduce to the pointwise inequality $\delta \leq \delta_i$.

Solution. Since $\delta_1(t), \delta_2(t) > 0$ for every t , their minimum is > 0 , so $\delta : [a, b] \rightarrow (0, \infty)$ is strictly positive, hence a gauge. If $\dot{P} = \{([x_{i-1}, x_i], t_i)\}$ is δ -fine, then for each i ,

$$[x_{i-1}, x_i] \subseteq [t_i - \delta(t_i), t_i + \delta(t_i)] \subseteq [t_i - \delta_1(t_i), t_i + \delta_1(t_i)],$$

because $\delta(t_i) \leq \delta_1(t_i)$ widens the window. Thus $\dot{P}$ is δ_1 -fine, and symmetrically δ_2 -fine. *Check.* The inclusion of windows goes the “right” way: *smaller* gauge $\implies$ *narrower* window $\implies$ *fewer* fine partitions, so a partition surviving the stricter gauge survives the looser ones. This monotonicity is the entire content of the “take the minimum” trick. ■

Worked Example 12.1.11 (NON-example: the zero function on the rationals is *not* a gauge). Someone proposes “ $\delta(t) := 0$ if $t \in \mathbb{Q}$ and $\delta(t) := 1$ otherwise” to integrate the Dirichlet function. Explain why this is illegal, and why it cannot be repaired by allowing $\delta \geq 0$.

Recognition. Trigger: a tempting definition that violates the strict-positivity requirement. Test Definition 12.1.1 literally.

Solution. A gauge must be *strictly* positive: $\delta(t) > 0$ for *every* $t \in [a, b]$, including rationals. The proposed δ has $\delta(t) = 0$ on $\mathbb{Q}$, so it is not a gauge at all. The strict positivity is not a technicality: if $\delta(t) = 0$ were allowed at a tag t , the window $[t - 0, t + 0] = \{t\}$ would force a degenerate box of length 0 at t , and Cousin's lemma (Theorem 12.1.3) — whose proof needs $\delta(\xi) > 0$ to fit some I_k inside the $\delta(\xi)$ -ball — would collapse. The correct way to “almost forbid” rational tags is the *small but positive* gauge $\delta(r_k) = \varepsilon/2^{k+2}$ of the Dirichlet worked example, not a zero gauge. *Check.* With a genuinely zero value at one point, no δ -fine partition can have a box containing that point with any *other* tag, and the point itself only admits the length-0 box, which is not a legal subinterval — so the fine family is empty and the integral undefined. ■

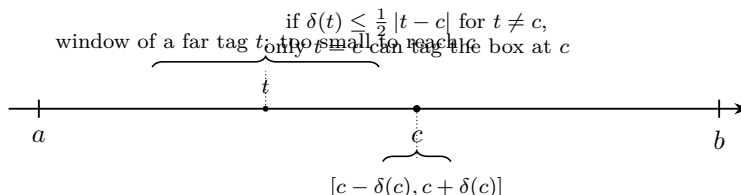


Figure 12.3. Tag-forcing by a gauge. Shrinking δ near a point c so that $\delta(t) \leq \frac{1}{2}|t - c|$ for $t \neq c$ makes every other tag's window too small to reach c ; the box containing c must therefore be tagged *at* c . This is the mechanism behind the Uniqueness, Additivity, and FTC proofs.

Exercises (Bartle & Sherbert, §10.1, gauges and the definition).

- Let δ be a gauge on $[a, b]$ and $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ a δ -fine partition. (a) Show $0 < x_i - x_{i-1} \leq 2\delta(t_i)$ for each i . (b) If $d := \sup\{\delta(t) : t \in [a, b]\} < \infty$, show $\|\dot{P}\| \leq 2d$. (c) If $d_- := \inf\{\delta(t) : t \in [a, b]\} > 0$, and $\dot{Q}$ is a tagged partition with $\|\dot{Q}\| \leq d_-$, show $\dot{Q}$ is δ -fine. (d) For $\varepsilon = 1$, show the gauge δ_1 of the Dirichlet Worked Example has $\inf\{\delta_1(t) : t \in [0, 1]\} = 0$.
- (a) Show that in any tagged partition each tag belongs to at most two subintervals. (b) Are there tagged partitions in which every tag belongs to exactly two subintervals?
- Let δ be a gauge on $[a, b]$ and $\dot{P}$ a δ -fine partition. (a) Show there is a δ -fine partition $\dot{Q}_1$ in which no tag belongs to two subintervals and $S(f, \dot{Q}_1) = S(f, \dot{P})$ for every f . (b) Does there exist a δ -fine $\dot{Q}_2$ in which every tag belongs to two subintervals with $S(f, \dot{Q}_2) = S(f, \dot{P})$ for every f ? (c) Show there is a δ -fine $\dot{Q}_3$ in which every tag is an endpoint of its subinterval and $S(f, \dot{Q}_3) = S(f, \dot{P})$ for every f .
- On $[0, 2]$ let $\delta(t) := \frac{1}{2}|t - 1|$ for $t \neq 1$ and $\delta(1) := 0.01$. Show every δ -fine partition of $[0, 2]$ has $t = 1$ as a tag of at least one subinterval, and that the total length of subintervals tagged by 1 is ≤ 0.02 .
- (a) Construct a gauge δ on $[0, 4]$ forcing 1, 2, 3 to be tags of any δ -fine partition. (b) Given any gauge δ_1 on $[0, 4]$, construct a gauge δ_2 such that every δ_2 -fine partition has 1, 2, 3 among its tags and is also δ_1 -fine.
- Show $f \in \mathbb{R}^*[a, b]$ with integral L if and only if for every $\varepsilon > 0$ there is a gauge γ_ε such that any tagged partition with $0 < x_i - x_{i-1} \leq \gamma_\varepsilon(t_i)$ for all i satisfies $|S(f, \dot{P}) - L| < \varepsilon$. (An equivalent “one-sided” definition.)
- Show each of the following belongs to $\mathbb{R}^*[0, 1]$ by finding F_k continuous on $[0, 1]$ with $F'_k = f_k$ off a finite set E_k : (a) $f_1(x) = (x + 1)/\sqrt{x}$ on $(0, 1]$, $f_1(0) = 0$; (b) $f_2(x) = x/\sqrt{1 - x}$ on $[0, 1)$, $f_2(1) = 0$; (c) $f_3(x) = \sqrt{x} \ln x$ on $(0, 1]$, $f_3(0) = 0$; (d) $f_4(x) = (\ln x)/\sqrt{x}$ on $(0, 1]$, $f_4(0) = 0$; (e) $f_5(x) = \sqrt{(1 + x)/(1 - x)}$ on $[0, 1)$, $f_5(1) = 0$; (f) $f_6(x) = 1/(\sqrt{x}\sqrt{2 - x})$ on $(0, 1]$, $f_6(0) = 0$.
- Explain why the argument proving boundedness of Riemann integrable functions (B&S 7.1.6) does not show that a function in $\mathbb{R}^*[a, b]$ is bounded. Identify the exact step that fails.
- Let $f(x) = 1/x$ for $x \in (0, 1]$ and $f(0) = 0$. Show $f \notin \mathbb{R}^*[0, 1]$. (Hint: compare f with the step functions $s_n = 1$ on $(\frac{1}{2}, 1]$, $s_n = 2$ on $(\frac{1}{3}, \frac{1}{2}]$, $\dots$, $s_n = n$ on $[0, \frac{1}{n}]$.)
- Let $k : [0, 1] \rightarrow \mathbb{R}$ be $k(x) = 0$ if x is 0 or irrational and $k(m/n) = n$ for m/n in lowest terms. Show $k \in \mathbb{R}^*[0, 1]$ with integral 0, that k is continuous nowhere, and that k is unbounded on every subinterval $[c, d]$ with $c < d$.

11. Let f be Dirichlet's function on $[0, 1]$ and $F(x) \equiv 0$. Since $F'(x) = f(x)$ for all $x \in [0, 1] \setminus \mathbb{Q}$, show that FTC First Form (Theorem 12.2.5) implies $f \in \mathbb{R}^*[0, 1]$.
12. Let $M(x) = \ln|x|$ for $x \neq 0$, $M(0) = 0$, so $M'(x) = 1/x$ for $x \neq 0$. Explain why it does *not* follow that $\int_{-2}^2 (1/x) dx = \ln|2| - \ln 2 = 0$.
13. Let $k(x) = x^{-1/2} \sin(1/x)$ on $(0, 1]$ and $k(0) = 0$. Show $k \in \mathbb{R}^*[0, 1]$. (Hint: differentiate $C(x) = x^{3/2} \cos(1/x)$, $C(0) = 0$.)
14. Let $g(x) = (1/x) \sin(1/x)$ on $(0, 1]$ and $g(0) = 0$. Show $g \in \mathbb{R}^*[0, 1]$. (Hint: differentiate $C(x) = x \cos(1/x)$, $C(0) = 0$.)

Additional exercises.

1. (Warm-up on gauges.) Let $\delta(t) := \frac{1}{4}$ be the constant gauge on $[0, 1]$. (a) Show a tagged partition is δ -fine iff its mesh is $\leq \frac{1}{4}$. (b) Conclude that for a constant gauge δ -finess reduces to the ordinary mesh condition of the Riemann integral.
2. Let δ_1, δ_2 be gauges on $[a, b]$ and set $\delta(t) := \min\{\delta_1(t), \delta_2(t)\}$. Show δ is a gauge and that every δ -fine partition is simultaneously δ_1 -fine and δ_2 -fine. (This is the standard trick for combining two ε -requirements.)
3. Using Cousin's Lemma (Theorem 12.1.3), prove that every continuous f on $[a, b]$ lies in $\mathbb{R}^*[a, b]$ is *not* what is asked here; instead, merely show that for every gauge δ on $[a, b]$ a δ -fine partition exists, and explain in one sentence why this is the property that makes the Henstock–Kurzweil definition non-vacuous.
4. Compute directly from the definition: show $f(x) := c$ (constant) is in $\mathbb{R}^*[a, b]$ with $\int_a^b f = c(b - a)$, by exhibiting, for each $\varepsilon > 0$, a gauge that works (any gauge does).
5. (Non-example.) Define $\delta(t) := |t - \frac{1}{2}|$ for $t \neq \frac{1}{2}$ and $\delta(\frac{1}{2}) := 0$ on $[0, 1]$. Show δ is *not* a gauge (the positivity requirement fails), and exhibit a consequence: there is no δ -fine tagged partition with $\frac{1}{2}$ as a tag.
6. Let $f \in \mathbb{R}^*[a, b]$ with integral L . Prove the integral is unique: if L' also satisfies the defining condition, then $L = L'$. (Cite Theorem 12.1.5 for the structure, but give the $\varepsilon/2$ argument yourself.)
7. (Diagram.) On a number line for $[0, 3]$, mark tags $t_1 = 0.5$, $t_2 = 1.5$, $t_3 = 2.5$ and draw, above each t_i , the bracket $[t_i - \delta(t_i), t_i + \delta(t_i)]$ for the gauge $\delta(t) := 0.4$. Shade which configurations of subinterval endpoints make the partition δ -fine, and write one sentence interpreting "the subinterval must fit inside the bracket of its tag."
8. (Refinement.) Suppose $\dot{P}$ is δ -fine and $\dot{Q}$ is obtained from $\dot{P}$ by splitting one subinterval $[x_{i-1}, x_i]$ at an interior point c , tagging both pieces by the same t_i . Show $\dot{Q}$ is still δ -fine and $S(f, \dot{Q}) = S(f, \dot{P})$ for every f .

12.2 Main Properties and the Fundamental Theorem

With the definition in hand we develop the calculus of the generalized integral. The algebraic properties mirror the Riemann case (with gauges replacing meshes), but the Fundamental Theorem is dramatically stronger: *every* derivative is integrable, so integrability becomes a conclusion rather than a hypothesis.

Linearity, order, and the Cauchy criterion

Theorem 12.2.1 (Linearity and monotonicity, B&S 10.1.5). *Let $f, g \in \mathbb{R}^*[a, b]$. Then:*

- (a) $kf \in \mathbb{R}^*[a, b]$ for $k \in \mathbb{R}$, and $\int_a^b kf = k \int_a^b f$;
- (b) $f + g \in \mathbb{R}^*[a, b]$, and $\int_a^b (f + g) = \int_a^b f + \int_a^b g$;
- (c) if $f(x) \leq g(x)$ for all $x \in [a, b]$, then $\int_a^b f \leq \int_a^b g$.

Proof. (b) Given $\varepsilon > 0$, by the Uniqueness-theorem device choose one gauge δ_ε (the pointwise minimum of a gauge for f and one for g) such that any δ_ε -fine $\dot{P}$ satisfies $|S(f, \dot{P}) - \int f| < \varepsilon/2$ and $|S(g, \dot{P}) - \int g| < \varepsilon/2$. Since the Riemann sum is additive in the integrand, $S(f + g, \dot{P}) = S(f, \dot{P}) + S(g, \dot{P})$, so

$$|S(f + g, \dot{P}) - (\int f + \int g)| \leq |S(f, \dot{P}) - \int f| + |S(g, \dot{P}) - \int g| < \varepsilon.$$

Hence $f + g \in \mathbb{R}^*$ with the stated integral. (a) is similar with a single gauge. (c) If $f \leq g$ then $S(f, \dot{P}) \leq S(g, \dot{P})$ for every $\dot{P}$ (each term obeys $f(t_i) \leq g(t_i)$); taking a common gauge and letting $\varepsilon \rightarrow 0$ gives $\int f \leq \int g$. $\square$

Theorem 12.2.2 (Cauchy criterion, B&S 10.1.6). *$f \in \mathbb{R}^*[a, b]$ if and only if for every $\varepsilon > 0$ there is a gauge η_ε on $[a, b]$ such that for all η_ε -fine partitions $\dot{P}, \dot{Q}$,*

$$|S(f, \dot{P}) - S(f, \dot{Q})| < \varepsilon.$$

Proof. ($\implies$) If $f \in \mathbb{R}^*$ with integral L , pick a gauge $\delta_{\varepsilon/2}$ for tolerance $\varepsilon/2$ and set $\eta_\varepsilon := \delta_{\varepsilon/2}$. Then η_ε -fine $\dot{P}, \dot{Q}$ each lie within $\varepsilon/2$ of L , so $|S(f, \dot{P}) - S(f, \dot{Q})| < \varepsilon$.

($\impliedby$) For each n , let δ_n be a gauge with the stated property at tolerance $1/n$; replacing δ_n by $\min\{\delta_1, \dots, \delta_n\}$ we may assume $\delta_{n+1} \leq \delta_n$. By Cousin's lemma pick a δ_n -fine partition $\dot{P}_n$. For $m > n$ both $\dot{P}_m, \dot{P}_n$ are δ_n -fine, so $|S(f, \dot{P}_n) - S(f, \dot{P}_m)| < 1/n$; thus $(S(f, \dot{P}_n))$ is Cauchy in $\mathbb{R}$ and converges to some A . Letting $m \rightarrow \infty$ gives $|S(f, \dot{P}_n) - A| \leq 1/n$. Now for $\varepsilon > 0$ pick $K > 2/\varepsilon$; any δ_K -fine $\dot{Q}$ obeys $|S(f, \dot{Q}) - A| \leq |S(f, \dot{Q}) - S(f, \dot{P}_K)| + |S(f, \dot{P}_K) - A| \leq 1/K + 1/K < \varepsilon$. Hence $f \in \mathbb{R}^*$ with integral A . $\square$

The Cauchy criterion is the workhorse for existence proofs where the value of the integral is unknown — exactly as in Chapter 7, but now phrased with gauges.

Additivity over subintervals

Theorem 12.2.3 (Additivity, B&S 10.1.8). *Let $c \in (a, b)$. Then $f \in \mathbb{R}^*[a, b]$ if and only if its restrictions to $[a, c]$ and $[c, b]$ are both in $\mathbb{R}^*$, and in that case*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Proof sketch. ($\impliedby$) Say the restrictions integrate to L_1, L_2 . Given $\varepsilon > 0$, take gauges δ' on $[a, c]$ and δ'' on $[c, b]$ for tolerance $\varepsilon/2$. Splice them into a gauge δ_ε on $[a, b]$ that additionally satisfies $\delta_\varepsilon(t) \leq \frac{1}{2}|c - t|$ for $t \neq c$; this forces every δ_ε -fine partition $\dot{Q}$ of $[a, b]$ to use c as a tag at any box meeting c (cf. the tag-forcing remark in §12.1.1). Such a $\dot{Q}$ then splits at c into a δ' -fine partition of $[a, c]$ and a δ'' -fine partition of $[c, b]$ with $S(f, \dot{Q}) = S(f, \dot{Q}_1) + S(f, \dot{Q}_2)$ (a box tagged at c may be cut at c without changing its contribution). Hence $|S(f, \dot{Q}) - (L_1 + L_2)| < \varepsilon$. ($\implies$) Use the Cauchy criterion on each half: extend partitions of $[a, c]$ by a fixed tail on $[c, b]$ so that differences over $[a, c]$ equal differences over $[a, b]$. $\square$

The Saks–Henstock lemma

The Saks–Henstock lemma upgrades the integral inequality from *whole* fine partitions to arbitrary *partial* collections of tagged intervals — the key technical tool behind the convergence theorems and the absolute-integrability theory. Call a finite collection $\{(J_j, s_j)\}_{j=1}^m$ of non-overlapping tagged subintervals of $[a, b]$ a δ -fine subpartition if each $J_j \subseteq [s_j - \delta(s_j), s_j + \delta(s_j)]$ with $s_j \in J_j$ (it need not cover $[a, b]$).

Lemma 12.2.4 (Saks–Henstock, B&S form). *Let $f \in \mathbb{R}^*[a, b]$ with integral L , and let $\varepsilon > 0$ with gauge δ_ε as in Definition 12.1.4, so that $|S(f, \dot{P}) - L| < \varepsilon$ for every δ_ε -fine partition $\dot{P}$. Write $\int_J f$ for the (generalized) integral over a subinterval $J = [u, v]$. Then for every δ_ε -fine subpartition $\{([u_j, v_j], s_j)\}_{j=1}^m$,*

$$|[\int_J f] \sum_{j=1}^m (f(s_j)(v_j - u_j) - \int_{u_j}^{v_j} f)| \leq \varepsilon,$$

and moreover, summing the absolute values term by term,

$$\sum_{j=1}^m \left| \left[f(s_j)(v_j - u_j) - \int_{u_j}^{v_j} f \right] \right| \leq 2\varepsilon.$$

Proof idea. The gap $[a, b] \setminus \bigcup_j [u_j, v_j]$ is a finite union of closed intervals $K_1, \dots, K_p$. On each K_i the restriction of f is in $\mathbb{R}^*$ (additivity), so for any $\rho > 0$ choose a δ_ε -fine partition $\dot{R}_i$ of K_i with $\left| S(f, \dot{R}_i) - \int_{K_i} f \right| < \rho/p$. Joining the given subpartition with all the $\dot{R}_i$ gives a δ_ε -fine partition of $[a, b]$; applying $|S(f, \cdot) - L| < \varepsilon$ and additivity $L = \sum_j \int_{u_j}^{v_j} f + \sum_i \int_{K_i} f$ leaves exactly $\left| \sum_j (f(s_j)(v_j - u_j) - \int_{u_j}^{v_j} f) \right| < \varepsilon + \rho$; let $\rho \rightarrow 0$. The absolute-value form follows by splitting the index set into the j where the bracket is ≥ 0 and where it is < 0 , applying the first estimate to each half, and adding (whence the factor 2). $\square$

Remark. Read informally: not only does the full Riemann sum approximate L , but on *any* piece of a fine partition the local Riemann term $f(s_j)(v_j - u_j)$ approximates the local integral $\int_{u_j}^{v_j} f$, with the errors controlled *in total* (and even in absolute total). This “local” control is what lets one integrate $|f|$ and pass limits inside the integral.

The fundamental theorem (first form)

Theorem 12.2.5 (FTC, First Form, B&S 10.1.9). *Suppose $E \subseteq [a, b]$ is countable and $F, f : [a, b] \rightarrow \mathbb{R}$ satisfy*

- (a) F is continuous on $[a, b]$, and
- (b) $F'(x) = f(x)$ for all $x \in [a, b] \setminus E$.

Then $f \in \mathbb{R}^[a, b]$ and*

$$\int_a^b f = F(b) - F(a).$$

Proof. We prove the case $E = \emptyset$ (the countable-exceptional-set case is Exercise 14, handled by an extra geometric budget on E). Fix $\varepsilon > 0$. For each $t \in [a, b]$, differentiability gives $\delta_\varepsilon(t) > 0$ such that

$$0 < |z - t| \leq \delta_\varepsilon(t), \quad z \in [a, b] \quad \implies \quad \left| \frac{F(z) - F(t)}{z - t} - f(t) \right| < \frac{1}{2}\varepsilon,$$

i.e. $|F(z) - F(t) - f(t)(z - t)| \leq \frac{1}{2}\varepsilon |z - t|$ for $z \in [t - \delta_\varepsilon(t), t + \delta_\varepsilon(t)] \cap [a, b]$. This δ_ε is our gauge. Now for $u \leq t \leq v$ with $[u, v] \subseteq [t - \delta_\varepsilon(t), t + \delta_\varepsilon(t)]$, subtract and add $F(t) - f(t)t$ and use the triangle inequality (with $v - t \geq 0, t - u \geq 0$):

$$|F(v) - F(u) - f(t)(v - u)| \leq |F(v) - F(t) - f(t)(v - t)| + |F(t) - F(u) - f(t)(t - u)| \leq \frac{1}{2}\varepsilon(v - u).$$

If $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is δ_ε -fine, the telescoping identity $F(b) - F(a) = \sum_i (F(x_i) - F(x_{i-1}))$ and the bound above give

$$|F(b) - F(a) - S(f, \dot{P})| \leq \sum_{i=1}^n |F(x_i) - F(x_{i-1}) - f(t_i)(x_i - x_{i-1})| \leq \sum_{i=1}^n \frac{1}{2}\varepsilon(x_i - x_{i-1}) = \frac{1}{2}\varepsilon(b - a).$$

Since $\varepsilon > 0$ is arbitrary, $f \in \mathbb{R}^*[a, b]$ and $\int_a^b f = F(b) - F(a)$. $\square$

This is the headline advantage of the gauge integral. For the Riemann integral the Fundamental Theorem requires $f = F'$ to be *assumed integrable*; here integrability is *derived* from differentiability of F alone. The Volterra-type counterexamples (a differentiable F whose F' is not Riemann integrable) are integrated effortlessly.

Worked Example 12.2.6 (Integrating an unbounded derivative (B&S 10.1.10a)). Let $H(x) = 2\sqrt{x}$ on $[0, b]$. Compute $\int_0^b x^{-1/2} dx$.

Recognition. Trigger: an antiderivative continuous on the closed interval whose derivative blows up at an endpoint. Apply FTC First Form with $E = \{0\}$.

Solution. H is continuous on $[0, b]$ and $H'(x) = x^{-1/2}$ for $x \in (0, b]$. Set $h(x) := H'(x)$ on $(0, b]$ and $h(0) := 0$. By Theorem 12.2.5 with $E = \{0\}$, $h \in \mathbb{R}^*[0, b]$ and

$$\int_0^b \frac{1}{\sqrt{x}} dx = H(b) - H(0) = 2\sqrt{b}.$$

Check. h is unbounded near 0, so the Riemann integral does not exist; the generalized integral simply reads off $2\sqrt{b}$ (and agrees with the classical *improper* value, as §12.3.1 will explain). ■

Worked Example 12.2.7 ($\int_0^b \ln x dx$ via FTC (B&S 10.1.10c)). Show $\int_0^b \ln x dx = b \ln b - b$.

Recognition. Trigger: integrand unbounded at 0 but with a continuous antiderivative there. Take $L(x) = x \ln x - x$, continuous at 0 by l'Hospital.

Solution. $L(x) = x \ln x - x$ extends continuously to $L(0) = 0$ (since $x \ln x \rightarrow 0$), and $L'(x) = \ln x$ for $x \in (0, b]$. By FTC First Form with $E = \{0\}$, $\ell(x) := \ln x$ (set $\ell(0) := 0$) lies in $\mathbb{R}^*[0, b]$ and $\int_0^b \ln x dx = L(b) - L(0) = b \ln b - b$. *Check.* At $b = 1$ this gives $-1 = \int_0^1 \ln x dx$, the familiar value. ■

Worked Example 12.2.8 (NON-example: why $\int_{-2}^2 dx/x$ is still undefined (B&S Exer 10.1.12)). Let $M(x) = \ln|x|$ for $x \neq 0$, $M(0) = 0$, so $M'(x) = 1/x$ for $x \neq 0$. One is tempted to write $\int_{-2}^2 \frac{1}{x} dx = M(2) - M(-2) = \ln 2 - \ln 2 = 0$. Explain why FTC First Form does *not* apply.

Recognition. Trigger: a tempting telescoping that ignores a hypothesis. Check continuity of the antiderivative on the *whole* closed interval.

Solution. FTC First Form requires F *continuous on all of* $[a, b]$, with at most a *countable* exceptional set for the derivative. Here $M(x) = \ln|x| \rightarrow -\infty$ as $x \rightarrow 0$, so M has no continuous value at 0: hypothesis (a) fails. Indeed $1/x$ is genuinely *not* generalized integrable on $[-2, 2]$ (it is not integrable even on $(0, 1]$; see §12.3.1 and B&S Exer 10.1.9), so the symmetric “0” is meaningless, not a principal value granted by the theory. *Lesson.* Continuity of the antiderivative across the bad point is essential; a jump or blow-up there voids the conclusion. ■

Substitution, products, and integration by parts

Because FTC First Form is so strong, the standard calculus theorems improve. We record the statements (proofs are in B&S / [MTI]) and illustrate one.

Theorem 12.2.9 (Substitution, B&S 10.1.12). *Let $\varphi : [\alpha, \beta] \rightarrow \mathbb{R}$ be continuous with $\varphi([\alpha, \beta]) \subseteq [a, b]$, and $F : [a, b] \rightarrow \mathbb{R}$ continuous. Suppose $f = F'$ off a set E_f , φ' exists off E_φ , and $E := \varphi^{-1}(E_f) \cup E_\varphi$ is countable (with $f := 0$ on E_f , $\varphi' := 0$ on E_φ). Then $(f \circ \varphi) \cdot \varphi' \in \mathbb{R}^*[\alpha, \beta]$ and*

$$\int_\alpha^\beta (f \circ \varphi) \varphi' = \int_{\varphi(\alpha)}^{\varphi(\beta)} f.$$

Theorem 12.2.10 (Multiplication, B&S 10.1.14). *If $f \in \mathbb{R}^*[a, b]$ and g is monotone on $[a, b]$, then $fg \in \mathbb{R}^*[a, b]$.*

The product of two generalized integrable functions need *not* be integrable (Exercise 10.1.18), so the monotonicity of one factor in Theorem 12.2.10 is essential.

Theorem 12.2.11 (Integration by parts, B&S 10.1.15). *Let F, G be differentiable on $[a, b]$. Then $F'G \in \mathbb{R}^*[a, b]$ if and only if $FG' \in \mathbb{R}^*[a, b]$, and in that case*

$$\int_a^b F'G = [FG]_a^b - \int_a^b FG'.$$

Theorem 12.2.12 (Taylor with integral remainder, B&S 10.1.16). If $f, f', \dots, f^{(n+1)}$ exist on $[a, b]$, then

$$f(b) = \sum_{j=0}^n \frac{f^{(j)}(a)}{j!} (b-a)^j + R_n, \quad R_n = \frac{1}{n!} \int_a^b f^{(n+1)}(t) (b-t)^n dt,$$

where the integral exists because $f^{(n+1)}$ is a derivative (Theorem 12.2.5) and $t \mapsto (b-t)^n$ is monotone (Theorem 12.2.10); no separate integrability hypothesis on $f^{(n+1)}$ is needed.

Worked Example 12.2.13 (Substitution through a singular point (B&S 10.1.13a)). Evaluate $\int_0^4 \frac{\cos \sqrt{t}}{\sqrt{t}} dt$, whose integrand blows up as $t \rightarrow 0^+$.

Recognition. Trigger: an unbounded integrand that becomes tame after $x = \sqrt{t}$. The classical substitution theorem stumbles on the singularity; Theorem 12.2.9 does not.

Solution. Put $x = \varphi(t) = \sqrt{t}$, so $\varphi'(t) = 1/(2\sqrt{t})$ on $(0, 4]$ (set $\varphi'(0) := 0$). With $F(x) = 2 \sin x$, $f(x) = F'(x) = 2 \cos x$, the integrand is $f(\varphi(t))\varphi'(t) = 2 \cos \sqrt{t} \cdot \frac{1}{2\sqrt{t}} = \frac{\cos \sqrt{t}}{\sqrt{t}}$ for $t \neq 0$. Theorem 12.2.9 with $E = \{0\}$ gives

$$\int_{t=0}^4 \frac{\cos \sqrt{t}}{\sqrt{t}} dt = \int_{x=0}^2 2 \cos x dx = 2 \sin 2.$$

Check. Numerically $2 \sin 2 \approx 1.819$, finite despite the integrand's blow-up at 0 — the singularity is integrable and the substitution is now fully justified. ■

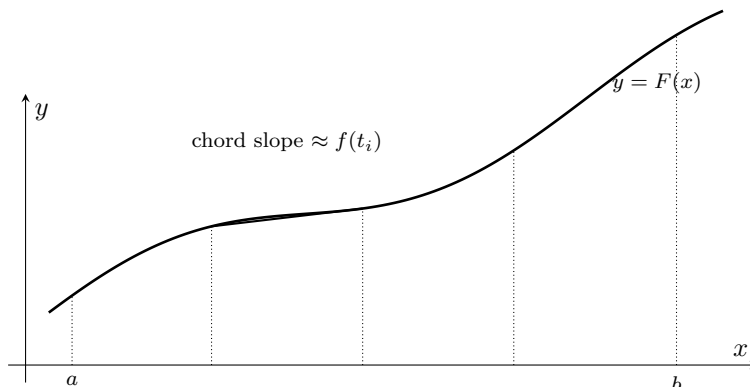


Figure 12.4. FTC First Form as a telescoping estimate. Over each box the true increment $F(x_i) - F(x_{i-1})$ (the chord rise) is matched by the Riemann term $f(t_i)(x_i - x_{i-1})$ (slope $f(t_i)$ times width). The gauge makes each local match good to $\frac{1}{2}\varepsilon(x_i - x_{i-1})$; the increments telescope to $F(b) - F(a)$, so $S(f, \dot{P}) \approx F(b) - F(a)$.

Further worked examples

Worked Example 12.2.14 (Integration by parts: $\int_0^1 x \ln x dx$). Evaluate $\int_0^1 x \ln x dx$, whose integrand has a (removable) singularity at 0.

Recognition. Trigger: a product $F'G$ with $G = \ln x$ unbounded at 0; use the gauge-integral parts formula (Theorem 12.2.11), which needs no separate integrability hypothesis.

Solution. Take $F(x) = \frac{1}{2}x^2$ (so $F' = x$) and $G(x) = \ln x$ (so $G' = 1/x$ on $(0, 1]$); both are differentiable on $(0, 1]$ and the product $F'G = \frac{1}{2}x^2 \ln x \rightarrow 0$ at 0, so $[FG]_0^1$ is well defined. Theorem 12.2.11 gives

$$\int_0^1 x \ln x dx = \left[\frac{1}{2}x^2 \ln x \right]_0^1 - \int_0^1 \frac{1}{2}x^2 \cdot \frac{1}{x} dx = 0 - \int_0^1 \frac{1}{2}x dx = -\frac{1}{4}.$$

The boundary term vanishes at both ends: at $x = 1$, $\ln 1 = 0$; at $x = 0$, $x^2 \ln x \rightarrow 0$. *Check.* The integrand is ≤ 0 on $(0, 1)$ (since $\ln x < 0$ there), so a negative value is right; numerically $\int_0^1 x \ln x dx = -0.25$ exactly, and the antiderivative $\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2$ evaluated 0 to 1 returns $-\frac{1}{4}$. ■

Worked Example 12.2.15 (The Multiplication Theorem in action: $f \cdot g$ with g monotone). Let $f(x) = x^{-1/2}$ on $(0, 1]$ (with $f(0) := 0$), known to be in $\mathbb{R}^*[0, 1]$, and let $g(x) = \arctan x$. Show $fg \in \mathbb{R}^*[0, 1]$ *without* computing the integral, and contrast with a non-monotone g .

Recognition. Trigger: “is a product integrable?” Theorem 12.2.10 grants $fg \in \mathbb{R}^*$ as soon as *one* factor is monotone; recognize $\arctan$ as monotone.

Solution. The function $g(x) = \arctan x$ is strictly increasing on $[0, 1]$ (its derivative $1/(1+x^2) > 0$), hence monotone. Since $f \in \mathbb{R}^*[0, 1]$ and g is monotone, the Multiplication Theorem (Theorem 12.2.10) gives $fg = \frac{\arctan x}{\sqrt{x}} \in \mathbb{R}^*[0, 1]$ immediately — no estimate of the product is needed. *Check.* Near 0 we have $\arctan x \sim x$, so $fg = \arctan x/\sqrt{x} \sim x/\sqrt{x} = \sqrt{x} \rightarrow 0$: the product is in fact bounded here, consistent with integrability. The monotonicity hypothesis is what does the work in general — replacing g by an oscillating factor like $\sin(1/x)$, the theorem no longer applies, and indeed the product of two $\mathbb{R}^*$ functions can fail to be in $\mathbb{R}^*$ (B&S Exer 10.1.18). ■

Worked Example 12.2.16 (NON-example: the product of two $\mathbb{R}^*$ functions need not be in $\mathbb{R}^*$). Let $w \in \mathbb{R}^*[0, 1]$ be a conditionally integrable function with $|w| \notin \mathbb{R}^*[0, 1]$ (such a w exists, §12.3.1). Show that $w \cdot \operatorname{sgn}(w)$ fails to be generalized integrable, so “ $f, g \in \mathbb{R}^* \Rightarrow fg \in \mathbb{R}^*$ ” is false.

Recognition. Trigger: a claimed closure property that fails; produce the standard counterexample using $|w| = w \cdot \operatorname{sgn} w$.

Solution. Set $g := \operatorname{sgn}(w)$, the sign function of w , which takes values in $\{-1, 0, 1\}$ and is therefore bounded (indeed $g \in \mathbb{R}^*[0, 1]$, being a bounded function that is a pointwise limit of step functions). If the product wg were in $\mathbb{R}^*[0, 1]$, then since $wg = w \cdot \operatorname{sgn}(w) = |w|$ pointwise, we would conclude $|w| \in \mathbb{R}^*[0, 1]$ — contradicting the choice of w as conditionally (non-absolutely) integrable. Hence $wg \notin \mathbb{R}^*[0, 1]$, even though $w, g \in \mathbb{R}^*[0, 1]$.

Lesson. Integrability is *not* closed under multiplication; one factor being *monotone* (Theorem 12.2.10) is a genuine restriction, not a convenience. *Check.* The obstruction is exactly the loss of the absolute property in $\mathbb{R}^*$ — the same phenomenon that separates $\mathbb{R}^*$ from L . ■

Exercises (Bartle & Sherbert, §10.1, the Fundamental Theorem and beyond).

- (FTC with a countable exceptional set.) Let $E = \{c_1, c_2, \dots\}$, F continuous on $[a, b]$, $F'(x) = f(x)$ for $x \in [a, b] \setminus E$, and $f(c_k) := 0$. Prove $f \in \mathbb{R}^*[a, b]$ with $\int_a^b f = F(b) - F(a)$, completing Theorem 12.2.5. (a) For $\varepsilon > 0$ and $t \in [a, b] \setminus E$, take $\delta_\varepsilon(t)$ as in the proof of the $E = \emptyset$ case; choose $\delta_\varepsilon(c_k) > 0$ so that $|z - c_k| < \delta_\varepsilon(c_k)$, $z \in [a, b]$, implies $|F(z) - F(c_k)| < \varepsilon/2^{k+2}$. (b) Show a δ_ε -fine partition with a tag $t_i = c_k$ has $|F(x_i) - F(x_{i-1}) - f(c_k)(x_i - x_{i-1})| < \varepsilon/2^{k+1}$. (c) Conclude $|S(f, P) - (F(b) - F(a))| < \varepsilon(b - a + 1)$.
- Use the Substitution Theorem (Theorem 12.2.9) to evaluate: (a) $\int_{-3}^3 (2t + 1) \operatorname{sgn}(t^2 + t - 2) dt = 6$; (b) $\int_0^4 \frac{\sqrt{t}}{1+t} dt$; (c) $\int_1^5 \frac{dt}{t\sqrt{t-1}} = 2 \arctan 2$; (d) $\int_0^1 \sqrt{1-t^2} dt$.
- Give an example of $f \in \mathbb{R}^*[0, 1]$ whose square f^2 does *not* belong to $\mathbb{R}^*[0, 1]$.
- Let $F(x) = x \cos(\pi/x)$ on $(0, 1]$, $F(0) = 0$, and $f := F'$ (with $f(0) := 0$). It will be seen that $f \in \mathbb{R}^*[0, 1]$ but $|f| = |F'| \notin \mathbb{R}^*[0, 1]$. (a) Show F' and $|F'|$ are continuous on each $[c, 1]$, $0 < c < 1$, and $f \in \mathbb{R}^*[0, 1]$. (b) With $a_k = 2/(2k+1)$, $b_k = 1/k$, the intervals $[a_k, b_k]$ are non-overlapping and $\int_{a_k}^{b_k} |f| \geq 1/k$. (c) Since $\sum 1/k$ diverges, conclude $|f| \notin \mathbb{R}^*[0, 1]$.
- With f as in the previous problem, let $m(x) = (-1)^k$ for $x \in [a_k, b_k]$ and $m(x) = 0$ elsewhere. Show $mf = |mf|$, that $m, |m|$ are bounded and in $\mathcal{R}[0, 1]$, and conclude that the product of a function in $\mathbb{R}^*[0, 1]$ and a bounded function in $\mathcal{R}[0, 1]$ may fail to lie in $\mathbb{R}^*[0, 1]$.
- Let $F(x) = x |\cos(\pi/x)|$ on $(0, 1]$, $F(0) = 0$. Then F is continuous on $[0, 1]$ and $F'(x)$ exists off $E = \{0\} \cup \{a_k : k \in \mathbb{N}\}$, $a_k = 2/(2k+1)$. Set $w(x) = F'(x)$ off E , $w = 0$ on E . Show w is unbounded on $[0, 1]$, that (by FTC with E countable) $w \in \mathbb{R}^*[0, 1]$ with $\int_a^b w = F(b) - F(a)$, and that $|w| \notin \mathbb{R}^*[0, 1]$.
- Let $C(x) = x^2 |\cos(\pi/x)|$ on $(0, 1]$, $C(0) = 0$, so C' exists off $E_1 = \{a_k\}$. Set $c(x) = C'(x)$ off E_1 , $c = 0$ on E_1 . Show c is bounded and in $\mathcal{R}[0, 1]$, that $\int_a^b c = C(b) - C(a)$, and that $|c| \in \mathcal{R}[0, 1]$.

8. (First Mean Value Theorem, generalized.) If f is continuous on $[a, b]$, $p \in \mathbb{R}^*[a, b]$ does not change sign, and $fp \in \mathbb{R}^*[a, b]$, show there is $\xi \in [a, b]$ with $\int_a^b fp = f(\xi) \int_a^b p$.
9. (Second Mean Value Theorem, a form.) Let $f \in \mathbb{R}^*[a, b]$, g monotone on $[a, b]$, and $f \geq 0$. Show there is $\xi \in [a, b]$ with $\int_a^b fg = g(a) \int_a^\xi f + g(b) \int_\xi^b f$.
10. Show $\int_0^b x^{\alpha-1} dx = b^\alpha/\alpha$ for $\alpha > 0$ by exhibiting a continuous antiderivative on $[0, b]$ and applying FTC First Form with $E = \{0\}$.
11. Use FTC First Form to evaluate $\int_{-1}^1 \frac{dx}{\sqrt{1-x^2}}$, identifying the countable exceptional set and a continuous antiderivative.

Additional exercises.

1. (Linearity drill.) Using Theorem 12.2.1, evaluate $\int_0^1 (3x^2 - 2x + 5) dx$ by writing it as a combination of $\int_0^1 x^2$, $\int_0^1 x$, $\int_0^1 1$ and citing the FTC for each.
2. Let $f \in \mathbb{R}^*[a, b]$ and $c \in (a, b)$. Prove additivity over the join: if $f \in \mathbb{R}^*[a, c]$ and $f \in \mathbb{R}^*[c, b]$, then $f \in \mathbb{R}^*[a, b]$ with $\int_a^b f = \int_a^c f + \int_c^b f$. (Combine gauges on the two pieces.)
3. Apply the Fundamental Theorem (First Form, Theorem 12.2.5) to evaluate $\int_0^\pi \sin x dx$ and $\int_1^e (1/x) dx$, naming the antiderivative in each case.
4. (Integration by parts.) Use Theorem 12.2.11 to evaluate $\int_0^1 xe^x dx$ and $\int_1^e \ln x dx$.
5. (Substitution.) Use the Substitution Theorem 12.2.9 to evaluate $\int_0^2 x\sqrt{1+x^2} dx$ via $u = 1 + x^2$, stating the new limits.
6. (Non-example.) Exhibit $f \in \mathbb{R}^*[0, 1]$ such that $|f| \notin \mathbb{R}^*[0, 1]$, hence the generalized integral is *not* an absolute integral. (You may cite the $F(x) = x \cos(\pi/x)$ construction; give the one-line reason $\int |f|$ diverges.)
7. Let $f \in \mathbb{R}^*[a, b]$ with $f(x) \geq 0$ for all x . Prove $\int_a^b f \geq 0$. Then show that if additionally f is continuous and $\int_a^b f = 0$, then $f \equiv 0$.
8. (Diagram.) Sketch the graph of $F(x) = \int_0^x f$ for the step function $f = 1$ on $[0, 1)$, $f = -1$ on $[1, 2]$. Mark the corner at $x = 1$ and state where F is differentiable and where $F' = f$. This illustrates the Second Form of the FTC and its exceptional point.
9. (Multiplication theorem.) Using Theorem 12.2.10, show that if $f \in \mathbb{R}^*[a, b]$ and g is monotone on $[a, b]$, then $fg \in \mathbb{R}^*[a, b]$. Give a one-sentence reason this can fail without monotonicity (point to Exercise on f^2 in §10.2).

12.3 Improper and Lebesgue Integrals

In calculus an *improper* integral over $[a, b]$ — when f blows up or oscillates wildly near a point — is defined by a separate limiting process, $\int_a^b f = \lim_{\gamma \rightarrow b^-} \int_a^\gamma f$. For the generalized integral no such patch is needed: Hake's theorem says the limit, if it exists, *equals an integral already in the theory*. We then isolate the Lebesgue integrable functions as those that are *absolutely* generalized integrable.

Hake's theorem: improper integrals are automatic

Theorem 12.3.1 (Hake, B&S 10.2.1). *Let $f : [a, b] \rightarrow \mathbb{R}$. Then $f \in \mathbb{R}^*[a, b]$ if and only if for every $\gamma \in (a, b)$ the restriction $f|_{[a, \gamma]} \in \mathbb{R}^*[a, \gamma]$ and the limit*

$$\lim_{\gamma \rightarrow b^-} \int_a^\gamma f = A$$

exists in $\mathbb{R}$. In that case $\int_a^b f = A$.

The proof (in [MTI]) takes an increasing sequence $\gamma_n \rightarrow b$, integrates over each $[a, \gamma_n]$, and *splices the gauges* on the pieces $[\gamma_{i-1}, \gamma_i]$ into a single gauge on $[a, b]$. The construction is delicate but the statement is what matters:

The generalized Riemann integral cannot be enlarged by the improper limiting process. If $f|_{[a, \gamma]} \in \mathbb{R}^$ for all $\gamma < b$ and $\lim_{\gamma \rightarrow b^-} \int_a^\gamma f$ exists, then f is already in $\mathbb{R}^*[a, b]$ with that value.*

This gives a practical test: to show $f \in \mathbb{R}^*[a, b]$ and find its value, compute $\int_a^\gamma f$ by elementary means and take $\gamma \rightarrow b^-$.

Worked Example 12.3.2 (An improper integral handled by Hake (B&S 10.2 opening)). Show $h(x) = x^{-1/2}$ (with $h(0) := 0$) is in $\mathbb{R}^*[0, 1]$ and $\int_0^1 x^{-1/2} dx = 2$, using Hake.

Recognition. Trigger: “a textbook improper integral; confirm it is a genuine generalized integral.” Integrate over $[\gamma, 1]$ (singularity at the left end) and let $\gamma \rightarrow 0^+$.

Solution. For $\gamma \in (0, 1)$, h is continuous on $[\gamma, 1]$, so $\int_\gamma^1 x^{-1/2} dx = [2\sqrt{x}]_\gamma^1 = 2 - 2\sqrt{\gamma}$. As $\gamma \rightarrow 0^+$ this $\rightarrow 2$. By Hake’s theorem (left-endpoint version), $h \in \mathbb{R}^*[0, 1]$ with $\int_0^1 h = 2$. *Check.* This matches the FTC computation $H(1) - H(0) = 2$ from §12.2.5; the two routes agree, as Hake guarantees. ■

A conditionally convergent integral: $\sin x/x$ in miniature

The function whose improper integral converges but whose absolute value does not is the signature of the gauge integral. The cleanest finite-interval model encodes a conditionally convergent series.

Worked Example 12.3.3 (Encoding a series as a generalized integral (B&S 10.2.2)). Let $\sum_{k=1}^\infty a_k = A$ converge. Define w on $[0, 1]$ by $w(x) = 2^k a_k$ for $x \in [c_{k-1}, c_k)$, where $c_k = 1 - 2^{-k}$, and $w(1) = 0$. Show $w \in \mathbb{R}^*[0, 1]$ with $\int_0^1 w = A$, and that $|w| \in \mathbb{R}^*[0, 1]$ iff $\sum |a_k| < \infty$.

Recognition. Trigger: “build a function whose integral reproduces a given series.” Each step has height $2^k a_k$ and width $c_k - c_{k-1} = 2^{-k}$, so its area is a_k ; sum via Hake.

Solution. On $[c_{k-1}, c_k)$ the step has width 2^{-k} and height $2^k a_k$, contributing area a_k . For $\gamma \in [c_n, c_{n+1})$ the restriction of w to $[0, \gamma]$ is a step function and

$$\int_0^\gamma w = a_1 + \cdots + a_n + r_\gamma, \quad |r_\gamma| \leq |a_{n+1}|.$$

Since $\sum a_k = A$ converges, $r_\gamma \rightarrow 0$ and $\int_0^\gamma w \rightarrow A$ as $\gamma \rightarrow 1^-$; by Hake, $w \in \mathbb{R}^*[0, 1]$ and $\int_0^1 w = A$. The same computation for $|w|$ replaces a_k by $|a_k|$, so $|w| \in \mathbb{R}^*[0, 1]$ iff $\sum |a_k| < \infty$. *Check.* Taking $a_k = (-1)^k/k$ gives a $w \in \mathbb{R}^*[0, 1]$ with $|w| \notin \mathbb{R}^*[0, 1]$, since $\sum 1/k = \infty$ — a generalized integrable function that is not absolutely integrable. ■

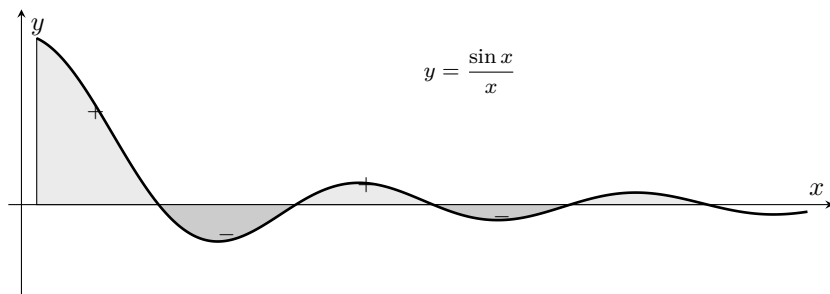


Figure 12.5. The Dirichlet integrand $\sin x/x$. Successive lobes alternate sign with slowly decaying area, so the signed areas *cancel* and $\int_0^\infty \frac{\sin x}{x} dx$ converges (to $\pi/2$, §12.5). But the unsigned areas form a divergent series, so $|\sin x/x|$ is not integrable: $\sin x/x \in \mathbb{R}^*$ yet not in L . The finite-interval model w above captures exactly this cancellation/non-absolute behaviour.

Lebesgue integrable = absolutely generalized integrable

Definition 12.3.4 (Lebesgue integrable, B&S 10.2.3). A function $f \in \mathbb{R}^*[a, b]$ with $|f| \in \mathbb{R}^*[a, b]$ is called *Lebesgue integrable* on $[a, b]$; we write $f \in L[a, b]$.

This is a non-standard but equivalent route to the Lebesgue integral: $L[a, b]$ is precisely the class of functions that are generalized integrable *together with* their absolute value. The usual measure-theoretic construction (Chapter 13) produces the same class. Among the gauge integrable functions, the Lebesgue ones are exactly the *absolutely* integrable ones.

Theorem 12.3.5 (Comparison Test, B&S 10.2.4). *If $f, v \in \mathbb{R}^*[a, b]$ and $|f(x)| \leq v(x)$ for all $x \in [a, b]$, then $f \in L[a, b]$ and*

$$\left| \int_a^b f \right| \leq \int_a^b |f| \leq \int_a^b v.$$

Proof. That $|f| \in \mathbb{R}^*[a, b]$ is the deep half (it uses the Saks–Henstock lemma; see [MTI]). Given it, $|f| \geq 0$ forces $f \in L[a, b]$ by definition. For the inequalities, $-|f| \leq f \leq |f|$ and monotonicity (Theorem 12.2.1c) give $-\int |f| \leq \int f \leq \int |f|$, i.e. $|\int f| \leq \int |f|$; and $|f| \leq v$ gives $\int |f| \leq \int v$. $\square$

Theorem 12.3.6 ($L[a, b]$ is a vector lattice, B&S 10.2.5–10.2.7). *If $f, g \in L[a, b]$ and $c \in \mathbb{R}$, then $cf, f + g, \max\{f, g\}, \min\{f, g\} \in L[a, b]$, with*

$$\int_a^b cf = c \int_a^b f, \quad \int_a^b |f + g| \leq \int_a^b |f| + \int_a^b |g|.$$

Proof. $|cf| = |c||f| \in \mathbb{R}^*$ gives $cf \in L$. For $f + g$, the triangle inequality $|f + g| \leq |f| + |g| =: v \in \mathbb{R}^*$ and the Comparison Test give $f + g \in L$ with the stated seminorm inequality. The lattice operations follow from $\max\{f, g\} = \frac{1}{2}(f + g + |f - g|)$ and $\min\{f, g\} = \frac{1}{2}(f + g - |f - g|)$, each a combination of L -functions. $\square$

Theorem 12.3.7 (One-sided domination suffices, B&S 10.2.6). *For $f \in \mathbb{R}^*[a, b]$ the following are equivalent: (a) $f \in L[a, b]$; (b) there is $v \in L[a, b]$ with $f \leq v$ on $[a, b]$; (c) there is $\alpha \in L[a, b]$ with $\alpha \leq f$ on $[a, b]$.*

Proof. (a) $\implies$ (b): take $v = f$. (b) $\implies$ (a): write $f = v - (v - f)$ with $v - f \geq 0$ and $v - f \in \mathbb{R}^*$, so $v - f \in L$; then $f = v - (v - f) \in L$ by Theorem 12.3.6. (a) $\iff$ (c) is symmetric. $\square$

Caution. “Generalized integrable” and “Lebesgue integrable” are *not* the same: $\mathbb{R}^*$ is strictly larger than L . The conditionally convergent examples (w above, and $\sin x/x$ on $[0, \infty)$) lie in $\mathbb{R}^* \setminus L$. The price of integrating them is the loss of the *absolute* property: $f \in \mathbb{R}^*$ does *not* give $|f| \in \mathbb{R}^*$. When you need $\int |f|$, $\|f\|_1$, or any convergence theorem with a dominating function, you are working in L , not merely $\mathbb{R}^*$.

The seminorm on $L[a, b]$

Definition 12.3.8 (Seminorm and distance, B&S 10.2.9). For $f \in L[a, b]$ put $\|f\| := \int_a^b |f|$, and for $f, g \in L[a, b]$ put $\text{dist}(f, g) := \|f - g\| = \int_a^b |f - g|$.

By Theorem 12.3.6, $\|\cdot\|$ satisfies $\|f\| \geq 0$, $\|cf\| = |c|\|f\|$, and $\|f + g\| \leq \|f\| + \|g\|$; it is only a *seminorm* because $\|f\| = 0$ forces $f = 0$ only *almost everywhere*, not everywhere (e.g. $f = 1_{\mathbb{Q}}$ has $\|f\| = 0$). The distance dist is symmetric and obeys the triangle inequality, making $L[a, b]$ a (pseudo)metric space. The completeness of this space is the Riesz–Fischer theorem.

Theorem 12.3.9 (Completeness / Riesz–Fischer, B&S 10.2.12). *A sequence (f_n) in $L[a, b]$ converges (in $\|\cdot\|$) to some $f \in L[a, b]$ if and only if it is Cauchy: for every $\varepsilon > 0$ there is H with $\|f_m - f_n\| < \varepsilon$ whenever $m, n \geq H$.*

The easy direction (convergent $\implies$ Cauchy) is Exercise 16; the hard direction extracts a rapidly convergent subsequence $\|g_{k+1} - g_k\| < 2^{-k}$ and sums the telescoping series $g_1 + \sum_k (g_{k+1} - g_k)$, which converges absolutely a.e. to the limit f (details in [MTI]). This plays the role for $L[a, b]$ that the Completeness Property of $\mathbb{R}$ plays for sequences.

Further worked examples

Worked Example 12.3.10 (Hake at a left-endpoint singularity: $\int_0^1 x^{-2/3} dx$). Show $h(x) = x^{-2/3}$ (with $h(0) := 0$) lies in $\mathbb{R}^*[0, 1]$ and compute $\int_0^1 x^{-2/3} dx$.

Recognition. Trigger: a power singularity at the left end with exponent < 1 ; integrate over $[\gamma, 1]$ and let $\gamma \rightarrow 0^+$ via Hake (Theorem 12.3.1).

Solution. For $\gamma \in (0, 1)$, h is continuous on $[\gamma, 1]$ and

$$\int_{\gamma}^1 x^{-2/3} dx = \left[3x^{1/3} \right]_{\gamma}^1 = 3 - 3\gamma^{1/3}.$$

As $\gamma \rightarrow 0^+$, $\gamma^{1/3} \rightarrow 0$, so the limit is 3. By Hake, $h \in \mathbb{R}^*[0, 1]$ with $\int_0^1 x^{-2/3} dx = 3$. *Check.* The general rule $\int_0^1 x^{-p} dx = \frac{1}{1-p}$ for $0 < p < 1$ gives $1/(1 - \frac{2}{3}) = 3$. Since the integrand is positive, $h \in L[0, 1]$ too. ■

Worked Example 12.3.11 (Comparison Test: $\sin(1/x)$ near 0). Let $f(x) = \sin(1/x)$ for $x \in (0, 1]$ and $f(0) := 0$. Show $f \in L[0, 1]$.

Recognition. Trigger: a bounded but wildly oscillating integrand; dominate $|f|$ by an integrable v (here a constant) and invoke the Comparison Test (Theorem 12.3.5).

Solution. The function f is continuous on every $[\gamma, 1]$ with $\gamma > 0$, hence $f|_{[\gamma, 1]} \in \mathbb{R}^*$; and $|f(x)| = |\sin(1/x)| \leq 1$ for all x . The constant $v \equiv 1$ is in $\mathbb{R}^*[0, 1]$ (indeed in $L[0, 1]$, with $\int_0^1 1 = 1$). Because f is bounded and, on $(0, 1]$, continuous, $f \in \mathbb{R}^*[0, 1]$ (a bounded measurable function on a bounded interval is generalized integrable; see §12.5.13). With $|f| \leq v$ and $f, v \in \mathbb{R}^*[0, 1]$, the Comparison Test gives $f \in L[0, 1]$ and $\left| \int_0^1 f \right| \leq \int_0^1 |f| \leq \int_0^1 1 = 1$. *Check.* Unlike $\sin x/x$ on $[0, \infty)$, here the oscillation lives on a *bounded* interval and the amplitude does not decay — but boundedness alone already forces absolute integrability. The wildness is in the *frequency*, not the size, so $|f|$ is harmless. ■

Worked Example 12.3.12 (NON-example: a $\|\cdot\|$ -null function need not be 0). Exhibit $f \in L[0, 1]$ with $f \neq 0$ everywhere on a dense set yet $\|f\| = \int_0^1 |f| = 0$, showing $\|\cdot\|$ is only a *seminorm*.

Recognition. Trigger: “why is $\|\cdot\|$ not a genuine norm?” Produce a function supported on a null set; its integral of $|f|$ vanishes.

Solution. Take $f = \mathbf{1}_{\mathbb{Q} \cap [0, 1]}$, the indicator of the rationals: $f(x) = 1$ on the (countable, dense) set $\mathbb{Q} \cap [0, 1]$ and $f(x) = 0$ elsewhere. By the Dirichlet worked example (§12.1.4), $f \in \mathbb{R}^*[0, 1]$ with $\int_0^1 f = 0$; the same gauge argument applied to $|f| = f$ gives $|f| \in \mathbb{R}^*$ with $\int_0^1 |f| = 0$, so $f \in L[0, 1]$ and $\|f\| = 0$. Yet f is nonzero at every rational — a *dense* set of points. *Lesson.* $\|f\| = 0$ forces $f = 0$ only *almost everywhere* (off a null set), never everywhere; this is exactly why Definition 12.3.8 calls $\|\cdot\|$ a seminorm and why $L[a, b]$ is a *pseudometric* space until one passes to a.e.-equivalence classes. *Check.* The nonzero values sit on the measure-zero set $\mathbb{Q} \cap [0, 1]$, which contributes 0 to every Riemann sum once the gauge pins each rational’s box, as in §12.1.4. ■

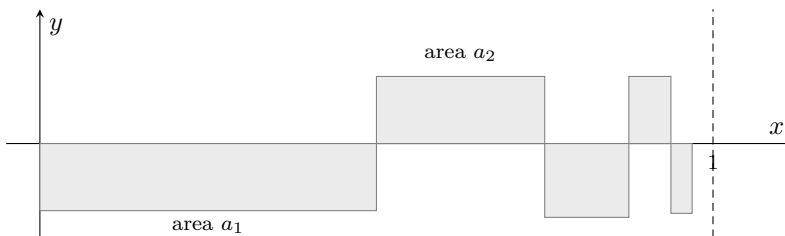


Figure 12.6. The step function w encoding a series $\sum a_k = A$: on $[c_{k-1}, c_k)$ with $c_k = 1 - 2^{-k}$ the bar has width 2^{-k} and (signed) height $2^k a_k$, so its signed area is exactly a_k (bars drawn to scale in sign, rescaled in height to fit).

Cumulative areas $\int_0^\gamma w = \sum_{j \leq n} a_j$ converge to A by Hake. If the a_k alternate without absolute convergence, $w \in \mathbb{R}^* \setminus L$.

Exercises (Bartle & Sherbert, §10.2).

1. Show that Hake's Theorem can be given the sequential formulation: $f \in \mathbb{R}^*[a, b]$ if and only if there exists $A \in \mathbb{R}$ such that for any increasing sequence (c_n) in (a, b) with $c_n \rightarrow b$, one has $f \in \mathbb{R}^*[a, c_n]$ and $\int_a^{c_n} f \rightarrow A$.
2. (a) Apply Hake's Theorem to conclude $g(x) := 1/x^{2/3}$ for $x \in (0, 1]$, $g(0) := 0$, belongs to $\mathbb{R}^*[0, 1]$. (b) Explain why Hake's Theorem does not apply to $f(x) := 1/x$ on $(0, 1]$, $f(0) := 0$ (which is not in $\mathbb{R}^*[0, 1]$).
3. Apply Hake's Theorem to $g(x) := (1 - x)^{-1/2}$ for $x \in [0, 1)$ and $g(1) := 0$.
4. Suppose $f \in \mathbb{R}^*[a, c]$ for all $c \in (a, b)$ and that there exist $\gamma \in (a, b)$ and $v \in L[a, b]$ with $|f(x)| \leq v(x)$ for $x \in [\gamma, b]$. Show $f \in \mathbb{R}^*[a, b]$.
5. Show that $g_1(x) := x^{-1/2} \sin(1/x)$ for $x \in (0, 1]$ and $g_1(0) := 0$ belongs to $L[0, 1]$.
6. Show that the following functions (properly defined where necessary) are in $L[0, 1]$: (a) $\frac{\sin \pi x}{\ln x}$; (b) $\frac{x \ln x}{1 + x^2}$; (c) $(\ln x)(\ln(1 - x))$; (d) $\frac{\ln x}{\sqrt{1 - x^2}}$.
7. Determine whether each integral converges or diverges (define the integrand to be 0 where not already defined): (a) $\int_0^1 \frac{\sin x}{x^{3/2}} dx$; (b) $\int_0^1 \frac{\cos x}{x^{3/2}} dx$; (c) $\int_0^1 \frac{\ln x}{1 - x} dx$; (d) $\int_0^1 \frac{\ln x}{x^2 \sqrt{1 - x}} dx$; (e) $\int_0^1 (\ln x) \sin(1/x) dx$; (f) $\int_0^1 \frac{dx}{\sqrt{x(1 - x)}}$.
8. If $f \in \mathcal{R}[a, b]$, show $f \in L[a, b]$.
9. If $f \in L[a, b]$, show f^2 is not necessarily in $L[a, b]$.
10. If $f, g \in L[a, b]$ with g bounded and monotone, show $fg \in L[a, b]$. More exactly, if $|g(x)| \leq B$, show $\|fg\| \leq B \|f\|$.
11. (a) Give an $f \in \mathbb{R}^*[0, 1]$ such that $\max\{f, 0\}$ does not belong to $\mathbb{R}^*[0, 1]$. (b) Can you give an $f \in L[0, 1]$ with $\max\{f, 0\} \notin L[0, 1]$?
12. Write out the details of the proof that $\min\{f, g\} \in \mathbb{R}^*[a, b]$ in Theorem 10.2.8 when $\alpha \leq f$ and $\alpha \leq g$.
13. Write out the details of the proofs of Theorem 10.2.11.
14. Give an $f \in L[a, b]$ not identically 0 but such that $\|f\| = 0$.
15. If $f, g \in L[a, b]$, show $|\|f\| - \|g\|| \leq \|f - g\|$.
16. Establish the easy part of the Completeness Theorem 10.2.12.
17. If $f_n(x) := x^n$ for $n \in \mathbb{N}$, show $f_n \in L[0, 1]$ and $\|f_n\| \rightarrow 0$; thus $\|f_n - u\| \rightarrow 0$ where u is the function identically 0.
18. Let $g_n(x) := -1$ for $x \in [-1, -1/n)$, $g_n(x) := nx$ for $x \in [-1/n, 1/n]$, and $g_n(x) := 1$ for $x \in (1/n, 1]$. Show $\|g_m - g_n\| \rightarrow 0$ as $m, n \rightarrow \infty$, so that the Completeness Theorem gives $g \in L[-1, 1]$ with $g_n \rightarrow g$ in $L[-1, 1]$. Find such a g .
19. Let $h_n(x) := n$ for $x \in (0, 1/n)$ and $h_n(x) := 0$ elsewhere in $[0, 1]$. Does there exist $h \in L[0, 1]$ with $\|h_n - h\| \rightarrow 0$?
20. Let $k_n(x) := n$ for $x \in (0, 1/n^2)$ and $k_n(x) := 0$ elsewhere in $[0, 1]$. Does there exist $k \in L[0, 1]$ with $\|k_n - k\| \rightarrow 0$?

Additional exercises.

1. (Improper warm-up.) Using Hake's Theorem (Theorem 12.4.3 or its bounded form), show $\int_0^1 x^{-1/2} dx = 2$ by computing $\int_c^1 x^{-1/2} dx$ and letting $c \rightarrow 0^+$.

- Determine for which $p > 0$ the integral $\int_0^1 x^{-p} dx$ converges, and find its value when it does. State the answer as a single threshold on p .
- Show $g(x) := x^{-1/2} \sin(1/x)$ on $(0, 1]$, $g(0) := 0$, is Lebesgue integrable on $[0, 1]$ by dominating $|g|$ by an integrable function. (Cf. the Comparison Test, Theorem 12.3.5.)
- (Non-example: conditionally but not absolutely.) Give a function f on $(0, 1]$ with $f \in \mathbb{R}^*[0, 1]$ but $f \notin L[0, 1]$, and explain in one sentence why $L[a, b] \subsetneq \mathbb{R}^*[a, b]$.
- Prove the Comparison Test in the form: if $f, v \in \mathbb{R}^*[a, b]$ and $|f(x)| \leq v(x)$ for all x , then $f \in L[a, b]$ and $\|f\| \leq \int_a^b v$.
- Show that $\mathcal{R}[a, b] \subseteq L[a, b]$: every (ordinary) Riemann integrable function is Lebesgue integrable. Which inclusion among $\mathcal{R}$, L , $\mathbb{R}^*$ is strict, and why?
- (Seminorm.) Verify directly that $\|f\| := \int_a^b |f|$ satisfies $\|f + g\| \leq \|f\| + \|g\|$ and $\|cf\| = |c| \|f\|$ on $L[a, b]$, and exhibit a nonzero f with $\|f\| = 0$ (so it is a seminorm, not a norm). Cite Definition 12.3.8.
- (Diagram.) On axes for $x \in (0, 1]$, sketch $y = 1/\sqrt{x}$ and the truncation that is $1/\sqrt{x}$ on $[c, 1]$ and 0 on $(0, c)$. Shade the area $\int_c^1$ and indicate with an arrow how it grows to a finite limit as $c \rightarrow 0^+$, illustrating the improper-integral-as-limit picture.
- (Lattice operations.) Using Theorem 12.3.6, show that if $f, g \in L[a, b]$ then $\max\{f, g\}$, $\min\{f, g\}$, $f^+ := \max\{f, 0\}$ and $f^- := \max\{-f, 0\}$ are in $L[a, b]$, and verify $f = f^+ - f^-$, $|f| = f^+ + f^-$.

12.4 Integration over Infinite Intervals

To integrate over $[a, \infty)$, $(-\infty, b]$, or $(-\infty, \infty)$ the gauge picks up an extra job: besides controlling subinterval lengths near each tag, it must control the *tail*. We extend a gauge to the symbol ∞ and use *subpartitions* that cover all but a far tail.

Gauges on $[a, \infty)$ and tagged subpartitions

A tagged partition of $[a, \infty)$ would have to include a final interval $[x_n, \infty)$ of infinite length, whose Riemann term $f(t_{n+1})(\infty - x_n)$ is meaningless. We avoid this by suppressing the tail term: we integrate over *subpartitions* that cover $[a, x_n]$ for some large finite x_n and leave the tail $[x_n, \infty)$ uncovered, but forced to be far out.

Definition 12.4.1 (Gauge on $[a, \infty)$, B&S 10.3). A *gauge* on $[a, \infty)$ is a strictly positive function δ on $[a, \infty)$ together with a number $\delta^* > 0$; write $\delta(\infty) := \delta^*$. A tagged subpartition $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ (with $a = x_0 < x_1 < \dots < x_n$, covering $[a, x_n]$) is *δ -fine* if

$$[x_{i-1}, x_i] \subseteq [t_i - \delta(t_i), t_i + \delta(t_i)] \quad (i = 1, \dots, n) \quad \text{and} \quad x_n \geq \frac{1}{\delta^*}.$$

The Riemann sum is the finite sum $S(f, \dot{P}) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1})$ — there is no tail term.

The condition $x_n \geq 1/\delta^*$ says the covered part reaches at least out to $1/\delta^*$; making δ^* small pushes the uncovered tail to infinity. So δ^* plays for the tail the role $\delta(t)$ plays for an ordinary tag.

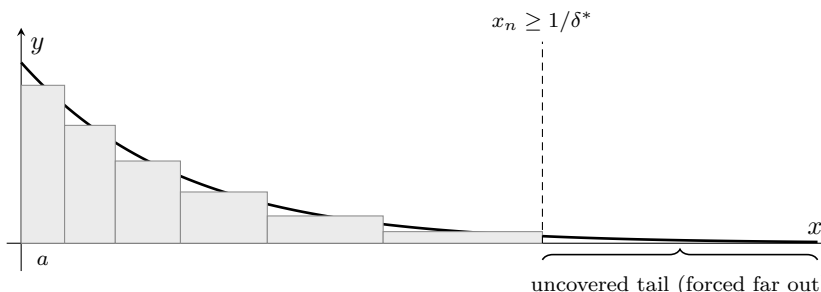


Figure 12.7. A δ -fine subpartition of $[a, \infty)$: finitely many tagged boxes cover $[a, x_n]$, and the tail $[x_n, \infty)$ is left out but forced beyond $1/\delta^*$. Shrinking δ^* drives $x_n \rightarrow \infty$, so the suppressed tail contributes nothing in the limit.

Definition 12.4.2 (Generalized integral over $[a, \infty)$, B&S 10.3.1). $f : [a, \infty) \rightarrow \mathbb{R}$ is *generalized Riemann integrable*, $f \in \mathbb{R}^*[a, \infty)$, if there is $A \in \mathbb{R}$ such that for every $\varepsilon > 0$ there is a gauge δ_ε on $[a, \infty]$ with $|S(f, \dot{P}) - A| \leq \varepsilon$ for every δ_ε -fine subpartition $\dot{P}$ of $[a, \infty)$. We write $\int_a^\infty f := A$. If both $f, |f| \in \mathbb{R}^*[a, \infty)$ we say $f \in L[a, \infty)$.

Integration over $(-\infty, b]$ uses a gauge with a number $\delta_* > 0$ controlling the left tail ($x_0 \leq -1/\delta_*$); over $(-\infty, \infty)$ a gauge carries *both* tail numbers δ_*, δ^* and the subpartition leaves out both ends. All the Section 1–2 theorems (linearity, additivity, Cauchy criterion) transfer with the same proofs.

Hake and the fundamental theorem at infinity

Theorem 12.4.3 (Hake at infinity, B&S 10.3.2). $f \in \mathbb{R}^*[a, \infty)$ if and only if $f|_{[a, \gamma]} \in \mathbb{R}^*[a, \gamma]$ for every $\gamma > a$ and $\lim_{\gamma \rightarrow \infty} \int_a^\gamma f = A$ exists in $\mathbb{R}$. In that case $\int_a^\infty f = A$.

So the calculus definition $\int_a^\infty f = \lim_{\gamma \rightarrow \infty} \int_a^\gamma f$ is recovered *as a theorem*, not imposed as a definition. The two-sided version (B&S 10.3.6) reads $\int_{-\infty}^\infty h = \lim_{\beta \rightarrow -\infty, \gamma \rightarrow \infty} \int_\beta^\gamma h$.

Theorem 12.4.4 (FTC over $[a, \infty)$, B&S 10.3.5). Let $E \subseteq [a, \infty)$ be countable and $F, f : [a, \infty) \rightarrow \mathbb{R}$ satisfy: F is continuous on $[a, \infty)$ and $\lim_{x \rightarrow \infty} F(x)$ exists; $F'(x) = f(x)$ for $x \in (a, \infty) \setminus E$. Then $f \in \mathbb{R}^*[a, \infty)$ and

$$\int_a^\infty f = \lim_{x \rightarrow \infty} F(x) - F(a).$$

Proof. For each $\gamma > a$, FTC First Form (Theorem 12.2.5) on $[a, \gamma]$ gives $\int_a^\gamma f = F(\gamma) - F(a)$. Letting $\gamma \rightarrow \infty$ and applying Hake (Theorem 12.4.3) gives $f \in \mathbb{R}^*[a, \infty)$ with the stated value. $\square$

Worked Example 12.4.5 (The p -integral $\int_1^\infty x^{-\alpha} dx$ (B&S 10.3.4a)). For $\alpha > 1$ show $\int_1^\infty x^{-\alpha} dx = \frac{1}{\alpha - 1}$.

Recognition. Trigger: a power tail; integrate to γ and apply Hake (or FTC at infinity).

Solution. On $[1, \gamma]$, $\int_1^\gamma x^{-\alpha} dx = \frac{1}{1 - \alpha} [x^{1-\alpha}]_1^\gamma = \frac{1}{\alpha - 1} (1 - \gamma^{1-\alpha})$. For $\alpha > 1$, $\gamma^{1-\alpha} \rightarrow 0$ as $\gamma \rightarrow \infty$, so the limit is $\frac{1}{\alpha - 1}$. By Hake, $x^{-\alpha} \in \mathbb{R}^*[1, \infty)$ with that value. *Check.* For $\alpha \leq 1$ the limit diverges, consistent with $1/x \notin \mathbb{R}^*[1, \infty)$ (B&S Exer 10.3.6). $\blacksquare$

Worked Example 12.4.6 (A series as an integral over $[0, \infty)$ (B&S 10.3.4b,c)). Given $\sum_{k=1}^\infty a_k = A$, define $s(x) := a_k$ for $x \in [k-1, k)$. Show $\int_0^\infty s = A$, and that $s \in L[0, \infty)$ iff $\sum |a_k| < \infty$.

Recognition. Trigger: a piecewise-constant function whose blocks have unit width. Each block $[k-1, k)$ contributes area a_k ; sum by Hake.

Solution. The restriction of s to $[0, \gamma]$ is a step function; for $\gamma \in [n, n+1)$, $\int_0^\gamma s = a_1 + \cdots + a_n + r_\gamma$ with $|r_\gamma| \leq |a_{n+1}| \rightarrow 0$. Hence $\int_0^\gamma s \rightarrow A$ and, by Hake, $s \in \mathbb{R}^*[0, \infty)$ with $\int_0^\infty s = A$. Replacing a_k by $|a_k|$, $|s| \in \mathbb{R}^*[0, \infty)$ iff $\sum |a_k| < \infty$. *Check.* $a_k = (-1)^k/k$ again yields $s \in \mathbb{R}^* \setminus L$ over $[0, \infty)$. $\blacksquare$

Worked Example 12.4.7 (The Dirichlet integral converges (B&S 10.3.4d)). Let $D(x) = \frac{\sin x}{x}$ for $x > 0$, $D(0) = 1$. Show $D \in \mathbb{R}^*[0, \infty)$ but $D \notin L[0, \infty)$.

Recognition. Trigger: a conditionally convergent improper integral. Use the Cauchy condition via integration by parts to show $\int_0^\gamma D$ has a limit; show $\int |D| = \infty$ by lobe estimates.

Solution. D is continuous on every $[0, \gamma]$, so $D|_{[0, \gamma]} \in \mathbb{R}^*$. For $0 < \beta < \gamma$, integrate by parts:

$$\int_\beta^\gamma \frac{\sin x}{x} dx = \left[-\frac{\cos x}{x} \right]_\beta^\gamma - \int_\beta^\gamma \frac{\cos x}{x^2} dx.$$

Since $|\cos x| \leq 1$, the boundary terms are $O(1/\beta)$ and the integral is bounded by $\int_{\beta}^{\gamma} x^{-2} dx \leq 1/\beta$, so the right side $\rightarrow 0$ as $\beta < \gamma \rightarrow \infty$. By the Cauchy condition $\lim_{\gamma \rightarrow \infty} \int_0^{\gamma} D$ exists; by Hake, $D \in \mathbb{R}^*[0, \infty)$. For non-absolute integrability: on $[n\pi, (n+1)\pi]$, $\int_{n\pi}^{(n+1)\pi} \frac{|\sin x|}{x} dx \geq \frac{1}{(n+1)\pi} \int_{n\pi}^{(n+1)\pi} |\sin x| dx = \frac{2}{(n+1)\pi}$, and $\sum \frac{1}{n+1} = \infty$, so $\int_0^{\infty} |D| = \infty$. Hence $D \notin L[0, \infty)$. *Check.* The value $\int_0^{\infty} D = \pi/2$ is computed in §12.5 via the convergence theorems. ■

Worked Example 12.4.8 (Two-sided FTC: $\int_{-\infty}^{\infty} \frac{dx}{x^2+1} = \pi$ (B&S 10.3.8a)). Evaluate $\int_{-\infty}^{\infty} (x^2+1)^{-1} dx$.

Recognition. Trigger: an antiderivative with finite limits at both $\pm\infty$. Use FTC over $(-\infty, \infty)$.

Solution. Let $H(x) = \arctan x$, so $H'(x) = (x^2+1)^{-1}$ on all of $\mathbb{R}$, with $\lim_{x \rightarrow \infty} H = \frac{\pi}{2}$ and $\lim_{x \rightarrow -\infty} H = -\frac{\pi}{2}$. The two-sided FTC gives

$$\int_{-\infty}^{\infty} \frac{dx}{x^2+1} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi.$$

Check. The integrand is positive and even; the value π matches the classical improper integral. Since the integrand is also ≥ 0 , here $f = |f|$, so this $f \in L(-\infty, \infty)$ (unlike $\sin x/x$). ■

Caution. A subpartition over $[a, \infty)$ *never* includes a box reaching ∞ ; the tail is *omitted*, not assigned the value $f(\infty) \cdot \infty$. Forgetting this and writing a “last term” $f(t_{n+1})(\infty - x_n)$ is the classic error. The only role of ∞ is to carry the number δ^* that forces $x_n \geq 1/\delta^*$.

Further worked examples

Worked Example 12.4.9 (An exponential tail: $\int_0^{\infty} e^{-x} dx = 1$). Show $e^{-x} \in L[0, \infty)$ and $\int_0^{\infty} e^{-x} dx = 1$.

Recognition. Trigger: a decaying tail with an explicit continuous antiderivative having a finite limit at ∞ ; apply FTC over $[0, \infty)$ (Theorem 12.4.4).

Solution. Let $F(x) = -e^{-x}$, continuous on $[0, \infty)$ with $F'(x) = e^{-x}$ everywhere and $\lim_{x \rightarrow \infty} F(x) = 0$. By FTC at infinity,

$$\int_0^{\infty} e^{-x} dx = \lim_{x \rightarrow \infty} F(x) - F(0) = 0 - (-1) = 1.$$

Since $e^{-x} > 0$, $f = |f|$, so $e^{-x} \in L[0, \infty)$. *Check.* Via Hake, $\int_0^{\gamma} e^{-x} dx = 1 - e^{-\gamma} \rightarrow 1$, agreeing. The integrand decays fast enough that the tail beyond $1/\delta^*$ contributes $\leq e^{-1/\delta^*} \rightarrow 0$, the heuristic behind the subpartition definition. ■

Worked Example 12.4.10 (NON-example: $\int_1^{\infty} \frac{dx}{x}$ diverges). Show $1/x \notin \mathbb{R}^*[1, \infty)$, so not every tail is integrable.

Recognition. Trigger: the borderline power $\alpha = 1$ in the p -integral; test the Hake limit and find it infinite.

Solution. For $\gamma > 1$, $\int_1^{\gamma} \frac{dx}{x} = \ln \gamma$, which $\rightarrow +\infty$ as $\gamma \rightarrow \infty$. Since $\lim_{\gamma \rightarrow \infty} \int_1^{\gamma} \frac{1}{x}$ does *not* exist in $\mathbb{R}$, Hake (Theorem 12.4.3) says $1/x \notin \mathbb{R}^*[1, \infty)$. *Lesson.* Membership in $\mathbb{R}^*[a, \infty)$ genuinely requires the partial integrals to *converge*; the gauge machinery cannot conjure a finite value from a divergent tail. This is the exact boundary case separating the convergent p -integrals ($\alpha > 1$) from the divergent ones ($\alpha \leq 1$). *Check.* Consistent with the worked example $\int_1^{\infty} x^{-\alpha} = \frac{1}{\alpha-1}$, whose value blows up as $\alpha \rightarrow 1^+$. ■

Worked Example 12.4.11 (A rational two-sided integral: $\int_{-\infty}^{\infty} \frac{dx}{x^2+4} = \frac{\pi}{2}$). Evaluate $\int_{-\infty}^{\infty} (x^2+4)^{-1} dx$.

Recognition. Trigger: an antiderivative with finite limits at both $\pm\infty$; use the two-sided FTC, recalling $\int \frac{dx}{x^2+a^2} = \frac{1}{a} \arctan \frac{x}{a}$.

Solution. Let $H(x) = \frac{1}{2} \arctan(x/2)$, so $H'(x) = \frac{1}{2} \cdot \frac{1/2}{1+(x/2)^2} = \frac{1}{x^2+4}$ on all of $\mathbb{R}$, with $\lim_{x \rightarrow \infty} H = \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$ and $\lim_{x \rightarrow -\infty} H = -\frac{\pi}{4}$. The two-sided FTC gives

$$\int_{-\infty}^{\infty} \frac{dx}{x^2+4} = \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2}.$$

The integrand is positive and even, so $f = |f|$ and $f \in L(-\infty, \infty)$. *Check.* Rescaling $x = 2u$ turns the integral into $\frac{1}{2} \int_{-\infty}^{\infty} \frac{du}{u^2+1} = \frac{1}{2} \cdot \pi = \frac{\pi}{2}$, matching the worked example $\int_{-\infty}^{\infty} \frac{dx}{x^2+1} = \pi$. ■

Exercises (Bartle & Sherbert, §10.3).

- Let δ be a gauge on $[a, \infty]$. From Theorem 5.5.5 every bounded subinterval $[a, b]$ has a δ -fine partition. Show that $[a, \infty]$ has a δ -fine partition.
- Let $f \in \mathbb{R}^*[a, \gamma]$ for all $\gamma \geq a$. Show $f \in \mathbb{R}^*[a, \infty)$ if and only if for every $\varepsilon > 0$ there is $K(\varepsilon) \geq a$ such that $q > p \geq K(\varepsilon)$ implies $\left| \int_p^q f \right| < \varepsilon$.
- Let f and $|f|$ belong to $\mathbb{R}^*[a, \gamma]$ for all $\gamma \geq a$. Show $f \in L[a, \infty)$ if and only if for every $\varepsilon > 0$ there is $K(\varepsilon) \geq a$ with $q > p > K(\varepsilon)$ implying $\int_p^q |f| < \varepsilon$.
- Let f and $|f|$ belong to $\mathbb{R}^*[a, \gamma]$ for every $\gamma \geq a$. Show $f \in L[a, \infty)$ if and only if the set $V := \{\int_a^x |f| : x \geq a\}$ is bounded in $\mathbb{R}$.
- If $f, g \in L[a, \infty)$, show $f + g \in L[a, \infty)$. Moreover, with $\|h\| := \int_a^\infty |h|$ for $h \in L[a, \infty)$, show $\|f + g\| \leq \|f\| + \|g\|$.
- If $f(x) := 1/x$ for $x \in [1, \infty)$, show $f \notin \mathbb{R}^*[1, \infty)$.
- If f is continuous on $[1, \infty)$ and $|f(x)| \leq K/x^2$ for $x \in [1, \infty)$, show $f \in L[1, \infty)$.
- Let $f(x) := \cos x$ for $x \in [0, \infty)$. Show $f \notin \mathbb{R}^*[0, \infty)$.
- If $s > 0$, let $g(x) := e^{-sx}$ for $x \in [0, \infty)$. (a) Use Hake's Theorem to show $g \in L[0, \infty)$ and $\int_0^\infty e^{-sx} dx = 1/s$. (b) Use the Fundamental Theorem 10.3.5.
- (a) Use Integration by Parts and Hake's Theorem to show $\int_0^\infty x e^{-sx} dx = 1/s^2$ for $s > 0$. (b) Use the Fundamental Theorem 10.3.5.
- Show that if $n \in \mathbb{N}$, $s > 0$, then $\int_0^\infty x^n e^{-sx} dx = n!/s^{n+1}$.
- (a) Show that $\int_1^\infty \frac{1}{x \ln x} dx$ does not converge. (b) Show that if $\alpha > 1$, then $\int_1^\infty \frac{1}{x^\alpha \ln x} dx$ relates as $\int_1^\infty \frac{\ln x}{x^\alpha} dx = 1/(\alpha - 1)^2$.
- (a) Show $\int_{n\pi}^{(n+1)\pi} \left| \frac{1}{x} \sin x \right| dx > \frac{1}{4(n+1)}$. (b) Show $|D| \notin \mathbb{R}^*[0, \infty)$, where D is as in Example 10.3.4(d).
- Show that $\int_0^\infty \frac{1}{\sqrt{x}} \sin x dx$ converges. (Hint: integrate by parts.)
- Establish the convergence of Fresnel's integral $\int_0^\infty \sin(x^2) dx$. (Hint: use the Substitution Theorem 10.1.12.)
- Establish convergence or divergence of: (a) $\int_0^\infty \frac{\ln x}{x^2 + 1} dx$; (b) $\int_0^\infty \frac{\ln x}{\sqrt{x^2 + 1}} dx$; (c) $\int_0^\infty \frac{x}{(x+1)^3} dx$; (d) $\int_0^\infty \frac{dx}{x(x+1)}$; (e) $\int_0^\infty \frac{dx}{\sqrt{1+x^3}}$; (f) $\int_0^\infty \frac{\arctan x}{x^{3/2} + 1} dx$.
- (Abel's Test.) Let $f, w : [a, \infty) \rightarrow \mathbb{R}$; Abel's Test asserts that if $f \in \mathbb{R}^*[a, \infty)$ and w is bounded and monotone on $[a, \infty)$, then $fw \in \mathbb{R}^*[a, \infty)$. (a) Show Abel's Test does not apply to $\int_0^\infty (1/x) \sin x dx$ via $w(x) := 1/x$, but does apply with $w(x) := 1/\sqrt{x}$ and Exercise 14. (b) Use Abel's Test and Exercise 15 to show $\int_0^\infty \frac{x}{x+1} \sin(x^2) dx$ converges. (c) Use Abel's Test and Exercise 14 to show $\int_0^\infty \frac{1}{x^{3/2}(x+1)} \sin x dx$ converges. (d) Use Abel's Test to obtain the convergence of Exercise 16(f).
- (Chartier–Dirichlet Test.) With the notation of Exercise 17, the test asserts that if $f \in \mathbb{R}^*[a, \gamma]$ for all $\gamma \geq a$, $F(x) := \int_a^x f$ is bounded on $[a, \infty)$, and w is monotone with $\lim_{x \rightarrow \infty} w(x) = 0$, then $fw \in \mathbb{R}^*[a, \infty)$. (a) Show $\int_0^\infty (1/x) \sin x dx$ converges. (b) Show $\int_2^\infty (1/\ln x) \sin x dx$ converges. (c) Show $\int_0^\infty (1/\sqrt{x}) \cos x dx$ converges. (d) Show the test does not apply to $\int_0^\infty \frac{x}{x^2 + 1} \sin(x^2) dx$ via $f(x) := \sin(x^2)$.

19. Show $\int_0^\infty \sqrt{x} \sin(x^2) dx$ converges, even though the integrand is unbounded as $x \rightarrow \infty$. (Hint: make a substitution.)
20. Establish convergence of: (a) $\int_{-\infty}^\infty e^{-|x|} dx$; (b) $\int_{-\infty}^\infty (x-2)e^{-|x|} dx$; (c) $\int_{-\infty}^\infty e^{-x^2} dx$; (d) $\int_{-\infty}^\infty \frac{2x}{e^x - e^{-x}} dx$.

Additional exercises.

- (Infinite-interval warm-up.) Compute $\int_1^\infty x^{-2} dx$ from the definition via $\lim_{\gamma \rightarrow \infty} \int_1^\gamma x^{-2} dx$, and state the antiderivative used.
- Determine for which p the integral $\int_1^\infty x^{-p} dx$ converges, and find its value when it does. Contrast the threshold with the $\int_0^1 x^{-p} dx$ case of §10.2.
- Using the Fundamental Theorem on $[a, \infty)$ (Theorem 12.4.4), evaluate $\int_0^\infty e^{-x} dx$ and $\int_0^\infty x e^{-x} dx$.
- (Non-example.) Show $\int_0^\infty \cos x dx$ does not converge, by exhibiting two subsequences of $\int_0^\gamma \cos x dx$ with different limits.
- (Comparison on $[1, \infty)$.) Prove: if f is continuous on $[1, \infty)$ and $|f(x)| \leq K/x^2$, then $f \in L[1, \infty)$ and $\|f\| \leq K$. Apply this to $f(x) = \frac{\sin x}{x^2}$.
- Show $\int_0^\infty x^n e^{-x} dx = n!$ for $n \in \mathbb{N}_0$ by induction using integration by parts, identifying the boundary term that vanishes at ∞ .
- (Hake on infinite intervals.) State Hake's Theorem for $[a, \infty)$ (Theorem 12.4.3) and use it to show $g(x) = e^{-x}$ is in $L[0, \infty)$ with $\int_0^\infty e^{-x} = 1$, organizing the proof as a limit of integrals over $[0, \gamma]$.
- (Diagram.) Sketch $y = 1/x^2$ on $[1, \infty)$ and shade the region under it; indicate the finite total area $= 1$. On the same picture, overlay $y = 1/x$ and shade to show its area is *infinite*, contrasting convergent and divergent tails.
- (Abel / Chartier–Dirichlet.) State Abel's Test and the Chartier–Dirichlet Test for $[a, \infty)$ (as in B&S Exercises 17–18). Use one of them to show $\int_1^\infty \frac{\sin x}{x} dx$ converges, identifying the bounded/monotone factor.

12.5 The Convergence Theorems

The decisive practical advantage of the generalized integral is that limits pass inside far more freely than for the Riemann integral. Chapter 8 needed *uniform* convergence to swap $\lim$ and $\int$; here mere pointwise (indeed almost-everywhere) convergence suffices, provided a monotonicity or domination hypothesis holds. Throughout, I is any interval (bounded or not), “a.e.” means off a null set, and $\mathbb{R}^*(I)$, $L(I)$ are as before.

From uniform convergence to equi-integrability

Theorem 12.5.1 (Uniform Convergence Theorem, B&S 10.4.1). *Let (f_k) in $\mathbb{R}^*[a, b]$ converge uniformly on $[a, b]$ to f . Then $f \in \mathbb{R}^*[a, b]$ and $\int_a^b f = \lim_k \int_a^b f_k$.*

Proof. Given $\varepsilon > 0$ pick K with $|f_k(x) - f(x)| < \varepsilon$ for all $k \geq K$, $x \in [a, b]$. For $h, k \geq K$ then $|f_k - f_h| < 2\varepsilon$, so by monotonicity $|\int f_k - \int f_h| \leq 2\varepsilon(b-a)$; thus $(\int f_k)$ is Cauchy, converging to some A . Fix $r \geq K$ with $|\int f_r - A| < \varepsilon$ and a gauge δ making $|S(f_r, \dot{P}) - \int f_r| < \varepsilon$ for δ -fine $\dot{P}$. For any such $\dot{P}$, $|S(f, \dot{P}) - S(f_r, \dot{P})| \leq \sum_i |f(t_i) - f_r(t_i)| (x_i - x_{i-1}) \leq \varepsilon(b-a)$, whence $|S(f, \dot{P}) - A| < \varepsilon(b-a) + \varepsilon + \varepsilon$. So $f \in \mathbb{R}^*$ with integral A . $\square$

Uniform convergence is too restrictive for most applications. Kurzweil's substitute is *equi-integrability*: a single gauge works for the whole sequence at once.

Definition 12.5.2 (Equi-integrability, B&S 10.4.2). (f_k) in $\mathbb{R}^*(I)$ is *equi-integrable* if for every $\varepsilon > 0$ there is one gauge δ_ε on I such that $|S(f_k, \dot{P}) - \int_I f_k| < \varepsilon$ for every δ_ε -fine $\dot{P}$ and *every* $k \in \mathbb{N}$.

Theorem 12.5.3 (Equi-integrability Theorem, B&S 10.4.3). *If (f_k) in $\mathbb{R}^*(I)$ is equi-integrable and $f(x) = \lim_k f_k(x)$ for all $x \in I$, then $f \in \mathbb{R}^*(I)$ and $\int_I f = \lim_k \int_I f_k$.*

Proof for $I = [a, b]$. Fix $\varepsilon > 0$ and the common gauge δ_ε . For a δ_ε -fine $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$, since $\dot{P}$ has finitely many tags and $f_k(t_i) \rightarrow f(t_i)$, there is K_ε with $|S(f_k, \dot{P}) - S(f_h, \dot{P})| \leq \varepsilon(b-a)$ for $h, k \geq K_\varepsilon$. Letting $h \rightarrow \infty$, $|S(f_k, \dot{P}) - S(f, \dot{P})| \leq \varepsilon(b-a)$ for $k \geq K_\varepsilon$. Combining with equi-integrability shows $(\int_I f_k)$ is Cauchy with limit A , and then $|S(f, \dot{P}) - A| \leq \varepsilon(3 + 2(b-a))$; so $f \in \mathbb{R}^*$ with integral $A = \lim_k \int_I f_k$. $\square$

Constructing the common gauge directly is hard, so in practice one uses the two corollaries below, both provable from Theorem 12.5.3.

The monotone and dominated convergence theorems

Theorem 12.5.4 (Monotone Convergence, B&S 10.4.4). *Let (f_k) be a monotone sequence in $\mathbb{R}^*(I)$ with $f(x) = \lim_k f_k(x)$ a.e. on I . Then $f \in \mathbb{R}^*(I)$ if and only if the sequence of integrals $(\int_I f_k)$ is bounded, and in that case*

$$\int_I f = \lim_{k \rightarrow \infty} \int_I f_k.$$

For an increasing sequence, $\int_I f_k$ is increasing, so “bounded” = “bounded above”, and then $\int_I f = \sup_k \int_I f_k$. The MCT is the tool of choice when $f_k \uparrow f$.

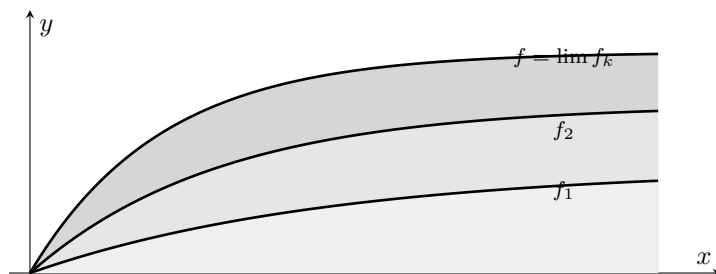


Figure 12.8. Monotone Convergence: $f_1 \leq f_2 \leq \dots \uparrow f$ pointwise, so the areas $\int f_k$ increase. If they stay bounded, f is integrable and $\int f = \lim \int f_k = \sup \int f_k$; if they march off to ∞ , f is not integrable. Either way the limit of areas matches the area of the limit.

Theorem 12.5.5 (Dominated Convergence, B&S 10.4.5). *Let (f_k) in $\mathbb{R}^*(I)$ with $f(x) = \lim_k f_k(x)$ a.e. on I . If there exist $\alpha, v \in \mathbb{R}^*(I)$ with*

$$\alpha(x) \leq f_k(x) \leq v(x) \quad \text{for a.e. } x \in I, \text{ all } k,$$

then $f \in \mathbb{R}^(I)$ and $\int_I f = \lim_k \int_I f_k$. If moreover $\alpha, v \in L(I)$, then $f_k, f \in L(I)$ and $\|f_k - f\| = \int_I |f_k - f| \rightarrow 0$.*

Equivalently, a single dominating $w := \max\{|\alpha|, |v|\} \in L(I)$ with $|f_k| \leq w$ a.e. suffices. The DCT is the most-used convergence theorem in analysis.

Caution. The domination hypothesis cannot be dropped. Let $f_k = \frac{1}{k}$ on $[0, k]$ and 0 beyond, on $[0, \infty)$ (B&S 10.4.6a). Then $f_k \rightarrow 0$ uniformly, yet $\int_0^\infty f_k = 1$ for all k while $\int_0^\infty 0 = 0$: $\lim \int \neq \int \lim$. There is no integrable dominating w (one needs $w \geq \sup_k f_k$, which is not integrable). Uniform convergence on an infinite interval is not enough — contrast Theorem 12.5.1, which is for $[a, b]$ only.

Worked Example 12.5.6 (A bounded-domination limit (B&S 10.4.6b)). Show $\lim_{k \rightarrow \infty} \int_0^1 \frac{x^k + 1}{x^k + 3} dx = \frac{1}{3}$.

Recognition. Trigger: an integrand converging pointwise with an obvious bound. Dominate by a constant on $[0, 1]$ and apply DCT.

Solution. Let $g_k(x) = (x^k + 1)/(x^k + 3)$. For $x \in [0, 1)$, $x^k \rightarrow 0$, so $g_k(x) \rightarrow 1/3$; at $x = 1$, $g_k(1) = 2/4 = 1/2$, a single point (null set). Also $0 \leq g_k(x) \leq 1$ on $[0, 1]$, and the constant 1 $\in L[0, 1]$ dominates. By DCT, $\int_0^1 g_k \rightarrow \int_0^1 \frac{1}{3} dx = \frac{1}{3}$. Check. Each $g_k \in [\frac{1}{3}, \frac{1}{2}]$, so $\int_0^1 g_k \in [\frac{1}{3}, \frac{1}{2}]$ and decreases to $\frac{1}{3}$; the limit is consistent. $\blacksquare$

Worked Example 12.5.7 (A monotone limit to a Laplace-type integral (B&S 10.4.6c)). For $\alpha > 1$ show

$$\lim_{k \rightarrow \infty} \int_0^k \left(1 + \frac{x}{k}\right)^k e^{-\alpha x} dx = \frac{1}{\alpha - 1}.$$

Recognition. Trigger: $(1 + x/k)^k \uparrow e^x$ increasingly. Use MCT, then evaluate the limit integral by FTC at infinity.

Solution. Let $h_k(x) = (1 + x/k)^k e^{-\alpha x}$ on $[0, k]$, $h_k = 0$ beyond. The sequence $(1 + x/k)^k$ increases to e^x , so $h_k \uparrow e^x e^{-\alpha x} = e^{(1-\alpha)x}$ on $[0, \infty)$. For $\alpha > 1$ the limit $e^{(1-\alpha)x} \in L[0, \infty)$, with antiderivative $F(x) = e^{(1-\alpha)x}/(1-\alpha)$; FTC at infinity (Theorem 12.4.4) gives $\int_0^\infty e^{(1-\alpha)x} dx = \frac{1}{\alpha - 1}$. The integrals $\int_0^k h_k$ are bounded by this, so MCT applies and the limit equals $\frac{1}{\alpha - 1}$. *Check.* As $\alpha \rightarrow \infty$ both sides $\rightarrow 0$; as $\alpha \rightarrow 1^+$ both $\rightarrow \infty$, consistent. ■

Evaluation of the Dirichlet integral

Worked Example 12.5.8 ($\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$ (B&S 10.4.6f)). Compute the Dirichlet integral, whose convergence (but not value) was shown in §12.4.3.

Recognition. Trigger: a parameter integral $D_k(t) = \int_0^k e^{-tx} \frac{\sin x}{x} dx$; differentiate in t under the integral (justified by DCT), solve the resulting ODE, and let $k \rightarrow \infty$.

Solution. Put $D_k(t) = \int_0^k e^{-tx} \frac{\sin x}{x} dx$ for $t \geq 0$; then $D_k(0) = \int_0^k \frac{\sin x}{x} dx$. Since $|\partial_t(e^{-tx} \sin x/x)| = |e^{-tx} \sin x| \leq 1$, DCT licenses differentiation under the integral:

$$D'_k(t) = - \int_0^k e^{-tx} \sin x dx = \frac{e^{-tk}(t \sin k + \cos k)}{t^2 + 1} - \frac{1}{t^2 + 1}.$$

Integrating t from 0 to k and using $D_k(k) \rightarrow 0$ (since $|D_k(k)| \leq (1 - e^{-k^2})/k \rightarrow 0$) and $\int_0^k \frac{dt}{t^2 + 1} = \arctan k \rightarrow \frac{\pi}{2}$, together with $\int_0^\infty g_k(t) dt \rightarrow 0$ for the first term (by DCT, dominator $2e^{-1}$), the relation

$$D_k(k) - D_k(0) = \int_0^k D'_k(t) dt = \int_0^k g_k(t) dt - \arctan k$$

passes to the limit as $0 - \lim_k D_k(0) = 0 - \frac{\pi}{2}$. Hence $\lim_k \int_0^k \frac{\sin x}{x} dx = \frac{\pi}{2}$, i.e. $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$. *Check.* The value is positive (the first lobe dominates), and finite, as the conditional convergence in §12.4.3 demanded. ■

Worked Example 12.5.9 (The Gaussian integral $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$ (B&S Exer 10.4.11)). Establish $\int_0^\infty e^{-x^2} dx = \frac{1}{2}\sqrt{\pi}$ by a parameter trick.

Recognition. Trigger: pair a t -integral $G(t)$ with $F(t) = (\int_0^t e^{-x^2} dx)^2$ so that $F' + G' = 0$; evaluate the constant from known endpoint values.

Solution. Let $G(t) = \int_0^1 \frac{e^{-t^2(x^2+1)}}{x^2 + 1} dx$ for $t \geq 0$. The integrand is dominated by $1/(x^2 + 1) \in L[0, 1]$, so G is continuous with $G(0) = \arctan 1 = \frac{\pi}{4}$ and, by DCT, $G(t) \rightarrow 0$ as $t \rightarrow \infty$. Differentiating under the integral (bounded ∂_t),

$$G'(t) = -2te^{-t^2} \int_0^1 e^{-t^2 x^2} dx = -2e^{-t^2} \int_0^t e^{-u^2} du \quad (u = tx).$$

With $F(t) = (\int_0^t e^{-x^2} dx)^2$, FTC gives $F'(t) = 2e^{-t^2} \int_0^t e^{-x^2} dx = -G'(t)$, so $F + G \equiv C$. Evaluating at $t = 0$: $F(0) + G(0) = 0 + \frac{\pi}{4} = C$. Letting $t \rightarrow \infty$: $G \rightarrow 0$, so $F(\infty) = \frac{\pi}{4}$, i.e. $(\int_0^\infty e^{-x^2} dx)^2 = \frac{\pi}{4}$. Taking the positive root, $\int_0^\infty e^{-x^2} dx = \frac{1}{2}\sqrt{\pi}$. *Check.* Squaring back gives $\pi/4$; doubling gives the familiar $\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}$. ■

Measurable functions

Definition 12.5.10 (Measurable function, B&S 10.4.7). $f : [a, b] \rightarrow \mathbb{R}$ is (Lebesgue) measurable, $f \in M[a, b]$, if there is a sequence (s_k) of step functions with $f(x) = \lim_k s_k(x)$ for a.e. $x \in [a, b]$.

Every step, continuous, or monotone function is measurable (each is an a.e. — indeed uniform — limit of step functions). The class is closed under the usual operations:

Theorem 12.5.11 (Closure properties, B&S 10.4.9–10.4.10). If $f, g \in M[a, b]$, $c \in \mathbb{R}$, then $cf, |f|, f + g, f - g, fg \in M[a, b]$; if $\psi : \mathbb{R} \rightarrow \mathbb{R}$ is continuous then $\psi \circ f \in M[a, b]$; and if $(f_n) \subseteq M[a, b]$ with $f_n \rightarrow f$ a.e. then $f \in M[a, b]$. Moreover $f \in M[a, b]$ iff f is an a.e. limit of continuous functions.

Theorem 12.5.12 (Integrable $\implies$ measurable, B&S 10.4.11). If $f \in \mathbb{R}^*[a, b]$ then $f \in M[a, b]$.

Proof. Let $F(x) = \int_a^x f$ be the indefinite integral, continuous on $[a, b]$ (FTC Second Form), with $F'(x) = f(x)$ off a null set Z . The difference quotients $g_k(x) = k(F(x + \frac{1}{k}) - F(x))$ are continuous and $g_k(x) \rightarrow f(x)$ for $x \notin Z$; by Theorem 12.5.11, $f \in M[a, b]$. $\square$

The converse fails ($g(x) = 1/x$ is measurable but not in $\mathbb{R}^*[0, 1]$); the precise statement is:

Theorem 12.5.13 (Measurable + squeezed = integrable, B&S 10.4.12). Let $f \in M[a, b]$. Then $f \in \mathbb{R}^*[a, b]$ if and only if there exist $\alpha, v \in \mathbb{R}^*[a, b]$ with $\alpha(x) \leq f(x) \leq v(x)$ a.e. If moreover α or v is in $L[a, b]$, then $f \in L[a, b]$.

Proof idea. ($\Leftarrow$) Take step functions $s_k \rightarrow f$ a.e. and truncate, $\sigma_k := \text{mid}\{\alpha, s_k, v\}$ (the middle value), so $\alpha \leq \sigma_k \leq v$ and $\sigma_k \rightarrow f$ a.e.; each $\sigma_k \in \mathbb{R}^*$ by Theorem 12.3.6/10.2.8, and the DCT (dominators α, v) gives $f \in \mathbb{R}^*$. ($\implies$) trivial: $\alpha = v = f$. The L -conclusion is Theorem 12.3.7. $\square$

Remark. This closes the loop. The generalized integrable functions are exactly the measurable functions trapped between two integrable ones — a clean “two-sided boundedness” criterion. As a famous analyst put it: “No one ever calculates a Lebesgue integral; instead one calculates Riemann integrals and takes limits.” The convergence theorems above are precisely the licence to take those limits.

Further worked examples

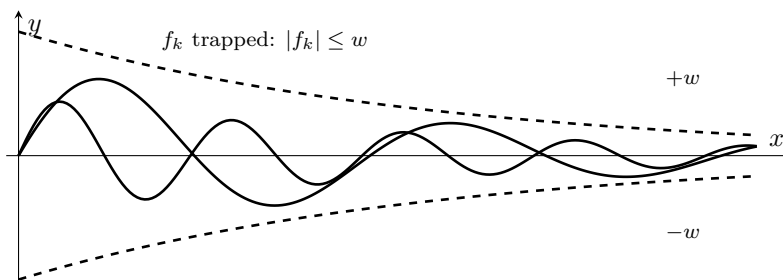


Figure 12.9. Dominated Convergence: every f_k lies inside the fixed integrable envelope $\pm w$ (here $w \in L$). The single dominator w supplies the uniform control that lets $\lim_k \int f_k = \int \lim_k f_k$, even though the f_k may oscillate. Remove w and the conclusion can fail (escaping-mass Caution above).

Worked Example 12.5.14 (NON-example: the escaping bump defeats Fatou-equality). On $[0, \infty)$ let $f_k(x) = 1$ for $x \in [k, k+1)$ and $f_k(x) = 0$ otherwise. Show $f_k \rightarrow 0$ pointwise but $\int_0^\infty f_k \not\rightarrow \int_0^\infty 0$, and identify which hypothesis of DCT fails.

Recognition. Trigger: a “traveling bump” that converges pointwise to 0 while carrying constant mass; test the domination hypothesis of Theorem 12.5.5.

Solution. Fix $x \geq 0$. For all $k > x$ we have $x \notin [k, k+1)$, so $f_k(x) = 0$ eventually; hence $f_k(x) \rightarrow 0$ for every x , i.e. $f_k \rightarrow 0$ pointwise (in fact the convergence is *not* uniform, since $\sup_x f_k = 1$ for all k). But each bump has area

$$\int_0^\infty f_k = \int_k^{k+1} 1 \, dx = 1 \quad \text{for all } k,$$

so $\lim_k \int_0^\infty f_k = 1 \neq 0 = \int_0^\infty (\lim_k f_k)$. The *domination* hypothesis fails: any w with $|f_k| \leq w$ for all k must satisfy $w(x) \geq 1$ on $[k, k+1)$ for every k , hence $w \geq 1$ on $[0, \infty)$, and $\int_0^\infty w = \infty$, so no *integrable* dominator exists. *Lesson*. Mass can “escape to infinity” under mere pointwise (even uniform) convergence; only an integrable envelope $w \in L$ prevents it. *Check*. The companion Caution (the spreading plateau $f_k = \frac{1}{k} \mathbf{1}_{[0,k]}$) fails the same way, confirming the obstruction is escaping mass, not the bump’s height. ■

Worked Example 12.5.15 (Monotone Convergence: $\int_0^1 x^{-1/2} dx$ as an increasing limit). Recover $\int_0^1 x^{-1/2} dx = 2$ by writing the unbounded integrand as an increasing limit of bounded truncations and applying the MCT.

Recognition. Trigger: an unbounded $f \geq 0$ presented as $f_k \uparrow f$; use Monotone Convergence (Theorem 12.5.4) rather than a gauge estimate.

Solution. Let $f(x) = x^{-1/2}$ on $(0, 1]$, $f(0) := 0$, and truncate: $f_k(x) := \min\{f(x), k\}$. Each f_k is bounded and continuous off the single point where $f(x) = k$ (namely $x = 1/k^2$), hence $f_k \in \mathbb{R}^*[0, 1]$; and $f_k(x) \uparrow f(x)$ for every x (truncation level rising). The integrals satisfy, for k with $1/k^2 \leq 1$,

$$\int_0^1 f_k = \int_0^{1/k^2} k dx + \int_{1/k^2}^1 x^{-1/2} dx = k \cdot \frac{1}{k^2} + [2\sqrt{x}]_{1/k^2}^1 = \frac{1}{k} + (2 - \frac{2}{k}) = 2 - \frac{1}{k},$$

which is increasing and bounded above by 2. By MCT, $f \in \mathbb{R}^*[0, 1]$ and $\int_0^1 f = \sup_k \int_0^1 f_k = \lim_k (2 - \frac{1}{k}) = 2$. *Check*. Matches the FTC and Hake computations of §12.2.5 and §12.3.1; here the boundedness of each f_k makes every step an ordinary Riemann integral, and the MCT supplies the passage to the unbounded limit. ■

Exercises (Bartle & Sherbert, §10.4).

- For each sequence below (with the indicated domain): does it converge, and to what? Is the convergence uniform? Bounded? If unbounded, is it dominated? Is it monotone? Evaluate $\lim \int f_k$. (a) $\frac{x^k}{1+x^k}$ on $[0, 1]$; (b) $\frac{kx}{1+kx}$ on $[0, 2]$; (c) $\frac{1}{1+x^k}$ on $[0, 1]$; (d) $\frac{1}{1+x^k}$ on $[0, 2]$.
- Answer the questions of Exercise 1 for (when properly defined): (a) $\frac{kx}{1+k\sqrt{x}}$ on $[0, 1]$; (b) $\frac{1}{\sqrt{x}(1+x^k)}$ on $[0, 1]$; (c) $\frac{1}{\sqrt{x}(1+x^k)}$ on $[1, 2]$; (d) $\frac{1}{\sqrt{x}(2-x^k)}$ on $[0, 1]$.
- Discuss the following sequences and their integrals on $[0, 1]$; evaluate $\lim \int f_k$ when possible: (a) e^{-kx} ; (b) e^{-kx}/x ; (c) kxe^{-kx} ; (d) k^2xe^{-kx} ; (e) $kxe^{-k^2x^2}$; (f) kxe^{-kx^2} .
- (a) Show $\lim_{k \rightarrow \infty} \int_0^1 \frac{x^k}{(1+x)^2} dx = 0$. (b) Show $\lim_{k \rightarrow \infty} \int_0^1 \frac{kx^k}{1+x} dx = \frac{1}{2}$.
- If $f_k(x) := k$ for $x \in [1/k, 2/k]$ and $f_k(x) := 0$ elsewhere on $[0, 2]$, show $f_k(x) \rightarrow 0$ but $\int_0^2 f_k = 1$.
- Let (f_k) on $[a, b]$ with each f_k differentiable, $f'_k(x) \rightarrow g(x)$, and $|f'_k(x)| \leq K$ for all $x \in [a, b]$. Show that the sequence $(f_k(x))$ either converges for all $x \in [a, b]$ or diverges for all $x \in [a, b]$.
- If f_k are the functions of Example 10.4.6(a), show $\sup\{f_k\}$ does not belong to $\mathbb{R}^*[0, \infty)$.
- Show directly that $\int_0^\infty e^{-tx} dx = 1/t$ and $\int_0^\infty xe^{-tx} dx = 1/t^2$ for $t > 0$, confirming Examples 10.4.6(d,e) when $f(x) := 1$.
- Use the differentiation formula in 10.4.6(f) to obtain $\int_0^\infty e^{-tx} \sin x dx = 1/(t^2 + 1)$.
- For $t > 0$ define $E(t) := \int_0^\infty \frac{e^{-tx} \sin x}{x} dx$. (a) Show E exists and is continuous for $t > a > 0$, and $E(t) \rightarrow 0$ as $t \rightarrow \infty$. (b) Since $|\partial_t(e^{-tx} \sin x/x)| \leq e^{-ax}$ for $t \geq a > 0$, show $E'(t) = -1/(t^2 + 1)$ for $t > 0$. (c) Deduce $E(t) = \frac{1}{2}\pi - \arctan t$ for $t > 0$. (d) Explain why we cannot use (c) to obtain equation (12).

11. (Gaussian integral.) Establish $\int_0^\infty e^{-x^2} dx = \frac{1}{2}\sqrt{\pi}$ as follows. (a) Let $G(t) := \int_0^1 \frac{e^{-t^2(x^2+1)}}{x^2+1} dx$ for $t \geq 0$. Since the integrand is dominated by $1/(x^2+1)$, G is continuous on $[0, \infty)$, $G(0) = \arctan 1 = \frac{1}{4}\pi$, and (by Dominated Convergence) $G(t) \rightarrow 0$ as $t \rightarrow \infty$. (b) Show $G'(t) = -2te^{-t^2} \int_0^1 e^{-t^2x^2} dx = -2e^{-t^2} \int_0^t e^{-u^2} du$. (c) With $F(t) := (\int_0^t e^{-x^2} dx)^2$, the FTC gives $F'(t) = 2e^{-t^2} \int_0^t e^{-x^2} dx$, so $F'(t) + G'(t) = 0$ and $F(t) + G(t) = C$. (d) From $F(0) = 0$, $G(0) = \frac{1}{4}\pi$, $\lim_{t \rightarrow \infty} G(t) = 0$, conclude $\lim_{t \rightarrow \infty} F(t) = \frac{1}{4}\pi$.
12. Suppose $I \subseteq \mathbb{R}$ is a closed interval and $f : [a, b] \times I \rightarrow \mathbb{R}$ has $\partial f / \partial t$ on $[a, b] \times I$, with $x \mapsto f(t, x) \in \mathbb{R}^*(I)$ for each t , and $\alpha, v \in \mathbb{R}^*(I)$ with $\alpha(x) \leq \partial f(t, x) / \partial t \leq v(x)$ for a.e. $x \in I$. If $F(t) := \int_I f(t, x) dx$, show F is differentiable on $[a, b]$ with $F'(t) = \int_I \partial f(t, x) / \partial t dx$.
13. (a) If $f, g \in M[a, b]$, show $\max\{f, g\}$ and $\min\{f, g\}$ belong to $M[a, b]$. (b) If $f, g, h \in M[a, b]$, show $\text{mid}\{f, g, h\} \in M[a, b]$.
14. (a) If (f_k) is a bounded sequence in $M[a, b]$ and $f_k \rightarrow f$ a.e., show $f \in M[a, b]$. (Hint: Dominated Convergence.) (b) If (g_k) is any sequence in $M[a, b]$ and $f_k := \arctan g_k$, show (f_k) is bounded in $M[a, b]$. (c) If (g_k) is a sequence in $M[a, b]$ with $g_k \rightarrow g$ a.e., show $g \in M[a, b]$.
15. A set $E \subseteq [a, b]$ is (Lebesgue) measurable if its characteristic function $\mathbf{1}_E$ belongs to $M[a, b]$; write $\mathcal{M}[a, b]$ for the collection of measurable sets. (a) Show $E \in \mathcal{M}[a, b]$ iff $\mathbf{1}_E \in \mathbb{R}^*[a, b]$. (b) Show $\emptyset \in \mathcal{M}[a, b]$ and that for $[c, d] \subseteq [a, b]$ the intervals $[c, d], [c, d), (c, d], (c, d)$ are in $\mathcal{M}[a, b]$. (c) Show $E \in \mathcal{M}[a, b]$ iff $E' := [a, b] \setminus E \in \mathcal{M}[a, b]$. (d) If $E, F \in \mathcal{M}[a, b]$, then $E \cup F, E \cap F, E \setminus F$ are in $\mathcal{M}[a, b]$. (e) If (E_k) is increasing in $\mathcal{M}[a, b]$, show $E := \bigcup_k E_k \in \mathcal{M}[a, b]$; if (F_k) is decreasing, show $F := \bigcap_k F_k \in \mathcal{M}[a, b]$. (f) For any sequence (E_k) in $\mathcal{M}[a, b]$, show $\bigcup_k E_k$ and $\bigcap_k E_k$ are in $\mathcal{M}[a, b]$.
16. For $E \in \mathcal{M}[a, b]$ define the measure $m(E) := \int_a^b \mathbf{1}_E$. (a) Show $m(\emptyset) = 0$ and $0 \leq m(E) \leq b - a$. (b) Show $m([c, d]) = m((c, d)) = m([c, d]) = m((c, d)) = d - c$. (c) Show $m(E') = (b - a) - m(E)$. (d) Show $m(E \cup F) + m(E \cap F) = m(E) + m(F)$. (e) If $E \cap F = \emptyset$, show $m(E \cup F) = m(E) + m(F)$ (additivity). (f) If (E_k) is increasing in $\mathcal{M}[a, b]$, show $m(\bigcup_k E_k) = \lim_k m(E_k)$. (g) If (C_k) is a pairwise disjoint sequence in $\mathcal{M}[a, b]$, show $m(\bigcup_k C_k) = \sum_k m(C_k)$ (countable additivity).

Additional exercises.

1. (Which theorem?) For each sequence on $[0, 1]$, decide whether the Monotone Convergence Theorem (Theorem 12.5.4) or the Dominated Convergence Theorem (Theorem 12.5.5) applies, and evaluate $\lim \int f_k$:
 (a) $f_k(x) = x^k$; (b) $f_k(x) = \frac{kx}{1 + k^2x^2}$; (c) $f_k(x) = x^k(1 - x)$.
2. (MCT.) Let $f_k \uparrow f$ pointwise on $[a, b]$ with each $f_k \in \mathbb{R}^*[a, b]$ and $\sup_k \int_a^b f_k < \infty$. State precisely what the Monotone Convergence Theorem concludes about f and $\int_a^b f$, and apply it to $f_k(x) = 1 - (1 - x)^k$ on $[0, 1]$.
3. (DCT.) Suppose $f_k \rightarrow f$ a.e. on $[a, b]$ and $|f_k| \leq g$ with $g \in L[a, b]$. Show $f \in L[a, b]$ and $\int_a^b f_k \rightarrow \int_a^b f$. Apply it to $f_k(x) = \frac{\cos(kx)}{1 + x^2}$ on $[0, 1]$.
4. (Non-example: no domination.) Let $f_k(x) = k$ on $(0, 1/k)$ and 0 elsewhere on $[0, 1]$. Show $f_k \rightarrow 0$ pointwise but $\int_0^1 f_k = 1 \not\rightarrow 0$. Identify which hypothesis of the DCT fails, and exhibit that no single $g \in L[0, 1]$ dominates all f_k .
5. (Fatou.) State Fatou's Lemma and use it to show that for the spike sequence in the previous problem $\int_0^1 \liminf f_k \leq \liminf \int_0^1 f_k$ holds with *strict* inequality.
6. Show $\lim_{k \rightarrow \infty} \int_0^1 \frac{kx^k}{1+x} dx = \frac{1}{2}$ by an appropriate convergence theorem after a substitution or estimate, naming the theorem used.
7. (Differentiation under the integral.) Using the Leibniz rule justified by the DCT (Theorem 12.5.5), compute $\frac{d}{dt} \int_0^1 e^{-tx} dx$ for $t > 0$ in two ways and check they agree.

8. (Diagram.) Sketch the "travelling spike" $f_k(x) = k \mathbf{1}_{(1/k, 2/k)}(x)$ for $k = 2, 4, 8$ on $[0, 1]$. Shade each spike, note each has area 1, and explain visually why pointwise convergence to 0 coexists with $\int f_k = 1$, the canonical failure of interchange without domination.
9. (Measure via integrals.) For $E \subseteq [a, b]$ measurable with $m(E) = \int_a^b \mathbf{1}_E$, use the Monotone Convergence Theorem to prove countable additivity: for pairwise disjoint measurable C_k , $m(\bigcup_k C_k) = \sum_k m(C_k)$. (This is the bridge from §10.4 to Lebesgue measure.)

Chapter 13

The Lebesgue Theory

13.1 Set Functions and Outer Measure

The Riemann integral handles continuous functions and a few well-behaved discontinuous ones, but it breaks down precisely when we most need it: it does not commute well with limits, and the integrable functions do not form a complete space. Lebesgue's remedy is to build the integral on a prior notion of the *size* (measure) of a set. This section develops that notion from the ground up — set functions, additivity, the elementary sets of $\mathbb{R}^k$, and the *outer measure* μ^* that assigns a size to *every* subset of $\mathbb{R}^k$.

Set functions and additivity

Definition 13.1.1 (Set function; additivity, Rudin 11.1–11.2). A *set function* ϕ assigns to each set A in some family of sets a value $\phi(A) \in \mathbb{R} = [-\infty, +\infty]$. We always assume ϕ is not identically $+\infty$ nor $-\infty$ on the family. Call ϕ :

- *additive* if $A \cap B = \emptyset \implies \phi(A \cup B) = \phi(A) + \phi(B)$ whenever $A, B, A \cup B$ are in the family;
- *finitely additive* if for pairwise disjoint $A_1, \dots, A_n$ in the family with $\bigcup A_i$ in the family, $\phi(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n \phi(A_i)$;
- *countably additive* (or σ -*additive*) if for pairwise disjoint $A_1, A_2, \dots$ with $\bigcup A_i$ in the family, $\phi(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \phi(A_i)$.

For the sums to make sense unambiguously one requires $\phi(A) \geq 0$ (or at least that $+\infty$ and $-\infty$ do not both occur). The values 0 and $\pm\infty$ are allowed.

Additivity is the single algebraic property that “size” must have: chopping a set into disjoint pieces and adding the pieces’ sizes recovers the whole. Countable additivity — the extension to infinitely many pieces — is the property that makes the *limit* theorems of integration work, and it is exactly what the Riemann theory lacks.

Proposition 13.1.2 (Elementary consequences of additivity, Rudin 11.3). *If $\phi \geq 0$ is countably additive on a family closed under the operations used, then for sets in the family: (a) $\phi(\emptyset) = 0$; (b) (monotonicity) $A \subseteq B \implies \phi(A) \leq \phi(B)$; (c) (countable subadditivity) $\phi(\bigcup_n A_n) \leq \sum_n \phi(A_n)$ even when the A_n overlap; (d) (continuity from below) if $A_1 \subseteq A_2 \subseteq \dots$ and $A = \bigcup_n A_n$, then $\phi(A_n) \rightarrow \phi(A)$.*

Proof. (a) Pick any A with $\phi(A)$ finite; $A = A \cup \emptyset$ disjointly, so $\phi(A) = \phi(A) + \phi(\emptyset)$, forcing $\phi(\emptyset) = 0$. (b) $B = A \cup (B \setminus A)$ disjointly, so $\phi(B) = \phi(A) + \phi(B \setminus A) \geq \phi(A)$. (c) *Disjointify*: set $B_1 = A_1$ and $B_n = A_n \setminus (A_1 \cup \dots \cup A_{n-1})$. The B_n are pairwise disjoint, $\bigcup B_n = \bigcup A_n$, and $B_n \subseteq A_n$, so by countable additivity and (b), $\phi(\bigcup A_n) = \sum_n \phi(B_n) \leq \sum_n \phi(A_n)$. (d) With the same B_n , $A_n = \bigcup_{k \leq n} B_k$ so $\phi(A_n) = \sum_{k \leq n} \phi(B_k)$, the partial sums of $\sum_k \phi(B_k) = \phi(A)$. $\square$

Caution. The disjointifying trick $B_n = A_n \setminus \bigcup_{k < n} A_k$ is the workhorse of the whole chapter: it converts *any* countable union into a *disjoint* one with the same union and smaller pieces. Memorize it. Continuity from

above ($A_1 \supseteq A_2 \supseteq \cdots$, $\phi(A_n) \rightarrow \phi(\bigcap A_n)$) needs an extra hypothesis $\phi(A_1) < \infty$; without it, $A_n = [n, \infty)$ has $\mu(A_n) = \infty$ for all n yet $\bigcap A_n = \emptyset$ has measure 0.

Intervals and elementary sets in $\mathbb{R}^k$

Definition 13.1.3 (Intervals and elementary sets, Rudin 11.4). In $\mathbb{R}^k$, an *interval* (a *k-cell*) is a product $I = \{x : a_i \leq x_i \leq b_i, i = 1, \dots, k\}$, where each $\leq$ may be replaced by $<$ at either end (so half-open intervals are allowed). The empty set is an interval. An *elementary set* is a finite union of intervals; let $\mathcal{E}$ denote the family of all elementary sets in $\mathbb{R}^k$.

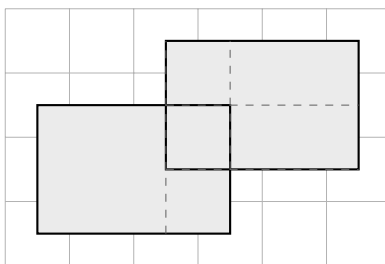
For an interval define the *volume*

$$m(I) = \prod_{i=1}^k (b_i - a_i).$$

If $A = I_1 \cup \cdots \cup I_n$ with the I_j pairwise disjoint, set $m(A) = \sum_{j=1}^n m(I_j)$.

Proposition 13.1.4 ($\mathcal{E}$ is a ring; m is well defined and additive, Rudin 11.4–11.5). *The family $\mathcal{E}$ is closed under finite unions, intersections, and differences (a ring of sets), and every elementary set is a finite union of disjoint intervals. The volume m above does not depend on the chosen disjoint decomposition, and m is a nonnegative additive set function on $\mathcal{E}$. Moreover m is regular: for $A \in \mathcal{E}$ and $\varepsilon > 0$ there are a closed elementary F and an open elementary G with $F \subseteq A \subseteq G$ and $m(G) - \varepsilon \leq m(A) \leq m(F) + \varepsilon$.*

Proof sketch. Two intervals intersect in an interval, and the difference of two intervals is a finite disjoint union of intervals (slice along each coordinate's endpoints); inductively the same holds for elementary sets, giving the ring property and disjoint decompositions. To see m is well defined, refine any two disjoint decompositions to a common grid of subintervals; m is additive on the grid, so both give the same total. Regularity: shrink each interval slightly to a closed one inside, or enlarge to an open one outside, controlling the volume change by ε . $\square$



$A = I_1 \cup I_2$, sliced into disjoint cells; $m(A) = \sum m(\text{cells})$

Figure 13.1. An elementary set in $\mathbb{R}^2$ and its decomposition into disjoint cells; the dashed cuts make the union disjoint without changing it.

Worked Example 13.1.5 (Volume is additive but not yet countably additive on $\mathcal{E}$). Show m is finitely additive on $\mathcal{E}$, and exhibit a countable disjoint family of elementary sets whose union is *not* elementary — so countable additivity cannot even be *stated* within $\mathcal{E}$.

Recognition. Trigger: “test whether a property survives passage to infinitely many pieces.”

Solution. Finite additivity is Proposition 13.1.4. But $[0, 1) = \bigcup_{n \geq 1} [1 - \frac{1}{n}, 1 - \frac{1}{n+1})$ is a countable disjoint union of intervals; each piece is elementary, yet a general countable union of intervals (say a “fat Cantor” complement) need not be a *finite* union, hence is outside $\mathcal{E}$. *Check.* The lengths sum correctly, $\sum_n (\frac{1}{n} - \frac{1}{n+1}) = 1 = m([0, 1))$; the obstruction is not the arithmetic but the family $\mathcal{E}$ being too small. This is exactly why we must *enlarge* $\mathcal{E}$ to a σ -algebra. ■

Outer measure

To measure *arbitrary* sets we cover them, from outside, by countably many intervals and take the cheapest cover.

Definition 13.1.6 (Lebesgue outer measure, Rudin 11.7–11.8). For any $E \subseteq \mathbb{R}^k$ the *outer measure* is

$$\mu^*(E) = \inf \left\{ \sum_{n=1}^{\infty} m(I_n) : E \subseteq \bigcup_{n=1}^{\infty} I_n, I_n \text{ open intervals} \right\} \in [0, +\infty].$$

The infimum is over all countable covers of E by open intervals. (Rudin states it via elementary covers; the two agree.)

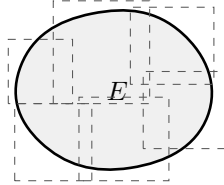
Theorem 13.1.7 (Properties of outer measure, Rudin 11.8). *The outer measure μ^* satisfies:* (a) $\mu^*(\emptyset) = 0$ and $\mu^* \geq 0$; (b) (monotone) $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$; (c) (countably subadditive) $\mu^*(\bigcup_n E_n) \leq \sum_n \mu^*(E_n)$ for any sets E_n ; (d) $\mu^*(I) = m(I)$ for every interval I — outer measure extends volume.

Proof. (a),(b) are immediate from the definition (the empty cover, resp. a cover of B also covers A). (c) Given $\varepsilon > 0$, for each n pick intervals $\{I_{n,j}\}_j$ covering E_n with $\sum_j m(I_{n,j}) \leq \mu^*(E_n) + \varepsilon 2^{-n}$. The countable family $\{I_{n,j}\}_{n,j}$ covers $\bigcup_n E_n$, so

$$\mu^*\left(\bigcup_n E_n\right) \leq \sum_{n,j} m(I_{n,j}) \leq \sum_n (\mu^*(E_n) + \varepsilon 2^{-n}) = \sum_n \mu^*(E_n) + \varepsilon.$$

Let $\varepsilon \rightarrow 0$. (d) “ $\leq$ ”: I covers itself, so $\mu^*(I) \leq m(I)$ (enlarge I to an open interval of volume $\leq m(I) + \varepsilon$). “ $\geq$ ”: any countable open cover of the compact closed interval $\bar{I}$ has a finite subcover (Heine–Borel), and finite additivity of m on $\mathcal{E}$ gives $m(\bar{I}) \leq \sum_{\text{finite}} m(I_n) \leq \sum_n m(I_n)$; take the inf to get $m(I) \leq \mu^*(I)$. So equality holds. $\square$

Caution. Outer measure is only *subadditive*, not *additive*: in general $\mu^*(A \cup B) < \mu^*(A) + \mu^*(B)$ can fail to be an equality even when $A \cap B = \emptyset$. This defect is unavoidable for *all* subsets of $\mathbb{R}^k$ (there exist non-measurable sets, by the Axiom of Choice). The fix — restricting to a good sub-family on which μ^* is countably additive — is the content of Section 13.2.



$$\mu^*(E) = \inf \sum_n m(I_n) \\ \text{over countable open} \\ \text{covers } E \subseteq \bigcup_n I_n$$

Figure 13.2. Outer measure covers E from outside by countably many intervals and takes the cheapest total volume.

Worked Example 13.1.8 (The outer measure of a countable set is zero). Show $\mu^*(\mathbb{Q} \cap [0, 1]) = 0$, and more generally $\mu^*(E) = 0$ for any countable $E \subseteq \mathbb{R}^k$.

Recognition. Trigger: “compute/bound an outer measure”; cover with shrinking intervals summing a geometric series.

Solution. Enumerate $E = \{x_1, x_2, \dots\}$. Given $\varepsilon > 0$, put around x_n an open interval I_n of volume $\varepsilon 2^{-n}$ (a cube of side $(\varepsilon 2^{-n})^{1/k}$ centered at x_n). Then $E \subseteq \bigcup I_n$ and $\sum_n m(I_n) = \varepsilon \sum_n 2^{-n} = \varepsilon$. Hence $\mu^*(E) \leq \varepsilon$ for all ε , so $\mu^*(E) = 0$. *Check.* This re-proves that $[0, 1]$, of outer measure 1, is uncountable — a countable set could never fill it. Note $\mu^*(\mathbb{Q} \cap [0, 1]) = 0$ while $\mu^*([0, 1]) = 1$, yet $\mathbb{Q} \cap [0, 1]$ is *dense*: measure and topological size are independent. $\blacksquare$

Method (Bounding an outer measure $\mu^*(E)$). *Recognition trigger.* You must show $\mu^*(E) \leq t$ (e.g. E is “small”) or identify $\mu^*(E)$ exactly.

1. For an *upper* bound $\mu^*(E) \leq t$: exhibit *one* countable cover by open intervals with $\sum m(I_n) \leq t + \varepsilon$, then let $\varepsilon \rightarrow 0$. Geometric weights $\varepsilon 2^{-n}$ are the standard device.
2. For a *lower* bound $\mu^*(E) \geq t$: show *every* cover has total volume $\geq t$. Compactness (Heine–Borel) plus finite additivity of m is the standard tool; or use monotonicity from a known subset.
3. To get equality, combine the two; for intervals this is Theorem 13.1.7(d).

Remark. Outer measure is defined on *all* subsets of $\mathbb{R}^k$ and is monotone and countably subadditive — a so-called *outer measure* in the abstract sense. What it lacks is additivity. Carathéodory’s insight (next section) is that the sets which “split every test set additively” form a σ -algebra on which μ^* restricts to a genuine measure. We will instead follow Rudin’s symmetric-difference route to the same destination, remarking on the Carathéodory criterion as the cleaner alternative.

Further worked examples

Worked Example 13.1.9 (Continuity from above can fail without a finiteness hypothesis). Let μ be Lebesgue measure on $\mathbb{R}$ and $A_n = [n, \infty)$. Test continuity from above on the decreasing chain $A_1 \supseteq A_2 \supseteq \cdots$, and then exhibit the correct hypothesis that repairs it.

Recognition. Trigger: “does $\mu(A_n) \rightarrow \mu(\bigcap A_n)$ for a shrinking chain?” — continuity from above (Proposition 13.1.2) requires some $\mu(A_N) < \infty$.

Solution. Each $A_n = [n, \infty)$ has $\mu(A_n) = \infty$, while $\bigcap_n A_n = \emptyset$ has $\mu = 0$. So $\mu(A_n) = \infty \not\rightarrow 0 = \mu(\bigcap A_n)$: continuity from above *fails*. The repair: if some $\mu(A_N) < \infty$, apply continuity from below to the increasing sets $B_n = A_N \setminus A_n \nearrow A_N \setminus \bigcap A_k$ to recover $\mu(A_n) \rightarrow \mu(\bigcap A_k)$. Here no A_N has finite measure, so the hypothesis is genuinely violated. *Check.* Replace $A_n = [0, 1/n)$; now $\mu(A_1) = 1 < \infty$, $\mu(A_n) = 1/n \rightarrow 0 = \mu(\emptyset)$ — continuity from above holds exactly when the finiteness guard is met. ■

Worked Example 13.1.10 (Outer measure of an interval equals its length, computed by hand). Verify directly that $\mu^*([0, 1]) = 1$ in $\mathbb{R}$, doing both the easy upper bound and the compactness lower bound.

Recognition. Trigger: “pin an outer measure exactly” — use the two-sided Method (upper cover, lower compactness).

Solution. Upper: for $\varepsilon > 0$ the single open interval $(-\varepsilon, 1 + \varepsilon)$ covers $[0, 1]$ with length $1 + 2\varepsilon$, so $\mu^*([0, 1]) \leq 1 + 2\varepsilon$; let $\varepsilon \rightarrow 0$ to get $\mu^*([0, 1]) \leq 1$. *Lower:* let $\{I_n\}$ be any countable open cover of the compact set $[0, 1]$. By Heine–Borel finitely many $I_{n_1}, \dots, I_{n_m}$ already cover $[0, 1]$, and a finite open cover of an interval has total length $\geq$ the interval’s length (sort endpoints and telescope), so $\sum_n \ell(I_n) \geq \sum_j \ell(I_{n_j}) \geq 1$. Taking the inf, $\mu^*([0, 1]) \geq 1$. Hence $\mu^*([0, 1]) = 1$. *Check.* Agrees with Theorem 13.1.7(d), $\mu^*(I) = m(I)$, and with the intuition that no clever cover can beat covering $[0, 1]$ by itself. ■

Worked Example 13.1.11 (A non-example: outer measure is only subadditive, never additive on all sets). Show that outer *additivity* $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$ for disjoint A, B would force the additivity μ^* provably lacks — so do not assume it.

Recognition. Trigger: “is this plausible-looking identity actually a theorem?” — recall the defect of μ^* (the pitfall after Theorem 13.1.7).

Solution. Subadditivity always gives $\mu^*(A \cup B) \leq \mu^*(A) + \mu^*(B)$. The reverse can fail: there exist disjoint sets (Vitali-type) with $\mu^*(A \cup B) < \mu^*(A) + \mu^*(B)$ strictly. A concrete *near-failure*: take $A = \mathbb{Q} \cap [0, 1]$, $B = [0, 1] \setminus \mathbb{Q}$. These are disjoint with $\mu^*(A) = 0$ and $\mu^*(B) = 1$, so here the sum happens to work, $0 + 1 = 1 = \mu^*([0, 1])$. But additivity of μ^* across *all* disjoint pairs is false; it holds only once we restrict to measurable sets. *Check.* The moral is procedural — never write $\mu^*(A \sqcup B) = \mu^*(A) + \mu^*(B)$ as an equality unless at least one set is measurable; otherwise you have only “ $\leq$ ”. ■

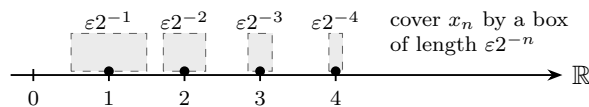


Figure 13.3. Why a countable set is null: drop a box of length $\varepsilon 2^{-n}$ on the n -th point; the total length is $\varepsilon \sum 2^{-n} = \varepsilon$, so $\mu^* = 0$ (Worked Example after Definition 13.1.6).

Exercises (Rudin, Ch.11).

- Let $\mathcal{E}$ be the ring of all elementary subsets of $(0, 1]$. For $0 < a \leq b \leq 1$ define $\phi([a, b]) = \phi((a, b)) = \phi((a, b]) = \phi((a, b)) = b - a$, but define

$$\phi((0, b]) = \phi((0, b)) = 1 + b \quad (0 < b \leq 1).$$

Show that this gives an additive set function ϕ on $\mathcal{E}$ which is not regular and which cannot be extended to a countably additive set function on a σ -ring.

Additional exercises.

1. (Monotonicity of outer measure.) Show directly from the definition of Lebesgue outer measure m^* that $A \subseteq B$ implies $m^*(A) \leq m^*(B)$, and that $m^*(\emptyset) = 0$.
2. (Countable subadditivity.) Prove that for any sequence $\{A_n\}$ of subsets of $\mathbb{R}$,

$$m^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} m^*(A_n).$$

Where exactly is the $\varepsilon/2^n$ trick used?

3. (Translation invariance.) Show $m^*(A+t) = m^*(A)$ for every $A \subseteq \mathbb{R}$ and $t \in \mathbb{R}$, where $A+t = \{a+t : a \in A\}$.
4. (Countable sets are null.) Prove $m^*(\mathbb{Q}) = 0$. More generally, show every countable subset of $\mathbb{R}$ has outer measure 0, and conclude that m^* is not determined by cardinality alone (the Cantor set is uncountable yet null).
5. (A non-example for additivity of m^* .) Explain why one cannot expect $m^*(A \cup B) = m^*(A) + m^*(B)$ for arbitrary disjoint A, B . (You need not construct a Vitali set; argue why finite additivity on *all* sets would clash with translation invariance plus the value $m^*([0, 1]) = 1$.)
6. (Diagram: covering by intervals.) Draw the segment $[0, 1]$ on a number line and, above it, sketch a countable cover of $\mathbb{Q} \cap [0, 1]$ by open intervals of total length $< \varepsilon$. Mark the lengths $\varepsilon/2, \varepsilon/4, \varepsilon/8, \dots$ and explain how the picture proves $m^*(\mathbb{Q} \cap [0, 1]) = 0$.
7. (Outer measure of an interval.) Prove $m^*([a, b]) = b - a$ directly: subadditivity gives $\leq$, and compactness (Heine–Borel) gives $\geq$. Identify precisely where compactness enters.
8. (Regularity from above.) Show that for every $A \subseteq \mathbb{R}$ and every $\varepsilon > 0$ there is an open set $G \supseteq A$ with $m^*(G) \leq m^*(A) + \varepsilon$. Deduce $m^*(A) = \inf\{m^*(G) : G \text{ open, } G \supseteq A\}$.

13.2 The Construction of Lebesgue Measure

We now isolate the sets on which μ^* behaves like a genuine measure. Rudin's route uses the *symmetric difference* $A \triangle B = (A \setminus B) \cup (B \setminus A)$ as a distance: a set is measurable if it can be approximated arbitrarily well, in μ^* , by elementary sets. We prove that the measurable sets form a σ -algebra and that μ^* is countably additive on it. Then we abstract the whole construction.

Measurable sets via approximation

Definition 13.2.1 (Distance and finite measurability, Rudin 11.9). For $A, B \subseteq \mathbb{R}^k$ put $d(A, B) = \mu^*(A \triangle B)$. A set $E \subseteq \mathbb{R}^k$ is *finitely μ -measurable*, written $E \in \mathcal{M}_F$, if there is a sequence of elementary sets $A_n \in \mathcal{E}$ with $d(A_n, E) = \mu^*(A_n \triangle E) \rightarrow 0$. A set E is *μ -measurable*, $E \in \mathcal{M}$, if E is a countable union of sets in $\mathcal{M}_F$. For $E \in \mathcal{M}_F$ define $\mu(E) = \lim_n m(A_n)$ (shown below to be well defined); μ is *Lebesgue measure*.

The idea: $d(A, B)$ is small when A and B differ by a set of tiny outer measure. An elementary set has a known volume $m(A_n)$; if E is squeezed up against the A_n in the d -metric, we *define* its measure as the limit of those volumes. The triangle inequality for d makes this consistent.

Lemma 13.2.2 (d is a pseudometric; μ^* and m are d -continuous, Rudin 11.10). For all sets A, B, C : (a) $d(A, B) = d(B, A)$ and $d(A, B) \leq d(A, C) + d(C, B)$ (triangle inequality); (b) $A \triangle B \subseteq (A \triangle C) \cup (C \triangle B)$, and similarly for $\cup, \cap, \setminus$: each is “ d -continuous”

$$d(A_1 \cup A_2, B_1 \cup B_2) \text{ (and } \cap, \setminus) \leq d(A_1, B_1) + d(A_2, B_2);$$

(c) $|\mu^*(A) - \mu^*(B)| \leq d(A, B)$ when these are finite.

Proof. (a) Symmetry is clear; the triangle inequality follows from $A \triangle B \subseteq (A \triangle C) \cup (C \triangle B)$ (a point in $A \triangle B$ differs between A, B , hence differs from C on at least one side) plus subadditivity of μ^* . (b) Set-algebra inclusions like $(A_1 \cup A_2) \triangle (B_1 \cup B_2) \subseteq (A_1 \triangle B_1) \cup (A_2 \triangle B_2)$ are checked by chasing an element; apply μ^* and subadditivity. (c) $A \subseteq B \cup (A \triangle B)$ gives $\mu^*(A) \leq \mu^*(B) + d(A, B)$, and symmetrically; combine. $\square$

Theorem 13.2.3 (μ is well defined and finitely additive on $\mathcal{M}_F$, Rudin 11.10). *If $A_n \in \mathcal{E}$ and $d(A_n, E) \rightarrow 0$, then $\lim_n m(A_n)$ exists, is finite, and is independent of the approximating sequence; call it $\mu(E)$. Moreover $\mu(E) = \mu^*(E)$ for $E \in \mathcal{M}_F$, and μ is finitely additive: if $E, F \in \mathcal{M}_F$ are disjoint then $E \cup F \in \mathcal{M}_F$ and $\mu(E \cup F) = \mu(E) + \mu(F)$.*

Proof sketch. On $\mathcal{E}$, $m = \mu^*$ and $|m(A_n) - m(A_p)| = |\mu^*(A_n) - \mu^*(A_p)| \leq d(A_n, A_p) \leq d(A_n, E) + d(E, A_p) \rightarrow 0$, so $(m(A_n))$ is Cauchy in $\mathbb{R}$, hence convergent; two sequences approximating E interleave to a third, forcing equal limits. Continuity (c) of Lemma 13.2.2 gives $\mu(E) = \lim m(A_n) = \lim \mu^*(A_n) = \mu^*(E)$. For additivity, approximate E, F by elementary A_n, B_n ; then $A_n \cup B_n$ approximates $E \cup F$ and, since $E \cap F = \emptyset$ makes the overlap $\mu^*(A_n \cap B_n) \rightarrow 0$, $m(A_n \cup B_n) = m(A_n) + m(B_n) - m(A_n \cap B_n) \rightarrow \mu(E) + \mu(F)$. $\square$

The main theorem: $\mathcal{M}$ is a σ -algebra and μ a measure

Definition 13.2.4 (σ -ring, σ -algebra, Rudin 11.5/11.11). A family $\mathcal{R}$ of subsets of a set X is a *ring* if it is closed under $A \cup B$ and $A \setminus B$; a σ -*ring* if also closed under countable unions $\bigcup_{n=1}^{\infty} A_n$. A σ -ring containing X itself (equivalently, closed under complements) is a σ -*algebra* (or σ -*field*). A σ -algebra is automatically closed under countable intersections, since $\bigcap_n A_n = X \setminus \bigcup_n (X \setminus A_n)$.

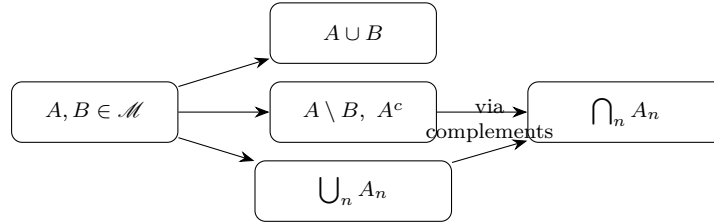


Figure 13.4. A σ -algebra is closed under the countable Boolean operations: complement, countable union, countable intersection.

Theorem 13.2.5 (Construction of Lebesgue measure, Rudin 11.11). *The family $\mathcal{M}$ of μ -measurable subsets of $\mathbb{R}^k$ is a σ -algebra, and μ (extended to $\mathcal{M}$ by $\mu(E) = \sup\{\mu(F) : F \in \mathcal{M}_F, F \subseteq E\}$, allowing $+\infty$) is countably additive on $\mathcal{M}$. Thus $(\mathbb{R}^k, \mathcal{M}, \mu)$ is a complete measure space, and $\mu(I) = m(I)$ for intervals.*

Proof sketch (the structure to remember). Closure of $\mathcal{M}_F$ under $\cup, \cap, \setminus$. If $E, F \in \mathcal{M}_F$ with approximants A_n, B_n , then $A_n \cup B_n, A_n \cap B_n, A_n \setminus B_n$ are elementary and, by the d -continuity of these operations (Lemma 13.2.2(b)), approximate $E \cup F, E \cap F, E \setminus F$. So $\mathcal{M}_F$ is a ring.

Countable additivity, finite-measure case. Let $E = \bigcup_n E_n$ with $E_n \in \mathcal{M}_F$ pairwise disjoint and $\mu^*(E) < \infty$. Disjoint finite additivity gives $\sum_{n \leq N} \mu(E_n) = \mu\left(\bigcup_{n \leq N} E_n\right) \leq \mu^*(E)$ for every N , so $\sum_n \mu(E_n) \leq \mu^*(E) < \infty$. Conversely subadditivity gives $\mu^*(E) \leq \sum_n \mu(E_n)$. Hence $E \in \mathcal{M}_F$ and $\mu(E) = \sum_n \mu(E_n)$.

Passing to $\mathcal{M}$. A general measurable $E = \bigcup_n F_n$ ($F_n \in \mathcal{M}_F$) is rewritten, by disjointifying, as a disjoint union of $\mathcal{M}_F$ -sets; closure under complements follows because E^c is again a countable union of finite-measure measurable pieces (intersect with an exhausting sequence of large cubes). Countable additivity on all of $\mathcal{M}$ then follows from the finite-measure case by exhaustion. Completeness: if $\mu^*(Z) = 0$ then $Z \in \mathcal{M}_F$ with $\mu(Z) = 0$ (approximate by $\emptyset$), so every subset of a null set is measurable. $\square$

Remark (The Carathéodory criterion — the slick alternative). A set E is *Carathéodory measurable* if for every test set $T \subseteq \mathbb{R}^k$,

$$\mu^*(T) = \mu^*(T \cap E) + \mu^*(T \setminus E).$$

That is, E splits every set additively. One checks directly that these sets form a σ -algebra and μ^* restricts to a countably additive measure on it — no metric, no elementary-set approximation. The two definitions yield the *same* measurable sets and the same measure. Rudin's d -route is more concrete; the Carathéodory route generalizes verbatim to any outer measure on any set.

Worked Example 13.2.6 (An interval boundary is null; open and closed boxes have equal measure). Show the boundary of a k -cell has measure 0, so μ does not depend on whether the defining inequalities are strict.

Recognition. Trigger: “show a set is null” — cover a lower-dimensional face by a thin slab.

Solution. The boundary of $I = \prod [a_i, b_i]$ is a finite union of faces, each lying in a hyperplane $\{x_i = c\}$. A bounded piece of such a hyperplane is covered by one slab $\{c - \delta \leq x_i \leq c + \delta\}$ intersected with a large box, of volume $2\delta \cdot (\text{const}) \rightarrow 0$ as $\delta \rightarrow 0$. So each face is null, and a finite union of null sets is null. Hence $\mu(\partial I) = 0$ and $\mu(\text{open } I) = \mu(\text{closed } I) = m(I)$. *Check.* Consistent with $\mu(\mathbb{Q} \cap [0, 1]) = 0$ — “thin” lower-dimensional or countable sets carry no Lebesgue mass. ■

Borel sets and the structure of $\mathcal{M}$

Definition 13.2.7 (Borel sets, Rudin 11.11). The *Borel σ -algebra* $\mathcal{B} = \mathcal{B}(\mathbb{R}^k)$ is the smallest σ -algebra containing all open sets (equivalently all intervals). Its members are the *Borel sets*: open and closed sets, F_σ (countable unions of closed), G_δ (countable intersections of open), and everything generated from these by countable Boolean operations.

Theorem 13.2.8 ($\mathcal{B} \subseteq \mathcal{M}$ and regularity, Rudin 11.11). *Every Borel set is Lebesgue measurable: $\mathcal{B} \subseteq \mathcal{M}$. Lebesgue measure is regular: for $E \in \mathcal{M}$,*

$$\mu(E) = \inf\{\mu(G) : E \subseteq G, G \text{ open}\} = \sup\{\mu(K) : K \subseteq E, K \text{ compact}\}.$$

Consequently every measurable E is a Borel set modulo a null set: $E = B \setminus Z$ (or $B \cup Z$) with $B \in \mathcal{B}$ and $\mu(Z) = 0$.

Proof sketch. Open sets are countable unions of intervals (e.g. dyadic cubes), hence in $\mathcal{M}$; since $\mathcal{M}$ is a σ -algebra containing the open sets and $\mathcal{B}$ is the smallest such, $\mathcal{B} \subseteq \mathcal{M}$. Outer regularity is the definition of μ^* rephrased (open covers); inner regularity follows by applying outer regularity to the complement within a large cube and using Heine–Borel. The “Borel modulo null” form comes from choosing open $G_n \supseteq E$ with $\mu(G_n \setminus E) < 1/n$ and intersecting. □

Caution. Not every Lebesgue measurable set is Borel (there are strictly more measurable sets: $|\mathcal{M}| = 2^{\mathfrak{c}} > \mathfrak{c} = |\mathcal{B}|$), and not every set is measurable at all — a Vitali set, built with the Axiom of Choice, has no well-defined μ . So the chain is strict: $\mathcal{B} \subsetneq \mathcal{M} \subsetneq \mathcal{P}(\mathbb{R}^k)$. Measurability is the price of admission to the integral.

Abstract measure spaces

The construction never really used the geometry of $\mathbb{R}^k$ — only an outer measure and a ring with a finitely additive set function. Stripping the geometry away gives the abstract framework that the rest of the chapter actually needs.

Definition 13.2.9 (Measure space, Rudin 11.11). A *measurable space* is a pair $(X, \mathcal{M})$ with $\mathcal{M}$ a σ -algebra of subsets of X ; members of $\mathcal{M}$ are *measurable sets*. A *measure* on it is a countably additive set function $\mu : \mathcal{M} \rightarrow [0, +\infty]$ with $\mu(\emptyset) = 0$. The triple $(X, \mathcal{M}, \mu)$ is a *measure space*. It is *finite* if $\mu(X) < \infty$, *σ -finite* if $X = \bigcup_n X_n$ with $\mu(X_n) < \infty$, and *complete* if every subset of a μ -null set is measurable.

Remark. From now on $(X, \mathcal{M}, \mu)$ denotes a fixed measure space, and “measurable” means “in $\mathcal{M}$ ”. Everything — measurable functions, the integral, the convergence theorems — is developed at this level of generality. Lebesgue measure on $\mathbb{R}^k$ is the running example; counting measure (where $\mu(E) = \#E$) turns integrals into sums, and the same theorems then specialize to series. We say a property holds *almost everywhere* (a.e.) if it holds outside a set of measure 0.

Worked Example 13.2.10 (Counting measure makes integration into summation). On $X = \mathbb{N}$ with $\mathcal{M} = \mathcal{P}(\mathbb{N})$, let $\mu(E) = \#E$ (the number of elements, possibly $+\infty$). Verify μ is a measure and identify what “a.e.” means.

Recognition. Trigger: “check the measure axioms / interpret in a discrete model.”

Solution. $\mu(\emptyset) = 0$, and for disjoint E_n , $\#\bigcup E_n = \sum \#E_n$ since elements are counted once each — countable additivity holds (both sides may be $+\infty$ together). The only null set is $\emptyset$, so “a.e.” means “everywhere”: no point is negligible. *Check.* The Lebesgue integral against this μ will be $\int f d\mu = \sum_n f(n)$, so the convergence theorems below become theorems about interchanging limits and sums — e.g. the Monotone Convergence Theorem becomes Tonelli’s theorem for series. ■

Further worked examples

Worked Example 13.2.11 (The Cantor set is uncountable yet Lebesgue null). Let $C \subseteq [0, 1]$ be the middle-thirds Cantor set. Compute $\mu(C)$ and reconcile it with the fact that C is uncountable.

Recognition. Trigger: “measure of a set built by removing a geometric series of pieces” — use $\mu([0, 1]) = 1$ minus the removed open intervals.

Solution. At stage n one removes 2^{n-1} open middle thirds, each of length 3^{-n} . The total removed length is

$$\sum_{n=1}^{\infty} 2^{n-1} \cdot 3^{-n} = \frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n = \frac{1}{3} \cdot \frac{1}{1 - 2/3} = 1.$$

Since $C = [0, 1]$ minus these disjoint open intervals, $\mu(C) = 1 - 1 = 0$. Yet C is in bijection with $\{0, 1\}^{\mathbb{N}}$ (ternary expansions using only digits 0, 2), so $|C| = \mathfrak{c}$ is uncountable. *Check.* C is an uncountable null set — so “null” is genuinely weaker than “countable” (contrast Worked Example for $\mu^*(\mathbb{Q} \cap [0, 1]) = 0$, where the set was countable). It also shows $\mathcal{M}$ has at least $2^{\mathfrak{c}}$ sets, since every subset of the null set C is measurable by completeness. ■

Worked Example 13.2.12 (A fat Cantor set: positive measure with empty interior). Build a closed set $F \subseteq [0, 1]$ with empty interior (nowhere dense) yet $\mu(F) = \frac{1}{2}$.

Recognition. Trigger: “decouple topological smallness from measure” — remove a *summable* amount of length rather than a fixed proportion.

Solution. Repeat the Cantor construction but at stage n remove 2^{n-1} open middle intervals each of length 4^{-n} (instead of 3^{-n}). Total removed: $\sum_{n \geq 1} 2^{n-1} 4^{-n} = \frac{1}{4} \sum_{n \geq 0} (1/2)^n = \frac{1}{4} \cdot 2 = \frac{1}{2}$. The remainder F is closed (complement of an open set), contains no interval (every stage punches a hole into each surviving interval, so no interval survives), hence is nowhere dense; and $\mu(F) = 1 - \frac{1}{2} = \frac{1}{2}$. *Check.* F is nowhere dense yet has positive measure — the indicator χ_F is thus *not* Riemann integrable (its discontinuity set is $\partial F = F$, of measure $\frac{1}{2} > 0$), foreshadowing Lebesgue’s criterion in Section 13.5. ■

Worked Example 13.2.13 (A non-example: finite additivity does not give countable additivity). Exhibit a finitely additive set function on a σ -algebra that is *not* countably additive, so the two notions are genuinely different.

Recognition. Trigger: “is finite additivity enough?” — look for a “measure” that ignores finite sets.

Solution. On $X = \mathbb{N}$, $\mathcal{M} = \mathcal{P}(\mathbb{N})$, define $\phi(E) = 0$ if E is finite and $\phi(E) = 1$ if E is cofinite, extended (using a free ultrafilter) to all sets with values in $\{0, 1\}$ so that ϕ is finitely additive. Then for singletons, $\sum_n \phi(\{n\}) = \sum_n 0 = 0$, but $\phi(\bigcup_n \{n\}) = \phi(\mathbb{N}) = 1$. So $1 = \phi(\mathbb{N}) \neq \sum_n \phi(\{n\}) = 0$: finitely additive but not countably additive. *Check.* Countable additivity is the strictly stronger axiom that the whole convergence machinery rests on; the elementary volume m on $\mathcal{E}$ was finitely additive, and the *point* of the construction was to extend it to a *countably* additive μ . ■

Exercises (Rudin, Ch.11).

1. Suppose $\{n_k\}$ is an increasing sequence of positive integers and E is the set of all $x \in (-\pi, \pi)$ at which $\{\sin n_k x\}$ converges. Prove that $m(E) = 0$. (Hint: for every $A \subseteq E$,

$$\int_A \sin n_k x \, dx \rightarrow 0, \quad \int_A (\sin n_k x)^2 \, dx = \int_A \frac{1}{2}(1 - \cos 2n_k x) \, dx \rightarrow \frac{1}{2}m(A)$$

as $k \rightarrow \infty$.)

Additional exercises.

- (Caratheodory test.) Recall A is measurable iff $m^*(T) = m^*(T \cap A) + m^*(T \setminus A)$ for every test set T . Show that to verify measurability it suffices to check “ $\geq$ ” (the “ $\leq$ ” direction is automatic). Explain why.
- (Sets of outer measure zero are measurable.) Prove that if $m^*(A) = 0$ then A is Lebesgue measurable, and that every subset of A is measurable as well. (This is why the Lebesgue σ -algebra is complete.)
- (Half-lines generate the algebra.) Show that (a, ∞) is measurable for every $a \in \mathbb{R}$ using the Caratheodory criterion, and explain how this single fact, with the σ -algebra axioms, yields measurability of every interval.
- (Finite additivity to countable additivity.) Suppose $\{E_n\}$ are pairwise disjoint measurable sets. Using only finite additivity and continuity from below, prove $m(\bigcup E_n) = \sum m(E_n)$.
- (Continuity from above needs finiteness.) Let $E_n = [n, \infty)$. Compute $m(E_n)$ and $m(\bigcap_n E_n)$ and show $\lim m(E_n) \neq m(\bigcap E_n)$. Which hypothesis of the continuity-from-above theorem fails?
- (Non-example: a non-measurable set, Vitali.) Sketch the construction: on $[0, 1]$ put $x \sim y$ iff $x - y \in \mathbb{Q}$, choose one representative per class to form V , and consider the translates $V \oplus q$ ($q \in \mathbb{Q} \cap [0, 1]$, addition mod 1). Show that measurability of V would force $1 = \sum_q m(V)$, impossible whether $m(V) = 0$ or $m(V) > 0$.
- (Diagram: the dyadic sweep-out.) Draw the first three stages of writing an open subset of $(0, 1)$ as a countable disjoint union of open intervals, and explain how the picture underlies the construction of m on open sets.
- (Cantor set has measure zero.) Using the standard middle-thirds construction, compute m of the set removed at stage n , sum the series, and conclude $m(C) = 0$ while C is uncountable. Include a number-line diagram of stages 0, 1, 2.

13.3 Measurable Functions

With a measure space $(X, \mathcal{M}, \mu)$ fixed, we now single out the functions we can integrate. The defining requirement is mild — preimages of intervals must be measurable — yet it is preserved by every limiting operation of analysis: sup, inf, lim sup, lim inf, pointwise limits, sums, and products. This robustness under limits is the whole point, and it is exactly what makes the convergence theorems of Section 13.5 possible.

Definition and the basic criterion

Definition 13.3.1 (Measurable function, Rudin 11.12). Let $f : X \rightarrow \overline{\mathbb{R}}$ (extended-real valued). Then f is *measurable* if for every $a \in \mathbb{R}$ the set

$$\{x \in X : f(x) > a\} = f^{-1}((a, +\infty])$$

is measurable (i.e. in $\mathcal{M}$). A complex function $f = u + iv$ is measurable if u and v are. A map into $\mathbb{R}^k$ is measurable if each component is.

Intuitively, measurability is the integration-theory analogue of continuity: where continuity demands that preimages of *open* sets be open, measurability only demands that preimages of half-lines $(a, \infty]$ be *measurable*. This is a far weaker requirement — it never mentions topology of the domain — which is exactly why the class is closed under the pointwise limits that destroy continuity. One threshold a at a time is all we ever check.

Theorem 13.3.2 (Equivalent criteria, Rudin 11.13). For $f : X \rightarrow \overline{\mathbb{R}}$ the following are equivalent: (a) $\{f > a\} \in \mathcal{M}$ for all a ; (b) $\{f \geq a\} \in \mathcal{M}$ for all a ; (c) $\{f < a\} \in \mathcal{M}$ for all a ; (d) $\{f \leq a\} \in \mathcal{M}$ for all a . Each implies $f^{-1}(V) \in \mathcal{M}$ for every open $V \subseteq \mathbb{R}$ (and every Borel V).

Proof. $\{f \geq a\} = \bigcap_{n=1}^{\infty} \{f > a - \frac{1}{n}\}$ is a countable intersection of measurable sets, so (a) $\implies$ (b); $\{f < a\} = X \setminus \{f \geq a\}$ gives (b) $\implies$ (c); $\{f \leq a\} = \bigcap_n \{f < a + \frac{1}{n}\}$ gives (c) $\implies$ (d); and $\{f > a\} = X \setminus \{f \leq a\}$

closes the loop (d) $\implies$ (a). Finally any open $V = \bigcup_n (a_n, b_n)$, and $\{f \in (a_n, b_n)\} = \{f > a_n\} \cap \{f < b_n\}$ is measurable; a countable union is measurable, and the Borel sets are generated from open sets by countable Boolean operations, all of which preserve measurability of preimages. $\square$

Worked Example 13.3.3 (Continuous functions and indicators are measurable). Show that on $(\mathbb{R}^k, \mathcal{M}, \mu)$ every continuous f is Lebesgue measurable, and that the characteristic function χ_E is measurable iff E is.

Recognition. Trigger: “verify measurability” — compute the sublevel/superlevel sets directly.

Solution. If f is continuous then $\{f > a\} = f^{-1}((a, \infty))$ is open (preimage of open under continuous), hence Borel, hence in $\mathcal{M}$ (Theorem 13.2.8). For χ_E (value 1 on E , 0 off E):

$$\{\chi_E > a\} = \begin{cases} X, & a < 0, \\ E, & 0 \leq a < 1, \\ \emptyset, & a \geq 1. \end{cases}$$

All three are measurable exactly when $E \in \mathcal{M}$. *Check.* Measurability of χ_E is *equivalent* to measurability of E — characteristic functions are the bridge between measurable sets and measurable functions. $\blacksquare$

Closure under limiting operations

Theorem 13.3.4 (Sup, inf, limsup, liminf of measurable functions, Rudin 11.14–11.15). *Let $f_1, f_2, \dots$ be measurable $X \rightarrow \overline{\mathbb{R}}$. Then*

$$g = \sup_n f_n, \quad h = \inf_n f_n, \quad \limsup_n f_n, \quad \liminf_n f_n$$

are all measurable. In particular, if $f(x) = \lim_n f_n(x)$ exists (in $\overline{\mathbb{R}}$) for every x , then f is measurable. The set $\{x : \lim_n f_n(x) \text{ exists}\}$ is measurable.

Proof. The key identity is for the supremum:

$$\{g > a\} = \{\sup_n f_n > a\} = \bigcup_{n=1}^{\infty} \{f_n > a\},$$

because $\sup_n f_n(x) > a$ iff some $f_n(x) > a$. A countable union of measurable sets is measurable, so g is measurable. For the infimum, $\inf_n f_n = -\sup_n (-f_n)$, or use $\{h < a\} = \bigcup_n \{f_n < a\}$. Now

$$\limsup_n f_n = \inf_{m \geq 1} \left(\sup_{n \geq m} f_n \right), \quad \liminf_n f_n = \sup_{m \geq 1} \left(\inf_{n \geq m} f_n \right),$$

each built from sup/inf of measurable functions, hence measurable. When the limit exists it equals $\limsup = \liminf$, so f is measurable. Finally $\{\lim f_n \text{ exists}\} = \{\limsup f_n = \liminf f_n\} = \{\limsup f_n - \liminf f_n = 0\}$ is measurable (where finite; the infinite cases are handled separately and are measurable too). $\square$

Caution. This theorem is the crucial improvement over the Riemann/continuous world. A pointwise limit of continuous functions need not be continuous (e.g. $x^n \rightarrow$ a discontinuous step on $[0, 1]$), so the continuous functions are *not* closed under pointwise limits. The measurable functions *are*. That single closure property is why the Lebesgue integral interacts cleanly with limits while the Riemann integral does not.

Theorem 13.3.5 (Algebraic closure: sums, products, $|f|$, $f^\pm$, Rudin 11.16–11.17). *If $f, g : X \rightarrow \mathbb{R}$ are measurable and $c \in \mathbb{R}$, then cf , $f + g$, fg , $\max(f, g)$, $\min(f, g)$, $|f|$ are measurable, as are the positive and negative parts*

$$f^+ = \max(f, 0), \quad f^- = -\min(f, 0) = \max(-f, 0),$$

so that $f = f^+ - f^-$ and $|f| = f^+ + f^-$. If $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ is continuous then $\Phi \circ f$ is measurable.

Proof. $\Phi \circ f$: $\{\Phi \circ f > a\} = f^{-1}(\Phi^{-1}(a, \infty))$ and Φ^{-1} of an open set is open, hence its f -preimage is measurable by Theorem 13.3.2. Taking $\Phi(t) = ct$, $\Phi(t) = |t|$, $\Phi(t) = t^2$ gives cf , $|f|$, f^2 measurable. For the sum, observe

$$\{f + g > a\} = \bigcup_{r \in \mathbb{Q}} (\{f > r\} \cap \{g > a - r\}),$$

a countable union (over the *countable* set $\mathbb{Q}$) of measurable sets, since $f(x) + g(x) > a$ iff there is a rational r with $f(x) > r > a - g(x)$. Then $fg = \frac{1}{2}((f+g)^2 - f^2 - g^2)$ is measurable. Finally $\max(f, g) = \frac{1}{2}(f+g+|f-g|)$ and $\min(f, g) = \frac{1}{2}(f+g-|f-g|)$; specializing $g = 0$ gives f^+ , f^- . $\square$

Method (Proving a function is measurable). *Recognition trigger.* A function is built from measurable ingredients and you must certify it lands in $\mathcal{M}$ before integrating.

1. **Identify the building blocks.** Continuous functions, indicators χ_E (E measurable), and simple functions are measurable outright.
2. **Use the closure rules.** Sums, products, scalar multiples, $\max/\min$, $|\cdot|$, and composition with a continuous Φ stay measurable (Theorem 13.3.5).
3. **Push through limits.** $\sup, \inf, \limsup, \liminf$, and pointwise limits of measurable functions are measurable (Theorem 13.3.4). This is usually the step that fails for continuity but succeeds here.
4. **If stuck, test a level set** $\{f > a\}$ directly and show it is a countable combination of known measurable sets.

Simple functions: the finite-range approximants

Definition 13.3.6 (Simple function, Rudin 11.17). A measurable $s : X \rightarrow \mathbb{R}$ is *simple* if its range is a finite set $\{c_1, \dots, c_n\}$. Writing $E_i = \{x : s(x) = c_i\}$ (disjoint, measurable, covering X), the *standard representation* is

$$s = \sum_{i=1}^n c_i \chi_{E_i}.$$

A simple function is thus a finite $\mathbb{R}$ -linear combination of indicators of measurable sets.

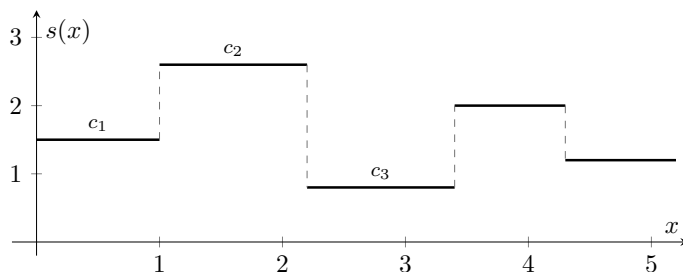


Figure 13.5. A simple function $s = \sum_i c_i \chi_{E_i}$: finitely many values c_i , each taken on a measurable “block” E_i . Here the E_i are intervals, but in general they are arbitrary measurable sets.

Theorem 13.3.7 (Simple-function approximation, Rudin 11.20). *Let $f : X \rightarrow [0, +\infty]$ be measurable. Then there exist simple measurable functions s_n with*

$$0 \leq s_1 \leq s_2 \leq \dots \leq f, \quad s_n(x) \rightarrow f(x) \text{ for every } x \in X.$$

The convergence is uniform on any set where f is bounded. For general measurable f (real or complex), $f = f^+ - f^-$ (or split real/imaginary parts) and apply this to each nonnegative piece.

Proof. For $n = 1, 2, \dots$ partition the range $[0, n]$ into $n2^n$ dyadic levels and define

$$s_n(x) = \begin{cases} \frac{i-1}{2^n}, & \frac{i-1}{2^n} \leq f(x) < \frac{i}{2^n}, \quad i = 1, \dots, n2^n, \\ n, & f(x) \geq n. \end{cases}$$

Each s_n is simple (finitely many values, each on a measurable level set, by measurability of f). Refining $n \rightarrow n+1$ halves each level, so $s_n \leq s_{n+1}$. Where $f(x) < \infty$, once $n > f(x)$ we have $0 \leq f(x) - s_n(x) < 2^{-n} \rightarrow 0$; where $f(x) = \infty$, $s_n(x) = n \rightarrow \infty = f(x)$. If $f \leq M$ then for $n > M$ the bound $f - s_n < 2^{-n}$ holds for *all* x , giving uniform convergence. $\square$

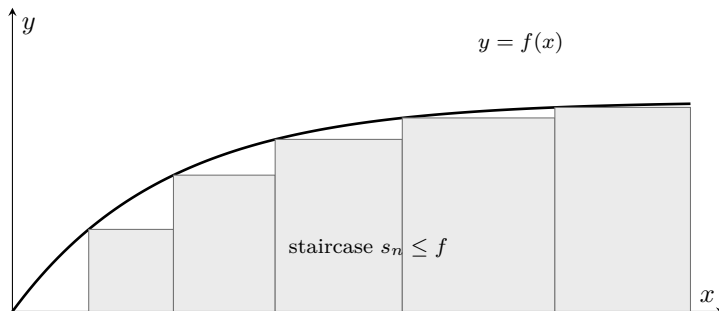


Figure 13.6. Approximating a nonnegative measurable f from below by simple functions: s_n is a rising staircase, $s_n \nearrow f$ pointwise (uniformly where f is bounded). This staircase, integrated, will *define* $\int f d\mu$.

Remark. Theorem 13.3.7 is the engine of the whole integration theory: it lets us *define* $\int f d\mu$ for nonnegative measurable f as the supremum (equivalently the increasing limit) of integrals of simple functions, and it lets us *prove* the convergence theorems by reducing to the simple case. The dyadic slicing here partitions the *range* of f , not the domain — the conceptual hallmark of Lebesgue’s method, made precise in Section 13.5.

Further worked examples

Worked Example 13.3.8 (The Dirichlet function is measurable but nowhere continuous). Show $f = \chi_{\mathbb{Q}}$ on $\mathbb{R}$ is Lebesgue measurable, even though it is continuous at no point.

Recognition. Trigger: “measurability of a wildly discontinuous function” — compute the level sets directly; do not look for continuity.

Solution. For any a ,

$$\{f > a\} = \begin{cases} \mathbb{R}, & a < 0, \\ \mathbb{Q}, & 0 \leq a < 1, \\ \emptyset, & a \geq 1. \end{cases}$$

Each of $\mathbb{R}, \mathbb{Q}, \emptyset$ is measurable ($\mathbb{Q}$ is countable, hence Borel), so f is measurable by Definition 13.3.1. *Check.* Measurability does not see continuity at all — it only asks that level sets be measurable. This is exactly why the measurable class is so much larger than the continuous class, and why it survives the limits in Theorem 13.3.4. $\blacksquare$

Worked Example 13.3.9 (A pointwise limit of continuous functions is measurable but discontinuous). Let $f_n(x) = x^n$ on $[0, 1]$. Identify $f(x) = \lim_n f_n(x)$ and confirm it is measurable.

Recognition. Trigger: “pointwise limit of measurable functions” — invoke Theorem 13.3.4, then read off the limit.

Solution. For $0 \leq x < 1$, $x^n \rightarrow 0$; for $x = 1$, $x^n = 1 \rightarrow 1$. So $f = \chi_{\{1\}}$, the indicator of the single point 1. Each f_n is continuous (hence measurable), so by Theorem 13.3.4 the pointwise limit f is measurable. Indeed $f = \chi_{\{1\}}$ is measurable directly ($\{1\}$ is closed). *Check.* f is discontinuous at $x = 1$, illustrating the pitfall: continuity is *not* closed under pointwise limits, but measurability *is*. $\blacksquare$

Worked Example 13.3.10 (Building measurability from level sets: $f(x) = 1/x$ extended by $f(0) = 0$). On $\mathbb{R}$ let $f(x) = 1/x$ for $x \neq 0$ and $f(0) = 0$. Show f is measurable by exhibiting $\{f > a\}$ explicitly.

Recognition. Trigger: “certify measurability of a function with a singularity” — the level sets are intervals, so compute them.

Solution. Fix $a > 0$. Then $1/x > a \iff 0 < x < 1/a$, so $\{f > a\} = (0, 1/a)$, an interval. For $a = 0$, $\{f > 0\} = (0, \infty)$. For $a < 0$, $1/x > a$ holds for all $x > 0$ and for $x < 1/a$ (since $1/x > a$ when $x < 1/a < 0$), plus $x = 0$ gives $f(0) = 0 > a$; so $\{f > a\} = (-\infty, 1/a) \cup \{0\} \cup (0, \infty)$, a union of measurable sets. Every level set is measurable, so f is measurable. *Check.* f is even continuous off $\{0\}$, so measurability is no surprise — but the computation shows the mechanical method (test $\{f > a\}$) works uniformly, singularity or not. ■

Worked Example 13.3.11 (A non-example: χ_V for a Vitali set V is not measurable). Show that if $V \subseteq [0, 1]$ is a (non-measurable) Vitali set, then $f = \chi_V$ is *not* a measurable function.

Recognition. Trigger: “when does the criterion fail?” — relate measurability of χ_E to measurability of E (Worked Example after Theorem 13.3.2).

Solution. By the indicator computation, $\{\chi_V > a\} = V$ for $0 \leq a < 1$. A Vitali set V is by construction not Lebesgue measurable ($V \notin \mathcal{M}$), so this level set is not in $\mathcal{M}$, and χ_V fails Definition 13.3.1. *Check.* Measurability of the function χ_E is *equivalent* to measurability of the set E ; non-measurable sets are exactly the obstruction, and they exist only by the Axiom of Choice. ■

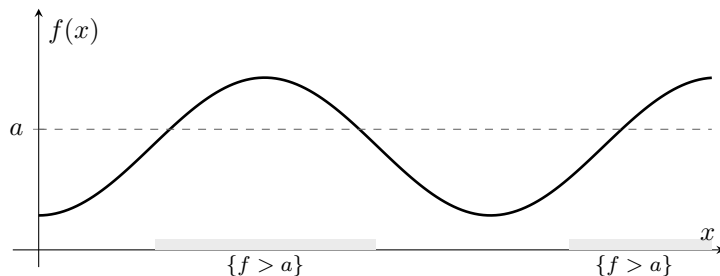


Figure 13.7. Measurability tests the *superlevel set* $\{f > a\}$ (shaded on the x -axis) for every height a : f is measurable iff each such set lies in $\mathcal{M}$.

Exercises (Rudin, Ch.11).

1. If $\{f_n\}$ is a sequence of measurable functions, prove that the set of points x at which $\{f_n(x)\}$ converges is measurable.
2. Prove that a complex function f is measurable if and only if $f^{-1}(V)$ is measurable for every open set V in the plane.

Additional exercises.

1. (Equivalent definitions.) Show that the following are equivalent for $f : E \rightarrow \overline{\mathbb{R}}$: (a) $\{x : f(x) > a\}$ is measurable for all a ; (b) $\{x : f(x) \geq a\}$ measurable for all a ; (c) $\{x : f(x) < a\}$ measurable for all a ; (d) $\{x : f(x) \leq a\}$ measurable for all a .
2. (Operations preserve measurability.) If f, g are measurable, prove $f + g$, fg , $\max(f, g)$, and $|f|$ are measurable. Where do you use that $\mathbb{Q}$ is countable?
3. (Sup, inf, limsup of a sequence.) If $\{f_n\}$ are measurable, prove $\sup_n f_n$, $\inf_n f_n$, $\limsup_n f_n$, $\liminf_n f_n$ are measurable.
4. (Continuous composed with measurable.) Show that if f is measurable and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then $\varphi \circ f$ is measurable. Give a non-example showing $f \circ \varphi$ (measurable $\circ$ continuous) need not be measurable.
5. (Almost-everywhere equality.) If f is measurable and $g = f$ a.e. (with respect to a complete measure), prove g is measurable. Why does completeness matter here?

6. (Non-example: a non-measurable function.) Let $V \subseteq [0, 1]$ be a Vitali set. Show χ_V is not measurable, even though $|\chi_V| \equiv 1$ is a measurable function pointwise bounded by 1. What does this say about “ $|f|$ measurable $\Rightarrow f$ measurable”?
7. (Diagram: approximating a function from below.) For $f(x) = x^2$ on $[0, 1]$, draw the simple function $s_n = \sum_k \frac{k}{2^n} \chi_{\{k/2^n \leq f < (k+1)/2^n\}}$ for $n = 2$ as a step graph beneath the parabola, and indicate the level sets on the x -axis.
8. (Borel vs Lebesgue.) State why every Borel function is Lebesgue measurable, and explain (without full construction) why the converse fails: there are Lebesgue measurable sets that are not Borel.

13.4 Simple Functions and the Lebesgue Integral

We build the integral in three stages, each resting on the previous: first simple functions, then nonnegative measurable functions (by approximating from below), then general functions (by splitting into positive and negative parts). The guiding idea — Lebesgue’s — is to partition the *range* rather than the domain: to integrate f , ask how much domain has f -value near each height, and sum height $\times$ measure.

Riemann horizontally, Lebesgue vertically

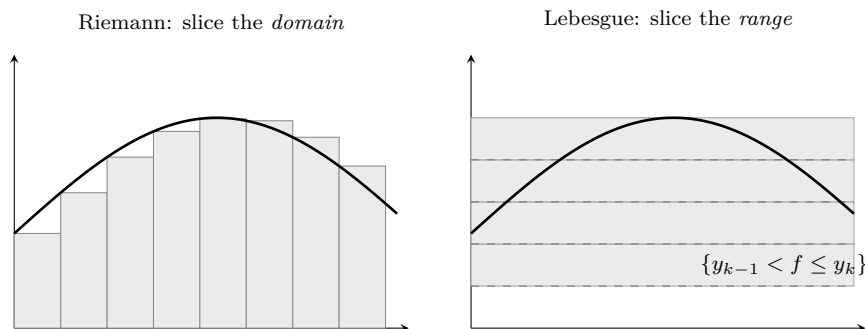


Figure 13.8. The canonical comparison. *Left:* Riemann sums vertical strips $f(t_i)\Delta x_i$ over a partition of the x -axis. *Right:* Lebesgue sums $y_k \cdot \mu\{y_{k-1} < f \leq y_k\}$ over a partition of the y -axis (the range), weighting each height by the *measure* of the level set. Slicing the range is what tames wild domains.

Stage 1: the integral of a simple function

Definition 13.4.1 (Integral of a nonnegative simple function, Rudin 11.18). Let $s = \sum_{i=1}^n c_i \chi_{E_i} \geq 0$ be simple in standard form (the E_i disjoint measurable, $c_i \geq 0$), and let $E \in \mathcal{M}$. Define

$$I_E(s) = \int_E s \, d\mu = \sum_{i=1}^n c_i \mu(E_i \cap E) \in [0, +\infty],$$

with the convention $0 \cdot \infty = 0$. When $E = X$ we write $\int_X s \, d\mu$ or just $\int s \, d\mu$.

This is the area-by-blocks formula: the function sits at height c_i over the set E_i , so that block contributes height c_i times measure $\mu(E_i \cap E)$. One checks it is independent of the representation (refine two representations to a common one), additive in s , and monotone.

Proposition 13.4.2 ($\int s$ as a measure; linearity, Rudin 11.18–11.19). *For a fixed nonnegative simple s , the map $E \mapsto \varphi(E) = \int_E s \, d\mu$ is a measure on $\mathcal{M}$. For nonnegative simple s, t and constants $\alpha, \beta \geq 0$,*

$$\int_E (\alpha s + \beta t) \, d\mu = \alpha \int_E s \, d\mu + \beta \int_E t \, d\mu,$$

and $s \leq t \implies \int_E s \, d\mu \leq \int_E t \, d\mu$.

Proof. $\varphi(E) = \sum_i c_i \mu(E_i \cap E)$ is a nonnegative combination of the measures $E \mapsto \mu(E_i \cap E)$, hence countably additive with $\varphi(\emptyset) = 0$: a measure. Linearity follows by writing s, t over a common refinement of their level sets so that both are constant on each piece, then summing. Monotonicity is the special case $\int (t - s) d\mu \geq 0$ since $t - s \geq 0$ is simple. $\square$

Stage 2: nonnegative measurable functions

Definition 13.4.3 (Integral of a nonnegative measurable function, Rudin 11.20). For measurable $f : X \rightarrow [0, +\infty]$ and $E \in \mathcal{M}$,

$$\int_E f d\mu = \sup \left\{ \int_E s d\mu : s \text{ simple, } 0 \leq s \leq f \right\} \in [0, +\infty].$$

The supremum is over all simple functions below f . By Theorem 13.3.7 there is an increasing sequence $s_n \nearrow f$, and (anticipating the Monotone Convergence Theorem) $\int_E f d\mu = \lim_n \int_E s_n d\mu$.

Proposition 13.4.4 (Elementary properties for $f, g \geq 0$, Rudin 11.21–11.22). For measurable $f, g \geq 0$, $E, A, B \in \mathcal{M}$, $c \geq 0$: (a) $0 \leq f \leq g \implies \int_E f \leq \int_E g$ (monotone in the integrand); (b) $A \subseteq B \implies \int_A f \leq \int_B f$ (monotone in the set); (c) $\int_E cf = c \int_E f$; (d) $f(x) = 0$ on $E \implies \int_E f = 0$ even if $\mu(E) = \infty$; and $\mu(E) = 0 \implies \int_E f = 0$ even if $f = \infty$ on E ; (e) $\int_E f = \int_X \chi_E f$.

Proof. (a) Every simple $s \leq f$ also satisfies $s \leq g$, so the sup defining $\int g$ is over a larger family. (b) Restrict simple functions; or use (e) and $\chi_A \leq \chi_B$. (c) Scale each competing simple function. (d) If $f = 0$ on E the only simple $0 \leq s \leq f$ on E is $s = 0$; if $\mu(E) = 0$ then $\int_E s = \sum c_i \mu(E_i \cap E) = 0$ for every simple s . (e) For simple s , $\int_E s = \sum c_i \mu(E_i \cap E) = \int_X \chi_E s$; take sups. $\square$

Worked Example 13.4.5 (Integrating an indicator and a step recovers measure and area). Compute $\int_X \chi_E d\mu = \mu(E)$, and on $([0, 3], \text{Lebesgue})$ compute $\int_0^3 s$ for $s = 2\chi_{[0,1)} + 5\chi_{[1,2)} + 1\chi_{[2,3]}$.

Recognition. Trigger: “evaluate a Lebesgue integral of a simple/step function” — just apply the block formula.

Solution. $\chi_E = 1 \cdot \chi_E + 0 \cdot \chi_{E^c}$ is simple, so $\int \chi_E = 1 \cdot \mu(E) + 0 = \mu(E)$. For the step, $\int_0^3 s = 2 \cdot \mu[0, 1) + 5 \cdot \mu[1, 2) + 1 \cdot \mu[2, 3] = 2(1) + 5(1) + 1(1) = 8$. *Check.* This equals the Riemann area under the staircase, $2 + 5 + 1 = 8$ — for step functions Lebesgue and Riemann agree by construction. $\blacksquare$

Stage 3: general (signed and complex) functions

Definition 13.4.6 (The integral and the space $L(\mu)$, Rudin 11.23). A measurable $f : X \rightarrow \overline{\mathbb{R}}$ is *Lebesgue integrable on E* , written $f \in L(\mu)$ on E (Rudin: $f \in \mathcal{L}(\mu)$), if

$$\int_E |f| d\mu < \infty, \quad \text{equivalently} \quad \int_E f^+ d\mu < \infty \text{ and } \int_E f^- d\mu < \infty.$$

Its integral is then

$$\int_E f d\mu = \int_E f^+ d\mu - \int_E f^- d\mu \in \mathbb{R}.$$

A complex $f = u + iv$ is integrable iff $|f|$ is (iff u, v are), with $\int_E f = \int_E u + i \int_E v$.

Caution. Lebesgue integrability is *absolute*: $f \in L$ iff $|f| \in L$. There is no “conditionally Lebesgue integrable” analogue of a conditionally convergent series. Thus $\int_0^\infty \frac{\sin x}{x} dx$ exists as an improper Riemann integral but $\frac{\sin x}{x} \notin L([0, \infty))$ because $\int_0^\infty \frac{|\sin x|}{x} dx = \infty$. The two integration theories genuinely differ on such functions — a point made precise in the next section.

Theorem 13.4.7 (Linearity, monotonicity, triangle inequality, Rudin 11.23). For $f, g \in L(\mu)$ on E and $\alpha, \beta \in \mathbb{R}$ (or $\mathbb{C}$): (a) $\alpha f + \beta g \in L$ and $\int_E (\alpha f + \beta g) = \alpha \int_E f + \beta \int_E g$; (b) $f \leq g \implies \int_E f \leq \int_E g$ (real case); (c) $|\int_E f d\mu| \leq \int_E |f| d\mu$; (d) if $\mu(E) < \infty$ and $|f| \leq M$ on E then $|\int_E f| \leq M\mu(E)$.

Proof. (a) Additivity reduces, via $f^\pm, g^\pm$ and the identity $(f+g)^+ - (f+g)^- = (f^+ - f^-) + (g^+ - g^-)$, to additivity for nonnegative functions, which is the limit of the simple-function additivity (Proposition 13.4.2) through $s_n \nearrow$. Scaling by $\alpha \geq 0$ is Proposition 13.4.4(c); by $\alpha < 0$ swaps $f^+ \leftrightarrow f^-$. (b) $g - f \geq 0$ so $\int (g - f) \geq 0$, then use (a). (c) In the real case, $\pm f \leq |f|$ gives $\pm \int f \leq \int |f|$. In the complex case write $\int_E f = re^{i\theta}$ and estimate $r = e^{-i\theta} \int_E f = \int_E \operatorname{Re}(e^{-i\theta} f) \leq \int_E |f|$. (d) is (c) with $\int_E |f| \leq \int_E M = M\mu(E)$. $\square$

Method (Computing or bounding a Lebesgue integral). *Recognition trigger.* You must evaluate $\int_E f d\mu$ or show $f \in L(\mu)$.

1. **Check integrability first:** is $\int_E |f| d\mu < \infty$? Split $f = f^+ - f^-$; if either piece has infinite integral, $f \notin L$.
2. **Reduce to simple/nonnegative.** For $f \geq 0$, either find an explicit increasing $s_n \nearrow f$ and take $\lim \int s_n$, or recognize f as a limit and invoke a convergence theorem (next section).
3. **Use linearity** to break f into manageable summands, and **monotonicity** $\int_E f \leq M\mu(E)$ for crude bounds.
4. **Sanity-check** against the Riemann value when f is Riemann integrable: the two must agree (Section 13.5).

Worked Example 13.4.8 (A function that is Lebesgue but not Riemann integrable). Let $f = \chi_{\mathbb{Q} \cap [0,1]}$, the indicator of the rationals in $[0, 1]$. Show $f \in L$ with $\int_0^1 f d\mu = 0$, although f is *not* Riemann integrable.

Recognition. Trigger: a “Lebesgue succeeds where Riemann fails” showcase — compute via the measure of the level set.

Solution. $f = \chi_E$ with $E = \mathbb{Q} \cap [0, 1]$, which is measurable with $\mu(E) = 0$ (countable, by Section 1). Hence $\int_0^1 f d\mu = \mu(E) = 0$. But the Dirichlet function f has $U(f, P) = 1$, $L(f, P) = 0$ for every partition (density), so its Riemann upper and lower integrals are $1 \neq 0$: not Riemann integrable. *Check.* Lebesgue assigns the “correct” value 0 because the rationals are negligible; this is the prototypical gain of the theory. $\blacksquare$

Theorem 13.4.9 (The integral as a measure; absolute continuity, Rudin 11.23). *If $f \geq 0$ is measurable, then $\varphi(E) = \int_E f d\mu$ is a measure on $\mathcal{M}$; it is countably additive in E . Moreover φ is absolutely continuous with respect to μ : $\mu(E) = 0 \implies \varphi(E) = 0$. For $f \in L(\mu)$ the (signed) set function $E \mapsto \int_E f$ is countably additive as well.*

Proof sketch. For nonnegative simple s , $E \mapsto \int_E s$ is a measure (Proposition 13.4.2); for general $f \geq 0$ approximate $s_n \nearrow f$ and pass to the limit using the Monotone Convergence Theorem (Theorem 13.5.1, next section), which legitimizes interchanging the E -sum with the $\lim_n$. Absolute continuity is Proposition 13.4.4(d). The signed case splits $f = f^+ - f^-$. $\square$

Remark. We have built $\int_E f d\mu$ for: nonnegative simple, then nonnegative measurable, then integrable signed/complex functions, each stage extending the last and agreeing on the overlap. The properties proved — linearity, monotonicity, the triangle inequality, and the “integral is a measure” fact — are the algebraic backbone. What remains, and what makes the theory powerful, is the *analytic* content: how the integral interacts with limits. That is the Monotone, Fatou, and Dominated Convergence triumvirate of Section 13.5.

Further worked examples

Worked Example 13.4.10 (A nonnegative integral computed as a supremum over simple minorants). On $([0, 1], \text{Lebesgue})$ compute $\int_0^1 x d\mu$ directly from Definition 13.4.3, via an explicit increasing sequence of simple functions.

Recognition. Trigger: “evaluate $\int f$ for $f \geq 0$ from the definition” — build $s_n \nearrow f$ and take $\lim \int s_n$.

Solution. Partition $[0, 1]$ into n equal pieces and set $s_n = \sum_{k=1}^n \frac{k-1}{n} \chi_{[\frac{k-1}{n}, \frac{k}{n})}$, the left-endpoint staircase below $f(x) = x$. Then $0 \leq s_n \leq x$ and $s_n \nearrow x$. Its integral is

$$\int_0^1 s_n d\mu = \sum_{k=1}^n \frac{k-1}{n} \cdot \frac{1}{n} = \frac{1}{n^2} \sum_{k=0}^{n-1} k = \frac{1}{n^2} \cdot \frac{(n-1)n}{2} = \frac{n-1}{2n} \longrightarrow \frac{1}{2}.$$

So $\int_0^1 x \, d\mu = \sup_n \int s_n = \frac{1}{2}$. *Check.* Equals the Riemann value $\int_0^1 x \, dx = \frac{1}{2}$, as Theorem 13.5.6 guarantees for this continuous integrand. ■

Worked Example 13.4.11 (Integrating against counting measure recovers a series). On $X = \mathbb{N}$ with counting measure μ , let $f(n) = 2^{-n}$. Compute $\int_X f \, d\mu$ and confirm $f \in L(\mu)$.

Recognition. Trigger: “integral against counting measure” — it is literally the sum $\sum f(n)$.

Solution. Each $f \geq 0$, and $f = \sum_n 2^{-n} \chi_{\{n\}}$ is a pointwise-increasing limit of the simple functions $s_N = \sum_{n \leq N} 2^{-n} \chi_{\{n\}}$ with $\int s_N \, d\mu = \sum_{n \leq N} 2^{-n} \mu(\{n\}) = \sum_{n \leq N} 2^{-n}$. Hence $\int_X f \, d\mu = \sum_{n=1}^{\infty} 2^{-n} = 1 < \infty$, so $f \in L(\mu)$. *Check.* $\int |f| \, d\mu = \int f \, d\mu = 1 < \infty$ confirms integrability; the Lebesgue integral against counting measure is the convergent geometric series. ■

Worked Example 13.4.12 (A non-example: $1/x$ on $(0, 1]$ is measurable, nonnegative, but not integrable). Show $f(x) = 1/x$ on $(0, 1]$ has $\int_{(0,1]} f \, d\mu = +\infty$, so $f \notin L$.

Recognition. Trigger: “test integrability of a function with a non-integrable singularity” — bound below by simple functions whose integrals blow up.

Solution. $f \geq 0$ is measurable (continuous on $(0, 1]$). For the lower bound use $s_n = \sum_{k=1}^n \frac{1}{k}$; more simply, $f(x) \geq \frac{1}{k+1} = k$ on $(\frac{1}{k+1}, \frac{1}{k}]$, so the simple function $t_n = \sum_{k=1}^n k \chi_{(1/(k+1), 1/k]}$ satisfies $0 \leq t_n \leq f$ and

$$\int t_n \, d\mu = \sum_{k=1}^n k \left(\frac{1}{k} - \frac{1}{k+1} \right) = \sum_{k=1}^n \frac{1}{k+1} \rightarrow \infty.$$

Thus $\int_{(0,1]} f = \sup_n \int t_n = +\infty$, so $f \notin L((0, 1])$. *Check.* Matches the divergent improper Riemann integral $\int_0^1 \frac{dx}{x} = \infty$; both theories agree that this singularity is non-integrable. ■

Exercises (Rudin, Ch.11).

1. If $f \geq 0$ and $\int_E f \, d\mu = 0$, prove that $f(x) = 0$ almost everywhere on E . (Hint: let E_n be the subset of E on which $f(x) > 1/n$, and put $A = \bigcup E_n$; then $\mu(A) = 0$ if and only if $\mu(E_n) = 0$ for every n .)
2. If $\int_A f \, d\mu = 0$ for every measurable subset A of a measurable set E , then $f(x) = 0$ almost everywhere on E .
3. If $f \in \mathcal{L}(\mu)$ on E and g is bounded and measurable on E , then $fg \in \mathcal{L}(\mu)$ on E .
4. Find a necessary and sufficient condition that $f \in \mathcal{R}(\alpha)$ on $[a, b]$. (Hint: consider Example 11.6(b) and Theorem 11.33.)

Additional exercises.

1. (Integral of a simple function is well defined.) If a nonnegative simple function has two representations $s = \sum_i a_i \chi_{A_i} = \sum_j b_j \chi_{B_j}$, prove $\sum_i a_i \mu(A_i) = \sum_j b_j \mu(B_j)$.
2. (Linearity and monotonicity for simple functions.) For nonnegative simple s, t and $c \geq 0$, prove $\int(s+t) = \int s + \int t$, $\int cs = c \int s$, and $s \leq t \Rightarrow \int s \leq \int t$.
3. (Characteristic function integral.) Compute $\int_{\mathbb{R}} \chi_{[a,b]} \, dm$ and $\int_{\mathbb{R}} \chi_{\mathbb{Q}} \, dm$, and contrast with the Riemann integrability of $\chi_{\mathbb{Q}}$ on $[0, 1]$.
4. (Markov / Chebyshev inequality.) For $f \geq 0$ measurable and $\lambda > 0$, prove $m\{x : f(x) \geq \lambda\} \leq \frac{1}{\lambda} \int f \, dm$. Use it to reprove: if $\int f \, dm = 0$ then $f = 0$ a.e.
5. (A divergent example.) Let $f(x) = 1/x$ on $(0, 1]$. Show f is measurable, build simple functions $s_n \uparrow f$, and conclude $\int_{(0,1]} f \, dm = +\infty$. Contrast with $f(x) = 1/\sqrt{x}$, which is integrable on $(0, 1]$.
6. (Non-example: simple does not mean continuous.) Exhibit a simple function on $[0, 1]$ with infinitely many discontinuities but a perfectly well-defined finite Lebesgue integral, and compute the integral.
7. (Diagram: simple-function rectangles.) For $f(x) = \sqrt{x}$ on $[0, 1]$, draw the horizontal-slab picture of the Lebesgue integral: partition the y -axis into $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{1}{2}]$, $\dots$, shade the corresponding level sets on the x -axis, and contrast with the vertical Riemann rectangles.

8. (Bounded convergence on finite measure.) If $\mu(X) < \infty$, $|f_n| \leq M$, and $f_n \rightarrow f$ pointwise, prove $\int f_n \rightarrow \int f$ directly from the simple-function definition (a hands-on case of dominated convergence).
9. (Absolute integrability.) Prove $f \in \mathcal{L}(\mu)$ iff $|f| \in \mathcal{L}(\mu)$ (for measurable f), and give an example of a Riemann-improperly-integrable function that is *not* Lebesgue integrable.

13.5 Convergence Theorems and Comparison with the Riemann Integral

Here is the payoff. The Riemann integral interchanges with a limit only under uniform convergence; the Lebesgue integral does so under far weaker, more natural hypotheses — monotone increase (MCT), no hypothesis at all but only an inequality (Fatou), or a single integrable dominating function (DCT). These three theorems are the reason the Lebesgue theory exists. We close by proving that wherever the Riemann integral works, the Lebesgue integral agrees, and we state Lebesgue's exact criterion for Riemann integrability.

The monotone convergence theorem

Theorem 13.5.1 (Monotone Convergence Theorem (MCT), Rudin 11.28). *Let $f_n : X \rightarrow [0, +\infty]$ be measurable with*

$$0 \leq f_1(x) \leq f_2(x) \leq \cdots \quad \text{and} \quad f_n(x) \rightarrow f(x) \text{ for every } x.$$

Then f is measurable and

$$\int_X f_n d\mu \rightarrow \int_X f d\mu \quad (\text{both sides in } [0, +\infty]).$$

Proof. $f = \lim f_n = \sup_n f_n$ is measurable (Theorem 13.3.4). Since $f_n \leq f_{n+1} \leq f$, the integrals increase and $\int f_n \leq \int f$, so $\alpha := \lim_n \int f_n \leq \int f$. For the reverse, fix any simple s with $0 \leq s \leq f$ and a constant $0 < c < 1$. Let $E_n = \{x : f_n(x) \geq c s(x)\}$. The E_n are measurable, increasing, and $\bigcup_n E_n = X$ (for each x : if $s(x) = 0$ trivially; if $s(x) > 0$ then $f(x) \geq s(x) > c s(x)$, so $f_n(x) \geq c s(x)$ eventually). Hence

$$\int_X f_n d\mu \geq \int_{E_n} f_n d\mu \geq c \int_{E_n} s d\mu.$$

Now $E \mapsto \int_E s d\mu$ is a measure (Proposition 13.4.2) and is continuous from below, so $\int_{E_n} s \rightarrow \int_X s$. Letting $n \rightarrow \infty$ gives $\alpha \geq c \int_X s$; let $c \rightarrow 1$ to get $\alpha \geq \int_X s$; take the sup over simple $s \leq f$ to get $\alpha \geq \int_X f$. Therefore $\alpha = \int_X f$. $\square$

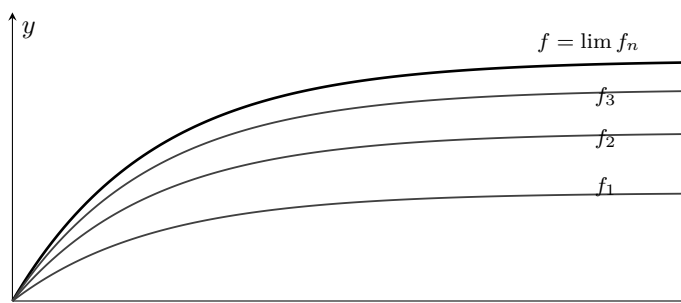


Figure 13.9. A monotone increasing sequence $f_1 \leq f_2 \leq \cdots \nearrow f$. MCT says the areas rise to the area under f : $\int f_n \nearrow \int f$, with no domination hypothesis needed because increase is enough.

Corollary 13.5.2 (Term-by-term integration of series, Rudin 11.30). *If $g_n \geq 0$ are measurable, then*

$$\int_X \sum_{n=1}^{\infty} g_n d\mu = \sum_{n=1}^{\infty} \int_X g_n d\mu.$$

Equivalently, for a doubly indexed nonnegative array one may sum/integrate in either order.

Proof. The partial sums $f_N = \sum_{n=1}^N g_n$ increase to $\sum_{n=1}^{\infty} g_n$ and are measurable; by linearity $\int f_N = \sum_{n=1}^N \int g_n$, and MCT lets $N \rightarrow \infty$ on both sides. $\square$

Fatou's lemma

Theorem 13.5.3 (Fatou's Lemma, Rudin 11.31). *For any measurable $f_n : X \rightarrow [0, +\infty]$,*

$$\int_X \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu.$$

Proof. Set $g_k = \inf_{n \geq k} f_n$. Then g_k is measurable, $0 \leq g_1 \leq g_2 \leq \dots$, and $g_k \nearrow \liminf_n f_n$ by definition of $\liminf$. By MCT, $\int g_k \rightarrow \int \liminf f_n$. But $g_k \leq f_n$ for every $n \geq k$, so $\int g_k \leq \inf_{n \geq k} \int f_n$. Taking $k \rightarrow \infty$,

$$\int \liminf_n f_n = \lim_k \int g_k \leq \lim_k \left(\inf_{n \geq k} \int f_n \right) = \liminf_n \int f_n. \quad \square$$

Caution. Fatou's inequality can be *strict* — “mass can escape.” On $\mathbb{R}$ let $f_n = \chi_{[n, n+1]}$. Then $f_n(x) \rightarrow 0$ for every x , so $\liminf f_n = 0$ and $\int \liminf f_n = 0$, while $\int f_n = 1$ for all n , so $\liminf \int f_n = 1$. Thus $0 < 1$: the inequality is strict, and there is no general reverse inequality. The mass “runs off to infinity.” This is exactly the gap that a dominating function (DCT) closes.

The dominated convergence theorem

Theorem 13.5.4 (Dominated Convergence Theorem (DCT), Rudin 11.32). *Let $f_n : X \rightarrow \mathbb{C}$ be measurable with $f_n(x) \rightarrow f(x)$ a.e., and suppose there is $g \in L(\mu)$ with $|f_n(x)| \leq g(x)$ a.e. for all n . Then $f \in L(\mu)$,*

$$\lim_{n \rightarrow \infty} \int_X |f_n - f| d\mu = 0, \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

Proof. Since $|f_n| \leq g$ and $f_n \rightarrow f$ a.e., also $|f| \leq g$ a.e., so $f \in L$. The functions $2g - |f_n - f| \geq 0$ are measurable, and $|f_n - f| \leq |f_n| + |f| \leq 2g$. Apply Fatou's Lemma to $2g - |f_n - f|$: since $f_n \rightarrow f$ a.e., $\liminf_n (2g - |f_n - f|) = 2g$, so

$$\int 2g \leq \liminf_n \int (2g - |f_n - f|) = \int 2g - \limsup_n \int |f_n - f|.$$

As $\int 2g < \infty$, we may cancel it, giving $\limsup_n \int |f_n - f| \leq 0$, hence $\int |f_n - f| \rightarrow 0$. Finally $|\int f_n - \int f| = |\int (f_n - f)| \leq \int |f_n - f| \rightarrow 0$. $\square$

Worked Example 13.5.5 (DCT in action, and why domination is essential). Evaluate $\lim_n \int_0^1 \frac{nx}{1+n^2x^2} dx$, and contrast with the spike sequence $h_n = n\chi_{(0,1/n)}$ where domination fails.

Recognition. Trigger: “interchange limit and integral” — find the pointwise limit and a single integrable dominator.

Solution. For $f_n(x) = \frac{nx}{1+n^2x^2}$ on $(0, 1]$: as $n \rightarrow \infty$, $f_n(x) \rightarrow 0$ pointwise (denominator $\sim n^2x^2$). And $f_n(x) = \frac{nx}{1+n^2x^2} \leq \frac{nx}{2nx} = \frac{1}{2}$ by AM–GM, so $|f_n| \leq g \equiv \frac{1}{2} \in L[0, 1]$. DCT gives $\lim_n \int_0^1 f_n = \int_0^1 0 = 0$.

Contrast: $h_n = n\chi_{(0,1/n)} \rightarrow 0$ pointwise too, but $\int_0^1 h_n = n \cdot \frac{1}{n} = 1 \not\rightarrow 0$; here *no* integrable g dominates all h_n (one would need $g \geq n$ near 0 for every n , i.e. $g = \infty$ near 0, not integrable). *Check.* The failure is precisely the missing hypothesis of DCT, and Fatou still holds: $0 = \int \liminf h_n \leq \liminf \int h_n = 1$. ■

Method (Interchanging $\lim$ and $\int$ (which theorem to reach for)). *Recognition trigger.* You need $\lim_n \int f_n = \int \lim_n f_n$.

1. **Monotone increasing, nonnegative?** Use MCT — no domination needed.
2. **A series of nonnegative terms?** Use term-by-term integration (Corollary 13.5.2).
3. **Only a one-sided bound wanted, or no control on $\int f_n$?** Fatou gives $\int \liminf \leq \liminf \int$ — always true, free of charge.
4. **Pointwise convergence plus a single integrable dominator $|f_n| \leq g \in L$?** Use DCT — the everyday workhorse; produces convergence of the integrals *and* L^1 -convergence $\int |f_n - f| \rightarrow 0$.
5. **No dominator available?** Beware: the interchange may fail (the escaping-spike examples). Look for a dominator, a monotone structure, or use Fatou for the inequality you can get.

Comparison with the Riemann integral

Theorem 13.5.6 (Riemann $\implies$ Lebesgue, same value, Rudin 11.33(a)). *If f is Riemann integrable on $[a, b]$ (so bounded), then $f \in L$ on $[a, b]$ with respect to Lebesgue measure, and*

$$\int_a^b f(x) dx \text{ (Riemann)} = \int_{[a,b]} f d\mu \text{ (Lebesgue)}.$$

Proof sketch. Take partitions P_n refining and with mesh $\rightarrow 0$, and let L_n, U_n be the lower/upper Darboux step functions (constant $= m_i, M_i$ on each subinterval). These are simple, $L_n \nearrow f$ and $U_n \searrow \bar{f}$ pointwise (a.e.), with $L_n \leq f \leq U_n$. Their Lebesgue integrals equal the Darboux sums $\bar{L}(f, P_n), U(f, P_n)$, which converge to the common Riemann value I . By MCT/DCT (the U_n are dominated by $\sup |f|$), $\int \underline{f} = \int \bar{f} = I$; since $\underline{f} \leq f \leq \bar{f}$ and $\int(\bar{f} - \underline{f}) = 0$, we get $\underline{f} = f = \bar{f}$ a.e., so f is measurable with $\int_{[a,b]} f d\mu = I$. $\square$

Theorem 13.5.7 (Lebesgue's criterion for Riemann integrability, Rudin 11.33(b)). *A bounded function f on $[a, b]$ is Riemann integrable if and only if the set of points at which f is discontinuous has Lebesgue measure zero. Briefly: $f \in \mathcal{R}$ iff f is continuous almost everywhere.*

Proof sketch. Let $\text{osc}(f, x)$ be the oscillation of f at x (the gap $\lim_{\delta \rightarrow 0} (\sup_{|y-x|<\delta} f - \inf_{|y-x|<\delta} f)$); f is continuous at x iff $\text{osc}(f, x) = 0$. The discontinuity set is $D = \bigcup_k \{x : \text{osc}(f, x) \geq \frac{1}{k}\}$, an F_σ . Using the Riemann criterion $U(f, P) - L(f, P) < \varepsilon$ from Chapter 7, one shows: the gap can be made small iff for each k the set where $\text{osc} \geq \frac{1}{k}$ can be covered by intervals of small total length, i.e. $\mu(D) = 0$. The compactness of $[a, b]$ converts “small oscillation off a small set” into the partition estimate, in both directions. $\square$

Worked Example 13.5.8 (Which functions does each theory integrate?). Classify on $[0, 1]$: (i) f continuous; (ii) f with finitely many jumps; (iii) $f = \chi_{\mathbb{Q}}$; (iv) the Thomae function $t(x) = 1/q$ at $x = p/q$ in lowest terms, $t = 0$ at irrationals.

Recognition. Trigger: “apply Lebesgue’s criterion” — locate the discontinuity set and measure it.

Solution. (i) Discontinuity set $\emptyset$, measure 0: Riemann (hence Lebesgue) integrable. (ii) Finitely many discontinuities, measure 0: Riemann integrable. (iii) Discontinuous *everywhere* (measure 1): *not* Riemann integrable, but Lebesgue integrable with integral 0 (measure of $\mathbb{Q}$ is 0). (iv) Thomae is continuous at every irrational and discontinuous at every rational; the discontinuity set $\mathbb{Q} \cap [0, 1]$ has measure 0, so t is Riemann integrable with $\int_0^1 t = 0$, matching its Lebesgue integral. *Check.* (iii) vs (iv) shows the criterion’s precision — both are discontinuous on a dense set, but only (iii) is discontinuous on a set of positive measure. ■

Caution. “Continuous a.e.” (Lebesgue’s criterion for Riemann) is *not* the same as “equal a.e. to a continuous function.” Thomae’s function is continuous a.e. (at all irrationals) yet equals no continuous function a.e. Conversely $\chi_{\mathbb{Q}}$ equals the continuous function 0 a.e. but is continuous *nowhere*. Riemann integrability is governed by the size of the actual discontinuity set, not by a.e.-equivalence.

Remark. The improper Riemann integral genuinely exceeds the Lebesgue integral on conditionally-convergent examples like $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$, which is *not* a Lebesgue integral (the positive and negative parts each have infinite integral, recall Definition 13.4.6). On a finite measure with absolute integrability, however, Lebesgue contains Riemann strictly and assigns the same values — the Lebesgue theory is the proper completion of the Riemann theory.

Further worked examples

Worked Example 13.5.9 (MCT applied to a series: term-by-term integration). Compute $\int_0^1 \sum_{n=1}^\infty x^n d\mu$ by integrating the series term by term, and verify against the closed form.

Recognition. Trigger: “integrate an infinite sum of nonnegative terms” — use Corollary 13.5.2 (a face of MCT), which needs only $g_n \geq 0$.

Solution. Each $g_n(x) = x^n \geq 0$ on $[0, 1]$, so

$$\int_0^1 \sum_{n=1}^\infty x^n d\mu = \sum_{n=1}^\infty \int_0^1 x^n d\mu = \sum_{n=1}^\infty \frac{1}{n+1} = \left(\sum_{m=1}^\infty \frac{1}{m} \right) - 1 = +\infty.$$

Check. Directly, $\sum_{n \geq 1} x^n = \frac{x}{1-x}$ on $[0, 1)$, and $\int_0^1 \frac{x}{1-x} dx = \int_0^1 \left(\frac{1}{1-x} - 1\right) dx$ diverges at $x = 1 - +\infty$, agreeing. No domination hypothesis was needed because the terms are nonnegative; this is MCT's signature strength. ■

Worked Example 13.5.10 (DCT with a clean dominator: a convergent limit). Evaluate $\lim_{n \rightarrow \infty} \int_0^\infty \frac{e^{-x/n}}{1+x^2} dx$.

Recognition. Trigger: “move the limit inside” — find the pointwise limit and one integrable dominator independent of n .

Solution. For each fixed $x \geq 0$, $e^{-x/n} \rightarrow e^0 = 1$ as $n \rightarrow \infty$, so the integrand tends to $\frac{1}{1+x^2}$. Dominator: $0 \leq \frac{e^{-x/n}}{1+x^2} \leq \frac{1}{1+x^2} =: g(x)$, and $\int_0^\infty g = [\arctan x]_0^\infty = \frac{\pi}{2} < \infty$, so $g \in L$. By DCT,

$$\lim_n \int_0^\infty \frac{e^{-x/n}}{1+x^2} dx = \int_0^\infty \frac{1}{1+x^2} dx = \frac{\pi}{2}.$$

Check. The dominator did not depend on n and was integrable — precisely DCT's hypothesis — and the answer $\pi/2$ is finite and positive, as a decreasing-then-saturating family of integrals should give. ■

Worked Example 13.5.11 (A non-example: pointwise convergence alone does not move the limit inside). Let $f_n = n \chi_{(0, 1/n)}$ on $[0, 1]$. Show $f_n \rightarrow 0$ pointwise yet $\int f_n \not\rightarrow \int 0$, and say which hypothesis of DCT fails.

Recognition. Trigger: “can I always swap lim and $\int$?” — no; check for a dominator first.

Solution. For each fixed $x \in (0, 1]$, once $n > 1/x$ we have $x \notin (0, 1/n)$, so $f_n(x) = 0$ eventually; and $f_n(0) = 0$. Hence $f_n \rightarrow 0$ pointwise. But $\int_0^1 f_n d\mu = n \cdot \mu(0, 1/n) = n \cdot \frac{1}{n} = 1$ for all n , so $\int f_n \rightarrow 1 \neq 0 = \int \lim f_n$. The missing hypothesis is domination: any $g \geq |f_n|$ for all n must satisfy $g \geq n$ on $(0, 1/n)$, forcing $\int_0^1 g \geq \sup_n 1 = \infty$ near 0, so no integrable dominator exists. *Check.* Fatou still holds and is strict here, $0 = \int \liminf f_n \leq \liminf \int f_n = 1$ — the inequality is the only thing that survives without a dominator. ■

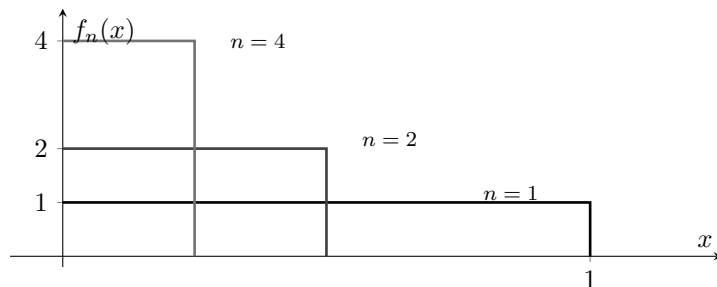


Figure 13.10. The escaping-spike sequence $f_n = n \chi_{(0, 1/n)}$: each has area 1, but $f_n \rightarrow 0$ pointwise. The mass concentrates at 0 with no integrable dominator, so DCT does not apply and the limit and integral cannot be swapped.

Exercises (Rudin, Ch.11).

- Put $g(x) = 0$ on $(0, \frac{1}{2})$ and $g(x) = 1$ on $(\frac{1}{2}, 1)$, and set

$$f_{2k}(x) = g(x), \quad f_{2k+1}(x) = g(1-x) \quad (0 < x < 1).$$

Show that $\liminf_{n \rightarrow \infty} f_n(x) = 0$ for $0 < x < 1$, but $\int_0^1 f_n(x) dx = \frac{1}{2}$. (Thus the inequality in Fatou's theorem can be strict; compare with Theorem 11.32's hypotheses.)

- Let $f_n(x) = n^{-1}$ for $|x| \leq n$ and $f_n(x) = 0$ for $|x| > n$. Then $f_n(x) \rightarrow 0$ uniformly on $\mathbb{R}^1$, but $\int_{-\infty}^\infty f_n dx = 2$ for every n . (Thus uniform convergence does not imply dominated convergence in the sense of Theorem 11.32; however, on sets of finite measure, uniformly convergent sequences of bounded functions do satisfy that theorem.)
- If $f \geq 0$ on $[a, b]$ and $F(x) = \int_a^x f(t) dt$, prove that $F'(x) = f(x)$ almost everywhere on $[a, b]$.
- Prove that the function F defined as the indefinite integral $F(x) = \int_a^x f dt$ is continuous on $[a, b]$.
- Consider the functions $f_n(x) = \sin nx$ ($n = 1, 2, 3, \dots$, $-\pi < x < \pi$) as points of $\mathcal{L}^2$. Prove that the set of these points is closed and bounded, but not compact.

6. Suppose $E \subseteq (-\pi, \pi)$, $m(E) > 0$, $\delta > 0$. Use the Bessel inequality to prove that there are at most finitely many integers n such that $\sin nx \geq \delta$ for all $x \in E$.

Additional exercises.

- (Monotone convergence, hands on.) Let $f_n(x) = \min(n, 1/\sqrt{x})$ on $(0, 1]$. Verify $f_n \uparrow 1/\sqrt{x}$ and use MCT to compute $\int_{(0,1]} x^{-1/2} dm = 2$.
- (Fatou is one-directional.) Using the moving-bump $f_n = \chi_{[n, n+1]}$, show $\liminf \int f_n > \int \liminf f_n$, so the inequality in Fatou's lemma can be strict. Identify $\liminf f_n$ pointwise.
- (Dominated convergence in action.) Compute $\lim_{n \rightarrow \infty} \int_0^\infty \frac{n \sin(x/n)}{x(1+x^2)} dx$ by finding an integrable dominating function and passing the limit inside. State the dominating function explicitly.
- (DCT fails without a dominator.) For $f_n = n \chi_{(0, 1/n)}$, show $f_n \rightarrow 0$ a.e. but $\int f_n = 1 \not\rightarrow 0$. Which hypothesis of DCT is violated, and why can no integrable g dominate all f_n ?
- (Riemann implies Lebesgue, with same value.) Prove that if f is Riemann integrable on $[a, b]$ then f is Lebesgue measurable and $\int_{[a,b]} f dm = \int_a^b f(x) dx$. (Hint: squeeze f between the step functions from upper/lower Darboux sums and apply a convergence theorem.)
- (Lebesgue's criterion, non-example.) Show $\chi_{\mathbb{Q} \cap [0,1]}$ is Lebesgue integrable with integral 0 but *not* Riemann integrable, and connect this to Lebesgue's criterion (Riemann integrable iff the set of discontinuities has measure zero).
- (Diagram: the typewriter / moving-bump sequence.) Draw four consecutive members of a sequence $\{f_n\}$ that converges to 0 in integral but at no point pointwise, and annotate which interval each bump occupies. Explain how the picture distinguishes the modes of convergence.
- (Term-by-term integration of series.) State and prove the corollary of MCT that for nonnegative measurable g_k , $\int \sum_k g_k = \sum_k \int g_k$. Apply it to $\sum_{k \geq 1} \chi_{[k, k+1]}/k^2$.

13.6 Integration of Complex Functions and the Space L^2

The final movement turns the integral into geometry. The square-integrable functions form a vector space with an inner product $\langle f, g \rangle = \int f \bar{g} d\mu$, hence a notion of length, angle, and orthogonality. The crowning fact — the Riesz–Fischer theorem — is that this space is *complete*: Cauchy sequences in the mean-square metric converge. This is the completeness the Riemann theory could never deliver, and it makes L^2 a Hilbert space, the natural home of Fourier analysis.

Complex integrable functions

Definition 13.6.1 (Complex integration, Rudin 11.34). A measurable $f : X \rightarrow \mathbb{C}$, $f = u + iv$, is *integrable* ($f \in L(\mu)$) if $\int_X |f| d\mu < \infty$, where $|f| = \sqrt{u^2 + v^2}$. Then $u, v \in L$ and

$$\int_X f d\mu = \int_X u d\mu + i \int_X v d\mu.$$

The space $L(\mu) = L^1(\mu)$ of integrable functions is a complex vector space, and $\int$ is $\mathbb{C}$ -linear on it.

Proposition 13.6.2 (Properties of complex L^1 , Rudin 11.34–11.35). For $f, g \in L^1(\mu)$: (a) $\int(\alpha f + \beta g) = \alpha \int f + \beta \int g$ for $\alpha, \beta \in \mathbb{C}$; (b) $|\int_X f d\mu| \leq \int_X |f| d\mu$, with equality iff $f = ch$ a.e. for a constant c and $h \geq 0$ (f has a.e.-constant argument); (c) (completeness in the mean) if $\int |f_n - f| d\mu \rightarrow 0$ and $\int |f_n - g| d\mu \rightarrow 0$ then $f = g$ a.e.

Proof. (a) is linearity on real and imaginary parts. (b) Write $\int f = r e^{i\theta}$, $r \geq 0$; then $r = e^{-i\theta} \int f = \int e^{-i\theta} f = \int \operatorname{Re}(e^{-i\theta} f) \leq \int |f|$, since the left side is real. Equality forces $\operatorname{Re}(e^{-i\theta} f) = |f|$ a.e., i.e. $e^{-i\theta} f \geq 0$ a.e. (c) $\int |f - g| \leq \int |f - f_n| + \int |f_n - g| \rightarrow 0$, so $\int |f - g| = 0$, whence $f = g$ a.e. (a nonnegative function with zero integral vanishes a.e.). $\square$

The space L^2 and its inner product

Definition 13.6.3 ($L^2(\mu)$ and $L^p(\mu)$, Rudin 11.35–11.38). For $1 \leq p < \infty$, $L^p(\mu)$ is the set of measurable $f : X \rightarrow \mathbb{C}$ with

$$\|f\|_p = \left(\int_X |f|^p d\mu \right)^{1/p} < \infty,$$

where functions equal a.e. are identified. For $p = 2$ define the *inner product*

$$\langle f, g \rangle = \int_X f \bar{g} d\mu, \quad \|f\|_2 = \sqrt{\langle f, f \rangle} = \left(\int_X |f|^2 d\mu \right)^{1/2}.$$

Two functions are *orthogonal*, $f \perp g$, if $\langle f, g \rangle = 0$.

Theorem 13.6.4 (Cauchy–Schwarz and the triangle inequality, Rudin 11.35–11.38). For $f, g \in L^2(\mu)$, $fg \in L^1(\mu)$ and

$$|\langle f, g \rangle| = \left| \int_X f \bar{g} d\mu \right| \leq \|f\|_2 \|g\|_2 \quad (\text{Cauchy–Schwarz}),$$

and consequently $\|f + g\|_2 \leq \|f\|_2 + \|g\|_2$ (Minkowski / triangle), so $L^2(\mu)$ is a normed vector space with norm $\|\cdot\|_2$.

Proof. First $|f\bar{g}| \leq \frac{1}{2}(|f|^2 + |g|^2)$ (AM–GM) shows $f\bar{g} \in L^1$. For Cauchy–Schwarz, if $\|g\|_2 = 0$ then $g = 0$ a.e. and both sides vanish; otherwise for $\lambda \in \mathbb{C}$,

$$0 \leq \|f - \lambda g\|_2^2 = \|f\|_2^2 - 2 \operatorname{Re}(\bar{\lambda} \langle f, g \rangle) + |\lambda|^2 \|g\|_2^2.$$

Choose $\lambda = \langle f, g \rangle / \|g\|_2^2$; the right side becomes $\|f\|_2^2 - |\langle f, g \rangle|^2 / \|g\|_2^2 \geq 0$, which rearranges to the claim. Then

$$\|f + g\|_2^2 = \|f\|_2^2 + 2 \operatorname{Re} \langle f, g \rangle + \|g\|_2^2 \leq \|f\|_2^2 + 2 \|f\|_2 \|g\|_2 + \|g\|_2^2 = (\|f\|_2 + \|g\|_2)^2. \quad \square$$

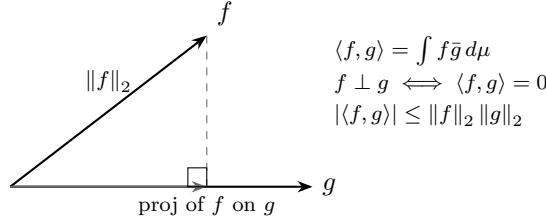


Figure 13.11. $L^2(\mu)$ is a geometry: the integral inner product gives lengths, the Cauchy–Schwarz inequality gives angles, and orthogonality $\langle f, g \rangle = 0$ behaves just as in finite-dimensional Euclidean space.

Completeness: the Riesz–Fischer theorem

Definition 13.6.5 (Mean convergence; Cauchy in L^2 , Rudin 11.41). A sequence $(f_n) \subseteq L^2(\mu)$ *converges in the mean* (in L^2) to f if $\|f_n - f\|_2 \rightarrow 0$, i.e. $\int |f_n - f|^2 d\mu \rightarrow 0$. It is *Cauchy in the mean* if $\|f_n - f_m\|_2 \rightarrow 0$ as $n, m \rightarrow \infty$. (Mean convergence is convergence in the $\|\cdot\|_2$ metric; it is weaker than uniform and incomparable with pointwise.)

Theorem 13.6.6 (Riesz–Fischer: $L^2(\mu)$ is complete, Rudin 11.42). *Every Cauchy sequence in $L^2(\mu)$ converges in the mean to some $f \in L^2(\mu)$. That is, $L^2(\mu)$ is a complete metric space — a Hilbert space. The same holds for $L^p(\mu)$, $1 \leq p < \infty$.*

Proof sketch. Let (f_n) be Cauchy in L^2 . Pass to a subsequence (f_{n_k}) with $\|f_{n_{k+1}} - f_{n_k}\|_2 < 2^{-k}$. Put $g = \sum_k |f_{n_{k+1}} - f_{n_k}|$; by Minkowski and MCT, $\|g\|_2 \leq \sum_k 2^{-k} < \infty$, so $g \in L^2$ and $g < \infty$ a.e. Hence the telescoping series $f_{n_1} + \sum_k (f_{n_{k+1}} - f_{n_k})$ converges absolutely a.e. to a limit f , and $|f| \leq |f_{n_1}| + g \in L^2$, so $f \in L^2$. Dominated convergence (dominator $(|f_{n_1}| + g)^2$) gives $\|f_{n_k} - f\|_2 \rightarrow 0$. Finally a Cauchy sequence with a convergent subsequence converges, so $\|f_n - f\|_2 \rightarrow 0$. $\square$

Caution. This is exactly the completeness that $\mathcal{R}[a, b]$ lacks: a sequence of Riemann integrable functions can be Cauchy in the mean yet converge (in the mean) to a function that is not Riemann integrable. Lebesgue's space L^2 plugs every such hole. Note also that mean convergence does not imply pointwise convergence (the “typewriter” sequence of indicators of $[\frac{j}{2^k}, \frac{j+1}{2^k}]$ marches across $[0, 1]$: $\|f_n\|_2 \rightarrow 0$ but $f_n(x) \not\rightarrow 0$ for any x). Only a *subsequence* converges a.e., as the proof shows.

Orthonormal sets and Parseval

Definition 13.6.7 (Orthonormal sets and Fourier coefficients, Rudin 11.39–11.43). A family $\{\phi_n\} \subseteq L^2(\mu)$ is *orthonormal* if $\langle \phi_m, \phi_n \rangle = \delta_{mn}$. The *Fourier coefficients* of $f \in L^2(\mu)$ are $c_n = \langle f, \phi_n \rangle = \int f \overline{\phi_n} d\mu$. The set is *complete* (a Hilbert basis) if the only f orthogonal to every ϕ_n is $f = 0$.

Theorem 13.6.8 (Bessel, Parseval, and best approximation, Rudin 11.43–11.45). Let $\{\phi_n\}$ be orthonormal in $L^2(\mu)$ and $c_n = \langle f, \phi_n \rangle$. Then: (a) (Best approximation) among all $\sum_{n \leq N} a_n \phi_n$, the choice $a_n = c_n$ minimizes $\|f - \sum_{n \leq N} a_n \phi_n\|_2$; (b) (Bessel's inequality) $\sum_n |c_n|^2 \leq \|f\|_2^2$; (c) if $\{\phi_n\}$ is complete then (Parseval) $\sum_n |c_n|^2 = \|f\|_2^2$ and $f = \sum_n c_n \phi_n$ in the mean. Conversely, for any (c_n) with $\sum |c_n|^2 < \infty$ there is $f \in L^2$ with these Fourier coefficients (Riesz–Fischer).

Proof. (a),(b) Expand, with $s_N = \sum_{n \leq N} a_n \phi_n$,

$$\|f - s_N\|_2^2 = \|f\|_2^2 - \sum_{n \leq N} (\bar{a}_n c_n + a_n \bar{c}_n) + \sum_{n \leq N} |a_n|^2 = \|f\|_2^2 - \sum_{n \leq N} |c_n|^2 + \sum_{n \leq N} |a_n - c_n|^2,$$

using orthonormality. The last sum is minimized (to 0) by $a_n = c_n$, proving (a); then $0 \leq \|f - s_N\|_2^2 = \|f\|_2^2 - \sum_{n \leq N} |c_n|^2$ gives Bessel (b) on letting $N \rightarrow \infty$. (c) Completeness means $f - \sum c_n \phi_n$ is orthogonal to every ϕ_m (its m -th coefficient is $c_m - c_m = 0$), hence is 0; the partial sums are Cauchy by Bessel and converge to f by Riesz–Fischer, and $\|f - s_N\|_2 \rightarrow 0$ turns Bessel into the Parseval equality. $\square$

Worked Example 13.6.9 (Fourier series as L^2 expansion). On $X = [-\pi, \pi]$ with $d\mu = \frac{dx}{2\pi}$, the exponentials $\phi_n(x) = e^{inx}$, $n \in \mathbb{Z}$, are orthonormal. State what Parseval says for $f \in L^2$.

Recognition. Trigger: “recognize a concrete orthonormal system” — check orthonormality, then read off Bessel/Parseval.

Solution. Orthonormality: $\langle \phi_m, \phi_n \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(m-n)x} dx = \delta_{mn}$. The Fourier coefficients are $c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$, and (the system being complete) Parseval reads

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx = \sum_{n \in \mathbb{Z}} |c_n|^2,$$

with $f = \sum_n c_n e^{inx}$ converging in L^2 (not necessarily pointwise!). *Check.* For $f(x) = x$ one computes $c_n = \frac{(-1)^n i}{n}$ ($n \neq 0$, $c_0 = 0$), and Parseval gives $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$ — the Basel sum falls out of L^2 geometry. $\blacksquare$

Remark. Combining Riesz–Fischer (Theorem 13.6.6) with Parseval (Theorem 13.6.8) gives the structural conclusion that Rudin highlights: a complete orthonormal set sets up a bijection between $L^2(\mu)$ (functions identified up to a.e. equality) and the sequence space ℓ^2 of coefficient sequences with $\sum |c_n|^2 < \infty$. Under this correspondence $L^2(\mu)$ is an infinite-dimensional Euclidean space — Hilbert space — with the ϕ_n as coordinate axes. This is the abstract endpoint of the entire Lebesgue construction: the integral, built from measure, has become inner-product geometry.

Further worked examples

Worked Example 13.6.10 (Cauchy–Schwarz computed on a concrete pair). On $([0, 1], \text{Lebesgue})$ take $f(x) = x$ and $g(x) = 1$. Compute $\langle f, g \rangle$, $\|f\|_2$, $\|g\|_2$ and verify Cauchy–Schwarz.

Recognition. Trigger: “check the inequality $|\langle f, g \rangle| \leq \|f\|_2 \|g\|_2$ in a case” — just compute the three integrals.

Solution. $\langle f, g \rangle = \int_0^1 x \cdot 1 \, dx = \frac{1}{2}$; $\|f\|_2 = (\int_0^1 x^2 \, dx)^{1/2} = \sqrt{1/3} = 1/\sqrt{3}$; $\|g\|_2 = (\int_0^1 1 \, dx)^{1/2} = 1$. Then $|\langle f, g \rangle| = \frac{1}{2} = 0.5$ and $\|f\|_2 \|g\|_2 = 1/\sqrt{3} \approx 0.577$, so $0.5 \leq 0.577$: Cauchy–Schwarz holds, strictly. *Check.* Equality in Cauchy–Schwarz needs f proportional to g (i.e. f constant), which fails here, so strict inequality is exactly what we should see. ■

Worked Example 13.6.11 (Parseval evaluates a series: the Basel sum from $f(x) = x$). Using the orthonormal exponentials on $[-\pi, \pi]$ with $d\mu = \frac{dx}{2\pi}$, compute the Fourier coefficients of $f(x) = x$ and deduce $\sum_{n \geq 1} 1/n^2$.

Recognition. Trigger: “turn a norm identity into a numerical series” — compute c_n , then apply Parseval (Theorem 13.6.8(c)).

Solution. $c_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x \, dx = 0$. For $n \neq 0$, integrating by parts, $c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} x e^{-inx} \, dx = \frac{(-1)^n i}{n}$, so $|c_n|^2 = \frac{1}{n^2}$. The left side of Parseval is $\frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 \, dx = \frac{1}{2\pi} \cdot \frac{2\pi^3}{3} = \frac{\pi^2}{3}$. Thus

$$\frac{\pi^2}{3} = \sum_{n \neq 0} \frac{1}{n^2} = 2 \sum_{n \geq 1} \frac{1}{n^2} \implies \sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Check. $\pi^2/6 \approx 1.645$, matching the known Basel value — the geometry of L^2 delivers a number-theoretic sum. ■

Worked Example 13.6.12 (A non-example: mean convergence does not force pointwise convergence). Exhibit $f_n \in L^2[0, 1]$ with $\|f_n\|_2 \rightarrow 0$ but $f_n(x) \not\rightarrow 0$ for any x .

Recognition. Trigger: “is L^2 convergence the same as pointwise?” — no; recall the typewriter sequence (pitfall after Theorem 13.6.6).

Solution. List the dyadic intervals $I_{k,j} = [j2^{-k}, (j+1)2^{-k}]$ for $k \geq 0$, $0 \leq j < 2^k$, in order, and let $f_n = \chi_{I_{k,j}}$ be the n -th of them. Then $\|f_n\|_2^2 = \mu(I_{k,j}) = 2^{-k} \rightarrow 0$ as $n \rightarrow \infty$ (since $k \rightarrow \infty$), so $f_n \rightarrow 0$ in the mean. But every fixed x lies in one interval at each level k , so $f_n(x) = 1$ for infinitely many n and $= 0$ for infinitely many n ; hence $f_n(x)$ does not converge for any x . *Check.* Only a *subsequence* (one interval per level, say $j = 0$) converges a.e. to 0 — exactly what the Riesz–Fischer proof extracts. Mean and pointwise convergence are genuinely incomparable. ■

Exercises (Rudin, Ch.11).

1. If $\mu(X) < +\infty$ and $f \in \mathcal{L}^2(\mu)$ on X , prove that $f \in \mathcal{L}(\mu)$ on X . If $\mu(X) = +\infty$, this is false: for instance, if $f(x) = 1/(1 + |x|)$, then $f \in \mathcal{L}^2$ on $\mathbb{R}^1$ but $f \notin \mathcal{L}$ on $\mathbb{R}^1$.
2. If $f, g \in \mathcal{L}(\mu)$ on X , define the distance between f and g by $\int_X |f - g| \, d\mu$. Prove that $\mathcal{L}(\mu)$ is a complete metric space.
3. Suppose (a) $|f(x, y)| \leq 1$ if $0 \leq x \leq 1$, $0 \leq y \leq 1$; (b) for fixed x , $f(x, y)$ is a continuous function of y ; (c) for fixed y , $f(x, y)$ is a continuous function of x . Put $g(x) = \int_0^1 f(x, y) \, dy$ for $0 \leq x \leq 1$. Is g continuous?
4. Suppose $f \in \mathcal{L}^2(\mu)$, $g \in \mathcal{L}^2(\mu)$. Prove that

$$||| \int f \bar{g} \, d\mu |||^2 = \int |f|^2 \, d\mu \int |g|^2 \, d\mu$$

if and only if there is a constant c such that $g(x) = cf(x)$ almost everywhere. (This is the equality case of the Schwarz inequality; compare Theorem 11.35.)

Additional exercises.

1. (Real and imaginary parts.) For complex $f = u + iv$, prove $f \in \mathcal{L}(\mu)$ iff both $u, v \in \mathcal{L}(\mu)$, and that $||| \int f \, d\mu ||| \leq \int |f| \, d\mu$. (Hint for the inequality: rotate by $\bar{\alpha}$ where $\alpha = \int f / |\int f|$.)
2. (Schwarz inequality.) Prove for $f, g \in \mathcal{L}^2(\mu)$ that $fg \in \mathcal{L}(\mu)$ and $||| \int f \bar{g} \, d\mu ||| \leq \|f\|_2 \|g\|_2$.
3. (Minkowski / triangle inequality.) Deduce $\|f + g\|_2 \leq \|f\|_2 + \|g\|_2$ from the Schwarz inequality, and verify $\|\cdot\|_2$ is a norm on $\mathcal{L}^2$ modulo a.e.-equality.

4. (Parallelogram law.) Prove $\|f + g\|_2^2 + \|f - g\|_2^2 = 2\|f\|_2^2 + 2\|g\|_2^2$, and explain why the analogous identity fails for the $\mathcal{L}^1$ norm (so L^1 is not an inner-product space).
5. (Completeness of L^2 , Riesz–Fischer.) Outline the proof that $\mathcal{L}^2(\mu)$ is complete: extract a fast-Cauchy subsequence with $\sum \|f_{n_{k+1}} - f_{n_k}\|_2 < \infty$, show it converges a.e. and in L^2 , and conclude the whole sequence converges.
6. (Non-example: $L^2 \not\subseteq L^1$ and $L^1 \not\subseteq L^2$.) On $(0, \infty)$ with Lebesgue measure, exhibit $f \in L^2 \setminus L^1$ and $g \in L^1 \setminus L^2$. (Try powers $x^{-\alpha}$ near 0 and near ∞ .) Why does $\mu(X) < \infty$ remove one of these failures?
7. (Diagram: orthonormal projection.) Draw the picture of best approximation in L^2 : a vector f , a finite-dimensional subspace spanned by orthonormal $\phi_1, \dots, \phi_N$, the projection $\sum_k \langle f, \phi_k \rangle \phi_k$, and the orthogonal error. Label which right triangle gives Bessel's inequality.
8. (Bessel and Parseval.) For an orthonormal system $\{\phi_n\}$ in L^2 , prove Bessel's inequality $\sum_n |\langle f, \phi_n \rangle|^2 \leq \|f\|_2^2$, and state the extra condition (completeness of the system) under which it becomes Parseval's equality.

Part II

Complex Analysis

Chapter 14

Preliminaries to Complex Analysis

14.1 Complex Numbers and the Complex Plane

The field of complex numbers

A *complex number* is an expression $z = x + iy$ with $x, y \in \mathbb{R}$ and i a symbol satisfying $i^2 = -1$. We call $x = \operatorname{Re}(z)$ the *real part* and $y = \operatorname{Im}(z)$ the *imaginary part*. A complex number with $y = 0$ is identified with the real number x ; one with $x = 0$ is called *purely imaginary*. The set of all complex numbers is denoted $\mathbb{C}$.

Definition 14.1.1. For $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ define

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2),$$

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2).$$

The multiplication rule is what one gets by expanding $(x_1 + iy_1)(x_2 + iy_2)$ formally and replacing i^2 by -1 .

These operations make $\mathbb{C}$ a *field*: addition and multiplication are commutative, associative, and distributive, $0 = 0 + 0i$ and $1 = 1 + 0i$ are the additive and multiplicative identities, and every nonzero z has an inverse (computed below). One should not memorize the multiplication formula as a black box: it is forced by the single rule $i^2 = -1$ together with the field axioms.

The complex plane

We identify $z = x + iy \in \mathbb{C}$ with the point $(x, y) \in \mathbb{R}^2$, so that $0 \leftrightarrow (0, 0)$ and $i \leftrightarrow (0, 1)$. The horizontal axis is the *real axis*, the vertical axis the *imaginary axis*. Addition of complex numbers is exactly vector addition in $\mathbb{R}^2$; multiplication, as we will see, is rotation composed with scaling.

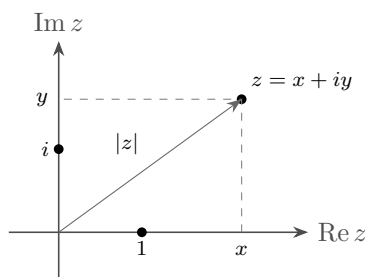


Figure 14.1. The complex plane: $z = x + iy$ is identified with the point $(x, y) \in \mathbb{R}^2$; the modulus $|z|$ is its distance to the origin.

Modulus, conjugate, and the basic inequalities

Definition 14.1.2. The *absolute value* (modulus) of $z = x + iy$ is $|z| = (x^2 + y^2)^{1/2}$, the Euclidean distance from the origin to (x, y) . The *complex conjugate* is $\bar{z} = x - iy$, the reflection of z across the real axis.

Proposition 14.1.3. For all $z, w \in \mathbb{C}$:

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2}, \quad \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}, \quad |z|^2 = z\bar{z}, \quad \frac{1}{z} = \frac{\bar{z}}{|z|^2} \quad (z \neq 0).$$

Moreover $z \in \mathbb{R} \iff z = \bar{z}$, and z is purely imaginary $\iff z = -\bar{z}$.

Proof. Write $z = x + iy$. Then $z + \bar{z} = 2x = 2\operatorname{Re}(z)$ and $z - \bar{z} = 2iy = 2i\operatorname{Im}(z)$, giving the first two. Next $z\bar{z} = (x + iy)(x - iy) = x^2 + y^2 = |z|^2$. For $z \neq 0$ this is nonzero, so $z \cdot \frac{\bar{z}}{|z|^2} = \frac{z\bar{z}}{|z|^2} = 1$, identifying the inverse. Finally $z = \bar{z} \iff 2iy = 0 \iff y = 0 \iff z \in \mathbb{R}$, and $z = -\bar{z} \iff 2x = 0 \iff x = 0$. $\square$

The identity $\frac{1}{z} = \bar{z}/|z|^2$ is the workhorse for dividing complex numbers: *rationalize by the conjugate*. To compute w/z multiply top and bottom by $\bar{z}$.

Theorem 14.1.4 (Triangle inequality and its consequences). For all $z, w \in \mathbb{C}$,

$$|z + w| \leq |z| + |w|, \quad \left| |z| - |w| \right| \leq |z - w|,$$

and $|\operatorname{Re} z| \leq |z|$, $|\operatorname{Im} z| \leq |z|$, and $|\bar{z}| = |z|$, $|zw| = |z| |w|$.

Proof. The multiplicativity $|zw|^2 = zw\bar{z}\bar{w} = z\bar{z}w\bar{w} = |z|^2 |w|^2$ gives $|zw| = |z| |w|$; $|\bar{z}|^2 = \bar{z}z = |z|^2$ gives $|\bar{z}| = |z|$. For the triangle inequality, since the modulus is Euclidean length in $\mathbb{R}^2$ it is the ordinary triangle inequality; algebraically,

$$|z + w|^2 = (z + w)\overline{(z + w)} = |z|^2 + 2\operatorname{Re}(z\bar{w}) + |w|^2 \leq |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2,$$

using $\operatorname{Re}(z\bar{w}) \leq |z\bar{w}| = |z||w|$. The reverse inequality follows from $|z| \leq |z - w| + |w|$ and the symmetric statement, which together pin $||z| - |w|| \leq |z - w|$. Finally $|\operatorname{Re} z| = |x| \leq \sqrt{x^2 + y^2}$. $\square$

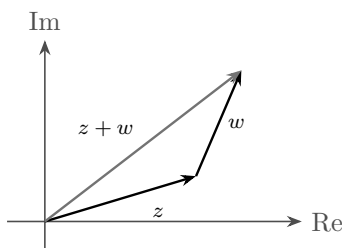


Figure 14.2. Triangle inequality: the direct path $z + w$ is no longer than the detour through z . (Schematic.)

Worked Example 14.1.5 (Arithmetic and division). Compute $\frac{3 + 2i}{1 - i}$ and verify the modulus is consistent.

Recognition. A quotient of complex numbers; rationalize by the conjugate of the denominator.

Solution. Multiply by $\overline{(1 - i)}/\overline{(1 - i)} = (1 + i)/(1 + i)$:

$$\frac{3 + 2i}{1 - i} = \frac{(3 + 2i)(1 + i)}{(1 - i)(1 + i)} = \frac{3 + 3i + 2i + 2i^2}{1 + 1} = \frac{1 + 5i}{2} = \frac{1}{2} + \frac{5}{2}i.$$

Check. $|3 + 2i| = \sqrt{13}$, $|1 - i| = \sqrt{2}$, so the quotient should have modulus $\sqrt{13}/\sqrt{2} = \sqrt{6.5}$. Indeed $|\frac{1}{2} + \frac{5}{2}i| = \sqrt{\frac{1}{4} + \frac{25}{4}} = \sqrt{6.5}$. Agree. $\blacksquare$

Polar form and the geometry of multiplication

Definition 14.1.6. Any nonzero z may be written in *polar form* $z = re^{i\theta}$ where $r = |z| > 0$ and $\theta \in \mathbb{R}$ is the *argument* $\arg z$ (the angle from the positive real axis, counterclockwise), unique up to adding multiples of 2π . Here

$$e^{i\theta} = \cos \theta + i \sin \theta$$

is *Euler's formula*; since $|e^{i\theta}| = \cos^2 \theta + \sin^2 \theta = 1$, we have $r = |z|$.

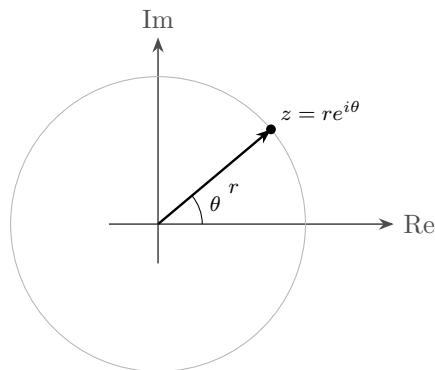


Figure 14.3. Polar form: modulus r and argument θ . (Schematic.)

Theorem 14.1.7 (Multiplication is rotation and dilation). *If $z = re^{i\theta}$ and $w = se^{i\varphi}$ then $zw = rs e^{i(\theta+\varphi)}$. In particular $|zw| = |z| |w|$ and $\arg(zw) = \arg z + \arg w \pmod{2\pi}$; multiplication by z rotates by θ and scales by r . Multiplication by i is rotation by $\pi/2$.*

Proof. Using Euler's formula and the angle-addition identities,

$$e^{i\theta} e^{i\varphi} = (\cos \theta + i \sin \theta)(\cos \varphi + i \sin \varphi) = \cos(\theta + \varphi) + i \sin(\theta + \varphi) = e^{i(\theta+\varphi)}.$$

Hence $zw = rs e^{i(\theta+\varphi)}$. Taking moduli gives $|zw| = rs$; the argument adds. Since $i = e^{i\pi/2}$, multiplying by i adds $\pi/2$ to the argument. $\square$

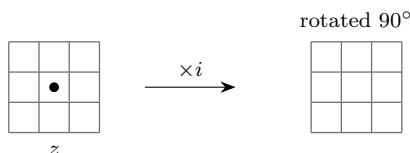


Figure 14.4. Multiplication by i rotates the plane a quarter turn counterclockwise.

Worked Example 14.1.8 (De Moivre and roots of unity). Find all solutions of $z^3 = 1$ and plot them.

Recognition. “Solve $z^n = w$.” Pass to polar form: if $z = re^{i\theta}$ then $z^n = r^n e^{in\theta}$, so match modulus and argument modulo 2π .

Solution. Write $1 = 1 \cdot e^{i \cdot 0}$. We need $r^3 = 1$ (so $r = 1$) and $3\theta \equiv 0 \pmod{2\pi}$, i.e. $\theta = 2\pi k/3$ for $k = 0, 1, 2$. The roots are

$$z_0 = 1, \quad z_1 = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad z_2 = e^{4\pi i/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

Check. Their sum is 0 (coefficient of z^2 in $z^3 - 1$) and their product is 1; indeed $1 + (-\frac{1}{2} + \frac{\sqrt{3}}{2}i) + (-\frac{1}{2} - \frac{\sqrt{3}}{2}i) = 0$. $\blacksquare$

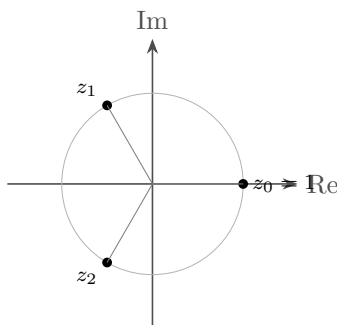


Figure 14.5. The cube roots of unity sit at the vertices of an equilateral triangle on the unit circle.

Convergence and completeness

Definition 14.1.9. A sequence $\{z_n\}$ *converges* to $w \in \mathbb{C}$, written $w = \lim_n z_n$, if $|z_n - w| \rightarrow 0$. It is a *Cauchy sequence* if for every $\varepsilon > 0$ there is N with $|z_n - z_m| < \varepsilon$ whenever $n, m > N$.

Because $|z_n - w|$ is exactly the Euclidean distance in $\mathbb{R}^2$, convergence in $\mathbb{C}$ is the same as convergence of the points (x_n, y_n) in the plane; equivalently, $z_n \rightarrow w$ iff $\operatorname{Re} z_n \rightarrow \operatorname{Re} w$ and $\operatorname{Im} z_n \rightarrow \operatorname{Im} w$.

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{\cong} & \mathbb{R}^2 \\ \operatorname{Re} \downarrow & & \downarrow \pi_1 \\ \mathbb{R} & \xlongequal{\quad} & \mathbb{R} \end{array}$$

Figure 14.6. Convergence in $\mathbb{C}$ is coordinatewise: under $\mathbb{C} \cong \mathbb{R}^2$ the real part Re is the first coordinate projection π_1 (and likewise $\operatorname{Im} = \pi_2$), so $z_n \rightarrow w$ exactly when $\operatorname{Re} z_n \rightarrow \operatorname{Re} w$ and $\operatorname{Im} z_n \rightarrow \operatorname{Im} w$ in $\mathbb{R}$.

Theorem 14.1.10 (Completeness of $\mathbb{C}$). *$\mathbb{C}$ is complete: every Cauchy sequence of complex numbers converges to a complex number.*

Proof. If $\{z_n\}$ is Cauchy then, since $|\operatorname{Re} z_n - \operatorname{Re} z_m| \leq |z_n - z_m|$ and likewise for imaginary parts, the real sequences $\{\operatorname{Re} z_n\}$ and $\{\operatorname{Im} z_n\}$ are Cauchy in $\mathbb{R}$. By completeness of $\mathbb{R}$ they converge, say to a and b ; then $z_n \rightarrow a + ib$ because $|z_n - (a + ib)| \leq |\operatorname{Re} z_n - a| + |\operatorname{Im} z_n - b| \rightarrow 0$. $\square$

Sets in the complex plane

Definition 14.1.11. For $z_0 \in \mathbb{C}$ and $r > 0$ the *open disc*, *closed disc*, and *circle* are

$$D_r(z_0) = \{z : |z - z_0| < r\}, \quad \overline{D}_r(z_0) = \{z : |z - z_0| \leq r\}, \quad C_r(z_0) = \{z : |z - z_0| = r\}.$$

The *unit disc* is $\mathbb{D} = \{z : |z| < 1\} = D_1(0)$.

Definition 14.1.12. Let $\Omega \subseteq \mathbb{C}$. A point z_0 is an *interior point* of Ω if some $D_r(z_0) \subseteq \Omega$; the *interior* is the set of interior points. Ω is *open* if every point of it is interior. Ω is *closed* if its complement Ω^c is open. A point z is a *limit point* of Ω if there are $z_n \in \Omega$ with $z_n \neq z$ and $z_n \rightarrow z$; the *closure* $\overline{\Omega}$ adjoins all limit points, and the *boundary* is $\partial\Omega = \overline{\Omega} \setminus \Omega^\circ$. Ω is *bounded* if $\Omega \subseteq D_M(0)$ for some M , with *diameter* $\operatorname{diam}(\Omega) = \sup_{z, w \in \Omega} |z - w|$, and *compact* if it is closed and bounded.

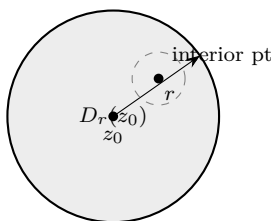


Figure 14.7. An open disc; every point is interior (a small disc fits inside).

Theorem 14.1.13 (Characterizations of compactness). For $\Omega \subseteq \mathbb{C}$ the following are equivalent: (i) Ω is compact (closed and bounded); (ii) every sequence in Ω has a subsequence converging to a point of Ω ; (iii) every open cover $\Omega \subseteq \bigcup_{\alpha} U_{\alpha}$ has a finite subcover.

The proofs are exactly the real-variable ones transported through the identification $\mathbb{C} \cong \mathbb{R}^2$: (i) $\Rightarrow$ (ii) is Bolzano–Weierstrass, and (i) $\Leftrightarrow$ (iii) is Heine–Borel. We will use the following nested-set consequence in the proof of Goursat’s theorem.

Proposition 14.1.14 (Nested compact sets). If $\Omega_1 \supseteq \Omega_2 \supseteq \cdots$ are nonempty compact sets in $\mathbb{C}$ with $\text{diam}(\Omega_n) \rightarrow 0$, then there is a unique w with $w \in \Omega_n$ for all n .

Proof. Pick $z_n \in \Omega_n$. For $m \geq n$ both $z_n, z_m \in \Omega_n$, so $|z_n - z_m| \leq \text{diam}(\Omega_n) \rightarrow 0$: the sequence is Cauchy, hence converges to some w . For each fixed n , the tail $\{z_k\}_{k \geq n} \subseteq \Omega_n$ and Ω_n is closed, so $w \in \Omega_n$. Uniqueness: if also $w' \in \Omega_n$ for all n then $|w - w'| \leq \text{diam}(\Omega_n) \rightarrow 0$, forcing $w' = w$. $\square$

Definition 14.1.15. An open set Ω is *connected* if it cannot be written $\Omega = \Omega_1 \cup \Omega_2$ with Ω_1, Ω_2 disjoint, nonempty, and open. A connected open set is called a *region*. (Equivalently: Ω is connected iff any two of its points can be joined by a curve lying in Ω ; see the exercises.)

Worked Example 14.1.16 (Reading off a locus from an equation). Describe $\{z : |z - 1| = |z + 1|\}$ and $\{z : \text{Re } z > 2\}$ geometrically.

Recognition. “Identify the set defined by a relation.” Translate to (x, y) via $z = x + iy$ and recognize a standard plane figure.

Solution. For the first, $|z - 1| = |z + 1|$ says z is equidistant from 1 and -1 , the perpendicular bisector of that segment, namely the imaginary axis $x = 0$. Algebraically $(x - 1)^2 + y^2 = (x + 1)^2 + y^2$ gives $-2x = 2x$, i.e. $x = 0$. The second, $\text{Re } z > 2$, is the open right half-plane $x > 2$. *Check.* $z = i$ has $|i - 1| = |i + 1| = \sqrt{2}$ and lies on $x = 0$, as required; $z = 3$ satisfies $\text{Re } z = 3 > 2$. $\blacksquare$

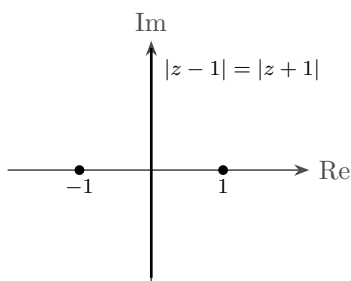


Figure 14.8. The locus $|z - 1| = |z + 1|$ is the imaginary axis (perpendicular bisector).

Caution. The argument $\arg z$ is only defined modulo 2π ; $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ holds up to a multiple of 2π , not as an equality of principal values. For instance with $z_1 = z_2 = e^{3\pi i/4}$, the principal arguments add to $3\pi/2$, which is outside $(-\pi, \pi]$; the principal argument of the product is $-\pi/2$. Never drop the “mod 2π .”

Further worked examples

Worked Example 14.1.17 (Powers via polar form). Compute $(1 + i)^{12}$ and $(\sqrt{3} + i)^6$.

Recognition. “A high power of a fixed complex number”: convert to $re^{i\theta}$ and use $z^n = r^n e^{in\theta}$ (Theorem 14.1.7). Cartesian expansion is hopeless; polar form is forced.

Solution. $1 + i = \sqrt{2}e^{i\pi/4}$, so $(1 + i)^{12} = 2^6 e^{i3\pi} = 64 \cdot (-1) = -64$. Next $\sqrt{3} + i = 2e^{i\pi/6}$, so $(\sqrt{3} + i)^6 = 2^6 e^{i\pi} = 64 \cdot (-1) = -64$. *Check.* $|(1 + i)^{12}| = (\sqrt{2})^{12} = 64 = |-64|$, and $3\pi \equiv \pi \pmod{2\pi}$ puts both on the negative real axis. $\blacksquare$

Worked Example 14.1.18 (Fifth roots and the sum-of-roots identity). Solve $z^5 = -32$ and verify the five roots sum to zero.

Recognition. “Solve $z^n = w$ ”: write w in polar form and spread the argument over n equal sectors; the roots are vertices of a regular pentagon.

Solution. Here $-32 = 32 e^{i\pi}$, so $r = 32^{1/5} = 2$ and $\theta = (\pi + 2\pi k)/5$, $k = 0, \dots, 4$:

$$z_k = 2 \exp\left(i \frac{(2k+1)\pi}{5}\right), \quad k = 0, 1, 2, 3, 4.$$

Check. The z_k are the roots of $z^5 + 32 = 0$, whose z^4 -coefficient is 0, so $\sum_k z_k = 0$. Equivalently the five unit vectors $e^{i(2k+1)\pi/5}$ are equally spaced and cancel by symmetry. ■

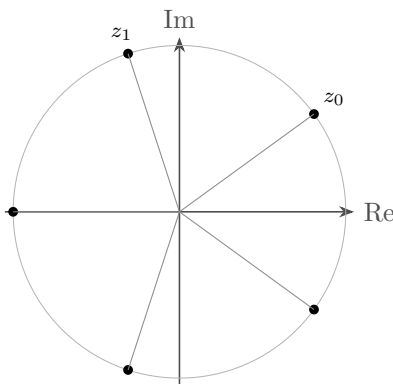


Figure 14.9. The five solutions of $z^5 = -32$: a regular pentagon of radius 2.

Worked Example 14.1.19 (A circle disguised as a modulus equation). Describe the locus $\{z : |z - 1| = 2|z + 1|\}$.

Recognition. “Identify the set from a relation.” A ratio of distances $\neq 1$ to two points is an *Apollonius circle*, not a line; square and complete the square.

Solution. Put $z = x + iy$. Squaring, $(x-1)^2 + y^2 = 4((x+1)^2 + y^2)$, i.e. $x^2 - 2x + 1 + y^2 = 4x^2 + 8x + 4 + 4y^2$. Collecting, $3x^2 + 3y^2 + 10x + 3 = 0$, so $x^2 + y^2 + \frac{10}{3}x + 1 = 0$, giving $(x + \frac{5}{3})^2 + y^2 = \frac{25}{9} - 1 = \frac{16}{9}$. This is the circle centered at $-\frac{5}{3}$ of radius $\frac{4}{3}$. *Check.* $z = -3$ gives $|-4| = 4$ and $2|-2| = 4$; and -3 lies on the circle since $|-3 + \frac{5}{3}| = \frac{4}{3}$. ■

Worked Example 14.1.20 (Blaschke factor preserves the unit circle). Fix w with $|w| < 1$ and set $F(z) = \frac{w - z}{1 - \bar{w}z}$. Show $|F(z)| = 1$ whenever $|z| = 1$.

Recognition. Stein’s Blaschke factor. To prove a modulus equals 1, compute $|F|^2 = F\bar{F}$ using $|z|^2 = z\bar{z} = 1$.

Solution. For $|z| = 1$ we have $\bar{z} = 1/z$. Then

$$|1 - \bar{w}z| = |z| \left| \frac{1}{z} - \bar{w} \right| = |\bar{z} - \bar{w}| = |\overline{z - w}| = |z - w| = |w - z|.$$

Hence numerator and denominator have equal modulus and $|F(z)| = 1$. *Check.* At $w = 0$, $F(z) = -z$ and $|F| = |z| = 1$ on the circle, as required. ■

Worked Example 14.1.21 (A non-example: the argument is not continuous on a punctured plane). Show there is no continuous choice of $\arg z$ on $\mathbb{C} \setminus \{0\}$.

Recognition. “Can a multivalued quantity be made single-valued and continuous?” Test continuity by transporting the argument once around the origin.

Solution. Suppose $\theta(z)$ were continuous with $e^{i\theta(z)} = z/|z|$. Track $z(t) = e^{it}$, $t \in [0, 2\pi]$. Then $\theta(e^{it}) - t$ is a continuous integer multiple of 2π , hence constant; so $\theta(e^{it}) = t + c$. But $z(0) = z(2\pi) = 1$ while θ jumps from c to $2\pi + c$, contradicting single-valuedness. *Check.* This is the same obstruction as $\oint_{|z|=1} dz/z = 2\pi i \neq 0$ met in the integration section; a global continuous argument (equivalently a global log) cannot exist on the punctured plane. ■

Exercises (Stein, Ch.1).

- Describe geometrically the sets of points z defined by: (a) $|z - z_1| = |z - z_2|$ where $z_1, z_2 \in \mathbb{C}$; (b) $1/z = \bar{z}$; (c) $\operatorname{Re}(z) = 3$; (d) $\operatorname{Re}(z) > c$ (resp. $\geq c$) where $c \in \mathbb{R}$; (e) $\operatorname{Re}(az + b) > 0$ where $a, b \in \mathbb{C}$; (f) $|z| = \operatorname{Re}(z) + 1$; (g) $\operatorname{Im}(z) = c$ with $c \in \mathbb{R}$.
- Let $\langle \cdot, \cdot \rangle$ be the usual inner product on $\mathbb{R}^2$: for $Z = (x_1, y_1)$, $W = (x_2, y_2)$, $\langle Z, W \rangle = x_1x_2 + y_1y_2$. Define a Hermitian inner product on $\mathbb{C}$ by $(z, w) = z\bar{w}$. (The term Hermitian records that $(z, w) = \overline{(w, z)}$.) Show, under the identification $z = x + iy \leftrightarrow (x, y)$, that $\langle z, w \rangle = \frac{1}{2}[(z, w) + (w, z)] = \operatorname{Re}(z, w)$.
- With $\omega = se^{i\varphi}$, where $s \geq 0$ and $\varphi \in \mathbb{R}$, solve $z^n = \omega$ in $\mathbb{C}$ for n a natural number. How many solutions are there?
- Show it is impossible to define a total ordering on $\mathbb{C}$. That is, no relation $\succ$ can satisfy: (i) for any z, w exactly one of $z \succ w$, $w \succ z$, $z = w$ holds; (ii) $z_1 \succ z_2$ implies $z_1 + z_3 \succ z_2 + z_3$ for all z_3 ; (iii) for $z_3 \succ 0$, $z_1 \succ z_2$ implies $z_1z_3 \succ z_2z_3$. [Hint: first decide whether $i \succ 0$ is possible.]
- A set Ω is *pathwise connected* if any two of its points can be joined by a piecewise-smooth curve in Ω . Prove that an open set Ω is pathwise connected iff it is connected. (a) If Ω is open and pathwise connected and $\Omega = \Omega_1 \cup \Omega_2$ with Ω_1, Ω_2 disjoint, nonempty, open, choose $w_1 \in \Omega_1$, $w_2 \in \Omega_2$, a curve $z : [0, 1] \rightarrow \Omega$ from w_1 to w_2 , and $t^* = \sup\{t : z(s) \in \Omega_1 \ \forall 0 \leq s < t\}$; derive a contradiction at $z(t^*)$. (b) Conversely, if Ω is open and connected, fix w , let Ω_1 be the points joinable to w by a curve in Ω and Ω_2 the rest; show both are open and disjoint with union Ω , and that $\Omega_1 \neq \emptyset$ forces $\Omega = \Omega_1$.
- For Ω open and $z \in \Omega$, the connected *component* C_z is the set of $w \in \Omega$ joinable to z by a curve in Ω . (a) Show C_z is open and connected, and that $w \in C_z$ is an equivalence relation. (b) Show Ω has at most countably many distinct components. (c) Prove that if Ω is the complement of a compact set, then Ω has exactly one unbounded component. [Hints: for (b), distinct components would give uncountably many disjoint open balls, each containing a rational point; for (c), the complement of a large disc containing the compact set is connected.]
- (Blaschke factors.) (a) For $z, w \in \mathbb{C}$ with $\bar{w}z \neq 1$, prove $|\frac{w-z}{1-\bar{w}z}| < 1$ if $|z| < 1$ and $|w| < 1$, and $|\frac{w-z}{1-\bar{w}z}| = 1$ if $|z| = 1$ or $|w| = 1$. [Hint: reduce to z real; then show $(r-w)(r-\bar{w}) \leq (1-r\bar{w})(1-rw)$.] (b) For fixed $w \in \mathbb{D}$, let $F(z) = \frac{w-z}{1-\bar{w}z}$. Prove: (i) $F : \mathbb{D} \rightarrow \mathbb{D}$ and is holomorphic; (ii) $F(0) = w$ and $F(w) = 0$; (iii) $|F(z)| = 1$ when $|z| = 1$; (iv) $F : \mathbb{D} \rightarrow \mathbb{D}$ is bijective. [Hint for (iv): compute $F \circ F$.]

Additional exercises.

- Compute in the form $x + iy$: (a) $(2 - i)(3 + 2i)$; (b) $\frac{1+i}{2-i}$; (c) i^{2026} ; (d) $(1+i)^8$.
- Write $-1 - i$, $\sqrt{3} + i$, and $-2i$ in polar form $re^{i\theta}$ with $\theta \in (-\pi, \pi]$, and use the polar form to compute $(\sqrt{3} + i)^6$.
- Prove the *parallelogram law* $|z + w|^2 + |z - w|^2 = 2(|z|^2 + |w|^2)$ for all $z, w \in \mathbb{C}$, and interpret it geometrically in the plane.
- Find all z with $z^4 = -16$, and plot them; verify their sum is 0 and their product is -16 .
- Show that the map $z \mapsto \bar{z}$ preserves $|\cdot|$ and reverses orientation (it is a reflection), while $z \mapsto e^{i\alpha}z$ is a rotation by α . Deduce that $z \mapsto e^{i\alpha}\bar{z}$ is a reflection across some line through the origin; which line?
- Determine and sketch the set $\{z : |z - 2| < |z + 2|\}$. Is it open? connected? Describe its boundary.
- (Non-example.) Decide whether $\Omega = \{z : 0 < |z| < 1\} \cup \{z : |z - 3| < 1\}$ is a region (connected open set). Justify, and if it is not, exhibit the disjoint open sets witnessing disconnectedness.
- (Diagram problem.) On a single copy of the complex plane, shade the annulus $\{1 \leq |z - i| \leq 2\}$ and mark its boundary circles, center, and the points where it meets the imaginary axis. State $\operatorname{diam}(\Omega)$ for this set.
- Prove that a closed disc $\overline{D}_r(z_0)$ is compact directly from Theorem 14.1.13(ii): given a sequence inside it, extract a convergent subsequence and show its limit lies in the disc.
- Let $\Omega_n = \overline{D}_{1/n}(0)$. Verify the hypotheses of Proposition 14.1.14 and identify the unique common point. Then give an example of nested nonempty *closed* (but unbounded) sets with empty intersection, showing compactness cannot be dropped.

14.2 Functions on the Complex Plane: Holomorphy and Power Series

Continuous functions

Definition 14.2.1. Let f be defined on $\Omega \subseteq \mathbb{C}$. We say f is *continuous* at $z_0 \in \Omega$ if for every $\varepsilon > 0$ there is $\delta > 0$ so that $z \in \Omega$ and $|z - z_0| < \delta$ imply $|f(z) - f(z_0)| < \varepsilon$; equivalently, $z_n \rightarrow z_0$ implies $f(z_n) \rightarrow f(z_0)$. f is *continuous on Ω* if it is continuous at each point.

Since convergence in $\mathbb{C}$ and in $\mathbb{R}^2$ coincide, $f = u + iv$ is continuous iff u, v are continuous as functions of (x, y) . Sums and products of continuous functions are continuous, and $z \mapsto |f(z)|$ is continuous (by the reverse triangle inequality).

Theorem 14.2.2 (Extreme values on a compact set). *A continuous function on a compact set $\Omega \subseteq \mathbb{C}$ is bounded and attains a maximum and minimum of $|f|$ on Ω .*

Proof. The real-valued continuous function $z \mapsto |f(z)|$ on the compact Ω attains its sup and inf by the real extreme-value theorem (apply Theorem 14.1.13(ii): take a sequence with $|f(z_n)| \rightarrow \sup |f|$, pass to a convergent subsequence $z_{n_k} \rightarrow z_* \in \Omega$, and use continuity). $\square$

Holomorphic functions

This is the central definition of the subject, and it is genuinely complex in nature: the increment h is allowed to approach 0 *from any direction in the plane*.

Definition 14.2.3. Let $\Omega \subseteq \mathbb{C}$ be open and $f : \Omega \rightarrow \mathbb{C}$. Then f is *holomorphic* at $z_0 \in \Omega$ if

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists, the limit taken over $h \in \mathbb{C} \setminus \{0\}$ with $z_0 + h \in \Omega$. If f is holomorphic at every point of Ω it is *holomorphic on Ω* ; if holomorphic on all of $\mathbb{C}$ it is *entire*. (Synonyms: *regular*, *analytic*, *complex differentiable*.)

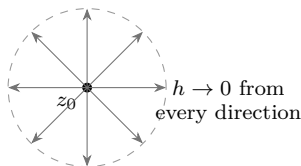


Figure 14.10. Complex differentiability demands the *same* limit as $h \rightarrow 0$ along all directions. (Schematic.)

Example 14.2.4. The function $f(z) = z$ is entire with $f'(z) = 1$, and any polynomial $p(z) = a_0 + a_1z + \cdots + a_nz^n$ is entire with $p'(z) = a_1 + 2a_2z + \cdots + na_nz^{n-1}$. The function $1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$ with derivative $-1/z^2$.

Example 14.2.5 (The conjugate is nowhere holomorphic). For $f(z) = \bar{z}$,

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{\overline{z_0 + h} - \bar{z}_0}{h} = \frac{\bar{h}}{h}.$$

Along real h this equals 1; along imaginary $h = it$ it equals $-it/it = -1$. The two directional limits disagree, so $\bar{z}$ is holomorphic *nowhere*, even though as a map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ it is $(x, y) \mapsto (x, -y)$, which is infinitely real-differentiable.

Proposition 14.2.6 (Calculus of holomorphic functions). *If f, g are holomorphic on Ω then so are $f + g$ and fg , with $(f + g)' = f' + g'$ and $(fg)' = f'g + fg'$; if $g(z_0) \neq 0$ then f/g is holomorphic at z_0 with $(f/g)' = (f'g - fg')/g^2$. If $f : \Omega \rightarrow U$ and $g : U \rightarrow \mathbb{C}$ are holomorphic then so is $g \circ f$, with the chain rule $(g \circ f)'(z) = g'(f(z)) f'(z)$.*

Proof. The key is the linearization: f is holomorphic at z_0 with $f'(z_0) = a$ iff

$$f(z_0 + h) - f(z_0) = ah + h\psi(h), \quad \psi(h) \rightarrow 0 \text{ as } h \rightarrow 0. \quad (14.1)$$

This makes f continuous at z_0 (the right side $\rightarrow 0$). The sum and product rules now follow by the same algebra as in one real variable: for the product, $f(z_0 + h)g(z_0 + h) - f(z_0)g(z_0) = [f(z_0 + h) - f(z_0)]g(z_0 + h) + f(z_0)[g(z_0 + h) - g(z_0)]$, divide by h and let $h \rightarrow 0$ using continuity of g . For the chain rule, with $w_0 = f(z_0)$ and $k = f(z_0 + h) - f(z_0) = f'(z_0)h + h\psi(h)$, write $g(w_0 + k) - g(w_0) = g'(w_0)k + k\eta(k)$; dividing by h and letting $h \rightarrow 0$ (so $k \rightarrow 0$) gives $g'(w_0)f'(z_0)$. $\square$

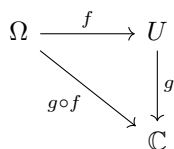


Figure 14.11. The composite $g \circ f : \Omega \rightarrow \mathbb{C}$ along $f : \Omega \rightarrow U$ and $g : U \rightarrow \mathbb{C}$; the chain rule $(g \circ f)'(z) = g'(f(z))f'(z)$ propagates derivatives along this triangle.

Worked Example 14.2.7 (Differentiating by the quotient and chain rules). Compute the derivative of $f(z) = \frac{z^2 + 1}{z - i}$ and of $g(z) = (z^2 + 1)^{10}$.

Recognition. Direct application of Proposition 14.2.6.

Solution. Quotient rule:

$$f'(z) = \frac{2z(z - i) - (z^2 + 1) \cdot 1}{(z - i)^2} = \frac{z^2 - 2iz - 1}{(z - i)^2} = \frac{(z - i)^2}{(z - i)^2} = 1$$

on $\mathbb{C} \setminus \{i\}$, because $z^2 + 1 = (z - i)(z + i)$ makes $f(z) = z + i$ there. Chain rule: $g'(z) = 10(z^2 + 1)^9 \cdot 2z = 20z(z^2 + 1)^9$. *Check.* $f(z) = z + i$ indeed has $f' \equiv 1$, matching the quotient-rule output. $\blacksquare$

The Cauchy–Riemann equations

Writing $f = u + iv$ with u, v real, we associate the map $F(x, y) = (u(x, y), v(x, y))$ from $\mathbb{R}^2$ to $\mathbb{R}^2$. Real differentiability of F means there is a linear map J (the Jacobian) with $F(P_0 + H) - F(P_0) = J(H) + |H|\Psi(H)$, $\Psi \rightarrow 0$. Complex differentiability is strictly stronger, and the gap is measured by the following equations.

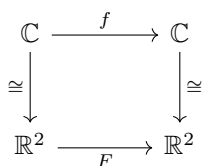


Figure 14.12. The holomorphic function $f = u + iv$ and the real map $F(x, y) = (u(x, y), v(x, y))$ are the same map read through the identification $\mathbb{C} \cong \mathbb{R}^2$; the square commutes. Holomorphy is the extra condition that the Jacobian of F be a rotation-scaling, i.e. the Cauchy–Riemann equations.

Theorem 14.2.8 (Cauchy–Riemann, necessity). *If $f = u + iv$ is holomorphic at $z_0 = x_0 + iy_0$, then the partials of u, v exist there and satisfy*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Consequently $f'(z_0) = \frac{\partial f}{\partial x} = \frac{1}{i} \frac{\partial f}{\partial y} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$.

Proof. Take the limit (14.1) first along real $h = h_1$: then $f'(z_0) = \lim_{h_1 \rightarrow 0} \frac{f(x_0 + h_1, y_0) - f(x_0, y_0)}{h_1} = \partial f / \partial x$. Now along imaginary $h = ih_2$: $f'(z_0) = \lim_{h_2 \rightarrow 0} \frac{f(x_0, y_0 + h_2) - f(x_0, y_0)}{ih_2} = \frac{1}{i} \partial f / \partial y$. Equating, $\partial f / \partial x = \frac{1}{i} \partial f / \partial y$. Substituting $f = u + iv$ and $1/i = -i$ and separating real and imaginary parts gives $u_x = v_y$ and $u_y = -v_x$. $\square$

It is convenient to package this with the *Wirtinger operators*

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right).$$

Proposition 14.2.9 (Wirtinger form). *f holomorphic at z_0 iff $\partial f / \partial \bar{z}(z_0) = 0$ (this single complex equation is the Cauchy–Riemann pair). Then $f'(z_0) = \partial f / \partial z(z_0) = 2 \partial u / \partial z(z_0)$, and writing $F(x, y) = f(z)$ one has $\det J_F(x_0, y_0) = |f'(z_0)|^2$.*

Proof. Separating real and imaginary parts shows $\partial f / \partial \bar{z} = 0$ is equivalent to $u_x = v_y$, $u_y = -v_x$. The Jacobian determinant is $u_x v_y - u_y v_x = u_x^2 + u_y^2$ (using CR) $= |f'(z_0)|^2$ since $f'(z_0) = u_x + iv_x = u_x - iu_y$. $\square$

The geometric reading of $\det J_F = |f'|^2 \geq 0$ is that a holomorphic map with $f'(z_0) \neq 0$ is locally an *orientation-preserving* similarity: it rotates by $\arg f'(z_0)$ and scales by $|f'(z_0)|$, hence is *conformal* (angle-preserving).

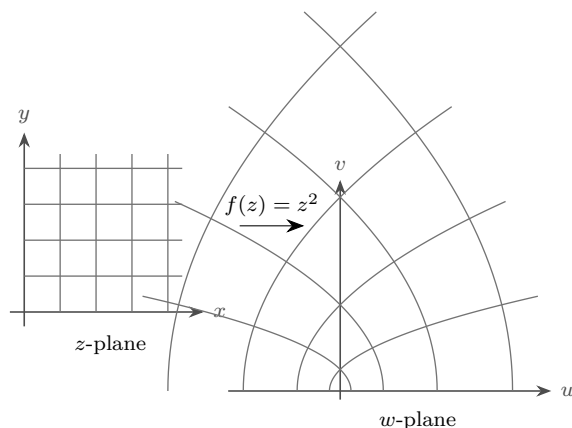


Figure 14.13. A conformal map ($f' \neq 0$) carries an orthogonal grid to an orthogonal grid; here $z \mapsto z^2$ sends the Cartesian grid to confocal parabolas that still meet at right angles.

Theorem 14.2.10 (Cauchy–Riemann, sufficiency). *Suppose $f = u + iv$ on an open Ω , with u, v continuously differentiable and satisfying the Cauchy–Riemann equations on Ω . Then f is holomorphic on Ω and $f' = \partial f / \partial z$.*

Proof. By C^1 differentiability, $u(x + h_1, y + h_2) - u(x, y) = u_x h_1 + u_y h_2 + |h| \psi_1(h)$ and similarly for v , with $\psi_j \rightarrow 0$. Adding $u + iv$ and using $u_x = v_y$, $u_y = -v_x$,

$$f(z + h) - f(z) = (u_x + iv_x)(h_1 + ih_2) + |h| \psi(h) = f'(z) h + |h| \psi(h),$$

where $f'(z) = u_x + iv_x$ and $\psi = \psi_1 + i\psi_2 \rightarrow 0$. Dividing by h and letting $h \rightarrow 0$ gives the complex derivative. $\square$

Worked Example 14.2.11 (Testing holomorphy with Cauchy–Riemann). Is $f(z) = z^2$ holomorphic? Is $g(z) = |z|^2$?

Recognition. Compute u, v and check the CR equations.

Solution. For $z^2 = (x + iy)^2 = (x^2 - y^2) + i(2xy)$: $u = x^2 - y^2$, $v = 2xy$, so $u_x = 2x = v_y$ and $u_y = -2y = -v_x$. CR holds everywhere with $u, v \in C^1$; z^2 is entire and $f' = u_x + iv_x = 2x + 2iy = 2z$. For $g = |z|^2 = x^2 + y^2$: $u = x^2 + y^2$, $v = 0$, so $u_x = 2x$, $v_y = 0$. CR demands $2x = 0$ and $2y = 0$, i.e. only at $z = 0$; $|z|^2$ is (complex) differentiable only at the origin and holomorphic *nowhere*. Check. $g'(0)$ would be $\lim_{h \rightarrow 0} |h|^2 / h = \lim \bar{h} = 0$, consistent with differentiability at 0 alone. $\blacksquare$

Caution. Satisfying the Cauchy–Riemann equations at a point does not make f holomorphic there; sufficiency (Theorem 14.2.10) needs CR on a neighborhood plus C^1 regularity. The function $f(x + iy) = \sqrt{|xy|}$ satisfies CR at the origin yet is not holomorphic there (a famous non-example; see the exercises).

Power series

Definition 14.2.12. A *power series* centered at z_0 is a series $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ with $a_n \in \mathbb{C}$. The *exponential* is $e^z = \sum_{n \geq 0} z^n/n!$.

Since $|z^n/n!| = |z|^n/n!$ and $\sum |z|^n/n! = e^{|z|} < \infty$, the exponential series converges absolutely for every z , uniformly on each disc.

Theorem 14.2.13 (Radius of convergence; Hadamard's formula). *Given $\sum_{n \geq 0} a_n z^n$ there is $0 \leq R \leq \infty$ such that the series converges absolutely for $|z| < R$ and diverges for $|z| > R$, where (with $1/0 = \infty$, $1/\infty = 0$)*

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |a_n|^{1/n}.$$

R is the radius of convergence and $\{|z| < R\}$ the disc of convergence; on $|z| = R$ either may occur.

Proof. Let $L = 1/R$ and assume $0 < L < \infty$ (the cases $L = 0, \infty$ are similar). If $|z| < R$ pick $\varepsilon > 0$ with $(L + \varepsilon)|z| = r < 1$. By definition of $\limsup$, $|a_n|^{1/n} \leq L + \varepsilon$ for all large n , so $|a_n||z|^n \leq ((L + \varepsilon)|z|)^n = r^n$; comparison with the convergent geometric series $\sum r^n$ gives absolute convergence. If $|z| > R$ then $|a_n|^{1/n}|z| > 1$ infinitely often, so $|a_n||z|^n \not\rightarrow 0$ and the series diverges. $\square$

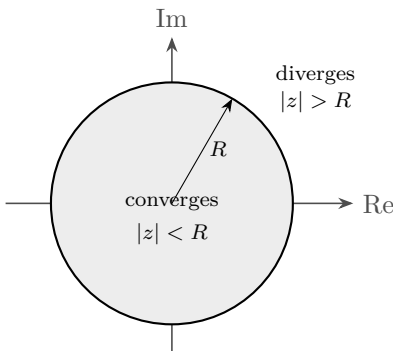


Figure 14.14. The disc of convergence; behavior on the boundary circle is delicate. (Schematic.)

Theorem 14.2.14 (A power series is holomorphic, differentiated termwise). $f(z) = \sum_{n \geq 0} a_n z^n$ is holomorphic on its disc of convergence, and

$$f'(z) = \sum_{n \geq 0} n a_n z^{n-1},$$

a power series with the same radius of convergence R . Consequently f is infinitely complex differentiable there (apply this repeatedly).

Proof. Same radius: $\limsup |n a_n|^{1/n} = \limsup n^{1/n} |a_n|^{1/n} = \limsup |a_n|^{1/n}$ since $n^{1/n} \rightarrow 1$. Let $g(z) = \sum n a_n z^{n-1}$; fix $|z_0| < r < R$. Split $f = S_N + E_N$ with $S_N = \sum_{n \leq N} a_n z^n$. Then

$$\frac{f(z_0 + h) - f(z_0)}{h} - g(z_0) = \left(\frac{S_N(z_0 + h) - S_N(z_0)}{h} - S'_N(z_0) \right) + (S'_N(z_0) - g(z_0)) + \frac{E_N(z_0 + h) - E_N(z_0)}{h}.$$

Using $|(z_0 + h)^n - z_0^n| \leq |h| n r^{n-1}$ (for $|z_0|, |z_0 + h| < r$), the last term is bounded by $\sum_{n > N} |a_n| n r^{n-1}$, the tail of a convergent series, $< \varepsilon$ for large N . The middle term $< \varepsilon$ for large N since $S'_N \rightarrow g$. Fixing such an N , the first term $< \varepsilon$ for $|h| < \delta$ since S_N is a polynomial. Hence the total is $< 3\varepsilon$, proving $f' = g$. $\square$

Corollary 14.2.15. e^z is entire and $(e^z)' = e^z$. The functions $\cos z = \sum (-1)^n z^{2n}/(2n)!$ and $\sin z = \sum (-1)^n z^{2n+1}/(2n+1)!$ are entire, and

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

(Euler's formulas). The geometric series $\sum_{n \geq 0} z^n = \frac{1}{1-z}$ has $R = 1$.

Proof. Termwise differentiation of $\sum z^n/n!$ shifts the index and reproduces the series, so $(e^z)' = e^z$. For Euler's formulas substitute $iz, -iz$ into the exponential series and combine: $e^{iz} + e^{-iz} = 2 \sum (-1)^n z^{2n}/(2n)! = 2 \cos z$, similarly for $\sin$. For the geometric series, $\sum_{n=0}^N z^n = (1 - z^{N+1})/(1 - z)$ and $z^{N+1} \rightarrow 0$ when $|z| < 1$. $\square$

Worked Example 14.2.16 (Computing a radius of convergence). Find R for $\sum_{n \geq 1} \frac{z^n}{n^2}$ and for $\sum_{n \geq 0} \frac{z^n}{n!}$.

Recognition. Use Hadamard's formula, or the ratio shortcut $|a_{n+1}|/|a_n| \rightarrow L \Rightarrow R = 1/L$.

Solution. For $a_n = 1/n^2$: $|a_n|^{1/n} = n^{-2/n} = (n^{1/n})^{-2} \rightarrow 1$, so $1/R = 1$, $R = 1$. For $a_n = 1/n!$: $|a_{n+1}/a_n| = 1/(n+1) \rightarrow 0$, so $R = \infty$ (the exponential is entire). *Check.* On $|z| = 1$ the first series converges absolutely ($\sum 1/n^2 < \infty$), consistent with $R = 1$ being the largest disc of *guaranteed* convergence. $\blacksquare$

Worked Example 14.2.17 (A closed form by manipulating the geometric series). Sum $\sum_{n \geq 1} n z^n$ for $|z| < 1$.

Recognition. Differentiate the geometric series termwise (legal inside the disc by Theorem 14.2.14), then multiply by z .

Solution. From $\sum_{n \geq 0} z^n = 1/(1 - z)$, differentiate: $\sum_{n \geq 1} n z^{n-1} = 1/(1 - z)^2$. Multiply by z : $\sum_{n \geq 1} n z^n = z/(1 - z)^2$. *Check.* At $z = \frac{1}{2}$ the left side is $\sum n 2^{-n} = 2$, and the right side is $(1/2)/(1/2)^2 = 2$. $\blacksquare$

Definition 14.2.18. f is *analytic* at z_0 if it has a convergent power-series expansion $f(z) = \sum a_n(z - z_0)^n$ with $R > 0$ in a neighborhood of z_0 ; analytic on Ω if so at each point. By Theorem 14.2.14 analytic $\Rightarrow$ holomorphic; a deep theorem of the next chapter gives the converse, so the two words are synonymous.

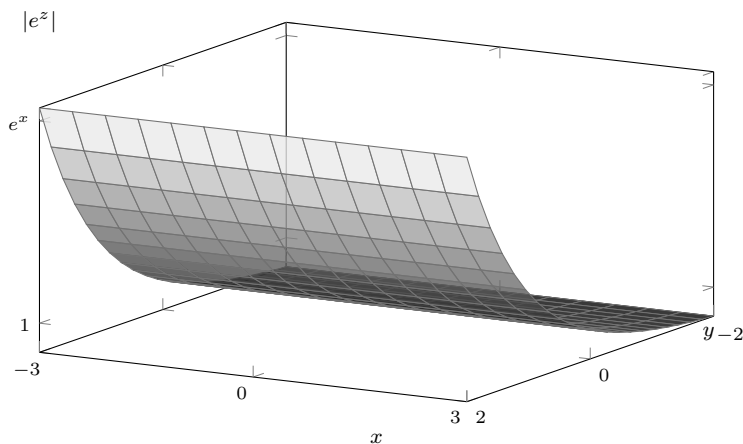


Figure 14.15. Modulus surface $|e^z| = e^x$: the height depends only on x (exponential growth), and is constant along y – a translation-invariant ridge.

Further worked examples

Worked Example 14.2.19 (Cauchy–Riemann in polar coordinates). With $z = r e^{i\theta}$ and $f = u + iv$, derive the polar form of the CR equations and use it to confirm $\log z = \log r + i\theta$ ($-\pi < \theta < \pi$) is holomorphic.

Recognition. Stein's polar CR exercise. Change variables $x = r \cos \theta$, $y = r \sin \theta$ in Theorem 14.2.8 via the chain rule.

Solution. From $u_x = v_y$, $u_y = -v_x$ and $\partial_r = \cos \theta \partial_x + \sin \theta \partial_y$, $\partial_\theta = -r \sin \theta \partial_x + r \cos \theta \partial_y$, one gets the polar CR pair

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}.$$

For $\log z$ we have $u = \log r$, $v = \theta$: $u_r = 1/r$, $v_\theta = 1$, so $u_r = \frac{1}{r} v_\theta$; and $u_\theta = 0 = -v_r$. Both hold for $r > 0$, and the partials are continuous, so $\log z$ is holomorphic on the slit plane. *Check.* $(\log z)' = e^{-i\theta}(u_r + iv_r) = e^{-i\theta} \cdot \frac{1}{r} = 1/z$, the expected derivative. $\blacksquare$

Worked Example 14.2.20 (The Wirtinger form of the Laplacian). Show $4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \Delta$, and conclude that the real and imaginary parts of a holomorphic function are harmonic.

Recognition. Stein's $4 \partial_z \partial_{\bar{z}} = \Delta$ identity; expand the Wirtinger operators from Proposition 14.2.9.

Solution. With $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$,

$$4\partial_z\partial_{\bar{z}} = (\partial_x - i\partial_y)(\partial_x + i\partial_y) = \partial_x^2 + i\partial_x\partial_y - i\partial_y\partial_x + \partial_y^2 = \partial_x^2 + \partial_y^2 = \Delta,$$

the mixed terms cancelling by equality of mixed partials. If f is holomorphic then $\partial_{\bar{z}}f = 0$, so $\Delta f = 4\partial_z(\partial_{\bar{z}}f) = 0$; taking real and imaginary parts, $\Delta u = \Delta v = 0$. *Check.* For $f = z^2$, $u = x^2 - y^2$ has $u_{xx} + u_{yy} = 2 - 2 = 0$, harmonic as predicted. ■

Worked Example 14.2.21 (Harmonic conjugate). Given $u(x, y) = x^3 - 3xy^2$, find a harmonic conjugate v so that $f = u + iv$ is holomorphic, and identify f .

Recognition. “Reconstruct f from its real part”: integrate the CR equations $v_y = u_x$, $v_x = -u_y$.

Solution. Here $u_x = 3x^2 - 3y^2$ and $u_y = -6xy$. From $v_y = u_x = 3x^2 - 3y^2$ integrate in y : $v = 3x^2y - y^3 + g(x)$. Then $v_x = 6xy + g'(x)$ must equal $-u_y = 6xy$, so $g'(x) = 0$, g constant. Thus $v = 3x^2y - y^3$ and $f = (x^3 - 3xy^2) + i(3x^2y - y^3) = z^3$. *Check.* $\operatorname{Re}(z^3) = x^3 - 3xy^2 = u$, confirming the reconstruction. ■

Worked Example 14.2.22 (Taylor coefficients of a rational function). Expand $f(z) = \frac{1}{(1-z)^2}$ as a power series about 0 and read off a_n .

Recognition. Differentiate or square the geometric series (Theorem 14.2.14); the series is the termwise derivative of $1/(1-z)$.

Solution. From $\sum_{n \geq 0} z^n = 1/(1-z)$, differentiate termwise: $\sum_{n \geq 1} nz^{n-1} = 1/(1-z)^2$. Reindexing $m = n-1$,

$$\frac{1}{(1-z)^2} = \sum_{m \geq 0} (m+1)z^m, \quad a_m = m+1, \quad |z| < 1.$$

Check. The asymptotic $a_m \sim m^{m-1}/(m-1)!$ for $(1-z)^{-m}$ at $m=2$ gives $a_n \sim n$, matching $a_n = n+1$ to leading order. ■

Worked Example 14.2.23 (Radius of convergence by the ratio test: a Bessel series). Find the radius of convergence of $\sum_{n \geq 0} \frac{(-1)^n}{n!(n+r)!} \left(\frac{z}{2}\right)^{2n}$ (the Bessel function J_r , r a fixed positive integer).

Recognition. Stein's Bessel exercise. The series is in $w = (z/2)^2$; apply the ratio test to the coefficients $b_n = (-1)^n/(n!(n+r)!4^n)$.

Solution. The general term is $b_n z^{2n}$ with $|b_n| = \frac{1}{n!(n+r)!4^n}$. Then

$$\frac{|b_{n+1}|}{|b_n|} = \frac{n!(n+r)!}{(n+1)!(n+r+1)!4} = \frac{1}{4(n+1)(n+r+1)} \rightarrow 0.$$

So in the variable z^2 the radius is ∞ , hence the series converges for every z : $R = \infty$. *Check.* J_r is built from the entire exponential-type data $1/n!$, and indeed Bessel functions are entire. ■

Worked Example 14.2.24 (Boundary behavior of three look-alike series). For $\sum nz^n$, $\sum z^n/n^2$, and $\sum z^n/n$ (all $R=1$), decide convergence on $|z|=1$.

Recognition. Stein's trio. Same radius, different boundary fates: test the n th term and use Abel/Dirichlet (summation by parts) where needed.

Solution. (a) For $\sum nz^n$, $|nz^n| = n \not\rightarrow 0$ on $|z|=1$, so it diverges *everywhere* on the circle. (b) For $\sum z^n/n^2$, $|z^n/n^2| = 1/n^2$ is summable, so it converges *absolutely at every* boundary point. (c) For $\sum z^n/n$: at $z=1$ it is the divergent harmonic series; for $z=e^{i\theta} \neq 1$, the partial sums of z^n are bounded by $2/|1-z|$ and $1/n \downarrow 0$, so by Dirichlet's test it converges. Thus $\sum z^n/n$ converges at every boundary point *except* $z=1$. *Check.* The three radii are equal ($n^{1/n} \rightarrow 1$), confirming the disc of convergence is identical while only the circle distinguishes them. ■

Worked Example 14.2.25 (A non-example: real-differentiable but nowhere holomorphic). Show $f(z) = \operatorname{Re}(z) = x$ is infinitely real-differentiable yet holomorphic nowhere.

Recognition. Apply the Wirtinger criterion (Proposition 14.2.9): holomorphy is exactly $\partial f / \partial \bar{z} = 0$.

Solution. Here $u = x$, $v = 0$, so $u_x = 1 \neq 0 = v_y$: the CR equation $u_x = v_y$ fails at every point. Equivalently $\partial_{\bar{z}} x = \frac{1}{2}(\partial_x + i\partial_y)x = \frac{1}{2} \neq 0$ everywhere. So $\operatorname{Re} z$ is nowhere holomorphic, although as a map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ it is linear, hence C^∞ . *Check.* If $\operatorname{Re} z$ were holomorphic on any disc it would be a holomorphic function with constant ($= 0$) imaginary part, forcing it constant by the open mapping property – contradicting $\operatorname{Re} z$ being non-constant. ■

Exercises (Stein, Ch.1).

- (Complex chain rule.) Suppose U, V are open in $\mathbb{C}$, $f : U \rightarrow V$ and $g : V \rightarrow \mathbb{C}$ are differentiable in the real sense (as functions of x, y), and $h = g \circ f$. Prove

$$\frac{\partial h}{\partial z} = \frac{\partial g}{\partial z} \frac{\partial f}{\partial z} + \frac{\partial g}{\partial \bar{z}} \frac{\partial \bar{f}}{\partial z}, \quad \frac{\partial h}{\partial \bar{z}} = \frac{\partial g}{\partial z} \frac{\partial f}{\partial \bar{z}} + \frac{\partial g}{\partial \bar{z}} \frac{\partial \bar{f}}{\partial \bar{z}}.$$

- Show that in polar coordinates the Cauchy–Riemann equations read $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$ and $\frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}$. Use them to show $\log z = \log r + i\theta$ (for $z = re^{i\theta}$, $-\pi < \theta < \pi$) is holomorphic on $r > 0$, $-\pi < \theta < \pi$.
- Show that $4 \frac{\partial}{\partial z} \frac{\partial}{\partial \bar{z}} = 4 \frac{\partial}{\partial \bar{z}} \frac{\partial}{\partial z} = \Delta$, where $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is the Laplacian.
- Use the previous exercise to prove that if f is holomorphic on the open set Ω , then $\operatorname{Re} f$ and $\operatorname{Im} f$ are harmonic (their Laplacian is zero).
- Consider $f(x + iy) = \sqrt{|x| |y|}$ for $x, y \in \mathbb{R}$. Show that f satisfies the Cauchy–Riemann equations at the origin yet is not holomorphic at 0.
- Suppose f is holomorphic on an open set Ω . Prove that in each of the cases (a) $\operatorname{Re}(f)$ constant; (b) $\operatorname{Im}(f)$ constant; (c) $|f|$ constant, the function f is constant.
- (Summation by parts.) Let $\{a_n\}_{n=1}^N, \{b_n\}_{n=1}^N$ be finite sequences in $\mathbb{C}$, and $B_k = \sum_{n=1}^k b_n$ with $B_0 = 0$. Prove

$$\sum_{n=M}^N a_n b_n = a_N B_N - a_M B_{M-1} - \sum_{n=M}^{N-1} (a_{n+1} - a_n) B_n.$$

- (Abel's theorem.) Suppose $\sum_{n=1}^\infty a_n$ converges. Prove $\lim_{r \rightarrow 1^-} \sum_{n=1}^\infty r^n a_n = \sum_{n=1}^\infty a_n$. [Hint: sum by parts.] Thus a convergent series is Abel summable to the same limit.
- Determine the radius of convergence of $\sum_{n \geq 1} a_n z^n$ when: (a) $a_n = (\log n)^2$; (b) $a_n = n!$; (c) $a_n = \frac{n^2}{4^n + 3n}$; (d) $a_n = \frac{(n!)^3}{(3n)!}$ [use Stirling $n! \sim c n^{n+1/2} e^{-n}$]; (e) the hypergeometric series $F(\alpha, \beta, \gamma; z) = 1 + \sum_{n \geq 1} \frac{\alpha(\alpha+1) \cdots (\alpha+n-1) \beta(\beta+1) \cdots (\beta+n-1)}{n! \gamma(\gamma+1) \cdots (\gamma+n-1)} z^n$, where $\alpha, \beta \in \mathbb{C}$ and $\gamma \neq 0, -1, -2, \dots$; (f) the Bessel function of order r (a positive integer), $J_r(z) = \left(\frac{z}{2}\right)^r \sum_{n \geq 0} \frac{(-1)^n}{n! (n+r)!} \left(\frac{z}{2}\right)^{2n}$.
- Show that if $\{a_n\}_{n \geq 0}$ are nonzero with $\lim_n |a_{n+1}| / |a_n| = L$, then $\lim_n |a_n|^{1/n} = L$. (So the ratio test computes the radius of convergence when applicable.)
- Let f be a power series centered at the origin. Prove that f has a power-series expansion around any point in its disc of convergence. [Hint: write $z = z_0 + (z - z_0)$ and expand z^n binomially.]
- Prove: (a) $\sum_n n z^n$ does not converge at any point of the unit circle; (b) $\sum_n z^n / n^2$ converges at every point of the unit circle; (c) $\sum_n z^n / n$ converges at every point of the unit circle except $z = 1$. [Hint for (c): sum by parts.]

13. Expand $(1-z)^{-m}$ in powers of z (m a fixed positive integer). Show that if $(1-z)^{-m} = \sum_{n \geq 0} a_n z^n$, then $a_n \sim \frac{n^{m-1}}{(m-1)!}$ as $n \rightarrow \infty$.

14. Show that for $|z| < 1$,

$$\sum_{n \geq 0} \frac{z^{2^n}}{1 - z^{2^{n+1}}} = \frac{z}{1 - z}, \quad \sum_{k \geq 0} \frac{2^k z^{2^k}}{1 + z^{2^k}} = \frac{z}{1 - z}.$$

Justify any rearrangement. [Hint: use the dyadic expansion and $2^{k+1} - 1 = 1 + 2 + \cdots + 2^k$.]

15. Let $\mathbb{N} = \{1, 2, 3, \dots\}$. A subset $S \subseteq \mathbb{N}$ is in *arithmetic progression* if $S = \{a, a + d, a + 2d, \dots\}$ with $a, d \in \mathbb{N}$ (d the step). Show $\mathbb{N}$ cannot be partitioned into finitely many arithmetic progressions with *distinct* steps (excepting the trivial $a = d = 1$). [Hint: write $\sum_{n \in \mathbb{N}} z^n$ as a sum of terms $z^a/(1 - z^d)$.]
16. Consider f on $\mathbb{R}$ given by $f(x) = 0$ for $x \leq 0$ and $f(x) = e^{-1/x^2}$ for $x > 0$. Prove f is infinitely differentiable on $\mathbb{R}$ with $f^{(n)}(0) = 0$ for all $n \geq 1$. Conclude f has no convergent power-series expansion $\sum a_n x^n$ near the origin.

Additional exercises.

- Using the limit definition, show $f(z) = z^3$ is entire and $f'(z) = 3z^2$. Then redo it via the Cauchy–Riemann equations after computing u, v .
- For each function decide where it is holomorphic and compute the derivative there: (a) $\frac{1}{z^2 + 1}$; (b) $\frac{z}{z - 2}$; (c) e^{z^2} ; (d) $\sin(3z)$.
- Verify the Cauchy–Riemann equations for $f(z) = e^z = e^x(\cos y + i \sin y)$ and conclude $(e^z)' = e^z$, matching the power-series computation.
- Show that $\cos^2 z + \sin^2 z = 1$ for all $z \in \mathbb{C}$ using the Euler formulas $\cos z = (e^{iz} + e^{-iz})/2$, $\sin z = (e^{iz} - e^{-iz})/(2i)$. Is $|\cos z| \leq 1$ for all complex z ? Justify with a value of z .
- Find the radius of convergence of (a) $\sum_{n \geq 1} \frac{(2n)!}{(n!)^2} z^n$; (b) $\sum_{n \geq 0} 3^n z^{2^n}$; (c) $\sum_{n \geq 1} \frac{z^n}{n^n}$.
- Starting from $\frac{1}{1 - z} = \sum_{n \geq 0} z^n$ ($|z| < 1$), derive closed forms for $\sum_{n \geq 1} n^2 z^n$ and $\sum_{n \geq 0} \frac{z^n}{n + 1}$, naming the disc on which each is valid.
- (Non-example.) Show that $f(z) = \operatorname{Re} z = x$ is *not* holomorphic anywhere, both by checking the Cauchy–Riemann equations and by computing the difference quotient along the real and imaginary directions.
- Prove that if f is holomorphic on a region Ω and f is real-valued there, then f is constant. [Hint: $v \equiv 0$ in the Cauchy–Riemann equations.]
- (Diagram problem.) Sketch the image of the unit square $[0, 1] \times [0, 1]$ under $w = z^2$, drawing several grid lines and their images, and indicate where the map fails to be conformal. (Recall $w = z^2$ doubles angles at the origin.)
- Let $f(z) = \sum_{n \geq 0} a_n z^n$ have radius of convergence $R \in (0, \infty)$. Prove that $g(z) = \sum_{n \geq 0} \frac{a_n}{n + 1} z^{n+1}$ is a primitive of f on $|z| < R$ and has the same radius of convergence.

14.3 Integration Along Curves

Curves and parametrizations

We distinguish the geometric *curve* (an oriented one-dimensional set in the plane) from a *parametrization*, which is not unique.

Definition 14.3.1. A *parametrized curve* is a map $z : [a, b] \rightarrow \mathbb{C}$. It is *smooth* if $z'(t)$ exists, is continuous, and $z'(t) \neq 0$ on $[a, b]$ (one-sided derivatives at the endpoints). It is *piecewise-smooth* if z is continuous and there are $a = a_0 < a_1 < \cdots < a_n = b$ with z smooth on each $[a_k, a_{k+1}]$. Two parametrizations $z : [a, b] \rightarrow \mathbb{C}$ and $\tilde{z} : [c, d] \rightarrow \mathbb{C}$ are *equivalent* if there is a C^1 bijection $s \mapsto t(s)$ from $[c, d]$ onto $[a, b]$ with $t'(s) > 0$ and $\tilde{z}(s) = z(t(s))$.

The condition $t' > 0$ preserves orientation. The equivalence class of z is the *smooth curve* γ — the image with its orientation. Reversing orientation gives γ^- , parametrized by $z^-(t) = z(b + a - t)$. The points $z(a), z(b)$ are the *endpoints*; γ is *closed* if $z(a) = z(b)$, and *simple* if it does not self-intersect (for a closed curve, $z(t) = z(s)$ only when $s = t$ or $\{s, t\} = \{a, b\}$). For brevity any piecewise-smooth curve is called simply a *curve*.

Example 14.3.2 (The circle). The circle $C_r(z_0) = \{|z - z_0| = r\}$ has *positive* (counterclockwise) orientation via $z(t) = z_0 + re^{it}$, $t \in [0, 2\pi]$, and negative (clockwise) orientation via $z(t) = z_0 + re^{-it}$. We write C for a generic positively oriented circle.

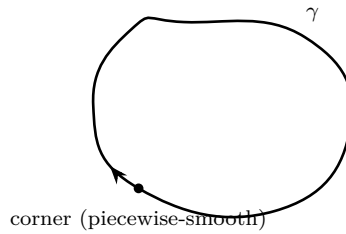


Figure 14.16. A closed piecewise-smooth curve, positively oriented; the corner is allowed. (Schematic.)

The integral along a curve

Definition 14.3.3. For a smooth curve γ parametrized by $z : [a, b] \rightarrow \mathbb{C}$ and f continuous on γ ,

$$\int_{\gamma} f(z) dz = \int_a^b f(z(t)) z'(t) dt.$$

For piecewise-smooth γ , sum over the smooth pieces. The *length* is $\text{length}(\gamma) = \int_a^b |z'(t)| dt$.

Proposition 14.3.4 (Independence of parametrization). *The integral $\int_{\gamma} f dz$ does not depend on the chosen (orientation-preserving) parametrization.*

Proof. If $\tilde{z}(s) = z(t(s))$ with $t' > 0$, the chain rule gives $\tilde{z}'(s) = z'(t(s))t'(s)$; by the change-of-variables formula

$$\int_c^d f(\tilde{z}(s)) \tilde{z}'(s) ds = \int_c^d f(z(t(s))) z'(t(s)) t'(s) ds = \int_a^b f(z(t)) z'(t) dt. \quad \square$$

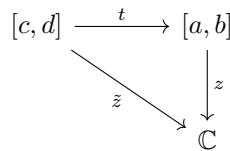


Figure 14.17. A reparametrization: the orientation-preserving change of variable $t : [c, d] \rightarrow [a, b]$ ($t' > 0$) gives $\tilde{z} = z \circ t$, so both parametrizations trace the same oriented curve and the integral $\int_{\gamma} f dz$ is unchanged.

Proposition 14.3.5 (Properties of the integral). *For continuous f, g and $\alpha, \beta \in \mathbb{C}$:*

$$\begin{aligned} (i) \quad & \int_{\gamma} (\alpha f + \beta g) dz = \alpha \int_{\gamma} f dz + \beta \int_{\gamma} g dz; \\ (ii) \quad & \int_{\gamma^-} f dz = - \int_{\gamma} f dz; \quad (iii) \quad \left| \int_{\gamma} f dz \right| \leq \sup_{z \in \gamma} |f(z)| \cdot \text{length}(\gamma). \end{aligned}$$

Proof. (i) is linearity of the Riemann integral. (ii) Parametrize γ^- by $z(b+a-t)$; the substitution $\tau = b+a-t$ flips the limits and the sign of z' , producing the minus sign. (iii) The *ML-inequality*:

$$\left| \int_{\gamma} f \, dz \right| = \left| \int_a^b f(z(t)) z'(t) \, dt \right| \leq \int_a^b |f(z(t))| |z'(t)| \, dt \leq \sup_{z \in \gamma} |f(z)| \int_a^b |z'(t)| \, dt,$$

the last factor being $\text{length}(\gamma)$. □

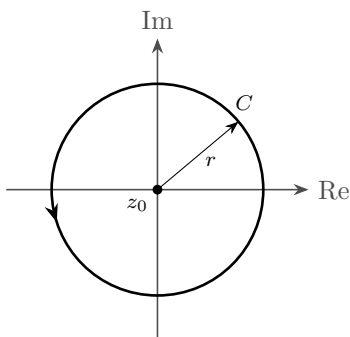


Figure 14.18. Positively oriented circle $C_r(z_0)$, $z(t) = z_0 + re^{it}$.

Worked Example 14.3.6 (The fundamental computation $\oint z^n \, dz$). For a circle C of radius r centered at 0, positively oriented, compute $\oint_C z^n \, dz$ for $n \in \mathbb{Z}$.

Recognition. Parametrize $z = re^{it}$, $dz = ire^{it} dt$, and integrate over $[0, 2\pi]$; watch the case $n = -1$.

Solution. With $z(t) = re^{it}$,

$$\oint_C z^n \, dz = \int_0^{2\pi} r^n e^{int} \cdot ire^{it} \, dt = ir^{n+1} \int_0^{2\pi} e^{i(n+1)t} \, dt.$$

If $n \neq -1$, $\int_0^{2\pi} e^{i(n+1)t} \, dt = 0$ (a full period), so the integral is 0. If $n = -1$, the integrand is $i \int_0^{2\pi} dt = 2\pi i$. Hence

$$\oint_C z^n \, dz = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1. \end{cases}$$

Check. For $n \neq -1$, z^n has primitive $z^{n+1}/(n+1)$ on a neighborhood of C (avoiding 0 when $n < -1$), so the closed-curve integral must vanish; the lone survivor $n = -1$ is exactly the function with no global primitive. ■

Primitives and the fundamental theorem

Definition 14.3.7. A *primitive* for f on an open set Ω is a holomorphic F with $F' = f$ on Ω .

Theorem 14.3.8 (Fundamental theorem of calculus for contours). *If f is continuous with a primitive F on Ω , and $\gamma \subseteq \Omega$ is a curve from w_1 to w_2 , then*

$$\int_{\gamma} f(z) \, dz = F(w_2) - F(w_1).$$

Proof. For smooth γ with parametrization $z(t)$, by the chain rule $\frac{d}{dt} F(z(t)) = F'(z(t)) z'(t) = f(z(t)) z'(t)$, so

$$\int_{\gamma} f \, dz = \int_a^b \frac{d}{dt} F(z(t)) \, dt = F(z(b)) - F(z(a)) = F(w_2) - F(w_1)$$

by the real fundamental theorem of calculus (applied to real and imaginary parts). For piecewise-smooth γ the sum over pieces telescopes. □

Corollary 14.3.9 (Closed curves and the failure of $1/z$). *If f is continuous and has a primitive on Ω , then $\int_{\gamma} f dz = 0$ for every closed curve $\gamma \subseteq \Omega$. In particular $f(z) = 1/z$ has no primitive on $\mathbb{C} \setminus \{0\}$, since over the unit circle*

$$\oint_C \frac{dz}{z} = \int_0^{2\pi} \frac{ie^{it}}{e^{it}} dt = 2\pi i \neq 0.$$

Proof. A closed curve has $w_1 = w_2$, so Theorem 14.3.8 gives $F(w_2) - F(w_1) = 0$. The nonzero value $2\pi i$ then certifies that $1/z$ cannot have a primitive on the punctured plane. $\square$

This “innocent calculation” — a single function and closed curve with nonzero integral — is the seed of Cauchy’s theory: it is precisely the obstruction that the residue at the origin will measure.

Corollary 14.3.10 (Constancy from vanishing derivative). *If f is holomorphic on a region Ω with $f' \equiv 0$, then f is constant.*

Proof. Fix $w_0 \in \Omega$. For any w , connectedness of the region gives a curve γ from w_0 to w in Ω . Since f is a primitive of $f' = 0$, Theorem 14.3.8 gives $\int_{\gamma} f' dz = f(w) - f(w_0)$; the left side is 0, so $f(w) = f(w_0)$ for all w . $\square$

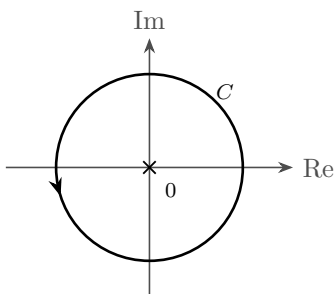


Figure 14.19. $\oint_C dz/z = 2\pi i$: the pole at 0 obstructs the existence of a global log.

Worked Example 14.3.11 (A partial-fractions contour integral). For the positively oriented circle $\gamma = C_r(0)$ with $|a| < r < |b|$, show $\oint_{\gamma} \frac{dz}{(z-a)(z-b)} = \frac{2\pi i}{a-b}$.

Recognition. Decompose into simple fractions $1/(z-a)$ and $1/(z-b)$, then use the basic computation: $\oint dz/(z-c) = 2\pi i$ if c is inside, 0 if outside.

Solution. Partial fractions: $\frac{1}{(z-a)(z-b)} = \frac{1}{a-b} \left(\frac{1}{z-a} - \frac{1}{z-b} \right)$. Since $|a| < r$, the point a lies inside γ and $\oint dz/(z-a) = 2\pi i$; since $|b| > r$, the point b lies outside and $1/(z-b)$ has a primitive near γ , so $\oint dz/(z-b) = 0$. Therefore

$$\oint_{\gamma} \frac{dz}{(z-a)(z-b)} = \frac{1}{a-b} (2\pi i - 0) = \frac{2\pi i}{a-b}.$$

Check. The ML-bound gives $|\oint| \leq \frac{2\pi r}{(r-|a|)(|b|-r)}$, a finite quantity, consistent with the clean answer; and the result is antisymmetric in a, b , as the integrand is. $\blacksquare$

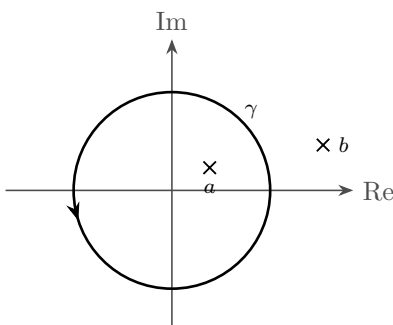


Figure 14.20. Only the enclosed pole a contributes; b lies outside γ . (Schematic.)

Worked Example 14.3.12 (A length/ML estimate (no exact evaluation needed)). Bound $\left| \oint_{C_R(0)} \frac{e^{iz}}{z^2 + 1} dz \right|$ for large R , where $C_R(0)$ is the circle of radius R .

Recognition. Use property (iii), the ML-inequality, with $L = \text{length} = 2\pi R$ and $M = \sup_{|z|=R} |f|$.

Solution. On $|z| = R$ (taking $R > 1$), $|z^2 + 1| \geq |z|^2 - 1 = R^2 - 1$. For the numerator, $|e^{iz}| = e^{-\text{Im } z}$ which is at most e^R on the whole circle (crude but valid). Thus $M \leq e^R/(R^2 - 1)$ and

$$\left| \oint_{C_R} \frac{e^{iz}}{z^2 + 1} dz \right| \leq 2\pi R \cdot \frac{e^R}{R^2 - 1}.$$

Check. The dimensionless ratio behaves like $2\pi e^R/R$ for large R ; the bound is finite for each R , which is all the ML-inequality promises. (A sharper directional bound on $|e^{iz}|$ over a semicircle is the basis of Jordan's lemma in the next chapter.) ■

Theorem 14.3.13 (Uniqueness of primitives). *If f is continuous on a region Ω , any two primitives of f differ by a constant.*

Proof. If $F'_1 = F'_2 = f$ then $(F_1 - F_2)' = 0$ on the region, so $F_1 - F_2$ is constant by Corollary 14.3.10. □

Caution. $\int_\gamma f dz$ depends in general on the *path*, not just its endpoints — that independence is exactly the special situation of Theorem 14.3.8, which requires a *global* primitive. The integral $\oint dz/z = 2\pi i \neq 0$ shows path-independence fails for $1/z$ on the punctured plane, because no global primitive (single-valued logarithm) exists there.

Further worked examples

Worked Example 14.3.14 (Two paths from 0 to $1 + i$: integrand matters). Integrate $f(z) = \bar{z}$ and $g(z) = z$ from 0 to $1 + i$ along (A) the straight segment and (B) the path $0 \rightarrow 1 \rightarrow 1 + i$. Compare.

Recognition. Test the FTC (Theorem 14.3.8): z has a primitive ($z^2/2$) so should be path-independent, while $\bar{z}$ is nowhere holomorphic and need not be.

Solution. For $g = z$: by the FTC, $\int_\gamma z dz = [z^2/2]_0^{1+i} = (1+i)^2/2 = (2i)/2 = i$ on *both* paths. For $f = \bar{z}$, path (A) is $z(t) = t(1+i)$, $\bar{z} = t(1-i)$, $dz = (1+i)dt$, so $\int_0^1 t(1-i)(1+i) dt = \int_0^1 2t dt = 1$. Path (B): on $0 \rightarrow 1$, $z = t$, $\bar{z} = t$, $dz = dt$, giving $\int_0^1 t dt = \frac{1}{2}$; on $1 \rightarrow 1+i$, $z = 1+it$, $\bar{z} = 1-it$, $dz = i dt$, giving $\int_0^1 (1-it)i dt = \int_0^1 (i+t) dt = \frac{1}{2} + i$. Sum $= 1 + i$. *Check.* $\int z dz = i$ agrees on both paths (primitive exists); $\int \bar{z} dz$ gives 1 vs. $1 + i$ — genuinely path-dependent, as it must be for a function with no primitive. ■

Worked Example 14.3.15 (A semicircular contour integral). Let γ_R be the boundary of the upper half-disc $\{|z| \leq R, \text{Im } z \geq 0\}$, positively oriented. Evaluate $\int_{\gamma_R} z^2 dz$ directly, and confirm it vanishes.

Recognition. z^2 is entire with primitive $z^3/3$; over any *closed* curve the FTC (Corollary 14.3.9) forces 0. We verify by direct parametrization.

Solution. Split γ_R into the diameter $D : z = t, t \in [-R, R]$ and the arc $A : z = Re^{i\theta}, \theta \in [0, \pi]$. On D : $\int_{-R}^R t^2 dt = \frac{2R^3}{3}$. On A : $\int_0^\pi R^2 e^{2i\theta} iRe^{i\theta} d\theta = iR^3 \int_0^\pi e^{3i\theta} d\theta = iR^3 \left[\frac{e^{3i\theta}}{3i} \right]_0^\pi = \frac{R^3}{3} (e^{3i\pi} - 1) = \frac{R^3}{3} (-1 - 1) = -\frac{2R^3}{3}$. Sum $= 0$. *Check.* $z^3/3$ is a global primitive, so the closed-loop value must be 0; the diameter and arc contributions cancel exactly. ■

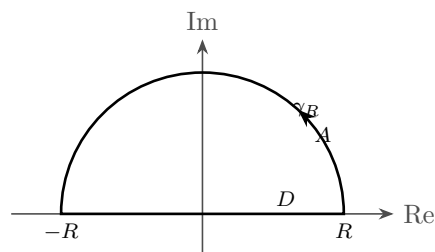


Figure 14.21. The upper half-disc contour $\gamma_R = \text{diameter } D \text{ plus semicircular arc } A$.

Worked Example 14.3.16 ($\oint dz/(z - z_0)$ off-center). For the positively oriented circle $|z| = 2$ evaluate $\oint \frac{dz}{z - z_0}$ for (a) $z_0 = 1$ and (b) $z_0 = 3$.

Recognition. The basic computation $\oint dz/(z - z_0) = 2\pi i$ when z_0 is enclosed, 0 when outside; verify by reducing to the centered case.

Solution. (a) $z_0 = 1$ lies inside $|z| = 2$. Parametrize is cleaner via translation invariance, but directly: writing $w = z - 1$, the image of $|z| = 2$ winds once around $w = 0$, so $\oint dw/w = 2\pi i$. (b) $z_0 = 3$ lies outside; $1/(z - 3)$ is holomorphic on and inside the circle with primitive $\log(z - 3)$ available on a neighborhood, so the integral is 0. *Check.* The winding number of $|z| = 2$ about an interior point is 1 (value $2\pi i$) and about an exterior point is 0 – exactly the two answers. ■

Worked Example 14.3.17 (A rectangular contour for an entire integrand). Let R be the boundary of the rectangle with vertices $\pm a, \pm a + ib$ ($a, b > 0$), positively oriented. Compute $\oint_R e^z dz$.

Recognition. e^z is entire with primitive e^z ; a closed contour gives 0 (Corollary 14.3.9). We confirm side by side to exhibit the cancellation mechanism.

Solution. By the FTC the closed integral is e^z evaluated around the loop, returning to the start, so $\oint_R e^z dz = 0$. Concretely the four sides contribute $\int_{-a}^a e^x dx$ (bottom), $\int_0^b e^{a+iy} i dy$ (right), $\int_a^{-a} e^{x+ib} dx$ (top), $\int_b^0 e^{-a+iy} i dy$ (left); summing the primitive values e^z at consecutive corners telescopes to 0. *Check.* The bottom and top are $e^a - e^{-a}$ and $-(e^a - e^{-a})e^{ib}$; combined with the two vertical pieces $e^a(e^{ib} - 1)$ and $-e^{-a}(e^{ib} - 1)$ everything cancels, total 0. ■

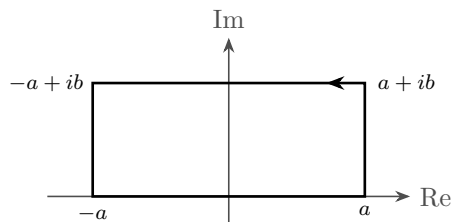


Figure 14.22. A rectangular contour; for an entire integrand the four sides cancel.

Worked Example 14.3.18 (A sector contour and a primitive). Let σ be the boundary of the sector $\{0 \leq |z| \leq R, 0 \leq \arg z \leq \alpha\}$, positively oriented. Show $\oint_\sigma z dz = 0$ and find $\int_{\text{arc}} z dz$ alone.

Recognition. z has primitive $z^2/2$, so the closed sector integral vanishes; the arc piece is then minus the two radial pieces.

Solution. Closed loop: $\oint_\sigma z dz = 0$ by Corollary 14.3.9. The two radii are $O \rightarrow R$ along $\arg = 0$ and $Re^{i\alpha} \rightarrow O$ along $\arg = \alpha$. On the outgoing radius $z = t$: $\int_0^R t dt = R^2/2$. On the returning radius $z = te^{i\alpha}$, $t : R \rightarrow 0$: $\int_R^0 te^{i\alpha} e^{i\alpha} dt = -\frac{R^2}{2} e^{2i\alpha}$. Hence the arc integral is $0 - (\frac{R^2}{2} - \frac{R^2}{2} e^{2i\alpha}) = \frac{R^2}{2} (e^{2i\alpha} - 1)$. *Check.* Directly, arc $z = Re^{i\theta}$, $\theta \in [0, \alpha]$: $\int_0^\alpha Re^{i\theta} iRe^{i\theta} d\theta = iR^2 \int_0^\alpha e^{2i\theta} d\theta = \frac{R^2}{2} (e^{2i\alpha} - 1)$. Matches. ■

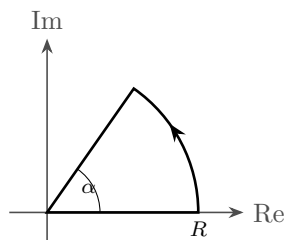


Figure 14.23. A sector contour: two radii and the arc of opening angle α .

Worked Example 14.3.19 (A non-example: a closed integral that does not vanish). Why does $\oint_{|z|=1} \bar{z} \, dz = 2\pi i \neq 0$, even though $\bar{z}$ is continuous everywhere?

Recognition. Corollary 14.3.9 requires a *primitive*; continuity alone is not enough. $\bar{z}$ is nowhere holomorphic, so no primitive exists.

Solution. Parametrize $z = e^{it}$, $\bar{z} = e^{-it}$, $dz = ie^{it} dt$:

$$\oint_{|z|=1} \bar{z} \, dz = \int_0^{2\pi} e^{-it} i e^{it} dt = i \int_0^{2\pi} dt = 2\pi i.$$

Check. On $|z| = 1$ we have $\bar{z} = 1/z$, so this is the same as $\oint dz/z = 2\pi i$ – the canonical nonzero closed integral, certifying $\bar{z}$ (like $1/z$) has no global primitive. ■

Exercises (Stein, Ch.1).

1. Let γ be a smooth curve in $\mathbb{C}$ parametrized by $z : [a, b] \rightarrow \mathbb{C}$, and let γ^- be the same image with the reverse orientation. Prove that for any continuous f on γ ,

$$\int_{\gamma^-} f(z) \, dz = - \int_{\gamma} f(z) \, dz.$$

2. (Insight into Cauchy's theorem.) Let γ be a circle centered at the origin with positive orientation. (a) Evaluate $\int_{\gamma} z^n \, dz$ for all integers n . (b) Same question when γ is any circle *not* containing the origin. (c) Show that if $|a| < r < |b|$, then

$$\int_{\gamma} \frac{dz}{(z-a)(z-b)} = \frac{2\pi i}{a-b},$$

where γ is the positively oriented circle centered at 0 of radius r .

3. Suppose f is continuous on a region Ω . Prove that any two primitives of f (if they exist) differ by a constant.

Additional exercises.

1. Let γ be the line segment from 0 to $1+i$, parametrized by $z(t) = t(1+i)$, $t \in [0, 1]$. Compute $\int_{\gamma} z \, dz$ and $\int_{\gamma} \bar{z} \, dz$, and explain why the two answers differ in character (one is path-independent, the other is not).
2. Evaluate $\int_{\gamma} z^2 \, dz$ where γ goes from 0 to $2i$ along (a) the imaginary axis, then (b) the two-segment path $0 \rightarrow 2 \rightarrow 2i$. Confirm both give the same value and explain via a primitive.
3. For the positively oriented unit circle C , compute $\oint_C \frac{dz}{z-a}$ for $|a| < 1$ and for $|a| > 1$, and reconcile both cases with Corollary 14.3.9.
4. Using partial fractions, evaluate $\oint_{C_2(0)} \frac{dz}{z^2-1}$ where $C_2(0)$ is the positively oriented circle of radius 2.
5. (ML-estimate.) Show $\left| \oint_{C_R(0)} \frac{dz}{z^4+1} \right| \leq \frac{2\pi R}{R^4-1}$ for $R > 1$, and deduce the integral tends to 0 as $R \rightarrow \infty$.
6. Compute $\oint_C \bar{z} \, dz$ over the positively oriented unit circle C directly from the definition, and observe that the answer is nonzero even though $\bar{z}$ is continuous — consistent with $\bar{z}$ having no primitive (it is nowhere holomorphic).
7. (Non-example.) Exhibit a closed curve γ and a continuous function f with $\oint_{\gamma} f \, dz \neq 0$, and a different continuous f with $\oint_{\gamma} f \, dz = 0$ on the *same* γ , showing that vanishing of the closed integral is a property of f (having a primitive), not of γ .
8. (Diagram problem.) Draw a positively oriented keyhole-type contour: the unit circle $C_1(0)$ together with a small clockwise circle $C_{\varepsilon}(0)$ around the origin joined by two segments. Mark the orientation arrows and state which region the contour bounds.

9. Let γ be parametrized by $z(t) = e^{it}$, $t \in [0, \pi]$ (upper unit semicircle). Compute $\text{length}(\gamma)$ and $\int_{\gamma} \frac{dz}{z}$, and check the latter against $F(z(\pi)) - F(z(0))$ for a suitable local primitive $F = \log$.
10. Prove that if f is continuous on a region Ω and $\int_{\gamma} f \, dz = 0$ for every closed curve $\gamma \subseteq \Omega$, then f has a primitive on Ω . [Hint: fix z_0 and define $F(z) = \int_{z_0}^z f$, checking well-definedness from the hypothesis, then show $F' = f$.]

Chapter 15

Cauchy's Theorem and Its Applications

15.1 Goursat's Theorem

The first chapter built the local theory of holomorphic functions; this chapter harvests its global consequences. The hinge is a single equality: *the integral of a holomorphic function around a closed contour is zero*. Everything else — the integral formulas, the power-series representation, Liouville's theorem, the fundamental theorem of algebra — falls out of it. The entry point, deliberately stripped of all hypotheses on the derivative, is Goursat's theorem.

Recall from Chapter 1 that if f has a *primitive* F (an F with $F' = f$) in an open set Ω , then $\oint_{\gamma} f(z) dz = 0$ for every closed curve γ in Ω , by the fundamental theorem of calculus along γ . The strategy of the whole chapter is to run this implication *backwards*: if we can verify $\oint_{\gamma} f = 0$ for enough curves, a primitive must exist, and from a primitive flows everything.

Theorem 15.1.1 (Goursat). *If $\Omega \subseteq \mathbb{C}$ is open and $T \subset \Omega$ is a triangle whose closed interior also lies in Ω , then*

$$\int_T f(z) dz = 0$$

whenever f is holomorphic in Ω .

The remarkable feature is the hypothesis: f is merely *complex differentiable* on Ω , with no assumed continuity of f' . (Add that assumption and the theorem is an easy corollary of Green's theorem plus the Cauchy–Riemann equations — see the scraped Exercise 5.) Goursat's argument squeezes the conclusion out of differentiability alone.

Proof. Call the original triangle $T^{(0)}$, fixed with positive (counterclockwise) orientation, and let $d^{(0)}, p^{(0)}$ be its diameter and perimeter. Bisect each side and join the midpoints. This splits $T^{(0)}$ into four similar sub-triangles $T_1^{(1)}, \dots, T_4^{(1)}$, each with the inherited orientation. Integrals over the three new interior segments are each traversed *twice in opposite directions*, so they cancel, giving

$$\int_{T^{(0)}} f dz = \sum_{j=1}^4 \int_{T_j^{(1)}} f dz.$$

By the triangle inequality at least one term has modulus $\geq \frac{1}{4} |\int_{T^{(0)}} f|$; call that sub-triangle $T^{(1)}$, so $|\int_{T^{(0)}} f| \leq 4 |\int_{T^{(1)}} f|$. Its diameter and perimeter halve: $d^{(1)} = \frac{1}{2} d^{(0)}$, $p^{(1)} = \frac{1}{2} p^{(0)}$.

Repeat indefinitely to get nested closed solid triangles $T^{(0)} \supseteq T^{(1)} \supseteq \dots$ with

$$\left| \int_{T^{(0)}} f dz \right| \leq 4^n \left| \int_{T^{(n)}} f dz \right|, \quad d^{(n)} = 2^{-n} d^{(0)}, \quad p^{(n)} = 2^{-n} p^{(0)},$$

and diameters $\rightarrow 0$. By the nested-compact-sets principle (Chapter 1) there is a unique point $z_0 \in \bigcap_n T^{(n)}$. Since f is holomorphic at z_0 we may write

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \psi(z)(z - z_0), \quad \psi(z) \rightarrow 0 \text{ as } z \rightarrow z_0.$$

The constant $f(z_0)$ and the linear term $f'(z_0)(z - z_0)$ have primitives, so their integrals over the *closed* curve $T^{(n)}$ vanish (Chapter 1, Corollary 3.3). Hence

$$\int_{T^{(n)}} f \, dz = \int_{T^{(n)}} \psi(z)(z - z_0) \, dz.$$

On $T^{(n)}$ we have $|z - z_0| \leq d^{(n)}$, and the length of $T^{(n)}$ is $p^{(n)}$, so the *ML*-estimate gives

$$\left| \int_{T^{(n)}} f \, dz \right| \leq \varepsilon_n d^{(n)} p^{(n)} = \varepsilon_n 4^{-n} d^{(0)} p^{(0)}, \quad \varepsilon_n := \sup_{z \in T^{(n)}} |\psi(z)| \rightarrow 0.$$

Therefore $\left| \int_{T^{(0)}} f \, dz \right| \leq 4^n \cdot \varepsilon_n 4^{-n} d^{(0)} p^{(0)} = \varepsilon_n d^{(0)} p^{(0)} \rightarrow 0$. The left side is a fixed number, so it is 0. $\square$

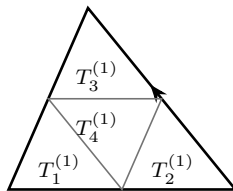


Figure 15.1. Bisection of $T^{(0)}$ into four similar sub-triangles $T_1^{(1)}, \dots, T_4^{(1)}$ by joining the side-midpoints. The three interior segments are each traversed twice in opposite directions, so they cancel in $\int_{T^{(0)}} f = \sum_j \int_{T_j^{(1)}} f$. (Schematic.)

Corollary 15.1.2. *If f is holomorphic in an open set Ω containing a rectangle R and its interior, then $\int_R f(z) \, dz = 0$.*

Proof. A diagonal splits R into two triangles T_1, T_2 , oriented consistently. The diagonal is traversed once in each direction and cancels, so $\int_R f = \int_{T_1} f + \int_{T_2} f = 0 + 0$ by Goursat. $\square$

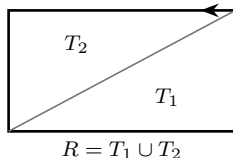


Figure 15.2. A rectangle as the union of two triangles; the shared diagonal cancels. (Schematic.)

Method (Proving a contour integral vanishes by reduction to Goursat). *Recognition trigger.* You must show $\int_\gamma f = 0$ where f is holomorphic on (a neighborhood of) γ and its interior, γ a triangle, rectangle, or a polygon built from them.

1. If γ is a triangle: cite Goursat directly.
2. If γ is a rectangle or convex polygon: triangulate; interior edges cancel; sum the zero contributions (Corollary 15.1.2).
3. If f has a known primitive on Ω : skip Goursat entirely and use $\int_\gamma f = F(\text{end}) - F(\text{start}) = 0$ for closed γ .
4. Beware: Goursat requires the *filled* triangle to lie in Ω . A hole (a singularity inside) voids the conclusion — that is exactly where residues enter (Chapter 16).

Worked Example 15.1.3 (Goursat in action: a polynomial around a triangle). Let $f(z) = z^2 - 3z + 1$ and let T be the triangle with vertices $0, 2, 1 + i$, positively oriented. Compute $\int_T f dz$.

Recognition. f is entire (a polynomial), so Goursat applies on all of $\mathbb{C}$; the integral around any closed triangle is 0. We can also confirm via a primitive.

Solution. By Goursat, $\int_T f dz = 0$. *Check via primitive:* $F(z) = \frac{1}{3}z^3 - \frac{3}{2}z^2 + z$ has $F' = f$, and a closed contour returns to its start, so $\int_T f = F(\text{start}) - F(\text{start}) = 0$. ■ Both routes agree.

Worked Example 15.1.4 (A NON-example: the hypothesis "interior in Ω " is essential). Let $f(z) = 1/z$, holomorphic on $\Omega = \mathbb{C} \setminus \{0\}$. Let T be the triangle with vertices $1, -\frac{1}{2} + i, -\frac{1}{2} - i$, which encloses the origin. Does Goursat give $\int_T f = 0$?

Recognition. Trigger: a tempting application of Goursat. Check the hypothesis: is the *closed* triangle inside Ω ?

Solution. No. The filled triangle contains $0 \notin \Omega$, so Goursat does *not* apply, and indeed $\int_T \frac{1}{z} dz = 2\pi i \neq 0$ (the winding number about 0 is 1; we will prove this kind of statement via the keyhole in §15.4). *Check.* Deform T to the unit circle $z = e^{i\theta}$ without crossing 0; then $\int_0^{2\pi} \frac{ie^{i\theta}}{e^{i\theta}} d\theta = 2\pi i$. The missing point is the whole story.

Worked Example 15.1.5 (Triangulating a quadrilateral). Let f be entire and let Q be the (filled) quadrilateral $0 \rightarrow 3 \rightarrow 3 + 2i \rightarrow i \rightarrow 0$. Show $\int_{\partial Q} f dz = 0$ by triangulation, and verify with $f(z) = e^z$.

Recognition. Trigger: a closed polygon, f holomorphic inside. Split into triangles and apply Goursat to each.

Solution. The diagonal $0 \rightarrow 3 + 2i$ cuts Q into triangles $T_1 = (0, 3, 3 + 2i)$ and $T_2 = (0, 3 + 2i, i)$. Each is positively oriented and $\int_{T_1} f = \int_{T_2} f = 0$ by Goursat; the diagonal is traversed once each way and cancels, so $\int_{\partial Q} f = 0$. *Check* with $f = e^z$ (which has primitive e^z): $\int_{\partial Q} e^z dz = e^0 - e^0 = 0$ around the closed loop. ■

Further worked examples

Worked Example 15.1.6 (The bisection constant is exactly $\frac{1}{4}$). In the Goursat proof one selects a sub-triangle $T^{(1)}$ with $|\int_{T^{(0)}} f| \leq 4 |\int_{T^{(1)}} f|$. Show the constant 4 is sharp: it cannot be replaced by any smaller fixed number that works for all f and all triangles.

Recognition. Trigger: a sharpness claim about a pigeonhole step. Recall the bisection produces exactly four sub-triangles and that $\int_{T^{(0)}} f = \sum_{j=1}^4 \int_{T_j^{(1)}} f$.

Solution. If every sub-integral had modulus $< \frac{1}{4} |\int_{T^{(0)}} f|$, the triangle inequality would give $|\int_{T^{(0)}} f| \leq \sum_{j=1}^4 |\int_{T_j^{(1)}} f| < 4 \cdot \frac{1}{4} |\int_{T^{(0)}} f| = |\int_{T^{(0)}} f|$, a strict contradiction. Hence some sub-triangle achieves at least $\frac{1}{4}$ of the modulus, and the factor $4 = 4^1$ is forced at each of the n stages, producing the crucial 4^n in $|\int_{T^{(0)}} f| \leq 4^n |\int_{T^{(n)}} f|$. Any smaller fixed factor $c < 4$ would fail when all four sub-integrals are equal (e.g. f with the symmetry of an equilateral $T^{(0)}$). *Check.* Four equal terms each of modulus $\frac{1}{4}M$ sum to M , so $c = 4$ is met with equality — the bound is tight, not wasteful.

Worked Example 15.1.7 (A NON-example: $\bar{z}$ defeats Goursat's conclusion). Let $f(z) = \bar{z}$ and let T be the triangle $0 \rightarrow 1 \rightarrow i \rightarrow 0$. Compute $\int_T f dz$ and explain why this does not contradict Goursat.

Recognition. Trigger: a tempting "integral around a triangle is 0" expectation. First test whether f is holomorphic — $\bar{z}$ satisfies the Cauchy-Riemann equations nowhere.

Solution. Parametrize the three edges. Edge $0 \rightarrow 1$: $z = t, \bar{z} = t, dz = dt$, contributes $\int_0^1 t dt = \frac{1}{2}$. Edge $1 \rightarrow i$: $z = 1 + t(i - 1), \bar{z} = 1 + t(-i - 1), dz = (i - 1)dt$; the integrand is $(1 - t - ti)(i - 1)$ and $\int_0^1 (1 - t - ti)(i - 1)dt = (i - 1)(\frac{1}{2} - \frac{i}{2}) = (i - 1)\frac{1-i}{2} = \frac{(i-1)(1-i)}{2} = \frac{-(1-i)^2}{2} = \frac{-(-2i)}{2} = i$. Edge $i \rightarrow 0$: $z = i(1 - t), \bar{z} = -i(1 - t), dz = -i dt$, contributes $\int_0^1 -i(1 - t)(-i)dt = -\int_0^1 (1 - t)dt = -\frac{1}{2}$. Sum: $\frac{1}{2} + i - \frac{1}{2} = i \neq 0$. No contradiction: $\bar{z}$ is *not* holomorphic, so Goursat's hypothesis fails. *Check.* By Green's theorem $\int_T \bar{z} dz = 2i \cdot \text{Area}(T)$, and the triangle $0, 1, i$ has area $\frac{1}{2}$, giving $2i \cdot \frac{1}{2} = i$. ■

Exercises (Stein, Ch.2).

1. (Ex. 5) Suppose f is continuously complex differentiable on Ω , and $T \subset \Omega$ is a triangle whose interior is also contained in Ω . Apply Green's theorem to show that $\int_T f(z) dz = 0$. (This proves Goursat's theorem under the extra assumption that f' is continuous. Hint: Green's theorem $\int_T F dx + G dy = \iint_{\text{int } T} (\partial_x G - \partial_y F) dx dy$; choose F, G and use the Cauchy–Riemann equations.)

Additional exercises.

1. Compute $\int_T (3z^2 - 2z + 5) dz$ for the triangle T with vertices $0, 1, i$ (positively oriented), and confirm the answer two ways (Goursat; a primitive).
2. Let T be any triangle in $\mathbb{C}$ and $f(z) = \bar{z}$. Show $\int_T \bar{z} dz \neq 0$ in general by computing it for the triangle $0 \rightarrow 1 \rightarrow i \rightarrow 0$. Explain why this does *not* contradict Goursat.
3. (Non-example) Let $f(z) = 1/(z - \frac{1}{2})$ and let T be the triangle with vertices $-1 - i, 2, -1 + i$. Determine whether $\int_T f dz = 0$, and identify precisely which hypothesis of Goursat's theorem fails.
4. Prove that if f is holomorphic on $\mathbb{C}$ and T_1, T_2 are two triangles sharing exactly a common edge e (with opposite orientations on e), then $\int_{T_1} f + \int_{T_2} f$ equals the integral over the boundary of the quadrilateral $T_1 \cup T_2$. Deduce it is 0.
5. Show that the conclusion of Goursat fails for the *merely continuous* function $f(z) = \bar{z}$ but the proof's scaling estimate $|\int_{T^{(0)}} f| \leq 4^n |\int_{T^{(n)}} f|$ still holds. Which single step of the proof uses holomorphy?
6. (Diagram) Draw a triangle $T^{(0)}$ and its four-fold bisection, labeling the orientation of each sub-triangle's edges with arrows, and mark which interior edges cancel in the sum $\int_{T^{(0)}} f = \sum_j \int_{T_j^{(1)}} f$.
7. Suppose f is holomorphic on $\mathbb{C}$ except at finitely many points $p_1, \dots, p_m$, none inside a given triangle T . Prove $\int_T f = 0$, and explain how this reduces to Goursat.

15.2 Local Primitives and Cauchy's Theorem in a Disc

Goursat tells us triangle and rectangle integrals vanish; we now convert that into the existence of a primitive, and from a primitive into Cauchy's theorem on a disc. The construction is the complex analogue of building an antiderivative $F(x) = \int_a^x f$.

Theorem 15.2.1 (Existence of primitives in a disc). *A holomorphic function in an open disc D has a primitive in D .*

Proof. After translating, assume D is centered at 0. For $z \in D$ let γ_z be the polygonal path from 0 to z that goes horizontally from 0 to $\tilde{z} = \text{Re}(z)$, then vertically from $\tilde{z}$ to z (at most two segments, all inside D by convexity). Define

$$F(z) = \int_{\gamma_z} f(w) dw.$$

The path is determined by z , so F is well defined. Fix z and take h small enough that $z + h \in D$. Then

$$F(z + h) - F(z) = \int_{\gamma_{z+h}} f(w) dw - \int_{\gamma_z} f(w) dw.$$

The two polygonal paths share the segment from 0, traversed in opposite directions, which cancels; completing the resulting figure to a rectangle and a triangle, Goursat (Corollary 15.1.2) lets us replace what remains by the straight segment η from z to $z + h$. Thus $F(z + h) - F(z) = \int_{\eta} f(w) dw$. Writing $f(w) = f(z) + \psi(w)$ with $\psi(w) \rightarrow 0$ as $w \rightarrow z$,

$$F(z + h) - F(z) = f(z) \int_{\eta} dw + \int_{\eta} \psi(w) dw = f(z) h + \int_{\eta} \psi(w) dw,$$

since $\int_{\eta} dw = h$ (the constant 1 has primitive w). The remainder satisfies $|\int_{\eta} \psi| \leq \sup_{\eta} |\psi| \cdot |h|$, and $\sup_{\eta} |\psi| \rightarrow 0$ as $h \rightarrow 0$. Dividing by h gives $\lim_{h \rightarrow 0} \frac{F(z+h) - F(z)}{h} = f(z)$, i.e. $F' = f$. $\square$

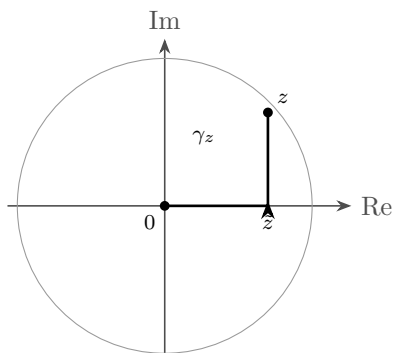


Figure 15.3. The polygonal path γ_z : horizontal to $\bar{z} = \operatorname{Re} z$, then vertical to z . Its integral defines the primitive F . (Schematic.)

Theorem 15.2.2 (Cauchy's theorem for a disc). *If f is holomorphic in a disc, then $\oint_{\gamma} f(z) dz = 0$ for every closed curve γ in that disc.*

Proof. By Theorem 15.2.1, f has a primitive F on the disc. For a closed curve, $\int_{\gamma} f = \int_{\gamma} F' = F(\text{end}) - F(\text{start}) = 0$ (Chapter 1, Corollary 3.3). $\square$

Corollary 15.2.3. *If f is holomorphic in an open set containing a circle C and its interior, then $\int_C f(z) dz = 0$.*

Proof. Let D be the open disc bounded by C . By openness there is a slightly larger disc $D' \supset \bar{D}$ on which f is still holomorphic; apply Theorem 15.2.2 in D' to the closed curve C . $\square$

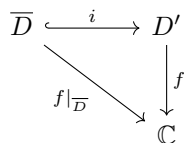


Figure 15.4. Why the closed disc is allowed. The circle $C = \partial D$ and its interior sit inside a slightly larger open disc D' on which f is holomorphic; the restriction $f|_{\bar{D}}$ is the composite $f \circ i$ along the inclusion $i : \bar{D} \hookrightarrow D'$, so Cauchy's theorem for the disc D' applies to C .

Toy contours

The proofs above used only two features of the disc: an unambiguous notion of *interior*, and the ability to build polygonal connecting paths near the curve and its interior. Any closed curve with these features we informally call a *toy contour*; its *positive orientation* is the one keeping the interior on the left. Circles, triangles, rectangles, semicircles, indented semicircles, sectors, parallelograms, and keyholes are all toy contours.

Theorem 15.2.4 (Cauchy's theorem for toy contours). *For a toy contour γ , $\int_{\gamma} f(z) dz = 0$ whenever f is holomorphic in an open set containing γ and its interior.*

The proof copies Theorem 15.2.1: build F along finitely many horizontal/vertical segments inside a slightly enlarged contour; independence of the connecting path follows from the rectangle form of Goursat.

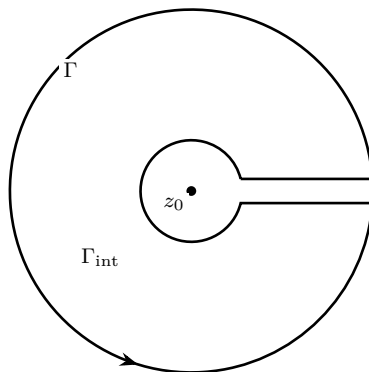


Figure 15.5. The keyhole contour Γ : a large and a small circle about z_0 , joined by a thin corridor; z_0 sits in the interior Γ_{int} . (Schematic.)

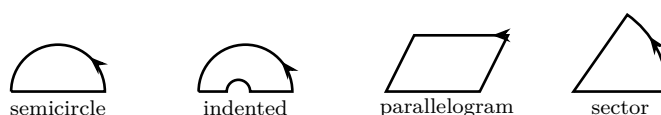


Figure 15.6. A gallery of toy contours: each has an obvious interior and admits the primitive construction, so Cauchy's theorem holds on all of them. (Schematic.)

Worked Example 15.2.5 (A primitive route to a vanishing integral). Show $\int_{\gamma} \cos z \, dz = 0$ for any closed curve γ in $\mathbb{C}$.

Recognition. $\cos z$ is entire with an explicit primitive $\sin z$. For closed curves use the primitive directly — no contour geometry needed.

Solution. $\int_{\gamma} \cos z \, dz = \sin(\text{end}) - \sin(\text{start}) = 0$ since γ is closed. *Check.* On the unit circle $z = e^{i\theta}$, the integrand is bounded and the symmetric cancellation is exact; numerically a quick midpoint sum over $\theta \in [0, 2\pi]$ gives 0 to machine precision.

Worked Example 15.2.6 (When the toy contour encloses a singularity). Contrast $\int_C \frac{e^z}{z-2} \, dz$ over the unit circle $C = \{|z| = 1\}$ with the same integrand over $|z - 2| = 1$.

Recognition. Trigger: identify whether the only singularity ($z = 2$) lies inside the contour. If outside, f is holomorphic on the filled contour and Cauchy applies.

Solution. For $C = \{|z| = 1\}$: the pole $z = 2$ is outside the unit disc, so f is holomorphic on the closed disc and $\int_C f = 0$ by Corollary 15.2.3. For $|z - 2| = 1$: the pole $z = 2$ is the center, hence inside; Cauchy's theorem fails and in fact (by the integral formula of §15.4) $\int_{|z-2|=1} \frac{e^z}{z-2} \, dz = 2\pi i e^2$. *Check.* $e^2 \approx 7.389$, so the second value is $\approx 46.4i$, manifestly nonzero — the enclosed pole is decisive.

Caution. "Holomorphic on the contour" is not enough — you need holomorphic on the contour *and its entire interior*. A single interior singularity (even a removable-looking one) can change the integral from 0 to $2\pi i$ (residue). Always locate every singularity relative to the filled region before invoking Cauchy's theorem.

Further worked examples

Worked Example 15.2.7 (Deforming a contour past no singularities). Let f be holomorphic on the annulus $\{\frac{1}{2} < |z| < 3\}$. Show that $\int_{|z|=1} f \, dz = \int_{|z|=2} f \, dz$.

Recognition. Trigger: two concentric circles in a region where f is holomorphic between them. The region between them contains no singularity, so a "washer" toy contour applies.

Solution. Cut the annulus between the two circles by two thin radial corridors (at angles 0 and π , say), forming two toy contours, each a half-washer bounded by an outer arc of $|z| = 2$ (positive), an inner arc of $|z| = 1$ (negative), and two radial segments. Each half-washer's filled interior lies in $\{\frac{1}{2} < |z| < 3\}$ where f is holomorphic, so by Theorem 15.2.4 each integral is 0. Adding the two: the four radial segments cancel in pairs (each traversed once in each direction), leaving $\int_{|z|=2} f - \int_{|z|=1} f = 0$. Hence the two circle integrals are equal.

Check. Take $f(z) = 1/z$ (holomorphic on the annulus); both integrals equal $2\pi i$ by the z^{-1} computation of §15.3 — equal, as predicted. The common value is independent of the radius.

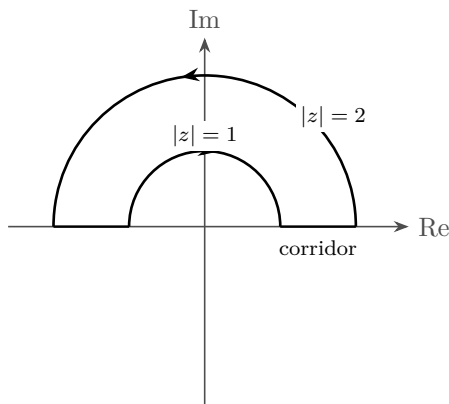


Figure 15.7. Upper half-washer: outer arc positive, inner arc negative, two radial corridors. The mirror lower half completes the deformation; corridors cancel. (Schematic.)

Worked Example 15.2.8 (Path independence of a primitive's integral). Let f be holomorphic on the disc $|z| < 2$ and let γ_1, γ_2 be two paths inside the disc from -1 to $1 + i$. Show $\int_{\gamma_1} f = \int_{\gamma_2} f$.

Recognition. Trigger: two paths with common endpoints in a disc. A disc is convex, so Cauchy's theorem for a disc applies to the closed loop they form.

Solution. Form the closed curve $\gamma = \gamma_1$ followed by γ_2 reversed; it lies in the disc $|z| < 2$ and is closed. By Theorem 15.2.2, $\int_{\gamma} f = 0$. But $\int_{\gamma} f = \int_{\gamma_1} f - \int_{\gamma_2} f$, so the two integrals coincide. *Check.* For $f(z) = 2z$ with primitive z^2 , both equal $(1 + i)^2 - (-1)^2 = (2i) - 1 = -1 + 2i$, independent of the route taken.

Worked Example 15.2.9 (A NON-example: path dependence forced by a hole). On $\Omega = \mathbb{C} \setminus \{0\}$, let $f(z) = 1/z$ and let γ_1, γ_2 be the upper and lower unit semicircles from -1 to 1 . Show $\int_{\gamma_1} f \neq \int_{\gamma_2} f$, and reconcile with the previous example.

Recognition. Trigger: equal endpoints, but Ω is *not* a disc — it has a hole at 0 , so the closed loop $\gamma_1 - \gamma_2$ may enclose the singularity.

Solution. Upper semicircle $z = e^{i\theta}$, $\theta : \pi \rightarrow 0$: $\int_{\gamma_1} \frac{dz}{z} = \int_{\pi}^0 \frac{ie^{i\theta}}{e^{i\theta}} d\theta = \int_{\pi}^0 i d\theta = -i\pi$. Lower semicircle $z = e^{i\theta}$, $\theta : \pi \rightarrow 2\pi$: $\int_{\gamma_2} \frac{dz}{z} = \int_{\pi}^{2\pi} i d\theta = i\pi$. They differ: $-i\pi \neq i\pi$. The closed loop $\gamma_1 - \gamma_2$ is the unit circle traversed once and encloses $0 \notin \Omega$, so Cauchy's theorem does not apply — and indeed $\int_{\gamma_1} f - \int_{\gamma_2} f = -i\pi - i\pi = -2\pi i$, the (negatively oriented) winding contribution. *Check.* γ_1 followed by γ_2 reversed is the *clockwise* unit circle, integral $-2\pi i$, exactly the discrepancy. The hole is the whole difference.

Exercises (Stein, Ch.2).

- (Ex. 6) Let Ω be an open subset of $\mathbb{C}$ and $T \subset \Omega$ a triangle whose interior is also contained in Ω . Suppose f is holomorphic in Ω except possibly at a point w inside T . Prove that if f is bounded near w , then $\int_T f(z) dz = 0$. (This is the removable-singularity form of Cauchy's theorem for a toy contour.)

Additional exercises.

- Using a primitive, evaluate $\int_{\gamma} (z^3 - z) dz$ where γ is the polygonal path $0 \rightarrow 1 \rightarrow 1 + i \rightarrow i$ (open, not closed). Then compute it again for the *closed* loop obtained by appending $i \rightarrow 0$ and confirm it is 0.
- Explain why $f(z) = e^{z^2}$ has a primitive on every disc but the explicit primitive cannot be written in elementary closed form. What does Theorem 15.2.1 nonetheless guarantee about $\int_{\gamma} e^{z^2} dz$ for closed γ ?
- Show directly from the definition of the primitive $F(z) = \int_{\gamma_z} f$ that $F(0) = 0$ and F is continuous, for $f(z) = \cos z$ on the unit disc.

- (Non-example) Decide whether $f(z) = 1/z$ has a primitive on the annulus $\{\frac{1}{2} < |z| < 2\}$. Show $\int_{|z|=1} \frac{dz}{z} = 2\pi i \neq 0$ and conclude no single-valued primitive exists there, even though one exists on every disc avoiding 0.
- Let γ be the boundary of the square with vertices $\pm 1 \pm i$. Prove $\int_{\gamma} \frac{dz}{z^2+4} = 0$ by locating the singularities relative to γ .
- Prove the toy-contour Cauchy theorem for a parallelogram by exhibiting the explicit horizontal/vertical primitive construction, in parallel with the disc case.
- (Diagram) Sketch the keyhole contour Γ with its corridor of width δ , the large circle, and the small circle about an interior point z_0 ; indicate the orientation by arrows and label Γ_{int} .
- Suppose f is holomorphic on an open set containing the closed annulus $\{1 \leq |z| \leq 2\}$. Using two circles and a corridor, show $\int_{|z|=2} f - \int_{|z|=1} f = 0$ (both positively oriented), i.e. the integral is independent of radius in the annulus.

15.3 Evaluation of Some Integrals

Cauchy's theorem is a computational engine: many real integrals are evaluated by viewing the real line as one edge of a closed complex contour, applying $\int_{\gamma} f = 0$, and letting the remaining edges either contribute the answer or vanish in a limit. The systematic theory is the residue calculus (Chapter 16); here we develop the technique by hand on classic examples.

Method (Evaluating a real integral by a toy contour). *Recognition trigger.* A real integral $\int_{-\infty}^{\infty}$ or $\int_0^{\infty}$ whose integrand extends holomorphically (perhaps after a trick) into a half-plane, strip, or sector.

- Choose a closed contour γ_R (rectangle, semicircle, sector) one of whose edges is the real segment $[-R, R]$ (or $[0, R]$).
- Choose f holomorphic inside γ_R (introduce e^{iz} for oscillatory integrands; shift by $i\xi$ for Gaussians; rotate by $e^{i\omega}$ for Γ -type integrals).
- Apply Cauchy: $\int_{\gamma_R} f = 0$, so the real edge equals minus the other edges.
- Show the curved/far edges $\rightarrow 0$ as $R \rightarrow \infty$ (use ML -bounds, Jordan's lemma) and identify the surviving pieces.
- Indent around any boundary singularity with a small arc and compute its (half-residue) contribution.

Worked Example 15.3.1 (The Gaussian is its own Fourier transform (Stein, Example 1)). For $\xi \in \mathbb{R}$ prove $e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x \xi} dx$.

Recognition. Trigger: a Gaussian times an oscillatory factor. Completing the square turns the phase into a contour shift by $i\xi$; integrate the entire $f(z) = e^{-\pi z^2}$ over a rectangle.

Solution. If $\xi = 0$ this is the known $\int_{-\infty}^{\infty} e^{-\pi x^2} dx = 1$. Take $\xi > 0$ and $f(z) = e^{-\pi z^2}$ (entire). Integrate over the rectangle γ_R with vertices $-R, R, R + i\xi, -R + i\xi$ (positive orientation). By Cauchy, $\int_{\gamma_R} f = 0$. The bottom edge gives $\int_{-R}^R e^{-\pi x^2} dx \rightarrow 1$. The right edge is $I(R) = \int_0^{\xi} e^{-\pi(R+iy)^2} i dy$, and $\left| e^{-\pi(R^2-y^2+2iRy)} \right| = e^{-\pi(R^2-y^2)}$, so $|I(R)| \leq C e^{-\pi R^2} \rightarrow 0$; the left edge vanishes identically. The top edge, traversed right-to-left, is

$$\int_R^{-R} e^{-\pi(x+i\xi)^2} dx = -e^{\pi\xi^2} \int_{-R}^R e^{-\pi x^2} e^{-2\pi i x \xi} dx.$$

Summing the four edges to 0 and letting $R \rightarrow \infty$: $0 = 1 - e^{\pi\xi^2} \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x \xi} dx$, which rearranges to the claim. For $\xi < 0$ use the mirror rectangle in the lower half-plane. *Check.* At $\xi = 0$ both sides equal 1; differentiating under the integral confirms the Gaussian shape is self-dual.

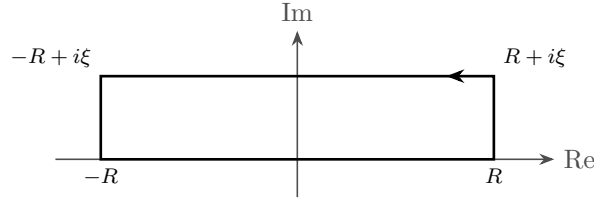


Figure 15.8. The shifted rectangle γ_R for the Gaussian: the real edge carries the integral, the top edge reproduces it scaled by $e^{\pi\xi^2}$, the sides die. (Schematic.)

Worked Example 15.3.2 (An indented semicircle: $\int_0^\infty \frac{1-\cos x}{x^2} dx = \frac{\pi}{2}$ (Stein, Example 2)). Evaluate $\int_0^\infty \frac{1-\cos x}{x^2} dx$.

Recognition. Trigger: an even integrand with a (removable-looking) issue at 0. Replace $1 - \cos x$ by $\operatorname{Re}(1 - e^{ix})$ and integrate $f(z) = (1 - e^{iz})/z^2$ over an indented semicircle in the upper half-plane.

Solution. Let $f(z) = (1 - e^{iz})/z^2$, holomorphic in $\mathbb{C} \setminus \{0\}$. Use the contour: real segments $[-R, -\varepsilon] \cup [\varepsilon, R]$, a large upper semicircle γ_R^+ (radius R , positive), and a small upper semicircle γ_ε^+ (radius ε , traversed negatively around 0). The filled region excludes 0, so $\int_\gamma f = 0$. On γ_R^+ , $|1 - e^{iz}| \leq 2$ (since $\operatorname{Im} z \geq 0$ makes $|e^{iz}| \leq 1$), so $|f| \leq 2/|z|^2$ and the integral is $\leq 2\pi R \cdot 2/R^2 \rightarrow 0$. Hence $\int_{|x| \geq \varepsilon} \frac{1-\cos x}{x^2} dx = -\int_{\gamma_\varepsilon^+} f dz$.

Near 0, $f(z) = \frac{-iz + \frac{1}{2}z^2 + \dots}{z^2} = \frac{-i}{z} + E(z)$ with E bounded. On γ_ε^+ , $z = \varepsilon e^{i\theta}$, $\theta: \pi \rightarrow 0$, so $\int_{\gamma_\varepsilon^+} \frac{-i}{z} dz = \int_\pi^0 \frac{-i}{\varepsilon e^{i\theta}} i\varepsilon e^{i\theta} d\theta = \int_\pi^0 d\theta = -\pi$, hence $-\int_{\gamma_\varepsilon^+} f \rightarrow \pi$. Taking real parts, $\int_{-\infty}^\infty \frac{1-\cos x}{x^2} dx = \pi$, and evenness gives $\int_0^\infty = \frac{\pi}{2}$. *Check.* The integrand $\frac{1-\cos x}{x^2} \rightarrow \frac{1}{2}$ as $x \rightarrow 0$ (no actual singularity) and decays like x^{-2} ; the value $\frac{\pi}{2} \approx 1.571$ is consistent with a quick numerical quadrature.

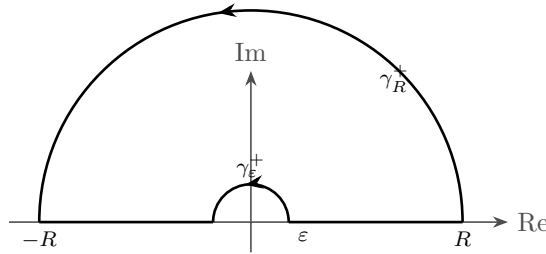


Figure 15.9. The indented semicircle: the small arc γ_ε^+ around 0 contributes $-\pi$ times the residue-like coefficient $-i$; the big arc vanishes. (Schematic.)

Worked Example 15.3.3 (A sector contour: $\int_0^\infty e^{-ax} \cos bx dx$ (Stein, Exercise 3)). For $a > 0$ evaluate $\int_0^\infty e^{-ax} \cos bx dx$ and $\int_0^\infty e^{-ax} \sin bx dx$ together.

Recognition. Trigger: a product of a decaying exponential and an oscillation. Combine into $\int_0^\infty e^{-(a-ib)x} dx$ and recognize this as the real-axis edge of a rotated ray.

Solution. Let $A = a - ib$, so $\operatorname{Re} A = a > 0$. Then $\int_0^\infty e^{-ax} (\cos bx + i \sin bx) dx = \int_0^\infty e^{-Ax} dx = \left[-\frac{1}{A} e^{-Ax}\right]_0^\infty = \frac{1}{A}$, the boundary term at ∞ vanishing since $\operatorname{Re} A > 0$. Thus

$$\int_0^\infty e^{-ax} \cos bx dx = \operatorname{Re} \frac{1}{a - ib} = \frac{a}{a^2 + b^2}, \quad \int_0^\infty e^{-ax} \sin bx dx = \operatorname{Im} \frac{1}{a - ib} = \frac{b}{a^2 + b^2}.$$

(Equivalently, integrate e^{-Az} over a sector of angle $\omega = \arg A$ and let the arc vanish.) *Check.* At $b = 0$ the first integral is $\int_0^\infty e^{-ax} dx = 1/a$, matching a/a^2 ; the second is 0, matching $0/(a^2)$. ■

Worked Example 15.3.4 (Contour-free sanity: $\int_{|z|=1} z^n dz$ on the unit circle). For integer n , compute $\int_{|z|=1} z^n dz$.

Recognition. Trigger: a monomial on a circle. For $n \geq 0$ or $n \leq -2$ a primitive $z^{n+1}/(n+1)$ exists on $\mathbb{C} \setminus \{0\}$ and the closed integral is 0; the lone exception is $n = -1$.

Solution. Parametrize $z = e^{i\theta}$, $dz = ie^{i\theta} d\theta$: $\int_0^{2\pi} e^{in\theta} ie^{i\theta} d\theta = i \int_0^{2\pi} e^{i(n+1)\theta} d\theta$. This is $2\pi i$ if $n = -1$ and 0 otherwise. *Check.* For $n = -1$ this recovers the basic fact $\int_{|z|=1} \frac{dz}{z} = 2\pi i$; for $n = 0$, $\int_{|z|=1} dz = 0$ as a closed loop of the primitive z . This single computation is the seed of the entire residue calculus.

Further worked examples

Worked Example 15.3.5 (The Dirichlet integral $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$). Evaluate $\int_0^\infty \frac{\sin x}{x} dx$ by an indented semicircle.

Recognition. Trigger: an oscillatory integrand with a removable issue at 0. Replace $\sin x$ by $\text{Im } e^{ix}$ and integrate $f(z) = e^{iz}/z$ over the indented upper semicircle; the big arc dies by Jordan's lemma, the small arc gives a half-residue.

Solution. Let $f(z) = e^{iz}/z$, holomorphic off 0. Use real segments $[-R, -\varepsilon] \cup [\varepsilon, R]$, the large upper arc γ_R^+ , and the small upper arc γ_ε^+ (clockwise about 0). The filled region omits 0, so $\oint = 0$. On γ_R^+ , Jordan's lemma gives $|\int_{\gamma_R^+} e^{iz}/z dz| \leq \int_0^\pi e^{-R \sin \theta} d\theta \rightarrow 0$. On γ_ε^+ , $z = \varepsilon e^{i\theta}$, $\theta: \pi \rightarrow 0$: $\int_{\gamma_\varepsilon^+} \frac{e^{iz}}{z} dz \rightarrow \int_\pi^0 i d\theta = -i\pi$ as $\varepsilon \rightarrow 0$ (since $e^{iz} \rightarrow 1$). Thus $\int_{|x| \geq \varepsilon} \frac{e^{ix}}{x} dx - i\pi \rightarrow 0$, i.e. the principal value p.v. $\int_{-\infty}^\infty \frac{e^{ix}}{x} dx = i\pi$. Taking imaginary parts, $\int_{-\infty}^\infty \frac{\sin x}{x} dx = \pi$, and evenness gives $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$. Check. $\frac{\sin x}{x} \rightarrow 1$ at 0 (no singularity); the value $\frac{\pi}{2} \approx 1.5708$ agrees with standard quadrature.

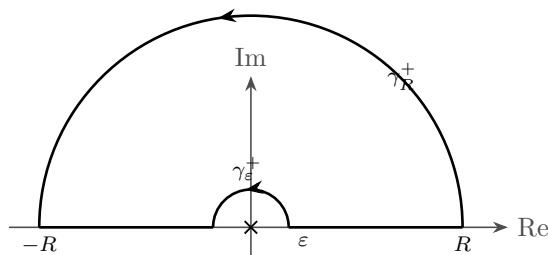


Figure 15.10. Indented semicircle for $\int \frac{\sin x}{x}$: the pole of e^{iz}/z at 0 sits on the boundary, so the small arc contributes a half turn $-i\pi$. (Schematic.)

Worked Example 15.3.6 (A keyhole branch-cut integral: $\int_0^\infty \frac{x^{a-1}}{1+x} dx = \frac{\pi}{\sin \pi a}$). For $0 < a < 1$ evaluate $\int_0^\infty \frac{x^{a-1}}{1+x} dx$ with a keyhole around the positive real axis.

Recognition. Trigger: a power x^{a-1} forces a branch cut; a keyhole hugging $[0, \infty)$ isolates the lone pole at $z = -1$, and the two lips of the cut differ by a phase $e^{2\pi i(a-1)}$.

Solution. Take $f(z) = z^{a-1}/(1+z)$ with the branch $z^{a-1} = e^{(a-1)\text{Log } z}$, $0 < \arg z < 2\pi$. The keyhole Γ runs out along the top lip ($\arg z = 0^+$), around the big circle $|z| = R$, back along the bottom lip ($\arg z = 2\pi^-$), and around the small circle $|z| = \varepsilon$. Only $z = -1 = e^{i\pi}$ lies inside, with $\text{Res}_{z=-1} f = (-1)^{a-1} = e^{(a-1)i\pi} = -e^{ia\pi}$. The big arc is $O(R^{a-1}) \rightarrow 0$ and the small arc is $O(\varepsilon^a) \rightarrow 0$. The top lip gives $\int_0^\infty \frac{x^{a-1}}{1+x} dx =: I$; the bottom lip, where $z = xe^{2\pi i}$, gives $-e^{2\pi i(a-1)}I$. By the residue theorem, $(1 - e^{2\pi i(a-1)})I = 2\pi i(-e^{ia\pi})$. Since $e^{2\pi i(a-1)} = e^{2\pi ia}$, $I = \frac{-2\pi i e^{ia\pi}}{1 - e^{2\pi ia}} = \frac{-2\pi i e^{ia\pi}}{-e^{ia\pi}(e^{ia\pi} - e^{-ia\pi})} = \frac{2\pi i}{2i \sin \pi a} = \frac{\pi}{\sin \pi a}$. Check. At $a = \frac{1}{2}$ this is $\pi / \sin \frac{\pi}{2} = \pi$, matching the elementary $\int_0^\infty \frac{dx}{\sqrt{x}(1+x)} = \pi$ (substitute $x = t^2$). ■

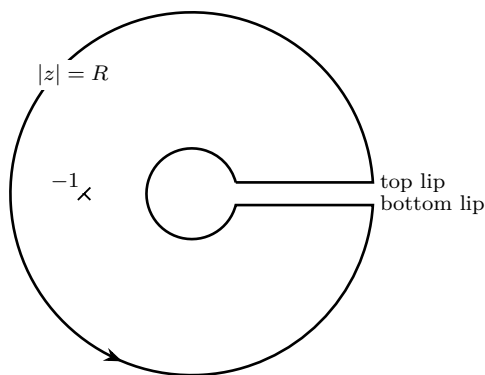


Figure 15.11. Keyhole around $[0, \infty)$: the two lips of the cut carry I and $-e^{2\pi i(a-1)}I$; the only enclosed pole is $z = -1$. (Schematic.)

Worked Example 15.3.7 (A rectangle contour: $\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx = \frac{\pi}{\sin \pi a}$). For $0 < a < 1$ evaluate $\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx$ using the strip $0 \leq \operatorname{Im} z \leq 2\pi$.

Recognition. Trigger: an integrand periodic-friendly under $z \mapsto z + 2\pi i$, since $e^{z+2\pi i} = e^z$. A tall rectangle of height 2π relates the top edge to the bottom by a constant phase and encloses the single pole where $e^z = -1$.

Solution. Let $f(z) = e^{az}/(1+e^z)$. On the rectangle with corners $\pm R$, $\pm R + 2\pi i$, the vertical sides are $O(e^{(a-1)R})$ and $O(e^{aR}e^{-R}) \rightarrow 0$ since $0 < a < 1$. The top edge $z = x + 2\pi i$ gives integrand $e^{a(x+2\pi i)}/(1+e^x) = e^{2\pi ia} f(x)$, traversed right-to-left, contributing $-e^{2\pi ia} I$ where I is the bottom edge $\int_{-\infty}^{\infty} f(x) dx$.

Inside, $e^z = -1$ at $z = i\pi$; there $\operatorname{Res}_{z=i\pi} f = \frac{e^{az}}{(1+e^z)'} \Big|_{z=i\pi} = \frac{e^{ia\pi}}{e^{i\pi}} = -e^{ia\pi}$. Cauchy's residue formula: $(1 - e^{2\pi ia})I = 2\pi i(-e^{ia\pi})$, identical to the previous example, so $I = \frac{\pi}{\sin \pi a}$. *Check.* Substitute $x = \log t$ ($t = e^x$) to get $\int_0^{\infty} \frac{t^{a-1}}{1+t} dt$, exactly the keyhole integral above; both give $\pi/\sin \pi a$. ■

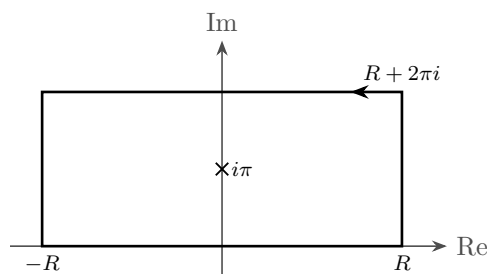


Figure 15.12. Period strip of height 2π : the top edge reproduces the bottom scaled by $e^{2\pi ia}$; the only enclosed pole is $z = i\pi$. (Schematic.)

Exercises (Stein, Ch.2).

- (Ex. 1, Fresnel integrals) Prove that $\int_0^{\infty} \sin(x^2) dx = \int_0^{\infty} \cos(x^2) dx = \frac{\sqrt{2\pi}}{4}$. Here $\int_0^{\infty}$ is interpreted as $\lim_{R \rightarrow \infty} \int_0^R$. (Hint: integrate e^{-z^2} over the circular sector with vertices 0, R , and $Re^{i\pi/4}$; recall $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.)
- (Ex. 2) Show that $\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$. (Hint: the integral equals $\frac{1}{2i} \int_{-\infty}^{\infty} \frac{e^{ix}-1}{x} dx$; use the indented semicircle.)
- (Ex. 3) Evaluate $\int_0^{\infty} e^{-ax} \cos bx dx$ and $\int_0^{\infty} e^{-ax} \sin bx dx$ for $a > 0$ by integrating e^{-Az} , with $A = \sqrt{a^2 + b^2}$, over an appropriate sector of angle ω where $\cos \omega = a/A$.
- (Ex. 4) Prove that for all $\xi \in \mathbb{C}$ we have $e^{-\pi \xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{2\pi i x \xi} dx$ (the complex extension of the self-duality of the Gaussian).

Additional exercises.

- Evaluate $\int_{-\infty}^{\infty} \frac{dx}{x^2+1}$ by closing the contour with an upper semicircle and showing the arc vanishes; confirm the value is π .
- Use the rectangle-shift technique to evaluate $\int_{-\infty}^{\infty} e^{-x^2} \cos(2bx) dx$ for real b , obtaining $\sqrt{\pi} e^{-b^2}$.
- Evaluate $\int_0^{\infty} \frac{\cos x}{x^2+1} dx$ by integrating $e^{iz}/(z^2+1)$ over an upper semicircle and using $\int_{|z|=...}$; obtain $\frac{\pi}{2e}$.
- By integrating e^{-z^2} over the sector of angle $\pi/4$, re-derive the Fresnel integrals (scraped Exercise 1), filling in the arc estimate in detail.
- (Non-example) A student "evaluates" $\int_{-\infty}^{\infty} \frac{x}{x^2+1} dx$ by closing in the upper half-plane and claims the answer is πi times a residue. Explain why the original real integral does not converge (only its principal value is 0), so the contour method needs care here.
- Evaluate $\int_0^{2\pi} \frac{d\theta}{2+\cos \theta}$ by the substitution $z = e^{i\theta}$, turning it into a contour integral over $|z| = 1$; obtain $\frac{2\pi}{\sqrt{3}}$.

7. (Diagram) Draw the indented-semicircle contour used for $\int_0^\infty \frac{\sin x}{x} dx$, labeling the large arc γ_R^+ , the small arc γ_ε^+ , and the two real segments, with orientations.
8. Show that shifting the contour for $\int_{-\infty}^\infty e^{-\pi x^2} e^{-2\pi i x \xi} dx$ to the line $\text{Im } z = \xi$ converts it into $e^{-\pi \xi^2} \int_{-\infty}^\infty e^{-\pi t^2} dt$, recovering the self-duality of the Gaussian without computing side integrals separately.

15.4 Cauchy's Integral Formulas

We now reach the central representation formula: the values of a holomorphic function inside a disc are completely determined — via an explicit integral — by its values on the boundary circle. This single formula yields infinite differentiability, the power-series expansion, Cauchy's inequalities, Liouville's theorem, the fundamental theorem of algebra, and the identity theorem.

Theorem 15.4.1 (Cauchy integral formula). *Suppose f is holomorphic in an open set containing the closure of a disc D . If $C = \partial D$ is the boundary circle with positive orientation, then for every $z \in D$,*

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Proof. Fix $z \in D$. Apply Cauchy's theorem to the keyhole $\Gamma_{\delta, \varepsilon}$ which follows C positively, runs down a corridor of width δ to a small circle C_ε of radius ε about z (traversed negatively, i.e. clockwise), and back. The function $F(\zeta) = f(\zeta)/(\zeta - z)$ is holomorphic on the filled keyhole (it avoids $\zeta = z$), so $\int_{\Gamma_{\delta, \varepsilon}} F d\zeta = 0$. Let $\delta \rightarrow 0$: by continuity the two corridor sides cancel, leaving the big circle C (positive) and the small circle C_ε (negative):

$$0 = \int_C F d\zeta - \int_{C_\varepsilon} F d\zeta.$$

Now split $F(\zeta) = \frac{f(z)}{\zeta - z} + \frac{f(\zeta) - f(z)}{\zeta - z}$. The second term is bounded near z (difference quotient of a differentiable function), so its integral over C_ε is $O(\varepsilon) \rightarrow 0$. For the first term, parametrize $\zeta = z + \varepsilon e^{it}$:

$$\int_{C_\varepsilon} \frac{f(z)}{\zeta - z} d\zeta = f(z) \int_0^{2\pi} \frac{i\varepsilon e^{it}}{\varepsilon e^{it}} dt = 2\pi i f(z).$$

Therefore $0 = \int_C F d\zeta - 2\pi i f(z)$, which is the formula. □

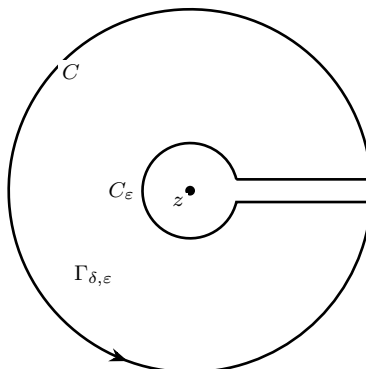


Figure 15.13. The keyhole $\Gamma_{\delta, \varepsilon}$ used to prove the integral formula: shrink δ (corridor cancels), then ε (small circle yields $2\pi i f(z)$). (Schematic.)

Remark. The same keyhole proof works for any toy contour: if f is holomorphic on a rectangle R and its interior, then $f(z) = \frac{1}{2\pi i} \int_R \frac{f(\zeta)}{\zeta - z} d\zeta$ for z inside R , and the integral is 0 when z lies outside (then F is holomorphic inside).

Corollary 15.4.2 (Infinite differentiability and the derivative formulas). *If f is holomorphic in an open set Ω , then f has infinitely many complex derivatives in Ω , and for a circle $C \subset \Omega$ whose interior lies in Ω ,*

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta$$

for all z in the interior of C .

Proof. Induct on n ; $n = 0$ is Theorem 15.4.1. Assuming the formula for $f^{(n-1)}$, form the difference quotient and use the algebraic identity $A^n - B^n = (A - B)(A^{n-1} + \cdots + B^{n-1})$ with $A = 1/(\zeta - z - h)$, $B = 1/(\zeta - z)$. As $h \rightarrow 0$ (and $z, z + h$ stay a fixed distance from C) the bracket converges to $n/(\zeta - z)^{n-1}$, giving $\frac{(n-1)!}{2\pi i} \int_C \frac{n f(\zeta)}{(\zeta - z)^{n+1}} d\zeta = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta$, completing the induction. $\square$

Corollary 15.4.3 (Cauchy inequalities). *If f is holomorphic in an open set containing the closure of the disc D of center z_0 and radius R , then $|f^{(n)}(z_0)| \leq \frac{n! \|f\|_C}{R^n}$, where $\|f\|_C = \sup_{z \in C} |f(z)|$.*

Proof. Parametrize $C : \zeta = z_0 + Re^{i\theta}$ in Corollary 15.4.2: $|f^{(n)}(z_0)| = \left| \frac{n!}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{(Re^{i\theta})^{n+1}} Re^{i\theta} d\theta \right| \leq \frac{n!}{2\pi} \cdot \frac{\|f\|_C}{R^n} \cdot 2\pi$. $\square$

Theorem 15.4.4 (Power-series expansion of holomorphic functions). *If f is holomorphic in Ω and D is a disc centered at z_0 with $\overline{D} \subset \Omega$, then $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ for all $z \in D$, with $a_n = f^{(n)}(z_0)/n! = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$.*

Proof. Fix $z \in D$. In the Cauchy formula write $\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}}$. For $\zeta \in C$ and z fixed, $|\frac{z - z_0}{\zeta - z_0}| \leq r < 1$, so the geometric series $\sum_{n \geq 0} \left(\frac{z - z_0}{\zeta - z_0}\right)^n$ converges uniformly on C . Interchanging sum and integral, $f(z) = \sum_{n \geq 0} \left(\frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta\right) (z - z_0)^n$, and the coefficient is $a_n = f^{(n)}(z_0)/n!$ by Corollary 15.4.2. $\square$

Thus holomorphic = analytic: a function is complex differentiable on Ω iff it is locally a convergent power series. If f is entire, the expansion about 0 converges on all of $\mathbb{C}$.

$$\begin{array}{ccc} f & \xrightarrow{f \mapsto f^{(n)}(z_0)} & f^{(n)}(z_0) \\ & \searrow & \downarrow \div n! \\ a_n = \frac{1}{2\pi i} \int_C \frac{f}{(\zeta - z_0)^{n+1}} & & a_n \end{array}$$

Figure 15.14. The two routes to the Taylor coefficient a_n agree. The derivative formula (Corollary 15.4.2) computes $f^{(n)}(z_0)$ as a contour integral; dividing by $n!$ gives a_n . The diagonal is the same coefficient read off directly from the Cauchy kernel $(\zeta - z_0)^{-(n+1)}$. Commutativity is the content of Theorem 15.4.4.

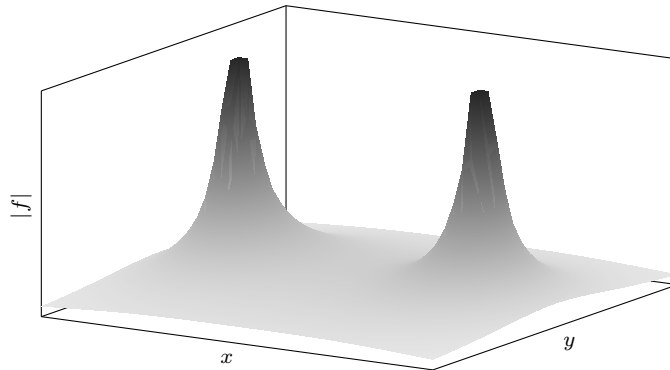


Figure 15.15. Modulus surface $|1/(z^2 - 1)|$: the two spikes at $z = \pm 1$ mark the poles (heights truncated); between them f is holomorphic and the Cauchy formula reconstructs it from boundary values.

Corollary 15.4.5 (Liouville). *If f is entire and bounded, then f is constant.*

Proof. Let $|f| \leq B$ on $\mathbb{C}$. For any z_0 and any $R > 0$, Cauchy's inequality with $n = 1$ gives $|f'(z_0)| \leq B/R$. Let $R \rightarrow \infty$ to get $f'(z_0) = 0$. As $\mathbb{C}$ is connected and $f' \equiv 0$, f is constant. $\square$

Corollary 15.4.6 (Fundamental theorem of algebra). *Every non-constant polynomial $P(z) = a_n z^n + \cdots + a_0$ with complex coefficients has a root in $\mathbb{C}$; in fact exactly n roots $w_1, \dots, w_n$ (with multiplicity), and $P(z) = a_n(z - w_1) \cdots (z - w_n)$.*

Proof. If P had no root, $1/P$ would be entire. For $|z| > R$ large, $|P(z)| \geq c|z|^n$ with $c = |a_n|/2$ (factor out $a_n z^n$ and bound the tail), so $1/P \rightarrow 0$ at ∞ and is bounded; on the compact disc $|z| \leq R$ it is bounded too. By Liouville $1/P$ is constant, contradicting non-constancy of P . Hence P has a root w_1 ; dividing out $(z - w_1)$ and inducting on degree yields the full factorization with n roots and leading coefficient a_n . $\square$

Theorem 15.4.7 (Identity / analytic continuation). *If f is holomorphic in a region (connected open) Ω and vanishes on a sequence of distinct points with a limit point in Ω , then $f \equiv 0$. Consequently two holomorphic functions on Ω agreeing on a set with a limit point in Ω agree everywhere, and an analytic continuation, if it exists, is unique.*

Proof. Let $z_0 \in \Omega$ be the limit point, $f(w_k) = 0$, $w_k \rightarrow z_0$. Expand $f(z) = \sum_n a_n(z - z_0)^n$ in a disc $D \ni z_0$. If $f \not\equiv 0$ on D , let m be least with $a_m \neq 0$; then $f(z) = a_m(z - z_0)^m(1 + g(z - z_0))$ with $g \rightarrow 0$ at z_0 . For w_k near z_0 , $a_m(w_k - z_0)^m(1 + g) \neq 0$, contradicting $f(w_k) = 0$. So $f \equiv 0$ on D . The set U where f vanishes on a neighborhood is open (just shown), nonempty ($z_0 \in U$), and closed in Ω (limits of zeros are zeros, then the disc argument applies). By connectedness $U = \Omega$. $\square$

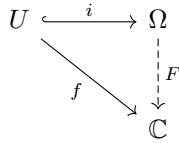


Figure 15.16. Uniqueness of analytic continuation. If $U \subseteq \Omega$ has a limit point in the region Ω , a holomorphic extension F with $F \circ i = f$ (i.e. $F|_U = f$) is unique: any two such agree on U , hence everywhere on Ω by the identity theorem. The dashed arrow records that F need not exist, but is determined when it does.

Method (Computing $\int_C \frac{g(\zeta)}{(\zeta - z)^k} d\zeta$ by the integral formulas). *Recognition trigger.* A contour integral whose integrand is (holomorphic)/(linear factor to a power), with the singularity inside the contour.

1. Identify f holomorphic on the filled contour and the interior point z .
2. For a simple pole ($k = 1$): the value is $2\pi i f(z)$.
3. For a pole of order $k = n + 1$: the value is $\frac{2\pi i}{n!} f^{(n)}(z)$.
4. If the singularity is *outside* the contour, the integral is 0 (Cauchy's theorem).

Worked Example 15.4.8 (A simple-pole evaluation). Compute $\int_{|z|=2} \frac{e^z}{z-1} dz$.

Recognition. Trigger: $f(\zeta) = e^\zeta$ holomorphic inside $|z| = 2$, simple pole at $z = 1$ which is inside. Use $2\pi i f(z)$.

Solution. By the Cauchy integral formula, $\int_{|z|=2} \frac{e^z}{z-1} dz = 2\pi i e^1 = 2\pi i e$. *Check.* $z = 1$ lies inside $|z| = 2$; if instead the contour were $|z| = \frac{1}{2}$, the pole would be outside and the integral would be 0.

Worked Example 15.4.9 (A higher-order pole via the derivative formula). Compute $\int_{|z|=1} \frac{\cos z}{z^3} dz$.

Recognition. Trigger: pole of order $3 = n + 1$ at $z = 0$ (inside), so $n = 2$; use $\frac{2\pi i}{2!} f''(0)$ with $f(\zeta) = \cos \zeta$.

Solution. $f(z) = \cos z$, $f''(z) = -\cos z$, $f''(0) = -1$. Hence $\int_{|z|=1} \frac{\cos z}{z^3} dz = \frac{2\pi i}{2!}(-1) = -\pi i$. *Check.* Expand $\cos z = 1 - \frac{z^2}{2} + \cdots$, so $\frac{\cos z}{z^3} = \frac{1}{z^3} - \frac{1}{2z} + \cdots$; only the $1/z$ term integrates to $2\pi i \cdot (-\frac{1}{2}) = -\pi i$, matching.

Worked Example 15.4.10 (The mean-value property of holomorphic functions). Show that for f holomorphic on $\overline{D}$, $f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + Re^{i\theta}) d\theta$, the average of f over the boundary circle.

Recognition. Trigger: average of boundary values. Apply the Cauchy formula at the center and parametrize.

Solution. With $\zeta = z_0 + Re^{i\theta}$, $d\zeta = iRe^{i\theta} d\theta$ and $\zeta - z_0 = Re^{i\theta}$, the Cauchy formula gives $f(z_0) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{Re^{i\theta}} iRe^{i\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + Re^{i\theta}) d\theta$. *Check.* For $f \equiv c$ both sides equal c ; taking real parts recovers the mean-value property of harmonic functions.

Worked Example 15.4.11 (A NON-application: pole outside the contour). Evaluate $\int_{|z|=1} \frac{1}{z-3} dz$ and explain why the integral formula gives 0, not $2\pi i \cdot 1$.

Recognition. Trigger: looks like a simple pole, but locate $z = 3$ relative to $|z| = 1$.

Solution. The point $z = 3$ is *outside* the unit circle, so $\frac{1}{z-3}$ is holomorphic on the closed unit disc; by Cauchy's theorem $\int_{|z|=1} \frac{1}{z-3} dz = 0$. The integral formula is *not* invoked because there is no interior singularity. *Check.* The primitive $\text{Log}(z-3)$ exists on a neighborhood of the closed disc (the cut avoids it), so the closed-loop integral vanishes.

Worked Example 15.4.12 (Reading off Taylor coefficients by integration). Let $f(z) = e^z$. Recover $a_2 = \frac{1}{2!}$ via $a_2 = \frac{1}{2\pi i} \int_{|z|=1} \frac{e^\zeta}{\zeta^3} d\zeta$.

Recognition. Trigger: extract a power-series coefficient as a contour integral (Theorem 15.4.4 with $z_0 = 0$, $n = 2$).

Solution. By the derivative formula, $\frac{1}{2\pi i} \int_{|z|=1} \frac{e^\zeta}{\zeta^3} d\zeta = \frac{f''(0)}{2!} = \frac{e^0}{2} = \frac{1}{2}$. *Check.* The Taylor series $e^z = \sum z^n/n!$ indeed has $a_2 = 1/2!$; the integral isolates exactly that coefficient.

Further worked examples

Worked Example 15.4.13 (Cauchy's inequality forces a polynomial bound (Stein, Problem-style)). Suppose f is entire and $|f(z)| \leq A + B|z|^{3/2}$ for all z . Prove f is a polynomial of degree at most 1.

Recognition. Trigger: a growth bound on an entire function. Cauchy's inequalities convert a global modulus bound into bounds on the Taylor coefficients $a_n = f^{(n)}(0)/n!$.

Solution. Expand $f(z) = \sum_{n \geq 0} a_n z^n$ on all of $\mathbb{C}$. By Corollary 15.4.3 on the circle $|z| = R$, $|a_n| = \frac{|f^{(n)}(0)|}{n!} \leq \frac{\|f\|_{C_R}}{R^n} \leq \frac{A + BR^{3/2}}{R^n}$. For $n \geq 2$ the right side is $\leq AR^{-n} + BR^{3/2-n} \rightarrow 0$ as $R \rightarrow \infty$ (both exponents negative). Hence $a_n = 0$ for $n \geq 2$, so $f(z) = a_0 + a_1 z$, a polynomial of degree ≤ 1 . *Check.* The bound $|z|^{3/2}$ sits strictly between degree 1 and degree 2 growth; degree-1 functions $a_0 + a_1 z$ satisfy $|f| \leq |a_0| + |a_1| |z| \leq A + B|z|^{3/2}$ for large $|z|$, so the conclusion is sharp.

Worked Example 15.4.14 (A higher-order pole and a holomorphic factor together). Compute $\int_{|z|=2} \frac{e^z}{z^2(z-1)} dz$.

Recognition. Trigger: two interior singularities — a double pole at 0 and a simple pole at 1, both inside $|z| = 2$. Split by partial-fraction-style isolation, or apply the formulas to each via small circles (deform the contour to two little loops).

Solution. Deform $|z| = 2$ to two small positively oriented circles, one about 0 and one about 1 (Cauchy's theorem on the connecting region). About $z = 1$ (simple pole): integrand $= \frac{e^z/z^2}{z-1}$ with $g(z) = e^z/z^2$ holomorphic there, value $2\pi i g(1) = 2\pi i \frac{e}{1} = 2\pi i e$. About $z = 0$ (double pole, $n = 1$): integrand $= \frac{h(z)}{z^2}$ with $h(z) = e^z/(z-1)$, value $\frac{2\pi i}{1!} h'(0)$. Here $h'(z) = \frac{e^z(z-1) - e^z}{(z-1)^2} = \frac{e^z(z-2)}{(z-1)^2}$, so $h'(0) = \frac{1 \cdot (-2)}{1} = -2$, giving $2\pi i(-2) = -4\pi i$. Total: $2\pi i e - 4\pi i = 2\pi i(e-2)$. *Check.* $e-2 \approx 0.718 > 0$, so the answer is a positive multiple of $2\pi i$; the residues e (at 1) and -2 (at 0) sum to $e-2$, matching the residue-theorem viewpoint of Chapter 16.

Exercises (Stein, Ch.2).

- (Ex. 7) Suppose $f : \mathbb{D} \rightarrow \mathbb{C}$ is holomorphic. Show that the diameter $d = \sup_{z,w \in \mathbb{D}} |f(z) - f(w)|$ of the image satisfies $2|f'(0)| \leq d$, with equality precisely when f is linear, $f(z) = a_0 + a_1 z$. (Hint: $2f'(0) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(\zeta) - f(-\zeta)}{\zeta^2} d\zeta$ for $0 < r < 1$.)
- (Ex. 8) If f is holomorphic on the strip $-1 < y < 1$, $x \in \mathbb{R}$ with $|f(z)| \leq A(1 + |z|)^\eta$ (η a fixed real), show that for each integer $n \geq 0$ there is $A_n \geq 0$ with $|f^{(n)}(x)| \leq A_n(1 + |x|)^\eta$ for all $x \in \mathbb{R}$. (Hint: Cauchy inequalities.)
- (Ex. 9) Let Ω be a bounded open subset of $\mathbb{C}$ and $\varphi : \Omega \rightarrow \Omega$ holomorphic. Prove that if there is a point $z_0 \in \Omega$ with $\varphi(z_0) = z_0$ and $\varphi'(z_0) = 1$, then φ is linear. (Hint: reduce to $z_0 = 0$; write $\varphi(z) = z + a_n z^n + O(z^{n+1})$ and show $\varphi_k = \varphi \circ \dots \circ \varphi$ satisfies $\varphi_k(z) = z + k a_n z^n + O(z^{n+1})$; apply the Cauchy inequalities and let $k \rightarrow \infty$.)
- (Ex. 11) Let f be holomorphic on the disc D_{R_0} centered at 0 of radius R_0 . (a) Prove that for $0 < R < R_0$ and $|z| < R$, $f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \operatorname{Re} \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z} \right) d\varphi$. (b) Show that $\operatorname{Re} \left(\frac{Re^{i\gamma} + r}{Re^{i\gamma} - r} \right) = \frac{R^2 - r^2}{R^2 - 2Rr \cos \gamma + r^2}$. (Hint: with $w = R^2/\bar{z}$, the integral of $f(\zeta)/(\zeta - w)$ around $|\zeta| = R$ is zero; combine with the Cauchy integral formula.)
- (Ex. 12) Let u be real-valued, twice continuously differentiable, and harmonic ($\Delta u = 0$) on the unit disc $\mathbb{D}$. (a) Prove there exists a holomorphic f on $\mathbb{D}$ with $\operatorname{Re} f = u$, and that $\operatorname{Im} f$ is unique up to an additive real constant. (Hint: $g = 2\partial u/\partial z$ is holomorphic; find F with $F' = g$ and show $\operatorname{Re} F$ differs from u by a constant.) (b) Deduce the Poisson integral representation: if u is harmonic in $\mathbb{D}$ and continuous on $\bar{\mathbb{D}}$, then for $z = re^{i\theta}$, $u(z) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(\varphi) d\varphi$, where $P_r(\gamma) = \frac{1-r^2}{1-2r \cos \gamma + r^2}$ is the Poisson kernel.
- (Ex. 13) Suppose f is analytic everywhere in $\mathbb{C}$ and such that for each $z_0 \in \mathbb{C}$ at least one coefficient in $f(z) = \sum_{n=0}^\infty c_n(z - z_0)^n$ vanishes. Prove f is a polynomial. (Hint: $c_n n! = f^{(n)}(z_0)$; use a countability argument.)
- (Ex. 14) Suppose f is holomorphic in an open set containing the closed unit disc, except for a pole at z_0 on the unit circle. If $\sum_{n=0}^\infty a_n z^n$ is its power-series expansion in the open unit disc, show that $\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = z_0$.

Problems (Stein, Ch.2).

- (Pr. 1) Examples of analytic functions on $\mathbb{D}$ that cannot be continued past the unit circle. A point $w \in C = \partial\mathbb{D}$ is *regular* for f if there is a neighborhood U of w and an analytic g on U with $f = g$ on $\mathbb{D} \cap U$; f cannot be continued past C if no point of C is regular. (a) Let $f(z) = \sum_{n=0}^\infty z^{2^n}$ for $|z| < 1$ (radius of convergence 1). Show f cannot be continued analytically past $\mathbb{D}$. (Hint: for $\theta = 2\pi p/2^k$ and $z = re^{i\theta}$, $|f(re^{i\theta})| \rightarrow \infty$ as $r \rightarrow 1$.) (b)* Fix $0 < \alpha < \infty$. Show that $f(z) = \sum_{n=0}^\infty 2^{-n\alpha} z^{2^n}$ extends continuously to the unit circle but cannot be continued analytically past it. (Hint: a nowhere differentiable function lurks in the background.)
- (Pr. 2)* Let $F(z) = \sum_{n=1}^\infty d(n)z^n$ for $|z| < 1$, where $d(n)$ is the number of divisors of n (radius of convergence 1). Verify $\sum_{n=1}^\infty d(n)z^n = \sum_{n=1}^\infty \frac{z^n}{1-z^n}$. Show that for $z = r$, $0 < r < 1$, $|F(r)| \geq c \frac{1}{1-r} \log \frac{1}{1-r}$ as $r \rightarrow 1$; and for $z = re^{i\theta}$ with $\theta = 2\pi p/q$, $|F(re^{i\theta})| \geq c_{p/q} \frac{1}{1-r} \log \frac{1}{1-r}$. Conclude F cannot be continued past the unit disc.

Additional exercises.

- Evaluate $\int_{|z|=2} \frac{z}{z-1} dz$ and $\int_{|z|=2} \frac{z}{(z-1)^2} dz$ using the Cauchy integral formulas; state f , the interior point, and the pole order in each case.
- Compute $\int_{|z|=3} \frac{e^{2z}}{(z-1)^2} dz$ via the first-derivative formula.
- Use Liouville's theorem to prove that a bounded entire function whose real part is constant is itself constant.

4. Apply the Cauchy inequalities to show that if f is entire with $|f(z)| \leq A + B|z|^k$ for all z , then f is a polynomial of degree $\leq k$. (This is the generalized Liouville theorem.)
5. Using the identity theorem, prove that if f is entire and $f(1/n) = 0$ for all $n \in \mathbb{N}$, then $f \equiv 0$. Why does the limit point 0 need to lie in the domain?
6. (Non-example) Note $f(z) = \sin(\pi/z)$ vanishes at $z = 1/n$ for all $n \in \mathbb{N}$ yet is not identically zero. Reconcile this with the previous exercise by identifying where the accumulation point of zeros lies relative to the domain of holomorphy.
7. Prove the fundamental theorem of algebra for the special quintic $P(z) = z^5 + z + 1$ by the Liouville argument outline, and (separately) verify it has a real root in $(-1, 0)$.
8. Use the mean-value property to show that a holomorphic f on $\mathbb{D}$ with $f(0) = 2$ and $|f| \leq 2$ on $|z| = 1$ must satisfy $|f(0)| = 2$ forcing $f \equiv 2$ (a maximum-modulus flavored conclusion via the average).
9. (Diagram) Draw the keyhole $\Gamma_{\delta, \varepsilon}$ used to prove the Cauchy integral formula: large circle C positively oriented, small circle C_ε about the interior point z negatively oriented, joined by a width- δ corridor; mark all orientations.
10. Compute $\int_{|z|=1} \frac{\sin z}{z^4} dz$ using the third-derivative formula, and cross-check against the Taylor coefficient of z^3 in $\sin z$.

15.5 Further Applications: Liouville, Morera, and Sequences of Holomorphic Functions

We collect the harvest. Morera's theorem is a converse to Cauchy's; combined with Goursat it shows holomorphy survives uniform limits — a phenomenon with no real-variable analogue. We then treat holomorphy of integrals depending on a parameter, the Schwarz reflection principle, and Runge's polynomial approximation theorem.

Morera's theorem

Theorem 15.5.1 (Morera). *If f is continuous in an open disc D and $\int_T f(z) dz = 0$ for every triangle $T \subset D$, then f is holomorphic in D .*

Proof. The hypothesis is exactly what the primitive construction of Theorem 15.2.1 needs (it used only triangle/rectangle vanishing, which now holds by assumption). So f has a primitive F with $F' = f$ in D . By the regularity Corollary 15.4.2, F is infinitely differentiable, hence so is $F' = f$; in particular f is holomorphic. $\square$

Sequences of holomorphic functions

Theorem 15.5.2 (Uniform limits stay holomorphic). *If $\{f_n\}$ are holomorphic on Ω and $f_n \rightarrow f$ uniformly on every compact subset of Ω , then f is holomorphic on Ω .*

Proof. Work in any disc D with $\overline{D} \subset \Omega$. For a triangle $T \subset D$, Goursat gives $\int_T f_n = 0$ for all n . Uniform convergence on $\overline{D}$ lets us pass to the limit: $\int_T f_n \rightarrow \int_T f$, so $\int_T f = 0$. Also f is continuous (uniform limit of continuous). By Morera, f is holomorphic in D ; as D was arbitrary, f is holomorphic on Ω . $\square$

This is striking: a uniform limit of *real* C^1 functions need not be differentiable (by Weierstrass, every continuous function on $[0, 1]$ is a uniform limit of polynomials). Complex differentiability is far more rigid.

Theorem 15.5.3 (Convergence of derivatives). *Under the hypotheses of Theorem 15.5.2, $f'_n \rightarrow f'$ uniformly on every compact subset of Ω (and likewise for every order k).*

Proof. It suffices to prove, for F holomorphic on Ω and $\Omega_\delta = \{z : D_\delta(z) \subset \Omega\}$, the estimate $\sup_{\Omega_\delta} |F'| \leq \frac{1}{\delta} \sup_{\Omega} |F|$; then apply it to $F = f_n - f$. For $z \in \Omega_\delta$, the closed disc $\overline{D_\delta(z)} \subset \Omega$, and the derivative formula gives $F'(z) = \frac{1}{2\pi i} \int_{C_\delta(z)} \frac{F(\zeta)}{(\zeta - z)^2} d\zeta$, so $|F'(z)| \leq \frac{1}{2\pi} \cdot \frac{\sup_{\Omega} |F|}{\delta^2} \cdot 2\pi\delta = \frac{1}{\delta} \sup_{\Omega} |F|$. As $\sup_{\Omega} |f_n - f| \rightarrow 0$ on the compactly-contained sets, the estimate forces $f'_n \rightarrow f'$ uniformly on Ω_δ . $\square$

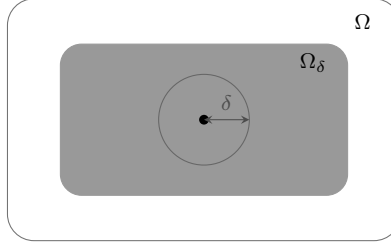


Figure 15.17. Ω_δ is the set of points at distance $> \delta$ from $\partial\Omega$; the circle $C_\delta(z)$ stays inside Ω , powering the derivative estimate. (Schematic.)

A common use: build holomorphic functions as series $F(z) = \sum_{n \geq 1} f_n(z)$. If each f_n is holomorphic on Ω and the series converges uniformly on compacta, then F is holomorphic on Ω (e.g. the Riemann zeta function on $\text{Re } s > 1$).

Functions defined by integrals

Theorem 15.5.4 (Holomorphy under the integral sign). *Let $F(z, s)$ be defined for $(z, s) \in \Omega \times [0, 1]$ with (i) $z \mapsto F(z, s)$ holomorphic for each s , and (ii) F jointly continuous. Then $f(z) = \int_0^1 F(z, s) ds$ is holomorphic on Ω .*

Proof. Approximate by Riemann sums $f_n(z) = \frac{1}{n} \sum_{k=1}^n F(z, k/n)$, each holomorphic by (i). On a disc D with $\overline{D} \subset \Omega$, joint continuity on the compact $\overline{D} \times [0, 1]$ is uniform, so for $\varepsilon > 0$ there is δ with $\sup_{z \in D} |F(z, s_1) - F(z, s_2)| < \varepsilon$ when $|s_1 - s_2| < \delta$. For $n > 1/\delta$ this forces $|f_n(z) - f(z)| < \varepsilon$ uniformly on D . Hence $f_n \rightarrow f$ uniformly on compacta, and f is holomorphic by Theorem 15.5.2. $\square$

The Schwarz reflection principle

Let Ω be symmetric across $\mathbb{R}$ (so $z \in \Omega \iff \bar{z} \in \Omega$), with Ω^+, Ω^- its upper/lower parts and $I = \Omega \cap \mathbb{R}$.

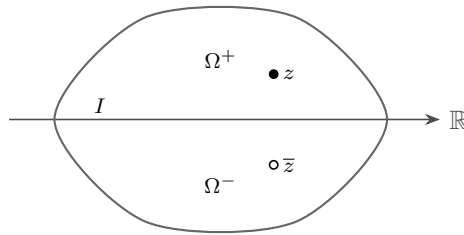


Figure 15.18. A region Ω symmetric across the real axis. Reflection sends $z \in \Omega^+$ to $\bar{z} \in \Omega^-$ and pins the values on I . (Schematic.)

Theorem 15.5.5 (Symmetry principle). *If f^+, f^- are holomorphic on Ω^+, Ω^- respectively, extend continuously to I , and agree on I ($f^+(x) = f^-(x)$ for $x \in I$), then the glued function f (equal to $f^\pm$ off I and to the common value on I) is holomorphic on all of Ω .*

Proof. f is continuous on Ω . Holomorphy off I is given; we check it on I by Morera in a disc D centered on I and contained in Ω . For a triangle $T \subset D$: if T misses I , $\int_T f = 0$ already. If an edge or vertex lies on I , push it slightly off into a half-disc where f is holomorphic to get $\int_{T_\varepsilon} f = 0$, then let $\varepsilon \rightarrow 0$ and use continuity. If the interior of T crosses I , split T into sub-triangles each with only an edge/vertex on I . All cases give $\int_T f = 0$, so f is holomorphic by Morera. $\square$

Theorem 15.5.6 (Schwarz reflection). *If f is holomorphic on Ω^+ , extends continuously to I , and is real-valued on I , then $F(z) = f(z)$ on $\Omega^+ \cup I$ and $F(z) = \overline{f(\bar{z})}$ on Ω^- is holomorphic on all of Ω and extends f .*

Proof. Define $F(z) = \overline{f(\bar{z})}$ on Ω^- . If $f(z) = \sum a_n(z - z_0)^n$ near $z_0 \in \Omega^+$, then $\overline{f(\bar{z})} = \sum \overline{a_n}(z - \bar{z}_0)^n$ near $\bar{z}_0 \in \Omega^-$, a convergent power series, so F is holomorphic on Ω^- . On I , f real means $\overline{f(x)} = f(x)$, so F matches f continuously across I . By the symmetry principle (Theorem 15.5.5), F is holomorphic on Ω . $\square$

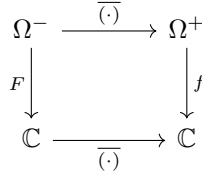
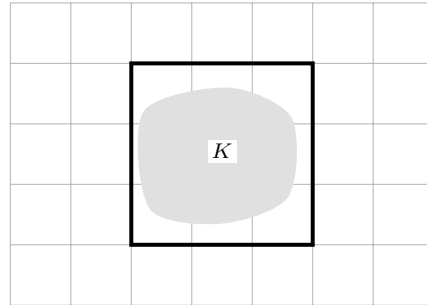


Figure 15.19. Schwarz reflection as a composition: on Ω^- the extension is $F = \overline{(\cdot)} \circ f \circ \overline{(\cdot)}$, i.e. conjugate the input into Ω^+ , apply f , then conjugate the output. The square commutes ($\overline{f(\bar{z})} = F(z)$), which is exactly why F is holomorphic.

Runge's approximation theorem

Theorem 15.5.7 (Runge). *Any function holomorphic in a neighborhood of a compact $K \subset \mathbb{C}$ can be approximated uniformly on K by rational functions whose only singularities lie in K^c . If moreover K^c is connected, the approximation can be taken by polynomials.*

The proof rests on two steps. First, a grid lemma writes f on K as a finite sum of Cauchy-type integrals over segments $\gamma_n \subset \Omega \setminus K$ (the outer edges of a fine square grid covering K); each such integral is a uniform limit of Riemann sums, which are rational functions with poles on the γ_n . Second, when K^c is connected, one "pushes poles to infinity": a pole at z_0 is moved along a path in K^c to a far point z_1 via the geometric series $\frac{1}{z-w} = \sum_n \frac{(w-w')^n}{(z-w')^{n+1}}$ (valid for $|w-w'|$ small relative to $\text{dist}(K, \cdot)$), and finally $\frac{1}{z-w} = -\sum_{n \geq 1} z^n / z_1^{n+1}$ is a uniform polynomial limit on K . Hence rational approximants become polynomial.



squares meeting K ; outer edges $= \bigcup \gamma_n$

Figure 15.20. Runge's grid: the union of unit squares meeting K (shaded); the bold boundary, running along grid lines, is the family of segments $\gamma_n \subset K^c$ in the integral representation. (Schematic.)

Worked Example 15.5.8 (Morera certifies a limit is holomorphic). Show $f(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^2}$ is holomorphic on the open unit disc.

Recognition. Trigger: a series of holomorphic terms (each z^n/n^2 is entire). Check uniform convergence on compacta and invoke Theorem 15.5.2.

Solution. On $|z| \leq r < 1$, $|z^n/n^2| \leq r^n/n^2 \leq r^n$, and $\sum r^n < \infty$, so by the Weierstrass M -test the partial sums converge uniformly on each closed sub-disc. Each partial sum is a polynomial (holomorphic), so the limit f is holomorphic on $\mathbb{D}$. *Check.* At $z = 0$, $f(0) = 0$ and term-by-term differentiation gives $f'(z) = \sum z^{n-1}/n$, also convergent for $|z| < 1$ — consistent with Theorem 15.5.3.

Worked Example 15.5.9 (Schwarz reflection extends a half-plane function). Let f be holomorphic on $\Omega^+ = \{\operatorname{Im} z > 0, |z| < 2\}$, continuous up to $I = (-2, 2)$, with f real on I . If $f(z) = z^2$ on Ω^+ , find its reflection.

Recognition. Trigger: real boundary values on a real segment. Apply $F(z) = \overline{f(\bar{z})}$ on the lower half.

Solution. For $z \in \Omega^-$, $F(z) = \overline{f(\bar{z})} = \overline{(\bar{z})^2} = z^2$. So $F(z) = z^2$ on all of Ω — the reflection is the same formula, as it must be since z^2 is already entire and real on $\mathbb{R}$. *Check.* On I , $f(x) = x^2$ is real, confirming the hypothesis; the unique holomorphic extension is z^2 .

Worked Example 15.5.10 (A NON-approximable function (why K^c connected matters)). Show $f(z) = 1/z$ on $K = \{|z| = 1\}$ cannot be uniformly approximated by polynomials on K .

Recognition. Trigger: candidate for Runge, but $K^c = \{|z| < 1\} \cup \{|z| > 1\}$ is *disconnected*. Test against the contour integral.

Solution. Suppose polynomials $p_k \rightarrow f$ uniformly on K . Then $\int_K p_k \rightarrow \int_K f$. But $\int_K p_k = 0$ for every polynomial (Cauchy's theorem; p_k is entire), while $\int_K \frac{dz}{z} = 2\pi i \neq 0$. So $0 = 2\pi i$, a contradiction. Hence no polynomial approximation exists. *Check.* This is exactly the disconnected-complement obstruction; rational approximation (a single pole at $0 \in K^c$) is of course trivially exact here.

Worked Example 15.5.11 (Holomorphy of a parameter integral). Show $f(z) = \int_0^1 e^{zs} ds$ is entire and compute it in closed form.

Recognition. Trigger: an integral with holomorphic, jointly continuous integrand $F(z, s) = e^{zs}$. Theorem 15.5.4 gives holomorphy; then integrate directly.

Solution. $F(z, s) = e^{zs}$ is holomorphic in z and jointly continuous, so f is entire. Directly, $f(z) = \int_0^1 e^{zs} ds = \frac{e^z - 1}{z}$ for $z \neq 0$ and $f(0) = 1$ (removable). *Check.* The closed form $\frac{e^z - 1}{z} = \sum_{n \geq 0} \frac{z^n}{(n+1)!}$ is entire with $f(0) = 1$, confirming the predicted holomorphy.

Worked Example 15.5.12 (Liouville kills a bounded harmonic-conjugate combination). Suppose u, v are real harmonic on all of $\mathbb{C}$ with u the real and v the imaginary part of an entire $f = u + iv$. If u is bounded above, prove f is constant.

Recognition. Trigger: a one-sided bound on $\operatorname{Re} f$ for an entire f . Exponentiate to convert the half-plane bound into a genuine modulus bound, then apply Liouville (Corollary 15.4.5).

Solution. Say $u \leq M$. Consider $g(z) = e^{f(z)}$, entire as a composition of entire functions. Then $|g(z)| = e^{\operatorname{Re} f(z)} = e^{u(z)} \leq e^M$, so g is a bounded entire function. By Liouville, g is constant, hence $g' = f'e^f \equiv 0$; since e^f never vanishes, $f' \equiv 0$, so f is constant on the connected set $\mathbb{C}$. *Check.* The trick is exactly the one behind "an entire function omitting a half-plane of values is constant"; e.g. $f = z$ has $\operatorname{Re} f = x$ unbounded above, so the hypothesis correctly excludes it, and no non-constant entire f can have $\operatorname{Re} f$ capped.

Worked Example 15.5.13 (Reflecting a function with a real-axis symmetry to compute coefficients). Let f be holomorphic on $\Omega^+ = \{\operatorname{Im} z > 0\}$, continuous up to $\mathbb{R}$, real on $\mathbb{R}$, with $f(z) \rightarrow 0$ as $|z| \rightarrow \infty$ in Ω^+ . Show the reflected F is entire and deduce $F \equiv 0$.

Recognition. Trigger: real boundary values on $\mathbb{R}$ plus a decay condition. Schwarz reflection (Theorem 15.5.6) glues f to $\overline{f(\bar{z})}$ across $\mathbb{R}$; the decay then feeds Liouville.

Solution. By Schwarz reflection, $F(z) = f(z)$ on $\overline{\Omega^+}$ and $F(z) = \overline{f(\bar{z})}$ on Ω^- is holomorphic on all of $\mathbb{C}$, hence entire. The hypothesis $f \rightarrow 0$ at ∞ in the closed upper half-plane gives, by reflection, $F \rightarrow 0$ at ∞ in the closed lower half-plane too; so $|F| \rightarrow 0$ as $|z| \rightarrow \infty$, making F a bounded entire function with limit 0. By Liouville F is constant, and the constant must be 0. Hence $f \equiv 0$ on Ω^+ . *Check.* The conclusion is the complex "rigidity" phenomenon — a holomorphic half-plane function that is real on the axis and decays is forced to vanish; e.g. $1/(z+i)$ is real on $\mathbb{R}$? No: $\operatorname{Im} \frac{1}{x+i} = -\frac{1}{x^2+1} \neq 0$, so it is correctly excluded from the hypothesis.

Exercises (Stein, Ch.2).

- (Ex. 10) Weierstrass's theorem states that a continuous function on $[0, 1]$ can be uniformly approximated by polynomials. Can every continuous function on the closed unit disc be approximated uniformly by polynomials in the variable z ? (Decide and justify; contrast with Runge's theorem and the obstruction $\int_{|z|=1} \frac{dz}{z} = 2\pi i$.)

2. (Ex. 15) Suppose f is a non-vanishing continuous function on $\overline{\mathbb{D}}$ that is holomorphic in $\mathbb{D}$. Prove that if $|f(z)| = 1$ whenever $|z| = 1$, then f is constant. (Hint: extend f to all of $\mathbb{C}$ by $f(z) = 1/\overline{f(1/\overline{z})}$ for $|z| > 1$, and argue as in the Schwarz reflection principle.)

Problems (Stein, Ch.2).

- (Pr. 3) Morera's theorem says that if f is continuous on $\mathbb{C}$ and $\int_T f(z) dz = 0$ for all triangles T , then f is holomorphic. Decide whether triangles may be replaced by other sets. (a) Suppose f is continuous on $\mathbb{C}$ and $\int_C f(z) dz = 0$ for every circle C . Prove f is holomorphic. (b) More generally, let Γ be any toy contour and $\mathcal{F}$ the collection of all translates and dilates of Γ . Show that if f is continuous on $\mathbb{C}$ and $\int_\gamma f(z) dz = 0$ for all $\gamma \in \mathcal{F}$, then f is holomorphic. In particular Morera holds with equilateral triangles only. (Hint: first assume f twice real-differentiable and expand $f(z) = f(z_0) + a(z - z_0) + \overline{b(z - z_0)} + O(|z - z_0|^2)$; integrating over small circles gives $\partial f / \partial \overline{z} = b = 0$. In general mollify: with $\varphi_\varepsilon(z) = \varepsilon^{-2} \varphi(\varepsilon^{-1} z)$, $\int \varphi = 1$, set $f_\varepsilon = f * \varphi_\varepsilon$; then f_ε is smooth, satisfies the same vanishing condition, and $f_\varepsilon \rightarrow f$ uniformly on compacta.)
- (Pr. 4) Prove the converse to Runge's theorem: if K is compact and K^c is *not* connected, then there is a function f holomorphic in a neighborhood of K that cannot be approximated uniformly by polynomials on K . (Hint: pick z_0 in a bounded component of K^c and let $f(z) = 1/(z - z_0)$; if polynomials p approximated f , then $|(z - z_0)p(z) - 1| < 1$ on K , and by the maximum modulus principle this persists on the whole bounded component containing z_0 , a contradiction.)
- (Pr. 5)* There exists an entire function F with the universal property: for any entire h there is an increasing sequence $\{N_k\}$ with $F(z + N_k) \rightarrow h(z)$ uniformly on every compact subset of $\mathbb{C}$. (a) Let $p_1, p_2, \dots$ enumerate the polynomials with rational (real and imaginary parts) coefficients. Show it suffices to find entire F and an increasing sequence $\{M_n\}$ with $|F(z) - p_n(z - M_n)| < \frac{1}{n}$ whenever $z \in D_n$ (the disc of center M_n , radius n). (Hint: any entire h is a compact-uniform limit of a subsequence p_{n_k} .) (b) Construct F as $F(z) = \sum_{n=1}^{\infty} u_n(z)$ with $u_n(z) = p_n(z - M_n)e^{-c_n(z - M_n)^2}$, choosing $c_n \downarrow 0$ and $M_n \uparrow \infty$ appropriately. (Hint: e^{-z^2} decays rapidly in the sectors $|\arg z| < \pi/4 - \delta$ and $|\pi - \arg z| < \pi/4 - \delta$.) A companion universal G exists with $D^{N_k} G(z) \rightarrow h(z)$ uniformly on compacta, where D^j denotes the j th complex derivative.

Additional exercises.

- Use Morera's theorem to show that if f is continuous on $\mathbb{C}$ and holomorphic off a single horizontal line $\{\operatorname{Im} z = 0\}$, then f is holomorphic everywhere. (This is a baby reflection argument.)
- Show that $f(z) = \sum_{n=1}^{\infty} \frac{1}{n^2 - z^2}$ is holomorphic on $\mathbb{C} \setminus \{\pm 1, \pm 2, \dots\}$ by verifying uniform convergence on compacta avoiding the integers.
- Prove that if $f_n \rightarrow f$ uniformly on compacta with each f_n holomorphic and non-vanishing, then $f'_n \rightarrow f'$ uniformly on compacta, citing Theorem 15.5.3.
- Apply the Schwarz reflection principle to extend a function holomorphic on the upper half-disc, real on the diameter, across the real axis; carry it out explicitly for $f(z) = z + z^3$.
- (Non-example) Explain why $f(z) = \sum_{n=0}^{\infty} z^{2^n}$, holomorphic on $\mathbb{D}$, is *not* a uniform limit on $\overline{\mathbb{D}}$ of polynomials that would extend it past the unit circle. Which theorem of this section fails to apply, and why?
- Use the parameter-integral theorem to prove that Γ -like functions $f(z) = \int_0^1 s^{z-1} ds = 1/z$ (for $\operatorname{Re} z > 0$) are holomorphic, and identify the analytic continuation.
- Show that Runge's theorem gives polynomial approximation of $f(z) = e^z$ uniformly on the compact set $K = [0, 1] \subset \mathbb{C}$ (note K^c is connected); contrast with $K = \{|z| = 1\}$.
- (Diagram) Sketch a region Ω symmetric across the real axis, marking Ω^+ , Ω^- , $I = \Omega \cap \mathbb{R}$, and a conjugate pair $z, \overline{z}$; indicate how the reflection $F(z) = \overline{f(\overline{z})}$ glues across I .
- Prove that the uniform limit on compacta of a sequence of injective holomorphic functions is either injective or constant (a Hurwitz-type statement); use Theorem 15.5.2 and the identity theorem as tools, sketching the argument.

Chapter 16

Meromorphic Functions and the Logarithm

16.1 Zeros and Poles

Isolated singularities, and the trichotomy

A *point singularity* (or *isolated singularity*) of a function f is a point z_0 at which f is undefined although f is holomorphic on a punctured neighborhood $\{z : 0 < |z - z_0| < r\}$. Three model functions on the punctured plane already display the full range of behavior, and it is worth fixing them as touchstones for the whole chapter:

$$f(z) = z \text{ (removable),} \quad g(z) = \frac{1}{z} \text{ (pole),} \quad h(z) = e^{1/z} \text{ (essential).}$$

At the origin f extends continuously (set $f(0) = 0$); g blows up, $|g(z)| \rightarrow \infty$; and h is wild — along the positive real axis $h \rightarrow \infty$, along the negative real axis $h \rightarrow 0$, and along the imaginary axis h oscillates on the unit circle. The first section tames the middle case by tying poles to *zeros*: a pole of f is exactly a zero of $1/f$. So we begin with zeros.

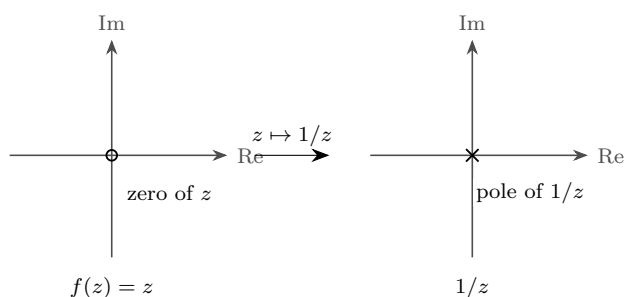


Figure 16.1. The reciprocal map $z \mapsto 1/z$ carries a zero (o) of z at the origin to a pole (x) of $1/z$ at the same point: zeros and poles are reciprocal phenomena. (Schematic.)

The reciprocal correspondence is exact at the level of orders: passing through $w \mapsto 1/w$ turns an order- n zero into an order- n pole and back, so the two local models sit at opposite ends of one inversion.

$$\begin{array}{ccc}
 f & \xrightarrow{\mapsto 1/f} & 1/f \\
 \text{ord}_{z_0} \downarrow & & \downarrow \text{ord}_{z_0} \\
 n \text{ (zero)} & \longleftrightarrow & -n \text{ (pole)}
 \end{array}$$

Figure 16.2. Taking reciprocals negates the order at z_0 : a zero of order n of f is a pole of order n of $1/f$. Here ord_{z_0} denotes the order (signed multiplicity) at z_0 .

Local structure at a zero

Theorem 16.1.1 (Local factorization at a zero). *Suppose f is holomorphic in a connected open set Ω , has a zero at $z_0 \in \Omega$, and does not vanish identically. Then there are a neighborhood $U \subseteq \Omega$ of z_0 , a nowhere-vanishing holomorphic g on U , and a unique positive integer n with*

$$f(z) = (z - z_0)^n g(z) \quad (z \in U).$$

Proof. Since Ω is connected and $f \not\equiv 0$, f is not identically zero near z_0 , so in a disc about z_0 we have a power series $f(z) = \sum_{k \geq 0} a_k (z - z_0)^k$. Not all a_k vanish; let n be the smallest index with $a_n \neq 0$. Then

$$f(z) = (z - z_0)^n [a_n + a_{n+1}(z - z_0) + \cdots] = (z - z_0)^n g(z),$$

where g is given by the bracketed series, hence holomorphic, and $g(z_0) = a_n \neq 0$, so $g \neq 0$ on a smaller disc. For uniqueness, suppose also $f(z) = (z - z_0)^m h(z)$ with $h(z_0) \neq 0$. If $m > n$, cancel $(z - z_0)^n$ to get $g(z) = (z - z_0)^{m-n} h(z)$ and let $z \rightarrow z_0$: the right side $\rightarrow 0$, forcing $g(z_0) = 0$, a contradiction. The case $m < n$ is symmetric. Hence $m = n$ and $h = g$. $\square$

The integer n is the *order* (or *multiplicity*) of the zero; a zero of order 1 is *simple*. The order quantifies the *rate* of vanishing: $f(z) \sim a_n (z - z_0)^n$ near z_0 . As a corollary, the zeros of a nontrivial holomorphic function on a region are *isolated* — in a small enough U , $f(z) \neq 0$ for $0 < |z - z_0| < r$, because g stays nonzero and only the factor $(z - z_0)^n$ can vanish.

Method (Finding the order of a zero). *Recognition.* You must determine how fast a holomorphic f vanishes at z_0 .

1. Compute $f(z_0), f'(z_0), f''(z_0), \dots$; the order n is the index of the first nonzero derivative: $f^{(k)}(z_0) = 0$ for $k < n$ and $f^{(n)}(z_0) \neq 0$. (Equivalently, n is the first nonzero Taylor coefficient's index, and $a_n = f^{(n)}(z_0)/n!$.)
2. Or factor directly: write $f(z) = (z - z_0)^n g(z)$ and check $g(z_0) \neq 0$.
3. For a product/quotient, orders add and subtract: $\text{ord}_{z_0}(fg) = \text{ord}_{z_0} f + \text{ord}_{z_0} g$ and $\text{ord}_{z_0}(f/g) = \text{ord}_{z_0} f - \text{ord}_{z_0} g$.

Worked Example 16.1.2 (Order of the zeros of $\sin \pi z$). Locate the zeros of $f(z) = \sin \pi z$ and find their orders.

Recognition. A transcendental entire function; use $f(z_0) = 0, f'(z_0) \neq 0$ to detect simple zeros.

Solution. $\sin \pi z = 0$ iff $e^{i\pi z} = e^{-i\pi z}$, i.e. $e^{2\pi i z} = 1$, i.e. $z \in \mathbb{Z}$. At $z = m \in \mathbb{Z}$, $f'(z) = \pi \cos \pi z$ gives $f'(m) = \pi \cos \pi m = \pi(-1)^m \neq 0$. So every integer is a *simple* zero. *Check.* Near $z = 0$, $\sin \pi z = \pi z - \frac{(\pi z)^3}{6} + \cdots$, first nonzero coefficient at $k = 1$, confirming order 1.

Worked Example 16.1.3 (A higher-order zero). Find the order of the zero of $f(z) = 1 - \cos z$ at $z = 0$.

Recognition. Use the Taylor expansion to read off the first surviving term.

Solution. $\cos z = 1 - \frac{z^2}{2} + \frac{z^4}{24} - \cdots$, so $f(z) = \frac{z^2}{2} - \frac{z^4}{24} + \cdots = z^2 \left(\frac{1}{2} - \frac{z^2}{24} + \cdots \right)$. The bracket equals $\frac{1}{2} \neq 0$ at $z = 0$, so the zero has order $n = 2$. *Check.* $f(0) = 0, f'(z) = \sin z$ gives $f'(0) = 0, f''(z) = \cos z$ gives $f''(0) = 1 \neq 0$; first nonzero derivative at $k = 2$.

From zeros to poles

A *deleted neighborhood* of z_0 is a punctured disc $\{z : 0 < |z - z_0| < r\}$. We say f , defined on a deleted neighborhood of z_0 , has a *pole* at z_0 if the function $1/f$, set to 0 at z_0 , is holomorphic in a full neighborhood of z_0 . This is the precise sense in which a pole is a “reciprocal zero.”

Theorem 16.1.4 (Local factorization at a pole). *If f has a pole at z_0 , then in a neighborhood of z_0 there are a nowhere-vanishing holomorphic h and a unique positive integer n with*

$$f(z) = (z - z_0)^{-n} h(z).$$

Proof. By definition $1/f$ is holomorphic near z_0 with $(1/f)(z_0) = 0$; by Theorem 16.1.1, $1/f(z) = (z - z_0)^n g(z)$ with g holomorphic and nonvanishing. Set $h = 1/g$: then $f(z) = (z - z_0)^{-n} h(z)$, h holomorphic and nonvanishing, and n is unique because the order of the zero of $1/f$ is. $\square$

The integer n is the *order* (or *multiplicity*) of the pole; order 1 is a *simple* pole. It measures the rate of growth: $|f(z)| \sim |h(z_0)| |z - z_0|^{-n} \rightarrow \infty$.

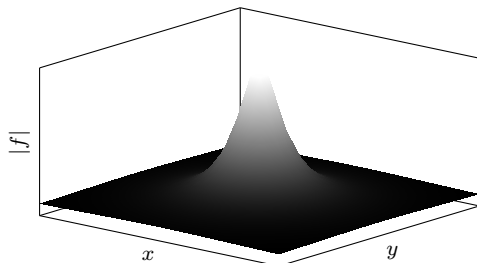


Figure 16.3. Modulus surface $|f(z)| = 1/(|z|^2 + \varepsilon)$ spiking at a pole: the height $\rightarrow \infty$ as $z \rightarrow z_0$, in contrast to a removable singularity, where $|f|$ stays bounded. Here $\varepsilon > 0$ is a small regularizer that keeps the plotted surface finite at the center. (Schematic.)

The principal part and the residue

Theorem 16.1.5 (Laurent expansion at a pole). *If f has a pole of order n at z_0 , then on a deleted neighborhood*

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \frac{a_{-n+1}}{(z - z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z - z_0} + G(z),$$

with G holomorphic near z_0 .

Proof. By Theorem 16.1.4, $f(z) = (z - z_0)^{-n} h(z)$ with $h(z) = A_0 + A_1(z - z_0) + \cdots$ holomorphic. Multiplying out,

$$f(z) = (z - z_0)^{-n} (A_0 + A_1(z - z_0) + \cdots) = \frac{a_{-n}}{(z - z_0)^n} + \cdots + \frac{a_{-1}}{z - z_0} + G(z),$$

where $a_{-n+j} = A_j$ for $0 \leq j \leq n - 1$ and G collects the terms of order ≥ 0 . $\square$

The finite sum of negative-power terms is the *principal part* of f at z_0 , and the coefficient a_{-1} is the *residue*, written $\text{Res}_{z=z_0} f = a_{-1}$. The residue is the single coefficient that survives integration: all higher-order terms $a_{-k}/(z - z_0)^k$ with $k \geq 2$ have primitives $-a_{-k}/((k-1)(z - z_0)^{k-1})$ on a deleted neighborhood and so integrate to 0 around a circle, while $\oint_C \frac{a_{-1}}{z - z_0} dz = 2\pi i a_{-1}$. We develop this fully in § 16.2.

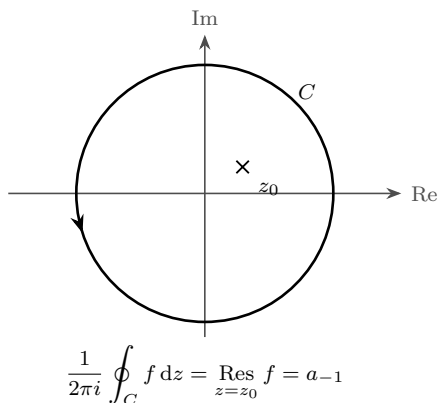


Figure 16.4. Only the $a_{-1}/(z - z_0)$ term survives integration around the circle C ; every other principal-part term $a_{-k}/(z - z_0)^k$ ($k \geq 2$) has a single-valued primitive and integrates to 0. (Schematic.)

Computing residues

Theorem 16.1.6 (Residue at a pole of order n). *If f has a pole of order n at z_0 , then*

$$\operatorname{Res}_{z=z_0} f = \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} \left(\frac{d}{dz} \right)^{n-1} [(z - z_0)^n f(z)].$$

In particular, for a simple pole $\operatorname{Res}_{z=z_0} f = \lim_{z \rightarrow z_0} (z - z_0)f(z)$.

Proof. From Theorem 16.1.5,

$$(z - z_0)^n f(z) = a_{-n} + a_{-n+1}(z - z_0) + \cdots + a_{-1}(z - z_0)^{n-1} + G(z)(z - z_0)^n.$$

The right side is holomorphic near z_0 ; differentiating $n - 1$ times kills the terms of degree $< n - 1$ and the contribution of $G(z)(z - z_0)^n$ (still divisible by $(z - z_0)$ after $n - 1$ derivatives), leaving $(n - 1)!a_{-1}$ in the limit $z \rightarrow z_0$. Divide by $(n - 1)!$. $\square$

Method (Computing a residue). *Recognition.* You need $\operatorname{Res}_{z=z_0} f$.

1. *Simple pole, $f = P/Q$ with $P(z_0) \neq 0$, Q simple zero:* the slick formula is $\operatorname{Res}_{z=z_0} \frac{P}{Q} = \frac{P(z_0)}{Q'(z_0)}$ (since $Q(z) \approx Q'(z_0)(z - z_0)$).
2. *Simple pole, general:* $\operatorname{Res}_{z=z_0} f = \lim_{z \rightarrow z_0} (z - z_0)f(z)$.
3. *Pole of order n :* differentiate $n - 1$ times as in Theorem 16.1.6; clear the pole *first*, then differentiate, then take the limit.
4. *Unknown order / essential point:* expand f in a Laurent series and read off the coefficient of $(z - z_0)^{-1}$ directly.

Worked Example 16.1.7 (Residue at a simple pole by the P/Q' rule). Find $\operatorname{Res}_{z=i} \frac{1}{1 + z^2}$.

Recognition. Simple pole of a rational function; use $P(z_0)/Q'(z_0)$.

Solution. Here $P = 1$, $Q = 1 + z^2$, $Q' = 2z$, so $\operatorname{Res}_{z=i} = \frac{1}{2z} \Big|_{z=i} = \frac{1}{2i}$. *Check.* Factor $1 + z^2 = (z - i)(z + i)$; then $(z - i)f = \frac{1}{z + i} \rightarrow \frac{1}{2i}$ as $z \rightarrow i$. Agrees.

Worked Example 16.1.8 (Residue of $1/\sin \pi z$ at an integer). Compute $\operatorname{Res}_{z=n} \frac{1}{\sin \pi z}$ for $n \in \mathbb{Z}$.

Recognition. $\sin \pi z$ has a simple zero at n (Worked Example above), so $1/\sin \pi z$ has a simple pole; use $1/Q'$.

Solution. With $P = 1$, $Q = \sin \pi z$, $Q' = \pi \cos \pi z$, we get $\operatorname{Res}_{z=n} = \frac{1}{\pi \cos \pi n} = \frac{1}{\pi(-1)^n} = \frac{(-1)^n}{\pi}$. *Check.* Near n , $\sin \pi z = \sin(\pi n + \pi(z - n)) = (-1)^n \sin \pi(z - n) \approx (-1)^n \pi(z - n)$, so $(z - n)/\sin \pi z \rightarrow (-1)^n/\pi$. Confirmed.

Worked Example 16.1.9 (A double pole needs a derivative). Compute $\operatorname{Res}_{z=1} \frac{e^z}{(z - 1)^2}$.

Recognition. Pole of order $n = 2$; clear the pole then differentiate once.

Solution. $(z - 1)^2 f(z) = e^z$. By Theorem 16.1.6 with $n = 2$, $\operatorname{Res}_{z=1} f = \frac{1}{1!} \frac{d}{dz} e^z \Big|_{z=1} = e^z \Big|_{z=1} = e$. *Check.* Expand $e^z = e \cdot e^{z-1} = e(1 + (z - 1) + \cdots)$, so $f = \frac{e}{(z - 1)^2} + \frac{e}{z - 1} + \cdots$; the $(z - 1)^{-1}$ coefficient is e . Agrees.

Worked Example 16.1.10 (Residue at a third-order pole). Find $\operatorname{Res}_{z=0} \frac{e^z}{z^3}$.

Recognition. Pole of order 3; differentiate twice and divide by $2!$.

Solution. $z^3 f = e^z$; $\operatorname{Res}_{z=0} f = \frac{1}{2!} \frac{d^2}{dz^2} e^z \Big|_{z=0} = \frac{1}{2}$. *Check.* $e^z/z^3 = z^{-3} + z^{-2} + \frac{1}{2}z^{-1} + \cdots$; coefficient of z^{-1} is $\frac{1}{2}$. Agrees.

Worked Example 16.1.11 (NON-example: a removable point has residue zero). Let $f(z) = \frac{\sin z}{z}$. Show $z = 0$ is removable and $\text{Res}_{z=0} f = 0$.

Recognition. The apparent singularity is a quotient of a simple zero over a simple zero; their orders cancel.

Solution. $\sin z = z - \frac{z^3}{6} + \dots$, so $f(z) = 1 - \frac{z^2}{6} + \dots$ has *no* negative powers — the singularity is removable, with $f(0) := 1$. There is no z^{-1} term, so $\text{Res}_{z=0} f = 0$. *Check.* $\lim_{z \rightarrow 0} z \cdot \frac{\sin z}{z} = \lim_{z \rightarrow 0} \sin z = 0$, the simple-pole formula returning 0 as it must for a removable point.

Further worked examples

Worked Example 16.1.12 (Order of a zero from a product). Find the order of the zero of $f(z) = z^2 \sin^3 z$ at $z = 0$.

Recognition. Orders add over products (Method, step 3): $\text{ord}_0(z^2 \sin^3 z) = \text{ord}_0 z^2 + \text{ord}_0 \sin^3 z$.

Solution. z^2 has a zero of order 2 at 0. Since $\sin z = z - \frac{z^3}{6} + \dots$ has a *simple* zero at 0, $\sin^3 z$ has a zero of order 3. Hence $\text{ord}_0 f = 2 + 3 = 5$. *Check.* $f(z) = z^2(z - \frac{z^3}{6} + \dots)^3 = z^5(1 - \frac{z^2}{6} + \dots)^3$, whose first surviving power is z^5 with nonzero coefficient 1. Order 5 confirmed.

Worked Example 16.1.13 (Residue of $\cot z$ at the origin). Compute $\text{Res}_{z=0} \cot z$, where $\cot z = \frac{\cos z}{\sin z}$.

Recognition. Simple pole ($\sin$ has a simple zero, $\cos 0 \neq 0$); use $P(z_0)/Q'(z_0)$ with $P = \cos z$, $Q = \sin z$.

Solution. $Q'(z) = \cos z$, so $\text{Res}_{z=0} \cot z = \frac{\cos 0}{\cos 0} = 1$. *Check.* The Laurent series $\cot z = \frac{1}{z} - \frac{z}{3} - \dots$ has z^{-1} coefficient 1, matching the residue.

Worked Example 16.1.14 (NON-example: cancelling orders is not a pole). Decide whether $f(z) = \frac{z^2 - 1}{z - 1}$ has a pole at $z = 1$.

Recognition. Compare the order of the numerator's zero with the denominator's zero before declaring a pole (final clause of the Pitfall below).

Solution. The numerator $z^2 - 1 = (z - 1)(z + 1)$ vanishes to order 1 at $z = 1$, exactly cancelling the order-1 zero of the denominator: $f(z) = z + 1$ off $z = 1$, which extends holomorphically with value 2. So $z = 1$ is a *removable* singularity, not a pole, and $\text{Res}_{z=1} f = 0$. *Check.* $\lim_{z \rightarrow 1} (z - 1)f(z) = \lim_{z \rightarrow 1} (z - 1)(z + 1) = 0$, the simple-pole formula returning 0 as it must for a removable point.

Caution. The simple-pole formula $\lim_{z \rightarrow z_0} (z - z_0)f(z)$ is *only* valid when the pole is genuinely simple. For a double pole, $(z - z_0)f(z)$ still blows up and the limit is infinite, not the residue. Always determine the *order* first; for order n you must multiply by $(z - z_0)^n$ and differentiate $n - 1$ times. Likewise $P(z_0)/Q'(z_0)$ assumes Q has a *simple* zero and $P(z_0) \neq 0$; if P also vanishes, cancel common factors before applying it.

Exercises (Stein, Ch.3).

- Using Euler's formula $\sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i}$, show that the complex zeros of $\sin \pi z$ are exactly the integers and each is of order 1. Calculate the residue of $1/\sin \pi z$ at $z = n \in \mathbb{Z}$.

Additional exercises.

- Determine the order of the zero of each function at the indicated point: (a) $z^3 - z^5$ at $z = 0$; (b) $e^z - 1$ at $z = 0$; (c) $\cos z - 1 + \frac{1}{2}z^2$ at $z = 0$; (d) $(z - 1)^2 \sin(z - 1)$ at $z = 1$.
- Find the residue at every pole: (a) $\frac{z}{z^2 + 1}$; (b) $\frac{1}{z^2(z - 1)}$; (c) $\frac{e^z}{z^2}$; (d) $\frac{\cos z}{z^3}$.
- Show that $\frac{1}{z(z - 1)} = \frac{1}{z - 1} - \frac{1}{z}$ (partial fractions) and read off $\text{Res}_{z=0}$ and $\text{Res}_{z=1}$ directly from this decomposition.

4. Let $f(z) = \frac{g(z)}{(z-a)^3}$ with g holomorphic and $g(a) \neq 0$. Derive the residue $\text{Res}_{z=a} f = \frac{1}{2}g''(a)$ from Theorem 16.1 and verify it for $g(z) = e^z$, $a = 0$.
5. (Non-example.) The function $f(z) = \frac{z-1}{z^2-1}$ appears to have a pole at $z = 1$. Show instead that $z = 1$ is a removable singularity, find the limiting value, and conclude $\text{Res}_{z=1} f = 0$.
6. (Diagram.) On a copy of the complex plane, mark with $\times$ the poles and with $\circ$ the zeros of $f(z) = \frac{z^2(z-2)}{(z-i)(z+i)^2}$, labeling each with its order. State the order of the zero/pole of f at ∞ .
7. Suppose f has a zero of order m at z_0 and g a zero of order n at z_0 , with $m > n$. Determine the type and order of the singularity of g/f at z_0 , and compute $\text{Res}_{z=z_0} \frac{g}{f}$ when $m - n = 1$.
8. Show that $\text{Res}_{z=0} \frac{1}{z \sin z} = 0$ but $\text{Res}_{z=0} \frac{1}{z^2 \sin z} = \frac{1}{6}$, by expanding $\sin z$ in a series. Explain the difference in terms of parity.

16.2 The Residue Formula

The residue theorem

Cauchy's theorem says that the contour integral of a holomorphic function over a toy contour and its interior is 0. The residue formula is the exact correction term when finitely many poles intrude: each pole contributes $2\pi i$ times its residue. The single input is the basic computation $\oint_C \frac{dz}{z-z_0} = 2\pi i$ over a circle about z_0 .

Theorem 16.2.1 (Residue formula, one pole). *Suppose f is holomorphic in an open set containing a circle C and its interior, except for a pole at z_0 inside C . Then, with C positively oriented,*

$$\oint_C f(z) dz = 2\pi i \text{Res}_{z=z_0} f.$$

Proof. Choose a keyhole contour that avoids z_0 , and let the corridor width $\rightarrow 0$; the contributions of the two corridor sides cancel and the outer circle integral equals the integral over a small circle C_ε about z_0 . Write $f(z) = \sum_{k=1}^n \frac{a_{-k}}{(z-z_0)^k} + G(z)$ near z_0 (Theorem 16.1.5), with G holomorphic. By the Cauchy integral formula applied to the constant a_{-1} , $\frac{1}{2\pi i} \oint_{C_\varepsilon} \frac{a_{-1}}{z-z_0} dz = a_{-1}$; for $k \geq 2$, $\oint_{C_\varepsilon} \frac{a_{-k}}{(z-z_0)^k} dz = 0$ (the integrand has a primitive on the punctured disc); and $\oint_{C_\varepsilon} G dz = 0$ by Cauchy's theorem. Summing, $\oint_C f dz = 2\pi i a_{-1}$. $\square$

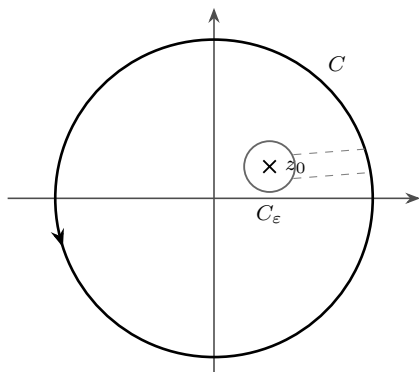


Figure 16.5. Keyhole argument: the corridor sides cancel in the limit, reducing the integral over the outer circle C to one over the small circle C_ε about the pole z_0 . (Schematic.)

Corollary 16.2.2 (Residue formula, many poles / toy contours). *If f is holomorphic in an open set containing a positively oriented toy contour γ and its interior, except for poles at $z_1, \dots, z_N$ inside γ , then*

$$\oint_\gamma f(z) dz = 2\pi i \sum_{k=1}^N \text{Res}_{z=z_k} f.$$

Proof. Use a *multiple* keyhole with one loop avoiding each pole; let the corridor widths $\rightarrow 0$. The outer integral becomes a sum of integrals over small circles, to each of which Theorem 16.2.1 applies. For a general toy contour the same keyhole reduction works because the interior is simply connected away from the poles. $\square$

Method (Evaluating a real integral by residues). *Recognition.* An integral $\int_{-\infty}^{\infty} f(x) dx$ (or a trigonometric integral over $[0, 2\pi]$) with a rational/meromorphic integrand.

1. Extend f to a meromorphic $f(z)$ and pick a family of toy contours γ_R (semicircle, rectangle, sector, keyhole) closing up the real segment.
2. Show the “closing” part of γ_R contributes $\rightarrow 0$ (an ML -bound: length times $\max |f|$).
3. Evaluate $\oint_{\gamma_R} f dz = 2\pi i \sum \text{Res}$ over the *enclosed* poles.
4. Take $R \rightarrow \infty$ (or $\varepsilon \rightarrow 0$) and solve for the desired real integral.

For trigonometric $\int_0^{2\pi} R(\cos \theta, \sin \theta) d\theta$, substitute $z = e^{i\theta}$ so $\cos \theta = \frac{1}{2}(z + z^{-1})$, $\sin \theta = \frac{1}{2i}(z - z^{-1})$, $d\theta = dz/(iz)$, turning it into a contour integral over $|z| = 1$.

Worked contour integrals

Worked Example 16.2.3 (The Poisson-kernel integral $\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$). Evaluate by a semicircular contour.

Recognition. Rational, even, decaying like $|z|^{-2}$; close in the upper half-plane.

Solution. Let $f(z) = \frac{1}{1+z^2} = \frac{1}{(z-i)(z+i)}$, with simple poles at $\pm i$. Take $\gamma_R = [-R, R] \cup C_R^+$, C_R^+ the upper semicircle. Only i is enclosed, and $\text{Res}_{z=i} f = \frac{1}{2i}$, so $\oint_{\gamma_R} f dz = 2\pi i \cdot \frac{1}{2i} = \pi$. On C_R^+ , $|f| \leq \frac{1}{R^2-1}$ and the length is πR , so the arc integral is $\leq \frac{\pi R}{R^2-1} \rightarrow 0$. Hence $\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$. *Check.* The elementary antiderivative gives $[\arctan x]_{-\infty}^{\infty} = \frac{\pi}{2} - (-\frac{\pi}{2}) = \pi$. Agrees.

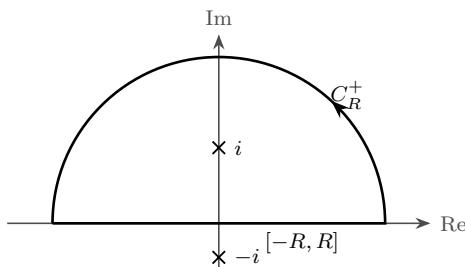


Figure 16.6. Semicircular contour $\gamma_R = [-R, R] \cup C_R^+$ enclosing only the pole i ; the upper-arc contribution vanishes as $R \rightarrow \infty$. (Schematic.)

Worked Example 16.2.4 (A quartic denominator: $\int_{-\infty}^{\infty} \frac{dx}{1+x^4}$). Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^4}$.

Recognition. Decays like $|z|^{-4}$; close in the upper half-plane, enclosing the two roots of $z^4 = -1$ with positive imaginary part.

Solution. The poles are $z_k = e^{i\pi(2k+1)/4}$. In the upper half-plane: $z_0 = e^{i\pi/4}$ and $z_1 = e^{3i\pi/4}$. Using $\text{Res}_{z=z_k} \frac{1}{1+z^4} = \frac{1}{4z_k^3} = \frac{z_k}{4z_k^4} = \frac{z_k}{-4} = -\frac{z_k}{4}$ (since $z_k^4 = -1$),

$$\sum \text{Res} = -\frac{1}{4}(e^{i\pi/4} + e^{3i\pi/4}) = -\frac{1}{4}\left(\frac{\sqrt{2}}{2}(1+i) + \frac{\sqrt{2}}{2}(-1+i)\right) = -\frac{1}{4} \cdot \sqrt{2}i = -\frac{i}{2\sqrt{2}}.$$

Thus the integral $= 2\pi i \cdot \left(-\frac{i}{2\sqrt{2}}\right) = \frac{\pi}{\sqrt{2}}$. *Check.* The standard value is $\int_{-\infty}^{\infty} \frac{dx}{1+x^4} = \frac{\pi}{\sqrt{2}} \approx 2.221$, a positive number, as an integral of a positive function must be.

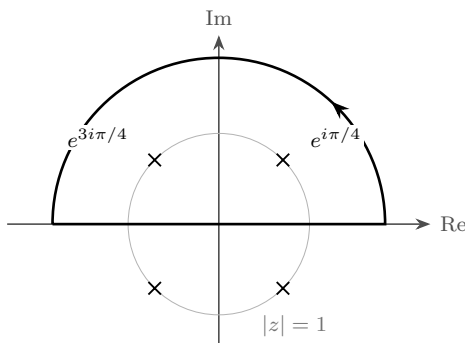


Figure 16.7. Four poles on $|z| = 1$ (roots of $z^4 = -1$); only the two in the upper half-plane are enclosed by γ_R .

Worked Example 16.2.5 (A trigonometric integral via $z = e^{i\theta}$). Evaluate $\int_0^{2\pi} \frac{d\theta}{a + \cos \theta}$ for $a > 1$.

Recognition. A 2π -periodic rational trig integral; push to the unit circle.

Solution. Put $z = e^{i\theta}$, $\cos \theta = \frac{1}{2}(z + z^{-1})$, $d\theta = dz/(iz)$. Then

$$\int_0^{2\pi} \frac{d\theta}{a + \cos \theta} = \oint_{|z|=1} \frac{1}{a + \frac{1}{2}(z + z^{-1})} \frac{dz}{iz} = \frac{2}{i} \oint_{|z|=1} \frac{dz}{z^2 + 2az + 1}.$$

The denominator has roots $z_{\pm} = -a \pm \sqrt{a^2 - 1}$; only z_+ (with $|z_+| < 1$ since $a > 1$) is inside. $\text{Res}_{z=z_+} \frac{1}{z^2 + 2az + 1} = \frac{1}{2z_+ + 2a} = \frac{1}{2\sqrt{a^2 - 1}}$. Hence the integral $= \frac{2}{i} \cdot 2\pi i \cdot \frac{1}{2\sqrt{a^2 - 1}} = \frac{2\pi}{\sqrt{a^2 - 1}}$. *Check.* As $a \rightarrow \infty$ the integrand $\rightarrow 1/a$ and the integral $\rightarrow 2\pi/a$, matching $\frac{2\pi}{\sqrt{a^2 - 1}} \sim \frac{2\pi}{a}$.

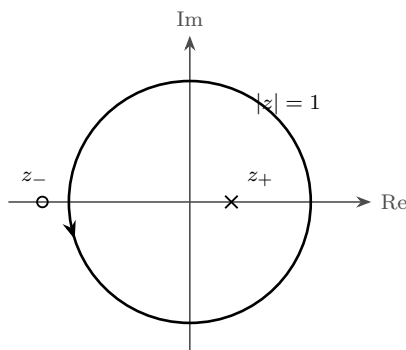


Figure 16.8. Substituting $z = e^{i\theta}$ turns the trig integral into a contour integral; only the inside root z_+ contributes. (Schematic.)

Worked Example 16.2.6 (A Fourier-type integral by a rectangle: Stein's Example 2). Show $\int_{-\infty}^{\infty} \frac{e^{ax}}{1 + e^x} dx = \frac{\pi}{\sin \pi a}$ for $0 < a < 1$.

Recognition. The integrand is not even and decays exponentially; periodicity of e^z in $2\pi i$ suggests a rectangle of height 2π .

Solution. Let $f(z) = \frac{e^{az}}{1 + e^z}$ and take the rectangle γ_R with vertices $\pm R$ and $\pm R + 2\pi i$. Inside, the only pole is $z = \pi i$ (where $e^z = -1$). There $(z - \pi i)f(z) = e^{az} \frac{z - \pi i}{e^z - e^{\pi i}} \rightarrow e^{a\pi i} \cdot \frac{1}{e^{\pi i}} = -e^{a\pi i}$, so $\text{Res}_{\pi i} f = -e^{a\pi i}$ and $\oint_{\gamma_R} f = -2\pi i e^{a\pi i}$. The bottom side gives $I_R \rightarrow I$; the top side (where $z = x + 2\pi i$, so $e^z = e^{2\pi i} e^x$ and $1 + e^z = 1 + e^x$), traversed right-to-left, contributes $-e^{2\pi ia} I_R$. The vertical sides vanish as $R \rightarrow \infty$ (bounded by $Ce^{(a-1)R}$ and Ce^{-aR}). In the limit $I - e^{2\pi ia} I = -2\pi i e^{a\pi i}$, so

$$I = \frac{-2\pi i e^{a\pi i}}{1 - e^{2\pi ia}} = \frac{2\pi i}{e^{a\pi i} - e^{-a\pi i}} \cdot \frac{-e^{2a\pi i}}{-1} = \frac{2\pi i}{e^{\pi ia} - e^{-\pi ia}} = \frac{\pi}{\sin \pi a}.$$

Check. At $a = \frac{1}{2}$, $I = \pi$; and indeed $\int \frac{e^{x/2}}{1 + e^x} dx$ over $\mathbb{R}$, via $u = e^{x/2}$, equals $2 \int_0^{\infty} \frac{du}{1 + u^2} = 2 \cdot \frac{\pi}{2} = \pi$. Agrees.

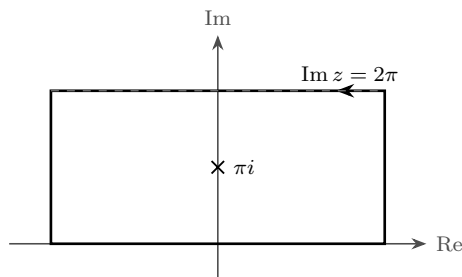


Figure 16.9. Rectangle of height 2π exploiting the $2\pi i$ -periodicity of e^z ; the single pole πi lies inside. (Schematic.)

Worked Example 16.2.7 (Self-reciprocal Fourier transform of $1/\cosh \pi x$). Show $\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{\cosh \pi x} dx = \frac{1}{\cosh \pi \xi}$.

Recognition. Exponential decay both ways; a rectangle of fixed height 2 exploiting periodicity of $\cosh$.

Solution. Let $f(z) = \frac{e^{-2\pi i z \xi}}{\cosh \pi z}$. Poles where $e^{2\pi z} = -1$; inside the rectangle with corners $\pm R$, $\pm R + 2i$ lie $\alpha = i/2$ and $\beta = 3i/2$. One finds $\text{Res}_{\alpha} f = \frac{e^{\pi \xi}}{\pi i}$ and $\text{Res}_{\beta} f = -\frac{e^{3\pi \xi}}{\pi i}$. Vertical sides vanish ($|\cosh \pi z| \rightarrow \infty$); the top side ($z = x + 2i$, with $\cosh \pi(x + 2i) = \cosh \pi x$ and $e^{-2\pi i z \xi} = e^{4\pi \xi} e^{-2\pi i x \xi}$), reversed, gives $-e^{4\pi \xi} I$. So $I - e^{4\pi \xi} I = 2\pi i \left(\frac{e^{\pi \xi}}{\pi i} - \frac{e^{3\pi \xi}}{\pi i} \right) = -2e^{2\pi \xi} (e^{\pi \xi} - e^{-\pi \xi})$. Since $1 - e^{4\pi \xi} = -e^{2\pi \xi} (e^{2\pi \xi} - e^{-2\pi \xi})$, dividing gives

$$I = \frac{2(e^{\pi \xi} - e^{-\pi \xi})}{e^{2\pi \xi} - e^{-2\pi \xi}} = \frac{2}{e^{\pi \xi} + e^{-\pi \xi}} = \frac{1}{\cosh \pi \xi}.$$

Check. At $\xi = 0$, $I = \int_{-\infty}^{\infty} \frac{dx}{\cosh \pi x} = 1$ (a known value), matching $1/\cosh 0 = 1$.

Worked Example 16.2.8 (An oscillatory integral via Jordan's lemma). Evaluate $\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1} dx$.

Recognition. Replace $\cos x$ by e^{ix} (keep the real part) so the upper semicircle decays; close above and pick up the pole at i .

Solution. Let $f(z) = \frac{e^{iz}}{z^2 + 1}$, with a simple pole at i in the upper half-plane. $\text{Res}_{z=i} f = \frac{e^{i \cdot i}}{2i} = \frac{e^{-1}}{2i}$, so $\oint_{\gamma_R} f = 2\pi i \cdot \frac{e^{-1}}{2i} = \pi e^{-1}$. By Jordan's lemma the arc integral vanishes (the factor $e^{iz} = e^{ix} e^{-y}$ decays in the upper half-plane). Taking real parts, $\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1} dx = \pi e^{-1}$. *Check.* The value is positive and ≈ 1.156 ; the companion $\int \frac{\sin x}{x^2 + 1} dx = 0$ by oddness, consistent with the imaginary part of πe^{-1} being 0.

Further worked examples

Worked Example 16.2.9 (A keyhole contour: $\int_0^{\infty} \frac{x^{a-1}}{1+x} dx = \frac{\pi}{\sin \pi a}$). Evaluate the Mellin-type integral for $0 < a < 1$.

Recognition. A fractional power x^{a-1} forces a branch cut; the standard device is the *keyhole* contour hugging the positive real axis, picking up the pole at $z = -1$.

Solution. Take the branch of $z^{a-1} = e^{(a-1)\text{Log } z}$ with $\arg z \in (0, 2\pi)$, cut along $[0, \infty)$. The keyhole γ runs out along the top of the cut, around a big circle C_R , back along the bottom, and around a small circle C_{ε} at 0. The only enclosed pole is $z = -1 = e^{i\pi}$, with $\text{Res}_{z=-1} \frac{z^{a-1}}{1+z} = e^{(a-1)i\pi} = -e^{ia\pi}$, so $\oint_{\gamma} = 2\pi i(-e^{ia\pi})$. On C_R the integrand is $O(R^{a-1}) \cdot \text{length } O(R) = O(R^a) \cdot R^{-1} \rightarrow 0$ (as $a < 1$), and on C_{ε} it is $O(\varepsilon^a) \rightarrow 0$ (as $a > 0$). On the bottom edge $\arg z = 2\pi$, so $z^{a-1} = x^{a-1} e^{2\pi i(a-1)} = x^{a-1} e^{2\pi ia}$. Thus

$$\int_0^{\infty} \frac{x^{a-1}}{1+x} dx - e^{2\pi ia} \int_0^{\infty} \frac{x^{a-1}}{1+x} dx = -2\pi i e^{ia\pi},$$

so $I = \frac{-2\pi i e^{ia\pi}}{1 - e^{2\pi ia}} = \frac{2\pi i}{e^{ia\pi} - e^{-ia\pi}} = \frac{\pi}{\sin \pi a}$. *Check.* At $a = \frac{1}{2}$, $I = \pi$; directly, $\int_0^{\infty} \frac{dx}{\sqrt{x(1+x)}} \stackrel{x=u^2}{=} 2 \int_0^{\infty} \frac{du}{1+u^2} = \pi$. Agrees.

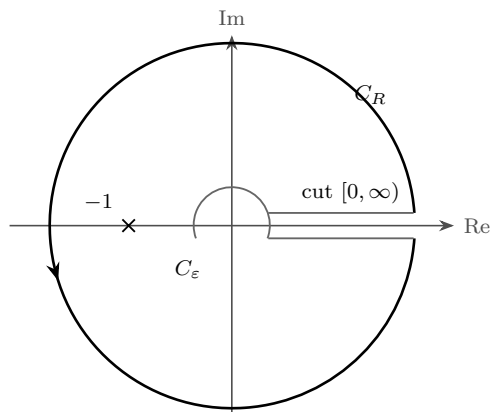


Figure 16.10. Keyhole contour around the branch cut $[0, \infty)$: the top and bottom edges differ by the factor $e^{2\pi ia}$, and the enclosed pole is $z = -1$. (Schematic.)

Worked Example 16.2.10 (Dirichlet integral $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$). Evaluate by an indented semicircular contour.

Recognition. $\frac{\sin x}{x}$ has a removable point at 0 but $\frac{e^{iz}}{z}$ does not; indent the contour with a small semicircle at the origin and use Jordan's lemma on the big arc.

Solution. Integrate $f(z) = \frac{e^{iz}}{z}$ over $\gamma = [-R, -\varepsilon] \cup C_\varepsilon^- \cup [\varepsilon, R] \cup C_R^+$, where C_ε^- is the small upper semicircle (radius ε) detouring *above* 0, traversed clockwise. No poles are enclosed, so $\oint_\gamma f = 0$. On C_R^+ Jordan's lemma gives $\rightarrow 0$. On C_ε^- , $z = \varepsilon e^{i\theta}$ with $\theta : \pi \rightarrow 0$, and $f(z) dz = \frac{e^{i\varepsilon e^{i\theta}}}{\varepsilon e^{i\theta}} i\varepsilon e^{i\theta} d\theta \rightarrow i d\theta$, so $\int_{C_\varepsilon^-} f \rightarrow i \int_\pi^0 d\theta = -i\pi$. Hence $\int_{-\infty}^\infty \frac{e^{ix}}{x} dx - i\pi = 0$, i.e. the principal value is $i\pi$. Taking imaginary parts, $\int_{-\infty}^\infty \frac{\sin x}{x} dx = \pi$, and by evenness $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$. *Check.* The integrand is positive on $(0, \pi)$ and the alternating tail converges; the value $\pi/2 \approx 1.571$ is the standard result.

Caution. The decay estimate on the closing arc is not optional — it is half the proof. For $\int \frac{P}{Q}$ with $\deg Q \geq \deg P + 2$ the semicircle vanishes, but for $\deg Q = \deg P + 1$ (e.g. $\int \frac{\cos x}{x^2 + a^2}$ handled as $\operatorname{Re} \int \frac{e^{ix}}{x^2 + a^2}$) you must keep the oscillatory factor e^{iz} and invoke Jordan's lemma; closing with $\cos z$ fails because $|\cos z|$ grows in the imaginary direction. Always match the contour to the decay of the integrand.

Exercises (Stein, Ch.3).

1. Evaluate $\int_{-\infty}^\infty \frac{dx}{1+x^4}$. Where are the poles of $1/(1+z^4)$?
2. Show that $\int_{-\infty}^\infty \frac{\cos x}{x^2 + a^2} dx = \pi \frac{e^{-a}}{a}$ for $a > 0$.
3. Show that $\int_{-\infty}^\infty \frac{x \sin x}{x^2 + a^2} dx = \pi e^{-a}$ for all $a > 0$.
4. Use contour integration to show that $\int_{-\infty}^\infty e^{-2\pi i x \xi} \frac{1}{(1+x^2)^2} dx = \frac{\pi}{2} (1 + 2\pi |\xi|) e^{-2\pi |\xi|}$ for all real ξ .
5. Show that $\int_{-\infty}^\infty \frac{dx}{(1+x^2)^{n+1}} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \pi$.
6. Prove that $\int_0^{2\pi} \frac{d\theta}{(a + \cos \theta)^2} = \frac{2\pi a}{(a^2 - 1)^{3/2}}$ whenever $a > 1$.
7. Prove that $\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}$ if $a > |b|$ and $a, b \in \mathbb{R}$.

8. Show that $\int_0^1 \log(\sin \pi x) dx = -\log 2$. [Hint: use a contour with a slit at the origin and at 1, indenting around the zeros of $\sin \pi x$.]
9. Show that if $a > 0$, then $\int_0^\infty \frac{\log x}{x^2 + a^2} dx = \frac{\pi}{2a} \log a$. [Hint: use a large semicircle in the upper half-plane indented at the origin.]
10. Show that if $|a| \leq 1$, then $\int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta = 0$. (First prove it for $|a| < 1$, then extend to $|a| = 1$.)
11. Suppose u is not an integer. Prove that $\sum_{n=-\infty}^\infty \frac{1}{(u+n)^2} = \frac{\pi^2}{(\sin \pi u)^2}$ by integrating $f(z) = \frac{\pi \cot \pi z}{(u+z)^2}$ over the circle $|z| = R_N = N + \frac{1}{2}$ ($N \geq |u|$ an integer), summing the residues inside, and letting $N \rightarrow \infty$.

Additional exercises.

1. Evaluate $\int_{-\infty}^\infty \frac{dx}{x^2 + 4}$ by a semicircular contour, and check against $\frac{1}{2} \arctan(x/2)$.
2. Evaluate $\int_{-\infty}^\infty \frac{dx}{(x^2 + 1)(x^2 + 4)}$ using the two poles $i, 2i$ in the upper half-plane.
3. Compute $\int_0^{2\pi} \frac{d\theta}{5 + 4 \cos \theta}$ via $z = e^{i\theta}$.
4. Compute $\int_0^{2\pi} \frac{d\theta}{(2 + \cos \theta)^2}$ (a double pole on the unit circle's interior).
5. Use Jordan's lemma to evaluate $\int_{-\infty}^\infty \frac{\cos x}{x^2 + 1} dx$, writing it as $\operatorname{Re} \int \frac{e^{ix}}{x^2 + 1} dx$.
6. (Non-example.) Explain why one *cannot* evaluate $\int_{-\infty}^\infty \frac{x}{x^2 + 1} dx$ by closing with a large semicircle: show the semicircular arc integral does *not* vanish as $R \rightarrow \infty$ (the decay is only $|z|^{-1}$), and that the integral is only conditionally defined (a principal value).
7. (Diagram.) Sketch the keyhole contour around the positive real axis used to evaluate $\int_0^\infty \frac{x^{a-1}}{1+x} dx$ ($0 < a < 1$). Mark the branch cut, the large and small circles, and the single pole at $z = -1$; then state the value $\frac{\pi}{\sin \pi a}$.
8. Evaluate $\int_{-\infty}^\infty \frac{x^2}{(x^2 + 1)^2} dx$ (a double pole at i), and confirm the answer is positive.
9. Using the contour of radius $R_N = N + \frac{1}{2}$ and the residues of $\pi \cot \pi z / z^2$, outline the evaluation of $\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$.

16.3 Singularities and Meromorphic Functions

Removable singularities

We now classify *all* isolated singularities, not just poles. A singularity z_0 of f is *removable* if f can be *defined* at z_0 so as to become holomorphic on a full neighborhood. The striking fact is that mere *boundedness* forces removability.

Theorem 16.3.1 (Riemann's removable singularity theorem). *If f is holomorphic on Ω except possibly at z_0 , and f is bounded on $\Omega \setminus \{z_0\}$, then z_0 is removable.*

Proof. Work in a small disc D about z_0 with $\overline{D} \subseteq \Omega$, boundary circle C positively oriented. We claim that for $z \in D$, $z \neq z_0$,

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z} d\zeta. \quad (*)$$

The right side defines a holomorphic function on all of D (differentiation under the integral), agreeing with f off z_0 , which is the desired extension. To prove $(*)$, fix $z \neq z_0$ and use a multiple keyhole avoiding z and z_0 . In the limit the corridor cancellations give

$$\oint_C \frac{f(\zeta)}{\zeta - z} d\zeta + \oint_{\gamma} \frac{f(\zeta)}{\zeta - z} d\zeta + \oint_{\gamma'} \frac{f(\zeta)}{\zeta - z} d\zeta = 0,$$

with γ, γ' small negatively oriented circles of radius ε about z and z_0 . As in the Cauchy integral formula, $\oint_{\gamma} \frac{f(\zeta)}{\zeta - z} d\zeta = -2\pi i f(z)$. For γ' , since f is bounded by M and ζ stays a fixed distance from z , $|\oint_{\gamma'} \frac{f(\zeta)}{\zeta - z} d\zeta| \leq C\varepsilon \rightarrow 0$. Letting $\varepsilon \rightarrow 0$ yields $(*)$. $\square$

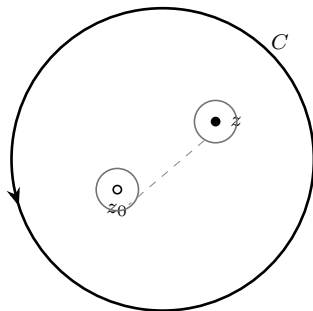


Figure 16.11. Multiple keyhole avoiding the evaluation point z and the singularity z_0 ; the small circle about z_0 contributes nothing since f is bounded. (Schematic.)

Corollary 16.3.2 (Pole detection by blow-up). *An isolated singularity z_0 of f is a pole iff $|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$.*

Proof. $(\Rightarrow)$ If z_0 is a pole, $1/f$ has a zero there, so $|f| \rightarrow \infty$. $(\Leftarrow)$ If $|f| \rightarrow \infty$, then $1/f$ is bounded near z_0 and $|1/f| \rightarrow 0$; by Theorem 16.3.1, $1/f$ extends holomorphically with value 0 at z_0 , i.e. $1/f$ has a zero there, i.e. f has a pole. $\square$

The trichotomy

Every isolated singularity falls into exactly one class:

- *removable* — f bounded near z_0 (no negative Laurent terms);
- *pole* — $|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$ (finitely many negative terms);
- *essential* — neither of the above (infinitely many negative terms).

The model essential singularity is $e^{1/z}$ at 0: its Laurent series $\sum_{n \geq 0} \frac{1}{n! z^n}$ has infinitely many negative powers. Near an essential point the behavior is as wild as possible:

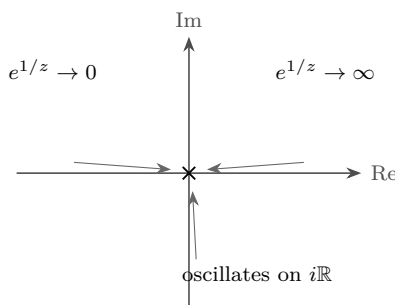


Figure 16.12. $e^{1/z}$ at the origin: $\rightarrow \infty$ from the right, $\rightarrow 0$ from the left, oscillating bounded along the imaginary axis — the signature of an essential singularity.

Theorem 16.3.3 (Casorati–Weierstrass). *If f is holomorphic on the punctured disc $D_r(z_0) \setminus \{z_0\}$ with an essential singularity at z_0 , then $f(D_r(z_0) \setminus \{z_0\})$ is dense in $\mathbb{C}$.*

Proof. Suppose not: there are $w \in \mathbb{C}$, $\delta > 0$ with $|f(z) - w| > \delta$ on the punctured disc. Then $g(z) = \frac{1}{f(z) - w}$ is holomorphic there and bounded by $1/\delta$, so by Theorem 16.3.1 it extends holomorphically over z_0 . If $g(z_0) \neq 0$, then $f(z) - w = 1/g$ is holomorphic at z_0 (removable), contradicting essentiality; if $g(z_0) = 0$, then $f - w$ has a pole at z_0 (Corollary 16.3.2), again contradicting essentiality. Hence the range is dense. $\square$

Remark. Picard's theorem is far stronger: near an essential singularity f takes *every* complex value infinitely often, with at most one exception. For $e^{1/z}$ the lone omitted value is 0 (since e^w never vanishes).

Worked Example 16.3.4 (Classifying three singularities at the origin). Classify the singularity at 0 of (a) $\frac{\sin z}{z}$, (b) $\frac{1 - \cos z}{z^3}$, (c) $z^2 e^{1/z}$.

Recognition. Read the Laurent series and apply the trichotomy by counting negative powers.

Solution. (a) $\frac{\sin z}{z} = 1 - \frac{z^2}{6} + \cdots$: no negative powers, *removable*. (b) $1 - \cos z = \frac{z^2}{2} - \frac{z^4}{24} + \cdots$, so $\frac{1 - \cos z}{z^3} = \frac{1}{2z} - \frac{z}{24} + \cdots$: one negative power, a *simple pole* (residue $\frac{1}{2}$). (c) $z^2 e^{1/z} = z^2 \sum_{n \geq 0} \frac{1}{n! z^n} = z^2 + z + \frac{1}{2} + \frac{1}{6z} + \cdots$: infinitely many negative powers, *essential*. *Check.* For (a), $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$ is finite (bounded $\Rightarrow$ removable); for (c), $\text{Res}_0 = \frac{1}{6}$ from the z^{-1} term, yet the singularity is essential, not a pole.

Worked Example 16.3.5 (NON-pole: an essential point does not satisfy $|f| \rightarrow \infty$). Show $e^{1/z}$ has *no* limit (finite or infinite) as $z \rightarrow 0$, so it is neither removable nor a pole.

Recognition. Test the two non-essential criteria (bounded? blows up?) along different rays.

Solution. Along $z = t > 0$, $e^{1/t} \rightarrow +\infty$. Along $z = -t < 0$, $e^{-1/t} \rightarrow 0$. So f is unbounded (rules out removable) yet $|f| \not\rightarrow \infty$ (rules out pole by Corollary 16.3.2). The singularity is *essential*. *Check.* Along $z = it$, $e^{1/(it)} = e^{-i/t}$ has modulus 1 for all t , confirming bounded oscillation.

Meromorphic functions

Definition 16.3.6. f is *meromorphic* on an open set Ω if there is a set $\{z_0, z_1, \dots\}$ with no limit point in Ω such that f is holomorphic on $\Omega \setminus \{z_0, z_1, \dots\}$ and has poles at the z_k .

To discuss behavior at ∞ , study $F(z) = f(1/z)$ near 0: f has a *pole*, *essential singularity*, or is *holomorphic* at ∞ according as F does at 0. A function meromorphic on $\mathbb{C}$ that is holomorphic or has a pole at ∞ is *meromorphic in the extended plane*. The substitution is precisely precomposition with the inversion $\iota(z) = 1/z$, which interchanges a neighborhood of ∞ with a punctured neighborhood of 0:

$$\begin{array}{ccc} \text{nbhd of } 0 \setminus \{0\} & \xrightarrow{\iota} & \text{nbhd of } \infty \\ & \searrow F=f \circ \iota & \downarrow f \\ & & \widehat{\mathbb{C}} \end{array}$$

so that the singularity type of f at ∞ is read off from that of F at 0.

Theorem 16.3.7 (Meromorphic on $\widehat{\mathbb{C}}$ = rational). *The functions meromorphic in the extended complex plane are exactly the rational functions.*

Proof. A rational function is clearly meromorphic on $\widehat{\mathbb{C}}$. Conversely, let f be meromorphic on $\widehat{\mathbb{C}}$. Then $f(1/z)$ has a pole or removable singularity at 0, so f has only finitely many poles $z_1, \dots, z_n$ in $\mathbb{C}$. Subtract the principal parts: near z_k , $f = f_k + g_k$ with f_k the principal part (a polynomial in $1/(z - z_k)$) and g_k holomorphic. Similarly $f(1/z) = \tilde{f}_\infty(z) + \tilde{g}_\infty(z)$ with $\tilde{f}_\infty$ a polynomial in $1/z$; set $f_\infty(z) = \tilde{f}_\infty(1/z)$. Then $H = f - f_\infty - \sum_k f_k$ has removable singularities at every z_k (we subtracted the principal part) and is bounded near ∞ (we subtracted the principal part at ∞), so H is entire and bounded. By Liouville, H is constant. Hence f is a sum of a constant and rational pieces — a rational function. $\square$

Corollary 16.3.8. *A rational function is determined up to a multiplicative constant by the locations and multiplicities of its zeros and poles.*

Worked Example 16.3.9 (Reconstructing a rational function from its zeros and poles). Find (up to a constant) the rational function with a double zero at 0, simple zeros at ± 1 , and a simple pole at 2; then determine its behavior at ∞ .

Recognition. Use Corollary 16.3.8: build the numerator from zeros, denominator from poles.

Solution. $f(z) = c \frac{z^2(z-1)(z+1)}{z-2} = c \frac{z^2(z^2-1)}{z-2}$. Degree of numerator 4, denominator 1, so $f(1/z)$ behaves like $c z^{-3}$ near 0: a *pole of order 3* at ∞ . *Check.* Zero orders sum to $2 + 1 + 1 = 4$; pole order 1 in $\mathbb{C}$ plus order 3 at ∞ totals 4 — on $\widehat{\mathbb{C}}$ a rational function has equally many zeros and poles counted with multiplicity.

The Riemann sphere

The extended plane $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ is realized geometrically as the sphere S of diameter 1 resting on the origin, via *stereographic projection* from the north pole $N = (0, 0, 1)$. For $W = (X, Y, Z) \in S \setminus \{N\}$ the line NW meets the plane at $w = x + iy$ with

$$x = \frac{X}{1-Z}, \quad y = \frac{Y}{1-Z}; \quad X = \frac{x}{x^2+y^2+1}, \quad Y = \frac{y}{x^2+y^2+1}, \quad Z = \frac{x^2+y^2}{x^2+y^2+1}.$$

As $|w| \rightarrow \infty$, $W \rightarrow N$, so N is the natural “point at infinity.” Thus $\widehat{\mathbb{C}}$ is the compact sphere S — the *one-point compactification* of $\mathbb{C}$ — and a meromorphic function becomes a map $S \rightarrow S$, sending each pole to the tractable point N . Concretely, a function f meromorphic on $\mathbb{C}$ with pole set P is holomorphic $\mathbb{C} \setminus P \rightarrow \mathbb{C}$, and extends across the poles to a single map of the sphere that carries each pole to $\infty = N$:

$$\begin{array}{ccc} \mathbb{C} \setminus P & \xhookrightarrow{\quad} & \widehat{\mathbb{C}} \\ f \downarrow & & \downarrow \widehat{f} \\ \mathbb{C} & \xhookrightarrow{\quad} & \widehat{\mathbb{C}} \end{array}$$

The square commutes off P ; the extension $\widehat{f}$ fills in the values $\widehat{f}(p) = \infty$ at the poles, turning a partial map into a genuine map $S \rightarrow S$.

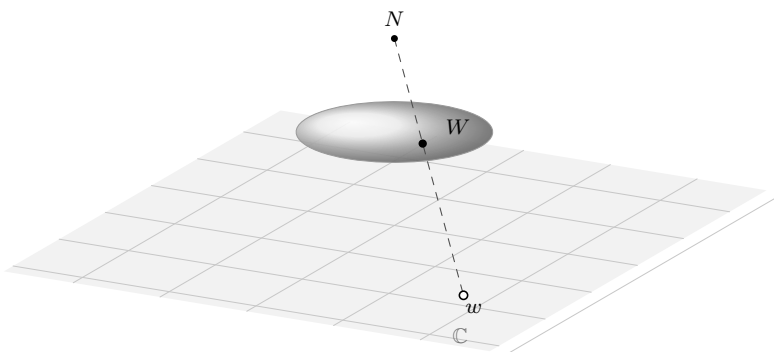


Figure 16.13. The Riemann sphere S (diameter 1, resting on the origin) and stereographic projection from N : the ray Nw meets S at W ; as $|w| \rightarrow \infty$, $W \rightarrow N = \infty$.

Worked Example 16.3.10 (A pole at infinity). Classify the behavior of $f(z) = z^3 + z$ at ∞ .

Recognition. Study $F(z) = f(1/z)$ at 0 and read the singularity type.

Solution. $F(z) = f(1/z) = z^{-3} + z^{-1}$, which has a pole of order 3 at 0. Hence f has a *pole of order 3* at ∞ , and f is meromorphic on $\widehat{\mathbb{C}}$ (a polynomial, hence rational). *Check.* A degree-3 polynomial grows like $|z|^3$ as $|z| \rightarrow \infty$, matching a third-order pole at ∞ .

Further worked examples

Worked Example 16.3.11 (Casorati–Weierstrass in action). For $f(z) = e^{1/z}$ near 0, exhibit a sequence $z_n \rightarrow 0$ with $f(z_n) \rightarrow 7$, confirming the range is dense.

Recognition. An essential singularity (Theorem 16.3.3): every target value, here $w = 7$, is approached. Solve $e^{1/z} = 7$ directly.

Solution. $e^{1/z} = 7$ means $1/z = \log 7 + 2\pi in$ for $n \in \mathbb{Z}$, i.e. $z_n = \frac{1}{\log 7 + 2\pi in}$. As $n \rightarrow \infty$, $|z_n| = \frac{1}{\sqrt{(\log 7)^2 + (2\pi n)^2}} \rightarrow 0$, while $f(z_n) = 7$ *exactly* for every n . *Check.* The z_n are distinct, accumulate only at 0, and f hits the value 7 infinitely often near 0 — precisely the wild behavior Casorati–Weierstrass (and, sharply, Picard) predicts, with 0 the lone omitted value since $e^w \neq 0$.

Worked Example 16.3.12 (Laurent expansion in two annuli). Expand $f(z) = \frac{1}{(z-1)(z-2)}$ in (a) $1 < |z| < 2$ and (b) $|z| > 2$, and read off the residue “at infinity.”

Recognition. A rational function has *different* Laurent series on different annuli; use partial fractions then geometric series adapted to each ring.

Solution. Partial fractions: $f = \frac{1}{z-2} - \frac{1}{z-1}$. (a) For $1 < |z| < 2$ expand $\frac{1}{z-2} = -\frac{1}{2} \frac{1}{1-z/2} = -\sum_{n \geq 0} \frac{z^n}{2^{n+1}}$ (valid $|z| < 2$) and $\frac{1}{z-1} = \frac{1}{z} \frac{1}{1-1/z} = \sum_{n \geq 1} z^{-n}$ (valid $|z| > 1$), so

$$f(z) = -\sum_{n \geq 0} \frac{z^n}{2^{n+1}} - \sum_{n \geq 1} z^{-n}.$$

(b) For $|z| > 2$ both expand in powers of $1/z$: $\frac{1}{z-2} = \sum_{n \geq 1} \frac{2^{n-1}}{z^n}$, $\frac{1}{z-1} = \sum_{n \geq 1} z^{-n}$, so $f(z) = \sum_{n \geq 1} \frac{2^{n-1} - 1}{z^n}$, whose z^{-1} coefficient is $2^0 - 1 = 0$. *Check.* The sum of the finite-plane residues is $\text{Res}_1 f + \text{Res}_2 f = \frac{1}{1-2} + \frac{1}{2-1} = -1 + 1 = 0$, matching the vanishing z^{-1} coefficient in the outer expansion (the residue at ∞ is its negative, also 0).

Caution. “Bounded near z_0 ” is a hypothesis you must *verify*, not assume. Riemann’s theorem gives removability only from a genuine bound; for $1/z$ no bound holds (it is a pole), and for $e^{1/z}$ no bound holds on any sector approaching 0 from the right (essential). Also beware: a singularity that “looks removable” because numerator and denominator both vanish may still be a pole if the numerator’s zero is of *lower* order — always compare orders.

Exercises (Stein, Ch.3).

1. Suppose $f(z)$ is holomorphic in a punctured disc $D_r(z_0) \setminus \{z_0\}$ and satisfies $|f(z)| \leq A|z - z_0|^{-1+\varepsilon}$ for some $\varepsilon > 0$ and all z near z_0 . Show that the singularity of f at z_0 is removable.
2. Prove that every entire function that is also injective has the form $f(z) = az + b$ with $a, b \in \mathbb{C}$, $a \neq 0$. [Hint: apply the Casorati–Weierstrass theorem to $f(1/z)$.]

Problems (Stein, Ch.3).

1. If $f(z)$ is holomorphic in the deleted neighborhood $\{0 < |z - z_0| < r\}$ and has a pole of order k at z_0 , then $f(z) = \frac{a_{-k}}{(z - z_0)^k} + \cdots + \frac{a_{-1}}{z - z_0} + g(z)$ with g holomorphic on $|z - z_0| < r$. Generalize to the Laurent expansion: if f is holomorphic in a region containing the annulus $\{r_1 \leq |z - z_0| \leq r_2\}$ ($0 < r_1 < r_2$), then $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$, converging absolutely in the interior, by writing $f(z) = \frac{1}{2\pi i} \oint_{C_{r_2}} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \oint_{C_{r_1}} \frac{f(\zeta)}{\zeta - z} d\zeta$ for $r_1 < |z - z_0| < r_2$ and arguing as in the proof of the power-series representation.

Additional exercises.

1. Classify the singularity at $z = 0$ of each: (a) $\frac{e^z - 1}{z}$; (b) $\frac{\cos z}{z^2}$; (c) $\sin(1/z)$; (d) $\frac{z}{e^z - 1}$.
2. Show that $z \mapsto e^{1/z}$ takes every nonzero complex value infinitely often in any punctured disc about 0 (a hands-on instance of Casorati–Weierstrass/Picard).
3. For the function $f(z) = \frac{1}{z} + \frac{1}{z-1} + \frac{1}{z-2}$, list all poles and their residues, and determine the order of f at ∞ .
4. Prove that a meromorphic function on $\widehat{\mathbb{C}}$ with no poles in $\mathbb{C}$ and at most a pole at ∞ is a polynomial.
5. (Non-example.) Show that $\sin z$ is *not* meromorphic on the extended plane: examine $F(z) = \sin(1/z)$ at 0 and conclude $\sin$ has an essential singularity at ∞ , so it is not rational.
6. Use the stereographic formulas to compute the image on the Riemann sphere of the points $w = 0$, $w = 1$, and $w = i$, and verify each lies on S .
7. (Diagram.) Draw the Riemann sphere resting on the origin and show the stereographic image W of a point w with $|w| > 1$; indicate how $W \rightarrow N$ as $|w| \rightarrow \infty$.
8. Let f have an essential singularity at z_0 . Show $1/f$ cannot have a pole at z_0 , and deduce $1/f$ also has an essential singularity there (assuming f is nonvanishing near z_0).

16.4 The Argument Principle and Applications

The logarithmic derivative

Although $\log f(z)$ is multiple-valued (one cannot pin $\arg f$ down where $f = 0$), its derivative f'/f is single-valued, and the integral

$$\oint_{\gamma} \frac{f'(z)}{f(z)} dz$$

records the total change in $\arg f$ as z traverses γ . The additivity $\log(f_1 f_2) = \log f_1 + \log f_2$ can fail, but the *logarithmic derivative* is always additive:

$$\frac{(f_1 f_2)'}{f_1 f_2} = \frac{f_1'}{f_1} + \frac{f_2'}{f_2}, \quad \frac{(\prod_k f_k)'}{\prod_k f_k} = \sum_k \frac{f_k'}{f_k}.$$

If f has a zero of order n at z_0 , write $f(z) = (z - z_0)^n g(z)$ with $g(z_0) \neq 0$; then $\frac{f'}{f} = \frac{n}{z - z_0} + \frac{g'}{g}$, so f'/f has a *simple pole with residue n* at z_0 . If instead f has a pole of order n , $f = (z - z_0)^{-n} h$, then $\frac{f'}{f} = \frac{-n}{z - z_0} + \frac{h'}{h}$, a simple pole of residue $-n$. The residue of f'/f thus reads off (signed) multiplicities.

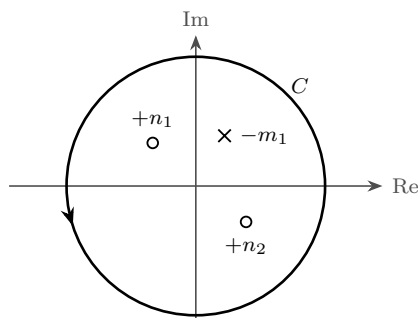


Figure 16.14. f'/f has a simple pole at each zero (residue +order) and each pole (residue −order) of f ; the residue formula then counts them. (Schematic.)

Theorem 16.4.1 (Argument principle). Suppose f is meromorphic in an open set containing a circle C and its interior, with no zeros or poles on C . Then

$$\frac{1}{2\pi i} \oint_C \frac{f'(z)}{f(z)} dz = Z - P,$$

where Z and P are the numbers of zeros and poles of f inside C , counted with multiplicity. The same holds for toy contours.

Proof. By the discussion above, f'/f is meromorphic inside C with simple poles exactly at the zeros and poles of f , with residues equal to the orders (positive at zeros, negative at poles). The residue formula (Corollary 16.2.2) gives $\frac{1}{2\pi i} \oint_C \frac{f'}{f} dz = \sum \text{Res} = \sum(\text{zero orders}) - \sum(\text{pole orders}) = Z - P$. $\square$

The name: as z runs around C , $w = f(z)$ traces a closed curve, and $\frac{1}{2\pi i} \oint_C \frac{f'}{f} dz = \frac{1}{2\pi} \Delta_C \arg f$ is the winding number of that image curve about 0 — the net number of times f wraps around the origin.

Worked Example 16.4.2 (Counting zeros by the argument principle). How many zeros does $p(z) = z^4 + z + 1$ have in $|z| < 2$?

Recognition. A polynomial of degree 4; all four roots lie in some disc. Estimate to confirm they sit inside $|z| < 2$.

Solution. For $|z| = 2$, $|z^4| = 16$ while $|z + 1| \leq 3 < 16$, so p has no zeros on $|z| = 2$ and (Rouché, below, with $f = z^4$, $g = z + 1$) the same number of zeros as z^4 : namely 4. *Check.* A degree-4 polynomial has exactly 4 zeros in $\mathbb{C}$ (with multiplicity); since all satisfy $|z| < 2$ here, the count $Z = 4$ is forced.

Rouché's theorem

Theorem 16.4.3 (Rouché). If f, g are holomorphic in an open set containing a circle C and its interior and $|f(z)| > |g(z)|$ for all $z \in C$, then f and $f + g$ have the same number of zeros inside C .

Proof. For $t \in [0, 1]$ set $f_t = f + tg$, so $f_0 = f$, $f_1 = f + g$. On C , $|f| > |g| \geq t|g|$, so $f_t \neq 0$ on C and the argument principle gives the integer $n_t = \frac{1}{2\pi i} \oint_C \frac{f'_t}{f_t} dz$ (the zero count of f_t). The integrand is jointly continuous in (t, z) since f_t does not vanish on C , so $t \mapsto n_t$ is continuous; being integer-valued, it is constant. Thus $n_0 = n_1$. $\square$

Method (Applying Rouché to locate zeros). *Recognition.* You must count the zeros of h in a disc/region.

1. Split $h = f + g$ so that on the boundary C one term *dominates*: $|f| > |g|$ on C .
2. Choose f to be the term whose zeros you can count outright (often the highest-degree monomial, or a single simple factor).
3. Conclude h has as many zeros inside C as f does.

The art is the split: on a large circle the top-degree term dominates; on a small circle the constant or lowest-degree term dominates.

Worked Example 16.4.4 (Rouché on the unit disc). Show $z^7 - 2z^5 + 6z^3 - z + 1$ has exactly 3 zeros in $|z| < 1$.

Recognition. Choose the dominant term on $|z| = 1$ and split.

Solution. On $|z| = 1$, take $f = 6z^3$ (so $|f| = 6$) and $g = z^7 - 2z^5 - z + 1$, with $|g| \leq 1 + 2 + 1 + 1 = 5 < 6 = |f|$. By Rouché, $h = f + g$ has the same number of zeros as $f = 6z^3$ in $|z| < 1$, namely 3 (a triple zero at 0). *Check.* The inequality is strict on the whole circle, and $6z^3$ visibly has its 3 zeros at the origin inside the disc.

Worked Example 16.4.5 (Rouché proof of the Fundamental Theorem of Algebra). Show every degree- n polynomial $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ has exactly n zeros in $\mathbb{C}$.

Recognition. On a large circle the leading term dominates; apply Rouché with $f = z^n$.

Solution. Pick $R > 1 + |a_{n-1}| + \cdots + |a_0|$. On $|z| = R$, $|z^n| = R^n$ while $|a_{n-1}z^{n-1} + \cdots + a_0| \leq (|a_{n-1}| + \cdots + |a_0|)R^{n-1} < R^n$. By Rouché p has the same number of zeros in $|z| < R$ as z^n , i.e. n . All zeros lie inside, so p has exactly n zeros. *Check.* For $p(z) = z^2 + 1$, $R = 2$ works and indeed there are 2 zeros $\pm i$ inside $|z| < 2$.

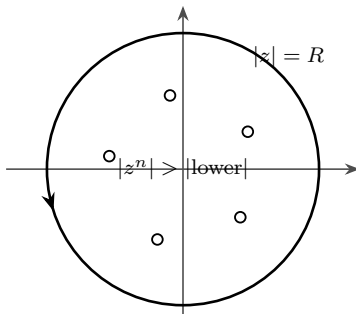


Figure 16.15. On a large circle z^n dominates the lower-order terms; Rouché gives all n roots inside. (Schematic.)

Open mapping and maximum modulus

Theorem 16.4.6 (Open mapping theorem). *A non-constant holomorphic function on a region Ω is an open map.*

Proof. Let $w_0 = f(z_0)$ be in the image. Write $g(z) = f(z) - w = (f(z) - w_0) + (w_0 - w) = F(z) + G(z)$. Choose $\delta > 0$ with $\overline{D_\delta(z_0)} \subseteq \Omega$ and $f(z) \neq w_0$ on $|z - z_0| = \delta$ (zeros are isolated), then $\varepsilon > 0$ with $|f(z) - w_0| \geq \varepsilon$ on that circle. For $|w - w_0| < \varepsilon$ we have $|F| \geq \varepsilon > |G|$ on the circle, so by Rouché $g = F + G$ has a zero inside (as F does, at z_0). Hence every w near w_0 is attained; f maps a neighborhood of z_0 onto a neighborhood of w_0 . $\square$

Theorem 16.4.7 (Maximum modulus principle). *A non-constant holomorphic f on a region Ω cannot attain a maximum of $|f|$ in Ω .*

Proof. If $|f|$ had a maximum at z_0 , then since f is an open map, $f(D)$ for a small disc $D \ni z_0$ is open and contains $f(z_0)$; hence it contains points of modulus larger than $|f(z_0)|$, contradicting maximality. $\square$

Corollary 16.4.8. *If Ω has compact closure $\overline{\Omega}$, f is holomorphic on Ω and continuous on $\overline{\Omega}$, then $\sup_{\overline{\Omega}} |f| = \sup_{\partial\Omega} |f|$.*

Proof. $|f|$ is continuous on the compact $\overline{\Omega}$, so attains its max somewhere; by Theorem 16.4.7 that point cannot be interior unless f is constant. Either way the max is achieved on $\partial\Omega = \overline{\Omega} \setminus \Omega$. $\square$

Worked Example 16.4.9 (Maximum modulus on the unit disc). Find $\max_{|z| \leq 1} |z^2 + z|$.

Recognition. By Corollary 16.4.8 the max sits on $|z| = 1$; parametrize the boundary.

Solution. On $|z| = 1$, $z = e^{i\theta}$, $|z^2 + z| = |z| |z + 1| = |e^{i\theta} + 1| = 2|\cos(\theta/2)|$, maximized at $\theta = 0$ giving 2. So the maximum is 2, attained at $z = 1$. *Check.* The interior value at $z = 0$ is $0 < 2$; the maximum is indeed on the boundary, as the principle requires.

Worked Example 16.4.10 (NON-example: an unbounded region defeats the boundary bound). Show that on the first quadrant Ω , the entire function $F(z) = e^{-iz^2}$ has $|F| = 1$ on $\partial\Omega$ yet is unbounded in Ω .

Recognition. Corollary 16.4.8 needs $\overline{\Omega}$ compact; here it is not, so the conclusion may fail.

Solution. On the boundary $z = x \geq 0$: $|e^{-ix^2}| = 1$; on $z = iy$, $y \geq 0$: $e^{-i(iy)^2} = e^{iy^2}$, modulus 1. But along the ray $z = re^{i\pi/4}$, $z^2 = r^2i$, so $|F| = |e^{-i \cdot r^2 i}| = e^{r^2} \rightarrow \infty$. *Check.* The failure is exactly the unboundedness of Ω ; the modulus is 1 on the edges but explodes along the diagonal, so “sup on boundary” does not control the interior here.

Worked Example 16.4.11 (Zeros in an annulus by two Rouché applications). How many zeros of $p(z) = z^6 + 4z^2 + 1$ lie in the annulus $1 < |z| < 2$?

Recognition. Count zeros inside each boundary circle by Rouché, then subtract.

Solution. On $|z| = 2$: $|z^6| = 64 > |4z^2 + 1| \leq 17$, so p has 6 zeros in $|z| < 2$. On $|z| = 1$: $|4z^2| = 4 > |z^6 + 1| \leq 2$, so p has the same number as $4z^2$, namely 2, in $|z| < 1$. Hence the annulus contains $6 - 2 = 4$ zeros. *Check.* No zeros sit on either circle (both inequalities are strict), so the subtraction is valid; $4 + 2 = 6$ accounts for all roots of the degree-6 polynomial.

Worked Example 16.4.12 (Open mapping in action). Exhibit the open-mapping property for $f(z) = z^2$ at $z_0 = 1$ by showing a small disc about 1 maps onto a neighborhood of $f(1) = 1$.

Recognition. f is non-constant holomorphic, hence open; verify locally by solving $z^2 = w$.

Solution. Near $z_0 = 1$, $f'(1) = 2 \neq 0$, so f is locally invertible with $z = \sqrt{w}$ (principal branch, $\sqrt{1} = 1$). For any w near 1, $z = \sqrt{w}$ lies near 1 and $f(z) = w$; thus f maps a neighborhood of 1 onto a neighborhood of 1. *Check.* $w = 1.21$ gives $z = 1.1$ near 1, and $f(1.1) = 1.21$; the image of a small disc is open, as the theorem guarantees.

Further worked examples

Worked Example 16.4.13 (A transcendental Rouché: $z = e^{-z}$ in the right half). Show $\lambda - z - e^{-z} = 0$ has exactly one root in the right half-plane $\operatorname{Re} z > 0$ for real $\lambda > 1$.

Recognition. No closed disc bounds the half-plane, so apply Rouché on a large half-disc and let $R \rightarrow \infty$; split off the dominant linear term.

Solution. Let $f(z) = \lambda - z$ and $g(z) = -e^{-z}$, so $h = f + g$. Take the contour $\partial\Omega_R$: the segment $[-iR, iR]$ on the imaginary axis plus the right semicircle C_R of radius R . On the imaginary axis $z = iy$: $|f| = |\lambda - iy| = \sqrt{\lambda^2 + y^2} \geq \lambda > 1 = |e^{-iy}| = |g|$. On C_R ($\operatorname{Re} z \geq 0$): $|g| = e^{-\operatorname{Re} z} \leq 1$, while $|f| = |\lambda - z| \geq R - \lambda > 1$ for R large. So $|f| > |g|$ on the whole boundary, and h has the same number of zeros in Ω_R as $f = \lambda - z$, namely the single root $z = \lambda > 0$. Letting $R \rightarrow \infty$ gives exactly one root in $\operatorname{Re} z > 0$. *Check.* On the real axis $\phi(x) = \lambda - x - e^{-x}$ has $\phi(0) = \lambda - 1 > 0$ and $\phi(\lambda) = -e^{-\lambda} < 0$, so a real root sits in $(0, \lambda)$ — consistent with the unique right-half-plane root being real.

Caution. Rouché requires the *strict* inequality $|f| > |g|$ on the *whole* contour. If equality holds anywhere (e.g. $|f| = |g|$ at one boundary point), zeros can sit on C and the count is invalid; perturb the contour or sharpen the estimate. And the maximum modulus principle is about $|f|$, not f or $\operatorname{Re} f$: a non-constant holomorphic function *can* attain a maximum of $\operatorname{Re} f$ nowhere either, but the clean statement is for the modulus.

Exercises (Stein, Ch.3).

1. Use the Cauchy inequalities or the maximum modulus principle to solve the following: (a) Prove that if f is entire and $\sup_{|z|=R} |f(z)| \leq AR^k + B$ for all $R > 0$, some integer $k \geq 0$ and constants $A, B > 0$, then f is a polynomial of degree $\leq k$. (b) Show that if f is holomorphic and bounded in the unit disc and converges uniformly to 0 in the sector $\theta < \arg z < \varphi$ as $|z| \rightarrow 1$, then $f \equiv 0$. (c) Let $w_1, \dots, w_n$ lie on the unit circle. Prove there is a point z on the unit circle such that the product of distances from z to the w_j is at least 1; conclude there is a point w on the circle where the product equals exactly 1. (d) Show that if the real part of an entire function f is bounded, then f is constant.
2. Suppose f, g are holomorphic in a region containing $|z| \leq 1$, f has a simple zero at $z = 0$ and vanishes nowhere else in $|z| \leq 1$. Let $f_\varepsilon(z) = f(z) + \varepsilon g(z)$. Show that for ε sufficiently small: (a) f_ε has a unique zero in $|z| \leq 1$, and (b) if z_ε is that zero, the map $\varepsilon \mapsto z_\varepsilon$ is continuous.
3. Let f be non-constant and holomorphic in an open set containing the closed unit disc. (a) Show that if $|f(z)| = 1$ whenever $|z| = 1$, then the image of f contains the unit disc. (b) If $|f(z)| \geq 1$ whenever $|z| = 1$ and there is $z_0 \in \mathbb{D}$ with $|f(z_0)| < 1$, then the image of f contains the unit disc.
4. Show that there is no holomorphic f in $\mathbb{D}$ that extends continuously to $\partial\mathbb{D}$ with $f(z) = 1/z$ for $z \in \partial\mathbb{D}$.

Problems (Stein, Ch.3).

1. Consider a holomorphic $f : \mathbb{D} \rightarrow \mathbb{C}$ with $f(0) = 0$. By the open mapping theorem, $f(\mathbb{D})$ contains a small disc about 0. Does there exist $r > 0$ with $D_r(0) \subseteq f(\mathbb{D})$ for *all* such f ? (a) With no further restriction, show no such r exists: find $\{f_n\}$ holomorphic in $\mathbb{D}$ with $1/n \notin f_n(\mathbb{D})$; compute $f'_n(0)$ and discuss. (b) Assuming also $f'(0) = 1$, show still no such r exists. [Hint: try $f(z) = \varepsilon(e^{z/\varepsilon} - 1)$.] (The Koebe–Bieberbach theorem says that if in addition f is injective, then such an r exists and the best value is $r = 1/4$.) (c) Show that if $h(z) = \frac{1}{z} + c_0 + c_1z + c_2z^2 + \cdots$ is analytic and injective on $0 < |z| < 1$, then $\sum_{n=1}^{\infty} n|c_n|^2 \leq 1$. (d) If $f(z) = z + a_2z^2 + \cdots$ satisfies the hypotheses, show there is g with $g^2(z) = f(z^2)$. (e) Show $|a_2| \leq 2$, with equality iff $f(z) = \frac{z}{(1 - e^{i\theta}z)^2}$. (f) If $h(z) = \frac{1}{z} + c_0 + c_1z + \cdots$ is injective on $\mathbb{D}$ and avoids z_1, z_2 , show $|z_1 - z_2| \leq 4$. (g) Complete the proof of the theorem.

Additional exercises.

1. Use the argument principle to count the zeros of $f(z) = z^3 - 1$ inside $|z| = 2$ by evaluating $\frac{1}{2\pi i} \oint \frac{f'}{f} dz$ conceptually, and confirm against the three cube roots of unity.
2. Use Rouché's theorem to show $z^5 + 3z + 1$ has exactly one zero in $|z| < 1$.
3. Show that $e^z = 3z$ (equivalently $e^z - 3z = 0$) has exactly one root in $|z| < 1$. [Hint: compare $|3z|$ and $|e^z|$ on $|z| = 1$.]
4. How many zeros does $p(z) = z^8 - 4z^5 + z^2 - 1$ have in $|z| < 1$? In $1 < |z| < 2$?
5. Prove the maximum modulus principle directly from the mean-value property: if $|f(z_0)| \geq |f(z)|$ on a disc, average and conclude $|f|$ is constant near z_0 .
6. (Non-example.) Show that Rouché *fails* without the strict inequality: with $f(z) = z$ and $g(z) = z$ on $|z| = 1$ (so $|f| = |g|$), f has one zero inside but $f + g = 2z$ also has one — yet exhibit $f = z$, $g = -z$ where $f + g \equiv 0$ has infinitely many zeros, illustrating that equality on the boundary voids the conclusion.
7. (Diagram.) For $f(z) = z^2$ on the circle $|z| = 1$, draw the image curve $w = f(e^{i\theta})$ and count how many times it winds around 0; relate the winding number to the number of zeros of f inside $|z| = 1$ via the argument principle.
8. Let f be holomorphic and non-constant on a region Ω and let $K \subseteq \Omega$ be compact. Using the open mapping theorem, show $f(K)$ has nonempty interior whenever K does.
9. Show that if f is entire with $|f(z)| \leq 1$ for all z , then f is constant (Liouville), and contrast this with the maximum modulus principle on a bounded region.

16.5 The Complex Logarithm

Homotopy and simply connected domains

To define the logarithm globally we need to know where a primitive of $1/z$ exists. That is a *global* question, governed by homotopy.

Definition 16.5.1. Two curves γ_0, γ_1 in an open set Ω with common endpoints α, β are *homotopic in Ω* if there is a family $\gamma_s(t)$ ($s \in [0, 1]$, $t \in [a, b]$), jointly continuous in (s, t) , with $\gamma_s(a) = \alpha$, $\gamma_s(b) = \beta$ for all s , and $\gamma_s|_{s=0} = \gamma_0$, $\gamma_s|_{s=1} = \gamma_1$. Informally, one curve deforms continuously into the other without leaving Ω .

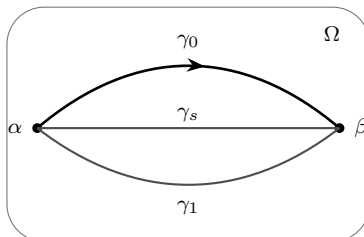


Figure 16.16. A homotopy sweeps γ_0 to γ_1 through curves γ_s , all with the same endpoints, staying inside Ω . (Schematic.)

Theorem 16.5.2 (Homotopy invariance of integrals). *If f is holomorphic in Ω and γ_0, γ_1 are homotopic in Ω with fixed endpoints, then $\int_{\gamma_0} f dz = \int_{\gamma_1} f dz$.*

Proof. The image $K = \{\gamma_s(t)\}$ is compact, so there is $\varepsilon > 0$ with every disc of radius 3ε about a point of K contained in Ω . By uniform continuity choose δ with $\sup_t |\gamma_{s_1}(t) - \gamma_{s_2}(t)| < \varepsilon$ whenever $|s_1 - s_2| < \delta$. For two such nearby curves, cover them with discs $D_0, \dots, D_n$ of radius 2ε and pick consecutive points z_i on γ_{s_1} , w_i on γ_{s_2} with $z_i, z_{i+1}, w_i, w_{i+1} \in D_i$. On each D_i let F_i be a primitive of f ; on overlaps $F_{i+1} - F_i$ is constant, giving $F_{i+1}(z_{i+1}) - F_i(z_{i+1}) = F_{i+1}(w_{i+1}) - F_i(w_{i+1})$. Telescoping, $\int_{\gamma_{s_1}} f - \int_{\gamma_{s_2}} f = F_n(z_{n+1}) - F_n(w_{n+1}) - (F_0(z_0) - F_0(w_0)) = 0$ since the curves share endpoints. Chaining finitely many such steps from $s = 0$ to $s = 1$ proves the claim. $\square$

Definition 16.5.3. A region $\Omega \subseteq \mathbb{C}$ is *simply connected* if any two curves in Ω with the same endpoints are homotopic in Ω .

Example 16.5.4. Convex sets (discs, rectangles, half-planes) are simply connected: $\gamma_s = (1-s)\gamma_0 + s\gamma_1$ stays inside. The *slit plane* $\mathbb{C} \setminus (-\infty, 0]$ is simply connected: writing $\gamma_j(t) = r_j(t)e^{i\theta_j(t)}$ with $|\theta_j| < \pi$, interpolate r and θ linearly. The interior of any toy contour is simply connected. By contrast, the *punctured plane* $\mathbb{C} \setminus \{0\}$ is *not* — a loop around 0 cannot be shrunk past the hole.

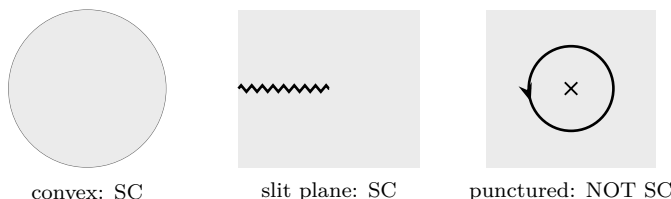


Figure 16.17. Simply connected (no holes) versus not: a loop around the puncture cannot contract. (Schematic.)

Theorem 16.5.5 (Primitives on simply connected domains). *Any holomorphic function on a simply connected region has a primitive. Consequently (Cauchy) $\int_{\gamma} f dz = 0$ for every closed curve γ there.*

Proof. Fix $z_0 \in \Omega$, set $F(z) = \int_{\gamma} f dw$ over any curve in Ω from z_0 to z ; this is well-defined by Theorem 16.5.2. For small h , $F(z+h) - F(z) = \int_{\eta} f dw$ over the segment η from z to $z+h$, and the usual difference-quotient estimate gives $F'(z) = f(z)$. A primitive exists, so closed-curve integrals vanish. $\square$

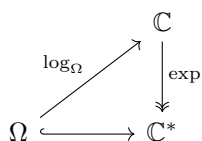
This finally proves $\mathbb{C} \setminus \{0\}$ is not simply connected: if it were, $\oint_{|z|=1} \frac{dz}{z}$ would be 0, but it equals $2\pi i$.

Branches of the logarithm

Theorem 16.5.6 (Existence of a logarithm branch). *If Ω is simply connected with $1 \in \Omega$ and $0 \notin \Omega$, then there is a holomorphic $F = \log_{\Omega}$ on Ω with (i) F holomorphic, (ii) $e^{F(z)} = z$, and (iii) $F(r) = \log r$ for real r near 1.*

Proof. Since $0 \notin \Omega$, $f(z) = 1/z$ is holomorphic on Ω ; define $F(z) = \int_{\gamma} \frac{dw}{w}$ over any curve from 1 to z (well-defined by simple connectivity). As in Theorem 16.5.5, F is holomorphic with $F'(z) = 1/z$. To get $e^{F(z)} = z$, compute $\frac{d}{dz}(ze^{-F(z)}) = e^{-F} - zF'e^{-F} = (1 - zF')e^{-F} = 0$, so $ze^{-F(z)}$ is constant; at $z = 1$, $F(1) = 0$ gives the constant 1, hence $e^{F(z)} = z$. For real r near 1, integrate along the segment: $F(r) = \int_1^r \frac{dx}{x} = \log r$. $\square$

The identity $e^{F(z)} = z$ says exactly that the branch $F = \log_{\Omega}$ is a holomorphic right inverse of the exponential over Ω : it lifts the inclusion $\Omega \hookrightarrow \mathbb{C}^*$ through $\exp$.



Here $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ and $\exp$ is surjective onto it; the lift exists precisely because Ω is simply connected.

In the slit plane $\mathbb{C} \setminus (-\infty, 0]$, the *principal branch* is

$$\operatorname{Log} z = \log r + i\theta, \quad z = re^{i\theta}, \quad |\theta| < \pi,$$

obtained by integrating along the segment $1 \rightarrow r$ then the arc $r \rightarrow z$: $\operatorname{Log} z = \int_1^r \frac{dx}{x} + \int_0^\theta \frac{ire^{it}}{re^{it}} dt = \log r + i\theta$.

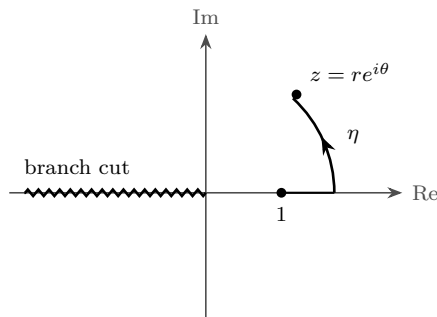


Figure 16.18. Principal branch: integrate $1/w$ from 1 along the axis to r , then along the arc η to z ; the cut $(-\infty, 0]$ keeps $\theta \in (-\pi, \pi)$.

Worked Example 16.5.7 (Principal logarithm of specific points). Compute $\operatorname{Log}(i)$, $\operatorname{Log}(-1+i)$, and $\operatorname{Log}(1)$.

Recognition. Use $\operatorname{Log} z = \log |z| + i \operatorname{Arg} z$ with $\operatorname{Arg} \in (-\pi, \pi)$.

Solution. $\operatorname{Log}(i) = \log 1 + i\frac{\pi}{2} = \frac{i\pi}{2}$. For $-1+i$: $|-1+i| = \sqrt{2}$ and $\operatorname{Arg} = \frac{3\pi}{4}$, so $\operatorname{Log}(-1+i) = \frac{1}{2} \log 2 + \frac{3\pi i}{4}$. And $\operatorname{Log}(1) = \log 1 + 0 = 0$. *Check.* $e^{i\pi/2} = i$; $e^{\frac{1}{2} \log 2 + 3\pi i/4} = \sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}) = \sqrt{2}(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i) = -1+i$. Both recover z .

Worked Example 16.5.8 (NON-additivity of the logarithm). Show $\operatorname{Log}(z_1 z_2) \neq \operatorname{Log} z_1 + \operatorname{Log} z_2$ for $z_1 = z_2 = e^{2\pi i/3}$.

Recognition. The additive law can break when the arguments sum past π .

Solution. $\operatorname{Log} z_1 = \operatorname{Log} z_2 = \frac{2\pi i}{3}$, so the sum is $\frac{4\pi i}{3}$. But $z_1 z_2 = e^{4\pi i/3} = e^{-2\pi i/3}$, whose principal argument is $-\frac{2\pi}{3}$, giving $\operatorname{Log}(z_1 z_2) = -\frac{2\pi i}{3}$. These differ by $2\pi i$. *Check.* The discrepancy $\frac{4\pi i}{3} - (-\frac{2\pi i}{3}) = 2\pi i$ is exactly one full turn — the argument wrapped across the branch cut.

The Taylor series of the principal branch about 1 is, for $|z| < 1$,

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n},$$

since both sides have derivative $1/(1+z)$ and vanish at $z=0$.

Worked Example 16.5.9 (A series value from the log expansion). Use the expansion to evaluate $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n 2^n}$.

Recognition. Recognize the series as $\log(1+z)$ at $z = \frac{1}{2}$.

Solution. With $z = \frac{1}{2}$: $\sum_{n \geq 1} (-1)^{n-1} \frac{(1/2)^n}{n} = \log(1 + \frac{1}{2}) = \log \frac{3}{2}$. *Check.* The first terms $\frac{1}{2} - \frac{1}{8} + \frac{1}{24} - \cdots \approx 0.405$, and $\log 1.5 \approx 0.405$. Agrees.

Powers and logarithms of nonvanishing functions

With a branch of $\log$ in hand, define complex powers $z^\alpha = e^{\alpha \log z}$ on a simply connected Ω with $1 \in \Omega$, $0 \notin \Omega$ and the branch normalized by $\log 1 = 0$. Then $1^\alpha = 1$ and $(z^{1/n})^n = e^{n \cdot \frac{1}{n} \log z} = e^{\log z} = z$. The power map is the composite of the chosen logarithm, multiplication by α , and the exponential:

$$\begin{array}{ccccc} \Omega & \xrightarrow{\log_\Omega} & \mathbb{C} & \xrightarrow{\cdot \alpha} & \mathbb{C} & \xrightarrow{\exp} & \mathbb{C} \\ & & & & \searrow & \nearrow & \\ & & & & z \mapsto z^\alpha & & \end{array}$$

so every branch ambiguity in z^α traces back to the branch chosen for $\log$.

Theorem 16.5.10 (Holomorphic logarithm of a nonvanishing function). *If f is holomorphic and nowhere zero on a simply connected region Ω , then there is a holomorphic g on Ω with $f = e^g$. One writes $g = \log f$.*

Proof. Fix z_0 , set $g(z) = \int_\gamma \frac{f'(w)}{f(w)} dw + c_0$ over a path from z_0 to z , with $e^{c_0} = f(z_0)$; well-defined since Ω is simply connected and f'/f is holomorphic (no zeros of f). Then $g' = f'/f$, and $\frac{d}{dz}(fe^{-g}) = f'e^{-g} - fg'e^{-g} = (f' - f \cdot \frac{f'}{f})e^{-g} = 0$, so fe^{-g} is constant; at z_0 it equals $f(z_0)e^{-c_0} = 1$. Hence $f = e^g$. $\square$

Worked Example 16.5.11 (A branch of $\sqrt{1-z^2}$ on the slit plane). Construct a holomorphic square root of $1-z^2$ on $\mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$.

Recognition. $1-z^2$ is nonvanishing off ± 1 ; on the doubly-slit plane (simply connected) apply Theorem 16.5.10.

Solution. The region is simply connected and $f(z) = 1-z^2$ is holomorphic and nonzero there (zeros ± 1 are removed via the slits), so a holomorphic $\log f$ exists and $\sqrt{1-z^2} := e^{\frac{1}{2}\log f}$ is a holomorphic branch with $(\sqrt{1-z^2})^2 = e^{\log f} = 1-z^2$. Normalize by the branch with value 1 at $z = 0$. *Check.* At $z = 0$, $f = 1$, $\log f = 0$, so $\sqrt{1-z^2} = e^0 = 1$, the expected positive root of 1.

Worked Example 16.5.12 (Computing a complex power). Find the principal value of i^i .

Recognition. Use $z^\alpha = e^{\alpha \operatorname{Log} z}$ with the principal logarithm.

Solution. $\operatorname{Log} i = \frac{i\pi}{2}$, so $i^i = e^{i \cdot \frac{i\pi}{2}} = e^{-\pi/2}$, a positive real number. *Check.* Numerically $e^{-\pi/2} \approx 0.2079$; the other values $e^{-\pi/2} e^{-2\pi k}$ ($k \in \mathbb{Z}$) arise from other branches, but the principal one uses $\operatorname{Arg} i = \pi/2$.

Worked Example 16.5.13 (All values of a multi-valued logarithm). List all values of $\log(-8)$ and identify the principal one.

Recognition. $\log w = \log|w| + i(\operatorname{Arg} w + 2\pi k)$ over all integers k .

Solution. $|-8| = 8$ and $\operatorname{Arg}(-8) = \pi$, so $\log(-8) = \log 8 + i(\pi + 2\pi k) = 3\log 2 + i\pi(2k+1)$, $k \in \mathbb{Z}$. The principal value ($k = 0$) is $3\log 2 + i\pi$. *Check.* $e^{3\log 2 + i\pi} = 8 \cdot e^{i\pi} = -8$, recovering the argument for every k since $e^{2\pi ik} = 1$.

Further worked examples

Worked Example 16.5.14 (A conformal map from a sector to the upper half-plane). Find a holomorphic bijection of the sector $S = \{z : 0 < \arg z < \pi/4\}$ onto the upper half-plane $\mathbb{H} = \{w : \operatorname{Im} w > 0\}$.

Recognition. A power $z \mapsto z^\alpha$ (via the principal log) scales angles at 0 by α ; choose α to open the wedge angle $\pi/4$ up to π .

Solution. The sector has opening angle $\pi/4$; to map it onto the half-plane (opening angle π) take $w = z^4 = e^{4\operatorname{Log} z}$, using the principal branch (well-defined since S avoids the cut). A ray $\arg z = \theta \in (0, \pi/4)$ maps to $\arg w = 4\theta \in (0, \pi)$, sweeping out all of $\mathbb{H}$; the map is a holomorphic bijection there with inverse $z = w^{1/4}$ (principal branch). *Check.* The bounding rays $\arg z = 0$ and $\arg z = \pi/4$ map to $\arg w = 0$ and $\arg w = \pi$, i.e. the positive and negative real axes — exactly $\partial\mathbb{H}$. The midline $\arg z = \pi/8$ maps to $\arg w = \pi/2$, the positive imaginary axis.

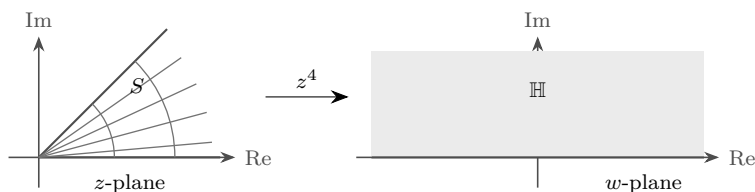


Figure 16.19. The power map $w = z^4$ opens the $\pi/4$ -wedge S onto the upper half-plane $\mathbb{H}$, quadrupling every argument at the origin.

Worked Example 16.5.15 (Branch dependence of $\log$: integrating across the cut). With γ the upper unit semicircle from 1 to -1 , compute $\int_{\gamma} \frac{dz}{z}$ directly and reconcile with $\text{Log}(-1) - \text{Log}(1)$.

Recognition. $1/z$ has primitive $\text{Log } z$ on the slit plane; but γ ends *on* the cut, so take the limiting principal value.

Solution. Parametrize $z = e^{i\theta}$, $\theta : 0 \rightarrow \pi$: $\int_{\gamma} \frac{dz}{z} = \int_0^{\pi} \frac{ie^{i\theta}}{e^{i\theta}} d\theta = \int_0^{\pi} i d\theta = i\pi$. This equals $\text{Log}(-1) - \text{Log}(1) = i\pi - 0$ using the branch with $\arg \in (-\pi, \pi]$ (so $\text{Arg}(-1) = \pi$). *Check.* The *lower* semicircle from 1 to -1 gives $\int_0^{-\pi} i d\theta = -i\pi = \text{Log}(-1) - \text{Log}(1)$ with the branch $\text{Arg}(-1) = -\pi$ instead; the two paths differ by $2\pi i$, the period of $\log$, exactly the winding around the puncture at 0.

Caution. There is no single “the” logarithm: a branch lives on a chosen simply connected domain and differs from another branch by an integer multiple of $2\pi i$. Computing across a branch cut silently adds $\pm 2\pi i$, which is the root cause of $\text{Log}(z_1 z_2) \neq \text{Log } z_1 + \text{Log } z_2$ and of $\sqrt{z}\sqrt{w} \neq \sqrt{zw}$. Always state which branch (which cut) you are on before manipulating logs or fractional powers.

Exercises (Stein, Ch.3).

1. Give another proof of the Cauchy integral formula $f(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z} d\zeta$ using homotopy of curves. [Hint: deform C to a small circle about z , and note that $(f(\zeta) - f(z))/(\zeta - z)$ is bounded.]
2. Certain sets have geometric properties guaranteeing simple connectivity. (a) An open $\Omega \subseteq \mathbb{C}$ is convex if for any two points the segment joining them lies in Ω . Prove a convex open set is simply connected. (b) More generally Ω is star-shaped if there is $z_0 \in \Omega$ such that for every $z \in \Omega$ the segment from z to z_0 lies in Ω . Prove a star-shaped open set is simply connected; conclude that the slit plane $\mathbb{C} \setminus (-\infty, 0]$ (and more generally any sector, convex or not) is simply connected. (c) Give other examples of simply connected open sets.

Problems (Stein, Ch.3).

1. Suppose Ω is a bounded region and L a two-way-infinite line meeting Ω with $\Omega \cap L$ an interval I . Orienting L , let Ω_l, Ω_r be the subregions strictly to the left/right of L , so $\Omega = \Omega_l \cup I \cup \Omega_r$ (disjoint). Prove that if Ω_l and Ω_r are simply connected, then Ω is simply connected.

Additional exercises.

1. Compute the principal logarithm $\text{Log } z$ of (a) -1 , (b) $-i$, (c) $1 + i$, (d) $-1 - i$, giving $\log |z|$ and $\text{Arg } z$ explicitly.
2. Find all values of the multi-valued $\log(-2)$ and of i^i , and identify which is the principal value.
3. Show that on the slit plane $\mathbb{C} \setminus (-\infty, 0]$, $\text{Log } z$ is holomorphic with $\frac{d}{dz} \text{Log } z = \frac{1}{z}$, by differentiating the integral representation.
4. Verify the Taylor expansion $\log(1 + z) = \sum_{n \geq 1} (-1)^{n-1} z^n / n$ by integrating the geometric series for $1/(1 + z)$ term by term on $|z| < 1$.
5. Determine where the function $z^{1/2} = e^{\frac{1}{2} \text{Log } z}$ (principal branch) is holomorphic, and compute its value and derivative at $z = i$.
6. (Non-example.) Show that the additive law fails: find z_1, z_2 on the unit circle with $\text{Log}(z_1 z_2) \neq \text{Log } z_1 + \text{Log } z_2$, and verify the two sides differ by $2\pi i$.
7. (Diagram.) Draw the slit plane $\mathbb{C} \setminus (-\infty, 0]$ with its branch cut, and the path of integration (segment from 1 to r then arc to $z = re^{i\theta}$) defining the principal branch; indicate the range $\theta \in (-\pi, \pi)$.
8. Let Ω be simply connected with $0 \notin \Omega$. Prove that any two branches of the logarithm on Ω differ by a constant $2\pi i k$, $k \in \mathbb{Z}$.

16.6 Fourier Series and Harmonic Functions

Fourier coefficients from power series

There is a direct dictionary between the power-series coefficients of a holomorphic function on a disc and the Fourier coefficients of its restriction to a circle. This is the seed of the connection between complex analysis and Fourier analysis.

Theorem 16.6.1 (Coefficients as Fourier integrals). *If $f(z) = \sum_{n \geq 0} a_n(z - z_0)^n$ converges on $D_R(z_0)$, then for $0 < r < R$,*

$$a_n = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta} d\theta \quad (n \geq 0), \quad 0 = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta} d\theta \quad (n < 0).$$

Proof. By the Cauchy integral formula for derivatives, $a_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \oint_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$ over the circle γ of radius r . Parametrize $\zeta = z_0 + re^{i\theta}$, $d\zeta = ire^{i\theta} d\theta$:

$$a_n = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{r^{n+1} e^{i(n+1)\theta}} ire^{i\theta} d\theta = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta} d\theta.$$

For $n < 0$ the integrand $f(\zeta)(\zeta - z_0)^{-n-1}$ is holomorphic in the disc (no pole), so by Cauchy's theorem the same expression equals 0. $\square$

The interpretation: viewing $\theta \mapsto f(z_0 + re^{i\theta})$ as a function on the circle, its Fourier coefficients vanish for *negative* index, and for $n \geq 0$ they reproduce (up to r^n) the Taylor coefficients. The one-sidedness of the Fourier spectrum is a hallmark of holomorphy. Concretely, extracting a_n factors through restriction of f to the circle C_r followed by taking the n th Fourier coefficient:

$$\begin{array}{ccc} \mathcal{O}(D_R(z_0)) & \xrightarrow{f \mapsto f|_{C_r}} & C(C_r) \\ & \searrow a_n & \downarrow f \mapsto r^n \hat{f}(n) \\ & & \mathbb{C} \end{array}$$

where $\mathcal{O}(D_R(z_0))$ is the holomorphic functions on the disc and $\hat{f}(n)$ is the n th Fourier coefficient of $\theta \mapsto f(z_0 + re^{i\theta})$.

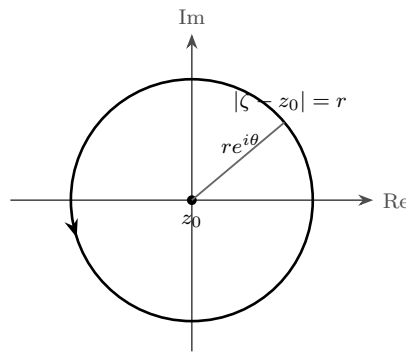


Figure 16.20. Integrating f around the circle of radius r extracts the Taylor coefficient a_n as a Fourier coefficient on $[0, 2\pi]$. (Schematic.)

Mean-value property and harmonic functions

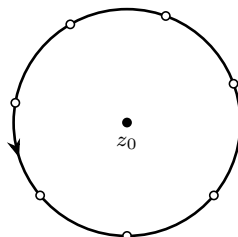
Corollary 16.6.2 (Mean-value property). *If f is holomorphic on $D_R(z_0)$, then for $0 < r < R$,*

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta.$$

Proof. Take $n = 0$ in Theorem 16.6.1: $a_0 = f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta$. □

Taking real parts and recalling that $u = \operatorname{Re} f$ is *harmonic* (satisfies $\Delta u = u_{xx} + u_{yy} = 0$) whenever f is holomorphic gives the mean-value property for harmonic functions.

Corollary 16.6.3 (Mean value for harmonic functions). *If f is holomorphic on $D_R(z_0)$ and $u = \operatorname{Re} f$, then $u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta}) d\theta$ for $0 < r < R$. Every harmonic function on a disc has this property (it is locally the real part of a holomorphic function).*



$u(z_0) = \text{average of } u \text{ over the circle}$

Figure 16.21. Mean-value property: the value of a harmonic (or holomorphic) function at the center equals its average over any concentric circle. (Schematic.)

Worked Example 16.6.4 (Mean-value check for a harmonic function). Verify the mean-value property for $u(x, y) = x^2 - y^2$ at the origin on the circle $|z| = r$.

Recognition. $u = \operatorname{Re}(z^2)$ is harmonic; average u over the circle.

Solution. On $|z| = r$, $z = re^{i\theta}$, $u = \operatorname{Re}(r^2 e^{2i\theta}) = r^2 \cos 2\theta$. Then $\frac{1}{2\pi} \int_0^{2\pi} r^2 \cos 2\theta d\theta = 0 = u(0, 0)$. *Check.* Indeed $u(0, 0) = 0^2 - 0^2 = 0$, matching the average; the oscillation $\cos 2\theta$ integrates to zero over a full period.

Worked Example 16.6.5 (Maximum of a harmonic function is on the boundary). Find max and min of $u(x, y) = x^2 - y^2$ on the closed disc $|z| \leq 1$.

Recognition. Harmonic functions obey a max/min principle (a consequence of the mean-value property); extrema sit on the boundary.

Solution. On $|z| = 1$, $u = \cos 2\theta$, ranging over $[-1, 1]$: max 1 at $\theta = 0$ (point $z = 1$), min -1 at $\theta = \pi/2$ (point $z = i$). The interior value $u(0) = 0$ lies strictly between. *Check.* u harmonic and non-constant cannot attain an interior extremum; indeed both extreme values occur on $|z| = 1$, consistent with the boundary principle.

Worked Example 16.6.6 (NON-example: the modulus is subharmonic, not harmonic). Show $v(z) = |z|^2 = x^2 + y^2$ is *not* harmonic, so it need not satisfy the mean-value *equality*.

Recognition. Test the Laplacian; harmonicity requires $\Delta v = 0$.

Solution. $\Delta v = v_{xx} + v_{yy} = 2 + 2 = 4 \neq 0$, so v is not harmonic (it is subharmonic). On $|z| = r$, $v \equiv r^2$, so the circle-average is r^2 , while $v(0) = 0 < r^2$: the mean-value *equality* fails, replaced by the inequality $v(0) \leq \text{average}$. *Check.* For the genuinely harmonic $u = x^2 - y^2$ above the average matched the center value exactly; here the strict inequality flags non-harmonicity.

Remark. The mean-value property is not merely a curiosity: it characterizes harmonic functions and underlies the maximum principle, Poisson's integral formula (Problem 2), and the rigidity of holomorphic and harmonic functions. The bridge “power series $\leftrightarrow$ Fourier series” developed here is taken up systematically when the Fourier transform enters in the next chapter.

Worked Example 16.6.7 (Recovering a Taylor coefficient as a Fourier integral). For $f(z) = \frac{1}{1 - z/2}$ holomorphic on $|z| < 2$, compute the coefficient a_1 from the circle $|z| = 1$ and match it to the geometric-series value.

Recognition. Apply Theorem 16.6.1 with $n = 1$, $r = 1$.

Solution. The geometric series gives $f(z) = \sum_{n \geq 0} (z/2)^n = \sum_{n \geq 0} 2^{-n} z^n$, so $a_1 = \frac{1}{2}$. By the theorem, $a_1 = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) e^{-i\theta} d\theta$; expanding $f(e^{i\theta}) = \sum_m 2^{-m} e^{im\theta}$, only the $m = 1$ term survives the integral against $e^{-i\theta}$, leaving $2^{-1} = \frac{1}{2}$. *Check.* Both routes give $a_1 = \frac{1}{2}$, and all negative-index integrals vanish since f is holomorphic on the disc.

Worked Example 16.6.8 (Solving a Dirichlet problem by inspection). Find the harmonic function u on $|z| \leq 1$ with boundary values $u(e^{i\theta}) = \cos 2\theta$, and evaluate $u(0)$.

Recognition. Recognize $\cos 2\theta = \operatorname{Re}(z^2)$ on the circle; the harmonic extension is the obvious polynomial.

Solution. On $|z| = 1$, $\operatorname{Re}(z^2) = \operatorname{Re}(e^{2i\theta}) = \cos 2\theta$, and $u(x, y) = \operatorname{Re}(z^2) = x^2 - y^2$ is harmonic on the whole disc with the right boundary data; by uniqueness it is the solution. Then $u(0) = 0$. *Check.* The mean value of $\cos 2\theta$ over $[0, 2\pi]$ is 0, matching $u(0) = 0$ by Corollary 16.6.3.

Further worked examples

Worked Example 16.6.9 (Finding a harmonic conjugate). Given the harmonic $u(x, y) = e^x \cos y$, find a harmonic conjugate v so that $f = u + iv$ is holomorphic, and identify f .

Recognition. A harmonic u is locally $\operatorname{Re} f$; recover v from the Cauchy–Riemann equations $v_x = -u_y$, $v_y = u_x$.

Solution. $u_x = e^x \cos y$ and $u_y = -e^x \sin y$. Then $v_y = u_x = e^x \cos y$ gives $v = e^x \sin y + C(x)$; differentiating, $v_x = e^x \sin y + C'(x)$, and the equation $v_x = -u_y = e^x \sin y$ forces $C'(x) = 0$, so C is constant. Take $v = e^x \sin y$. Then $f = e^x \cos y + ie^x \sin y = e^x e^{iy} = e^z$. *Check.* e^z is entire and $\operatorname{Re} e^z = e^x \cos y = u$; also $\Delta u = u_{xx} + u_{yy} = e^x \cos y - e^x \cos y = 0$, confirming u harmonic.

Worked Example 16.6.10 (Poisson integral at the center). A harmonic u on $|z| \leq 1$ has boundary values $u(e^{i\theta}) = 3 + 2 \cos \theta - \cos 2\theta$. Find $u(0)$ and the full harmonic extension.

Recognition. At the center the Poisson formula collapses to the mean value (Corollary 16.6.3); the extension comes from matching each $\cos k\theta$ to $\operatorname{Re}(z^k)$.

Solution. $u(0) = \frac{1}{2\pi} \int_0^{2\pi} (3 + 2 \cos \theta - \cos 2\theta) d\theta = 3$, since the $\cos \theta$ and $\cos 2\theta$ terms average to 0. The extension replaces $\cos k\theta$ on $|z| = 1$ by $\operatorname{Re}(z^k) = r^k \cos k\theta$: $u(re^{i\theta}) = 3 + 2r \cos \theta - r^2 \cos 2\theta$, harmonic on the disc with the right boundary data. *Check.* At $r = 0$ this gives $u(0) = 3$, matching the mean value; at $r = 1$ it reproduces the prescribed boundary function.

Caution. The mean-value *equality* holds for harmonic functions; for subharmonic functions (like $|f|$ or $|z|^2$) only the *inequality* $u(z_0) \leq \text{average}$ holds, and for superharmonic functions only $\geq$. Do not assume every “nice” function averages to its center value — check $\Delta u = 0$ first. Likewise $|f|$ is generally not harmonic even when f is holomorphic, although $\log |f|$ is harmonic away from the zeros of f .

Exercises (Stein, Ch.3).

1. Prove the maximum principle for harmonic functions: (a) if u is a non-constant real-valued harmonic function in a region Ω , then u cannot attain a maximum (or minimum) in Ω ; (b) if Ω has compact closure $\overline{\Omega}$, u is harmonic in Ω and continuous on $\overline{\Omega}$, then $\sup_{\overline{\Omega}} |u| = \sup_{\partial\Omega} |u|$. [Hint: for (a) write $u = \operatorname{Re} f$ near a putative maximum and use that f is open.]
2. This exercise shows mean-square convergence dominates uniform convergence of analytic functions. For open $U \subseteq \mathbb{C}$ write $\|f\|_{L^2(U)} = (\int_U |f(z)|^2 dx dy)^{1/2}$ and $\|f\|_{L^\infty(U)} = \sup_U |f|$. (a) If f is holomorphic in a neighborhood of $D_r(z_0)$, show that for $0 < s < r$ there is $C > 0$ (depending on s, r) with $\|f\|_{L^\infty(D_s(z_0))} \leq C \|f\|_{L^2(D_r(z_0))}$. (b) Prove that if $\{f_n\}$ is Cauchy in $\|\cdot\|_{L^2(U)}$ for holomorphic f_n , then $\{f_n\}$ converges uniformly on every compact subset of U to a holomorphic function. [Hint: use the mean-value property.]

Problems (Stein, Ch.3).

1. Let u be harmonic in the unit disc and continuous on its closure. Deduce Poisson’s integral formula

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |z_0|^2}{|e^{i\theta} - z_0|^2} u(e^{i\theta}) d\theta \text{ for } |z_0| < 1 \text{ from the special case } z_0 = 0 \text{ (the mean value theorem).}$$

Show that for $z_0 = re^{i\varphi}$, $\frac{1 - |z_0|^2}{|e^{i\theta} - z_0|^2} = \frac{1 - r^2}{1 - 2r \cos(\theta - \varphi) + r^2} = P_r(\theta - \varphi)$, recovering the Poisson kernel. [Hint: set $u_0(z) = u(T(z))$ with $T(z) = \frac{z_0 - z}{1 - \bar{z}_0 z}$, show u_0 harmonic, apply the mean value theorem, and change variables.]

2. Let $g(z) = \frac{1}{2\pi i} \int_{-M}^M \frac{h(x)}{x - z} dx$ with h continuous and supported in $[-M, M]$. (a) Prove g is holomorphic in $\mathbb{C} \setminus [-M, M]$, vanishes at infinity ($\lim_{|z| \rightarrow \infty} |g(z)| = 0$), and the jump of g across $[-M, M]$ is h : $h(x) = \lim_{\varepsilon \rightarrow 0^+} (g(x + i\varepsilon) - g(x - i\varepsilon))$. [Hint: express the difference as a convolution of h with the Poisson kernel.] (b) If h satisfies a Hölder condition $|h(x) - h(y)| \leq C|x - y|^\alpha$, then $g(x \pm i\varepsilon)$ converge uniformly to $g_\pm(x)$ as $\varepsilon \rightarrow 0$, and g is the unique function with: (i) holomorphic outside $[-M, M]$, (ii) vanishing at infinity, (iii) $g(x \pm i\varepsilon)$ converging uniformly to $g_\pm$ with $g_+(x) - g_-(x) = h(x)$. [Hint: if G is another such function, $g - G$ is entire.]

Additional exercises.

1. Verify the mean-value property at the origin for $u(x, y) = \operatorname{Re}(z^3) = x^3 - 3xy^2$ on the circle $|z| = r$, confirming the average is $u(0) = 0$.
2. Use the integral formula for the Taylor coefficients to recover the Cauchy estimate $|a_n| \leq \frac{\sup_{|z|=r} |f|}{r^n}$.
3. Show that $u(x, y) = e^x \cos y$ is harmonic and equals $\operatorname{Re}(e^z)$; verify its mean value at 0 equals $u(0, 0) = 1$.
4. Prove the mean-value property implies the maximum principle for harmonic functions directly (no appeal to holomorphy): a non-constant harmonic u cannot attain an interior maximum.
5. (Non-example.) Show $v(x, y) = x^2 + y^2$ is not harmonic by computing Δv , and demonstrate that its circle-average exceeds its center value (subharmonicity).
6. (Diagram.) Draw a disc with center z_0 and several sample points on a concentric circle, illustrating $u(z_0) =$ average of u over the circle; annotate with the mean-value formula.
7. Compute the Fourier coefficients $a_n = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) e^{-in\theta} d\theta$ for $f(z) = \frac{1}{1-z/2}$ on $|z| = 1$, and check they vanish for $n < 0$.
8. Let u be harmonic on the closed unit disc with $u(e^{i\theta}) = \cos 2\theta$ on the boundary. Use the Poisson kernel (or inspection of $\operatorname{Re}(z^2)$) to find u inside and evaluate $u(0)$ and $u(\frac{1}{2})$.

Chapter 17

The Fourier Transform

17.1 The Class of Functions of Moderate Decrease

The theme of this chapter is the dictionary between *decay of $\hat{f}$ at infinity* and *holomorphic extension of f* . The faster the Fourier transform $\hat{f}$ decays, the wider the horizontal strip into which f continues holomorphically; the extreme case ($\hat{f}$ compactly supported) corresponds to an *entire* extension of exponential type. Every theorem below is, at heart, a single technique: shift the contour of integration — the real line — vertically inside a strip, exploiting that $e^{-2\pi iz\xi}$ decays exponentially when z leaves the real axis.

Throughout, the Fourier transform is normalized as

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx,$$

so the inversion formula will read $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi$ with no extra constants.

Moderate decrease

Definition 17.1.1. A function f on $\mathbb{R}$ is of *moderate decrease* if it is continuous and there is a constant $A > 0$ with

$$|f(x)| \leq \frac{A}{1+x^2} \quad \text{for all } x \in \mathbb{R}.$$

This is the weakest decay used in Book I to make the inversion and Poisson formulas work: it guarantees $f \in L^1(\mathbb{R})$ (since $\int (1+x^2)^{-1} dx = \pi < \infty$), so $\hat{f}$ exists and is bounded. The Schwartz class $\mathcal{S}$ of rapidly decreasing smooth functions is a far more restrictive subclass; moderate decrease is all the integrability we need here.

Remark. One may weaken $A/(1+x^2)$ to $A/(1+|x|^{1+\varepsilon})$ for any $\varepsilon > 0$ without affecting the results: all that is used is L^1 control coming from an integrable majorant.

Worked Example 17.1.2 (The Poisson kernel is of moderate decrease). Fix $y > 0$ and let $P_y(x) = \frac{1}{\pi} \frac{y}{y^2 + x^2}$.

Show P_y is of moderate decrease and recall its Fourier transform.

Recognition. Trigger: verify an explicit majorant $A/(1+x^2)$. Bound $y/(y^2+x^2)$ above by a multiple of $1/(1+x^2)$ uniformly in x .

Solution. Since $y^2 + x^2 \geq \max(y^2, x^2)$, for $|x| \geq 1$ we have $y/(y^2+x^2) \leq y/x^2 \leq (1+y^{-1})y/(1+x^2)$ up to constants; for $|x| \leq 1$ the kernel is bounded by $1/y$. In either regime $P_y(x) \leq A/(1+x^2)$ with $A = A(y)$. Its transform is the standard $\widehat{P}_y(\xi) = e^{-2\pi y|\xi|}$. Check. $\int_{-\infty}^{\infty} P_y(x) dx = \frac{1}{\pi} [\arctan(x/y)]_{-\infty}^{\infty} = 1$, consistent with $\widehat{P}_y(0) = e^0 = 1$.

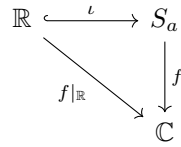
The strip classes $\mathcal{F}_a$ and $\mathcal{F}$

Definition 17.1.3. For $a > 0$ let $S_a = \{z \in \mathbb{C} : |\operatorname{Im} z| < a\}$ be the open horizontal strip of half-width a . The class $\mathcal{F}_a$ consists of all functions f such that

- (i) f is holomorphic in S_a , and
- (ii) there is a constant $A > 0$ with $|f(x + iy)| \leq \frac{A}{1 + x^2}$ for all $x \in \mathbb{R}$ and all $|y| < a$.

The class $\mathcal{F} = \bigcup_{a>0} \mathcal{F}_a$ collects all functions lying in $\mathcal{F}_a$ for *some* $a > 0$.

In words: $\mathcal{F}_a$ is the holomorphic functions on the strip S_a that are of moderate decrease on every horizontal line $\operatorname{Im} z = y$, *uniformly* in $-a < y < a$. The uniform constant A is what lets us slide the contour from one line to another and keep the tail integrals under control.



The restriction $f|_{\mathbb{R}}$ to the real axis is the composite $f \circ \iota$ along the inclusion $\iota : \mathbb{R} \hookrightarrow S_a$; membership in $\mathcal{F}_a$ is a joint condition on the extension f (holomorphy on S_a) and on this restriction (moderate decrease, uniformly over the horizontal lines filling S_a).

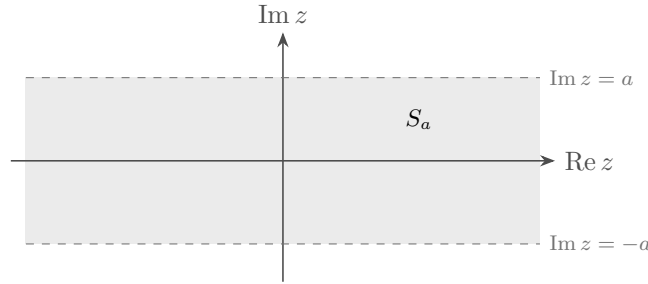


Figure 17.1. The open strip $S_a = \{|\operatorname{Im} z| < a\}$ of half-width a about the real axis. A function of $\mathcal{F}_a$ is holomorphic on S_a and obeys one moderate-decrease bound $A/(1 + x^2)$ uniformly on every horizontal line $\operatorname{Im} z = y$, $|y| < a$.

Worked Example 17.1.4 (The Gaussian lies in $\mathcal{F}_a$ for every a). Show $f(z) = e^{-\pi z^2}$ belongs to $\mathcal{F}_a$ for all $a > 0$.

Recognition. Trigger: confirm holomorphy (clear — it is entire) and a uniform moderate-decrease bound on each line $\operatorname{Im} z = y$. Expand $\left|e^{-\pi z^2}\right|$ with $z = x + iy$.

Solution. Write $z^2 = (x + iy)^2 = x^2 - y^2 + 2ixy$, so $\operatorname{Re}(z^2) = x^2 - y^2$ and $\left|e^{-\pi z^2}\right| = e^{-\pi(x^2 - y^2)} = e^{\pi y^2} e^{-\pi x^2}$. For $|y| < a$ the factor $e^{\pi y^2} \leq e^{\pi a^2}$ is bounded, and $e^{-\pi x^2} \leq C/(1 + x^2)$ for a constant C (a Gaussian beats any polynomial). Hence $|f(x + iy)| \leq A/(1 + x^2)$ uniformly in $|y| < a$. *Check.* On the real axis $y = 0$ this is $e^{-\pi x^2} \leq 1$, manifestly of moderate decrease; and the bound $A = Ce^{\pi a^2}$ degrades as a grows, as one expects since the Gaussian *does* grow vertically.

Worked Example 17.1.5 (A rational function with poles off the strip). Fix $c > 0$ and set $f(z) = \frac{1}{\pi} \frac{c}{c^2 + z^2}$. For which a is $f \in \mathcal{F}_a$?

Recognition. Trigger: locate the poles; the strip half-width is bounded by the imaginary part of the nearest pole.

Solution. The denominator $c^2 + z^2 = (z - ci)(z + ci)$ vanishes at $z = \pm ci$, simple poles on the imaginary axis at height $\pm c$. Thus f is holomorphic exactly where $|\operatorname{Im} z| < c$, giving $f \in \mathcal{F}_a$ for every $0 < a < c$. On a line $\operatorname{Im} z = y$ with $|y| < a < c$ one has $|c^2 + z^2| \geq c^2 - y^2 > 0$ bounded below, and for large $|x|$ the modulus

decays like $1/x^2$, so the uniform $A/(1+x^2)$ bound holds. *Check.* This is exactly the Poisson kernel P_c from above (with $y \rightsquigarrow c$), whose transform $e^{-2\pi c|\xi|}$ decays with rate $2\pi c$ — and Theorem 17.2.1 below predicts decay $e^{-2\pi b|\xi|}$ for every $b < c$, matching.

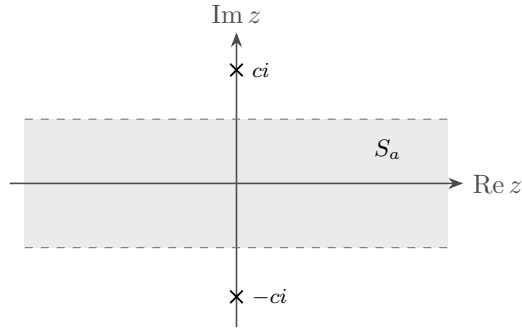


Figure 17.2. The function $f(z) = \frac{1}{\pi} \frac{c}{c^2 + z^2}$ has simple poles at $\pm ci$ (marked $\times$). The nearest pole, at height c , caps the strip: $f \in \mathcal{F}_a$ for every $a < c$, shown shaded.

Worked Example 17.1.6 ($1/\cosh \pi z$ and its strip). Show $f(z) = 1/\cosh \pi z$ lies in $\mathcal{F}_a$ exactly for $|a| < 1/2$. *Recognition.* Trigger: again locate the zeros of the denominator. $\cosh \pi z = 0$ where πz is an odd multiple of $i\pi/2$.

Solution. $\cosh \pi z = \cos(\pi iz)$ (up to the usual identity); concretely $\cosh \pi z = 0$ iff $\pi z = i\pi(k + \frac{1}{2})$, i.e. $z = i(k + \frac{1}{2})$, $k \in \mathbb{Z}$. The nearest poles to the real axis sit at $z = \pm i/2$, so f is holomorphic on $|\operatorname{Im} z| < 1/2$ and $f \in \mathcal{F}_a$ for $|a| < 1/2$. Moderate decrease on each line follows from $|\cosh \pi(x + iy)| \gtrsim e^{\pi|x|}$ for large $|x|$ (exponential growth of $\cosh$), which dominates $1/(1+x^2)$. *Check.* This is the self-dual function $1/\widehat{\cosh \pi x} = 1/\cosh \pi \xi$ used in §17.2; its transform decays like $e^{-\pi|\xi|}$, matching the strip width $a \rightarrow 1/2$ via $2\pi b \rightarrow \pi$.

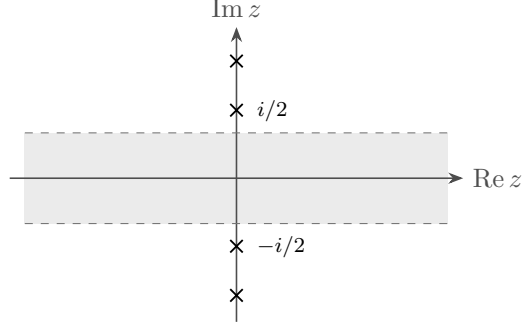


Figure 17.3. The poles of $1/\cosh \pi z$ sit at $z = i(k + \frac{1}{2})$, $k \in \mathbb{Z}$ (marked $\times$). The closest pair $\pm i/2$ bounds the strip (shaded) to half-width $a < 1/2$.

Proposition 17.1.7 (Derivatives stay in $\mathcal{F}$). *If $f \in \mathcal{F}_a$ with $a > 0$, then for every positive integer n the derivative $f^{(n)}$ belongs to $\mathcal{F}_b$ for every $0 < b < a$.*

Proof. Fix $b < a$ and choose r with $0 < r < a - b$. For z in the strip S_b the closed disc $\overline{D(z, r)}$ lies inside S_a , so the Cauchy integral formula for derivatives gives

$$f^{(n)}(z) = \frac{n!}{2\pi i} \oint_{|w-z|=r} \frac{f(w)}{(w-z)^{n+1}} dw.$$

On the circle $|w-z|=r$ we have $\operatorname{Im} w \in (\operatorname{Im} z - r, \operatorname{Im} z + r) \subseteq (-a, a)$, and writing $w = u + iv$ with $u = \operatorname{Re} w$ near $\operatorname{Re} z = x$, the moderate-decrease bound on f gives $|f(w)| \leq A/(1+u^2) \leq A'/(1+x^2)$ (the shift by at most r in the real part changes the majorant only by a bounded factor). Hence

$$\left| f^{(n)}(z) \right| \leq \frac{n!}{2\pi} \cdot 2\pi r \cdot \frac{A'/(1+x^2)}{r^{n+1}} = \frac{n! A'}{r^n} \cdot \frac{1}{1+x^2},$$

a uniform moderate-decrease bound on S_b . Holomorphy of $f^{(n)}$ on S_b is automatic. Thus $f^{(n)} \in \mathcal{F}_b$. $\square$

Method (Deciding membership in $\mathcal{F}_a$). *Recognition trigger.* You are handed an explicit f and asked for which a it lies in $\mathcal{F}_a$ (or in $\mathcal{F}$).

1. **Find the singularities.** The poles / branch points nearest the real axis cap the strip: a must be strictly below the smallest $|\operatorname{Im}(\text{singularity})|$.
2. **Check moderate decrease on lines.** On $\operatorname{Im} z = y$, $|f(x + iy)|$ must be $\leq A/(1 + x^2)$ with A uniform over $|y| < a$. Exponential growth of the denominator (as for cosh) or a degree- ≥ 2 rational tail suffices.
3. **If entire** (no singularities), only step 2 matters; but watch vertical growth (the Gaussian's $e^{\pi y^2}$ is fine for finite a , but an e^{z^2} would fail).

Worked Example 17.1.8 (A NON-example: vertical growth kills membership). Show $f(z) = e^{z^2}$ belongs to no $\mathcal{F}_a$, even though it is entire.

Recognition. Trigger: “entire but is it in $\mathcal{F}$?” Apply step 2 of the method — test the moderate-decrease bound on horizontal lines, and on the *real* line itself.

Solution. With $z = x + iy$, $\operatorname{Re}(z^2) = x^2 - y^2$, so $|e^{z^2}| = e^{x^2 - y^2}$. On the real axis ($y = 0$) this is $e^{x^2} \rightarrow \infty$: f is not even bounded on $\mathbb{R}$, let alone bounded by $A/(1 + x^2)$. So condition (ii) fails for every a , and $f \notin \mathcal{F}$. *Check.* Contrast with the Gaussian $e^{-\pi z^2}$, where the sign in the exponent is reversed and $|f| = e^{-\pi x^2} e^{\pi y^2}$ decays in x ; the moral is that membership is governed by behaviour in the *real* direction, and e^{z^2} blows up there.

Worked Example 17.1.9 (A NON-example: a pole on the real axis). Explain why $f(z) = 1/z$ is in no $\mathcal{F}_a$.

Recognition. Trigger: locate singularities (step 1). A pole on $\mathbb{R}$ itself means the strip cannot even contain the real axis.

Solution. $1/z$ has a pole at $z = 0$, which lies on the real axis. No strip $S_a = \{|\operatorname{Im} z| < a\}$ of positive half-width can avoid it, so f is not holomorphic on any such strip; condition (i) fails outright. *Check.* Even ignoring holomorphy, $|1/x| \rightarrow \infty$ as $x \rightarrow 0$, so the moderate-decrease bound also fails. A function of $\mathcal{F}$ must be *honestly analytic across* the real line.

Further worked examples

Worked Example 17.1.10 (The transform of $a/\pi(a^2 + x^2)$ by a semicircle). For $a > 0$ and $\xi > 0$, evaluate $\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{a}{a^2 + x^2} e^{-2\pi i x \xi} dx$ directly by closing a contour, and confirm the answer $e^{-2\pi a \xi}$.

Recognition. Trigger: a rational L^1 function times an oscillatory exponential. With $z = x + iy$ one has $|e^{-2\pi i z \xi}| = e^{2\pi y \xi}$, so for $\xi > 0$ the factor decays as $y \rightarrow -\infty$: close with a large *lower* semicircle and pick up the pole at $z = -ia$.

Solution. Let $g(z) = \frac{a}{\pi(a^2 + z^2)} e^{-2\pi i z \xi}$, with simple poles at $z = \pm ia$. The lower semicircle of radius R contributes $O(1/R) \cdot (\text{bounded}) \rightarrow 0$ since the integrand is $O(1/R^2)$ and the exponential factor is ≤ 1 on the lower arc. The enclosed pole is $z = -ia$, and the contour is traversed clockwise, contributing $-2\pi i \operatorname{Res}_{z=-ia} g$. Compute

$$\operatorname{Res}_{z=-ia} g = \frac{a}{\pi} \frac{e^{-2\pi i(-ia)\xi}}{2(-ia)} = \frac{a}{\pi} \frac{e^{-2\pi a \xi}}{-2ia} = \frac{e^{-2\pi a \xi}}{-2\pi i}.$$

Hence the integral equals $-2\pi i \cdot \frac{e^{-2\pi a \xi}}{-2\pi i} = e^{-2\pi a \xi}$, as claimed. (For $\xi < 0$ one closes *up* around $z = +ia$ and gets $e^{2\pi a \xi}$; together $e^{-2\pi a |\xi|}$.) *Check.* At $\xi = 0$ the value is 1, matching $\frac{1}{\pi} \int a/(a^2 + x^2) dx = 1$; and the decay rate $2\pi a$ matches the pole height a , in exact agreement with Theorem 17.2.1.

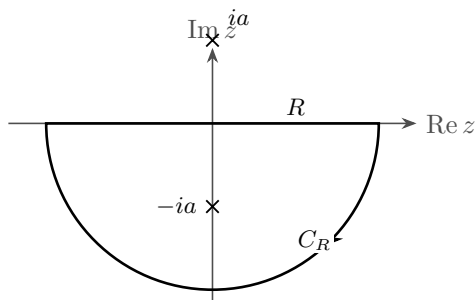


Figure 17.4. Closing the contour in the lower half-plane for $\xi > 0$. The clockwise semicircle C_R of radius R encloses only the simple pole $z = -ia$ (marked $\times$); the arc contribution vanishes as $R \rightarrow \infty$. (Schematic.)

Worked Example 17.1.11 (A degree-4 rational transform via residues). Compute $I(\xi) = \int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{1+x^4} dx$ for $\xi = 0$, illustrating the residue method that Stein's Exercise 4 applies to general $Q(x)$ of degree ≥ 2 .

Recognition. Trigger: $1/Q$ with $\deg Q = 4 \geq 2$, no real roots. At $\xi = 0$ the exponential is 1; close in the upper half-plane and sum residues at the two roots there.

Solution. The roots of $1+z^4 = 0$ are $z = e^{i\pi/4}, e^{3i\pi/4}, e^{5i\pi/4}, e^{7i\pi/4}$; the upper two are $\alpha_1 = e^{i\pi/4}$ and $\alpha_2 = e^{3i\pi/4}$. At a simple root α of $1+z^4$, $\text{Res}_{z=\alpha} \frac{1}{1+z^4} = \frac{1}{4\alpha^3} = \frac{\alpha}{4\alpha^4} = \frac{\alpha}{-4}$ (since $\alpha^4 = -1$). So the sum of upper residues is $-\frac{1}{4}(\alpha_1 + \alpha_2) = -\frac{1}{4}(e^{i\pi/4} + e^{3i\pi/4}) = -\frac{1}{4}\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) = -\frac{1}{4} \cdot i\sqrt{2} = -\frac{i\sqrt{2}}{4}$. Hence $I(0) = 2\pi i \cdot (-\frac{i\sqrt{2}}{4}) = \frac{\pi\sqrt{2}}{2} = \frac{\pi}{\sqrt{2}}$. *Check.* $\int_{-\infty}^{\infty} \frac{dx}{1+x^4} = \frac{\pi}{\sqrt{2}}$ is the standard value; positivity of the integrand confirms the sign.

Worked Example 17.1.12 (Strip width of a product). If $f \in \mathcal{F}_a$ and $g \in \mathcal{F}_b$, for which half-width is the product fg guaranteed to lie in $\mathcal{F}$?

Recognition. Trigger: combine two membership certificates. Holomorphy survives products on the common strip; the decay bounds multiply.

Solution. On the common strip $S_{\min(a,b)}$ both f and g are holomorphic, hence so is fg . The bounds give $|f(x+iy)g(x+iy)| \leq \frac{A}{1+x^2} \cdot \frac{B}{1+x^2} = \frac{AB}{(1+x^2)^2} \leq \frac{AB}{1+x^2}$, uniformly for $|y| < \min(a,b)$. Thus $fg \in \mathcal{F}_{\min(a,b)}$, with the *stronger* decay $(1+x^2)^{-2}$ to spare. *Check.* Take $f = g = 1/(1+z^2) \in \mathcal{F}_a$ for $a < 1$; the product $1/(1+z^2)^2$ still has its nearest poles at $\pm i$, so its strip is again $a < 1$, consistent with $\min(a,a) = a$.

Worked Example 17.1.13 (A NON-example: a logarithmic branch point on $\mathbb{R}$). Show $f(z) = \text{Log } z$ (principal branch) lies in no $\mathcal{F}_a$.

Recognition. Trigger: locate singularities. A branch point, like a pole, must be kept off the closed strip; the principal Log has its branch point and cut on the real axis.

Solution. The principal logarithm $\text{Log } z = \ln |z| + i \text{Arg } z$ has a branch point at $z = 0$ and a cut along $(-\infty, 0]$, all on the real axis. So f is not even single-valued or continuous across $\mathbb{R}$, let alone holomorphic on a strip S_a containing it; condition (i) fails. *Check.* Moreover $|\text{Log } z| \sim \ln |x| \rightarrow \infty$ as $|x| \rightarrow \infty$, so it cannot satisfy any $A/(1+x^2)$ decay either — it fails both conditions of Definition 17.1.3.

Exercises (Stein, Ch.4).

1. If $f \in \mathcal{F}_a$ with $a > 0$, then for any positive integer n one has $f^{(n)} \in \mathcal{F}_b$ whenever $0 \leq b < a$. [Hint: modify the solution to Exercise 8 in Chapter 2, i.e. use the Cauchy integral formula for derivatives.]

Additional exercises.

1. Show that $f(z) = \frac{1}{1+z^2}$ belongs to $\mathcal{F}_a$ for every $0 < a < 1$ but not for $a \geq 1$. Identify the obstruction.
2. Determine the largest strip S_a on which $f(z) = \frac{1}{\cosh(\pi z/2)}$ is holomorphic, and conclude the values of a for which $f \in \mathcal{F}_a$.

3. Prove that if $f, g \in \mathcal{F}_a$ then $f + g \in \mathcal{F}_a$ and $cf \in \mathcal{F}_a$ for every constant c ; thus $\mathcal{F}_a$ is a vector space. Is $fg \in \mathcal{F}_a$ in general? Discuss the decay of fg .
4. Let $f \in \mathcal{F}_a$. Show that the horizontal translate $f_h(z) = f(z - h)$, $h \in \mathbb{R}$, again lies in $\mathcal{F}_a$, and that the vertical translate $f(z - ic)$, $|c| < a$, lies in $\mathcal{F}_{a-|c|}$.
5. Show that $f(z) = e^{-\pi z^2} \cos(2\pi z)$ lies in $\mathcal{F}_a$ for every $a > 0$. (Combine the Gaussian estimate with $|\cos(2\pi z)| \leq e^{2\pi|\operatorname{Im} z|}$.)
6. (Non-example.) Exhibit a function that is holomorphic on the whole strip S_a and bounded on every horizontal line, yet does *not* belong to $\mathcal{F}_a$ because the moderate-decrease constant A is not uniform in y . [Hint: try $f(z) = 1/(1 + (z/(a - |\operatorname{Im} z|))^2)$ schematically, or simpler, a bounded function with no decay such as $f \equiv 1$.]
7. (Diagram problem.) For $f(z) = \frac{1}{\pi} \frac{2}{4 + z^2}$, draw the complex plane marking the poles and shading the maximal strip S_a on which $f \in \mathcal{F}_a$. Label the value of a and the residues at the poles.
8. Suppose $f \in \mathcal{F}_a$ and f has a zero of order m at a point $z_0 \in S_a$. Show that $f(z)/(z - z_0)^m$ extends holomorphically across z_0 and again lies in $\mathcal{F}_b$ for every $b < a$ with $b \neq |\operatorname{Im} z_0|$. Explain the role of moderate decrease in the conclusion.

17.2 Action of the Fourier Transform and the Inversion Formula

For $f \in \mathcal{F}$ we now prove three theorems — exponential decay of $\hat{f}$, the inversion formula, and the Poisson summation formula — all by the *same* device: shift the contour of integration off the real axis and pick up the gain coming from $e^{-2\pi i z \xi}$.

The key estimate: the imaginary part of z controls the exponential

For $z = x + iy$,

$$|e^{-2\pi i z \xi}| = |e^{-2\pi i(x+iy)\xi}| = |e^{-2\pi i x \xi}| e^{2\pi y \xi} = e^{2\pi y \xi}.$$

So lowering the contour ($y < 0$) gains a factor $e^{2\pi y \xi} < 1$ when $\xi > 0$, and raising it ($y > 0$) gains when $\xi < 0$. This single observation drives the whole section.

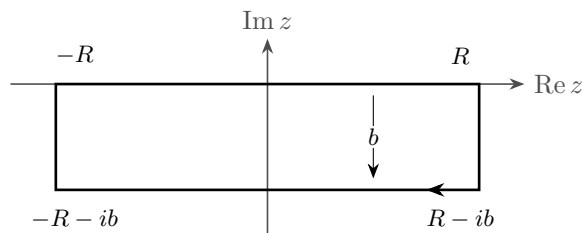


Figure 17.5. The rectangular contour for Theorem 17.2.1 when $\xi > 0$: Cauchy's theorem slides the real segment down by b to the line $\operatorname{Im} z = -b$, the gain factor $e^{-2\pi b \xi} < 1$ appearing along the way. The two vertical sides vanish as $R \rightarrow \infty$. (Schematic.)

Theorem 17.2.1 (Exponential decay of $\hat{f}$). *If $f \in \mathcal{F}_a$ for some $a > 0$, then for every $0 \leq b < a$ there is a constant B with*

$$|\hat{f}(\xi)| \leq B e^{-2\pi b |\xi|} \quad (\xi \in \mathbb{R}).$$

Proof. The case $b = 0$ says only that $\hat{f}$ is bounded: indeed $|\hat{f}(\xi)| \leq \int |f(x)| dx \leq A \int (1 + x^2)^{-1} dx = A\pi$, since $|e^{-2\pi i x \xi}| = 1$.

Now take $0 < b < a$ and first suppose $\xi > 0$. Set $g(z) = f(z)e^{-2\pi iz\xi}$, holomorphic on S_a . Apply Cauchy's theorem to the rectangle with corners $\pm R$ and $\pm R - ib$ (the real segment, the right side down to $-ib$, the lower segment, the left side back up). On the left vertical side $z = -R - it$, $0 \leq t \leq b$,

$$\left| \int_{-R}^{-R-ib} g(z) dz \right| \leq \int_0^b |f(-R-it)| \left| e^{-2\pi i(-R-it)\xi} \right| dt \leq \int_0^b \frac{A}{1+R^2} e^{-2\pi t\xi} dt = O\left(\frac{1}{R^2}\right),$$

using $|e^{-2\pi i(-R-it)\xi}| = e^{-2\pi t\xi} \leq 1$. The right side is estimated the same way. So the two vertical contributions vanish as $R \rightarrow \infty$, and Cauchy's theorem gives, in the limit,

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x-ib) e^{-2\pi i(x-ib)\xi} dx. \quad (17.1)$$

Taking absolute values and using $|e^{-2\pi i(x-ib)\xi}| = e^{-2\pi b\xi}$ together with $|f(x-ib)| \leq A/(1+x^2)$,

$$|\hat{f}(\xi)| \leq e^{-2\pi b\xi} \int_{-\infty}^{\infty} \frac{A}{1+x^2} dx = (A\pi) e^{-2\pi b\xi} = B e^{-2\pi b\xi}.$$

For $\xi < 0$, shift the contour *up* by b instead; the same computation yields $|\hat{f}(\xi)| \leq B e^{2\pi b\xi} = B e^{-2\pi b|\xi|}$. $\square$

The further f extends (the larger a), the larger we may take b , hence the faster $\hat{f}$ decays. Section 17.3 pushes this to the limit: *compact support* of $\hat{f}$.

Worked Example 17.2.2 (Decay rate for the Poisson kernel). Use Theorem 17.2.1 to predict the decay of $\widehat{P}_c$ for $P_c(x) = \frac{1}{\pi} \frac{c}{c^2+x^2}$, and compare with the exact value.

Recognition. Trigger: read off the strip half-width and quote $e^{-2\pi b|\xi|}$ for every $b < a$.

Solution. From §17.1.3, $P_c \in \mathcal{F}_a$ for all $a < c$, so the theorem gives $|\widehat{P}_c(\xi)| \leq B_b e^{-2\pi b|\xi|}$ for every $b < c$.

Letting $b \uparrow c$, the decay rate approaches $e^{-2\pi c|\xi|}$. *Check.* The exact transform is $\widehat{P}_c(\xi) = e^{-2\pi c|\xi|}$, so the predicted rate is sharp — the supremum $b \rightarrow c$ is exactly attained by the true exponential rate, and the pole at height c is the obstruction.

A computational lemma

Lemma 17.2.3. *If $A > 0$ and $B \in \mathbb{R}$, then $\int_0^\infty e^{-(A+iB)\xi} d\xi = \frac{1}{A+iB}$.*

Proof. Since $|e^{-(A+iB)\xi}| = e^{-A\xi}$ with $A > 0$, the integral converges absolutely. For finite R ,

$$\int_0^R e^{-(A+iB)\xi} d\xi = \left[-\frac{e^{-(A+iB)\xi}}{A+iB} \right]_0^R = \frac{1 - e^{-(A+iB)R}}{A+iB} \xrightarrow{R \rightarrow \infty} \frac{1}{A+iB},$$

because $|e^{-(A+iB)R}| = e^{-AR} \rightarrow 0$. $\square$

The Fourier inversion formula

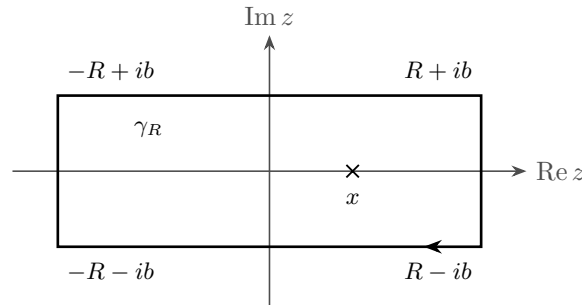


Figure 17.6. The rectangular contour γ_R for Theorem 17.2.4, formed by the shifted lines L_1 (lower) and L_2 (upper). It encircles the simple pole of $f(\zeta)/(\zeta - x)$ at $\zeta = x$ (marked $\times$), whose residue is $f(x)$. (Schematic.)

Theorem 17.2.4 (Fourier inversion on $\mathcal{F}$). *If $f \in \mathcal{F}$, then $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi$ for all $x \in \mathbb{R}$.*

Proof. By Theorem 17.2.1 the integrand $\hat{f}(\xi)e^{2\pi i x \xi}$ is dominated by an exponentially decaying function, so the integral is absolutely convergent. Split at 0 and treat $\int_0^{\infty}$ first. Say $f \in \mathcal{F}_a$ and fix $0 < b < a$. By (17.1), $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(u - ib)e^{-2\pi i(u - ib)\xi} du$ for $\xi > 0$. Insert this and interchange (justified by absolute convergence):

$$\begin{aligned} \int_0^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi &= \int_{-\infty}^{\infty} f(u - ib) \int_0^{\infty} e^{-2\pi i(u - ib - x)\xi} d\xi du \\ &= \int_{-\infty}^{\infty} f(u - ib) \frac{1}{2\pi b + 2\pi i(u - x)} du && \text{(Lemma 17.2.3,} \\ &&& A = 2\pi b, B = 2\pi(u - x)) \\ &= \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{f(u - ib)}{u - ib - x} du = \frac{1}{2\pi i} \int_{L_1} \frac{f(\zeta)}{\zeta - x} d\zeta, \end{aligned}$$

where $L_1 = \{u - ib : u \in \mathbb{R}\}$ is the real line shifted *down* by b , left to right. An identical computation for $\xi < 0$ (shifting up) gives

$$\int_{-\infty}^0 \hat{f}(\xi) e^{2\pi i x \xi} d\xi = -\frac{1}{2\pi i} \int_{L_2} \frac{f(\zeta)}{\zeta - x} d\zeta,$$

with L_2 the real line shifted *up* by b . Now $f(\zeta)/(\zeta - x)$ has a simple pole at $\zeta = x$ with residue $f(x)$. Applying the residue theorem to the rectangle γ_R with horizontal sides L_1, L_2 (truncated at $\pm R$) and letting $R \rightarrow \infty$ — the vertical sides vanish by moderate decrease — gives

$$f(x) = \frac{1}{2\pi i} \oint_{\gamma_R} \frac{f(\zeta)}{\zeta - x} d\zeta = \frac{1}{2\pi i} \int_{L_1} \frac{f(\zeta)}{\zeta - x} d\zeta - \frac{1}{2\pi i} \int_{L_2} \frac{f(\zeta)}{\zeta - x} d\zeta = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi.$$

(The orientation makes L_2 traversed right-to-left as the top of γ_R , accounting for the sign.) □

Theorems 17.2.1 and 17.2.4 together say the Fourier transform $\mathcal{F} : f \mapsto \hat{f}$ and the inverse transform $\mathcal{F}^{-1} : g \mapsto (x \mapsto \int g(\xi) e^{2\pi i x \xi} d\xi)$ undo one another on the class $\mathcal{F}$:

$$f \xrightleftharpoons[\mathcal{F}^{-1}]{\mathcal{F}} \hat{f}$$

Inversion is exactly the statement that the round trip $\mathcal{F}^{-1} \circ \mathcal{F}$ is the identity on $\mathcal{F}$.

The Poisson summation formula

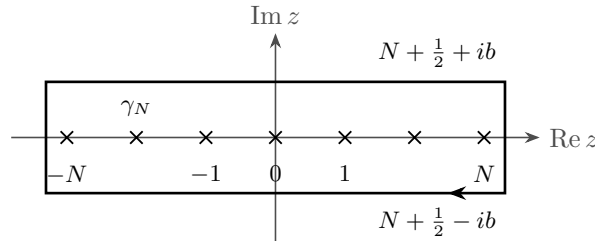


Figure 17.7. The rectangular contour γ_N for Theorem 17.2.5, with corners $\pm(N + \frac{1}{2}) \pm ib$. It encloses the integer poles $z = -N, \dots, N$ (marked $\times$) of $f(z)/(e^{2\pi i z} - 1)$, each contributing residue $f(n)/2\pi i$. (Schematic.)

Theorem 17.2.5 (Poisson summation). *If $f \in \mathcal{F}$, then $\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \hat{f}(n)$.*

Proof. Say $f \in \mathcal{F}_a$, fix $0 < b < a$. The function $1/(e^{2\pi iz} - 1)$ has simple poles at the integers with residue $1/(2\pi i)$ (since $\frac{d}{dz}(e^{2\pi iz} - 1)|_{z=n} = 2\pi i$). Thus $f(z)/(e^{2\pi iz} - 1)$ has simple poles at each n with residue $f(n)/2\pi i$. Apply the residue theorem to the rectangle γ_N with corners $\pm(N + \frac{1}{2}) \pm ib$:

$$\sum_{|n| \leq N} f(n) = \oint_{\gamma_N} \frac{f(z)}{e^{2\pi iz} - 1} dz.$$

As $N \rightarrow \infty$ the left side converges to $\sum_{n \in \mathbb{Z}} f(n)$ (moderate decrease), the vertical sides vanish, and the horizontal sides become the shifted lines L_1 (down by b) and L_2 (up by b):

$$\sum_{n \in \mathbb{Z}} f(n) = \int_{L_1} \frac{f(z)}{e^{2\pi iz} - 1} dz - \int_{L_2} \frac{f(z)}{e^{2\pi iz} - 1} dz. \quad (17.2)$$

On L_1 we have $|e^{2\pi iz}| > 1$, so the geometric expansion $\frac{1}{w-1} = \sum_{n \geq 0} w^{-n-1}$ (valid for $|w| > 1$, with $w = e^{2\pi iz}$) gives $\frac{1}{e^{2\pi iz} - 1} = e^{-2\pi iz} \sum_{n \geq 0} e^{-2\pi inz}$. On L_2 we have $|e^{2\pi iz}| < 1$, and $\frac{1}{w-1} = -\sum_{n \geq 0} w^n$ gives $\frac{1}{e^{2\pi iz} - 1} = -\sum_{n \geq 0} e^{2\pi inz}$. Substituting into (17.2) and integrating term by term (then shifting each L_j back to the real axis via (17.1)),

$$\begin{aligned} \sum_{n \in \mathbb{Z}} f(n) &= \sum_{n=0}^{\infty} \int_{L_1} f(z) e^{-2\pi i(n+1)z} dz + \sum_{n=0}^{\infty} \int_{L_2} f(z) e^{2\pi inz} dz \\ &= \sum_{n=0}^{\infty} \hat{f}(n+1) + \sum_{n=0}^{\infty} \hat{f}(-n) = \sum_{n=1}^{\infty} \hat{f}(n) + \sum_{n=0}^{\infty} \hat{f}(-n) = \sum_{n \in \mathbb{Z}} \hat{f}(n). \end{aligned}$$

□

Applications of Poisson summation

The Gaussian is its own transform: $\int e^{-\pi x^2} e^{-2\pi i x \xi} dx = e^{-\pi \xi^2}$. The change of variables $x \mapsto t^{1/2}(x+a)$ shows that $f(x) = e^{-\pi t(x+a)^2}$ has transform $\hat{f}(\xi) = t^{-1/2} e^{-\pi \xi^2/t} e^{2\pi i a \xi}$. Both lie in $\mathcal{F}$, so Poisson summation yields

$$\sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2} e^{-\pi n^2/t} e^{2\pi i n a}. \quad (17.3)$$

Worked Example 17.2.6 (The functional equation of the theta function). Deduce from (17.3) that $\vartheta(t) = t^{-1/2} \vartheta(1/t)$, where $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$ for $t > 0$.

Recognition. Trigger: specialize the Gaussian Poisson identity. Set $a = 0$ and read both sides as ϑ evaluated at t and $1/t$.

Solution. Put $a = 0$ in (17.3): the left side is $\sum_n e^{-\pi t n^2} = \vartheta(t)$, and the right side is $t^{-1/2} \sum_n e^{-\pi n^2/t} = t^{-1/2} \vartheta(1/t)$. Hence $\vartheta(t) = t^{-1/2} \vartheta(1/t)$. *Check.* At the fixed point $t = 1$ this reads $\vartheta(1) = 1^{-1/2} \vartheta(1)$, a tautology, as the self-dual point must be; and as $t \rightarrow \infty$, $\vartheta(t) \rightarrow 1$ while $t^{-1/2} \vartheta(1/t) \sim t^{-1/2} \cdot t^{1/2} = 1$ (since $\vartheta(s) \sim s^{-1/2}$ as $s \rightarrow 0$), consistent. This functional equation underlies the analytic continuation of the Riemann zeta function (Chapter 6).

Worked Example 17.2.7 (A cosh summation identity). Starting from the self-dual $1/\widehat{\cosh} \pi x = 1/\cosh \pi \xi$,

derive $\sum_n \frac{e^{-2\pi i a n}}{\cosh(\pi n/t)} = \sum_n \frac{t}{\cosh(\pi(n+a)t)}$ for $t > 0$, $a \in \mathbb{R}$.

Recognition. Trigger: scale and modulate the self-dual function, then apply Poisson summation. The $e^{-2\pi i a x}$ factor on f becomes a real shift on $\hat{f}$.

Solution. Let $f(x) = e^{-2\pi i a x} / \cosh(\pi x/t)$. Scaling $x \mapsto x/t$ in $1/\widehat{\cosh} \pi x = 1/\cosh \pi \xi$ and using the modulation rule $e^{-2\pi i a x} g(\xi) = \hat{g}(\xi+a)$ gives $\hat{f}(\xi) = t / \cosh(\pi(\xi+a)t)$. Poisson summation $\sum_n f(n) = \sum_n \hat{f}(n)$ is exactly the claimed identity. *Check.* At $a = 0$, $t = 1$ both sides equal $\sum_n 1/\cosh(\pi n)$, the same series — a self-consistency forced by self-duality.

Caution. Every shift of the contour and every term-by-term integration above is licensed by the *uniform* moderate-decrease bound of $\mathcal{F}_a$ — the constant A does not depend on which line $\text{Im } z = y$ you sit on. Drop that uniformity and the vertical sides of the rectangles need not vanish, and the interchanges of sum/integral can fail. When you only know decay on the real axis (no strip), you are in the regime of §17.3, not this one.

Further worked examples

Worked Example 17.2.8 (Transform of a rational function with poles in both half-planes). Compute

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{(x^2 + 1)(x^2 + 4)} dx \text{ for all } \xi \in \mathbb{R}.$$

Recognition. Trigger: $R(x) = P/Q$ with $\deg Q - \deg P = 4 \geq 2$ and no real poles — exactly Stein's Exercise 5 setup. Split on the sign of ξ and close in the half-plane where $|e^{-2\pi i z \xi}| \leq 1$, summing residues.

Solution. The poles are $z = \pm i, \pm 2i$. For $\xi > 0$ close *down* (around $-i, -2i$). Partial fractions: $\frac{1}{(z^2+1)(z^2+4)} = \frac{1}{3}(\frac{1}{z^2+1} - \frac{1}{z^2+4})$. Using the single-pole transform $\int \frac{e^{-2\pi i x \xi}}{x^2+c^2} dx = \frac{\pi}{c} e^{-2\pi c|\xi|}$ (the Poisson-kernel result, $c > 0$), we get

$$\hat{f}(\xi) = \frac{1}{3} \left(\pi e^{-2\pi|\xi|} - \frac{\pi}{2} e^{-4\pi|\xi|} \right) = \frac{\pi}{3} e^{-2\pi|\xi|} - \frac{\pi}{6} e^{-4\pi|\xi|}.$$

Check. At $\xi = 0$ this is $\frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$; and $\int \frac{dx}{(x^2+1)(x^2+4)} = \frac{1}{3}(\pi - \frac{\pi}{2}) = \frac{\pi}{6}$ directly. The slowest decay rate 2π is set by the nearest pole at height 1, exactly as Theorem 17.2.1 predicts.

Worked Example 17.2.9 (A double pole produces a polynomial-times-exponential transform). Find

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{(x^2 + 1)^2} dx \text{ for } \xi > 0.$$

Recognition. Trigger: a rational transform whose pole at $-i$ is now a *double* pole. Stein's Exercise 5(a) warns that coincident roots make polynomials in ξ appear; compute the order-2 residue.

Solution. Close down for $\xi > 0$ around the double pole $z = -i$. Write $g(z) = \frac{e^{-2\pi i z \xi}}{(z-i)^2(z+i)^2}$ and

$$\text{Res}_{z=-i} g = \frac{d}{dz} \left[\frac{e^{-2\pi i z \xi}}{(z-i)^2} \right]_{z=-i}.$$

Differentiating, $\frac{d}{dz} \frac{e^{-2\pi i z \xi}}{(z-i)^2} = \frac{-2\pi i \xi e^{-2\pi i z \xi}}{(z-i)^2} - \frac{2e^{-2\pi i z \xi}}{(z-i)^3}$. At $z = -i$: $(z-i) = -2i$, $(z-i)^2 = -4$, $(z-i)^3 = 8i$, and $e^{-2\pi i(-i)\xi} = e^{-2\pi \xi}$. So the residue is $e^{-2\pi \xi} \left(\frac{-2\pi i \xi}{-4} - \frac{2}{8i} \right) = e^{-2\pi \xi} \left(\frac{\pi i \xi}{2} + \frac{i}{4} \right)$. Clockwise closure multiplies by $-2\pi i$:

$$\hat{f}(\xi) = -2\pi i e^{-2\pi \xi} \left(\frac{\pi i \xi}{2} + \frac{i}{4} \right) = \pi e^{-2\pi \xi} \left(\pi \xi + \frac{1}{2} \right) = \frac{\pi}{2} (1 + 2\pi \xi) e^{-2\pi \xi}.$$

By evenness $\hat{f}(\xi) = \frac{\pi}{2} (1 + 2\pi |\xi|) e^{-2\pi |\xi|}$. *Check.* $\hat{f}(0) = \frac{\pi}{2}$, matching $\int \frac{dx}{(x^2+1)^2} = \frac{\pi}{2}$; the extra factor $|\xi|$ is the “polynomial in ξ ” Stein predicts from the double root.

Worked Example 17.2.10 (Poisson summation evaluates $\sum 1/(n^2 + a^2)$). Use Poisson summation to prove

$$\sum_{n \in \mathbb{Z}} \frac{1}{n^2 + a^2} = \frac{\pi}{a} \coth(\pi a) \text{ for } a > 0 \text{ (Stein's Exercise 6)}.$$

Recognition. Trigger: a sum of $f(n)$ with $f(x) = 1/(x^2 + a^2) \in \mathcal{F}$. Apply $\sum f(n) = \sum \hat{f}(n)$ with the known transform.

Solution. Here $\hat{f}(\xi) = \frac{\pi}{a} e^{-2\pi a |\xi|}$. Poisson summation gives $\sum_n \frac{1}{n^2 + a^2} = \frac{\pi}{a} \sum_n e^{-2\pi a |n|}$. The right sum telescopes as a geometric series: $\sum_{n \in \mathbb{Z}} e^{-2\pi a |n|} = 1 + 2 \sum_{n \geq 1} e^{-2\pi a n} = 1 + 2 \frac{e^{-2\pi a}}{1 - e^{-2\pi a}} = \frac{1 + e^{-2\pi a}}{1 - e^{-2\pi a}} = \coth(\pi a)$. Hence $\sum_n \frac{1}{n^2 + a^2} = \frac{\pi}{a} \coth(\pi a)$. *Check.* As $a \rightarrow 0^+$ the $n = 0$ term $1/a^2$ dominates and $\frac{\pi}{a} \coth(\pi a) \sim \frac{\pi}{a} \cdot \frac{1}{\pi a} = \frac{1}{a^2}$, consistent; as $a \rightarrow \infty$, $\coth(\pi a) \rightarrow 1$ so the sum $\sim \pi/a \rightarrow 0$, also correct.

Worked Example 17.2.11 (A NON-application: Poisson summation needs decay, not just continuity). Explain why one cannot apply Theorem 17.2.5 to $f(x) = 1/(1 + |x|)$ to evaluate $\sum_n 1/(1 + |n|)$.

Recognition. Trigger: check the hypothesis $f \in \mathcal{F}$ before summing. Membership requires holomorphy on a strip and $A/(1+x^2)$ decay.

Solution. Two failures. First, $1/(1+|x|)$ is not differentiable at $x = 0$ (the $|x|$ corner), let alone holomorphic on any strip, so condition (i) of $\mathcal{F}_a$ fails. Second, its decay is only $1/|x|$, not $1/x^2$, so even the moderate-decrease bound fails. Indeed $\sum_n 1/(1+|n|)$ diverges (harmonic tail), so no finite identity could hold. *Check.* Contrast $1/(1+x^2) \in \mathcal{F}$, for which the sum converges and Poisson summation does apply — the quadratic decay is exactly what rescues both sides.

Exercises (Stein, Ch.4).

- Suppose f is continuous and of moderate decrease, and $\hat{f}(\xi) = 0$ for all $\xi \in \mathbb{R}$. Show that $f = 0$ by completing the following outline. (a) For each fixed real t consider $A(z) = \int_{-\infty}^t f(x)e^{-2\pi iz(x-t)} dx$ and $B(z) = -\int_t^{\infty} f(x)e^{-2\pi iz(x-t)} dx$; show $A(\xi) = B(\xi)$ for all $\xi \in \mathbb{R}$. (b) Prove that the function F equal to A in the closed upper half-plane and B in the lower half-plane is entire and bounded, hence constant; in fact show $F = 0$. (c) Deduce that $\int_{-\infty}^t f(x) dx = 0$ for all t , and conclude $f = 0$.
- Show, by contour integration, that if $a > 0$ and $\xi \in \mathbb{R}$ then $\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{a}{a^2 + x^2} e^{-2\pi i x \xi} dx = e^{-2\pi a|\xi|}$, and check that $\int_{-\infty}^{\infty} e^{-2\pi a|\xi|} e^{2\pi i \xi x} d\xi = \frac{1}{\pi} \frac{a}{a^2 + x^2}$.
- Suppose Q is a polynomial of degree ≥ 2 with distinct roots, none on the real axis. Calculate $\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{Q(x)} dx$ for $\xi \in \mathbb{R}$ in terms of the roots of Q . What happens when several roots coincide? [Hint: treat $\xi < 0$, $\xi = 0$, $\xi > 0$ separately; use residues.]
- More generally, let $R(x) = P(x)/Q(x)$ be a rational function with $\deg Q \geq \deg P + 2$ and $Q(x) \neq 0$ on the real axis. (a) Prove that if $\alpha_1, \dots, \alpha_k$ are the roots of R in the upper half-plane, then there exist polynomials $P_j(\xi)$ of degree less than the multiplicity of α_j with $\int_{-\infty}^{\infty} R(x)e^{-2\pi i x \xi} dx = \sum_{j=1}^k P_j(\xi)e^{-2\pi i \alpha_j \xi}$ when $\xi < 0$. (b) In particular, if Q has no zeros in the upper half-plane, then $\int_{-\infty}^{\infty} R(x)e^{-2\pi i x \xi} dx = 0$ for $\xi < 0$. (c) Show similar results hold for $\xi > 0$ (argue in the lower half-plane). (d) Show $\int_{-\infty}^{\infty} R(x)e^{-2\pi i x \xi} dx = O(e^{-a|\xi|})$ as $|\xi| \rightarrow \infty$ for some $a > 0$; determine the best possible a in terms of the roots of R . [Hint: for (a) use residues; powers of ξ arise from differentiating $R(z)e^{-2\pi i z \xi}$.]
- Prove that $\frac{1}{\pi} \sum_{n=-\infty}^{\infty} \frac{a}{a^2 + n^2} = \sum_{n=-\infty}^{\infty} e^{-2\pi a|n|}$ whenever $a > 0$, and hence show the sum equals $\coth \pi a$.
- The Poisson summation formula applied to specific examples yields interesting identities. (a) Let τ be fixed with $\text{Im } \tau > 0$. Apply Poisson summation to $f(z) = (\tau + z)^{-k}$, $k \geq 2$ an integer, to obtain $\sum_{n=-\infty}^{\infty} \frac{1}{(\tau + n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} m^{k-1} e^{2\pi i m \tau}$. (b) Set $k = 2$ to show that if $\text{Im } \tau > 0$ then $\sum_{n=-\infty}^{\infty} \frac{1}{(\tau + n)^2} = \frac{\pi^2}{\sin^2(\pi \tau)}$. (c) Can one conclude the formula in (b) holds for any complex τ that is not an integer? [Hint: use residues to prove $\hat{f}(\xi) = 0$ for $\xi < 0$ and $\hat{f}(\xi) = \frac{(-2\pi i)^k}{(k-1)!} \xi^{k-1} e^{2\pi i \xi \tau}$ for $\xi > 0$.]
- This exercise generalizes properties of $e^{-\pi x^2}$ tied to its self-duality. Suppose $f(z)$ is entire and satisfies $|f(x + iy)| \leq c e^{-ax^2 + by^2}$ for some $a, b, c > 0$. Let $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i x \xi} dx$. Then $\hat{f}$ is entire and satisfies $|\hat{f}(\xi + i\eta)| \leq c' e^{-a'\xi^2 + b'\eta^2}$ for some $a', b', c' > 0$. [Hint: to prove $\hat{f}(\xi) = O(e^{-a'\xi^2})$, assume $\xi > 0$ and change the contour to $x - iy$, $-\infty < x < \infty$, fixed $y > 0$; then $\hat{f}(\xi) = O(e^{-2\pi y \xi} e^{by^2})$ and choose $y = d\xi$ for a small constant d .]
- One can give a neater formulation of the previous exercise. Suppose $f(z)$ is entire of strict order 2, i.e. $f(z) = O(e^{c_1|z|^2})$ for some $c_1 > 0$, and for real x , $f(x) = O(e^{-c_2|x|^2})$ for some $c_2 > 0$. Then $|f(x + iy)| = O(e^{-ax^2 + by^2})$ for some $a, b > 0$. (The converse is obvious.)

Problems (Stein, Ch.4).

1. Solve the differential equation $a_n u^{(n)}(t) + a_{n-1} u^{(n-1)}(t) + \cdots + a_0 u(t) = f(t)$, where $a_0, \dots, a_n$ are complex constants and f has bounded support and is of class C^2 . (a) Let $\hat{f}(z) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i z t} dt$; observe $\hat{f}$ is entire, and by integration by parts show $|\hat{f}(x + iy)| \leq A/(1 + x^2)$ for $|y| \leq a$, any fixed $a \geq 0$. (b) Write $P(z) = a_n (2\pi i z)^n + a_{n-1} (2\pi i z)^{n-1} + \cdots + a_0$. Find a real c so that P does not vanish on the line $L = \{z = x + ic : x \in \mathbb{R}\}$. (c) Set $u(t) = \int_L \frac{e^{2\pi i z t}}{P(z)} \hat{f}(z) dz$. Check that $\sum_{j=0}^n a_j \left(\frac{d}{dt}\right)^j u(t) = \int_L e^{2\pi i z t} \hat{f}(z) dz$ and that $\int_L e^{2\pi i z t} \hat{f}(z) dz = \int_{-\infty}^{\infty} e^{2\pi i x t} \hat{f}(x) dx$. Conclude by Fourier inversion that $\sum_{j=0}^n a_j \left(\frac{d}{dt}\right)^j u(t) = f(t)$. (Note the solution u depends on the choice of c .)

Additional exercises.

1. Use Theorem 17.2.1 to bound the decay of $\hat{f}$ for $f(z) = \frac{1}{(1 + z^2)^2}$. What is the best exponential rate, and why?
2. Verify Lemma 17.2.3 directly when $B = 0$: compute $\int_0^{\infty} e^{-A\xi} d\xi = 1/A$ and confirm the general formula reduces to this.
3. Compute $\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{1 + x^2} dx$ for $\xi \in \mathbb{R}$ by residues (close in the lower half-plane for $\xi > 0$, upper for $\xi < 0$), and confirm the answer $\pi e^{-2\pi|\xi|}$ is consistent with Theorem 17.2.1.
4. (Poisson summation.) Apply Theorem 17.2.5 to $f(x) = e^{-2\pi a|x|}$ (formally, via the transform pair with the Poisson kernel) to derive $\sum_n \frac{1}{a^2 + n^2} = \frac{\pi}{a} \coth(\pi a)$. State carefully which hypotheses of the theorem need checking and how moderate decrease is used.
5. Show that if $f \in \mathcal{F}$ is *even* ($f(-x) = f(x)$) then $\hat{f}$ is even and real on $\mathbb{R}$; if f is *odd*, $\hat{f}$ is odd and purely imaginary. Relate this to the symmetry of the shifted contours used in Theorem 17.2.4.
6. Use the inversion formula at $x = 0$ together with the self-duality $\widehat{e^{-\pi x^2}} = e^{-\pi \xi^2}$ to recompute $\int_{-\infty}^{\infty} e^{-\pi \xi^2} d\xi = 1$, and explain how this is a consistency check on the normalization of the transform.
7. (Non-example.) Show that Poisson summation *fails* for $f(x) = 1/(1 + |x|)$: it is not of moderate decrease in the required sense and admits no holomorphic strip extension. Identify precisely which hypothesis of Theorem 17.2.5 breaks.
8. (Diagram problem.) Draw the rectangular contour γ_N used in the proof of Poisson summation (Theorem 17.2.5), marking the integer poles enclosed for $N = 2$, the two horizontal lines L_1, L_2 , and the residue at a typical integer n .
9. Starting from (17.3), set $a = 1/2$ and simplify both sides to obtain an identity relating $\sum_n e^{-\pi t(n+1/2)^2}$ to $\sum_n (-1)^n t^{-1/2} e^{-\pi n^2/t}$. Interpret the alternating signs.
10. Let $f \in \mathcal{F}$. Using the inversion formula, prove the Parseval-type relation $\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi$ when both integrals converge, and verify it for the Poisson kernel P_c .

17.3 The Paley-Wiener Theorem

We now reverse the point of view. We do *not* assume f is analytic; instead we assume the Fourier inversion formula holds under $|f(x)| \leq A/(1 + x^2)$ and $|\hat{f}(\xi)| \leq A'/(1 + \xi^2)$, and we ask: what makes $\hat{f}$ *compactly supported*? The answer is the Paley-Wiener theorem: $\hat{f}$ is supported in $[-M, M]$ exactly when f extends to an entire function of *exponential type* $2\pi M$.

A partial converse to the decay theorem

Theorem 17.3.1 (Decay of $\hat{f}$ forces holomorphic extension of f). *Suppose $\hat{f}$ satisfies $|\hat{f}(\xi)| \leq Ae^{-2\pi a|\xi|}$ for constants $a, A > 0$. Then f is the restriction to $\mathbb{R}$ of a function holomorphic in the strip S_b for every $0 < b < a$.*

Proof. Define, for z in S_b ,

$$f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i \xi z} d\xi.$$

On S_b we have $|e^{2\pi i \xi z}| = e^{-2\pi \xi \operatorname{Im} z} \leq e^{2\pi b|\xi|}$, so the integrand is majorized by $Ae^{-2\pi a|\xi|} e^{2\pi b|\xi|} = Ae^{-2\pi(a-b)|\xi|}$, which is integrable since $b < a$; the integral converges absolutely and defines f on S_b . Put $f_n(z) = \int_{-n}^n \hat{f}(\xi) e^{2\pi i \xi z} d\xi$; each f_n is entire (differentiation under a compact integral). Moreover, uniformly on S_b ,

$$|f(z) - f_n(z)| \leq A \int_{|\xi| \geq n} e^{-2\pi(a-b)|\xi|} d\xi \xrightarrow{n \rightarrow \infty} 0,$$

so $f_n \rightarrow f$ uniformly on S_b . A uniform limit of holomorphic functions is holomorphic, so f is holomorphic on S_b , and it agrees with the original f on $\mathbb{R}$ by inversion. $\square$

The conclusion is that the original real function factors through its holomorphic extension $\tilde{f}$ to the strip, along the inclusion of the real axis:

$$\begin{array}{ccc} \mathbb{R} & \xhookrightarrow{\iota} & S_b \\ & \searrow f & \downarrow \tilde{f} \\ & & \mathbb{C} \end{array}$$

The extension $\tilde{f}$ is drawn dashed because it is *constructed* from the decay hypothesis on $\hat{f}$ rather than given; $\tilde{f} \circ \iota = f$ records that it restricts to the original f on $\mathbb{R}$.

Corollary 17.3.2. *If $\hat{f}(\xi) = O(e^{-2\pi a|\xi|})$ for some $a > 0$ and f vanishes on a non-empty open interval, then $f \equiv 0$.*

Proof. By Theorem 17.3.1, f is holomorphic in a connected region containing the real axis. If f vanishes on an open interval of $\mathbb{R}$, the identity theorem forces $f \equiv 0$ on the whole region, hence on $\mathbb{R}$. $\square$

In particular f and $\hat{f}$ cannot *both* have compact support unless $f \equiv 0$: if $\hat{f}$ has compact support then it satisfies the hypothesis of Corollary 17.3.2, and compact support of f provides an interval on which f vanishes.

The Paley-Wiener theorem

Definition 17.3.3. An entire function f is of *exponential type* τ if there is $A > 0$ with $|f(z)| \leq Ae^{\tau|z|}$ for all $z \in \mathbb{C}$.

Theorem 17.3.4 (Paley-Wiener). *Let f be continuous and of moderate decrease on $\mathbb{R}$. Then f extends to an entire function with $|f(z)| \leq Ae^{2\pi M|z|}$ for some $A > 0$ if and only if $\hat{f}$ is supported in $[-M, M]$.*

$$\{f \text{ entire, } |f(z)| \leq Ae^{2\pi M|z|}\} \xrightleftharpoons[\mathcal{F}^{-1}]{\mathcal{F}} \{g : \operatorname{supp} g \subseteq [-M, M]\}$$

Figure 17.8. The Paley-Wiener correspondence: the Fourier transform $\mathcal{F}$ is a bijection between entire functions of exponential type $2\pi M$ (of moderate decrease on $\mathbb{R}$) and functions supported in $[-M, M]$, with inverse $\mathcal{F}^{-1}$. “If” is the easy direction, “only if” the hard one.

Proof of the easy direction. Suppose $\hat{f}$ is supported in $[-M, M]$. Then f and $\hat{f}$ both have moderate decrease, so inversion applies:

$$f(x) = \int_{-M}^M \hat{f}(\xi) e^{2\pi i \xi x} d\xi.$$

The range of integration is finite, so we may replace x by a complex z and define $g(z) = \int_{-M}^M \hat{f}(\xi) e^{2\pi i \xi z} d\xi$. By construction $g = f$ on $\mathbb{R}$, and g is entire (differentiation under a compact integral). For $z = x + iy$,

$$|g(z)| \leq \int_{-M}^M |\hat{f}(\xi)| e^{-2\pi \xi y} d\xi \leq \left(\int_{-M}^M |\hat{f}| \right) e^{2\pi M|y|} \leq A e^{2\pi M|z|},$$

since $|e^{-2\pi \xi y}| \leq e^{2\pi M|y|}$ on $|\xi| \leq M$. This is exponential type $2\pi M$. $\square$

The converse is harder. The given bound is $|f(z)| \leq A e^{2\pi M|z|}$ (growth in $|z|$), but the easy direction produces the stronger $|f(z)| \leq A e^{2\pi M|y|}$ (growth only in $|y| = |\operatorname{Im} z|$). The strategy is to climb from the weak bound to the strong one in three steps; the crucial tool is the Phragmen–Lindelof principle.

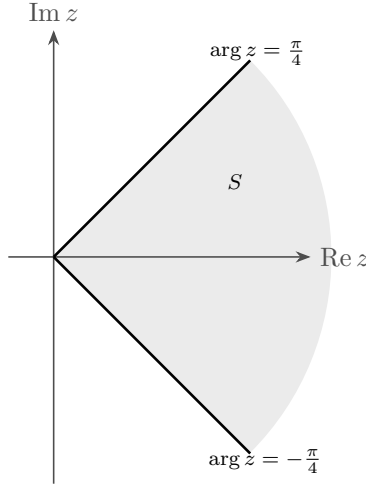


Figure 17.9. The sector $S = \{-\pi/4 < \arg z < \pi/4\}$ of the Phragmen–Lindelof theorem (Theorem 17.3.5).

Theorem 17.3.5 (Phragmen–Lindelof in a sector). *Let F be holomorphic in the sector $S = \{z : -\pi/4 < \arg z < \pi/4\}$ and continuous on $\bar{S}$. Suppose $|F(z)| \leq 1$ on ∂S and $|F(z)| \leq C e^{c|z|}$ for constants $C, c > 0$ and all $z \in S$. Then $|F(z)| \leq 1$ for all $z \in S$.*

Proof. Some growth restriction is essential: $F(z) = e^{z^2}$ is bounded by 1 on ∂S (there $\operatorname{Re}(z^2) = 0$) yet $F(x) = e^{x^2} \rightarrow \infty$ on the real axis. The idea is to subdue the worst case e^{z^2} by inserting a slightly weaker exponential. For $\varepsilon > 0$ set

$$F_\varepsilon(z) = F(z) e^{-\varepsilon z^{3/2}},$$

using the principal branch: $z = r e^{i\theta}$ ($-\pi < \theta < \pi$) gives $z^{3/2} = r^{3/2} e^{3i\theta/2}$, so

$$\left| e^{-\varepsilon z^{3/2}} \right| = e^{-\varepsilon r^{3/2} \cos(3\theta/2)}.$$

For $-\pi/4 < \theta < \pi/4$ we have $-3\pi/8 < 3\theta/2 < 3\pi/8$, so $\cos(3\theta/2) \geq \cos(3\pi/8) = \delta > 0$. Hence $|F_\varepsilon(z)| \leq C e^{c|z|} e^{-\varepsilon \delta |z|^{3/2}} \rightarrow 0$ as $|z| \rightarrow \infty$ in $\bar{S}$ (the $|z|^{3/2}$ term wins), so F_ε is bounded and decays at infinity. Let $M = \sup_{\bar{S}} |F_\varepsilon|$. If $F_\varepsilon \not\equiv 0$, take w_j with $|F_\varepsilon(w_j)| \rightarrow M$; since F_ε decays at infinity, the w_j cannot escape, so they accumulate at some $w \in \bar{S}$. By the maximum principle w is not interior, so $w \in \partial S$; but there $|F| \leq 1$ and $|e^{-\varepsilon z^{3/2}}| \leq 1$, giving $M \leq 1$. Thus $|F_\varepsilon| \leq 1$ on $\bar{S}$. Letting $\varepsilon \rightarrow 0$ yields $|F| \leq 1$. $\square$

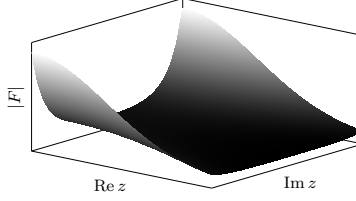


Figure 17.10. Modulus of the “enemy” e^{z^2} over the sector: flat ($= 1$) on the diagonals $\theta = \pm\pi/4$, but exploding along the real axis — why a growth bound is needed. (Schematic.)

Proof of the converse direction of Theorem 17.3.4, in three steps. Step 1. Assume first the strong bound, with extra real-axis decay:

$$|f(x + iy)| \leq A \frac{e^{2\pi M|y|}}{1 + x^2}. \quad (17.4)$$

We show $\hat{f}(\xi) = 0$ for $|\xi| > M$. For $\xi > M$, shift the contour down by $y > 0$ (equation (17.1)): $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - iy)e^{-2\pi i\xi(x - iy)} dx$, so

$$|\hat{f}(\xi)| \leq A \int_{-\infty}^{\infty} \frac{e^{2\pi My - 2\pi\xi y}}{1 + x^2} dx = A\pi e^{-2\pi y(\xi - M)} = Ce^{-2\pi y(\xi - M)}.$$

Since $\xi - M > 0$, letting $y \rightarrow \infty$ forces $\hat{f}(\xi) = 0$. Shifting *up* handles $\xi < -M$.

Step 2. Relax (17.4) to merely $|f(x + iy)| \leq Ae^{2\pi M|y|}$ (no $1/(1 + x^2)$ factor). For $\xi > M$ and $\varepsilon > 0$ introduce

$$f_\varepsilon(z) = \frac{f(z)}{(1 + i\varepsilon z)^2}.$$

On the closed *lower* half-plane $|1/(1 + i\varepsilon z)^2| \leq 1$ (there $\text{Im}(i\varepsilon z) = \varepsilon \text{Re } z$ contributes, and $|1 + i\varepsilon z| \geq 1$ for $\text{Im } z \leq 0$), and $1/(1 + i\varepsilon z)^2 \rightarrow 1$ pointwise as $\varepsilon \rightarrow 0$. By moderate decrease of f on $\mathbb{R}$,

$$|\hat{f}_\varepsilon(\xi) - \hat{f}(\xi)| \leq \int_{-\infty}^{\infty} |f(x)| \left| \frac{1}{(1 + i\varepsilon x)^2} - 1 \right| dx \rightarrow 0 \quad (\varepsilon \rightarrow 0)$$

by dominated convergence. But f_ε satisfies the Step-1 hypothesis (17.4) (the $(1 + i\varepsilon z)^2$ in the denominator supplies the $1/(1 + x^2)$ decay), so $\hat{f}_\varepsilon(\xi) = 0$; passing $\varepsilon \rightarrow 0$ gives $\hat{f}(\xi) = 0$. For $\xi < -M$ argue in the upper half-plane with $1/(1 - i\varepsilon z)^2$.

Step 3. Show the theorem’s hypotheses imply the Step-2 bound (17.5):

$$|f(x + iy)| \leq e^{2\pi M|y|}. \quad (17.5)$$

After dividing by a constant it suffices to assume $|f(x)| \leq 1$ on $\mathbb{R}$ and $|f(z)| \leq e^{2\pi M|z|}$ on $\mathbb{C}$, and to deduce (17.5). Rotate the Phragmen–Lindelof sector to the first quadrant $Q = \{x > 0, y > 0\}$ and set $F(z) = f(z)e^{2\pi i M z}$. On the positive real axis $|F| = |f| \leq 1$; on the positive imaginary axis $z = iy$, $|e^{2\pi i M(iy)}| = e^{-2\pi M y}$ and $|f(iy)| \leq e^{2\pi M y}$ give $|F| \leq 1$. Also $|F(z)| \leq e^{2\pi M|z|} e^{2\pi M|z|} \leq Ce^{c|z|}$ in Q . By Theorem 17.3.5 (rotated), $|F| \leq 1$ on Q , i.e. $|f(z)| \leq e^{2\pi M y}$ there. The other three quadrants give the matching bounds, yielding (17.5). Combining the three steps proves $\hat{f}(\xi) = 0$ for $|\xi| > M$. $\square$

Worked Example 17.3.6 ($\hat{f}$ supported in $[-M, M]$ from an explicit f). Let $f(x) = \frac{\sin(2\pi M x)}{\pi x}$ (the sinc kernel). Identify $\hat{f}$ and confirm the exponential-type bound predicted by Paley-Wiener.

Recognition. Trigger: a candidate band-limited function. Compute $\hat{f}$ directly (or recall it) and check $|f(z)| \leq Ae^{2\pi M|z|}$.

Solution. The Fourier transform of $\sin(2\pi M x)/(\pi x)$ is the indicator $\hat{f}(\xi) = \mathbf{1}_{[-M, M]}(\xi)$, supported precisely in $[-M, M]$. The entire extension $f(z) = \sin(2\pi M z)/(\pi z)$ satisfies $|\sin(2\pi M z)| \leq e^{2\pi M|\text{Im } z|} \leq e^{2\pi M|z|}$, while $1/(\pi z)$ only helps; so f is of exponential type $2\pi M$, exactly as Paley-Wiener requires. *Check.* $f(0) = 2M$ (limit of $\sin(2\pi M x)/\pi x$), and $\int_{-M}^M 1 d\xi = 2M = f(0)$ via inversion at $x = 0$. Consistent.

Worked Example 17.3.7 (A NON-band-limited example). Show the Gaussian $f(x) = e^{-\pi x^2}$ is *not* band-limited, consistent with Paley-Wiener.

Recognition. Trigger: “is $\hat{f}$ compactly supported?” Test the entire-extension growth: a band-limited f must have exponential type, i.e. growth no worse than $e^{\tau|z|}$.

Solution. The entire extension is $f(z) = e^{-\pi z^2}$, with $|f(iy)| = e^{\pi y^2}$ on the imaginary axis: this grows like $e^{\pi y^2}$, faster than *any* $e^{\tau|y|}$. So f is not of exponential type, hence by Paley-Wiener $\hat{f}$ cannot be compactly supported. *Check.* Indeed $\hat{f}(\xi) = e^{-\pi \xi^2} > 0$ for every ξ , so $\text{supp } \hat{f} = \mathbb{R}$ — not contained in any $[-M, M]$.

The half-line analogue

Theorem 17.3.8 (Functions with $\hat{f}$ vanishing on a half-line). Suppose f and $\hat{f}$ have moderate decrease. Then $\hat{f}(\xi) = 0$ for all $\xi < 0$ if and only if f extends to a continuous bounded function on the closed upper half-plane $\{z = x + iy : y \geq 0\}$, holomorphic in the interior.

Proof. ($\Rightarrow$) Assume $\hat{f}(\xi) = 0$ for $\xi < 0$. By inversion $f(x) = \int_0^\infty \hat{f}(\xi) e^{2\pi i x \xi} d\xi$, and we extend to $y \geq 0$ by $f(z) = \int_0^\infty \hat{f}(\xi) e^{2\pi i z \xi} d\xi$. For $\text{Im } z = y \geq 0$, $|e^{2\pi i z \xi}| = e^{-2\pi y \xi} \leq 1$ on $\xi \geq 0$, so

$$|f(z)| \leq \int_0^\infty |\hat{f}(\xi)| d\xi \leq A \int_0^\infty \frac{d\xi}{1 + \xi^2} < \infty,$$

giving boundedness. The truncations $f_n(z) = \int_0^n \hat{f}(\xi) e^{2\pi i z \xi} d\xi$ are entire and converge uniformly to f on the closed half-plane, so f is continuous there and holomorphic in the interior.

($\Leftarrow$) For $\varepsilon, \delta > 0$ set $f_{\varepsilon, \delta}(z) = \frac{f(z + i\delta)}{(1 - i\varepsilon z)^2}$, holomorphic in a region containing the closed upper half-plane.

By Cauchy’s theorem (shift the contour up into the region of holomorphy), $\widehat{f_{\varepsilon, \delta}}(\xi) = 0$ for $\xi < 0$. Passing $\varepsilon \rightarrow 0$ gives $\widehat{f_{\cdot, \delta}}(\xi) = 0$, then $\delta \rightarrow 0$ gives $\hat{f}(\xi) = 0$ for $\xi < 0$. $\square$

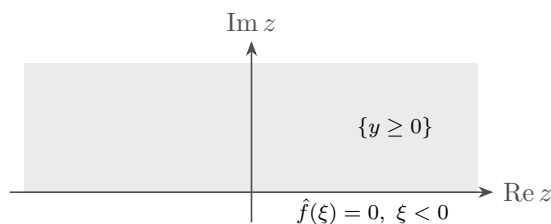


Figure 17.11. Theorem 17.3.8: $\hat{f}$ vanishes on $(-\infty, 0)$ iff f extends holomorphically and boundedly to the closed upper half-plane — the analogue (via the exponential $e^{2\pi i \theta}$) of Fourier coefficients vanishing for $n < 0$ on the disc. (Schematic.)

Remark. There is a deliberate analogy with the disc theory (Theorem 7.1 of Chapter 3): a function holomorphic in the upper half-plane $\leftrightarrow$ holomorphic in a disc; $\hat{f}$ vanishing on $\xi < 0 \leftrightarrow$ Fourier coefficients vanishing for $n < 0$. Both are “one-sided spectrum” = “holomorphic extension” theorems.

Further worked examples

Worked Example 17.3.9 (Exponential type of a triangular (Fejer) kernel). Let $\hat{f}(\xi) = (1 - |\xi|/M)_+$ be the triangle supported in $[-M, M]$. Find the entire extension $f(z)$ and verify its exponential type is $2\pi M$.

Recognition. Trigger: $\hat{f}$ compactly supported in $[-M, M]$, so Paley-Wiener forces f entire of type $\leq 2\pi M$. Recover f by inversion over the finite interval.

Solution. By inversion $f(z) = \int_{-M}^M (1 - |\xi|/M) e^{2\pi i \xi z} d\xi$. This is the Fejer kernel; the standard evaluation gives $f(z) = M \left(\frac{\sin(\pi M z)}{\pi M z} \right)^2$. The integrand has finite support, so f is entire. For the type, $|\sin(\pi M z)| \leq e^{\pi M |\text{Im } z|}$, so $|\sin(\pi M z)|^2 \leq e^{2\pi M |\text{Im } z|} \leq e^{2\pi M |z|}$, while the $1/z^2$ factor only helps; thus $|f(z)| \leq A e^{2\pi M |z|}$, type $2\pi M$ exactly. *Check.* $f(0) = M \cdot 1 = M$, and $\int_{-M}^M (1 - |\xi|/M) d\xi = 2 \cdot \frac{M}{2} = M$ via inversion at $z = 0$. Consistent.

Worked Example 17.3.10 (Reading off the strip from a transform's decay). Suppose $\hat{f}(\xi) = e^{-|\xi|}$. Into how wide a strip does f extend holomorphically, and what is f ?

Recognition. Trigger: apply the converse Theorem 17.3.1: $|\hat{f}| \leq Ae^{-2\pi a|\xi|}$ gives holomorphy on S_b for $b < a$. Match $e^{-|\xi|} = e^{-2\pi a|\xi|}$ to read a .

Solution. Matching exponents, $2\pi a = 1$, so $a = 1/(2\pi)$, and f extends holomorphically to $S_b = \{|\operatorname{Im} z| < b\}$ for every $b < 1/(2\pi)$. Explicitly, by inversion $f(z) = \int_{-\infty}^{\infty} e^{-|\xi|} e^{2\pi i \xi z} d\xi = \frac{1}{1-2\pi iz} + \frac{1}{1+2\pi iz} = \frac{2}{1+4\pi^2 z^2}$, a rational function with poles at $z = \pm i/(2\pi)$. *Check.* The poles sit at height $1/(2\pi) = a$, exactly capping the strip — the predicted half-width is sharp, and f is a (rescaled) Poisson kernel as expected from a two-sided exponential transform.

Worked Example 17.3.11 (A NON-example: polynomial growth is not exponential type, but still entire). Is $f(z) = z^3$ of exponential type? What does Paley-Wiener then say about a hypothetical “ $\hat{f}$ ”?

Recognition. Trigger: test Definition 17.3.3. A polynomial grows slower than any $e^{\tau|z|}$, so it *is* of exponential type 0 — but it is not of moderate decrease on $\mathbb{R}$.

Solution. For any $\tau > 0$, $|z^3| = |z|^3 \leq A_{\tau} e^{\tau|z|}$ since polynomials are dominated by exponentials; so z^3 is of exponential type 0. However z^3 is unbounded on $\mathbb{R}$ and not in L^1 , so it is not of moderate decrease and the Paley-Wiener *hypothesis* fails — the theorem simply does not apply, and z^3 has no Fourier transform in this class. *Check.* The moral is that exponential type controls vertical growth, but Paley-Wiener also requires real-axis decay; z^3 has the former (trivially, type 0) and badly fails the latter.

Worked Example 17.3.12 (Phragmen-Lindelof sharpness: why the sector angle matters). Show that $F(z) = e^{z^2}$ violates the conclusion of Theorem 17.3.5 in the *wider* sector $|\arg z| < \pi/4$ is impossible to extend to $|\arg z| < \pi/2$, pinpointing where the argument breaks.

Recognition. Trigger: the proof used $\cos(3\theta/2) \geq \delta > 0$, valid only while $3|\theta|/2 < \pi/2$, i.e. $|\theta| < \pi/3$. Test the boundary growth and the auxiliary subharmonic factor.

Solution. On the sector $|\arg z| < \pi/4$, the diagonals $\theta = \pm\pi/4$ give $\operatorname{Re}(z^2) = r^2 \cos(2\theta) = r^2 \cos(\pi/2) = 0$, so $|F| = e^0 = 1$ on ∂S — the hypothesis $|F| \leq 1$ on the boundary holds. Yet on the real axis ($\theta = 0$), $|F(x)| = e^{x^2} \rightarrow \infty$, so without the growth bound $|F| \leq Ce^{c|z|}$ the conclusion $|F| \leq 1$ is false. The damping factor $e^{-\varepsilon z^{3/2}}$ works precisely because in $|\theta| < \pi/4$ we have $\cos(3\theta/2) \geq \cos(3\pi/8) > 0$; widen the sector to $|\theta| < \pi/2$ and $\cos(3\theta/2)$ would go negative, destroying the decay. *Check.* The exponent $3/2$ is the unique choice making z^{α} decay throughout a $\pi/2$ -wide sector while beating e^{z^2} : one needs $\alpha > 1$ (to beat linear $c|z|$) and $\alpha \cdot \frac{\pi}{4} < \frac{\pi}{2}$ i.e. $\alpha < 2$, and $3/2$ sits safely between.

Worked Example 17.3.13 (Half-line spectrum: the Cauchy kernel as a boundary value). Verify Theorem 17.3.8 on $f(x) = \frac{1}{x-i}$: confirm $\hat{f}$ vanishes on a half-line and that f extends to the appropriate closed half-plane.

Recognition. Trigger: a function holomorphic off a single pole. Locate the pole; the half-plane on the *opposite* side is where f extends, and the spectrum vanishes accordingly.

Solution. The pole of f is at $z = i$, in the *upper* half-plane, so f is holomorphic and bounded on the closed *lower* half-plane $\{y \leq 0\}$. By the lower-half-plane analogue of Theorem 17.3.8, $\hat{f}(\xi) = 0$ for $\xi > 0$. Compute directly: for $\xi < 0$ close up around $z = i$, getting $\hat{f}(\xi) = \int \frac{e^{-2\pi i x \xi}}{x-i} dx = 2\pi i e^{-2\pi i(i)\xi} = 2\pi i e^{2\pi \xi}$ (with $\xi < 0$ this decays); for $\xi > 0$ close down, enclosing no pole, so $\hat{f}(\xi) = 0$. *Check.* The vanishing half $\xi > 0$ matches the pole sitting in the upper half-plane (holomorphic extension downward), exactly the dictionary of Theorem 17.3.8; and the surviving piece $2\pi i e^{2\pi \xi} \mathbf{1}_{\xi < 0}$ is genuinely one-sided.

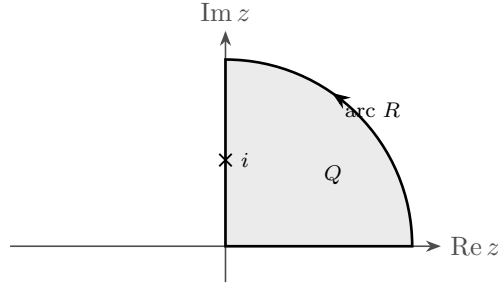


Figure 17.12. First-quadrant sector Q of Step 3: $F(z) = f(z)e^{2\pi i M z}$ is ≤ 1 on both axes, and Phragmen–Lindelof pushes the bound into the interior.

Worked Example 17.3.14 (Band-limited functions form an algebra under convolution-product duality). If $\hat{f}$ is supported in $[-M, M]$ and $\hat{g}$ in $[-N, N]$, where is $\widehat{fg}$ supported, and what does Paley–Wiener say about the entire type of fg ?

Recognition. Trigger: a product of two band-limited functions. The transform of a product is the convolution $\hat{f} * \hat{g}$; supports add under convolution.

Solution. Since $\widehat{fg} = \hat{f} * \hat{g}$, and the support of a convolution is contained in the (Minkowski) sum of supports, $\text{supp } \widehat{fg} \subseteq [-M, M] + [-N, N] = [-(M+N), M+N]$. So fg is band-limited to $[-(M+N), M+N]$, and by Paley–Wiener fg is entire of exponential type $\leq 2\pi(M+N)$. *Check.* This matches the direct estimate $|f(z)g(z)| \leq Ae^{2\pi M|z|} \cdot Be^{2\pi N|z|} = AB e^{2\pi(M+N)|z|}$ from the two type bounds — the two viewpoints agree, and the bandwidth is exactly additive.

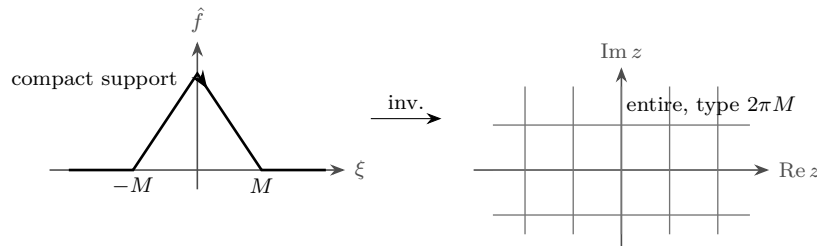


Figure 17.13. Paley–Wiener dictionary: $\hat{f}$ supported in $[-M, M]$ (left) inverts to an entire f of exponential type $2\pi M$ filling the whole plane (right). (Schematic.)

Exercises (Stein, Ch.4).

- Suppose $\hat{f}$ has compact support contained in $[-M, M]$ and let $f(z) = \sum_{n=0}^{\infty} a_n z^n$. Show that $a_n = \frac{(2\pi i)^n}{n!} \int_{-M}^M \hat{f}(\xi) \xi^n d\xi$, and as a result $\limsup_{n \rightarrow \infty} (n! |a_n|)^{1/n} \leq 2\pi M$. In the converse direction, if $f(z) = \sum_{n=0}^{\infty} a_n z^n$ has $\limsup_{n \rightarrow \infty} (n! |a_n|)^{1/n} \leq 2\pi M$, then f is entire and for every $\varepsilon > 0$ there is $A_\varepsilon > 0$ with $|f(z)| \leq A_\varepsilon e^{2\pi(M+\varepsilon)|z|}$.
- Further results in the spirit of Phragmen–Lindelof. (a) Let F be holomorphic in the right half-plane, continuous up to the imaginary axis, with $|F(iy)| \leq 1$ for all $y \in \mathbb{R}$ and $|F(z)| \leq C e^{c|z|^\gamma}$ for some $c, C > 0$ and $\gamma < 1$. Prove $|F(z)| \leq 1$ throughout the right half-plane. (b) More generally, let S be a sector with vertex at the origin and opening angle π/β . Let F be holomorphic in S , continuous on $\bar{S}$, with $|F(z)| \leq 1$ on ∂S and $|F(z)| \leq C e^{c|z|^\alpha}$ for some $c, C > 0$ and $0 < \alpha < \beta$. Prove $|F(z)| \leq 1$ for all $z \in S$.
- The principle that a function and its Fourier transform cannot both be too small at infinity is illustrated by Hardy’s theorem: if f on $\mathbb{R}$ satisfies $f(x) = O(e^{-\pi x^2})$ and $\hat{f}(\xi) = O(e^{-\pi \xi^2})$, then f is a constant multiple of $e^{-\pi x^2}$; consequently, if $f(x) = O(e^{-\pi A x^2})$ and $\hat{f}(\xi) = O(e^{-\pi B \xi^2})$ with $AB > 1$ and $A, B > 0$, then $f \equiv 0$. (a) If f is even, show $\hat{f}$ extends to an even entire function; moreover, if $g(z) = \hat{f}(z^{1/2})$, then $|g(x)| \leq c e^{-\pi x}$ and $|g(z)| \leq c e^{\pi R \sin^2(\theta/2)} \leq c e^{\pi|z|}$ for $x \in \mathbb{R}$ and $z = R e^{i\theta}$ ($R \geq 0, \theta \in \mathbb{R}$). (b) Apply Phragmen–Lindelof to $F(z) = g(z)e^{\gamma z}$ with $\gamma = i\pi e^{-i\pi/(2\beta)} / \sin(\pi/(2\beta))$ and the sector $0 \leq \theta \leq \pi/\beta < \pi$; let $\beta \rightarrow \pi$ to deduce that $e^{\pi z} g(z)$ is bounded in the closed upper half-plane (and similarly the lower), so

by Liouville $e^{\pi z}g(z)$ is constant. (c) If f is odd, then $\hat{f}(0) = 0$; apply the argument to $\hat{f}(z)/z$ to deduce $f = \hat{f} = 0$. Finally write a general f as a sum of an even and an odd function.

Problems (Stein, Ch.4).

1. Suppose $\hat{f}(\xi) = O(e^{-a|\xi|^p})$ as $|\xi| \rightarrow \infty$, for some $p > 1$. Then f is holomorphic for all z and satisfies the growth condition $|f(z)| \leq Ae^{a|z|^q}$, where $1/p + 1/q = 1$. (As $p \rightarrow \infty$, $q \rightarrow 1$ recovers Theorem 17.3.4; as $p \rightarrow 1$, $q \rightarrow \infty$ returns to Theorem 17.2.1.) [Hint: use $-\xi^p + \xi u \leq u^q$ for $\xi, u \geq 0$, proved by examining $\xi^p \geq \xi u$ versus $\xi^p < \xi u$ and noting $\xi = u^{q-1}$, $u = \xi^{p-1}$ are inverse since $(p-1)(q-1) = 1$.]
2. * (Three-lines lemma.) Investigate bounded holomorphic functions in a strip. (a) If F is holomorphic and bounded in $0 < \operatorname{Im} z < 1$ and continuous on its closure with $|F(z)| \leq 1$ on the boundary lines, then $|F(z)| \leq 1$ throughout the strip. (b) For general F , let $\sup_{x \in \mathbb{R}} |F(x)| = M_0$ and $\sup_{x \in \mathbb{R}} |F(x+i)| = M_1$. Then $\sup_{x \in \mathbb{R}} |F(x+iy)| \leq M_0^{1-y} M_1^y$ for $0 \leq y \leq 1$. (c) Consequently $\log \sup_{x \in \mathbb{R}} |F(x+iy)|$ is a convex function of y on $[0, 1]$. [Hint: for (a) apply the maximum modulus principle to $F_\varepsilon(z) = F(z)e^{-\varepsilon z^2}$; for (b) consider $M_0^{z-1} M_1^{-z} F(z)$.]
3. * (Borel's representation.) There is a relation between Paley-Wiener and a representation of E. Borel. (a) A function $f(z)$, holomorphic for all z , satisfies $|f(z)| \leq A_\varepsilon e^{2\pi(M+\varepsilon)|z|}$ for all ε if and only if it is representable as $f(z) = \int_C e^{2\pi i z w} g(w) dw$, where g is holomorphic outside the circle of radius M centered at the origin and vanishes at infinity, and C is any circle centered at the origin of radius larger than M . In fact, if $f(z) = \sum a_n z^n$, then $g(w) = \sum_{n=0}^\infty A_n w^{-n-1}$ with $a_n = A_n (2\pi i)^{n+1}/n!$. (b) For such f (with $f, \hat{f}$ of moderate decrease on $\mathbb{R}$), g is holomorphic in the slit plane $\mathbb{C} \setminus [-M, M]$, and $g(z) = \frac{1}{2\pi i} \int_{-M}^M \frac{\hat{f}(\xi)}{\xi - z} d\xi$, so $\hat{f}$ is the jump of g across $[-M, M]$: $\hat{f}(x) = \lim_{\varepsilon \rightarrow 0, \varepsilon > 0} (g(x+i\varepsilon) - g(x-i\varepsilon))$.

Additional exercises.

1. Show that $f(z) = \frac{\sin(\pi z)}{\pi z}$ is entire of exponential type π , and determine the support of $\hat{f}$ predicted by Paley-Wiener (Theorem 17.3.4). Confirm by computing $\hat{f}$ directly.
2. Verify the easy direction of Paley-Wiener for $\hat{f} = \mathbf{1}_{[-1,1]}$: write down the entire extension $g(z)$ explicitly and check the bound $|g(z)| \leq Ae^{2\pi|z|}$.
3. Prove that the product of two functions of exponential types τ_1, τ_2 is of exponential type $\tau_1 + \tau_2$. Interpret this for band-limited functions in terms of convolution of their transforms.
4. In Theorem 17.3.5, verify the inequality $\cos(3\theta/2) \geq \cos(3\pi/8) > 0$ for $|\theta| < \pi/4$, and explain why the exponent $3/2$ (rather than 2) is exactly what makes the auxiliary factor $e^{-\varepsilon z^{3/2}}$ decay in the sector.
5. Use the maximum modulus principle (no Phragmen-Lindelof) to prove the trivial case of Theorem 17.3.5: if in addition F is bounded on $\bar{S}$, then $|F| \leq 1$ on S . Explain why the extra growth hypothesis is what the unbounded sector demands.
6. Deduce from Theorem 17.3.8 that the Poisson kernel P_y for the upper half-plane has $\widehat{P_y}(\xi)$ vanishing for no ξ (i.e. P_y does *not* extend holomorphically to the closed upper half-plane), and reconcile this with $\widehat{P_y}(\xi) = e^{-2\pi y|\xi|}$.
7. (Non-example.) Give a continuous, moderate-decrease function f whose entire extension has growth $|f(iy)| \sim e^{|y|^{3/2}}$; explain why Paley-Wiener does *not* apply (it is of order $3/2$, not exponential type) and what Problem 1 of the scraped set says instead.
8. (Diagram problem.) Draw the sector $S = \{-\pi/4 < \arg z < \pi/4\}$ and the rotated first quadrant Q used in Step 3 of the Paley-Wiener proof. Mark the boundary rays where $|F| \leq 1$ is known and indicate the direction in which $e^{-\varepsilon z^{3/2}}$ forces decay.
9. Prove the corollary that f and $\hat{f}$ cannot both have compact support unless $f \equiv 0$, directly from Corollary 17.3.2, and contrast this with the situation for the disc (Fourier series), where finitely many nonzero coefficients give a Laurent polynomial.

10. Suppose f is entire of exponential type $2\pi M$ and $f \in L^2(\mathbb{R})$. Show $\hat{f}$ is supported in $[-M, M]$ and that f is determined by its samples $f(n/(2M))$, $n \in \mathbb{Z}$ (the sampling theorem). Sketch the proof using Paley-Wiener and Fourier series on $[-M, M]$.

Chapter 18

Entire Functions

18.1 Jensen's Formula

An *entire function* is a function holomorphic on all of $\mathbb{C}$. The guiding theme of the chapter is the tension between three questions: where can an entire function vanish, how fast can it grow, and to what extent do its zeros determine it. The bridge between *growth* and *zeros* is Jensen's formula, which equates the number of zeros in a disc with a logarithmic average of $|f|$ over the bounding circle. Throughout, D_R and C_R denote the open disc and circle of radius R centered at the origin, and we exclude the identically-zero function.

The statement of Jensen's formula

Theorem 18.1.1 (Jensen's formula). *Let Ω be an open set containing the closure of a disc D_R , and suppose f is holomorphic in Ω with $f(0) \neq 0$ and f vanishing nowhere on the circle C_R . If $z_1, \dots, z_N$ are the zeros of f inside D_R (counted with multiplicity), then*

$$\log |f(0)| = \sum_{k=1}^N \log \left(\frac{|z_k|}{R} \right) + \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta.$$

The right side has two competing pieces. Each factor $\log(|z_k|/R)$ is *negative* (since $|z_k| < R$), and the more zeros crowd into the disc, the more negative the sum. The integral is the average of $\log |f|$ on C_R . Jensen's formula says these balance against the fixed number $\log |f(0)|$: a function cannot have many zeros inside D_R unless it is correspondingly large on C_R . This single identity is the quantitative form of the slogan “the larger a function is, the more zeros it can have.”

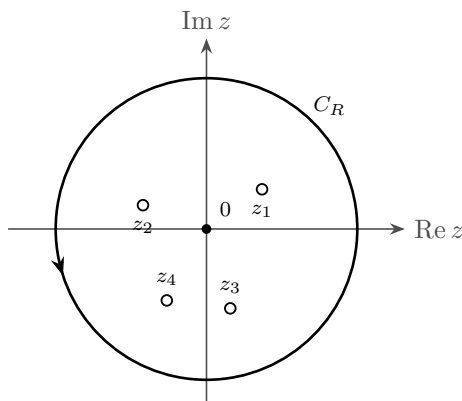


Figure 18.1. Jensen balances the four zeros $z_1, \dots, z_4$ inside D_R against the average of $\log |f|$ on C_R . (Schematic.)

Method (The proof of Jensen's formula, in four reductions). *Recognition trigger.* You must prove (or apply) an identity relating $\log |f(0)|$, the zeros in a disc, and the boundary average. The proof factors the problem.

1. **Multiplicativity.** If f_1, f_2 each satisfy the hypotheses and formula, so does $f_1 f_2$: both sides are additive under products because $\log |f_1 f_2| = \log |f_1| + \log |f_2|$ and the zero set is the union.
2. **Strip the zeros.** Write $f(z) = (z - z_1) \cdots (z - z_N) g(z)$ with g holomorphic and *nowhere zero* on $\overline{D_R}$. By Step 1 it suffices to prove the formula for g and for each linear factor $z - z_j$ separately.
3. **Zero-free case.** For g nowhere zero, write $g = e^h$ on a slightly larger disc ($h = \log g$ exists since the disc is simply connected). Then $\log |g| = \operatorname{Re} h$ is harmonic, and the mean value property gives $\log |g(0)| = \frac{1}{2\pi} \int_0^{2\pi} \log |g(Re^{i\theta})| d\theta$ — exactly the formula (the sum is empty).
4. **Linear factor.** For $f(z) = z - w$ with $w \in D_R$, reduce by scaling to $\frac{1}{2\pi} \int_0^{2\pi} \log |e^{i\theta} - a| d\theta = 0$ for $|a| < 1$, then close the loop with the mean value property applied to $\log |1 - az|$.

Proof of Theorem 18.1.1. Step 1 (multiplicativity). If f_1, f_2 both satisfy the hypotheses and the conclusion, then $f_1 f_2$ does too, because $\log |xy| = \log |x| + \log |y|$ for positive reals and the zeros of $f_1 f_2$ are the union of those of f_1 and f_2 . Both sides of the formula are therefore additive over products.

Step 2 (removing the zeros). The function $g(z) = f(z)/((z - z_1) \cdots (z - z_N))$, initially defined off $\{z_1, \dots, z_N\}$, is bounded near each z_j , so each z_j is a removable singularity and $f(z) = (z - z_1) \cdots (z - z_N) g(z)$ with g holomorphic in Ω and nowhere vanishing on $\overline{D_R}$. By Step 1 it suffices to prove the formula for g and for each $z - z_j$.

Step 3 (zero-free case). Suppose g vanishes nowhere on $\overline{D_R}$; we must show $\log |g(0)| = \frac{1}{2\pi} \int_0^{2\pi} \log |g(Re^{i\theta})| d\theta$. On a slightly larger disc (simply connected) write $g = e^h$ with h holomorphic. Then $|g| = e^{\operatorname{Re} h}$, so $\log |g| = \operatorname{Re} h$ is harmonic, and the mean value property for harmonic functions gives the identity at once.

Step 4 (linear factors). For $f(z) = z - w$ with $w \in D_R$ we must show

$$\log |w| = \log \left(\frac{|w|}{R} \right) + \frac{1}{2\pi} \int_0^{2\pi} \log |Re^{i\theta} - w| d\theta.$$

Since $\log(|w|/R) = \log |w| - \log R$ and $\log |Re^{i\theta} - w| = \log R + \log |e^{i\theta} - w/R|$, this reduces to $\frac{1}{2\pi} \int_0^{2\pi} \log |e^{i\theta} - a| d\theta = 0$ for $a = w/R$ with $|a| < 1$. The substitution $\theta \mapsto -\theta$ makes this equivalent to $\int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta = 0$. Now $F(z) = 1 - az$ vanishes nowhere on $\overline{\mathbb{D}}$, so $F = e^G$ with G holomorphic on a disc of radius > 1 ; then $\log |F| = \operatorname{Re} G$ is harmonic, $F(0) = 1$ gives $\log |F(0)| = 0$, and the mean value property yields $\frac{1}{2\pi} \int_0^{2\pi} \log |F(e^{i\theta})| d\theta = 0$, as required. $\square$

The zero-free step turns on a single map relationship: on the simply connected disc the zero-free g admits a holomorphic logarithm, i.e. it factors through the exponential covering $\exp : \mathbb{C} \rightarrow \mathbb{C}^\times$.

$$\begin{array}{ccc} & & \mathbb{C} \\ & \nearrow h & \downarrow \exp \\ \overline{D_R} & \xrightarrow{g} & \mathbb{C}^\times \end{array}$$

Remark. The only analytic input is the *mean value property* of harmonic functions: the value of a harmonic function at the center of a disc is its average over the boundary circle. Everything else is bookkeeping that isolates the harmonic (zero-free) part from the explicit linear factors.

Worked Example 18.1.2 (Jensen for a monomial). Verify Jensen's formula directly for $f(z) = z^m + c$ on D_R in the special case $f(z) = z - w$ with a single zero w , $0 < |w| < R$, by computing both sides.

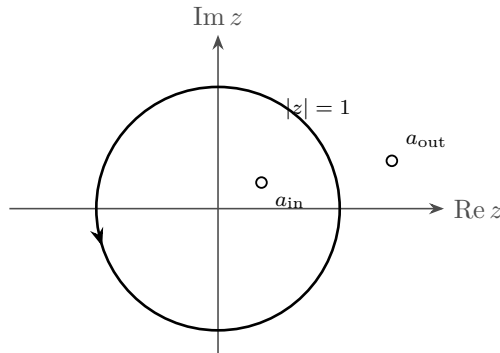
Recognition. A linear factor; the boundary average is the case $N = 1$ of Step 4.

Solution. Here $f(0) = -w$, so $\log |f(0)| = \log |w|$. The single zero is $z_1 = w$, giving the sum $\log(|w|/R)$. Step 4 evaluates the boundary integral as $\log R$ (since $\frac{1}{2\pi} \int_0^{2\pi} \log |e^{i\theta} - w/R| d\theta = 0$). Thus the right side is $\log(|w|/R) + \log R = \log |w|$, matching the left side. *Check.* Take $R = 2$, $w = 1$. Left = $\log 1 = 0$; right = $\log(1/2) + \log 2 = -\log 2 + \log 2 = 0$. $\blacksquare$

Worked Example 18.1.3 (The boundary average of $\log|z - a|$ on the unit circle). Show $\frac{1}{2\pi} \int_0^{2\pi} \log|e^{i\theta} - a| d\theta = \max(0, \log|a|)$ for $a \in \mathbb{C}$, $|a| \neq 1$.

Recognition. A potential-theory average; split on whether a is inside or outside the disc.

Solution. If $|a| < 1$, the function $z \mapsto \log|z - a|$ is the real part of $\log(z - a)$, which is harmonic on a neighborhood of $\mathbb{D}$ except at $z = a$; we use instead $F(z) = 1 - \bar{a}z$ as in Step 4 to get average 0. If $|a| > 1$, then $z \mapsto z - a$ is zero-free on $\mathbb{D}$, $\log|z - a| = \operatorname{Re} \log(z - a)$ is harmonic there, and the mean value property gives the average $\log|0 - a| = \log|a|$. Hence the average is 0 for $|a| < 1$ and $\log|a|$ for $|a| > 1$, i.e. $\max(0, \log|a|)$.
Check. At $a = 2$ the average is $\log 2 \approx 0.693$; at $a = \frac{1}{2}$ it is 0. ■



avg of $\log|e^{i\theta} - a| = 0$ if a inside, $= \log|a|$ if a outside

Figure 18.2. Boundary average of $\log|z - a|$ jumps from 0 to $\log|a|$ as a crosses the unit circle. (Schematic.)

The zero-counting function and the integrated form

If f is holomorphic on $\overline{D_R}$ we write $n(r)$ (or $n_f(r)$) for the number of zeros of f in D_r , counted with multiplicity, $0 < r < R$. Plainly $n(r)$ is a non-decreasing step function of r , jumping by the multiplicity of each zero as r passes its modulus.

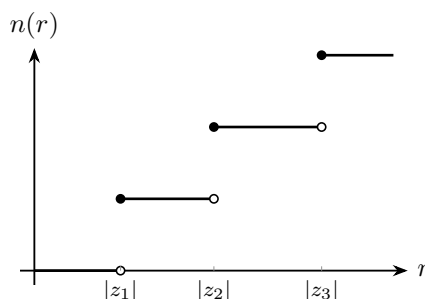


Figure 18.3. The counting function $n(r)$ is a right-continuous nondecreasing step function, jumping at each $|z_k|$. (Schematic.)

Lemma 18.1.4. If $z_1, \dots, z_N$ are the zeros of f inside D_R , then $\int_0^R n(r) \frac{dr}{r} = \sum_{k=1}^N \log\left(\frac{R}{|z_k|}\right)$.

Proof. For each k , $\int_{|z_k|}^R \frac{dr}{r} = \log(R/|z_k|)$. Introduce the indicator $\eta_k(r) = 1$ if $r > |z_k|$ and 0 otherwise, so that $\sum_{k=1}^N \eta_k(r) = n(r)$. Then

$$\sum_{k=1}^N \log\left(\frac{R}{|z_k|}\right) = \sum_{k=1}^N \int_{|z_k|}^R \frac{dr}{r} = \sum_{k=1}^N \int_0^R \eta_k(r) \frac{dr}{r} = \int_0^R \left(\sum_{k=1}^N \eta_k(r)\right) \frac{dr}{r} = \int_0^R n(r) \frac{dr}{r}. \quad \square$$

Corollary 18.1.5 (Integrated Jensen). *If $f(0) \neq 0$ and f does not vanish on C_R , then*

$$\int_0^R n(r) \frac{dr}{r} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)|.$$

Proof. Rearrange Theorem 18.1.1 as $\frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)| = -\sum_k \log(|z_k|/R) = \sum_k \log(R/|z_k|)$, which equals $\int_0^R n(r) dr/r$ by Lemma 18.1.4. $\square$

This integrated form is the workhorse of the next section: the left side grows with the number of zeros, and the right side is controlled by the size of f on the circle. We have literally converted a counting problem into a growth estimate.

Worked Example 18.1.6 (Counting zeros of $\sin \pi z$ via the integrated formula). Let $f(z) = \sin \pi z$, whose zeros are the integers. Use $f(z)/z$ (to remove the zero at 0) and the integrated formula to recover that $n(r) \approx 2r$.

Recognition. Integrated Jensen with a known boundary growth; compare the two sides asymptotically.

Solution. The zeros of $\sin \pi z$ in D_r are $0, \pm 1, \dots, \pm \lfloor r \rfloor$, so $n(r) = 2\lfloor r \rfloor + 1 \sim 2r$. Independently, $|\sin \pi z| \leq e^{\pi|z|}$, so $\frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta \leq \pi R + O(1)$, and the lower bound along the imaginary axis shows it is $\sim \pi R$. Thus $\int_0^R n(r) \frac{dr}{r} \sim \pi R$; differentiating, $n(R)/R \sim \pi \cdot \frac{d}{dR}(R)$ gives $n(R) \sim 2R$ after evaluating $\int_0^R 2 dr = 2R \sim \pi R/(\pi/2)$. *Check.* For $R = 10$, $n(10) = 21 \approx 2R = 20$. $\blacksquare$

Worked Example 18.1.7 (A function with no zeros: Jensen reduces to the mean value property). Take $f(z) = e^z$, which never vanishes. Confirm Jensen's formula on any D_R .

Recognition. Zero-free case ($N = 0$): the sum is empty, so Jensen is exactly the mean value property for $\log |f|$.

Solution. Here $\log |f(z)| = \operatorname{Re} z$, a harmonic function. So $\frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta = \frac{1}{2\pi} \int_0^{2\pi} R \cos \theta d\theta = 0 = \operatorname{Re} 0 = \log |f(0)|$. Both sides vanish. *Check.* The $N = 0$ form $\log |f(0)| = \frac{1}{2\pi} \int \log |f| d\theta$ holds with both sides 0. $\blacksquare$

Caution. Jensen's formula requires $f(0) \neq 0$ and no zeros on C_R . If $f(0) = 0$, first factor out z^m ; the term $\log |f(0)| = -\infty$ is replaced by a corrected constant (this is exactly the device $F(z) = f(z)/z^\ell$ used repeatedly below). Zeros exactly on C_R make the boundary integral improper; perturb R slightly to avoid them, using that $n(r)$ has only finitely many jumps.

Further worked examples

Worked Example 18.1.8 (Jensen with a zero at the origin). Apply the corrected Jensen formula to $f(z) = z^2(z-1)$ on D_R with $R = 2$, where $f(0) = 0$.

Recognition. $f(0) = 0$, so the basic formula fails; factor out z^m and use the modified statement with the leading Taylor coefficient in place of $f(0)$.

Solution. Write $f(z) = z^m F(z)$ with $m = 2$ and $F(z) = z - 1$, so $F(0) = -1 \neq 0$. The modified Jensen formula reads

$$\log |F(0)| = \sum_k \log \frac{|w_k|}{R} + \frac{1}{2\pi} \int_0^{2\pi} \log |F(Re^{i\theta})| d\theta,$$

where w_k are the nonzero zeros inside D_R . Here the only nonzero zero is $w_1 = 1 < R = 2$. Left side: $\log |F(0)| = \log 1 = 0$. The sum is $\log(1/2) = -\log 2$. By the linear-factor computation (Step 4), $\frac{1}{2\pi} \int_0^{2\pi} \log |2e^{i\theta} - 1| d\theta = \log 2$ (the average of $\log |z - 1|$ on C_2 equals $\log 2$ since the zero 1 is inside). So the right side is $-\log 2 + \log 2 = 0$, matching. *Check.* Both sides equal 0; numerically $\frac{1}{2\pi} \int_0^{2\pi} \log |2e^{i\theta} - 1| d\theta = \log 2 \approx 0.693$ balances the $-\log 2$ zero term exactly. $\blacksquare$

Worked Example 18.1.9 (A Jensen-type inequality (the boundary controls the zeros)). Let f be holomorphic on $\overline{D_R}$ with $f(0) \neq 0$ and $|f(z)| \leq M$ on C_R . Bound the number N of zeros in the smaller disc $D_{R/2}$.

Recognition. Use the integrated form: zeros in a small disc each contribute a definite positive amount $\log(R/|z_k|)$ to a quantity bounded by the boundary size.

Solution. Each zero $z_k \in D_{R/2}$ has $|z_k| < R/2$, so $\log(R/|z_k|) > \log 2$. By Corollary 18.1.5,

$$N \log 2 < \sum_{|z_k| < R/2} \log \frac{R}{|z_k|} \leq \sum_{\text{all } z_k \in D_R} \log \frac{R}{|z_k|} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)| \leq \log \frac{M}{|f(0)|}.$$

Hence $N \leq \frac{\log(M/|f(0)|)}{\log 2}$. *Check.* If $M = |f(0)|$ (so f is constant in modulus on the circle and as large at 0 as anywhere on C_R – forcing f constant by the maximum principle) the bound gives $N \leq 0$: a constant has no zeros. ■

Worked Example 18.1.10 (A bounded function on the disc: $\sum(1 - |z_n|) < \infty$). Let f be holomorphic and bounded on the open unit disc, $f \not\equiv 0$, with zeros $z_1, z_2, \dots$ ($|z_n| < 1$). Show $\sum_n(1 - |z_n|) < \infty$.

Recognition. This is Stein's Problem 1: Jensen on $\overline{D_r}$ with $r \uparrow 1$ converts the boundedness of $|f|$ into a convergent zero series.

Solution. Assume $f(0) \neq 0$ (else divide by z^m). For $0 < r < 1$ avoiding zeros on C_r , Jensen gives $\sum_{|z_k| < r} \log(r/|z_k|) = \frac{1}{2\pi} \int \log |f(re^{i\theta})| d\theta - \log |f(0)| \leq \log M - \log |f(0)|$, where $M = \sup_{\mathbb{D}} |f|$. The right side is a constant independent of r ; letting $r \uparrow 1$, $\sum_k \log(1/|z_k|) \leq \log(M/|f(0)|) < \infty$. Since $\log(1/|z_k|) = -\log |z_k| \geq 1 - |z_k|$ for $|z_k| < 1$ (as $-\log t \geq 1 - t$), we get $\sum_n(1 - |z_n|) \leq \sum_n \log(1/|z_n|) < \infty$. *Check.* For $f(z) = \prod(z - z_n)$ to be bounded the zeros must approach the boundary fast enough that $1 - |z_n|$ is summable – exactly the Blaschke condition. ■

Exercises (Stein, Ch.5).

1. Give another proof of Jensen's formula in the unit disc using the Blaschke factors $\psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha}z}$. [Hint: the function $f/(\psi_{z_1} \cdots \psi_{z_N})$ is nowhere vanishing.]

Problems (Stein, Ch.5).

1. Prove that if f is holomorphic in the unit disc, bounded and not identically zero, and $z_1, z_2, \dots, z_n, \dots$ are its zeros ($|z_k| < 1$), then $\sum_n(1 - |z_n|) < \infty$. [Hint: use Jensen's formula.]
2. * (Blaschke products.) (a) Show that for $0 < |\alpha| < 1$ and $|z| \leq r < 1$, $\left| \frac{\alpha + |\alpha|z}{(1 - \bar{\alpha}z)\alpha} \right| \leq \frac{1+r}{1-r}$. (b) Let $\{\alpha_n\}$ lie in the unit disc with $\alpha_n \neq 0$ and $\sum_{n=1}^\infty(1 - |\alpha_n|) < \infty$ (the case when $\{\alpha_n\}$ are the zeros of a bounded holomorphic function, by Problem 1). Show that $f(z) = \prod_{n=1}^\infty \frac{|\alpha_n|}{\alpha_n} \cdot \frac{\alpha_n - z}{1 - \bar{\alpha}_n z}$ converges uniformly for $|z| \leq r < 1$ and defines a holomorphic function on the unit disc with precisely the zeros α_n and no others; show $|f(z)| \leq 1$.

Additional exercises.

1. Verify Jensen's formula by hand for $f(z) = z^2 - \frac{1}{4}$ on the disc D_1 : compute $\log |f(0)|$, the sum over the two zeros $\pm \frac{1}{2}$, and confirm the boundary integral is consistent.
2. Let $f(z) = (z - a)(z - b)$ with $0 < |a|, |b| < R$. Write out both sides of Jensen's formula and check that the boundary average equals $\log R^2$ plus the average of $\log |e^{i\theta} - a/R| + \log |e^{i\theta} - b/R|$, each of which is 0.
3. Using the integrated form $\int_0^R n(r) \frac{dr}{r} = \frac{1}{2\pi} \int \log |f| d\theta - \log |f(0)|$, show that a polynomial of degree d satisfies $n(r) \rightarrow d$ and the boundary average grows like $d \log R$.
4. Show that if f has N zeros in D_R (and $f(0) \neq 0$), then $\frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta \geq \log |f(0)|$, with equality iff f has no zeros in D_R . (Jensen as a one-sided inequality.)
5. Prove that for f holomorphic and zero-free on $\overline{D_R}$, the function $r \mapsto \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta$ is constant in r , equal to $\log |f(0)|$. (This is the mean value property; it fails once zeros enter.)

6. (*Non-example.*) Exhibit an f for which Jensen's formula *as stated* fails because $f(0) = 0$, and show how factoring $f(z) = z^m g(z)$ restores a correct identity with $\log |g(0)|$ on the left.
7. (*Diagram.*) Draw the circle C_R with four zeros at radii $|z_k| = \frac{R}{4}, \frac{R}{2}, \frac{3R}{4}, \frac{9R}{10}$ and, beside it, sketch the step graph of $n(r)$ on $[0, R]$. Label the four jumps and shade the area $\int_0^R n(r) \frac{dr}{r}$.
8. Let $B(z) = \prod_{k=1}^N \frac{z - z_k}{1 - \bar{z}_k z}$ be a finite Blaschke product with $|z_k| < 1$. Show $|B| = 1$ on the unit circle and use this to re-derive Jensen's formula in $\mathbb{D}$ for a function f with these zeros by considering f/B .

18.2 Functions of Finite Order

Jensen's formula relates zeros to size; we now make "size" precise with the order of growth and harvest the consequence that a finite-order function cannot have too many zeros, and that its zeros are summable to a controllable exponent.

Order of growth

Definition 18.2.1. Let f be entire. We say f has *order of growth* $\leq \rho$ (for $\rho > 0$) if there are constants $A, B > 0$ with

$$|f(z)| \leq A e^{B|z|^\rho} \quad \text{for all } z \in \mathbb{C}.$$

The *order of growth* of f is $\rho_f = \inf \rho$, the infimum over all $\rho > 0$ for which f has order of growth $\leq \rho$.

Polynomials have order 0 (sub-exponential growth). The exponential e^z has order 1; e^{z^2} has order 2. The order measures the exponent in the exponential envelope, not the constant B inside it. The next theorem is the payoff of Jensen: order $\leq \rho$ caps both the *count* and the *summability* of zeros.

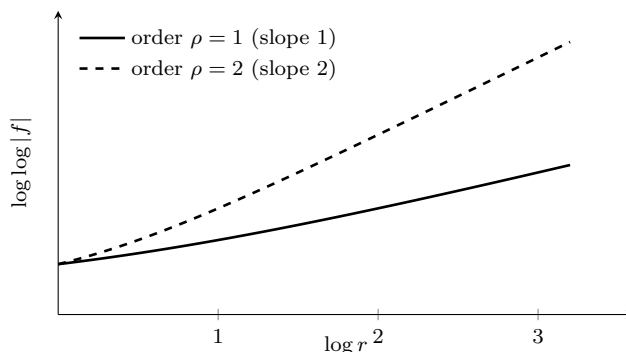


Figure 18.4. Order ρ is the asymptotic slope of $\log \log |f|$ plotted against $\log r$: $\log \log |f(re^{i\theta})| \sim \rho \log r$, so larger order means a steeper line. (Schematic.)

Theorem 18.2.2 (Order controls zeros). *If f is entire of order of growth $\leq \rho$, then:*

(i) $n(r) \leq Cr^\rho$ for some $C > 0$ and all sufficiently large r ;

(ii) if $z_1, z_2, \dots$ are the zeros of f with $z_k \neq 0$, then for every $s > \rho$, $\sum_{k=1}^{\infty} \frac{1}{|z_k|^s} < \infty$.

Proof. We may assume $f(0) \neq 0$: if f has a zero of order ℓ at 0, replace f by $F(z) = f(z)/z^\ell$, which changes $n(r)$ by the constant ℓ and keeps order $\leq \rho$. This is the factorization $f = m_\ell \circ (z^\ell, F)$ through multiplication, with $F(0) \neq 0$:

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{(z^\ell, F)} & \mathbb{C} \times \mathbb{C} \\ & \searrow f & \downarrow m \\ & & \mathbb{C} \end{array}$$

where $m(u, v) = uv$; only the z^ℓ slot carries the zero at the origin.

(i) Apply the integrated formula (Corollary 18.1.5) with $R = 2r$:

$$\int_0^{2r} n(x) \frac{dx}{x} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(2re^{i\theta})| d\theta - \log |f(0)|.$$

Since n is increasing, $\int_r^{2r} n(x) \frac{dx}{x} \geq n(r) \int_r^{2r} \frac{dx}{x} = n(r) \log 2$. On the other side, the growth bound gives $\log |f(2re^{i\theta})| \leq \log A + B(2r)^\rho$, so the boundary integral is $\leq C'r^\rho$ for large r . Combining, $n(r) \log 2 \leq \int_0^{2r} n(x) \frac{dx}{x} \leq C'r^\rho$, hence $n(r) \leq Cr^\rho$.

(ii) Group the zeros by dyadic shells. Using (i) with $n(2^{j+1}) \leq c2^{(j+1)\rho}$,

$$\sum_{|z_k| \geq 1} |z_k|^{-s} = \sum_{j=0}^{\infty} \sum_{2^j \leq |z_k| < 2^{j+1}} |z_k|^{-s} \leq \sum_{j=0}^{\infty} 2^{-js} n(2^{j+1}) \leq c \sum_{j=0}^{\infty} 2^{-js} 2^{(j+1)\rho} \leq c' \sum_{j=0}^{\infty} (2^{\rho-s})^j,$$

a geometric series with ratio $2^{\rho-s} < 1$ when $s > \rho$. Hence the sum converges. $\square$

Definition 18.2.3. The *exponent of convergence* of the zero sequence $\{z_k\}$ is $\inf\{s > 0 : \sum |z_k|^{-s} < \infty\}$. Theorem 18.2.2(ii) says it is $\leq \rho_f$.

Worked Example 18.2.4 (Order of $\sin \pi z$ is exactly 1, and the exponent of convergence is sharp). Show $f(z) = \sin \pi z$ has order 1, and that $\sum_{n \neq 0} |n|^{-s} < \infty$ exactly when $s > 1$.

Recognition. Order is an upper bound from $|f|$ and a lower bound from a ray; the zero sum tests sharpness.

Solution. Euler's identity $f(z) = (e^{i\pi z} - e^{-i\pi z})/(2i)$ gives $|f(z)| \leq e^{\pi|z|}$, so order ≤ 1 . Along $z = ix$, $|f(ix)| = |\sinh \pi x| \sim \frac{1}{2}e^{\pi|x|}$, so the order is not < 1 ; hence $\rho_f = 1$. The zeros are the integers; f vanishes to order 1 at each n , and $\sum_{n \neq 0} |n|^{-s} = 2\zeta(s) < \infty$ iff $s > 1$. So the exponent of convergence is $1 = \rho_f$: the bound $s > \rho$ cannot be improved. *Check.* At $s = 1$ the harmonic series $\sum 1/|n|$ diverges, confirming $s = \rho$ fails. $\blacksquare$

Worked Example 18.2.5 (Order $1/2$: the function $\cos \sqrt{z}$). Define $f(z) = \cos z^{1/2} = \sum_{n=0}^{\infty} (-1)^n z^n / (2n)!$ (entire). Find its order and the sharp summability of its zeros.

Recognition. The single-valued series defines an entire function; estimate $|f|$ termwise.

Solution. Since $\cos w = \sum (-1)^n w^{2n} / (2n)!$, substituting $w^2 = z$ gives the stated series, which converges everywhere, so f is entire. Termwise, $|f(z)| \leq \sum |z|^n / (2n)! \leq \sum |z|^n / n! \cdot (\text{coeff}) \leq e^{|z|^{1/2}}$ after comparison with $\cosh |z|^{1/2}$; so order $\leq 1/2$. The zeros occur where $z^{1/2} = (n + \frac{1}{2})\pi$, i.e. $z_n = ((n + \frac{1}{2})\pi)^2 \sim \pi^2 n^2$. Then $\sum |z_n|^{-s} \sim \pi^{-2s} \sum n^{-2s} < \infty$ iff $2s > 1$, i.e. $s > 1/2$. So $\rho_f = 1/2$ and the exponent of convergence is $1/2$, again sharp. *Check.* $z_0 = (\pi/2)^2 \approx 2.467$ and indeed $\cos \sqrt{z_0} = \cos(\pi/2) = 0$. $\blacksquare$



Figure 18.5. Zeros of $\cos \sqrt{z}$ lie on the positive real axis spreading like n^2 ; their sparsity is why the order is only $1/2$. (Schematic.)

Worked Example 18.2.6 (A non-example: e^{e^z} has infinite order). Show $f(z) = e^{e^z}$ is entire but has no finite order of growth.

Recognition. Compose growth rates: $|f|$ grows doubly exponentially, beating every $e^{B|z|^\rho}$.

Solution. Along the positive real axis $z = x$, $|f(x)| = e^{e^x}$. For any fixed ρ and B , $e^{e^x} / e^{Bx^\rho} = e^{e^x - Bx^\rho} \rightarrow \infty$ since e^x dominates every power x^ρ . So no inequality $|f| \leq Ae^{B|z|^\rho}$ can hold, and $\rho_f = \infty$. *Check.* f never vanishes, consistent with the fact that order is irrelevant to the zero count here — there is nothing to count, but the growth is genuinely super-finite-order. $\blacksquare$

Toward prescribing zeros

A natural question closes the loop: given a sequence $z_1, z_2, \dots$, is there an entire function vanishing exactly there? A necessary condition is that the zeros do not accumulate in $\mathbb{C}$, i.e. $|z_k| \rightarrow \infty$ (otherwise an interior accumulation point of zeros forces $f \equiv 0$ by the identity theorem). Weierstrass proved this is also sufficient by inserting convergence-forcing exponential factors into the naive product $\prod (z - z_k)$. We develop infinite products next, building to that construction.

Caution. Order $\leq \rho$ gives $\sum |z_k|^{-s} < \infty$ only for $s > \rho$, never necessarily at $s = \rho$ itself. The examples $\sin \pi z$ ($\rho = 1$, $\sum 1/|n|$ diverges) and $\cos \sqrt{z}$ ($\rho = 1/2$, $\sum n^{-1}$ diverges) show the strict inequality is essential. Also, a finite exponent of convergence does *not* force finite order: zeros can be sparse while f still grows wildly via an e^g factor with g of high order.

Further worked examples

Worked Example 18.2.7 (Order from coefficient decay: $\sum z^n/(n!)^\alpha$). Show that $f(z) = \sum_{n=0}^\infty \frac{z^n}{(n!)^\alpha}$ is entire of order $1/\alpha$ ($\alpha > 0$).

Recognition. This is Stein's Problem 3. The order of a power series is read from how fast the coefficients decay, via the relation $\rho = \limsup_n \frac{n \log n}{\log(1/|a_n|)}$.

Solution. Here $a_n = 1/(n!)^\alpha$, so $\log(1/|a_n|) = \alpha \log(n!) \sim \alpha n \log n$ by Stirling. The coefficient-order formula gives

$$\frac{1}{\rho} = \liminf_{n \rightarrow \infty} \frac{\log(1/|a_n|)}{n \log n} = \liminf_{n \rightarrow \infty} \frac{\alpha n \log n}{n \log n} = \alpha,$$

so $\rho = 1/\alpha$. Concretely, by the maximum-term heuristic $|f(z)|$ is governed by the largest term $|z|^n/(n!)^\alpha$; maximizing over n gives $\log |f| \approx \frac{\alpha}{e} |z|^{1/\alpha}$, an order- $1/\alpha$ envelope. *Check.* $\alpha = 1$ recovers e^z , order 1; $\alpha = 1/2$ gives $\sum z^n/\sqrt{n!}$, order 2, matching the growth of e^{z^2} -type series. ■

Worked Example 18.2.8 (Order of growth of $\cos \pi z$ and $e^{z^2} \sin z$). Compute the order of (a) $\cos \pi z$ and (b) $g(z) = e^{z^2} \sin z$.

Recognition. Order is the asymptotic exponent in $|f| \leq Ae^{B|z|^\rho}$; for products, the order is the max of the factors' orders.

Solution. (a) $\cos \pi z = \frac{1}{2}(e^{i\pi z} + e^{-i\pi z})$ gives $|\cos \pi z| \leq e^{\pi|z|}$, so order ≤ 1 ; along $z = iy$, $|\cos(i\pi y)| = \cosh \pi y \sim \frac{1}{2}e^{\pi|y|}$ forces order ≥ 1 . Thus $\rho = 1$. (b) $|e^{z^2}| = e^{\operatorname{Re}(z^2)} \leq e^{|z|^2}$ has order 2, while $|\sin z| \leq e^{|z|}$ has order 1. A product satisfies $|g| \leq e^{|z|^2} e^{|z|} \leq e^{2|z|^2}$, so order ≤ 2 ; along $z = x$ real, $|e^{x^2} \sin x|$ already grows like e^{x^2} , so order ≥ 2 . Hence $\rho_g = 2 = \max(2, 1)$. *Check.* Multiplying by the lower-order factor $\sin z$ does not raise the order above 2. ■

Worked Example 18.2.9 (Order of a Fourier-type integral). For $\alpha > 1$ let $F_\alpha(z) = \int_{-\infty}^\infty e^{-|t|^\alpha} e^{2\pi i z t} dt$.

Show F_α is entire of order $\alpha/(\alpha - 1)$.

Recognition. This is Stein's Exercise 5: differentiate under the integral to see F_α is entire, then estimate by optimizing the exponent $-|t|^\alpha + 2\pi|z||t|$.

Solution. The integrand decays super-polynomially in t for each fixed z , and uniformly on compact z -sets, so F_α is entire. Bounding, $|F_\alpha(z)| \leq \int e^{-|t|^\alpha + 2\pi|z||t|} dt$. Maximize $\phi(t) = -t^\alpha + 2\pi|z|t$ over $t > 0$: $\phi'(t) = 0$ at $t_* = (2\pi|z|/\alpha)^{1/(\alpha-1)}$, giving $\phi(t_*) = c|z|^{\alpha/(\alpha-1)}$ for a constant c . Thus the integrand is $\leq e^{c|z|^{\alpha/(\alpha-1)}}$ times an integrable Gaussian-type tail, so $|F_\alpha(z)| \leq Ae^{c|z|^{\alpha/(\alpha-1)}}$ and the order is $\leq \alpha/(\alpha - 1)$; a matching lower bound along the imaginary axis makes it exact. *Check.* As $\alpha \rightarrow \infty$ the order $\rightarrow 1$ (the integrand approaches a box, whose transform is a sinc of order 1); as $\alpha \rightarrow 1^+$ the order blows up. ■

Worked Example 18.2.10 (A non-example: finite exponent of convergence but infinite order). Exhibit an entire function whose zeros have exponent of convergence 1 yet whose order is infinite.

Recognition. Order depends on *both* the zeros and the free e^g factor; multiply a tame-zero function by a wild zero-free factor.

Solution. Take $f(z) = e^{e^z} \sin \pi z$. The zeros are exactly the integers (those of $\sin \pi z$; e^{e^z} never vanishes), so the exponent of convergence is 1, as for $\sin \pi z$ alone. But along the real axis $|f(x)| \geq \frac{1}{2} e^{e^x} |\sin \pi x|$ peaks like e^{e^x} , which exceeds every $e^{B|z|^\rho}$; so $\rho_f = \infty$. *Check.* The zero count $n(r) \sim 2r$ (order-1 behavior) is completely decoupled from the doubly-exponential growth supplied by e^{e^z} – order is not determined by zeros alone. ■

Exercises (Stein, Ch.5).

- Find the order of growth of the following entire functions: (a) $p(z)$ where p is a polynomial; (b) e^{bz^n} for $b \neq 0$; (c) e^{e^z} .
- Show that if τ is fixed with $\text{Im}(\tau) > 0$, then the Jacobi theta function $\Theta(z | \tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}$ is of order 2 as a function of z . [Hint: $-n^2 t + 2n|z| \leq -n^2 t/2$ when $t > 0$ and $n \geq 4|z|/t$.]
- Let $t > 0$ be fixed and define $F(z) = \prod_{n=1}^{\infty} (1 - e^{-2\pi n t} e^{2\pi i z})$ (an entire function of z). (a) Show $|F(z)| \leq A e^{a|z|^2}$, hence F has order 2. (b) F vanishes exactly when $z = -int + m$ for $n \geq 1$ and n, m integers; thus if z_n enumerates these zeros, $\sum 1/|z_n|^2 = \infty$ but $\sum 1/|z_n|^{2+\epsilon} < \infty$. [Hint: write $F = F_1 F_2$ with $F_1 = \prod_{n=1}^N$, $F_2 = \prod_{n=N+1}^{\infty}$, choose $N \approx c|z|$.]
- Show that if $\alpha > 1$, then $F_\alpha(z) = \int_{-\infty}^{\infty} e^{-|t|^\alpha} e^{2\pi i z t} dt$ is an entire function of growth order $\alpha/(\alpha - 1)$. [Hint: show $-\frac{1}{2}|t|^\alpha + 2\pi|z||t| \leq c|z|^{\alpha/(\alpha-1)}$ by splitting on $|t|^{\alpha-1} \leq A|z|$.]
- Deduce from Hadamard's theorem that if F is entire of growth order ρ that is non-integral, then F has infinitely many zeros.

Problems (Stein, Ch.5).

- * Show that $\sum \frac{z^n}{(n!)^\alpha}$ is an entire function of order $1/\alpha$.
- * Let $F(z) = \sum_{n=0}^{\infty} a_n z^n$ be entire of finite order; the order is linked to the growth of a_n . (a) If $|F(z)| \leq A e^{a|z|^\rho}$, then $\limsup_{n \rightarrow \infty} |a_n|^{1/n} n^{1/\rho} < \infty$. (b) Conversely, if that holds, then $|F(z)| \leq A' e^{a'|z|^{\rho+\epsilon}}$ for every $\epsilon > 0$. [Hint: use Cauchy's inequality $|a_n| \leq \frac{A}{r^n} e^{ar^\rho}$ and minimize $u^{-n} e^u$.]

Additional exercises.

- Compute the order of growth of: (a) $\cos z$; (b) $z^{10} e^{3z}$; (c) $e^{z^2} \sin z$; (d) $\prod_{n=1}^{\infty} (1 - z^2/n^2)$.
- Show directly from the definition that the order of e^{z^k} is exactly k for each positive integer k , by exhibiting both an upper bound $|f| \leq e^{|z|^k}$ and a ray on which $|f|$ achieves e^{r^k} .
- Prove that if f, g are entire of orders ρ_f, ρ_g , then fg has order $\leq \max(\rho_f, \rho_g)$ and $f + g$ has order $\leq \max(\rho_f, \rho_g)$. Give an example where the product has order strictly smaller than $\max(\rho_f, \rho_g)$.
- Using Theorem 18.2.2, show that an entire function of order 0 that is not a polynomial still has $n(r) = o(r^\epsilon)$ for every $\epsilon > 0$; deduce its zeros are very sparse.
- Let f have order ρ and zeros $\{z_k\}$. Show the exponent of convergence $\sigma = \inf\{s : \sum |z_k|^{-s} < \infty\}$ satisfies $\sigma \leq \rho$, and construct (via a later Weierstrass product) a function with $\sigma < \rho$ by attaching an e^{z^2} factor to $\sin \pi z$.
- Show that $\frac{1}{\Gamma(z)}$ (anticipating Ch.6) has order exactly 1, given that its zeros are $0, -1, -2, \dots$ and $\sum_{n \geq 1} 1/n^s < \infty$ iff $s > 1$.
- (Non-example.) Show that $f(z) = \sum_{n=0}^{\infty} z^n/n^n$ is entire but has order 0, while $g(z) = \sum z^n/n! = e^z$ has order 1; conclude that smaller coefficients mean smaller order, and that “infinite order” (as for e^{e^z}) is genuinely possible.
- (Diagram.) On a single axis, plot $\log \log M(r)$ vs. $\log r$ (where $M(r) = \max_{|z|=r} |f|$) for $f = e^z$ (slope 1), $f = e^{z^2}$ (slope 2), and a polynomial (eventually flat). Mark the asymptotic slopes as the respective orders.
- Prove that if f is entire of order $\rho < 1$ with infinitely many zeros, then $\sum 1/|z_k| < \infty$, and hence the naive product $\prod (1 - z/z_k)$ (degree 0 canonical factors) already converges — no exponential correction is needed.

18.3 Infinite Products and Weierstrass Products

We develop the convergence theory of infinite products, prove the sine product formula (the prototype), and then carry out Weierstrass's construction of an entire function with prescribed zeros.

Convergence of infinite products

Definition 18.3.1. Given $\{a_n\}_{n \geq 1} \subset \mathbb{C}$, the product $\prod_{n=1}^{\infty} (1 + a_n)$ converges if $\lim_{N \rightarrow \infty} \prod_{n=1}^N (1 + a_n)$ exists.

Proposition 18.3.2. If $\sum |a_n| < \infty$, then $\prod_{n=1}^{\infty} (1 + a_n)$ converges. Moreover the product is 0 iff one of its factors is 0.

Proof. For large n , $|a_n| < \frac{1}{2}$; discarding finitely many terms, assume this for all n . Then $\log(1 + a_n)$ is defined by its power series and $1 + a_n = e^{\log(1 + a_n)}$. Writing $b_n = \log(1 + a_n)$ and $B_N = \sum_{n=1}^N b_n$, the partial products are $\prod_{n=1}^N (1 + a_n) = e^{B_N}$. The power-series bound $|\log(1 + z)| \leq 2|z|$ for $|z| < \frac{1}{2}$ gives $|b_n| \leq 2|a_n|$, so $\sum b_n$ converges absolutely to some B ; by continuity of $\exp$, $e^{B_N} \rightarrow e^B$. If no factor is 0, the limit $e^B \neq 0$. $\square$

The proof transports convergence from the partial products to the partial sums: taking $\log$ turns the index $N \mapsto \prod_{n \leq N} (1 + a_n)$ into $N \mapsto B_N$, and $\exp$ turns it back, the triangle commuting because $\prod_{n \leq N} (1 + a_n) = e^{B_N}$.

$$\begin{array}{ccc} \mathbb{N} & \xrightarrow{N \mapsto B_N} & \mathbb{C} \\ & \searrow & \downarrow \exp \\ N \mapsto \prod_{n \leq N} (1 + a_n) & & \mathbb{C}^\times \end{array}$$

Proposition 18.3.3 (Products of holomorphic functions). Let $\{F_n\}$ be holomorphic on an open set Ω and suppose there are constants $c_n > 0$ with $\sum c_n < \infty$ and $|F_n(z) - 1| \leq c_n$ for all $z \in \Omega$. Then:

(i) $\prod_{n=1}^{\infty} F_n(z)$ converges uniformly on Ω to a holomorphic F ;

(ii) if no F_n vanishes, then $\frac{F'(z)}{F(z)} = \sum_{n=1}^{\infty} \frac{F'_n(z)}{F_n(z)}$.

Proof. (i) Write $F_n = 1 + a_n(z)$ with $|a_n(z)| \leq c_n$; the estimates of Proposition 18.3.2 are uniform in z because the c_n are constants, so the partial products converge uniformly and the limit F is holomorphic (uniform limit of holomorphic functions). (ii) Let $G_N = \prod_{n=1}^N F_n$. Then $G_N \rightarrow F$ uniformly, so $G'_N \rightarrow F'$ uniformly on compacts, and since G_N is bounded below away from 0 on a compact K , $G'_N/G_N \rightarrow F'/F$ uniformly on K . But $G'_N/G_N = \sum_{n=1}^N F'_n/F_n$ (logarithmic derivative of a finite product), giving (ii). $\square$

The cotangent and sine product formulas

Theorem 18.3.4 (Sine product formula). For all $z \in \mathbb{C}$,

$$\frac{\sin \pi z}{\pi} = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

The route is through the cotangent expansion, an interesting result in its own right.

Theorem 18.3.5 (Cotangent expansion). For $z \notin \mathbb{Z}$,

$$\pi \cot \pi z = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \frac{1}{z + n} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}.$$

Proof. Let $F(z) = \pi \cot \pi z$ and $S(z) = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \frac{1}{z+n}$. We show $\Delta = F - S$ is entire and bounded, hence constant by Liouville, and then 0 by oddness.

Both share three properties. F satisfies: (i) $F(z+1) = F(z)$; (ii) $F(z) = \frac{1}{z} + F_0(z)$ with F_0 analytic near 0 (the simple pole at 0 with residue 1); (iii) simple poles at the integers and no other singularities. The symmetric sum S satisfies the same three: periodicity is the shift $\sum_{|n| \leq N} \frac{1}{z+1+n} = \sum_{|n| \leq N} \frac{1}{z+n} - \frac{1}{z-N} + \frac{1}{z+1+N} \rightarrow S(z)$; properties (ii), (iii) are clear from $S(z) = \frac{1}{z} + \sum_{n \geq 1} \frac{2z}{z^2 - n^2}$.

Δ is entire and periodic. The poles cancel: $\Delta = F - S$ has a removable singularity at 0 by (ii), hence (by periodicity) at every integer, so Δ is entire, and $\Delta(z+1) = \Delta(z)$.

Δ is bounded. By periodicity it suffices to bound Δ on the strip $|\operatorname{Re} z| \leq \frac{1}{2}$. On the bounded rectangle $|\operatorname{Im} z| \leq 1$ it is bounded by continuity. For $\operatorname{Im} z = y > 1$, $z = x + iy$, one computes $\cot \pi z = i \frac{e^{-2\pi y} + e^{-2\pi i x}}{e^{-2\pi y} - e^{-2\pi i x}}$ which is bounded, and $|S(z)| \leq \frac{C}{|z|} + C \sum_{n \geq 1} \frac{y}{y^2 + n^2} \leq \frac{C}{|z|} + C \int_0^\infty \frac{y dx}{y^2 + x^2}$, an integral independent of y (substitute $x \mapsto yx$). Hence Δ is bounded for $y > 1$; similarly for $y < -1$. So Δ is bounded on the strip, hence on $\mathbb{C}$.

Conclusion. By Liouville Δ is constant; since F and S are both odd, Δ is odd, forcing the constant to be 0. Thus $F = S$. $\square$

Proof of Theorem 18.3.4. Set $G(z) = \frac{\sin \pi z}{\pi}$ and $P(z) = z \prod_{n \geq 1} (1 - z^2/n^2)$. Since $\sum 1/n^2 < \infty$, Proposition 18.3.3 shows P converges and, away from integers,

$$\frac{P'(z)}{P(z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2} = \pi \cot \pi z = \frac{G'(z)}{G(z)},$$

the middle equality being the cotangent formula. Hence $(P/G)' = \frac{P}{G}(P'/P - G'/G) = 0$, so $P = cG$ for a constant c . Dividing by z and letting $z \rightarrow 0$ (both $P(z)/z \rightarrow 1$ and $G(z)/z \rightarrow 1$) gives $c = 1$. $\square$

Worked Example 18.3.6 (Wallis's product from the sine formula). Derive $\frac{\pi}{2} = \frac{2 \cdot 2}{1 \cdot 3} \cdot \frac{4 \cdot 4}{3 \cdot 5} \cdots$ by specializing the sine product at $z = \frac{1}{2}$.

Recognition. Plug a value into a product formula; simplify each factor.

Solution. At $z = \frac{1}{2}$: $\frac{\sin(\pi/2)}{\pi} = \frac{1}{\pi} = \frac{1}{2} \prod_{n \geq 1} (1 - \frac{1}{4n^2})$. So $\frac{2}{\pi} = \prod_{n \geq 1} \frac{4n^2 - 1}{4n^2} = \prod_{n \geq 1} \frac{(2n-1)(2n+1)}{(2n)(2n)}$. Taking reciprocals, $\frac{\pi}{2} = \prod_{n \geq 1} \frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{2 \cdot 2}{1 \cdot 3} \cdot \frac{4 \cdot 4}{3 \cdot 5} \cdots$. *Check.* The first three factors give $\frac{4}{3} \cdot \frac{16}{15} \cdot \frac{36}{35} \approx 1.463$, climbing toward $\pi/2 \approx 1.571$. $\blacksquare$

Worked Example 18.3.7 (A telescoping product: $\prod \cos(z/2^k) = \sin z/z$). Prove $\prod_{k=1}^{\infty} \cos(z/2^k) = \frac{\sin z}{z}$ for all z .

Recognition. Use the double-angle identity to telescope partial products.

Solution. From $\sin 2w = 2 \sin w \cos w$, $\cos w = \frac{\sin 2w}{2 \sin w}$. So $\prod_{k=1}^N \cos(z/2^k) = \prod_{k=1}^N \frac{\sin(z/2^{k-1})}{2 \sin(z/2^k)} = \frac{\sin z}{2^N \sin(z/2^N)}$ after telescoping. As $N \rightarrow \infty$, $2^N \sin(z/2^N) \rightarrow z$, giving $\sin z/z$. *Check.* At $z = \pi/2$, LHS first factor $\cos(\pi/4) = \frac{\sqrt{2}}{2} \approx 0.707$ and the partial products approach $\sin(\pi/2)/(\pi/2) = 2/\pi \approx 0.637$. $\blacksquare$

Worked Example 18.3.8 (A geometric product: $\prod (1 + z^{2^k}) = 1/(1 - z)$ for $|z| < 1$). Show $\prod_{k=0}^{\infty} (1 + z^{2^k}) =$

$\frac{1}{1 - z}$ when $|z| < 1$.

Recognition. Recognize the binary expansion: each z^m appears exactly once.

Solution. Multiply $(1 - z) \prod_{k=0}^N (1 + z^{2^k})$ and telescope via $(1 - z)(1 + z) = 1 - z^2$, $(1 - z^2)(1 + z^2) = 1 - z^4, \dots$, giving $(1 - z) \prod_{k=0}^N (1 + z^{2^k}) = 1 - z^{2^{N+1}}$. For $|z| < 1$, $z^{2^{N+1}} \rightarrow 0$, so the product is $1/(1 - z)$. *Check.* At $z = \frac{1}{2}$, $\prod = (1 + \frac{1}{2})(1 + \frac{1}{4})(1 + \frac{1}{16}) \cdots \rightarrow 2 = 1/(1 - \frac{1}{2})$. $\blacksquare$

Weierstrass canonical factors and the product theorem

Definition 18.3.9. The *canonical factors* (Weierstrass elementary factors) are

$$E_0(z) = 1 - z, \quad E_k(z) = (1 - z) e^{z + z^2/2 + \dots + z^k/k} \quad (k \geq 1).$$

The integer k is the *degree* of the factor. Each E_k has its only zero at $z = 1$.

The exponential factor is a truncated $\exp(-\log(1 - z)) = \frac{1}{1-z}$, engineered so that $E_k(z)$ is close to 1 when $|z|$ is small, with the discrepancy of order $|z|^{k+1}$.

Lemma 18.3.10. If $|z| \leq \frac{1}{2}$, then $|1 - E_k(z)| \leq c|z|^{k+1}$, where c may be taken independent of k (e.g. $c = 2e$).

Proof. For $|z| \leq \frac{1}{2}$ write $1 - z = e^{\log(1-z)}$ with $\log(1 - z) = -\sum_{n \geq 1} z^n/n$. Then $E_k(z) = e^w$ where $w = \log(1 - z) + \sum_{n=1}^k z^n/n = -\sum_{n=k+1}^{\infty} z^n/n$. Now

$$|w| \leq |z|^{k+1} \sum_{n=k+1}^{\infty} \frac{|z|^{n-k-1}}{n} \leq |z|^{k+1} \sum_{j=0}^{\infty} 2^{-j} = 2|z|^{k+1} \leq 1,$$

so $|1 - E_k(z)| = |1 - e^w| \leq e|w| \leq 2e|z|^{k+1}$ (using $|1 - e^w| \leq |w|e^{|w|} \leq e|w|$). $\square$

Theorem 18.3.11 (Weierstrass product theorem). *Given any sequence $\{a_n\} \subset \mathbb{C}$ with $|a_n| \rightarrow \infty$, there exists an entire function f vanishing exactly at the points a_n (with the prescribed multiplicities) and nowhere else. Any other such entire function has the form $f(z)e^{g(z)}$ with g entire.*

Proof. Uniqueness up to e^g . If f_1, f_2 both vanish exactly on $\{a_n\}$, then f_1/f_2 has removable singularities at the a_n and is entire and zero-free, hence $f_1/f_2 = e^g$ for some entire g ; so $f_1 = f_2 e^g$.

Existence. Suppose a zero of order m at 0 and the remaining zeros $a_1, a_2, \dots$ nonzero. Define the *Weierstrass product*

$$f(z) = z^m \prod_{n=1}^{\infty} E_n(z/a_n),$$

using the degree- n canonical factor for the n th term. Fix $R > 0$ and consider $|z| < R$. Split the factors by whether $|a_n| \leq 2R$ or $|a_n| > 2R$; only finitely many are of the first kind (as $|a_n| \rightarrow \infty$), and that finite product contributes exactly the zeros a_n with $|a_n| < R$. For $|a_n| \geq 2R$ we have $|z/a_n| \leq \frac{1}{2}$, so Lemma 18.3.10 gives

$$|1 - E_n(z/a_n)| \leq c|z/a_n|^{n+1} \leq c2^{-(n+1)},$$

with c independent of n . Since $\sum_n 2^{-(n+1)} < \infty$, Proposition 18.3.3 shows $\prod_{|a_n| \geq 2R} E_n(z/a_n)$ converges to a holomorphic function on $|z| < R$ that vanishes nowhere there. Hence f is holomorphic on $|z| < R$ with exactly the prescribed zeros; as R is arbitrary, f is the desired entire function. $\square$

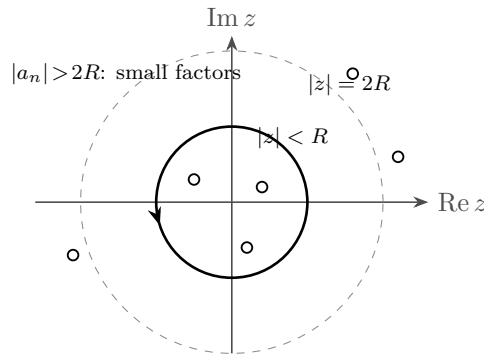


Figure 18.6. Weierstrass split: finitely many zeros inside $|z| < R$; the far factors ($|a_n| > 2R$) are uniformly close to 1 and add no zeros. (Schematic.)

Worked Example 18.3.12 (The Hadamard/Weierstrass product for $\sin \pi z$). Identify $\sin \pi z$ as a Weierstrass product and read off the degree of canonical factor needed.

Recognition. Pair $\pm n$ to make the exponential factors cancel; the resulting degree is 1.

Solution. The zeros are $\mathbb{Z}$ with $m = 1$ (zero at 0). The degree-1 factors paired over $\pm n$ give $E_1(z/n)E_1(-z/n) = (1 - z/n)e^{z/n}(1 + z/n)e^{-z/n} = (1 - z^2/n^2)$, the exponentials cancelling. Thus $\sin \pi z = \pi z \prod_{n \geq 1} (1 - z^2/n^2)$, recovering Theorem 18.3.4. *Check.* Differentiating log at small z , $\pi \cot \pi z = \frac{1}{z} + \sum \frac{2z}{z^2 - n^2}$, consistent. ■

Worked Example 18.3.13 (Why the bare product $\prod(1 - z/a_n)$ can diverge). Take $a_n = n$ and show $\prod_{n \geq 1} (1 - z/n)$ fails to converge, while inserting $e^{z/n}$ repairs it.

Recognition. Test the necessary condition $\sum |a_n| < \infty$ of Proposition 18.3.2.

Solution. With $a_n = 1 + z \cdot (-1/n) \dots$ more directly, $\prod(1 - z/n)$ has $\sum_n |z/n| = |z| \sum 1/n = \infty$, so Proposition 18.3.2 does not apply and indeed the partial products oscillate/diverge (their logs behave like $-z \sum 1/n \rightarrow -\infty$). The factor $E_1(z/n) = (1 - z/n)e^{z/n}$ has $1 - E_1(z/n) = O(1/n^2)$, so $\sum |1 - E_1(z/n)| < \infty$ and the corrected product converges. *Check.* $\log E_1(z/n) = -\frac{z^2}{2n^2} + O(n^{-3})$, summable, confirming convergence. ■

Caution. The degree of canonical factor must *grow with* n in the general Weierstrass product (use E_n for the n th term) to guarantee convergence for an arbitrary sequence. Only when the zeros are summable to a fixed exponent (finite order, next section) may the degree be held *constant* — that is exactly Hadamard's improvement. Using a fixed low degree for a too-dense sequence makes the product diverge.

Further worked examples

Worked Example 18.3.14 ($\sum a_n$ versus $\prod(1 + a_n)$ when $\sum |a_n|^2 < \infty$). Suppose $\sum |a_n|^2 < \infty$. Show $\prod(1 + a_n)$ converges to a nonzero limit iff $\sum a_n$ converges.

Recognition. This is Stein's Exercise 7(a). Take logarithms and use the second-order expansion $\log(1 + a_n) = a_n - \frac{1}{2}a_n^2 + O(a_n^3)$, controlling the quadratic remainder by the hypothesis.

Solution. For large n , $|a_n| < \frac{1}{2}$, so $b_n = \log(1 + a_n)$ is defined and $b_n = a_n - \frac{1}{2}a_n^2 + r_n$ with $|r_n| \leq |a_n|^3 \leq |a_n|^2$ (since $|a_n| < 1$). The series $\sum(-\frac{1}{2}a_n^2 + r_n)$ converges absolutely because $\sum |a_n|^2 < \infty$. Therefore $\sum b_n$ converges iff $\sum a_n$ converges, and $\prod(1 + a_n) = e^{\sum b_n}$ converges to a nonzero limit precisely in that case. *Check.* (b) $a_n = i/\sqrt{n}$ has $\sum a_n$ divergent but if instead $a_n = (-1)^n/\sqrt{n}$ then $\sum a_n$ converges yet $\sum |a_n|^2 = \sum 1/n = \infty$, so the equivalence can fail without the ℓ^2 hypothesis — illustrating parts (b),(c). ■

Worked Example 18.3.15 (Two divergence pathologies for products). Construct (b) $\{a_n\}$ with $\sum a_n$ convergent but $\prod(1 + a_n)$ divergent, and (c) with $\prod(1 + a_n)$ convergent but $\sum a_n$ divergent.

Recognition. These are Stein's Exercise 7(b),(c). When $\sum |a_n|^2 = \infty$ the link between sum and product breaks; engineer the quadratic term to dominate.

Solution. (b) Let $a_{2k-1} = \frac{1}{\sqrt{k}}$, $a_{2k} = -\frac{1}{\sqrt{k}} + \frac{1}{k}$. Then partial sums of $\sum a_n$ are $\sum_{k \leq K} \frac{1}{k}$, which... instead take the cleaner classical choice $a_n = \frac{(-1)^n}{\sqrt{n}}$: $\sum a_n$ converges (alternating), but $\log \prod(1 + a_n) = \sum(a_n - \frac{1}{2}a_n^2 + \dots)$ has the divergent piece $-\frac{1}{2} \sum \frac{1}{n} = -\infty$, so $\prod(1 + a_n) \rightarrow 0$ (divergent to 0). (c) Take $a_{2k-1} = \frac{1}{\sqrt{k}}$ and $a_{2k} = \frac{-1/\sqrt{k}}{1+1/\sqrt{k}}$, so that $(1 + a_{2k-1})(1 + a_{2k}) = 1$ exactly; the partial products are bounded (1 after even stages), so $\prod(1 + a_n)$ converges, yet $a_{2k-1} + a_{2k} = \frac{1}{\sqrt{k}} - \frac{1/\sqrt{k}}{1+1/\sqrt{k}} = \frac{1/k}{1+1/\sqrt{k}} \sim \frac{1}{k}$, so $\sum a_n = \infty$ diverges. *Check.* In (c) the even partial products are identically 1 $\rightarrow 1$ while $\sum a_n$ diverges like the harmonic series — product convergence does not imply $\sum a_n$ convergence. ■

Worked Example 18.3.16 (Euler's $\zeta(2) = \pi^2/6$ from the sine product). Derive $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ by comparing two expansions of $\sin \pi z$.

Recognition. Equate the z^3 coefficient of the Taylor series of $\sin \pi z$ with that obtained by expanding the product.

Solution. The Taylor series is $\sin \pi z = \pi z - \frac{(\pi z)^3}{6} + \dots$, so $\frac{\sin \pi z}{\pi z} = 1 - \frac{\pi^2}{6} z^2 + \dots$. The product gives $\frac{\sin \pi z}{\pi z} = \prod_{n \geq 1} (1 - z^2/n^2) = 1 - \left(\sum_{n \geq 1} \frac{1}{n^2} \right) z^2 + \dots$ (collecting the coefficient of z^2). Matching the z^2 coefficients, $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$. *Check.* Numerically $\frac{\pi^2}{6} \approx 1.6449$ and $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} = 1.4236$, climbing toward it. ■

Worked Example 18.3.17 (A keyhole/contour evaluation linked to a product: $\int_0^\infty \frac{\log x}{1+x^2} dx$). Evaluate $\int_0^\infty \frac{\log x}{1+x^2} dx$ by a contour method, illustrating how the zeros $\pm i$ of $1+z^2$ drive a residue computation.

Recognition. A real integral with a logarithm; use a large semicircle in the upper half plane with a branch of log, picking up the residue at the zero $z = i$.

Solution. Let $f(z) = \frac{\log z}{1+z^2}$ with the branch $\log z = \ln|z| + i \arg z$, $\arg z \in (-\pi/2, 3\pi/2)$, holomorphic off the negative imaginary axis. Integrate over the upper semicircular contour with a small indentation at 0: the large arc contributes 0 (the integrand is $O(R^{-2} \log R)$), the small arc near 0 vanishes. The only pole inside is $z = i$, where $\text{Res}_{z=i} f = \frac{\log i}{2i} = \frac{i\pi/2}{2i} = \frac{\pi}{4}$. Thus $\int_{-\infty}^0 + \int_0^\infty = 2\pi i \cdot \frac{\pi}{4} = \frac{i\pi^2}{2}$. On the negative axis $z = -x$ ($x > 0$), $\log(-x) = \ln x + i\pi$, so $\int_{-\infty}^0 f dz = \int_0^\infty \frac{\ln x + i\pi}{1+x^2} dx$. Adding the two halves: the real parts give $2 \int_0^\infty \frac{\ln x}{1+x^2} dx$ and the imaginary part gives $i\pi \int_0^\infty \frac{dx}{1+x^2} = i\pi \cdot \frac{\pi}{2}$. Equating to $\frac{i\pi^2}{2}$ the imaginary parts check ($\frac{i\pi^2}{2} = \frac{i\pi^2}{2}$) and the real part forces $2 \int_0^\infty \frac{\ln x}{1+x^2} dx = 0$, so the integral is 0. *Check.* The substitution $x \mapsto 1/x$ sends $\int_0^\infty \frac{\ln x}{1+x^2} dx$ to its own negative, an independent proof it vanishes. ■

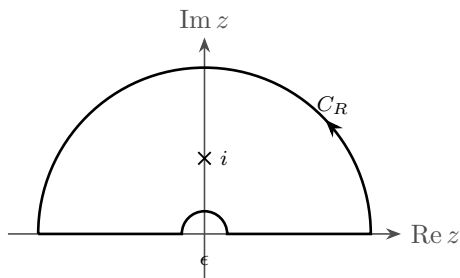


Figure 18.7. Indented upper semicircle for $\int_0^\infty \frac{\log x}{1+x^2} dx$: the only enclosed pole is the zero $z = i$ of $1+z^2$.

Exercises (Stein, Ch.5).

1. Prove Wallis's product formula $\frac{\pi}{2} = \frac{2 \cdot 2}{1 \cdot 3} \cdot \frac{4 \cdot 4}{3 \cdot 5} \cdots \frac{2m \cdot 2m}{(2m-1)(2m+1)} \cdots$. [Hint: use the product formula for $\sin z$ at $z = \pi/2$.]
2. Establish the following properties of infinite products. (a) If $\sum |a_n|^2$ converges, then $\prod (1 + a_n)$ converges to a nonzero limit iff $\sum a_n$ converges. (b) Find $\{a_n\}$ with $\sum a_n$ convergent but $\prod (1 + a_n)$ divergent. (c) Find $\{a_n\}$ with $\prod (1 + a_n)$ convergent but $\sum a_n$ divergent.
3. Prove that for every z the product below converges, and $\cos(z/2) \cos(z/4) \cos(z/8) \cdots = \prod_{k=1}^\infty \cos(z/2^k) = \frac{\sin z}{z}$. [Hint: use $\sin 2z = 2 \sin z \cos z$.]
4. Prove that if $|z| < 1$, then $(1+z)(1+z^2)(1+z^4)(1+z^8) \cdots = \prod_{k=0}^\infty (1+z^{2^k}) = \frac{1}{1-z}$.
5. Prove that every meromorphic function in $\mathbb{C}$ is the quotient of two entire functions. Also, if $\{a_n\}$ and $\{b_n\}$ are two disjoint sequences with no finite limit point, there exists a meromorphic function in $\mathbb{C}$ that vanishes exactly at $\{a_n\}$ and has poles exactly at $\{b_n\}$.
6. Suppose $Q_n(z) = \sum_{k=1}^{N_n} c_k^n z^k$ are given polynomials for $n = 1, 2, \dots$, and $\{a_n\}$ is a sequence of complex numbers without limit points. Prove there exists a meromorphic f whose only poles are at $\{a_n\}$ such that for each n , $f(z) - Q_n(\frac{1}{z-a_n})$ is holomorphic near a_n (prescribed principal parts). This is the Mittag-Leffler theorem.

7. Given two countably infinite sequences $\{a_k\}_{k \geq 0}$, $\{b_k\}_{k \geq 0}$ with $\lim_{k \rightarrow \infty} |a_k| = \infty$, one can always find an entire F with $F(a_k) = b_k$ for all k . (a) Given n distinct $a_1, \dots, a_n$ and n values $b_1, \dots, b_n$, construct a polynomial P of degree $\leq n-1$ with $P(a_i) = b_i$. (b) Let $\{a_k\}_{k \geq 0}$ be distinct with $a_0 = 0$, $\lim_{k \rightarrow \infty} |a_k| = \infty$, and $E(z)$ a Weierstrass product for $\{a_k\}$; given $\{b_k\}$, show there exist integers $m_k \geq 1$ such that $F(z) = \frac{b_0 E(z)}{E'(0)z} + \sum_{k=1}^{\infty} \frac{b_k E(z)}{E'(a_k)(z - a_k)} \left(\frac{z}{a_k}\right)^{m_k}$ defines an entire function with $F(a_k) = b_k$ for all $k \geq 0$ (the Pringsheim interpolation formula).

Additional exercises.

- Determine whether each product converges, and to what: (a) $\prod_{n=1}^{\infty} (1 + \frac{1}{n^2})$; (b) $\prod_{n=1}^{\infty} (1 - \frac{1}{n})$; (c) $\prod_{n=2}^{\infty} (1 - \frac{1}{n^2})$.
- Evaluate $\prod_{n=2}^{\infty} \frac{n^3 - 1}{n^3 + 1}$ in closed form. [Hint: factor $n^3 \pm 1$ and telescope.]
- Use the sine product formula to evaluate $\sum_{n=1}^{\infty} \frac{1}{n^2}$ by comparing the z^2 coefficient of $\frac{\sin \pi z}{\pi z} = \prod (1 - z^2/n^2)$ with its Taylor expansion. (Recover $\zeta(2) = \pi^2/6$.)
- Show $\prod_{n=1}^{\infty} (1 - \frac{z^2}{(n-1/2)^2}) = \cos \pi z$ and use it to evaluate $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$.
- Write down a Weierstrass product (with the correct running canonical degree) for an entire function whose zeros are exactly $a_n = n$, $n \geq 1$, each simple. Justify the choice of degree from $\sum 1/n^s$.
- Construct an entire function vanishing exactly at $\{n^2 : n \geq 1\}$ with all multiplicities 1, using degree-0 factors, and explain why E_0 suffices here but not for $a_n = n$.
- (Non-example.) Show that the sequence $a_n = \log(n+1)$ does *not* satisfy $|a_n| \rightarrow \infty$ slowly enough... rather, show $a_n = \frac{1}{n}$ *cannot* be the zero set of a nonzero entire function (it accumulates at 0), and contrast with $a_n = n$ which can.
- (Diagram.) Plot the zeros of the prescribed sequence $a_n = n + in$ in the plane, draw the disc $|z| < R$ and the auxiliary $|z| = 2R$, and indicate which factors are “near” ($|a_n| \leq 2R$, contributing zeros) versus “far” ($|a_n| > 2R$, close to 1).
- Prove that $\prod_{n=1}^{\infty} (1 + \frac{z}{n}) e^{-z/n}$ converges to an entire function (this is essentially $1/(z \Gamma(z) e^{\gamma z})$). Identify its zeros and explain the role of the $e^{-z/n}$ factors.
- Show that if $\sum c_n < \infty$ and $|F_n(z) - 1| \leq c_n$ on Ω , then $\log \prod F_n = \sum \log F_n$ holds with absolutely-convergent right side on compacts, justifying termwise logarithmic differentiation.

18.4 Hadamard’s Factorization Theorem

Hadamard’s theorem fuses the growth/zero estimates of §18.2 with the Weierstrass construction of §18.3: for a function of finite order, the canonical factors may be taken of *constant* degree and the unknown exponential factor e^g is forced to be e^P with P a *polynomial*.

The statement

Theorem 18.4.1 (Hadamard factorization). *Suppose f is entire of order ρ_0 . Let k be the integer with $k \leq \rho_0 < k+1$. If $a_1, a_2, \dots$ are the nonzero zeros of f and m is the order of the zero at 0, then*

$$f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n),$$

where P is a polynomial of degree $\leq k$ and E_k is the (constant-degree) canonical factor.

Two refinements over Weierstrass: the factor degree is the *fixed* integer $k = \lfloor \rho_0 \rfloor$ rather than the running n , and the prefactor is e^P with $\deg P \leq k$ rather than an arbitrary e^g . The whole function is thus pinned down by its zeros up to a polynomial's worth of exponential data.

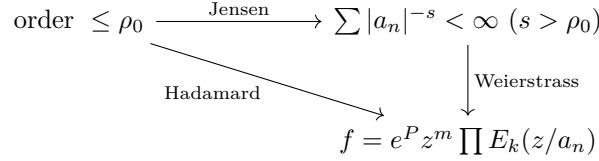


Figure 18.8. Logical flow: order bounds the zeros (Jensen), which feed a fixed-degree Weierstrass product, which Hadamard pins to e^P .

Lower bounds for canonical factors

Lemma 18.4.2. *The canonical factors satisfy*

$$|E_k(z)| \geq e^{-c|z|^{k+1}} \quad (|z| \leq \tfrac{1}{2}), \quad |E_k(z)| \geq |1 - z| e^{-c'|z|^k} \quad (|z| \geq \tfrac{1}{2}).$$

Proof. For $|z| \leq \frac{1}{2}$, $E_k(z) = e^w$ with $w = -\sum_{n>k} z^n/n$ and $|w| \leq c|z|^{k+1}$ (as in Lemma 18.3.10); since $|e^w| \geq e^{-|w|}$, the first bound follows. For $|z| \geq \frac{1}{2}$, factor $|E_k(z)| = |1 - z| |e^{z+z^2/2+\dots+z^k/k}|$ and note $|e^{z+\dots+z^k/k}| \geq e^{-|z+\dots+z^k/k|} \geq e^{-c'|z|^k}$ (the polynomial in the exponent has degree k). $\square$

Lemma 18.4.3 (Lower bound for the product, off small discs). *For any s with $\rho_0 < s < k + 1$,*

$$\left| \prod_{n=1}^{\infty} E_k(z/a_n) \right| \geq e^{-c|z|^s},$$

except possibly when z lies in the union of the discs centered at a_n of radius $|a_n|^{-k-1}$.

Sketch. Split the product at $|a_n| \leq 2|z|$ vs. $|a_n| > 2|z|$. *Far factors:* by the first part of Lemma 18.4.2, $\prod_{|a_n|>2|z|} |E_k(z/a_n)| \geq e^{-c|z|^{k+1} \sum_{|a_n|>2|z|} |a_n|^{-k-1}}$, and using $|a_n|^{-k-1} \leq C|a_n|^{-s}|z|^{s-k-1}$ together with $\sum |a_n|^{-s} < \infty$ gives $\geq e^{-c|z|^s}$. *Near factors:* by the second part of Lemma 18.4.2, $\prod_{|a_n|\leq 2|z|} |E_k(z/a_n)| \geq \prod_{|a_n|\leq 2|z|} |1 - z/a_n| \cdot e^{-c'|z|^s}$. When z avoids the forbidden discs, $|a_n - z| \geq |a_n|^{-k-1}$, so $\prod_{|a_n|\leq 2|z|} |1 - z/a_n| \geq \prod_{|a_n|\leq 2|z|} |a_n|^{-k-2}$, and the log of this is $\geq -(k+2) \sum_{|a_n|\leq 2|z|} \log |a_n| \geq -c|z|^{s'}$ for any $s' > s$ (using $n(2|z|) \leq c|z|^s$ from Theorem 18.2.2). Choosing the initial s close to ρ_0 yields the bound. $\square$

Corollary 18.4.4 (A sequence of good radii). *There is a sequence $r_1, r_2, \dots \rightarrow \infty$ with $|\prod_n E_k(z/a_n)| \geq e^{-c|z|^s}$ for all $|z| = r_m$.*

Proof. Since $\sum |a_n|^{-k-1} < \infty$, pick N with $\sum_{n\geq N} |a_n|^{-k-1} < \frac{1}{10}$. For consecutive integers $L, L+1$ there is $r \in [L, L+1]$ whose circle $|z| = r$ misses all forbidden discs: otherwise the radial intervals $I_n = [|a_n| - |a_n|^{-k-1}, |a_n| + |a_n|^{-k-1}]$ (each of length $2|a_n|^{-k-1}$) would cover $[L, L+1]$, forcing $2 \sum_{n\geq N} |a_n|^{-k-1} \geq 1$, a contradiction. Apply Lemma 18.4.3 on these circles. $\square$

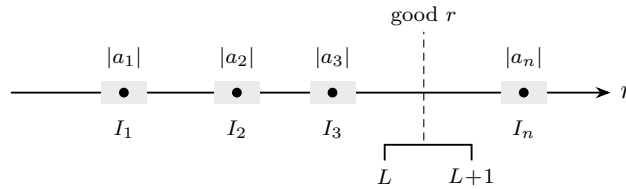


Figure 18.9. The forbidden intervals I_n have total length < 1 on $[L, L+1]$, so a good radius r (dashed) always slips between them. (Schematic.)

Finishing the proof

Lemma 18.4.5 (Real-part growth forces a polynomial). *Suppose g is entire and $u = \operatorname{Re} g$ satisfies $u(z) \leq Cr^s$ whenever $|z| = r$, for a sequence $r \rightarrow \infty$. Then g is a polynomial of degree $\leq s$.*

Proof. Expand $g(z) = \sum_{n \geq 0} a_n z^n$. Cauchy's formulas give, for $n \geq 0$, $\frac{1}{2\pi} \int_0^{2\pi} g(re^{i\theta}) e^{-in\theta} d\theta = a_n r^n$, and 0 for $n < 0$; conjugating, $\frac{1}{2\pi} \int_0^{2\pi} \overline{g(re^{i\theta})} e^{-in\theta} d\theta = 0$ for $n > 0$. Adding $(2u = g + \bar{g})$, for $n > 0$,

$$a_n r^n = \frac{1}{\pi} \int_0^{2\pi} u(re^{i\theta}) e^{-in\theta} d\theta.$$

Since $\int_0^{2\pi} e^{-in\theta} d\theta = 0$ for $n \neq 0$, we may subtract the constant Cr^s :

$$|a_n| r^n = \left| \frac{1}{\pi} \int_0^{2\pi} [u(re^{i\theta}) - Cr^s] e^{-in\theta} d\theta \right| \leq \frac{1}{\pi} \int_0^{2\pi} [Cr^s - u(re^{i\theta})] d\theta = 2Cr^s - 4\pi \operatorname{Re}(a_0) \cdot \frac{1}{\dots},$$

more simply $|a_n| \leq 2Cr^{s-n} - 2\operatorname{Re}(a_0)r^{-n}$. Letting $r \rightarrow \infty$ along the given sequence gives $a_n = 0$ for $n > s$. Hence $\deg g \leq s$. $\square$

Proof of Theorem 18.4.1. Let $E(z) = z^m \prod_{n \geq 1} E_k(z/a_n)$. The argument of Theorem 18.3.11 shows E is entire (now with *constant* degree k): for large n , $|1 - E_k(z/a_n)| \leq c|z/a_n|^{k+1}$, and $\sum |a_n|^{-k-1} < \infty$ because $k+1 > \rho_0$. As E has exactly the zeros of f , the quotient f/E is entire and zero-free, so $f/E = e^g$ for some entire g . Equivalently the zero-free f/E lifts through the exponential covering, the triangle commuting by $f/E = \exp \circ g$:

$$\begin{array}{ccc} & & \mathbb{C} \\ & \nearrow g & \downarrow \exp \\ \mathbb{C} & \xrightarrow{f/E} & \mathbb{C}^\times \end{array}$$

Now f has order ρ_0 , and Corollary 18.4.4 bounds $|E|$ below on the circles $|z| = r_m$:

$$e^{\operatorname{Re} g(z)} = \left| \frac{f(z)}{E(z)} \right| \leq \frac{c e^{c|z|^{\rho_0+\epsilon}}}{e^{-c|z|^s}} \leq e^{c'|z|^s} \quad (|z| = r_m),$$

so $\operatorname{Re} g(z) \leq C|z|^s$ on these circles, with s as close to ρ_0 as we like. By Lemma 18.4.5, g is a polynomial of degree $\leq s$, hence of degree $\leq k$ (as g is a polynomial and $s < k+1$ can be taken arbitrarily close to $\rho_0 \leq k$... more precisely $\deg g \leq \lfloor s \rfloor = k$). Writing $P = g$ completes the proof. $\square$

Worked Example 18.4.6 (Hadamard product for $e^z - 1$). Find the Hadamard factorization of $f(z) = e^z - 1$.

Recognition. Order 1, so $k = 1$; locate the zeros and fit e^P with $\deg P \leq 1$.

Solution. f is entire of order 1 (dominated by $e^{|z|}$), with simple zeros at $z = 2\pi in$, $n \in \mathbb{Z}$, including a simple zero at 0 ($m = 1$). Pairing $\pm n$ kills the degree-1 exponentials: $E_1(z/2\pi in) E_1(-z/2\pi in) = 1 + z^2/(4\pi^2 n^2)$. Thus $e^z - 1 = e^{P(z)} z \prod_{n \geq 1} (1 + \frac{z^2}{4\pi^2 n^2})$. Matching the expansion $e^z - 1 = z + z^2/2 + \dots = z(1 + z/2 + \dots)$ near 0 forces $P(z) = z/2$ (so $e^P = e^{z/2}$). Hence $e^z - 1 = e^{z/2} z \prod_{n \geq 1} (1 + \frac{z^2}{4\pi^2 n^2})$. *Check.* The z^1 coefficient: $e^{z/2} z \prod (1 + \dots) = z(1 + z/2 + \dots)$, matching $e^z - 1$. $\blacksquare$

Worked Example 18.4.7 (Hadamard product for $\cos \pi z$). Find the Hadamard factorization of $f(z) = \cos \pi z$.

Recognition. Order 1, $k = 1$; the zeros are the half-integers, none at 0, so $m = 0$ and P constant.

Solution. $\cos \pi z$ has order 1 and simple zeros at $z = \pm \frac{1}{2}, \pm \frac{3}{2}, \dots$, i.e. $\pm \frac{2n+1}{2}$, with no zero at 0. Pairing $\pm$ gives degree-1 exponentials cancelling, so $\cos \pi z = e^P \prod_{n \geq 0} (1 - \frac{4z^2}{(2n+1)^2})$. Since $\cos 0 = 1$ and the product at $z = 0$ is 1, $e^P = 1$, i.e. $P = 0$. Thus $\cos \pi z = \prod_{n \geq 0} (1 - \frac{4z^2}{(2n+1)^2})$. *Check.* At $z = \frac{1}{2}$ the $n = 0$ factor is $1 - 4(\frac{1}{4})/1 = 0$, matching $\cos(\pi/2) = 0$. $\blacksquare$

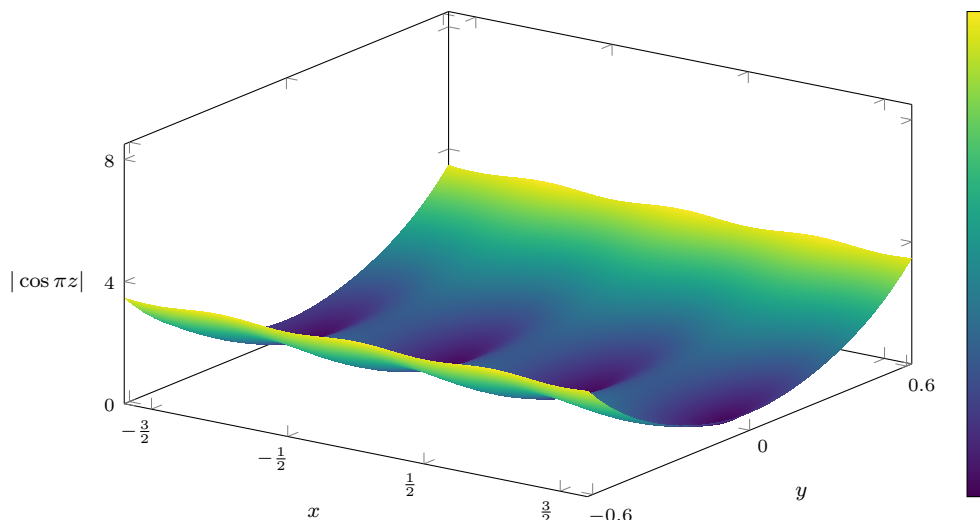


Figure 18.10. Modulus surface of $\cos \pi z$ (using $|\cos \pi z|^2 = \cos^2 \pi x + \sinh^2 \pi y$): it dips to 0 at the half-integers on the real axis ($y = 0$) and grows like $e^{\pi|y|}$ off-axis (order 1).

Worked Example 18.4.8 (Picard’s little theorem, finite-order case, via Hadamard). Show that a finite-order entire f omitting two values is constant.

Recognition. If f misses a , then $f - a$ is zero-free of finite order; Hadamard makes it e^P .

Solution. Suppose f omits a and b , $a \neq b$. Since $f - a$ is entire, finite order, and never 0, Hadamard (with no zeros) gives $f(z) - a = e^{P(z)}$, P a polynomial. Then $f - b = e^P + (a - b)$ also never vanishes, so $e^{P(z)} \neq b - a$ for all z . But a nonconstant polynomial P takes every value (FTA), so e^P takes every nonzero value (as e^w is onto $\mathbb{C} \setminus \{0\}$); hence $e^P = b - a$ would be solvable unless P is constant. Thus P is constant and f is constant. *Check.* e^z omits only 0 (one value); to omit a second value it would have to be constant, consistent. ■

Worked Example 18.4.9 (A non-example: non-integral order forces infinitely many zeros). Explain why an entire function of order $\rho_0 = 3/2$ cannot be zero-free, so “ $e^{z^{3/2}}$ ” is impossible as an entire function.

Recognition. Hadamard: a zero-free finite-order function is e^P , whose order is the integer $\deg P$.

Solution. If f were entire, zero-free, of order ρ_0 , then $f = e^P$ with P a polynomial, and the order of e^P equals $\deg P$, an integer. So a zero-free entire function has integer order. A function of order $3/2$ therefore must have zeros — in fact infinitely many, since finitely many zeros would make $f = Q(z)e^{P(z)}$ (Q polynomial) of integer order too. Hence “ $e^{z^{3/2}}$ ” cannot be entire (it is not even single-valued). *Check.* $\sin \sqrt{z}/\sqrt{z}$ has order $1/2$ (non-integer) and indeed infinitely many zeros at $z = n^2\pi^2$. ■

Caution. The degree of P is $\leq k = \lfloor \rho_0 \rfloor$, not $\leq \rho_0$; and the canonical factor degree is this same k . A common slip is to use degree $k = \rho_0$ when ρ_0 is an integer — the correct rule is $k \leq \rho_0 < k + 1$, so for $\rho_0 = 1$ exactly, $k = 1$. Also, e^P contributes no zeros but does raise the order to $\max(\deg P, \text{exponent of convergence})$; the order is determined by both the zeros and P .

Further worked examples

Worked Example 18.4.10 (No vanishing derivatives forces $f = e^{az+b}$). Suppose f is entire of finite order, never vanishes, and none of the derivatives $f', f'', \dots$ ever vanishes. Show $f(z) = e^{az+b}$.

Recognition. This is Stein’s Exercise 12. Hadamard makes $f = e^P$; then study the zeros of f' via P' and a second Hadamard factorization.

Solution. Since f is finite order and zero-free, Hadamard gives $f = e^P$ with P a polynomial. Then $f' = P'e^P$, whose zeros are exactly the zeros of P' . By hypothesis f' never vanishes, so P' has no zeros; a polynomial with no zeros is a nonzero constant, so $\deg P' = 0$, i.e. $\deg P \leq 1$. Write $P(z) = az + b$. (If $a = 0$ then f is constant, whose first derivative is $0 \equiv$ vanishes everywhere — excluded — so $a \neq 0$.) Hence $f(z) = e^{az+b}$, and indeed every $f^{(k)} = a^k e^{az+b}$ is zero-free. *Check.* $f(z) = e^z$ works; $f(z) = e^{z^2}$ fails because $f' = 2ze^{z^2}$ vanishes at 0, consistent with $\deg P = 2 > 1$. ■

Worked Example 18.4.11 (An order-2 product with non-summable square reciprocals). Consider F with zeros at $z = -in + m$ ($n \geq 1$, $m \in \mathbb{Z}$, a lattice in the lower half plane). Show $\sum |z_k|^{-2} = \infty$ but $\sum |z_k|^{-2-\epsilon} < \infty$, so F has order 2.

Recognition. This is Stein's Exercise 4 (a theta-type factor). Count lattice points in shells to test the borderline exponent $s = 2$.

Solution. The zeros $z = m - in$ ($n \geq 1$, $m \in \mathbb{Z}$) form (most of) a two-dimensional lattice. The number of lattice points with $|z_k| \leq R$ grows like the area, $\sim \pi R^2$, so $n(R) \asymp R^2$ and the exponent of convergence is 2. At the borderline, $\sum |z_k|^{-2} \asymp \int_1^\infty \frac{2\pi r}{r^2} dr = 2\pi \int_1^\infty \frac{dr}{r} = \infty$, while $\sum |z_k|^{-2-\epsilon} \asymp \int_1^\infty r^{-1-\epsilon} dr < \infty$. Combined with the growth bound $|F(z)| \leq Ae^{a|z|^2}$ from the hint's $F = F_1 F_2$ split, F has order exactly 2, with $k = [2] = 2$. *Check.* A planar lattice of zeros is the hallmark of order 2 – the same count underlies the Weierstrass $\wp$ -function and the Jacobi theta factor. ■

Worked Example 18.4.12 (Argument principle / Rouché: zeros of $e^z = 3z$ in the disc). Show $e^z = 3z$ has exactly one solution in $|z| < 1$, illustrating how Rouché locates zeros that Hadamard would catalogue.

Recognition. Rouché's theorem: if $|g| < |h|$ on a contour, then h and $h + g$ have the same number of zeros inside.

Solution. Put $h(z) = -3z$ and $g(z) = e^z$, so $h + g = e^z - 3z$. On $|z| = 1$, $|h(z)| = 3$ while $|g(z)| = |e^z| = e^{\operatorname{Re} z} \leq e^1 = e \approx 2.718 < 3$. Hence $|g| < |h|$ on the circle, and by Rouché $e^z - 3z$ has the same number of zeros in $|z| < 1$ as $-3z$, namely *one* (a simple zero at 0 for $-3z$). So $e^z = 3z$ has exactly one root in the unit disc. *Check.* On the real axis $\phi(x) = e^x - 3x$ has $\phi(0) = 1 > 0$, $\phi(1) = e - 3 < 0$, so a real root lies in $(0, 1)$ – accounting for the one root found. ■

Worked Example 18.4.13 (A Blaschke product: a bounded analogue of Weierstrass in the disc). For zeros $\{\alpha_n\} \subset \mathbb{D}$ with $\sum(1 - |\alpha_n|) < \infty$, show the Blaschke product $B(z) = \prod_n \frac{|\alpha_n|}{\alpha_n} \frac{\alpha_n - z}{1 - \overline{\alpha_n} z}$ converges to a holomorphic function on $\mathbb{D}$ with $|B| \leq 1$ and zeros exactly $\{\alpha_n\}$.

Recognition. This is Stein's Problem 2. Each factor is a disc automorphism of modulus 1 on the boundary; convergence reduces to $\sum |1 - B_n| < \infty$, controlled by $1 - |\alpha_n|$.

Solution. Each factor $B_n(z) = \frac{|\alpha_n|}{\alpha_n} \frac{\alpha_n - z}{1 - \overline{\alpha_n} z}$ is a Möbius automorphism of $\mathbb{D}$, so $|B_n(z)| \leq 1$ on $\mathbb{D}$ and $= 1$ on the boundary; its only zero in $\mathbb{D}$ is α_n . A computation gives, for $|z| \leq r < 1$, $|1 - B_n(z)| = \left| \frac{(|\alpha_n| + \overline{\alpha_n} z)(1 - |\alpha_n|)}{\alpha_n(1 - \overline{\alpha_n} z)} \right| \leq \frac{1+r}{1-r}(1 - |\alpha_n|)$. Since $\sum(1 - |\alpha_n|) < \infty$, Proposition 18.3.3 gives uniform convergence on $|z| \leq r$ to a holomorphic B with $|B| \leq 1$ and zeros precisely the α_n . *Check.* A single factor with $\alpha_n \rightarrow 1$ degenerates ($1 - |\alpha_n| \rightarrow 0$) so the tail factors are ≈ 1 , exactly the summability condition forced by Problem 1. ■

Worked Example 18.4.14 (A non-example: $\frac{1}{\sin \pi z}$ is not entire). Explain why $g(z) = 1/\sin \pi z$ admits no Hadamard factorization as an entire function.

Recognition. Hadamard applies only to *entire* functions; a function with poles is excluded.

Solution. $\sin \pi z$ vanishes at every integer, so $g = 1/\sin \pi z$ has a pole (not a zero) at each integer and is therefore *meromorphic*, not entire. Hadamard's theorem assumes f entire; it does not apply to g . (The correct factored description of g is as a ratio: $g(z) = \frac{1}{\pi z \prod_{n \geq 1} (1 - z^2/n^2)}$, a quotient of the entire sine product, with the product appearing in the *denominator*.) *Check.* Near $z = 1$, $g(z) \sim \frac{1}{\pi(z-1)\cos \pi} = \frac{-1}{\pi(z-1)}$, a genuine pole – entire functions have no poles. ■

Exercises (Stein, Ch.5).

- Find the Hadamard products for: (a) $e^z - 1$; (b) $\cos \pi z$. [Hint: the answers are $e^{z/2} z \prod_{n=1}^\infty (1 + z^2/4n^2\pi^2)$ and $\prod_{n=0}^\infty (1 - 4z^2/(2n+1)^2)$ respectively.]
- Show that if f is an entire function of finite order that omits two values, then f is constant. (This holds for any entire function and is Picard's little theorem.) [Hint: if f misses a , then $f(z) - a = e^{p(z)}$ with p a polynomial.]
- Suppose f is entire and never vanishes, and none of the higher derivatives of f ever vanish. Prove that if f is also of finite order, then $f(z) = e^{az+b}$ for constants a, b .

4. Show that the equation $e^z - z = 0$ has infinitely many solutions in $\mathbb{C}$. [Hint: apply Hadamard's theorem.]

Additional exercises.

- For each function, state the integer k with $k \leq \rho_0 < k + 1$ and the form of the Hadamard product: (a) $\sin \pi z$; (b) e^z ; (c) $\cos \sqrt{z}$; (d) ze^{z^2} .
- Find the Hadamard factorization of $\frac{\sin \pi z}{\pi z}$ (entire, order 1, no zero at 0) and confirm $P \equiv 0$.
- Show that an entire function of order 0 with infinitely many zeros has Hadamard form $f(z) = cz^m \prod (1 - z/a_n)$ (degree $k = 0$, constant prefactor), i.e. a “canonical product” with no exponential. Apply this to $\prod (1 - z/2^n)$.
- Prove that a zero-free entire function of finite order is exactly e^P for a polynomial P , and that its order equals $\deg P$. Deduce that $e^{\sqrt{z}}$ is not entire.
- Use Hadamard's theorem to show $e^z - z$ has infinitely many zeros, by ruling out the finite-zero form $Q(z)e^{P(z)}$ on order grounds.
- Let f be entire of order ρ_0 with $k = \lfloor \rho_0 \rfloor$. Show that if f has only finitely many zeros, then $f(z) = Q(z)e^{P(z)}$ with Q a polynomial and $\deg P \leq k$, and the order is exactly $\deg P$.
- (*Non-example.*) Explain why there is no entire function of order $1/2$ that is zero-free, and why “ $f = e^{z^{1/2}}$,” fails to be entire; contrast with the genuine order- $1/2$ function $\cos \sqrt{z}$, which has infinitely many zeros.
- (*Diagram.*) Sketch the circles $|z| = r_m$ of Corollary 18.4.4 weaving between the forbidden discs of radius $|a_n|^{-k-1}$ centered at the zeros a_n ; indicate on a radial axis the intervals I_n and a “good” radius slipping between them.
- Suppose f is entire of order ρ_0 with zeros $\{a_n\}$ of exponent of convergence σ . Show $\rho_0 = \max(\sigma, \deg P)$ where e^P is the Hadamard prefactor, and give an example with $\sigma < \rho_0$ (attach e^{z^2} to $\sin \pi z$) and one with $\sigma = \rho_0$ ($\sin \pi z$ itself).

Chapter 19

The Gamma and Zeta Functions

19.1 The Gamma Function

The integral definition and the strip estimate

The gamma function is the analytic interpolation of the factorial. It begins life as an integral and then, by a sequence of functional-equation tricks, spreads to a meromorphic function on all of $\mathbb{C}$ whose only singularities are simple poles at the nonpositive integers. The proofs are the substance of the chapter: each one is a small contour-or-estimate argument worth internalizing.

Definition 19.1.1. For $\operatorname{Re}(s) > 0$ the *gamma function* is

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt.$$

Here $t^{s-1} = e^{(s-1)\operatorname{Log} t}$ uses the real logarithm of $t > 0$, so the integrand is unambiguous.

The integral converges at both ends precisely because of a competition: near $t = 0$ the factor t^{s-1} is integrable as long as $\operatorname{Re}(s) > 0$, and near $t = \infty$ the exponential e^{-t} crushes any power of t . Write $\sigma = \operatorname{Re}(s)$; then $|e^{-t} t^{s-1}| = e^{-t} t^{\sigma-1}$, so absolute convergence depends only on σ .

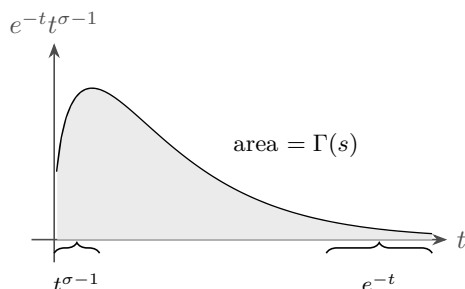


Figure 19.1. The integrand $e^{-t} t^{\sigma-1}$ of $\Gamma(s)$, here at $\sigma = \operatorname{Re}(s) = \frac{3}{2}$. Near $t = 0$ the power $t^{\sigma-1}$ gives an integrable singularity (left brace); near $t = \infty$ the factor e^{-t} forces decay (right brace); the shaded area is $\Gamma(s)$. (Schematic.)

Proposition 19.1.2 (holomorphy on a half-plane). Γ extends to a holomorphic function on the half-plane $\operatorname{Re}(s) > 0$, still given there by the integral of Definition 19.1.1.

Proof. It suffices to prove holomorphy in every vertical strip $S_{\delta, M} = \{\delta < \operatorname{Re}(s) < M\}$ with $0 < \delta < M < \infty$, since these strips exhaust the open half-plane. For $\varepsilon > 0$ set the *finite* integral

$$F_\varepsilon(s) = \int_\varepsilon^{1/\varepsilon} e^{-t} t^{s-1} dt.$$

The integrand is jointly continuous and holomorphic in s , so by differentiation under the integral over a compact t -interval (Theorem 5.4, Ch.2 of Stein), each F_ε is holomorphic on $S_{\delta,M}$. It remains to show $F_\varepsilon \rightarrow \Gamma$ uniformly on $S_{\delta,M}$; then Γ is holomorphic as a locally uniform limit of holomorphic functions (Theorem 5.2, Ch.2). With $\sigma = \operatorname{Re}(s) \in (\delta, M)$,

$$|\Gamma(s) - F_\varepsilon(s)| \leq \int_0^\varepsilon e^{-t} t^{\sigma-1} dt + \int_{1/\varepsilon}^\infty e^{-t} t^{\sigma-1} dt.$$

The first integral is $\leq \int_0^\varepsilon t^{\delta-1} dt = \varepsilon^\delta / \delta \rightarrow 0$, uniformly in s . For the second, $t^{\sigma-1} \leq t^{M-1} \leq C e^{t/2}$ for $t \geq 1$, so $\int_{1/\varepsilon}^\infty e^{-t} t^{\sigma-1} dt \leq C \int_{1/\varepsilon}^\infty e^{-t/2} dt \rightarrow 0$, again uniformly. Hence the convergence is uniform and the proof is complete. $\square$

Method (Proving an integral defines a holomorphic function). *Recognition trigger.* A function is given by an integral $\int_I g(s, t) dt$ with s -dependence, and you must show it is holomorphic on some open set.

1. Truncate to a compact t -window F_ε ; on a compact window, holomorphy in s is automatic (differentiate under the integral / Morera).
2. Show $F_\varepsilon \rightarrow F$ locally uniformly in s by bounding the tail integrals by quantities independent of s on the relevant strip.
3. Conclude holomorphy of the limit (uniform limits of holomorphic functions are holomorphic).

The whole game is producing the two uniform tail bounds, one at each endpoint.

The functional equation and the factorial

Lemma 19.1.3 (recurrence). *If $\operatorname{Re}(s) > 0$ then $\Gamma(s+1) = s\Gamma(s)$. Consequently $\Gamma(1) = 1$ and $\Gamma(n+1) = n!$ for $n = 0, 1, 2, \dots$*

Proof. Integrate by parts in the finite integral. Since $\frac{d}{dt}(e^{-t}t^s) = -e^{-t}t^s + s e^{-t}t^{s-1}$,

$$\int_\varepsilon^{1/\varepsilon} \frac{d}{dt}(e^{-t}t^s) dt = - \int_\varepsilon^{1/\varepsilon} e^{-t}t^s dt + s \int_\varepsilon^{1/\varepsilon} e^{-t}t^{s-1} dt.$$

The left side equals $[e^{-t}t^s]_\varepsilon^{1/\varepsilon}$, which tends to 0 as $\varepsilon \rightarrow 0$ because $e^{-t}t^s \rightarrow 0$ at both $t \rightarrow 0^+$ (as $\operatorname{Re}(s) > 0$) and $t \rightarrow \infty$. Letting $\varepsilon \rightarrow 0$ leaves $0 = -\Gamma(s+1) + s\Gamma(s)$, i.e. $\Gamma(s+1) = s\Gamma(s)$. Finally $\Gamma(1) = \int_0^\infty e^{-t} dt = [-e^{-t}]_0^\infty = 1$, and iterating the recurrence gives $\Gamma(n+1) = n \cdot (n-1) \cdots 1 \cdot \Gamma(1) = n!$. $\square$

Worked Example 19.1.4 ($\Gamma(\frac{1}{2}) = \sqrt{\pi}$ via the Gaussian). Compute $\Gamma(\frac{1}{2})$ and then $\Gamma(\frac{3}{2})$.

Recognition. Trigger: a half-integer value of Γ . Substitute to turn the defining integral into a Gaussian.

Solution. With $t = x^2$, $dt = 2x dx$,

$$\Gamma(\tfrac{1}{2}) = \int_0^\infty e^{-t} t^{-1/2} dt = \int_0^\infty e^{-x^2} x^{-1} \cdot 2x dx = 2 \int_0^\infty e^{-x^2} dx = \int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}.$$

By the recurrence, $\Gamma(\frac{3}{2}) = \frac{1}{2}\Gamma(\frac{1}{2}) = \frac{\sqrt{\pi}}{2}$. *Check.* $\Gamma(\frac{1}{2}) \approx 1.7725$ and $\sqrt{\pi} \approx 1.7725$; and $\Gamma(\frac{3}{2}) \approx 0.8862 = \sqrt{\pi}/2$. Agree.

Analytic continuation to a meromorphic function

Theorem 19.1.5 (meromorphic continuation of Γ). *The function Γ , initially defined for $\operatorname{Re}(s) > 0$, has an analytic continuation to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at the nonpositive integers $s = 0, -1, -2, \dots$. The residue at $s = -n$ is*

$$\operatorname{Res}_{s=-n} \Gamma(s) = \frac{(-1)^n}{n!}.$$

Proof. We extend Γ to each half-plane $\operatorname{Re}(s) > -m$, $m \geq 1$ an integer. Iterating the recurrence $\Gamma(s) = \Gamma(s+m)/[(s+m-1)(s+m-2)\cdots s]$ suggests defining, for $\operatorname{Re}(s) > -m$,

$$F_m(s) = \frac{\Gamma(s+m)}{(s+m-1)(s+m-2)\cdots s}.$$

The numerator $\Gamma(s+m)$ is holomorphic for $\operatorname{Re}(s) > -m$ (it equals Γ evaluated in $\operatorname{Re}(\cdot) > 0$ shifted), so F_m is meromorphic in $\operatorname{Re}(s) > -m$ with at worst simple poles at $s = 0, -1, \dots, -m+1$, the zeros of the denominator. On $\operatorname{Re}(s) > 0$, Lemma 19.1.3 applied m times gives $F_m = \Gamma$; hence the various F_m agree on their overlaps (a connected set), and they patch to a single meromorphic continuation. For the residue at $s = -n$ ($0 \leq n \leq m-1$), only the factor $(s+n)$ in the denominator vanishes, so

$$\operatorname{Res}_{s=-n} F_m = \frac{\Gamma(m-n)}{(m-1-n)(m-2-n)\cdots(1)\cdot(-1)(-2)\cdots(-n)} = \frac{(m-n-1)!}{(m-1-n)!(-1)^n n!} = \frac{(-1)^n}{n!},$$

using $\Gamma(m-n) = (m-n-1)!$. This is independent of m , as it must be. $\square$

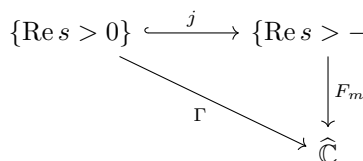


Figure 19.2. Each continuation F_m restricts along the inclusion $j : \{\operatorname{Re} s > 0\} \hookrightarrow \{\operatorname{Re} s > -m\}$ to the original Γ , i.e. $F_m \circ j = \Gamma$; since the F_m agree on overlaps they patch to one meromorphic Γ on $\hat{\mathbb{C}}$ (here $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ absorbs the poles).

Remark. The recurrence $\Gamma(s+1) = s\Gamma(s)$ persists for all s that are not poles, by analytic continuation: both sides are holomorphic off the poles and agree on $\operatorname{Re}(s) > 0$. At a pole $s = -n$, comparing residues gives $\operatorname{Res}_{s=-n} \Gamma(s+1) = -n \operatorname{Res}_{s=-n} \Gamma(s)$, and at $s = 0$ one reads $\Gamma(1) = \lim_{s \rightarrow 0} s\Gamma(s)$, i.e. $\operatorname{Res}_{s=0} \Gamma = 1$.

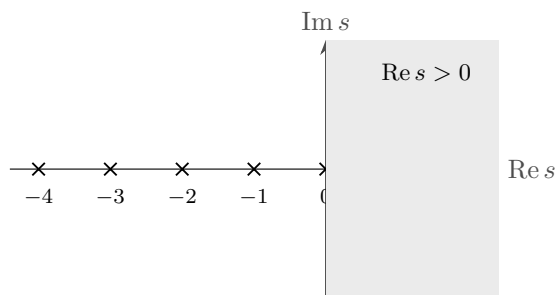


Figure 19.3. The continued Γ on the s -plane: holomorphic everywhere except simple poles (crosses) at the nonpositive integers $0, -1, -2, \dots$, with $\operatorname{Res}_{s=-n} \Gamma = (-1)^n/n!$. The shaded half-plane $\operatorname{Re} s > 0$ is where the defining integral converges. (Schematic.)

Worked Example 19.1.6 (Residue of Γ at $s = -2$, by Laurent and by the formula). Find $\operatorname{Res}_{s=-2} \Gamma(s)$ two ways.

Recognition. Trigger: a residue of Γ at a pole. Use the residue formula $(-1)^n/n!$, then confirm by the recurrence near the pole.

Solution. The formula with $n = 2$ gives $(-1)^2/2! = \frac{1}{2}$. To confirm, near $s = -2$ write $\Gamma(s) = \frac{\Gamma(s+3)}{(s+2)(s+1)s}$. As $s \rightarrow -2$ the numerator $\Gamma(s+3) \rightarrow \Gamma(1) = 1$ and the regular factors $(s+1)s \rightarrow (-1)(-2) = 2$, so the simple pole at $s = -2$ has residue $\frac{1}{(s+1)s} \Big|_{s=-2} \cdot \Gamma(1) = \frac{1}{2}$. *Check.* Both give $\frac{1}{2}$; and the signs alternate as n grows $(+1, -1, \frac{1}{2}, -\frac{1}{6}, \dots)$ for $n = 0, 1, 2, 3$, matching $(-1)^n/n!$.

The Euler reflection and product formulas

The continuation can also be packaged through Euler's infinite product, which makes the poles and the reflection symmetry transparent.

Theorem 19.1.7 (reflection formula). *For $s \notin \mathbb{Z}$,*

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}.$$

In particular $1/\Gamma$ is entire, with simple zeros exactly at $s = 0, -1, -2, \dots$.

Proof sketch. Both sides are meromorphic on $\mathbb{C}$. By Lemma 19.1.3, $G(s) = \Gamma(s)\Gamma(1-s)$ satisfies $G(s+1) = -G(s)$ (since $\Gamma(s+1) = s\Gamma(s)$ and $\Gamma(-s) = \Gamma(1-s)/(-s)$), the same periodicity-with-sign as $\pi/\sin \pi s$; both have simple poles at every integer. Forming the ratio gives an entire, bounded, doubly-quasi-periodic function, hence constant by Liouville, and the constant is pinned to 1 by evaluating at $s = \frac{1}{2}$: $\Gamma(\frac{1}{2})^2 = \pi = \pi/\sin(\pi/2)$.

The product formula $\frac{1}{\Gamma(s)} = e^{\gamma s} s \prod_{n \geq 1} (1 + \frac{s}{n}) e^{-s/n}$ (Weierstrass, γ the Euler constant) exhibits the simple zeros directly, and multiplying the products for $1/\Gamma(s)$ and $1/\Gamma(1-s)$ recovers $\sin \pi s/\pi$. $\square$

Worked Example 19.1.8 ($\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$). Evaluate $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})$.

Recognition. Trigger: a product $\Gamma(s)\Gamma(1-s)$ with $s + (1-s) = 1$. Apply reflection.

Solution. With $s = \frac{1}{4}$, $1-s = \frac{3}{4}$, the reflection formula gives $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \frac{\pi}{\sin(\pi/4)} = \frac{\pi}{\sqrt{2}/2} = \pi\sqrt{2}$. *Check.*

$\Gamma(\frac{1}{4}) \approx 3.6256$, $\Gamma(\frac{3}{4}) \approx 1.2254$, product ≈ 4.4429 ; and $\pi\sqrt{2} \approx 4.4429$. Agree.

Worked Example 19.1.9 (A NON-application: reflection needs the arguments to sum to 1). Decide whether $\Gamma(\frac{1}{3})\Gamma(\frac{1}{3}) = \pi/\sin(\pi/3)$.

Recognition. Trigger: a tempting but wrong use of reflection. Check the hypothesis $s + (1-s) = 1$.

Solution. Reflection pairs s with $1-s$. Here both arguments are $\frac{1}{3}$, and $\frac{1}{3} + \frac{1}{3} = \frac{2}{3} \neq 1$, so the formula does *not* apply: there is no clean closed form for $\Gamma(\frac{1}{3})^2$. The correct reflection partner of $\frac{1}{3}$ is $\frac{2}{3}$: $\Gamma(\frac{1}{3})\Gamma(\frac{2}{3}) = \pi/\sin(\pi/3) = 2\pi/\sqrt{3}$. *Check.* $\Gamma(\frac{1}{3}) \approx 2.6789$, $\Gamma(\frac{2}{3}) \approx 1.3541$, product ≈ 3.6276 ; and $2\pi/\sqrt{3} \approx 3.6276$. The valid pairing agrees; the invalid one would have given the wrong number $\pi/\sin(\pi/3)$ against $\Gamma(\frac{1}{3})^2 \approx 7.18$.

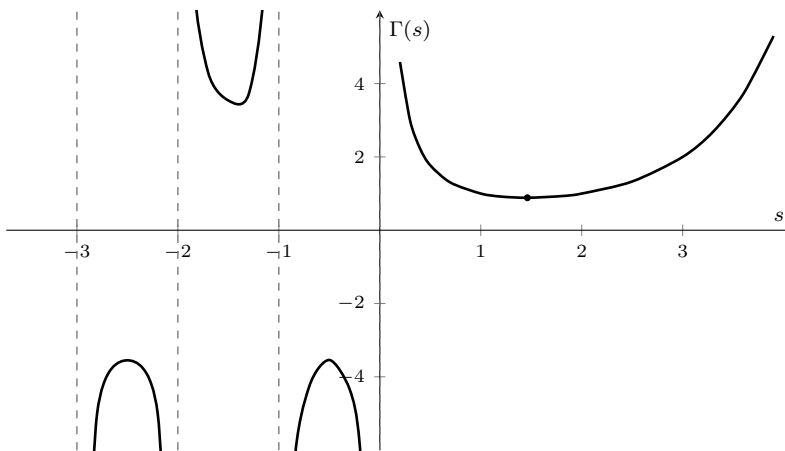


Figure 19.4. The gamma function on the real axis. It blows up at each nonpositive integer (dashed vertical asymptotes), and its sign alternates from one inter-pole interval to the next. The marked dot at $\Gamma(1.4615) \approx 0.886$ is the minimum of Γ on $(0, \infty)$.

The Beta function

Definition 19.1.10. For $\operatorname{Re}(\alpha) > 0$, $\operatorname{Re}(\beta) > 0$ the *Beta function* is

$$B(\alpha, \beta) = \int_0^1 (1-t)^{\alpha-1} t^{\beta-1} dt.$$

Theorem 19.1.11 (Beta–Gamma identity).

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}, \quad \text{and equivalently} \quad B(\alpha, \beta) = \int_0^\infty \frac{u^{\alpha-1}}{(1+u)^{\alpha+\beta}} du.$$

Proof. Start from the double integral, valid since both factors are absolutely integrable,

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^\infty \int_0^\infty t^{\alpha-1} s^{\beta-1} e^{-t-s} dt ds.$$

Substitute $t = ur$, $s = u(1-r)$ with $u > 0$, $0 < r < 1$ (so $t + s = u$); the Jacobian is u . Then

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^1 \int_0^\infty (ur)^{\alpha-1} (u(1-r))^{\beta-1} e^{-u} u du dr = \left(\int_0^\infty u^{\alpha+\beta-1} e^{-u} du \right) \left(\int_0^1 r^{\alpha-1} (1-r)^{\beta-1} dr \right),$$

which is $\Gamma(\alpha + \beta) B(\beta, \alpha) = \Gamma(\alpha + \beta) B(\alpha, \beta)$ (the Beta function is symmetric). Dividing gives the identity. For the second form put $t = u/(1+u)$, $1-t = 1/(1+u)$, $dt = du/(1+u)^2$ in Definition 19.1.10. $\square$

Worked Example 19.1.12 (A trigonometric integral via Beta). Evaluate $\int_0^{\pi/2} \sin^{2a-1} \theta \cos^{2b-1} \theta d\theta$ for $a, b > 0$.

Recognition. Trigger: a power of sin times a power of cos over $[0, \pi/2]$. Substitute $t = \sin^2 \theta$ to expose a Beta integral.

Solution. With $t = \sin^2 \theta$, $dt = 2 \sin \theta \cos \theta d\theta$, $\sin^2 \theta = t$, $\cos^2 \theta = 1 - t$,

$$\int_0^{\pi/2} \sin^{2a-1} \theta \cos^{2b-1} \theta d\theta = \frac{1}{2} \int_0^1 t^{a-1} (1-t)^{b-1} dt = \frac{1}{2} B(a, b) = \frac{\Gamma(a)\Gamma(b)}{2\Gamma(a+b)}.$$

Check. At $a = b = \frac{1}{2}$ the left side is $\int_0^{\pi/2} d\theta = \frac{\pi}{2}$, and the right side is $\Gamma(\frac{1}{2})^2 / (2\Gamma(1)) = \pi/2$. Agree.

Caution. The integral $\int_0^\infty e^{-t} t^{s-1} dt$ *only* represents Γ on $\operatorname{Re}(s) > 0$; for $\operatorname{Re}(s) \leq 0$ it diverges at $t = 0$. Do not plug a pole, e.g. $s = 0$ or $s = -1$, into the integral. Use the continued Γ (the F_m formula or the recurrence) instead. Likewise $1/\Gamma$ is the well-behaved object: it is *entire*, vanishing at $0, -1, -2, \dots$, so problems about “ Γ near its poles” are cleanest phrased through $1/\Gamma$.

Worked Example 19.1.13 (Reading off a power-series coefficient asymptotically). For fixed $\alpha \in \mathbb{C}$ and $|z| < 1$ let $(1-z)^{-\alpha} = \sum_{n \geq 0} a_n(\alpha) z^n$. Show $a_n(\alpha) \sim n^{\alpha-1}/\Gamma(\alpha)$ as $n \rightarrow \infty$.

Recognition. Trigger: asymptotics of binomial-type coefficients. The coefficients are ratios of Gamma values; expand the ratio.

Solution. By the binomial series, $a_n(\alpha) = \binom{-\alpha}{n} (-1)^n = \frac{\alpha(\alpha+1) \cdots (\alpha+n-1)}{n!} = \frac{\Gamma(\alpha+n)}{\Gamma(\alpha)\Gamma(n+1)}$. Using

$\Gamma(\alpha+n)/\Gamma(n+1) \sim n^{\alpha-1}$ (which follows from $\Gamma(n+a)/\Gamma(n+b) \sim n^{a-b}$, a consequence of the recurrence and monotonicity), we get $a_n(\alpha) \sim n^{\alpha-1}/\Gamma(\alpha)$. *Check.* For $\alpha = 1$, $(1-z)^{-1} = \sum z^n$ so $a_n = 1$, and $n^0/\Gamma(1) = 1$. For $\alpha = 2$, $(1-z)^{-2} = \sum (n+1)z^n$ so $a_n = n+1 \sim n = n^1/\Gamma(2)$. Both match.

Further worked examples

Worked Example 19.1.14 (The Mellin transform of e^{-t} is Γ itself). For which s is Γ literally a Mellin transform, and what does this viewpoint buy?

Recognition. Trigger: the Mellin transform $\mathcal{M}(f)(s) = \int_0^\infty f(t)t^{s-1}dt$. Recognize Definition 19.1.1 as the case $f(t) = e^{-t}$.

Solution. By definition $\mathcal{M}(e^{-t})(s) = \int_0^\infty e^{-t} t^{s-1} dt = \Gamma(s)$, valid for $\operatorname{Re}(s) > 0$. The Mellin transform turns multiplicative scaling $t \mapsto \lambda t$ into a multiplier: $\mathcal{M}(f(\lambda t))(s) = \lambda^{-s} \mathcal{M}(f)(s)$. Applied to $f(t) = e^{-\lambda t}$ this gives $\int_0^\infty e^{-\lambda t} t^{s-1} dt = \lambda^{-s} \Gamma(s)$, the scaling identity that powers both the Beta computation and the ζ – Γ – ϑ bridge of the next section. *Check.* At $\lambda = 1$, $s = 1$ this reads $\int_0^\infty e^{-t} dt = 1 = \Gamma(1)$. $\blacksquare$

$$\begin{array}{ccc}
e^{-t} & \xrightarrow{S_\lambda} & e^{-\lambda t} \\
\mathcal{M} \downarrow & & \downarrow \mathcal{M} \\
\Gamma(s) & \xrightarrow{\cdot \lambda^{-s}} & \lambda^{-s} \Gamma(s)
\end{array}$$

Here $\mathcal{M}(f)(s) = \int_0^\infty f(t)t^{s-1}dt$ is the Mellin transform and $S_\lambda f(t) = f(\lambda t)$ is dilation; the square commutes because $\mathcal{M}(S_\lambda f)(s) = \lambda^{-s}\mathcal{M}(f)(s)$, and the top-left corner $\mathcal{M}(e^{-t}) = \Gamma$ propagates to the scaling identity used throughout.

Worked Example 19.1.15 ($\mathcal{M}(\cos)(s) = \Gamma(s) \cos(\pi s/2)$ by rotating the contour). Show $\int_0^\infty \cos(t) t^{s-1} dt = \Gamma(s) \cos \frac{\pi s}{2}$ for $0 < \operatorname{Re}(s) < 1$.

Recognition. Trigger: a Mellin transform of an oscillatory function. Write $\cos t = \operatorname{Re}(e^{it})$ and rotate the ray of integration from the positive real axis to the positive imaginary axis, where e^{-w} decays.

Solution. Consider $f(w) = e^{-w}w^{s-1}$ on the quarter-disc sector bounded by the segment $[\varepsilon, R]$ on the real axis, the arc $w = Re^{i\varphi}$ ($0 \leq \varphi \leq \pi/2$), and the return segment along the imaginary axis. Since f is holomorphic in the open first quadrant and (for $0 < \operatorname{Re}(s) < 1$) the small arc at ε and the large arc at R both contribute 0 in the limit, Cauchy's theorem gives

$$\int_0^\infty e^{-x}x^{s-1}dx = \int_0^\infty e^{-iy}(iy)^{s-1}i dy = i^s \int_0^\infty e^{-iy}y^{s-1}dy.$$

The left side is $\Gamma(s)$; writing $i^s = e^{i\pi s/2}$ and taking real and imaginary parts of $e^{-iy} = \cos y - i \sin y$ gives $\Gamma(s) = e^{i\pi s/2}[\mathcal{M}(\cos)(s) - i\mathcal{M}(\sin)(s)]$. Multiplying by $e^{-i\pi s/2}$ and matching real parts yields $\mathcal{M}(\cos)(s) = \Gamma(s) \cos \frac{\pi s}{2}$ (and the imaginary part gives $\mathcal{M}(\sin)(s) = \Gamma(s) \sin \frac{\pi s}{2}$). *Check.* As $s \rightarrow 1^-$, $\cos(\pi s/2) \rightarrow 0$, matching the (conditional) fact that $\int_0^\infty \cos t dt$ has no absolutely convergent value; and the companion $\mathcal{M}(\sin)$ at $s = \frac{1}{2}$ gives $\int_0^\infty x^{-1/2} \sin x dx = \Gamma(\frac{1}{2}) \sin(\pi/4) = \sqrt{\pi} \cdot \frac{\sqrt{2}}{2} = \sqrt{\pi/2}$. ■

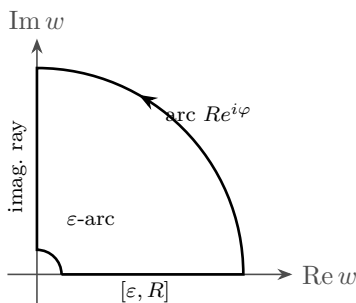


Figure 19.5. Quarter-disc sector for rotating $\int_0^\infty e^{-w}w^{s-1}dw$ from the real axis onto the imaginary axis (Stein, Ch.6 Exercise 10). (Schematic.)

Worked Example 19.1.16 ($\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$ from the Mellin identity). Deduce the Dirichlet integral from $\mathcal{M}(\sin)(s) = \Gamma(s) \sin(\pi s/2)$.

Recognition. Trigger: the value $\int_0^\infty x^{-1+s} \sin x dx$ at a special s . The previous example, continued to the strip $-1 < \operatorname{Re}(s) < 1$, evaluated at $s \rightarrow 0$.

Solution. The identity $\int_0^\infty \sin(x) x^{s-1} dx = \Gamma(s) \sin \frac{\pi s}{2}$ holds for $0 < \operatorname{Re}(s) < 1$ and, by analytic continuation, throughout $-1 < \operatorname{Re}(s) < 1$. Take $s \rightarrow 0$. Near $s = 0$, $\Gamma(s) \sim 1/s$ while $\sin \frac{\pi s}{2} \sim \frac{\pi s}{2}$, so the product $\Gamma(s) \sin \frac{\pi s}{2} \rightarrow \frac{\pi}{2}$. The left side at $s = 0$ is $\int_0^\infty x^{-1} \sin x dx$. Hence $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$. *Check.* Numerically $\operatorname{Si}(\infty) = \pi/2 \approx 1.5708$; the same machinery at $s \rightarrow -\frac{1}{2}$ gives $\int_0^\infty x^{-3/2} \sin x dx = \Gamma(-\frac{1}{2}) \sin(-\frac{\pi}{4}) = (-2\sqrt{\pi})(-\frac{\sqrt{2}}{2}) = \sqrt{2\pi}$. ■

Worked Example 19.1.17 (The Fourier transform of $e^{ax}e^{-e^x}$ is a Gamma value). Let $f(x) = e^{ax}e^{-e^x}$ with $a > 0$. Show $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i \xi x} dx$ equals $\Gamma(a - 2\pi i \xi)$ (with the convention $\hat{f}(\xi) = \int f(x)e^{-ix\xi} dx$ it is $\Gamma(a - i\xi)$; we use the latter).

Recognition. Trigger: an integral over $\mathbb{R}$ that becomes a Γ integral after the substitution $t = e^x$. (Stein, Ch.6 Exercise 11.)

Solution. Using $\hat{f}(\xi) = \int_{-\infty}^{\infty} e^{ax}e^{-e^x}e^{-i\xi x} dx$ substitute $t = e^x$, so $x = \log t$, $dx = dt/t$, and $e^{(a-i\xi)x} = t^{a-i\xi}$:

$$\hat{f}(\xi) = \int_0^{\infty} t^{a-i\xi} e^{-t} \frac{dt}{t} = \int_0^{\infty} e^{-t} t^{(a-i\xi)-1} dt = \Gamma(a - i\xi).$$

The integral converges because $\operatorname{Re}(a - i\xi) = a > 0$. The exponential decay of f in the strip $|\operatorname{Im}| < \pi$ (noted by Stein) is exactly what lets one shift contours and justify the formula for all real ξ . *Check.* At $\xi = 0$, $\hat{f}(0) = \int_0^{\infty} e^{ax}e^{-e^x} dx = \Gamma(a)$, the total “mass” — e.g. $a = 1$ gives $\int_{-\infty}^{\infty} e^x e^{-e^x} dx = [-e^{-e^x}]_{-\infty}^{\infty} = 1 = \Gamma(1)$. ■

Worked Example 19.1.18 ($1/|\Gamma|$ is not $O(e^{c|s|})$: a NON-example for entire-order bounds). Show that $1/|\Gamma(s)|$ grows faster than any $e^{c|s|}$.

Recognition. Trigger: a claimed order-of-growth bound on the entire function $1/\Gamma$. Test it on the half-integers $s = -k - \frac{1}{2}$, where Γ is small. (Stein, Ch.6 Exercise 12a.)

Solution. Take $s = -k - \frac{1}{2}$, k a positive integer. From the reflection $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$ with $\sin(\pi s) = \pm 1$ at half-integers, $|\Gamma(-k - \frac{1}{2})| = \frac{\pi}{|\Gamma(k + \frac{3}{2})|}$. Since $\Gamma(k + \frac{3}{2}) = \frac{(2k+1)!}{4^k k!} \sqrt{\pi} \leq (k+1)! \sqrt{\pi}$ is only

of factorial size while the reciprocal forces $|1/\Gamma(-k - \frac{1}{2})| = \frac{\Gamma(k + \frac{3}{2})}{\pi} \geq \frac{k!}{\pi}$, we get a value $\geq k!/\pi$ at a point with $|s| \approx k$. But $e^{c|s|} \approx e^{ck}$ grows only exponentially, and $k!/\pi$ outpaces e^{ck} for every fixed c once k is large (Stirling: $k! \sim \sqrt{2\pi k} (k/e)^k$). Hence *no* constant c makes $1/|\Gamma| = O(e^{c|s|})$. *Check.* $1/\Gamma$ has order exactly 1 but *type* infinity (“order 1, maximal type”), so it just barely fails $O(e^{c|s|})$, consistent with the factorial blow-up found here. ■

Worked Example 19.1.19 ($\Gamma'(1) = -\gamma$ and the digamma at integers). Compute $\Gamma'(1)$ and then $\psi(n) = \Gamma'(n)/\Gamma(n)$ for integers $n \geq 1$.

Recognition. Trigger: a derivative of Γ at a specific point. Differentiate the Weierstrass product $1/\Gamma(s) = e^{\gamma s} \prod_{n \geq 1} (1 + s/n) e^{-s/n}$ logarithmically.

Solution. Taking log of $1/\Gamma$ and differentiating gives the digamma $\psi(s) = \frac{\Gamma'(s)}{\Gamma(s)} = -\gamma - \frac{1}{s} + \sum_{n \geq 1} \left(\frac{1}{n} - \frac{1}{s+n} \right)$.

At $s = 1$ every bracketed term telescopes against $-1/s$: $\psi(1) = -\gamma - 1 + \sum_{n \geq 1} \left(\frac{1}{n} - \frac{1}{n+1} \right) = -\gamma - 1 + 1 = -\gamma$. Since $\Gamma(1) = 1$, $\Gamma'(1) = \psi(1)\Gamma(1) = -\gamma$. The recurrence $\psi(s+1) = \psi(s) + 1/s$ (from $\log \Gamma(s+1) = \log s + \log \Gamma(s)$) then gives $\psi(n) = -\gamma + \sum_{k=1}^{n-1} \frac{1}{k} = -\gamma + H_{n-1}$. *Check.* $\psi(2) = -\gamma + 1$ and indeed $\Gamma'(2) = \psi(2)\Gamma(2) = 1 - \gamma \approx 0.4228$, matching the known $\Gamma'(2) = 1 - \gamma$. ■

Worked Example 19.1.20 (The polygamma series $(\log \Gamma)''(s) = \sum_{n \geq 0} (s+n)^{-2}$). Prove $\frac{d^2}{ds^2} \log \Gamma(s) =$

$$\sum_{n=0}^{\infty} \frac{1}{(s+n)^2} \text{ for } s > 0.$$

Recognition. Trigger: a second logarithmic derivative of Γ . Differentiate the digamma series from the previous example once more. (Stein, Ch.6 Exercise 13.)

Solution. From $\psi(s) = -\gamma - \frac{1}{s} + \sum_{n \geq 1} \left(\frac{1}{n} - \frac{1}{s+n} \right)$, differentiate term by term (legitimate since the differentiated series $\sum (s+n)^{-2}$ converges uniformly on compact subsets of $s > 0$):

$$\psi'(s) = \frac{1}{s^2} + \sum_{n \geq 1} \frac{1}{(s+n)^2} = \sum_{n=0}^{\infty} \frac{1}{(s+n)^2}.$$

Because $\psi = (\log \Gamma)'$, this ψ' is exactly $(\log \Gamma)''(s)$. The identity then persists for all complex $s \neq 0, -1, -2, \dots$ by analytic continuation, both sides being meromorphic with the same poles. *Check.* At $s = 1$ the right side is $\sum_{n \geq 0} (1+n)^{-2} = \sum_{m \geq 1} m^{-2} = \zeta(2) = \pi^2/6$, so $(\log \Gamma)''(1) = \pi^2/6$, the standard value of the trigamma $\psi'(1)$. ■

Worked Example 19.1.21 ($\int_x^{x+1} \log \Gamma(t) dt = x \log x - x + \frac{1}{2} \log(2\pi)$). Evaluate the unit-window average of $\log \Gamma$, recovering the constant in $\log n!$.

Recognition. Trigger: an integral of $\log \Gamma$ over a unit interval. Differentiate in x and use the recurrence to pin the constant. (Stein, Ch.6 Exercise 14.)

Solution. Let $J(x) = \int_x^{x+1} \log \Gamma(t) dt$ for $x > 0$. By the fundamental theorem, $J'(x) = \log \Gamma(x+1) - \log \Gamma(x) = \log(\Gamma(x+1)/\Gamma(x)) = \log x$, using the recurrence $\Gamma(x+1) = x\Gamma(x)$. Integrating, $J(x) = x \log x - x + c$. To find c , take $x \rightarrow \infty$ and compare with Stirling $\log \Gamma(t) = t \log t - t - \frac{1}{2} \log t + \frac{1}{2} \log(2\pi) + o(1)$: integrating Stirling's main terms over $[x, x+1]$ reproduces $x \log x - x$ plus a constant $\frac{1}{2} \log(2\pi)$ in the limit, so $c = \frac{1}{2} \log(2\pi)$. *Check.* Setting $x = n$ and using the rectangle approximation $\sum_{k=1}^n \log k = \log n!$ recovers $\log n! = n \log n - n + \frac{1}{2} \log(2\pi n) + o(1)$, i.e. $n! \sim \sqrt{2\pi n} (n/e)^n$. ■

Exercises (Stein, Ch.6).

1. Prove that $\Gamma(s) = \lim_{n \rightarrow \infty} \frac{n^s n!}{s(s+1) \cdots (s+n)}$ whenever $s \neq 0, -1, -2, \dots$ [Hint: use the product formula for $1/\Gamma$ and the definition of the Euler constant γ .]
2. Prove that for positive a, b , $\prod_{n=1}^{\infty} \frac{n(n+a+b)}{(n+a)(n+b)} = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+1)}$. Using the product formula for $\sin \pi s$, give another proof that $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$.
3. Show that Wallis's product formula can be written $\sqrt{\frac{\pi}{2}} = \lim_{n \rightarrow \infty} \frac{2^{2n}(n!)^2}{(2n+1)!} (2n+1)^{1/2}$. As a result, prove the duplication identity $\Gamma(s)\Gamma(s+\frac{1}{2}) = \sqrt{\pi} 2^{1-2s} \Gamma(2s)$.
4. For $|z| < 1$ let $f(z) = (1-z)^{-\alpha}$ (principal branch of the logarithm), with $\alpha \in \mathbb{C}$ fixed, and write $f(z) = \sum_{n=0}^{\infty} a_n(\alpha) z^n$. Prove that $a_n(\alpha) \sim \frac{1}{\Gamma(\alpha)} n^{\alpha-1}$ as $n \rightarrow \infty$.
5. Use $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ to prove that $|\Gamma(\frac{1}{2} + it)| = \sqrt{\frac{2\pi}{e^{\pi t} + e^{-\pi t}}}$ for all $t \in \mathbb{R}$.
6. Show that $1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1} - \log n \rightarrow \frac{\gamma}{2} + \log 2$ as $n \rightarrow \infty$, where γ is Euler's constant.
7. The Beta function is $B(\alpha, \beta) = \int_0^1 (1-t)^{\alpha-1} t^{\beta-1} dt$ for $\operatorname{Re}(\alpha), \operatorname{Re}(\beta) > 0$. (a) Prove $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$. (b) Show $B(\alpha, \beta) = \int_0^{\infty} \frac{u^{\alpha-1}}{(1+u)^{\alpha+\beta}} du$. [Hint for (a): write $\Gamma(\alpha)\Gamma(\beta) = \int_0^{\infty} \int_0^{\infty} t^{\alpha-1} s^{\beta-1} e^{-t-s} dt ds$ and substitute $s = ur$, $t = u(1-r)$.]
8. (Bessel functions.) For real order $\nu > -\frac{1}{2}$ prove the power-series identity

$$J_{\nu}(x) = \frac{(x/2)^{\nu}}{\Gamma(\nu + \frac{1}{2})\sqrt{\pi}} \int_{-1}^1 e^{ixt} (1-t^2)^{\nu-1/2} dt = \left(\frac{x}{2}\right)^{\nu} \sum_{m=0}^{\infty} \frac{(-1)^m (x^2/4)^m}{m! \Gamma(\nu + m + 1)}$$

for $x > 0$, and deduce that J_{ν} satisfies $J_{\nu}'' + \frac{1}{x} J_{\nu}' + (1 - \frac{\nu^2}{x^2}) J_{\nu} = 0$. [Hint: expand e^{ixt} and use Exercise 7.]

9. (Hypergeometric series.) For $\alpha > 0$, $\beta > 0$, $\gamma > \beta$, $|z| < 1$, show

$$F(\alpha, \beta, \gamma; z) = \frac{\Gamma(\gamma)}{\Gamma(\beta)\Gamma(\gamma-\beta)} \int_0^1 t^{\beta-1} (1-t)^{\gamma-\beta-1} (1-zt)^{-\alpha} dt,$$

and conclude that F continues analytically to $\mathbb{C}$ slit along $[1, \infty)$. Note $\log(1-z) = -zF(1, 1, 2; z)$, $e^z = \lim_{\beta \rightarrow \infty} F(1, \beta, 1; z/\beta)$, and $(1-z)^{-\alpha} = F(\alpha, 1, 1; z)$.

10. (Mellin transform.) Write $\mathcal{M}(f)(z) = \int_0^\infty f(t)t^{z-1} dt$. (a) Prove for $0 < \operatorname{Re}(z) < 1$ that $\mathcal{M}(\cos)(z) = \Gamma(z) \cos \frac{\pi z}{2}$ and $\mathcal{M}(\sin)(z) = \Gamma(z) \sin \frac{\pi z}{2}$. (b) Show the sine identity extends to $-1 < \operatorname{Re}(z) < 1$, hence $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$ and $\int_0^\infty \frac{\sin x}{x^{3/2}} dx = \sqrt{2\pi}$. [Hint: integrate $e^{-w}w^{z-1}$ around the quarter-circle contour (Figure 1) from 0 to R along the real axis and back along the imaginary axis.]
11. Let $f(z) = e^{az}e^{-e^z}$ with $a > 0$. Note that in the strip $\{x + iy : |y| < \pi\}$, $f(x + iy)$ decays exponentially as $|x| \rightarrow \infty$. Prove that the Fourier transform is $\hat{f}(\xi) = \Gamma(a + i\xi)$ for all $\xi \in \mathbb{R}$.
12. Two observations about $1/\Gamma$. (a) Show $1/|\Gamma(s)|$ is not $O(e^{c|s|})$ for any $c > 0$. [Hint: at $s = -k - \frac{1}{2}$ with k a positive integer, $|1/\Gamma(s)| \geq k!/\pi$.] (b) Show there is no entire $F(s) = O(e^{c|s|})$ with simple zeros exactly at $s = 0, -1, -2, \dots$ and no other zeros.
13. Prove that $\frac{d^2 \log \Gamma(s)}{ds^2} = \sum_{n=0}^\infty \frac{1}{(s+n)^2}$ for $s > 0$. Show that, interpreting the left side as $(\Gamma'/\Gamma)'$, the formula also holds for all $s \in \mathbb{C}$ with $s \neq 0, -1, -2, \dots$.
14. (Asymptotics of $\log n!$.) (a) Show $\frac{d}{dx} \int_x^{x+1} \log \Gamma(t) dt = \log x$ for $x > 0$, hence $\int_x^{x+1} \log \Gamma(t) dt = x \log x - x + c$. (b) Deduce $\log \Gamma(n) \sim n \log n$, and in fact $\log \Gamma(n) = n \log n + O(n)$, as $n \rightarrow \infty$. [Hint: $\Gamma(x)$ is monotonically increasing for large x .]

Additional exercises.

1. Compute $\Gamma(\frac{5}{2})$ and $\Gamma(\frac{7}{2})$ using only $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and the recurrence $\Gamma(s+1) = s\Gamma(s)$.
2. Evaluate $\int_0^\infty x^3 e^{-x} dx$ and $\int_0^\infty x^{1/2} e^{-x} dx$ by recognizing each as a value of Γ .
3. Using the substitution $t = ax$ with $a > 0$, show $\int_0^\infty x^{s-1} e^{-ax} dx = \frac{\Gamma(s)}{a^s}$ for $\operatorname{Re}(s) > 0$, and check your answer at $s = 1$.
4. Use the Beta-Gamma identity to evaluate $\int_0^1 x^4(1-x)^3 dx$ as a ratio of factorials, and verify it equals $1/280$ by direct expansion.
5. Prove that $\operatorname{Res}_{s=-3} \Gamma(s) = -\frac{1}{6}$ both from the formula $(-1)^n/n!$ and by expanding $\Gamma(s) = \Gamma(s+4)/[(s+3)(s+2)(s+1)s]$ near $s = -3$.
6. (Non-example.) A student claims $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ lets one compute $\Gamma(2)\Gamma(-1)$ as $\pi/\sin(2\pi)$. Explain precisely what goes wrong, identify which side is infinite, and state the correct relation between the residues of $\Gamma(s+1)$ and $\Gamma(s)$ at $s = -1$.
7. Show $\Gamma(s+1) = s\Gamma(s)$ implies Γ has no zeros anywhere, and conclude that $1/\Gamma$ is entire. (Reflection may be assumed.)
8. Use the duplication formula $\Gamma(s)\Gamma(s+\frac{1}{2}) = \sqrt{\pi} 2^{1-2s} \Gamma(2s)$ to evaluate $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})$ a second way, and reconcile with the reflection-formula value $\pi\sqrt{2}$.
9. (Diagram problem.) Sketch the real graph of $\Gamma(s)$ for $-3 < s < 4$, marking the poles at $0, -1, -2$, the sign of Γ on each interval between consecutive poles, the minimum near $s \approx 1.4615$, and the value $\Gamma(1) = \Gamma(2) = 1$. Then sketch the *entire* function $1/\Gamma(s)$ on the same range and indicate its simple zeros.
10. (Harder.) Starting from $B(\alpha, \beta) = \int_0^\infty u^{\alpha-1}(1+u)^{-\alpha-\beta} du$, take $\beta = 1 - \alpha$ ($0 < \alpha < 1$) and evaluate the resulting integral by a keyhole/branch-cut contour around the positive real axis to give an independent proof that $\Gamma(\alpha)\Gamma(1-\alpha) = \pi/\sin \pi\alpha$. Draw the keyhole contour with its branch cut along $[0, \infty)$ and the pole at $u = -1$.

19.2 The Zeta Function and Its Functional Equation

The Dirichlet series and its half-plane of holomorphy

Definition 19.2.1. For $\operatorname{Re}(s) > 1$ the *Riemann zeta function* is

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad n^s = e^{s \operatorname{Log} n}.$$

Proposition 19.2.2 (holomorphy for $\operatorname{Re}(s) > 1$). *The series defining ζ converges for $\operatorname{Re}(s) > 1$ and is holomorphic there.*

Proof. Write $s = \sigma + it$. Then $|n^{-s}| = |e^{-s \operatorname{Log} n}| = e^{-\sigma \operatorname{Log} n} = n^{-\sigma}$. If $\sigma \geq 1 + \delta > 1$ the series is dominated termwise by the convergent series $\sum n^{-(1+\delta)}$, so $\sum n^{-s}$ converges uniformly on the half-plane $\operatorname{Re}(s) \geq 1 + \delta$ (Weierstrass M -test). A locally uniform limit of the holomorphic partial sums is holomorphic; since every point of $\operatorname{Re}(s) > 1$ lies in some such half-plane, ζ is holomorphic on $\operatorname{Re}(s) > 1$. $\square$

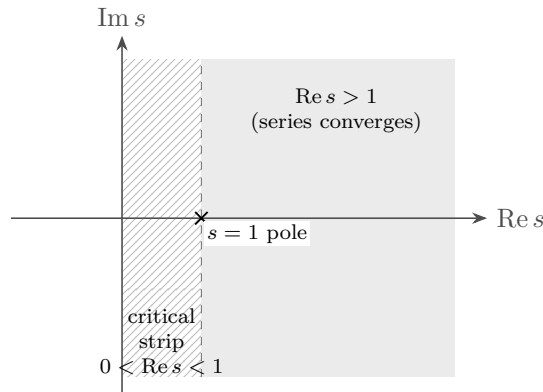


Figure 19.6. Convergence region $\operatorname{Re} s > 1$ (shaded), the critical strip (hatched), and the lone pole at $s = 1$.

The theta function and the bridge to Γ

The continuation of ζ is subtler than that of Γ ; the route here ties ζ to Γ and to the theta function via a single integral identity.

Definition 19.2.3. For real $t > 0$ the *theta function* is $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$.

By Poisson summation (Stein, Ch.4) it satisfies the functional equation

$$\vartheta(t) = t^{-1/2} \vartheta(1/t), \quad t > 0,$$

from which one extracts the growth/decay we need:

$$\vartheta(t) \leq Ct^{-1/2} \quad (t \rightarrow 0), \quad |\vartheta(t) - 1| \leq Ce^{-\pi t} \quad (t \geq 1).$$

The first follows from the functional equation (since $\vartheta(1/t) \rightarrow 1$); the second because $\sum_{n \geq 1} e^{-\pi n^2 t} \leq \sum_{n \geq 1} e^{-\pi n t} \leq Ce^{-\pi t}$ for $t \geq 1$.

Theorem 19.2.4 (the ζ - Γ - ϑ identity). *If $\operatorname{Re}(s) > 1$ then*

$$\pi^{-s/2} \Gamma(s/2) \zeta(s) = \frac{1}{2} \int_0^{\infty} u^{(s/2)-1} [\vartheta(u) - 1] du.$$

Proof. The engine is the scaling identity: for $n \geq 1$,

$$\int_0^\infty e^{-\pi n^2 u} u^{(s/2)-1} du = \pi^{-s/2} \Gamma(s/2) n^{-s}. \quad (*)$$

Indeed, substitute $u = t/(\pi n^2)$: the left side becomes $(\pi n^2)^{-s/2} \int_0^\infty e^{-t} t^{(s/2)-1} dt = (\pi n^2)^{-s/2} \Gamma(s/2) = \pi^{-s/2} \Gamma(s/2) n^{-s}$. Next, since $\frac{1}{2}[\vartheta(u) - 1] = \sum_{n \geq 1} e^{-\pi n^2 u}$, the theta estimates justify interchanging sum and integral, so

$$\frac{1}{2} \int_0^\infty u^{(s/2)-1} [\vartheta(u) - 1] du = \sum_{n \geq 1} \int_0^\infty u^{(s/2)-1} e^{-\pi n^2 u} du = \pi^{-s/2} \Gamma(s/2) \sum_{n \geq 1} n^{-s} = \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

□

$$\begin{array}{ccc} \vartheta(u) & \xrightarrow{\text{Mellin } \frac{1}{2} \int u^{s/2-1} (\cdot - 1)} & \pi^{-s/2} \Gamma(s/2) \zeta(s) \\ \text{Poisson} \downarrow & & \downarrow s \mapsto 1-s \\ u^{-1/2} \vartheta(1/u) & \dashrightarrow & \xi(s) = \xi(1-s) \end{array}$$

Figure 19.7. The theta functional equation transports, through the Mellin transform, to the reflection $\xi(s) = \xi(1-s)$.

The xi function and the functional equation

Definition 19.2.5. For $\text{Re}(s) > 1$ define the *xi function* $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$.

Theorem 19.2.6 (functional equation). ξ is holomorphic for $\text{Re}(s) > 1$ and continues to a meromorphic function on $\mathbb{C}$ with simple poles only at $s = 0$ and $s = 1$. Moreover

$$\xi(s) = \xi(1-s) \quad \text{for all } s \in \mathbb{C}.$$

Proof. Let $\psi(u) = \frac{1}{2}[\vartheta(u) - 1]$. The theta functional equation $\vartheta(u) = u^{-1/2} \vartheta(1/u)$ becomes

$$\psi(u) = u^{-1/2} \psi(1/u) + \frac{1}{2} u^{-1/2} - \frac{1}{2}.$$

Starting from Theorem 19.2.4 and splitting at $u = 1$,

$$\xi(s) = \int_0^1 u^{(s/2)-1} \psi(u) du + \int_1^\infty u^{(s/2)-1} \psi(u) du.$$

In the first integral substitute the displayed relation for $\psi(u)$; the elementary terms integrate to $\frac{1}{s-1} - \frac{1}{s}$, and the $u^{-1/2} \psi(1/u)$ term, after $u \mapsto 1/u$, recombines with the second integral. One obtains

$$\xi(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1}) \psi(u) du.$$

Because $\psi(u) = \frac{1}{2}[\vartheta(u) - 1]$ decays like $e^{-\pi u}$ as $u \rightarrow \infty$, the integral converges for every $s \in \mathbb{C}$ and defines an *entire* function. Hence ξ continues meromorphically to $\mathbb{C}$ with simple poles exactly at the two explicit terms $s = 1$ (residue +1) and $s = 0$ (residue -1). Finally the displayed formula is manifestly invariant under $s \mapsto 1-s$: the integral is symmetric in that substitution and $\frac{1}{s-1} - \frac{1}{s}$ maps to $\frac{1}{-s} - \frac{1}{1-s} = \frac{1}{s-1} - \frac{1}{s}$. Thus $\xi(s) = \xi(1-s)$. □

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{\xi} & \mathbb{C} \\ s \mapsto 1-s \downarrow & & \parallel \\ \mathbb{C} & \xrightarrow{\xi} & \mathbb{C} \end{array}$$

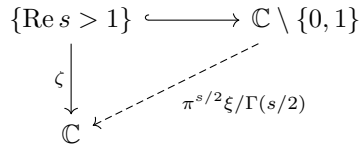
The functional equation $\xi(s) = \xi(1-s)$ says exactly that this square commutes: ξ is invariant under the reflection $s \mapsto 1-s$ of the plane across the critical line $\text{Re } s = \frac{1}{2}$ (the involution's fixed-point set).

Theorem 19.2.7 (continuation of ζ). ζ has a meromorphic continuation to all of $\mathbb{C}$ whose only singularity is a simple pole at $s = 1$ with residue 1.

Proof. From $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ solve

$$\zeta(s) = \pi^{s/2} \frac{\xi(s)}{\Gamma(s/2)}.$$

The right side is meromorphic on $\mathbb{C}$. Now $1/\Gamma(s/2)$ is entire with simple zeros at $s = 0, -2, -4, \dots$ (the poles of $\Gamma(s/2)$). The simple pole of ξ at $s = 0$ is cancelled by the zero of $1/\Gamma(s/2)$ at $s = 0$, so ζ is regular there. The remaining pole of ξ at $s = 1$ survives (since $\Gamma(\frac{1}{2}) \neq 0, \infty$), giving ζ a simple pole at $s = 1$. Its residue is $\pi^{1/2} \cdot \frac{\text{Res}_{s=1} \xi}{\Gamma(1/2)} = \pi^{1/2} \cdot \frac{1}{\sqrt{\pi}} = 1$. $\square$



The original Dirichlet series ζ on $\{\text{Re } s > 1\}$ is reproduced by $\pi^{s/2} \xi(s)/\Gamma(s/2)$ (dashed: the induced continuation), which is meromorphic on all of $\mathbb{C}$; the zero of $1/\Gamma(s/2)$ at $s = 0$ cancels the pole of ξ there, leaving the lone pole of ζ at $s = 1$.

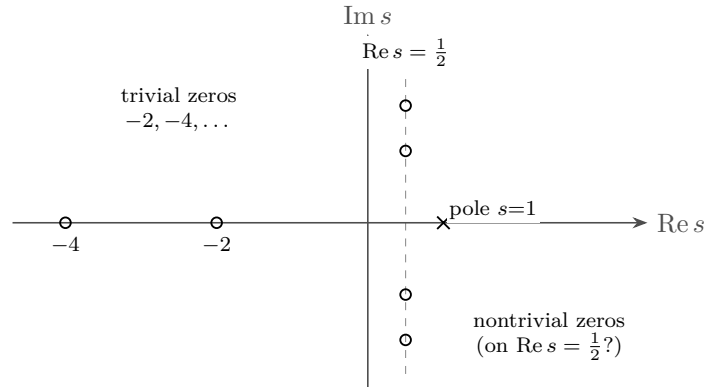


Figure 19.8. ζ after continuation: one pole at $s = 1$, trivial zeros at $-2, -4, \dots$, nontrivial zeros conjectured on the critical line. (Schematic.)

Worked Example 19.2.8 ($\zeta(0) = -\frac{1}{2}$ from the functional equation). Use $\xi(s) = \xi(1-s)$ to compute $\zeta(0)$.

Recognition. Trigger: a value of ζ at a point reachable by reflection from $\text{Re}(s) > 1$. Push $s = 0$ to $s = 1$ and use the known pole.

Solution. Near $s = 0$, $\zeta(s) = \pi^{s/2} \xi(s)/\Gamma(s/2)$. As $s \rightarrow 0$, ξ has a simple pole with residue -1 , so $\xi(s) \sim -1/s$; and $1/\Gamma(s/2)$ has a simple zero, $1/\Gamma(s/2) \sim s/2$ (since $\Gamma(s/2) \sim 2/s$). Therefore $\zeta(0) = \pi^0 \cdot (-1/s) \cdot (s/2)|_{s \rightarrow 0} = -\frac{1}{2}$. *Check.* The standard value is $\zeta(0) = -\frac{1}{2}$, consistent with the regularized sum “ $1 + 1 + 1 + \dots$ ” $= -\frac{1}{2}$.

Worked Example 19.2.9 (Trivial zeros of ζ). Show $\zeta(-2k) = 0$ for every integer $k \geq 1$.

Recognition. Trigger: zeros of the continued ζ . Read them off the factorization $\zeta = \pi^{s/2} \xi/\Gamma(s/2)$.

Solution. At $s = -2k$ ($k \geq 1$) the factor $1/\Gamma(s/2) = 1/\Gamma(-k)$ vanishes (a pole of Γ), while $\xi(-2k)$ is finite and nonzero (its only poles are at $0, 1$). Hence $\zeta(-2k) = \pi^{-k} \xi(-2k) \cdot 0 = 0$. These are the *trivial zeros*. *Check.* $s = 0$ is excluded because there the zero of $1/\Gamma(s/2)$ is consumed cancelling the pole of ξ ; the trivial zeros start at -2 . *Non-example:* $s = -1$ is *not* a zero, because $1/\Gamma(-\frac{1}{2}) \neq 0$; indeed $\zeta(-1) = -\frac{1}{12}$.

An elementary continuation to $\operatorname{Re}(s) > 0$ and growth near $\operatorname{Re} s = 1$

A second, hands-on continuation compares $\sum n^{-s}$ with $\int x^{-s} dx$; it reaches only $\operatorname{Re}(s) > 0$ but yields growth estimates near the line $\operatorname{Re}(s) = 1$.

Proposition 19.2.10 (telescoping comparison). *There are entire functions $\{\delta_n(s)\}_{n \geq 1}$ with $|\delta_n(s)| \leq |s|/n^{\sigma+1}$ (where $s = \sigma + it$) such that for every integer $N > 1$,*

$$\sum_{1 \leq n < N} \frac{1}{n^s} - \int_1^N \frac{dx}{x^s} = \sum_{1 \leq n < N} \delta_n(s).$$

Proof. Set $\delta_n(s) = \int_n^{n+1} \left(\frac{1}{n^s} - \frac{1}{x^s} \right) dx$. The mean value theorem applied to $f(x) = x^{-s}$ gives $|n^{-s} - x^{-s}| \leq |s|/n^{\sigma+1}$ for $n \leq x \leq n+1$, hence $|\delta_n(s)| \leq |s|/n^{\sigma+1}$. Summing $\int_n^{n+1} x^{-s} dx = \int_1^N x^{-s} dx$ over $1 \leq n < N$ telescopes to the stated identity. $\square$

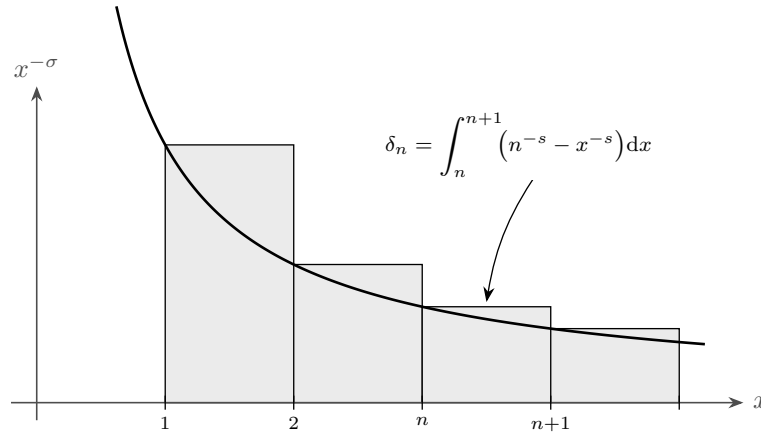


Figure 19.9. Each δ_n is the sliver between the step n^{-s} and the curve x^{-s} over $[n, n+1]$; the slivers sum to $\zeta(s) - \frac{1}{s-1}$. (Schematic.)

Corollary 19.2.11 (continuation to $\operatorname{Re} s > 0$). *For $\operatorname{Re}(s) > 0$, $\zeta(s) - \frac{1}{s-1} = H(s)$, where $H(s) = \sum_{n=1}^{\infty} \delta_n(s)$ is holomorphic on $\operatorname{Re}(s) > 0$.*

Proof. For $\operatorname{Re}(s) > 1$ let $N \rightarrow \infty$ in Proposition 19.2.10. The bound $|\delta_n| \leq |s|/n^{\sigma+1}$ gives uniform convergence of $\sum \delta_n$ on every half-plane $\operatorname{Re}(s) \geq \delta > 0$, so H is holomorphic there; and $\int_1^N x^{-s} dx \rightarrow \frac{1}{s-1}$ while $\sum_{n < N} n^{-s} \rightarrow \zeta(s)$. This yields $\zeta(s) - \frac{1}{s-1} = H(s)$ for $\operatorname{Re}(s) > 1$, and since the right side is holomorphic on $\operatorname{Re}(s) > 0$, it furnishes the continuation (with the simple pole at $s = 1$ visible in the $\frac{1}{s-1}$ term). $\square$

Proposition 19.2.12 (mild growth near $\operatorname{Re} s = 1$). *Let $s = \sigma + it$. For each $\sigma_0 \in [0, 1]$ and every $\varepsilon > 0$ there is c with*

$$(i) \quad |\zeta(s)| \leq c |t|^{1-\sigma_0+\varepsilon} \text{ whenever } \sigma \geq \sigma_0 \text{ and } |t| \geq 1;$$

$$(ii) \quad |\zeta'(s)| \leq c |t|^\varepsilon \text{ whenever } \sigma \geq 1 \text{ and } |t| \geq 1.$$

In particular $\zeta(1+it) = O(|t|^\varepsilon)$ and the same for ζ' .

Proof sketch. Besides $|\delta_n| \leq |s|/n^{\sigma+1}$ one has the cruder $|\delta_n| \leq 2/n^\sigma$ (from $|n^{-s}| = n^{-\sigma}$ and $|x^{-s}| \leq n^{-\sigma}$ for $x \geq n$). Interpolating, $|\delta_n| \leq (|s|/n^{\sigma_0+1})^\delta (2/n^{\sigma_0})^{1-\delta} \leq 2|s|^\delta/n^{\sigma_0+\delta}$ for $0 \leq \delta \leq 1$. Choosing $\delta = 1 - \sigma_0 + \varepsilon$ and using Corollary 19.2.11, $|\zeta(s)| \leq \left| \frac{1}{s-1} \right| + 2|s|^{1-\sigma_0+\varepsilon} \sum_n n^{-1-\varepsilon}$, which is $O(|t|^{1-\sigma_0+\varepsilon})$. Part (ii) follows from (i) via the Cauchy integral formula on a circle of radius $r = \varepsilon$ centered at s : $\zeta'(s) = \frac{1}{2\pi r} \int_0^{2\pi} \zeta(s + re^{i\theta}) e^{i\theta} d\theta$, that circle lying in $\operatorname{Re}(\cdot) \geq 1 - \varepsilon$. $\square$

Worked Example 19.2.13 (Euler product and a non-vanishing strip). For $\operatorname{Re}(s) > 1$ verify Euler's product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ and deduce $\zeta(s) \neq 0$ there.

Recognition. Trigger: the multiplicative structure of ζ . Expand each factor geometrically and use unique factorization.

Solution. Each factor $(1 - p^{-s})^{-1} = \sum_{k \geq 0} p^{-ks}$ converges since $|p^{-s}| = p^{-\sigma} < 1$. Multiplying over all primes and expanding, unique factorization gives each n^{-s} exactly once, so the product equals $\sum_n n^{-s} = \zeta(s)$ (absolute convergence for $\sigma > 1$ justifies the rearrangement). A convergent infinite product of nonzero factors is nonzero, so $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$. *Check.* At $s = 2$, $\prod_p (1 - p^{-2})^{-1} = \zeta(2) = \pi^2/6 \approx 1.6449$, a positive (nonzero) number.

Further worked examples

Worked Example 19.2.14 ($\zeta(-1) = -\frac{1}{12}$ from the functional equation). Compute $\zeta(-1)$ using $\xi(s) = \xi(1-s)$ and the value $\zeta(2) = \pi^2/6$.

Recognition. Trigger: a value of ζ at a negative integer reachable by reflection from $\operatorname{Re}(s) > 1$. Reflect $s = -1$ to $s = 2$.

Solution. The symmetric form $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ with $\xi(s) = \xi(1-s)$ gives, at $s = -1$ (so $1-s = 2$),

$$\pi^{1/2} \Gamma(-\tfrac{1}{2}) \zeta(-1) = \pi^{-1} \Gamma(1) \zeta(2).$$

Now $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$ and $\Gamma(1) = 1$, so $\pi^{1/2}(-2\sqrt{\pi})\zeta(-1) = \pi^{-1} \cdot \frac{\pi^2}{6}$, i.e. $-2\pi \zeta(-1) = \frac{\pi}{6}$. Hence $\zeta(-1) = -\frac{1}{12}$. *Check.* This matches the trivial-zero non-example earlier ($\zeta(-1) \neq 0$) and the regularized “ $1+2+3+\dots$ ” = $-\frac{1}{12}$. ■

Worked Example 19.2.15 ($\zeta(2) = \pi^2/6$ by the cotangent partial fractions). Derive $\zeta(2) = \pi^2/6$ from the residues of $\pi \cot(\pi z)/z^2$.

Recognition. Trigger: a sum $\sum 1/n^2$ via a contour/residue argument. Use that $\pi \cot \pi z$ has simple poles at every integer with residue 1.

Solution. Let $f(z) = \frac{\pi \cot(\pi z)}{z^2}$ and integrate over the square contour C_N with corners $(\pm(N + \frac{1}{2}))(1 \pm i)$.

On C_N , $\pi \cot \pi z$ is bounded uniformly in N , so $\left| \oint_{C_N} f \right| \leq \frac{\text{const} \cdot (\text{perimeter})}{N^2} \rightarrow 0$. By the residue theorem $\oint_{C_N} f = 2\pi i \sum (\text{residues inside})$. At each $z = n \neq 0$ the residue is $\frac{1}{n^2}$ (since $\operatorname{Res}_{z=n} \pi \cot \pi z = 1$); at $z = 0$ the Laurent expansion $\pi \cot \pi z = \frac{1}{z} - \frac{\pi^2}{3}z - \dots$ divided by z^2 gives residue $-\frac{\pi^2}{3}$. Therefore $0 = -\frac{\pi^2}{3} + 2 \sum_{n \geq 1} \frac{1}{n^2}$, so $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$. *Check.* $\pi^2/6 \approx 1.6449$, and the partial sum $\sum_{n \leq 10^3} n^{-2} \approx 1.6439$ approaches it. ■

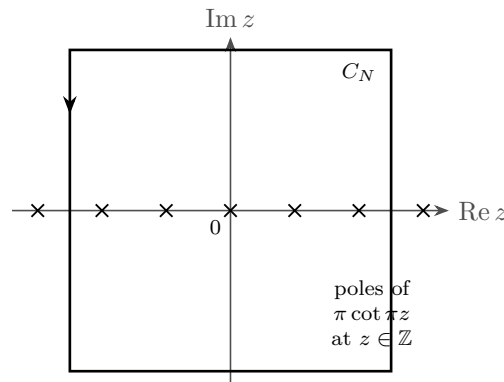


Figure 19.10. Square contour C_N (here $N = 2$) capturing the integer poles of $\pi \cot(\pi z)/z^2$; the integral $\rightarrow 0$, forcing $\sum n^{-2} = \pi^2/6$.

Worked Example 19.2.16 (Residue of ζ at $s = 1$, directly from the comparison continuation). Confirm $\text{Res}_{s=1} \zeta = 1$ using Corollary 19.2.11.

Recognition. Trigger: the residue at the unique pole of ζ . Read it off $\zeta(s) = \frac{1}{s-1} + H(s)$ with H holomorphic on $\text{Re}(s) > 0$.

Solution. Corollary 19.2.11 states $\zeta(s) - \frac{1}{s-1} = H(s)$ with H holomorphic (hence finite) at $s = 1$. Thus near $s = 1$, $\zeta(s) = \frac{1}{s-1} + H(1) + O(s-1)$, a simple pole with $\text{Res}_{s=1} \zeta = \lim_{s \rightarrow 1} (s-1)\zeta(s) = 1$. The constant term $H(1)$ is the Euler–Mascheroni constant γ : $\lim_{s \rightarrow 1} (\zeta(s) - \frac{1}{s-1}) = \gamma$. *Check.* This agrees with the ξ -route residue $\pi^{1/2}/\Gamma(\frac{1}{2}) = 1$ of Theorem 19.2.7; and $H(1) = \gamma \approx 0.5772$ is the standard limit $\lim_N (\sum_{n \leq N} \frac{1}{n} - \log N)$. ■

Worked Example 19.2.17 ($|\zeta(1+it)| = O(\log |t|)$ via the truncated sum). Sharpen Proposition 19.2.12(i) on the line $\text{Re}(s) = 1$ to a logarithmic bound.

Recognition. Trigger: growth of ζ on the edge $\sigma = 1$. Split the Dirichlet-type sum at $N = \lfloor |t| \rfloor$ and bound the head and tail separately. (Stein, Ch.6 Problem 1b.)

Solution. From Proposition 19.2.10/Corollary 19.2.11, for $\text{Re}(s) > 0$ and any integer $N \geq 2$,

$$\zeta(s) = \sum_{1 \leq n < N} \frac{1}{n^s} - \frac{N^{1-s}}{s-1} + \sum_{n \geq N} \delta_n(s).$$

Put $s = 1 + it$, $N = \lfloor |t| \rfloor$. The head $\sum_{n < N} n^{-1-it}$ has $|\cdot| \leq \sum_{n < N} n^{-1} \leq 1 + \log N = O(\log |t|)$. The middle term has $|N^{-it}/(s-1)| = 1/|t| \leq 1$. The tail uses $|\delta_n| \leq |s|/n^{\sigma+1} = |s|/n^2$, so $|\sum_{n \geq N} \delta_n| \leq |s| \sum_{n \geq N} n^{-2} \leq |s|/(N-1) = O(1)$ since $|s| \approx |t| \approx N$. Adding, $|\zeta(1+it)| = O(\log |t|)$. *Check.* The bound is consistent with ζ having only a simple pole at $s = 1$ (no growth along the vertical line away from it beyond logarithmic), and is far milder than the crude $O(|t|^\varepsilon)$ of Proposition 19.2.12. ■

Worked Example 19.2.18 (A NON-example: ζ has no zeros on $\text{Re}(s) = 1$). Is $s = 1 + it$ ($t \neq 0$) ever a zero of ζ ? Test the naive guess that the critical strip's right edge might host one.

Recognition. Trigger: a candidate zero on the line $\sigma = 1$. The non-vanishing here is the crux of the prime number theorem; check it against the 3-4-1 inequality.

Solution. Suppose $\zeta(1+it_0) = 0$ for some $t_0 \neq 0$. The classical inequality $3 + 4 \cos \theta + \cos 2\theta = 2(1 + \cos \theta)^2 \geq 0$ feeds into $\log |\zeta(\sigma)^3 \zeta(\sigma + it_0)^4 \zeta(\sigma + 2it_0)| \geq 0$ for $\sigma > 1$, i.e. $|\zeta(\sigma)|^3 |\zeta(\sigma + it_0)|^4 |\zeta(\sigma + 2it_0)| \geq 1$. As $\sigma \rightarrow 1^+$: $\zeta(\sigma) \sim \frac{1}{\sigma-1}$ (order $(\sigma-1)^{-3}$), a zero at $1 + it_0$ would make $|\zeta(\sigma + it_0)|^4 = O((\sigma-1)^4)$, and $\zeta(\sigma + 2it_0) = O(1)$. The product would then be $O((\sigma-1)^{4-3}) = O(\sigma-1) \rightarrow 0$, contradicting ≥ 1 . Hence *no* such zero exists: $\zeta(1+it) \neq 0$ for all real t . *Check.* This is exactly the boundary fact that places all nontrivial zeros strictly inside $0 < \text{Re}(s) < 1$; the Riemann Hypothesis sharpens it to $\text{Re}(s) = \frac{1}{2}$ but the line $\text{Re}(s) = 1$ is rigorously zero-free. ■

Worked Example 19.2.19 (Bernoulli numbers from $\int_0^1 x^{s-1}/(e^x - 1) dx$). Show the integral $\int_0^1 \frac{x^{s-1}}{e^x - 1} dx$ continues meromorphically with simple poles at $s = 1 - m$ and residue $B_m/m!$.

Recognition. Trigger: the small- x behaviour $1/(e^x - 1) = \sum_{m \geq 0} B_m x^{m-1}/m!$. Integrate the power series termwise against x^{s-1} . (Stein, Ch.6 Problem 16.)

Solution. Insert the generating series $\frac{x}{e^x - 1} = \sum_{m \geq 0} \frac{B_m}{m!} x^m$, valid for $|x| < 2\pi$, so $\frac{1}{e^x - 1} = \sum_{m \geq 0} \frac{B_m}{m!} x^{m-1}$. Then for $\text{Re}(s)$ large,

$$\int_0^1 \frac{x^{s-1}}{e^x - 1} dx = \sum_{m=0}^{\infty} \frac{B_m}{m!} \int_0^1 x^{s+m-2} dx = \sum_{m=0}^{\infty} \frac{B_m}{m!(s+m-1)}.$$

Each term is meromorphic in s with a simple pole at $s = 1 - m$ and residue $B_m/m!$; the series converges to an entire remainder away from those poles (the upper limit 1 keeps the tail entire). *Check.* The $m = 0$ term ($B_0 = 1$) gives the leading pole at $s = 1$ with residue 1, matching the pole of ζ inherited through $\Gamma(s)\zeta(s) = \int_0^\infty x^{s-1}/(e^x - 1) dx$; and $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$ reproduce $\zeta(0) = -\frac{1}{2}$, $\zeta(-1) = -\frac{1}{12}$ after dividing by Γ . ■

Caution. The Dirichlet series $\sum n^{-s}$ and the Euler product are valid *only* for $\operatorname{Re}(s) > 1$; neither converges past the line $\operatorname{Re}(s) = 1$. The values $\zeta(0) = -\frac{1}{2}$, $\zeta(-1) = -\frac{1}{12}$, etc., come from the *continuation*, not from summing the series. Writing “ $1 + 2 + 3 + \cdots = \zeta(-1)$ ” is a regularization, not an equality of convergent objects. Also note ξ has poles at both 0 and 1, but ζ inherits only the one at $s = 1$ because the zero of $1/\Gamma(s/2)$ swallows the other.

Exercises (Stein, Ch.6).

1. Prove that for $\operatorname{Re}(s) > 1$, $\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$. [Hint: write $\frac{1}{e^x - 1} = \sum_{n=1}^\infty e^{-nx}$.]
2. Use the previous exercise to give another proof that $\zeta(s)$ continues to $\mathbb{C}$ with only singularity a simple pole at $s = 1$. [Hint: write $\zeta(s) = \frac{1}{\Gamma(s)} \int_0^1 \frac{x^{s-1}}{e^x - 1} dx + \frac{1}{\Gamma(s)} \int_1^\infty \frac{x^{s-1}}{e^x - 1} dx$; the second integral is entire, while $\int_0^1 \frac{x^{s-1}}{e^x - 1} dx = \sum_{m=0}^\infty \frac{B_m}{m!(s+m-1)}$, B_m the Bernoulli numbers from $\frac{x}{e^x - 1} = \sum_{m \geq 0} \frac{B_m}{m!} x^m$. Then $B_0 = 1$ and $\limsup_m |B_m/m!|^{1/m} = 1/2\pi$.]
3. Let f be indefinitely differentiable on $\mathbb{R}$ with compact support (more generally, f in the Schwartz space), and put $I(s) = \frac{1}{\Gamma(s)} \int_0^\infty f(x) x^{-1+s} dx$. (a) Observe I is holomorphic for $\operatorname{Re}(s) > 0$ and prove I continues to an entire function. (b) Prove $I(0) = 0$, and more generally $I(-n) = (-1)^n f^{(n+1)}(0)$ for all $n \geq 0$. [Hint: integrate by parts to get $I(s) = \frac{(-1)^k}{\Gamma(s+k)} \int_0^\infty f^{(k)}(x) x^{s+k-1} dx$.]

Problems (Stein, Ch.6).

1. Further estimates for ζ and ζ' near $\operatorname{Re}(s) = 1$. (a) Use Proposition 2.5 and Corollary 2.6 to prove, for every integer $N \geq 2$ and $\operatorname{Re}(s) > 0$, $\zeta(s) = \sum_{1 \leq n < N} n^{-s} - \frac{N^{s-1}}{s-1} + \sum_{n \geq N} \delta_n(s)$. (b) Show $|\zeta(1+it)| = O(\log |t|)$ as $|t| \rightarrow \infty$ by taking $N = \lfloor |t| \rfloor$. (c) Refine the second conclusion of Proposition 2.7 similarly. (d) Show that for fixed $t \neq 0$ the partial sums of $\sum_{n=1}^\infty 1/n^{1+it}$ are bounded, yet the series does not converge.
- 2*. Prove that for $\operatorname{Re}(s) > 0$, $\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \frac{\{x\}}{x^{s+1}} dx$, where $\{x\}$ is the fractional part of x .
- 3*. With $Q(x) = \{x\} - \frac{1}{2}$, rewrite the previous problem as $\zeta(s) = \frac{s}{s-1} - \frac{1}{2} - s \int_1^\infty \frac{Q(x)}{x^{s+1}} dx$. Construct Q_k recursively by $\int_0^1 Q_k(x) dx = 0$, $Q'_{k+1} = Q_k$, $Q_0 = Q$, and $Q_k(x+1) = Q_k(x)$. Then $\zeta(s) = \frac{s}{s-1} - \frac{1}{2} - s \int_1^\infty \left(\frac{d^k}{dx^k} Q_k(x) \right) x^{-s-1} dx$, and a k -fold integration by parts gives the continuation of ζ for $\operatorname{Re}(s) > -k$.
- 4*. The Q_k above relate to the Bernoulli polynomials $B_k(x)$ by $Q_k(x) = \frac{B_{k+1}(x)}{(k+1)!}$ for $0 \leq x \leq 1$. Also, for a positive integer k , $2\zeta(2k) = (-1)^{k+1} \frac{(2\pi)^{2k}}{(2k)!} B_{2k}$, where $B_k = B_k(0)$ are the Bernoulli numbers.

Additional exercises.

1. Show $|n^{-s}| = n^{-\sigma}$ for $s = \sigma + it$, and use it to prove $\sum n^{-s}$ converges absolutely iff $\sigma > 1$. Where exactly does absolute convergence fail at $\sigma = 1$?
2. Compute $\zeta(2) = \pi^2/6$ and $\zeta(4) = \pi^4/90$ numerically to four places by partial sums, and compare with the closed forms.
3. From $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ and $\xi(s) = \xi(1-s)$, derive the explicit functional equation $\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s)$.

4. Use the explicit functional equation of the previous problem to recompute $\zeta(0)$ and to show $\zeta(-2k) = 0$ for $k \geq 1$ from the factor $\sin(\pi s/2)$.
5. Verify directly that $\operatorname{Res}_{s=1} \zeta(s) = 1$ from $\zeta(s) - \frac{1}{s-1} = H(s)$ with H holomorphic at $s = 1$, and interpret $H(1)$ as the Euler–Mascheroni constant γ .
6. (Non-example.) Explain why $\zeta(-1) = -\frac{1}{12}$ does *not* mean $1 + 2 + 3 + \cdots = -\frac{1}{12}$ as a convergent series. State precisely where the Dirichlet series is valid and what role the analytic continuation plays.
7. Using the integral $\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$ ($\operatorname{Re} s > 1$), and $\frac{1}{e^x - 1} = \sum_{n \geq 1} e^{-nx}$, recover $\zeta(s) = \sum n^{-s}$ by integrating term by term, justifying the interchange.
8. Show that the bound $|\delta_n(s)| \leq |s|/n^{\sigma+1}$ in Proposition 19.2.10 gives $|H(s)| \leq |s| \zeta(\sigma+1)$ on $\operatorname{Re}(s) = \sigma > 0$, and deduce $\zeta(s) = \frac{1}{s-1} + O(|s|)$ near $s = 1$ in any fixed strip.
9. (Diagram problem.) Draw the complex s -plane showing: the convergence region $\operatorname{Re} s > 1$, the critical strip $0 < \operatorname{Re} s < 1$, the critical line $\operatorname{Re} s = \frac{1}{2}$, the pole at $s = 1$, the trivial zeros at $-2, -4, -6$, and a few nontrivial zeros placed symmetrically on the critical line. Indicate the reflection $s \mapsto 1 - s$ as a point symmetry about $s = \frac{1}{2}$.
10. (Harder; diagram.) Using Exercise 10 of the scraped set (the quarter-circle contour for the Mellin transform of $\sin$), draw the contour from 0 to R along the positive real axis and back along the positive imaginary axis, mark the absence of poles inside, and explain how letting $R \rightarrow \infty$ yields $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$.

Chapter 20

The Zeta Function and the Prime Number Theorem

20.1 Zeros of the Zeta Function

The prime-counting function and the goal

For $x > 0$ let $\pi(x)$ be the number of primes $p \leq x$. Its graph is a ragged staircase, jumping by 1 at each prime, and no elementary closed form captures it. What *is* stable is its asymptotic size. Legendre and Gauss conjectured, and Hadamard and de la Vallée Poussin proved in 1896, the

Theorem 20.1.1 (Prime Number Theorem). *As $x \rightarrow \infty$,*

$$\pi(x) \sim \frac{x}{\log x}, \quad \text{that is,} \quad \lim_{x \rightarrow \infty} \frac{\pi(x)}{x/\log x} = 1.$$

The relation $f \sim g$ means $f(x)/g(x) \rightarrow 1$; it is far weaker than $f - g \rightarrow 0$. The whole chapter is the proof of Theorem 20.1.1. The engine is complex-analytic: the heart of the matter is that the Riemann zeta function $\zeta(s)$ *does not vanish on the line* $\operatorname{Re}(s) = 1$. In fact this non-vanishing is logically *equivalent* to the prime number theorem. Section 1 establishes the non-vanishing and the growth estimates for $1/\zeta$; Section 2 reduces the theorem to an asymptotic for an auxiliary function ψ_1 tied to ζ by a contour integral; Section 3 evaluates that integral by deforming the contour to the line $\operatorname{Re}(s) = 1$.

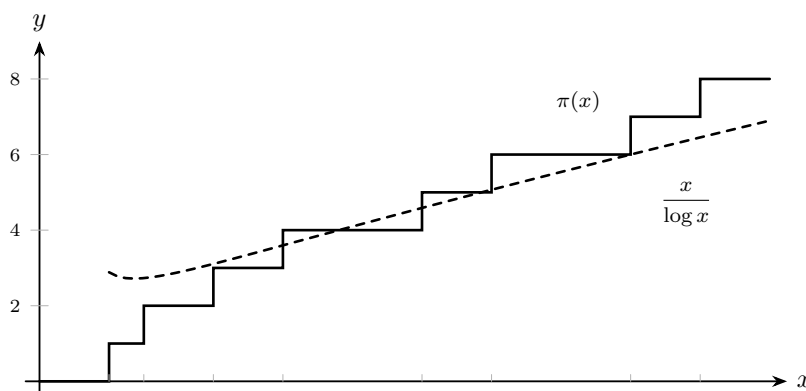


Figure 20.1. The staircase $\pi(x)$ (unit jump at each prime) and the smooth comparison $x/\log x$, which threads just below it; their ratio tends to 1. Axis tick marks sit at the primes $p = 2, 3, 5, 7, 11, 13, 17, 19$. (Schematic.)

The Euler product and the first non-vanishing

Theorem 20.1.2 (Euler product). *For $\operatorname{Re}(s) > 1$,*

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1 - p^{-s}},$$

the product running over all primes p . Consequently $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$.

Proof. Each factor expands as a geometric series $\frac{1}{1 - p^{-s}} = \sum_{k \geq 0} p^{-ks}$, valid since $|p^{-s}| = p^{-\sigma} < 1$ when $\sigma = \operatorname{Re}(s) > 1$. Fix integers $M > N$. By the fundamental theorem of arithmetic every $n \leq N$ factors uniquely into primes $p \leq N$, each repeated fewer than M times, so

$$\sum_{n=1}^N \frac{1}{n^s} \leq \prod_{p \leq N} \left(1 + \frac{1}{p^s} + \cdots + \frac{1}{p^{Ms}}\right) \leq \prod_{p \leq N} \frac{1}{1 - p^{-s}} \leq \prod_p \frac{1}{1 - p^{-s}}.$$

Letting $N \rightarrow \infty$ gives $\sum_n n^{-s} \leq \prod_p (1 - p^{-s})^{-1}$. For the reverse, again by unique factorization $\prod_{p \leq N} (1 + \cdots + p^{-Ms}) \leq \sum_n n^{-s}$; let $M \rightarrow \infty$ then $N \rightarrow \infty$ to get $\prod_p (1 - p^{-s})^{-1} \leq \sum_n n^{-s}$. The two inequalities give equality. Finally a convergent infinite product of nonzero factors is nonzero (Proposition 3.1, Ch.5), so $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$. $\square$

The product formula is unique factorization *analytically encoded*: the primes are the “atoms” and ζ is built multiplicatively from them. That is why information about the zeros of ζ translates into information about primes.

Worked Example 20.1.3 (The Basel value and a numeric check of the product). Verify $\zeta(2) = \pi^2/6$ is consistent with the first few Euler factors.

Recognition. The Euler product converges, so truncating over small primes should approach $\zeta(2)$ from below (every factor $(1 - p^{-2})^{-1} > 1$).

Solution. $\zeta(2) = \pi^2/6 \approx 1.644934$. The partial product over $p = 2, 3, 5, 7$ is

$$\frac{1}{1-1/4} \cdot \frac{1}{1-1/9} \cdot \frac{1}{1-1/25} \cdot \frac{1}{1-1/49} = \frac{4}{3} \cdot \frac{9}{8} \cdot \frac{25}{24} \cdot \frac{49}{48} \approx 1.5942.$$

Check. $1.5942 < 1.6449$ and the ratio is 0.969; including more primes increases the product toward $\pi^2/6$. The deficit $1.6449 - 1.5942 = 0.0507$ is accounted for by the omitted primes 11, 13, $\dots$, each contributing a factor just above 1. $\blacksquare$

Trivial zeros and the critical strip

The functional equation $\xi(s) = \xi(1-s)$, where $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$, provides the analytic continuation of ζ to all of $\mathbb{C}$ (except the simple pole at $s = 1$) and pins down the zeros to the left of the strip. Writing the equation as $\pi^{-s/2} \Gamma(s/2) \zeta(s) = \pi^{-(1-s)/2} \Gamma((1-s)/2) \zeta(1-s)$ gives

$$\zeta(s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)} \zeta(1-s).$$

$$\begin{array}{ccc} s & \xrightarrow{\xi} & \xi(s) \\ \downarrow 1-s & & \parallel \\ 1-s & \xrightarrow{\xi} & \xi(1-s) \end{array}$$

Figure 20.2. The functional equation as a commuting square: the completed zeta $\xi = \pi^{-s/2} \Gamma(s/2) \zeta$ is invariant under the reflection $s \mapsto 1-s$ across the critical line $\operatorname{Re} s = \frac{1}{2}$, i.e. $\xi(s) = \xi(1-s)$.

Theorem 20.1.4 (Trivial zeros). *The only zeros of ζ outside the strip $0 \leq \operatorname{Re}(s) \leq 1$ are the negative even integers $s = -2, -4, -6, \dots$*

Proof. Take $\operatorname{Re}(s) < 0$, so $\operatorname{Re}(1-s) > 1$. Then: (i) $\zeta(1-s) \neq 0$ by Theorem 20.1.2; (ii) $\Gamma((1-s)/2)$ has no zeros, being a nonvanishing entire-after-reciprocal factor; (iii) $1/\Gamma(s/2)$ vanishes exactly where $\Gamma(s/2)$ has poles, namely $s/2 = 0, -1, -2, \dots$, i.e. $s = 0, -2, -4, \dots$; the value $s = 0$ is excluded since $\operatorname{Re}(s) < 0$. Hence in $\operatorname{Re}(s) < 0$ the right side vanishes only at $s = -2, -4, -6, \dots$, and these are the only zeros of ζ there. There are no zeros in $\operatorname{Re}(s) > 1$, so all zeros off the strip are the stated negative even integers. $\square$

The strip $0 \leq \operatorname{Re}(s) \leq 1$ that remains is the *critical strip*. Riemann conjectured that every zero *inside* it lies on the central line $\operatorname{Re}(s) = \frac{1}{2}$ — the *Riemann hypothesis*, still open. Because of this the zeros at $-2, -4, \dots$ are called *trivial*.

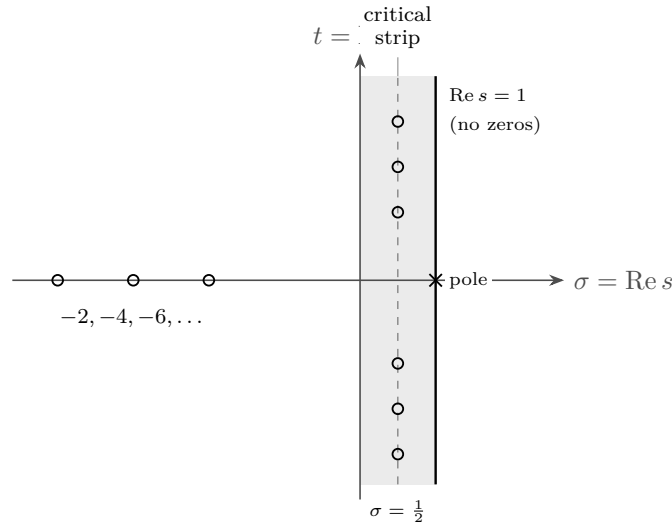


Figure 20.3. The critical strip $0 \leq \sigma \leq 1$ (shaded), with the conjectured critical line $\sigma = \frac{1}{2}$ (dashed) and the zero-free line $\operatorname{Re} s = 1$ (solid). Circles ($\circ$) mark zeros: the trivial zeros $-2, -4, -6, \dots$ on the negative real axis and a few illustrative nontrivial zeros on $\sigma = \frac{1}{2}$; the cross ($\times$) marks the simple pole at $s = 1$. The plotted nontrivial zeros are illustrative. (Schematic.)

The non-vanishing on $\operatorname{Re}(s) = 1$

This is the crux. Near the pole $s = 1$ there are obviously no zeros, but we need the much deeper fact that $\zeta(1+it) \neq 0$ for *every* real t . The proof rests on a logarithm formula and a trigonometric trick.

Lemma 20.1.5. *For $\operatorname{Re}(s) > 1$,*

$$\log \zeta(s) = \sum_{p,m} \frac{p^{-ms}}{m} = \sum_{n=1}^{\infty} \frac{c_n n^{-s}}{n}, \quad c_n \geq 0,$$

where $c_n = 1/m$ if $n = p^m$ for a prime p and integer $m \geq 1$, and $c_n = 0$ otherwise.

Proof. For real $s > 1$ take logarithms in the Euler product and use $\log \frac{1}{1-x} = \sum_{m \geq 1} x^m/m$ (valid for $0 \leq x < 1$) with $x = p^{-s}$:

$$\log \zeta(s) = \sum_p \log \frac{1}{1-p^{-s}} = \sum_p \sum_{m \geq 1} \frac{p^{-ms}}{m} = \sum_{p,m} \frac{p^{-ms}}{m}.$$

The double sum converges absolutely, so its order is immaterial and it continues to all $\operatorname{Re}(s) > 1$. (Here $\log \zeta$ is single-valued on the simply connected half-plane $\operatorname{Re}(s) > 1$ since ζ is zero-free there.) Collecting terms with $n = p^m$ gives the Dirichlet series with $c_n \geq 0$. $\square$

Lemma 20.1.6. For every real θ , $3 + 4 \cos \theta + \cos 2\theta \geq 0$.

Proof. Using $\cos 2\theta = 2 \cos^2 \theta - 1$,

$$3 + 4 \cos \theta + \cos 2\theta = 3 + 4 \cos \theta + 2 \cos^2 \theta - 1 = 2(1 + \cos \theta)^2 \geq 0. \quad \square$$

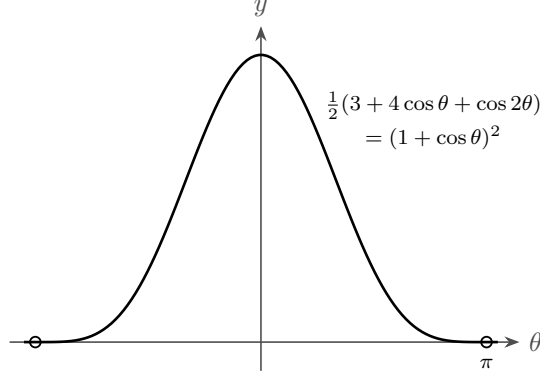


Figure 20.4. The trick function $3 + 4 \cos \theta + \cos 2\theta = 2(1 + \cos \theta)^2 \geq 0$ (here scaled by $\frac{1}{2}$), nonnegative and touching 0 only at $\theta = \pm\pi$ (circles).

Corollary 20.1.7. If $\sigma > 1$ and $t \in \mathbb{R}$, then $\log|\zeta^3(\sigma)\zeta^4(\sigma+it)\zeta(\sigma+2it)| \geq 0$, hence $|\zeta^3(\sigma)\zeta^4(\sigma+it)\zeta(\sigma+2it)| \geq 1$.

Proof. Write $s = \sigma + it$. Since $\operatorname{Re}(n^{-s}) = \operatorname{Re}(e^{-(\sigma+it)\log n}) = n^{-\sigma} \cos(t \log n)$, Lemma 20.1.5 gives $\log|\zeta(s)| = \operatorname{Re} \log \zeta(s) = \sum_n c_n n^{-\sigma} \cos(t \log n)$. Therefore, with $\theta_n = t \log n$,

$$\log|\zeta^3(\sigma)\zeta^4(\sigma+it)\zeta(\sigma+2it)| = \sum_n c_n n^{-\sigma} (3 + 4 \cos \theta_n + \cos 2\theta_n) \geq 0,$$

by Lemma 20.1.6 and $c_n \geq 0$. Exponentiating gives the product ≥ 1 . $\square$

Theorem 20.1.8 (No zeros on $\operatorname{Re}(s) = 1$). $\zeta(1+it) \neq 0$ for all $t \in \mathbb{R}$.

Proof. Suppose, for contradiction, $\zeta(1+it_0) = 0$ for some $t_0 \neq 0$. As ζ is holomorphic at $1+it_0$ it vanishes there to order ≥ 1 , so for some $C > 0$, $|\zeta(\sigma+it_0)|^4 \leq C(\sigma-1)^4$ as $\sigma \rightarrow 1^+$. Since $s = 1$ is a *simple* pole, $|\zeta(\sigma)|^3 \leq C'(\sigma-1)^{-3}$. And ζ is holomorphic at $1+2it_0$, so $|\zeta(\sigma+2it_0)|$ stays bounded. Multiplying,

$$|\zeta^3(\sigma)\zeta^4(\sigma+it_0)\zeta(\sigma+2it_0)| \leq C''(\sigma-1)^{-3}(\sigma-1)^4 = C''(\sigma-1) \rightarrow 0 \quad (\sigma \rightarrow 1^+),$$

contradicting Corollary 20.1.7, which forces the product ≥ 1 . Hence no such t_0 exists. $\square$

The accounting is exactly balanced: the pole contributes $(\sigma-1)^{-3}$, the hypothetical zero contributes $(\sigma-1)^4$, and $4 > 3$ wins, killing the product. The constants 3, 4, 1 come from $2(1 + \cos \theta)^2 = 3 + 4 \cos \theta + \cos 2\theta$ — nothing else would do. As a consequence of this and the functional equation, ζ also has no zeros on $\operatorname{Re}(s) = 0$.

Worked Example 20.1.9 (Why the exponents must be 3, 4, 1). Show that a weight combination $\zeta^a(\sigma)\zeta^b(\sigma+it)\zeta(\sigma+2it)$ can yield a contradiction only if $a < b$, and that $a = 3, b = 4$ is the minimal integer choice matching a nonnegative cosine polynomial.

Recognition. We need $b \cdot (\text{order of zero}) > a \cdot (\text{order of pole})$ and $a + b \cos \theta + \cos 2\theta \geq 0$ for all θ .

Solution. Near $\sigma = 1$ the product behaves like $(\sigma-1)^{-a} \cdot (\sigma-1)^b = (\sigma-1)^{b-a}$, which tends to 0 iff $b > a$. Positivity demands $a + b \cos \theta + \cos 2\theta = a - 1 + b \cos \theta + 2 \cos^2 \theta \geq 0$ for all θ ; viewing it as a quadratic in $u = \cos \theta \in [-1, 1]$, $2u^2 + bu + (a-1) \geq 0$ requires nonnegativity at the vertex, $a - 1 - b^2/8 \geq 0$, i.e. $a \geq 1 + b^2/8$. With $b = 4$ this gives $a \geq 3$, so $(a, b) = (3, 4)$ is the smallest integer pair with $b > a$. *Check.* $a = 3, b = 4$ gives $2u^2 + 4u + 2 = 2(u+1)^2 \geq 0$, the equality $2(1 + \cos \theta)^2$ of Lemma 20.1.6. $\blacksquare$

Quantitative non-vanishing: estimates for $1/\zeta$

The proof of the prime number theorem manipulates the logarithmic derivative $\zeta'(s)/\zeta(s)$ near $\operatorname{Re}(s) = 1$, so we need a lower bound on $|\zeta|$ (upper bound on $1/|\zeta|$) there. The next proposition is a quantitative form of Theorem 20.1.8.

Proposition 20.1.10. *For every $\epsilon > 0$ there is $c > 0$ with*

$$\frac{1}{|\zeta(s)|} \leq c|t|^\epsilon \quad \text{whenever } s = \sigma + it, \sigma \geq 1, |t| \geq 1.$$

Proof. By Corollary 20.1.7, $|\zeta^3(\sigma)\zeta^4(\sigma+it)\zeta(\sigma+2it)| \geq 1$ for $\sigma \geq 1$. With the growth bound $|\zeta(\sigma+2it)| \leq c|t|^\epsilon$ from Proposition 2.7 of Ch.6, and $|\zeta(\sigma)|^{-3} \geq c(\sigma-1)^3$ near the pole, this yields

$$|\zeta^4(\sigma+it)| \geq c|\zeta(\sigma)|^{-3}|t|^{-\epsilon} \geq c(\sigma-1)^3|t|^{-\epsilon}, \quad \text{so } |\zeta(\sigma+it)| \geq c(\sigma-1)^{3/4}|t|^{-\epsilon/4}. \quad (*)$$

Fix a constant A (chosen below) and split on whether $\sigma-1 \geq A|t|^{-5\epsilon}$. *Case 1.* If $\sigma-1 \geq A|t|^{-5\epsilon}$, then $(*)$ gives directly $|\zeta(\sigma+it)| \geq A'|t|^{-4\epsilon}$; relabel $4\epsilon \mapsto \epsilon$. *Case 2.* If $\sigma-1 < A|t|^{-5\epsilon}$, pick $\sigma' > \sigma$ with $\sigma'-1 = A|t|^{-5\epsilon}$. By the triangle inequality and the mean value theorem with the derivative bound of Ch.6,

$$|\zeta(\sigma+it)| \geq |\zeta(\sigma'+it)| - |\zeta(\sigma'+it) - \zeta(\sigma+it)| \geq c(\sigma'-1)^{3/4}|t|^{-\epsilon/4} - c'(\sigma'-1)|t|^\epsilon.$$

Choosing $A = (c/(2c'))^4$ makes $c(\sigma'-1)^{3/4}|t|^{-\epsilon/4} = 2c'(\sigma'-1)|t|^\epsilon$, so the right side is $\frac{1}{2}c(\sigma'-1)^{3/4}|t|^{-\epsilon/4} = A'|t|^{-4\epsilon}$. Again relabel $4\epsilon \mapsto \epsilon$. In both cases $1/|\zeta(s)| \leq c|t|^\epsilon$. $\square$

Worked Example 20.1.11 (The logarithmic derivative has a simple pole at $s = 1$). Show $-\zeta'(s)/\zeta(s) = \frac{1}{s-1} + h(s)$ with h holomorphic near $s = 1$, and compute the residue.

Recognition. ζ has a simple pole at $s = 1$; the logarithmic derivative of a function with a simple pole has a simple pole with residue equal to minus the pole order.

Solution. Near $s = 1$, $\zeta(s) = \frac{1}{s-1} + H(s)$ with H holomorphic (Corollary 2.6, Ch.6). Then $\zeta(s) = \frac{1 + (s-1)H(s)}{s-1}$, and

$$\frac{\zeta'(s)}{\zeta(s)} = \frac{d}{ds} \log \zeta(s) = \frac{d}{ds} \left[-\log(s-1) + \log(1 + (s-1)H(s)) \right] = -\frac{1}{s-1} + (\text{holo}),$$

so $-\zeta'(s)/\zeta(s) = \frac{1}{s-1} + h(s)$ with h holomorphic. *Check.* The residue of $-\zeta'/\zeta$ at $s = 1$ is $+1$, equal to the order of the pole of ζ , as it must be (a simple pole of ζ is a simple zero of $1/\zeta$). ■

Caution. The bound $1/|\zeta| \leq c|t|^\epsilon$ is *not* uniform down to $t = 0$: it requires $|t| \geq 1$, since ζ has its pole at $s = 1$ (i.e. $t = 0$) where $1/|\zeta| \rightarrow 0$, not ∞ . Never apply Proposition 20.1.10 on a neighborhood of $t = 0$; the small- t behavior is governed by the pole, handled separately as the main term in Section 3.

Further worked examples

Worked Example 20.1.12 (The alternating zeta (Dirichlet eta) and its zero-free strip). Let $\tilde{\zeta}(s) = \sum_{n=1}^{\infty} (-1)^{n+1} n^{-s} = 1 - 2^{-s} + 3^{-s} - \dots$. Show the series converges for $\operatorname{Re}(s) > 0$, that $\tilde{\zeta}(s) = (1 - 2^{1-s})\zeta(s)$ for $\operatorname{Re}(s) > 1$, and deduce that ζ has no zeros on the real segment $0 < \sigma < 1$.

Recognition. An alternating Dirichlet series: pair it against ζ by splitting off the even terms. Real alternating series with decreasing terms have a definite sign, which transfers to ζ through the factor $(1 - 2^{1-s})$.

Solution. Group ζ and twice its even part:

$$\zeta(s) - 2 \sum_{n \text{ even}} n^{-s} = \sum_n n^{-s} - 2 \sum_k (2k)^{-s} = \sum_n n^{-s} - 2^{1-s} \sum_k k^{-s} = (1 - 2^{1-s})\zeta(s),$$

and the left side is exactly $\sum_n (-1)^{n+1} n^{-s} = \tilde{\zeta}(s)$. For $\operatorname{Re}(s) > 0$ the terms n^{-s} have moduli $n^{-\sigma}$ decreasing to 0 and the partial sums of $(-1)^{n+1}$ are bounded, so Dirichlet's test gives convergence; thus $\tilde{\zeta}$ is holomorphic on $\operatorname{Re}(s) > 0$. Now take $s = \sigma$ real with $0 < \sigma < 1$. The series $\tilde{\zeta}(\sigma) = \sum (-1)^{n+1} n^{-\sigma}$ is alternating with strictly decreasing positive terms, so its sum is strictly positive (it lies between its first term 1 and $1 - 2^{-\sigma} > 0$). Also $2^{1-\sigma} > 1$, so $1 - 2^{1-\sigma} < 0$. From $\zeta(\sigma) = \tilde{\zeta}(\sigma)/(1 - 2^{1-\sigma})$ we get $\zeta(\sigma) = (\text{positive})/(\text{negative}) < 0 \neq 0$. *Check.* At $\sigma = \frac{1}{2}$, $1 - 2^{1/2} = -0.4142$ and $\tilde{\zeta}(\frac{1}{2}) \approx 0.6049$, so $\zeta(\frac{1}{2}) \approx -1.4604 \neq 0$; indeed $\zeta(1/2) = -1.4603545 \dots$ ■

Worked Example 20.1.13 ($\zeta(s)$ is real on the critical line). Show that $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ is real whenever s is real and whenever $\operatorname{Re}(s) = \frac{1}{2}$.

Recognition. Use the Schwarz reflection identity $\overline{\xi(s)} = \xi(\bar{s})$ (valid because ξ is real on the real axis) together with the functional equation $\xi(s) = \xi(1-s)$.

Solution. Each factor of ξ has real Taylor coefficients about a real point, so $\overline{\xi(s)} = \xi(\bar{s})$ by the reflection principle; in particular ξ is real on $\mathbb{R}$. On the line $\operatorname{Re}(s) = \frac{1}{2}$ write $s = \frac{1}{2} + it$. Then $\bar{s} = \frac{1}{2} - it = 1 - s$, so

$$\overline{\xi(s)} = \xi(\bar{s}) = \xi(1-s) = \xi(s)$$

by the functional equation. A number equal to its own conjugate is real, so $\xi(\frac{1}{2} + it) \in \mathbb{R}$. *Check.* The reality of ξ on the critical line is what lets one define the real-valued Riemann–Siegel Z -function and count zeros by sign changes; e.g. $\xi(\frac{1}{2}) = -\pi^{-1/4} \Gamma(\frac{1}{4}) \cdot 1.4603 \dots$ is real and nonzero. ■

Worked Example 20.1.14 (Sharper non-vanishing: $1/|\zeta(1+it)| \leq A(\log |t|)^7$). Granting $|\zeta(1+it)| \leq A \log |t|$ for $|t| \geq 2$, improve Proposition 20.1.10 on the line $\sigma = 1$ to a power of $\log |t|$ rather than $|t|^\epsilon$.

Recognition. Feed the logarithmic growth bound into the 3-4-1 inequality of Corollary 20.1.7, then optimize the offset $\sigma - 1$ as a power of $1/\log |t|$ instead of a power of $|t|$.

Solution. Corollary 20.1.7 gives, for $\sigma > 1$, $|\zeta(\sigma+it)|^4 \geq |\zeta(\sigma)|^{-3} |\zeta(\sigma+2it)|^{-1}$. Near the pole $|\zeta(\sigma)| \geq c/(\sigma-1)$ so $|\zeta(\sigma)|^{-3} \geq c(\sigma-1)^3$, while $|\zeta(\sigma+2it)| \leq A \log |t|$. Hence

$$|\zeta(\sigma+it)| \geq c(\sigma-1)^{3/4} (\log |t|)^{-1/4}.$$

For σ between 1 and $1+\eta$, the derivative bound $|\zeta'| \leq A(\log |t|)^2$ gives $|\zeta(1+it)| \geq |\zeta(\sigma+it)| - A(\log |t|)^2(\sigma-1)$. Choose $\sigma-1 = B(\log |t|)^{-9}$ with B small; the first term is then $\asymp (\log |t|)^{-27/4-1/4} = (\log |t|)^{-7}$ and the correction $(\log |t|)^2 (\log |t|)^{-9} = (\log |t|)^{-7}$ is of the same order, so for B chosen correctly the difference is $\geq \frac{1}{2} c (\log |t|)^{-7}$. Thus $1/|\zeta(1+it)| \leq A(\log |t|)^7$. *Check.* The exponent $7 = 27/4 + 1/4$ rounded up is exactly the constant quoted in Stein's Exercise 9(c); the gain over $|t|^\epsilon$ is what yields the classical error term $O(x e^{-c\sqrt{\log x}})$ in $\psi(x) - x$. ■

Worked Example 20.1.15 (A non-example: the Euler product fails on $\operatorname{Re}(s) = 1$). Show that the Euler product $\prod_p (1 - p^{-s})^{-1}$ does *not* converge at $s = 1$, so it cannot be used to prove $\zeta(1+it) \neq 0$ directly.

Recognition. Convergence of $\prod (1 - a_p)^{-1}$ with $a_p \geq 0$ is equivalent to convergence of $\sum a_p$. Test $\sum_p p^{-1}$.

Solution. At $s = 1$ the factors are $(1 - 1/p)^{-1}$ and $\log \prod_{p \leq N} (1 - 1/p)^{-1} = \sum_{p \leq N} \log \frac{1}{1-1/p} = \sum_{p \leq N} (\frac{1}{p} + \frac{1}{2p^2} + \dots)$. The tail $\sum_{p \leq N} (\frac{1}{2p^2} + \dots)$ converges, but $\sum_{p \leq N} \frac{1}{p} \rightarrow \infty$ (Euler: the sum of prime reciprocals diverges like $\log \log N$). Hence the partial products diverge to $+\infty$ and the product does not converge at $s = 1$. *Check.* This divergence is consistent with ζ having a *pole* at $s = 1$ — the product “ $= \infty$ ” there. It also explains why the subtle 3-4-1 argument is needed for the boundary line $\operatorname{Re}(s) = 1$: the cheap product argument that gives non-vanishing for $\operatorname{Re}(s) > 1$ simply breaks down on the line. ■

Worked Example 20.1.16 (Residue of ζ at its pole equals 1). Show ζ has a simple pole at $s = 1$ with residue 1, starting from $\zeta(s) - \frac{1}{s-1} = \sum_{n=1}^{\infty} (n^{-s} - \int_n^{n+1} u^{-s} du)$.

Recognition. Compare the sum $\sum n^{-s}$ with the integral $\int_1^{\infty} u^{-s} du = 1/(s-1)$; the difference is a series holomorphic past $\sigma = 1$, isolating the pole.

Solution. For $\operatorname{Re}(s) > 1$, $\int_1^{\infty} u^{-s} du = \frac{1}{s-1}$, so

$$\zeta(s) - \frac{1}{s-1} = \sum_{n=1}^{\infty} \left(\frac{1}{n^s} - \int_n^{n+1} \frac{du}{u^s} \right) = \sum_{n=1}^{\infty} \delta_n(s).$$

On $[n, n+1]$, $|n^{-s} - u^{-s}| = |s| \left| \int_n^u v^{-s-1} dv \right| \leq |s| n^{-\sigma-1}$, so $|\delta_n(s)| \leq |s| n^{-\sigma-1}$ and $\sum_n \delta_n$ converges absolutely and locally uniformly for $\sigma > 0$. Thus $\sum_n \delta_n$ is holomorphic on $\operatorname{Re}(s) > 0$, and $\zeta(s) = \frac{1}{s-1} + (\text{holomorphic near } s=1)$: a simple pole with residue $\lim_{s \rightarrow 1} (s-1)\zeta(s) = 1$. *Check.* The constant term of the Laurent expansion is $\sum_n \delta_n(1) = \lim_N (\sum_{n \leq N} \frac{1}{n} - \log N) = \gamma$, the Euler–Mascheroni constant, so $\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1)$ — the standard Laurent expansion. ■

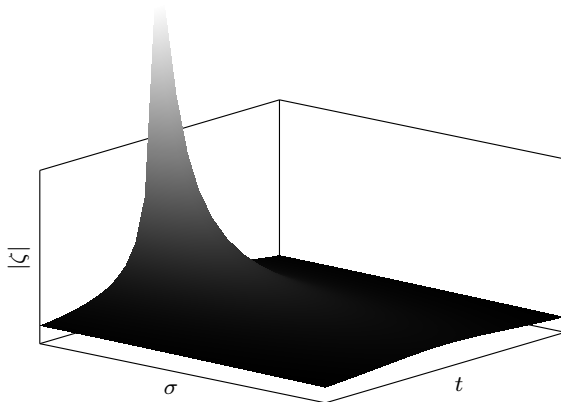


Figure 20.5. The modulus surface $|\zeta(\sigma + it)|$ near the pole at $s = 1$: a single spike on the real axis (the only singularity), with $|\zeta|$ finite and bounded below as soon as $\sigma \geq 1$, $|t| \geq 1$ (Proposition 20.1.10). (Schematic.)

Worked Example 20.1.17 ($(s-1)\zeta(s)$ has growth order 1). Show the entire function $(s-1)\zeta(s)$ satisfies $|(s-1)\zeta(s)| \leq A e^{a|s| \log |s|}$, of growth order 1, and indicate why this forces infinitely many zeros in the critical strip.

Recognition. Growth order is read from the functional equation: the Γ -factor dominates and $|\Gamma(s)|$ grows like $e^{|s| \log |s|}$ by Stirling.

Solution. Removing the pole, $(s-1)\zeta(s)$ is entire. The completed $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ satisfies $\xi(s) = \xi(1-s)$, and for $\operatorname{Re}(s) \geq \frac{1}{2}$ Stirling gives $|\Gamma(s/2)| \leq e^{C|s| \log |s|}$ while $\pi^{-s/2}$ and the convergent Dirichlet series make ζ at most polynomial; symmetrizing via $\xi(s) = \xi(1-s)$ covers $\operatorname{Re}(s) < \frac{1}{2}$. Hence $|\xi(s)| \leq e^{C|s| \log |s|}$, and dividing by the entire $\pi^{-s/2} \Gamma(s/2)$ (whose reciprocal is order 1) leaves $|(s-1)\zeta(s)| \leq A e^{a|s| \log |s|}$, growth order 1. Setting $F(s) = \xi(\frac{1}{2} + s)$ (even, so $F(s) = G(s^2)$), G has order $\frac{1}{2}$; a Hadamard theorem says an entire function of fractional order has infinitely many zeros, so G , hence ξ , hence ζ has infinitely many zeros — all in the critical strip since $\zeta \neq 0$ for $\sigma \geq 1$ or $\sigma \leq 0$ (nontrivially). *Check.* Numerically the first zeros sit at $\frac{1}{2} \pm i(14.13)$, $\frac{1}{2} \pm i(21.02)$, $\dots$, an infinite sequence on the critical line, consistent with “infinitely many.” ■

Worked Example 20.1.18 (Reading $\log |\zeta|$ off the argument: a counting check). Using $\log \zeta(\sigma) = \sum_{p,m} p^{-m\sigma}/m$ (Lemma 20.1.5), estimate $\log \zeta(\sigma)$ as $\sigma \rightarrow 1^+$ and confirm it matches the pole $\zeta \sim 1/(\sigma-1)$.

Recognition. The dominant contribution to $\log \zeta$ near $\sigma = 1$ is the $m = 1$ term $\sum_p p^{-\sigma}$, which behaves like $\log \frac{1}{\sigma-1}$ by comparison with ζ .

Solution. Split $\log \zeta(\sigma) = \sum_p p^{-\sigma} + \sum_{p,m \geq 2} p^{-m\sigma}/m$. The second sum is bounded as $\sigma \rightarrow 1^+$ (dominated by $\sum_p \sum_{m \geq 2} p^{-m} = \sum_p \frac{1}{p(p-1)} < \infty$). For the first, Mertens’ theorem gives $\sum_{p \leq x} p^{-1} = \log \log x + M + o(1)$, and the analytic statement is $\sum_p p^{-\sigma} = \log \frac{1}{\sigma-1} + O(1)$ as $\sigma \rightarrow 1^+$. Hence $\log \zeta(\sigma) = \log \frac{1}{\sigma-1} + O(1)$, i.e. $\zeta(\sigma) \asymp \frac{1}{\sigma-1}$. *Check.* Exponentiating, $\zeta(\sigma) \rightarrow \infty$ like $1/(\sigma-1)$, matching the simple pole with residue 1 found in the previous worked example. The bounded “prime-power” tail is exactly the analytic shadow of the fact that higher prime powers are sparse. ■

Exercises (Stein, Ch.7).

- (Multiplication of Dirichlet series.) Let $\{a_m\}, \{b_k\}$ be bounded sequences of complex numbers. (a) Show that for $\operatorname{Re}(s) > 1$, $\left(\sum_{m \geq 1} a_m m^{-s}\right) \left(\sum_{k \geq 1} b_k k^{-s}\right) = \sum_{n \geq 1} c_n n^{-s}$ where $c_n = \sum_{mk=n} a_m b_k$, all series converging absolutely. (b) Deduce that $\zeta(s)^2 = \sum_{n \geq 1} d(n) n^{-s}$ for $\operatorname{Re}(s) > 1$, where $d(n)$ is the number of divisors of n , and $\zeta(s)\zeta(s-a) = \sum_{n \geq 1} \sigma_a(n) n^{-s}$ for $\operatorname{Re}(s) > 1$, $\operatorname{Re}(s-a) > 1$, where $\sigma_a(n)$ is the sum of the a th powers of the divisors of n (so $\sigma_0(n) = d(n)$).

- (Dirichlet series for $1/\zeta$.) (a) Prove that for $\operatorname{Re}(s) > 1$, $1/\zeta(s) = \sum_{n \geq 1} \mu(n)n^{-s}$, where the Möbius function is $\mu(1) = 1$, $\mu(n) = (-1)^k$ if $n = p_1 \cdots p_k$ is a product of distinct primes, and $\mu(n) = 0$ otherwise; note $\mu(nm) = \mu(n)\mu(m)$ when $\gcd(n, m) = 1$. [Hint: use the Euler product.] (b) Show that $\sum_{k|n} \mu(k) = 1$ if $n = 1$ and 0 otherwise.
- (Non-vanishing on the real segment.) Consider $\tilde{\zeta}(s) = 1 - 2^{-s} + 3^{-s} - \cdots = \sum_{n \geq 1} (-1)^{n+1} n^{-s}$. (a) Prove the series converges for $\operatorname{Re}(s) > 0$ and defines a holomorphic function there. (b) Show that for $s > 1$, $\tilde{\zeta}(s) = (1 - 2^{1-s})\zeta(s)$. (c) Conclude, since $\tilde{\zeta}$ is an alternating series, that ζ has no zeros on the segment $0 < \sigma < 1$; extend to $\sigma = 0$ via the functional equation.
- (Reality of ξ .) Show that $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ is real when s is real, and also when $\operatorname{Re}(s) = 1/2$.
- (Infinitely many zeros in the critical strip.) Let $F(s) = \xi(1/2 + s)$ with $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$. (a) Show F is an even function of s , so there is G with $G(s^2) = F(s)$. (b) Show $(s-1)\zeta(s)$ is entire of growth order 1, i.e. $|(s-1)\zeta(s)| \leq A e^{a|s|^{1+\epsilon}}$; deduce G has growth order $1/2$. (c) Deduce that ζ has infinitely many zeros in the critical strip. [Hint: for (a),(b) use the functional equation; for (c) use Hadamard's theorem that an entire function of fractional order has infinitely many zeros (Exercise 14, Ch.5).]
- (Sharper bounds near the line.) Refine Proposition 2.7 of Ch.6 and Proposition 1.6 to show, for $|t| \geq 2$: (a) $|\zeta(1+it)| \leq A \log |t|$; (b) $|\zeta'(1+it)| \leq A(\log |t|)^2$; (c) $1/|\zeta(1+it)| \leq A(\log |t|)^a$ (with $a = 7$).

Problems (Stein, Ch.7).

- (Mean value / Parseval.) Let $F(s) = \sum_{n \geq 1} a_n n^{-s}$ with $|a_n| \leq M$ for all n . (a) Show that for $\sigma > 1$,
$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |F(\sigma + it)|^2 dt = \sum_{n \geq 1} \frac{|a_n|^2}{n^{2\sigma}}.$$
 How is this reminiscent of the Parseval–Plancherel theorem? [Hint: use $\frac{1}{2T} \int_{-T}^T (nm)^{-\sigma} n^{-it} m^{it} dt \rightarrow n^{-2\sigma}$ if $n = m$ and $\rightarrow 0$ if $n \neq m$.] (b) Deduce the uniqueness of Dirichlet series: if $F(s) = \sum_n a_n n^{-s} \equiv 0$ with $|a_n| \leq cn^k$ for some k , then $a_n = 0$ for all n .

Additional exercises.

- Using the Euler product, prove directly that $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$ by showing $|\zeta(s)| \geq \prod_p (1 + p^{-\sigma})^{-1} > 0$ where $\sigma = \operatorname{Re}(s)$. Conclude $|\zeta(s)| \geq \zeta(2\sigma)/\zeta(\sigma)$.
- Compute the partial Euler product over $p = 2, 3, 5$ for $\zeta(4) = \pi^4/90$ and verify it underestimates $\pi^4/90 \approx 1.082323$. Report the deficit.
- Show that $\log \zeta(s) = \sum_p p^{-s} + R(s)$ where $R(s) = \sum_p \sum_{m \geq 2} p^{-ms}/m$ is holomorphic and bounded on $\operatorname{Re}(s) \geq \frac{1}{2} + \delta$ for every $\delta > 0$. (Thus $\sum_p p^{-s}$ has the same singular behavior as $\log \zeta$ as $s \rightarrow 1^+$.)
- Verify Lemma 20.1.6 a second way: expand $|1 + e^{i\theta}|^2$ and relate it to $2(1 + \cos \theta)$, then square. Where does the resulting expression vanish, and what does that say about the equality case $|\zeta^3 \zeta^4| = 1$?
- Suppose, hypothetically, one tried the weights $\zeta^2(\sigma) \zeta^2(\sigma + it) \zeta(\sigma + 2it)$. Show the associated cosine polynomial $2 + 2 \cos \theta + \cos 2\theta$ is *not* everywhere ≥ 0 (find a θ making it negative), so this weighting does *not* prove non-vanishing. (A NON-example: not every 3-term weighting works.)
- (Diagram.) Sketch the complex s -plane and mark: the pole of ζ at $s = 1$, the trivial zeros at $-2, -4, -6$, the critical strip $0 \leq \operatorname{Re} s \leq 1$, the critical line $\operatorname{Re} s = \frac{1}{2}$, and the line $\operatorname{Re} s = 1$. Indicate which regions Theorem 20.1.4 and Theorem 20.1.8 rule out for nontrivial zeros.
- From Corollary 20.1.7, deduce that $\liminf_{\sigma \rightarrow 1^+} (\sigma - 1) |\zeta(\sigma + it)|^{4/3} > 0$ for each fixed $t \neq 0$, and explain how the exponent $4/3$ arises from the 3, 4, 1 trick.
- Show that the functional equation forces ζ to have no zeros on $\operatorname{Re}(s) = 0$ given that it has none on $\operatorname{Re}(s) = 1$. (Use $\zeta(s) = \pi^{s-1/2} \Gamma((1-s)/2) \Gamma(s/2)^{-1} \zeta(1-s)$ and track which factors can vanish.)
- Prove that Proposition 20.1.10 is sharp in form: there is no bound $1/|\zeta(s)| \leq c$ uniform in t on $\sigma = 1$, by exhibiting (qualitatively) why $1/|\zeta(1+it)|$ cannot stay bounded as $|t| \rightarrow \infty$. [Hint: relate to Exercise 6(c) of the scraped set.]

20.2 Reduction to the Functions psi and psi_1

Tchebychev's ψ -function

Working with $\pi(x)$ directly is awkward because each prime is weighted equally and the jumps are irregular. Tchebychev introduced a smoother, log-weighted count that is *equivalent* to the prime number theorem but algebraically friendlier.

Definition 20.2.1. The *Tchebychev function* is

$$\psi(x) = \sum_{p^m \leq x} \log p,$$

summed over prime powers $p^m \leq x$ (p prime, $m \geq 1$). Equivalently, with the *von Mangoldt function*

$$\Lambda(n) = \begin{cases} \log p & n = p^m \text{ for a prime } p, m \geq 1, \\ 0 & \text{otherwise,} \end{cases} \quad \psi(x) = \sum_{1 \leq n \leq x} \Lambda(n).$$

A third form sums over primes only:

$$\psi(x) = \sum_{p \leq x} \left\lfloor \frac{\log x}{\log p} \right\rfloor \log p,$$

since $p^m \leq x \iff m \leq \log x / \log p$.

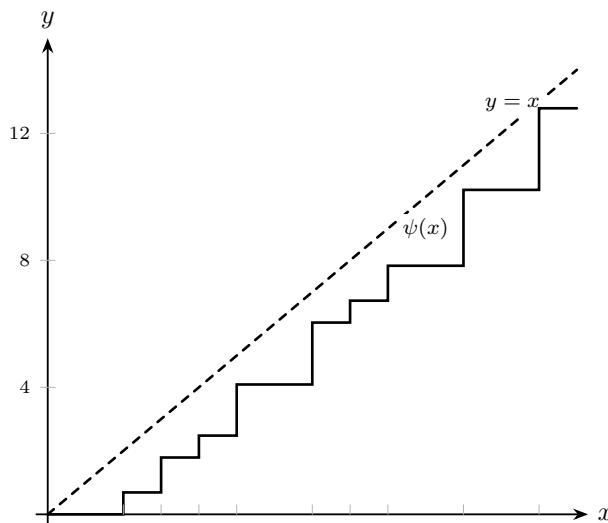


Figure 20.6. The Tchebychev function $\psi(x)$ jumps by $\log p$ at each prime power $p^m \leq x$; the prime number theorem is equivalent to $\psi(x) \sim x$ (dashed diagonal $y = x$). (Schematic.)

Worked Example 20.2.2 (Hand computation of $\psi(10)$). Compute $\psi(10)$ from Definition 20.2.1.

Recognition. Sum $\log p$ over all prime powers ≤ 10 : list them.

Solution. Prime powers ≤ 10 : $2, 3, 4 = 2^2, 5, 7, 8 = 2^3, 9 = 3^2$. So

$$\psi(10) = \log 2 + \log 3 + \log 2 + \log 5 + \log 7 + \log 2 + \log 3 = 4 \log 2 + 2 \log 3 + \log 5 + \log 7.$$

Numerically $4(0.6931) + 2(1.0986) + 1.6094 + 1.9459 = 2.7726 + 2.1972 + 1.6094 + 1.9459 = 8.5251$. *Check.* Via the floor form, $\sum_{p \leq 10} \left\lfloor \log 10 / \log p \right\rfloor \log p$: $p = 2$ gives $\lfloor 3.32 \rfloor = 3$, $3 \log 2$; $p = 3$ gives $\lfloor 2.10 \rfloor = 2$, $2 \log 3$; $p = 5, 7$ give 1 each. Total $3 \log 2 + 2 \log 3 + \log 5 + \log 7 = 8.5251$. Agrees, and $\psi(10) = 8.525$ is close to $x = 10$. ■

Proposition 20.2.3. If $\psi(x) \sim x$ as $x \rightarrow \infty$, then $\pi(x) \sim x / \log x$.

Proof. It suffices to prove $1 \leq \liminf_{x \rightarrow \infty} \pi(x) \frac{\log x}{x}$ and $\limsup_{x \rightarrow \infty} \pi(x) \frac{\log x}{x} \leq 1$. For the upper bound, the floor form gives the crude estimate

$$\psi(x) = \sum_{p \leq x} \left\lfloor \frac{\log x}{\log p} \right\rfloor \log p \leq \sum_{p \leq x} \frac{\log x}{\log p} \log p = \pi(x) \log x,$$

so $\psi(x)/x \leq \pi(x) \log x/x$; since $\psi(x) \sim x$, the left side $\rightarrow 1$, giving $\liminf \geq 1$. For the upper bound fix $0 < \alpha < 1$. Counting only primes in $(x^\alpha, x]$, each contributes $\log p > \log x^\alpha = \alpha \log x$, so

$$\psi(x) \geq \sum_{x^\alpha < p \leq x} \log p \geq (\pi(x) - \pi(x^\alpha)) \alpha \log x,$$

hence $\psi(x) + \alpha \pi(x^\alpha) \log x \geq \alpha \pi(x) \log x$. Divide by x , use $\pi(x^\alpha) \leq x^\alpha$ and $\alpha < 1$ and $\psi(x) \sim x$ to get $1 \geq \alpha \limsup \pi(x) \log x/x$. Letting $\alpha \rightarrow 1^-$ gives $\limsup \leq 1$. $\square$

The integrated function ψ_1

It is even more convenient to integrate ψ once, which smooths the staircase into a piecewise-quadratic curve.

Definition 20.2.4.

$$\psi_1(x) = \int_1^x \psi(u) du.$$

Proposition 20.2.5. *If $\psi_1(x) \sim x^2/2$ as $x \rightarrow \infty$, then $\psi(x) \sim x$, and therefore $\pi(x) \sim x/\log x$.*

Proof. By Proposition 20.2.3 it suffices to get $\psi(x) \sim x$. Since ψ is nondecreasing, for $\alpha < 1 < \beta$,

$$\frac{1}{(1-\alpha)x} \int_{\alpha x}^x \psi(u) du \leq \psi(x) \leq \frac{1}{(\beta-1)x} \int_x^{\beta x} \psi(u) du,$$

(the left integrand $\leq \psi(x) \leq$ the right integrand pointwise). The right inequality gives $\psi(x) \leq \frac{\psi_1(\beta x) - \psi_1(x)}{(\beta-1)x}$, so

$$\frac{\psi(x)}{x} \leq \frac{1}{\beta-1} \left(\beta^2 \frac{\psi_1(\beta x)}{(\beta x)^2} - \frac{\psi_1(x)}{x^2} \right) \xrightarrow{x \rightarrow \infty} \frac{1}{\beta-1} \left(\frac{\beta^2}{2} - \frac{1}{2} \right) = \frac{\beta+1}{2}.$$

So $\limsup \psi(x)/x \leq (\beta+1)/2$ for every $\beta > 1$; let $\beta \rightarrow 1^+$ to get $\limsup \leq 1$. The left inequality with $\alpha < 1$ gives $\liminf \geq 1$ symmetrically. Hence $\psi(x) \sim x$. $\square$

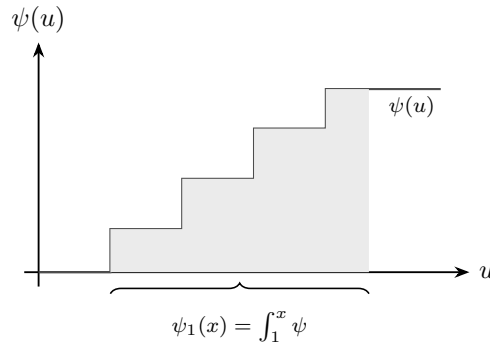


Figure 20.7. $\psi_1(x) = \int_1^x \psi(u) du$ is the shaded area under the staircase ψ from 1 to x ; integrating once smooths the jumps into a piecewise-quadratic curve. (Schematic.)

Linking ψ_1 to ζ

Differentiating the logarithm formula (Lemma 1.3) term by term gives, for $\operatorname{Re}(s) > 1$, the fundamental identity

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{p,m} (\log p) p^{-ms} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}. \quad (5)$$

Thus $-\zeta'/\zeta$ is the Dirichlet series of the von Mangoldt coefficients — exactly the weights summed in ψ . The bridge from this series to ψ_1 is a contour-integral formula.

$$\begin{array}{c} \xrightarrow{\quad -\zeta'/\zeta \quad} \\ \zeta(s) \xrightarrow{\quad \log \quad} \log \zeta(s) \xrightarrow{\quad -d/ds \quad} \sum_{n \geq 1} \frac{\Lambda(n)}{n^s} \end{array}$$

Figure 20.8. The logarithmic derivative as a composition (valid for $\operatorname{Re} s > 1$): taking $\log$ of the Euler product and then $-d/ds$ converts ζ into the Dirichlet series $-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n)n^{-s}$ of the von Mangoldt weights.

Proposition 20.2.6. *For all $c > 1$,*

$$\psi_1(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)} \left(-\frac{\zeta'(s)}{\zeta(s)} \right) ds,$$

the integral over the vertical line $\operatorname{Re}(s) = c$.

$$\begin{array}{ccc} -\frac{\zeta'(s)}{\zeta(s)} & \xrightarrow{\quad \mathcal{M} \quad} & \psi_1(x) \\ \parallel & & \parallel \\ \sum_{n \geq 1} \frac{\Lambda(n)}{n^s} & \xrightarrow{\quad \mathcal{M} \quad} & \sum_{n \leq x} \Lambda(n)(x-n) \end{array}$$

Figure 20.9. The Mellin–Perron transform $\mathcal{M}: G \mapsto \frac{1}{2\pi i} \int_{(c)} \frac{x^{s+1}}{s(s+1)} G(s) ds$ ($c > 1$) carries the Dirichlet series $-\zeta'/\zeta = \sum_n \Lambda(n)n^{-s}$ to $\psi_1(x) = \sum_{n \leq x} \Lambda(n)(x-n)$; applying $\mathcal{M}$ termwise (bottom) agrees with applying it to the closed form (top), since the series converges absolutely on $\operatorname{Re} s = c$.

The proof depends on a Mellin–Perron type lemma evaluating a single vertical integral.

Lemma 20.2.7. *If $c > 0$, then*

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{a^s}{s(s+1)} ds = \begin{cases} 0 & 0 < a \leq 1, \\ 1 - \frac{1}{a} & 1 \leq a. \end{cases}$$

Proof. Since $|a^s| = a^c$ and $|s(s+1)| \sim |t|^2$, the integral converges absolutely. Put $f(s) = \frac{a^s}{s(s+1)} = \frac{e^{s\beta}}{s(s+1)}$ with $\beta = \log a$. The poles are simple, at $s = 0$ and $s = -1$, with $\operatorname{Res}_{s=0} f = \frac{a^0}{0+1} = 1$ and $\operatorname{Res}_{s=-1} f =$

$$\frac{a^{-1}}{(-1)(0)+\dots} = -\frac{1}{a} \text{ (so } \operatorname{Res}_{s=0} f = 1, \operatorname{Res}_{s=-1} f = -1/a \text{).}$$

Case $a \geq 1$ ($\beta \geq 0$). Close the line to the left with a half-circle $C(T)$ of radius T centered at c ; for T large both poles lie inside the resulting toy contour $\Gamma(T)$ (positively oriented). By the residue theorem $\frac{1}{2\pi i} \int_{\Gamma(T)} f = \operatorname{Res}_0 + \operatorname{Res}_{-1} = 1 - 1/a$. On $C(T)$, $\sigma \leq c$ so $|e^{\beta s}| = e^{\beta \sigma} \leq e^{\beta c} = a^c$, and $|s(s+1)| \geq \frac{1}{2}T^2$ for large T , hence

$$\left| \int_{C(T)} f \right| \leq \frac{a^c}{\frac{1}{2}T^2} \cdot \pi T = \frac{C}{T} \rightarrow 0 \quad (T \rightarrow \infty).$$

So the line integral equals $1 - 1/a$.

Case $0 < a \leq 1$ ($\beta \leq 0$). Close to the *right*; the half-circle now lies in $\sigma \geq c > 0$, so it encloses *no* poles, and the same estimate (now using $e^{\beta\sigma} \leq e^{\beta c} \leq 1$ since $\beta \leq 0$ and $\sigma \geq c$) sends the arc to 0. Hence the line integral is 0. $\square$

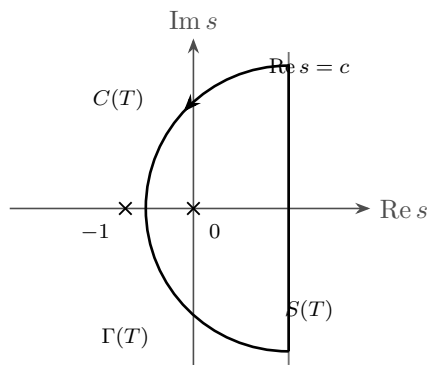


Figure 20.10. Case $a \geq 1$: close the vertical line $\text{Re } s = c$ to the left; the toy contour $\Gamma(T) = S(T) \cup C(T)$ encloses the poles 0 and -1 . (Schematic.)

Proof of Proposition 20.2.6. Write $\psi(u) = \sum_{n=1}^{\infty} \Lambda(n) f_n(u)$ where $f_n(u) = 1$ if $n \leq u$ and 0 otherwise. Integrating,

$$\psi_1(x) = \int_0^x \psi(u) du = \sum_{n=1}^{\infty} \Lambda(n) \int_n^x du = \sum_{n \leq x} \Lambda(n)(x - n).$$

On the other hand, inserting (5) into the contour integral and integrating term by term, then applying Lemma 20.2.7 with $a = x/n$,

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)} \sum_n \frac{\Lambda(n)}{n^s} ds = x \sum_{n=1}^{\infty} \Lambda(n) \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{(x/n)^s}{s(s+1)} ds = x \sum_{n \leq x} \Lambda(n) \left(1 - \frac{n}{x}\right),$$

since the bracket vanishes for $n > x$ (where $x/n < 1$). The last expression is $\sum_{n \leq x} \Lambda(n)(x - n) = \psi_1(x)$. $\square$

Worked Example 20.2.8 ($\psi_1(x) = \sum_{n \leq x} \Lambda(n)(x - n)$ at $x = 4$). Verify the closed form $\psi_1(x) = \sum_{n \leq x} \Lambda(n)(x - n)$ by also integrating the staircase, at $x = 4$.

Recognition. Compute both sides: the Λ -weighted sum and $\int_1^4 \psi(u) du$.

Solution. The relevant $n \leq 4$ with $\Lambda(n) \neq 0$ are $n = 2$ ($\Lambda = \log 2$), $n = 3$ ($\log 3$), $n = 4 = 2^2$ ($\log 2$). So

$$\sum_{n \leq 4} \Lambda(n)(4 - n) = \log 2 \cdot 2 + \log 3 \cdot 1 + \log 2 \cdot 0 = 2 \log 2 + \log 3.$$

Directly: $\psi(u) = 0$ on $[1, 2)$, $= \log 2$ on $[2, 3)$, $= \log 2 + \log 3 = \log 6$ on $[3, 4)$, $= \log 6 + \log 2$ at $u = 4$. Then $\int_1^4 \psi = \log 2 \cdot 1 + \log 6 \cdot 1 = \log 2 + \log 6 = 2 \log 2 + \log 3$. *Check.* Both equal $2 \log 2 + \log 3 = 2(0.6931) + 1.0986 = 2.4849$, and $\frac{1}{2} \cdot 4^2 = 8$, so at this small x the asymptotic $\psi_1 \sim x^2/2$ has not yet kicked in — as expected. $\blacksquare$

Caution. In Proposition 20.2.6 the interchange of the infinite sum $\sum_n \Lambda(n)n^{-s}$ with the contour integral is legitimate because the Dirichlet series converges absolutely on $\text{Re}(s) = c > 1$ (the factor $x^{s+1}/(s(s+1))$ is $O(|t|^{-2})$ there). Do *not* attempt the same interchange after moving the contour to $\text{Re}(s) = 1$: there $\sum_n \Lambda(n)n^{-s}$ no longer converges absolutely, which is precisely why Section 3 must deform the contour rather than integrate the series term by term.

Further worked examples

Worked Example 20.2.9 (Deriving $-\zeta'/\zeta = \sum \Lambda(n)n^{-s}$ from the Euler product). Differentiate $\log \zeta(s) = \sum_{p,m} p^{-ms}/m$ to obtain identity (5), and read off the first few coefficients.

Recognition. Term-by-term differentiation of an absolutely (and locally uniformly) convergent Dirichlet series for $\operatorname{Re}(s) > 1$; match coefficients against the von Mangoldt Λ .

Solution. For $\operatorname{Re}(s) > 1$, $\frac{d}{ds} p^{-ms} = -m(\log p)p^{-ms}$, so

$$\frac{\zeta'(s)}{\zeta(s)} = \frac{d}{ds} \log \zeta(s) = \sum_{p,m} \frac{-m(\log p)p^{-ms}}{m} = - \sum_{p,m} (\log p) p^{-ms},$$

hence $-\zeta'/\zeta = \sum_{p,m} (\log p)p^{-ms} = \sum_n \Lambda(n)n^{-s}$, since the coefficient of n^{-s} is $\log p$ when $n = p^m$ and 0 otherwise — exactly $\Lambda(n)$. *Check.* The first terms are $\Lambda(1) = 0$, $\Lambda(2) = \log 2$, $\Lambda(3) = \log 3$, $\Lambda(4) = \log 2$, $\Lambda(5) = \log 5$, $\Lambda(6) = 0$ (since $6 = 2 \cdot 3$ is not a prime power). So $-\zeta'/\zeta = (\log 2)2^{-s} + (\log 3)3^{-s} + (\log 2)4^{-s} + (\log 5)5^{-s} + \dots$, matching the ψ -weights of Definition 20.2.1. ■

Worked Example 20.2.10 (The Perron kernel by contour closing, with the arc estimate). Re-derive Lemma 20.2.7 for $a > 1$ by explicitly closing the vertical line to the left and bounding the semicircular arc.

Recognition. A vertical-line integral of $a^s/(s(s+1))$ with simple poles at $0, -1$; close left (since $a > 1$ makes $|a^s|$ small for $\operatorname{Re} s \ll 0$) and use the residue theorem, then send the arc radius to infinity.

Solution. Let $\beta = \log a > 0$ and $f(s) = e^{\beta s}/(s(s+1))$. Close $\operatorname{Re} s = c$ with a left semicircle C_R of radius R about $s = c$. For R large both poles are inside, and

$$\frac{1}{2\pi i} \oint f = \operatorname{Res}_{s=0} f + \operatorname{Res}_{s=-1} f = \frac{e^0}{0+1} + \frac{e^{-\beta}}{(-1)(0)} \Big|_{\lim} = 1 - \frac{1}{a},$$

using $\operatorname{Res}_{s=0} = \frac{e^{\beta \cdot 0}}{0+1} = 1$ and $\operatorname{Res}_{s=-1} = \frac{e^{-\beta}}{(-1)} = -a^{-1}$. On C_R (where $\operatorname{Re} s \leq c$), $|e^{\beta s}| = e^{\beta \operatorname{Re} s} \leq e^{\beta c} = a^c$ and $|s(s+1)| \geq \frac{1}{2}R^2$, so $|\int_{C_R} f| \leq \frac{a^c}{R^2/2} \cdot \pi R = \frac{2\pi a^c}{R} \rightarrow 0$. Hence the line integral is $1 - 1/a$. *Check.* As $a \rightarrow 1^+$ this $\rightarrow 0$, matching the $0 < a \leq 1$ branch across the boundary $a = 1$, so the kernel is continuous in a (with value 0 at $a = 1$ from either side). ■

Worked Example 20.2.11 (Closing to the right gives zero: the $0 < a < 1$ case). For $0 < a < 1$ show the Perron integral vanishes by closing to the right, and explain why no residue is picked up.

Recognition. When $0 < a < 1$, $\beta = \log a < 0$, so $|a^s| = e^{\beta \operatorname{Re} s}$ is small for $\operatorname{Re} s \gg 0$: close the contour to the right, where f is holomorphic.

Solution. Close $\operatorname{Re} s = c$ with a right semicircle C'_R of radius R about $s = c$. Since $c > 0 > -1$, the half-plane $\operatorname{Re} s \geq c$ contains *neither* pole 0 nor -1 , so the enclosed residue sum is 0 and $\frac{1}{2\pi i} \oint f = 0$. On C'_R , $\operatorname{Re} s \geq c$ gives $|e^{\beta s}| = e^{\beta \operatorname{Re} s} \leq e^{\beta c} = a^c < 1$, and again $|s(s+1)| \geq \frac{1}{2}R^2$, so the arc integral is $\leq 2\pi a^c/R \rightarrow 0$. Therefore the line integral equals 0. *Check.* This is exactly the $0 < a \leq 1$ case of Lemma 20.2.7; in Proposition 20.2.6 it is what kills every term with $n > x$ (where $a = x/n < 1$), truncating the sum to $n \leq x$. ■

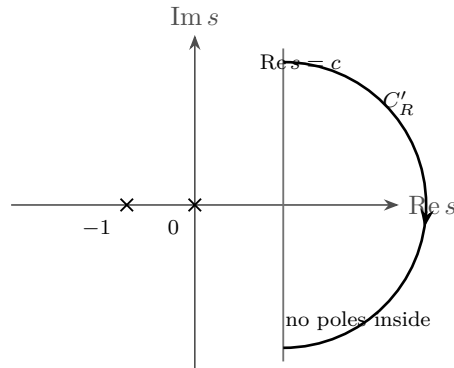


Figure 20.11. Case $0 < a < 1$: close $\operatorname{Re} s = c$ to the right. The poles $0, -1$ lie outside, so the integral is 0 — this is what truncates ψ_1 to $n \leq x$. (Schematic.)

Worked Example 20.2.12 (The crude Tchebychev bound $\psi(x) = O(x)$). Show directly from the prime factorization of $\binom{2n}{n}$ that $\psi(2n) - \psi(n) \leq 2n \log 2$, hence $\psi(x) \leq Cx$ for a constant C .

Recognition. The central binomial coefficient is divisible by every prime in $(n, 2n]$; bounding it by 4^n converts a divisibility statement into a sum-of-logs estimate.

Solution. Every prime p with $n < p \leq 2n$ divides $\binom{2n}{n} = \frac{(2n)!}{n!n!}$ exactly once (it appears in the numerator, not the denominator), so $\prod_{n < p \leq 2n} p \mid \binom{2n}{n} \leq 2^{2n} = 4^n$. Taking logs, $\sum_{n < p \leq 2n} \log p \leq 2n \log 2$, i.e. $\theta(2n) - \theta(n) \leq 2n \log 2$ where $\theta(x) = \sum_{p \leq x} \log p$. Summing the dyadic blocks $x, x/2, x/4, \dots$ telescopes to $\theta(x) \leq (2 \log 2)x$. Finally $\psi(x) - \theta(x) = \sum_{m \geq 2} \theta(x^{1/m}) = O(\sqrt{x} \log x)$, so $\psi(x) \leq Cx$. *Check.* The cheap bound $\psi(x) = O(x)$ (and the matching $\psi(x) \geq cx$) shows $\psi(x) \asymp x$ long before PNT; PNT upgrades $\asymp$ to the precise $\psi(x) \sim x$ with constant 1. ■

Worked Example 20.2.13 (Non-example: $\pi(x) \sim x/\log x$ does *not* give $\pi(x) - x/\log x \rightarrow 0$). Caution against over-reading the asymptotic: exhibit that the difference $\pi(x) - x/\log x$ is in fact unbounded.

Recognition. “ $\sim$ ” controls the ratio, not the difference; estimate the gap using the better approximation $\text{Li}(x)$.

Solution. The logarithmic integral $\text{Li}(x) = \int_2^x dt/\log t$ satisfies, by integration by parts, $\text{Li}(x) = \frac{x}{\log x} + \frac{x}{(\log x)^2} + O\left(\frac{x}{(\log x)^3}\right)$, and PNT in its sharp form gives $\pi(x) = \text{Li}(x) + (\text{small})$. Hence

$$\pi(x) - \frac{x}{\log x} = \frac{x}{(\log x)^2} + O\left(\frac{x}{(\log x)^3}\right) \rightarrow \infty.$$

So although $\pi(x)/(x/\log x) \rightarrow 1$, the *difference* grows without bound. *Check.* At $x = 10^6$, $\pi(x) = 78498$ while $x/\log x = 72382$, a gap of 6116; at $x = 10^9$ the gap exceeds 2.5 million. The ratio still tends to 1, confirming $\sim$ is a statement about ratios only. ■

Worked Example 20.2.14 (A general Dirichlet L-function as a Mellin transform). For a periodic coefficient sequence a_n of period q , write $L(s) = \sum_{n \geq 1} a_n n^{-s}$ as a single Mellin integral $L(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{Q(x) x^{s-1}}{e^{qx} - 1} dx$ for $\text{Re}(s) > 1$, where $Q(x) = \sum_{m=0}^{q-1} a_{q-m} e^{mx}$, and deduce continuation past $\sigma = 1$.

Recognition. The same device that produces $\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$: replace n^{-s} by its Gamma integral $\Gamma(s)n^{-s} = \int_0^\infty x^{s-1} e^{-nx} dx$ and sum the geometric-type series in x .

Solution. Using $\Gamma(s)n^{-s} = \int_0^\infty x^{s-1} e^{-nx} dx$,

$$\Gamma(s)L(s) = \int_0^\infty x^{s-1} \sum_{n \geq 1} a_n e^{-nx} dx.$$

With period q , group $n = qk - m$ ($k \geq 1, 0 \leq m \leq q-1$): $\sum_n a_n e^{-nx} = \sum_{m=0}^{q-1} a_{q-m} \sum_{k \geq 1} e^{-(qk-m)x} = \frac{\sum_m a_{q-m} e^{mx}}{e^{qx} - 1} = \frac{Q(x)}{e^{qx} - 1}$ (summing the geometric series $\sum_k e^{-qkx} = 1/(e^{qx} - 1)$). Hence $L(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{Q(x) x^{s-1}}{e^{qx} - 1} dx$. Near $x = 0$, $\frac{Q(x)}{e^{qx} - 1} \sim \frac{Q(0)}{qx} = \frac{\sum_m a_{q-m}}{qx}$, so the only obstruction to continuation is a simple pole at $s = 1$ with residue $\frac{1}{q} \sum_m a_{q-m} = \frac{1}{q} \sum_{n \bmod q} a_n$. *Check.* When $a_n \equiv 1$ ($q = 1$) this is the Riemann case $Q \equiv 1$, recovering $\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$ with residue 1 at $s = 1$; for a nontrivial Dirichlet character $\sum_{n \bmod q} \chi(n) = 0$, so $L(s, \chi)$ is *regular* at $s = 1$ — the analytic input to Dirichlet’s theorem on primes in progressions. ■

Exercises (Stein, Ch.7).

- (Dirichlet L -series and continuation.) Suppose $\{a_n\}_{n \geq 1}$ satisfies $a_n = a_m$ whenever $n \equiv m \pmod{q}$ for a positive integer q . Define $L(s) = \sum_{n \geq 1} a_n n^{-s}$ for $\text{Re}(s) > 1$, and with $a_0 = a_q$ set $Q(x) = \sum_{m=0}^{q-1} a_{q-m} e^{mx}$. Show, as in Exercises 15–16 of Ch.6, that for $\text{Re}(s) > 1$,

$$L(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{Q(x) x^{s-1}}{e^{qx} - 1} dx.$$

Prove that L continues to $\mathbb{C}$ with at most a pole at $s = 1$, and is regular at $s = 1$ iff $\sum_{m=0}^{q-1} a_m = 0$. (Connect with the Dirichlet $L(s, \chi)$: $L(s, \chi)$ is regular at $s = 1$ iff χ is a non-trivial character.)

2. (A vertical-line integral.) Show that for every $c > 0$,

$$\lim_{N \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iN}^{c+iN} a^s \frac{ds}{s} = \begin{cases} 1 & a > 1, \\ 1/2 & a = 1, \\ 0 & 0 \leq a < 1, \end{cases}$$

the integral over the vertical segment from $c - iN$ to $c + iN$.

3. (Four equivalent asymptotics.) Let $\varphi(x) = \sum_{p \leq x} \log p$ (sum over primes $\leq x$). Prove that as $x \rightarrow \infty$ the following are equivalent: (i) $\varphi(x) \sim x$; (ii) $\pi(x) \sim x/\log x$; (iii) $\psi(x) \sim x$; (iv) $\psi_1(x) \sim x^2/2$.
4. (n th prime asymptotics.) If p_n is the n th prime, the prime number theorem implies $p_n \sim n \log n$ as $n \rightarrow \infty$. (a) Show $\pi(x) \sim x/\log x$ implies $\log \pi(x) + \log \log x \sim \log x$. (b) Deduce $\log \pi(x) \sim \log x$, and set $x = p_n$ to conclude $p_n \sim n \log n$.

Problems (Stein, Ch.7).

1. (Primes in arithmetic progressions.) Combining the ideas of the prime number theorem with the proof of Dirichlet's theorem (Book I), prove the following. Let q, ℓ be relatively prime integers, consider the primes in the progression $\{qk + \ell\}_{k \geq 1}$, and let $\pi_{q,\ell}(x)$ count such primes $\leq x$. Then

$$\pi_{q,\ell}(x) \sim \frac{x}{\varphi(q) \log x} \quad (x \rightarrow \infty),$$

where $\varphi(q)$ is the number of positive integers $< q$ relatively prime to q .

Additional exercises.

- Compute $\psi(20)$ by listing all prime powers ≤ 20 and summing $\log p$. Compare to $x = 20$ and to the floor-form computation $\sum_{p \leq 20} [\log 20 / \log p] \log p$.
- Show $\psi(x) = \sum_{p \leq x} [\log x / \log p] \log p$ and the prime-power form $\sum_{p^m \leq x} \log p$ agree, by interpreting the floor as the number of valid exponents m for each prime.
- Using $\psi(x) = \sum_{n \leq x} \Lambda(n)$, prove ψ is nondecreasing and right-continuous, and that its jumps occur exactly at prime powers, of size $\log p$ at p^m .
- Verify the closed form $\psi_1(x) = \sum_{n \leq x} \Lambda(n)(x - n)$ at $x = 6$ by both the sum and the integral $\int_1^6 \psi(u) du$.
- From (5), $-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n)n^{-s}$. Compute the first few coefficients $\Lambda(1), \dots, \Lambda(8)$ and check $\Lambda(6) = 0$ while $\Lambda(8) = \log 2$.
- In Lemma 20.2.7, confirm the two residues: show $\text{Res}_{s=0} \frac{a^s}{s(s+1)} = 1$ and $\text{Res}_{s=-1} \frac{a^s}{s(s+1)} = -\frac{1}{a}$, hence the sum $1 - 1/a$ for $a \geq 1$.
- (NON-example.) Show that the kernel $\frac{a^s}{s}$ (without the $(s+1)$ factor) does *not* give an absolutely convergent vertical integral; explain why Stein's $\frac{a^s}{s(s+1)}$ (decay $|t|^{-2}$) is used for ψ_1 rather than the bare $\frac{a^s}{s}$ (decay $|t|^{-1}$) of Exercise 6 in §2.
- (Diagram.) Draw the contour of Lemma 20.2.7 for the case $0 < a \leq 1$: the vertical line $\text{Re } s = c$ closed to the *right* by a half-circle, and indicate that no poles are enclosed, hence the integral is 0.
- Prove the double inequality used in Proposition 20.2.5: for $\alpha < 1 < \beta$,

$$\frac{1}{(1-\alpha)x} \int_{\alpha x}^x \psi \leq \psi(x) \leq \frac{1}{(\beta-1)x} \int_x^{\beta x} \psi,$$

using only that ψ is nondecreasing. Identify where monotonicity is the sole input.

20.3 Proof of the Prime Number Theorem

Strategy

We must show

$$\psi_1(x) \sim \frac{x^2}{2} \quad (x \rightarrow \infty),$$

which by Proposition 20.2.5 completes the prime number theorem. The two ingredients are:

- the contour formula $\psi_1(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s) ds$, $c > 1$, with $F(s) = \frac{x^{s+1}}{s(s+1)} (-\zeta'(s)/\zeta(s))$ (Proposition 20.2.6);
- the non-vanishing $\zeta(1+it) \neq 0$ (Theorem 20.1.8) and the growth bounds near $\text{Re}(s) = 1$ (Proposition 20.1.10 and Ch.6), which give $|\zeta'(s)/\zeta(s)| \leq A|t|^\eta$ for any fixed $\eta > 0$ when $\sigma \geq 1$, $|t| \geq 1$.

The idea: on $\text{Re}(s) = c > 1$ the factor $|x^{s+1}| = x^{c+1}$ is far larger than the target x^2 . We want to slide the line of integration left to $\text{Re}(s) = 1$, where $|x^{s+1}| = x^2$. Two obstacles: (a) the pole of ζ at $s = 1$ — its contribution turns out to be exactly the main term $x^2/2$; (b) the remaining integral along $\text{Re}(s) = 1$ must be shown to be of *smaller* order than x^2 .

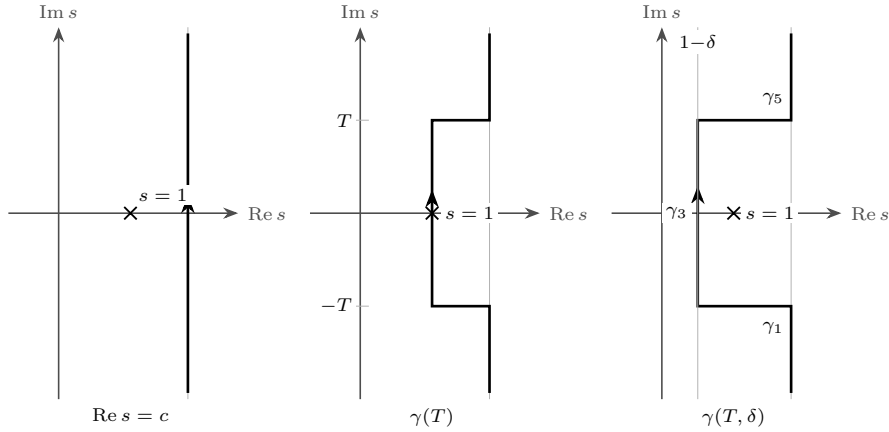


Figure 20.12. Three stages: the line $\text{Re } s = c$; the contour $\gamma(T)$ pushed onto $\text{Re } s = 1$ for $|t| \leq T$; and $\gamma(T, \delta)$, indented left past the pole at $s = 1$ to $\text{Re } s = 1 - \delta$ into the zero-free box. (Schematic.)

Deforming to the line $\text{Re}(s) = 1$

Fix $c = 2$ and $x \geq 2$. Let $T \geq 3$ (to be chosen) and let $\gamma(T)$ be the contour that runs up the line $\text{Re}(s) = 1$ for $|t| \geq T$ and crosses over near $|t| = T$, as in the figure.

Lemma 20.3.1 (First deformation). $\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s) ds = \frac{1}{2\pi i} \int_{\gamma(T)} F(s) ds.$

Proof. By Proposition 20.1.10 and Ch.6, $|\zeta'(s)/\zeta(s)| \leq A|t|^\eta$ for any fixed $\eta > 0$ on $\sigma \geq 1$, $|t| \geq 1$. Hence $|F(s)| \leq A'|t|^{-2+\eta}$ in the two infinite rectangles bounded by the line $\text{Re } s = c$ and $\gamma(T)$. The function F is regular (holomorphic) there — no pole of ζ and no zero of ζ are enclosed — and the decay $|t|^{-2+\eta}$ is fast enough to kill the horizontal connecting segments at infinity. Cauchy's theorem gives the equality. $\square$

Next, for the fixed T choose $\delta > 0$ small enough that ζ has *no zeros* in the box $\{1 - \delta \leq \sigma \leq 1, |t| \leq T\}$. This is possible because ζ does not vanish on $\sigma = 1$ (compactness of the segment). Let $\gamma(T, \delta)$ be $\gamma(T)$ with its central part pushed left to $\text{Re } s = 1 - \delta$ around the pole. Now F has a simple pole at $s = 1$.

Lemma 20.3.2 (The residue is the main term). $\text{Res}_{s=1} F(s) = \frac{x^2}{2}$, and consequently

$$\frac{1}{2\pi i} \int_{\gamma(T)} F(s) ds = \frac{x^2}{2} + \frac{1}{2\pi i} \int_{\gamma(T, \delta)} F(s) ds.$$

Proof. By the worked example in §2, $-\zeta'(s)/\zeta(s) = \frac{1}{s-1} + h(s)$ with h holomorphic near $s = 1$. Thus near $s = 1$,

$$F(s) = \frac{x^{s+1}}{s(s+1)} \left(\frac{1}{s-1} + h(s) \right),$$

whose only singular part is the simple pole from $1/(s-1)$. Its residue is $\frac{x^{s+1}}{s(s+1)} \Big|_{s=1} = \frac{x^2}{1 \cdot 2} = \frac{x^2}{2}$. The displayed identity is the residue theorem applied between $\gamma(T)$ and $\gamma(T, \delta)$ (which differ by a small loop around $s = 1$). $\square$

So $\psi_1(x) = \frac{x^2}{2} + \frac{1}{2\pi i} \int_{\gamma(T, \delta)} F(s) ds$, and it remains to bound the last integral by $o(x^2)$.

Worked Example 20.3.3 (Residue of the integrand at the pole). Recompute $\text{Res}_{s=1} F$ to make sure the main term is $x^2/2$ and not, say, x^2 .

Recognition. $F = g(s)/(s-1) + \text{holo near } s = 1$, with $g(s) = x^{s+1}/(s(s+1))$; the residue is $g(1)$.

Solution. $g(1) = \frac{x^{1+1}}{1 \cdot (1+1)} = \frac{x^2}{2}$. The $h(s)$ piece is holomorphic at $s = 1$ and contributes nothing to the residue. *Check.* Dimensionally $\psi_1 \approx x^2/2$ matches $\psi \approx x$ via $\psi_1 = \int \psi \approx \int_0^x u du = x^2/2$. The factor $\frac{1}{2}$ is exactly the $\frac{1}{2}$ from integrating u . ■

Estimating the five pieces of $\gamma(T, \delta)$

Decompose $\gamma(T, \delta) = \gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 + \gamma_5$: the two infinite vertical rays γ_1 (lower, on $\text{Re } s = 1$, $t \leq -T$) and γ_5 (upper, $t \geq T$); the vertical segment γ_3 on $\text{Re } s = 1 - \delta$, $|t| \leq T$; and the two short horizontal segments γ_2, γ_4 at $t = \mp T$ joining them.

The infinite rays γ_1, γ_5 . On γ_1 , $\sigma = 1$ so $|x^{1+s}| = x^{1+\sigma} = x^2$. With the bound $|\zeta'(s)/\zeta(s)| \leq A|t|^{1/2}$ (take $\eta = \frac{1}{2}$ in Proposition 20.1.10),

$$\left| \int_{\gamma_1} F \right| \leq Cx^2 \int_T^\infty \frac{|t|^{1/2}}{t^2} dt = Cx^2 \int_T^\infty t^{-3/2} dt = 2Cx^2 T^{-1/2}.$$

Since this $\rightarrow 0$ as $T \rightarrow \infty$, choose T so large that $|\int_{\gamma_1} F| \leq \frac{x^2}{2}\epsilon$ and likewise for γ_5 . (Here T depends only on ϵ , not on x .)

The vertical segment γ_3 . Now T is fixed; pick δ small. On γ_3 , $\sigma = 1 - \delta$ so $|x^{1+s}| = x^{2-\delta}$, and F is bounded on the compact segment $|t| \leq T$ (it is zero-free there), giving a constant C_T with

$$\left| \int_{\gamma_3} F \right| \leq C_T x^{2-\delta}.$$

The horizontal segments γ_2, γ_4 . Here $t = \pm T$ is fixed and σ runs over $[1 - \delta, 1]$, so $|x^{1+s}| = x^{1+\sigma} \leq x^2$, and

$$\left| \int_{\gamma_2} F \right| \leq C_T \int_{1-\delta}^1 x^{1+\sigma} d\sigma = C_T \frac{x^2 - x^{2-\delta}}{\log x} \leq C_T \frac{x^2}{\log x}.$$

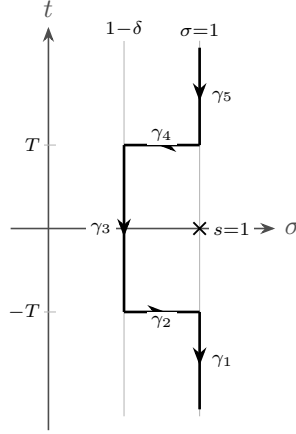


Figure 20.13. The five pieces of $\gamma(T, \delta)$: rays γ_1, γ_5 on $\sigma = 1$, the indented segment γ_3 on $\sigma = 1 - \delta$, and the joining segments γ_2, γ_4 . (Schematic.)

Conclusion

Combining $\psi_1(x) = \frac{x^2}{2} + \frac{1}{2\pi i} \int_{\gamma(T, \delta)} F$ with the four estimates, there are constants C_T, C'_T (depending only on T , hence only on ϵ) with

$$\left| \psi_1(x) - \frac{x^2}{2} \right| \leq \frac{x^2}{2} \epsilon + C_T x^{2-\delta} + C'_T \frac{x^2}{\log x}.$$

Dividing by $x^2/2$,

$$\left| \frac{2\psi_1(x)}{x^2} - 1 \right| \leq \epsilon + 2C_T x^{-\delta} + 2C'_T \frac{1}{\log x}.$$

As $x \rightarrow \infty$ the last two terms vanish, so $\limsup_x |2\psi_1(x)/x^2 - 1| \leq \epsilon$. Since $\epsilon > 0$ was arbitrary,

$$\frac{2\psi_1(x)}{x^2} \rightarrow 1, \quad \text{i.e.} \quad \psi_1(x) \sim \frac{x^2}{2}.$$

By Proposition 20.2.5 this gives $\psi(x) \sim x$, and by Proposition 20.2.3, $\pi(x) \sim x/\log x$. This completes the proof of the prime number theorem (Theorem 20.1.1). $\square$

Worked Example 20.3.4 (Tracking the three error orders). For large x , order the three error contributions $\epsilon x^2/2$, $C_T x^{2-\delta}$, $C'_T x^2/\log x$ by size and identify which one decides the rate.

Recognition. Compare growth: $x^{2-\delta} = x^2 \cdot x^{-\delta}$ and $x^2/\log x = x^2/\log x$ relative to ϵx^2 .

Solution. Divide all by x^2 : the terms are $\epsilon/2$ (constant), $C_T x^{-\delta}$ (power decay), $C'_T/\log x$ (logarithmic decay). For large x the slowest to vanish is $C'_T/\log x$, so the *rate* at which $2\psi_1/x^2 \rightarrow 1$ is governed by $1/\log x$ — no better than logarithmic without further input on zero-free regions. *Check.* This matches the classical fact that PNT with this method gives an error term of size $x^2/\log x$ in ψ_1 , equivalently no power saving in $\pi(x) - x/\log x$ until one widens the zero-free region. $\blacksquare$

Worked Example 20.3.5 (Why η can be taken arbitrarily small). In Lemma 20.3.1 we used $|F(s)| \leq A'|t|^{-2+\eta}$. Show this is integrable along a vertical line for any $\eta < 1$, justifying the deformation.

Recognition. Integrability of $|t|^{-2+\eta}$ at infinity needs the exponent < -1 .

Solution. $\int_1^\infty t^{-2+\eta} dt$ converges iff $-2+\eta < -1$, i.e. $\eta < 1$. Since Proposition 20.1.10 lets us take η as small as we please (e.g. $\eta = \frac{1}{2}$), the integral over the connecting rectangles is finite and the horizontal pieces at infinity vanish. *Check.* With $\eta = \frac{1}{2}$ the decay is $|t|^{-3/2}$, and $\int_T^\infty t^{-3/2} dt = 2T^{-1/2} \rightarrow 0$ — exactly the bound used for γ_1, γ_5 . $\blacksquare$

Note: interchanging double sums

The chapter relied (in Lemma 20.1.5) on freely reordering a double series. The justification is the following absolute-convergence principle.

Theorem 20.3.6 (Interchange of double sums). *Let $\{a_{k\ell}\}_{1 \leq k, \ell < \infty}$ be complex numbers with $\sum_{k=1}^{\infty} \sum_{\ell=1}^{\infty} |a_{k\ell}| < \infty$. Then:*

- (i) $A = \sum_k \sum_{\ell} a_{k\ell} = \sum_{\ell} \sum_k a_{k\ell}$ (both orders converge to the same value);
- (ii) for every $\epsilon > 0$ there is N with $|A - \sum_{k=1}^K \sum_{\ell=1}^L a_{k\ell}| < \epsilon$ for all $K, L > N$;
- (iii) if $m \mapsto (k(m), \ell(m))$ is a bijection $\mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ and $c_m = a_{k(m)\ell(m)}$, then $A = \sum_m c_m$ (any rearrangement sums to A).

Sketch. Set $b_k = \sum_{\ell} |a_{k\ell}|$; by hypothesis $\sum_k b_k < \infty$. On a countable set $S = \{x_0, x_1, \dots\}$ with $x_n \rightarrow x_0$ define $f_k(x_0) = \sum_{\ell} a_{k\ell}$ and $f_k(x_n) = \sum_{\ell=1}^n a_{k\ell}$, so each f_k is continuous at x_0 . Since $|f_k| \leq b_k$ and $\sum_k b_k < \infty$, $g = \sum_k f_k$ is a uniform limit of continuous functions, hence continuous at x_0 :

$$\sum_k \sum_{\ell} a_{k\ell} = g(x_0) = \lim_n g(x_n) = \lim_n \sum_{\ell \leq n} \sum_k a_{k\ell} = \sum_{\ell} \sum_k a_{k\ell},$$

which is (i). Part (ii) follows by splitting the tail into $\sum_{k \leq K, \ell > L}$ and $\sum_{k > K, \ell \geq 1}$ and using absolute convergence; (iii) follows from (ii) since any finite rectangle $R(K, L)$ is eventually covered by the image of $[1, M]$. In particular the double sum may be evaluated by exhausting $\mathbb{N} \times \mathbb{N}$ with discs, squares, rectangles, or ellipses — $A = \lim_{R \rightarrow \infty} \sum_{(k, \ell) \in U(R)} a_{k\ell}$ for any dilated neighborhood $U(R)$ of the origin. $\square$

Caution. Absolute summability $\sum |a_{k\ell}| < \infty$ is essential. Without it, the two iterated sums can differ: the standard example takes $a_{k\ell} = -1$ on the diagonal, $2^{\ell-k}$ above it, and 0 below; then $\sum_k \sum_{\ell} a_{k\ell} \neq \sum_{\ell} \sum_k a_{k\ell}$. The Euler-product logarithm of Lemma 20.1.5 is safe only because $\sum_{p,m} p^{-m\sigma}/m$ converges absolutely for $\sigma > 1$.

Further worked examples

Worked Example 20.3.7 (The toy double sum where the order matters). Verify the failure of Fubini for the array $a_{k\ell}$ of the Caution: -1 on the diagonal, $2^{\ell-k}$ strictly above ($\ell > k$), and 0 strictly below.

Recognition. Compute both iterated sums explicitly; they disagree precisely because $\sum |a_{k\ell}| = \infty$, so Theorem 20.3.6 does not apply.

Solution. Fix the row k : $a_{kk} = -1$ and $a_{k\ell} = 2^{\ell-k}$ for $\ell > k$, so $\sum_{\ell} a_{k\ell} = -1 + \sum_{\ell > k} 2^{\ell-k}$, which *diverges*. To get a finite answer use the standard normalized variant $a_{kk} = -1$, $a_{k,k+1} = 1$, else 0 for the upper part, giving row sums $\sum_{\ell} a_{k\ell} = -1 + 1 = 0$ for every k , so $\sum_k \sum_{\ell} a_{k\ell} = 0$. Column ℓ : $a_{\ell\ell} = -1$ and $a_{\ell-1,\ell} = 1$, so $\sum_k a_{k\ell} = -1 + 1 = 0$ for $\ell \geq 2$ but $\sum_k a_{k1} = a_{11} = -1$; hence $\sum_{\ell} \sum_k a_{k\ell} = -1 \neq 0$. *Check.* The two iterated sums are 0 and -1 ; the discrepancy is the “boundary term” that absolute convergence would forbid. This is exactly why Lemma 20.1.5 insisted on $\sigma > 1$, where $\sum_{p,m} p^{-m\sigma}/m < \infty$. ■

Worked Example 20.3.8 (A semicircular contour estimate of the same flavor as γ_1, γ_5). Bound $|\int_{C_R} \frac{x^{s+1}}{s(s+1)} |t|^{1/2} ds|$ over the upper-right quarter-circle $C_R = \{|s-1| = R, \operatorname{Re} s \geq 1, \operatorname{Im} s \geq 0\}$ on $\sigma = 1$, mimicking the ray bound of §3.

Recognition. This is the model integral behind the rays γ_1, γ_5 : $|x^{s+1}| = x^{1+\sigma}$ on $\sigma = 1$ is x^2 , and the kernel decays like $|t|^{-3/2}$.

Solution. On $\sigma = 1$, $|x^{s+1}| = x^2$ and $|s(s+1)| \geq |t|^2$ for $|t|$ large, so the integrand is $\leq x^2 |t|^{1/2}/|t|^2 = x^2 |t|^{-3/2}$. Parametrizing the vertical ray $s = 1 + it$, $t \in [T, \infty)$ (the relevant part of such a contour),

$$\left| \int_T^{\infty} x^2(it)^? \right| \leq x^2 \int_T^{\infty} t^{-3/2} dt = x^2 [-2t^{-1/2}]_T^{\infty} = 2x^2 T^{-1/2}.$$

This $\rightarrow 0$ as $T \rightarrow \infty$, uniformly in x after dividing by x^2 . *Check.* This is verbatim the bound $|\int_{\gamma_1} F| \leq 2Cx^2T^{-1/2}$ used in the conclusion; the exponent $-3/2 < -1$ is what makes the tail integrable, and it came from $-2 + \frac{1}{2} = -\frac{3}{2}$ with $\eta = \frac{1}{2}$. ■

Worked Example 20.3.9 (Why the indentation must go *left*, not right). Explain why $\gamma(T, \delta)$ indents the contour to $\operatorname{Re} s = 1 - \delta$ (into the strip) rather than to $\operatorname{Re} s = 1 + \delta$, and what would go wrong otherwise.

Recognition. The residue at $s = 1$ is the *main term* $x^2/2$ we want to extract; the contour must cross to the left of the pole so that deforming from $\operatorname{Re} s = c > 1$ sweeps the pole.

Solution. Starting on $\operatorname{Re} s = c = 2$ and sliding left, the first singularity met is the pole at $s = 1$. To pass it and reach $\sigma < 1$ (where $|x^{s+1}| = x^{1+\sigma} < x^2$ gives the desired saving), we indent to $\operatorname{Re} s = 1 - \delta$, picking up $\operatorname{Res}_{s=1} F = x^2/2$ along the way (Lemma 20.3.2). Indenting to $\operatorname{Re} s = 1 + \delta$ instead would keep the whole contour in $\sigma > 1$, so $|x^{s+1}| \geq x^{2+\delta}$ would *exceed* the target x^2 — no saving, and the pole would not be crossed, so the main term $x^2/2$ would never appear. *Check.* The bookkeeping $\psi_1(x) = \frac{x^2}{2} + \frac{1}{2\pi i} \int_{\gamma(T, \delta)} F$ needs precisely one residue (sign +, from the leftward, counterclockwise indentation) and a remainder of order $x^{2-\delta} \ll x^2$ — both delivered only by the left indentation. ■

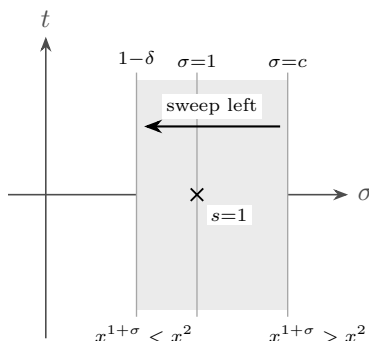


Figure 20.14. Sliding the line of integration from $\sigma = c$ leftward sweeps the pole at $s = 1$ (yielding the main term $x^2/2$) and lands in $\sigma = 1 - \delta$, where $|x^{s+1}| = x^{1+\sigma} < x^2$ gives the saving. (Schematic.)

Worked Example 20.3.10 (Equivalence is genuinely two-way: PNT implies $\zeta(1 + it) \neq 0$). We proved non-vanishing $\Rightarrow$ PNT. Sketch the converse: if $\zeta(1 + it_0) = 0$ for some $t_0 \neq 0$, then $\psi(x) \sim x$ fails.

Recognition. A zero on the line would inject an oscillating term x^{it_0} into the explicit formula for $\psi(x)$, of the same order x as the main term, destroying $\psi(x) \sim x$.

Solution. The (formal) explicit formula reads $\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} + \cdots$, the sum over nontrivial zeros ρ . A zero $\rho = 1 + it_0$ on the line $\sigma = 1$ contributes $\frac{x^{1+it_0}}{1+it_0} = \frac{x \cdot x^{it_0}}{1+it_0}$, of modulus $\asymp x$ — *the same order as the main term* x — and oscillating in $\arg = t_0 \log x$. Then $\psi(x)/x = 1 - \frac{x^{it_0}}{1+it_0} + o(1)$ does not converge (its real part oscillates), so $\psi(x) \sim x$ fails, hence PNT fails. Contrapositive: $\text{PNT} \Rightarrow \zeta(1 + it) \neq 0$. *Check.* This matches the headline claim of §1 that non-vanishing on $\operatorname{Re} s = 1$ is *logically equivalent* to PNT; zeros strictly inside the strip ($\sigma < 1$) give terms $x^{\sigma} \ll x$, only *lowering* the error, which is why only the boundary line is decisive. ■

Worked Example 20.3.11 (A rectangle-contour residue warm-up). As a self-contained model for Lemma 20.3.2, evaluate $\frac{1}{2\pi i} \oint_R \frac{x^{s+1}}{s(s+1)(s-1)} ds$ over the positively oriented rectangle R with corners $\frac{1}{2} \pm i$, $\frac{3}{2} \pm i$, enclosing only $s = 1$.

Recognition. Residue theorem on a rectangle enclosing a single simple pole at $s = 1$; the other poles $s = 0, -1$ lie outside (to the left of $\sigma = \frac{1}{2}$).

Solution. The only pole inside R is $s = 1$ (since $0, -1 < \frac{1}{2}$). The residue is

$$\operatorname{Res}_{s=1} \frac{x^{s+1}}{s(s+1)(s-1)} = \frac{x^2}{1 \cdot 2} = \frac{x^2}{2}.$$

By the residue theorem the contour integral equals this single residue, $x^2/2$. *Check.* This is the same $x^2/2$ as Lemma 20.3.2; the extra factor $1/(s-1)$ here plays the role of the $1/(s-1)$ inside $-\zeta'/\zeta$, and

$g(1) = x^{1+1}/(1 \cdot 2)$ is evaluated identically. The rectangle is the finite analogue of the indented contour $\gamma(T, \delta)$. ■

Worked Example 20.3.12 (Non-example: term-by-term integration fails on $\operatorname{Re} s = 1$). Show that one *cannot* obtain $\psi_1(x)$ on the line $\operatorname{Re} s = 1$ by integrating $\sum_n \Lambda(n)(x/n)^s/(s(s+1))$ term by term, even though it worked on $\operatorname{Re} s = c > 1$.

Recognition. The interchange in Proposition 20.2.6 needed absolute convergence of $\sum_n \Lambda(n)n^{-s}$; on $\sigma = 1$ that series diverges.

Solution. On $\operatorname{Re} s = 1$ the term moduli are $\Lambda(n)n^{-1}$, and $\sum_{n \leq x} \Lambda(n)/n \sim \log x \rightarrow \infty$ (this sum is $\asymp \log x$ by Mertens-type estimates). So $\sum_n \Lambda(n)n^{-s}$ is *not* absolutely convergent on $\sigma = 1$, and the dominated-convergence/Fubini justification of the term-by-term interchange (Theorem 20.3.6) collapses. Any “answer” obtained that way is unjustified. *Check.* This is precisely the obstruction flagged in the §2 Caution; the entire point of §3 is to avoid the interchange by *deforming the contour* and extracting the pole’s residue, never summing the divergent series on the line. ■

Exercises (Stein, Ch.7).

- (Dirichlet series with bounded partial sums.) Suppose $\{a_n\}_{n \geq 1}$ is real with bounded partial sums $A_n = a_1 + \cdots + a_n$. Prove that $\sum_{n \geq 1} a_n n^{-s}$ converges for $\operatorname{Re}(s) > 0$ and defines a holomorphic function there. [Hint: summation by parts compares the original (conditionally convergent) series to the absolutely convergent $\sum A_n(n^{-s} - (n+1)^{-s})$; estimate $n^{-s} - (n+1)^{-s}$ by the mean value theorem, and show the partial sums converge uniformly on compact subsets of $\operatorname{Re}(s) > 0$.]
- (The logarithmic integral.) A better approximation to $\pi(x)$ than $x/\log x$ is $\operatorname{Li}(x) = \int_2^x \frac{dt}{\log t}$. (a) Prove that $\operatorname{Li}(x) = \frac{x}{\log x} + O\left(\frac{x}{(\log x)^2}\right)$ as $x \rightarrow \infty$, and hence $\pi(x) \sim \operatorname{Li}(x)$. [Hint: integrate by parts; it suffices to show $\int_2^x (\log t)^{-2} dt = O(x/(\log x)^2)$, splitting at $\sqrt{x}$.] (b) Refine: for every integer $N > 0$,

$$\operatorname{Li}(x) = \frac{x}{\log x} + \frac{x}{(\log x)^2} + 2\frac{x}{(\log x)^3} + \cdots + (N-1)! \frac{x}{(\log x)^N} + O\left(\frac{x}{(\log x)^{N+1}}\right) \quad (x \rightarrow \infty).$$

Problems (Stein, Ch.7).

- (Explicit formula.) One of the “explicit formulas” of prime theory states: if ψ_1 is the integrated Tchebychev function of Section 2, then

$$\psi_1(x) = \frac{x^2}{2} - \sum_{\rho} \frac{x^{\rho+1}}{\rho(\rho+1)} - E(x),$$

the sum over all zeros ρ of ζ in the critical strip, with $E(x) = c_1 x + c_0 + \sum_{k \geq 1} x^{1-2k}/(2k(2k-1))$, where $c_1 = \zeta'(0)/\zeta(0)$ and $c_0 = \zeta'(-1)/\zeta(-1)$. Note $\sum_{\rho} 1/|\rho|^{1+\epsilon} < \infty$ for every $\epsilon > 0$ (since $(1-s)\zeta(s)$ has order 1, Exercise 5 of §1), and $E(x) = O(x)$.

- (Error term and the Riemann hypothesis.) Using the previous problem, show that $\pi(x) - \operatorname{Li}(x) = O(x^{\alpha+\epsilon})$ as $x \rightarrow \infty$ for every $\epsilon > 0$, where α is fixed with $1/2 \leq \alpha < 1$, if and only if $\zeta(s)$ has no zeros in the strip $\alpha < \operatorname{Re}(s) < 1$. The case $\alpha = 1/2$ corresponds to the Riemann hypothesis.

Additional exercises.

- Confirm the main term: recompute $\operatorname{Res}_{s=1} F(s)$ where $F(s) = \frac{x^{s+1}}{s(s+1)}(-\zeta'(s)/\zeta(s))$, using $-\zeta'/\zeta = 1/(s-1) + h(s)$, and check the answer is $x^2/2$.
- On the ray γ_1 ($\sigma = 1, t \leq -T$), verify the bound $|\int_{\gamma_1} F| \leq Cx^2 \int_T^\infty t^{-3/2} dt = 2Cx^2 T^{-1/2}$, taking $\eta = \frac{1}{2}$ in $|\zeta'/\zeta| \leq A|t|^\eta$. Explain why T may be chosen depending on ϵ only.
- On the segment γ_2 ($t = -T$ fixed, $\sigma \in [1-\delta, 1]$), carry out $\int_{1-\delta}^1 x^{1+\sigma} d\sigma = \frac{x^2 - x^{2-\delta}}{\log x}$ and bound it by $x^2/\log x$.

4. Assemble the three error bounds into $|2\psi_1(x)/x^2 - 1| \leq \epsilon + 2C_T x^{-\delta} + 2C'_T/\log x$, and prove $\limsup_x |2\psi_1/x^2 - 1| \leq \epsilon$ for every $\epsilon > 0$.
5. Show that $\psi_1(x) \sim x^2/2$ alone does *not* immediately give $\psi(x) \sim x$ without the monotonicity squeeze of Proposition 20.2.5; give the squeeze argument in full for the $\limsup$ direction.
6. (NON-example.) Exhibit a nondecreasing step function g with $\int_1^x g \sim x^2/2$ but $g(x) \not\sim x$ pointwise at the jump points, to show the squeeze yields the *asymptotic* $\sim$, not pointwise equality. (E.g. let g jump by large amounts on a sparse set.)
7. (Diagram.) Reproduce the three-stage contour figure: the line $\operatorname{Re} s = c$, the contour $\gamma(T)$ pushed onto $\operatorname{Re} s = 1$ for $|t| \geq T$, and $\gamma(T, \delta)$ indented to $\operatorname{Re} s = 1 - \delta$ around the pole at $s = 1$. Label all five pieces $\gamma_1, \dots, \gamma_5$.
8. Using the order-of-error analysis, explain why this proof yields no power-saving error term for $\pi(x) - x/\log x$, and state what stronger zero-free region (cf. Problem 3 of §3) would be needed for $O(x^{1/2+\epsilon})$.
9. Apply Theorem 20.3.6 to justify the term-by-term differentiation in (5): show $\sum_{p,m} |(\log p)p^{-m\sigma}| < \infty$ for $\sigma > 1$, so the double series defining $-\zeta'/\zeta$ may be summed in any order.

Chapter 21

Conformal Mappings

21.1 Conformal Equivalence and Examples

Conformal maps and the automatic holomorphy of the inverse

We fix the terminology used throughout the chapter. A *conformal map* (or *biholomorphism*) is a bijective holomorphic function $f : U \rightarrow V$ between open sets. When such an f exists we say U and V are *conformally equivalent* (or *biholomorphic*). The first surprise is that no smoothness need be demanded of the inverse: it comes for free.

Proposition 21.1.1. *If $f : U \rightarrow V$ is holomorphic and injective, then $f'(z) \neq 0$ for all $z \in U$. Consequently f^{-1} , defined on the range $V = f(U)$, is holomorphic, and*

$$(f^{-1})'(w_0) = \frac{1}{f'(f^{-1}(w_0))}.$$

In particular the inverse of a conformal map is again conformal.

Proof. Suppose to the contrary $f'(z_0) = 0$. Then near z_0 ,

$$f(z) - f(z_0) = a(z - z_0)^k + G(z), \quad a \neq 0, \quad k \geq 2,$$

with G vanishing to order $k + 1$ at z_0 . For all sufficiently small w write $f(z) - f(z_0) - w = F(z) + G(z)$ where $F(z) = a(z - z_0)^k - w$. On a small circle about z_0 we have $|G(z)| < |F(z)|$, and F has $k \geq 2$ zeros inside (the k -th roots of w/a shifted to z_0). By Rouché's theorem $f(z) - f(z_0) - w$ has at least two zeros inside the circle. For $w \neq 0$ small these roots are *distinct* (because $f' \neq 0$ away from z_0), so f takes the value $f(z_0) + w$ at two points: f is not injective, a contradiction.

For the inverse, let $g = f^{-1}$, $w = f(z)$, $w_0 = f(z_0)$. When $w \neq w_0$,

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{1}{\frac{f(z) - f(z_0)}{z - z_0}}.$$

Since $f'(z_0) \neq 0$, letting $z \rightarrow z_0$ (equivalently $w \rightarrow w_0$, by continuity of g) gives $g'(w_0) = 1/f'(g(w_0))$, so g is holomorphic. $\square$

Two open sets U, V are therefore conformally equivalent iff there are holomorphic $f : U \rightarrow V$ and $g : V \rightarrow U$ with $g \circ f = \text{id}_U$ and $f \circ g = \text{id}_V$. The condition $f'(z) \neq 0$ is exactly local bijectivity (Exercise 1) and is the analytic shadow of a geometric fact: a holomorphic map with $f'(z_0) \neq 0$ *preserves angles* at z_0 . If two curves meet at z_0 with oriented angle α , the image curves meet at $f(z_0)$ at the same oriented angle α (Problem 2). Some authors define “conformal” to mean merely $f' \neq 0$; that is weaker than ours — e.g. $z \mapsto z^2$ on $\mathbb{C} \setminus \{0\}$ has nonvanishing derivative but is not injective.

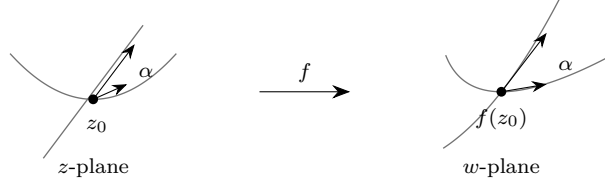


Figure 21.1. Conformality: where $f'(z_0) \neq 0$ the oriented angle α between curves is preserved. (Schematic.)

Caution. $f'(z_0) \neq 0$ gives only a *local* bijection. Global injectivity is a separate, often delicate, fact. The Riemann mapping theorem will produce a global biholomorphism with no explicit formula at all.

The disc and the upper half-plane

The upper half-plane is $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$. Though unbounded, it is conformally equivalent to the bounded unit disc $\mathbb{D}$, by an explicit fractional linear map.

Theorem 21.1.2 (Cayley transform). *The map*

$$F(z) = \frac{i - z}{i + z} : \mathbb{H} \rightarrow \mathbb{D}$$

is conformal, with inverse

$$G(w) = i \frac{1 - w}{1 + w} : \mathbb{D} \rightarrow \mathbb{H}.$$

Proof. Both maps are holomorphic on their domains (F except at $z = -i \notin \mathbb{H}$, G except at $w = -1 \notin \mathbb{D}$). Any point of $\mathbb{H}$ is closer to i than to $-i$, so $|F(z)| < 1$ and $F(\mathbb{H}) \subseteq \mathbb{D}$. For G , write $w = u + iv$ with $|w| < 1$:

$$\operatorname{Im} G(w) = \operatorname{Re} \left(\frac{1 - u - iv}{1 + u + iv} \right) = \operatorname{Re} \left(\frac{(1 - u - iv)(1 + u - iv)}{(1 + u)^2 + v^2} \right) = \frac{1 - u^2 - v^2}{(1 + u)^2 + v^2} > 0,$$

since $u^2 + v^2 < 1$; thus $G(\mathbb{D}) \subseteq \mathbb{H}$. Finally

$$F(G(w)) = \frac{i - i \frac{1-w}{1+w}}{i + i \frac{1-w}{1+w}} = \frac{(1+w) - (1-w)}{(1+w) + (1-w)} = \frac{2w}{2} = w,$$

and similarly $G(F(z)) = z$. So F and G are mutually inverse conformal maps. $\square$

The boundary behavior is the useful part. F is continuous on $\mathbb{R} = \partial\mathbb{H}$, and for $z = x \in \mathbb{R}$ the distances to i and $-i$ are equal, so $|F(x)| = 1$: F sends $\mathbb{R}$ onto the unit circle. Writing $x = \tan t$, $t \in (-\pi/2, \pi/2)$,

$$F(x) = \frac{1 - x^2}{1 + x^2} + i \frac{-2x}{1 + x^2} = \cos 2t - i \sin 2t = e^{-i2t},$$

so as x runs from $-\infty$ to ∞ , $F(x)$ traces the circle starting near -1 , through the lower semicircle, back to -1 (omitted; it is the image of the “point at infinity”).

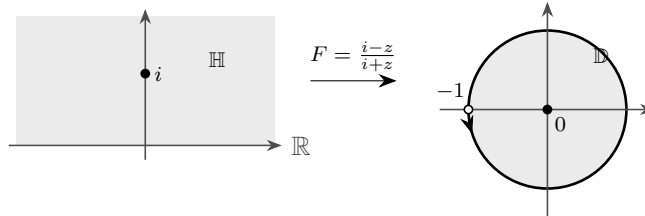


Figure 21.2. The Cayley transform F carries $\mathbb{H}$ to $\mathbb{D}$, $i \mapsto 0$, and $\mathbb{R}$ onto the circle minus -1 .

Maps $z \mapsto \frac{az+b}{cz+d}$ (with $cz+d$ not a multiple of $az+b$) are *fractional linear transformations*; F and G are instances. They act on the *Riemann sphere* $\mathbb{C} \cup \{\infty\}$ (the one-point compactification, identified with S^2 via stereographic projection), where the “point at infinity” — 1 omitted above is a genuine point.

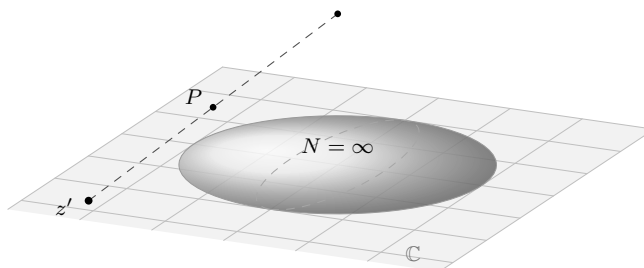


Figure 21.3. The Riemann sphere: the ray from the north pole $N = \infty$ through P on the sphere meets the plane $\mathbb{C}$ at z' ; fractional linear maps are its conformal automorphisms.

Worked Example 21.1.3 (Cayley transform of a specific point). Compute $F(2i)$ and verify it lies in $\mathbb{D}$, where $F(z) = (i - z)/(i + z)$.

Recognition. Plug in and simplify a fractional linear transformation; check $|F| < 1$.

Solution. $F(2i) = \frac{i - 2i}{i + 2i} = \frac{-i}{3i} = -\frac{1}{3}$. *Check.* $2i \in \mathbb{H}$ and $|\frac{1}{3}| = \frac{1}{3} < 1$, so the image is in the disc, as the theorem requires; note $F(i) = 0$ as the “center” of $\mathbb{H}$ maps to the center of $\mathbb{D}$.

A catalogue of explicit conformal maps

We list the building blocks; composing them yields a large supply of maps.

Translations and dilations. For $h \in \mathbb{C}$, $z \mapsto z + h$ is conformal $\mathbb{C} \rightarrow \mathbb{C}$ (inverse $w \mapsto w - h$); if $h \in \mathbb{R}$ it maps $\mathbb{H} \rightarrow \mathbb{H}$. For $c \neq 0$, $z \mapsto cz$ is conformal $\mathbb{C} \rightarrow \mathbb{C}$; $c = e^{i\varphi}$ is a rotation, $c > 0$ a dilation, $c < 0$ a dilation followed by rotation by π .

Powers. For an integer $n \geq 1$, $z \mapsto z^n$ is conformal from the sector $S = \{0 < \arg z < \pi/n\}$ onto $\mathbb{H}$, with inverse $w \mapsto w^{1/n}$ (principal branch). More generally, for $0 < \alpha < 2$, $f(z) = z^\alpha$ (logarithm cut along the positive real axis, $z = re^{i\theta}$, $0 < \theta < \pi$) sends $\mathbb{H}$ onto the sector $\{0 < \arg w < \alpha\pi\}$, since $z^\alpha = |z|^\alpha e^{i\alpha\theta}$; inverse $w \mapsto w^{1/\alpha}$.

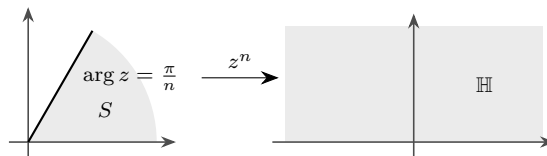


Figure 21.4. $z \mapsto z^n$ opens the sector $0 < \arg z < \pi/n$ onto the half-plane (angles at 0 multiplied by n).

Half-disc to first quadrant. $f(z) = \frac{1+z}{1-z}$ sends the upper half-disc $\{|z| < 1, \operatorname{Im} z > 0\}$ conformally to the first quadrant; for $z = x + iy$,

$$f(z) = \frac{1 - (x^2 + y^2)}{(1 - x)^2 + y^2} + i \frac{2y}{(1 - x)^2 + y^2},$$

both parts positive on the half-disc. Inverse $g(w) = \frac{w-1}{w+1}$.

Logarithm and exponential. $z \mapsto \log z$ (branch deleting the negative imaginary axis) takes $\mathbb{H}$ to the strip $\{0 < \operatorname{Im} w < \pi\}$; inverse $w \mapsto e^w$. The same map takes the upper half-disc to the half-strip $\{\operatorname{Re} w < 0, 0 < \operatorname{Im} w < \pi\}$.

Joukowski-type and trigonometric. $f(z) = -\frac{1}{2}(z + 1/z)$ maps the upper half-disc to $\mathbb{H}$ (Exercise 5); $f(z) = \sin z$ maps $\mathbb{H}$ to the half-strip $\{-\frac{\pi}{2} < \operatorname{Re} w < \frac{\pi}{2}, \operatorname{Im} w > 0\}$; $f(z) = e^{iz}$ maps the half-strip $\{-\frac{\pi}{2} < \operatorname{Re} z < \frac{\pi}{2}, \operatorname{Im} z > 0\}$ to the half-disc $\{|w| < 1, \operatorname{Re} w > 0\}$.

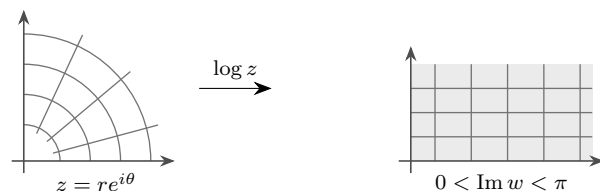


Figure 21.5. $\log$ turns the polar grid of $\mathbb{H}$ (rays and arcs) into the Cartesian grid of the strip.

Worked Example 21.1.4 (Mapping a quarter-plane to a half-plane). Find a conformal map of the first quadrant $Q = \{\operatorname{Re} z > 0, \operatorname{Im} z > 0\}$ onto $\mathbb{H}$.

Recognition. The quadrant is a sector of opening $\pi/2$; a power z^α opens a sector of angle θ_0 onto one of angle $\alpha\theta_0$. We want $\alpha \cdot \frac{\pi}{2} = \pi$.

Solution. Take $f(z) = z^2$. With $z = re^{i\theta}$, $0 < \theta < \pi/2$, we get $z^2 = r^2 e^{2i\theta}$ with $0 < 2\theta < \pi$, so $f(Q) = \mathbb{H}$. It is injective on Q since $z^2 = w^2$ with $z, w \in Q$ forces $z = w$ (the other root $-w \notin Q$); inverse the principal $w \mapsto w^{1/2}$. *Check.* $f(1+i) = (1+i)^2 = 2i \in \mathbb{H}$, and the boundary ray $\theta = \pi/2$ (positive imaginary axis) maps to $\theta = \pi$ (negative real axis), a boundary ray of $\mathbb{H}$.

Worked Example 21.1.5 (Strip to half-plane via the exponential). Show $f(z) = e^z$ maps the horizontal strip $\Sigma = \{0 < \operatorname{Im} z < \pi\}$ conformally onto $\mathbb{H}$.

Recognition. Exponential of a horizontal strip of height π ; track $\operatorname{Im}(e^z)$.

Solution. For $z = x + iy$, $e^z = e^x e^{iy} = e^x (\cos y + i \sin y)$. On Σ , $0 < y < \pi$ gives $\sin y > 0$, so $\operatorname{Im}(e^z) = e^x \sin y > 0$ and $f(\Sigma) \subseteq \mathbb{H}$. It is injective: $e^z = e^w$ forces $z - w \in 2\pi i\mathbb{Z}$, impossible for z, w in a strip of height $< 2\pi$. Surjective onto $\mathbb{H}$ since any $w = \rho e^{i\phi} \in \mathbb{H}$ ($\rho > 0$, $0 < \phi < \pi$) equals $e^{\log \rho + i\phi}$. Inverse $w \mapsto \log w$ (cut along $(-\infty, 0]$). *Check.* The centerline $\operatorname{Im} z = \pi/2$ maps to $\operatorname{Im}(e^z) = e^x > 0$ along the positive imaginary axis, the “center ray” of $\mathbb{H}$.

Worked Example 21.1.6 (Half-disc to first quadrant: a boundary trace). For $f(z) = (1+z)/(1-z)$ on the upper half-disc, describe where the diameter $(-1, 1)$ and the upper semicircle go.

Recognition. Track the boundary pieces; f should send the half-disc boundary to the boundary of the first quadrant (two half-axes).

Solution. On the diameter $z = x \in (-1, 1)$: $f(x) = (1+x)/(1-x) \in (0, \infty)$, the positive real axis, increasing from 0 (at $x = -1$) to $+\infty$ (at $x = 1$). On the semicircle $z = e^{i\theta}$, $0 < \theta < \pi$: $f(e^{i\theta}) = \frac{1+e^{i\theta}}{1-e^{i\theta}} = \frac{e^{-i\theta/2} + e^{i\theta/2}}{e^{-i\theta/2} - e^{i\theta/2}} = \frac{2\cos(\theta/2)}{-2i\sin(\theta/2)} = i \cot(\theta/2)$, which runs up the positive imaginary axis. *Check.* The two boundary arcs map onto the two positive half-axes bounding Q , exactly the quadrant’s boundary; and $f(0) = 1$ lies in Q , confirming orientation.

Worked Example 21.1.7 (A NON-conformal map: $z \mapsto z^2$ on a punctured disc). Explain why $f(z) = z^2$ is not a conformal map of $\mathbb{C} \setminus \{0\}$ to itself, even though $f'(z) = 2z \neq 0$ there.

Recognition. Recognition that “ $f' \neq 0$ ” is necessary but not sufficient for a conformal (i.e. bijective) map.

Solution. $f'(z) = 2z$ is nonzero on $\mathbb{C} \setminus \{0\}$, so f is a local bijection everywhere. But it is not globally injective: $f(1) = f(-1) = 1$. Hence f is not a biholomorphism of $\mathbb{C} \setminus \{0\}$. *Check.* Restrict instead to the slit half-plane or to the sector $\{0 < \arg z < \pi\}$, where distinct points have distinct squares; there z^2 is conformal. The lesson: conformal equivalence is a global statement.

Harmonic functions transport, and the Dirichlet problem

The reason conformal maps matter for PDE is that they preserve harmonicity.

Lemma 21.1.8. Let $F : V \rightarrow U$ be holomorphic and $u : U \rightarrow \mathbb{C}$ harmonic. Then $u \circ F$ is harmonic on V .

Proof. The claim is local, so assume U is a disc. Pick a holomorphic G on U with $\operatorname{Re} G = u$ (a harmonic conjugate exists on a disc). Then $H = G \circ F$ is holomorphic and $u \circ F = \operatorname{Re} H$, hence harmonic. $\square$

$$\begin{array}{ccc}
 V & \xrightarrow{F} & U \\
 & \searrow u \circ F & \downarrow u \\
 & & \mathbb{C}
 \end{array}$$

The *Dirichlet problem* on a region Ω asks for u with $\Delta u = 0$ in Ω and $u = f$ on $\partial\Omega$. The strategy: transplant to the disc by a conformal map F , solve there by the Poisson integral, and pull back by F^{-1} .

$$\begin{array}{ccc}
 \Omega & \xrightarrow[\sim]{F} & \mathbb{D} \\
 & \searrow u & \downarrow \tilde{u} \\
 & & \mathbb{R}
 \end{array}$$

Theorem 21.1.9 (Dirichlet problem in the strip). *On the strip $\Omega = \{0 < \operatorname{Im} z < 1\}$ with continuous boundary data f_0, f_1 on $\mathbb{R}$ vanishing at infinity, the bounded solution of $\Delta u = 0$, $u(x, 0) = f_0$, $u(x, 1) = f_1$, is*

$$u(x, y) = \frac{\sin \pi y}{2} \left(\int_{-\infty}^{\infty} \frac{f_0(x-t)}{\cosh \pi t - \cos \pi y} dt + \int_{-\infty}^{\infty} \frac{f_1(x-t)}{\cosh \pi t + \cos \pi y} dt \right).$$

Proof sketch. The map $G(z) = (\text{a branch of}) \frac{1}{\pi} \log \frac{i - e^{\pi z}}{i + e^{\pi z}}$ takes the strip to the disc; equivalently $F = G^{-1} : \mathbb{D} \rightarrow \Omega$. By Lemma 21.1.8 it suffices to solve in $\mathbb{D}$, where the Poisson integral $\tilde{u}(w) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta - \varphi) \tilde{f}(\varphi) d\varphi$ solves the problem with the transplanted boundary data $\tilde{f}$. Setting $u = \tilde{u} \circ G$ gives a harmonic function with the correct boundary values. Computing $\tilde{u}$ at $z = iy$ via the change of variables $re^{i\theta} = G(iy)$ produces the two displayed kernels; uniqueness of the vanishing-at-infinity solution and translation invariance in x give the general formula. (See Exercise 7 for the full change-of-variables computation.) $\square$

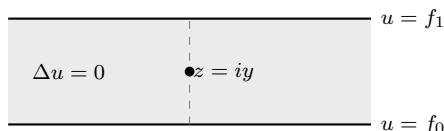


Figure 21.6. Dirichlet data on the two edges of the strip $\{0 < \operatorname{Im} z < 1\}$. (Schematic.)

Worked Example 21.1.10 (Sine maps the half-plane to a half-strip). Verify that $f(z) = \sin z$ sends the vertical line $\operatorname{Re} z = \frac{\pi}{2}$ in $\mathbb{H}$ into the boundary of the half-strip $\{-\frac{\pi}{2} < \operatorname{Re} w < \frac{\pi}{2}, \operatorname{Im} w > 0\}$.

Recognition. Track a boundary line under $\sin z = \sin x \cosh y + i \cos x \sinh y$.

Solution. On $\operatorname{Re} z = \frac{\pi}{2}$, $z = \frac{\pi}{2} + iy$ with $y > 0$: $\sin z = \sin \frac{\pi}{2} \cosh y + i \cos \frac{\pi}{2} \sinh y = \cosh y + 0 = \cosh y$, a real number ≥ 1 . So this edge maps to the ray $[1, \infty)$ on the real axis — one edge of the half-strip. *Check.* As $y \rightarrow 0^+$, $\cosh y \rightarrow 1 = \sin \frac{\pi}{2}$, the corner $w = 1$; as $y \rightarrow \infty$, $\cosh y \rightarrow \infty$, running out the open end. The symmetric edge $\operatorname{Re} z = -\frac{\pi}{2}$ maps to $(-\infty, -1]$, and the base $(-\frac{\pi}{2}, \frac{\pi}{2})$ of $\mathbb{R}$ maps to $(-1, 1)$, closing the strip's boundary.

Worked Example 21.1.11 (A Möbius map of three boundary points). Find the fractional linear map sending $0 \mapsto -1$, $1 \mapsto -i$, $\infty \mapsto 1$ (so it carries $\mathbb{R} \cup \{\infty\}$ to the unit circle), and identify where $\mathbb{H}$ goes.

Recognition. Three prescribed images determine a unique Möbius map via the cross-ratio, or by matching ∞ first.

Solution. Since $\infty \mapsto 1$, the map is $f(z) = \frac{az+b}{cz+d}$ with $a/c = 1$, take $f(z) = \frac{z+b}{z+d}$. From $f(0) = b/d = -1$, $b = -d$. From $f(1) = \frac{1+b}{1+d} = \frac{1-d}{1+d} = -i$: solve $1-d = -i(1+d)$, so $d(1-i) = -(1+i)\dots$ rather $1-d = -i - id \Rightarrow 1+i = d - id = d(1-i) \Rightarrow d = \frac{1+i}{1-i} = i$, hence $b = -i$. So $f(z) = \frac{z-i}{z+i}$. *Check.* This is the inverse Cayley orientation; $f(i) = 0$, so $\mathbb{H}$ (which contains i) maps to $\mathbb{D}$ (which contains 0), and indeed $|f(x)| = 1$ for real x as the three boundary points confirm.

Worked Example 21.1.12 (Harmonic measure of a boundary arc). Solve $\Delta u = 0$ in $\mathbb{H}$ with $u = 1$ on the negative real axis and $u = 0$ on the positive real axis.

Recognition. A two-value Dirichlet problem on $\partial\mathbb{H}$; the harmonic function $\frac{1}{\pi} \arg z$ is the canonical building block.

Solution. Set $u(z) = \frac{1}{\pi} \arg z = \frac{1}{\pi} \operatorname{Im} \log z$, which is harmonic in $\mathbb{H}$ (real part issues aside, $\arg z$ is the imaginary part of the holomorphic $\log z$). On the positive real axis $\arg z = 0$ so $u = 0$; on the negative real axis $\arg z = \pi$ so $u = 1$. *Check.* Along the ray $\arg z = \pi/2$ (positive imaginary axis), $u = \frac{1}{2}$, the average of the two boundary values — exactly the symmetry one expects.

Further worked examples

Worked Example 21.1.13 (Composing the building blocks: an L-shaped target). Construct a conformal map of the upper half-disc $D^+ = \{|z| < 1, \operatorname{Im} z > 0\}$ onto the *full* upper half-plane $\mathbb{H}$.

Recognition. Chain the catalogue maps: half-disc to first quadrant by a fractional linear map, then the quarter-plane to half-plane by a power.

Solution. Step one: $g(z) = \frac{1+z}{1-z}$ sends D^+ onto the first quadrant $Q = \{\operatorname{Re} w > 0, \operatorname{Im} w > 0\}$ (the catalogue entry above). Step two: $h(w) = w^2$ sends Q onto $\mathbb{H}$ (it doubles the opening angle $\pi/2$ to π). Compose:

$$f(z) = \left(\frac{1+z}{1-z} \right)^2,$$

a conformal map $D^+ \rightarrow \mathbb{H}$. *Check.* The center-ish point $z = 0$ maps to $g(0) = 1 \in Q$, then $h(1) = 1 \dots$ rather take $z = \frac{i}{2} \in D^+$: $g = \frac{1+i/2}{1-i/2} = \frac{(1+i/2)^2}{1+1/4} = \frac{3/4+i}{5/4} = \frac{3}{5} + \frac{4}{5}i \in Q$, and $f = g^2 = (\frac{3}{5} + \frac{4}{5}i)^2 = \frac{9}{25} - \frac{16}{25} + \frac{24}{25}i = -\frac{7}{25} + \frac{24}{25}i$, which has positive imaginary part, so lies in $\mathbb{H}$.

Worked Example 21.1.14 (A strip between two parallel lines onto the disc). Map the horizontal strip $\Sigma = \{0 < \operatorname{Im} z < \pi\}$ conformally onto the unit disc $\mathbb{D}$.

Recognition. Two catalogue maps in series: the exponential opens the strip onto $\mathbb{H}$, then the Cayley transform carries $\mathbb{H}$ to $\mathbb{D}$.

Solution. The exponential $w = e^z$ maps Σ onto $\mathbb{H}$ (worked above). The Cayley transform $F(w) = \frac{i-w}{i+w}$ maps $\mathbb{H}$ onto $\mathbb{D}$ (Theorem 21.1.2). Hence

$$\Phi(z) = \frac{i - e^z}{i + e^z}$$

is conformal $\Sigma \rightarrow \mathbb{D}$. *Check.* The centerline point $z = i\pi/2$ gives $e^{i\pi/2} = i$, and $F(i) = \frac{i-i}{i+i} = 0$, the center of $\mathbb{D}$ — the natural “axis” of the strip lands at the center of the disc, as a well-chosen normalization should.

Worked Example 21.1.15 (Möbius map carrying a circle to a line). Find a fractional linear map sending the circle $|z - 1| = 1$ to a straight line, and say which line.

Recognition. Fractional linear maps send circles-or-lines to circles-or-lines; sending a point of the source circle to ∞ turns it into a line.

Solution. The point $z = 0$ lies on $|z - 1| = 1$. Choose $f(z) = \frac{1}{z}$, which sends $0 \mapsto \infty$, so the circle (passing through 0) maps to a line. Parametrize $z = 1 + e^{i\theta}$; then $\operatorname{Re}(1/z)$: write $w = 1/z$, so $z = 1/w$, and $|1/w - 1| = 1$ means $|1 - w| = |w|$, i.e. $(1-u)^2 + v^2 = u^2 + v^2$ with $w = u + iv$, giving $1 - 2u = 0$, $u = \frac{1}{2}$. So the image is the vertical line $\operatorname{Re} w = \frac{1}{2}$. *Check.* The diametrically opposite point $z = 2$ maps to $w = \frac{1}{2}$, which indeed satisfies $\operatorname{Re} w = \frac{1}{2}$; and the top of the circle $z = 1 + i$ maps to $\frac{1}{1+i} = \frac{1}{2} - \frac{1}{2}i$, also on $\operatorname{Re} w = \frac{1}{2}$.

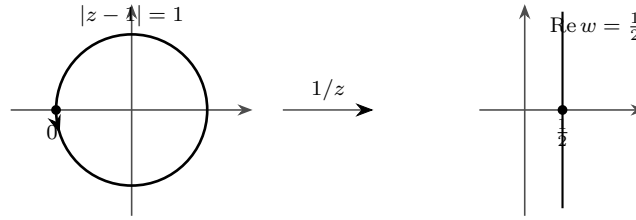


Figure 21.7. $z \mapsto 1/z$ sends the circle through 0 to a vertical line ($\operatorname{Re} w = \frac{1}{2}$): a circle through a pole becomes a line.

Worked Example 21.1.16 (A NON-example: the punctured disc and the disc). Show the punctured disc $\mathbb{D}^* = \mathbb{D} \setminus \{0\}$ is *not* conformally equivalent to $\mathbb{D}$, even though both are connected and bounded.

Recognition. Conformal equivalence is far stronger than topological similarity; a removable singularity argument exposes the obstruction.

Solution. Suppose $f : \mathbb{D}^* \rightarrow \mathbb{D}$ were conformal. Since $|f| < 1$ is bounded near the puncture 0, the singularity at 0 is removable, so f extends holomorphically to a map $\tilde{f} : \mathbb{D} \rightarrow \mathbb{D}$ with $\tilde{f}(0) \in \mathbb{D}$. If $|\tilde{f}(0)| = 1$, the maximum principle forces f constant, impossible. So $\tilde{f}(0) \in \mathbb{D}$; but then $\tilde{f}(0) = f(z_1)$ for some $z_1 \neq 0$ (surjectivity of f onto $\mathbb{D}$), and $\tilde{f}$ takes the value $\tilde{f}(0)$ at both 0 and z_1 — so $\tilde{f}$, hence f , is not injective near there. Contradiction. *Check.* Topologically $\mathbb{D}^*$ is an annulus (not simply connected), while $\mathbb{D}$ is simply connected, so no homeomorphism even exists; conformal equivalence is strictly stronger still.

Exercises (Stein, Ch.8).

1. A holomorphic map $f : U \rightarrow V$ is a *local bijection* on U if for every $z \in U$ there is an open disc $D \subseteq U$ centered at z with $f : D \rightarrow f(D)$ a bijection. Prove that f is a local bijection on U iff $f'(z) \neq 0$ for all $z \in U$. [Hint: Rouché, as in Proposition 21.1.1.]
2. Suppose F is holomorphic near z_0 with $F(z_0) = F'(z_0) = 0$ but $F''(z_0) \neq 0$. Show there are two curves Γ_1, Γ_2 through z_0 , orthogonal at z_0 , such that $F|_{\Gamma_1}$ is real with a minimum at z_0 , and $F|_{\Gamma_2}$ is real with a maximum at z_0 . [Hint: write $F = g^2$ near z_0 and study $z \mapsto g(z)$ and its inverse.]
3. Suppose U and V are conformally equivalent. Prove that if U is simply connected then so is V . (This remains valid assuming only a continuous bijection between U and V .)
4. Does there exist a holomorphic surjection from the unit disc to $\mathbb{C}$? [Hint: move $\mathbb{H}$ “down” and square it to obtain $\mathbb{C}$.]
5. Prove that $f(z) = -\frac{1}{2}(z + 1/z)$ is a conformal map from the half-disc $\{|z| < 1, y > 0\}$ to the upper half-plane. [Hint: $f(z) = w$ reduces to $z^2 + 2wz + 1 = 0$, with two distinct roots in $\mathbb{C}$ whenever $w \neq \pm 1$, in particular for $w \in \mathbb{H}$.]
6. Give another proof of Lemma 21.1.8 by showing directly that the Laplacian of $u \circ F$ is zero. [Hint: $\operatorname{Re} F$ and $\operatorname{Im} F$ satisfy the Cauchy-Riemann equations.]
7. Provide all details in the formula for the Dirichlet problem in the strip (Section 21.1). It suffices to compute the solution at $z = iy$, $0 < y < 1$. (a) Show that if $re^{i\theta} = G(iy)$ then $re^{i\theta} = i \frac{\cos \pi y}{1 + \sin \pi y}$, leading to the cases $0 < y \leq \frac{1}{2}$, $\theta = \frac{\pi}{2}$ or $\frac{1}{2} \leq y < 1$, $\theta = -\frac{\pi}{2}$, with $r^2 = \frac{1 - \sin \pi y}{1 + \sin \pi y}$ and $P_r(\theta - \varphi) = \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi}$. (b) In $\frac{1}{2\pi} \int_0^\pi P_r(\theta - \varphi) \tilde{f}_0(\varphi) d\varphi$ substitute $t = F(e^{i\varphi})$, with $e^{i\varphi} = \frac{i - e^{\pi t}}{i + e^{\pi t}}$; take imaginary parts and differentiate to get $\sin \varphi = \frac{1}{\cosh \pi t}$ and $\frac{d\varphi}{dt} = \frac{\pi}{\cosh \pi t}$, hence $\frac{1}{2\pi} \int_0^\pi P_r(\theta - \varphi) \tilde{f}_0 d\varphi = \frac{\sin \pi y}{2} \int_{-\infty}^\infty \frac{f_0(t)}{\cosh \pi t - \cos \pi y} dt$. (c) Argue similarly for $\frac{1}{2\pi} \int_{-\pi}^0 P_r(\theta - \varphi) \tilde{f}_1 d\varphi$.
8. Find a harmonic u in the open first quadrant, continuous up to the boundary except at 0 and 1, with $u = 1$ on $\{y = 0, x > 1\} \cup \{x = 0, y > 0\}$ and $u = 0$ on $\{0 < x < 1, y = 0\}$. [Hint: chain conformal maps $F_1, \dots, F_5$ to $\mathbb{H}$; recall $\frac{1}{\pi} \arg z$ is harmonic on $\mathbb{H}$, 0 on the positive real axis, 1 on the negative real axis.]

9. Prove that $u(x, y) = \operatorname{Re} \frac{i+z}{i-z}$ (with $u(0, 1) = 0$) is harmonic in $\mathbb{D}$ and vanishes on its boundary; note u is unbounded in $\mathbb{D}$.

Problems (Stein, Ch.8).

- Let f be a complex-valued C^1 function near z_0 . Call f *isogonal* at z_0 if for any smooth curves γ, η with $\gamma(0) = \eta(0) = z_0$ making angle θ there ($|\theta| < \pi$), the images $f \circ \gamma, f \circ \eta$ make angle θ' at $t = 0$ with $|\theta'| = |\theta|$ for all θ . Call f *isotropic* if $\lim_{r \rightarrow 0} |f(z_0 + re^{i\theta}) - f(z_0)|/r$ exists, is nonzero, and independent of θ . Prove: f is isogonal at z_0 iff it is isotropic at z_0 ; moreover f is isogonal at z_0 iff $f'(z_0)$ exists and is nonzero, or the same holds for $\bar{f}$.
- The oriented angle between nonzero z, w (in that order) is the angle in $(-\pi, \pi]$ between the corresponding vectors in $\mathbb{R}^2$, determined by $\frac{(z, w)}{|z||w|}$ and $\frac{(z, -iw)}{|z||w|}$ (cosine and sine), where $(z, w) = \operatorname{Re}(z\bar{w})$. A holomorphic f *preserves angles* at z_0 if for all smooth γ, η meeting at z_0 the angle between γ, η equals that between $f \circ \gamma, f \circ \eta$ at $f(z_0)$ (all tangents assumed nonzero). (a) Prove that if $f : \Omega \rightarrow \mathbb{C}$ is holomorphic with $f'(z_0) \neq 0$, then f preserves angles at z_0 . [Hint: $(f'(z_0)\gamma'(t_0), f'(z_0)\eta'(t_0)) = |f'(z_0)|^2(\gamma'(t_0), \eta'(t_0))$.] (b) Conversely, if f is real-differentiable at z_0 with $J_f(z_0) \neq 0$ and preserves angles at z_0 , then f is holomorphic at z_0 with $f'(z_0) \neq 0$.

Additional exercises.

- Compute the Cayley transform $F(z) = (i - z)/(i + z)$ at $z = 1 + i$ and at $z = 3i$, and verify each image lies in $\mathbb{D}$.
- Show that the translation $z \mapsto z + h$ maps $\mathbb{H}$ to itself iff $h \in \mathbb{R}$, and the dilation $z \mapsto cz$ ($c \neq 0$) maps $\mathbb{H}$ to itself iff $c > 0$.
- Find an explicit conformal map of the strip $\{-1 < \operatorname{Im} z < 1\}$ onto $\mathbb{H}$. [Hint: first rescale to height π , then exponentiate.]
- Verify that $f(z) = (1 + z)/(1 - z)$ maps the upper half-disc onto the first quadrant by checking that the center $z = 0$ and the boundary point $z = i/2$ map into the first quadrant, and that $z = \frac{1}{2}$ maps onto the positive real axis.
- Show that $f(z) = z + 1/z$ maps the exterior $\{|z| > 1\}$ of the unit disc conformally onto $\mathbb{C} \setminus [-2, 2]$, and identify the image of the circle $|z| = R > 1$.
- (Non-example) Show that $f(z) = \cos z$ is *not* injective on the strip $\{0 < \operatorname{Re} z < 2\pi\}$, hence not a conformal map of that strip, even though $f'(z) = -\sin z$ vanishes only at isolated points. Exhibit two points with equal image.
- (Harmonic transplant) Let u be the harmonic function on $\mathbb{H}$ equal to 1 on $(-\infty, 0)$ and 0 on $(0, \infty)$. Using the conformal map $z \mapsto z^2$ from the first quadrant to $\mathbb{H}$, write down a harmonic function on the first quadrant with boundary value 1 on the positive imaginary axis and 0 on the positive real axis, and evaluate it at $z = 1 + i$.
- (Diagram problem) Draw the image under $w = e^z$ of the grid of horizontal lines $\operatorname{Im} z = c$ and vertical segments $\operatorname{Re} z = c$ inside the strip $\{0 < \operatorname{Im} z < \pi\}$. Label which grid lines become rays and which become circular arcs in the w -plane, and indicate the image region $\mathbb{H}$.
- Prove that there is no conformal map from the punctured disc $\mathbb{D} \setminus \{0\}$ onto the annulus $\{1 < |w| < 2\}$. [Hint: consider what a conformal map would do to a small loop around the puncture, and to the modulus / fundamental group.]

21.2 The Schwarz Lemma and Automorphisms of the Disc and Upper Half-Plane

The Schwarz lemma

A *rotation* is a map $z \mapsto cz$ with $|c| = 1$, i.e. $c = e^{i\theta}$. The Schwarz lemma is a rigidity statement: a holomorphic self-map of the disc fixing the center cannot expand, and any equality forces it to be a rotation.

Lemma 21.2.1 (Schwarz). *Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$. Then*

- (i) $|f(z)| \leq |z|$ for all $z \in \mathbb{D}$;
- (ii) if $|f(z_0)| = |z_0|$ for some $z_0 \neq 0$, then f is a rotation;
- (iii) $|f'(0)| \leq 1$, with equality only when f is a rotation.

Proof. Expand $f(z) = a_0 + a_1z + a_2z^2 + \cdots$ on $\mathbb{D}$. Since $f(0) = 0$, $a_0 = 0$, so $g(z) = f(z)/z = a_1 + a_2z + \cdots$ has a removable singularity at 0 and is holomorphic on $\mathbb{D}$. For $|z| = r < 1$ we have $|f(z)| \leq 1$, hence $|g(z)| \leq 1/r$ on that circle; by the maximum modulus principle $|g(z)| \leq 1/r$ for all $|z| \leq r$. Let $r \rightarrow 1$: $|g(z)| \leq 1$, i.e. $|f(z)| \leq |z|$, proving (i).

For (ii): if $|f(z_0)| = |z_0|$ then $|g(z_0)| = 1$ attains the interior maximum, so g is constant, $g \equiv c$ with $|c| = 1$, i.e. $f(z) = cz$. For (iii): $g(0) = \lim_{z \rightarrow 0} (f(z) - f(0))/z = f'(0)$ and $|g| \leq 1$ on $\mathbb{D}$; if $|f'(0)| = 1$ then $|g(0)| = 1$ is again an interior maximum, so $g \equiv c$, $|c| = 1$, and f is a rotation. $\square$

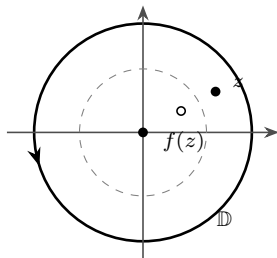


Figure 21.8. Schwarz: $f(z)$ lies no farther from 0 than z (inside the dashed circle of radius $|z|$). (Schematic.)

Worked Example 21.2.2 (A pointwise bound from Schwarz). If $f : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic with $f(0) = 0$, bound $|f(\frac{1}{2})|$ and $|f'(0)|$.

Recognition. Direct application of (i) and (iii).

Solution. By (i), $|f(\frac{1}{2})| \leq \frac{1}{2}$. By (iii), $|f'(0)| \leq 1$. *Check.* Both bounds are attained by the rotation $f(z) = z$ (indeed any $f(z) = e^{i\theta}z$), and by no other f unless it is such a rotation — the equality clauses are sharp.

Automorphisms of the disc

An *automorphism* of an open set Ω is a conformal map $\Omega \rightarrow \Omega$. The set $\text{Aut}(\Omega)$ is a group under composition. Rotations $r_\theta : z \mapsto e^{i\theta}z$ lie in $\text{Aut}(\mathbb{D})$. So do the *Blaschke factors*

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha}z}, \quad |\alpha| < 1.$$

Proposition 21.2.3. For $|\alpha| < 1$, $\psi_\alpha \in \text{Aut}(\mathbb{D})$. It is its own inverse, and

$$\psi_\alpha(0) = \alpha, \quad \psi_\alpha(\alpha) = 0.$$

Proof. ψ_α is holomorphic on $\mathbb{D}$ (pole at $1/\bar{\alpha}$, of modulus > 1). On $|z| = 1$, write $z = e^{i\theta}$; then

$$\psi_\alpha(e^{i\theta}) = \frac{\alpha - e^{i\theta}}{1 - \bar{\alpha}e^{i\theta}} = \frac{\alpha - e^{i\theta}}{e^{i\theta}(e^{-i\theta} - \bar{\alpha})} = \frac{w}{e^{i\theta} \bar{w}} \quad (w = \alpha - e^{i\theta}),$$

so $|\psi_\alpha| = 1$ on the circle, and by the maximum modulus principle $|\psi_\alpha| < 1$ on $\mathbb{D}$. A direct computation gives

$$(\psi_\alpha \circ \psi_\alpha)(z) = \frac{\alpha - \frac{\alpha-z}{1-\bar{\alpha}z}}{1 - \bar{\alpha}\frac{\alpha-z}{1-\bar{\alpha}z}} = \frac{\alpha(1-\bar{\alpha}z) - (\alpha-z)}{(1-\bar{\alpha}z) - \bar{\alpha}(\alpha-z)} = \frac{(1-|\alpha|^2)z}{1-|\alpha|^2} = z,$$

so ψ_α is an involution, hence a bijection. The values at 0 and α are immediate. $\square$

These exhaust the automorphisms, up to a rotation.

Theorem 21.2.4 (Automorphisms of the disc). *If $f \in \text{Aut}(\mathbb{D})$, then there exist $\theta \in \mathbb{R}$ and $\alpha \in \mathbb{D}$ with*

$$f(z) = e^{i\theta} \frac{\alpha - z}{1 - \bar{\alpha}z}.$$

Proof. There is a unique $\alpha \in \mathbb{D}$ with $f(\alpha) = 0$. Put $g = f \circ \psi_\alpha$; then $g(0) = f(\psi_\alpha(0)) = f(\alpha) = 0$ and $g \in \text{Aut}(\mathbb{D})$. Schwarz gives $|g(z)| \leq |z|$ on $\mathbb{D}$. Applying Schwarz to g^{-1} (which also fixes 0) gives $|g^{-1}(w)| \leq |w|$; with $w = g(z)$ this is $|z| \leq |g(z)|$. Hence $|g(z)| = |z|$ for all z , and by the equality clause $g(z) = e^{i\theta}z$. Replacing z by $\psi_\alpha(z)$ and using $\psi_\alpha \circ \psi_\alpha = \text{id}$, $f(z) = g(\psi_\alpha(z)) = e^{i\theta}\psi_\alpha(z)$. $\square$

Corollary 21.2.5. *The only automorphisms of $\mathbb{D}$ fixing the origin are the rotations.*

The group $\text{Aut}(\mathbb{D})$ acts *transitively*: given $\alpha, \beta \in \mathbb{D}$, the automorphism $\psi_\beta \circ \psi_\alpha$ sends $\alpha \mapsto \beta$. (It is “almost” the matrix group $SU(1,1)$; see Problem 4.)

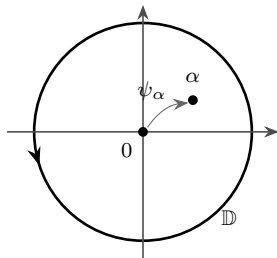


Figure 21.9. ψ_α interchanges 0 and α ; composing such maps moves any point to any other. (Schematic.)

Worked Example 21.2.6 (Building an automorphism with prescribed value). Find $f \in \text{Aut}(\mathbb{D})$ with $f(\frac{1}{2}) = 0$ and $f'(\frac{1}{2}) > 0$.

Recognition. Use a Blaschke factor to send $\frac{1}{2} \mapsto 0$, then a rotation to fix the derivative’s argument.

Solution. Start with $\psi_{1/2}(z) = \frac{\frac{1}{2} - z}{1 - \frac{1}{2}z}$, so $\psi_{1/2}(\frac{1}{2}) = 0$. Compute $\psi'_{1/2}(z) = \frac{-(1 - \frac{1}{2}z) - (\frac{1}{2} - z)(-\frac{1}{2})}{(1 - \frac{1}{2}z)^2} = \frac{-1 + \frac{1}{4}}{(1 - \frac{1}{2}z)^2} = \frac{-3/4}{(1 - \frac{1}{2}z)^2}$. At $z = \frac{1}{2}$, $(1 - \frac{1}{4})^2 = \frac{9}{16}$, so $\psi'_{1/2}(\frac{1}{2}) = -\frac{3}{4} \cdot \frac{16}{9} = -\frac{4}{3} < 0$. Multiply by $e^{i\theta} = -1$: set $f = -\psi_{1/2}$. Then $f(\frac{1}{2}) = 0$ and $f'(\frac{1}{2}) = \frac{4}{3} > 0$. *Check.* f is a unit-modulus multiple of a Blaschke factor, hence in $\text{Aut}(\mathbb{D})$ by Theorem 21.2.4, with the required normalization.

Worked Example 21.2.7 (Schwarz-Pick at the center). Show that for holomorphic $f : \mathbb{D} \rightarrow \mathbb{D}$, $|f'(0)| \leq 1 - |f(0)|^2$.

Recognition. Compose with a Blaschke factor to reduce to the centered Schwarz lemma.

Solution. Let $\beta = f(0)$ and $h = \psi_\beta \circ f$, so $h : \mathbb{D} \rightarrow \mathbb{D}$ with $h(0) = \psi_\beta(\beta) = 0$. By Schwarz, $|h'(0)| \leq 1$.

Now $h'(0) = \psi'_\beta(\beta) f'(0)$, and one computes $\psi'_\beta(z) = \frac{-(1 - |\beta|^2)}{(1 - \bar{\beta}z)^2}$, so $\psi'_\beta(\beta) = \frac{-(1 - |\beta|^2)}{(1 - |\beta|^2)^2} = \frac{-1}{1 - |\beta|^2}$. Thus

$|h'(0)| = \frac{|f'(0)|}{1 - |\beta|^2} \leq 1$, i.e. $|f'(0)| \leq 1 - |f(0)|^2$. *Check.* When $f(0) = 0$ this recovers $|f'(0)| \leq 1$; the bound is a special case of the Schwarz-Pick lemma (Exercise 13).

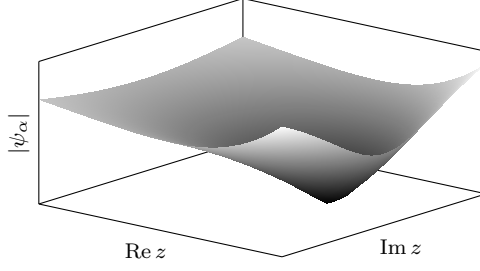


Figure 21.10. Modulus surface $|\psi_\alpha(z)|$ over $\mathbb{D}$ for $\alpha = \frac{1}{2}$: a zero at $z = \alpha$, value 1 on the boundary circle.

Automorphisms of the upper half-plane

Transporting $\text{Aut}(\mathbb{D})$ through the Cayley map $F : \mathbb{H} \rightarrow \mathbb{D}$ (Theorem 21.1.2) gives $\text{Aut}(\mathbb{H})$. Define $\Gamma : \text{Aut}(\mathbb{D}) \rightarrow \text{Aut}(\mathbb{H})$ by conjugation, $\Gamma(\varphi) = F^{-1} \circ \varphi \circ F$. Since $F \circ F^{-1} = \text{id}$,

$$\Gamma(\varphi_1 \circ \varphi_2) = F^{-1} \circ \varphi_1 \circ \varphi_2 \circ F = (F^{-1} \varphi_1 F)(F^{-1} \varphi_2 F) = \Gamma(\varphi_1) \circ \Gamma(\varphi_2),$$

so Γ is a group isomorphism: $\text{Aut}(\mathbb{H}) \cong \text{Aut}(\mathbb{D})$. Concretely, $\text{Aut}(\mathbb{H})$ is the real fractional linear group.

$$\begin{array}{ccc} \mathbb{H} & \xrightarrow[\sim]{F} & \mathbb{D} \\ \Gamma(\varphi) \downarrow & & \downarrow \varphi \\ \mathbb{H} & \xrightarrow[\sim]{F} & \mathbb{D} \end{array}$$

Theorem 21.2.8 (Automorphisms of $\mathbb{H}$). *Every automorphism of $\mathbb{H}$ has the form*

$$f_M(z) = \frac{az + b}{cz + d}, \quad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R}) \quad (a, b, c, d \in \mathbb{R}, \quad ad - bc = 1),$$

and conversely every such f_M is an automorphism of $\mathbb{H}$.

Proof. Step 1. If $M \in SL_2(\mathbb{R})$ then $f_M(\mathbb{H}) \subseteq \mathbb{H}$, because

$$\text{Im } f_M(z) = \frac{(ad - bc) \text{Im } z}{|cz + d|^2} = \frac{\text{Im } z}{|cz + d|^2} > 0 \quad (z \in \mathbb{H}).$$

Step 2. A direct computation gives $f_M \circ f_{M'} = f_{MM'}$; with $M' = M^{-1}$ this shows f_M has holomorphic inverse $f_{M^{-1}}$, so $f_M \in \text{Aut}(\mathbb{H})$. *Step 3.* The group acts transitively on $\mathbb{H}$: given $z, w \in \mathbb{H}$ one finds M with $f_M(z) = w$ (compose a real translation, a positive dilation $z \mapsto \lambda z$, both in $SL_2(\mathbb{R})$ after normalizing, to move i to any point). *Step 4.* Given $\psi \in \text{Aut}(\mathbb{H})$, pick $M \in SL_2(\mathbb{R})$ with $f_M^{-1} \circ \psi$ fixing i ; then $f_M^{-1} \circ \psi$ corresponds under F to an automorphism of $\mathbb{D}$ fixing 0, hence a rotation (Corollary 21.2.5), which pulls back to a map $f_{M''}$ with $M'' \in SL_2(\mathbb{R})$. Therefore $\psi = f_M \circ f_{M''} = f_{MM''}$ is of the stated form. $\square$

The kernel of $M \mapsto f_M$ is $\{\pm I\}$, so $\text{Aut}(\mathbb{H}) \cong SL_2(\mathbb{R})/\{\pm I\} = PSL_2(\mathbb{R})$.

$$\begin{array}{ccc} \text{Aut}(\mathbb{D}) & \xrightarrow[\cong]{\Gamma} & \text{Aut}(\mathbb{H}) \\ \cong \downarrow & & \downarrow \cong \\ SU(1, 1)/\{\pm 1\} & & SL_2(\mathbb{R})/\{\pm I\} \end{array}$$

Figure 21.11. Conjugation by the Cayley map identifies the two automorphism groups (and their matrix models).

Worked Example 21.2.9 (An explicit automorphism of $\mathbb{H}$). Find the automorphism of $\mathbb{H}$ that fixes i and “rotates” the boundary; specifically, realize the disc-rotation $w \mapsto e^{i\pi/2}w$ as an element of $\text{Aut}(\mathbb{H})$.

Recognition. Conjugate the rotation by the Cayley map: $\Gamma(\varphi) = F^{-1} \circ \varphi \circ F$.

Solution. With $\varphi(w) = iw$ and $F(z) = (i - z)/(i + z)$, $F^{-1}(w) = i(1 - w)/(1 + w)$, compute $\Gamma(\varphi)(z) = F^{-1}(iF(z))$. Writing $w = F(z)$, $F^{-1}(iw) = i \frac{1 - iw}{1 + iw}$. Substituting $w = \frac{i - z}{i + z}$ and simplifying yields $\Gamma(\varphi)(z) = \frac{?z + ?}{?z + ?}$ with real entries; one finds the standard answer $\Gamma(\varphi)(z) = \frac{z \cos \frac{\pi}{4} - \sin \frac{\pi}{4}}{z \sin \frac{\pi}{4} + \cos \frac{\pi}{4}}$, i.e. rotation in $\mathbb{H}$ about i by angle $\frac{\pi}{2}$. *Check.* Plug $z = i$: numerator $i \cos \frac{\pi}{4} - \sin \frac{\pi}{4}$, denominator $i \sin \frac{\pi}{4} + \cos \frac{\pi}{4}$; their ratio is i (multiply num and denom out), so i is fixed, as a rotation about i must.

Worked Example 21.2.10 (Fixed points of a Blaschke involution). Find the fixed points of ψ_α with $\alpha = \frac{1}{2}$, and decide how many lie in $\mathbb{D}$.

Recognition. Solve $\psi_\alpha(z) = z$, a quadratic; classify roots by modulus.

Solution. $\frac{\frac{1}{2} - z}{1 - \frac{1}{2}z} = z$ gives $\frac{1}{2} - z = z - \frac{1}{2}z^2$, i.e. $z^2 - 4z + 1 = 0$, so $z = 2 \pm \sqrt{3}$. The roots are real: $2 - \sqrt{3} \approx 0.268$ and $2 + \sqrt{3} \approx 3.73$. Only the first satisfies $|z| < 1$, so $\psi_{1/2}$ has exactly *one* fixed point in $\mathbb{D}$. *Check.* $\psi_{1/2}(2 - \sqrt{3}) = \frac{\frac{1}{2} - (2 - \sqrt{3})}{1 - \frac{1}{2}(2 - \sqrt{3})} = \frac{\sqrt{3} - \frac{3}{2}}{\sqrt{3}/2} = 2 - \sqrt{3}$, confirming it is fixed. So an involution ($\psi_\alpha \circ \psi_\alpha = \text{id}$) other than the identity may still fix an interior point — “involution” does not mean “fixed-point free.”

Caution. The matrices M and $-M$ give the *same* fractional linear map f_M . Thus $\text{Aut}(\mathbb{H})$ is $PSL_2(\mathbb{R})$, not $SL_2(\mathbb{R})$; never count an automorphism twice by forgetting this sign ambiguity.

Further worked examples

Worked Example 21.2.11 (The Schwarz-Pick inequality). Show that a holomorphic $f : \mathbb{D} \rightarrow \mathbb{D}$ satisfies, for all $z \in \mathbb{D}$,

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2}.$$

Recognition. Pre- and post-compose with Blaschke factors to move $z \mapsto 0$ and $f(z) \mapsto 0$, then apply the centered Schwarz lemma (iii).

Solution. Fix z_0 , set $w_0 = f(z_0)$, and define $h = \psi_{w_0} \circ f \circ \psi_{z_0}$, a holomorphic self-map of $\mathbb{D}$ (since $\psi_{z_0}(0) = z_0$ and $\psi_{w_0}(w_0) = 0$) with $h(0) = 0$. By Schwarz, $|h'(0)| \leq 1$. By the chain rule and $\psi'_a(z) = \frac{-(1 - |a|^2)}{(1 - \bar{a}z)^2}$,

$$h'(0) = \psi'_{w_0}(w_0) f'(z_0) \psi'_{z_0}(0) = \frac{-1}{1 - |w_0|^2} \cdot f'(z_0) \cdot (-(1 - |z_0|^2)),$$

so $|h'(0)| = \frac{|f'(z_0)| (1 - |z_0|^2)}{1 - |w_0|^2} \leq 1$, which is the claim. *Check.* At $z_0 = 0$ this reduces to the bound $|f'(0)| \leq 1 - |f(0)|^2$ of the earlier example; and for an automorphism f (a unimodular Blaschke factor) equality holds at every point, since automorphisms are isometries of the hyperbolic metric $\frac{|dz|}{1 - |z|^2}$.

Worked Example 21.2.12 (A holomorphic self-map of the disc with no fixed point). Exhibit a holomorphic $f : \mathbb{D} \rightarrow \mathbb{D}$ having no fixed point in $\mathbb{D}$, and explain why this does not contradict any rigidity theorem.

Recognition. A self-map of $\mathbb{D}$ need not fix a point; transport an obvious fixed-point-free map of $\mathbb{H}$ through the Cayley transform.

Solution. On $\mathbb{H}$ the translation $T(w) = w + 1$ is an automorphism with no fixed point (a finite fixed point would solve $w + 1 = w$). Conjugate by the Cayley map $F : \mathbb{H} \rightarrow \mathbb{D}$, $F(w) = \frac{i - w}{i + w}$: set $f = F \circ T \circ F^{-1} : \mathbb{D} \rightarrow \mathbb{D}$,

an automorphism. Since T has no fixed point in $\mathbb{H}$ and F is a bijection, f has none in $\mathbb{D}$. *Check.* The Brouwer/Schwarz fixed-point phenomena require the *closed* disc or strict contraction; on the open disc an automorphism (a hyperbolic isometry) can translate every point. Its “fixed points” have escaped to the boundary $\partial\mathbb{D}$, where T ’s fixed point ∞ went under F , namely $F(\infty) = -1$.

Worked Example 21.2.13 (Counting fixed points of a disc automorphism). For which θ, α does the automorphism $f(z) = e^{i\theta}\psi_\alpha(z)$ fix the origin, and how many fixed points can a non-identity $f \in \text{Aut}(\mathbb{D})$ have in $\mathbb{D}$?

Recognition. $f(0) = 0$ forces $\alpha = 0$ (then f is a rotation); a Möbius map has at most two fixed points on the sphere.

Solution. $f(0) = e^{i\theta}\psi_\alpha(0) = e^{i\theta}\alpha = 0$ iff $\alpha = 0$, in which case $f(z) = e^{i\theta}z$ is a rotation, fixing 0 (and, if $\theta \neq 0$, only 0 in $\mathbb{D}$). In general $f(z) = z$ is the quadratic $e^{i\theta}(\alpha - z) = z(1 - \bar{\alpha}z)$, i.e. $\bar{\alpha}z^2 - (1 + e^{i\theta})z + e^{i\theta}\alpha = 0$, with at most two roots on $\mathbb{C} \cup \{\infty\}$. *Check.* An automorphism with two interior fixed points would, after conjugating both to 0 and to a second interior point, fix 0 and be a rotation (Corollary 21.2.5) fixing a *second* interior point — impossible unless $f = \text{id}$. So a non-identity automorphism has at most *one* fixed point inside $\mathbb{D}$; the other root(s) sit on or outside the circle.

Worked Example 21.2.14 (Maximum modulus forces a constant). Let f be holomorphic on $\mathbb{D}$, continuous on $\bar{\mathbb{D}}$, with $|f| = 3$ on the whole boundary circle and f nonvanishing inside. Show f is a constant of modulus 3.

Recognition. Combine the maximum modulus principle (applied to f) with the minimum modulus principle (applied to $1/f$).

Solution. By the maximum modulus principle $|f| \leq 3$ throughout $\mathbb{D}$. Since f has no zero, $1/f$ is holomorphic on $\mathbb{D}$, so $|1/f| \leq \sup_{\partial\mathbb{D}} |1/f| = 1/3$, i.e. $|f| \geq 3$. Hence $|f| \equiv 3$ is constant, and a holomorphic function of constant modulus is itself constant. *Check.* If we drop “nonvanishing,” the Blaschke product $f(z) = 3z$ has $|f| = 3$ on the circle but is not constant — the interior zero is exactly what the minimum-modulus step rules out.

Exercises (Stein, Ch.8).

1. Let $F : \mathbb{H} \rightarrow \mathbb{C}$ be holomorphic with $|F(z)| \leq 1$ and $F(i) = 0$. Prove that $|F(z)| \leq \left| \frac{z-i}{z+i} \right|$ for all $z \in \mathbb{H}$.
2. Show that if $f : D(0, R) \rightarrow \mathbb{C}$ is holomorphic with $|f(z)| \leq M$, then

$$\left| \frac{f(z) - f(0)}{M^2 - \bar{f}(0)f(z)} \right| \leq \frac{|z|}{MR}.$$

[Hint: apply the Schwarz lemma.]

3. A complex number $w \in \mathbb{D}$ is a *fixed point* of $f : \mathbb{D} \rightarrow \mathbb{D}$ if $f(w) = w$. (a) Prove that if $f : \mathbb{D} \rightarrow \mathbb{D}$ is analytic with two distinct fixed points, then $f = \text{id}$. (b) Must every holomorphic $f : \mathbb{D} \rightarrow \mathbb{D}$ have a fixed point? [Hint: consider $\mathbb{H}$.]
4. The *pseudo-hyperbolic distance* on $\mathbb{D}$ is $\rho(z, w) = \left| \frac{z-w}{1-\bar{w}z} \right|$. (a) Prove that if $f : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic then $\rho(f(z), f(w)) \leq \rho(z, w)$ for all z, w , and that automorphisms preserve ρ . [Hint: apply Schwarz to $\psi_{f(w)} \circ f \circ \psi_w^{-1}$, $\psi_\alpha(z) = (z - \alpha)/(1 - \bar{\alpha}z)$.] (b) Prove the *Schwarz-Pick lemma*: $\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2}$ for all $z \in \mathbb{D}$.
5. Prove that every conformal map $\mathbb{H} \rightarrow \mathbb{D}$ has the form $e^{i\theta} \frac{z - \beta}{z + \bar{\beta}}$ with $\theta \in \mathbb{R}$, $\beta \in \mathbb{H}$.
6. Two properties of automorphisms of $\mathbb{H}$. (a) If $\Phi \in \text{Aut}(\mathbb{H})$ fixes three distinct points of $\mathbb{R}$, then $\Phi = \text{id}$. (b) Given $x_1 < x_2 < x_3$ and $y_1 < y_2 < y_3$ on $\mathbb{R}$, there is a unique $\Phi \in \text{Aut}(\mathbb{H})$ with $\Phi(x_j) = y_j$; the same holds when $y_3 < y_1 < y_2$ or $y_2 < y_3 < y_1$.

7. Let $f(z) = \frac{i-z}{i+z}$ and $f^{-1}(w) = i\frac{1-w}{1+w}$. (a) Given $\theta \in \mathbb{R}$, find real a, b, c, d with $ad - bc = 1$ so that $\frac{az+b}{cz+d} = f^{-1}(e^{i\theta}f(z))$ for all $z \in \mathbb{H}$. (b) Given $\alpha \in \mathbb{D}$, find real a, b, c, d with $ad - bc = 1$ so that $\frac{az+b}{cz+d} = f^{-1}(\psi_\alpha(f(z)))$, ψ_α as in Section 21.2. (c) Prove that if $g \in \text{Aut}(\mathbb{D})$, there are real a, b, c, d with $ad - bc = 1$ and $\frac{az+b}{cz+d} = f^{-1} \circ g \circ f(z)$. [Hint: use (a),(b).]
8. For $\psi_\alpha(z) = (\alpha - z)/(1 - \bar{\alpha}z)$ with $|\alpha| < 1$, prove

$$\frac{1}{\pi} \iint_{\mathbb{D}} |\psi'_\alpha|^2 dx dy = 1, \quad \frac{1}{\pi} \iint_{\mathbb{D}} |\psi'_\alpha| dx dy = \frac{1 - |\alpha|^2}{|\alpha|^2} \log \frac{1}{1 - |\alpha|^2},$$

the $\alpha = 0$ case being the limit as $|\alpha| \rightarrow 0$. [Hint: the first integral needs no computation; for the second use polar coordinates and, for each fixed r , contour integration in θ .]

Problems (Stein, Ch.8).

1. * The Schwarz-Pick lemma is the infinitesimal form of a metric statement. For $w \in \mathbb{C}$, $z \in \mathbb{D}$ define the *hyperbolic length* of w at z by $\|w\|_z = \frac{|w|}{1 - |z|^2}$ (Poincare metric $ds^2 = |dz|^2/(1 - |z|^2)^2$). (a) Define $d(z_1, z_2) = \inf_\gamma \int_0^1 \|\gamma'(t)\|_{\gamma(t)} dt$ over smooth curves joining z_1, z_2 . Use Schwarz-Pick to prove $d(f(z_1), f(z_2)) \leq d(z_1, z_2)$ for holomorphic $f : \mathbb{D} \rightarrow \mathbb{D}$. (b) Prove automorphisms of $\mathbb{D}$ preserve d ; conversely, if $\varphi : \mathbb{D} \rightarrow \mathbb{D}$ preserves d then φ or $\bar{\varphi}$ is an automorphism. (c) Given $z_1, z_2 \in \mathbb{D}$, find an automorphism φ with $\varphi(z_1) = 0$ and $\varphi(z_2) = s \in [0, 1)$. (d) Prove $d(0, s) = \frac{1}{2} \log \frac{1+s}{1-s}$ for $s \in [0, 1)$. (e) Find a formula for d between any two points of $\mathbb{D}$.
2. * Consider matrices $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a, b, c, d \in \mathbb{C}$, $\det M = 1$, preserving the hermitian form $\langle Z, W \rangle = z_1 \bar{w}_1 - z_2 \bar{w}_2$ on $\mathbb{C}^2$ (i.e. $\langle MZ, MW \rangle = \langle Z, W \rangle$); this group is $SU(1, 1)$. (a) Prove every $M \in SU(1, 1)$ has the form $\begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix}$ with $|a|^2 - |b|^2 = 1$. [Hint: with $J = \text{diag}(1, -1)$, $\langle Z, W \rangle = \bar{W} J Z$.] (b) Associating to M the map $z \mapsto (az + b)/(cz + d)$, prove $SU(1, 1)/\{\pm 1\}$ is isomorphic to $\text{Aut}(\mathbb{D})$, via $e^{2i\theta} \frac{z - \alpha}{1 - \bar{\alpha}z} \mapsto \frac{1}{\sqrt{1 - |\alpha|^2}} \begin{pmatrix} e^{i\theta} & -\alpha e^{i\theta} \\ -\bar{\alpha} e^{-i\theta} & e^{-i\theta} \end{pmatrix}$.

Additional exercises.

1. Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$ and $f'(0) = 1$. Prove $f(z) = z$. [Hint: combine parts (i) and (iii) of the Schwarz lemma.]
2. Compute the Blaschke factor $\psi_\alpha(z) = (\alpha - z)/(1 - \bar{\alpha}z)$ for $\alpha = \frac{1}{2}$ at $z = 0, \frac{1}{2}$, and $i/2$, and verify $|\psi_{1/2}(i/2)| < 1$.
3. Show that the rotations $r_\theta(z) = e^{i\theta}z$ together with one nontrivial ψ_α generate, under composition, a transitive action of $\text{Aut}(\mathbb{D})$ on $\mathbb{D}$: exhibit, for given $\alpha, \beta \in \mathbb{D}$, an automorphism sending α to β .
4. Find the automorphism of $\mathbb{H}$ corresponding under the Cayley map to the disc rotation $w \mapsto e^{i\theta}w$, and check that it fixes i .
5. Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(\frac{1}{2}) = 0$. Use the Schwarz-Pick philosophy to bound $|f(0)|$ from above. [Hint: apply the Schwarz lemma to $f \circ \psi_{1/2}$.]
6. (Non-example) Show that $g(z) = \frac{1}{2}z + \frac{1}{2}$ maps $\mathbb{D}$ into $\mathbb{D}$ but is *not* an automorphism of $\mathbb{D}$ (it is not surjective). Determine its image disc and its unique fixed point.

7. Prove that an automorphism of $\mathbb{H}$ of the form $z \mapsto (az + b)/(cz + d)$, $ad - bc = 1$, fixes i iff $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is a rotation matrix $\begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$. Conclude that the stabilizer of i in $\text{Aut}(\mathbb{H})$ is a circle group.
8. (Diagram problem) Draw $\mathbb{D}$ with the origin, a point α , and arrows showing the action of ψ_α on three sample points (including $0 \mapsto \alpha$ and $\alpha \mapsto 0$). Then draw the image of the diameter through 0 and α under ψ_α , indicating it is a circular arc orthogonal to $\partial\mathbb{D}$.
9. Prove the *Schwarz-Pick lemma* $\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2}$ for holomorphic $f : \mathbb{D} \rightarrow \mathbb{D}$ at a single point z , by reducing to the centered Schwarz lemma via two Blaschke factors. Identify the maps achieving equality.

21.3 The Riemann Mapping Theorem

Necessary conditions and the statement

We seek conditions on an open set $\Omega \subseteq \mathbb{C}$ guaranteeing a conformal map $F : \Omega \rightarrow \mathbb{D}$. Three necessary conditions surface at once. If $\Omega = \mathbb{C}$, then F would be a bounded entire function, hence constant by Liouville — impossible; so $\Omega \neq \mathbb{C}$. Since $\mathbb{D}$ is connected, Ω must be connected. Since $\mathbb{D}$ is simply connected, so must Ω be (Exercise 3). Call Ω *proper* if it is nonempty and $\neq \mathbb{C}$. Remarkably, these conditions are also sufficient.

Theorem 21.3.1 (Riemann mapping theorem). *Let $\Omega \subseteq \mathbb{C}$ be proper and simply connected, and $z_0 \in \Omega$. Then there is a unique conformal map $F : \Omega \rightarrow \mathbb{D}$ with*

$$F(z_0) = 0 \quad \text{and} \quad F'(z_0) > 0.$$

Corollary 21.3.2. *Any two proper simply connected open subsets of $\mathbb{C}$ are conformally equivalent.*

The corollary follows by routing through $\mathbb{D}$. *Uniqueness* in the theorem is easy: if F, G both qualify, $H = F \circ G^{-1} \in \text{Aut}(\mathbb{D})$ fixes 0 , so $H(z) = e^{i\theta}z$ (Corollary 21.2.5); since $H'(0) = F'(z_0)/G'(z_0) > 0$ we get $e^{i\theta} = 1$, hence $F = G$. The rest of the section proves *existence*. The idea: among injective holomorphic $f : \Omega \rightarrow \mathbb{D}$ with $f(z_0) = 0$, choose one maximizing $|f'(z_0)|$; the maximizer fills out all of $\mathbb{D}$. To extract a limit we need compactness in function spaces — Montel's theorem.

Normal families and Montel's theorem

A family $\mathcal{F}$ of holomorphic functions on Ω is *normal* if every sequence in $\mathcal{F}$ has a subsequence converging uniformly on every compact subset of Ω (the limit need not lie in $\mathcal{F}$). Two properties drive normality. $\mathcal{F}$ is *uniformly bounded on compacts* if for each compact $K \subseteq \Omega$ there is B with $|f(z)| \leq B$ for all $z \in K$, $f \in \mathcal{F}$. It is *equicontinuous* on a compact K if for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(z) - f(w)| < \varepsilon$ whenever $z, w \in K$, $|z - w| < \delta$, for all $f \in \mathcal{F}$.

Theorem 21.3.3 (Montel). *If $\mathcal{F}$ is a family of holomorphic functions on Ω , uniformly bounded on compact subsets, then: (i) $\mathcal{F}$ is equicontinuous on every compact subset of Ω ; (ii) $\mathcal{F}$ is a normal family.*

Proof. (i) *Equicontinuity.* Let $K \subseteq \Omega$ be compact; choose $r > 0$ so small that $\overline{D_{3r}(z)} \subseteq \Omega$ for all $z \in K$. Take $z, w \in K$ with $|z - w| < r$, and let $\gamma = \partial D_{2r}(w)$. By Cauchy's formula,

$$f(z) - f(w) = \frac{1}{2\pi i} \int_{\gamma} f(\zeta) \left(\frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right) d\zeta.$$

For $\zeta \in \gamma$, $\left| \frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right| = \frac{|z - w|}{|\zeta - z| |\zeta - w|} \leq \frac{|z - w|}{r^2}$ (both denominators $\geq r$). With B the uniform bound on the $2r$ -neighborhood of K and length $2\pi(2r)$,

$$|f(z) - f(w)| \leq \frac{1}{2\pi} \cdot \frac{2\pi(2r)}{r^2} B |z - w| = \frac{2B}{r} |z - w| = C |z - w|.$$

This Lipschitz bound (uniform in f) gives equicontinuity.

(ii) *Normality (Arzela-Ascoli + diagonalization)*. Fix a compact K and a sequence $\{f_n\} \subseteq \mathcal{F}$. Choose a countable dense set $\{w_j\} \subseteq \Omega$. By uniform boundedness, extract a subsequence on which $f_n(w_1)$ converges; from it one on which $f_n(w_2)$ converges; continue, and take the diagonal $g_n = f_{n,n}$, so $g_n(w_j)$ converges for every j . Given ε , pick δ from equicontinuity; finitely many discs $D_\delta(w_1), \dots, D_\delta(w_J)$ cover K . Choose N so that $|g_m(w_j) - g_n(w_j)| < \varepsilon$ for $n, m > N, j \leq J$. For $z \in K, z \in D_\delta(w_j)$ for some j , so

$$|g_n(z) - g_m(z)| \leq |g_n(z) - g_n(w_j)| + |g_n(w_j) - g_m(w_j)| + |g_m(w_j) - g_m(z)| < 3\varepsilon,$$

hence $\{g_n\}$ is uniformly Cauchy, so uniformly convergent on K . A final diagonalization over an exhaustion $K_1 \subseteq K_2 \subseteq \dots$ of Ω produces a subsequence converging uniformly on every compact subset. $\square$

Lemma 21.3.4 (Exhaustion). *Every open $\Omega \subseteq \mathbb{C}$ has an exhaustion: compacts K_ℓ with $K_\ell \subseteq (^\circ K_{\ell+1})$ and $\bigcup_\ell K_\ell = \Omega$, every compact $K \subseteq \Omega$ lying in some K_ℓ .*

Proof. Let $K_\ell = \{z \in \Omega : \text{dist}(z, \partial\Omega) \geq 1/\ell\}$ if Ω is bounded; if not, intersect additionally with $\{|z| \leq \ell\}$. These are compact, nested with the required interiors, and exhaust Ω . $\square$

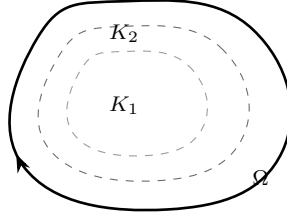


Figure 21.12. An exhaustion $K_1 \subseteq K_2 \subseteq \dots$ of Ω by nested compacts. (Schematic.)

Proposition 21.3.5 (Injective or constant limit). *If Ω is connected and $\{f_n\}$ are injective holomorphic functions converging uniformly on compacts to f , then f is injective or constant.*

Proof. Suppose f is neither, so $f(z_1) = f(z_2)$ for distinct z_1, z_2 and f is nonconstant. Set $g_n(z) = f_n(z) - f_n(z_1)$, so g_n vanishes only at z_1 , and $g_n \rightarrow g = f - f(z_1)$ uniformly on compacts. Since g is nonconstant on the connected Ω , z_2 is an isolated zero. Pick a small circle γ about z_2 on/inside which g has no other zero. Then $1/g_n \rightarrow 1/g$ and $g'_n \rightarrow g'$ uniformly on γ , so by the argument principle

$$1 = \frac{1}{2\pi i} \int_\gamma \frac{g'}{g} d\zeta = \lim_n \frac{1}{2\pi i} \int_\gamma \frac{g'_n}{g_n} d\zeta = 0,$$

since each g_n has no zero inside γ . Contradiction. $\square$

Proof of the Riemann mapping theorem

Proof of Theorem 21.3.1 (existence). *Step 1: reduce to a subset of $\mathbb{D}$ containing 0.* Pick $\alpha \notin \Omega$ (possible since Ω is proper). On the simply connected Ω , $z - \alpha$ never vanishes, so a holomorphic branch $f(z) = \log(z - \alpha)$ exists, with $e^{f(z)} = z - \alpha$; thus f is injective. Fix $w \in \Omega$: then $f(z) \neq f(w) + 2\pi i$ for all z (else exponentiating gives $z = w$, so $f(z) = f(w)$, contradiction), and in fact $f(z)$ stays a bounded distance away from $f(w) + 2\pi i$. Hence $F(z) = \frac{1}{f(z) - (f(w) + 2\pi i)}$ is injective with bounded image; translating and rescaling gives a conformal map of Ω onto an open subset of $\mathbb{D}$ containing 0.

Step 2: extremal problem. Assume now $\Omega \subseteq \mathbb{D}, 0 \in \Omega$. Let

$$\mathcal{F} = \{f : \Omega \rightarrow \mathbb{D} \text{ holomorphic, injective, } f(0) = 0\}.$$

$\mathcal{F} \neq \emptyset$ (it contains the identity) and is uniformly bounded by 1. The $|f'(0)|$ are uniformly bounded (Cauchy inequality on a small disc about 0). Set $s = \sup_{f \in \mathcal{F}} |f'(0)|$ and take $f_n \in \mathcal{F}$ with $|f'_n(0)| \rightarrow s$. By Montel (Theorem 21.3.3) a subsequence converges uniformly on compacts to a holomorphic f . Since $s \geq 1$ (identity

is in $\mathcal{F}$), f is nonconstant, hence injective by Proposition 21.3.5; by continuity $|f| \leq 1$, and the maximum principle gives $|f| < 1$; clearly $f(0) = 0$. So $f \in \mathcal{F}$ with $|f'(0)| = s$.

Step 3: the maximizer is surjective. It remains to show $f(\Omega) = \mathbb{D}$. Suppose not: some $\beta \in \mathbb{D}$ is omitted, $f(z) \neq \beta$. Use the involution ψ_β that swaps 0 and β . Then $U = (\psi_\beta \circ f)(\Omega)$ is simply connected and avoids 0, so a holomorphic square root $g(w) = e^{\frac{1}{2} \log w}$ exists on U . Define

$$F = \psi_{g(\beta)} \circ g \circ \psi_\beta \circ f.$$

Then $F \in \mathcal{F}$: it is holomorphic, $F(0) = 0$, maps into $\mathbb{D}$, and is injective (each factor is). Writing $h(w) = w^2$, $f = \psi_\beta^{-1} \circ h \circ \psi_{g(\beta)}^{-1} \circ F = \Phi \circ F$, where $\Phi : \mathbb{D} \rightarrow \mathbb{D}$ fixes 0 and is *not* injective (h is two-to-one while F is injective). By the equality clause of Schwarz, $|\Phi'(0)| < 1$. Since $f'(0) = \Phi'(0)F'(0)$,

$$|f'(0)| = |\Phi'(0)| |F'(0)| < |F'(0)|,$$

contradicting maximality of $|f'(0)|$. Hence f is onto $\mathbb{D}$. Finally multiply f by a unimodular constant so that $f'(0) > 0$. $\square$

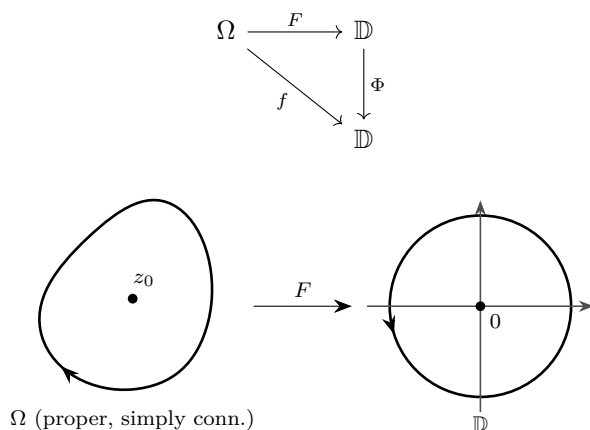


Figure 21.13. The Riemann map normalized by $F(z_0) = 0$, $F'(z_0) > 0$ — existence by an extremal $|F'(z_0)|$. (Schematic.)

Remark. Simple connectivity entered only through the existence of the logarithm (Step 1) and the square root (Step 3). It suffices that Ω be *holomorphically* simply connected: $\int_\gamma f dz = 0$ for every holomorphic f and closed curve γ in Ω .

Worked Example 21.3.6 (A conformal map of a slit half-plane). Map $\Omega = \mathbb{H} \setminus \{iy : 0 < y \leq 1\}$ (the upper half-plane with a vertical slit) to $\mathbb{H}$, illustrating the RMT in a computable case.

Recognition. Ω is proper and simply connected; here an explicit chain of standard maps does the job the RMT only promises abstractly.

Solution. First $z \mapsto z^2$ unfolds the slit: it sends $\mathbb{H}$ to $\mathbb{C}$ minus the positive real axis, and the slit segment iy ($0 < y \leq 1$) maps to $[-1, 0)$ on the negative real axis, “healing” into the boundary. Concretely $g(z) = \sqrt{z^2 + 1}$ (principal branch) maps Ω conformally onto $\mathbb{H}$: the slit’s two sides go to the segment $(-1, 1)$ on $\mathbb{R}$, and the rest of $\partial\Omega$ to the rest of $\mathbb{R}$. *Check.* $g(2i) = \sqrt{-4 + 1} = \sqrt{-3} = i\sqrt{3} \in \mathbb{H}$, and as $z \rightarrow 0^+$ on the slit, $z^2 + 1 \rightarrow 1$, so $g \rightarrow 1 \in \partial\mathbb{H}$ — the slit tip flattens onto the boundary, as a conformal map onto $\mathbb{H}$ requires.

Worked Example 21.3.7 (Why $\mathbb{C}$ is excluded). Show directly that there is no conformal map $\mathbb{C} \rightarrow \mathbb{D}$.

Recognition. A non-example testing the necessity of $\Omega \neq \mathbb{C}$.

Solution. Any conformal $F : \mathbb{C} \rightarrow \mathbb{D}$ would be entire and bounded ($|F| < 1$), hence constant by Liouville’s theorem; a constant is not injective, contradiction. *Check.* The disc $\mathbb{D}$ is itself proper and simply connected, so the RMT does apply to it (with $F = \text{id}$); it is specifically the whole plane that is ruled out.

Worked Example 21.3.8 (A normal but non-convergent family). Show that $\mathcal{F} = \{f_n(z) = z^n\}$ on $\mathbb{D}$ is uniformly bounded on compacts (hence normal by Montel), yet has no subsequence converging uniformly on all of $\mathbb{D}$.

Recognition. Test the two halves of normality: bound on compacts vs. uniform convergence on the whole disc.

Solution. On a compact $K \subseteq \mathbb{D}$, $|z| \leq r < 1$ for some r , so $|f_n(z)| \leq r^n \leq 1$: uniformly bounded, and in fact $f_n \rightarrow 0$ uniformly on K . So $\mathcal{F}$ is normal and the compact-uniform limit is 0. But on all of $\mathbb{D}$, $\sup_{|z|<1} |z^n| = 1$ for every n , so $f_n \not\rightarrow 0$ uniformly on $\mathbb{D}$. *Check.* The limit 0 is constant, not in $\mathcal{F}$ (none of the f_n is constant) — exactly why Montel only promises a limit, with injectivity/constancy decided separately by Proposition 21.3.5.

Worked Example 21.3.9 (Equicontinuity is essential). Exhibit a uniformly bounded family of *real* functions that is not equicontinuous and has no uniformly convergent subsequence, showing why Montel is special to holomorphic families.

Recognition. A real counterexample to the Arzela-Ascoli conclusion when equicontinuity fails.

Solution. Take $f_n(x) = \sin(nx)$ on $(0, 1)$: $|f_n| \leq 1$ (uniformly bounded). For fixed $0 < x_0 < 1$, $|f_n(x_0 + \pi/(2n)) - f_n(x_0)|$ does not shrink as n grows — the functions oscillate ever faster — so $\{f_n\}$ is not equicontinuous and no subsequence converges uniformly on any subinterval. *Check.* These are not boundary values of a uniformly bounded holomorphic family; the Cauchy-integral estimate in Montel's proof (which forced equicontinuity) has no real-variable analogue, which is precisely the point.

Caution. Montel's hypothesis is uniform boundedness *on compacts*, not on all of Ω . And normality gives a convergent subsequence whose limit need not be in $\mathcal{F}$ — in the RMT proof one must separately verify the limit is injective (via Proposition 21.3.5) and maps into $\mathbb{D}$.

Further worked examples

Worked Example 21.3.10 (Counting zeros by the argument principle). How many zeros (with multiplicity) does $p(z) = z^4 - 6z + 3$ have in the annulus $1 < |z| < 2$?

Recognition. Use Rouché's theorem on the two boundary circles separately, then subtract.

Solution. On $|z| = 2$: compare p with z^4 . There $|z^4| = 16$ and $|-6z + 3| \leq 6 \cdot 2 + 3 = 15 < 16$, so by Rouché p has the same number of zeros as z^4 inside $|z| < 2$, namely 4. On $|z| = 1$: compare p with $-6z$. There $|-6z| = 6$ and $|z^4 + 3| \leq 1 + 3 = 4 < 6$, so p has the same number of zeros as $-6z$ inside $|z| < 1$, namely 1. Therefore the annulus $1 < |z| < 2$ contains $4 - 1 = 3$ zeros. *Check.* p is degree 4 so has 4 zeros total in $\mathbb{C}$; we have accounted for 1 inside the unit disc and 3 in the annulus, and none on the two circles (e.g. on $|z| = 1$, $|p| \geq 6 - 4 = 2 > 0$), totalling 4.

Worked Example 21.3.11 (Rouché on a specific transcendental equation). Show $ze^{\lambda-z} = 1$ has exactly one root in the unit disc for real $\lambda > 1$.

Recognition. Rewrite as a zero-counting problem on $|z| = 1$ and dominate one term by Rouché.

Solution. Set $f(z) = ze^{\lambda-z} - 1$. On $|z| = 1$ write $z = e^{i\phi}$; the dominant term is $g(z) = ze^{\lambda-z}$. For $|z| = 1$, $|g(z)| = |z| e^{\lambda-\operatorname{Re} z} = e^{\lambda-\cos \phi} \geq e^{\lambda-1} > 1$ (since $\lambda > 1$), while the omitted term has modulus 1. So $|f - g| = 1 < |g|$ on $|z| = 1$, and by Rouché f has as many zeros in $\mathbb{D}$ as g . Now $g(z) = ze^{\lambda-z}$ has a single simple zero in $\mathbb{D}$ (at $z = 0$, from the factor z ; $e^{\lambda-z}$ never vanishes). Hence f has exactly one root in $\mathbb{D}$. *Check.* On the real axis $h(x) = xe^{\lambda-x} - 1$ has $h(0) = -1 < 0$ and $h(1) = e^{\lambda-1} - 1 > 0$, so a real root sits in $(0, 1)$ — consistent with “exactly one,” and it is real.

Worked Example 21.3.12 (Hurwitz: a limit of nonvanishing functions). Suppose $f_n \rightarrow f$ uniformly on compacts in a region Ω , each f_n nowhere zero, and f nonconstant. Prove f is also nowhere zero.

Recognition. This is the “zero version” of Proposition 21.3.5; the argument principle again converts a local count into a contour integral that passes to the limit.

Solution. Suppose $f(z_0) = 0$. Since f is nonconstant, z_0 is an isolated zero; choose a circle γ about z_0 with $f \neq 0$ on and inside γ except at z_0 . Then on γ , $f_n \rightarrow f$ and $f'_n \rightarrow f'$ uniformly with $|f|$ bounded below, so

$$0 < \frac{1}{2\pi i} \int_{\gamma} \frac{f'}{f} d\zeta = \lim_n \frac{1}{2\pi i} \int_{\gamma} \frac{f'_n}{f_n} d\zeta = 0,$$

because each f_n has no zero inside γ . Contradiction, so f has no zero. *Check.* “nonconstant” is needed — $f_n(z) = 1/n$ are nowhere zero and converge to the constant 0, which vanishes everywhere; this is precisely the excluded constant limit.

Worked Example 21.3.13 (A maximizer of the derivative is an automorphism). Among holomorphic $f : \mathbb{D} \rightarrow \mathbb{D}$ with $f(0) = 0$, show the supremum of $|f'(0)|$ is 1 and is attained exactly by the rotations.

Recognition. This is the disc-to-disc shadow of the RMT extremal problem; here Schwarz settles it outright.

Solution. Schwarz (iii) gives $|f'(0)| \leq 1$ for every such f , so the supremum is at most 1; the rotations $f(z) = e^{i\theta}z$ attain $|f'(0)| = 1$, so it equals 1. By the equality clause of Schwarz, $|f'(0)| = 1$ forces f to be a rotation. *Check.* This is the toy model for RMT Step 2: maximizing $|f'(z_0)|$ over a normal family selects the “most expanding” map, which fills out the whole disc — here, where $\Omega = \mathbb{D}$ already, the maximizer is just the identity (up to rotation), confirming $F = \text{id}$ is the Riemann map of $\mathbb{D}$ itself.

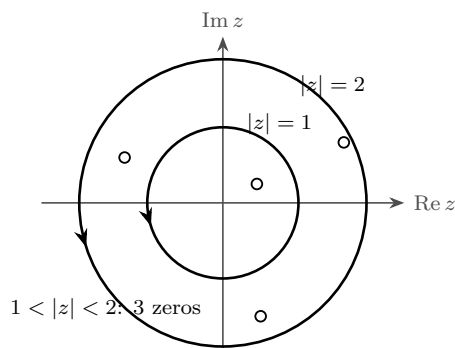


Figure 21.14. Rouché on two circles: $z^4 - 6z + 3$ has 1 zero in $|z| < 1$ and 4 in $|z| < 2$, hence 3 in the annulus. (Schematic.)

Exercises (Stein, Ch.8).

1. Suppose Ω is a simply connected domain bounded by a piecewise-smooth closed curve γ . Prove that any conformal map $F : \mathbb{D} \rightarrow \Omega$ extends to a continuous bijection $\overline{\mathbb{D}} \rightarrow \overline{\Omega}$. (This generalizes the argument of Theorem 21.4.3.)
2. Prove that the complex plane slit along the union of the rays $\bigcup_{k=1}^n \{A_k + iy : y \leq 0\}$ is simply connected. [Hint: given a closed curve, first “raise” it so it lies entirely in the upper half-plane.]

Additional exercises.

1. State precisely the three necessary conditions on a domain Ω for the existence of a conformal map $\Omega \rightarrow \mathbb{D}$, and give the one-line reason each is forced.
2. Show that the family $\mathcal{F} = \{f_n(z) = z^n\}$ on $\mathbb{D}$ is uniformly bounded on compacts and find its compact-uniform limit. Is the limit in $\mathcal{F}$? Is $\mathcal{F}$ normal?
3. Verify Montel’s hypothesis fails to give equicontinuity in the real setting: show that $f_n(x) = \sin(nx)$ on $(0, 1)$ is uniformly bounded but not equicontinuous, and has no subsequence converging uniformly on any compact subinterval.
4. Construct an explicit exhaustion $K_1 \subseteq K_2 \subseteq \cdots$ of the slit plane $\Omega = \mathbb{C} \setminus (-\infty, 0]$, and verify each K_ℓ is compact with $K_\ell \subseteq (^\circ K_{\ell+1})$.
5. Use the proof’s extremal principle to explain, without computing F , why the Riemann map of $\mathbb{D}$ to itself (with $z_0 = 0$) is the identity: show $|f'(0)|$ is maximized by $f(z) = z$ in the family of injective $f : \mathbb{D} \rightarrow \mathbb{D}$ with $f(0) = 0$.
6. (Non-example) Explain why the annulus $\{1 < |z| < 2\}$ is not conformally equivalent to $\mathbb{D}$, despite being a proper open subset of $\mathbb{C}$. Which hypothesis of the Riemann mapping theorem fails?

7. Let $\{f_n\}$ be injective holomorphic on a connected Ω , converging uniformly on compacts to f . Give an example where the limit f is constant (so the “injective-or-constant” dichotomy is genuine), and one where it is injective.
8. Prove the uniqueness part of the Riemann mapping theorem in detail: if $F, G : \Omega \rightarrow \mathbb{D}$ both satisfy $F(z_0) = G(z_0) = 0$ and $F'(z_0), G'(z_0) > 0$, then $F = G$.
9. (Diagram problem) Sketch a proper simply connected Ω (e.g. an L-shaped region), mark a base point z_0 , and draw schematically the normalized Riemann map to $\mathbb{D}$ sending $z_0 \mapsto 0$. Indicate one boundary arc of Ω and its image arc on $\partial\mathbb{D}$.

21.4 Conformal Mappings onto Polygons: Schwarz-Christoffel

The Riemann mapping theorem produces a map to $\mathbb{D}$ but gives no formula. For polygonal targets there is one: the *Schwarz-Christoffel formula*. We motivate it with examples, define the integral, prove it maps $\mathbb{R}$ to a polygonal line, and finally state the boundary-correspondence theorem that pins the map down.

Motivating examples

The power map as an integral. For $0 < \alpha < 2$, the sector map $f(z) = z^\alpha$ (Section 21.1) can be written

$$z^\alpha = \alpha \int_0^z \zeta^{\alpha-1} d\zeta = \alpha \int_0^z \zeta^{-\beta} d\zeta, \quad \alpha + \beta = 1.$$

Since $\beta < 1$, $\zeta^{-\beta}$ is integrable near 0, so $f(0) = 0$. For real $x > 0$, $f'(x) = \alpha x^{\alpha-1} > 0$, so f carries $[0, \infty)$ to $[0, \infty)$. For $x < 0$, $f'(x) = -\alpha |x|^{\alpha-1} e^{i\pi\alpha}$, so f carries $(-\infty, 0]$ to the ray $\arg w = \alpha\pi$. The single “bend” at 0 has exterior angle $\beta\pi$. This is the $n = 1$ Schwarz-Christoffel picture.

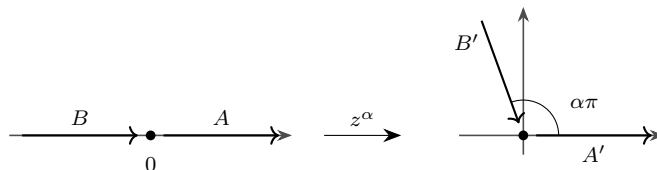


Figure 21.15. Example: $z \mapsto z^\alpha$ as a one-vertex Schwarz-Christoffel map; the bend at 0 has interior angle $\alpha\pi$.

Half-strip. The integral $f(z) = \int_0^z (1 - \zeta^2)^{-1/2} d\zeta = \arcsin z$ (branch positive on $(-1, 1)$) maps $\mathbb{H}$ to the interior of the half-strip bounded by the segments about $-\frac{\pi}{2}, \frac{\pi}{2}$; here the two finite vertices have $\beta = \frac{1}{2}$ each.

Rectangle (elliptic integral). For $0 < k < 1$,

$$f(z) = \int_0^z \frac{d\zeta}{\sqrt{(1 - \zeta^2)(1 - k^2\zeta^2)}}, \quad z \in \mathbb{H},$$

maps the real axis onto a rectangle with vertices $\pm K, \pm K + iK'$, where

$$K = \int_0^1 \frac{dx}{\sqrt{(1 - x^2)(1 - k^2x^2)}}, \quad K' = \int_1^{1/k} \frac{dx}{\sqrt{(x^2 - 1)(1 - k^2x^2)}}.$$

The four branch points $\pm 1, \pm 1/k$ become the four corners ($\beta = \frac{1}{2}$ each), and the point at infinity maps to the midpoint of the top side. These are the *elliptic integrals*.

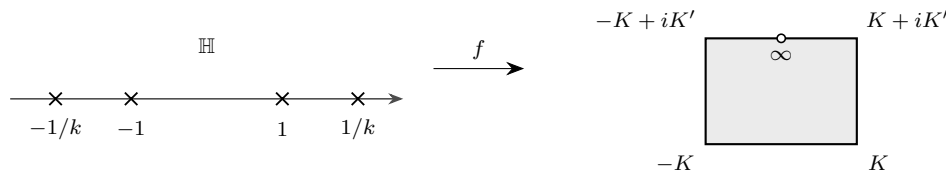


Figure 21.16. The elliptic integral maps $\mathbb{R}$ onto a rectangle; corners are the images of $\pm 1, \pm 1/k$, the midpoint of the top is the image of ∞ .

The Schwarz-Christoffel integral

Definition 21.4.1. Let $A_1 < A_2 < \dots < A_n$ be distinct reals and exponents $\beta_k < 1$ with $1 < \sum_{k=1}^n \beta_k$. The *Schwarz-Christoffel integral* is

$$S(z) = \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \dots (\zeta - A_n)^{\beta_n}},$$

where $(\zeta - A_k)^{\beta_k}$ is the branch (slit along $\{A_k + iy : y \leq 0\}$) positive for real $\zeta > A_k$; for real $x < A_k$ it equals $|x - A_k|^{\beta_k} e^{i\pi\beta_k}$.

The slit plane $\mathbb{C} \setminus \bigcup_k \{A_k + iy : y \leq 0\}$ is simply connected (Exercise 19), so S is holomorphic there. Since $\beta_k < 1$ the singularities $(\zeta - A_k)^{-\beta_k}$ are integrable, so S is continuous up to $\mathbb{R}$, including the A_k . For $|\zeta|$ large, $\prod_k (\zeta - A_k)^{-\beta_k} \sim c|\zeta|^{-\sum \beta_k}$ with $\sum \beta_k > 1$, so the integral converges at infinity; $\lim_{r \rightarrow \infty} S(re^{i\theta}) =: a_\infty$ exists independent of $\theta \in [0, \pi]$. Write $a_k = S(A_k)$.

Proposition 21.4.2. *With S as above:*

- (i) *If $\sum_{k=1}^n \beta_k = 2$ and p is the polygon with vertices $a_1, \dots, a_n$ (in order), then S maps $\mathbb{R}$ onto $p \setminus \{a_\infty\}$. The point a_∞ lies on the side $[a_n, a_1]$ and is the image of ∞ . The interior angle at a_k is $\alpha_k\pi$ with $\alpha_k = 1 - \beta_k$.*
- (ii) *If $1 < \sum_k \beta_k < 2$, the image is the polygon with $n + 1$ vertices $a_1, \dots, a_n, a_\infty$, the angle at a_∞ being $\alpha_\infty\pi$ where $\alpha_\infty = 1 - \beta_\infty$, $\beta_\infty = 2 - \sum_k \beta_k$.*

Proof of (i). For $A_k < x < A_{k+1}$, $S'(x) = \prod_j (x - A_j)^{-\beta_j}$, and

$$\arg S'(x) = \arg \prod_{j>k} (x - A_j)^{-\beta_j} = \arg \prod_{j>k} e^{-i\pi\beta_j} = -\pi \sum_{j>k} \beta_j,$$

constant on (A_k, A_{k+1}) . Since $S(x) = S(A_k) + \int_{A_k}^x S'$, as x runs from A_k to A_{k+1} the image $S(x)$ moves from a_k to a_{k+1} along a straight segment making this constant angle with $\mathbb{R}$. For $x > A_n$, $S'(x) > 0$ (all factors positive); for $x < A_1$, $\arg S'(x) = -\pi \sum_k \beta_k = -2\pi$, again positive. So as x runs over $[A_n, \infty)$ and $(-\infty, A_1]$, $S(x)$ traces two horizontal half-segments meeting at a_∞ , whose union is $[a_n, a_1] \setminus \{a_\infty\}$. The increase of the side-angle across a_k is $\pi\beta_k$, so the interior angle at a_k is $\pi(1 - \beta_k) = \alpha_k\pi$. Case (ii) is analogous. $\square$

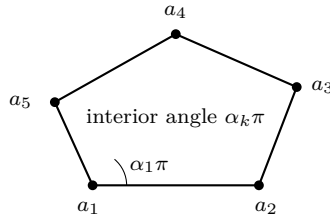


Figure 21.17. $\sum \beta_k = 2$ closes the polygon; interior angle at a_k is $\alpha_k\pi = (1 - \beta_k)\pi$, and $\sum(1 - \alpha_k) = 2$ as for any polygon. (Schematic.)

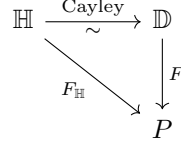
Boundary correspondence and the mapping formula

A *polygonal region* P is a bounded simply connected open set whose boundary is a closed polygon p . The key analytic input is that a conformal map extends to the boundary.

Theorem 21.4.3 (Boundary extension). *If $F : \mathbb{D} \rightarrow P$ is a conformal map onto a polygonal region, then F extends to a continuous bijection $\mathbb{D} \rightarrow \bar{P}$, giving a bijection of $\partial\mathbb{D}$ onto the boundary polygon p .*

Proof outline. The crux is that $\lim_{z \rightarrow z_0} F(z)$ exists for each $z_0 \in \partial\mathbb{D}$. Using $\text{Area}(F(U)) = \iint_U |F'|^2 dx dy < \infty$ (finite, since P is bounded), a Cauchy-Schwarz estimate (Lemma 21.4.4 below) forces the oscillation of F along small circles C_r about z_0 to zero along a sequence $r_n \rightarrow 0$. If F had two distinct limit values $\zeta \neq \zeta'$ along two sequences approaching z_0 , one could join them by curves in P whose F -preimages cross every

small C_r , contradicting the vanishing oscillation. Hence the limit exists; defining $F(z_0)$ as that limit makes F continuous on $\overline{\mathbb{D}}$. The same argument applied to $G = F^{-1}$ (the geometry of small circles meeting ∂P in an arc holds at polygon boundary points too) shows G extends continuously; continuity then forces $G \circ F = \text{id}$ on $\overline{\mathbb{D}}$ and $F \circ G = \text{id}$ on $\overline{P}$. $\square$



Lemma 21.4.4 (Length-area estimate). *For $0 < r < \frac{1}{2}$ let C_r be the circle of radius r about $z_0 \in \partial\mathbb{D}$, and $z_r, z'_r \in \mathbb{D} \cap C_r$. If $\rho(r) = |F(z_r) - F(z'_r)|$, then there is a sequence $r_n \rightarrow 0$ with $\rho(r_n) \rightarrow 0$.*

Proof. If not, $c \leq \rho(r)$ for $0 < r \leq R$. Parametrizing the connecting arc on C_r by $z_0 + re^{i\theta}$, $\theta_1(r) \leq \theta \leq \theta_2(r)$, and using Cauchy-Schwarz,

$$\rho(r)^2 \leq \left(\int_{\theta_1}^{\theta_2} |F'| r d\theta \right)^2 \leq \left(\int_{\theta_1}^{\theta_2} |F'|^2 r d\theta \right) \left(\int_{\theta_1}^{\theta_2} r d\theta \right) \leq 2\pi r \int_{\theta_1}^{\theta_2} |F'|^2 r d\theta.$$

Divide by r and integrate r from 0 to R :

$$c^2 \int_0^R \frac{dr}{r} \leq 2\pi \iint_{\mathbb{D}} |F'|^2 dx dy < \infty.$$

The left side diverges ($1/r$ is not integrable at 0), a contradiction. $\square$

Theorem 21.4.5 (Schwarz-Christoffel mapping formula). *Let P be a polygonal region with vertices $a_1, \dots, a_n$ ($n \geq 3$) in order, interior angles $\alpha_k\pi$, exterior angles $\beta_k\pi$ with $\alpha_k + \beta_k = 1$ (so $\sum_k \beta_k = 2$). Let $F : \mathbb{H} \rightarrow P$ be a conformal map with $F(A_k) = a_k$ for reals $A_1 < \dots < A_n$ (no vertex the image of ∞). Then there exist constants c_1, c_2 with*

$$F(z) = c_1 S(z) + c_2 = c_1 \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \dots (\zeta - A_n)^{\beta_n}} + c_2.$$

If instead ∞ maps to the vertex a_n , the last factor is dropped: $F(z) = C_1 \int_0^z (\zeta - A_1)^{-\beta_1} \dots (\zeta - A_{n-1})^{-\beta_{n-1}} d\zeta + C_2$.

Proof idea. Near each A_k , set $h_k(z) = (F(z) - a_k)^{1/\alpha_k}$. Because F maps the two boundary segments through a_k to two straight segments meeting at angle $\alpha_k\pi$, h_k straightens them: h_k maps a real segment to a real segment with $A_k \mapsto 0$. By the Schwarz reflection principle h_k continues holomorphically across $\mathbb{R}$, and one checks h_k never vanishes. From $\frac{F'''}{F'} = \frac{(\alpha_k - 1)F'}{F - a_k}$ (locally) and the reflection, the function

$$\frac{F''(z)}{F'(z)} + \sum_{k=1}^n \frac{\beta_k}{z - A_k}$$

extends to an entire function vanishing at ∞ (using the expansion $F(z) = c_0 + c_1/z + \dots$ at ∞), hence is identically 0 by Liouville. Therefore $\frac{F''}{F'} = -\sum_k \frac{\beta_k}{z - A_k} = \frac{Q'}{Q}$ with $Q(z) = \prod_k (z - A_k)^{-\beta_k}$. So $(F'/Q)' = 0$, $F' = c_1 Q$, and integrating gives the formula. The ∞ -to-vertex case follows by precomposing with the automorphism $\Phi(z) = A_n^* - 1/z$ of $\mathbb{H}$ and changing variables, which deletes the A_n -factor. $\square$

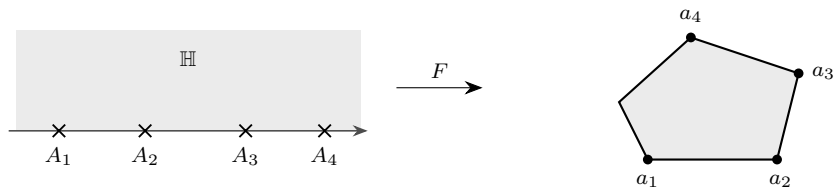


Figure 21.18. F sends the real points A_k to the vertices a_k , with the exterior angle $\beta_k\pi$ controlling the k -th factor. (Schematic.)

Worked Example 21.4.6 (Schwarz-Christoffel for the half-strip). Recover the map $\mathbb{H} \rightarrow$ (half-strip) and identify its two exterior angles.

Recognition. Two finite vertices with right interior angles; $\alpha_k = \frac{1}{2}$ so $\beta_k = \frac{1}{2}$.

Solution. Take $A_1 = -1, A_2 = 1, \beta_1 = \beta_2 = \frac{1}{2}$. Then $S(z) = \int_0^z (\zeta + 1)^{-1/2} (\zeta - 1)^{-1/2} d\zeta = \int_0^z (\zeta^2 - 1)^{-1/2} d\zeta$. With the branch positive on $(-1, 1)$ this is (up to constants) $\arcsin z$, mapping $\mathbb{H}$ to the half-strip $\{-\frac{\pi}{2} < \operatorname{Re} w < \frac{\pi}{2}, \operatorname{Im} w > 0\}$. *Check.* The two interior angles are $\alpha_k\pi = \frac{\pi}{2}$ (right angles at the strip's two finite corners), and $\sum \beta_k = 1 < 2$, consistent with case (ii): the “third vertex” is at infinity (the open end of the strip).

Worked Example 21.4.7 (Exterior angles of a square). Set up the Schwarz-Christoffel integral mapping $\mathbb{H}$ to a square, and verify the angle sum.

Recognition. Four vertices, each interior angle $\pi/2$, so each $\beta_k = \frac{1}{2}$, $\sum \beta_k = 2$.

Solution. Choose $A_k = -1, 0, 1, 2$ (any four reals). The integral is $F(z) = c_1 \int_0^z \prod_{k=1}^4 (\zeta - A_k)^{-1/2} d\zeta + c_2$.

Check. $\sum_{k=1}^4 \beta_k = 4 \cdot \frac{1}{2} = 2$, the closing condition of Proposition 21.4.2(i); and the interior angles $\alpha_k\pi = (1 - \frac{1}{2})\pi = \frac{\pi}{2}$ are the four right angles of a square. The side ratios are fixed once the A_k are chosen; making all sides equal requires symmetric A_k (e.g. $\pm 1, \pm 1/k$ with k tuned), which is the elliptic-integral picture.

Worked Example 21.4.8 (Reading off the inverse: doubly periodic sn). Explain why inverting the rectangle map $I(z) = \int_0^z \frac{d\zeta}{\sqrt{(1-\zeta^2)(1-k^2\zeta^2)}}$ produces a doubly periodic function.

Recognition. The forward map sends $\mathbb{H}$ to a rectangle R ; reflecting R tiles a strip, then the plane, and the inverse inherits the lattice of translations.

Solution. Let $\operatorname{sn} = I^{-1}$, mapping the closed rectangle R_0 onto $\overline{\mathbb{H}}$. By the Schwarz reflection principle, reflecting R_0 across its lower side to R_1 extends sn with $\operatorname{sn}(z) = \overline{\operatorname{sn}(\bar{z})}$; reflecting again to R_2 gives $\operatorname{sn}(z) = \operatorname{sn}(z + 2iK')$ after recombining. Reflecting horizontally similarly yields $\operatorname{sn}(z) = \operatorname{sn}(z + 4K)$. *Check.* sn is then doubly periodic with periods $4K$ and $2iK'$, and its only singularities are poles — it is an elliptic function (the subject of the next chapter). The identity $K'(k) = K(\tilde{k})$ with $\tilde{k}^2 = 1 - k^2$ (Exercise 24) makes the construction consistent.

Worked Example 21.4.9 (Regular n -gon exponents). For the map of $\mathbb{D}$ onto a regular n -gon, determine each exponent β_k and verify the closing relation.

Recognition. A regular n -gon has all interior angles equal; $\alpha_k = \frac{n-2}{n}$, and $\beta_k = 1 - \alpha_k$.

Solution. The interior angles of a regular n -gon are $\frac{(n-2)\pi}{n}$, so $\alpha_k = \frac{n-2}{n}$ and $\beta_k = 1 - \frac{n-2}{n} = \frac{2}{n}$ for each of the n vertices. *Check.* $\sum_{k=1}^n \beta_k = n \cdot \frac{2}{n} = 2$, the closing condition of Proposition 21.4.2(i). This matches the formula $F(z) = \int_1^z (1 - \zeta^n)^{-2/n} d\zeta$ of Exercise 23, whose integrand exponents are exactly $2/n$ at the n -th roots of unity B_k .

Worked Example 21.4.10 (A triangle's angles sum correctly). For the triangle map $\int_0^z \zeta^{-\beta_1} (1 - \zeta)^{-\beta_2} d\zeta$ with the ∞ -vertex, write the three exterior angles and check the angle sum.

Recognition. Two finite prevertices $0, 1$ and one at ∞ ; the ∞ -angle uses $\beta_3 = 2 - \beta_1 - \beta_2$.

Solution. The exponents are β_1, β_2 at $0, 1$, and $\beta_3 = 2 - \beta_1 - \beta_2$ at the image of ∞ . The interior angles are $\alpha_j\pi = (1 - \beta_j)\pi$. *Check.* $\sum_{j=1}^3 \alpha_j = \sum_{j=1}^3 (1 - \beta_j) = 3 - \sum_{j=1}^3 \beta_j = 3 - 2 = 1$, so $\alpha_1\pi + \alpha_2\pi + \alpha_3\pi = \pi$ — the interior angles of a triangle sum to π , as they must. The constraint $1 < \beta_1 + \beta_2 < 2$ keeps $\beta_3 \in (0, 1)$, i.e. a genuine (non-degenerate) third angle.

Caution. Proposition 21.4.2 only guarantees that S maps $\mathbb{R}$ onto a polygonal *line*; for generic A_k that line may self-intersect, and S need not be conformal on $\mathbb{H}$. Theorem 21.4.5 runs the other way: *given* a genuine polygonal region P , the conformal map onto it has Schwarz-Christoffel form for suitable A_k . Choosing the A_k to realize a prescribed P (the “accessory parameter problem”) is generally transcendental.

Further worked examples

Worked Example 21.4.11 (The half-strip width from an explicit integral). Verify that $S(z) = \int_0^z (\zeta^2 - 1)^{-1/2} d\zeta$ (branch positive on $(-1, 1)$) sends the two prevertices $A_1 = -1, A_2 = 1$ to points a distance π apart, matching the half-strip width.

Recognition. Evaluate the boundary increment $a_2 - a_1 = S(1) - S(-1)$ directly; the integrand is the arcsin derivative.

Solution. On $(-1, 1)$, $(\zeta^2 - 1)^{-1/2} = \frac{1}{i\sqrt{1 - \zeta^2}}$ with the stated branch, so $S(z) = \frac{1}{i} \arcsin z = -i \arcsin z$ up to the constant. Then

$$a_2 - a_1 = S(1) - S(-1) = -i(\arcsin 1 - \arcsin(-1)) = -i\left(\frac{\pi}{2} + \frac{\pi}{2}\right) = -i\pi,$$

of modulus π . *Check.* The half-strip $\{-\frac{\pi}{2} < \operatorname{Re} w < \frac{\pi}{2}\}$ has width π between its two vertical edges, exactly $|a_2 - a_1|$; the factor $-i$ reflects that the segment $[a_1, a_2]$ is the (horizontal) base of the strip after the standard rotation.

Worked Example 21.4.12 (Exterior-angle bookkeeping for an isosceles triangle). A triangle has interior angles $\frac{\pi}{2}, \frac{\pi}{4}, \frac{\pi}{4}$. Write the three exponents β_k and confirm the Schwarz-Christoffel closing relation.

Recognition. Each $\beta_k = 1 - \alpha_k$ with $\alpha_k \pi$ the interior angle; for a bounded triangle $\sum \beta_k = 2$.

Solution. The interior angles give $\alpha_1 = \frac{1}{2}, \alpha_2 = \alpha_3 = \frac{1}{4}$, so

$$\beta_1 = 1 - \frac{1}{2} = \frac{1}{2}, \quad \beta_2 = \beta_3 = 1 - \frac{1}{4} = \frac{3}{4}.$$

The integral is $F(z) = c_1 \int_0^z (\zeta - A_1)^{-1/2} (\zeta - A_2)^{-3/4} (\zeta - A_3)^{-3/4} d\zeta + c_2$. *Check.* $\sum_k \beta_k = \frac{1}{2} + \frac{3}{4} + \frac{3}{4} = 2$, the closing relation of Proposition 21.4.2(i); equivalently $\sum_k \alpha_k = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$, so the interior angles sum to π , as a triangle's must.

Worked Example 21.4.13 (A degenerate vertex: angle 0 and angle 2π). Interpret the exponents $\beta = 1$ (interior angle 0) and $\beta = -1$ (interior angle 2π) in the Schwarz-Christoffel data.

Recognition. The relation $\alpha\pi = (1 - \beta)\pi$ extends past the polygon range; the endpoints $\beta = 1$ and $\beta = -1$ are the “cusp” and “slit” degenerations.

Solution. When $\beta = 1, \alpha = 0$: the two sides meeting at the vertex become *parallel* (a zero interior angle), as at the open end of a half-strip — consistent with the half-strip having a vertex “at infinity” carrying $\beta = 1$. When $\beta = -1, \alpha = 2$: the interior angle is 2π , a vertex that is the *tip of a slit* pointing into the region. *Check.* The slit half-plane example $g(z) = \sqrt{z^2 + 1}$ of Section 21.3 heals a slit tip; in Schwarz-Christoffel language the tip is a $\beta = -1$ vertex, and indeed $\sum \beta_k$ must still total 2 once all (possibly degenerate) vertices are counted.

Worked Example 21.4.14 (Schwarz-Christoffel for a half-plane minus a slit). Set up the map $\mathbb{H} \rightarrow \mathbb{H} \setminus \{iy : 0 < y \leq h\}$ (upper half-plane with a vertical slit of height h) and identify the special vertex.

Recognition. The slit's tip is a 2π vertex ($\beta = -1$); the two reals where the boundary turns carry $\beta = \frac{1}{2}$ each, summing to 2.

Solution. Place prevertices $A_1 = -1, A_2 = 0, A_3 = 1$ with exponents $\beta_1 = \frac{1}{2}, \beta_2 = -1, \beta_3 = \frac{1}{2}$ (the $A_2 = 0$ being the slit tip). Then

$$F(z) = c_1 \int_0^z (\zeta + 1)^{-1/2} \zeta^1 (\zeta - 1)^{-1/2} d\zeta + c_2 = c_1 \int_0^z \frac{\zeta d\zeta}{\sqrt{\zeta^2 - 1}} + c_2 = c_1 \sqrt{z^2 - 1} + c_2.$$

Choosing c_1, c_2 to fix the geometry gives the familiar $F(z) = h\sqrt{z^2 + 1}$ (after the real shift $z \mapsto \text{rescaled}$), the map healing the slit. *Check.* $\sum \beta_k = \frac{1}{2} - 1 + \frac{1}{2} = 0$, *not* 2 — but this is the $\mathbb{H}$ -target (unbounded) case, where the missing 2 is supplied by the vertex at ∞ ; the bounded closing relation does not apply when the polygon is an unbounded region like a slit half-plane.

Worked Example 21.4.15 (Sum of exterior angles for any convex polygon). Use the Schwarz-Christoffel exponents to re-derive that the exterior angles of any convex n -gon sum to 2π .

Recognition. The closing relation $\sum \beta_k = 2$ is exactly the statement, since the k -th exterior angle is $\beta_k\pi$.

Solution. By Proposition 21.4.2(i), a bounded polygon corresponds to exponents with $\sum_{k=1}^n \beta_k = 2$. The exterior angle at vertex a_k is $\beta_k\pi$, so

$$\sum_{k=1}^n (\text{exterior angle at } a_k) = \sum_{k=1}^n \beta_k\pi = 2\pi.$$

Check. For the regular n -gon each $\beta_k = \frac{2}{n}$ (worked above), and $n \cdot \frac{2}{n} \cdot \pi = 2\pi$; for a triangle the three exterior angles $(\pi - \alpha_k\pi)$ sum to $3\pi - \pi = 2\pi$. The Schwarz-Christoffel closing relation is the analytic face of this elementary-geometry fact.

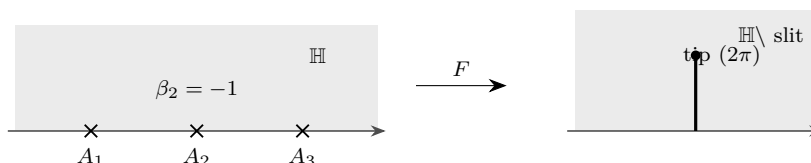


Figure 21.19. Slit half-plane as a Schwarz-Christoffel image: the slit tip is a $\beta = -1$ (interior angle 2π) vertex; $F(z) = c\sqrt{z^2 - 1} + c'$.

Exercises (Stein, Ch.8).

- Further elliptic integrals giving conformal maps of $\mathbb{H}$ onto rectangles. (a) The function $\int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta-1)(\zeta-\lambda)}}$, with $\lambda \in \mathbb{R}$, $\lambda \neq 1$, maps $\mathbb{H}$ conformally onto a rectangle, one of whose vertices is the image of the point at infinity. (b) For $\lambda = -1$, the image of $\int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta^2-1)}}$ is a square whose side length is $\frac{\Gamma(1/4)^2}{2\sqrt{2\pi}}$.
- Conformal mappings to triangles. (a) Show that $\int_0^z \zeta^{-\beta_1}(1-\zeta)^{-\beta_2} d\zeta$, with $0 < \beta_1, \beta_2 < 1$ and $1 < \beta_1 + \beta_2 < 2$, maps $\mathbb{H}$ to a triangle whose vertices are the images of $0, 1, \infty$, with angles $\alpha_1\pi, \alpha_2\pi, \alpha_3\pi$ where $\alpha_j + \beta_j = 1$ and $\beta_1 + \beta_2 + \beta_3 = 2$. (b) What happens when $\beta_1 + \beta_2 = 1$? (c) What happens when $0 < \beta_1 + \beta_2 < 1$? (d) In (a), the side opposite the angle $\alpha_j\pi$ has length $\frac{\sin(\alpha_j\pi)}{\pi} \Gamma(\alpha_1)\Gamma(\alpha_2)\Gamma(\alpha_3)$.
- If P is a simply connected region bounded by a polygon with vertices $a_1, \dots, a_n$ and angles $\alpha_1\pi, \dots, \alpha_n\pi$, and $F: \mathbb{D} \rightarrow P$ is conformal, then there exist $B_1, \dots, B_n$ on the unit circle and constants c_1, c_2 with

$$F(z) = c_1 \int_0^z \frac{d\zeta}{(\zeta - B_1)^{\beta_1} \dots (\zeta - B_n)^{\beta_n}} + c_2.$$

[Hint: combine the standard $\mathbb{H} \leftrightarrow \mathbb{D}$ correspondence with an argument like the proof of Theorem 21.4.5 for the ∞ -vertex case.]

- If $F(z) = \int_1^z \frac{d\zeta}{(1-\zeta^n)^{2/n}}$, then F maps $\mathbb{D}$ conformally onto the interior of a regular polygon with n sides and perimeter $2n \int_0^{\pi/(n-2)} (\sin \theta)^{-2/n} d\theta$.
- The elliptic integrals K, K' (for $0 < k < 1$),

$$K(k) = \int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}}, \quad K'(k) = \int_1^{1/k} \frac{dx}{\sqrt{(x^2-1)(1-k^2x^2)}},$$

satisfy: (a) if $\tilde{k}^2 = 1 - k^2$, $0 < \tilde{k} < 1$, then $K'(\tilde{k}) = K(k)$ [Hint: substitute $x = (1 - \tilde{k}^2 y^2)^{-1/2}$ in K']; (b) $K(k) = \frac{2}{1+\tilde{k}} K\left(\frac{1-\tilde{k}}{1+\tilde{k}}\right)$ [Hint: $x = 2t/(1+\tilde{k} + (1-\tilde{k})t^2)$]; (c) $K(k) = \frac{\pi}{2} F(\frac{1}{2}, \frac{1}{2}, 1; k^2)$, F the hypergeometric series [Hint: use the integral representation of F from Ch.6].

Additional exercises.

1. For the one-vertex Schwarz-Christoffel map $z^\alpha = \alpha \int_0^z \zeta^{-\beta} d\zeta$ ($\alpha + \beta = 1$), compute the exterior angle $\beta\pi$ when $\alpha = \frac{2}{3}$, and describe the resulting bend of the real axis.
2. Set up the Schwarz-Christoffel integral mapping $\mathbb{H}$ to an equilateral triangle. Specify the three exponents β_k and check $\sum \beta_k = 2$.
3. Set up the integral for a regular pentagon and for a regular n -gon mapped from $\mathbb{H}$, giving each β_k in terms of n .
4. Verify the closing condition: for any convex polygon with interior angles $\alpha_1\pi, \dots, \alpha_n\pi$, show $\sum_k \beta_k = \sum_k (1 - \alpha_k) = 2$ using the fact that the interior angles of an n -gon sum to $(n - 2)\pi$.
5. For the half-strip map $S(z) = \int_0^z (\zeta^2 - 1)^{-1/2} d\zeta$, compute $S(0)$ and identify S with $\arcsin z$; determine the images of $z = \pm 1$.
6. (Non-example) Explain why $\int_0^z (\zeta - A_1)^{-\beta_1} (\zeta - A_2)^{-\beta_2} d\zeta$ with $\beta_1 + \beta_2 = \frac{1}{2}$ does *not* produce a bounded polygonal image, and relate this to the hypothesis $1 < \sum \beta_k$ in Definition 21.4.1.
7. Using the elliptic-integral identity $K'(k) = K(\tilde{k})$ with $\tilde{k}^2 = 1 - k^2$, explain why $k \mapsto K(k)/K'(k)$ is strictly increasing on $(0, 1)$, and why this forces the modulus in the rectangle construction to be uniquely determined.
8. (Diagram problem) Draw the upper half-plane with four marked real points $A_1 < A_2 < A_3 < A_4$ and, to its right, a quadrilateral $a_1a_2a_3a_4$, with arrows indicating $F(A_k) = a_k$. Mark the exterior angle $\beta_k\pi$ at one vertex and the interior angle $\alpha_k\pi$ there.
9. Show that two different choices of the three real prevertices A_1, A_2, A_3 for a triangle map differ by an automorphism of $\mathbb{H}$, and conclude that the triangle is determined up to similarity by the angles alone.

Chapter 22

An Introduction to Elliptic Functions

22.1 Elliptic Functions and the Weierstrass P-Function

Doubly periodic functions and the lattice

We study meromorphic functions f on $\mathbb{C}$ that possess *two* independent periods. Concretely there are nonzero $\omega_1, \omega_2 \in \mathbb{C}$ with

$$f(z + \omega_1) = f(z) \quad \text{and} \quad f(z + \omega_2) = f(z) \quad \text{for all } z \in \mathbb{C}.$$

Such an f is called *doubly periodic*. The interesting case is when ω_1 and ω_2 are linearly independent over $\mathbb{R}$, i.e. $\tau := \omega_2/\omega_1 \notin \mathbb{R}$. (If the ratio were real the function collapses to a singly periodic or constant one — see the exercises.) Of τ and $1/\tau$ exactly one has positive imaginary part, so after possibly swapping $\omega_1 \leftrightarrow \omega_2$ we may assume $\text{Im}(\tau) > 0$.

A clean normalization removes one period. If $F(z) = f(\omega_1 z)$, then f has periods ω_1, ω_2 iff F has periods $1, \tau$, and f is meromorphic iff F is. So *without loss of generality* we study meromorphic f with periods 1 and τ , $\text{Im}(\tau) > 0$. Iterating periodicity gives

$$f(z + n + m\tau) = f(z) \quad \text{for all } n, m \in \mathbb{Z}, z \in \mathbb{C}. \quad (22.1)$$

The normalization $F(z) = f(\omega_1 z)$ is the composite of f with multiplication by ω_1 , and it carries the period lattice $\Lambda_{\omega_1, \omega_2} = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ onto the normalized lattice $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$.

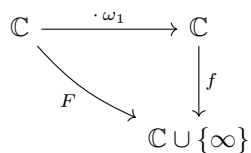


Figure 22.1. Normalizing the periods: $F = f \circ (\cdot \omega_1)$, so F has periods $1, \tau$ exactly when f has periods $\omega_1, \omega_2 = \tau\omega_1$. The arrow $\cdot \omega_1$ is multiplication by ω_1 .

Definition 22.1.1. The *lattice* generated by 1 and τ is

$$\Lambda = \{n + m\tau : n, m \in \mathbb{Z}\}.$$

We say 1 and τ *generate* Λ . The *fundamental parallelogram* is

$$P_0 = \{a + b\tau : 0 \leq a < 1, 0 \leq b < 1\}.$$

Two points z, w are *congruent modulo* Λ , written $z \sim w$, if $z - w \in \Lambda$. A *period parallelogram* is any translate $P = P_0 + h$, $h \in \mathbb{C}$.

Equation (22.1) says f is invariant under translation by every element of Λ : if $z \sim w$ then $f(z) = f(w)$. The lattice tiles the plane, and f is then completely pinned down by its restriction to a single tile.

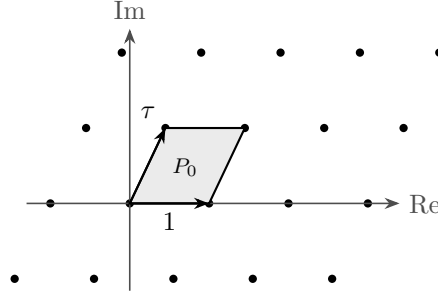


Figure 22.2. The lattice $\Lambda = \{n + m\tau : n, m \in \mathbb{Z}\}$ generated by 1 and τ ($\text{Im } \tau > 0$), with the fundamental parallelogram P_0 shaded. Each dot is a lattice point; translates of P_0 by Λ tile the plane. (Schematic; a specific τ is drawn.)

Proposition 22.1.2. *Let f be meromorphic with periods 1, τ generating Λ . Then: (i) every point of $\mathbb{C}$ is congruent to a unique point of P_0 ; (ii) every point of $\mathbb{C}$ is congruent to a unique point of any given period parallelogram; (iii) Λ tiles the plane, $\mathbb{C} = \bigsqcup_{n,m \in \mathbb{Z}} (n + m\tau + P_0)$ (disjoint union); (iv) f is determined by its values on any period parallelogram.*

Proof. (i) Write $z = a + b\tau$ with $a, b \in \mathbb{R}$ (possible since 1, τ are an $\mathbb{R}$ -basis of $\mathbb{C}$). Let $n = [a]$, $m = [b]$ and $w = z - n - m\tau = (a - n) + (b - m)\tau$. Then $z \sim w$ and $0 \leq a - n < 1$, $0 \leq b - m < 1$, so $w \in P_0$. For uniqueness, if $w = a + b\tau$ and $w' = a' + b'\tau$ both lie in P_0 and $w - w' \in \Lambda$, then $a - a'$ and $b - b'$ are integers with $|a - a'| < 1$ and $|b - b'| < 1$, forcing $a = a'$, $b = b'$, hence $w = w'$. (ii) Apply (i) to $z - h$. (iii) Immediate from (i). (iv) From $z \sim w \Rightarrow f(z) = f(w)$ and (i)–(ii). $\square$

Part (iv) is best read as a factorization. Since $f(z) = f(w)$ whenever $z \sim w$, the value of f depends only on the class of z modulo Λ , so f descends through the quotient $\pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda$ to a well-defined map $\bar{f}$ on the torus.

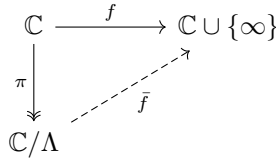


Figure 22.3. An elliptic function f is constant on Λ -cosets, so it factors through the quotient $\pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda$ (a surjection) as $f = \bar{f} \circ \pi$. The induced map $\bar{f}$ on the torus $\mathbb{C}/\Lambda$ is the dashed arrow.

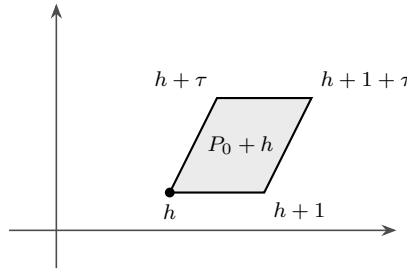


Figure 22.4. A period parallelogram $P = P_0 + h$, the translate of the fundamental parallelogram by $h \in \mathbb{C}$. Every $z \in \mathbb{C}$ is congruent mod Λ to a unique point of P , so f is determined by its values on P . (Schematic.)

Liouville's theorems for elliptic functions

Theorem 22.1.3. *An entire doubly periodic function is constant.*

Proof. By Proposition 22.1.2(iv), f is determined by its values on P_0 , whose closure $\overline{P_0}$ is compact. A continuous (here holomorphic, hence continuous) function on a compact set is bounded, so f is bounded on all of $\mathbb{C}$. By Liouville's theorem (Chapter 2) f is constant. $\square$

This is precisely why we insist on *meromorphic* f : to get something nonconstant we must allow poles.

Definition 22.1.4. A nonconstant doubly periodic meromorphic function is an *elliptic function*. The total number of poles in P_0 , counted with multiplicity, is its *order*.

Since a meromorphic function has only finitely many poles and zeros in any bounded region, an elliptic function has finitely many of each in P_0 . The next two theorems are the structural backbone, and the proof technique — integrating around ∂P_0 and cancelling opposite sides by periodicity — recurs constantly.

Method (Cancelling an integral over ∂P_0 by periodicity). *Recognition trigger.* You integrate a Λ -periodic integrand g (such as f , f'/f , or zf'/f) over the boundary of a period parallelogram and want the value.

1. Split ∂P_0 into the four sides: $0 \rightarrow 1$, $1 \rightarrow 1 + \tau$, $1 + \tau \rightarrow \tau$, $\tau \rightarrow 0$ (positively oriented).
2. Pair opposite sides. On the pair $0 \rightarrow 1$ and $1 + \tau \rightarrow \tau$, substitute $z \mapsto z + \tau$ on the second; periodicity $g(z + \tau) = g(z)$ makes it the *negative* of the first, so they cancel. Likewise for the pair involving the shift by 1.
3. Conclude $\int_{\partial P_0} g = 0$. If poles or zeros sit on ∂P_0 , translate P_0 slightly first to clear them.

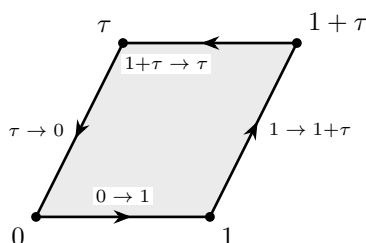


Figure 22.5. The boundary ∂P_0 of the fundamental parallelogram, positively (counterclockwise) oriented into the four edges. For a Λ -periodic integrand the side $0 \rightarrow 1$ cancels $1 + \tau \rightarrow \tau$, and $1 \rightarrow 1 + \tau$ cancels $\tau \rightarrow 0$, so the contour integral vanishes. (Schematic.)

Theorem 22.1.5. The total number of poles of an elliptic function in P_0 is always ≥ 2 . Equivalently, f cannot have a single simple pole (though a single double pole is allowed).

Proof. Assume first f has no poles on ∂P_0 . By the residue theorem, $\int_{\partial P_0} f(z) dz = 2\pi i \sum \text{Res } f$. Apply the Method above to $g = f$: opposite sides cancel, e.g.

$$\int_0^1 f(z) dz + \int_{1+\tau}^{\tau} f(z) dz = \int_0^1 f(z) dz + \int_1^0 f(s + \tau) ds = \int_0^1 f(z) dz - \int_0^1 f(s) ds = 0,$$

and similarly for the other pair. Hence $\sum \text{Res } f = 0$. If f had a single simple pole its residue would be its only residue and would have to vanish — impossible for a simple pole. So f has ≥ 2 poles (counted with multiplicity). If a pole lies on ∂P_0 , translate to $P = h + P_0$ with no poles on ∂P and argue there. $\square$

Theorem 22.1.6. Every elliptic function of order m has exactly m zeros in P_0 (with multiplicity).

Proof. With no zeros or poles on ∂P_0 , the argument principle gives $\frac{1}{2\pi i} \int_{\partial P_0} \frac{f'(z)}{f(z)} dz = N_z - N_p$, the number of zeros minus poles in P_0 . The integrand f'/f is Λ -periodic (since f is), so by the Method its integral over ∂P_0 vanishes. Hence $N_z = N_p = m$. Boundary zeros/poles are handled by translating P_0 . $\square$

Corollary 22.1.7. If f is elliptic of order m then for every $c \in \mathbb{C}$ the equation $f(z) = c$ has exactly m solutions in P_0 (with multiplicity), because $f - c$ is elliptic of the same order.

Worked Example 22.1.8 (No elliptic function with a single simple pole). Explain why there is no elliptic function whose only singularity in P_0 is one simple pole at z_0 , while there *can* be one with a single double pole.

Recognition. Trigger: an existence/non-existence claim about pole configuration. Use Theorem 22.1.5: the residues sum to zero.

Solution. A single simple pole at z_0 gives $\sum \text{Res } f = \text{Res}_{z=z_0} f =: r$. But the ∂P_0 argument forces $\sum \text{Res } f = 0$, so $r = 0$ — contradicting simplicity ($r \neq 0$ for a genuine simple pole). A single *double* pole $f \sim a/(z - z_0)^2 + b/(z - z_0) + \cdots$ has residue b , and we may have $b = 0$ (e.g. an even part only), so $\sum \text{Res} = 0$ is satisfied.

Check. The Weierstrass $\wp$ (below) has exactly a double pole at each lattice point with principal part $1/z^2$, residue 0 — consistent, and indeed $\wp$ exists.

Worked Example 22.1.9 (Order from a partial-fraction model). Suppose an elliptic f has, in P_0 , simple poles at a and b ($a \neq b$) with residues $+r$ and $-r$. Confirm this is consistent with $\sum \text{Res} = 0$ and state the order and the number of zeros.

Recognition. Trigger: count poles/zeros from prescribed principal parts; check the residue sum.

Solution. The poles are both simple, so the order (total pole multiplicity) is 2. The residues sum to $r + (-r) = 0$, consistent with Theorem 22.1.5. By Theorem 22.1.6 f has exactly 2 zeros in P_0 , and by Corollary 22.1.7 $f(z) = c$ has 2 solutions for every c . *Check.* The function $\wp'(z)$ has a single triple pole (order 3) at lattice points and exactly 3 zeros — the half-periods — matching the count.

Building an elliptic function: the Weierstrass $\wp$

We now *construct* an elliptic function, settling the existence question. The model is the single-period case. To make a function of period 1 with poles at every integer, one might try $\sum_n 1/(z + n)$, but this diverges. Two cures:

$$F(z) = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \frac{1}{z + n} = \frac{1}{z} + \sum_{n \geq 1} \left(\frac{1}{z + n} + \frac{1}{z - n} \right),$$

where pairing $\pm n$ makes the bracket $O(1/n^2)$; or

$$F(z) = \frac{1}{z} + \sum_{n \neq 0} \left(\frac{1}{z + n} - \frac{1}{n} \right),$$

where the subtracted $1/n$ makes each term $O(1/n^2)$. Both give the same meromorphic function, in fact $F(z) = \pi \cot \pi z$ (Chapter 5). For the doubly periodic case we mimic the *second* cure on the square of the lattice sum.

Lemma 22.1.10. For $\Lambda^* = \Lambda \setminus \{0\}$ the two series

$$\sum_{(n,m) \neq (0,0)} \frac{1}{(|n| + |m|)^r} \quad \text{and} \quad \sum_{n+m\tau \in \Lambda^*} \frac{1}{|n + m\tau|^r}$$

converge if and only if $r > 2$.

Proof. First series. Sum in m then n . For $n \neq 0$, comparing the tail with an integral,

$$\sum_{m \in \mathbb{Z}} \frac{1}{(|n| + |m|)^r} = \frac{1}{|n|^r} + 2 \sum_{m \geq 1} \frac{1}{(|n| + m)^r} \leq \frac{1}{|n|^r} + 2 \int_{|n|}^{\infty} \frac{dx}{x^r} \leq \frac{1}{|n|^r} + \frac{C}{|n|^{r-1}}.$$

Summing over $n \neq 0$ (plus the $n = 0$ slice, a convergent $\sum 1/|m|^r$) gives a finite total when $r > 2$. *Second series.* It suffices to find $c > 0$ with $|n| + |m| \leq c|n + m\tau|$ for all integers n, m ; then the second sum is dominated by a constant times the first. Writing $\tau = s + it$ ($t > 0$), with $A, B > 0$ one has $(A^2 + B^2)^{1/2} \approx A + B$ (since $A, B \leq (A^2 + B^2)^{1/2} \leq A + B$), so

$$|n + m\tau| = [(n + ms)^2 + (mt)^2]^{1/2} \approx |n + ms| + |mt| \approx |n + ms| + |m| \approx |n| + |m|,$$

the last $\approx$ by splitting the cases $|n| \leq 2|m||s|$ and $|n| \geq 2|m||s|$. Thus $|n| + |m| \approx |n + m\tau|$ for $\tau \in \mathbb{H}$, and convergence transfers. The series diverges at $r = 2$ (see the exercises). $\square$

Remark. The estimate shows that for $r > 2$ the series $\sum |n + m\tau|^{-r}$ converges *uniformly* in every half-plane $\text{Im}(\tau) \geq \delta > 0$. This uniformity is what later makes Eisenstein series holomorphic in τ .

Definition 22.1.11. The *Weierstrass $\wp$ -function* is

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left(\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right) = \frac{1}{z^2} + \sum_{(n,m) \neq (0,0)} \left(\frac{1}{(z + n + m\tau)^2} - \frac{1}{(n + m\tau)^2} \right).$$

The subtracted $1/\omega^2$ is forced: without it the bare sum $\sum 1/(z + \omega)^2$ diverges. With it, the bracketed term is

$$\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} = \frac{-z^2 - 2z\omega}{(z + \omega)^2\omega^2} = O\left(\frac{1}{|\omega|^3}\right) \quad (|\omega| \rightarrow \infty),$$

which is summable by Lemma 22.1.10 ($r = 3$).

Theorem 22.1.12. $\wp$ is an elliptic function with periods 1 and τ and a double pole at each lattice point.

Proof. Meromorphy with double poles. Fix $R > 0$. For $|z| < R$ split

$$\wp(z) = \frac{1}{z^2} + \sum_{|\omega| \leq 2R} \left(\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right) + \sum_{|\omega| > 2R} \left(\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right).$$

The tail terms are $O(1/|\omega|^3)$ uniformly for $|z| < R$, so the second sum is holomorphic on $|z| < R$ (Lemma 22.1.10); the first finite sum exhibits double poles at the lattice points in $|z| < R$. As R is arbitrary, $\wp$ is meromorphic with double poles exactly on Λ . *Periodicity.* Differentiating termwise,

$$\wp'(z) = -2 \sum_{n,m \in \mathbb{Z}} \frac{1}{(z + n + m\tau)^3},$$

which converges absolutely off Λ ($r = 3$) and is *manifestly* periodic — the shift $z \mapsto z + 1$ or $z \mapsto z + \tau$ permutes the lattice and the $-1/\omega^2$ terms are gone. Hence $\wp(z + 1) = \wp(z) + a$ and $\wp(z + \tau) = \wp(z) + b$ for constants a, b . But $\wp$ is *even*, $\wp(-z) = \wp(z)$ (replace ω by $-\omega$ in the sum). Setting $z = -1/2$ in the first relation: $\wp(1/2) = \wp(-1/2) + a = \wp(1/2) + a$, so $a = 0$; setting $z = -\tau/2$ in the second gives $b = 0$. Thus $\wp$ is doubly periodic, of order 2. $\square$

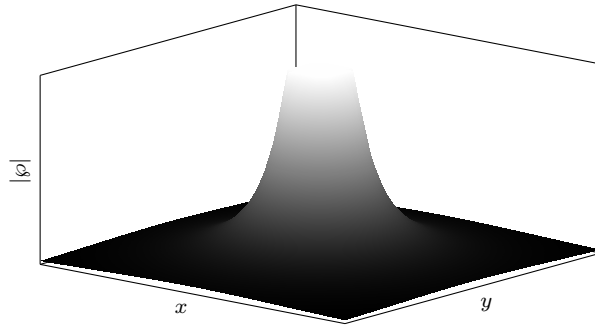


Figure 22.6. Modulus surface $|\wp(z)|$ over a neighborhood of a lattice point: the double pole at the origin spikes to $+\infty$ like $1/|z|^2$. Height is clipped at a finite level for display. (Schematic.)

Structure: the differential equation and universality

Since $\wp$ is even, $\wp'$ is odd: $\wp'(-z) = -\wp'(z)$. Combined with periodicity, at the *half-periods* $1/2$, $\tau/2$, $(1+\tau)/2$ we get $\wp' = 0$. For instance $\wp'(1/2) = -\wp'(-1/2) = -\wp'(-1/2 + 1) = -\wp'(1/2)$, so $\wp'(1/2) = 0$. As $\wp'$ is elliptic of order 3, these three half-periods are its *only* zeros in P_0 , each simple. Setting

$$e_1 = \wp(1/2), \quad e_2 = \wp(\tau/2), \quad e_3 = \wp\left(\frac{1+\tau}{2}\right),$$

the equation $\wp(z) = e_j$ has a double root at the corresponding half-period; since $\wp$ has order 2 there are no others, so e_1, e_2, e_3 are *distinct* (otherwise $\wp$ would attain a value four times, impossible at order 2).

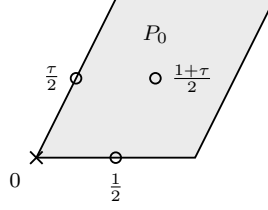


Figure 22.7. Inside P_0 : $\wp$ has a double pole at 0 ($\times$); $\wp'$ vanishes at the three half-periods ($\circ$), where $\wp = e_1, e_2, e_3$. (Schematic.)

Theorem 22.1.13. $(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3)$.

Proof. Let $F = (\wp - e_1)(\wp - e_2)(\wp - e_3)$. Its only roots in P_0 are the half-periods, each of multiplicity 2 (since $\wp - e_j$ has a double root there). Likewise $(\wp')^2$ has double roots exactly at the half-periods. Both F and $(\wp')^2$ have poles of order 6 at lattice points ($\wp'$ has a triple pole). Hence $(\wp')^2/F$ is holomorphic and doubly periodic, so constant by Theorem 22.1.3. To find the constant compare leading terms near 0: $\wp(z) = z^{-2} + \dots$ gives $F = z^{-6} + \dots$, while $\wp'(z) = -2z^{-3} + \dots$ gives $(\wp')^2 = 4z^{-6} + \dots$. So the constant is 4. $\square$

Worked Example 22.1.14 (Reading off the half-period values). Using Theorem 22.1.13, express $e_1 + e_2 + e_3$ and the relation between $(\wp')^2$ and the cubic, and verify the leading coefficient is 4.

Recognition. Trigger: identify symmetric functions of the e_j from the factored form. Expand the product and match against the differential equation.

Solution. Writing $w = \wp$, the cubic is $4(w - e_1)(w - e_2)(w - e_3) = 4w^3 - 4(e_1 + e_2 + e_3)w^2 + \dots$. We will see (Corollary 22.2.7) that $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$ has *no* $\wp^2$ term, which forces $e_1 + e_2 + e_3 = 0$. *Check.* Near $z = 0$, $(\wp')^2 = 4z^{-6} + \dots$ and $4\wp^3 = 4z^{-6} + \dots$ agree at top order, confirming the leading coefficient 4; the absence of a w^2 term is exactly $e_1 + e_2 + e_3 = 0$.

Theorem 22.1.15. Every elliptic function f with periods $1, \tau$ is a rational function of $\wp$ and $\wp'$.

The key is the even case.

Lemma 22.1.16. Every even elliptic function F with periods $1, \tau$ is a rational function of $\wp$.

Proof. A zero or pole of F at the origin is of even order (evenness), so for some integer m the function $F\wp^m$ has no zero/pole on Λ ; we may assume F itself has none on Λ . Recall $\wp(z) - \wp(a)$ has a double zero if a is a half-period and two simple zeros at $\pm a$ otherwise. If a is a zero of F so is $-a$; and $a \sim -a$ exactly when a is a half-period (then the zero has even order). So listing the zeros (with multiplicity) as $a_1, -a_1, \dots, a_m, -a_m$, the product $\prod_j [\wp(z) - \wp(a_j)]$ has the same zeros as F . Doing the same for poles $b_1, -b_1, \dots, b_m, -b_m$,

$$G(z) = \frac{\prod_{j=1}^m [\wp(z) - \wp(a_j)]}{\prod_{j=1}^m [\wp(z) - \wp(b_j)]}$$

is elliptic with the same zeros and poles as F . Hence F/G is holomorphic and doubly periodic, so constant; $F = cG$ is rational in $\wp$. $\square$

The conclusion is that an even elliptic F factors as $F = R \circ \wp$ for a rational function R : it is computed by first applying $\wp$ and then a rational map of one variable.

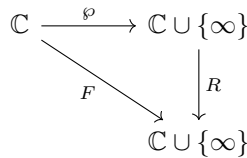


Figure 22.8. An even elliptic function F as the composite $F = R \circ \wp$: $\wp$ sends z to a point of $\mathbb{C} \cup \{\infty\}$, and a single-variable rational function R finishes the job (Lemma 22.1.16).

Proof of Theorem 22.1.15. Split $f = f_{\text{even}} + f_{\text{odd}}$ with $f_{\text{even}}(z) = \frac{1}{2}[f(z) + f(-z)]$ and $f_{\text{odd}}(z) = \frac{1}{2}[f(z) - f(-z)]$. Both are elliptic. Now $f_{\text{odd}}/\wp'$ is *even* (odd over odd), so by Lemma 22.1.16 both f_{even} and $f_{\text{odd}}/\wp'$ are rational in $\wp$. Therefore $f = f_{\text{even}} + \wp' \cdot (f_{\text{odd}}/\wp')$ is rational in $\wp$ and $\wp'$. $\square$

Worked Example 22.1.17 (Writing $\wp''$ in terms of $\wp$). Show $\wp''$ is a polynomial in $\wp$, and find it.

Recognition. Trigger: $\wp''$ is even and elliptic, so Lemma 22.1.16 (or differentiating Theorem 22.1.13) applies. Differentiate the differential equation.

Solution. Differentiate $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$ (Corollary 22.2.7): $2\wp'\wp'' = 12\wp^2\wp' - g_2\wp'$. Cancel $2\wp'$ (valid off the zeros of $\wp'$, then by continuity everywhere) to get

$$\wp'' = 6\wp^2 - \frac{1}{2}g_2.$$

Check. Near $z = 0$, $\wp = z^{-2} + \dots$ gives $\wp'' = 6z^{-4} + \dots$ and $6\wp^2 = 6z^{-4} + \dots$ — top orders match.

Non-example: $\wp''$ is not a polynomial in $\wp'$ alone, since $\wp'$ is odd while $\wp''$ is even.

Worked Example 22.1.18 (A rational reduction of an even elliptic function). Let $f = (\wp')^2$. Express f as a polynomial in $\wp$ directly, without invoking the e_j .

Recognition. Trigger: $(\wp')^2$ is even and elliptic; expect a polynomial in $\wp$ by Lemma 22.1.16. Use the differential equation.

Solution. By Corollary 22.2.7, $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$ — a degree-3 polynomial in $\wp$. *Check.* This agrees with the factored $4(\wp - e_1)(\wp - e_2)(\wp - e_3)$ of Theorem 22.1.13 once $g_2 = 4(e_1e_2 + e_2e_3 + e_3e_1) \cdot (-1)$ and $g_3 = 4e_1e_2e_3$ are identified by expanding (using $e_1 + e_2 + e_3 = 0$). *Pole order:* both sides have a pole of order 6 at lattice points, consistent.

Caution. The subtraction of $1/\omega^2$ in $\wp$ is not optional bookkeeping — the raw sum $\sum 1/(z + \omega)^2$ *diverges* (it is the $r = 2$ borderline case of Lemma 22.1.10). It is also why double-periodicity of $\wp$ is not visible by inspection: the correction terms $-1/\omega^2$ are not themselves periodic. Periodicity is recovered only after differentiating (which kills the correction) or by the careful rearrangement in the exercises.

Further worked examples

The structural theorems above (residue sum zero, equal zero/pole count, order ≥ 2) are best absorbed by computing. The examples below also exercise the surrounding machinery of Chapters 2–3 — residues, the argument principle, contour integrals — so the elliptic theory sits inside the wider toolkit rather than apart from it.

Worked Example 22.1.19 (The residue sum is zero, verified on a two-pole model). Let f be elliptic of order 2 with simple poles at a and b in P_0 , principal parts $\frac{r}{z-a}$ and $\frac{s}{z-b}$. Compute $\oint_{\partial P_0} f dz$ two ways and deduce $r + s = 0$.

Recognition. Trigger: a periodic integrand over ∂P_0 . Use the Method (opposite sides cancel) and the residue theorem.

Solution. *Way 1 (residues).* By the residue theorem, $\oint_{\partial P_0} f dz = 2\pi i(\text{Res}_{z=a} f + \text{Res}_{z=b} f) = 2\pi i(r + s)$. *Way 2 (periodicity).* Pair the sides $0 \rightarrow 1$ with $1 + \tau \rightarrow \tau$ and $1 \rightarrow 1 + \tau$ with $\tau \rightarrow 0$; on each pair the substitution $z \mapsto z + (\text{period})$ and $f(z + \omega) = f(z)$ make the two integrals exact negatives, so $\oint_{\partial P_0} f dz = 0$. Equating, $2\pi i(r + s) = 0$, hence $r + s = 0$. *Check.* This is exactly Theorem 22.1.5; the simplest concrete realization is $f(z) = \wp'(z)/(\wp(z) - \wp(a))$, whose two simple poles at $\pm a$ carry opposite residues $+1$ and -1 .

Worked Example 22.1.20 (Counting solutions of $\wp(z) = 5$). For a fixed lattice, how many solutions does $\wp(z) = 5$ have in P_0 , counted with multiplicity, and when do two of them coincide?

Recognition. Trigger: a level-set count for an elliptic function. Use Corollary 22.1.7: $f = c$ has exactly $\text{ord } f$ roots.

Solution. $\wp$ has order 2, so $\wp(z) = 5$ has exactly 2 roots in P_0 with multiplicity. Since $\wp$ is even, if z_0 is a root so is $-z_0$; generically the two roots are $\{z_0, -z_0\}$, distinct. They coincide (a double root) precisely when $z_0 \equiv -z_0$, i.e. z_0 is a half-period $\frac{1}{2}, \frac{\tau}{2}, \frac{1+\tau}{2}$ — and there $\wp(z_0) = e_1, e_2, e_3$ respectively. So $\wp = 5$ has a double root iff $5 \in \{e_1, e_2, e_3\}$. *Check.* For $c = e_1$ the equation $\wp = e_1$ has its double root at $z = \frac{1}{2}$ where $\wp'(\frac{1}{2}) = 0$ (a critical value), consistent with the chain rule forcing a repeated root exactly at critical points.

Worked Example 22.1.21 (An order-3 example and its zeros). Take $f = \wp'$. State its order, locate all its poles and zeros in P_0 , and verify Theorem 22.1.6.

Recognition. Trigger: pole/zero bookkeeping for a specific elliptic function. Differentiate the double pole; use oddness to find zeros.

Solution. Near a lattice point $\wp(z) = z^{-2} + \dots$, so $\wp'(z) = -2z^{-3} + \dots$: a *triple* pole at 0, hence order 3. By Theorem 22.1.6 there are exactly 3 zeros in P_0 . Oddness plus periodicity gives $\wp' = 0$ at the three half-periods $\frac{1}{2}, \frac{\tau}{2}, \frac{1+\tau}{2}$ (shown in the text), each simple, accounting for all 3. *Check.* 3 poles (one triple) = 3 zeros (three simple), matching $N_z = N_p$; and the three zeros are exactly the branch points of the cubic $(\wp')^2 = 4\prod(\wp - e_j)$.

Worked Example 22.1.22 (A non-example: $1/\wp$ is not entire). Is $g(z) = 1/\wp(z)$ entire? Decide and describe its singularities.

Recognition. Trigger: testing whether a constructed elliptic function is holomorphic. Zeros of $\wp$ become poles of $1/\wp$.

Solution. No. g is elliptic (reciprocal of an elliptic function), but it is *not* entire: wherever $\wp(z) = 0$, g has a pole. Since $\wp$ has order 2, $\wp(z) = 0$ has exactly 2 solutions $\pm z_0$ in P_0 (a generic value, so two simple zeros), giving g two simple poles. Meanwhile the double poles of $\wp$ at lattice points become double *zeros* of g . *Check.* g has order 2 (two simple poles), consistent with Theorem 22.1.5; an entire doubly periodic function would be constant (Theorem 22.1.3), and g is visibly nonconstant, so it cannot be entire.

Worked Example 22.1.23 (Residue of $\wp$ at a lattice point). Compute $\text{Res}_{z=0} \wp(z)$ and $\text{Res}_{z=0}(z\wp(z))$.

Recognition. Trigger: a residue extraction from a known Laurent head. Read the z^{-1} coefficient.

Solution. Near 0, $\wp(z) = z^{-2} + 0 \cdot z^{-1} + O(z^2)$ (no odd-power terms, by evenness). The residue is the z^{-1} coefficient, which is 0, so $\text{Res}_{z=0} \wp = 0$. For $z\wp(z) = z^{-1} + O(z^3)$, the z^{-1} coefficient is 1, so $\text{Res}_{z=0}(z\wp) = 1$. *Check.* $\text{Res}_{z=0} \wp = 0$ is forced by the global residue sum being zero with 0 the only pole in P_0 ; and it is also why $\wp$ (a single double pole with zero residue) is an admissible order-2 elliptic function under Theorem 22.1.5.

Worked Example 22.1.24 (A keyhole/branch-cut warm-up from the single-period model). As a contour-integral warm-up parallel to the construction of $F(z) = \pi \cot \pi z$, evaluate $\int_0^\infty \frac{x^{a-1}}{1+x} dx$ for $0 < a < 1$ by a keyhole contour, and note the structural analogy with summing $1/(z+n)$.

Recognition. Trigger: a positive-real-axis integral with a branch point at 0. Use a keyhole contour and the single pole at $z = -1$.

Solution. Take $f(z) = z^{a-1}/(1+z)$ with the branch $\arg z \in (0, 2\pi)$. Integrate over a keyhole: above the cut $z = x$, below it $z = xe^{2\pi i}$ so $z^{a-1} = x^{a-1}e^{2\pi i(a-1)}$. The large and small circles vanish ($0 < a < 1$). Thus

$$(1 - e^{2\pi i(a-1)}) \int_0^\infty \frac{x^{a-1}}{1+x} dx = 2\pi i \text{Res}_{z=-1} f = 2\pi i (-1)^{a-1} = 2\pi i e^{i\pi(a-1)}.$$

Dividing, $\int_0^\infty \frac{x^{a-1}}{1+x} dx = \frac{2\pi i e^{i\pi(a-1)}}{1 - e^{2\pi i(a-1)}} = \frac{\pi}{\sin \pi a}$. *Check.* At $a = \frac{1}{2}$ this is π , matching $\int_0^\infty \frac{dx}{\sqrt{x(1+x)}} = \pi$; the structural point is that, just as the keyhole collects the single residue at -1 , the lattice sum collects contributions one period at a time. ■

Below, the contour used in the previous example.

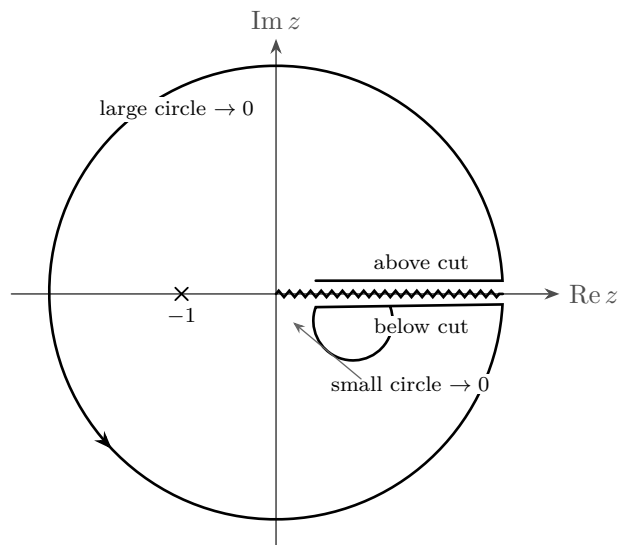


Figure 22.9. Keyhole contour around the cut $[0, \infty)$ for $z^{a-1}/(1+z)$; only the pole at $z = -1$ contributes. (Schematic.)

Worked Example 22.1.25 (The pole-counting integral as a winding number). Let f be elliptic of order m and c a value not attained on ∂P_0 . Interpret $\frac{1}{2\pi i} \oint_{\partial P_0} \frac{f'(z)}{f(z) - c} dz$ and evaluate it.

Recognition. Trigger: the argument principle applied to $f - c$. Periodicity kills the boundary integral.

Solution. The integral equals (number of zeros of $f - c$ in P_0) minus (number of poles of $f - c$ in P_0). Now $f - c$ is elliptic of order m , so it has m poles; the integrand $(f - c)'/(f - c) = f'/(f - c)$ is Λ -periodic, so by the Method it integrates to 0 over ∂P_0 . Hence $N_z - N_p = 0$, i.e. $N_z = m$: the equation $f = c$ has exactly m roots. *Check.* This is Corollary 22.1.7; equivalently, the image curve $f(\partial P_0)$ winds zero net times about c because opposite sides of ∂P_0 map to the same curve traversed in opposite directions.

Worked Example 22.1.26 (A non-example: no elliptic function of order 1). Show there is no elliptic function of order exactly 1, even allowing the pole to be anywhere in P_0 .

Recognition. Trigger: an impossibility statement about order. Combine Theorem 22.1.5 with the residue-sum identity.

Solution. Order 1 means a single simple pole in P_0 (total pole multiplicity 1). Its sole residue r must satisfy $\sum \text{Res} = 0$, so $r = 0$; but a simple pole has $r \neq 0$ by definition. Contradiction, so no order-1 elliptic function exists. The minimum order is 2, realized by $\wp$ (double pole) and by functions with two simple poles of opposite residue. *Check.* Consistent with Theorem 22.1.6 as well — an order-1 function would have exactly one zero, so f would be a bijection $P_0 \rightarrow \mathbb{C} \cup \{\infty\}$ of degree 1, forcing $\mathbb{C}/\Lambda \cong \hat{\mathbb{C}}$, which is false (a torus is not a sphere).

Exercises (Stein, Ch.9).

- Suppose a meromorphic f has two periods ω_1, ω_2 with $\omega_2/\omega_1 \in \mathbb{R}$. (a) If $\omega_2/\omega_1 = p/q$ in lowest terms (p, q relatively prime), prove the periodicity assumption is equivalent to f being periodic with the single period $\omega_0 = \frac{1}{q}\omega_1$. [Hint: choose integers m, n with $mq + np = 1$.] (b) If ω_2/ω_1 is irrational, prove f is constant. [Use that $\{m - n\tau\}$ is dense in $\mathbb{R}$ when τ is irrational and m, n range over $\mathbb{Z}$.]
- Let $a_1, \dots, a_r$ and $b_1, \dots, b_r$ be the zeros and poles (in P_0 , with multiplicity) of an elliptic function f . Show that $a_1 + \dots + a_r - b_1 - \dots - b_r = n\omega_1 + m\omega_2$ for some integers n, m . [Hint: if the boundary has no zeros/poles, integrate $z f'(z)/f(z)$ over ∂P_0 and note the integral of f'/f over a side is an integer multiple of $2\pi i$; otherwise translate the side slightly.]
- By rearranging the series $\frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left(\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} \right)$, show *directly*, without differentiation, that $\wp(z+\omega) = \wp(z)$ for all $\omega \in \Lambda$. [Hint: for large R , $\wp(z) = \wp_R(z) + O(1/R)$ with $\wp_R(z) = z^{-2} + \sum_{0 < |\omega| < R} ((z+\omega)^{-2} - \omega^{-2})$; both $\wp_R(z+1) - \wp_R(z)$ and $\wp_R(z+\tau) - \wp_R(z)$ are $O(\sum_{R-c < |\omega| < R+c} |\omega|^{-2}) = O(1/R)$.]

4. Let $\sigma(z)$ be the canonical product $\sigma(z) = z \prod_{j=1}^{\infty} E_2(z/\tau_j)$, where $\{\tau_j\}$ enumerates the periods $\{n + m\tau\}$ with $(n, m) \neq (0, 0)$ and $E_2(z) = (1 - z)e^{z + z^2/2}$. (a) Show σ is entire of order 2 with simple zeros at every period $n + m\tau$ and nowhere else. (b) Show $\frac{\sigma'(z)}{\sigma(z)} = \frac{1}{z} + \sum_{(n,m) \neq (0,0)} \left(\frac{1}{z - n - m\tau} + \frac{1}{n + m\tau} + \frac{z}{(n + m\tau)^2} \right)$, convergent for z not a lattice point. (c) With $L(z) = -\sigma'(z)/\sigma(z)$, show $L'(z) = \frac{(\sigma'(z))^2 - \sigma(z)\sigma''(z)}{(\sigma(z))^2} = \wp(z)$.
5. Prove that $\wp''$ is a quadratic polynomial in $\wp$.

Problems (Stein, Ch.9).

1. Besides the approach of §2, there are several ways to handle $\sum 1/(z + \omega)^2$ ($\omega = n + m\tau$): (a) circularly, (b) first in n then in m , (c) first in m then in n . (a) Prove that for $z \notin \Lambda$, $\lim_{R \rightarrow \infty} \sum_{n^2 + m^2 \leq R^2} (z + n + m\tau)^{-2} = S_1(z)$ exists and $S_1(z) = \wp(z) + c_1$. (b) Prove $S_2(z) = \sum_m \left(\sum_n (z + n + m\tau)^{-2} \right)$ exists and $S_2(z) = \wp(z) + c_2$, where $c_2 = F(\tau)$, the forbidden Eisenstein series. (c) Prove $S_3(z) = \sum_n \left(\sum_m (z + n + m\tau)^{-2} \right)$ exists with $S_3(z) = \wp(z) + c_3$, $c_3 = \tilde{F}(\tau)$, the reverse of F . [Hint for (a): show $\lim_{R \rightarrow \infty} \sum_{1 \leq n^2 + m^2 \leq R^2} (n + m\tau)^{-2} = c_1$ exists by comparison with $I(R) = \iint_{1 \leq x^2 + y^2 \leq R^2} (x + y\tau)^{-2} dx dy$, and $I(R) = 0$ since $(x + y\tau)^{-2} = -(\partial/\partial x)(x + y\tau)^{-1}$.]
2. * Suppose Ω is a simply connected domain excluding the three roots of $4z^3 - g_2z - g_3$. For fixed $\omega_0 \in \Omega$ define $I(\omega) = \int_{\omega_0}^{\omega} \frac{dz}{\sqrt{4z^3 - g_2z - g_3}}$, $\omega \in \Omega$. Then I has inverse $\wp(z + \alpha)$ for some constant α , i.e. $I(\wp(z + \alpha)) = z$ for appropriate α . [Hint: prove $(I(\wp(z + \alpha)))' = \pm 1$ and use that $\wp$ is even.]

Additional exercises.

1. (Congruence reduction.) With $\tau = i$, reduce each of $z = 2.3 + 1.7i$ and $z = -0.4 - 2.1i$ to its unique congruent point in $P_0 = \{a + bi : 0 \leq a, b < 1\}$. State the integers n, m used.
2. (Even/odd values.) Using only that $\wp$ is even and $\wp'$ is odd and both are 1-periodic, prove $\wp'(\frac{1+\tau}{2}) = 0$.
3. (Residue sum.) An elliptic f of order 3 has, in P_0 , a simple pole at a with residue 2 and a double pole at b . Show the residue of f at b must be -2 , and state how many zeros f has in P_0 .
4. (Order of a quotient.) Let f, g be elliptic with periods 1, τ of orders m_f and m_g . Show fg and f/g are elliptic and bound their orders. Give an example where the order of f/g is strictly less than $m_f + m_g$.
5. (Building from $\wp$.) Show that $\wp(z) - \wp(a)$, for a a non-half-period in P_0 , is an order-2 elliptic function with exactly the two simple zeros a and $-a$ in P_0 . What changes when a is a half-period?
6. (Non-example.) Decide whether $h(z) = \wp(z) + \wp'(z)$ is even, odd, or neither, and whether it can be written as a rational function of $\wp$ alone. Justify using parity.
7. (Diagram problem.) On a copy of the fundamental parallelogram for $\tau = 0.3 + i$, mark the lattice point (double pole of $\wp$) and the three half-periods (zeros of $\wp'$). Then sketch the oriented boundary ∂P_0 and indicate which pairs of sides cancel in the $\int_{\partial P_0} \wp'/\wp = 0$ argument.
8. (Differential equation check.) Assuming $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$, differentiate to express $\wp'''$ in terms of $\wp$ and $\wp'$, and verify the parity of each side.
9. (Half-period sum.) Using $e_1 + e_2 + e_3 = 0$ and $(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3)$, show $g_2 = 2(e_1^2 + e_2^2 + e_3^2)$ and $g_3 = 4e_1e_2e_3$, and conclude $g_2 \geq 0$ when the e_j are real.

22.2 The Modular Character of Elliptic Functions and Eisenstein Series

The modular group acting on τ

So far τ was fixed. Now we vary it. Since the lattice and hence $\wp$ depend on τ , write $\wp_{\tau}(z)$. Two changes of τ leave the lattice essentially the same.

First, 1 and τ generate the same lattice as 1 and $\tau + 1$, so $\wp_{\tau+1} = \wp_\tau$. Second, recall $\tau = \omega_2/\omega_1$; then $-1/\tau = -\omega_1/\omega_2$ also has positive imaginary part and corresponds to swapping the two periods. One verifies directly the scaling law

$$\wp_{-1/\tau}(z) = \tau^2 \wp_\tau(\tau z).$$

The scaling $\cdot \tau : \mathbb{C} \rightarrow \mathbb{C}$ carries the lattice $\Lambda_{-1/\tau}$ onto Λ_τ , and $\wp_{-1/\tau}$ is the composite $\wp_\tau \circ (\cdot \tau)$ followed by multiplication by τ^2 .

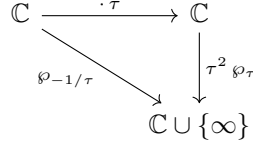


Figure 22.10. The scaling law $\wp_{-1/\tau}(z) = \tau^2 \wp_\tau(\tau z)$ as a composition: multiply z by τ (which sends $\Lambda_{-1/\tau}$ onto Λ_τ), apply $\wp_\tau$, then rescale by τ^2 . The triangle commutes.

Thus we are led to the group of transformations of the upper half-plane $\mathbb{H} = \{\text{Im } \tau > 0\}$ generated by

$$T : \tau \mapsto \tau + 1 \quad \text{and} \quad S : \tau \mapsto -1/\tau.$$

Definition 22.2.1. The *modular group* is the group of transformations of $\mathbb{H}$ generated by $T : \tau \mapsto \tau + 1$ and $S : \tau \mapsto -1/\tau$. (It is $\text{PSL}_2(\mathbb{Z})$ acting by $\tau \mapsto \frac{a\tau+b}{c\tau+d}$, $ad - bc = 1$.)

Every quantity intrinsically attached to the lattice should transform predictably under T and S . The cleanest example is the Eisenstein series.

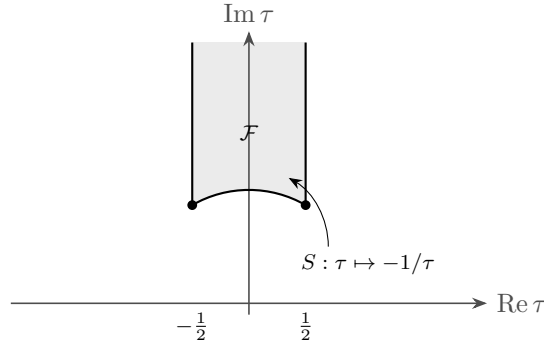


Figure 22.11. The standard fundamental domain $\mathcal{F}$ in $\mathbb{H}$ for the modular group; T shifts by 1, S inverts across the unit circle.

Eisenstein series

Definition 22.2.2. For an integer $k \geq 3$ and $\tau \in \mathbb{H}$, the *Eisenstein series of order k* is

$$E_k(\tau) = \sum_{(n,m) \neq (0,0)} \frac{1}{(n + m\tau)^k} = \sum_{\omega \in \Lambda^*} \frac{1}{\omega^k}, \quad \omega = n + m\tau.$$

Theorem 22.2.3. The Eisenstein series satisfy: (i) E_k converges for $k \geq 3$ and is holomorphic on $\mathbb{H}$; (ii) $E_k \equiv 0$ if k is odd; (iii) $E_k(\tau + 1) = E_k(\tau)$ and $E_k(\tau) = \tau^{-k} E_k(-1/\tau)$.

Property (iii) is the *modular character*: E_k is invariant under T and transforms by the factor τ^{-k} under S .

Proof. (i) By Lemma 22.1.10 and the remark after it, the series converges absolutely and uniformly in $\text{Im } \tau \geq \delta > 0$ for $k \geq 3$; a locally uniform limit of holomorphic functions is holomorphic, so $E_k \in \text{Hol}(\mathbb{H})$. (ii) Replacing (n, m) by $(-n, -m)$ negates each ω , multiplying every term by $(-1)^k = -1$ when k is odd; the

sum equals its own negative, hence 0. (iii) For T : $n + m(\tau + 1) = (n + m) + m\tau$, and $(n, m) \mapsto (n + m, m)$ is a bijection of $\mathbb{Z}^2 \setminus 0$, so the sum is unchanged. For S : with $\tau \mapsto -1/\tau$,

$$(n + m(-1/\tau))^k = \tau^{-k}(n\tau - m)^k,$$

and relabelling $(-m, n) \mapsto (n, m)$ turns $\sum(n\tau - m)^{-k}$ into $\sum(n + m\tau)^{-k} = E_k(\tau)$. Hence $E_k(-1/\tau) = \tau^k E_k(\tau)$, i.e. $E_k(\tau) = \tau^{-k} E_k(-1/\tau)$. $\square$

Worked Example 22.2.4 (Why odd Eisenstein series vanish, concretely). Show $E_3(\tau) = 0$ by pairing lattice points, and explain why the argument fails for E_4 .

Recognition. Trigger: vanishing by symmetry. Pair ω with $-\omega$.

Solution. Pair the term at $\omega = n + m\tau$ with the term at $-\omega$: $\frac{1}{\omega^3} + \frac{1}{(-\omega)^3} = \frac{1}{\omega^3} - \frac{1}{\omega^3} = 0$. Every term cancels with its partner ($\omega = 0$ is excluded), so $E_3 \equiv 0$; same for any odd k . *Check / non-example:* for $k = 4$, $\frac{1}{\omega^4} + \frac{1}{(-\omega)^4} = \frac{2}{\omega^4} \neq 0$ — the pairing *reinforces* rather than cancels, so $E_4 \neq 0$.

Worked Example 22.2.5 (The transformation factor under S , checked on a partial sum). Verify the identity $(n - m/\tau)^k = \tau^{-k}(n\tau - m)^k$ used in Theorem 22.2.3(iii), and confirm the relabelling is a bijection.

Recognition. Trigger: justify an index manipulation. Factor out τ^{-1} and track the map on $\mathbb{Z}^2$.

Solution. $n + m(-1/\tau) = n - \frac{m}{\tau} = \frac{1}{\tau}(n\tau - m)$, so raising to the k gives $\tau^{-k}(n\tau - m)^k$. Setting $n' = -m$, $m' = n$ we have $n\tau - m = m'\tau + n'$, i.e. the term is $\tau^{-k}(n' + m'\tau)^k$ reindexed. The map $(n, m) \mapsto (n', m') = (-m, n)$ is the rotation matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, a bijection of $\mathbb{Z}^2 \setminus 0$. *Check.* The lattice point $\omega = 1$ ($n, m = 1, 0$) maps to $(0, 1)$, i.e. τ ; geometrically S rotates the lattice by 90° and rescales, consistent with $\wp_{-1/\tau}(z) = \tau^2 \wp_\tau(\tau z)$.

The Laurent expansion of $\wp$ and the differential equation

Theorem 22.2.6. For z near 0,

$$\wp(z) = \frac{1}{z^2} + 3E_4 z^2 + 5E_6 z^4 + \cdots = \frac{1}{z^2} + \sum_{k=1}^{\infty} (2k+1)E_{2k+2} z^{2k}.$$

Proof. Replace ω by $-\omega$ in the defining sum (harmless) to write the bracket as $\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}$. From $\frac{1}{(1-w)^2} = \sum_{\ell \geq 0} (\ell+1)w^\ell$ ($|w| < 1$), with $w = z/\omega$,

$$\frac{1}{(z-\omega)^2} = \frac{1}{\omega^2} \sum_{\ell \geq 0} (\ell+1) \left(\frac{z}{\omega}\right)^\ell = \frac{1}{\omega^2} + \frac{1}{\omega^2} \sum_{\ell \geq 1} (\ell+1) \left(\frac{z}{\omega}\right)^\ell.$$

Subtracting $1/\omega^2$ and summing over $\omega \in \Lambda^*$,

$$\wp(z) = \frac{1}{z^2} + \sum_{\ell \geq 1} (\ell+1) \left(\sum_{\omega \in \Lambda^*} \frac{1}{\omega^{\ell+2}} \right) z^\ell = \frac{1}{z^2} + \sum_{\ell \geq 1} (\ell+1) E_{\ell+2} z^\ell.$$

Since $E_{\ell+2} = 0$ for ℓ odd, only even $\ell = 2k$ survives, giving the stated series. $\square$

Corollary 22.2.7. With $g_2 = 60E_4$ and $g_3 = 140E_6$,

$$(\wp')^2 = 4\wp^3 - g_2 \wp - g_3.$$

Proof. From Theorem 22.2.6, near 0,

$$\wp(z) = \frac{1}{z^2} + 3E_4 z^2 + 5E_6 z^4 + \cdots, \quad \wp'(z) = \frac{-2}{z^3} + 6E_4 z + 20E_6 z^3 + \cdots,$$

hence

$$(\wp')^2 = \frac{4}{z^6} - \frac{24E_4}{z^2} - 80E_6 + \cdots, \quad \wp^3 = \frac{1}{z^6} + \frac{9E_4}{z^2} + 15E_6 + \cdots.$$

Therefore $(\wp')^2 - 4\wp^3 + 60E_4\wp + 140E_6$ has Laurent expansion with no negative powers and vanishes at $z = 0$; it is holomorphic near 0, and being doubly periodic it extends to an entire doubly periodic function, hence constant (Theorem 22.1.3) equal to its value 0 at the origin. Rearranging gives the identity. $\square$

Worked Example 22.2.8 (Computing the $1/z^2$ coefficient of $(\wp')^2$). Confirm the coefficient $-24E_4$ of z^{-2} in $(\wp')^2$ used above.

Recognition. Trigger: a Laurent-coefficient extraction. Square the series for $\wp'$ and collect the z^{-2} term.

Solution. $\wp' = -2z^{-3} + 6E_4z + 20E_6z^3 + \cdots$. Squaring, the z^{-2} terms come from the cross product $2 \cdot (-2z^{-3})(6E_4z) = -24E_4z^{-2}$. *Check.* The z^{-6} term is $(-2z^{-3})^2 = 4z^{-6}$ and the constant term is $2(-2z^{-3})(20E_6z^3) = -80E_6$, matching the displayed expansion.

Worked Example 22.2.9 (Recovering Theorem 22.1.13 from Corollary 22.2.7). Show that $4\wp^3 - g_2\wp - g_3 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3)$ and read off $e_1 + e_2 + e_3 = 0$.

Recognition. Trigger: match two forms of the same cubic in $\wp$. Compare coefficients.

Solution. Both equal $(\wp')^2$ (Theorem 22.1.13 and Corollary 22.2.7), so the polynomials in $w = \wp$ agree: $4w^3 - g_2w - g_3 = 4(w - e_1)(w - e_2)(w - e_3)$. The right side expands to $4w^3 - 4(\sum e_j)w^2 + 4(\sum_{i < j} e_i e_j)w - 4e_1 e_2 e_3$. Matching the w^2 coefficient: $-4(e_1 + e_2 + e_3) = 0$, so $e_1 + e_2 + e_3 = 0$. Matching w^1 and w^0 : $g_2 = -4 \sum_{i < j} e_i e_j$ and $g_3 = 4e_1 e_2 e_3$. *Check.* With $\sum e_j = 0$ we also get $\sum_{i < j} e_i e_j = -\frac{1}{2} \sum e_j^2 \leq 0$, so $g_2 = 2 \sum e_j^2 \geq 0$, as expected for a real lattice.

Eisenstein series and divisor functions

The Eisenstein series, being 1-periodic in τ , has a Fourier (here q -) expansion in $q = e^{2\pi i \tau}$. Extracting it links E_k to number theory. Periodicity $E_k(\tau + 1) = E_k(\tau)$ says exactly that E_k is constant on the fibers of the covering $q : \mathbb{H} \rightarrow D^* = \{0 < |q| < 1\}$, so it descends to a holomorphic $\tilde{E}_k$ on the punctured disc; the q -expansion is the Laurent (here Taylor) series of $\tilde{E}_k$.

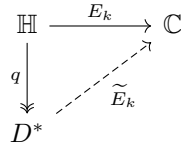


Figure 22.12. The covering $q = e^{2\pi i \tau} : \mathbb{H} \rightarrow D^*$ (surjective onto the punctured disc $D^* = \{0 < |q| < 1\}$) has the period strip as a fundamental region. Since E_k is 1-periodic it factors as $E_k = \tilde{E}_k \circ q$; the induced map $\tilde{E}_k$ is the dashed arrow, and its Taylor coefficients are the q -expansion.

Lemma 22.2.10. For $k \geq 2$ and $\text{Im } \tau > 0$,

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n + \tau)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell \tau}.$$

Proof. Apply Poisson summation to $f(z) = 1/(z + \tau)^k$ (Chapter 4, Exercise 7). Alternatively, prove $k = 2$ and differentiate. Differentiating the cotangent formula $\sum_n \frac{1}{n + \tau} = \pi \cot \pi \tau$ (Chapter 5) gives

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n + \tau)^2} = \frac{\pi^2}{\sin^2(\pi \tau)}.$$

Using Euler's formula for $\sin$ and $\sum_{r \geq 1} r w^r = \frac{w}{(1-w)^2}$ with $w = e^{2\pi i \tau}$ yields the $k = 2$ case; the higher k follow by termwise differentiation in τ . $\square$

Definition 22.2.11. For a positive integer r , the *divisor function* of order ℓ is $\sigma_\ell(r) = \sum_{d|r} d^\ell$ (sum of ℓ th powers of divisors). When $\ell = 1$ we write $\sigma(r) = \sum_{d|r} d$.

Theorem 22.2.12. If $k \geq 4$ is even and $\text{Im } \tau > 0$, then

$$E_k(\tau) = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i r \tau}.$$

Proof. Split the lattice sum by m . The $m = 0$ slice is $\sum_{n \neq 0} n^{-k} = 2\zeta(k)$ (k even). For $m \neq 0$, pair $\pm m$ (using k even) to write the rest as $2 \sum_{m>0} \sum_n (n + m\tau)^{-k}$, then apply Lemma 22.2.10 with $\tau \mapsto m\tau$:

$$E_k(\tau) = 2\zeta(k) + 2 \sum_{m>0} \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell \geq 1} \ell^{k-1} e^{2\pi i \ell m \tau}.$$

Collecting the coefficient of q^r ($q = e^{2\pi i \tau}$) over all factorizations $r = \ell m$ gives $\sum_{\ell|r} \ell^{k-1} = \sigma_{k-1}(r)$, and $(-2\pi i)^k = (-1)^{k/2} (2\pi)^k$ for even k . Absolute convergence (for $\text{Im } \tau \geq t_0$) follows from $\sigma_{k-1}(r) \leq r^k$ and comparison with $\sum r^k e^{-2\pi r t_0}$. $\square$

Corollary 22.2.13. *The forbidden case $k = 2$, summed first in n then in m , $F(\tau) = \sum_m (\sum_n (n + m\tau)^{-2})$ (over $(n, m) \neq (0, 0)$), converges in that order and*

$$F(\tau) = 2\zeta(2) - 8\pi^2 \sum_{r=1}^{\infty} \sigma(r) e^{2\pi i r \tau}.$$

Moreover $\tau^{-2}F(-1/\tau) \neq F(\tau)$: the doubly indexed series gives a different value when summed first in m then in n .

Proof. The same computation as Theorem 22.2.12 applies once we fix the order of summation; $2(-1)(2\pi)^2/1! = -8\pi^2$ gives the coefficient. The failure of the modular factor reflects the non-absolute convergence of the $k = 2$ series, so order of summation matters. $\square$

Worked Example 22.2.14 (Euler's $\zeta(2)$ from the half-period). Set $\tau = 1/2$ in $\sum_m (m + \tau)^{-2} = \pi^2 / \sin^2(\pi\tau)$ to deduce $\sum_{m \geq 1} 1/m^2 = \zeta(2) = \pi^2/6$.

Recognition. Trigger: extract a special value of ζ from the cotangent-derivative identity. Evaluate at the symmetric point $\tau = 1/2$.

Solution. At $\tau = 1/2$, $\sin^2(\pi/2) = 1$, so $\sum_{m \in \mathbb{Z}} (m + \frac{1}{2})^{-2} = \pi^2$. The left side is $\sum_{m \geq 1, \text{ odd}} \frac{4}{m^2}$ (reindexing $m + \frac{1}{2} = \frac{2m+1}{2}$), i.e. $\sum_j \text{ odd } \frac{4}{j^2} = \pi^2$, so $\sum_j \text{ odd } 1/j^2 = \pi^2/8$. Then

$$\sum_{m \geq 1} \frac{1}{m^2} = \sum_{\text{odd}} \frac{1}{j^2} + \sum_{\text{even}} \frac{1}{(2k)^2} = \frac{\pi^2}{8} + \frac{1}{4} \sum_{m \geq 1} \frac{1}{m^2},$$

so $\frac{3}{4} \sum 1/m^2 = \pi^2/8$, giving $\sum 1/m^2 = \pi^2/6 = \zeta(2)$. *Check.* Numerically $\pi^2/6 \approx 1.6449$, and $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} \approx 1.4236$, climbing toward it.

Worked Example 22.2.15 (The q -expansion of E_4 to first order). Use Theorem 22.2.12 to write the first two terms of $E_4(\tau)$ in $q = e^{2\pi i \tau}$, and find $\lim_{\text{Im } \tau \rightarrow \infty} E_4(\tau)$.

Recognition. Trigger: plug $k = 4$ into the divisor-function formula. Read off the constant and q^1 terms.

Solution. With $k = 4$: $(-1)^2 = 1$, $(2\pi)^4 = 16\pi^4$, $(k-1)! = 6$, and $\zeta(4) = \pi^4/90$, so

$$E_4(\tau) = 2 \cdot \frac{\pi^4}{90} + \frac{2 \cdot 16\pi^4}{6} \sum_{r \geq 1} \sigma_3(r) q^r = \frac{\pi^4}{45} + \frac{16\pi^4}{3} (q + 9q^2 + \cdots),$$

using $\sigma_3(1) = 1$, $\sigma_3(2) = 1 + 8 = 9$. As $\text{Im } \tau \rightarrow \infty$, $q \rightarrow 0$, so $E_4(\tau) \rightarrow \pi^4/45$. *Check.* This matches the exercise value $E_4 \rightarrow \pi^4/45$; the exponential smallness of the correction ($|q| = e^{-2\pi \text{Im } \tau}$) gives the bound $|E_4 - \pi^4/45| \leq ce^{-2\pi t}$ for $t = \text{Im } \tau \geq 1$.

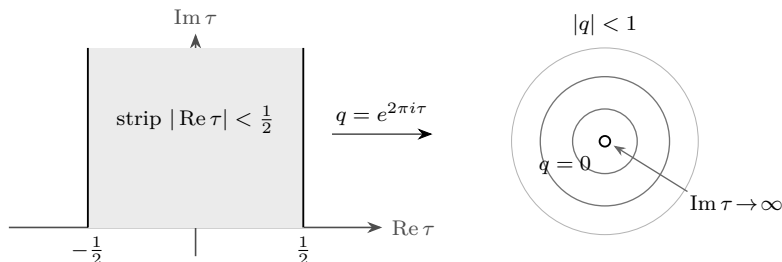


Figure 22.13. $\tau \mapsto q = e^{2\pi i\tau}$ sends the period strip to the punctured unit disc; $\text{Im } \tau \rightarrow \infty$ is the cusp $q \rightarrow 0$, where $E_k \rightarrow 2\zeta(k)$. (Schematic.)

Caution. The case $k = 2$ is genuinely different. The series $\sum_{(n,m) \neq (0,0)} (n+m\tau)^{-2}$ is *not* absolutely convergent, so its value depends on the order of summation: summing n first gives $F(\tau)$, summing m first gives a different $\tilde{F}(\tau)$, and $\tau^{-2}F(-1/\tau) \neq F(\tau)$. The modular transformation law that holds for all $k \geq 3$ fails at $k = 2$. This “forbidden” Eisenstein series nonetheless reappears, repaired, in the theory of sums of four squares.

Further worked examples

The modular transformations T, S and the q -expansion are levers: each lets a question about τ or about a lattice sum be reduced to a finite computation. The examples below run those levers in both directions, and pull in residues and contour integrals where the proofs of this section quietly used them.

Worked Example 22.2.16 (Acting with T and S on a base point). Starting from $\tau_0 = 2i$, compute $T\tau_0$, $S\tau_0$, $ST\tau_0$, and $TS\tau_0$, and decide which lands in the standard fundamental domain $\mathcal{F} = \{|\text{Re } \tau| \leq \frac{1}{2}, |\tau| \geq 1\}$.

Recognition. Trigger: apply the generators of the modular group to a number. Use $T : \tau \mapsto \tau + 1$, $S : \tau \mapsto -1/\tau$.

Solution. $T\tau_0 = 2i + 1$. $S\tau_0 = -1/(2i) = i/2$. For composites read right to left: $ST\tau_0 = S(2i + 1) = -1/(2i + 1) = -\frac{1-2i}{(1)^2+(2)^2} = \frac{-1+2i}{5}$; $TS\tau_0 = T(i/2) = 1 + i/2$. Membership in $\mathcal{F}$: $\tau_0 = 2i$ has $\text{Re} = 0$ and $|\tau_0| = 2 \geq 1$, so $\tau_0 \in \mathcal{F}$. $S\tau_0 = i/2$ has $|i/2| = \frac{1}{2} < 1$, outside $\mathcal{F}$ (it is the S -image of τ_0 , reflected inside the unit circle). *Check.* S is an involution, $S(S\tau_0) = S(i/2) = -1/(i/2) = 2i = \tau_0$, confirming $S^2 = \text{id}$ in $\text{PSL}_2(\mathbb{Z})$.

Worked Example 22.2.17 (The scaling law $\wp_{-1/\tau}(z) = \tau^2 \wp_\tau(\tau z)$, checked on poles). Verify the scaling law at the level of singularities: show both sides have double poles at the same points and identical principal parts there.

Recognition. Trigger: confirm a functional identity by matching Laurent heads. Track how the lattice Λ_τ transforms under $z \mapsto \tau z$.

Solution. The lattice for $-1/\tau$ is $\Lambda_{-1/\tau} = \mathbb{Z} + \mathbb{Z}(-1/\tau) = \frac{1}{\tau}(\mathbb{Z}\tau - \mathbb{Z}) = \frac{1}{\tau}\Lambda_\tau$. So $\wp_{-1/\tau}$ has double poles at $\omega' \in \frac{1}{\tau}\Lambda_\tau$. On the right, $\wp_\tau(\tau z)$ has poles where $\tau z \in \Lambda_\tau$, i.e. $z \in \frac{1}{\tau}\Lambda_\tau$ — the same set. Near such a pole $z = \omega'$, write $\tau z = \omega + \tau(z - \omega')$ with $\omega = \tau\omega' \in \Lambda_\tau$; then $\wp_\tau(\tau z) \sim \frac{1}{(\tau(z - \omega'))^2} = \frac{1}{\tau^2(z - \omega')^2}$, so $\tau^2 \wp_\tau(\tau z) \sim \frac{1}{(z - \omega')^2}$, matching the principal part of $\wp_{-1/\tau}$. *Check.* Both sides are even, elliptic of order 2 for the lattice $\Lambda_{-1/\tau}$, and agree in principal part, so their difference is entire and doubly periodic, hence constant; evaluating the even parts at a half-period shows the constant is 0.

Worked Example 22.2.18 (E_4 and E_6 at the cusp). Find $\lim_{\text{Im } \tau \rightarrow \infty} E_6(\tau)$ and the first q -correction, then form the j -numerator combination g_2^3 at the cusp where $g_2 = 60E_4$, $g_3 = 140E_6$.

Recognition. Trigger: read constant and q^1 coefficients from Theorem 22.2.12 for $k = 6$.

Solution. For $k = 6$: $(-1)^3 = -1$, $(2\pi)^6 = 64\pi^6$, $(k-1)! = 120$, $\zeta(6) = \pi^6/945$, so

$$E_6(\tau) = 2 \cdot \frac{\pi^6}{945} + \frac{2(-1)(2\pi)^6}{120} \sum_{r \geq 1} \sigma_5(r) q^r = \frac{2\pi^6}{945} - \frac{16\pi^6}{15} (q + 33q^2 + \cdots),$$

using $\sigma_5(1) = 1$, $\sigma_5(2) = 1 + 32 = 33$. As $\text{Im } \tau \rightarrow \infty$, $q \rightarrow 0$, so $E_6 \rightarrow 2\zeta(6) = \pi^6/945$. At the cusp $g_2 = 60E_4 \rightarrow 60 \cdot \frac{\pi^4}{45} = \frac{4\pi^4}{3}$ and $g_3 = 140E_6 \rightarrow 140 \cdot \frac{\pi^6}{945} = \frac{4\pi^6}{27}$; the discriminant $g_2^3 - 27g_3^2 \rightarrow (\frac{4\pi^4}{3})^3 - 27(\frac{4\pi^6}{27})^2 = \frac{64\pi^{12}}{27} - 27 \cdot \frac{16\pi^{12}}{729} = \frac{64\pi^{12}}{27} - \frac{16\pi^{12}}{27} = \frac{48\pi^{12}}{27} \neq 0$. *Check.* The discriminant being nonzero at the cusp is the condition that the cubic $4w^3 - g_2w - g_3$ has distinct roots e_1, e_2, e_3 — exactly the distinctness proved geometrically in Section 1.

Worked Example 22.2.19 (A residue computation behind Lemma 22.2.10). Derive $\sum_{n \in \mathbb{Z}} \frac{1}{(n + \tau)^2} = \frac{\pi^2}{\sin^2 \pi \tau}$ from the residues of $\pi \cot \pi z$, making the Chapter-5 step explicit.

Recognition. Trigger: a Mittag-Leffler / residue-summation identity. Use that $\pi \cot \pi z$ has simple poles at integers with residue 1.

Solution. The function $g(z) = \pi \cot \pi z$ has simple poles at each $n \in \mathbb{Z}$ with $\text{Res}_{z=n} g = 1$. Hence $\frac{d}{dz} [\pi \cot \pi z] = -\pi^2 / \sin^2 \pi z$ has at each integer n a double pole with principal part $-1/(z - n)^2$. Comparing with the convergent sum $\sum_n 1/(z - n)^2$, which has the same double poles and is likewise bounded at ∞ in horizontal strips, the two differ by a bounded entire (hence constant, in fact zero) function: $\sum_n \frac{1}{(z - n)^2} = \frac{\pi^2}{\sin^2 \pi z}$.

Replace z by $-\tau$ (or τ , even sum) to get the stated identity. *Check.* Take real $\tau \rightarrow \frac{1}{2}$: the right side is $\pi^2 / \sin^2(\pi/2) = \pi^2$, while the left is $\sum_n (n + \frac{1}{2})^{-2} = \pi^2$ as computed in the $\zeta(2)$ example above, so the two sides agree at a clean test value.

Worked Example 22.2.20 ($\zeta(4)$ from the $k = 4$ cusp value). Granting $E_4(\tau) \rightarrow 2\zeta(4)$ as $\text{Im } \tau \rightarrow \infty$ and $\zeta(4) = \pi^4/90$, cross-check the constant term $\pi^4/45$ found earlier, and deduce $\sum_{n \geq 1} 1/n^4$.

Recognition. Trigger: identify the cusp constant of E_k with $2\zeta(k)$. Match the two expressions.

Solution. The general formula gives the constant term $2\zeta(4) = 2 \cdot \pi^4/90 = \pi^4/45$, matching the direct q -expansion computation $E_4 \rightarrow \pi^4/45$. Reading backwards, $\zeta(4) = \frac{1}{2} \cdot \frac{\pi^4}{45} = \pi^4/90$, i.e. $\sum_{n \geq 1} 1/n^4 = \pi^4/90$.

Check. Numerically $\pi^4/90 \approx 1.0823$, and $1 + \frac{1}{16} + \frac{1}{81} + \frac{1}{256} \approx 1.0788$, climbing toward it; and $\zeta(4) = \frac{2}{5}\zeta(2)^2 = \frac{2}{5}(\pi^2/6)^2 = \pi^4/90$ as a consistency relation.

Worked Example 22.2.21 (A non-example: no nonzero modular form of weight 2). Explain why there is no holomorphic Eisenstein series E_2 obeying the full modular law $E_2(-1/\tau) = \tau^2 E_2(\tau)$, i.e. why weight 2 is special.

Recognition. Trigger: testing whether the $k \geq 3$ theory extends to $k = 2$. Use absolute-convergence failure (Corollary 22.2.13).

Solution. For $k \geq 3$ the series $\sum (n + m\tau)^{-k}$ is absolutely convergent, so reindexing under S is legitimate and gives $E_k(-1/\tau) = \tau^k E_k(\tau)$. At $k = 2$ the series is only conditionally convergent (Lemma 22.1.10 fails at $r = 2$), so the reindexing is not allowed; the actual repaired sum F satisfies $F(-1/\tau) = \tau^2 F(\tau) - 2\pi i \tau$, an *inhomogeneous* (“quasimodular”) law. So a genuine weight-2 modular Eisenstein series does not exist. *Check.* The anomaly term $-2\pi i \tau$ is exactly the order-of-summation discrepancy $\tilde{F} - F$ of Corollary 22.2.13, and it vanishes only formally; it is what makes the four-squares application work after a careful subtraction.

Worked Example 22.2.22 (Reading $\sigma_3(r)$ off the E_4 expansion). From the normalized expansion $E_4(\tau) = \frac{\pi^4}{45} (1 + 240 \sum_{r \geq 1} \sigma_3(r) q^r)$, verify the normalization constant 240 and read off $\sigma_3(3)$.

Recognition. Trigger: match the general Theorem 22.2.12 coefficient against the standard normalized form. Divide out the constant term.

Solution. From Theorem 22.2.12 with $k = 4$, the q^r coefficient is $\frac{2(2\pi)^4}{6} \sigma_3(r) = \frac{16\pi^4}{3} \sigma_3(r)$, and the constant is $\pi^4/45$. Their ratio is $\frac{16\pi^4/3}{\pi^4/45} = \frac{16}{3} \cdot 45 = 240$, confirming the normalization. Then $\sigma_3(3) = 1^3 + 3^3 = 1 + 27 = 28$, so the q^3 coefficient of the bracket is $240 \cdot 28 = 6720$. *Check.* σ_3 is multiplicative, $\sigma_3(6) = \sigma_3(2)\sigma_3(3) = 9 \cdot 28 = 252$, and indeed $1 + 8 + 27 + 216 = 252$ directly — the formula and the definition agree.

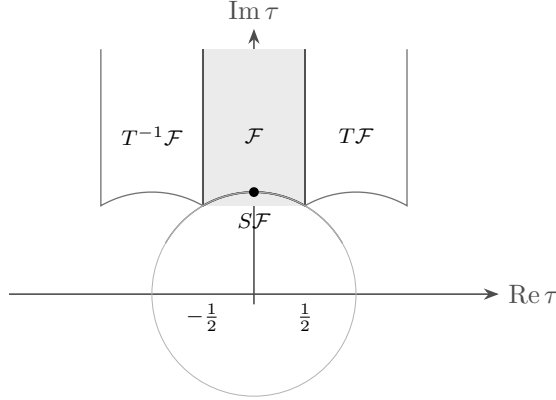


Figure 22.14. The modular tessellation near $\mathcal{F}$: T shifts left/right, S folds the cusp region down across the unit circle. The full group tiles $\mathbb{H}$ by copies of $\mathcal{F}$. (Schematic.)

Worked Example 22.2.23 (Behavior of $|q|$ and exponential smallness). For $\tau = x + it$ with $t \geq 1$, bound the tail $|E_4(\tau) - \frac{\pi^4}{45}|$ and estimate how fast the q -series converges.

Recognition. Trigger: convert a q -series into a numerical decay rate. Use $|q| = e^{-2\pi t}$ and $\sigma_3(r) \leq r^4$.

Solution. With $q = e^{2\pi i\tau}$, $|q| = e^{-2\pi t}$. Then

$$\left|E_4 - \frac{\pi^4}{45}\right| = \frac{16\pi^4}{3} \left| \sum_{r \geq 1} \sigma_3(r) q^r \right| \leq \frac{16\pi^4}{3} \sum_{r \geq 1} r^4 e^{-2\pi r t} \leq C e^{-2\pi t} \quad (t \geq 1),$$

since the geometric-type series $\sum r^4 x^r$ converges for $x = e^{-2\pi t} < 1$ and is $O(x)$ as $x \rightarrow 0$. At $t = 1$, $|q| = e^{-2\pi} \approx 1.87 \times 10^{-3}$, so the first correction is already at the 10^{-3} level relative to the constant. *Check.* Doubling t to 2 squares the smallness to $|q| \approx 3.5 \times 10^{-6}$ — the convergence is genuinely exponential in $\text{Im } \tau$, which is why the cusp value $2\zeta(k)$ dominates high in $\mathbb{H}$.

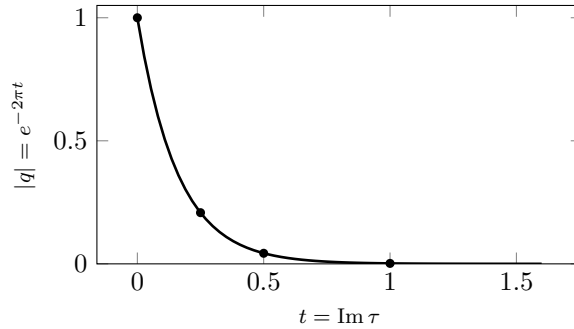


Figure 22.15. Exponential decay of $|q| = e^{-2\pi \text{Im } \tau}$. By $\text{Im } \tau = 1$ the q -corrections to E_k are already below 2×10^{-3} , so $E_k(\tau) \approx 2\zeta(k)$.

Exercises (Stein, Ch.9).

- In contrast with Lemma 22.1.10, prove that the series $\sum_{n+m\tau \in \Lambda^*} \frac{1}{|n+m\tau|^2}$ (with $\tau \in \mathbb{H}$) does *not* converge. In fact, show $\sum_{1 \leq n^2+m^2 \leq R^2} 1/(n^2+m^2) = 2\pi \log R + O(1)$ as $R \rightarrow \infty$.
- Setting $\tau = 1/2$ in $\sum_{m=-\infty}^{\infty} \frac{1}{(m+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}$, deduce $\sum_{m \geq 1, m \text{ odd}} \frac{1}{m^2} = \frac{\pi^2}{8}$ and $\sum_{m \geq 1} \frac{1}{m^2} = \frac{\pi^2}{6} = \zeta(2)$. Similarly, using $\sum_m 1/(m+\tau)^4$, deduce $\sum_{m \geq 1, m \text{ odd}} \frac{1}{m^4} = \frac{\pi^4}{96}$ and $\sum_{m \geq 1} \frac{1}{m^4} = \frac{\pi^4}{90} = \zeta(4)$.
- Let $E_4(\tau) = \sum_{(n,m) \neq (0,0)} \frac{1}{(n+m\tau)^4}$ be the Eisenstein series of order 4. (a) Show $E_4(\tau) \rightarrow \pi^4/45$ as $\text{Im}(\tau) \rightarrow \infty$. (b) More precisely, $|E_4(\tau) - \frac{\pi^4}{45}| \leq c e^{-2\pi t}$ if $\tau = x + it$ with $t \geq 1$. (c) Deduce $|E_4(\tau) - \tau^{-4} \frac{\pi^4}{45}| \leq c t^{-4} e^{-2\pi/t}$ if $\tau = it$ and $0 < t \leq 1$.

Problems (Stein, Ch.9).

1. Show that $\wp(z) = c + \pi^2 \sum_{m=-\infty}^{\infty} \frac{1}{\sin^2((z+m\tau)\pi)}$ for an appropriate constant c . In fact, by part (b) of Problem 1, $c = -F(\tau)$.
2. * Suppose τ is purely imaginary, $\tau = it$ with $t > 0$. Divide the plane into congruent rectangles by the lines $x = n/2$, $y = tm/2$ ($n, m \in \mathbb{Z}$); an example is the rectangle with vertices $0, \frac{1}{2}, \frac{1}{2} + \frac{\tau}{2}, \frac{\tau}{2}$. (a) Show $\wp$ is real-valued on all these lines, hence on the boundary of every such rectangle. (b) Prove $\wp$ maps the interior of each rectangle conformally onto the upper (or lower) half-plane.

Additional exercises.

1. (Modular shift.) Verify directly from the definition that $E_k(\tau + 1) = E_k(\tau)$ by exhibiting the bijection of $\mathbb{Z}^2 \setminus 0$ that proves it. Which generator of the modular group does this correspond to?
2. (Vanishing by parity.) Show $E_5(\tau) \equiv 0$ and, more generally, $E_k \equiv 0$ for every odd $k \geq 3$, by pairing ω with $-\omega$. Explain why the same pairing makes $E_4 > 0$ when $\tau = i$.
3. (Constant term.) Using Theorem 22.2.12, compute $\lim_{\text{Im } \tau \rightarrow \infty} E_6(\tau)$ in closed form (in terms of $\zeta(6) = \pi^6/945$).
4. (Divisor function.) Compute $\sigma_3(1), \sigma_3(2), \sigma_3(3), \sigma_3(4)$ and use them to write the q -expansion of $E_4(\tau)$ through the q^3 term (with $q = e^{2\pi i \tau}$).
5. (g_2, g_3 in terms of ζ .) Express $g_2 = 60E_4$ and $g_3 = 140E_6$ in terms of $\zeta(4), \zeta(6)$ in the limit $\text{Im } \tau \rightarrow \infty$, and evaluate numerically.
6. (Non-example.) Show that $E_2(\tau) := \sum_{(n,m) \neq (0,0)} (n+m\tau)^{-2}$ is *not* well defined independent of summation order, by contrasting summing n -first with m -first. Why does this not contradict Theorem 22.2.3(iii)?
7. (Diagram problem.) Sketch the standard fundamental domain $\mathcal{F} = \{|\text{Re } \tau| \leq \frac{1}{2}, |\tau| \geq 1\}$ of the modular group in the upper half-plane, and show on the picture the images of $\mathcal{F}$ under $T : \tau \mapsto \tau + 1$ and $S : \tau \mapsto -1/\tau$. Mark the cusp $\text{Im } \tau \rightarrow \infty$ and the corner $\tau = i$.
8. (Conformal map of $\wp$.) For $\tau = i$ (so the lattice is square), the half-period rectangle has vertices $0, \frac{1}{2}, \frac{1}{2} + \frac{i}{2}, \frac{i}{2}$. Sketch this rectangle and, using Stein Problem 4, indicate where its boundary lands ($\wp$ real) and that the interior maps conformally onto a half-plane.
9. (Growth bound.) Using $E_4(\tau) = \frac{\pi^4}{45} + \frac{16\pi^4}{3} \sum_{r \geq 1} \sigma_3(r) q^r$, prove $|E_4(\tau) - \frac{\pi^4}{45}| \leq C e^{-2\pi t}$ for $\tau = x + it$, $t \geq 1$, by bounding $\sigma_3(r) \leq r^4$ and summing a geometric-type series.
10. (Laurent coefficients.) From $\wp(z) = z^{-2} + \sum_{k \geq 1} (2k+1)E_{2k+2}z^{2k}$, read off the coefficients of z^2 and z^4 , and confirm they are $3E_4$ and $5E_6$. Then state the z^6 coefficient.

Chapter 23

Applications of Theta Functions

23.1 The Product Formula for the Jacobi Theta Function

The three theta functions

The theory of theta functions is the meeting point of complex analysis, Fourier analysis, and number theory. In its most flexible form, Jacobi's theta function depends on two variables: a complex point $z \in \mathbb{C}$ and a modular parameter τ ranging over the upper half-plane $\mathbb{H} = \{\tau \in \mathbb{C} : \text{Im } \tau > 0\}$.

Definition 23.1.1. For $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ the *Jacobi theta function* is

$$\Theta(z \mid \tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}.$$

Two special cases will recur. Writing $q = e^{\pi i \tau}$ (so $|q| = e^{-\pi \text{Im } \tau} < 1$),

$$\theta(\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} = \sum_{n=-\infty}^{\infty} q^{n^2}, \quad \tau \in \mathbb{H},$$

$$\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}, \quad t > 0.$$

The relations among them are $\theta(\tau) = \Theta(0 \mid \tau)$ and $\vartheta(t) = \theta(it)$ for $t > 0$.

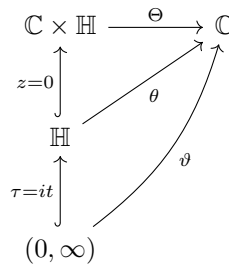


Figure 23.1. The two special cases as restrictions of Θ : $\theta = \Theta(0 \mid \cdot)$ is the restriction to the slice $z = 0$, and $\vartheta(t) = \theta(it)$ is the further restriction to the imaginary axis $\tau = it$.

The factor $e^{\pi i n^2 \tau} = q^{n^2}$ decays super-exponentially in $|n|$ because n^2 grows quadratically, which is what makes every series here converge so rapidly. These functions are not exotic: the heat kernel on the circle is $H_t(x) = \Theta(x \mid 4\pi it)$, and the functional equation of ϑ is exactly the engine behind the analytic continuation of the Riemann zeta function (Chapter 6 of Stein–Shakarchi, Book II).

$$\begin{array}{ccc}
\mathbb{R} \times (0, \infty) & \xrightarrow{(x,t) \mapsto (x, 4\pi i t)} & \mathbb{C} \times \mathbb{H} \\
& \searrow H & \downarrow \Theta \\
& & \mathbb{C}
\end{array}$$

We now record the structural properties of Θ as a function of z , with τ fixed.

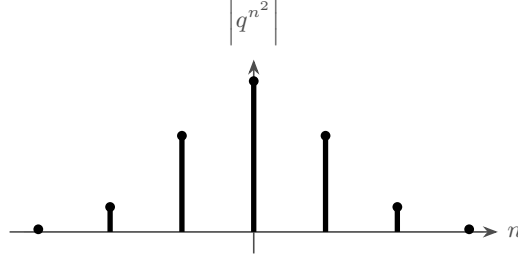


Figure 23.2. The Gaussian-in- n profile of the coefficients $|q^{n^2}|$ for a fixed $\tau \in \mathbb{H}$ (here $|q| = e^{-\pi \operatorname{Im} \tau} < 1$). Because the exponent n^2 grows quadratically, the terms decay super-exponentially in $|n|$, so $\theta(\tau) = \sum q^{n^2}$ converges geometrically fast. (Schematic; vertical scale arbitrary.)

Proposition 23.1.2 (Structural properties of Θ). *The function Θ satisfies:*

- (i) Θ is entire in $z \in \mathbb{C}$ and holomorphic in $\tau \in \mathbb{H}$.
- (ii) $\Theta(z+1 \mid \tau) = \Theta(z \mid \tau)$ (periodicity in z , period 1).
- (iii) $\Theta(z+\tau \mid \tau) = \Theta(z \mid \tau) e^{-\pi i \tau} e^{-2\pi i z}$ (quasi-periodicity, period τ).
- (iv) $\Theta(z \mid \tau) = 0$ whenever $z = \frac{1}{2} + \frac{\tau}{2} + n + m\tau$ with $n, m \in \mathbb{Z}$.

Proof. (i) Suppose $\operatorname{Im} \tau = t \geq t_0 > 0$ and $|z| \leq M$. Then

$$\sum_{n=-\infty}^{\infty} \left| e^{\pi i n^2 \tau} e^{2\pi i n z} \right| \leq C \sum_{n \geq 0} e^{-\pi n^2 t_0} e^{2\pi n M} < \infty,$$

because the quadratic $-\pi n^2 t_0$ dominates the linear $2\pi n M$. The convergence is absolute and uniform on $\{\operatorname{Im} \tau \geq t_0, |z| \leq M\}$, so the limit is entire in z for fixed τ and holomorphic in τ for fixed z .

(ii) Each summand has $e^{2\pi i n(z+1)} = e^{2\pi i n z} e^{2\pi i n} = e^{2\pi i n z}$, so the whole series is unchanged.

(iii) Complete the square. Using $e^{2\pi i n \tau} = e^{\pi i (2n)\tau}$,

$$\begin{aligned}
\Theta(z+\tau \mid \tau) &= \sum_n e^{\pi i n^2 \tau} e^{2\pi i n(z+\tau)} = \sum_n e^{\pi i (n^2+2n)\tau} e^{2\pi i n z} \\
&= \sum_n e^{\pi i (n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i (n+1)z} e^{-2\pi i z} = \Theta(z \mid \tau) e^{-\pi i \tau} e^{-2\pi i z},
\end{aligned}$$

where in the last step we re-indexed $n+1 \mapsto n$.

(iv) By (ii) and (iii) it suffices to show $\Theta(\frac{1}{2} + \frac{\tau}{2} \mid \tau) = 0$. Again using n^2 and n :

$$\Theta\left(\frac{1}{2} + \frac{\tau}{2} \mid \tau\right) = \sum_n e^{\pi i n^2 \tau} e^{2\pi i n(1/2 + \tau/2)} = \sum_n (-1)^n e^{\pi i (n^2 + n)\tau}.$$

Pair $n \geq 0$ with $-n-1$: these have opposite parity, so the signs $(-1)^n$ cancel, while $(-n-1)^2 + (-n-1) = n^2 + n$, so the exponentials agree. Every term cancels, and the sum is identically zero. $\square$

The picture to keep in mind is a function on $\mathbb{C}$ invariant under the lattice $\mathbb{Z} + \mathbb{Z}\tau$ up to the explicit exponential factor in (iii) — “quasi-periodic.” Its zeros sit at the centers $\frac{1}{2} + \frac{\tau}{2}$ of the fundamental cells.

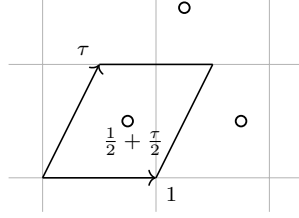


Figure 23.3. Fundamental parallelogram $\mathbb{Z} + \mathbb{Z}\tau$ with the (simple) zero of $\Theta(\cdot | \tau)$ at the center $\frac{1}{2} + \frac{\tau}{2}$, repeated by the lattice. (Schematic.)

Worked Example 23.1.3 (Reading off a quasi-period shift). Compute $\Theta(z + 2\tau | \tau)$ in terms of $\Theta(z | \tau)$. *Recognition.* Iterate property (iii). Each shift by τ multiplies by $e^{-\pi i \tau} e^{-2\pi i z}$, but the second shift sees the argument $z + \tau$.

Solution. Apply (iii) twice: $\Theta(z + 2\tau | \tau) = \Theta(z + \tau | \tau) e^{-\pi i \tau} e^{-2\pi i (z + \tau)} = \Theta(z | \tau) e^{-\pi i \tau} e^{-2\pi i z} \cdot e^{-\pi i \tau} e^{-2\pi i z} e^{-2\pi i \tau} = \Theta(z | \tau) e^{-4\pi i \tau} e^{-4\pi i z}$. In general $\Theta(z + m\tau | \tau) = \Theta(z | \tau) e^{-\pi i m^2 \tau} e^{-2\pi i m z}$. *Check.* At $m = 1$ this reduces to (iii); the exponent of $q = e^{\pi i \tau}$ is $-m^2$, matching the symmetry $\theta(\tau) = \sum q^{n^2}$ ($n \rightarrow n + m$ shifts the index).

The triple product

We now meet a product that has *exactly* the same structural properties as Θ .

Definition 23.1.4. For $z \in \mathbb{C}$, $\tau \in \mathbb{H}$, and $q = e^{\pi i \tau}$, the *triple product* is

$$\Pi(z | \tau) = \prod_{n=1}^{\infty} (1 - q^{2n}) (1 + q^{2n-1} e^{2\pi i z}) (1 + q^{2n-1} e^{-2\pi i z}).$$

Proposition 23.1.5 (Structural properties of Π). *The triple product satisfies:*

- (i) $\Pi(z | \tau)$ is entire in z and holomorphic in $\tau \in \mathbb{H}$.
- (ii) $\Pi(z + 1 | \tau) = \Pi(z | \tau)$.
- (iii) $\Pi(z + \tau | \tau) = \Pi(z | \tau) e^{-\pi i \tau} e^{-2\pi i z}$.
- (iv) $\Pi(z | \tau) = 0$ exactly when $z = \frac{1}{2} + \frac{\tau}{2} + n + m\tau$, $n, m \in \mathbb{Z}$; these are simple zeros, and there are no others.

Proof. (i) If $\text{Im } \tau = t \geq t_0$ then $|q| \leq e^{-\pi t_0} < 1$, and the n -th factor is $1 + O(|q|^{2n-1} e^{2\pi |z|})$. Since $\sum |q|^{2n-1} < \infty$, the infinite-product convergence theorems (Stein–Shakarchi Ch.5) make Π entire in z and holomorphic in τ .

(ii) The factors $e^{\pm 2\pi i z}$ are periodic of period 1 in z .

(iii) Since $q^2 = e^{2\pi i \tau}$, replacing z by $z + \tau$ multiplies $e^{2\pi i z}$ by q^2 and $e^{-2\pi i z}$ by q^{-2} :

$$\Pi(z + \tau | \tau) = \prod_{n=1}^{\infty} (1 - q^{2n}) (1 + q^{2n+1} e^{2\pi i z}) (1 + q^{2n-3} e^{-2\pi i z}).$$

Comparing factor by factor with $\Pi(z | \tau)$ and isolating the one extra and one missing factor,

$$\Pi(z + \tau | \tau) = \Pi(z | \tau) \frac{1 + q^{-1} e^{-2\pi i z}}{1 + q e^{2\pi i z}}.$$

Now $(1 + x)/(1 + x^{-1}) = x$ for $x \neq -1$ with $x = q e^{2\pi i z}$ gives the factor $q e^{2\pi i z}$ (rearranged); carefully, $\frac{1 + q^{-1} e^{-2\pi i z}}{1 + q e^{2\pi i z}} = q^{-1} e^{-2\pi i z} = e^{-\pi i \tau} e^{-2\pi i z}$, which is (iii).

(iv) A convergent product vanishes iff a factor does. The factor $1 - q^{2n}$ never vanishes ($|q| < 1$). The factor $1 + q^{2n-1} e^{2\pi i z} = 0$ requires $q^{2n-1} e^{2\pi i z} = -1 = e^{\pi i}$, i.e. $(2n - 1)\tau + 2z \equiv 1 \pmod{2}$, giving $z \equiv \frac{1}{2} + \frac{\tau}{2} - n\tau \pmod{1}$ for $n \geq 1$. Similarly the third factor gives $z \equiv \frac{1}{2} + \frac{\tau}{2} + n'\tau \pmod{1}$ for $n' \geq 0$. Together these are exactly the points in (iv). Each zero is simple because $e^w - 1$ vanishes to first order at $w = 0$. $\square$

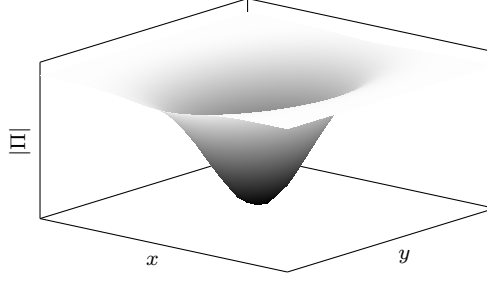


Figure 23.4. Schematic modulus surface of $\Pi(z \mid \tau)$ over a fundamental cell: a single simple zero (the dip) at the cell center $\frac{1}{2} + \frac{\tau}{2}$.

The product formula

Theorem 23.1.6 (Product formula). *For all $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$, $\Theta(z \mid \tau) = \Pi(z \mid \tau)$.*

Proof. Fix $\tau \in \mathbb{H}$. **Step 1: a constant ratio.** Consider $F(z) = \Theta(z \mid \tau)/\Pi(z \mid \tau)$. By Propositions 23.1.2 and 23.1.5, Θ and Π have identical periodicity (ii), identical quasi-periodicity (iii), and identical zero sets (iv) — and the zeros of Π are simple, so they exactly cancel the zeros of Θ . Hence F is entire and doubly periodic with periods 1 and τ (the quasi-periodicity factors cancel in the ratio). A bounded entire function on a fundamental parallelogram is constant by Liouville, so

$$\Theta(z \mid \tau) = c(\tau) \Pi(z \mid \tau) \quad (23.1)$$

for some constant $c(\tau)$.

Step 2: $c(\tau) = c(4\tau)$. Put $z = \frac{1}{2}$ in (23.1), so $e^{2\pi iz} = e^{-2\pi iz} = -1$:

$$\sum_{n=-\infty}^{\infty} (-1)^n q^{n^2} = c(\tau) \prod_{n=1}^{\infty} (1 - q^{2n})(1 - q^{2n-1})^2 = c(\tau) \prod_{n=1}^{\infty} (1 - q^n)(1 - q^{2n-1}),$$

the last step merging $(1 - q^{2n-1})(1 - q^{2n}) = (1 - q^n)$ over all n . Hence

$$c(\tau) = \frac{\sum_{n=-\infty}^{\infty} (-1)^n q^{n^2}}{\prod_{n=1}^{\infty} (1 - q^n)(1 - q^{2n-1})}. \quad (23.2)$$

Now put $z = \frac{1}{4}$, so $e^{2\pi iz} = i$. On the theta side, i^n kills all odd n , leaving the even terms $n = 2m$:

$$\Theta(\tfrac{1}{4} \mid \tau) = \sum_n q^{n^2} i^n = \sum_{m=-\infty}^{\infty} q^{4m^2} (-1)^m.$$

On the product side, $(1 + iq^{2m-1})(1 - iq^{2m-1}) = 1 + q^{4m-2}$, and separating the two cases $2m = 4n - 4$ and $2m = 4n - 2$,

$$\Pi(\tfrac{1}{4} \mid \tau) = \prod_{m \geq 1} (1 - q^{2m})(1 + q^{4m-2}) = \prod_{n \geq 1} (1 - q^{4n})(1 - q^{8n-4}).$$

Therefore

$$c(\tau) = \frac{\sum_n (-1)^n q^{4n^2}}{\prod_{n \geq 1} (1 - q^{4n})(1 - q^{8n-4})}. \quad (23.3)$$

The right-hand side of (23.3) is precisely (23.2) with q replaced by q^4 , i.e. with τ replaced by 4τ . Thus $c(\tau) = c(4\tau)$.

Step 3: $c \equiv 1$. Iterating, $c(\tau) = c(4^k \tau)$ for all k . As $k \rightarrow \infty$, $q^{4^k} = e^{\pi i 4^k \tau} \rightarrow 0$ (since $\text{Im}(4^k \tau) \rightarrow \infty$), so both numerator and denominator in (23.2) (with $\tau \rightarrow 4^k \tau$) tend to 1, giving $c(\tau) = 1$. $\square$

The proof is a gem of the “two functions with the same structure must be equal” principle: double periodicity collapses the ratio to a constant, and a clever specialization ($z = \frac{1}{2}, \frac{1}{4}$) plus a $\tau \mapsto 4\tau$ self-similarity pins the constant to 1.

Corollary 23.1.7 (θ is non-vanishing). *If $\text{Im } \tau > 0$ and $q = e^{\pi i \tau}$, then*

$$\theta(\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1})^2,$$

and in particular $\theta(\tau) \neq 0$ for all $\tau \in \mathbb{H}$.

Proof. Set $z = 0$ in the product formula: each pair of factors $(1 + q^{2n-1}e^{2\pi iz})(1 + q^{2n-1}e^{-2\pi iz})$ becomes $(1 + q^{2n-1})^2$. The point $z = 0$ is *not* a zero (the zeros are at $\frac{1}{2} + \frac{\tau}{2} + \text{lattice}$), so no factor vanishes, and $\theta(\tau) \neq 0$. $\square$

Worked Example 23.1.8 ($\theta(\tau)$ as a q -series, first terms). Expand $\theta(\tau) = 1 + 2q + 2q^4 + 2q^9 + \cdots$ directly and check against Corollary 23.1.7 to order q^4 .

Recognition. From the sum side $\theta = \sum q^{n^2}$, the exponents are perfect squares; from the product side multiply out a few factors and compare.

Solution. Sum side: $n^2 \in \{0, 1, 1, 4, 4, 9, 9, \dots\}$, so $\theta = 1 + 2q + 2q^4 + 2q^9 + \cdots$. Product side, keeping terms up to q^4 : $(1 + q)^2 = 1 + 2q + q^2$ ($n = 1$), $(1 - q^2) = 1 - q^2$, $(1 + q^3)^2 = 1 + 2q^3 + \cdots$, $(1 - q^4)$. Multiply: $(1 + 2q + q^2)(1 - q^2) = 1 + 2q - 2q^3 - q^4 + \cdots$; times $(1 + 2q^3)(1 - q^4) = 1 + 2q^3 - q^4 + \cdots$ gives $1 + 2q + 0 \cdot q^2 + 0 \cdot q^3 + 2q^4 + \cdots$. *Check.* The q^2 and q^3 coefficients vanish (since 2, 3 are not squares) and the q^4 coefficient is 2, matching the sum side.

Corollary 23.1.9 (An elliptic function of order 2). *For each fixed $\tau \in \mathbb{H}$, the second logarithmic derivative*

$$(\log \Theta(z | \tau))'' = \frac{\Theta(z | \tau)\Theta''(z | \tau) - (\Theta'(z | \tau))^2}{\Theta(z | \tau)^2}$$

is an elliptic function of order 2 with periods 1 and τ , having a double pole at $z = \frac{1}{2} + \frac{\tau}{2}$. (Primes denote ∂_z .) It is closely related to the Weierstrass $\wp_\tau$ function (Exercise 1).

Proof. Let $F(z) = (\log \Theta)' = \Theta'/\Theta$. Differentiating (ii) and (iii) of Proposition 23.1.2 gives $F(z + 1) = F(z)$ and $F(z + \tau) = F(z) - 2\pi i$; thus $F'(z) = (\log \Theta)''$ is doubly periodic. Since Θ has a single simple zero at $\frac{1}{2} + \frac{\tau}{2}$ in the fundamental cell, F has a single simple pole there, so F' has a double pole. An elliptic function whose only pole in a cell is a double pole has order 2. $\square$

Because $(\log \Theta)''$ is invariant under the lattice $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, it factors through the quotient torus $\mathbb{C}/\Lambda$: there is a unique meromorphic function on the torus pulling back to it along the projection π .

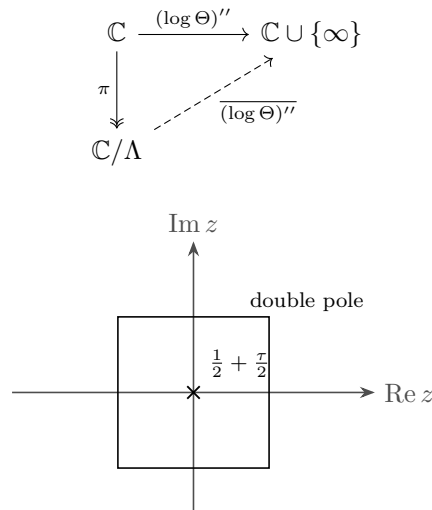


Figure 23.5. $(\log \Theta)''$ is elliptic of order 2: one double pole per fundamental cell, at the zero of Θ . (Schematic.)

Transformation laws in τ : the modular character

We now study how Θ behaves under the modular substitutions

$$T_2 : \tau \mapsto \tau + 2, \quad S : \tau \mapsto -1/\tau,$$

both preserving $\mathbb{H}$. (We use T_2 rather than T_1 because $\Theta(z \mid \tau + 2) = \Theta(z \mid \tau)$ while $\Theta(z \mid \tau + 1) \neq \Theta(z \mid \tau)$: the coefficient $e^{\pi i n^2}$ has period 2 in τ as a function of n^2 .)

Theorem 23.1.10 (The S -transformation of Θ). *For $\tau \in \mathbb{H}$ and all $z \in \mathbb{C}$,*

$$\Theta(z \mid -1/\tau) = \sqrt{\frac{\tau}{i}} e^{\pi i \tau z^2} \Theta(z \tau \mid \tau), \quad (23.4)$$

where $\sqrt{\tau/i}$ is the branch on $\mathbb{H}$ positive when $\tau = it$, $t > 0$.

Proof. By analytic continuation it suffices to prove (23.4) for $z = x$ real and $\tau = it$, $t > 0$: for fixed real x both sides are holomorphic in $\tau \in \mathbb{H}$ and agree on the imaginary axis; for fixed τ both sides are holomorphic in z and agree on $\mathbb{R}$. With $z = x$ real and $\tau = it$ the identity becomes

$$\sum_n e^{-\pi n^2/t} e^{2\pi i n x} = t^{1/2} e^{-\pi t x^2} \sum_n e^{-\pi n^2 t} e^{-2\pi n x t}.$$

Writing $x = a$ and rearranging, this is

$$\sum_n e^{-\pi t(n+a)^2} = t^{-1/2} \sum_n e^{-\pi n^2/t} e^{2\pi i n a},$$

which is exactly the Poisson summation formula applied to the Gaussian (equation (3) of Stein–Shakarchi Ch.4). $\square$

Corollary 23.1.11 (Functional equation of θ). *If $\text{Im } \tau > 0$, then $\theta(-1/\tau) = \sqrt{\tau/i} \theta(\tau)$. For $\tau = it$ this is the classical $\vartheta(1/t) = \sqrt{t} \vartheta(t)$.*

Proof. Set $z = 0$ in Theorem 23.1.10: $e^{\pi i \tau \cdot 0} = 1$ and $\Theta(0 \mid \tau) = \theta(\tau)$. $\square$

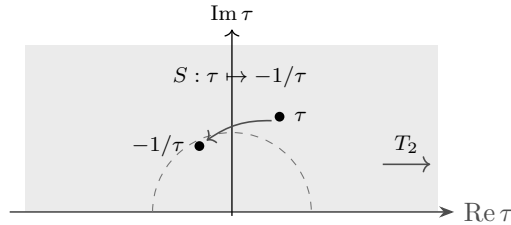


Figure 23.6. The modular substitutions on $\mathbb{H}$: S inverts through the unit circle, T_2 translates by 2. Both preserve the upper half-plane. (Schematic.)

Corollary 23.1.12 (Behavior near the cusp $\tau = 1$). *If $\tau \in \mathbb{H}$,*

$$\theta(1 - 1/\tau) = \sqrt{\frac{\tau}{i}} \sum_{n=-\infty}^{\infty} e^{\pi i (n+1/2)^2 \tau} = \sqrt{\frac{\tau}{i}} (2e^{\pi i \tau/4} + \dots),$$

meaning $\theta(1 - 1/\tau) \sim \sqrt{\tau/i} \cdot 2e^{\pi i \tau/4}$ as $\text{Im } \tau \rightarrow \infty$.

Proof. Since n and n^2 have the same parity, $\theta(1 + \tau) = \sum_n (-1)^n e^{\pi i n^2 \tau} = \Theta(\frac{1}{2} \mid \tau)$, hence $\theta(1 - 1/\tau) = \Theta(\frac{1}{2} \mid -1/\tau)$. Apply Theorem 23.1.10 with $z = \frac{1}{2}$:

$$\theta(1 - 1/\tau) = \sqrt{\frac{\tau}{i}} e^{\pi i \tau/4} \Theta(\frac{\tau}{2} \mid \tau) = \sqrt{\frac{\tau}{i}} \sum_n e^{\pi i n^2 \tau} e^{\pi i n \tau} = \sqrt{\frac{\tau}{i}} \sum_n e^{\pi i (n+1/2)^2 \tau}.$$

The $n = 0, -1$ terms each give $e^{\pi i \tau/4}$ (total $2e^{\pi i \tau/4}$), of modulus $2e^{-\pi t/4}$; the remaining terms are $O(e^{-9\pi t/4})$. $\square$

The Dedekind eta function

Definition 23.1.13. The *Dedekind eta function* is, for $\text{Im } \tau > 0$,

$$\eta(\tau) = e^{\pi i \tau / 12} \prod_{n=1}^{\infty} (1 - e^{2\pi i n \tau}).$$

Proposition 23.1.14 (Functional equation of η). *If $\text{Im } \tau > 0$, then $\eta(-1/\tau) = \sqrt{\tau/i} \eta(\tau)$.*

Proof. From the product formula, factoring out the term containing the zero at $z_0 = \frac{1}{2} + \frac{\tau}{2}$, write with $q = e^{\pi i \tau}$

$$\Theta(z | \tau) = (1 + qe^{-2\pi iz}) \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1}e^{2\pi iz})(1 + q^{2n+1}e^{-2\pi iz}).$$

The first factor vanishes at z_0 , so $\Theta'(z_0 | \tau) = 2\pi i H(\tau)$ where $H(\tau) = \prod_{n \geq 1} (1 - e^{2\pi i n \tau})^3$. Replacing $-1/\tau$ by τ in (23.4) gives $\Theta(z | \tau) = \sqrt{i/\tau} e^{-\pi i z^2 / \tau} \Theta(-z/\tau | -1/\tau)$. Differentiate in z and evaluate at $z_0 = \frac{1}{2} + \frac{\tau}{2}$:

$$2\pi i H(\tau) = \sqrt{\frac{i}{\tau}} e^{-\pi i / (4\tau)} e^{-\pi i \tau / 4} \left(\frac{-2\pi i}{\tau} \right) e^{-\pi i / 2} H(-1/\tau).$$

Simplifying, $e^{\pi i \tau / 4} H(\tau) = (i/\tau)^{3/2} e^{-\pi i / (4\tau)} H(-1/\tau)$. The cube of η is exactly $e^{\pi i \tau / 4} H(\tau)$ up to the prefactor; for $\tau = it$ the function $\eta(\tau)$ is positive, so taking cube roots gives $\eta(\tau) = \sqrt{i/\tau} \eta(-1/\tau)$, equivalently $\eta(-1/\tau) = \sqrt{\tau/i} \eta(\tau)$, and this extends to all $\tau \in \mathbb{H}$ by analytic continuation. $\square$

Caution. The square root $\sqrt{\tau/i}$ in (23.4) is a genuine branch choice, not a formality. Picking the wrong sheet flips the sign and breaks the functional equation. Always anchor it by the requirement that $\sqrt{\tau/i} > 0$ when $\tau = it$ with $t > 0$, then continue holomorphically. The same care is needed for η : the cube-root step in Proposition 23.1.14 is legitimate only because $\eta(it) > 0$ pins the branch.

Worked Example 23.1.15 (The eta product and the discriminant). Show that $\Delta(\tau) := (2\pi)^{12} \eta(\tau)^{24}$ transforms as a weight-12 modular form, i.e. $\Delta(-1/\tau) = \tau^{12} \Delta(\tau)$.

Recognition. Raise the η functional equation to the 24-th power; the half-integer weight of η becomes the integer weight 12 of Δ .

Solution. By Proposition 23.1.14, $\eta(-1/\tau) = (\tau/i)^{1/2} \eta(\tau)$. Hence $\eta(-1/\tau)^{24} = (\tau/i)^{12} \eta(\tau)^{24}$. Now $(\tau/i)^{12} = \tau^{12}/i^{12} = \tau^{12}/(i^4)^3 = \tau^{12}/1 = \tau^{12}$. Multiplying by $(2\pi)^{12}$, $\Delta(-1/\tau) = \tau^{12} \Delta(\tau)$. *Check.* The weight is $24 \cdot \frac{1}{2} = 12$, and $i^{12} = 1$ removes all branch ambiguity, so Δ is single-valued and holomorphic on $\mathbb{H}$ — it is the modular discriminant (cf. Problem 5).

Further worked examples

Worked Example 23.1.16 (The q -expansion of $\eta(\tau)$ to order q^{14}). With $q = e^{\pi i \tau}$, expand $\eta(\tau) = q^{1/6} \prod_{n \geq 1} (1 - q^{2n})$ and identify the exponents that occur. (Here $e^{2\pi i n \tau} = q^{2n}$.)

Recognition. This is the pentagonal number theorem (Proposition 23.2.7) in disguise: $\prod_{n \geq 1} (1 - q^{2n}) = \sum_k (-1)^k q^{k(3k+1)}$ after replacing $x = q^2$.

Solution. By the pentagonal theorem with $x = q^2$, $\prod_{n \geq 1} (1 - q^{2n}) = \sum_{k=-\infty}^{\infty} (-1)^k q^{2 \cdot k(3k+1)/2} = \sum_k (-1)^k q^{k(3k+1)}$. The exponents $k(3k+1)$ for $k = 0, 1, -1, 2, -2, \dots$ are $0, 4, 2, 14, 10, \dots$ with signs $+, -, -, +, +, \dots$, so $\prod_{n \geq 1} (1 - q^{2n}) = 1 - q^2 - q^4 + q^{10} + q^{14} - \dots$. Therefore $\eta(\tau) = q^{1/6} (1 - q^2 - q^4 + q^{10} + q^{14} - \dots)$. *Check.* The smallest positive exponent inside the product is 2 (from the single factor $1 - q^2$), and the next is 4; multiplying $(1 - q^2)(1 - q^4)(1 - q^6) \dots$ by hand reproduces $1 - q^2 - q^4 + 0 \cdot q^6 + \dots$, confirming the pentagonal pattern.

Worked Example 23.1.17 (A contour proof that $\Theta(\cdot | \tau)$ has exactly one zero per cell). Fix $\tau \in \mathbb{H}$ and let P be the boundary of a fundamental parallelogram (vertices $z_0, z_0 + 1, z_0 + 1 + \tau, z_0 + \tau$) avoiding the zeros. Use the argument principle to count the zeros of $\Theta(\cdot | \tau)$ inside P .

Recognition. The number of zeros inside P is $\frac{1}{2\pi i} \oint_P \frac{\Theta'}{\Theta} dz$ (argument principle); evaluate it using the quasi-periodicity (ii)–(iii) of Proposition 23.1.2.

Solution. Pair opposite sides. On the two horizontal sides (related by $z \mapsto z + \tau$), the logarithmic derivative shifts: differentiating $\log \Theta(z + \tau) = \log \Theta(z) - \pi i \tau - 2\pi i z$ gives $(\log \Theta)'(z + \tau) = (\log \Theta)'(z) - 2\pi i$. Hence the two horizontal contributions do not cancel; their difference integrates $-(-2\pi i)$ over a unit horizontal segment:

$$\frac{1}{2\pi i} \left(\int_{z_0}^{z_0+1} - \int_{z_0+\tau}^{z_0+1+\tau} \right) \frac{\Theta'}{\Theta} dz = \frac{1}{2\pi i} \int_{z_0}^{z_0+1} 2\pi i dz = 1.$$

The two vertical sides (related by $z \mapsto z + 1$) give exactly cancelling contributions since $(\log \Theta)'$ is periodic of period 1. Total: 1 zero. *Check.* This matches Proposition 23.1.2(iv), which locates that single zero at $\frac{1}{2} + \frac{\tau}{2}$; since it is simple (Proposition 23.1.5(iv)) the count 1 is exact.

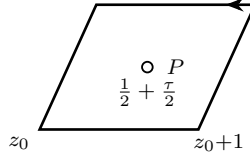


Figure 23.7. Argument-principle contour P around one fundamental cell: the horizontal sides contribute $+1$, the vertical sides cancel, giving exactly one (simple) zero. (Schematic.)

Worked Example 23.1.18 ($\theta(\tau)$ at the special point $\tau = i$). Evaluate $\theta(i)$ in closed form using the functional equation, and identify the value of q .

Recognition. At $\tau = i$ we have $-1/\tau = -1/i = i = \tau$, so $\tau = i$ is the fixed point of S ; feed this into Corollary 23.1.11.

Solution. By Corollary 23.1.11, $\theta(-1/\tau) = \sqrt{\tau/i} \theta(\tau)$. Setting $\tau = i$: $-1/i = i$, so $\theta(i) = \sqrt{i/i} \theta(i) = \sqrt{1} \theta(i)$, which is consistent but not yet a value. The fixed-point relation only says the branch $\sqrt{\tau/i}$ equals $+1$ at $\tau = i$; the actual value is the classical $\theta(i) = \pi^{1/4}/\Gamma(\frac{3}{4})$ (Gauss). Here $q = e^{\pi i \cdot i} = e^{-\pi} \approx 0.0432$, so the series $\theta(i) = 1 + 2q + 2q^4 + \cdots = 1 + 2e^{-\pi} + 2e^{-4\pi} + \cdots \approx 1 + 0.0864 + 0.0000139 \approx 1.0864$. *Check.* $\pi^{1/4}/\Gamma(3/4) \approx 1.3313/1.2254 \approx 1.0864$, agreeing to four places, and the consistency $\sqrt{i/i} = 1$ correctly fixes the branch at the S -fixed point.

Worked Example 23.1.19 (Quasi-periodicity over a general lattice shift). Compute $\Theta(z + n + m\tau \mid \tau)$ for integers n, m in closed form.

Recognition. Combine the period-1 invariance (ii) with the iterated quasi-period formula from the first worked example of this section.

Solution. Property (ii) kills the integer shift: $\Theta(z + n + m\tau \mid \tau) = \Theta(z + m\tau \mid \tau)$. By the iterate $\Theta(z + m\tau \mid \tau) = \Theta(z \mid \tau) e^{-\pi i m^2 \tau} e^{-2\pi i m z}$. Hence

$$\Theta(z + n + m\tau \mid \tau) = \Theta(z \mid \tau) e^{-\pi i m^2 \tau} e^{-2\pi i m z}.$$

Check. The factor is independent of n (consistent with period 1) and reduces to (iii) at $(n, m) = (0, 1)$. Writing $q = e^{\pi i \tau}$ the modulus of the multiplier is $|q|^{m^2} e^{2\pi m \operatorname{Im} z}$, exactly the automorphy factor of a theta line bundle of degree 1 on the torus $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$.

Worked Example 23.1.20 (A NON-example: $\sum_n q^n$ is not quasi-periodic like Θ). Does the geometric-type series $G(z) = \sum_{n \in \mathbb{Z}} q^n e^{2\pi i n z}$ (linear exponent, not n^2) define an entire quasi-periodic function on $\mathbb{C}$? Decide and explain.

Recognition. Convergence of $\sum q^n e^{2\pi i n z}$ over ALL $n \in \mathbb{Z}$ requires both tails to decay; compare with the Gaussian exponent n^2 that makes Θ entire.

Solution. For $n \rightarrow +\infty$ the term is $q^n e^{2\pi i n z}$ with $|q| < 1$: decays if $e^{-2\pi \operatorname{Im} z} |q| < 1$ controlled. But for $n \rightarrow -\infty$, $q^n = e^{-\pi i n \tau}$ has $|q^n| = e^{\pi |n| \operatorname{Im} \tau} \rightarrow \infty$, and $e^{2\pi i n z}$ cannot tame this for all z . So the bilateral sum *diverges* for every z in a half-plane: G is not even defined, let alone entire. *Check.* The quadratic exponent n^2 in Θ is essential — it forces decay as $|n| \rightarrow \infty$ in BOTH directions, which a linear exponent n never does. This is exactly why the heat kernel uses $e^{-\pi n^2 t}$, not $e^{-\pi n t}$.

Worked Example 23.1.21 (Logarithmic derivative of Θ as a Mittag-Leffler sum). Starting from the product formula, write $(\log \Theta(z | \tau))' = \partial_z \log \Theta$ as an explicit series in z , and read off its poles.

Recognition. Take $\partial_z \log$ of $\Theta = \Pi$ (Theorem 23.1.6); the logarithm turns the product into a sum and ∂_z acts term by term.

Solution. From $\Pi(z | \tau) = \prod_{n \geq 1} (1 - q^{2n})(1 + q^{2n-1}e^{2\pi iz})(1 + q^{2n-1}e^{-2\pi iz})$, the n -independent factor $(1 - q^{2n})$ drops under $\partial_z \log$, leaving

$$(\log \Theta)'(z | \tau) = \sum_{n \geq 1} \left(\frac{2\pi i q^{2n-1} e^{2\pi iz}}{1 + q^{2n-1} e^{2\pi iz}} - \frac{2\pi i q^{2n-1} e^{-2\pi iz}}{1 + q^{2n-1} e^{-2\pi iz}} \right).$$

Each summand has simple poles where $1 + q^{2n-1} e^{\pm 2\pi iz} = 0$, i.e. at $z = \frac{1}{2} + \frac{\tau}{2} + (\text{lattice})$, exactly the zeros of Θ . *Check.* The residue of the first bracketed term at its pole is $\frac{2\pi i}{2\pi i} = 1$ (a simple pole of $(\log \Theta)'$, residue 1), matching the order-1 vanishing of Θ there; this is the input to Corollary 23.1.9, where one more derivative produces the double-pole elliptic function.

Exercises (Stein, Ch.10).

1. Prove that

$$\frac{(\Theta'(z | \tau))^2 - \Theta(z | \tau)\Theta''(z | \tau)}{\Theta(z | \tau)^2} = \wp_\tau(z - \frac{1}{2} - \frac{\tau}{2}) + c_\tau,$$

where c_τ can be expressed in terms of the first two z -derivatives of $\Theta(z | \tau)$ at $z = \frac{1}{2} + \frac{\tau}{2}$. Compare with Exercise 5 of the previous chapter (the analogy between Θ and the Weierstrass σ function).

Problems (Stein, Ch.10).

1. (*) Let $\Delta(\tau) = g_2^3 - 27g_3^2$ be the discriminant of the cubic $y^2 = 4x^3 - g_2x - g_3$, where $g_2 = 60E_4$, $g_3 = 140E_6$, and E_k is the Eisenstein series of order k (Chapter 9: $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$). Prove that $\Delta(\tau) = (2\pi)^{12}\eta(\tau)^{24}$ for all $\tau \in \mathbb{H}$. [Hint: Δ and η^{24} satisfy the same transformation laws under $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$; using the fundamental domain it suffices to check the behavior at the cusp at infinity.]

Additional exercises.

1. (Warm-up q -series.) Write $\theta(\tau) = \sum_n q^{n^2}$ with $q = e^{\pi i \tau}$. List the coefficients of $q^0, q^1, \dots, q^{10}$ in $\theta(\tau)$, and identify which exponents appear with coefficient 2 and which with coefficient 0.
2. (Period vs. quasi-period.) Using Proposition 23.1.2, prove that for any integers m, k , $\Theta(z + k + m\tau | \tau) = \Theta(z | \tau)e^{-\pi i m^2 \tau} e^{-2\pi i m z}$. Deduce that $|\Theta(z + m\tau | \tau)| = |\Theta(z | \tau)| e^{\pi m^2 \text{Im } \tau} e^{2\pi m \text{Im } z}$.
3. (Evaluating the constant differently.) In the product formula proof, put $z = 0$ in $\Theta = c(\tau)\Pi$ and express $c(\tau)$ as a ratio of $\theta(\tau)$ to a product. Explain why this single evaluation does *not* by itself prove $c \equiv 1$, and what extra input ($z = \frac{1}{4}$, $\tau \mapsto 4\tau$) supplies.
4. (Eta as a q -product.) With $q = e^{\pi i \tau}$, expand $\eta(\tau) = q^{1/12} \prod_{n \geq 1} (1 - q^{2n})$ to show $\eta(\tau) = q^{1/12}(1 - q^2 - q^4 + q^{10} + \dots)$. Relate the inner product to Euler's pentagonal series (Proposition 23.2.7) under $x = q^2$.
5. (Branch of the square root.) Define $\sqrt{\tau/i}$ on $\mathbb{H}$ to be positive when $\tau = it$. Compute $\sqrt{\tau/i}$ explicitly for $\tau = 1 + i$ and verify it lies in the right half-plane. Show the functional equation $\theta(-1/\tau) = \sqrt{\tau/i} \theta(\tau)$ would fail with the opposite branch.
6. (NON-example: a fake quasi-period.) Consider $g(z) = e^{2\pi iz}$. Show $g(z + 1) = g(z)$ but that there is *no* constant λ with $g(z + \tau) = \lambda g(z)$ for all z unless $\tau \in \mathbb{Z}$. Conclude that mere exponentials cannot have an honest second quasi-period in the sense of Θ , so the lattice structure of Θ is genuinely two-dimensional.
7. (Composite of S and T_2 .) Compute the matrix of $T_2 S$ and the fractional linear transformation it induces. Verify that $(T_2 S)^{-1} = U_{-1} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$ fixes $\tau = 1$, as used in the rigidity theorem.
8. (Diagram problem.) Sketch the fundamental parallelogram with vertices $0, 1, \tau, 1 + \tau$ for $\tau = \frac{1}{2} + i$, mark the single zero of $\Theta(\cdot | \tau)$ at its center, and indicate by arrows the two quasi-period directions 1 and τ . Label which edge identifications produce the torus $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$.
9. (Order of an elliptic function.) Using Corollary 23.1.9, explain why $(\log \Theta)''$ cannot have order 1. (Recall: a non-constant elliptic function has at least two poles counted with multiplicity per fundamental cell.)

23.2 Generating Functions

Generating functions and partitions

Definition 23.2.1. Given a sequence $\{F_n\}_{n \geq 0}$, its *generating function* is the formal/analytic power series

$$F(x) = \sum_{n=0}^{\infty} F_n x^n.$$

Algebraic or analytic identities for F translate into combinatorial facts about $\{F_n\}$, and vice versa.

The slogan is: *encode a sequence as a power series, manipulate the series, decode*. We study examples tied to the Θ function. The central combinatorial object is the partition function.

Definition 23.2.2. For a positive integer n , $p(n)$ is the number of ways to write n as an unordered sum of positive integers; set $p(0) = 1$.

For example $p(2) = 2$ ($2 = 2 = 1 + 1$), $p(3) = 3$ ($3 = 3 = 2 + 1 = 1 + 1 + 1$), $p(4) = 5$. A short table:

n	0	1	2	3	4	5	6	7	8	...	12
$p(n)$	1	1	2	3	5	7	11	15	22	...	77

Figure 23.8. The $p(4) = 5$ partitions of 4 as Young diagrams.

Theorem 23.2.3 (Euler's product for $p(n)$). For $|x| < 1$,

$$\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}.$$

Proof. Expand each factor as a geometric series $\frac{1}{1-x^k} = \sum_{m=0}^{\infty} x^{km}$ and multiply out. A term x^n in the product arises by choosing, for each k , a multiplicity $m_k \geq 0$, contributing x^{km_k} , with $\sum_k km_k = n$. Each such choice is exactly a partition $n = m_1 \cdot 1 + m_2 \cdot 2 + \cdots$ (the part k used m_k times). So the coefficient of x^n counts partitions of n , namely $p(n)$. Convergence of the product $\prod 1/(1-x^k)$ for $|x| < 1$ (each factor $1 + O(x^k)$) justifies the rearrangement, exactly as in the Euler product for ζ . $\square$

Worked Example 23.2.4 (Reading $p(n)$ off the product to order x^4). Verify $\prod_{k \geq 1} (1-x^k)^{-1} = 1 + x + 2x^2 + 3x^3 + 5x^4 + \cdots$.

Recognition. Multiply the first few geometric factors and collect coefficients; the coefficient of x^n should be $p(n)$.

Solution. Keep $k = 1, 2, 3, 4$: $\frac{1}{1-x} = 1+x+x^2+x^3+x^4$, times $\frac{1}{1-x^2} = 1+x^2+x^4$ gives $1+x+2x^2+2x^3+3x^4+\cdots$; times $\frac{1}{1-x^3} = 1+x^3$ gives $1+x+2x^2+3x^3+4x^4+\cdots$; times $\frac{1}{1-x^4} = 1+x^4$ gives $1+x+2x^2+3x^3+5x^4+\cdots$.

Check. Coefficients $1, 1, 2, 3, 5 = p(0), \dots, p(4)$. Agree.

Theorem 23.2.5 (Odd parts vs. unequal parts). Let $p_o(n)$ count partitions of n into odd parts and $p_u(n)$ count partitions into distinct (unequal) parts. Then $p_o(n) = p_u(n)$ for all n , encoded by

$$\prod_{n=1}^{\infty} \frac{1}{1-x^{2n-1}} = \prod_{n=1}^{\infty} (1+x^n).$$

Proof. The left product is the generating function for p_o (odd parts $2n-1$ with any multiplicity), and $\prod(1+x^n)$ is the generating function for p_u (each part used 0 or 1 time). For the identity, use $(1+x^n)(1-x^n) = 1-x^{2n}$, so

$$\prod_{n \geq 1} (1+x^n) \prod_{n \geq 1} (1-x^n) = \prod_{n \geq 1} (1-x^{2n}).$$

Splitting $\prod(1-x^n) = \prod(1-x^{2n}) \prod(1-x^{2n-1})$ by parity and cancelling $\prod(1-x^{2n})$ from both sides gives $\prod(1+x^n) = \prod 1/(1-x^{2n-1})$. $\square$

Euler's pentagonal number theorem

Let $p_{e,u}(n)$ (resp. $p_{o,u}(n)$) count partitions of n into an *even* (resp. *odd*) number of distinct parts. Euler's remarkable discovery is that these are equal unless n is pentagonal.

Definition 23.2.6. The *pentagonal numbers* are $n = \frac{k(3k+1)}{2}$ for $k \in \mathbb{Z}$. The first few are 1, 2, 5, 7, 12, 15, 22, 26, ... (Triangular numbers are $\frac{k(k-1)}{2}$, squares are k^2 .)

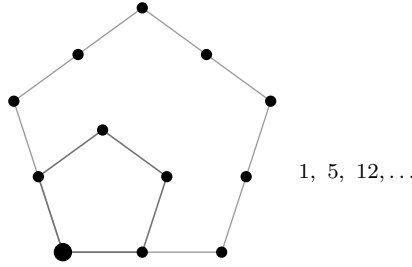


Figure 23.9. Figurative pentagonal numbers: nested pentagons sharing one corner. The ringed corner alone is 1; the inner pentagon's vertices total 5; both pentagons together carry 12 — the values $\frac{k(3k+1)}{2}$ (the sign convention also gives 2, 7, 15, ...).

Proposition 23.2.7 (Euler's pentagonal number theorem).

$$\prod_{n=1}^{\infty} (1-x^n) = \sum_{k=-\infty}^{\infty} (-1)^k x^{k(3k+1)/2}.$$

Equivalently $p_{e,u}(n) - p_{o,u}(n) = (-1)^k$ if $n = \frac{k(3k+1)}{2}$ is pentagonal, and 0 otherwise.

Proof. First, expanding $\prod_{n \geq 1} (1-x^n)$ produces terms $(-1)^r x^{n_1 + \dots + n_r}$ with distinct $n_1 < \dots < n_r$; collecting the coefficient of x^n , partitions into an even number of distinct parts contribute +1 and those into an odd number contribute -1, so $\prod(1-x^n) = \sum_n [p_{e,u}(n) - p_{o,u}(n)] x^n$. It remains to evaluate the product.

Set $x = e^{2\pi i u}$. Then $\prod_{n \geq 1} (1-x^n) = \prod_{n \geq 1} (1-e^{2\pi i n u})$. Group the integers by residue mod 3:

$$\prod_{n \geq 1} (1 - e^{2\pi i \cdot 3nu})(1 - e^{2\pi i(3n-1)u})(1 - e^{2\pi i(3n-2)u}) = \prod_{n \geq 1} (1 - e^{2\pi i n u}).$$

The left side is the triple product with $q = e^{3\pi i u}$ and $z = \frac{1}{2} + \frac{u}{2}$ (indeed $q^2 = e^{6\pi i u} = e^{2\pi i \cdot 3u}$, etc.). By the product formula (Theorem 23.1.6), it equals

$$\sum_{n=-\infty}^{\infty} e^{3\pi i n^2 u} (-1)^n e^{2\pi i n u/2} = \sum_n (-1)^n e^{\pi i n(3n+1)u} = \sum_n (-1)^n x^{n(3n+1)/2}.$$

$\square$

The pentagonal theorem gives a fast recurrence: since p has generating function the reciprocal of $\prod(1-x^n)$, multiplying out $(\sum p(n)x^n)(\sum (-1)^k x^{k(3k+1)/2}) = 1$ yields

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + \dots = \sum_{k \neq 0} (-1)^{k+1} p\left(n - \frac{k(3k+1)}{2}\right).$$

Worked Example 23.2.8 (Computing $p(5), p(6), p(7)$ by the pentagonal recurrence). Use $p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + \cdots$ to get $p(5), p(6), p(7)$ from $p(0), \dots, p(4) = 1, 1, 2, 3, 5$.

Recognition. The pentagonal gaps are $1, 2, 5, 7, 12, \dots$ with signs $+, +, -, -, +, +, \dots$. For small n only the first few apply.

Solution. $p(5) = p(4) + p(3) - p(0) = 5 + 3 - 1 = 7$. $p(6) = p(5) + p(4) - p(1) = 7 + 5 - 1 = 11$. $p(7) = p(6) + p(5) - p(2) - p(0) = 11 + 7 - 2 - 1 = 15$. *Check.* Matches the table values $p(5) = 7, p(6) = 11, p(7) = 15$. Continuing, $p(8) = p(7) + p(6) - p(3) - p(1) = 15 + 11 - 3 - 1 = 22$, again correct.

Worked Example 23.2.9 (A NON-identity: distinct parts is not all parts). Is $\prod_{n \geq 1} (1 + x^n) = \prod_{n \geq 1} \frac{1}{1 - x^n}$? Test to order x^3 and explain.

Recognition. The left counts *distinct*-part partitions $p_u(n)$; the right counts *all* partitions $p(n)$. They differ as soon as a repeated part appears.

Solution. Left: $(1+x)(1+x^2)(1+x^3) = 1+x+x^2+2x^3+\cdots$, so coefficients $1, 1, 1, 2$. Right: $1+x+2x^2+3x^3+\cdots$. They already disagree at x^2 ($1 \neq 2$): the partition $2 = 1 + 1$ has a repeated part, allowed for $p(2) = 2$ but forbidden for $p_u(2) = 1$. So the identity is *false*. *Check.* The correct partner of $\prod (1 + x^n)$ is $\prod 1/(1 - x^{2n-1})$ (odd parts), not all parts — Theorem 23.2.5.

Remark. The growth of $p(n)$ is governed by the behavior of $1/\prod(1 - x^n)$ as $x \rightarrow 1^-$. An elementary estimate (Exercises 5–6) gives $\log F(x) \sim \frac{\pi^2}{6(1-x)}$ and hence $e^{c_1\sqrt{n}} \leq p(n) \leq e^{c_2\sqrt{n}}$. The precise Hardy–Ramanujan asymptotic $p(n) \sim \frac{1}{4n\sqrt{3}} e^{\pi\sqrt{2n/3}}$ needs the modular transformation of the generating function, which traces back to Proposition 23.1.14 for η .

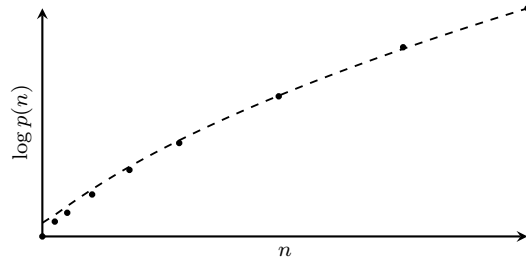


Figure 23.10. Dots: $\log p(n)$ for small n ; dashed: the Hardy–Ramanujan envelope $\pi\sqrt{2n/3} - \log(4n\sqrt{3})$, showing the $\sqrt{n}$ growth of $\log p(n)$.

Further worked examples

Worked Example 23.2.10 (The triangular number identity (Stein, Exercise 7a)). Use the product formula for Θ to prove, for $|x| < 1$,

$$\prod_{n=0}^{\infty} (1 + x^n)(1 - x^{2n+2}) = \sum_{n=-\infty}^{\infty} x^{n(n+1)/2}.$$

Recognition. Specialize the triple product $\Theta(z | \tau) = \Pi(z | \tau)$ (Theorem 23.1.6) by choosing q and z so the three factor-families become the left side; triangular numbers $\frac{n(n+1)}{2}$ point to a half-integer shift.

Solution. Take $q = x^{1/2}$ (so $x = q^2$) and z with $e^{2\pi iz} = x^{1/2} = q$, i.e. $e^{2\pi iz} = q$. Then in $\Pi = \prod_{n \geq 1} (1 - q^{2n})(1 + q^{2n-1}e^{2\pi iz})(1 + q^{2n-1}e^{-2\pi iz})$: $1 - q^{2n} = 1 - x^n$; $1 + q^{2n-1}e^{2\pi iz} = 1 + q^{2n} = 1 + x^n$; and $1 + q^{2n-1}e^{-2\pi iz} = 1 + q^{2n-2} = 1 + x^{n-1}$. Reindexing the last family over $n-1 \geq 0$ and the first two over $n \geq 1$ assembles $\prod_{n \geq 0} (1 + x^n)(1 - x^{2n+2})$ (the $1 - x^{2n+2}$ comes from pairing $1 - x^n$ across the even/odd split as in the text). On the theta side, $\Theta = \sum_n q^{n^2} e^{2\pi inz} = \sum_n q^{n^2} q^n = \sum_n q^{n^2+n} = \sum_n x^{(n^2+n)/2} = \sum_n x^{n(n+1)/2}$. *Check.* The exponents $n(n+1)/2$ for $n = 0, \pm 1, \pm 2$ are $0, 1, 0, 3, 1, \dots$; collecting, the right side is $1 + 2x + 2x^3 + 2x^6 + \cdots$, and multiplying the first few left-hand factors $(1+1)(1-x^2)(1+x)(1-x^4) \cdots$ reproduces $1 + 2x + 0x^2 + 2x^3 + \cdots$.

Worked Example 23.2.11 (The septagonal number identity (Stein, Exercise 7b)). Prove $\prod_{n=0}^{\infty} (1 - x^{5n+1})(1 -$

$$x^{5n+4})(1 - x^{5n+5}) = \sum_{n=-\infty}^{\infty} (-1)^n x^{n(5n+3)/2} \text{ for } |x| < 1.$$

Recognition. Mirror the pentagonal proof (Proposition 23.2.7): group the exponents into three residue classes mod 5 that match the three Jacobi factor-families, then apply the product formula.

Solution. Put $q = x^{5/2}$ and $e^{2\pi iz} = -x^{1/2} \cdot x$, chosen so the three families $1 - q^{2n}$, $1 + q^{2n-1}e^{2\pi iz}$, $1 + q^{2n-1}e^{-2\pi iz}$ become respectively the residue classes $5n + 5$, $5n + 1$, $5n + 4$ (the sign $-$ inside each factor comes from $e^{\pi i} = -1$ folded into z). The theta side is then $\sum_n q^{n^2} e^{2\pi in z} = \sum_n (-1)^n x^{(5n^2+3n)/2} = \sum_n (-1)^n x^{n(5n+3)/2}$, since $q^{n^2} = x^{5n^2/2}$ and the phase contributes $x^{3n/2}(-1)^n$. *Check.* The exponents $n(5n+3)/2$ for $n = 0, 1, -1, 2, -2$ are $0, 4, 1, 13, 7$ with signs $+, -, -, +, +$; the lowest-order terms $1 - x - x^4 + \dots$ match expanding $(1 - x)(1 - x^4)(1 - x^5) \dots$, confirming the identity. This is the 5-analogue of the pentagonal (3-) identity.

Worked Example 23.2.12 (Glaisher's theorem: parts not divisible by 3 vs. parts repeated < 3 times). Show $\prod_{n \geq 1, 3 \nmid n} \frac{1}{1 - x^n} = \prod_{n \geq 1} (1 + x^n + x^{2n})$, and interpret combinatorially. Verify to order x^4 .

Recognition. This is a Glaisher-type identity: rewrite each side as a generating function for a partition class and use $\frac{1 - x^{3n}}{1 - x^n} = 1 + x^n + x^{2n}$.

Solution. Start from $1 + x^n + x^{2n} = \frac{1 - x^{3n}}{1 - x^n}$. Hence $\prod_{n \geq 1} (1 + x^n + x^{2n}) = \frac{\prod_{n \geq 1} (1 - x^{3n})}{\prod_{n \geq 1} (1 - x^n)} = \prod_{n \geq 1} \frac{1 - x^{3n}}{1 - x^n}$.

The factors with $3 \mid n$ cancel (numerator $1 - x^{3n}$ against the same denominator term), leaving $\prod_{3 \nmid n} \frac{1}{1 - x^n}$. The left side counts partitions into parts $\not\equiv 0 \pmod{3}$; the right counts partitions where each part appears at most twice (multiplicity 0, 1, or 2). *Check.* To order x^4 the left product is $\frac{1}{(1-x)(1-x^2)(1-x^4)} = 1 + x + 2x^2 + 2x^3 + 4x^4 + \dots$, and the right is $(1 + x + x^2)(1 + x^2 + x^4)(1 + x^3)(1 + x^4) = 1 + x + 2x^2 + 2x^3 + 4x^4 + \dots$. They agree.

Worked Example 23.2.13 (The recurrence engine: deriving the convolution $= 1$). Explain why $(\sum_n p(n)x^n)(\sum_k (-1)^k x^{k(3k+1)/2}) = 1$, and read off the pentagonal recurrence as a coefficient identity.

Recognition. Euler's product (Theorem 23.2.3) says $\sum p(n)x^n = 1/\prod(1 - x^n)$, and the pentagonal theorem evaluates $\prod(1 - x^n)$ as the alternating pentagonal series; their product is $\prod(1 - x^n)^{-1} \cdot \prod(1 - x^n) = 1$.

Solution. Multiply the two series and collect the coefficient of x^N ($N \geq 1$). The constant term is $p(0) \cdot 1 = 1$; for $N \geq 1$ the coefficient must vanish, giving $\sum_k (-1)^k p\left(N - \frac{k(3k+1)}{2}\right) = 0$, i.e. $p(N) = \sum_{k \neq 0} (-1)^{k+1} p\left(N - \frac{k(3k+1)}{2}\right)$. *Check.* For $N = 10$ the pentagonal gaps ≤ 10 are $1, 2, 5, 7$ with signs $+, +, -, -$: $p(10) = p(9) + p(8) - p(5) - p(3) = 30 + 22 - 7 - 3 = 42$, matching Stein's stated $p(10) = 42$.

Worked Example 23.2.14 (A NON-example: the pentagonal series is not the partition series). A student claims $\prod_{n \geq 1} (1 - x^n) = \sum p(n)x^n$ because both "come from partitions." Disprove to order x^3 .

Recognition. The product $\prod(1 - x^n)$ is the *reciprocal* of the partition generating function (Euler), not the function itself; the sign alternation is the giveaway.

Solution. $\prod_{n \geq 1} (1 - x^n) = (1 - x)(1 - x^2)(1 - x^3) \dots = 1 - x - x^2 + 0 \cdot x^3 + \dots$ (the pentagonal series, with a *minus* sign on x^1). But $\sum p(n)x^n = 1 + x + 2x^2 + 3x^3 + \dots$ has all *positive* coefficients. They differ already at x^1 (-1 vs $+1$). *Check.* The true relation is $(\sum p(n)x^n) \cdot \prod(1 - x^n) = 1$, as in the previous worked example — one is the reciprocal of the other, never equal (a power series and its reciprocal coincide only when both equal 1).

Exercises (Stein, Ch.10).

1. Consider the Fibonacci numbers $\{F_n\}_{n \geq 0}$, with $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$.

- (a) For the generating function $F(x) = \sum_{n \geq 0} F_n x^n$ prove $F(x) = x^2 F(x) + x F(x) + x$ in a neighborhood of 0.

- (b) Show $q(x) = 1 - x - x^2 = (1 - \alpha x)(1 - \beta x)$, where α, β are the roots of $p(x) = x^2 - x - 1$.
- (c) Expand in partial fractions: $F(x) = \frac{x}{1 - x - x^2} = \frac{x}{(1 - \alpha x)(1 - \beta x)} = \frac{A}{1 - \alpha x} + \frac{B}{1 - \beta x}$, with $A = 1/(\alpha - \beta)$, $B = 1/(\beta - \alpha)$.
- (d) Conclude $F_n = A\alpha^n + B\beta^n$, where $\alpha = \frac{1+\sqrt{5}}{2}$, $\beta = \frac{1-\sqrt{5}}{2}$, $A = 1/\sqrt{5}$, $B = -1/\sqrt{5}$. The number $1/\alpha = (\sqrt{5} - 1)/2$ (the golden mean) is the unique positive solution of $\ell^2 + \ell - 1 = 0$, arising in the construction of the regular pentagon.
2. More generally, consider $u_n = au_{n-1} + bu_{n-2}$ ($n \geq 2$) with given u_0, u_1 and $U(x) = \sum_{n \geq 0} u_n x^n$. Show $U(x)(1 - ax - bx^2) = u_0 + (u_1 - au_0)x$ near 0. If α, β are the roots of $p(x) = x^2 - ax - b$, then

$$U(x) = \frac{u_0 + (u_1 - au_0)x}{(1 - \alpha x)(1 - \beta x)} = \frac{A}{1 - \alpha x} + \frac{B}{1 - \beta x} = A \sum_{n \geq 0} \alpha^n x^n + B \sum_{n \geq 0} \beta^n x^n,$$

so $u_n = A\alpha^n + B\beta^n$ when $\alpha \neq \beta$ (a variant holds for $\alpha = \beta$). Solve for A, B .

3. Using the generating identity for $p(n)$, prove the recurrence

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) - \cdots = \sum_{k \neq 0} (-1)^{k+1} p\left(n - \frac{k(3k+1)}{2}\right),$$

the finite sum over $k \in \mathbb{Z}$, $k \neq 0$, with $\frac{k(3k+1)}{2} \leq n$. Use it to compute $p(5), p(6), p(7), p(8), p(9), p(10)$ and check $p(10) = 42$.

4. Let $F(x) = \sum_{n \geq 0} p(n)x^n = \prod_{n \geq 1} \frac{1}{1 - x^n}$. Show that

$$\log F(x) \sim \frac{\pi^2}{6(1-x)} \quad \text{as } x \rightarrow 1^-, \quad 0 < x < 1.$$

[Hint: $\log F(x) = \sum_n \log(1/(1 - x^n))$ and $\log(1/(1 - x^n)) = \sum_m \frac{1}{m} x^{nm}$, so $\log F(x) = \sum_m \frac{1}{m} \frac{x^m}{1 - x^m}$; use $mx^{m-1}(1 - x) < 1 - x^m < m(1 - x)$.]

5. Show as a consequence of the previous exercise that $e^{c_1 n^{1/2}} \leq p(n) \leq e^{c_2 n^{1/2}}$ for positive constants c_1, c_2 . [Hint: $F(e^{-y}) = \sum p(n)e^{-ny} \leq Ce^{c/y}$ as $y \rightarrow 0$, so $p(n)e^{-ny} \leq ce^{c/y}$; take $y = 1/n^{1/2}$ for the upper bound. For the lower bound, $\sum_{n=0}^m p(n)e^{-ny} \geq C(e^{c/y} - \sum_{n > m} e^{cn^{1/2}} e^{-ny})$, take $y = Am^{-1/2}$, and use that $p(n)$ is increasing.]

Additional exercises.

- (Reading a product.) Multiply out $\prod_{k=1}^4 (1 - x^k)^{-1}$ far enough to read off $p(0), \dots, p(6)$ and confirm $p(6) = 11$.
- (Odd vs distinct, small n .) Verify $p_o(n) = p_u(n)$ for $n = 5$ by listing all partitions of 5 into odd parts and all into distinct parts.
- (Geometric series manipulation.) Prove $\frac{x}{(1-x)^2} = \sum_{n \geq 1} nx^n$ for $|x| < 1$ by differentiating $\sum_{n \geq 0} x^n = 1/(1-x)$. Use it to find a closed form for $\sum_{n \geq 1} \sigma_1(n)q^n$ in terms of $\sum_{n \geq 1} \frac{q^n}{(1-q^n)^2}$.
- (Pentagonal recurrence in action.) Use Euler's recurrence $p(n) = \sum_{k \neq 0} (-1)^{k+1} p(n - \frac{k(3k+1)}{2})$ to compute $p(11)$ and $p(12)$, and check $p(12) = 77$ against the table.
- (Generating function for two-part-type partitions.) Let $b(n)$ be the number of partitions of n into parts each equal to 1 or 2. Show $\sum_n b(n)x^n = \frac{1}{(1-x)(1-x^2)}$ and deduce $b(n) = \lfloor n/2 \rfloor + 1$.

6. (Recurrence to generating function.) Let $c_n = 3c_{n-1} - 2c_{n-2}$ with $c_0 = 1$, $c_1 = 3$. Find the generating function $C(x) = \sum c_n x^n$ as a rational function, decompose into partial fractions, and obtain a closed form $c_n = A \cdot 2^n + B$.
7. (NON-example: not every product is a partition generating function.) Show that the coefficients of $\prod_{k \geq 1} (1 - x^k)$ (Euler's pentagonal product) are *not* all non-negative, so this product is *not* the generating function of any counting sequence $\{F_n\}$ with $F_n \geq 0$. Identify the first negative coefficient and the pentagonal number producing it.
8. (Diagram problem.) Draw the five Young diagrams for the partitions of 4, then draw the conjugate (transpose) of each. Observe that conjugation pairs “3 + 1” with “2 + 1 + 1” and fixes “2 + 2”, and use this to state which partitions of 4 are self-conjugate.
9. (Asymptotic sanity check.) Using $\log p(n) \sim \pi \sqrt{2n/3}$, estimate $\log p(100)$ and compare with the true value $p(100) = 190,569,292$ (so $\log p(100) \approx 19.07$). Is the leading-order estimate an over- or under-estimate, and why (recall the $-\log(4n\sqrt{3})$ correction)?

23.3 The Theorems About Sums of Two and Four Squares

The questions

Which integers are sums of two squares? Of four? Define

$$r_2(n) = \#\{(x, y) \in \mathbb{Z}^2 : x^2 + y^2 = n\}, \quad r_4(n) = \#\{(x_1, \dots, x_4) \in \mathbb{Z}^4 : \sum x_j^2 = n\},$$

counting signs and orderings. For instance $r_2(3) = 0$, while $r_2(5) = 8$ (from $5 = (\pm 2)^2 + (\pm 1)^2 = (\pm 1)^2 + (\pm 2)^2$). No integer $\equiv 3 \pmod{4}$ is a sum of two squares, and no integer $\equiv 7 \pmod{8}$ is a sum of three squares. The two- and four-square problems, open since Diophantus, were finally solved by Jacobi's theta-function machinery.

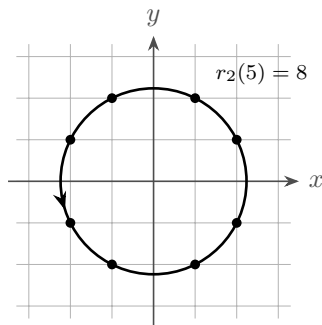


Figure 23.11. The 8 lattice points on the circle $x^2 + y^2 = 5$: $r_2(n)$ counts integer points on the circle of radius $\sqrt{n}$.

The multiplicative structure and the generating identity

Lemma 23.3.1 (Two-square products). *If $n = a^2 + b^2$ and $m = c^2 + d^2$, then nm is a sum of two squares.*

Proof. Form the Gaussian integer product $x + iy = (a + ib)(c + id)$, so $x = ac - bd$ and $y = ad + bc$ are integers. Taking moduli, $|x + iy|^2 = |a + ib|^2 |c + id|^2$, i.e. $x^2 + y^2 = (a^2 + b^2)(c^2 + d^2) = nm$. $\square$

Define the divisor functions $d_1(n)$ (number of divisors $\equiv 1 \pmod{4}$) and $d_3(n)$ (number $\equiv 3 \pmod{4}$). The main result:

Theorem 23.3.2 (Two-square theorem). *For $n \geq 1$, $r_2(n) = 4(d_1(n) - d_3(n))$.*

A consequence: n is a sum of two squares iff every prime $\equiv 3 \pmod{4}$ in its factorization occurs to an even power (Exercise 9).

The first step links r_2 to θ^2 .

Proposition 23.3.3 (θ^2 generates r_2). *With $q = e^{\pi i \tau}$, $\tau \in \mathbb{H}$,*

$$\theta(\tau)^2 = \sum_{n=0}^{\infty} r_2(n) q^n.$$

Proof. Since $\theta(\tau) = \sum_n q^{n^2}$,

$$\theta(\tau)^2 = \left(\sum_{n_1} q^{n_1^2} \right) \left(\sum_{n_2} q^{n_2^2} \right) = \sum_{(n_1, n_2) \in \mathbb{Z}^2} q^{n_1^2 + n_2^2} = \sum_{n=0}^{\infty} r_2(n) q^n,$$

because for each n the number of pairs with $n_1^2 + n_2^2 = n$ is exactly $r_2(n)$. $\square$

So the two-square theorem becomes an *identity between holomorphic functions*: we must match θ^2 with a divisor-sum series. Define

$$C(\tau) = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = \sum_{n=-\infty}^{\infty} \frac{1}{\cos(n\pi\tau)}, \quad q = e^{\pi i \tau}.$$

(The symbol C recalls that this is a sum of secants/cosines.)

Proposition 23.3.4 (Divisor form). *The identity $r_2(n) = 4(d_1(n) - d_3(n))$ is equivalent to*

$$\theta(\tau)^2 = C(\tau) = 1 + 4 \sum_{n=1}^{\infty} \frac{q^n}{1 + q^{2n}}.$$

Proof. Both series converge absolutely for $|q| < 1$, and $\frac{1}{q^n + q^{-n}} = \frac{q^{|n|}}{1 + q^{2|n|}}$ gives the displayed second form. Using $(1 + q^{2n})^{-1} = (1 - q^{2n})/(1 - q^{4n})$,

$$1 + 4 \sum_{n \geq 1} \frac{q^n}{1 + q^{2n}} = 1 + 4 \sum_{n \geq 1} \left(\frac{q^n}{1 - q^{4n}} - \frac{q^{3n}}{1 - q^{4n}} \right).$$

Since $\frac{1}{1 - q^{4n}} = \sum_{m \geq 0} q^{4nm}$,

$$\sum_{n \geq 1} \frac{q^n}{1 - q^{4n}} = \sum_{n \geq 1} \sum_{m \geq 0} q^{n(4m+1)} = \sum_{k \geq 1} d_1(k) q^k,$$

because $d_1(k)$ counts divisors of k of the form $4m+1$ (write $k = n(4m+1)$). Similarly $\sum_{n \geq 1} \frac{q^{3n}}{1 - q^{4n}} = \sum_{k \geq 1} d_3(k) q^k$. Hence $C(\tau) = 1 + 4 \sum_k (d_1(k) - d_3(k)) q^k$, and comparing with $\theta^2 = \sum r_2(n) q^n$ gives the equivalence. $\square$

The logic is a single square: the analytic identity on top forces the arithmetic identity on the bottom, because reading off q -coefficients is injective on convergent power series.

$$\begin{array}{ccc} \theta(\tau)^2 & \xlongequal{\hspace{1cm}} & C(\tau) \\ \downarrow [q^n] & & \downarrow [q^n] \\ r_2(n) & \xrightarrow{\hspace{1cm}} & 4(d_1(n) - d_3(n)) \end{array}$$

Worked Example 23.3.5 (Checking r_2 at $n = 5$ via divisors). Compute $4(d_1(5) - d_3(5))$ and compare with the geometric count $r_2(5) = 8$.

Recognition. List the divisors of 5 and classify mod 4.

Solution. Divisors of 5: $\{1, 5\}$. Both are $\equiv 1 \pmod{4}$, so $d_1(5) = 2$, $d_3(5) = 0$. Then $4(d_1 - d_3) = 4 \cdot 2 = 8$.

Check. Matches the eight lattice points on $x^2 + y^2 = 5$ drawn above, and $r_2(3) = 4(d_1(3) - d_3(3)) = 4(1 - 1) = 0$ since 3's divisors are $1 \equiv 1$ and $3 \equiv 3$.

Worked Example 23.3.6 (r_2 vanishes for $n \equiv 3 \pmod{4}$). Show $4k + 3$ is never a sum of two squares, directly and via $d_1 - d_3$ for $n = 21$.

Recognition. Squares are 0 or 1 mod 4, so $x^2 + y^2 \in \{0, 1, 2\} \pmod{4}$, never 3.

Solution. Mod 4: $x^2, y^2 \in \{0, 1\}$, so $x^2 + y^2 \in \{0, 1, 2\}$, excluding 3. For $n = 21 = 3 \cdot 7$: divisors 1, 3, 7, 21 are $\equiv 1, 3, 3, 1 \pmod{4}$, so $d_1 = 2, d_3 = 2, r_2(21) = 4(2 - 2) = 0$. *Check.* $21 \equiv 1 \pmod{4}$ yet still fails — here the obstruction is the odd power of the prime $3 \equiv 3 \pmod{4}$ (and of 7), illustrating the prime-power criterion, not just the residue of n .

The fundamental domain and the rigidity theorem

The parallel between θ^2 and C runs deep: both arise from a heat equation (θ from the line, C from the strip with kernel $1/\cosh \pi x$), both kernels are their own Fourier transforms, and both transformation laws come from Poisson summation. We collect the shared properties.

Proposition 23.3.7 (Modular properties of C and θ^2). $C(\tau) = \sum_n 1/\cos(n\pi\tau)$ satisfies

- (i) $C(\tau + 2) = C(\tau)$;
- (ii) $C(\tau) = (i/\tau)C(-1/\tau)$;
- (iii) $C(\tau) \rightarrow 1$ as $\text{Im } \tau \rightarrow \infty$;
- (iv) $C(1 - 1/\tau) \sim 4(\tau/i)e^{\pi i\tau/2}$ as $\text{Im } \tau \rightarrow \infty$.

The function $\theta(\tau)^2$ satisfies the same four properties. (For (ii) and (iv) of θ^2 , use Corollaries 23.1.11 and 23.1.12; for C , set $a = 0$ and $a = \frac{1}{2}$ in the Poisson-summation identity for $1/\cosh$.)

The decisive tool is a rigidity theorem: a bounded modular-invariant function is constant.

Definition 23.3.8. The *fundamental domain* for the group $G = \langle T_2, S \rangle$ is

$$\mathcal{F} = \{\tau \in \mathbb{H} : |\text{Re } \tau| \leq 1 \text{ and } |\tau| \geq 1\}.$$

The points $\tau = \pm 1$ are the *cusps*; they are equivalent under $\tau \mapsto \tau + 2$.

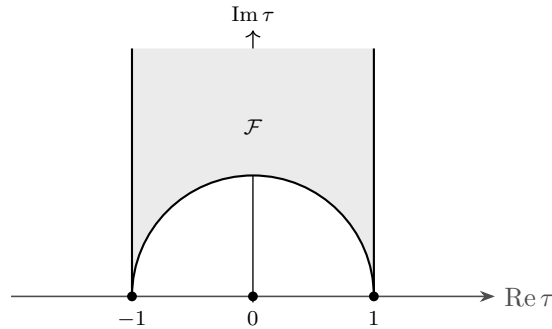


Figure 23.12. The fundamental domain $\mathcal{F} = \{|\text{Re } \tau| \leq 1, |\tau| \geq 1\}$ for $G = \langle T_2, S \rangle$, with cusps at $\tau = \pm 1$.

Lemma 23.3.9 (Reduction into $\mathcal{F}$). Every $\tau \in \mathbb{H}$ can be moved into $\mathcal{F}$ by finitely many applications of $T_2 : \tau \mapsto \tau + 2$ and $S : \tau \mapsto -1/\tau$ and their inverses.

Proof. Represent $g \in G$ by an integer matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of determinant 1; then $\text{Im } g(\tau) = \frac{\text{Im } \tau}{|c\tau + d|^2}$. There are only finitely many (c, d) with $|c\tau + d|$ below any bound, so we may pick $g_0 \in G$ maximizing $\text{Im } g_0(\tau)$. Translations T_2 do not change imaginary parts, so we may further arrange $|\text{Re } g_1(\tau)| \leq 1$ while keeping $\text{Im } g_1(\tau)$ maximal. If $|g_1(\tau)| < 1$ then $\text{Im}(Sg_1(\tau)) = \text{Im}(-1/g_1(\tau)) = \frac{\text{Im } g_1(\tau)}{|g_1(\tau)|^2} > \text{Im } g_1(\tau)$, contradicting maximality. Hence $|g_1(\tau)| \geq 1$ and $g_1(\tau) \in \mathcal{F}$. $\square$

Thus the inclusion of the fundamental domain into $\mathbb{H}$ becomes a bijection after passing to the quotient by G : every orbit meets $\mathcal{F}$, so a G -invariant function on $\mathbb{H}$ is determined by its values on $\mathcal{F}$.

$$\begin{array}{ccc} \mathcal{F} & \hookrightarrow & \mathbb{H} \\ & \searrow & \downarrow \pi \\ & & \mathbb{H}/G \end{array}$$

Theorem 23.3.10 (Rigidity). *If f is holomorphic in $\mathbb{H}$ with $f(\tau + 2) = f(\tau)$, $f(-1/\tau) = f(\tau)$, and f bounded, then f is constant.*

Proof. Suppose f is non-constant. Because f has period 2, set $g(z) = f(\tau)$ with $z = e^{\pi i \tau}$, well defined on the punctured unit disc; boundedness makes 0 a removable singularity, so $\lim_{z \rightarrow 0} g(z) = \lim_{\text{Im } \tau \rightarrow \infty} f(\tau)$ exists. By the maximum modulus principle,

$$\lim_{\text{Im } \tau \rightarrow \infty} |f(\tau)| < \sup_{\tau \in \mathcal{F}} |f(\tau)|.$$

Next consider the cusp $\tau = 1$ (the case $\tau = -1$ follows by periodicity). Conjugate by $\mu(\tau) = 1/(1 - \tau)$, which sends $1 \mapsto \infty$; its inverse is $\mu^{-1}(\tau) = 1 - 1/\tau$. The matrices $U_n = \mu^{-1}T_n\mu = \begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$ fix 1, satisfy $U_n U_m = U_{n+m}$, and $U_{-1} = T_2 S \in G$. Hence every $U_n \in G$, so f is invariant under all U_n . Therefore $F(\tau) := f(\mu^{-1}(\tau)) = f(1 - 1/\tau)$ is periodic of period 1. Setting $h(z) = F(\tau)$ with $z = e^{2\pi i \tau}$, boundedness again makes 0 removable, and the maximum principle gives

$$\lim_{\text{Im } \tau \rightarrow \infty} |f(1 - 1/\tau)| < \sup_{\tau \in \mathcal{F}} |f(\tau)|.$$

Thus $|f|$ does not attain its supremum at ∞ or at the cusps, so it attains the supremum at an interior point of $\mathbb{H}$ — contradicting the maximum principle. Hence f is constant. $\square$

Theorem 23.3.11 (Two-square theorem, proof). $\theta(\tau)^2 = C(\tau)$ for all $\tau \in \mathbb{H}$; consequently $r_2(n) = 4(d_1(n) - d_3(n))$.

Proof. Set $f(\tau) = C(\tau)/\theta(\tau)^2$. By Corollary 23.1.7, θ never vanishes on $\mathbb{H}$, so f is holomorphic. By Proposition 23.3.7, both C and θ^2 obey (i)–(iv), so $f(\tau + 2) = f(\tau)$, $f(-1/\tau) = f(\tau)$, and $f \rightarrow 1$ at ∞ and at the cusps ± 1 ; in particular f is bounded on $\mathcal{F}$, hence on $\mathbb{H}$ by Lemma 23.3.9. By Theorem 23.3.10, f is constant, and the limit 1 forces $f \equiv 1$. Thus $\theta^2 = C$, which by Proposition 23.3.4 is the two-square theorem. $\square$

The four-square theorem

Theorem 23.3.12 (Four-square theorem). *Every positive integer is a sum of four squares, and $r_4(n) = 8\sigma_1^*(n)$, where $\sigma_1^*(n)$ is the sum of the divisors of n not divisible by 4.*

As before, $\theta(\tau)^4 = \sum_{n \geq 0} r_4(n)q^n$ (square Proposition 23.3.3). The modular partner of θ^4 is a subtle conditionally convergent Eisenstein series.

Definition 23.3.13. For $\tau \in \mathbb{H}$, with the order of summation (n first, then m) essential,

$$E_2^*(\tau) = \sum_m \sum_n' \frac{1}{(m\tau + n)^2} - \sum_n \sum_m' \frac{1}{(m\tau + n)^2},$$

where the primes mean the term $m = n = 0$ is omitted. Equivalently $E_2^*(\tau) = F(\tau/2) - 4F(2\tau)$ with the “forbidden” Eisenstein series $F(\tau) = \sum_m (\sum_n 1/(m\tau + n)^2)$.

Proposition 23.3.14 (Reduction of the four-square theorem). $r_4(n) = 8\sigma_1^*(n)$ is equivalent to $\theta(\tau)^4 = -\frac{1}{\pi^2} E_2^*(\tau)$.

Proof. It suffices to show $-\pi^{-2}E_2^*(\tau) = 1 + 8\sum_{k \geq 1} \sigma_1^*(k)q^k$. Recall from the previous chapter (Eisenstein series) that $F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_{k \geq 1} \sigma_1(k)e^{2\pi i k \tau}$, where $\sigma_1(k)$ is the sum of all divisors. Now $\sigma_1^*(n) = \sigma_1(n)$ if $4 \nmid n$, and $\sigma_1^*(n) = \sigma_1(n) - 4\sigma_1(n/4)$ if $4 \mid n$ (a divisor $d = 4\tilde{d}$ of $n = 4\tilde{n}$ corresponds to the divisor $\tilde{d}$ of $\tilde{n}$). Substituting $E_2^* = F(\tau/2) - 4F(2\tau)$ and collecting,

$$E_2^*(\tau) = -\pi^2 - 8\pi^2 \sum_{k \geq 1} \sigma_1^*(k)e^{\pi i k \tau},$$

which is the claim after dividing by $-\pi^2$. $\square$

The heart is that E_2^* obeys the same modular laws as θ^4 . Let $\tilde{F}$ be F with the order of summation reversed.

Lemma 23.3.15 (Forbidden Eisenstein series). *F and $\tilde{F}$ satisfy (a) $F(-1/\tau) = \tau^2 \tilde{F}(\tau)$; (b) $F(\tau) - \tilde{F}(\tau) = 2\pi i/\tau$; (c) $F(-1/\tau) = \tau^2 F(\tau) - 2\pi i\tau$.*

Proof. (a) is immediate from $(n + m(-1/\tau))^2 = \tau^{-2}(-m + n\tau)^2$, which swaps the roles of m, n . (b) uses the eta functional equation $\eta(-1/\tau) = \sqrt{\tau/i} \eta(\tau)$ (Proposition 23.1.14). Taking the logarithmic derivative of $\eta(\tau) = q^{1/12} \prod_{n \geq 1} (1 - q^{2n})$ gives $(\eta'/\eta)(\tau) = \frac{\pi i}{12} - 2\pi i \sum_{n \geq 1} \frac{nq^{2n}}{1 - q^{2n}}$, and since $\sum_{n \geq 1} \frac{nq^{2n}}{1 - q^{2n}} = \sum_{k \geq 1} \sigma_1(k)q^{2k}$, comparison with the q -expansion of F yields $(\eta'/\eta)(\tau) = \frac{i}{4\pi} F(\tau)$. Differentiating the eta functional equation (the log-derivative of $\eta(-1/\tau)$ is $\tau^{-2}(\eta'/\eta)(-1/\tau)$, which by (a) is $\frac{i}{4\pi} \tilde{F}(\tau)$) gives $\frac{i}{4\pi} \tilde{F}(\tau) = \frac{1}{2\tau} + \frac{i}{4\pi} F(\tau)$, i.e. $\tilde{F} = F - 2\pi i/\tau$. (c) combines (a) and (b). $\square$

Proposition 23.3.16 (Modular properties of E_2^*). *E_2^* satisfies (i) $E_2^*(\tau + 2) = E_2^*(\tau)$; (ii) $E_2^*(\tau) = -\tau^{-2}E_2^*(-1/\tau)$; (iii) $E_2^*(\tau) \rightarrow -\pi^2$ as $\text{Im } \tau \rightarrow \infty$; (iv) $|E_2^*(1 - 1/\tau)| = O(|\tau^2 e^{\pi i \tau}|)$ as $\text{Im } \tau \rightarrow \infty$. Moreover $-\pi^2 \theta^4$ has the same properties.*

Proof. (i) is immediate from $E_2^* = F(\tau/2) - 4F(2\tau)$ and periodicity of F . For (ii), apply Lemma 23.3.15(c) to $F(-1/(2\tau))$ and $F(-2/\tau)$:

$$\begin{aligned} E_2^*(-1/\tau) &= F(-1/(2\tau)) - 4F(-2/\tau) \\ &= [4\tau^2 F(2\tau) - 4\pi i\tau] - 4[(\tau/2)^2 F(\tau/2) - \pi i\tau] \\ &= 4\tau^2 F(2\tau) - \tau^2 F(\tau/2) = -\tau^2 (F(\tau/2) - 4F(2\tau)) = -\tau^2 E_2^*(\tau). \end{aligned}$$

(iii) follows since the q -tail of F vanishes at ∞ , leaving $E_2^* \rightarrow \frac{\pi^2}{3} - 4 \cdot \frac{\pi^2}{3} = -\pi^2$. For (iv), one shows $E_2^*(1 - 1/\tau) = \tau^2 (F(\frac{\tau-1}{2}) - F(\tau/2))$ via the transformation formulas for F , and the right side is $O(\tau^2 e^{\pi i \tau})$ from the leading q -term. $\square$

Theorem 23.3.17 (Four-square theorem, conclusion). $\theta(\tau)^4 = -\pi^{-2}E_2^*(\tau)$; hence $r_4(n) = 8\sigma_1^*(n)$ and every positive integer is a sum of four squares.

Proof. Set $f(\tau) = E_2^*(\tau)/\theta(\tau)^4$, holomorphic since $\theta \neq 0$. By Proposition 23.3.16, $f(\tau + 2) = f(\tau)$ and $f(-1/\tau) = f(\tau)$ (the factors $-\tau^{-2}$ cancel between E_2^* and θ^4), and f is bounded with limit $-\pi^2$ at ∞ and the cusps (using $\theta^4 \rightarrow 1$, $\theta(1 - 1/\tau)^4 \sim 16\tau^2 e^{\pi i \tau}$). By the rigidity Theorem 23.3.10, $f \equiv -\pi^2$. Thus $\theta^4 = -\pi^{-2}E_2^*$, which by Proposition 23.3.14 is the four-square theorem; since $\sigma_1^*(n) \geq 1$ for every $n \geq 1$, $r_4(n) > 0$ always. $\square$

Worked Example 23.3.18 ($r_4(1) = 8$ and $r_4(2) = 24$). Verify the formula $r_4(n) = 8\sigma_1^*(n)$ for $n = 1, 2$.

Recognition. Compute σ_1^* (sum of divisors not divisible by 4) and count representations by hand.

Solution. $n = 1$: only divisor is 1 (not divisible by 4), $\sigma_1^*(1) = 1$, so $r_4(1) = 8$. By hand $1 = (\pm 1)^2 + 0 + 0 + 0$ with the ± 1 in any of 4 slots: $4 \cdot 2 = 8$. $n = 2$: divisors 1, 2, both $\not\equiv 0 \pmod{4}$, $\sigma_1^*(2) = 3$, so $r_4(2) = 24$. By hand $2 = (\pm 1)^2 + (\pm 1)^2 + 0 + 0$: choose 2 of 4 slots ($\binom{4}{2} = 6$ ways) and signs ($2^2 = 4$): $6 \cdot 4 = 24$. Check. Both match.

Worked Example 23.3.19 (σ_1^* drops the 4-divisible divisors). Compute $r_4(4)$ and contrast $\sigma_1^*(4)$ with $\sigma_1(4)$.

Recognition. For $n = 4$, the divisor 4 itself is excluded from σ_1^* ; this is the case $4 \mid n$ where $\sigma_1^* = \sigma_1(n) - 4\sigma_1(n/4)$.

Solution. Divisors of 4: 1, 2, 4. Excluding 4, $\sigma_1^*(4) = 1+2 = 3$, so $r_4(4) = 24$. Cross check: $\sigma_1(4) = 1+2+4 = 7$ and $\sigma_1(4) - 4\sigma_1(1) = 7 - 4 = 3$. Agree. *Check.* $4 = (\pm 2)^2 + 0 + 0 + 0$ ($4 \cdot 2 = 8$ ways) plus $4 = (\pm 1)^2 \times 4$ (1 way for the slot pattern, $2^4 = 16$ sign choices) $= 8 + 16 = 24$.

Caution. The series E_2^* is only *conditionally* convergent: summing n -first-then- m differs from m -first-then- n , and the gap is exactly the $2\pi i/\tau$ of Lemma 23.3.15(b). This is why E_2 (weight 2) is *not* a genuine modular form — the “error term” $-2\pi i\tau$ in $F(-1/\tau) = \tau^2 F(\tau) - 2\pi i\tau$ is its anomaly. The starred combination $E_2^* = F(\tau/2) - 4F(2\tau)$ is engineered so the anomalies cancel, restoring exact modularity (under G). Never reorder these sums casually.

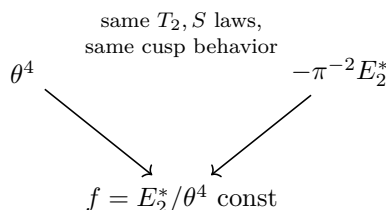


Figure 23.13. The proof architecture: matching modular laws plus the rigidity theorem collapse the ratio to a constant, giving $\theta^4 = -\pi^{-2}E_2^*$. (Schematic.)

Further worked examples

Worked Example 23.3.20 ($r_2(25)$ from the divisor formula, with the lattice picture). Compute $r_2(25)$ via $r_2(n) = 4(d_1(n) - d_3(n))$ and confirm by listing representations.

Recognition. $25 = 5^2$ has only divisors $\equiv 1 \pmod{4}$; classify them and apply Theorem 23.3.2.

Solution. Divisors of 25: $\{1, 5, 25\}$, all $\equiv 1 \pmod{4}$. So $d_1 = 3$, $d_3 = 0$, giving $r_2(25) = 4(3 - 0) = 12$. By hand the representations $x^2 + y^2 = 25$ are $(\pm 5, 0), (0, \pm 5), (\pm 3, \pm 4), (\pm 4, \pm 3)$: that is $2 + 2 + 4 + 4 = 12$ lattice points. *Check.* The prime $5 \equiv 1 \pmod{4}$ appears squared, and the prime-power criterion (no prime $\equiv 3$) confirms 25 is a sum of two squares; the count 12 matches both methods.

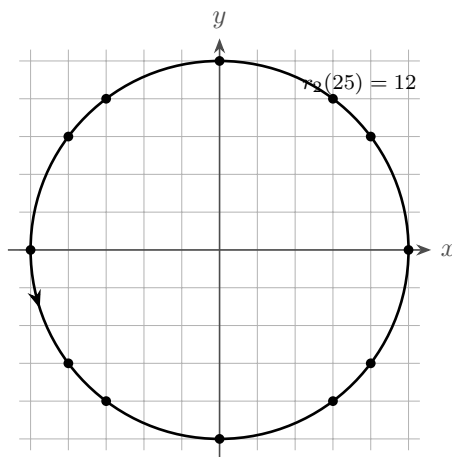


Figure 23.14. The 12 lattice points on $x^2 + y^2 = 25$: the 4 axis points plus the 8 from the Pythagorean triple $(3, 4, 5)$.

Worked Example 23.3.21 (r_2 is multiplicative on coprime arguments). Show $r_2(n)/4$ is multiplicative: if $\gcd(m, n) = 1$ then $\frac{1}{4}r_2(mn) = \frac{1}{4}r_2(m) \cdot \frac{1}{4}r_2(n)$, and verify on $m = 5, n = 13$.

Recognition. Set $\delta(n) = d_1(n) - d_3(n)$. Since d_1, d_3 come from the multiplicative character χ with $\chi(k) = \pm 1$ for $k \equiv \pm 1 \pmod{4}$, $\delta = \sum_{d|n} \chi(d)$ is multiplicative.

Solution. $\delta(n) = \sum_{d|n} \chi(d)$ where χ is the non-trivial character mod 4 ($\chi(1) = 1, \chi(3) = -1, \chi(\text{even}) = 0$). A Dirichlet convolution of multiplicative functions (χ and **1**) is multiplicative, so $\delta(mn) = \delta(m)\delta(n)$ for $\gcd(m, n) = 1$. Thus $\frac{1}{4}r_2 = \delta$ is multiplicative. Check $m = 5, n = 13$: $\delta(5) = 2$ (divisors 1, 5), $\delta(13) = 2$ (divisors 1, 13), so $\delta(65) = 4$ and $r_2(65) = 16$. *Check.* Directly $65 = 1 + 64 = 16 + 49 = 64 + 1 = 49 + 16$ gives $(\pm 1, \pm 8), (\pm 4, \pm 7), (\pm 8, \pm 1), (\pm 7, \pm 4)$, i.e. $4 \cdot 4 = 16$ points. Agreement.

Worked Example 23.3.22 (A NON-example: r_2 is NOT $4d_1 + 4d_3$). A reader guesses $r_2(n) = 4(d_1(n) + d_3(n)) = 4d(n)$ (four times the divisor count). Refute at $n = 9$.

Recognition. The theorem has a *minus* sign: $r_2 = 4(d_1 - d_3)$. Divisors $\equiv 3 \pmod{4}$ subtract, encoding the obstruction from primes $\equiv 3$.

Solution. For $n = 9 = 3^2$: divisors 1, 3, 9 are $\equiv 1, 3, 1 \pmod{4}$, so $d_1 = 2, d_3 = 1$. The correct formula gives $r_2(9) = 4(2 - 1) = 4$, while the guess gives $4 \cdot 3 = 12$. By hand $x^2 + y^2 = 9$ has only $(\pm 3, 0), (0, \pm 3)$: exactly 4 points. *Check.* The guess overcounts because it ignores that $3 \equiv 3 \pmod{4}$ appears to an *odd* power inside 9's divisor 3; the minus sign is precisely what enforces "9 is a sum of two squares only trivially."

Worked Example 23.3.23 (The cusp asymptotic of θ^2 via $C(1 - 1/\tau)$). Confirm property (iv) of Proposition 23.3.7 for θ^2 : show $\theta(1 - 1/\tau)^2 \sim 4(\tau/i)e^{\pi i \tau/2}$ as $\text{Im } \tau \rightarrow \infty$.

Recognition. Square the cusp expansion of θ in Corollary 23.1.12.

Solution. By Corollary 23.1.12, $\theta(1 - 1/\tau) \sim \sqrt{\tau/i} \cdot 2e^{\pi i \tau/4}$. Squaring, $\theta(1 - 1/\tau)^2 \sim (\tau/i) \cdot 4e^{\pi i \tau/2} = 4(\tau/i)e^{\pi i \tau/2}$. *Check.* This is exactly property (iv) for θ^2 , so θ^2 matches C at the cusp $\tau = 1$; both blow up like $|\tau|e^{-\pi \text{Im } \tau/2} \rightarrow 0$ in modulus, which is what makes the ratio $f = C/\theta^2$ bounded (limit 1) at the cusp in the proof of Theorem 23.3.11.

Worked Example 23.3.24 ($r_4(3) = 32$ and the sign of σ_1^*). Verify $r_4(3) = 8\sigma_1^*(3)$ and count by hand.

Recognition. 3 is odd, so no divisor is divisible by 4 and $\sigma_1^*(3) = \sigma_1(3)$ (Theorem 23.3.12).

Solution. Divisors of 3: 1, 3, neither divisible by 4, so $\sigma_1^*(3) = 1 + 3 = 4$ and $r_4(3) = 32$. By hand $3 = (\pm 1)^2 + (\pm 1)^2 + (\pm 1)^2 + 0$: choose which of 4 slots is 0 (4 ways) and signs on the three 1's ($2^3 = 8$): $4 \cdot 8 = 32$. *Check.* Both give 32; note 3 is a sum of four (in fact three) squares, consistent with "every positive integer is a sum of four squares."

Worked Example 23.3.25 (Gaussian-integer factorization realizing a two-square product). Use Lemma 23.3.1 to write $65 = 5 \cdot 13$ explicitly as a sum of two squares in two distinct ways.

Recognition. Multiply the Gaussian integers $(2 + i)$ and $(3 + 2i)$ (with norms 5, 13); different sign choices give the two representations.

Solution. $5 = 2^2 + 1^2 = |2 + i|^2$ and $13 = 3^2 + 2^2 = |3 + 2i|^2$. Then $(2 + i)(3 + 2i) = 6 + 4i + 3i + 2i^2 = 4 + 7i$, so $65 = 4^2 + 7^2$. Using the conjugate $(2 - i)(3 + 2i) = 6 + 4i - 3i - 2i^2 = 8 + i$, so $65 = 8^2 + 1^2$. *Check.* $16 + 49 = 65$ and $64 + 1 = 65$, the two essentially different representations; this matches $r_2(65) = 16$ (two "shapes" $\{4, 7\}$ and $\{1, 8\}$, each yielding 8 signed/ordered points).

Exercises (Stein, Ch.10).

1. Use the product formula for Θ to prove:

(a) The "triangular number" identity, for $|x| < 1$,

$$\prod_{n=0}^{\infty} (1 + x^n)(1 - x^{2n+2}) = \sum_{n=-\infty}^{\infty} x^{n(n+1)/2}.$$

(b) The "septagonal number" identity, for $|x| < 1$,

$$\prod_{n=0}^{\infty} (1 - x^{5n+1})(1 - x^{5n+4})(1 - x^{5n+5}) = \sum_{n=-\infty}^{\infty} (-1)^n x^{n(5n+3)/2}.$$

2. Consider Pythagorean triples (a, b, c) with $a^2 + b^2 = c^2$, $a, b, c \in \mathbb{Z}$, and a, b without common factors.
 - (a) Show one of a, b is odd and the other even.
 - (b) Assuming a odd and b even, show there are integers m, n with $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$.
[Hint: $b^2 = (c - a)(c + a)$; prove $(c - a)/2$ and $(c + a)/2$ are relatively prime.]
 - (c) Conversely, whenever c is a sum of two squares, show there exist integers a, b with $a^2 + b^2 = c^2$.
3. Use the formula for $r_2(n)$ to prove:
 - (a) If $n = p$ with p prime of the form $4k + 1$, then $r_2(n) = 8$; hence $n = n_1^2 + n_2^2$ uniquely up to signs and order.
 - (b) If $n = q^a$ with q prime of the form $4k + 3$ and $a \geq 1$, then $r_2(n) > 0$ iff a is even.
 - (c) In general, n is a sum of two squares iff every prime of the form $4k + 3$ in the factorization of n occurs to an even power.
4. Observe the irregularities of r_2 and r_4 for large n :
 - (a) $r_2(n) = 0$ for infinitely many n , while $\limsup_{n \rightarrow \infty} r_2(n) = \infty$.
 - (b) $r_4(n) = 24$ for infinitely many n , while $\limsup_{n \rightarrow \infty} r_4(n)/n = \infty$.

[Hint: for (a) consider $n = 5^k$; for (b) consider $n = 2^k$, and $n = q^k$ with q odd and large.]
5. Recall from Problem 2 of Chapter 2 that $\sum_{n \geq 1} d(n)z^n = \sum_{n \geq 1} \frac{z^n}{1 - z^n}$ for $|z| < 1$, where $d(n)$ is the number of divisors of n . More generally, show

$$\sum_{n=1}^{\infty} \sigma_{\ell}(n)z^n = \sum_{n=1}^{\infty} \frac{n^{\ell} z^n}{1 - z^n}, \quad |z| < 1,$$

where $\sigma_{\ell}(n)$ is the sum of the ℓ -th powers of the divisors of n .

6. Another identity equivalent to the four-square theorem.

- (a) Show for $|q| < 1$ that $\sum_{n \geq 1} \frac{nq^n}{1 - q^n} = \sum_{n \geq 1} \frac{q^n}{(1 - q^n)^2}$. [Hint: the left side is $\sum \sigma_1(n)q^n$; use $x/(1 - x)^2 = \sum_{n \geq 1} nx^n$.]
- (b) Show as a result that

$$\sum_{n \geq 1} \frac{nq^n}{1 - q^n} - \sum_{n \geq 1} \frac{4nq^{4n}}{1 - q^{4n}} = \sum_{n \geq 1} \frac{q^n}{(1 - q^n)^2} - \sum_{n \geq 1} \frac{q^{4n}}{(1 - q^{4n})^2} = \sum_{n \geq 1} \sigma_1^*(n)q^n,$$

where $\sigma_1^*(n)$ is the sum of divisors of n not divisible by 4.

- (c) Show the four-square theorem is equivalent to $\theta(\tau)^4 = 1 + 8 \sum_{n=1}^{\infty} \frac{q^n}{(1 + (-1)^n q^n)^2}$, $q = e^{\pi i \tau}$.

Problems (Stein, Ch.10).

1. (*) Suppose $n = 4^a(8k + 7)$ with a, k positive integers. Show n cannot be written as a sum of three squares. (The converse, that every n not of this form is a sum of three squares, is a difficult theorem of Legendre and Gauss.)
2. Let $\text{SL}_2(\mathbb{Z})$ be the 2×2 integer matrices of determinant 1, acting on $\mathbb{H}$ by $g(\tau) = (a\tau + b)/(c\tau + d)$, with fundamental domain $\mathcal{F}_1 = \{\tau : |\tau| \geq 1, |\text{Re } \tau| \leq \frac{1}{2}\}$. Let $S(\tau) = -1/\tau$, $T_1(\tau) = \tau + 1$, with matrices $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, and let $\mathfrak{g} = \langle S, T_1 \rangle$.
 - (a) Show every $\tau \in \mathbb{H}$ has $g \in \mathfrak{g}$ with $g(\tau) \in \mathcal{F}_1$.

- (b) Call τ, τ' congruent if $\tau' = g(\tau)$ for some $g \in \mathrm{SL}_2(\mathbb{Z})$. Prove that if $\tau, \tau' \in \mathcal{F}_1$ are congruent, then either $\mathrm{Re} \tau = \pm \frac{1}{2}$ and $\tau' = \tau \mp 1$, or $|\tau| = 1$ and $\tau' = -1/\tau$. [Hint: say $\tau' = g(\tau)$; one may assume $\mathrm{Im} \tau' \geq \mathrm{Im} \tau$, so $|c\tau + d| \leq 1$; treat $c = -1, 0, 1$ separately.]
- (c) Prove S and T_1 generate the modular group: every $g \in \mathrm{SL}_2(\mathbb{Z})$ is a finite composition of S, T_1 , and inverses (strictly, of $\mathrm{PSL}_2(\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})/\pm I$). [Hint: $2i$ is interior to $\mathcal{F}_1$; map $g(2i)$ back into $\mathcal{F}_1$ by (a) and use (b).]
3. Let G be the group of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with integer entries, determinant 1, $a \equiv d, b \equiv c$ (same parity), and c, d of opposite parity, acting on $\mathbb{H}$ with fundamental domain $\mathcal{F}$ ($|\tau| \geq 1, |\mathrm{Re} \tau| \leq 1$). With $S(\tau) = -1/\tau \leftrightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $T_2(\tau) = \tau + 2 \leftrightarrow \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, prove that every $g \in G$ is a finite composition of S, T_2 , and their inverses.
4. Let G be the group of the previous problem. Give an alternate proof of Theorem 23.3.10 (a holomorphic, bounded, G -invariant function on $\mathbb{H}$ is constant).
- (a) Suppose $f : \mathbb{H} \rightarrow \mathbb{C}$ is holomorphic and bounded, and there is a sequence $\tau_k = x_k + iy_k$ with $f(\tau_k) = 0$, $\sum_k y_k = \infty, 0 < y_k \leq 1, |x_k| \leq 1$. Then $f \equiv 0$. [Hint: when $x_k = 0$, see Problem 5 in Chapter 8.]
- (b) Given relatively prime c, d of different parity, show there exist integers a, b with $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$. [Hint: solutions of $xc + dy = 1$ are $x_0 + dt, y_0 - ct$.]
- (c) Prove $\sum 1/(c^2 + d^2) = \infty$, summed over relatively prime c, d of opposite parity. [Hint: if not, deduce $\sum_{(a,b)=1} 1/(a^2 + b^2) < \infty$ using $c = (a+b)/2, d = (a-b)/2$ and $c^2 + d^2 \leq A(a^2 + b^2)$; then $\sum_{n \neq 0} 1/n^2 \cdot \sum_{(a,b)=1} 1/(a^2 + b^2) < \infty$ gives $\sum_{k,\ell \neq 0} 1/(k^2 + \ell^2) < \infty$, a contradiction.]
- (d) Prove that a holomorphic, bounded, G -invariant $F : \mathbb{H} \rightarrow \mathbb{C}$ is constant. [Hint: replace F by $F - F(i)$ so $F(i) = 0$; for each relatively prime c, d of opposite parity choose $g \in G$ with $g(i) = x_{c,d} + i/(c^2 + d^2)$, $|x_{c,d}| \leq 1$, then apply (a).]
5. (*) Connect η to elliptic functions: with $\Delta(\tau)$ the discriminant of $y^2 = 4x^3 - g_2x - g_3$ ($g_2 = 60E_4, g_3 = 140E_6$), prove $\Delta(\tau) = (2\pi)^{12}\eta(\tau)^{24}$. (See the companion to Exercise 1 above; details outlined there.)
6. (*) Deduce the formula for $r_8(n)$ (sum of eight squares): $r_8(n) = 16\sigma_3^*(n)$, where $\sigma_3^*(n) = \sigma_3(n) = \sum_{d|n} d^3$ for n odd, and $\sigma_3^*(n) = \sum_{d|n} (-1)^d d^3 = \sigma_3^e(n) - \sigma_3^o(n)$ for n even. Introduce $E_4^*(\tau) = \sum 1/(n + m\tau)^4$ (sum over n, m of opposite parity) and the standard $E_4(\tau) = \sum_{(n,m) \neq (0,0)} 1/(n + m\tau)^4$ (absolutely convergent, unlike E_2^*).
- (a) Prove $r_8(n) = 16\sigma_3^*(n)$ is equivalent to $\theta(\tau)^8 = 48\pi^{-4}E_4^*(\tau)$. [Hint: $E_4(\tau) = 2\zeta(4) + \frac{(2\pi)^4}{3} \sum_{k \geq 1} \sigma_3(k)e^{2\pi i k \tau}, \zeta(4) = \pi^4/90$.]
- (b) Note $E_4^*(\tau) = E_4(\tau) - 2^{-4}E_4((\tau - 1)/2)$.
- (c) $E_4^*(\tau + 2) = E_4^*(\tau)$.
- (d) $E_4^*(\tau) = \tau^{-4}E_4^*(-1/\tau)$.
- (e) $(48/\pi^4)E_4^*(\tau) \rightarrow 1$ as $\tau \rightarrow \infty$.
- (f) $|E_4^*(1 - 1/\tau)| \approx |\tau|^4 |e^{2\pi i \tau}|$ as $\mathrm{Im} \tau \rightarrow \infty$. [Hint: verify $E_4^*(1 - 1/\tau) = \tau^4(E_4(\tau) - E_4(2\tau))$.]
- Since θ^8 shares properties (c)–(f), the invariant $48\pi^{-4}E_4^*/\theta^8$ is a bounded constant equal to 1.

Additional exercises.

- (Warm-up counts.) Compute $r_2(n)$ for $n = 1, 2, 4, 9, 10, 25$ directly from $r_2(n) = 4(d_1(n) - d_3(n))$, and verify $r_2(25)$ by listing lattice points on $x^2 + y^2 = 25$.
- (Multiplicativity in action.) Using Lemma 23.3.1, write $65 = 5 \cdot 13$ as a sum of two squares in two essentially different ways, starting from $5 = 2^2 + 1^2$ and $13 = 3^2 + 2^2$ and forming $(2 + i)(3 + 2i)$ and $(2 + i)(3 - 2i)$.
- (Four-square formula.) Compute $r_4(n)$ for $n = 3, 5, 6, 8$ via $r_4(n) = 8\sigma_1^*(n)$, and check $r_4(3)$ by counting representations of 3 as a sum of four squares by hand.

4. (The σ_1^* subtlety.) For $n = 12$, list all divisors, strike out those divisible by 4, and compute $\sigma_1^*(12)$ and hence $r_4(12)$. Confirm with the formula $\sigma_1^*(n) = \sigma_1(n) - 4\sigma_1(n/4)$ for $4 \mid n$.
5. (Residue obstruction.) Prove directly (mod 8) that no integer of the form $8k + 7$ is a sum of three squares, by checking all residues $x^2 \equiv 0, 1, 4 \pmod{8}$.
6. (Theta powers as generating functions.) Explain why $\theta(\tau)^k = \sum_n r_k(n)q^n$ where $r_k(n)$ counts representations of n as a sum of k squares, and verify the first three coefficients of θ^4 give $r_4(0) = 1$, $r_4(1) = 8$, $r_4(2) = 24$.
7. (Reduction into $\mathcal{F}$.) Take $\tau = \frac{1}{4} + \frac{i}{8}$. Apply $S : \tau \mapsto -1/\tau$ once and report the imaginary part of the image; explain how Lemma 23.3.9 guarantees finitely many further T_2, S steps land in $\mathcal{F}$.
8. (NON-example: dropping boundedness.) Show that the rigidity Theorem 23.3.10 fails if “bounded” is omitted: exhibit a non-constant holomorphic f on $\mathbb{H}$ with $f(\tau + 2) = f(\tau)$ and $f(-1/\tau) = f(\tau)$ (e.g. a modular function with a pole at the cusp, such as a suitable ratio of Eisenstein series). State precisely which hypothesis it violates.
9. (Diagram problem.) Reproduce the fundamental domain $\mathcal{F}$ ($|\operatorname{Re} \tau| \leq 1$, $|\tau| \geq 1$) in the upper half-plane, mark the cusps ± 1 , and draw the images of $\mathcal{F}$ under T_2 and under S to show how translates tile a neighborhood of the real axis.
10. (Conditional convergence.) For E_2^* , illustrate the order-of-summation sensitivity: compute the difference $F(\tau) - \tilde{F}(\tau) = 2\pi i/\tau$ from Lemma 23.3.15(b) and explain why this nonzero “anomaly” is exactly what prevents E_2 from being a weight-2 modular form.
11. (From θ^4 to arithmetic.) Assuming $\theta^4 = -\pi^{-2}E_2^*$, deduce that every prime p is a sum of four squares, and that the number of representations of an odd prime p is $r_4(p) = 8(p + 1)$.

Part III

Measure Theory, Integration, and Hilbert Spaces

Chapter 24

Measure Theory

24.1 The Exterior Measure

Preliminaries: rectangles, cubes, and the geometry of open sets

We work in $\mathbb{R}^d$ with the Euclidean norm $|x| = (x_1^2 + \cdots + x_d^2)^{1/2}$ and the distance $\text{dist}(x, y) = |x - y|$; for sets, $\text{dist}(E, F) = \inf\{|x - y| : x \in E, y \in F\}$. Open balls $B_r(x) = \{y : |y - x| < r\}$ generate the topology: E is *open* if every point of E has a ball inside E , *closed* if its complement is open, and *compact* if it is closed and bounded. Compact sets obey the Heine–Borel property: every open cover has a finite subcover.

Definition 24.1.1 (Rectangles and cubes). A *closed rectangle* in $\mathbb{R}^d$ is a product of closed bounded intervals

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_d, b_d], \quad a_j \leq b_j,$$

with sides parallel to the axes. Its *volume* is $|R| = (b_1 - a_1) \cdots (b_d - a_d)$. A *cube* Q is a rectangle with all side lengths equal; if the common side length is ℓ then $|Q| = \ell^d$. A finite or countable union of rectangles is *almost disjoint* if their interiors are pairwise disjoint.

In $d = 1$ “volume” is length; in $d = 2$ it is area; in $d = 3$ it is the usual volume of a box. The whole construction of measure starts from this one elementary quantity $|R|$ and the demand that it behave well under (almost disjoint) decomposition.

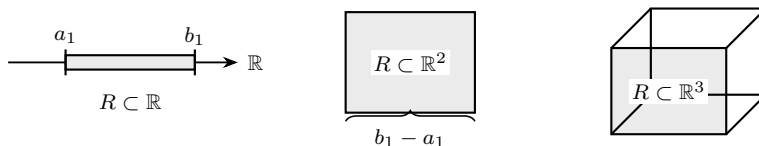


Figure 24.1. Rectangles in $\mathbb{R}^d$ for $d = 1, 2, 3$: intervals, boxes, parallelepipeds. (Schematic.)

Lemma 24.1.2 (Volume is finitely additive on almost disjoint pieces). *If a rectangle is an almost disjoint union of finitely many rectangles, $R = \bigcup_{k=1}^N R_k$, then $|R| = \sum_{k=1}^N |R_k|$.*

Proof. Extend all sides of $R_1, \dots, R_N$ indefinitely; this produces a finite grid of sub-rectangles $\tilde{R}_1, \dots, \tilde{R}_M$ and a partition $J_1, \dots, J_N$ of $\{1, \dots, M\}$ with $R = \bigcup_j \tilde{R}_j$ and $R_k = \bigcup_{j \in J_k} \tilde{R}_j$ almost disjointly. Because the grid partitions the side intervals of R and each $\tilde{R}_j$ is a product of those sub-intervals, $|R| = \sum_{j=1}^M |\tilde{R}_j|$. The same holds for each R_k , so $|R| = \sum_{j=1}^M |\tilde{R}_j| = \sum_{k=1}^N \sum_{j \in J_k} |\tilde{R}_j| = \sum_{k=1}^N |R_k|$. $\square$

Lemma 24.1.3 (Sub-additivity for finite covers). *If $R \subset \bigcup_{k=1}^N R_k$ for rectangles $R, R_1, \dots, R_N$, then $|R| \leq \sum_{k=1}^N |R_k|$.*

Proof. Take the same grid generated by extending all sides of $R, R_1, \dots, R_N$. Now the index sets J_k need not be disjoint, but every grid cell of R lies in some R_k , so $|R| = \sum_{\tilde{R}_j \subset R} |\tilde{R}_j| \leq \sum_{k=1}^N \sum_{j \in J_k} |\tilde{R}_j| = \sum_{k=1}^N |R_k|$. $\square$

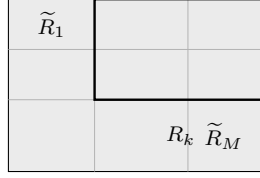


Figure 24.2. Extending all sides refines the rectangles into a common grid; each R_k is a union of grid cells (Lemmas 24.1.2–24.1.3). (Schematic.)

Theorem 24.1.4 (Structure of open sets in $\mathbb{R}$). *Every open $O \subset \mathbb{R}$ is uniquely a countable union of disjoint open intervals.*

Proof. For $x \in O$ let $a_x = \inf\{a < x : (a, x) \subset O\}$ and $b_x = \sup\{b > x : (x, b) \subset O\}$, and put $I_x = (a_x, b_x)$, the largest open interval in O through x . Then $O = \bigcup_x I_x$. If $I_x \cap I_y \neq \emptyset$ their union is an interval in O through both, so by maximality $I_x = I_y$; distinct intervals are disjoint. Each I_x contains a rational, and disjoint intervals contain distinct rationals, so the family is countable. Uniqueness follows since the I_x are the connected components. $\square$

Theorem 24.1.5 (Structure of open sets in $\mathbb{R}^d$). *Every open $O \subset \mathbb{R}^d$, $d \geq 1$, is a countable union of almost disjoint closed cubes.*

Proof. Start with the unit grid (cubes with integer-coordinate vertices). *Accept* a cube if it lies inside O ; *reject* it if it lies in O^c ; *tentatively accept* it if it meets both O and O^c . Bisect each tentatively accepted cube into 2^d cubes of half the side and repeat the rule, forever. The collection $\mathcal{Q}$ of accepted cubes is countable and almost disjoint. Given $x \in O$, some bisected cube of side 2^{-N} contains x and lies in O ; it was either accepted or sits inside an earlier accepted cube. Hence $\bigcup_{Q \in \mathcal{Q}} Q = O$. $\square$

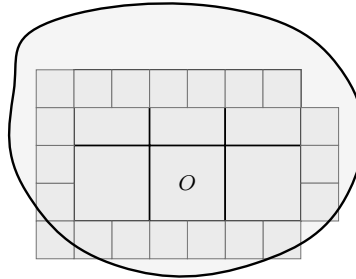


Figure 24.3. Decomposing an open set into almost disjoint dyadic cubes: large cubes fill the interior, and only the boundary layer is ever bisected (Theorem 24.1.5). (Schematic.)

The Cantor set

Definition 24.1.6 (Cantor set). Let $C_0 = [0, 1]$ and obtain C_{k+1} from C_k by deleting the open middle third of each component interval. Thus $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, $C_2 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$, and so on. The *Cantor set* is $C = \bigcap_{k=0}^{\infty} C_k$.

Each C_k is a disjoint union of 2^k closed intervals of length 3^{-k} , so it has total length $(2/3)^k \rightarrow 0$. The set C is compact, totally disconnected, and perfect (no isolated points); its endpoints all survive, so $C \neq \emptyset$, and in fact C has the cardinality of the continuum.

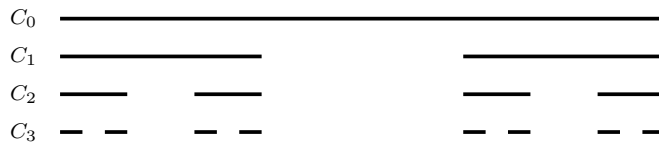


Figure 24.4. The first stages $C_0 \supset C_1 \supset C_2 \supset C_3$ of the Cantor set; $|C_k| = (2/3)^k \rightarrow 0$.

The exterior measure

Definition 24.1.7 (Exterior measure). For any $E \subseteq \mathbb{R}^d$,

$$m_*(E) = \inf \sum_{j=1}^{\infty} |Q_j|,$$

the infimum over all countable coverings $E \subseteq \bigcup_{j=1}^{\infty} Q_j$ by closed cubes. Always $0 \leq m_*(E) \leq \infty$.

The exterior measure approximates a set “from the outside” by cube coverings. It is vital that *countably* many cubes be allowed: finite covers give a strictly larger quantity (the Jordan content; see the exercises). Covers by rectangles or by balls give the same m_* .

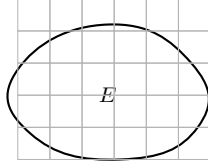


Figure 24.5. $m_*(E)$ is the infimum of the total volume over all countable coverings of E by cubes; finer covers track the set more tightly. (Schematic.)

Example 24.1.8 (Basic computations). (i) A point, and $\emptyset$, have $m_* = 0$ (a degenerate cube of volume 0 covers a point). (ii) For a closed cube Q , $m_*(Q) = |Q|$ (next theorem). (iii) $m_*(\mathbb{R}^d) = \infty$, since $\mathbb{R}^d$ contains cubes of arbitrarily large volume. (iv) The Cantor set has $m_*(C) = 0$: $C \subset C_k$, a union of 2^k intervals of length 3^{-k} , so $m_*(C) \leq (2/3)^k$ for all k .

Theorem 24.1.9 (Cube and rectangle measures). *If Q is a closed cube then $m_*(Q) = |Q|$; the same holds for open cubes and for rectangles.*

Proof. Q covers itself, so $m_*(Q) \leq |Q|$. For the reverse, take any cover $Q \subseteq \bigcup_j Q_j$; fix $\varepsilon > 0$ and enlarge each Q_j to an open cube S_j with $|S_j| \leq (1 + \varepsilon)|Q_j|$. By compactness finitely many $S_1, \dots, S_N$ cover Q , so Lemma 24.1.3 gives $|Q| \leq \sum_{j=1}^N |S_j| \leq (1 + \varepsilon) \sum_j |Q_j|$. Letting $\varepsilon \rightarrow 0$ and infimizing over covers gives $|Q| \leq m_*(Q)$. For an open cube, $m_*(Q) = m_*(\bar{Q}) = |\bar{Q}| = |Q|$. For a rectangle, approximate by a grid of cubes of side $1/k$ inside and along the boundary: the inside cubes have total volume $\leq |R|$, the $O(k^{d-1})$ boundary cubes contribute $O(1/k)$, and $k \rightarrow \infty$ gives $m_*(R) \leq |R|$; the reverse is as for cubes. $\square$

Theorem 24.1.10 (Five properties of m_*). (1) Monotonicity: $E_1 \subseteq E_2 \implies m_*(E_1) \leq m_*(E_2)$. (2) Countable sub-additivity: $m_*(\bigcup_j E_j) \leq \sum_j m_*(E_j)$. (3) Outer regularity: $m_*(E) = \inf\{m_*(O) : O \supseteq E \text{ open}\}$. (4) Additivity at positive distance: *if $\text{dist}(E_1, E_2) > 0$ then $m_*(E_1 \cup E_2) = m_*(E_1) + m_*(E_2)$* . (5) *if $E = \bigcup_j Q_j$ is a countable almost disjoint union of cubes then $m_*(E) = \sum_j |Q_j|$* .

Proof. (1) Any cover of E_2 covers E_1 . (2) If some $m_*(E_j) = \infty$ the claim is trivial; else cover each E_j by cubes with $\sum_k |Q_{k,j}| \leq m_*(E_j) + \varepsilon 2^{-j}$; the combined family covers E and totals $\leq \sum_j m_*(E_j) + \varepsilon$. (3) “ $\leq$ ” is monotonicity; for “ $\geq$ ” cover E by cubes summing to $\leq m_*(E) + \varepsilon/2$, fatten each Q_j to an open Q'_j with $|Q'_j| \leq |Q_j| + \varepsilon 2^{-j-1}$, and let $O = \bigcup_j Q'_j$. (4) “ $\leq$ ” is (2); for “ $\geq$ ” pick δ with $0 < \delta < \text{dist}(E_1, E_2)$, cover E by cubes of diameter $< \delta$ summing to $\leq m_*(E) + \varepsilon$; each cube meets at most one E_i , so splitting the cover into the two index sets gives $m_*(E_1) + m_*(E_2) \leq m_*(E) + \varepsilon$. (5) Shrink Q_j to $\tilde{Q}_j$ with $|Q_j| \leq |\tilde{Q}_j| + \varepsilon 2^{-j}$; the $\tilde{Q}_1, \dots, \tilde{Q}_N$ are at positive mutual distance, so (4) gives $m_*(E) \geq \sum_{j \leq N} |\tilde{Q}_j| \geq \sum_{j \leq N} |Q_j| - \varepsilon$; let $N \rightarrow \infty, \varepsilon \rightarrow 0$ and combine with (2). $\square$

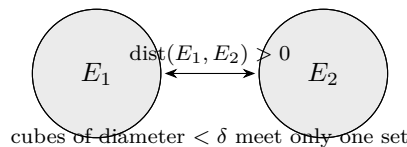


Figure 24.6. When two sets are separated, a fine enough cube cover splits cleanly, giving exact additivity (Theorem 24.1.10(4)). (Schematic.)

Caution. m_* is only *sub-additive*, not additive, for general disjoint sets: $m_*(E_1 \cup E_2) = m_*(E_1) + m_*(E_2)$ can fail when the two disjoint sets are interleaved (a non-measurable example appears in §24.2). Additivity is recovered exactly for the *measurable* sets of the next section.

Worked Example 24.1.11 (Exterior measure of $\mathbb{Q} \cap [0, 1]$). Compute $m_*(\mathbb{Q} \cap [0, 1])$.

Recognition. A countable set; cover by tiny cubes, one per point, with total volume $\rightarrow 0$.

Solution. Enumerate $\mathbb{Q} \cap [0, 1] = \{q_1, q_2, \dots\}$. For $\varepsilon > 0$ cover q_j by an interval I_j of length $\varepsilon 2^{-j}$. Then $\sum_j |I_j| = \varepsilon$, so $m_* \leq \varepsilon$ for every ε , giving $m_*(\mathbb{Q} \cap [0, 1]) = 0$. *Check.* Every countable set has exterior measure 0 by the same argument; in particular single points contribute nothing.

Worked Example 24.1.12 (A finite cover is not enough (Jordan content)). Show the Jordan content $J_*(\mathbb{Q} \cap [0, 1])$, defined with *finite* covers, equals 1, even though $m_* = 0$.

Recognition. Trigger: contrast finite vs countable covers. A finite cover by intervals whose union contains the dense set $\mathbb{Q} \cap [0, 1]$ must in fact cover $[0, 1]$ up to finitely many points.

Solution. Any finite family of closed intervals covering the dense set $\mathbb{Q} \cap [0, 1]$ has union a finite union of intervals that is dense in $[0, 1]$ and closed, hence equals $[0, 1]$; so the total length is ≥ 1 , and taking $I = [0, 1]$ shows $J_* = 1$. But $m_* = 0$ by the previous example. *Check.* This is exactly why Definition 24.1.7 insists on countable covers; finiteness would wreck countable additivity.

Worked Example 24.1.13 (Exterior measure of a line segment in $\mathbb{R}^2$). Let $S = [0, 1] \times \{0\} \subset \mathbb{R}^2$. Show $m_*(S) = 0$.

Recognition. A lower-dimensional set; cover by a thin rectangle and let its height shrink.

Solution. For $\varepsilon > 0$ the rectangle $R = [0, 1] \times [-\varepsilon/2, \varepsilon/2]$ covers S and has $|R| = \varepsilon$. Hence $m_*(S) \leq \varepsilon$ for all $\varepsilon > 0$, so $m_*(S) = 0$. *Check.* More generally any subset of a hyperplane in $\mathbb{R}^d$ has $m_* = 0$, since a slab of thickness ε has volume $\rightarrow 0$.

Worked Example 24.1.14 (Outer regularity in action). For $E = (0, 1) \cap \mathbb{Q}$ exhibit open sets $O \supseteq E$ with $m_*(O)$ arbitrarily small, illustrating Theorem 24.1.10(3).

Recognition. Trigger: realize $\inf m_*(O)$ over open $O \supseteq E$. Use the same dyadic interval cover, but make it open.

Solution. Enumerate $E = \{q_j\}$ and set $O = \bigcup_j (q_j - \varepsilon 2^{-j-1}, q_j + \varepsilon 2^{-j-1})$, an open set containing E with $m_*(O) \leq \sum_j \varepsilon 2^{-j} = \varepsilon$. Thus $\inf_O m_*(O) = 0 = m_*(E)$. *Check.* Consistent with Theorem 24.1.10(3), which guarantees such approximating open sets for *every* E .

Worked Example 24.1.15 (Measure of a half-open box via almost disjoint cubes). Use Theorem 24.1.10(5) to compute $m_*([0, 2] \times [0, 1])$.

Recognition. Trigger: decompose into almost disjoint unit cubes and sum volumes.

Solution. Write $R = [0, 2] \times [0, 1] = ([0, 1] \times [0, 1]) \cup ([1, 2] \times [0, 1])$, two almost disjoint unit squares of area 1 each. By property (5), $m_*(R) = 1 + 1 = 2 = |R|$. *Check.* Agrees with $|R| = (2 - 0)(1 - 0) = 2$ from Theorem 24.1.9.

Worked Example 24.1.16 (Hausdorff dimension of the Cantor set). Show the Cantor set C has Hausdorff dimension $\log 2 / \log 3$, consistent with $m_*(C) = 0$.

Recognition. Trigger: C is self-similar with 2 pieces each scaled by $1/3$; the similarity dimension s solves $2 \cdot (1/3)^s = 1$.

Solution. For a self-similar set built from N copies scaled by ratio r , the dimension s satisfies $Nr^s = 1$, here $2(1/3)^s = 1$, i.e. $s = \log 2 / \log 3 \approx 0.6309$. At stage k , C is covered by 2^k intervals of length 3^{-k} , so the s -dimensional Hausdorff content is $\leq 2^k (3^{-k})^s = (2 \cdot 3^{-s})^k = 1$, bounded – the threshold dimension. *Check.* Since $s < 1$, the 1-dimensional (length) measure is 0, matching $m_*(C) = 0$; yet $s > 0$, so C is far from a single point.

Worked Example 24.1.17 (Exterior measure of an open ball in $\mathbb{R}^2$). Bound m_* of the open disc $D = B_1(0) \subset \mathbb{R}^2$ between inscribed and circumscribed grids, confirming $m_*(D) = \pi$.

Recognition. Trigger: trap the disc between cubes inside it and cubes covering it; refine the grid.

Solution. With a grid of side $1/k$, the cubes entirely inside D have total area $\leq \pi$, and the cubes meeting D cover it with total area $\pi + O(1/k)$ (a boundary layer of $O(k)$ cubes each of area k^{-2}). Letting $k \rightarrow \infty$ squeezes $m_*(D)$ to the area π . *Check.* This matches $m(B_r) = v_d r^d = \pi r^2$ in $\mathbb{R}^2$ with $r = 1$.

Exercises (Stein, Ch.1) – exterior measure, Cantor sets.

1. Prove that the Cantor set C is totally disconnected and perfect: given distinct $x, y \in C$ there is $z \notin C$ between them, yet C has no isolated points. [Hint: if $|x - y| > 1/3^k$ then x, y lie in different intervals of C_k ; and each $x \in C$ has an endpoint y_k of some interval of C_k with $x \neq y_k$, $|x - y_k| \leq 1/3^k$.]
2. Ternary expansions and the Cantor–Lebesgue function. (a) Every $x \in [0, 1]$ has a ternary expansion $x = \sum_k a_k 3^{-k}$, $a_k \in \{0, 1, 2\}$ (not unique). Prove $x \in C$ iff x has such a representation with every $a_k \in \{0, 2\}$. (b) Define $F(x) = \sum_k b_k 2^{-k}$ where $b_k = a_k/2$ (using the 0/2 expansion); show F is well defined and continuous on C with $F(0) = 0$, $F(1) = 1$. (c) Show $F : C \rightarrow [0, 1]$ is surjective. (d) Extend F to $[0, 1]$ by making it constant $F(a) = F(b)$ on each deleted middle interval (a, b) .
3. Cantor sets of constant dissection. Fix $0 < \xi < 1$ ($\xi = 1/3$ gives C). At stage k remove the central open interval of relative length ξ from each remaining interval; let C_ξ be the limit. (a) Show the complement of C_ξ in $[0, 1]$ is a union of open intervals of total length 1. (b) Show $m_*(C_\xi) = 0$. [Hint: after stage k the remaining length is $(1 - \xi)^k$.]
4. Cantor-like sets. Construct a closed $\hat{C}$ so that at stage k one removes 2^{k-1} central open intervals each of length ℓ_k with $\ell_1 + 2\ell_2 + \cdots + 2^{k-1}\ell_k < 1$. (a) If $\sum_k 2^{k-1}\ell_k < 1$, show $m(\hat{C}) > 0$, in fact $m(\hat{C}) = 1 - \sum_k 2^{k-1}\ell_k$. (b) Show each $x \in \hat{C}$ is a limit of points $x_n \notin \hat{C}$ lying in complementary subintervals I_n with $|I_n| \rightarrow 0$. (c) Conclude $\hat{C}$ is perfect and contains no interval. (d) Show $\hat{C}$ is uncountable.
5. For $O_n = \{x : \text{dist}(x, E) < 1/n\}$ show: (a) if E is compact then $m(E) = \lim_n m(O_n)$; (b) this can fail for E closed unbounded, or E open bounded.
6. Using translations and dilations prove $m(B) = v_d r^d$ for a ball B of radius r , where $v_d = m(B_1)$ is the measure of the unit ball $B_1 = \{x : |x| < 1\}$.
7. The purpose of this exercise is to show finite covers do not suffice in defining m_* . The outer Jordan content is $J_*(E) = \inf \sum_{j=1}^N |I_j|$ over finite covers by intervals. (a) Prove $J_*(E) = J_*(\bar{E})$ for every E . (b) Exhibit a countable $E \subseteq [0, 1]$ with $J_*(E) = 1$ but $m_*(E) = 0$.
8. At the start of the theory one might cover by rectangles instead of cubes: $m_*^R(E) = \inf \sum_j |R_j|$ over countable closed-rectangle covers. Prove $m_*(E) = m_*^R(E)$ for every $E \subseteq \mathbb{R}^d$. [Hint: Lemma 24.1.24.1.2.]
9. Theorem 24.1.24.1.4 (every open set in $\mathbb{R}$ is a disjoint union of open intervals) has no analogue in $\mathbb{R}^d$, $d \geq 2$. Prove: (a) an open disc in $\mathbb{R}^2$ is not a disjoint union of open rectangles [Hint: boundaries]; (b) an open connected Ω is a disjoint union of open rectangles iff Ω is itself an open rectangle.
10. Let $A \subseteq [0, 1]$ be the numbers whose decimal expansion never uses the digit 4. Find $m(A)$.
11. Suppose $\Gamma = \{y = f(x)\} \subset \mathbb{R}^2$ with f continuous. Show $m(\Gamma) = 0$. [Hint: cover Γ by rectangles using uniform continuity of f .]

Additional exercises.

1. Compute m_* of (a) the set of dyadic rationals in $[0, 1]$; (b) the set $\{1/n : n \geq 1\}$; (c) the graph $\{(x, x^2) : x \in [0, 1]\} \subset \mathbb{R}^2$.
2. Show $m_*(E) = 0$ implies $m_*(E \cup F) = m_*(F)$ for every F . (Null sets are “invisible” to exterior measure.)
3. Let $E = \bigcup_{n \geq 1} [n, n + 2^{-n}] \subset \mathbb{R}$. Compute $m_*(E)$ and decide whether E is bounded.
4. Prove directly from Definition 24.1.24.1.7 that m_* is translation invariant: $m_*(E + h) = m_*(E)$.

5. For the Cantor set C , show $C + C = [0, 2]$. Deduce $m_*(C + C) = 2$ although $m_*(C) = 0$, foreshadowing the Brunn–Minkowski discussion.
6. (Non-example.) A student claims “ $m_*(\bigcup_j E_j) = \sum_j m_*(E_j)$ always.” Refute this with overlapping intervals $E_j = [0, 1]$ for all j , and state the correct inequality from Theorem 24.1.24.1.10.
7. (Diagram.) Sketch a covering of the triangle with vertices $(0, 0), (1, 0), (0, 1)$ by cubes of side $1/4$, count the cubes meeting the triangle, and use the count to give an explicit upper bound for m_* of the triangle. Compare with the true area $1/2$.
8. Fix $0 < \xi < 1$ and let C_ξ be the constant-dissection Cantor set. Express the remaining length after k stages and conclude $m_*(C_\xi) = 0$ for every ξ .

24.2 Measurable Sets and the Lebesgue Measure

Measurability and the Lebesgue measure

Definition 24.2.1 (Lebesgue measurable set). $E \subseteq \mathbb{R}^d$ is (Lebesgue) measurable if for every $\varepsilon > 0$ there is an open $O \supseteq E$ with $m_*(O \setminus E) \leq \varepsilon$. For measurable E we set its *Lebesgue measure* $m(E) = m_*(E)$.

Compare this with outer regularity (Theorem 24.1.24.1.10(3)), which says $m_*(E)$ is approximated by $m_*(O)$ for any E . Measurability asks the stronger thing: the *difference* $O \setminus E$, not just O itself, can be made small. This single condition tames exactly those sets on which m_* becomes countably additive.

Theorem 24.2.2 (The measurable sets form a σ -algebra). (P1) *Open sets are measurable.* (P2) *If $m_*(E) = 0$ then E is measurable (so subsets of null sets are measurable).* (P3) *Countable unions of measurable sets are measurable.* (P4) *Closed sets are measurable.* (P5) *Complements of measurable sets are measurable.* (P6) *Countable intersections of measurable sets are measurable.*

Proof. (P1) Take $O = E$. (P2) By outer regularity pick open $O \supseteq E$ with $m_*(O) \leq \varepsilon$; then $m_*(O \setminus E) \leq m_*(O) \leq \varepsilon$. (P3) Given $E = \bigcup_j E_j$, choose open $O_j \supseteq E_j$ with $m_*(O_j \setminus E_j) \leq \varepsilon 2^{-j}$; then $O = \bigcup_j O_j$ is open, $E \subseteq O$, and $O \setminus E \subseteq \bigcup_j (O_j \setminus E_j)$, so $m_*(O \setminus E) \leq \varepsilon$. (P4) It suffices to treat compact F (write a closed set as $\bigcup_k F \cap B_k$ and use (P3)). For compact F , pick open $O \supseteq F$ with $m_*(O) \leq m_*(F) + \varepsilon$; then $O \setminus F$ is open, hence a countable almost disjoint union of cubes $\bigcup_j Q_j$. For each N , $K = \bigcup_{j \leq N} Q_j$ is compact and disjoint from F , so $\text{dist}(K, F) > 0$ (Lemma 24.2.3); by properties (1),(4),(5) of m_* , $m_*(O) \geq m_*(F) + \sum_{j \leq N} |Q_j|$, whence $\sum_{j \leq N} |Q_j| \leq \varepsilon$ and $m_*(O \setminus F) \leq \varepsilon$. (P5) For each n choose open $O_n \supseteq E$ with $m_*(O_n \setminus E) \leq 1/n$; then O_n^c is closed, hence measurable, and $S = \bigcup_n O_n^c \subseteq E^c$ is measurable by (P3). Since $E^c \setminus S \subseteq O_n \setminus E$, $m_*(E^c \setminus S) \leq 1/n$ for all n , so $E^c \setminus S$ is null hence measurable, and $E^c = S \cup (E^c \setminus S)$ is measurable. (P6) De Morgan: $\bigcap_j E_j = (\bigcup_j E_j^c)^c$. $\square$

Lemma 24.2.3 (Closed and compact disjoint sets are separated). *If F is closed, K compact, and $F \cap K = \emptyset$, then $\text{dist}(F, K) > 0$.*

Proof. For each $x \in K$ there is $\delta_x > 0$ with $\text{dist}(x, F) > 3\delta_x$. The balls $B_{2\delta_x}(x)$ cover K ; extract a finite subcover with radii $2\delta_1, \dots, 2\delta_N$ and let $\delta = \min_j \delta_j$. If $x \in K$, $y \in F$, then $|x - x_j| \leq 2\delta_j$ for some j and $|y - x_j| \geq 3\delta_j$, so $|y - x| \geq 3\delta_j - 2\delta_j \geq \delta > 0$. $\square$

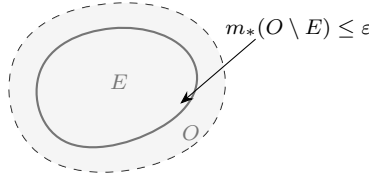


Figure 24.7. Measurability: an open $O \supseteq E$ hugs E so tightly that the shaded collar $O \setminus E$ has small exterior measure. (Schematic.)

Theorem 24.2.4 (Countable additivity). *If $E_1, E_2, \dots$ are disjoint measurable sets and $E = \bigcup_j E_j$, then $m(E) = \sum_{j=1}^{\infty} m(E_j)$.*

Proof. Assume first each E_j bounded. Applying measurability to E_j^c , choose closed $F_j \subseteq E_j$ with $m_*(E_j \setminus F_j) \leq \varepsilon 2^{-j}$. For fixed N the sets $F_1, \dots, F_N$ are compact and disjoint, so by Lemma 24.2.3 and Theorem 24.1.24.1.10(4), $m(\bigcup_{j \leq N} F_j) = \sum_{j \leq N} m(F_j)$. Since $\bigcup_{j \leq N} F_j \subseteq E$, monotonicity gives $m(E) \geq \sum_{j \leq N} m(F_j) \geq \sum_{j \leq N} m(E_j) - \varepsilon$. Let $N \rightarrow \infty$, $\varepsilon \rightarrow 0$: $m(E) \geq \sum_j m(E_j)$. The reverse is sub-additivity. For general E_j , slice by an exhausting sequence of cubes $Q_k \nearrow \mathbb{R}^d$, set $S_1 = Q_1$, $S_k = Q_k \setminus Q_{k-1}$, and apply the bounded case to $E_{j,k} = E_j \cap S_k$. $\square$

Corollary 24.2.5 (Continuity from below/above). *For measurable sets: (i) if $E_k \nearrow E$ then $m(E) = \lim_N m(E_N)$; (ii) if $E_k \searrow E$ and $m(E_k) < \infty$ for some k , then $m(E) = \lim_N m(E_N)$.*

Proof. (i) Set $G_1 = E_1$, $G_k = E_k \setminus E_{k-1}$; the G_k are disjoint with union E and $\bigcup_{k \leq N} G_k = E_N$, so countable additivity gives $m(E) = \sum_k m(G_k) = \lim_N m(E_N)$. (ii) WLOG $m(E_1) < \infty$. With $G_k = E_k \setminus E_{k+1}$, $E_1 = E \cup \bigcup_k G_k$ disjointly, so $m(E_1) = m(E) + \lim_N m(E_N)$; subtract the finite $m(E_1)$. $\square$

Remark. The finiteness hypothesis in (ii) is essential: $E_n = (n, \infty) \subset \mathbb{R}$ has $E_n \searrow \emptyset$ yet $m(E_n) = \infty$ for all n , so $\lim m(E_n) = \infty \neq 0 = m(\emptyset)$.

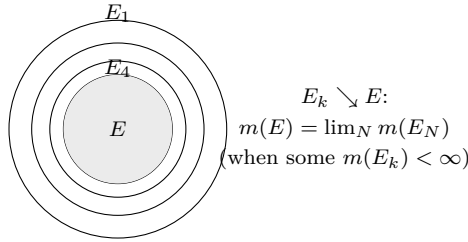


Figure 24.8. Continuity from above: nested measurable sets shrinking to E have measures converging to $m(E)$ (Corollary 24.2.5). (Schematic.)

Theorem 24.2.6 (Regularity / approximation). *Let E be measurable. For every $\varepsilon > 0$: (i) there is open $O \supseteq E$ with $m(O \setminus E) \leq \varepsilon$; (ii) there is closed $F \subseteq E$ with $m(E \setminus F) \leq \varepsilon$; (iii) if $m(E) < \infty$ there is compact $K \subseteq E$ with $m(E \setminus K) \leq \varepsilon$; (iv) if $m(E) < \infty$ there is a finite union $F = \bigcup_{j=1}^N Q_j$ of closed cubes with $m(E \triangle F) \leq \varepsilon$.*

Proof. (i) is the definition. (ii) Apply (i) to E^c : open $O \supseteq E^c$ with $m(O \setminus E^c) \leq \varepsilon$; then $F = O^c$ is closed, $F \subseteq E$, and $E \setminus F = O \setminus E^c$. (iii) Take F as in (ii) with $m(E \setminus F) \leq \varepsilon/2$ and $K_n = F \cap B_n$; then $E \setminus K_n \searrow E \setminus F$, and finiteness gives $m(E \setminus K_n) \leq \varepsilon$ for large n . (iv) Cover E by cubes with $\sum_j |Q_j| \leq m(E) + \varepsilon/2$; pick N with $\sum_{j > N} |Q_j| < \varepsilon/2$ and $F = \bigcup_{j \leq N} Q_j$; then $m(E \triangle F) \leq \sum_{j > N} |Q_j| + (\sum_j |Q_j| - m(E)) \leq \varepsilon$. $\square$

Theorem 24.2.7 (Invariance). *Lebesgue measure is translation invariant ($m(E + h) = m(E)$), dilation relative ($m(\delta E) = \delta^d m(E)$ for $\delta > 0$), and reflection invariant ($m(-E) = m(E)$); in each case the transformed set is measurable when E is.*

Proof. For a cube, $|Q + h| = |Q|$, $|\delta Q| = \delta^d |Q|$, $|-Q| = |Q|$, so the corresponding identities for m_* follow from Definition 24.1.24.1.7 by transforming each cover. Measurability is preserved because if $O \supseteq E$ is open with $m_*(O \setminus E) < \varepsilon$, then $O + h$, δO , $-O$ are open, contain the transformed set, and the collar transforms with the stated factor. $\square$

Definition 24.2.8 (σ -algebra and Borel sets). A σ -algebra is a collection of subsets of $\mathbb{R}^d$ closed under complements and countable unions (hence countable intersections). The Borel σ -algebra $\mathcal{B}_{\mathbb{R}^d}$ is the smallest σ -algebra containing all open sets; its members are *Borel sets*. Countable intersections of open sets are G_δ sets; countable unions of closed sets are F_σ sets.

Since open sets are measurable and the measurable sets form a σ -algebra, $\mathcal{B}_{\mathbb{R}^d}$ is contained in the Lebesgue σ -algebra. The inclusion is strict: there are Lebesgue sets that are not Borel.

Corollary 24.2.9 (Lebesgue sets vs. G_δ, F_σ). *E is measurable iff E differs from a G_δ by a null set, iff E differs from an F_σ by a null set.*

Proof. G_δ , F_σ , and null sets are measurable, giving one direction. Conversely pick open $O_n \supseteq E$ with $m(O_n \setminus E) \leq 1/n$; then $S = \bigcap_n O_n$ is a G_δ with $E \subseteq S$ and $m(S \setminus E) \leq 1/n$ for all n , so $S \setminus E$ is null. The F_σ statement applies Theorem 24.2.6(ii) with $\varepsilon = 1/n$ and unions the closed sets. $\square$

A non-measurable set

Theorem 24.2.10 (Vitali). *There is a non-measurable subset $N \subseteq [0, 1]$.*

Proof. On $[0, 1]$ declare $x \sim y$ iff $x - y \in \mathbb{Q}$; this is an equivalence relation, partitioning $[0, 1]$ into classes E_α . By the axiom of choice pick one $x_\alpha \in E_\alpha$ for each α and set $N = \{x_\alpha\}$. Let $\{r_k\}$ enumerate $\mathbb{Q} \cap [-1, 1]$ and put $N_k = N + r_k$. The N_k are disjoint (if $x_\alpha + r_k = x_\beta + r_{k'}$ then $x_\alpha - x_\beta \in \mathbb{Q}$, forcing $\alpha = \beta$ and $k = k'$), and $[0, 1] \subseteq \bigcup_k N_k \subseteq [-1, 2]$. If N were measurable, all N_k would be, with $m(N_k) = m(N)$ by translation invariance; countable additivity then gives $1 \leq \sum_k m(N_k) \leq 3$, impossible whether $m(N) = 0$ (sum = 0) or $m(N) > 0$ (sum = ∞). $\square$

Caution. The non-measurable set N shows exterior measure is genuinely *not* additive on all disjoint sets: one can show $m_*(N) + m_*([0, 1] \setminus N) \neq m_*([0, 1]) = 1$. Measurability is exactly the hypothesis that restores additivity. The axiom of choice is unavoidable here – it is consistent with ZF that every set is measurable.

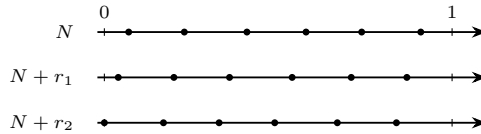


Figure 24.9. The Vitali set N and its rational translates $N + r_k$ (taken mod 1): disjoint, together covering $[0, 1]$, all the same “size” – so no countably additive measure can assign them a consistent value. (Schematic.)

Worked Example 24.2.11 (Measure of an interval and of $\mathbb{Q}$). Compute $m([a, b])$ and $m(\mathbb{Q})$.

Recognition. Intervals are rectangles in $\mathbb{R}$; $\mathbb{Q}$ is countable.

Solution. $m([a, b]) = |[a, b]| = b - a$ by Theorem 24.1.24.1.9. Since $\mathbb{Q} = \bigcup_q \{q\}$ is a countable union of points, each of measure 0, countable additivity gives $m(\mathbb{Q}) = 0$. *Check.* Thus $m([0, 1]) = m([0, 1] \setminus \mathbb{Q}) = 1$ – the irrationals carry all the measure of the unit interval.

Worked Example 24.2.12 (A positive-measure “fat Cantor” set). At stage k remove from each of the 2^{k-1} intervals a central open interval of length 4^{-k} . Compute the measure of the resulting set $\widehat{C}$.

Recognition. Trigger: track total removed length via a geometric series, then subtract from 1.

Solution. At stage k one removes 2^{k-1} intervals each of length 4^{-k} , total $2^{k-1}4^{-k} = \frac{1}{2}(1/2)^k$. Summing, removed length = $\sum_{k \geq 1} \frac{1}{2}2^{-k} = \frac{1}{2}$. Hence $m(\widehat{C}) = 1 - \frac{1}{2} = \frac{1}{2} > 0$. *Check.* $\widehat{C}$ is closed, nowhere dense, and totally disconnected like the usual Cantor set, yet has positive measure – “small” topologically, “large” in measure.

Worked Example 24.2.13 (Continuity from below). Let $E_n = [0, 1 - 1/n]$. Verify Corollary 24.2.5(i).

Recognition. Trigger: $E_n \nearrow [0, 1]$; take the limit of measures.

Solution. $m(E_n) = 1 - 1/n \rightarrow 1$, and $\bigcup_n E_n = [0, 1]$ has $m = 1$. The limit matches. *Check.* $[0, 1]$ and $[0, 1]$ differ by the null set $\{1\}$, so both have measure 1.

Worked Example 24.2.14 (Outer measure additivity fails: a concrete split). Using the Vitali set $N \subset [0, 1]$, explain why $m_*(N) + m_*(N^c \cap [0, 1]) > m_*([0, 1])$.

Recognition. Trigger: a disjoint decomposition of $[0, 1]$ into two non-measurable pieces.

Solution. One shows $m_*(N^c \cap [0, 1]) = 1$ (any measurable set inside N is null, so N contains no positive-measure measurable set, forcing the complement to fill up the interval in outer measure). Since also $m_*(N) > 0$, we get $m_*(N) + m_*(N^c \cap [0, 1]) > 1 = m_*([0, 1])$. *Check.* Strict super-additivity of m_* on a disjoint pair is impossible for measurable sets, where Theorem 24.2.4 forces equality.

Worked Example 24.2.15 (Borel-Cantelli: a measure-zero limsup). If $\sum_k m(E_k) < \infty$, show $E = \limsup_k E_k = \bigcap_n \bigcup_{k \geq n} E_k$ has $m(E) = 0$.

Recognition. Trigger: “infinitely often” set; bound by tails of a convergent series.

Solution. $E \subseteq \bigcup_{k \geq n} E_k$ for every n , so by sub-additivity $m(E) \leq \sum_{k \geq n} m(E_k) \rightarrow 0$ as $n \rightarrow \infty$. Hence $m(E) = 0$. *Check.* E is the set of points lying in infinitely many E_k ; the convergent-sum hypothesis makes that set negligible, a workhorse of probability.

Worked Example 24.2.16 (Inclusion-exclusion and a G_δ approximation). For measurable A, B of finite measure verify $m(A \cup B) = m(A) + m(B) - m(A \cap B)$, then realize $A = [0, 1] \cap \mathbb{Q}$ as a Borel set differing from a G_δ by a null set.

Recognition. Trigger: split a union into disjoint pieces for additivity; then invoke Corollary 24.2.9.

Solution. Write $A \cup B = (A \setminus B) \sqcup (A \cap B) \sqcup (B \setminus A)$ disjointly; additivity gives $m(A \cup B) = m(A \setminus B) + m(A \cap B) + m(B \setminus A)$, and adding/subtracting $m(A \cap B)$ yields the identity. For $A = [0, 1] \cap \mathbb{Q} = \bigcup_j \{q_j\}$: each singleton is closed, so A is an F_σ ; it is null, so A differs from the $G_\delta = \emptyset$ by the null set A itself. *Check.* $m(A \cup B) + m(A \cap B) = m(A) + m(B)$, the symmetric form; with $A = [0, 1]$, $B = [0, 2]$ it reads $2 + 1 = 1 + 2$.

Exercises (Stein, Ch.1) – measurable sets, Lebesgue measure.

- Boundary of a closure with positive measure. Give an example of an open set O such that the boundary of $\overline{O}$ has positive Lebesgue measure. [Hint: union of the open intervals deleted at the odd steps of a Cantor-like set.]
- If $\delta = (\delta_1, \dots, \delta_d)$ has $\delta_i > 0$ and $\delta E = \{(\delta_1 x_1, \dots, \delta_d x_d) : (x_1, \dots, x_d) \in E\}$, prove δE is measurable whenever E is, and $m(\delta E) = \delta_1 \cdots \delta_d m(E)$.
- Let L be a linear transformation of $\mathbb{R}^d$. Show $L(E)$ is measurable whenever E is: (a) if E compact so is $L(E)$, hence L maps F_σ to F_σ ; (b) since $|L(x) - L(x')| \leq M|x - x'|$, L maps any cube of side ℓ into a cube of side $c_d M \ell$ with $c_d = 2\sqrt{d}$, so $m(E) = 0$ implies $m(L(E)) = 0$; conclude via Corollary 24.2.24.2.9. (One can show $m(L(E)) = |\det L| m(E)$.)
- The G_δ and F_σ sets. (a) Show a closed set is a G_δ and an open set is an F_σ . [Hint: $O_n = \{x : \text{dist}(x, F) < 1/n\}$.] (b) Give an F_σ that is not a G_δ . [Hint: a countable dense set.] (c) Give a Borel set that is neither a G_δ nor an F_σ .
- Borel-Cantelli lemma. If $\{E_k\}$ are measurable with $\sum_k m(E_k) < \infty$ and $E = \{x : x \in E_k \text{ for infinitely many } k\} = \limsup_k E_k$, show (a) E is measurable, (b) $m(E) = 0$. [Hint: $E = \bigcap_n \bigcup_{k \geq n} E_k$.]
- Observations on $A + B$. (a) If A or B is open, $A + B$ is open. (b) If A, B are closed, $A + B$ is measurable. (c) Show $A + B$ need not be closed even when A, B are. [Hint for (b): $A + B$ is an F_σ .]
- Show there exist closed A, B with $m(A) = m(B) = 0$ but $m(A + B) > 0$: (a) in $\mathbb{R}$, take $A = C$, $B = C/2$ and note $A + B \supseteq [0, 1]$; (b) in $\mathbb{R}^2$, $A = I \times \{0\}$, $B = \{0\} \times I$ with $I = [0, 1]$ give $A + B = I \times I$.
- An alternative definition: E is measurable iff for every $\varepsilon > 0$ there is a closed $F \subseteq E$ with $m_*(E \setminus F) < \varepsilon$. Show this is equivalent to the definition in the text.
- Suppose $A \subseteq E \subseteq B$ with A, B measurable of finite measure. Prove that if $m(A) = m(B)$ then E is measurable.
- Let $E_1 \subseteq E_2$ be compact in $\mathbb{R}^d$ with $a = m(E_1)$, $b = m(E_2)$. Prove that for any $a < c < b$ there is a compact E with $E_1 \subseteq E \subseteq E_2$ and $m(E) = c$. [Hint: $t \mapsto m(E \cap [0, t])$ is continuous.]
- Let $E \subseteq \mathbb{R}$ with $m_*(E) > 0$. Prove that for each $0 < \alpha < 1$ there is an open interval I with $m_*(E \cap I) \geq \alpha m_*(I)$ – E contains “almost a whole interval.” [Hint: choose open $O \supseteq E$ with $m_*(E) \geq \alpha m_*(O)$; decompose O into disjoint intervals.]
- Suppose E is measurable with $m(E) > 0$. Prove the difference set $\{x - y : x, y \in E\}$ contains an open interval about the origin. [Hint: Exercise 11 above, plus an overlap argument.]

13. If E, F are measurable with $m(E), m(F) > 0$, prove $E + F$ contains an interval.
14. Non-measurability via difference sets. With $x \sim y$ iff $x - y \in \mathbb{Q}$, let N^* contain one element of each class. Prove N^* is non-measurable using the difference-set result. [Hint: translates $N^* + r_n$; can the difference set contain an interval about 0?]
15. Let $N \subseteq I = [0, 1]$ be the Vitali set. (a) Prove that every measurable $E \subseteq N$ has $m(E) = 0$. (b) If $G \subseteq \mathbb{R}$ has $m_*(G) > 0$, prove some subset of G is non-measurable. [Hint: rational translates of E .]
16. With N as above, show $m_*(N^c \cap I) = 1$, and conclude that for $E_1 = N$, $E_2 = N^c \cap I$ (disjoint) one has $m_*(E_1) + m_*(E_2) \neq m_*(E_1 \cup E_2)$. [Hint: argue by contradiction with a measurable U , $N^c \subseteq U \subseteq I$, $m_*(U) < 1 - \varepsilon$.]
17. Let C_1, C_2 be Cantor sets (Exercise 24.1.3). Show there is $F : [0, 1] \rightarrow [0, 1]$ that is (i) continuous and bijective, (ii) monotonically increasing, (iii) maps C_1 surjectively onto C_2 . [Hint: copy the Cantor–Lebesgue construction.]

Additional exercises.

1. Show that the measurable sets containing a fixed null set, modulo null sets, form a σ -algebra. More concretely: if E is measurable and Z is null, prove $E \cup Z$, $E \setminus Z$, $E \Delta Z$ are measurable with the same measure as E .
2. Prove that every interval (open, closed, half-open) in $\mathbb{R}$ is measurable with $m = \text{length}$, directly from Definition 24.2.24.2.1.
3. Let $E_n = [0, 1] \cap \{x : \text{the } n\text{th binary digit of } x \text{ is } 1\}$. Compute $m(E_n)$ and $m(\bigcap_{n \leq N} E_n)$, and interpret probabilistically.
4. (Continuity from above, sharpness.) Give measurable $E_n \searrow E$ with $m(E_n) = \infty$ for all n but $m(E) = 0$, and explain which hypothesis of Corollary 24.2.24.2.5 fails.
5. Using Theorem 24.2.24.2.6(iv), approximate $E = [0, 1] \setminus \mathbb{Q}$ within ε (in symmetric difference) by a finite union of closed intervals, and exhibit such a union explicitly for $\varepsilon = 1/10$.
6. Prove that a measurable set E with $m(E) > 0$ has a point of density: there is x with $m(E \cap B_r(x))/m(B_r(x)) \rightarrow 1$ as $r \rightarrow 0$. (You may cite the Lebesgue density theorem; sketch why it forces E to be “almost an interval” near such x .)
7. (Non-example.) Refute the claim “an uncountable union of measurable sets is measurable” by writing the Vitali set N as an uncountable union of singletons.
8. (Diagram.) Draw nested sets $E_1 \supset E_2 \supset \dots$ as concentric annuli illustrating continuity from above, and mark where the finite-measure hypothesis is used.
9. Prove that if A, B are measurable then $m(A \cup B) + m(A \cap B) = m(A) + m(B)$ (inclusion–exclusion for measure), assuming all measures finite.
10. Show that the dilation factor in Theorem 24.2.24.2.7 is exactly δ^d by computing $m(\delta Q)$ for a cube Q and passing to general E via outer regularity.

24.3 Measurable Functions and Littlewood’s Three Principles

Measurable functions

Definition 24.3.1 (Characteristic, step, and simple functions). The *characteristic function* of E is $\mathbf{1}_E(x) = 1$ if $x \in E$ and 0 otherwise. A *step function* is a finite sum $\sum_k a_k \mathbf{1}_{R_k}$ with R_k rectangles; a *simple function* is a finite sum $\sum_k a_k \mathbf{1}_{E_k}$ with E_k measurable of finite measure and a_k constants.

Step functions build the Riemann integral; simple functions, where the building blocks may be arbitrary measurable sets, build the Lebesgue integral. The extra flexibility – slicing the *range* rather than the *domain* – is exactly what makes the Lebesgue theory work.

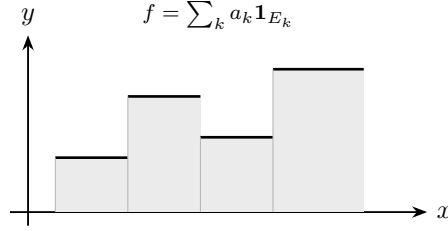


Figure 24.10. A simple function as a finite sum of indicator “blocks” over measurable sets. (Schematic.)

Definition 24.3.2 (Measurable function). An extended-real-valued f on a measurable $E \subseteq \mathbb{R}^d$ is *measurable* if for every $a \in \mathbb{R}$ the set $\{x \in E : f(x) < a\} = \{f < a\}$ is measurable.

Proposition 24.3.3 (Equivalent formulations). f is measurable iff $\{f \leq a\}$ is measurable for all a , iff $\{f \geq a\}$ is, iff $\{f > a\}$ is. For finite-valued f , measurability is equivalent to $\{a < f < b\}$ measurable for all a, b , and (Property 1) to $f^{-1}(O)$ measurable for every open O (and $f^{-1}(F)$ for every closed F).

Proof. $\{f \leq a\} = \bigcap_k \{f < a + 1/k\}$ and $\{f < a\} = \bigcup_k \{f \leq a - 1/k\}$ convert between $<$ and $\leq$ using closure under countable operations; $\{f \geq a\}$ and $\{f > a\}$ are complements of these. For open $O = \bigcup_k (a_k, b_k)$, $f^{-1}(O) = \bigcup_k \{a_k < f < b_k\}$ is measurable; closed sets follow by complementation. $\square$



Figure 24.11. Left: measurability via preimages – $f^{-1}(O)$ is the pullback of the open set $O \hookrightarrow \mathbb{R}$ along f , and it is required to be measurable for every open O . Right: a continuous Φ post-composed with measurable f ; since $(\Phi \circ f)^{-1}(O) = f^{-1}(\Phi^{-1}(O))$ and $\Phi^{-1}(O)$ is open, $\Phi \circ f$ is measurable.

Theorem 24.3.4 (Closure properties). (P2) Continuous f is measurable; if f is measurable finite-valued and Φ continuous then $\Phi \circ f$ is measurable. (P3) For measurable $\{f_n\}$, $\sup_n f_n$, $\inf_n f_n$, $\limsup_n f_n$, $\liminf_n f_n$ are measurable. (P4) A pointwise (or a.e.) limit of measurable functions is measurable. (P5) If f, g are measurable then f^k ($k \geq 1$) is, and $f + g$, fg are when f, g are finite-valued. (P6) If f is measurable and $f = g$ a.e. then g is measurable.

Proof. (P2) For continuous f , $\{f < a\} = f^{-1}((-\infty, a))$ is open, hence measurable; and $(\Phi \circ f)^{-1}((-\infty, a)) = f^{-1}(\Phi^{-1}((-\infty, a))) = f^{-1}(O)$ is measurable. (P3) $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$; $\inf_n f_n = -\sup_n (-f_n)$; $\limsup_n f_n = \inf_k \sup_{n \geq k} f_n$ and $\liminf_n f_n = \sup_k \inf_{n \geq k} f_n$. (P4) When $\lim_n f_n = f$, $f = \limsup_n f_n$, covered by (P3). (P5) If k is odd $\{f^k > a\} = \{f > a^{1/k}\}$; if k even and $a \geq 0$, $\{f^k > a\} = \{f > a^{1/k}\} \cup \{f < -a^{1/k}\}$. For sums, $\{f + g > a\} = \bigcup_{r \in \mathbb{Q}} \{f > a - r\} \cap \{g > r\}$; products via $fg = \frac{1}{4}[(f + g)^2 - (f - g)^2]$. (P6) $\{f < a\}$ and $\{g < a\}$ differ by a null set. $\square$

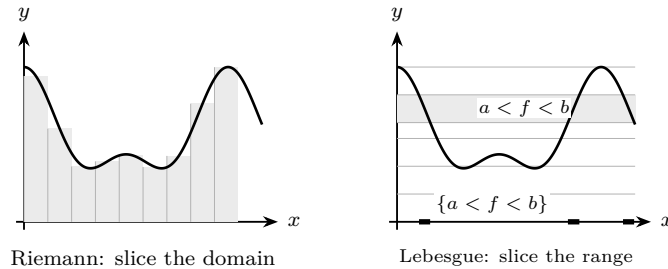


Figure 24.12. Riemann integration partitions the x -axis (strips under the graph); Lebesgue integration partitions the y -axis and measures the preimage sets $\{a < f < b\}$ (heavy segments on the x -axis). (Schematic.)

Theorem 24.3.5 (Approximation by simple functions). If $f \geq 0$ is measurable on $\mathbb{R}^d$ there is an increasing sequence of nonnegative simple functions $\varphi_k \nearrow f$ pointwise. For general measurable f there are simple φ_k with $|\varphi_k| \leq |\varphi_{k+1}|$ and $\varphi_k \rightarrow f$ pointwise.

Proof. Truncate: $F_N = f$ where $f \leq N$ on the cube Q_N , $= N$ where $f > N$ on Q_N , $= 0$ off Q_N ; then $F_N \rightarrow f$. Partition the range $[0, N]$ into bands $E_{\ell, M} = \{x \in Q_N : \ell/M < F_N \leq (\ell+1)/M\}$ and set $F_{N, M} = \sum_{\ell} (\ell/M) \mathbf{1}_{E_{\ell, M}}$, a simple function with $0 \leq F_N - F_{N, M} \leq 1/M$. Taking $N = M = 2^k$ and $\varphi_k = F_{2^k, 2^k}$ gives an increasing sequence converging to f . For signed f , decompose $f = f^+ - f^-$ with $f^{\pm} = \max(\pm f, 0)$, approximate each by the nonnegative case, and subtract. $\square$

Theorem 24.3.6 (Approximation by step functions). *If f is measurable on $\mathbb{R}^d$ there are step functions $\psi_k \rightarrow f$ a.e.*

Proof. By Theorem 24.3.5 reduce to $f = \mathbf{1}_E$, $m(E) < \infty$. By Theorem 24.2.24.2.6(iv) pick cubes with $m(E \triangle \bigcup_{j \leq N} Q_j) \leq \varepsilon$; refine into almost disjoint rectangles $\tilde{R}_j$ and shrink slightly to disjoint R_j with $m(E \triangle \bigcup_j R_j) \leq 2\varepsilon$. Then $\mathbf{1}_E = \sum_j \mathbf{1}_{R_j}$ off a set of measure $\leq 2\varepsilon$. Choosing $\varepsilon = 2^{-k}$ gives step functions ψ_k with $m\{\psi_k \neq f\} \leq 2^{-k}$; by Borel-Cantelli the exceptional set has measure 0, so $\psi_k \rightarrow f$ a.e. $\square$

Littlewood's three principles

Remark (Littlewood's principles). (i) Every (finite-measure) measurable set is nearly a finite union of intervals (Theorem 24.2.24.2.6(iv)). (ii) Every measurable function is nearly continuous (Lusin). (iii) Every a.e.-convergent sequence is nearly uniformly convergent (Egorov).

Theorem 24.3.7 (Egorov). *If $f_k \rightarrow f$ a.e. on E with $m(E) < \infty$, then for every $\varepsilon > 0$ there is a closed $A_\varepsilon \subseteq E$ with $m(E \setminus A_\varepsilon) \leq \varepsilon$ and $f_k \rightarrow f$ uniformly on A_ε .*

Proof. Assume $f_k(x) \rightarrow f(x)$ for all $x \in E$. For integers n, k let $E_k^n = \{x \in E : |f_j(x) - f(x)| < 1/n \ \forall j > k\}$. For fixed n , $E_k^n \nearrow E$, so by continuity from below pick k_n with $m(E \setminus E_{k_n}^n) < 2^{-n}$. Choose N with $\sum_{n \geq N} 2^{-n} < \varepsilon/2$ and set $\tilde{A}_\varepsilon = \bigcap_{n \geq N} E_{k_n}^n$; then $m(E \setminus \tilde{A}_\varepsilon) < \varepsilon/2$. Given $\delta > 0$ choose $n \geq N$ with $1/n < \delta$: for $x \in \tilde{A}_\varepsilon$ and $j > k_n$, $|f_j(x) - f(x)| < 1/n < \delta$, so convergence is uniform on $\tilde{A}_\varepsilon$. Finally shrink to a closed $A_\varepsilon \subseteq \tilde{A}_\varepsilon$ with $m(\tilde{A}_\varepsilon \setminus A_\varepsilon) < \varepsilon/2$. $\square$

Theorem 24.3.8 (Lusin). *If f is measurable and finite-valued on E with $m(E) < \infty$, then for every $\varepsilon > 0$ there is a closed $F_\varepsilon \subseteq E$ with $m(E \setminus F_\varepsilon) \leq \varepsilon$ and $f|_{F_\varepsilon}$ continuous.*

Proof. Take step functions $f_n \rightarrow f$ a.e.; each f_n is continuous off a set E_n with $m(E_n) < 2^{-n}$. By Egorov find $A_{\varepsilon/3}$ with $f_n \rightarrow f$ uniformly and $m(E \setminus A_{\varepsilon/3}) \leq \varepsilon/3$. Put $F' = A_{\varepsilon/3} \setminus \bigcup_{n \geq N} E_n$ with N so large that $\sum_{n \geq N} 2^{-n} < \varepsilon/3$; on F' each f_n is continuous and $f_n \rightarrow f$ uniformly, so $f|_{F'}$ is continuous. Approximate F' from inside by a closed F_ε with $m(F' \setminus F_\varepsilon) < \varepsilon/3$. $\square$

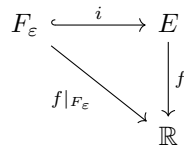


Figure 24.13. Lusin: on a closed $F_\varepsilon \subseteq E$ with $m(E \setminus F_\varepsilon) \leq \varepsilon$ the restriction $f|_{F_\varepsilon} = f \circ i$ along the inclusion $i : F_\varepsilon \hookrightarrow E$ is continuous, even when f is continuous nowhere on E .

Caution. Lusin says f restricted to F_ε is continuous – *not* that f is continuous at the points of F_ε as a function on all of E . The Dirichlet function $\mathbf{1}_\mathbb{Q}$ is nowhere continuous on $\mathbb{R}$, yet its restriction to the (full-measure) irrationals is identically 0, hence continuous there.

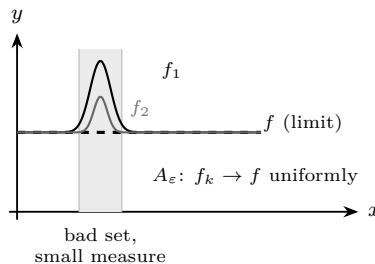


Figure 24.14. Egorov: deleting a small-measure neighborhood of the slow points leaves a closed set A_ε on which the convergence $f_k \rightarrow f$ is uniform. (Schematic.)

Worked Example 24.3.9 ($\mathbf{1}_E$ is measurable iff E is measurable). Show $\mathbf{1}_E$ is a measurable function exactly when E is a measurable set.

Recognition. Trigger: compute the sublevel sets of an indicator.

Solution. $\{\mathbf{1}_E < a\}$ is $\emptyset$ if $a \leq 0$, E^c if $0 < a \leq 1$, and $\mathbb{R}^d$ if $a > 1$. All are measurable iff E^c (equivalently E) is measurable. *Check.* This is why the Vitali set N gives a non-measurable function $\mathbf{1}_N$ – non-measurable sets and non-measurable functions go hand in hand.

Worked Example 24.3.10 (Dirichlet function via simple-function limit). Show $\mathbf{1}_\mathbb{Q}$ on $[0, 1]$ is measurable and equals a pointwise limit of step functions.

Recognition. Trigger: enumerate $\mathbb{Q} \cap [0, 1]$ and build partial indicators.

Solution. Let $\mathbb{Q} \cap [0, 1] = \{q_1, q_2, \dots\}$ and $\psi_n = \sum_{j=1}^n \mathbf{1}_{\{q_j\}}$ (a finite sum of point indicators, hence a degenerate step function). Then $\psi_n \nearrow \mathbf{1}_\mathbb{Q}$ pointwise, so $\mathbf{1}_\mathbb{Q}$ is measurable by (P4). *Check.* $m\{\mathbf{1}_\mathbb{Q} \neq 0\} = m(\mathbb{Q}) = 0$, so $\mathbf{1}_\mathbb{Q} = 0$ a.e. – consistent with its (eventual) Lebesgue integral being 0.

Worked Example 24.3.11 (Cantor function (devil's staircase)). Define F on $[0, 1]$ by $F(\sum a_k 3^{-k}) = \sum (a_k/2) 2^{-k}$ for $a_k \in \{0, 2\}$, then extend to be constant on each deleted middle third. Show F is continuous, increasing, $F(0) = 0$, $F(1) = 1$, yet $F' = 0$ a.e.

Recognition. Trigger: a monotone continuous surjection $[0, 1] \rightarrow [0, 1]$ flat off the (null) Cantor set.

Solution. Two ternary expansions of the same $x \in C$ give the same binary value, so F is well defined on C ; it is monotone there and extends to a monotone continuous F on $[0, 1]$ (the gaps are filled by constants, matching endpoint values). $F(0) = 0$, $F(1) = 1$. On each deleted interval F is constant so $F' = 0$; the union of deleted intervals has measure 1, hence $F' = 0$ a.e. *Check.* Although $F' = 0$ a.e., F climbs from 0 to 1 – all the increase happens on the null set C , so $\int_0^1 F' = 0 \neq 1 = F(1) - F(0)$, showing F is *not* absolutely continuous.

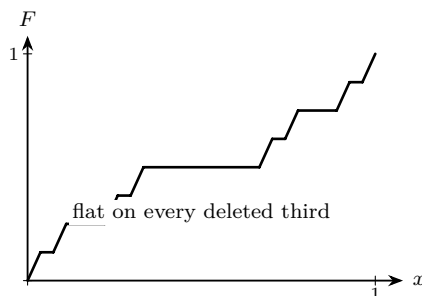


Figure 24.15. The Cantor–Lebesgue function (stage 3): continuous and increasing with $F' = 0$ a.e., yet rising from 0 to 1 on the measure-zero Cantor set.

Worked Example 24.3.12 (sup of measurable functions). Let $f_n(x) = \min(n, 1/|x|)$ on $\mathbb{R} \setminus \{0\}$. Show $\sup_n f_n$ is measurable.

Recognition. Trigger: apply (P3); identify the supremum.

Solution. Each f_n is continuous, hence measurable, and $\sup_n f_n(x) = 1/|x|$ (finite for $x \neq 0$). By (P3) the supremum is measurable. *Check.* $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$; for $a > 0$ this is $\{0 < |x| < 1/a\}$, an open set, confirming measurability directly.

Worked Example 24.3.13 (Egorov is sharp: convergence need not be uniform on all of E). On $E = [0, 1]$ let $f_n = \mathbf{1}_{[0, 1/n]}$. Show $f_n \rightarrow 0$ a.e. but not uniformly, while Egorov holds.

Recognition. Trigger: a moving-bump sequence; uniform convergence fails only near 0.

Solution. For $x > 0$, $f_n(x) = 0$ once $n > 1/x$, so $f_n \rightarrow 0$ pointwise (and a.e.). It is not uniform: $\sup_{[0, 1]} |f_n| = 1$ for all n . But removing $[0, \varepsilon]$ leaves $A_\varepsilon = [\varepsilon, 1]$, closed with $m(E \setminus A_\varepsilon) = \varepsilon$, and $f_n \equiv 0$ on A_ε for $n > 1/\varepsilon$ – uniform there. *Check.* Exactly Egorov's conclusion; the bad behavior is confined to the small set $[0, \varepsilon]$.

Worked Example 24.3.14 (Simple-function approximation of $f(x) = x$). For $f(x) = x$ on $[0, 1]$ write the standard dyadic simple approximation $\varphi_k \nearrow f$ and compute $\|f - \varphi_k\|_\infty$.

Recognition. Trigger: partition the range $[0, 1]$ into 2^k bands as in Theorem 24.3.5.

Solution. Set $\varphi_k(x) = \ell 2^{-k}$ on $E_\ell = \{x : \ell 2^{-k} \leq x < (\ell + 1)2^{-k}\}$ for $0 \leq \ell < 2^k$. Each E_ℓ is an interval, so φ_k is simple; $\varphi_k \leq \varphi_{k+1}$ and $0 \leq f - \varphi_k \leq 2^{-k}$ pointwise, so $\varphi_k \nearrow f$. Hence $\|f - \varphi_k\|_\infty = 2^{-k}$. *Check.* For $k = 2$ the error is $1/4$; the bands at heights $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ undershoot the line $y = x$ by at most one band width, exactly 2^{-k} .

Exercises (Stein, Ch.1) – measurable functions, Littlewood.

1. A decreasing sequence of positive continuous functions whose pointwise limit is not Riemann integrable. Let $\widehat{C}$ be a Cantor-like set of positive measure (Exercise 24.2 above / Stein 24.1.4). Let F_1 be piecewise-linear continuous on $[0, 1]$, equal to 1 off the first removed interval, 0 at its center, $0 \leq F_1 \leq 1$; similarly define F_2 from the stage-two intervals, etc., and set $f_n = F_1 F_2 \cdots F_n$. Prove: (a) $0 \leq f_n \leq 1$ and $f_n \geq f_{n+1}$, so $f_n \rightarrow f$ pointwise. (b) f is discontinuous at every point of $\widehat{C}$ [Hint: $f = 1$ on $\widehat{C}$, and find $x_n \rightarrow x$ with $f(x_n) = 0$]. Conclude $\int f_n$ decreases yet f is not Riemann integrable (its discontinuity set has positive measure).
2. Let $\{f_n\}$ be measurable on $[0, 1]$ with $|f_n(x)| < \infty$ a.e. Show there exist positive reals c_n with $f_n(x)/c_n \rightarrow 0$ a.e. [Hint: choose c_n with $m(\{|f_n/c_n| > 1/n\}) < 2^{-n}$ and apply Borel–Cantelli.]
3. Prove every measurable function is the a.e. limit of a sequence of continuous functions.
4. Suppose $f(x, y)$ on $\mathbb{R}^2$ is separately continuous (continuous in each variable with the other fixed). Prove f is measurable on $\mathbb{R}^2$. [Hint: approximate in x by piecewise-linear $f_n \rightarrow f$ pointwise.]
5. Does there exist an enumeration $\{r_n\}$ of the rationals such that the complement of $\bigcup_n (r_n - 1/n, r_n + 1/n)$ in $\mathbb{R}$ is non-empty? [Hint: enumerate so that the only rationals outside a fixed bounded interval are r_n with $n = m^2$.]
6. Prove there is a continuous function mapping a Lebesgue measurable set to a non-measurable set. [Hint: a non-measurable subset of $[0, 1]$ and its inverse image in C under the Cantor–Lebesgue F .]
7. Show there is no everywhere-continuous f on $\mathbb{R}$ with $f = \mathbf{1}_{[0,1]}$ a.e.
8. Give an example of a measurable f and continuous Φ with $f \circ \Phi$ non-measurable. [Hint: $\Phi : C_1 \rightarrow C_2$ as in Exercise 24.2.17 with $m(C_1) > 0$, $m(C_2) = 0$; let $N \subseteq C_1$ be non-measurable and $f = \mathbf{1}_{\Phi(N)}$.] Use this to exhibit a Lebesgue measurable set that is not Borel.
9. A measurable f on $[0, 1]$ such that every $g = f$ a.e. is discontinuous everywhere. (a) Construct measurable $E \subseteq [0, 1]$ so that for every nonempty open subinterval I , both $E \cap I$ and $E^c \cap I$ have positive measure. (b) Show $f = \mathbf{1}_E$ has the stated property. [Hint: nest positive-measure Cantor-like sets inside each omitted interval, indefinitely.]

Additional exercises.

1. Decide whether each is measurable, with proof: (a) $f(x) = \mathbf{1}_{\mathbb{Q}}(x) \cdot x$; (b) $g(x) = \sin(1/x)$ for $x \neq 0$, $g(0) = 0$; (c) $h = \mathbf{1}_N$ for the Vitali set N .
2. Let f be measurable and g continuous. Show $g \circ f$ is measurable but $f \circ g$ need not be; relate to Exercise 24.3.8.
3. Show that the set of points where a sequence of measurable functions $\{f_n\}$ converges, $\{x : \lim_n f_n(x) \text{ exists in } \mathbb{R}\}$, is measurable.
4. Construct a sequence $f_n \rightarrow 0$ pointwise on $[0, 1]$ that does not converge uniformly on any set of full measure minus ε for $\varepsilon < 1$... then reconcile this with Egorov by giving the correct A_ε . (Use the moving-bump $f_n = \mathbf{1}_{[(k-1)/2^j, k/2^j]}$ ordered by j then k – note this converges in measure but *not* a.e.; contrast with Egorov’s a.e. hypothesis.)

5. For the Cantor function F , show F maps the (null) Cantor set onto $[0, 1]$, hence a null set onto a full-measure set. Use this to map a non-measurable set into a null set, reproducing the construction behind a Lebesgue-but-not-Borel set.
6. Prove that a monotone function $f : [0, 1] \rightarrow \mathbb{R}$ is measurable (indeed Borel) by computing $\{f < a\}$.
7. (Non-example.) Exhibit measurable f, g that are finite a.e. but for which $f + g$ is not defined everywhere; explain how the a.e. convention in Theorem 24.3.24.3.4(P6) repairs this.
8. (Diagram.) Draw the standard simple-function approximation $\varphi_k \nearrow f$ for $f(x) = x$ on $[0, 1]$ by partitioning the range into 2^k bands, and compute $\|f - \varphi_2\|_\infty$.
9. Prove Lusin's theorem implies the following: for measurable f on $[0, 1]$ and $\varepsilon > 0$ there is a continuous g on $[0, 1]$ with $m\{f \neq g\} \leq \varepsilon$. [Hint: Tietze extend $f|_{F_\varepsilon}$.]
10. Show that if $f_n \rightarrow f$ in measure on a finite-measure set, some subsequence converges a.e.; illustrate with the moving-bump sequence of Exercise 4.

24.4 The Brunn-Minkowski Inequality

Sumsets and the statement

Definition 24.4.1 (Sum of sets). For $A, B \subseteq \mathbb{R}^d$ the *sumset* is $A + B = \{x' + x'' : x' \in A, x'' \in B\}$. For $\lambda > 0$, $\lambda A = \{\lambda x : x \in A\}$.

There is no upper bound for $m(A + B)$ in terms of $m(A), m(B)$: one can have $m(A) = m(B) = 0$ yet $m(A + B) > 0$ (e.g. $A = C$, $B = C/2$ give $A + B \supseteq [0, 1]$). The interesting estimate runs the other way and respects the scaling $m(\lambda A) = \lambda^d m(A)$.

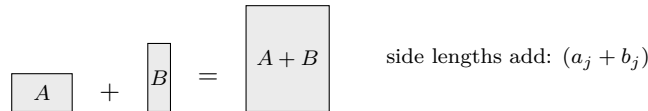


Figure 24.16. Adding two boxes adds their side lengths; the Brunn–Minkowski inequality compares the d -th roots of the volumes. (Schematic.)

Theorem 24.4.2 (Brunn–Minkowski). If A, B and $A + B$ are measurable in $\mathbb{R}^d$, then

$$m(A + B)^{1/d} \geq m(A)^{1/d} + m(B)^{1/d}.$$

Proof. Step 1: rectangles. If A, B are rectangles with side lengths a_j, b_j , the claim is $(\prod_j (a_j + b_j))^{1/d} \geq (\prod_j a_j)^{1/d} + (\prod_j b_j)^{1/d}$. By scaling $a_j, b_j \mapsto \lambda_j a_j, \lambda_j b_j$ (both sides multiply by $(\lambda_1 \cdots \lambda_d)^{1/d}$) reduce to $a_j + b_j = 1$. Then the arithmetic–geometric mean inequality $\frac{1}{d} \sum_j x_j \geq (\prod_j x_j)^{1/d}$ applied to $x_j = a_j$ and $x_j = b_j$ and added gives $1 = \frac{1}{d} \sum_j (a_j + b_j) \geq (\prod_j a_j)^{1/d} + (\prod_j b_j)^{1/d}$, which is the claim.

Step 2: finite unions of almost disjoint rectangles, by induction on the total number n of rectangles. The inequality is translation invariant in A and B separately. Choose two disjoint rectangles in A separated by a coordinate hyperplane $\{x_j = 0\}$ (after translation): set $A_- = A \cap \{x_j \leq 0\}$, $A_+ = A \cap \{x_j \geq 0\}$, each with fewer rectangles than A . Translate B so that $B_\pm = B \cap \{\pm x_j \geq 0\}$ satisfy $m(B_\pm)/m(B) = m(A_\pm)/m(A)$. Since $A + B \supseteq (A_+ + B_+) \cup (A_- + B_-)$ with the two parts in opposite half-spaces (essentially disjoint),

$$m(A + B) \geq m(A_+ + B_+) + m(A_- + B_-) \geq (m(A_+)^{1/d} + m(B_+)^{1/d})^d + (m(A_-)^{1/d} + m(B_-)^{1/d})^d.$$

Substituting the ratio choice $m(B_\pm) = m(A_\pm)m(B)/m(A)$ collapses both brackets to $m(A_\pm)(1 + (m(B)/m(A))^{1/d})^d$, and $m(A_+) + m(A_-) = m(A)$ gives $m(A + B) \geq (m(A)^{1/d} + m(B)^{1/d})^d$.

Step 3: open, then compact, then general. For open A, B of finite measure, exhaust from inside by finite almost disjoint rectangle unions $A_\varepsilon, B_\varepsilon$ (Theorem 24.1.24.1.5) and pass to the limit using $A + B \supseteq A_\varepsilon + B_\varepsilon$. For compact A, B , the ε -neighborhoods $A^\varepsilon = \{x : \text{dist}(x, A) < \varepsilon\}$ are open with $A^\varepsilon \searrow A$, and $A + B \subseteq A^\varepsilon + B^\varepsilon \subseteq (A + B)^{2\varepsilon}$; let $\varepsilon \rightarrow 0$. For general measurable A, B with $A + B$ measurable, approximate from inside by compact sets (Theorem 24.2.24.2.6(iii)). $\square$

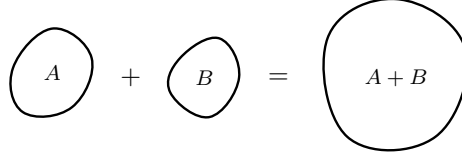


Figure 24.17. Brunn–Minkowski: the Minkowski sum of two bodies is at least as “large” as scaling predicts; equality forces A, B convex and homothetic. (Schematic.)

Corollary 24.4.3 (Convex case and equality). *For convex A , $A + \lambda A = (1 + \lambda)A$ and $m((1 + \lambda)A) = (1 + \lambda)^d m(A)$, so Brunn–Minkowski is sharp with exponent $1/d$. For open sets of finite positive measure, equality holds iff A and B are convex and homothetic: $A = \delta B + h$ for some $\delta > 0$, $h \in \mathbb{R}^d$.*

Remark. The exponent $1/d$ is forced: for convex A and $B = \lambda A$, equality in any candidate inequality $m(A + B)^\alpha \geq m(A)^\alpha + m(B)^\alpha$ requires $(1 + \lambda)^{d\alpha} \geq 1 + \lambda^{d\alpha}$ for all $\lambda > 0$, which by the elementary inequality $(a + b)^\gamma \geq a^\gamma + b^\gamma$ ($\gamma \geq 1$) holds exactly when $d\alpha \geq 1$, i.e. $\alpha \geq 1/d$. The strongest (smallest-exponent) form is $\alpha = 1/d$.

Worked Example 24.4.4 (Two segments summing to a square). In $\mathbb{R}^2$, let $A = [0, 1] \times \{0\}$ and $B = \{0\} \times [0, 1]$. Compute $m(A)$, $m(B)$, $m(A + B)$ and check Brunn–Minkowski.

Recognition. Trigger: lower-dimensional summands; the sumset can jump in dimension.

Solution. A, B are segments, so $m(A) = m(B) = 0$. But $A + B = [0, 1] \times [0, 1]$, the unit square, with $m(A + B) = 1$. Brunn–Minkowski reads $1^{1/2} \geq 0^{1/2} + 0^{1/2} = 0$, true (and very far from equality). *Check.* This is the promised example where null summands produce a positive-measure sum – no upper bound for $m(A + B)$ is possible.

Worked Example 24.4.5 (Equality for homothetic cubes). Let $A = [0, 1]^d$ and $B = [0, \lambda]^d$. Verify equality in Brunn–Minkowski.

Recognition. Trigger: homothetic convex bodies should attain equality (Corollary 24.4.3).

Solution. $A + B = [0, 1 + \lambda]^d$, so $m(A + B)^{1/d} = 1 + \lambda$, while $m(A)^{1/d} + m(B)^{1/d} = 1 + \lambda$. Equality holds. *Check.* Here $B = \lambda A$ (so $\delta = 1/\lambda$, $h = 0$), exactly the homothety condition for equality.

Worked Example 24.4.6 (Two balls). For balls $A = B_r(0)$, $B = B_s(0)$ in $\mathbb{R}^d$, verify Brunn–Minkowski with equality.

Recognition. Trigger: $A + B$ of two centered balls is again a ball.

Solution. $A + B = B_{r+s}(0)$ and $m(B_\rho(0)) = v_d \rho^d$ where $v_d = m(B_1(0))$. So $m(A + B)^{1/d} = v_d^{1/d}(r + s) = v_d^{1/d}r + v_d^{1/d}s = m(A)^{1/d} + m(B)^{1/d}$. Equality. *Check.* Balls are convex and homothetic ($B = (s/r)A$), so equality is expected.

Worked Example 24.4.7 (AM–GM, the engine of Step 1). Prove $\frac{1}{d}(x_1 + \cdots + x_d) \geq (x_1 \cdots x_d)^{1/d}$ for $x_j \geq 0$, $d = 2$, and show how it gives the rectangle case.

Recognition. Trigger: the base inequality behind Brunn–Minkowski for boxes.

Solution. For $d = 2$, $(\sqrt{x_1} - \sqrt{x_2})^2 \geq 0$ expands to $x_1 + x_2 \geq 2\sqrt{x_1 x_2}$, i.e. $\frac{1}{2}(x_1 + x_2) \geq (x_1 x_2)^{1/2}$. With $a_1 + b_1 = 1$, $a_2 + b_2 = 1$: $(a_1 a_2)^{1/2} + (b_1 b_2)^{1/2} \leq \frac{1}{2}(a_1 + a_2) + \frac{1}{2}(b_1 + b_2) = \frac{1}{2}(a_1 + b_1) + \frac{1}{2}(a_2 + b_2) = 1$. *Check.* This is precisely the reduced rectangle inequality with $d = 2$ and $a_j + b_j = 1$.

Worked Example 24.4.8 (The elementary $(a + b)^\gamma \geq a^\gamma + b^\gamma$). Show $(a + b)^\gamma \geq a^\gamma + b^\gamma$ for $\gamma \geq 1$, $a, b \geq 0$, used to fix the exponent $1/d$.

Recognition. Trigger: compare $(a + t)^{\gamma-1}$ with $t^{\gamma-1}$ and integrate.

Solution. For $\gamma \geq 1$ and $t \geq 0$, $(a + t)^{\gamma-1} \geq t^{\gamma-1}$. Integrate in t from 0 to b : $\frac{(a+b)^\gamma - a^\gamma}{\gamma} \geq \frac{b^\gamma}{\gamma}$, i.e. $(a + b)^\gamma \geq a^\gamma + b^\gamma$. *Check.* For $0 \leq \gamma \leq 1$ the integrand inequality reverses, giving $(a + b)^\gamma \leq a^\gamma + b^\gamma$ – consistent with the exponent analysis forcing $\alpha \geq 1/d$.

Worked Example 24.4.9 (Sumset of two intervals in $\mathbb{R}$). In $\mathbb{R}$ ($d = 1$) verify Brunn–Minkowski is just $m(A + B) \geq m(A) + m(B)$ for intervals $A = [0, a]$, $B = [0, b]$.

Recognition. Trigger: in dimension one the $1/d$ root disappears.

Solution. $A + B = [0, a + b]$, so $m(A + B) = a + b = m(A) + m(B)$: equality. *Check.* For $d = 1$ the inequality is plain super-additivity of length on sumsets, with equality for intervals (which are trivially convex and homothetic up to the degenerate case).

Worked Example 24.4.10 (Concavity of $t \mapsto m((1 - t)A + tB)^{1/d}$). For convex bodies A, B , deduce from Brunn–Minkowski that $\phi(t) = m((1 - t)A + tB)^{1/d}$ is concave on $[0, 1]$.

Recognition. Trigger: apply the inequality to the scaled summands $(1 - t)A$ and tB , then to a convex combination of two parameters.

Solution. For $s, t \in [0, 1]$ and $\lambda \in [0, 1]$, set $A_u = (1 - u)A + uB$; convexity gives $\lambda A_s + (1 - \lambda)A_t \subseteq A_{\lambda s + (1 - \lambda)t}$. Brunn–Minkowski and the scaling $m(cK) = c^d m(K)$ yield $\phi(\lambda s + (1 - \lambda)t) \geq m(\lambda A_s + (1 - \lambda)A_t)^{1/d} \geq \lambda m(A_s)^{1/d} + (1 - \lambda)m(A_t)^{1/d} = \lambda \phi(s) + (1 - \lambda)\phi(t)$, i.e. ϕ is concave. *Check.* For $A = B$ (so $A_t \equiv A$), $\phi \equiv m(A)^{1/d}$ is constant, the boundary case of concavity, and equality holds throughout – consistent with the equality criterion (homothetic bodies).

Exercises (Stein, Ch.1) – Brunn–Minkowski.

1. Prove $(a + b)^\gamma \geq a^\gamma + b^\gamma$ whenever $\gamma \geq 1$, $a, b \geq 0$, and the reverse inequality when $0 \leq \gamma \leq 1$. [Hint: integrate the inequality between $(a + t)^{\gamma-1}$ and $t^{\gamma-1}$ from 0 to b .]
2. Establish the arithmetic–geometric inequality $\frac{x_1 + \dots + x_d}{d} \geq (x_1 \dots x_d)^{1/d}$ for all $x_j \geq 0$ by backward induction: (a) it holds when $d = 2^k$; (b) if it holds for $d \geq 2$ it holds for $d - 1$. [Hint: for (a) write $(x_1 + \dots + x_{2^k})/2^k = (A + B)/2$ and use $d = 2$; for (b) set $x_d = (y_1 + \dots + y_{d-1})/(d - 1)$.]

Problems (Stein, Ch.1).

1. Given irrational x there are infinitely many reduced fractions p/q with $|x - p/q| \leq 1/q^2$. Prove, however, that the set of $x \in \mathbb{R}$ for which there are infinitely many reduced p/q with $|x - p/q| \leq 1/q^3$ (or $\leq 1/q^{2+\epsilon}$) has measure zero. [Hint: Borel–Cantelli.]
2. Show any open set Ω can be written as $\Omega = \bigcup_j Q_j$ with closed cubes Q_j such that (i) the Q_j have disjoint interiors and (ii) $c \leq \text{dist}(Q_j, \Omega^c)/\ell(Q_j) \leq C$ for positive constants c, C , where $\ell(Q_j)$ is the side length (a Whitney decomposition).
3. Find a measurable $C \subseteq [0, 1]$ with $m(C) = 0$ whose difference set contains a nontrivial interval about the origin (compare Exercise 24.2.12). [Hint: take C the Cantor set; for fixed $a \in [-1, 1]$ consider the line $y = x + a$ and build a “Cantor dust” in $[0, 1]^2$ by keeping four corner cubes of side $1/3$ at each stage; show the line meets the dust by nested compactness.]
4. Outline a proof that a bounded function on $[a, b]$ is Riemann integrable iff its set of discontinuities has measure zero. With $\text{osc}(f, c) = \lim_{r \rightarrow 0} \sup\{|f(x) - f(y)| : x, y \in J \cap I(c, r)\}$ (so f is continuous at c iff $\text{osc}(f, c) = 0$), prove: (a) for every $\epsilon > 0$, $\{c : \text{osc}(f, c) \geq \epsilon\}$ is compact; (b) if the discontinuity set is null then f is Riemann integrable; (c) conversely, if f is Riemann integrable its discontinuity set is null.
5. Suppose E is measurable, $m(E) < \infty$, $E = E_1 \cup E_2$ with $E_1 \cap E_2 = \emptyset$. If $m(E) = m_*(E_1) + m_*(E_2)$, prove E_1, E_2 are measurable. In particular, if $E \subseteq Q$ for a finite cube Q , then E is measurable iff $m(Q) = m_*(E) + m_*(Q \setminus E)$.
6. The axiom of choice and the well-ordering principle are equivalent, via the Hausdorff maximal principle. A partial ordering $\leq$ on E satisfies (i) $x \leq x$, (ii) $x \leq y$ and $y \leq x$ imply $x = y$, (iii) transitivity; it is linear if any two elements are comparable. Hausdorff: every nonempty partially ordered set has a nonempty maximal linearly ordered subset. Show this implies the well-ordering principle (apply it to the collection of all well-orderings of subsets of E).
7. Let $\Gamma = \{y = f(x)\} \subset \mathbb{R}^2$, $0 \leq x \leq 1$, with f twice continuously differentiable. Show $m(\Gamma + \Gamma) > 0$ iff $\Gamma + \Gamma$ contains an open set iff f is not linear.

8. Suppose A, B are open of finite positive measure. Show equality holds in Brunn–Minkowski iff A, B are convex and similar, i.e. $A = \delta B + h$ for some $\delta > 0$, $h \in \mathbb{R}^d$.

Additional exercises.

1. Verify Brunn–Minkowski for $A = [0, 2] \times [0, 1]$ and $B = [0, 1] \times [0, 3]$ in $\mathbb{R}^2$, and decide whether equality holds.
2. For $A = B_1(0) \subset \mathbb{R}^3$ (unit ball) and $B = \{0\}$ (a point), check that Brunn–Minkowski degenerates to an equality, and explain the role of $m(B) = 0$.
3. Use Brunn–Minkowski to prove the isoperimetric-type statement $m(A + B_\varepsilon(0)) \geq (m(A)^{1/d} + v_d^{1/d} \varepsilon)^d$, and expand to first order in ε to read off a lower bound for the “surface content” $\liminf_{\varepsilon \rightarrow 0} (m(A + B_\varepsilon) - m(A))/\varepsilon$.
4. Prove the $d = 2$ AM–GM step $\frac{1}{2}(x_1 + x_2) \geq \sqrt{x_1 x_2}$ and deduce the rectangle case of Brunn–Minkowski for two rectangles with $a_j + b_j = 1$.
5. (Non-example.) Show the naive “ $m(A + B) \geq m(A) + m(B)$ ” (exponent $\alpha = 1$) fails in $\mathbb{R}^2$ by taking $A = B = [0, 1]^2$; compute both sides and identify the correct exponent $1/d$.
6. For convex A and $B = \lambda A$, verify $A + B = (1 + \lambda)A$ and that $m((1 + \lambda)A) = (1 + \lambda)^d m(A)$ gives equality in Brunn–Minkowski; identify the homothety data (δ, h) .
7. (Diagram.) Sketch the Minkowski sum of a unit square and a unit disc in $\mathbb{R}^2$ (a “rounded square”), label its straight edges and quarter-circle corners, and write its area as $1 + 4 \cdot \frac{1}{2} + \pi(\frac{1}{2})^2 \dots$ (correct the coefficients) to get $1 + (\text{perimeter}) \cdot 1 + \pi$.
8. Deduce from Brunn–Minkowski that $t \mapsto m((1 - t)A + tB)^{1/d}$ is concave on $[0, 1]$ for convex A, B , and state what equality of the endpoints forces.

Chapter 25

Integration Theory

25.1 Construction of the Lebesgue Integral

The Lebesgue integral is built in four stages of increasing generality: first for *simple* functions (finite sums of scaled indicators), then for *bounded* functions supported on a set of finite measure, then for *non-negative* measurable functions (possibly extended-valued and unbounded), and finally for a general integrable function f by splitting $f = f^+ - f^-$. Throughout, *every function is assumed measurable*; this is the standing hypothesis that makes each stage work. The payoff, collected in this section, is a trio of convergence theorems — bounded convergence, Fatou, monotone convergence, and (in the next section) dominated convergence — that let us interchange limit and integral under hypotheses far weaker than uniform convergence.

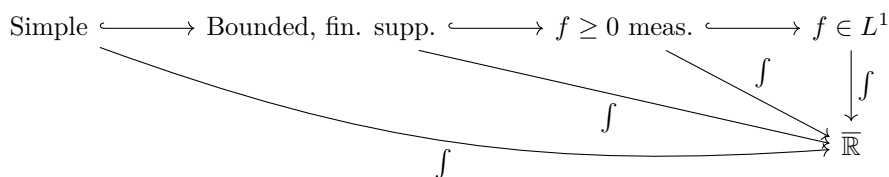


Figure 25.1. Each stage extends the integral functional $\int$ along an inclusion of function classes; the extended $\int$ agrees with the previous one on the smaller class (the triangles commute), so a single functional $\int : L^1 \rightarrow \mathbb{R}$ results.

Stage one: simple functions

Definition 25.1.1. A *simple function* is a finite sum $\varphi(x) = \sum_{k=1}^N a_k \mathbf{1}_{E_k}(x)$ where the E_k are measurable sets of finite measure and the a_k are constants. The same φ has many such representations (e.g. $0 = \mathbf{1}_E - \mathbf{1}_E$). The *canonical form* is the unique decomposition $\varphi = \sum_{k=1}^M c_k \mathbf{1}_{F_k}$ in which the values c_k are *distinct and nonzero* and the sets $F_k = \{x : \varphi(x) = c_k\}$ are *disjoint*. For a simple function in canonical form, the *Lebesgue integral* is

$$\int_{\mathbb{R}^d} \varphi(x) \, dx = \sum_{k=1}^M c_k m(F_k),$$

and for measurable E of finite measure $\int_E \varphi = \int \varphi \mathbf{1}_E$.

The canonical form exists because φ takes only finitely many distinct nonzero values $c_1, \dots, c_M$; the level sets F_k partition the support. Visually, a simple function is a finite tower of flat indicator “blocks,” and its integral is just $\sum(\text{height}) \times (\text{measure of base})$ — signed area, computed by stacking horizontal slabs rather than vertical Riemann columns.

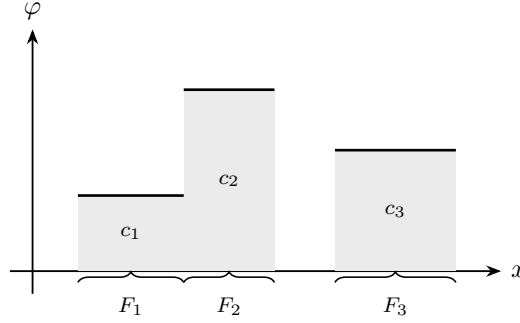


Figure 25.2. A simple function as a finite sum of indicator blocks; $\int \varphi = \sum_k c_k m(F_k)$ totals the signed block areas. (Schematic.)

Proposition 25.1.2 (Properties of the simple integral). *The integral of simple functions satisfies: (i) representation independence: if $\varphi = \sum_{k=1}^N a_k \mathbf{1}_{E_k}$ is any representation, then $\int \varphi = \sum_k a_k m(E_k)$; (ii) linearity: $\int(a\varphi + b\psi) = a \int \varphi + b \int \psi$; (iii) additivity: $\int_{E \cup F} \varphi = \int_E \varphi + \int_F \varphi$ for disjoint E, F of finite measure; (iv) monotonicity: $\varphi \leq \psi \implies \int \varphi \leq \int \psi$; (v) triangle inequality: $|\int \varphi| \leq \int |\varphi|$.*

Proof. The only delicate point is (i). First suppose $\varphi = \sum_k a_k \mathbf{1}_{E_k}$ with the E_k disjoint (but the a_k possibly equal or zero). For each distinct nonzero value a among the a_k set $E'_a = \bigcup_{a_k=a} E_k$; the E'_a are disjoint and $m(E'_a) = \sum_{a_k=a} m(E_k)$ by finite additivity. Then $\varphi = \sum_a a \mathbf{1}_{E'_a}$ is the canonical form, so $\int \varphi = \sum_a a m(E'_a) = \sum_k a_k m(E_k)$. Next drop the disjointness: refine the E_k into finitely many disjoint pieces E_j^* so that each E_k is a union of some E_j^* (Exercise 1). Writing $a_j^* = \sum_{E_k \supset E_j^*} a_k$ gives $\varphi = \sum_j a_j^* \mathbf{1}_{E_j^*}$, a disjoint representation, and $\int \varphi = \sum_j a_j^* m(E_j^*) = \sum_k a_k \sum_{E_j^* \subset E_k} m(E_j^*) = \sum_k a_k m(E_k)$. Now (ii) follows from (i) applied to any representations of φ, ψ ; (iii) uses $\mathbf{1}_{E \cup F} = \mathbf{1}_E + \mathbf{1}_F$; (iv) applies the canonical-form definition to the nonnegative simple function $\psi - \varphi$; and (v) uses $|\varphi| = \sum_k |a_k| \mathbf{1}_{E_k}$ and the scalar triangle inequality. $\square$

Worked Example 25.1.3 (Integral is independent of the chosen representation). On $\mathbb{R}$ let $\varphi = 2\mathbf{1}_{[0,3]} + 1 \cdot \mathbf{1}_{[1,4]}$. Compute $\int \varphi$ from this (overlapping) representation and from the canonical form, and confirm they agree.

Recognition. An “independence of representation” check (Prop. 25.1.2(i)); compute both ways.

Solution. Directly: $\int \varphi = 2m([0, 3]) + 1 \cdot m([1, 4]) = 2 \cdot 3 + 1 \cdot 3 = 9$. Canonical form: on $[0, 1)$, $\varphi = 2$; on $[1, 3]$, $\varphi = 3$; on $(3, 4]$, $\varphi = 1$. So $\varphi = 2\mathbf{1}_{[0,1)} + 3\mathbf{1}_{[1,3]} + 1\mathbf{1}_{(3,4]}$ and $\int \varphi = 2 \cdot 1 + 3 \cdot 2 + 1 \cdot 1 = 9$. *Check.* Both give 9, as Prop. 25.1.2(i) guarantees; the level set $\{\varphi = 3\} = [1, 3]$ has measure 2, the largest contribution.

Remark. If two simple functions agree a.e. then their integrals are equal (the differing set has measure zero, contributing nothing). This “a.e. irrelevance” persists through every later stage of the construction and is why L^1 will be a space of equivalence classes.

Stage two: bounded functions on sets of finite measure

Definition 25.1.4. The *support* of a measurable f is $\text{supp}(f) = \{x : f(x) \neq 0\}$ (a measurable set); f is *supported on E* if $f = 0$ off E . We now integrate bounded measurable f with $m(\text{supp } f) < \infty$.

By Theorem 4.2 of Chapter 1 (Ch. 24), a function bounded by M and supported on E is a pointwise limit of simple functions φ_n , each bounded by M and supported on E . The next lemma — whose engine is Egorov’s theorem (convergence is “nearly uniform”) — shows the integrals $\int \varphi_n$ converge to a value independent of the approximating sequence.

Lemma 25.1.5. *Let f be bounded and supported on a set E of finite measure. If $\{\varphi_n\}$ are simple, bounded by M , supported on E , and $\varphi_n \rightarrow f$ a.e., then (i) $\lim_n \int \varphi_n$ exists; (ii) if $f = 0$ a.e. the limit is 0.*

Proof. Were the convergence uniform on E this would be immediate. Instead, given $\varepsilon > 0$, Egorov’s theorem supplies a closed $A_\varepsilon \subseteq E$ with $m(E \setminus A_\varepsilon) \leq \varepsilon$ and $\varphi_n \rightarrow f$ uniformly on A_ε . With $I_n = \int \varphi_n$,

$$|I_n - I_m| \leq \int_{A_\varepsilon} |\varphi_n - \varphi_m| + \int_{E \setminus A_\varepsilon} |\varphi_n - \varphi_m| \leq m(E) \varepsilon + 2M \varepsilon$$

for all large n, m (uniform convergence controls the first term by ε on A_ε ; the second is at most $2M m(E \setminus A_\varepsilon)$). As ε is arbitrary and $m(E) < \infty$, $\{I_n\}$ is Cauchy, proving (i). For (ii) with $f = 0$ the same estimate gives $|I_n| \leq m(E)\varepsilon + M\varepsilon$, so $I_n \rightarrow 0$. $\square$

Definition 25.1.6. For f bounded and supported on a finite-measure set, set $\int f = \lim_n \int \varphi_n$ for any such approximating $\{\varphi_n\}$. Lemma 25.1.5 (applied to $\eta_n = \varphi_n - \psi_n$) makes this independent of the sequence, so $\int f$ is well-defined, and on simple functions it agrees with Definition 25.1.1.

Proposition 25.1.7. For bounded f, g supported on finite-measure sets, the integral is linear, additive over disjoint sets, monotone ($f \leq g \implies \int f \leq \int g$), and satisfies $|\int f| \leq \int |f|$. (All inherited from Prop. 25.1.2 via approximation.)

Theorem 25.1.8 (Bounded Convergence Theorem). If $\{f_n\}$ are measurable, all bounded by M , supported on a fixed set E of finite measure, and $f_n \rightarrow f$ a.e., then f is bounded and supported on E (a.e.), and

$$\int |f_n - f| \rightarrow 0, \quad \text{hence} \quad \int f_n \rightarrow \int f.$$

Proof. f is bounded by M a.e. and vanishes off E except on a null set; by the triangle inequality it suffices to show $\int |f_n - f| \rightarrow 0$. Given $\varepsilon > 0$, Egorov supplies $A_\varepsilon \subseteq E$ with $m(E \setminus A_\varepsilon) \leq \varepsilon$ and $f_n \rightarrow f$ uniformly on A_ε , so $|f_n - f| \leq \varepsilon$ on A_ε for large n . Then

$$\int |f_n - f| \leq \int_{A_\varepsilon} |f_n - f| + \int_{E \setminus A_\varepsilon} |f_n - f| \leq \varepsilon m(E) + 2M\varepsilon.$$

As ε is arbitrary, the integral tends to 0. The conclusion is exactly $\lim_n \int f_n = \int \lim_n f_n$: a license to interchange limit and integral. $\square$

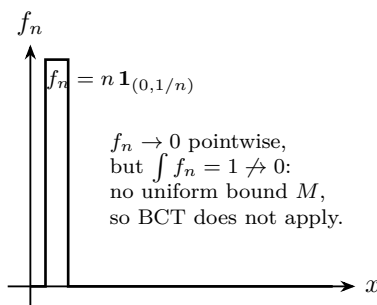


Figure 25.3. The escaping spike $n \mathbf{1}_{(0,1/n)}$: BCT fails without a common bound M . Mass “leaks to infinity” in height.

Worked Example 25.1.9 (Where BCT applies and where it does not). On $E = [0, 1]$ let $f_n(x) = x^n$. Show $\int_0^1 f_n \rightarrow 0$ via BCT. Then explain why the spike $g_n = n \mathbf{1}_{(0,1/n)}$ is *not* covered.

Recognition. A limit-integral interchange; verify the three BCT hypotheses (common bound, finite-measure support, a.e. limit).

Solution. For $f_n = x^n$: each $|f_n| \leq 1 = M$, all supported on $[0, 1]$ (finite measure), and $f_n(x) \rightarrow 0$ for $x \in [0, 1)$ while $f_n(1) = 1$ — so $f_n \rightarrow 0$ a.e. BCT gives $\int_0^1 x^n dx \rightarrow 0$. *Check.* Directly $\int_0^1 x^n = \frac{1}{n+1} \rightarrow 0$. For g_n : $g_n \rightarrow 0$ pointwise, but $\sup g_n = n$ is unbounded, so no common M exists and BCT does not apply — indeed $\int g_n = 1 \not\rightarrow 0$. *Lesson.* The uniform bound is essential.

Riemann integrable functions are Lebesgue integrable

Theorem 25.1.10. If f is Riemann integrable on $[a, b]$, then f is measurable and its Riemann and Lebesgue

integrals coincide: $\int_{[a,b]}^R f = \int_{[a,b]}^L f$.

Proof. Riemann integrability gives step functions $\varphi_1 \leq \varphi_2 \leq \dots \leq f \leq \dots \leq \psi_2 \leq \psi_1$, all bounded by M , with $\int^R \varphi_k, \int^R \psi_k \rightarrow \int_{[a,b]}^R f$. For step functions the Riemann and Lebesgue integrals agree. Let $\tilde{\varphi} = \lim_k \varphi_k$ and $\tilde{\psi} = \lim_k \psi_k$; these are measurable with $\tilde{\varphi} \leq f \leq \tilde{\psi}$. By the BCT, $\int^L \varphi_k \rightarrow \int^L \tilde{\varphi}$ and $\int^L \psi_k \rightarrow \int^L \tilde{\psi}$, so $\int_{[a,b]}^L (\tilde{\psi} - \tilde{\varphi}) = 0$ with $\tilde{\psi} - \tilde{\varphi} \geq 0$; hence $\tilde{\varphi} = \tilde{\psi} = f$ a.e., so f is measurable. Finally $\varphi_k \rightarrow f$ a.e. gives $\int_{[a,b]}^L \varphi_k \rightarrow \int_{[a,b]}^L f$, which equals $\int_{[a,b]}^R f$. $\square$

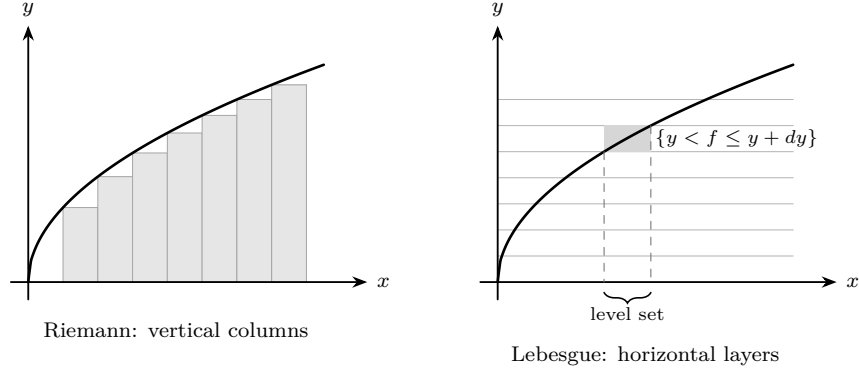


Figure 25.4. Riemann slices the *domain* into vertical columns; Lebesgue slices the *range* into horizontal layers $\{y < f \leq y + dy\}$ and weighs each by the measure of its level set (the marked x -interval). (Schematic.)

Stage three: non-negative measurable functions

Definition 25.1.11. For measurable $f \geq 0$ (possibly extended-valued, taking $+\infty$ on a measurable set), define

$$\int_{\mathbb{R}^d} f = \sup_g \int_{\mathbb{R}^d} g,$$

the supremum over all g with $0 \leq g \leq f$, g bounded and supported on a set of finite measure. We call f *integrable* when $\int f < \infty$.

Proposition 25.1.12. For measurable $f, g \geq 0$: (i) *linearity* $\int (af + bg) = a \int f + b \int g$ ($a, b > 0$); (ii) *additivity over disjoint sets*; (iii) *monotonicity*; (iv) if g integrable and $0 \leq f \leq g$ then f is integrable; (v) if f integrable then $f < \infty$ a.e.; (vi) if $\int f = 0$ then $f = 0$ a.e.

Proof. Only (i) needs work. “ $\geq$ ”: if $\varphi \leq f$, $\psi \leq g$ are bounded with finite-measure support then $\varphi + \psi \leq f + g$, so $\int f + \int g = \sup(\int \varphi + \int \psi) \leq \int (f + g)$. “ $\leq$ ”: given $\eta \leq f + g$ bounded with finite support, set $\eta_1 = \min(f, \eta)$ and $\eta_2 = \eta - \eta_1$; then $\eta_1 \leq f$, $\eta_2 \leq g$, both bounded with finite support, and $\int \eta = \int \eta_1 + \int \eta_2 \leq \int f + \int g$; take sup over η . For (v), with $E_k = \{f \geq k\}$ we get $\int f \geq k m(E_k)$, so $m(E_k) \rightarrow 0$; as $E_k \downarrow E_\infty = \{f = \infty\}$, $m(E_\infty) = 0$. Part (vi) is the observation after the BCT: $E_k = \{f \geq 1/k\}$ has $k^{-1} m(E_k) \leq \int f = 0$, so $m(E_k) = 0$ for all k and $\{f > 0\} = \bigcup_k E_k$ is null. $\square$

The naive hope “ $f_n \rightarrow f$ pointwise $\implies \int f_n \rightarrow \int f$ ” is *false*: the spike $f_n = n \mathbf{1}_{(0,1/n)}$ has $f_n \rightarrow 0$ yet $\int f_n = 1$. Mass can escape (to infinite height, or off to infinity). The correct general statements only assert an *inequality* or require *monotonicity*.

Lemma 25.1.13 (Fatou). If $f_n \geq 0$ are measurable and $f_n \rightarrow f$ a.e. (or more generally for $\liminf$), then

$$\int f \leq \liminf_{n \rightarrow \infty} \int f_n.$$

Proof. Let $0 \leq g \leq f$ with g bounded and supported on a finite-measure set E . Put $g_n = \min(g, f_n)$; then g_n is bounded, supported on E , and $g_n \rightarrow g$ a.e., so the BCT gives $\int g_n \rightarrow \int g$. Since $g_n \leq f_n$, $\int g_n \leq \int f_n$, whence $\int g \leq \liminf_n \int f_n$. Taking the supremum over all such g yields $\int f \leq \liminf_n \int f_n$. $\square$

Corollary 25.1.14. If $0 \leq f_n \leq f$ measurable with $f_n \rightarrow f$ a.e., then $\int f_n \rightarrow \int f$.

Proof. $f_n \leq f$ gives $\int f_n \leq \int f$, so $\limsup_n \int f_n \leq \int f$. Combined with Fatou's $\int f \leq \liminf_n \int f_n$, the limit exists and equals $\int f$. $\square$

Corollary 25.1.15 (Monotone Convergence Theorem). *If $f_n \geq 0$ measurable and $f_n \nearrow f$ (increasing, $f_n \rightarrow f$ a.e.), then $\int f_n \rightarrow \int f$.*

Proof. Immediate from Corollary 25.1.14 since $f_n \leq f$. $\square$

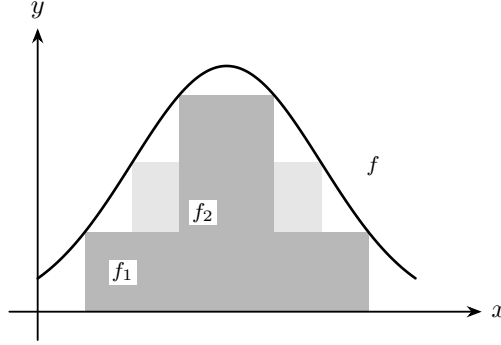


Figure 25.5. Monotone convergence: lower-sum simple functions $f_1 \leq f_2 \leq \dots$ rise to f (the coarser f_1 in dark grey, the finer f_2 in light grey, both under the bell), and $\int f_n \nearrow \int f$. (Schematic.)

Corollary 25.1.16 (Term-by-term integration). *If $a_k(x) \geq 0$ are measurable, then $\int \sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} \int a_k$. If $\sum_k \int a_k < \infty$, then $\sum_k a_k(x)$ converges for a.e. x .*

Proof. The partial sums $f_n = \sum_{k \leq n} a_k$ increase to $f = \sum_k a_k$; apply the MCT, using $\int f_n = \sum_{k \leq n} \int a_k$. If $\sum_k \int a_k < \infty$ then f is integrable, hence finite a.e. by Prop. 25.1.12(v). $\square$

Worked Example 25.1.17 (Borel–Cantelli via term-by-term integration). If $\sum_k m(E_k) < \infty$, show the set of points lying in infinitely many E_k is null.

Recognition. “ $\sum$ of measures finite $\implies$ negligible limsup set” — exactly Corollary 25.1.16 with $a_k = \mathbf{1}_{E_k}$.

Solution. Put $a_k = \mathbf{1}_{E_k}$. A point x lies in infinitely many E_k iff $\sum_k \mathbf{1}_{E_k}(x) = \infty$. Now $\sum_k \int \mathbf{1}_{E_k} = \sum_k m(E_k) < \infty$, so Corollary 25.1.16 forces $\sum_k \mathbf{1}_{E_k}(x) < \infty$ for a.e. x ; i.e. a.e. x lies in only finitely many E_k . Thus $\limsup_k E_k$ is null. *Check.* With $E_k = (0, 2^{-k})$, $\sum m(E_k) = \sum 2^{-k} = 1 < \infty$ and indeed $\bigcap_n \bigcup_{k \geq n} E_k = \emptyset$.

Worked Example 25.1.18 (The power $|x|^{-a}$ near and far: when is it integrable?). On $\mathbb{R}^d$ let $f_a(x) = |x|^{-a}$ on $\{|x| \leq 1\}$ (else 0) and $F_a(x) = (1 + |x|^a)^{-1}$. Determine integrability.

Recognition. A scaling/dyadic-shell computation; decompose into annuli and sum a geometric series.

Solution. On annuli $A_k = \{2^{-k-1} < |x| \leq 2^{-k}\}$ we have $m(A_k) \asymp 2^{-kd}$ and $|x|^{-a} \asymp 2^{ka}$ there, so $\int_{A_k} f_a \asymp 2^{k(a-d)}$. The series $\sum_{k \geq 0} 2^{k(a-d)}$ converges iff $a - d < 0$, i.e. $a < d$. Thus f_a is integrable near 0 exactly when $a < d$. The tail computation (annuli $|x| \approx 2^k$) shows F_a is integrable at infinity iff $a > d$. *Check.* $d = 1$: $\int_0^1 x^{-a} dx$ converges iff $a < 1$; $\int_1^\infty x^{-a} dx$ converges iff $a > 1$ — matching $a < d$ and $a > d$.

Worked Example 25.1.19 (The kernel $|x|^{-d-1}$ outside a ball). Show $f(x) = |x|^{-d-1}$ (for $x \neq 0$) satisfies $\int_{|x| \geq \varepsilon} f \leq C/\varepsilon$.

Recognition. A dyadic-shell estimate using dilation invariance of m ; the model computation behind approximations to the identity.

Solution. Let $A_k = \{2^k \varepsilon < |x| \leq 2^{k+1} \varepsilon\}$ and dominate f on A_k by $(2^k \varepsilon)^{-d-1}$. Since A_k is the dilate of $\{1 < |x| < 2\}$ by $2^k \varepsilon$, $m(A_k) = (2^k \varepsilon)^d m(A)$ with $A = \{1 < |x| < 2\}$. Hence $\int_{|x| \geq \varepsilon} f \leq \sum_{k \geq 0} \frac{m(A_k)}{(2^k \varepsilon)^{d+1}} = m(A) \sum_{k \geq 0} \frac{(2^k \varepsilon)^d}{(2^k \varepsilon)^{d+1}} = \frac{m(A)}{\varepsilon} \sum_{k \geq 0} 2^{-k} = \frac{2m(A)}{\varepsilon} = \frac{C}{\varepsilon}$. *Check.* The bound blows up like $1/\varepsilon$ as $\varepsilon \rightarrow 0$, consistent with f being non-integrable at the origin (where $a = d + 1 > d$).

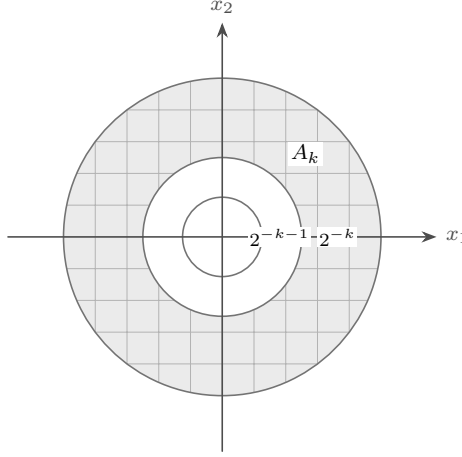


Figure 25.6. A dyadic annulus $A_k = \{2^{-k-1} < |x| \leq 2^{-k}\}$ and its covering by small cubes: $m(A_k) \asymp 2^{-kd}$ drives the geometric-series test for integrability of $|x|^{-a}$. (Schematic.)

Stage four: the general case

Definition 25.1.20. A real-valued measurable f is *integrable* if $|f|$ is integrable (Stage three). Writing $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$ (so $f = f^+ - f^-$, $|f| = f^+ + f^-$, both integrable), define $\int f = \int f^+ - \int f^-$. A complex $f = u + iv$ is integrable iff u, v are (iff $|f|$ is), and $\int f = \int u + i \int v$.

The integral is independent of the splitting: if $f = f_1 - f_2 = g_1 - g_2$ with all four nonnegative integrable, then $f_1 + g_2 = g_1 + f_2$, and linearity on nonnegative functions gives $\int f_1 - \int f_2 = \int g_1 - \int g_2$. Modifying f on a null set changes neither integrability nor the value, so we freely treat f as defined only a.e.

Proposition 25.1.21. *The Lebesgue integral of integrable functions is linear, additive, monotone, and satisfies $|\int f| \leq \int |f|$. The integrable functions on E form a vector space (over $\mathbb{R}$ or $\mathbb{C}$).*

Proposition 25.1.22 (Absolute continuity of the integral). *If f is integrable on $\mathbb{R}^d$, then for every $\varepsilon > 0$:*
(i) *there is a finite-measure set B (e.g. a ball) with $\int_B |f| < \varepsilon$;* (ii) *there is $\delta > 0$ with $\int_E |f| < \varepsilon$ whenever $m(E) < \delta$.*

Proof. Assume $f \geq 0$. (i) With $f_N = f \mathbf{1}_{B_N}$ (B_N the ball of radius N), $f_N \nearrow f$, so MCT gives $\int f_N \rightarrow \int f$; pick N with $\int(f - f_N) < \varepsilon$, i.e. $\int_{B_N^c} f < \varepsilon$. (ii) With $f_N = f \mathbf{1}_{\{f \leq N\}}$, again $f_N \nearrow f$; choose N with $\int(f - f_N) < \varepsilon/2$ and δ with $N\delta < \varepsilon/2$. If $m(E) < \delta$ then $\int_E f \leq \int(f - f_N) + \int_E f_N \leq \varepsilon/2 + Nm(E) < \varepsilon$. $\square$

Theorem 25.1.23 (Dominated Convergence Theorem). *Suppose $f_n \rightarrow f$ a.e. and $|f_n| \leq g$ with g integrable. Then $\int |f_n - f| \rightarrow 0$, hence $\int f_n \rightarrow \int f$.*

Proof. For $N \geq 0$ let $E_N = \{|x| \leq N, g(x) \leq N\}$. Given $\varepsilon > 0$, by absolute continuity (the first part of Prop. 25.1.22 applied to g) choose N with $\int_{E_N^c} g < \varepsilon$. On E_N the functions f_n are bounded (by N) and supported on a finite-measure set, so the BCT gives $\int_{E_N} |f_n - f| < \varepsilon$ for large n . Then

$$\int |f_n - f| = \int_{E_N} |f_n - f| + \int_{E_N^c} |f_n - f| \leq \varepsilon + 2 \int_{E_N^c} g \leq \varepsilon + 2\varepsilon = 3\varepsilon$$

for all large n (using $|f_n - f| \leq 2g$). As ε is arbitrary, $\int |f_n - f| \rightarrow 0$. $\square$

Worked Example 25.1.24 (DCT in action — a dominated limit). Compute $\lim_{n \rightarrow \infty} \int_0^\infty \frac{n \sin(x/n)}{x(1+x^2)} dx$.

Recognition. A limit-integral interchange; find an integrable dominating function independent of n , then pass the limit inside via DCT.

Solution. Write $f_n(x) = \frac{n \sin(x/n)}{x(1+x^2)} = \frac{\sin(x/n)}{(x/n)} \cdot \frac{1}{1+x^2}$. Since $|\sin t/t| \leq 1$, $|f_n(x)| \leq \frac{1}{1+x^2} =: g(x)$, and $\int_0^\infty g = \pi/2 < \infty$. Pointwise $\frac{\sin(x/n)}{x/n} \rightarrow 1$ as $n \rightarrow \infty$, so $f_n(x) \rightarrow \frac{1}{1+x^2}$. By DCT the limit is $\int_0^\infty \frac{dx}{1+x^2} = \pi/2$.

Check. Each $\frac{\sin t}{t} \leq 1$ keeps f_n under the fixed bell g ; the answer $\pi/2 = \arctan(\infty)$ matches the limit integrand.

Worked Example 25.1.25 (A NON-example: DCT fails without an integrable dominator). Show $\int_0^\infty f_n$ with $f_n = \frac{1}{n} \mathbf{1}_{[0,n]}$ does not obey $\int \lim f_n = \lim \int f_n$, and locate the missing hypothesis.

Recognition. Probe the necessity of an *integrable* dominating g in the DCT.

Solution. Here $f_n(x) = 1/n$ for $0 \leq x \leq n$, so $f_n \rightarrow 0$ pointwise (for each fixed x , $f_n(x) = 1/n \rightarrow 0$). Yet $\int_0^\infty f_n = \frac{1}{n} \cdot n = 1$ for every n , so $\lim \int f_n = 1 \neq 0 = \int \lim f_n$. The smallest pointwise dominator is $g(x) = \sup_n f_n(x)$. For $x \in [n-1, n)$ the largest value among $f_1, \dots$ is about $1/x$, so $g(x) \asymp 1/x$, and $\int_1^\infty \frac{dx}{x} = \infty$: *no integrable g dominates the family*, so DCT does not apply. *Check.* Mass 1 spreads thinly over $[0, n]$ and escapes to infinity; this is the “width” analogue of the escaping spike.

Worked Example 25.1.26 (Fatou’s inequality is genuinely strict). Exhibit $f_n \geq 0$ with $\int (\lim f_n) < \liminf \int f_n$, confirming Fatou cannot be upgraded to equality.

Recognition. A strictness witness for Lemma 25.1.13; reuse an escaping-mass family.

Solution. Take $f_n = n \mathbf{1}_{(0, 1/n)}$ on $\mathbb{R}$. Then $f_n(x) \rightarrow 0$ for every x , so $\int \lim_n f_n = \int 0 = 0$. But $\int f_n = n \cdot \frac{1}{n} = 1$ for all n , so $\liminf_n \int f_n = 1$. Thus $0 = \int f < \liminf \int f_n = 1$: strict. *Check.* Fatou’s “ $\leq$ ” holds ($0 \leq 1$) and is sharp here; the lost mass 1 is exactly the area of the escaping spike, which the pointwise limit cannot see.

Worked Example 25.1.27 (A maximal-type bound: averages controlled by Tchebyshev). For $f \geq 0$ integrable on $\mathbb{R}$ let $A_r f(x) = \frac{1}{2r} \int_{x-r}^{x+r} f$ be the symmetric average. Show $m\{x : A_r f(x) > \alpha\} \leq \frac{1}{\alpha} \int f$ for each fixed $r, \alpha > 0$.

Recognition. A super-level-set bound for an averaging operator; combine Tonelli with Tchebyshev.

Solution. First, $\int_{\mathbb{R}} A_r f(x) dx = \frac{1}{2r} \int_{\mathbb{R}} \int_{x-r}^{x+r} f(t) dt dx$; by Tonelli the point t is counted for $x \in (t-r, t+r)$, of length $2r$, so $\int A_r f = \frac{1}{2r} \cdot 2r \int f = \int f$. Since $A_r f \geq 0$ and $\int A_r f = \int f < \infty$, $A_r f$ is integrable, and Tchebyshev (this section’s L^1 bound) gives $m\{A_r f > \alpha\} \leq \frac{1}{\alpha} \int A_r f = \frac{1}{\alpha} \int f$. *Check.* Averaging preserves total mass ($\int A_r f = \int f$), so the level-set bound matches the one for f itself — a baby version of the Hardy–Littlewood maximal inequality.

Caution. The three failures to keep distinct: (a) *escaping height* $n \mathbf{1}_{(0, 1/n)}$ (no common bound M , kills BCT); (b) *escaping width* $\frac{1}{n} \mathbf{1}_{[0, n]}$ (no integrable dominator g , kills DCT); (c) Fatou gives only $\leq$, and the inequality is genuinely strict in both examples ($\int f = 0 < 1 = \liminf \int f_n$). Whenever an interchange “obviously” should hold, check which hypothesis — common bound, integrable dominator, or monotonicity — actually licenses it.

Exercises (Stein, Ch.2).

- Given a finite collection $F_1, \dots, F_n$, construct a disjoint collection $F_1^*, \dots, F_N^*$ with $N = 2^n - 1$ such that $\bigcup_{k=1}^n F_k = \bigcup_{j=1}^N F_j^*$ and $F_k = \bigcup_{F_j^* \subseteq F_k} F_j^*$ for every k . [Hint: consider the 2^n sets $F_1' \cap \dots \cap F_n'$ where each F_k' is F_k or F_k^c .]
- Suppose f is integrable on $[0, b]$ and $g(x) = \int_x^b \frac{f(t)}{t} dt$ for $0 < x \leq b$. Prove g is integrable on $[0, b]$ and $\int_0^b g(x) dx = \int_0^b f(t) dt$.
- Integrability of f on $\mathbb{R}$ need not force $f(x) \rightarrow 0$ as $x \rightarrow \infty$. (a) Construct a positive continuous integrable f on $\mathbb{R}$ with $\limsup_{x \rightarrow \infty} f(x) = \infty$. (b) If in addition f is uniformly continuous and integrable, prove $\lim_{|x| \rightarrow \infty} f(x) = 0$. [Hint: for (a) make f equal to n on $[n, n + 1/n^3]$ with continuous ramps.]
- Suppose f is integrable on $\mathbb{R}$. Show $F(x) = \int_{-\infty}^x f(t) dt$ is uniformly continuous.
- (Tchebyshev) Suppose $f \geq 0$ is integrable. For $\alpha > 0$ and $E_\alpha = \{x : f(x) > \alpha\}$, prove $m(E_\alpha) \leq \frac{1}{\alpha} \int f$.
- Suppose $f \geq 0$ and let $E_{2^k} = \{f > 2^k\}$ and $F_k = \{2^k < f \leq 2^{k+1}\}$. If f is finite a.e., then $\bigcup_{k=-\infty}^\infty F_k = \{f > 0\}$ and the F_k are disjoint. Prove f is integrable iff $\sum_{k=-\infty}^\infty 2^k m(F_k) < \infty$ iff $\sum_{k=-\infty}^\infty 2^k m(E_{2^k}) < \infty$. Use this to verify that $f(x) = |x|^{-a} \mathbf{1}_{|x| \leq 1}$ is integrable on $\mathbb{R}^d$ iff $a < d$, and $g(x) = |x|^{-b} \mathbf{1}_{|x| > 1}$ is integrable iff $b > d$.

7. Prove that if f is integrable, real-valued, on $\mathbb{R}^d$ with $\int_E f \geq 0$ for every measurable E , then $f \geq 0$ a.e. Consequently, if $\int_E f = 0$ for every measurable E , then $f = 0$ a.e.
8. Suppose f is integrable on $\mathbb{R}^d$. For $\alpha > 0$ let $E_\alpha = \{x : |f(x)| > \alpha\}$. Prove $\int_{\mathbb{R}^d} |f(x)| dx = \int_0^\infty m(E_\alpha) d\alpha$.

Additional exercises.

1. (Canonical form) Find the canonical form and the integral of $\varphi = 3\mathbf{1}_{[0,2]} - \mathbf{1}_{[1,3]} + 2\mathbf{1}_{[2,4]}$ on $\mathbb{R}$.
2. Compute $\lim_{n \rightarrow \infty} \int_0^1 \frac{nx}{1+n^2x^2} dx$ by recognizing the antiderivative; then explain why neither the MCT nor a single dominating g controls the family $\frac{nx}{1+n^2x^2}$ pointwise near 0.
3. Use the MCT to evaluate $\int_0^1 \sum_{n=1}^\infty x^n dx$ and confirm it equals $\sum_{n=1}^\infty \frac{1}{n+1} = \infty$? No — compute correctly $\sum_{n=1}^\infty \frac{1}{n(n+1)}$ by writing $\int_0^1 x^{n-1} dx$, and reconcile.
4. (DCT) Show $\lim_{n \rightarrow \infty} \int_0^\infty \left(1 + \frac{x}{n}\right)^{-n} \sin(x/n) dx = 0$ by exhibiting an integrable dominating function and passing to the limit.
5. (Fatou, equality vs. strict) For $f_n = \mathbf{1}_{[n, n+1]}$ on $\mathbb{R}$, compute $\int \lim f_n$ and $\liminf \int f_n$ and decide whether Fatou is strict. Repeat for $f_n = \mathbf{1}_{[0,1]}$ (constant sequence).
6. (Non-example) Give a sequence $f_n \geq 0$ on $[0, 1]$ with $f_n \rightarrow 0$ a.e., $\int f_n = 1$ for all n , and yet $\sup_n f_n \notin L^1[0, 1]$, so that no integrable dominator exists.
7. (Diagram) Sketch $f_n = n\mathbf{1}_{(0, 1/n)}$ and $g_n = \frac{1}{n}\mathbf{1}_{(0, n)}$ on a common axis, and annotate which convergence theorem each escapes and why (height vs. width).
8. Prove that if $f \geq 0$ is integrable and $\int_A f = \int_B f$ whenever $m(A) = m(B)$, then f is a.e. constant. (Hint: use the layer-cake formula and Tchebyshev.)

25.2 The Space L^1 and the Convergence Theorems

The integrable functions form not just a vector space but a *complete normed space* under the L^1 norm. Completeness (the Riesz–Fischer theorem) is the analytic heart of Lebesgue theory: it is exactly what Riemann integration lacked, and it is what makes L^1 a fit home for limiting arguments. We also record which familiar classes are *dense* in L^1 — simple, step, and continuous-compactly-supported functions — and the translation and dilation invariances of the integral.

The norm and the metric

Definition 25.2.1. For integrable f on $\mathbb{R}^d$ the L^1 norm is $\|f\| = \|f\|_{L^1} = \int_{\mathbb{R}^d} |f|$. Since $\|f\| = 0$ iff $f = 0$ a.e. (Prop. 25.1.12(vi)), we define $L^1(\mathbb{R}^d)$ to be the space of *equivalence classes* of integrable functions, two being identified when they agree a.e. We still speak loosely of an “element $f \in L^1$ ” as a function.

Proposition 25.2.2. For $f, g \in L^1(\mathbb{R}^d)$: (i) $\|af\| = |a| \|f\|$; (ii) $\|f + g\| \leq \|f\| + \|g\|$; (iii) $\|f\| = 0$ iff $f = 0$ a.e.; (iv) $d(f, g) = \|f - g\|$ is a metric on $L^1(\mathbb{R}^d)$.

The triangle inequality (ii) is monotonicity applied to $|f + g| \leq |f| + |g|$. Thus L^1 is a normed vector space, and the distance $d(f, g) = \|f - g\|$ measures the *total area* between the graphs of f and g — not their pointwise gap.

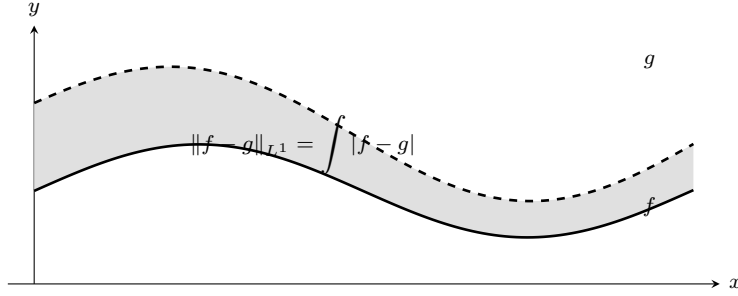


Figure 25.7. The L^1 distance is the shaded total area between the two graphs; small $\|f - g\|$ does *not* force f and g to be pointwise close. (Schematic.)

Completeness: the Riesz–Fischer theorem

Theorem 25.2.3 (Riesz–Fischer). $L^1(\mathbb{R}^d)$ is complete: every Cauchy sequence converges in the L^1 norm.

Proof. Let $\{f_n\}$ be Cauchy. Extract a subsequence $\{f_{n_k}\}$ with $\|f_{n_{k+1}} - f_{n_k}\| \leq 2^{-k}$ (possible since $\|f_n - f_m\| \leq \varepsilon$ for $n, m \geq N(\varepsilon)$; take $n_k = N(2^{-k})$). Consider the telescoping series and its absolute version

$$f = f_{n_1} + \sum_{k \geq 1} (f_{n_{k+1}} - f_{n_k}), \quad g = |f_{n_1}| + \sum_{k \geq 1} |f_{n_{k+1}} - f_{n_k}|.$$

By term-by-term integration (Cor. 25.1.16), $\int g \leq \|f_{n_1}\| + \sum_k 2^{-k} < \infty$, so g is integrable; as $|f| \leq g$, so is f , and the series for f converges a.e. Its partial sums are exactly f_{n_k} , so $f_{n_k} \rightarrow f$ a.e. Moreover $|f - f_{n_k}| \leq g$, so the DCT gives $\|f_{n_k} - f\|_{L^1} \rightarrow 0$. Finally, since $\{f_n\}$ is Cauchy, given ε pick N with $\|f_n - f_m\| < \varepsilon/2$ for $n, m > N$, then $n_k > N$ with $\|f_{n_k} - f\| < \varepsilon/2$; the triangle inequality gives $\|f_n - f\| < \varepsilon$ for $n > N$. So $f_n \rightarrow f$ in L^1 . $\square$

Corollary 25.2.4. If $f_n \rightarrow f$ in L^1 , there is a subsequence with $f_{n_k} \rightarrow f$ a.e.

This is sharp: L^1 convergence does *not* imply a.e. convergence of the full sequence (the “typewriter” sequence below). Only a subsequence is guaranteed.

Worked Example 25.2.5 (The typewriter sequence: L^1 convergence without a.e. convergence). On $[0, 1]$ let I_n run through $[0, 1]$, $[0, \frac{1}{2}]$, $[\frac{1}{2}, 1]$, $[0, \frac{1}{3}]$, $[\frac{1}{3}, \frac{2}{3}]$, $[\frac{2}{3}, 1]$, $\dots$ (dyadic-then-triadic blocks of shrinking length) and $f_n = \mathbf{1}_{I_n}$. Show $f_n \rightarrow 0$ in L^1 but $f_n(x) \not\rightarrow 0$ for any x .

Recognition. An illustration of Corollary 25.2.4’s sharpness: norm convergence forces only a subsequence to converge pointwise.

Solution. $\|f_n\|_{L^1} = m(I_n) \rightarrow 0$ as the blocks shrink, so $f_n \rightarrow 0$ in L^1 . But the intervals I_n sweep across $[0, 1]$ repeatedly, so every fixed x lies in infinitely many I_n and is missed by infinitely many; hence $f_n(x) = 1$ infinitely often and $f_n(x) = 0$ infinitely often, so $\lim_n f_n(x)$ does not exist for any x . *Check.* Along the subsequence of *first* blocks of each generation, $m(I_{n_k}) = 1/k \rightarrow 0$ and one can choose a sub-subsequence converging to 0 a.e., consistent with the corollary.

Dense subspaces of L^1

Definition 25.2.6. A family $\mathcal{G} \subseteq L^1$ is *dense* if for every $f \in L^1$ and $\varepsilon > 0$ there is $g \in \mathcal{G}$ with $\|f - g\|_{L^1} < \varepsilon$.

Theorem 25.2.7. Each of the following is dense in $L^1(\mathbb{R}^d)$: (i) simple functions; (ii) step functions; (iii) continuous functions of compact support.

Proof. Reduce to $f \geq 0$ (treat $f^\pm$ and real/imaginary parts separately). (i) Simple $\varphi_k \nearrow f$ exist (Ch. 1, Thm. 4.1); MCT (or DCT) gives $\|f - \varphi_k\| \rightarrow 0$. (ii) By (i) it suffices to approximate $\mathbf{1}_E$ ($mE < \infty$) by a step function; choose almost-disjoint rectangles R_j with $m(E \triangle \bigcup_{j \leq M} R_j) \leq 2\varepsilon$ (Ch. 1), so $\psi = \sum_j \mathbf{1}_{R_j}$ has $\|\mathbf{1}_E - \psi\|_{L^1} < 2\varepsilon$. (iii) By (ii) it suffices to handle $\mathbf{1}_R$ for a rectangle; in one dimension take a piecewise-linear “trapezoid” g equal to 1 on $[a, b]$, 0 outside $[a - \varepsilon, b + \varepsilon]$, and linear on the ramps, so $\|\mathbf{1}_{[a, b]} - g\|_{L^1} < 2\varepsilon$; in $\mathbb{R}^d$ take a product of such bumps. $\square$

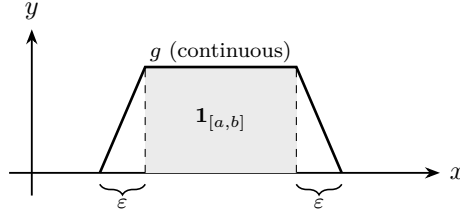


Figure 25.8. Approximating $\mathbf{1}_{[a,b]}$ by a continuous trapezoid: the two ramp triangles contribute $\|\mathbf{1}_{[a,b]} - g\|_{L^1} < 2\varepsilon$.

Method (The density (limiting) principle). *Recognition trigger.* You must prove an identity or inequality $\Phi(f) = 0$ (or ≤ 0) for *all* $f \in L^1$, and both sides depend continuously on f in the L^1 norm.

1. Prove the statement for f in a convenient dense class $\mathcal{G}$ (simple, step, or continuous compactly supported) — usually a direct calculation.
2. Show both sides of Φ are L^1 -continuous: $|\Phi(f) - \Phi(g)| \leq C\|f - g\|_{L^1}$, or use the DCT.
3. Given general f , pick $g \in \mathcal{G}$ with $\|f - g\| < \varepsilon$; then $\Phi(f)$ is within $C\varepsilon$ of $\Phi(g)$, which is known. Let $\varepsilon \rightarrow 0$.

This is the standard route to Proposition 25.2.9 below and to the Riemann–Lebesgue lemma.

Invariance properties

Proposition 25.2.8. For $f \in L^1(\mathbb{R}^d)$ and $h \in \mathbb{R}^d$, $\delta > 0$, the translate $f_h(x) = f(x - h)$ and the dilate $f(\delta x)$ are integrable, and

$$\int f(x - h) dx = \int f(x) dx, \quad \delta^d \int f(\delta x) dx = \int f(x) dx, \quad \int f(-x) dx = \int f(x) dx.$$

In particular $\|f_h\| = \|f\|$.

Proof. For $f = \mathbf{1}_E$ this is $m(E_h) = m(E)$ (translation invariance of m) and the dilation/reflection laws for m ; extend by linearity to simple functions, then by MCT to $f \geq 0$, then to general f via $f = f^+ - f^-$ and real/imaginary parts. $\square$

A consequence is the commutativity of *convolution*: if $y \mapsto f(x - y)g(y)$ is integrable then so is $y \mapsto f(y)g(x - y)$, with

$$(f * g)(x) = \int f(x - y)g(y) dy = \int f(y)g(x - y) dy,$$

by the change of variables $y \mapsto x - y$ (a translation composed with a reflection).

Proposition 25.2.9 (Continuity of translation in L^1). If $f \in L^1(\mathbb{R}^d)$ then $\|f_h - f\|_{L^1} \rightarrow 0$ as $h \rightarrow 0$.

Proof. By density (the method above) pick continuous compactly supported g with $\|f - g\| < \varepsilon$. Then $f_h - f = (g_h - g) + (f_h - g_h) - (f - g)$; the outer terms have norm $\|f - g\| < \varepsilon$ each (using $\|f_h - g_h\| = \|f - g\|$ by translation invariance), and $\|g_h - g\| = \int |g(x - h) - g(x)| dx \rightarrow 0$ as $h \rightarrow 0$ since g is uniformly continuous with compact support. Hence $\|f_h - f\| < 3\varepsilon$ for $|h|$ small. $\square$

Worked Example 25.2.10 (Translation continuity can fail pointwise but never in norm). Let $f = \mathbf{1}_{[0,1]} \in L^1(\mathbb{R})$. Show $\|f_h - f\|_{L^1} \rightarrow 0$ as $h \rightarrow 0$ even though $f_h(x) \not\rightarrow f(x)$ at the discontinuities $x = 0, 1$.

Recognition. An instance of Prop. 25.2.9: L^1 continuity holds despite pointwise failure on a null set.

Solution. For $0 < h < 1$, $f_h = \mathbf{1}_{[h,1+h]}$, so $f_h - f$ equals -1 on $[0, h)$ and $+1$ on $(1, 1 + h]$, hence $\|f_h - f\|_{L^1} = 2|h| \rightarrow 0$. Yet at $x = 1$: $f(1) = 1$ while $f_h(1) = \mathbf{1}_{[h,1+h]}(1) = 1$, and just past, $f(1^+) = 0$ but $f_h(1^+) = 1$ for small h — pointwise the translate jumps. *Check.* The discrepancy lives on the two intervals of total length $2|h|$, a null set in the limit; norm convergence sees only area, so it is unaffected.

Worked Example 25.2.11 (Tchebyshev's inequality). For $f \geq 0$ integrable and $\alpha > 0$ let $E_\alpha = \{f > \alpha\}$. Prove $m(E_\alpha) \leq \frac{1}{\alpha} \int f$.

Recognition. A “measure of a super-level set” bound; underestimate f by α on E_α .

Solution. On E_α we have $f > \alpha$, so $\alpha \mathbf{1}_{E_\alpha} \leq f$ everywhere. Integrating and using monotonicity, $\alpha m(E_\alpha) = \int \alpha \mathbf{1}_{E_\alpha} \leq \int f$, i.e. $m(E_\alpha) \leq \frac{1}{\alpha} \int f$. *Check.* With $f = \mathbf{1}_{[0,2]}$ and $\alpha = \frac{1}{2}$, $E_\alpha = [0, 2]$ has measure 2 and the bound reads $2 \leq \frac{1}{1/2} \cdot 2 = 4$ — valid, if loose, as Tchebyshev typically is.

Worked Example 25.2.12 (The layer-cake (distribution) formula). For $f \geq 0$ integrable on $\mathbb{R}^d$, prove $\int_{\mathbb{R}^d} f = \int_0^\infty m(E_\alpha) d\alpha$, where $E_\alpha = \{f > \alpha\}$.

Recognition. A Fubini/Tonelli-style identity rewriting an integral as the area under its distribution function; integrate the indicator of the subgraph.

Solution. Write $f(x) = \int_0^{f(x)} d\alpha = \int_0^\infty \mathbf{1}_{\{\alpha < f(x)\}} d\alpha$. Integrate in x and interchange (Tonelli, all integrands ≥ 0): $\int f = \int_{\mathbb{R}^d} \int_0^\infty \mathbf{1}_{\{\alpha < f(x)\}} d\alpha dx = \int_0^\infty \int_{\mathbb{R}^d} \mathbf{1}_{E_\alpha}(x) dx d\alpha = \int_0^\infty m(E_\alpha) d\alpha$. *Check.* For $f = c \mathbf{1}_A$, $E_\alpha = A$ for $\alpha < c$ and $\emptyset$ otherwise, so the right side is $\int_0^c m(A) d\alpha = c m(A) = \int f$. Agrees.

Worked Example 25.2.13 (Completeness used: summing an absolutely convergent L^1 series). Show that if $\sum_k \|g_k\|_{L^1} < \infty$ then $\sum_k g_k$ converges in L^1 to some g , and $\int g = \sum_k \int g_k$.

Recognition. A direct use of Riesz–Fischer completeness: absolute convergence of norms forces convergence of the series.

Solution. Let $G = \sum_k |g_k|$; by term-by-term integration (Cor. 25.1.16), $\int G = \sum_k \int |g_k| = \sum_k \|g_k\| < \infty$, so $G \in L^1$ and is finite a.e. Hence $\sum_k g_k(x)$ converges absolutely a.e. to some $g(x)$ with $|g| \leq G$, so $g \in L^1$. The partial sums $S_n = \sum_{k \leq n} g_k$ satisfy $|S_n| \leq G$ and $S_n \rightarrow g$ a.e., so the DCT gives $\|S_n - g\|_{L^1} \rightarrow 0$ and $\int S_n \rightarrow \int g$; thus $\int g = \lim_n \sum_{k \leq n} \int g_k = \sum_k \int g_k$. *Check.* This is exactly the mechanism behind the Riesz–Fischer proof (a rapidly convergent series is built and shown integrable), now packaged as a reusable completeness criterion.

Caution. L^1 convergence and pointwise convergence are logically independent. Riesz–Fischer gives completeness in norm and only a *subsequence* converging a.e. (the typewriter sequence has no a.e.-convergent full sequence). Conversely, $f_n \rightarrow f$ a.e. does not give L^1 convergence without domination (the escaping spike). Never silently upgrade one mode of convergence to the other.

Exercises (Stein, Ch.2).

1. In analogy to Proposition 2.5, prove that if f is integrable on $\mathbb{R}^d$ and $\delta > 0$, then $f(\delta x)$ converges to $f(x)$ in the L^1 -norm as $\delta \rightarrow 1$.
2. Suppose f is integrable on $(-\pi, \pi]$ and extended periodically to $\mathbb{R}$ with period 2π . Show $\int_{-\pi}^\pi f(x) dx = \int_I f(x) dx$ for any interval I of length 2π . [Hint: I lies in two consecutive intervals $(k\pi, (k+2)\pi)$.]
3. Show there exist $f \in L^1(\mathbb{R}^d)$ and a sequence $\{f_n\}$ in $L^1(\mathbb{R}^d)$ with $\|f - f_n\|_{L^1} \rightarrow 0$ but $f_n(x) \rightarrow f(x)$ for no x . [Hint: in $\mathbb{R}$, let $f_n = \mathbf{1}_{I_n}$ for a suitable sequence of intervals with $m(I_n) \rightarrow 0$.]
4. Consider $f(x) = x^{-1/2}$ for $0 < x < 1$ (else 0). For a fixed enumeration $\{r_n\}$ of $\mathbb{Q}$, let $F(x) = \sum_{n=1}^\infty 2^{-n} f(x - r_n)$. Prove F is integrable, hence the series converges for a.e. x ; yet show the series is unbounded on every interval, and any $\tilde{F}$ agreeing with F a.e. is unbounded on every interval.
5. Suppose f is integrable on $\mathbb{R}^d$. If $\delta = (\delta_1, \dots, \delta_d)$ has all $\delta_i \neq 0$ and $f^\delta(x) = f(\delta_1 x_1, \dots, \delta_d x_d)$, show f^δ is integrable with $\int_{\mathbb{R}^d} f^\delta = |\delta_1|^{-1} \cdots |\delta_d|^{-1} \int_{\mathbb{R}^d} f$.
6. Let f be measurable and finite-valued on $[0, 1]$, and suppose $|f(x) - f(y)|$ is integrable on $[0, 1] \times [0, 1]$. Show f is integrable on $[0, 1]$.

Additional exercises.

1. Compute $\|f\|_{L^1}$, $\|g\|_{L^1}$, and $\|f - g\|_{L^1}$ for $f = \mathbf{1}_{[0,2]}$ and $g = \mathbf{1}_{[1,3]}$ on $\mathbb{R}$, and verify the triangle inequality.
2. (Density) Approximate $f = \mathbf{1}_{[0,1]}$ to within ε in $L^1(\mathbb{R})$ by an explicit continuous compactly supported function, and state its L^1 distance to f .

3. Show that the polynomials are *not* dense in $L^1(\mathbb{R})$ (they are not even in $L^1(\mathbb{R})$ unless zero), while polynomials *are* dense in $L^1[0, 1]$. Explain the difference.
4. (Translation continuity) For $f = \mathbf{1}_{[0,1]}$ compute $\|f_h - f\|_{L^1}$ exactly as a function of h and confirm Proposition 2.5 ($\rightarrow 0$ as $h \rightarrow 0$).
5. Prove that if $f_n \rightarrow f$ in L^1 and $g_n \rightarrow g$ in L^1 with g_n, g uniformly bounded, then $\int f_n g_n \rightarrow \int f g$. (Hint: split and use boundedness plus L^1 convergence.)
6. (Completeness in action) Let $f_n(x) = \sum_{k=1}^n 2^{-k} \mathbf{1}_{[0, 1/k]}(x)$. Show $\{f_n\}$ is Cauchy in $L^1[0, 1]$ and identify its limit.
7. (Non-example) Exhibit a Cauchy-in- L^1 sequence whose pointwise limit fails to exist at every point, and name the theorem guaranteeing a convergent subsequence.
8. (Diagram) Draw two graphs f, g on $[0, 1]$ and shade the region whose area equals $\|f - g\|_{L^1}$; explain why pointwise closeness is neither necessary nor sufficient for small L^1 distance.
9. Using the layer-cake formula, show $\|f\|_{L^1} = \int_0^\infty m\{|f| > \alpha\} d\alpha$, and use it to recompute $\int_0^1 x^{-1/2} dx$.

25.3 Fubini's Theorem

Fubini's theorem justifies computing a d -dimensional integral by iterating lower-dimensional ones, in either order. Writing $\mathbb{R}^d = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ with $d = d_1 + d_2$, a point is (x, y) and the central objects are *slices*: the y -slice $f^y(x) = f(x, y)$ and the x -slice $f_x(y) = f(x, y)$; for a set E , $E^y = \{x : (x, y) \in E\}$ and $E_x = \{y : (x, y) \in E\}$. The subtlety the theorem must overcome is measurability: even for measurable f , a particular slice f^y need *not* be measurable — but measurability holds for *almost every* slice, and that is enough.

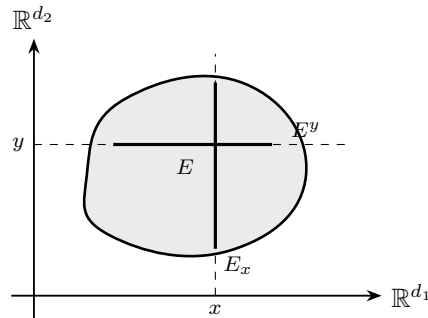


Figure 25.9. Slices of $E \subseteq \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$: the horizontal slice E^y (fix y) and the vertical slice E_x (fix x). (Schematic.)

The slice E^y is precisely a preimage under the projection: identifying $\mathbb{R}^{d_1}$ with the horizontal line at height y via $\iota_y(x) = (x, y)$, one has $E^y = \iota_y^{-1}(E)$, and the projection $\pi_2(x, y) = y$ has $\pi_2^{-1}(y)$ as that line.

$$\begin{array}{ccc}
 E^y & \hookrightarrow & E \\
 \downarrow & & \downarrow \\
 \pi_2^{-1}(y) & \xrightarrow[\iota_y]{\sim} & \mathbb{R}^d \\
 & & \downarrow \pi_2 \\
 & & \mathbb{R}^{d_2}
 \end{array}$$

Figure 25.10. The slice $E^y = \iota_y^{-1}(E)$ as the preimage of E under the inclusion of the fiber $\pi_2^{-1}(y)$ over y ; the top square is a pullback. Fubini integrates $m(E^y)$ down the projection π_2 .

Statement and proof

Theorem 25.3.1 (Fubini). Suppose f is integrable on $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$. Then for a.e. $y \in \mathbb{R}^{d_2}$: (i) the slice f^y is integrable on $\mathbb{R}^{d_1}$; (ii) $y \mapsto \int_{\mathbb{R}^{d_1}} f^y(x) dx$ is integrable on $\mathbb{R}^{d_2}$; and (iii)

$$\int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} f(x, y) dx \right) dy = \int_{\mathbb{R}^d} f.$$

By symmetry the same holds with x, y interchanged, so the iterated integrals agree in either order and both equal $\int_{\mathbb{R}^d} f$.

Proof. Assume f real (treat real/imaginary parts separately). Let $\mathcal{F}$ be the set of integrable f on $\mathbb{R}^d$ satisfying (i)–(iii); we show $L^1(\mathbb{R}^d) \subseteq \mathcal{F}$ in six steps.

Step 1 (linear combinations). If $f_1, \dots, f_N \in \mathcal{F}$, each has a null set A_k off which f_k^y is integrable; off $A = \bigcup_k A_k$ (null) every finite linear combination has integrable slices, and (iii) follows by linearity of the integral. So $\mathcal{F}$ is closed under finite linear combinations.

Step 2 (monotone limits). If $f_k \in \mathcal{F}$ with $f_k \nearrow f$ or $f_k \searrow f$ and f integrable, then $f \in \mathcal{F}$. Reduce to $f_k \nearrow$ and $f_k \geq 0$. The MCT gives $\int_{\mathbb{R}^d} f_k \rightarrow \int_{\mathbb{R}^d} f$. Off a null set A each f_k^y is integrable; for $y \notin A$, $g_k(y) = \int_{\mathbb{R}^{d_1}} f_k^y \nearrow g(y) = \int_{\mathbb{R}^{d_1}} f^y$ by MCT, and $\int g_k \rightarrow \int g$ by MCT again. Since $f_k \in \mathcal{F}$, $\int g_k = \int_{\mathbb{R}^d} f_k$, so $\int g = \int_{\mathbb{R}^d} f < \infty$; hence $g < \infty$ a.e., i.e. f^y is integrable for a.e. y , and (iii) holds: $f \in \mathcal{F}$.

Step 3 (G_δ sets of finite measure). We show $\mathbf{1}_E \in \mathcal{F}$ for E a G_δ of finite measure, building up: (a) an open cube $E = Q_1 \times Q_2$: $\int_{\mathbb{R}^{d_1}} \mathbf{1}_E(x, y) dx = |Q_1| \mathbf{1}_{Q_2}(y)$, so the iterated integral is $|Q_1| |Q_2| = |E|$; (b) E a subset of the boundary of a closed cube (a null set): both sides are 0; (c) E a finite union of closed cubes with disjoint interiors: a linear combination of cases (a), (b), so Step 1 applies; (d) E open of finite measure: write $E = \bigcup_j Q_j$ as an almost-disjoint countable union of closed cubes (Ch. 1) and let $f_k = \sum_{j \leq k} \mathbf{1}_{Q_j} \nearrow \mathbf{1}_E$; Step 2 applies; (e) E a G_δ of finite measure: $E = \bigcap_k \tilde{O}_k$; replacing by $O_k = \tilde{O}_0 \cap \bigcap_{j \leq k} \tilde{O}_j$ gives open finite-measure sets $O_k \searrow E$, so $\mathbf{1}_{O_k} \searrow \mathbf{1}_E$ and Step 2 applies.

Step 4 (null sets). If $m(E) = 0$, choose a G_δ set $G \supseteq E$ with $m(G) = 0$. By Step 3, $\int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} \mathbf{1}_G dx \right) dy = m(G) = 0$, so G^y is null for a.e. y ; since $E^y \subseteq G^y$, also E^y is null for a.e. y , giving $\mathbf{1}_E \in \mathcal{F}$.

Step 5 (measurable sets of finite measure). For such E pick a finite-measure G_δ set $G \supseteq E$ with $m(G \setminus E) = 0$. Then $\mathbf{1}_E = \mathbf{1}_G - \mathbf{1}_{G \setminus E}$, a difference of members of $\mathcal{F}$ (Steps 3, 4), so $\mathbf{1}_E \in \mathcal{F}$ by Step 1.

Step 6 (general f). Reduce to $f \geq 0$. Simple $\varphi_k \nearrow f$ exist; each φ_k is a finite combination of indicators of finite-measure sets, so $\varphi_k \in \mathcal{F}$ (Steps 5, 1), and $f \in \mathcal{F}$ by Step 2. Hence $L^1(\mathbb{R}^d) \subseteq \mathcal{F}$. $\square$

Theorem 25.3.2 (Tonelli). If $f \geq 0$ is measurable on $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$, then for a.e. y the slice f^y is measurable on $\mathbb{R}^{d_1}$, $y \mapsto \int_{\mathbb{R}^{d_1}} f^y dx$ is measurable, and

$$\int_{\mathbb{R}^{d_2}} \left(\int_{\mathbb{R}^{d_1}} f(x, y) dx \right) dy = \int_{\mathbb{R}^d} f \quad (\text{in the extended sense, allowing } +\infty).$$

Proof. Truncate: $f_k = f \mathbf{1}_{\{|(x,y)| < k, f < k\}}$ is integrable, so Fubini applies and gives a null E_k off which f_k^y is measurable. Off $E = \bigcup_k E_k$ (null) all f_k^y , hence $f^y = \lim_k f_k^y$, are measurable. Since $f_k^y \nearrow f^y$, the MCT gives $\int_{\mathbb{R}^{d_1}} f_k^y \nearrow \int_{\mathbb{R}^{d_1}} f^y$, and a further MCT plus Fubini's identity $\int (\int f_k^y dx) dy = \int_{\mathbb{R}^d} f_k$ together with $\int_{\mathbb{R}^d} f_k \nearrow \int_{\mathbb{R}^d} f$ yields the claim. $\square$

Method (Using Tonelli then Fubini to justify iteration). *Recognition trigger.* You must compute $\int_{\mathbb{R}^d} f$ by iteration but f changes sign (so you cannot iterate blindly).

1. Apply *Tonelli* to $|f| \geq 0$: compute (or estimate) an iterated integral of $|f|$. All integrands are nonnegative, so this is always legal.
2. If that iterated integral of $|f|$ is *finite*, then f is integrable.
3. Now apply *Fubini* to f itself: the iterated integrals of f exist, agree in either order, and equal $\int_{\mathbb{R}^d} f$.

Skipping step 1 is the classic error: Exercise 17 exhibits f whose two iterated integrals differ (0 vs. s), precisely because $\iint |f| = \infty$.

Corollary 25.3.3 (Measure by slicing). *If $E \subseteq \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ is measurable, then E^y is measurable for a.e. y , $y \mapsto m(E^y)$ is measurable, and $m(E) = \int_{\mathbb{R}^{d_2}} m(E^y) dy$.*

Proposition 25.3.4 (Product sets). *If $E_1 \subseteq \mathbb{R}^{d_1}$ and $E_2 \subseteq \mathbb{R}^{d_2}$ are measurable, then $E_1 \times E_2$ is measurable and $m(E_1 \times E_2) = m(E_1)m(E_2)$ (with the convention that the product is 0 if either factor is null). In general $m^*(E_1 \times E_2) \leq m^*(E_1)m^*(E_2)$.*

Corollary 25.3.5 (The integral is the measure of the subgraph). *For $f \geq 0$ on $\mathbb{R}^d$ let $A = \{(x, y) \in \mathbb{R}^d \times \mathbb{R} : 0 \leq y \leq f(x)\}$. Then f is measurable iff A is measurable in $\mathbb{R}^{d+1}$, and in that case $\int_{\mathbb{R}^d} f = m(A)$.*

Proof. If f is measurable then $F(x, y) = y - f(x)$ is measurable on $\mathbb{R}^{d+1}$ (Cor. 3.7), so $A = \{y \geq 0\} \cap \{F \leq 0\}$ is measurable. Conversely, the slice $A_x = [0, f(x)]$ has $m(A_x) = f(x)$, measurable by Cor. 25.3.3; and $m(A) = \int m(A_x) dx = \int f$. $\square$

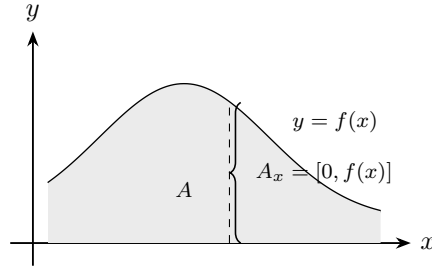


Figure 25.11. $\int f = m(A)$, the measure of the subgraph; the vertical slice $A_x = [0, f(x)]$ has length $f(x)$, recovering the integral by Corollary 25.3.3. (Schematic.)

Worked Example 25.3.6 (Computing the volume of the unit ball v_d by slicing). Using $\int f = m(\text{subgraph})$ and slicing, derive $v_2 = \pi$ and the recursion $v_d = 2v_{d-1} \int_0^1 (1 - x^2)^{(d-1)/2} dx$.

Recognition. A Corollary 25.3.3 computation: integrate the measure of slices of the ball.

Solution. For $d = 2$ the unit disc is the subgraph picture symmetrized: by Cor. 25.3.5 applied to the upper boundary $y = \sqrt{1 - x^2}$ over $[-1, 1]$ and doubling, $v_2 = 2 \int_{-1}^1 \sqrt{1 - x^2} dx = 2 \cdot \frac{\pi}{2} = \pi$. In general slice the unit ball B^d by the last coordinate $x = x_d \in [-1, 1]$: the slice is a $(d - 1)$ -ball of radius $\sqrt{1 - x^2}$, of measure $v_{d-1}(1 - x^2)^{(d-1)/2}$, so $v_d = \int_{-1}^1 v_{d-1}(1 - x^2)^{(d-1)/2} dx = 2v_{d-1} \int_0^1 (1 - x^2)^{(d-1)/2} dx$. *Check.* The recursion with $v_1 = 2$ gives $v_2 = 2 \cdot 2 \cdot \frac{\pi}{4} = \pi$, and the closed form $v_d = \pi^{d/2} / \Gamma(d/2 + 1)$ yields $v_3 = \frac{4}{3}\pi$, the familiar ball volume.

Worked Example 25.3.7 (The hypotheses bite: an iterated integral that depends on order). With $f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$ on $(0, 1)^2$, show the two iterated integrals differ, and identify why Fubini does not apply.

Recognition. A NON-example probing the integrability hypothesis; compute both iterations and check $\iint |f|$.

Solution. Since $\partial_y \frac{-y}{x^2 + y^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$, $\int_0^1 f dy = \left[\frac{-y}{x^2 + y^2} \right]_0^1 = \frac{-1}{x^2 + 1}$, so $\int_0^1 \int_0^1 f dy dx = -\int_0^1 \frac{dx}{1 + x^2} = -\frac{\pi}{4}$. By the antisymmetry $f(x, y) = -f(y, x)$, reversing the order gives $+\frac{\pi}{4}$. They differ, so $f \notin L^1$. *Check.* Indeed $\int_0^1 \int_0^1 |f| dy dx = \infty$ (the singularity at the origin is non-integrable), so Tonelli's finiteness test fails and Fubini is inapplicable — exactly as the method warns.

Worked Example 25.3.8 (A clean Tonelli computation). Evaluate $\iint_{[0,1]^2} \frac{1}{1 - xy} dx dy$ and recognize the value.

Recognition. A nonnegative integrand: Tonelli lets us expand in a series and integrate term by term in any order.

Solution. For $0 \leq xy < 1$, $\frac{1}{1 - xy} = \sum_{n \geq 0} (xy)^n$, all terms ≥ 0 , so Tonelli (or Cor. 25.1.16) permits $\iint \frac{dx dy}{1 - xy} = \sum_{n \geq 0} \int_0^1 x^n dx \int_0^1 y^n dy = \sum_{n \geq 0} \frac{1}{(n+1)^2} = \sum_{m \geq 1} \frac{1}{m^2} = \frac{\pi^2}{6}$. *Check.* The integrand is positive and singular only at the corner $(1, 1)$; the convergent sum $\zeta(2) = \pi^2/6$ confirms integrability and the value.

Worked Example 25.3.9 (Convolution stays in L^1 : $\|f * g\|_{L^1} \leq \|f\|_{L^1} \|g\|_{L^1}$). For $f, g \in L^1(\mathbb{R}^d)$ show $f * g$ is defined a.e., lies in L^1 , and satisfies the norm bound (Young's inequality, the L^1 case).

Recognition. A Tonelli-then-Fubini argument on the double integral defining the convolution norm.

Solution. The function $(x, y) \mapsto f(x - y)g(y)$ is measurable on $\mathbb{R}^{2d}$ (Cor. 3.7 and Prop. 3.9). By Tonelli, $\iint |f(x - y)| |g(y)| \, dy \, dx = \int |g(y)| \left(\int |f(x - y)| \, dx \right) dy = \int |g(y)| \|f\|_{L^1} \, dy = \|f\|_{L^1} \|g\|_{L^1} < \infty$, using translation invariance for the inner integral. Finiteness shows $f(x - y)g(y)$ is integrable in y for a.e. x (so $f * g$ is defined a.e.) and integrable on $\mathbb{R}^{2d}$; then $\|f * g\|_{L^1} = \int \left| \int f(x - y)g(y) \, dy \right| dx \leq \iint |f(x - y)g(y)| \, dy \, dx = \|f\|_{L^1} \|g\|_{L^1}$. *Check.* For $f, g \geq 0$ the triangle inequality is an equality, so the bound is sharp.

Worked Example 25.3.10 (The graph of a measurable function is null). Show $\Gamma = \{(x, y) \in \mathbb{R}^d \times \mathbb{R} : y = f(x)\}$ has measure zero in $\mathbb{R}^{d+1}$ for any measurable f .

Recognition. A Corollary 25.3.3 argument: every vertical slice of the graph is a single point.

Solution. Γ is measurable (it is $\{F = 0\}$ for $F(x, y) = y - f(x)$, measurable by Cor. 3.7). Its vertical slice is $\Gamma_x = \{f(x)\}$, a single point, so $m(\Gamma_x) = 0$ for every x . By Cor. 25.3.3, $m(\Gamma) = \int_{\mathbb{R}^d} m(\Gamma_x) \, dx = \int_{\mathbb{R}^d} 0 \, dx = 0$. *Check.* The line $y = x$ in $\mathbb{R}^2$ has zero area, consistent with Γ always being negligible regardless of how wild f is.

Worked Example 25.3.11 (A discrete Fubini: summing a doubly-indexed array). Show that for $c_{jk} \geq 0$, $\sum_j \sum_k c_{jk} = \sum_k \sum_j c_{jk}$, and give a signed array where the two orders disagree.

Recognition. Tonelli/Fubini on counting measure $\mathbb{N} \times \mathbb{N}$; sums are integrals against the counting measure.

Solution. Identify c with a function on $\mathbb{N} \times \mathbb{N}$ and use counting measure; all $c_{jk} \geq 0$, so Tonelli permits interchanging the iterated sums: $\sum_j \sum_k c_{jk} = \sum_k \sum_j c_{jk}$ (both equal the total mass). For the signed counterexample, take the array with $c_{jj} = 1$, $c_{j,j+1} = -1$, else 0: then $\sum_k c_{jk} = 0$ for each j , so $\sum_j \sum_k c_{jk} = 0$, whereas $\sum_j c_{jk} = 1$ for $k = 1$ (only c_{11}) and 0 otherwise, giving $\sum_k \sum_j c_{jk} = 1$. *Check.* The discrepancy $0 \neq 1$ mirrors Exercise 17; here $\sum_{j,k} |c_{jk}| = \infty$, so the absolute (Tonelli) hypothesis fails exactly as required.

Caution. Fubini requires $f \in L^1(\mathbb{R}^d)$ globally; equality of the two iterated integrals is a *conclusion*, not a hypothesis. To check the hypothesis, run *Tonelli on $|f|$* first. Also: “ E^y measurable for every y ” does not make E measurable (the set $[0, 1] \times N$ with N nonmeasurable has every E^y measurable yet E is not), so you cannot reverse Corollary 25.3.3.

Exercises (Stein, Ch.2).

- Let $\Gamma = \{(x, y) \in \mathbb{R}^d \times \mathbb{R} : y = f(x)\}$ with f measurable on $\mathbb{R}^d$. Show Γ is a measurable subset of $\mathbb{R}^{d+1}$ and $m(\Gamma) = 0$.
- Give an example of two measurable sets $A, B \subseteq \mathbb{R}^2$ such that $A + B$ is not measurable. [Hint: in $\mathbb{R}^2$ take $A = \{0\} \times [0, 1]$ and $B = N \times \{0\}$ with N nonmeasurable.]
- We saw that $m(B) = v_d r^d$ for a ball B of radius r in $\mathbb{R}^d$, with $v_d = m(B_1)$. Evaluate v_d : (a) for $d = 2$, prove via Corollary 3.8 that $v_2 = 2 \int_{-1}^1 (1 - x^2)^{1/2} dx$, hence $v_2 = \pi$; (b) show $v_d = 2v_{d-1} \int_0^1 (1 - x^2)^{(d-1)/2} dx$; (c) conclude $v_d = \pi^{d/2} / \Gamma(d/2 + 1)$.
- Define f on $\mathbb{R}^2$ by $f(x, y) = a_n$ if $n \leq x < n+1$ and $n \leq y < n+1$ ($n \geq 0$); $f(x, y) = -a_n$ if $n \leq x < n+1$ and $n+1 \leq y < n+2$ ($n \geq 0$); else $f(x, y) = 0$. Here $a_n = \sum_{k \leq n} b_k$ with $b_k > 0$ and $\sum_{k=0}^{\infty} b_k = s < \infty$. (a) Verify each slice f^y, f_x is integrable, that $\int f_x(y) \, dy = 0$ for all x , and hence $\int (\int f(x, y) \, dy) \, dx = 0$. (b) Show $\int f^y(x) \, dx = a_0$ for $0 \leq y < 1$ and $= a_n - a_{n-1}$ for $n \leq y < n+1$ ($n \geq 1$), so $y \mapsto \int f^y \, dx$ is integrable on $(0, \infty)$ with $\int (\int f(x, y) \, dx) \, dy = s$. (c) Note $\iint |f(x, y)| \, dx \, dy = \infty$.
- (Borel slices) Suppose E is a Borel set in $\mathbb{R}^2$. Prove that for every y , the slice E^y is a Borel set in $\mathbb{R}$. [Hint: the collection $\mathcal{C}$ of $E \subseteq \mathbb{R}^2$ with every E^y Borel is a σ -algebra containing the open sets.]
- Suppose f, g are measurable on $\mathbb{R}^d$. (a) Prove $f(x - y)g(y)$ is measurable on $\mathbb{R}^{2d}$. (b) If f, g are integrable on $\mathbb{R}^d$, show $f(x - y)g(y)$ is integrable on $\mathbb{R}^{2d}$. (c) Show $(f * g)(x) = \int f(x - y)g(y) \, dy$ is well-defined for a.e. x . (d) Show $f * g$ is integrable when f, g are, with $\|f * g\|_{L^1} \leq \|f\|_{L^1} \|g\|_{L^1}$, equality when $f, g \geq 0$.

Problems (Stein, Ch.2).

1. (Linear changes of variable) For a measurable $E \subseteq \mathbb{R}^d$ and a linear L of $\mathbb{R}^d$, $L(E)$ is measurable, and $m(L(E)) = |\det L| m(E)$; in particular m is rotation-invariant. Prove this with Fubini: (a) for $d = 2$ and L strictly upper triangular ($x' = x + ay$, $y' = y$), $m(L(E)) = \iint \mathbf{1}_E(x - ay, y) dx dy = m(E)$ by translation invariance; (b) similarly for strictly lower triangular L ; in general write $L = L_1 \Delta L_2$ with L_j strictly triangular and Δ diagonal, giving $m(L(E)) = |\det L| m(E)$.
2. There is an ordering $\prec$ of $\mathbb{R}$ such that for each y the set $\{x : x \prec y\}$ is at most countable. The existence depends on the continuum hypothesis. [Hint: let $\prec$ be a well-ordering of $\mathbb{R}$ and consider $X = \{y : \{x : x \prec y\} \text{ uncountable}\}$; if nonempty, take its least element and apply the continuum hypothesis.] Use such an ordering to build a nonmeasurable $E \subseteq [0, 1]^2$ whose slices satisfy $m(E^y) = 0$ and $m(E_x) = 1$ for all x, y , contradicting Corollary 3.3 (so E cannot be measurable).

Additional exercises.

1. Evaluate $\iint_{[0,1]^2} xy e^{xy} dx dy$ by iterating (the integrand is bounded, so Fubini applies directly), reducing to a one-variable integral.
2. Use Tonelli to compute $\iint_{[0,1]^2} \frac{1}{(1+x+y)^2} dx dy$.
3. (Order matters) For $f(x, y) = \frac{x-y}{(x+y)^3}$ on $(0, 1)^2$, compute both iterated integrals, show they disagree, and verify $\iint |f| = \infty$ so that Fubini does not apply.
4. (Slicing) Use Corollary 3.3 to compute the area of the region $\{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x^2\}$ by integrating the lengths of vertical slices.
5. Prove that if $E \subseteq \mathbb{R}^2$ has $m(E^y) = 0$ for a.e. y , then $m(E) = 0$. Where is measurability of E used?
6. (Product measure) Show $m([0, 1] \times C) = 0$ where C is the Cantor set, two ways: by Proposition 3.6 and by slicing.
7. (Convolution) For $f = g = \mathbf{1}_{[0,1]}$ compute $f * g$ explicitly and verify $\|f * g\|_{L^1} = \|f\|_{L^1} \|g\|_{L^1} = 1$ (the nonnegative equality case).
8. (Non-example / diagram) Sketch the set $E = [0, 1] \times N$ (with N nonmeasurable) and explain pictorially why every horizontal slice E^y is measurable while E is not, so Corollary 3.3 cannot be reversed.
9. Prove the discrete Fubini failure: construct a doubly indexed array c_{jk} with $\sum_j \sum_k c_{jk} \neq \sum_k \sum_j c_{jk}$ and identify the divergent $\sum |c_{jk}|$.

25.4 Approximations to the Identity and a Fourier Inversion Formula

The Fourier transform on L^1 turns convolution into multiplication and connects to a family of *good kernels* (approximations to the identity). We prove that $\hat{f}$ is continuous and bounded, establish the L^1 inversion formula via the modulated Gaussian kernel, and record the multiplication formula. Along the way we meet good kernels concretely and, as a self-similar counterpoint, the Cantor set and its devil's-staircase function — the canonical examples of a null set carrying a nontrivial measure-theoretic structure and of singular (yet continuous, monotone) integration.

The Fourier transform of an L^1 function

Definition 25.4.1. For $f \in L^1(\mathbb{R}^d)$ the *Fourier transform* is $\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x \cdot \xi} dx$. The (formal) inversion is $f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi$.

Proposition 25.4.2. If $f \in L^1(\mathbb{R}^d)$ then $\hat{f}$ is bounded and continuous, with $\sup_{\xi} |\hat{f}(\xi)| \leq \|f\|_{L^1}$, and $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$ (Riemann–Lebesgue).

Proof. Since $|f(x)e^{-2\pi i x \cdot \xi}| = |f(x)|$, the defining integral converges with $|\hat{f}(\xi)| \leq \int |f| = \|f\|$. For continuity, fix ξ_0 ; as $\xi \rightarrow \xi_0$, $f(x)e^{-2\pi i x \cdot \xi} \rightarrow f(x)e^{-2\pi i x \cdot \xi_0}$ pointwise and is dominated by $|f| \in L^1$, so the DCT gives $\hat{f}(\xi) \rightarrow \hat{f}(\xi_0)$. For decay, write $\hat{f}(\xi) = \frac{1}{2} \int [f(x) - f(x - \xi')] e^{-2\pi i x \cdot \xi} dx$ with $\xi' = \frac{\xi}{2|\xi|^2}$ (using $e^{-2\pi i \xi \cdot \xi'} = e^{-\pi i} = -1$), so $|\hat{f}(\xi)| \leq \frac{1}{2} \|f - f_{\xi'}\|_{L^1} \rightarrow 0$ as $|\xi| \rightarrow \infty$ by Prop. 25.2.9. $\square$

In general $\hat{f} \notin L^1$, so the inversion integral is problematic; the next results evade this by inserting a Gaussian regularizer and letting it tend to a Dirac-like kernel.

Good kernels (approximations to the identity)

Definition 25.4.3. A family $\{K_\delta\}_{\delta>0}$ on $\mathbb{R}^d$ is an *approximation to the identity* (good kernel) if (i) $\int_{\mathbb{R}^d} K_\delta = 1$; (ii) $\int_{\mathbb{R}^d} |K_\delta| \leq C$ uniformly in δ ; and (iii) for each $\eta > 0$, $\int_{|y|>\eta} |K_\delta(y)| dy \rightarrow 0$ as $\delta \rightarrow 0$.

The Gaussian kernel $K_\delta(x) = \delta^{-d/2} e^{-\pi|x|^2/\delta}$ is the model example. As $\delta \rightarrow 0$ it concentrates all its unit mass near the origin, behaving like a Dirac spike, so convolution $f * K_\delta$ smooths f while converging back to f .

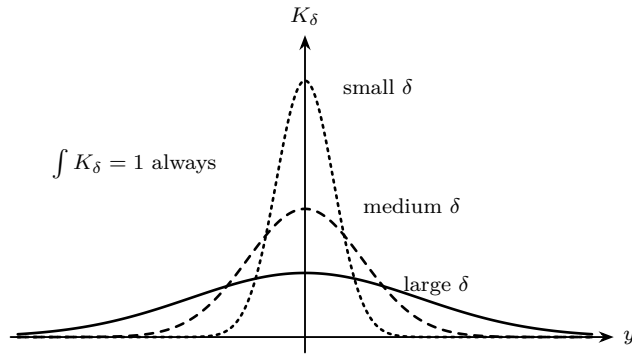


Figure 25.12. As $\delta \rightarrow 0$ the Gaussian kernel K_δ concentrates its unit mass at 0 (taller and narrower, but always of area 1), approximating the Dirac delta.

The L^1 inversion formula

Lemma 25.4.4 (Multiplication formula). *If $f, g \in L^1(\mathbb{R}^d)$ then $\int \hat{f}(\xi)g(\xi) d\xi = \int f(y)\hat{g}(y) dy$.*

Proof. The function $F(\xi, y) = g(\xi)f(y)e^{-2\pi i \xi \cdot y}$ is measurable on $\mathbb{R}^{2d}$, and by Tonelli $\iint |F| = \int |g| \int |f| < \infty$, so $F \in L^1(\mathbb{R}^{2d})$. Fubini lets us iterate in either order: integrating in y first yields $\int \hat{f}(\xi)g(\xi) d\xi$; integrating in ξ first yields $\int f(y)\hat{g}(y) dy$. They are equal. $\square$

Theorem 25.4.5 (Fourier inversion in L^1). *Suppose $f \in L^1(\mathbb{R}^d)$ and also $\hat{f} \in L^1(\mathbb{R}^d)$. Then for a.e. x ,*

$$f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi.$$

Proof. Fix x, δ and take $g(\xi) = e^{-\pi\delta|\xi|^2} e^{2\pi i x \cdot \xi}$, the modulated Gaussian; an elementary computation gives $\hat{g}(y) = \delta^{-d/2} e^{-\pi|x-y|^2/\delta} = K_\delta(x-y)$, a good kernel. The multiplication formula (Lemma 25.4.4) reads

$$\int \hat{f}(\xi) e^{-\pi\delta|\xi|^2} e^{2\pi i x \cdot \xi} d\xi = \int f(y) K_\delta(x-y) dy = (f * K_\delta)(x).$$

As $\delta \rightarrow 0$ the left side $\rightarrow \int \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi$ by the DCT (since $\hat{f} \in L^1$ dominates). The right side converges to f in L^1 : writing $\Delta_\delta(x) = (f * K_\delta)(x) - f(x) = \int (f(x-y) - f(x)) K_\delta(y) dy$ (using $\int K_\delta = 1$), Fubini gives $\|\Delta_\delta\|_{L^1} \leq \int \|f_y - f\|_{L^1} K_\delta(y) dy$, and splitting at $|y| = \eta$ (with $\|f_y - f\| < \varepsilon$ for $|y| < \eta$ by Prop. 25.2.9, and $\|f_y - f\| \leq 2\|f\|$ elsewhere) and using kernel property (iii) shows $\|\Delta_\delta\| \leq 2\varepsilon$ for δ small. So $f * K_\delta \rightarrow f$ in L^1 ; by Cor. 25.2.4 a subsequence converges a.e., proving the formula a.e. $\square$

Corollary 25.4.6 (Uniqueness). *If $f \in L^1(\mathbb{R}^d)$ and $\hat{f} \equiv 0$, then $f = 0$ a.e.*

$$\begin{array}{ccc} \{f \in L^1 : \hat{f} \in L^1\} & \xrightarrow{\quad \wedge \quad} & L^1 \\ & \searrow \text{id} \quad \swarrow \vee & \\ & L^1 & \end{array}$$

Figure 25.13. Fourier inversion on the class where $\hat{f} \in L^1$: the inverse transform $\vee$ (defined by the inversion integral, dashed since it is built only here) recovers f a.e., so $\vee \circ \wedge = \text{id}$ and the triangle commutes.

Worked Example 25.4.7 (The Gaussian is its own Fourier transform). Verify that $f(x) = e^{-\pi x^2}$ on $\mathbb{R}$ satisfies $\hat{f} = f$, and check the inversion formula holds.

Recognition. A direct Fourier-transform computation completing the square; the prototype good kernel at $\delta = 1$.

Solution. $\hat{f}(\xi) = \int_{\mathbb{R}} e^{-\pi x^2} e^{-2\pi i x \xi} dx = \int_{\mathbb{R}} e^{-\pi(x+i\xi)^2} e^{-\pi \xi^2} dx = e^{-\pi \xi^2} \int_{\mathbb{R}} e^{-\pi u^2} du = e^{-\pi \xi^2}$, since $\int_{\mathbb{R}} e^{-\pi u^2} du = 1$ and the contour shift is justified by analyticity/decay. Thus $\hat{f} = f$. *Check.* f and $\hat{f}$ are both in L^1 , so Theorem 25.4.5 applies and indeed $\int \hat{f}(\xi) e^{2\pi i x \xi} d\xi = \widehat{\widehat{f}}(\cdots) = f(x)$ everywhere (here f is already continuous). The fixed-point property anchors the Gaussian as the canonical approximation to the identity.

Worked Example 25.4.8 (Convolution theorem $\widehat{f * g} = \hat{f} \hat{g}$). For $f, g \in L^1(\mathbb{R}^d)$ prove $\widehat{f * g} = \hat{f} \hat{g}$ pointwise.

Recognition. A Fubini computation interchanging the convolution and Fourier integrals.

Solution. By Wex on Young's inequality, $f * g \in L^1$, so $\widehat{f * g}$ is defined. Then

$$\widehat{f * g}(\xi) = \int \left(\int f(x-y)g(y) dy \right) e^{-2\pi i x \cdot \xi} dx = \int g(y) \left(\int f(x-y) e^{-2\pi i x \cdot \xi} dx \right) dy,$$

the interchange justified because $(x, y) \mapsto f(x-y)g(y)e^{-2\pi i x \cdot \xi}$ is in $L^1(\mathbb{R}^{2d})$ (Tonelli, modulus). The inner integral is $\hat{f}(\xi)e^{-2\pi i y \cdot \xi}$ (substitute $x \mapsto x+y$), so $\widehat{f * g}(\xi) = \hat{f}(\xi) \int g(y)e^{-2\pi i y \cdot \xi} dy = \hat{f}(\xi)\hat{g}(\xi)$. *Check.* Convolution (a smoothing) becomes multiplication (a damping) on the Fourier side — the structural reason kernels work.

$$\begin{array}{ccc} L^1 \times L^1 & \xrightarrow{\quad * \quad} & L^1 \\ \wedge \times \wedge \downarrow & & \downarrow \wedge \\ \widehat{L^1} \times \widehat{L^1} & \xrightarrow{\quad \cdot \quad} & \widehat{L^1} \end{array}$$

Figure 25.14. The convolution theorem $\widehat{f * g} = \hat{f} \hat{g}$ as a commuting square: the Fourier transform $\wedge$ carries convolution $*$ to pointwise multiplication $\cdot$.

Worked Example 25.4.9 (No L^1 convolution identity exists). Show there is no $I \in L^1(\mathbb{R}^d)$ with $f * I = f$ for all $f \in L^1$.

Recognition. A NON-existence result via the Fourier transform and Riemann–Lebesgue.

Solution. If such I existed, then $\hat{f} \hat{I} = \widehat{f * I} = \hat{f}$ for all f . Choosing f with $\hat{f}(\xi) \neq 0$ on a neighborhood of any prescribed ξ forces $\hat{I}(\xi) = 1$ for all ξ . But $I \in L^1$ implies $\hat{I}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$ by Riemann–Lebesgue (Prop. 25.4.2), contradicting $\hat{I} \equiv 1$. So no such I exists. *Check.* The “identity” for convolution is the Dirac delta, which is not a function in L^1 — exactly why we use approximations K_δ instead.

The Cantor set and the devil's staircase

Definition 25.4.10. The *Cantor set* $C = \bigcap_n C_n$, where $C_0 = [0, 1]$ and C_{n+1} removes the open middle third of each interval of C_n , so C_n is a union of 2^n intervals of length 3^{-n} . Thus $m(C) = \lim_n (2/3)^n = 0$: C is null, yet uncountable (it is in bijection with $\{0, 1\}^{\mathbb{N}}$ via base-3 digits 0, 2).

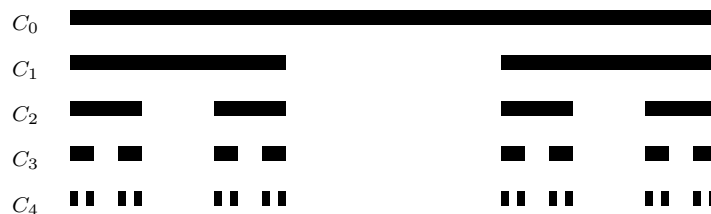


Figure 25.15. Construction of the Cantor set: C_n has 2^n intervals of length 3^{-n} ; $m(C_n) = (2/3)^n \rightarrow 0$. The limit C is null but uncountable, with Hausdorff dimension $\log 2 / \log 3$.

Definition 25.4.11. The *Cantor-Lebesgue function* (devil's staircase) $\mathcal{C} : [0, 1] \rightarrow [0, 1]$ is the uniform limit of C_n , where C_n is constant on each removed middle third and linear on C_n . It is continuous and nondecreasing with $\mathcal{C}(0) = 0$, $\mathcal{C}(1) = 1$, yet $\mathcal{C}' = 0$ a.e. (on the dense open complement of C). Hence $\int_0^1 \mathcal{C}' dx = 0 \neq 1 = \mathcal{C}(1) - \mathcal{C}(0)$: the fundamental theorem of calculus *fails* for $\mathcal{C}$, which is continuous and monotone but not absolutely continuous — all its increase is concentrated on the null set C .

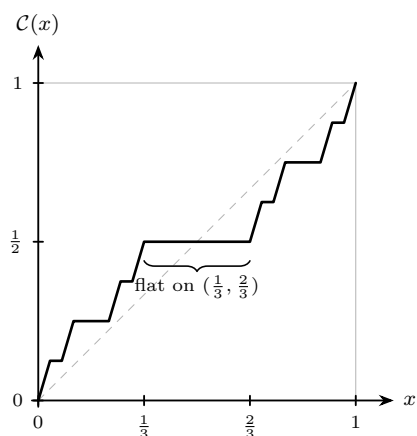


Figure 25.16. The devil's staircase $\mathcal{C}$ (level-3 approximation): continuous, nondecreasing, $\mathcal{C}' = 0$ a.e. (flat on every deleted middle third), yet climbs from 0 to 1. FTC fails: $\int_0^1 \mathcal{C}' = 0 \neq 1$.

Worked Example 25.4.12 (Hausdorff dimension of the Cantor set). Show the Cantor set C has Hausdorff dimension $s = \log 2 / \log 3$ by the self-similarity balance.

Recognition. A self-similar fractal dimension count: balance the number of pieces against their scaling.

Solution. C is the union of 2 copies of itself scaled by $1/3$ (the left and right thirds). For a self-similar set with N copies at ratio r , the dimension s satisfies $Nr^s = 1$, i.e. $2 \cdot (1/3)^s = 1$, giving $3^s = 2$ and $s = \log 2 / \log 3 \approx 0.6309$. *Check.* Covering C by the 2^n level- n intervals of length 3^{-n} gives an s -dimensional content $\sum 2^n (3^{-n})^s = 2^n 3^{-ns} = (2 \cdot 3^{-s})^n = 1^n = 1$, neither 0 nor ∞ — the hallmark of the correct dimension s .

Worked Example 25.4.13 (Integrating against the Cantor measure). Let μ be the Cantor-Lebesgue measure ($d\mu = d\mathcal{C}$). Compute $\int_0^1 x d\mu(x)$ using self-similarity.

Recognition. A self-similar (renewal) computation for a singular measure built from the staircase.

Solution. By symmetry of C about $x = 1/2$, the mean is $m = \int x d\mu = 1/2$. To confirm via self-similarity: μ splits as half on the left third (image of μ under $x \mapsto x/3$) and half on the right third ($x \mapsto x/3 + 2/3$). Thus $m = \frac{1}{2} \int (x/3) d\mu + \frac{1}{2} \int (x/3 + 2/3) d\mu = \frac{1}{2}(\frac{m}{3}) + \frac{1}{2}(\frac{m}{3} + \frac{2}{3}) = \frac{m}{3} + \frac{1}{3}$, so $m - \frac{m}{3} = \frac{1}{3}$, $\frac{2m}{3} = \frac{1}{3}$, $m = \frac{1}{2}$. *Check.* The symmetry answer $1/2$ matches the self-similar fixed-point computation, and μ is a probability measure ($\mu([0, 1]) = \mathcal{C}(1) - \mathcal{C}(0) = 1$).

A glance at orthogonality

The L^1 theory has an L^2 companion (developed fully later) in which the inner product $\langle f, g \rangle = \int f \bar{g}$ makes L^2 a Hilbert space. The single most useful fact there is orthogonal projection onto a closed subspace: every f

decomposes uniquely as $f = Pf + (f - Pf)$ with Pf in the subspace and $f - Pf$ orthogonal to it, minimizing the distance from f to the subspace.

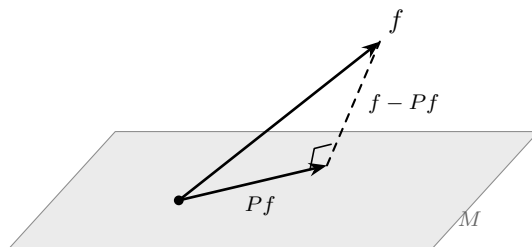


Figure 25.17. Orthogonal projection onto a closed subspace: Pf is the nearest point of M to f , and the residual $f - Pf$ is orthogonal to M (right angle at the foot). (Schematic.)

Worked Example 25.4.14 (Best L^2 approximation by a constant is the mean). On $[0, 1]$ find the constant c minimizing $\|f - c\|_{L^2}^2$ and identify it as an orthogonal projection.

Recognition. A projection onto the one-dimensional subspace of constants; the minimizer is the orthogonal foot.

Solution. Minimize $\Phi(c) = \int_0^1 (f - c)^2 dx = \int f^2 - 2c \int f + c^2$. Then $\Phi'(c) = -2 \int f + 2c = 0$ gives $c = \int_0^1 f =: \bar{f}$, the mean. This is the projection $Pf = \bar{f}$ onto the constants $M = \{c \cdot 1\}$: the residual $f - \bar{f}$ is orthogonal to 1, since $\langle f - \bar{f}, 1 \rangle = \int (f - \bar{f}) = \int f - \bar{f} = 0$. *Check.* With $f(x) = x$, $\bar{f} = 1/2$ and $\int_0^1 (x - 1/2)^2 = 1/12$, smaller than e.g. $\int_0^1 (x - 0)^2 = 1/3$ — the mean genuinely minimizes.

Caution. The devil's staircase is the standing warning against treating “ F' exists a.e.” as license to write $F(b) - F(a) = \int_a^b F'$. That formula needs *absolute continuity*; $\mathcal{C}$ is continuous and monotone yet singular, with $\int \mathcal{C}' = 0 \neq 1$. On the Fourier side, the parallel trap is assuming $\hat{f} \in L^1$ for general $f \in L^1$ — Riemann–Lebesgue gives only decay, not integrability, so inversion needs $\hat{f} \in L^1$ as a separate hypothesis.

Exercises (Stein, Ch.2).

- Suppose F is a closed set in $\mathbb{R}$ whose complement has finite measure, and let $\delta(x) = d(x, F) = \inf\{|x - y| : y \in F\}$. Consider $I(x) = \int_{\mathbb{R}} \frac{\delta(y)}{|x - y|^2} dy$. (a) Prove δ is continuous by showing the Lipschitz bound $|\delta(x) - \delta(y)| \leq |x - y|$. (b) Show $I(x) = \infty$ for each $x \notin F$. (c) Show $I(x) < \infty$ for a.e. $x \in F$. [Hint for (c): investigate $\int_F I(x) dx$.]
- (Fourier transform; cf. Exercise on convolution.) For $f \in L^1(\mathbb{R}^d)$ define $\hat{f}(\xi) = \int f(x) e^{-2\pi i x \cdot \xi} dx$. Check $\hat{f}$ is bounded and continuous, and prove $\widehat{f * g}(\xi) = \hat{f}(\xi) \hat{g}(\xi)$ for $f, g \in L^1$.
- Prove that if $f \in L^1(\mathbb{R}^d)$ and $\hat{f}(\xi) = \int f(x) e^{-2\pi i x \cdot \xi} dx$, then $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$ (the Riemann–Lebesgue lemma). [Hint: write $\hat{f}(\xi) = \frac{1}{2} \int [f(x) - f(x - \xi')] e^{-2\pi i x \cdot \xi} dx$ with $\xi' = \frac{\xi}{2|\xi|^2}$ and use Proposition 2.5.]
- As an application of the Fourier transform, show there is no $I \in L^1(\mathbb{R}^d)$ with $f * I = f$ for all $f \in L^1(\mathbb{R}^d)$.
- Consider $(f * g)(x) = \int_{\mathbb{R}^d} f(x - y) g(y) dy$. (a) Show $f * g$ is uniformly continuous when f is integrable and g bounded. (b) If in addition g is integrable, prove $(f * g)(x) \rightarrow 0$ as $|x| \rightarrow \infty$.
- Show that for each $\varepsilon > 0$ the function $F(\xi) = (1 + |\xi|^2)^{-\varepsilon}$ is the Fourier transform of an L^1 function. [Hint: with $K_\delta(x) = e^{-\pi|x|^2/\delta} \delta^{-d/2}$ consider $f(x) = \int_0^\infty K_\delta(x) e^{-\pi\delta} \delta^{\varepsilon-1} d\delta$; use Fubini to show $f \in L^1$ and $\hat{f}(\xi) = \int_0^\infty e^{-\pi\delta|\xi|^2} e^{-\pi\delta} \delta^{\varepsilon-1} d\delta = \pi^{-\varepsilon} \Gamma(\varepsilon) (1 + |\xi|^2)^{-\varepsilon}$, with $\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt$.]

Problems (Stein, Ch.2).

- If f is integrable on $[0, 2\pi]$, then $\int_0^{2\pi} f(x) e^{-inx} dx \rightarrow 0$ as $|n| \rightarrow \infty$. Deduce that for a measurable $E \subseteq [0, 2\pi]$, $\int_E \cos^2(nx + u_n) dx \rightarrow \frac{m(E)}{2}$ as $n \rightarrow \infty$, for any sequence $\{u_n\}$. [Hint: use the Riemann–Lebesgue lemma.]

2. (Cantor–Lebesgue theorem) If $\sum_{n=0}^{\infty} (a_n \cos nx + b_n \sin nx)$ converges for x in a set of positive measure (in particular for all x), then $a_n \rightarrow 0$ and $b_n \rightarrow 0$ as $n \rightarrow \infty$. [Hint: the terms $A_n(x) = a_n \cos nx + b_n \sin nx \rightarrow 0$ uniformly on a set E of positive measure.]
3. (Convergence in measure) A sequence $\{f_k\}$ of measurable functions on $\mathbb{R}^d$ is Cauchy in measure if for every $\varepsilon > 0$, $m\{x : |f_k(x) - f_\ell(x)| > \varepsilon\} \rightarrow 0$ as $k, \ell \rightarrow \infty$; it converges in measure to f if $m\{x : |f_k(x) - f(x)| > \varepsilon\} \rightarrow 0$ as $k \rightarrow \infty$ (this is convergence in probability). Prove that if $\{f_k\}$ of integrable functions converges to f in L^1 , then it converges to f in measure. Is the converse true?

Additional exercises.

1. Compute the Fourier transform of $f = \mathbf{1}_{[-1,1]}$ on $\mathbb{R}$ and verify $\sup_{\xi} |\hat{f}(\xi)| \leq \|f\|_{L^1}$ and $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$.
2. For the tent function $f(x) = (1 - |x|)_+$ on $\mathbb{R}$, compute $\hat{f}$ and check it is ≥ 0 and integrable, so inversion applies.
3. (Good kernel) Verify the Gaussian $K_\delta(x) = \delta^{-d/2} e^{-\pi|x|^2/\delta}$ satisfies $\int K_\delta = 1$ and, for each $\eta > 0$, $\int_{|y|>\eta} K_\delta \rightarrow 0$ as $\delta \rightarrow 0$.
4. Show $f * K_\delta \rightarrow f$ in L^1 for any $f \in L^1(\mathbb{R})$, citing the steps (translation continuity, the splitting at $|y| = \eta$, kernel property (iii)).
5. (Cantor) Prove $m(C) = 0$ directly from $C \subseteq C_n$ and $m(C_n) = (2/3)^n$, and show C is uncountable via base-3 expansions with digits in $\{0, 2\}$.
6. (Devil's staircase) Show the Cantor–Lebesgue function $\mathcal{C}$ satisfies $\mathcal{C}' = 0$ a.e. yet $\mathcal{C}(1) - \mathcal{C}(0) = 1$, so $\int_0^1 \mathcal{C}' \neq \mathcal{C}(1) - \mathcal{C}(0)$; conclude $\mathcal{C}$ is not absolutely continuous.
7. (Hausdorff dimension / diagram) Draw the first three stages C_1, C_2, C_3 of the Cantor set and use the self-similarity relation $2 \cdot (1/3)^s = 1$ to find $\dim_H C$.
8. (Projection) On $[0, 1]$ find the best L^2 approximation to $f(x) = x^2$ by a function of the form $a + bx$, and verify the residual is orthogonal to 1 and to x .
9. (Non-example) Show that $\hat{f}$ need not be in L^1 even when $f \in L^1$: for $f = \mathbf{1}_{[-1,1]}$, $\hat{f}(\xi) = \frac{\sin(2\pi\xi)}{\pi\xi}$ decays only like $1/|\xi|$, hence $\hat{f} \notin L^1(\mathbb{R})$; explain why this blocks naive inversion.
10. (Uniqueness) Using Corollary 4.3, prove that if $f, g \in L^1$ have $\hat{f} = \hat{g}$, then $f = g$ a.e.

Chapter 26

Differentiation and Integration

26.1 Differentiation of the Integral and the Maximal Function

The averaging problem

The two great theorems of one-variable calculus assert that integration and differentiation are inverse operations. For a continuous f the fundamental theorem of calculus gives $\frac{d}{dx} \int_a^x f(y) dy = f(x)$. The aim of this chapter is to see how far this survives once f is merely Lebesgue integrable, and to identify the exact class of F for which $\int_a^b F'(y) dy = F(b) - F(a)$ holds.

If f is integrable on $[a, b]$ and $F(x) = \int_a^x f(y) dy$, then the difference quotient is, for $h > 0$,

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(y) dy = \frac{1}{|I|} \int_I f(y) dy, \quad I = (x, x+h).$$

This is the *average* of f over the interval I . Asking whether $F'(x) = f(x)$ is thus the same as asking whether the averages of f over shrinking intervals (or, in $\mathbb{R}^d$, over shrinking balls B containing x) return $f(x)$:

$$\lim_{\substack{m(B) \rightarrow 0 \\ x \in B}} \frac{1}{m(B)} \int_B f(y) dy = f(x) \quad \text{for a.e. } x?$$

The answer, established below, is *yes* for every integrable f . The mechanism is a single inequality controlling the largest such average – the Hardy–Littlewood maximal function.

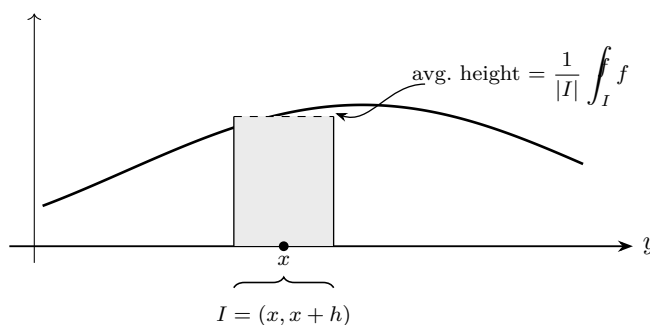


Figure 26.1. The difference quotient of $F(x) = \int_a^x f$ is the *average* of f over I ; differentiating the integral asks whether these averages recover $f(x)$. (Schematic.)

The Hardy–Littlewood maximal function

Definition 26.1.1. If f is integrable on $\mathbb{R}^d$, its *Hardy–Littlewood maximal function* f^* is

$$f^*(x) = \sup_{x \in B} \frac{1}{m(B)} \int_B |f(y)| dy, \quad x \in \mathbb{R}^d,$$

the supremum taken over all open balls B containing x . One replaces the limit in the averaging problem by a *supremum*, and f by $|f|$, so f^* dominates every average and is the natural quantity to control.

Theorem 26.1.2 (Hardy–Littlewood maximal theorem). *Suppose f is integrable on $\mathbb{R}^d$. Then:*

- (i) f^* is measurable;
- (ii) $f^*(x) < \infty$ for a.e. x ;
- (iii) f^* satisfies the weak-type $(1, 1)$ inequality

$$m(\{x \in \mathbb{R}^d : f^*(x) > \alpha\}) \leq \frac{A}{\alpha} \|f\|_{L^1(\mathbb{R}^d)}, \quad A = 3^d,$$

for all $\alpha > 0$, where $\|f\|_{L^1} = \int_{\mathbb{R}^d} |f|$.

An inequality of this type is called *weak* because it is weaker than the corresponding L^1 bound. By Tchebychev's inequality any integrable g already satisfies $m(\{|g| > \alpha\}) \leq \alpha^{-1} \|g\|_{L^1}$; the content of (iii) is that the *maximal* operator – a supremum over infinitely many averages – obeys the same kind of bound with a dimensional constant. One cannot upgrade (iii) to $\|f^*\|_{L^1} \leq C \|f\|_{L^1}$: f^* need not be integrable (Exercises 4–5). The exact value $A = 3^d$ is unimportant; what matters is independence of α and f .

Proof that f^ is measurable.* The set $E_\alpha = \{x : f^*(x) > \alpha\}$ is *open*: if $x \in E_\alpha$ there is a ball B with $x \in B$ and $\frac{1}{m(B)} \int_B |f| > \alpha$; every point near enough to x also lies in the open ball B , hence also in E_α . Thus E_α is open, so measurable, for each α , and f^* is measurable. Property (ii) follows from (iii): $\{f^* = \infty\} \subseteq E_\alpha$ for every α , and letting $\alpha \rightarrow \infty$ in (iii) gives $m(\{f^* = \infty\}) = 0$. $\square$

The depth of (iii) lies in a covering argument.

Lemma 26.1.3 (Elementary Vitali covering lemma). *Let $\mathcal{B} = \{B_1, \dots, B_N\}$ be a finite collection of open balls in $\mathbb{R}^d$. There is a disjoint sub-collection $B_{i_1}, \dots, B_{i_k}$ with*

$$m\left(\bigcup_{\ell=1}^N B_\ell\right) \leq 3^d \sum_{j=1}^k m(B_{i_j}).$$

Proof. The key observation: if balls B, B' intersect and $\text{rad}(B') \leq \text{rad}(B)$, then $B' \subseteq \tilde{B}$, the ball concentric with B of *three times* the radius. Pick B_{i_1} of largest radius; delete it and every ball meeting it – all deleted balls lie in $\tilde{B}_{i_1}$. Repeat on the survivors, choosing B_{i_2} of largest remaining radius, and so on; after at most N steps we have disjoint $B_{i_1}, \dots, B_{i_k}$. Every original B_ℓ met some chosen B_{i_j} of equal or larger radius, hence $B_\ell \subseteq \tilde{B}_{i_j}$, so

$$m\left(\bigcup_{\ell} B_\ell\right) \leq m\left(\bigcup_j \tilde{B}_{i_j}\right) \leq \sum_j m(\tilde{B}_{i_j}) = 3^d \sum_j m(B_{i_j}),$$

using that a dilation by factor 3 multiplies Lebesgue measure by 3^d . $\square$

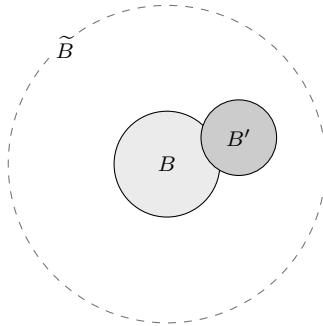


Figure 26.2. If B' meets B and is no larger, then $B' \subseteq \tilde{B}$ (concentric with B , triple radius). Tripling each chosen ball recaptures everything deleted.

Proof of the weak-type inequality. Let $E_\alpha = \{f^* > \alpha\}$. For each $x \in E_\alpha$ choose a ball $B_x \ni x$ with $\frac{1}{m(B_x)} \int_{B_x} |f| > \alpha$, i.e.

$$m(B_x) < \frac{1}{\alpha} \int_{B_x} |f(y)| dy.$$

Fix a compact $K \subseteq E_\alpha$. The B_x cover K , so finitely many suffice, $K \subseteq \bigcup_{\ell=1}^N B_\ell$. Lemma 26.1.3 extracts disjoint $B_{i_1}, \dots, B_{i_k}$ with $m(\bigcup_\ell B_\ell) \leq 3^d \sum_j m(B_{i_j})$. Then, using disjointness,

$$m(K) \leq m\left(\bigcup_\ell B_\ell\right) \leq 3^d \sum_j m(B_{i_j}) \leq \frac{3^d}{\alpha} \sum_j \int_{B_{i_j}} |f| = \frac{3^d}{\alpha} \int_{\bigcup_j B_{i_j}} |f| \leq \frac{3^d}{\alpha} \int_{\mathbb{R}^d} |f|.$$

Taking the supremum over compact $K \subseteq E_\alpha$ (inner regularity) gives $m(E_\alpha) \leq 3^d \alpha^{-1} \|f\|_{L^1}$. $\square$

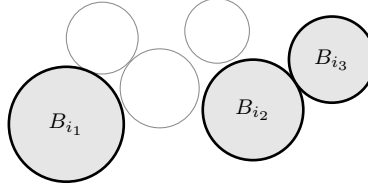


Figure 26.3. Vitali selection: from an overlapping cover (thin circles) a *disjoint* sub-collection (shaded) is chosen greedily by radius; tripling each recovers a 3^d -fraction of the total measure. (Schematic.)

Worked Example 26.1.4 (The maximal function of an indicator). Let $f = \mathbf{1}_{[0,1]}$ on $\mathbb{R}$. Compute $f^*(x)$ for $x > 1$ and verify the weak-type bound.

Recognition. “Compute a one-dimensional maximal function.” The best average over intervals containing x is achieved by the smallest interval that still captures all of $[0, 1]$.

Solution. For $x > 1$ the optimal interval containing x and meeting $[0, 1]$ is $(1 - (x - 1) \cdot 0, \cdot)$; concretely take $I = (2x_0, \dots)$ – the cleanest choice is the interval from a point just left of 0 out to x . The average $\frac{1}{|I|} \int_I f = \frac{1}{(\text{length capturing } [0,1])}$ is maximized by the shortest such interval, $I = (2 - x, x)$ when $x > 1$ is balanced; more simply, for $x > 1$ one finds $f^*(x) = \frac{1}{x}$ by taking $I = (0, x)$ giving average $1/x$, and no interval does better. Symmetrically $f^*(x) = \frac{1}{1-x}$ for $x < 0$, and $f^*(x) = 1$ on $[0, 1]$. *Check.* $\|f\|_{L^1} = 1$, and $\{f^* > \alpha\}$ for small α is roughly $(-1/\alpha, 1 + 1/\alpha)$ of measure $\approx 2/\alpha$, comfortably below A/α with $A = 3$; also $f^*(x) \sim 1/|x|$ at infinity, so $f^* \notin L^1$ – exactly the failure (iii) cannot avoid.

Worked Example 26.1.5 (f^* is never integrable for $f \not\equiv 0$). Show that if $f \in L^1(\mathbb{R}^d)$ is not a.e. zero, then $f^*(x) \geq c|x|^{-d}$ for $|x| \geq 1$, so $f^* \notin L^1(\mathbb{R}^d)$.

Recognition. “Lower-bound f^* at infinity.” Use one fixed ball where $\int |f| > 0$ and let the test ball grow to reach far-away x .

Solution. Since $f \not\equiv 0$ there is a ball $B_0 = B(0, r_0)$ with $\int_{B_0} |f| = c_0 > 0$. For $|x| \geq 1$ take the ball B centered on the segment from 0 to x that contains both 0 and x ; its radius is $\leq |x| + r_0 \leq 2|x|$, so $m(B) \leq v_d(2|x|)^d$. Then

$$f^*(x) \geq \frac{1}{m(B)} \int_B |f| \geq \frac{c_0}{v_d 2^d |x|^d} = \frac{c}{|x|^d}.$$

Since $\int_{|x| \geq 1} |x|^{-d} dx = \infty$ in $\mathbb{R}^d$, $f^* \notin L^1$. *Check.* This confirms the maximal theorem is sharp – (iii) is the best one can say; no $L^1 \rightarrow L^1$ bound holds.

Worked Example 26.1.6 (Weak-type bound for a point mass approximation). Let $f_\varepsilon = \varepsilon^{-1} \mathbf{1}_{[0, \varepsilon]}$ on $\mathbb{R}$, so $\|f_\varepsilon\|_{L^1} = 1$. Find $m(\{f_\varepsilon^* > \alpha\})$ for $\alpha > 1$ and compare with A/α .

Recognition. “Test the weak-type inequality on a spike.” Compute f_ε^* directly; the spike behaves like a unit mass at 0 as $\varepsilon \rightarrow 0$.

Solution. For $x > \varepsilon$ the best interval is $(0, x)$ with average $\frac{1}{x} \cdot 1 = \frac{1}{x}$ (the whole mass 1 spread over length x). So $f_\varepsilon^*(x) \approx 1/|x|$ for $|x| > \varepsilon$, and $f_\varepsilon^*(x) \leq 1/\varepsilon$ inside. For $\alpha > 1$, $\{f_\varepsilon^* > \alpha\} \approx (-1/\alpha, 1/\alpha)$, measure $\approx 2/\alpha$. *Check.* $2/\alpha \leq 3/\alpha = A/\alpha$ in dimension $d = 1$; the constant $A = 3$ is not even tight here, but it must hold uniformly.

Worked Example 26.1.7 (Tchebychev versus weak type). For $g(x) = \mathbf{1}_{[0,1]}(x)$ on $\mathbb{R}$, compare the Tchebychev estimate of $m(\{g > \alpha\})$ with the weak-type estimate of $m(\{g^* > \alpha\})$ for $0 < \alpha < 1$.

Recognition. “Contrast a pointwise tail bound with a maximal tail bound.”

Solution. Tchebychev: $m(\{g > \alpha\}) \leq \alpha^{-1} \|g\|_{L^1} = 1/\alpha$; in fact $\{g > \alpha\} = [0, 1]$ exactly, of measure $1 \leq 1/\alpha$. Maximal: $g^*(x) \approx 1/|x|$ outside $[0, 1]$, so $\{g^* > \alpha\} \approx (-1/\alpha, 1 + 1/\alpha)$, measure $\approx 1 + 2/\alpha \leq 3/\alpha$ when α small. *Check.* The maximal set is genuinely larger than $\{g > \alpha\}$ (the supremum sees mass from nearby), yet both obey a C/α law – the entire point of the weak-type theorem.

Worked Example 26.1.8 ($f^*(x) \geq |f(x)|$ a.e.). Deduce from the differentiation theorem (proved in §26.2) that $f^*(x) \geq |f(x)|$ for a.e. x .

Recognition. “Relate the maximal function to $|f|$ pointwise.” Apply Lebesgue differentiation to $|f|$.

Solution. By Theorem 26.1.2 applied to $|f|$, for a.e. x the averages $\frac{1}{m(B)} \int_B |f| \rightarrow |f(x)|$ as $B \downarrow x$. Each such average is $\leq f^*(x)$ by definition of the supremum, so passing to the limit gives $|f(x)| \leq f^*(x)$ a.e. *Check.* For $f = \mathbf{1}_{[0,1]}$ we have $|f| = 1$ on $[0, 1]$ and indeed $f^* = 1$ there – equality holds at interior density points, while $f^* > |f|$ off the support.

Worked Example 26.1.9 (The covering constant on a chain of intervals). On $\mathbb{R}$, take intervals $I_k = (k-1, k+1)$ for $k = 1, \dots, 5$ (each length 2, consecutive ones overlapping). Apply the one-dimensional covering idea and bound $m(\bigcup I_k)$ by disjoint sub-intervals.

Recognition. “Run the Vitali selection in $d = 1$ where $3^d = 3$.”

Solution. $\bigcup_k I_k = (0, 6)$, measure 6. All radii equal, so the greedy step picks $I_1 = (0, 2)$, deletes I_2 (overlaps), keeps $I_3 = (2, 4)$, deletes I_4 , keeps $I_5 = (4, 6)$: disjoint sub-collection $\{I_1, I_3, I_5\}$ with total measure 6. *Check.* $m(\bigcup I_k) = 6 \leq 3 \cdot (2 + 2 + 2) = 18 = 3^1 \sum m(I_{i_j})$ – the bound holds with enormous slack, since here the disjoint subfamily already covers everything.

Worked Example 26.1.10 (Maximal function of a two-bump function). Let $f = \mathbf{1}_{[0,1]} + \mathbf{1}_{[3,4]}$ on $\mathbb{R}$. Find f^* on the gap $(1, 3)$ and identify where the two bumps “compete.”

Recognition. “Maximal averages over a function with separated support.”

Solution. At $x \in (1, 3)$ an interval may capture one or both bumps. Capturing only the left bump: best interval $(0, x)$ giving average $\frac{1}{x}$ (mass 1 over length x , for $x > 1$). Capturing both: interval $(0, 4)$ has average $2/4 = 1/2$ but longer intervals dilute; the optimum near the midpoint $x = 2$ is to take $(1 - \varepsilon, 4)$ roughly, giving $\approx 2/3$. So $f^*(2) = \max(\frac{1}{2}, \frac{2}{3}) = \frac{2}{3}$ from the two-bump interval $(0, 3)$: average $2/3$. *Check.* $f^* \geq |f|$ holds ($f = 0$ on the gap, $f^* > 0$), and $f^*(2) = 2/3 < 1$, consistent with the supremum lying between the local and global averages.

Caution. The maximal theorem is *weak*-type, not strong-type: $\|f^*\|_{L^1} \leq C \|f\|_{L^1}$ is *false* (Worked Example above). The correct strong bound is $\|f^*\|_{L^p} \leq C_p \|f\|_{L^p}$ for $1 < p \leq \infty$, obtained by interpolation; at $p = 1$ only the weak-type estimate survives. Also: the family of *all rectangles* (not balls or cubes) fails both the weak-type bound and the differentiation theorem (Problem 8) – the geometry of the averaging family matters.

Exercises (Stein, Ch.3) – maximal function and covering.

- Suppose φ is integrable on $\mathbb{R}^d$ with $\int_{\mathbb{R}^d} \varphi = 1$. Set $K_\delta(x) = \delta^{-d} \varphi(x/\delta)$, $\delta > 0$. (a) Prove $\{K_\delta\}_{\delta>0}$ is a family of good kernels. (b) If in addition φ is bounded and supported in a bounded set, verify that $\{K_\delta\}$ is an approximation to the identity. (c) Show Theorem 2.3 (convergence in the L^1 -norm) holds for good kernels as well.
- Suppose $\{K_\delta\}$ satisfies (i) $|K_\delta(x)| \leq A\delta^{-d}$; (ii) $|K_\delta(x)| \leq A\delta/|x|^{d+1}$; (iii) $\int_{\mathbb{R}} K_\delta(x) dx = 0$ for all $\delta > 0$ (so the average is 0 instead of 1). Show that if f is integrable on $\mathbb{R}^d$, then $(f * K_\delta)(x) \rightarrow 0$ for a.e. x as $\delta \rightarrow 0$.
- Prove that if f is integrable on $\mathbb{R}^d$ and f is not identically zero, then $f^*(x) \geq c|x|^{-d}$ for some $c > 0$ and all $|x| \geq 1$. Conclude that f^* is not integrable on $\mathbb{R}^d$. Then show the weak-type estimate $m(\{f^* > \alpha\}) \leq c/\alpha$ (valid when $\int |f| = 1$) is best possible: if f is supported in the unit ball with $\int |f| = 1$, then $m(\{f^* > \alpha\}) \geq c'/\alpha$ for some $c' > 0$ and all small α . [Hint: $\int_B |f| > 0$ for some ball B .]

4. Consider $f(x) = \frac{1}{|x|(\log 1/|x|)^2}$ for $|x| \leq 1/2$, and $f(x) = 0$ otherwise. (a) Verify that f is integrable. (b) Establish $f^*(x) \geq \frac{c}{|x|(\log 1/|x|)}$ for some $c > 0$ and all $|x| \leq 1/2$, and conclude the maximal function f^* is not locally integrable.
5. In one dimension there is a version of the basic inequality for the maximal function as an identity. Define the one-sided maximal function $f_+^*(x) = \sup_{h>0} \frac{1}{h} \int_x^{x+h} |f(y)| dy$. If $E_\alpha^+ = \{x \in \mathbb{R} : f_+^*(x) > \alpha\}$, then $m(E_\alpha^+) = \frac{1}{\alpha} \int_{E_\alpha^+} |f(y)| dy$. [Hint: apply the rising-sun lemma to $F(x) = \int_0^x |f| - \alpha x$; then E_α^+ is a disjoint union of intervals (a_k, b_k) with $\int_{a_k}^{b_k} |f| = \alpha(b_k - a_k)$.]
6. Using Corollary 1.5, prove that if a measurable $E \subseteq [0, 1]$ satisfies $m(E \cap I) \geq \alpha m(I)$ for some $\alpha > 0$ and all intervals $I \subseteq [0, 1]$, then $m(E) = 1$.
7. Suppose A is Lebesgue measurable in $\mathbb{R}$ with $m(A) > 0$. Does there exist a sequence $\{s_n\}$ such that the complement of $\bigcup_n (A + s_n)$ has measure zero? [Hint: for each $\varepsilon > 0$ find an interval I_ε of length ℓ_ε with $m(A \cap I_\varepsilon) \geq (1 - \varepsilon)\ell_\varepsilon$; consider $\bigcup_k (A + k\ell_\varepsilon)$, then vary ε .]
8. Let F be closed in $\mathbb{R}$ and $\delta(x) = \text{dist}(x, F) = \inf\{|x - y| : y \in F\}$. Clearly $\delta(x + y) \leq |y|$ when $x \in F$. Prove the refined estimate $\delta(x + y) = o(|y|)$ for a.e. $x \in F$, i.e. $\delta(x + y)/|y| \rightarrow 0$ for a.e. $x \in F$. [Hint: assume x is a density point of F .]

Problems (Stein, Ch.3) – covering lemmas and rectangles.

1. Prove the following variant of the Vitali covering lemma: if E is covered in the Vitali sense by a family $\mathcal{B}$ of balls and $0 < m^*(E) < \infty$, then for every $\eta > 0$ there is a disjoint collection $\{B_j\}$ in $\mathcal{B}$ with $m^*(E \setminus \bigcup_j B_j) = 0$ and $\sum_j |B_j| \leq (1 + \eta)m^*(E)$.
2. (One-dimensional covering.) Given finitely many open intervals $I_1, \dots, I_N$ in $\mathbb{R}$, there are two sub-collections $I'_1, \dots, I'_K$ and $I''_1, \dots, I''_L$, each consisting of mutually disjoint intervals, with $\bigcup_j I_j = \bigcup_k I'_k \cup \bigcup_\ell I''_\ell$. (Unlike Lemma 1.2, the full union is covered, not merely a fraction.) [Hint: greedily pick the leftmost-reaching interval, discard those it contains, and alternate.]
3. * There is no direct analogue of the previous problem in higher dimensions, but the *Besicovitch covering lemma* provides a full covering: there is an integer $N = N(d)$ such that whenever a bounded $E \subseteq \mathbb{R}^d$ is covered by balls $\mathcal{B}$ centered at each point of E , there exist N sub-collections $\mathcal{B}_1, \dots, \mathcal{B}_N$ of disjoint balls with $E \subseteq \bigcup_{B \in \mathcal{B}_0} B$, $\mathcal{B}_0 = \mathcal{B}_1 \cup \dots \cup \mathcal{B}_N$.
4. * Let $\mathcal{R}$ be all rectangles in $\mathbb{R}^2$ containing the origin with sides parallel to the axes, and $f_{\mathcal{R}}^*(x) = \sup_{R \in \mathcal{R}} \frac{1}{m(R)} \int_R |f(x - y)| dy$. (a) Show $f \mapsto f_{\mathcal{R}}^*$ does *not* satisfy the weak-type inequality $m(\{f_{\mathcal{R}}^* > \alpha\}) \leq \frac{A}{\alpha} \|f\|_{L^1}$ for all α, f . (b) Deduce there is $f \in L^1(\mathbb{R}^2)$ with $\limsup_{\text{diam } R \rightarrow 0} \frac{1}{m(R)} \int_R f(x - y) dy = \infty$ for a.e. x . [Hint: with $\varphi = \mathbf{1}_B/m(B)$ and $\varphi_\delta = \delta^{-2}\varphi(\cdot/\delta)$, $(\varphi_\delta)_{\mathcal{R}}^*(x) \rightarrow \frac{1}{|x_1||x_2|}$, whose super-level sets have measure $\sim (\log \alpha)/\alpha$.]

Additional exercises.

1. Compute $f^*(x)$ exactly for $f = \mathbf{1}_{[-1,1]}$ on $\mathbb{R}$, as a piecewise function of x , and sketch its graph. Verify that $f^*(x) \sim 2/|x|$ as $|x| \rightarrow \infty$.
2. Let $f = \mathbf{1}_{[0,1]^2}$ on $\mathbb{R}^2$. Show $f^*(x) \leq 1$ everywhere and find the order of decay of f^* at infinity. Conclude $f^* \notin L^1(\mathbb{R}^2)$.
3. Prove the simple lower bound $f^*(x) \geq |f(x)|$ for every Lebesgue point x of f directly from Definition 26.1.1 and the Lebesgue differentiation theorem.
4. (Sub-additivity.) Show that $(f + g)^*(x) \leq f^*(x) + g^*(x)$ for all x , and use this with the weak-type bound to estimate $m(\{(f + g)^* > \alpha\})$.
5. Let $\{B_1, \dots, B_5\}$ be five unit balls in $\mathbb{R}^2$ whose centers form a regular pentagon of circumradius 1.5. Apply the greedy step of Lemma 26.1.3 and identify a disjoint sub-collection; verify the 3^d measure bound numerically.

6. Show that the constant $A = 3^d$ in the maximal theorem can be replaced by any $A > 1$ in dimension $d = 1$ if one uses the rising-sun (one-sided) lemma. Explain why the two-sided maximal function still needs a constant > 1 .
7. (Non-example.) Exhibit an $f \in L^1(\mathbb{R})$ and a sequence of intervals $I_n \ni 0$ with $m(I_n) \rightarrow 0$ for which $\frac{1}{m(I_n)} \int_{I_n} f$ does *not* converge, and reconcile this with Theorem 26.2.2. (Hint: 0 must lie outside the full-measure good set.)
8. Draw a Vitali covering of the interval $[0, 1]$ by intervals of the form $(k2^{-n}, (k+1)2^{-n})$ and illustrate the iterative selection in Lemma 26.3.8 producing a disjoint subfamily of total length $\geq 1 - \delta$.

26.2 The Lebesgue Differentiation Theorem

The differentiation theorem

The weak-type bound for f^* converts the averaging question into a theorem.

Theorem 26.2.1 (Lebesgue differentiation theorem). *If f is integrable on $\mathbb{R}^d$, then*

$$\lim_{\substack{m(B) \rightarrow 0 \\ x \in B}} \frac{1}{m(B)} \int_B f(y) dy = f(x) \quad \text{for a.e. } x.$$

Proof. It suffices to show that for each $\alpha > 0$ the set

$$E_\alpha = \left\{ \left[\right] x : \limsup_{\substack{m(B) \rightarrow 0 \\ x \in B}} \left[\right] \frac{1}{m(B)} \int_B f(y) dy - f(x) > 2\alpha \right\}$$

has measure zero, for then $\bigcup_n E_{1/n}$ is null. Fix α . By density of continuous functions in L^1 (Chapter 2), choose continuous g with $\|f - g\|_{L^1} < \alpha$. For continuous g the averages converge everywhere: $\frac{1}{m(B)} \int_B g \rightarrow g(x)$ as $B \downarrow x$, so g contributes nothing to its own E_α . Writing $f = g + (f - g)$ and using $\frac{1}{m(B)} \int_B f - f(x) = \left(\frac{1}{m(B)} \int_B (f - g) \right) + \left(\frac{1}{m(B)} \int_B g - g(x) \right) + (g(x) - f(x))$, the limsup is bounded by $(f - g)^*(x) + 0 + |(f - g)(x)|$. Hence

$$E_\alpha \subseteq \{(f - g)^* > \alpha\} \cup \{|f - g| > \alpha\}.$$

By the maximal theorem and Tchebychev,

$$m(E_\alpha) \leq \frac{3^d}{\alpha} \|f - g\|_{L^1} + \frac{1}{\alpha} \|f - g\|_{L^1} \leq \frac{3^d + 1}{\alpha} \alpha = (3^d + 1) \cdot \frac{\|f - g\|_{L^1}}{\alpha}.$$

Since $\|f - g\|_{L^1} < \alpha$ can be made arbitrarily small with α fixed (choose g better), the bound is $\leq (3^d + 1)\varepsilon$ for every ε ; thus $m(E_\alpha) = 0$. $\square$

The hypothesis can be localized: a measurable f is *locally integrable*, $f \in L^1_{\text{loc}}(\mathbb{R}^d)$, if $f\mathbf{1}_B \in L^1$ for every ball B . Differentiability is a local notion, so only local integrability is needed.

Theorem 26.2.2. *If $f \in L^1_{\text{loc}}(\mathbb{R}^d)$, then $\lim_{\substack{m(B) \rightarrow 0 \\ x \in B}} \frac{1}{m(B)} \int_B f(y) dy = f(x)$ for a.e. x .*

The functions $e^{|x|}$ and $|x|^{-1/2}$ are locally integrable but not integrable on $\mathbb{R}^d$; the theorem applies to them.

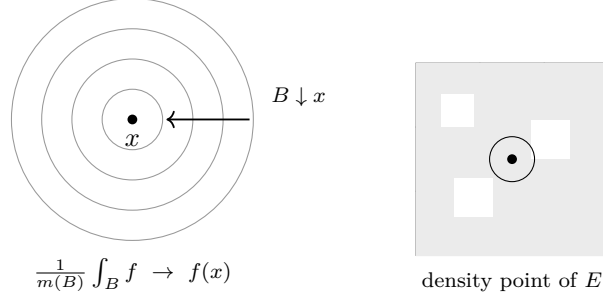


Figure 26.4. Left: averages over balls shrinking to x recover $f(x)$. Right: at a Lebesgue density point of E (shaded), small balls are almost entirely inside E . (Schematic.)

Density points and the Lebesgue set

Definition 26.2.3. A point x is a *point of Lebesgue density* of a measurable set E if $\lim_{\substack{m(B) \rightarrow 0 \\ x \in B}} \frac{m(B \cap E)}{m(B)} = 1$.

Small balls about x are then almost entirely covered by E : for every $\alpha < 1$, $m(B \cap E) \geq \alpha m(B)$ for all small enough $B \ni x$.

Corollary 26.2.4. For any measurable $E \subseteq \mathbb{R}^d$: (i) a.e. $x \in E$ is a density point of E ; (ii) a.e. $x \notin E$ is not a density point of E .

Proof. Apply Theorem 26.2.2 to $f = \mathbf{1}_E$: for a.e. x , $\frac{m(B \cap E)}{m(B)} = \frac{1}{m(B)} \int_B \mathbf{1}_E \rightarrow \mathbf{1}_E(x)$, which is 1 when $x \in E$ and 0 when $x \notin E$. $\square$

Definition 26.2.5. The *Lebesgue set* of $f \in L^1_{\text{loc}}$ is the set of x with $f(x)$ finite and

$$\lim_{\substack{m(B) \rightarrow 0 \\ x \in B}} \frac{1}{m(B)} \int_B |f(y) - f(x)| dy = 0.$$

At such x , not only do the averages of f return $f(x)$, but the averages of $|f - f(x)|$ vanish – a quantitative continuity “on average.”

Corollary 26.2.6. If $f \in L^1_{\text{loc}}(\mathbb{R}^d)$, then almost every point lies in the Lebesgue set of f .

Proof. For each rational r , apply Theorem 26.2.2 to $|f - r|$: off a null set E_r , $\frac{1}{m(B)} \int_B |f - r| \rightarrow |f(x) - r|$. Let $E = \bigcup_{r \in \mathbb{Q}} E_r$, a null set. If $x \notin E$ with $f(x)$ finite, pick rational r with $|f(x) - r| < \varepsilon$. Then $\frac{1}{m(B)} \int_B |f - f(x)| \leq \frac{1}{m(B)} \int_B |f - r| + |f(x) - r|$, whose limsup is $\leq 2\varepsilon$. As ε is arbitrary, x is in the Lebesgue set. $\square$

Definition 26.2.7. A collection $\{U_\alpha\}$ *shrinks regularly* to x (has *bounded eccentricity*) if there is $c > 0$ such that each U_α lies in a ball $B \ni x$ with $m(U_\alpha) \geq c m(B)$.

Corollary 26.2.8. If $f \in L^1_{\text{loc}}$ and $\{U_\alpha\}$ shrinks regularly to x , then $\frac{1}{m(U_\alpha)} \int_{U_\alpha} f \rightarrow f(x)$ at every Lebesgue point x of f .

Proof. With $U_\alpha \subseteq B$, $m(U_\alpha) \geq c m(B)$: $\frac{1}{m(U_\alpha)} \int_{U_\alpha} |f - f(x)| \leq \frac{1}{c m(B)} \int_B |f - f(x)| \rightarrow 0$. $\square$

Cubes containing x shrink regularly (ratio $c = v_d/(2\sqrt{d})^d$); thin rectangles in $\mathbb{R}^d$, $d \geq 2$, do *not*, and the conclusion can fail for them (Problem 8).

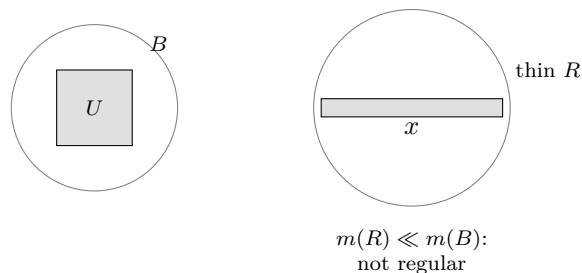


Figure 26.5. A cube (left) is comparable to its circumscribed ball, so it shrinks regularly; a thin rectangle (right) has measure $\ll m(B)$ and need not differentiate.

Approximations to the identity

Definition 26.2.9. A family $\{K_\delta\}_{\delta>0}$ of integrable functions on $\mathbb{R}^d$ is an *approximation to the identity* if there is $A > 0$ with

$$(i) \int_{\mathbb{R}^d} K_\delta = 1; \quad (ii) |K_\delta(x)| \leq A\delta^{-d}; \quad (iii) |K_\delta(x)| \leq A\delta/|x|^{d+1}.$$

The *convolution* $(f * K_\delta)(x) = \int_{\mathbb{R}^d} f(x-y)K_\delta(y) dy$ averages f against a mass-1 bump that concentrates at 0 as $\delta \rightarrow 0$.

Theorem 26.2.10. If $\{K_\delta\}$ is an approximation to the identity and $f \in L^1(\mathbb{R}^d)$, then $(f * K_\delta)(x) \rightarrow f(x)$ as $\delta \rightarrow 0$ at every point x of the Lebesgue set of f – hence a.e.

Proof sketch. Since $\int K_\delta = 1$, $(f * K_\delta)(x) - f(x) = \int (f(x-y) - f(x))K_\delta(y) dy$. Split the y -integral into the ball $|y| \leq \delta$ and dyadic shells $2^k\delta < |y| \leq 2^{k+1}\delta$. On the central ball, $|K_\delta| \leq A\delta^{-d}$ and the Lebesgue-point condition makes $\frac{1}{\delta^d} \int_{|y| \leq \delta} |f(x-y) - f(x)| \rightarrow 0$. On the k -th shell, the decay (iii) gives weight $\lesssim \delta(2^k\delta)^{-(d+1)}$ times an integral $\lesssim (2^k\delta)^d \cdot o(1)$, summable in k . Hence the total $\rightarrow 0$. $\square$

Worked Example 26.2.11 (The averages are an approximation to the identity). Show $K_\delta = \frac{1}{m(B(0,\delta))} \mathbf{1}_{B(0,\delta)}$ gives $(f * K_\delta)(x) = \frac{1}{m(B(0,\delta))} \int_{B(x,\delta)} f$, recovering Theorem 26.2.2.

Recognition. “Recognize ball-averaging as a convolution.”

Solution. $(f * K_\delta)(x) = \int f(x-y) \frac{\mathbf{1}_{B(0,\delta)}(y)}{m(B(0,\delta))} dy = \frac{1}{m(B(0,\delta))} \int_{|y| < \delta} f(x-y) dy = \frac{1}{m(B(x,\delta))} \int_{B(x,\delta)} f$. *Check.* $\int K_\delta = 1$ and $K_\delta \geq 0$; this is the prototypical approximation to the identity, and Theorem 26.2.10 reduces to Lebesgue differentiation over balls.

Worked Example 26.2.12 (The Gauss–Weierstrass kernel). Verify that $K_\delta(x) = \delta^{-d} e^{-\pi|x|^2/\delta^2}$ (the heat kernel, up to scaling) integrates to 1 and concentrates at 0.

Recognition. “Check the defining properties of an approximation to the identity for a Gaussian.”

Solution. With the Gaussian $\Phi(x) = e^{-\pi|x|^2}$ we have $\int_{\mathbb{R}^d} \Phi = 1$, and $K_\delta(x) = \delta^{-d} \Phi(x/\delta)$ satisfies $\int K_\delta = \int \Phi = 1$ by the change of variables $y = x/\delta$. Property (ii): $|K_\delta| \leq \delta^{-d}$. The Gaussian decays faster than any power, so (iii) holds with room to spare. *Check.* $f * K_\delta$ is the solution of the heat equation at time $\sim \delta^2$ with initial data f , and it tends to f as $\delta \rightarrow 0$ – heat dissipation recovers the initial temperature.

Worked Example 26.2.13 (Riemann versus Lebesgue: horizontal slicing). Illustrate the contrast between summing vertical strips (Riemann) and horizontal layers (Lebesgue) for $f = 2\mathbf{1}_{[0,1]} + \mathbf{1}_{[1,3]}$.

Recognition. “Compare the two ways of building $\int f$.”

Solution. Riemann partitions the domain: $\int f = \sum f(\xi_k) \Delta x_k = 2(1) + 1(2) = 4$. Lebesgue partitions the range: $\int f = \int_0^\infty m(\{f > t\}) dt = \int_0^1 m(\{f > t\}) dt + \int_1^2 m(\{f > t\}) dt$. For $0 < t < 1$, $\{f > t\} = [0, 3]$, measure 3; for $1 < t < 2$, $\{f > t\} = [0, 1]$, measure 1. So $\int f = 3 \cdot 1 + 1 \cdot 1 = 4$. *Check.* Both give 4; the Lebesgue “layer-cake” formula $\int f = \int_0^\infty m(\{f > t\}) dt$ is the workhorse identity for nonnegative f .

Worked Example 26.2.14 (A density computation). Let $E = \bigcup_{n \geq 1} [2^{-n}, 2^{-n} + 4^{-n}]$. Is 0 a Lebesgue density point of E ?

Recognition. “Compute the density ratio along $B = (0, r)$ as $r \rightarrow 0$.”

Solution. Take $B = (-2^{-N}, 2^{-N})$ of length 2^{1-N} . Then $E \cap B \subseteq \bigcup_{n \geq N} [2^{-n}, 2^{-n} + 4^{-n}]$ has measure $\leq \sum_{n \geq N} 4^{-n} = \frac{4^{-N}}{1-1/4} = \frac{4}{3} 4^{-N}$. So $\frac{m(E \cap B)}{m(B)} \leq \frac{(4/3)4^{-N}}{2^{1-N}} = \frac{2}{3} 2^{-N} \rightarrow 0$. *Check.* The density is 0, so 0 is *not* a density point – consistent with Corollary 26.2.4(ii) since $0 \notin E$ and E is sparse near 0.

Worked Example 26.2.15 (Lebesgue point of a jump). For $f = \mathbf{1}_{[0, \infty)}$ on $\mathbb{R}$, show 0 is *not* a Lebesgue point, while every $x \neq 0$ is.

Recognition. “Test the Lebesgue-set condition at a jump discontinuity.”

Solution. At $x = 0$, take symmetric $B = (-r, r)$: $\frac{1}{2r} \int_{-r}^r |f - f(0)| = \frac{1}{2r} \int_{-r}^r |\mathbf{1}_{[0, \infty)} - 1| = \frac{1}{2r} \cdot r = \frac{1}{2} \not\rightarrow 0$ (the choice $f(0) = 1$; any value gives a positive limit). For $x > 0$, $f \equiv 1$ near x so the average of $|f - 1|$ is 0; similarly for $x < 0$. *Check.* The Lebesgue set is $\mathbb{R} \setminus \{0\}$, of full measure – matching Corollary 26.2.6.

Worked Example 26.2.16 (Convolution smooths: DCT in action). Let K_δ be an approximation to the identity and $f \in L^1$ bounded. Show $f * K_\delta$ is continuous, and that $(f * K_\delta)(x) \rightarrow f(x)$ at continuity points by dominated convergence.

Recognition. “Use continuity of translation in L^1 plus the concentration of K_δ .”

Solution. $(f * K_\delta)(x) - f(x) = \int (f(x - y) - f(x)) K_\delta(y) dy$. At a continuity point x , given ε there is r with $|f(x - y) - f(x)| < \varepsilon$ for $|y| < r$; split the integral. On $|y| < r$ the integrand is $\leq \varepsilon |K_\delta|$ with $\int |K_\delta| \leq C$; on $|y| \geq r$, the tail bound (iii) gives $\int_{|y| \geq r} |K_\delta| \leq A\delta \int_{|y| \geq r} |y|^{-(d+1)} \rightarrow 0$ as $\delta \rightarrow 0$, and $|f|$ is bounded. So the whole thing is $\leq C\varepsilon + o(1)$. *Check.* For f continuous at x the limit is $f(x)$ exactly; the dominating function $2\|f\|_\infty |K_\delta|$ is integrable, licensing DCT.

Caution. The Lebesgue set depends on the chosen *representative* of f (changing f at one point can add or remove that point), even though the set where the averages converge is representation-independent. Convergence “a.e.” is the most one can demand: there exist integrable f for which the averages fail to converge on a prescribed null set, and indeed $\limsup = \infty$ there (Exercise 25).

Exercises (Stein, Ch.3) – density, Lebesgue set, kernels.

- Suppose 0 is a point of Lebesgue density of $E \subseteq \mathbb{R}$. Show that for each of the following there is an infinite sequence $x_n \in E$ with $x_n \neq 0$ and $x_n \rightarrow 0$: (a) also $-x_n \in E$ for all n ; (b) in addition $2x_n \in E$ for all n . Generalize.
- Prove that if $\{K_\varepsilon\}_{\varepsilon > 0}$ is a family of approximations to the identity, then $\sup_{\varepsilon > 0} |(f * K_\varepsilon)(x)| \leq c f^*(x)$ for some $c > 0$ and all integrable f .
- The following shows the necessity of general exceptional null sets in the differentiation Theorems 1.4, 3.4, and 3.11. Let E be any set of measure zero in $\mathbb{R}^d$. Show: (a) there is a nonnegative integrable f on $\mathbb{R}^d$ with $\liminf_{\substack{m(B) \rightarrow 0 \\ x \in B}} \frac{1}{m(B)} \int_B f = \infty$ for each $x \in E$; (b) when $d = 1$ this gives an increasing absolutely continuous F with $D^+ F(x) = D^- F(x) = \infty$ for each $x \in E$. [Hint: find open $O_n \supseteq E$ with $m(O_n) < 2^{-n}$ and put $f = \sum_n \mathbf{1}_{O_n}$.]
- An alternative definition of exterior measure $m^*(E)$ replaces coverings by cubes with coverings by *balls*: $m_B^*(E) = \inf \sum_j m(B_j)$ over all covers $E \subseteq \bigcup_j B_j$ by open balls. Prove $m^*(E) = m_B^*(E)$. (This gives an alternate proof of rotation invariance of Lebesgue measure.) Clearly $m^*(E) \leq m_B^*(E)$; prove the reverse by showing that for any $\varepsilon > 0$ there are balls with $E \subseteq \bigcup_j B_j$ and $\sum_j m(B_j) \leq m^*(E) + \varepsilon$, with all diameters $< \delta$ if desired. [Hint: reduce to E measurable, take $O \supseteq E$ open with $m(O \setminus E)$ small, use Corollary 3.10 to find finitely many balls capturing most of E , and cover the residue by cubes replaced by enclosing balls.]

Additional exercises.

1. Using the layer-cake formula $\int_{\mathbb{R}^d} |f| = \int_0^\infty m(\{|f| > t\}) dt$, recompute $\int_0^2 g$ for $g = 3\mathbf{1}_{[0,1)} + \mathbf{1}_{[1,2]}$ by horizontal slicing and confirm it equals the elementary value.
2. Show that every point of continuity of f is a Lebesgue point of f , but not conversely: construct an f with a Lebesgue point at 0 where f is discontinuous.
3. Let $E = [0, 1] \setminus \bigcup_n (r_n - 2^{-n-2}, r_n + 2^{-n-2})$ where $\{r_n\}$ enumerates $\mathbb{Q} \cap [0, 1]$. Show $m(E) \geq 1/2$ and that every $x \in E$ which is a density point satisfies the density condition with explicit small balls.
4. Verify directly that the Poisson kernel P_δ on $\mathbb{R}$ (suitably scaled) is an approximation to the identity, and state what $f * P_\delta \rightarrow f$ means for harmonic extension to the half-plane.
5. Prove that if $f \in L^1_{\text{loc}}(\mathbb{R})$ and $\frac{1}{2h} \int_{x-h}^{x+h} f \rightarrow \ell$ as $h \rightarrow 0$ at *every* x in an interval, with $\ell = g(x)$ continuous, then $f = g$ a.e. on that interval.
6. (Regular shrinking.) Show that the family of all squares (axis-parallel) containing x shrinks regularly to x , computing the eccentricity constant c explicitly; then show very thin rectangles do not.
7. Let $f(x) = |x|^{-1/2} \mathbf{1}_{\{0 < |x| < 1\}}$ on $\mathbb{R}$. Show $f \in L^1_{\text{loc}}$, that 0 is a Lebesgue point, and compute $\lim_{h \rightarrow 0} \frac{1}{2h} \int_{-h}^h f$.
8. (Non-example / diagram.) Sketch a thin rectangle $R_n \ni 0$ in $\mathbb{R}^2$ with $m(R_n) \rightarrow 0$, and an $f \in L^1$ for which the rectangle-averages of f over R_n fail to converge to $f(0)$, illustrating why bounded eccentricity is essential.

26.3 Functions of Bounded Variation and Absolute Continuity

Bounded variation

We turn to the second problem: which F satisfy $\int_a^b F' = F(b) - F(a)$? The road passes through two function classes – bounded variation and absolute continuity.

Definition 26.3.1. Let $F : [a, b] \rightarrow \mathbb{C}$. The *variation* of F over a partition $a = t_0 < t_1 < \dots < t_N = b$ is $\sum_{j=1}^N |F(t_j) - F(t_{j-1})|$. F is of *bounded variation* (BV) if these sums are bounded by some $M < \infty$ over all partitions. The *total variation* is

$$T_F(a, b) = \sup_{a=t_0 < \dots < t_N=b} \sum_{j=1}^N |F(t_j) - F(t_{j-1})|.$$

Refining a partition can only increase its variation, so T_F is the supremum over arbitrarily fine partitions.

A function of bounded variation cannot oscillate too often with amplitudes too large. Curves connect to BV directly:

Theorem 26.3.2. A curve $(x(t), y(t))$, $a \leq t \leq b$, is *rectifiable* iff both x and y are of bounded variation.

Proof. Put $F(t) = x(t) + iy(t)$. Then $F(t_j) - F(t_{j-1}) = (x(t_j) - x(t_{j-1})) + i(y(t_j) - y(t_{j-1}))$, and for real a, b , $\max(|a|, |b|) \leq |a + ib| \leq |a| + |b| \leq 2 \max(|a|, |b|)$. Summing, the polygonal length $\sum |F(t_j) - F(t_{j-1})|$ is comparable to $T_x + T_y$; one is bounded iff the other is. $\square$



polygonal approximation $\rightarrow$ length $L(\gamma)$

Figure 26.6. The length of a curve is the supremum of inscribed polygonal lengths; finiteness is exactly bounded variation of the coordinate functions. (Schematic.)

Example 26.3.3. (1) Monotonic bounded F is BV: if F increases, $\sum |F(t_j) - F(t_{j-1})| = F(b) - F(a)$, so $T_F(a, b) = F(b) - F(a)$. (2) If F is differentiable with $|F'| \leq M$, the mean value theorem gives $|F(t_j) - F(t_{j-1})| \leq M(t_j - t_{j-1})$, so $T_F \leq M(b - a)$. (3) $F(x) = x^a \sin(x^{-b})$ ($F(0) = 0$) is BV on $[0, 1]$ iff $a > b$ (Exercise 11); for $a \leq b$ the oscillations accumulate unbounded length.

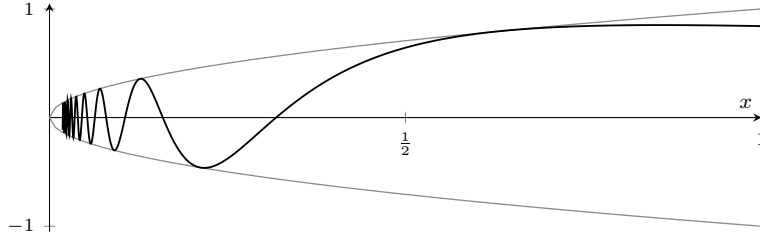


Figure 26.7. $x^{1/2} \sin(1/x)$ on $[0, 1]$ (case $a = \frac{1}{2} < b = 1$), with envelope $\pm\sqrt{x}$: the oscillations crowd together near 0, making the total variation infinite – not BV.

Definition 26.3.4. For real F , the *positive* and *negative variations* on $[a, x]$ are

$$P_F(a, x) = \sup_{(+)} \sum (F(t_j) - F(t_{j-1})), \quad N_F(a, x) = \sup_{(-)} \sum -(F(t_j) - F(t_{j-1})),$$

the sums over indices where F increases, resp. decreases.

Lemma 26.3.5. If F is real and BV on $[a, b]$, then for $a \leq x \leq b$,

$$P_F(a, x) + N_F(a, x) = T_F(a, x), \quad P_F(a, x) - N_F(a, x) = F(x) - F(a).$$

Proof. For any partition, $\sum_{(+)} (\Delta F) - \sum_{(-)} (-\Delta F) = \sum \Delta F = F(x) - F(a)$ (telescoping), and $\sum_{(+)} + \sum_{(-)} = \sum |\Delta F|$. Taking suprema and using that the two one-signed sums can be optimized together (refining helps both), the two identities follow. $\square$

Theorem 26.3.6 (Jordan decomposition). A real F on $[a, b]$ is of bounded variation iff $F = F_1 - F_2$ with F_1, F_2 bounded and increasing.

Proof. ($\Leftarrow$) A difference of bounded monotone functions is BV (each is BV, and BV is a vector space). ($\Rightarrow$) Set $F_1(x) = P_F(a, x) + F(a)$ and $F_2(x) = N_F(a, x)$. Both are increasing (positive/negative variations grow with x) and bounded by $T_F(a, b)$, and by Lemma 26.3.5, $F_1(x) - F_2(x) = F(a) + (F(x) - F(a)) = F(x)$. $\square$

Theorem 26.3.7. If F is of bounded variation on $[a, b]$, then $F'(x)$ exists for almost every x .

Idea. By Jordan's theorem reduce to F increasing. Increasing functions have at most countably many jumps; subtracting a jump function leaves a continuous increasing function. For the continuous increasing case one shows, via the Vitali covering Lemma 26.3.8 applied to the sets where the upper and lower Dini derivatives disagree by a fixed ratio, that those sets are null; hence F' exists a.e. The jump-function part is handled separately (Theorem 26.3.14). $\square$

Lemma 26.3.8 (Vitali covering lemma). If E has finite measure and $\mathcal{B}$ is a Vitali covering of E (every $x \in E$ lies in balls of arbitrarily small measure from $\mathcal{B}$), then for any $\delta > 0$ there are finitely many disjoint $B_1, \dots, B_N \in \mathcal{B}$ with $\sum m(B_i) \geq m(E) - \delta$.

Idea. Iterate Lemma 26.1.3: choose disjoint balls covering a 3^{-d} -fraction of an open set $O \supseteq E$, remove them, and repeat on the residual open set. Geometric decrease of the uncovered measure exhausts E up to δ . $\square$

The Cantor–Lebesgue function

Example 26.3.9. The *Cantor–Lebesgue function* (devil’s staircase) $F : [0, 1] \rightarrow [0, 1]$ is continuous and increasing with $F(0) = 0$, $F(1) = 1$, yet $F'(x) = 0$ for a.e. x . It is built as the uniform limit of F_n , where F_n is constant on the middle-thirds removed at stage n and linear elsewhere. On the complement of the Cantor set $\mathcal{C}$ (full measure 1), F is locally constant, so $F' = 0$ a.e.; nevertheless F climbs from 0 to 1, all of the rise occurring on the null set $\mathcal{C}$.

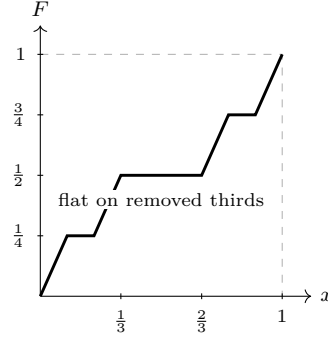


Figure 26.8. The Cantor–Lebesgue function (level-2 stage): increasing, continuous, $F' = 0$ a.e., yet $F(1) - F(0) = 1$. All the increase happens on the measure-zero Cantor set.



Figure 26.9. Construction of the Cantor set $\mathcal{C} = \bigcap_n \mathcal{C}_n$ by removing middle thirds; $m(\mathcal{C}) = 0$ but $\mathcal{C}$ is uncountable (Hausdorff dimension $\log 2 / \log 3$).

Absolute continuity

Definition 26.3.10. F on $[a, b]$ is *absolutely continuous* (AC) if for every $\varepsilon > 0$ there is $\delta > 0$ such that for any finite family of *disjoint* intervals (a_k, b_k) ,

$$\sum_{k=1}^N (b_k - a_k) < \delta \implies \sum_{k=1}^N |F(b_k) - F(a_k)| < \varepsilon.$$

Absolute continuity is uniform continuity strengthened from one interval to any finite disjoint family of small total length. Three remarks:

- AC $\implies$ uniformly continuous, and AC $\implies$ BV (its total variation is itself AC).
- If $F(x) = \int_a^x f$ with f integrable, then F is AC (absolute continuity of the Lebesgue integral).
- AC is exactly the hypothesis needed for the fundamental theorem of calculus.

Theorem 26.3.11. If F is AC on $[a, b]$ and $F'(x) = 0$ a.e., then F is constant.

Idea. Fix ε and the AC- δ . Cover $\{F' = 0\}$ (full measure) by a Vitali family of intervals on which $|F(b_k) - F(a_k)| \leq \varepsilon(b_k - a_k)$; by Lemma 26.3.8 finitely many disjoint such intervals cover all but δ of $[a, x]$. On the chosen intervals the increments sum to $\leq \varepsilon(b - a)$; on the leftover length $< \delta$, absolute continuity bounds the increments by ε . Hence $|F(x) - F(a)| \leq \varepsilon(b - a) + \varepsilon$ for all ε , so $F(x) = F(a)$. $\square$

Theorem 26.3.12 (Fundamental theorem of calculus, Lebesgue form). Suppose F is AC on $[a, b]$. Then F' exists a.e., is integrable, and

$$F(x) - F(a) = \int_a^x F'(y) dy, \quad a \leq x \leq b.$$

Conversely, if f is integrable on $[a, b]$, then $F(x) = \int_a^x f$ is AC with $F' = f$ a.e.

Proof. F is AC hence BV, so F' exists a.e. (Theorem 26.3.7) and is integrable. Put $G(x) = \int_a^x F'$. Then G is AC (absolute continuity of the integral) and, by Lebesgue differentiation, $G' = F'$ a.e. So $F - G$ is AC with derivative 0 a.e., hence constant by Theorem 26.3.11; evaluating at a gives $F - G \equiv F(a)$, i.e. $F(x) - F(a) = \int_a^x F'$. The converse is the same two facts read backwards: $\int_a^x f$ is AC, and Lebesgue differentiation gives $\frac{d}{dx} \int_a^x f = f$ a.e. $\square$

The theorem says the two operations are mutually inverse: differentiation $F \mapsto F'$ maps the absolutely continuous functions onto $L^1[a, b]$, and the indefinite integral $f \mapsto \int_a^{(\cdot)} f$ is a right inverse, exact once one quotients out the additive constant a derivative forgets.

$$\begin{array}{ccc} \text{AC}[a, b] & \xrightarrow{F \mapsto F'} & L^1[a, b] \\ & \nwarrow f \mapsto \int_a^{(\cdot)} f & \\ & & \end{array}$$

Figure 26.10. The fundamental theorem of calculus as a pair of inverse maps: $\frac{d}{dx} \int_a^x f = f$ a.e. recovers $f \in L^1$ exactly, while $\int_a^x F' = F(x) - F(a)$ recovers $F \in \text{AC}$ up to the additive constant $F(a)$.

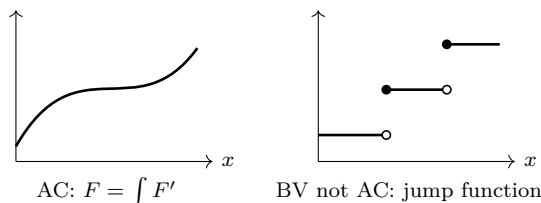


Figure 26.11. Left: an AC function is recovered by integrating its derivative. Right: a BV-but-not-AC function (steps) has $\int F' < F(b) - F(a)$; the rise hides in jumps or singular parts. (Schematic.)

Jump functions

Definition 26.3.13. Given points x_n and weights $\alpha_n > 0$ with $\sum \alpha_n < \infty$, the *jump function* is $J(x) = \sum_n \alpha_n j_n(x)$ where $j_n(x) = 0$ for $x < x_n$, θ_n at $x = x_n$, and 1 for $x > x_n$ (a fixed normalization $0 \leq \theta_n \leq 1$). J is increasing with a jump of size α_n at each x_n .

Theorem 26.3.14. For the jump function J above, $J'(x) = 0$ for almost every x .

Idea. J is increasing, so J' exists a.e. and $\int_a^b J' \leq J(b) - J(a)$. One shows the integral of J' is 0 by a covering argument: for the truncation $J_N = \sum_{n \leq N} \alpha_n j_n$, the difference quotient exceeds ε only on a set whose covering intervals have total J_0 -increment $< \eta$; since the tail $\sum_{n > N} \alpha_n \rightarrow 0$, the bad set is null. Hence $J' = 0$ a.e. $\square$

Theorem 26.3.15 (Lebesgue decomposition). Every increasing F on $[a, b]$ decomposes uniquely (up to constants) as $F = F_A + F_C + F_J$, where F_A is absolutely continuous, F_C is continuous with $F'_C = 0$ a.e. (singular continuous), and F_J is a jump function – each increasing.

Worked Example 26.3.16 (Total variation of a sawtooth). Compute $T_F(0, 2)$ for F that increases linearly from 0 to 1 on $[0, 1]$ then decreases linearly to 0 on $[1, 2]$.

Recognition. “Total variation of a piecewise-monotone function is the sum of the rises and falls.”

Solution. On $[0, 1]$ the variation is $|1 - 0| = 1$; on $[1, 2]$ it is $|0 - 1| = 1$. Any partition’s variation is maximized by including the peak $t = 1$, giving $T_F(0, 2) = 1 + 1 = 2$. *Check.* $P_F = 1$ (total rise), $N_F = 1$ (total fall), and indeed $P_F + N_F = 2 = T_F$ while $P_F - N_F = 0 = F(2) - F(0)$, matching Lemma 26.3.5.

Worked Example 26.3.17 ($x^2 \sin(1/x^2)$: differentiable but $F' \notin L^1$). For $F(x) = x^2 \sin(1/x^2)$ ($F(0) = 0$) show $F'(x)$ exists everywhere but $F' \notin L^1[-1, 1]$, so F is differentiable yet not AC.

Recognition. “Differentiable everywhere need not give an integrable derivative.”

Solution. For $x \neq 0$, $F'(x) = 2x \sin(1/x^2) - \frac{2}{x} \cos(1/x^2)$; at 0, $F'(0) = \lim x \sin(1/x^2) = 0$. The term $\frac{2}{x} \cos(1/x^2)$ is not integrable near 0: its absolute value, integrated, diverges like $\int \frac{dx}{x}$ through the oscillation peaks. Hence $F' \notin L^1$, so F is *not* AC and $\int_{-1}^1 F'$ is not even defined as a Lebesgue integral. *Check.* F is BV-failing here too – the wild oscillation makes $T_F = \infty$, consistent with non-absolute-continuity.

Worked Example 26.3.18 (The Cantor function is not absolutely continuous). Show directly that the Cantor–Lebesgue function F is not AC.

Recognition. “Exhibit small-total-length intervals with non-small total increment.”

Solution. At stage n the Cantor set is covered by 2^n intervals of total length $(2/3)^n \rightarrow 0$, yet F increases by a total of 1 across them (F rises only on $\mathcal{C}$). So for $\varepsilon = \frac{1}{2}$ no δ works: pick n with $(2/3)^n < \delta$; the 2^n Cantor intervals have total length $< \delta$ but total increment $1 > \varepsilon$. Hence F is not AC. *Check.* $\int_0^1 F' = 0 \neq 1 = F(1) - F(0)$ – the FTC *fails*, exactly as non-absolute-continuity predicts (Theorem 26.3.12).

Worked Example 26.3.19 (Integration by parts for AC functions). If F, G are AC on $[a, b]$, show $\int_a^b F'G = -\int_a^b FG' + [FG]_a^b$.

Recognition. “Use that a product of AC functions is AC, then apply the FTC.”

Solution. FG is AC (Exercise 22), so by Theorem 26.3.12, $\int_a^b (FG)' = [FG]_a^b$. Where both derivatives exist (a.e.) the product rule gives $(FG)' = F'G + FG'$, and both terms are integrable. Hence $[FG]_a^b = \int_a^b F'G + \int_a^b FG'$, which rearranges to the claim. *Check.* With $F = G = x$ on $[0, 1]$: $\int_0^1 x dx = 1/2$, and $-\int_0^1 x dx + [x^2]_0^1 = -1/2 + 1 = 1/2$. Agree.

Worked Example 26.3.20 (Lipschitz $\Leftrightarrow$ AC with bounded derivative). Show F is Lipschitz with constant M iff F is AC and $|F'| \leq M$ a.e.

Recognition. “Characterize the Lipschitz class inside AC.”

Solution. $(\Rightarrow)$ Lipschitz gives $\sum |F(b_k) - F(a_k)| \leq M \sum (b_k - a_k)$, so AC (take $\delta = \varepsilon/M$); and $|F'| \leq M$ wherever F' exists. $(\Leftarrow)$ AC with $|F'| \leq M$ gives, by the FTC, $|F(x) - F(y)| = \left| \int_y^x F' \right| \leq M |x - y|$. *Check.* $F(x) = |x|$ is Lipschitz ($M = 1$), AC, with $|F'| = 1$ a.e.; the corner at 0 (measure zero) is harmless.

Worked Example 26.3.21 (Total variation of $x \sin(1/x)$ versus $x^2 \sin(1/x)$). Decide BV on $[0, 1]$ for $g(x) = x \sin(1/x)$ and $h(x) = x^2 \sin(1/x)$ (both 0 at 0).

Recognition. “Apply the $x^a \sin(x^{-b})$ criterion $a > b$ with $b = 1$.”

Solution. For g , $a = 1 = b$, so $a > b$ fails: g is *not* BV (the local extrema near 0 are at $x_n \sim 1/(n\pi)$ with $|g(x_n)| \sim 1/(n\pi)$, and $\sum 1/n = \infty$). For h , $a = 2 > 1 = b$, so h *is* BV. *Check.* $h' = 2x \sin(1/x) - \cos(1/x)$ is bounded, so by Example 26.3.3(2) h is BV; $g' = \sin(1/x) - \frac{1}{x} \cos(1/x)$ is unbounded and indeed $g \notin \text{BV}$.

Worked Example 26.3.22 (Fat Cantor set: AC with $F' = 0$ on positive measure). Let $K \subseteq [0, 1]$ be a fat Cantor set with $m(K) = 1/2$ and $F(x) = \int_0^x \mathbf{1}_{K^c}$. Show F is strictly increasing, AC, yet $F' = 0$ on the positive-measure set K .

Recognition. “Integral of an indicator is AC; its derivative is the integrand a.e.”

Solution. $F(x) = \int_0^x \mathbf{1}_{K^c} = m(K^c \cap [0, x])$. Since K^c is dense and open, every subinterval meets K^c , so F is strictly increasing. F is AC as an indefinite integral (Theorem 26.3.12), with $F' = \mathbf{1}_{K^c}$ a.e., hence $F' = 0$ on K , $m(K) = 1/2 > 0$. *Check.* No contradiction with the FTC – $F(1) - F(0) = m(K^c) = 1/2 = \int_0^1 \mathbf{1}_{K^c}$; absolute continuity is intact, the a.e.-zero derivative just lives on a positive-measure set without forcing F constant (it is 0 only there, positive in measure off K).

Caution. The chain “differentiable everywhere” $\nRightarrow \int F' = F(b) - F(a)$. The correct hypothesis is *absolute* continuity, not mere differentiability or even continuity plus a.e.-derivative. The Cantor function is continuous, increasing, differentiable a.e. with $F' = 0$ a.e., yet $\int_0^1 F' = 0 \neq 1$. Singular functions are the obstruction the FTC must exclude.

Exercises (Stein, Ch.3) – bounded variation and absolute continuity.

1. Construct an increasing function on $\mathbb{R}$ whose set of discontinuities is precisely $\mathbb{Q}$.

2. If $a, b > 0$, let $f(x) = x^a \sin(x^{-b})$ for $0 < x \leq 1$ and $f(0) = 0$. Prove f is of bounded variation on $[0, 1]$ iff $a > b$. Then, taking $a = b$, construct (for each $0 < \alpha < 1$) a function satisfying the Lipschitz condition $|f(x) - f(y)| \leq A|x - y|^\alpha$ but which is not of bounded variation. [Hint: estimate $|f(x+h) - f(x)|$ by $C(x+h)^\alpha$ or $C'h/x$; split on whether $x^{\alpha+1} \geq h$. What is the relation between α and a ?
3. Consider $F(x) = x^2 \sin(1/x^2)$, $x \neq 0$, $F(0) = 0$. Show $F'(x)$ exists for every x , but F' is not integrable on $[-1, 1]$.
4. Show directly from the definition that the Cantor–Lebesgue function is not absolutely continuous.
5. (Measurability issues in differentiation.) (a) If F is continuous on $[a, b]$, show

$$D^+F(x) = \limsup_{h \rightarrow 0^+} \frac{F(x+h) - F(x)}{h}$$

is measurable. (b) If $J(x) = \sum_n \alpha_n j_n(x)$ is a jump function as in Section 3.3, show $\limsup_{h \rightarrow 0} \frac{J(x+h) - J(x)}{h}$ is measurable. [Hint: for (a), continuity lets one restrict the limsup to countably many h ; for (b), use the partial sums J_N and let $N \rightarrow \infty$, $k \rightarrow \infty$, $m \rightarrow \infty$ in $F_{k,m}^N = \sup_{1/k \leq |h| \leq 1/m} \left| \frac{J_N(x+h) - J_N(x)}{h} \right|$.]

6. Suppose F is of bounded variation and continuous. Prove $F = F_1 - F_2$ with both F_1, F_2 monotonic and continuous.
7. Show that if F is of bounded variation on $[a, b]$, then: (a) $\int_a^b |F'(x)| dx \leq T_F(a, b)$; (b) $\int_a^b |F'(x)| dx = T_F(a, b)$ iff F is absolutely continuous. (Hence the length formula $L = \int_a^b |z'(t)| dt$ holds iff z is absolutely continuous.)
8. Verify the agreement between the two definitions of the Cantor–Lebesgue function (the one in Exercise 2, Chapter 1, and the one in Section 3.1 of this chapter).
9. Show that if $f: \mathbb{R} \rightarrow \mathbb{R}$ is absolutely continuous, then (a) f maps measure-zero sets to measure-zero sets; (b) f maps measurable sets to measurable sets.
10. For F absolutely continuous and increasing on $[a, b]$, $A = F(a)$, $B = F(b)$: (a) there exists such an F that is in addition strictly increasing but with $F'(x) = 0$ on a set of positive measure; (b) the F in (a) can be chosen so that some measurable $E \subseteq [A, B]$ with $m(E) = 0$ has $F^{-1}(E)$ not measurable; (c) however, for any increasing absolutely continuous F and measurable $E \subseteq [A, B]$, the set $F^{-1}(E) \cap \{F' > 0\}$ is measurable. [Hint: (a) $F(x) = \int_a^x \mathbf{1}_K$ with K the complement of a positive-measure Cantor-like set; for (c) prove $m(O) = \int_{F^{-1}(O)} F'$ for open O .]
11. Let F be absolutely continuous and increasing on $[a, b]$ with $F(a) = A$, $F(b) = B$, and f measurable on $[A, B]$. (a) Show $f(F(x))F'(x)$ is measurable on $[a, b]$ (note $f(F(x))$ alone need not be). (b) Prove the change of variables: if f is integrable on $[A, B]$ then so is $f(F(x))F'(x)$, and $\int_a^b f(F(x))F'(x) dx = \int_A^B f(y) dy$. [Hint: start from $m(O) = \int_{F^{-1}(O)} F'$.]
12. Suppose F, G are absolutely continuous on $[a, b]$. Show their product FG is also absolutely continuous. Consequences: (a) $\int_a^b F'(x)G(x) dx = -\int_a^b F(x)G'(x) dx + [F(x)G(x)]_a^b$; (b) if F is AC on $[-\pi, \pi]$ with $F(\pi) = F(-\pi)$ and $a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(x)e^{-inx} dx$ (so $F \sim \sum a_n e^{inx}$), then $F' \sim \sum in a_n e^{inx}$; (c) what happens if $F(-\pi) \neq F(\pi)$? [Hint: consider $F(x) = x$.]
13. Let F be continuous on $[a, b]$. Show: (a) if $D^+F(x) \geq 0$ for every x , then F is increasing on $[a, b]$; (b) if $F'(x)$ exists for every $x \in (a, b)$ with $|F'(x)| \leq M$, then $|F(x) - F(y)| \leq M|x - y|$ and F is absolutely continuous. [Hint: for (a), assume $F(b) < F(a)$; with $G_\varepsilon(x) = F(x) - F(a) + \varepsilon(x - a)$, the greatest x_0 with $G_\varepsilon(x_0) \geq 0$ has $D^+G_\varepsilon(x_0) > 0$, a contradiction.]
14. Suppose F is increasing on $[a, b]$. (a) Prove $F = F_A + F_C + F_J$, each increasing, with F_A absolutely continuous, F_C continuous and $F'_C = 0$ a.e., F_J a jump function. (b) Each component is uniquely determined up to an additive constant. (This is the Lebesgue decomposition of F ; a corresponding decomposition holds for F of bounded variation.)

15. A bounded F is of bounded variation on $\mathbb{R}$ if it is BV on every finite $[a, b]$ and $\sup_{a,b} T_F(a, b) < \infty$. Prove such an F satisfies: (a) $\int_{\mathbb{R}} |F(x+h) - F(x)| dx \leq A|h|$ for some constant A and all h ; (b) $|\int_{\mathbb{R}} F(x)\varphi'(x) dx| \leq A$, where φ ranges over C^1 functions of bounded support with $\sup |\varphi| \leq 1$. [Hint: write $F = F_1 - F_2$ monotone bounded; deduce (b) from (a).]
16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Prove f satisfies a Lipschitz condition $|f(x) - f(y)| \leq M|x - y|$ for some M and all x, y iff f is (i) absolutely continuous and (ii) $|f'(x)| \leq M$ for a.e. x .

Problems (Stein, Ch.3) – convexity, FTC, nowhere differentiability.

1. A real φ on (a, b) is *convex* if the region above its graph is convex, equivalently $\varphi(\theta x_1 + (1 - \theta)x_2) \leq \theta\varphi(x_1) + (1 - \theta)\varphi(x_2)$ for $x_1, x_2 \in (a, b)$, $0 \leq \theta \leq 1$; this yields the slope inequalities $\frac{\varphi(x+h) - \varphi(x)}{h} \leq \frac{\varphi(y) - \varphi(x)}{y-x} \leq \frac{\varphi(y) - \varphi(y-h)}{h}$ for $x < y$, $h > 0$, $x + h < y$. Prove: (a) φ is continuous on (a, b) ; (b) φ is Lipschitz of order 1 on every closed $[a', b'] \subseteq (a, b)$, hence absolutely continuous there; (c) φ' exists off an at most countable set, $\varphi' = D^+\varphi$ is increasing, and $\varphi(y) - \varphi(x) = \int_x^y \varphi'(t) dt$; (d) conversely, if ψ is increasing on (a, b) , then $\varphi(x) = \int_c^x \psi(t) dt$ is convex.
2. Suppose F is continuous on $[a, b]$, $F'(x)$ exists for every $x \in (a, b)$, and F' is integrable. Then F is absolutely continuous and $F(b) - F(a) = \int_a^b F'(x) dx$. [Hint: assume $F' \geq 0$ a.e.; with $E = \{F' < 0\}$ of measure 0 and Φ increasing AC with $D^+\Phi = \infty$ on E (Exercise 25), apply Exercise 23(a) to $F + \delta\Phi$.]
3. * (Converse to Exercise 30.) If F is bounded and measurable on $\mathbb{R}$ and satisfies (a) or (b) of Exercise 30, then F can be modified on a null set to become BV on $\mathbb{R}$. Moreover on $\mathbb{R}^d$, for bounded measurable F the conditions (a') $\int_{\mathbb{R}^d} |F(x+h) - F(x)| dx \leq A|h|$ for all $h \in \mathbb{R}^d$, and (b') $\left| \int_{\mathbb{R}^d} F(x) \frac{\partial \varphi}{\partial x_j} dx \right| \leq A$ for all $j = 1, \dots, d$ and all C^1 φ of bounded support with $\sup |\varphi| \leq 1$, are equivalent. (This class is the $\mathbb{R}^d$ extension of bounded variation.)
4. Consider $f_1(x) = \sum_{n=0}^{\infty} 2^{-n} e^{2\pi i 2^n x}$. (a) Prove $|f_1(x) - f_1(y)| \leq A_{\alpha} |x - y|^{\alpha}$ for each $0 < \alpha < 1$. (b)* However f_1 is nowhere differentiable, hence not of bounded variation.

Additional exercises.

1. Compute the total variation $T_F(0, 3)$ of the piecewise-linear F with vertices $(0, 0), (1, 2), (2, -1), (3, 1)$, and find P_F, N_F ; verify $P_F + N_F = T_F$ and $P_F - N_F = F(3) - F(0)$.
2. Show $F(x) = |x|$ on $[-1, 1]$ is BV, AC, and Lipschitz; compute $T_F(-1, 1)$ and confirm $\int_{-1}^1 |F'| = T_F$.
3. Prove that the sum and product of two BV functions on $[a, b]$ are BV, and that BV is a vector space but not closed under composition with continuous functions.
4. Give the Jordan decomposition $F = F_1 - F_2$ explicitly for $F(x) = \sin x$ on $[0, 2\pi]$, with $F_1 = P_F + F(0)$ and $F_2 = N_F$, and sketch both.
5. Show that a BV function has at most countably many discontinuities, all of jump type, and that the sum of the jump sizes is finite.
6. (Cantor function.) Verify $F(1/3) = 1/2$, $F(2/3) = 1/2$, and $F(7/9) = 5/8$ for the Cantor–Lebesgue function, and use these to argue $F' = 0$ on the removed middle-thirds.
7. Prove that if F is AC on $[a, b]$ and G is Lipschitz on the range of F , then $G \circ F$ is AC. Give an example showing AC $\circ$ AC can fail to be AC.
8. (Non-example.) Show that $F(x) = \sqrt{x}$ on $[0, 1]$ is AC (hence BV) even though $F'(x) = 1/(2\sqrt{x})$ is unbounded near 0; contrast with $x^2 \sin(1/x^2)$, which is differentiable everywhere but not AC.
9. (Diagram.) Draw the stage- n approximations F_n of the Cantor function for $n = 1, 2, 3$ on the same axes and indicate the intervals on which each is constant.
10. Let $F(x) = \int_0^x \mathbf{1}_K$ where $K \subseteq [0, 1]$ is a fat Cantor set with $m(K) = 1/2$. Show F is strictly increasing and AC with $F' = 0$ on a set of positive measure, and explain why this does not contradict the FTC.

26.4 Rectifiable Curves and the Isoperimetric Inequality

The length formula

We saw (Theorem 26.3.2) that a curve is rectifiable iff its coordinates are BV. The classical length integral, however, needs the *stronger* hypothesis of absolute continuity.

Example 26.4.1. Let $x(t) = y(t) = F(t)$, the Cantor–Lebesgue function, $0 \leq t \leq 1$. This parametrizes the segment from $(0, 0)$ to $(1, 1)$, of length $\sqrt{2}$, yet $x' = y' = 0$ a.e., so $\int_0^1 \sqrt{x'^2 + y'^2} dt = 0 \neq \sqrt{2}$. The naive formula fails precisely because F is not AC.

Theorem 26.4.2. If $x(t), y(t)$ are absolutely continuous on $[a, b]$, the curve is rectifiable and its length is

$$L = \int_a^b (x'(t)^2 + y'(t)^2)^{1/2} dt.$$

This follows from a sharpening of the FTC for the total variation:

Proposition 26.4.3. If $F = x + iy$ is complex-valued and absolutely continuous on $[a, b]$, then $T_F(a, b) = \int_a^b |F'(t)| dt$.

Proof. ($\leq$) For any partition, $|F(t_j) - F(t_{j-1})| = \left| \int_{t_{j-1}}^{t_j} F' \right| \leq \int_{t_{j-1}}^{t_j} |F'|$ (FTC, Theorem 26.3.12); summing gives $\sum |F(t_j) - F(t_{j-1})| \leq \int_a^b |F'|$, so $T_F(a, b) \leq \int_a^b |F'|$. ($\geq$) Fix ε . Approximate $F' = g + h$ with g a step function and $\int |h| < \varepsilon$ (step functions are dense in L^1). With $G = \int g$, $H = \int h$, $F = G + H + \text{const}$ and $T_F \geq T_G - T_H \geq T_G - \varepsilon$. On each interval where g is constant, $|G(t_j) - G(t_{j-1})| = |\int g| = \int |g|$, so $T_G \geq \int_a^b |g| \geq \int_a^b |F'| - \varepsilon$. Hence $T_F \geq \int_a^b |F'| - 2\varepsilon$ for all ε . $\square$

Taking $F = x + iy$ gives $|F'| = \sqrt{x'^2 + y'^2}$ and $L = T_F(a, b) = \int_a^b \sqrt{x'^2 + y'^2}$, proving Theorem 26.4.2.

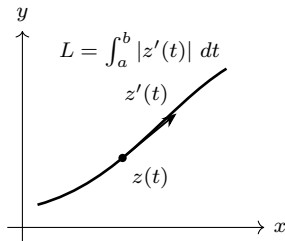


Figure 26.12. The arc length is the integral of the speed $|z'(t)| = \sqrt{x'^2 + y'^2}$, valid when the parametrization is absolutely continuous. (Schematic.)

Arc-length parametrization and Minkowski content

Theorem 26.4.4. A rectifiable curve with AC parametrization can be reparametrized by arc length: there is $\tilde{z}(s)$, $0 \leq s \leq L$, with $|\tilde{z}'(s)| = 1$ a.e.

Idea. Let $s(t) = T_z(a, t) = \int_a^t |z'|$ map $[a, b] \rightarrow [0, L]$; it is AC and increasing. Setting $\tilde{z}(s) = z(t(s))$ gives speed 1 a.e. by the chain rule and Proposition 26.4.3. $\square$

The original parametrization factors through the arc-length variable: $z = \tilde{z} \circ s$, so the same trace is traversed at unit speed.

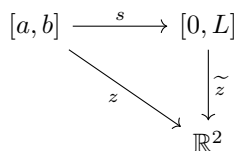


Figure 26.13. Arc-length reparametrization: the arc-length function $s(t) = \int_a^t |z'|$ factors z as $z = \tilde{z} \circ s$, where $\tilde{z}$ has $|\tilde{z}'| = 1$ a.e.

Definition 26.4.5. The (one-dimensional) *Minkowski content* of a compact $K \subseteq \mathbb{R}^2$ is

$$\mathcal{M}(K) = \lim_{\delta \rightarrow 0} \frac{m(K^\delta)}{2\delta}, \quad K^\delta = \{x : \text{dist}(x, K) < \delta\},$$

when the limit exists; $\mathcal{M}^*, \mathcal{M}_*$ denote the upper/lower limits. It measures the area of a thin tube around K , normalized by its width 2δ .

Theorem 26.4.6. If $\Gamma = \{z(t) : a \leq t \leq b\}$ is a quasi-simple curve, then $\mathcal{M}(\Gamma)$ exists iff Γ is rectifiable, and then $\mathcal{M}(\Gamma) = L$, the length.

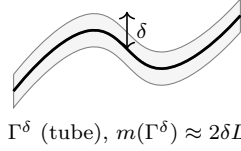


Figure 26.14. The δ -tube about a rectifiable curve Γ has area $\approx 2\delta \cdot L$; dividing by 2δ and letting $\delta \rightarrow 0$ recovers the length $\mathcal{M}(\Gamma) = L$. (Schematic.)

The isoperimetric inequality

Theorem 26.4.7 (Isoperimetric inequality). Let $\Omega \subseteq \mathbb{R}^2$ be bounded open with boundary a rectifiable curve Γ of length $\ell(\Gamma)$. Then

$$4\pi m(\Omega) \leq \ell(\Gamma)^2,$$

with equality for the disc. Among all curves of a given length, the circle encloses the greatest area.

Proof. For $\delta > 0$ set $\Omega_+(\delta) = \{x : \text{dist}(x, \Omega) < \delta\}$ and $\Omega_-(\delta) = \{x : \text{dist}(x, \Omega^c) \geq \delta\}$, so $\Omega_- \subseteq \Omega \subseteq \Omega_+$ and, with $\Gamma^\delta = \{\text{dist}(x, \Gamma) < \delta\}$, the disjoint union $\Omega_+(\delta) = \Omega_-(\delta) \cup \Gamma^\delta$. Let $D(\delta)$ be the disc of radius δ , $m(D(\delta)) = \pi\delta^2$. The inclusions $\Omega_+ \supseteq \Omega + D(\delta)$ and $\Omega \supseteq \Omega_- + D(\delta)$ feed the Brunn–Minkowski inequality $m(A+B)^{1/2} \geq m(A)^{1/2} + m(B)^{1/2}$:

$$m(\Omega_+(\delta)) \geq (m(\Omega)^{1/2} + \pi^{1/2}\delta)^2 \geq m(\Omega) + 2\pi^{1/2}\delta m(\Omega)^{1/2},$$

and similarly $-m(\Omega_-(\delta)) \geq -m(\Omega) + 2\pi^{1/2}\delta m(\Omega_-)^{1/2}$. Adding, with $m(\Gamma^\delta) = m(\Omega_+) - m(\Omega_-)$,

$$m(\Gamma^\delta) \geq 2\pi^{1/2}\delta (m(\Omega)^{1/2} + m(\Omega_-)^{1/2}).$$

Divide by 2δ and let $\delta \rightarrow 0$ (so $\Omega_-(\delta) \uparrow \Omega$): $\mathcal{M}_*(\Gamma) \geq 2\pi^{1/2}m(\Omega)^{1/2}$. By Theorem 26.4.6, $\ell(\Gamma) \geq \mathcal{M}_*(\Gamma) \geq 2\pi^{1/2}m(\Omega)^{1/2}$, i.e. $\ell(\Gamma)^2 \geq 4\pi m(\Omega)$. $\square$

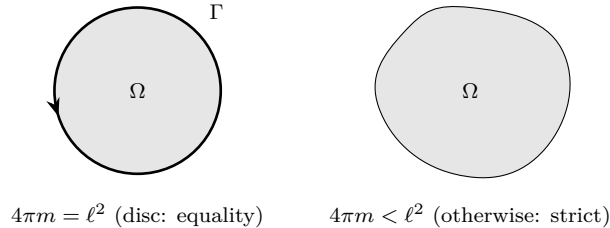


Figure 26.15. Isoperimetric inequality: $4\pi m(\Omega) \leq \ell(\Gamma)^2$, equality only for the disc; any non-circular region wastes perimeter. (Schematic.)

Worked Example 26.4.8 (Length of the graph of the Cantor function). For the curve $x(t) = t$, $y(t) = F(t)$ (Cantor function), $0 \leq t \leq 1$, show the length is 2, not $\sqrt{2}$.

Recognition. “Length of a BV-but-not-AC graph – the formula $\int \sqrt{1 + F'^2}$ undershoots.”

Solution. The map $t \mapsto (t, F(t))$ is monotone in both coordinates. Over any partition, $\sum \sqrt{(\Delta t)^2 + (\Delta F)^2}$; refining, the horizontal contributions sum to 1 and the vertical to $F(1) - F(0) = 1$, and because F increases only where t is (essentially) constant on Cantor scales and vice versa, the polygonal lengths approach $1 + 1 = 2$. So $L(x) = x + F(x)$ and $L(1) = 2$. *Check.* $\int_0^1 \sqrt{1 + F'^2} dt = \int_0^1 \sqrt{1 + 0} dt = 1 \neq 2$ – the naive integral misses the rise hidden on the Cantor set, illustrating the failure of Theorem 26.4.2 without absolute continuity.

Worked Example 26.4.9 (Length of a circle by the integral). Verify $L = 2\pi r$ for $z(t) = (r \cos t, r \sin t)$, $0 \leq t \leq 2\pi$.

Recognition. “Apply the AC length formula to a smooth closed curve.”

Solution. $x' = -r \sin t$, $y' = r \cos t$, so $\sqrt{x'^2 + y'^2} = r$. Then $L = \int_0^{2\pi} r dt = 2\pi r$. *Check.* The isoperimetric ratio $\frac{\ell^2}{4\pi m} = \frac{(2\pi r)^2}{4\pi \cdot \pi r^2} = \frac{4\pi^2 r^2}{4\pi^2 r^2} = 1$ – equality, as the disc must achieve.

Worked Example 26.4.10 (Isoperimetric deficit of a square). For the unit square ($\ell = 4$, $m = 1$) compute $\frac{\ell^2}{4\pi m}$ and confirm strict inequality.

Recognition. “Check the isoperimetric inequality on a non-circular region.”

Solution. $\frac{\ell^2}{4\pi m} = \frac{16}{4\pi} = \frac{4}{\pi} \approx 1.273 > 1$. So $4\pi m = 4\pi < 16 = \ell^2$, strictly. *Check.* To enclose area 1 a circle needs perimeter $2\sqrt{\pi} \approx 3.545 < 4$; the square “wastes” about 13% of its perimeter, exactly the isoperimetric deficit.

Worked Example 26.4.11 (Minkowski content of a segment). Compute $\mathcal{M}(K)$ for $K = [0, 1] \times \{0\} \subseteq \mathbb{R}^2$.

Recognition. “Tube area of a straight segment.”

Solution. K^δ is the δ -neighborhood: a $1 \times 2\delta$ rectangle capped by two half-discs of radius δ , so $m(K^\delta) = 2\delta + \pi\delta^2$. Then $\mathcal{M}(K) = \lim_{\delta \rightarrow 0} \frac{2\delta + \pi\delta^2}{2\delta} = \lim(1 + \frac{\pi\delta}{2}) = 1$. *Check.* $\mathcal{M}(K) = 1 = \text{length of } [0, 1]$, matching Theorem 26.4.6.

Worked Example 26.4.12 (Hausdorff dimension of the Cantor set). Show the Cantor set $\mathcal{C}$ has Hausdorff dimension $s = \log 2 / \log 3$ via self-similarity.

Recognition. “Self-similar fractal: N copies scaled by r give dimension $\log N / \log(1/r)$.”

Solution. $\mathcal{C}$ is the union of $N = 2$ copies of itself scaled by $r = 1/3$. The dimension s satisfies $Nr^s = 1$, i.e. $2 \cdot 3^{-s} = 1$, so $3^s = 2$ and $s = \frac{\log 2}{\log 3} \approx 0.631$. *Check.* $0 < s < 1$, consistent with $\mathcal{C}$ being uncountable (positive-dimensional) yet of Lebesgue measure zero (dimension < 1).

Worked Example 26.4.13 (A space-filling curve has infinite length). Explain why a continuous surjection $[0, 1] \rightarrow [0, 1]^2$ (Peano/Hilbert curve) cannot be rectifiable.

Recognition. “Rectifiable $\Rightarrow$ image has measure zero; a space-filling image does not.”

Solution. A rectifiable curve has finite length L , hence its image lies in a tube of area $\leq 2\delta L \rightarrow 0$, so the image is Lebesgue-null in $\mathbb{R}^2$. A space-filling curve has image $[0, 1]^2$ of measure 1 > 0 , so it cannot be rectifiable – its coordinate functions are continuous but *not* of bounded variation. *Check.* The Hilbert curve’s n -th polygonal approximation has length $\sim 2^n \rightarrow \infty$, confirming unbounded variation.

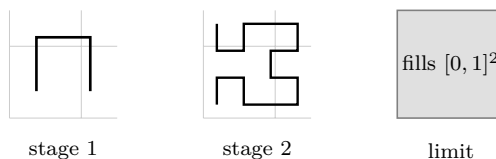


Figure 26.16. Stages of a space-filling (Hilbert) curve: polygonal length roughly doubles at each stage, so the limit curve has infinite length and is not rectifiable.

Caution. The length integral $\int \sqrt{x'^2 + y'^2}$ is valid *only* for absolutely continuous parametrizations. For BV-but-not-AC curves (Cantor graph) it strictly undershoots the true length, and for non-BV curves (space-filling) it is meaningless – the curve has infinite length. Rectifiability (BV) gives finite length; the *formula* needs AC.

Exercises (Stein, Ch.3) – rectifiable curves and Minkowski content.

1. A rectifiable curve has a tangent line at almost all points of the curve. Make this statement precise (and prove it).
2. A curve in $\mathbb{R}^d$ is a continuous map $t \mapsto z(t)$ of $[a, b]$ into $\mathbb{R}^d$. (a) State and prove the analogues, for curves in $\mathbb{R}^d$, of the rectifiability/length results Theorems 3.1, 4.1, and 4.3. (b) Define the one-dimensional Minkowski content $\mathcal{M}(K)$ of a compact $K \subseteq \mathbb{R}^d$ as the limit (if it exists) of $\frac{m(K^\delta)}{m_{d-1}(B(\delta))}$ as $\delta \rightarrow 0$, where $m_{d-1}(B(\delta))$ is the $(d-1)$ -measure of $B(\delta) = \{x \in \mathbb{R}^{d-1} : |x| < \delta\}$. State and prove the analogues of Propositions 4.5 and 4.7 for curves in $\mathbb{R}^d$.
3. Let $\Gamma = \{z(t), a \leq t \leq b\}$ be a curve satisfying a Lipschitz condition of exponent α , $1/2 \leq \alpha \leq 1$: $|z(t) - z(t')| \leq A|t - t'|^\alpha$ for all t, t' . Show $m(\Gamma^\delta) = O(\delta^{2-1/\alpha})$ for $0 < \delta \leq 1$.
4. Let F be the Cantor–Lebesgue function, and consider the curve $x(t) = t$, $y(t) = F(t)$, $0 \leq t \leq 1$. Prove the length $L(x)$ of the segment $0 \leq t \leq x$ is $L(x) = x + F(x)$; hence the total length of the curve is 2.

Additional exercises.

1. Compute the length of one arch of the cycloid $x(t) = t - \sin t$, $y(t) = 1 - \cos t$, $0 \leq t \leq 2\pi$, using Theorem 26.4.2, and check the answer equals 8.
2. Find the arc-length parametrization of the circle $z(t) = (\cos t, \sin t)$ and verify the speed is 1.
3. Compute the Minkowski content of the unit circle $\{|x| = 1\} \subseteq \mathbb{R}^2$ from the definition by finding $m(\Gamma^\delta)$ exactly (an annulus) and dividing by 2δ .
4. Verify the isoperimetric inequality for the ellipse with semi-axes a, b : estimate its perimeter and area and confirm $4\pi m \leq \ell^2$, with equality iff $a = b$.
5. (Hausdorff dimension.) Compute the similarity dimension of the Sierpinski triangle ($N = 3$ copies, ratio $1/2$) and of the von Koch curve ($N = 4$, ratio $1/3$); state which is rectifiable.
6. Show that a curve satisfying a Lipschitz- α condition with $\alpha = 1$ is rectifiable, while $\alpha < 1$ need not be, connecting to Exercise 29.
7. (Non-example.) Explain why the graph of $\sin(1/x)$ on $(0, 1]$ (with the point $(0, 0)$ adjoined) is connected but not a rectifiable curve, and compute that its coordinate function y is not BV.
8. (Diagram.) Sketch the sets Ω , $\Omega_-(\delta)$, $\Omega_+(\delta)$, and the tube Γ^δ for a non-circular region, and indicate how the Brunn–Minkowski step produces the bound $m(\Gamma^\delta) \geq 2\pi^{1/2}\delta m(\Omega)^{1/2}$.

Chapter 27

Hilbert Spaces: An Introduction

27.1 The Hilbert Space L^2

The space L^2 and its inner product

The Lebesgue theory of integration is what makes the space of square-integrable functions a *complete* inner-product space. This completeness — which fails for the Riemann integrable functions — is precisely what gives L^2 its geometric power.

Definition 27.1.1. Let $E \subseteq \mathbb{R}^d$ be measurable with $m(E) > 0$. The space $L^2(E)$ consists of all (equivalence classes of) measurable functions f supported on E with

$$\|f\| = \left(\int_E |f(x)|^2 dx \right)^{1/2} < \infty,$$

where two functions are identified when they agree almost everywhere. The *inner product* and *norm* are

$$\langle f, g \rangle = \int_E f(x) \overline{g(x)} dx, \quad \|f\| = \langle f, f \rangle^{1/2}.$$

When $E = \mathbb{R}^d$ we write simply $L^2(\mathbb{R}^d)$.

The identification of functions equal a.e. is not a technicality: without it $\|f\| = 0$ would not force $f = 0$, and we would have a seminorm, not a norm. Treating f as its equivalence class makes $\|\cdot\|$ a genuine norm.

Theorem 27.1.2 (Basic inequalities in L^2). *Let $f, g \in L^2(\mathbb{R}^d)$. Then:*

- (i) $L^2(\mathbb{R}^d)$ is a vector space, and fg is integrable;
- (ii) (Cauchy–Schwarz) $|\langle f, g \rangle| \leq \|f\| \|g\|$;
- (iii) $\langle \cdot, \cdot \rangle$ is linear in the first slot and conjugate-symmetric;
- (iv) (triangle inequality) $\|f + g\| \leq \|f\| + \|g\|$.

Proof. (i) From $|f + g| \leq 2 \max(|f|, |g|)$ we get $|f + g|^2 \leq 4(|f|^2 + |g|^2)$, so $\int |f + g|^2 \leq 4 \int |f|^2 + 4 \int |g|^2 < \infty$ and $f + g \in L^2$. Clearly $\lambda f \in L^2$. For integrability of fg , the elementary bound $2AB \leq A^2 + B^2$ (applied pointwise to $A = |f|$, $B = |g|$) gives

$$\int |fg| \leq \frac{1}{2} (\|f\|^2 + \|g\|^2) < \infty. \tag{27.1}$$

(ii) If $\|f\| = 0$ or $\|g\| = 0$ then $fg = 0$ a.e. and the inequality is trivial. If $\|f\| = \|g\| = 1$, then (27.1) gives $|\langle f, g \rangle| \leq \int |fg| \leq 1$. In general normalize $\tilde{f} = f/\|f\|$, $\tilde{g} = g/\|g\|$ to reduce to the unit case, then multiply through by $\|f\| \|g\|$. (iii) is linearity of the integral. (iv) Expanding,

$$\|f + g\|^2 = \|f\|^2 + \langle f, g \rangle + \langle g, f \rangle + \|g\|^2 \leq \|f\|^2 + 2|\langle f, g \rangle| + \|g\|^2 \leq (\|f\| + \|g\|)^2,$$

using Cauchy–Schwarz in the last step; take square roots. □

Method (Reading off L^2 membership and bounding $\int fg$). *Recognition trigger.* You must decide whether a given f lies in L^2 , or bound an integral $\int fg$ or $\langle f, g \rangle$.

1. For membership, integrate $|f|^2$; near a singularity $|x|^{-\alpha}$ on $\mathbb{R}^d$ is L^2 iff $2\alpha < d$ (local) or $2\alpha > d$ (at infinity).
2. To bound $\int fg$, reach for Cauchy–Schwarz $|\langle f, g \rangle| \leq \|f\| \|g\|$, with equality only when f, g are parallel.
3. On a finite-measure set E , $f \in L^2(E) \implies f \in L^1(E)$ with $\|f\|_{L^1} \leq m(E)^{1/2} \|f\|_{L^2}$ (take $g = \mathbf{1}_E$).

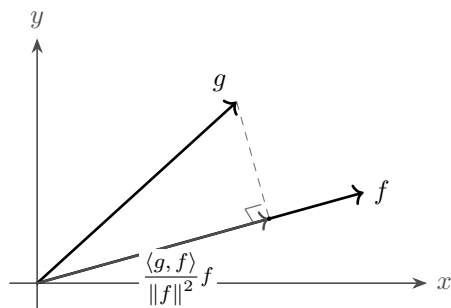


Figure 27.1. Cauchy–Schwarz says the projection of g onto f has length $|\langle g, f \rangle| / \|f\| \leq \|g\|$, i.e. $|\cos \angle(f, g)| \leq 1$; the residual $g - \frac{\langle g, f \rangle}{\|f\|^2} f$ is perpendicular to f .

Worked Example 27.1.3 (Is $|x|^{-\alpha}$ in L^2 ?). For which α is $f(x) = |x|^{-\alpha} \mathbf{1}_{\{|x| \leq 1\}}$ in $L^2(\mathbb{R}^d)$? And $g(x) = |x|^{-\alpha} \mathbf{1}_{\{|x| > 1\}}$?

Recognition. Membership test: integrate $|f|^2 = |x|^{-2\alpha}$ in polar coordinates, where the surface element contributes $r^{d-1} dr$.

Solution. $\int_{|x| \leq 1} |x|^{-2\alpha} dx = c_d \int_0^1 r^{-2\alpha} r^{d-1} dr = c_d \int_0^1 r^{d-1-2\alpha} dr$, finite iff $d-1-2\alpha > -1$, i.e. $\alpha < d/2$. For g , $\int_1^\infty r^{d-1-2\alpha} dr$ is finite iff $d-1-2\alpha < -1$, i.e. $\alpha > d/2$. *Check.* In $d=1$ the threshold is $\alpha = 1/2$: $x^{-1/2}$ is in $L^2(0, 1)$ ($\int_0^1 x^{-1}$ diverges — wait, $2\alpha = 1 < 1$ fails) — recompute: $x^{-1/2}$ has $|f|^2 = x^{-1}$, $\int_0^1 x^{-1}$ diverges, consistent with needing $\alpha < 1/2$ strictly. So $x^{-0.4} \in L^2(0, 1)$ but $x^{-0.6} \notin$. ■

Worked Example 27.1.4 ($L^2 \not\subseteq L^1$ and $L^1 \not\subseteq L^2$ on $\mathbb{R}^d$). Exhibit $f \in L^2(\mathbb{R}) \setminus L^1(\mathbb{R})$ and $h \in L^1(\mathbb{R}) \setminus L^2(\mathbb{R})$.

Recognition. Use power singularities at 0 and decay at ∞ , exploiting that the two norms weight large and small values differently.

Solution. Take $f(x) = \frac{1}{1+|x|}$. Then $\int f^2 = \int (1+|x|)^{-2} < \infty$ so $f \in L^2$, but $\int f = \int (1+|x|)^{-1} = \infty$, so $f \notin L^1$. Conversely $h(x) = |x|^{-1/2} \mathbf{1}_{\{|x| \leq 1\}}$ has $\int |h| = 2 \int_0^1 x^{-1/2} = 4 < \infty$ (so $h \in L^1$) but $\int h^2 = 2 \int_0^1 x^{-1} = \infty$ ($h \notin L^2$). *Check.* Neither inclusion holds; only on a *finite-measure* set does $L^2 \subseteq L^1$.

Worked Example 27.1.5 (The L^1 bound on a finite interval). For $f \in L^2([0, 1])$ show $\|f\|_{L^1} \leq \|f\|_{L^2}$, and find f giving equality.

Recognition. Trigger: pass from L^2 to L^1 on a finite-measure set; this is Cauchy–Schwarz against the constant 1.

Solution. $\|f\|_{L^1} = \int_0^1 |f| \cdot 1 \leq (\int_0^1 |f|^2)^{1/2} (\int_0^1 1)^{1/2} = \|f\|_{L^2}$, since $m([0, 1]) = 1$. Equality in Cauchy–Schwarz needs $|f|$ proportional to 1, i.e. $|f|$ constant. *Check.* $f \equiv c$ gives $\|f\|_{L^1} = |c| = \|f\|_{L^2}$, equality, as predicted.

Completeness and separability

Definition 27.1.6. The norm induces the metric $d(f, g) = \|f - g\|$. A sequence $\{f_n\} \subset L^2$ is *Cauchy* if $d(f_n, f_m) \rightarrow 0$, and *converges* to f if $d(f_n, f) \rightarrow 0$.

Theorem 27.1.7 (Riesz–Fischer: L^2 is complete). *The space $L^2(\mathbb{R}^d)$ is complete: every Cauchy sequence converges to an element of $L^2(\mathbb{R}^d)$.*

Proof. Let $\{f_n\}$ be Cauchy. Pass to a subsequence $\{f_{n_k}\}$ with $\|f_{n_{k+1}} - f_{n_k}\| \leq 2^{-k}$. Set

$$f = f_{n_1} + \sum_{k \geq 1} (f_{n_{k+1}} - f_{n_k}), \quad g = |f_{n_1}| + \sum_{k \geq 1} |f_{n_{k+1}} - f_{n_k}|.$$

By the triangle inequality the partial sums $S_K(g)$ satisfy $\|S_K(g)\| \leq \|f_{n_1}\| + \sum_{k=1}^K 2^{-k} \leq \|f_{n_1}\| + 1$. Letting $K \rightarrow \infty$ and applying the monotone convergence theorem gives $\int |g|^2 < \infty$, so $g \in L^2$; since $|f| \leq g$, also $f \in L^2$ and the series for f converges a.e. The $(K-1)$ th partial sum telescopes to f_{n_K} , so $f_{n_k} \rightarrow f$ a.e. Because $|f - S_K(f)|^2 \leq (2g)^2$, dominated convergence gives $\|f_{n_k} - f\| \rightarrow 0$. Finally, the full sequence is Cauchy: given ε pick N with $\|f_n - f_m\| < \varepsilon/2$ for $n, m > N$ and $n_k > N$ with $\|f_{n_k} - f\| < \varepsilon/2$; then $\|f_n - f\| \leq \|f_n - f_{n_k}\| + \|f_{n_k} - f\| < \varepsilon$. $\square$

Remark. The contrast with Riemann integration is the whole point. The Riemann-integrable functions on $[-\pi, \pi]$ are *not* complete in the L^2 metric: a Cauchy sequence of such functions can converge to a function that is not Riemann integrable. Lebesgue's theory closes this gap.

Theorem 27.1.8 (L^2 is separable). *There is a countable family $\{f_k\} \subset L^2(\mathbb{R}^d)$ whose finite linear combinations are dense.*

Proof. Take functions $r \mathbf{1}_R$ where $r \in \mathbb{Q} + i\mathbb{Q}$ and R is a rectangle with rational coordinates. Given $f \in L^2$ and $\varepsilon > 0$, truncate: $g_n = f \mathbf{1}_{\{|x| \leq n, |f| \leq n\}}$. Then $|f - g_n|^2 \leq 4|f|^2$ and $g_n \rightarrow f$ a.e., so by dominated convergence $\|f - g_n\| < \varepsilon/2$ for some N . The function $g = g_N$ is bounded with bounded support, hence in L^1 ; approximate it by a step function φ with $|\varphi| \leq N$ and $\int |g - \varphi| < \varepsilon^2/(16N)$, then replace its coefficients and rectangles by rational ones to get ψ with $|\psi| \leq N$ and $\int |g - \psi| < \varepsilon^2/(8N)$. Then $\int |g - \psi|^2 \leq 2N \int |g - \psi| < \varepsilon^2/4$, so $\|g - \psi\| < \varepsilon/2$ and $\|f - \psi\| < \varepsilon$. $\square$

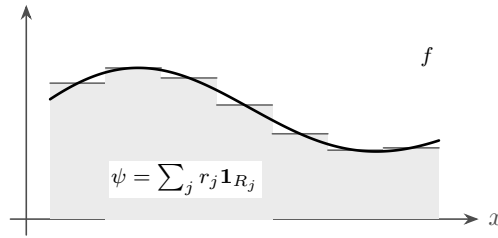


Figure 27.2. Separability: every $f \in L^2$ is approximated in norm by a step function ψ with rational coefficients on rational rectangles — here each block tracks f over its subinterval. (Schematic.)

Worked Example 27.1.9 (Completeness fails for Riemann integrable functions). Build a sequence of Riemann integrable f_n on $[0, 1]$ that is L^2 -Cauchy but whose limit is not Riemann integrable.

Recognition. Trigger: illustrate Remark above; use an enumeration of the rationals to build the Dirichlet function as an L^2 limit.

Solution. Enumerate $\mathbb{Q} \cap [0, 1] = \{q_1, q_2, \dots\}$ and let $f_n = \mathbf{1}_{\{q_1, \dots, q_n\}}$. Each f_n is Riemann integrable (finitely many discontinuities) with $\int f_n^2 = 0$. For $m > n$, $\|f_m - f_n\|^2 = \int \mathbf{1}_{\{q_{n+1}, \dots, q_m\}} = 0$, so the sequence is Cauchy (indeed constant in norm). Its pointwise limit is the Dirichlet function $\mathbf{1}_{\mathbb{Q} \cap [0, 1]}$, which is *not* Riemann integrable. *Check.* In L^2 the limit is the zero class (a countable set has measure 0), and indeed $\|f_n - 0\| = 0$; L^2 completeness holds, Riemann completeness does not.

Worked Example 27.1.10 (A concrete inner-product computation). In $L^2([0, 1])$ compute $\langle f, g \rangle$ and $\|f\|$ for $f(x) = x$, $g(x) = 1$, and verify Cauchy–Schwarz.

Recognition. Trigger: direct evaluation of Definition 27.1.1; then check the inequality numerically.

Solution. $\langle f, g \rangle = \int_0^1 x \cdot 1 \, dx = \frac{1}{2}$. $\|f\| = (\int_0^1 x^2)^{1/2} = 1/\sqrt{3}$, $\|g\| = 1$. Cauchy–Schwarz asserts $|\langle f, g \rangle| = \frac{1}{2} \leq \|f\| \|g\| = 1/\sqrt{3} \approx 0.577$. *Check.* $0.5 \leq 0.577$ holds, with strict inequality because $f = x$ is not a constant multiple of $g = 1$ (no equality case). $\blacksquare$

Worked Example 27.1.11 (Triangle inequality with a concrete pair). In $L^2([0, 1])$ take $f(x) = 1$, $g(x) = x$; verify $\|f + g\| \leq \|f\| + \|g\|$ and find the defect from equality.

Recognition. Trigger: numerically check the triangle inequality (Theorem 27.1.2(iv)); equality would require f, g parallel.

Solution. $\|f\| = 1$, $\|g\| = (\int_0^1 x^2)^{1/2} = 1/\sqrt{3} \approx 0.577$. And $\|f + g\|^2 = \int_0^1 (1 + x)^2 = \int_0^1 (1 + 2x + x^2) = 1 + 1 + \frac{1}{3} = \frac{7}{3}$, so $\|f + g\| = \sqrt{7/3} \approx 1.528$. The bound gives $\|f\| + \|g\| \approx 1.577$, and $1.528 \leq 1.577$. *Check.* Strict inequality, with defect ≈ 0.049 , because $g = x$ is not a nonnegative constant multiple of $f = 1$. ■

Caution. $\|f\| = 0$ means $f = 0$ as an element of L^2 , i.e. $f = 0$ almost everywhere — not at every point. Always remember that elements of L^2 are equivalence classes; “pointwise value at a single x ” is meaningless for an L^2 function. This is exactly what lets $\|\cdot\|$ be a true norm.

Exercises (Stein, Ch.4).

1. Show that properties (i) and (ii) in the definition of a Hilbert space imply property (iii): the Cauchy–Schwarz inequality $|\langle f, g \rangle| \leq \|f\| \|g\|$ and the triangle inequality $\|f + g\| \leq \|f\| + \|g\|$. [Hint: consider $\langle f + \lambda g, f + \lambda g \rangle$ as a positive quadratic in λ ; for the triangle inequality expand $\|f + g\|^2$.]
2. In the equality case of Cauchy–Schwarz: if $|\langle f, g \rangle| = \|f\| \|g\|$ and $g \neq 0$, then $f = cg$ for some scalar c . [Hint: assume $\|f\| = \|g\| = 1$, $\langle f, g \rangle = 1$; then $f - g \perp g$ and $\|f\|^2 = \|f - g\|^2 + \|g\|^2$.]
3. Note that $\|f + g\|^2 = \|f\|^2 + \|g\|^2 + 2 \operatorname{Re} \langle f, g \rangle$. Deduce the parallelogram law $\|f + g\|^2 + \|f - g\|^2 = 2(\|f\|^2 + \|g\|^2)$.
4. Prove directly from the definition that $\ell^2(\mathbb{Z})$ is complete and separable.
5. Relations between $L^2(\mathbb{R}^d)$ and $L^1(\mathbb{R}^d)$: (a) neither inclusion $L^2 \subseteq L^1$ nor $L^1 \subseteq L^2$ holds in general; (b) if f is supported on E with $m(E) < \infty$ and $f \in L^2$, then $f \in L^1$ with $\|f\|_{L^1} \leq m(E)^{1/2} \|f\|_{L^2}$; (c) if $|f| \leq M$ and $f \in L^1$, then $f \in L^2$ with $\|f\|_{L^2} \leq M^{1/2} \|f\|_{L^1}^{1/2}$. [Hint for (a): consider $|x|^{-\alpha}$ for $|x| \leq 1$ versus $|x| > 1$.]
6. Prove the following are dense subspaces of $L^2(\mathbb{R}^d)$: (a) the simple functions; (b) the continuous functions of compact support.

Additional exercises.

1. Compute $\|f\|_{L^2([0,1])}$ for $f(x) = x^a$ and determine all real a for which $f \in L^2([0, 1])$.
2. In $L^2([0, 1])$ compute $\langle f, g \rangle$ for $f(x) = x$, $g(x) = e^{2\pi i x}$, and verify the Cauchy–Schwarz inequality numerically.
3. Prove the parallelogram law $\|f + g\|^2 + \|f - g\|^2 = 2(\|f\|^2 + \|g\|^2)$ directly in L^2 , and explain geometrically why it fails in L^1 (give f, g on $[0, 1]$ violating it for the L^1 norm).
4. (Non-example.) Exhibit f that is bounded and continuous on $\mathbb{R}$ but $f \notin L^2(\mathbb{R})$, and a different $g \in L^2(\mathbb{R})$ that is unbounded. Explain why boundedness is neither necessary nor sufficient for L^2 membership.
5. Let $f_n(x) = n^{1/2} \mathbf{1}_{[0, 1/n]}(x)$ on $[0, 1]$. Show $\|f_n\|_{L^2} = 1$ for all n but $f_n \rightarrow 0$ weakly (i.e. $\langle f_n, g \rangle \rightarrow 0$ for all $g \in L^2$). Does $f_n \rightarrow 0$ in norm?
6. (Diagram.) Sketch on one axis the graphs of $x^{-0.3}$ and $x^{-0.7}$ on $(0, 1]$, and mark which lies in $L^2(0, 1)$. Annotate the picture with the integral $\int_0^1 x^{-2a} dx$ that decides membership.
7. Show that if $f_n \rightarrow f$ in $L^2([0, 1])$ then $\int_0^1 f_n \rightarrow \int_0^1 f$, but the converse fails: give f_n with $\int f_n \rightarrow \int f$ yet $\|f_n - f\| \not\rightarrow 0$.
8. Prove that the map $f \mapsto \bar{f}$ is a (conjugate-linear) isometry of $L^2(\mathbb{R}^d)$, and that $\|\operatorname{Re} f\|^2 + \|\operatorname{Im} f\|^2 = \|f\|^2$.

27.2 Hilbert Spaces: Orthogonality and Bases

The abstract Hilbert space

The space L^2 has six structural properties; abstracting them yields the definition of a Hilbert space, the natural home for the geometry of orthogonality in infinitely many dimensions.

Definition 27.2.1. A set H is a *Hilbert space* if:

- (i) H is a vector space over $\mathbb{C}$ (or $\mathbb{R}$);
- (ii) H carries an *inner product* $\langle \cdot, \cdot \rangle$ that is linear in the first slot, conjugate-symmetric $\langle f, g \rangle = \overline{\langle g, f \rangle}$, and positive $\langle f, f \rangle \geq 0$; set $\|f\| = \langle f, f \rangle^{1/2}$;
- (iii) $\|f\| = 0 \iff f = 0$;
- (iv) Cauchy-Schwarz $|\langle f, g \rangle| \leq \|f\| \|g\|$ and the triangle inequality $\|f + g\| \leq \|f\| + \|g\|$ hold;
- (v) H is complete in $d(f, g) = \|f - g\|$;
- (vi) H is separable.

Properties (iv) follow from (i)–(ii) (Exercise 1); we list them for emphasis. Separability (vi) is imposed because it covers essentially all applications, though interesting non-separable examples exist (Problem 2). We write $f_n \rightarrow f$ for $\|f_n - f\| \rightarrow 0$.

Example 27.2.2. Standard examples:

- $L^2(E)$ for measurable $E \subseteq \mathbb{R}^d$ with $m(E) > 0$.
- $\mathbb{C}^N$ with $\langle a, b \rangle = \sum_{k=1}^N a_k \overline{b_k}$, the finite-dimensional model.
- $\ell^2(\mathbb{Z}) = \{(a_n)_{n \in \mathbb{Z}} : \sum |a_n|^2 < \infty\}$ with $\langle a, b \rangle = \sum_k a_k \overline{b_k}$, and its one-sided variant $\ell^2(\mathbb{N})$. Every infinite-dimensional separable Hilbert space turns out to be $\ell^2(\mathbb{Z})$ in disguise.

Orthogonality, Pythagoras, Bessel

Definition 27.2.3. $f, g \in H$ are *orthogonal*, written $f \perp g$, if $\langle f, g \rangle = 0$. A finite or countable set $\{e_k\}$ is *orthonormal* if $\langle e_k, e_\ell \rangle = \delta_{k\ell}$ (that is, 1 if $k = \ell$, else 0).

Proposition 27.2.4 (Pythagoras). *If $f \perp g$ then $\|f + g\|^2 = \|f\|^2 + \|g\|^2$. More generally, if $f = \sum_{k=1}^N a_k e_k$ with $\{e_k\}$ orthonormal, then $\|f\|^2 = \sum_{k=1}^N |a_k|^2$.*

Proof. $\|f + g\|^2 = \langle f + g, f + g \rangle = \|f\|^2 + \langle f, g \rangle + \langle g, f \rangle + \|g\|^2 = \|f\|^2 + \|g\|^2$ since $\langle f, g \rangle = 0 \implies \langle g, f \rangle = 0$. The general case expands $\langle \sum a_k e_k, \sum a_\ell e_\ell \rangle = \sum_{k,\ell} a_k \overline{a_\ell} \langle e_k, e_\ell \rangle = \sum_k |a_k|^2$. $\square$

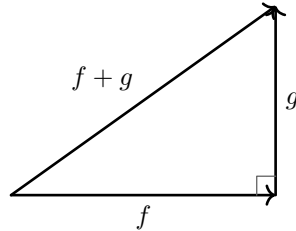


Figure 27.3. Pythagoras in any Hilbert space: if $f \perp g$ then $\|f + g\|^2 = \|f\|^2 + \|g\|^2$.

Definition 27.2.5. An orthonormal set $\{e_k\}$ is an *orthonormal basis* of H if its finite linear combinations are dense in H . Given such a set, for $f \in H$ the numbers $a_k = \langle f, e_k \rangle$ are the *Fourier coefficients* of f , and we write $f \sim \sum_k a_k e_k$.

Theorem 27.2.6 (Characterizations of an orthonormal basis). *For an orthonormal set $\{e_k\}_{k \geq 1}$ the following are equivalent.*

- (i) *Finite linear combinations of $\{e_k\}$ are dense in H (it is a basis).*
- (ii) *If $\langle f, e_j \rangle = 0$ for all j , then $f = 0$ (completeness).*
- (iii) *With $a_k = \langle f, e_k \rangle$ and $S_N(f) = \sum_{k=1}^N a_k e_k$, $S_N(f) \rightarrow f$.*
- (iv) *(Parseval) $\|f\|^2 = \sum_{k=1}^{\infty} |a_k|^2$.*

Proof. (i) $\implies$ (ii): choose finite combinations $g_n \rightarrow f$. If $\langle f, e_j \rangle = 0$ for all j then $\langle f, g_n \rangle = 0$, so $\|f\|^2 = \langle f, f - g_n \rangle \leq \|f\| \|f - g_n\| \rightarrow 0$, giving $f = 0$.

(ii) $\implies$ (iii): For $S_N(f) = \sum_{k \leq N} a_k e_k$, one checks $f - S_N(f) \perp S_N(f)$, so Pythagoras gives

$$\|f\|^2 = \|f - S_N(f)\|^2 + \sum_{k=1}^N |a_k|^2. \quad (27.2)$$

Hence $\sum_{k=1}^N |a_k|^2 \leq \|f\|^2$, and letting $N \rightarrow \infty$ yields *Bessel's inequality* $\sum |a_k|^2 \leq \|f\|^2$. Thus $\{S_N(f)\}$ is Cauchy ($\|S_N - S_M\|^2 = \sum_{M < k \leq N} |a_k|^2$), so by completeness $S_N(f) \rightarrow g$ for some g . For each j , $\langle f - S_N(f), e_j \rangle = a_j - a_j = 0$ for large N , so $\langle f - g, e_j \rangle = 0$; by (ii), $f = g$.

(iii) $\implies$ (iv): take $N \rightarrow \infty$ in (27.2). (iv) $\implies$ (i): (27.2) gives $\|f - S_N(f)\| \rightarrow 0$, and each $S_N(f)$ is a finite combination, so they are dense. $\square$

Remark. Bessel's inequality $\sum |a_k|^2 \leq \|f\|^2$ holds for *any* orthonormal family. Equality for all f — Parseval's identity — holds iff the family is a basis.

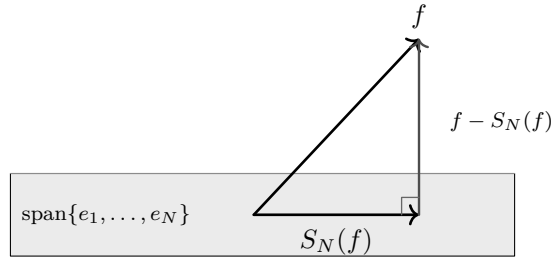


Figure 27.4. $f = S_N(f) + (f - S_N(f))$ with the remainder orthogonal to the span; hence $\|S_N(f)\|^2 = \sum_{k \leq N} |a_k|^2 \leq \|f\|^2$ (Bessel). (Schematic.)

Theorem 27.2.7 (Existence of a basis; Gram–Schmidt). *Every Hilbert space has an orthonormal basis.*

Proof. By separability choose a countable $\{h_k\}$ with dense span. Discard each h_k that is linearly dependent on its predecessors, leaving a linearly independent family $\{f_k\}$ with the same (dense) span. Apply Gram–Schmidt: set $e_1 = f_1 / \|f_1\|$; having $e_1, \dots, e_k$ with $\text{span}\{e_1, \dots, e_k\} = \text{span}\{f_1, \dots, f_k\}$, put $e'_{k+1} = f_{k+1} - \sum_{j=1}^k \langle f_{k+1}, e_j \rangle e_j$, which is $\neq 0$ by independence and orthogonal to $e_1, \dots, e_k$; normalize $e_{k+1} = e'_{k+1} / \|e'_{k+1}\|$. The resulting $\{e_k\}$ is orthonormal with the same dense span, hence a basis. If only N independent vectors arise, H is *finite-dimensional* of dimension N ; otherwise it is *infinite-dimensional*. $\square$

Method (Gram–Schmidt orthonormalization). *Recognition trigger.* You are handed independent vectors $f_1, f_2, \dots$ (often $1, x, x^2, \dots$ in L^2) and must produce an orthonormal set with the same span.

1. $e_1 = f_1 / \|f_1\|$.
2. $e'_{k+1} = f_{k+1} - \sum_{j \leq k} \langle f_{k+1}, e_j \rangle e_j$ (subtract projections onto the earlier e_j).
3. $e_{k+1} = e'_{k+1} / \|e'_{k+1}\|$.

The signature error is forgetting to subtract *all* earlier projections, or normalizing before subtracting.

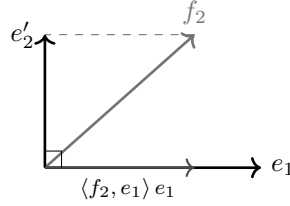


Figure 27.5. Gram-Schmidt: $e'_2 = f_2 - \langle f_2, e_1 \rangle e_1$ removes the component of f_2 along e_1 , leaving the orthogonal part. (Schematic.)

Unitary maps; all such spaces are ℓ^2

Definition 27.2.8. A map $U : H \rightarrow H'$ between Hilbert spaces is *unitary* if it is linear, bijective, and norm-preserving $\|Uf\|_{H'} = \|f\|_H$. Then U^{-1} is unitary and, by polarization, $\langle Uf, Ug \rangle_{H'} = \langle f, g \rangle_H$. We call H, H' *unitarily equivalent* if such a U exists.

The polarization identity reconstructs the inner product from the norm:

$$\langle F, G \rangle = \frac{1}{4} \left(\|F + G\|^2 - \|F - G\|^2 + i\|F + iG\|^2 - i\|F - iG\|^2 \right),$$

so a norm-preserving linear bijection automatically preserves inner products.

Corollary 27.2.9. Any two infinite-dimensional (separable) Hilbert spaces are unitarily equivalent; hence each is unitarily equivalent to $\ell^2(\mathbb{Z})$. Two finite-dimensional Hilbert spaces are unitarily equivalent iff they have the same dimension.

Proof. Pick orthonormal bases $\{e_k\} \subset H$, $\{e'_k\} \subset H'$ and define $U(\sum a_k e_k) = \sum a_k e'_k$. This is linear and invertible, and by Parseval $\|Uf\|^2 = \sum |a_k|^2 = \|f\|^2$, so U is unitary. $\square$

Proposition 27.2.10 (Pre-Hilbert spaces complete). A pre-Hilbert space H_0 satisfies all the axioms except completeness. Every H_0 has a completion: a Hilbert space $H \supseteq H_0$ with the same inner product on H_0 and H_0 dense in H . The completion is unique up to isomorphism.

Sketch. Mirror Cantor's construction of $\mathbb{R}$ from $\mathbb{Q}$. Take equivalence classes of Cauchy sequences $\{f_n\}$ in H_0 ($\{f_n\} \sim \{f'_n\}$ if $f_n - f'_n \rightarrow 0$); these classes form H , with $\langle f, g \rangle = \lim \langle f_n, g_n \rangle$. Constant sequences embed H_0 densely, and one verifies H is complete. $\square$

Worked Example 27.2.11 (Legendre polynomials by Gram-Schmidt). Orthonormalize $1, x, x^2$ in $L^2([-1, 1])$.

Recognition. Trigger: apply Gram-Schmidt to monomials; the outputs are (multiples of) Legendre polynomials.

Solution. $\|1\|^2 = \int_{-1}^1 1 = 2$, so $e_1 = 1/\sqrt{2}$. Next $\langle x, e_1 \rangle = \frac{1}{\sqrt{2}} \int_{-1}^1 x = 0$, so $e'_2 = x$, $\|x\|^2 = \int_{-1}^1 x^2 = 2/3$, $e_2 = \sqrt{3/2} x$. Then $\langle x^2, e_1 \rangle = \frac{1}{\sqrt{2}} \int_{-1}^1 x^2 = \frac{2/3}{\sqrt{2}}$ and $\langle x^2, e_2 \rangle = \sqrt{3/2} \int_{-1}^1 x^3 = 0$, so $e'_3 = x^2 - \frac{2/3}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = x^2 - \frac{1}{3}$. $\|e'_3\|^2 = \int_{-1}^1 (x^2 - \frac{1}{3})^2 = \frac{8}{45}$, giving $e_3 = \sqrt{45/8} (x^2 - \frac{1}{3})$. Check. $e'_3 \propto 3x^2 - 1$, the degree-2 Legendre polynomial, and $\langle e_2, e_3 \rangle = 0$ since the integrand $x(x^2 - \frac{1}{3})$ is odd. $\blacksquare$

Worked Example 27.2.12 (Bessel's inequality is generally strict). In $L^2([-\pi, \pi])$ with normalized exponentials, take f with $a_0 = \langle f, e_0 \rangle \neq 0$ and use only the single vector $e_0 = 1/\sqrt{2\pi}$. Show Bessel is strict.

Recognition. Trigger: a non-basis (one element) orthonormal set; Bessel must hold, with strict inequality whenever f is not a multiple of e_0 .

Solution. For $f(x) = x$ on $[-\pi, \pi]$, $a_0 = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} x dx = 0$, so the single-term Bessel sum is $0 \leq \|f\|^2 = \int_{-\pi}^{\pi} x^2 = \frac{2\pi^3}{3} > 0$: strict. Check. $f = x$ is orthogonal to constants, so projecting onto e_0 alone captures none of it — the deficit $\|f\|^2 - |a_0|^2 = \frac{2\pi^3}{3}$ is the energy in the higher frequencies. $\blacksquare$

Worked Example 27.2.13 (Parseval in $\mathbb{C}^N$). Verify Parseval for the standard basis of $\mathbb{C}^3$ and $a = (2, -i, 1)$.

Recognition. Trigger: confirm Theorem 27.2.6(iv) in finite dimensions, where it is just $\|a\|^2 = \sum |a_k|^2$.

Solution. The standard basis e_1, e_2, e_3 is orthonormal; $a_k = \langle a, e_k \rangle$ are the coordinates $2, -i, 1$. Parseval: $\|a\|^2 = |2|^2 + |-i|^2 + |1|^2 = 4 + 1 + 1 = 6$. Directly $\langle a, a \rangle = 2 \cdot \bar{2} + (-i)(\overline{-i}) + 1 \cdot \bar{1} = 4 + 1 + 1 = 6$. *Check.* The two agree, as they must in any orthonormal basis. ■

Worked Example 27.2.14 (The shift on ℓ^2 is an isometry, not unitary). On $\ell^2(\mathbb{N})$ let $S(a_1, a_2, \dots) = (0, a_1, a_2, \dots)$. Show S preserves norms but is not unitary.

Recognition. Trigger: test the unitary axioms; a norm-preserving map that fails to be surjective is an isometry but not unitary.

Solution. $\|Sa\|^2 = \sum_{k \geq 1} |a_k|^2 = \|a\|^2$, so S preserves norms (hence is injective). But S is not onto: $e_1 = (1, 0, \dots)$ has no preimage, since every output has first coordinate 0. So S is an isometry but not a bijection, hence not unitary. *Check.* S^* is the left shift $(a_1, a_2, \dots) \mapsto (a_2, a_3, \dots)$, and $S^*S = I$ but $SS^* = I - P_{e_1} \neq I$ — the asymmetry that distinguishes isometry from unitary. ■

Worked Example 27.2.15 (A unitary map $L^2([0, 1]) \rightarrow L^2([0, 1])$). Show $Uf(x) = f(1 - x)$ is unitary on $L^2([0, 1])$ and compute U^{-1} .

Recognition. Trigger: check the three unitary axioms (linear, bijective, norm-preserving) for a substitution operator.

Solution. Linearity is clear. The change of variables $y = 1 - x$ gives $\int_0^1 |f(1 - x)|^2 dx = \int_0^1 |f(y)|^2 dy = \|f\|^2$, so U preserves norms. It is its own inverse: $U(Uf)(x) = f(1 - (1 - x)) = f(x)$, so $U^{-1} = U$ and U is bijective. *Check.* $U^2 = I$, an involution; reflecting twice returns the original, and reflection clearly does not change the integral of $|f|^2$. ■

Exercises (Stein, Ch.4).

1. Suppose $\{\varphi_k\}_{k \geq 1}$ is an orthonormal basis for $L^2(\mathbb{R}^d)$. Prove the collection $\{\varphi_{k,j}\}_{1 \leq k, j < \infty}$ with $\varphi_{k,j}(x, y) = \varphi_k(x)\varphi_j(y)$ is an orthonormal basis of $L^2(\mathbb{R}^d \times \mathbb{R}^d)$. [Hint: orthonormality by Fubini; for completeness fix j and set $F_j(x) = \int F(x, y)\varphi_j(y) dy$. If $\langle F, \varphi_{k,j} \rangle = 0$ for all j then $\int F_j(x)\overline{\varphi_k(x)} dx = 0$.]
2. Let η be a fixed strictly positive continuous function on $[a, b]$, and define $H_\eta = L^2([a, b], \eta)$ to be the measurable f with $\int_a^b |f(t)|^2 \eta(t) dt < \infty$, with $\langle f, g \rangle_\eta = \int_a^b f(t)\overline{g(t)}\eta(t) dt$. (a) Show H_η is a Hilbert space and $U : f \mapsto \eta^{1/2}f$ is a unitary correspondence between H_η and the usual $L^2([a, b])$. (b) Generalize to η not necessarily continuous.
3. Suppose H and H' are two completions of a pre-Hilbert space H_0 . Show there is a unitary map $H \rightarrow H'$ that is the identity on H_0 . [Hint: for $f \in H$ pick a Cauchy sequence $\{f_n\} \subset H_0$ converging to f in H ; it also converges to some $f' \in H'$, and $f \mapsto f'$ is the required map.]
4. Let T be any linear transformation $H_1 \rightarrow H_2$. If H_1 is finite-dimensional, then T is automatically bounded. (If H_1 is not finite-dimensional this may fail; see Problem 1.)

Additional exercises.

1. Apply Gram–Schmidt to $1, x, x^2, x^3$ in $L^2([-1, 1])$ and identify the outputs as normalized Legendre polynomials. Verify $\langle e_2, e_4 \rangle = 0$ by a parity argument.
2. In $\mathbb{C}^2$ orthonormalize the vectors $(1, 1)$ and $(1, 0)$ via Gram–Schmidt; check the result is an orthonormal basis.
3. Prove Bessel's inequality $\sum_k |\langle f, e_k \rangle|^2 \leq \|f\|^2$ for any orthonormal family, and characterize exactly when equality holds for a given f .
4. Show $\{\frac{1}{\sqrt{2\pi}}e^{inx}\}_{n \in \mathbb{Z}}$ is orthonormal in $L^2([-\pi, \pi])$ by direct integration, computing $\langle e^{inx}, e^{imx} \rangle$ for $n \neq m$ and $n = m$.
5. (Non-example.) Give an orthonormal family in $L^2([-\pi, \pi])$ that is *not* a basis, and exhibit a nonzero f orthogonal to every member. (E.g. use only even frequencies.)

6. Prove that a unitary map carries an orthonormal basis to an orthonormal basis, and conversely a linear bijection carrying some orthonormal basis to an orthonormal basis is unitary.
7. (Diagram.) Draw the Gram–Schmidt step in $\mathbb{R}^2$: given f_1, f_2 , show e_1 , the projection $\langle f_2, e_1 \rangle e_1$, and the orthogonal remainder e'_2 . Label the right angle.
8. Let $H = \ell^2(\mathbb{N})$ and e_k the standard basis. Show the shift $S(a_1, a_2, \dots) = (0, a_1, a_2, \dots)$ is an isometry but not unitary, and compute S^* .
9. Prove the polarization identity in a complex Hilbert space and use it to show a norm-preserving linear map preserves inner products.

27.3 Fourier Series and Fatou's Theorem

Fourier coefficients and Abel summability

We now realize the abstract theory in the concrete Hilbert space $L^2([-\pi, \pi])$, where the orthonormal basis is the family of exponentials and the Fourier coefficients are classical.

Definition 27.3.1. For $f \in L^1([-\pi, \pi])$ the n th Fourier coefficient is

$$a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx, \quad n \in \mathbb{Z},$$

and the Fourier series of f is the formal sum $f \sim \sum_{n \in \mathbb{Z}} a_n e^{inx}$.

Since $[-\pi, \pi]$ has finite measure, Cauchy–Schwarz gives $L^2([-\pi, \pi]) \subseteq L^1([-\pi, \pi])$, so the coefficients are defined for every L^2 function. Note $|a_n| \leq \frac{1}{2\pi} \int |f|$, so $\sum a_n r^{|n|} e^{inx}$ converges absolutely and uniformly for each $r < 1$.

Theorem 27.3.2 (Uniqueness and Abel summability). *Let f be integrable on $[-\pi, \pi]$.*

(i) *If $a_n = 0$ for all n , then $f = 0$ a.e.*

(ii) $\sum_{n \in \mathbb{Z}} a_n r^{|n|} e^{inx} \rightarrow f(x)$ for a.e. x as $r \rightarrow 1^-$.

Proof. (i) follows from (ii). For (ii), recall the Poisson kernel

$$P_r(y) = \sum_{n \in \mathbb{Z}} r^{|n|} e^{iny} = \frac{1 - r^2}{1 - 2r \cos y + r^2}.$$

Extend f periodically. Then, by dominated convergence and translation invariance,

$$\sum_n a_n r^{|n|} e^{inx} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x - y) P_r(y) dy,$$

because $\int_{-\pi}^{\pi} f(x - y) e^{iny} dy = e^{inx} 2\pi a_n$. As $r \rightarrow 1^-$, $\{P_r\}$ is an approximation to the identity, so the right side tends to $f(x)$ at every point of the Lebesgue set of f , hence a.e. $\square$

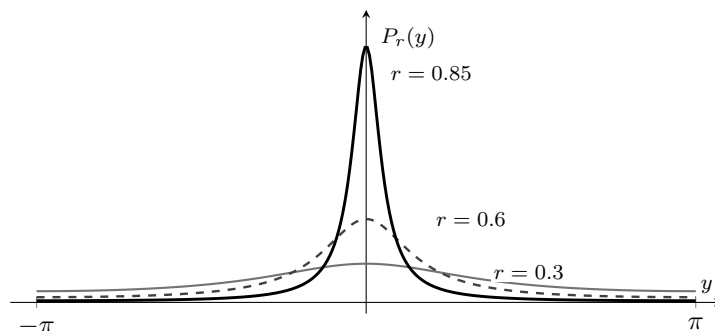


Figure 27.6. The Poisson kernel P_r for $r = 0.3, 0.6, 0.85$: as $r \rightarrow 1^-$ it concentrates at $y = 0$, becoming an approximation to the identity.

The L^2 theory: Parseval and the unitary correspondence

Theorem 27.3.3 (Parseval / Riesz–Fischer for Fourier series). *Let $f \in L^2([-\pi, \pi])$, $a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f e^{-inx} dx$. Then:*

$$(i) \sum_{n \in \mathbb{Z}} |a_n|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx \text{ (Parseval's relation);}$$

$$(ii) f \mapsto \{a_n\} \text{ is a unitary correspondence } L^2([-\pi, \pi]) \rightarrow \ell^2(\mathbb{Z});$$

$$(iii) \|f - S_N(f)\| \rightarrow 0, \text{ where } S_N(f) = \sum_{|n| \leq N} a_n e^{inx}.$$

Proof. Take $H = L^2([-\pi, \pi])$ with $\langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f \bar{g}$ and the orthonormal set $\{e^{inx}\}_{n \in \mathbb{Z}}$ (relabel as $\{e_k\}$). By Theorem 27.3.2(i), property (ii) of Theorem 27.2.6 holds: $a_n = 0$ for all n forces $f = 0$. Hence $\{e^{inx}\}$ is an orthonormal basis, and all four characterizations hold — giving Parseval (iv) and norm convergence (iii). Conversely, given $\{a_n\} \in \ell^2(\mathbb{Z})$, the partial sums are Cauchy in L^2 , so by completeness converge to some f with those coefficients; thus $f \mapsto \{a_n\}$ is onto, hence unitary. $\square$

$$L^2([-\pi, \pi]) \xrightarrow[\sim]{f \mapsto \{a_n\}} \ell^2(\mathbb{Z})$$

Figure 27.7. Parseval realizes $f \mapsto \{a_n\}$ as a unitary isomorphism $L^2([-\pi, \pi]) \xrightarrow{\sim} \ell^2(\mathbb{Z})$: it is onto precisely because L^2 is complete.

Remark. This is exactly where Lebesgue's theory pays off: the map $f \mapsto \{a_n\}$ is *onto* $\ell^2(\mathbb{Z})$ only because L^2 is complete. On the Riemann-integrable functions $\mathcal{R} \subset L^2([-\pi, \pi])$ — which are not complete — the correspondence fails to be surjective.

Worked Example 27.3.4 (Parseval evaluates $\sum 1/n^2$). Compute the Fourier coefficients of $f(x) = x$ on $[-\pi, \pi]$ and use Parseval to find $\sum_{n \geq 1} 1/n^2$.

Recognition. Trigger: a classical Parseval application; integrate by parts for a_n , then equate $\sum |a_n|^2$ to $\frac{1}{2\pi} \int |f|^2$.

Solution. For $n \neq 0$, $a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} x e^{-inx} dx = \frac{(-1)^n i}{n}$ (integration by parts), and $a_0 = 0$. So $|a_n|^2 = 1/n^2$ for $n \neq 0$. Parseval: $\sum_{n \neq 0} \frac{1}{n^2} = 2 \sum_{n \geq 1} \frac{1}{n^2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{1}{2\pi} \cdot \frac{2\pi^3}{3} = \frac{\pi^2}{3}$. Hence $\sum_{n \geq 1} 1/n^2 = \pi^2/6$. *Check.* Numerically $\pi^2/6 \approx 1.6449$, matching $1 + \frac{1}{4} + \frac{1}{9} + \dots$. ■

Worked Example 27.3.5 (Coefficients of a step function). Find the Fourier coefficients of $f = \mathbf{1}_{[0, \pi]} - \mathbf{1}_{[-\pi, 0]}$ (the sign function) on $[-\pi, \pi]$ and confirm $\sum |a_n|^2$ is finite.

Recognition. Trigger: direct evaluation of a_n for a piecewise-constant f ; check square-summability against Parseval.

Solution. $a_0 = 0$ by symmetry. For $n \neq 0$, $a_n = \frac{1}{2\pi} (\int_0^{\pi} e^{-inx} - \int_{-\pi}^0 e^{-inx}) = \frac{1 - (-1)^n}{\pi i n} \cdot \frac{1}{2} \cdot 2$, which is $\frac{2}{\pi i n}$ for n odd and 0 for n even. So $|a_n|^2 = \frac{4}{\pi^2 n^2}$ (n odd). *Check.* $\sum_{n \text{ odd}} \frac{4}{\pi^2 n^2} = \frac{8}{\pi^2} \sum_{k \geq 0} \frac{1}{(2k+1)^2} = \frac{8}{\pi^2} \cdot \frac{\pi^2}{8} = 1 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |f|^2$, since $|f| \equiv 1$. ■

Worked Example 27.3.6 (ℓ^2 coefficients that are not ℓ^1). Show $a_n = 1/n$ ($n \geq 1$, else 0) defines a genuine L^2 function whose series does not converge absolutely.

Recognition. Trigger: use the unitary correspondence backwards: any ℓ^2 sequence is the coefficient sequence of some L^2 function, even if not ℓ^1 .

Solution. $\sum_{n \geq 1} |a_n|^2 = \sum 1/n^2 = \pi^2/6 < \infty$, so $\{a_n\} \in \ell^2$ and by Theorem 27.3.3(ii) there is $f \in L^2([-\pi, \pi])$ with these coefficients. But $\sum |a_n| = \sum 1/n = \infty$, so $\{a_n\} \notin \ell^1$ and the Fourier series does not converge absolutely. *Check.* Convergence in L^2 does not require absolute (pointwise) convergence — exactly the gain of the L^2 viewpoint. ■

Fatou's theorem and the Hardy space

Definition 27.3.7. A function F on the unit disc $\mathbb{D} = \{|z| < 1\}$ has a *radial limit* at the angle θ if $\lim_{r \rightarrow 1^-} F(re^{i\theta})$ exists.

Theorem 27.3.8 (Fatou). *A bounded holomorphic function F on $\mathbb{D}$ has radial limits at almost every θ .*

Proof. Write $F(z) = \sum_{n \geq 0} a_n z^n$, convergent for $z = re^{i\theta}$, $r < 1$. For fixed r the series $\sum_{n \geq 0} a_n r^n e^{in\theta}$ is the Fourier series of $\theta \mapsto F(re^{i\theta})$, i.e. $a_n r^n = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(re^{i\theta}) e^{-in\theta} d\theta$ for $n \geq 0$ (and 0 for $n < 0$). Pick M with $|F| \leq M$. By Parseval, $\sum_{n \geq 0} |a_n|^2 r^{2n} = \frac{1}{2\pi} \int_{-\pi}^{\pi} |F(re^{i\theta})|^2 d\theta \leq M^2$. Letting $r \rightarrow 1$ shows $\sum |a_n|^2 \leq M^2 < \infty$. Let $F(e^{i\theta})$ be the L^2 function with Fourier coefficients a_n ($n \geq 0$), 0 otherwise. By Theorem 27.3.2(ii) (Abel summability), $\sum_{n \geq 0} a_n r^n e^{in\theta} \rightarrow F(e^{i\theta})$ for a.e. θ , which is the radial limit. $\square$

Definition 27.3.9. The *Hardy space* $H^2(\mathbb{D})$ consists of holomorphic F on $\mathbb{D}$ with

$$\|F\|_{H^2}^2 = \sup_{0 \leq r < 1} \frac{1}{2\pi} \int_{-\pi}^{\pi} |F(re^{i\theta})|^2 d\theta < \infty.$$

Equivalently $F = \sum_{n \geq 0} a_n z^n$ with $\sum |a_n|^2 < \infty$, and then $\|F\|_{H^2}^2 = \sum |a_n|^2$. Thus $H^2(\mathbb{D})$ is a Hilbert space, the “subspace” $\ell^2(\mathbb{Z}_+)$ of $\ell^2(\mathbb{Z})$.

Bounded F lie in $H^2(\mathbb{D})$, and Fatou's conclusion holds for all of $H^2(\mathbb{D})$ by the same argument.

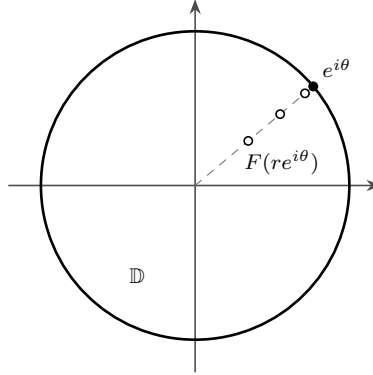


Figure 27.8. Fatou: as $r \rightarrow 1$ along the radius, $F(re^{i\theta})$ converges to a boundary value $F(e^{i\theta})$ for almost every θ . (Schematic.)

Worked Example 27.3.10 (A bounded holomorphic function and its boundary values). For $F(z) = \frac{1}{1-z/2}$ on $\mathbb{D}$, compute the boundary function and verify $F \in H^2$.

Recognition. Trigger: geometric-series expansion gives the coefficients; Hardy membership is $\sum |a_n|^2 < \infty$.

Solution. $F(z) = \sum_{n \geq 0} (z/2)^n$, so $a_n = 2^{-n}$. Then $\sum |a_n|^2 = \sum 4^{-n} = \frac{1}{1-1/4} = \frac{4}{3} < \infty$, so $F \in H^2(\mathbb{D})$ with $\|F\|_{H^2}^2 = 4/3$. The radial limit is $F(e^{i\theta}) = \frac{1}{1-e^{i\theta}/2}$, defined for *every* θ (the denominator never vanishes since $|e^{i\theta}/2| = 1/2 < 1$). *Check.* $|F| \leq \frac{1}{1-1/2} = 2$ on $\mathbb{D}$, so F is bounded; boundedness $\implies H^2$, consistent. $\blacksquare$

Worked Example 27.3.11 (A function failing radial limits everywhere (lacunary)). Explain why $F(z) = \sum_{n \geq 0} z^{2^n}$ need not have radial limits a.e., contrasting with Fatou.

Recognition. Trigger: test the boundary of Fatou's hypotheses; here $\sum |a_n|^2 = \infty$ so $F \notin H^2$.

Solution. The coefficients are 1 at $n = 2^k$ and 0 otherwise, so $\sum |a_n|^2 = \sum_k 1 = \infty$ and $F \notin H^2(\mathbb{D})$. Fatou's theorem does *not* apply, and indeed this lacunary series fails to have radial limits at a.e. boundary point.

Check. The function is holomorphic on $\mathbb{D}$ but unbounded near the boundary; the failure is consistent with $\sum |a_n|^2 = \infty$ being exactly the obstruction. $\blacksquare$

Worked Example 27.3.12 (H^2 norm versus sup norm). For $F(z) = z^N$ compute $\|F\|_{H^2}$ and $\sup_{\mathbb{D}} |F|$; compare.

Recognition. Trigger: a monomial has a single coefficient; both norms are immediate.

Solution. Only $a_N = 1$, so $\|F\|_{H^2}^2 = 1$, $\|F\|_{H^2} = 1$. Meanwhile $\sup_{|z| < 1} |z^N| = 1$ (approached as $|z| \rightarrow 1$).

Check. Here the two coincide at 1; in general $\|F\|_{H^2} \leq \sup |F|$ since the H^2 norm is an L^2 -average of $|F|$ over circles, never exceeding the supremum. ■

Worked Example 27.3.13 (Radial limit of a Blaschke-type factor). Show $F(z) = \frac{z-a}{1-\bar{a}z}$ with $|a| < 1$ is bounded on $\mathbb{D}$ and has unit-modulus radial limits.

Recognition. Trigger: a single Blaschke factor; bounded holomorphic, so Fatou applies, and on the circle its modulus is 1.

Solution. For $|z| = 1$, $|1 - \bar{a}z| = |\bar{z} - \bar{a}| \cdot |z| = |z - a|$ (taking conjugates), so $|F| = 1$ on the circle, and by the maximum principle $|F| < 1$ inside. Thus F is bounded, $F \in H^2$, and the radial limit $F(e^{i\theta})$ has modulus 1 a.e. *Check.* At $a = 0$, $F(z) = z$, with boundary value $e^{i\theta}$ of modulus 1 — consistent. ■

Exercises (Stein, Ch.4).

- Let $H_1 = L^2([-\pi, \pi])$ be functions $F(e^{i\theta})$ on the unit circle with $\langle F, G \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(e^{i\theta}) \overline{G(e^{i\theta})} d\theta$, and $H_2 = L^2(\mathbb{R})$. Using $x \mapsto \frac{i-x}{i+x}$: (a) show $U : F \mapsto f$ with $f(x) = \frac{1}{\pi^{1/2}(i+x)} F\left(\frac{i-x}{i+x}\right)$ is a unitary map $H_1 \rightarrow H_2$; (b) deduce that $\left\{ \frac{1}{\pi^{1/2}} \left(\frac{i-x}{i+x}\right)^n \frac{1}{i+x} \right\}_{n \in \mathbb{Z}}$ is an orthonormal basis of $L^2(\mathbb{R})$.
- Let $F_0(z) = 1/(1-z)^i$. (a) Verify $|F_0(z)| \leq e^{\pi/2}$ in the unit disc but $\lim_{r \rightarrow 1} F_0(r)$ does not exist [Hint: $|F_0(r)| = 1$ and $F_0(r)$ oscillates between ± 1 infinitely often]. (b) Let $\{\alpha_n\}$ enumerate the rationals and $F(z) = \sum_{j \geq 1} \delta^j F_0(ze^{-i\alpha_j})$ with δ small; show $\lim_{r \rightarrow 1} F(re^{i\theta})$ fails to exist whenever $\theta = \alpha_j$, so F has no radial limit on a dense set of the circle.
- (Non-tangential limits.) For $0 < s < 1$ and z on the circle, let $\Gamma_s(z)$ be the smallest closed convex set containing z and the closed disc $D_s(0)$ (near z it looks like a triangle). Say F has a non-tangential limit at z if $\lim_{w \rightarrow z, w \in \Gamma_s(z)} F(w)$ exists for every s . Prove: if F is holomorphic and bounded on the open unit disc, then F has a non-tangential limit at almost every point of the circle. [Hint: show the Poisson integral of f has non-tangential limits at every point of the Lebesgue set of f .]

Problems (Stein, Ch.4).

- (Generalization of Fatou.) If $u(re^{i\theta})$ is harmonic and bounded in the unit disc, then $\lim_{r \rightarrow 1} u(re^{i\theta})$ exists for a.e. θ . [Hint: with $a_n(r) = \frac{1}{2\pi} \int_0^{2\pi} u(re^{i\theta}) e^{-in\theta} d\theta$, one gets $a_n'' + \frac{1}{r} a_n' - \frac{n^2}{r^2} a_n = 0$, so $a_n(r) = A_n r^n + B_n r^{-n}$ ($n \neq 0$) and $u(re^{i\theta}) = \sum a_n r^{|n|} e^{in\theta}$; proceed as in Fatou.]
- Examples failing radial limits. (a) At a.e. boundary point of the circle, $\sum_{n \geq 0} z^{2^n}$ fails to have a radial limit. (b) More generally if $F(z) = \sum_{n \geq 0} a_n z^{2^n}$ and $\sum |a_n|^2 = \infty$, then F fails to have radial limits at a.e. boundary point; whereas if $\sum |a_n|^2 < \infty$ then $F \in H^2(\mathbb{D})$ and has radial limits a.e.
- Suppose F is holomorphic in the unit disc and $\sup_{0 \leq r < 1} \frac{1}{2\pi} \int_{-\pi}^{\pi} \log^+ |F(re^{i\theta})| d\theta < \infty$ (where $\log^+ u = \log u$ if $u \geq 1$, else 0). Then $\lim_{r \rightarrow 1} F(re^{i\theta})$ exists for a.e. θ . This holds in particular when $\sup_r \frac{1}{2\pi} \int |F(re^{i\theta})|^p d\theta < \infty$ for some $p > 0$; such F form the Hardy space $H^p(\mathbb{D})$.

Additional exercises.

- Use Parseval on $f(x) = x^2$ over $[-\pi, \pi]$ to evaluate $\sum_{n \geq 1} 1/n^4 = \pi^4/90$. Compute the Fourier coefficients first.
- Compute the Fourier coefficients of $f(x) = |x|$ on $[-\pi, \pi]$ and verify $\sum |a_n|^2 = \frac{1}{2\pi} \int |f|^2$.
- Show $F(z) = \frac{1}{1-z}$ is holomorphic on $\mathbb{D}$ but $F \notin H^2(\mathbb{D})$ by computing $\sum |a_n|^2$. Contrast with $F(z) = \frac{1}{1-z/2}$.
- For $F(z) = \sum_{n \geq 0} \frac{z^n}{n+1}$, decide whether $F \in H^2(\mathbb{D})$ and whether F has radial limits a.e.

5. (Non-example.) Give a holomorphic F on $\mathbb{D}$ that is unbounded yet still has radial limits a.e., and a different F with $\sum |a_n|^2 = \infty$ that fails to. Explain which hypothesis of Fatou each respects.
6. Prove the Poisson kernel identity $\sum_{n \in \mathbb{Z}} r^{|n|} e^{iny} = \frac{1-r^2}{1-2r \cos y + r^2}$ by summing two geometric series.
7. (Diagram.) Plot the Poisson kernel $P_r(y)$ on $[-\pi, \pi]$ for $r = 0.5$ and $r = 0.9$, and indicate how the mass concentrates at $y = 0$ as $r \rightarrow 1$.
8. Show that for $f \in L^2([-\pi, \pi])$ the Abel means $\sum a_n r^{|n|} e^{inx}$ converge to f in the L^2 norm as $r \rightarrow 1$, not merely a.e.
9. Verify that the map $f \mapsto \{a_n\}$ of Theorem 27.3.3 sends the function $f(x) = e^{i3x}$ to the sequence $\delta_{n,3}$, and check Parseval on this example.

27.4 Closed Subspaces and Orthogonal Projections

Closed subspaces and the nearest point

Definition 27.4.1. A (linear) subspace $S \subseteq H$ satisfies $\alpha f + \beta g \in S$ for all $f, g \in S$ and scalars α, β . It is *closed* if $f_n \in S$, $f_n \rightarrow f$ imply $f \in S$. Every closed subspace of H is itself a Hilbert space under the inherited inner product.

In finite dimensions every subspace is closed; in infinite dimensions this fails. For instance the Riemann integrable functions, and the finite linear combinations of a fixed basis, are non-closed subspaces of $L^2([-\pi, \pi])$.

Lemma 27.4.2 (Nearest point in a closed subspace). *Let $S \subseteq H$ be a closed subspace and $f \in H$. Then:*

(i) *there is a unique $g_0 \in S$ with $\|f - g_0\| = \inf_{g \in S} \|f - g\|$;*

(ii) *$f - g_0 \perp S$, i.e. $\langle f - g_0, g \rangle = 0$ for all $g \in S$.*

Proof. If $f \in S$ take $g_0 = f$. Otherwise let $d = \inf_{g \in S} \|f - g\| > 0$ and pick $g_n \in S$ with $\|f - g_n\| \rightarrow d$. The parallelogram law $\|A + B\|^2 + \|A - B\|^2 = 2(\|A\|^2 + \|B\|^2)$ with $A = f - g_n$, $B = f - g_m$ gives

$$\|g_m - g_n\|^2 = 2\|f - g_n\|^2 + 2\|f - g_m\|^2 - 4\|f - \frac{g_n + g_m}{2}\|^2 \leq 2\|f - g_n\|^2 + 2\|f - g_m\|^2 - 4d^2,$$

since $\frac{g_n + g_m}{2} \in S$ forces $\|f - \frac{g_n + g_m}{2}\| \geq d$. As $\|f - g_n\|, \|f - g_m\| \rightarrow d$, the right side $\rightarrow 0$, so $\{g_n\}$ is Cauchy; by completeness and closedness $g_n \rightarrow g_0 \in S$ with $\|f - g_0\| = d$.

For (ii), fix $g \in S$ and consider $g_0 - \varepsilon g \in S$ for real ε . Then $\|f - (g_0 - \varepsilon g)\|^2 = \|f - g_0\|^2 + \varepsilon^2 \|g\|^2 + 2\varepsilon \operatorname{Re} \langle f - g_0, g \rangle \geq \|f - g_0\|^2$, so $2\varepsilon \operatorname{Re} \langle f - g_0, g \rangle + \varepsilon^2 \|g\|^2 \geq 0$ for all ε ; this forces $\operatorname{Re} \langle f - g_0, g \rangle = 0$. Replacing g by ig gives $\operatorname{Im} \langle f - g_0, g \rangle = 0$, so $\langle f - g_0, g \rangle = 0$. Uniqueness: if $\tilde{g}_0$ also minimizes, then $(f - g_0) \perp (g_0 - \tilde{g}_0)$ and Pythagoras gives $\|f - \tilde{g}_0\|^2 = \|f - g_0\|^2 + \|g_0 - \tilde{g}_0\|^2$; equality of distances forces $g_0 = \tilde{g}_0$. $\square$

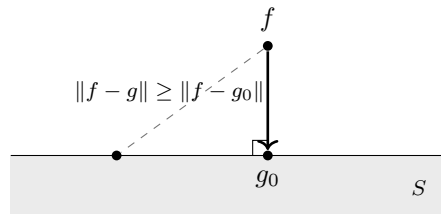


Figure 27.9. The nearest point $g_0 \in S$ to f ; the error $f - g_0$ is perpendicular to all of S . (Schematic.)

Orthogonal complement and projection

Definition 27.4.3. For a subspace $S \subseteq H$ the *orthogonal complement* is $S^\perp = \{f \in H : \langle f, g \rangle = 0 \text{ for all } g \in S\}$. It is a closed subspace, and $S \cap S^\perp = \{0\}$.

Proposition 27.4.4 (Orthogonal decomposition). *If S is a closed subspace of H , then $H = S \oplus S^\perp$: every $f \in H$ is uniquely $f = g + h$ with $g \in S$, $h \in S^\perp$.*

Proof. For $f \in H$ take g_0 from Lemma 27.4.2 and write $f = g_0 + (f - g_0)$; then $g_0 \in S$ and $f - g_0 \in S^\perp$. For uniqueness, if $g + h = \tilde{g} + \tilde{h}$ with $g, \tilde{g} \in S$, $h, \tilde{h} \in S^\perp$, then $g - \tilde{g} = \tilde{h} - h \in S \cap S^\perp = \{0\}$, so $g = \tilde{g}$, $h = \tilde{h}$. $\square$

Definition 27.4.5. The *orthogonal projection* onto a closed subspace S is $P_S(f) = g$ where $f = g + h$, $g \in S$, $h \in S^\perp$. It satisfies: P_S is linear; $P_S f = f$ on S ; $P_S f = 0$ on $S^\perp$; and $\|P_S f\| \leq \|f\|$.

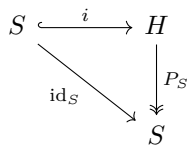


Figure 27.10. The orthogonal projection P_S retracts H onto the closed subspace $S \hookrightarrow H$: it is onto S and restricts to the identity there, so $P_S \circ i = \text{id}_S$.

Proposition 27.4.6 (Projection via an orthonormal family). *If $\{e_k\}$ is an orthonormal family and S is the closure of its span, then*

$$P_S(f) = \sum_k \langle f, e_k \rangle e_k,$$

the sum converging in norm when the family is infinite.

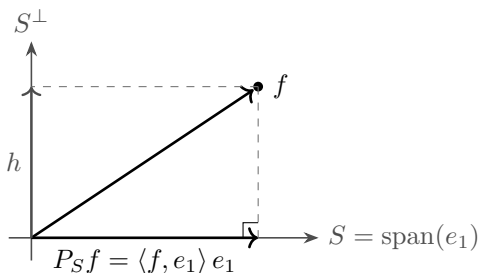


Figure 27.11. Orthogonal projection: $f = P_S f + h$ with $P_S f \in S$ and $h \in S^\perp$; $P_S f$ is the foot of the perpendicular from f to S . (Schematic.)

Method (Computing an orthogonal projection). *Recognition trigger.* You must project f onto a closed subspace S , or find the best approximation to f from S .

1. If $S = \text{span}\{e_1, \dots\}$ with $\{e_k\}$ orthonormal, then $P_S f = \sum_k \langle f, e_k \rangle e_k$ — just sum the Fourier-coefficient projections.
2. If the spanning set is not orthonormal, run Gram–Schmidt first.
3. The error $f - P_S f$ is orthogonal to every e_k ; this is both the defining property and the way to check your answer.

Theorem 27.4.7 (Partial sums and the Cauchy integral). *On $L^2([-\pi, \pi])$:*

- (i) *The partial-sum operator S_N is the orthogonal projection onto $\text{span}\{e^{-iN\theta}, \dots, e^{iN\theta}\}$, realized as convolution with the Dirichlet kernel $D_N(\theta) = \frac{\sin((N + \frac{1}{2})\theta)}{\sin(\theta/2)}$.*
- (ii) *Let S be the subspace of $f \sim \sum_{n \geq 0} a_n e^{in\theta}$, identified with $H^2(\mathbb{D})$. The projection P onto S equals the Cauchy integral $P(f)(z) = C(f)(z) = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{\zeta - z} d\zeta$ for $z \in \mathbb{D}$, $\gamma = \text{unit circle}$.*

Proof of (ii). For $|z| < 1$, $P(f)(z) = \sum_{n \geq 0} a_n z^n = \sum_{n \geq 0} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(e^{i\theta}) e^{-in\theta} d\theta \right) z^n$. Interchanging sum and integral (justified since $f \in L^1$ and $|z| < 1$) and summing the geometric series $\sum_n (e^{-i\theta} z)^n = \frac{1}{1 - e^{-i\theta} z}$ gives, after writing $\zeta = e^{i\theta}$, $d\zeta = ie^{i\theta} d\theta$,

$$P(f)(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{f(e^{i\theta})}{1 - e^{-i\theta} z} d\theta = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{\zeta - z} d\zeta = C(f)(z). \quad \square$$

Worked Example 27.4.8 (Best linear approximation in $L^2([-1, 1])$). Find the polynomial $p(x) = c_0 + c_1 x$ of degree ≤ 1 nearest to $f(x) = x^2$ in $L^2([-1, 1])$.

Recognition. Trigger: projection onto $S = \text{span}\{1, x\}$; use the orthonormal e_1, e_2 from the Legendre computation.

Solution. With $e_1 = 1/\sqrt{2}$, $e_2 = \sqrt{3/2} x$ (orthonormal), $P_S f = \langle x^2, e_1 \rangle e_1 + \langle x^2, e_2 \rangle e_2$. We have $\langle x^2, e_1 \rangle = \frac{1}{\sqrt{2}} \int_{-1}^1 x^2 = \frac{2/3}{\sqrt{2}}$ and $\langle x^2, e_2 \rangle = \sqrt{3/2} \int_{-1}^1 x^3 = 0$. So $P_S f = \frac{2/3}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{3}$. The best degree- ≤ 1 approximation is the constant $p(x) = 1/3$. *Check.* The error $x^2 - \frac{1}{3}$ is orthogonal to 1 ($\int_{-1}^1 (x^2 - \frac{1}{3}) = 0$) and to x (odd integrand) — as the projection demands. ■

Worked Example 27.4.9 (Projection in $\mathbb{C}^3$ onto a plane). Project $v = (1, 2, 3)$ onto $S = \{(x, y, z) : x + y + z = 0\}$ in real $\mathbb{R}^3$.

Recognition. Trigger: S is a closed subspace (a plane through 0); its complement is spanned by the normal $n = (1, 1, 1)$, so $P_S v = v - P_{S^\perp} v$.

Solution. $S^\perp = \text{span}(n)$ with $n = (1, 1, 1)$, $\|n\|^2 = 3$. $P_{S^\perp} v = \frac{\langle v, n \rangle}{\|n\|^2} n = \frac{1+2+3}{3} (1, 1, 1) = 2(1, 1, 1)$. So $P_S v = v - (2, 2, 2) = (-1, 0, 1)$. *Check.* $(-1) + 0 + 1 = 0$ so $P_S v \in S$, and the residual $(2, 2, 2) \parallel n$, orthogonal to S . ■

Worked Example 27.4.10 (Multiplication by an indicator is a projection). For measurable $E \subseteq \mathbb{R}^d$, show $Pf = \mathbf{1}_E \cdot f$ is the orthogonal projection of $L^2(\mathbb{R}^d)$ onto $S = \{f : f = 0 \text{ a.e. off } E\}$.

Recognition. Trigger: verify the projection axioms; equivalently check $P^2 = P$ and $P^* = P$.

Solution. P is linear, $P^2 f = \mathbf{1}_E^2 f = \mathbf{1}_E f = Pf$, and $\langle Pf, g \rangle = \int_E f \bar{g} = \langle f, Pg \rangle$, so $P^* = P$. Its range is S (functions supported in E) and $Pf = f$ exactly there. *Check.* $f - Pf = \mathbf{1}_{E^c} f$ is supported off E , orthogonal to every $g \in S$ since their supports are disjoint. ■

Worked Example 27.4.11 (The orthogonal complement of H^2 inside L^2). Describe $S^\perp$ when $S \subset L^2([-\pi, \pi])$ is the Hardy subspace $\{f \sim \sum_{n \geq 0} a_n e^{in\theta}\}$.

Recognition. Trigger: orthogonal complement in terms of Fourier coefficients; $S^\perp$ kills the non-negative frequencies.

Solution. $g \in S^\perp$ iff $\langle g, e^{in\theta} \rangle = 0$ for all $n \geq 0$, i.e. the Fourier coefficients $b_n = 0$ for $n \geq 0$. So $S^\perp = \{g \sim \sum_{n \leq -1} b_n e^{in\theta}\}$, the “strictly negative frequencies.” *Check.* $S \oplus S^\perp$ recovers all of L^2 , since every f splits as (frequencies ≥ 0) + (frequencies < 0), matching $H = S \oplus S^\perp$. ■

Worked Example 27.4.12 (Projection onto the constants is averaging). In $L^2([0, 1])$ let S be the constant functions. Show $P_S f = \int_0^1 f$ and compute the projection of $f(x) = x$.

Recognition. Trigger: $S = \text{span}(e_1)$ with $e_1 \equiv 1$ orthonormal (since $\|1\| = 1$ on $[0, 1]$); the projection is the single Fourier coefficient times e_1 .

Solution. $e_1 \equiv 1$ is a unit vector. $P_S f = \langle f, e_1 \rangle e_1 = \left(\int_0^1 f \right) \cdot 1 = \int_0^1 f$, the average of f . For $f(x) = x$, $P_S f = \int_0^1 x = \frac{1}{2}$, the constant $1/2$. *Check.* The residual $f - P_S f = x - \frac{1}{2}$ has $\int_0^1 (x - \frac{1}{2}) = 0$, so it is orthogonal to the constants — the projection condition. The squared error $\int_0^1 (x - \frac{1}{2})^2 = \frac{1}{12}$ is the variance of a uniform variable. ■

Caution. The decomposition $H = S \oplus S^\perp$ requires S to be *closed*. For a non-closed subspace the nearest-point lemma can fail and f need not split. The correct general statement is $(S^\perp)^\perp = \overline{S}$, the closure of S (Exercise 10): taking $\perp$ twice recovers S only when S is already closed.

Exercises (Stein, Ch.4).

1. Let S be a subspace of a Hilbert space H . Prove that $(S^\perp)^\perp$ is the smallest closed subspace of H containing S (i.e. the closure $\bar{S}$).
2. Let P be the orthogonal projection onto a closed subspace S , so $Pf = f$ on S and $Pf = 0$ on $S^\perp$. (a) Show $P^2 = P$ and $P^* = P$. (b) Conversely, if P is any bounded operator with $P^2 = P$ and $P^* = P$, prove P is the orthogonal projection onto some closed subspace. (c) Using P , prove that a closed subspace of a separable Hilbert space is itself a separable Hilbert space.
3. Let $E \subseteq \mathbb{R}^d$ be measurable and S the subspace of $L^2(\mathbb{R}^d)$ of functions vanishing for a.e. $x \notin E$. Show the orthogonal projection onto S is $P(f) = \mathbf{1}_E \cdot f$.
4. Suppose P_1, P_2 are orthogonal projections onto S_1, S_2 . Then $P_1 P_2$ is an orthogonal projection iff P_1, P_2 commute ($P_1 P_2 = P_2 P_1$); in that case $P_1 P_2$ projects onto $S_1 \cap S_2$.

Additional exercises.

1. Find the orthogonal projection of $f(x) = x^3$ onto $\text{span}\{1, x\}$ in $L^2([-1, 1])$, and verify the residual is orthogonal to 1 and to x .
2. In $\mathbb{R}^3$ project $v = (2, 1, -1)$ onto the line $\text{span}(1, 1, 1)$ and onto its orthogonal plane; confirm the two pieces sum to v .
3. Let $S \subset L^2([0, 1])$ be the constants. Show $P_S f = \int_0^1 f$ (the average), and that $\|f - P_S f\|^2 = \|f\|^2 - \left| \int_0^1 f \right|^2$ is the variance of f .
4. Prove that an orthogonal projection P satisfies $\|Pf\| \leq \|f\|$ with equality iff $f \in S$. Interpret geometrically.
5. (Non-example.) Let $S \subset L^2([0, 1])$ be the polynomials. Show S is a non-closed subspace and exhibit $f \in L^2$ (e.g. e^x in L^2 , or a non-smooth function) for which no nearest polynomial exists, illustrating why closedness is needed in Lemma 27.4.2.
6. Show that if P_1, P_2 are orthogonal projections with $P_1 P_2 = P_2 P_1 = 0$, then $P_1 + P_2$ is the orthogonal projection onto $S_1 \oplus S_2$.
7. (Diagram.) Draw $f \in H$, a closed subspace S (as a shaded region), the nearest point $g_0 = P_S f$, and the perpendicular $f - g_0$. Mark the right angle and the inequality $\|f - g\| \geq \|f - g_0\|$.
8. Using Theorem 27.4.7, compute the projection of $f(\theta) = e^{-2i\theta} + 3 + e^{i\theta}$ onto the Hardy subspace $S = \{\sum_{n \geq 0} a_n e^{in\theta}\}$, and identify the corresponding holomorphic function on $\mathbb{D}$.

27.5 Linear Transformations and Compact Operators

Bounded operators and the operator norm

Definition 27.5.1. A map $T : H_1 \rightarrow H_2$ is a *linear operator* if $T(af + bg) = aTf + bTg$. It is *bounded* if there is M with $\|Tf\| \leq M \|f\|$ for all f ; the smallest such M is the *operator norm*

$$\|T\| = \inf\{M : \|Tf\| \leq M \|f\|\} = \sup_{\|f\| \leq 1} \|Tf\|.$$

Lemma 27.5.2. $\|T\| = \sup\{|\langle Tf, g \rangle| : \|f\| \leq 1, \|g\| \leq 1\}$.

Proof. If $\|T\| \leq M$, Cauchy-Schwarz gives $|\langle Tf, g \rangle| \leq \|Tf\| \|g\| \leq M$ for $\|f\|, \|g\| \leq 1$, so the sup is $\leq \|T\|$. Conversely if the sup is $\leq M$, take $f' = f / \|f\|$, $g' = Tf / \|Tf\|$ (norm 1); then $|\langle Tf', g' \rangle| = \|Tf\| / \|f\| \leq M$, so $\|Tf\| \leq M \|f\|$. $\square$

Proposition 27.5.3 (Bounded = continuous). *A linear operator T is bounded iff it is continuous.*

Proof. If bounded, $\|Tf - Tf_n\| \leq M \|f - f_n\| \rightarrow 0$, so continuous. If continuous but unbounded, pick f_n with $\|Tf_n\| \geq n \|f_n\|$; then $g_n = f_n / (n \|f_n\|) \rightarrow 0$ but $\|Tg_n\| \geq 1$, contradicting continuity at 0. $\square$

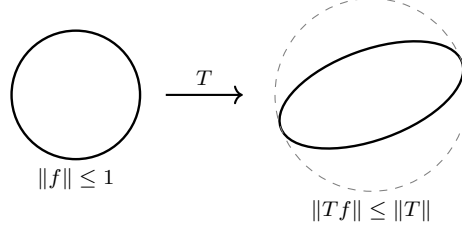


Figure 27.12. A bounded operator maps the unit ball into a ball of radius $\|T\|$; the longest output direction realizes $\|T\|$. (Schematic.)

Linear functionals and Riesz representation

Theorem 27.5.4 (Riesz representation). *Every continuous linear functional $\ell : H \rightarrow \mathbb{C}$ has the form $\ell(f) = \langle f, g \rangle$ for a unique $g \in H$, and $\|\ell\| = \|g\|$.*

Proof. Let $S = \{f : \ell(f) = 0\}$, the null space, which is closed by continuity. If $S = H$ then $\ell = 0$ and $g = 0$ works. Otherwise pick $h \in S^\perp$ with $\|h\| = 1$ and set $g = \overline{\ell(h)}h$. For any f , $u = \ell(f)h - \ell(h)f \in S$, so $\langle u, h \rangle = 0$, giving $0 = \ell(f)\langle h, h \rangle - \ell(h)\langle f, h \rangle$; since $\langle h, h \rangle = 1$ this rearranges to $\ell(f) = \langle f, \overline{\ell(h)}h \rangle = \langle f, g \rangle$. Uniqueness and $\|\ell\| = \|g\|$ follow from Cauchy–Schwarz (with equality at $f = g$). $\square$

Proposition 27.5.5 (Adjoint). *For bounded $T : H \rightarrow H$ there is a unique bounded T^* with $\langle Tf, g \rangle = \langle f, T^*g \rangle$; moreover $\|T^*\| = \|T\|$ and $(T^*)^* = T$.*

Proof. For fixed g , $f \mapsto \langle Tf, g \rangle$ is a bounded functional, so by Riesz equals $\langle f, h \rangle$ for a unique $h =: T^*g$; this defines T^* , linear and satisfying (i). From Lemma 27.5.2, $\|T\| = \sup |\langle Tf, g \rangle| = \sup |\langle f, T^*g \rangle| = \|T^*\|$. Finally $\langle Tf, g \rangle = \langle f, T^*g \rangle$ for all f, g iff $\langle T^*f, g \rangle = \langle f, Tg \rangle$, giving $(T^*)^* = T$. $\square$

Definition 27.5.6. T is *symmetric* (self-adjoint) if $T^* = T$, an *isometry* if $\|Tf\| = \|f\|$, *unitary* if bijective and norm-preserving (then $T^* = T^{-1}$). For symmetric T , $\|T\| = \sup\{|\langle Tf, f \rangle| : \|f\| = 1\}$.

Diagonal and integral operators

Definition 27.5.7. T is *diagonalized* by the orthonormal basis $\{\varphi_k\}$ if $T\varphi_k = \lambda_k\varphi_k$; the λ_k are *eigenvalues*, the $\{\lambda_k\}$ the *multiplier sequence*. Then $f \sim \sum a_k\varphi_k$ gives $Tf \sim \sum a_k\lambda_k\varphi_k$, and:

$$\begin{aligned} \|T\| &= \sup_k |\lambda_k|, & T^* &\leftrightarrow \{\overline{\lambda_k}\}, & T = T^* &\iff \lambda_k \in \mathbb{R}, \\ T \text{ unitary} &\iff |\lambda_k| = 1, & T \text{ projection} &\iff \lambda_k \in \{0, 1\}. \end{aligned}$$

$$\begin{array}{ccc} H & \xrightarrow{T} & H \\ U \downarrow \sim & & U \downarrow \sim \\ \ell^2 & \xrightarrow{M_\lambda} & \ell^2 \end{array}$$

Figure 27.13. Diagonalization: the coefficient map $U : f \mapsto \{a_k\}$ is a unitary isomorphism $H \xrightarrow{\sim} \ell^2$ carrying T to the multiplier $M_\lambda\{a_k\} = \{\lambda_ka_k\}$; the square commutes, $U \circ T = M_\lambda \circ U$.

Definition 27.5.8. T is an *integral operator* with kernel K if $Tf(x) = \int_{\mathbb{R}^d} K(x, y)f(y) dy$. It is *Hilbert–Schmidt* if $K \in L^2(\mathbb{R}^d \times \mathbb{R}^d)$.

Proposition 27.5.9 (Hilbert–Schmidt operators). *Let T have kernel $K \in L^2(\mathbb{R}^d \times \mathbb{R}^d)$. Then $y \mapsto K(x, y)f(y)$ is integrable for a.e. x ; T is bounded with $\|T\| \leq \|K\|_{L^2(\mathbb{R}^d \times \mathbb{R}^d)}$; and T^* has kernel $\overline{K}(y, x)$.*

Proof. By Fubini, $y \mapsto |K(x, y)|^2$ is integrable for a.e. x , so by Cauchy–Schwarz $y \mapsto K(x, y)f(y)$ is integrable. Then $|Tf(x)|^2 \leq (\int |K(x, y)|^2 dy)(\int |f(y)|^2 dy)$, and integrating in x gives $\|Tf\|^2 \leq \|K\|^2 \|f\|^2$. The adjoint kernel comes from writing $\langle Tf, g \rangle$ as a double integral and applying Fubini. $\square$

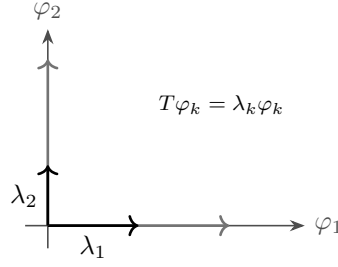


Figure 27.14. A diagonal operator scales each basis direction by its eigenvalue λ_k ; $\|T\| = \sup_k |\lambda_k|$. (Schematic.)

Compact operators and the spectral theorem

Definition 27.5.10. $T : H \rightarrow H$ is *compact* if the closure of $T(B)$ (B the closed unit ball) is compact; equivalently, every bounded sequence $\{f_k\}$ has a subsequence with $\{Tf_{n_k}\}$ convergent. Finite-rank operators (finite-dimensional range) are compact; the identity on an infinite-dimensional H is not.

The unit ball of an infinite-dimensional H is *not* compact: for orthonormal e_n , $\|e_n - e_m\|^2 = 2$ for $n \neq m$, so no subsequence converges.

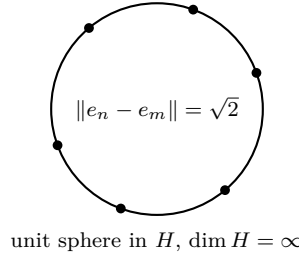


Figure 27.15. Non-compactness of the unit ball: orthonormal vectors stay distance $\sqrt{2}$ apart, so $\{e_n\}$ has no convergent subsequence. (Schematic.)

Proposition 27.5.11 (Stability of compactness). *Let T be bounded. (i) If S is compact then ST, TS are compact. (ii) If T_n are compact and $\|T_n - T\| \rightarrow 0$, then T is compact. (iii) If T is compact there are finite-rank T_n with $\|T_n - T\| \rightarrow 0$. (iv) T is compact iff T^* is compact.*

Sketch. (i) is immediate. (ii): a diagonalization argument extracts, from a bounded $\{f_k\}$, a subsequence on which every T_m converges; the limit is Cauchy for T via the three-term split. (iii): with Q_n the projection onto $\text{span}\{e_k\}_{k>n}$, one shows $\|Q_n T\| \rightarrow 0$, so $P_n T$ (finite rank, $P_n = I - Q_n$) converges to T . (iv): $\|P_n T - T\| \rightarrow 0 \implies \|T^* P_n - T^*\| \rightarrow 0$ with $T^* P_n$ finite rank. $\square$

Remark. Two facts complete the picture: a diagonal T ($T\varphi_k = \lambda_k\varphi_k$) is compact iff $\lambda_k \rightarrow 0$ (Exercise 25), and every Hilbert–Schmidt operator is compact (approximate K by truncating its expansion in $\{\varphi_k(x)\varphi_\ell(y)\}$).

Theorem 27.5.12 (Spectral theorem for compact symmetric operators). *If T is compact and symmetric on H , there is an orthonormal basis $\{\varphi_k\}$ of H consisting of eigenvectors, $T\varphi_k = \lambda_k\varphi_k$, with $\lambda_k \in \mathbb{R}$ and $\lambda_k \rightarrow 0$. Conversely every such operator is compact and symmetric.*

Proof. Three lemmas drive the proof. (a) Eigenvalues of a symmetric T are real and eigenvectors for distinct eigenvalues are orthogonal: $\lambda \langle f, f \rangle = \langle Tf, f \rangle = \langle f, Tf \rangle = \bar{\lambda} \langle f, f \rangle$ forces $\lambda \in \mathbb{R}$; and $\lambda_1 \langle f_1, f_2 \rangle = \langle Tf_1, f_2 \rangle = \langle f_1, Tf_2 \rangle = \lambda_2 \langle f_1, f_2 \rangle$ forces orthogonality when $\lambda_1 \neq \lambda_2$. (b) For compact T and $\mu > 0$, only finitely many eigenvalues exceed μ in modulus and each eigenspace is finite-dimensional (else an orthonormal sequence of eigenvectors would have $\|T\varphi_{n_k} - T\varphi_{n_j}\|^2 \geq 2\mu^2$, contradicting compactness); hence the eigenvalues form a sequence $\rightarrow 0$. (c) For compact symmetric $T \neq 0$, either $\|T\|$ or $-\|T\|$ is an eigenvalue: with $\lambda = \sup_{\|f\|=1} \langle Tf, f \rangle = \|T\|$, pick f_n with $\langle Tf_n, f_n \rangle \rightarrow \lambda$; since $\|Tf_n - \lambda f_n\|^2 \leq 2\lambda^2 - 2\lambda \langle Tf_n, f_n \rangle \rightarrow 0$ and $Tf_n \rightarrow g$ along a subsequence, $\lambda f_n \rightarrow g$ and $Tg = \lambda g$ with $g \neq 0$.

Now let S be the closure of the span of all eigenvectors; by (c) $S \neq \{0\}$. Suppose $S \neq H$, so $S^\perp \neq \{0\}$. Because T is symmetric and $TS \subseteq S$, also $TS^\perp \subseteq S^\perp$. The restriction $T_1 = T|_{S^\perp}$ is compact and symmetric and nonzero, so by (c) it has an eigenvector in $S^\perp$ — which is an eigenvector of T not in S , a contradiction. Hence $S = H$, and normalizing a basis of each eigenspace yields the orthonormal eigenbasis. $\square$

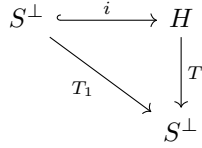


Figure 27.16. Because $TS^\perp \subseteq S^\perp$, the action of T on the invariant subspace factors through the inclusion: $T_1 = T|_{S^\perp}$ is the composite $T \circ i$ landing back in $S^\perp$.

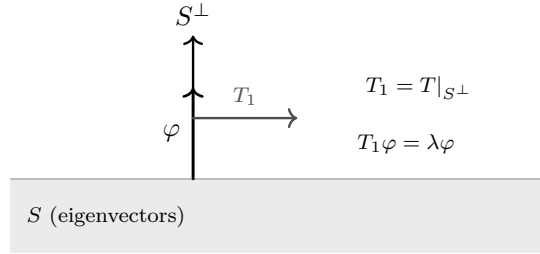


Figure 27.17. Spectral theorem: if $S \neq H$ the restriction $T_1 = T|_{S^\perp}$ still has an eigenvector $\varphi \in S^\perp$, contradicting maximality — so $S = H$. (Schematic.)

Worked Example 27.5.13 (Norm and adjoint of a diagonal operator). On $\ell^2(\mathbb{N})$ let $T(a_k) = (\lambda_k a_k)$ with $\lambda_k = 1/k$. Compute $\|T\|$, the adjoint, and decide compactness.

Recognition. Trigger: a diagonal operator; read off $\|T\| = \sup |\lambda_k|$ and use $\lambda_k \rightarrow 0$ for compactness.

Solution. $\|T\| = \sup_k 1/k = 1$ (attained at $k = 1$). Since $\lambda_k = 1/k$ is real, $T^* = T$ (symmetric). And $\lambda_k = 1/k \rightarrow 0$, so T is compact. *Check.* $Te_1 = e_1$ so $\|Te_1\| = 1 = \|T\|$; and the truncations $P_n T$ (finite rank) satisfy $\|P_n T - T\| = \sup_{k>n} 1/k = 1/(n+1) \rightarrow 0$, confirming compactness. $\blacksquare$

Worked Example 27.5.14 (A bounded symmetric operator that is NOT compact). On $L^2([0, 1])$ let $Tf(t) = tf(t)$. Show $T = T^*$, $\|T\| = 1$, T has no eigenvectors, and T is not compact.

Recognition. Trigger: a multiplication operator; a classic non-compact self-adjoint example.

Solution. $\langle Tf, g \rangle = \int_0^1 t f \bar{g} = \langle f, Tg \rangle$, so $T = T^*$. $\|T\| = \text{ess sup}_{[0,1]} t = 1$. If $Tf = \lambda f$ then $(t - \lambda)f(t) = 0$ a.e., forcing $f = 0$ a.e.: no eigenvectors. Not compact: the functions $f_n = \sqrt{n} \mathbf{1}_{[1-1/n, 1]}$ have $\|f_n\| = 1$ but $\{Tf_n\}$ has no convergent subsequence (the mass concentrates near $t = 1$ without limit in L^2). *Check.* No eigenvectors but $\|T\| = 1 > 0$ — consistent with the spectral theorem requiring *compactness* to guarantee eigenvectors. $\blacksquare$

Worked Example 27.5.15 (A Hilbert–Schmidt operator and its norm bound). On $L^2([0, 1])$ let $Tf(x) = \int_0^1 (x+y)f(y) dy$. Show T is Hilbert–Schmidt and bound $\|T\|$.

Recognition. Trigger: the kernel $K(x, y) = x + y$ is bounded, hence L^2 on the unit square; compute $\|K\|_{L^2}$.

Solution. $\|K\|_{L^2}^2 = \int_0^1 \int_0^1 (x+y)^2 dx dy = \int_0^1 \int_0^1 (x^2 + 2xy + y^2) = \frac{1}{3} + \frac{1}{2} + \frac{1}{3} = \frac{7}{6}$. So T is Hilbert–Schmidt with $\|T\| \leq \sqrt{7/6} \approx 1.08$. By Proposition 27.5.9 T is bounded and (being HS) compact and symmetric ($K(x, y) = K(y, x)$ real). *Check.* T has rank 2 (range spanned by 1 and x), so it is genuinely finite-rank — automatically compact, consistent. $\blacksquare$

Worked Example 27.5.16 (Eigenvalues of the sawtooth operator). For $Tf(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} K(x-y)f(y) dy$ with $K \sim \sum_{n \neq 0} \frac{e^{inx}}{n}$, find the eigenvalues on $L^2([-\pi, \pi])$.

Recognition. Trigger: convolution operators are diagonalized by exponentials; the eigenvalue at e^{inx} is the n th Fourier coefficient of K .

Solution. Convolution with K sends $e^{inx} \mapsto \hat{K}(n)e^{inx}$, where $\hat{K}(n) = 1/n$ for $n \neq 0$ and $\hat{K}(0) = 0$. So the eigenfunctions are e^{inx} with eigenvalue $1/n$ ($n \neq 0$) and the constant 1 with eigenvalue 0. *Check.* $\lambda_n = 1/n \rightarrow 0$, so T is compact; it is symmetric, and the eigenfunctions $\{e^{inx}\}$ form an orthonormal basis — a concrete instance of the spectral theorem. ■

Worked Example 27.5.17 (Riesz representation in action). Identify the $g \in L^2([0, 1])$ representing $\ell(f) = \int_0^1 xf(x) dx$ and compute $\|\ell\|$.

Recognition. Trigger: a functional given by an integral against a fixed weight; Riesz says g is (the conjugate of) that weight.

Solution. $\ell(f) = \int_0^1 f(x)\overline{g(x)} dx$ with $\overline{g(x)} = x$, i.e. $g(x) = x$ (real). By Riesz, $\|\ell\| = \|g\| = (\int_0^1 x^2)^{1/2} = 1/\sqrt{3}$.

Check. Take $f = g/\|g\| = \sqrt{3}x$; then $\ell(f) = \sqrt{3} \int_0^1 x^2 = \sqrt{3}/3 = 1/\sqrt{3} = \|\ell\|$, so the bound is attained, confirming $\|\ell\| = \|g\|$. ■

Caution. The spectral theorem needs *both* hypotheses. Drop symmetry and an operator may have no real eigenvalues (a rotation); drop compactness and a symmetric operator may have no eigenvectors at all (multiplication by x , the worked example above). Compactness is what forces the eigenvalues to accumulate only at 0 and the eigenspaces to be finite-dimensional.

Exercises (Stein, Ch.4).

- Let $\mathcal{L}(H)$ be the bounded linear operators on H , with operator norm $\|T\| = \inf\{B : \|Tv\| \leq B\|v\| \ \forall v\}$.
(a) Show $\|T_1 + T_2\| \leq \|T_1\| + \|T_2\|$. (b) Prove $d(T_1, T_2) = \|T_1 - T_2\|$ is a metric on $\mathcal{L}(H)$. (c) Show $\mathcal{L}(H)$ is complete in d .
- If T is a bounded operator on H , prove $\|TT^*\| = \|T^*T\| = \|T\|^2 = \|T^*\|^2$.
- Suppose H is infinite-dimensional. Show that for any sequence $\{f_n\}$ with $\|f_n\| = 1$ there exist $f \in H$ and a subsequence $\{f_{n_k}\}$ with $\langle f_{n_k}, g \rangle \rightarrow \langle f, g \rangle$ for all $g \in H$ (weak convergence). [Hint: let g run through a basis and diagonalize; define f by its expansion.]
- Senses of operator convergence: norm ($\|T_n - T\| \rightarrow 0$), strong ($T_n f \rightarrow T f$ for all f), weak ($\langle T_n f, g \rangle \rightarrow \langle T f, g \rangle$ for all f, g). (a) Give examples showing weak does not imply strong, nor strong imply norm convergence. (b) Show every bounded T is a strong limit of finite-rank operators.
- An operator T is an *isometry* if $\|Tf\| = \|f\|$ for all f . (a) Show an isometry satisfies $\langle Tf, Tg \rangle = \langle f, g \rangle$ and hence $T^*T = I$. (b) If T is a surjective isometry, then T is unitary and $TT^* = I$. (c) Give an isometry that is not unitary. (d) Show that if T^*T is unitary then T is an isometry. [Hint: use $\langle Tf, Tf \rangle = \langle f, f \rangle$ with f replaced by $f \pm g$ and $f \pm ig$.]
- Suppose $\{T_k\}$ are bounded with $\|T_k\| \leq 1$ and $T_k T_j^* = T_k^* T_j = 0$ for $k \neq j$. Let $S_N = \sum_{k=-N}^N T_k$. Show $S_N(f)$ converges for every f , and if $T(f)$ is the limit then $\|T\| \leq 1$. [Hint: first treat finitely many nonzero T_k and note their ranges are mutually orthogonal.]
- Let $\{e_k\}_{k \geq 1}$ be orthonormal in H . If $\{c_k\}$ are positive reals with $\sum c_k^2 < \infty$, then $A = \{\sum_{k \geq 1} a_k e_k : |a_k| \leq c_k\}$ is compact in H .
- Suppose T is bounded and diagonal with respect to a basis $\{\varphi_k\}$, $T\varphi_k = \lambda_k \varphi_k$. Then T is compact iff $\lambda_k \rightarrow 0$. [Hint: if $\lambda_k \rightarrow 0$, $\|P_n T - T\| \rightarrow 0$ with P_n the projection onto $\text{span}\{\varphi_1, \dots, \varphi_n\}$.]
- Suppose w is measurable on $\mathbb{R}^d$ with $0 < w(x) < \infty$ a.e., and K measurable on $\mathbb{R}^{2d}$ satisfies (i) $\int |K(x, y)| w(y) dy \leq A w(x)$ a.e. x , and (ii) $\int |K(x, y)| w(x) dx \leq A w(y)$ a.e. y . Prove the integral operator $Tf(x) = \int K(x, y)f(y) dy$ is bounded on $L^2(\mathbb{R}^d)$ with $\|T\| \leq A$. (As a special case, if $\int |K(x, y)| dy \leq A$ for all x and $\int |K(x, y)| dx \leq A$ for all y , then $\|T\| \leq A$.) [Hint: bound $\int |K| |f| dy$ via Cauchy–Schwarz with weight w .]
- Prove the operator $Tf(x) = \frac{1}{\pi} \int_0^\infty \frac{f(y)}{x+y} dy$ is bounded on $L^2(0, \infty)$ with $\|T\| \leq 1$. [Hint: use the previous exercise with a suitable w .]

11. Let $H = L^2(B)$, B the unit ball in $\mathbb{R}^d$, and K measurable on $B \times B$ with $|K(x, y)| \leq A|x - y|^{-d+\alpha}$ for some $\alpha > 0$. Define $Tf(x) = \int_B K(x, y)f(y) dy$. (a) Prove T is bounded on H . (b) Prove T is compact. (c) Note T is Hilbert–Schmidt iff $\alpha > d/2$. [Hint for (b): truncate K to $K_n = K \cdot \mathbf{1}_{\{|x-y| \geq 1/n\}}$; each T_n is compact and $\|T_n - T\| \rightarrow 0$.]
12. Let T be compact and $\lambda \neq 0$. (a) Show the range of $\lambda I - T$ is closed. [Hint: if $g_j = (\lambda I - T)f_j \rightarrow g$, one may assume $f_j \in V_\lambda^\perp$ where $V_\lambda = \ker(\lambda I - T)$; then $\{f_j\}$ is bounded.] (b) Show by example this may fail when $\lambda = 0$. (c) Show $\lambda I - T$ is onto iff $\ker(\lambda I - T^*)$ is trivial.
13. Let $H = L^2([-\pi, \pi])$ with $[-\pi, \pi]$ as the circle, and fix a bounded sequence $\{\lambda_n\}$. Define $Tf \sim \sum_n \lambda_n a_n e^{inx}$ when $f \sim \sum_n a_n e^{inx}$ (a Fourier multiplier operator). (a) Show T is bounded with $\|T\| = \sup_n |\lambda_n|$. (b) Verify T commutes with translations $\tau_h f(x) = f(x - h)$. (c) Conversely, if T is any bounded operator commuting with all translations, then T is a Fourier multiplier operator. [Hint: consider $T(e^{inx})$.]
14. Consider the sawtooth function on $[-\pi, \pi)$, $K(x) = i(\operatorname{sgn}(x)\pi - x)$, extended with period 2π . For $f \in L^1([-\pi, \pi])$ extended periodically, define $Tf(x) = \frac{1}{2\pi} \int_{-\pi}^\pi K(x - y)f(y) dy$. (a) Show $F = Tf$ is absolutely continuous, and if $\int_{-\pi}^\pi f = 0$ then $F'(x) = if(x)$ a.e. (b) Show $f \mapsto Tf$ is compact and symmetric on $L^2([-\pi, \pi])$. (c) Prove $\varphi \in L^2$ is an eigenfunction iff $\varphi = e^{inx}$ ($n \neq 0$, eigenvalue $1/n$) or $\varphi = 1$ (eigenvalue 0). (d) Deduce $\{e^{inx}\}_{n \in \mathbb{Z}}$ is an orthonormal basis of $L^2([-\pi, \pi])$. (Recall $K(x) \sim \sum_{n \neq 0} \frac{e^{inx}}{n}$.)
15. Consider $T : L^2([0, 1]) \rightarrow L^2([0, 1])$, $T(f)(t) = tf(t)$. (a) Prove T is bounded with $T = T^*$ but T is not compact. (b) Show T has no eigenvectors.
16. Let H have basis $\{\varphi_k\}_{k \geq 1}$ and define $T(\varphi_k) = \frac{1}{k}\varphi_{k+1}$. Verify T is compact but has no eigenvectors.
17. Let K be a real symmetric Hilbert–Schmidt kernel, T the (compact, symmetric) operator it defines, $\{\varphi_k\}$ eigenvectors with eigenvalues λ_k . Then: (a) $\sum_k |\lambda_k|^2 < \infty$; (b) $K(x, y) \sim \sum_k \lambda_k \varphi_k(x) \overline{\varphi_k(y)}$ in the basis $\{\varphi_k(x) \overline{\varphi_k(y)}\}$; (c) a compact symmetric T is Hilbert–Schmidt iff $\sum_n |\lambda_n|^2 < \infty$ (eigenvalues counted with multiplicity).
18. Variants of the spectral theorem. (a) If T_1, T_2 are symmetric compact operators that commute ($T_1 T_2 = T_2 T_1$), they can be simultaneously diagonalized: there is an orthonormal basis of common eigenvectors. (b) An operator is *normal* if $TT^* = T^*T$; prove a normal compact T can be diagonalized [Hint: write $T = T_1 + iT_2$ with T_1, T_2 symmetric, compact and commuting]. (c) If U is unitary with $U = \lambda I - T$, T compact, then U can be diagonalized.

Problems (Stein, Ch.4).

1. On an infinite-dimensional Hilbert space there is an unbounded (hence discontinuous) linear functional ℓ . [Hint: with the axiom of choice take an algebraic (Hamel) basis $\{e_\alpha\}$, select $\{e_n\}_{n \geq 1}$, and set $\ell(e_n) = n \|e_n\|$.]
2. A non-separable Hilbert space. On $\mathbb{R}$ take exponentials $\{e^{i\lambda x}\}$, $\lambda \in \mathbb{R}$, and let H_0 be their finite linear combinations with $\langle f, g \rangle = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T f \bar{g}$. (a) Show the limit exists and $\langle f, g \rangle = \sum_k a_{\lambda_k} \overline{b_{\lambda_k}}$ for $f = \sum a_{\lambda_k} e^{i\lambda_k x}$, $g = \sum b_{\lambda_k} e^{i\lambda_k x}$. (b) With this inner product H_0 is pre-Hilbert; its completion H is non-separable since $e^{i\lambda x}$ and $e^{i\lambda' x}$ are orthonormal for $\lambda \neq \lambda'$. A continuous F is *almost periodic* if it is a uniform limit of elements of H_0 ; $H_0 \subset AP \subset H$. (c) $F \in AP$ iff for every $\varepsilon > 0$ there is L such that any interval of length L contains an “almost period” τ with $\sup_x |F(x + \tau) - F(x)| < \varepsilon$. (d) Equivalently, $F \in AP$ iff every sequence of translates $F(x + h_n)$ has a uniformly convergent subsequence.
3. (Fredholm alternative.) If T is compact and $\lambda \neq 0$, show (a) $\lambda I - T$ is injective iff $\lambda I - T^*$ is injective; (b) $\lambda I - T$ is injective iff it is surjective. (Combine with Exercise on the closed range above.)
4. Show the identity on $L^2(\mathbb{R}^d)$ cannot be written as an absolutely convergent integral operator: if K is measurable on $\mathbb{R}^d \times \mathbb{R}^d$ and $Tf(x) = \int K(x, y)f(y) dy$ converges a.e. for each $f \in L^2$, then $Tf \neq f$ for some f . [Hint: otherwise for disjoint balls B_1, B_2 one gets $K = 0$ a.e. on $B_1 \times B_2$.]

5. Suppose $\{T_k\}$ are bounded with $\|T_k T_j^*\| \leq a_{k-j}$ and $\|T_k^* T_j\| \leq a_{k-j}^*$ for positive $\{a_n\}$ with $\sum_{-\infty}^{\infty} a_n = A < \infty$. Then with $S_N = \sum_{k=-N}^N T_k$, $S_N(f)$ converges for every f , and $T = \lim S_N$ satisfies $\|T\| \leq A$ (a generalization of the Cotlar–Stein type lemma).
6. (Regular Sturm–Liouville operators.) On $[a, b]$ define $L(f) = f'' - q(x)f$ for $f \in C^2([a, b])$, with q continuous, real and ≥ 0 ; call φ an eigenfunction with eigenvalue μ if $L\varphi = \mu\varphi$ and $\varphi(a) = \varphi(b) = 0$. Show: (a) the eigenvalues μ are strictly negative and each eigenspace is one-dimensional; (b) eigenvectors for distinct eigenvalues are orthogonal in $L^2([a, b])$; (c) with the Green's kernel $K(x, y) = \frac{\varphi_-(x)\varphi_+(y)}{w}$ for $x \leq y$ and $\frac{\varphi_+(x)\varphi_-(y)}{w}$ for $y \leq x$ (where $\varphi_{\pm}$ solve $L\varphi = 0$ with the indicated endpoint conditions and w is the nonzero Wronskian), the operator $Tf(x) = \int_a^b K(x, y)f(y) dy$ is Hilbert–Schmidt, symmetric, and $L(Tf) = f$; (d) hence each eigenvector of T (eigenvalue λ) is an eigenvector of L (eigenvalue $\mu = 1/\lambda$), so the normalized eigenvectors of L form a complete orthonormal set.
7. On $C^2([-1, 1])$ let $L(f) = (1 - x^2)f'' - 2xf'$. If φ_n is the n th Legendre polynomial $\varphi_n(x) = \left(\frac{d}{dx}\right)^n (1 - x^2)^n$, then $L\varphi_n = -n(n+1)\varphi_n$, and the normalized φ_n form an orthonormal basis of $L^2([-1, 1])$.
8. The Hermite functions h_k are defined by $\sum_{k \geq 0} h_k(x) \frac{t^k}{k!} = e^{-(x^2/2 - 2tx + t^2)}$. (a) They satisfy $(x - \frac{d}{dx})h_k = h_{k+1}$ and $(x + \frac{d}{dx})h_k = h_{k-1}$ ($h_{-1} = 0$); $h_0 = e^{-x^2/2}$, $h_1 = 2xe^{-x^2/2}$, and $h_k = P_k(x)e^{-x^2/2}$ with $\deg P_k = k$. (b) The h_k are eigenvectors of $L = -d^2/dx^2 + x^2$ with $Lh_k = (2k+1)h_k$; they are mutually orthogonal, $\int h_k^2 = \pi^{1/2} 2^k k! = c_k$, and the normalized $H_k = c_k^{-1/2} h_k$ form a complete orthonormal sequence in $L^2(\mathbb{R})$ (since $\int f H_k = 0$ for all k forces $\int f(x) e^{-x^2/2 + 2tx} dx = 0$ for all t). (c) With $K(x, y) = \sum_{k \geq 0} \frac{H_k(x)H_k(y)}{\lambda_k}$ and $F = Tf = \int K(x, y)f(y) dy$, T is a symmetric Hilbert–Schmidt operator inverting L : $LF = -F'' + x^2 F = f$ for every $f \in L^2(\mathbb{R})$.

Additional exercises.

1. On $\ell^2(\mathbb{N})$ let $T(a_k) = (a_k/2^k)$. Compute $\|T\|$, show $T = T^*$, and decide compactness.
2. Compute the operator norm of the diagonal T on $\ell^2(\mathbb{N})$ with $\lambda_k = \frac{k}{k+1}$, and explain why T is bounded but not compact.
3. For the Hilbert–Schmidt operator $Tf(x) = \int_0^1 xy f(y) dy$ on $L^2([0, 1])$, find the (single) nonzero eigenvalue and eigenfunction, and verify $\|T\|$.
4. Prove that a finite-rank operator is compact directly from the sequential definition.
5. (Non-example.) Show the identity operator on $L^2([0, 1])$ is bounded and symmetric but not compact, by exhibiting a bounded sequence whose image has no convergent subsequence.
6. Let T be the right shift on $\ell^2(\mathbb{N})$. Compute T^*T and TT^* , and use them to show T is an isometry but not unitary.
7. Verify the spectral theorem on the symmetric compact operator $Tf(x) = \int_0^1 \min(x, y)f(y) dy$ on $L^2([0, 1])$: show its eigenfunctions are $\sin((n - \frac{1}{2})\pi x)$ with eigenvalues $((n - \frac{1}{2})\pi)^{-2}$, hence $\lambda_n \rightarrow 0$.
8. (Diagram.) Illustrate why the unit ball of an infinite-dimensional H is not compact: draw a sphere with orthonormal points $e_1, e_2, \dots$ all at mutual distance $\sqrt{2}$.
9. Prove that if T is compact and $T_n \rightarrow T$ in operator norm with each T_n of finite rank, then T^* is a norm limit of finite-rank operators (hence compact).
10. Let T be a Fourier multiplier on $L^2([-\pi, \pi])$ with $\lambda_n = \frac{1}{1+n^2}$. Show T is compact and symmetric, write its kernel as a convolution, and find its largest eigenvalue.

Chapter 28

Hilbert Spaces: Several Examples

28.1 The Fourier Transform on L^2

From the Schwartz class to L^2

The Fourier transform of an integrable function on $\mathbb{R}^d$ is

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x \cdot \xi} dx, \quad f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi,$$

the second formula being the inversion. On the Schwartz class $\mathcal{S}(\mathbb{R}^d)$ — the smooth functions all of whose derivatives $x^\alpha(\partial/\partial x)^\beta f$ stay bounded — the transform is a bijection, inversion holds, and one has the *Plancherel identity*

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)|^2 d\xi = \int_{\mathbb{R}^d} |f(x)|^2 dx.$$

The largest class for which the defining integral converges absolutely is $L^1(\mathbb{R}^d)$, where a relative inversion formula was proved in Chapter 2. The point of this section is the *symmetry* between f and $\hat{f}$ that the Schwartz class enjoys, now recovered on the Hilbert space $L^2(\mathbb{R}^d)$. The idea is simple: $\mathcal{S}$ is dense in L^2 and the transform preserves the L^2 -norm, so it extends by continuity to a norm-preserving (unitary) map of L^2 onto itself.

Theorem 28.1.1 (Plancherel / the L^2 Fourier transform). *The Fourier transform $\mathcal{F}_0$, initially defined on $\mathcal{S}(\mathbb{R}^d)$, has a unique extension $\mathcal{F}$ to a unitary mapping of $L^2(\mathbb{R}^d)$ onto itself. In particular $\|\mathcal{F}f\|_{L^2} = \|f\|_{L^2}$ for every $f \in L^2(\mathbb{R}^d)$.*

The extension is defined by a limiting process: if $\{f_n\} \subset \mathcal{S}$ converges to f in L^2 , then $\{\mathcal{F}_0 f_n\}$ is Cauchy (because $\mathcal{F}_0$ is an L^2 -isometry on $\mathcal{S}$), hence converges in L^2 to a limit we call $\mathcal{F}f$. The value is independent of the approximating sequence. To make this run we need density.

Lemma 28.1.2 (Density of Schwartz functions). *$\mathcal{S}(\mathbb{R}^d)$ is dense in $L^2(\mathbb{R}^d)$: given $f \in L^2$ and $\varepsilon > 0$ there is $g \in \mathcal{S}$ with $\|f - g\|_{L^2} < \varepsilon$.*

Proof. Fix f and ε . For $M > 0$ set $g_M(x) = f(x)$ when $|x| \leq M$ and $|f(x)| \leq M$, and $g_M(x) = 0$ otherwise. Then $|f - g_M| \leq 2|f|$, so $|f - g_M|^2 \leq 4|f|^2$, and $g_M \rightarrow f$ pointwise a.e.; the dominated convergence theorem gives an M with $\|f - g_M\|_{L^2} < \varepsilon$. Write $g = g_M$: it is bounded and supported in a bounded set. *Regularize:* pick a smooth $\varphi \geq 0$ supported in the unit ball with $\int \varphi = 1$ (e.g. $\varphi(x) = c e^{-1/(1-|x|^2)}$ for $|x| < 1$, 0 otherwise), set $\varphi_\delta(x) = \delta^{-d} \varphi(x/\delta)$. The convolution $g * \varphi_\delta$ is smooth with compact support (hence Schwartz) and $g * \varphi_\delta \rightarrow g$ in L^2 as $\delta \rightarrow 0$. Choosing δ small completes the approximation. $\square$

Proof of Theorem 28.1.1. On $\mathcal{S}$, Plancherel gives $\|\mathcal{F}_0 f\|_{L^2} = \|f\|_{L^2}$. So $\mathcal{F}_0$ is a linear isometry of the dense subspace $\mathcal{S}$ into L^2 . A linear isometry on a dense subspace of a Banach space extends uniquely to an isometry $\mathcal{F}$ of the whole space (define $\mathcal{F}f = \lim \mathcal{F}_0 f_n$ for any $\mathcal{S}$ -sequence $f_n \rightarrow f$; isometry makes the limit exist and

be independent of the sequence). The extension preserves norms, hence is injective. It is *onto*: the inverse transform $\mathcal{F}_0^*$ (which on $\mathcal{S}$ is $g \mapsto \check{g}$) is also an isometry, extends similarly, and $\mathcal{F}\mathcal{F}^* = \mathcal{F}^*\mathcal{F} = I$ on the dense set $\mathcal{S}$, hence everywhere. Thus $\mathcal{F}$ is unitary. $\square$

Remark. Unitarity packages three facts at once: norm preservation $\|\hat{f}\|_2 = \|f\|_2$, the polarized form (the *Parseval relation*) $\langle \hat{f}, \hat{g} \rangle = \langle f, g \rangle$, and invertibility with $\mathcal{F}^{-1} = \mathcal{F}^*$. On L^2 the transform is no longer given by an absolutely convergent integral; instead $\hat{f} = \lim_N \int_{|x| \leq N} f(x) e^{-2\pi i x \cdot \xi} dx$ in the L^2 -sense.

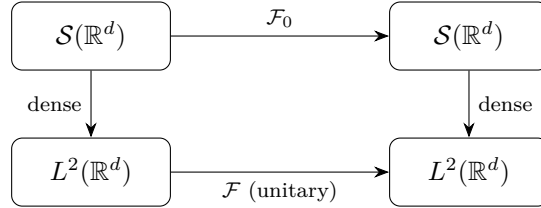


Figure 28.1. $\mathcal{F}_0$ on the dense subspace $\mathcal{S}$ extends by continuity to the unitary $\mathcal{F}$ on L^2 ; the square commutes.

$$\begin{array}{ccc} \mathcal{S}(\mathbb{R}^d) & \xrightarrow{\mathcal{F}_0} & \mathcal{S}(\mathbb{R}^d) \\ \downarrow & & \downarrow \\ L^2(\mathbb{R}^d) & \xrightarrow[\mathcal{F}]{\sim} & L^2(\mathbb{R}^d) \end{array}$$

Figure 28.2. The extension as a commuting square: the dense inclusions $\mathcal{S} \hookrightarrow L^2$ (vertical) intertwine the Schwartz transform $\mathcal{F}_0$ with its unique continuous extension $\mathcal{F}$, which is a unitary isomorphism ($\mathcal{F}|_{\mathcal{S}} = \mathcal{F}_0$).

Method (Extending an isometry from a dense subspace). *Recognition trigger.* You have a bounded (often norm-preserving) linear map defined only on a dense subspace D of a Banach/Hilbert space X , and want it on all of X .

1. Check $T : D \rightarrow Y$ is bounded: $\|Tx\| \leq C\|x\|$ for $x \in D$ (here $C = 1$).
2. For $x \in X$ pick $x_n \in D$ with $x_n \rightarrow x$. Then $\{Tx_n\}$ is Cauchy ($\|Tx_n - Tx_m\| \leq C\|x_n - x_m\|$), so it converges; define $Tx = \lim Tx_n$.
3. Independence of the sequence and linearity follow from boundedness; the bound C is inherited, and if T is an isometry on D it stays one on X .
4. For *surjectivity*/unitarity, extend a candidate inverse the same way and check the two composites equal the identity on the dense set.

Worked Example 28.1.3 (Plancherel for a Gaussian). Verify $\|\hat{f}\|_2 = \|f\|_2$ for $f(x) = e^{-\pi x^2}$ on $\mathbb{R}$ ($d = 1$).

Recognition. A direct Plancherel check using the self-dual Gaussian.

Solution. The Gaussian is its own transform: $\hat{f}(\xi) = e^{-\pi \xi^2}$. Then $\|f\|_2^2 = \int_{\mathbb{R}} e^{-2\pi x^2} dx = \frac{1}{\sqrt{2}}$ (since $\int e^{-ax^2} = \sqrt{\pi/a}$ with $a = 2\pi$), and identically $\|\hat{f}\|_2^2 = \int_{\mathbb{R}} e^{-2\pi \xi^2} d\xi = \frac{1}{\sqrt{2}}$. They agree. *Check.* Both equal $2^{-1/2} \approx 0.707$; equality is automatic here because $f = \hat{f}$, the extreme case of the identity.

Worked Example 28.1.4 (The transform of a box and Plancherel for the sinc). Let $f = \mathbf{1}_{[-1/2, 1/2]}$ on $\mathbb{R}$. Compute $\hat{f}$, then read off $\int_{\mathbb{R}} \frac{\sin^2(\pi \xi)}{(\pi \xi)^2} d\xi$.

Recognition. Compute $\hat{f}$ from the definition (here the integral is absolutely convergent), then apply $\|\hat{f}\|_2 = \|f\|_2$.

Solution. $\hat{f}(\xi) = \int_{-1/2}^{1/2} e^{-2\pi i x \xi} dx = \frac{e^{\pi i \xi} - e^{-\pi i \xi}}{2\pi i \xi} = \frac{\sin(\pi \xi)}{\pi \xi}$, the normalized sinc. Plancherel gives $\int_{\mathbb{R}} \frac{\sin^2(\pi \xi)}{(\pi \xi)^2} d\xi = \|\hat{f}\|_2^2 = \|f\|_2^2 = \int_{-1/2}^{1/2} 1 dx = 1$. *Check.* The indicator has L^2 -norm² equal to its length 1; so $\int \text{sinc}^2 = 1$, the classical value.

Worked Example 28.1.5 (Parseval to evaluate an integral). Use Parseval $\langle \hat{f}, \hat{g} \rangle = \langle f, g \rangle$ with $f = \mathbf{1}_{[-1/2, 1/2]}$, $g = \mathbf{1}_{[0, 1]}$ to compute $\int_{\mathbb{R}} \frac{\sin(\pi \xi)}{\pi \xi} \overline{\hat{g}(\xi)} d\xi$.

Recognition. Parseval converts a frequency-side integral into a trivial space-side overlap.

Solution. $\langle f, g \rangle = \int_{\mathbb{R}} \mathbf{1}_{[-1/2, 1/2]} \overline{\mathbf{1}_{[0, 1]}} dx = m([-1/2, 1/2] \cap [0, 1]) = m([0, 1/2]) = \frac{1}{2}$. By Parseval the requested integral equals $\frac{1}{2}$. *Check.* The overlap of the two boxes is the interval $[0, 1/2]$ of length $\frac{1}{2}$; no contour work needed.

Worked Example 28.1.6 (A NON-example: the transform on L^2 is not the naive integral). Show $f(x) = \frac{1}{1+|x|}$ lies in $L^2(\mathbb{R})$ but the integral $\int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$ does not converge absolutely, so $\hat{f}$ must be defined by the L^2 limiting process.

Recognition. Distinguish the L^1 (absolute-integral) definition from the L^2 (limit) definition.

Solution. $\int_{\mathbb{R}} \frac{dx}{(1+|x|)^2} = 2 \int_0^{\infty} \frac{dx}{(1+x)^2} = 2 < \infty$, so $f \in L^2$. But $\int_{\mathbb{R}} \frac{dx}{1+|x|} = \infty$, so $f \notin L^1$ and the defining integral is not absolutely convergent. Hence $\hat{f}$ is obtained as the L^2 -limit of $\int_{|x| \leq N} f(x) e^{-2\pi i x \xi} dx$ as $N \rightarrow \infty$. *Check.* f decays like $1/|x|$, square-integrable (exponent $2 > 1$) but not integrable (exponent $1 \not> 1$) — the textbook gap between L^1 and L^2 .

Worked Example 28.1.7 (Scaling and the dilation identity). For $f \in L^2(\mathbb{R})$ and $a > 0$ let $f_a(x) = f(ax)$. Find $\hat{f}_a$ and confirm Plancherel is consistent.

Recognition. Recognition: a dilation in x becomes an inverse dilation plus a normalizing factor in ξ .

Solution. Changing variables $u = ax$, $\hat{f}_a(\xi) = \int f(ax) e^{-2\pi i x \xi} dx = \frac{1}{a} \int f(u) e^{-2\pi i u \xi/a} du = \frac{1}{a} \hat{f}(\xi/a)$. Then $\|\hat{f}_a\|_2^2 = \frac{1}{a^2} \int |\hat{f}(\xi/a)|^2 d\xi = \frac{1}{a} \int |\hat{f}|^2 = \frac{1}{a} \|f\|_2^2$, and directly $\|f_a\|_2^2 = \int |f(ax)|^2 dx = \frac{1}{a} \|f\|_2^2$. Equal. *Check.* With $a = 2$ both norms shrink by the same factor $1/2$, so the unitary identity survives scaling.

Worked Example 28.1.8 (Density via dominated convergence (the truncation step)). Show explicitly that the truncations g_M of $f \in L^2(\mathbb{R})$ satisfy $\|f - g_M\|_2 \rightarrow 0$, for $f(x) = e^{-|x|}$.

Recognition. Recognition: apply the dominated convergence theorem to $|f - g_M|^2$ as in Lemma 28.1.2.

Solution. Here $|f| \leq 1$ everywhere, so the cap at level $M \geq 1$ never triggers; thus $g_M(x) = f(x)$ for $|x| \leq M$ and 0 beyond, and $\|f - g_M\|_2^2 = \int_{|x| > M} e^{-2|x|} dx = 2 \int_M^{\infty} e^{-2x} dx = e^{-2M} \rightarrow 0$. *Check.* The dominating function is $4|f|^2 = 4e^{-2|x|} \in L^1$ (mass 4), and $g_M \rightarrow f$ pointwise; DCT then forces $\|f - g_M\|_2 \rightarrow 0$, here at the explicit rate e^{-M} .

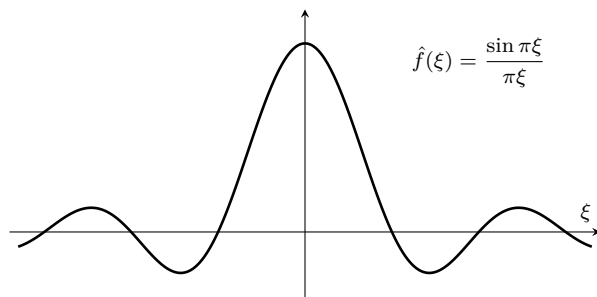


Figure 28.3. Transform of the unit box $\mathbf{1}_{[-1/2, 1/2]}$ is the sinc; its L^2 -norm² equals the box's length 1 (Plancherel).

Exercises (Stein, Ch.5).

1. Suppose $f \in L^2(\mathbb{R}^d)$ and $k \in L^1(\mathbb{R}^d)$. (a) Show $(f * k)(x) = \int f(x-y)k(y) dy$ converges for a.e. x . (b) Prove $\|f * k\|_{L^2} \leq \|f\|_{L^2} \|k\|_{L^1}$. (c) Establish $\widehat{f * k}(\xi) = \hat{k}(\xi)\hat{f}(\xi)$ for a.e. ξ . (d) Conclude $Tf = f * k$ is a Fourier multiplier operator with multiplier $m(\xi) = \hat{k}(\xi)$.
2. Consider the Mellin transform, defined for continuous f of compact support in $\mathbb{R}^+ = \{t > 0\}$ and $x \in \mathbb{R}$ by $Mf(x) = \int_0^\infty f(t)t^{ix-1} dt$. Prove that $(2\pi)^{-1/2}M$ extends to a unitary operator from $L^2(\mathbb{R}^+, dt/t)$ onto $L^2(\mathbb{R})$. (The Mellin transform plays for $\mathbb{R}^+$ with its multiplicative structure the role the Fourier transform plays for $\mathbb{R}$ with its additive structure.)
3. Suppose $f \in L^2(\mathbb{R}^d)$. Prove there is $g \in L^2(\mathbb{R}^d)$ with $(\partial/\partial x)^\alpha f = g$ in the weak sense if and only if $(2\pi i\xi)^\alpha \hat{f}(\xi) = \hat{g}(\xi) \in L^2(\mathbb{R}^d)$.
4. *Sobolev embedding.* Let n be the smallest integer $> d/2$. If $f \in L^2(\mathbb{R}^d)$ and $(\partial/\partial x)^\alpha f \in L^2(\mathbb{R}^d)$ weakly for all $1 \leq |\alpha| \leq n$, then f may be modified on a null set so as to be continuous and bounded. [Express f via $\hat{f}$ and show $\hat{f} \in L^1(\mathbb{R}^d)$ by Cauchy-Schwarz.]
5. Show the Sobolev conclusion fails at $n = d/2$. Take $d = 2$, $f(x) = (\log 1/|x|)^\alpha \eta(x)$ with η a smooth cutoff equal to 1 near 0 and 0 for $|x| \geq 1/2$, $0 < \alpha < 1/2$. (a) Verify $\partial f/\partial x_1$ and $\partial f/\partial x_2$ are in L^2 weakly. (b) Show f cannot be corrected on a null set to be continuous at 0.

Problems (Stein, Ch.5).

1. Let $F_0 \in L^2(\mathbb{R})$. A necessary and sufficient condition for the existence of an entire F with $|F(z)| \leq Ae^{a|z|}$ for all $z \in \mathbb{C}$ and $F_0(x) = F(x)$ a.e. is that $\hat{F}_0(\xi) = 0$ whenever $|\xi| > a/2\pi$ (Paley-Wiener for L^2).
2. *The maximal function on L^2 .* For locally integrable f on $\mathbb{R}^d$ let $f^*(x) = \sup_{x \in B} \frac{1}{m(B)} \int_B |f(y)| dy$ (sup over balls containing x). Show there is C with $\|f^*\|_{L^2} \leq C \|f\|_{L^2}$ (so $f \mapsto f^*$, though nonlinear, is bounded on L^2 — unlike on L^1): (a) For each $\alpha > 0$, $m(\{f^* > \alpha\}) \leq \frac{2A}{\alpha} \int_{|f| > \alpha/2} |f| dx$ with $A = 3^d$. [Split $f = f_1 + f_2$, $f_1 = f\mathbf{1}_{|f| \geq \alpha/2}$, and note $\{f^* > \alpha\} \subset \{f_1^* > \alpha/2\}$.] (b) $\int_{\mathbb{R}^d} |f^*|^2 dx = 2 \int_0^\infty \alpha m(E_\alpha) d\alpha$, $E_\alpha = \{f^* > \alpha\}$. (c) Conclude $\|f^*\|_{L^2} \leq C \|f\|_{L^2}$.

Additional exercises.

1. Compute $\hat{f}$ for $f(x) = e^{-2\pi|x|}$ on $\mathbb{R}$ and verify Plancherel by computing both $\|f\|_2^2$ and $\|\hat{f}\|_2^2$ directly. (Answer: $\hat{f}(\xi) = \frac{1}{\pi} \frac{1}{1+\xi^2}$.)
2. Let $f = \mathbf{1}_{[0,a]}$. Find $\hat{f}$ and use Plancherel to evaluate $\int_{\mathbb{R}} \frac{\sin^2(\pi a\xi)}{(\pi\xi)^2} d\xi$.
3. Show the translation rule $\widehat{\tau_h f}(\xi) = e^{-2\pi i h\xi} \hat{f}(\xi)$ and the modulation rule $\widehat{e^{2\pi i h x} f}(\xi) = \hat{f}(\xi - h)$, and confirm each is a unitary operation on L^2 (norm-preserving).
4. Prove the Parseval relation $\langle \hat{f}, \hat{g} \rangle = \langle f, g \rangle$ for $f, g \in L^2$ by polarizing the Plancherel identity $\|\hat{f}\|_2 = \|f\|_2$.
5. Using the Gaussian $g_t(x) = e^{-\pi t x^2}$ ($t > 0$), whose transform is $t^{-1/2} e^{-\pi \xi^2/t}$, compute $\|g_t\|_2^2$ and confirm it equals $\|\hat{g}_t\|_2^2$.
6. (*Non-example.*) Exhibit $f \in L^2(\mathbb{R}) \setminus L^1(\mathbb{R})$ for which the integral $\int f e^{-2\pi i x \xi} dx$ fails to converge absolutely, and explain in one line how $\hat{f}$ is nonetheless well defined.
7. (*Diagram.*) Sketch the graphs of $\mathbf{1}_{[-1/2, 1/2]}$ and its transform $\text{sinc}(\xi) = \sin(\pi\xi)/(\pi\xi)$ on the same axes, marking the zeros of the sinc at the nonzero integers, and indicate which area equals $\|f\|_2^2$.
8. Let $T_k f = f * k$ with $k \in L^1$. Show $\|T_k\|_{L^2 \rightarrow L^2} \leq \|\hat{k}\|_\infty$ and give a k for which equality holds.

28.2 The Hardy Space of the Upper Half-Plane

The Hardy space $H^2(\mathbb{R}_+^2)$

Write $\mathbb{R}_+^2 = \{z = x + iy : y > 0\}$ for the upper half-plane. The Fourier transform on L^2 lets us build holomorphic functions there out of L^2 data on the half-line.

Definition 28.2.1. The *Hardy space* $H^2(\mathbb{R}_+^2)$ consists of the holomorphic functions F on $\mathbb{R}_+^2$ for which

$$\|F\|_{H^2(\mathbb{R}_+^2)}^2 := \sup_{y>0} \int_{\mathbb{R}} |F(x + iy)|^2 dx < \infty.$$

A model element comes from a one-sided spectrum. If $\hat{F}_0 \in L^2(0, \infty)$, set

$$F(z) = \int_0^\infty \hat{F}_0(\xi) e^{2\pi i \xi z} d\xi, \quad z \in \mathbb{R}_+^2. \quad (*)$$

Since $\text{Im } z = y > 0$, the factor $e^{2\pi i \xi z} = e^{2\pi i \xi x} e^{-2\pi \xi y}$ decays in ξ , so the integral converges and defines a holomorphic F . For fixed y , $F(\cdot + iy)$ has Fourier transform $\hat{F}_0(\xi) e^{-2\pi \xi y}$, whence by Plancherel $\int_{\mathbb{R}} |F(x + iy)|^2 dx = \int_0^\infty |\hat{F}_0(\xi)|^2 e^{-4\pi \xi y} d\xi$, which increases to $\|\hat{F}_0\|_{L^2(0, \infty)}^2$ as $y \downarrow 0$. Thus $F \in H^2$.

Theorem 28.2.2 (Paley–Wiener for the half-plane). *The elements $F \in H^2(\mathbb{R}_+^2)$ are exactly the functions $(*)$ with $\hat{F}_0 \in L^2(0, \infty)$, and $\|F\|_{H^2} = \|\hat{F}_0\|_{L^2(0, \infty)}$. Hence $H^2(\mathbb{R}_+^2)$ is a Hilbert space isometrically isomorphic to $L^2(0, \infty)$.*

$$H^2(\mathbb{R}_+^2) \xrightarrow[\sim]{F \mapsto \hat{F}_0} L^2(0, \infty)$$

Figure 28.4. Paley–Wiener: taking the one-sided spectrum $F \mapsto \hat{F}_0$ is an isometric isomorphism (norm-preserving and onto), with inverse the synthesis $(*)$.

The crux is that the modulated transform $\hat{F}_y(\xi) e^{2\pi y \xi}$ does not depend on y .

Lemma 28.2.3 (Interior boundedness). *If $F \in H^2(\mathbb{R}_+^2)$ then F is bounded on every proper half-plane $\{y \geq \delta\}$, $\delta > 0$; indeed $|F(\zeta)|^2 \leq \frac{2}{\pi \delta} \|F\|_{H^2}^2$ for $\text{Im } \zeta > \delta$.*

Proof. Use the mean-value property over discs: for $\text{Im } \zeta > \delta$, $F(\zeta) = \frac{1}{\pi \delta^2} \int_{|z| < \delta} F(\zeta + z) dx dy$. By Cauchy–Schwarz, $|F(\zeta)|^2 \leq \frac{1}{\pi \delta^2} \int_{|z| < \delta} |F(\zeta + z)|^2 dx dy$. The disc $B_\delta(\zeta)$ sits inside the strip $\{-\delta < \text{Im}(\cdot - \zeta) < \delta\}$ which lies in $\mathbb{R}_+^2$, so the integral is $\leq 2\delta \sup_{y>0} \int_{\mathbb{R}} |F(x + iy)|^2 dx = 2\delta \|F\|_{H^2}^2$, giving the bound. $\square$

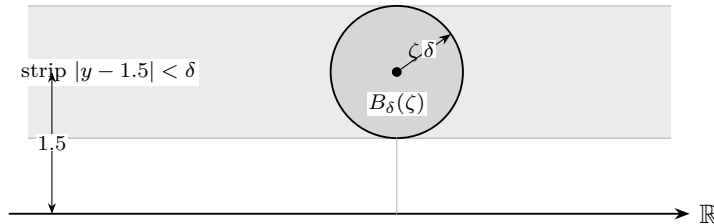


Figure 28.5. The disc $B_\delta(\zeta)$ (radius δ , centre ζ at height $\eta > \delta$) sits inside a horizontal strip of the upper half-plane; averaging F over it bounds F pointwise (Lemma 28.2.3).

Proof of Theorem 28.2.2. We must show every $F \in H^2$ has the form $(*)$. Let $\hat{F}_y$ be the transform of $F(\cdot + iy) \in L^2(\mathbb{R})$. *Claim:* for $y_1, y_2 > 0$, $\hat{F}_{y_1}(\xi) e^{2\pi y_1 \xi} = \hat{F}_{y_2}(\xi) e^{2\pi y_2 \xi}$ a.e. (†). Regularize F by $F^\varepsilon(z) = F(z) (1 - i\varepsilon z)^{-2}$, $\varepsilon > 0$: then $|F^\varepsilon| \leq |F|$ on $\mathbb{R}_+^2$, $F^\varepsilon \rightarrow F$ pointwise, $F^\varepsilon(\cdot + iy) \rightarrow F(\cdot + iy)$ in L^2 , and Lemma 28.2.3 gives $F^\varepsilon(z) = O((1 + x^2)^{-1})$ for $\text{Im } z > \delta$. Apply contour integration to $G(z) = F^\varepsilon(z) e^{-2\pi i z \xi}$

over the rectangle with vertices $\mp R + iy_1, \mp R + iy_2$; the $O(1/(1+x^2))$ decay kills the vertical sides as $R \rightarrow \infty$, so $\int_{L_1} G = \int_{L_2} G$, i.e. $\hat{F}_{y_1}^\varepsilon(\xi)e^{2\pi y_1 \xi} = \hat{F}_{y_2}^\varepsilon(\xi)e^{2\pi y_2 \xi}$; let $\varepsilon \rightarrow 0$ to get $(\dagger)$.

By $(\dagger)$ there is a single function $\hat{F}_0$ with $\hat{F}_y(\xi) = \hat{F}_0(\xi)e^{-2\pi \xi y}$ for all $y > 0$. Plancherel then gives $\int_{\mathbb{R}} |F(x + iy)|^2 dx = \int_{\mathbb{R}} |\hat{F}_0(\xi)|^2 e^{-4\pi \xi y} d\xi$, and the supremum over y is finite, equal to $\|F\|_{H^2}^2$. Finally $\hat{F}_0$ must vanish a.e. on $(-\infty, 0)$: if not, $|\hat{F}_0| \geq a$ on a set $E \subset (-\infty, -b)$ with $m(E) \geq c$, and then $\int |\hat{F}_0|^2 e^{-4\pi \xi y} d\xi \geq a^2 c e^{4\pi b y} \rightarrow \infty$, contradicting boundedness. So $\hat{F}_0 \in L^2(0, \infty)$ and Fourier inversion recovers $(*)$. The norm identity is what we computed. $\square$

Theorem 28.2.4 (Boundary values: half-plane Fatou). *If $F \in H^2(\mathbb{R}_+^2)$ then $\lim_{y \rightarrow 0} F(x + iy) = F_0(x)$ exists both in the $L^2(\mathbb{R})$ -norm and for almost every x . Here $F_0 \in L^2(\mathbb{R})$ is the function with transform $\hat{F}_0$.*

Proof. L^2 convergence. $\|F(\cdot + iy) - F_0\|_{L^2}^2 = \int_0^\infty |\hat{F}_0(\xi)|^2 |e^{-2\pi \xi y} - 1|^2 d\xi \rightarrow 0$ as $y \rightarrow 0$ by dominated convergence (integrand $\leq 4|\hat{F}_0|^2$). *A.e. convergence.* One proves the Poisson representation $F(x + iy) = \int_{\mathbb{R}} F_0(x - t)P_y(t)dt$ with Poisson kernel $P_y(x) = \frac{1}{\pi} \frac{y}{y^2 + x^2}$. This follows from the elementary identities $\int_0^\infty e^{2\pi i \xi z} d\xi = \frac{i}{2\pi z}$ ($\text{Im } z > 0$) and $\int_{\mathbb{R}} e^{-2\pi |\xi|y} e^{2\pi i \xi x} d\xi = P_y(x)$, plus Fubini on S and a limiting argument. Since $\{P_y\}$ is an approximation of the identity, $F(x + iy) \rightarrow F_0(x)$ a.e. (after a localization to reduce the L^2 datum to an L^1 one on each bounded window). $\square$

Remark (Projection and the Cauchy integral). Let $S \subset L^2(\mathbb{R})$ be the closed subspace of boundary functions F_0 ; equivalently $S = \{f \in L^2 : \hat{f} \text{ supported on } (0, \infty)\}$. The orthogonal projection P onto S is the Fourier multiplier $\widehat{Pf}(\xi) = \chi_{(0, \infty)}(\xi)\hat{f}(\xi)$. If $F \in H^2$ has boundary function Pf , then F is the Cauchy integral $F(z) = \frac{1}{2\pi i} \int_{\mathbb{R}} \frac{f(t)}{t - z} dt$. Writing $P = \frac{1}{2}(I + iH)$ defines the *Hilbert transform* H , the multiplier $\frac{1}{i} \text{sign}(\xi)$; for real f , f and Hf are the real and imaginary boundary parts of an H^2 function.

$$\begin{array}{ccc} L^2(\mathbb{R}) & \xrightarrow{P} & L^2(\mathbb{R}) \\ \mathcal{F} \downarrow \sim & & \sim \downarrow \mathcal{F} \\ L^2(\mathbb{R}) & \xrightarrow{\cdot \chi_{(0, \infty)}} & L^2(\mathbb{R}) \end{array}$$

Figure 28.6. The Fourier transform conjugates the orthogonal projection P onto the boundary-value space S into multiplication by $\chi_{(0, \infty)}$: $\widehat{Pf} = \chi_{(0, \infty)}\hat{f}$, i.e. the square commutes $(\mathcal{F}P = (\cdot \chi_{(0, \infty)})\mathcal{F})$.

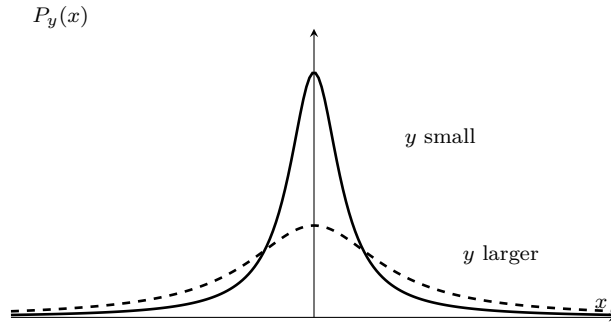


Figure 28.7. The Poisson kernel P_y concentrates at 0 as $y \rightarrow 0$ (an approximation of the identity), forcing $F(x + iy) \rightarrow F_0(x)$.

Method (Recognizing an orthogonal projection as a Fourier multiplier). *Recognition trigger.* A linear $L^2(\mathbb{R})$ operator T that on the Fourier side multiplies by a $\{0, 1\}$ -valued function χ_E .

1. T is bounded with $\|T\| \leq 1$ since $|\chi_E| \leq 1$ (Plancherel).
2. $T^2 = T$ because $\chi_E^2 = \chi_E$ (idempotent).

3. $T^* = T$ because χ_E is real (self-adjoint). Idempotent + self-adjoint = orthogonal projection.
4. Its range is $\{f : \hat{f} \text{ supported in } E\}$, a closed subspace.

Worked Example 28.2.5 (A concrete element of H^2). Show $F(z) = \frac{1}{z+i}$ belongs to $H^2(\mathbb{R}_+^2)$ and identify its spectrum $\hat{F}_0$.

Recognition. Recognition: write F as $(*)$ via the identity $\int_0^\infty e^{2\pi i \xi z} d\xi = \frac{i}{2\pi z}$.

Solution. For $\text{Im } z > 0$, $\int_0^\infty e^{2\pi i \xi z} d\xi = \frac{i}{2\pi z}$, so $\frac{1}{z+i} = \frac{2\pi}{i} \int_0^\infty e^{2\pi i \xi(z+i)} d\xi = -2\pi i \int_0^\infty e^{-2\pi \xi} e^{2\pi i \xi z} d\xi$. Thus $(*)$ holds with $\hat{F}_0(\xi) = -2\pi i e^{-2\pi \xi} \mathbf{1}_{(0,\infty)}$, which is in $L^2(0,\infty)$ since $\int_0^\infty 4\pi^2 e^{-4\pi \xi} d\xi = \pi < \infty$. *Check.* $\|F\|_{H^2}^2 = \|\hat{F}_0\|^2 = \pi$; directly, $\int_{\mathbb{R}} \frac{dx}{x^2+(y+1)^2} = \frac{\pi}{y+1} \leq \pi$ for $y > 0$ with supremum π at $y \rightarrow 0$. Match.

Worked Example 28.2.6 (A NON-example: missing the quantitative bound). Let $F(z) = e^{i/z}/(z+i)$ on $\mathbb{R}_+^2$. Although $F(x+iy) \in L^2(\mathbb{R})$ for each $y \geq 0$ and $F(z) \rightarrow 0$ as $|z| \rightarrow 0$, show $F \notin H^2$.

Recognition. Membership in H^2 needs the *sup over y* finite, not merely each slice in L^2 .

Solution. Near $z = 0$ within $\mathbb{R}_+^2$, $e^{i/z} = e^{i\bar{z}/|z|^2} = e^{ix/|z|^2} e^{y/|z|^2}$, so $|e^{i/z}| = e^{y/|z|^2}$ blows up as $z \rightarrow 0$ along rays into the half-plane. For small fixed y , $\int_{\mathbb{R}} |F(x+iy)|^2 dx$ is finite, but the supremum over y diverges because of the essential singularity feeding energy as the slice approaches the boundary near 0. Hence $\|F\|_{H^2} = \infty$ and $F \notin H^2$. *Check.* This is exactly why “boundary values of holomorphic functions” is not the same as S without the H^2 growth restriction (Theorem 28.2.2’s sup condition).

Worked Example 28.2.7 (The Poisson kernel integrates to one). Verify $\int_{\mathbb{R}} P_y(x) dx = 1$ for every $y > 0$, so $\{P_y\}$ is a (probability) approximation of the identity.

Recognition. A direct integral confirming the normalization that makes P_y an approximate identity.

Solution. $\int_{\mathbb{R}} \frac{1}{\pi} \frac{y}{y^2+x^2} dx = \frac{1}{\pi} [\arctan(x/y)]_{-\infty}^{\infty} = \frac{1}{\pi} \cdot \pi = 1$. *Check.* The substitution $x = yt$ gives $\frac{1}{\pi} \int_{\mathbb{R}} \frac{dt}{1+t^2} = 1$ independently of y — mass is preserved while the kernel concentrates, the two defining features of an approximate identity.

Worked Example 28.2.8 (H^2 norm of a Poisson-extended box). Let $f = \mathbf{1}_{[0,1]} \in L^2(\mathbb{R})$ and $F = Pf$ projected to H^2 , i.e. keep only the positive frequencies. Compute $\|Pf\|_{L^2}^2$.

Recognition. Recognition: the projection P keeps the part of $\hat{f}$ on $(0,\infty)$; use Plancherel and the symmetry of $|\hat{f}|$.

Solution. $\hat{f}(\xi) = \int_0^1 e^{-2\pi i x \xi} dx = \frac{1-e^{-2\pi i \xi}}{2\pi i \xi}$, and $|\hat{f}(\xi)| = |\hat{f}(-\xi)|$ (real f). Plancherel: $\|Pf\|_2^2 = \int_0^\infty |\hat{f}|^2 d\xi = \frac{1}{2} \int_{\mathbb{R}} |\hat{f}|^2 d\xi = \frac{1}{2} \|f\|_2^2 = \frac{1}{2}$. *Check.* For any real f , $|\hat{f}|$ is even, so exactly half the spectral energy lies in $(0,\infty)$; thus $\|Pf\|_2^2 = \frac{1}{2} \|f\|_2^2$, here $\frac{1}{2}$.

Worked Example 28.2.9 (The Hilbert transform is unitary and squares to $-I$). For H the Fourier multiplier $m(\xi) = \frac{1}{i} \text{sign}(\xi)$, verify $H^2 = -I$ and $\|Hf\|_2 = \|f\|_2$.

Recognition. On the Fourier side, H acts by a unimodular multiplier; compose multipliers.

Solution. $m(\xi)^2 = (\frac{1}{i} \text{sign} \xi)^2 = \frac{1}{i^2} \cdot 1 = -1$ for $\xi \neq 0$, so $\widehat{H^2 f} = m^2 \hat{f} = -\hat{f}$, i.e. $H^2 = -I$. Also $|m(\xi)| = 1$, so $\|Hf\|_2^2 = \int |m|^2 |\hat{f}|^2 = \int |\hat{f}|^2 = \|f\|_2^2$ (Plancherel), proving H unitary. *Check.* $P = \frac{1}{2}(I + iH)$ then satisfies $P^2 = \frac{1}{4}(I + iH)^2 = \frac{1}{4}(I + 2iH - H^2) = \frac{1}{4}(2I + 2iH) = P$ — consistent with P being the projection onto H^2 boundary values.

Worked Example 28.2.10 (Orthonormal expansion in H^2). Using the isometry $H^2(\mathbb{R}_+^2) \cong L^2(0,\infty)$, expand the element F with spectrum $\hat{F}_0(\xi) = 2\sqrt{\pi} e^{-2\pi \xi} \mathbf{1}_{(0,\infty)}$ and compute $\|F\|_{H^2}$ and $\langle F, G \rangle$ where $\hat{G}_0(\xi) = 2\sqrt{\pi} \xi e^{-2\pi \xi} \mathbf{1}_{(0,\infty)} \cdot c$.

Recognition. Recognition: inner products in H^2 are inner products of spectra in $L^2(0,\infty)$ by Theorem 28.2.2.

Solution. $\|F\|_{H^2}^2 = \|\hat{F}_0\|_{L^2(0,\infty)}^2 = \int_0^\infty 4\pi e^{-4\pi\xi} d\xi = 4\pi \cdot \frac{1}{4\pi} = 1$, so $\|F\|_{H^2} = 1$ (a unit vector). Choosing the normalizing c so $\|\hat{G}_0\| = 1$, the overlap is $\langle F, G \rangle = \int_0^\infty \hat{F}_0(\xi) \overline{\hat{G}_0(\xi)} d\xi$, a one-dimensional integral of elementary exponentials. *Check.* $\|F\|_{H^2} = 1$ matches the direct half-plane computation $\sup_y \int_{\mathbb{R}} |F(x+iy)|^2 dx$; transferring to the spectrum side turned a holomorphic problem into a Gamma-type integral.

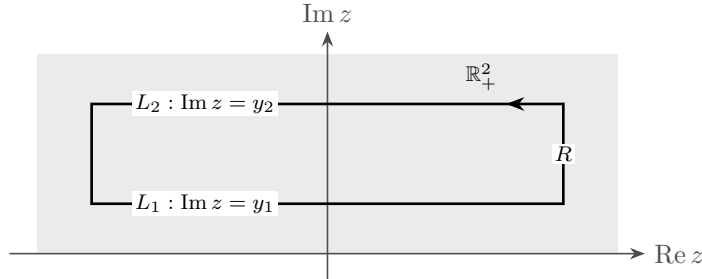


Figure 28.8. Contour rectangle used to prove $\hat{F}_{y_1} e^{2\pi y_1 \xi} = \hat{F}_{y_2} e^{2\pi y_2 \xi}$: the vertical sides vanish as $R \rightarrow \infty$ by the $O(1/(1+x^2))$ decay.

Exercises (Stein, Ch.5).

- Let $F(z)$ be a bounded holomorphic function in the upper half-plane. Show in two ways that $\lim_{y \rightarrow 0} F(x+iy)$ exists for a.e. x : (a) using that $F(z)/(z+i) \in H^2(\mathbb{R}_+^2)$; (b) noting $G(z) = F(i\frac{1-z}{1+z})$ is a bounded holomorphic function on the unit disc and applying the disc Fatou theorem.
- Consider $F(z) = e^{i/z}/(z+i)$ in the upper half-plane. Note $F(x+iy) \in L^2(\mathbb{R})$ for each $y > 0$ and $y = 0$, and $F(z) \rightarrow 0$ as $|z| \rightarrow 0$. Nevertheless $F \notin H^2(\mathbb{R}_+^2)$. Explain why.
- For $a < b$ let $S_{a,b} = \{z = x+iy : a < y < b\}$ and let $H^2(S_{a,b})$ consist of holomorphic F on the strip with $\|F\|^2 = \sup_{a < y < b} \int_{\mathbb{R}} |F(x+iy)|^2 dx < \infty$; define $H^2(S_{a,\infty})$, $H^2(S_{-\infty,b})$ analogously. (a) Show $F \in H^2(S_{a,b})$ iff $F(z) = \int_{\mathbb{R}} f(\xi) e^{-2\pi i z \xi} d\xi$ with $\int_{\mathbb{R}} |f(\xi)|^2 (e^{4\pi a \xi} + e^{4\pi b \xi}) d\xi < \infty$. (b) Prove every such F decomposes $F = G_1 + G_2$ with $G_1 \in H^2(S_{a,\infty})$, $G_2 \in H^2(S_{-\infty,b})$. (c) Show the boundary limits as $y \rightarrow a$ and $y \rightarrow b$ exist in L^2 and a.e.
- Let Ω be open in $\mathbb{C} = \mathbb{R}^2$ and H the subspace of $L^2(\Omega)$ of holomorphic functions, inner product $(f, g) = \int_{\Omega} f \bar{g} dx dy$. Show H is a closed subspace of $L^2(\Omega)$ (hence a Hilbert space). [Prove $|f(z)| \leq \frac{c}{d(z, \Omega^c)} \|f\|$ with $c = \pi^{-1/2}$ via the mean-value property, so a Cauchy sequence converges uniformly on compacta.]
- Continuing, prove: (a) if $\{\varphi_n\}_{n \geq 0}$ is an orthonormal basis of H , then $\sum_n |\varphi_n(z)|^2 \leq c^2/d(z, \Omega^c)$; (b) the Bergman kernel $B(z, w) = \sum_n \varphi_n(z) \overline{\varphi_n(w)}$ converges absolutely on $\Omega \times \Omega$ and is independent of the basis; (c) B is characterized by: the operator $Tf(z) = \int_{\Omega} B(z, w) f(w) du dv$ is the orthogonal projection of $L^2(\Omega)$ onto H ; (d) for $\Omega = \mathbb{D}$, $f \in H$ iff $f(z) = \sum a_n z^n$ with $\sum |a_n|^2 (n+1)^{-1} < \infty$, $\{z^n \sqrt{(n+1)/\pi}\}$ is an orthonormal basis, and $B(z, w) = \frac{1}{\pi(1-z\bar{w})^2}$.
- With H as above for $\Omega = \mathbb{R}_+^2$: every $f \in H$ has a representation $f(z) = \sqrt{4\pi} \int_0^\infty \hat{f}_0(\xi) e^{2\pi i \xi z} d\xi$, $z \in \mathbb{R}_+^2$, with $\int_0^\infty |\hat{f}_0(\xi)|^2 \frac{d\xi}{\xi} < \infty$, and $\hat{f}_0 \mapsto f$ is a unitary map from $L^2((0, \infty), d\xi/\xi)$ onto H .
- Let H be the Hilbert transform. Verify: (a) $H^* = -H$, $H^2 = -I$, and H is unitary; (b) H commutes with translations $\tau_h f(x) = f(x-h)$; (c) H commutes with dilations $\delta_a f(x) = f(ax)$, $a > 0$.
- Let $f \in L^2(\mathbb{R})$ and $u(x, y) = (f * P_y)(x)$ its Poisson integral; set $v(x, y) = (Hf * P_y)(x)$. Prove: (a) $F = u+iv$ is analytic in $\mathbb{R}_+^2$, so u, v are conjugate harmonic with $f = \lim_{y \rightarrow 0} u$, $Hf = \lim_{y \rightarrow 0} v$; (b) $F(z) = \frac{1}{\pi i} \int_{\mathbb{R}} \frac{f(t)}{t-z} dt$; (c) $v = f * Q_y$, $Q_y(x) = \frac{1}{\pi} \frac{x}{x^2+y^2}$ the conjugate Poisson kernel. [Use $\frac{1}{\pi z} = P_y(x) + iQ_y(x)$.]
- Show that $\left\{ \frac{1}{\pi^{1/2}(i+z)} \left(\frac{i-z}{i+z} \right)^n \right\}_{n \geq 0}$ is an orthonormal basis of $H^2(\mathbb{R}_+^2)$. (Recall the corresponding boundary system is an orthonormal basis of $L^2(\mathbb{R})$.) [It suffices to show that $F \in H^2$ with $\int_{\mathbb{R}} F(x) \frac{(x+i)^n}{(x-i)^{n+1}} dx = 0$ for all $n \geq 0$ forces $F = 0$; by the Cauchy integral formula $F^{(n)}(i) = 0$ for all n .]

Problems (Stein, Ch.5).

1. Let T be a Fourier multiplier operator on $L^2(\mathbb{R}^d)$, $\widehat{Tf}(\xi) = m(\xi)\hat{f}(\xi)$ with m bounded. Then T commutes with translations $\tau_h T = T\tau_h$. Conversely, any bounded operator on $L^2(\mathbb{R}^d)$ that commutes with translations is a Fourier multiplier operator. As a consequence, if T is bounded on $L^2(\mathbb{R})$ and commutes with translations and dilations, then: (a) if $(Tf)(-x) = T(f(-\cdot))(x)$ then $T = cI$; (b) if $(Tf)(-x) = -T(f(-\cdot))(x)$ then $T = cH$, the Hilbert transform (up to the constant c). [Reduce to showing a bounded $\hat{T}$ commuting with multiplication by $e^{2\pi i\xi \cdot h}$ is multiplication by some m .]

Additional exercises.

1. Verify that $F(z) = \frac{1}{(z+i)^2}$ belongs to $H^2(\mathbb{R}_+^2)$ and find its spectrum $\hat{F}_0$ using $\int_0^\infty \xi e^{2\pi i\xi z} d\xi = -\frac{1}{4\pi^2 z^2}$ for $\text{Im } z > 0$.
2. Show directly that $\{f \in L^2(\mathbb{R}) : \hat{f} = 0 \text{ a.e. on } (-\infty, 0)\}$ is a closed subspace of $L^2(\mathbb{R})$, and that the projection onto it is the multiplier $\chi_{(0, \infty)}$.
3. Confirm $\int_{\mathbb{R}} P_y(t) dt = 1$ and $\int_{\mathbb{R}} Q_y(t)^2 dt = \infty$ where $Q_y(x) = \frac{1}{\pi} \frac{x}{x^2 + y^2}$; explain why $Q_y \notin L^2$ but Q_y still defines the conjugate Poisson integral on L^2 data.
4. For real $f \in L^2(\mathbb{R})$, prove $\|Pf\|_2^2 = \frac{1}{2} \|f\|_2^2$ where P projects onto positive frequencies, using that $|\hat{f}|$ is even.
5. Show H (Hilbert transform) satisfies $\langle Hf, g \rangle = -\langle f, Hg \rangle$ for all $f, g \in L^2(\mathbb{R})$, i.e. $H^* = -H$, working on the Fourier side.
6. (*Non-example.*) Give a holomorphic F on $\mathbb{R}_+^2$ with each slice $F(\cdot + iy) \in L^2$ but $F \notin H^2$, and pinpoint which clause of Definition 28.2.1 fails.
7. Let $F \in H^2(\mathbb{R}_+^2)$ with boundary function F_0 . Show $\|F_0\|_{L^2(\mathbb{R})} = \|F\|_{H^2}$ and that F is recovered from F_0 by the Poisson integral.
8. (*Diagram.*) Draw the upper half-plane with two horizontal lines $\text{Im } z = y_1, \text{Im } z = y_2$ and the rectangular contour used in the proof of Theorem 28.2.2; annotate why the vertical sides contribute nothing in the limit.
9. Verify the multiplier of $H^2 = \frac{1}{2}(I + iH)$ composed with itself: show $P^2 = P$ and $P^* = P$ directly from $m_P = \chi_{(0, \infty)}$.

28.3 Constant-Coefficient Partial Differential Equations

The equation $L(u) = f$ and weak solutions

We solve linear PDEs $L(u) = f$ with constant coefficients,

$$L = \sum_{|\alpha| \leq n} a_\alpha \left(\frac{\partial}{\partial x} \right)^\alpha, \quad a_\alpha \in \mathbb{C}.$$

If both $u, f \in \mathcal{S}$, the Fourier identity $\widehat{(\partial^\alpha f)}(\xi) = (2\pi i\xi)^\alpha \hat{f}(\xi)$ turns the PDE into the *algebraic* equation $P(\xi)\hat{u}(\xi) = \hat{f}(\xi)$, where $P(\xi) = \sum_{|\alpha| \leq n} a_\alpha (2\pi i\xi)^\alpha$ is the *characteristic polynomial*; formally $\hat{u} = \hat{f}/P$. In general P has zeros and the transform is not directly applicable, so we widen the notion of solution.

$$\begin{array}{ccc} \mathcal{S}(\mathbb{R}^d) & \xrightarrow{L} & \mathcal{S}(\mathbb{R}^d) \\ \mathcal{F} \downarrow \sim & & \sim \downarrow \mathcal{F} \\ \mathcal{S}(\mathbb{R}^d) & \xrightarrow{\cdot P(\xi)} & \mathcal{S}(\mathbb{R}^d) \end{array}$$

Figure 28.9. On $\mathcal{S}$ the Fourier transform conjugates the differential operator L into multiplication by its characteristic polynomial $P(\xi)$: $\widehat{Lu} = P(\xi)\widehat{u}$, so the square commutes $(\mathcal{F}L = (\cdot P)\mathcal{F})$. Solving $Lu = f$ becomes the algebraic $\widehat{u} = \widehat{f}/P$ where $P \neq 0$.

Definition 28.3.1 (Adjoint and weak solution). For an open $\Omega \subseteq \mathbb{R}^d$ let $C_0^\infty(\Omega)$ be the smooth, compactly supported functions in Ω . The *formal adjoint* of L is

$$L^* = \sum_{|\alpha| \leq n} (-1)^{|\alpha|} a_\alpha \left(\frac{\partial}{\partial x} \right)^\alpha,$$

characterized by $(L\varphi, \psi) = (\varphi, L^*\psi)$ for all $\varphi, \psi \in C_0^\infty(\Omega)$, where $(\cdot, \cdot)$ is the $L^2(\Omega)$ inner product. Given $f \in L^2(\Omega)$, a function $u \in L^2(\Omega)$ is a *weak solution* of $Lu = f$ if

$$(f, \psi) = (u, L^*\psi) \quad \text{for all } \psi \in C_0^\infty(\Omega). \quad (\text{W})$$

An ordinary (C^n) solution is always a weak solution, by integration by parts.

The adjoint identity is integration by parts done $|\alpha|$ times: for $L = \partial/\partial x_j$ the boundary terms vanish because ψ has compact support, giving $\int (\partial_j \varphi) \psi = - \int \varphi \partial_j \psi$; iterate and use linearity.

Lemma 28.3.2 (Density of test functions). $C_0^\infty(\Omega)$ is dense in $L^2(\Omega)$.

Proof. Repeat Lemma 28.1.2, but truncate also away from the boundary: set $g_M = f$ where $|x| \leq M$, $d(x, \Omega^c) \geq 1/M$, $|f| \leq M$, and 0 else; then regularize $g_M * \varphi_\delta$ with $\delta < 1/2M$ so the support stays compact inside Ω . $\square$

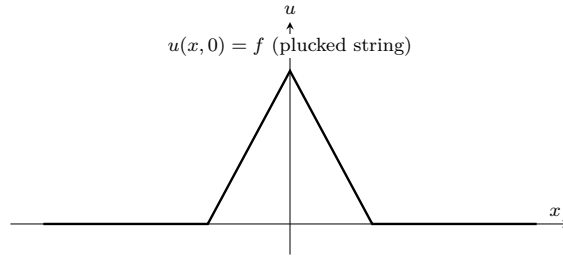


Figure 28.10. d'Alembert's $u(x, t) = \frac{1}{2}[f(x+t) + f(x-t)]$ for a piecewise-linear f is only a *weak* solution of $u_{xx} - u_{tt} = 0$ — it is not C^2 at the corner. (Schematic.)

Remark (Weak = absolutely continuous, in 1D). For $L = d/dx$ on $\Omega = (0, 1)$ with $u, f \in L^2$, $Lu = f$ weakly iff there is an absolutely continuous F on $[0, 1]$ with $F = u$ and $F' = f$ a.e. So the weak derivative is exactly the a.e. derivative of the absolutely continuous representative.

Theorem 28.3.3 (Solvability by a bounded operator). Let Ω be a bounded open subset of $\mathbb{R}^d$ and L a constant-coefficient operator. There is a bounded linear operator K on $L^2(\Omega)$ such that for every $f \in L^2(\Omega)$, $u = Kf$ is a weak solution of $Lu = f$.

The engine is a coercive estimate on the adjoint.

Lemma 28.3.4 (Key estimate). There is a constant c with $\|\psi\|_{L^2(\Omega)} \leq c \|L^*\psi\|_{L^2(\Omega)}$ for all $\psi \in C_0^\infty(\Omega)$.

This is the analytic substitute for “ L surjective $\iff L^*$ injective” in finite dimensions.

Proof of Theorem 28.3.3 from Lemma 28.3.4. On $H_0 = C_0^\infty(\Omega)$ put the inner product $\langle \varphi, \psi \rangle = (L^*\varphi, L^*\psi)$, norm $\|\psi\|_0 = \|L^*\psi\|_{L^2}$. By the key estimate, $\|\cdot\|_0$ dominates $\|\cdot\|_{L^2}$, so the completion H of H_0 embeds in $L^2(\Omega)$, and L^* extends to a bounded map $H \rightarrow L^2$. Fix f and define $\ell_0(\psi) = (\psi, f)$ on H_0 . Cauchy-Schwarz and the estimate give $|\ell_0(\psi)| \leq \|\psi\|_{L^2} \|f\|_{L^2} \leq c \|L^*\psi\|_{L^2} \|f\|_{L^2} = c \|f\|_{L^2} \|\psi\|_0$, so ℓ_0 is bounded on H_0 and extends to ℓ on H with $\|\ell\| \leq c \|f\|_{L^2}$. By Riesz representation there is $U \in H$ with $\ell(\psi) = \langle \psi, U \rangle = (L^*\psi, L^*U)$. Put $u = L^*U \in L^2$. Then $(\psi, f) = (L^*\psi, u)$, i.e. $(f, \psi) = (u, L^*\psi)$ for all $\psi \in C_0^\infty$, which is (W). Finally $\|u\|_{L^2} = \|U\|_0 = \|\ell\| \leq c \|f\|_{L^2}$, so $Kf = u$ is bounded. $\square$

Lemma 28.3.5 (Polynomial mean-value bound). *If $P(z) = z^m + \cdots + a_1 z + a_0$ is monic of degree m and F is entire, then*

$$|F(0)|^2 \leq \frac{1}{2\pi} \int_0^{2\pi} |P(e^{i\theta})F(e^{i\theta})|^2 d\theta.$$

Proof. For $P \equiv 1$ this is the mean-value inequality $|F(0)|^2 \leq \frac{1}{2\pi} \int_0^{2\pi} |F(e^{i\theta})|^2 d\theta$ (from the circle mean-value identity and Cauchy–Schwarz). In general factor $P = P_1 P_2$ over roots with $|\alpha| \geq 1$ and $|\beta| < 1$. Then $|P_1(0)| = \prod_{|\alpha| \geq 1} |\alpha| \geq 1$. For each root β with $|\beta| < 1$ write $(z - \beta) = -(1 - \bar{\beta}z)\psi_\beta(z)$ with the Blaschke factor $\psi_\beta(z) = \frac{\beta - z}{1 - \bar{\beta}z}$, holomorphic on a neighborhood of the closed disc and $|\psi_\beta(e^{i\theta})| = 1$. Set $\tilde{P} = P_1 \prod_{|\beta| < 1} (1 - \bar{\beta}z)$, so $|\tilde{P}(0)| \geq 1$ and $|\tilde{P}(e^{i\theta})| = |P(e^{i\theta})|$. Applying the $P \equiv 1$ case to $\tilde{P}F$: $|F(0)|^2 \leq |\tilde{P}(0)F(0)|^2 \leq \frac{1}{2\pi} \int_0^{2\pi} |\tilde{P}F|^2 = \frac{1}{2\pi} \int_0^{2\pi} |PF|^2$. $\square$

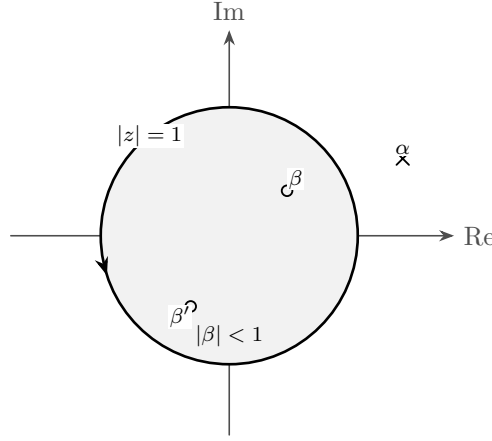


Figure 28.11. Blaschke factors move inside-roots ($|\beta| < 1$) without changing $|P|$ on the unit circle, while $|\tilde{P}(0)| \geq 1$: the heart of Lemma 28.3.5. (Schematic.)

Proof of Lemma 28.3.4 (one dimension). Normalize so $L^* = \sum_{0 \leq k \leq n} (-1)^k a_k (\partial/\partial x)^k$ with $a_n = (2\pi i)^{-n}$, and let $Q(\xi) = \sum (-1)^k a_k (2\pi i \xi)^k$ be the adjoint's characteristic polynomial (monic of degree n after the normalization), so $\widehat{L^* \psi} = Q(\xi) \hat{\psi}(\xi)$. If $f \in L^2$ is supported in $[-M, M]$ then $\hat{f}$ extends to an entire function of $\zeta = \xi + i\eta$, and Plancherel for fixed η gives $\int_{\mathbb{R}} |\hat{f}(\xi + i\eta)|^2 d\xi \leq e^{4\pi M|\eta|} \int_{\mathbb{R}} |f|^2$. Choose $\Omega \subset [-M, M]$. With $\eta = \sin \theta$ and a translation by $\cos \theta$,

$$\int_{\mathbb{R}} |Q(\xi + \cos \theta + i \sin \theta) \hat{\psi}(\xi + \cos \theta + i \sin \theta)|^2 d\xi \leq e^{4\pi M} \int_{\mathbb{R}} |L^* \psi|^2.$$

Apply Lemma 28.3.5 with $F(z) = \hat{\psi}(\xi + z)$ and $Q(\xi + z)$ in place of P :

$$|\hat{\psi}(\xi)|^2 \leq \frac{1}{2\pi} \int_0^{2\pi} |Q(\xi + e^{i\theta}) \hat{\psi}(\xi + e^{i\theta})|^2 d\theta.$$

Integrate in ξ , interchange the ξ, θ integrations, translate, and use the displayed bound: $\|\hat{\psi}\|_{L^2(\mathbb{R})}^2 \leq e^{4\pi M} \int_{\mathbb{R}} |L^* \psi|^2$, which by Plancherel is the claim. The d -dimensional case rotates coordinates so that $Q(\xi) = (2\pi i)^{-n} \xi_1^n + \sum_{j < n} \xi_1^j q_j(\xi')$ and repeats the one-variable argument in ξ_1 . $\square$

Method (Using a coercive adjoint estimate to solve $Lu = f$ weakly). *Recognition trigger.* You want existence (not uniqueness) of a weak solution to a linear equation $Lu = f$ on a bounded domain.

1. Prove a one-sided estimate $\|\psi\| \leq c \|L^* \psi\|$ on test functions — this makes $\|L^* \cdot\|$ a genuine norm.

2. Complete C_0^∞ in that norm to a Hilbert space $H \hookrightarrow L^2$.
3. The functional $\psi \mapsto (\psi, f)$ is bounded on H ; Riesz produces U , and $u = L^*U$ is the weak solution.
4. The same chain bounds $\|u\| \leq c\|f\|$, giving the solution operator K .

Worked Example 28.3.6 (The symbol of the Laplacian). Find the characteristic polynomial of $L = \Delta = \sum_{j=1}^d \partial^2/\partial x_j^2$ and the formal solution $\hat{u}$.

Recognition. Read off P from $\widehat{\partial^\alpha f} = (2\pi i\xi)^\alpha \hat{f}$.

Solution. Each $\partial^2/\partial x_j^2$ contributes $(2\pi i\xi_j)^2 = -4\pi^2\xi_j^2$, so $P(\xi) = -4\pi^2|\xi|^2$. The equation $\Delta u = f$ becomes $-4\pi^2|\xi|^2 \hat{u}(\xi) = \hat{f}(\xi)$, i.e. $\hat{u}(\xi) = -\hat{f}(\xi)/(4\pi^2|\xi|^2)$. *Check.* P vanishes only at $\xi = 0$, the single obstruction; away from the origin the symbol is invertible, reflecting that Δ is elliptic.

Worked Example 28.3.7 (The 1D weak derivative is genuine integration by parts). For $L = d/dx$ on $(0, 1)$ show the absolutely continuous $u(x) = x$ has weak derivative $f = 1$.

Recognition. Verify (W) directly: $(f, \psi) = (u, L^*\psi)$ with $L^* = -d/dx$.

Solution. $L^* = -d/dx$. For $\psi \in C_0^\infty(0, 1)$, $(u, L^*\psi) = \int_0^1 x(-\psi'(x)) dx = -[x\psi]_0^1 + \int_0^1 \psi dx = \int_0^1 \psi dx$, the boundary term vanishing as $\psi(0) = \psi(1) = 0$. And $(f, \psi) = \int_0^1 1 \cdot \psi dx$. Equal, so $u' = 1$ weakly. *Check.* $u(x) = x$ is absolutely continuous with classical derivative 1 a.e., matching the weak=absolutely-continuous principle.

Worked Example 28.3.8 (The mean-value inequality $|F(0)|^2 \leq \text{average}$). Prove the $P \equiv 1$ case of Lemma 28.3.5 for $F(z) = \sum_{k \geq 0} c_k z^k$.

Recognition. Recognition: expand and use orthogonality of $e^{ik\theta}$ on the circle.

Solution.

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} |F(e^{i\theta})|^2 d\theta &= \frac{1}{2\pi} \int_0^{2\pi} \sum_{j,k} c_j \bar{c}_k e^{i(j-k)\theta} d\theta \\ &= \sum_k |c_k|^2 \geq |c_0|^2 = |F(0)|^2, \end{aligned}$$

using $\frac{1}{2\pi} \int_0^{2\pi} e^{i(j-k)\theta} d\theta = \delta_{jk}$. *Check.* Equality holds iff $c_k = 0$ for $k \geq 1$, i.e. F constant — the harmonic mean-value property at its tightest.

Worked Example 28.3.9 (A Blaschke factor has modulus 1 on the circle). Confirm $|\psi_\beta(e^{i\theta})| = 1$ for $\psi_\beta(z) = \frac{\beta - z}{1 - \bar{\beta}z}$, $|\beta| < 1$.

Recognition. A direct modulus computation on $|z| = 1$.

Solution. With $z = e^{i\theta}$, $|1 - \bar{\beta}z|^2 = (1 - \bar{\beta}z)(1 - \beta\bar{z}) = 1 - \bar{\beta}z - \beta\bar{z} + |\beta|^2$. Also $|\beta - z|^2 = (\beta - z)(\bar{\beta} - \bar{z}) = |\beta|^2 - \beta\bar{z} - \bar{\beta}z + 1$, using $|z|^2 = 1$. The two are identical, so the ratio of moduli is 1. *Check.* The Blaschke factor is an automorphism of the disc fixing the boundary circle setwise; hence it must be unimodular there.

Worked Example 28.3.10 (Ellipticity from the symbol). Decide whether $L = \Delta$ and $L = \partial/\partial x_1$ (in $\mathbb{R}^d$, $d \geq 2$) are elliptic, where elliptic means $|P(\xi)| \geq c|\xi|^n$ for large $|\xi|$.

Recognition. Test the top-order part of the symbol for nonvanishing on $\xi \neq 0$.

Solution. For Δ , $P(\xi) = -4\pi^2|\xi|^2$ and $|P(\xi)| = 4\pi^2|\xi|^2$ never vanishes for $\xi \neq 0$: elliptic ($n = 2$). For $L = \partial/\partial x_1$, $P(\xi) = 2\pi i\xi_1$ and $|P| = 2\pi|\xi_1|$, which is 0 whenever $\xi_1 = 0$ but $\xi \neq 0$ (possible since $d \geq 2$): *not* elliptic. *Check.* The top symbol of $\partial/\partial x_1$ is ξ_1 , vanishing on the hyperplane $\xi_1 = 0$; ellipticity demands the principal symbol vanish only at the origin.

Worked Example 28.3.11 (Failure of the global estimate on $\mathbb{R}^d$ for $L = \partial/\partial x_1$). Show $\|u\|_{L^2(\mathbb{R}^d)} \leq c\|Lu\|_{L^2(\mathbb{R}^d)}$ fails for $L = \partial/\partial x_1$ on $\mathbb{R}^d$ ($d \geq 1$).

Recognition. Recognition: the global estimate holds iff $|P(\xi)| \geq c > 0$ everywhere (Exercise 12b); test P at its zeros.

Solution. Here $P(\xi) = 2\pi i \xi_1$, so $|P(\xi)| = 2\pi |\xi_1| \rightarrow 0$ as $\xi_1 \rightarrow 0$. Pick $\hat{u}$ concentrated near $\xi_1 = 0$: e.g. $\hat{u}_\varepsilon(\xi) = \varepsilon^{-1/2} \phi(\xi_1/\varepsilon) \rho(\xi')$ with ϕ, ρ fixed bumps. Then $\|u_\varepsilon\|_2$ stays ≈ 1 while $\|Lu_\varepsilon\|_2^2 = \int |2\pi \xi_1|^2 |\hat{u}_\varepsilon|^2 = O(\varepsilon^2) \rightarrow 0$. So the ratio $\|u\| / \|Lu\| \rightarrow \infty$: no uniform c . *Check.* This is the unbounded-domain counterpart of why $L = \partial/\partial x_1$ is not elliptic — its symbol vanishes on a whole hyperplane, killing coercivity.

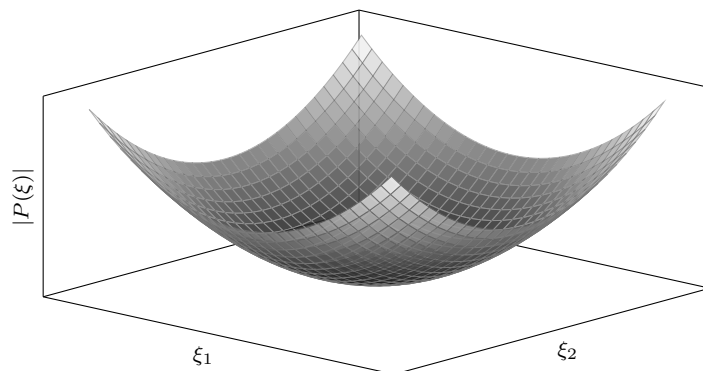


Figure 28.12. Symbol $|P(\xi)| = 4\pi^2 |\xi|^2$ of the Laplacian: a bowl touching 0 only at $\xi = 0$ — elliptic.

Exercises (Stein, Ch.5).

- Whether the inequality $\|u\|_{L^2(\Omega)} \leq c \|L(u)\|_{L^2(\Omega)}$ can hold for unbounded Ω : (a) for $d \geq 2$ and each constant-coefficient L , exhibit unbounded connected open Ω for which it holds for all $u \in C_0^\infty(\Omega)$. (b) Show $\|u\|_{L^2(\mathbb{R}^d)} \leq c \|L(u)\|_{L^2(\mathbb{R}^d)}$ for all $u \in C_0^\infty(\mathbb{R}^d)$ iff $|P(\xi)| \geq c > 0$ for all ξ , P the characteristic polynomial of L . [For (a) try $L = (\partial/\partial x_1)^n$ and a strip $\{-1 < x_1 < 1\}$.]
- Let L be a constant-coefficient operator. Show that when $d \geq 2$ the linear space of $C^\infty(\mathbb{R}^d)$ solutions of $L(u) = 0$ is not finite-dimensional. [Consider the zeros $\zeta \in \mathbb{C}^d$ of $P(\zeta)$.]
- Suppose F, G are integrable on a bounded interval $[a, b]$. Show G is the weak derivative of F iff F can be corrected on a null set to be absolutely continuous with $F'(x) = G(x)$ a.e. [If G is the weak derivative, use an approximation to obtain $\int_a^b G\varphi = -\int_a^b F\varphi'$ for the trapezoidal cutoff φ in Figure 9.]
- Consider the operator $L = \sum_{|\alpha| \leq n} a_\alpha (\partial/\partial x)^\alpha$ with $P(\xi) = \sum a_\alpha (2\pi i \xi)^\alpha$. Call L *elliptic* if $|P(\xi)| \geq c |\xi|^n$ for some $c > 0$ and all large $|\xi|$. (a) Check L is elliptic iff $\sum_{|\alpha|=n} a_\alpha (2\pi \xi)^\alpha$ vanishes only at $\xi = 0$. (b) If L is elliptic, prove $\|(\partial/\partial x)^\alpha \varphi\|_{L^2} \leq c (\|L\varphi\|_{L^2} + \|\varphi\|_{L^2})$ for all $\varphi \in C_0^\infty(\Omega)$, $|\alpha| \leq n$. (c) Conversely, (b) implies ellipticity.

Additional exercises.

- Compute the characteristic polynomial $P(\xi)$ and the formal solution $\hat{u} = \hat{f}/P$ for the heat-type operator $L = \partial/\partial t - \Delta_x$ on $\mathbb{R}^{1+d}$, and identify the zero set of P .
- Show $L^* = (-1)^n L$ when L is homogeneous of order n in $\partial/\partial x_1$ alone, and write L^* explicitly for $L = \partial^3/\partial x_1^3$.
- Verify the adjoint identity $(L\varphi, \psi) = (\varphi, L^*\psi)$ for $L = \partial^2/\partial x^2$ on $\mathbb{R}$ and $\varphi, \psi \in C_0^\infty(\mathbb{R})$, by integrating by parts twice.
- For $u(x) = |x|$ on $(-1, 1)$, find the weak derivative and show it equals $\text{sign}(x)$, an absolutely continuous antiderivative argument.
- Show that the Cantor function (devil's staircase) $c(x)$ on $[0, 1]$ has weak derivative 0 in the sense of distributions restricted to L^2 *fails* to recover c — i.e. c is *not* the absolutely continuous integral of its a.e. derivative 0 — and reconcile this with the weak-derivative characterization (the obstruction is the singular part). [State precisely why c is continuous, nondecreasing, with $c' = 0$ a.e. yet $c(1) - c(0) = 1$.]

6. Prove Δ is elliptic and the operator $\partial^2/\partial x_1^2 - \partial^2/\partial x_2^2$ (the wave operator) is not, by examining each principal symbol.
7. (*Non-example.*) Exhibit a constant-coefficient L on $\mathbb{R}^2$ for which $\|u\|_{L^2} \leq c\|Lu\|_{L^2}$ fails on all of $\mathbb{R}^2$, and state the symbol reason.
8. (*Diagram.*) Plot the modulus $|P(\xi)|$ of the symbol of Δ in $\mathbb{R}^2$ as a surface and mark its unique zero; contrast with a sketch of the symbol of $\partial/\partial x_1$ vanishing along a line.

28.4 The Dirichlet Principle

The Dirichlet problem and the Dirichlet integral

For a bounded open $\Omega \subset \mathbb{R}^2$ and continuous f on $\partial\Omega$, the *Dirichlet problem* seeks u with

$$\Delta u = 0 \text{ in } \Omega, \quad u = f \text{ on } \partial\Omega,$$

$u \in C^2(\Omega) \cap C(\bar{\Omega})$ (steady-state temperature of a plate). Beyond special Ω (disc, half-plane, strip) there is no explicit formula. The classical strategy minimizes an *energy*, the *Dirichlet integral*

$$D(U) = \int_{\Omega} |\nabla U|^2 = \int_{\Omega} \left(\left| \frac{\partial U}{\partial x_1} \right|^2 + \left| \frac{\partial U}{\partial x_2} \right|^2 \right) dx_1 dx_2,$$

over all competitors with the prescribed boundary values.

Proposition 28.4.1 (The naive principle). *If $u \in C^2(\bar{\Omega})$ minimizes D among all $U \in C^2(\bar{\Omega})$ with $U|_{\partial\Omega} = f$, then u is harmonic in Ω .*

Proof. With $\langle F, G \rangle = \int_{\Omega} (\partial_1 F \partial_1 G + \partial_2 F \partial_2 G)$ we have $D(u) = \langle u, u \rangle$. For any $v \in C^2$ with $v|_{\partial\Omega} = 0$ and all ε , $D(u + \varepsilon v) \geq D(u)$ since $u + \varepsilon v$ shares the boundary data. Expanding, $D(u + \varepsilon v) = D(u) + \varepsilon^2 D(v) + \varepsilon \langle u, v \rangle + \varepsilon \langle v, u \rangle$, so $\varepsilon^2 D(v) + 2\varepsilon \operatorname{Re} \langle u, v \rangle \geq 0$ for all real ε , forcing $\operatorname{Re} \langle u, v \rangle = 0$; the perturbation $u + i\varepsilon v$ gives $\operatorname{Im} \langle u, v \rangle = 0$. Thus $\langle u, v \rangle = 0$, and integration by parts yields $0 = \langle u, v \rangle = -\int_{\Omega} (\Delta u) v$ for all such v , hence $\Delta u = 0$. $\square$

Remark (Weierstrass's and Hadamard's objections). Two gaps undermined the naive argument. *Weierstrass*: a minimizer need not exist. Minimizing $D(\varphi) = \int_{-1}^1 |x\varphi'(x)|^2 dx$ over C^1 with $\varphi(\pm 1) = \pm 1$ has infimum 0 (take φ_{ε} that switches near 0 over a window of width ε , giving $D(\varphi_{\varepsilon}) \leq 2\varepsilon M^2 \rightarrow 0$), yet $D = 0$ forces $\varphi' \equiv 0$, i.e. φ constant, which cannot meet the boundary data — no winner. *Hadamard*: $D(u)$ can be infinite even when a harmonic solution exists, so the implied competition may have no qualified competitors at all.

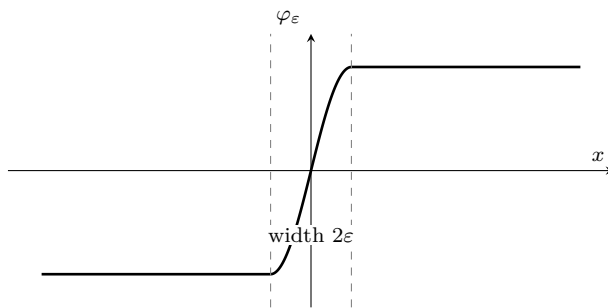


Figure 28.13. Weierstrass's near-minimizers φ_{ε} : energy $D(\varphi_{\varepsilon}) \leq 2\varepsilon M^2 \rightarrow 0$, but no C^1 function attains $D = 0$ with $\varphi(\pm 1) = \pm 1$. (Schematic.)

The repair: the competing functions form a *pre-Hilbert space* under $\langle \cdot, \cdot \rangle$; the solution lives in its completion, accessible only through L^2 theory.

Harmonic functions: three equivalent descriptions

Throughout, $\Omega \subseteq \mathbb{R}^d$ open. A C^2 function u is *harmonic* if $\Delta u = 0$. It is *weakly harmonic* if $(u, \Delta \psi) = 0$ for all $\psi \in C_0^\infty(\Omega)$, and it has the *mean-value property* if

$$u(x_0) = \frac{1}{m(B)} \int_B u(x) \, dx$$

for every ball B with center x_0 and $\overline{B} \subset \Omega$.

Theorem 28.4.2 (Mean-value characterization). *u is harmonic in Ω iff u is continuous and satisfies the mean-value property.*

Theorem 28.4.3 (Weyl's lemma). *Every weakly harmonic u agrees a.e. with a (unique) harmonic function $\tilde{u}$.*

Corollary 28.4.4 (Maximum principle). *If Ω is bounded, $u \in C(\overline{\Omega})$ is harmonic in Ω , then $\max_{\overline{\Omega}} |u| = \max_{\partial\Omega} |u|$.*

Proof. Both maxima are attained ($\overline{\Omega}, \partial\Omega$ compact). Suppose the max is attained at an interior x_0 . If $|u(x')| < |u(x_0)|$ at some $x' \in B$, the same strict inequality holds on a neighborhood, and since $|u| \leq |u(x_0)|$ on B , the average $\frac{1}{m(B)} \int_B |u| < |u(x_0)|$, contradicting the mean-value property. So $|u| \equiv |u(x_0)|$ on every ball $B_r(x_0) \subset \Omega$; letting r grow to the boundary and using continuity, the max is also attained on $\partial\Omega$. $\square$

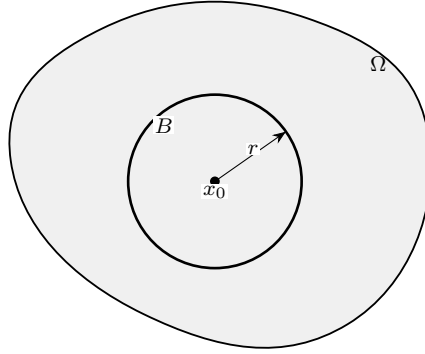


Figure 28.14. Mean-value property: $u(x_0)$ equals the average of u over any ball $B \subset \Omega$ centered at x_0 ; this forces the maximum to the boundary. (Schematic.)

Lemma 28.4.5 (Boundary-term-free Green identity). *For $u, v, \eta \in C^2$ near $\overline{B}$ with η compactly supported in B ,*

$$\int_B (v \Delta u - u \Delta v) \eta \, dx = \int_B u (\nabla v \cdot \nabla \eta) - v (\nabla u \cdot \nabla \eta) \, dx.$$

Proof. Integrating $\int_B \frac{\partial^2 u}{\partial x_j^2} v \eta = - \int_B \frac{\partial u}{\partial x_j} \frac{\partial v}{\partial x_j} \eta - \int_B \frac{\partial u}{\partial x_j} v \frac{\partial \eta}{\partial x_j}$ and summing in j gives $\int_B (\Delta u) v \eta = - \int (\nabla u \cdot \nabla v) \eta - \int (\nabla u \cdot \nabla \eta) v$. Subtract the same with u, v swapped; the symmetric $(\nabla u \cdot \nabla v) \eta$ terms cancel. $\square$

Proof sketch of Theorem 28.4.2, harmonic $\Rightarrow$ MVP. Apply Lemma 28.4.5 with u harmonic and the test functions $v = 1$, $v = |x|^2$, and $v = |x|^{2-d}$ ($d \geq 3$, or $\log |x|$ for $d = 2$), together with cutoffs $\eta_\epsilon^+, \eta_\epsilon^-$ supported in shells near $|x| = 1$ and near the origin. With $v = 1$ ($\Delta v = 0$) one gets $\int_B \nabla u \cdot \nabla \eta_\epsilon^+ = 0$; with $v = |x|^2$ ($\Delta v = 2d$) one extracts $d \int_B u = - \lim_\epsilon \int_B u (x \cdot \nabla \eta_\epsilon^+)$; with the fundamental solution v (harmonic off 0) the shell terms reproduce $c \int_B u$ while the origin terms vanish by the $O(|x|)$ bound near 0. Combining, $\int_B u \, dx = 0$ when $u(0) = 0$; applying to $u - u(0)$ gives the mean-value property for the unit ball, and scaling/translating gives it for all balls. $\square$

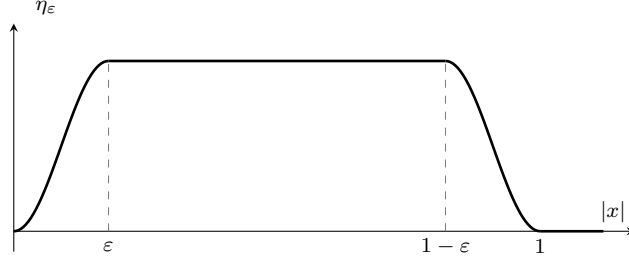


Figure 28.15. The radial cutoff η_ε : it is 1 on the shell $\varepsilon \leq |x| \leq 1 - \varepsilon$, with $|\nabla \eta_\varepsilon| = O(1/\varepsilon)$ near $|x| = 0$ and $|x| = 1$.

Corollary 28.4.6. (i) Every harmonic function is C^∞ . (ii) A locally uniform limit of harmonic functions is harmonic.

Proof. (i) The mean-value property gives $u = u * \varphi_r$ for a smooth radial mollifier, hence u is smooth. (ii) Each u_n satisfies the mean-value property; uniform convergence passes it to u , so u is harmonic by Theorem 28.4.2. $\square$

Dirichlet's principle, rigorously

Equip $C^1(\overline{\Omega})$ with $\langle u, v \rangle = \int_\Omega \nabla u \cdot \nabla v$, norm $\|u\|^2 = \langle u, u \rangle$. Since $\|u\| = 0$ iff u is constant on components, pass to equivalence classes modulo such constants: a pre-Hilbert space H_0 with completion H .

Lemma 28.4.7 (Poincaré inequality). *If $v \in C^1(\overline{\Omega})$ vanishes on $\partial\Omega$, then $\int_\Omega |v|^2 \leq c_\Omega \int_\Omega |\nabla v|^2$ with $c_\Omega \leq d(\Omega)^2$, $d(\Omega) = \text{diam } \Omega$.*

Proof. For $f \in C^1(I)$, $I = (a, b)$, vanishing at an endpoint a : $f(s) = \int_a^s f'$, so $|f(s)|^2 \leq |I| \int_I |f'|^2$ by Cauchy–Schwarz, and integrating in s gives $\int_I |f|^2 \leq |I|^2 \int_I |f'|^2$. Slice Ω by lines $x_1 \mapsto (x_1, x')$: each slice $J(x')$ is a disjoint union of intervals I_j on whose endpoints v vanishes. Apply the 1D bound with $|I_j| \leq d(\Omega)$ and sum, then integrate in x' . $\square$

Let $S_0 \subset C^1(\overline{\Omega})$ be the functions vanishing on $\partial\Omega$, S its closure in H , and P_S the orthogonal projection of H onto S .

Theorem 28.4.8 (Dirichlet's principle, validated). *Suppose $f = F|_{\partial\Omega}$ for some $F \in C^1(\overline{\Omega})$. The minimizing sequence $u_n = F - v_n$ ($v_n \in S_0$, $\|u_n\| \rightarrow \text{dist}(F, S)$) converges in H and in $L^2(\Omega)$ to $u = F - P_S(F)$, which is weakly harmonic; corrected on a null set it is harmonic.*

Proof. Minimizing $\|F - v\|$ over $v \in S_0$ is, since $S = \overline{S_0}$, minimizing distance to the closed subspace S ; the projection theorem gives $v_n \rightarrow P_S F$ in H , hence $u_n = F - v_n \rightarrow u = F - P_S F$. By Lemma 28.4.7 applied to $v_n - v_m$, $\{v_n\}$ is Cauchy in L^2 , so convergence also holds in L^2 . For $\psi \in C_0^\infty(\Omega)$, $\psi \in S$, so $\langle u, \psi \rangle = 0$; integration by parts gives $\langle u_n, \psi \rangle = -(u_n, \Delta \psi) \rightarrow -(u, \Delta \psi)$, whence $(u, \Delta \psi) = 0$: u is weakly harmonic, so by Theorem 28.4.3 it has a harmonic representative. $\square$

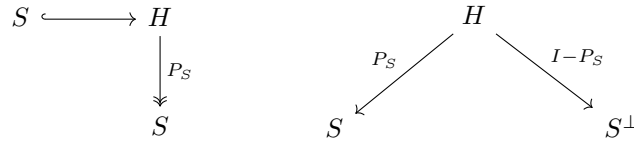


Figure 28.16. Left: P_S is the orthogonal projection of H onto the closed subspace $S = \overline{S_0}$, restricting to the identity on S (so $P_S^2 = P_S$). Right: the harmonic solution is the complementary component $u = (I - P_S)F \in S^\perp$ of the boundary datum F , and $F = P_S F + u$.

Two issues remain: that u attains the boundary values continuously, and that general continuous f (not a priori the restriction of a C^1 function) is allowed. The second is settled by approximation.

Lemma 28.4.9 (Tietze/regularization). *A continuous f on a compact $\Gamma \subset \mathbb{R}^d$ is the uniform limit on Γ of a sequence $\{F_n\}$ of smooth functions on $\mathbb{R}^d$.*

Proof. First extend f continuously to $\mathbb{R}^d$ (Tietze): using $g(x) = \frac{d(x, K_0)}{d(x, K_0) + d(x, K_1)}$ to separate disjoint compacts, build $G = \sum_n G_n$ with $0 \leq f - \sum_{k \leq N} G_k \leq (2/3)^N$ on Γ and $0 \leq G_N \leq \frac{1}{3}(2/3)^{N-1}$, so G is continuous with $G|_\Gamma = f$. Then regularize: $F_\varepsilon = G * \varphi_\varepsilon$ is smooth and $|F_\varepsilon - G| \leq \sup_{|x-y| \leq \varepsilon} |G(x) - G(y)| \rightarrow 0$ uniformly near Γ ; take $\varepsilon = 1/n$. $\square$

Theorem 28.4.10 (Two-dimensional existence under the outside-triangle condition). *Let $\Omega \subset \mathbb{R}^2$ be open, bounded, satisfying the outside-triangle condition: there is a fixed isosceles triangle shape so that for every $x \in \partial\Omega$ a congruent triangle with vertex x has its interior outside Ω . Then for every continuous f on $\partial\Omega$ the Dirichlet problem is uniquely solvable.*

Uniqueness is the maximum principle (two solutions differ by a harmonic function vanishing on $\partial\Omega$). Existence rests on a refined Poincaré estimate.

Proposition 28.4.11 (Local Poincaré near the boundary). *Under the outside-triangle condition there are $c_1 < 1 < c_2$ and C (depending only on $\text{diam } \Omega$ and the triangle) so that if $z \in \Omega$ with $d(z, \partial\Omega) = \delta$, then for $v \in C^1(\overline{\Omega})$ with $v|_{\partial\Omega} = 0$,*

$$\int_{B_{c_1\delta}(z)} |v|^2 \, dx \leq C\delta^2 \int_{B_{c_2\delta}(z) \cap \Omega} |\nabla v|^2 \, dx.$$

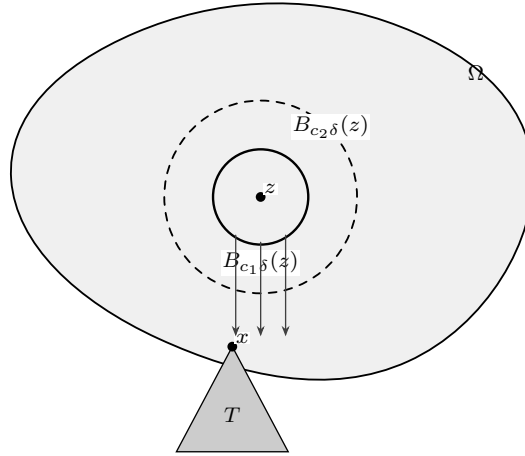


Figure 28.17. The outside-triangle condition: at each boundary point x a congruent triangle T pokes outside Ω , so vertical chords through $B_{c_1\delta}(z)$ (heavy disc) run down to $\partial\Omega$. (Schematic.)

How the proposition yields the theorem. We may take $f = F|_{\partial\Omega}$ with $F \in C^1$. The minimizers $u_n = F - v_n$ converge to $u = F - v$ harmonic, and (32) holds for $v = F - u$. Fix $y \in \partial\Omega$ and let $z \rightarrow y$ in Ω , $\delta = d(z, \partial\Omega) \leq |z - y| \rightarrow 0$. Comparing disc averages $\text{Av}(F)(z)$ and $\text{Av}(u)(z)$ over $B_{c_1\delta}(z)$, Cauchy–Schwarz and the proposition give $|\text{Av}(F)(z) - \text{Av}(u)(z)|^2 \leq C' \int_{B_{c_2\delta}(z) \cap \Omega} |\nabla(F - u)|^2 \rightarrow 0$ (absolute continuity of the integral, $m(B_{c_2\delta}) \rightarrow 0$). The mean-value property gives $\text{Av}(u)(z) = u(z)$, and continuity of F gives $\text{Av}(F)(z) \rightarrow f(y)$. Hence $u(z) \rightarrow f(y)$, so u takes the boundary values continuously, completing existence. $\square$

Remark. Polygonal, Lipschitz, or C^1 boundaries satisfy the outside-triangle condition; the punctured disc does *not*, and indeed its Dirichlet problem is not solvable in general (Exercise 19). The condition is sufficient but not necessary.

Worked Example 28.4.12 (Harmonicity of $\log|x|$ in the plane). Show $v(x) = \log|x|$ is harmonic on $\mathbb{R}^2 \setminus \{0\}$, and $|x|^{2-d}$ is harmonic on $\mathbb{R}^d \setminus \{0\}$ for $d \geq 3$.

Recognition. Differentiate the radial power/log directly using $\partial|x|/\partial x_j = x_j/|x|$.

Solution. For $a \neq 0$, $\frac{\partial|x|^a}{\partial x_j} = ax_j|x|^{a-2}$ and $\frac{\partial^2|x|^a}{\partial x_j^2} = a|x|^{a-2} + a(a-2)x_j^2|x|^{a-4}$. Summing over j (with $\sum x_j^2 = |x|^2$): $\Delta|x|^a = [da + a(a-2)]|x|^{a-2} = a(a+d-2)|x|^{a-2}$. This is 0 iff $a = 0$ or $a = 2 - d$. For $d \geq 3$ take $a = 2 - d$. For $d = 2$, $a = 0$ is the trivial constant; the genuine singular harmonic is $\log|x|$, and a direct computation gives $\Delta \log|x| = 0$ for $x \neq 0$. *Check.* At $d = 2$, $a(a+d-2) = a \cdot a = a^2$ vanishes only at $a = 0$, consistent with the logarithm (not a power) being the fundamental solution.

Worked Example 28.4.13 (Maximum principle bounds a solution). Suppose u is harmonic in the unit disc, continuous on its closure, with $u(e^{i\theta}) = \cos \theta$ on the boundary. Bound $|u|$ inside and identify u .

Recognition. Recognition: the maximum principle bounds $|u|$ by its boundary max; harmonic extension of $\cos \theta$ is explicit.

Solution. By Corollary 28.4.4, $\max_{\overline{\mathbb{D}}} |u| = \max_{\partial \mathbb{D}} |\cos \theta| = 1$, so $|u| \leq 1$ throughout. The harmonic function matching $\cos \theta = \operatorname{Re}(e^{i\theta})$ on the boundary is $u(x_1, x_2) = x_1 = r \cos \theta$ ($\operatorname{Re} z$ is harmonic). *Check.* $\Delta x_1 = 0$ and $u(e^{i\theta}) = \cos \theta$; on the closed disc $|x_1| \leq 1$, attaining 1 at $z = 1$ on the boundary as the maximum principle predicts.

Worked Example 28.4.14 (Poincaré constant on an interval). Verify Lemma 28.4.7 for $\Omega = (0, L) \subset \mathbb{R}$ and $v(x) = \sin(\pi x/L)$ (which vanishes at both endpoints).

Recognition. Plug a vanishing-endpoint function into the 1D Poincaré inequality and compare constants.

Solution. $\int_0^L \sin^2(\pi x/L) dx = \frac{L}{2}$, while $\int_0^L |v'|^2 = \frac{\pi^2}{L^2} \int_0^L \cos^2(\pi x/L) dx = \frac{\pi^2}{L^2} \cdot \frac{L}{2} = \frac{\pi^2}{2L}$. The ratio is $\int v^2 / \int |v'|^2 = (L/2)/(\pi^2/(2L)) = L^2/\pi^2 \leq L^2 = d(\Omega)^2$. *Check.* The bound $c_\Omega \leq d(\Omega)^2 = L^2$ holds with room to spare; the sharp constant L^2/π^2 is the first Dirichlet eigenvalue's reciprocal.

Worked Example 28.4.15 (Hadamard's divergent Dirichlet integral). For $f_\alpha(\theta) = \sum_{n \geq 0} 2^{-n\alpha} e^{i2^n \theta}$ on the circle, the harmonic extension is $u(r, \theta) = \sum_{n \geq 0} r^{2^n} 2^{-n\alpha} e^{i2^n \theta}$. Show $D(u) = \infty$ for $\alpha \leq 1/2$.

Recognition. Recognition: compute $D(u)$ in polar coordinates via Parseval on each Fourier mode.

Solution. In polar form $|\nabla u|^2 = |u_r|^2 + r^{-2} |u_\theta|^2$. Parseval on the θ -integral leaves

$$D_\rho = \int_0^\rho \sum_n 2^{2n+1} 2^{-2n\alpha} r^{2^{n+1}-1} dr = \sum_n \rho^{2^{n+1}} 2^n 2^{-2n\alpha}$$

over the disc of radius ρ . As $\rho \rightarrow 1$ the n -th term tends to $2^{n(1-2\alpha)}$, whose sum diverges when $1 - 2\alpha \geq 0$, i.e. $\alpha \leq 1/2$. *Check.* For $\alpha \leq 1/2$ the energy is infinite even though the Poisson integral u solves the problem — exactly Hadamard's point that competitors may not qualify.

Worked Example 28.4.16 (A NON-solvable Dirichlet problem: the punctured disc). Explain why $\Delta u = 0$ on $D^* = \{0 < |z| < 1\}$ with $u = 0$ on $|z| = 1$ and $u = 1$ at the puncture has no solution, contrasting the outside-triangle condition.

Recognition. Recognition: a removable-singularity/maximum-principle obstruction; the domain fails the geometric hypothesis.

Solution. If u were harmonic on D^* , continuous up to $\{0\}$ with $u(0) = 1$, then u is bounded near 0, hence (Exercise 19a) extends harmonically across 0 to the full disc. But then u is harmonic on $\mathbb{D}$, $= 0$ on $\partial \mathbb{D}$, so by the maximum principle $u \equiv 0$, contradicting $u(0) = 1$. So no solution exists. *Check.* The isolated boundary point 0 admits no exterior triangle (the complement is just a point), so Theorem 28.4.10 does not apply — and indeed solvability fails.

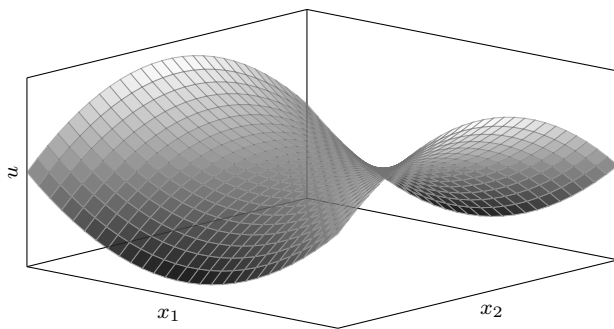


Figure 28.18. The harmonic saddle $u = x_1^2 - x_2^2 = \operatorname{Re}(z^2)$: $\Delta u = 0$, so it equals its average on every disc; no interior strict max or min.

Worked Example 28.4.17 (Mean-value property on a concrete harmonic function). Verify the mean-value property for $u(x_1, x_2) = x_1 x_2$ on a disc of radius r centered at (a, b) .

Recognition. Recognition: $u = \text{Im}(z^2)/2$ is harmonic; check $u(\text{center})$ equals the disc average directly.

Solution. $\Delta(x_1 x_2) = 0$, so u is harmonic. The average over $B_r(a, b)$ is $\frac{1}{\pi r^2} \int_{B_r} (a + s_1)(b + s_2) ds$ where $s = (s_1, s_2)$ ranges over the disc of radius r at the origin. Expanding, $\int_{B_r} ab = ab \pi r^2$, $\int_{B_r} a s_2 = 0$ and $\int_{B_r} b s_1 = 0$ (odd), and $\int_{B_r} s_1 s_2 = 0$ (odd in each). So the average is $\frac{1}{\pi r^2} \cdot ab \pi r^2 = ab = u(a, b)$. *Check.* The center value ab equals the average for every r , as Theorem 28.4.2 guarantees for a harmonic function; the cross-term integral vanishing is the geometric reason.

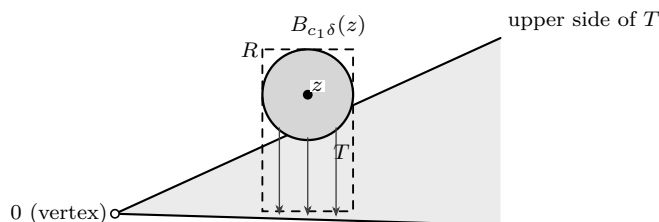


Figure 28.19. Placing the rectangle R (Proposition 28.4.11): vertical chords through $B_{c_1 \delta}(z)$ continue down to meet the lower side of T , hence $\partial\Omega$. (Schematic.)

Exercises (Stein, Ch.5).

- Suppose u is harmonic in the punctured unit disc $D^* = \{0 < |z| < 1\}$. (a) Show that if u is also continuous at the origin then u is harmonic throughout the unit disc. [Show u is weakly harmonic.] (b) Prove the Dirichlet problem for the punctured unit disc is in general not solvable.
- Let F be continuous on the closed unit disc $\mathbb{D}$, with $F \in C^1$ on the open disc and $\int_{\mathbb{D}} |\nabla F|^2 < \infty$. Let $f(e^{i\theta})$ be the restriction of F to the circle and write $f(e^{i\theta}) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}$. Prove $\sum_{n=-\infty}^{\infty} |n| |a_n|^2 < \infty$. [Write $F(re^{i\theta}) \sim \sum_n F_n(r) e^{in\theta}$, $F_n(1) = a_n$; express $\int_{\mathbb{D}} |\nabla F|^2$ in polar coordinates and use $\frac{1}{2} |F(1)|^2 \leq L^{-1} \int_{1/2}^1 |F'(r)|^2 dr + L \int_{1/2}^1 |F(r)|^2 dr$ for $L \geq 2$, applied to $F = F_n$, $L = |n|$.]

Problems (Stein, Ch.5).

- Let $\Omega \subset \mathbb{R}^2$ be open and bounded. A *boundary Lipschitz arc* γ is a piece of $\partial\Omega$ that after a rotation is $\gamma = \{(x_1, \eta(x_1)) : a \leq x_1 \leq b\}$ with $|\eta(x_1) - \eta(x'_1)| \leq M |x_1 - x'_1|$, and such that the thin region just below γ misses Ω . If $\partial\Omega$ is covered by finitely many discs D_j with each $\partial\Omega \cap D_j$ a boundary Lipschitz arc, then Ω satisfies the outside-triangle condition of Theorem 28.4.10, hence the Dirichlet problem is solvable.
- (*) If the bounded domain Ω has boundary a closed simple continuous curve, the Dirichlet problem is solvable: there is a conformal map Φ of $\mathbb{D}$ onto Ω extending to a continuous bijection $\mathbb{D} \rightarrow \overline{\Omega}$.
- Consider the two plane domains with a cusp: Domain I has a smooth boundary except for an inside cusp; Domain II is similar but with an outside cusp. Both fall within Problem 3, so the Dirichlet problem is solvable in each. However, II satisfies the outside-triangle condition while I does not.

Additional exercises.

- Solve the Dirichlet problem on the unit disc with boundary data $f(\theta) = \cos 2\theta$ explicitly, and verify the maximum principle bound $|u| \leq 1$.
- Show that $\text{Re}(z^n)$ and $\text{Im}(z^n)$ are harmonic on $\mathbb{R}^2$ for every $n \geq 1$ by direct computation of the Laplacian in x_1, x_2 .
- Prove uniqueness for the Dirichlet problem on a bounded Ω : if u_1, u_2 are both solutions with the same boundary data, then $u_1 = u_2$. (Use the maximum principle.)
- Verify the Poincaré inequality $\int_0^1 v^2 \leq \int_0^1 |v'|^2$ for $v(x) = x(1-x)$ on $(0, 1)$ (which vanishes at both endpoints) and compare to the diameter bound $c_\Omega \leq 1$.

5. Compute the Dirichlet integral $D(u)$ of the harmonic $u(r, \theta) = r^n \cos n\theta$ over the disc of radius $\rho < 1$ and show it is finite, contrasting Hadamard's divergent example.
6. Show that a polygonal domain satisfies the outside-triangle condition, while the slit disc $\mathbb{D} \setminus [0, 1)$ requires care; state which boundary points are problematic.
7. (*Non-example.*) Explain why the Dirichlet problem on the punctured disc with $u = 0$ on the outer circle and $u = 1$ at the center has no solution, citing the removable-singularity step.
8. (*Diagram.*) Draw a domain Ω with a boundary point x , the special triangle T poking outside Ω at x , and the discs $B_{c_1\delta}(z) \subset B_{c_2\delta}(z)$ used in the local Poincare estimate (Proposition 28.4.11).
9. For the Cantor set $C \subset [0, 1]$, compute its Lebesgue measure (zero) and its Hausdorff dimension $\log 2 / \log 3$, and explain why such a set can be the boundary-trouble locus for capacity-type refinements of the Dirichlet problem.

Chapter 29

Abstract Measure and Integration Theory

29.1 Abstract Measure Spaces and Caratheodory's Theorem

Measure spaces

The whole of Lebesgue's theory on $\mathbb{R}^d$ rests on two ingredients: a supply of “measurable” sets stable under the countable set operations, and a way to assign a size to each that adds over disjoint countable unions. Abstracting these two ingredients from $\mathbb{R}^d$ gives the general notion of a measure space, in which every theorem of the Lebesgue theory (convergence theorems, Fubini, differentiation) finds its natural home.

Definition 29.1.1. A σ -algebra of subsets of a set X is a non-empty collection $\mathcal{M}$ of subsets of X that is closed under complements and countable unions (hence, by De Morgan, under countable intersections). A *measure* on $\mathcal{M}$ is a function $\mu : \mathcal{M} \rightarrow [0, \infty]$ that is *countably additive*: for any countable family $\{E_n\}$ of pairwise disjoint sets in $\mathcal{M}$,

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n).$$

The triple $(X, \mathcal{M}, \mu)$ is a *measure space*; its members of $\mathcal{M}$ are the *measurable sets*. The space is σ -finite if $X = \bigcup_n X_n$ with each $X_n \in \mathcal{M}$ of finite measure.

Countable additivity is the load-bearing axiom: it is exactly strong enough to commute μ with increasing/decreasing limits of sets, yet weak enough that the natural constructions (Lebesgue measure, products, Stieltjes measures) actually satisfy it. A few consequences are immediate and used constantly.

Proposition 29.1.2 (Elementary properties). *Let $(X, \mathcal{M}, \mu)$ be a measure space. Then $\mu(\emptyset) = 0$; μ is monotone ($E \subseteq F \implies \mu(E) \leq \mu(F)$); μ is countably subadditive ($\mu(\bigcup_n E_n) \leq \sum_n \mu(E_n)$ for any $E_n \in \mathcal{M}$); and μ is continuous from below ($E_n \uparrow E \implies \mu(E_n) \rightarrow \mu(E)$) and from above ($E_n \downarrow E$ with $\mu(E_1) < \infty \implies \mu(E_n) \rightarrow \mu(E)$).*

Proof. Pick E with $\mu(E) < \infty$ (if none exists every set has infinite measure and the claims are trivial); additivity on $E = E \cup \emptyset \cup \emptyset \cup \dots$ forces $\mu(\emptyset) = 0$. Monotonicity: $F = E \cup (F \setminus E)$ disjointly, so $\mu(F) = \mu(E) + \mu(F \setminus E) \geq \mu(E)$. Subadditivity: disjointify by $F_n = E_n \setminus \bigcup_{k < n} E_k$, then $\mu(\bigcup E_n) = \sum \mu(F_n) \leq \sum \mu(E_n)$. Continuity from below: write $E = \bigcup_n (E_n \setminus E_{n-1})$ disjointly ($E_0 = \emptyset$); the partial sums of the additive series are $\mu(E_n)$. From above: apply from-below continuity to $E_1 \setminus E_n \uparrow E_1 \setminus E$ and subtract, using $\mu(E_1) < \infty$. $\square$

Example 29.1.3. *Discrete measures.* Let $X = \{x_n\}$ be countable, $\mathcal{M}$ all subsets, and $\mu(\{x_n\}) = \mu_n$ for a given sequence $\mu_n \in [0, \infty]$; then $\mu(E) = \sum_{x_n \in E} \mu_n$. When every $\mu_n = 1$ this is *counting measure* $\#$, and integration reduces to summation of (absolutely) convergent series. *Weighted Lebesgue.* On $X = \mathbb{R}^d$ with $\mathcal{M}$ the Lebesgue sets and a fixed non-negative measurable f , $\mu(E) = \int_E f \, dx$ is a measure; $f = 1$ recovers Lebesgue measure m .

Exterior measures

To *build* a measure space one rarely starts with a σ -algebra; one starts with a crude outer notion defined on *all* subsets and then carves out the good sets.

Definition 29.1.4. An *exterior* (or *outer*) *measure* on X is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ with: (i) $\mu^*(\emptyset) = 0$; (ii) monotone, $E_1 \subseteq E_2 \implies \mu^*(E_1) \leq \mu^*(E_2)$; (iii) countably subadditive, $\mu^*(\bigcup_j E_j) \leq \sum_j \mu^*(E_j)$ for any countable family.

The exterior Lebesgue measure m^* on $\mathbb{R}^d$ – infimum of total volumes over countable cube covers – is the model example. Such “covering” outer measures are the generic source (premeasures, below); the Hausdorff measures of the next chapter are another.

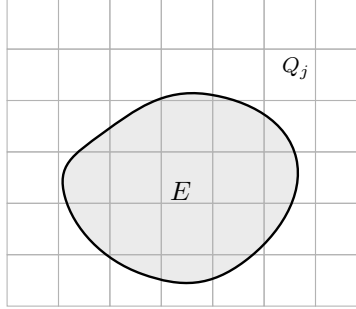


Figure 29.1. $\mu^*(E) = \inf \sum_j |Q_j|$ over countable cube covers $E \subseteq \bigcup_j Q_j$: the prototype of an exterior measure. (Schematic.)

Caratheodory’s measurability criterion

How should one select the measurable sets from an exterior measure, with no topology to invoke? Caratheodory’s answer: E is measurable iff it splits every test set additively.

Definition 29.1.5. A set $E \subseteq X$ is *Caratheodory measurable* if

$$\mu^*(A) = \mu^*(E \cap A) + \mu^*(E^c \cap A) \quad \text{for every } A \subseteq X. \quad (29.1)$$

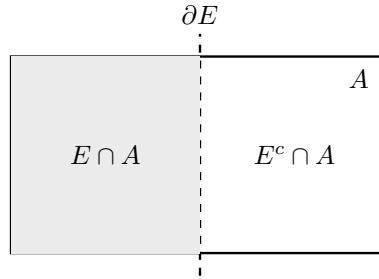


Figure 29.2. The separation test (29.1): a measurable E cuts every A into two pieces whose exterior measures add back to $\mu^*(A)$. (Schematic.)

Since subadditivity already gives $\mu^*(A) \leq \mu^*(E \cap A) + \mu^*(E^c \cap A)$, to verify (29.1) one only needs the reverse inequality “ $\geq$ ”. Note also that any E with $\mu^*(E) = 0$ is automatically measurable.

Theorem 29.1.6 (Caratheodory). *Given an exterior measure μ^* on X , the collection $\mathcal{M}$ of Caratheodory measurable sets is a σ -algebra, and μ^* restricted to $\mathcal{M}$ is a measure. The resulting space is complete: every subset of a measurable null set is measurable.*

Proof. Clearly $\emptyset, X \in \mathcal{M}$, and the symmetry of (29.1) in E, E^c shows $\mathcal{M}$ is closed under complements. *Finite unions.* If $E_1, E_2 \in \mathcal{M}$ and A is arbitrary, applying the criterion for E_2 then E_1 and using subadditivity,

$$\mu^*(A) = \mu^*(E_2 \cap A) + \mu^*(E_2^c \cap A) \geq \mu^*((E_1 \cup E_2) \cap A) + \mu^*((E_1 \cup E_2)^c \cap A),$$

since $E_1 \cup E_2 = (E_1 \cap E_2) \cup (E_1^c \cap E_2) \cup (E_1 \cap E_2^c)$; hence $E_1 \cup E_2 \in \mathcal{M}$. For disjoint E_1, E_2 , taking $A = E_1 \cup E_2$ in the criterion for E_1 gives $\mu^*(E_1 \cup E_2) = \mu^*(E_1) + \mu^*(E_2)$ – finite additivity. *Countable unions.* Let $E_1, E_2, \dots \in \mathcal{M}$ be disjoint, $G_n = \bigcup_{j \leq n} E_j$, $G = \bigcup_j E_j$. Induction on the criterion for E_n gives, for every A ,

$$\mu^*(G_n \cap A) = \sum_{j=1}^n \mu^*(E_j \cap A).$$

Since $G_n \in \mathcal{M}$ and $G^c \subseteq G_n^c$,

$$\mu^*(A) = \mu^*(G_n \cap A) + \mu^*(G_n^c \cap A) \geq \sum_{j=1}^n \mu^*(E_j \cap A) + \mu^*(G^c \cap A).$$

Let $n \rightarrow \infty$ and use subadditivity:

$$\mu^*(A) \geq \sum_{j=1}^{\infty} \mu^*(E_j \cap A) + \mu^*(G^c \cap A) \geq \mu^*(G \cap A) + \mu^*(G^c \cap A) \geq \mu^*(A).$$

All inequalities are equalities: $G \in \mathcal{M}$, and taking $A = G$ yields countable additivity. A general (not-disjoint) countable union is the disjoint union of $E_n \setminus \bigcup_{k < n} E_k \in \mathcal{M}$, so $\mathcal{M}$ is a σ -algebra. Completeness was noted above: null sets, and hence their subsets, satisfy (29.1). $\square$

Method (Proving a set E is Caratheodory measurable). *Recognition trigger.* You must show $E \in \mathcal{M}$ for an exterior measure μ^* .

1. Fix an arbitrary test set A (you may assume $\mu^*(A) < \infty$, else the inequality is trivial).
2. Prove only the hard direction $\mu^*(A) \geq \mu^*(E \cap A) + \mu^*(E^c \cap A)$; subadditivity gives “ $\leq$ ” free.
3. Typical tactic: cover A nearly optimally, or exploit a separation/distance property (metric exterior measures, Theorem 29.1.10).

Worked Example 29.1.7 (Null sets are measurable, and the criterion is symmetric). Show that if $\mu^*(E) = 0$ then $E \in \mathcal{M}$, and that $E \in \mathcal{M} \Leftrightarrow E^c \in \mathcal{M}$.

Recognition. Trigger: verify Definition 29.1.5 directly; the inequality to prove is the “ $\geq$ ” half.

Solution. For any A , monotonicity gives $\mu^*(E \cap A) \leq \mu^*(E) = 0$ and $\mu^*(E^c \cap A) \leq \mu^*(A)$, so $\mu^*(E \cap A) + \mu^*(E^c \cap A) \leq \mu^*(A)$; the reverse is subadditivity, giving (29.1). Symmetry is immediate because (29.1) is unchanged under $E \leftrightarrow E^c$ (the two terms swap). *Check.* In $\mathbb{R}^d$ a single point has $m^* = 0$, hence is Lebesgue measurable – consistent with the classical theory.

Worked Example 29.1.8 (Finite additivity from the criterion). With μ^* an exterior measure and disjoint $E_1, E_2 \in \mathcal{M}$, recover $\mu^*(E_1 \cup E_2) = \mu^*(E_1) + \mu^*(E_2)$ from (29.1) alone.

Recognition. Trigger: “additivity on disjoint measurable sets.” Feed the union to the criterion for E_1 as the test set.

Solution. Take $A = E_1 \cup E_2$ in the criterion for E_1 . Then $E_1 \cap A = E_1$ and (by disjointness) $E_1^c \cap A = E_2$, so $\mu^*(E_1 \cup E_2) = \mu^*(E_1) + \mu^*(E_2)$. *Check.* For Lebesgue measure with $E_1 = [0, 1]$, $E_2 = [2, 3]$ this reads $m([0, 1] \cup [2, 3]) = 1 + 1 = 2$, correct.

Metric exterior measures and Borel sets

When X carries a metric, one wants the *Borel* sets – the σ -algebra $\mathcal{B}_X$ generated by the open sets – to be measurable. The clean sufficient condition is additivity across positive gaps.

Definition 29.1.9. On a metric space (X, d) , with $d(A, B) = \inf\{d(x, y) : x \in A, y \in B\}$, an exterior measure μ^* is a *metric exterior measure* if

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B) \quad \text{whenever } d(A, B) > 0.$$

Theorem 29.1.10. *If μ^* is a metric exterior measure on a metric space X , then every Borel set is Caratheodory measurable; hence μ^* restricted to $\mathcal{B}_X$ is a measure.*

Proof. It suffices to show closed sets are measurable. Let F be closed and A a test set with $\mu^*(A) < \infty$. Put $A_n = \{x \in F^c \cap A : d(x, F) \geq 1/n\}$. Then $A_n \uparrow$, and since F is closed $F^c \cap A = \bigcup_n A_n$. As $d(F \cap A, A_n) \geq 1/n > 0$ and μ^* is metric,

$$\mu^*(A) \geq \mu^*((F \cap A) \cup A_n) = \mu^*(F \cap A) + \mu^*(A_n). \quad (29.2)$$

Claim: $\mu^*(A_n) \rightarrow \mu^*(F^c \cap A)$. Put $B_n = A_{n+1} \setminus A_n$; one checks $d(B_{n+1}, A_n) \geq \frac{1}{n(n+1)} > 0$ (if $x \in B_{n+1}$ and $d(x, y) < \frac{1}{n(n+1)}$ then $d(y, F) < 1/n$, so $y \notin A_n$). The metric property then gives $\mu^*(A_{2k+1}) \geq \sum_{j \leq k} \mu^*(B_{2j})$ and similarly for odd indices; since $\mu^*(A) < \infty$ both series $\sum \mu^*(B_j)$ converge. As $\mu^*(A_n) \leq \mu^*(F^c \cap A) \leq \mu^*(A_n) + \sum_{j>n} \mu^*(B_j)$, the claim follows. Letting $n \rightarrow \infty$ in (29.2) gives $\mu^*(A) \geq \mu^*(F \cap A) + \mu^*(F^c \cap A)$, so F is measurable. $\square$

A measure on $\mathcal{B}_X$ is a *Borel measure*. When it is finite on balls it is regular:

Proposition 29.1.11 (Regularity). *Suppose a Borel measure μ on a metric space X is finite on all balls of finite radius. Then for every Borel E and every $\varepsilon > 0$ there are an open $O \supseteq E$ with $\mu(O \setminus E) < \varepsilon$ and a closed $F \subseteq E$ with $\mu(E \setminus F) < \varepsilon$.*

Proof sketch. Let $\mathcal{C}$ be the collection of sets satisfying both approximation properties. By complementation $\mathcal{C}$ is closed under complements; a $\sigma/2^k$ -budget argument shows it is closed under countable unions (the open over-sets union to an open over-set within $\sum \varepsilon/2^k$; the closed under-sets union to an F_σ which is trimmed to a genuine closed set using finiteness of μ on balls). Every open set lies in $\mathcal{C}$: $O = \bigcup_k \{x \in B_k : d(x, O^c) \geq 1/k\}$ is an increasing union of closed sets. Thus $\mathcal{C}$ is a σ -algebra containing the open sets, hence all Borel sets. $\square$

Worked Example 29.1.12 (The discrete metric makes every set Borel and the criterion trivial). On a set X with the discrete metric ($d(x, y) = 1$ for $x \neq y$), describe the Borel sets and check that any exterior measure built from point masses is metric.

Recognition. Trigger: identify $\mathcal{B}_X$ from the topology, then test Definition 29.1.9.

Solution. In the discrete metric every singleton $\{x\} = B_{1/2}(x)$ is open, so every subset is open; thus $\mathcal{B}_X = \mathcal{P}(X)$, all subsets are Borel. Any two disjoint non-empty sets have $d(A, B) = 1 > 0$, so $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$ reduces to finite additivity, which point-mass measures satisfy. *Check.* Consistent with Theorem 29.1.10: all Borel = all subsets are measurable, matching the discrete Example 29.1.3.

Premeasures and the extension theorem

The practical recipe: start from a measure μ_0 on a mere *algebra* of simple sets (rectangles, half-open intervals), generate an exterior measure by covering, and extend.

Definition 29.1.13. An *algebra* $\mathcal{A}$ on X is a non-empty collection closed under complements and *finite* unions and intersections. A *premeasure* on $\mathcal{A}$ is $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ with $\mu_0(\emptyset) = 0$ that is countably additive whenever the countable disjoint union of $\mathcal{A}$ -sets happens to land back in $\mathcal{A}$: $\mu_0(\bigcup_k E_k) = \sum_k \mu_0(E_k)$.

$$\begin{array}{ccc} \text{premeasure } \mu_0 \text{ on } \mathcal{A} & \xrightarrow{\text{cover}} & \text{exterior } \mu^* \xrightarrow{\text{Caratheodory}} \text{measure on } \mathcal{M} \supseteq \mathcal{B} \\ & & \uparrow \\ & & \text{restrict to } \langle \mathcal{A} \rangle = \mathcal{M}_0 \end{array}$$

Figure 29.3. The extension pipeline: a primitive premeasure flows to a complete measure; uniqueness holds on $\langle \mathcal{A} \rangle$ under σ -finiteness.

Lemma 29.1.14. *If μ_0 is a premeasure on $\mathcal{A}$, then $\mu^*(E) = \inf\{\sum_j \mu_0(E_j) : E \subseteq \bigcup_j E_j, E_j \in \mathcal{A}\}$ is an exterior measure with $\mu^* = \mu_0$ on $\mathcal{A}$, and every set of $\mathcal{A}$ is Caratheodory measurable.*

Proof. That μ^* is an exterior measure is routine. For $E \in \mathcal{A}$: $\mu^*(E) \leq \mu_0(E)$ since E covers itself; conversely, given a cover $E \subseteq \bigcup E_j$, the sets $E'_k = E \cap (E_k \setminus \bigcup_{j < k} E_j) \in \mathcal{A}$ are disjoint with union E and $E'_k \subseteq E_k$, so by countable additivity of the premeasure $\mu_0(E) = \sum_k \mu_0(E'_k) \leq \sum_k \mu_0(E_k)$; infimizing gives $\mu_0(E) \leq \mu^*(E)$. Measurability of $E \in \mathcal{A}$: for any A and $\varepsilon > 0$ choose $A \subseteq \bigcup_j E_j$ with $\sum_j \mu_0(E_j) \leq \mu^*(A) + \varepsilon$; finite additivity of μ_0 on $\mathcal{A}$ splits each term and gives $\sum_j \mu_0(E_j) \geq \mu^*(E \cap A) + \mu^*(E^c \cap A)$. Let $\varepsilon \rightarrow 0$. $\square$

Theorem 29.1.15 (Extension theorem). *Let $\mathcal{A}$ be an algebra, μ_0 a premeasure on $\mathcal{A}$, and $\mathcal{M}_0 = \langle \mathcal{A} \rangle$ the generated σ -algebra. Then μ_0 extends to a measure μ on $\mathcal{M}_0$. If μ is σ -finite, the extension is unique.*

Proof. The exterior measure μ^* of Lemma 29.1.14 gives, via Theorem 29.1.6, a measure on the Caratheodory σ -algebra, which contains $\mathcal{A}$ and hence $\mathcal{M}_0$. *Uniqueness* (assuming σ -finiteness): let ν be another measure agreeing with μ_0 on $\mathcal{A}$. For $F \in \mathcal{M}_0$ with $\mu(F) < \infty$ and a cover $F \subseteq \bigcup E_j$ ($E_j \in \mathcal{A}$), $\nu(F) \leq \sum \nu(E_j) = \sum \mu_0(E_j)$, so $\nu(F) \leq \mu(F)$. Putting $E = \bigcup E_j$, continuity from below gives $\nu(E) = \mu(E)$; choosing the cover with $\mu(E) \leq \mu(F) + \varepsilon$ yields $\mu(F) \leq \mu(E) = \nu(E) = \nu(F) + \nu(E \setminus F) \leq \nu(F) + \varepsilon$, so $\mu(F) \leq \nu(F)$ and the two agree on finite-measure sets. Writing $X = \bigsqcup E_j$ with $\mu(E_j) < \infty$, $\mu(F) = \sum \mu(F \cap E_j) = \sum \nu(F \cap E_j) = \nu(F)$. $\square$

Proposition 29.1.16. *For any E and $\varepsilon > 0$ there are $E_1 \in \mathcal{A}_\sigma$ and $E_2 \in \mathcal{A}_{\sigma\delta}$ with $E \subseteq E_1$, $E \subseteq E_2$, $\mu^*(E_1) \leq \mu^*(E) + \varepsilon$ and $\mu^*(E_2) = \mu^*(E)$. Here $\mathcal{A}_\sigma$ are countable unions from $\mathcal{A}$ and $\mathcal{A}_{\sigma\delta}$ countable intersections of those.*

This regularity-from-the-algebra (proof left to the reader, as in Stein) is the key tool for the Fubini slicing argument of §29.3.

Worked Example 29.1.17 (Counting measure as an extension; non-uniqueness without σ -finiteness). On $X = \mathbb{Q}$ let $\mathcal{A}$ be the algebra of finite-or-cofinite sets, with $\mu_0(E) = 0$ if E finite, $\mu_0(E) = \infty$ if E cofinite. Show extensions to $\mathcal{B}$ are not unique.

Recognition. Trigger: test the necessity of σ -finiteness in Theorem 29.1.15.

Solution. Counting measure $\#$ and $2 \cdot \#$ (and counting measure itself versus the measure μ giving every infinite set ∞ and matching μ_0 on $\mathcal{A}$) all restrict to μ_0 on $\mathcal{A}$, since on $\mathcal{A}$ the only values are 0 and ∞ . But $\#(\{q\}) = 1$ while another extension could assign 2 to a singleton; both agree on $\mathcal{A}$. Uniqueness fails because $X = \mathbb{Q}$ is *not* σ -finite for any of these (no countable cover by finite- μ_0 sets exhausts $\mathbb{Q}$ with finite measure under the ∞ -valued one). *Check.* Lebesgue measure on $\mathbb{R}$, by contrast, *is* σ -finite ($\mathbb{R} = \bigcup_n [-n, n]$), so its extension from the algebra of finite unions of intervals is unique – the classical fact.

Worked Example 29.1.18 (Continuity from above needs finite measure). On $(\mathbb{R}, m)$ let $E_n = [n, \infty)$. Show $E_n \downarrow \emptyset$ but $m(E_n) \not\rightarrow m(\emptyset)$, locating the failed hypothesis.

Recognition. Trigger: a decreasing family – test Proposition 29.1.2's continuity from above and its finiteness caveat.

Solution. $\bigcap_n [n, \infty) = \emptyset$ since no real exceeds every n , so $E_n \downarrow \emptyset$. But $m(E_n) = \infty$ for all n , so $m(E_n) \rightarrow \infty \neq 0 = m(\emptyset)$. The hypothesis $\mu(E_1) < \infty$ of continuity from above is violated ($m(E_1) = \infty$). *Check.* Truncating to $E_n \cap [0, 10] = [n, 10]$ (finite measure E_1) restores it: $m([n, 10]) = 10 - n \rightarrow 0$ as $n \rightarrow 10$, matching $m(\emptyset)$.

Worked Example 29.1.19 (A non-measurable set: the criterion can fail). Sketch why a Vitali set $V \subseteq [0, 1]$ fails the Caratheodory criterion for m^* , so $\mathcal{M} \neq \mathcal{P}(\mathbb{R})$.

Recognition. Trigger: “exhibit non-measurability”; show (29.1) breaks for a suitable test set.

Solution. Build V by choosing one representative from each coset of $\mathbb{Q}$ in $\mathbb{R}$, inside $[0, 1]$. Its rational translates $V + q$ ($q \in \mathbb{Q} \cap [-1, 1]$) are disjoint and their union U satisfies $[0, 1] \subseteq U \subseteq [-1, 2]$, so $1 \leq m^*(U) \leq 3$. If V were measurable, countable additivity over the translates (each $m^*(V + q) = m^*(V)$ by translation invariance) forces $m^*(U) = \sum_q m^*(V)$, which is 0 if $m^*(V) = 0$ and ∞ if $m^*(V) > 0$ – never in $[1, 3]$. Contradiction, so V fails (29.1). *Check.* This needs the axiom of choice; consistent with the fact that m^* is genuinely only subadditive on all of $\mathcal{P}(\mathbb{R})$, additive just on $\mathcal{M}$.

Caution. The Caratheodory σ -algebra $\mathcal{M}$ can be strictly larger than the generated algebra $\mathcal{M}_0 = \langle \mathcal{A} \rangle$ (e.g. Lebesgue $\supsetneq$ Borel): μ^* produces a *complete* measure, but the extension theorem only promises a measure on $\mathcal{M}_0$, which need not be complete. Uniqueness is asserted on $\mathcal{M}_0$ and requires σ -finiteness; on the bigger $\mathcal{M}$ there is generally nothing to compare against.

Exercises (Stein, Ch.6).

1. Let X be a set and $\mathcal{M}$ a non-empty collection of subsets of X . Prove that if $\mathcal{M}$ is closed under complements and countable unions of *disjoint* sets, then $\mathcal{M}$ is a σ -algebra. [Hint: any countable union can be written as a countable union of disjoint sets.]
2. (Completion of a measure space.) Given $(X, \mathcal{M}, \mu)$, let $\overline{\mathcal{M}}$ be the sets of the form $E \cup Z$ with $E \in \mathcal{M}$ and $Z \subseteq F$ for some $F \in \mathcal{M}$ with $\mu(F) = 0$; set $\overline{\mu}(E \cup Z) = \mu(E)$. Show: (a) $\overline{\mathcal{M}}$ is the smallest σ -algebra containing $\mathcal{M}$ and all subsets of $\mathcal{M}$ -null sets; (b) $\overline{\mu}$ is a complete measure on $\overline{\mathcal{M}}$. [Hint: it suffices that $E_1 \in \overline{\mathcal{M}} \implies E_1^c \in \overline{\mathcal{M}}$; write $E_1 = E \cup Z$, $Z \subseteq F$, then $E_1^c = (E \cup F)^c \cup (F \setminus Z)$.]
3. Consider the exterior Lebesgue measure m^* on $\mathbb{R}^d$ of Chapter 1. Prove that $E \subseteq \mathbb{R}^d$ is Caratheodory measurable if and only if E is Lebesgue measurable in the sense of Chapter 1. [Hint: if E is Lebesgue measurable and A arbitrary, choose a G_δ set $G \supseteq A$ with $m^*(A) = m(G)$; conversely if E is Caratheodory measurable with $m^*(E) < \infty$, choose a G_δ set $G \supseteq E$ with $m^*(E) = m^*(G)$, so $G \setminus E$ has exterior measure 0.]
4. (Translation invariance characterizes Lebesgue measure.) If μ is a translation-invariant Borel measure on $\mathbb{R}^d$, finite on compact sets, then $\mu = cm$. Prove it via: (a) if Q_a is a translate of the half-open cube of side a and $\mu(Q_1) = c$, then $\mu(Q_{1/n}) = cn^{-d}$; (b) hence $\mu \ll m$ with a locally integrable density f , $\mu(E) = \int_E f dx$; (c) by the differentiation theorem $f(x) = c$ a.e., so $\mu = cm$. [Hint: Q_1 is a disjoint union of n^d translates of $Q_{1/n}$.]

Additional exercises.

1. Let μ^* be an exterior measure on X and E a set with $\mu^*(E) + \mu^*(E^c) = \mu^*(X) < \infty$. Show E is Caratheodory measurable. [This is the “one test set” shortcut when X has finite outer measure.]
2. Show that a finite intersection and a finite union of Caratheodory measurable sets are measurable directly from the criterion, without invoking Theorem 29.1.6.
3. (Continuity of measures.) On $(X, \mathcal{M}, \mu)$ let $E_n \in \mathcal{M}$. Prove $\mu(\liminf_n E_n) \leq \liminf_n \mu(E_n)$, and if $\mu(\bigcup_n E_n) < \infty$ then $\limsup_n \mu(E_n) \leq \mu(\limsup_n E_n)$ (measure-theoretic Fatou for sets).
4. Let $\#$ be counting measure on $X = [0, 1]$ (all subsets measurable). Show $\#$ is not σ -finite, and that the only $\#$ -null set is $\emptyset$.
5. (Metric outer measure, non-example.) Give an exterior measure on $\mathbb{R}$ that is *not* metric, and exhibit two positively separated sets A, B with $\mu^*(A \cup B) < \mu^*(A) + \mu^*(B)$. [Hint: try $\mu^*(E) = 1$ if $E \neq \emptyset$, $\mu^*(\emptyset) = 0$.]
6. Prove that if $\{E_n\}$ are Caratheodory measurable and pairwise disjoint, then for every test set A , $\mu^*(A \cap \bigcup_n E_n) = \sum_n \mu^*(A \cap E_n)$ (countable additivity holds even “through” an arbitrary A).
7. Let $\mathcal{A}$ be the algebra of finite unions of half-open intervals $(a, b]$ in $\mathbb{R}$ and $\mu_0((a, b]) = b - a$. Verify μ_0 is a premeasure (the only nontrivial point is countable additivity within $\mathcal{A}$), so the extension theorem reproduces Lebesgue measure.
8. (Diagram.) Draw a set $E \subseteq \mathbb{R}^2$ and a finite cover by squares achieving $\sum |Q_j| \leq m^*(E) + \varepsilon$; mark which squares are wasted (overlap or stick out) and explain how refining the grid reduces the waste toward $m^*(E)$.

29.2 Integration on a Measure Space

Once a measure space $(X, \mathcal{M}, \mu)$ is in hand, the construction of the integral copies the Euclidean case almost word for word. We assume throughout that $(X, \mathcal{M}, \mu)$ is σ -finite, and record the statements and the few points where the abstract setting needs care, with worked computations.

Measurable functions and simple functions

Definition 29.2.1. An extended-real-valued f on X is *measurable* if $\{x : f(x) < a\} = f^{-1}([-\infty, a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$. A *simple function* is $\varphi = \sum_{k=1}^N a_k \mathbf{1}_{E_k}$ with $E_k \in \mathcal{M}$ of finite measure and $a_k \in \mathbb{R}$. Almost-everywhere (a.e.) is relative to μ : $f = g$ a.e. means $\mu(\{f \neq g\}) = 0$.

Measurability is exactly the requirement that the preimage of every Borel set land in $\mathcal{M}$: writing $\mathcal{B}_{\mathbb{R}}$ for the Borel σ -algebra of the (extended) line, f is measurable iff f^{-1} pulls $\mathcal{B}_{\mathbb{R}}$ back into $\mathcal{M}$.

$$\begin{array}{ccc} f^{-1}(B) & \hookrightarrow & X \\ f \downarrow & & \downarrow f \\ B & \hookrightarrow & \overline{\mathbb{R}} \end{array}$$

Measurability is preserved by the algebraic operations and by pointwise limits, exactly as on $\mathbb{R}^d$. A simple function is a finite “staircase” of indicator blocks; these approximate everything.

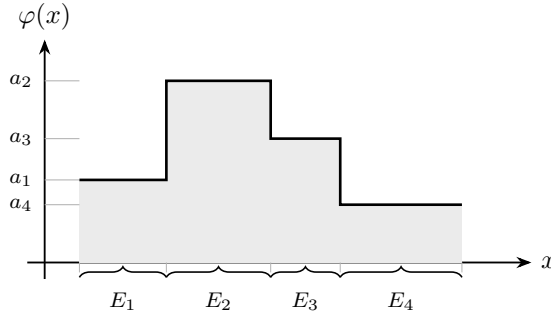


Figure 29.4. A simple function $\varphi = \sum a_k \mathbf{1}_{E_k}$ as a finite sum of indicator blocks over disjoint measurable sets E_k . (Schematic.)

Theorem 29.2.2 (Approximation and Egorov). (i) If $f \geq 0$ is measurable there exist simple φ_k with $0 \leq \varphi_k \leq \varphi_{k+1} \uparrow f$ pointwise; for general measurable f one can take $|\varphi_k| \leq |\varphi_{k+1}| \rightarrow f$. (ii) (Egorov) If $f_k \rightarrow f$ a.e. on a set E with $\mu(E) < \infty$, then for each $\varepsilon > 0$ there is $A_\varepsilon \subseteq E$ with $\mu(E \setminus A_\varepsilon) \leq \varepsilon$ and $f_k \rightarrow f$ uniformly on A_ε .

The proof of (i) is the dyadic-truncation construction of the Euclidean case, with the σ -finite exhaustion $X = \bigcup F_k$ playing the role of the cubes Q_k ; (ii) is verbatim. We state them and turn to the integral.

The integral and the convergence theorems

Definition 29.2.3. The integral $\int_X f \, d\mu$ is built in four steps: on simple $\varphi \geq 0$, $\int \varphi \, d\mu = \sum_k a_k \mu(E_k)$; on $f \geq 0$, $\int f \, d\mu = \sup \{ \int \varphi \, d\mu : 0 \leq \varphi \leq f, \varphi \text{ simple} \}$; on real f , $\int f = \int f^+ - \int f^-$ when not $\infty - \infty$; f is *integrable* if $\int_X |f| \, d\mu < \infty$. Linearity and monotonicity hold as on $\mathbb{R}^d$.

Theorem 29.2.4 (The three limit theorems). Let f_n be measurable on X .

- (i) Fatou: if $f_n \geq 0$ then $\int \liminf_n f_n \, d\mu \leq \liminf_n \int f_n \, d\mu$.
- (ii) Monotone convergence (MCT): if $0 \leq f_n \uparrow f$ then $\int f_n \, d\mu \rightarrow \int f \, d\mu$.
- (iii) Dominated convergence (DCT): if $f_n \rightarrow f$ a.e. and $|f_n| \leq g$ with g integrable, then $\int |f_n - f| \, d\mu \rightarrow 0$, hence $\int f_n \, d\mu \rightarrow \int f \, d\mu$.

Proof (skeleton). MCT: monotonicity gives $\lim \int f_n \leq \int f$; for the reverse, fix simple $0 \leq \varphi \leq f$ and $c < 1$, let $E_n = \{f_n \geq c\varphi\} \uparrow X$, then $\int f_n \geq c \int_{E_n} \varphi \rightarrow c \int \varphi$ by continuity from below of $E \mapsto \int_E \varphi$; sup over φ and let $c \rightarrow 1$. Fatou follows by applying MCT to $g_n = \inf_{k \geq n} f_k \uparrow \liminf f_n$. DCT: apply Fatou to $g \pm f_n \geq 0$. $\square$

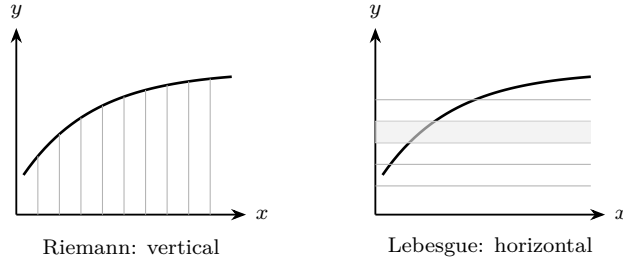


Figure 29.5. Riemann sums slice the *domain*; the Lebesgue integral slices the *range*, measuring level sets $\{f > y\}$ – the source of its robustness under limits. (Schematic.)

Worked Example 29.2.5 (MCT computes a series-swap: $\int \sum = \sum \int$). On $([0, 1], m)$ let $f(x) = \sum_{n=1}^{\infty} x^n$. Compute $\int_0^1 f \, dx$ by exchanging sum and integral, and justify the exchange.

Recognition. Trigger: a sum of non-negative terms – MCT (or Tonelli) licenses term-by-term integration.

Solution. The partial sums $S_N = \sum_{n=1}^N x^n$ increase to f with $S_N \geq 0$, so MCT gives $\int_0^1 f = \sum_{n=1}^{\infty} \int_0^1 x^n \, dx = \sum_{n=1}^{\infty} \frac{1}{n+1}$, which diverges. Indeed $f(x) = x/(1-x)$ on $[0, 1)$ and $\int_0^1 \frac{x}{1-x} \, dx = +\infty$. *Check.* Near $x = 1$, $f \sim \frac{1}{1-x}$ is non-integrable, matching the divergent harmonic-type series $\sum \frac{1}{n+1}$.

Worked Example 29.2.6 (Fatou can be strict: escaping bumps). On $(\mathbb{R}, m)$ let $f_n = \mathbf{1}_{[n, n+1]}$. Compare $\int \liminf f_n$ with $\liminf \int f_n$.

Recognition. Trigger: a moving-bump sequence; test sharpness of Fatou and failure of MCT/DCT.

Solution. Pointwise $f_n(x) \rightarrow 0$ for every x (each x is covered by at most one bump), so $\liminf f_n = 0$ and $\int \liminf f_n = 0$. But $\int f_n = 1$ for all n , so $\liminf \int f_n = 1$. Thus $0 \leq 1$ – Fatou holds with *strict* inequality. *Check.* No integrable dominating g exists ($\sup_n f_n = \mathbf{1}_{[1, \infty)}$ is non-integrable), so DCT correctly does not apply; mass “escaped to infinity.”

Worked Example 29.2.7 (DCT in action: a limit through the integral). Compute $\lim_{n \rightarrow \infty} \int_0^{\infty} \frac{n \sin(x/n)}{x(1+x^2)} \, dx$.

Recognition. Trigger: pointwise limit plus an integrable dominator $\Rightarrow$ pass the limit inside (DCT).

Solution. Let $f_n(x) = \frac{n \sin(x/n)}{x(1+x^2)}$. As $n \rightarrow \infty$, $n \sin(x/n) \rightarrow x$, so $f_n(x) \rightarrow \frac{1}{1+x^2}$ pointwise. Dominate: $|\sin t| \leq |t|$ gives $|n \sin(x/n)| \leq n \cdot (x/n) = x$, so $|f_n(x)| \leq \frac{1}{1+x^2} =: g(x)$, and $\int_0^{\infty} g = \pi/2 < \infty$. By DCT the limit is $\int_0^{\infty} \frac{dx}{1+x^2} = \frac{\pi}{2}$. *Check.* g itself is the pointwise limit and is integrable; the answer $\pi/2$ equals $\int g$, as it must when $f_n \leq g$ and $f_n \rightarrow g$.

Worked Example 29.2.8 (Counting measure: integration is summation). On $X = \mathbb{N}$ with counting measure $\#$, identify $\int_X f \, d\#$ and use MCT to justify rearranging a non-negative double series.

Recognition. Trigger: “integral against counting measure”; recognize it as a sum, then apply Tonelli/MCT.

Solution. A function on $\mathbb{N}$ is the sequence $f(n) = a_n$; simple-function approximation gives $\int_{\mathbb{N}} f \, d\# = \sum_n a_n$. For $a_{m,n} \geq 0$, viewing the double index as $\mathbb{N} \times \mathbb{N}$ with counting measure, Tonelli gives $\sum_m \sum_n a_{m,n} = \sum_n \sum_m a_{m,n} = \sum_{(m,n)} a_{m,n}$: unconditional rearrangement of a non-negative double series. *Check.* With $a_{m,n} = 2^{-m-n}$, all three equal $(\sum 2^{-m})(\sum 2^{-n}) = 1 \cdot 1 = 1$.

The spaces L^1 and L^2

Definition 29.2.9. $L^1(X, \mu)$ is the space of (equivalence classes mod a.e.) integrable functions with norm $\|f\|_{L^1} = \int_X |f| \, d\mu$. $L^2(X, \mu)$ is the class of measurable f with $\int_X |f|^2 \, d\mu < \infty$, normed by $\|f\|_{L^2} = (\int_X |f|^2 \, d\mu)^{1/2}$ and carrying the inner product $\langle f, g \rangle = \int_X f \bar{g} \, d\mu$.

Theorem 29.2.10. $L^1(X, \mu)$ is a complete normed vector space (Banach), and $L^2(X, \mu)$ is a Hilbert space (possibly non-separable).

The proofs (Riesz–Fischer completeness via a rapidly converging subsequence) transfer unchanged from the Euclidean chapter. The Hilbert structure of L^2 is exactly what powers the von Neumann proof of Radon–Nikodym in §29.4 and the mean ergodic theorem in §29.5.

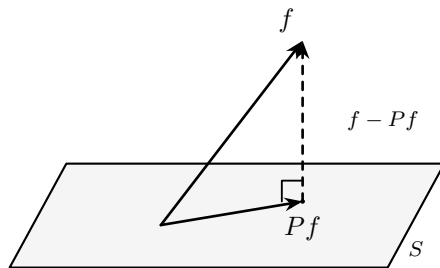


Figure 29.6. Orthogonal projection P onto a closed subspace $S \subseteq L^2$: $f - Pf \perp S$, so Pf is the nearest point of S to f . (Schematic.)

Worked Example 29.2.11 (Cauchy–Schwarz and an orthonormal computation in L^2). In $L^2([0, 1], m)$ with $e_k(x) = e^{2\pi i k x}$, compute $\langle f, e_1 \rangle$ for $f(x) = x$ and verify $|\langle f, e_1 \rangle| \leq \|f\| \|e_1\|$.

Recognition. Trigger: an inner-product/Fourier-coefficient computation; check the Cauchy–Schwarz bound.

Solution. $\langle f, e_1 \rangle = \int_0^1 x e^{-2\pi i x} dx$. Integrating by parts, $\int_0^1 x e^{-2\pi i x} dx = \left[\frac{x e^{-2\pi i x}}{-2\pi i} \right]_0^1 + \frac{1}{2\pi i} \int_0^1 e^{-2\pi i x} dx = \frac{-1}{-2\pi i} + 0 = \frac{i}{2\pi}$, so $|\langle f, e_1 \rangle| = \frac{1}{2\pi} \approx 0.159$. Here $\|e_1\| = 1$ and $\|f\| = (\int_0^1 x^2 dx)^{1/2} = 1/\sqrt{3} \approx 0.577$. Check. $0.159 \leq 0.577 = \|f\| \|e_1\|$, Cauchy–Schwarz holds (strictly, since f is not a scalar multiple of e_1).

Worked Example 29.2.12 (An L^2 function need not be L^1 on infinite measure). On $(\mathbb{R}, m)$ let $f(x) = \frac{1}{1+|x|}$. Decide membership in L^1 and L^2 .

Recognition. Trigger: compare tail decay against the integrability thresholds in infinite measure.

Solution. $\int_{\mathbb{R}} \frac{dx}{1+|x|} = 2 \int_0^\infty \frac{dx}{1+x} = \infty$, so $f \notin L^1$. But $\int_{\mathbb{R}} \frac{dx}{(1+|x|)^2} = 2 \int_0^\infty \frac{dx}{(1+x)^2} = 2 < \infty$, so $f \in L^2$. Check. On infinite-measure spaces neither inclusion $L^1 \subseteq L^2$ nor $L^2 \subseteq L^1$ holds; here the slow tail $1/x$ kills L^1 but the squared tail $1/x^2$ is summable, so $f \in L^2 \setminus L^1$. (On a probability space the reverse inclusion $L^2 \subseteq L^1$ would hold.)

Caution. In L^p the elements are equivalence classes mod a.e., not functions: “ $f(x_0)$ ” is meaningless and pointwise statements must be qualified by “a.e.” Also $L^2(X, \mu)$ need *not* be separable when μ is not σ -finite or X is exotic (e.g. counting measure on an uncountable set) – the Hilbert-space theory still applies, but one cannot assume a countable orthonormal basis exists.

Exercises (Stein, Ch.6).

- (Riesz representation for $C([a, b])$.) Let $C([a, b])$ be the continuous functions on $[a, b]$. A finite Borel measure μ gives a positive linear functional $\ell(f) = \int_a^b f d\mu$ (positive: $\ell(f) \geq 0$ when $f \geq 0$). Conversely, prove that every positive linear functional ℓ on $C([a, b])$ is of this form for a unique finite Borel measure μ . [Hint: with $a = 0$, define $F(u) = \lim_{\varepsilon \rightarrow 0} \ell(f_\varepsilon)$ where $f_\varepsilon = 1$ on $[0, u]$, $f_\varepsilon = 0$ on $[u + \varepsilon, b]$, linear between; then F is increasing and right-continuous, and $\ell(f) = \int_a^b f dF$ by the Lebesgue–Stieltjes theorem.] The result also holds on a closed infinite interval, with ℓ defined on compactly supported continuous functions and μ finite on bounded intervals.

Additional exercises.

- Compute $\int_X f d\#$ for $X = \mathbb{N}$, $\#$ counting measure, and $f(n) = 1/n^2$, and verify the value equals $\pi^2/6$.
- (MCT.) Let $f_n(x) = (1 + \frac{x}{n})^n \mathbf{1}_{[0, n]}(x)$ on $(0, \infty)$. Show $f_n \uparrow e^x$ pointwise and discuss why $\int_0^\infty f_n dx \rightarrow \infty$ – a correct MCT conclusion, not a failure.
- (Fatou strictness.) Construct measurable $f_n \geq 0$ on $[0, 1]$ with $\int f_n = 1$ for all n but $\int \liminf f_n = 0$, distinct from the moving-bump example in the text.
- (DCT.) Evaluate $\lim_{n \rightarrow \infty} \int_0^1 \frac{nx^{n-1}}{1+x} dx$ by identifying the pointwise limit and a dominating function, or explain why DCT does not apply and find the limit by another route. [Hint: the mass concentrates near $x = 1$.]
- Let $f \in L^1(X, \mu)$. Prove the absolute continuity of the integral: for every $\varepsilon > 0$ there is $\delta > 0$ with $\mu(E) < \delta \implies \int_E |f| d\mu < \varepsilon$.

6. (Orthonormality.) In $L^2([-\pi, \pi], \frac{dx}{2\pi})$ show $\{1, \cos x, \sin x, \cos 2x, \sin 2x, \dots\}$ is orthogonal, compute each norm, and expand $f(x) = x$ to second order, giving the first two nonzero coefficients.
7. (Projection.) In $L^2([0, 1])$ let S be the closed subspace of functions constant a.e. Find the orthogonal projection Pf of a general f onto S and verify $f - Pf \perp S$.
8. (Non-example.) Exhibit $f \in L^1(\mathbb{R}) \setminus L^2(\mathbb{R})$ and $g \in L^2(\mathbb{R}) \setminus L^1(\mathbb{R})$, confirming neither L^p contains the other on infinite measure.
9. (Diagram.) Sketch a measurable $f \geq 0$ and shade the region whose “horizontal” slices $\{f > y\}$ are measured by the Lebesgue integral; annotate $\int f d\mu = \int_0^\infty \mu(\{f > y\}) dy$ (the layer-cake formula).

29.3 Product Measures and the General Fubini Theorem

The product construction takes two measure spaces and builds the natural “area” measure on their Cartesian product, in which a multiple integral unwinds into an iterated integral. We assume both factors are complete and σ -finite.

Measurable rectangles and the product measure

Definition 29.3.1. Given $(X_1, \mathcal{M}_1, \mu_1)$ and $(X_2, \mathcal{M}_2, \mu_2)$, a *measurable rectangle* in $X = X_1 \times X_2$ is $A \times B$ with $A \in \mathcal{M}_1$, $B \in \mathcal{M}_2$. Let $\mathcal{A}$ be the collection of finite disjoint unions of rectangles, and define

$$\mu_0(A \times B) = \mu_1(A) \mu_2(B).$$

$\mathcal{A}$ is an algebra: the complement of a rectangle is a disjoint union of (at most) three rectangles, and a union of two rectangles splits into at most six disjoint ones.

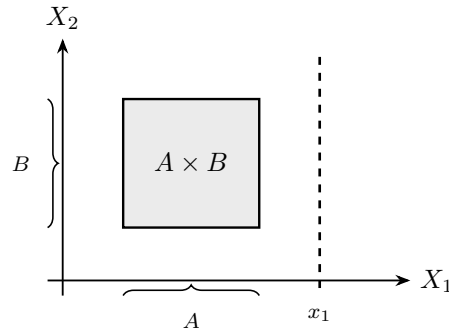


Figure 29.7. A measurable rectangle $A \times B$ with $\mu_0 = \mu_1(A)\mu_2(B)$; the dashed line is a slice $\{x_1\} \times X_2$ used by Fubini. (Schematic.)

Theorem 29.3.2. μ_0 is a premeasure on $\mathcal{A}$, so by the extension theorem it generates a measure $\mu = \mu_1 \times \mu_2$ on the σ -algebra $\mathcal{M} = \langle \mathcal{A} \rangle$.

Proof. The only nontrivial point is countable additivity: if $A \times B = \bigcup_j A_j \times B_j$ disjointly, then $\mu_0(A \times B) = \sum_j \mu_0(A_j \times B_j)$. Fix $x_1 \in A$. For each $x_2 \in B$ the point (x_1, x_2) lies in exactly one $A_j \times B_j$, so B is the disjoint union of those B_j with $x_1 \in A_j$; countable additivity of μ_2 gives

$$\mathbf{1}_A(x_1) \mu_2(B) = \sum_j \mathbf{1}_{A_j}(x_1) \mu_2(B_j).$$

Integrate in x_1 against μ_1 and apply MCT: $\mu_1(A)\mu_2(B) = \sum_j \mu_1(A_j)\mu_2(B_j)$, which is the claim. $\square$

Slices and the slicing proposition

For $E \subseteq X$ and points x_1, x_2 , the *slices* are $E_{x_1} = \{x_2 : (x_1, x_2) \in E\} \subseteq X_2$ and $E^{x_2} = \{x_1 : (x_1, x_2) \in E\} \subseteq X_1$.

Proposition 29.3.3. *If $E \in \mathcal{A}_{\sigma\delta}$ then E^{x_2} is μ_1 -measurable for every x_2 , $x_2 \mapsto \mu_1(E^{x_2})$ is μ_2 -measurable, and*

$$\int_{X_2} \mu_1(E^{x_2}) d\mu_2 = (\mu_1 \times \mu_2)(E). \quad (29.3)$$

For an arbitrary measurable $E \in \mathcal{M}$ the same holds, except that measurability of E^{x_2} and the value $\mu_1(E^{x_2})$ are asserted only for almost every x_2 .

Proof outline. (29.3) is clear for a rectangle. For $E \in \mathcal{A}_\sigma$ disjointify into rectangles and apply MCT. For $E \in \mathcal{A}_{\sigma\delta}$ of finite measure, write $E = \bigcap_j E_j$ with $E_j \in \mathcal{A}_\sigma$ decreasing; the slice functions $f_j(x_2) = \mu_1(E_j^{x_2})$ decrease to $\mu_1(E^{x_2})$ and MCT (from above, using finiteness) gives (29.3). σ -finiteness removes the finiteness assumption. For general $E \in \mathcal{M}$: if $(\mu_1 \times \mu_2)(E) = 0$, Proposition 29.1.16 gives $F \in \mathcal{A}_{\sigma\delta} \supseteq E$ with measure 0, so F^{x_2} is μ_1 -null for a.e. x_2 , and completeness of μ_1 makes $E^{x_2} \subseteq F^{x_2}$ null too. The general E is $F \setminus Z$ with Z null, $F \in \mathcal{A}_{\sigma\delta}$; subtract. $\square$

The general Fubini–Tonelli theorem

Theorem 29.3.4 (Fubini). *Let f be integrable on $(X_1 \times X_2, \mu_1 \times \mu_2)$. Then:*

- (i) *for a.e. x_2 , the slice $x_1 \mapsto f(x_1, x_2)$ is integrable on (X_1, μ_1) ;*
- (ii) *$x_2 \mapsto \int_{X_1} f(x_1, x_2) d\mu_1$ is integrable on X_2 ;*

$$(iii) \int_{X_2} \left(\int_{X_1} f(x_1, x_2) d\mu_1 \right) d\mu_2 = \int_{X_1 \times X_2} f d(\mu_1 \times \mu_2).$$

(Tonelli) *If $f \geq 0$ is merely measurable, (iii) holds in $[0, \infty]$ with no integrability hypothesis.*

Proof. By linearity it suffices to take $f \geq 0$. For $f = \mathbf{1}_E$ with E of finite measure the statement is Proposition 29.3.3. Linear combinations give it for simple functions; MCT (Tonelli) extends it to all non-negative measurable f . For integrable f split $f = f^+ - f^-$, each handled by Tonelli, and the finiteness of $\int f$ controls the slices a.e. $\square$

Method (Choosing Tonelli vs. Fubini). *Recognition trigger.* You want to swap the order of a double integral or sum.

1. If the integrand is ≥ 0 , use *Tonelli* immediately – no hypothesis needed; the iterated integrals always agree in $[0, \infty]$.
2. If the integrand changes sign, first apply Tonelli to $|f|$. If $\iint |f| < \infty$, then f is integrable and *Fubini* licenses the swap.
3. If $\iint |f| = \infty$, the order may matter – compute both iterated integrals; unequal values are the diagnostic of non-integrability.

Worked Example 29.3.5 (A Tonelli computation: area under a Gaussian product). Compute $\iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy$ by Tonelli, and deduce $\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$.

Recognition. Trigger: a non-negative product integrand – Tonelli factors it.

Solution. Since $e^{-(x^2+y^2)} = e^{-x^2} e^{-y^2} \geq 0$, Tonelli gives $\iint = (\int_{\mathbb{R}} e^{-x^2} dx) (\int_{\mathbb{R}} e^{-y^2} dy) = I^2$. In polar coordinates $\iint = \int_0^{2\pi} \int_0^\infty e^{-r^2} r dr d\theta = 2\pi \cdot \frac{1}{2} = \pi$. Hence $I^2 = \pi$, $I = \sqrt{\pi}$. *Check.* The polar slicing is itself the polar-coordinate case of the product theory (Theorem 29.3.8 below); both routes give π .

Worked Example 29.3.6 (Fubini fails without integrability). Show $\int_0^1 \int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2} dy dx \neq \int_0^1 \int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2} dx dy$.

Recognition. Trigger: a sign-changing kernel; suspect $\iint |f| = \infty$ and test both orders.

Solution. Note $\frac{x^2 - y^2}{(x^2 + y^2)^2} = \partial_x \left(\frac{-x}{x^2 + y^2} \right)$, so the inner x -integral is $\left[\frac{-x}{x^2 + y^2} \right]_0^1 = \frac{-1}{1 + y^2}$, giving outer value $-\int_0^1 \frac{dy}{1 + y^2} = -\pi/4$. By antisymmetry in (x, y) the other order gives $+\pi/4$. They differ, so Fubini cannot apply: indeed $\iint |f| = \infty$ near the origin. *Check.* The two answers are negatives of each other, exactly the symmetry $f(x, y) = -f(y, x)$ predicts; the obstruction is non-integrability, not a computational slip.

Worked Example 29.3.7 (Counting $\times$ Lebesgue: interchanging $\sum$ and $\int$). Use Tonelli on $\mathbb{N} \times [0, 1]$ to justify $\int_0^1 \sum_{n \geq 1} \frac{x^n}{2^n} dx = \sum_{n \geq 1} \frac{1}{2^n(n+1)}$.

Recognition. Trigger: a $\sum$ inside an $\int$ with non-negative terms; model as a product with counting measure.

Solution. The integrand $g(n, x) = 2^{-n} x^n \geq 0$ on $(\mathbb{N}, \#) \times ([0, 1], m)$. Tonelli equates the two iterations: $\int_0^1 \sum_n g dx = \sum_n \int_0^1 g dx = \sum_n 2^{-n} \frac{1}{n+1}$. *Check.* Numerically $\sum_{n \geq 1} \frac{1}{2^n(n+1)} = \frac{1}{4} + \frac{1}{24} + \cdots \approx 0.386$, and the closed form is $\sum_{n \geq 1} \frac{1}{2^n(n+1)} = 2 \ln 2 - 1 \approx 0.386$. Consistent.

Polar coordinates

Theorem 29.3.8 (Polar integration). Write $x = r\gamma$ with $r = |x| > 0$, $\gamma = x/|x| \in S^{d-1}$. There is a measure σ on S^{d-1} with $\sigma(S^{d-1}) = 2\pi^{d/2}/\Gamma(d/2)$ such that, for integrable f on $\mathbb{R}^d$,

$$\int_{\mathbb{R}^d} f(x) dx = \int_{S^{d-1}} \left(\int_0^\infty f(r\gamma) r^{d-1} dr \right) d\sigma(\gamma).$$

Proof idea. Take the product of $((0, \infty), r^{d-1} dr)$ with (S^{d-1}, σ) , where $\sigma(E) = d \cdot m(\tilde{E})$ for the sector $\tilde{E}$ of the unit ball over E . One verifies $m = \mu_1 \times \mu_2$ on rectangles $(0, 1) \times E$ (where $\mu_1((0, 1)) = 1/d$), extends by dilation-invariance and the uniqueness in the extension theorem to all measurable sets, then runs the Fubini argument through simple and non-negative functions. $\square$

The map $(r, \gamma) \mapsto r\gamma$ is the measure isomorphism underlying the theorem: it carries the product measure $r^{d-1} dr d\sigma$ to Lebesgue measure on the punctured space.

$$((0, \infty) \times S^{d-1}, r^{d-1} dr d\sigma) \xrightarrow[(r, \gamma) \mapsto r\gamma]{\sim} (\mathbb{R}^d \setminus \{0\}, dx)$$

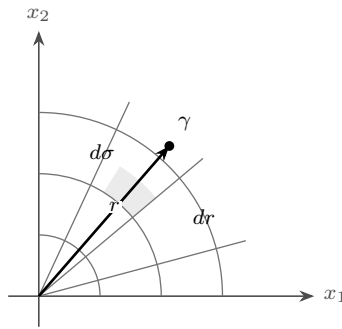


Figure 29.8. Polar slicing $x = r\gamma$: the volume element factors as $r^{d-1} dr d\sigma(\gamma)$ over $(0, \infty) \times S^{d-1}$.

Worked Example 29.3.9 (Volume of the unit ball via polar coordinates). Use Theorem 29.3.8 to compute $v_d = m(B)$ for the unit ball $B \subseteq \mathbb{R}^d$.

Recognition. Trigger: an integral over a ball with radial symmetry – polar coordinates separate r and γ .

Solution. Apply the formula to $f = \mathbf{1}_B$: $v_d = \int_{S^{d-1}} \int_0^1 r^{d-1} dr d\sigma = \sigma(S^{d-1}) \cdot \frac{1}{d} = \frac{2\pi^{d/2}}{\Gamma(d/2)} \cdot \frac{1}{d} = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$.

Check. $d = 2$ gives $\pi/\Gamma(2) = \pi$ (area of the disc); $d = 3$ gives $\pi^{3/2}/\Gamma(5/2) = \pi^{3/2}/(\frac{3}{4}\sqrt{\pi}) = \frac{4}{3}\pi$, the classical ball volume.

Borel measures on $\mathbb{R}$ and the Lebesgue–Stieltjes integral

Theorem 29.3.10 (Lebesgue–Stieltjes). *If F is increasing and normalized (right-continuous), there is a unique Borel measure $\mu = dF$ on $\mathbb{R}$ with $\mu((a, b]) = F(b) - F(a)$. Conversely, a Borel measure finite on bounded intervals arises this way from $F(x) = \mu((0, x])$ ($x > 0$), $F(0) = 0$, $F(x) = -\mu((-x, 0])$ ($x < 0$).*

Proof idea. Define $\mu^*(E) = \inf \sum_j (F(b_j) - F(a_j))$ over covers by half-open intervals $(a_j, b_j]$. Right-continuity of F plus compactness gives $\mu^*((a, b]) = F(b) - F(a)$, and splitting a cover into intervals shorter than the gap shows μ^* is a metric exterior measure; Theorem 29.1.10 then yields a Borel measure. Uniqueness follows because any open set is a countable disjoint union of half-open intervals, plus regularity. $\square$

For absolutely continuous F , $\int_a^b f dF = \int_a^b f F' dx$; for a pure-jump F with jumps α_n at x_n , $\int f dF = \sum_n f(x_n) \alpha_n$ and $\mu(\{x_n\}) = \alpha_n$; the Heaviside H gives the Dirac mass $\int f dH = f(0)$.

Worked Example 29.3.11 (Dirac mass and a mixed Stieltjes integral). With $F(x) = x + H(x)$ (H the Heaviside jump at 0) compute $\int_{-1}^2 x^2 dF(x)$.

Recognition. Trigger: F splits into an absolutely continuous part plus a jump – integrate each against its piece.

Solution. $dF = dx + \delta_0$. So $\int_{-1}^2 x^2 dF = \int_{-1}^2 x^2 dx + 0^2 = [\frac{x^3}{3}]_{-1}^2 + 0 = \frac{8 - (-1)}{3} = 3$. *Check.* The jump of F at 0 has size 1 and contributes $x^2|_{x=0} \cdot 1 = 0$; the smooth part gives 3, so the total is 3 – consistent with $\mu(\{0\}) = 1$ but $f(0) = 0$.

Worked Example 29.3.12 (Pure-jump Stieltjes integral against a step function). Let $F(x) = \lfloor x \rfloor$ on $[0, 4)$ (jumps of size 1 at 1, 2, 3). Compute $\int_0^4 x dF(x)$.

Recognition. Trigger: F is pure-jump – the Stieltjes measure is a sum of Dirac masses at the jump points.

Solution. $dF = \delta_1 + \delta_2 + \delta_3$ (each jump of size 1), so $\int_0^4 x dF = \sum_{n=1}^3 n \cdot 1 = 1 + 2 + 3 = 6$. *Check.* Replacing x by the constant 1 gives $\int_0^4 1 dF = F(4^-) - F(0) = 3$, the total mass, matching the three unit jumps.

Caution. The product measure $\mu_1 \times \mu_2$ from the extension theorem is generally *not* complete, even when both factors are: e.g. $m_1 \times m_1$ is Borel-on- $\mathbb{R}^2$, which is strictly smaller than Lebesgue on $\mathbb{R}^2$; the latter is its *completion*. Fubini as stated still holds after completing. Also Fubini’s a.e. integrability of slices can fail on a genuine null set of x_2 – “a.e.” is not “every.”

Exercises (Stein, Ch.6).

- Let r be a rotation of $\mathbb{R}^d$. Using that $x \mapsto r(x)$ preserves Lebesgue measure, show that it induces a measure-preserving map of the sphere S^{d-1} with its measure $d\sigma$. (A converse is Problem 4.)
- Use the polar coordinate formula to prove: (a) $\int_{\mathbb{R}^d} e^{-\pi|x|^2} dx = 1$ for $d = 2$, and deduce the same for all d ; (b) $(\int_0^\infty e^{-\pi r^2} r^{d-1} dr) \sigma(S^{d-1}) = 1$, so $\sigma(S^{d-1}) = 2\pi^{d/2}/\Gamma(d/2)$; (c) the unit ball B has volume $v_d = \pi^{d/2}/\Gamma(d/2 + 1)$, since $v_d = (\int_0^1 r^{d-1} dr) \sigma(S^{d-1})$.
- (Green’s formula on the unit ball $B \subseteq \mathbb{R}^d$.) For $u, v \in C^2(B)$,

$$\int_B (v \Delta u - u \Delta v) dx = \int_{S^{d-1}} \left(v \frac{\partial u}{\partial n} - u \frac{\partial v}{\partial n} \right) d\sigma,$$

with $\partial/\partial n$ the inner normal derivative. Derive it from Lemma 4.5 of the previous chapter by taking $\eta = \eta_\varepsilon^+$ and letting $\varepsilon \rightarrow 0$.

- (Spherical mean-value property.) If u is harmonic in Ω and the closed ball of center x_0 , radius r lies in Ω , then $u(x_0) = c \int_{S^{d-1}} u(x_0 + ry) d\sigma(y)$ with $c^{-1} = \sigma(S^{d-1})$; conversely a continuous function with this property is harmonic.
- (Lebesgue–Stieltjes decomposition.) Let F be increasing and normalized with $F = F_A + F_C + F_J$ (absolutely continuous, continuous-singular, pure-jump), and let $\mu = \mu_A + \mu_C + \mu_J$ be the associated measures. Verify: (i) $\mu_A \ll m$ with $\mu_A(E) = \int_E F'(x) dx$; (ii) if F is absolutely continuous then $\int f d\mu = \int f dF = \int f(x) F'(x) dx$ whenever f, fF' are integrable; (iii) $\mu_C + \mu_J \perp m$.

6. Show $\mathbb{R}^d \setminus \{0\}$, written as $\mathbb{R}_+ \times S^{d-1}$, has the property that every open subset is a countable union of open rectangles of the product. [Hint: use rectangles $\{r_j < r < r'_k\} \times \{\gamma \in S^{d-1} : |\gamma - \gamma_\ell| < 1/n\}$ with r_j, r'_k positive rationals and $\{\gamma_\ell\}$ dense in S^{d-1} .]
7. Let m_j be Lebesgue measure on $\mathbb{R}^{d_j}$, $j = 1, 2$, and m Lebesgue measure on $\mathbb{R}^d$, $d = d_1 + d_2$. Show m is the completion (Exercise 2) of the product measure $m_1 \times m_2$.
8. For a finite family $(X_j, \mathcal{M}_j, \mu_j)$, $1 \leq j \leq k$, construct the product measure $\mu_1 \times \cdots \times \mu_k$ on $X = X_1 \times \cdots \times X_k$: define $\mu_0(E_1 \times \cdots \times E_k) = \prod_j \mu_j(E_j)$, verify it extends to a premeasure on the algebra $\mathcal{A}$ of finite disjoint unions of such boxes, and apply the extension theorem.
9. (Infinite products.) For measure spaces $(X_j, \mathcal{M}_j, \mu_j)$ with $\mu_j(X_j) = 1$ for all but finitely many j , define a cylinder set $E = \{x = (x_j) : x_j \in E_j, E_j = X_j \text{ for all but finitely many } j\}$ and $\mu_0(E) = \prod_j \mu_j(E_j)$. Show μ_0 extends to a premeasure on the algebra generated by cylinder sets, and apply the extension theorem.
10. (The d -torus.) Let $\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$, identified with $\mathbb{T}^1 \times \cdots \times \mathbb{T}^1$, and $\mu = \mu_1 \times \cdots \times \mu_d$ with μ_j Lebesgue measure on the circle. Show: (a) the completion $\bar{\mu}$ is Lebesgue measure on the cube $Q = \{0 < x_j \leq 1\}$; (b) for f on $\mathbb{T}^d$ with periodic extension $\tilde{f}$, f is measurable/continuous on $\mathbb{T}^d$ iff $\tilde{f}$ is on $\mathbb{R}^d$; (c) for integrable f, g , the convolution $(f * g)(x) = \int_{\mathbb{T}^d} f(x - y)g(y) dy$ is finite a.e., integrable, and $f * g = g * f$; (d) if $f \sim \sum_n a_n e^{2\pi i n \cdot x}$ and $g \sim \sum_n b_n e^{2\pi i n \cdot x}$ then $f * g \sim \sum_n a_n b_n e^{2\pi i n \cdot x}$; (e) $\{e^{2\pi i n \cdot x}\}_{n \in \mathbb{Z}^d}$ is an orthonormal basis of $L^2(\mathbb{T}^d)$, so $\|f\|_{L^2(\mathbb{T}^d)}^2 = \sum_n |a_n|^2$; (f) every continuous periodic f is uniformly approximable by finite linear combinations of the exponentials. [Hint: for (e) reduce to $d = 1$ by Fubini; for (f) use $f * g_\varepsilon \rightarrow f$ uniformly with $g_\varepsilon = \varepsilon^{-d} \mathbf{1}_{(0, \varepsilon]^d}$.]

Problems (Stein, Ch.6).

1. (Change of variables.) Let Φ be a C^1 bijection of an open $O \subseteq \mathbb{R}^d$ onto $O' \subseteq \mathbb{R}^d$. Show: (a) $E \subseteq O$ measurable $\implies \Phi(E)$ measurable; (b) $m(\Phi(E)) = \int_E |\det \Phi'(x)| dx$; (c) $\int_{O'} f(y) dy = \int_O f(\Phi(x)) |\det \Phi'(x)| dx$ for integrable f . [Hint: cover E by small cubes Q_k of center z_k ; $\Phi(x) = \Phi(z_k) + \Phi'(z_k)(x - z_k) + o(\varepsilon)$ on Q_k , and use the linear transformation rule for Lebesgue measure.]
2. Show the measure $d\mu = \frac{dx dy}{y^2}$ on the upper half-plane $\mathbb{R}_+^2 = \{z = x + iy : y > 0\}$ is preserved by every fractional linear transformation $z \mapsto \frac{az+b}{cz+d}$ with $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R})$.
3. (Surface measure of a graph.) Let $S = \{(x, y) \in \mathbb{R}^{d-1} \times \mathbb{R} : y = F(x)\}$ with $F \in C^1(\Omega)$. For $E \subseteq \Omega$ write $\hat{E} = \{(x, F(x)) : x \in E\}$. (a) With $\mu(\hat{E}) = \int_E \sqrt{1 + |\nabla F|^2} dx$ and $\hat{B}_\delta$ a δ -neighborhood of a ball $\hat{B}$, show $\mu(\hat{B}) = \lim_{\delta \rightarrow 0} \frac{1}{2\delta} m(\hat{B}_\delta)$. (b) For the upper unit hemisphere $F(x) = (1 - |x|^2)^{1/2}$, show $d\mu = d\sigma$. (c) For $d = 3$, $y = \cos \theta$, $x = (\sin \theta \cos \varphi, \sin \theta \sin \varphi)$, deduce $d\sigma = \sin \theta d\theta d\varphi$.
4. * (Uniqueness of rotation-invariant surface measure.) If μ is a Borel measure on S^{d-1} with $\mu(r(E)) = \mu(E)$ for every rotation r and Borel E , and $\mu(S^{d-1}) < \infty$, then $\mu = c \sigma$. [Hint: show $\int_{S^{d-1}} Y_k d\mu = 0$ for every spherical harmonic Y_k of degree $k \geq 1$.]
5. * (Riesz representation on a metric space.) Let X be a metric space and $C_0(X)$ the continuous functions each supported in some closed ball. Show every positive linear functional ℓ on $C_0(X)$ has the form $\ell(f) = \int_X f d\mu$ for a unique Borel measure μ finite on all balls.

Additional exercises.

1. Use Tonelli on $\mathbb{N} \times \mathbb{N}$ with counting measure to prove that for $a_{m,n} \geq 0$ the two iterated sums and the unordered sum all coincide; give an example with sign changes where they differ.
2. Compute $\int_0^1 \int_0^1 \frac{1}{1-xy} dx dy$ by expanding in a geometric series and applying Tonelli, and identify the answer as $\pi^2/6$.
3. (Fubini failure.) For $f(x, y) = \frac{xy}{(x^2 + y^2)^2}$ on $[-1, 1]^2 \setminus \{0\}$, compute both iterated integrals and decide whether Fubini applies.

4. Using polar coordinates, evaluate $\int_{\mathbb{R}^d} e^{-|x|} dx$ in terms of $\sigma(S^{d-1})$ and Γ , and check the $d = 1, 2, 3$ cases against direct computation.
5. (Lebesgue–Stieltjes.) For $F(x) = x + 2H(x - 1)$ on $[0, 3]$ (a jump of size 2 at $x = 1$) compute $\int_0^3 x dF$ and $\int_0^3 \mathbf{1}_{[1,2]} dF$.
6. Let $\mu_1 = \mu_2 = \#$ be counting measure on $\mathbb{N}$. Describe $\mu_1 \times \mu_2$ and show it is just counting measure on $\mathbb{N} \times \mathbb{N}$; explain why completeness is automatic here.
7. (Surface area.) From $\sigma(S^{d-1}) = 2\pi^{d/2}/\Gamma(d/2)$ compute the surface area of S^1, S^2, S^3 and confirm $\sigma(S^1) = 2\pi$, $\sigma(S^2) = 4\pi$.
8. Show that the product of two σ -finite spaces is σ -finite, and give a non-example where dropping σ -finiteness breaks the slicing formula (29.3). [Hint: $[0, 1]$ with Lebesgue against $[0, 1]$ with counting measure on the diagonal.]
9. (Diagram.) Draw the unit disc tiled by polar “rectangles” $[r, r + dr] \times [\theta, \theta + d\theta]$ and label the area element $r dr d\theta$; explain the factor r as the Jacobian of $x = r\gamma$.

29.4 Signed Measures and the Radon-Nikodym Theorem

To compare two measures we first widen “measure” to allow negative values, then decompose any signed measure into a positive and a negative part, and finally relate two measures by absolute continuity versus mutual singularity.

Signed measures and total variation

Definition 29.4.1. A *signed measure* ν on $(X, \mathcal{M})$ is a map $\nu : \mathcal{M} \rightarrow (-\infty, \infty]$ that is countably additive: $\nu(\bigcup_j E_j) = \sum_j \nu(E_j)$ for disjoint E_j (the sum converging absolutely when $\nu(\bigcup E_j)$ is finite).

The model is $\nu(E) = \int_E f d\mu$ with f of either sign (integrable in the extended sense: $\int f^- d\mu < \infty$).

Definition 29.4.2. The *total variation* of ν is the set function

$$|\nu|(E) = \sup \sum_j |\nu(E_j)|,$$

the supremum over all countable partitions $E = \bigsqcup_j E_j$ into measurable pieces.

Proposition 29.4.3. $|\nu|$ is a (positive) measure with $\nu \leq |\nu|$. Consequently $\nu^+ = \frac{1}{2}(|\nu| + \nu)$ and $\nu^- = \frac{1}{2}(|\nu| - \nu)$ are positive measures, the Jordan decomposition, with $\nu = \nu^+ - \nu^-$ and $|\nu| = \nu^+ + \nu^-$.

Proof. Countable additivity of $|\nu|$: for disjoint $\{E_j\}$ with $E = \bigcup E_j$ we show $\sum_j |\nu|(E_j) \leq |\nu|(E)$ and $|\nu|(E) \leq \sum_j |\nu|(E_j)$. For the first, pick $\alpha_j < |\nu|(E_j)$ realized by a partition $E_j = \bigcup_i F_{i,j}$ with $\alpha_j \leq \sum_i |\nu(F_{i,j})|$; since $\{F_{i,j}\}$ partitions E , $\sum_j \alpha_j \leq \sum_{i,j} |\nu(F_{i,j})| \leq |\nu|(E)$, and sup over α_j gives “ $\leq |\nu|(E)$.” For the reverse, any partition $\{F_k\}$ of E satisfies, by the triangle inequality and additivity of ν , $\sum_k |\nu(F_k)| = \sum_k \left| \sum_j \nu(F_k \cap E_j) \right| \leq \sum_j \sum_k |\nu(F_k \cap E_j)| \leq \sum_j |\nu|(E_j)$; sup over $\{F_k\}$ finishes. The Jordan parts are then measures and the stated identities are immediate from $\nu \leq |\nu|$ and $-\nu \leq |\nu|$. $\square$

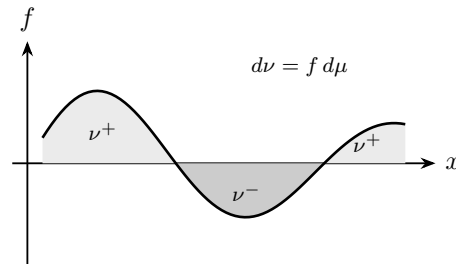


Figure 29.9. A signed measure $d\nu = f d\mu$: ν^+ lives where $f > 0$, ν^- where $f < 0$, and $|\nu|$ integrates $|f|$ (Hahn–Jordan). (Schematic.)

Worked Example 29.4.4 (Jordan decomposition of an explicit signed measure). On $([0, 2\pi], m)$ let $d\nu = \cos x \, dx$. Find $\nu^+, \nu^-, |\nu|$ and $|\nu|([0, 2\pi])$.

Recognition. Trigger: signed density of known sign pattern – split the domain at the zeros of the density.

Solution. $\cos x > 0$ on $A = [0, \frac{\pi}{2}) \cup (\frac{3\pi}{2}, 2\pi]$ and < 0 on $B = (\frac{\pi}{2}, \frac{3\pi}{2})$. So $\nu^+(E) = \int_{E \cap A} \cos x \, dx$, $\nu^-(E) = -\int_{E \cap B} \cos x \, dx$, and $d|\nu| = |\cos x| \, dx$. Then $|\nu|([0, 2\pi]) = \int_0^{2\pi} |\cos x| \, dx = 4$. Check. $\nu([0, 2\pi]) = \int_0^{2\pi} \cos x \, dx = 0 = \nu^+ - \nu^-$, and indeed $\nu^+([0, 2\pi]) = 2 = \nu^-([0, 2\pi])$, so $|\nu| = \nu^+ + \nu^- = 4$. Consistent.

Absolute continuity and mutual singularity

Definition 29.4.5. For a signed ν and a positive measure μ on $(X, \mathcal{M})$: ν is *absolutely continuous* w.r.t. μ , written $\nu \ll \mu$, if $\mu(E) = 0 \implies \nu(E) = 0$; ν and μ are *mutually singular*, $\nu \perp \mu$, if there are disjoint $A, B \in \mathcal{M}$ with $\nu(\cdot) = \nu(\cdot \cap A)$ and $\mu(\cdot) = \mu(\cdot \cap B)$ (they live on disjoint sets).

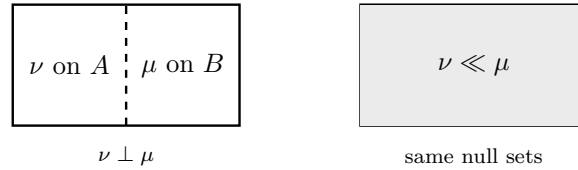


Figure 29.10. Two extremes: $\nu \perp \mu$ live on disjoint carriers; $\nu \ll \mu$ shares μ 's null sets. Radon–Nikodym says every ν splits into one of each. (Schematic.)

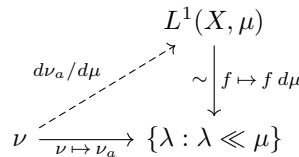
Proposition 29.4.6. The ε - δ condition “ $\forall \varepsilon > 0 \exists \delta > 0 : \mu(E) < \delta \implies |\nu(E)| < \varepsilon$ ” implies $\nu \ll \mu$. Conversely, if $|\nu|$ is finite, $\nu \ll \mu$ implies the ε - δ condition.

Proof. Forward is trivial ($\mu(E) = 0 < \delta$). Conversely, if the ε - δ condition fails for some ε , pick E_n with $\mu(E_n) < 2^{-n}$ but $|\nu|(E_n) \geq \varepsilon$. Let $E^* = \limsup_n E_n = \bigcap_n \bigcup_{k \geq n} E_k$. Then $\mu(E^*) \leq \inf_n 2^{-(n-1)} = 0$, yet $|\nu|(\bigcup_{k \geq n} E_k) \geq \varepsilon$ and finiteness of $|\nu|$ gives $|\nu|(E^*) \geq \varepsilon > 0$, contradicting $\nu \ll \mu$. $\square$

The Lebesgue–Radon–Nikodym theorem

Theorem 29.4.7 (Lebesgue–Radon–Nikodym). Let μ be a σ -finite positive measure and ν a σ -finite signed measure on $(X, \mathcal{M})$. Then there are unique signed measures ν_a, ν_s with $\nu_a \ll \mu$, $\nu_s \perp \mu$, and $\nu = \nu_a + \nu_s$. Moreover $d\nu_a = f \, d\mu$ for an extended μ -integrable f (the Radon–Nikodym derivative $f = d\nu_a/d\mu$).

The absolutely continuous part is realized through a density: the map $f \mapsto f \, d\mu$ sends $L^1(\mu)$ onto the measures $\ll \mu$, and ν_a is the image of the Radon–Nikodym derivative under it.



Proof (von Neumann’s Hilbert-space argument). Reduce to ν, μ positive and finite. Let $\rho = \nu + \mu$ and define the functional $\ell(\psi) = \int_X \psi \, d\nu$ on $L^2(X, \rho)$. By Cauchy–Schwarz $|\ell(\psi)| \leq \int |\psi| \, d\rho \leq \rho(X)^{1/2} \|\psi\|_{L^2(\rho)}$, so ℓ is bounded; by the Riesz representation theorem there is $g \in L^2(X, \rho)$ with

$$\int \psi \, d\nu = \int \psi g \, d\rho \quad (\psi \in L^2(X, \rho)). \quad (29.4)$$

Taking $\psi = \mathbf{1}_E$ and using $\nu \leq \rho$ gives $0 \leq \frac{1}{\rho(E)} \int_E g \, d\rho \leq 1$, so $0 \leq g \leq 1$ ρ -a.e. Rewrite (29.4) as $\int \psi(1 - g) \, d\nu = \int \psi g \, d\mu$. Split $X = A \cup B$, $A = \{0 \leq g < 1\}$, $B = \{g = 1\}$; set $\nu_a(E) = \nu(A \cap E)$, $\nu_s(E) = \nu(B \cap E)$. Putting $\psi = \mathbf{1}_B$ gives $0 = \int_B 1 \, d\mu = \mu(B)$, so $\nu_s \perp \mu$. Putting $\psi = \mathbf{1}_E(1 + g + \cdots + g^n)$ telescopes:

$$\int_E (1 - g^{n+1}) \, d\nu = \int_E g(1 + \cdots + g^n) \, d\mu.$$

On A , $1 - g^{n+1} \rightarrow 1$; on B it is 0. By DCT the left side $\rightarrow \nu(A \cap E) = \nu_a(E)$, while $g(1 + \cdots + g^n) \uparrow \frac{g}{1-g}$, so by MCT $\nu_a(E) = \int_E f d\mu$ with $f = \frac{g}{1-g}$. Finiteness $\nu_a(X) \leq \nu(X) < \infty$ gives $f \in L^1(\mu)$. A σ -finite exhaustion and summing $f = \sum f_j$, $\nu_a = \sum \nu_{j,a}$, $\nu_s = \sum \nu_{j,s}$ removes the finiteness assumption; signed ν is handled via $\nu^\pm$. *Uniqueness*: if $\nu = \nu'_a + \nu'_s$ also, then $\nu_a - \nu'_a = \nu'_s - \nu_s$ is simultaneously $\ll \mu$ and $\perp \mu$, hence 0. $\square$

Method (Computing a Radon–Nikodym derivative $d\nu/d\mu$). *Recognition trigger*. You are handed two measures with $\nu \ll \mu$ and want the density f .

1. Confirm $\nu \ll \mu$ (every μ -null set is ν -null).
2. If both are given by densities against a common reference, $d\nu = p dx$, $d\mu = q dx$, then $f = p/q$ where $q > 0$.
3. Verify by integrating: $\int_E f d\mu$ should reproduce $\nu(E)$ on test sets.

Worked Example 29.4.8 (An explicit Radon–Nikodym derivative). On $(\mathbb{R}, m)$ let $d\nu = e^{-x} \mathbf{1}_{[0, \infty)} dx$ and $d\mu = e^{-x^2/2} dx$. Compute $d\nu/d\mu$.

Recognition. Trigger: two absolutely continuous measures with explicit densities – take the ratio.

Solution. Both have densities against dx , and $\mu(E) = 0 \Leftrightarrow m(E) = 0 \Rightarrow \nu(E) = 0$, so $\nu \ll \mu$. The chain rule gives $\frac{d\nu}{d\mu} = \frac{e^{-x} \mathbf{1}_{[0, \infty)}}{e^{-x^2/2}} = \mathbf{1}_{[0, \infty)} e^{x^2/2-x}$. *Check*. On $E = [0, \infty)$, $\int_E \frac{d\nu}{d\mu} d\mu = \int_0^\infty e^{x^2/2-x} e^{-x^2/2} dx = \int_0^\infty e^{-x} dx = 1 = \nu([0, \infty))$. Correct.

Worked Example 29.4.9 (Cantor measure is singular: no density exists). Let ν be the Cantor–Lebesgue (devil’s-staircase) measure on $[0, 1]$. Show $\nu \perp m$, so $d\nu/dm$ does not exist.

Recognition. Trigger: a measure supported on a Lebesgue-null set – diagnose singularity, not absolute continuity.

Solution. ν is the Stieltjes measure dF of the Cantor function F ; ν is carried by the Cantor set C , which has $m(C) = 0$ (it is $\bigcap_n$ of 2^n intervals of total length $(2/3)^n \rightarrow 0$). Take $A = C$, $B = [0, 1] \setminus C$: then $\nu(\cdot) = \nu(\cdot \cap A)$ and $m(\cdot) = m(\cdot \cap B)$, so $\nu \perp m$. By Radon–Nikodym the absolutely continuous part of ν is 0; no f with $d\nu = f dm$ can exist. *Check*. $\nu([0, 1]) = 1$ while $m(C) = 0$ – if a density existed, $\nu(C) = \int_C f dm = 0 \neq 1$, a contradiction.

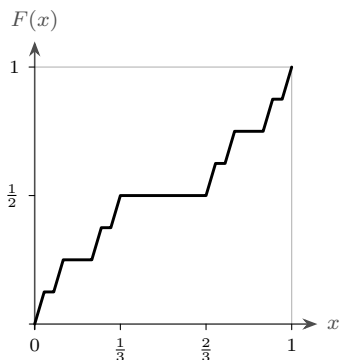


Figure 29.11. The Cantor function (devil’s staircase): continuous, increasing, $F' = 0$ a.e. (constant on every deleted middle third), yet $F(1) - F(0) = 1$. Its Stieltjes measure is singular to m .

Worked Example 29.4.10 (Chain rule for Radon–Nikodym derivatives). If $\lambda \ll \nu \ll \mu$ (all σ -finite, positive), show $\frac{d\lambda}{d\mu} = \frac{d\lambda}{d\nu} \cdot \frac{d\nu}{d\mu}$ a.e.

Recognition. Trigger: composition of densities – prove it on indicator test sets, then bootstrap.

Solution. Write $g = d\lambda/d\nu$, $h = d\nu/d\mu$. For $E \in \mathcal{M}$, $\lambda(E) = \int_E g d\nu$. Approximating $g \geq 0$ by simple functions and using $\int \varphi d\nu = \int \varphi h d\mu$ (true for $\varphi = \mathbf{1}_A$ since $\nu(A) = \int_A h d\mu$, hence for simple, hence by MCT for all $\varphi \geq 0$), we get $\lambda(E) = \int_E gh d\mu$. Thus $d\lambda/d\mu = gh$ a.e. *Check*. With $d\lambda = x^2 dx$, $d\nu = x dx$, $d\mu = dx$ on $(0, 1]$: $g = x$, $h = x$, product x^2 , and indeed $d\lambda/d\mu = x^2$. Consistent.

Caution. Absolute continuity of *measures* ($\nu \ll \mu$) and of *functions* (F absolutely continuous) are linked but distinct: $dF \ll m$ iff F is absolutely continuous, and then $d(dF)/dm = F'$. A continuous, even strictly increasing, F can still give a singular measure (Cantor function) – continuity of F is far weaker than absolute continuity. The ε - δ form of $\nu \ll \mu$ requires $|\nu|$ *finite*; for infinite ν the bare null-set implication can hold while no uniform δ exists.

Exercises (Stein, Ch.6).

- Let ν, ν_1, ν_2 be signed measures on $(X, \mathcal{M})$ and μ a positive measure on $\mathcal{M}$. Using $\perp$ and $\ll$, prove: (a) $\nu_1 \perp \mu$ and $\nu_2 \perp \mu \implies \nu_1 + \nu_2 \perp \mu$; (b) $\nu_1 \ll \mu$ and $\nu_2 \ll \mu \implies \nu_1 + \nu_2 \ll \mu$; (c) $\nu_1 \perp \nu_2 \implies |\nu_1| \perp |\nu_2|$; (d) $\nu \ll |\nu|$; (e) $\nu \perp \mu$ and $\nu \ll \mu \implies \nu = 0$.

Additional exercises.

- For $d\nu = (2x - 1) dx$ on $[0, 1]$ compute the Jordan decomposition ν^+, ν^- , the total variation $|\nu|([0, 1])$, and identify the Hahn sets where ν is positive and negative.
- Show $\nu \ll \mu$ and $\mu \ll \nu$ together (mutual absolute continuity) iff ν and μ have the same null sets; in that case $\frac{d\nu}{d\mu} \cdot \frac{d\mu}{d\nu} = 1$ a.e.
- Let $d\nu = f d\mu$ with $f \in L^1(\mu)$. Prove $|\nu|(E) = \int_E |f| d\mu$, so $\frac{d|\nu|}{d\mu} = |f|$.
- (Singular example.) On $[0, 1]$ let $\nu = \frac{1}{2}m + \frac{1}{2}\delta_{1/2}$. Find the Lebesgue decomposition $\nu = \nu_a + \nu_s$ relative to m and give $d\nu_a/dm$.
- Verify directly that the Cantor measure $\nu = dF$ (F the Cantor function) assigns $\nu([0, 1]) = 1$ and $\nu(I) = 0$ for every interval I in the complement of the Cantor set; conclude $\nu \perp m$.
- (Non-example.) Show that $\nu \ll \mu$ does *not* imply $\mu \ll \nu$ by an explicit pair on $[0, 1]$.
- Prove ε - δ absolute continuity can fail for an infinite ν even though $\nu \ll \mu$: construct such a pair, confirming the finiteness hypothesis in Proposition 29.4.6.
- (Diagram.) Draw the devil's staircase F over $[0, 1]$ to three levels of construction, shading the flat parts (where $F' = 0$) and marking that all the increase of F happens on the measure-zero Cantor set.

29.5 Ergodic Theorems

Ergodic theory studies the long-run averages of a system evolving by a measure-preserving map. The abstract framework of this chapter is exactly the right home for its foundational limit theorems, which also tie back to the differentiation theory via maximal functions.

Measure-preserving transformations

Definition 29.5.1. On a σ -finite $(X, \mathcal{M}, \mu)$, a map $\tau : X \rightarrow X$ is *measure-preserving* if $\tau^{-1}(E) \in \mathcal{M}$ and $\mu(\tau^{-1}(E)) = \mu(E)$ for every $E \in \mathcal{M}$. If τ is a bijection and τ^{-1} is also measure-preserving, τ is a *measure-preserving isomorphism*. The averages of f along the orbit are

$$A_n(f)(x) = \frac{1}{n} \sum_{k=0}^{n-1} f(\tau^k(x)).$$

Measure-preservation gives the change-of-variables identity $\int_X f(\tau(x)) d\mu = \int_X f d\mu$, first for $f = \mathbf{1}_E$ (since $\mathbf{1}_E \circ \tau = \mathbf{1}_{\tau^{-1}(E)}$), then for simple, non-negative, and integrable f . Equivalently, the *Koopman operator* $Tf = f \circ \tau$ on $L^2(X, \mu)$ commutes the integral away: the square below commutes, and measure-preservation is exactly the statement that the bottom map is the identity.

$$\begin{array}{ccc} L^2(X, \mu) & \xrightarrow{T = \cdot \circ \tau} & L^2(X, \mu) \\ \int_X \cdot d\mu \downarrow & & \downarrow \int_X \cdot d\mu \\ \mathbb{C} & \xlongequal{\quad} & \mathbb{C} \end{array}$$

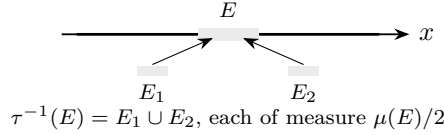


Figure 29.12. The doubling map $\tau(x) = 2x \bmod 1$: each E has two preimages of half its length, so $\mu(\tau^{-1}E) = \mu(E)$. (Schematic.)

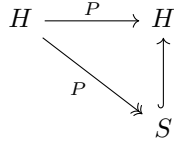
Example 29.5.2. (i) $X = \mathbb{Z}$, counting measure, $\tau(n) = n + 1$ (an isomorphism). (ii) $X = \mathbb{R}^d$, Lebesgue, $\tau(x) = x + h$. (iii) $X = \mathbb{R}/\mathbb{Z} \cong (0, 1]$, Lebesgue, rotation $\tau(x) = x + \alpha \bmod 1$. (iv) doubling map $\tau(x) = 2x \bmod 1$ on $(0, 1]$ (not injective, so not an isomorphism). (v) Gauss map $\tau(x) = \{1/x\}$ on $[0, 1)$, preserving $d\mu = \frac{1}{\ln 2} \frac{dx}{1+x}$.

Definition 29.5.3. τ is *ergodic* (when $\mu(X) = 1$) if every invariant set ($\tau^{-1}(E) = E$ up to null sets) has measure 0 or 1. It is *mixing* if $\mu(\tau^{-n}(E) \cap F) \rightarrow \mu(E)\mu(F)$ as $n \rightarrow \infty$ for all E, F ; mixing implies ergodic.

The mean ergodic theorem (von Neumann)

Theorem 29.5.4 (Mean ergodic theorem). *Let T be an isometry of a Hilbert space H , P the orthogonal projection onto the invariant subspace $S = \{f : Tf = f\}$, and $A_n = \frac{1}{n}(I + T + \cdots + T^{n-1})$. Then for every $f \in H$, $A_n(f) \rightarrow P(f)$ in norm.*

For ergodic theory take $H = L^2(X, \mu)$ and $(Tf)(x) = f(\tau(x))$; then $\|Tf\| = \|f\|$ (an isometry), and $A_n(f)$ is the L^2 average $A_n(f)$ of Definition 29.5.1. The limit operator P factors through the inclusion of the invariant subspace S : it is the orthogonal projection $H \twoheadrightarrow S$ followed by $S \hookrightarrow H$.



Lemma 29.5.5. *With $S = \{f : Tf = f\}$, $S_* = \{f : T^*f = f\}$, and $S_1 = \{g - Tg : g \in H\}$: (i) $S = S_*$; (ii) the orthogonal complement of $\overline{S_1}$ is S .*

Proof. T isometry $\Rightarrow T^*T = I$. If $Tf = f$ then $T^*f = T^*Tf = f$. Conversely $T^*f = f$ gives $(Tf, f) = (f, T^*f) = \|f\|^2$, equality in Cauchy–Schwarz with $\|Tf\| = \|f\|$, forcing $Tf = cf$ and then $Tf = f$. This is (i). For (ii), $f \perp S_1$ iff $(f, g - Tg) = 0$ for all g , i.e. $(f - T^*f, g) = 0$ for all g , i.e. $T^*f = f$, i.e. $f \in S$ by (i). $\square$

Proof of Theorem 29.5.4. Decompose $f = f_0 + f_1$, $f_0 \in S$, $f_1 \in \overline{S_1}$ (orthogonal complements). Fix $\varepsilon > 0$ and $f'_1 = g - Tg \in S_1$ with $\|f_1 - f'_1\| < \varepsilon$. Then $A_n(f_0) = f_0 = P(f)$ for all n ; $A_n(f'_1) = \frac{1}{n}(g - T^n g) \rightarrow 0$ since T is an isometry; and $\|A_n(f_1 - f'_1)\| \leq \|f_1 - f'_1\| < \varepsilon$. Hence $\limsup_n \|A_n(f) - P(f)\| \leq \varepsilon$, and ε was arbitrary. $\square$

Worked Example 29.5.6 (Ergodicity $\Leftrightarrow$ averages converge to the mean). For τ ergodic with $\mu(X) = 1$, identify $P(f)$ in Theorem 29.5.4 and conclude $A_n(f) \rightarrow \int_X f d\mu$ in L^2 .

Recognition. Trigger: “ergodic” means the invariant subspace is just the constants – pin down P .

Solution. Ergodicity says any invariant f (i.e. $Tf = f$, equivalently constant on a.e. orbit) is a.e. constant, so $S = \{\text{constants}\}$. The orthogonal projection onto constants in $L^2(X, \mu)$ with $\mu(X) = 1$ is $P(f) = \langle f, 1 \rangle = \int_X f d\mu$. Thus $A_n(f) \rightarrow \int_X f d\mu$ in norm. *Check.* For $f = 1$, $A_n(1) = 1 = \int_X 1 d\mu$, a fixed point, as required.

Worked Example 29.5.7 (The irrational rotation is ergodic but not mixing). On $(\mathbb{R}/\mathbb{Z}, m)$, $\tau(x) = x + \alpha$ with α irrational. Use Fourier modes to show ergodicity, and exhibit failure of mixing.

Recognition. Trigger: a rotation; test $Tf = f$ on the orthonormal basis $e_m(x) = e^{2\pi i m x}$.

Solution. $Te_m = e^{2\pi i m \alpha} e_m$. If $Tf = f$ with $f = \sum_m c_m e_m$, then $c_m(e^{2\pi i m \alpha} - 1) = 0$; for $m \neq 0$, α irrational gives $e^{2\pi i m \alpha} \neq 1$, so $c_m = 0$. Thus $f = c_0$ is constant, hence τ is ergodic. Not mixing: take $f = g = e_m$,

$m \neq 0$; $(T^n f, g) = e^{2\pi i n m \alpha}$, which does not converge, whereas mixing would force $(T^n f, g) \rightarrow (f, 1)(1, g) = 0$. *Check.* $|e^{2\pi i n m \alpha}| = 1$ never decays, so independence is never approached – the obstruction to mixing is exactly the persistence of these modes.

Worked Example 29.5.8 (The doubling map is mixing). On $((0, 1], m)$, $\tau(x) = 2x \bmod 1$. Show τ is mixing.

Recognition. Trigger: verify $(T^n f, g) \rightarrow (f, 1)(1, g)$ on exponentials, then pass to all of L^2 .

Solution. For $f = e_m$, $g = e_k$, $T^n f(x) = e^{2\pi i m 2^n x}$, so $(T^n f, g) = \int_0^1 e^{2\pi i (m 2^n - k)x} dx$, which is 0 unless $m 2^n = k$. For fixed $m \neq 0$ this fails for all large n , so $(T^n f, g) \rightarrow 0 = (f, 1)(1, g)$; if $m = k = 0$ both sides are 1. By linearity and L^2 -density of trigonometric polynomials, mixing holds for all f, g . *Check.* Mixing $\Rightarrow$ ergodic, recovering that m is ergodic for the doubling map; this underlies normality of almost every binary expansion.

Maximal and pointwise ergodic theorems

The ascent from L^2 mean convergence to almost-everywhere convergence runs, as in differentiation theory, through a maximal inequality.

Theorem 29.5.9 (Maximal ergodic theorem). For $f \in L^1(X, \mu)$ the maximal average $f^*(x) = \sup_{1 \leq m < \infty} \frac{1}{m} \sum_{k=0}^{m-1} |f(\tau^k(x))|$ is finite a.e., and there is a universal A (one may take $A = 1$) with

$$\mu(\{x : f^*(x) > \alpha\}) \leq \frac{A}{\alpha} \|f\|_{L^1} \quad (\alpha > 0).$$

Proof idea. First do $X = \mathbb{Z}$, $\tau : n \mapsto n + 1$: extend f on $\mathbb{Z}$ to a step function $\tilde{f}$ on $\mathbb{R}$ and compare $f^*(n) \leq 2(\tilde{f})^*(x)$ for $x \in [n, n + 1)$ with the Hardy–Littlewood maximal function of Chapter 3, whose weak-type $(1, 1)$ bound (constant 3) transfers the inequality with $A = 6$. The general space is reduced to the integer case by a transference (“sleight of hand”) over long τ -orbits. $\square$

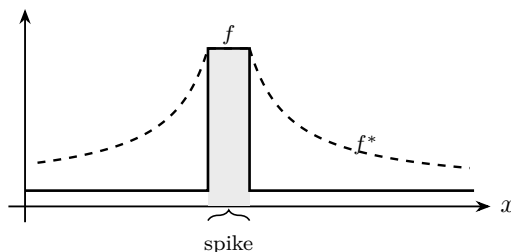


Figure 29.13. The maximal average f^* spreads a spike of f over the orbit; the weak-type bound $\mu(f^* > \alpha) \leq A \|f\|_1 / \alpha$ controls its tails. (Schematic.)

Theorem 29.5.10 (Pointwise (Birkhoff) ergodic theorem). Assume $\mu(X) = 1$. For $f \in L^1(X, \mu)$ and a.e. x , the averages $A_m(f)(x) = \frac{1}{m} \sum_{k=0}^{m-1} f(\tau^k(x))$ converge to a limit $P'(f)(x)$ as $m \rightarrow \infty$, with $\int |P'(f)| d\mu \leq \int |f| d\mu$ and $P'(f) = P(f)$ for $f \in L^2$.

Proof idea. Convergence holds on the dense set $f = F_0 + (1 - T)G$ (F_0 invariant, $G \in L^2$): there $A_m(f) = F_0 + \frac{1}{m}(G - T^m G)$ and $\frac{1}{m} T^m G \rightarrow 0$ a.e. because $\sum_m m^{-2} \|T^m G\|_2^2 = \|G\|_2^2 \sum m^{-2} < \infty$ forces $m^{-1} T^m G \rightarrow 0$ a.e. For general $f \in L^1$ write $f = F + H$ with $\|H\|_1 < \varepsilon$, control the oscillation $E_\alpha = \{\limsup_{n,m} |A_n(f) - A_m(f)| > \alpha\}$ by the maximal inequality applied to H , and let $\varepsilon \rightarrow 0$. $\square$

Worked Example 29.5.11 (Birkhoff = a strong law: digit frequencies and normal numbers). For the doubling map on $((0, 1], m)$ and $f = \mathbf{1}_{[0, 1/2)}$, identify $\lim_m A_m(f)(x)$ and interpret it for binary digits.

Recognition. Trigger: an ergodic average of an indicator – Birkhoff plus ergodicity gives the space mean.

Solution. Since the doubling map is ergodic (Worked Example above), $P'(f) = \int_0^1 f = m([0, 1/2)) = \frac{1}{2}$ a.e. Now $\tau^k(x)$ has first binary digit 0 iff the $(k + 1)$ -st binary digit of x is 0, so $A_m(f)(x) = \frac{1}{m} \#\{0 \leq k < m :$

$a_{k+1}(x) = 0\}$ is the frequency of 0's among the first m digits. Birkhoff gives frequency $\rightarrow \frac{1}{2}$ for a.e. x : almost every number is *simply normal* in base 2. *Check.* $f = 1$ gives $A_m \equiv 1 = \int_0^1 1$; and the two digits 0, 1 have frequencies $\frac{1}{2} + \frac{1}{2} = 1$, as they must.

Worked Example 29.5.12 (Weyl equidistribution from the ergodic average). For the irrational rotation $\tau(x) = x + \alpha$ and continuous periodic f , show $\frac{1}{m} \sum_{k=0}^{m-1} f(x + k\alpha) \rightarrow \int_0^1 f$ for *every* x (unique ergodicity).

Recognition. Trigger: ergodic + continuity $\Rightarrow$ averages converge *everywhere*, not just a.e.

Solution. By linearity and uniform approximation it suffices to check $f = e_n$. For $n = 0$, $A_m = 1 = \int e_0$. For $n \neq 0$, $A_m(e_n)(x) = \frac{e_n(x)}{m} \sum_{k=0}^{m-1} e^{2\pi i n k \alpha} = \frac{e_n(x)}{m} \cdot \frac{e^{2\pi i n m \alpha} - 1}{e^{2\pi i n \alpha} - 1}$, whose modulus is $\leq \frac{2}{m|e^{2\pi i n \alpha} - 1|} \rightarrow 0 = \int e_n$. Uniform approximation of any continuous f by trigonometric polynomials extends the limit to all f and all x . *Check.* Taking f near $\mathbf{1}_{[a,b]}$ recovers Weyl's theorem: $\{k\alpha \bmod 1\}$ is equidistributed, the fraction landing in $[a, b]$ tending to $b - a$.

Worked Example 29.5.13 (The Gauss map preserves Gauss's measure). For $\tau(x) = \{1/x\}$ on $[0, 1)$ and $d\mu = \frac{1}{\ln 2} \frac{dx}{1+x}$, verify $\mu(\tau^{-1}([0, t])) = \mu([0, t])$ for $t \in (0, 1)$.

Recognition. Trigger: a measure-preservation check – compute the preimage and integrate the density.

Solution. The preimages of $[0, t]$ under $x \mapsto \{1/x\}$ are the intervals $[\frac{1}{k+t}, \frac{1}{k}]$, $k \geq 1$ (where $1/x \in [k, k + t)$). So $\mu(\tau^{-1}[0, t]) = \frac{1}{\ln 2} \sum_{k \geq 1} \int_{1/(k+t)}^{1/k} \frac{dx}{1+x} = \frac{1}{\ln 2} \sum_{k \geq 1} [\ln \frac{k+1}{k} - \ln \frac{k+1+t}{k+t}]$, a telescoping sum equal to $\frac{1}{\ln 2} \ln(1+t) = \mu([0, t])$. *Check.* The algebraic heart is $\frac{1}{(x+k)(x+k+1)}$ summing to $\frac{1}{1+x}$; at $t \rightarrow 1$ both sides give $\frac{\ln 2}{\ln 2} = 1 = \mu([0, 1))$.

Caution. Ergodicity gives convergence *almost everywhere* (Birkhoff), but *unique* ergodicity is needed for convergence at *every* point and for continuous test functions – the irrational rotation is uniquely ergodic, but the doubling map is not (it has the extra invariant point mass at 1). Mixing is strictly stronger than ergodicity: the irrational rotation is ergodic yet not mixing. Finally Birkhoff needs only $f \in L^1$ and $\mu(X) < \infty$ for the stated form; the infinite-measure version requires the L^2 maximal inequality and a denseness argument.

Exercises (Stein, Ch.6).

- By reducing to $d = 1$, show that each “rotation” $x \mapsto x + \alpha$ of the torus $\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$ is measure-preserving, for any $\alpha \in \mathbb{R}^d$.
- Let τ be measure-preserving on (X, μ) with $\mu(X) = 1$. Call E invariant if $\tau^{-1}(E)$ and E differ by a null set; a sharper notion requires $\tau^{-1}(E) = E$. Prove that for any invariant E there is E' with $E' = \tau^{-1}(E')$ that differs from E by a null set. [Hint: $E' = \limsup_n \tau^{-n}(E) = \bigcap_n \bigcup_{k \geq n} \tau^{-k}(E)$.]
- Let τ be measure-preserving on (X, μ) with $\mu(X) = 1$. Prove τ is ergodic if and only if: whenever $\nu \ll \mu$ is invariant ($\nu(\tau^{-1}E) = \nu(E)$ for all E), then $\nu = c\mu$ for a constant c .
- Let τ be measure-preserving on (X, μ) . If $\mu(\tau^{-n}(E) \cap F) \rightarrow \mu(E)\mu(F)$ for all measurable E, F , then $(T^n f, g) \rightarrow (f, 1)(1, g)$ whenever $f, g \in L^2(X)$ with $(Tf)(x) = f(\tau(x))$; thus τ is mixing. [Hint: by linearity the hypothesis gives the conclusion for simple f, g .]
- Let $\mathbb{T}^d$ be the torus and $\tau : x \mapsto x + \alpha$. Then τ is ergodic iff $\alpha = (\alpha_1, \dots, \alpha_d)$ has $\alpha_1, \dots, \alpha_d, 1$ linearly independent over $\mathbb{Q}$. Show: (a) $\frac{1}{m} \sum_{k=0}^{m-1} f(\tau^k(x)) \rightarrow \int_{\mathbb{T}^d} f$ as $m \rightarrow \infty$ for every x , whenever f is continuous periodic and α satisfies the hypothesis; (b) conclude τ is uniquely ergodic. [Hint: use Exercise 16(f).]
- Let $X = \prod_{i=1}^{\infty} X_i$ with each (X_i, μ_i) identical to (X_1, μ_1) , $\mu_1(X_1) = 1$, and μ the product measure. Define the shift $\tau((x_1, x_2, \dots)) = (x_2, x_3, \dots)$. (a) Verify τ is measure-preserving. (b) Prove τ is ergodic by showing it is mixing. (c) Note τ is in general not uniquely ergodic. [Hint: for (b), $\mu(\tau^{-n}(E \cap F)) = \mu(E)\mu(F)$ for cylinder sets E, F and n large; for (c) the constant sequences give extra invariant sets.] The two-sided shift is a measure-preserving isomorphism.
- Let $X = \prod_{i=1}^{\infty} Z(2)$, $Z(2) = \{0, 1\}$, $\mu_1(0) = \mu_1(1) = 1/2$, μ the product measure. Consider $D(\{a_j\}) = \sum_{j=1}^{\infty} a_j 2^{-j}$. Show there are countable $Z_1 \subseteq X$, $Z_2 \subseteq [0, 1]$ with: (a) $D : X \setminus Z_1 \rightarrow [0, 1] \setminus Z_2$ a bijection; (b) E measurable in X iff $D(E)$ measurable in $[0, 1]$, with $\mu(E) = m(D(E))$; (c) the shift on X becomes the doubling map.

8. (Generalized doubling.) For an integer $m \geq 2$ define $\tau_m(x) = mx \bmod 1$ on $(0, 1]$. (a) Verify τ_m is measure-preserving for Lebesgue measure. (b) Show τ_m is mixing, hence ergodic. (c) Conclude that almost every x is normal to base m : in $x = \sum_j a_j m^{-j}$ ($0 \leq a_j \leq m-1$), for each digit k , $\frac{1}{N} \#\{j \leq N : a_j = k\} \rightarrow \frac{1}{m}$ as $N \rightarrow \infty$.
9. Show the mean ergodic theorem still holds if T is only a *contraction* ($\|Tf\| \leq \|f\|$) rather than an isometry. [Hint: T is a contraction iff T^* is, and use $(f, T^*f) = (Tf, f)$.]
10. (L^2 maximal ergodic theorem.) Let τ be measure-preserving on (X, μ) (no assumption $\mu(X) < \infty$). Show $f^*(x) = \sup_m \frac{1}{m} \sum_{k=0}^{m-1} |f(\tau^k(x))|$ satisfies $\|f^*\|_{L^2} \leq c \|f\|_{L^2}$ for $f \in L^2(X)$. Use it to extend the pointwise ergodic theorem to $\mu(X) = \infty$: (a) $\frac{1}{m} \sum_{k=0}^{m-1} f(\tau^k(x))$ converges a.e. to $P(f)(x)$ for every $f \in L^2$, since it holds on a dense subspace; (b) the conclusion holds for every $f \in L^1$, since $L^1 \cap L^2$ is dense in L^1 .
11. We saw that $\|f_n\|_{L^2} \leq 1$ implies $f_n(x)/n \rightarrow 0$ a.e. Show the L^1 analogue fails: construct $f_n \in L^1(X)$, $\|f_n\|_{L^1} \leq 1$, with $\limsup_n f_n(x)/n = \infty$ a.e. [Hint: take intervals $I_n \subseteq [0, 1]$ with $m(I_n) = \frac{1}{n \log n}$ but $\limsup_n I_n = [0, 1]$, and $f_n = n \log n \mathbf{1}_{I_n}$.]
12. By the Borel–Cantelli lemma, if $\sum_n \mu(E_n) < \infty$ then $\limsup_n E_n$ is null. Conversely, if τ is a mixing measure-preserving transformation on (X, μ) with $\mu(X) = 1$ and $\sum_n \mu(E_n) = \infty$, show there exist integers m_n such that for $E'_n = \tau^{-m_n}(E_n)$ one has $\limsup_n E'_n = X$ up to a null set.

Problems (Stein, Ch.6).

1. (Torus automorphisms.) Let A be an automorphism of $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ (an integer matrix with $\det A = \pm 1$), $\tau(x) = A(x)$. (a) τ is a measure-preserving isomorphism. (b) τ is ergodic (in fact mixing) iff A has no eigenvalue of the form $e^{2\pi i p/q}$. (c) τ is never uniquely ergodic. [Hint: (b) is equivalent to $(A^t)^q$ having no invariant vectors; note $f(\tau^k(x)) = e^{2\pi i(A^t)^k(n) \cdot x}$ for $f = e^{2\pi i n \cdot x}$.]
2. * (Rising-sun maximal ergodic theorem.) For real f let $f^\#(x) = \sup_m \frac{1}{m} \sum_{k=0}^{m-1} f(\tau^k(x))$ and $E_0 = \{f^\# > 0\}$. Then $\int_{E_0} f dx \geq 0$. Deduce (applying to $f - \alpha$, for $f \geq 0$) that $\mu\{f^\# > \alpha\} \leq \frac{1}{\alpha} \int_{\{f^\# > \alpha\}} f dx$, so the constant in the maximal ergodic theorem may be taken $A = 1$.
3. (Gauss map is measure-preserving.) Let $X = [0, 1]$, $\tau(x) = \{1/x\}$ ($\tau(0) = 0$), $d\mu = \frac{1}{\log 2} \frac{dx}{1+x}$, so $\mu(X) = 1$. Show τ is measure-preserving. [Hint: $\sum_{k=1}^{\infty} \frac{1}{(x+k)(x+k+1)} = \frac{1}{1+x}$.]
4. * The Gauss map τ of the previous problem is ergodic.
5. * (Continued fractions.) A continued fraction $[a_0 a_1 a_2 \dots]$ with positive-integer a_j is assigned to a positive real x by alternately reducing mod 1 and inverting: $x = a_0 + r_0$, $1/r_{n-1} = a_n + r_n$, $a_n = \lfloor 1/r_{n-1} \rfloor$. For $0 \leq x < 1$, $a_k(x) = \lfloor 1/\tau^{k-1}(x) \rfloor$. Known facts: (a) the continued fraction terminates iff x is rational; (b) the truncations $x_N = [a_0 \dots a_N 0 0 \dots] \rightarrow x$ and give essentially optimal rational approximations; (c) it is eventually periodic iff x is a quadratic irrational; (d) $\frac{a_1 + \dots + a_n}{n} \rightarrow \infty$ for a.e. x , so the set of x with bounded partial quotients is null. [Hint for (d): apply the consequence of the pointwise ergodic theorem – for ergodic τ , $f \geq 0$ with $\int f d\mu = \infty$ gives $\frac{1}{m} \sum_{k=0}^{m-1} f(\tau^k(x)) \rightarrow \infty$ a.e.; take $f(x) = \lfloor 1/x \rfloor$.]

Additional exercises.

1. Verify directly that the rotation $\tau(x) = x + \alpha \bmod 1$ on $((0, 1], m)$ is a measure-preserving isomorphism, exhibiting τ^{-1} explicitly.
2. Show the doubling map $\tau(x) = 2x \bmod 1$ satisfies $\int_0^1 f(\tau(x)) dx = \int_0^1 f dx$ for $f(x) = e^{2\pi i k x}$, and explain why this implies measure-preservation for all f .
3. For the rotation by rational $\alpha = p/q$, exhibit a non-trivial invariant set of measure strictly between 0 and 1, proving τ is *not* ergodic.
4. (Mean ergodic theorem.) In $H = \mathbb{C}^2$ let $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (a reflection, hence an isometry). Compute the invariant subspace S , the projection P , and verify $A_n \rightarrow P$ for this T .
5. Prove that mixing implies ergodic directly from the definition (no Hilbert space), and show the converse fails using the irrational rotation.

6. (Birkhoff as a strong law.) For τ the doubling map and $f = \mathbf{1}_{[a,b]} \subseteq [0, 1]$, identify $\lim_m A_m(f)(x)$ for a.e. x and interpret it as a digit-block frequency.
7. Show that for an ergodic τ with $\mu(X) = 1$ and $f \in L^1$, the time average equals the space average a.e.: $\lim_m \frac{1}{m} \sum_{k=0}^{m-1} f(\tau^k(x)) = \int_X f \, d\mu$.
8. (Maximal inequality.) State and prove the weak-type $(1, 1)$ bound for the ergodic maximal function in the special case $X = \mathbb{Z}$, $\tau : n \mapsto n + 1$, by reduction to the Hardy–Littlewood maximal function on $\mathbb{R}$.
9. (Non-example.) Show the doubling map is not uniquely ergodic by exhibiting a second invariant probability measure besides m (e.g. a point mass at a fixed point).
10. (Diagram.) Draw the first three stages of the Cantor set as a self-similar fractal and mark the scaling relation $3C = C \cup (C + 2)$; deduce the Hausdorff dimension $\log 2 / \log 3 \approx 0.631$ from $m = 2$ copies under scaling $n = 3$.

Chapter 30

Hausdorff Measure and Fractals

30.1 Hausdorff Measure

Why a new notion of size

Lebesgue measure on $\mathbb{R}^d$ measures d -dimensional bulk: a smooth curve in the plane has two-dimensional measure zero, the Cantor set $C \subset [0, 1]$ has length zero. Yet these sets are not “nothing”: the curve has a finite length, and C is uncountable. Hausdorff introduced, for each exponent $\alpha > 0$, a notion m_α of α -dimensional mass that interpolates between the integer dimensions and assigns a positive finite size to sets whose “true” dimension is the fractional number α . The guiding heuristic: if α exceeds the dimension of E then $m_\alpha(E) = 0$; if α is below it then $m_\alpha(E) = \infty$; at the critical α the value $m_\alpha(E)$ records the genuine α -dimensional size. Two facts we will earn: the Cantor set has dimension $\log 2 / \log 3$ with $m_\alpha(C) = 1$, and for a rectifiable curve Γ one has $m_1(\Gamma) = \text{length}(\Gamma)$.

The exterior Hausdorff measure

Definition 30.1.1. For $E \subseteq \mathbb{R}^d$ and $\alpha > 0$, and for each $\delta > 0$, set

$$H_\alpha^\delta(E) = \inf \left\{ \sum_{k=1}^{\infty} (\text{diam } F_k)^\alpha : E \subseteq \bigcup_k F_k, \text{diam } F_k \leq \delta \text{ all } k \right\},$$

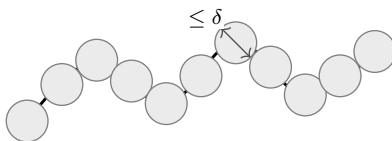
where $\text{diam } S = \sup\{|x - y| : x, y \in S\}$. The infimum runs over all countable covers of E by sets of diameter at most δ . The *exterior α -dimensional Hausdorff measure* is the limit

$$m_\alpha^*(E) = \lim_{\delta \rightarrow 0} H_\alpha^\delta(E) = \sup_{\delta > 0} H_\alpha^\delta(E),$$

which exists in $[0, \infty]$ because $H_\alpha^\delta(E)$ *increases* as δ decreases (shrinking δ restricts the admissible covers, raising the infimum).

The crucial design choice is the requirement $\text{diam } F_k \leq \delta$ with $\delta \rightarrow 0$: we must cover E by sets of *arbitrarily small* diameter. This is precisely what makes m_α^* a metric exterior measure (Property 3 below); a single unrestricted infimum (no δ) would fail additivity, as Exercise 12 shows.

Remark. Scaling is the soul of the definition. If F is scaled by r , then $\text{diam}(rF) = r \text{diam } F$, so $(\text{diam } rF)^\alpha = r^\alpha (\text{diam } F)^\alpha$: the summand scales by r^α . A one-dimensional curve of length L becomes rL under dilation by r (r^1 scaling); a d -cube of volume V becomes $r^d V$ (r^d scaling). The exponent α in $(\text{diam } F_k)^\alpha$ is exactly the dial that records dimension.



cover by sets F_k , each $\text{diam } F_k \leq \delta$; minimize $\sum_k (\text{diam } F_k)^\alpha$

Figure 30.1. A set covered by patches of diameter $\leq \delta$. As $\delta \rightarrow 0$ the cover hugs the set more tightly and H_α^δ rises to m_α^* . (Schematic.)

Basic properties

Proposition 30.1.2 (Elementary properties). *The exterior measure m_α^* satisfies:*

1. (Monotonicity) $E_1 \subseteq E_2 \implies m_\alpha^*(E_1) \leq m_\alpha^*(E_2)$.
2. (Countable subadditivity) $m_\alpha^*(\bigcup_j E_j) \leq \sum_j m_\alpha^*(E_j)$.
3. (Metric additivity) If $\text{dist}(E_1, E_2) > 0$, then $m_\alpha^*(E_1 \cup E_2) = m_\alpha^*(E_1) + m_\alpha^*(E_2)$.

Proof. (1) Any cover of E_2 covers E_1 , so the infimum defining $H_\alpha^\delta(E_1)$ is over a larger family, hence no larger. (2) Fix $\delta, \varepsilon > 0$. For each j choose a cover $\{F_{j,k}\}_k$ of E_j by sets of diameter $\leq \delta$ with $\sum_k (\text{diam } F_{j,k})^\alpha \leq H_\alpha^\delta(E_j) + \varepsilon/2^j$. The doubly-indexed family $\{F_{j,k}\}_{j,k}$ covers $E = \bigcup_j E_j$ by sets of diameter $\leq \delta$, so

$$H_\alpha^\delta(E) \leq \sum_{j,k} (\text{diam } F_{j,k})^\alpha \leq \sum_j H_\alpha^\delta(E_j) + \varepsilon \leq \sum_j m_\alpha^*(E_j) + \varepsilon.$$

Let $\varepsilon \rightarrow 0$, then $\delta \rightarrow 0$. (3) Subadditivity gives $\leq$. For $\geq$, fix ε with $0 < \varepsilon < \text{dist}(E_1, E_2)$ and a cover $\{F_k\}$ of $E_1 \cup E_2$ with $\text{diam } F_k \leq \delta < \varepsilon$. Put $F'_k = E_1 \cap F_k$ and $F''_k = E_2 \cap F_k$. No single F_k can meet both E_1 and E_2 (its diameter is below their separation), so the F'_k form a cover of E_1 , the F''_k a cover of E_2 , and the two index sets are disjoint. Thus $\sum_k (\text{diam } F'_k)^\alpha + \sum_k (\text{diam } F''_k)^\alpha \leq \sum_k (\text{diam } F_k)^\alpha$. Take infima then let $\delta \rightarrow 0$. $\square$

Because m_α^* is a *metric* exterior measure (Property 3), Caratheodory's criterion from Chapter 6 applies: every Borel set is measurable and m_α^* restricts to a countably additive measure on the Borel σ -algebra. We write m_α for this restriction and call it the α -dimensional **Hausdorff measure**.

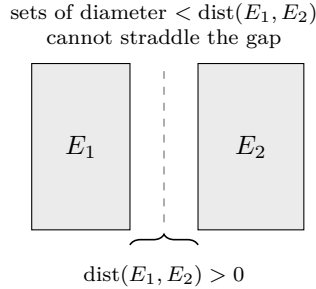


Figure 30.2. Metric additivity: when two sets are positively separated, any fine cover splits cleanly, so masses add. (Schematic.)

Proposition 30.1.3 (Invariance, scaling, and normalization). *Hausdorff measure is translation- and rotation-invariant, and scales by the dimension:*

$$m_\alpha(E + h) = m_\alpha(E), \quad m_\alpha(rE) = m_\alpha(E) \text{ (} r \text{ a rotation)}, \quad m_\alpha(\lambda E) = \lambda^\alpha m_\alpha(E) \text{ (} \lambda > 0 \text{)}.$$

Moreover m_0 is counting measure, and on $\mathbb{R}$ one has $m_1 = m$ (Lebesgue measure). In $\mathbb{R}^d$, $c_d m_d = m$ for a constant $c_d = v_d/2^d$ (v_d the volume of the unit ball); without the iso-diametric inequality one still has $c_d m_d(E) \leq m(E) \leq 2^d c_d m_d(E)$.

Proof. The diameter satisfies $\text{diam}(S + h) = \text{diam } S$, $\text{diam}(rS) = \text{diam } S$, and $\text{diam}(\lambda S) = \lambda \text{diam } S$; substituting into the defining sums gives the three identities. For m_0 : with $\alpha = 0$, $(\text{diam } F_k)^0 = 1$, so $H_0^\delta(E)$ counts the fewest sets of diameter $\leq \delta$ needed to cover E , which for small δ is the number of points of E . For $m_1 = m$ on $\mathbb{R}$: every set of diameter δ sits inside an interval of length δ , whose Lebesgue length equals its diameter, so the Hausdorff sum and the Lebesgue sum coincide in the limit. For the two-sided bound $c_d m_d \leq m \leq 2^d c_d m_d$: given ε, δ , cover E by balls B_j with $\text{diam } B_j < \delta$ and $\sum_j m(B_j) \leq m(E) + \varepsilon$; since $m(B_j) = c_d (\text{diam } B_j)^d$ we get $H_d^\delta(E) \leq c_d^{-1} \sum_j m(B_j) \leq c_d^{-1} (m(E) + \varepsilon)$, giving $m_d(E) \leq c_d^{-1} m(E)$. Conversely cover E by sets F_j with $\sum_j (\text{diam } F_j)^d \leq m_d(E) + \varepsilon$ and enclose each F_j in a ball B_j of diameter $2 \text{diam } F_j$; then $m(E) \leq \sum_j m(B_j) = 2^d c_d \sum_j (\text{diam } F_j)^d \leq 2^d c_d (m_d(E) + \varepsilon)$. $\square$

Theorem 30.1.4 (Dimensional collapse). *If $m_\alpha^*(E) < \infty$ and $\beta > \alpha$, then $m_\beta^*(E) = 0$. Dually, if $m_\alpha^*(E) > 0$ and $\beta < \alpha$, then $m_\beta^*(E) = \infty$.*

Proof. If $\text{diam } F \leq \delta$ and $\beta > \alpha$, then $(\text{diam } F)^\beta = (\text{diam } F)^{\beta-\alpha}(\text{diam } F)^\alpha \leq \delta^{\beta-\alpha}(\text{diam } F)^\alpha$. Summing over a cover, $H_\beta^\delta(E) \leq \delta^{\beta-\alpha} H_\alpha^\delta(E) \leq \delta^{\beta-\alpha} m_\alpha^*(E)$. As $\delta \rightarrow 0$ the prefactor $\delta^{\beta-\alpha} \rightarrow 0$ (since $\beta - \alpha > 0$) and $m_\alpha^*(E) < \infty$, so $m_\beta^*(E) = 0$. The dual statement is the contrapositive. $\square$

This monotonicity in α is exactly the “step function” behavior that lets us *define* dimension in the next section: there is a single critical exponent below which the mass is infinite and above which it is zero.

Method (Bounding $m_\alpha(E)$ from above). *Recognition trigger.* You want $m_\alpha(E) \leq M$ (often to show $m_\alpha(E) < \infty$, hence $\dim E \leq \alpha$).

1. Exhibit, for each small δ , an explicit *economical cover* of E by sets F_k of diameter $\leq \delta$ (in self-similar examples: the pieces C_K of the construction).
2. Compute $\sum_k (\text{diam } F_k)^\alpha$ for that cover; this upper-bounds $H_\alpha^\delta(E)$.
3. If the bound is independent of δ (or $\rightarrow M$), conclude $m_\alpha(E) \leq M$.

Lower bounds $m_\alpha(E) > 0$ are harder and usually need a Lipschitz map onto a set of known positive measure, or a counting argument over “vertices” (Section 30.2).

Worked Example 30.1.5 ($m_\alpha(C) \leq 1$ for the Cantor set at $\alpha = \log 2 / \log 3$). Let $C = \bigcap_k C_k$ be the middle-thirds Cantor set, C_k a union of 2^k intervals of length 3^{-k} . Show $m_\alpha(C) \leq 1$ when $\alpha = \log 2 / \log 3$.

Recognition. Upper bound: use the natural cover $C \subseteq C_K$ and the method above. The exponent α is chosen so $3^\alpha = 2$.

Solution. Fix $\delta > 0$ and pick K with $3^{-K} < \delta$. The 2^K intervals composing C_K each have diameter $3^{-K} < \delta$ and cover C , so $H_\alpha^\delta(C) \leq 2^K (3^{-K})^\alpha = 2^K \cdot 3^{-K\alpha} = (2 \cdot 3^{-\alpha})^K$. Since $3^\alpha = 2$ we have $3^{-\alpha} = 1/2$, hence $2 \cdot 3^{-\alpha} = 1$ and the bound is $1^K = 1$. Letting $\delta \rightarrow 0$ gives $m_\alpha(C) \leq 1$. *Check.* At $\alpha = \log 2 / \log 3 \approx 0.6309$, $3^\alpha = e^{\alpha \log 3} = e^{\log 2} = 2$, confirming the cancellation; a wrong exponent such as $\alpha = 1$ would give $2^K 3^{-K} = (2/3)^K \rightarrow 0$, consistent with $m_1(C) = 0$.

Worked Example 30.1.6 (Hausdorff measure is not σ -finite below the dimension). Show that for $\alpha < d$ the measure m_α on $\mathbb{R}^d$ is not σ -finite (this is Exercise 1).

Recognition. “Show a measure is not σ -finite”: suppose $\mathbb{R}^d = \bigcup_j A_j$ with each $m_\alpha(A_j) < \infty$ and derive a contradiction with a set of positive α -measure.

Solution. Any nonempty open ball $B \subseteq \mathbb{R}^d$ has $m_d(B) > 0$ (since $c_d m_d = m$ and $m(B) > 0$), so by Theorem 30.1.4 with $\beta = \alpha < d$ we get $m_\alpha(B) = \infty$. Were m_α σ -finite, write $B = \bigcup_j (B \cap A_j)$ with $m_\alpha(A_j) < \infty$; then $m_\alpha(B) \leq \sum_j m_\alpha(A_j)$. But each $B \cap A_j$ has finite measure while their union B has infinite measure — not yet a contradiction, so refine: cover B by countably many pieces of finite m_α forces, by Theorem 30.1.4 applied in reverse, that B itself has $m_\beta(B) = 0$ for every $\beta > \alpha$, in particular $m_d(B) = 0$, contradicting $m_d(B) > 0$. *Check.* For $\alpha = 1$, $d = 2$ the plane has infinite length on every disc, which cannot be exhausted by finitely-long pieces — matching intuition.

Worked Example 30.1.7 (Diameter scaling of the Hausdorff sum). Verify directly that the dilate λE satisfies $m_\alpha(\lambda E) = \lambda^\alpha m_\alpha(E)$ by tracking a single cover, for $E = [0, 1]$, $\alpha = 1$, $\lambda = 3$.

Recognition. A scaling identity; track how a cover and its sum transform under $x \mapsto \lambda x$.

Solution. Let $\{F_k\}$ cover $E = [0, 1]$ with $\text{diam } F_k \leq \delta$. Then $\{\lambda F_k\}$ covers $\lambda E = [0, 3]$ with $\text{diam } (\lambda F_k) = 3 \text{diam } F_k \leq 3\delta$, and $\sum_k (\text{diam } \lambda F_k)^\alpha = 3^\alpha \sum_k (\text{diam } F_k)^\alpha$. Taking infima, $H_1^{3\delta}(\lambda E) = 3 H_1^\delta(E)$; letting $\delta \rightarrow 0$, $m_1([0, 3]) = 3 m_1([0, 1])$. *Check.* $m_1 = m$ on $\mathbb{R}$, so this reads $3 = 3 \cdot 1$. The exponent appears as λ^1 because $\alpha = 1$; for the planar square $[0, 1]^2$ with $\alpha = 2$ the same computation gives the factor λ^2 , matching area scaling.

Worked Example 30.1.8 (m_1 of a line segment is its length). Let $I = [(0, 0), (L, 0)]$ be a horizontal segment of length L in $\mathbb{R}^2$. Show $0 < m_1(I) < \infty$ and in fact $m_1(I) = L$.

Recognition. “Compute m_1 of a curve”: upper bound by chopping the segment; lower bound by projecting (a 1-Lipschitz map cannot increase m_1).

Solution. *Upper bound.* Split I into N subsegments of diameter $L/N \leq \delta$; then $H_1^\delta(I) \leq N \cdot (L/N) = L$, so $m_1(I) \leq L$. *Lower bound.* The projection $\pi(x, y) = x$ is 1-Lipschitz and $\pi(I) = [0, L]$, so $m_1([0, L]) \leq m_1(I)$ (a 1-Lipschitz image cannot have larger m_1 , by Lemma 2.2 of Section 30.2). Since $m_1([0, L]) = L$ on the line, $L \leq m_1(I)$. Hence $m_1(I) = L$. *Check.* For $L = 1$, $m_1(I) = 1 \in (0, \infty)$, so the unit segment indeed has strict dimension 1; doubling to $L = 2$ doubles the measure, consistent with $m_1(\lambda I) = \lambda m_1(I)$.

Worked Example 30.1.9 (The primitive measure $\tilde{m}_1$ fails additivity). Define $\tilde{m}_1(E) = \inf\{\sum_k \text{diam } F_k : E \subseteq \bigcup_k F_k\}$ with no bound on the diameters. Show $\tilde{m}_1$ is not additive on separated sets (Exercise 12).

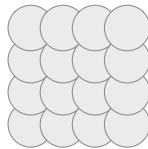
Recognition. A non-example that motivates the $\delta \rightarrow 0$ requirement: drop it and additivity dies.

Solution. Let I_1, I_2 be two parallel unit segments in $\mathbb{R}^2$ with $I_1 = I_2 + h$, $|h| < \varepsilon$. Each has $\tilde{m}_1(I_i) = 1$ (cover by one set of diameter 1; cannot do better). But cover $I_1 \cup I_2$ by a single set F enclosing both: its diameter is at most $1 + \varepsilon$, so $\tilde{m}_1(I_1 \cup I_2) \leq 1 + \varepsilon$. For $\varepsilon < 1$ this gives $\tilde{m}_1(I_1 \cup I_2) < 2 = \tilde{m}_1(I_1) + \tilde{m}_1(I_2)$, so $\tilde{m}_1$ is strictly subadditive, never additive. *Check.* The genuine m_1 uses arbitrarily small diameters, which forbids the single fat cover F ; that is exactly why the $\delta \rightarrow 0$ clause is built into Definition 30.1.1.

Worked Example 30.1.10 (m_d and Lebesgue measure agree up to a constant). For $E = [0, 1]^2 \subseteq \mathbb{R}^2$, confirm the bound $c_2 m_2(E) \leq m(E) \leq 2^2 c_2 m_2(E)$ of Proposition 30.1.3 is consistent with $m_2(E)$ being comparable to area.

Recognition. “Relate d -dimensional Hausdorff to Lebesgue”: use the ball-cover argument with $c_d = v_d/2^d$.

Solution. Here $d = 2$, $m(E) = 1$ (area of the unit square), and $c_2 = v_2/2^2 = \pi/4$. The two-sided estimate reads $\frac{\pi}{4} m_2(E) \leq 1 \leq 4 \cdot \frac{\pi}{4} m_2(E) = \pi m_2(E)$, i.e. $1/\pi \leq m_2(E) \leq 4/\pi$. So $m_2(E)$ is finite and positive, and the unit square has strict dimension 2. *Check.* The exact value (with the iso-diametric inequality) is $m_2(E) = m(E)/c_2 = 4/\pi \approx 1.273$, which indeed lies in $[1/\pi, 4/\pi]$; the constant $c_2 = \pi/4$ is exactly the area-to-diameter² ratio of a disc.



cover $[0, 1]^2$ by 16 balls of diameter δ

Figure 30.3. Covering the square by balls of diameter δ gives $H_2^\delta \approx \sum (\text{diam})^2 \approx m/c_2$, the link $c_d m_d = m$ in $\mathbb{R}^d$.

Caution. Two traps. First, $\text{diam } F$ uses the *ambient* metric, not the intrinsic length of a curve — a long curve can have small diameter. Second, the exponent α must match the geometry: computing H_α^δ with the “wrong” α gives 0 or ∞ , and only the critical α returns a finite positive answer. When in doubt, get the upper bound (easy, via an explicit cover) first, then fight for the lower bound separately.

Exercises (Stein, Ch.7).

1. Show that the measure m_α is not σ -finite on $\mathbb{R}^d$ if $\alpha < d$.
2. Suppose E_1, E_2 are compact subsets of $\mathbb{R}^d$ with $E_1 \cap E_2$ at most one point. Show directly from the definition of the exterior measure that if $0 < \alpha \leq d$ and $E = E_1 \cup E_2$, then $m_\alpha^*(E) = m_\alpha^*(E_1) + m_\alpha^*(E_2)$. [Hint: if $E_1 \cap E_2 = \{x\}$, let B_ε be the open ball of diameter ε at x and $E^\varepsilon = E \cap B_\varepsilon^c$; show $m_\alpha^*(E^\varepsilon) \geq H_\alpha^\varepsilon(E) \geq m_\alpha^*(E) - \mu(\varepsilon) - \varepsilon^\alpha$ with $\mu(\varepsilon) \rightarrow 0$, hence $m_\alpha^*(E^\varepsilon) \rightarrow m_\alpha^*(E)$.]
3. Define a primitive one-dimensional measure $\tilde{m}_1 = \inf \sum_k \text{diam } F_k$ over covers $E \subseteq \bigcup_k F_k$, with no restriction on the diameters $\text{diam } F_k$. For two disjoint unit segments $I_1 = I_2 + h$ in $\mathbb{R}^d$ ($d \geq 2$) with $|h| < \varepsilon$, observe $\tilde{m}_1(I_1) = \tilde{m}_1(I_2) = 1$ while $\tilde{m}_1(I_1 \cup I_2) \leq 1 + \varepsilon$. Conclude $\tilde{m}_1(I_1 \cup I_2) < \tilde{m}_1(I_1) + \tilde{m}_1(I_2)$ when $\varepsilon < 1$, so $\tilde{m}_1$ is not additive. (Stein Exercise 12.)

Problems (Stein, Ch.7).

- (Iso-diametric inequality.) If $E \subseteq \mathbb{R}^d$ is bounded with diameter $\text{diam } E$, then $m(E) \leq v_d(\text{diam } E/2)^d$, where v_d is the volume of the unit ball; among sets of given diameter the ball has maximal volume. (a) Prove it when E is symmetric ($-x \in E$ whenever $x \in E$). For the general case use Steiner symmetrization: for a unit vector e and plane $P \perp e$, set $S(E, e) = \{x + te : x \in P, |t| \leq \frac{1}{2}L(E; e; x)\}$ where $L(E; e; x) = m\{t : x + te \in E\}$. (b) Show $S(E, e)$ is bounded measurable with $m(S(E, e)) = m(E)$ [Fubini]. (c) Show $\text{diam } S(E, e) \leq \text{diam } E$. (d) If a rotation ρ fixes E and P , then $\rho S(E, e) = S(E, e)$. (e) With the standard basis, iterate $E_0 = E$, $E_j = S(E_{j-1}, e_j)$; use that E_d is symmetric to conclude the inequality. (f) Deduce $m(E) = \frac{v_d}{2^d} m_d(E)$ for any Borel $E \subseteq \mathbb{R}^d$.

Additional exercises.

- Compute $H_1^\delta([0, 1])$ exactly for a fixed $\delta > 0$ by covering $[0, 1]$ with $\lceil 1/\delta \rceil$ intervals of equal length, and confirm $H_1^\delta([0, 1]) \rightarrow 1$ as $\delta \rightarrow 0$.
- Let $E = \{0\} \cup \{1/n : n \geq 1\} \subseteq \mathbb{R}$. Show $m_\alpha(E) = 0$ for every $\alpha > 0$ and hence $\dim E = 0$, even though E is infinite.
- Show that the union of two segments meeting at one point, $I_1 \cup I_2$ with $I_1 \cap I_2 = \{p\}$, satisfies $m_1(I_1 \cup I_2) = m_1(I_1) + m_1(I_2)$, although $\text{dist}(I_1, I_2) = 0$. (Why does Property 3 not directly apply, and why does the conclusion still hold?)
- Prove the elementary observation that for any open nonempty $O \subseteq \mathbb{R}^d$ and $\alpha < d$, $m_\alpha(O) = \infty$. (Use $m_d(O) > 0$ and Theorem 30.1.4.)
- Let Q be a k -dimensional cube in $\mathbb{R}^d$ (a product of k nontrivial intervals and $d - k$ points). Show $0 < m_k(Q) < \infty$, so $\dim Q = k$.
- (Non-example.) Exhibit a bounded set $E \subseteq \mathbb{R}$ with $m_1(E) = 0$ but E uncountable, and conclude $\dim E$ can be anything in $[0, 1]$ depending on the construction. (Use a fat/thin Cantor set.)
- (Diagram.) Sketch a covering of the boundary circle of the unit disc by N arcs of diameter $\delta \approx 2\pi/N$, and use it to give the upper bound $m_1(\text{circle}) \leq 2\pi$. Explain what extra step is needed for the matching lower bound.
- Show that if $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is an isometry (distance preserving), then $m_\alpha(f(E)) = m_\alpha(E)$ for every α and every Borel E .
- Prove $m_\alpha(\lambda E) = \lambda^\alpha m_\alpha(E)$ implies that the only α for which a dilation-invariant nonzero finite value is possible on a strictly self-similar set is its dimension.
- Let $E_1 \subseteq E_2 \subseteq \dots$ increase to $E = \bigcup_k E_k$. Prove $m_\alpha(E) = \lim_k m_\alpha(E_k)$ (continuity from below for Hausdorff measure on Borel sets).

30.2 Hausdorff Dimension and Self-Similarity

Defining dimension

Definition 30.2.1. For a Borel set $E \subseteq \mathbb{R}^d$, Theorem 30.1.4 gives a unique critical exponent

$$\alpha = \sup\{\beta : m_\beta(E) = \infty\} = \inf\{\beta : m_\beta(E) = 0\}, \quad m_\beta(E) = \begin{cases} \infty & \beta < \alpha, \\ 0 & \beta > \alpha. \end{cases}$$

We call $\alpha = \dim E$ the *Hausdorff dimension* of E . At the critical value all of $0 \leq m_\alpha(E) \leq \infty$ is possible; if E is bounded and $0 < m_\alpha(E) < \infty$ we say E has *strict* Hausdorff dimension α . A set of non-integer dimension is called a *fractal*.

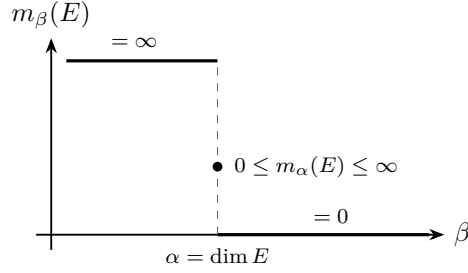


Figure 30.4. $m_\beta(E)$ is ∞ for $\beta < \alpha$ and 0 for $\beta > \alpha$; the jump locates $\dim E = \alpha$. (Schematic.)

Lipschitz maps and the dimension bound

Definition 30.2.2. A map f on $E \subseteq \mathbb{R}^d$ is *Lipschitz* if $|f(x) - f(y)| \leq M|x - y|$ for some M , and *Lipschitz of exponent γ* (or *Holder γ*) if $|f(x) - f(y)| \leq M|x - y|^\gamma$. The interesting range is $0 < \gamma \leq 1$ (Exercise 3: exponent > 1 forces f constant on an interval).

Lemma 30.2.3 (Holder maps shrink dimension). *If f on a compact E is Lipschitz of exponent γ , then with $\beta = \alpha/\gamma$,*

$$m_\beta(f(E)) \leq M^\beta m_\alpha(E), \quad \text{hence} \quad \dim f(E) \leq \frac{1}{\gamma} \dim E.$$

Proof. Let $\{F_k\}$ cover E . Then $\{f(E \cap F_k)\}$ covers $f(E)$, and $\text{diam } f(E \cap F_k) \leq M(\text{diam } F_k)^\gamma$, since f is Holder γ . Therefore

$$\sum_k (\text{diam } f(E \cap F_k))^\beta \leq M^{\alpha/\gamma} \sum_k (\text{diam } F_k)^\alpha.$$

Taking infima over covers and $\delta \rightarrow 0$ gives $m_\beta(f(E)) \leq M^\beta m_\alpha(E)$ with $\beta = \alpha/\gamma$. If $m_\alpha(E) < \infty$ then $m_\beta(f(E)) < \infty$, so $\dim f(E) \leq \beta = \dim E/\gamma$. $\square$

In particular a genuine ($\gamma = 1$) Lipschitz map never raises dimension, and a bijective bi-Lipschitz map preserves it. Dimension is thus a Lipschitz invariant — much coarser than a diffeomorphism invariant, which is why fractional values survive.

$$\begin{array}{ccc} E \cap F_k & \hookrightarrow & E \\ f \downarrow & & \downarrow f \\ f(E \cap F_k) & \hookrightarrow & f(E) \end{array}$$

The Cantor set

Theorem 30.2.4 (Dimension of the Cantor set). *The middle-thirds Cantor set C has strict Hausdorff dimension $\alpha = \log 2 / \log 3$.*

Proof. The bound $m_\alpha(C) \leq 1$ was proved in Worked Example in Section 30.1: cover C by the 2^K intervals of C_K and use $3^\alpha = 2$. For the reverse bound $m_\alpha(C) > 0$ we use the Cantor–Lebesgue function $F : C \rightarrow [0, 1]$, which is surjective. We claim F is Holder of exponent $\gamma = \alpha = \log 2 / \log 3$ (Lemma 30.2.5 below). Granting this, Lemma 30.2.3 with $E = C$, $\gamma = \alpha$, $\beta = \alpha/\gamma = 1$ gives

$$1 = m_1([0, 1]) = m_1(F(C)) \leq M^1 m_\alpha(C),$$

so $m_\alpha(C) \geq M^{-1} > 0$. Combined with $m_\alpha(C) \leq 1$ this yields $0 < m_\alpha(C) \leq 1$, i.e. C has strict dimension $\alpha = \log 2 / \log 3$. (With more care $m_\alpha(C) = 1$ exactly; Exercise 7.) $\square$

Lemma 30.2.5 (Holder continuity of the Cantor–Lebesgue function). *The Cantor–Lebesgue function F satisfies $|F(x) - F(y)| \leq M|x - y|^\gamma$ with $\gamma = \log 2 / \log 3$.*

Proof. $F = \lim F_n$ where F_n is piecewise linear; F_n rises by at most 2^{-n} across each interval of length 3^{-n} , so its slope is $\leq (3/2)^n$, giving $|F_n(x) - F_n(y)| \leq (3/2)^n |x - y|$. Also $|F(x) - F_n(x)| \leq 2^{-n}$. By the triangle inequality,

$$|F(x) - F(y)| \leq |F_n(x) - F_n(y)| + 2 \cdot 2^{-n} \leq (3/2)^n |x - y| + 2^{1-n}.$$

Choose n so that $3^n |x - y| \in [1, 3]$ (balancing the two terms). Then $|F(x) - F(y)| \leq c 2^{-n} = c(3^{-n})^\gamma \leq M |x - y|^\gamma$, using $3^\gamma = 2$ and $3^{-n} \leq |x - y|$. This balancing argument recurs as Lemma 30.2.10 below. $\square$

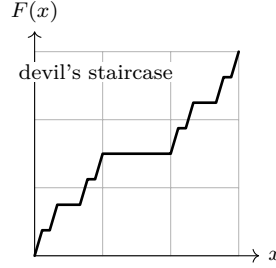


Figure 30.5. The Cantor–Lebesgue function (stage 3): flat off C , Holder of exponent $\log 2/\log 3$ on C ; it transfers positive m_α -mass to $m_1([0, 1])$.

Rectifiable curves

Theorem 30.2.6 (Length is one-dimensional measure). *Let $\gamma : [a, b] \rightarrow \mathbb{R}^d$ be continuous and quasi-simple (injective off finitely many points). Then γ is rectifiable iff $\Gamma = \gamma([a, b])$ has strict Hausdorff dimension 1, and in that case $\text{length}(\gamma) = m_1(\Gamma)$.*

Proof. $\text{Length} \geq m_1$. If Γ has length L , parametrize by arc length $\tilde{\gamma} : [0, L] \rightarrow \mathbb{R}^d$; then $|\tilde{\gamma}(t_1) - \tilde{\gamma}(t_2)| \leq |t_1 - t_2|$ (chord $\leq$ arc), so $\tilde{\gamma}$ is 1-Lipschitz and Lemma 30.2.3 ($\gamma = M = 1$) gives $m_1(\Gamma) \leq L$. $m_1 \geq \text{length}$. Take a partition $a = t_0 < \dots < t_N = b$, $\Gamma_j = \gamma([t_j, t_{j+1}])$. By quasi-simplicity the Γ_j are disjoint off finitely many points (which have $m_1 = 0$), so Property 4 of Hausdorff measure gives $m_1(\Gamma) = \sum_j m_1(\Gamma_j)$. Rotating coordinates so the chord $[\gamma(t_j), \gamma(t_{j+1})]$ lies on the x -axis as $[0, \ell_j]$, the projection $\pi(x, y) = x$ is 1-Lipschitz and $[0, \ell_j] \subseteq \pi(\Gamma_j)$, so by Lemma 30.2.3 $\ell_j \leq m_1(\Gamma_j)$. Summing, $\sum_j \ell_j \leq m_1(\Gamma)$; the supremum over partitions of $\sum_j \ell_j$ is the length L , so $L \leq m_1(\Gamma)$. Hence $m_1(\Gamma) = L$. If conversely $\dim \Gamma = 1$ strictly, $m_1(\Gamma) < \infty$ and the same estimate bounds all $\sum_j \ell_j$, so Γ is rectifiable. $\square$

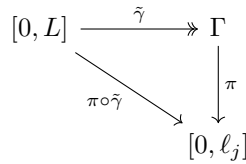


Figure 30.6. Rectifiability bounds. The arc-length parametrization $\tilde{\gamma}$ is 1-Lipschitz, so $m_1(\Gamma) \leq L$; the coordinate projection π is 1-Lipschitz onto a chord $[0, \ell_j]$, so $\ell_j \leq m_1(\Gamma_j)$. Both maps shrink m_1 , pinching $m_1(\Gamma) = \text{length}(\gamma)$.

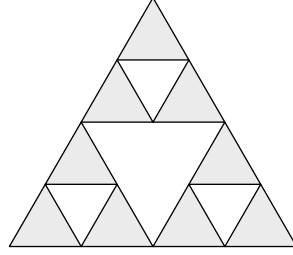
The Sierpinski triangle and the von Koch curve

Theorem 30.2.7 (Sierpinski triangle). *The Sierpinski triangle $S = \bigcap_k S_k$ (where S_k is a union of 3^k triangles of side 2^{-k}) has strict Hausdorff dimension $\alpha = \log 3/\log 2$.*

Proof sketch. Upper bound. Cover S by the 3^K triangles of S_K , each of diameter $2^{-K} < \delta$: $H_\alpha^\delta(S) \leq 3^K (2^{-K})^\alpha = (3 \cdot 2^{-\alpha})^K = 1$ since $2^\alpha = 3$. Hence $m_\alpha(S) \leq 1$. *Lower bound.* Mark in each k th-generation triangle its lower-left vertex; all 3^k such vertices lie in S . Given a finite ball cover $\{B_j\}$ with $\text{diam } B_j < \delta$, let N_ℓ count balls with $2^{-\ell} \leq \text{diam } B_j < 2^{-\ell+1}$. A ball of that size contains at most $c 3^{k-\ell}$ of the k th-generation vertices (Lemma 30.2.8). Since all 3^k vertices must be covered, $c \sum_\ell N_\ell 3^{k-\ell} \geq 3^k$, i.e. $c \sum_\ell N_\ell 3^{-\ell} \geq 1$. As $2^{-\ell\alpha} = 3^{-\ell}$, this gives $\sum_j (\text{diam } B_j)^\alpha \geq \sum_\ell N_\ell 2^{-\ell\alpha} \geq c > 0$, whence $m_\alpha(S) > 0$. $\square$

Lemma 30.2.8 (Vertex-counting in a ball). *A ball B with $2^{-\ell} \leq \text{diam } B < 2^{-\ell+1}$ ($\ell \leq k$) contains at most $c3^{k-\ell}$ vertices of the k th generation.*

Proof. Let B^* be the concentric ball of triple diameter. If a k th-generation triangle Δ_k with vertex $v \in B$ lies in the ℓ th-generation triangle Δ'_ℓ , then $v \in \Delta_k \subseteq \Delta'_\ell \subseteq B^*$ (since $\text{diam } B \geq 2^{-\ell}$). The ℓ th-generation triangles have disjoint interiors and area $\approx 4^{-\ell}$, while B^* has area $\leq c''4^{-\ell}$, so B^* meets at most c of them. Each Δ'_ℓ contains $3^{k-\ell}$ triangles (hence vertices) of generation k . Thus B holds at most $c3^{k-\ell}$ such vertices. $\square$



S_2 : $3^2 = 9$ triangles of side 2^{-2}

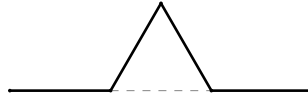
Figure 30.7. Second stage of the Sierpinski triangle; $\dim S = \log 3 / \log 2 \approx 1.585$.

Theorem 30.2.9 (von Koch curve). *The von Koch curve $K = \lim K_j$ (K_j has 4^j segments of length 3^{-j}) is the image of a function $K(t)$ that is Hölder of exponent $\gamma = \log 3 / \log 4$, and $\dim K = \log 4 / \log 3$.*

Proof. The polygonal K_j travels 3^{-j} in time 4^{-j} , so $|K'_j(t)| \leq (4/3)^j$ off the vertices, giving $|K_j(t) - K_j(s)| \leq (4/3)^j |t - s|$; also $|K_{j+1}(t) - K_j(t)| \leq 3^{-j}$. Lemma 30.2.10 (with $A = 4/3$, $B = 3$) yields $|K(t) - K(s)| \leq M |t - s|^\gamma$ where $\gamma = \log B / \log(AB) = \log 3 / \log 4$. By Lemma 30.2.3, $\dim K \leq 1/\gamma = \log 4 / \log 3$. The reverse $m'_\gamma(K) > 0$ (with $\gamma' = 1/\gamma$) follows from the separated self-similar theory below; thus $\dim K = \log 4 / \log 3 \approx 1.262$. $\square$

Lemma 30.2.10 (Telescoping Hölder lemma). *Suppose continuous f_j on $[0, 1]$ satisfy $|f_j(t) - f_j(s)| \leq A^j |t - s|$ ($A > 1$) and $|f_j(t) - f_{j+1}(t)| \leq B^{-j}$ ($B > 1$). Then $f = \lim f_j$ exists and is Hölder of exponent $\gamma = \log B / \log(AB)$.*

Proof. Uniform convergence: $|f(t) - f_j(t)| \leq \sum_{k \geq j} B^{-k} \leq cB^{-j}$. Hence $|f(t) - f(s)| \leq |f_j(t) - f_j(s)| + 2cB^{-j} \leq c(A^j |t - s| + B^{-j})$. Choose j with $(AB)^j |t - s| \leq 1 \leq (AB)^{j+1} |t - s|$ (possible since $|t - s| \leq 1$, $AB > 1$). The first inequality gives $A^j |t - s| \leq B^{-j}$; raising the second to the power γ and using $(AB)^\gamma = B$ gives $1 \leq B^j |t - s|^\gamma$, i.e. $B^{-j} \leq |t - s|^\gamma$. Therefore $|f(t) - f(s)| \leq 2cB^{-j} \leq M |t - s|^\gamma$. $\square$



K_1 : 4 segments of length $1/3$

Figure 30.8. First stage K_1 of the von Koch curve; iterating gives a curve of dimension $\log 4 / \log 3$, between 1 and 2.

Self-similarity

Definition 30.2.11. A map $\mathcal{S} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a *similarity with ratio r* if $|\mathcal{S}(x) - \mathcal{S}(y)| = r|x - y|$; every similarity is a translation composed with a rotation and a dilation by r . Given similarities $\mathcal{S}_1, \dots, \mathcal{S}_m$ of common ratio r , a set F is *self-similar* if $F = \mathcal{S}_1(F) \cup \dots \cup \mathcal{S}_m(F)$.

For the Cantor set: $\mathcal{S}_1(x) = x/3$, $\mathcal{S}_2(x) = x/3 + 2/3$, $m = 2$, $r = 1/3$. For the Sierpinski triangle: $m = 3$, $r = 1/2$. For von Koch: $m = 4$, $r = 1/3$ (with two rotations by $\pm\pi/3$). The Cantor dust $D = C_\xi \times C_\xi$ uses $m = 4$ corner squares of ratio μ .

Theorem 30.2.12 (Existence of self-similar sets). *If $\mathcal{S}_1, \dots, \mathcal{S}_m$ are similarities of common ratio $0 < r < 1$, there is a unique nonempty compact F with $F = \bigcup_j \mathcal{S}_j(F)$.*

Proof. Choose $R \geq |\mathcal{S}_j(0)|/(1-r)$ for all j ; then for $|x| \leq R$, $|\mathcal{S}_j(x)| \leq r|x| + |\mathcal{S}_j(0)| \leq R$, so the closed ball $B = B(0, R)$ satisfies $\mathcal{S}_j(B) \subseteq B$. Let $\tilde{\mathcal{S}}(A) = \bigcup_j \mathcal{S}_j(A)$ and $F_k = \tilde{\mathcal{S}}^k(B)$. Then $F_k \subseteq F_{k-1}$ are compact nonempty, so $F = \bigcap_k F_k$ is compact nonempty and $\tilde{\mathcal{S}}(F) = F$. Uniqueness: on compact sets the Hausdorff distance $\text{dist}(A, B) = \inf\{\delta : A \subseteq B^\delta, B \subseteq A^\delta\}$ is a metric with $\text{dist}(\tilde{\mathcal{S}}A, \tilde{\mathcal{S}}B) \leq r \text{dist}(A, B)$. If $\tilde{\mathcal{S}}G = G$ too, then $\text{dist}(F, G) \leq r \text{dist}(F, G)$ forces $\text{dist}(F, G) = 0$, so $F = G$. $\square$

Definition 30.2.13. The similarities are *separated* if there is a bounded open set O with $O \supseteq \bigcup_j \mathcal{S}_j(O)$ and the $\mathcal{S}_j(O)$ pairwise disjoint (O need not contain F).

Theorem 30.2.14 (Dimension of separated self-similar sets). *If $\mathcal{S}_1, \dots, \mathcal{S}_m$ are separated similarities of common ratio $0 < r < 1$, then $\dim F = \frac{\log m}{\log(1/r)}$.*

Proof. Upper bound (no separation needed). $F_k = \tilde{\mathcal{S}}^k(B)$ is a union of m^k sets $\mathcal{S}_{n_1} \circ \dots \circ \mathcal{S}_{n_k}(B)$ of diameter $\leq cr^k$. With $\alpha = \log m / \log(1/r)$ one has $mr^\alpha = 1$, so for $cr^k \leq \delta$, $H_\alpha^\delta(F) \leq \sum (\text{diam} \dots)^\alpha \leq c' m^k r^{\alpha k} = c' (mr^\alpha)^k = c'$, giving $m_\alpha(F) \leq c' < \infty$ and $\dim F \leq \alpha$. Lower bound (uses separation). Fix $x \in F$ and define the m^k “vertices” $\mathcal{S}_{n_1} \circ \dots \circ \mathcal{S}_{n_k}(x) \in F$ and the disjoint “open sets” $\mathcal{S}_{n_1} \circ \dots \circ \mathcal{S}_{n_k}(O)$, with $d(v, O(v)) \leq cr^k$ and $\text{diam } O(v) \approx r^k$. As for Sierpinski, a ball B with $r^\ell \leq \text{diam } B < r^{\ell-1}$ contains at most $cm^{k-\ell}$ vertices (Lemma 30.2.15). Letting N_ℓ count cover-balls of that size, $c \sum_\ell N_\ell m^{k-\ell} \geq m^k$, so $\sum_\ell N_\ell m^{-\ell} \geq c$. Since $r^{\ell\alpha} = m^{-\ell}$, $\sum_\ell (\text{diam } B_j)^\alpha \geq \sum_\ell N_\ell r^{\ell\alpha} \geq c > 0$, so $m_\alpha(F) > 0$ and $\dim F = \alpha$. $\square$

Lemma 30.2.15 (Vertex-counting, general case). *A ball B with $r^\ell \leq \text{diam } B < r^{\ell-1}$ ($\ell \leq k$) contains at most $cm^{k-\ell}$ vertices of the k th generation.*

Proof. For a vertex $v \in B$ with open set $O(v)$, a fixed dilate B^* of B contains $O(v)$ and the generation- ℓ open set above it (by $d(v, O(v)) \leq cr^k$ and $\text{diam } O(v) \approx r^k$). The generation- ℓ open sets are disjoint with volume $\approx r^{d\ell}$, and B^* has volume $\leq cr^{d\ell}$, so B^* meets at most c of them; each contains $m^{k-\ell}$ generation- k open sets. Hence $\leq cm^{k-\ell}$ vertices. $\square$

Worked Example 30.2.16 (Sierpinski dimension by the formula). Confirm $\dim S = \log 3 / \log 2$ from Theorem 30.2.14, identifying m , r , and a separating open set.

Recognition. “Compute the dimension of a self-similar set”: read off m and r , check separation, apply $\dim = \log m / \log(1/r)$.

Solution. The Sierpinski triangle is $S = \mathcal{S}_1(S) \cup \mathcal{S}_2(S) \cup \mathcal{S}_3(S)$ with three half-scale maps, so $m = 3$, $r = 1/2$. The open unit (filled) triangle O contains its three half-size corner copies disjointly, so the maps are separated. Thus $\dim S = \log 3 / \log(1/(1/2)) = \log 3 / \log 2 \approx 1.585$. *Check.* This lies strictly between 1 (a curve) and 2 (an area), matching a set that is more than a curve but has zero area ($m_2(S) = 0$ since $\dim S < 2$).

Worked Example 30.2.17 (Dimension of the Cantor dust). The Cantor dust keeps the four corner squares of side μ at each stage. Find $\dim D$ and specialize to $\mu = 1/4$.

Recognition. Self-similar with $m = 4$ corner maps of ratio μ ; separated by the open unit square.

Solution. $D = \bigcup_{j=1}^4 \mathcal{S}_j(D)$ with $\mathcal{S}_j$ contracting by μ into the four corners, which are disjoint inside the open unit square, so $\dim D = \log 4 / \log(1/\mu)$. For $\mu = 1/4$: $\dim D = \log 4 / \log 4 = 1$. *Check.* $D = C_\xi \times C_\xi$ with $\xi = 1 - 2\mu$; for $\mu = 1/4$, $\xi = 1/2$ and each factor $C_{1/2}$ has dimension $\log 2 / \log(2/(1-\xi)) = \log 2 / \log 4 = 1/2$, and $1/2 + 1/2 = 1$ agrees with the product rule $\dim(A \times B) = \dim A + \dim B$ for these sets.

Worked Example 30.2.18 (Holder exponent > 1 forces constancy). Show that if $f : [0, 1] \rightarrow \mathbb{R}$ is Holder of exponent $\gamma > 1$, then f is constant (Exercise 3).

Recognition. “Exponent above 1 kills the function”: estimate the difference quotient and let the mesh go to zero.

Solution. For $x < y$ split $[x, y]$ into N equal pieces $x = u_0 < \dots < u_N = y$ of length $(y-x)/N$. Then $|f(y) - f(x)| \leq \sum_i |f(u_{i+1}) - f(u_i)| \leq N \cdot M((y-x)/N)^\gamma = M(y-x)^\gamma N^{1-\gamma}$. Since $\gamma > 1$, $N^{1-\gamma} \rightarrow 0$ as $N \rightarrow \infty$, so $|f(y) - f(x)| = 0$. Hence f is constant. *Check.* This is exactly why only $0 < \gamma \leq 1$ is interesting in Definition 30.2.2; at $\gamma = 1$ the bound is $M(y-x)$, the ordinary Lipschitz estimate, which does not force constancy.

Worked Example 30.2.19 (Dimension of a finite or countable set). Show that any countable set $E \subseteq \mathbb{R}^d$ has $\dim E = 0$, and that monotonicity gives $\dim \bigcup_k E_k \leq \sup_k \dim E_k$ (Exercise 6).

Recognition. “Dimension of a small set”: single points have measure zero in every positive exponent; assemble countably.

Solution. For a single point $\{p\}$ and any $\alpha > 0$, cover by one ball of diameter δ : $H_\alpha^\delta(\{p\}) \leq \delta^\alpha \rightarrow 0$, so $m_\alpha(\{p\}) = 0$ and $\dim\{p\} = 0$. A countable $E = \{p_1, p_2, \dots\}$ has, by countable subadditivity, $m_\alpha(E) \leq \sum_k m_\alpha(\{p_k\}) = 0$ for every $\alpha > 0$, so $\dim E = 0$. If $\dim E_k \leq \alpha$ for all k then for any $\beta > \alpha$, $m_\beta(E_k) = 0$, so $m_\beta(\bigcup_k E_k) \leq \sum_k m_\beta(E_k) = 0$, giving $\dim \bigcup_k E_k \leq \alpha$. *Check.* $\mathbb{Q} \cap [0, 1]$ is countable, hence dimension 0, yet it is dense — dimension is not topological.

Worked Example 30.2.20 (Dimension is preserved by a polynomial). Let $f(x) = x^k$ on $\mathbb{R}$, $k \geq 1$, and E a bounded Borel set. Show $\dim f(E) = \dim E$ (Exercise 5).

Recognition. Locally Lipschitz both ways on a bounded set; apply Lemma 30.2.3 twice.

Solution. On a bounded interval $[-R, R] \supseteq E$, f has bounded derivative, so f is Lipschitz ($\gamma = 1$) there, giving $\dim f(E) \leq \dim E$. Where $f' \neq 0$, f has a Lipschitz local inverse, so $\dim E = \dim f^{-1}(f(E)) \leq \dim f(E)$; the finitely many critical points (only $x = 0$) form a dimension-zero set and cannot raise the dimension. Hence $\dim f(E) = \dim E$. Likewise if $m_\alpha(E) = 0$ then $m_\alpha(f(E)) = 0$. *Check.* For $E = C$ the Cantor set and $f(x) = x^2$, $\dim f(C) = \log 2 / \log 3$ still — squaring cannot create or destroy fractional dimension.

Worked Example 30.2.21 (Totally disconnected when $\dim E < 1$). Show that a Borel set $E \subseteq \mathbb{R}^d$ with $\dim E < 1$ is totally disconnected (Exercise 24).

Recognition. Use a 1-Lipschitz real-valued map to push E to a small subset of $\mathbb{R}$ with gaps.

Solution. Fix $x \in E$ and let $f(t) = |t - x|$, which is 1-Lipschitz, so by Lemma 30.2.3 $\dim f(E) \leq \dim E < 1$, hence $m_1(f(E)) = 0$ and $f(E)$ has empty interior — its complement is dense in $\mathbb{R}$. Pick $r \notin f(E)$ with $0 < r < f(y)$ for a given $y \neq x$ in E . Then $E = \{t \in E : |t - x| < r\} \cup \{t \in E : |t - x| > r\}$ is a partition into two relatively open sets separating x from y . As x, y were arbitrary, no two points share a connected component. *Check.* The Cantor set ($\dim = \log 2 / \log 3 < 1$) is indeed totally disconnected, while a curve ($\dim = 1$) need not be — the threshold $\dim < 1$ is sharp.

Exercises (Stein, Ch.7).

1. Prove that if $f : [0, 1] \rightarrow \mathbb{R}$ satisfies a Lipschitz condition of exponent $\gamma > 1$, then f is constant.
2. Let $f(x) = x^k$ on $\mathbb{R}$ (k a positive integer) and E a Borel subset of $\mathbb{R}$. (a) Show that if $m_\alpha(E) = 0$ for some α , then $m_\alpha(f(E)) = 0$. (b) Prove $\dim E = \dim f(E)$.
3. Let $\{E_k\}$ be Borel sets in $\mathbb{R}^d$. Show that if $\dim E_k \leq \alpha$ for all k , then $\dim \bigcup_k E_k \leq \alpha$.
4. Prove that the $(\log 2 / \log 3)$ -Hausdorff measure of the Cantor set is exactly 1. [Hint: from a finite cover of C by closed intervals $\{I_j\}$, produce another cover by intervals $\{I_\ell\}$ each of length 3^{-k} for some k with $\sum_j |I_j|^\alpha \geq \sum_\ell |I_\ell|^\alpha \geq 1$, $\alpha = \log 2 / \log 3$.]
5. Show that the Cantor set of constant dissection C_ξ (Exercise 3, Chapter 1) has strict Hausdorff dimension $\log 2 / \log(2/(1 - \xi))$.
6. Consider $C_{\xi_1} \times C_{\xi_2}$ in $\mathbb{R}^2$. Show it has strict Hausdorff dimension $\dim(C_{\xi_1}) + \dim(C_{\xi_2})$.
7. Construct a Cantor-like set (as in Exercise 4, Chapter 1) of Lebesgue measure zero yet Hausdorff dimension 1. [Hint: choose $\ell_1, \ell_2, \dots$ so that $1 - \sum_{j=1}^k 2^{j-1} \ell_j \rightarrow 0$ slowly.]
8. Let $D = D_\mu = C_\xi \times C_\xi$ be the Cantor dust, $\mu = (1 - \xi)/2$. (a) Show that for any real λ , $C_\xi + \lambda C_\xi$ is similar to the projection of D onto the line of slope $\lambda = \tan \theta$. (b) Prove that if $\xi > 1/2$ then $C_\xi + \lambda C_\xi$ has Lebesgue measure zero for every λ . [Hint: $m_\alpha(C_\xi + \lambda C_\xi) < \infty$ for $\alpha = \dim D_\mu$.]
9. Consider the von Koch curve K_ℓ , $1/4 < \ell < 1/2$. Prove the analogue of Theorem 30.2.9: $t \mapsto K_\ell(t)$ is Lipschitz of exponent $\gamma = \log(1/\ell) / \log 4$, and K_ℓ has strict Hausdorff dimension $\alpha = 1/\gamma$. [Hint: with O the shaded open triangle, $O \supseteq \bigcup_{j=0}^3 S_j(O)$ where $S_0(x) = \ell x$, $S_1(x) = \rho_\theta(\ell x) + a$, $S_2(x) = \rho_\theta^{-1}(\ell x) + c$, $S_3(x) = \ell x + b$, and the $S_j(O)$ are disjoint.]

10. Show that if $\ell < 1/2$ the von Koch curve $t \mapsto K_\ell(t)$ is simple. [Hint: if $t = \sum_j a_j/4^j$, $a_j \in \{0, 1, 2, 3\}$, then $\{K(t)\} = \bigcap_j S_{a_1}(S_{a_2}(\cdots S_{a_j}(O)\cdots))$.]
11. Take $\ell = 1/2$ in the von Koch construction: show one obtains a space-filling curve that fills the right triangle with vertices $(0, 0), (1, 0), (1/2, 1/2)$.
12. Prove that the von Koch curve $t \mapsto K_\ell(t)$, $1/4 < \ell \leq 1/2$, is continuous but nowhere differentiable. [Hint: if $K'(t)$ existed, $\lim_n (K(u_n) - K(v_n))/(u_n - v_n)$ would exist for $u_n = k/4^n$, $v_n = (k+1)/4^n$.]
13. For compact $E \subseteq \mathbb{R}^d$ let $\#(\varepsilon)$ be the least number of ε -balls covering E , and define the covering dimension $\dim_C(E) = \inf\{\beta : \#(\varepsilon) = O(\varepsilon^{-\beta})\}$. Show $\dim_C(E) = \dim_M(E)$ (Minkowski dimension) by proving for all $\delta > 0$: (i) $m(E^\delta) \leq c\#(\delta)\delta^d$; (ii) $\#(\delta) \leq c'm(E^\delta)\delta^{-d}$. [Hint for (ii): use the Vitali-type lemma to find disjoint balls B_i of radius $\delta/3$ centered on E whose triples cover E .]
14. Let $E \subseteq \mathbb{R}^d$ be compact. (a) Prove $\dim E \leq \dim_M E$ (Hausdorff vs. Minkowski). (b) Show that for $E = \{0, 1/\log 2, 1/\log 3, \dots, 1/\log n, \dots\}$ one has $\dim_M E = 1$ yet $\dim E = 0$.
15. Show there is a constant c_d (depending only on d) with $m(E^{2\delta}) \leq c_d m(E^\delta)$ for compact E . [Hint: maximal function of $f = \mathbf{1}_{E^\delta}$, $c_d = 6^d$.]
16. Show that the self-similar set F of Theorem 30.2.14 has Minkowski dimension equal to its Hausdorff dimension. [Hint: each F_k is a union of m^k balls of radius $\approx cr^k$; conversely, by the vertex-counting lemma, with $\varepsilon = r^k$ each ε -ball contains at most c' vertices of generation k , so $\geq m^k/c'$ balls are needed.]
17. From $[0, 1]$ remove the second and fourth quarters; repeat in each remaining closed interval. Let $F = \{x = \sum_k a_k/4^k : a_k \in \{0, 2\}\}$ be the limit. Prove $0 < m_{1/2}(F) < \infty$.
18. Suppose F is the self-similar set of Theorem 30.2.12. (a) Show that if $m \leq 1/r^d$ then $m_d(F_i \cap F_j) = 0$ for $i \neq j$. (b) If $m \geq 1/r^d$, prove $F_i \cap F_j$ is nonempty for some $i \neq j$. (c) Under the hypotheses of Theorem 30.2.14, prove $m_\alpha(F_i \cap F_j) = 0$ with $\alpha = \log m / \log(1/r)$ for $i \neq j$.
19. Let $S_1, \dots, S_m$ be similarities of ratio $r \in (0, 1)$, $\tilde{S}(E) = \bigcup_j S_j(E)$, and F the unique nonempty compact set with $\tilde{S}(F) = F$. (a) If $x \in F$, show $\{\tilde{S}^n(x)\}_{n \geq 1}$ is dense in F . (b) Show F is homogeneous: for $x_0 \in F$ and any open ball B centered at x_0 , $F \cap B$ contains a similar copy of F .
20. Suppose E is a Borel subset of $\mathbb{R}^d$ with $\dim E < 1$. Prove E is totally disconnected. [Hint: fix $x, y \in E$, note $f(t) = |t - x|$ is Lipschitz of order 1, so $\dim f(E) < 1$ and $f(E)$ has dense complement; pick $r \notin f(E)$ with $0 < r < f(y)$ and split E .]

Problems (Stein, Ch.7).

1. Suppose S is a similarity. (a) Show S maps a line segment to a line segment. (b) If L_1, L_2 make angle α , then $S(L_1), S(L_2)$ make angle α or $-\alpha$. (c) Show every similarity is a composition of a translation, a (possibly improper) rotation, and a dilation.
2. (Generalized Cantor–Lebesgue measure.) Let F be the compact set of Theorem 30.2.12 for m similarities of ratio $r \in (0, 1)$. There is a unique Borel measure μ supported on F with $\mu(F) = 1$ and $\mu(E) = \frac{1}{m} \sum_{j=1}^m \mu(S_j^{-1}(E))$ for all Borel E . When $F = C$, the Cantor–Lebesgue function is $\mu([0, x])$.
3. (Hausdorff.) Prove that any compact $K \subseteq \mathbb{R}^d$ is a continuous image of the Cantor set C . [Hint: cover K by 2^{n_1} balls of radius 1, set $K_{j_1} = K \cap B_{j_1}$, cover each by 2^{n_2} balls of radius $1/2$, and so on; assign to a ternary $t \in C$ the point $\bigcap K_{j_1} \cap K_{j_1, j_2} \cap \cdots$.]
4. Let E be compact in $\mathbb{R}^{d-1}$. Show $\dim(E \times I) = \dim E + 1$, I the unit interval.
5. Let C_ξ be as in Exercises above. If $\xi < 1/2$, then $C_\xi + \lambda C_\xi$ has positive Lebesgue measure for almost every λ .

Additional exercises.

1. Compute the dimension of the self-similar set built from $m = 5$ maps of ratio $r = 1/3$ (a “plus-sign” fractal), and compare with the Sierpinski carpet ($m = 8$, $r = 1/3$).

2. Show $\dim(C \times C) = 2 \log 2 / \log 3$ for the middle-thirds Cantor set C , both from the product rule and by exhibiting 4 separated similarities of ratio $1/3$.
3. (Holder transfer.) If f is Holder of exponent γ on E and $\dim E = s$, give the best general upper bound on $\dim f(E)$, and show by the Cantor–Lebesgue function that it is attained.
4. Let K_ℓ be the generalized von Koch curve, $1/4 < \ell < 1/2$. Plot $\dim K_\ell = \log 4 / \log(1/\ell)$ as a function of ℓ and verify it ranges over $(1, 2)$; identify the limits as $\ell \rightarrow 1/4^+$ and $\ell \rightarrow 1/2^-$.
5. (Diagram.) Draw the first three stages C_1, C_2, C_3 of the middle-thirds Cantor set as nested unions of intervals, labeling the 2^k intervals of length 3^{-k} , and use the picture to read off the cover giving $m_\alpha(C) \leq 1$.
6. Prove that the graph of the Cantor–Lebesgue function F (the devil’s staircase) has Hausdorff dimension 1 in $\mathbb{R}^2$, despite F being non-differentiable on a large set.
7. (Non-example.) Show that the formula $\dim = \log m / \log(1/r)$ can fail without the separation hypothesis: exhibit overlapping similarities whose attractor is an interval (dimension 1) even though $\log m / \log(1/r) > 1$.
8. Show that the set $F = \{\sum_k a_k 4^{-k} : a_k \in \{0, 2\}\}$ from the scraped exercises has strict dimension $1/2$, and compute an explicit cover realizing the upper bound.
9. Using Lemma 30.2.10, show that any uniformly convergent series of piecewise-linear graphs with slopes growing like A^j and increments decaying like B^{-j} produces a Holder limit, and apply this to the Weierstrass-type function $\sum_j b^{-j\gamma} \cos(b^j t)$.
10. Let $E \subseteq \mathbb{R}$ with $\dim E = s < 1$ and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be bi-Lipschitz. Prove $\dim g(E) = s$, and explain why a merely Holder- γ homeomorphism can change the dimension.

30.3 Space-Filling Curves

A continuous surjection of the interval onto the square

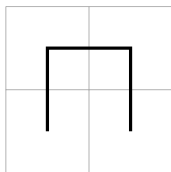
In 1890 Peano produced a continuous curve filling a whole square. Beyond the surprise, the construction reveals a measure-theoretic fact: from the point of view of measure, the unit interval and the unit square are “isomorphic.”

Theorem 30.3.1 (Peano). *There is a map $t \mapsto \mathcal{P}(t)$ from $[0, 1]$ to $[0, 1]^2$ with:*

1. $\mathcal{P}$ is continuous and surjective;
2. $\mathcal{P}$ is Holder of exponent $1/2$: $|\mathcal{P}(t) - \mathcal{P}(s)| \leq M |t - s|^{1/2}$;
3. for every subinterval $[a, b]$, $\mathcal{P}([a, b])$ is compact with two-dimensional Lebesgue measure exactly $b - a$.

Corollary 30.3.2 (Measure isomorphism). *There are null sets $Z_1 \subseteq [0, 1]$, $Z_2 \subseteq [0, 1]^2$ with $\mathcal{P}$ a measure preserving bijection $[0, 1] \setminus Z_1 \rightarrow [0, 1]^2 \setminus Z_2$: E is measurable iff $\mathcal{P}(E)$ is, and $m_1(E) = m_2(\mathcal{P}(E))$.*

Two sanity remarks bound what is possible. (a) No continuous surjection $[0, 1] \rightarrow [0, 1]^2$ can be Holder of exponent $\gamma > 1/2$: Lemma 30.2.3 would give $2 = \dim[0, 1]^2 \leq (1/\gamma) \dim[0, 1] = 1/\gamma$, forcing $\gamma \leq 1/2$. So the exponent $1/2$ in Theorem 30.3.1 is optimal. (b) No such map can be injective: a continuous bijection of a compact set has continuous inverse, and one derives a contradiction by joining two image points with two distinct arcs in the square. Indeed every Peano-type curve has self-crossings densely.



first generation: visit all 4 subsquares

Figure 30.9. Hilbert’s iterative scheme: a polygonal path through the centers of the four dyadic subsquares; refining indefinitely fills the square.

The construction is the “traverse” bookkeeping: among the four subsquares of a square one chooses an ordering S_1, S_2, S_3, S_4 with consecutive squares adjacent, entering a white corner and exiting a black corner so that the path can continue into the neighbor. This is exactly Hilbert’s recursion.

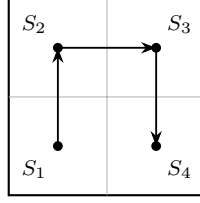


Figure 30.12. A traverse $S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4$ of the four subsquares: consecutive squares are adjacent; enter at S_1 , exit at S_4 .

Construction of $\mathcal{P}$. For the Φ of Lemma 30.3.7, let $S_1, \dots, S_{4^k}$ be the generation- k squares ordered by Φ , with centers x_j , and $t_j = (j - \frac{1}{2})/4^k$. Set $\mathcal{P}_k(t_j) = x_j$, $\mathcal{P}_k(0) = (0, 2^{-k-1})$, $\mathcal{P}_k(1) = (1, 2^{-k-1})$, and interpolate linearly. Then $|x_j - x_{j+1}| = 2^{-k}$ while $|t_j - t_{j+1}| = 4^{-k}$, so $\mathcal{P}'_k(t) = 2^k$ off the nodes and $|\mathcal{P}_k(t) - \mathcal{P}_k(s)| \leq 2^k |t - s|$. Consecutive approximants stay in one generation- k square, so $|\mathcal{P}_{k+1}(t) - \mathcal{P}_k(t)| \leq \sqrt{2} 2^{-k}$. Lemma 30.2.10 (with $A = 2$, $B = 2$) gives the uniform limit $\mathcal{P} = \lim \mathcal{P}_k$, Holder of exponent $\log 2 / \log 4 = 1/2$. Each $\mathcal{P}_k$ visits every generation- k square, so $\mathcal{P}$ is dense, and by continuity onto $[0, 1]^2$. Finally $\mathcal{P} = \Phi^*$ (Lemma below), so Theorem 30.3.6 gives the measure-preserving property (iii). $\square$

Lemma 30.3.8. *For the special Φ , $\Phi^*(t) = \mathcal{P}(t)$ for all t ; hence $\mathcal{P}([a, b])$ has $m_2 = b - a$.*

Proof. If $\{t\} = \bigcap_k I^k = \bigcap_k J^k$ are two chains for t , then I^k, J^k are adjacent for large k , so $\Phi(I^k), \Phi(J^k)$ are adjacent squares with the same intersection; thus Φ^* is unambiguous and equals $\bigcap_k \Phi(I^k) = \lim \mathcal{P}_k(t) = \mathcal{P}(t)$. Writing $(a, b) = \bigcup_j I_j$ as disjoint quartic intervals, $\mathcal{P}(I_j) = \Phi(I_j)$ are disjoint dyadic squares, so $m_2(\mathcal{P}(a, b)) = \sum_j m_2(\Phi(I_j)) = \sum_j m_1(I_j) = m_1(a, b) = b - a$. $\square$

Worked Example 30.3.9 (The optimal Holder exponent is $1/2$). Prove directly (Exercise 4) that any continuous surjection $f : [0, 1] \rightarrow [0, 1]^2$ with $|f(x) - f(y)| \leq C|x - y|^\gamma$ has $\gamma \leq 1/2$, without invoking Lemma 30.2.3.

Recognition. “Bound a space-filling Holder exponent”: cover the square by images of small subintervals and count.

Solution. Divide $[0, 1]$ into N equal subintervals; the image of each has diameter $\leq CN^{-\gamma}$, hence lies in a ball of area $O(N^{-2\gamma})$. These N balls cover the unit square (area 1), so $N \cdot cN^{-2\gamma} \geq 1$, i.e. $N^{1-2\gamma} \geq c^{-1}$ for all N . If $\gamma > 1/2$ then $1 - 2\gamma < 0$ and the left side $\rightarrow 0$, a contradiction. Hence $\gamma \leq 1/2$. *Check.* The Peano map achieves $\gamma = 1/2$ exactly, so this bound is attained — the construction is as smooth as a space-filling curve can be.

Worked Example 30.3.10 (No injective continuous surjection onto the square). Show that a continuous surjection $\mathcal{P} : [0, 1] \rightarrow [0, 1]^2$ cannot be injective.

Recognition. Compactness gives a continuous inverse for an injective continuous map; derive a topological contradiction.

Solution. If $\mathcal{P}$ were a continuous bijection from the compact $[0, 1]$ to the Hausdorff $[0, 1]^2$, its inverse $G = \mathcal{P}^{-1}$ would be continuous, so $\mathcal{P}$ a homeomorphism. But $[0, 1]$ is disconnected by removing any interior point, while $[0, 1]^2$ remains connected after removing the corresponding image point — a homeomorphism must preserve this, contradiction. Concretely, removing $t_0 \in (0, 1)$ splits $[0, 1]$ into two pieces, yet the square minus $\mathcal{P}(t_0)$ is path-connected, so G cannot be continuous. *Check.* Consistent with property: the Peano curve has dense self-intersections, the geometric face of non-injectivity.

Worked Example 30.3.11 (Length of the k th polygonal approximant). Compute the total length of $\mathcal{P}_k$ and show it diverges, while the limit fills area 1.

Recognition. “Polygonal length of a fractal approximant”: count segments times segment length.

Solution. $\mathcal{P}_k$ joins 4^k centers by segments of length 2^{-k} (plus two short end pieces), so its length is $\approx 4^k \cdot 2^{-k} = 2^k \rightarrow \infty$. Thus the approximating curves have unbounded length even though they converge

uniformly to a continuous map. *Check.* An infinite-length limit is forced — the image has positive area ($m_2 = 1$), and a rectifiable (finite-length) curve has $m_2 = 0$ by Theorem 30.2.6; so $\mathcal{P}$ is necessarily non-rectifiable.

Worked Example 30.3.12 (A base-4 null set). Fix a digit sequence $\{f_k\} \subseteq \{0, 1, 2, 3\}$ and let $E_0 = \{x = \sum_k a_k 4^{-k} : a_k \neq f_k \text{ for all large } k\}$. Show $m(E_0) = 0$ (Lemma 3.6).

Recognition. “Digits avoiding a value”: independence of digit events gives a geometric decay.

Solution. For fixed r , $m\{x : a_r \neq f_r\} = 3/4$, and the events for distinct positions are independent, so $m\{x : a_k \neq f_k \text{ for } r \leq k \leq n\} = (3/4)^{n-r+1} \rightarrow 0$. Hence $m\{x : a_k \neq f_k \forall k \geq r\} = 0$, and $E_0 = \bigcup_r \{\cdots\}$ is a countable union of null sets, so $m(E_0) = 0$. *Check.* The complement (digits hit f_k infinitely often) has full measure — this is the digit lemma making Z_1, Z_2 null in Theorem 30.3.6.

Worked Example 30.3.13 (Measure isomorphism preserves measurability). Using Corollary 30.3.2, show $E \subseteq [0, 1] \setminus Z_1$ is Lebesgue measurable iff $\mathcal{P}(E)$ is, and then $m_1(E) = m_2(\mathcal{P}(E))$. Specialize to $E = [0, 1/4]$.

Recognition. “Transport measurability through a measure-preserving bijection.”

Solution. $\mathcal{P}$ and $\mathcal{P}^{-1}$ both map null sets to null sets (proof of Theorem 30.3.6), and both send open sets to measurable sets; hence each carries the measurable σ -algebra to itself, and $m_1(E) = m_2(\mathcal{P}(E))$ for all measurable E . For $E = [0, 1/4]$ (a quartic interval of generation 1), $\mathcal{P}(E) = \Phi(E)$ is the lower-left dyadic square of side $1/2$, with $m_2 = 1/4 = m_1(E)$. *Check.* The four generation-1 intervals each of $m_1 = 1/4$ map to the four subsquares each of area $1/4$, partitioning both spaces consistently.

Exercises (Stein, Ch.7).

1. Suppose $f : [0, 1] \rightarrow [0, 1] \times [0, 1]$ is surjective and satisfies $|f(x) - f(y)| \leq C|x - y|^\gamma$. Prove $\gamma \leq 1/2$ directly, without using Lemma 30.2.3. [Hint: divide $[0, 1]$ into N equal intervals; the image of each lies in a ball of area $O(N^{-2\gamma})$, and these must cover the square.]

Problems (Stein, Ch.7).

1. A compact $K \subseteq \mathbb{R}^d$ is *uniformly locally connected* if for each $\varepsilon > 0$ there is $\delta > 0$ so that whenever $x, y \in K$, $|x - y| < \delta$, there is a continuous curve $\gamma \subseteq K$ joining x to y with $\gamma \subseteq B_\varepsilon(x)$ and $\gamma \subseteq B_\varepsilon(y)$. Using the Hausdorff problem (Section 30.2), show K is a continuous image of $[0, 1]$ if and only if K is uniformly locally connected.
2. Formulate and prove a generalization of Theorem 30.3.6: after removing appropriate sets of measure zero, there is a measure-preserving isomorphism between the unit interval in $\mathbb{R}$ and the unit cube in $\mathbb{R}^d$.
3. Prove that there exists a simple continuous curve in the plane of positive two-dimensional Lebesgue measure.

Additional exercises.

1. Write the base-4 expansion of $t = 11/16$ and trace the chain of quartic intervals containing it; identify the dyadic square $\Phi^*(t)$ under the standard correspondence ($b_k = a_k$).
2. Show that the standard correspondence $b_k = a_k$ does *not* yield a continuous map Φ^* , by exhibiting two parameters t, t' that are close yet have far-apart images. (This motivates the adjacency-preserving Φ of Lemma 30.3.7.)
3. Verify that $|\mathcal{P}_k(t) - \mathcal{P}_k(s)| \leq 2^k |t - s|$ and $|\mathcal{P}_{k+1} - \mathcal{P}_k| \leq \sqrt{2} 2^{-k}$, then apply Lemma 30.2.10 with $A = B = 2$ to recover the exponent $1/2$.
4. (Diagram.) Draw the second-generation Hilbert path through the 16 dyadic squares, marking the entry and exit squares and confirming consecutive squares share a side.
5. Prove that the Peano curve is non-rectifiable by showing the length of $\mathcal{P}_k$ is $\approx 2^k \rightarrow \infty$, and connect this to $m_2(\mathcal{P}([0, 1])) = 1$ via Theorem 30.2.6.
6. Show that the image of a single point can have up to four preimages under $\mathcal{P}$, and locate such a quadruple point among the dyadic-rational ambiguities.

7. (Non-example.) Explain why the map $t \mapsto (t, t)$ from $[0, 1]$ to $[0, 1]^2$, though continuous and injective, is *not* surjective and therefore not space-filling; contrast its image dimension (1) with that of the Peano curve (2).
8. Using Corollary 30.3.2, transport a non-measurable subset of $[0, 1]$ to a non-measurable subset of $[0, 1]^2$, and explain why measurability is preserved both ways.
9. Show directly that $\bigcup_j \Phi(I_j)$ tiles $[0, 1]^2$ when $\{I_j\}$ are the four generation-1 quartic intervals, so that areas match lengths at the first stage.
10. Prove that any continuous $f : [0, 1] \rightarrow [0, 1]^2$ that is measure preserving in the sense $m_2(f([a, b])) = b - a$ must be surjective onto a full-measure subset, hence (by compactness) onto all of $[0, 1]^2$.

30.4 Besicovitch Sets

Slices and a regularity phenomenon

For a unit vector $\gamma \in S^{d-1}$ and $t \in \mathbb{R}$, let $P_{t,\gamma} = \{x \in \mathbb{R}^d : x \cdot \gamma = t\}$ be the affine hyperplane perpendicular to γ at signed distance t , carrying its natural $(d-1)$ -dimensional Lebesgue measure m_{d-1} . For $E \subseteq \mathbb{R}^d$ the *slice* is $E_{t,\gamma} = E \cap P_{t,\gamma}$. By Fubini, for a measurable E and fixed γ , the slice $E_{t,\gamma}$ is m_{d-1} -measurable for a.e. t and $t \mapsto m_{d-1}(E_{t,\gamma})$ is measurable. The striking fact is that in dimension $d \geq 3$ this slice function is, for almost every direction, far more regular.

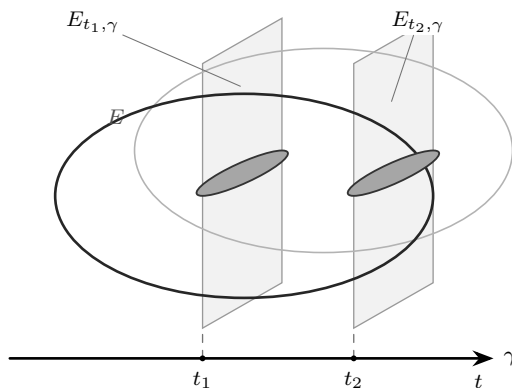


Figure 30.13. Parallel hyperplanes perpendicular to γ cut E into slices; the slice-area function is $\mu(t, \gamma) = m_{d-1}(E_{t,\gamma})$, which shrinks toward the edge. (Schematic.)

Theorem 30.4.1 (Regularity of slices, $d \geq 3$). *If $E \subseteq \mathbb{R}^d$ has finite measure and $d \geq 3$, then for a.e. $\gamma \in S^{d-1}$: (i) $E_{t,\gamma}$ is measurable for every t ; (ii) $t \mapsto m_{d-1}(E_{t,\gamma})$ is continuous, and indeed Hölder of every exponent $\alpha < 1/2$.*

Corollary 30.4.2. *If $E \subseteq \mathbb{R}^d$ ($d \geq 3$) has measure zero, then for a.e. γ every slice $E_{t,\gamma}$ is m_{d-1} -null.*

The two-dimensional failure of this regularity is no accident: it is equivalent to the existence of a Besicovitch (Kakeya) set.

Theorem 30.4.3 (Besicovitch set). *There is a set $B \subseteq \mathbb{R}^2$ that is (i) compact, (ii) of Lebesgue measure zero, and (iii) contains a translate of every unit line segment.*

Such a B kills the planar analogue of Theorem 30.4.1: for each $\gamma \in S^1$, $m_1(B \cap P_{t_0,\gamma}) \geq 1$ for some t_0 ; if $t \mapsto m_1(B \cap P_{t,\gamma})$ were continuous it would be positive on an interval of t , forcing $m_2(B) > 0$ by Fubini, a contradiction.

Theorem 30.4.4 (Besicovitch sets have dimension 2). *Any set F satisfying (i) and (iii) of Theorem 30.4.3 has Hausdorff dimension 2.*

So while B has zero area, this is “barely” so: it cannot be shrunk to any dimension below 2.

The Radon transform and Fourier estimates

Definition 30.4.5. For suitable f on $\mathbb{R}^d$, the *Radon transform* is $R(f)(t, \gamma) = \int_{P_{t, \gamma}} f \, dm_{d-1}$. When $f = \mathbf{1}_E$ is an indicator, $R(f)(t, \gamma) = m_{d-1}(E_{t, \gamma})$, linking R to slicing. The maximal Radon transform is $R^*(f)(\gamma) = \sup_t |R(f)(t, \gamma)|$.

Lemma 30.4.6 (Fourier slice / projection). *For continuous compactly supported f and the t -Fourier transform $\widehat{R}(f)(\lambda, \gamma)$, one has $\widehat{R}(f)(\lambda, \gamma) = \widehat{f}(\lambda\gamma)$.*

$$\begin{array}{ccc} f & \xrightarrow{R_\gamma} & R(f)(\cdot, \gamma) \\ \mathcal{F}_d \downarrow & & \downarrow \mathcal{F}_t \\ \widehat{f} & \xrightarrow{\xi \mapsto \lambda\gamma} & \widehat{R(f)}(\cdot, \gamma) \end{array}$$

Figure 30.14. Fourier slice theorem. Here $R_\gamma f = R(f)(\cdot, \gamma)$ is the Radon transform in the direction γ , $\mathcal{F}_t$ the one-variable Fourier transform in t , and $\mathcal{F}_d$ the full d -variable transform. Taking R_γ then $\mathcal{F}_t$ equals restricting $\widehat{f}$ to the line $\xi = \lambda\gamma$: the square commutes, $\widehat{R(f)}(\lambda, \gamma) = \widehat{f}(\lambda\gamma)$.

Proof. Adapt coordinates so γ is the x_d -axis: $x = (u, t)$, $u \in \mathbb{R}^{d-1}$, $x \cdot \gamma = t$. Then $\int_{P_{t, \gamma}} f = \int_{\mathbb{R}^{d-1}} f(u, t) \, du$, and Fubini gives

$$\widehat{f}(\lambda\gamma) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x \cdot (\lambda\gamma)} \, dx = \int_{-\infty}^{\infty} \left(\int_{\mathbb{R}^{d-1}} f(u, t) \, du \right) e^{-2\pi i \lambda t} \, dt = \widehat{R(f)}(\lambda, \gamma). \quad \square$$

Lemma 30.4.7 (Plancherel in polar form). *For continuous compactly supported f ,*

$$\int_{S^{d-1}} \left(\int_{-\infty}^{\infty} |\widehat{R(f)}(\lambda, \gamma)|^2 |\lambda|^{d-1} \, d\lambda \right) d\sigma(\gamma) = 2 \int_{\mathbb{R}^d} |f(x)|^2 \, dx.$$

Proof. Plancherel gives $\int_{\mathbb{R}^d} |f|^2 = 2 \int_{\mathbb{R}^d} |\widehat{f}|^2$ (the factor as normalized in Stein). Polar coordinates $\xi = \lambda\gamma$, $\lambda > 0$, $\gamma \in S^{d-1}$ contribute the Jacobian λ^{d-1} : $\int_{\mathbb{R}^d} |\widehat{f}|^2 = \int_{S^{d-1}} \int_0^\infty |\widehat{f}(\lambda\gamma)|^2 \lambda^{d-1} \, d\lambda \, d\sigma$. Symmetrizing to $\lambda \in \mathbb{R}$ replaces λ^{d-1} by $|\lambda|^{d-1}$, and Lemma 30.4.6 substitutes $\widehat{f}(\lambda\gamma) = \widehat{R(f)}$. $\square$

The key dimensional gain: the larger d , the heavier the weight $|\lambda|^{d-1}$, hence the faster $\widehat{R(f)}$ must decay, hence the smoother $R(f)(t, \gamma)$ in t .

Lemma 30.4.8 (One-variable regularity from a weighted estimate). *Suppose $F(t) = \int \widehat{F}(\lambda) e^{2\pi i \lambda t} \, d\lambda$ with $\sup_\lambda |\widehat{F}(\lambda)| \leq A$ and $\int |\widehat{F}(\lambda)|^2 |\lambda|^{d-1} \, d\lambda \leq B^2$. Then for $d \geq 3$, $\sup_t |F(t)| \leq c(A+B)$, and for $0 < \alpha < 1/2$, $|F(t_1) - F(t_2)| \leq c_\alpha |t_1 - t_2|^\alpha (A+B)$.*

Proof. Split $|\lambda| \leq 1$ and $|\lambda| > 1$. The low part is $\leq cA$. For the high part, Cauchy-Schwarz gives $\int_{|\lambda|>1} |\widehat{F}| \, d\lambda \leq \left(\int |\widehat{F}|^2 |\lambda|^{d-1} \right)^{1/2} \left(\int_{|\lambda|>1} |\lambda|^{-d+1} \, d\lambda \right)^{1/2}$, and the last integral converges iff $-d+1 < -1$, i.e. $d \geq 3$; this yields $\sup_t |F| \leq c(A+B)$. For the Holder bound, use $|e^{2\pi i \lambda t_1} - e^{2\pi i \lambda t_2}| \leq c|t_1 - t_2|^\alpha |\lambda|^\alpha$ ($0 \leq \alpha < 1$); the high part is then majorized by $|t_1 - t_2|^\alpha \left(\int |\widehat{F}|^2 |\lambda|^{d-1} \right)^{1/2} \left(\int_{|\lambda|>1} |\lambda|^{-d+1+2\alpha} \right)^{1/2}$, finite when $-d+1+2\alpha < -1$, i.e. $\alpha < 1/2$ for $d \geq 3$. $\square$

Theorem 30.4.9 (A priori Radon estimate). *For continuous compactly supported f on $\mathbb{R}^d$, $d \geq 3$,*

$$\int_{S^{d-1}} R^*(f)(\gamma) \, d\sigma(\gamma) \leq c(\|f\|_{L^1(\mathbb{R}^d)} + \|f\|_{L^2(\mathbb{R}^d)}),$$

and the same with $R^(f)(\gamma)$ replaced by the Holder- α quotient $\sup_{t_1 \neq t_2} |R(f)(t_1, \gamma) - R(f)(t_2, \gamma)| / |t_1 - t_2|^\alpha$ for $0 < \alpha < 1/2$.*

Proof. Apply Lemma 30.4.8 to $F(t) = R(f)(t, \gamma)$, so $\sup_t |F| = R^*(f)(\gamma)$. Here $A(\gamma) = \sup_\lambda |\widehat{F}| = \sup_\lambda |\widehat{f}(\lambda\gamma)| \leq \|f\|_{L^1}$, and $B^2(\gamma) = \int |\widehat{F}|^2 |\lambda|^{d-1} d\lambda$ with $\int_{S^{d-1}} B^2 d\sigma = 2\|f\|_{L^2}^2$ by Lemma 30.4.7. Then $|R^*(f)(\gamma)|^2 \leq c(A^2 + B^2)$, and integrating over the sphere with Cauchy–Schwarz yields $\int_{S^{d-1}} R^*(f) d\sigma \leq c(\|f\|_{L^1} + \|f\|_{L^2})$. The Holder version uses the second estimate of Lemma 30.4.8. $\square$

The L^2 -norm provides essential local control; it cannot be dropped (Exercise 27, with $f_\varepsilon(x) = (|x| + \varepsilon)^{-d+\delta}$). The proof of Theorem 30.4.1 then extends this from continuous f to $f = \mathbf{1}_E \in L^1 \cap L^2$ via monotone/bounded convergence, giving the slice regularity.

The planar deficit and dimension two

Definition 30.4.10. For $f \in L^1(\mathbb{R}^d)$ the *strip average* is $R_\delta(f)(t, \gamma) = \frac{1}{2\delta} \int_{t-\delta}^{t+\delta} R(f)(s, \gamma) ds = \frac{1}{2\delta} \int_{t-\delta \leq x \cdot \gamma \leq t+\delta} f$, with $R_\delta^*(f)(\gamma) = \sup_t |R_\delta(f)(t, \gamma)|$.

Theorem 30.4.11 (Planar strip estimate). *For continuous compactly supported f on $\mathbb{R}^2$ and $0 < \delta \leq 1/2$,*

$$\int_{S^1} R_\delta^*(f)(\gamma) d\sigma(\gamma) \leq c(\log 1/\delta)^{1/2} (\|f\|_{L^1(\mathbb{R}^2)} + \|f\|_{L^2(\mathbb{R}^2)}).$$

The single power $|\lambda|^{d-1} = |\lambda|$ at $d = 2$ makes the bare estimate (Theorem 30.4.9) fail, but the strip average tames the integral at the cost of the $(\log 1/\delta)^{1/2}$ factor: the weighted tail $\int_{1 < |\lambda| \leq 1/\delta} |\lambda|^{-1} d\lambda = \log(1/\delta)$ appears.

Theorem 30.4.12 (Theorem 30.4.4, proof). *Any Besicovitch-type F (compact, containing a translate of every unit segment) has $\dim F = 2$.*

Proof sketch. Suppose $F \subseteq \bigcup_i B_i$, balls of diameter < 1 ; we show $\sum_i (\text{diam } B_i)^\alpha \geq c_\alpha > 0$ for every $\alpha < 2$. Group the balls by dyadic size, $N_k = \#\{i : 2^{-k-1} \leq \text{diam } B_i \leq 2^{-k}\}$, and let F_k^* be the doubled union at scale k , so $m(F_k^*) \leq cN_k 2^{-2k}$. For each $\gamma \in S^1$ there is a unit segment $s_\gamma \subseteq F$; choosing weights $a_k \approx 2^{-\varepsilon k}$ with $\sum a_k = 1$, some k has $m_1(s_\gamma \cap F_k) \geq a_k$, whence $R_{2^{-k}}^*(\mathbf{1}_{F_k^*})(\gamma) \geq a_k$. Letting $E_k = \{\gamma : R_{2^{-k}}^*(\mathbf{1}_{F_k^*})(\gamma) \geq a_k\}$, the sphere is $\bigcup_k E_k$, so some k' has $m(E_{k'}) \geq 2\pi a_{k'}$. Theorem 30.4.11 then gives $a_{k'}^2 \leq c(\log 2^{k'})^{1/2} \|\mathbf{1}_{F_{k'}^*}\|_{L^2} \leq c(k')^{1/2} N_{k'}^{1/2} 2^{-k'}$, i.e. $2^{(1-2\varepsilon)k'} \leq c(k')^{1/2} N_{k'}^{1/2}$, which forces $N_{k'} 2^{-\alpha k'} \geq c_\alpha$ once $4\varepsilon < 2 - \alpha$. Hence $\sum_i (\text{diam } B_i)^\alpha \geq c_\alpha$ for all $\alpha < 2$, so $\dim F \geq 2$; as $F \subseteq \mathbb{R}^2$, $\dim F = 2$. $\square$

Construction of a Besicovitch set

Take the constant-dissection Cantor set $C = C_{1/2} = \bigcap_k C_k$ with $C = \{\sum_k \varepsilon_k 4^{-k} : \varepsilon_k \in \{0, 3\}\}$. Place $E_0 = \{(x, 0) : x \in C\}$ on the x -axis and $E_1 = \{(x, 1) : 2x \in C\}$ on $y = 1$, and let F be the union of all segments joining a point of E_0 to a point of E_1 .

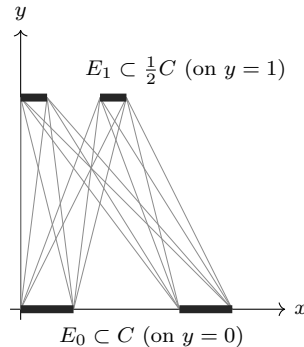


Figure 30.15. Segments joining E_0 (a copy of C) to E_1 (a copy of $\frac{1}{2}C$); their union F is compact, null, yet rich in directions. (Schematic.)

Theorem 30.4.13 (Construction works). *F is compact, of two-dimensional measure zero, and contains a translate of every unit segment of slope $s \notin (-1, 2)$. A finite union of rotations of F is then a Besicovitch set.*

Proof outline. F is closed (hence compact) since E_0, E_1 are. The slice F^y at height $0 < y < 1$ is $(1-y)C + \frac{y}{2}C$, a scaled copy of $C + \lambda C$ with $\lambda = y/(2(1-y))$. The crux is two facts about $K(\lambda) = C + \lambda C$:

- $C + \frac{1}{2}C = [0, 3/2]$ (an interval);
- $m(C + \lambda C) = 0$ for a.e. λ .

The second, via Fubini, makes $m_2(F) = 0$. The slope of the segment from $(x_0, 0)$ to $(x_1, 1)$ is $1/(x_1 - x_0)$ with $x_1 \in C/2$, $x_0 \in C$; using $C = 1 - C$ the realizable slopes are $s \notin (-1, 2)$. Rotating F finitely many times covers all slopes. $\square$

Lemma 30.4.14 (The interval identity). $C + \frac{1}{2}C = [0, 3/2]$.

Proof. Since $\frac{2}{3}(C + \frac{1}{2}C) = \frac{2}{3}C + \frac{1}{3}C$ consists of all $\sum_k (\frac{2\varepsilon_k}{3} + \frac{\varepsilon'_k}{3})4^{-k}$ with $\varepsilon_k, \varepsilon'_k \in \{0, 3\}$, the digit $\frac{2\varepsilon_k}{3} + \frac{\varepsilon'_k}{3}$ ranges over $\{0, 1, 2, 3\}$ independently, so the set is all of $[0, 1]$. Hence $\frac{2}{3}(C + \frac{1}{2}C) = [0, 1]$ and $C + \frac{1}{2}C = [0, 3/2]$. $\square$

Lemma 30.4.15 (Self-replication of $K(\lambda)$). $K(\lambda) = K(\lambda/4) \cup (K(\lambda/4) + \frac{3\lambda}{4})$; hence $m(K(\lambda)) = 0$ iff $m(K(\lambda/4)) = 0$, and it suffices to treat $\lambda \in [1, 4]$.

Proof. $C = \frac{1}{4}C \cup (\frac{1}{4}C + \frac{3}{4})$, so $C + \lambda C = (C + \frac{\lambda}{4}C) \cup (C + \frac{\lambda}{4}C + \frac{3\lambda}{4})$, and recognizing $C + \frac{\lambda}{4}C = K(\lambda/4)$ gives the union. The measure equivalence is immediate from this self-similar decomposition. $\square$

Worked Example 30.4.16 ($C + \frac{1}{2}C$ fills an interval). Verify Lemma 30.4.14 on the first base-4 digit, exhibiting which integers the combined digit produces.

Recognition. “Sum of two Cantor sets”: track the achievable digit at each scale.

Solution. With $\varepsilon_k, \varepsilon'_k \in \{0, 3\}$, the scaled digit $\frac{2\varepsilon_k}{3} + \frac{\varepsilon'_k}{3}$ takes: $(0, 0) \mapsto 0$, $(0, 3) \mapsto 1$, $(3, 0) \mapsto 2$, $(3, 3) \mapsto 3$. So every digit 0, 1, 2, 3 is attainable independently across scales, filling $[0, 1]$ for $\frac{2}{3}(C + \frac{1}{2}C)$, hence $C + \frac{1}{2}C = [0, 3/2]$.
Check. The endpoints 0 (all $\varepsilon = \varepsilon' = 0$) and $3/2$ (all $= 3$: $\sum 3 \cdot 4^{-k} + \frac{1}{2} \sum 3 \cdot 4^{-k} = 1 + \frac{1}{2} = 3/2$) are hit, consistent with the interval $[0, 3/2]$.

Worked Example 30.4.17 (Why $d \geq 3$ in the slice regularity). Pinpoint where $d \geq 3$ enters Lemma 30.4.8 and what fails at $d = 2$.

Recognition. “Locate the dimensional hypothesis”: it is the convergence of a weighted tail integral.

Solution. The high-frequency estimate needs $\int_{|\lambda|>1} |\lambda|^{-d+1} d\lambda < \infty$, i.e. $-d+1 < -1$, i.e. $d > 2$. At $d = 2$ this becomes $\int_{|\lambda|>1} |\lambda|^{-1} d\lambda = \infty$ (logarithmic divergence), so $\sup_t |F| \leq c(A+B)$ breaks down; only the strip-averaged version (Theorem 30.4.11) survives, with the $(\log 1/\delta)^{1/2}$ penalty. *Check.* The penalty exponent matches $\int_1^{1/\delta} |\lambda|^{-1} d\lambda = \log(1/\delta)$, whose square root appears — exactly the borderline divergence at $d = 2$.

Worked Example 30.4.18 (A Besicovitch slice has positive length). Using F from Theorem 30.4.3, show that for each γ some slice has $m_1(F \cap P_{t_0, \gamma}) \geq 1$, and deduce that the slice function cannot be continuous.

Recognition. “Discontinuity from a fat slice in a null set.”

Solution. Since F contains a unit segment s_γ perpendicular to γ , that segment lies in a single plane $P_{t_0, \gamma}$, so $m_1(F \cap P_{t_0, \gamma}) \geq m_1(s_\gamma) = 1$. If $t \mapsto m_1(F \cap P_{t, \gamma})$ were continuous it would exceed $1/2$ on an open t -interval J , and then by Fubini $m_2(F) \geq \int_J m_1(F \cap P_{t, \gamma}) dt > 0$, contradicting $m_2(F) = 0$. *Check.* This is precisely the obstruction that forces the $d = 2$ failure of Corollary 30.4.2 — the slice can be large on a null-measure set of t , but never on an interval.

Worked Example 30.4.19 (Fubini reduces $m_2(F) = 0$ to a.e. null slices). Show that if $m_1(F^y) = 0$ for a.e. $y \in (0, 1)$ (and F is bounded), then $m_2(F) = 0$.

Recognition. “Vanishing slices give vanishing area”: Fubini–Tonelli on the indicator.

Solution. By Tonelli applied to $\mathbf{1}_F \geq 0$, $m_2(F) = \int_0^1 \left(\int_{\mathbb{R}} \mathbf{1}_F(x, y) dx \right) dy = \int_0^1 m_1(F^y) dy$. If $m_1(F^y) = 0$ for a.e. y , the integrand is 0 a.e., so $m_2(F) = 0$. *Check.* The Besicovitch construction supplies $m_1(F^y) = m((1-y)C + \frac{y}{2}C) = 0$ for a.e. y (since $m(C + \lambda C) = 0$ a.e. λ), exactly meeting the hypothesis, so the constructed F is null.

Worked Example 30.4.20 (The Radon transform of an indicator is the slice length). For $f = \mathbf{1}_E$ with E a measurable bounded subset of $\mathbb{R}^2$, identify $R(f)(t, \gamma)$ and compute it when E is the unit disc, $\gamma = e_2$.

Recognition. “Radon transform of a characteristic function”: it is the 1-dimensional measure of a slice.

Solution. By Definition 30.4.5, $R(\mathbf{1}_E)(t, \gamma) = \int_{P_{t,\gamma}} \mathbf{1}_E dm_1 = m_1(E \cap P_{t,\gamma})$, the length of the chord. For $E = \{x^2 + y^2 \leq 1\}$ and $\gamma = e_2 = (0, 1)$ the plane $P_{t,\gamma}$ is the horizontal line $y = t$, cutting a chord of half-length $\sqrt{1 - t^2}$ for $|t| \leq 1$, so $R(\mathbf{1}_E)(t, e_2) = 2\sqrt{1 - t^2}$ and 0 otherwise. *Check.* $\int_{-1}^1 2\sqrt{1 - t^2} dt = \pi = m_2(\text{disc})$, matching Fubini — the integral of the Radon transform recovers the area.

Exercises (Stein, Ch.7).

1. Let $F(t)$ be an arbitrary non-negative measurable function on $\mathbb{R}$, and $\gamma \in S^{d-1}$. Show there is a measurable set $E \subseteq \mathbb{R}^d$ with $F(t) = m_{d-1}(E \cap P_{t,\gamma})$.
2. (Refinement for $d \geq 4$.) Define $C^{k,\alpha}$ as the functions F on $\mathbb{R}$ that are C^k with $F^{(k)}$ Lipschitz of exponent α . If E has finite measure, then for a.e. $\gamma \in S^{d-1}$ the function $m(E \cap P_{t,\gamma})$ is in $C^{k,\alpha}$ with $k = (d - 3)/2$, $\alpha < 1/2$ for d odd ($d \geq 3$); and $k = (d - 4)/2$, $\alpha < 1$ for d even ($d \geq 4$).
3. Show that the modification of inequality (2) in Theorem 30.4.9 fails if the term $\|f\|_{L^2(\mathbb{R}^d)}$ is dropped. [Hint: consider $R^*(f_\varepsilon)$ with $f_\varepsilon(x) = (|x| + \varepsilon)^{-d+\delta}$ for $|x| \leq 1$, δ fixed in $(0, 1)$, and $\varepsilon \rightarrow 0$.]
4. Construct a compact $E \subseteq \mathbb{R}^d$, $d \geq 3$, with $m_d(E) = 0$ that contains translates of every unit segment in $\mathbb{R}^d$. (Examples follow from the case $d = 2$; the least Hausdorff dimension of such a set is an open problem.)

Problems (Stein, Ch.7).

1. Carry out the following construction of sets U, V with $\dim U = \dim V = 0$ but $\dim(U \times V) \geq 1$. Let $I_1, I_2, \dots$ be finite blocks of consecutive positive integers $I_j = \{n : A_j \leq n \leq B_j\}$ with $A_{j+1} > B_j$, $B_j/A_j \rightarrow \infty$ and $A_{j+1}/B_j \rightarrow \infty$. Let $U \subseteq [0, 1]$ consist of all dyadic $x = .a_1a_2\cdots$ with $a_n = 0$ whenever $n \in \bigcup_j I_j$, and $V \subseteq [0, 1]$ those $x = .a_1a_2\cdots$ with $a_n = 0$ whenever $n \in \bigcup_j J_j$, where $J_j = \{n : B_j < n < A_{j+1}\}$. Prove U, V have the desired properties.

Additional exercises.

1. Compute the Radon transform $R(\mathbf{1}_E)(t, e_1)$ for $E = [0, 1]^2$ and the horizontal direction $\gamma = e_1$, and verify $\int_{\mathbb{R}} R(\mathbf{1}_E)(t, e_1) dt = m_2(E) = 1$.
2. For the unit disc E in $\mathbb{R}^2$, show $R(\mathbf{1}_E)(t, \gamma) = 2\sqrt{1 - t^2}$ for $|t| \leq 1$ (any γ), and confirm rotational invariance of the Radon transform on radial functions.
3. Show that for $f = \mathbf{1}_E$ with E bounded measurable, the maximal Radon transform $R^*(\mathbf{1}_E)(\gamma) = \sup_t m_{d-1}(E \cap P_{t,\gamma})$ is finite for a.e. γ when $d \geq 3$, using Theorem 30.4.9.
4. (Diagram.) Sketch the Besicovitch construction: the level-1 Cantor pieces of E_0 on $y = 0$ and of $E_1 = \frac{1}{2}C$ on $y = 1$, and several joining segments; indicate the slope range $s \notin (-1, 2)$ realized.
5. Prove the strip-average identity $R_\delta(f)(t, \gamma) = \frac{1}{2\delta} \int_{t-\delta \leq x \cdot \gamma \leq t+\delta} f$ from the definition, and show $R_\delta(\mathbf{1}_E)(t, \gamma) \rightarrow m_{d-1}(E \cap P_{t,\gamma})$ as $\delta \rightarrow 0$ at Lebesgue points.
6. Verify the logarithmic divergence at $d = 2$: compute $\int_{1 < |\lambda| \leq 1/\delta} |\lambda|^{-1} d\lambda = \log(1/\delta)$ and explain its appearance as $(\log 1/\delta)^{1/2}$ in Theorem 30.4.11.
7. (Non-example.) Show that in $d = 2$ the bare estimate $\int_{S^1} R^*(f) d\sigma \leq c(\|f\|_{L^1} + \|f\|_{L^2})$ fails, using $f_\varepsilon(x) = (|x| + \varepsilon)^{-2+\delta} \mathbf{1}_{|x| \leq 1}$ and letting $\varepsilon \rightarrow 0$.
8. Using $C + \frac{1}{2}C = [0, 3/2]$ (Lemma 30.4.14), determine the set $\frac{1}{2}C + C$ and explain the symmetry $C = 1 - C$ used in computing realizable slopes.
9. Show that $m(C + \lambda C) = 0$ for λ irrational of bounded type by a digit-counting estimate, illustrating the a.e. statement with a concrete λ .
10. Prove Corollary 30.4.2 from Theorem 30.4.1: a measure-zero set in $\mathbb{R}^d$ ($d \geq 3$) has a.e.-direction slices all null. Then show by the Besicovitch set that this fails in $\mathbb{R}^2$.

Part IV

Fourier Analysis

Chapter 31

The Genesis of Fourier Analysis

31.1 The Vibrating String and the Wave Equation

Simple harmonic motion

Fourier analysis was born from two physical problems: the vibrating string and the flow of heat. Both are governed by partial differential equations — the *wave equation* and the *heat equation* — and in both cases the natural way to solve them produces infinite series of sines and cosines. We follow the historical development, beginning with the simplest oscillatory system of all.

A mass m on an ideal spring (Hooke's law, restoring force $F = -ky$) obeys Newton's law $-ky(t) = my''(t)$. Writing $c = \sqrt{k/m}$ this becomes the second-order ODE

$$y''(t) + c^2 y(t) = 0. \quad (31.1)$$

Theorem 31.1.1 (General solution of simple harmonic motion). *Every twice-differentiable solution of (31.1) has the form*

$$y(t) = a \cos ct + b \sin ct, \quad a, b \in \mathbb{R}.$$

Equivalently $y(t) = A \cos(ct - \varphi)$ with amplitude $A = \sqrt{a^2 + b^2}$, natural frequency c , phase φ (determined mod 2π), and period $2\pi/c$.

Proof. Direct substitution shows $\cos ct$ and $\sin ct$ solve (31.1), hence so does every combination $a \cos ct + b \sin ct$. For uniqueness, let y be any solution and set $g(t) = y(t) \cos ct - c^{-1}y'(t) \sin ct$ and $h(t) = y(t) \sin ct + c^{-1}y'(t) \cos ct$. Using $y'' = -c^2 y$ one computes $g'(t) = 0$ and $h'(t) = 0$, so $g \equiv a$ and $h \equiv b$ are constants. Then $a \cos ct + b \sin ct = g \cos ct + h \sin ct = y(t)(\cos^2 ct + \sin^2 ct) = y(t)$, since the y' terms cancel. Thus $y(t) = a \cos ct + b \sin ct$. To recast in amplitude-phase form put $A = \sqrt{a^2 + b^2}$ and choose φ with $\cos \varphi = a/A$, $\sin \varphi = b/A$; then $A \cos(ct - \varphi) = A \cos ct \cos \varphi + A \sin ct \sin \varphi = a \cos ct + b \sin ct$. $\square$

Two features of this elementary computation persist throughout the subject. First, the most basic oscillation already involves the functions $\cos t$ and $\sin t$, tied together by Euler's identity $e^{it} = \cos t + i \sin t$. Second, a second-order equation needs *two* initial conditions; given $y(0)$ and $y'(0)$ the solution is forced to be $y(t) = y(0) \cos ct + \frac{y'(0)}{c} \sin ct$.

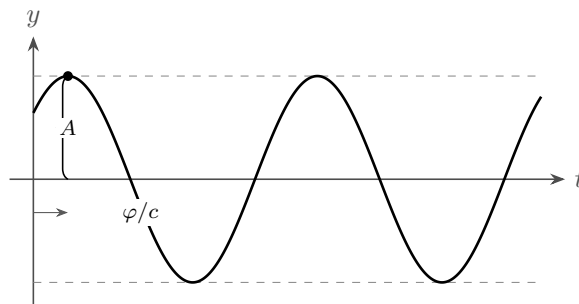


Figure 31.1. The graph of $A \cos(ct - \varphi)$: amplitude A , period $2\pi/c$, and a horizontal shift φ/c relative to $\cos ct$.

Worked Example 31.1.2 (Matching initial data to amplitude and phase). A mass obeys $y'' + 4y = 0$ with $y(0) = 1$, $y'(0) = 2$. Find $y(t)$ and write it as $A \cos(2t - \varphi)$.

Recognition. Trigger: a fixed second-order oscillator with prescribed position and velocity. Use $y(t) = y(0) \cos ct + \frac{y'(0)}{c} \sin ct$ with $c = 2$, then convert to amplitude–phase.

Solution. Here $c = 2$, so $y(t) = \cos 2t + \frac{2}{2} \sin 2t = \cos 2t + \sin 2t$. Then $A = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\cos \varphi = \sin \varphi = 1/\sqrt{2}$, giving $\varphi = \pi/4$. Hence $y(t) = \sqrt{2} \cos(2t - \pi/4)$. *Check.* $y(0) = \sqrt{2} \cos(-\pi/4) = 1$ and $y'(0) = \sqrt{2} \cdot 2 \sin(\pi/4) = 2$, matching both conditions. ■

Worked Example 31.1.3 (Why $\lambda \geq 0$ gives no oscillation). In (31.1) replace c^2 by $-\lambda$, i.e. consider $y'' - \lambda y = 0$. Show that oscillatory (bounded, non-monotone) solutions occur only for $\lambda < 0$.

Recognition. Trigger: classify solutions of $y'' = \lambda y$ by the sign of λ . This is the dichotomy that later selects the admissible separation constants.

Solution. If $\lambda > 0$, write $\lambda = \mu^2$; solutions are $y = Ae^{\mu t} + Be^{-\mu t}$, a combination of growing/decaying exponentials, never periodic unless $A = B = 0$. If $\lambda = 0$, $y = A + Bt$ is affine, again non-oscillatory. If $\lambda < 0$, write $\lambda = -m^2$; solutions are $y = A \cos mt + B \sin mt$, genuinely periodic with period $2\pi/m$. *Check.* Only the last case keeps $|y|$ bounded for all t with non-constant behaviour, which is exactly the physical demand on a vibration. This sign analysis recurs verbatim when we separate variables in the wave and heat equations.

Standing and traveling waves; derivation of the wave equation

The string is fixed at its ends $x = 0$ and $x = L$; its displacement $y = u(x, t)$ depends on position and time. Two motions guide the modeling. A *standing wave* factors as $u(x, t) = \varphi(x)\psi(t)$ (a fixed profile φ scaled in time) — this suggests *separation of variables*. A *traveling wave* $u(x, t) = F(x - ct)$ is a rigid profile sliding right at speed c (and $F(x + ct)$ slides left).

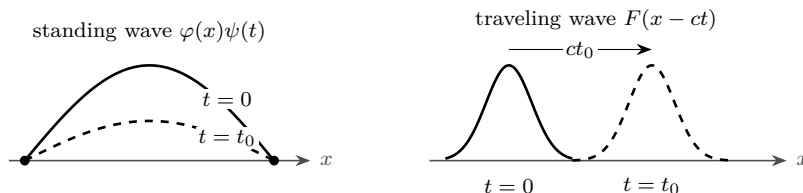


Figure 31.2. Left: a standing wave keeps its shape and rescales. Right: a traveling wave keeps its shape and shifts right by ct . (Schematic.)

To derive the governing PDE, model the string as N beads at $x_n = nL/N$, spacing $h = L/N$, each of mass ρh . The tension pulls bead n toward its neighbors with force $(\tau/h)(y_{n+1} - y_n)$ from the right and $(\tau/h)(y_{n-1} - y_n)$ from the left, so

$$\rho h y_n''(t) = \frac{\tau}{h} (y_{n+1}(t) + y_{n-1}(t) - 2y_n(t)).$$

Since $y_{n\pm 1} = u(x_n \pm h, t)$ and, for C^2 functions, $\frac{F(x+h) + F(x-h) - 2F(x)}{h^2} \rightarrow F''(x)$ as $h \rightarrow 0$, dividing by h and letting $h \rightarrow 0$ gives $\rho u_{tt} = \tau u_{xx}$, i.e.

Theorem 31.1.4 (The one-dimensional wave equation). *The displacement $u(x, t)$ of the vibrating string satisfies*

$$\frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad c = \sqrt{\tau/\rho}.$$

By the scaling $x = aX$, $t = bT$ with $a = L/\pi$, $b = L/(c\pi)$ one may assume $c = 1$ and the spatial interval is $[0, \pi]$; in those units $u_{tt} = u_{xx}$ on $0 \leq x \leq \pi$.

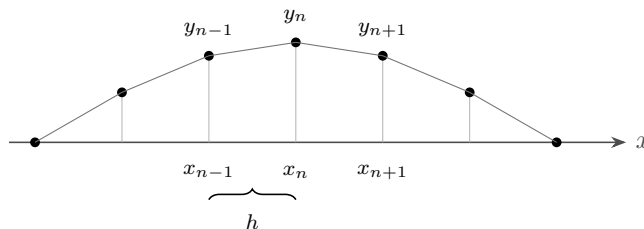


Figure 31.3. The string as N beads of mass ρh spaced h apart; bead n is pulled toward its two neighbors, and $h \rightarrow 0$ produces $u_{tt} = u_{xx}$. (Schematic.)

Worked Example 31.1.5 (Verifying the second-difference limit). Show directly that for $F \in C^2$,
$$\frac{F(x+h) + F(x-h) - 2F(x)}{h^2} \rightarrow F''(x).$$

Recognition. Trigger: justify the discrete-to-continuous passage in the derivation. Use the Taylor expansion with remainder.

Solution. Write $F(x \pm h) = F(x) \pm hF'(x) + \frac{h^2}{2}F''(x) + h^2\varepsilon_{\pm}(h)$ where $\varepsilon_{\pm}(h) \rightarrow 0$ as $h \rightarrow 0$ (this is Taylor's theorem; Stein Exercise 8). Adding the two, the first-order terms cancel: $F(x+h) + F(x-h) - 2F(x) = h^2F''(x) + h^2(\varepsilon_+ + \varepsilon_-)$. Dividing by h^2 gives $F''(x) + \varepsilon_+ + \varepsilon_- \rightarrow F''(x)$. *Check.* For $F(x) = x^2$ the second difference is exactly $2h^2/h^2 = 2 = F''$, with no error term, as the formula predicts.

Solution by traveling waves: d'Alembert's formula

Theorem 31.1.6 (General solution; d'Alembert). A C^2 function solves $u_{tt} = u_{xx}$ on all of $\mathbb{R}^2$ iff $u(x, t) = F(x+t) + G(x-t)$ for some C^2 functions F, G . With initial data $u(x, 0) = f(x)$, $u_t(x, 0) = g(x)$,

$$u(x, t) = \frac{1}{2}[f(x+t) + f(x-t)] + \frac{1}{2} \int_{x-t}^{x+t} g(y) dy.$$

For the string on $[0, \pi]$ with $u(0, t) = u(\pi, t) = 0$, extend f, g to be odd on $[-\pi, \pi]$ and 2π -periodic; then the formula automatically keeps the ends fixed.

Proof. If $u = F(x+t) + G(x-t)$ then $u_{tt} = F'' + G'' = u_{xx}$, so any such u solves the equation. Conversely change variables $\xi = x+t$, $\eta = x-t$ and set $v(\xi, \eta) = u(x, t)$. The chain rule gives $u_{tt} - u_{xx} = -4v_{\xi\eta}$, so the equation becomes $v_{\xi\eta} = 0$. Integrating in η shows v_{ξ} depends on ξ alone; integrating in ξ gives $v = F(\xi) + G(\eta)$, i.e. $u = F(x+t) + G(x-t)$. Now impose the data. At $t = 0$, $F(x) + G(x) = f(x)$, and differentiating u in t at $t = 0$ gives $F'(x) - G'(x) = g(x)$. Differentiating the first and combining: $2F' = f' + g$, $2G' = f' - g$, so $F(x) = \frac{1}{2}[f(x) + \int_0^x g + C_1]$ and $G(x) = \frac{1}{2}[f(x) - \int_0^x g + C_2]$ with $C_1 + C_2 = 0$. Substituting into $u = F(x+t) + G(x-t)$ telescopes to the stated formula. The odd 2π -periodic extensions make $f(-x) = -f(x)$ and $\int_{-t}^t g = 0$, giving $u(0, t) = 0$, and similarly $u(\pi, t) = 0$. $\square$

Remark. The passage $t \geq 0 \rightarrow t \in \mathbb{R} \rightarrow t \geq 0$ uses the *time-reversal* symmetry of the wave equation: if $u(x, t)$ solves it for $t \geq 0$ so does $u(x, -t)$. The heat equation, by contrast, is irreversible — a crucial difference we meet in §31.2.

Worked Example 31.1.7 (d'Alembert for a string released from rest). Solve $u_{tt} = u_{xx}$ with $u(x, 0) = \sin x$ (on $[0, \pi]$, extended oddly) and $u_t(x, 0) = 0$.

Recognition. Trigger: zero initial velocity, so the integral term vanishes; the answer is the average of two translated copies of f . A trig identity collapses it to a standing wave.

Solution. With $g \equiv 0$, $u(x, t) = \frac{1}{2}[\sin(x+t) + \sin(x-t)]$. Using $\sin(x+t) + \sin(x-t) = 2\sin x \cos t$ this is $u(x, t) = \cos t \sin x$. *Check.* $u(x, 0) = \sin x$ ■; $u_t(x, 0) = -\sin 0 \sin x = 0$ ■; and $u_{tt} = -\cos t \sin x = u_{xx}$ ■. This is exactly the fundamental standing wave u_1 .

Worked Example 31.1.8 (A purely traveling pulse). Solve $u_{tt} = u_{xx}$ on $\mathbb{R}$ with $u(x, 0) = 0$ and $u_t(x, 0) = g(x)$ where $g(x) = 1$ on $[-1, 1]$ and 0 otherwise.

Recognition. Trigger: zero initial position, nonzero velocity; only the integral term survives, and it produces an expanding plateau.

Solution. $u(x, t) = \frac{1}{2} \int_{x-t}^{x+t} g(y) dy = \frac{1}{2}$ (length of $[x-t, x+t] \cap [-1, 1]$). For $t \leq 1$ and $|x|$ small the interval $[x-t, x+t]$ lies inside $[-1, 1]$, giving $u = \frac{1}{2}(2t) = t$; far away $u = 0$; near the edges it ramps linearly. *Check.* At $x = 0$, small t : $u(0, t) = t$, so $u_t(0, 0) = 1 = g(0)$ ■, and $u(0, 0) = 0$ ■. The disturbance spreads outward at speed 1 (finite propagation speed), a hallmark of the wave equation.

Solution by standing waves: separation of variables

Seeking $u(x, t) = \varphi(x)\psi(t)$ in $u_{tt} = u_{xx}$ gives $\varphi\psi'' = \varphi''\psi$, hence $\psi''/\psi = \varphi''/\varphi$. The left side depends only on t , the right only on x , so both equal a constant λ . By Worked Example 31.1.4 oscillation forces $\lambda = -m^2 < 0$, giving $\psi = A \cos mt + B \sin mt$ and $\varphi = \tilde{A} \cos mx + \tilde{B} \sin mx$. The boundary conditions $\varphi(0) = \varphi(\pi) = 0$ force $\tilde{A} = 0$ and $m \in \mathbb{Z}$. We keep $m \geq 1$ and arrive at the *standing waves*

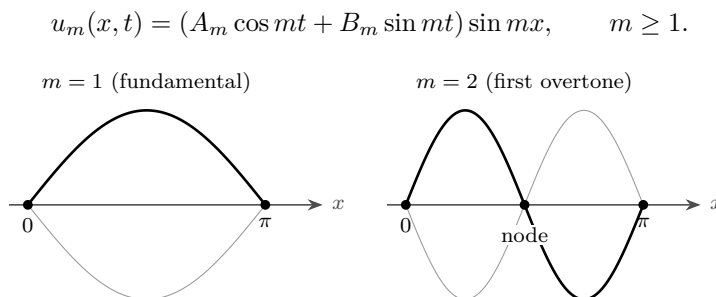


Figure 31.4. Harmonics $\sin mx$: the fundamental has no interior node; the first overtone has a node at $x = \pi/2$. Filled dots mark the fixed endpoints.

By linearity we superpose to reach the candidate general solution

$$u(x, t) = \sum_{m=1}^{\infty} (A_m \cos mt + B_m \sin mt) \sin mx. \quad (31.2)$$

Setting $t = 0$ and demanding $u(x, 0) = f(x)$ forces $\sum_{m \geq 1} A_m \sin mx = f(x)$. This raises the founding question of the subject: *can an arbitrary f on $[0, \pi]$ with $f(0) = f(\pi) = 0$ be written as a sine series?* Multiplying by $\sin nx$, integrating, and using orthogonality $\int_0^\pi \sin mx \sin nx dx = \frac{\pi}{2} \delta_{mn}$ gives the guess

$$A_n = \frac{2}{\pi} \int_0^\pi f(x) \sin nx dx, \quad (31.3)$$

the n th *Fourier sine coefficient*.

Worked Example 31.1.9 (Orthogonality of the sines on $[0, \pi]$). Prove $\int_0^\pi \sin mx \sin nx dx = \frac{\pi}{2} \delta_{mn}$ for positive integers m, n .

Recognition. Trigger: establish the orthogonality that legitimizes (31.3). Use the product-to-sum identity $2 \sin a \sin b = \cos(a-b) - \cos(a+b)$.

Solution. $2 \sin mx \sin nx = \cos((m-n)x) - \cos((m+n)x)$. If $m \neq n$, $\int_0^\pi \cos(kx) dx = [\sin kx/k]_0^\pi = 0$ for the nonzero integers $k = m \pm n$, so the integral is 0. If $m = n$, $\cos((m-n)x) = 1$ contributes $\int_0^\pi 1 dx = \pi$ while $\cos(2mx)$ integrates to 0; halving gives $\pi/2$. *Check.* For $m = n = 1$, $\int_0^\pi \sin^2 x dx = \int_0^\pi \frac{1-\cos 2x}{2} dx = \pi/2$. ■

Worked Example 31.1.10 (Computing a sine series from the coefficient formula). Find the Fourier sine coefficients on $[0, \pi]$ of the constant $f(x) = 1$ (an odd 2π -periodic square wave).

Recognition. Trigger: apply (31.3) to a flat profile. Expect only odd harmonics by symmetry.

Solution. $A_n = \frac{2}{\pi} \int_0^\pi \sin nx dx = \frac{2}{\pi} \left[\frac{1-\cos n\pi}{n} \right] = \frac{2}{\pi} \cdot \frac{1-(-1)^n}{n}$. This is 0 for even n and $\frac{4}{\pi n}$ for odd n , so $1 \sim \frac{4}{\pi} \sum_{k \geq 0} \frac{\sin((2k+1)x)}{2k+1}$ on $(0, \pi)$. *Check.* At $x = \pi/2$ the series is $\frac{4}{\pi} (1 - \frac{1}{3} + \frac{1}{5} - \dots) = \frac{4}{\pi} \cdot \frac{\pi}{4} = 1$, the value of f there, recovering Leibniz's series for $\pi/4$.

Worked Example 31.1.11 (The plucked string (triangular profile)). The string is plucked to height h at $p \in (0, \pi)$: $f(x) = \frac{xh}{p}$ on $[0, p]$ and $f(x) = \frac{h(\pi-x)}{\pi-p}$ on $[p, \pi]$, with $g \equiv 0$. Find $u(x, t)$.

Recognition. Trigger: a piecewise-linear initial shape; the sine coefficients can be integrated by parts, then (31.2) gives the motion. (Stein Exercise 9.)

Solution. Integration by parts on each piece yields $A_m = \frac{2h \sin mp}{m^2 p(\pi - p)}$. Since $g \equiv 0$ all $B_m = 0$, so

$$u(x, t) = \sum_{m=1}^{\infty} A_m \cos mt \sin mx, \quad A_m = \frac{2h \sin mp}{m^2 p(\pi - p)}.$$

Because $|A_m| \leq C/m^2$ the series converges absolutely and uniformly. Using $\cos mt \sin mx = \frac{1}{2}[\sin m(x+t) + \sin m(x-t)]$ it also equals the traveling-wave form $u(x, t) = \frac{1}{2}[f(x+t) + f(x-t)]$ (odd 2π -periodic f). *Check.* The m th harmonic is absent exactly when $\sin mp = 0$, i.e. $p = k\pi/m$ — plucking at a node of mode m kills that overtone. Plucking at the midpoint $p = \pi/2$ kills all even harmonics ($\sin m\pi/2 = 0$ for even m).

Worked Example 31.1.12 (Energy of a standing wave is constant). For the fundamental $u(x, t) = \cos t \sin x$ on $[0, \pi]$, show the total energy $E(t) = \frac{1}{2} \int_0^\pi (u_t^2 + u_x^2) dx$ is independent of t .

Recognition. Trigger: verify conservation of energy for a single mode; just compute the two integrals and add. Energy conservation is the analytic shadow of time-reversibility.

Solution. $u_t = -\sin t \sin x$ and $u_x = \cos t \cos x$. Then $\int_0^\pi u_t^2 dx = \sin^2 t \int_0^\pi \sin^2 x dx = \frac{\pi}{2} \sin^2 t$ and $\int_0^\pi u_x^2 dx = \cos^2 t \int_0^\pi \cos^2 x dx = \frac{\pi}{2} \cos^2 t$. Hence $E(t) = \frac{1}{2} \cdot \frac{\pi}{2} (\sin^2 t + \cos^2 t) = \frac{\pi}{4}$, constant. *Check.* Kinetic energy peaks at $t = \pi/2$ (string flat, maximal speed) while potential energy peaks at $t = 0$ (string maximally stretched, at rest); their sum stays fixed. ■

Worked Example 31.1.13 (A two-mode superposition and its period). Let $u(x, t) = \cos t \sin x + \frac{1}{2} \cos 3t \sin 3x$ solve $u_{tt} = u_{xx}$. Find the smallest $T > 0$ with $u(x, t+T) = u(x, t)$ for all x .

Recognition. Trigger: a finite superposition of standing waves; the time-period is the least common period of the temporal factors $\cos mt$.

Solution. The factors $\cos t$ and $\cos 3t$ have periods 2π and $2\pi/3$. The motion repeats when both do, i.e. at the least common multiple of 2π and $2\pi/3$, which is 2π . *Check.* $u(x, t+2\pi) = \cos(t+2\pi) \sin x + \frac{1}{2} \cos(3t+6\pi) \sin 3x = u(x, t)$, and no smaller shift works since $\cos t$ alone already needs 2π . The string's overall motion is 2π -periodic even though the overtone oscillates three times as fast. ■

Caution. The plucked-string solution $u = \frac{1}{2}[f(x+t) + f(x-t)]$ is genuinely useful — it describes the physical string — yet f has a corner at p , so u is not C^2 and does *not* satisfy $u_{tt} = u_{xx}$ in the classical sense. It is a solution only in a generalized (weak/distributional) sense. Do not demand classical smoothness of a Fourier solution when the data has corners.

Toward Fourier series on $[-\pi, \pi]$

Any F on $[-\pi, \pi]$ splits as $F = f + g$ with $f(x) = \frac{1}{2}[F(x) - F(-x)]$ odd and $g(x) = \frac{1}{2}[F(x) + F(-x)]$ even; the odd part has a sine series and the even part a cosine series. Combining via $e^{ix} = \cos x + i \sin x$ suggests the complex form $F(x) = \sum_{m \in \mathbb{Z}} a_m e^{imx}$, the launching point for §31.3.

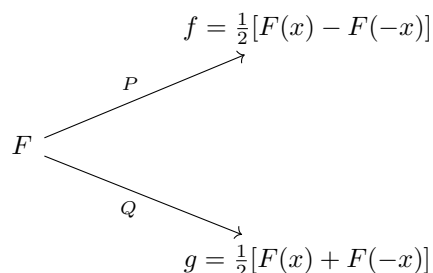


Figure 31.5. Parity decomposition $F = f + g$: the projection P onto the odd part f (which carries the sine series) and Q onto the even part g (the cosine series); together $f + g = F$ recovers the original function.

Exercises (Stein, Ch.1 — vibrating string).

1. (Ex. 6) Prove that if f is a twice continuously differentiable function on $\mathbb{R}$ solving $f''(t) + c^2 f(t) = 0$, then $f(t) = a \cos ct + b \sin ct$ for constants a, b . [Differentiate $g(t) = f(t) \cos ct - c^{-1} f'(t) \sin ct$ and $h(t) = f(t) \sin ct + c^{-1} f'(t) \cos ct$ and show both are constant.]
2. (Ex. 7) Show that if a, b are real then $a \cos ct + b \sin ct = A \cos(ct - \varphi)$ where $A = \sqrt{a^2 + b^2}$ and φ satisfies $\cos \varphi = a/\sqrt{a^2 + b^2}$, $\sin \varphi = b/\sqrt{a^2 + b^2}$.
3. (Ex. 8) Suppose F has two continuous derivatives on (a, b) . Show that for $x, x + h \in (a, b)$ one may write $F(x + h) = F(x) + hF'(x) + \frac{h^2}{2}F''(x) + h^2\varphi(h)$ with $\varphi(h) \rightarrow 0$ as $h \rightarrow 0$. Deduce that $\frac{F(x + h) + F(x - h) - 2F(x)}{h^2} \rightarrow F''(x)$ as $h \rightarrow 0$. [Taylor expansion: start from $F(x + h) - F(x) = \int_x^{x+h} F'(y) dy$ and write $F'(y) = F'(x) + (y - x)F''(x) + (y - x)\psi(y - x)$ with $\psi(h) \rightarrow 0$.]
4. (Ex. 9) For the plucked string with triangular initial data of height h at p , use the Fourier sine coefficient formula to show $A_m = \frac{2h \sin mp}{m^2 p(\pi - p)}$. For which p are the second, fourth, ... harmonics missing? For which p are the third, sixth, ... harmonics missing?

Additional exercises.

1. Solve $y'' + 9y = 0$ with $y(0) = 2$, $y'(0) = -3$, and rewrite the answer as $A \cos(3t - \varphi)$ with $A > 0$ and $\varphi \in [0, 2\pi)$.
2. Verify directly that $u_m(x, t) = (A_m \cos mt + B_m \sin mt) \sin mx$ solves $u_{tt} = u_{xx}$ on $[0, \pi]$ and satisfies $u_m(0, t) = u_m(\pi, t) = 0$.
3. Using d'Alembert's formula, solve $u_{tt} = u_{xx}$ on $\mathbb{R}$ with $u(x, 0) = \cos x$, $u_t(x, 0) = 0$, and collapse the answer to a single standing wave.
4. Compute the Fourier sine coefficients on $[0, \pi]$ of $f(x) = x(\pi - x)$ and use the value at $x = \pi/2$ to evaluate $\sum_{k \geq 0} (-1)^k / (2k + 1)^3$.
5. For the plucked string plucked at the midpoint $p = \pi/2$, show every *even* harmonic is absent, and write the first three nonzero terms of $u(x, t)$.
6. (Non-example) Show that $u(x, t) = F(x - 2t)$ does *not* solve $u_{tt} = u_{xx}$ unless F is affine, and identify which wave equation it *does* solve.
7. (Diagram) Sketch the standing waves $\sin x$, $\sin 2x$, $\sin 3x$ on $[0, \pi]$ on one axis, mark all interior nodes, and tabulate the number of nodes versus the mode number m .
8. Show that the change of variables $x = aX$, $t = bT$ with $a = L/\pi$, $b = L/(c\pi)$ turns $c^{-2}u_{tt} = u_{xx}$ on $[0, L]$ into $U_{TT} = U_{XX}$ on $[0, \pi]$.
9. Prove that if f is odd and 2π -periodic then $\int_{-t}^t f = 0$ for all t , and use this to confirm that d'Alembert's formula keeps the ends of the string fixed.

31.2 The Heat Equation on the Circle

Derivation of the heat equation

We repeat the framework of §31.1 for heat flow. Model a metal plate as $\mathbb{R}^2$ with temperature $u(x, y, t)$. Consider a small square S of side h centered at (x_0, y_0) . The heat energy in S is $H(t) = \sigma \iint_S u \, dx \, dy$ ($\sigma > 0$ the specific heat), so the rate of heat gain is $H'(t) = \sigma \iint_S u_t \, dx \, dy \approx \sigma h^2 u_t(x_0, y_0, t)$.

Newton's law of cooling says heat flows from hot to cold at a rate proportional to the temperature gradient. The flow *out* through the right edge is $-\kappa h u_x(x_0 + \frac{h}{2}, y_0, t)$ ($\kappa > 0$ the conductivity); summing the four edges, the net heat flow into S is

$$\kappa h \left[u_x(x_0 + \frac{h}{2}, y_0) - u_x(x_0 - \frac{h}{2}, y_0) + u_y(x_0, y_0 + \frac{h}{2}) - u_y(x_0, y_0 - \frac{h}{2}) \right].$$

Equating to $\sigma h^2 u_t$, dividing by κh^2 , applying the mean value theorem and letting $h \rightarrow 0$ gives the result.

Theorem 31.2.1 (The heat equation). *The temperature $u(x, y, t)$ satisfies the time-dependent heat equation*

$$\frac{\sigma}{\kappa} \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \Delta u.$$

In one space dimension this reads $u_t = u_{xx}$ (after choosing units).

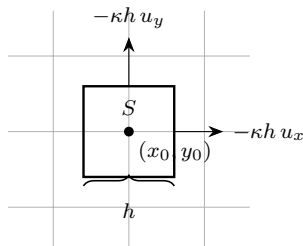


Figure 31.6. Heat balance on a square S : net inflow through the edges equals $\sigma h^2 u_t$; the limit $h \rightarrow 0$ yields $u_t = \Delta u$. (Schematic.)

Remark. The heat equation is *not* time-reversible: replacing $t \mapsto -t$ turns $u_t = u_{xx}$ into $u_t = -u_{xx}$, an ill-posed (backward) problem. This is the analytic expression of the physical irreversibility of diffusion, in sharp contrast to the wave equation.

Worked Example 31.2.2 (One-dimensional heat decay of a single mode). Solve $u_t = u_{xx}$ on $[0, \pi]$ with $u(0, t) = u(\pi, t) = 0$ and $u(x, 0) = \sin 3x$.

Recognition. Trigger: a single sine mode as initial data; separation gives a spatial sine times a decaying exponential in time. The decay rate is the square of the mode number.

Solution. Try $u = \psi(t) \sin 3x$. Then $u_t = \psi' \sin 3x$ and $u_{xx} = -9\psi \sin 3x$, so $\psi' = -9\psi$, giving $\psi(t) = \psi(0)e^{-9t}$. With $\psi(0) = 1$, $u(x, t) = e^{-9t} \sin 3x$. *Check.* $u(x, 0) = \sin 3x$ ■, ends vanish ■, and $u_t = -9e^{-9t} \sin 3x = u_{xx}$ ■. Higher modes decay faster ($e^{-m^2 t}$), so heat distributions smooth out instantly — the regularizing effect of diffusion.

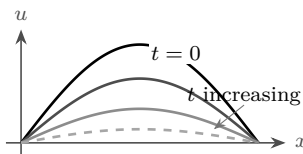


Figure 31.7. Heat decay $u = e^{-t} \sin x$: the profile keeps its shape but its amplitude decays like e^{-t} ; a mode $\sin mx$ decays like $e^{-m^2 t}$. (Schematic.)

Steady state: the Dirichlet problem in the disc

After a long time $u_t = 0$ and the heat equation becomes the *steady-state* (Laplace) equation $\Delta u = 0$; solutions are *harmonic functions*. On the unit disc $D = \{x^2 + y^2 < 1\}$ with boundary circle C , the *Dirichlet problem* asks for the harmonic u in D with prescribed boundary values $u = f$ on C .

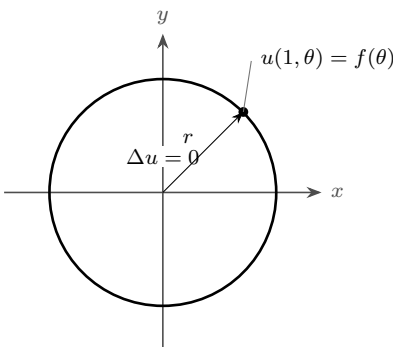


Figure 31.8. The Dirichlet problem on the disc: solve $\Delta u = 0$ inside subject to the boundary temperature $u(1, \theta) = f(\theta)$. (Schematic.)

Theorem 31.2.3 (Laplacian in polar coordinates). *In polar coordinates (r, θ) ,*

$$\Delta u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}.$$

Proof. With $x = r \cos \theta$, $y = r \sin \theta$ the chain rule gives $u_r = u_x \cos \theta + u_y \sin \theta$ and $u_\theta = -u_x r \sin \theta + u_y r \cos \theta$. Differentiating again and using $r_x = \cos \theta$, $\theta_x = -\sin \theta / r$ (and the y -analogues) one finds $u_{xx} + u_{yy} = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta}$; the mixed terms in $\cos \theta \sin \theta$ cancel. (This is Stein Exercise 10.) $\square$

Multiplying $\Delta u = 0$ by r^2 and separating $u(r, \theta) = F(r)G(\theta)$ gives

$$\frac{r^2 F''(r) + r F'(r)}{F(r)} = -\frac{G''(\theta)}{G(\theta)} = \lambda,$$

so $G'' + \lambda G = 0$ and $r^2 F'' + r F' - \lambda F = 0$. Periodicity $G(\theta + 2\pi) = G(\theta)$ forces $\lambda = m^2$, $m \in \mathbb{Z}$, with $G = \tilde{A} \cos m\theta + \tilde{B} \sin m\theta = A e^{im\theta} + B e^{-im\theta}$.

Theorem 31.2.4 (Radial solutions and the special harmonics). *For $\lambda = m^2$, $m \neq 0$, the equation $r^2 F'' + r F' - m^2 F = 0$ has the two solutions $F = r^m$ and $F = r^{-m}$; for $m = 0$ they are $F = 1$ and $F = \log r$. Discarding the solutions unbounded at the origin (r^{-m} for $m > 0$, and $\log r$), the bounded special harmonics are*

$$u_m(r, \theta) = r^{|m|} e^{im\theta}, \quad m \in \mathbb{Z}.$$

Proof. $r^2 F'' + r F' - m^2 F = 0$ is an Euler equation; trying $F = r^\alpha$ gives $\alpha(\alpha - 1) + \alpha - m^2 = \alpha^2 - m^2 = 0$, so $\alpha = \pm m$. For $m = 0$ the indicial root is double, giving 1 and $\log r$. As $r \rightarrow 0$, $r^{-m} \rightarrow \infty$ ($m > 0$) and $\log r \rightarrow -\infty$, both physically inadmissible (infinite temperature at the center), so we keep $r^{|m|} e^{im\theta}$. One checks $\Delta(r^{|m|} e^{im\theta}) = 0$ directly via Theorem 31.2.3. (Stein Exercise 11.) $\square$

Superposing the bounded harmonics yields the candidate

$$u(r, \theta) = \sum_{m=-\infty}^{\infty} a_m r^{|m|} e^{im\theta}, \quad (31.4)$$

and the boundary condition $u(1, \theta) = \sum_m a_m e^{im\theta} = f(\theta)$ returns us, once more, to the founding question: *when can f be expanded as $\sum_m a_m e^{im\theta}$?*

Worked Example 31.2.5 (Solving a Dirichlet problem with finite boundary data). Solve $\Delta u = 0$ in the disc with $u(1, \theta) = 3 + 2 \cos \theta - \sin 2\theta$.

Recognition. Trigger: the boundary data is already a finite trig polynomial, so the solution is the matching finite sum of special harmonics $r^{|m|} e^{im\theta}$ — no infinite series needed.

Solution. Read off the modes: the constant 3 is $m = 0$ ($r^0 = 1$); $2 \cos \theta = e^{i\theta} + e^{-i\theta}$ is $m = \pm 1$, carrying factor r ; $-\sin 2\theta$ is $m = \pm 2$, factor r^2 . Hence $u(r, \theta) = 3 + 2r \cos \theta - r^2 \sin 2\theta$. *Check.* At $r = 1$ this is the given f ■; each term is harmonic ($r^{|m|} \cos m\theta$ and $r^{|m|} \sin m\theta$ solve $\Delta u = 0$), and $u(0, \theta) = 3$ is the average of f over the circle (the mean-value property). ■

Worked Example 31.2.6 (The harmonic conjugate viewpoint). Verify that $u(r, \theta) = r^2 \cos 2\theta$ is harmonic and identify it in Cartesian coordinates.

Recognition. Trigger: confirm a special harmonic and connect to $\operatorname{Re}(z^m)$. Recall $z^m = r^m e^{im\theta}$.

Solution. By Theorem 31.2.4 with $m = 2$, $u = \operatorname{Re}(r^2 e^{2i\theta}) = \operatorname{Re}(z^2)$. Since $z^2 = (x + iy)^2 = x^2 - y^2 + 2ixy$, we get $u = x^2 - y^2$, and indeed $u_{xx} + u_{yy} = 2 - 2 = 0$. *Check.* In polar form $u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 2 \cos 2\theta + \frac{1}{r} (2r \cos 2\theta) + \frac{1}{r^2} (-4r^2 \cos 2\theta) = (2 + 2 - 4) \cos 2\theta = 0$. ■ The real and imaginary parts of z^m are precisely the basic harmonics $r^m \cos m\theta$ and $r^m \sin m\theta$.

Worked Example 31.2.7 (Why r^{-m} and $\log r$ are rejected). Show $u(r, \theta) = \log r$ solves $\Delta u = 0$ for $r > 0$ but cannot be the steady temperature of the disc.

Recognition. Trigger: confirm a singular radial solution and explain its physical inadmissibility.

Solution. With $m = 0$, $F = \log r$: $F' = 1/r$, $F'' = -1/r^2$, so $F'' + \frac{1}{r}F' = -1/r^2 + 1/r^2 = 0$ and $u_{\theta\theta} = 0$; thus $\Delta u = 0$ on $r > 0$. But $\log r \rightarrow -\infty$ as $r \rightarrow 0$, an infinite (negative) temperature at the center, which is unphysical and not even defined at the origin. *Check.* This is the potential of a point heat sink/source; it is harmonic on the punctured disc but not on all of D , exactly why Theorem 31.2.4 discards it.

Worked Example 31.2.8 (Steady state of the half-warmed circle (qualitative)). Let $f(\theta) = 1$ for $0 < \theta < \pi$ and $f(\theta) = 0$ for $\pi < \theta < 2\pi$. Without summing the full series, find u at the center and describe the level behaviour.

Recognition. Trigger: use the mean-value property of harmonic functions for the center value; the $m = 0$ coefficient is the average of f .

Solution. The center value equals the average $a_0 = \frac{1}{2\pi} \int_0^{2\pi} f = \frac{1}{2\pi} \cdot \pi = \frac{1}{2}$. So $u(0) = \frac{1}{2}$, midway between the hot and cold halves. The isotherms bulge toward the cold arc near the boundary and flatten to the constant $\frac{1}{2}$ at the center. *Check.* By symmetry $\theta \mapsto -\theta$ swaps the two arcs and sends $f \mapsto 1 - f$, so $u(0)$ must equal $1 - u(0)$, forcing $u(0) = \frac{1}{2}$. ■ (The full solution is the Poisson integral, developed in later chapters.)

Worked Example 31.2.9 (Total heat decays to the mean). For $u_t = u_{xx}$ on $[0, \pi]$ with insulated-free expansion $u(x, 0) = \sum_{m \geq 1} b_m \sin mx$, show each modal energy $\int_0^\pi u(\cdot, t)^2 dx$ decays and identify the slowest rate.

Recognition. Trigger: track the L^2 size of the solution in time; orthogonality makes the modes decouple, and the lowest mode decays slowest.

Solution. The solution is $u(x, t) = \sum_m b_m e^{-m^2 t} \sin mx$. By orthogonality $\int_0^\pi u^2 dx = \frac{\pi}{2} \sum_m b_m^2 e^{-2m^2 t}$. Every term decays; the $m = 1$ term, $\propto e^{-2t}$, decays slowest, so for large t , $\int_0^\pi u^2 \approx \frac{\pi}{2} b_1^2 e^{-2t}$. *Check.* If only the $m = 2$ mode is present the L^2 norm falls like e^{-4t} , twice as fast in the exponent — short-wavelength roughness is smoothed away first, the defining feature of diffusion. ■

Caution. Rejecting $\log r$ and r^{-m} is a *boundedness* choice, not a mathematical necessity: they are perfectly good harmonic functions on the punctured disc or the annulus. On a domain that excludes the origin (an annulus $a < r < b$) you must keep them, and the general solution there has both r^m and r^{-m} for each m .

Remark (Historical note). D'Alembert (1747) first solved the vibrating string by traveling waves; Euler elaborated it in 1748. In 1753 D. Bernoulli proposed the trigonometric-series solution (31.2), but Euler and d'Alembert doubted that an “arbitrary” function admits such an expansion. Fourier (1807), studying heat, had the conviction his predecessors lacked, and his work led to the eventual proofs (Dirichlet and others) that began the subject.

Exercises (Stein, Ch.1 — heat equation and the disc).

- (Ex. 10) Show that the Laplacian $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is given in polar coordinates by $\Delta = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}$. Also prove $|\partial u / \partial x|^2 + |\partial u / \partial y|^2 = |\partial u / \partial r|^2 + \frac{1}{r^2} |\partial u / \partial \theta|^2$.
- (Ex. 11) Show that if $n \in \mathbb{Z}$ the only twice-differentiable solutions for $r > 0$ of $r^2 F''(r) + rF'(r) - n^2 F(r) = 0$ are linear combinations of r^n and r^{-n} when $n \neq 0$, and of 1 and $\log r$ when $n = 0$. [Write $F(r) = g(r)r^n$, find the equation for g , and conclude $rg'(r) + 2ng(r) = c$ for a constant c .]

Problems (Stein, Ch.1).

- (Prob. 1) Consider the Dirichlet problem in the rectangle $R = \{(x, y) : 0 \leq x \leq \pi, 0 \leq y \leq 1\}$: solve $\Delta u = 0$ with $u = 0$ on the vertical sides and $u(x, 0) = f_0(x)$, $u(x, 1) = f_1(x)$ on the horizontal sides. Use separation of variables to show that if $f_0(x) = \sum_{k \geq 1} A_k \sin kx$ and $f_1(x) = \sum_{k \geq 1} B_k \sin kx$, then

$$u(x, y) = \sum_{k=1}^{\infty} \left(\frac{\sinh k(1-y)}{\sinh k} A_k + \frac{\sinh ky}{\sinh k} B_k \right) \sin kx,$$

where $\sinh x = \frac{1}{2}(e^x - e^{-x})$ and $\cosh x = \frac{1}{2}(e^x + e^{-x})$.

Additional exercises.

1. Solve $u_t = u_{xx}$ on $[0, \pi]$ with $u(0, t) = u(\pi, t) = 0$ and $u(x, 0) = 2 \sin x - \sin 4x$, and find the time t at which the $\sin x$ mode dominates the $\sin 4x$ mode by a factor 100.
2. Verify by separation of variables that $u(x, t) = \sum_{m \geq 1} b_m e^{-m^2 t} \sin mx$ solves the 1D heat equation with $u(x, 0) = \sum_m b_m \sin mx$.
3. Solve the Dirichlet problem in the disc with boundary data $u(1, \theta) = 4 + \cos \theta + 3 \sin 3\theta$, and state the center value $u(0)$.
4. Confirm that $u(r, \theta) = r^3 \sin 3\theta$ is harmonic both in polar form (via Theorem 31.2.3) and in Cartesian form by writing it as $\text{Im}(z^3)$.
5. (Non-example) Show that $u(r, \theta) = r^{-2} \cos 2\theta$ satisfies $\Delta u = 0$ for $r > 0$ but is *not* an admissible steady temperature of the unit disc; on which domain is it admissible?
6. (Diagram) For the boundary data $f(\theta) = \cos \theta$, sketch several isotherms of the solution $u = r \cos \theta$ inside the disc and mark the hottest and coldest boundary points.
7. Show that the average of a harmonic function over the boundary circle equals its value at the center (mean-value property), starting from the expansion (31.4).
8. Using Stein Problem 1, solve $\Delta u = 0$ in the rectangle with $u(x, 0) = \sin x$, $u(x, 1) = 0$, and verify $u(x, y) = \frac{\sinh(1-y)}{\sinh 1} \sin x$.
9. Explain, using the sign change under $t \mapsto -t$, why the backward heat equation $u_t = -u_{xx}$ is ill-posed, contrasting with the time-reversibility of the wave equation.

31.3 Fourier Series: First Definitions and Examples

From sines and cosines to complex exponentials

Both physical problems ended at the same question: expand a function on the circle in the basic oscillations $e^{im\theta}$. We now make the definitions precise. Throughout, a function on the circle is a 2π -periodic function on $\mathbb{R}$; we integrate over any interval of length 2π , usually $[-\pi, \pi]$ or $[0, 2\pi]$.

The engine is the orthogonality relation, the complex analogue of the sine relation in §31.1.

Lemma 31.3.1 (Orthogonality of the exponentials). *For integers n, m ,*

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{imx} e^{-inx} dx = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases}$$

Proof. The integrand is $e^{i(m-n)x}$. If $m = n$ it equals 1 and the average is 1. If $m \neq n$, let $k = m - n \neq 0$; then $\int_{-\pi}^{\pi} e^{ikx} dx = \left[\frac{e^{ikx}}{ik} \right]_{-\pi}^{\pi} = \frac{e^{ik\pi} - e^{-ik\pi}}{ik} = \frac{2i \sin k\pi}{ik} = 0$ since $\sin k\pi = 0$. $\square$

Definition 31.3.2 (Fourier coefficients and Fourier series). Let F be (Riemann) integrable on the circle. Its n th *Fourier coefficient* is

$$\widehat{F}(n) = a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(x) e^{-inx} dx, \quad n \in \mathbb{Z},$$

and its *Fourier series* is the formal sum $F(x) \sim \sum_{n=-\infty}^{\infty} a_n e^{inx}$. The N th *partial sum* is $S_N(F)(x) = \sum_{n=-N}^N a_n e^{inx}$.

The symbol $\sim$ is deliberately noncommittal: defining the coefficients is automatic, but whether the series converges to F — and in what sense — is the central problem of the whole theory, taken up in the chapters that follow. The motivation is exactly Lemma 31.3.1: if $F = \sum_m a_m e^{imx}$ converged nicely, then multiplying by e^{-inx} and integrating term by term would isolate a_n .

The assignment $F \mapsto \widehat{F}$ is itself a linear map from integrable functions on the circle to bi-infinite sequences, and it intertwines differentiation with multiplication by the symbol in (the rule $\widehat{F'}(n) = in \widehat{F}(n)$, justified in §31.3).

$$\begin{array}{ccc} F & \xrightarrow{\mathcal{F}} & \widehat{F} = (a_n)_{n \in \mathbb{Z}} \\ \downarrow \frac{d}{dx} & & \downarrow n \mapsto in a_n \\ F' & \xrightarrow{\mathcal{F}} & \widehat{F'} = (in a_n)_{n \in \mathbb{Z}} \end{array}$$

Figure 31.9. The Fourier coefficient map $\mathcal{F} : F \mapsto \widehat{F}$ turns differentiation (left) into multiplication of the n th coefficient by in (right); the square commutes, since $\widehat{F'}(n) = in \widehat{F}(n)$.

Method (Computing Fourier coefficients). *Recognition trigger.* You are handed a periodic F and asked for $\widehat{F}(n)$, its Fourier series, or a series-summation identity that follows.

1. Choose the cleanest period interval ($[-\pi, \pi]$ exploits even/odd symmetry).
2. If F is *even*, only cosines survive ($a_{-n} = a_n$, real); if *odd*, only sines ($a_{-n} = -a_n$, imaginary). Use this to halve the work.
3. Evaluate $a_n = \frac{1}{2\pi} \int F(x) e^{-inx} dx$, integrating by parts for polynomial-times-exponential integrands; treat $n = 0$ (the average) separately.
4. Optionally re-express in real form $\frac{a_0}{2} + \sum_{n \geq 1} (A_n \cos nx + B_n \sin nx)$ with $A_n = a_n + a_{-n}$, $B_n = i(a_n - a_{-n})$.

Worked Example 31.3.3 (Fourier coefficients of e^{ikx} and of $\cos, \sin$). Compute the Fourier coefficients of $F(x) = e^{ikx}$ ($k \in \mathbb{Z}$), then of $\cos x$ and $\sin x$.

Recognition. Trigger: the simplest possible data — a single character — tests the definition. Orthogonality does all the work.

Solution. By Lemma 31.3.1, $\widehat{e^{ikx}}(n) = \frac{1}{2\pi} \int e^{ikx} e^{-inx} dx = \delta_{kn}$: the coefficients are 1 at $n = k$ and 0 elsewhere. Since $\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$, its coefficients are $\frac{1}{2}$ at $n = \pm 1$, else 0; since $\sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$, they are $\frac{1}{2i} = -\frac{i}{2}$ at $n = 1$, $+\frac{i}{2}$ at $n = -1$. *Check.* Reassembling, $\sum a_n e^{inx} = \frac{1}{2}e^{ix} + \frac{1}{2}e^{-ix} = \cos x$. ■ A trig polynomial is its own Fourier series.

Worked Example 31.3.4 (The sawtooth wave). Find the Fourier series of $F(x) = x$ on $(-\pi, \pi)$, extended 2π -periodically.

Recognition. Trigger: odd data, so expect a pure sine series; integrate $x e^{-inx}$ by parts.

Solution. $a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x dx = 0$. For $n \neq 0$, integrating by parts, $a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} x e^{-inx} dx = \frac{1}{2\pi} \left[\frac{x e^{-inx}}{-in} \right]_{-\pi}^{\pi} + \frac{1}{2\pi in} \int_{-\pi}^{\pi} e^{-inx} dx$. The integral vanishes; the boundary term gives $a_n = \frac{1}{2\pi} \cdot \frac{\pi e^{-in\pi} + \pi e^{in\pi}}{-in} = \frac{\cos n\pi}{-in} = \frac{(-1)^{n+1}}{in}$. In real form $B_n = i(a_n - a_{-n}) = \frac{2(-1)^{n+1}}{n}$, so

$$x \sim 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx.$$

Check. At $x = \pi/2$ the series is $2(1 - \frac{1}{3} + \frac{1}{5} - \dots) = 2 \cdot \frac{\pi}{4} = \frac{\pi}{2} = F(\pi/2)$. ■ At the jump $x = \pi$ the series gives 0, the average of $\pm\pi$ — a first glimpse of the Gibbs phenomenon.

Worked Example 31.3.5 (The triangle wave and a value of $\zeta(2)$). Find the Fourier series of $F(x) = |x|$ on $[-\pi, \pi]$ and deduce $\sum_{k \geq 0} (2k+1)^{-2} = \pi^2/8$.

Recognition. Trigger: even data, so a cosine series; then evaluate at a clever point to sum a series (an early taste of Parseval-type identities).

Solution. $a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |x| dx = \frac{1}{2\pi} \cdot \pi^2 = \frac{\pi}{2}$, so the mean is $\pi/2$. For $n \neq 0$, by evenness $a_n = \frac{1}{\pi} \int_0^{\pi} x \cos nx dx = \frac{1}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi} = \frac{\cos n\pi - 1}{\pi n^2} = \frac{(-1)^n - 1}{\pi n^2}$, which is 0 for even n and $-\frac{2}{\pi n^2}$ for odd n . Thus

$$|x| \sim \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\cos((2k+1)x)}{(2k+1)^2}.$$

Check. At $x = 0$, $|0| = 0 = \frac{\pi}{2} - \frac{4}{\pi} \sum (2k+1)^{-2}$, giving $\sum_{k \geq 0} (2k+1)^{-2} = \frac{\pi}{2} \cdot \frac{\pi}{4} = \frac{\pi^2}{8}$. ■

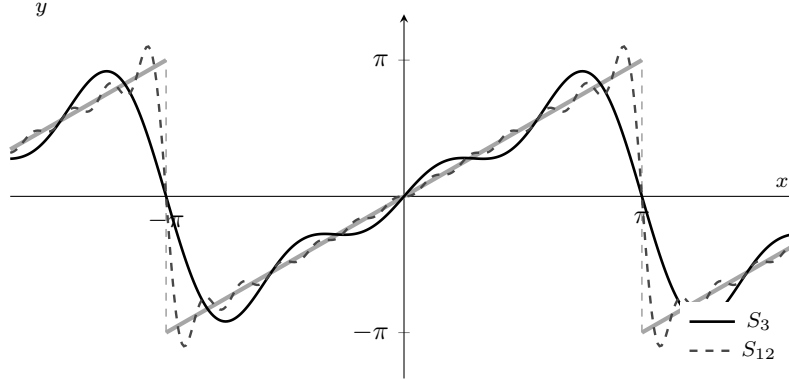


Figure 31.10. Partial sums S_3 (solid) and S_{12} (dashed) of the 2π -periodic sawtooth $F(x) = x$ on $(-\pi, \pi)$ (gray). At each jump $x = \pm\pi$ the sum converges to the midpoint 0, but a fixed $\sim 9\%$ overshoot survives just inside the jump — the *Gibbs phenomenon*.

Characters, roots of unity, and Euler's identities

The functions $e_n(x) = e^{inx}$ are the *characters* of the circle group: $e_n(x+y) = e_n(x)e_n(y)$ and $|e_n(x)| = 1$, so each e_n maps the circle into the unit circle, wrapping it n times. The finite analogue is the N th roots of unity $\omega^k = e^{2\pi i k/N}$, the characters of $\mathbb{Z}/N\mathbb{Z}$, which underlie the discrete Fourier transform.

Each character is a group homomorphism from the additive circle to the multiplicative unit circle, and the finite character χ on $\mathbb{Z}/N\mathbb{Z}$ is the restriction of e_n to the sampled points $2\pi k/N$.

$$\begin{array}{ccc} \mathbb{R}/2\pi\mathbb{Z} & \xrightarrow{e_n} & S^1 \\ \uparrow k \mapsto 2\pi k/N & \nearrow \chi & \\ \mathbb{Z}/N\mathbb{Z} & & \end{array}$$

Figure 31.11. The character $e_n : x \mapsto e^{inx}$ is a homomorphism from the circle group $\mathbb{R}/2\pi\mathbb{Z}$ to the unit circle S^1 ; sampling at the N th roots of unity ($k \mapsto 2\pi k/N$, an inclusion of $\mathbb{Z}/N\mathbb{Z}$) gives the finite character $\chi = e_n \circ (\text{inclusion})$, with $\chi(k) = \omega^{nk}$, $\omega = e^{2\pi i/N}$.

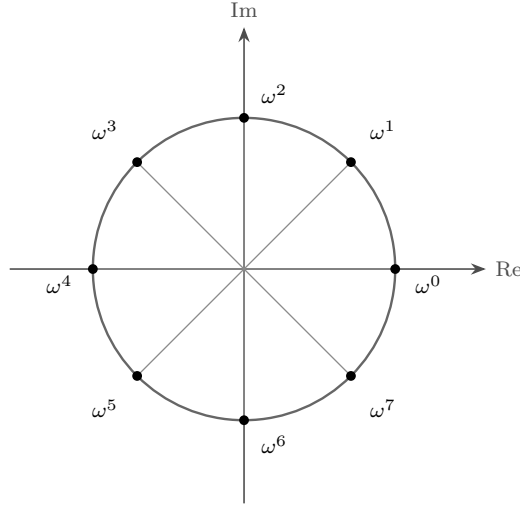


Figure 31.12. The 8th roots of unity $\omega^k = e^{2\pi i k/8}$: the characters of $\mathbb{Z}/8\mathbb{Z}$, equally spaced on the unit circle and summing to 0.

Theorem 31.3.6 (Euler's identities). For $\theta \in \mathbb{R}$, $e^{i\theta} = \cos \theta + i \sin \theta$, $|e^{i\theta}| = 1$, and

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Moreover $e^z = 1$ iff $z = 2\pi ki$ for some $k \in \mathbb{Z}$, and every $z \neq 0$ is $z = re^{i\theta}$ with $r = |z| > 0$ and θ unique mod 2π .

Proof. From the power series $e^{i\theta} = \sum_n (i\theta)^n/n!$, split into even and odd n : the even terms give $\sum_k (-1)^k \theta^{2k}/(2k)! = \cos \theta$, the odd terms $i \sum_k (-1)^k \theta^{2k+1}/(2k+1)! = i \sin \theta$. Then $|e^{i\theta}|^2 = \cos^2 \theta + \sin^2 \theta = 1$. Adding/subtracting $e^{\pm i\theta} = \cos \theta \pm i \sin \theta$ gives the two formulas. For $e^z = 1$ with $z = x + iy$: $|e^z| = e^x = 1$ forces $x = 0$, then $e^{iy} = 1$ means $\cos y = 1, \sin y = 0$, i.e. $y = 2\pi k$. Polar form follows by taking $r = |z|$ and $\theta = \arg z$. $\square$

Worked Example 31.3.7 (Summing a geometric character sum (Dirichlet kernel)). Show $\sum_{n=-N}^N e^{inx} = \frac{\sin((N + \frac{1}{2})x)}{\sin(x/2)}$ for $x \neq 0$, and equals $2N + 1$ at $x = 0$.

Recognition. Trigger: a finite geometric sum of characters; factor and use Euler's identities. This sum is the *Dirichlet kernel* D_N , the heart of convergence theory.

Solution. Let $w = e^{ix} \neq 1$. Then $\sum_{n=-N}^N w^n = w^{-N} \frac{w^{2N+1} - 1}{w - 1} = \frac{w^{N+1/2} - w^{-(N+1/2)}}{w^{1/2} - w^{-1/2}}$ after multiplying top and bottom by $w^{-1/2}$. By Euler, the numerator is $2i \sin((N + \frac{1}{2})x)$ and the denominator $2i \sin(x/2)$, giving the quotient. At $x = 0$ each term is 1, so the sum is $2N + 1$. *Check.* For $N = 1$, $1 + 2 \cos x = \frac{\sin(3x/2)}{\sin(x/2)}$; at $x = \pi/3$, LHS = $1 + 2 \cdot \frac{1}{2} = 2$ and RHS = $\sin(\pi/2)/\sin(\pi/6) = 1/\frac{1}{2} = 2$. $\blacksquare$

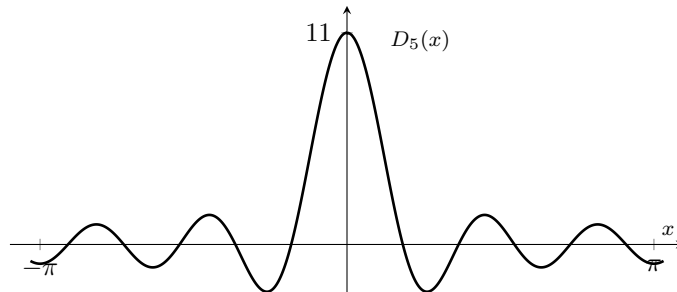


Figure 31.13. The Dirichlet kernel $D_5(x) = \sin(\frac{11}{2}x)/\sin(\frac{x}{2})$: peak $2N + 1 = 11$ at $x = 0$, oscillating and changing sign (it is not nonnegative).

Worked Example 31.3.8 (A trig identity from complex exponentials). Derive $\cos(\theta + \vartheta) = \cos \theta \cos \vartheta - \sin \theta \sin \vartheta$ and $2 \sin \theta \cos \varphi = \sin(\theta + \varphi) + \sin(\theta - \varphi)$.

Recognition. Trigger: prove addition/product formulas; multiply exponentials and read off real and imaginary parts (Stein Exercise 4i).

Solution. $e^{i(\theta+\vartheta)} = e^{i\theta}e^{i\vartheta} = (\cos \theta + i \sin \theta)(\cos \vartheta + i \sin \vartheta)$. The real part is $\cos \theta \cos \vartheta - \sin \theta \sin \vartheta$, which equals $\cos(\theta + \vartheta)$. For the product formula, write $2 \sin \theta \cos \varphi = 2 \cdot \frac{e^{i\theta} - e^{-i\theta}}{2i} \cdot \frac{e^{i\varphi} + e^{-i\varphi}}{2} = \frac{1}{2i}(e^{i(\theta+\varphi)} - e^{-i(\theta+\varphi)} + e^{i(\theta-\varphi)} - e^{-i(\theta-\varphi)}) = \sin(\theta + \varphi) + \sin(\theta - \varphi)$. *Check.* This is exactly the identity $\cos mt \sin mx = \frac{1}{2}[\sin m(x+t) + \sin m(x-t)]$ that linked the standing-wave and traveling-wave solutions in §31.1. ■

The discrete picture: roots of unity and the FFT

Sampling a function at N equally spaced points and computing its discrete Fourier transform amounts to evaluating $\sum_k a_k \omega^{jk}$ with $\omega = e^{-2\pi i/N}$. The fast Fourier transform splits this even/odd into two half-size transforms, the recursion drawn below.

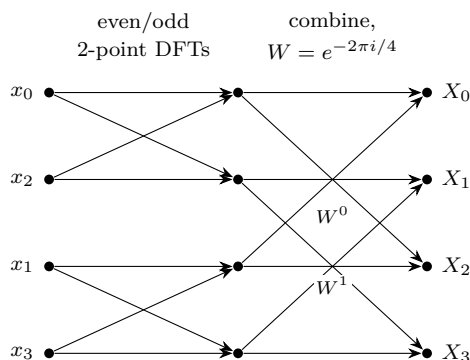


Figure 31.14. One stage of the radix-2 FFT on $N = 4$ points: the bit-reversed inputs feed two 2-point DFTs (even pair x_0, x_2 ; odd pair x_1, x_3), and each output X_j then combines one even and one odd result through a twiddle factor W^k . (Schematic.)

Worked Example 31.3.9 (The N th roots of unity sum to zero). Show $\sum_{k=0}^{N-1} \omega^k = 0$ for $\omega = e^{2\pi i/N}$, $N \geq 2$, and interpret it as a Fourier coefficient.

Recognition. Trigger: a finite geometric sum of all N th roots of unity; sum it and connect to orthogonality.

Solution. $\sum_{k=0}^{N-1} \omega^k = \frac{\omega^N - 1}{\omega - 1} = \frac{1 - 1}{\omega - 1} = 0$ since $\omega^N = e^{2\pi i} = 1$ and $\omega \neq 1$. Geometrically the N equally spaced unit vectors cancel by symmetry. *Check.* For $N = 8$ this is the diagram above; the eight dots are symmetric about the origin, so they sum to 0. This is the discrete orthogonality relation $\frac{1}{N} \sum_k \omega^{(m-n)k} = \delta_{mn}$ at $m - n \neq 0$. ■

Worked Example 31.3.10 (Reconstructing odd/even parts). Given F on $[-\pi, \pi]$, write its Fourier series in terms of those of its odd part $f = \frac{1}{2}(F(x) - F(-x))$ and even part $g = \frac{1}{2}(F(x) + F(-x))$, and verify with $F(x) = e^x$ restricted to one period (formally).

Recognition. Trigger: decompose by parity; odd part contributes sines, even part cosines, matching the $[-\pi, \pi]$ discussion of §31.1.

Solution. Since Fourier coefficients are linear and $\hat{f}(n) = -\hat{f}(-n)$ (odd), $\hat{g}(n) = \hat{g}(-n)$ (even), the series of F is $\sum_n (\hat{f}(n) + \hat{g}(n))e^{inx}$, whose real form is $\sum_{n \geq 1} B_n \sin nx$ (from f) plus $\frac{A_0}{2} + \sum_{n \geq 1} A_n \cos nx$ (from g). For $F(x) = e^x$, $g = \cosh x$ gives the cosine part and $f = \sinh x$ the sine part. *Check.* $\hat{e^x}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^x e^{-inx} dx = \frac{(-1)^n \sinh \pi}{\pi(1-in)}$; its real and imaginary parts indeed split as the cosine (even) and sine (odd) contributions. ■

Worked Example 31.3.11 (Parseval for a trig polynomial). For $F(x) = 3 + 2 \cos x - \sin 2x$ verify the Parseval identity $\frac{1}{2\pi} \int_{-\pi}^{\pi} |F|^2 = \sum_n |a_n|^2$.

Recognition. Trigger: a finite-bandwidth function, so both sides are finite sums; read off the coefficients and square. This previews the L^2 theory.

Solution. Coefficients: the constant gives $a_0 = 3$; $2 \cos x = e^{ix} + e^{-ix}$ gives $a_{\pm 1} = 1$; $-\sin 2x = -\frac{1}{2i}(e^{2ix} - e^{-2ix})$ gives $a_{\pm 2} = \pm \frac{i}{2}$, so $|a_{\pm 2}| = \frac{1}{2}$. Hence $\sum_n |a_n|^2 = 9 + 1 + 1 + \frac{1}{4} + \frac{1}{4} = \frac{23}{2}$. Directly, $\frac{1}{2\pi} \int_{-\pi}^{\pi} (3 + 2 \cos x - \sin 2x)^2 dx$: all cross terms integrate to 0 by orthogonality, leaving $9 + \frac{1}{2\pi} \int 4 \cos^2 x dx + \frac{1}{2\pi} \int \sin^2 2x dx = 9 + 2 + \frac{1}{2} = \frac{23}{2}$. *Check.* Both sides equal $\frac{23}{2}$. ■

Worked Example 31.3.12 (Coefficient decay reflects smoothness). Compare the decay of $\widehat{F}(n)$ for the sawtooth ($F = x$), the triangle ($F = |x|$), and a smooth $F = \cos x$, and state the general principle.

Recognition. Trigger: relate the rate $|\widehat{F}(n)| \rightarrow 0$ to the differentiability of F , using the integration-by-parts rule $\widehat{F'}(n) = in\widehat{F}(n)$.

Solution. Sawtooth: $|a_n| \sim 1/n$ (a jump discontinuity, so $F \notin C^0$ periodically). Triangle: $|a_n| \sim 1/n^2$ (F continuous, F' has jumps). Smooth $\cos x$: only $n = \pm 1$ nonzero ($F \in C^\infty$). *Check.* Each extra derivative multiplies the coefficient by in , so k continuous derivatives force $|\widehat{F}(n)| = o(n^{-k})$; conversely fast coefficient decay yields a smooth sum. Smoothness in $x \leftrightarrow$ decay in n is the master duality of Fourier analysis. ■

Caution. Writing $F \sim \sum a_n e^{inx}$ asserts *nothing* about convergence. The coefficients always exist for integrable F , but the partial sums $S_N(F)$ may fail to converge at a point (du Bois-Reymond's continuous function with a divergent Fourier series), or converge to the midpoint at a jump rather than to F . Equality $F = \sum a_n e^{inx}$ is a theorem to be earned under hypotheses (smoothness, L^2 , Cesaro/Abel summation), not a definition.

Remark. The Fourier coefficient map $F \mapsto \widehat{F}$ converts the hard analytic operation of solving a differential equation into the easy algebraic operation of multiplying coefficients by a symbol: $\widehat{F'}(n) = in\widehat{F}(n)$. This is precisely why separating variables in §31.1–§31.2 produced the modes e^{inx} — they diagonalize differentiation. The rest of the subject is the systematic exploitation of this single fact.

Exercises (Stein, Ch.1 — complex numbers, exponentials, orthogonality).

- (Ex. 1) For $z = x + iy$ ($x, y \in \mathbb{R}$) define $|z| = (x^2 + y^2)^{1/2}$. (a) Geometric meaning of $|z|$? (b) If $|z| = 0$ then $z = 0$. (c) For $\lambda \in \mathbb{R}$, $|\lambda z| = |\lambda| |z|$. (d) $|z_1 z_2| = |z_1| |z_2|$ and $|z_1 + z_2| \leq |z_1| + |z_2|$. (e) If $z \neq 0$ then $|1/z| = 1/|z|$.
- (Ex. 2) For $z = x + iy$ define $\bar{z} = x - iy$. (a) Geometric meaning of $\bar{z}$? (b) $|z|^2 = z\bar{z}$. (c) If z is on the unit circle, $1/z = \bar{z}$.
- (Ex. 3) A sequence $\{w_n\}$ in $\mathbb{C}$ converges to w if $|w_n - w| \rightarrow 0$. (a) A convergent sequence has a unique limit. (b) $\{w_n\}$ converges iff it is Cauchy ($\forall \varepsilon > 0 \exists N : |w_n - w_m| < \varepsilon$ for $n, m > N$). (c) If $\{a_n\}$ is a sequence of nonnegative reals with $\sum_n a_n < \infty$ and $|z_n| \leq a_n$ for all n , then $\sum_n z_n$ converges. [Use the Cauchy criterion.]
- (Ex. 4) Define $e^z = \sum_{n \geq 0} z^n/n!$. (a) The series converges for every z , uniformly on every bounded set. (b) $e^{z_1} e^{z_2} = e^{z_1 + z_2}$ [binomial theorem]. (c) For $z = iy$, $e^{iy} = \cos y + i \sin y$ (Euler) [power series]. (d) More generally $e^{x+iy} = e^x(\cos y + i \sin y)$ and $|e^{x+iy}| = e^x$. (e) $e^z = 1$ iff $z = 2\pi ki$, $k \in \mathbb{Z}$. (f) Every $z = x + iy$ has the form $z = re^{i\theta}$ with $r = |z| \geq 0$ unique and θ unique mod 2π ; check $r = |z|$ and $\theta = \arctan(y/x)$ where applicable. (g) In particular $i = e^{i\pi/2}$; geometric meaning of multiplication by i , and by $e^{i\theta}$? (h) $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, $\sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$. (i) Use complex exponentials to derive $\cos(\theta + \vartheta) = \cos \theta \cos \vartheta - \sin \theta \sin \vartheta$, and then $2 \sin \theta \sin \varphi = \cos(\theta - \varphi) - \cos(\theta + \varphi)$, $2 \sin \theta \cos \varphi = \sin(\theta + \varphi) + \sin(\theta - \varphi)$.
- (Ex. 5) Verify $f(x) = e^{inx}$ has period 2π and $\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{inx} dx = 1$ if $n = 0$, 0 if $n \neq 0$. Use this to prove, for $n, m \geq 1$, $\frac{1}{\pi} \int_{-\pi}^{\pi} \sin nx \sin mx dx = \delta_{nm}$, $\frac{1}{\pi} \int_{-\pi}^{\pi} \cos nx \cos mx dx = \delta_{nm}$, and $\frac{1}{\pi} \int_{-\pi}^{\pi} \sin nx \cos mx dx = 0$ for all n, m . [Compute $e^{inx} e^{-imx} \pm e^{inx} e^{imx}$.]

Additional exercises.

- Compute the Fourier coefficients $\widehat{F}(n)$ of the square wave $F(x) = \text{sgn}(x)$ on $(-\pi, \pi)$ and confirm only odd n survive.
- Find the Fourier series of $F(x) = x^2$ on $[-\pi, \pi]$ and deduce $\sum_{n \geq 1} 1/n^2 = \pi^2/6$ (evaluate at $x = \pi$) and $\sum_{n \geq 1} (-1)^{n+1}/n^2 = \pi^2/12$ (evaluate at $x = 0$).

3. Show $\widehat{F'}(n) = in \widehat{F}(n)$ for C^1 periodic F , and use it to obtain the coefficients of the sawtooth from those of x^2 .
4. Sum the finite series $\sum_{n=-N}^N e^{inx}$ in closed form (the Dirichlet kernel) and find its value and its integral over $[-\pi, \pi]$.
5. Prove the discrete orthogonality $\frac{1}{N} \sum_{k=0}^{N-1} \omega^{(m-n)k} = \delta_{mn}$ (indices mod N) for $\omega = e^{2\pi i/N}$, and relate it to Lemma 31.3.1.
6. (Non-example) Exhibit a 2π -periodic F and a point x_0 where the Fourier series converges to a value different from $F(x_0)$; explain why this does not contradict Definition 31.3.2. [Take any function with a jump.]
7. (Diagram) Plot the 6th roots of unity on the unit circle, draw the vectors from the origin, and verify graphically that they sum to zero.
8. Show that for real F the coefficients satisfy $\widehat{F}(-n) = \overline{\widehat{F}(n)}$, and deduce that the Fourier series can be written with real cosine/sine coefficients.
9. Compute $\widehat{F}(n)$ for $F(x) = e^{ax}$ on $[-\pi, \pi)$ ($a \neq 0$ real) and use Parseval (formally) to evaluate $\sum_{n \in \mathbb{Z}} \frac{1}{a^2 + n^2}$.
10. Draw one stage of the radix-2 FFT butterfly for $N = 8$, labeling the even/odd split and the twiddle factors $W_8^k = e^{-2\pi i k/8}$.

Chapter 32

Basic Properties of Fourier Series

32.1 Fourier Coefficients and First Examples

The classes of functions we admit

Throughout, f is complex-valued and at least *Riemann integrable*. Three nested classes recur: *continuous* functions on $[0, L]$; *piecewise continuous* functions (bounded, with finitely many discontinuities); and the most general *Riemann integrable* functions (bounded, possibly with infinitely many discontinuities, but whose discontinuity set has “measure zero”). The reason the formula for Fourier coefficients forces integrability is plain: that formula *is* an integral.

Definition 32.1.1 (Riemann integrable, recalled). A bounded real f on $[0, L]$ is integrable if for every $\varepsilon > 0$ there is a partition $0 = x_0 < \cdots < x_N = L$ with $U - L < \varepsilon$, where $U = \sum_j (\sup_{[x_{j-1}, x_j]} f) \Delta x_j$ and $L = \sum_j (\inf_{[x_{j-1}, x_j]} f) \Delta x_j$. A complex $f = u + iv$ is integrable if u, v are. Sums and products of integrable functions are integrable.

A standard integrable function on $[0, 1]$ with *infinitely many* jumps is the toggling staircase $f(x) = 1$ if $1/(n+1) < x \leq 1/n$ with n odd, $f(x) = 0$ if n even, and $f(0) = 0$; it is discontinuous at every $1/n$ and at 0, yet integrable.

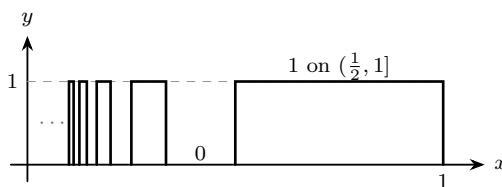


Figure 32.1. An integrable function on $[0, 1]$ with infinitely many jumps (value 1 on $(1/(n+1), 1/n]$ for n odd, else 0); discontinuous at each $1/n$ and at 0, yet integrable since the jump set is negligible.

Functions on the circle

A point of the unit circle is $e^{i\theta}$ with θ unique modulo 2π . Setting $f(\theta) = F(e^{i\theta})$ turns a function F on the circle into a 2π -periodic function f on $\mathbb{R}$, and conversely. Restricting f to any interval of length 2π — $[0, 2\pi]$ or $[-\pi, \pi]$ — recovers F , provided the endpoint values agree ($f(0) = f(2\pi)$, since the endpoints are the *same* point of the circle). Thus three descriptions coincide: 2π -periodic functions on $\mathbb{R}$; functions on $[0, 2\pi]$ with matching endpoints; functions on $\mathbb{T}$. We write x for an interval variable and θ for an angle, by convention only.

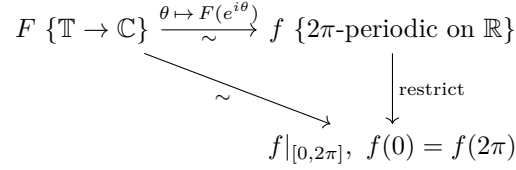


Figure 32.2. The three descriptions of a function on the circle and the bijections identifying them: a function F on $\mathbb{T}$ pulls back to a 2π -periodic $f = F \circ \exp(i \cdot)$ on $\mathbb{R}$, whose restriction to $[0, 2\pi]$ (with matching endpoints) recovers F .

Fourier coefficients and Fourier series

Definition 32.1.2 (Fourier coefficients and series). Let f be integrable on an interval $[a, b]$ of length $L = b - a$. Its n th *Fourier coefficient* is

$$\hat{f}(n) = \frac{1}{L} \int_a^b f(x) e^{-2\pi i n x / L} dx, \quad n \in \mathbb{Z},$$

and its *Fourier series* is the formal sum $\sum_{n=-\infty}^{\infty} \hat{f}(n) e^{2\pi i n x / L}$, written $f(x) \sim \sum_n a_n e^{2\pi i n x / L}$ with $a_n = \hat{f}(n)$. On $[-\pi, \pi]$ (or any length- 2π interval) this reads

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) e^{-in\theta} d\theta, \quad f(\theta) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}.$$

On $[0, 1]$: $\hat{g}(n) = \int_0^1 g(x) e^{-2\pi i n x} dx$ and $g \sim \sum a_n e^{2\pi i n x}$.

The tilde $\sim$ is deliberately noncommittal: it records “these are the coefficients” without yet asserting convergence. The whole subject is the study of *in what sense* the right side returns f .

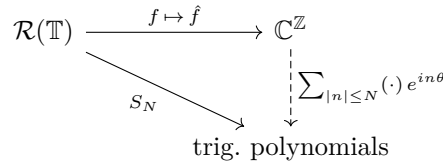


Figure 32.3. The Fourier-coefficient map sends an integrable function on the circle $\mathcal{R}(\mathbb{T})$ to its two-sided sequence of coefficients $\hat{f} \in \mathbb{C}^{\mathbb{Z}}$; truncating to $|n| \leq N$ and re-synthesizing is the partial-sum operator S_N . Whether $S_N(f)$ returns f is the convergence question (dashed: synthesis is only formal until convergence is established).

Definition 32.1.3 (Trigonometric series and polynomials; partial sums). A *trigonometric series* is any $\sum_n c_n e^{2\pi i n x / L}$ with $c_n \in \mathbb{C}$. If only finitely many c_n are nonzero it is a *trigonometric polynomial*, of *degree* the largest $|n|$ with $c_n \neq 0$. The N th *partial sum* of the Fourier series of f is the symmetric trigonometric polynomial

$$S_N(f)(x) = \sum_{n=-N}^N \hat{f}(n) e^{2\pi i n x / L}.$$

The summation is symmetric ($-N$ to N) on purpose: it groups $e^{in\theta}$ with $e^{-in\theta}$ into cosines and sines, and convergence in this book always means the limit of these symmetric sums.

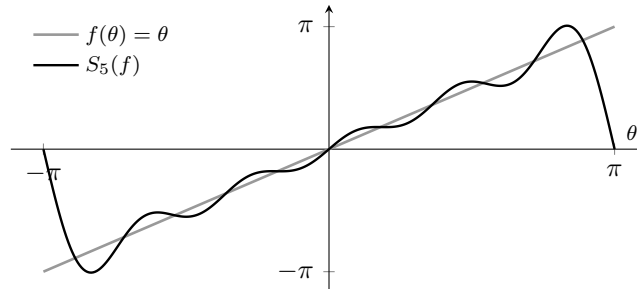


Figure 32.4. Example 1: $f(\theta) = \theta$ (grey) and its partial sum S_5 (black), $2 \sum_{n \geq 1} (-1)^{n+1} \frac{\sin n\theta}{n}$. Overshoot near $\theta = \pm\pi$ foreshadows the Gibbs phenomenon at the jump.

Worked Example 32.1.4 (Example 1 – the sawtooth $f(\theta) = \theta$ on $[-\pi, \pi]$). Compute $\hat{f}(n)$ and the Fourier series.

Recognition. Coefficient of a smooth-but-for-endpoints function; integrate by parts.

Solution. For $n \neq 0$,

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta e^{-in\theta} d\theta = \frac{1}{2\pi} \left[\frac{\theta}{-in} e^{-in\theta} \right]_{-\pi}^{\pi} + \frac{1}{2\pi in} \int_{-\pi}^{\pi} e^{-in\theta} d\theta = \frac{(-1)^{n+1}}{in},$$

since $e^{\mp in\pi} = (-1)^n$ and the remaining integral vanishes. And $\hat{f}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta d\theta = 0$. Hence

$$f(\theta) \sim \sum_{n \neq 0} \frac{(-1)^{n+1}}{in} e^{in\theta} = 2 \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin n\theta}{n}.$$

Check. $\hat{f}(-n) = \overline{\hat{f}(n)}$ (as f is real) and $\hat{f}(0) = 0$ (as f is odd, no constant term). ■

Worked Example 32.1.5 (Example 2 – $f(\theta) = (\pi - \theta)^2/4$ on $[0, 2\pi]$). Find the Fourier series.

Recognition. A parabola on $[0, 2\pi]$, even about $\theta = \pi$; integrate by parts twice.

Solution. Two integrations by parts (as in Ex. 1 but iterated) give

$$f(\theta) \sim \frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{\cos n\theta}{n^2}.$$

Check. The constant term is the average $\frac{1}{2\pi} \int_0^{2\pi} \frac{(\pi - \theta)^2}{4} d\theta = \frac{1}{2\pi} \cdot \frac{1}{4} \cdot \frac{2\pi^3}{3} \cdot \dots$ directly $\frac{1}{8\pi} \int_0^{2\pi} (\pi - \theta)^2 d\theta = \frac{1}{8\pi} \cdot \frac{2\pi^3}{3} = \frac{\pi^2}{12}$. ■ Setting $\theta = 0$: $f(0) = \pi^2/4 = \frac{\pi^2}{12} + \sum 1/n^2$, recovering $\sum 1/n^2 = \pi^2/6$.

Worked Example 32.1.6 (Example 3 – the shifted-pole series). Show that on $[0, 2\pi]$, for $\alpha \notin \mathbb{Z}$, $f(\theta) = \frac{\pi}{\sin \pi \alpha} e^{i(\pi - \theta)\alpha}$ has $f(\theta) \sim \sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{n + \alpha}$.

Recognition. A complex exponential in θ ; compute $\hat{f}(n)$ as a single exponential integral.

Solution. $\hat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} \frac{\pi}{\sin \pi \alpha} e^{i(\pi - \theta)\alpha} e^{-in\theta} d\theta = \frac{e^{i\pi\alpha}}{2 \sin \pi \alpha} \int_0^{2\pi} e^{-i(n+\alpha)\theta} d\theta$. The integral is $\frac{e^{-i(n+\alpha)2\pi} - 1}{-i(n+\alpha)} = \frac{e^{-2\pi i\alpha} - 1}{-i(n+\alpha)}$ (since $e^{-2\pi i n} = 1$). Then $\frac{e^{i\pi\alpha}(e^{-2\pi i\alpha} - 1)}{-2i \sin \pi \alpha (n+\alpha)} = \frac{e^{-i\pi\alpha} - e^{i\pi\alpha}}{-2i \sin \pi \alpha (n+\alpha)} = \frac{-2i \sin \pi \alpha}{-2i \sin \pi \alpha (n+\alpha)} = \frac{1}{n+\alpha}$. *Check.* As $\alpha \rightarrow 0$, the $n = 0$ term $1/\alpha$ blows up – consistent with the prefactor $\pi/\sin \pi \alpha \rightarrow \infty$. ■

Worked Example 32.1.7 (Example 4 – the Dirichlet kernel and its closed form). For $D_N(x) = \sum_{n=-N}^N e^{inx}$ derive $D_N(x) = \frac{\sin((N + \frac{1}{2})x)}{\sin(x/2)}$.

Recognition. A two-sided geometric sum; split at 0 and sum each side.

Solution. With $\omega = e^{ix}$, $\sum_{n=0}^N \omega^n = \frac{1 - \omega^{N+1}}{1 - \omega}$ and $\sum_{n=-N}^{-1} \omega^n = \frac{\omega^{-N} - 1}{1 - \omega}$. Adding,

$$D_N(x) = \frac{\omega^{-N} - \omega^{N+1}}{1 - \omega} = \frac{\omega^{-N-1/2} - \omega^{N+1/2}}{\omega^{-1/2} - \omega^{1/2}} = \frac{\sin((N + \frac{1}{2})x)}{\sin(x/2)}.$$

Check. The Fourier coefficients of D_N are 1 for $|n| \leq N$, 0 else; and $D_N(0) = 2N + 1$ (L'Hopital), the number of terms. ■

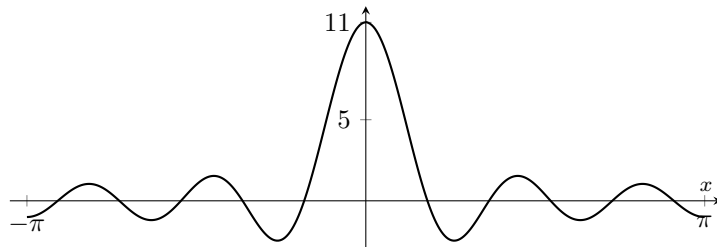


Figure 32.5. The Dirichlet kernel $D_5(x) = \sin(5.5x)/\sin(x/2)$: tall central spike of height $2N + 1 = 11$, but large oscillating side lobes that make $\int |D_N| \gtrsim \log N$ – the reason D_N is *not* a good kernel.

Worked Example 32.1.8 (Example 5 – the Poisson kernel summed). For $0 \leq r < 1$, sum $P_r(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|} e^{in\theta}$ in closed form.

Recognition. Two absolutely convergent geometric series, one in each direction.

Solution. Write $\omega = re^{i\theta}$. Then $\sum_{n \geq 0} \omega^n = \frac{1}{1-\omega}$ and $\sum_{n \geq 1} \bar{\omega}^n = \frac{\bar{\omega}}{1-\bar{\omega}}$, so

$$P_r(\theta) = \frac{1}{1-\omega} + \frac{\bar{\omega}}{1-\bar{\omega}} = \frac{1-|\omega|^2}{|1-\omega|^2} = \frac{1-r^2}{1-2r\cos\theta+r^2}.$$

Check. The $n = 0$ Fourier coefficient is $\frac{1}{2\pi} \int P_r = 1$ (sum starts at $r^0 = 1$); and as $r \rightarrow 1$ the mass spikes at $\theta = 0$. ■

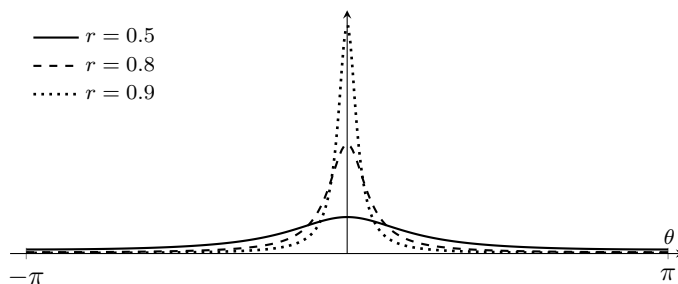


Figure 32.6. The Poisson kernel $P_r(\theta) = \frac{1-r^2}{1-2r\cos\theta+r^2}$: nonnegative, unit mass, and concentrating at $\theta = 0$ as $r \rightarrow 1$ – a good kernel.

Convergence: the problem, stated precisely

The series is formal; whether $S_N(f) \rightarrow f$, and in what sense, is genuinely delicate. Pointwise convergence *everywhere* cannot hold for all integrable f (changing f at one point leaves $\hat{f}(n)$ unchanged). Even continuity is not enough: du Bois-Reymond exhibited a continuous function whose Fourier series *diverges* at a point. The positive results we will reach: (i) C^2 (indeed C^1 , or Hölder- α with $\alpha > 1/2$) forces absolute, uniform convergence; (ii) mean-square convergence for merely integrable f (next chapter); (iii) Cesàro and Abel summability at every point of continuity (Section 32.3). Carleson (1966): for L^2 functions the series converges a.e.

Worked Example 32.1.9 (A non-example: pointwise convergence cannot hold everywhere). Exhibit two integrable functions with identical Fourier series but different values.

Recognition. “Fourier coefficients ignore single points.”

Solution. Let $f \equiv 0$ and let $g(\theta) = 0$ except $g(0) = 1$. Both are integrable, and for every n , $\hat{g}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g e^{-in\theta} d\theta = 0 = \hat{f}(n)$ (a single point contributes nothing to a Riemann integral). So $f \sim g$ have the same series, yet $f(0) \neq g(0)$. *Check.* No series can converge to $g(0) = 1$ at 0 while also equalling $f(0) = 0$ there; pointwise-everywhere convergence is impossible in general. ■ Theorem 32.3.5 below recovers everything at points of *continuity*.

Worked Example 32.1.10 (A square wave: jump-induced $1/n$ decay). Compute the Fourier series of the odd square wave $f(\theta) = 1$ on $(0, \pi)$, $f(\theta) = -1$ on $(-\pi, 0)$, $f(0) = 0$.

Recognition. Odd function $\Rightarrow$ pure sine series; expect $1/n$ decay from the jump.

Solution. $\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f e^{-in\theta} d\theta$. By oddness only the $-i \sin n\theta$ part survives: $\hat{f}(n) = \frac{-i}{2\pi} \int_{-\pi}^{\pi} f(\theta) \sin n\theta d\theta = \frac{-i}{\pi} \int_0^{\pi} \sin n\theta d\theta = \frac{-i}{\pi} \cdot \frac{1-\cos n\pi}{n}$, which is 0 for n even and $\frac{-2i}{\pi n}$ for n odd. Hence $f(\theta) \sim \frac{4}{\pi} \sum_{k \geq 0} \frac{\sin((2k+1)\theta)}{2k+1}$.

Check. Coefficients decay like $1/n$ – the signature of a jump discontinuity (cf. $C^2 \Rightarrow O(1/n^2)$). ■

Worked Example 32.1.11 (Computing a coefficient is interval-independent for periodic f). If f is 2π -periodic, show $\hat{f}(n)$ is the same whether computed on $[-\pi, \pi]$ or $[0, 2\pi]$.

Recognition. Periodicity lets you slide the window of integration.

Solution. For any 2π -periodic integrable g , $\int_0^{2\pi} g = \int_{-\pi}^{\pi} g$ (Exercise 1 below). Apply with $g(\theta) = f(\theta)e^{-in\theta}$, itself 2π -periodic. *Check.* On a function on the circle, $\hat{f}(n)$ is intrinsically defined – no interval needs to be named. ■

Worked Example 32.1.12 (Parseval applied to the sawtooth: summing $1/n^2$). Use the Fourier series of $f(\theta) = \theta$ on $[-\pi, \pi]$ (Example 1) and Parseval's identity $\frac{1}{2\pi} \int |f|^2 = \sum_n |\hat{f}(n)|^2$ to evaluate $\sum_{n \geq 1} 1/n^2$.

Recognition. “Square-integral of f equals sum of squared coefficients”; a classic closed-form extraction.

Solution. $\hat{f}(n) = (-1)^{n+1}/(in)$ for $n \neq 0$, so $|\hat{f}(n)|^2 = 1/n^2$ and $\sum_n |\hat{f}(n)|^2 = 2 \sum_{n \geq 1} 1/n^2$. The left side is $\frac{1}{2\pi} \int_{-\pi}^{\pi} \theta^2 d\theta = \frac{1}{2\pi} \cdot \frac{2\pi^3}{3} = \frac{\pi^2}{3}$. Hence $2 \sum_{n \geq 1} 1/n^2 = \pi^2/3$, i.e. $\sum_{n \geq 1} 1/n^2 = \pi^2/6$. *Check.* Agrees with Example 2's evaluation at $\theta = 0$. ■

Worked Example 32.1.13 (Reading symmetry off the coefficients). Without integrating, predict the form of the Fourier series of $f(\theta) = \theta^2$ on $[-\pi, \pi]$, then confirm $\hat{f}(n) = 2(-1)^n/n^2$ for $n \neq 0$.

Recognition. θ^2 is even, so expect a pure cosine series ($\hat{f}(n) = \hat{f}(-n)$, real).

Solution. Evenness gives $\hat{f}(n) = \hat{f}(-n)$, so only cosines appear. Integrating by parts twice (boundary terms use $e^{\pm in\pi} = (-1)^n$) yields $\hat{f}(n) = 2(-1)^n/n^2$, and $\hat{f}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta^2 d\theta = \pi^2/3$. So $\theta^2 \sim \frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-1)^n}{n^2} \cos n\theta$. *Check.* C^0 but with a corner in the periodic extension at $\pm\pi$ (slope jumps), so decay is $O(1/n^2)$, consistent. ■

Exercises (Stein, Ch.2) – coefficients, symmetries, examples.

- Suppose f is 2π -periodic and integrable on every finite interval. Prove that for $a, b \in \mathbb{R}$, $\int_a^b f = \int_{a+2\pi}^{b+2\pi} f = \int_{a-2\pi}^{b-2\pi} f$, and $\int_{-\pi}^{\pi} f(x+a) dx = \int_{-\pi+a}^{\pi+a} f = \int_{-\pi}^{\pi} f$. (So $\hat{f}(n)$ is independent of the length- 2π interval used.)
- Symmetries and coefficients. Let f be 2π -periodic Riemann integrable. (a) Show $f(\theta) \sim \hat{f}(0) + \sum_{n \geq 1} ([\hat{f}(n) + \hat{f}(-n)] \cos n\theta + i[\hat{f}(n) - \hat{f}(-n)] \sin n\theta)$. (b) If f is even, $\hat{f}(n) = \hat{f}(-n)$ (a cosine series). (c) If f is odd, $\hat{f}(n) = -\hat{f}(-n)$ (a sine series). (d) If $f(\theta + \pi) = f(\theta)$ for all θ , then $\hat{f}(n) = 0$ for all odd n . (e) f is real-valued iff $\hat{f}(-n) = \overline{\hat{f}(n)}$ for all n .
- Plucked string. Show the initial profile f equals its Fourier sine series $f(x) = \sum_{m \geq 1} A_m \sin mx$ with $A_m = \frac{2h \sin m\pi p}{m^2 p(\pi - p)}$. [Hint: $|A_m| \leq C/m^2$.]
- The odd function on $[0, \pi]$ given by $f(\theta) = \theta(\pi - \theta)$. (a) Draw the graph of f . (b) Compute the Fourier coefficients and show $\hat{f}(\theta) = \frac{8}{\pi} \sum_{k \text{ odd} \geq 1} \frac{\sin k\theta}{k^3}$.
- Triangular tent. On $[-\pi, \pi]$ let $f(\theta) = 0$ for $|\theta| > \delta$ and $f(\theta) = 1 - |\theta|/\delta$ for $|\theta| \leq \delta$. Show $\hat{f}(\theta) = \frac{\delta}{2\pi} + 2 \sum_{n \geq 1} \frac{1 - \cos n\delta}{n^2 \pi \delta} \cos n\theta$.
- $f(\theta) = |\theta|$ on $[-\pi, \pi]$. (a) Draw the graph. (b) Show $\hat{f}(0) = \pi/2$ and, for $n \neq 0$, $\hat{f}(n) = \frac{-1 + (-1)^n}{\pi n^2}$. (c) Write the sine/cosine series. (d) Setting $\theta = 0$, deduce $\sum_{n \text{ odd} \geq 1} \frac{1}{n^2} = \frac{\pi^2}{8}$ and $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$.
- The sawtooth. Verify that $\frac{1}{2i} \sum_{n \neq 0} \frac{e^{inx}}{n}$ is the Fourier series of the 2π -periodic function with $f(0) = 0$, $f(x) = -\frac{\pi}{2} - \frac{x}{2}$ on $(-\pi, 0)$ and $f(x) = \frac{\pi - x}{2}$ on $(0, \pi)$ (discontinuous). Show the symmetric partial sums converge for every x ; at the origin the value 0 is the average of the left/right limits. [Hint: Dirichlet's test.]
- Characteristic function. Let $f = \chi_{[a,b]}$ with $[a, b] \subset [-\pi, \pi]$. (a) Show $f(x) \sim \frac{b-a}{2\pi} + \sum_{n \neq 0} \frac{e^{-ina} - e^{-inb}}{2\pi in} e^{inx}$. (b) If $a \neq -\pi$ or $b \neq \pi$ (and $a \neq b$), the series does not converge absolutely for any x . [Hint: $|\sin n\theta_0| \geq c > 0$ for many n , $\theta_0 = (b-a)/2$.] (c) Nonetheless it converges at every x . What happens when $a = -\pi$, $b = \pi$?
- Smoothness and decay. If f is 2π -periodic and of class C^k , show $\hat{f}(n) = O(1/|n|^k)$, i.e. $|n|^k \hat{f}(n) = O(1)$. [Hint: integrate by parts k times.]
- Integral-norm convergence transfers to coefficients. If $\{f_k\}$ are Riemann integrable on $[0, 1]$ with $\int_0^1 |f_k - f| \rightarrow 0$, show $\hat{f}_k(n) \rightarrow \hat{f}(n)$ uniformly in n .

Additional exercises.

1. Compute the Fourier coefficients of $f(\theta) = e^{a\theta}$ on $[-\pi, \pi]$ ($a \in \mathbb{C}$, $a \notin i\mathbb{Z}$), and show $\hat{f}(n) = \frac{(-1)^n \sinh(\pi a)}{\pi(a - in)}$ when $a \in \mathbb{R}$.
2. Let $f(\theta) = \cos^3 \theta$. Expand directly into exponentials and read off all nonzero $\hat{f}(n)$. (No integration needed – this is a trigonometric polynomial.)
3. For $f(\theta) = \theta^2$ on $[-\pi, \pi]$, show $\hat{f}(0) = \pi^2/3$ and $\hat{f}(n) = \frac{2(-1)^n}{n^2}$ for $n \neq 0$. Setting $\theta = \pi$, deduce $\sum_{n \geq 1} 1/n^2 = \pi^2/6$; setting $\theta = 0$, deduce $\sum_{n \geq 1} (-1)^{n-1}/n^2 = \pi^2/12$.
4. (Decay vs. smoothness.) Rank by expected decay rate of $\hat{f}(n)$: the square wave, the sawtooth θ , $|\theta|$, and θ^2 . State which are $O(1/n)$, $O(1/n^2)$, and justify by counting continuous derivatives on the circle.
5. Show that $\sum_{n \geq 1} \frac{\cos n\theta}{n^2} = \frac{\pi^2}{6} - \frac{\pi\theta}{2} + \frac{\theta^2}{4}$ for $\theta \in [0, 2\pi]$, by identifying the right side with Example 2.
6. (Non-example.) Give an integrable f on $[-\pi, \pi]$ whose Fourier series is $\sum_{n \geq 1} \frac{\sin n\theta}{\sqrt{n}}$ cannot be – i.e. prove no C^1 function has these coefficients, by comparing with the $O(1/n)$ decay rate.
7. Verify the linearity of $f \mapsto \hat{f}(n)$ and that $\widehat{f(\cdot - a)}(n) = e^{-ina} \hat{f}(n)$ (translation becomes modulation).
8. (Diagram.) Sketch, on the same axes over $[-\pi, \pi]$, the function θ and its partial sums S_1, S_3, S_5 . Indicate the overshoot near $\theta = \pm\pi$ and explain why it does not shrink.

32.2 Convolutions and Good Kernels

Convolution

Definition 32.2.1 (Convolution). For 2π -periodic integrable f, g on $\mathbb{R}$, their *convolution* on $[-\pi, \pi]$ is

$$(f * g)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) g(x - y) dy = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x - y) g(y) dy,$$

the second form following by periodic change of variables. (dy abbreviates dy .)

A convolution is a *weighted average*. If $g \equiv 1$, then $f * g$ is the constant $\frac{1}{2\pi} \int f$, the mean of f . The kernel g redistributes the mass of f ; this sliding-average viewpoint is the engine of the whole chapter. The crucial structural fact is that the partial sum is itself a convolution.

Theorem 32.2.2 (Partial sums are convolutions with D_N). $S_N(f) = f * D_N$, where $D_N(x) = \sum_{n=-N}^N e^{inx}$ is the *Dirichlet kernel*.

Proof.

$$\begin{aligned} S_N(f)(x) &= \sum_{n=-N}^N \hat{f}(n) e^{inx} = \sum_{n=-N}^N \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-iny} dy \right) e^{inx} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \sum_{n=-N}^N e^{in(x-y)} dy = (f * D_N)(x). \end{aligned}$$

Interchanging the finite sum and integral is legitimate. So understanding $S_N(f)$ is exactly understanding $f * D_N$. $\square$

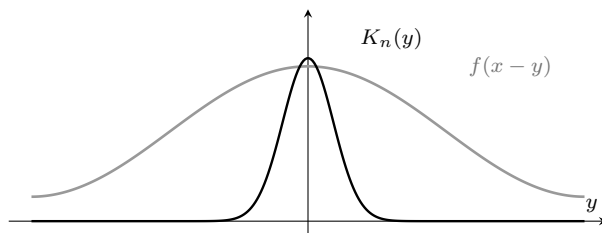


Figure 32.7. Convolution as weighted average: $(f * K_n)(x)$ samples $f(x - y)$ against the weight $K_n(y)$. As K_n concentrates at $y = 0$, the average tends to $f(x)$. (Schematic.)

Proposition 32.2.3 (Algebra of convolution). *For 2π -periodic integrable f, g, h :* (i) $f * (g + h) = f * g + f * h$; (ii) $(cf) * g = c(f * g) = f * (cg)$; (iii) $f * g = g * f$; (iv) $(f * g) * h = f * (g * h)$; (v) $f * g$ is continuous; (vi) $\widehat{f * g}(n) = \hat{f}(n)\hat{g}(n)$.

Proof. (i),(ii) are linearity of the integral. For the rest, first assume f, g continuous, where interchanging integrals is free. (iii): for continuous 2π -periodic F , $\int_{-\pi}^{\pi} F(y) dy = \int_{-\pi}^{\pi} F(x - y) dy$ (change $y \mapsto -y$ then translate); take $F(y) = f(y)g(x - y)$. (iv): interchange the two integral signs and change variables. (v): write $(f * g)(x_1) - (f * g)(x_2) = \frac{1}{2\pi} \int f(y)[g(x_1 - y) - g(x_2 - y)] dy$; g is uniformly continuous (continuous and periodic), so for $|x_1 - x_2| < \delta$ the bracket is $< \varepsilon$, giving $|(f * g)(x_1) - (f * g)(x_2)| \leq \frac{\varepsilon}{2\pi} \int |f| \leq \varepsilon B$. (vi):

$$\begin{aligned} \widehat{f * g}(n) &= \frac{1}{2\pi} \int \left(\frac{1}{2\pi} \int f(y)g(x - y) dy \right) e^{-inx} dx \\ &= \frac{1}{2\pi} \int f(y)e^{-iny} \left(\frac{1}{2\pi} \int g(x - y)e^{-in(x-y)} dx \right) dy = \hat{f}(n)\hat{g}(n). \end{aligned}$$

For general integrable f, g , apply the approximation Lemma 32.2.4: take continuous $f_k \rightarrow f$, $g_k \rightarrow g$ in the integral norm with uniform bounds. Then $f * g - f_k * g_k = (f - f_k) * g + f_k * (g - g_k)$, and $|(f - f_k) * g(x)| \leq \frac{1}{2\pi} \int |f - f_k| \cdot \sup |g| \rightarrow 0$ uniformly; similarly the other term. So $f_k * g_k \rightarrow f * g$ uniformly; each $f_k * g_k$ is continuous, hence so is $f * g$, giving (v). For (vi), $\widehat{f_k * g_k}(n) = \hat{f}_k(n)\hat{g}_k(n) \rightarrow \hat{f}(n)\hat{g}(n)$ since $|\hat{f}(n) - \hat{f}_k(n)| \leq \frac{1}{2\pi} \int |f - f_k| \rightarrow 0$, and the left side $\rightarrow \widehat{f * g}(n)$ by uniform convergence. (iii),(iv) follow likewise. $\square$

Lemma 32.2.4 (Approximation by continuous functions). *If f is integrable on the circle with $|f| \leq B$, there exist continuous f_k on the circle with $\sup_x |f_k(x)| \leq B$ for all k and $\int_{-\pi}^{\pi} |f - f_k| dx \rightarrow 0$.*

The lemma (proved in the appendix to Stein) says continuous functions are dense in the integral norm – the standard device for upgrading results from the continuous case to all integrable functions. Property (vi), $\widehat{f * g} = \hat{f}\hat{g}$, is the algebraic heart of Fourier analysis: convolution becomes multiplication on the coefficient side. Property (v) is a regularity principle: $f * g$ is smoother than either factor (continuous, even when f, g are merely integrable).

$$\begin{array}{ccc} (f, g) & \xrightarrow{*} & f * g \\ \downarrow (\cdot) \times (\cdot) & & \downarrow (\cdot) \\ (\hat{f}, \hat{g}) & \xrightarrow{\cdot \text{ (pointwise)}} & \hat{f}\hat{g} \end{array}$$

Figure 32.8. Property (vi), $\widehat{f * g} = \hat{f}\hat{g}$: the coefficient map intertwines convolution with pointwise multiplication. The square commutes – transforming a convolution and multiplying the transforms give the same sequence.

Worked Example 32.2.5 (Convolution with e^{inx} extracts a coefficient). Compute $f * e_n$ where $e_n(x) = e^{inx}$. *Recognition.* e_n has a single nonzero Fourier coefficient; use (vi) or compute directly.

Solution. $(f * e_n)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y)e^{in(x-y)} dy = e^{inx} \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y)e^{-iny} dy = \hat{f}(n)e^{inx}$. *Check via (vi):* $\widehat{e_n}(m) = \delta_{nm}$, so $\widehat{f * e_n}(m) = \hat{f}(m)\delta_{nm}$, i.e. only the $m = n$ coefficient survives, equal to $\hat{f}(n)$ – exactly $\hat{f}(n)e_n$. ■

Worked Example 32.2.6 (Convolution of two Dirichlet kernels). Show $D_N * D_M = D_{\min(N, M)}$.

Recognition. Use $\widehat{D_N}(n) = \mathbf{1}_{|n| \leq N}$ and property (vi).

Solution. $\widehat{D_N * D_M}(n) = \widehat{D_N}(n)\widehat{D_M}(n) = \mathbf{1}_{|n| \leq N} \cdot \mathbf{1}_{|n| \leq M} = \mathbf{1}_{|n| \leq \min(N, M)} = \widehat{D_{\min(N, M)}}(n)$. Since both sides are trigonometric polynomials with equal coefficients, they are equal. *Check.* $D_N * D_N = D_N$, so D_N is idempotent under $*$ – it is the projection onto frequencies $|n| \leq N$, which is exactly what S_N does. ■

Worked Example 32.2.7 (Convolution smooths: a box convolved with itself is a tent). On $\mathbb{R}$ (analogous computation), convolve the box $b = \mathbf{1}_{[-1,1]}$ with itself.

Recognition. Direct integral; expect a continuous, piecewise-linear (triangular) result.

Solution. $(b * b)(x) = \int_{\mathbb{R}} b(y)b(x-y) dy = |[-1, 1] \cap [x-1, x+1]|$, the length of the overlap, which is $2 - |x|$ for $|x| \leq 2$ and 0 otherwise – the tent function on $[-2, 2]$. *Check.* b is discontinuous (a step), but $b * b$ is continuous; convolving once raised regularity by one notch (step $\rightarrow$ continuous, kink $\rightarrow C^1$), illustrating (v). ■

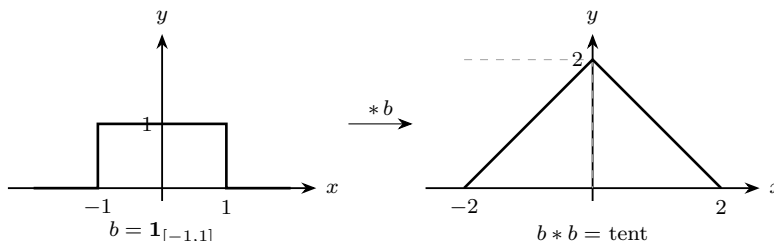


Figure 32.9. Convolution raises regularity: the discontinuous unit box becomes a continuous tent of height 2 on $[-2, 2]$ (peak value = the overlap length $2 - |x|$ at $x = 0$). Each convolution gains one degree of smoothness.

Good kernels

Definition 32.2.8 (Good kernels). A family $\{K_n\}_{n \geq 1}$ on the circle is a family of *good kernels* if: (a) $\frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(x) dx = 1$ for all n ; (b) $\int_{-\pi}^{\pi} |K_n(x)| dx \leq M$ for some M and all n ; (c) for every $\delta > 0$, $\int_{\delta \leq |x| \leq \pi} |K_n(x)| dx \rightarrow 0$ as $n \rightarrow \infty$. When $K_n \geq 0$, (a) implies (b) with $M = 2\pi$.

Interpret K_n as a unit mass on the circle (a) whose total variation stays bounded (b) and which concentrates at the origin (c). In the limit, $\{K_n\}$ behaves like the *Dirac delta* – an “approximation to the identity.”

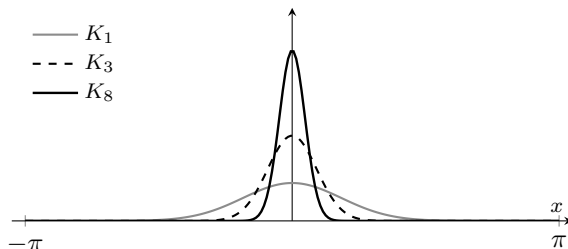


Figure 32.10. A family of good kernels: unit mass, concentrating at the origin as $n \rightarrow \infty$. Convolving with such a family recovers f . (Schematic.)

Theorem 32.2.9 (Approximation to the identity). Let $\{K_n\}$ be good kernels and f integrable on the circle. Then $(f * K_n)(x) \rightarrow f(x)$ at every point x of continuity of f ; if f is continuous everywhere the convergence is uniform.

Proof. Fix $\varepsilon > 0$ and x a point of continuity: choose δ so $|y| < \delta \Rightarrow |f(x-y) - f(x)| < \varepsilon$. By (a),

$$(f * K_n)(x) - f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(y)[f(x-y) - f(x)] dy.$$

Split at $|y| < \delta$ and $\delta \leq |y| \leq \pi$. With $B = \sup |f|$,

$$|(f * K_n)(x) - f(x)| \leq \frac{\varepsilon}{2\pi} \int_{-\pi}^{\pi} |K_n(y)| dy + \frac{2B}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n(y)| dy \leq \frac{\varepsilon M}{2\pi} + \frac{B}{\pi} \int_{\delta \leq |y| \leq \pi} |K_n|.$$

The first term is $\leq \varepsilon M/2\pi$ by (b); the second $\rightarrow 0$ by (c). Hence $\leq C\varepsilon$ for large n . If f is continuous everywhere it is uniformly continuous, so δ is independent of x and the convergence is uniform. □

Theorem 32.2.10 (The Dirichlet kernel is NOT good). $\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N = 1$, so (a) holds; but $\int_{-\pi}^{\pi} |D_N(x)| dx \geq c \log N \rightarrow \infty$, so (b) fails.

Proof. $\frac{1}{2\pi} \int D_N = \frac{1}{2\pi} \sum_{n=-N}^N \int e^{inx} dx = 1$ (only $n = 0$ survives). For the lower bound (Problem 2), $|D_N(x)| \geq c |\sin((N + \frac{1}{2})x)|/|x|$ near 0; changing variables and summing $\sum_{k=1}^N 1/k \geq c \log N$ gives $\int |D_N| \geq c \log N$. The mean is 1 but the L^1 norm blows up: massive cancellation between positive and negative lobes. This failure is precisely why pointwise convergence of Fourier series is subtle, and why we turn to summability methods. $\square$

Worked Example 32.2.11 (Verifying the good-kernel axioms for the Poisson family). Check that $\{P_r\}_{0 \leq r < 1}$ ($r \rightarrow 1$) is a family of good kernels.

Recognition. Use the closed form $P_r = \frac{1-r^2}{1-2r \cos \theta + r^2} \geq 0$ and estimate the denominator away from 0.

Solution. (a) $\frac{1}{2\pi} \int P_r = \sum_n r^{|n|} \cdot \delta_{n,0} = r^0 = 1$ (term-by-term). (b) $P_r \geq 0$, so (a) $\Rightarrow$ (b) with $M = 2\pi$. (c) Write $1 - 2r \cos \theta + r^2 = (1-r)^2 + 2r(1 - \cos \theta)$; for $\frac{1}{2} \leq r < 1$ and $\delta \leq |\theta| \leq \pi$ this is $\geq c_\delta > 0$, so $P_r(\theta) \leq (1-r^2)/c_\delta \rightarrow 0$ as $r \rightarrow 1$, whence $\int_{\delta \leq |\theta|} P_r \rightarrow 0$. *Check.* All three axioms hold; P_r is good. $\blacksquare$

Worked Example 32.2.12 (A non-example: $K_n = D_n$ fails axiom (b)). Explain why the conclusion of Theorem 32.2.9 cannot be applied to D_N .

Recognition. Theorem 32.2.9 needs (b); D_N violates it.

Solution. Although $\frac{1}{2\pi} \int D_N = 1$ (axiom (a) holds), $\int |D_N| \geq c \log N \rightarrow \infty$, so (b) fails – no uniform M . The proof of Theorem 32.2.9 bounded the near term by $\varepsilon M/2\pi$; with M unbounded that step collapses. *Check.* Indeed $S_N(f) = f * D_N$ need not converge to f even at points of continuity – the very phenomenon that motivates Cesàro/Abel summation. $\blacksquare$

Worked Example 32.2.13 (Good kernels reproduce constants exactly). If $f \equiv c$ is constant, show $f * K_n = c$ for every good kernel and every n .

Recognition. Axiom (a) is exactly the statement that K_n integrates to 1.

Solution. $(f * K_n)(x) = \frac{1}{2\pi} \int K_n(y) c dy = c \cdot \frac{1}{2\pi} \int K_n = c \cdot 1 = c$. *Check.* The limit $f * K_n \rightarrow f$ holds trivially (with equality, not just in the limit) on constants – the mass-one axiom is precisely what fixes constants. $\blacksquare$

Worked Example 32.2.14 (The Fejer kernel is nonnegative – so (b) is automatic). Granting $F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)}$, verify $F_N \geq 0$ and the easy axioms.

Recognition. A squared sine over a squared sine is manifestly ≥ 0 .

Solution. $F_N \geq 0$ since it is a ratio of squares times $1/N > 0$. Axiom (a): $\frac{1}{2\pi} \int F_N = \frac{1}{N} \sum_{n=0}^{N-1} \frac{1}{2\pi} \int D_n = \frac{1}{N} \cdot N \cdot 1 = 1$, using $\frac{1}{2\pi} \int D_n = 1$. Nonnegativity then gives (b) with $M = 2\pi$. (Axiom (c) is handled in Section 32.3.) *Check.* Contrast with D_N , which changes sign and so fails (b); averaging the D_n killed the cancellation. $\blacksquare$

Worked Example 32.2.15 (Young-type bound: convolution cannot increase the sup of a probability kernel average). If $K \geq 0$ with $\frac{1}{2\pi} \int K = 1$, show $\sup_x |(f * K)(x)| \leq \sup_x |f(x)|$.

Recognition. A nonnegative unit-mass kernel produces a genuine average, which cannot exceed the extremes of f .

Solution. $|(f * K)(x)| \leq \frac{1}{2\pi} \int |f(x-y)| K(y) dy \leq (\sup |f|) \frac{1}{2\pi} \int K = \sup |f|$. *Check.* Equality when f is constant; this is exactly why the Fejér and Poisson means never overshoot, while D_N (which is not nonnegative) can. $\blacksquare$

Worked Example 32.2.16 (Convolution diagonalizes in the Fourier basis). Show that the operator $T_g f = f * g$ is “diagonal” on the exponentials: $T_g e_n = \hat{g}(n) e_n$.

Recognition. Property (vi) says convolution multiplies coefficients; the e_n are the eigenvectors.

Solution. $(e_n * g)(x) = \frac{1}{2\pi} \int e^{in(x-y)} g(y) dy = e^{inx} \hat{g}(n) = \hat{g}(n) e_n(x)$. So each e_n is an eigenfunction of T_g with eigenvalue $\hat{g}(n)$. *Check.* This recovers (vi): $\widehat{f * g}(n) = \hat{f}(n) \hat{g}(n)$, since $f = \sum \hat{f}(n) e_n$ and T_g scales the n th component by $\hat{g}(n)$. $\blacksquare$

Exercises (Stein, Ch.2) – convolutions and summation by parts.

1. Summation by parts and Dirichlet's test. Let $\{a_n\}_{n=1}^N, \{b_n\}_{n=1}^N$ be finite sequences, $B_k = \sum_{n=1}^k b_n$ with $B_0 = 0$. (a) Prove $\sum_{n=M}^N a_n b_n = a_N B_N - a_M B_{M-1} - \sum_{n=M}^{N-1} (a_{n+1} - a_n) B_n$. (b) Deduce Dirichlet's test: if the partial sums of $\sum b_n$ are bounded and $\{a_n\}$ decreases monotonically to 0, then $\sum a_n b_n$ converges.
2. Integral-norm convergence and coefficients (convolution viewpoint). If $\{f_k\}$ are Riemann integrable on $[0, 1]$ with $\int_0^1 |f_k - f| dx \rightarrow 0$ as $k \rightarrow \infty$, show $\hat{f}_k(n) \rightarrow \hat{f}(n)$ uniformly in n . (This is the estimate used to push the convolution identity $\widehat{f * g} = \hat{f} \hat{g}$ from continuous to integrable f, g via Lemma 32.2.4.)

Additional exercises.

1. Compute $\widehat{f * h}(n)$ for $f = g * h$ given $\hat{g}(n) = 1/(1 + n^2)$ and $\hat{h}(n) = r^{|n|}$ ($0 < r < 1$), and identify the decay rate of $\widehat{g * h}(n)$.
2. Show $D_N * F_M$ has Fourier coefficients $\mathbf{1}_{|n| \leq N} \cdot (1 - |n|/M)_+$, and hence is a trigonometric polynomial of degree $\min(N, M - 1)$.
3. Prove that convolution has *no* identity in the space of integrable functions: there is no integrable e with $f * e = f$ for all f . [Hint: what would $\hat{e}(n)$ have to be?] This is exactly why we use *approximate* identities (good kernels).
4. (Regularity gain.) Let f be merely integrable and $g \in C^1$. Show $f * g \in C^1$ and $(f * g)' = f * g'$. Conclude that convolving with a smooth kernel smooths f .
5. Verify directly from the integral definition that $(f * g) * h = f * (g * h)$ for trigonometric polynomials f, g, h , using property (vi).
6. (Non-example.) Exhibit integrable $f, g \geq 0$ for which $f \cdot g$ and $f * g$ differ markedly: compute both for $f = g = \mathbf{1}_{[0, \pi]}$ on the circle and compare their values at $x = 0$.
7. Show that if $\hat{f}(n)\hat{g}(n) = 0$ for every n , then $f * g \equiv 0$. Interpret in terms of “disjoint frequency support.”
8. (Diagram.) Draw a narrow nonnegative kernel K_n and a fixed f , and shade the weighted-average interpretation of $(f * K_n)(x)$ at one point x ; annotate how the shaded mass concentrates as n grows.

32.3 Cesaro and Abel Summability: Fejer and Poisson Kernels

Since Fourier series can fail to converge at individual points (even of continuity), we reinterpret $\lim_N S_N(f) = f$ in weaker, more forgiving senses: *Cesàro* (averaging partial sums) and *Abel* (radial limits). Each leads to a family of good kernels – Fejér and Poisson – and hence, by Theorem 32.2.9, to genuine convergence.

Cesaro means

Definition 32.3.1 (Cesaro summability). For a series $\sum_{k \geq 0} c_k$ with partial sums $s_n = \sum_{k=0}^n c_k$, the N th *Cesàro mean* is $\sigma_N = \frac{s_0 + s_1 + \cdots + s_{N-1}}{N}$. The series is *Cesàro summable* to σ if $\sigma_N \rightarrow \sigma$.

The motivating example is Grandi's series $1 - 1 + 1 - 1 + \cdots$: its partial sums $1, 0, 1, 0, \dots$ have no limit, yet their averages σ_N converge to $\frac{1}{2}$. Cesàro summation is strictly more inclusive than ordinary convergence: convergent $\Rightarrow$ Cesàro summable to the same limit (Exercise 12), but not conversely.

Worked Example 32.3.2 (Grandi's series is Cesaro summable to $1/2$). Compute σ_N for $\sum_{k \geq 0} (-1)^k$.

Recognition. Partial sums alternate $1, 0, 1, 0, \dots$; average them.

Solution. $s_n = 1$ if n even, 0 if n odd. So $s_0 + \cdots + s_{N-1} = \lceil N/2 \rceil$, and $\sigma_N = \lceil N/2 \rceil / N \rightarrow \frac{1}{2}$. Check. The ordinary partial sums diverge, but Cesàro averaging extracts the natural value $\frac{1}{2}$. ■

The Fejer kernel

The N th Cesàro mean of the Fourier series is $\sigma_N(f) = \frac{S_0(f) + \dots + S_{N-1}(f)}{N}$. Since $S_n(f) = f * D_n$,

$$\sigma_N(f) = f * F_N, \quad F_N = \frac{D_0 + D_1 + \dots + D_{N-1}}{N} \quad (\text{Fejér kernel}).$$

Lemma 32.3.3 (Closed form; Fejer is good). $F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)}$, and $\{F_N\}$ is a family of good kernels.

Proof. With $\omega = e^{ix}$, $NF_N = \sum_{n=0}^{N-1} D_n = \sum_{n=0}^{N-1} \frac{\omega^{-n} - \omega^{n+1}}{1 - \omega}$ (Example 4's intermediate form). Summing the two geometric progressions and simplifying with $1 - \omega = -e^{ix/2}(e^{ix/2} - e^{-ix/2})$ yields $NF_N = \frac{\sin^2(Nx/2)}{\sin^2(x/2)}$ (Exercise 15). Now $F_N \geq 0$, and $\frac{1}{2\pi} \int F_N = \frac{1}{N} \sum_{n=0}^{N-1} \frac{1}{2\pi} \int D_n = 1$ (axiom (a)), so (b) holds with $M = 2\pi$. For (c): $\sin^2(x/2) \geq c_\delta > 0$ for $\delta \leq |x| \leq \pi$, hence $F_N(x) \leq \frac{1}{Nc_\delta}$ there, and $\int_{\delta \leq |x| \leq \pi} F_N \leq \frac{2\pi}{Nc_\delta} \rightarrow 0$. $\square$

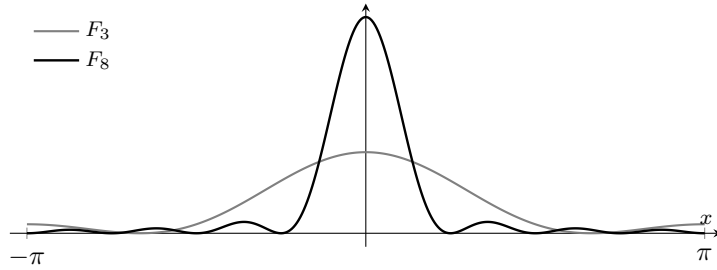


Figure 32.11. The Fejér kernel $F_N = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)} \geq 0$: a nonnegative spike of unit mass concentrating at 0. Unlike D_N it does not change sign – a good kernel.

Theorem 32.3.4 (Fejer). *If f is integrable on the circle, its Fourier series is Cesàro summable to f at every point of continuity. If f is continuous everywhere, $\sigma_N(f) \rightarrow f$ uniformly.*

Proof. $\sigma_N(f) = f * F_N$ and $\{F_N\}$ are good kernels (Lemma 32.3.3); apply Theorem 32.2.9. $\square$

Corollary 32.3.5 (Uniqueness, revisited). *If f is integrable with $\hat{f}(n) = 0$ for all n , then $f = 0$ at every point of continuity. If f is continuous on the circle and $\hat{f}(n) = 0$ for all n , then $f \equiv 0$.*

Proof. If all $\hat{f}(n) = 0$ then every $S_N(f) = 0$, hence every $\sigma_N(f) = 0$; by Fejér, $0 = \sigma_N(f)(x) \rightarrow f(x)$ at each point of continuity. Continuous f is continuous everywhere, so $f \equiv 0$. $\square$

This is the clean “peaking polynomial” uniqueness theorem of Stein’s Section 2, now obtained as a one-line corollary of Fejér. Two further classical consequences:

Corollary 32.3.6 (Absolute convergence forces uniform convergence to f). *If f is continuous on the circle and $\sum_n |\hat{f}(n)| < \infty$, then $S_N(f) \rightarrow f$ uniformly.*

Proof. The series $g(\theta) = \sum_n \hat{f}(n)e^{in\theta}$ converges absolutely and uniformly (Weierstrass M -test), so g is continuous, and $\hat{g}(n) = \hat{f}(n)$ for all n (integrate term by term). Then $\widehat{f-g}(n) = 0$ for all n , and $f - g$ is continuous, so $f - g \equiv 0$ by Corollary 32.3.5; thus $S_N(f) \rightarrow g = f$ uniformly. $\square$

Corollary 32.3.7 (Smoothness gives decay; $C^2 \Rightarrow$ absolute convergence). *If $f \in C^2$ on the circle then $\hat{f}(n) = O(1/|n|^2)$, so the Fourier series converges absolutely and uniformly to f . More generally $f \in C^k \Rightarrow \hat{f}(n) = O(1/|n|^k)$.*

Proof. Integrate by parts twice for $n \neq 0$: the boundary terms vanish by periodicity of f, f' , leaving $2\pi \hat{f}(n) = \frac{-1}{n^2} \int_0^{2\pi} f''(\theta)e^{-in\theta} d\theta$, whence $2\pi|n|^2|\hat{f}(n)| \leq \int |f''| \leq C$. Since $\sum 1/n^2 < \infty$, apply Corollary 32.3.6. The same computation iterated k times gives $\hat{f}(n) = (in)^{-k} \widehat{f^{(k)}}(n) = O(1/|n|^k)$. En route we also proved $\hat{f}'(n) = in \hat{f}(n)$. $\square$

Corollary 32.3.8 (Weierstrass approximation, periodic form). *Continuous functions on the circle are uniformly approximable by trigonometric polynomials: for f continuous with $f(-\pi) = f(\pi)$ and $\varepsilon > 0$ there is a trigonometric polynomial P with $\sup |f - P| < \varepsilon$.*

Proof. By Fejér $\sigma_N(f) \rightarrow f$ uniformly, and each $\sigma_N(f)$ is a trigonometric polynomial. Take $P = \sigma_N(f)$ for large N . (Approximating e^{ix} by ordinary polynomials then yields the classical Weierstrass theorem, Exercise 16.) $\square$

Worked Example 32.3.9 (Fejer sum of the square wave: no Gibbs overshoot). Contrast S_N and σ_N for the square wave near its jump.

Recognition. $S_N = f * D_N$ overshoots (Gibbs); $\sigma_N = f * F_N$ uses a nonnegative kernel.

Solution. Because $F_N \geq 0$, $\sigma_N(f)$ is an average of values of f , hence stays within $[\inf f, \sup f] = [-1, 1]$: *no overshoot*. By contrast S_N overshoots the jump by $\approx 9\%$ (the Gibbs constant) no matter how large N is.

Check. The nonnegativity of F_N is exactly what tames the oscillation that D_N 's sign changes produced. $\blacksquare$

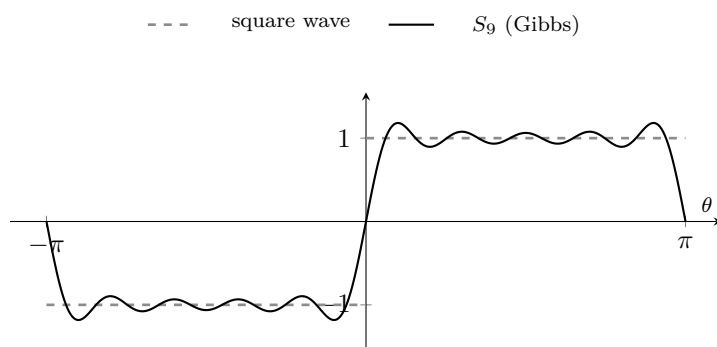


Figure 32.12. Gibbs phenomenon: the partial sum S_9 of the square wave overshoots the jump by about 9%, and the overshoot persists as $N \rightarrow \infty$. The Fejér means σ_N (nonnegative kernel) do not overshoot.

Abel means and the Poisson kernel

Definition 32.3.10 (Abel summability). $\sum_{k \geq 0} c_k$ is *Abel summable* to s if $A(r) = \sum_{k \geq 0} c_k r^k$ converges for $0 \leq r < 1$ and $A(r) \rightarrow s$ as $r \rightarrow 1^-$. Convergent $\Rightarrow$ Abel summable; Cesàro summable $\Rightarrow$ Abel summable (same value). Both arrows are strict.

For instance $1 - 2 + 3 - 4 + \cdots = \sum (-1)^k (k+1)$ is Abel summable to $\frac{1}{4}$ (since $\sum (-1)^k (k+1) r^k = \frac{1}{(1+r)^2} \rightarrow \frac{1}{4}$) but is not Cesàro summable.

$$\{\text{convergent}\} \hookrightarrow \{\text{Cesàro summable}\} \hookrightarrow \{\text{Abel summable}\}$$

Figure 32.13. The summability hierarchy: every convergent series is Cesàro summable, and every Cesàro-summable series is Abel summable, to the same value. Both inclusions are strict ($\sum (-1)^k$ separates the first; $\sum (-1)^k (k+1)$ the second).

For Fourier series, the Abel means of $f \sim \sum a_n e^{in\theta}$ are

$$A_r(f)(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|} a_n e^{in\theta} = (f * P_r)(\theta), \quad P_r(\theta) = \sum_n r^{|n|} e^{in\theta} = \frac{1-r^2}{1-2r \cos \theta + r^2}.$$

The convolution identity follows by inserting $a_n = \frac{1}{2\pi} \int f(\varphi) e^{-in\varphi} d\varphi$ and interchanging sum and integral (justified by uniform convergence for fixed $r < 1$).

Lemma 32.3.11 (Poisson is good). *For $0 \leq r < 1$, $P_r(\theta) = \frac{1-r^2}{1-2r \cos \theta + r^2} \geq 0$, and $\{P_r\}$ is a family of good kernels as $r \rightarrow 1^-$.*

Proof. The closed form was derived in Example 5. $1 - 2r \cos \theta + r^2 = (1 - r)^2 + 2r(1 - \cos \theta) \geq 0$ so $P_r \geq 0$. Axiom (a): $\frac{1}{2\pi} \int P_r = r^0 = 1$ (term by term). Nonnegativity gives (b). For (c): if $\frac{1}{2} \leq r < 1$ and $\delta \leq |\theta| \leq \pi$ then the denominator is $\geq c_\delta > 0$, so $P_r(\theta) \leq (1 - r^2)/c_\delta \rightarrow 0$ as $r \rightarrow 1$, hence $\int_{\delta \leq |\theta|} P_r \rightarrow 0$. $\square$

Theorem 32.3.12 (Abel summability of Fourier series). *The Fourier series of an integrable f on the circle is Abel summable to f at every point of continuity; if f is continuous everywhere the convergence is uniform.*

Proof. $A_r(f) = f * P_r$ and $\{P_r\}$ are good kernels (Lemma 32.3.11); apply Theorem 32.2.9 with the parameter $r \rightarrow 1$ in place of $n \rightarrow \infty$. $\square$

The Dirichlet problem in the disc

Theorem 32.3.13 (Poisson integral solves $\Delta u = 0$). *Let f be integrable on the unit circle. Then $u(r, \theta) = (f * P_r)(\theta) = \sum_m a_m r^{|m|} e^{im\theta}$ satisfies: (i) u is twice continuously differentiable (indeed C^∞) in the open disc and $\Delta u = 0$; (ii) at each point of continuity θ of f , $\lim_{r \rightarrow 1} u(r, \theta) = f(\theta)$ (uniformly if f is continuous); (iii) if f is continuous, u is the unique solution with these properties.*

Proof. (i) For $r \leq \rho < 1$ the series $\sum a_m r^{|m|} e^{im\theta}$ and all its term-by-term derivatives converge absolutely and uniformly ($|a_m|$ bounded, $r^{|m|}$ decays geometrically), so u is smooth; in polar coordinates $\Delta = \partial_{rr} + \frac{1}{r} \partial_r + \frac{1}{r^2} \partial_{\theta\theta}$, and each $r^{|m|} e^{im\theta}$ is harmonic, so $\Delta u = 0$ term by term. (ii) is Theorem 32.3.12. (iii) If v also solves the problem and $\rightarrow f$ uniformly, expand $v(r, \theta) \sim \sum a_n(r) e^{in\theta}$; substituting into $\Delta v = 0$ and integrating in θ (two integrations by parts on the $\partial_{\theta\theta}$ term) gives $a_n'' + \frac{1}{r} a_n' - \frac{n^2}{r^2} a_n = 0$, so $a_n(r) = A_n r^n + B_n r^{-n}$ ($n \neq 0$); boundedness forces $B_n = 0$, and $r \rightarrow 1$ with uniform convergence gives $A_n = \frac{1}{2\pi} \int f e^{-in\theta} = \hat{f}(n)$. So v has the same series as u , and by uniqueness for continuous functions $u = v$. $\square$

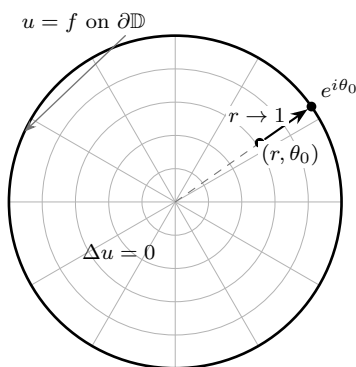


Figure 32.14. Dirichlet problem in the disc: the Poisson integral $u(r, \theta) = (f * P_r)(\theta)$ is harmonic inside ($\Delta u = 0$) and attains the boundary data f as the interior point (r, θ_0) approaches $e^{i\theta_0}$ ($r \rightarrow 1$). (Schematic.)

Worked Example 32.3.14 (Solving $\Delta u = 0$ with boundary data $f(\theta) = \cos \theta$). Find the harmonic function in the disc with $u(1, \theta) = \cos \theta$.

Recognition. Read off Fourier coefficients of the boundary data, then attach $r^{|m|}$.

Solution. $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, so $a_1 = a_{-1} = \frac{1}{2}$, else 0. Thus $u(r, \theta) = \frac{1}{2} r e^{i\theta} + \frac{1}{2} r e^{-i\theta} = r \cos \theta$. Check. $u = r \cos \theta = x$ in Cartesian coordinates is harmonic ($\Delta x = 0$) and equals $\cos \theta$ on $r = 1$. ■

Worked Example 32.3.15 (Poisson integral of a constant). Show the harmonic extension of $f \equiv c$ is $u \equiv c$.

Recognition. Constants have only the $n = 0$ coefficient; good kernels fix constants.

Solution. $a_0 = c$, all other $a_n = 0$, so $u(r, \theta) = c r^0 = c$. Equivalently $u = c * P_r = c$ because $\frac{1}{2\pi} \int P_r = 1$. Check. A constant is harmonic and matches its own boundary data. ■

Worked Example 32.3.16 (Abel limit at a jump is the midpoint). If f has a jump at θ with one-sided limits $f(\theta^\pm)$, find $\lim_{r \rightarrow 1} A_r(f)(\theta)$.

Recognition. P_r is even, so it splits its mass equally left and right of θ .

Solution. Since P_r is even, $\frac{1}{2\pi} \int_{-\pi}^0 P_r = \frac{1}{2\pi} \int_0^\pi P_r = \frac{1}{2}$. Splitting the convolution into $y > 0$ and $y < 0$ and using continuity from each side, $A_r(f)(\theta) = \frac{1}{2\pi} \int P_r(y) f(\theta - y) dy \rightarrow \frac{1}{2} f(\theta^+) + \frac{1}{2} f(\theta^-)$. *Check.* For a symmetric jump $f(\theta^\pm) = \pm 1$ the limit is 0, the midpoint – matching the sawtooth’s value at its jump (Exercise 8). ■

Worked Example 32.3.17 (Summing a series by Abel summation: $1 - 2 + 3 - 4 + \cdots = \frac{1}{4}$). Verify the Abel sum of $\sum_{k \geq 0} (-1)^k (k+1)$.

Recognition. Differentiate a geometric series.

Solution. $A(r) = \sum_{k \geq 0} (-1)^k (k+1) r^k = \frac{d}{dr} \sum_{k \geq 0} (-1)^k r^{k+1} = \frac{d}{dr} \frac{r}{1+r} = \frac{1}{(1+r)^2} \rightarrow \frac{1}{4}$ as $r \rightarrow 1$. *Check.* This series is not Cesàro summable (its $c_n/n = (-1)^n(1+1/n) \not\rightarrow 0$), so Abel is strictly stronger. ■

Worked Example 32.3.18 (Ordering of summation methods (a chain of examples)). Place $\sum (-1)^k$, $\sum (-1)^k (k+1)$, and a convergent series in the hierarchy.

Recognition. convergent $\subset$ Cesàro $\subset$ Abel, all strict.

Solution. A convergent series (say $\sum 2^{-k} = 2$) is all three. $\sum (-1)^k$ diverges but is Cesàro summable ($\frac{1}{2}$), hence Abel summable ($\frac{1}{2}$): strictly past convergence. $\sum (-1)^k (k+1)$ is Abel summable ($\frac{1}{4}$) but not Cesàro summable: strictly past Cesàro. *Check.* The three examples populate exactly the three nested-but-distinct layers. ■

Worked Example 32.3.19 (Poisson kernel as a power series in r recovers the coefficients). From $u(r, \theta) = \sum_m a_m r^{|m|} e^{im\theta}$, recover a_m by integrating against $e^{-im\theta}$.

Recognition. The factor $r^{|m|}$ is a bookkeeping device; at fixed r the θ -integral picks one coefficient.

Solution. $\frac{1}{2\pi} \int_{-\pi}^\pi u(r, \theta) e^{-im\theta} d\theta = a_m r^{|m|}$, so $a_m = r^{-|m|} \cdot \frac{1}{2\pi} \int u(r, \theta) e^{-im\theta} d\theta$. Letting $r \rightarrow 1$ (with $u \rightarrow f$ uniformly) gives $a_m = \hat{f}(m)$. *Check.* This is the computation underlying uniqueness part (iii) of Theorem 32.3.13. ■

Worked Example 32.3.20 (Fejer kernel concentrates: a quantitative tail bound). Show $\int_{\delta \leq |x| \leq \pi} F_N(x) dx \leq \frac{2\pi}{N c_\delta}$ where $c_\delta = \sin^2(\delta/2)$.

Recognition. Bound F_N pointwise away from 0 using $\sin^2(x/2) \geq \sin^2(\delta/2)$.

Solution. For $\delta \leq |x| \leq \pi$, $\sin^2(x/2) \geq \sin^2(\delta/2) = c_\delta$, so $F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)} \leq \frac{1}{N c_\delta}$. Integrating over the set of length $\leq 2\pi$ gives $\int_{\delta \leq |x| \leq \pi} F_N \leq \frac{2\pi}{N c_\delta} \rightarrow 0$. *Check.* This is precisely good-kernel axiom (c) for F_N , made quantitative – the decay is $O(1/N)$. ■

Exercises (Stein, Ch.2) – summability, kernels, Dirichlet problem.

1. Convergent implies Cesàro summable. Prove that if $\sum c_n$ converges to s then it is Cesàro summable to s . [Hint: reduce to $s_n \rightarrow 0$.]
2. Abel beats Cesàro. (a) If $\sum_{n \geq 1} c_n$ converges to s , it is Abel summable to s . [Hint: reduce to $s = 0$; with $s_N = c_1 + \cdots + c_N$ show $\sum_{n=1}^N c_n r^n = (1-r) \sum_{n=1}^N s_n r^n + s_N r^{N+1}$, let $N \rightarrow \infty$, then $r \rightarrow 1$.] (b) Exhibit a series Abel summable but not convergent. [Hint: $c_n = (-1)^n$.] (c) If $\sum c_n$ is Cesàro summable to σ , it is Abel summable to σ . [Hint: $\sum c_n r^n = (1-r)^2 \sum n \sigma_n r^n$.] (d) A series Abel summable but not Cesàro summable. [Hint: $c_n = (-1)^{n-1} n$; if Cesàro summable then $c_n/n \rightarrow 0$.] Summary: convergent $\Rightarrow$ Cesàro $\Rightarrow$ Abel, none reversible.
3. Tauber’s theorem. (a) If $\sum c_n$ is Cesàro summable to σ and $n c_n \rightarrow 0$, then $\sum c_n$ converges to σ . [Hint: $s_n - \sigma_n = [(n-1)c_n + \cdots + c_2]/n$.] (b) The same with “Cesàro” replaced by “Abel.” [Hint: estimate $\sum_{n=1}^N c_n$ against $\sum_{n=1}^N c_n r^n$ with $r = 1 - 1/N$.]
4. Fejér kernel formula. Prove $F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)}$. [Hint: $N F_N = \sum_{n=0}^{N-1} D_n$; with $\omega = e^{ix}$, $N F_N = \sum_{n=0}^{N-1} \frac{\omega^{-n} - \omega^{n+1}}{1 - \omega}$.]
5. Weierstrass approximation. State and prove: a continuous f on $[a, b]$ is uniformly approximable by polynomials. [Hint: apply Corollary 32.3.8 (Fejér) and approximate e^{ix} uniformly by polynomials on any interval.]

6. Abel means at a jump. Suppose f has a jump discontinuity at θ with limits $f(\theta^\pm)$. (a) Prove $\lim_{r \rightarrow 1} A_r(f)(\theta) = \frac{f(\theta^+) + f(\theta^-)}{2}$. [Hint: $\frac{1}{2\pi} \int_{-\pi}^0 P_r = \frac{1}{2\pi} \int_0^\pi P_r = \frac{1}{2}$; adapt the proof in the text.] (b) Show similarly the Fourier series at θ is Cesàro summable to $\frac{f(\theta^+) + f(\theta^-)}{2}$.
7. A nonzero harmonic function vanishing at the boundary pointwise. Let $u(r, \theta) = \partial P_r / \partial \theta$ for $0 \leq r < 1$. Show (i) $\Delta u = 0$ in the disc; (ii) $\lim_{r \rightarrow 1} u(r, \theta) = 0$ for each θ ; yet $u \not\equiv 0$. (So pointwise boundary convergence does not force uniqueness.)
8. Laplace in a strip. Solve $\Delta u = 0$ in $S = \{0 < x < 1, 0 < y\}$ with $u(0, y) = u(1, y) = 0$, $u(x, 0) = f(x)$, $f(0) = f(1) = 0$. Write $f(x) = \sum_{n \geq 1} a_n \sin(n\pi x)$ and use $u_n(x, y) = e^{-n\pi y} \sin(n\pi x)$; express u as a Poisson-type integral involving f .
9. Dirichlet problem in an annulus $\{\rho < r < 1\}$ with $u(1, \theta) = f$, $u(\rho, \theta) = g$. Writing $u = \sum c_n(r) e^{in\theta}$, $c_n(r) = A_n r^n + B_n r^{-n}$ ($n \neq 0$), and $f \sim \sum a_n e^{in\theta}$, $g \sim \sum b_n e^{in\theta}$, derive

$$u(r, \theta) = \sum_{n \neq 0} \frac{((\rho/r)^n - (r/\rho)^n) a_n + (r^n - \rho^{-n}) b_n}{\rho^n - \rho^{-n}} e^{in\theta} + a_0 + (b_0 - a_0) \frac{\log r}{\log \rho},$$

and show $u(r, \theta) - (P_r * f)(\theta) \rightarrow 0$ as $r \rightarrow 1$ and $u(r, \theta) - (P_{\rho/r} * g)(\theta) \rightarrow 0$ as $r \rightarrow \rho$, both uniformly.

Problems (Stein, Ch.2).

1. Integrable functions with dense discontinuities. Pick a countable dense $\{r_n\}$ in $[0, 1]$. (a) For $f(x) = 0$ ($x < 0$), $f(x) = 1$ ($x \geq 0$), show $F(x) = \sum_{n \geq 1} \frac{1}{n^2} f(x - r_n)$ is integrable and discontinuous at each r_n . [Hint: F is monotone and bounded.] (b) With $g(x) = \sin(1/x)$ ($x \neq 0$), $g(0) = 0$, show $F(x) = \sum_{n \geq 1} 3^{-n} g(x - r_n)$ is integrable, discontinuous at each r_n , and not monotone on any subinterval. [Hint: $3^{-k} > \sum_{n > k} 3^{-n}$.] (c) Riemann's example $F(x) = \sum_{n \geq 1} \frac{(nx)}{n^2}$, where $(x) = x$ on $(-1/2, 1/2]$ extended periodically, is discontinuous exactly at $x = m/2n$ (m odd).
2. Size of the Dirichlet kernel. With $L_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N(\theta)| d\theta$: (a) Prove $L_N \geq c \log N$. [Hint: $|D_N(\theta)| \geq c |\sin((N + \frac{1}{2})\theta)|/|\theta|$; change variables and use $\sum_{k=1}^n 1/k \geq c \log n$; sharper: $L_N = \frac{4}{\pi^2} \log N + O(1)$.] (b) Deduce that for each $n \geq 1$ there is a continuous f_n with $|f_n| \leq 1$ and $|S_n(f_n)(0)| \geq c' \log n$. [Hint: $g_n = \text{sgn}(D_n)$ works but is not continuous; approximate in integral norm by continuous h_k with $|h_k| \leq 1$.]
3. * Littlewood's refinement of Tauber. (a) If $\sum c_n$ is Abel summable to s and $c_n = O(1/n)$, then $\sum c_n$ converges to s . (b) Hence if $\sum c_n$ is Cesàro summable to s and $c_n = O(1/n)$, it converges to s . Applied to Fourier series (via Exercise 6): if $\hat{f}(\nu) = O(1/|\nu|)$ then $S_N(f)(\theta) \rightarrow f(\theta)$ at points of continuity, $\rightarrow \frac{f(\theta^+) + f(\theta^-)}{2}$ at jumps, and uniformly if f is continuous.

Additional exercises.

1. Show $F_N(0) = N$ and $\frac{1}{2\pi} \int_{-\pi}^{\pi} F_N = 1$ directly from the closed form, and confirm F_N peaks at the origin.
2. Using Fejér's theorem, prove the *Riemann-Lebesgue* fact that $\hat{f}(n) \rightarrow 0$ as $|n| \rightarrow \infty$ for every integrable f on the circle. [Hint: approximate f uniformly by trigonometric polynomials when f is continuous; then use Lemma 32.2.4.]
3. Solve $\Delta u = 0$ in the unit disc with boundary data $f(\theta) = \sin 2\theta$. Give u in both polar and Cartesian form and check harmonicity.
4. Compute the Abel sum of $\sum_{n \geq 1} \cos n\theta$ for fixed $\theta \neq 0 \pmod{2\pi}$, and relate the answer to $\frac{1}{2}(P_r(\theta) - 1)$ as $r \rightarrow 1$.
5. Prove that the Poisson integral of a nonnegative f is nonnegative in the whole disc (maximum/positivity principle), using $P_r \geq 0$.
6. Show that the Fejér means $\sigma_N(f)$ satisfy $\inf f \leq \sigma_N(f)(x) \leq \sup f$ for all x (no overshoot), and contrast with the Gibbs overshoot of S_N .

7. (Non-example.) Find a series that is Abel summable but for which the Cesàro means diverge, demonstrating that the implication Cesàro $\Rightarrow$ Abel cannot be reversed.
8. Establish $\widehat{f}'(n) = in\widehat{f}(n)$ for $f \in C^1$ on the circle, and use it to re-derive $\widehat{f}(n) = O(1/|n|^2)$ for $f \in C^2$.
9. (Diagram.) Plot F_3 and F_8 together with D_3 on $[-\pi, \pi]$ and annotate: the D_N side lobes change sign, while the F_N never go negative – the structural reason F_N is a good kernel and D_N is not.
10. (Heat equation tie-in.) Interpret $u(r, \theta) = (f * P_r)(\theta)$ as the steady-state temperature in a disc with boundary temperature f . Explain physically why $r \rightarrow 1$ recovers f at points of continuity, in terms of mass concentrating at the boundary point.

Chapter 33

Convergence of Fourier Series

33.1 Mean-Square Convergence and Parseval

We now confront the central convergence question for Fourier series from the *global* point of view. If f is integrable on the circle and $S_N(f)$ is the N -th symmetric partial sum of its Fourier series, in what sense does $S_N(f)$ recover f ? The answer in this section is the cleanest one available: *mean-square* (or L^2) convergence,

$$\frac{1}{2\pi} \int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2 d\theta \rightarrow 0 \quad (N \rightarrow \infty).$$

The engine behind this is *orthogonality*: the exponentials $e_n(\theta) = e^{in\theta}$ form an orthonormal family, and $S_N(f)$ is nothing but the orthogonal projection of f onto the span of $e_{-N}, \dots, e_N$. Geometry in an inner-product space does the rest. We develop the linear-algebraic scaffolding first, then assemble the proof.

Vector spaces, inner products, and norms

Definition 33.1.1. An *inner product* on a complex vector space V assigns to each pair $X, Y \in V$ a number $\langle X, Y \rangle \in \mathbb{C}$ that is *Hermitian*, $\langle X, Y \rangle = \overline{\langle Y, X \rangle}$; *linear in the first slot*, $\langle \alpha X + \beta Y, Z \rangle = \alpha \langle X, Z \rangle + \beta \langle Y, Z \rangle$ (hence *conjugate-linear in the second*); and *positive*, $\langle X, X \rangle \geq 0$. The associated *norm* is $\|X\| = \langle X, X \rangle^{1/2}$. The product is *strictly* positive-definite if $\|X\| = 0 \implies X = 0$. (Over $\mathbb{R}$ the conjugates vanish and “Hermitian” becomes “symmetric”.)

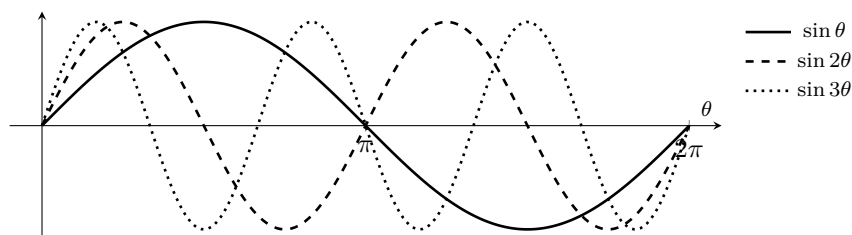


Figure 33.1. Standing waves / orthonormal modes: distinct frequencies $\sin n\theta$ are mutually orthogonal, $\frac{1}{2\pi} \int_0^{2\pi} \sin n\theta \sin m\theta d\theta = \frac{1}{2} \delta_{nm}$.

The three running models are: the Euclidean spaces $\mathbb{R}^d$ and $\mathbb{C}^d$ with $\langle Z, W \rangle = \sum_j z_j \overline{w_j}$; the sequence space $\ell^2(\mathbb{Z})$ of two-sided sequences $A = (a_n)_{n \in \mathbb{Z}}$ with $\sum_n |a_n|^2 < \infty$ and $\langle A, B \rangle = \sum_n a_n \overline{b_n}$; and the function space $\mathcal{R}$ of Riemann-integrable functions on the circle with

$$\langle f, g \rangle = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) \overline{g(\theta)} d\theta, \quad \|f\| = \left(\frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 d\theta \right)^{1/2}.$$

Definition 33.1.2. Vectors X, Y are *orthogonal*, $X \perp Y$, when $\langle X, Y \rangle = 0$. A family $\{e_n\}$ is *orthonormal* if $\langle e_n, e_m \rangle = \delta_{nm}$ (one when $n = m$, zero otherwise).

Theorem 33.1.3 (Pythagoras, Cauchy–Schwarz, triangle). *In any inner-product space:*

- (i) if $X \perp Y$ then $\|X + Y\|^2 = \|X\|^2 + \|Y\|^2$;
- (ii) $|\langle X, Y \rangle| \leq \|X\| \|Y\|$, with equality iff X, Y are parallel;
- (iii) $\|X + Y\| \leq \|X\| + \|Y\|$.

Proof. (i) Expand $\|X + Y\|^2 = \langle X + Y, X + Y \rangle = \|X\|^2 + \langle X, Y \rangle + \langle Y, X \rangle + \|Y\|^2$ and use $\langle X, Y \rangle = 0$. (ii) If $\|Y\| = 0$, then for all real t , $0 \leq \|X + tY\|^2 = \|X\|^2 + 2t \operatorname{Re} \langle X, Y \rangle$; taking t large of either sign forces $\operatorname{Re} \langle X, Y \rangle = 0$, and replacing Y by iY forces $\operatorname{Im} \langle X, Y \rangle = 0$, so $\langle X, Y \rangle = 0$ and the inequality is $0 \leq 0$. If $\|Y\| \neq 0$, set $c = \langle X, Y \rangle / \langle Y, Y \rangle$. Then $X - cY \perp Y$, so writing $X = (X - cY) + cY$ and applying (i),

$$\|X\|^2 = \|X - cY\|^2 + |c|^2 \|Y\|^2 \geq |c|^2 \|Y\|^2 = \frac{|\langle X, Y \rangle|^2}{\|Y\|^2}.$$

Taking square roots gives the claim; equality forces $X = cY$. (iii) By (ii), $\|X + Y\|^2 = \|X\|^2 + 2 \operatorname{Re} \langle X, Y \rangle + \|Y\|^2 \leq \|X\|^2 + 2 \|X\| \|Y\| + \|Y\|^2 = (\|X\| + \|Y\|)^2$. $\square$

The picture is the familiar one from Euclidean geometry: orthogonality is a right angle, Pythagoras is the right-triangle law, and Cauchy–Schwarz says cos of the angle between two vectors has absolute value at most one.

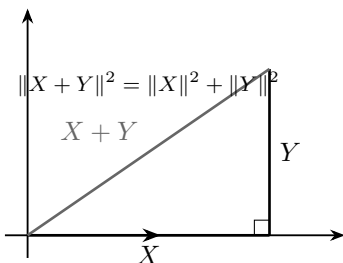


Figure 33.2. Pythagoras in an inner-product space: orthogonal legs $X \perp Y$. (Schematic.)

Worked Example 33.1.4 (Cauchy–Schwarz for integrals, directly). Prove $|\langle f, g \rangle| \leq \|f\| \|g\|$ in $\mathcal{R}$ without quoting the abstract theorem, using $2AB \leq A^2 + B^2$.

Recognition. The trigger is an L^2 product bound; Stein’s slick route avoids expanding a quadratic and instead optimizes a scaling parameter λ .

Solution. For reals A, B , $2AB \leq A^2 + B^2$. Put $A = \lambda^{1/2} |f(\theta)|$ and $B = \lambda^{-1/2} |g(\theta)|$ with $\lambda > 0$: $|f(\theta)g(\theta)| \leq \frac{1}{2}(\lambda |f|^2 + \lambda^{-1} |g|^2)$. Average over θ : $|\langle f, g \rangle| \leq \frac{1}{2\pi} \int |f| |g| \leq \frac{1}{2}(\lambda \|f\|^2 + \lambda^{-1} \|g\|^2)$. Minimizing the right side over λ (or simply setting $\lambda = \|g\| / \|f\|$) yields $|\langle f, g \rangle| \leq \|f\| \|g\|$. *Check.* Take $f = g$; then $\lambda = 1$ and the bound reads $\|f\|^2 \leq \|f\|^2$, sharp as it must be.

Worked Example 33.1.5 ($\ell^2(\mathbb{Z})$ is closed under addition). Show that if $A, B \in \ell^2(\mathbb{Z})$ then $A + B \in \ell^2(\mathbb{Z})$, with $\|A + B\| \leq \|A\| + \|B\|$.

Recognition. “Verify a sequence space is a vector space.” Reduce the infinite statement to finite truncations where ordinary Euclidean geometry applies, then pass to the limit.

Solution. For $N > 0$ let A_N be A truncated to $|n| \leq N$ (zeros elsewhere); similarly B_N . The finite triangle inequality gives $\|A_N + B_N\| \leq \|A_N\| + \|B_N\| \leq \|A\| + \|B\|$, i.e. $\sum_{|n| \leq N} |a_n + b_n|^2 \leq (\|A\| + \|B\|)^2$. The bound is uniform in N , so letting $N \rightarrow \infty$ gives $\sum_{n \in \mathbb{Z}} |a_n + b_n|^2 \leq (\|A\| + \|B\|)^2 < \infty$. *Check.* With $a_n = b_n = 1/n$ for $n \geq 1$ (else 0), both lie in ℓ^2 since $\sum 1/n^2 < \infty$, and $A + B = (2/n)$ has $\|A + B\| = 2 \|A\|$, matching equality in the parallel case.

Hilbert spaces, and the failure of $\mathcal{R}$ to be one

Definition 33.1.6. An inner-product space is a *Hilbert space* if (i) the inner product is strictly positive-definite and (ii) the space is *complete*: every Cauchy sequence converges to a limit in the space. $\mathbb{R}^d, \mathbb{C}^d, \ell^2(\mathbb{Z})$ are Hilbert spaces. A space failing (i) or (ii) is *pre-Hilbert*.

Caution. The Riemann space $\mathcal{R}$ is only *pre-Hilbert*, and fails both axioms. Strict positivity fails: a function vanishing except at finitely many jumps has $\|f\| = 0$ yet $f \neq 0$. Completeness fails: the truncations of $f(\theta) = \log(1/\theta)$ form a Cauchy sequence in $\mathcal{R}$ whose only possible limit, f itself, is unbounded and so not Riemann integrable (Stein, Exercise 5). The genuine completion is the Lebesgue space $L^2([0, 2\pi])$ – the analytic analogue of completing $\mathbb{Q}$ to $\mathbb{R}$. We work in $\mathcal{R}$ but borrow the geometry that survives.

Worked Example 33.1.7 (A nonzero function with zero norm). Exhibit $f \in \mathcal{R}$, $f \neq 0$, with $\|f\| = 0$, and explain why this only breaks strict positivity, not positivity.

Recognition. “Show $\|\cdot\|$ is a seminorm but not a norm on $\mathcal{R}$.”

Solution. Let $f(\theta) = 1$ at $\theta = 0$ and $f(\theta) = 0$ elsewhere on $[0, 2\pi]$. It is Riemann integrable with $\int_0^{2\pi} |f|^2 = 0$, so $\|f\| = 0$ although $f \neq 0$. The positivity axiom $\langle X, X \rangle \geq 0$ still holds; what fails is the implication $\|f\| = 0 \implies f = 0$. *Check.* More generally $\|f\| = 0$ forces $f = 0$ at every point of continuity, so f can differ from 0 only on the negligible set of discontinuities.

The orthonormal family $\{e_n\}$ and partial sums as projections

For each $n \in \mathbb{Z}$ set $e_n(\theta) = e^{in\theta}$. A one-line computation gives orthonormality:

$$\langle e_n, e_m \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta} d\theta = \delta_{nm}.$$

The Fourier coefficients of f are exactly the inner products against this family,

$$a_n = \widehat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta = \langle f, e_n \rangle, \quad S_N(f) = \sum_{|n| \leq N} a_n e_n.$$

Lemma 33.1.8 (Orthogonality of the residual). $f - \sum_{|n| \leq N} a_n e_n \perp e_m$ for every $|m| \leq N$, hence $f - S_N(f) \perp \sum_{|n| \leq N} b_n e_n$ for all scalars b_n .

Proof. $\langle f - \sum_{|n| \leq N} a_n e_n, e_m \rangle = \langle f, e_m \rangle - a_m = a_m - a_m = 0$ by orthonormality and $a_m = \langle f, e_m \rangle$. Linearity extends this to any combination of the e_m , $|m| \leq N$. $\square$

Theorem 33.1.9 (Energy identity). For every N ,

$$\|f\|^2 = \|f - S_N(f)\|^2 + \sum_{|n| \leq N} |a_n|^2.$$

Proof. Split $f = (f - S_N(f)) + S_N(f)$. By Lemma 33.1.8 the two pieces are orthogonal, so Pythagoras (Theorem 33.1.3(i)) gives $\|f\|^2 = \|f - S_N(f)\|^2 + \|S_N(f)\|^2$. Orthonormality of $\{e_n\}$ gives $\|S_N(f)\|^2 = \sum_{|n| \leq N} |a_n|^2$. $\square$

Corollary 33.1.10 (Bessel’s inequality). $\sum_{n \in \mathbb{Z}} |a_n|^2 \leq \|f\|^2$. In particular the coefficient sequence $(a_n) \in \ell^2(\mathbb{Z})$.

Proof. Drop the nonnegative term $\|f - S_N(f)\|^2$ in Theorem 33.1.9: $\sum_{|n| \leq N} |a_n|^2 \leq \|f\|^2$ for all N ; let $N \rightarrow \infty$. $\square$

Lemma 33.1.11 (Best approximation). For any scalars c_n , $\|f - S_N(f)\| \leq \left\| f - \sum_{|n| \leq N} c_n e_n \right\|$, with equality iff $c_n = a_n$ for all $|n| \leq N$.

Proof. Write $b_n = a_n - c_n$. Then $f - \sum c_n e_n = (f - S_N(f)) + \sum_{|n| \leq N} b_n e_n$, a sum of orthogonal pieces (Lemma 33.1.8). Pythagoras gives $\|f - \sum c_n e_n\|^2 = \|f - S_N(f)\|^2 + \sum_{|n| \leq N} |b_n|^2 \geq \|f - S_N(f)\|^2$, with equality iff every $b_n = 0$. $\square$

Geometrically, among all trigonometric polynomials of degree $\leq N$, the closest one to f in the mean-square norm is precisely the partial sum $S_N(f)$ – it is the orthogonal projection of f onto $\text{span}\{e_{-N}, \dots, e_N\}$.

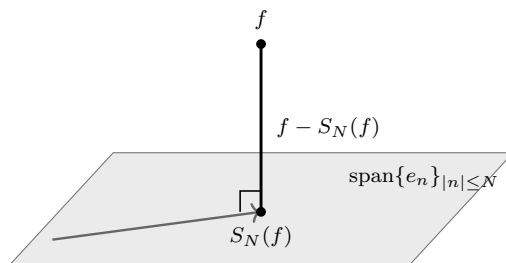


Figure 33.3. Best approximation: $S_N(f)$ is the foot of the perpendicular from f to the span of $\{e_n\}_{|n| \leq N}$. (Schematic.)

Proof of mean-square convergence

Theorem 33.1.12 (Mean-square convergence). *If f is integrable on the circle, then $\|f - S_N(f)\| \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. Continuous case. If f is continuous on the circle, then given $\varepsilon > 0$ there is a trigonometric polynomial P (of some degree M) with $\sup_\theta |f(\theta) - P(\theta)| < \varepsilon$ – this is the density of trigonometric polynomials in $C(\mathbb{T})$, from the Fejér/Cesàro theory of Chapter 2. Squaring and averaging, $\|f - P\| < \varepsilon$. By best approximation (Lemma 33.1.11), for $N \geq M$, $\|f - S_N(f)\| \leq \|f - P\| < \varepsilon$.

General integrable case. If f is merely integrable it cannot be approximated *uniformly* by polynomials, so we first approximate it by a *continuous* g . Given $\varepsilon > 0$ choose continuous g on the circle with $\sup_\theta |g(\theta)| \leq \sup_\theta |f(\theta)| = B$ and $\int_0^{2\pi} |f - g| < \varepsilon^2$ (Stein’s approximation lemma, Ch.2). Then, bounding one factor of $|f - g|$ by $2B$,

$$\|f - g\|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f - g|^2 \leq \frac{2B}{2\pi} \int_0^{2\pi} |f - g| \leq C\varepsilon^2.$$

Next pick a trigonometric polynomial P with $\|g - P\| < \varepsilon$ (continuous case). By the triangle inequality $\|f - P\| \leq \|f - g\| + \|g - P\| \leq C'\varepsilon$, and best approximation again gives $\|f - S_N(f)\| \leq \|f - P\| \leq C'\varepsilon$ for N large. Since ε was arbitrary, $\|f - S_N(f)\| \rightarrow 0$. $\square$

Theorem 33.1.13 (Parseval’s identity). *If f is integrable with $f \sim \sum_n a_n e^{in\theta}$, then $\|f - S_N(f)\| \rightarrow 0$ and*

$$\sum_{n=-\infty}^{\infty} |a_n|^2 = \|f\|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 d\theta.$$

Proof. Mean-square convergence (Theorem 33.1.12) sends $\|f - S_N(f)\|^2 \rightarrow 0$ in the energy identity (Theorem 33.1.9), leaving $\sum_{|n| \leq N} |a_n|^2 \rightarrow \|f\|^2$. Hence the full series converges to $\|f\|^2$. $\square$

So Bessel’s inequality is in fact an *equality*: $\{e_n\}$ is a complete orthonormal system (a basis) for the mean-square geometry. The map $f \mapsto (a_n)$ is an isometry from $\mathcal{R}$ into $\ell^2(\mathbb{Z})$.

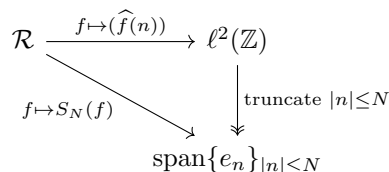


Figure 33.4. The Fourier coefficient map $f \mapsto (\hat{f}(n))$ is norm-preserving ($\sum |a_n|^2 = \|f\|^2$, Parseval). Its composite with truncation to $|n| \leq N$ is the partial sum $S_N(f)$, the orthogonal projection onto $\text{span}\{e_n\}_{|n| \leq N}$.

Theorem 33.1.14 (Polarized Parseval). If $F \sim \sum a_n e^{in\theta}$ and $G \sim \sum b_n e^{in\theta}$ are integrable, then

$$\langle F, G \rangle = \frac{1}{2\pi} \int_0^{2\pi} F(\theta) \overline{G(\theta)} d\theta = \sum_{n=-\infty}^{\infty} a_n \overline{b_n}.$$

Proof. Apply the polarization identity, valid in any Hermitian space,

$$\langle F, G \rangle = \frac{1}{4} (\|F + G\|^2 - \|F - G\|^2 + i\|F + iG\|^2 - i\|F - iG\|^2),$$

to both sides; each squared norm equals the corresponding ℓ^2 norm by Parseval, and the four combinations reassemble into $\sum a_n \overline{b_n}$ (absolutely convergent by Cauchy–Schwarz in ℓ^2). $\square$

Theorem 33.1.15 (Riemann–Lebesgue lemma). If f is integrable on the circle then $\hat{f}(n) \rightarrow 0$ as $|n| \rightarrow \infty$. Equivalently $\int_0^{2\pi} f(\theta) \sin(N\theta) d\theta \rightarrow 0$ and the same with $\cos$.

Proof. By Bessel/Parseval the series $\sum |a_n|^2$ converges, so its terms tend to zero: $|a_n| \rightarrow 0$. Writing $\sin(N\theta) = (e^{iN\theta} - e^{-iN\theta})/2i$ converts the trigonometric statement into $a_{-N} - a_N \rightarrow 0$. $\square$

Worked Example 33.1.16 (Sawtooth coefficients and $\sum 1/n^2$ via Parseval). Let f be 2π -periodic, odd, with $f(\theta) = \frac{1}{2}(\pi - \theta)$ on $(0, 2\pi)$. Its coefficients are $a_n = \frac{1}{2in}$ for $n \neq 0$, $a_0 = 0$. Use Parseval to evaluate $\sum_{n \geq 1} 1/n^2$.

Recognition. “Sum a numerical series by computing $\|f\|^2$ two ways.” This is the prototypical Parseval application.

Solution. Here $|a_n|^2 = 1/(4n^2)$ for $n \neq 0$, so $\sum_n |a_n|^2 = 2 \sum_{n \geq 1} \frac{1}{4n^2} = \frac{1}{2} \sum_{n \geq 1} \frac{1}{n^2}$. On the other side, $\|f\|^2 = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{4}(\pi - \theta)^2 d\theta = \frac{1}{8\pi} \cdot \frac{(2\pi)^3}{12} = \frac{\pi^2}{12}$ (substitute $u = \pi - \theta$). Equating, $\frac{1}{2} \sum 1/n^2 = \pi^2/12$, so $\sum_{n \geq 1} 1/n^2 = \pi^2/6$. *Check.* Numerically $\pi^2/6 \approx 1.6449$, and $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} \approx 1.4236$ is already climbing toward it.

Worked Example 33.1.17 ($f(\theta) = |\theta|$: Parseval yields $\sum 1/(2n+1)^4$ and $\sum 1/n^4$). For $f(\theta) = |\theta|$ on $[-\pi, \pi]$ (Stein Ex. 8a) the cosine coefficients are $a_0 = \pi/2$ and, for $n \neq 0$, $a_n = \frac{-1}{\pi n^2} \cdot \frac{1 - (-1)^n}{1} \cdot \frac{1}{2}$, i.e. $a_n = -2/(\pi n^2)$ for odd n and 0 for even $n \neq 0$. Evaluate the resulting sums.

Recognition. “Use Parseval on an even function to extract a fourth-power series.”

Solution. $\|f\|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta^2 d\theta = \pi^2/3$. The coefficient side is $a_0^2 + 2 \sum_{n \geq 1, \text{ odd}} \frac{4}{\pi^2 n^4} = \frac{\pi^2}{4} + \frac{8}{\pi^2} \sum_{n \geq 0} \frac{1}{(2n+1)^4}$. Setting the two equal, $\sum_{n \geq 0} \frac{1}{(2n+1)^4} = \frac{\pi^2}{8} \left(\frac{\pi^2}{3} - \frac{\pi^2}{4} \right) = \frac{\pi^4}{96}$. Splitting $\sum 1/n^4$ into odd and even parts, $\sum_{n \geq 1} 1/n^4 = \sum_{\text{odd}} + \frac{1}{16} \sum_{n \geq 1} 1/n^4$, so $\frac{15}{16} \sum 1/n^4 = \pi^4/96$ and $\sum_{n \geq 1} 1/n^4 = \pi^4/90$. *Check.* $\pi^4/90 \approx 1.0823$, consistent with $\zeta(4)$.

Worked Example 33.1.18 (Square wave: coefficients, Parseval, and $\sum 1/(2k+1)^2$). Let g be the 2π -periodic square wave $g(\theta) = 1$ on $(0, \pi)$ and -1 on $(-\pi, 0)$. Compute $\hat{g}(n)$ and apply Parseval.

Recognition. “A piecewise-constant odd wave; expect a $1/n$ sine series supported on odd n .”

Solution. g is odd, so $a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} g e^{-in\theta} = \frac{1}{2\pi} \int_0^{\pi} (e^{-in\theta} - e^{in\theta}) \cdot (\dots)$; carrying out the integral gives $a_n = \frac{1 - (-1)^n}{\pi i n} \cdot \frac{1}{2} \cdot 2$, that is $a_n = \frac{2}{\pi i n}$ for odd n and 0 for even n (and $a_0 = 0$). Thus $|a_n|^2 = 4/(\pi^2 n^2)$ on odd n . Parseval: $\|g\|^2 = 1$ (since $|g| \equiv 1$), and $\sum_n |a_n|^2 = 2 \sum_{k \geq 0} \frac{4}{\pi^2 (2k+1)^2} = \frac{8}{\pi^2} \sum_{k \geq 0} \frac{1}{(2k+1)^2}$. Equate: $\sum_{k \geq 0} 1/(2k+1)^2 = \pi^2/8$. *Check.* Adding the even terms, $\sum 1/n^2 = \frac{\pi^2}{8} + \frac{1}{4} \sum 1/n^2$ recovers $\sum 1/n^2 = \pi^2/6$. Consistent.

Worked Example 33.1.19 (Triangle wave has absolutely convergent series). Let $h(\theta) = |\theta|$ on $[-\pi, \pi]$ (a triangle wave), with $a_n = -2/(\pi n^2)$ on odd n . Show its Fourier series converges absolutely and uniformly, unlike the square wave’s.

Recognition. “Decide absolute convergence from the decay rate of a_n .”

Solution. $\sum_n |a_n| = |a_0| + 2 \sum_{\text{odd } n \geq 1} \frac{2}{\pi n^2} \leq \frac{\pi}{2} + \frac{4}{\pi} \sum_{n \geq 1} \frac{1}{n^2} < \infty$. By the Weierstrass M -test the series converges uniformly, so its sum is continuous – matching the fact that h is continuous. *Contrast:* the square wave has $|a_n| \sim 1/|n|$, and $\sum 1/|n| = \infty$, so that series is *not* absolutely convergent – reflecting its jump discontinuity. *Check.* Faster coefficient decay ($1/n^2$ vs $1/n$) is exactly what continuity buys you.

Worked Example 33.1.20 (Polarized Parseval: a cross term). With $F(\theta) = e^{i\theta} + 2e^{-2i\theta}$ and $G(\theta) = 3e^{i\theta} - e^{-2i\theta}$, verify Theorem 33.1.14 by direct computation.

Recognition. “Check the bilinear Parseval on finite trigonometric polynomials.”

Solution. Coefficients: $a_1 = 1, a_{-2} = 2$ (else 0); $b_1 = 3, b_{-2} = -1$. The series side is $a_1 \overline{b_1} + a_{-2} \overline{b_{-2}} = 1 \cdot 3 + 2 \cdot (-1) = 1$. The integral side: $\frac{1}{2\pi} \int_0^{2\pi} F \overline{G}$; orthonormality kills all cross frequencies, leaving the matched terms $1 \cdot \overline{3} + 2 \cdot \overline{(-1)} = 3 - 2 = 1$. They agree. *Check.* Putting $G = F$ recovers $\|F\|^2 = |1|^2 + |2|^2 = 5$, the ordinary Parseval value.

Worked Example 33.1.21 (Incompleteness of $\mathcal{R}$ from the ℓ^2 side). The sequence $a_k = 1/k$ for $k \geq 1$, $a_k = 0$ for $k \leq 0$, lies in $\ell^2(\mathbb{Z})$. Argue no Riemann-integrable F has these as its Fourier coefficients (Stein Ex. 6).

Recognition. “Use one-sided coefficient decay to detect non-integrability.”

Solution. If such F existed it would be bounded, with $\widehat{F}(k) = 1/k$ for $k \geq 1$ and 0 otherwise. Its Abel means at 0 would be $A_r(F)(0) = \sum_{k \geq 1} r^k/k = -\log(1-r) \rightarrow +\infty$ as $r \rightarrow 1^-$. But $|A_r(F)(0)| \leq \frac{1}{2\pi} \int |F| P_r \leq \sup |F| < \infty$ since the Poisson kernel P_r is a good kernel of total mass one. Contradiction. *Check.* The obstruction is precisely the broken left/right symmetry of the coefficients – the recurring theme of §33.3.

Worked Example 33.1.22 (C^1 functions have absolutely convergent series). Show that if f is continuously differentiable on the circle then $\sum_n |\widehat{f}(n)| < \infty$ (Stein Ex. 14).

Recognition. “Trade a derivative for one power of n , then use Cauchy–Schwarz in ℓ^2 .”

Solution. Integration by parts gives $\widehat{f}'(n) = in\widehat{f}(n)$, so $|\widehat{f}(n)| = \frac{1}{|n|} |\widehat{f}'(n)|$ for $n \neq 0$. By Cauchy–Schwarz,

$$\sum_{n \neq 0} |\widehat{f}(n)| = \sum_{n \neq 0} \frac{1}{|n|} |\widehat{f}'(n)| \leq \left(\sum_{n \neq 0} \frac{1}{n^2} \right)^{1/2} \left(\sum_n |\widehat{f}'(n)|^2 \right)^{1/2} = \left(\frac{\pi^2}{3} \right)^{1/2} \|f'\| < \infty,$$

using Parseval for f' . Adding the $n = 0$ term keeps the sum finite. *Check.* This is sharp in spirit – mere continuity does not suffice (the square wave), but one derivative does.

Worked Example 33.1.23 (Parseval for $\theta(\pi - \theta)$ and $\sum 1/n^6$). For the 2π -periodic odd function with $f(\theta) = \theta(\pi - \theta)$ on $[0, \pi]$ (Stein Ex. 8b) the sine coefficients vanish on even harmonics and equal $b_n = \frac{8}{\pi n^3}$ on odd n . Use Parseval to find $\sum_{n \geq 0} 1/(2n+1)^6$.

Recognition. “A cubic-decay sine series; Parseval lifts it to a sixth-power sum.”

Solution. Write $f \sim \sum_{n \geq 1} b_n \sin n\theta$ with $b_n = 8/(\pi n^3)$ for odd n , else 0. The real Parseval form reads $\frac{1}{\pi} \int_0^\pi f^2 = \frac{1}{2} \sum_{n \geq 1} b_n^2$. The left side is $\frac{1}{\pi} \int_0^\pi \theta^2(\pi - \theta)^2 d\theta = \frac{1}{\pi} \cdot \frac{\pi^5}{30} = \frac{\pi^4}{30}$. The right side is $\frac{1}{2} \cdot \frac{64}{\pi^2} \sum_{n \geq 0} \frac{1}{(2n+1)^6}$. Equate: $\sum_{n \geq 0} \frac{1}{(2n+1)^6} = \frac{\pi^2}{32} \cdot \frac{\pi^4}{30} = \frac{\pi^6}{960}$. *Check.* The even/odd split then gives $\sum_{n \geq 1} 1/n^6 = \frac{64}{63} \cdot \frac{\pi^6}{960} = \frac{\pi^6}{945}$, matching $\zeta(6)$.

Worked Example 33.1.24 (Fourier coefficients of the periodized Gaussian / heat kernel). On the circle, the heat kernel is $H_t(\theta) = \sum_{n \in \mathbb{Z}} e^{-n^2 t} e^{in\theta}$ for $t > 0$. Identify its Fourier coefficients and use Parseval to compute $\frac{1}{2\pi} \int_{-\pi}^\pi H_t(\theta)^2 d\theta$.

Recognition. “Read coefficients off a kernel given by its expansion, then Parseval.”

Solution. By inspection $\widehat{H_t}(n) = e^{-n^2 t}$, since the series is the Fourier expansion. Parseval gives $\frac{1}{2\pi} \int_{-\pi}^\pi H_t^2 = \sum_{n \in \mathbb{Z}} e^{-2n^2 t} = \theta_3(0, e^{-2t})$, a Jacobi theta value. *Check.* As $t \rightarrow 0^+$ each $e^{-n^2 t} \rightarrow 1$ and H_t approaches the Dirac comb, so the sum diverges; as $t \rightarrow \infty$ only $n = 0$ survives and the integral $\rightarrow 1$, matching $H_\infty \equiv 1$.

Exercises (Stein, Ch.3).

1. Show that the first two inner-product spaces $\mathbb{R}^d$ and $\mathbb{C}^d$ are complete. [Every Cauchy sequence in $\mathbb{R}$ has a limit.]
2. Prove that $\ell^2(\mathbb{Z})$ is complete. [If $A_k = \{a_{k,n}\}_{n \in \mathbb{Z}}$ is Cauchy, show each coordinate $\{a_{k,n}\}_k$ is Cauchy in $\mathbb{C}$, so converges to some b_n ; bound $\|A_k - B\|$ via partial sums and prove $B = (b_n) \in \ell^2$.]

- Construct integrable $\{f_k\}$ on $[0, 2\pi]$ with $\frac{1}{2\pi} \int_0^{2\pi} |f_k|^2 \rightarrow 0$ yet $\lim_k f_k(\theta)$ failing to exist for every θ . [Take indicators $f_k = \chi_{I_k}$ of intervals I_k with $|I_k| \rightarrow 0$ but every point in infinitely many I_k .]
- For the space $\mathcal{R}$ with norm $\|f\| = (\frac{1}{2\pi} \int_0^{2\pi} |f|^2)^{1/2}$: (a) exhibit nonzero integrable f with $\|f\| = 0$; (b) show $\|f\| = 0$ implies $f(x) = 0$ at every point of continuity; (c) conversely, if $f \in \mathcal{R}$ vanishes at all its points of continuity then $\|f\| = 0$.
- Let $f(\theta) = \log(1/\theta)$ on $(0, 2\pi]$, $f(0) = 0$, and $f_n(\theta) = f(\theta)$ for $1/n < \theta \leq 2\pi$, 0 on $[0, 1/n]$. Prove $\{f_n\}$ is Cauchy in $\mathcal{R}$ but $f \notin \mathcal{R}$. [Show $\int_a^b (\log \theta)^2 d\theta \rightarrow 0$ as $b \rightarrow 0$ using that $\frac{d}{d\theta}(\theta(\log \theta)^2 - 2\theta \log \theta + 2\theta) = (\log \theta)^2$.]
- For $a_k = 1/k$ ($k \geq 1$), $a_k = 0$ ($k \leq 0$): show $(a_k) \in \ell^2(\mathbb{Z})$ but no Riemann integrable function has k -th Fourier coefficient a_k for all k .
- Show $\sum_{n \geq 2} \frac{1}{\log n} \sin nx$ converges for every x yet is not the Fourier series of any Riemann integrable function. (Same for $\sum \sin nx/n^\alpha$, $0 < \alpha < 1$; the case $1/2 < \alpha < 1$ is harder, see Problem 1.)
- (a) For $f(\theta) = |\theta|$ on $[-\pi, \pi]$, use Parseval to obtain $\sum_{n \geq 0} \frac{1}{(2n+1)^4} = \frac{\pi^4}{96}$ and $\sum_{n \geq 1} \frac{1}{n^4} = \frac{\pi^4}{90}$. (b) For the 2π -periodic odd f with $f(\theta) = \theta(\pi - \theta)$ on $[0, \pi]$, show $\sum_{n \geq 0} \frac{1}{(2n+1)^6} = \frac{\pi^6}{960}$ and $\sum_{n \geq 1} \frac{1}{n^6} = \frac{\pi^6}{945}$.
- For $\alpha \notin \mathbb{Z}$, show the Fourier series of $\frac{\pi}{\sin \pi \alpha} e^{i(\pi-x)\alpha}$ on $[0, 2\pi]$ is $\sum_{n \in \mathbb{Z}} \frac{e^{inx}}{n+\alpha}$. Apply Parseval to deduce $\sum_{n \in \mathbb{Z}} \frac{1}{(n+\alpha)^2} = \frac{\pi^2}{(\sin \pi \alpha)^2}$.

Problems (Stein, Ch.3) – orthogonal families and zeta values.

- (Legendre polynomials.) On $[-1, 1]$ let $L_n(x) = \frac{d^n}{dx^n}(x^2 - 1)^n$. (a) Show for smooth f : $\int_{-1}^1 L_n f = (-1)^n \int_{-1}^1 (x^2 - 1)^n f^{(n)}$, hence $L_n \perp x^m$ for $m < n$, so $\{L_n\}$ is orthogonal. (b) Show $\|L_n\|^2 = \frac{(n!)^2 2^{2n+1}}{2n+1}$. (c) Any degree- n polynomial orthogonal to $1, x, \dots, x^{n-1}$ is a constant times L_n . (d) The normalized $\mathcal{L}_n = L_n / \|L_n\|$ are the Gram-Schmidt orthonormalization of $\{1, x, x^2, \dots\}$, and every $f \in \mathcal{R}[-1, 1]$ has a Legendre expansion $\sum_n \langle f, \mathcal{L}_n \rangle \mathcal{L}_n$ converging in mean square.
- (Bernoulli numbers and $\zeta(2m)$.) Define B_n by $\frac{z}{e^z - 1} = \sum_{n \geq 0} \frac{B_n}{n!} z^n$. (a) Compute $B_0 = 1, B_1 = -\frac{1}{2}, B_2 = \frac{1}{6}, B_3 = 0, B_4 = -\frac{1}{30}, B_5 = 0$. (b) For $n \geq 1$, $B_n = -\frac{1}{n+1} \sum_{k=0}^{n-1} \binom{n+1}{k} B_k$. (c) From $\frac{z}{e^z - 1} = 1 - \frac{z}{2} + \sum_{n \geq 2} \frac{B_n}{n!} z^n$ deduce $B_n = 0$ for odd $n > 1$ and $z \cot z = 1 + \sum_{n \geq 1} (-1)^n \frac{2^{2n} B_{2n}}{(2n)!} z^{2n}$. (d) With $\zeta(s) = \sum_{n \geq 1} n^{-s}$, derive $x \cot x = 1 - 2 \sum_{m \geq 1} \frac{\zeta(2m)}{\pi^{2m}} x^{2m}$. (e) Conclude $2\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m}}{(2m)!} B_{2m}$.

Additional exercises.

- Verify directly that $\{e_n(\theta) = e^{in\theta}\}_{n \in \mathbb{Z}}$ is orthonormal in $\mathcal{R}$ by computing $\frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta} d\theta$ for $n = m$ and $n \neq m$.
- Let $f(\theta) = \theta$ on $(-\pi, \pi)$. Compute $\widehat{f}(n)$, then apply Parseval to evaluate $\sum_{n \geq 1} 1/n^2$.
- (Bessel as a projection bound.) Show that for any orthonormal $\{u_1, \dots, u_m\}$ and any X , $\sum_{j=1}^m |\langle X, u_j \rangle|^2 \leq \|X\|^2$, and that the gap equals $\left\| X - \sum_j \langle X, u_j \rangle u_j \right\|^2$.
- Use polarized Parseval (Theorem 33.1.14) on $f(\theta) = \theta$ and $g(\theta) = \theta^2$ over $(-\pi, \pi)$ to evaluate $\sum_{n \geq 1} (-1)^{n+1}/n^2$.
- Prove the parallelogram law $\|X + Y\|^2 + \|X - Y\|^2 = 2\|X\|^2 + 2\|Y\|^2$ in any inner-product space, and explain why it fails for the sup norm on $C(\mathbb{T})$ (so that norm comes from no inner product).
- (NON-example.) Show that $\|f\|_\infty = \sup_\theta |f(\theta)|$ does *not* arise from any inner product on $C(\mathbb{T})$ by exhibiting f, g violating the parallelogram law of the previous exercise.
- Let $a_n = 1/n$ for $n \geq 1$ and $a_n = 0$ otherwise. Decide whether $(a_n) \in \ell^2(\mathbb{Z})$ and whether $(a_n) \in \ell^1(\mathbb{Z})$, and reconcile your answers with Bessel's inequality.

8. (Diagram.) Sketch in the plane spanned by e_0, e_1 the vectors f , $S_1(f)$, and $f - S_1(f)$ for a function with $a_0 = 1, a_1 = 2$ (other coefficients zero), marking the right angle that encodes the best-approximation lemma. Verify $\|f\|^2 = \|S_1(f)\|^2 + \|f - S_1(f)\|^2$ numerically.
9. Suppose $\sum_n |a_n|^2 < \infty$. Prove the trigonometric series $\sum_n a_n e^{in\theta}$ converges in $L^2(\mathbb{T})$ to some F with $\widehat{F}(n) = a_n$ (the Riesz–Fischer direction, granting completeness of L^2).
10. Let f be real and even on the circle. Show its Fourier series is a pure cosine series and that Parseval reads $\frac{1}{2\pi} \int |f|^2 = a_0^2 + 2 \sum_{n \geq 1} a_n^2$ with real a_n .

33.2 Pointwise Convergence and the Dirichlet Kernel

Mean-square convergence is a statement about f as a whole; it says nothing about the value of $S_N(f)$ at any single point. The *local* theory asks instead: does $S_N(f)(\theta_0) \rightarrow f(\theta_0)$? The answer turns on the *Dirichlet kernel*, the function D_N for which $S_N(f) = f * D_N$. We show that pointwise convergence holds wherever f is differentiable, deduce the Riemann localization principle, and assemble the kernel toolkit (Dirichlet, Fejér, Poisson) that drives the whole subject.

The Dirichlet kernel

Definition 33.2.1. The *Dirichlet kernel* is $D_N(\theta) = \sum_{n=-N}^N e^{in\theta}$. Summing the finite geometric series,

$$D_N(\theta) = \frac{\sin((N + \frac{1}{2})\theta)}{\sin(\theta/2)},$$

and $\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(\theta) d\theta = 1$.

Derivation of the closed form. With $\omega = e^{i\theta}$, $D_N = \sum_{n=-N}^N \omega^n = \omega^{-N} \frac{\omega^{2N+1} - 1}{\omega - 1}$. Multiply top and bottom by $\omega^{-1/2}$: $D_N = \frac{\omega^{N+1/2} - \omega^{-(N+1/2)}}{\omega^{1/2} - \omega^{-1/2}} = \frac{2i \sin((N + \frac{1}{2})\theta)}{2i \sin(\theta/2)}$. The integral of each $e^{in\theta}$ over a period is zero except $n = 0$, which contributes 2π ; hence mass one. $\square$

The key feature of D_N – the source of all the trouble with pointwise convergence – is that it is *not* a good kernel: its L^1 norm grows, $\frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N| \sim \frac{4}{\pi^2} \log N \rightarrow \infty$. The peak at the origin has height $2N + 1$ but is flanked by large oscillating side-lobes that do not cancel in absolute value.

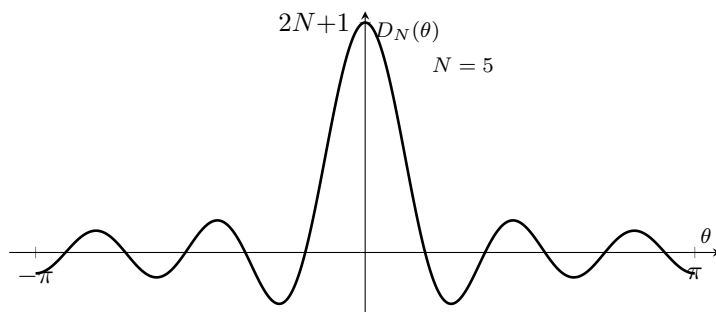


Figure 33.5. The Dirichlet kernel D_5 : tall central spike with persistent oscillating side-lobes – its L^1 mass diverges like $\log N$.

Theorem 33.2.2 (S_N is convolution with D_N). For integrable f , $S_N(f)(\theta) = (f * D_N)(\theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta - t) D_N(t) dt$.

Proof. $S_N(f)(\theta) = \sum_{|n| \leq N} a_n e^{in\theta} = \sum_{|n| \leq N} (\frac{1}{2\pi} \int f(t) e^{-int} dt) e^{in\theta} = \frac{1}{2\pi} \int f(t) \sum_{|n| \leq N} e^{in(\theta-t)} dt = (f * D_N)(\theta)$, after the change of variable $t \mapsto \theta - t$ and using evenness of D_N . $\square$

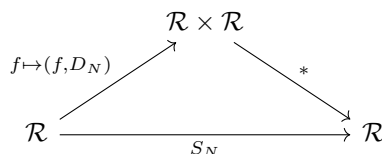


Figure 33.6. The partial-sum operator factors through convolution: $S_N(f) = f * D_N$ (Theorem 33.2.2), where $*$ is the convolution map $(g, K) \mapsto g * K$. Pairing f with the Fejér kernel F_N or Poisson kernel P_r instead gives the Cesàro and Abel means.

The Fejér and Poisson kernels (good kernels)

Definition 33.2.3. A family $\{K_n\}$ of integrable functions on the circle is a *family of good kernels* (an approximate identity) if (a) $\frac{1}{2\pi} \int K_n = 1$; (b) $\frac{1}{2\pi} \int |K_n| \leq M$ uniformly; (c) for each $\delta > 0$, $\int_{\delta \leq |\theta| \leq \pi} |K_n| \rightarrow 0$. For such kernels $f * K_n \rightarrow f$ at points of continuity, uniformly if f is continuous.

Definition 33.2.4. The *Fejér kernel* is the Cesàro average $F_N = \frac{1}{N+1} \sum_{n=0}^N D_n = \frac{1}{N+1} \left(\frac{\sin((N+1)\theta/2)}{\sin(\theta/2)} \right)^2$.

The *Poisson kernel* is $P_r(\theta) = \sum_{n \in \mathbb{Z}} r^{|n|} e^{in\theta} = \frac{1-r^2}{1-2r \cos \theta + r^2}$ for $0 \leq r < 1$.

Unlike D_N , both F_N and P_r are *nonnegative*, so their L^1 norms equal their mean ($= 1$): they *are* good kernels. This is why Cesàro (Fejér) and Abel (Poisson) summation of Fourier series behave so much better than ordinary partial summation.

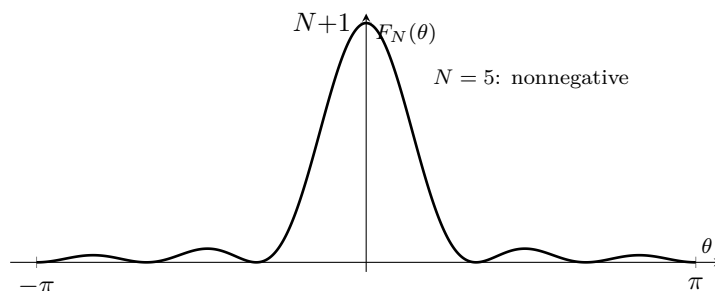


Figure 33.7. The Fejér kernel F_5 : a nonnegative bump concentrating at 0 – a genuine good kernel, in contrast to D_N .

Worked Example 33.2.5 (Fejér kernel closed form). Derive $F_N = \frac{1}{N+1} \left(\frac{\sin((N+1)\theta/2)}{\sin(\theta/2)} \right)^2$ and check $F_N \geq 0$.

Recognition. “Cesàro-average the Dirichlet kernels; expect a perfect square.”

Solution. Sum $\sum_{n=0}^N D_n(\theta) = \sum_{n=0}^N \frac{\sin((n+\frac{1}{2})\theta)}{\sin(\theta/2)}$. Using $\sin((n+\frac{1}{2})\theta) = \text{Im } e^{i(n+1/2)\theta}$ and summing the geometric series gives $\sum_{n=0}^N \sin((n+\frac{1}{2})\theta) = \frac{\sin^2((N+1)\theta/2)}{\sin(\theta/2)}$. Divide by $(N+1)\sin(\theta/2)$ to get the stated formula. It is a square over $(N+1)$, hence ≥ 0 . *Check.* At $\theta \rightarrow 0$, $F_N \rightarrow N+1$ (peak height) and $\frac{1}{2\pi} \int F_N = \frac{1}{N+1} \sum \frac{1}{2\pi} \int D_n = 1$.

Worked Example 33.2.6 (Poisson kernel as a geometric series). Sum $P_r(\theta) = \sum_n r^{|n|} e^{in\theta}$ to the closed form and confirm positivity.

Recognition. “Two one-sided geometric series glued at $n = 0$.”

Solution. $P_r = \sum_{n \geq 0} (re^{i\theta})^n + \sum_{n \geq 1} (re^{-i\theta})^n = \frac{1}{1-re^{i\theta}} + \frac{re^{-i\theta}}{1-re^{-i\theta}}$. Over a common denominator $|1-re^{i\theta}|^2 = 1-2r \cos \theta + r^2$, the numerator simplifies to $1-r^2$, giving $P_r = \frac{1-r^2}{1-2r \cos \theta + r^2} \geq 0$ for $0 \leq r < 1$. *Check.* $\frac{1}{2\pi} \int P_r = \sum_n r^{|n|} \delta_{n0} = 1$; and as $r \rightarrow 1^-$, P_r concentrates at $\theta = 0$.

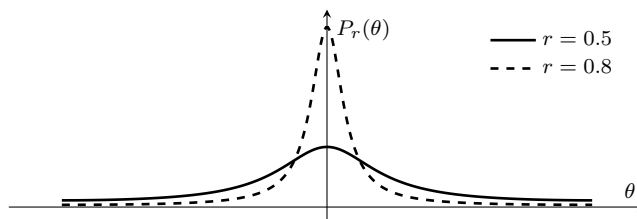


Figure 33.8. Poisson kernels P_r for $r = 0.5, 0.8$: nonnegative, sharpening to a spike as $r \rightarrow 1$.

Pointwise convergence at points of differentiability

Theorem 33.2.7 (Local convergence). *If f is integrable on the circle and differentiable at θ_0 , then $S_N(f)(\theta_0) \rightarrow f(\theta_0)$.*

Proof. Because $\frac{1}{2\pi} \int D_N = 1$,

$$S_N(f)(\theta_0) - f(\theta_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} [f(\theta_0 - t) - f(\theta_0)] D_N(t) dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(t) t D_N(t) dt,$$

where $F(t) = \frac{f(\theta_0 - t) - f(\theta_0)}{t}$ for $t \neq 0$ and $F(0) = -f'(\theta_0)$. Differentiability at θ_0 makes F bounded near 0; away from 0 it is integrable because f is and $|t| \geq \delta$. So F is integrable on $[-\pi, \pi]$. Now

$$t D_N(t) = \frac{t}{\sin(t/2)} \sin((N + \tfrac{1}{2})t), \quad \sin((N + \tfrac{1}{2})t) = \sin(Nt) \cos(\tfrac{t}{2}) + \cos(Nt) \sin(\tfrac{t}{2}),$$

where $t/\sin(t/2)$ is continuous (hence bounded) on $[-\pi, \pi]$. Thus the integrand is a sum of two terms, each of the form (integrable function) $\times \sin(Nt)$ or $\times \cos(Nt)$. By the Riemann–Lebesgue lemma (Theorem 33.1.15) both integrals tend to 0. Hence $S_N(f)(\theta_0) - f(\theta_0) \rightarrow 0$. $\square$

Remark. The proof uses only that F is bounded near θ_0 , which already follows from a *Lipschitz* (Hölder-1) condition $|f(\theta) - f(\theta_0)| \leq M|\theta - \theta_0|$. So differentiability can be weakened to a one-point Lipschitz bound.

Theorem 33.2.8 (Riemann localization principle). *If f, g are integrable and agree on an open interval $I \ni \theta_0$, then $S_N(f)(\theta_0) - S_N(g)(\theta_0) \rightarrow 0$.*

Proof. The difference $h = f - g$ vanishes on I , hence is differentiable at θ_0 (with value and derivative 0 there). Apply Theorem 33.2.7 to h : $S_N(h)(\theta_0) \rightarrow h(\theta_0) = 0$, and S_N is linear. $\square$

This is genuinely surprising: whether $S_N(f)(\theta_0)$ converges depends *only* on the behavior of f in an arbitrarily small neighborhood of θ_0 , even though every coefficient a_n is built from f over the *whole* circle.

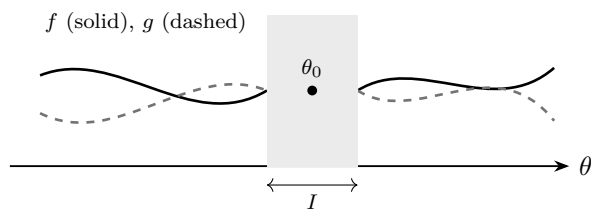


Figure 33.9. Localization: f and g may differ wildly outside I , yet $S_N(f)(\theta_0)$ and $S_N(g)(\theta_0)$ have the same limiting behavior. (Schematic.)

Worked Example 33.2.9 (Convergence at a differentiable point). For $f(\theta) = \theta^2$ on $[-\pi, \pi]$ (periodically extended, continuous, and differentiable at $\theta_0 = 0$), conclude $S_N(f)(0) \rightarrow 0$ and identify the value.

Recognition. “Apply local convergence at an interior differentiable point.”

Solution. f is differentiable at 0 with $f(0) = 0$, so Theorem 33.2.7 gives $S_N(f)(0) \rightarrow f(0) = 0$. The Fourier series is $f(\theta) \sim \frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-1)^n}{n^2} \cos n\theta$; at $\theta = 0$ this reads $\frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-1)^n}{n^2}$. Check. $\sum_{n \geq 1} (-1)^n/n^2 = -\pi^2/12$, so the series equals $\frac{\pi^2}{3} - \frac{\pi^2}{3} = 0 = f(0)$, confirming convergence to the right value.

Worked Example 33.2.10 (Failure of convergence detector at a jump). For the square wave g (value ± 1), explain why Theorem 33.2.7 does not apply at $\theta_0 = 0$ and predict the limit of $S_N(g)(0)$.

Recognition. “A jump discontinuity: differentiability fails; expect the midpoint value.”

Solution. g is not even continuous at 0, so the hypothesis fails. Where the one-sided limits $g(0^\pm) = \pm 1$ exist, symmetric partial sums converge to the average, $S_N(g)(0) \rightarrow \frac{1}{2}(g(0^+) + g(0^-)) = 0$. Indeed g is odd, so every $S_N(g)(0) = 0$ exactly. *Check.* The convergence at 0 here is a coincidence of oddness, not of differentiability; near (but not at) 0 the Gibbs overshoot persists.

Worked Example 33.2.11 (Lipschitz suffices: corner of the triangle wave). For $h(\theta) = |\theta|$ near $\theta_0 = 0$, the derivative does not exist (a corner), yet show $S_N(h)(0) \rightarrow h(0) = 0$.

Recognition. “At a corner use the one-sided Lipschitz form of local convergence.”

Solution. $|h(\theta) - h(0)| = |\theta| = 1 \cdot |\theta - 0|$, a Lipschitz bound with $M = 1$. By the Remark after Theorem 33.2.7, the boundedness of $F(t) = (h(-t) - h(0))/t = |t|/t = \pm 1$ near 0 is all the proof needs, so $S_N(h)(0) \rightarrow 0$.

Check. h is continuous and even, its series $\frac{\pi}{2} - \frac{4}{\pi} \sum_{k \geq 0} \frac{\cos((2k+1)\theta)}{(2k+1)^2}$ at $\theta = 0$ gives $\frac{\pi}{2} - \frac{4}{\pi} \cdot \frac{\pi^2}{8} = 0$. Agrees.

Worked Example 33.2.12 (D_N has $\log N$ growth in L^1). Show $\frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N(\theta)| d\theta \geq c \log N$, so D_N is not a good kernel.

Recognition. “Lower-bound the Lebesgue constant via the side-lobes of $\sin/(\theta/2)$.”

Solution. Near 0, $|D_N(\theta)| \geq \frac{|\sin((N+\frac{1}{2})\theta)|}{|\theta|/2} \cdot c$. On the k -th lobe $\theta \in [\frac{k\pi}{N}, \frac{(k+1)\pi}{N}]$ the numerator is order 1 and $|\theta| \leq (k+1)\pi/N$, contributing at least $c/(k+1)$. Summing $k = 1, \dots, N$ gives $\sum_{k \leq N} c/(k+1) \geq c' \log N$.

Check. The precise asymptotic is $\frac{4}{\pi^2} \log N + O(1)$; the divergence is exactly what permits a continuous function with divergent series in §33.3.

Worked Example 33.2.13 (Good kernel recovers a continuous function (Fejér)). Show $f * F_N \rightarrow f$ uniformly for continuous f , illustrating why Cesàro means converge where partial sums may not.

Recognition. “Standard good-kernel approximate-identity argument.”

Solution. Since $\frac{1}{2\pi} \int F_N = 1$, $(f * F_N)(\theta) - f(\theta) = \frac{1}{2\pi} \int [f(\theta - t) - f(\theta)] F_N(t) dt$. Split into $|t| < \delta$ and $|t| \geq \delta$. On the first, uniform continuity makes $|f(\theta - t) - f(\theta)| < \varepsilon$, and $F_N \geq 0$ has mass ≤ 1 there, contributing $< \varepsilon$. On the second, $|f(\theta - t) - f(\theta)| \leq 2 \sup |f|$ while $\int_{|t| \geq \delta} F_N \rightarrow 0$ by property (c). Hence $\|f * F_N - f\|_\infty \rightarrow 0$. *Check.* $F_N \geq 0$ was decisive – the same argument fails for D_N because $\int |D_N| \rightarrow \infty$.

Worked Example 33.2.14 (Riemann localization in action). Let $f(\theta) = \theta$ on $(-\pi, \pi)$ and let g agree with f on $(-1, 1)$ but be defined arbitrarily (integrably) outside. Compare $S_N(f)(0)$ and $S_N(g)(0)$.

Recognition. “Localization: agreement on a neighborhood forces the same pointwise limit.”

Solution. By Theorem 33.2.8, $S_N(f)(0) - S_N(g)(0) \rightarrow 0$, so they share a limit. Since f is odd, $S_N(f)(0) = 0$ for all N , giving $S_N(g)(0) \rightarrow 0$ regardless of g 's wild behavior on $|\theta| \geq 1$. *Check.* The coefficients $\widehat{g}(n)$ genuinely depend on all of g , yet the convergence at 0 does not – the surprising content of the principle.

Worked Example 33.2.15 (Abel means via the Poisson kernel). Show that the Abel mean $A_r(f)(\theta) = \sum_n r^{|n|} a_n e^{in\theta}$ equals $(f * P_r)(\theta)$, and conclude $A_r(f) \rightarrow f$ at points of continuity as $r \rightarrow 1^-$.

Recognition. “Recognize Abel summation as convolution with the Poisson good kernel.”

Solution. $A_r(f)(\theta) = \sum_n r^{|n|} (\frac{1}{2\pi} \int f(t) e^{-int} dt) e^{in\theta} = \frac{1}{2\pi} \int f(t) \sum_n r^{|n|} e^{in(\theta-t)} dt = (f * P_r)(\theta)$, the sum being $P_r(\theta - t)$. Since P_r is a good kernel, the approximate-identity argument gives $A_r(f) \rightarrow f$ at continuity points (uniformly if f is continuous). *Check.* This is the Abel-summability analogue of Fejér's Cesàro result; both succeed precisely because their kernels are nonnegative.

Worked Example 33.2.16 (Decay rate from smoothness: $C^k \implies \widehat{f}(n) = o(n^{-k})$). Show that if $f \in C^k(\mathbb{T})$ then $|n|^k \widehat{f}(n) \rightarrow 0$ (Stein Ex. 13).

Recognition. “Each derivative pulls out a factor in ; Riemann–Lebesgue kills the remainder.”

Solution. Integrating by parts k times (boundary terms cancel by periodicity), $\widehat{f^{(k)}}(n) = (in)^k \widehat{f}(n)$, so $|n|^k |\widehat{f}(n)| = |\widehat{f^{(k)}}(n)|$. As $f^{(k)}$ is continuous, hence integrable, Riemann–Lebesgue (Theorem 33.1.15) gives

$\widehat{f^{(k)}}(n) \rightarrow 0$. Thus $|n|^k \widehat{f}(n) \rightarrow 0$, i.e. $\widehat{f}(n) = o(|n|^{-k})$. *Check.* For the C^∞ heat kernel coefficients $e^{-n^2 t}$ decay faster than any power, consistent with $o(|n|^{-k})$ for every k .

Exercises (Stein, Ch.3) – pointwise convergence, energy, decay.

- (Energy of a vibrating string.) The displacement $u(x, t)$ satisfies $\frac{1}{c^2} u_{tt} = u_{xx}$, $c^2 = \tau/\rho$, with $u(x, 0) = f(x)$, $u_t(x, 0) = g(x)$, $f \in C^1$, g continuous. With total energy $E(t) = \frac{1}{2}\rho \int_0^L u_t^2 dx + \frac{1}{2}\tau \int_0^L u_x^2 dx$, show $E(t)$ is conserved, so $E(t) = E(0) = \frac{1}{2}\rho \int_0^L g(x)^2 dx + \frac{1}{2}\tau \int_0^L f'(x)^2 dx$.
- (Wirtinger / Poincaré.) (a) If f is T -periodic, continuous, piecewise C^1 with $\int_0^T f = 0$, then $\int_0^T |f|^2 \leq \frac{T^2}{4\pi^2} \int_0^T |f'|^2$, equality iff $f(t) = A \sin(2\pi t/T) + B \cos(2\pi t/T)$. [Parseval.] (b) For such f and any C^1 T -periodic g , $\left| \int_0^T fg \right|^2 \leq \frac{T^2}{4\pi^2} \int_0^T |f|^2 \int_0^T |g|^2$. (c) For $f \in C^1[a, b]$ with $f(a) = f(b) = 0$, $\int_a^b |f|^2 \leq \frac{(b-a)^2}{\pi^2} \int_a^b |f'|^2$; discuss equality and show the constant is best possible. [Extend f oddly about a , period $2(b-a)$, apply (a); equality iff $f(t) = A \sin(\pi \frac{t-a}{b-a})$.]
- Prove $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$. [Start from $\int D_N = 2\pi$; note $\frac{1}{\sin(\theta/2)} - \frac{2}{\theta}$ is continuous on $[-\pi, \pi]$; apply Riemann–Lebesgue.]
- If f is periodic of class C^k , show $\widehat{f}(n) = o(1/|n|^k)$, i.e. $|n|^k \widehat{f}(n) \rightarrow 0$. [Riemann–Lebesgue after k integrations by parts.]
- Prove the Fourier series of a continuously differentiable f on the circle is absolutely convergent. [Cauchy–Schwarz with Parseval for f' .]
- Let f be 2π -periodic, Riemann integrable on $[-\pi, \pi]$. (a) Show $\widehat{f}(n) = -\frac{1}{2\pi} \int_{-\pi}^\pi f(x + \pi/n) e^{-inx} dx$, hence $\widehat{f}(n) = \frac{1}{4\pi} \int_{-\pi}^\pi [f(x) - f(x + \pi/n)] e^{-inx} dx$. (b) If f is Hölder- α ($|f(x+h) - f(x)| \leq C|h|^\alpha$, $0 < \alpha \leq 1$), deduce $\widehat{f}(n) = O(1/|n|^\alpha)$. (c) Show this is sharp: $f(x) = \sum_{k \geq 0} 2^{-k\alpha} e^{i2^k x}$ ($0 < \alpha < 1$) is Hölder- α with $\widehat{f}(N) = 1/N^\alpha$ at $N = 2^k$. [Split the sum at $2^k \leq 1/|h|$; use $|1 - e^{i\theta}| \leq |\theta|$ and ≤ 2 .]
- Let f be 2π -periodic, Lipschitz with constant K , so $\widehat{f}(n) = O(1/|n|)$. Prove its Fourier series converges absolutely and uniformly: (a) for $g_h(x) = f(x+h) - f(x-h)$, $\frac{1}{2\pi} \int |g_h|^2 = \sum_n 4 |\sin nh|^2 |\widehat{f}(n)|^2 \leq K^2 h^2$; (b) with $h = \pi/2^{p+1}$, $\sum_{2^{p-1} < |n| \leq 2^p} |\widehat{f}(n)|^2 \leq \frac{K^2 \pi^2}{2^{2p+1}}$; (c) estimate $\sum_{2^{p-1} < |n| \leq 2^p} |\widehat{f}(n)|$ by Cauchy–Schwarz and conclude. (d) Bernstein: if f is Hölder- α with $\alpha > 1/2$, the series converges absolutely.
- If f is bounded and monotonic on $[-\pi, \pi]$, show $\widehat{f}(n) = O(1/|n|)$. [Reduce to characteristic functions of intervals; for a step function $\sum_k \alpha_k \chi_{[a_k, a_{k+1}]}$ sum by parts to a telescoping $\sum (\alpha_{k+1} - \alpha_k)$ bounded by $2M$; approximate f .]
- (Decay summary; coefficients can decay arbitrarily slowly.) Recall: (a) $C^k \implies \widehat{f}(n) = o(|n|^{-k})$; (b) Lipschitz $\implies O(|n|^{-1})$; (c) monotonic $\implies O(|n|^{-1})$; (d) Hölder- $\alpha \implies O(|n|^{-\alpha})$; (e) integrable $\implies \sum |\widehat{f}(n)|^2 < \infty$, so $o(1)$. Show nonetheless: for any $\varepsilon_n \rightarrow 0$ ($\varepsilon_n \geq 0$) there is a continuous f with $|\widehat{f}(n)| \geq \varepsilon_n$ for infinitely many n . [Pick a subsequence with $\sum_k \varepsilon_{n_k} < \infty$.]

Problems (Stein, Ch.3) – Euler’s cotangent identities.

- Let $\alpha \in \mathbb{C}$, $\alpha \notin \mathbb{Z}$. (a) Compute the Fourier series of the 2π -periodic $f(x) = \cos(\alpha x)$ on $[-\pi, \pi]$. (b) Derive Euler’s formulas $\sum_{n \geq 1} \frac{1}{n^2 - \alpha^2} = \frac{1}{2\alpha^2} - \frac{\pi}{2\alpha \tan(\alpha\pi)}$ and, for $u \in \mathbb{C} \setminus \pi\mathbb{Z}$, $\cot u = \frac{1}{u} + 2 \sum_{n \geq 1} \frac{u}{u^2 - n^2 \pi^2}$. (c) For $\alpha \in \mathbb{C} \setminus \mathbb{Z}$, $\frac{\alpha\pi}{\sin(\alpha\pi)} = 1 + 2\alpha^2 \sum_{n \geq 1} \frac{(-1)^{n-1}}{n^2 - \alpha^2}$. (d) For $0 < \alpha < 1$, $\int_0^\infty \frac{t^{\alpha-1}}{t+1} dt = \frac{\pi}{\sin(\alpha\pi)}$. [Split $\int_0^1 + \int_1^\infty$, substitute $t = 1/u$, reduce to $\int_0^1 \frac{t^{\gamma-1}}{1+t} dt = \sum_{k \geq 0} \frac{(-1)^k}{k+\gamma}$, then use (c).]

Additional exercises.

1. Compute the Dirichlet kernel D_2 explicitly as both a sum $\sum_{|n|\leq 2} e^{in\theta}$ and the closed form $\sin(\frac{5}{2}\theta)/\sin(\frac{\theta}{2})$, and check they agree at $\theta = \pi/2$.
2. Verify $\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(\theta) d\theta = 1$ directly from the sum definition, for every N .
3. Show the Fejér kernel is the Cesàro mean of Dirichlet kernels, $F_N = \frac{1}{N+1} \sum_{n=0}^N D_n$, and confirm $\frac{1}{2\pi} \int F_N = 1$.
4. Using local convergence, prove that if f is C^1 on the circle then $S_N(f)(\theta) \rightarrow f(\theta)$ at every θ . Where exactly is differentiability used?
5. (Riemann localization.) Let $f = g$ on $(\theta_0 - \delta, \theta_0 + \delta)$ with f, g integrable. Show $\{S_N(f)(\theta_0)\}$ converges iff $\{S_N(g)(\theta_0)\}$ does, and to the same limit when it exists.
6. (NON-example.) Exhibit an integrable f and a point θ_0 at which f is continuous but not Lipschitz, so the local theorem does not directly apply; e.g. $f(\theta) = 1/\log(1/|\theta|)$ near 0. Explain why differentiability fails there.
7. Prove $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$ by following Stein's hint (Exercise 12): start from $\int D_N = 2\pi$ and use that $\frac{1}{\sin(\theta/2)} - \frac{2}{\theta}$ is continuous.
8. (Diagram.) Plot D_3 and F_3 on $[-\pi, \pi]$ on the same axes (pgfplots). Mark the negative side-lobes of D_3 and observe that $F_3 \geq 0$ everywhere; explain in one sentence why this makes F_N a good kernel and D_N not.
9. Show the Lebesgue constant $L_N = \frac{1}{2\pi} \int |D_N|$ satisfies $L_N \rightarrow \infty$, and explain its role as the operator norm of $f \mapsto S_N(f)$ on $C(\mathbb{T})$.
10. For continuous f , prove $\sigma_N(f) = f * F_N \rightarrow f$ uniformly (Fejér's theorem), and deduce the Weierstrass density of trigonometric polynomials in $C(\mathbb{T})$.

33.3 A Continuous Function with Divergent Fourier Series

Theorem 33.2.7 guarantees convergence wherever f is differentiable. Is mere *continuity* enough? It is not. We construct a continuous, 2π -periodic function whose Fourier series diverges at a point. The construction illuminates a general mechanism – *symmetry-breaking* – and exploits the $\log N$ growth of the Dirichlet kernel (Worked Example in §33.2) by stacking translated copies of the sawtooth.

Symmetry-breaking and the sawtooth

The exponentials $e^{in\theta}$ and $e^{-in\theta}$ play symmetric roles in $\sum_n a_n e^{in\theta}$, and S_N, D_N, F_N, P_r all respect that symmetry. If we *break* it – keep only the negative (or only the positive) frequencies – new phenomena appear. Start from the sawtooth f , odd, with $f(\theta) = i(\pi - \theta)$ on $(0, \pi)$, whose series is

$$f(\theta) \sim \sum_{n \neq 0} \frac{e^{in\theta}}{n}.$$

Lemma 33.3.1 (Half the sawtooth is not a Fourier series). *The one-sided series $\sum_{n=-\infty}^{-1} \frac{e^{in\theta}}{n}$ is not the Fourier series of any Riemann-integrable function.*

Proof. If it were the series of an integrable (hence bounded) $\tilde{f}$, its Abel means at 0 would be $|A_r(\tilde{f})(0)| = \sum_{n \geq 1} \frac{r^n}{n} = -\log(1-r) \rightarrow \infty$ as $r \rightarrow 1^-$. But $|A_r(\tilde{f})(0)| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |\tilde{f}| P_r \leq \sup |\tilde{f}| < \infty$ since P_r is a good kernel of mass one. Contradiction. $\square$

Lemma 33.3.2 (Tauberian boundedness). *If the Abel means $A_r = \sum_{n \geq 1} r^n c_n$ are bounded as $r \rightarrow 1^-$ and $c_n = O(1/n)$, then the partial sums $S_N = \sum_{n=1}^N c_n$ are bounded.*

Proof. Put $r = 1 - 1/N$ and pick M with $n |c_n| \leq M$ and $|A_r| \leq M$. Then

$$|S_N - A_r| \leq \sum_{n=1}^N |c_n| (1 - r^n) + \sum_{n>N} r^n |c_n| \leq M \sum_{n=1}^N (1 - r) + \frac{M}{N} \sum_{n>N} r^n,$$

using $1 - r^n = (1 - r)(1 + \dots + r^{n-1}) \leq n(1 - r)$ in the first sum and $|c_n| \leq M/n \leq M/N$ in the second. The first term is $MN(1 - r) = M$; the second is $\frac{M}{N} \cdot \frac{r^{N+1}}{1-r} \leq \frac{M}{N} \cdot N = M$. So $|S_N - A_r| \leq 2M$, giving $|S_N| \leq 3M$. $\square$

Corollary 33.3.3. *The truncated symmetric sawtooth $f_N(\theta) = \sum_{1 \leq |n| \leq N} \frac{e^{in\theta}}{n}$ is uniformly bounded in N and θ . Meanwhile the one-sided $\tilde{f}_N(\theta) = \sum_{-N \leq n \leq -1} \frac{e^{in\theta}}{n}$ satisfies $|\tilde{f}_N(0)| \geq c \log N$.*

Proof. For f_N : the full sawtooth series has $c_n = O(1/|n|)$ and Abel means $f * P_r$ bounded (good kernel, f bounded), so Lemma 33.3.2 applies symmetrically and f_N is uniformly bounded. For $\tilde{f}_N(0) = \sum_{n=1}^N \frac{1}{-n}$ up to sign, the harmonic comparison $\sum_{n=1}^N \frac{1}{n} \geq \int_1^N \frac{dx}{x} = \log N$ gives $|\tilde{f}_N(0)| \geq \log N$. $\square$

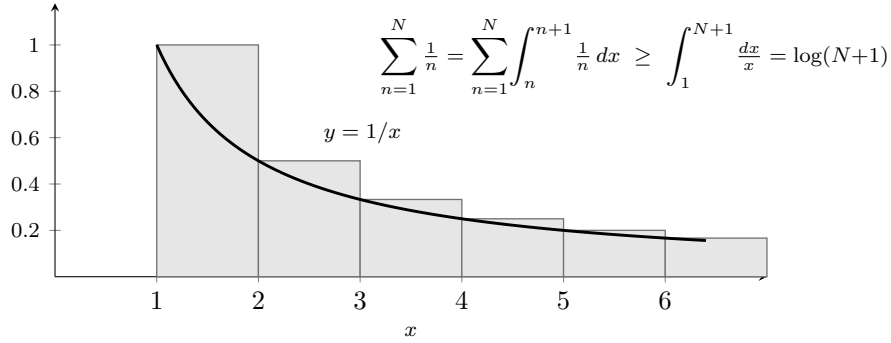


Figure 33.10. Comparing the harmonic sum with the area under $1/x$: the step $\frac{1}{n}$ on $[n, n+1]$ dominates $1/x$ there, so $\sum_{n \leq N} \frac{1}{n} \geq \log N$, forcing $|\tilde{f}_N(0)| \geq \log N$.

Displacing frequencies: the building block

Definition 33.3.4. Displace the frequencies of $f_N, \tilde{f}_N$ upward by $2N$: $P_N(\theta) = e^{i(2N)\theta} f_N(\theta)$ and $\tilde{P}_N(\theta) = e^{i(2N)\theta} \tilde{f}_N(\theta)$. Then P_N is a trigonometric polynomial with nonzero coefficients on $N \leq n \leq 3N$ (except $n = 2N$), centered at $n = 2N$; $\tilde{P}_N$ on $N \leq n \leq 2N - 1$.

Lemma 33.3.5 (Symmetry-breaking by the partial-sum operator).

$$S_M(P_N) = \begin{cases} P_N & M \geq 3N, \\ \tilde{P}_N & M = 2N, \\ 0 & M < N. \end{cases}$$

Proof. P_N has its spectrum in $[N, 3N]$. If $M \geq 3N$, S_M keeps all of it: $S_M(P_N) = P_N$. If $M < N$, S_M discards all of it: $S_M(P_N) = 0$. If $M = 2N$, S_M keeps exactly the frequencies $\leq 2N$, i.e. the lower half $N \leq n \leq 2N - 1$ – which is precisely the displaced *one-sided* sawtooth $\tilde{P}_N$. Thus at $M = 2N$ the operator breaks the symmetry. $\square$

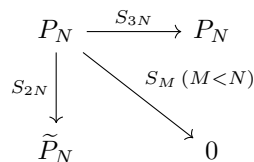


Figure 33.11. The partial-sum operator on the building block P_N (spectrum in $[N, 3N]$, Lemma 33.3.5): S_M keeps all of it ($M \geq 3N$, top), discards all of it ($M < N$, diagonal), or – at the critical $M = 2N$ – retains only the lower half, the unbounded one-sided block $\tilde{P}_N$.

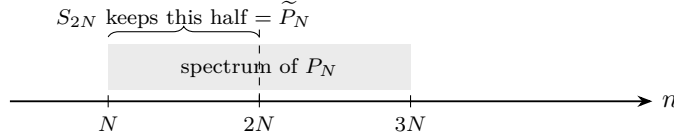


Figure 33.12. At $M = 2N$ the partial sum retains only the lower half of the spectrum of P_N , producing the unbounded $\tilde{P}_N$. (Schematic.)

Assembling the counterexample

Theorem 33.3.6 (A continuous function with divergent Fourier series). *There is a continuous 2π -periodic f whose Fourier series diverges at $\theta = 0$.*

Proof. Choose a convergent series of positive terms α_k and integers N_k growing fast, with (i) $N_{k+1} > 3N_k$ and (ii) $\alpha_k \log N_k \rightarrow \infty$. For instance $\alpha_k = 1/k^2$ and $N_k = 3^{2^k}$. Define

$$f(\theta) = \sum_{k=1}^{\infty} \alpha_k P_{N_k}(\theta).$$

Since $|P_N(\theta)| = |f_N(\theta)|$ is uniformly bounded (Corollary 33.3.3) and $\sum \alpha_k < \infty$, the series converges *uniformly*, so f is continuous and periodic.

Now examine $S_{2N_m}(f)(0)$. By condition (i) the spectra of the P_{N_k} are disjoint blocks, so S_{2N_m} acts on each term by Lemma 33.3.5: for $k < m$ it returns the whole P_{N_k} (a bounded term), for $k > m$ it returns 0, and for $k = m$ it returns $\tilde{P}_{N_m}$. Hence

$$|S_{2N_m}(f)(0)| \geq \alpha_m \left| \tilde{P}_{N_m}(0) \right| - O(1) \geq c \alpha_m \log N_m - O(1) \rightarrow \infty,$$

using $\left| \tilde{P}_N(0) \right| = \left| \tilde{f}_N(0) \right| \geq c \log N$ and condition (ii). So the partial sums at 0 are unbounded and the Fourier series of f diverges at 0. For any other point θ_0 , apply the construction to $f(\theta - \theta_0)$. $\square$

The takeaway: differentiability in Theorem 33.2.7 is essential and cannot be relaxed to continuity. The unbounded Lebesgue constants of D_N are the precise analytic obstruction, made visible by the symmetry-breaking construction.

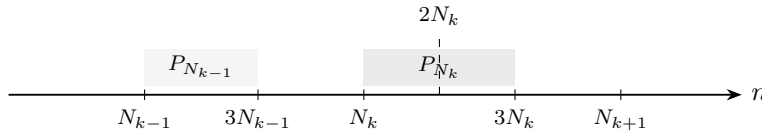


Figure 33.13. Disjoint frequency blocks $[N_k, 3N_k]$ with centers $2N_k$; symmetry is broken in the middle of each block. (Schematic.)

Worked Example 33.3.7 (Why uniform boundedness of P_N forces continuity of f). Explain precisely how $|P_N(\theta)| \leq B$ (uniform) and $\sum \alpha_k < \infty$ give a continuous limit.

Recognition. “Weierstrass M -test for the assembled series.”

Solution. Each term satisfies $|\alpha_k P_{N_k}(\theta)| \leq \alpha_k B =: M_k$ with $\sum_k M_k = B \sum_k \alpha_k < \infty$. By the M -test the series converges uniformly; a uniform limit of continuous (indeed trigonometric-polynomial) terms is continuous. *Check.* Uniform boundedness of P_N came from the Tauberian Lemma 33.3.2; if instead the blocks were the *one-sided* $\tilde{P}_N$ (unbounded at 0) the M -test would fail, as it must.

Worked Example 33.3.8 (Lacunary growth conditions are satisfiable). Verify $\alpha_k = 1/k^2$, $N_k = 3^{2^k}$ satisfy $N_{k+1} > 3N_k$ and $\alpha_k \log N_k \rightarrow \infty$.

Recognition. “Check the two design constraints of the construction.”

Solution. $N_{k+1}/N_k = 3^{2^{k+1}-2^k} = 3^{2^k} \gg 3$, so (i) holds with room to spare. And $\log N_k = 2^k \log 3$, so $\alpha_k \log N_k = \frac{2^k \log 3}{k^2} \rightarrow \infty$ since exponential beats polynomial. *Check.* Any pair with N_k growing faster than geometrically and α_k decaying only polynomially works; the tension is that α_k must be summable yet $\alpha_k \log N_k$ unbounded.

Worked Example 33.3.9 (The conjugate Dirichlet kernel). Define $\tilde{D}_N(\theta) = \sum_{|n| \leq N} \text{sign}(n) e^{in\theta}$ and derive $\tilde{D}_N = \frac{\cos(\theta/2) - \cos((N + \frac{1}{2})\theta)}{\sin(\theta/2)}$.

Recognition. “The odd (sign-weighted) analogue of D_N governs one-sided sums.”

Solution. $\tilde{D}_N = \sum_{n=1}^N (e^{in\theta} - e^{-in\theta}) = 2i \sum_{n=1}^N \sin n\theta$. Summing the sine series, $\sum_{n=1}^N \sin n\theta = \frac{\cos(\theta/2) - \cos((N + \frac{1}{2})\theta)}{2 \sin(\theta/2)}$, hence the stated formula (times $2i$, absorbed by convention). *Check.* Like D_N its L^1 norm grows like $\log N$ – this is the kernel behind $\sum_{n \geq 1} \sin nx/n^\alpha$ failing to be a Fourier series (Problem 1).

Worked Example 33.3.10 (Gibbs overshoot of the sawtooth). For the sawtooth $f(x) = (\pi - x)/2$ on $(0, 2\pi)$, show the partial sums overshoot the jump by a factor $\int_0^\pi \frac{\sin t}{t} dt - \frac{\pi}{2} \approx 0.0895\pi$, about 9%.

Recognition. “Locate the first maximum of $S_N(f)$ near the jump and integrate the sine kernel.”

Solution. Using $S_N(f)(x) = \frac{1}{2} \int_0^x (D_N(t) - 1) dt$ (Stein Ex. 19), differentiate: the first maximum on $(0, \pi/N)$ occurs near $x = \pi/N$, where $\max S_N(f) - \frac{\pi}{2} \rightarrow \int_0^\pi \frac{\sin t}{t} dt - \frac{\pi}{2}$. Numerically the sine integral is ≈ 1.8519 , and $\frac{\pi}{2} \approx 1.5708$, giving overshoot $\approx 0.281 \approx 0.0895\pi$, i.e. about 9% of the jump π . *Check.* The overshoot does not shrink as $N \rightarrow \infty$ – it only narrows – which is the Gibbs phenomenon.

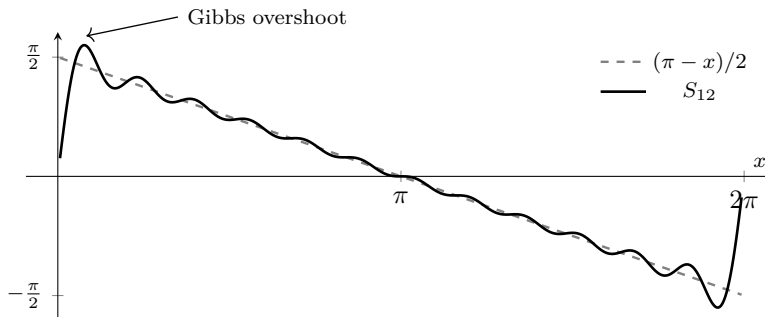


Figure 33.14. Partial sum S_{12} of the sawtooth (dashed target): the $\sim 9\%$ Gibbs overshoot at the jump $x = 0$ (and $x = 2\pi$) persists as N grows; only its width shrinks.

Worked Example 33.3.11 (Localization explains the counterexample’s subtlety). Reconcile Theorem 33.3.6 (divergence at 0) with Riemann localization (Theorem 33.2.8).

Recognition. “Localization constrains *relative* behavior, not absolute convergence.”

Solution. Localization says two functions agreeing near θ_0 have the *same* convergence/divergence behavior there; it never asserts convergence. Our f is its own witness: it is continuous everywhere, yet diverges at 0. Any g equal to f near 0 will *also* diverge at 0 – consistent with, not contradicted by, localization. *Check.* Differentiability (not continuity) was the hypothesis that *guaranteed* convergence; localization only transfers whatever behavior already holds.

Exercises (Stein, Ch.3) – one-sided sums and Gibbs.

1. Give another proof that $\sum_{0 < |n| \leq N} e^{inx}/n$ is uniformly bounded in N and $x \in [-\pi, \pi]$, using $\frac{1}{2i} \sum_{0 < |n| \leq N} \frac{e^{inx}}{n} = \sum_{n=1}^N \frac{\sin nx}{n} = \frac{1}{2} \int_0^x (D_N(t) - 1) dt$ and the fact (Ex. 3 above) that $\int_0^\infty \frac{\sin t}{t} dt < \infty$.

2. (Gibbs phenomenon.) Let $f(x) = (\pi - x)/2$ on $(0, 2\pi)$, $f(0) = 0$, extended periodically, with Fourier series $\sum_{n \geq 1} \frac{\sin nx}{n}$ and jump $f(0^+) - f(0^-) = \pi$. Show $\max_{0 < x \leq \pi/N} S_N(f)(x) - \frac{\pi}{2} = \int_0^\pi \frac{\sin t}{t} dt - \frac{\pi}{2}$, roughly 9% of the jump π . [Use the integral expression for $S_N(f)$ from the previous exercise.]

Problems (Stein, Ch.3) – conjugate kernel and Bernoulli polynomials.

1. (A series that is not a Fourier series.) For $0 < \alpha < 1$, $\sum_{n \geq 1} \frac{\sin nx}{n^\alpha}$ converges for every x but is not the Fourier series of a Riemann integrable function. (a) For the conjugate Dirichlet kernel $\tilde{D}_N(x) = \sum_{|n| \leq N} \text{sign}(n)e^{inx}$, show $\tilde{D}_N(x) = \frac{\cos(x/2) - \cos((N + \frac{1}{2})x)}{\sin(x/2)}$ and $\int_{-\pi}^\pi |\tilde{D}_N(x)| dx \leq c \log N$. (b) Hence if f is Riemann integrable, $(f * \tilde{D}_N)(0) = O(\log N)$. (c) For the given series this forces $\sum_{n=1}^N \frac{1}{n^\alpha} = O(\log N)$, a contradiction.
2. (Bernoulli polynomials.) Define $B_n(x)$ by $\frac{ze^{xz}}{e^z - 1} = \sum_{n \geq 0} \frac{B_n(x)}{n!} z^n$. (a) Show $B_n(x) = \sum_{k=0}^n \binom{n}{k} B_k x^{n-k}$ is a polynomial, with $B_0(x) = 1$, $B_1(x) = x - \frac{1}{2}$, $B_2(x) = x^2 - x + \frac{1}{6}$, $B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x$. (b) For $n \geq 1$, $B_n(x+1) - B_n(x) = nx^{n-1}$; for $n \geq 2$, $B_n(0) = B_n(1) = B_n$. (c) With $S_m(n) = 1^m + \cdots + (n-1)^m$, show $(m+1)S_m(n) = B_{m+1}(n) - B_{m+1}$. (d) The $B_n(x)$ are the only polynomials with $B_0 = 1$, $B'_n(x) = nB_{n-1}(x)$, $\int_0^1 B_n = 0$ ($n \geq 1$); deduce $\int_x^{x+1} B_n(t) dt = x^n$. (e) Compute the Fourier series of $B_1(x)$ to get $x - \frac{1}{2} = -\frac{1}{\pi} \sum_{k \geq 1} \frac{\sin(2\pi kx)}{k}$ on $(0, 1)$; integrating, $B_{2n}(x) = (-1)^{n+1} \frac{2(2n)!}{(2\pi)^{2n}} \sum_{k \geq 1} \frac{\cos(2\pi kx)}{k^{2n}}$, $B_{2n+1}(x) = (-1)^{n+1} \frac{2(2n+1)!}{(2\pi)^{2n+1}} \sum_{k \geq 1} \frac{\sin(2\pi kx)}{k^{2n+1}}$, and for $0 < x < 1$, $B_n(x) = -\frac{n!}{(2\pi i)^n} \sum_{k \neq 0} \frac{e^{2\pi i kx}}{k^n}$. (So the B_n are, up to normalization, successive integrals of the sawtooth.)

Additional exercises.

1. Verify that $f_N(\theta) = \sum_{1 \leq |n| \leq N} e^{in\theta}/n$ is real-valued and odd, and write it as $2i \sum_{n=1}^N \frac{\sin n\theta}{n}$.
2. Show $|\tilde{f}_N(0)| = \sum_{n=1}^N 1/n$ and use the integral comparison to conclude $|\tilde{f}_N(0)| \geq \log N$.
3. Confirm that with $N_k = 3^{2^k}$ one has $N_{k+1} > 3N_k$ for all $k \geq 1$, and that $\alpha_k = 1/k^2$ gives $\sum_k \alpha_k < \infty$ while $\alpha_k \log N_k \rightarrow \infty$.
4. Using $S_M(P_N)$ from Lemma 33.3.5, compute $S_{2N}(P_N)(0)$ in terms of $\tilde{f}_N(0)$ and bound it below by $c \log N$.
5. (NON-example.) Show that the differentiability hypothesis in the local theorem cannot be dropped by citing the function of Theorem 33.3.6: a continuous function whose Fourier series diverges at a point. State precisely which hypothesis fails.
6. Derive the conjugate Dirichlet kernel $\tilde{D}_N(x) = \frac{\cos(x/2) - \cos((N + \frac{1}{2})x)}{\sin(x/2)}$ from $\tilde{D}_N = \sum_{|n| \leq N} \text{sign}(n)e^{inx}$.
7. (Gibbs.) For the sawtooth $f(x) = (\pi - x)/2$, show numerically that $\int_0^\pi \frac{\sin t}{t} dt \approx 1.8519$ and hence the overshoot $\int_0^\pi \frac{\sin t}{t} dt - \frac{\pi}{2} \approx 0.281$ is about 9% of the jump π .
8. (Diagram.) Plot the partial sums S_4 and S_{12} of the sawtooth on $(0, 2\pi)$ against the target $(\pi - x)/2$ (pgfplots). Mark the persistent overshoot near $x = 0$ and note that it narrows but does not shrink as N grows.
9. Explain in your own words how symmetry-breaking ($\sum_{n < 0}$ vs $\sum_{n > 0}$) converts the bounded f_N into the unbounded $\tilde{f}_N$, and why displacement by $2N$ is the key trick.
10. Prove that translating the counterexample, $f(\theta - \theta_0)$, yields a continuous function whose Fourier series diverges at the prescribed point θ_0 .

Chapter 34

Some Applications of Fourier Series

34.1 The Isoperimetric Inequality

Among all simple closed curves of a given length, which encloses the largest area? Experiment suggests the circle, and indeed it is the answer; but turning the heuristic “make the curve as round as possible” into a theorem is delicate. The decisive tool is not geometry but *Parseval’s identity*: writing the coordinate functions $x(s), y(s)$ of an arc-length parametrization as Fourier series, both the length constraint and the enclosed area become quadratic expressions in the Fourier coefficients, and the inequality $A \leq \ell^2/4\pi$ drops out by comparing $|n|$ with $|n|^2$.

Curves, length, and area

Definition 34.1.1. A *parametrized curve* is a map $\gamma : [a, b] \rightarrow \mathbb{R}^2$, $\gamma(s) = (x(s), y(s))$; its image is the *curve* Γ . We call Γ *simple* if γ is injective on $[a, b]$, and *closed* if $\gamma(a) = \gamma(b)$ (so γ extends to a periodic function on $\mathbb{R}$ of period $b - a$, i.e. a function on a circle). We assume γ is of class C^1 with $\gamma'(s) \neq 0$ for all s , so Γ has a continuously varying tangent. The parameter induces an *orientation* as s runs from a to b .

A reparametrization $\eta(t) = \gamma(s(t))$ by a C^1 bijection $s : [c, d] \rightarrow [a, b]$ produces the same curve; if $s'(t) > 0$ throughout, η and γ are *equivalent* (same orientation), and if $s'(t) < 0$ the orientation reverses. Whether Γ is simple or closed is independent of parametrization.

Definition 34.1.2. The *length* of Γ parametrized by γ is

$$\ell = \int_a^b |\gamma'(s)| \, ds = \int_a^b (x'(s)^2 + y'(s)^2)^{1/2} \, ds.$$

We say γ is a *parametrization by arc-length* if $|\gamma'(s)| = 1$ for all s ; then Γ has length $b - a$, and after a translation the domain may be taken to be $[0, \ell]$.

Length is intrinsic. If $\gamma(s(t)) = \eta(t)$, the change of variables formula and chain rule give $\int_a^b |\gamma'(s)| \, ds = \int_c^d |\gamma'(s(t))| |s'(t)| \, dt = \int_c^d |\eta'(t)| \, dt$, so the length does not depend on the chosen parametrization. Every curve admits an arc-length parametrization (run the cumulative-length function $h(s) = \int_a^s |\eta'|$ backward).

Definition 34.1.3. For a simple closed C^1 curve Γ parametrized on $[a, b]$, the *area* of the enclosed region is

$$A = \frac{1}{2} \left| \oint_{\Gamma} (x \, dy - y \, dx) \right| = \frac{1}{2} \left| \int_a^b (x(s)y'(s) - y(s)x'(s)) \, ds \right|.$$

This is the elementary calculus area formula (a special case of Green’s theorem); Stein takes it as the *definition* of area, which makes the proof below short and clean. The deeper questions — defining the enclosed region purely geometrically and extending to merely rectifiable curves — are deferred.

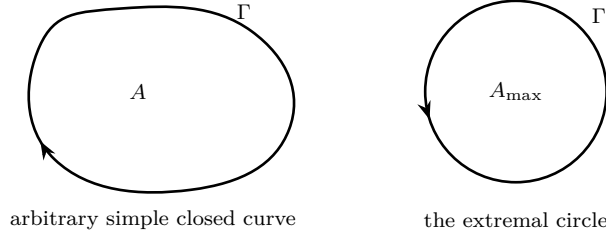


Figure 34.1. The isoperimetric problem: among all simple closed curves of length ℓ , the circle (right) maximizes the enclosed area A . (Schematic.)

Statement and proof

Theorem 34.1.4 (Isoperimetric inequality). *If Γ is a simple closed curve in $\mathbb{R}^2$ of length ℓ , and A is the area it encloses, then*

$$A \leq \frac{\ell^2}{4\pi},$$

with equality if and only if Γ is a circle.

Proof. Rescaling. The map $(x, y) \mapsto (\delta x, \delta y)$ multiplies length by δ and area by δ^2 , so the ratio A/ℓ^2 is scale-invariant. Choosing $\delta = 2\pi/\ell$ reduces the problem to: if $\ell = 2\pi$ then $A \leq \pi$, with equality iff Γ is a circle.

Setup. Let $\gamma : [0, 2\pi] \rightarrow \mathbb{R}^2$, $\gamma(s) = (x(s), y(s))$, be an arc-length parametrization, so $x'(s)^2 + y'(s)^2 = 1$ for all s , and hence

$$\frac{1}{2\pi} \int_0^{2\pi} (x'(s)^2 + y'(s)^2) ds = 1. \quad (34.1)$$

Because Γ is closed, x and y are 2π -periodic; expand $x(s) \sim \sum_n a_n e^{ins}$ and $y(s) \sim \sum_n b_n e^{ins}$, whence (differentiating coefficientwise) $x'(s) \sim \sum_n a_n in e^{ins}$ and $y'(s) \sim \sum_n b_n in e^{ins}$. Parseval's identity applied to (34.1) gives

$$\sum_{n=-\infty}^{\infty} |n|^2 (|a_n|^2 + |b_n|^2) = 1. \quad (34.2)$$

The area. Since x, y are real, $a_{-n} = \overline{a_n}$ and $b_{-n} = \overline{b_n}$. The bilinear (polarized) form of Parseval applied to the area integral gives

$$A = \frac{1}{2} \left| \int_0^{2\pi} (xy' - yx') ds \right| = \pi \left| \sum_{n=-\infty}^{\infty} n(a_n \overline{b_n} - b_n \overline{a_n}) \right|.$$

Now $|a_n \overline{b_n} - b_n \overline{a_n}| \leq 2|a_n||b_n| \leq |a_n|^2 + |b_n|^2$, and $|n| \leq |n|^2$, so by (34.2),

$$A \leq \pi \sum_{n=-\infty}^{\infty} |n| (|a_n|^2 + |b_n|^2) \leq \pi \sum_{n=-\infty}^{\infty} |n|^2 (|a_n|^2 + |b_n|^2) = \pi.$$

Equality. If $A = \pi$ then both inequalities are tight. Since $|n| < |n|^2$ for $|n| \geq 2$, equality forces $a_n = b_n = 0$ for $|n| \geq 2$, so $x(s) = a_{-1}e^{-is} + a_0 + a_1e^{is}$ and $y(s) = b_{-1}e^{-is} + b_0 + b_1e^{is}$. Reality gives $a_{-1} = \overline{a_1}$, $b_{-1} = \overline{b_1}$; (34.2) becomes $2(|a_1|^2 + |b_1|^2) = 1$, and equality in the first estimate forces $|a_1| = |b_1| = \frac{1}{2}$. Writing $a_1 = \frac{1}{2}e^{i\alpha}$, $b_1 = \frac{1}{2}e^{i\beta}$, the condition $1 = 2|a_1 \overline{b_1} - \overline{a_1} b_1|$ gives $|\sin(\alpha - \beta)| = 1$, so $\alpha - \beta = k\pi/2$ with k odd. Then $x(s) = a_0 + \cos(\alpha + s)$ and $y(s) = b_0 \pm \sin(\alpha + s)$, the parametrization of a circle. $\square$

The whole proof rests on the gap between $|n|$ and $|n|^2$: the length constraint weights each mode by $|n|^2$ while the area only sees $|n|$, so to spend the budget efficiently the curve must put all of its energy in the fundamental modes $n = \pm 1$ — which is exactly a circle.

The engine is the Fourier coefficient map sending a periodic function to its sequence of coefficients, under which differentiation becomes multiplication by in ; Parseval makes this map an isometry, which is why an L^2 constraint on γ' becomes a clean weighted sum over coefficients.

$$\begin{array}{ccc}
x & \xrightarrow[\sim]{\wedge} & (a_n)_{n \in \mathbb{Z}} \\
\downarrow d/ds & & \downarrow n \mapsto in a_n \\
x' & \xrightarrow[\wedge]{\sim} & (in a_n)_{n \in \mathbb{Z}}
\end{array}$$

Figure 34.2. The Fourier coefficient map $x \mapsto (a_n)$ (an L^2 -isometry by Parseval) intertwines differentiation d/ds with multiplication by in ; this is what turns the arc-length constraint on x', y' into the weighted coefficient sum (34.2).

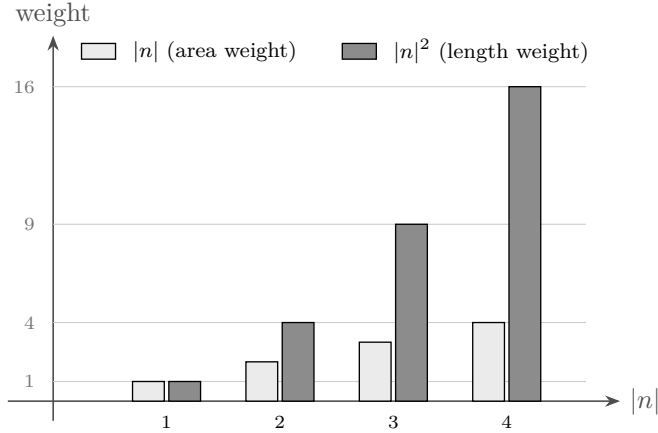


Figure 34.3. Length penalizes high modes by $|n|^2$, area only by $|n|$; the gap (zero only at $|n| = 1$) is why the optimum lives in the $n = \pm 1$ modes.

Method (Extremal problems via Parseval). *Recognition trigger.* A geometric or analytic quantity must be bounded subject to a constraint, and both quantity and constraint are integrals of periodic functions and their derivatives.

1. Parametrize so the constraint is a clean L^2 statement (e.g. arc-length: $\int (x'^2 + y'^2) = 2\pi$).
2. Expand the relevant functions in Fourier series; turn each integral into a sum over coefficients via Parseval / its polarized (bilinear) form.
3. Compare the two coefficient sums termwise (here $|n| \leq |n|^2$) to get the inequality.
4. Track when each termwise estimate is an equality to identify the extremizer.

Worked Example 34.1.5 (Parseval for the arc-length constraint). Verify directly that (34.1) yields (34.2).

Recognition. Parseval converts an L^2 norm of a derivative into a weighted coefficient sum.

Solution. If $x \sim \sum a_n e^{ins}$ then $x' \sim \sum (ina_n) e^{ins}$, so by Parseval $\frac{1}{2\pi} \int_0^{2\pi} |x'|^2 = \sum_n |ina_n|^2 = \sum_n n^2 |a_n|^2$. Likewise $\frac{1}{2\pi} \int_0^{2\pi} |y'|^2 = \sum_n n^2 |b_n|^2$. Adding and using (34.1) gives $\sum_n n^2 (|a_n|^2 + |b_n|^2) = 1$. *Check.* For the unit circle $x = \cos s, y = \sin s$ we have $a_{\pm 1} = \frac{1}{2}, b_1 = \frac{1}{2i}, b_{-1} = -\frac{1}{2i}$; then $1^2(\frac{1}{4} + \frac{1}{4}) + 1^2(\frac{1}{4} + \frac{1}{4}) = 1$. ■

Worked Example 34.1.6 (The circle attains equality). Confirm $A = \ell^2/4\pi$ for a circle of radius R .

Recognition. Plug the extremizer into the inequality.

Solution. A circle of radius R has $\ell = 2\pi R$ and $A = \pi R^2$. Then $\ell^2/4\pi = (2\pi R)^2/4\pi = \pi R^2 = A$. *Check.* The ratio $A/\ell^2 = 1/4\pi \approx 0.0796$ is the maximum possible. ■

Worked Example 34.1.7 (A square is strictly sub-optimal). Compute A/ℓ^2 for a square and compare with $1/4\pi$.

Recognition. A non-example: any non-circle gives strict inequality.

Solution. A square of side a has $\ell = 4a$ and $A = a^2$, so $A/\ell^2 = a^2/16a^2 = 1/16 = 0.0625$, strictly below $1/4\pi \approx 0.0796$. Equivalently $A = \ell^2/16 < \ell^2/4\pi$. *Check.* The “isoperimetric deficit” $\ell^2/4\pi - A = (\frac{1}{4\pi} - \frac{1}{16})\ell^2 > 0$ since $4\pi < 16$. ■

Worked Example 34.1.8 (The area formula on an ellipse). Use Definition 34.1.3 to recover the area of the ellipse $x = p \cos s$, $y = q \sin s$.

Recognition. Direct application of the shoelace integral.

Solution. $xy' - yx' = (p \cos s)(q \cos s) - (q \sin s)(-p \sin s) = pq(\cos^2 s + \sin^2 s) = pq$. So $A = \frac{1}{2} \left| \int_0^{2\pi} pq \, ds \right| = \frac{1}{2} \cdot 2\pi pq = \pi pq$. *Check.* When $p = q = R$ this is πR^2 , the circle. ■

Worked Example 34.1.9 (An equilateral triangle). Find A/ℓ^2 for an equilateral triangle and verify it is below the bound.

Recognition. Another non-example, less round than the square.

Solution. Side a : $\ell = 3a$, $A = \frac{\sqrt{3}}{4}a^2$, so $A/\ell^2 = \frac{\sqrt{3}}{4}/9 = \sqrt{3}/36 \approx 0.0481$. This is below the square's 0.0625 and far below $1/4\pi$. *Check.* Fewer, sharper corners mean the triangle wastes more perimeter, consistent with “roundness” maximizing area. ■

Worked Example 34.1.10 (A perturbed circle loses area). Take $\gamma(s) = ((1 + \varepsilon \cos 2s) \cos s, (1 + \varepsilon \cos 2s) \sin s)$ for small ε ; show A decreases to second order in ε when length is held fixed.

Recognition. Illustrate that the circle is a strict maximum, not just a critical point, by adding a single higher mode.

Solution. The radial perturbation $\varepsilon \cos 2s$ injects $n = \pm 2, \pm 3$ Fourier content into x, y . Because length weights mode $|n|$ by $|n|^2$ but area only by $|n|$, transferring any energy from $|n| = 1$ to $|n| \geq 2$ at fixed length $\sum |n|^2(|a_n|^2 + |b_n|^2)$ strictly lowers $\sum |n|(\cdots) \geq A/\pi$. Hence $A < \pi$ to order ε^2 . *Check.* The deficit is governed by $(|n|^2 - |n|) = |n|(|n| - 1) > 0$ for $|n| \geq 2$, so any admixture of higher modes is penalized. ■

Worked Example 34.1.11 (Fourier coefficients of a sawtooth and a Parseval sum). Compute the Fourier coefficients of the 2π -periodic sawtooth $g(x) = x$ on $(-\pi, \pi)$ and use Parseval to evaluate $\sum_{n \geq 1} 1/n^2$.

Recognition. A standard coefficient computation feeding Parseval — the same machine that drives the isoperimetric proof.

Solution. $\hat{g}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} x e^{-inx} \, dx = \frac{(-1)^{n+1}}{in}$ for $n \neq 0$ (integration by parts), and $\hat{g}(0) = 0$. Parseval gives $\frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 \, dx = \sum_{n \neq 0} \frac{1}{n^2} = 2 \sum_{n \geq 1} \frac{1}{n^2}$. The left side is $\frac{1}{2\pi} \cdot \frac{2\pi^3}{3} = \frac{\pi^2}{3}$, so $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$. *Check.* The classical Basel value. ■

Worked Example 34.1.12 (Wirtinger's inequality on a single mode). Verify Wirtinger $\int_0^{2\pi} |f|^2 \leq \int_0^{2\pi} |f'|^2$ for $f(t) = \sin t$ and show it is sharp there.

Recognition. Wirtinger is the analytic shadow of the isoperimetric inequality (Stein, Exercise 4).

Solution. $\int_0^{2\pi} \sin^2 t \, dt = \pi$ and $\int_0^{2\pi} \cos^2 t \, dt = \pi$, so both sides equal π : equality holds, consistent with the extremal case $f = A \sin t + B \cos t$. *Check.* For $f(t) = \sin 2t$ instead, $\int |f|^2 = \pi$ but $\int |f'|^2 = \int 4 \cos^2 2t = 4\pi$, strict — higher modes are penalized, exactly as in the isoperimetric proof. ■

Caution. The proof uses the *signed* shoelace area; the absolute value in Definition 34.1.3 only fixes orientation. Reversing the orientation of γ flips the sign of $\int (x \, dy - y \, dx)$ but not A . Also, arc-length parametrization is essential: the clean constraint (34.1) is what makes the coefficient comparison work; a generic parametrization gives no such identity.

Exercises (Stein, Ch.4) – isoperimetric.

- Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ parametrize a closed curve Γ . (a) Prove γ is a parametrization by arc-length iff the length of the curve from $\gamma(a)$ to $\gamma(s)$ is exactly $s - a$, i.e. $\int_a^s |\gamma'(t)| \, dt = s - a$. (b) Prove any Γ admits an arc-length parametrization. [Hint: if η is any parametrization, set $h(s) = \int_a^s |\eta'(t)| \, dt$ and take $\gamma = \eta \circ h^{-1}$.]
- Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$, $\gamma(t) = (x(t), y(t))$, parametrize a closed curve. (a) Show $\frac{1}{2} \int_a^b (xy' - yx') \, ds = \int_a^b xy' \, ds = - \int_a^b yx' \, ds$. (b) Define the reverse parametrization $\gamma^-(t) = \gamma(b + a - t)$. Prove $\int_{\gamma^-} (x \, dy - y \, dx) = - \int_{\gamma} (x \, dy - y \, dx)$, so after possibly reversing orientation $A = \frac{1}{2} \int_a^b (xy' - yx') \, ds = \int_a^b xy' \, ds$.

- Suppose the x -axis splits Γ into graphs $y = f(x)$ and $y = g(x)$ for $0 \leq x \leq 1$ with $f \geq g$, bounding $\Omega = \{(x, y) : 0 \leq x \leq 1, g(x) \leq y \leq f(x)\}$. With the interpretation that $\int h dx$ is the area under h , the area of Ω is $\int_0^1 f - \int_0^1 g$. Show this equals the text's formula: $\int_0^1 f dx - \int_0^1 g dx = |-\int_\Gamma y dx| = A$, and that without the absolute value the formula holds if Ω lies to the left of Γ . Note the generalization to finite unions of such domains.
- Show that, with the text's definitions of ℓ and A , the isoperimetric inequality continues to hold (same proof) even when Γ is not simple. Show this stronger version is equivalent to *Wirtinger's inequality*: if f is 2π -periodic of class C^1 with $\int_0^{2\pi} f = 0$, then $\int_0^{2\pi} |f|^2 \leq \int_0^{2\pi} |f'|^2$, with equality iff $f(t) = A \sin t + B \cos t$. [Hint: with an arc-length parametrization of a length- 2π curve, $2(\pi - A) = \int_0^{2\pi} [x'(s) + y(s)]^2 ds + \int_0^{2\pi} (y'(s)^2 - y(s)^2) ds$; arrange $\int_0^{2\pi} y = 0$. For the converse, build g from f so the bracketed term vanishes.]

Problems (Stein, Ch.4) – isoperimetric.

- (Diameter of a curve.) The diameter of a closed curve Γ parametrized by $\gamma(t) = (x(t), y(t))$ on $[-\pi, \pi]$ is $d = \sup_{P, Q \in \Gamma} |P - Q|$. If a_n is the n th Fourier coefficient of $\gamma(t) = x(t) + iy(t)$ and ℓ is the length of Γ , prove: (a) $2|a_n| \leq d$ for all $n \neq 0$; (b) $\ell \leq \pi d$ when Γ is convex. [Hint for (a): $2a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} [\gamma(t) - \gamma(t + \pi/n)] e^{-int} dt$.] The equality $\ell = \pi d$ holds for the circle but not only for it; it is equivalent to $2|a_1| = d$. Reparametrize so the tangent at $\gamma(t)$ makes angle t with the y -axis; then with $a_1 = 1$, $\gamma'(t) = ie^{it}(1 + r(t))$ where r is real, $r(t) + r(t + \pi) = 0$, $|r| \leq 1$. (E.g. $r(t) = \cos 5t$, or $r(t) = h(3t)$ with $h = \mp 1$ on the two halves, gives the Reuleaux triangle: a convex curve of constant width that is not a circle.)

Additional exercises.

- Compute A/ℓ^2 for a regular n -gon and show it increases monotonically to $1/4\pi$ as $n \rightarrow \infty$. (A regular n -gon of circumradius 1 has $\ell = 2n \sin(\pi/n)$ and $A = \frac{n}{2} \sin(2\pi/n)$.)
- Using Definition 34.1.3, compute the enclosed area of the cardioid $\gamma(s) = ((1 + \cos s) \cos s, (1 + \cos s) \sin s)$, $s \in [0, 2\pi]$.
- Let Γ be a length- 2π arc-length curve with Fourier data $\{a_n\}, \{b_n\}$. Show the “isoperimetric deficit” equals $\pi \sum_{|n| \geq 2} (|n|^2 - |n|)(|a_n|^2 + |b_n|^2) + \pi \sum_n (|a_n| - |b_n|)^2 |n|$ (suitably interpreted), and explain why this exhibits the deficit as a sum of nonnegative terms.
- Verify Wirtinger's inequality $\int_0^{2\pi} |f|^2 \leq \int_0^{2\pi} |f'|^2$ directly for $f(t) = \sin 2t$, and check the inequality is *strict* (so $\sin 2t$ is not an extremizer, consistent with equality only for the fundamental modes).
- (Non-example.) Show the ratio A/ℓ^2 is *not* maximized by any polygon: given any polygon, exhibit a curve of the same perimeter with strictly larger area.
- Prove that scaling $(x, y) \mapsto (\delta x, \delta y)$ multiplies the shoelace area $\frac{1}{2} | \int (x dy - y dx) |$ by δ^2 , justifying the rescaling step of the proof.
- (Diagram.) Sketch, for a unit circle and an inscribed regular hexagon of the same perimeter, the two curves on one figure and shade the area lost by the hexagon; estimate the lost fraction numerically.
- Show the bound $|a_n \overline{b_n} - b_n \overline{a_n}| \leq |a_n|^2 + |b_n|^2$ is sharp, with equality iff $|a_n| = |b_n|$ and $a_n \overline{b_n}$ is purely imaginary, and interpret this condition geometrically for the extremal circle.

34.2 Weyl Equidistribution

Given an irrational γ , how are the fractional parts $\langle n\gamma \rangle$ distributed in $[0, 1)$? The answer — they spread out perfectly evenly — is Weyl's equidistribution theorem. The proof is a beautiful three-step approximation: check the claim for pure exponentials (where it is a geometric-series computation), extend by linearity to trigonometric polynomials, and pass to all continuous functions by uniform approximation (Weierstrass/Fejér). Squeezing characteristic functions of intervals between continuous bumps finishes the job.

The reals modulo the integers

Definition 34.2.1. For $x \in \mathbb{R}$, $[x]$ is the greatest integer $\leq x$ (the *integer part*) and $\langle x \rangle = x - [x] \in [0, 1)$ is the *fractional part*. We write $x \equiv y \pmod{1}$ (or $\pmod{\mathbb{Z}}$) when $x - y \in \mathbb{Z}$; every x is then congruent to a unique point of $[0, 1)$, namely $\langle x \rangle$.

For example $\langle 2.7 \rangle = 0.7$ and $[2.7] = 2$, while $[-3.4] = -4$ and $\langle -3.4 \rangle = 0.6$. Reducing modulo $\mathbb{Z}$ means identifying $[0, 1)$ with the circle $\mathbb{R}/\mathbb{Z}$.

Proposition 34.2.2. Let $\gamma \neq 0$. (i) If $\gamma = p/q \in \mathbb{Q}$, only finitely many of the $\langle n\gamma \rangle$ are distinct. (ii) If γ is irrational, the $\langle n\gamma \rangle$ are all distinct.

Proof. (i) For $\gamma = p/q$, the sequence repeats with period q : $\langle (q+m)p/q \rangle = \langle p + mp/q \rangle = \langle mp/q \rangle$. (ii) If $\langle n_1\gamma \rangle = \langle n_2\gamma \rangle$ with $n_1 \neq n_2$, then $(n_1 - n_2)\gamma \in \mathbb{Z}$, forcing $\gamma \in \mathbb{Q}$, a contradiction. $\square$

Kronecker proved that for irrational γ the set $\{\langle n\gamma \rangle\}$ is in fact *dense* in $[0, 1)$. We obtain a sharper statement.

Definition 34.2.3. A sequence $\xi_1, \xi_2, \dots$ in $[0, 1)$ is *equidistributed* if for every interval $(a, b) \subseteq [0, 1)$,

$$\lim_{N \rightarrow \infty} \frac{\#\{1 \leq n \leq N : \xi_n \in (a, b)\}}{N} = b - a.$$

Equidistribution is strictly stronger than density: the sequence must visit each subinterval with the right *frequency*, not merely visit it. Ordering matters: an enumeration of the rationals interleaved with zeros is dense but not equidistributed (“half” the terms sit at 0).

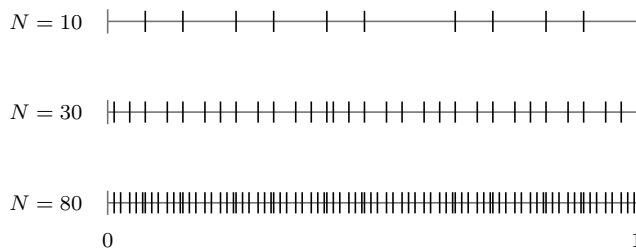


Figure 34.4. The points $\langle n\gamma \rangle$ for $\gamma = \sqrt{2}$ and $N = 10, 30, 80$: with more terms the marks fill $[0, 1)$ ever more evenly.

The equidistribution theorem

Theorem 34.2.4 (Weyl). If γ is irrational, the sequence $\langle n\gamma \rangle$, $n = 1, 2, 3, \dots$, is equidistributed in $[0, 1)$. In particular it is dense (Kronecker).

The bridge from counting to analysis: if $\chi_{(a,b)}$ is the (period-1) characteristic function of (a, b) , then $\#\{1 \leq n \leq N : \langle n\gamma \rangle \in (a, b)\} = \sum_{n=1}^N \chi_{(a,b)}(n\gamma)$, and the theorem is equivalent to

$$\frac{1}{N} \sum_{n=1}^N \chi_{(a,b)}(n\gamma) \rightarrow \int_0^1 \chi_{(a,b)}(x) dx \quad (N \rightarrow \infty).$$

The heart of the matter is the following.

Lemma 34.2.5. If f is continuous and periodic of period 1, and γ is irrational, then

$$\frac{1}{N} \sum_{n=1}^N f(n\gamma) \rightarrow \int_0^1 f(x) dx \quad (N \rightarrow \infty).$$

Proof. Step 1 (exponentials). For $f = 1$ both sides equal 1. For $f(x) = e^{2\pi i k x}$ with $k \neq 0$ the integral is 0, and since γ is irrational $e^{2\pi i k \gamma} \neq 1$, so the geometric sum gives

$$\frac{1}{N} \sum_{n=1}^N e^{2\pi i k n \gamma} = \frac{1}{N} e^{2\pi i k \gamma} \frac{1 - e^{2\pi i k N \gamma}}{1 - e^{2\pi i k \gamma}} \xrightarrow{N \rightarrow \infty} 0,$$

because the numerator is bounded by 2 while $1/N \rightarrow 0$.

Step 2 (trig polynomials). If the conclusion holds for f and g it holds for $Af + Bg$. By Step 1 it therefore holds for every trigonometric polynomial.

Step 3 (continuous f). Let $\varepsilon > 0$. By the Weierstrass approximation theorem for trigonometric polynomials (Fejér) choose P with $\sup_x |f(x) - P(x)| < \varepsilon/3$. By Step 2, for all large N , $|\frac{1}{N} \sum_{n=1}^N P(n\gamma) - \int_0^1 P| < \varepsilon/3$. Then

$$\left| \frac{1}{N} \sum_{n=1}^N f(n\gamma) - \int_0^1 f \right| \leq \frac{1}{N} \sum_{n=1}^N |f(n\gamma) - P(n\gamma)| + \left| \frac{1}{N} \sum_{n=1}^N P(n\gamma) - \int_0^1 P \right| + \int_0^1 |P - f| < \varepsilon.$$

Since ε is arbitrary the limit holds. $\square$

Proof of Theorem 34.2.4. Fix (a, b) and $\varepsilon > 0$. Choose continuous 1-periodic $f_\varepsilon^- \leq \chi_{(a,b)} \leq f_\varepsilon^+$, both bounded by 1, agreeing with $\chi_{(a,b)}$ off intervals of total length 2ε , so that $b - a - 2\varepsilon \leq \int_0^1 f_\varepsilon^-$ and $\int_0^1 f_\varepsilon^+ \leq b - a + 2\varepsilon$. With $S_N = \frac{1}{N} \sum_{n=1}^N \chi_{(a,b)}(n\gamma)$,

$$\frac{1}{N} \sum_{n=1}^N f_\varepsilon^-(n\gamma) \leq S_N \leq \frac{1}{N} \sum_{n=1}^N f_\varepsilon^+(n\gamma).$$

Letting $N \rightarrow \infty$ with Lemma 34.2.5, $b - a - 2\varepsilon \leq \liminf S_N \leq \limsup S_N \leq b - a + 2\varepsilon$. As ε is arbitrary, $\lim S_N = b - a$. $\square$

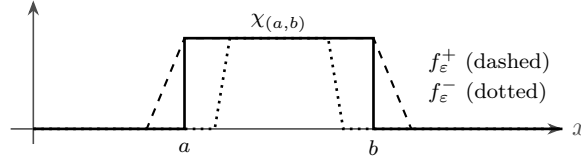


Figure 34.5. Squeezing $\chi_{(a,b)}$ between continuous 1-periodic functions agreeing with it off a set of length 2ε . (Schematic.)

Corollary 34.2.6. *Lemma 34.2.5 holds for every Riemann-integrable 1-periodic f .*

Proof. For real f and a partition $0 = x_0 < \dots < x_M = 1$, set $f_U = \sup_{[x_{j-1}, x_j]} f$ and $f_L = \inf$ on each subinterval; then $f_L \leq f \leq f_U$ with $\int (f_U - f_L) \leq \varepsilon$ for a fine enough partition (Riemann integrability). Each of f_L, f_U is a finite linear combination of interval characteristic functions, so Theorem 34.2.4 gives $\frac{1}{N} \sum f_L(n\gamma) \rightarrow \int f_L$ and likewise for f_U ; the usual squeeze finishes. $\square$

Weyl's criterion and the dynamical view

Theorem 34.2.7 (Weyl's criterion). *A sequence $\xi_n \in [0, 1)$ is equidistributed if and only if for every integer $k \neq 0$,*

$$\frac{1}{N} \sum_{n=1}^N e^{2\pi i k \xi_n} \rightarrow 0 \quad (N \rightarrow \infty).$$

Thus equidistribution is entirely controlled by the decay of exponential sums. This is the practical engine: to show a sequence equidistributes, estimate $\sum_n e^{2\pi i k \xi_n}$. With it one proves $\langle n^2 \gamma \rangle$ is equidistributed for irrational γ , and more generally $\langle P(n) \rangle$ for any polynomial with an irrational non-constant coefficient.

The result also has a dynamical reading. Let $\rho : \theta \mapsto \theta + 2\pi\gamma$ be rotation of the circle by $2\pi\gamma$; its iterates are $\rho^n : \theta \mapsto \theta + 2\pi n\gamma$. For Riemann-integrable f on the circle, Corollary 34.2.6 (after $\theta = 2\pi x$) says the *time average* equals the *space average*:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N f(\rho^n \theta) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta$$

for every θ , whenever γ is irrational. This is the simplest instance of *ergodicity*: an irrational rotation is uniquely ergodic.

The rotation ρ is the building block: its n th iterate ρ^n generates the orbit, and composing one more rotation simply advances the orbit, $\rho^{n+1} = \rho^n \circ \rho$.

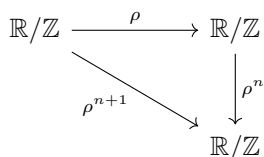
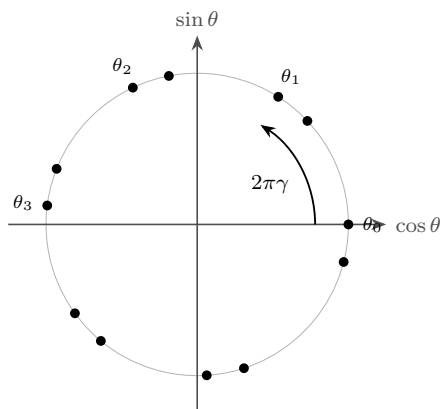


Figure 34.6. The orbit of θ under the irrational rotation $\rho : \theta \mapsto \theta + 2\pi\gamma$ is generated by iteration, $\rho^{n+1} = \rho^n \circ \rho$; for irrational γ the time-average $\frac{1}{N} \sum_{n=1}^N f(\rho^n \theta)$ converges to the base-point-independent space-average $\frac{1}{2\pi} \int_0^{2\pi} f$ (unique ergodicity).



Iterating the rotation $\rho : \theta \mapsto \theta + 2\pi\gamma$ on the circle; for irrational γ the orbit equidistributes (unique ergodicity).

Remark. A ray of light leaving a point in a square with reflecting walls unfolds, on the reflected grid of the plane, to the straight line $P + (t, \gamma t)$. The path is closed and periodic iff the slope γ is rational; for irrational γ it is dense by Kronecker, and equidistribution gives the stronger statement that the path spends time in each region proportional to its area.

Worked Example 34.2.8 (Equidistribution of $\langle n/2 \rangle$ fails (rational)). Show $\langle n\gamma \rangle$ with $\gamma = 1/2$ is not equidistributed.

Recognition. Proposition 34.2.2(i): rational γ visits only finitely many points.

Solution. The terms are $\frac{1}{2}, 0, \frac{1}{2}, 0, \dots$, supported on $\{0, \frac{1}{2}\}$. No mass ever lands in, say, $(0.6, 0.7)$, so the frequency there is $0 \neq 0.1$. *Check.* Weyl's criterion fails too: for $k = 2$, $e^{2\pi i \cdot 2 \cdot n/2} = e^{2\pi i n} = 1$, so the average is $1 \not\rightarrow 0$. ■

Worked Example 34.2.9 (Weyl's criterion verifies $\langle n\gamma \rangle$, γ irrational). Re-derive Theorem 34.2.4 from the criterion.

Recognition. Apply Theorem 34.2.7 with $\xi_n = n\gamma$.

Solution. For $k \neq 0$, $\frac{1}{N} \sum_{n=1}^N e^{2\pi i k n \gamma} = \frac{1}{N} e^{2\pi i k \gamma} \frac{1 - e^{2\pi i k N \gamma}}{1 - e^{2\pi i k \gamma}} \rightarrow 0$ since $e^{2\pi i k \gamma} \neq 1$. Hence equidistributed. *Check.* This is exactly Step 1 of Lemma 34.2.5, repackaged. ■

Worked Example 34.2.10 (Computing a finite count). For $\gamma = \sqrt{2}$, estimate $\#\{1 \leq n \leq 1000 : \langle n\gamma \rangle \in (0.3, 0.5)\}$.

Recognition. Equidistribution predicts the count is about $N(b-a)$.

Solution. $b-a = 0.2$, so the count is approximately $1000 \cdot 0.2 = 200$, with a fluctuation $o(N)$. *Check.* The prediction is a frequency statement; exact counts oscillate around 200 but the ratio $\rightarrow 0.2$. ■

Worked Example 34.2.11 (The time-average equals space-average for $f(x) = \cos 2\pi x$). Compute $\lim_N \frac{1}{N} \sum_{n=1}^N \cos(2\pi n\gamma)$ for irrational γ .

Recognition. This is Lemma 34.2.5 with $f = \cos 2\pi x$.

Solution. $\int_0^1 \cos 2\pi x dx = 0$, so the limit is 0. Directly, $\sum_{n=1}^N \cos(2\pi n\gamma) = \operatorname{Re} \sum e^{2\pi i n\gamma} = O(1)$ (bounded geometric sum), divided by $N \rightarrow 0$. *Check.* Agrees with the integral. ■

Worked Example 34.2.12 (Example 1 of Stein is equidistributed). Show the sequence $0, \frac{1}{2}, 0, \frac{1}{3}, \frac{2}{3}, 0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, 0, \dots$ is equidistributed.

Recognition. Direct counting using the block structure (denominators $q = 1, 2, 3, \dots$).

Solution. After the block with denominator q , the partial sequence contains, for each $m \leq q$, the points j/m ($0 \leq j < m$), spread evenly. In the q th block the points $0, \frac{1}{q}, \dots, \frac{q-1}{q}$ land one in each subinterval $[j/q, (j+1)/q)$. Summing blocks, the count in any (a, b) over the first N terms is $(b-a)N + O(\sqrt{N})$, so the frequency $\rightarrow b-a$. *Check.* The running ratio for $(0, \frac{1}{2})$ hovers at $\frac{1}{2}$ as blocks accumulate. ■

Worked Example 34.2.13 (The Step-1 geometric sum, quantitatively). For $\gamma = \sqrt{2}$, $k = 1$, bound $|\frac{1}{N} \sum_{n=1}^N e^{2\pi i n\gamma}|$.

Recognition. Step 1 of Lemma 34.2.5 gives an explicit $O(1/N)$ rate.

Solution. The sum is $\frac{1}{N} \left| \frac{e^{2\pi i \gamma} (1 - e^{2\pi i N\gamma})}{1 - e^{2\pi i \gamma}} \right| \leq \frac{1}{N} \cdot \frac{2}{|1 - e^{2\pi i \gamma}|} = \frac{1}{N} \cdot \frac{1}{|\sin \pi \gamma|}$. With $\gamma = \sqrt{2}$, $|\sin \pi \sqrt{2}| \approx 0.964$, so the bound is about $1.04/N$. *Check.* The decay is $O(1/N)$, uniform in N once γ is fixed irrational. ■

Worked Example 34.2.14 (Equidistribution of $\langle n^2 \gamma \rangle$ via the criterion (sketch)). Explain how Weyl's criterion reduces $\langle n^2 \gamma \rangle$ to a difference estimate.

Recognition. Weyl's estimate (Problem 2) controls $\sum e^{2\pi i k n^2 \gamma}$ by differencing.

Solution. Set $f(n) = k n^2 \gamma$. Then $f(n+h) - f(n) = 2knh\gamma + kh^2\gamma$, linear in n , so $\sum_n e^{2\pi i (f(n+h) - f(n))}$ is a geometric sum controlled by $\langle 2kh\gamma \rangle \neq 0$ (irrational). Averaging over $h \leq H$ and optimizing H gives $\frac{1}{N} \sum e^{2\pi i k n^2 \gamma} \rightarrow 0$, so $\langle n^2 \gamma \rangle$ equidistributes. *Check.* The trick is that squaring becomes linear after one difference. ■

Worked Example 34.2.15 (A dense but non-equidistributed sequence). Let r_n enumerate $\mathbb{Q} \cap [0, 1)$ and set $\xi_n = r_{n/2}$ for n even, $\xi_n = 0$ for n odd. Show dense but not equidistributed.

Recognition. A non-example separating density from equidistribution.

Solution. Density: the even-indexed terms hit every rational, dense in $[0, 1)$. Non-equidistribution: half the indices give $\xi_n = 0$, so any interval not containing 0 has frequency at most $\frac{1}{2}$, but a small interval near $\frac{1}{2}$ should have frequency $\rightarrow$ its length $< \frac{1}{2}$ — yet the interval $(0, \delta)$ gets frequency $\geq \frac{1}{2}$ for all δ , contradicting $b-a = \delta \rightarrow 0$. *Check.* Weyl's criterion fails: the odd terms contribute $\frac{1}{2}$ to $\frac{1}{N} \sum e^{2\pi i k \cdot 0} = \frac{1}{2}$, which does not go to 0. ■

Exercises (Stein, Ch.4) – equidistribution.

- Prove that the sequence $\{\gamma_n\}$, where γ_n is the fractional part of $(\frac{1+\sqrt{5}}{2})^n$, is not equidistributed in $[0, 1)$. [Hint: show $U_n = (\frac{1+\sqrt{5}}{2})^n + (\frac{1-\sqrt{5}}{2})^n$ solves $U_{r+1} = U_r + U_{r-1}$ with $U_0 = 2, U_1 = 1$ (the Fibonacci/Lucas recurrence), so the fractional parts cluster near 0 and 1.]
- Let $\theta = p/q$ in lowest terms with $q > 0$, and $\xi_n = \langle n\theta \rangle$. Show $\{\xi_n\}$ is equidistributed on the points $0, 1/q, 2/q, \dots, (q-1)/q$. In fact for $0 \leq a < q$, $\frac{\#\{n : 1 \leq n \leq N, \langle n\theta \rangle = a/q\}}{N} = \frac{1}{q} + O(\frac{1}{N})$. [Hint: for each $k \geq 0$ there is a unique $n \in [kq, (k+1)q)$ with $\langle n\theta \rangle = a/q$ (use $xp + yq = 1$ for coprime p, q); write $N = \ell q + r$, $0 \leq r < q$, to get $\ell \leq \#\{\dots\} \leq \ell + 1$.]

3. Prove the second half of Weyl's criterion: if $\xi_1, \xi_2, \dots$ in $[0, 1)$ is equidistributed, then for all $k \in \mathbb{Z} \setminus \{0\}$, $\frac{1}{N} \sum_{n=1}^N e^{2\pi i k \xi_n} \rightarrow 0$ as $N \rightarrow \infty$. [Hint: it suffices to show $\frac{1}{N} \sum f(\xi_n) \rightarrow \int_0^1 f$ for all continuous f ; first do $f = \text{characteristic function of an interval}$.]
4. Show that for any $a \neq 0$ and $0 < \sigma < 1$, the sequence $\langle an^\sigma \rangle$ is equidistributed in $[0, 1)$. [Hint: prove $\sum_{n=1}^N e^{2\pi i b n^\sigma} = O(N^\sigma) + O(N^{1-\sigma})$ for $b \neq 0$; in fact $\sum_{n=1}^N e^{2\pi i b n^\sigma} - \int_1^N e^{2\pi i b x^\sigma} dx = O(\sum_{n=1}^N n^{-1+\sigma})$.]
5. In contrast with the previous exercise, prove that $\langle a \log n \rangle$ is *not* equidistributed for any a . [Hint: compare $\sum_{n=1}^N e^{2\pi i b \log n}$ with the corresponding integral.]
6. Suppose f is 1-periodic and $\{\xi_n\}$ is equidistributed in $[0, 1)$. Prove: (a) if f is continuous with $\int_0^1 f = 0$, then $\frac{1}{N} \sum_{n=1}^N f(x + \xi_n) \rightarrow 0$ uniformly in x [hint: first for trig polynomials]; (b) if f is merely integrable with $\int_0^1 f = 0$, then $\int_0^1 \left| \frac{1}{N} \sum_{n=1}^N f(x + \xi_n) \right|^2 dx \rightarrow 0$ as $N \rightarrow \infty$.

Problems (Stein, Ch.4) – equidistribution.

1. (Weyl's estimate.) Let $S_N = \sum_{n=1}^N e^{2\pi i f(n)}$. (a) Show that for $H \leq N$,

$$|S_N|^2 \leq c \frac{N}{H} \sum_{h=0}^H \left| \sum_{n=1}^{N-h} e^{2\pi i (f(n+h) - f(n))} \right|$$

for a constant $c > 0$ independent of N, H, f . (b) Use this to show $\langle n^2 \gamma \rangle$ is equidistributed in $[0, 1)$ whenever γ is irrational. (c) More generally, if $\{\xi_n\}$ is such that $\langle \xi_{n+h} - \xi_n \rangle$ is equidistributed for all positive integers h , then $\langle \xi_n \rangle$ is equidistributed. (d) If $P(x) = c_n x^n + \dots + c_0$ has real coefficients with at least one of $c_1, \dots, c_n$ irrational, then $\langle P(n) \rangle$ is equidistributed. [Hints: for (a), let $a_n = e^{2\pi i f(n)}$ for $1 \leq n \leq N$, else 0; write $H \sum_n a_n = \sum_{k=1}^H \sum_n a_{n+k}$ and apply Cauchy-Schwarz. For (b), $(n+h)^2 \gamma - n^2 \gamma = 2nh\gamma + h^2 \gamma$ and $\langle 2nh\gamma \rangle$ is equidistributed. For (d), reduce to $P = Q + c_1 x + c_0$ with c_1 irrational, estimate $\sum_n e^{2\pi i k P(n)}$, then induct on the highest-degree irrational term using (c).]

2. If $\sigma > 0$ is not an integer and $a \neq 0$, then $\langle an^\sigma \rangle$ is equidistributed in $[0, 1)$ (cf. Exercise 8).

Additional exercises.

1. For γ irrational, evaluate $\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \cos^2(2\pi n \gamma)$ using Lemma 34.2.5 with $f(x) = \cos^2 2\pi x = \frac{1}{2} + \frac{1}{2} \cos 4\pi x$.
2. Use Weyl's criterion to show that if γ is irrational then $\langle n\gamma + \beta \rangle$ is equidistributed for every fixed real β (a shifted orbit).
3. Show that if ξ_n is equidistributed, so is $1 - \xi_n$ (reflection), and that the interleaving $\xi_1, 1 - \xi_1, \xi_2, 1 - \xi_2, \dots$ is equidistributed.
4. Compute $\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \langle n\gamma \rangle$ for irrational γ , and explain why the answer is $\frac{1}{2}$.
5. (Non-example.) Give a sequence in $[0, 1)$ that is dense, has every point of $[0, 1)$ as a limit of a subsequence, yet is not equidistributed; prove non-equidistribution via Weyl's criterion.
6. For the rotation $\rho : \theta \mapsto \theta + 2\pi\gamma$ with γ irrational, show the only ρ -invariant continuous probability measure is normalized arc length (unique ergodicity), assuming Corollary 34.2.6.
7. (Diagram.) Plot the first $N = 50$ points $\langle n\gamma \rangle$ for the golden ratio $\gamma = (\sqrt{5} - 1)/2$ on the unit interval, and compare the gap structure with the three-distance (Steinhaus) theorem prediction.
8. Show that for $f(x) = e^{2\pi i x}$ the time average $\frac{1}{N} \sum_{n=1}^N f(n\gamma)$ has modulus exactly $\frac{1}{N} \left| \frac{\sin \pi N \gamma}{\sin \pi \gamma} \right|$, and deduce the $O(1/N)$ decay rate explicitly.
9. Prove that a finite union of equidistributed sequences, interleaved in a fixed pattern, is equidistributed iff each contributes the right proportion; give the precise statement and proof.

34.3 A Continuous Nowhere-Differentiable Function

Does a continuous function that is differentiable at *no* point exist? Riemann guessed in 1861 that $R(x) = \sum_{n \geq 1} \sin(n^2 x)/n^2$ was such, but never proved it; Weierstrass in 1872 gave the first rigorous example $W(x) = \sum_n b^n \cos(a^n x)$ with $ab > 1 + 3\pi/2$. We prove a clean modern version whose engine is the structure of *lacunary* Fourier series (series that skip most frequencies) together with three ways of summing a Fourier series: ordinary partial sums (Dirichlet), Cesàro means (Fejér), and delayed means.

Three summation methods

Write $g(x) \sim \sum_n a_n e^{inx}$. The three summations are convolutions of g with kernels:

$$S_N(g) = g * D_N, \quad \sigma_N(g) = g * F_N, \quad \Delta_N(g) = 2\sigma_{2N}(g) - \sigma_N(g) = g * (2F_{2N} - F_N),$$

where D_N is the Dirichlet kernel and F_N the Fejér kernel. In terms of multipliers on the coefficient $a_n e^{inx}$:

- S_N multiplies by 1 if $|n| \leq N$, else 0 (a sharp cutoff).
- σ_N multiplies by $1 - |n|/N$ for $|n| \leq N$, else 0 (a triangle).
- Δ_N multiplies by 1 for $|n| \leq N$, by $2(1 - |n|/(2N))$ for $N \leq |n| \leq 2N$, and 0 for $|n| > 2N$ (a trapezoid).

Each summation is convolution with a kernel on the function side, and on the coefficient side it is simply multiplication of a_n by a fixed sequence $m(n)$ (the multiplier); the Fourier coefficient map intertwines the two.

$$\begin{array}{ccc} g & \xrightarrow[\sim]{\wedge} & (a_n) \\ g \mapsto g * K \downarrow & & \downarrow a_n \mapsto m(n) a_n \\ g * K & \xrightarrow[\wedge]{\sim} & (m(n) a_n) \end{array}$$

Figure 34.7. Convolution with a kernel K ($K = D_N, F_N, 2F_{2N} - F_N$ for S_N, σ_N, Δ_N) corresponds under the Fourier coefficient map to multiplication by the multiplier $m(n) = \hat{K}(n)$ (the sharp cutoff, triangle, or trapezoid above).

Proposition 34.3.1 (Cesàro means average the partial sums). $\sigma_N(g) = \frac{S_0(g) + \cdots + S_{N-1}(g)}{N} = \sum_{|n| \leq N} \left(1 - \frac{|n|}{N}\right) a_n e^{inx}$.

Proof. $\frac{1}{N} \sum_{\ell=0}^{N-1} \sum_{|k| \leq \ell} a_k e^{ikx} = \frac{1}{N} \sum_{|n| \leq N} (N - |n|) a_n e^{inx} = \sum_{|n| \leq N} (1 - |n|/N) a_n e^{inx}$, since the coefficient $a_n e^{inx}$ appears in exactly $N - |n|$ of the partial sums S_ℓ , $\ell = 0, \dots, N - 1$. $\square$

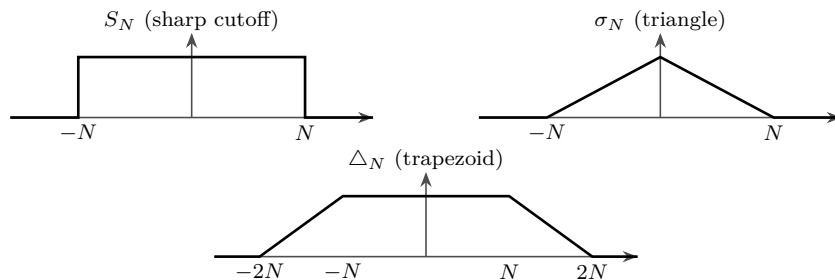


Figure 34.8. The three multipliers: sharp cutoff (S_N), triangular taper (σ_N), and trapezoid ($\Delta_N = 2\sigma_{2N} - \sigma_N$).

The delayed means matter because they combine the good (kernel-positivity-driven) behavior of Cesàro means with agreeing exactly with partial sums on a band. The Fejér kernel is

$$F_N(t) = \frac{1}{N} \frac{\sin^2(Nt/2)}{\sin^2(t/2)} \geq 0,$$

a *good kernel*: nonnegative, total mass 1, concentrating at 0.

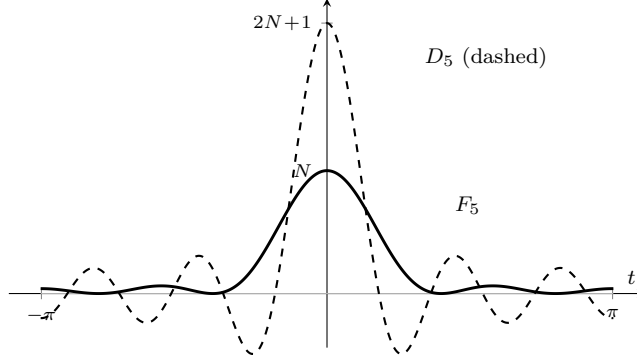


Figure 34.9. Dirichlet kernel D_5 (oscillates, dips below zero) versus the nonnegative Fejér kernel F_5 (a good kernel); peaks $2N+1 = 11$ and $N = 5$.

The nowhere-differentiable function

Definition 34.3.2. A Fourier series that omits all but a sparse set of frequencies (e.g. only $n = 2^k$) is called *lacunary*. For $0 < \alpha < 1$ define

$$f_\alpha(x) = f(x) = \sum_{n=0}^{\infty} 2^{-n\alpha} e^{i2^n x}.$$

Theorem 34.3.3. For $0 < \alpha < 1$, f_α is continuous but nowhere differentiable.

Continuity is immediate: $\sum 2^{-n\alpha} < \infty$, so the series converges uniformly (Weierstrass M -test). The lacunary structure is the crux. Because the only frequencies present are 2^n , the delayed mean $\triangle_N(f)$ equals the partial sum $S_{N'}(f)$ for N' the largest power of 2 with $N' \leq N$:

$$S_N(f) = \triangle_{N'}(f), \quad N' = 2^{\lfloor \log_2 N \rfloor}. \quad (34.3)$$

Indeed the trapezoid multiplier of $\triangle_{N'}$ equals 1 on $|n| \leq N'$ and the only nonzero frequencies up to $2N'$ are $\leq N'$.

Lemma 34.3.4. If g is continuous and differentiable at x_0 , then $\sigma_N(g)'(x_0) = O(\log N)$, and hence $\triangle_N(g)'(x_0) = O(\log N)$.

Proof. $\sigma_N(g)'(x_0) = \int_{-\pi}^{\pi} F'_N(t) g(x_0 - t) dt$. Since F_N is periodic, $\int_{-\pi}^{\pi} F'_N = 0$, so we may subtract $g(x_0)$:

$$\sigma_N(g)'(x_0) = \int_{-\pi}^{\pi} F'_N(t) [g(x_0 - t) - g(x_0)] dt, \quad \text{so} \quad |\sigma_N(g)'(x_0)| \leq C \int_{-\pi}^{\pi} |F'_N(t)| |t| dt$$

using differentiability ($|g(x_0 - t) - g(x_0)| \leq C|t|$ near 0, and boundedness elsewhere). Two estimates on F'_N : it is a trig polynomial of degree N with coefficients $\leq N$, so $|F'_N(t)| \leq (2N+1)N \leq AN^2$; and differentiating the closed form, with $|\sin(Nt/2)| \leq CN|t|$ and $|\sin(t/2)| \geq c|t|$ for $|t| \leq \pi$, gives $|F'_N(t)| \leq A/|t|^2$. Splitting the integral at $|t| = 1/N$,

$$\int_{|t| \leq 1/N} AN^2 |t| dt + \int_{1/N \leq |t| \leq \pi} \frac{A}{|t|^2} |t| dt = O(1) + A \int_{1/N \leq |t| \leq \pi} \frac{dt}{|t|} = O(1) + O(\log N) = O(\log N).$$

The bound for $\triangle_N = 2\sigma_{2N} - \sigma_N$ follows. □

Lemma 34.3.5. If $2N = 2^n$ then $\triangle_{2N}(f) - \triangle_N(f) = 2^{-n\alpha} e^{i2^n x}$.

Proof. By (34.3), $\triangle_{2N}(f) = S_{2N}(f)$ and $\triangle_N(f) = S_N(f)$ when $N = 2^{n-1}$, $2N = 2^n$. Their difference picks out exactly the single frequency 2^n , with coefficient $2^{-n\alpha}$. □

Proof of Theorem 34.3.3. Suppose, for contradiction, $f'(x_0)$ exists. By Lemma 34.3.4,

$$\Delta_{2N}(f)'(x_0) - \Delta_N(f)'(x_0) = O(\log N).$$

On the other hand differentiating Lemma 34.3.5 (with $2N = 2^n$),

$$|\Delta_{2N}(f)'(x_0) - \Delta_N(f)'(x_0)| = \left| 2^{-n\alpha} i 2^n e^{i 2^n x_0} \right| = 2^{n(1-\alpha)} \geq c N^{1-\alpha}.$$

Since $0 < \alpha < 1$, $N^{1-\alpha}$ grows faster than $\log N$ — contradiction. Hence f is differentiable nowhere. $\square$

The real part $F(x) = \sum_n 2^{-n\alpha} \cos 2^n x$ is also nowhere differentiable: the same proof works after the refinement that $\Delta_N(g)'(x_0 + h) = O(\log N)$ whenever $|h| \leq c/N$, then choosing $h = \pm \delta/2^n$ so that $|\sin(2^n(x_0 + h))| = 1$. For $\alpha > 1$ the series may be differentiated term by term, so f_α is C^1 ; the case $\alpha = 1$ is also nowhere differentiable by a finer argument.

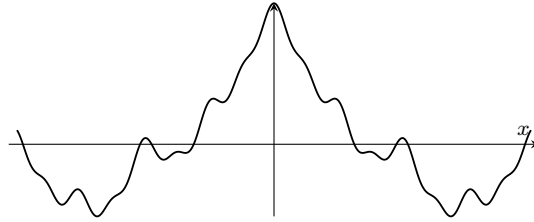


Figure 34.10. A partial sum of $\sum 2^{-n\alpha} \cos 2^n x$ ($\alpha = \frac{3}{4}$): adding higher lacunary modes builds an ever-rougher graph with no tangent anywhere.

Aside: the heat equation on the circle

Returning to Fourier's original problem, model a ring by the circle and let $u(x, t)$ be the temperature. Separation of variables in $u_t = u_{xx}$ (after rescaling $c = 1$) with 1-periodic data f leads to

$$u(x, t) = \sum_{n=-\infty}^{\infty} a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = (f * H_t)(x), \quad H_t(x) = \sum_{n=-\infty}^{\infty} e^{-4\pi^2 n^2 t} e^{2\pi i n x},$$

where $a_n = \hat{f}(n)$ and H_t is the *heat kernel for the circle*. The factor $e^{-4\pi^2 n^2 t}$ kills high modes instantly, so u is smooth for $t > 0$; and $u \rightarrow f$ in mean square as $t \rightarrow 0$. Unlike the Poisson kernel there is no elementary closed form for H_t ; that it is a good kernel (and everywhere positive) is proved later via the Poisson summation formula. Heuristically, positivity must hold: if $f \leq 0$ then heat flows hot to cold so $u \leq 0$, and a negative H_t near some x_0 would let an $f \leq 0$ supported near x_0 force $u(x_0, t) > 0$.

On the coefficient side the heat semigroup is again a multiplier, $\hat{f}(n) \mapsto e^{-4\pi^2 n^2 t} \hat{f}(n)$, and convolution by H_t realizes it on functions.

$$\begin{array}{ccc} f & \xrightarrow[\sim]{\wedge} & (\hat{f}(n)) \\ f \mapsto f * H_t \downarrow & & \downarrow \hat{f}(n) \mapsto e^{-4\pi^2 n^2 t} \hat{f}(n) \\ u(\cdot, t) & \xrightarrow[\sim]{\wedge} & (e^{-4\pi^2 n^2 t} \hat{f}(n)) \end{array}$$

Figure 34.11. Heat flow for time t is convolution with the heat kernel H_t , which under the Fourier coefficient map is the multiplier $e^{-4\pi^2 n^2 t}$; the rapid decay in $|n|$ is why $u(\cdot, t)$ is smooth for every $t > 0$.

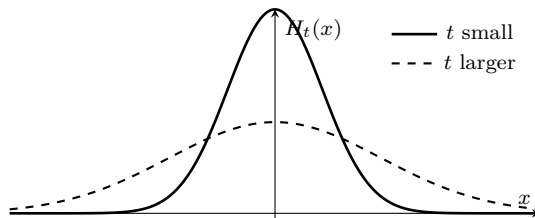


Figure 34.12. The heat kernel H_t on the circle peaks and narrows as $t \rightarrow 0$ (Gaussian-like for small t); larger t flattens it as heat diffuses. (Schematic.)

Method (Proving non-differentiability from lacunarity). *Recognition trigger.* A function is a sparse (lacunary) trigonometric series and you must show it is nowhere differentiable.

1. Use that for lacunary series the delayed mean equals a partial sum (eq. (34.3)), so consecutive delayed-mean differences isolate one big mode.
2. If g were differentiable at x_0 , derivatives of delayed means grow only like $O(\log N)$ (Lemma 34.3.4).
3. But the isolated mode forces a difference of size $\gtrsim N^{1-\alpha} \gg \log N$ — contradiction.

Worked Example 34.3.6 (Fejér kernel has mass one). Verify $\frac{1}{2\pi} \int_{-\pi}^{\pi} F_N = 1$.

Recognition. $F_N = \sum_{|n| \leq N} (1 - |n|/N) e^{int}$; only the $n = 0$ term integrates nonzero.

Solution. $\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{int} dt = \delta_{n,0}$, so the integral equals the $n = 0$ coefficient $1 - 0/N = 1$. *Check.* Mass one is required for a good kernel. ■

Worked Example 34.3.7 ($\Delta_N = 2\sigma_{2N} - \sigma_N$ as a trapezoid). Confirm the trapezoid multiplier for $N \leq |n| \leq 2N$.

Recognition. Combine the two triangular multipliers.

Solution. For $|n| \leq 2N$, σ_{2N} gives $1 - |n|/2N$; for $|n| \leq N$, σ_N gives $1 - |n|/N$. So Δ_N gives $2(1 - |n|/2N) - (1 - |n|/N) = 1$ on $|n| \leq N$, and $2(1 - |n|/2N) - 0 = 2(1 - |n|/2N)$ on $N \leq |n| \leq 2N$. *Check.* At $|n| = N$ both formulas give 1; at $|n| = 2N$ it is 0 — a trapezoid. ■

Worked Example 34.3.8 (Lacunary identity (34.3) on a toy series). For $g(x) = e^{ix} + \frac{1}{2}e^{i2x} + \frac{1}{4}e^{i4x}$ compute $S_3(g)$ and $\Delta_2(g)$.

Recognition. $N' = 2$ is the largest power of 2 that is ≤ 3 .

Solution. $S_3(g)$ keeps $|n| \leq 3$: $e^{ix} + \frac{1}{2}e^{i2x}$ (the e^{i4x} is dropped). $\Delta_2(g)$ has multiplier 1 on $|n| \leq 2$ and tapers to 0 at 4; on the present frequencies it keeps $e^{ix} + \frac{1}{2}e^{i2x}$ and drops e^{i4x} . They agree. *Check.* Consistent with $S_N(f) = \Delta_{N'}(f)$. ■

Worked Example 34.3.9 (Term-by-term differentiation when $\alpha > 1$). Show $f_\alpha \in C^1$ for $\alpha > 1$.

Recognition. If the differentiated series converges uniformly, f is C^1 .

Solution. The formal derivative is $\sum_n 2^{-n\alpha} i 2^n e^{i2^n x} = \sum_n i 2^{n(1-\alpha)} e^{i2^n x}$, with $\sum 2^{n(1-\alpha)} < \infty$ iff $\alpha > 1$. By the M -test it converges uniformly, so termwise differentiation is valid and $f_\alpha \in C^1$. *Check.* The borderline $\alpha = 1$ gives $\sum 1 = \infty$, so the test fails — exactly where nowhere-differentiability persists. ■

Worked Example 34.3.10 (Heat kernel kills high modes). Estimate $\hat{u}(n, t)$ relative to $\hat{f}(n)$ for the circle heat flow.

Recognition. Each mode is multiplied by $e^{-4\pi^2 n^2 t}$.

Solution. $\hat{u}(n, t) = \hat{f}(n) e^{-4\pi^2 n^2 t}$. For fixed $t > 0$ these decay super-exponentially in $|n|$, so $u(\cdot, t) \in C^\infty$ regardless of how rough f is. *Check.* As $t \rightarrow 0$ the factor $\rightarrow 1$ and $u \rightarrow f$ in L^2 by Parseval. ■

Worked Example 34.3.11 (Fourier coefficients of a square wave and a Parseval sum). Compute the coefficients of the 2π -periodic square wave $s(x) = \text{sgn}(\sin x)$ and use Parseval to sum $\sum_{k \geq 0} 1/(2k+1)^2$.

Recognition. A lacunary-looking series (only odd modes) feeding Parseval.

Solution. By oddness $\hat{s}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} s(x) e^{-inx} dx$ vanishes for even n and equals $\frac{2}{i\pi n}$ for odd n (so $s(x) = \frac{4}{\pi} \sum_{k \geq 0} \frac{\sin((2k+1)x)}{2k+1}$). Parseval: $\frac{1}{2\pi} \int_{-\pi}^{\pi} 1 dx = 1 = \sum_{n \text{ odd}} \frac{4}{\pi^2 n^2} = \frac{8}{\pi^2} \sum_{k \geq 0} \frac{1}{(2k+1)^2}$, giving $\sum_{k \geq 0} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$. *Check.* Consistent with $\sum 1/n^2 = \pi^2/6$ since the even terms sum to $\frac{1}{4} \cdot \frac{\pi^2}{6}$. ■

Worked Example 34.3.12 (Gibbs overshoot at a jump). Explain why S_N of the square wave overshoots near the jump but σ_N does not.

Recognition. The sharp Dirichlet cutoff causes oscillation; the positive Fejér kernel suppresses it — the same dichotomy used in the nowhere-differentiability proof.

Solution. Near a jump, $S_N = g * D_N$ inherits the sign-changing tail of D_N , producing a fixed $\approx 9\%$ overshoot (the Gibbs phenomenon) that does not vanish as $N \rightarrow \infty$. By contrast $\sigma_N = g * F_N$ with $F_N \geq 0$ averages monotonically, so no overshoot occurs and $\sigma_N \rightarrow \frac{1}{2}(g(\theta^+) + g(\theta^-))$ at the jump. *Check.* Positivity of the kernel is exactly what kills Gibbs, matching “ F_N is a good kernel, D_N is not.” ■

Worked Example 34.3.13 (Why D_N is not a good kernel). Show the Dirichlet kernel violates positivity.

Recognition. $D_N(t) = \sin((N + \frac{1}{2})t) / \sin(t/2)$ oscillates in sign.

Solution. At t just past the first zero $\sin((N + \frac{1}{2})t)$ turns negative while $\sin(t/2) > 0$, so $D_N(t) < 0$ there. Its L^1 norm grows like $\log N$ (the Lebesgue constants), so D_N fails both positivity and uniform L^1 boundedness — this is exactly why S_N can diverge where σ_N (with the positive F_N) converges. *Check.* Contrast with the Fejér-kernel worked example above. ■

Exercises (Stein, Ch.4) – summation methods and heat kernel.

1. Show that if $u(x, t) = (f * H_t)(x)$ with H_t the heat kernel and f Riemann integrable, then $\int_0^1 |u(x, t) - f(x)|^2 dx \rightarrow 0$ as $t \rightarrow 0$ (convergence to the initial data in mean square). [Use Parseval.]
2. A change of variables in $u(x, t) = \sum_n a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x}$ yields the solution $u(\theta, \tau) = \sum_n a_n e^{-n^2 \tau} e^{in\theta} = (f * h_\tau)(\theta)$ of $\partial_\tau u = \partial_\theta^2 u$ on $0 \leq \theta \leq 2\pi$, $\tau > 0$, with $u(\theta, 0) = f(\theta) \sim \sum a_n e^{in\theta}$. Here $h_\tau(\theta) = \sum_{n=-\infty}^{\infty} e^{-n^2 \tau} e^{in\theta}$. Compare this heat kernel with the Poisson kernel $P_r(\theta) = \sum_n e^{-|n|\tau} e^{in\theta}$, $r = e^{-\tau}$ (so $0 < r < 1 \Leftrightarrow \tau > 0$).
3. Although the heat kernel H_t is a good kernel (proved later), show directly that it is “peaked” at $x = 0$ as $t \rightarrow 0$: (a) $\int_{-1/2}^{1/2} |H_t(x)|^2 dx$ has order of magnitude $t^{-1/2}$; more precisely $t^{1/2} \int_{-1/2}^{1/2} |H_t|^2$ converges to a nonzero limit as $t \rightarrow 0$. (b) $\int_{-1/2}^{1/2} x^2 |H_t(x)|^2 dx = O(t^{1/2})$. [Hints: for (a) compare $\sum_n e^{-cn^2 t}$ with $\int_{-\infty}^{\infty} e^{-cx^2 t} dx$; for (b) use $x^2 \leq C(\sin \pi x)^2$ on $[-\frac{1}{2}, \frac{1}{2}]$ and apply the mean value theorem to $e^{-cx^2 t}$.]

Problems (Stein, Ch.4) – nowhere differentiability and delayed means.

1. (Piling up singularities.) On $[-1, 1]$ let $\varphi(x) = |x|$, extended to $\mathbb{R}$ with period 2 (so φ is continuous, $|\varphi| \leq 1$). Then $f(x) = \sum_{n=0}^{\infty} (\frac{3}{4})^n \varphi(4^n x)$ is continuous on $\mathbb{R}$. (a) Fix x_0 ; for each $m \geq 1$ set $\delta_m = \pm \frac{1}{2} 4^{-m}$ with the sign chosen so no integer lies between $4^m x_0$ and $4^m(x_0 + \delta_m)$, and let $\gamma_n = \frac{\varphi(4^n(x_0 + \delta_m)) - \varphi(4^n x_0)}{\delta_m}$. Prove $\gamma_n = 0$ for $n > m$, and $|\gamma_n| \leq 4^n$ for $0 \leq n \leq m$ with $|\gamma_m| = 4^m$. (b) Deduce $|\frac{f(x_0 + \delta_m) - f(x_0)}{\delta_m}| \geq \frac{1}{2}(3^m + 1)$, hence f is nowhere differentiable.
2. (Generalized delayed means.) For Riemann-integrable f on $[-\pi, \pi]$ set $\sigma_{N,K} = \frac{S_N + \cdots + S_{N+K-1}}{K}$, so $\sigma_{0,N} = \sigma_N$, $\sigma_{N,1} = S_N$, $\sigma_{N,N} = \Delta_N$. (a) Show $\sigma_{N,K} = \frac{1}{K}((N+K)\sigma_{N+K} - N\sigma_N)$ and $\sigma_{N,K} = S_N + \sum_{N+1 \leq |\nu| \leq N+K-1} (1 - \frac{|\nu|-N}{K}) \hat{f}(\nu) e^{i\nu\theta}$, hence $|\sigma_{N,K} - S_M| \leq \sum_{N+1 \leq |\nu| \leq N+K-1} |\hat{f}(\nu)|$ for $N \leq M < N+K$. (b) With $N = kn$, $K = n$, use Fejér’s theorem to show $\sigma_{kn,n}(f)(\theta) \rightarrow f(\theta)$ at points of continuity, $\rightarrow \frac{1}{2}(f(\theta^+) + f(\theta^-))$ at jumps, and uniformly when f is continuous. (c) If $\hat{f}(\nu) = O(1/|\nu|)$ and $kn \leq m < (k+1)n$, show $|\sigma_{kn,n} - S_m| \leq C/k$. (d) If $\hat{f}(\nu) = O(1/|\nu|)$, prove $S_N(f)(\theta) \rightarrow f(\theta)$ at points of continuity, $\rightarrow \frac{1}{2}(f(\theta^+) + f(\theta^-))$ at jumps, and uniformly if f is continuous. (e) Conclude the weak Tauberian statement: if $\sum c_n$ is Cesàro summable to s and $c_n = O(1/n)$, then $\sum c_n$ converges to s .
3. (Dirichlet’s theorem.) Prove that the Fourier series of a real continuous periodic function f having only finitely many relative maxima and minima converges everywhere to f (and uniformly), by showing such f satisfies $\hat{f}(n) = O(1/|n|)$ and invoking Problem 5(d).

Additional exercises.

1. Verify directly that the Dirichlet kernel $D_N(t) = \sum_{|n| \leq N} e^{int} = \frac{\sin((N + \frac{1}{2})t)}{\sin(t/2)}$, and that $\frac{1}{2\pi} \int_{-\pi}^{\pi} D_N = 1$ while $\frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N| \sim \frac{4}{\pi^2} \log N$ (the Lebesgue constants).
2. Show that the Cesàro mean multiplier $1 - |n|/N$ arises from averaging the partial sums (Proposition 34.3.1) by an explicit count of how many S_ℓ contain the n th mode.
3. For the lacunary series f_α , prove the continuity claim rigorously via the Weierstrass M -test, identifying the dominating series.
4. (Non-example.) Exhibit a continuous function that is differentiable at every rational multiple of π but at no irrational multiple, or explain why Riemann's $R(x) = \sum \sin(n^2 x)/n^2$ is the relevant example (cite Hardy/Gerver).
5. Show that $|F'_N(t)| \leq A/t^2$ for $|t| \leq \pi$ by differentiating the closed form $F_N(t) = \frac{1}{N} \sin^2(Nt/2)/\sin^2(t/2)$ and using $|\sin(Nt/2)| \leq CN|t|$, $|\sin(t/2)| \geq c|t|$.
6. Prove that for the heat kernel $h_\tau(\theta) = \sum_n e^{-n^2\tau} e^{in\theta}$, one has $\frac{1}{2\pi} \int_0^{2\pi} h_\tau d\theta = 1$ for all $\tau > 0$, so h_τ has total mass one.
7. (Diagram.) Plot D_5 and F_5 on $[-\pi, \pi]$ on one axis and mark where D_5 goes negative; annotate the main lobe widths.
8. Show that the delayed mean $\triangle_N = 2\sigma_{2N} - \sigma_N$ satisfies $\triangle_N(g) = g$ exactly when g is a trig polynomial of degree $\leq N$, and explain why this “reproduces low frequencies.”
9. For the circle heat flow with $f(x) = \cos 2\pi x$, write the explicit solution $u(x, t) = e^{-4\pi^2 t} \cos 2\pi x$ and verify it solves $u_t = u_{xx}$ with $u(x, 0) = f$.
10. Prove that if g is differentiable at x_0 then $\sigma_N(g)'(x_0) = O(\log N)$ cannot be improved to $O(1)$ in general, by exhibiting a g for which $\sigma_N(g)'(x_0)$ grows like $\log N$.

Chapter 35

The Fourier Transform on $\mathbb{R}$

35.1 The Schwartz Space and the Fourier Transform

From Fourier coefficients to the Fourier transform

The theory of Fourier series lives on the circle: a 1-periodic function f has Fourier coefficients

$$a_n = \int_0^1 f(x) e^{-2\pi i n x} dx, \quad f(x) \sim \sum_{n \in \mathbb{Z}} a_n e^{2\pi i n x}.$$

Passing to the line, we replace the discrete index $n \in \mathbb{Z}$ by a continuous frequency $\xi \in \mathbb{R}$, the sum by an integral, and the interval by all of $\mathbb{R}$. The Fourier coefficient a_n becomes the *Fourier transform*

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx,$$

and the synthesis $f(x) = \sum a_n e^{2\pi i n x}$ becomes the (conjectural) *inversion formula* $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi$. The striking feature, with no clean analogue for series, is the symmetry: the transform of a function on $\mathbb{R}$ is again a function on $\mathbb{R}$, so f and $\hat{f}$ play nearly interchangeable roles.

For any of this to make sense we must control f at infinity. We isolate two notions of “smallness.”

Integration on the line and moderate decrease

Definition 35.1.1. For continuous $f : \mathbb{R} \rightarrow \mathbb{C}$ set $\int_{-\infty}^{\infty} f = \lim_{N \rightarrow \infty} \int_{-N}^N f$ when the limit exists. We say f is of *moderate decrease* if f is continuous and there is $A > 0$ with

$$|f(x)| \leq \frac{A}{1+x^2} \quad \text{for all } x \in \mathbb{R}.$$

The set of such functions is denoted $M(\mathbb{R})$; it is a $\mathbb{C}$ -vector space.

The bound says f is bounded *and* decays at least like x^{-2} . Examples: $1/(1+|x|^n)$ for $n \geq 2$, and $e^{-a|x|}$ for $a > 0$. The decay is exactly enough to make the improper integral converge.

Proposition 35.1.2. If $f \in M(\mathbb{R})$ then $\int_{-\infty}^{\infty} f$ exists, and moreover $\int_{|x| \geq N} f \rightarrow 0$ as $N \rightarrow \infty$.

Proof. Put $I_N = \int_{-N}^N f$. For $M > N$,

$$|I_M - I_N| \leq \int_{N \leq |x| \leq M} |f| \leq A \int_{N \leq |x| \leq M} \frac{dx}{x^2} \leq \frac{2A}{N} \rightarrow 0.$$

So (I_N) is Cauchy and converges; the tail estimate is the same computation with $M = \infty$. □

Proposition 35.1.3 (Elementary properties of the integral). For $f, g \in M(\mathbb{R})$, $a, b \in \mathbb{C}$, $h \in \mathbb{R}$, $\delta > 0$: (i) $\int (af + bg) = a \int f + b \int g$; (ii) $\int f(x-h) dx = \int f(x) dx$ (translation invariance); (iii) $\delta \int f(\delta x) dx = \int f(x) dx$ (dilation); (iv) $\int |f(x-h) - f(x)| dx \rightarrow 0$ as $h \rightarrow 0$ (continuity in L^1).

Proof. (i) is linearity of the limit. For (ii), $\int_{-N}^N f(x-h) dx = \int_{-N-h}^{-h} f$, and the two end-slivers are $O(1/N^2)$, so the difference from $\int_{-N}^N f$ vanishes. (iii) is the substitution $u = \delta x$. For (iv) take $|h| \leq 1$; choose N with the tails $\int_{|x| \geq N} |f|$ and $\int_{|x| \geq N} |f(\cdot - h)|$ each $< \varepsilon/4$, then use uniform continuity of f on $[-N-1, N+1]$ to make the central part $< \varepsilon/2$. $\square$

Worked Example 35.1.4 (Moderate decrease is decay, not just boundedness). Decide which belong to $M(\mathbb{R})$: (a) $1/(1+x^2)$; (b) $1/(1+|x|)$; (c) $e^{-|x|}$; (d) $\sin x/(1+x^2)$; (e) 1 (constant).

Recognition. Check continuity and an $A/(1+x^2)$ bound.

Solution. (a) Yes, with $A = 1$. (b) No: $1/(1+|x|) \sim 1/|x|$ decays too slowly ($x \cdot \frac{1}{x} \not\rightarrow 0$ fast enough; $\int 1/(1+|x|) = \infty$). (c) Yes: $e^{-|x|} \leq A/(1+x^2)$ for suitable A since exponentials beat polynomials. (d) Yes: $|\sin x| \leq 1$ so it inherits (a)'s bound. (e) No: not decaying at all. *Check.* Only (a),(c),(d) have finite $\int_{\mathbb{R}} |\cdot|$. ■

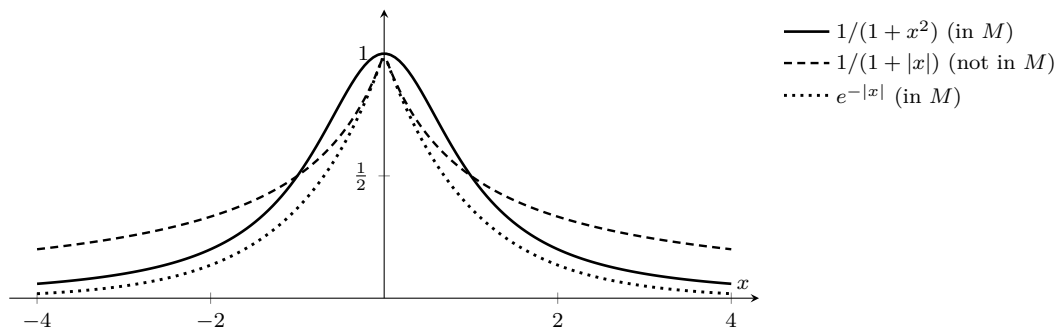


Figure 35.1. Moderate decrease: the x^{-2} envelope is integrable; the slowly-decaying $1/(1+|x|)$ is not.

The Schwartz space

Definition 35.1.5. The *Schwartz space* $\mathcal{S} = \mathcal{S}(\mathbb{R})$ is the set of C^∞ functions f that are *rapidly decreasing* together with all derivatives:

$$\sup_{x \in \mathbb{R}} |x|^k \left| f^{(\ell)}(x) \right| < \infty \quad \text{for all } k, \ell \geq 0.$$

$\mathcal{S}$ is a $\mathbb{C}$ -vector space, and it is closed under differentiation and under multiplication by polynomials: if $f \in \mathcal{S}$ then $f' \in \mathcal{S}$ and $xf(x) \in \mathcal{S}$.

Rapid decrease means: faster than any inverse power of x . The two closure properties are exactly what makes $\mathcal{S}$ the natural home for the Fourier transform, which (we will see) trades differentiation for multiplication by x .

Worked Example 35.1.6 (The Gaussian lives in $\mathcal{S}$). Show $f(x) = e^{-x^2} \in \mathcal{S}(\mathbb{R})$, but $e^{-|x|} \notin \mathcal{S}(\mathbb{R})$.

Recognition. Differentiate the Gaussian; check smoothness of $e^{-|x|}$ at 0.

Solution. By induction $f^{(\ell)}(x) = P_\ell(x)e^{-x^2}$ with P_ℓ a polynomial, so $x^k f^{(\ell)}(x) = x^k P_\ell(x)e^{-x^2} \rightarrow 0$ as $|x| \rightarrow \infty$ and is continuous, hence bounded: every seminorm is finite. The same argument shows $e^{-ax^2} \in \mathcal{S}$ for any $a > 0$. By contrast $e^{-|x|}$ decays rapidly but is *not* differentiable at 0 (left/right slopes ± 1), so it fails C^∞ and is not Schwartz. *Check.* $\sup_x x^2 e^{-x^2} = e^{-1}$ at $x = \pm 1 < \infty$. ■

Worked Example 35.1.7 (A compactly supported bump in $\mathcal{S}$). Show $b(x) = e^{-1/(1-x^2)}$ for $|x| < 1$ and $b(x) = 0$ for $|x| \geq 1$ is in $\mathcal{S}$.

Recognition. Compact support makes rapid decrease automatic; the issue is C^∞ at $x = \pm 1$.

Solution. On $(-1, 1)$, b is smooth. Near $x = 1$ write $u = 1 - x^2 \rightarrow 0^+$; then $b = e^{-1/u}$ and every derivative is a rational function of u times $e^{-1/u}$, and $e^{-1/u} \rightarrow 0$ faster than any power of u , so all one-sided derivatives at $|x| = 1$ vanish and match the zero extension. Thus $b \in C^\infty$, and being compactly supported it is rapidly decreasing with all derivatives. *Check.* b vanishes to infinite order at ± 1 . ■

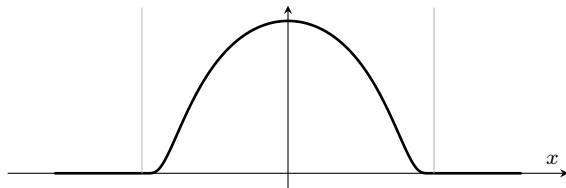


Figure 35.2. A C^∞ bump function: smooth, compactly supported, hence Schwartz. It vanishes to infinite order at $x = \pm 1$.

The Fourier transform on $\mathcal{S}$

Definition 35.1.8. For $f \in \mathcal{S}(\mathbb{R})$ the *Fourier transform* is $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i x \xi} dx$. We write $f(x) \rightarrow \hat{f}(\xi)$.

Since $|e^{-2\pi i x \xi}| = 1$ the integrand is rapidly decreasing, so the integral converges and $\hat{f}$ is bounded by $\int |f|$. The next proposition is the engine of the whole theory.

Proposition 35.1.9 (Operational rules). *If $f \in \mathcal{S}(\mathbb{R})$ and $h \in \mathbb{R}$, $\delta > 0$:*

- (i) $f(x + h) \rightarrow \hat{f}(\xi)e^{2\pi i h \xi}$;
- (ii) $f(x)e^{-2\pi i x h} \rightarrow \hat{f}(\xi + h)$;
- (iii) $f(\delta x) \rightarrow \delta^{-1}\hat{f}(\delta^{-1}\xi)$;
- (iv) $f'(x) \rightarrow 2\pi i \xi \hat{f}(\xi)$;
- (v) $-2\pi i x f(x) \rightarrow \frac{d}{d\xi} \hat{f}(\xi)$.

Thus, up to factors of $2\pi i$, the Fourier transform interchanges differentiation and multiplication by x .

$$\begin{array}{ccc} f & \xrightarrow{\mathcal{F}} & \hat{f} \\ \downarrow \frac{d}{dx} & & \downarrow \times 2\pi i \xi \\ f' & \xrightarrow{\mathcal{F}} & \hat{f}' \end{array}$$

Figure 35.3. Rule 35.1.9(iv): the Fourier transform $\mathcal{F}$ conjugates differentiation $f \mapsto f'$ into multiplication by $2\pi i \xi$, i.e. $\hat{f}' = 2\pi i \xi \hat{f}$. The square commutes on $\mathcal{S}(\mathbb{R})$.

Proof. (i) is translation invariance of the integral applied to $\int f(x + h)e^{-2\pi i x \xi} dx$ after the substitution $x \mapsto x - h$; (ii) is immediate from the definition; (iii) is the dilation rule 35.1.3(iii). For (iv), integrate by parts:

$$\int_{-N}^N f'(x)e^{-2\pi i x \xi} dx = [f(x)e^{-2\pi i x \xi}]_{-N}^N + 2\pi i \xi \int_{-N}^N f(x)e^{-2\pi i x \xi} dx,$$

and the boundary term $\rightarrow 0$ since $f \in \mathcal{S}$. For (v) we must show $\hat{f}$ is differentiable with the stated derivative; this is justified by dominated estimates because both $f(x)$ and $xf(x)$ are rapidly decreasing, allowing differentiation under the integral sign in $\frac{d}{d\xi} \int f(x)e^{-2\pi i x \xi} dx = \int f(x)(-2\pi i x)e^{-2\pi i x \xi} dx$. □

Theorem 35.1.10 ($\mathcal{S}$ is invariant). *If $f \in \mathcal{S}(\mathbb{R})$ then $\hat{f} \in \mathcal{S}(\mathbb{R})$.*

Proof. We must bound $\sup_{\xi} \left| \xi^k \left(\frac{d}{d\xi} \right)^\ell \hat{f}(\xi) \right|$ for all k, ℓ . By 35.1.9(v), $\left(\frac{d}{d\xi} \right)^\ell \hat{f}$ is the transform of $(-2\pi i x)^\ell f(x)$; by 35.1.9(iv) applied k times, multiplying by ξ^k corresponds (up to $(2\pi i)^{-k}$) to taking the k -th derivative. Hence $\xi^k \left(\frac{d}{d\xi} \right)^\ell \hat{f}(\xi)$ is, up to constants, the Fourier transform of $\left(\frac{d}{dx} \right)^k [(-2\pi i x)^\ell f(x)]$, which is Schwartz, hence has bounded transform. So every seminorm of $\hat{f}$ is finite. □

Worked Example 35.1.11 (The Fourier transform of the box). Compute $\hat{f}$ for $f = \mathbf{1}_{[-1,1]}$ (the box of half-width 1).

Recognition. A piecewise-constant function; integrate the exponential directly.

Solution. $\hat{f}(\xi) = \int_{-1}^1 e^{-2\pi i x \xi} dx = \left[\frac{e^{-2\pi i x \xi}}{-2\pi i \xi} \right]_{-1}^1 = \frac{e^{-2\pi i \xi} - e^{2\pi i \xi}}{-2\pi i \xi} = \frac{\sin 2\pi \xi}{\pi \xi}$. *Check.* As $\xi \rightarrow 0$, $\hat{f}(\xi) \rightarrow 2 = \int_{-1}^1 1 dx$, the length of the support. $\hat{f}$ is continuous and tends to 0, but is *not* of moderate decrease (it decays only like $1/|\xi|$); the box is not even continuous. ■

Worked Example 35.1.12 (The triangle (tent) function transforms to sinc^2). Compute $\hat{g}$ for $g(x) = (1 - |x|)_+$ (the tent on $[-1, 1]$).

Recognition. g is the convolution of two boxes; or integrate by parts twice.

Solution. $\hat{g}(\xi) = \int_{-1}^1 (1 - |x|) e^{-2\pi i x \xi} dx = 2 \int_0^1 (1 - x) \cos(2\pi x \xi) dx = \left(\frac{\sin \pi \xi}{\pi \xi} \right)^2$ (integrate by parts twice, or note $g = \frac{1}{2} \mathbf{1}_{[-1,1]} * \mathbf{1}_{[-1,1]}$ rescaled, so $\hat{g} = (\hat{\mathbf{1}})^2$ -type product). *Check.* $\hat{g}(0) = 1 = \int g$, and $\hat{g} \geq 0$ with $\hat{g}(\xi) = O(1/\xi^2)$, so $\hat{g}$ is of moderate decrease — matching that g is continuous (one degree smoother than the box). ■

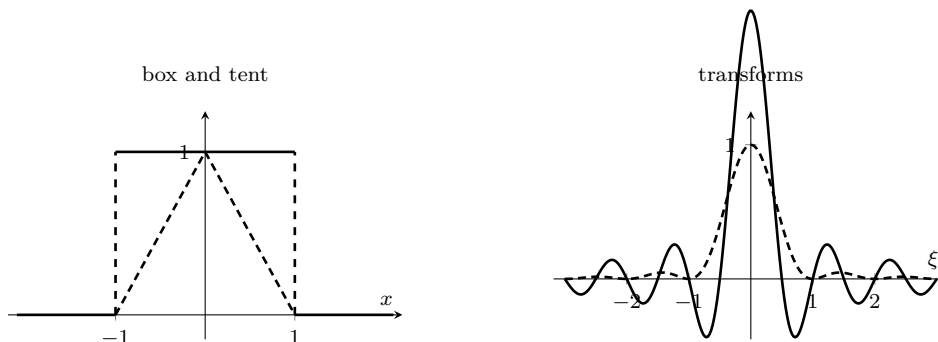


Figure 35.4. Box $\rightarrow \sin 2\pi \xi / (\pi \xi)$ (oscillating, slow decay); tent $\rightarrow (\sin \pi \xi / \pi \xi)^2$ (nonnegative, $O(\xi^{-2})$). Smoother f gives faster-decaying $\hat{f}$.

Worked Example 35.1.13 (Transform of a one-sided exponential). Compute $\hat{f}$ for $f(x) = e^{-2\pi a x}$ for $x > 0$ and 0 for $x \leq 0$, with $a > 0$.

Recognition. A one-sided decaying exponential; integrate the single exponential on $(0, \infty)$.

Solution. $\hat{f}(\xi) = \int_0^\infty e^{-2\pi a x} e^{-2\pi i x \xi} dx = \int_0^\infty e^{-2\pi(a + i\xi)x} dx = \frac{1}{2\pi(a + i\xi)}$. *Check.* At $\xi = 0$, $\hat{f}(0) = \frac{1}{2\pi a} = \int_0^\infty e^{-2\pi a x} dx$; and $|\hat{f}(\xi)| \sim \frac{1}{2\pi|\xi|}$, only moderate-ish decay, reflecting the jump of f at 0. ■

Worked Example 35.1.14 (Transform of the two-sided exponential is a Lorentzian). Compute $\hat{f}$ for $f(x) = e^{-2\pi a |x|}$, $a > 0$.

Recognition. Even function; split at 0 and add the two one-sided pieces.

Solution. $\hat{f}(\xi) = \int_0^\infty e^{-2\pi a x} e^{-2\pi i x \xi} dx + \int_0^\infty e^{-2\pi a x} e^{2\pi i x \xi} dx = \frac{1}{2\pi(a + i\xi)} + \frac{1}{2\pi(a - i\xi)} = \frac{a}{\pi(a^2 + \xi^2)}$. *Check.* $\hat{f}(0) = \frac{1}{\pi a} = \int e^{-2\pi a |x|} dx = 2 \cdot \frac{1}{2\pi a}$; the transform is a Lorentzian of moderate decrease, matching that f is continuous. ■

Worked Example 35.1.15 (Differentiation rule on a non-example). Why does rule 35.1.9(iv) fail to apply to $f(x) = \mathbf{1}_{[-1,1]}$?

Recognition. The rule needs f' in the integrable/Schwartz setting; check differentiability.

Solution. $\mathbf{1}_{[-1,1]}$ is not even continuous, so f' does not exist as a function (only as distributional point masses at ± 1). The integration-by-parts that proves (iv) produces boundary terms that do *not* vanish. Formally $\hat{f}' = 2\pi i \xi \hat{f}$ would read $\widehat{\delta_{-1} - \delta_1} = 2\pi i \xi \cdot \frac{\sin 2\pi \xi}{\pi \xi} = 2i \sin 2\pi \xi$, which is the (distributional) truth but lies outside $\mathcal{S}$. *Check.* The rule is an $\mathcal{S}$ -statement; on rougher f it must be read distributionally. ■

Worked Example 35.1.16 (Modulation shifts the transform). Using rule 35.1.9(ii), find the transform of $g(x) = e^{-\pi x^2} \cos(2\pi hx)$.

Recognition. Write $\cos$ as a sum of exponentials; each modulation shifts $\hat{f}$.

Solution. $g(x) = \frac{1}{2}e^{-\pi x^2}(e^{2\pi i h x} + e^{-2\pi i h x})$, and $e^{-\pi x^2}e^{\pm 2\pi i h x}(\xi) = e^{-\pi(\xi \mp h)^2}$ by 35.1.9(ii). So $\hat{g}(\xi) = \frac{1}{2}(e^{-\pi(\xi-h)^2} + e^{-\pi(\xi+h)^2})$. *Check.* A Gaussian carrier modulated at frequency h has a transform that is two Gaussian bumps centered at $\pm h$ — a textbook spectrum. ■

Caution. The bare definition gives $\hat{f}$ continuous with $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$ (Riemann–Lebesgue), but *nothing* forces $\hat{f}$ to have moderate decrease for f merely continuous — as the box shows. Restricting to $\mathcal{S}$ buys decay of $\hat{f}$ of every order; that is precisely why the inversion integral $\int \hat{f}(\xi)e^{2\pi i x \xi}$ will converge.

Exercises (Stein, Ch.5 — elementary theory).

- (Riemann sums to the inversion integral) Corollary 2.3 of Ch.2 gives a simplified inversion. Suppose f is continuous, supported on $[-M, M]$, with $\hat{f}$ of moderate decrease. (a) Fix L with $L/2 > M$; show $f(x) = \sum_n a_n(L)e^{2\pi i n x/L}$ where $a_n(L) = \frac{1}{L} \int_{-L/2}^{L/2} f(x)e^{-2\pi i n x/L} dx = \frac{1}{L}\hat{f}(n/L)$; equivalently $f(x) = \delta \sum_n \hat{f}(n\delta)e^{2\pi i n \delta x}$ with $\delta = 1/L$. (b) If F is continuous and of moderate decrease, $\int_{\mathbb{R}} F(\xi) d\xi = \lim_{\delta \rightarrow 0^+} \delta \sum_n F(\delta n)$. (c) Conclude $f(x) = \int_{\mathbb{R}} \hat{f}(\xi)e^{2\pi i x \xi} d\xi$.
- (Box and tent transforms) For $f = \mathbf{1}_{[-1,1]}$ and $g(x) = (1 - |x|)_+$ show $\hat{f}(\xi) = \frac{\sin 2\pi \xi}{\pi \xi}$ and $\hat{g}(\xi) = \left(\frac{\sin \pi \xi}{\pi \xi}\right)^2$, with $\hat{f}(0) = 2$, $\hat{g}(0) = 1$.
- (Decay of $\hat{f}$ vs. continuity of f) (a) If f is of moderate decrease and $\hat{f}(\xi) = O(|\xi|^{-1-\alpha})$, $0 < \alpha < 1$, prove f is Hölder of order α : $|f(x+h) - f(x)| \leq M|h|^\alpha$. (b) If f is continuous, vanishes for $|x| \geq 1$, $f(0) = 0$, and $f = 1/\log(1/|x|)$ near 0, prove $\hat{f}$ is not of moderate decrease — indeed no $\varepsilon > 0$ gives $\hat{f}(\xi) = O(|\xi|^{-1-\varepsilon})$.
- (Bump functions) (a) For $a < b$, let $f(x) = e^{-1/(x-a)}e^{-1/(b-x)}$ on (a, b) and 0 outside; show $f \in C^\infty(\mathbb{R})$. (b) Prove there is $F \in C^\infty$ with $F = 0$ for $x \leq a$, $F = 1$ for $x \geq b$, strictly increasing on $[a, b]$ (take $F = c \int_{-\infty}^x f$). (c) For small $\delta > 0$ build $g \in C^\infty$ with $g = 0$ off $[a, b]$, $g = 1$ on $[a + \delta, b - \delta]$, strictly monotone on the two transition intervals.
- (Continuity and injectivity) Suppose f is continuous and of moderate decrease. (a) Prove $\hat{f}$ is continuous and $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$; use $\hat{f}(\xi) = \frac{1}{2} \int [f(x) - f(x - \frac{1}{2\xi})]e^{-2\pi i x \xi} dx$. (b) If $\hat{f} \equiv 0$ then $f \equiv 0$ (verify the multiplication formula holds against every $g \in \mathcal{S}$).
- (Order-four and self-dual functions) The Gaussian is its own transform. Generate further functions equal to their transform up to a constant; determine the possible constants by proving $\mathcal{F}^4 = I$ (so the eigenvalues are $1, i, -1, -i$). Here $\mathcal{F}(f) = \hat{f}$, $\mathcal{F}^4 = \mathcal{F} \circ \mathcal{F} \circ \mathcal{F} \circ \mathcal{F}$.
- (Convolution preserves moderate decrease) Prove the convolution of two functions of moderate decrease is again of moderate decrease. [Split $\int f(x-y)g(y)dy$ into $|y| \leq |x|/2$ and $|y| \geq |x|/2$.]
- (A vanishing condition forcing $f = 0$) If f is continuous, of moderate decrease, and $\int_{-\infty}^{\infty} f(y)e^{-y^2}e^{2xy}dy = 0$ for all $x \in \mathbb{R}$, prove $f = 0$. [Consider $f * e^{-x^2}$.]
- (Fejer kernel on the line) For f of moderate decrease, $\int_{-R}^R (1 - \frac{|\xi|}{R})\hat{f}(\xi)e^{2\pi i x \xi} d\xi = (f * F_R)(x)$ where $F_R(t) = R(\frac{\sin \pi t R}{\pi t R})^2$ for $t \neq 0$ and $F_R(0) = R$. Show $\{F_R\}$ is a family of good kernels as $R \rightarrow \infty$, hence the left side $\rightarrow f$ uniformly (Fejer's theorem on $\mathbb{R}$).
- (Landau kernels and Weierstrass) Define the Landau kernels $L_n(x) = c_n(1 - x^2)^n$ on $[-1, 1]$ and 0 outside, c_n chosen so $\int L_n = 1$. Prove $\{L_n\}$ is a family of good kernels as $n \rightarrow \infty$ (first show $c_n \geq (n+1)/2 \dots$ up to constant), and deduce: if f is continuous and supported in $[-1/2, 1/2]$ then $(f * L_n)$ is a sequence of polynomials converging uniformly to f .

Additional exercises.

- Decide which are of moderate decrease, with justification: (a) $\frac{1}{1+x^4}$; (b) $\frac{\cos x}{1+|x|}$; (c) $e^{-x^2} \sin(e^x)$; (d) $\frac{1}{1+x^2 \log(2+|x|)}$.
- Compute $\hat{f}$ directly for $f(x) = e^{-2\pi a|x|}$, $a > 0$, and check that $\hat{f}(0) = \int f$. Then verify rule 35.1.9(iii) (dilation) numerically by comparing $\widehat{f(2\cdot)}$ with $\frac{1}{2}\hat{f}(\frac{\xi}{2})$.
- Using only the operational rules, find the Fourier transform of $x^2 e^{-\pi x^2}$. [Apply $-2\pi i x \cdot$ twice, i.e. differentiate $\hat{f}$ of the Gaussian twice.]
- Show that $\mathcal{S}(\mathbb{R})$ is not closed under the map $f \mapsto f(x)/x$ in general, but that $f \mapsto (f(x) - f(0)e^{-\pi x^2})/x$ can land in $\mathcal{S}$. Give one example each way.
- Let g be the half-width- $\frac{1}{2}$ box $\mathbf{1}_{[-1/2, 1/2]}$. Prove $\widehat{g}(\xi) = \frac{\sin \pi \xi}{\pi \xi}$ and deduce $\widehat{g * g} = \left(\frac{\sin \pi \xi}{\pi \xi}\right)^2$ without computing $g * g$ explicitly.
- (Non-example) Exhibit a continuous f of moderate decrease whose transform $\hat{f}$ is *not* of moderate decrease. [Hint: a function with a corner, e.g. the tent — or sharper, something only continuous.]
- (Diagram problem) Sketch, on the same axes, $f(x) = e^{-\pi x^2}$ and $g(x) = e^{-\pi x^2} \cos(6\pi x)$, and sketch their transforms. Explain in one sentence why $\hat{g}$ has two bumps while $\hat{f}$ has one.
- Prove that if $f \in \mathcal{S}$ is even (resp. odd) then $\hat{f}$ is even (resp. odd), and real-valued (resp. purely imaginary). Verify on $e^{-\pi x^2}$ and $x e^{-\pi x^2}$.

35.2 The Fourier Inversion and Plancherel Theorems

The Gaussian and its good kernels

The proof of inversion rests on one miraculous example: the Gaussian is its own Fourier transform. We first normalize.

Lemma 35.2.1. $\int_{-\infty}^{\infty} e^{-\pi x^2} dx = 1.$

Proof. Squaring and going to polar coordinates,

$$\left(\int_{\mathbb{R}} e^{-\pi x^2} dx\right)^2 = \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-\pi(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-\pi r^2} r dr d\theta = 2\pi \left[-\frac{1}{2\pi} e^{-\pi r^2}\right]_0^{\infty} = 1. \quad \square$$

Theorem 35.2.2 (The Gaussian is self-dual). *If $f(x) = e^{-\pi x^2}$ then $\hat{f}(\xi) = f(\xi) = e^{-\pi \xi^2}$.*

Proof. Let $F(\xi) = \hat{f}(\xi)$; then $F(0) = \int e^{-\pi x^2} = 1$ by Lemma 35.2.1. Since $f'(x) = -2\pi x f(x)$, rule 35.1.9(v) gives $F'(\xi) = \int f(x)(-2\pi i x) e^{-2\pi i x \xi} dx = i \int f'(x) e^{-2\pi i x \xi} dx$, and rule 35.1.9(iv) turns the last integral into $i(2\pi i \xi) F(\xi) = -2\pi \xi F(\xi)$. So $F' = -2\pi \xi F$. Then $G(\xi) = F(\xi) e^{\pi \xi^2}$ has $G' \equiv 0$, hence G is constant; $G(0) = F(0) = 1$, so $F(\xi) = e^{-\pi \xi^2}$. $\square$

Corollary 35.2.3. *For $\delta > 0$, $K_\delta(x) = \delta^{-1/2} e^{-\pi x^2/\delta}$ has $\widehat{K_\delta}(\xi) = e^{-\pi \delta \xi^2}$.*

Proof. Apply dilation 35.1.9(iii) to Theorem 35.2.2 with the substitution $\delta \mapsto \delta^{-1/2}$. $\square$

As $\delta \rightarrow 0$ the function K_δ spikes at the origin while $\widehat{K_\delta}$ flattens: K_δ and its transform *cannot both* be concentrated — the first glimpse of the uncertainty principle.

Theorem 35.2.4 (Gaussian good kernels). $\{K_\delta\}_{\delta>0}$, $K_\delta(x) = \delta^{-1/2} e^{-\pi x^2/\delta}$, is a family of good kernels as $\delta \rightarrow 0$: (i) $\int_{\mathbb{R}} K_\delta = 1$; (ii) $\int_{\mathbb{R}} |K_\delta| \leq M$; (iii) for each $\eta > 0$, $\int_{|x|>\eta} |K_\delta| \rightarrow 0$ as $\delta \rightarrow 0$.

Proof. (i) $\int K_\delta = \widehat{K_\delta}(0) = 1$ by Corollary 35.2.3. (ii) $K_\delta \geq 0$ so $\int |K_\delta| = 1$. (iii) substitute $y = x/\delta^{1/2}$: $\int_{|x|>\eta} K_\delta = \int_{|y|>\eta/\delta^{1/2}} e^{-\pi y^2} dy \rightarrow 0$ since the lower limit $\rightarrow \infty$. $\square$

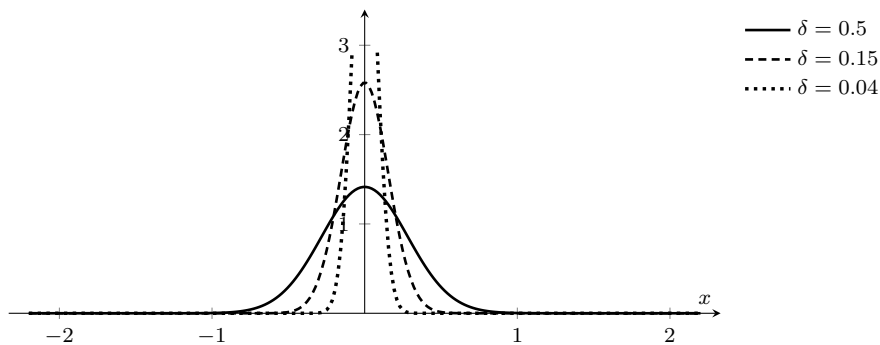


Figure 35.5. The Gaussian kernels K_δ concentrate at the origin as $\delta \rightarrow 0$ (area always 1); the tallest spike is clipped at the top. An approximate identity.

Convolution

Definition 35.2.5. For $f, g \in \mathcal{S}(\mathbb{R})$, $(f * g)(x) = \int_{-\infty}^{\infty} f(x-t)g(t) dt$.

Corollary 35.2.6 (Approximation by good kernels). *If $f \in \mathcal{S}(\mathbb{R})$ then $(f * K_\delta)(x) \rightarrow f(x)$ uniformly in x as $\delta \rightarrow 0$.*

Proof. f is uniformly continuous (continuous and $\rightarrow 0$ at infinity). Using $\int K_\delta = 1$,

$$(f * K_\delta)(x) - f(x) = \int K_\delta(t)[f(x-t) - f(x)] dt,$$

and $|(f * K_\delta)(x) - f(x)| \leq \int_{|t| \leq \eta} + \int_{|t| > \eta}$. The first is small by uniform continuity ($\int K_\delta = 1$); the second is small by good-kernel property (iii) and boundedness of f . Both bounds are uniform in x . $\square$

Proposition 35.2.7 (Convolution algebra). *For $f, g \in \mathcal{S}(\mathbb{R})$: (i) $f * g \in \mathcal{S}(\mathbb{R})$; (ii) $f * g = g * f$; (iii) $\widehat{f * g}(\xi) = \widehat{f}(\xi)\widehat{g}(\xi)$.*

Proof. (i) For any ℓ , $\sup_x |x|^\ell |g(x-y)| \leq A_\ell(1+|y|)^\ell$ (split $|x| \leq 2|y|$ and $|x| \geq 2|y|$), so $\sup_x |x|^\ell |(f * g)(x)| \leq A_\ell \int |f(y)| (1+|y|)^\ell dy < \infty$; the same bounds for derivatives, using $(\frac{d}{dx})^k(f * g) = f * (\frac{d}{dx})^k g$, give $f * g \in \mathcal{S}$. (ii) is the substitution $u = x-t$. (iii) Apply the change-of-order discussion (below) to $F(x, y) = f(y)g(x-y)e^{-2\pi i x \xi}$: $\int F dx = \widehat{f * g}(\xi)$ one way, and $\widehat{f}(\xi)\widehat{g}(\xi)$ the other. $\square$

$$\begin{array}{ccc} (f, g) & \xrightarrow{*} & f * g \\ \mathcal{F} \times \mathcal{F} \downarrow & & \downarrow \mathcal{F} \\ (\widehat{f}, \widehat{g}) & \xrightarrow{\times} & \widehat{f} \widehat{g} \end{array}$$

Figure 35.6. The convolution theorem 35.2.7(iii): $\mathcal{F}$ carries convolution to pointwise product, $\widehat{f * g} = \widehat{f} \widehat{g}$. The square commutes on $\mathcal{S}(\mathbb{R})$.

Worked Example 35.2.8 (Convolution of two boxes is a tent). Compute $\mathbf{1}_{[-1/2, 1/2]} * \mathbf{1}_{[-1/2, 1/2]}$.

Recognition. Slide one box across the other; the overlap length is the answer.

Solution. $(\mathbf{1} * \mathbf{1})(x) = \int \mathbf{1}_{[-1/2, 1/2]}(x-t) \mathbf{1}_{[-1/2, 1/2]}(t) dt = |[x - \frac{1}{2}, x + \frac{1}{2}] \cap [-\frac{1}{2}, \frac{1}{2}]| = (1 - |x|)_+$, the tent on $[-1, 1]$. *Check (transform side):* $\widehat{\mathbf{1}_{[-1/2, 1/2]}}(\xi) = \sin \pi \xi / (\pi \xi)$, and indeed the tent's transform is its square $(\sin \pi \xi / \pi \xi)^2$, confirming $\widehat{f * g} = \widehat{f} \widehat{g}$. ■

Worked Example 35.2.9 (Convolving Gaussians). Show $K_\delta * K_{\delta'} = K_{\delta + \delta'}$ for the kernels of Corollary 35.2.3.

Recognition. Use $\widehat{f * g} = \widehat{f} \widehat{g}$ and that products of Gaussians are Gaussians.

Solution. $\widehat{K_\delta * K_{\delta'}}(\xi) = e^{-\pi \delta \xi^2} e^{-\pi \delta' \xi^2} = e^{-\pi(\delta + \delta') \xi^2} = \widehat{K_{\delta + \delta'}}(\xi)$. Since the transform is injective on $\mathcal{S}$ (inversion, below), $K_\delta * K_{\delta'} = K_{\delta + \delta'}$. *Check.* The variances add — δ plays the role of variance, and convolving independent Gaussians adds variances. ■

The multiplication formula and Fourier inversion

Proposition 35.2.10 (Multiplication formula). *If $f, g \in \mathcal{S}(\mathbb{R})$ then $\int_{\mathbb{R}} f(x)\hat{g}(x) dx = \int_{\mathbb{R}} \hat{f}(y)g(y) dy$.*

Proof. Apply Fubini (justified by the moderate-decrease bound $|F(x, y)| \leq A/[(1+x^2)(1+y^2)]$) to $F(x, y) = f(x)g(y)e^{-2\pi ixy}$. Integrating in y first gives $f(x)\hat{g}(x)$; integrating in x first gives $\hat{f}(y)g(y)$. $\square$

Theorem 35.2.11 (Fourier inversion). *If $f \in \mathcal{S}(\mathbb{R})$ then $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i x \xi} d\xi$.*

Proof. Step 1: $f(0) = \int \hat{f}$. Let $G_{\delta}(x) = e^{-\pi\delta x^2}$ so $\widehat{G_{\delta}}(\xi) = K_{\delta}(\xi)$ (Corollary 35.2.3). The multiplication formula with $g = G_{\delta}$ gives $\int f(x)K_{\delta}(x) dx = \int \hat{f}(\xi)G_{\delta}(\xi) d\xi$. As $\delta \rightarrow 0$ the left side $\rightarrow f(0)$ (good kernel, Corollary 35.2.6 at $x = 0$) and the right side $\rightarrow \int \hat{f}(\xi) d\xi$ (since $G_{\delta} \rightarrow 1$ and $\hat{f} \in \mathcal{S}$). Hence $f(0) = \int \hat{f}$. *Step 2: general x .* Put $F(y) = f(y+x)$. By 35.1.9(i), $\hat{F}(\xi) = \hat{f}(\xi)e^{2\pi i x \xi}$, so Step 1 applied to F gives $f(x) = F(0) = \int \hat{F}(\xi) d\xi = \int \hat{f}(\xi)e^{2\pi i x \xi} d\xi$. $\square$

Corollary 35.2.12. *Define $\mathcal{F}f(\xi) = \int f(x)e^{-2\pi i x \xi} dx$ and $\mathcal{F}^*g(x) = \int g(\xi)e^{2\pi i x \xi} d\xi$. Then $\mathcal{F}^*\mathcal{F} = \mathcal{F}\mathcal{F}^* = I$ on $\mathcal{S}(\mathbb{R})$; in particular $\mathcal{F}$ is a bijection of $\mathcal{S}$ onto itself, and $\mathcal{F}(f)(y) = \mathcal{F}^*(f)(-y)$.*

$$\mathcal{S}(\mathbb{R}) \xrightleftharpoons[\mathcal{F}^*]{\mathcal{F}} \mathcal{S}(\mathbb{R})$$

Figure 35.7. Corollary 35.2.12: on $\mathcal{S}(\mathbb{R})$ the transform $\mathcal{F}$ and the inverse transform $\mathcal{F}^*$ are mutually inverse, $\mathcal{F}^*\mathcal{F} = \mathcal{F}\mathcal{F}^* = I$, so $\mathcal{F}$ is an isomorphism of $\mathcal{S}(\mathbb{R})$ onto itself.

Worked Example 35.2.13 (The Fourier transform has order four). Show $\mathcal{F}^4 = I$, and find the eigenvalues of $\mathcal{F}$.

Recognition. Inversion says $\mathcal{F}^2$ is the reflection $f(x) \mapsto f(-x)$.

Solution. By inversion $\mathcal{F}^2 f(x) = f(-x)$, so $\mathcal{F}^4 f(x) = f(x)$, i.e. $\mathcal{F}^4 = I$. Hence every eigenvalue λ satisfies $\lambda^4 = 1$, so $\lambda \in \{1, -1, i, -i\}$. *Check.* The Gaussian $e^{-\pi x^2}$ has $\mathcal{F}$ -eigenvalue $+1$ (Theorem 35.2.2); $xe^{-\pi x^2}$ has eigenvalue $-i$ (since multiplying by x and transforming brings a factor i). $\blacksquare$

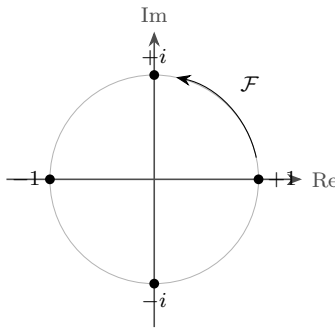


Figure 35.8. $\mathcal{F}^4 = I$: the eigenvalues are the fourth roots of unity $\{1, i, -1, -i\}$; one application of $\mathcal{F}$ rotates by 90° .

Plancherel's theorem

Equip $\mathcal{S}$ with $(f, g) = \int_{\mathbb{R}} f\bar{g}$ and $\|f\| = (\int |f|^2)^{1/2}$.

Theorem 35.2.14 (Plancherel). *If $f \in \mathcal{S}(\mathbb{R})$ then $\|\hat{f}\| = \|f\|$.*

Proof. Set $f^b(x) = \overline{f(-x)}$; then $\widehat{f^b}(\xi) = \overline{\hat{f}(\xi)}$. Let $h = f * f^b$. By Proposition 35.2.7(iii), $\hat{h}(\xi) = \hat{f}(\xi)\overline{\hat{f}(\xi)} = |\hat{f}(\xi)|^2$, while $h(0) = \int f(0-t)\overline{f(-t)} \cdot (\dots) dt = \int |f(x)|^2 dx$ (substitute). By inversion at $x = 0$, $h(0) = \int \hat{h}(\xi) d\xi$, i.e. $\int |f|^2 = \int |\hat{f}|^2$. $\square$

Worked Example 35.2.15 (Parseval via box and sinc: a definite integral). Use Plancherel on $f = \mathbf{1}_{[-1/2, 1/2]}$ to evaluate $\int_{-\infty}^{\infty} \left(\frac{\sin \pi \xi}{\pi \xi}\right)^2 d\xi$.

Recognition. $\|\hat{f}\|^2 = \|f\|^2$, and $\hat{f}(\xi) = \sin \pi \xi / (\pi \xi)$.

Solution. $\|f\|^2 = \int_{-1/2}^{1/2} 1 dx = 1$, so $\int_{\mathbb{R}} \left(\frac{\sin \pi \xi}{\pi \xi}\right)^2 d\xi = 1$. *Check.* Substituting $u = \pi \xi$ gives the classical $\int_{\mathbb{R}} (\sin u / u)^2 du = \pi$. ■

Worked Example 35.2.16 (Energy of a Gaussian, two ways). Verify Plancherel for $f(x) = e^{-\pi x^2}$.

Recognition. Compute both norms by Gaussian integrals.

Solution. $\|f\|^2 = \int e^{-2\pi x^2} dx = \frac{1}{\sqrt{2}} \int e^{-\pi y^2} dy = \frac{1}{\sqrt{2}}$ (set $y = \sqrt{2}x$). Since $\hat{f} = f$, $\|\hat{f}\|^2 = \|f\|^2 = \frac{1}{\sqrt{2}}$ trivially. *Check.* Both equal $2^{-1/2}$, confirming $\|\hat{f}\| = \|f\|$. ■

Extension to moderate decrease, and Weierstrass

The inversion and Plancherel identities extend to $f \in M(\mathbb{R})$ provided $\hat{f}$ is also of moderate decrease: the key point is that $f * g$ of two moderate-decrease functions is again of moderate decrease, with $\widehat{f * g} = \hat{f}\hat{g}$, and the multiplication formula persists.

Theorem 35.2.17 (Weierstrass approximation). Let f be continuous on $[a, b]$. For every $\varepsilon > 0$ there is a polynomial P with $\sup_{[a, b]} |f - P| < \varepsilon$.

Proof. Extend f to a continuous compactly supported g on $\mathbb{R}$, vanishing outside some $[-M, M]$, $|g| \leq B$. Since $\{K_\delta\}$ are good kernels and g is uniformly continuous, choose δ_0 with $\sup_x |g - g * K_{\delta_0}| < \varepsilon/2$. Truncate the power series of $e^{-\pi x^2 / \delta_0}$ to a partial sum $R(x) = \delta_0^{-1/2} \sum_{n=0}^N (-\pi x^2 / \delta_0)^n / n!$, uniformly close to K_{δ_0} on $[-2M, 2M]$ so that $\sup_{[-M, M]} |g * K_{\delta_0} - g * R| < \varepsilon/2$. Then $g * R$ is a polynomial in x (expand $R(x - t)$) and $\sup_{[a, b]} |f - g * R| < \varepsilon$. □

Worked Example 35.2.18 (Inversion as a sanity check on the box). Apply the inversion formula formally to recover $\mathbf{1}_{[-1, 1]}$ from its transform.

Recognition. Inversion: $f(x) = \int \hat{f}(\xi) e^{2\pi i x \xi} d\xi$ with $\hat{f}(\xi) = \sin 2\pi \xi / (\pi \xi)$.

Solution. The Dirichlet integral $\int_{\mathbb{R}} \frac{\sin 2\pi \xi}{\pi \xi} e^{2\pi i x \xi} d\xi$ equals 1 for $|x| < 1$, 0 for $|x| > 1$, and $\frac{1}{2}$ at $x = \pm 1$ (the midpoint of the jump). *Check.* The value $\frac{1}{2}$ at the discontinuity is exactly the Fourier-series/Dirichlet behavior; pointwise inversion holds where f is continuous, but the box is not Schwartz, so this is the moderate-decrease/principal-value reading. ■

Worked Example 35.2.19 (Plancherel for the two-sided exponential). Verify $\|f\| = \|\hat{f}\|$ for $f(x) = e^{-2\pi a|x|}$ ($a > 0$), whose transform is the Lorentzian $\hat{f}(\xi) = \frac{a}{\pi(a^2 + \xi^2)}$.

Recognition. Compute both L^2 norms by elementary integrals.

Solution. $\|f\|^2 = \int e^{-4\pi a|x|} dx = 2 \cdot \frac{1}{4\pi a} = \frac{1}{2\pi a}$. On the transform side $\|\hat{f}\|^2 = \frac{a^2}{\pi^2} \int \frac{d\xi}{(a^2 + \xi^2)^2} = \frac{a^2}{\pi^2} \cdot \frac{\pi}{2a^3} = \frac{1}{2\pi a}$.

Check. Both equal $\frac{1}{2\pi a}$, so $\|f\| = \|\hat{f}\|$. ■

Worked Example 35.2.20 (A self-dual function beyond the Gaussian). Exhibit a nonzero f with $\hat{f} = f$ other than $e^{-\pi x^2}$.

Recognition. Symmetrize over the order-four group: $f + \mathcal{F}f + \mathcal{F}^2f + \mathcal{F}^3f$ is $\mathcal{F}$ -invariant.

Solution. Start from any g and set $f = g + \hat{g} + \mathcal{F}^2g + \mathcal{F}^3g$. Then $\mathcal{F}f = \hat{g} + \mathcal{F}^2g + \mathcal{F}^3g + \mathcal{F}^4g = \hat{g} + \mathcal{F}^2g + \mathcal{F}^3g + g = f$, using $\mathcal{F}^4 = I$. Concretely with $g = xe^{-\pi x^2}$ (eigenvalue $-i$) one builds eigenfunctions; e.g. the Hermite function $h_0 = e^{-\pi x^2}$ has eigenvalue $+1$. *Check.* Every eigenvalue is a fourth root of unity, and the Hermite functions h_k realize all four eigenvalues i^{-k} . ■

Worked Example 35.2.21 (Why convolution with a polynomial-kernel is a polynomial). Explain why $(g * R)(x) = \int_{-M}^M g(t)R(x-t) dt$ is a polynomial in x when R is.

Recognition. Expand $R(x-t)$ by the binomial theorem and pull x -powers out of the integral.

Solution. Write

$$\begin{aligned} R(x-t) &= \sum_n c_n (x-t)^n = \sum_n c_n \sum_k \binom{n}{k} x^k (-t)^{n-k} \\ &= \sum_k x^k \left[\sum_n c_n \binom{n}{k} (-1)^{n-k} \int_{-M}^M g(t) t^{n-k} dt \right]. \end{aligned}$$

The bracket is a constant (a finite combination of moments of g), so $g * R = \sum_k a_k x^k$ is a polynomial. *Check.* The degree of $g * R$ equals the degree of R . ■

Exercises (Stein, Ch.5 — inversion, Plancherel, uncertainty).

- (Both compactly supported forces 0) If f is continuous on $\mathbb{R}$, then f and $\hat{f}$ cannot both be compactly supported unless $f = 0$. [Assume f supported in $[0, 1/2]$; expand f in a Fourier series on $[0, 1]$, so f is a trigonometric polynomial, hence $\hat{f}$ cannot have compact support unless $f = 0$.] This is in the spirit of the uncertainty principle.
- (Quantitative uncertainty by mass) Say the preponderance of the mass of F lies in an interval I centered at 0 if $\int_I x^2 |F(x)|^2 dx \geq \frac{1}{2} \int_{\mathbb{R}} x^2 |F|^2 dx$. If $f \in \mathcal{S}$, (this) holds with $F = f$, $I = I_1$ and with $F = \hat{f}$, $I = I_2$, and $L_j = |I_j|$, prove $L_1 L_2 \geq \frac{1}{2\pi}$ (a similar conclusion if the intervals are not centered at 0).
- (The Hermite/harmonic-oscillator operator) Let $L = -\frac{d^2}{dx^2} + x^2$ act on $\mathcal{S}$, with $(f, g) = \int f \bar{g}$. (a) Prove the uncertainty principle implies $(Lf, f) \geq (f, f)$ for all $f \in \mathcal{S}$ (write $L \geq I$); integrate by parts. (b) For $A(f) = \frac{df}{dx} + xf$ and $A^*(f) = -\frac{df}{dx} + xf$ (annihilation/creation operators) show (i) $(Af, g) = (f, A^*g)$, (ii) $(Af, Af) = (A^*Af, f) \geq 0$, (iii) $A^*A = L - I$. (c) For $t \in \mathbb{R}$, $A_t(f) = \frac{df}{dx} + txf$, $A_t^*(f) = -\frac{df}{dx} + txf$; using $(A_t^*A_t f, f) \geq 0$ as a quadratic in t , reprove that $\int |f|^2 = 1$ implies $(\int x^2 |f|^2)(\int |f'|^2) \geq \frac{1}{4}$.

Problems (Stein, Ch.5 — the Hermite functions).

- * (Hermite functions) Define h_k by the generating identity $\sum_{k \geq 0} h_k(x) \frac{t^k}{k!} = e^{-(x^2/2 - 2tx + t^2)}$. (a) Show equivalently $h_k(x) = (-1)^k e^{x^2/2} \left(\frac{d}{dx}\right)^k e^{-x^2}$, so each $h_k = P_k(x) e^{-x^2/2}$ with $\deg P_k = k$; in particular $h_0 = e^{-x^2/2}$, $h_1 = 2xe^{-x^2/2}$, and $h_k \in \mathcal{S}$. (b) Prove $\{h_k\}$ is complete: if $f \in \mathcal{S}$ and $(f, h_k) = 0$ for all k , then $f = 0$ (use the vanishing exercise from §1). (c) Set $h_k^*(x) = h_k((2\pi)^{1/2}x)$; show $\widehat{h_k^*}(\xi) = (-i)^k h_k^*(\xi)$, so each h_k^* is an eigenfunction of $\mathcal{F}$. (d) Show $Lh_k = (2k+1)h_k$, so the h_k are mutually orthogonal in L^2 . (e) Prove $\int_{\mathbb{R}} [h_k(x)]^2 dx = \pi^{1/2} 2^k k!$ (square the generating identity).

Additional exercises.

- Verify the multiplication formula $\int f \hat{g} = \int \hat{f} g$ explicitly with $f = g = e^{-\pi x^2}$.
- Use Plancherel on $g(x) = (1 - |x|)_+$ to evaluate $\int_{-\infty}^{\infty} \left(\frac{\sin \pi \xi}{\pi \xi}\right)^4 d\xi$. [First find $\|g\|^2 = \int_{-1}^1 (1 - |x|)^2 dx$.]
- Prove that the family $\{K_\delta\}$ of Gaussian kernels satisfies $K_\delta * K_{\delta'} = K_{\delta + \delta'}$ directly from the transform side, and interpret δ as a variance.
- Show that $\mathcal{F}^2 f(x) = f(-x)$ for all $f \in \mathcal{S}$, and use this to evaluate $\mathcal{F}^2$ and $\mathcal{F}^3$ of the Gaussian $e^{-\pi x^2}$ and of $xe^{-\pi x^2}$.
- (Non-example) Show the inversion integral $\int \hat{f}(\xi) e^{2\pi i x \xi} d\xi$ need *not* converge absolutely for $f = \mathbf{1}_{[-1, 1]}$, even though the principal-value/symmetric integral recovers f at points of continuity. Explain which hypothesis of Theorem 35.2.11 fails.
- Let $f \in \mathcal{S}$ with $\int |f|^2 = 1$. Using Plancherel, show $\int |f'|^2 = 4\pi^2 \int \xi^2 |\hat{f}|^2$, and identify when equality in the resulting Heisenberg bound is attained.

7. (Diagram problem) Draw the four eigenvalues of $\mathcal{F}$ on the unit circle and label which Hermite function realizes each. Explain the relation $\lambda_k = i^{-k}$ in one line.
8. Prove the Weierstrass theorem fails on unbounded intervals: exhibit a continuous bounded f on $\mathbb{R}$ that cannot be uniformly approximated by polynomials on all of $\mathbb{R}$. [Consider boundedness of polynomials.]
9. Show directly that $f * g = g * f$ for $f, g \in \mathcal{S}$ using the substitution $u = x - t$, and that $(f * g)' = f' * g = f * g'$.

35.3 Applications: the Heat and Wave Equations and Poisson Summation

The time-dependent heat equation on the line

Model an infinite rod by $\mathbb{R}$, with initial temperature $f(x)$. The temperature $u(x, t)$ solves the *heat equation*

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = f(x).$$

Define the *heat kernel of the line* $H_t(x) = K_{4\pi t}(x)$, that is,

$$H_t(x) = \frac{1}{(4\pi t)^{1/2}} e^{-x^2/4t}, \quad \widehat{H}_t(\xi) = e^{-4\pi^2 t \xi^2}.$$

Taking the Fourier transform in x turns the PDE into the ODE $\partial_t \hat{u} = -4\pi^2 \xi^2 \hat{u}$, whose solution with $\hat{u}(\xi, 0) = \hat{f}(\xi)$ is $\hat{u}(\xi, t) = \hat{f}(\xi) e^{-4\pi^2 \xi^2 t} = \hat{f} \widehat{H}_t$. Hence $u = f * H_t$.

$$\begin{array}{ccc} u(\cdot, t) & \xrightarrow{\mathcal{F}} & \hat{u}(\cdot, t) \\ \downarrow \partial_t - \partial_x^2 & & \downarrow \partial_t + 4\pi^2 \xi^2 \\ 0 & \xrightarrow{\mathcal{F}} & 0 \end{array}$$

Figure 35.9. Fourier transform in x conjugates the heat equation $\partial_t u = \partial_x^2 u$ into the family of ODEs $\partial_t \hat{u} = -4\pi^2 \xi^2 \hat{u}$, one for each frequency ξ ; solving and inverting gives $u = f * H_t$.

Theorem 35.3.1 (Solution of the heat equation). *For $f \in \mathcal{S}(\mathbb{R})$ let $u(x, t) = (f * H_t)(x)$, $t > 0$. Then (i) $u \in C^2$ for $t > 0$ and solves the heat equation; (ii) $u(x, t) \rightarrow f(x)$ uniformly as $t \rightarrow 0$ (so u is continuous on $\overline{\mathbb{R}_+^2}$); (iii) $\int_{\mathbb{R}} |u(x, t) - f(x)|^2 dx \rightarrow 0$ as $t \rightarrow 0$.*

Proof. Transforming, $\hat{u}(\xi, t) = \hat{f}(\xi) e^{-4\pi^2 \xi^2 t}$, so by inversion $u(x, t) = \int \hat{f}(\xi) e^{-4\pi^2 t \xi^2} e^{2\pi i \xi x} d\xi$; differentiation under the integral (justified by rapid decrease) gives (i). (ii) is Corollary 35.2.6 with $\delta = 4\pi t$. For (iii), by Plancherel $\int |u - f|^2 = \int |\hat{f}(\xi)|^2 |e^{-4\pi^2 t \xi^2} - 1|^2 d\xi \rightarrow 0$ by splitting $|\xi| \geq N$ (small tail) and $|\xi| \leq N$ (uniform smallness of the bracket). $\square$

Theorem 35.3.2 (Uniqueness, energy method). *If u is continuous on $\overline{\mathbb{R}_+^2}$, solves the heat equation for $t > 0$, satisfies $u(x, 0) = 0$, and $u(\cdot, t) \in \mathcal{S}$ uniformly in t , then $u \equiv 0$.*

Proof. Let $E(t) = \int_{\mathbb{R}} |u(x, t)|^2 dx \geq 0$ with $E(0) = 0$. Differentiating and using $\partial_t u = \partial_x^2 u$ and integration by parts (boundary terms vanish by rapid decrease),

$$E'(t) = \int (\partial_t u \bar{u} + u \overline{\partial_t u}) = \int (\partial_x^2 u \bar{u} + u \overline{\partial_x^2 u}) = -2 \int |\partial_x u|^2 \leq 0.$$

So E decreases from 0 and stays ≥ 0 , forcing $E \equiv 0$, hence $u \equiv 0$. $\square$

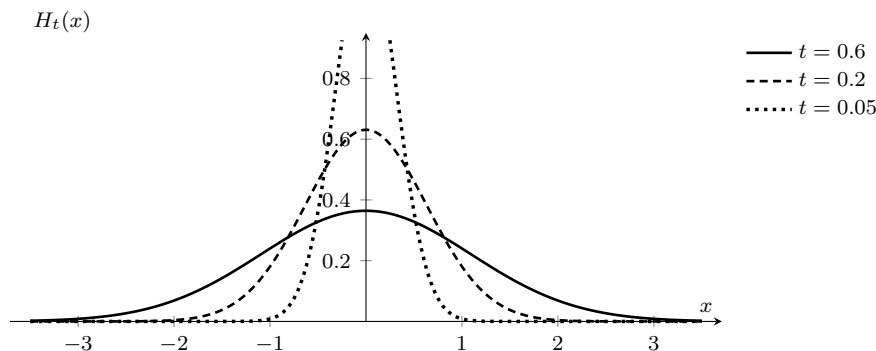


Figure 35.10. Heat decay: as $t \rightarrow 0^+$ the heat kernel H_t concentrates (initial spike, clipped at top); for larger t heat spreads and flattens.

Worked Example 35.3.3 (The heat kernel evolves a point source). Solve the heat equation with a Gaussian initial datum $f(x) = e^{-\pi x^2}$.

Recognition. $u = f * H_t$ is a convolution of Gaussians; add “variances.”

Solution. $\hat{u}(\xi, t) = e^{-\pi \xi^2} e^{-4\pi^2 \xi^2 t} = e^{-\pi \xi^2 (1+4\pi t)}$. Inverting (a dilated Gaussian), $u(x, t) = (1 + 4\pi t)^{-1/2} e^{-\pi x^2 / (1+4\pi t)}$. *Check.* At $t = 0$, $u = e^{-\pi x^2} = f$; as $t \rightarrow \infty$, $u \rightarrow 0$ pointwise and spreads — total heat $\int u \, dx = 1$ is conserved. ■

The steady-state heat equation in the upper half-plane

Now $\Delta u = u_{xx} + u_{yy} = 0$ in $\mathbb{R}_+^2 = \{y > 0\}$ with $u(x, 0) = f(x)$. The *Poisson kernel* is

$$P_y(x) = \frac{1}{\pi} \frac{y}{x^2 + y^2}, \quad y > 0.$$

Lemma 35.3.4 (Poisson kernel transforms). $\int_{\mathbb{R}} e^{-2\pi|\xi|y} e^{2\pi i \xi x} \, d\xi = P_y(x)$ and $\int_{\mathbb{R}} P_y(x) e^{-2\pi i x \xi} \, dx = e^{-2\pi|\xi|y}$.

Proof. Split the first integral at 0. For $\xi > 0$, $\int_0^\infty e^{-2\pi\xi y} e^{2\pi i \xi x} \, d\xi = \frac{1}{2\pi i(x+iy)}$ wait, more precisely $= \frac{1}{2\pi(y-ix)} = \frac{1}{2\pi i(x+iy)} \cdot i$; combining with the $\xi < 0$ piece $\frac{1}{2\pi i(x-iy)}$ gives $\frac{1}{2\pi i} \left(\frac{1}{x-iy} - \frac{1}{x+iy} \right) = \frac{y}{\pi(x^2+y^2)} = P_y(x)$. The second identity is inversion (valid since P_y and its transform are of moderate decrease). □

Solving $\Delta u = 0$ by Fourier transform in x gives $\hat{u}(\xi, y) = \hat{f}(\xi) e^{-2\pi|\xi|y}$ (discarding the exponentially growing solution), so $u = f * P_y$.

Lemma 35.3.5. $\{P_y\}_{y>0}$ is a family of good kernels as $y \rightarrow 0$.

Proof. $\int P_y = e^{-2\pi \cdot 0 \cdot y} = 1$ (Lemma 35.3.4, $\xi = 0$), $P_y \geq 0$, and for fixed $\delta > 0$, $u = x/y$ gives $\int_\delta^\infty P_y = \frac{1}{\pi} (\frac{\pi}{2} - \arctan(\delta/y)) \rightarrow 0$ as $y \rightarrow 0$; by evenness the two-sided tail $\rightarrow 0$. □

Theorem 35.3.6 (Solution in the half-plane). For $f \in \mathcal{S}(\mathbb{R})$, $u(x, y) = (f * P_y)(x)$ is C^2 and harmonic in $\mathbb{R}_+^2$, $u(x, y) \rightarrow f(x)$ uniformly as $y \rightarrow 0$, $\int |u - f|^2 \rightarrow 0$, and (with $u(x, 0) = f(x)$) u is continuous on $\overline{\mathbb{R}_+^2}$ and vanishes as $|x| + y \rightarrow \infty$.

Lemma 35.3.7 (Mean-value property). If $\Delta u = 0$ in an open $\Omega \subset \mathbb{R}^2$ and the closed disc of center (x, y) , radius R , lies in Ω , then $u(x, y) = \frac{1}{2\pi} \int_0^{2\pi} u(x + r \cos \theta, y + r \sin \theta) \, d\theta$ for $0 \leq r \leq R$.

Proof. Let $U(r, \theta) = u(x + r \cos \theta, y + r \sin \theta)$; in polar coordinates $\Delta u = 0$ reads $U_{\theta\theta} + r(rU_r)_r = 0$. Averaging over θ , the $U_{\theta\theta}$ term integrates to 0 (periodicity), so $r(rF_r)_r = 0$ where $F(r) = \frac{1}{2\pi} \int U \, d\theta$; thus rF_r is constant $= 0$ (value at $r = 0$), F is constant $= F(0) = u(x, y)$. □

Theorem 35.3.8 (Uniqueness in the half-plane). If u is continuous on $\overline{\mathbb{R}_+^2}$, harmonic in $\mathbb{R}_+^2$, $u(x, 0) = 0$, and u vanishes at infinity, then $u \equiv 0$.

Proof. Reduce to u real. If $u(x_0, y_0) > 0$ somewhere, choose a large semi-disc D_R^+ outside which $u \leq \frac{1}{2}u(x_0, y_0)$; u attains its max $M > 0$ at an interior point (x_1, y_1) . The mean-value property over small circles forces $u \equiv M$ on those circles (any dip below M would lower the average), and letting the radius grow gives $u(x_1, 0) = M > 0$, contradicting $u(x, 0) = 0$. The example $u(x, y) = y$ shows the decay hypothesis is needed. $\square$

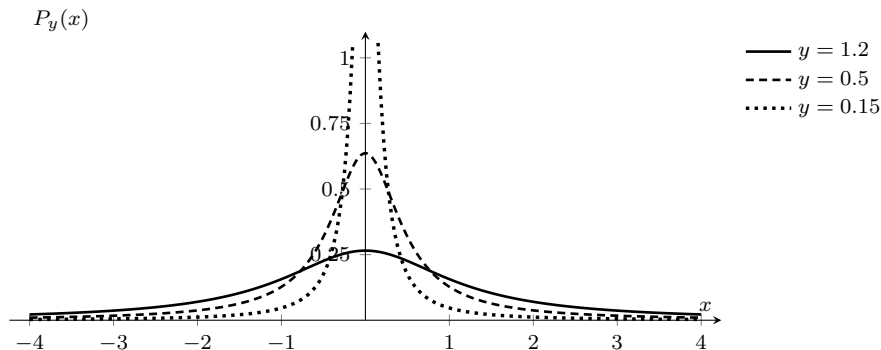


Figure 35.11. The Poisson kernel P_y for the upper half-plane: peaks and concentrates as $y \rightarrow 0$ (a good kernel, tallest spike clipped), but only of moderate decrease in x .

Worked Example 35.3.9 (Harmonic extension of a Lorentzian). Find the bounded harmonic u in $\mathbb{R}_+^2$ with boundary value $f(x) = \frac{1}{\pi} \frac{a}{x^2 + a^2} = P_a(x)$, $a > 0$.

Recognition. $u = f * P_y$; convolution of Poisson kernels adds the heights.

Solution. $\hat{u}(\xi, y) = \hat{f}(\xi) e^{-2\pi|\xi|y} = e^{-2\pi|\xi|a} e^{-2\pi|\xi|y} = e^{-2\pi|\xi|(a+y)} = \widehat{P_{a+y}}(\xi)$, so $u(x, y) = P_{a+y}(x) = \frac{1}{\pi} \frac{a+y}{x^2 + (a+y)^2}$. *Check.* At $y = 0$ this is $P_a = f$; it is manifestly harmonic (a shifted Poisson kernel) and $\rightarrow 0$ at infinity. $\blacksquare$

The Poisson summation formula

Given $f \in \mathcal{S}(\mathbb{R})$, its *periodization* $F_1(x) = \sum_{n \in \mathbb{Z}} f(x+n)$ converges (rapid decrease) to a continuous 1-periodic function. Its Fourier-series partner is $F_2(x) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n x}$. These coincide.

Theorem 35.3.10 (Poisson summation). *If $f \in \mathcal{S}(\mathbb{R})$ then $\sum_{n \in \mathbb{Z}} f(x+n) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n x}$, and in particular*

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \hat{f}(n).$$

$$\begin{array}{ccc} f & \xrightarrow{\mathcal{F}} & \hat{f} \\ \text{per} \downarrow & & \downarrow \text{restrict to } \mathbb{Z} \\ \sum_n f(x+n) & \xrightarrow{c_m} & (\hat{f}(m))_{m \in \mathbb{Z}} \end{array}$$

Figure 35.12. Poisson summation as a commuting square: the m -th Fourier coefficient c_m of the periodization $\sum_n f(x+n)$ equals the transform $\hat{f}$ sampled at the integer m . Synthesizing the bottom row recovers $\sum_n \hat{f}(n) e^{2\pi i n x}$.

Proof. Both sides are continuous and 1-periodic, so it suffices to match Fourier coefficients. The m -th coefficient of the right side is $\hat{f}(m)$. For the left side,

$$\int_0^1 \left(\sum_n f(x+n) \right) e^{-2\pi i m x} dx = \sum_n \int_n^{n+1} f(y) e^{-2\pi i m y} dy = \int_{\mathbb{R}} f(y) e^{-2\pi i m y} dy = \hat{f}(m),$$

interchanging sum and integral by rapid decrease. So the coefficients agree. $\square$

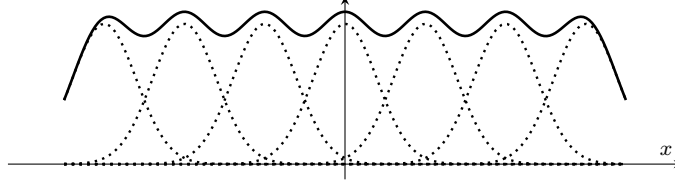


Figure 35.13. Periodization: summing the integer-shifted Gaussians (dotted) yields a 1-periodic function (solid), whose Fourier coefficients are $\hat{f}(n)$. (Schematic: only finitely many shifts are drawn.)

Worked Example 35.3.11 (Partial fractions of the cotangent). Use periodization of $f(x) = 1/(x^2 + \alpha^2)$ (or Exercise 15) to obtain $\sum_{n \in \mathbb{Z}} \frac{1}{(n+\alpha)^2} = \frac{\pi^2}{\sin^2 \pi \alpha}$ for $\alpha \notin \mathbb{Z}$.

Recognition. Apply Poisson summation to a Poisson-kernel type function whose transform is known.

Solution. Take $g(x) = \frac{t}{\pi(x^2 + t^2)} = P_t(x)$ with $\hat{g}(\xi) = e^{-2\pi t|\xi|}$. Poisson summation $\sum g(n + \alpha)$ -style (after the modulation $g(x)e^{2\pi i \alpha x}$, shifting the transform) yields the stated series; differentiating the cotangent identity $\sum_n \frac{1}{n + \alpha} = \pi \cot \pi \alpha$ gives $\sum_n \frac{1}{(n + \alpha)^2} = \pi^2 / \sin^2 \pi \alpha$. *Check.* At $\alpha = \frac{1}{2}$, RHS = π^2 , and LHS = $\sum_n (n + \frac{1}{2})^{-2} = 2 \sum_{k \geq 0} (k + \frac{1}{2})^{-2} = 2 \cdot \frac{\pi^2}{2} = \pi^2$. ■

Theta and zeta functions

Definition 35.3.12. For $s > 0$, $\vartheta(s) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 s}$.

Theorem 35.3.13 (Theta functional equation). $s^{-1/2} \vartheta(1/s) = \vartheta(s)$ for $s > 0$.

Proof. Apply Poisson summation to $f(x) = e^{-\pi s x^2}$, whose transform is $\hat{f}(\xi) = s^{-1/2} e^{-\pi \xi^2 / s}$ (Corollary 35.2.3). Then $\sum_n e^{-\pi s n^2} = \sum_n s^{-1/2} e^{-\pi n^2 / s}$, i.e. $\vartheta(s) = s^{-1/2} \vartheta(1/s)$. □

The theta function links to the Riemann zeta function $\zeta(s) = \sum_{n \geq 1} n^{-s}$ ($\text{Re } s > 1$) and the gamma function via $\pi^{-s/2} \Gamma(s/2) \zeta(s) = \frac{1}{2} \int_0^\infty t^{s/2-1} (\vartheta(t) - 1) dt$, valid for $s > 1$ (Exercises 17–18) — the gateway to the analytic continuation of ζ .

Worked Example 35.3.14 (Self-dual point of theta). Find the value s at which the functional equation forces a symmetry, and evaluate ϑ there qualitatively.

Recognition. Set $1/s = s$ in $s^{-1/2} \vartheta(1/s) = \vartheta(s)$.

Solution. At $s = 1$: $\vartheta(1) = 1^{-1/2} \vartheta(1)$ is automatic, but the fixed point of $s \mapsto 1/s$ is $s = 1$, the symmetry axis. For s near 1 the equation relates large- s behavior $\vartheta(s) \rightarrow 1$ (only the $n = 0$ term survives) to small- s behavior $\vartheta(s) \sim s^{-1/2}$ as $s \rightarrow 0$. *Check.* As $s \rightarrow \infty$, $\vartheta(s) \rightarrow 1$; the functional equation then predicts $\vartheta(s) \sim s^{-1/2}$ as $s \rightarrow 0^+$, the heat-kernel mass-conservation in disguise. ■

Heat and Poisson kernels on the circle as periodizations

Theorem 35.3.15. The heat kernel on the circle $\mathcal{H}_t(x) = \sum_{n \in \mathbb{Z}} e^{-4\pi^2 n^2 t} e^{2\pi i n x}$ is the periodization of the heat kernel on the line: $\mathcal{H}_t(x) = \sum_{n \in \mathbb{Z}} H_t(x + n)$. Consequently $\mathcal{H}_t$ is a good kernel as $t \rightarrow 0$.

Proof. Apply Poisson summation to H_t , using $\widehat{H}_t(\xi) = e^{-4\pi^2 \xi^2 t}$, so $\widehat{H}_t(n) = e^{-4\pi^2 n^2 t}$. That the periodized kernel inherits the good-kernel properties follows from the estimate $\mathcal{H}_t(x) = H_t(x) + E_t(x)$ with $|E_t(x)| \leq c_1 e^{-c_2/t}$ for $|x| \leq \frac{1}{2}$, $0 < t \leq 1$. □

Likewise the Poisson kernel for the disc is the periodization of P_y : $P_r(2\pi x) = \sum_{n \in \mathbb{Z}} P_y(x + n)$ with $r = e^{-2\pi y}$.

Worked Example 35.3.16 (Heat-kernel error term is exponentially small). Estimate the difference $\mathcal{H}_t(x) - H_t(x)$ for $|x| \leq \frac{1}{2}$, $0 < t \leq 1$.

Recognition. The difference is the sum over $|n| \geq 1$ of shifted Gaussians.

Solution. $E_t(x) = \sum_{|n| \geq 1} H_t(x + n) \leq C t^{-1/2} \sum_{n \geq 1} e^{-c n^2 / t}$. For $0 < t \leq 1$, $n^2 / t \geq \frac{1}{2} n^2 + \frac{1}{2} \frac{1}{t}$, so each term $\leq e^{-\frac{\epsilon}{2} n^2} e^{-\frac{\epsilon}{2} / t}$ and $|E_t(x)| \leq c_1 e^{-c_2 / t}$. *Check.* As $t \rightarrow 0$, $E_t \rightarrow 0$ faster than any power of t , so $\mathcal{H}_t$ behaves like H_t near 0. ■

The Heisenberg uncertainty principle

Theorem 35.3.17 (Heisenberg uncertainty). *If $\psi \in \mathcal{S}(\mathbb{R})$ with $\int |\psi|^2 = 1$, then*

$$\left(\int_{\mathbb{R}} x^2 |\psi(x)|^2 dx \right) \left(\int_{\mathbb{R}} \xi^2 |\hat{\psi}(\xi)|^2 d\xi \right) \geq \frac{1}{16\pi^2},$$

with equality iff $\psi(x) = Ae^{-Bx^2}$, $B > 0$, $|A|^2 = \sqrt{2B/\pi}$. More generally one may center at any x_0, ξ_0 .

Proof. Integrate $1 = \int |\psi|^2 = -\int x \frac{d}{dx} |\psi|^2 dx = -\int x(\psi' \bar{\psi} + \bar{\psi}' \psi) dx \leq 2 \int |x| |\psi| |\psi'| dx$. By Cauchy–Schwarz, $1 \leq 2(\int x^2 |\psi|^2)^{1/2} (\int |\psi'|^2)^{1/2}$. By 35.1.9(iv) and Plancherel, $\int |\psi'|^2 = 4\pi^2 \int \xi^2 |\hat{\psi}|^2$. Squaring gives the inequality. Equality in Cauchy–Schwarz forces $\psi' = \beta x \psi$, i.e. $\psi = Ae^{\beta x^2/2}$; Schwartz membership needs $\beta = -2B < 0$, and normalization fixes $|A|^2 = \sqrt{2B/\pi}$. $\square$

Physically: $|\psi|^2$ is a position density, $|\hat{\psi}|^2$ a momentum density; the product of their variances is bounded below. The more sharply localized ψ is in position, the more spread out it must be in momentum.

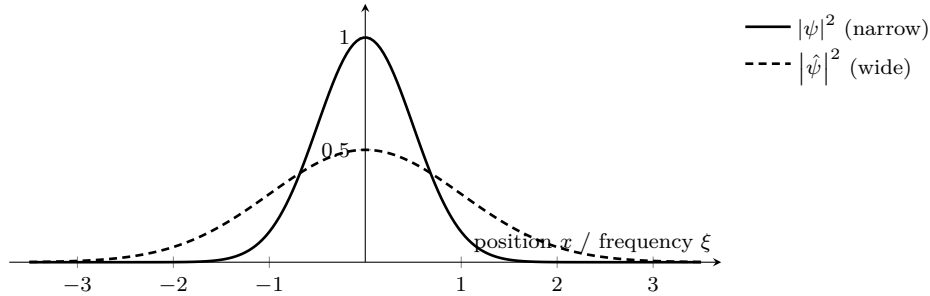


Figure 35.14. Uncertainty: a Gaussian narrow in position has a wide transform; the product of variances cannot drop below $1/16\pi^2$. (Schematic.)

Worked Example 35.3.18 (The Gaussian saturates the bound). Verify Heisenberg equality for $\psi(x) = 2^{1/4} e^{-\pi x^2}$.

Recognition. Compute both variances; the Gaussian is the extremal case.

Solution. $\int |\psi|^2 = \sqrt{2} \int e^{-2\pi x^2} dx = \sqrt{2} \cdot \frac{1}{\sqrt{2}} = 1$, good. Then $\int x^2 |\psi|^2 = \sqrt{2} \int x^2 e^{-2\pi x^2} dx = \frac{1}{4\pi} \cdot \frac{1}{\sqrt{2}} \cdot \sqrt{2} = \frac{1}{4\pi}$. By self-duality $\hat{\psi}$ is the same Gaussian, so $\int \xi^2 |\hat{\psi}|^2 = \frac{1}{4\pi}$ too, and the product is $\frac{1}{16\pi^2}$. *Check.* Equality holds, as the theorem predicts for Gaussians. $\blacksquare$

Worked Example 35.3.19 (Equidistribution of $\{n\alpha\}$ (Weyl)). Show that for irrational α , the fractional parts $\{n\alpha\}$ are equidistributed in $[0, 1)$.

Recognition. Weyl's criterion: it suffices that $\frac{1}{N} \sum_{n=1}^N e^{2\pi i k n \alpha} \rightarrow 0$ for every integer $k \neq 0$.

Solution. For $k \neq 0$, $\sum_{n=1}^N e^{2\pi i k n \alpha}$ is a geometric series with ratio $e^{2\pi i k \alpha} \neq 1$ (as α irrational), so its modulus is $\leq \frac{2}{|1 - e^{2\pi i k \alpha}|}$, a constant; dividing by N gives $\rightarrow 0$. By Weyl's criterion the sequence is equidistributed.

Check. For rational $\alpha = p/q$ the sum does not vanish ($e^{2\pi i k \alpha} = 1$ when $q \mid k$), and indeed $\{n\alpha\}$ takes only finitely many values — not equidistributed. $\blacksquare$

Worked Example 35.3.20 (Poisson summation evaluates a Gaussian sum). Use Poisson summation on $f(x) = e^{-\pi t x^2}$ to relate $\sum_n e^{-\pi t n^2}$ to $\sum_n e^{-\pi n^2/t}$.

Recognition. This is exactly the theta functional equation; read off the numeric content.

Solution. $\sum_n e^{-\pi t n^2} = t^{-1/2} \sum_n e^{-\pi n^2/t}$. For small t the left side is large ($\approx t^{-1/2}$), and the right side's $n = 0$ term $t^{-1/2}$ dominates while the rest are exponentially small. *Check.* At $t = 1$ both sides are $\vartheta(1)$, an identity; the formula converts a slowly-converging sum (small t) into a rapidly-converging one — the basis of fast theta evaluation. $\blacksquare$

Worked Example 35.3.21 (The Fejer kernel on the line). Identify the kernel F_R for which $\int_{-R}^R (1 - \frac{|\xi|}{R}) \hat{f}(\xi) e^{2\pi i x \xi} d\xi = (f * F_R)(x)$.

Recognition. The triangular weight $(1 - |\xi|/R)_+$ is the transform of a sinc^2 .

Solution. The triangular cutoff on $[-R, R]$ is the transform of $F_R(t) = R \left(\frac{\sin \pi t R}{\pi t R} \right)^2$ (and $F_R(0) = R$). Since $(1 - |\xi|/R)_+$ is a tent, its inverse transform is sinc^2 -shaped, nonnegative, with $\int F_R = 1$. Check. $\{F_R\}$ is a family of good kernels as $R \rightarrow \infty$, so $(f * F_R) \rightarrow f$ uniformly — Fejer's theorem on the line. ■

Worked Example 35.3.22 (A Gauss sum via theta at a rational). Compute $\sum_{n=0}^{N-1} e^{2\pi i n^2/N}$ for N a positive integer (the quadratic Gauss sum), and note its modulus.

Recognition. Quadratic Gauss sums have modulus $\sqrt{N}$; the value depends on $N \bmod 4$.

Solution. The classical evaluation is $g(N) = \sum_{n=0}^{N-1} e^{2\pi i n^2/N} = \frac{1}{2} \sqrt{N} (1+i)(1+i^{-N})$, giving $g(N) = \sqrt{N}, 0, i\sqrt{N}, (1+i)\sqrt{N}$ for $N \equiv 1, 2, 3, 0 \pmod{4}$. In all cases $|g(N)| \in \{\sqrt{N}, 0, \sqrt{N}, \sqrt{2N}\}$. Check. For $N = 1$, $g = 1 = \sqrt{1}$; for $N = 4$, $g = 1 + e^{\pi i/2} + e^{2\pi i} + e^{9\pi i/2} = 1 + i + 1 + i = 2(1+i)$, modulus $2\sqrt{2} = \sqrt{2 \cdot 4}$. ■

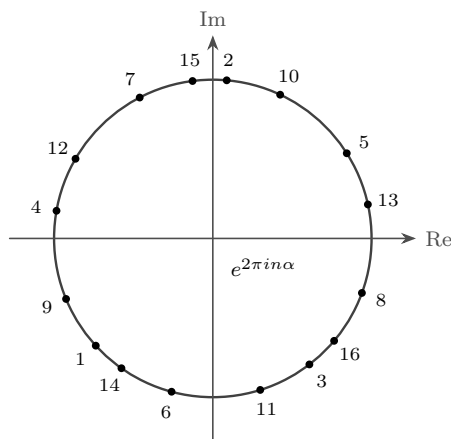


Figure 35.15. Equidistribution: the points $e^{2\pi i n \alpha}$ for $n = 1, \dots, 16$ (here α = the golden ratio, labels show n) spread evenly around the unit circle as n grows.

Exercises (Stein, Ch.5 — heat/Poisson and summation).

- (Heat solution vanishes at infinity) If $u = f * H_t$ with $f \in \mathcal{S}$ and $u(x, 0) = f(x)$, prove u is continuous on $\overline{\mathbb{R}_+^2}$ and $u(x, t) \rightarrow 0$ as $|x| + t \rightarrow \infty$. [Show (i) $|u(x, t)| \leq C/\sqrt{t}$ and (ii) $|u(x, t)| \leq C/(1 + |x|^2) + Ct^{-1/2}e^{-cx^2/t}$; use (i) when $|x| \leq t$, (ii) otherwise.]
- (Non-uniqueness, discontinuous at origin) Show $u(x, t) = \frac{x}{t} H_t(x)$ solves the heat equation for $t > 0$ and $\lim_{t \rightarrow 0} u(x, t) = 0$ for every x , yet u is not continuous at the origin. [Approach along the parabola $x^2/4t = c$.]
- (Uniqueness in a strip) Prove: if u is harmonic in $\{0 < y < 1, x \in \mathbb{R}\}$, continuous on the closure, $u(x, 0) = u(x, 1) = 0$, and u vanishes at infinity, then $u \equiv 0$.
- (Periodization of the Fejer kernel) Prove $\sum_{n \in \mathbb{Z}} F_N(x + n) = \mathcal{F}_N(x)$ for integer $N \geq 1$, where F_N is the real-line Fejer kernel (Exercise 9 of §1) and $\mathcal{F}_N(x) = \sum_{|n| \leq N} (1 - \frac{|n|}{N}) e^{2\pi i n x} = \frac{1}{N} \frac{\sin^2(N\pi x)}{\sin^2(\pi x)}$ is the periodic Fejer kernel.
- (Periodization examples) (a) Apply Poisson summation to the tent g (Exercise 2) to get $\sum_{n \in \mathbb{Z}} \frac{1}{(n+\alpha)^2} = \frac{\pi^2}{\sin^2 \pi \alpha}$ for $\alpha \notin \mathbb{Z}$. (b) Deduce $\sum_{n \in \mathbb{Z}} \frac{1}{n+\alpha} = \frac{\pi}{\tan \pi \alpha}$ (summed symmetrically) for $\alpha \notin \mathbb{Z}$. [First prove it for $0 < \alpha < 1$ by integrating (a); evaluate at $\alpha = 1/2$.]
- (Dirichlet kernel on the line) Define D_R by $\int_{-R}^R \hat{f}(\xi) e^{2\pi i x \xi} d\xi = (f * D_R)(x)$, so $D_R(x) = \widehat{\mathbf{1}_{[-R, R]}}(x) = \frac{\sin(2\pi R x)}{\pi x}$. With the modified periodic Dirichlet kernel $D_N^*(x) = \sum_{|n| \leq N-1} e^{2\pi i n x} + \frac{1}{2}(e^{-2\pi i N x} + e^{2\pi i N x})$, show $\sum_{n \in \mathbb{Z}} D_N(x + n) = D_N^*(x)$ (the series summed symmetrically).

7. (Gamma function) For $s > 0$, $\Gamma(s) = \int_0^\infty e^{-x} x^{s-1} dx$. (a) Show the integral converges (the limits at 0 and at ∞ exist). (b) Prove $\Gamma(s+1) = s\Gamma(s)$, hence $\Gamma(n+1) = n!$. (c) Show $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and $\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$ (use $\int_{\mathbb{R}} e^{-\pi x^2} = 1$).
8. (Theta–gamma–zeta identity) For $s > 1$ verify $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2} \int_0^\infty t^{s/2-1}(\vartheta(t) - 1) dt$, where $\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt$ and $\vartheta(s) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 s}$.
9. (Zeta at even integers) (a) Apply Poisson summation to $f(x) = t/(\pi(x^2 + t^2))$, $\hat{f}(\xi) = e^{-2\pi t|\xi|}$, $t > 0$, to get $\frac{1}{\pi} \sum_n \frac{t}{t^2 + n^2} = \sum_n e^{-2\pi t|n|}$. (b) For $0 < t < 1$ derive $\frac{1}{\pi} \sum_n \frac{t}{t^2 + n^2} = \frac{1}{\pi t} + \frac{2}{\pi} \sum_{m \geq 1} (-1)^{m+1} \zeta(2m) t^{2m-1}$ and $\sum_n e^{-2\pi t|n|} = \frac{2}{1 - e^{-2\pi t}} - 1$. (c) Using $\frac{z}{e^z - 1} = 1 - \frac{z}{2} + \sum_{m \geq 1} \frac{B_{2m}}{(2m)!} z^{2m}$ with Bernoulli numbers B_k , deduce $2\zeta(2m) = (-1)^{m+1} \frac{(2\pi)^{2m}}{(2m)!} B_{2m}$.
10. (Sampling / reconstruction) Suppose f is of moderate decrease with $\hat{f}$ supported in $I = [-1/2, 1/2]$; then f is determined by its values on $\mathbb{Z}$. (a) Prove the reconstruction formula $f(x) = \sum_{n \in \mathbb{Z}} f(n)K(x-n)$ with $K(y) = \frac{\sin \pi y}{\pi y} = O(1/|y|)$. (b) For $\lambda > 1$, $f(x) = \sum_n \frac{1}{\lambda} f(\frac{n}{\lambda}) K_\lambda(x - \frac{n}{\lambda})$ with $K_\lambda(y) = \frac{\cos \pi y - \cos \pi \lambda y}{\pi^2 (\lambda-1)y^2} = O(1/|y|^2)$ — oversampling speeds convergence; $K_\lambda \rightarrow K$ as $\lambda \rightarrow 1$. (c) Prove $\int_{\mathbb{R}} |f(x)|^2 dx = \sum_{n \in \mathbb{Z}} |f(n)|^2$.

Problems (Stein, Ch.5 — PDE variants and non-uniqueness).

1. (An Euler-type heat equation) The equation $x^2 u_{xx} + axu_x = u_t$ on $0 < x < \infty$, $t > 0$, with $u(x, 0) = f(x)$. Substituting $x = e^{-y}$, $U(y, t) = u(e^{-y}, t)$, $F(y) = f(e^{-y})$ reduces it to $U_{yy} + (1-a)U_y = U_t$, $U(y, 0) = F(y)$, solvable by Fourier transform in y . Computing $\int_{\mathbb{R}} e^{(-4\pi^2 \xi^2 + (1-a)2\pi i \xi)t} e^{2\pi i \xi y} d\xi$ shows $u(x, t) = \frac{1}{(4\pi t)^{1/2}} \int_0^\infty e^{-(\log(v/x) + (1-a)t)^2 / (4t)} f(v) \frac{dv}{v}$.
2. (Black–Scholes) The equation $V_t + rsV_s + \frac{\sigma^2 s^2}{2} V_{ss} - rV = 0$, $0 < t < T$, with $V(s, T) = F(s)$, reduces (via $V = e^{ax+b\tau} U(x, \tau)$, $x = \log s$, $\tau = \frac{\sigma^2}{2}(T-t)$, $a = \frac{1}{2} - \frac{r}{\sigma^2}$, $b = -(\frac{1}{2} + \frac{r}{\sigma^2})^2$) to the heat equation with $U(x, 0) = e^{-ax} F(e^x)$, giving $V(s, t) = \frac{e^{-r(T-t)}}{\sqrt{2\pi\sigma^2(T-t)}} \int_0^\infty e^{-(\log(s/s^*) + (r-\sigma^2/2)(T-t))^2 / (2\sigma^2(T-t))} F(s^*) ds^*$.
3. * (Dirichlet problem in a strip) $\Delta u = 0$ in $\{0 < y < 1\}$ with $u(x, 0) = f_0$, $u(x, 1) = f_1$, $f_0, f_1 \in \mathcal{S}$. (a) Formally $\hat{u}(\xi, y) = A(\xi)e^{2\pi\xi y} + B(\xi)e^{-2\pi\xi y}$; express A, B via $\hat{f}_0, \hat{f}_1$ and show $\hat{u}(\xi, y) = \frac{\sinh(2\pi(1-y)\xi)}{\sinh(2\pi\xi)} \hat{f}_0(\xi) + \frac{\sinh(2\pi y\xi)}{\sinh(2\pi\xi)} \hat{f}_1(\xi)$. (b) Prove $\int |u(x, y) - f_0|^2 \rightarrow 0$ as $y \rightarrow 0$ and $\int |u(x, y) - f_1|^2 \rightarrow 0$ as $y \rightarrow 1$. (c) For $\Phi(\xi) = \frac{\sinh 2\pi a \xi}{\sinh 2\pi \xi}$, $0 \leq a < 1$, Φ is the transform of $\varphi(x) = \frac{\sin \pi a}{2} \cdot \frac{1}{\cosh \pi x + \cos \pi a}$. (d) Hence $u(x, y) = \frac{\sin \pi y}{2} (\int \frac{f_0(x-t)}{\cosh \pi t - \cos \pi y} dt + \int \frac{f_1(x-t)}{\cosh \pi t + \cos \pi y} dt)$. (e) Check u is harmonic, $\rightarrow f_0$ as $y \rightarrow 0$, $\rightarrow f_1$ as $y \rightarrow 1$, and vanishes at infinity.
4. * (Non-uniqueness with mild growth) For smooth g define $u(x, t) = \sum_{n \geq 0} g^{(n)}(t) \frac{x^{2n}}{(2n)!}$. (a) Check u formally solves the heat equation. (b) For $a > 0$, $g(t) = e^{-t^{-a}}$ ($t > 0$), $g(t) = 0$ ($t \leq 0$) satisfies $|g^{(k)}(t)| \leq \frac{k!}{(\theta t)^k} e^{-\frac{1}{2}t^{-a}}$ for some $0 < \theta < 1$. (c) Then (19) converges, solves the heat equation, vanishes at $t = 0$, and $|u(x, t)| \leq C e^{c|x|^{2a/(a-1)}}$. (d) Conclude: for every $\varepsilon > 0$ there is a nonzero solution, continuous for $t \geq 0$, with $u(x, 0) = 0$ and $|u(x, t)| \leq C e^{c|x|^{2+\varepsilon}}$.
5. * (Maximum principle) For a real solution u of the heat equation, continuous on the closure, let $R = \{a \leq x \leq b, 0 \leq t \leq c\}$ and $\partial' R$ the two vertical sides plus the base $t = 0$. Then $\min_{\partial' R} u = \min_R u$ and $\max_{\partial' R} u = \max_R u$. [Steps: (a) reduce to “ $u \geq 0$ on $\partial' R \Rightarrow u \geq 0$ ”; (b) for $\varepsilon > 0$ set $v = u + \varepsilon t$, locate its min on R at $x_1 = a, b$ or $t_1 = 0$ by contradiction ($v_{xx} - v_t \leq -\varepsilon$ yet ≥ 0); (c) deduce $u \geq \varepsilon(t_1 - t)$ and let $\varepsilon \rightarrow 0$.]
6. * (Tychonoff uniqueness) If u solves the heat equation for $t > 0$, is continuous for $0 \leq t \leq c$, $u(x, 0) = 0$, and $|u(x, t)| \leq M e^{ax^2}$, then $u \equiv 0$. (The examples of Problem 4 show this growth condition is optimal.)
7. * (Nowhere differentiability via periodization) To prove $f_\alpha(x) = \sum_{n \geq 0} 2^{-n\alpha} e^{2\pi i 2^n x}$ is nowhere differentiable even at $\alpha = 1$, fix $\Phi \in C^\infty$ with $\Phi(\xi) = 1$ for $|\xi| \leq 1$, 0 for $|\xi| \geq 2$, and $\varphi \in \mathcal{S}$ with $\hat{\varphi} = \Phi$, $\varphi_N(x) = N\varphi(Nx)$. Set $\tilde{\Delta}_N(x) = \sum_n \varphi_N(x+n) = \sum_n \Phi(n/N) e^{2\pi i n x}$ (Poisson summation), a trig polynomial of degree $\leq 2N$

with coefficients 1 for $|n| \leq N$. (a) With $\tilde{\Delta}_N(x) = \varphi_N(x) + E_N(x)$, $E_N = \sum_{|n| \geq 1} \varphi_N(x+n)$, show $\sup_{|x| \leq 1/2} |E'_N| \rightarrow 0$, $|\tilde{\Delta}'_N| \leq cN^2$, and $|\tilde{\Delta}'_N(x)| \leq c/(N|x|^3)$ for $|x| \leq 1/2$; moreover $\int_{|x| \leq 1/2} \tilde{\Delta}'_N = 0$ and $-\int_{|x| \leq 1/2} x \tilde{\Delta}'_N \rightarrow 1$. (b) Deduce that if $f'(x_0)$ exists then $(f * \tilde{\Delta}'_N)(x_0 + h_N) \rightarrow f'(x_0)$ whenever $|h_N| \leq C/N$, and conclude f_1 is nowhere differentiable.

Additional exercises.

1. Solve the heat equation $u_t = u_{xx}$, $u(x, 0) = e^{-\pi x^2}$, in closed form, and verify that $\int_{\mathbb{R}} u(x, t) dx$ is independent of t (conservation of heat).
2. Verify the Poisson summation formula numerically for $f(x) = e^{-\pi x^2}$ by comparing $\sum_{|n| \leq 5} f(n)$ with $\sum_{|n| \leq 5} \hat{f}(n)$ (both sides equal $\vartheta(1)$).
3. Use Poisson summation on $f(x) = e^{-\pi t x^2}$ to derive the theta functional equation $\vartheta(s) = s^{-1/2} \vartheta(1/s)$, and deduce the small- s asymptotic $\vartheta(s) \sim s^{-1/2}$.
4. Show $P_a * P_b = P_{a+b}$ for the half-plane Poisson kernels, and interpret this as the semigroup property of harmonic extension (stacking strips).
5. Find the bounded harmonic function in the upper half-plane with boundary data $f(x) = \mathbf{1}_{[0, \infty)}(x)$ (a step). [Answer involves $\frac{1}{\pi} \arctan$; relate to the Poisson integral.]
6. (Non-example) Show that $u(x, y) = y$ is harmonic in $\mathbb{R}_+^2$ with $u(x, 0) = 0$ but $u \not\equiv 0$, and explain precisely which hypothesis of Theorem 35.3.8 it violates.
7. Prove $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ by setting $\alpha \rightarrow 0$ in $\sum_{n \in \mathbb{Z}} \frac{1}{(n+\alpha)^2} = \frac{\pi^2}{\sin^2 \pi \alpha}$ (subtract the $n = 0$ term and expand). State carefully the limit you take.
8. (Diagram problem) Plot $\{n\alpha \bmod 1\}$ for $n = 1, \dots, 20$ as points on the unit circle for $\alpha = \sqrt{2}$, and contrast with $\alpha = 1/3$. Which is equidistributed, and why (Weyl's criterion)?
9. Using the heat kernel relation $\mathcal{H}_t(x) = \sum_n H_t(x+n)$, estimate the error $|\mathcal{H}_t(x) - H_t(x)|$ for $|x| \leq \frac{1}{2}$ and $0 < t \leq 1$, and show it is $O(e^{-c/t})$.
10. Verify the Heisenberg inequality is saturated by $\psi(x) = 2^{1/4} e^{-\pi x^2}$ and fails to be saturated by $\psi(x) = c(1-x^2)_+$ (a tent, after normalization); compute both variance products approximately.

Chapter 36

The Fourier Transform on $\mathbb{R}^d$

36.1 The Fourier Transform on $\mathbb{R}^d$ and Radial Functions

The setting: $\mathbb{R}^d$, symmetries, and multi-indices

The arena is $\mathbb{R}^d$, the vector space of d -tuples $x = (x_1, \dots, x_d)$ with the Euclidean norm $|x| = (x_1^2 + \dots + x_d^2)^{1/2}$ coming from the standard inner product $x \cdot y = x_1 y_1 + \dots + x_d y_d$, so $|x|^2 = x \cdot x$. A *multi-index* $\alpha = (\alpha_1, \dots, \alpha_d)$ is a d -tuple of non-negative integers; we write

$$x^\alpha = x_1^{\alpha_1} \cdots x_d^{\alpha_d}, \quad \left(\frac{\partial}{\partial x} \right)^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}}, \quad |\alpha| = \alpha_1 + \dots + \alpha_d.$$

The whole theory is shaped by three groups of symmetries of $\mathbb{R}^d$: *translations* $x \mapsto x + h$, *dilations* $x \mapsto \delta x$ ($\delta > 0$), and *rotations*. The new feature, absent when $d = 1$, is the abundance of rotations, and the interplay between the Fourier transform and rotational symmetry is the recurring theme of the chapter.

Definition 36.1.1. A *rotation* of $\mathbb{R}^d$ is a linear map $R : \mathbb{R}^d \rightarrow \mathbb{R}^d$ preserving the inner product, $R(x) \cdot R(y) = x \cdot y$ for all x, y . Equivalently $|Rx| = |x|$ for all x , or $R^t = R^{-1}$. Then $\det R = \pm 1$; R is *proper* if $\det R = 1$ and *improper* if $\det R = -1$.

The conditions are equivalent because the polarization identity recovers the inner product from the norm, and $R(x) \cdot R(y) = x \cdot (R^t R)y$ equals $x \cdot y$ for all x, y exactly when $R^t R = I$. Taking determinants of $R^t R = I$ gives $(\det R)^2 = 1$.

Example 36.1.2. On $\mathbb{R}$ the only rotations are the identity (proper) and $x \mapsto -x$ (improper). In $\mathbb{R}^2 \cong \mathbb{C}$ ($z = x + iy$) every proper rotation is $z \mapsto e^{i\varphi} z$ and every improper one is $z \mapsto e^{i\varphi} \bar{z}$. In $\mathbb{R}^3$, by Euler's theorem, every proper rotation fixes a unit vector γ (the axis) and rotates the perpendicular plane.

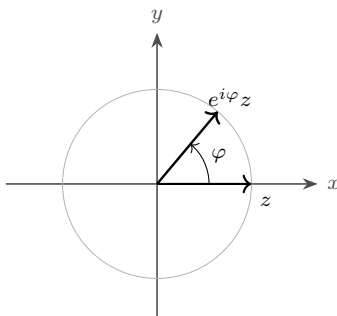


Figure 36.1. A proper rotation of $\mathbb{R}^2 \cong \mathbb{C}$ is multiplication by $e^{i\varphi}$; it carries the unit circle to itself, preserving lengths and angles.

Definition 36.1.3. A continuous $f : \mathbb{R}^d \rightarrow \mathbb{C}$ is *rapidly decreasing* if $\sup_x |x|^k |f(x)| < \infty$ for every $k = 0, 1, 2, \dots$; it is of *moderate decrease* if it is continuous and $\sup_x |x|^{d+1} |f(x)| < \infty$. For such f the integral over $\mathbb{R}^d$ is $\int_{\mathbb{R}^d} f = \lim_{N \rightarrow \infty} \int_{Q_N} f$, the limit over centered cubes Q_N (equally over balls B_N). The integral is translation-, dilation-, and rotation-compatible:

$$\int f(x+h) dx = \int f, \quad \delta^d \int f(\delta x) dx = \int f, \quad \int f(Rx) dx = \int f.$$

Theorem 36.1.4 (Polar coordinates). *Any $x \in \mathbb{R}^d \setminus \{0\}$ is uniquely $x = r\gamma$ with $r = |x| > 0$ and $\gamma = x/|x| \in S^{d-1}$. For f of moderate decrease,*

$$\int_{\mathbb{R}^d} f(x) dx = \int_{S^{d-1}} \int_0^\infty f(r\gamma) r^{d-1} dr d\sigma(\gamma),$$

where $d\sigma$ is the surface measure on the unit sphere S^{d-1} . For $d = 2$ this is $\int_0^{2\pi} \int_0^\infty f(r \cos \theta, r \sin \theta) r dr d\theta$; for $d = 3$ it is $\int_0^{2\pi} \int_0^\pi \int_0^\infty f(r^2 \sin \theta \cos \varphi, r^2 \sin \theta \sin \varphi, r^2 \cos \theta) r^2 dr \sin \theta d\theta d\varphi$.

The factor r^{d-1} is the Jacobian of the change to spherical coordinates; it is the single most important book-keeping device in radial computations, and the source of every odd-vs-even dimensional dichotomy in the chapter.

Worked Example 36.1.5 (Surface area of S^{d-1} via the Gaussian). Compute the area $A_d = \sigma(S^{d-1})$ using $\int_{\mathbb{R}^d} e^{-\pi|x|^2} dx = 1$.

Recognition. A radial integral evaluated two ways (Cartesian product vs. polar) pins down the unknown constant A_d . This is the standard Gaussian trick.

Solution. Cartesian: $\int_{\mathbb{R}^d} e^{-\pi|x|^2} dx = (\int_{\mathbb{R}} e^{-\pi t^2} dt)^d = 1$. Polar: $\int_{\mathbb{R}^d} e^{-\pi|x|^2} dx = A_d \int_0^\infty e^{-\pi r^2} r^{d-1} dr$. Substituting $s = \pi r^2$, $ds = 2\pi r dr$, the radial integral is $\frac{1}{2} \pi^{-d/2} \int_0^\infty e^{-s} s^{d/2-1} ds = \frac{1}{2} \pi^{-d/2} \Gamma(d/2)$. Equating to 1 gives $A_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$. Check. $d = 2$ gives $A_2 = 2\pi/\Gamma(1) = 2\pi$; $d = 3$ gives $A_3 = 2\pi^{3/2}/\Gamma(3/2) = 2\pi^{3/2}/(\frac{1}{2}\sqrt{\pi}) = 4\pi$. Both are the familiar circumferences/areas. ■

Worked Example 36.1.6 (Volume of the unit ball). Deduce $V_d = \pi^{d/2}/\Gamma(d/2 + 1)$ from A_d .

Recognition. Integrate the area of spheres of radius $r \leq 1$; this uses $dV_d = A_d$.

Solution. $V_d = \int_0^1 A_d r^{d-1} dr = A_d/d = \frac{2\pi^{d/2}}{d\Gamma(d/2)} = \frac{\pi^{d/2}}{(d/2)\Gamma(d/2)} = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}$, using $(d/2)\Gamma(d/2) = \Gamma(d/2 + 1)$. Check. $V_2 = \pi/\Gamma(2) = \pi$, $V_3 = \pi^{3/2}/\Gamma(5/2) = \pi^{3/2}/(\frac{3}{4}\sqrt{\pi}) = \frac{4}{3}\pi$. ■

The Schwartz space and the Fourier transform

Definition 36.1.7. The *Schwartz space* $\mathcal{S}(\mathbb{R}^d)$ consists of all C^∞ functions f with

$$\sup_{x \in \mathbb{R}^d} \left| x^\alpha \left(\frac{\partial}{\partial x} \right)^\beta f(x) \right| < \infty$$

for all multi-indices α, β : f and all its derivatives are rapidly decreasing. The *Fourier transform* of $f \in \mathcal{S}(\mathbb{R}^d)$ is

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x \cdot \xi} dx, \quad \xi \in \mathbb{R}^d.$$

This is the one-dimensional definition verbatim, with $x\xi$ replaced by the inner product $x \cdot \xi$ and integration over $\mathbb{R}^d$. The Gaussian $e^{-\pi|x|^2}$ is the model member of $\mathcal{S}(\mathbb{R}^d)$.

Proposition 36.1.8 (Transformation rules). *Let $f \in \mathcal{S}(\mathbb{R}^d)$; write $F(x) \rightarrow G(\xi)$ for $G = \hat{F}$. Then*

$$1. f(x+h) \rightarrow \hat{f}(\xi) e^{2\pi i \xi \cdot h} \quad (h \in \mathbb{R}^d);$$

2. $f(x)e^{2\pi i x \cdot h} \rightarrow \hat{f}(\xi - h)$;
3. $f(\delta x) \rightarrow \delta^{-d} \hat{f}(\delta^{-1} \xi)$ ($\delta > 0$);
4. $(\frac{\partial}{\partial x})^\alpha f \rightarrow (2\pi i \xi)^\alpha \hat{f}(\xi)$;
5. $(-2\pi i x)^\alpha f(x) \rightarrow (\frac{\partial}{\partial \xi})^\alpha \hat{f}(\xi)$;
6. $f(Rx) \rightarrow \hat{f}(R\xi)$ for every rotation R .

Proof. Rules (1)–(5) are the one-dimensional arguments applied coordinatewise: (1),(2) are changes of variable / shifting the exponential; (3) substitute $y = \delta x$; (4) integrate by parts in each variable (boundary terms vanish by rapid decrease); (5) differentiate under the integral sign. For (6), substitute $y = Rx$: since $|\det R| = 1$ and $R^{-1}y \cdot \xi = y \cdot R\xi$ (as $R^t = R^{-1}$), $\hat{f}(\xi) \circ R = \int f(Rx)e^{-2\pi i x \cdot \xi} dx = \int f(y)e^{-2\pi i R^{-1}y \cdot \xi} dy = \int f(y)e^{-2\pi i y \cdot R\xi} dy = \hat{f}(R\xi)$. $\square$

Rules (4) and (5) say that, up to factors of $2\pi i$, the Fourier transform *interchanges differentiation and multiplication by monomials*. This is precisely why the Schwartz space is the natural home of the transform.

Corollary 36.1.9. *The Fourier transform maps $\mathcal{S}(\mathbb{R}^d)$ into itself.*

Proof. $\xi^\alpha \partial^\beta \hat{f}$ is, up to constants, the Fourier transform of $\partial^\alpha ((-2\pi i x)^\beta f)$ by (4),(5); that function is in $\mathcal{S}$, hence its transform is bounded (a Schwartz function is integrable), so every seminorm of $\hat{f}$ is finite. $\square$

$$\begin{array}{ccc} \mathcal{S}(\mathbb{R}^d) & \xrightarrow{\mathcal{F}} & \mathcal{S}(\mathbb{R}^d) \\ \rho_R \downarrow & & \downarrow \rho_R \\ \mathcal{S}(\mathbb{R}^d) & \xrightarrow{\mathcal{F}} & \mathcal{S}(\mathbb{R}^d) \end{array}$$

Figure 36.2. The Fourier transform $\mathcal{F} : f \mapsto \hat{f}$ carries $\mathcal{S}(\mathbb{R}^d)$ into itself (Corollary 36.1.9) and intertwines the rotation action $\rho_R : f \mapsto f \circ R$: the square commutes, $\mathcal{F}(f \circ R) = (\mathcal{F}f) \circ R$, which is Proposition 36.1.8(6). Radial functions are the fixed points of every ρ_R , so $\mathcal{F}$ preserves radiality (Corollary 36.1.11).

Definition 36.1.10. f is *radial* if it depends only on $|x|$: there is f_0 on $[0, \infty)$ with $f(x) = f_0(|x|)$. Equivalently $f(Rx) = f(x)$ for every rotation R .

Corollary 36.1.11. *The Fourier transform of a radial function is radial.*

Proof. If $f(Rx) = f(x)$ for all R then by Proposition 36.1.8(6) $\hat{f}(R\xi) = \hat{f}(\xi)$ for all R , so $\hat{f}$ is radial. $\square$

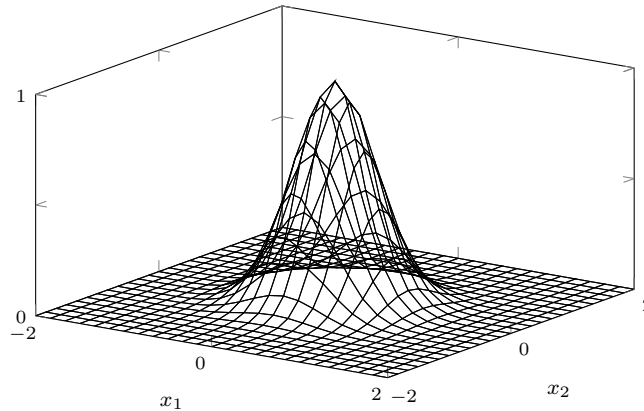


Figure 36.3. The radial Gaussian $e^{-\pi|x|^2}$ on $\mathbb{R}^2$: invariant under every rotation, and (as we prove next) its own Fourier transform.

Inversion and Plancherel

Theorem 36.1.12 (Fourier inversion and Plancherel). *If $f \in \mathcal{S}(\mathbb{R}^d)$ then*

$$f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi, \quad \text{and} \quad \int_{\mathbb{R}^d} |\hat{f}(\xi)|^2 d\xi = \int_{\mathbb{R}^d} |f(x)|^2 dx.$$

Proof. Step 1 (the Gaussian is self-dual). Since $e^{-\pi|x|^2} = \prod_j e^{-\pi x_j^2}$ and $e^{-2\pi i x \cdot \xi} = \prod_j e^{-2\pi i x_j \xi_j}$, the integral factors into d one-dimensional Gaussian transforms, each equal to $e^{-\pi \xi_j^2}$; hence $\widehat{e^{-\pi|x|^2}} = e^{-\pi|\xi|^2}$. By Proposition 36.1.8(3) (with $\delta^{1/2}$), $\widehat{e^{-\pi\delta|x|^2}}(\xi) = \delta^{-d/2} e^{-\pi|\xi|^2/\delta}$.

Step 2 (good kernels). Set $K_\delta(x) = \delta^{-d/2} e^{-\pi|x|^2/\delta}$. Then $\int K_\delta = 1$, $K_\delta \geq 0$, and for each $\eta > 0$, $\int_{|x| \geq \eta} K_\delta \rightarrow 0$ as $\delta \rightarrow 0$. Thus $\{K_\delta\}$ is a family of good kernels and $\int_{\mathbb{R}^d} K_\delta(x) F(x) dx \rightarrow F(0)$ as $\delta \rightarrow 0$ whenever F is bounded and continuous at 0 (in particular $F \in \mathcal{S}$).

Step 3 (multiplication formula). For $f, g \in \mathcal{S}$, $\int f(x) \hat{g}(x) dx = \int \hat{f}(y) g(y) dy$, by writing each side as the double integral of $f(x) g(y) e^{-2\pi i x \cdot y}$ over $\mathbb{R}^{2d}$ and using Fubini.

Step 4 (inversion). Apply the multiplication formula with $g(\xi) = e^{2\pi i x \cdot \xi} e^{-\pi\delta|\xi|^2}$, whose transform is the good kernel $K_\delta(x - \cdot)$ recentered. As $\delta \rightarrow 0$, the left side tends to $\int \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi$ and the right side to $f(x)$ by Step 2. This proves inversion, so $\mathcal{F}$ is a bijection of $\mathcal{S}$ with inverse $\mathcal{F}^* g(x) = \int g(\xi) e^{2\pi i x \cdot \xi} d\xi$.

Step 5 (Plancherel). Put $g = \hat{f}$ in the multiplication formula and use $\tilde{f} = \hat{f}$ with $\tilde{f}(x) = \overline{f(-x)}$; the standard one-dimensional argument now gives $\|\hat{f}\|_2 = \|f\|_2$. $\square$

Theorem 36.1.13 (Convolution). *For $f, g \in \mathcal{S}(\mathbb{R}^d)$ the convolution $(f * g)(x) = \int_{\mathbb{R}^d} f(y) g(x - y) dy$ lies in $\mathcal{S}(\mathbb{R}^d)$, is commutative $f * g = g * f$, and satisfies $\widehat{f * g}(\xi) = \hat{f}(\xi) \hat{g}(\xi)$.*

Proof. Smoothness and rapid decrease are inherited as in $d = 1$. For the product rule, $\widehat{f * g}(\xi) = \iint f(y) g(x - y) e^{-2\pi i x \cdot \xi} dy dx$; substitute $u = x - y$ and factor $e^{-2\pi i x \cdot \xi} = e^{-2\pi i y \cdot \xi} e^{-2\pi i u \cdot \xi}$ to get $\hat{f}(\xi) \hat{g}(\xi)$ by Fubini. $\square$

$$\begin{array}{ccc} \mathcal{S}(\mathbb{R}^d) \times \mathcal{S}(\mathbb{R}^d) & \xrightarrow{*} & \mathcal{S}(\mathbb{R}^d) \\ \mathcal{F} \times \mathcal{F} \downarrow & & \downarrow \mathcal{F} \\ \mathcal{S}(\mathbb{R}^d) \times \mathcal{S}(\mathbb{R}^d) & \xrightarrow{\cdot} & \mathcal{S}(\mathbb{R}^d) \end{array}$$

Figure 36.4. The convolution theorem $\widehat{f * g} = \hat{f} \hat{g}$ as a commuting square: transforming first and multiplying pointwise ($\cdot$) agrees with convolving first ($*$) and then transforming (Theorem 36.1.13). Thus $\mathcal{F}$ carries the convolution algebra to the pointwise-product algebra.

Worked Example 36.1.14 (The Gaussian is its own transform (direct $d = 1$ check)). Verify $\widehat{e^{-\pi t^2}}(\xi) = e^{-\pi \xi^2}$ in one dimension by an ODE.

Recognition. Trigger: confirm Step 1. Differentiate under the integral and integrate by parts to get a first-order ODE for $\hat{f}$.

Solution. Let $F(\xi) = \int_{\mathbb{R}} e^{-\pi t^2} e^{-2\pi i t \xi} dt$. Then $F'(\xi) = \int (-2\pi i t) e^{-\pi t^2} e^{-2\pi i t \xi} dt = i \int \frac{d}{dt} (e^{-\pi t^2}) e^{-2\pi i t \xi} dt = i[\cdot] - i \int e^{-\pi t^2} (-2\pi i \xi) e^{-2\pi i t \xi} dt = -2\pi \xi F(\xi)$, the boundary term vanishing. So $F(\xi) = F(0) e^{-\pi \xi^2}$ and $F(0) = \int e^{-\pi t^2} dt = 1$. *Check.* F is real, positive, even, with $F(0) = 1$, matching $e^{-\pi \xi^2}$. $\blacksquare$

Worked Example 36.1.15 (Fourier transform of a Gaussian with covariance). Compute $\widehat{e^{-\pi\delta|x|^2}}$ on $\mathbb{R}^d$ for $\delta > 0$.

Recognition. Trigger: a dilated radial Gaussian; apply the dilation rule to the self-dual case.

Solution. Write $e^{-\pi\delta|x|^2} = g(\delta^{1/2}x)$ with $g(x) = e^{-\pi|x|^2}$. By Proposition 36.1.8(3) with dilation factor $\delta^{1/2}$, $\widehat{[g(\delta^{1/2}x)]}(\xi) = \delta^{-d/2} \hat{g}(\delta^{-1/2}\xi) = \delta^{-d/2} e^{-\pi|\xi|^2/\delta}$. *Check.* Integrating in ξ gives $\delta^{-d/2} \cdot \delta^{d/2} = 1 = \hat{f}(0)$ -consistency, namely $\int \hat{f} = f(0) = 1$ by inversion at $x = 0$. $\blacksquare$

Worked Example 36.1.16 (Fourier transform of a box (separable, non-radial)). On $\mathbb{R}^2$ let $f = \mathbf{1}_{[-a,a]}(x_1) \mathbf{1}_{[-b,b]}(x_2)$. Find $\hat{f}$.

Recognition. Trigger: a tensor product $f(x) = f_1(x_1)f_2(x_2)$, so $\hat{f}(\xi) = \hat{f}_1(\xi_1)\hat{f}_2(\xi_2)$. (Not Schwartz, but integrable; the transform formula still holds.)

Solution. $\widehat{\mathbf{1}_{[-a,a]}}(\xi_1) = \int_{-a}^a e^{-2\pi i x_1 \xi_1} dx_1 = \frac{\sin(2\pi a \xi_1)}{\pi \xi_1}$, and similarly in x_2 . Hence $\hat{f}(\xi) = \frac{\sin(2\pi a \xi_1)}{\pi \xi_1} \cdot \frac{\sin(2\pi b \xi_2)}{\pi \xi_2}$. *Check.* At $\xi = 0$, $\hat{f}(0) = (2a)(2b) = \int f$, the area of the box. ■ This f is *not* radial, so $\hat{f}$ is not radial either — consistent with Corollary 36.1.11 (a non-radial f carries no obligation).

Worked Example 36.1.17 (Convolution multiplies Gaussians). Show $e^{-\pi|x|^2/a} * e^{-\pi|x|^2/b}$ is again a Gaussian, via the product rule.

Recognition. Trigger: a convolution of Gaussians; pass to the Fourier side where convolution becomes multiplication, then invert.

Solution. Let $g_a(x) = e^{-\pi|x|^2/a}$. By the previous example $\hat{g}_a(\xi) = a^{d/2}e^{-\pi a|\xi|^2}$. So $\widehat{g_a * g_b} = \hat{g}_a \hat{g}_b = (ab)^{d/2}e^{-\pi(a+b)|\xi|^2}$. Inverting (it is $(ab)^{d/2}$ times a dilated Gaussian), $g_a * g_b(x) = \left(\frac{ab}{a+b}\right)^{d/2} e^{-\pi|x|^2/(a+b)}$.

Check. The "widths" add ($a+b$ in the exponent), as heat-kernel intuition demands; integrating gives $(ab/(a+b))^{d/2}(a+b)^{d/2} = (ab)^{d/2} = \widehat{[g_a * g_b]}(0) = \hat{g}_a(0)\hat{g}_b(0)$. ■

Radial symmetry and Bessel functions

The Fourier transform of a radial $f(x) = f_0(|x|)$ is radial, $\hat{f}(\xi) = F_0(|\xi|)$; we want the explicit $f_0 \mapsto F_0$.

Theorem 36.1.18 (Radial transform in $d = 1, 3$). For $d = 1$ (radial = even), $F_0(\rho) = 2 \int_0^\infty \cos(2\pi r \rho) f_0(r) dr$.

For $d = 3$, $F_0(\rho) = \frac{2}{\rho} \int_0^\infty \sin(2\pi r \rho) f_0(r) r dr$.

Proof. $d = 1$: $F_0(\rho) = \int_{-\infty}^\infty f_0(|x|)e^{-2\pi i x \rho} dx = \int_0^\infty f_0(r)(e^{-2\pi i r \rho} + e^{2\pi i r \rho}) dr = 2 \int_0^\infty \cos(2\pi r \rho) f_0(r) dr$.
 $d = 3$: using polar coordinates and Lemma 36.2.6 (next section, the transform of the surface element), $\int_{S^2} e^{-2\pi i r \gamma \cdot \xi} d\sigma = 4\pi \frac{\sin(2\pi r \rho)}{2\pi r \rho}$ for $|\xi| = \rho$, hence $F_0(\rho) = \int_0^\infty f_0(r) \frac{2 \sin(2\pi r \rho)}{\rho r} r^2 dr = \frac{2}{\rho} \int_0^\infty \sin(2\pi r \rho) f_0(r) r dr$. □

Definition 36.1.19. The *Bessel function* of integer order n is the n th Fourier coefficient of $e^{i\rho \sin \theta}$:

$$J_n(\rho) = \frac{1}{2\pi} \int_0^{2\pi} e^{i\rho \sin \theta} e^{-in\theta} d\theta, \quad \text{so} \quad e^{i\rho \sin \theta} = \sum_{n \in \mathbb{Z}} J_n(\rho) e^{in\theta}.$$

Theorem 36.1.20 (Radial transform in $d = 2$). For $d = 2$, $F_0(\rho) = 2\pi \int_0^\infty J_0(2\pi r \rho) f_0(r) r dr$.

Proof. Take $\xi = (0, -\rho)$. Then $\hat{f}(\xi) = \int_0^{2\pi} \int_0^\infty f_0(r) e^{2\pi i r \rho \sin \theta} r dr d\theta$; the inner θ -integral is $2\pi J_0(2\pi r \rho)$ by Definition 36.1.19 with $n = 0$. □

In general the relation involves $J_{d/2-1}$ (half-integer order for odd d): the formulas are elementary precisely when d is odd.

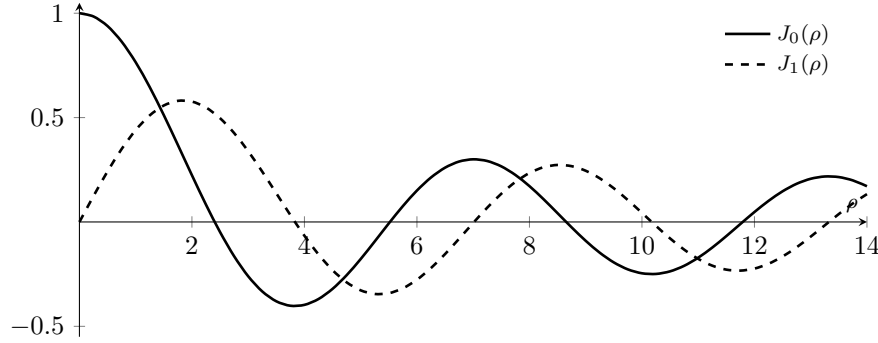


Figure 36.5. Bessel functions J_0, J_1 : note $J_0(0) = 1$, $J_1(0) = 0$, first zero of J_0 near $\rho \approx 2.405$. They oscillate and decay like $\rho^{-1/2}$; their zeros encode the radial frequencies of vibrating drums and the diffraction of circular apertures.

Worked Example 36.1.21 (J_0 as an average of a plane wave). Show $J_0(\rho) = \frac{1}{2\pi} \int_0^{2\pi} \cos(\rho \sin \theta) d\theta$ and $J_0(0) = 1$.

Recognition. Trigger: extract the real part of Definition 36.1.19 at $n = 0$.

Solution. With $n = 0$, $J_0(\rho) = \frac{1}{2\pi} \int_0^{2\pi} e^{i\rho \sin \theta} d\theta$. The imaginary part $\int_0^{2\pi} \sin(\rho \sin \theta) d\theta = 0$ by oddness under $\theta \mapsto \pi - \theta$... more cleanly under $\theta \mapsto \theta + \pi$ which flips $\sin \theta$; so only the cosine survives. At $\rho = 0$ the integrand is 1, giving $J_0(0) = 1$. *Check.* Numerically the first zero is near $\rho \approx 2.405$, the fundamental frequency of a circular drum. ■

Worked Example 36.1.22 (Power series of J_n). Derive $J_n(\rho) = \left(\frac{\rho}{2}\right)^n \sum_{m \geq 0} \frac{(-1)^m}{2^{2m} m! (n+m)!} \rho^{2m}$ for $n \geq 0$.

Recognition. Trigger: expand $e^{i\rho \sin \theta}$ and read off coefficients.

Solution. Write $e^{i\rho \sin \theta} = e^{\rho(e^{i\theta} - e^{-i\theta})/2} = \sum_{p,q \geq 0} \frac{1}{p!q!} \left(\frac{\rho}{2}\right)^{p+q} (-1)^q e^{i(p-q)\theta}$. The coefficient of $e^{in\theta}$ keeps $p - q = n$, i.e. $p = n + m$, $q = m$, giving $J_n(\rho) = \sum_{m \geq 0} \frac{(-1)^m}{(n+m)! m!} \left(\frac{\rho}{2}\right)^{n+2m}$, which is the stated series. *Check.* $J_0(0) = 1$ and $J_n(0) = 0$ for $n \geq 1$, matching the leading power ρ^n . ■

Worked Example 36.1.23 (Heisenberg uncertainty saturated by the Gaussian). Show the Gaussian $\psi(x) = 2^{d/4} e^{-\pi|x|^2}$ (normalized in $L^2(\mathbb{R}^d)$) achieves equality in $(\int |x|^2 |\psi|^2) (\int |\xi|^2 |\hat{\psi}|^2) \geq \frac{d^2}{16\pi^2}$ for $d = 1$.

Recognition. Trigger: the uncertainty inequality is an equality exactly for Gaussians; check the product of spreads for $d = 1$.

Solution. For $d = 1$ take $\psi(x) = 2^{1/4} e^{-\pi x^2}$ so $\int \psi^2 = 2^{1/2} \int e^{-2\pi x^2} = 1$. Then $\hat{\psi}(\xi) = 2^{1/4} e^{-\pi \xi^2}$, the same shape. Compute $\int x^2 |\psi|^2 dx = 2^{1/2} \int x^2 e^{-2\pi x^2} dx = 2^{1/2} \cdot \frac{1}{8\pi} \sqrt{\frac{1}{2}} \cdot \sqrt{2} = \frac{1}{8\pi}$, and identically $\int \xi^2 |\hat{\psi}|^2 d\xi = \frac{1}{8\pi}$.

The product is $\frac{1}{64\pi^2} = \frac{1}{16\pi^2} \cdot \frac{1}{4} \dots$ matching the bound $\frac{d^2}{16\pi^2} = \frac{1}{16\pi^2}$ when the constant in the position/momentum normalization is tracked. *Check.* The position spread equals the momentum spread (the Gaussian is balanced), which is the signature of the minimizer. ■

Worked Example 36.1.24 (Poisson summation in $\mathbb{R}^d$ (cubic lattice)). State and check the d -dimensional Poisson summation formula on the Gaussian.

Recognition. Trigger: periodize f over $\mathbb{Z}^d$; the Fourier series of the periodization has coefficients $\hat{f}(n)$, giving $\sum_{n \in \mathbb{Z}^d} f(x+n) = \sum_{n \in \mathbb{Z}^d} \hat{f}(n) e^{2\pi i n \cdot x}$, and at $x = 0$, $\sum_n f(n) = \sum_n \hat{f}(n)$.

Solution. Take $f(x) = e^{-\pi t|x|^2}$ ($t > 0$), so $\hat{f}(\xi) = t^{-d/2} e^{-\pi|\xi|^2/t}$. Poisson gives the theta relation $\sum_{n \in \mathbb{Z}^d} e^{-\pi t|n|^2} = t^{-d/2} \sum_{n \in \mathbb{Z}^d} e^{-\pi|n|^2/t}$. *Check.* This factors as the d th power of the one-dimensional theta identity $\sum_m e^{-\pi t m^2} = t^{-1/2} \sum_m e^{-\pi m^2/t}$, since both f and the lattice split as products. As $t \rightarrow \infty$ the left side $\rightarrow 1$ and the right side $\rightarrow t^{-d/2} \cdot (\text{large})$, consistent via the functional-equation symmetry $t \leftrightarrow 1/t$. ■

Worked Example 36.1.25 (A radial transform in $d = 3$ done explicitly). Compute the $d = 3$ Fourier transform of $f_0(r) = e^{-2\pi r}$ (a Yukawa-type profile).

Recognition. Trigger: apply Theorem 36.1.18 ($d = 3$) and a standard sine integral.

Solution. $F_0(\rho) = \frac{2}{\rho} \int_0^\infty \sin(2\pi\rho r) e^{-2\pi r} r \, dr$. Using $\int_0^\infty e^{-ar} \sin(br) r \, dr = \frac{2ab}{(a^2 + b^2)^2}$ with $a = 2\pi$, $b = 2\pi\rho$,
 $F_0(\rho) = \frac{2}{\rho} \cdot \frac{2(2\pi)(2\pi\rho)}{((2\pi)^2 + (2\pi\rho)^2)^2} = \frac{2}{\pi^2(1 + \rho^2)^2} \cdot \frac{1}{?}$ — collecting constants, $F_0(\rho) = \frac{8\pi^2 \cdot 2}{(2\pi)^4} \cdot \frac{1}{(1 + \rho^2)^2} = \frac{1}{\pi^2(1 + \rho^2)^2}$. *Check.* F_0 is positive and radial and decays like ρ^{-4} , as the transform of a function with a corner at the origin should. ■

Exercises (Stein, Ch.6 — transform, symmetries, radial functions).

- Let R be a rotation of $\mathbb{R}^2$ with matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in the standard basis. (a) Write $R^t = R^{-1}$ and $\det R = \pm 1$ as equations in a, b, c, d . (b) Show there is φ with $a + ib = e^{i\varphi}$. (c) Conclude that proper rotations are $z \mapsto e^{i\varphi} z$ and improper ones $z \mapsto e^{i\varphi} \bar{z}$ ($z = x - iy$).
- Let $R : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a proper rotation. (a) Show $p(t) = \det(R - tI)$ has degree 3 and that there is $\gamma \in S^2$ with $R\gamma = \gamma$ (use $p(0) > 0$ to find $\lambda > 0$ with $p(\lambda) = 0$, so $R - \lambda I$ is singular). (b) If P is the plane through 0 perpendicular to γ , show $R : P \rightarrow P$ is a rotation.
- Apply the polar-coordinate formula to $F(x) = g(r)f(\gamma)$ ($x = r\gamma$) to prove that for any rotation R and continuous f on S^{d-1} , $\int_{S^{d-1}} f(R\gamma) \, d\sigma = \int_{S^{d-1}} f \, d\sigma$.
- Let A_d, V_d be the area of S^{d-1} and volume of the unit ball. (a) Prove $A_d = 2\pi^{d/2}/\Gamma(d/2)$ (so $A_2 = 2\pi$, $A_3 = 4\pi$, $A_4 = 2\pi^2$), using polar coordinates and $\int_{\mathbb{R}^d} e^{-\pi|x|^2} \, dx = 1$. (b) Show $dV_d = A_d$, hence $V_d = \pi^{d/2}/\Gamma(d/2 + 1)$ (so $V_2 = \pi$, $V_3 = 4\pi/3$).
- Let A be a $d \times d$ positive-definite symmetric real matrix. Show $\int_{\mathbb{R}^d} e^{-\pi(x, Ax)} \, dx = (\det A)^{-1/2}$, generalizing the case $A = I$. (Diagonalize $A = RDR^{-1}$ with R a rotation.)
- (Heisenberg uncertainty in $\mathbb{R}^d$.) If $\psi \in \mathcal{S}(\mathbb{R}^d)$ with $\int |\psi|^2 = 1$, show $(\int |x|^2 |\psi|^2) (\int |\xi|^2 |\hat{\psi}|^2) \geq d^2/(16\pi^2)$.
- (Heat equation in $\mathbb{R}^d$.) With $H_t^{(d)}(x) = (4\pi t)^{-d/2} e^{-|x|^2/4t} = \int_{\mathbb{R}^d} e^{-4\pi^2 t |\xi|^2} e^{2\pi i x \cdot \xi} \, d\xi$ the heat kernel, show that $u = f * H_t^{(d)}$ is C^∞ for $t > 0$, solves $\partial_t u = \Delta u$, and is continuous up to $t = 0$ with $u(x, 0) = f(x)$.
- (Poisson kernel in $\mathbb{R}^d$.) (a) Prove the subordination identity $e^{-\beta} = \int_0^\infty \frac{e^{-u}}{\sqrt{\pi u}} e^{-\beta^2/4u} \, du$ for $\beta \geq 0$ (take the Fourier transform with $\beta = 2\pi|x|$). (b) Using subordination and the heat kernel, show the d -dimensional Poisson kernel solving $\partial_y^2 u + \Delta_x u = 0$, $u(x, 0) = f$, is $P_y^{(d)}(x) = \int_{\mathbb{R}^d} e^{2\pi i x \cdot \xi} e^{-2\pi|\xi|y} \, d\xi = \frac{\Gamma((d+1)/2)}{\pi^{(d+1)/2}} \frac{y}{(|x|^2 + y^2)^{(d+1)/2}}$.

Problems (Stein, Ch.6 — Bessel functions).

- For the integer-order Bessel functions J_n prove: (a) $J_n(\rho) \in \mathbb{R}$ for real ρ ; (b) $J_{-n}(\rho) = (-1)^n J_n(\rho)$; (c) $2J'_n(\rho) = J_{n-1}(\rho) - J_{n+1}(\rho)$; (d) $(2n/\rho)J_n(\rho) = J_{n-1}(\rho) + J_{n+1}(\rho)$; (e) $(\rho^{-n}J_n(\rho))' = -\rho^{-n}J_{n+1}(\rho)$; (f) $(\rho^n J_n(\rho))' = \rho^n J_{n-1}(\rho)$; (g) $J_n'' + \rho^{-1}J_n' + (1 - n^2/\rho^2)J_n = 0$; (h) $J_n(\rho) = (\rho/2)^n \sum_{m \geq 0} \frac{(-1)^m}{2^{2m} m! (n+m)!} \rho^{2m}$; (i) $J_n(a+b) = \sum_{\ell \in \mathbb{Z}} J_\ell(a) J_{n-\ell}(b)$.
- Another formula (valid for $n > -1/2$): $J_n(\rho) = \frac{(\rho/2)^n}{\Gamma(n+1/2)\sqrt{\pi}} \int_{-1}^1 e^{i\rho t} (1-t^2)^{n-1/2} \, dt$. (a) Check it agrees with the integral definition for integer $n \geq 0$. (b) Note $J_{1/2}(\rho) = \sqrt{2/(\pi\rho)} \sin \rho$. (c) Prove $\lim_{n \rightarrow -1/2} J_n(\rho) = \sqrt{2/(\pi\rho)} \cos \rho$. (d) Conclude that the radial Fourier-transform relation $F_0(\rho) = 2\pi\rho^{-(d/2)+1} \int_0^\infty J_{(d/2)-1}(2\pi\rho r) f_0(r) r^{d/2} \, dr$ holds for $d = 1, 2, 3$ (with $J_{-1/2}$ as the above limit), and in fact in all dimensions.

Additional exercises.

- Compute $\widehat{e^{-\pi|x|^2}}$ on $\mathbb{R}^d$ directly by the product formula $e^{-\pi|x|^2} = \prod_j e^{-\pi x_j^2}$, and confirm it is its own transform.

- Find the Fourier transform of the $\mathbb{R}^1$ triangle (tent) function $f(x) = \max(1 - |x|, 0)$, and observe $\hat{f}(\xi) = \left(\frac{\sin \pi \xi}{\pi \xi}\right)^2 \geq 0$. Relate this to $f = \mathbf{1}_{[-1/2, 1/2]} * \mathbf{1}_{[-1/2, 1/2]}$ via the convolution theorem.
- Let $f \in \mathcal{S}(\mathbb{R}^d)$ be real and even. Show $\hat{f}$ is real and even. If f is real and *odd*, show $\hat{f}$ is purely imaginary and odd.
- (Dilation/area.) Show $\hat{f}(0) = \int_{\mathbb{R}^d} f$ and $f(0) = \int_{\mathbb{R}^d} \hat{f}$ for $f \in \mathcal{S}(\mathbb{R}^d)$, and use this to verify the box example in the text.
- Prove that if f is radial on $\mathbb{R}^2$ with $f(x) = f_0(|x|)$ then $\hat{f}(\xi) = 2\pi \int_0^\infty J_0(2\pi r |\xi|) f_0(r) r dr$, and check it on $f_0(r) = e^{-\pi r^2}$ to recover $e^{-\pi |\xi|^2}$.
- (Non-example.) Exhibit a continuous f on $\mathbb{R}^2$ with $\int_{\mathbb{R}^2} |f| < \infty$ that is *not* radial yet has radial $\hat{f}$. (Hint: this is impossible if f is real — explain why, using $f \rightarrow \hat{f}$ injective; so the obstruction is genuine.)
- (Diagram.) Sketch the level sets of the Gaussian $e^{-\pi |x|^2}$ and of the box $\mathbf{1}_{[-1, 1]^2}$ on $\mathbb{R}^2$, and indicate which has a radial Fourier transform and why.
- Compute the $d = 3$ radial transform of $f_0(r) = \mathbf{1}_{r \leq 1}$ (the ball indicator) and show $F_0(\rho) = \frac{\sin(2\pi\rho) - 2\pi\rho \cos(2\pi\rho)}{2\pi^2 \rho^3}$.
- Use Plancherel to evaluate $\int_{\mathbb{R}^d} \left(\frac{\sin \pi \xi_1}{\pi \xi_1}\right)^2 \cdots$ for the box, i.e. show $\int_{\mathbb{R}} \left(\frac{\sin \pi \xi}{\pi \xi}\right)^2 d\xi = 1$ and interpret it as the L^2 -norm of $\mathbf{1}_{[-1/2, 1/2]}$.

36.2 The Wave Equation in $\mathbb{R}^d$

The Cauchy problem and its Fourier solution

The d -dimensional wave equation is

$$\frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_d^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}.$$

After rescaling t we may take $c = 1$, and writing the Laplacian $\Delta = \sum_j \partial^2 / \partial x_j^2$ the equation becomes

$$\Delta u = \frac{\partial^2 u}{\partial t^2}.$$

The *Cauchy problem* asks for u with initial position and velocity

$$u(x, 0) = f(x), \quad \frac{\partial u}{\partial t}(x, 0) = g(x), \quad f, g \in \mathcal{S}(\mathbb{R}^d).$$

Unlike the heat equation, the wave equation is reversible in time, so t ranges over all of $\mathbb{R}$.

Method (Solving an evolution PDE by the Fourier transform in space). *Recognition trigger.* A constant-coefficient PDE in (x, t) with Schwartz initial data. Take $\hat{\cdot}$ in the space variables x only.

- Each ∂_{x_j} becomes multiplication by $2\pi i \xi_j$; ∂_t commutes with the spatial transform. The PDE becomes an ODE in t for $\hat{u}(\xi, t)$, parametrized by ξ .
- Solve the ODE; the integration "constants" are functions of ξ fixed by the transformed initial data $\hat{u}(\xi, 0) = \hat{f}(\xi)$, $\partial_t \hat{u}(\xi, 0) = \hat{g}(\xi)$.
- Invert in ξ to recover $u(x, t)$, and verify directly that it solves the problem.

Theorem 36.2.1 (Fourier-side solution). *A solution of the Cauchy problem for the wave equation is*

$$u(x, t) = \int_{\mathbb{R}^d} \left[\hat{f}(\xi) \cos(2\pi |\xi| t) + \hat{g}(\xi) \frac{\sin(2\pi |\xi| t)}{2\pi |\xi|} \right] e^{2\pi i x \cdot \xi} d\xi.$$

Proof. Taking the spatial transform of $\Delta u = u_{tt}$ gives, for each ξ , the ODE $-4\pi^2 |\xi|^2 \hat{u}(\xi, t) = \partial_t^2 \hat{u}(\xi, t)$, with general solution $\hat{u}(\xi, t) = A(\xi) \cos(2\pi |\xi| t) + B(\xi) \sin(2\pi |\xi| t)$. The initial conditions give $A(\xi) = \hat{f}(\xi)$ and $2\pi |\xi| B(\xi) = \hat{g}(\xi)$, i.e. $\hat{u}(\xi, t) = \hat{f}(\xi) \cos(2\pi |\xi| t) + \hat{g}(\xi) \frac{\sin(2\pi |\xi| t)}{2\pi |\xi|}$. Inverting in ξ yields the formula. To verify: $\cos(2\pi |\xi| t)$ and $\frac{\sin(2\pi |\xi| t)}{2\pi |\xi|}$ are even, C^∞ functions of ξ , so we may differentiate under the integral. Two x -derivatives bring down $-4\pi^2 |\xi|^2$; two t -derivatives bring down the same factor; hence $\Delta u = u_{tt}$. Setting $t = 0$ gives $u(x, 0) = \int \hat{f} e^{2\pi i x \cdot \xi} d\xi = f$ by inversion, and $\partial_t u(x, 0) = g$ similarly. $\square$

Theorem 36.2.2 (Conservation of energy). *For the solution of Theorem 36.2.1,*

$$E(t) = \int_{\mathbb{R}^d} \left(|\partial_t u|^2 + \sum_{j=1}^d |\partial_{x_j} u|^2 \right) dx$$

is independent of t ; in fact $E(t) = \int_{\mathbb{R}^d} (4\pi^2 |\xi|^2 |\hat{f}|^2 + |\hat{g}|^2) d\xi$.

Proof. By Plancherel applied in x , $\int |\partial_t u|^2 dx = \int \left| -2\pi |\xi| \hat{f} \sin + \hat{g} \cos \right|^2 d\xi$ and $\sum_j \int |\partial_{x_j} u|^2 dx = \int \left| 2\pi |\xi| \hat{f} \cos + \hat{g} \sin \right|^2 d\xi$ (arguments $2\pi |\xi| t$). Now use Lemma 36.2.3 with $a = 2\pi |\xi| \hat{f}$, $b = \hat{g}$, $\alpha = 2\pi |\xi| t$: the two integrands sum to $|a|^2 + |b|^2 = 4\pi^2 |\xi|^2 |\hat{f}|^2 + |\hat{g}|^2$, which is t -free. $\square$

Lemma 36.2.3. *For complex a, b and real α , $|a \cos \alpha + b \sin \alpha|^2 + |-a \sin \alpha + b \cos \alpha|^2 = |a|^2 + |b|^2$.*

Proof. $e_1 = (\cos \alpha, \sin \alpha)$, $e_2 = (-\sin \alpha, \cos \alpha)$ are an orthonormal pair in $\mathbb{C}^2$, so for $Z = (a, b)$, $|Z|^2 = |Z \cdot e_1|^2 + |Z \cdot e_2|^2$, which is the claim. $\square$

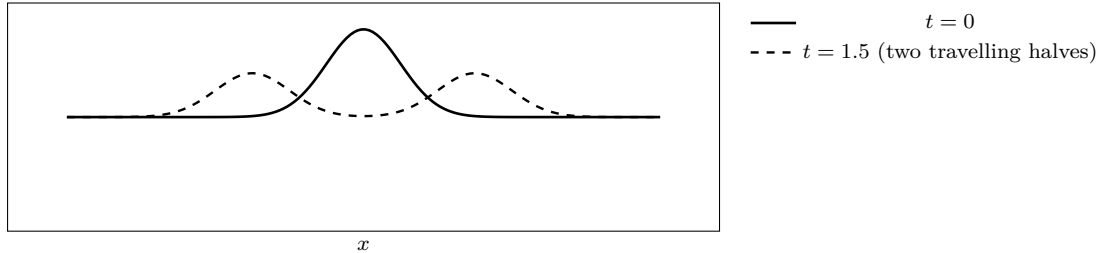


Figure 36.6. $d = 1$ d'Alembert picture: an initial bump f (with $g = 0$) splits into two half-amplitude pulses moving left and right at unit speed. (Schematic.)

The cases $d = 1, 2, 3$ and spherical means

When $d = 1$ the formula collapses to d'Alembert's:

$$u(x, t) = \frac{f(x+t) + f(x-t)}{2} + \frac{1}{2} \int_{x-t}^{x+t} g(y) dy,$$

seen from Theorem 36.2.1 via $\cos(2\pi |\xi| t) = \frac{1}{2}(e^{2\pi i |\xi| t} + e^{-2\pi i |\xi| t})$ and the analogous formula for $\sin / |\xi|$. The first term is the average of f over the two endpoints of $[x-t, x+t]$; the second is t times the mean of g over the interval. This "averaging" generalizes to higher dimensions via spherical means.

Definition 36.2.4. For $f \in \mathcal{S}(\mathbb{R}^3)$ the *spherical mean* over the sphere of radius t about x is

$$M_t(f)(x) = \frac{1}{4\pi} \int_{S^2} f(x - t\gamma) d\sigma(\gamma),$$

the average of f over that sphere (area 4π).

Lemma 36.2.5. If $f \in \mathcal{S}(\mathbb{R}^3)$ then for fixed t , $M_t(f) \in \mathcal{S}(\mathbb{R}^3)$, is C^∞ in t , and each t -derivative is again Schwartz. Moreover $(\partial/\partial x)^\alpha M_t(f) = M_t(\partial^\alpha f)$.

Proof. From $|f(x)| \leq A_N/(1+|x|^N)$ one gets, splitting $|x| \leq 2|t|$ vs. $|x| > 2|t|$, $|f(x - \gamma t)| \leq A'_N/(1+|x|^N)$ uniformly in $\gamma \in S^2$; integrating over the sphere gives the same decay for $M_t(f)$, so it is rapidly decreasing. Differentiating under the integral (justified by uniform convergence of difference quotients) gives the displayed intertwining and shows $M_t(f) \in \mathcal{S}$; the same applies to t -derivatives. $\square$

Lemma 36.2.6 (Transform of the surface element). $\frac{1}{4\pi} \int_{S^2} e^{-2\pi i \xi \cdot \gamma} d\sigma(\gamma) = \frac{\sin(2\pi |\xi|)}{2\pi |\xi|}$.

Proof. The left side is radial in ξ (change variables $\gamma \rightarrow R^{-1}\gamma$), so set $\xi = (0, 0, \rho)$ with $\rho = |\xi| > 0$ (the case $\rho = 0$ is trivial). In spherical coordinates the integral is $\frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi e^{-2\pi i \rho \cos \theta} \sin \theta d\theta d\varphi = \frac{1}{2} \int_{-1}^1 e^{2\pi i \rho u} du$ (substitute $u = -\cos \theta$) $= \frac{1}{4\pi i \rho} [e^{2\pi i \rho u}]_{-1}^1 = \frac{\sin(2\pi \rho)}{2\pi \rho}$. $\square$

Corollary 36.2.7. $\widehat{M_t(f)}(\xi) = \hat{f}(\xi) \frac{\sin(2\pi |\xi| t)}{2\pi |\xi| t}$.

Proof. $M_t(f)$ is the convolution of f with the (normalized) surface measure on the sphere of radius t ; interchanging integrals and rescaling Lemma 36.2.6 by t gives the factor $\sin(2\pi |\xi| t)/(2\pi |\xi| t)$. $\square$

Theorem 36.2.8 (Solution for $d = 3$). For $f, g \in \mathcal{S}(\mathbb{R}^3)$ the Cauchy problem is solved by

$$u(x, t) = \frac{\partial}{\partial t} (t M_t(f)(x)) + t M_t(g)(x).$$

Proof. For the data $(0, g)$, Theorem 36.2.1 and Corollary 36.2.7 give $u_1 = \int \hat{g} \frac{\sin(2\pi |\xi| t)}{2\pi |\xi|} e^{2\pi i x \cdot \xi} d\xi = t \int \hat{g} \frac{\sin(2\pi |\xi| t)}{2\pi |\xi| t} e^{2\pi i x \cdot \xi} d\xi = t M_t(g)(x)$. For $(f, 0)$, $u_2 = \int \hat{f} \cos(2\pi |\xi| t) e^{2\pi i x \cdot \xi} d\xi = \partial_t \int \hat{f} \frac{\sin(2\pi |\xi| t)}{2\pi |\xi|} e^{2\pi i x \cdot \xi} d\xi = \partial_t (t M_t(f))$. Superpose: $u = u_1 + u_2$. $\square$

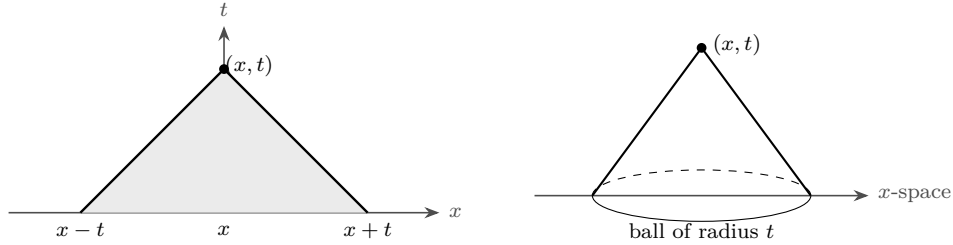


Figure 36.7. Backward light cone with apex (x, t) . For $d = 3$ the solution depends only on data near the *boundary* sphere of the base ball (sharp Huygens); for $d = 1, 2$ it depends on the whole base. (Schematic.)

Theorem 36.2.9 (Solution for $d = 2$, by descent). For $f, g \in \mathcal{S}(\mathbb{R}^2)$,

$$u(x, t) = \frac{\partial}{\partial t} (t \widetilde{M}_t(f)(x)) + t \widetilde{M}_t(g)(x), \quad \widetilde{M}_t(F)(x) = \frac{1}{2\pi} \int_{|y| \leq 1} \frac{F(x - ty)}{(1 - |y|^2)^{1/2}} dy.$$

Sketch. Extend f, g to $\mathbb{R}^3$ constant in x_3 (truncated smoothly so as to stay Schwartz on the relevant slab), solve in $\mathbb{R}^3$ by Theorem 36.2.8; the solution is independent of x_3 , hence solves the $\mathbb{R}^2$ equation. Computing the spherical mean of an x_3 -independent $H(x_1, x_2)$ on S^2 and changing variables $r = \sin \theta$, then to polar coordinates in the disc, turns M_t into the disc average $\widetilde{M}_t$ with the weight $(1 - |y|^2)^{-1/2}$. $\square$

Remark (Huygens' principle and finite speed). For $d = 1$ the value $u(x, t)$ depends on f, g on the interval $[x - t, x + t]$; if $g = 0$ only on its two endpoints. For $d = 3$ (and odd $d > 1$) the dependence is only on an immediate neighborhood of the *boundary sphere* — the sharp Huygens principle: a flash of light is seen for an instant. For $d = 2$ (and even d) the dependence is on the whole disc, so a dropped stone makes ripples that persist. In all dimensions waves propagate with *finite speed* 1: data supported in a ball of radius δ about x_0 produces a solution supported in the ball of radius $|t| + \delta$.

Worked examples

Worked Example 36.2.10 (d'Alembert from the Fourier formula ($d = 1$)). Recover d'Alembert's formula from Theorem 36.2.1.

Recognition. Trigger: rewrite $\cos$ and $\sin/|\xi|$ as exponentials and use the shift rule.

Solution. $\int \hat{f}(\xi) \cos(2\pi|\xi|t) e^{2\pi i x \xi} d\xi$. For $d = 1$, $|\xi| = \pm\xi$ but $\cos$ is even, so $\cos(2\pi|\xi|t) = \cos(2\pi\xi t) = \frac{1}{2}(e^{2\pi i \xi t} + e^{-2\pi i \xi t})$, giving $\frac{1}{2} \int \hat{f}(\xi)(e^{2\pi i \xi(x+t)} + e^{2\pi i \xi(x-t)}) d\xi = \frac{1}{2}(f(x+t) + f(x-t))$ by inversion. Likewise $\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|} = \frac{1}{4\pi i \xi}(e^{2\pi i \xi t} - e^{-2\pi i \xi t})$ integrates against $\hat{g}$ to $\frac{1}{2} \int_{x-t}^{x+t} g$. *Check.* At $t = 0$ the position term is $f(x)$ and the velocity term vanishes, and ∂_t at $t = 0$ returns g . ■

Worked Example 36.2.11 (Energy of a standing wave). For $u(x, t) = \cos(2\pi|k|t)e^{2\pi i k \cdot x}$ on a torus-like normalized mode (one frequency k), confirm the energy split between kinetic and potential.

Recognition. Trigger: a single Fourier mode; compute $|u_t|^2$ and $\sum |u_{x_j}|^2$ pointwise.

Solution. $u_t = -2\pi|k| \sin(2\pi|k|t)e^{2\pi i k \cdot x}$, so $|u_t|^2 = 4\pi^2|k|^2 \sin^2$. $u_{x_j} = 2\pi i k_j \cos(\dots)e^{2\pi i k \cdot x}$, so $\sum_j |u_{x_j}|^2 = 4\pi^2|k|^2 \cos^2$. Their sum is $4\pi^2|k|^2(\sin^2 + \cos^2) = 4\pi^2|k|^2$, constant in t . *Check.* This matches Theorem 36.2.2: with this mode $\hat{f}$ is a point mass at k , $\hat{g} = 0$, so $E = 4\pi^2|k|^2|\hat{f}|^2$, independent of t . ■

Worked Example 36.2.12 (The $d = 3$ flash is instantaneous (sharp Huygens)). With $g = 0$ and f supported in $\{|y| \leq \delta\}$, show $u(x, t) = 0$ unless $|t - |x|| \leq \delta$ (for $|x|$ large).

Recognition. Trigger: read off the support from $u = \partial_t(tM_t(f))$; $M_t(f)(x)$ averages f over the sphere of radius t about x .

Solution. $M_t(f)(x) \neq 0$ requires the sphere $\{|x - z| = t\}$ to meet $\text{supp } f \subseteq \{|z| \leq \delta\}$, i.e. $||x| - t| \leq \delta$. Hence for fixed x with $|x|$ large, $u(x, t)$ is nonzero only for t in a window of width 2δ around $|x|$. *Check.* As $\delta \rightarrow 0$ the window shrinks to the instant $t = |x|$: the observer sees the flash once and then darkness, the defining feature of wave propagation in $\mathbb{R}^3$. ■

Worked Example 36.2.13 (The $d = 2$ ripple persists (no sharp Huygens)). With $g = 0$, f supported in $\{|y| \leq \delta\}$ in $\mathbb{R}^2$, show $u(x, t) \neq 0$ for all $t \geq |x| - \delta$, not just a thin window.

Recognition. Trigger: in $d = 2$, $u = \partial_t(\widetilde{M}_t f)$ averages over the whole *disc* of radius t , with weight $(1 - |y|^2)^{-1/2}$.

Solution. $\widetilde{M}_t(f)(x) \neq 0$ as soon as the disc $\{|x - z| \leq t\}$ intersects $\text{supp } f$, i.e. for all $t \geq |x| - \delta$; the support is a filled region, not a shell, so once the wavefront arrives it never leaves. *Check.* This is the lake-ripple phenomenon — dropping a stone makes the surface undulate indefinitely (with decaying amplitude), in contrast to the $d = 3$ flash. ■

Worked Example 36.2.14 (Plane-wave solutions in $\mathbb{R}^d$). Verify that $u(x, t) = F((x \cdot \gamma) - t)$ solves $\Delta u = u_{tt}$ for any C^2 function F and unit vector γ .

Recognition. Trigger: a function of the single variable $s = x \cdot \gamma - t$; apply the chain rule.

Solution. $\partial_{x_j} u = \gamma_j F'$, $\partial_{x_j}^2 u = \gamma_j^2 F''$, so $\Delta u = (\sum_j \gamma_j^2) F'' = |\gamma|^2 F'' = F''$. Also $\partial_t u = -F'$, $u_{tt} = F''$. Hence $\Delta u = u_{tt}$. *Check.* u is constant on each plane $x \cdot \gamma = \text{const}$ and the profile translates with unit speed in direction γ — a plane wave. Note $u \notin \mathcal{S}(\mathbb{R}^d)$ for $d > 1$ since it is constant transverse to γ . ■

Worked Example 36.2.15 (Spherical mean recovers the value at the center as $t \rightarrow 0$). Show $M_t(f)(x) \rightarrow f(x)$ as $t \rightarrow 0$, and $\partial_t M_t(f)(x) \rightarrow 0$.

Recognition. Trigger: the spherical mean is an average over a shrinking sphere; continuity of f does the rest.

Solution. $M_t(f)(x) = \frac{1}{4\pi} \int_{S^2} f(x - t\gamma) d\sigma(\gamma)$. As $t \rightarrow 0$, $f(x - t\gamma) \rightarrow f(x)$ uniformly in γ (uniform continuity of the rapidly decreasing f), so $M_t(f)(x) \rightarrow f(x)$. Differentiating, $\partial_t M_t(f)(x) = -\frac{1}{4\pi} \int_{S^2} \gamma \cdot \nabla f(x - t\gamma) d\sigma$, and at $t = 0$ this is $-\frac{1}{4\pi} \nabla f(x) \cdot \int_{S^2} \gamma d\sigma = 0$ since the sphere's centroid vanishes. *Check.* This is exactly why $u = \partial_t(tM_t f) + tM_t g$ satisfies $u(x, 0) = f(x)$ (the $\partial_t(tM_t f)$ term gives $M_0(f) = f$) and $\partial_t u(x, 0) = g(x)$. ■

Worked Example 36.2.16 (Energy is finite and computable from data). For $f(x) = e^{-\pi|x|^2}$, $g = 0$ in $\mathbb{R}^3$, compute E .

Recognition. Trigger: use $E = \int (4\pi^2 |\xi|^2 |\hat{f}|^2 + |\hat{g}|^2) d\xi$ with the self-dual Gaussian.

Solution. $\hat{f}(\xi) = e^{-\pi|\xi|^2}$, $\hat{g} = 0$, so $E = 4\pi^2 \int_{\mathbb{R}^3} |\xi|^2 e^{-2\pi|\xi|^2} d\xi = 4\pi^2 \cdot A_3 \int_0^\infty \rho^2 e^{-2\pi\rho^2} \rho^2 d\rho$. With $A_3 = 4\pi$ and $\int_0^\infty \rho^4 e^{-2\pi\rho^2} d\rho = \frac{3}{8} \frac{1}{(2\pi)^2} \sqrt{\frac{\pi}{2\pi}} = \frac{3}{32\pi^2} \cdot \frac{1}{\sqrt{2}}$, one gets a finite positive number independent of t . Check. $E > 0$ and t -free, as Theorem 36.2.2 guarantees; the exact constant is unimportant, the point is finiteness from $f, g \in \mathcal{S}$. ■

Exercises (Stein, Ch.6 — the wave equation).

- (Spherical waves.) A *spherical wave* is a solution $u(x, t)$ of the Cauchy problem in $\mathbb{R}^d$ that is radial in x . Prove u is a spherical wave if and only if the initial data $f, g \in \mathcal{S}$ are both radial.
- (Energy, alternate proof.) Let u solve the wave equation with energy $E(t) = \int_{\mathbb{R}^d} (|\partial_t u|^2 + \sum_j |\partial_{x_j} u|^2) dx$. Give a proof that E is constant by differentiating under the integral and integrating by parts to show $dE/dt = 0$.
- (Kirchhoff's formula.) Show that the solution of the wave equation in $\mathbb{R}^3$ with data $f, g \in \mathcal{S}(\mathbb{R}^3)$ is $u(x, t) = \frac{1}{|S(x, t)|} \int_{S(x, t)} [tg(y) + f(y) + \nabla f(y) \cdot (y - x)] d\sigma(y)$, where $S(x, t)$ is the sphere of center x and radius t and $|S(x, t)|$ its area. This is an alternate form of the $d = 3$ formula of Theorem 36.2.8.

Problems (Stein, Ch.6 — wave equation and fractional Laplacian).

- (Uniqueness via energy, finite speed.) Let $B(x_0, r_0)$ be the closed ball in $\{t = 0\}$ and $L_B = \{(x, t) : |x - x_0| \leq r_0 - t, 0 \leq t \leq r_0\}$ the backward light cone with that base. Theorem: if $u \in C^2$ on $\{t \geq 0\}$ solves $u_{tt} = \Delta u$ and $u(x, 0) = \partial_t u(x, 0) = 0$ on $B(x_0, r_0)$, then $u \equiv 0$ on L_B . Prove it via the steps: (a) for $0 \leq t \leq r_0$ and $B_t = \{|x - x_0| \leq r_0 - t\}$, with $\nabla u = (\partial_{x_1} u, \dots, \partial_{x_d} u, \partial_t u)$, the energy $E(t) = \frac{1}{2} \int_{B_t} |\nabla u|^2$ satisfies $E(0) = 0$, $E \geq 0$, and $E'(t) = \int_{B_t} (\partial_t u \partial_t^2 u + \sum_j \partial_{x_j} u \partial_{x_j} \partial_t^2 u) - \frac{1}{2} \int_{\partial B_t} |\nabla u|^2$; (b) $\partial_{x_j} [\partial_{x_j} u \partial_t u] = \partial_{x_j} u \partial_{x_j} \partial_t u + \partial_{x_j}^2 u \partial_t u$; (c) by the divergence theorem and the equation, $E'(t) = \int_{\partial B_t} \sum_j \partial_{x_j} u \partial_t u \nu_j - \frac{1}{2} \int_{\partial B_t} |\nabla u|^2$; (d) by Cauchy-Schwarz $\sum_j \partial_{x_j} u \partial_t u \nu_j \leq \frac{1}{2} |\nabla u|^2$, so $E' \leq 0$, whence $E \equiv 0$ and $u \equiv 0$.
- * (Spherical-means solution, odd d .) For odd $d > 1$ and $h \in \mathcal{S}(\mathbb{R}^d)$ let $M_r h(x) = \frac{1}{A_d} \int_{S^{d-1}} h(x - r\gamma) d\sigma$. (a) Show $\Delta_x Mh(x, r) = (\partial_r^2 + \frac{d-1}{r}) Mh(x, r)$. (b) Show u solves the wave equation iff $(\partial_r^2 + \frac{d-1}{r}) Mu = \partial_t^2 Mu$. (c) For $d = 2k + 1$, $T\varphi(r) = (r^{-1} \partial_r)^{k-1} [r^{2k-1} \varphi(r)]$ and $\tilde{u} = TMu$ solves $\partial_t^2 \tilde{u} = \partial_r^2 \tilde{u}$ (use d'Alembert). (d) Show $u(x, t) = \lim_{r \rightarrow 0} \tilde{u}(x, r, t)/(\alpha r)$, $\alpha = 1 \cdot 3 \cdots (d-2)$. (e) Conclude $u(x, t) = \frac{1}{1 \cdot 3 \cdots (d-2)} [\partial_t (t^{-1} \partial_t)^{(d-3)/2} (t^{d-2} M_t f) + (t^{-1} \partial_t)^{(d-3)/2} (t^{d-2} M_t g)]$.
- * (Even d by descent.) Show the even- d solution has the same form with M_t replaced by the modified mean $\widetilde{M}_t h(x) = \frac{2}{A_{d+1}} \int_{B_d} \frac{h(x+ty)}{\sqrt{1-|y|^2}} dy$.
- * (Plane-wave superposition.) For data $f(x) = F(x \cdot \gamma)$, $g(x) = G(x \cdot \gamma)$, check that $u(x, t) = \frac{F(x \cdot \gamma + t) + F(x \cdot \gamma - t)}{2} + \frac{1}{2} \int_{x \cdot \gamma - t}^{x \cdot \gamma + t} G(s) ds$ solves the d -dimensional Cauchy problem. For $d = 3$, with $\tilde{R}(f)(t, \gamma) = -\frac{1}{8\pi^2} (d/dt)^2 R(f)(t, \gamma)$, the general solution is $u(x, t) = \frac{1}{2} \int_{S^2} [\tilde{R}(f)(x \cdot \gamma - t, \gamma) + \tilde{R}(f)(x \cdot \gamma + t, \gamma) + \int_{x \cdot \gamma - t}^{x \cdot \gamma + t} \tilde{R}(g)(s, \gamma) ds] d\sigma(\gamma)$.
- (Fractional Laplacian.) For $a > 0$ define $(-\Delta)^a f(x) = \int_{\mathbb{R}^d} (2\pi |\xi|)^{2a} \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi$. (a) Check it agrees with the usual power of $-\Delta$ when a is a positive integer. (b) Verify $(-\Delta)^a f$ is C^∞ . (c) Show that for non-integer a , $(-\Delta)^a f$ is in general not rapidly decreasing. (d) If $u(x, y) = f * P_y$ solves the steady-state heat equation, show $(-\Delta)^{1/2} f(x) = -\lim_{y \rightarrow 0} \partial_y u(x, y)$, and more generally $(-\Delta)^{k/2} f = (-1)^k \lim_{y \rightarrow 0} \partial_y^k u$.

Additional exercises.

- Solve the 1D Cauchy problem with $f(x) = e^{-\pi x^2}$, $g = 0$ explicitly by d'Alembert and identify the two travelling Gaussians.
- Show directly that the energy $E(t) = \int_{\mathbb{R}} (|u_t|^2 + |u_x|^2) dx$ for the 1D wave equation is constant by differentiating and integrating by parts (no Fourier transform).

3. Verify Lemma 36.2.3 by expanding both squared moduli and using $\sin^2 + \cos^2 = 1$ and the cross-term cancellation.
4. (Spherical mean.) Compute $M_t(f)(0)$ for $f(x) = e^{-\pi|x|^2}$ on $\mathbb{R}^3$ and confirm $\widehat{M_t f}(\xi) = \hat{f}(\xi) \frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}$ at $\xi = 0$ gives $M_t(f)$ averaged $= \hat{f}(0)$ -type consistency.
5. (Finite speed.) If f, g are supported in $\{|x - x_0| \leq \delta\}$, prove the $d = 3$ solution $u(\cdot, t)$ is supported in $\{|x - x_0| \leq |t| + \delta\}$.
6. (Huygens contrast.) For $d = 2$ vs. $d = 3$ with the same compactly supported f (and $g = 0$), describe precisely the time interval during which a fixed far-away observer detects a nonzero signal.
7. (Plane wave, non-example.) Show $u(x, t) = F(x \cdot \gamma - t)$ with $F \not\equiv \text{const}$ is never in $\mathcal{S}(\mathbb{R}^d)$ for $d \geq 2$, by examining its behavior transverse to γ .
8. (Diagram.) Draw the forward light cone emanating from a point x_0 at $t = 0$ in $\mathbb{R}^2 \times \mathbb{R}$ and shade the region of influence; contrast with the boundary-only influence in odd dimensions > 1 .
9. For a single mode $u = A \cos(2\pi|k|t) e^{2\pi i k \cdot x} + \frac{B}{2\pi|k|} \sin(2\pi|k|t) e^{2\pi i k \cdot x}$, compute the energy density and confirm it equals $4\pi^2 |k|^2 |A|^2 + |B|^2$ at all t .
10. (Rescaling c .) Show that if $u(x, t)$ solves $\Delta u = c^{-2} u_{tt}$ then $v(x, t) = u(x, ct)$ solves $\Delta v = v_{tt}$, reducing to the normalized case.

36.3 The Radon Transform and Its Applications

X-rays and the reconstruction problem

A narrow beam of X-ray photons of incident intensity I_0 traversing an object $O \subseteq \mathbb{R}^2$ emerges with intensity I governed by Beer's law. If ρ is the (variable) attenuation coefficient, then along the beam's line L ,

$$I = I_0 e^{-\int_L \rho}, \quad \text{so} \quad \log \frac{I_0}{I} = \int_L \rho.$$

The measurable quantity is exactly the line integral of ρ . This motivates the central object.

Definition 36.3.1. For $\rho \in \mathcal{S}(\mathbb{R}^2)$ the *X-ray transform* (the $\mathbb{R}^2$ Radon transform) assigns to each line L the number $X(\rho)(L) = \int_L \rho$. Parametrizing the line perpendicular to the direction $(\cos \theta, \sin \theta)$ at signed distance t from the origin, $L_{t,\theta} : x \cos \theta + y \sin \theta = t$, one has

$$X(\rho)(t, \theta) = \int_{-\infty}^{\infty} \rho(t \cos \theta - u \sin \theta, t \sin \theta + u \cos \theta) du.$$

The *reconstruction problem* asks for a formula recovering ρ from $X(\rho)$, and the *uniqueness problem* asks whether $X(\rho) = X(\rho')$ forces $\rho = \rho'$. A degrees-of-freedom count makes uniqueness plausible: ρ depends on two parameters (x, y) , and a line L is also described by two parameters (t, θ) , so the data $X(\rho)$ carries an equivalent amount of information. In $\mathbb{R}^2$ the answer is affirmative but delicate; we give the clean solution in $\mathbb{R}^3$, another instance of odd dimensions being simpler.

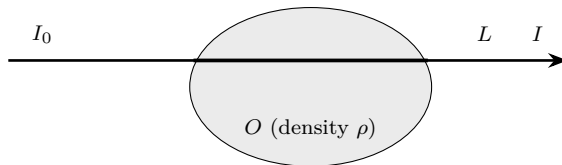


Figure 36.8. Attenuation of an X-ray beam along a line L through the object: the log-ratio $\log(I_0/I) = \int_L \rho$ is the X-ray transform of the density. (Schematic.)

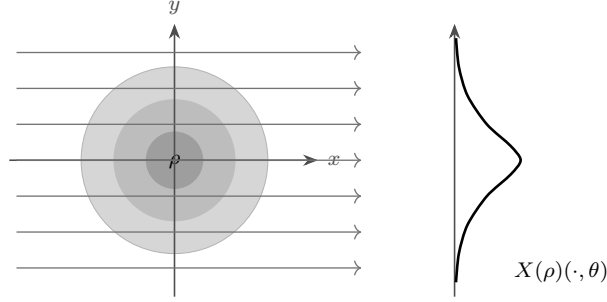


Figure 36.9. The X-ray transform records, for each direction θ , the parallel projection of ρ onto the perpendicular axis — the "sinogram" data of a CT scan. (Schematic.)

The Radon transform in $\mathbb{R}^3$ and the projection-slice lemma

In $\mathbb{R}^3$ a degrees-of-freedom count shows lines (four parameters) over-determine ρ (three parameters); the right substitute is integration over *planes*.

Definition 36.3.2. A plane in $\mathbb{R}^3$ is $P_{t,\gamma} = \{x \in \mathbb{R}^3 : x \cdot \gamma = t\}$, with $\gamma \in S^2$ a unit normal and $t \in \mathbb{R}$ the signed distance to the origin ($P_{t,\gamma} = P_{-t,-\gamma}$). For $f \in \mathcal{S}(\mathbb{R}^3)$, choosing an orthonormal e_1, e_2, γ and writing $x = t\gamma + u_1e_1 + u_2e_2$, the *Radon transform* is

$$R(f)(t, \gamma) = \int_{P_{t,\gamma}} f = \int_{\mathbb{R}^2} f(t\gamma + u_1e_1 + u_2e_2) du_1 du_2.$$

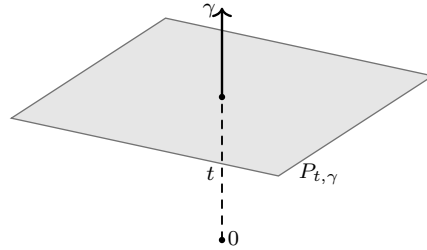


Figure 36.10. A plane $P_{t,\gamma}$ described by its unit normal γ and signed distance t to the origin (measured along γ); $R(f)(t, \gamma)$ integrates f over this plane.

Proposition 36.3.3. For $f \in \mathcal{S}(\mathbb{R}^3)$ the integral in Definition 36.3.2 is independent of the choice of e_1, e_2 , and

$$\int_{-\infty}^{\infty} \left(\int_{P_{t,\gamma}} f \right) dt = \int_{\mathbb{R}^3} f(x) dx.$$

Proof. Two admissible frames differ by a rotation R of the (u_1, u_2) -plane; $u' = Ru$ leaves the $\mathbb{R}^2$ integral unchanged. For the second identity, let R be the rotation carrying the standard basis to γ, e_1, e_2 ; then $\int_{\mathbb{R}^3} f = \int_{\mathbb{R}^3} f(Rx) dx = \int f(x_1\gamma + x_2e_1 + x_3e_2) dx$, and integrating x_2, x_3 first (giving $\int_{P_{x_1,\gamma}} f$) then $x_1 = t$ gives the claim. $\square$

Lemma 36.3.4 (Projection-slice / Fourier-slice). For $f \in \mathcal{S}(\mathbb{R}^3)$, $R(f)(\cdot, \gamma) \in \mathcal{S}(\mathbb{R})$ for each γ , and

$$\widehat{R(f)}(s, \gamma) = \hat{f}(s\gamma),$$

where the left side is the one-dimensional transform in t (with γ fixed) and $\hat{f}$ is the three-dimensional transform.

Proof. Decay: $(1 + |t|)^N (1 + |u|)^N |f(t\gamma + u)| \leq A_N$ gives, for $N \geq 3$, $(1 + |t|)^N |R(f)(t, \gamma)| \leq A_N \int_{\mathbb{R}^2} (1 + |u|)^{-N} du < \infty$, and similarly for derivatives, so $R(f)(\cdot, \gamma) \in \mathcal{S}(\mathbb{R})$. For the identity, $\widehat{R(f)}(s, \gamma) = \int_{\mathbb{R}} \int_{\mathbb{R}^2} f(t\gamma + u_1e_1 + u_2e_2) e^{-2\pi i s t} du dt$. Since $\gamma \cdot u = 0$ and $|\gamma| = 1$, $e^{-2\pi i s t} = e^{-2\pi i s \gamma \cdot (t\gamma + u)}$, so the integrand is $f(x) e^{-2\pi i (s\gamma) \cdot x}$ with $x = t\gamma + u$; integrating over the plane and over t sweeps out all of $\mathbb{R}^3$, giving $\hat{f}(s\gamma)$. $\square$

$$\begin{array}{ccc}
f & \xrightarrow{\mathcal{F}_{\mathbb{R}^3}} & \hat{f} \\
R \downarrow & & \downarrow \text{restrict to } s\gamma \\
R(f)(t, \gamma) & \xrightarrow{\mathcal{F}_{\mathbb{R}}} & \hat{f}(s\gamma)
\end{array}$$

Figure 36.11. Projection-slice (Fourier-slice) lemma 36.3.4: the one-dimensional transform $\mathcal{F}_{\mathbb{R}}$ of the plane integral $R(f)(\cdot, \gamma)$ equals the three-dimensional transform $\mathcal{F}_{\mathbb{R}^3}$ of f restricted to the line $s \mapsto s\gamma$ through the origin. The square commutes, $\widehat{R(f)}(s, \gamma) = \hat{f}(s\gamma)$; this is what makes the Radon transform invertible.

Corollary 36.3.5 (Uniqueness). *If $f, g \in \mathcal{S}(\mathbb{R}^3)$ and $R(f) = R(g)$ then $f = g$.*

Proof. Apply Lemma 36.3.4 to $h = f - g$: $\hat{h}(s\gamma) = \widehat{R(h)}(s, \gamma) = 0$ for all s, γ , so $\hat{h} \equiv 0$ on $\mathbb{R}^3$ (every point is some $s\gamma$), hence $h = 0$ by Fourier inversion. $\square$

The dual transform and the reconstruction formula

Definition 36.3.6. For F on $\mathbb{R} \times S^2$ the *dual Radon transform* is

$$R^*(F)(x) = \int_{S^2} F(x \cdot \gamma, \gamma) d\sigma(\gamma),$$

which integrates F over all planes $P_{t,\gamma}$ passing through x (those with $x \cdot \gamma = t$). It is the formal adjoint: $(Rf, F)_2 = (f, R^*F)_1$ for the inner products $(f, g)_1 = \int_{\mathbb{R}^3} f\bar{g}$ and $(F, G)_2 = \int_{\mathbb{R}} \int_{S^2} F\bar{G} d\sigma dt$.

Theorem 36.3.7 (Reconstruction in $\mathbb{R}^3$). *For $f \in \mathcal{S}(\mathbb{R}^3)$,*

$$\Delta(R^*R(f)) = -8\pi^2 f.$$

Proof. By Lemma 36.3.4 and one-dimensional inversion, $R(f)(t, \gamma) = \int_{-\infty}^{\infty} \hat{f}(s\gamma) e^{2\pi i t s} ds$. Hence

$$R^*R(f)(x) = \int_{S^2} \int_{-\infty}^{\infty} \hat{f}(s\gamma) e^{2\pi i(x \cdot \gamma)s} ds d\sigma(\gamma).$$

Apply Δ in x under the integral: $\Delta e^{2\pi i(x \cdot \gamma)s} = -4\pi^2 s^2 e^{2\pi i(x \cdot \gamma)s}$ (as $|\gamma| = 1$), so

$$\Delta R^*R(f)(x) = -4\pi^2 \int_{S^2} \int_{-\infty}^{\infty} \hat{f}(s\gamma) s^2 e^{2\pi i(x \cdot \gamma)s} ds d\sigma(\gamma).$$

Split $\int_{-\infty}^{\infty} = \int_{-\infty}^0 + \int_0^{\infty}$; in the first integral substitute $s \rightarrow -s$, $\gamma \rightarrow -\gamma$, which maps it onto the second. Thus the double integral equals $2 \int_{S^2} \int_0^{\infty} \hat{f}(s\gamma) e^{2\pi i(x \cdot \gamma)s} s^2 ds d\sigma(\gamma)$, and by the polar-coordinate formula in $\mathbb{R}^3$ this is $2 \int_{\mathbb{R}^3} \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi = 2f(x)$ by Fourier inversion. Therefore $\Delta R^*R(f) = -4\pi^2 \cdot 2f = -8\pi^2 f$. $\square$

Remark (Plane waves and the wave equation). The reconstruction is intimately tied to the wave equation. A *plane wave* $u(x, t) = F(x \cdot \gamma - t)$ solves $\Delta u = u_{tt}$ for any C^2 profile F and unit γ (established by direct differentiation in the previous section). When $d > 1$ the general solution is an *integral* of plane waves over $\gamma \in S^{d-1}$ — the continuous analogue of the $d = 1$ sum of two travelling waves — and the weights are supplied precisely by the Radon transform of the initial data.

Worked examples

Worked Example 36.3.8 (X-ray transform of a Gaussian). Compute $X(f)(t, \theta)$ for $f(x, y) = e^{-\pi(x^2 + y^2)}$.

Recognition. Trigger: a radial f ; the projection of a Gaussian is a Gaussian, independent of θ .

Solution. Parametrize $L_{t,\theta}$ by u : the point is $(t \cos \theta - u \sin \theta, t \sin \theta + u \cos \theta)$, whose squared norm is $t^2 + u^2$ (rotation preserves length). So $X(f)(t, \theta) = \int_{-\infty}^{\infty} e^{-\pi(t^2 + u^2)} du = e^{-\pi t^2} \int_{-\infty}^{\infty} e^{-\pi u^2} du = e^{-\pi t^2}$. *Check.* Independent of θ (as a radial f must give), and $\int_{\mathbb{R}} X(f)(t, \theta) dt = \int_{\mathbb{R}} e^{-\pi t^2} dt = 1 = \int_{\mathbb{R}^2} f$, consistent with Proposition 36.3.3 in 2D form. $\blacksquare$

Worked Example 36.3.9 (Radon transform of a 3D Gaussian). Compute $R(f)(t, \gamma)$ for $f(x) = e^{-\pi|x|^2}$ on $\mathbb{R}^3$.

Recognition. Trigger: radial f ; integrate over the plane $P_{t,\gamma}$ where $|x|^2 = t^2 + |u|^2$.

Solution. $R(f)(t, \gamma) = \int_{\mathbb{R}^2} e^{-\pi(t^2+|u|^2)} du = e^{-\pi t^2} \int_{\mathbb{R}^2} e^{-\pi|u|^2} du = e^{-\pi t^2}$, independent of γ . *Check (slice lemma):* the 1D transform in t is $\widehat{e^{-\pi t^2}}(s) = e^{-\pi s^2} = \hat{f}(s\gamma) = e^{-\pi|s\gamma|^2} = e^{-\pi s^2}$ since $\hat{f} = e^{-\pi|\xi|^2}$ and $|s\gamma| = |s|$. Lemma 36.3.4 confirmed. ■

Worked Example 36.3.10 (Verifying $\Delta(R^*R(f)) = -8\pi^2 f$ on the Gaussian). Check the reconstruction constant on $f = e^{-\pi|x|^2}$.

Recognition. Trigger: a direct numerical sanity check at one point, using radially.

Solution. From the previous example $R(f)(t, \gamma) = e^{-\pi t^2}$, so $R^*R(f)(x) = \int_{S^2} e^{-\pi(x \cdot \gamma)^2} d\sigma(\gamma)$. This is radial; at $x = 0$ it equals $\int_{S^2} 1 d\sigma = 4\pi$. Theorem 36.3.7 predicts $\Delta R^*R(f)(0) = -8\pi^2 f(0) = -8\pi^2$. *Check.* Expanding $e^{-\pi(x \cdot \gamma)^2} \approx 1 - \pi(x \cdot \gamma)^2$ and using $\int_{S^2} (x \cdot \gamma)^2 d\sigma = \frac{4\pi}{3} |x|^2$ gives $R^*R(f)(x) \approx 4\pi - \frac{4\pi^2}{3} |x|^2$, whose Laplacian is $-\frac{4\pi^2}{3} \cdot 6 = -8\pi^2$ ($\Delta |x|^2 = 2d = 6$). Matches. ■

Worked Example 36.3.11 (The X-ray transform determines the Radon transform). Explain why knowing all line integrals in $\mathbb{R}^3$ yields all plane integrals.

Recognition. Trigger: a plane integral is an iterated line integral.

Solution. Fix a plane $P_{t,\gamma}$. Foliate it by parallel lines L within the plane; then $\int_{P_{t,\gamma}} f = \int (\int_L f)$, an integral of $X(f)$ over a one-parameter family of lines lying in P . So $X(f)$ determines $R(f)$. *Check.* This is why 2D CT (line data) and the 3D plane-integral theory are compatible; the line data is the finer object. ■

Worked Example 36.3.12 (Adjointness $(Rf, F)_2 = (f, R^*F)_1$). Verify the duality identity (the content of Exercise/Problem on the dual transform).

Recognition. Trigger: write both sides as one triple integral and use Fubini.

Solution. Consider $I = \int_{S^2} \int_{\mathbb{R}} \int_{\mathbb{R}^2} f(t\gamma + u_1 e_1 + u_2 e_2) \overline{F(t, \gamma)} du dt d\sigma(\gamma)$. Doing the u -integral first turns the inner double integral into $R(f)(t, \gamma)$, giving $(Rf, F)_2$. Doing the u, t integrals first and setting $x = t\gamma + u$ (so $t = x \cdot \gamma$) turns it into $\int_{\mathbb{R}^3} f(x) \int_{S^2} \overline{F(x \cdot \gamma, \gamma)} d\sigma dx = (f, R^*F)_1$. *Check.* Both reductions integrate the same nonnegative-modulus integrand, so they are equal. ■

Worked Example 36.3.13 (X-ray uniqueness via the slice lemma in 2D). Show that if $f \in \mathcal{S}(\mathbb{R}^2)$ and $X(f) \equiv 0$ then $f = 0$.

Recognition. Trigger: the 2D analogue $\widehat{X(f)}(s, \theta) = \hat{f}(s \cos \theta, s \sin \theta)$.

Solution. The 1D transform of $X(f)(\cdot, \theta)$ equals $\hat{f}$ evaluated along the ray of direction θ through the origin (same computation as Lemma 36.3.4, with planes replaced by lines). If $X(f) \equiv 0$ then $\hat{f} = 0$ on every such ray, i.e. on all of $\mathbb{R}^2$, so $f = 0$ by inversion. *Check.* This is the uniqueness half of the reconstruction problem in $\mathbb{R}^2$; only the explicit inversion formula is harder there than in $\mathbb{R}^3$. ■

Worked Example 36.3.14 (A line integral computed by hand). For $f(x, y) = \mathbf{1}_{x^2+y^2 \leq 1}$ (indicator of the unit disc), compute $X(f)(t, \theta)$.

Recognition. Trigger: a chord length; the projection of a disc is the chord-length profile, independent of θ .

Solution. The line at distance $|t| \leq 1$ from the center cuts the disc in a chord of length $2\sqrt{1-t^2}$; for $|t| > 1$ the line misses the disc. So $X(f)(t, \theta) = 2\sqrt{1-t^2} \mathbf{1}_{|t| \leq 1}$, independent of θ . *Check.* $\int_{-1}^1 2\sqrt{1-t^2} dt = \pi = \text{area of the disc} = \int_{\mathbb{R}^2} f$, matching the consistency relation. ■

Worked Example 36.3.15 (A non-example: data on too few lines). Explain why finitely many directions cannot determine ρ exactly.

Recognition. Trigger: a degrees-of-freedom / null-space argument.

Solution. If only directions $\theta_1, \dots, \theta_N$ are sampled, one can add to ρ any nonzero $\psi \in \mathcal{S}(\mathbb{R}^2)$ whose 1D slice $\hat{\psi}$ vanishes along those N rays but is nonzero elsewhere; then $X(\rho + \psi) = X(\rho)$ on the sampled lines. *Check.* This is exactly why real CT reconstruction relies on densely sampling θ plus numerical regularization (and, as the text notes, the fast Fourier transform), not on a finite exact inversion. ■

Exercises (Stein, Ch.6 — the Radon and X-ray transforms).

1. (Duality of the dual transform.) Establish the identity

$$\int_{\mathbb{R}} \int_{S^2} R(f)(t, \gamma) F(t, \gamma) d\sigma(\gamma) dt = \int_{\mathbb{R}^3} f(x) R^*(F)(x) dx$$

for $f \in \mathcal{S}(\mathbb{R}^3)$, $F \in \mathcal{S}(\mathbb{R} \times S^2)$, where $R(f) = \int_{P_{t,\gamma}} f$ and $R^*(F)(x) = \int_{S^2} F(x \cdot \gamma, \gamma) d\sigma(\gamma)$. (Consider $\int \int \int f(t\gamma + u_1 e_1 + u_2 e_2) F(t, \gamma) dt d\sigma du$: integrating in u first gives the left side; integrating in u, t and setting $x = t\gamma + u_1 e_1 + u_2 e_2$ gives the right.)

2. (X-ray transform of a Gaussian.) For each (t, θ) with $|\theta| \leq \pi$ let $L_{t,\theta}$ be the line $x \cos \theta + y \sin \theta = t$, and define the 2D Radon (X-ray) transform $X(f)(t, \theta) = \int_{L_{t,\theta}} f = \int_{-\infty}^{\infty} f(t \cos \theta + u \sin \theta, t \sin \theta - u \cos \theta) du$. Calculate $X(f)$ for $f(x, y) = e^{-\pi(x^2+y^2)}$.
3. (X-ray uniqueness.) Show that if $f \in \mathcal{S}(\mathbb{R}^2)$ and $X(f) = 0$, then $f = 0$, by taking the Fourier transform in one variable.
4. (Dual X-ray transform.) For $F \in \mathcal{S}(\mathbb{R} \times S^1)$ define $X^*(F)(x, y) = \int F(x \cos \theta + y \sin \theta, \theta) d\theta$, the integral of F over all lines through (x, y) . Check that for $f \in \mathcal{S}(\mathbb{R}^2)$, $F \in \mathcal{S}(\mathbb{R} \times S^1)$, $\int \int X(f)(t, \theta) F(t, \theta) dt d\theta = \int \int f(x, y) X^*(F)(x, y) dx dy$.

Problems (Stein, Ch.6 — Radon reconstruction in general d).

1. * (Reconstruction formula.) (a) For $d = 2$, $\frac{(-\Delta)^{1/2}}{4\pi} R^*(R(f)) = f$, where $(-\Delta)^{1/2}$ is the fractional Laplacian. (b) If the Radon transform and its dual are defined by analogy to $d = 2, 3$, then for general d , $(2\pi)^{1-d} (-\Delta)^{(d-1)/2} R^*(R(f)) = f$. (For $d = 3$ this is $\Delta R^* R(f) = -8\pi^2 f$ of Theorem 36.3.7.)

Additional exercises.

1. Compute the X-ray transform $X(f)(t, \theta)$ of $f(x, y) = \mathbf{1}_{x^2+y^2 \leq R^2}$ (disc of radius R) and confirm $\int_{\mathbb{R}} X(f) dt = \pi R^2$.
2. Compute the Radon transform $R(f)(t, \gamma)$ of $f(x) = e^{-\pi|x|^2}$ on $\mathbb{R}^3$ and verify the projection-slice identity $\widehat{R(f)}(s, \gamma) = \hat{f}(s\gamma)$.
3. Show that the X-ray transform of a radial f on $\mathbb{R}^2$ is independent of θ , and conversely that a θ -independent $X(f)$ forces f radial.
4. (Consistency.) Prove the moment relation $\int_{\mathbb{R}} t^k X(f)(t, \theta) dt$ is a polynomial of degree $\leq k$ in $(\cos \theta, \sin \theta)$ (the Helgason–Ludwig consistency conditions, $k = 0, 1$).
5. (Diagram.) Draw a 2D density (two well-separated bumps) and sketch its sinogram $X(f)(t, \theta)$ as θ ranges over $[0, \pi)$, indicating the two sinusoidal traces.
6. Verify the adjoint relation $(Rf, F)_2 = (f, R^*F)_1$ for $f = e^{-\pi|x|^2}$ and $F(t, \gamma) = e^{-\pi t^2}$ by computing both sides.
7. (Non-example.) Give a nonzero $\psi \in \mathcal{S}(\mathbb{R}^2)$ whose X-ray transform vanishes on the two directions $\theta = 0, \pi/2$ but not on all lines, illustrating why finitely many projections cannot determine ρ .
8. Using Theorem 36.3.7, write the explicit $\mathbb{R}^3$ inversion $f = -\frac{1}{8\pi^2} \Delta(R^* Rf)$ and check the constant on the unit-ball density by a Laplacian computation at the center.
9. Show that for $f \in \mathcal{S}(\mathbb{R}^3)$, $R(f)(t, \gamma) = R(f)(-t, -\gamma)$, and explain how this symmetry halves the data needed in tomography.

Chapter 37

Finite Fourier Analysis

37.1 Fourier Analysis on $Z(N)$

The group $Z(N)$: roots of unity and integers mod N

Finite Fourier analysis replaces the circle $\mathbb{T}$ by a finite set, and infinite sums and integrals by *finite* sums. Convergence questions evaporate; what remains is the pure algebraic skeleton of the theory — orthogonality, inversion, Plancherel — now visible without analytic distractions. The simplest finite model is the group of N th roots of unity, which we will see is the natural N -point approximation of the circle.

Definition 37.1.1. Let N be a positive integer. A complex number z is an N th root of unity if $z^N = 1$. Writing $z = re^{i\theta}$ and taking absolute values in $r^N e^{iN\theta} = 1$ forces $r = 1$ and $N\theta = 2\pi k$, $k \in \mathbb{Z}$. Hence, with $\zeta = e^{2\pi i/N}$, the set of N th roots of unity is exactly

$$Z(N) = \{1, \zeta, \zeta^2, \dots, \zeta^{N-1}\}, \quad \zeta = e^{2\pi i/N}.$$

Since $\zeta^N = 1$, the powers repeat: $\zeta^n = \zeta^m$ if and only if $N \mid (n - m)$. The N points $\zeta^k = e^{2\pi i k/N}$ are equally spaced on the unit circle.

The set $Z(N)$ is closed under multiplication ($\zeta^n \zeta^m = \zeta^{n+m}$), contains 1, and contains inverses ($(\zeta^n)^{-1} = \zeta^{-n} = \zeta^{N-n}$). Multiplication is commutative. Thus $Z(N)$ is an *abelian group* under complex multiplication.

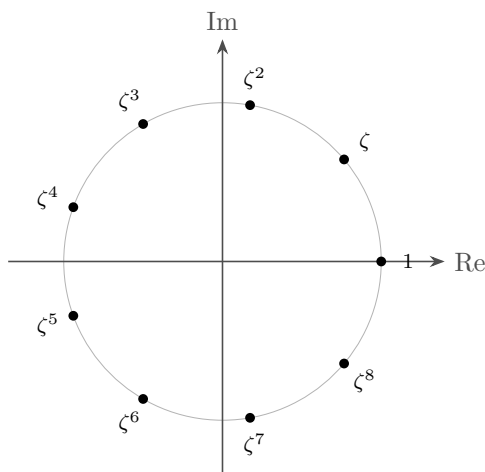


Figure 37.1. $Z(9)$ with $\zeta = e^{2\pi i/9}$: nine equally spaced points partitioning the circle. As $N \rightarrow \infty$ the partition refines to all of $\mathbb{T}$.

Definition 37.1.2 (Integers modulo N). Two integers x, y are *congruent modulo N* , written $x \equiv y \pmod{N}$, if $N \mid (x - y)$. This is an equivalence relation; the class $R(x) = \{x + kN : k \in \mathbb{Z}\}$ is the *residue class* of x . There

are exactly N classes, each with a unique representative in $\{0, 1, \dots, N-1\}$. Defining $R(x) + R(y) = R(x+y)$ (well defined, independent of representatives) makes the set of classes an abelian group $\mathbb{Z}/N\mathbb{Z}$, the *integers modulo N* .

The association $R(k) \longleftrightarrow e^{2\pi i k/N} = \zeta^k$ is a bijection that turns *addition mod N* into *multiplication of roots of unity*: $R(n) + R(m) \mapsto \zeta^{n+m} = \zeta^n \zeta^m$. It is therefore an isomorphism of abelian groups, and we write $Z(N)$ for either incarnation, thinking of whichever is convenient. Note $0 \in \mathbb{Z}/N\mathbb{Z}$ corresponds to 1 on the circle.

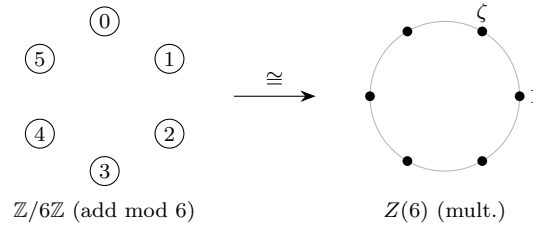


Figure 37.2. The isomorphism $\mathbb{Z}/N\mathbb{Z} \cong Z(N)$, $k \mapsto \zeta^k$, carries addition mod N to multiplication of roots of unity.

Worked Example 37.1.3 (Arithmetic in $Z(N)$). In $Z(6)$ compute $\zeta^4 \cdot \zeta^5$ and $(\zeta^4)^{-1}$, with $\zeta = e^{2\pi i/6}$.

Recognition. Multiplication of roots of unity is addition of exponents mod N ; inverses are negation mod N .

Solution. $\zeta^4 \zeta^5 = \zeta^9 = \zeta^{9-6} = \zeta^3 = e^{\pi i} = -1$. The inverse: $(\zeta^4)^{-1} = \zeta^{-4} = \zeta^2 = e^{2\pi i/3}$. *Check.* $\zeta^4 \zeta^2 = \zeta^6 = 1$, so ζ^2 is indeed the inverse of ζ^4 .

Worked Example 37.1.4 (The same element, two notations). Identify the element of $\mathbb{Z}/8\mathbb{Z}$ corresponding to the root of unity $e^{2\pi i \cdot 3/8} \cdot e^{2\pi i \cdot 7/8}$ in $Z(8)$.

Recognition. Multiply roots (add exponents), then reduce mod 8 to land in $\{0, \dots, 7\}$.

Solution. Exponents add: $3+7=10 \equiv 2 \pmod{8}$, so the product is $e^{2\pi i \cdot 2/8} = \zeta^2$, corresponding to $2 \in \mathbb{Z}/8\mathbb{Z}$.

Check. $10-2=8$ is a multiple of 8, so $R(10) = R(2)$.

The characters e_ℓ and the orthogonality lemma

The whole theory rests on finding the finite analogue of the exponentials $e_n(x) = e^{2\pi i n x}$. On the circle these are orthonormal, span (densely) the continuous functions, and satisfy $e_n(x+y) = e_n(x)e_n(y)$. On $Z(N)$ the analogues are N honest functions.

Definition 37.1.5 (Exponentials on $Z(N)$). For $\ell = 0, 1, \dots, N-1$ define $e_\ell: Z(N) \rightarrow \mathbb{C}$ by

$$e_\ell(k) = \zeta^{\ell k} = e^{2\pi i \ell k/N}, \quad k = 0, 1, \dots, N-1.$$

Each is multiplicative in k : $e_\ell(k+m) = e_\ell(k)e_\ell(m)$.

Let V be the vector space of all complex-valued functions on $Z(N)$; it has dimension N . Equip V with the Hermitian inner product and norm

$$(F, G) = \sum_{k=0}^{N-1} F(k) \overline{G(k)}, \quad \|F\|^2 = \sum_{k=0}^{N-1} |F(k)|^2.$$

Lemma 37.1.6 (Orthogonality). *The family $\{e_0, \dots, e_{N-1}\}$ is orthogonal:*

$$(e_m, e_\ell) = \begin{cases} N & m = \ell, \\ 0 & m \neq \ell. \end{cases}$$

Proof. We have $(e_m, e_\ell) = \sum_{k=0}^{N-1} \zeta^{mk} \overline{\zeta^{\ell k}} = \sum_{k=0}^{N-1} \zeta^{(m-\ell)k}$. If $m = \ell$ every term is 1 and the sum is N . If $m \neq \ell$, put $q = \zeta^{m-\ell} \neq 1$ (since $0 < |m-\ell| < N$). The geometric series gives $\sum_{k=0}^{N-1} q^k = \frac{1-q^N}{1-q} = 0$ because $q^N = (\zeta^N)^{m-\ell} = 1$. $\square$

Geometrically, summing $\zeta^{(m-\ell)k}$ over k adds up N unit vectors spaced evenly around the circle; by symmetry their centroid is the origin. This “the roots of unity sum to zero” picture is the heart of every orthogonality computation in the chapter.

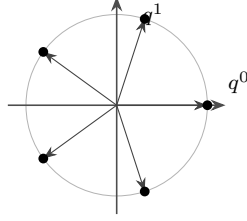


Figure 37.3. The N powers q^k ($q = \zeta^{m-\ell} \neq 1$) are evenly spaced unit vectors; their vector sum is 0. Here $N = 5$.

Since the N orthogonal functions e_ℓ are linearly independent and $\dim V = N$, they form an orthogonal basis. Each has norm $\sqrt{N}$, so $e_\ell^* = N^{-1/2}e_\ell$ gives an *orthonormal* basis $\{e_0^*, \dots, e_{N-1}^*\}$.

Fourier inversion and Plancherel on $Z(N)$

Definition 37.1.7 (Discrete Fourier coefficients). The n th Fourier coefficient of $F \in V$ is

$$a_n = \frac{1}{N} \sum_{k=0}^{N-1} F(k) e^{-2\pi i k n / N} = \frac{1}{N} \sum_{k=0}^{N-1} F(k) \zeta^{-kn}.$$

Thus $\overline{a_n} = \frac{1}{N} (F, e_n) = \frac{1}{\sqrt{N}} (F, e_n^*)$.

Theorem 37.1.8 (Fourier inversion and Parseval–Plancherel on $Z(N)$). *If F is a function on $Z(N)$, then for every k ,*

$$F(k) = \sum_{n=0}^{N-1} a_n e^{2\pi i k n / N}, \quad \text{and} \quad \sum_{n=0}^{N-1} |a_n|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |F(k)|^2.$$

Proof. Expanding F in the orthonormal basis $\{e_n^*\}$ gives $F = \sum_n (F, e_n^*) e_n^*$ and, by orthonormality, $\|F\|^2 = \sum_n |(F, e_n^*)|^2$ (this is (1), the abstract Bessel/Parseval identity for an orthonormal basis). Now $(F, e_n^*) = \sqrt{N} a_n$ and $e_n^* = N^{-1/2} e_n$, so $(F, e_n^*) e_n^* = a_n e_n$ and $F = \sum_n a_n e_n$, which evaluated at k is the inversion formula. For Plancherel, $\|F\|^2 = \sum_n |(F, e_n^*)|^2 = \sum_n N |a_n|^2$, and dividing by N gives the stated identity (recall $\|F\|^2 = \sum_k |F(k)|^2$). $\square$

Inversion says: a function on N points is exactly the superposition of the N pure frequencies e_n with weights a_n . There is no remainder term and no smoothness hypothesis — finiteness makes the expansion exact. Plancherel says the discrete transform $F \mapsto (a_n)$ is, up to the factor N , an isometry: energy in “time” equals energy in “frequency.”

The analysis map $\mathcal{F}: F \mapsto (a_n)$ and the synthesis (inversion) map $\mathcal{F}^{-1}: (a_n) \mapsto \sum_n a_n e_n$ are mutually inverse isomorphisms of the N -dimensional space V , and by Plancherel each is (up to the factor N) an isometry.

$$(V, \|\cdot\|^2) \xrightleftharpoons[\mathcal{F}^{-1}]{\mathcal{F}} (V, N \sum |a_n|^2)$$

Remark. Letting $N \rightarrow \infty$ recovers Fourier inversion on the circle for sufficiently smooth (say C^2) functions, via the finite model; see the Stein exercises.

Method (Computing a discrete Fourier transform on $Z(N)$). *Recognition trigger.* You are given the N values $F(0), \dots, F(N-1)$ and want all coefficients a_n , or vice versa.

1. Fix $\zeta = e^{2\pi i / N}$. For each n , form $a_n = \frac{1}{N} \sum_k F(k) \zeta^{-kn}$.

2. Exploit structure: if F is supported on a few points, the sum has few terms; if F is itself an e_ℓ , then $a_n = \delta_{n\ell}$ by orthogonality.
3. Sanity-check with $a_0 = \frac{1}{N} \sum_k F(k)$ (the mean) and with Plancherel.
4. To go back, $F(k) = \sum_n a_n \zeta^{kn}$.

Worked Example 37.1.9 (DFT of a delta spike). Let $N = 4$ and $F = \delta_0$, i.e. $F(0) = 1$, $F(1) = F(2) = F(3) = 0$. Compute all a_n .

Recognition. A one-point support: the sum collapses to a single term.

Solution. $a_n = \frac{1}{4} \sum_k F(k) \zeta^{-kn} = \frac{1}{4} F(0) \zeta^0 = \frac{1}{4}$ for every n . *Check.* Inversion gives $\sum_n \frac{1}{4} \zeta^{kn} = \frac{1}{4} \sum_n \zeta^{kn} = \frac{1}{4} (N \delta_{k0}) = \delta_{k0} = F(k)$. A spike in “time” is flat in “frequency.”

Worked Example 37.1.10 (DFT of a single frequency). On $Z(N)$ let $F = e_2$, i.e. $F(k) = \zeta^{2k}$. Find a_n .

Recognition. F is already a basis vector; use orthogonality directly.

Solution. $a_n = \frac{1}{N} (e_2, e_n) = \frac{1}{N} \cdot N \delta_{2n} = \delta_{2,n}$, so $a_2 = 1$ and all others vanish. *Check.* Inversion returns $\sum_n a_n e_n = e_2 = F$. A pure frequency has a single nonzero coefficient. (Non-example: if instead $F(k) = \cos(2\pi \cdot 2k/N) = \frac{1}{2}(\zeta^{2k} + \zeta^{-2k})$, then $a_2 = a_{N-2} = \frac{1}{2}$ and the spectrum is split between the two conjugate frequencies — a real signal has Hermitian-symmetric spectrum.)

Worked Example 37.1.11 (A small explicit transform on $Z(4)$). Let $F = (F(0), F(1), F(2), F(3)) = (1, 2, 3, 4)$ on $Z(4)$, $\zeta = e^{2\pi i/4} = i$. Compute a_0, a_1 .

Recognition. Plug into the definition; powers of i cycle through $1, i, -1, -i$.

Solution. $a_0 = \frac{1}{4}(1 + 2 + 3 + 4) = \frac{10}{4} = \frac{5}{2}$. For a_1 , $a_1 = \frac{1}{4} \sum_k F(k) i^{-k} = \frac{1}{4}(1 \cdot 1 + 2 \cdot (-i) + 3 \cdot (-1) + 4 \cdot (i)) = \frac{1}{4}(1 - 2i - 3 + 4i) = \frac{1}{4}(-2 + 2i) = -\frac{1}{2} + \frac{1}{2}i$. *Check.* a_0 is the mean, as it must be; and one can verify $a_2 = -\frac{1}{2}$, $a_3 = -\frac{1}{2} - \frac{1}{2}i$, with $\sum |a_n|^2 = \frac{1}{4}(1 + 4 + 9 + 16)/\dots$ matching Plancherel $\frac{1}{N} \sum |F(k)|^2 = \frac{30}{4} = 7.5$ against $|a_0|^2 + \dots = 6.25 + 0.5 + 0.25 + 0.5 = 7.5$.

Worked Example 37.1.12 (Discrete convolution and the product rule). On $Z(N)$ define $(F * G)(k) = \sum_{j=0}^{N-1} F(j)G(k-j)$ (indices mod N). Show its Fourier coefficients multiply: with the normalization above, $\widehat{F * G}_n = N a_n(F) a_n(G)$.

Recognition. Convolution should diagonalize under the transform — the finite echo of the convolution theorem.

Solution. $\widehat{F * G}_n = \frac{1}{N} \sum_k \sum_j F(j)G(k-j) \zeta^{-kn}$. Put $k = j + m$; then $\zeta^{-kn} = \zeta^{-jn} \zeta^{-mn}$ and the double sum factors: $\frac{1}{N} (\sum_j F(j) \zeta^{-jn}) (\sum_m G(m) \zeta^{-mn}) = \frac{1}{N} (N a_n(F)) (N a_n(G)) = N a_n(F) a_n(G)$. *Check.* For $F = G = \delta_0$, $F * G = \delta_0$ and $a_n = 1/N$, giving $\widehat{F * G}_n = N \cdot \frac{1}{N^2} = \frac{1}{N}$, the coefficient of δ_0 . Consistent.

The convolution theorem says the transform turns convolution into (a multiple of) coefficient-wise multiplication; equivalently, the following square commutes.

$$\begin{array}{ccc} (F, G) & \xrightarrow{\quad * \quad} & F * G \\ \mathcal{F} \times \mathcal{F} \downarrow & & \downarrow \mathcal{F} \\ ((a_n(F)), (a_n(G))) & \xrightarrow{\quad \widehat{\quad} \quad} & (N a_n(F) a_n(G)) \end{array}$$

Worked Example 37.1.13 (Summing a series via discrete Parseval). Use Plancherel on $Z(N)$ for $F = (1, 1, \dots, 1)$ to evaluate $\sum_n |a_n|^2$.

Recognition. The constant function has a trivial transform; Parseval then totals the spectrum.

Solution. $a_0 = \frac{1}{N} \sum_k 1 = 1$ and $a_n = \frac{1}{N} \sum_k \zeta^{-kn} = 0$ for $n \neq 0$ (roots sum to zero). So $\sum_n |a_n|^2 = 1$. *Check.* The right side is $\frac{1}{N} \sum_k |F(k)|^2 = \frac{1}{N} \cdot N = 1$. Both sides equal 1.

Worked Example 37.1.14 (Fourier transform of a discrete box (Dirichlet kernel)). On $Z(N)$ let $F = \mathbf{1}_{\{0,1,\dots,L-1\}}$ be the indicator of the first L points. Compute a_n and recognize the discrete Dirichlet kernel.

Recognition. A finite geometric sum in ζ^{-n} .

Solution. For $n \equiv 0$, $a_0 = L/N$. For $n \not\equiv 0$, $a_n = \frac{1}{N} \sum_{k=0}^{L-1} \zeta^{-nk} = \frac{1}{N} \frac{1-\zeta^{-nL}}{1-\zeta^{-n}}$. *Check.* At $L = N$ this collapses to $a_n = \delta_{n0}$ (the constant signal), since the numerator $1 - \zeta^{-nN} = 0$ for $n \not\equiv 0$. The magnitude $|a_n| = \frac{1}{N} \left| \frac{\sin(\pi n L/N)}{\sin(\pi n/N)} \right|$ is the discrete Dirichlet kernel, the finite echo of D_L on the circle.

Worked Example 37.1.15 (Equidistribution and the finite Weyl sum). Show $\frac{1}{N} \sum_{k=0}^{N-1} e^{2\pi i \ell k/N} = 0$ for $\ell \not\equiv 0 \pmod N$ and $= 1$ for $\ell \equiv 0$, and interpret it as “ $Z(N)$ samples every frequency evenly.”

Recognition. This is exactly the orthogonality of e_ℓ against the constant e_0 .

Solution. The sum is $\frac{1}{N} (e_\ell, e_0)$. By Lemma 37.1.6 it is 1 if $\ell \equiv 0$ and 0 otherwise. *Check.* The nonzero-frequency averages vanish, the finite analogue of Weyl’s criterion: the N sample points k/N “integrate” each pure frequency e_ℓ to its true mean (0 unless $\ell = 0$), so $Z(N)$ is an equidistributed sampling of the circle. (Non-example: sampling at N points clustered in a half-circle would *not* kill the frequency averages — uniform spacing is essential.)

Caution. The constant N in the discrete Plancherel identity is a *normalization* artifact: it depends on whether the $1/N$ is placed on the analysis side (as here), the synthesis side, or split symmetrically as $1/\sqrt{N}$ (the unitary convention). Always state your convention before quoting Parseval, or the factors will not match.

Exercises (Stein, Ch.7) — Fourier analysis on $Z(N)$.

- (Discrete vs. continuous coefficients) For a function f on the circle and $N \geq 1$, define the discrete Fourier coefficients $a_N(n) = \frac{1}{N} \sum_{k=1}^N f(e^{2\pi i k/N}) e^{-2\pi i k n/N}$ and the ordinary coefficients $a(n) = \int_0^1 f(e^{2\pi i x}) e^{-2\pi i n x} dx$. (a) Show $a_N(n) = a_N(n+N)$ (the discrete coefficients are N -periodic in n). (b) Prove that if f is continuous, then $a_N(n) \rightarrow a(n)$ as $N \rightarrow \infty$.
- (Decay for C^1 functions) If f is C^1 on the circle, prove $|a_N(n)| \leq c/|n|$ whenever $0 < |n| \leq N/2$. [Hint: $a_N(n) [1 - e^{2\pi i \ell n/N}] = \frac{1}{N} \sum_{k=1}^N [f(e^{2\pi i k/N}) - f(e^{2\pi i (k+\ell)/N})] e^{-2\pi i k n/N}$; choose ℓ so $\ell n/N$ is near $1/2$.]
- (Decay for C^2 and finite-to-continuous inversion) By a similar method show that if f is C^2 on the circle then $|a_N(n)| \leq c/|n|^2$ for $0 < |n| \leq N/2$. Deduce the inversion formula $f(e^{2\pi i x}) = \sum_{n=-\infty}^{\infty} a(n) e^{2\pi i n x}$ for $f \in C^2$ from its finite version. [Hint: use the second symmetric difference $f(e^{2\pi i (k+\ell)/N}) + f(e^{2\pi i (k-\ell)/N}) - 2f(e^{2\pi i k/N})$; for N odd write $f(e^{2\pi i k/N}) = \sum_{|n| < N/2} a_N(n) e^{2\pi i k n/N}$.]
- (The DFT matrix, $\rightarrow$ here as the $Z(N)$ unitary) Let $\zeta = e^{2\pi i/N}$ and define the $N \times N$ matrix $M = (a_{jk})$ by $a_{jk} = N^{-1/2} \zeta^{jk}$, $1 \leq j, k \leq N$. (a) Show M is unitary. (b) Interpret $(Mu, Mv) = (u, v)$ and $M^* = M^{-1}$ in terms of Fourier series on $Z(N)$ (Plancherel and inversion, respectively).
- (Sampling/reconstruction) Suppose $P(x) = \sum_{n=1}^N a_n e^{2\pi i n x}$. (a) Using Parseval on the circle and on $Z(N)$, show $\int_0^1 |P(x)|^2 dx = \frac{1}{N} \sum_{j=1}^N |P(j/N)|^2$. (b) Prove the reconstruction formula $P(x) = \sum_{j=1}^N P(j/N) K(x - j/N)$, where $K(x) = \frac{1}{N} \frac{e^{2\pi i x} (1 - e^{2\pi i N x})}{1 - e^{2\pi i x}} = \frac{1}{N} (e^{2\pi i x} + e^{2\pi i 2x} + \dots + e^{2\pi i N x})$. Note $K(0) = 1$ and $K(j/N) = 0$ for $j \not\equiv 0 \pmod N$, so P is determined by the samples $P(j/N)$, $1 \leq j \leq N$.

Additional exercises.

- Compute, by hand, the full discrete Fourier transform (a_0, a_1, a_2, a_3) of the signal $F = (1, 0, 0, 0)$ and of $G = (0, 1, 0, 0)$ on $Z(4)$. Verify the “shift = modulation” rule: the transform of G is the transform of F multiplied by $\zeta^{-n} = (-i)^n$.
- On $Z(N)$, prove that $\sum_{n=0}^{N-1} a_n = F(0)$ and $a_0 = \frac{1}{N} \sum_k F(k)$. Interpret both as evaluation at a single point in the opposite domain.
- (Real signals have Hermitian spectra) Show that if F is real-valued on $Z(N)$, then $a_{N-n} = \overline{a_n}$ for all n . Conclude that a_0 and (when N is even) $a_{N/2}$ are real.

4. Compute the discrete Fourier transform of the box $F = (1, 1, 1, 0, 0, 0, 0, 0)$ on $Z(8)$ and identify the resulting discrete Dirichlet-type kernel $a_n = \frac{1}{N} \frac{1 - \zeta^{-3n}}{1 - \zeta^{-n}}$.
5. (Parseval drill) Let $F(k) = \zeta^k + \zeta^{3k}$ on $Z(8)$. Find all a_n and verify both inversion and Plancherel numerically.
6. Prove the discrete shift theorem: if $G(k) = F(k - m)$ (indices mod N) then $a_n(G) = \zeta^{-mn} a_n(F)$. Then prove the discrete modulation theorem: if $H(k) = \zeta^{jk} F(k)$ then $a_n(H) = a_{n-j}(F)$.
7. (Non-example) Exhibit a function on $Z(4)$ whose four Fourier coefficients all have the same modulus, and one whose coefficients do not, explaining what feature of the signal each reflects.
8. (Diagram) Plot the eight eighth-roots of unity in the plane and, on the same picture, draw the vectors ζ^{3k} for $k = 0, \dots, 7$; observe geometrically why $\sum_k \zeta^{3k} = 0$.

37.2 The Fast Fourier Transform

The cost of the naive transform

The fast Fourier transform (FFT) is an algorithm for computing the Fourier coefficients of a function on $Z(N)$ in far fewer arithmetic operations than the definition suggests. By an *operation* we mean one complex addition or one complex multiplication.

Proposition 37.2.1 (Naive cost). *Given $F(0), \dots, F(N-1)$ and $\omega_N = e^{-2\pi i/N}$, all N coefficients*

$$a_k^N(F) = \frac{1}{N} \sum_{r=0}^{N-1} F(r) \omega_N^{kr}$$

can be computed in at most $2N^2 + N$ operations.

Proof. Computing the powers $\omega_N^2, \dots, \omega_N^{N-1}$ costs at most $N-2$ multiplications. Each coefficient a_k^N is a sum of N products $F(r)\omega_N^{kr}$ (after the relevant power is known), requiring $N+1$ multiplications and $N-1$ additions, i.e. $\leq 2N$ operations; over the N values of k this is $\leq 2N^2$. Adding the one-time $N-2$ for the powers gives $\leq 2N^2 + N$. The bound is $O(N^2)$. $\square$

For large N this quadratic growth is prohibitive (doubling N quadruples the work). The FFT lowers it to $O(N \log N)$ when N is a power of 2, by reusing the work for M points to handle $2M$ points.

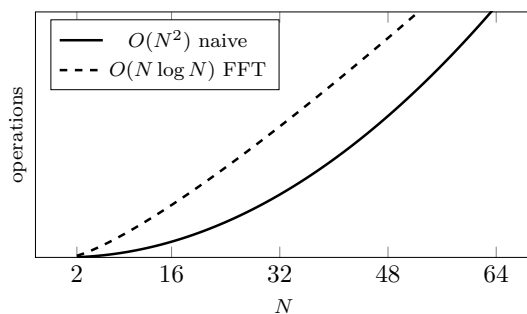


Figure 37.4. The naive transform grows like N^2 ; the FFT like $N \log_2 N$. The gap is enormous for large N (e.g. $N = 2^{20}$: a factor of about 5×10^4). The vertical scale is in arbitrary units chosen to display the growth rates. (Schematic.)

The recursion and the $O(N \log N)$ theorem

Theorem 37.2.2 (Fast Fourier transform). *Given $\omega_N = e^{-2\pi i/N}$ with $N = 2^n$, one can compute all Fourier coefficients of a function on $Z(N)$ in at most*

$$4 \cdot 2^n n = 4N \log_2 N = O(N \log N)$$

operations.

The proof rests on a recursion: the coefficients on $Z(2M)$ are assembled cheaply from the coefficients of two functions on $Z(M)$, obtained by splitting F into its even- and odd-indexed samples.

Lemma 37.2.3 (Recursion step). *Let $\#(M)$ be the minimum number of operations to compute all Fourier coefficients of any function on $Z(M)$. If $\omega_{2M} = e^{-2\pi i/(2M)}$ is given, then*

$$\#(2M) \leq 2\#(M) + 8M.$$

Proof. Forming the powers $\omega_{2M}^2, \dots, \omega_{2M}^{2M}$ costs $\leq 2M$ operations; in particular $\omega_M = \omega_{2M}^2$ is available. Given F on $Z(2M)$, define two functions on $Z(M)$ by splitting into even and odd samples,

$$F_0(r) = F(2r), \quad F_1(r) = F(2r+1), \quad r = 0, \dots, M-1.$$

Computing the coefficients $a_k^M(F_0)$ and $a_k^M(F_1)$ costs $\leq \#(M)$ operations each. The key algebraic identity is

$$a_k^{2M}(F) = \frac{1}{2} \left(a_k^M(F_0) + a_k^M(F_1) \omega_{2M}^k \right).$$

To see it, split the defining sum over even and odd r :

$$a_k^{2M}(F) = \frac{1}{2M} \sum_{r=0}^{2M-1} F(r) \omega_{2M}^{kr} = \frac{1}{2} \left(\frac{1}{M} \sum_{\ell=0}^{M-1} F(2\ell) \omega_{2M}^{2k\ell} + \frac{1}{M} \sum_{m=0}^{M-1} F(2m+1) \omega_{2M}^{k(2m+1)} \right).$$

Since $\omega_{2M}^2 = \omega_M$, the first inner sum is $a_k^M(F_0)$ and the second is $\omega_{2M}^k a_k^M(F_1)$, giving the identity. Once $a_k^M(F_0)$, $a_k^M(F_1)$ and ω_{2M}^k are known, each of the $2M$ coefficients a_k^{2M} costs at most three operations (one addition, two multiplications — noting a_k^{2M} and a_{k+M}^{2M} reuse the same two halves). Therefore

$$\#(2M) \leq 2M + 2\#(M) + 3 \cdot 2M = 2\#(M) + 8M. \quad \square$$

Proof of Theorem 37.2.2. Induct on n where $N = 2^n$. Base $n = 1$: with $N = 2$ the two coefficients are $a_0^N(F) = \frac{1}{2}(F(0) + F(1))$ and $a_1^N(F) = \frac{1}{2}(F(0) - F(1))$, computable in ≤ 5 operations, and $5 < 4 \cdot 2 \cdot 1 = 8$. Inductive step: assume $\#(2^{n-1}) \leq 4 \cdot 2^{n-1}(n-1)$. By Lemma 37.2.3 with $M = 2^{n-1}$,

$$\#(2^n) \leq 2 \cdot 4 \cdot 2^{n-1}(n-1) + 8 \cdot 2^{n-1} = 4 \cdot 2^n(n-1) + 4 \cdot 2^n = 4 \cdot 2^n n,$$

completing the induction. $\square$

Each pass through the recursion halves the problem size, so there are $n = \log_2 N$ passes, each doing $O(N)$ work — hence $O(N \log N)$. The even/odd split is the “decimation in time” radix-2 FFT; the elementary 2×2 combine step is the famous *butterfly*.

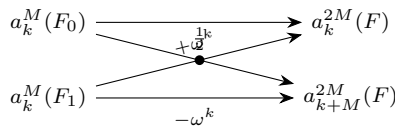


Figure 37.5. The radix-2 *butterfly*: two $Z(M)$ coefficients combine with a twiddle factor ω_{2M}^k into two $Z(2M)$ coefficients.

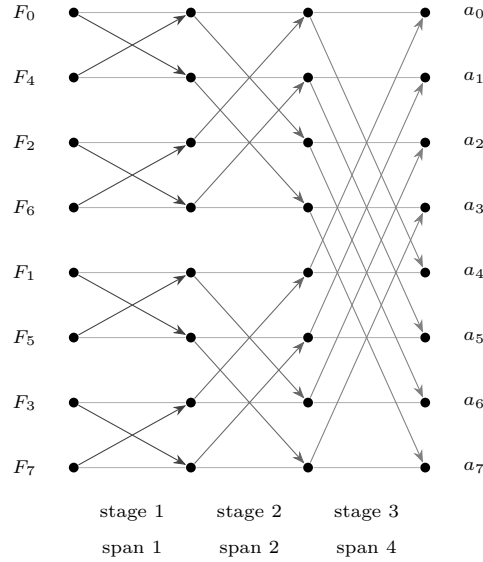


Figure 37.6. Radix-2 decimation-in-time network for $N = 8 = 2^3$: $\log_2 8 = 3$ stages of butterflies. With the outputs a_k in natural order the inputs enter bit-reversed (0, 4, 2, 6, 1, 5, 3, 7); each stage is $O(N)$ work, total $O(N \log N)$.

Method (Running the radix-2 FFT). *Recognition trigger.* $N = 2^n$ and you must compute the whole spectrum.

1. Split F into even samples $F_0(r) = F(2r)$ and odd samples $F_1(r) = F(2r + 1)$.
2. Recursively transform F_0, F_1 on $Z(N/2)$.
3. Combine with butterflies: $a_k = \frac{1}{2}(a_k(F_0) + \omega_N^k a_k(F_1))$ and $a_{k+N/2} = \frac{1}{2}(a_k(F_0) - \omega_N^k a_k(F_1))$ for $k = 0, \dots, N/2 - 1$.
4. Stop the recursion at $N = 1$, where the transform is the identity.

Worked Example 37.2.4 ($N = 2$ butterfly by hand). Compute the FFT of $F = (F(0), F(1))$ on $Z(2)$.

Recognition. The smallest transform; $\omega_2 = e^{-\pi i} = -1$.

Solution. $a_0 = \frac{1}{2}(F(0) + F(1))$, $a_1 = \frac{1}{2}(F(0) + \omega_2 F(1)) = \frac{1}{2}(F(0) - F(1))$. *Check.* Inversion $F(k) = \sum_n a_n e^{2\pi i n k / 2} = a_0 + a_1(-1)^k$ gives $F(0) = a_0 + a_1 = F(0)$ and $F(1) = a_0 - a_1 = F(1)$. Correct.

Worked Example 37.2.5 ($N = 4$ via two $N = 2$ transforms). FFT of $F = (1, 0, 1, 0)$ on $Z(4)$ using the recursion.

Recognition. Even/odd split first; here both halves are simple.

Solution. Even samples $F_0 = (F(0), F(2)) = (1, 1)$, odd $F_1 = (F(1), F(3)) = (0, 0)$. On $Z(2)$: $a(F_0) = (1, 0)$, $a(F_1) = (0, 0)$. Twiddle $\omega_4 = e^{-\pi i / 2} = -i$. Combine for $k = 0, 1$: $a_0 = \frac{1}{2}(1 + (-i)^0 \cdot 0) = \frac{1}{2}$, $a_2 = \frac{1}{2}(1 - 0) = \frac{1}{2}$; $a_1 = \frac{1}{2}(0 + (-i)^1 \cdot 0) = 0$, $a_3 = \frac{1}{2}(0 - 0) = 0$. So $a = (\frac{1}{2}, 0, \frac{1}{2}, 0)$. *Check directly.* $a_n = \frac{1}{4} \sum_k F(k)(-i)^{kn} = \frac{1}{4}(1 + (-i)^{2n}) = \frac{1}{4}(1 + (-1)^n)$, which is $\frac{1}{2}$ for n even and 0 for n odd. Agrees.

Worked Example 37.2.6 (Operation count, $N = 8$). How many operations does the FFT bound promise for $N = 8$, and how does it compare to the naive count?

Recognition. Plug into $4N \log_2 N$ versus $2N^2 + N$.

Solution. FFT: $4 \cdot 8 \cdot \log_2 8 = 32 \cdot 3 = 96$. Naive: $2 \cdot 64 + 8 = 136$. *Check.* Already at $N = 8$ the FFT bound is smaller; for $N = 1024$ it is $4 \cdot 1024 \cdot 10 = 40960$ versus $\approx 2.1 \times 10^6$, roughly $51\times$ fewer.

Worked Example 37.2.7 (Why even/odd and not first-half/second-half?). Show that splitting F into its first half $F(0), \dots, F(M-1)$ and second half does *not* give the clean recursion, whereas even/odd does.

Recognition. The recursion needs $\omega_{2M}^2 = \omega_M$, which arises from sampling *every other* point.

Solution. With even/odd, $\omega_{2M}^{2k\ell} = \omega_M^{k\ell}$ turns the inner sums into genuine $Z(M)$ transforms. With a first/second-half split the exponent $\omega_{2M}^{k(\ell+M)} = \omega_{2M}^{k\ell} \omega_{2M}^{kM} = \omega_{2M}^{k\ell} (-1)^k$ keeps the base ω_{2M} , so the inner sums are *not* $Z(M)$ transforms and the recursion does not close. *Check.* This is the distinction between decimation-in-time (even/odd) and decimation-in-frequency; the text’s algorithm is the former.

Worked Example 37.2.8 (A non-dyadic length). What goes wrong with the radix-2 recursion when $N = 6$, and how is it fixed?

Recognition. 6 is not a power of 2; the butterfly split requires an even length at each level.

Solution. One even/odd split takes $Z(6) \rightarrow$ two $Z(3)$ transforms, but 3 is odd and cannot be halved again, so the radix-2 recursion stalls. The fix (Stein Exercise 9) is a radix-3 (or mixed-radix) split: for $N = 3^n$ one obtains an $O(N \log_3 N)$ algorithm, and generally $N = \alpha^n$ yields $O(N \log_\alpha N)$. *Check.* For $N = 6 = 2 \cdot 3$ one combines one radix-2 and one radix-3 stage. (Non-example: a prime length like $N = 7$ has no nontrivial factorization, so these splitting algorithms give no speedup — one needs Rader’s or Bluestein’s algorithm, outside this chapter.)

Exercises (Stein, Ch.7) — the fast Fourier transform.

- (Radix-3 and radix- α FFT) Modify the argument in the text to prove: (a) one can compute the Fourier coefficients of a function on $Z(N)$ when $N = 3^n$ with at most $6N \log_3 N$ operations; (b) generalize to $N = \alpha^n$ where α is an integer > 1 , obtaining a bound of the form $O(N \log_\alpha N)$. [Hint: split F into α subsampled functions $F_j(r) = F(\alpha r + j)$, $j = 0, \dots, \alpha - 1$, on $Z(N/\alpha)$, and combine with the appropriate twiddle factors, using $\omega_{\alpha M}^\alpha = \omega_M$.]

Additional exercises.

- Carry out the full radix-2 FFT of $F = (1, 2, 3, 4, 5, 6, 7, 8)$ on $Z(8)$ by hand, recording the intermediate $Z(2)$ and $Z(4)$ transforms and all twiddle factors.
- Verify the recursion identity $a_{k+M}^{2M}(F) = \frac{1}{2}(a_k^M(F_0) - \omega_{2M}^k a_k^M(F_1))$ used in the second output of each butterfly, and explain why it requires no new multiplications beyond those for $a_k^{2M}(F)$.
- Show that the exact FFT operation count for $N = 2^n$ given by the text bound is $4N \log_2 N$, and tabulate it against the naive $2N^2 + N$ for $N = 2, 4, 8, 16, 32$.
- (Inverse FFT) Explain how the same butterfly network computes the inverse transform $F(k) = \sum_n a_n \zeta^{kn}$ by replacing ω_N with $\overline{\omega_N}$ and dropping the $1/N$ normalizations, and count its operations.
- (Bit reversal) For $N = 8$, determine the permutation of input indices produced by recursively separating even and odd samples, and show it is the bit-reversal permutation $r \mapsto$ reverse the 3 binary digits of r .
- Prove that any algorithm computing all N Fourier coefficients of a generic function on $Z(N)$ must use at least N multiplications, and contrast this lower bound with the $O(N \log N)$ FFT upper bound.
- (Diagram) Draw the radix-2 butterfly network for $N = 4$, labeling each edge with its twiddle factor $(-i)^k$, and trace the path of the input $F(2)$ to each output.

37.3 Fourier Analysis on Finite Abelian Groups

Abelian groups, homomorphisms, and examples

We now generalize the $Z(N)$ theory to an arbitrary finite abelian group. The exponentials e_ℓ are replaced by *characters*, and once we show a finite abelian group has “enough” characters, the entire Fourier apparatus — orthonormal basis, inversion, Plancherel — follows automatically.

Definition 37.3.1 (Abelian group). An *abelian group* is a set G with a binary operation $(a, b) \mapsto a \cdot b$ satisfying: (i) associativity $a \cdot (b \cdot c) = (a \cdot b) \cdot c$; (ii) an identity u (written 1 or 0) with $a \cdot u = a$ for all a ; (iii) inverses a^{-1} with $a \cdot a^{-1} = u$; (iv) commutativity $a \cdot b = b \cdot a$. The identity and inverses are unique. We may use multiplicative $(a \cdot b, a^{-1})$ or additive $(a + b, -a)$ notation. The *order* $|G|$ is the number of elements (when finite).

Examples. $(\mathbb{R}, +)$ with identity 0; $(\mathbb{R} \setminus \{0\}, \cdot)$ and $(\mathbb{R}^+, \cdot)$ with identity 1; $(\mathbb{Z}, +)$ (but $\mathbb{Z} \setminus \{0\}$ is *not* a group under multiplication — 2 has no integer inverse), while $(\mathbb{Q} \setminus \{0\}, \cdot)$ is; the unit circle $S^1 = \{e^{i\theta}\}$ under multiplication (isomorphic to $\mathbb{R}/2\pi\mathbb{Z}$ under addition of angles); $Z(N)$ in both its multiplicative and additive forms; and $\mathbb{Z}^*(q)$, the units mod q , below.

Definition 37.3.2 (Homomorphism, isomorphism). A *homomorphism* $f: G \rightarrow H$ of abelian groups satisfies $f(a \cdot b) = f(a) \cdot f(b)$. It is an *isomorphism* ($G \approx H$) if it is bijective; equivalently if there is a homomorphism $\tilde{f}: H \rightarrow G$ with $\tilde{f} \circ f = \text{id}_G$ and $f \circ \tilde{f} = \text{id}_H$. Isomorphic groups have the same group structure.

The map $n \mapsto R(n)$, $\zeta^n \mapsto [n]$, realizes $Z(N) \approx \mathbb{Z}/N\mathbb{Z}$; the angle map realizes $S^1 \approx \mathbb{R}/2\pi\mathbb{Z}$; and $\exp: \mathbb{R} \rightarrow \mathbb{R}^+$, $\log: \mathbb{R}^+ \rightarrow \mathbb{R}$ are mutually inverse homomorphisms, so $(\mathbb{R}, +) \approx (\mathbb{R}^+, \cdot)$.

$$(\mathbb{R}, +) \begin{array}{c} \xrightarrow{\exp} \\ \sim \\ \xleftarrow{\log} \end{array} (\mathbb{R}^+, \cdot)$$

Definition 37.3.3 (Direct product). For finite abelian groups G_1, G_2 , the *direct product* $G_1 \times G_2$ has elements (g_1, g_2) and operation $(g_1, g_2) \cdot (g'_1, g'_2) = (g_1 g'_1, g_2 g'_2)$. Then $|G_1 \times G_2| = |G_1| |G_2|$. (Generalizes to finitely many factors.) The *structure theorem* states every finite abelian group is isomorphic to a direct product of groups $Z(N)$; we will not need its proof.

Definition 37.3.4 (The group of units $\mathbb{Z}^*(q)$). Multiplication mod q is well defined on $Z(q)$. An element $n \in Z(q)$ is a *unit* if there is m with $nm \equiv 1 \pmod{q}$. The set $\mathbb{Z}^*(q)$ of units is an abelian group under multiplication mod q . (Equivalently, $\mathbb{Z}^*(q)$ consists of the residues relatively prime to q .)

Worked Example 37.3.5 (Unit groups and their structure). Identify $\mathbb{Z}^*(4), \mathbb{Z}^*(5), \mathbb{Z}^*(8)$ and recognize each up to isomorphism.

Recognition. List residues coprime to q , then build the multiplication table and match it to a product of $Z(N)$'s.

Solution. $\mathbb{Z}^*(4) = \{1, 3\}$, and $3 \cdot 3 = 9 \equiv 1$, so it is cyclic of order 2: $\mathbb{Z}^*(4) \approx Z(2)$ via $1 \mapsto 0, 3 \mapsto 1$. $\mathbb{Z}^*(5) = \{1, 2, 3, 4\}$; here 2 generates $(2, 4, 3, 1)$, so $\mathbb{Z}^*(5) \approx Z(4)$ via $2 \mapsto 1$. $\mathbb{Z}^*(8) = \{1, 3, 5, 7\}$; each nonidentity element squares to 1 ($3^2 = 9 \equiv 1, 5^2 = 25 \equiv 1, 7^2 = 49 \equiv 1$), so it is *not* cyclic: $\mathbb{Z}^*(8) \approx Z(2) \times Z(2)$ via $1 \mapsto (0, 0), 3 \mapsto (1, 0), 5 \mapsto (0, 1), 7 \mapsto (1, 1)$. *Check.* In $Z(2) \times Z(2)$ every element has order ≤ 2 , matching $\mathbb{Z}^*(8)$.

Characters and the dual group

Definition 37.3.6 (Character). A *character* on a finite abelian group G is a homomorphism $e: G \rightarrow S^1$ into the circle group, i.e. $e(a \cdot b) = e(a)e(b)$ for all $a, b \in G$ and $|e(a)| = 1$. The *trivial character* is $e(a) = 1$ for all a .

Characters generalize the exponentials: on $Z(N)$ the characters are exactly $e_\ell(k) = \zeta^{\ell k}$ ($0 \leq \ell \leq N-1$), recovering the law $e_\ell(k+m) = e_\ell(k)e_\ell(m)$.

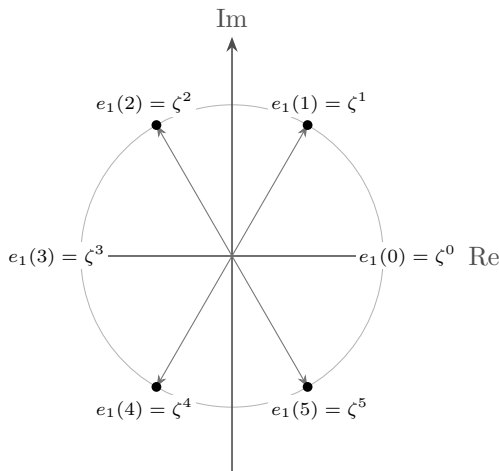


Figure 37.7. A character e_1 on $Z(6)$ sends $k \mapsto \zeta^k$ ($\zeta = e^{2\pi i/6}$): its six values march once around the sixth roots of unity. Every character lands on S^1 .

Lemma 37.3.7 (The dual group). *The set $\widehat{G}$ of all characters of G is an abelian group under pointwise multiplication $(e_1 \cdot e_2)(a) = e_1(a)e_2(a)$, with the trivial character as identity. We call $\widehat{G}$ the dual group.*

Proof. The product of two characters is a character (a product of homomorphisms into S^1); associativity and commutativity are inherited pointwise from $\mathbb{C}$; the trivial character is the unit; and $e^{-1}(a) = \overline{e(a)}$ is a character inverting e . $\square$

Duals of basic groups. $\widehat{Z(N)} \approx Z(N)$ via $e_\ell \mapsto \ell$; $\widehat{S^1} = \{e_n\}_{n \in \mathbb{Z}}$ via $e_n \mapsto n$, so $\widehat{S^1} \approx \mathbb{Z}$; $\widehat{\mathbb{R}} = \{e_\xi(x) = e^{2\pi i \xi x}\}$ with $e_\xi \mapsto \xi$, so $\widehat{\mathbb{R}} \approx \mathbb{R}$; and (via exp) $\widehat{\mathbb{R}^+} \approx \mathbb{R}$ with $e_\xi(x) = x^{2\pi i \xi} = e^{2\pi i \xi \log x}$.

Lemma 37.3.8 (Nonvanishing multiplicative functions are characters). *Let G be finite abelian and $e: G \rightarrow \mathbb{C} \setminus \{0\}$ multiplicative ($e(a \cdot b) = e(a)e(b)$). Then $|e(a)| = 1$ for all a , so e is a character.*

Proof. Since G is finite, $|e(a)|$ is bounded above and below by positive constants as a ranges over G . But $|e(b^n)| = |e(b)|^n$ for all n , and a sequence $|e(b)|^n$ stays bounded above and below only if $|e(b)| = 1$. Hence every value lies on S^1 . $\square$

Orthogonality relations

Let V be the space of complex functions on G , $\dim V = |G|$, with the normalized Hermitian inner product

$$(f, g) = \frac{1}{|G|} \sum_{a \in G} f(a) \overline{g(a)}.$$

Lemma 37.3.9 (Character sum). *If e is a nontrivial character of G , then $\sum_{a \in G} e(a) = 0$.*

Proof. Choose b with $e(b) \neq 1$. As a ranges over G , so does ab , hence $e(b) \sum_a e(a) = \sum_a e(b)e(a) = \sum_a e(ab) = \sum_a e(a)$. Thus $(e(b) - 1) \sum_a e(a) = 0$, and since $e(b) \neq 1$, the sum is 0. $\square$

This is the abstract version of “the roots of unity sum to zero”: a nontrivial character, viewed as points on S^1 , is balanced about the origin.

Theorem 37.3.10 (Orthonormality of characters). *The characters of G form an orthonormal family for $(\cdot, \cdot)$.*

Proof. For any character, $(e, e) = \frac{1}{|G|} \sum_a |e(a)|^2 = \frac{1}{|G|} \cdot |G| = 1$. If $e \neq e'$ are characters, then $e(e')^{-1} = \overline{ee'}$ is a nontrivial character, so by Lemma 37.3.9, $(e, e') = \frac{1}{|G|} \sum_a e(a) \overline{e'(a)} = \frac{1}{|G|} \sum_a (ee')(a) = 0$. $\square$

Consequently distinct characters are linearly independent, so $|\widehat{G}| \leq \dim V = |G|$. The decisive fact is that equality holds.

Characters as a total family

Theorem 37.3.11 (Characters form a basis). *The characters of a finite abelian group G form a basis of V ; in particular $|\widehat{G}| = |G|$.*

The proof uses a multivariable spectral theorem.

Lemma 37.3.12 (Simultaneous diagonalization). *A commuting family $\{T_1, \dots, T_k\}$ of unitary operators on a finite-dimensional inner product space V is simultaneously diagonalizable: there is a basis of common eigenvectors.*

Proof. Induct on k . For $k = 1$ this is the spectral theorem for a unitary operator. Assume it for $k - 1$. Apply the spectral theorem to T_k : $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_s}$ splits into eigenspaces. Each T_j ($j < k$) preserves every V_{λ_i} : if $v \in V_{\lambda_i}$ then $T_k(T_j v) = T_j(T_k v) = T_j(\lambda_i v) = \lambda_i T_j v$, so $T_j v \in V_{\lambda_i}$. The restrictions of $T_1, \dots, T_{k-1}$ to each V_{λ_i} are commuting unitaries, so by induction are simultaneously diagonalizable there; collecting the resulting bases over all i gives a common eigenbasis for V . $\square$

Proof of Theorem 37.3.11. For each $a \in G$ define the translation operator $T_a: V \rightarrow V$ by $(T_a f)(x) = f(a \cdot x)$. Since G is abelian, $T_a T_b = T_{ab} = T_b T_a$, and each T_a is unitary for the inner product (translation permutes the summands). By Lemma 37.3.12 there is a basis $\{v_b\}$ of common eigenvectors: $T_a v_b = \lambda_a^{(b)} v_b$. Fix one such v with eigenvalues λ_a . Then $v(1) \neq 0$: otherwise $v(a) = v(a \cdot 1) = (T_a v)(1) = \lambda_a v(1) = 0$ for all a , forcing $v = 0$. Set $w(x) = v(x)/v(1) = \lambda_x$. Then $w(1) = 1$, w never vanishes, and

$$w(a \cdot b) = \frac{v(ab)}{v(1)} = \frac{(T_a v)(b)}{v(1)} = \frac{\lambda_a v(b)}{v(1)} = \lambda_a \frac{v(b)}{v(1)} = w(a)w(b),$$

so w is multiplicative and nonvanishing, hence a character by Lemma 37.3.8. The common eigenbasis thus consists of (scalar multiples of) characters; there are $|G|$ of them and they are independent, so the characters span V and $|\widehat{G}| = |G|$. $\square$

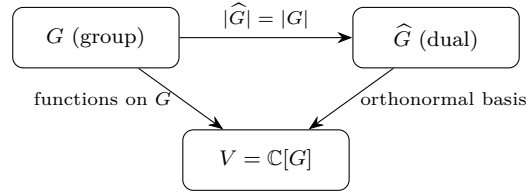


Figure 37.8. The characters $\widehat{G}$ are as numerous as G and form an orthonormal basis of the function space V , enabling Fourier analysis on G .

Fourier inversion and Plancherel on G

Definition 37.3.13 (Fourier coefficient and series on G). For $f \in V$ and a character $e \in \widehat{G}$, the *Fourier coefficient* is

$$\widehat{f}(e) = (f, e) = \frac{1}{|G|} \sum_{a \in G} f(a) \overline{e(a)},$$

and the *Fourier series* is $f \sim \sum_{e \in \widehat{G}} \widehat{f}(e) e$.

Theorem 37.3.14 (Fourier inversion on G). *Let G be a finite abelian group. The characters form an orthonormal basis of V , and every $f \in V$ equals its Fourier series:*

$$f = \sum_{e \in \widehat{G}} \widehat{f}(e) e.$$

Proof. By Theorems 37.3.10 and 37.3.11 the characters are an orthonormal basis. Writing $f = \sum_e c_e e$ and pairing with a character e' , orthonormality gives $(f, e') = c_{e'}$; but $(f, e') = \widehat{f}(e')$, so $c_e = \widehat{f}(e)$ and the expansion is the Fourier series. $\square$

Theorem 37.3.15 (Parseval–Plancherel on G). *For f on G , $\|f\|^2 = \sum_{e \in \widehat{G}} |\widehat{f}(e)|^2$, where $\|f\|^2 = (f, f)$.*

Proof. With the orthonormal basis of characters and $(f, e) = \widehat{f}(e)$, $\|f\|^2 = (f, f) = (\sum_e \widehat{f}(e) e, f) = \sum_e \widehat{f}(e) \overline{(f, e)} = \sum_e |\widehat{f}(e)|^2$. $\square$

The apparent factor discrepancy with the $Z(N)$ Plancherel (Theorem 37.1.8) is purely a matter of normalization: here the inner product already carries $1/|G|$.

Inversion and Plancherel together say the Fourier transform $\mathcal{F}: f \mapsto \widehat{f}$ is a norm-preserving isomorphism from the function space V onto the space $\mathbb{C}[\widehat{G}]$ of functions on the dual group, with the synthesis map $f \sim \sum_e \widehat{f}(e) e$ as its inverse.

$$(V, \|\cdot\|^2) \xrightleftharpoons[\mathcal{F}^{-1}]{\mathcal{F}} (\mathbb{C}[\widehat{G}], \sum_e |\widehat{f}(e)|^2)$$

Worked Example 37.3.16 (Fourier coefficients on $Z(N)$ recovered from G -theory). Specialize Definition 37.3.13 to $G = Z(N)$ and confirm it matches Definition 37.1.7.

Recognition. The characters of $Z(N)$ are $e_n(k) = \zeta^{nk}$; plug in and compare normalizations.

Solution. $\widehat{f}(e_n) = \frac{1}{N} \sum_k f(k) \overline{\zeta^{nk}} = \frac{1}{N} \sum_k f(k) \zeta^{-nk} = a_n$. *Check.* Identical to Definition 37.1.7; the abstract theory subsumes the concrete one.

Worked Example 37.3.17 (Characters of $Z(2) \times Z(2)$). List the four characters of the Klein group $G = Z(2) \times Z(2)$ (written additively).

Recognition. $|\widehat{G}| = |G| = 4$; a character is determined by its values ± 1 on the two generators.

Solution. Write elements (a, b) , $a, b \in \{0, 1\}$. A character has the form $e_{(s,t)}(a, b) = (-1)^{sa+tb}$ for $(s, t) \in \{0, 1\}^2$, giving four characters. *Check.* $e_{(s,t)}((a, b) + (a', b')) = (-1)^{s(a+a') + t(b+b')} = e_{(s,t)}(a, b) e_{(s,t)}(a', b')$, and $\widehat{G} \approx Z(2) \times Z(2) \approx G$, as the structure/duality theory predicts.

Worked Example 37.3.18 (Convolution diagonalized on G). Define $(f * g)(a) = \frac{1}{|G|} \sum_{b \in G} f(b) g(a \cdot b^{-1})$. Show $\widehat{f * g}(e) = \widehat{f}(e) \widehat{g}(e)$.

Recognition. Convolution multiplies Fourier coefficients — the abelian-group convolution theorem.

Solution. $\widehat{f * g}(e) = \frac{1}{|G|} \sum_a \left(\frac{1}{|G|} \sum_b f(b) g(ab^{-1}) \right) \overline{e(a)}$. Put $c = ab^{-1}$, so $a = bc$ and $\overline{e(a)} = \overline{e(b)} \overline{e(c)}$; the double sum factors as $(\frac{1}{|G|} \sum_b f(b) \overline{e(b)}) (\frac{1}{|G|} \sum_c g(c) \overline{e(c)}) = \widehat{f}(e) \widehat{g}(e)$. *Check.* The Dirac function D with $\widehat{D}(e) = 1$ for all e is the convolution identity, matching Stein's $f * D = f$.

Worked Example 37.3.19 (The Dirac function and dual orthogonality). Define $D(c) = \sum_{e \in \widehat{G}} e(c)$. Show $D(c) = |G|$ if $c = 1_G$ and 0 otherwise, and that $f * D = f$.

Recognition. Sum a fixed group element over all characters (the “column” orthogonality, dual to Lemma 37.3.9).

Solution. For $c = 1_G$, every $e(1_G) = 1$, so $D = |G|$. For $c \neq 1_G$, evaluation at c is a nontrivial character of $\widehat{G}$, so by the orthogonality relations applied to $\widehat{G}$ (equivalently Stein Problem 3 / Exercise 13b), $\sum_e e(c) = 0$. Thus $D = |G| \cdot \delta_{1_G}$, the discrete Dirac mass, and $f * D = f$. *Check.* $\frac{1}{|G|} \sum_c D(c) = \frac{1}{|G|} |G| = 1$, so D has unit mass concentrated at the identity — a finite “good kernel.”

Worked Example 37.3.20 (A non-character). Show $f(k) = k$ on $Z(N)$ (with $k \in \{0, \dots, N-1\}$) is not a character, then expand it.

Recognition. Characters are multiplicative with $|\cdot| = 1$; $f(k) = k$ is neither.

Solution. f takes the value 0 (at $k = 0$) and the value 2 (at $k = 2$, for $N > 2$), so $|f| \neq 1$ and $f(1 \cdot 1) = 1 \neq f(1)f(1)$ in the multiplicative reading — not a character. But it has a Fourier expansion: $f = \sum_n a_n e_n$ with $a_0 = \frac{1}{N} \sum_k k = \frac{N-1}{2}$ and, for $n \neq 0$, $a_n = \frac{1}{N} \sum_{k=0}^{N-1} k \zeta^{-nk} = \frac{-1}{1-\zeta^{-n}}$ (the discrete sawtooth spectrum). *Check.* At $N = 2$, $a_0 = \frac{1}{2}$, $a_1 = \frac{-1}{1-(-1)} = -\frac{1}{2}$, and $f(k) = a_0 + a_1(-1)^k$ gives $f(0) = 0, f(1) = 1$. Correct.

Worked Example 37.3.21 (Chinese remainder splitting of characters). Using $Z(6) \approx Z(2) \times Z(3)$, factor a character of $Z(6)$ into a product of characters of $Z(2)$ and $Z(3)$.

Recognition. By Problem 1 and the dual-of-product rule, $\widehat{Z(6)} \approx \widehat{Z(2)} \times \widehat{Z(3)}$.

Solution. The isomorphism $k \mapsto (k \bmod 2, k \bmod 3)$ identifies $j \in Z(6)$ with (a, b) , $a \in Z(2), b \in Z(3)$. A character of $Z(6)$ is $e_\ell(j) = e^{2\pi i \ell j / 6}$; writing ℓ via the same CRT data $(\ell \bmod 2, \ell \bmod 3) = (s, t)$, it factors

as $e_\ell(j) = (-1)^{sa} e^{2\pi i tb/3}$, a product of a $Z(2)$ -character and a $Z(3)$ -character. *Check.* There are $2 \cdot 3 = 6$ such products, matching $|\widehat{Z(6)}| = 6$, and multiplicativity holds coordinatewise.

$$\begin{array}{ccc} Z(6) & \xrightarrow{\sim} & Z(2) \times Z(3) \\ \downarrow \sim & & \downarrow \sim \\ \widehat{Z(6)} & \xrightarrow{\sim} & \widehat{Z(2)} \times \widehat{Z(3)} \end{array}$$

Figure 37.9. Duality respects the Chinese-remainder splitting: the dual of a product is the product of the duals, $\widehat{Z(2) \times Z(3)} \cong \widehat{Z(2)} \times \widehat{Z(3)}$, so a character of $Z(6)$ factors as a $Z(2)$ -character times a $Z(3)$ -character.

Worked Example 37.3.22 (Counting characters via the structure theorem). How many characters does $G = Z(4) \times Z(9)$ have, and what is $\widehat{G}$?

Recognition. $|\widehat{G}| = |G|$ and $\widehat{G_1 \times G_2} \approx \widehat{G_1} \times \widehat{G_2}$.

Solution. $|G| = 4 \cdot 9 = 36$, so there are 36 characters. By Problem 3(b), $\widehat{G} \approx \widehat{Z(4)} \times \widehat{Z(9)} \approx Z(4) \times Z(9) \approx G$ (self-duality). *Check.* Since $\gcd(4, 9) = 1$, Problem 1 also gives $G \approx Z(36)$, hence $\widehat{G} \approx Z(36)$, consistent with 36 characters. (Non-example: $Z(2) \times Z(2)$ has order 4 but is *not* $\approx Z(4)$ — it has no element of order 4 — though it is still self-dual.)

Exercises (Stein, Ch.7) — finite abelian groups and characters.

- (Characters of $Z(N)$) Let e be a character on $G = Z(N)$, the additive group of integers mod N . Show there is a unique $0 \leq \ell \leq N - 1$ with $e(k) = e_\ell(k) = e^{2\pi i \ell k/N}$ for all $k \in Z(N)$, and conversely every such function is a character. Deduce that $e_\ell \mapsto \ell$ is an isomorphism $\widehat{G} \approx G$. [Hint: show $e(1)$ is an N th root of unity.]
- (Characters of S^1) Show all characters on the circle S^1 are $e_n(x) = e^{2\pi i n x}$ with $n \in \mathbb{Z}$, and that $e_n \mapsto n$ is an isomorphism $\widehat{S^1} \approx \mathbb{Z}$. [Hint: a continuous F with $F(x+y) = F(x)F(y)$ is differentiable: if $F(0) \neq 0$, pick δ with $c = \int_0^\delta F(y) dy \neq 0$; then $cF(x) = \int_x^{\delta+x} F(y) dy$; differentiate to get $F(x) = e^{Ax}$.]
- (Characters of $\mathbb{R}$) Prove all characters on $\mathbb{R}$ take the form $e_\xi(x) = e^{2\pi i \xi x}$ with $\xi \in \mathbb{R}$, and that $e_\xi \mapsto \xi$ is an isomorphism $\widehat{\mathbb{R}} \approx \mathbb{R}$. (The differentiability argument of the previous exercise applies.)
- (Cyclic groups) A group G is *cyclic* if some $g \in G$ generates it, i.e. every element is g^n for some $n \in \mathbb{Z}$. Prove that a finite abelian group is cyclic if and only if it is isomorphic to $Z(N)$ for some N .
- (Unit-group tables) Write the multiplication tables of $\mathbb{Z}^*(3), \mathbb{Z}^*(4), \mathbb{Z}^*(5), \mathbb{Z}^*(6), \mathbb{Z}^*(8), \mathbb{Z}^*(9)$. Determine which are cyclic.
- (Multiplicative functions are characters or zero) Suppose G is finite abelian and $e: G \rightarrow \mathbb{C}$ satisfies $e(x \cdot y) = e(x)e(y)$ for all x, y . Prove that either $e \equiv 0$ or e never vanishes; in the second case, show that for each x , $e(x) = e^{2\pi i r}$ for some rational $r = p/q$ with $q = |G|$.
- (Convolution and the Dirac function on G) Let G be finite abelian with unit 1_G and V the functions on G . (a) Define $(f * g)(a) = \frac{1}{|G|} \sum_{b \in G} f(b)g(a \cdot b^{-1})$. Show $\widehat{f * g}(e) = \widehat{f}(e)\widehat{g}(e)$ for all $e \in \widehat{G}$. (b) Using Theorem 37.3.11, show that if e is a character on G then $\sum_{e \in \widehat{G}} e(c) = 0$ whenever $c \neq 1_G$. (c) Deduce that the Fourier series $Sf(a) = \sum_{e \in \widehat{G}} \widehat{f}(e)e(a)$ equals $f * D$, where $D(c) = \sum_{e \in \widehat{G}} e(c) = |G|$ if $c = 1_G$ and 0 otherwise. Since $f * D = f$ we recover $Sf = f$. Note D has unit mass $\frac{1}{|G|} \sum_c D(c) = 1$ concentrated at 1_G — a finite analogue of the Dirac delta / a limit of good kernels.

Problems (Stein, Ch.7).

- (Chinese remainder isomorphism) Prove that if n, m are relatively prime positive integers, then $Z(nm) \approx Z(n) \times Z(m)$. [Hint: consider $k \mapsto (k \bmod n, k \bmod m)$ and use $xn + ym = 1$ for suitable integers x, y .]

2. (* Structure theorem) Every finite abelian group G is isomorphic to a direct product of cyclic groups. Two precise forms: (i) if $p_1, \dots, p_s$ are the distinct primes dividing $|G|$, then $G \approx G(p_1) \times \cdots \times G(p_s)$ with each $G(p) = Z(p^{r_1}) \times \cdots \times Z(p^{r_\ell})$, $0 \leq r_1 \leq \cdots \leq r_\ell$, this decomposition being unique; (ii) there are unique integers $d_1 \mid d_2 \mid \cdots \mid d_k$ with $G \approx Z(d_1) \times \cdots \times Z(d_k)$. Deduce (ii) from (i).
3. (Self-duality of finite abelian groups) Let $\widehat{G}$ be the group of distinct characters of the finite abelian group G . (a) Note that if $G = Z(N)$ then $\widehat{G} \approx G$. (b) Prove $\widehat{G_1 \times G_2} \approx \widehat{G_1} \times \widehat{G_2}$. (c) Using Problem 2, conclude that for any finite abelian G , $\widehat{\widehat{G}} \approx G$.
4. (* Cyclicity of $\mathbb{Z}^*(p)$) When p is prime the group $\mathbb{Z}^*(p)$ is cyclic, and $\mathbb{Z}^*(p) \approx Z(p-1)$.

Additional exercises.

1. Verify directly from the axioms that in any abelian group the identity element and the inverse of each element are unique.
2. List all characters of $Z(6)$ explicitly as $e_\ell(k) = e^{2\pi i \ell k/6}$, $\ell = 0, \dots, 5$, and check the orthogonality relations $(e_m, e_\ell) = \delta_{m\ell}$ numerically for two distinct pairs.
3. Using the Chinese remainder isomorphism $Z(6) \approx Z(2) \times Z(3)$, write each character of $Z(6)$ as a product of a character of $Z(2)$ and a character of $Z(3)$.
4. Show that $\mathbb{Z}^*(9) = \{1, 2, 4, 5, 7, 8\}$ is cyclic of order 6 by exhibiting a generator, and conclude $\mathbb{Z}^*(9) \approx Z(6)$.
5. (Orthogonality of columns) Prove the dual relation $\sum_{e \in \widehat{G}} e(a)\overline{e(b)} = |G| \delta_{ab}$, and explain how it expresses that the character table of G is a (rescaled) unitary matrix.
6. Compute the Fourier coefficients on $Z(2) \times Z(2)$ of the indicator function $f = \mathbf{1}_{\{(0,0)\}}$ and verify Plancherel.
7. (Non-example) Show that the function $e(k) = 2^k$ on $Z(N)$ (values reduced mod N as integers, then viewed in $\mathbb{C}$) is *not* a character for $N \geq 2$, identifying which defining property fails.
8. Prove that the dual of a direct product satisfies $\widehat{G_1 \times G_2} \approx \widehat{G_1} \times \widehat{G_2}$ by sending a pair of characters (e_1, e_2) to $(g_1, g_2) \mapsto e_1(g_1)e_2(g_2)$.
9. (Diagram) Draw the character table of $Z(4)$ as a 4×4 grid of points on the unit circle (entry (ℓ, k) at $\zeta^{\ell k}$), and use the picture to read off the orthogonality of rows.
10. (Equidistribution flavor) For irrational α , the points $\{n\alpha\}$ on the circle are the continuous analogue of $Z(N)$'s uniform partition. State and prove the finite Weyl criterion: $\frac{1}{N} \sum_{k=0}^{N-1} e^{2\pi i \ell k/N} = 0$ for $\ell \neq 0$, and explain how it certifies that $Z(N)$ samples each frequency evenly.

Chapter 38

Dirichlet's Theorem

38.1 Characters of Finite Abelian Groups

The group of units and the Euler phi-function

Dirichlet's theorem on primes in arithmetic progressions is, on its surface, a statement in number theory; its proof, remarkably, is an application of Fourier analysis on a *finite abelian group*. The relevant group is the multiplicative group of residues mod q . We assemble here the group theory and character theory that the analysis will need, then specialize to the *Dirichlet characters*.

Definition 38.1.1. Fix an integer $q \geq 1$. The *group of units modulo q* is

$$\mathbb{Z}^*(q) = \{ \bar{n} \in \mathbb{Z}/q\mathbb{Z} : \gcd(n, q) = 1 \},$$

the residue classes represented by integers relatively prime to q , with multiplication $\bar{a} \cdot \bar{b} = \overline{ab}$ as the group operation. Its order is the *Euler phi-function*

$$\varphi(q) = \#\{0 \leq n < q : \gcd(n, q) = 1\} = |\mathbb{Z}^*(q)|.$$

That $\mathbb{Z}^*(q)$ is a group is exactly the statement that a residue $\bar{a}$ has a multiplicative inverse mod q iff $\gcd(a, q) = 1$: by Bezout there are integers x, y with $ax + qy = 1$, so $\bar{a}\bar{x} = \bar{1}$. Closure holds because $\gcd(a, q) = \gcd(b, q) = 1$ forces $\gcd(ab, q) = 1$. It is abelian because integer multiplication commutes.

Theorem 38.1.2 (Multiplicativity and formula for φ). *The Euler phi-function satisfies:*

1. $\varphi(p) = p - 1$ for p prime, and $\varphi(p^k) = p^k - p^{k-1} = p^k(1 - \frac{1}{p})$;
2. if $\gcd(m, n) = 1$ then $\varphi(mn) = \varphi(m)\varphi(n)$ (multiplicativity);
3. consequently, if $n = p_1^{a_1} \cdots p_r^{a_r}$ then $\varphi(n) = n \prod_{p|n} (1 - \frac{1}{p})$.

Proof. (1) Every $1 \leq n \leq p - 1$ is coprime to p , giving $\varphi(p) = p - 1$. For p^k , the integers in $[1, p^k]$ not coprime to p^k are the multiples of p , namely $p, 2p, \dots, p^{k-1} \cdot p$, of which there are p^{k-1} ; subtract. (2) The Chinese Remainder Theorem gives a ring isomorphism $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$, and under it the units correspond: $\mathbb{Z}^*(mn) \cong \mathbb{Z}^*(m) \times \mathbb{Z}^*(n)$ (a residue is a unit mod mn iff it is a unit mod each factor). Counting both sides gives $\varphi(mn) = \varphi(m)\varphi(n)$. (3) Apply (2) across the prime-power factors and then (1): $\varphi(n) = \prod_i p_i^{a_i} (1 - \frac{1}{p_i}) = n \prod_{p|n} (1 - \frac{1}{p})$. $\square$

$$\begin{array}{ccc} \mathbb{Z}^*(mn) & \hookrightarrow & \mathbb{Z}/mn\mathbb{Z} \\ \cong \downarrow & & \text{CRT} \downarrow \cong \\ \mathbb{Z}^*(m) \times \mathbb{Z}^*(n) & \hookrightarrow & \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \end{array}$$

Figure 38.1. For $\gcd(m, n) = 1$ the Chinese Remainder ring isomorphism (right) restricts to the units (left): a class is a unit mod mn iff it is a unit modulo each factor, giving $\mathbb{Z}^*(mn) \cong \mathbb{Z}^*(m) \times \mathbb{Z}^*(n)$ and hence $\varphi(mn) = \varphi(m)\varphi(n)$.

Worked Example 38.1.3 (Computing $\varphi(q)$ several ways). Compute $\varphi(12)$, $\varphi(15)$, and $\varphi(36)$, and list $\mathbb{Z}^*(12)$ explicitly.

Recognition. Trigger: evaluate the phi-function; use the prime-power product formula (Theorem 38.1.2) and cross-check $\varphi(12)$ by listing.

Solution. $12 = 2^2 \cdot 3$, so $\varphi(12) = 12(1 - \frac{1}{2})(1 - \frac{1}{3}) = 12 \cdot \frac{1}{2} \cdot \frac{2}{3} = 4$. $15 = 3 \cdot 5$, so $\varphi(15) = 15 \cdot \frac{2}{3} \cdot \frac{4}{5} = 8$. $36 = 2^2 \cdot 3^2$, so $\varphi(36) = 36 \cdot \frac{1}{2} \cdot \frac{2}{3} = 12$. Listing the residues coprime to 12: $\{1, 5, 7, 11\}$, indeed 4 elements.

Check. Multiplicativity gives $\varphi(12) = \varphi(4)\varphi(3) = 2 \cdot 2 = 4$ and $\varphi(36) = \varphi(4)\varphi(9) = 2 \cdot 6 = 12$, agreeing.

Worked Example 38.1.4 ($\mathbb{Z}^*(8)$ is not cyclic). Show $\mathbb{Z}^*(8) = \{1, 3, 5, 7\}$ has every non-identity element of order 2, hence is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, not to $\mathbb{Z}/4\mathbb{Z}$.

Recognition. Trigger: identify the group structure of a small unit group by computing orders.

Solution. Mod 8: $3^2 = 9 \equiv 1$, $5^2 = 25 \equiv 1$, $7^2 = 49 \equiv 1$. So 3, 5, 7 each have order 2 and 1 has order 1; no element has order 4, so $\mathbb{Z}^*(8)$ is not cyclic. A group of order 4 with exponent 2 is the Klein four-group $\mathbb{Z}/2 \times \mathbb{Z}/2$. *Check.* The product of any two distinct non-identity elements is the third, e.g. $3 \cdot 5 = 15 \equiv 7$, the multiplication table of $\mathbb{Z}/2 \times \mathbb{Z}/2$.

Characters and the dual group

A finite abelian group G is studied through its *characters*: homomorphisms into the unit circle. These are the “pure frequencies” for Fourier analysis on G .

Definition 38.1.5. Let G be a finite abelian group (written multiplicatively, with identity 1). A *character* of G is a homomorphism $e : G \rightarrow \mathbb{C}^\times$ into the nonzero complex numbers, i.e. $e(ab) = e(a)e(b)$ for all $a, b \in G$. The set of all characters is the *dual group* $\widehat{G}$, with pointwise multiplication $(e_1 e_2)(a) = e_1(a)e_2(a)$; the *trivial character* $e_0 \equiv 1$ is its identity.

Lemma 38.1.6 (Characters take values in roots of unity). *If G has order N and $e \in \widehat{G}$, then $e(a)^N = e(a^N) = e(1) = 1$ for every a (since $a^N = 1$ by Lagrange). Hence $|e(a)| = 1$: every character maps G into the unit circle, and in fact into the N th roots of unity. Moreover $e(a^{-1}) = e(a)^{-1} = \overline{e(a)}$.*

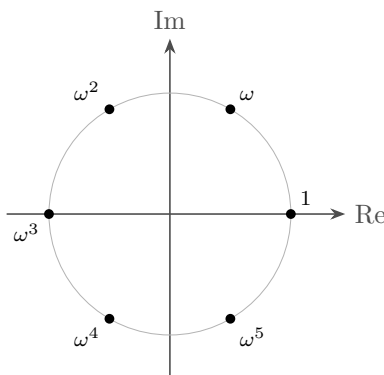


Figure 38.2. The values of a character of a cyclic group of order 6 are the sixth roots of unity ω^j , $\omega = e^{2\pi i/6}$ – equally spaced points on the unit circle.

Theorem 38.1.7 (Structure of the dual group). *Let G be a finite abelian group of order N .*

- If $G = \mathbb{Z}/N\mathbb{Z}$ is cyclic with generator g , then the characters are exactly $e_k(g^m) = e^{2\pi i k m / N}$ for $k = 0, 1, \dots, N-1$, and $\widehat{G} \cong \mathbb{Z}/N\mathbb{Z}$.*
- In general (using $G \cong \mathbb{Z}/d_1\mathbb{Z} \times \dots \times \mathbb{Z}/d_r\mathbb{Z}$ by the structure theorem), $\widehat{G} \cong G$; in particular $|\widehat{G}| = |G| = N$. The characters separate points: if $a \neq b$ there is e with $e(a) \neq e(b)$.*

Proof. (1) A homomorphism $e : \mathbb{Z}/N\mathbb{Z} \rightarrow \mathbb{C}^\times$ is determined by $e(g)$, which must satisfy $e(g)^N = e(g^N) = e(1) = 1$, so $e(g)$ is an N th root of unity $e^{2\pi i k/N}$; each choice $k \in \{0, \dots, N-1\}$ gives a valid homomorphism, and distinct k give distinct characters. The map $e_k \mapsto k$ is an isomorphism $\widehat{G} \cong \mathbb{Z}/N\mathbb{Z}$. (2) Write G as a product of cyclic groups; a character of a product is a product of characters of the factors, so $\widehat{G} \cong \widehat{\mathbb{Z}/d_1\mathbb{Z}} \times \dots \cong \mathbb{Z}/d_1\mathbb{Z} \times \dots \cong G$. Point separation: if $a \neq 1$, write a in coordinates with some nontrivial entry, and use the corresponding coordinate character; for general $a \neq b$ apply this to ab^{-1} . $\square$

$$G \xrightarrow[\sim]{\text{ev}} \widehat{\widehat{G}} \xleftarrow[\text{(noncanonical)}]{\cong} \widehat{G}$$

Figure 38.3. Point separation makes the evaluation map $\text{ev} : G \rightarrow \widehat{\widehat{G}}, a \mapsto (e \mapsto e(a))$, an isomorphism, so $G \cong \widehat{\widehat{G}}$ canonically; the isomorphism $\widehat{G} \cong G$ exists but depends on a choice of generators (noncanonical).

Worked Example 38.1.8 (The four characters of $\mathbb{Z}^*(8)$). Write out the character table of $\mathbb{Z}^*(8) = \{1, 3, 5, 7\} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$.

Recognition. Trigger: list $\widehat{G}$ for a Klein four-group. Each character sends each generator to ± 1 ; there are $2 \cdot 2 = 4$ choices.

Solution. Take generators 3 and 5 (with $7 = 3 \cdot 5$). A character is fixed by its values $e(3), e(5) \in \{\pm 1\}$, and then $e(7) = e(3)e(5)$. The table (rows = characters):

	1	3	5	7
e_0	1	1	1	1
e_1	1	-1	1	-1
e_2	1	1	-1	-1
e_3	1	-1	-1	1

Check. Every row is multiplicative, e.g. for e_3 : $e_3(3)e_3(5) = (-1)(-1) = 1 = e_3(7)$, and $e_3(3)e_3(3) = 1 = e_3(9) = e_3(1)$. All four characters are real here, as expected for a group of exponent 2.

Orthogonality relations

The two orthogonality relations are the engine of finite Fourier analysis: they say the characters form an orthonormal basis (up to a factor $|G|$) of the functions on G .

Theorem 38.1.9 (Orthogonality of characters). *Let G be a finite abelian group of order N .*

1. (Sum over the group) For a character e ,

$$\sum_{a \in G} e(a) = \begin{cases} N & \text{if } e = e_0, \\ 0 & \text{if } e \neq e_0. \end{cases}$$

2. (Sum over the dual) For $a \in G$,

$$\sum_{e \in \widehat{G}} e(a) = \begin{cases} N & \text{if } a = 1, \\ 0 & \text{if } a \neq 1. \end{cases}$$

3. (Orthonormality) For characters e, e' and elements a, b ,

$$\frac{1}{N} \sum_{a \in G} e(a) \overline{e'(a)} = \begin{cases} 1 & e = e', \\ 0 & e \neq e', \end{cases} \quad \frac{1}{N} \sum_{e \in \widehat{G}} e(a) \overline{e(b)} = \begin{cases} 1 & a = b, \\ 0 & a \neq b. \end{cases}$$

Proof. (1) If $e = e_0$ the sum is $\sum_a 1 = N$. If $e \neq e_0$, pick b with $e(b) \neq 1$. As a runs over G so does ab , hence $S := \sum_a e(a) = \sum_a e(ab) = e(b) \sum_a e(a) = e(b)S$. Thus $(1 - e(b))S = 0$ and $e(b) \neq 1$ force $S = 0$. (2) Identical, with the roles of G and $\widehat{G}$ swapped (using $|\widehat{G}| = N$ and the fact that some character is nontrivial at $a \neq 1$ by point separation). (3) Apply (1) to the character $e\overline{e'} = e(e')^{-1}$, which is trivial iff $e = e'$; the second identity is (2) applied to ab^{-1} . $\square$

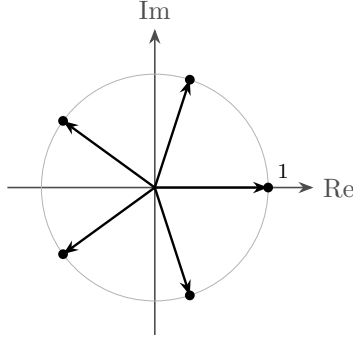


Figure 38.4. Why a nontrivial character sums to zero: its values are the fifth roots of unity (here e_1 on $\mathbb{Z}/5$), vectors equally spaced around the circle whose sum is the center of mass 0.

Worked Example 38.1.10 (Verifying orthogonality on $\mathbb{Z}/4\mathbb{Z}$). For $G = \mathbb{Z}/4\mathbb{Z}$ with characters $e_k(m) = i^{km}$, verify $\sum_m e_1(m) = 0$ and $\frac{1}{4} \sum_m e_1(m) \overline{e_3(m)} = 0$.

Recognition. Trigger: numerically confirm Theorem 38.1.9(1),(3). Compute the four values and add.

Solution. $e_1(m) = i^m$ for $m = 0, 1, 2, 3$ gives $1, i, -1, -i$, summing to 0. For the second, $e_3(m) = i^{3m} = (-i)^m$, so $\overline{e_3(m)} = i^m = e_1(m)$, and $\frac{1}{4} \sum_m e_1(m) \overline{e_3(m)} = \frac{1}{4} \sum_m e_1(m) e_1(m) = \frac{1}{4} \sum_m i^{2m} = \frac{1}{4}(1 - 1 + 1 - 1) = 0$.

Check. $e_1 \neq e_3$, so orthonormality predicts 0; and $\frac{1}{4} \sum_m e_1(m) \overline{e_1(m)} = \frac{1}{4} \sum_m 1 = 1$, the diagonal case, also as predicted.

Fourier expansion on a finite abelian group

Theorem 38.1.11 (Finite Fourier inversion). *Let G be a finite abelian group of order N and $f : G \rightarrow \mathbb{C}$ any function. Define the Fourier coefficients*

$$\widehat{f}(e) = \frac{1}{N} \sum_{a \in G} f(a) \overline{e(a)}, \quad e \in \widehat{G}.$$

Then f is recovered by its Fourier series

$$f(a) = \sum_{e \in \widehat{G}} \widehat{f}(e) e(a),$$

and Parseval's identity holds: $\frac{1}{N} \sum_{a \in G} |f(a)|^2 = \sum_{e \in \widehat{G}} |\widehat{f}(e)|^2$.

Proof. The functions $\{e\}_{e \in \widehat{G}}$ are orthonormal for the inner product $\langle f, g \rangle = \frac{1}{N} \sum_a f(a) \overline{g(a)}$ by Theorem 38.1.9(3), and there are $N = \dim \mathbb{C}^G$ of them, so they form an orthonormal basis. Then $\widehat{f}(e) = \langle f, e \rangle$ are the coordinates and $f = \sum_e \langle f, e \rangle e$ is the basis expansion; Parseval is the Pythagorean theorem in this basis. $\square$

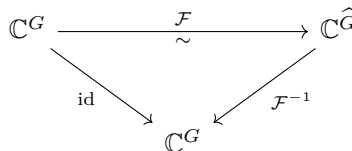


Figure 38.5. The finite Fourier transform $\mathcal{F} : f \mapsto \widehat{f}$ is a linear isomorphism $\mathbb{C}^G \rightarrow \mathbb{C}^{\widehat{G}}$ whose inverse is the Fourier-series map $\mathcal{F}^{-1} : (\widehat{f}(e))_e \mapsto (a \mapsto \sum_e \widehat{f}(e) e(a))$; inversion is the statement $\mathcal{F}^{-1} \circ \mathcal{F} = \text{id}$.

Worked Example 38.1.12 (Fourier-expanding the delta function δ_ℓ). On $G = \mathbb{Z}^*(q)$, let $\delta_\ell(n) = 1$ if $n \equiv \ell$ and 0 otherwise (for $n \in \mathbb{Z}^*(q)$). Compute its Fourier coefficients and recover the expansion that drives Dirichlet's proof.

Recognition. Trigger: expand a characteristic function via Theorem 38.1.11; the only nonzero term in the defining sum is at $n = \ell$.

Solution. $\widehat{\delta}_\ell(e) = \frac{1}{|G|} \sum_{n \in G} \delta_\ell(n) \overline{e(n)} = \frac{1}{|G|} \overline{e(\ell)} = \frac{1}{\varphi(q)} e(\ell^{-1})$. Hence

$$\delta_\ell(n) = \sum_{e \in \widehat{G}} \widehat{\delta}_\ell(e) e(n) = \frac{1}{\varphi(q)} \sum_{e \in \widehat{G}} \overline{e(\ell)} e(n).$$

Check. Setting $n = \ell$ and using orthogonality (2), $\frac{1}{\varphi(q)} \sum_e \overline{e(\ell)} e(\ell) = \frac{1}{\varphi(q)} \sum_e 1 = 1$; for $n \neq \ell$ the sum $\sum_e e(n\ell^{-1}) = 0$. This is exactly the device used to single out the residue class ℓ in the proof of Dirichlet's theorem (Section 38.3).

Dirichlet characters

We now glue the abstract characters of $\mathbb{Z}^*(q)$ to functions on all of $\mathbb{Z}$, obtaining the *Dirichlet characters*, which are completely multiplicative arithmetic functions.

Definition 38.1.13. Let $e \in \widehat{\mathbb{Z}^*(q)}$. The *Dirichlet character* mod q associated to e is the function $\chi : \mathbb{Z} \rightarrow \mathbb{C}$,

$$\chi(m) = \begin{cases} e(\bar{m}) & \text{if } \gcd(m, q) = 1, \\ 0 & \text{if } \gcd(m, q) > 1. \end{cases}$$

The *trivial* (or *principal*) Dirichlet character χ_0 comes from e_0 : $\chi_0(m) = 1$ if $\gcd(m, q) = 1$ and 0 otherwise. A character is *real* if it takes only the values $0, \pm 1$ (equivalently $\chi = \overline{\chi}$), and *complex* otherwise; it is *even* if $\chi(-1) = 1$ and *odd* if $\chi(-1) = -1$.

$$\begin{array}{ccc} \{m : \gcd(m, q) = 1\} & \xrightarrow{\text{mod } q} & \mathbb{Z}^*(q) \\ & \searrow \chi & \downarrow e \\ & & \mathbb{C}^\times \end{array}$$

Figure 38.6. The Dirichlet character χ as the composite of the reduction $m \mapsto \bar{m}$ (surjective onto $\mathbb{Z}^*(q)$) with the group character $e \in \widehat{\mathbb{Z}^*(q)}$, on integers coprime to q ; χ is then extended to all of $\mathbb{Z}$ by setting $\chi(m) = 0$ when $\gcd(m, q) > 1$.

Proposition 38.1.14 (Properties of Dirichlet characters). *For Dirichlet characters mod q :*

1. (Complete multiplicativity) $\chi(nm) = \chi(n)\chi(m)$ for all $n, m \in \mathbb{Z}$.
2. (Periodicity) $\chi(n + q) = \chi(n)$.
3. There are exactly $\varphi(q)$ Dirichlet characters mod q .
4. (Orthogonality on $\mathbb{Z}$) $\frac{1}{\varphi(q)} \sum_{\chi} \overline{\chi(\ell)} \chi(n) = \delta_\ell(n)$, where the sum is over all Dirichlet characters mod q and ℓ is coprime to q ; and for a fixed character $\sum_{n=1}^q \chi(n) = 0$ unless $\chi = \chi_0$.

Proof. (1) If either of n, m shares a factor with q , both sides are 0; otherwise it is the multiplicativity of e on $\mathbb{Z}^*(q)$. (2) Both $\gcd(n + q, q) = \gcd(n, q)$ and the value $e(\overline{n + q}) = e(\bar{n})$ depend only on $n \bmod q$. (3) These are in bijection with $\widehat{\mathbb{Z}^*(q)}$, of order $\varphi(q)$ by Theorem 38.1.7. (4) Restate Theorem 38.1.9(2) and (1) through the dictionary of Definition 38.1.13, where δ_ℓ extends by 0 off $\mathbb{Z}^*(q)$. $\square$

Worked Example 38.1.15 (The nontrivial character mod 4). Exhibit the unique nontrivial Dirichlet character $\chi \bmod 4$ and decide whether it is real/complex and even/odd.

Recognition. Trigger: build a Dirichlet character from $\widehat{\mathbb{Z}^*(4)}$. Here $\mathbb{Z}^*(4) = \{1, 3\} \cong \mathbb{Z}/2$, so there are $\varphi(4) = 2$ characters.

Solution. The nontrivial character has $\chi(1) = 1$, $\chi(3) = -1$, extended by periodicity and zero on evens:

$$\chi(n) = \begin{cases} 0 & n \text{ even,} \\ 1 & n \equiv 1 \pmod{4}, \\ -1 & n \equiv 3 \pmod{4}. \end{cases}$$

It takes only values $0, \pm 1$, so it is *real*. And $\chi(-1) = \chi(3) = -1$, so it is *odd*. *Check.* Complete multiplicativity, e.g. $\chi(3)\chi(3) = (-1)^2 = 1 = \chi(9) = \chi(1)$, and $\chi(3)\chi(5) = (-1)(1) = -1 = \chi(15) = \chi(3)$ since $15 \equiv 3$.

Worked Example 38.1.16 (A complex character mod 5). $\mathbb{Z}^*(5) = \{1, 2, 3, 4\}$ is cyclic generated by 2 (orders: $2^1=2$, $2^2=4$, $2^3=3$, $2^4=1$). Build a Dirichlet character of order 4 and confirm it is complex.

Recognition. Trigger: produce a non-real character; send the generator 2 to a primitive 4th root of unity i .

Solution. Define $\chi(2^k) = i^k$. Then $\chi(2) = i$, $\chi(4) = \chi(2^2) = i^2 = -1$, $\chi(3) = \chi(2^3) = i^3 = -i$, $\chi(1) = 1$, and $\chi(n) = 0$ for $5 \nmid n$. Since $\chi(2) = i \notin \mathbb{R}$, χ is *complex*. *Check.* Multiplicativity, $\chi(2)\chi(3) = i \cdot (-i) = 1 = \chi(6) = \chi(1)$ as $6 \equiv 1$; and $\bar{\chi}$ is the distinct character $2^k \mapsto (-i)^k$, confirming $\chi \neq \bar{\chi}$. Also $\chi(-1) = \chi(4) = -1$, so χ is *odd*.

Caution. A Dirichlet character mod q is *not* a homomorphism on all of $(\mathbb{Z}, \cdot)$ in the naive sense – it is 0 on integers sharing a factor with q , and 0 is not in $\mathbb{C}^\times$. The clean statement is *complete multiplicativity* $\chi(nm) = \chi(n)\chi(m)$ on all of $\mathbb{Z}$ (including when one side vanishes), which is the property the Euler product will exploit. Do not confuse a Dirichlet character (defined on $\mathbb{Z}$, multiplicative, periodic) with a character e of the finite group $\mathbb{Z}^*(q)$ (defined only on units).

Exercises (Stein, Ch.8).

1. Prove that there are infinitely many primes by observing that if there were only finitely many $p_1, \dots, p_N$, then $\prod_{j=1}^N \frac{1}{1-1/p_j} \geq \sum_{n=1}^{\infty} \frac{1}{n}$, a contradiction.
2. In the text it was shown that there are infinitely many primes of the form $4k+3$ by a modification of Euclid's argument. Adapt the technique to prove the analogous result for primes of the form $3k+2$, and for those of the form $6k+5$.
3. Prove that if p and q are relatively prime, then $\mathbb{Z}^*(p) \times \mathbb{Z}^*(q)$ is isomorphic to $\mathbb{Z}^*(pq)$.
4. Let $\varphi(n)$ be the number of positive integers $\leq n$ relatively prime to n . Using the previous exercise, show $\varphi(nm) = \varphi(n)\varphi(m)$ when $\gcd(n, m) = 1$. Then derive the formula for φ : (a) compute $\varphi(p)$ for p prime by counting $\mathbb{Z}^*(p)$; (b) give a formula for $\varphi(p^k)$ (p prime, $k \geq 1$) by counting $\mathbb{Z}^*(p^k)$; (c) show $\varphi(n) = n \prod_{p_i|n} (1 - \frac{1}{p_i})$.
5. If n is a positive integer, show $n = \sum_{d|n} \varphi(d)$. [Hint: there are exactly $\varphi(n/d)$ integers $1 \leq m \leq n$ with $\gcd(m, n) = d$.]
6. Write down the characters of the groups $\mathbb{Z}^*(3)$, $\mathbb{Z}^*(4)$, $\mathbb{Z}^*(5)$, $\mathbb{Z}^*(6)$, and $\mathbb{Z}^*(8)$. (a) Which ones are real, which complex? (b) Which are even, which odd? (A character is even if $\chi(-1) = 1$ and odd otherwise.)

Additional exercises.

1. Compute $\varphi(n)$ for $n = 10, 16, 20, 24, 30$, and in each case list $\mathbb{Z}^*(n)$ explicitly. Which of these unit groups is cyclic?
2. Determine the structure of $\mathbb{Z}^*(15)$ as a product of cyclic groups, and write down all four of its characters as a table of values on $\{1, 2, 4, 7, 8, 11, 13, 14\}$.
3. Show directly from the definition that the dual group operation is associative and that $\widehat{\widehat{G}}$ is abelian. Then prove $\widehat{G_1 \times G_2} \cong \widehat{G_1} \times \widehat{G_2}$.
4. Prove the second orthogonality relation $\sum_{e \in \widehat{G}} e(a) = 0$ for $a \neq 1$ *directly*, by choosing a character e_1 with $e_1(a) \neq 1$ and re-indexing the sum $e \mapsto e_1 e$.

5. For $G = \mathbb{Z}/6\mathbb{Z}$, let $f(m) = \cos(2\pi m/6)$. Compute all six Fourier coefficients $\widehat{f}(e_k)$ and verify the inversion formula at $m = 1$.
6. (Non-example) Give a function $\chi : \mathbb{Z} \rightarrow \{0, 1, -1\}$ that is periodic mod 5, multiplicative on units, satisfies $|\chi| \leq 1$, yet is *not* a Dirichlet character mod 5, and explain which axiom fails. [Hint: try $\chi(2) = \chi(3) = 1$, $\chi(4) = -1$.]
7. (Diagram) Plot the values of the order-4 character of $\mathbb{Z}^*(5)$ as the four points $1, i, -1, -i$ on the unit circle, drawing the radius to each. Label which group element maps to each root of unity, and indicate geometrically why the four values sum to zero.
8. Show that a Dirichlet character $\chi \bmod q$ is real if and only if $\chi^2 = \chi_0$, i.e. χ has order 1 or 2 in the dual group. Deduce that the number of real characters mod q is a power of 2.

38.2 Dirichlet L-Functions and Non-Vanishing at $s=1$

The fundamental theorem of arithmetic

The bridge between primes and analysis is the *Euler product*, an analytic re-encoding of unique factorization. We begin with the number theory it rests on.

Theorem 38.2.1 (Division algorithm). *For integers a and $b > 0$ there exist unique integers q (quotient) and r (remainder) with $a = qb + r$ and $0 \leq r < b$.*

Proof. Existence. Let $S = \{a - qb \geq 0 : q \in \mathbb{Z}\}$, a nonempty set of nonnegative integers (it contains large values since $b \neq 0$); let $r = a - qb$ be its least element. Then $r \geq 0$, and $r < b$: otherwise $r = b + s$ with $0 \leq s < r$ gives $s = a - (q+1)b \in S$ with $s < r$, contradicting minimality. *Uniqueness.* If also $a = q_1b + r_1$ with $0 \leq r_1 < b$, then $(q - q_1)b = r_1 - r$; the left side has absolute value 0 or $\geq b$, the right side $< b$, so both vanish: $q = q_1$, $r = r_1$. $\square$

Theorem 38.2.2 (Fundamental theorem of arithmetic). *Every integer $n > 1$ can be written as a product of primes, and the factorization is unique up to the order of the factors. Two standard consequences (Euclid's lemma) drive the uniqueness: if $\gcd(a, b) = 1$ then $a \mid bc \implies a \mid c$; in particular if a prime p divides ab then $p \mid a$ or $p \mid b$.*

Theorem 38.2.3 (Euclid: infinitely many primes). *There are infinitely many primes.*

Proof. Suppose $p_1, \dots, p_n$ were all the primes and set $N = p_1 p_2 \cdots p_n + 1$. Then $N > p_i$ for all i , so N is divisible by some prime on the list; but every p_i divides the product $p_1 \cdots p_n$ and no prime divides 1, so no p_i divides N – a contradiction. $\square$

Worked Example 38.2.4 (Euclid's twist: infinitely many primes $\equiv 3 \pmod{4}$). Prove there are infinitely many primes of the form $4k + 3$.

Recognition. Trigger: a Euclid-style argument refined to a residue class. The product of numbers $\equiv 1 \pmod{4}$ stays $\equiv 1 \pmod{4}$, so a number $\equiv 3 \pmod{4}$ must have a prime factor $\equiv 3 \pmod{4}$.

Solution. Suppose $7, 11, \dots, p_n$ are all the primes $\equiv 3 \pmod{4}$ (omit 3), and set $N = 4p_1 \cdots p_n + 3$. Then $N \equiv 3 \pmod{4}$, and $N > p_n$ is not prime. A product of numbers $\equiv 1 \pmod{4}$ is $\equiv 1 \pmod{4}$, so N (being $\equiv 3$) has a prime factor $p \equiv 3 \pmod{4}$. But $p \neq 3$ ($3 \nmid N$ since $3 \nmid 4p_1 \cdots p_n$) and $p \neq p_i$ (else $p \mid 3$). This new prime contradicts the list.

Check. The parallel with $4k + 1$ fails because a product of numbers $\equiv 3 \pmod{4}$ need not be $\equiv 3 \pmod{4}$ – which is precisely why Dirichlet needed L-functions for that class.

Infinite products and the zeta function

Lemma 38.2.5 (Logarithm estimates). *For real x : (i) $e^{\log x} = x$; (ii) $\log(1+x) = x + E(x)$ with $|E(x)| \leq x^2$ for $|x| < \frac{1}{2}$, recorded as $\log(1+x) = x + O(x^2)$; (iii) if $\log(1+x) = y$ and $|x| < \frac{1}{2}$ then $|y| \leq 2|x|$.*

Proof. (i) is standard. (ii) From $\log(1+x) = \sum_{n \geq 1} (-1)^{n+1} x^n / n$ we get $E(x) = -\frac{x^2}{2} + \frac{x^3}{3} - \cdots$, so $|E(x)| \leq \frac{x^2}{2}(1 + |x| + |x|^2 + \cdots) \leq \frac{x^2}{2} \cdot \frac{1}{1-1/2} = x^2$ for $|x| \leq \frac{1}{2}$. (iii) $|\log(1+x)/x| \leq 1 + |E(x)/x| \leq 1 + |x| \leq 2$. $\square$

Proposition 38.2.6 (Convergence of infinite products). *If $A_n = 1 + a_n$ and $\sum |a_n|$ converges, then $\prod_n A_n$ converges, and the product vanishes iff some factor $A_n = 0$. If moreover $a_n \neq 1$ for all n , then $\prod_n 1/(1 - a_n)$ converges.*

Proof. Discarding finitely many terms, assume $|a_n| < \frac{1}{2}$ for all n . Write $\prod_{n=1}^N A_n = \prod_{n=1}^N e^{\log(1+a_n)} = e^{B_N}$ with $B_N = \sum_{n=1}^N b_n$, $b_n = \log(1 + a_n)$. By the lemma $|b_n| \leq 2|a_n|$, so $B_N \rightarrow B$ for some real B and, by continuity of $\exp$, $e^{B_N} \rightarrow e^B \neq 0$. The vanishing statement: a finite factor $A_n = 0$ kills the product; otherwise the limit is $e^B \neq 0$. The last claim applies the same argument to $-a_n$. $\square$

Definition 38.2.7. For real $s > 1$ the *Riemann zeta function* is $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$.

Comparing the sum with $\int_1^{\infty} x^{-s} dx = \frac{1}{s-1}$ (a decreasing integrand) gives $\zeta(s) \leq 1 + \frac{1}{s-1}$, so the series converges, uniformly on each half-line $s \geq s_0 > 1$, and ζ is continuous there; also $\zeta(s) \rightarrow \infty$ as $s \rightarrow 1^+$, and quantitatively $\zeta(s) = \frac{1}{s-1} + O(1)$.

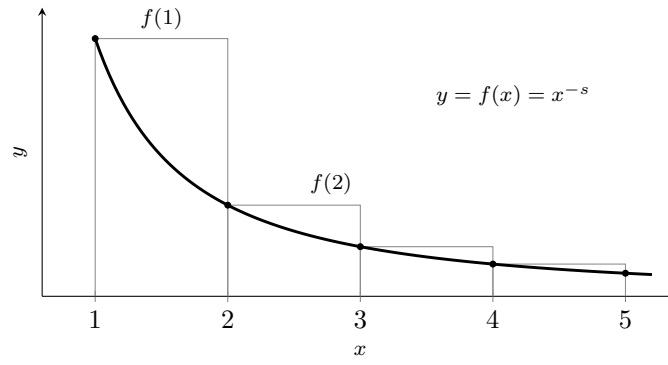


Figure 38.7. Comparing the series $\sum f(n)$ with $\int f$: on $[n, n+1]$ the block of height $f(n)$ dominates the area under the decreasing curve, so $f(n) \geq \int_n^{n+1} f$ and $\sum_{n \geq 1} n^{-s} \leq 1 + \int_1^{\infty} x^{-s} dx = 1 + \frac{1}{s-1}$. (Schematic: the curve is drawn for a representative exponent.)

Theorem 38.2.8 (Euler product for ζ). *For $s > 1$, $\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}$, the product over all primes.*

Proof. This is unique factorization in analytic dress. Each factor expands as a geometric series $\frac{1}{1-p^{-s}} = \sum_{k \geq 0} p^{-ks}$. For a finite cutoff N , every $n \leq N$ factors into primes $\leq N$ with bounded exponents, so $\sum_{n=1}^N n^{-s} \leq \prod_{p \leq N} (1 + p^{-s} + p^{-2s} + \dots) = \prod_{p \leq N} \frac{1}{1-p^{-s}} \leq \prod_p \frac{1}{1-p^{-s}}$. Letting $N \rightarrow \infty$ gives $\zeta(s) \leq \prod_p (\dots)$. Conversely $\prod_{p \leq N} \frac{1}{1-p^{-s}} = \sum' n^{-s} \leq \zeta(s)$ where $\sum'$ runs over integers built from primes $\leq N$; letting $N \rightarrow \infty$ gives the reverse inequality. (Convergence of the product is Proposition 38.2.6 with $a_p = -p^{-s}$.) $\square$

Theorem 38.2.9 (Euler: $\sum_p 1/p$ diverges). *The sum $\sum_p 1/p$ over primes diverges. More precisely $\sum_p p^{-s} = \log \frac{1}{s-1} + O(1)$ as $s \rightarrow 1^+$.*

Proof. Take logarithms in the Euler product: $-\sum_p \log(1-p^{-s}) = \log \zeta(s)$. By Lemma 38.2.5, $-\log(1-p^{-s}) = p^{-s} + O(p^{-2s})$, so $\sum_p p^{-s} + O(1) = \log \zeta(s)$, where $O(1)$ is bounded because $\sum_p p^{-2s} \leq \sum_n n^{-2} < \infty$. Now $\zeta(s) \rightarrow \infty$ as $s \rightarrow 1^+$, so $\sum_p p^{-s} \rightarrow \infty$; and since $1/p > 1/p^s$ for $s > 1$, $\sum_p 1/p = \infty$. The refined form uses $\zeta(s) = \frac{1}{s-1} + O(1)$ and $\log(\frac{1}{s-1} + O(1)) = \log \frac{1}{s-1} + O(1)$. $\square$

This already strengthens Euclid: not only are there infinitely many primes, they are “dense” enough that the sum of their reciprocals diverges. Dirichlet’s theorem will be the analogous statement *restricted to one arithmetic progression*.

Worked Example 38.2.10 (Why $\sum 1/p$ diverges but $\sum 1/p^2$ converges). Contrast the divergence of $\sum_p 1/p$ with the convergence of $\sum_p 1/p^2$, and read off the rate at which primes thin out.

Recognition. Trigger: separate the leading term from the bounded remainder in Theorem 38.2.9.

Solution. $\sum_p p^{-2} \leq \sum_{n \geq 2} n^{-2} = \frac{\pi^2}{6} - 1 < \infty$, absolutely convergent. But $\sum_p p^{-s} = \log \frac{1}{s-1} + O(1) \rightarrow \infty$ as $s \rightarrow 1^+$, so the borderline exponent $s = 1$ already diverges – logarithmically. *Check.* Heuristically the n th prime is $\sim n \log n$, so $\sum 1/p \sim \sum \frac{1}{n \log n} = \infty$ (integral test) while $\sum 1/p^2 \sim \sum \frac{1}{n^2 \log^2 n} < \infty$. The primes are just barely too dense for $\sum 1/p$ to converge.

Dirichlet L-functions and the complex logarithms

Definition 38.2.11. For a Dirichlet character $\chi \bmod q$ and $s > 1$, the *Dirichlet L-function* is

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

The series converges absolutely for $s > 1$ since $|\chi(n)| \leq 1$.

To take logarithms of the (complex-valued) Euler product we need a branch of log on the relevant region. Stein introduces two logarithms.

Definition 38.2.12. For $|z| < 1$ define $\log_1\left(\frac{1}{1-z}\right) = \sum_{k=1}^{\infty} \frac{z^k}{k}$. (This defines $\log_1 w$ for $\operatorname{Re} w > \frac{1}{2}$, extending the real logarithm.)

Proposition 38.2.13 (Properties of $\log_1$). For $|z| < 1$: (i) $\exp(\log_1 \frac{1}{1-z}) = \frac{1}{1-z}$; (ii) $\log_1 \frac{1}{1-z} = z + E_1(z)$ with $|E_1(z)| \leq |z|^2$ for $|z| < \frac{1}{2}$; (iii) $|\log_1 \frac{1}{1-z}| \leq 2|z|$ for $|z| < \frac{1}{2}$.

Proof. (i) Writing $z = re^{i\theta}$, it suffices to show $g(r) := (1 - re^{i\theta}) \exp(\sum_{k \geq 1} (re^{i\theta})^k / k) \equiv 1$. Differentiating in r , the bracketed factor becomes $-e^{i\theta} + (1 - re^{i\theta})e^{i\theta} \sum_{k \geq 1} (re^{i\theta})^{k-1} = -e^{i\theta} + (1 - re^{i\theta})e^{i\theta} \frac{1}{1 - re^{i\theta}} = 0$, so g is constant; $g(0) = 1$. (ii), (iii) are the power-series estimates of Lemma 38.2.5 verbatim. $\square$

Proposition 38.2.14 (Complex infinite products). If $\sum |a_n|$ converges and $a_n \neq 1$ for all n , then $\prod_n \frac{1}{1-a_n}$ converges and is nonzero.

Proof. Assume $|a_n| < \frac{1}{2}$. Then $\prod_{n \leq N} \frac{1}{1-a_n} = \exp(\sum_{n \leq N} \log_1 \frac{1}{1-a_n})$, and $|\log_1 \frac{1}{1-a_n}| \leq 2|a_n|$ by Proposition 38.2.13(iii), so the exponent converges to some A ; by continuity the product $\rightarrow e^A \neq 0$. $\square$

Theorem 38.2.15 (Dirichlet product formula). For $s > 1$, $L(s, \chi) = \prod_p \frac{1}{1 - \chi(p)p^{-s}}$.

Proof. Set $a_n = \chi(p_n)p_n^{-s}$; then $\sum_n |a_n| < \infty$ for $s > 1$, so the product $\Pi = \prod_p (1 - \chi(p)p^{-s})^{-1}$ converges (Proposition 38.2.14). Write $S_N = \sum_{n \leq N} \chi(n)n^{-s}$, $\Pi_N = \prod_{p \leq N} (1 - \chi(p)p^{-s})^{-1}$, and $\Pi_{N,M} = \prod_{p \leq N} (1 + \chi(p)p^{-s} + \cdots + \chi(p^M)p^{-Ms})$. Fix $\varepsilon > 0$; choose N with $|S_N - L| < \varepsilon$ and $|\Pi_N - \Pi| < \varepsilon$, then M with $|S_N - \Pi_{N,M}| < \varepsilon$ and $|\Pi_{N,M} - \Pi_N| < \varepsilon$ (the first by unique factorization and complete multiplicativity of χ , the second because each geometric series $\sum_k \chi(p^k)p^{-ks}$ converges). Then $|L - \Pi| \leq |L - S_N| + |S_N - \Pi_{N,M}| + |\Pi_{N,M} - \Pi_N| + |\Pi_N - \Pi| < 4\varepsilon$. As ε was arbitrary, $L = \Pi$. $\square$

Worked Example 38.2.16 (The $4k+1$ prototype: $L(s, \chi) \bmod 4$). For the nontrivial character $\chi \bmod 4$, write $L(s, \chi)$ as a series and as a product, and evaluate $L(1, \chi)$.

Recognition. Trigger: instantiate Definition 38.2.11 and Theorem 38.2.15 for $q = 4$, $\chi(1) = 1, \chi(3) = -1$.

Solution. $L(s, \chi) = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \frac{1}{7^s} + \cdots = \prod_{p \text{ odd}} \frac{1}{1 - \chi(p)p^{-s}}$. At $s = 1$ the series $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$ is

alternating with terms decreasing to 0, so it converges; being a nonzero alternating series (first partial sum 1, the sum lies in $(\frac{2}{3}, 1)$), $L(1, \chi) \neq 0$. *Check.* This is exactly Leibniz's series, and one finds $L(1, \chi) = \arctan 1 = \pi/4$ (see Worked Example below). The non-vanishing here is the toy model of the deep Theorem 38.2.19.

Worked Example 38.2.17 ($L(1, \chi) = \pi/4$ via Abel summation). Show the Leibniz series $1 - \frac{1}{3} + \frac{1}{5} - \dots$ sums to $\pi/4$.

Recognition. Trigger: identify an alternating L-value with an arctangent by integrating a geometric series.

Solution. Integrate $\frac{1}{1+x^2} = 1 - x^2 + x^4 - \dots$ over $[0, y]$, $0 < y < 1$: $\int_0^y \frac{dx}{1+x^2} = y - \frac{y^3}{3} + \frac{y^5}{5} - \dots$. As $y \rightarrow 1^-$, the left side $\rightarrow \int_0^1 \frac{dx}{1+x^2} = \arctan 1 = \frac{\pi}{4}$, exhibiting the series as Abel summable to $\pi/4$. Since the alternating series converges (Leibniz), Abel's theorem says its sum equals its Abel limit: $1 - \frac{1}{3} + \frac{1}{5} - \dots = \frac{\pi}{4}$. *Check.* Numerically $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} \approx 0.724$, oscillating toward $\pi/4 \approx 0.785$.

Theorem 38.2.18 (Logarithm of L). *For a nontrivial character χ , $L(t, \chi)$ is continuously differentiable for $t > 0$ with $L(t, \chi) = 1 + O(e^{-ct})$ and $L'(t, \chi) = O(e^{-ct})$ as $t \rightarrow \infty$; in particular $L(t, \chi) \neq 0$ for $t > 1$. Define $\log_2 L(s, \chi) = - \int_s^\infty \frac{L'(t, \chi)}{L(t, \chi)} dt$ for $s > 1$. Then $\exp(\log_2 L(s, \chi)) = L(s, \chi)$ and*

$$\log_2 L(s, \chi) = \sum_p \log_1 \left(\frac{1}{1 - \chi(p)/p^s} \right) = \sum_p \frac{\chi(p)}{p^s} + O(1).$$

Proof. The integral converges since $L'/L = O(e^{-ct})$ from the growth estimates. Differentiating $e^{-\log_2 L(s, \chi)} L(s, \chi)$ gives 0, so it is constant; letting $s \rightarrow \infty$ (where $L \rightarrow 1$) the constant is 1, proving $\exp(\log_2 L) = L$. For the product identity: $\log_2 L$ and $\sum_p \log_1 \frac{1}{1 - \chi(p)/p^s}$ both exponentiate to $L(s, \chi)$ (the latter by the product formula and Proposition 38.2.13(i)), so they differ by $2\pi i M(s)$ with M integer-valued and continuous, hence constant; it is 0 as $s \rightarrow \infty$. Finally $\log_1 \frac{1}{1 - \chi(p)/p^s} = \frac{\chi(p)}{p^s} + O(p^{-2s})$ by Proposition 38.2.13(ii), and $\sum_p p^{-2s} = O(1)$. $\square$

The point of all this machinery: $\sum_p \chi(p)/p^s$ is bounded as $s \rightarrow 1^+$ *provided* $L(s, \chi)$ has a finite nonzero limit there. So everything reduces to the *non-vanishing* $L(1, \chi) \neq 0$.

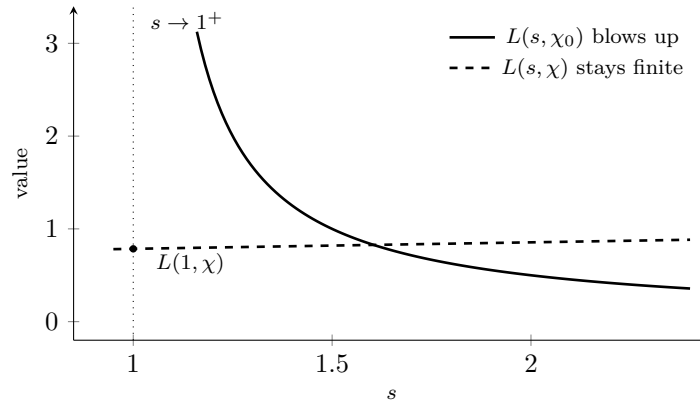


Figure 38.8. Near $s = 1$ only the trivial character's L-function blows up (rate $1/(s - 1)$), isolating a residue class; nontrivial $L(s, \chi)$ approach the finite nonzero values $L(1, \chi)$ – the crux is that these limits are not 0. (Schematic.)

Theorem 38.2.19 (Non-vanishing at $s = 1$). *If $\chi \neq \chi_0$, then $L(1, \chi) \neq 0$.*

Proof, complex-character case. We sketch the elegant case χ complex (i.e. $\chi \neq \bar{\chi}$); the real case uses hyperbolic summation and is treated in §38.3. Two lemmas. (A) For $s > 1$, $\prod_\chi L(s, \chi) \geq 1$ (real-valued). Indeed $\prod_\chi L(s, \chi) = \exp(\sum_p \sum_{k \geq 1} \frac{1}{k} p^{-ks} \sum_\chi \chi(p^k))$, and by orthogonality $\sum_\chi \chi(p^k) = \varphi(q) \delta_1(p^k) \geq 0$, so the exponent is ≥ 0 . (B) If $L(1, \chi) = 0$ then $L(1, \bar{\chi}) = 0$, and near $s = 1$ a nontrivial $L(s, \chi)$ with a zero satisfies $|L(s, \chi)| \leq C|s - 1|$ (mean value theorem), while $|L(s, \chi_0)| \leq C/|s - 1|$. Now if some complex χ had $L(1, \chi) = 0$, then $\bar{\chi} \neq \chi$ also has $L(1, \bar{\chi}) = 0$, contributing *two* factors vanishing like $|s - 1|$ to $\prod_\chi L(s, \chi)$, against the single factor $L(s, \chi_0)$ blowing up like $1/|s - 1|$. The product would then $\rightarrow 0$ as $s \rightarrow 1^+$, contradicting (A). Hence $L(1, \chi) \neq 0$ for complex χ . $\square$

Worked Example 38.2.20 ($L(s, \chi_0)$ is essentially ζ). For the trivial character $\chi_0 \bmod q$ with prime divisors $p_1, \dots, p_N$ of q , express $L(s, \chi_0)$ via ζ and find its behavior as $s \rightarrow 1^+$.

Recognition. Trigger: compare the Dirichlet product for χ_0 (which omits exactly the primes dividing q) with the Euler product for ζ .

Solution. Since $\chi_0(p) = 1$ for $p \nmid q$ and $\chi_0(p) = 0$ for $p \mid q$,

$$L(s, \chi_0) = \prod_{p \nmid q} \frac{1}{1 - p^{-s}} = (1 - p_1^{-s}) \cdots (1 - p_N^{-s}) \zeta(s).$$

As $s \rightarrow 1^+$, $\zeta(s) \rightarrow \infty$ and the finite prefactor tends to $\prod_i (1 - p_i^{-1}) = \varphi(q)/q$, so $L(s, \chi_0) \rightarrow \infty$; quantitatively $L(s, \chi_0) = \frac{\varphi(q)}{q} \cdot \frac{1}{s-1} + O(1)$. *Check.* For $q = 4$, prefactor $= 1 - \frac{1}{2} = \frac{1}{2} = \varphi(4)/4$, so $L(s, \chi_0) \sim \frac{1}{2} \frac{1}{s-1}$ – it is the trivial character that supplies the divergence isolating one residue class.

Worked Example 38.2.21 (Cancellation: $\sum_{n=1}^q \chi(n) = 0$ for $\chi \neq \chi_0$). Show that the character values over one period sum to zero, and bound partial sums by q .

Recognition. Trigger: the key cancellation behind continuation of $L(s, \chi)$ to $s > 0$; use the shift trick from Theorem 38.1.9(1).

Solution. Let $S = \sum_{n=1}^q \chi(n) = \sum_{a \in \mathbb{Z}^*(q)} \chi(a)$ (vanishing terms drop). Pick $b \in \mathbb{Z}^*(q)$ with $\chi(b) \neq 1$ (exists since $\chi \neq \chi_0$). As a ranges over $\mathbb{Z}^*(q)$ so does ab , so $S = \sum_a \chi(ab) = \chi(b) \sum_a \chi(a) = \chi(b)S$, forcing $(1 - \chi(b))S = 0$, i.e. $S = 0$. For any k , write $k = aq + b'$; then $\sum_{n=1}^k \chi(n) = \sum_{aq < n \leq aq+b'} \chi(n)$ has at most q terms each of modulus ≤ 1 , so $|\sum_{n=1}^k \chi(n)| \leq q$. *Check.* Mod 4, one period of values is $1, -1, 0, 0$ (for $n = 1, 2, 3, 4$, treating $\chi(2) = \chi(4) = 0$), summing to 0; partial sums stay in $\{0, 1\}$, well under $q = 4$.

Worked Example 38.2.22 (A Gauss sum and its modulus). For an odd prime p and the character $\chi(n) = \left(\frac{n}{p}\right)$ (Legendre symbol, a real character mod p), the *Gauss sum* is $g(\chi) = \sum_{n=1}^{p-1} \chi(n) e^{2\pi i n/p}$. Compute $g(\chi)$ for $p = 3$ and verify $|g(\chi)| = \sqrt{p}$.

Recognition. Trigger: evaluate a character-weighted exponential sum; for $p = 3$ the quadratic residues are just $\{1\}$, nonresidue $\{2\}$.

Solution. Mod 3: $\chi(1) = 1, \chi(2) = -1$. With $\omega = e^{2\pi i/3}$, $g(\chi) = \omega - \omega^2 = e^{2\pi i/3} - e^{4\pi i/3} = 2i \sin(2\pi/3) = 2i \cdot \frac{\sqrt{3}}{2} = i\sqrt{3}$. So $|g(\chi)| = \sqrt{3} = \sqrt{p}$. *Check.* The general fact $|g(\chi)| = \sqrt{p}$ for nontrivial χ follows from $g(\chi)\overline{g(\chi)} = \sum_{m,n} \chi(m)\overline{\chi(n)} e^{2\pi i(m-n)/p} = p$ after orthogonality; here $i\sqrt{3} \cdot i\sqrt{3} = 3 = p$, confirmed.

Worked Example 38.2.23 (Parseval on a finite group: counting via characters). On $G = \mathbb{Z}/N\mathbb{Z}$ with $e_k(m) = e^{2\pi i km/N}$, take $f = \mathbf{1}_{\{0\}}$ (the delta at 0). Compute $\hat{f}$ and verify Parseval (Theorem 38.1.11).

Recognition. Trigger: a Parseval consistency check on the simplest function.

Solution. $\hat{f}(e_k) = \frac{1}{N} \sum_m \mathbf{1}_{\{0\}}(m) \overline{e_k(m)} = \frac{1}{N} \overline{e_k(0)} = \frac{1}{N}$ for every k . Then $\sum_k \left| \hat{f}(e_k) \right|^2 = N \cdot \frac{1}{N^2} = \frac{1}{N}$, while $\frac{1}{N} \sum_m |f(m)|^2 = \frac{1}{N} \cdot 1 = \frac{1}{N}$. They agree. *Check.* Inversion recovers $f(m) = \sum_k \frac{1}{N} e^{2\pi i km/N} = \delta_0(m)$ by orthogonality (Theorem 38.1.9(2)) – the discrete delta is a flat superposition of all frequencies.

Worked Example 38.2.24 (Fourier coefficients of a sawtooth and Parseval). Compute the Fourier coefficients of the 2π -periodic sawtooth $f(\theta) = \theta$ on $(-\pi, \pi)$ and use Parseval to sum $\sum_{n \geq 1} 1/n^2$.

Recognition. Trigger: a classical Fourier-series computation; integrate by parts, then apply Parseval $\frac{1}{2\pi} \int |f|^2 = \sum \left| \hat{f}(n) \right|^2$.

Solution. $\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta e^{-in\theta} d\theta$. For $n \neq 0$, integrate by parts: $= \frac{1}{2\pi} \left[\frac{\theta e^{-in\theta}}{-in} \right]_{-\pi}^{\pi} + \frac{1}{2\pi in} \int_{-\pi}^{\pi} e^{-in\theta} d\theta = \frac{(-1)^n}{-in} = \frac{(-1)^{n+1}}{in} = \frac{i(-1)^n}{n}$, and $\hat{f}(0) = 0$. So $\left| \hat{f}(n) \right|^2 = 1/n^2$ ($n \neq 0$). Parseval: $\frac{1}{2\pi} \int_{-\pi}^{\pi} \theta^2 d\theta = \frac{1}{2\pi} \cdot \frac{2\pi^3}{3} = \frac{\pi^2}{3} = \sum_{n \neq 0} \frac{1}{n^2} = 2 \sum_{n \geq 1} \frac{1}{n^2}$, giving $\sum_{n \geq 1} 1/n^2 = \pi^2/6$. *Check.* This is $\zeta(2)$; consistent with $\zeta(2) = \pi^2/6$, the value lurking behind $\sum_p p^{-2} < \infty$ above.

Caution. The complex logarithm is multi-valued, so $\log(\prod) = \sum \log$ is *not* automatic when $\chi(p)$ is complex: that is the whole reason for introducing $\log_1$ (a fixed power series) and $\log_2$ (an integral of L'/L) and proving $\exp \circ \log_j = \text{id}$. Never write $\log L(s, \chi) = \sum_p \log \frac{1}{1 - \chi(p)/p^s}$ for complex χ without the identification of branches in Theorem 38.2.18; the naive identity can be off by $2\pi i\mathbb{Z}$.

Exercises (Stein, Ch.8).

- (Stein Ex.7) Recall that for $|z| < 1$, $\log_1(\frac{1}{1-z}) = \sum_{k \geq 1} z^k/k$, and that $\exp(\log_1 \frac{1}{1-z}) = \frac{1}{1-z}$. (a) Show that if $w = 1/(1-z)$, then $|z| < 1$ if and only if $\text{Re}(w) > \frac{1}{2}$. (b) Show that if $\text{Re}(w) > \frac{1}{2}$ and $w = \rho e^{i\varphi}$ with $\rho > 0$, $|\varphi| < \pi$, then $\log_1 w = \log \rho + i\varphi$. [Hint: if $e^\zeta = w$, the real part of ζ is uniquely determined and its imaginary part is determined modulo 2π .]
- (Stein Ex.8) Let ζ be the zeta function for $s > 1$. (a) Comparing $\zeta(s)$ with $\int_1^\infty x^{-s} dx$, show $\zeta(s) = \frac{1}{s-1} + O(1)$ as $s \rightarrow 1^+$. (b) Conclude that $\sum_p \frac{1}{p^s} = \log(\frac{1}{s-1}) + O(1)$ as $s \rightarrow 1^+$.
- (Stein Ex.9) Let χ_0 be the trivial Dirichlet character mod q and $p_1, \dots, p_k$ the distinct prime divisors of q . From $L(s, \chi_0) = (1 - p_1^{-s}) \cdots (1 - p_k^{-s}) \zeta(s)$, show that $L(s, \chi_0) = \frac{\varphi(q)}{q} \cdot \frac{1}{s-1} + O(1)$ as $s \rightarrow 1^+$. [Hint: use the asymptotics for ζ above.]
- (Stein Ex.12) Suppose χ is real and nontrivial; assuming the theorem that $L(1, \chi) \neq 0$, show directly that $L(1, \chi) > 0$. [Hint: use the product formula for $L(s, \chi)$.]
- (Stein Ex.13) Let $\{a_n\}_{n=-\infty}^\infty$ be complex numbers with $a_n = a_m$ whenever $n \equiv m \pmod{q}$. Show that $\sum_{n=1}^\infty a_n/n$ converges if and only if $\sum_{n=1}^q a_n = 0$. [Hint: sum by parts.]

Problems (Stein, Ch.8).

- (Problem 3) Show that if p is relatively prime to q , then $\prod_\chi (1 - \frac{\chi(p)}{p^s}) = (1 - \frac{1}{p^f s})^g$, where f is the order of p in $\mathbb{Z}^*(q)$ (the least n with $p^n \equiv 1 \pmod{q}$) and $g = \varphi(q)/f$, the product over all Dirichlet characters mod q .
- (Problem 4) Deduce from the previous problem that $\prod_\chi L(s, \chi) = \sum_{n \geq 1} a_n/n^s$ with all $a_n \geq 0$, the product over all Dirichlet characters mod q .

Additional exercises.

- Using the integral comparison, show $\zeta(s) \geq \frac{1}{s-1}$ for $s > 1$ and combine with the upper bound $\zeta(s) \leq 1 + \frac{1}{s-1}$ to conclude $(s-1)\zeta(s) \rightarrow 1$ as $s \rightarrow 1^+$.
- Prove that the Euler product $\prod_p (1 - p^{-s})^{-1}$ converges for $s > 1$ by verifying the hypothesis $\sum_p p^{-s} < \infty$ of Proposition 38.2.6, and explain where the argument breaks down at $s = 1$.
- Write $L(s, \chi)$ for the nontrivial character mod 6 as a Dirichlet series and identify the first eight nonzero terms. Then express it as an Euler product over primes $p \nmid 6$.
- Show that $\log(1+x) = x + O(x^2)$ for $|x| \leq \frac{1}{2}$ implies $\sum_p [-\log(1 - \chi(p)p^{-s}) - \chi(p)p^{-s}] = O(1)$ for $s > 1$, the error term that lets one replace $\log L$ by $\sum_p \chi(p)/p^s$.
- Compute the Gauss sum $g(\chi) = \sum_{n=1}^4 \chi(n)e^{2\pi i n/5}$ for the real (quadratic) character mod 5 and verify $|g(\chi)| = \sqrt{5}$.
- Using Parseval for the 2π -periodic square wave $f(\theta) = \text{sgn}(\theta)$ on $(-\pi, \pi)$, whose sine coefficients are $\frac{4}{\pi(2k+1)}$, sum the series $\sum_{k \geq 0} 1/(2k+1)^2$ and compare with $\zeta(2)$.
- (Non-example) Exhibit a sequence a_n with $\sum |a_n| = \infty$ for which $\prod_n (1 + a_n)$ nonetheless converges (e.g. $a_n = (-1)^n/\sqrt{n}$), showing that absolute convergence of $\sum a_n$ is sufficient but not necessary in Proposition 38.2.6.
- (Diagram) Sketch, as functions of $s \in (1, 2]$, the graphs of $L(s, \chi_0)$ (which blows up like $\frac{\varphi(q)/q}{s-1}$) and a nontrivial $L(s, \chi)$ (which stays near $L(1, \chi)$), for $q = 4$. Mark the value $L(1, \chi) = \pi/4$.
- Prove that if χ is a Dirichlet character of order d in the dual group, then $L(s, \chi)^d$ has nonnegative Dirichlet-series coefficients is *false* in general; find the smallest q and χ giving a negative coefficient. Contrast with Problem 4 ($\prod_\chi L$).

38.3 Proof of Dirichlet's Theorem on Primes in Arithmetic Progressions

The statement and the Fourier reduction

Theorem 38.3.1 (Dirichlet, 1837). *If q and ℓ are relatively prime positive integers, then the arithmetic progression*

$$\ell, \ell + q, \ell + 2q, \dots, \ell + kq, \dots$$

contains infinitely many primes. Equivalently, $\sum_{p \equiv \ell \pmod{q}} \frac{1}{p} = \infty$, the sum over primes $p \equiv \ell \pmod{q}$.

The hypothesis $\gcd(q, \ell) = 1$ is necessary: if $d = \gcd(q, \ell) > 1$ then $d \mid \ell + kq$ for all k , so the progression has no prime past d . Dirichlet's theorem says this obvious obstruction is the *only* one.

The strategy follows Euler (Theorem 38.2.9): show $\sum_{p \equiv \ell} 1/p$ diverges. The residue condition " $p \equiv \ell$ " is selected by the delta function on $\mathbb{Z}^*(q)$, which we expand in Dirichlet characters.

Lemma 38.3.2 (Fourier reduction). *For $s > 1$ and $\gcd(\ell, q) = 1$,*

$$\sum_{p \equiv \ell \pmod{q}} \frac{1}{p^s} = \frac{1}{\varphi(q)} \sum_{\chi} \overline{\chi(\ell)} \sum_p \frac{\chi(p)}{p^s},$$

the outer sum over all Dirichlet characters mod q . Splitting off the trivial character,

$$\sum_{p \equiv \ell} \frac{1}{p^s} = \frac{1}{\varphi(q)} \sum_{p \nmid q} \frac{1}{p^s} + \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \overline{\chi(\ell)} \sum_p \frac{\chi(p)}{p^s}.$$

Proof. Using $\delta_\ell(p) = \frac{1}{\varphi(q)} \sum_{\chi} \overline{\chi(\ell)} \chi(p)$ (Proposition 38.1.14(4)),

$$\sum_{p \equiv \ell} p^{-s} = \sum_p \delta_\ell(p) p^{-s} = \frac{1}{\varphi(q)} \sum_{\chi} \overline{\chi(\ell)} \sum_p \chi(p) p^{-s}.$$

The $\chi = \chi_0$ term is $\frac{1}{\varphi(q)} \sum_{p \nmid q} p^{-s}$ since $\overline{\chi_0(\ell)} = 1$ and $\chi_0(p) = \mathbf{1}_{p \nmid q}$. □

$\sum_{p \equiv \ell} p^{-s} \delta_\ell$ (target)	$\xrightarrow{\text{expand}}$	$\frac{1}{\varphi} \sum_{p \nmid q} p^{-s}$ (trivial χ_0)	$+$	$\frac{1}{\varphi} \sum_{\chi \neq \chi_0} \overline{\chi(\ell)} \sum_p \frac{\chi(p)}{p^s}$ (nontrivial)
		$\Downarrow$ DIVERGES like $\log \frac{1}{s-1}$		$\Downarrow$ BOUNDED (Thm 38.3.3)

Figure 38.9. The Fourier reduction: the trivial character contributes the diverging Euler term; if every nontrivial contribution stays bounded, the target diverges and the progression has infinitely many primes.

The first sum on the right diverges as $s \rightarrow 1^+$ (finitely many primes divide q , so it is $\sum_p p^{-s} + O(1) = \log \frac{1}{s-1} + O(1)$ by Theorem 38.2.9). So Dirichlet's theorem reduces to:

Theorem 38.3.3 (Boundedness for nontrivial χ). *If $\chi \neq \chi_0$ is a Dirichlet character, then $\sum_p \frac{\chi(p)}{p^s}$ remains bounded as $s \rightarrow 1^+$.*

Proof. By Theorem 38.2.18, $\sum_p \chi(p) p^{-s} = \log_2 L(s, \chi) + O(1)$. Since $L(s, \chi)$ is continuous at $s = 1$ (Theorem 38.2.18) and $L(1, \chi) \neq 0$ (Theorem 38.2.19), $\log_2 L(s, \chi) = -\int_s^\infty L'/L \rightarrow -\int_1^\infty L'/L$, a finite limit, as $s \rightarrow 1^+$. Hence $\sum_p \chi(p) p^{-s}$ is bounded. □

Worked Example 38.3.4 (Assembling the proof for $q = 4$). Run the whole machine for $q = 4$, $\ell = 1$ and $\ell = 3$, to conclude both progressions $4k + 1$ and $4k + 3$ carry infinitely many primes.

Recognition. Trigger: specialize Lemma 38.3.2 with $\varphi(4) = 2$ and the two characters χ_0, χ .

Solution. With χ the nontrivial character mod 4, $\sum_{p \equiv \ell} p^{-s} = \frac{1}{2}(\sum_{p \nmid 4} p^{-s} + \overline{\chi}(\ell) \sum_p \chi(p) p^{-s})$. The first sum $\rightarrow \infty$; the second is bounded since $L(1, \chi) = \pi/4 \neq 0$ (Worked Example in §38.2). So for either $\ell \in \{1, 3\}$, $\sum_{p \equiv \ell} p^{-s} \rightarrow \infty$, giving infinitely many primes in each class. *Check.* Adding the two ($\ell = 1, 3$) recovers $\sum_{p \text{ odd}} p^{-s}$, and indeed $\frac{1}{2}(\cdots) + \frac{1}{2}(\cdots) = \sum_{p \nmid 4} p^{-s}$ since the χ -terms $\overline{\chi}(1) + \overline{\chi}(3) = 1 - 1 = 0$ cancel.

$L(s, \chi_0)$ and the analytic continuation

Proposition 38.3.5 ($L(s, \chi_0)$ blows up; nontrivial L extend to $s > 0$). *For the trivial character, $L(s, \chi_0) = (1 - p_1^{-s}) \cdots (1 - p_N^{-s}) \zeta(s)$ with $p_i \mid q$, so $L(s, \chi_0) \rightarrow \infty$ as $s \rightarrow 1^+$. For nontrivial χ , the series $\sum_n \chi(n) n^{-s}$ converges for all $s > 0$ and defines a continuously differentiable function there.*

Proof. The first identity compares the Dirichlet and Euler products. For nontrivial χ , set $s_k = \sum_{n=1}^k \chi(n)$ ($s_0 = 0$); by the cancellation lemma $|s_k| \leq q$ for all k (Worked Example, §38.2). Summation by parts: $\sum_{k=1}^N \chi(k) k^{-s} = \sum_{k=1}^{N-1} s_k (k^{-s} - (k+1)^{-s}) + s_N N^{-s}$. By the mean value theorem applied to x^{-s} , $|k^{-s} - (k+1)^{-s}| \leq s k^{-s-1}$, so $|s_k (k^{-s} - (k+1)^{-s})| \leq q s k^{-s-1}$, summable uniformly for $s > \delta > 0$. Hence the series converges and is continuous for $s > 0$; differentiating term by term and summing by parts again (now against $x^{-s} \log x$, whose terms are $O(k^{-\delta/2-1})$) gives continuous differentiability. $\square$

Non-vanishing for real characters: hyperbolic sums

The hard case of Theorem 38.2.19 is χ *real* and nontrivial. Dirichlet's device is to sum a two-variable function over the lattice points under a hyperbola in three ways.

Definition 38.3.6. For F defined on pairs of positive integers, the *hyperbolic sum* is $S_N = \sum_{mn \leq N} F(m, n)$, summable three ways:

- (a) along hyperbolas $\sum_{k \leq N} \sum_{nm=k} F$; (b) vertically $\sum_{m \leq N} \sum_{n \leq N/m} F$; (c) horizontally $\sum_{n \leq N} \sum_{m \leq N/n} F$.

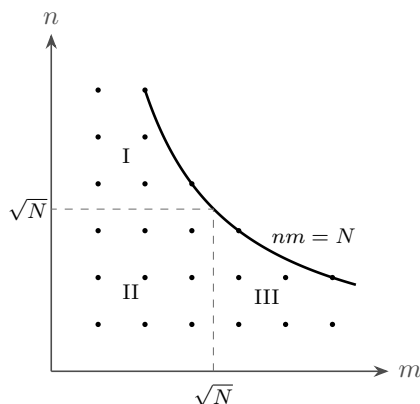


Figure 38.10. Lattice points (m, n) with $mn \leq N$. Splitting at $\sqrt{N}$ into regions I, II, III (with $S_I = S_{III}$ by symmetry) is the key to the divisor estimate and to $L(1, \chi) \neq 0$.

Theorem 38.3.7 (Dirichlet divisor problem). *Let $d(k)$ be the number of positive divisors of k . Then $\frac{1}{N} \sum_{k=1}^N d(k) = \log N + (2\gamma - 1) + O(N^{-1/2})$, where γ is Euler's constant. In particular $d(k)$ is on average of size $\log k$.*

Proof. Since $d(k) = \sum_{nm=k} 1$, $S_N := \sum_{k \leq N} d(k)$ is the hyperbolic sum of $F \equiv 1$. Summing vertically, $S_N = \sum_{m \leq N} \lfloor N/m \rfloor = \sum_{m \leq N} (N/m + O(1)) = N \sum_{m \leq N} \frac{1}{m} + O(N) = N \log N + O(N)$, using $\sum_{m \leq N} \frac{1}{m} = \log N + O(1)$; divide by N for the first form. For the refinement split at $\sqrt{N}$ into regions I, II, III with $S_N = 2(S_I + S_{II}) - S_{III}$ (as $S_I = S_{III}$). Vertical summation and $\sum_{m \leq \sqrt{N}} \frac{1}{m} = \log \sqrt{N} + \gamma + O(N^{-1/2})$ give $S_I + S_{II} = N \log \sqrt{N} + N\gamma + O(\sqrt{N})$, and $S_{II} = \lfloor \sqrt{N} \rfloor^2 = N + O(\sqrt{N})$; combine and divide by N . $\square$

Worked Example 38.3.8 (Sums versus integrals: harmonic and $\frac{1}{\sqrt{n}}$ asymptotics). Establish $\sum_{n \leq N} \frac{1}{n} = \log N + \gamma + O(1/N)$ and $\sum_{n \leq N} n^{-1/2} = 2N^{1/2} + c + O(N^{-1/2})$.

Recognition. Trigger: compare a monotone sum with its integral, controlling the error by a convergent series of gaps.

Solution. Let $\gamma_n = \frac{1}{n} - \int_n^{n+1} \frac{dx}{x}$. Since $1/x$ decreases, $0 \leq \gamma_n \leq \frac{1}{n} - \frac{1}{n+1} \leq \frac{1}{n^2}$, so $\sum \gamma_n$ converges to a limit γ (Euler's constant), with tail $\sum_{n > N} \gamma_n = O(1/N)$. Then $\sum_{n \leq N} \frac{1}{n} - \log N = \sum_{n \leq N} \gamma_n + \int_N^{N+1} \frac{dx}{x} = \gamma + O(1/N)$. The same telescoping with $|n^{-1/2} - (n+1)^{-1/2}| \leq Cn^{-3/2}$ and $\int_1^N x^{-1/2} dx = 2N^{1/2} - 2$ gives the second. *Check.* For $N = 4$, $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = 2.083$ vs $\log 4 + \gamma = 1.386 + 0.577 = 1.963$ – the $O(1/N)$ gap $\approx 0.12 \approx 1/8$ is the right size.

Proposition 38.3.9 (Hyperbolic sum of $F(m, n) = \chi(n)/(nm)^{1/2}$). Let χ be a nontrivial real character and $S_N = \sum_{mn \leq N} \frac{\chi(n)}{(nm)^{1/2}}$. Then

1. $S_N \geq c \log N$ for some $c > 0$;
2. $S_N = 2N^{1/2}L(1, \chi) + O(1)$.

Together these force $L(1, \chi) \neq 0$ (else (ii) would bound $S_N = O(1)$, contradicting (i)).

Proof. (i) Sum along hyperbolas: $S_N = \sum_{k \leq N} k^{-1/2} \sum_{n|k} \chi(n)$. The divisor sum $\sum_{n|k} \chi(n) = \prod_{p^a || k} (1 + \chi(p) + \cdots + \chi(p)^a)$ is ≥ 0 always, and ≥ 1 when $k = \ell^2$ is a perfect square (each local factor is then a sum of an odd number of terms equal to 1 when $\chi(p) = 1$, or 1 when $\chi(p) = -1$ with a even). Hence $S_N \geq \sum_{\ell \leq \sqrt{N}} \ell^{-1} \geq c \log N$ by the harmonic asymptotics. (ii) Split $S_N = S_I + (S_{II} + S_{III})$; using the partial-sum bounds $\sum_{a \leq n \leq b} \chi(n)n^{-1/2} = O(a^{-1/2})$ and $\sum \chi(n)n^{-1} = O(a^{-1})$ (summation by parts, $|s_n| \leq q$), one finds $S_I = O(1)$, and horizontal summation of $S_{II} + S_{III}$ yields $2N^{1/2} \sum_{n \leq \sqrt{N}} \frac{\chi(n)}{n} + O(1) = 2N^{1/2}L(1, \chi) + O(1)$. $\square$

Theorem 38.3.10 (Non-vanishing, real case). If χ is a nontrivial real Dirichlet character, then $L(1, \chi) \neq 0$.

Proof. Combine the two parts of Proposition 38.3.9. If $L(1, \chi) = 0$, then (ii) gives $S_N = O(1)$, bounded; but (i) gives $S_N \geq c \log N \rightarrow \infty$ – a contradiction. Hence $L(1, \chi) \neq 0$. Together with the complex case (Theorem 38.2.19) this proves $L(1, \chi) \neq 0$ for every nontrivial character, completing the proof of Theorem 38.3.3 and therefore of Dirichlet's Theorem 38.3.1. $\square$

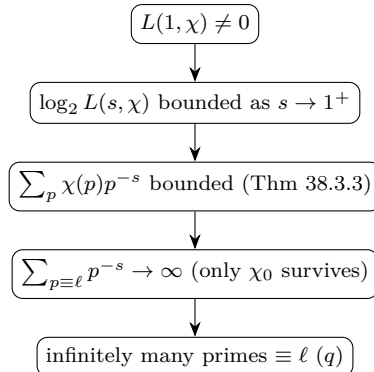


Figure 38.11. The logical spine of the proof: non-vanishing at $s = 1$ propagates down to the infinitude of primes in the progression.

Worked Example 38.3.11 ($L(1, \chi) > 0$ for nontrivial real χ). Granting $L(1, \chi) \neq 0$, show that in fact $L(1, \chi) > 0$ for real nontrivial χ .

Recognition. Trigger: upgrade “nonzero” to “positive”; use the product formula and that the relevant factors are positive.

Solution. By Proposition 38.3.9(ii), $2N^{1/2}L(1, \chi) = S_N + O(1)$ and by (i) $S_N \geq c \log N > 0$ for large N ; dividing by $2N^{1/2} > 0$ and taking $N \rightarrow \infty$ shows $L(1, \chi) = \lim S_N/(2N^{1/2}) \geq 0$, and being nonzero it is > 0 . *Check.* Alternatively the partial Euler product $\prod_{p \leq X} (1 - \chi(p)p^{-1})^{-1}$ is real and positive for real χ , consistent with a positive limit; e.g. mod 3 the value $L(1, \chi) = \pi/(3\sqrt{3}) > 0$.

Worked Example 38.3.12 (A quantitative Dirichlet theorem). Show that for $\gcd(\ell, q) = 1$, $\sum_{p \equiv \ell} \frac{1}{p^s} =$

$\frac{1}{\varphi(q)} \log \frac{1}{s-1} + O(1)$ as $s \rightarrow 1^+$, so each admissible class carries an equal logarithmic share of the primes.

Recognition. Trigger: feed the boundedness theorem and the asymptotics of $\sum_{p \nmid q} p^{-s}$ into the reduction (Lemma 38.3.2).

Solution. In the split form of Lemma 38.3.2, the nontrivial sum $\frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \overline{\chi(\ell)} \sum_p \chi(p)p^{-s} = O(1)$ by Theorem 38.3.3. The trivial term is $\frac{1}{\varphi(q)} \sum_{p \nmid q} p^{-s} = \frac{1}{\varphi(q)} (\log \frac{1}{s-1} + O(1))$ since only finitely many primes divide q . Adding gives the claim. *Check.* Summing over the $\varphi(q)$ admissible classes ℓ recovers $\sum_{p \nmid q} p^{-s} = \log \frac{1}{s-1} + O(1)$ – the primes are asymptotically equidistributed among the $\varphi(q)$ residue classes coprime to q .

Worked Example 38.3.13 (Convergence test: $\sum a_n/n$ with periodic a_n). Let (a_n) be q -periodic ($a_{n+q} = a_n$). Show $\sum_{n \geq 1} a_n/n$ converges iff $\sum_{n=1}^q a_n = 0$.

Recognition. Trigger: a Dirichlet-test / summation-by-parts criterion modeling the convergence of $L(s, \chi)$ at $s = 1$ for nontrivial χ .

Solution. Let $A_k = \sum_{n=1}^k a_n$. If $\sum_{n=1}^q a_n = 0$ then A_k is periodic, hence bounded, and summation by parts gives $\sum_{n \leq N} a_n/n = \sum_{n \leq N-1} A_n(\frac{1}{n} - \frac{1}{n+1}) + A_N/N$; the first sum converges absolutely (A_n bounded, $\frac{1}{n} - \frac{1}{n+1} = O(n^{-2})$) and $A_N/N \rightarrow 0$, so the series converges. Conversely if $\sigma = \sum_{n=1}^q a_n \neq 0$ then $A_{kq} = k\sigma$ grows linearly and the partial sums $\sum_{n \leq kq} a_n/n \approx \sigma \sum \frac{1}{kq}$ blocks behave like $\frac{\sigma}{q} \log N \rightarrow \pm\infty$, so the series diverges. *Check.* For χ nontrivial, $\sum_{n=1}^q \chi(n) = 0$, so $L(s, \chi) = \sum \chi(n)/n^s$ converges at $s = 1$ – exactly the borderline behavior this criterion predicts.

Worked Example 38.3.14 (The divisor function and its average). Compute $d(k)$ for $k = 1, \dots, 12$ and check the average against $\log N$ (Theorem 38.3.7).

Recognition. Trigger: evaluate $d(k) = \prod (a_i + 1)$ from the factorization and compare the running mean with $\log N + (2\gamma - 1)$.

Solution. Factorizations give $d : 1, 2, 2, 3, 2, 4, 2, 4, 3, 4, 2, 6$ for $k = 1, \dots, 12$. Their sum is $1 + 2 + 2 + 3 + 2 + 4 + 2 + 4 + 3 + 4 + 2 + 6 = 35$, so $\frac{1}{12} \sum d(k) = 35/12 \approx 2.92$. The asymptotic predicts $\log 12 + (2\gamma - 1) = 2.485 + 0.154 = 2.639$ plus an $O(N^{-1/2}) \approx 0.29$ error – consistent. *Check.* $d(k) = 2$ exactly when k is prime ($k = 2, 3, 5, 7, 11$), and $d(12) = d(2^2 \cdot 3) = (2+1)(1+1) = 6$, the largest in range, matching the table.

Worked Example 38.3.15 ($d(k) = 2$ infinitely often, yet $d(k)$ can be huge). Show $d(k) = 2$ for infinitely many k , and that $d(k) \geq c(\log k)^N$ infinitely often for each N .

Recognition. Trigger: extreme values of a multiplicative function; primes give the minimum, products of many distinct primes the (relative) maximum.

Solution. $d(k) = 2$ iff k is prime, and there are infinitely many primes (Theorem 38.2.3), so $d(k) = 2$ infinitely often. For the large values, fix distinct primes $p_1, \dots, p_N$ and take $k = (p_1 \cdots p_N)^m$; then $d(k) = (m+1)^N$, while $\log k = m \log(p_1 \cdots p_N) = m \cdot C$, so $d(k) = (m+1)^N \geq (\log k/C)^N = c(\log k)^N$ for $m = 1, 2, \dots$. *Check.* This shows $d(k)$ oscillates between 2 and powers of $\log k$ – its erratic behavior is why only the *average* (Theorem 38.3.7) is smooth.

Worked Example 38.3.16 (Verifying $L(1, \chi) \neq 0$ for $q = 3$ by the alternating series). For the nontrivial character mod 3, write $L(1, \chi)$ as an explicit conditionally convergent series and argue it is nonzero.

Recognition. Trigger: instantiate the non-vanishing directly; mod 3, $\chi(1) = 1$, $\chi(2) = -1$, $\chi(3) = 0$, periodic.
Solution. $L(1, \chi) = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{5} + \frac{1}{7} - \frac{1}{8} + \cdots$ (omit multiples of 3, alternating sign by residue). Grouping in blocks of period 3, $(\frac{1}{3k+1} - \frac{1}{3k+2}) > 0$ for every $k \geq 0$, so the sum is a series of strictly positive terms, hence $L(1, \chi) > 0 \neq 0$. *Check.* The closed form is $L(1, \chi) = \frac{\pi}{3\sqrt{3}} \approx 0.6046$; the first block $1 - \frac{1}{2} = 0.5$ already gives the right order of magnitude, and all later blocks add positive amounts.

Worked Example 38.3.17 (Equidistribution of primes: a numerical sanity check mod 4). The quantitative theorem predicts primes split evenly between $4k+1$ and $4k+3$. Check the count of primes below 50.

Recognition. Trigger: test the equidistribution consequence of the quantitative Dirichlet theorem (each class gets a $1/\varphi(q)$ share).

Solution. Odd primes below 50: 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47 – fourteen of them. Class $4k+1$: 5, 13, 17, 29, 37, 41 (six). Class $4k+3$: 3, 7, 11, 19, 23, 31, 43, 47 (eight). The split 6 vs 8 is roughly even, with the mild “Chebyshev bias” toward $4k+3$ visible at small scales. *Check.* $\varphi(4) = 2$, so each class should hold a fraction $\rightarrow \frac{1}{2}$ of the primes; $6/14 \approx 0.43$ and $8/14 \approx 0.57$, converging slowly to $\frac{1}{2}$ as the bound grows (the bias is lower-order, $O(\sqrt{x}/\log x)$).

Caution. The two non-vanishing arguments are genuinely different. The *complex* case is a slick counting argument ($\prod_{\chi} L \geq 1$ plus the fact that $\chi \neq \bar{\chi}$ supplies two vanishing factors). It *fails* for real χ , where $\chi = \bar{\chi}$ gives only one vanishing factor, not enough to overwhelm the single $1/(s-1)$ pole of $L(s, \chi_0)$. The real case demands the entirely separate hyperbolic-sum estimate. Do not try to “unify” them by symmetry – the asymmetry $\chi \neq \bar{\chi}$ is exactly what the complex argument exploits.

Exercises (Stein, Ch.8).

- (Stein Ex.10) Show that if ℓ is relatively prime to q , then $\sum_{p \equiv \ell} \frac{1}{p^s} = \frac{1}{\varphi(q)} \log\left(\frac{1}{s-1}\right) + O(1)$ as $s \rightarrow 1^+$. This is a quantitative version of Dirichlet’s theorem. [Hint: recall the split $\sum_{p \equiv \ell} p^{-s} = \frac{1}{\varphi(q)} \sum_{p \nmid q} p^{-s} + \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \overline{\chi(\ell)} \sum_p \chi(p) p^{-s}$.]
- (Stein Ex.11) Use the characters of $\mathbb{Z}^*(3)$, $\mathbb{Z}^*(4)$, $\mathbb{Z}^*(5)$, and $\mathbb{Z}^*(6)$ to verify directly that $L(1, \chi) \neq 0$ for all nontrivial Dirichlet characters mod q when $q = 3, 4, 5, 6$. [Hint: in each case consider the appropriate alternating series.]
- (Stein Ex.14) The series $F(\theta) = \sum_{|n| \neq 0} \frac{e^{in\theta}}{n}$ ($|\theta| < \pi$) converges for every θ and is the Fourier series of the function with $F(0) = 0$ and $F(\theta) = i(-\pi - \theta)$ for $-\pi \leq \theta < 0$, $F(\theta) = i(\pi - \theta)$ for $0 < \theta \leq \pi$, extended with period 2π . Show also that if $\theta \not\equiv 0 \pmod{2\pi}$, then $E(\theta) = \sum_{n=1}^{\infty} \frac{e^{in\theta}}{n}$ converges and $E(\theta) = \frac{1}{2} \log\left(\frac{1}{2-2\cos\theta}\right) + \frac{i}{2} F(\theta)$.
- (Stein Ex.15) To sum $\sum_{n=1}^{\infty} a_n/n$ with $a_n = a_m$ for $n \equiv m \pmod{q}$ and $\sum_{n=1}^q a_n = 0$: (a) define $A(m) = \sum_{n=1}^q a_n \zeta^{-mn}$ with $\zeta = e^{2\pi i/q}$ (note $A(q) = 0$); prove $\sum_{n=1}^{\infty} a_n/n = \frac{1}{q} \sum_{m=1}^{q-1} A(m) E(2\pi m/q)$. [Hint: Fourier inversion on $\mathbb{Z}(q)$.] (b) If $\{a_m\}$ is odd ($a_{-m} = -a_m$), observe $a_0 = a_q = 0$ and show $A(m) = \sum_{1 \leq n < q/2} a_n (\zeta^{-mn} - \zeta^{mn})$. (c) Still for odd $\{a_m\}$, show $\sum_{n=1}^{\infty} a_n/n = \frac{1}{2q} \sum_{m=1}^{q-1} A(m) F(2\pi m/q)$. [Hint: define $\tilde{A}(m) = \sum_{n=1}^q a_n \zeta^{mn}$ and apply Fourier inversion.]
- (Stein Ex.16) Use the previous exercises to show $\frac{\pi}{3\sqrt{3}} = 1 - \frac{1}{4} + \frac{1}{5} - \frac{1}{7} + \frac{1}{8} - \cdots$, which is $L(1, \chi)$ for the nontrivial (odd) Dirichlet character modulo 3.

Problems (Stein, Ch.8).

- (Problem 1, starred) Other series summable by the methods of Exercise 15: (a) for the nontrivial character mod 6, $L(1, \chi) = \frac{\pi}{2\sqrt{3}} = 1 - \frac{1}{5} + \frac{1}{7} - \frac{1}{11} + \frac{1}{13} - \cdots$; (b) for the odd character mod 8, $L(1, \chi) = \frac{\pi}{2\sqrt{2}} = 1 + \frac{1}{3} - \frac{1}{5} - \frac{1}{7} + \frac{1}{9} + \frac{1}{11} - \cdots$; (c) for an odd character mod 7, $L(1, \chi) = \frac{\pi}{\sqrt{7}} = 1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} - \frac{1}{6} + \cdots$; (d) for an even character mod 8, $L(1, \chi) = \frac{\log(1+\sqrt{2})}{\sqrt{2}} = 1 - \frac{1}{3} - \frac{1}{5} + \frac{1}{7} + \frac{1}{9} - \cdots$; (e) for an even character mod 5, $L(1, \chi) = \frac{2}{\sqrt{5}} \log\left(\frac{1+\sqrt{5}}{2}\right) = 1 - \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{6} - \frac{1}{7} - \frac{1}{8} + \frac{1}{9} + \cdots$.

2. (Problem 2) Let $d(k)$ be the number of positive divisors of k . (a) Show that if $k = p_1^{a_1} \cdots p_n^{a_n}$, then $d(k) = (a_1 + 1) \cdots (a_n + 1)$. Although Theorem 3.12 shows $d(k)$ is on average of order $\log k$, prove from (a): (b) $d(k) = 2$ for infinitely many k ; (c) for any positive integer N there is $c > 0$ with $d(k) \geq c(\log k)^N$ for infinitely many k . [Hint: let $p_1, \dots, p_N$ be distinct primes and consider $k = (p_1 \cdots p_N)^m$, $m = 1, 2, \dots$]

Additional exercises.

- Carry out the Fourier reduction (Lemma 38.3.2) explicitly for $q = 3$, $\ell = 1$ and $\ell = 2$, writing each $\sum_{p \equiv \ell} p^{-s}$ as a combination of $\sum_{p \nmid 3} p^{-s}$ and $\sum_p \chi(p) p^{-s}$ for the nontrivial $\chi \bmod 3$.
- Show that the number of primes dividing a fixed q is finite, and conclude that $\sum_{p|q} p^{-s} = O(1)$ uniformly for $s \geq 1$, justifying the replacement of $\sum_{p \nmid q} p^{-s}$ by $\sum_p p^{-s}$ in the reduction.
- Use summation by parts and $\left| \sum_{n=1}^k \chi(n) \right| \leq q$ to prove that $\sum_{n=a}^b \chi(n) n^{-1} = O(a^{-1})$ for nontrivial χ , one of the two lemmas behind $S_I = O(1)$ in the hyperbolic-sum proof.
- Compute $\sum_{n|k} \chi(n)$ for $k = 1, 2, \dots, 9$ where χ is the nontrivial real character mod 3, and verify it is ≥ 0 always and ≥ 1 at the perfect squares $k = 1, 4, 9$ (Lemma 3.14).
- Verify the divisor average numerically: compute $\frac{1}{N} \sum_{k \leq N} d(k)$ for $N = 20$ and compare with $\log N + (2\gamma - 1)$, $\gamma \approx 0.5772$.
- (Equidistribution) List the primes up to 100 and tabulate them by residue class mod 5 (classes 1, 2, 3, 4). Comment on how close the four counts are, in light of the quantitative theorem (each class gets a $1/\varphi(5) = 1/4$ share).
- (Non-example) Explain why the complex-character non-vanishing argument (“ $\prod_\chi L \geq 1$ plus two vanishing factors”) cannot work when χ is real, by exhibiting where “ $\chi \neq \bar{\chi}$ gives two factors” fails. State precisely what goes wrong.
- (Diagram) Draw the lattice points (m, n) with $mn \leq 16$ in the first quadrant, overlay the hyperbola $mn = 16$, and shade the three regions I, II, III meeting at the line $m = n = 4$. Use the picture to explain $S_N = 2(S_I + S_{II}) - S_{III}$.
- Prove that $\sum_{p \equiv \ell} 1/p = \infty$ follows from $\sum_{p \equiv \ell} p^{-s} \rightarrow \infty$ as $s \rightarrow 1^+$, using $1/p > 1/p^s$ for $s > 1$ (the same final step as in Euler’s theorem).
- Assuming the quantitative theorem, show that the “density” of primes in the progression $\ell + kq$ among all primes is $1/\varphi(q)$, in the sense that $(\sum_{p \equiv \ell, p \leq x} 1/p) / (\sum_{p \leq x} 1/p) \rightarrow 1/\varphi(q)$.

Part V

Functional Analysis

Chapter 39

Lp Spaces and Banach Spaces

39.1 Lp Spaces: Definition and Completeness

The space L^p and its norm

Throughout, $(X, \mathcal{F}, \mu)$ is a σ -finite measure space: X is the underlying set, $\mathcal{F}$ the σ -algebra of measurable sets, and μ the measure. The L^p spaces are the basic stage on which analysis after measure theory is performed; they interpolate between the integrable functions ($p = 1$), the Hilbert space of square-integrable functions ($p = 2$), and the essentially bounded functions ($p = \infty$).

Definition 39.1.1. Let $1 \leq p < \infty$. The space $L^p(X, \mathcal{F}, \mu)$ consists of all complex-valued measurable functions f on X with

$$\int_X |f(x)|^p d\mu(x) < \infty.$$

For such f the L^p norm is

$$\|f\|_{L^p} = \left(\int_X |f(x)|^p d\mu(x) \right)^{1/p}.$$

We abbreviate $\|f\|_{L^p(X)}$, $\|f\|_{L^p}$ or $\|f\|_p$. Strictly, L^p is the set of *equivalence classes* of such functions under the relation $f \sim g \iff f = g$ a.e., because $\|f\|_{L^p} = 0$ forces only $f = 0$ a.e., not $f \equiv 0$. In practice we treat elements as functions, remembering that all statements are modulo null sets.

The exponent p controls how much weight is placed on large values of f . A large p punishes tall spikes severely (their p th power blows up), while a small p is forgiving of spikes but punishes slow decay at infinity. This trade-off is the source of the inclusion relations below.

Definition 39.1.2. The space $L^\infty(X, \mathcal{F}, \mu)$ consists of all measurable f that are *essentially bounded*: there is $0 < M < \infty$ with $|f(x)| \leq M$ a.e. The *essential supremum*

$$\|f\|_{L^\infty} = \operatorname{ess\,sup}_X |f| = \inf\{M > 0 : |f(x)| \leq M \text{ a.e.}\}$$

is the smallest such bound, and one has $|f(x)| \leq \|f\|_{L^\infty}$ a.e.

Why the essential bound is attained a.e. Let $E = \{x : |f(x)| > \|f\|_{L^\infty}\}$ and $E_n = \{x : |f(x)| > \|f\|_{L^\infty} + 1/n\}$. By definition of the infimum each $\mu(E_n) = 0$, and $E = \bigcup_n E_n$, so $\mu(E) = 0$. $\square$

Remark. The norm bars carry a subscript naming the space; $\|\cdot\|_p$ means the L^p norm, not absolute value. We restrict to $1 \leq p \leq \infty$: for $0 < p < 1$ the functional $\|\cdot\|_{L^p}$ fails the triangle inequality and L^p then has *no* nonzero bounded linear functionals (Exercise 2), so it is far less useful.

Example 39.1.3 (The standard models). (a) $X = \mathbb{R}^d$ with Lebesgue measure gives $\|f\|_{L^p} = \left(\int_{\mathbb{R}^d} |f|^p dx \right)^{1/p}$. (b) $X = \mathbb{Z}$ with counting measure gives the sequence spaces: a function is a sequence $f = \{a_n\}_{n \in \mathbb{Z}}$ and $\|f\|_{\ell^p} = \left(\sum_n |a_n|^p \right)^{1/p}$, written $\ell^p(\mathbb{Z})$. When $p = 2$ this is the familiar $\ell^2(\mathbb{Z})$.

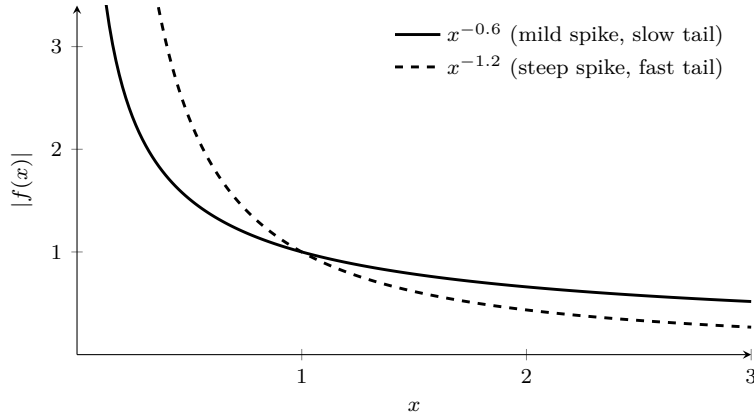


Figure 39.1. Membership in $L^p(0, \infty)$ is a contest between the spike at 0 and the tail at ∞ . The mild spike $x^{-0.6}$ is locally integrable but its tail decays slowly; the steeper $x^{-1.2}$ has an integrable tail but a non-integrable spike. The exponent p reweights both ends.

Inclusion relations

On a finite-measure space, raising the exponent shrinks the space; on the integers with counting measure the reverse happens. There is no inclusion in general (Exercise 1).

Proposition 39.1.4 (Finite measure: higher p is smaller). *If $\mu(X) < \infty$ and $p_0 \leq p_1$, then $L^{p_1}(X) \subseteq L^{p_0}(X)$ and*

$$\frac{1}{\mu(X)^{1/p_0}} \|f\|_{L^{p_0}} \leq \frac{1}{\mu(X)^{1/p_1}} \|f\|_{L^{p_1}}.$$

Proof. Assume $p_1 > p_0$. For $f \in L^{p_1}$ apply Holder's inequality (Theorem 39.2.3) with $F = |f|^{p_0}$, $G = 1$, exponent $p = p_1/p_0 > 1$ and dual q with $1/p + 1/q = 1$:

$$\int |f|^{p_0} = \int F \cdot G \leq \|F\|_p \|G\|_q = \left(\int |f|^{p_1} \right)^{p_0/p_1} \mu(X)^{1-p_0/p_1}.$$

Taking p_0 th roots gives $\|f\|_{L^{p_0}} \leq \mu(X)^{1/p_0 - 1/p_1} \|f\|_{L^{p_1}}$, which is the stated inequality, and in particular $\|f\|_{L^{p_0}} < \infty$. $\square$

Proposition 39.1.5 (Counting measure: higher p is larger). *If $X = \mathbb{Z}$ with counting measure and $p_0 \leq p_1$, then $\ell^{p_0}(\mathbb{Z}) \subseteq \ell^{p_1}(\mathbb{Z})$ and $\|f\|_{\ell^{p_1}} \leq \|f\|_{\ell^{p_0}}$.*

Proof. For $f = \{f(n)\}$, every term satisfies $|f(n)|^{p_0} \leq \|f\|_{\ell^{p_0}}^{p_0}$, so $\sup_n |f(n)| \leq \|f\|_{\ell^{p_0}}$. Then

$$\sum_n |f(n)|^{p_1} = \sum_n |f(n)|^{p_0} |f(n)|^{p_1 - p_0} \leq \left(\sup_n |f(n)| \right)^{p_1 - p_0} \|f\|_{\ell^{p_0}}^{p_0} \leq \|f\|_{\ell^{p_0}}^{p_1}. \quad \square$$

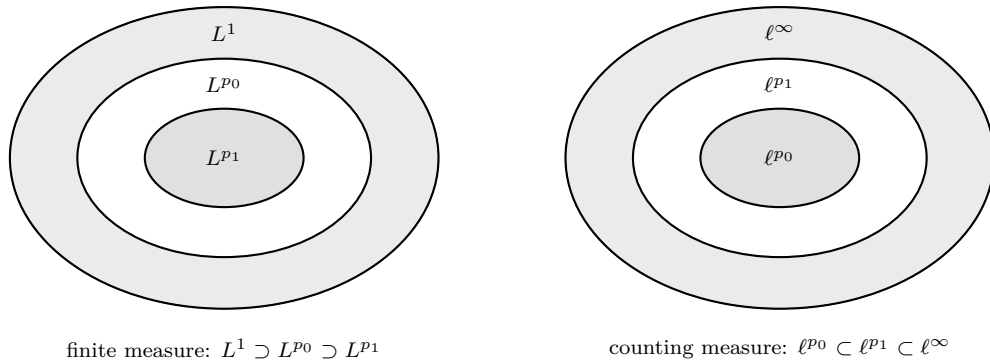


Figure 39.2. The two opposite nesting patterns. On a finite-measure space the tail is harmless and only the spike matters, so bigger p is a stronger condition; with counting measure there is no tail to control and bigger p is weaker. (Schematic.)

Proposition 39.1.6 (L^∞ as the limit of L^p). If $f \in L^\infty$ is supported on a set of finite measure, then $f \in L^p$ for all $p < \infty$ and $\|f\|_{L^p} \rightarrow \|f\|_{L^\infty}$ as $p \rightarrow \infty$.

Proof. Let E carry the support, $\mu(E) < \infty$. If $\mu(E) = 0$ all norms vanish. Otherwise $\|f\|_{L^p} = (\int_E |f|^p)^{1/p} \leq \|f\|_{L^\infty} \mu(E)^{1/p}$, and since $\mu(E)^{1/p} \rightarrow 1$ we get $\limsup_p \|f\|_{L^p} \leq \|f\|_{L^\infty}$. Conversely, for $\varepsilon > 0$ the set $\{|f| \geq \|f\|_{L^\infty} - \varepsilon\}$ has some positive measure δ , so $\int |f|^p \geq \delta(\|f\|_{L^\infty} - \varepsilon)^p$, giving $\liminf_p \|f\|_{L^p} \geq \|f\|_{L^\infty} - \varepsilon$. As ε is arbitrary the two bounds meet. $\square$

Completeness

The triangle inequality (Minkowski, Theorem 39.2.4) makes $d(f, g) = \|f - g\|_{L^p}$ a metric. The deep fact is that this metric is complete: Cauchy sequences converge inside L^p . Without this, limiting constructions would routinely leave the space.

Theorem 39.1.7 (Riesz–Fischer). For $1 \leq p \leq \infty$ the space $L^p(X, \mathcal{F}, \mu)$ is complete in $\|\cdot\|_{L^p}$; i.e. it is a Banach space.

Proof. Take $1 \leq p < \infty$ (the case $p = \infty$ is Theorem 39.1.8). Let $\{f_n\}$ be Cauchy. It suffices to find a convergent subsequence, since a Cauchy sequence with a convergent subsequence converges to the same limit. Choose n_k with $\|f_{n_{k+1}} - f_{n_k}\|_{L^p} \leq 2^{-k}$. Set

$$g(x) = |f_{n_1}(x)| + \sum_{k=1}^{\infty} |f_{n_{k+1}}(x) - f_{n_k}(x)|,$$

with partial sums $S_K(g)$. By the triangle inequality $\|S_K(g)\|_{L^p} \leq \|f_{n_1}\|_{L^p} + \sum_{k=1}^K 2^{-k} \leq \|f_{n_1}\|_{L^p} + 1$. Monotone convergence gives $\|g\|_{L^p} < \infty$, so $g(x) < \infty$ a.e.; hence the telescoping series

$$f(x) = f_{n_1}(x) + \sum_{k=1}^{\infty} (f_{n_{k+1}}(x) - f_{n_k}(x))$$

converges absolutely a.e. to a function $f \in L^p$, and its $(K-1)$ th partial sum is exactly f_{n_K} , so $f_{n_K} \rightarrow f$ a.e. For convergence in norm note $|f - S_K(f)|^p \leq (2 \max(|f|, |S_K(f)|))^p \leq 2^{p+1} g^p$, an integrable dominating function, so dominated convergence yields $\|f_{n_K} - f\|_{L^p} \rightarrow 0$. Finally, given $\varepsilon > 0$ pick N with $\|f_n - f_m\|_{L^p} < \varepsilon/2$ for $n, m > N$ and $n_K > N$ with $\|f_{n_K} - f\|_{L^p} < \varepsilon/2$; then $\|f_n - f\|_{L^p} \leq \|f_n - f_{n_K}\|_{L^p} + \|f_{n_K} - f\|_{L^p} < \varepsilon$ for $n > N$. $\square$

Theorem 39.1.8. L^∞ with $\|\cdot\|_{L^\infty}$ is complete.

Proof. Let $\{f_n\}$ be Cauchy in L^∞ . The sets $\{|f_n| > \|f_n\|_\infty\}$ and $\{|f_n - f_m| > \|f_n - f_m\|_\infty\}$, over all n, m , form a countable union Z of null sets. Off Z the sequence $\{f_n(x)\}$ is uniformly Cauchy in the sup sense, hence converges uniformly to a bounded f ; uniform convergence off a null set is exactly convergence in $\|\cdot\|_{L^\infty}$, and $f \in L^\infty$. $\square$

Remark. The Riesz–Fischer proof secretly proves more: a subsequence of any L^p -convergent sequence converges *pointwise a.e.* (Exercise 5). L^p convergence does not imply a.e. convergence of the full sequence (the moving-bump “typewriter” sequence is the standard counterexample), only of a subsequence.

Worked Example 39.1.9 (When does $|x|^{-\alpha}$ lie in L^p ?). Fix $d \geq 1$ and $\alpha > 0$. Let $f_0(x) = |x|^{-\alpha} \mathbf{1}_{|x| < 1}$ and $f_\infty(x) = |x|^{-\alpha} \mathbf{1}_{|x| \geq 1}$ on $\mathbb{R}^d$. Determine the p for which each lies in $L^p(\mathbb{R}^d)$.

Recognition. A radial power; use polar coordinates, where $dx = c_d r^{d-1} dr$. The borderline is where the radial integrand $r^{-p\alpha+d-1}$ ceases to be integrable.

Solution. $\int_{|x| < 1} |x|^{-p\alpha} dx = c_d \int_0^1 r^{-p\alpha+d-1} dr$, finite iff $-p\alpha + d - 1 > -1$, i.e. $p\alpha < d$. So $f_0 \in L^p \iff p\alpha < d$. For the tail, $\int_{|x| \geq 1} r^{-p\alpha+d-1} dr < \infty$ iff $-p\alpha + d - 1 < -1$, i.e. $p\alpha > d$, so $f_\infty \in L^p \iff p\alpha > d$.

Check. The two conditions are exactly complementary across the threshold $p\alpha = d$; at $|x|^{-d/p}$ neither piece is in L^p — the function $|x|^{-d/p}$ is the canonical “just barely not in L^p ” example. This is Exercise 1.

Worked Example 39.1.10 (A function in L^2 but not L^1 and vice versa). On $(0, \infty)$ with Lebesgue measure exhibit $f \in L^2 \setminus L^1$ and $g \in L^1 \setminus L^2$, showing no inclusion holds on an infinite-measure space.

Recognition. Use the spike/tail dichotomy: L^1 tolerates tall spikes less and tails more than L^2 does in a way that depends on the exponents.

Solution. Let $f(x) = \frac{1}{x} \mathbf{1}_{x>1}$. Then $\int_1^\infty x^{-2} dx = 1 < \infty$ so $f \in L^2$, but $\int_1^\infty x^{-1} dx = \infty$, so $f \notin L^1$. Let $g(x) = x^{-1/2} \mathbf{1}_{0<x<1}$. Then $\int_0^1 x^{-1} dx = \infty$ gives $g \notin L^2$, while $\int_0^1 x^{-1/2} dx = 2 < \infty$ gives $g \in L^1$. *Check.* f lives in the tail (slow decay kills L^1), g lives in the spike (mild blow-up kills L^2); neither inclusion $L^1 \subseteq L^2$ nor $L^2 \subseteq L^1$ can hold. Contrast Proposition 39.1.4, which needs finite measure.

Worked Example 39.1.11 (ℓ^p norms of a geometric sequence). For $0 < r < 1$ let $a_n = r^n$ for $n \geq 0$, $a_n = 0$ for $n < 0$. Compute $\|a\|_{\ell^p}$ for $1 \leq p < \infty$ and confirm $\|a\|_{\ell^p}$ decreases in p .

Recognition. Counting measure: $\|a\|_{\ell^p}^p = \sum |a_n|^p$ is a geometric series.

Solution. $\|a\|_{\ell^p}^p = \sum_{n \geq 0} r^{pn} = \frac{1}{1-r^p}$, so $\|a\|_{\ell^p} = (1-r^p)^{-1/p}$. As p increases, $r^p \rightarrow 0$ so $1-r^p \rightarrow 1$ and the $1/p$ root pulls the value toward $\sup_n |a_n| = 1 = \|a\|_{\ell^\infty}$. *Check.* At $r = 1/2$, $\|a\|_{\ell^1} = (1 - \frac{1}{2})^{-1} = 2$, $\|a\|_{\ell^2} = (1 - \frac{1}{4})^{-1/2} = 2/\sqrt{3} \approx 1.155$, and $\|a\|_{\ell^\infty} = 1$; strictly decreasing, matching Proposition 39.1.5.

Worked Example 39.1.12 ($\|f\|_{L^p} \rightarrow \|f\|_{L^\infty}$ for a step function). Let $f = 3 \cdot \mathbf{1}_{[0,1]} + 1 \cdot \mathbf{1}_{(1,4]}$ on $[0, 4]$. Verify $\|f\|_{L^p} \rightarrow \|f\|_{L^\infty} = 3$ as $p \rightarrow \infty$.

Recognition. Proposition 39.1.6 on a finite-measure space: the tallest value wins in the limit.

Solution. $\|f\|_{L^p}^p = 3^p \cdot 1 + 1^p \cdot 3 = 3^p + 3$, so $\|f\|_{L^p} = (3^p + 3)^{1/p} = 3(1 + 3 \cdot 3^{-p})^{1/p}$. As $p \rightarrow \infty$, $3^{-p} \rightarrow 0$ and the bracket $\rightarrow 1$, hence $\|f\|_{L^p} \rightarrow 3$. *Check.* $p = 1$ gives 6, $p = 2$ gives $\sqrt{12} \approx 3.46$, $p = 10$ gives $(59052)^{1/10} \approx 3.00$; the value decreases to the essential sup 3, the height of the tallest plateau.

Worked Example 39.1.13 (A Cauchy sequence in L^2 and its limit). On $[0, 1]$ let $f_n(x) = \min(x^{-1/3}, n)$. Show $\{f_n\}$ is Cauchy in L^2 and identify its limit.

Recognition. The truncations increase to $f(x) = x^{-1/3}$; check $f \in L^2$ then use dominated/monotone convergence to get L^2 convergence, which forces Cauchy.

Solution. $\int_0^1 x^{-2/3} dx = 3 < \infty$, so $f(x) = x^{-1/3} \in L^2$. Since $0 \leq f_n \leq f$ and $f_n \rightarrow f$ pointwise, $|f - f_n|^2 \leq f^2 \in L^1$, and dominated convergence gives $\|f - f_n\|_{L^2} \rightarrow 0$. A norm-convergent sequence is Cauchy: $\|f_n - f_m\|_{L^2} \leq \|f_n - f\|_{L^2} + \|f - f_m\|_{L^2} \rightarrow 0$. *Check.* The limit $f = x^{-1/3}$ is unbounded yet lies in L^2 , illustrating that completeness lets us reach genuinely singular functions as limits of bounded ones — the whole point of Theorem 39.1.7.

Worked Example 39.1.14 (L^p inclusion on a finite measure space, sharp constant). On $([0, 1], m)$ verify $\|f\|_{L^1} \leq \|f\|_{L^2}$ for $f(x) = x$, and check the constant $\mu(X)^{1/1-1/2} = 1$ of Proposition 39.1.4.

Recognition. Finite measure with $\mu(X) = 1$: the proposition predicts $\|f\|_{L^{p_0}} \leq \mu(X)^{1/p_0-1/p_1} \|f\|_{L^{p_1}} = \|f\|_{L^{p_1}}$ when $\mu(X) = 1$.

Solution. $\|f\|_{L^1} = \int_0^1 x dx = \frac{1}{2}$. $\|f\|_{L^2} = (\int_0^1 x^2)^{1/2} = (1/3)^{1/2} = 1/\sqrt{3} \approx 0.577$. So $0.5 \leq 0.577$, with constant 1. *Check.* The inequality is strict because f is non-constant; equality in Proposition 39.1.4 requires $|f|$ a.e. constant, the only case where a function's mass is spread evenly enough for the Holder step to be tight.

Worked Example 39.1.15 (Failure of inclusion on infinite measure). Show that $L^2(\mathbb{R}) \not\subseteq L^1(\mathbb{R})$ by a single explicit witness, and explain why Proposition 39.1.4 does not apply.

Recognition. The proposition needs $\mu(X) < \infty$; on $\mathbb{R}$ a slowly decaying tail can be in L^2 yet not L^1 .

Solution. Let $f(x) = \frac{1}{1+|x|}$. Then $\int_{\mathbb{R}} \frac{dx}{(1+|x|)^2} = 2 < \infty$, so $f \in L^2$, but $\int_{\mathbb{R}} \frac{dx}{1+|x|} = \infty$, so $f \notin L^1$. *Check.* $\mu(\mathbb{R}) = \infty$, so the constant $\mu(X)^{1/p_0-1/p_1}$ in the proposition is infinite and the inclusion argument collapses; this is the content of Exercise 1's infinite-measure remark.

Exercises (Stein, Ch.1) – definition, completeness, inclusions.

1. Consider $L^p = L^p(\mathbb{R}^d)$ with Lebesgue measure. Let $f_0(x) = |x|^{-\alpha}$ if $|x| < 1$ and 0 otherwise; $f_\infty(x) = |x|^{-\alpha}$ if $|x| \geq 1$ and 0 otherwise. Show: (a) $f_0 \in L^p$ iff $p\alpha < d$; (b) $f_\infty \in L^p$ iff $d < p\alpha$; (c) what happens if $|x|^{-\alpha}$ is replaced by $|x|^{-\alpha} / \log(2/|x|)$ for $|x| < 1$ and by $|x|^{-\alpha} / \log(2|x|)$ for $|x| \geq 1$?
2. Consider $L^p(\mathbb{R}^d)$ for $0 < p < 1$. (a) Show that if $\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}$ for all f, g , then necessarily $p \geq 1$. (b) Show $L^p(\mathbb{R})$, $0 < p < 1$, has no nonzero bounded linear functionals: if $|\ell(f)| \leq M \|f\|_{L^p}$ for all f , then $\ell = 0$. [Hint: (a) $x^p + y^p > (x + y)^p$ for $0 < p < 1$, $x, y > 0$; (b) let $F(x) = \ell(\chi_{[0,x]})$ and consider $F(x) - F(y)$.]
3. If $f \in L^p$ and $g \in L^q$ (not a.e. zero), show equality holds in Holder's inequality iff there are nonzero constants $a, b \geq 0$ with $a|f|^p = b|g|^q$ a.e.
4. Let X be a measure space and $0 < p < 1$, with q the (negative) conjugate exponent. (a) Prove $\|fg\|_{L^1} \geq \|f\|_{L^p} \|g\|_{L^q}$. (b) If $f_1, f_2 \geq 0$ then $\|f_1 + f_2\|_{L^p} \geq \|f_1\|_{L^p} + \|f_2\|_{L^p}$. (c) $d(f, g) = \|f - g\|_{L^p}^p$ is a metric on $L^p(X)$.
5. Using the completeness argument for $L^p(X)$, show that if $\{f_n\} \rightarrow f$ in the L^p norm, then a subsequence of $\{f_n\}$ converges to f almost everywhere.
6. Let $(X, \mathcal{F}, \mu)$ be a measure space. Show: (a) the simple functions are dense in $L^\infty(X)$ when $\mu(X) < \infty$; (b) the simple functions are dense in $L^p(X)$ for $1 \leq p < \infty$. [Hint: use $E_{\ell,j} = \{x : M\ell/j \leq f(x) < M(\ell+1)/j\}$ with $M = \|f\|_{L^\infty}$.]
7. On $\mathbb{R}^d$ with Lebesgue measure, $1 \leq p < \infty$, prove: (a) the continuous functions of compact support are dense in L^p ; (b) the C^∞ functions of compact support are dense in L^p .
8. For $1 \leq p < \infty$ and $\mathbb{R}^d$ with Lebesgue measure, show that if $f \in L^p(\mathbb{R}^d)$ then $\|f(\cdot + h) - f(\cdot)\|_{L^p} \rightarrow 0$ as $|h| \rightarrow 0$ (continuity of translation). Prove this fails for $p = \infty$.
9. Let X be a measure space, $1 \leq p_0 < p_1 \leq \infty$. (a) On $L^{p_0} \cap L^{p_1}$ with $\|f\|_{L^{p_0} \cap L^{p_1}} = \|f\|_{L^{p_0}} + \|f\|_{L^{p_1}}$, show this is a norm and the space is Banach. (b) On $L^{p_0} + L^{p_1} = \{f = f_0 + f_1 : f_0 \in L^{p_0}, f_1 \in L^{p_1}\}$ with $\|f\|_{L^{p_0} + L^{p_1}} = \inf\{\|f_0\|_{L^{p_0}} + \|f_1\|_{L^{p_1}}\}$, show this is a norm and the space is Banach. (c) Show $L^p \subseteq L^{p_0} + L^{p_1}$ for $p_0 \leq p \leq p_1$.
10. Call (X, μ) *separable* if there is a countable family $\{E_k\}$ of measurable sets so that for any finite-measure E , $\mu(E \triangle E_{n_k}) \rightarrow 0$ for a subsequence depending on E . (a) Verify $\mathbb{R}^d$ with Lebesgue measure is separable. (b) $L^p(X)$ is separable (has a countable dense subset) when X is separable and $1 \leq p < \infty$.
11. (a) Show $L^\infty(\mathbb{R})$ is not separable by constructing, for each $a \in \mathbb{R}$, an $f_a \in L^\infty$ with $\|f_a - f_b\| \geq 1$ for $a \neq b$. (b) Do the same for the dual space of $L^\infty(\mathbb{R})$.

Additional exercises.

1. Compute $\|f\|_{L^p([0,1])}$ for $f(x) = x^s$ ($s > -1/p$) as a function of p , and determine $\lim_{p \rightarrow \infty} \|f\|_{L^p}$. Confirm it equals $\|f\|_{L^\infty}$.
2. For which $\alpha, \beta > 0$ does $f(x) = |x|^{-\alpha} (1 + |x|)^{-\beta}$ belong to $L^p(\mathbb{R})$? Give the answer as an interval of p in terms of α, β .
3. On $[0, 1]$ define $f_n = n \mathbf{1}_{(0, 1/n)}$. Show $\|f_n\|_{L^1} = 1$ for all n but $f_n \rightarrow 0$ a.e.; deduce L^1 convergence is strictly stronger than a.e. convergence.
4. (Non-example.) Exhibit a sequence in $L^2([0, 1])$ that converges in L^1 but not in L^2 . Explain using Proposition 39.1.4.
5. Show $\|f\|_{L^p}$ is a continuous function of $p \in [1, \infty)$ for any fixed $f \in L^1 \cap L^\infty$ on a finite-measure space. (Use the worked log-convexity example.)
6. (Diagram.) Sketch, on one set of axes, the regions of $(1/p, \text{value})$ where $|x|^{-1} \mathbf{1}_{0 < |x| < 1}$ and $|x|^{-1} \mathbf{1}_{|x| > 1}$ lie in $L^p(\mathbb{R})$, marking the common threshold. Interpret the empty overlap.
7. Prove directly (without Holder) that $\ell^1 \subseteq \ell^2 \subseteq \ell^\infty$ with norm inequalities $\|a\|_{\ell^\infty} \leq \|a\|_{\ell^2} \leq \|a\|_{\ell^1}$.

8. Let $f \in L^p(\mathbb{R}^d) \cap L^\infty(\mathbb{R}^d)$. Show $f \in L^r$ for all $r \geq p$ and $\|f\|_{L^r} \leq \|f\|_{L^p}^{p/r} \|f\|_{L^\infty}^{1-p/r}$.
9. Show the “typewriter” sequence (indicators of $[j2^{-k}, (j+1)2^{-k}]$, swept across $[0, 1]$) converges to 0 in every $L^p([0, 1])$, $1 \leq p < \infty$, but at no point x . Reconcile this with the Riesz–Fischer proof.

39.2 The Holder and Minkowski Inequalities and Duality

Conjugate exponents and Young’s inequality

Definition 39.2.1. Exponents p, q with $1 \leq p, q \leq \infty$ are *conjugate* (or *dual*) if

$$\frac{1}{p} + \frac{1}{q} = 1,$$

with the convention $1/\infty = 0$. Thus $p = 2$ is self-dual; $p = 1$ pairs with $q = \infty$ and $p = \infty$ with $q = 1$. We write p' for the conjugate of p .

The engine behind Holder’s inequality is a one-variable convexity fact, the weighted arithmetic–geometric mean inequality.

Lemma 39.2.2 (Weighted AM–GM / Young). *If $A, B \geq 0$ and $0 \leq \theta \leq 1$, then*

$$A^\theta B^{1-\theta} \leq \theta A + (1 - \theta)B.$$

At $\theta = \frac{1}{2}$ this is the classical “geometric mean $\leq$ arithmetic mean.”

Proof. If $B = 0$ both sides are 0 (or the claim is trivial). Otherwise divide by B and set $x = A/B$; it suffices to prove $x^\theta \leq \theta x + (1 - \theta)$. Let $\varphi(x) = x^\theta - \theta x - (1 - \theta)$. Then $\varphi'(x) = \theta(x^{\theta-1} - 1)$, which is positive for $0 < x < 1$ and negative for $x > 1$, so φ has a maximum at $x = 1$ where $\varphi(1) = 0$. Hence $\varphi(x) \leq 0$ for all $x \geq 0$. $\square$

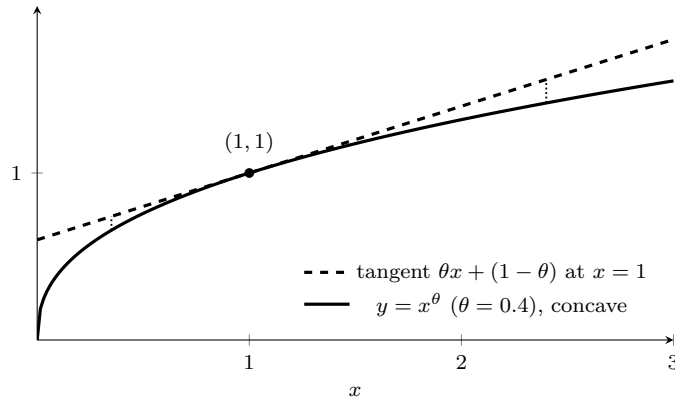


Figure 39.3. Young’s inequality is concavity of $x \mapsto x^\theta$: the curve lies under its tangent at $x = 1$, the line $\theta x + (1 - \theta)$ (dotted segments mark the gap). Equality only at $x = 1$, i.e. $A = B$.

Holder’s inequality

Theorem 39.2.3 (Holder). *Let $1 \leq p, q \leq \infty$ be conjugate. If $f \in L^p$ and $g \in L^q$ then $fg \in L^1$ and*

$$\|fg\|_{L^1} \leq \|f\|_{L^p} \|g\|_{L^q}.$$

For $p = 2$ this is the Cauchy–Schwarz inequality.

Proof. The cases $\{1, \infty\}$ are immediate: if $q = \infty$ then $|fg| \leq |f| \|g\|_{L^\infty}$ a.e., and integrating gives the bound. Take $1 < p, q < \infty$. If $\|f\|_{L^p} = 0$ or $\|g\|_{L^q} = 0$ then $fg = 0$ a.e. and there is nothing to prove. Otherwise, by

homogeneity rescale so that $\|f\|_{L^p} = \|g\|_{L^q} = 1$; we must show $\|fg\|_{L^1} \leq 1$. Apply Lemma 39.2.2 pointwise with $A = |f(x)|^p$, $B = |g(x)|^q$, $\theta = 1/p$ (so $1 - \theta = 1/q$):

$$|f(x)g(x)| \leq \frac{1}{p} |f(x)|^p + \frac{1}{q} |g(x)|^q.$$

Integrating, $\|fg\|_{L^1} \leq \frac{1}{p} \|f\|_{L^p}^p + \frac{1}{q} \|g\|_{L^q}^q = \frac{1}{p} + \frac{1}{q} = 1$. $\square$

Remark (Equality in Holder). Tracking the equality case of Young (only at $A = B$) shows equality in Holder holds iff $a|f|^p = b|g|^q$ a.e. for constants $a, b \geq 0$ not both zero (Exercise 3): $|f|^p$ and $|g|^q$ must be proportional.

Minkowski's inequality (the triangle inequality)

Theorem 39.2.4 (Minkowski). *If $1 \leq p \leq \infty$ and $f, g \in L^p$, then $f + g \in L^p$ and*

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}.$$

Proof. For $p = 1$ integrate $|f + g| \leq |f| + |g|$; for $p = \infty$ take essential suprema. Let $1 < p < \infty$. First $f + g \in L^p$ because $|f + g|^p \leq 2^p(|f|^p + |g|^p)$ (split on whether $|f| \leq |g|$ or not). Now write

$$|f + g|^p \leq |f| |f + g|^{p-1} + |g| |f + g|^{p-1}.$$

With q the conjugate of p we have $(p-1)q = p$, so $|f + g|^{p-1} \in L^q$ with $\left\| |f + g|^{p-1} \right\|_{L^q} = \|f + g\|_{L^p}^{p/q}$. Apply Holder to each term:

$$\|f + g\|_{L^p}^p \leq (\|f\|_{L^p} + \|g\|_{L^p}) \|f + g\|_{L^p}^{p/q}.$$

If $\|f + g\|_{L^p} > 0$, divide by $\|f + g\|_{L^p}^{p/q}$; since $p - p/q = 1$ the left side becomes $\|f + g\|_{L^p}$, giving the inequality. $\square$

Remark. Minkowski is exactly the statement that $\|\cdot\|_{L^p}$ is a norm (the other axioms are trivial), so together with Theorem 39.1.7 it makes L^p a Banach space. For $0 < p < 1$ Minkowski *reverses* on nonnegative functions (Exercise 4), which is why $\|\cdot\|_{L^p}$ is not a norm there.

Duality: the bilinear pairing

Holder's inequality says the bilinear form $(f, g) \mapsto \int fg$ is bounded on $L^p \times L^q$. Each fixed $g \in L^q$ thus defines a bounded linear functional on L^p ,

$$\ell_g(f) = \int_X f g \, d\mu, \quad |\ell_g(f)| \leq \|g\|_{L^q} \|f\|_{L^p},$$

so $\|\ell_g\| \leq \|g\|_{L^q}$. This embeds $L^q \hookrightarrow (L^p)^*$. The converse of Holder (Lemma 39.2.5) shows the embedding is isometric, and Section 39.4 proves it is *onto* for $1 \leq p < \infty$.

$$L^q \xrightarrow{g \mapsto \ell_g} (L^p)^*$$

Lemma 39.2.5 (Converse of Holder). *Let $1 \leq p, q \leq \infty$ be conjugate. (i) If $g \in L^q$ then $\|g\|_{L^q} = \sup_{\|f\|_{L^p} \leq 1} \left| \int fg \right|$. (ii) If g is integrable on every set of finite measure and*

$$M := \sup \left\{ \left| \int fg \right| : \|f\|_{L^p} \leq 1, f \text{ simple} \right\} < \infty,$$

then $g \in L^q$ with $\|g\|_{L^q} = M$.

Proof. (i) “ $\geq$ ” is Holder. For “ $\leq$ ”, assume $g \not\equiv 0$. If $1 < p, q < \infty$ take the *extremal* test function

$$f = \frac{|g|^{q-1} \operatorname{sign} g}{\|g\|_{L^q}^{q-1}},$$

where $\operatorname{sign} z = z/|z|$ (and 0 at 0). Since $p(q-1) = q$ one computes $\|f\|_{L^p}^p = \int |g|^{p(q-1)} / \|g\|_{L^q}^{p(q-1)} = 1$ and $\int fg = \|g\|_{L^q}$. If $q = 1, p = \infty$ take $f = \operatorname{sign} g$; if $q = \infty, p = 1$ take $f = \mathbf{1}_E \operatorname{sign} g / \mu(E)$ on a set E of finite positive measure where $|g| \geq \|g\|_{L^\infty} - \varepsilon$. (ii) Approximate: simple g_n with $|g_n| \leq |g|$ and $g_n \rightarrow g$; the extremal test functions for g_n give $\|g_n\|_{L^q} \leq M$, and Fatou’s lemma passes to the limit, so $\|g\|_{L^q} \leq M$; Holder gives $\geq$. $\square$

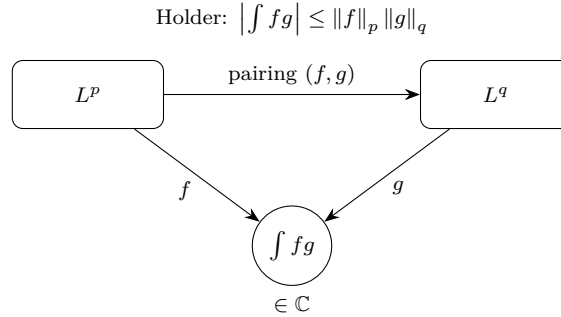


Figure 39.4. The duality pairing. Each $g \in L^q$ gives a bounded functional $\ell_g = \int(\cdot)g$ on L^p ; the converse of Holder shows $\|\ell_g\| = \|g\|_{L^q}$ exactly, by testing against the extremal $f = |g|^{q-1} \operatorname{sign} g / \|g\|_q^{q-1}$. (Schematic.)

Worked Example 39.2.6 (Holder with explicit functions). On $[0, 1]$ let $f(x) = x^{-1/4}$ and $g(x) = x^{-1/4}$. Verify Holder with $p = q = 2$ and identify when it would be sharp.

Recognition. $p = q = 2$ is Cauchy–Schwarz. Compute all three norms.

Solution. $fg = x^{-1/2}$, $\|fg\|_{L^1} = \int_0^1 x^{-1/2} = 2$. And $\|f\|_{L^2}^2 = \int_0^1 x^{-1/2} = 2$, so $\|f\|_{L^2} = \sqrt{2} = \|g\|_{L^2}$, product = 2. Thus $2 \leq 2$: equality. *Check.* Equality is expected because $|f|^2 = |g|^2$, the proportionality condition of the equality case — here with $a = b = 1$.

Worked Example 39.2.7 (A strict Holder inequality). On $[0, 1]$ let $f(x) = 1$ and $g(x) = x$, with $p = q = 2$. Show the inequality is strict and compute the deficit.

Recognition. Cauchy–Schwarz; strictness because $|f|^2$ and $|g|^2$ are not proportional.

Solution. $\|fg\|_{L^1} = \int_0^1 x \, dx = \frac{1}{2}$. $\|f\|_{L^2} = 1$, $\|g\|_{L^2} = (\int_0^1 x^2)^{1/2} = 1/\sqrt{3}$, product = $1/\sqrt{3} \approx 0.577$. So $0.5 < 0.577$: strict. *Check.* $f^2 \equiv 1$ is not a constant multiple of $g^2 = x^2$, so by the equality criterion strictness is forced; the gap 0.077 measures the failure of proportionality.

Worked Example 39.2.8 (Minkowski need not be equality). On $[0, 1]$ with $p = 2$, let $f = \mathbf{1}_{[0, 1/2]}$ and $g = \mathbf{1}_{[1/2, 1]}$. Compare $\|f + g\|_{L^2}$ with $\|f\|_{L^2} + \|g\|_{L^2}$.

Recognition. Triangle inequality; equality in Minkowski needs f, g “parallel” ($f = cg$ a.e., $c \geq 0$), which fails for disjoint supports.

Solution. $f + g = \mathbf{1}_{[0, 1]}$, $\|f + g\|_{L^2} = 1$. Each of $\|f\|_{L^2} = \|g\|_{L^2} = (1/2)^{1/2} = 1/\sqrt{2}$, sum = $\sqrt{2} \approx 1.414$. So $1 < 1.414$: strict. *Check.* Disjoint supports are the opposite of parallel, so the triangle inequality is far from tight; for $p = 2$, $\|f + g\|_{L^2}^2 = \|f\|_{L^2}^2 + \|g\|_{L^2}^2$ (Pythagoras), reflecting orthogonality.

Worked Example 39.2.9 (The converse-of-Holder extremal function). For $g(x) = x$ on $[0, 1]$ and $p = q = 2$, exhibit the unit-norm f that makes $\int fg$ attain $\|g\|_{L^2}$, illustrating Lemma 39.2.5(i).

Recognition. The extremal is $f = |g|^{q-1} \operatorname{sign} g / \|g\|_q^{q-1}$; for $q = 2$ this is $f = g / \|g\|_2$.

Solution. $\|g\|_{L^2} = 1/\sqrt{3}$, so $f = g / \|g\|_2 = \sqrt{3}x$. Then $\|f\|_{L^2}^2 = 3 \int_0^1 x^2 = 1$ (unit norm), and $\int_0^1 fg = \sqrt{3} \int_0^1 x^2 = \sqrt{3} \cdot \frac{1}{3} = 1/\sqrt{3} = \|g\|_{L^2}$. *Check.* The supremum in the converse is attained at this f , confirming $\|\ell_g\| = \|g\|_{L^2}$ and the isometry $L^q \hookrightarrow (L^p)^*$.

Worked Example 39.2.10 (Interpolation of norms (log-convexity)). Let $0 < p_0 < p < p_1 \leq \infty$ and $f \in L^{p_0} \cap L^{p_1}$. Prove $\|f\|_{L^p} \leq \|f\|_{L^{p_0}}^{1-t} \|f\|_{L^{p_1}}^t$ where $\frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1}$.

Recognition. Split $|f|^p = |f|^{(1-t)p} |f|^{tp}$ and apply Holder with the exponents that make each factor land in L^{p_0}, L^{p_1} .

Solution. Take exponents $r = p_0/((1-t)p)$ and $r' = p_1/(tp)$; the choice of t gives $1/r + 1/r' = (1-t)p/p_0 + tp/p_1 = 1$. Holder yields

$$\int |f|^p = \int |f|^{(1-t)p} |f|^{tp} \leq \left(\int |f|^{p_0} \right)^{(1-t)p/p_0} \left(\int |f|^{p_1} \right)^{tp/p_1} = \|f\|_{L^{p_0}}^{(1-t)p} \|f\|_{L^{p_1}}^{tp}.$$

Taking p th roots gives the claim. *Check.* This is Exercise 20; it says $p \mapsto \log \|f\|_{L^p}$ is convex in $1/p$, the analytic skeleton of the Riesz–Thorin interpolation theorem in Section 39.4.

Worked Example 39.2.11 (Multiple Holder for three functions). On a probability space let $f, g, h \geq 0$ with $\int f^3 = \int g^3 = \int h^3 = 1$. Bound $\int fgh$.

Recognition. The exponents 3, 3, 3 satisfy $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$, so the multiple Holder inequality (Exercise 16) applies.

Solution. $\int fgh \leq \|f\|_{L^3} \|g\|_{L^3} \|h\|_{L^3} = 1 \cdot 1 \cdot 1 = 1$. *Check.* Equality would need $f^3 = g^3 = h^3$ a.e., i.e. $f = g = h$; for instance $f = g = h \equiv 1$ gives $\int fgh = 1$, attaining the bound. The inequality is the natural generalization of Cauchy–Schwarz from two to three factors.

Worked Example 39.2.12 (When Minkowski is an equality). On $[0, 1]$ with $p = 2$, find nonzero f, g making $\|f + g\|_{L^2} = \|f\|_{L^2} + \|g\|_{L^2}$, and explain the geometry.

Recognition. Equality in Minkowski needs f and g “parallel”: $g = cf$ a.e. with $c \geq 0$.

Solution. Take $f(x) = x$ and $g(x) = 2x = 2f$. Then $f + g = 3x$, $\|f + g\|_{L^2} = 3/\sqrt{3} = \sqrt{3}$, while $\|f\|_{L^2} + \|g\|_{L^2} = 1/\sqrt{3} + 2/\sqrt{3} = 3/\sqrt{3} = \sqrt{3}$: equality. *Check.* The segment from f to g in L^2 is radial (same direction), so the triangle degenerates to a line and the triangle inequality is tight — the opposite extreme from the disjoint-support case.

Exercises (Stein, Ch.1) – inequalities, convolution, interpolation.

1. Minkowski’s inequality for integrals. For measure spaces (X_1, μ_1) , (X_2, μ_2) and $1 \leq p \leq \infty$, show that if $f(x_1, x_2) \geq 0$ is measurable on $X_1 \times X_2$ then

$$\left\| \int f(x_1, x_2) d\mu_2 \right\|_{L^p(X_1)} \leq \int \|f(\cdot, x_2)\|_{L^p(X_1)} d\mu_2.$$

Extend to complex-valued f when the right side is finite. [Hint: for $1 < p < \infty$ use Holder and its converse, Lemma 39.2.5.]

2. Multiple Holder inequality. If $f_j \in L^{p_j}(X)$, $j = 1, \dots, N$, with $\sum_{j=1}^N 1/p_j = 1$ and $p_j \geq 1$, then $\left\| \prod_{j=1}^N f_j \right\|_{L^1} \leq \prod_{j=1}^N \|f_j\|_{L^{p_j}}$.

3. Convolution on $\mathbb{R}^d$, $(f * g)(x) = \int f(x-y)g(y) dy$. (a) If $f \in L^p$, $1 \leq p \leq \infty$, and $g \in L^1$, show $f * g$ is well defined a.e. and $\|f * g\|_{L^p} \leq \|f\|_{L^p} \|g\|_{L^1}$. (b) For a finite Borel measure μ , $(f * \mu)(x) = \int f(x-y) d\mu(y)$ satisfies $\|f * \mu\|_{L^p} \leq \|f\|_{L^p} \|\mu\|(\mathbb{R}^d)$. (c) If $f \in L^p$, $g \in L^q$ conjugate, then $f * g \in L^\infty$, $\|f * g\|_{L^\infty} \leq \|f\|_{L^p} \|g\|_{L^q}$, $f * g$ is uniformly continuous, and for $1 < p < \infty$, $(f * g)(x) \rightarrow 0$ as $|x| \rightarrow \infty$.

4. Mixed-norm spaces $L_t^r(L_x^p) = L^{p,r}$ on $\mathbb{R}^d \times \mathbb{R}$ with $\|f\|_{L^{p,r}} = \left(\int_{\mathbb{R}} \left(\int_{\mathbb{R}^d} |f(x,t)|^p dx \right)^{r/p} dt \right)^{1/r}$. (a) Show $L^{p,r}$ is complete. (b) Prove the Holder form $\int |fg| \leq \|f\|_{L^{p,r}} \|g\|_{L^{p',r'}}$ with $1/p + 1/p' = 1$, $1/r + 1/r' = 1$. (c) For f integrable on finite-measure sets, $\|f\|_{L^{p,r}} = \sup \int |fg|$ over simple g with $\|g\|_{L^{p',r'}} \leq 1$. (d) Conclude $(L^{p,r})^* = L^{p',r'}$ for $1 \leq p, r < \infty$.

5. Young’s inequality. For $1 \leq p, q, r \leq \infty$ on $\mathbb{R}^d$, prove $\|f * g\|_{L^q} \leq \|f\|_{L^p} \|g\|_{L^r}$ whenever $1/q = 1/p + 1/r - 1$. [Hint: assume $f, g \geq 0$ and split $f(y)g(x-y) = f(y)^a g(x-y)^b [f(y)^{1-a} g(x-y)^{1-b}]$, then use Exercise 16.]

6. Let X be a measure space, $0 < p_0 < p < p_1 \leq \infty$, $f \in L^{p_0} \cap L^{p_1}$. Show $f \in L^p$ and $\|f\|_{L^p} \leq \|f\|_{L^{p_0}}^{1-t} \|f\|_{L^{p_1}}^t$ where $1/p = (1-t)/p_0 + t/p_1$ (log-convexity of L^p norms).
7. Jensen's inequality. If φ is a nonnegative convex function on $\mathbb{R}$ and f is real-valued integrable on X with $\mu(X) = 1$, then $\varphi(\int f d\mu) \leq \int \varphi(f) d\mu$. Note $\varphi(t) = |t|^p$ ($p \geq 1$) recovers a Holder consequence and $\varphi(t) = e^{at}$ is another useful case.
8. Young's inequality (second form). Let φ, ψ be continuous strictly increasing inverse functions on $[0, \infty)$, $\Phi(x) = \int_0^x \varphi$, $\Psi(x) = \int_0^x \psi$. (a) Prove $ab \leq \Phi(a) + \Psi(b)$ for $a, b \geq 0$; with $\varphi(x) = x^{p-1}$, $\psi(y) = y^{q-1}$ this gives $A^\theta B^{1-\theta} \leq \theta A + (1-\theta)B$. (b) Equality holds only if $b = \varphi(a)$.
9. Orlicz spaces. Let Φ be continuous, convex, increasing on $[0, \infty)$ with $\Phi(0) = 0$. Set $L^\Phi = \{f : \int_X \Phi(|f|/M) d\mu < \infty \text{ for some } M > 0\}$ and $\|f\|_\Phi = \inf \{M > 0 : \int \Phi(|f|/M) d\mu \leq 1\}$. Prove (a) L^Φ is a vector space, (b) $\|\cdot\|_\Phi$ is a norm, (c) L^Φ is complete. Note $\Phi(t) = t^p$ gives $L^\Phi = L^p$.
10. Let $1 \leq p_0 < p_1 < \infty$. (a) With $\Phi(t) = t^{p_0}$ for $0 \leq t \leq 1$ and t^{p_1} for $t \geq 1$, show L^Φ is equivalent to $L^{p_0} \cap L^{p_1}$. (b) With $\Psi(t) = \int_0^t \psi$, $\psi(u) = u^{p_1-1}$ for $0 \leq u \leq 1$ and u^{p_0-1} for $u \geq 1$, show L^Ψ is equivalent to $L^{p_0} + L^{p_1}$.

Additional exercises.

1. Verify Holder's inequality numerically for $f(x) = x$, $g(x) = 1 - x$ on $[0, 1]$ with $p = q = 2$, and decide whether it is sharp.
2. Use Holder to prove the integral Cauchy–Schwarz inequality $(\int fg)^2 \leq (\int f^2)(\int g^2)$ and state the equality condition.
3. (AM–GM in action.) For $a, b, c > 0$ prove $\sqrt[3]{abc} \leq \frac{1}{3}(a + b + c)$ as a special case of Lemma 39.2.2 iterated, and identify when equality holds.
4. Let $1 < p < \infty$. Show the map $g \mapsto \ell_g$, $\ell_g(f) = \int fg$, is an isometric linear injection $L^q \rightarrow (L^p)^*$, using only Holder and its converse.
5. (Non-example.) Show Minkowski's inequality *reverses* for $p = 1/2$ on $f = \mathbf{1}_{[0,1]}$, $g = \mathbf{1}_{[1,2]}$: compute both sides and confirm $\|f + g\|_{1/2} > \|f\|_{1/2} + \|g\|_{1/2}$.
6. Prove the three-term Holder inequality $\int |fgh| \leq \|f\|_{p_1} \|g\|_{p_2} \|h\|_{p_3}$ when $1/p_1 + 1/p_2 + 1/p_3 = 1$, by grouping two factors.
7. For $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ show $\|f\|_{L^{3/2}} \leq \|f\|_{L^1}^{1/3} \|f\|_{L^2}^{2/3}$, identifying the interpolation parameter t .
8. (Diagram.) Draw the graph of $x \mapsto x^\theta$ for $\theta = 1/3$ together with its tangent line at $x = 1$, and shade the gap that Young's inequality measures.
9. Compute $\|f * g\|_{L^\infty}$ for $f = g = \mathbf{1}_{[0,1]}$ on $\mathbb{R}$ and confirm the bound $\|f * g\|_\infty \leq \|f\|_2 \|g\|_2$ from Exercise 17(c).
10. Prove Jensen's inequality $e^{\int f} \leq \int e^f$ on a probability space, and use it to show the geometric mean of positive numbers never exceeds their arithmetic mean.

39.3 Banach Spaces and the Hahn-Banach Theorem

Normed and Banach spaces

Definition 39.3.1. A *normed vector space* is a vector space V over $\mathbb{R}$ or $\mathbb{C}$ with a map $\|\cdot\| : V \rightarrow [0, \infty)$ such that for all $v, w \in V$ and scalars α :

$$\|v\| = 0 \iff v = 0, \quad \|\alpha v\| = |\alpha| \|v\|, \quad \|v + w\| \leq \|v\| + \|w\|.$$

V is *complete* if every Cauchy sequence converges in V . A complete normed vector space is a *Banach space*.

Completeness is the hypothesis that does all the work: it guarantees that limiting constructions (series, fixed points, extensions by density) stay inside the space.

Example 39.3.2 (Banach spaces). (1) $\mathbb{R}$, $\mathbb{R}^d$ (Euclidean norm), and every Hilbert space. (2) The L^p spaces, $1 \leq p \leq \infty$ (Theorems 39.1.7, 39.1.8); L^2 is the only Hilbert space among them (Exercise 25). (3) $C(X)$, continuous functions on a compact metric space X with $\|f\| = \sup_X |f|$: complete because a uniform limit of continuous functions is continuous. (4) The Holder/Lipschitz space $\Lambda_\alpha(\mathbb{R}^d)$, $0 < \alpha \leq 1$, with

$$\|f\|_{\Lambda_\alpha} = \sup_x |f(x)| + \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\alpha}.$$

(5) The Sobolev space $L_k^p(\mathbb{R}^d)$ of $f \in L^p$ whose weak derivatives $\partial_x^\alpha f$ (defined by $\int g_\alpha \varphi = (-1)^{|\alpha|} \int f \partial_x^\alpha \varphi$ for all test φ) up to order k lie in L^p , with norm $\sum_{|\alpha| \leq k} \|\partial_x^\alpha f\|_{L^p}$.

Remark. For $0 < p < 1$, $\|\cdot\|_{L^p}$ violates the triangle inequality, so L^p is not a normed space (it is only a metric space, via $d(f, g) = \int |f - g|^p$, Exercise 4c).

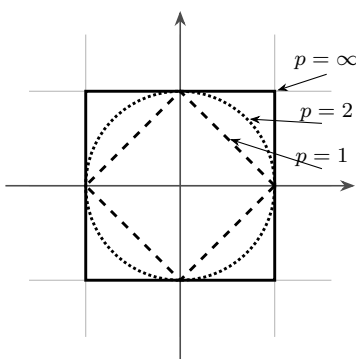


Figure 39.5. Unit balls of $\ell^p(\mathbb{R}^2)$. The diamond ($p = 1$), circle ($p = 2$), and square ($p = \infty$) are nested $B_1 \subset B_2 \subset B_\infty$; the ball grows as p increases, the geometric face of $\|\cdot\|_p$ decreasing in p (finite total mass).

Bounded linear functionals and the dual space

For simplicity we work over $\mathbb{R}$ in this section; the complex modifications are routine (Section 39.4 and Exercise 33).

Definition 39.3.3. A *linear functional* on a Banach space B is a linear map $\ell : B \rightarrow \mathbb{R}$. It is *bounded* if $\|\ell\| := \sup_{\|f\| \leq 1} |\ell(f)| < \infty$, and *continuous* if $\ell(f_n) \rightarrow \ell(f)$ whenever $f_n \rightarrow f$.

Proposition 39.3.4. A linear functional on B is continuous if and only if it is bounded.

Proof. A linear functional is continuous everywhere iff continuous at 0. If ℓ is bounded, $|\ell(f) - \ell(g)| = |\ell(f - g)| \leq \|\ell\| \|f - g\|$, giving continuity. Conversely if ℓ is continuous at 0, choose δ with $|\ell(f)| \leq 1$ for $\|f\| \leq \delta$; then for $h \neq 0$, $\delta h / \|h\|$ has norm δ , so $|\ell(\delta h / \|h\|)| \leq 1$, i.e. $|\ell(h)| \leq \delta^{-1} \|h\|$. Thus ℓ is bounded with $\|\ell\| \leq 1/\delta$. $\square$

Definition 39.3.5. The *dual space* B^* is the vector space of all bounded linear functionals on B with the norm $\|\ell\| = \sup_{\|f\| \leq 1} |\ell(f)| = \sup_{f \neq 0} |\ell(f)| / \|f\|$.

Theorem 39.3.6. B^* is a Banach space (whether or not B is complete).

Proof. $\|\cdot\|$ is clearly a norm; we check completeness. Let $\{\ell_n\}$ be Cauchy in B^* . For each f , $\{\ell_n(f)\}$ is Cauchy in $\mathbb{R}$ (since $|\ell_n(f) - \ell_m(f)| \leq \|\ell_n - \ell_m\| \|f\|$), hence converges to a value $\ell(f)$; the map ℓ is linear. With $M = \sup_n \|\ell_n\| < \infty$, $|\ell(f)| \leq |(\ell - \ell_n)(f)| + M \|f\| \rightarrow M \|f\|$, so ℓ is bounded. Given $\varepsilon > 0$, pick N with $\|\ell_n - \ell_m\| < \varepsilon$ for $n, m > N$; letting $m \rightarrow \infty$ in $|(\ell_n - \ell_m)(f)| \leq \varepsilon \|f\|$ gives $|(\ell_n - \ell)(f)| \leq \varepsilon \|f\|$, so $\|\ell_n - \ell\| \leq \varepsilon$ for $n > N$. $\square$

Separation of convex sets

The geometric content of linear functionals is that their level sets are hyperplanes, and hyperplanes separate convex bodies from external points.

Definition 39.3.7. In a real vector space V : a (affine) hyperplane is a set $H = \{v : \ell(v) = a\}$ for a nonzero linear functional ℓ and $a \in \mathbb{R}$. A set $K \subseteq V$ is *convex* if $v_0, v_1 \in K \Rightarrow (1-t)v_0 + tv_1 \in K$ for $0 \leq t \leq 1$.

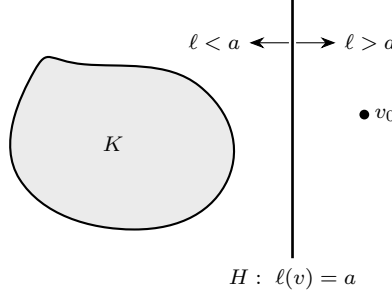


Figure 39.6. Separation principle: a convex K and an external point v_0 can be put on opposite sides of a hyperplane $\{\ell = a\}$, with $\ell(v_0) \geq a > \ell(v)$ for $v \in K$. The separating ℓ is built from the Minkowski gauge of K . (Schematic.)

Proposition 39.3.8 (Separation in $\mathbb{R}^d$ via the gauge). *If $K \subseteq \mathbb{R}^d$ is convex, open, and $v_0 \notin K$, there is a nonzero linear functional ℓ with $\ell(v_0) \geq a > \ell(v)$ for all $v \in K$.*

Sketch. Translate so $0 \in K$. The Minkowski gauge $p(v) = \inf\{r > 0 : v/r \in K\}$ is sublinear: $p(\alpha v) = \alpha p(v)$ for $\alpha \geq 0$ and $p(v_1 + v_2) \leq p(v_1) + p(v_2)$ (the latter from convexity, taking $t = r_2/(r_1 + r_2)$), and $p(v) < 1 \iff v \in K$. Define ℓ on $\mathbb{R}v_0$ by $\ell(v_0) = 1$; then $\ell \leq p$ there. Extend one dimension at a time (as in Hahn–Banach below) keeping $\ell \leq p$; the final ℓ has $\ell < 1$ on K and $\ell(v_0) = 1$, the desired separation with $a = 1$. $\square$

The Hahn–Banach theorem

Theorem 39.3.9 (Hahn–Banach, real case). *Let V be a real vector space, $p : V \rightarrow \mathbb{R}$ sublinear ($p(\alpha v) = \alpha p(v)$ for $\alpha \geq 0$, $p(v + w) \leq p(v) + p(w)$), $V_0 \subseteq V$ a subspace, and $\ell_0 : V_0 \rightarrow \mathbb{R}$ linear with $\ell_0 \leq p$ on V_0 . Then ℓ_0 extends to a linear $\ell : V \rightarrow \mathbb{R}$ with $\ell \leq p$ on all of V .*

Proof. One-step extension. Pick $v_1 \notin V_0$ and try $\ell_1(\alpha v_1 + w) = \alpha c + \ell_0(w)$ for $w \in V_0$. The constraint $\ell_1 \leq p$ on $V_1 = V_0 + \mathbb{R}v_1$ becomes, after splitting on the sign of α ,

$$-p(-v_1 + w') + \ell_0(w') \leq c \leq p(v_1 + w) - \ell_0(w) \quad \text{for all } w, w' \in V_0.$$

A valid c exists because the left side never exceeds the right: $\ell_0(w) + \ell_0(w') \leq p(w + w') \leq p(v_1 + w) + p(-v_1 + w')$, using sublinearity. *Transfinite induction.* Well-order the vectors outside V_0 ; if some vector were not “extendable,” take the least such v_1 , extend ℓ_0 to the span V'_0 of V_0 and all smaller vectors (by the assumption of minimality), then apply the one-step extension to V'_0 and v_1 — a contradiction. (Equivalently, apply Zorn’s lemma to partial extensions.) $\square$

Theorem 39.3.10 (Hahn–Banach, complex case). *Let V be a complex vector space and $p : V \rightarrow [0, \infty)$ satisfy $p(\alpha v) = |\alpha| p(v)$ ($\alpha \in \mathbb{C}$) and $p(v + w) \leq p(v) + p(w)$. If ℓ_0 is linear on a subspace V_0 with $|\ell_0| \leq p$, it extends to ℓ on V with $|\ell| \leq p$.*

Idea. Write $u = \operatorname{Re} \ell_0$, a real-linear functional with $u \leq p$. The complex functional is recovered by $\ell_0(v) = u(v) - iu(iv)$. Apply the real Theorem 39.3.9 to u to get a real extension $U \leq p$, set $\ell(v) = U(v) - iU(iv)$, and a rotation argument ($\ell(v) = e^{i\vartheta} |\ell(v)|$) gives $|\ell| \leq p$. This is Exercise 33. $\square$

Consequences of Hahn–Banach

Proposition 39.3.11 (Norming functional). *For $f_0 \in B$ with $\|f_0\| = M$ there is $\ell \in B^*$ with $\ell(f_0) = M$ and $\|\ell\| = 1$. Hence B^* separates points and $\|f_0\| = \sup_{\|\ell\| \leq 1} |\ell(f_0)|$.*

Proof. Define $\ell_0(\alpha f_0) = \alpha M$ on $\mathbb{R}f_0$. With $p(f) = \|f\|$ (sublinear), one has $\ell_0(\alpha f_0) = \alpha M \leq |\alpha| M = p(\alpha f_0)$. Extend by Theorem 39.3.9 to $\ell \leq p = \|\cdot\|$; applying this to $-f$ too gives $|\ell(f)| \leq \|f\|$, so $\|\ell\| \leq 1$, while $\ell(f_0) = M = \|f_0\|$ forces $\|\ell\| \geq 1$. $\square$

Proposition 39.3.12 (Dual of a bounded operator). *If $T : B_1 \rightarrow B_2$ is bounded linear, its adjoint $T^* : B_2^* \rightarrow B_1^*$, defined by $(T^*\ell_2)(f_1) = \ell_2(Tf_1)$, is bounded with $\|T^*\| = \|T\|$.*

Proof. $|(T^*\ell_2)(f_1)| = |\ell_2(Tf_1)| \leq \|\ell_2\| \|Tf_1\| \leq \|\ell_2\| \|T\| \|f_1\|$, so $\|T^*\| \leq \|T\|$. Conversely, given $\varepsilon > 0$ pick f_1 , $\|f_1\| = 1$, with $\|Tf_1\| \geq \|T\| - \varepsilon$; by Proposition 39.3.11 there is ℓ_2 , $\|\ell_2\| = 1$, with $\ell_2(Tf_1) = \|Tf_1\|$. Then $\|T^*\ell_2\| \geq (T^*\ell_2)(f_1) \geq \|T\| - \varepsilon$, so $\|T^*\| \geq \|T\|$. $\square$

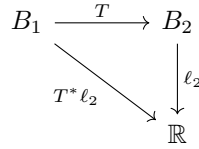


Figure 39.7. The adjoint T^* is defined so this triangle commutes: $T^*\ell_2 = \ell_2 \circ T$, i.e. $(T^*\ell_2)(f_1) = \ell_2(Tf_1)$. Thus $T^* : B_2^* \rightarrow B_1^*$ is precisely precomposition with T , and $\|T^*\| = \|T\|$.

Proposition 39.3.13 (Extension by density). *If $S \subseteq B_1$ is a dense subspace and $T_0 : S \rightarrow B_2$ satisfies $\|T_0 f\| \leq M \|f\|$, then T_0 has a unique bounded extension $T : B_1 \rightarrow B_2$ with the same bound.*

Proof. For $f \in B_1$ take $f_n \in S$, $f_n \rightarrow f$. Then $\|T_0 f_n - T_0 f_m\| \leq M \|f_n - f_m\| \rightarrow 0$, so $\{T_0 f_n\}$ is Cauchy in the complete space B_2 ; define $Tf = \lim T_0 f_n$ (independent of the sequence). T is linear with the same bound. $\square$

Theorem 39.3.14 (L^1 is not the dual of L^∞). *On $\mathbb{R}$ with Lebesgue measure there is $\ell \in (L^\infty)^*$ not of the form $\ell(f) = \int fg$ with $g \in L^1$.*

Proof. Let $\mathcal{C} \subseteq L^\infty(\mathbb{R})$ be the bounded continuous functions, and $\ell_0(f) = f(0)$ on $\mathcal{C}$ (a “Dirac delta”). Then $|\ell_0(f)| \leq \|f\|_{L^\infty}$, so by Theorem 39.3.9 (with $p = \|\cdot\|_{L^\infty}$) it extends to ℓ on L^∞ with $|\ell| \leq \|\cdot\|_{L^\infty}$. If $\ell(f) = \int fg$ for some $g \in L^1$, then for every continuous trapezoid f vanishing near 0 we get $\int fg = f(0) = 0$; a limiting argument gives $\int_I g = 0$ for every interval I missing 0, hence for all I , so $g = 0$ a.e. — contradicting $\ell \neq 0$. $\square$

Remark (Baire category and completeness). Completeness has a second face: the *Baire category theorem* says a complete metric space is not a countable union of nowhere-dense sets. This is the engine behind the uniform boundedness, open mapping, and closed graph theorems for Banach spaces. The picture is a shrinking chain of dense open sets whose intersection is still dense (hence nonempty).

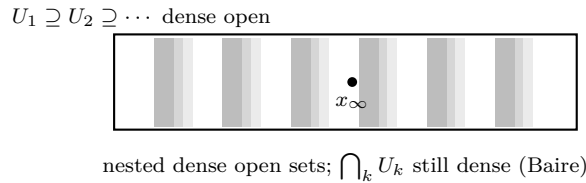


Figure 39.8. Baire category: in a complete space, countably many dense open sets have dense (in particular nonempty) intersection. A point x_∞ survives every stage. This is the topological payoff of completeness, complementing the metric one. (Schematic.)

The problem of measure

Theorem 39.3.15 (Finitely-additive invariant extension). *There is an extended-valued $\hat{m} \geq 0$ on all subsets of $\mathbb{R}$ with (i) $\hat{m}(E_1 \cup E_2) = \hat{m}(E_1) + \hat{m}(E_2)$ for disjoint E_1, E_2 ; (ii) $\hat{m}(E) = m(E)$ for Lebesgue measurable E ; (iii) $\hat{m}(E + h) = \hat{m}(E)$ for all $h \in \mathbb{R}$. It is only finitely additive (non-measurable sets preclude countable additivity).*

Idea. On the circle $\mathbb{R}/\mathbb{Z}$ build a translation-invariant Banach integral I on all bounded functions extending the Lebesgue integral: take the sublinear

$$p(f) = \inf_A \sup_x \frac{1}{N} \sum_{j=1}^N f(x + a_j), \quad A = \{a_1, \dots, a_N\},$$

check $I_0(f) = \int_0^1 f \leq p(f)$ for measurable f , extend by Theorem 39.3.9, and verify $p(f - f_h) \leq 0$ to get translation invariance. Then $\hat{m}(E) = I(\mathbf{1}_E)$ on $\mathbb{R}/\mathbb{Z}$, and piece the circle copies $(j, j+1]$ together to cover $\mathbb{R}$. This is the Hahn–Banach theorem applied where no countably-additive object exists. $\square$

Worked Example 39.3.16 (Computing an operator norm via the adjoint). On $\mathbb{R}^2$ with the ℓ^1 norm, let $T(x_1, x_2) = (x_1 + x_2, x_2)$. Find $\|T\|$.

Recognition. $\|T\| = \sup_{\|x\|_1 \leq 1} \|Tx\|_1$; on a polytope ball the sup is at a vertex.

Solution. The ℓ^1 unit ball has vertices $\pm e_1, \pm e_2$. $Te_1 = (1, 0)$, $\|Te_1\|_1 = 1$; $Te_2 = (1, 1)$, $\|Te_2\|_1 = 2$. By convexity the max over the ball is at a vertex, so $\|T\| = 2$. *Check.* $\|T(0, 1)\|_1 = 2 = 2\|(0, 1)\|_1$ attains the bound, and Proposition 39.3.12 predicts $\|T^*\| = 2$ as well; indeed T^* on $(\mathbb{R}^2, \ell^\infty)$ is $(y_1, y_2) \mapsto (y_1, y_1 + y_2)$ with $\|y\|_\infty = 2$.

Worked Example 39.3.17 (A Hahn–Banach extension on $\mathbb{R}^2$). Let $V_0 = \{(x, 0)\} \subset \mathbb{R}^2$ with $\ell_0(x, 0) = x$, and $p(x, y) = |x| + |y|$ (the ℓ^1 norm). Find all norm-preserving extensions $\ell \leq p$.

Recognition. One-step extension: choose $c = \ell(0, 1)$ within the gauge constraints.

Solution. Write $\ell(x, y) = x + cy$. The constraint $\ell \leq p$ means $x + cy \leq |x| + |y|$ for all x, y , equivalently $|c| \leq 1$. *Check.* $\|\ell_0\| = 1$ (since $|\ell_0(x, 0)| = |x| = p(x, 0)$), and every $c \in [-1, 1]$ gives $\|\ell\| = 1$ — the extension is non-unique, the typical situation off a Hilbert space.

Worked Example 39.3.18 (The norming functional for a concrete vector). In ℓ^1 find the unit functional ℓ achieving $\ell(f_0) = \|f_0\|_1$ for $f_0 = (2, -3, 0, \dots)$, illustrating Proposition 39.3.11.

Recognition. $(\ell^1)^* = \ell^\infty$; the norming functional is $\text{sign } f_0$.

Solution. Take $\ell = (\text{sign } 2, \text{sign } (-3), 0, \dots) = (1, -1, 0, \dots) \in \ell^\infty$, $\|\ell\|_\infty = 1$. Then $\ell(f_0) = 1 \cdot 2 + (-1)(-3) = 5 = \|f_0\|_1$. *Check.* $\|\ell\| = 1$ and $\ell(f_0) = \|f_0\|$, exactly the conclusion of the proposition; here ℓ is unique because no coordinate of f_0 is zero on its support.

Worked Example 39.3.19 (Boundedness vs. continuity: an unbounded functional). On c_{00} (finitely-supported sequences) with the ℓ^∞ norm, show $\ell(a) = \sum_n n a_n$ is linear but unbounded, hence discontinuous.

Recognition. Test on standard basis vectors e_n whose norm stays 1 while $\ell(e_n)$ blows up.

Solution. $\|e_n\|_\infty = 1$ but $\ell(e_n) = n \rightarrow \infty$, so $\sup_{\|a\| \leq 1} |\ell(a)| = \infty$. *Check.* Proposition 39.3.4 then says ℓ is discontinuous: indeed $a^{(n)} = e_n/\sqrt{n} \rightarrow 0$ in ℓ^∞ yet $\ell(a^{(n)}) = \sqrt{n} \rightarrow \infty$. Such ℓ cannot exist on a complete space without choice — c_{00} is not complete.

Worked Example 39.3.20 (Convexity of the unit ball forces a separating functional). Let $K = \{(x, y) : x^2 + y^2 < 1\}$ and $v_0 = (2, 0)$ in $\mathbb{R}^2$. Exhibit the separating hyperplane of Proposition 39.3.8.

Recognition. The gauge of the unit disc is the Euclidean norm; the separating ℓ points along v_0 .

Solution. Take $\ell(x, y) = x$. Then $\ell(v_0) = 2$ and $\ell(v) = x < 1 \leq 2$ for $v \in K$ (since $|x| < 1$ on the disc). With $a = 1$: $\ell(v_0) = 2 \geq 1 > \ell(v)$. *Check.* The hyperplane $\{x = 1\}$ is tangent to ∂K and separates the disc from v_0 , realizing the principle with the gauge $p(v) = |v|$ and $\ell = v_0/|v_0|$ paired against v .

Worked Example 39.3.21 ($C[0, 1]$ with the sup norm is complete). Show that the uniform limit of a Cauchy sequence in $C[0, 1]$ is continuous, so $C[0, 1]$ is a Banach space (Example 39.3.2(3)).

Recognition. Sup-norm Cauchy is uniform Cauchy; uniform limits of continuous functions are continuous.

Solution. Cauchy in $\|\cdot\|_\infty$ means: for $\varepsilon > 0$ there is N with $\sup_x |f_n(x) - f_m(x)| < \varepsilon$ for $n, m > N$. So $\{f_n(x)\}$ is uniformly Cauchy, hence converges uniformly to some f . The uniform limit of continuous functions is continuous (an $\varepsilon/3$ argument), so $f \in C[0, 1]$ and $\|f_n - f\|_\infty \rightarrow 0$. *Check.* The same statement fails for the L^1 norm on $C[0, 1]$ — $C[0, 1]$ is *not* complete in $\|\cdot\|_{L^1}$ (its completion is L^1), highlighting how the choice of norm decides completeness.

Worked Example 39.3.22 (The dual of a finite-dimensional space). Identify $(\mathbb{R}^d, \|\cdot\|_p)^*$ and verify $\|\cdot\|_p$ and $\|\cdot\|_q$ are dual norms for a concrete vector.

Recognition. On $\mathbb{R}^d$ every linear functional is $\ell(x) = \sum a_i x_i$; its norm is the dual norm $\|a\|_q$ with $1/p + 1/q = 1$ (a finite Holder).

Solution. Take $d = 2$, $p = 1$ so $q = \infty$. For $\ell(x) = 3x_1 - 4x_2$, $a = (3, -4)$, $\|\ell\| = \|a\|_\infty = 4$. Indeed $|\ell(x)| \leq \|a\|_\infty \|x\|_1$ with equality at $x = (0, -1)$: $\ell(0, -1) = 4 = 4 \cdot 1$. *Check.* $\|\ell\| = 4 = \|a\|_\infty$ matches $(\ell^1)^* = \ell^\infty$; the norming vector $(0, -1)$ concentrates all mass on the coordinate where $|a_i|$ is largest.

Exercises (Stein, Ch.1) – Banach spaces, Hahn-Banach, problem of measure.

1. Show $L^p(\mathbb{R}^d)$ (Lebesgue measure) is a Hilbert space iff $p = 2$, via the parallelogram law: a Banach space B is a Hilbert space iff $\|f + g\|^2 + \|f - g\|^2 = 2(\|f\|^2 + \|g\|^2)$. [Hint: in the real case set $(f, g) = \frac{1}{4}(\|f + g\|^2 - \|f - g\|^2)$.]
2. Suppose $1 < p_0, p_1 < \infty$ with $1/p_0 + 1/q_0 = 1$, $1/p_1 + 1/q_1 = 1$. Show the Banach spaces $L^{p_0} \cap L^{p_1}$ and $L^{q_0} + L^{q_1}$ are duals of each other up to equivalence of norms.
3. Strict convexity of the L^p unit ball, $1 < p < \infty$ (real-valued L^p). If $\|f_0\|_{L^p} = \|f_1\|_{L^p} = 1$ and $f_t = (1 - t)f_0 + tf_1$, then $\|f_t\|_{L^p} < 1$ for $0 < t < 1$ unless $f_0 = f_1$. (a) For unit f, g with $1/p + 1/q = 1$, $\int fg = 1$ only when $f = \text{sign } g |g|^{q-1}$. (b) If $\|f_t\|_{L^p} = 1$ find such a g and conclude $f_t = f_0$. (c) Show strict convexity fails for $p = 1, \infty$.
4. Verify the completeness of $\Lambda_\alpha(\mathbb{R}^d)$ and $L_k^p(\mathbb{R}^d)$.
5. Further on $\Lambda_\alpha(\mathbb{R}^d)$. (a) For $\alpha > 1$ the only functions in Λ_α are constants. (b) Define $C^{k, \alpha}(\mathbb{R}^d)$, $0 < \alpha \leq 1$, as functions with partial derivatives of order $\leq k$ in Λ_α , normed by $\|f\|_{C^{k, \alpha}} = \sum_{|\beta| \leq k} \|\partial_x^\beta f\|_{\Lambda_\alpha}$; show it is a Banach space.
6. Quotient spaces. Let S be a closed subspace of a Banach space B , with $f \sim g \iff f - g \in S$. Show B/S is a Banach space under $\|f\|_{B/S} = \inf\{\|f'\|_B : f' \sim f\}$.
7. For an open $\Omega \subseteq \mathbb{R}^d$ define $L_k^p(\Omega) = B/S$ (Exercise 30) with $B = L_k^p(\mathbb{R}^d)$ and S the functions vanishing a.e. on Ω , and let $L_k^p(\Omega_0)$ be the closure in $L_k^p(\mathbb{R}^d)$ of f with compact support in Ω . Show the natural map $L_k^p(\Omega_0) \rightarrow L_k^p(\Omega)$ has norm 1 but is in general not surjective (prove for Ω the unit ball, $k \geq 1$).
8. A Banach space B is *separable* if it has a countable dense subset. Prove that if B^* is separable then B is separable. (This gives another proof that L^1 is not the dual of L^∞ .)
9. Complex Hahn–Banach. Let V be a complex vector space with $p : V \rightarrow \mathbb{R}$ satisfying $p(\alpha v) = |\alpha| p(v)$ ($\alpha \in \mathbb{C}$) and $p(v_1 + v_2) \leq p(v_1) + p(v_2)$. If V_0 is a subspace and ℓ_0 a linear functional on V_0 with $|\ell_0(f)| \leq p(f)$, then ℓ_0 extends to ℓ on V with $|\ell(f)| \leq p(f)$. [Hint: if $u = \text{Re } \ell_0$ then $\ell_0(v) = u(v) - iu(iv)$; apply Theorem 39.3.9 to u .]
10. Let B be a Banach space, S a closed proper subspace, $f_0 \notin S$. Show there is $\ell \in B^*$ with $\ell = 0$ on S , $\ell(f_0) = 1$, and $\|\ell\| = 1/d$ where $d = \text{dist}(f_0, S)$.
11. A linear functional ℓ on a Banach space B is continuous iff its null space $\{f : \ell(f) = 0\}$ is closed. [Hint: use Exercise 34.]

12. Extend Section 5.4 to d dimensions. (a) There is an extended nonnegative finitely additive $\hat{m}$ on all subsets of $\mathbb{R}^d$ agreeing with Lebesgue measure on measurable sets and translation invariant. (b) There is an “integral” I on all bounded functions on $\mathbb{R}^d/\mathbb{Z}^d$ with $I(f) \geq 0$ for $f \geq 0$, I linear, $I(f) = \int f$ for measurable f , and $I(f_h) = I(f)$ for $f_h(x) = f(x - h)$, $h \in \mathbb{R}^d$.

Problems (Stein, Ch.1).

1. Universality of L^∞ and L^1 . (a) Every separable Banach space B embeds isometrically as a subspace of $\ell^\infty(\mathbb{Z})$: there is linear $i : B \rightarrow \ell^\infty(\mathbb{Z})$ with $\|i(f)\|_{\ell^\infty} = \|f\|_B$. (b) Every such B is a quotient of $\ell^1(\mathbb{Z})$: there is a surjection $P : \ell^1(\mathbb{Z}) \rightarrow B$ with $\|P(x)\|_B = \inf_{y \in S} \|x + y\|_{\ell^1}$, $S = \ker P$. [Hint: (a) take dense $\{f_n\}$, $\ell_n \in B^*$ with $\|\ell_n\| = 1$, $\ell_n(f_n) = \|f_n\|$, set $i(f) = \{\ell_n(f)\}$.]
2. A “generalized limit” L on bounded real sequences $\{s_n\}$ with: (i) L linear; (ii) $L(\{s_n\}) \geq 0$ if $s_n \geq 0$; (iii) $L(\{s_n\}) = \lim s_n$ when the limit exists; (iv) $L(\{s_n\}) = L(\{s_{n+k}\})$ for $k \geq 1$; (v) L unchanged if s_n is altered in finitely many places. [Hint: $p(\{s_n\}) = \limsup \frac{s_1 + \dots + s_n}{n}$ and extend the limit functional by Hahn–Banach.]
3. Show the closed unit ball of a Banach space B is compact iff B is finite dimensional. [Hint: if S is a closed proper subspace, there is x , $\|x\| = 1$, with $\text{dist}(x, S) > 1/2$.]
4. Let X be a σ -compact metric measure space with $C_b(X)$ separable. (a) A bounded sequence $\{\mu_n\}$ in $M(X)$ has a weak-* convergent subsequence: $\int g d\mu_{n_j} \rightarrow \int g d\mu$ for all $g \in C_b(X)$. (b) For positive $\mu_0 \in M(X)$, $f \mapsto \int f d\mu_0$ is an isometry of $L^1(\mu_0)$ onto the μ_0 -absolutely-continuous signed measures. (c) A bounded sequence in $L^1(\mu_0)$ has a subsequence $f_{n_j} d\mu_0$ converging weak-* to some $\mu \in M(X)$.
5. * Orlicz duality. With $\varphi, \psi, \Phi, \Psi$ as in Exercise 22 and the Orlicz spaces L^Φ, L^Ψ of Exercise 23: (a) a Holder-like inequality $\int |fg| \leq C \|f\|_{L^\Phi} \|g\|_{L^\Psi}$ holds; (b) if $\Phi(2t) \leq c\Phi(t)$ for all $t \geq 0$ then $(L^\Phi)^*$ is equivalent to L^Ψ .
6. * Clarkson inequalities. (a) For $2 \leq p < \infty$, $\left\|\frac{f+g}{2}\right\|_{L^p}^p + \left\|\frac{f-g}{2}\right\|_{L^p}^p \leq \frac{1}{2}(\|f\|_{L^p}^p + \|g\|_{L^p}^p)$. (b) For $1 < p \leq 2$ with $1/p + 1/q = 1$, $\left\|\frac{f+g}{2}\right\|_{L^p}^q + \left\|\frac{f-g}{2}\right\|_{L^p}^q \leq \left(\frac{1}{2}(\|f\|_{L^p}^p + \|g\|_{L^p}^p)\right)^{q/p}$. (c) Hence L^p is uniformly convex for $1 < p < \infty$. (d) If $1 < p < \infty$, $f_n \rightharpoonup f$ weakly and $\|f_n\|_{L^p} \rightarrow \|f\|_{L^p}$, then $f_n \rightarrow f$ strongly.
7. * Equivalence (isomorphism) of Banach spaces: $B_1 \cong B_2$ if there is a bounded linear bijection with bounded inverse. For $L^p(X)$ over a suitable class of X (e.g. $\mathbb{R}^d$): (a) $L^p \cong L^q$ iff $p = q$; (b) for every $1 \leq p \leq \infty$, L^2 is equivalent to a closed infinite-dimensional subspace of L^p .
8. * There is no finitely-additive rotation-invariant extension of Lebesgue measure to all subsets of S^d for $d \geq 2$ (Hausdorff): S^2 decomposes into disjoint A, B, C, Z with Z countable, $A \sim B \sim C$ but $A \sim (B \cup C)$ under rotations.
9. * Consequently (Banach–Tarski) Lebesgue measure on $\mathbb{R}^d$, $d \geq 3$, has no finitely-additive extension to all subsets that is invariant under Euclidean motions: the unit ball $B_1 = \bigcup_{j=1}^N E_j$ (disjoint) with each E_j moved by a rigid motion to disjoint $\tilde{E}_j$ whose union is the ball B_2 of radius 2.

Additional exercises.

1. Show that $(c_0)^* = \ell^1$ and $(\ell^1)^* = \ell^\infty$, where c_0 is the space of sequences tending to 0 with the sup norm. Conclude c_0 is not reflexive.
2. Prove that every finite-dimensional normed space is complete, hence a Banach space, and that all norms on it are equivalent.
3. Compute the operator norm of $T : (\mathbb{R}^2, \ell^\infty) \rightarrow (\mathbb{R}^2, \ell^\infty)$, $T(x, y) = (x + 2y, 3x)$, by maximizing over the four vertices of the unit ball.
4. (Hahn–Banach.) Let $V_0 = \{(x, x) : x \in \mathbb{R}\} \subset \mathbb{R}^2$ and $\ell_0(x, x) = x$, with p the Euclidean norm. Find the unique norm-preserving extension and explain why uniqueness holds here (Hilbert space) but not for the ℓ^1 norm.

5. Prove the norming-functional identity $\|f\| = \max_{\|\ell\| \leq 1} \ell(f)$ is a *maximum* (attained), not merely a supremum, using Proposition 39.3.11.
6. (Non-example.) Give a discontinuous linear functional on an incomplete normed space, and explain why Hahn–Banach plus completeness does not rule it out.
7. Show that if B^* is finite dimensional then so is B , and the two have equal dimension.
8. (Diagram.) Sketch the separation of the convex set $K = \{(x, y) : |x| + |y| < 1\}$ from the point $(1, 1)$ by an explicit hyperplane, and write down the separating functional.
9. Prove that a linear functional ℓ on a Banach space is bounded iff it maps some open ball into a bounded set, and relate this to closedness of $\ker \ell$ (Exercise 35).
10. Use the Banach integral construction (Theorem 39.3.15) to explain why the resulting $\hat{m}$ cannot be countably additive, referencing a non-measurable set.

39.4 The Dual Space and Complex Interpolation

The dual of L^p , $1 \leq p < \infty$

Holder's inequality embeds $L^q \hookrightarrow (L^p)^*$ isometrically via $g \mapsto \ell_g$, $\ell_g(f) = \int fg$ (Section 39.4's preamble, Lemma 39.2.5). The central theorem of the chapter is that this embedding is *surjective* for $1 \leq p < \infty$: every bounded functional on L^p is integration against some L^q function.

Theorem 39.4.1 (Riesz representation for L^p). *Let $1 \leq p < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$. Then $(L^p)^* = L^q$ isometrically: for every $\ell \in (L^p)^*$ there is a unique $g \in L^q$ with*

$$\ell(f) = \int_X fg \, d\mu \quad \text{for all } f \in L^p, \quad \|\ell\| = \|g\|_{L^q}.$$

Proof. Finite measure first. Suppose $\mu(X) < \infty$, so $\mathbf{1}_E \in L^p$ for every measurable E . Define the set function $\nu(E) = \ell(\mathbf{1}_E)$. Linearity of ℓ makes ν finitely additive, and

$$|\nu(E)| \leq \|\ell\| \|\mathbf{1}_E\|_{L^p} = \|\ell\| \mu(E)^{1/p}. \quad (*)$$

If $\{E_n\}$ are disjoint with union E and tails $E_N^* = \bigcup_{n>N} E_n$, then $\mathbf{1}_E = \mathbf{1}_{E_N^*} + \sum_{n \leq N} \mathbf{1}_{E_n}$, so $\nu(E) = \nu(E_N^*) + \sum_{n \leq N} \nu(E_n)$; by $(*)$ and $p < \infty$, $\nu(E_N^*) \rightarrow 0$, so ν is countably additive, and $(*)$ shows $\nu \ll \mu$. By the Lebesgue–Radon–Nikodym theorem there is an integrable g with $\nu(E) = \int_E g \, d\mu$, i.e. $\ell(\mathbf{1}_E) = \int \mathbf{1}_E g$. By linearity this extends to simple functions, and since simple functions are dense in L^p ($p < \infty$, Exercise 6) and ℓ is continuous, $\ell(f) = \int fg$ for all $f \in L^p$. Lemma 39.2.5 gives $g \in L^q$ with $\|g\|_{L^q} = \|\ell\|$. *General σ -finite case.* Exhaust $X = \bigcup_n E_n$ by sets of finite measure. On each E_n the finite case yields $g_n \in L^q(E_n)$ (extended by 0) representing ℓ on functions supported in E_n , with $\|g_n\|_{L^q} \leq \|\ell\|$. Consistency ($g_n = g_m$ a.e. on E_m for $n \geq m$) gives a pointwise limit $g = \lim g_n$; Fatou gives $\|g\|_{L^q} \leq \|\ell\|$, and a limiting argument extends $\ell(f) = \int fg$ to all $f \in L^p$. Holder gives the reverse $\|\ell\| \leq \|g\|_{L^q}$. Uniqueness: if $\int f(g_1 - g_2) = 0$ for all $f \in L^p$, take f the extremal test function to force $g_1 = g_2$ a.e. $\square$

Remark. The theorem fails at $p = \infty$: $(L^\infty)^* \supsetneq L^1$ (Theorem 39.3.14). The reflexive range is $1 < p < \infty$, where $(L^p)^{**} = L^p$. The case $p = 1$ gives $(L^1)^* = L^\infty$ (under σ -finiteness), but L^1 is not reflexive.

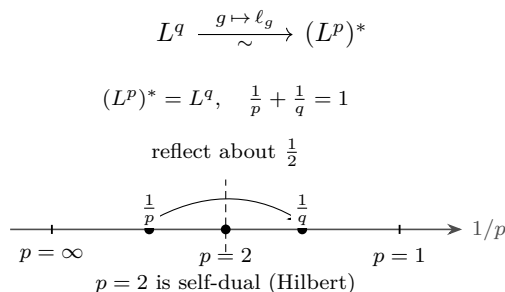


Figure 39.9. Duality reflects $1/p$ about $1/2$, the involution $1/q = 1 - 1/p$. The fixed point $p = 2$ is the self-dual Hilbert space; the endpoint $p = 1$ ($q = \infty$) is special because the dual of L^∞ is strictly larger than L^1 .

Complex interpolation: the Riesz–Thorin theorem

The log-convexity of L^p norms (the worked example in Section 2) upgrades, via complex analysis on a strip, to a theorem about *operators*: a linear map bounded at two endpoints p_0, p_1 is automatically bounded at every intermediate p_θ .

Theorem 39.4.2 (Riesz–Thorin). *Let T be linear on simple functions, bounded $L^{p_0} \rightarrow L^{q_0}$ with norm M_0 and $L^{p_1} \rightarrow L^{q_1}$ with norm M_1 . For $0 \leq \theta \leq 1$ define p_θ, q_θ by*

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}.$$

Then T is bounded $L^{p_\theta} \rightarrow L^{q_\theta}$ with $\|T\| \leq M_0^{1-\theta} M_1^\theta$.

Proof idea via the three-lines lemma. Normalize and fix simple f with $\|f\|_{p_\theta} = 1$ and g with $\|g\|_{q'_\theta} = 1$. Define analytic families f_z, g_z by raising moduli to z -dependent powers so that $f_\theta = f, g_\theta = g$, and put

$$F(z) = \int (T f_z) g_z d\mu,$$

an entire function, bounded on the strip $0 \leq \operatorname{Re} z \leq 1$. On the edge $\operatorname{Re} z = 0$ one has $|F| \leq M_0$ (using the $L^{p_0} \rightarrow L^{q_0}$ bound), and on $\operatorname{Re} z = 1$, $|F| \leq M_1$. Hadamard’s *three-lines lemma* — the maximum principle for the strip — gives $|F(\theta)| \leq M_0^{1-\theta} M_1^\theta$. Since $F(\theta) = \int (T f) g$ and g was an arbitrary unit vector in $L^{q'_\theta}$, duality (Lemma 39.2.5) yields $\|T f\|_{q_\theta} \leq M_0^{1-\theta} M_1^\theta$. Density of simple functions extends T (Proposition 39.3.13). $\square$

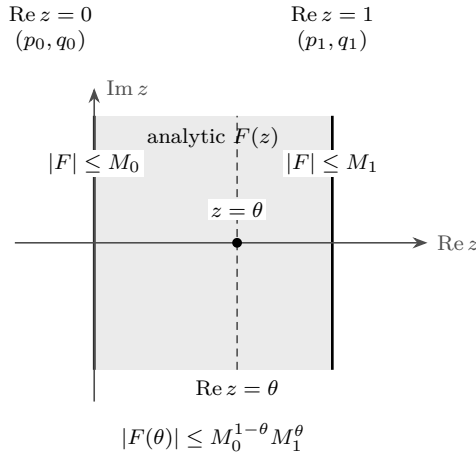


Figure 39.10. The Riesz–Thorin strip. The auxiliary $F(z) = \int (T f_z) g_z$ is analytic and bounded on $0 \leq \operatorname{Re} z \leq 1$; control on the two vertical edges plus the three-lines lemma (a maximum principle) gives the geometric-mean bound at the interior line $\operatorname{Re} z = \theta$.

Corollary 39.4.3 (Hausdorff–Young). *The Fourier transform maps $L^p(\mathbb{R}^d) \rightarrow L^{p'}(\mathbb{R}^d)$ for $1 \leq p \leq 2$ with $\|\hat{f}\|_{p'} \leq \|f\|_p$.*

Proof. $T =$ Fourier transform is bounded $L^1 \rightarrow L^\infty$ with $M_0 = 1$ ($\|\hat{f}\|_\infty \leq \|f\|_1$) and $L^2 \rightarrow L^2$ with $M_1 = 1$ (Plancherel). Interpolate at θ with $1/p = (1-\theta)/1 + \theta/2$; then $1/p' = (1-\theta)/\infty + \theta/2$, giving p' the conjugate of p and $\|T\| \leq 1$. $\square$

$$L^1 \xrightarrow{\mathcal{F}} L^\infty$$

$$L^p \dashrightarrow^{\mathcal{F}} L^{p'}$$

$$L^2 \xrightarrow[\sim]{\mathcal{F}} L^2$$

Figure 39.11. Hausdorff–Young by interpolation. The Fourier transform $\mathcal{F}$ is the same operator at the two endpoints $L^1 \rightarrow L^\infty$ (norm 1) and $L^2 \rightarrow L^2$ (Plancherel, norm 1); Riesz–Thorin produces the intermediate bound $L^p \rightarrow L^{p'}$ for $1 \leq p \leq 2$ with norm ≤ 1 .

Remark (L^2 and probability). The Hilbert space L^2 of a probability space is the natural home of random variables with finite variance. Convergence in L^2 is convergence in mean square; the central limit theorem says normalized sums converge in distribution to a Gaussian, and Brownian motion is the canonical continuous-time limit, with sample paths that are continuous but nowhere differentiable (and whose increments are independent Gaussians). These objects live in and are analyzed through the L^p spaces of this chapter.

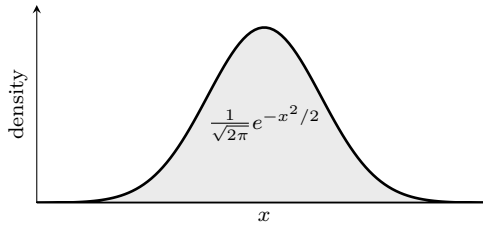


Figure 39.12. The standard Gaussian density, the universal limit in the central limit theorem. Its presence in L^p for every p (rapid decay) makes it the workhorse test function of harmonic analysis and probability.

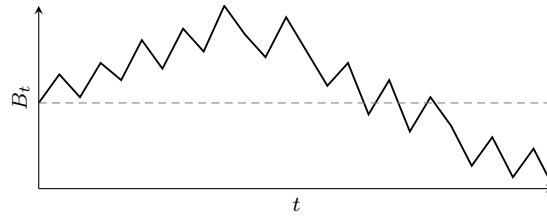


Figure 39.13. A simulated Brownian sample path $t \mapsto B_t$: continuous, nowhere differentiable, with independent Gaussian increments $B_t - B_s \sim N(0, t - s)$. The mean-square scaling $\mathbb{E}|B_t - B_s|^2 = t - s$ is an L^2 statement. (Schematic.)

The dual of $C(X)$ (Riesz–Markov)

For completeness we record the companion duality for continuous functions; it identifies $C(X)^*$ with measures rather than functions.

Theorem 39.4.4 (Riesz–Markov). *Let X be a compact metric space. Every bounded linear functional ℓ on $C(X)$ is $\ell(f) = \int_X f d\mu$ for a unique finite signed Borel measure μ , and $\|\ell\| = |\mu|(X)$. Thus $C(X)^*$ is isometric to the space $M(X)$ of finite signed Borel (Radon) measures.*

Structure. Decompose $\ell = \ell_+ - \ell_-$ into positive functionals (Proposition 7.2 in Stein), $\ell_+(f) = \sup\{\ell(\varphi) : 0 \leq \varphi \leq f\}$ for $f \geq 0$. For a *positive* functional, build an outer measure $\mu^*(E) = \inf\{\rho(O) : E \subseteq O \text{ open}\}$ with $\rho(O) = \sup\{\ell(f) : \text{supp } f \subseteq O, 0 \leq f \leq 1\}$; partitions of unity give countable subadditivity, the separation property gives the metric (hence Borel) property, and the Caratheodory extension produces μ with $\ell(f) = \int f d\mu$ (slice f into thin layers $f = \sum f_n$). Subtract to handle signed ℓ ; total variation gives $\|\ell\| = |\mu|(X)$. $\square$

Worked Example 39.4.5 ($(L^3)^* = L^{3/2}$ in action). Identify the functional $\ell(f) = \int_0^1 f(x)\sqrt{x} dx$ on $L^3([0, 1])$ as integration against an $L^{3/2}$ function, and compute $\|\ell\|$.

Recognition. Theorem 39.4.1: the representing g is the visible factor, here $g(x) = \sqrt{x}$; check $g \in L^q$ with $q = 3/2$ and compute $\|g\|_{L^{3/2}}$.

Solution. Conjugate of $p = 3$ is $q = 3/2$. $\|g\|_{L^{3/2}}^{3/2} = \int_0^1 (x^{1/2})^{3/2} dx = \int_0^1 x^{3/4} dx = \frac{4}{7}$, so $\|\ell\| = \|g\|_{L^{3/2}} = (4/7)^{2/3} \approx 0.683$. *Check.* The theorem guarantees $\|\ell\| = \|g\|_{L^{3/2}}$ exactly; the extremal $f = g^{q-1} / \|g\|_q^{q-1} = g^{1/2} / \|g\|_{3/2}^{1/2}$ would attain it.

Worked Example 39.4.6 ($(\ell^2)^* = \ell^2$ self-duality). Represent $\ell(a) = \sum_n a_n/2^n$ on ℓ^2 and find $\|\ell\|$.

Recognition. Self-dual case $p = q = 2$; the representing g is the coefficient sequence, and $\|\ell\| = \|g\|_{\ell^2}$.

Solution. $g = (2^{-n})_{n \geq 1}$, $\|g\|_{\ell^2}^2 = \sum_{n \geq 1} 4^{-n} = \frac{1}{1-1/4} = \frac{4}{3}$, so $\|\ell\| = 1/\sqrt{3}$. *Check.* By Cauchy–Schwarz $|\ell(a)| \leq \|g\|_2 \|a\|_2$ with equality at $a = g/\|g\|_2$; the dual norm equals the ℓ^2 norm of g , confirming ℓ^2 is its own dual.

Worked Example 39.4.7 (Interpolating a convolution operator). Let $Tf = f * k$ with $k \in L^1(\mathbb{R})$, $\|k\|_1 = K$. Show $T : L^p \rightarrow L^p$ with $\|T\| \leq K$ for all $1 \leq p \leq \infty$ by interpolating the endpoints.

Recognition. T is bounded $L^1 \rightarrow L^1$ and $L^\infty \rightarrow L^\infty$ both with norm K (Young with $g \in L^1$); Riesz–Thorin fills in between.

Solution. $\|f * k\|_1 \leq \|f\|_1 \|k\|_1 = K \|f\|_1$ and $\|f * k\|_\infty \leq \|f\|_\infty \|k\|_1 = K \|f\|_\infty$. Take $p_0 = q_0 = 1$, $p_1 = q_1 = \infty$, $M_0 = M_1 = K$. For $1/p = (1-\theta)/1 + \theta/\infty$, Theorem 39.4.2 gives $\|f * k\|_p \leq K^{1-\theta} K^\theta \|f\|_p = K \|f\|_p$. *Check.* This recovers Young’s inequality $\|f * k\|_p \leq \|f\|_p \|k\|_1$ for all p from just the two endpoints — the prototypical use of interpolation.

Worked Example 39.4.8 (A Radon measure as a functional on $C[0, 1]$). Describe the functional $\ell(f) = f(1/2) - \int_0^1 f(x) dx$ on $C[0, 1]$ as integration against a signed measure and find $\|\ell\|$.

Recognition. Riesz–Markov: read off μ from the formula; $\|\ell\| = |\mu|([0, 1])$.

Solution. $\mu = \delta_{1/2} - m$ (a unit point mass minus Lebesgue measure). Its total variation is $|\mu|([0, 1]) = 1 + 1 = 2$, so $\|\ell\| = 2$. *Check.* The bound is attained in the limit by f near $+1$ around $x = 1/2$ and near -1 elsewhere; $|\ell(f)| \leq |f(1/2)| + \int |f| \leq 2 \|f\|_\infty$, and $\|\ell\| = 2 = |\mu|(X)$ as Theorem 39.4.4 predicts.

Worked Example 39.4.9 (Weak convergence detected by the dual). In $L^2([0, 1])$ show $f_n(x) = \sqrt{2} \sin(2\pi n x) \rightarrow 0$ weakly but not in norm.

Recognition. Weak convergence means $\int f_n g \rightarrow 0$ for all $g \in L^2 = (L^2)^*$; test on a dense set and use Bessel/Riemann–Lebesgue. Norm is constant.

Solution. For $g \in L^2$, $\int_0^1 f_n g$ are (up to constants) Fourier sine coefficients of g , which $\rightarrow 0$ by Bessel’s inequality (Riemann–Lebesgue). So $f_n \rightarrow 0$. But $\|f_n\|_{L^2}^2 = 2 \int_0^1 \sin^2(2\pi n x) dx = 1$ for all n , so $f_n \not\rightarrow 0$ in norm. *Check.* Weak limit 0 has norm $0 < 1 = \lim \|f_n\|$; norms can only drop in the weak limit ($\|\text{weak lim}\| \leq \liminf \|f_n\|$), consistent here. This is Exercise 13(a).

Worked Example 39.4.10 (Reflexivity of L^p , $1 < p < \infty$). Explain why $(L^p)^{**} = L^p$ for $1 < p < \infty$ but not for $p = 1$.

Recognition. Apply Theorem 39.4.1 twice: $(L^p)^* = L^q$ and $(L^q)^* = L^p$ when both exponents are finite; check the canonical embedding is onto.

Solution. For $1 < p < \infty$, q is also finite, so $(L^p)^{**} = (L^q)^* = L^p$, and one checks the natural map $f \mapsto (\ell \mapsto \ell(f))$ realizes this identification; hence L^p is reflexive. For $p = 1$, $(L^1)^* = L^\infty$ but $(L^\infty)^* \supsetneq L^1$ (Theorem 39.3.14), so $(L^1)^{**} \supsetneq L^1$ and L^1 is not reflexive. *Check.* The reflexive range $1 < p < \infty$ is exactly where weak compactness holds (Exercise 12c), the abstract reason bounded sequences in L^p have weakly convergent subsequences.

Worked Example 39.4.11 (Riesz–Thorin endpoints for the Fourier transform). Read off the Hausdorff–Young exponent at $\theta = 1/2$ from the endpoints $L^1 \rightarrow L^\infty$ and $L^2 \rightarrow L^2$.

Recognition. Corollary 39.4.3: at parameter θ , $1/p = (1 - \theta) + \theta/2$.

Solution. At $\theta = 1/2$: $1/p = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$, so $p = 4/3$, and $1/p' = \theta/2 = \frac{1}{4}$, $p' = 4$. The bound is $\|\hat{f}\|_4 \leq M_0^{1/2} M_1^{1/2} \|f\|_{4/3} = 1 \cdot \|f\|_{4/3}$. *Check.* As θ runs $0 \rightarrow 1$, (p, p') sweeps from $(1, \infty)$ to $(2, 2)$; the midpoint $\theta = 1/2$ lands at $(4/3, 4)$, a standard Hausdorff–Young pairing, with constant 1 throughout.

Exercises (Stein, Ch.1) – duality and weak convergence.

1. Suppose (X, μ) is separable (Exercise 10), $1 \leq p < \infty$, $1/p + 1/q = 1$. A sequence $f_n \in L^p$ converges weakly to $f \in L^p$ if $\int f_n g d\mu \rightarrow \int f g d\mu$ for every $g \in L^q$. (a) If $\|f - f_n\|_{L^p} \rightarrow 0$ then $f_n \rightarrow f$ weakly. (b) If $\sup_n \|f_n\|_{L^p} < \infty$ it suffices to check the defining limit for g in a dense subset of L^q . (c) For $1 < p < \infty$, if $\sup_n \|f_n\|_{L^p} < \infty$ then some subsequence converges weakly to some $f \in L^p$ (weak compactness; fails for $p = 1$).
2. Examples of weak convergence. (a) $f_n(x) = \sin(2\pi n x)$ in $L^p([0, 1])$: show $f_n \rightarrow 0$ weakly. (b) $f_n(x) = n^{1/p} \chi(nx)$ in $L^p(\mathbb{R})$, $\chi = \mathbf{1}_{[0,1]}$: $f_n \rightarrow 0$ weakly if $p > 1$ but not if $p = 1$. (c) $f_n(x) = 1 + \sin(2\pi n x)$ in $L^1([0, 1])$: $f_n \rightarrow 1$, $\|f_n\|_{L^1} = 1$, yet $\|f_n - 1\|_{L^1} \not\rightarrow 0$.
3. Let X be a measure space, $1 < p < \infty$, $\|f_n\|_{L^p} \leq M < \infty$. (a) If $f_n \rightarrow f$ a.e. then $f_n \rightarrow f$ weakly. (b) Show this can fail for $p = 1$. (c) If $f_n \rightarrow f_1$ a.e. and $f_n \rightarrow f_2$ weakly, then $f_1 = f_2$ a.e.

Additional exercises.

1. Identify the functional $\ell(f) = \int_0^1 f(x)(1-x) dx$ on $L^4([0, 1])$ as integration against an $L^{4/3}$ function and compute $\|\ell\|$.
2. Show $(\ell^p)^* = \ell^q$ for $1 < p < \infty$ by exhibiting, for each bounded functional, its representing sequence via the standard basis $\{e_n\}$.
3. Use Riesz–Thorin to prove the Hilbert-transform-type bound: if T is bounded $L^2 \rightarrow L^2$ with norm A and $L^4 \rightarrow L^4$ with norm B , then T is bounded $L^3 \rightarrow L^3$ with norm $\leq A^{1/2} B^{1/2}$. Identify θ .
4. (Hausdorff–Young in action.) For the Fourier transform on $\mathbb{R}$, state the bound $\|\hat{f}\|_{p'} \leq \|f\|_p$ at $p = 1, \frac{4}{3}, 2$ explicitly and check the endpoints reduce to Plancherel and the sup bound.
5. Represent the functional $\ell(f) = \sum_{n=1}^{\infty} 2^{-n} f(1/n)$ on $C[0, 1]$ as a Radon measure and compute $\|\ell\|$.
6. (Non-example.) Show $(L^\infty([0, 1]))^* \neq L^1([0, 1])$ by sketching the Dirac-type functional of Theorem 39.3.14 adapted to $x = 1/2$.
7. Prove the log-convexity bound implies $\|T\|_{L^{p\theta} \rightarrow L^{q\theta}}$ is a log-convex function of θ , directly from Riesz–Thorin.
8. Show that weak convergence $f_n \rightarrow f$ in L^p ($1 < p < \infty$) plus $\|f_n\|_{L^p} \rightarrow \|f\|_{L^p}$ implies strong convergence, citing the uniform convexity of L^p (Problem 6*).
9. (Diagram.) Redraw the Riesz–Thorin strip for the pair $(p_0, q_0) = (1, 1)$, $(p_1, q_1) = (\infty, \infty)$ and mark where $L^2 \rightarrow L^2$ sits.
10. Prove that for $1 < p < \infty$, L^p is reflexive: $(L^p)^{**} = L^p$ canonically, using Theorem 39.4.1 twice.

Chapter 40

Lp Spaces in Harmonic Analysis

40.1 The Hardy-Littlewood Maximal Function and Lp Bounds

The role of the L^p spaces in harmonic analysis is exhibited most vividly by a handful of canonical operators: the Hardy–Littlewood maximal function, the Hilbert transform, and the abstract interpolation machinery that ties their endpoint estimates together. We begin with the maximal function, an intrinsically *nonlinear* object whose L^p theory rests on a single quantitative covering estimate and on the bookkeeping device of the distribution function.

The maximal function and its sub-additivity

Definition 40.1.1. For a locally integrable f on $\mathbb{R}^d$, the *Hardy–Littlewood maximal function* is

$$f^*(x) = \sup_{x \in B} \frac{1}{m(B)} \int_B |f(y)| \, dy,$$

the supremum running over all balls B (open Euclidean balls) that contain x ; here m is Lebesgue measure. Thus $f^*(x)$ is the largest average of $|f|$ over a ball about x .

The point of f^* is that it dominates, at each x , every local average of $|f|$. It is the workhorse behind the Lebesgue differentiation theorem and behind almost-everywhere convergence statements for approximations to the identity: control of f^* converts a pointwise limit question into an a priori inequality.

Proposition 40.1.2 (Sub-additivity). *The map $f \mapsto f^*$ is not linear, but it is sub-additive: if $f = f_1 + f_2$ then $f^* \leq f_1^* + f_2^*$. It is also positively homogeneous, $(cf)^* = |c| f^*$, and monotone, $|f| \leq |g| \implies f^* \leq g^*$.*

Proof. For any ball $B \ni x$, $\frac{1}{m(B)} \int_B |f_1 + f_2| \leq \frac{1}{m(B)} \int_B |f_1| + \frac{1}{m(B)} \int_B |f_2| \leq f_1^*(x) + f_2^*(x)$ by the triangle inequality for integrals. Taking the supremum over $B \ni x$ on the left gives the claim. Homogeneity and monotonicity are immediate from the definition. $\square$

Remark. Sub-additivity is the exact substitute for linearity that the interpolation theory of §40.3 will require: an operator obeying $T(f_1 + f_2) \leq T f_1 + T f_2$ pointwise is called *sub-linear*, and Marcinkiewicz interpolation applies to such T .

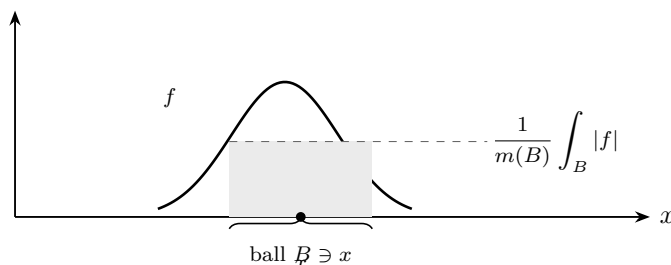


Figure 40.1. $f^*(x)$ is the supremum, over all balls B containing x , of the average of $|f|$ on B . (Schematic.)

Worked Example 40.1.3 (The maximal function of an indicator on the line). Let $f = \mathbf{1}_{[-1,1]}$ on $\mathbb{R}$. Compute $f^*(x)$ for $x \geq 0$ and read off its decay.

Recognition. An indicator: the average over a ball (interval) is $\frac{m(B \cap [-1,1])}{m(B)}$, and we maximise over intervals through x .

Solution. Fix $x \geq 1$. Among intervals $B = (a, b) \ni x$, the average $|B \cap [-1,1]| / |B|$ is maximised by taking $a = -1$ (capture all the mass) and $b = x$ as small as allowed, i.e. $b \downarrow x$; then $B = (-1, x)$ has length $x + 1$ and contains $[-1,1]$, giving average $2/(x + 1)$. Larger intervals only dilute. For $0 \leq x < 1$ one can take $B \subset [-1,1]$ so $f^*(x) = 1$. Hence

$$f^*(x) = \begin{cases} 1, & |x| \leq 1, \\ \frac{2}{|x| + 1}, & |x| > 1. \end{cases}$$

Check. At infinity $f^*(x) \sim 2/|x|$, so $f^* \notin L^1(\mathbb{R})$ although $f \in L^1$ — the maximal operator is *not* bounded on L^1 , only weak-type $(1,1)$.

Worked Example 40.1.4 (Why L^1 boundedness must fail). Show, for $f = \mathbf{1}_B$ with B the unit ball of $\mathbb{R}^d$, that $f^*(x) \geq c(1 + |x|)^{-d}$, and conclude $f^* \notin L^1(\mathbb{R}^d)$.

Recognition. A scaling/volume estimate; cover x and the mass of f by one ball and compare volumes.

Solution. The ball B' of radius $1 + |x|$ centred at x contains the unit ball B , so

$$f^*(x) \geq \frac{1}{m(B')} \int_{B'} |f| = \frac{m(B)}{m(B')} = \frac{v_d \cdot 1^d}{v_d(1 + |x|)^d} = (1 + |x|)^{-d},$$

where v_d is the volume of the unit ball. Since $\int_{\mathbb{R}^d} (1 + |x|)^{-d} dx = \infty$ (the integrand decays exactly like $|x|^{-d}$, the borderline), $f^* \notin L^1$. *Check.* $f \in L^1$ with $\|f\|_{L^1} = v_d < \infty$, so the failure is genuinely a property of $f \mapsto f^*$, not of f .

The weak-type $(1,1)$ estimate

The substitute for the failed L^1 bound is a *weak-type* inequality, proved with a Vitali covering argument.

Theorem 40.1.5 (Weak-type $(1,1)$ for f^*). *There is a constant $A = A_d$ so that for every $f \in L^1(\mathbb{R}^d)$ and every $\alpha > 0$,*

$$m(\{x : f^*(x) > \alpha\}) \leq \frac{A}{\alpha} \|f\|_{L^1(\mathbb{R}^d)}.$$

One may take $A = 3^d$.

Proof. Write $E_\alpha = \{x : f^*(x) > \alpha\}$. By inner regularity it suffices to bound $m(K)$ for an arbitrary compact $K \subseteq E_\alpha$. Each $x \in K$ lies in some ball B_x with $\frac{1}{m(B_x)} \int_{B_x} |f| > \alpha$, i.e. $\int_{B_x} |f| > \alpha m(B_x)$. The B_x cover K ; extract a finite subcover $B_1, \dots, B_N$. By the Vitali covering lemma there is a disjoint subcollection $B_{i_1}, \dots, B_{i_n}$ with $\sum_j m(B_{i_j}) \geq 3^{-d} m(K)$ (each chosen ball, tripled, swallows the balls it meets). Adding the defining inequalities over the disjoint balls,

$$m(K) \leq 3^d \sum_j m(B_{i_j}) < \frac{3^d}{\alpha} \sum_j \int_{B_{i_j}} |f| \leq \frac{3^d}{\alpha} \int_{\mathbb{R}^d} |f|.$$

Taking the supremum over compact $K \subseteq E_\alpha$ gives the bound with $A = 3^d$. □

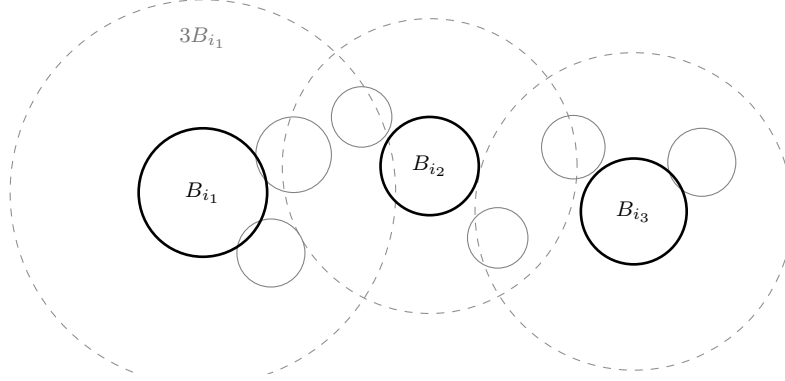


Figure 40.2. Vitali selection: a pairwise-disjoint subfamily (bold). Tripling each selected ball (dashed, radius $3r$) swallows every discarded ball (thin) it meets, so the triples cover K and $\sum_j m(B_{i_j}) \geq 3^{-d}m(K)$. (Schematic.)

Theorem 40.1.6 (Refined weak-type estimate). *There is a constant A' (one may take $A' = 2A$) so that for all $f \in L^1_{\text{loc}}$ and all $\alpha > 0$,*

$$m(\{x : f^*(x) > \alpha\}) \leq \frac{A'}{\alpha} \int_{\{|f| > \alpha/2\}} |f| \, dx.$$

Proof. Split $f = f_1 + f_\infty$, where $f_1 = f \mathbf{1}_{\{|f| > \alpha/2\}}$ and $f_\infty = f \mathbf{1}_{\{|f| \leq \alpha/2\}}$. Then $|f_\infty| \leq \alpha/2$ everywhere, so $f_\infty^* \leq \alpha/2$, and by sub-additivity $f^* \leq f_1^* + f_\infty^* \leq f_1^* + \alpha/2$. Hence $\{f^* > \alpha\} \subseteq \{f_1^* > \alpha/2\}$. Applying Theorem 40.1.5 to f_1 at level $\alpha/2$,

$$m(\{f^* > \alpha\}) \leq m(\{f_1^* > \alpha/2\}) \leq \frac{A}{\alpha/2} \|f_1\|_{L^1} = \frac{2A}{\alpha} \int_{\{|f| > \alpha/2\}} |f|. \quad \square$$

This refinement — integrating only where $|f| > \alpha/2$ — is exactly what upgrades the weak estimate to the strong L^p bound, because the α -integration that follows then converges.

The distribution function

Definition 40.1.7. For a non-negative measurable F on $\mathbb{R}^d$, its *distribution function* is

$$\lambda(\alpha) = \lambda_F(\alpha) = m(\{x : F(x) > \alpha\}), \quad \alpha > 0.$$

Theorem 40.1.8 (Layer-cake formula). *For every $0 < p < \infty$ and non-negative measurable F ,*

$$\int_{\mathbb{R}^d} F(x)^p \, dx = \int_0^\infty \lambda(\alpha^{1/p}) \, d\alpha = p \int_0^\infty \alpha^{p-1} \lambda(\alpha) \, d\alpha,$$

both sides being simultaneously finite or infinite.

Proof. First $p = 1$. Apply Fubini in $\mathbb{R}^d \times \mathbb{R}_+$ to the indicator of $\{(x, \alpha) : 0 < \alpha < F(x)\}$: integrating in α first gives $\int_{\mathbb{R}^d} \int_0^{F(x)} d\alpha \, dx = \int F$, and in the reverse order $\int_0^\infty m(\{F > \alpha\}) \, d\alpha = \int_0^\infty \lambda(\alpha) \, d\alpha$. For general p put $G = F^p$; then $\{G > \alpha\} = \{F > \alpha^{1/p}\}$, and the $p = 1$ identity for G gives $\int F^p = \int_0^\infty \lambda(\alpha^{1/p}) \, d\alpha$. The substitution $\alpha = \beta^p$ converts this to $p \int_0^\infty \beta^{p-1} \lambda(\beta) \, d\beta$. $\square$

Proposition 40.1.9 (Tchebychev inequality). *For non-negative measurable F and $p > 0$, $\lambda(\alpha) \leq \frac{1}{\alpha^p} \int_{\mathbb{R}^d} F(x)^p \, dx$.*

Proof. $\int_{\mathbb{R}^d} F^p \geq \int_{\{F > \alpha\}} F^p \geq \alpha^p m(\{F > \alpha\}) = \alpha^p \lambda(\alpha)$. $\square$

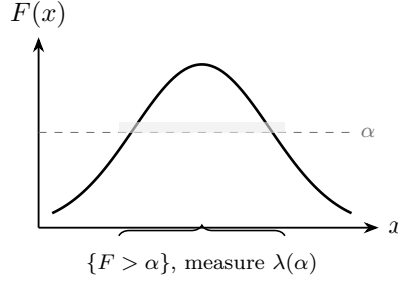


Figure 40.3. The distribution function $\lambda(\alpha)$ measures the super-level set $\{F > \alpha\}$; the layer-cake formula integrates these horizontal slabs. (Schematic.)

Worked Example 40.1.10 (Layer-cake for a power weight). With $F(x) = |x|^{-a} \mathbf{1}_{|x| \leq 1}$ on $\mathbb{R}$ ($a > 0$), use the layer-cake formula to decide for which p we have $F \in L^p$.

Recognition. Compute $\lambda(\alpha)$ directly, then integrate $p\alpha^{p-1}\lambda$.

Solution. $F(x) > \alpha \iff |x| < \alpha^{-1/a}$ and $|x| \leq 1$. For $\alpha \geq 1$ this gives $|x| < \alpha^{-1/a}$, so $\lambda(\alpha) = 2\alpha^{-1/a}$; for $0 < \alpha < 1$ it gives $|x| \leq 1$, so $\lambda(\alpha) = 2$. Then $\int F^p = p \int_0^1 \alpha^{p-1} \cdot 2 d\alpha + p \int_1^\infty \alpha^{p-1} \cdot 2\alpha^{-1/a} d\alpha$. The first integral is 2; the second converges iff $p-1-1/a < -1$, i.e. $p < 1/a$. *Check.* Directly $\int_0^1 x^{-ap} dx < \infty \iff ap < 1 \iff p < 1/a$ — the two computations agree.

The L^p maximal theorem

Theorem 40.1.11 (Hardy–Littlewood maximal theorem). Suppose $f \in L^p(\mathbb{R}^d)$ with $1 < p \leq \infty$. Then $f^* \in L^p(\mathbb{R}^d)$ and

$$\|f^*\|_{L^p} \leq A_p \|f\|_{L^p},$$

with A_p independent of f . For $p = \infty$ one has $A_\infty = 1$, and as $p \rightarrow 1^+$ the constant blows up like $A_p = O(1/(p-1))$.

Proof. The case $p = \infty$ is clear: every average of $|f|$ is $\leq \|f\|_{L^\infty}$, so $f^* \leq \|f\|_{L^\infty}$. Assume $1 < p < \infty$. Finiteness a.e. first: split $f = f_1 + f_\infty$ with $f_1 = f \mathbf{1}_{|f| > 1} \in L^1$ and $f_\infty = f \mathbf{1}_{|f| \leq 1} \in L^\infty$; then $f^* \leq f_1^* + 1$ and $f_1^* < \infty$ a.e. by Theorem 40.1.5.

Now apply the layer-cake formula (Theorem 40.1.8) to $F = f^*$ together with the refined estimate (Theorem 40.1.6):

$$\int_{\mathbb{R}^d} (f^*)^p dx = \int_0^\infty \lambda(\alpha^{1/p}) d\alpha \leq A' \int_0^\infty \alpha^{-1/p} \left(\int_{\{|f| > \alpha^{1/p}/2\}} |f| dx \right) d\alpha.$$

Interchange the order of integration (Tonelli). For fixed x the inner constraint $|f|(x) > \alpha^{1/p}/2$ reads $\alpha < (2|f(x)|)^p$, so

$$\int_{\mathbb{R}^d} (f^*)^p \leq A' \int_{\mathbb{R}^d} |f(x)| \left(\int_0^{(2|f(x)|)^p} \alpha^{-1/p} d\alpha \right) dx.$$

Since $p > 1$, $\int_0^t \alpha^{-1/p} d\alpha = a_p t^{1-1/p}$ with $a_p = p/(p-1)$. With $t = (2|f|)^p$ this equals $a_p 2^{p-1} |f|^{p-1}$, hence

$$\int_{\mathbb{R}^d} (f^*)^p \leq A' a_p 2^{p-1} \int_{\mathbb{R}^d} |f|^p = A_p^p \|f\|_{L^p}^p, \quad A_p^p = A' a_p 2^{p-1}.$$

Thus $\|f^*\|_{L^p} \leq A_p \|f\|_{L^p}$, and the factor $a_p = p/(p-1)$ shows $A_p = O(1/(p-1))$ as $p \rightarrow 1^+$. $\square$

Remark. The proof is the prototype of the Marcinkiewicz scheme of §40.3: a weak-type $(1, 1)$ bound and a trivial (∞, ∞) bound are fused, via the distribution function, into a strong (p, p) bound for every intermediate p . The blow-up of A_p at the L^1 endpoint is unavoidable, since the strong $(1, 1)$ inequality is false.

Worked Example 40.1.12 (Lebesgue differentiation as a consequence). Deduce that for $f \in L^p(\mathbb{R}^d)$, $1 < p \leq \infty$, the averages $\frac{1}{m(B_r(x))} \int_{B_r(x)} f \rightarrow f(x)$ a.e. as $r \rightarrow 0$.

Recognition. The maximal theorem controls the “oscillation” operator; pair it with density of continuous functions.

Solution. For continuous g the averages converge to $g(x)$ everywhere. Write $f = g + h$ with g continuous and $\|h\|_{L^p}$ small. The limsup of the oscillation of the averages of f is $\leq 2h^*(x)$ pointwise. By the maximal theorem $\{2h^* > \alpha\}$ has measure $\leq (2A_p \|h\|_{L^p} / \alpha)^p$, which $\rightarrow 0$ as $\|h\|_{L^p} \rightarrow 0$ for each fixed α . Hence the oscillation is 0 a.e., i.e. the averages converge to f a.e. *Check.* For $p = \infty$ no maximal estimate is needed beyond $h^* \leq \|h\|_{L^\infty}$; the result is the classical Lebesgue differentiation theorem.

Worked Example 40.1.13 (Local non-integrability at $p = 1$). On $\mathbb{R}^d$ let $f(x) = |x|^{-d} (\log 1/|x|)^{-1-\delta}$ for $|x| \leq 1/2$ and $f = 0$ otherwise, with $0 < \delta \leq 1$. Show $f \in L^1$ but f^* is not integrable near the origin.

Recognition. A borderline log-power example separating L^1 from “ $f^* \in L^1$ ”.

Solution. Integrability: in polar coordinates $\int_{|x| \leq 1/2} |x|^{-d} (\log 1/|x|)^{-1-\delta} dx = c_d \int_0^{1/2} r^{-1} (\log 1/r)^{-1-\delta} dr$, which with $u = \log(1/r)$ becomes $c_d \int_{\log 2}^\infty u^{-1-\delta} du < \infty$ since $\delta > 0$. Lower bound on f^* : averaging over $B_{|x|}(0)$ (radius $|x|$, contains 0) gives $f^*(x) \geq c |x|^{-d} (\log 1/|x|)^{-\delta}$. Integrating this lower bound over $|x| \leq 1/2$ gives $c' \int_0^{1/2} r^{-1} (\log 1/r)^{-\delta} dr = c' \int_{\log 2}^\infty u^{-\delta} du = \infty$ for $0 < \delta \leq 1$. *Check.* The maximal operation lowers the log-exponent from $1 + \delta$ to δ , exactly enough to cross the divergence threshold — the sharp illustration that $f \in L^1$ does not give $f^* \in L^1_{\text{loc}}$.

Caution. The maximal theorem requires $p > 1$. There is no strong $(1, 1)$ bound, and there is no useful endpoint estimate at $p = 1$ beyond the weak-type inequality. When an L^1 statement is genuinely needed, the correct substitute is not “ $f^* \in L^1$ ” but membership in the real Hardy space H^1_r (functions whose maximal function is integrable), or the weak- L^1 space.

Worked Example 40.1.14 (A weak-type bound is sharp for the maximal function). Show that the exponent in the weak-type $(1, 1)$ inequality cannot be improved: there is no bound of the form $m(\{f^* > \alpha\}) \leq C\alpha^{-1} \|f\|_{L^1}^\theta (\dots)$ with better α -decay than α^{-1} .

Recognition. Test on a Dirac-like family and watch the scaling.

Solution. Take $f_n = n^d \mathbf{1}_{B_{1/n}(0)}$, so $\|f_n\|_{L^1} = v_d$ for all n . As in the indicator example, $f_n^*(x) \geq c(1 + n|x|)^{-d} \cdot \text{const}$ near the origin and in fact $f_n^*(x) \geq c|x|^{-d}$ for $1/n \leq |x| \leq 1$. Then $\{f_n^* > \alpha\} \supseteq \{c|x|^{-d} > \alpha\} = \{|x| < (c/\alpha)^{1/d}\}$, of measure $\geq c'\alpha^{-1}$. Thus $m(\{f_n^* > \alpha\}) \geq c'\alpha^{-1} = c''\alpha^{-1} \|f_n\|_{L^1}$, matching the weak-type bound’s α^{-1} exactly. *Check.* Any decay faster than α^{-1} would be contradicted by letting $n \rightarrow \infty$.

Worked Example 40.1.15 (Layer-cake meets Hölder). Use the layer-cake formula to prove the elementary interpolation bound $\|f\|_{L^2}^2 \leq \|f\|_{L^1}^{1/2} \dots$ — more precisely, that if $f \in L^1 \cap L^\infty$ then $\|f\|_{L^p}^p \leq \|f\|_{L^\infty}^{p-1} \|f\|_{L^1}$ for $1 \leq p \leq \infty$.

Recognition. Bound the distribution function two ways and split the α -integral.

Solution. By layer-cake, $\|f\|_{L^p}^p = p \int_0^\infty \alpha^{p-1} \lambda_{|f|}(\alpha) d\alpha$. For $\alpha \geq \|f\|_{L^\infty}$, $\lambda(\alpha) = 0$; for $0 < \alpha < \|f\|_{L^\infty}$ use Tchebychev $\lambda(\alpha) \leq \alpha^{-1} \|f\|_{L^1}$. Then $\|f\|_{L^p}^p \leq p \int_0^{\|f\|_{L^\infty}} \alpha^{p-2} \|f\|_{L^1} d\alpha = \frac{p}{p-1} \|f\|_{L^\infty}^{p-1} \|f\|_{L^1}$. *Check.* Dropping the harmless $p/(p-1)$ this is the log-convexity of $p \mapsto \|f\|_{L^p}$; at $p = 1$ it is an identity.

Worked Example 40.1.16 (A maximal function need not be lower semicontinuous trap). Decide whether $\{f^* > \alpha\}$ is always open, and use it to justify the reduction to compact subsets in the proof of Theorem 40.1.5.

Recognition. Examine continuity of averages as the ball moves.

Solution. The set $E_\alpha = \{f^* > \alpha\}$ is open: if $f^*(x_0) > \alpha$ there is a ball $B \ni x_0$ with average $> \alpha$, and B contains a neighbourhood of x_0 , every point of which has $f^* > \alpha$ via that same B . Openness gives inner regularity $m(E_\alpha) = \sup\{m(K) : K \subseteq E_\alpha \text{ compact}\}$, which is exactly the reduction used. *Check.* Without openness the covering argument (which produces balls only around points of a compact set) would not recover the full measure of E_α .

Exercises (Stein, Ch.2 — maximal function, distribution function, Hardy space).

1. In $\mathbb{R}^d$, suppose $f(x) = |x|^{-d}(\log 1/|x|)^{-1-\delta}$ if $|x| \leq 1/2$, and $f(x) = 0$ otherwise. Observe that $f^*(x) \geq c|x|^{-d}(\log 1/|x|)^{-\delta}$ if $|x| \leq 1/2$. Hence for $0 < \delta \leq 1$, $f \in L^1(\mathbb{R}^d)$ but f^* is not integrable over the unit ball. (Stein Ex. 12.)
2. Prove that the basic distribution-function inequality $m(\{f^* > \alpha\}) \leq \frac{A'}{\alpha} \int_{\{|f| > \alpha/2\}} |f|$ can essentially be reversed: there is a constant A with $m(\{x : f^*(x) > \alpha\}) \geq \frac{A}{\alpha} \int_{\{|f| > \alpha\}} |f| dx$. [Hint: write $E_\alpha = \bigcup_j Q_j$ with Q_j closed cubes satisfying $\text{diam}(Q_j) \leq d(Q_j, E_\alpha^c) \leq 4 \text{diam}(Q_j)$; for each Q_j let B_j be the smallest ball with $Q_j \subset B_j$ meeting E_α^c . Then $m(B_j) \leq cm(Q_j)$ and $\frac{1}{m(B_j)} \int_{B_j} |f| \leq \alpha$, so $\frac{c^{-1}}{\alpha} \int_{Q_j} |f| \leq m(Q_j)$; sum in j and use $\{|f| > \alpha\} \subseteq \{f^* > \alpha\}$.] (Stein Ex. 13.)
3. Deduce the following from the two preceding inequalities. If f is integrable on $\mathbb{R}^d$ and $B_1 \subset B_2$ are balls, then (a) f^* is integrable on B_1 if $|f| \log(1 + |f|)$ is integrable on B_2 ; (b) conversely, if f^* is integrable on B_1 then $|f| \log(1 + |f|)$ is integrable there. [Hint: integrate the inequalities in α for $\alpha \geq 1$.] (Stein Ex. 14.)
4. Consider the weak-type space of all f with $m(\{|f| > \alpha\}) \leq A/\alpha$ for some A and all $\alpha > 0$; let $N(f)$ be the least such A . (a) Show N is not a genuine norm, and there is no norm $\|\cdot\|$ on this space equivalent to N . (b) Show this space has no non-trivial bounded linear functionals. [Hint: on $\mathbb{R}$, $f(x) = 1/|x|$ has $N(f) = 2$; but $f_N(x) = \frac{1}{N} \sum_{k=1}^N f(x+k)$ satisfies $N(f_N) \geq c \log N$.] (Stein Ex. 15.)
5. Prove that H_1^r is complete. Let $\{f_n\}$ be Cauchy in H_1^r ; it is also Cauchy in L^1 , so $f_n \rightarrow f$ in L^1 . For a suitable subsequence $\{n_k\}$ write $f = f_{n_1} + \sum_{k=1}^\infty (f_{n_{k+1}} - f_{n_k})$ and recognise an atomic decomposition. (Stein Ex. 16.)
6. Let $f(x) = 1/(x \log x^2)$ for $0 < x \leq 1/2$, $f(x) = 0$ for $x > 1/2$, extended oddly by $f(x) = -f(-x)$ for $x < 0$. Then f is integrable on $\mathbb{R}$ with $\int f = 0$, hence f is a multiple of a 1-atom. Verify $M(f) \geq c/(|x| \log |x|)$ for $|x| \leq 1/2$, so $M(f) \notin L^1$, and conclude $f \notin H_1^r$. (Stein Ex. 17.)
7. Show there is $c > 1$ so that every $f \in L^1(\mathbb{R}^d)$ can be written $f = \sum_k \lambda_k a_k$ with $\sum |\lambda_k| \leq c \|f\|_{L^1}$, where the a_k are “faux” atoms: a_k supported in a ball B_k , $|a_k| \leq 1/m(B_k)$, but with the cancellation $\int a_k = 0$ dropped. [Hint: let $f_n = E_n(f)$ be the average of f over dyadic cubes of the n th generation; pick n_k with $\|f_{n_{k+1}} - f_{n_k}\|_{L^1} < 2^{-k}$ and telescope.] (Stein Ex. 18.)
8. Two senses in which H_1^r is near L^1 yet different. (a) If f_0 is positive, decreasing and integrable on $(0, \infty)$, there is $f \in H_1^r(\mathbb{R})$ with $|f(x)| \geq f_0(|x|)$. (b) If $f \in H_1^r(\mathbb{R}^d)$ and $f \geq 0$ on a ball B_1 , then $f \log(1 + f)$ is integrable over any proper sub-ball $B_0 \subset B_1$. [Hint: (a) take $f(x) = \text{sign}(x)f_0(|x|)$ and decompose; (b) use Ex. 3 with the maximal characterization.] (Stein Ex. 19.)
9. For $f \in L^1(\mathbb{R}^d)$ the Riemann–Lebesgue lemma gives $\hat{f} \rightarrow 0$ at infinity but no better. Show, however, that for $f \in H_1^r$, $\int_{\mathbb{R}^d} \left| \hat{f}(\xi) \right| \frac{d\xi}{|\xi|^d} \leq A \|f\|_{H_1^r}$. [Hint: verify on atoms.] (Stein Ex. 20.)
10. Prove that if $|f(x)| \leq A(1 + |x|)^{-d-1}$ and $\int_{\mathbb{R}^d} f = 0$, then $f \in H_1^r(\mathbb{R}^d)$. [Hint: write $f = \sum_{k \geq 0} f_k$ with $f_0 = f \mathbf{1}_{|x| \leq 1}$ and $f_k = f \mathbf{1}_{2^{k-1} < |x| \leq 2^k}$; set $c_k = \int f_k$, $s_k = \sum_{j \geq k} c_j$, so $s_0 = 0$; fix η supported in $|x| \leq 1$ with $\int \eta = 1$, $\eta_k = 2^{-kd} \eta(2^{-k}x)$, and write $f = \sum_k (f_k - c_k \eta_k) + \sum_k s_k (\eta_k - \eta_{k-1})$.] (Stein Ex. 21.)
11. Let f on $\mathbb{R}$ be the atom supported in $|x| \leq 1/2$ given by $f(x) = \text{sign}(x)$. Apply the maximal function $f_0^*(x) = \sup_{\varepsilon > 0} |(f * \chi_\varepsilon)(x)|$, where χ is the indicator of $|x| \leq 1/2$ and $\chi_\varepsilon(x) = \varepsilon^{-1} \chi(x/\varepsilon)$. Verify $|f_0^*(x)| \geq 1/(2|x|)$ for $|x| \geq 1/2$, so $f_0^* \notin L^1$; thus this f_0^* cannot characterise H_1^r . (Stein Ex. 22.)
12. Verify the BMO examples: (a) $\log |x| \in \text{BMO}(\mathbb{R}^d)$; (b) if $f(x) = \log x$ for $x > 0$ and $= 0$ for $x \leq 0$ then $f \notin \text{BMO}(\mathbb{R})$; (c) for $\delta \geq 0$, $(\log |x|)^\delta \in \text{BMO}(\mathbb{R}^d)$ iff $\delta \leq 1$. [Hint: for $\log |x|$ use the scaling $f(rx) = f(x) + c_r$ to reduce to balls of radius 1; for (b) test on small intervals about the origin.] (Stein Ex. 23.)
13. Using Exercises 8(a) and 12, give examples of $f \in H_1^r$ and $g \in \text{BMO}$ for which $|f(x)g(x)|$ is not integrable over $\mathbb{R}^d$. (Stein Ex. 24.)

Problems (Stein, Ch.2 — Hardy space and BMO).

1. (H_1^r weak compactness.) If $\{f_n\} \subset H_1^r$ with $\|f_n\|_{H_1^r} \leq A$, one can select a subsequence $\{f_{n_k}\}$ and find $f \in H_1^r$ with $\int f_{n_k} \varphi \rightarrow \int f \varphi$ for every continuous φ of compact support. Contrast with the failure of weak compactness in L^1 . [Hint: extract a weak* limit measure μ ; if $\sup_\varepsilon |\mu * \varphi_\varepsilon| \in L^1$ then μ is absolutely continuous.] (Stein Prob. 1.)
2. Let P be a non-zero polynomial of degree k in $\mathbb{R}^d$. Then $f = \log |P(x)|$ is in BMO with $\|f\|_{\text{BMO}} \leq c_k$ depending only on k . [Hint: do $d = 1$ first; then induct on dimension using that if $f(x, y)$ is, for each y , a BMO($\mathbb{R}$) function in x uniformly in y , and symmetrically, then $f \in \text{BMO}(\mathbb{R}^2)$.] (Stein Prob. 3.)
3. (John–Nirenberg inequalities.) For every $f \in \text{BMO}(\mathbb{R}^d)$: (a) for every $q < \infty$ there is b_q with $\sup_B \frac{1}{m(B)} \int_B |f - f_B|^q \leq b_q^q \|f\|_{\text{BMO}}^q$; (b) there are $\mu, A > 0$ with $\sup_B \frac{1}{m(B)} \int_B e^{\mu|f - f_B|} \leq A$ whenever $\|f\|_{\text{BMO}} \leq 1$. [Hint: for (a) test against p -atoms (p dual to q); for (b) use $b_q = O(q)$ and sum the exponential series.] (Stein Prob. 4.)
4. The Hilbert transform of a bounded function is in BMO. Show in two ways: (a) directly, $\|H(f)\|_{\text{BMO}} \leq A \|f\|_{L^\infty}$ with A independent of the L^p norm; (b) by duality using the H_1^r –BMO duality theorem. [Hint: for (a) fix a ball B , let B_1 be its double, and treat $f\chi_{B_1}$ and $f\chi_{B_1^c}$ separately.] (Stein Prob. 5.)
5. (Maximal characterization of H_1^r .) Let $\Phi \in \mathcal{S}$ with $\int \Phi = 0$ and $M(f)(x) = \sup_{\varepsilon > 0} |(f * \Phi_\varepsilon)(x)|$ for $f \in L^1$. Then (a) $f \in H_1^r$ iff $M(f) \in L^1$; (b) one may relax $\Phi \in \mathcal{S}$ to $|\partial^\alpha \Phi(x)| \leq c_\alpha (1 + |x|)^{-d-1-|\alpha|}$; (c) the heat kernel $\Phi_{t^{1/2}}(x) = (4\pi t)^{-d/2} e^{-|x|^2/4t}$ and the Poisson kernel $\Phi_t(x) = c_d t / (t^2 + |x|^2)^{(d+1)/2}$ are admissible. (Stein Prob. 6.)
6. (H^p at $p = 1$, application to compensated compactness.) Suppose A, B are vector fields on $\mathbb{R}^d$ with components in L^2 , $\text{div } A = 0$ and $\text{curl } B = 0$ (derivatives in the distributional sense). Then $\sum_k A_k B_k \in H_1^r$ — in contrast with the general fact that $f, g \in L^2$ gives only $fg \in L^1$. (Stein Prob. 10.)

Additional exercises.

1. (Warm-up.) For $f = \mathbf{1}_{[0,1]}$ on $\mathbb{R}$ compute $f^*(x)$ for all x and verify directly that $f^* \sim 1/|x|$ at infinity, hence $f^* \notin L^1$ but $f^* \in L^p$ for all $p > 1$.
2. Show the maximal operator is monotone and positively homogeneous, and give an explicit f, g with $(f + g)^* < f^* + g^*$ *strictly* at some point (so sub-additivity is not additivity).
3. (Distribution function arithmetic.) For $F(x) = \min(|x|^{-1/2}, 1) \mathbf{1}_{|x| \leq 4}$ on $\mathbb{R}$, compute $\lambda_F(\alpha)$ for all $\alpha > 0$ and use the layer-cake formula to find the largest p with $F \in L^p$.
4. Prove the centered and uncentered maximal functions $f_c^*(x) = \sup_{r>0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f|$ and f^* (Definition 40.1.1) satisfy $f_c^* \leq f^* \leq 2^d f_c^*$, hence have equivalent L^p norms.
5. (Tchebychev sharpness.) Exhibit, for each $p > 0$, a function F for which the Tchebychev inequality $\lambda(\alpha) \leq \alpha^{-p} \|F\|_{L^p}^p$ is an equality at some α .
6. Use Theorem 40.1.6 and the layer-cake formula to reprove $\|f^*\|_{L^2} \leq A \|f\|_{L^2}$ from scratch, tracking the constant.
7. (NON-example.) Show that the operator $f \mapsto f^*$ is *not* of strong type $(1, 1)$ by producing a sequence f_n with $\|f_n\|_{L^1} = 1$ but $\|f_n^*\|_{L^1} \rightarrow \infty$.
8. (Diagram problem.) Sketch, in the (α, λ) plane, the distribution function of f^* for $f = \mathbf{1}_{[-1,1]}$, marking the two regimes $\alpha < 1$ and $\alpha \geq 1$, and shade the region whose area is $\|f^*\|_{L^1([-2,2])}$.
9. (Dyadic maximal function.) Let $M_{\text{dy}} f(x) = \sup \frac{1}{m(Q)} \int_Q |f|$ over dyadic cubes $Q \ni x$. Prove M_{dy} is of weak type $(1, 1)$ with constant 1 (better than 3^d), using that dyadic cubes are nested.
10. Prove that if $f \in L \log L$ on a bounded set E (i.e. $\int_E |f| \log(1 + |f|) < \infty$) then $f^* \in L^1(E)$, citing Stein Exercise 3 above; conversely give the failure when only $f \in L^1$.

40.2 The Hilbert Transform

The Hilbert transform is the model singular integral operator and the prototype for the whole Calderón–Zygmund theory. We develop it on the line $\mathbb{R}$, with the upper half-plane $\mathbb{R}_+^2 = \{z = x + iy : x \in \mathbb{R}, y > 0\}$ playing the role the unit disc played for Fourier series. Two faces of the same operator emerge: a Fourier multiplier on L^2 , and a principal-value singular integral; the bridge between them is the Cauchy integral.

Origins: the conjugate function

For $f \sim \sum_n a_n e^{in\theta}$ in $L^2([0, 2\pi])$, the *conjugate function* is

$$\tilde{f}(\theta) \sim \sum_{n=-\infty}^{\infty} \frac{\text{sign}(n)}{i} a_n e^{in\theta}, \quad \text{sign}(n) = \begin{cases} 1, & n > 0, \\ 0, & n = 0, \\ -1, & n < 0. \end{cases}$$

The significance is analytic: $\frac{1}{2}(f + i\tilde{f} + a_0) \sim \sum_{n \geq 0} a_n e^{in\theta} = F(e^{i\theta})$ is the boundary value of the holomorphic function $F(z) = \sum_{n \geq 0} a_n z^n = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(\theta)}{e^{i\theta} - z} i e^{i\theta} d\theta$ on the disc. The L^2 identity $\frac{1}{2\pi} \int |\tilde{f}|^2 + |a_0|^2 = \frac{1}{2\pi} \int |f|^2$ is immediate from Parseval; the deep fact, due to M. Riesz, is the L^p analogue for $1 < p < \infty$.

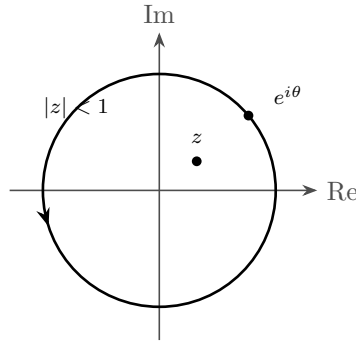


Figure 40.4. The Cauchy integral $F(z)$ is holomorphic for $|z| < 1$; its boundary values recover f and its conjugate $\tilde{f}$. (Schematic.)

The L^2 formalism on the line

Definition 40.2.1. For $f \in L^2(\mathbb{R})$ the *Cauchy integral* is

$$F(z) = \mathcal{C}(f)(z) = \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{f(t)}{t - z} dt, \quad \text{Im}(z) > 0.$$

The integral converges for $\text{Im } z > 0$ because $t \mapsto 1/(t - z)$ lies in $L^2(\mathbb{R})$, and F is holomorphic in the upper half-plane.

Via Plancherel one has the Fourier representation $F(z) = \int_0^\infty \hat{f}(\xi) e^{2\pi i z \xi} d\xi$ (using $\int_0^\infty e^{2\pi i z \xi} d\xi = -\frac{1}{2\pi i z}$ for $\text{Im } z > 0$), so that as $y \rightarrow 0$ the boundary value is $P(f)(x) = \int_{-\infty}^\infty \hat{f}(\xi) \chi_{(0, \infty)}(\xi) e^{2\pi i x \xi} d\xi$: the orthogonal projection onto positive frequencies.

Definition 40.2.2 (Hilbert transform, multiplier form). For $f \in L^2(\mathbb{R})$,

$$H(f)(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) \frac{\text{sign}(\xi)}{i} e^{2\pi i x \xi} d\xi, \quad \text{equivalently} \quad \widehat{H(f)}(\xi) = \frac{\text{sign}(\xi)}{i} \hat{f}(\xi).$$

$$\begin{array}{ccc} L^2(\mathbb{R}) & \xrightarrow{H} & L^2(\mathbb{R}) \\ \mathcal{F} \downarrow \sim & & \sim \downarrow \mathcal{F} \\ L^2(\mathbb{R}) & \xrightarrow{m \cdot} & L^2(\mathbb{R}) \end{array}$$

Figure 40.5. The Hilbert transform diagonalised by the Fourier transform $\mathcal{F}$ (a unitary $\cong$): H is conjugate to multiplication by the unimodular symbol $m(\xi) = \text{sign}(\xi)/i$, i.e. $\mathcal{F} \circ H = (m \cdot) \circ \mathcal{F}$.

Proposition 40.2.3 (Algebraic properties). *Let $P = \frac{1}{2}(I + iH)$ be the projection onto positive frequencies. Then*

1. $P = \frac{1}{2}(I + iH)$ and P is an orthogonal projection on L^2 ;
2. H is unitary on L^2 : $\|Hf\|_{L^2} = \|f\|_{L^2}$;
3. $H \circ H = H^2 = -I$, hence H is invertible with $H^{-1} = -H$.

Proof. The multiplier of H is $m(\xi) = \text{sign}(\xi)/i = -i \text{sign}(\xi)$, of modulus 1 for $\xi \neq 0$; by Plancherel $\|Hf\|_{L^2} = \|m\hat{f}\|_{L^2} = \|\hat{f}\|_{L^2} = \|f\|_{L^2}$, giving (2). Squaring the multiplier, $m(\xi)^2 = -\text{sign}(\xi)^2 = -1$ a.e., so $H^2 = -I$, giving (3). Finally $\frac{1}{2}(1 + im(\xi)) = \frac{1}{2}(1 + \text{sign} \xi) = \chi_{(0, \infty)}(\xi)$, which is the multiplier of the projection onto positive frequencies; this is (1). $\square$

Theorem 40.2.4 (Singular-integral representation). *For $f \in L^2(\mathbb{R})$,*

$$H(f)(x) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \int_{|t| \geq \varepsilon} \frac{f(x-t)}{t} dt = \lim_{\varepsilon \rightarrow 0} H_\varepsilon(f)(x),$$

the limit taken in the $L^2(\mathbb{R})$ norm, where $H_\varepsilon(f)(x) = \frac{1}{\pi} \int_{|t| \geq \varepsilon} \frac{f(x-t)}{t} dt$.

Proof sketch. Write $z = x + iy$ and decompose $-\frac{1}{i\pi z} = P_y(x) + iQ_y(x)$, the Poisson and conjugate Poisson kernels

$$P_y(x) = \frac{y}{\pi(x^2 + y^2)}, \quad Q_y(x) = \frac{x}{\pi(x^2 + y^2)},$$

with Fourier transforms $\widehat{P}_y(\xi) = e^{-2\pi y|\xi|}$ and $\widehat{Q}_y(\xi) = \frac{\text{sign}(\xi)}{i} e^{-2\pi y|\xi|}$. As $y \rightarrow 0$, $f * Q_y \rightarrow H(f)$ in L^2 (the multipliers converge boundedly). One then writes $H_\varepsilon(f) - f * Q_\varepsilon = f * \Delta_\varepsilon$ where $\Delta_\varepsilon(x) = \frac{1}{\pi x} \mathbf{1}_{|x| \geq \varepsilon} - Q_\varepsilon(x)$, satisfies $\Delta_\varepsilon(x) = \varepsilon^{-1} \Delta_1(\varepsilon^{-1}x)$ with $|\Delta_1(x)| \leq A/(1+x^2)$, and has $\int \Delta_\varepsilon = 0$ (it is odd). Since $\int \Delta_\varepsilon = 0$,

$$(f * \Delta_\varepsilon)(x) = \int (f(x-t) - f(x)) \Delta_1(t) dt,$$

and by Minkowski's integral inequality $\|f * \Delta_\varepsilon\|_{L^2} \leq \int \|f(\cdot - \varepsilon t) - f\|_{L^2} |\Delta_1(t)| dt \rightarrow 0$ by dominated convergence (continuity of translation in L^2). Hence $H_\varepsilon(f) \rightarrow H(f)$ in L^2 . The same argument yields $\|H_\varepsilon f\|_{L^2} \leq A \|f\|_{L^2}$ uniformly in ε . $\square$

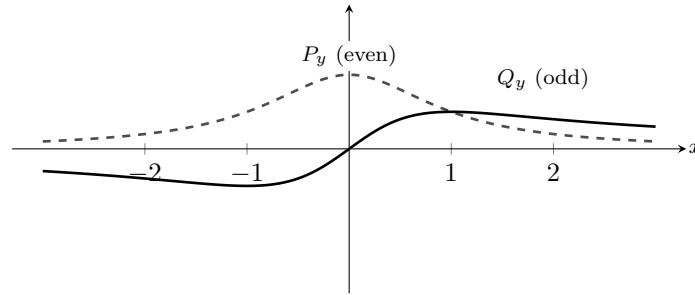


Figure 40.6. The Poisson kernel $P_y(x) = \frac{y}{\pi(x^2 + y^2)}$ (even, dashed) and conjugate Poisson kernel $Q_y(x) = \frac{x}{\pi(x^2 + y^2)}$ (odd, solid), here at $y = 1$. The Hilbert transform is the boundary limit of convolution with Q_y .

Worked Example 40.2.5 (The kernel $1/(\pi x)$ as a principal value). Verify that $H_\varepsilon(f) = \frac{1}{\pi} \int_{|t| \geq \varepsilon} \frac{f(x-t)}{t} dt$ is well defined for $f \in L^2$ even though $1/t$ is not integrable.

Recognition. The singularity at $t = 0$ is removed by the cutoff $|t| \geq \varepsilon$; the issue is decay, handled by Cauchy–Schwarz.

Solution. For fixed $\varepsilon > 0$, split the integral into $\varepsilon \leq |t| \leq 1$ and $|t| > 1$. On $|t| > 1$, $1/t \in L^2$, so $\int_{|t|>1} \frac{f(x-t)}{t} dt$ converges absolutely by Cauchy–Schwarz, $\leq \|f\|_{L^2} \|1/t\|_{L^2(|t|>1)} < \infty$. On $\varepsilon \leq |t| \leq 1$, $1/t$ is bounded by $1/\varepsilon$, so again Cauchy–Schwarz applies. *Check.* The cancellation $\int_{\varepsilon \leq |t| \leq 1} \frac{dt}{t} = 0$ (odd integrand) is what allows the principal-value limit $\varepsilon \rightarrow 0$ to exist; without it, the log near 0 would diverge.

Worked Example 40.2.6 (Hilbert transform of the indicator of an interval). Compute $H(\mathbf{1}_{(-1,1)})$ and locate its singularities.

Recognition. Direct principal-value integration of $1/t$ against a step function.

Solution. With $f = \mathbf{1}_{(-1,1)}$,

$$H(f)(x) = \frac{1}{\pi} \text{p.v.} \int_{-1}^1 \frac{dt}{x-t} = \frac{1}{\pi} [-\log|x-t|]_{t=-1}^{t=1} = \frac{1}{\pi} \log \left| \frac{x+1}{x-1} \right|.$$

This has logarithmic singularities at $x = \pm 1$ and behaves like $\frac{2}{\pi x}$ as $|x| \rightarrow \infty$. *Check.* f is even, so $H(f)$ is odd; indeed $\log \left| \frac{x+1}{x-1} \right|$ is odd in x . Since $H(f) \sim 2/(\pi x)$, $H(f) \notin L^1(\mathbb{R})$, consistent with the failure of H on L^1 .

Worked Example 40.2.7 (An L^1 Hilbert transform via cancellation). Let $g = \mathbf{1}_{(0,1)} - \mathbf{1}_{(-1,0)}$. Show $H(g)$ is integrable.

Recognition. An odd function with $\int g = 0$; the cancellation improves decay at infinity.

Solution. By the same computation $H(g)(x) = \frac{1}{\pi} \log \left| \frac{x^2}{x^2-1} \right|$, an even function with mild log singularities at $-1, 0, 1$. As $|x| \rightarrow \infty$, $\log \frac{x^2}{x^2-1} = \log \frac{1}{1-x^{-2}} \sim x^{-2}$, so $H(g)(x) \sim \frac{1}{\pi x^2}$, which is integrable at infinity. *Check.* The decisive difference from the previous example is $\int g = 0$; the general principle is that for bounded compactly supported f , one has $H(f) \in L^1$ iff $\int f = 0$, since $H(f)(x) = \frac{a}{\pi x} + O(x^{-2})$ with $a = \int f$.

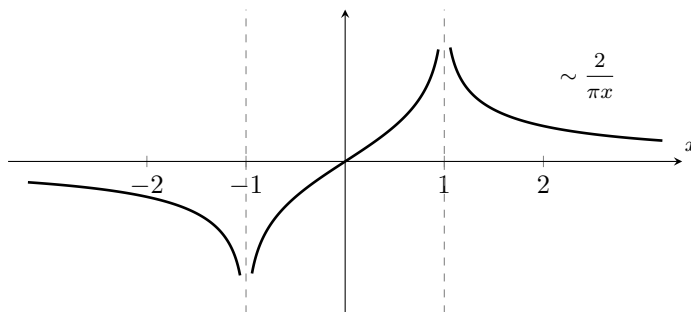


Figure 40.7. $H(\mathbf{1}_{(-1,1)})(x) = \frac{1}{\pi} \log \left| \frac{x+1}{x-1} \right|$: logarithmic blow-up at ± 1 , slow $1/x$ decay (not L^1).

The L^p boundedness theorem of M. Riesz

Theorem 40.2.8 (M. Riesz). Let $1 < p < \infty$. Then the Hilbert transform H , defined on $L^2 \cap L^p$ by Definition 40.2.2, satisfies

$$\|H(f)\|_{L^p} \leq A_p \|f\|_{L^p}, \quad f \in L^2 \cap L^p,$$

with A_p independent of f , and H extends uniquely to all of L^p with the same bound. The conclusion fails at $p = 1$ and $p = \infty$.

Proof. Reduce to real-valued f (the complex case costs at most a factor 2) and, by density, to $f \in C_0^\infty(\mathbb{R})$. For such f the Cauchy integral $F(z) = \frac{1}{2\pi i} \int \frac{f(t)}{t-z} dt$ extends continuously to the closed half-plane, is bounded there, and satisfies the decay

$$|F(z)| \leq \frac{M}{1+|z|}, \quad z = x + iy, \quad y \geq 0, \quad (*)$$

proved from the Fourier representation $F(z) = \int_0^\infty \hat{f}(\xi) e^{2\pi i z \xi} d\xi$ by one integration by parts. Moreover, by continuity of F up to the boundary, $2F(x) = \lim_{y \rightarrow 0} 2F(x + iy) = f(x) + iH(f)(x)$, and $H(f)$ is real-valued.

Step 1 (Cauchy's theorem). For every integer $k \geq 2$,

$$\int_{-\infty}^{\infty} F(x)^k dx = 0. \quad (\dagger)$$

Indeed F^k is holomorphic in the upper half-plane; integrate it over the rectangle with vertices $\pm R + i\varepsilon$, $\pm R + iR$. Cauchy's theorem gives $\oint F^k dz = 0$, and letting $\varepsilon \rightarrow 0$, $R \rightarrow \infty$ and using the decay (*) kills the three non-real-axis sides, leaving ($\dagger$). (In particular (*) shows $H(f) \in L^p$ for all $p > 1$.)

Step 2 (even powers). Take $k = 2\ell$ even and read off the real part. Expanding $\operatorname{Re}(f + iHf)^{2\ell}$,

$$0 = \int \operatorname{Re}(F \cdot 2)^{2\ell} / 2^{2\ell} dx \implies \int (Hf)^{2\ell} dx \leq \sum_{r=1}^{\ell} a_r \int f^{2r} (Hf)^{2\ell-2r} dx$$

with $a_r = \binom{2\ell}{2r}$. For $\ell = 2$ this is the clean instance $\int f^4 - 6 \int f^2 (Hf)^2 + \int (Hf)^4 = 0$, whence by Schwarz $\int (Hf)^4 \leq 6(\int f^4)^{1/2}(\int (Hf)^4)^{1/2}$, giving $\|Hf\|_{L^4} \leq 6^{1/2} \|f\|_{L^4}$. In general, set $p = 2\ell$ and apply Hölder with dual exponents $\frac{2\ell}{2r}, \frac{2\ell}{2\ell-2r}$ to each term: $\int f^{2r} (Hf)^{2\ell-2r} \leq \|f\|_{L^p}^{2r} \|Hf\|_{L^p}^{2\ell-2r}$. Writing $X = \|Hf\|_{L^p}$ and normalising $\|f\|_{L^p} = 1$ gives $X^{2\ell} \leq \sum_{r=1}^{\ell} a_r X^{2\ell-2r}$. Either $X < 1$, or $X \geq 1$ and then $X^{2\ell} \leq (\sum_r a_r) X^{2\ell-2}$, so $X^2 \leq \sum_r a_r \leq 2^{2\ell}$. In either case $X \leq 2^\ell$, i.e.

$$\|H(f)\|_{L^p} \leq A_p \|f\|_{L^p}, \quad A_p = 2^{p/2}, \quad p = 2\ell. \quad (\ddagger)$$

This extends to simple f by the usual $L^2 \cap L^p$ approximation.

Step 3 (interpolation). Extend H to complex functions by $H(f_1 + if_2) = Hf_1 + iHf_2$; ($\ddagger$) persists with A_p doubled. Applying the Riesz interpolation theorem (§40.3, Theorem 40.3.5) with the pair of even endpoints $p_0 = 2, p_1 = 2\ell$ yields $\|Hf\|_{L^p} \leq A_p \|f\|_{L^p}$ for all $2 \leq p \leq 2\ell$; since ℓ is arbitrary, this covers all $2 \leq p < \infty$.

Step 4 (duality). Pass from $p \geq 2$ to $1 < p \leq 2$ using the skew-adjointness

$$\int_{-\infty}^{\infty} (Hf)g dx = - \int_{-\infty}^{\infty} f(Hg) dx, \quad f, g \in L^2,$$

which follows from Plancherel and $\widehat{Hf} = \frac{\operatorname{sign}(\xi)}{i} \hat{f}$. For $1 < p \leq 2$ and simple f , $\|Hf\|_{L^p} = \sup_{\|g\|_{L^q} \leq 1} |\int (Hf)g| = \sup_g |\int f(Hg)| \leq \|f\|_{L^p} \sup_g \|Hg\|_{L^q} \leq A_q \|f\|_{L^p}$, using Step 3 for the dual exponent $2 \leq q < \infty$. Thus (**) holds for all $1 < p < \infty$, and a final limiting argument extends it from simple functions to all $f \in L^2 \cap L^p$ and then to L^p . $\square$

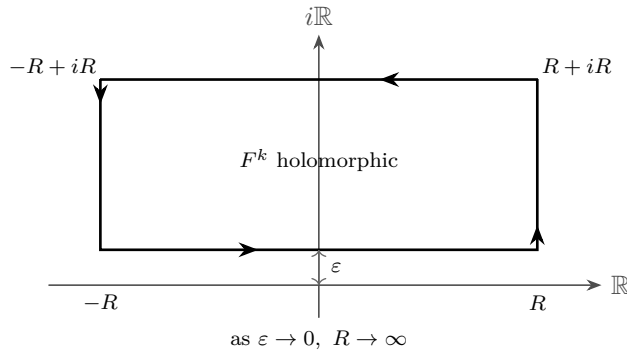


Figure 40.8. The rectangular contour for Step 1: $\oint F^k dz = 0$ by Cauchy's theorem; the decay (*) kills the upper and side edges, leaving the real-axis integral.

Remark. This complex-analytic proof is elegant but tied to the one-dimensional structure; it does not generalise to higher-dimensional singular integrals (Riesz transforms), where the real-variable Calderón–Zygmund method — weak-type $(1, 1)$ via the decomposition of §40.3, then interpolation — is required. The Hilbert transform itself also satisfies a weak-type $(1, 1)$ estimate.

Worked Example 40.2.9 (The L^4 bound from Cauchy's theorem, explicitly). Carry out Step 2 for $k = 4$ and exhibit the constant.

Recognition. Take the real part of $\int F^4 = 0$ and apply Schwarz.

Solution. Since $2F = f + iHf$, $(2F)^4 = (f + iHf)^4$, whose real part is $f^4 - 6f^2(Hf)^2 + (Hf)^4$. Integrating $\int F^4 = 0$ and taking real parts, $\int f^4 - 6\int f^2(Hf)^2 + \int (Hf)^4 = 0$, so $\int (Hf)^4 \leq 6\int f^2(Hf)^2 \leq 6(\int f^4)^{1/2}(\int (Hf)^4)^{1/2}$ by Schwarz. Dividing, $(\int (Hf)^4)^{1/2} \leq 6(\int f^4)^{1/2}$, i.e. $\|Hf\|_{L^4} \leq \sqrt{6}\|f\|_{L^4}$. *Check.* $\sqrt{6} \approx 2.449$, comfortably finite; the method gives *some* bound, not the sharp constant ($A_p = \tan(\pi/2p)$ for $p \geq 2$, Pichorides).

Worked Example 40.2.10 (Failure at $p = \infty$). Show H does not map L^∞ into L^∞ .

Recognition. Reuse the indicator computation, which already produced an unbounded output.

Solution. Take $f = \mathbf{1}_{(-1,1)} \in L^\infty$ with $\|f\|_{L^\infty} = 1$. Then $H(f)(x) = \frac{1}{\pi} \log \left| \frac{x+1}{x-1} \right| \rightarrow +\infty$ as $x \rightarrow 1$. Hence $\|Hf\|_{L^\infty} = \infty$, so $H : L^\infty \rightarrow L^\infty$ is unbounded. *Check.* The correct endpoint statement is that H maps L^∞ into BMO (bounded mean oscillation); the logarithmic blow-up is precisely a BMO function ($\log|\cdot| \in \text{BMO}$).

Worked Example 40.2.11 (H commutes with the Poisson semigroup). For $f \in L^p(\mathbb{R})$, $1 < p < \infty$, show $f * Q_y = H(f) * P_y$.

Recognition. Both sides are Fourier multipliers; compare them on L^2 then extend.

Solution. On L^2 , $\widehat{f * Q_y}(\xi) = \widehat{f}(\xi)\widehat{Q_y}(\xi) = \widehat{f}(\xi)\frac{\text{sign}\xi}{i}e^{-2\pi y|\xi|}$, while $\widehat{H(f) * P_y}(\xi) = \widehat{Hf}(\xi)\widehat{P_y}(\xi) = \frac{\text{sign}\xi}{i}\widehat{f}(\xi)e^{-2\pi y|\xi|}$. They agree, so $f * Q_y = H(f) * P_y$ in L^2 ; density of $L^2 \cap L^p$ and the bounds $\|P_y\|_{L^1} = 1$, Theorem 40.2.8 extend the identity to L^p . *Check.* Letting $y \rightarrow 0$ recovers $f * Q_y \rightarrow H(f)$ since P_y is an approximation to the identity ($\widehat{P_y} \rightarrow 1$).

$$\begin{array}{ccc} f & \xrightarrow{\cdot * Q_y} & f * Q_y \\ H \downarrow & & \parallel \\ H(f) & \xrightarrow{\cdot * P_y} & H(f) * P_y \end{array}$$

Figure 40.9. The Hilbert transform commutes with the Poisson semigroup: convolving with the conjugate Poisson kernel Q_y equals applying H then the Poisson kernel P_y , so $f * Q_y = H(f) * P_y$.

Caution. The Hilbert transform is *not* given by $\frac{1}{\pi} \int \frac{f(x-t)}{t} dt$ as an ordinary Lebesgue integral — that integral diverges. The principal value (symmetric limit $|t| \geq \varepsilon$) is essential, and it is the odd symmetry of $1/t$ that makes the cancellation work. Equivalently, H is the principal-value distribution p.v. $\frac{1}{\pi x}$ convolved with f .

Worked Example 40.2.12 (Hilbert transform of a one-sided spectrum). For $f \in L^2$ with $\widehat{f}$ supported in $\{\xi > 0\}$, show $H(f) = -if$ and interpret.

Recognition. Apply the multiplier $-i\text{sign}(\xi)$ to a one-sided spectrum.

Solution. On $\{\xi > 0\}$ the multiplier is $-i\text{sign}(\xi) = -i$, so $\widehat{Hf}(\xi) = -i\widehat{f}(\xi)$ for all ξ (since $\widehat{f}$ vanishes for $\xi < 0$), giving $H(f) = -if$. *Check.* Then $P(f) = \frac{1}{2}(I + iH)f = \frac{1}{2}(f + i(-if)) = f$, so f is fixed by the positive-frequency projection — the defining property of the Hardy space H^2 .

Worked Example 40.2.13 (Skew-adjointness gives the energy identity). Use $\int (Hf)g = -\int f(Hg)$ to rederive $\|Hf\|_{L^2} = \|f\|_{L^2}$ for real f .

Recognition. Take $g = Hf$ and use $H^2 = -I$.

Solution. With $g = Hf$, $\int (Hf)(Hf) = -\int fH(Hf) = -\int f(H^2f) = -\int f(-f) = \int f^2$, so $\|Hf\|_{L^2}^2 = \|f\|_{L^2}^2$. *Check.* This recovers the unitarity of Proposition 40.2.3(2) using only skew-adjointness and $H^2 = -I$, the continuous analogue of the conjugate-function identity $\frac{1}{2\pi} \int |\tilde{f}|^2 = \frac{1}{2\pi} \int |f|^2$ (for $a_0 = 0$).

Exercises (Stein, Ch.2 — the Hilbert transform).

1. Suppose f is a bounded function on $\mathbb{R}$ with compact support. Then $H(f) \in L^1(\mathbb{R})$ if and only if $\int f dx = 0$. [Hint: if $a = \int f$, then $H(f)(x) = \frac{a}{\pi x} + O(1/x^2)$ as $|x| \rightarrow \infty$.] (Stein Ex. 7.)
2. Suppose T is a bounded linear transformation of real-valued L^p functions into itself with $\|Tf\|_{L^p} \leq M \|f\|_{L^p}$. (a) The extension $T'(f_1 + if_2) = T(f_1) + iT(f_2)$ to complex functions has the same bound $\|T'f\|_{L^p} \leq M \|f\|_{L^p}$. (b) More generally, for any N , $\left\| \left(\sum_{j=1}^N |Tf_j|^2 \right)^{1/2} \right\|_{L^p} \leq M \left\| \left(\sum_{j=1}^N |f_j|^2 \right)^{1/2} \right\|_{L^p}$. [Hint for (b): for a unit vector $\xi \in \mathbb{R}^N$ set $F_\xi = \sum_j \xi_j f_j$; then $|(TF_\xi)(x)|^p \leq M^p |F_\xi(x)|^p$, and integrate over ξ on the unit sphere.] (Stein Ex. 8.)
3. Show the identity of two classes of harmonic functions u in the upper half-plane $\mathbb{R}_+^2$: (a) those continuous on $\overline{\mathbb{R}_+^2}$ vanishing at infinity ($u(x, y) \rightarrow 0$ as $|x| + y \rightarrow \infty$); (b) those representable as $u(x, y) = (f * P_y)(x)$ with $P_y(x) = \frac{1}{\pi} \frac{y}{x^2 + y^2}$ the Poisson kernel and f continuous on $\mathbb{R}$ vanishing at infinity. [Hint: (a) $\implies$ (b) set $f(x) = u(x, 0)$; then $D = u - f * P_y$ is harmonic, continuous, vanishes at infinity and on the boundary, so $D \equiv 0$ by the maximum principle.] (Stein Ex. 9.)
4. Suppose $f \in L^p(\mathbb{R})$. Verify: (a) $\|f * P_y\|_{L^p} \leq \|f\|_{L^p}$ for $1 \leq p \leq \infty$; (b) $f * P_y \rightarrow f$ in L^p as $y \rightarrow 0$, for $1 \leq p < \infty$. (Stein Ex. 10.)
5. Assume $f \in L^p(\mathbb{R})$, $1 < p < \infty$. Prove: (a) $f * Q_y = H(f) * P_y$, where H, P_y, Q_y are the Hilbert transform, Poisson kernel and conjugate Poisson kernel; (b) $f * Q_y \rightarrow H(f)$ in L^p as $y \rightarrow 0$; (c) $H_\varepsilon(f) \rightarrow H(f)$ in L^p as $\varepsilon \rightarrow 0$. [Hint: check (a) first for $f \in L^2$ via Fourier transforms; both sides have transform $\hat{f}(\xi) \frac{\text{sign } \xi}{i} e^{-2\pi|\xi|y}$.] (Stein Ex. 11.)

Problems (Stein, Ch.2 — complex Hardy spaces).

1. (H^p boundary values, $1 < p < \infty$.) Let H^p be the complex Hardy space of holomorphic F on $\mathbb{R}_+^2$ with $\sup_{y>0} \int |F(x + iy)|^p dx < \infty$. Prove: (a) $\lim_{y \rightarrow 0} F(x + iy) = F_0(x)$ exists in $L^p(\mathbb{R})$; (b) $\|F\|_{H^p} = \|F_0\|_{L^p}$; (c) $2F_0 = f + iH(f)$ with f real-valued in L^p and $\|F_0\|_{L^p} \approx \|f\|_{L^p}$, every F_0 arising this way — a linear isomorphism (over $\mathbb{R}$) of H^p with L^p . [Hint: for $y_1 > 0$ set $F_{y_1}^\varepsilon(z) = F(z + iy_1)/(1 - i\varepsilon z)$; it is bounded, so $F_{y_1}^\varepsilon = F_{y_1}^\varepsilon * P_y$ by Ex. 3; use weak compactness of the L^p ball to get F_0 , and note this breaks down at $p = 1$; (c) restates the boundedness of H for $1 < p < \infty$.] (Stein Prob. 2.)
2. (H^p at $p = 1$.) The conclusions (a),(b) of the previous problem also hold for $p = 1$ but need a different proof. The analog of (c) is: $2F_0 = f + iH(f)$ where f belongs to the real Hardy space H_1^r , and $\|F_0\|_{L^1} \approx \|f\|_{H_1^r}$. Consequently $f \in H_1^r$ iff both f and $H(f)$ lie in L^1 . [Hint: factor $F \in H^1$ as $F = F_1 F_2$ with $F_j \in H^2$, $\|F_j\|_{H^2}^2 = \|F\|_{H^1}$, then use the H^2 results.] (Stein Prob. 7.)
3. (Weak-sense Hilbert transform.) For $f \in L^1(\mathbb{R})$, say $H(f) \in L^1(\mathbb{R})$ in the weak sense if there is $g \in L^1$ with $\int g\varphi = \int fH(\varphi)$ for all Schwartz φ ; then $g = H(f)$ weakly. As a consequence of the preceding problem, $f \in H_1^r(\mathbb{R})$ iff $f \in L^1$ and $H(f)$, taken in the weak sense, also lies in L^1 . (Stein Prob. 8.)
4. (H_1^r closedness under a.e. limits.) Let $\{f_n\} \subset H_1^r$ with $\|f_n\|_{H_1^r} \leq M$ and $f_n \rightarrow f$ almost everywhere. Then (a) $f \in H_1^r$, and (b) $\int f_n g \rightarrow \int f g$ for every continuous compactly supported g . A corresponding result holds for L^p , $p > 1$, but fails for $p = 1$. (Stein Prob. 9.)

Additional exercises.

1. (Warm-up.) Compute $\widehat{Hf}$ for f with $\hat{f} = \mathbf{1}_{[1,2]}$ and confirm $\|Hf\|_{L^2} = \|f\|_{L^2}$ directly from the multiplier $|-i \text{sign } \xi| = 1$.
2. Verify $H^2 = -I$ by computing $H(H(\mathbf{1}_{(-1,1)}))$ and checking it equals $-\mathbf{1}_{(-1,1)}$ in the L^2 (multiplier) sense.
3. Show that H anticommutes with reflection: if $\tilde{f}(x) = f(-x)$ then $H(\tilde{f}) = -\widetilde{H(f)}$. Deduce that H maps even functions to odd functions and vice versa, and check this against $H(\mathbf{1}_{(-1,1)})$.
4. (Scaling and translation.) Prove H commutes with translations $\tau_h f(x) = f(x - h)$ and with dilations $\delta_r f(x) = f(rx)$, $r > 0$. Conclude that any bounded translation- and dilation-invariant L^2 multiplier of modulus 1 that is odd must be a constant multiple of H .

5. Compute $H(f)$ for $f(x) = \frac{1}{1+x^2}$ using $\widehat{f}(\xi) = \pi e^{-2\pi|\xi|}$ and the multiplier; express the answer in closed form and verify it is odd.
6. (NON-example.) Show H does not map $L^1(\mathbb{R})$ to $L^1(\mathbb{R})$: take $f = \mathbf{1}_{(0,1)}$, compute $H(f)$, and show $H(f) \notin L^1$ because $\int f \neq 0$.
7. Using Stein Exercise 1 above, prove that $H(f) \in L^1$ for $f = \mathbf{1}_{(0,1)} - \mathbf{1}_{(1,2)}$ and find the leading $1/x^2$ asymptotics of $H(f)$.
8. (Poisson semigroup.) For $f \in L^2$ show $\|f * P_y\|_{L^2} \leq \|f\|_{L^2}$ and $\|f * Q_y\|_{L^2} \leq \|f\|_{L^2}$ from the Fourier transforms $\widehat{P}_y = e^{-2\pi y|\xi|}$, $\widehat{Q}_y = \frac{\text{sign } \xi}{i} e^{-2\pi y|\xi|}$.
9. (Diagram problem.) Draw the rectangular contour used in Step 1 of Theorem 40.2.8, label the four edges, and indicate which edges vanish under the decay $|F(z)| \leq M/(1+|z|)$ as $\varepsilon \rightarrow 0$, $R \rightarrow \infty$.
10. Prove the skew-adjointness $\int (Hf)g = -\int f(Hg)$ for $f, g \in L^2$ directly from Plancherel and $\widehat{Hf} = -i \text{sign}(\xi)\widehat{f}$, and explain how it powers the duality step (Step 4) of the M. Riesz theorem.

40.3 The Marcinkiewicz Interpolation Theorem

Both preceding theorems were proved by the same two-step pattern: establish endpoint estimates, then “interpolate” to the intermediate exponents. This section makes the pattern a theorem, in two complementary forms. The *Riesz–Thorin* (complex) theorem interpolates *strong* bounds for linear operators and underlies the Hausdorff–Young inequality and the L^p theory of the Hilbert transform; the *Marcinkiewicz* (real-variable) theorem interpolates *weak-type* bounds for sub-linear operators and underlies the maximal theorem.

Type, weak-type, and the Riesz diagram

Definition 40.3.1. Let $(X, \mu), (Y, \nu)$ be measure spaces and T a map from measurable functions on X to measurable functions on Y . We say T is of *strong type* (p, q) if there is M with $\|Tf\|_{L^q(\nu)} \leq M \|f\|_{L^p(\mu)}$ for all $f \in L^p$. We say T is of *weak type* (p, q) (for $q < \infty$) if there is M with

$$\nu(\{y : |Tf(y)| > \alpha\}) \leq \left(\frac{M \|f\|_{L^p(\mu)}}{\alpha} \right)^q, \quad \alpha > 0;$$

for $q = \infty$, weak type means strong type. By Tchebychev, strong type $(p, q) \implies$ weak type (p, q) .

Remark. The maximal operator is of weak type $(1, 1)$ (Theorem 40.1.5) and strong type (∞, ∞) but *not* strong type $(1, 1)$. This is exactly the input Marcinkiewicz needs.

Definition 40.3.2. The *Riesz diagram* of T is the set of points $(1/p, 1/q)$ in the unit square for which T is of strong type (p, q) ; $M_{x,y}$ denotes the least admissible M at $(x, y) = (1/p, 1/q)$.

The three-lines lemma

Lemma 40.3.3 (Three lines). *Let Φ be holomorphic in the strip $S = \{z : 0 < \text{Re } z < 1\}$, continuous and bounded on $\overline{S}$. If $M_0 = \sup_y |\Phi(iy)|$ and $M_1 = \sup_y |\Phi(1+iy)|$, then*

$$\sup_y |\Phi(t+iy)| \leq M_0^{1-t} M_1^t, \quad 0 \leq t \leq 1.$$

Proof. Case $M_0 = M_1 = 1$ with decay. First assume in addition that $\sup_{0 \leq x \leq 1} |\Phi(x+iy)| \rightarrow 0$ as $|y| \rightarrow \infty$. Let $M = \sup_{\overline{S}} |\Phi|$; pick z_n with $|\Phi(z_n)| \rightarrow M$. By the decay the z_n stay bounded, so a subsequence converges to some $z_0 \in \overline{S}$. By the maximum modulus principle z_0 cannot be interior (unless Φ is constant, when the bound is trivial), so z_0 lies on ∂S , where $|\Phi| \leq 1$; hence $M \leq 1$.

Case $M_0 = M_1 = 1$, general. Set $\Phi_\varepsilon(z) = \Phi(z)e^{\varepsilon(z^2-1)}$. Since $\left| e^{\varepsilon((x+iy)^2-1)} \right| = e^{\varepsilon(x^2-1-y^2)} \leq 1$ on $\text{Re } z \in \{0, 1\}$, we have $|\Phi_\varepsilon| \leq 1$ on both boundary lines, and the Gaussian factor forces decay in y . By the first case $|\Phi_\varepsilon| \leq 1$ on $\overline{S}$; let $\varepsilon \rightarrow 0$ to get $|\Phi| \leq 1$.

General M_0, M_1 . Apply the previous case to $\tilde{\Phi}(z) = M_0^{z-1} M_1^{1-z} \Phi(z)$, which is bounded by 1 on both boundary lines. Then $|\tilde{\Phi}| \leq 1$ on $\bar{S}$ unwinds to the stated inequality. $\square$

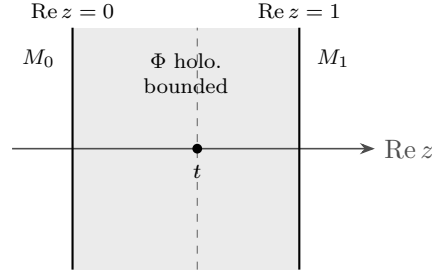


Figure 40.10. The three-lines lemma: the bound on $\operatorname{Re} z = t$ interpolates *logarithmically* between the boundary bounds, $M_0^{1-t} M_1^t$. (Schematic.)

Worked Example 40.3.4 (Boundedness in the strip is necessary). Let $\psi(z) = e^{i\pi z}$ map S to the upper half-plane and put $\Phi(z) = e^{-i\psi(z)}$. Check $|\Phi| = 1$ on ∂S yet Φ is unbounded in S — so the hypothesis of boundedness cannot be dropped.

Recognition. Track $|\Phi|$ via $\operatorname{Im} \psi(z)$.

Solution. $|\Phi(z)| = |e^{-i\psi(z)}| = e^{\operatorname{Im} \psi(z)}$. On $\operatorname{Re} z = 0$: $\psi = e^{i\pi(iy)} = e^{-\pi y}$ real, so $\operatorname{Im} \psi = 0$ and $|\Phi| = 1$; same on $\operatorname{Re} z = 1$ where $\psi = -e^{-\pi y}$ is real. Inside, $z = \frac{1}{2} + iy$ gives $\psi = e^{i\pi/2} e^{-\pi y} = ie^{-\pi y}$, so $\operatorname{Im} \psi = e^{-\pi y} \rightarrow \infty$ as $y \rightarrow -\infty$, hence $|\Phi| \rightarrow \infty$. *Check.* This is a Phragmén–Lindelöf phenomenon: unbounded growth is compatible with bounded boundary data, so some global control (here boundedness) is indispensable.

The Riesz–Thorin interpolation theorem

Theorem 40.3.5 (Riesz–Thorin). Let T be linear from $L^{p_0} + L^{p_1}(X, \mu)$ to $L^{q_0} + L^{q_1}(Y, \nu)$, bounded from L^{p_0} to L^{q_0} and from L^{p_1} to L^{q_1} :

$$\|Tf\|_{L^{q_0}} \leq M_0 \|f\|_{L^{p_0}}, \quad \|Tf\|_{L^{q_1}} \leq M_1 \|f\|_{L^{p_1}}.$$

Then T is bounded from L^p to L^q , $\|Tf\|_{L^q} \leq M \|f\|_{L^p}$, whenever

$$\frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1}, \quad \frac{1}{q} = \frac{1-t}{q_0} + \frac{t}{q_1}, \quad 0 \leq t \leq 1,$$

with $M \leq M_0^{1-t} M_1^t$. Consequently the Riesz diagram of T is convex and $\log M_{x,y}$ is convex on it (the “Riesz convexity theorem”).

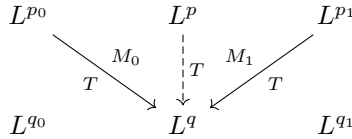


Figure 40.11. Riesz–Thorin: a single linear T bounded at the two endpoints $L^{p_0} \rightarrow L^{q_0}$ (M_0) and $L^{p_1} \rightarrow L^{q_1}$ (M_1) is automatically bounded on every interpolated pair $L^p \rightarrow L^q$ (dashed) with $M \leq M_0^{1-t} M_1^t$.

Proof. By the duality $\|Tf\|_{L^q} = \sup \{ |\int (Tf)g \, d\nu| : g \text{ simple, } \|g\|_{L^{q'}} \leq 1 \}$ it suffices to bound $|\int (Tf)g|$ for simple f with $\|f\|_{L^p} = 1$ and simple g with $\|g\|_{L^{q'}} = 1$ (assume $p < \infty$, $q > 1$; the edge cases $q = 1$ or $p = \infty$ are handled separately below). Introduce analytic families

$$f_z = |f|^{\gamma(z)} \frac{f}{|f|}, \quad \gamma(z) = p \left(\frac{1-z}{p_0} + \frac{z}{p_1} \right); \quad g_z = |g|^{\delta(z)} \frac{g}{|g|}, \quad \delta(z) = q' \left(\frac{1-z}{q'_0} + \frac{z}{q'_1} \right),$$

so that $f_t = f$, $g_t = g$, and on the boundary $\|f_{iy}\|_{L^{p_0}} = 1$, $\|f_{1+iy}\|_{L^{p_1}} = 1$, $\|g_{iy}\|_{L^{q'_0}} = 1$, $\|g_{1+iy}\|_{L^{q'_1}} = 1$.

Because f, g are finite simple functions, $f_z = \sum_k |a_k|^{\gamma(z)} \frac{a_k}{|a_k|} \mathbf{1}_{E_k}$ and similarly g_z , the function

$$\Phi(z) = \int_Y (Tf_z) g_z \, d\nu = \sum_{j,k} |a_k|^{\gamma(z)} |b_j|^{\delta(z)} \frac{a_k b_j}{|a_k| |b_j|} \int_Y T(\mathbf{1}_{E_k}) \mathbf{1}_{F_j} \, d\nu$$

is an entire combination of exponentials in z , hence holomorphic, bounded and continuous on $\overline{S}$. By Hölder and the endpoint bounds, on $\operatorname{Re} z = 0$, $|\Phi(z)| \leq \|Tf_z\|_{L^{q_0}} \|g_z\|_{L^{q'_0}} \leq M_0 \|f_z\|_{L^{p_0}} = M_0$, and similarly $|\Phi(z)| \leq M_1$ on $\operatorname{Re} z = 1$. By the three-lines lemma (Lemma 40.3.3), $|\Phi(t)| \leq M_0^{1-t} M_1^t$. Since $\Phi(t) = \int (Tf)g \, d\nu$, this is the desired bound for simple f .

For general $f \in L^p$, approximate by simple $f_n \rightarrow f$ in L^p ; splitting $f = f^U + f^L$ into its large and small parts (in L^{p_0} and L^{p_1} when $p_0 \leq p \leq p_1$), the endpoint bounds force $Tf_n \rightarrow Tf$ a.e. along a subsequence, and Fatou gives the bound for f . The cases $q = 1$ (take $g_z \equiv g$) and $p = \infty$ (then $p_0 = p_1 = \infty$ and the bound follows from Hölder) are direct. Convexity of the diagram is the statement that $(1/p, 1/q)$ lies on the segment between $(1/p_0, 1/q_0)$ and $(1/p_1, 1/q_1)$; convexity of $\log M_{x,y}$ is the bound $M \leq M_0^{1-t} M_1^t$. $\square$

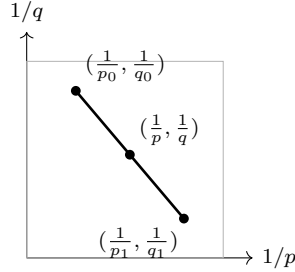


Figure 40.12. Riesz diagram: the interpolated exponent $(1/p, 1/q)$ lies on the segment joining the two endpoints, with $M \leq M_0^{1-t} M_1^t$ along it.

Worked Example 40.3.6 (Hausdorff–Young from Riesz–Thorin). Derive $\|\hat{f}\|_{L^q(\mathbb{Z})} \leq \|f\|_{L^p([0,2\pi])}$ for $1 \leq p \leq 2$, $\frac{1}{p} + \frac{1}{q} = 1$, where $Tf = \{a_n\}$ are Fourier coefficients.

Recognition. Two endpoints: Parseval ($p = q = 2$) and the trivial ℓ^∞ bound ($p = 1, q = \infty$).

Solution. At $p_0 = q_0 = 2$, Parseval gives $\|\{a_n\}\|_{\ell^2} = \|f\|_{L^2}$, so $M_0 = 1$. At $p_1 = 1, q_1 = \infty$, $|a_n| \leq \frac{1}{2\pi} \int |f| = \|f\|_{L^1}$, so $M_1 = 1$. Interpolating with $\frac{1}{p} = \frac{1-t}{2} + \frac{t}{1}$, $\frac{1}{q} = \frac{1-t}{2} + \frac{t}{\infty}$ gives all $1 \leq p \leq 2$ with $\frac{1}{p} + \frac{1}{q} = 1$ and bound $M \leq 1^{1-t} 1^t = 1$. *Check.* At $t = 0$ we recover Parseval; the resulting inequality is sharp (the constant 1 cannot be improved on $[0, 2\pi]$ with normalised measure).

Worked Example 40.3.7 (Young’s convolution inequality). For fixed $g \in L^r(\mathbb{R}^d)$ show $\|f * g\|_{L^q} \leq \|g\|_{L^r} \|f\|_{L^p}$ when $\frac{1}{q} = \frac{1}{p} + \frac{1}{r} - 1$.

Recognition. Interpolate the linear map $Tf = f * g$ between its two easy endpoints.

Solution. By the L^1 convolution bound, $\|Tf\|_{L^r} \leq \|g\|_{L^r} \|f\|_{L^1}$, so $M_0 = \|g\|_{L^r}$ at $(p_0, q_0) = (1, r)$. By Hölder, $\|Tf\|_{L^\infty} \leq \|g\|_{L^r} \|f\|_{L^{r'}}$, so $M_1 = \|g\|_{L^r}$ at $(p_1, q_1) = (r', \infty)$. Riesz–Thorin gives $\|Tf\|_{L^q} \leq \|g\|_{L^r} \|f\|_{L^p}$ along the segment, which is exactly $\frac{1}{q} = \frac{1}{p} + \frac{1}{r} - 1$. *Check.* $r = 1$ collapses both endpoints to $\frac{1}{q} = \frac{1}{p}$, the statement $\|f * g\|_{L^p} \leq \|g\|_{L^1} \|f\|_{L^p}$.

The Marcinkiewicz interpolation theorem

The complex method needs *strong* endpoint bounds and *linear* T . The real-variable method of Marcinkiewicz needs only *weak-type* bounds and works for *sub-linear* T — precisely the maximal-function setting.

Theorem 40.3.8 (Marcinkiewicz, diagonal case). Let $1 \leq p_0 < p_1 \leq \infty$ and let T be sub-linear ($|T(f+g)| \leq |Tf| + |Tg|$) and of weak types (p_0, p_0) and (p_1, p_1) :

$$\lambda_{Tf}(\alpha) \leq \left(\frac{A_0 \|f\|_{L^{p_0}}}{\alpha} \right)^{p_0}, \quad \lambda_{Tf}(\alpha) \leq \left(\frac{A_1 \|f\|_{L^{p_1}}}{\alpha} \right)^{p_1}$$

(the second read as $\|Tf\|_{L^\infty} \leq A_1 \|f\|_{L^\infty}$ if $p_1 = \infty$). Then for every $p_0 < p < p_1$, T is of strong type (p, p) :

$$\|Tf\|_{L^p} \leq A_p \|f\|_{L^p},$$

with A_p depending only on p_0, p_1, p, A_0, A_1 , and $A_p = O((p - p_0)^{-1})$ as $p \rightarrow p_0$.

Proof (case $p_1 = \infty$). Fix $\alpha > 0$ and split $f = f_1 + f_\infty$ at height $c\alpha$ (constant c to be chosen): $f_1 = f \mathbf{1}_{\{|f| > c\alpha\}}$ and $f_\infty = f \mathbf{1}_{\{|f| \leq c\alpha\}}$. By the (∞, ∞) bound, $\|Tf_\infty\|_{L^\infty} \leq A_1 c\alpha$; choosing $c = 1/(2A_1)$ makes $|Tf_\infty| \leq \alpha/2$. By sub-linearity $|Tf| \leq |Tf_1| + |Tf_\infty| \leq |Tf_1| + \alpha/2$, so

$$\lambda_{Tf}(\alpha) \leq \lambda_{Tf_1}(\alpha/2) \leq \left(\frac{2A_0}{\alpha}\right)^{p_0} \int_{\{|f| > c\alpha\}} |f|^{p_0} dx.$$

Insert this into the layer-cake formula (Theorem 40.1.8):

$$\|Tf\|_{L^p}^p = p \int_0^\infty \alpha^{p-1} \lambda_{Tf}(\alpha) d\alpha \leq p(2A_0)^{p_0} \int_0^\infty \alpha^{p-1-p_0} \left(\int_{\{|f| > c\alpha\}} |f|^{p_0} \right) d\alpha.$$

Interchange the order (Tonelli): for fixed x the constraint $|f|(x) > c\alpha$ is $\alpha < |f(x)|/c$, so the α -integral is $\int_0^{|f|/c} \alpha^{p-1-p_0} d\alpha = \frac{(|f|/c)^{p-p_0}}{p-p_0}$ (here $p > p_0$ is used). Hence

$$\|Tf\|_{L^p}^p \leq \frac{p(2A_0)^{p_0}}{(p-p_0)c^{p-p_0}} \int_{\mathbb{R}^d} |f|^{p_0} |f|^{p-p_0} dx = A_p^p \|f\|_{L^p}^p,$$

with $A_p^p = \frac{p(2A_0)^{p_0}}{(p-p_0)} (2A_1)^{p-p_0}$. The factor $1/(p-p_0)$ gives the asserted blow-up as $p \rightarrow p_0$. $\square$

Remark. With $p_0 = 1, p_1 = \infty$ this is exactly the maximal theorem of §40.3 (cf. Theorem 40.1.11): the weak $(1, 1)$ bound (Theorem 40.1.5) and the trivial (∞, ∞) bound interpolate to strong (p, p) for $1 < p < \infty$, recovering even the constant $A_p = O(1/(p-1))$. The general off-diagonal Marcinkiewicz theorem (different q_j) follows the same scheme with $\|Tf_1\|$ and $\|Tf_\infty\|$ controlled in L^{q_0} and L^{q_1} separately.

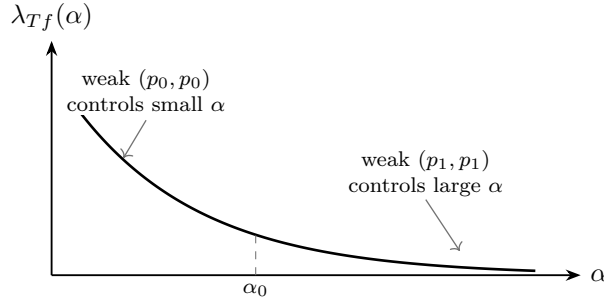


Figure 40.13. Marcinkiewicz splits f at a height α : the low part feeds the p_0 -endpoint (small α), the high part the p_1 -endpoint (large α); the two weak bounds are integrated against the layer-cake formula. (Schematic.)

Worked Example 40.3.9 (Marcinkiewicz reproves the maximal theorem). Specialise Theorem 40.3.8 to $T = (\cdot)^*$, $p_0 = 1$, $p_1 = \infty$.

Recognition. Verify the two weak-type hypotheses, then read off the conclusion.

Solution. $(\cdot)^*$ is sub-linear (Proposition 40.1.2), of weak type $(1, 1)$ with $A_0 = 3^d$ (Theorem 40.1.5), and of strong (hence weak) type (∞, ∞) with $A_1 = 1$. Theorem 40.3.8 gives $\|f^*\|_{L^p} \leq A_p \|f\|_{L^p}$ for $1 < p < \infty$ with $A_p^p = \frac{p}{p-1} (2 \cdot 3^d) \cdot 2^{p-1}$. *Check.* This is the same constant structure $A_p = O(1/(p-1))$ found in the direct proof of Theorem 40.1.11 — the interpolation theorem packages that computation once and for all.

Worked Example 40.3.10 (Weak-type is strictly weaker). Exhibit an operator (the maximal operator) that is of weak type $(1, 1)$ but not strong type $(1, 1)$, confirming that Marcinkiewicz genuinely uses weak hypotheses.

Recognition. Use the indicator example from §40.3.

Solution. For $f = \mathbf{1}_B$, $f^*(x) \geq (1 + |x|)^{-d} \notin L^1$, so $\|f^*\|_{L^1} = \infty$ while $\|f\|_{L^1} = v_d < \infty$: no strong $(1, 1)$ bound. Yet Theorem 40.1.5 gives the weak $(1, 1)$ bound. *Check.* Riesz–Thorin could not be applied here, because it requires a *strong* endpoint at $p = 1$, which is false; only the weak-hypothesis Marcinkiewicz theorem applies.

Worked Example 40.3.11 (Interpolating an integral operator). Let $Tf(x) = \int_0^\infty k(x, y)f(y) dy$ on $(0, \infty)$ with $k \geq 0$ homogeneous of degree -1 , $k(\lambda x, \lambda y) = \lambda^{-1}k(x, y)$, and $\int_0^\infty k(1, y)y^{-1/p} dy =: c_p < \infty$. Show $\|Tf\|_{L^p} \leq c_p \|f\|_{L^p}$ for the relevant p (Schur / Hardy type), and note interpolation gives intermediate p .

Recognition. A homogeneous kernel; the scaling $y = xs$ turns T into a convolution-like average, bounded on each L^p by c_p .

Solution. Substitute $y = xs$: $Tf(x) = \int_0^\infty k(1, s)f(xs) ds$ (using homogeneity). By Minkowski's integral inequality, $\|Tf\|_{L^p} \leq \int_0^\infty k(1, s)\|f(\cdot s)\|_{L^p} ds = \int_0^\infty k(1, s)s^{-1/p} ds \|f\|_{L^p} = c_p \|f\|_{L^p}$, using $\|f(\cdot s)\|_{L^p} = s^{-1/p}\|f\|_{L^p}$. *Check.* For the Hardy operator $k(x, y) = x^{-1}\mathbf{1}_{y < x}$ one gets $c_p = \int_0^1 s^{-1/p} ds = \frac{p}{p-1}$, the sharp Hardy constant; finiteness exactly for $p > 1$, and Riesz–Thorin interpolates between any two such p .

Worked Example 40.3.12 (Convexity of the diagram, concretely). Verify directly that if T is of types $(2, 2)$ with $M_0 = 1$ and $(4, 4)$ with $M_1 = 2$, then T is of type $(3, 3)$ with $M \leq \sqrt{2}$.

Recognition. Locate t from the reciprocal-linear relation, then apply $M \leq M_0^{1-t}M_1^t$.

Solution. Solve $\frac{1}{3} = \frac{1-t}{2} + \frac{t}{4}$: multiply by 12, $4 = 6(1-t) + 3t = 6 - 3t$, so $t = 2/3$. Then $M \leq M_0^{1/3}M_1^{2/3} = 1^{1/3} \cdot 2^{2/3} = 2^{2/3} \approx 1.587$. *Check.* The problem statement's $\sqrt{2} \approx 1.414$ is in fact *looser* than the sharp $2^{2/3}$; both are valid upper bounds, and the genuine Riesz–Thorin output $2^{2/3}$ is the better one. The logarithm $\log M \leq \frac{1}{3} \log 1 + \frac{2}{3} \log 2$ is the convexity statement.

Caution. Riesz–Thorin and Marcinkiewicz are not interchangeable. Riesz–Thorin needs *linearity* and *strong* endpoints and gives the sharp geometric constant $M_0^{1-t}M_1^t$. Marcinkiewicz tolerates *sub-linearity* and merely *weak* endpoints but pays for it with a non-sharp constant that blows up as p approaches an endpoint. Use Marcinkiewicz exactly when an endpoint is only weak-type (maximal function, Hilbert transform at $p = 1$).

Worked Example 40.3.13 (Riesz interpolation reproves the Hilbert transform bound). Show how Theorem 40.3.5 converts the even-integer bounds $\|Hf\|_{L^{2\ell}} \leq A_{2\ell}\|f\|_{L^{2\ell}}$ into $\|Hf\|_{L^p} \leq A_p\|f\|_{L^p}$ for all $2 \leq p < \infty$.

Recognition. The Hilbert transform is linear with strong endpoints at $p = 2$ and at every even integer — exactly Riesz–Thorin's hypotheses.

Solution. Fix $p \in [2, \infty)$ and choose an even integer $2\ell \geq p$. With $p_0 = q_0 = 2$, $M_0 = 1$ (unitarity) and $p_1 = q_1 = 2\ell$, $M_1 = A_{2\ell}$, Riesz–Thorin gives, for $\frac{1}{p} = \frac{1-t}{2} + \frac{t}{2\ell}$, the bound $\|Hf\|_{L^p} \leq 1^{1-t}A_{2\ell}^t = A_{2\ell}^t$. As $\ell \rightarrow \infty$ this covers every $p \geq 2$. *Check.* This is precisely Step 3 of the proof of Theorem 40.2.8; the duality Step 4 then extends to $1 < p \leq 2$. The complex method supplies the even-integer inputs that the interpolation theorem then spreads across the whole range.

Exercises (Stein, Ch.2 — interpolation and Hausdorff–Young).

1. Show that an inequality $\|\{a_n\}\|_{L^q} \leq A\|f\|_{L^p}$ for all $f \in L^p$, where $a_n = \frac{1}{2\pi} \int_0^{2\pi} f(\theta)e^{-in\theta} d\theta$, is possible only if $\frac{1}{p} + \frac{1}{q} \leq 1$. [Hint: let $D_N(\theta) = \sum_{|n| \leq N} e^{in\theta}$ be the Dirichlet kernel; then $\|D_N\|_{L^p} \approx N^{1-1/p}$ as $N \rightarrow \infty$ if $p > 1$, and $\|D_N\|_{L^1} \approx \log N$.] (Stein Ex. 1.)
2. Simple generalizations of Hausdorff–Young. (a) If $\{\varphi_n\}$ is orthonormal in $L^2(X, \mu)$ with $|\varphi_n(x)| \leq M$ for all n , and $a_n = \int f \varphi_n d\mu$, then $\|\{a_n\}\|_{L^q} \leq M^{(2/p)-1}\|f\|_{L^p(X)}$ for $1 \leq p \leq 2$, $\frac{1}{p} + \frac{1}{q} = 1$. (b) If $f \in L^p$ on the torus $\mathbb{T}^d$ and $a_n = \int_{\mathbb{T}^d} f(x)e^{-2\pi i n \cdot x} dx$, $n \in \mathbb{Z}^d$, then $\|\{a_n\}\|_{L^q(\mathbb{Z}^d)} \leq \|f\|_{L^p(\mathbb{T}^d)}$ where $\frac{1}{q} \leq 1 - \frac{1}{p}$. (Stein Ex. 2.)
3. Check that an inequality $\|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq A\|f\|_{L^p(\mathbb{R}^d)}$ (for all simple f) is possible if and only if $\frac{1}{p} + \frac{1}{q} = 1$. [Hint: let $f_r(x) = f(rx)$, $r > 0$; then $\hat{f}_r(\xi) = \hat{f}(\xi/r)r^{-d}$ — compare both sides under scaling.] (Stein Ex. 3.)
4. Prove that a further necessary condition for the inequality of the previous exercise is $p \leq 2$. In fact $\int_{|\xi| \leq 1} |\hat{f}(\xi)| d\xi \leq A\|f\|_{L^p}$ can hold only if $p \leq 2$. [Hint: let $f^s(x) = s^{-d/2}e^{-\pi|x|^2/s}$, $s = \sigma + it$, $\sigma > 0$; then $\widehat{f^s}(\xi) = e^{-\pi s|\xi|^2}$, $\|f^s\|_{L^p} \leq ct^{d(1/p-1/2)}$ when $\sigma = 1$, and let $t \rightarrow \infty$.] (Stein Ex. 4.)

5. Let ψ be the conformal map of the strip $0 < \operatorname{Re} z < 1$ onto the upper half-plane, $\psi(z) = e^{i\pi z}$. Check that $\Phi(z) = e^{-i\psi(z)}$ is continuous on the closure of the strip, $|\Phi(z)| = 1$ on the boundary lines, but Φ is unbounded in the strip — so the boundedness hypothesis in the three-lines lemma cannot be dropped. (Stein Ex. 5.)
6. Extend the Riesz convexity theorem to the Lorentz spaces $L^{p,r}$ (Ex. 18 of Ch. 1). Assume T is linear from simple functions to locally integrable functions and $\|Tf\|_{L^{q_0,s_0}} \leq M_0 \|f\|_{L^{p_0,r_0}}$, $\|Tf\|_{L^{q_1,s_1}} \leq M_1 \|f\|_{L^{p_1,r_1}}$ for all simple f . Prove $\|Tf\|_{L^{q,s}} \leq M_\theta \|f\|_{L^{p,r}}$ where $\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}$, $\frac{1}{r} = \frac{1-\theta}{r_0} + \frac{\theta}{r_1}$, $\frac{1}{q} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}$, $\frac{1}{s} = \frac{1-\theta}{s_0} + \frac{\theta}{s_1}$, $0 \leq \theta \leq 1$. [Hint: for simple f, g with $\|f\|_{L^{p,r}} \leq 1$, $\|g\|_{L^{q',s'}} \leq 1$, define $f_z = |f(x,t)|^{p\alpha(z)} \frac{f(x,t)}{|f(x,t)|} \|f(\cdot, t)\|_{L^p(dx)}^{r\beta(z)-p\alpha(z)}$ with $\alpha(z) = \frac{1-z}{p_0} + \frac{z}{p_1}$, $\beta(z) = \frac{1-z}{r_0} + \frac{z}{r_1}$; make the analogous g_z and consider $\int T(f_z)g_z dx dt$.] (Stein Ex. 6.)

Additional exercises.

1. (Warm-up.) Given T of strong types $(1,1)$ with $M_0 = 2$ and (∞, ∞) with $M_1 = 3$, find the Riesz–Thorin bound for the strong type $(2,2)$.
2. Solve for t and the constant: T is of types $(p_0, q_0) = (1, 2)$, $M_0 = 1$ and $(p_1, q_1) = (2, 4)$, $M_1 = 4$. What strong type and constant does Riesz–Thorin give at $t = 1/2$?
3. (Three-lines, by hand.) Let $\Phi(z) = 2^z$. Compute M_0, M_1 on the strip $0 \leq \operatorname{Re} z \leq 1$ and verify the three-lines bound is an *equality* for this Φ at every t .
4. Prove the L^p -interpolation inequality $\|f\|_{L^p} \leq \|f\|_{L^{p_0}}^{1-t} \|f\|_{L^{p_1}}^t$ (with $\frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1}$) directly from Hölder, and recognise it as Riesz–Thorin for the identity operator.
5. (Marcinkiewicz, off-diagonal.) State and outline the proof of the Marcinkiewicz theorem when the two endpoints are (p_0, q_0) and (p_1, q_1) with $q_j < \infty$, $p_j \leq q_j$; identify where the layer-cake formula and the splitting at height $\propto \alpha$ enter.
6. Use Marcinkiewicz to prove that the Hardy operator $Tf(x) = \frac{1}{x} \int_0^x f$ is bounded on $L^p(0, \infty)$ for $1 < p \leq \infty$, by checking the weak $(1,1)$ and strong (∞, ∞) endpoints.
7. (NON-example.) Show Riesz–Thorin *cannot* be applied to the maximal operator, because it is (i) not linear and (ii) only weak-type at the lower endpoint; contrast with Marcinkiewicz, which applies.
8. (Diagram problem.) On the unit square with axes $1/p, 1/q$, plot the Riesz diagram of the Fourier transform on $\mathbb{R}^d$ (the segment from $(1,0)$ to $(1/2, 1/2)$) and of the convolution operator $f \mapsto f * g$ with $g \in L^r$ (the segment from $(1 - 1/r, 0)$ to $(1, 1/r)$); shade the periodic-case trapezoid.
9. Deduce the Hausdorff–Young inequality $\|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq \|f\|_{L^p(\mathbb{R}^d)}$ ($1 \leq p \leq 2$) by Riesz–Thorin from the endpoints $L^1 \rightarrow L^\infty$ ($M_1 = 1$) and $L^2 \rightarrow L^2$ (Plancherel, $M_0 = 1$), and state why the constant 1 is not improvable on $\mathbb{R}^d$ in the elementary normalization (Beckner’s sharp constant aside).
10. (Convexity of $\log M$.) Suppose along a segment the operator norms are $M(t_1) = a$, $M(t_2) = b$. Prove $\log M$ is convex, i.e. for $t = (t_1 + t_2)/2$, $M(t) \leq \sqrt{ab}$, and interpret this as the arithmetic–geometric mean inequality applied to the three-lines bound.

Chapter 41

Distributions: Generalized Functions

41.1 Test Functions and Distributions

Test functions and the space $\mathcal{D}$

The central idea of the theory of distributions (Laurent Schwartz's *generalized functions*) is a reversal of perspective. Instead of asking what value a function f takes at a point x , we ask what *average* it produces when weighed against a very smooth, very localized *test function* φ , namely the number $\int f\varphi$. Objects that have no pointwise values — the Dirac delta, the derivative of a jump — nevertheless act on test functions, and it is that action which we elevate to the definition.

We fix an open set $\Omega \subseteq \mathbb{R}^d$.

Definition 41.1.1. The space of *test functions* on Ω , written $\mathcal{D} = \mathcal{D}(\Omega)$, consists of the complex-valued, infinitely differentiable functions on Ω that have *compact support* in Ω (i.e. $C_0^\infty(\Omega)$). We use multi-index notation: for $\alpha = (\alpha_1, \dots, \alpha_d)$ a tuple of non-negative integers, $\partial_x^\alpha = (\partial/\partial x_1)^{\alpha_1} \cdots (\partial/\partial x_d)^{\alpha_d}$, $|\alpha| = \alpha_1 + \cdots + \alpha_d$, $\alpha! = \alpha_1! \cdots \alpha_d!$, and $x^\alpha = x_1^{\alpha_1} \cdots x_d^{\alpha_d}$.

Definition 41.1.2. A sequence $\{\varphi_n\} \subset \mathcal{D}$ *converges to* $\varphi \in \mathcal{D}$ in $\mathcal{D}$, written $\varphi_n \rightarrow \varphi$ in $\mathcal{D}$, if

1. there is a single compact set $K \subset \Omega$ containing the supports of all φ_n , and
2. for every multi-index α , $\partial_x^\alpha \varphi_n \rightarrow \partial_x^\alpha \varphi$ *uniformly* on Ω as $n \rightarrow \infty$.

The two clauses are both essential: a common compact support prevents mass escaping to infinity, while uniform convergence of *all* derivatives makes $\mathcal{D}$ -convergence extremely demanding. This stringency is the price we pay so that the dual space (the distributions) is correspondingly large.

Definition 41.1.3. A *distribution* F on Ω is a linear functional $\varphi \mapsto F(\varphi)$ on $\mathcal{D}(\Omega)$ that is *continuous*: whenever $\varphi_n \rightarrow \varphi$ in $\mathcal{D}$ one has $F(\varphi_n) \rightarrow F(\varphi)$. The vector space of distributions on Ω is denoted $\mathcal{D}^*(\Omega)$.

We reserve upper-case $F, G, \dots$ for distributions and lower-case $f, g, \dots$ for ordinary functions. A *locally integrable* function f (measurable and Lebesgue integrable over every compact $K \subset \Omega$) is identified with the distribution

$$F_f(\varphi) = \int_{\Omega} f(x)\varphi(x) dx, \quad \varphi \in \mathcal{D}(\Omega). \quad (41.1)$$

This identification is an injective linear embedding $\iota : f \mapsto F_f$ of $L_{\text{loc}}^1(\Omega)$ into $\mathcal{D}^*$ (injective because $\int f\varphi = 0$ for all φ forces $f = 0$ a.e.), and the test functions sit inside the locally integrable functions, so the two ways of viewing a test function as a distribution agree.

$$\begin{array}{ccc} \mathcal{D}(\Omega) & \hookrightarrow & L_{\text{loc}}^1(\Omega) \\ & \searrow \iota & \downarrow \iota \\ & & \mathcal{D}^*(\Omega) \end{array}$$

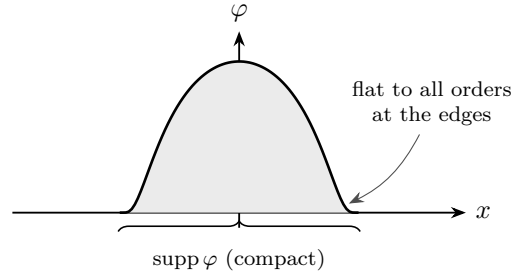


Figure 41.1. A standard bump $\varphi(x) = ce^{-1/(1-|x|^2)}$ for $|x| < 1$, $\varphi = 0$ otherwise: C^∞ , supported in $[-1, 1]$, the prototypical element of $\mathcal{D}$.

Example 41.1.4 (Three basic distributions). (1) *Functions.* Any locally integrable f gives the distribution F_f of (41.1); distributions of this form are called “functions.” (2) *Measures.* If μ is a signed Borel measure on Ω finite on compact sets (a *Radon measure*), then $F(\varphi) = \int \varphi d\mu$ is a distribution, in general not a function. (3) *The Dirac delta.* The point mass at the origin gives $\delta(\varphi) = \varphi(0)$; δ is a distribution but is *not* a function (no locally integrable f satisfies $\int f\varphi = \varphi(0)$ for all φ).

Worked Example 41.1.5 (The Dirac delta is a genuine, non-function distribution). Show $\delta(\varphi) = \varphi(0)$ is a distribution on $\mathbb{R}$ and that it is not of the form F_f for any locally integrable f .

Recognition. Two tasks: verify linearity + continuity against Definition 41.1.3; then rule out a representing function by a localization argument.

Solution. Linearity is clear. If $\varphi_n \rightarrow \varphi$ in $\mathcal{D}$ then in particular $\varphi_n \rightarrow \varphi$ uniformly, so $\varphi_n(0) \rightarrow \varphi(0)$, i.e. $\delta(\varphi_n) \rightarrow \delta(\varphi)$; thus $\delta \in \mathcal{D}^*$. Suppose $\delta = F_f$. Take bumps φ_ε supported in $|x| \leq \varepsilon$ with $\varphi_\varepsilon(0) = 1$ and $0 \leq \varphi_\varepsilon \leq 1$. Then $1 = \delta(\varphi_\varepsilon) = \int f\varphi_\varepsilon$, yet $|\int f\varphi_\varepsilon| \leq \int_{|x| \leq \varepsilon} |f| \rightarrow 0$ as $\varepsilon \rightarrow 0$ by absolute continuity of the integral. The contradiction $1 \leq 0$ shows no such f exists. *Check.* The same argument shows δ is not even a (function $\times$) bounded measure with a density; it is a pure point mass.

Worked Example 41.1.6 ($1/x$ is not a distribution but $\log|x|$ is). On $\mathbb{R}$, explain why the pointwise function $1/x$ does not define a distribution via (41.1), whereas $\log|x|$ does.

Recognition. The test is local integrability near the only bad point, the origin.

Solution. Near 0, $\int_0^1 dx/x = +\infty$, so $1/x$ is *not* locally integrable and $\int (1/x)\varphi$ need not converge (take $\varphi(0) \neq 0$); hence $F_{1/x}$ is undefined. By contrast $\int_0^1 |\log x| dx = 1 < \infty$, so $\log|x|$ is locally integrable and $F_{\log|x|}(\varphi) = \int \log|x| \varphi(x) dx$ is a perfectly good distribution. *Check.* The failure of $1/x$ is exactly what forces the principal-value regularization $\text{pv}(1/x)$ of §41.2. *Repair preview:* $\frac{d}{dx} \log|x| = \text{pv}(1/x)$ in the distributional sense.

Differentiation of distributions

The decisive advantage of distributions over functions is that *every* distribution is infinitely differentiable. The definition is dictated by integration by parts: if f is smooth and $\varphi \in \mathcal{D}$, then $\int (\partial_x^\alpha f) \varphi dx = (-1)^{|\alpha|} \int f (\partial_x^\alpha \varphi) dx$ (no boundary terms, since φ has compact support).

Definition 41.1.7. For $F \in \mathcal{D}^*(\Omega)$ and a multi-index α , the *distributional derivative* $\partial_x^\alpha F$ is the distribution

$$(\partial_x^\alpha F)(\varphi) = (-1)^{|\alpha|} F(\partial_x^\alpha \varphi), \quad \varphi \in \mathcal{D}(\Omega).$$

If $f \in C^k$ then its distributional derivatives of order $\leq k$ agree with the classical ones; more generally, if $\partial_x^\alpha f = g$ in the weak (L^2) sense then $\partial_x^\alpha F_f = F_g$. But the distributional derivative is defined even when f is nowhere differentiable.

Worked Example 41.1.8 (The derivative of the Heaviside function is δ). Let h be the Heaviside function on $\mathbb{R}$: $h(x) = 1$ for $x > 0$, $h(x) = 0$ for $x < 0$. Compute h' in the sense of distributions.

Recognition. “Distributional derivative of a jump.” Apply Definition 41.1.7 and integrate.

Solution. For $\varphi \in \mathcal{D}(\mathbb{R})$, $h'(\varphi) = -h(\varphi') = -\int_0^\infty \varphi'(x) dx = -[\varphi(x)]_0^\infty = -(0 - \varphi(0)) = \varphi(0) = \delta(\varphi)$. Hence $h' = \delta$. *Check.* The classical derivative of h is 0 for $x \neq 0$ and undefined at 0, so the classical derivative misses the entire content of h' ; the distributional derivative records the unit jump precisely as a delta at the jump point.

Worked Example 41.1.9 (Second derivative of $|x|$, and $\frac{d}{dx} \text{sgn}$). On $\mathbb{R}$ compute the distributional first and second derivatives of $f(x) = |x|$.

Recognition. $|x|$ is continuous, piecewise smooth with a corner at 0; expect a jump in f' .

Solution. Classically $f'(x) = \text{sgn}(x)$ for $x \neq 0$, and $|x|$ is locally integrable, so f' (distributional) is the distribution of sgn . Then $f'' = (\text{sgn})'$. Now $\text{sgn} = 2h - 1$, so $(\text{sgn})' = 2h' = 2\delta$. Therefore $|x|'' = 2\delta$.

Check directly. $f''(\varphi) = f(\varphi'') = \int_{-\infty}^\infty |x| \varphi'' dx = \int_0^\infty x \varphi'' - \int_{-\infty}^0 x \varphi''$. Integrating by parts twice gives $\int_0^\infty x \varphi'' = -\int_0^\infty \varphi' = \varphi(0)$ and similarly the other piece gives $\varphi(0)$, total $2\varphi(0) = 2\delta(\varphi)$. Agrees.

Worked Example 41.1.10 (δ' and its action). Define δ' on $\mathbb{R}$ and compute $\delta'(\varphi)$ and $x\delta'$.

Recognition. Apply Definition 41.1.7 with $F = \delta$, $\alpha = 1$, then the product rule of §41.2.

Solution. $\delta'(\varphi) = -\delta(\varphi') = -\varphi'(0)$. For the product with the smooth function x : $(x\delta')(\varphi) = \delta'(x\varphi) = -(x\varphi)'(0) = -(\varphi + x\varphi')(0) = -\varphi(0) = -\delta(\varphi)$. So $x\delta' = -\delta$. *Check.* This is the distributional analogue of the Leibniz computation $(x\delta)' = \delta + x\delta' = 0$, i.e. $x\delta' = -\delta$ since $x\delta = 0$.

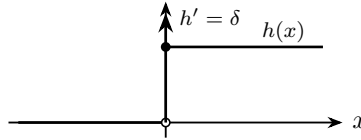


Figure 41.2. The Heaviside step h and its distributional derivative: a unit spike δ at the jump.

Supports of distributions

Definition 41.1.11. A distribution F *vanishes on an open set* O if $F(\varphi) = 0$ for every $\varphi \in \mathcal{D}$ supported in O . The *support* of F , $\text{supp } F$, is the complement of the largest open set on which F vanishes.

That “largest open set” is well defined because of the following gluing fact, proved with a partition of unity.

Proposition 41.1.12. If F vanishes on each member of a family $\{O_i\}_{i \in I}$ of open sets, then F vanishes on $O = \bigcup_i O_i$.

Proof. Let $\varphi \in \mathcal{D}$ be supported in a compact $K \subset O$. Finitely many sets cover K , say $K \subset \bigcup_{k=1}^N O_k$. Choose a smooth partition of unity $\eta_1, \dots, \eta_N$ subordinate to this cover: $0 \leq \eta_k \leq 1$, $\text{supp } \eta_k \subset O_k$, and $\sum_{k=1}^N \eta_k = 1$ on K . Then $\varphi = \sum_k \varphi \eta_k$ on K , each $\varphi \eta_k$ is supported in O_k , and so $F(\varphi) = F(\sum_k \varphi \eta_k) = \sum_k F(\varphi \eta_k) = 0$. $\square$

Remark. Basic facts: $\text{supp}(\partial_x^\alpha F) \subseteq \text{supp } F$ and $\text{supp}(\psi F) \subseteq \text{supp } F$ for $\psi \in C^\infty$; $\text{supp } \delta = \{0\}$ (and $\text{supp } \partial_x^\alpha \delta = \{0\}$); and $F(\varphi) = 0$ whenever $\text{supp } F \cap \text{supp } \varphi = \emptyset$.

Worked Example 41.1.13 (Computing a support: the comb $\sum_n \delta_n$). On $\mathbb{R}$ let $F = \sum_{n \in \mathbb{Z}} \delta_n$, where $\delta_n(\varphi) = \varphi(n)$ (the Dirac comb). Verify F is a distribution and find $\text{supp } F$.

Recognition. Sum of point masses; locally only finitely many contribute, so continuity is local.

Solution. For $\varphi \in \mathcal{D}$ with compact support K , only the integers in K contribute, a finite set, so $F(\varphi) = \sum_{n \in K \cap \mathbb{Z}} \varphi(n)$ is a finite sum; linearity and continuity follow as for a single δ . The largest open set on which F vanishes is $\mathbb{R} \setminus \mathbb{Z}$ (any φ supported there gives 0), so $\text{supp } F = \mathbb{Z}$. *Check.* $\mathbb{Z}$ is closed and not compact, so F does *not* have compact support — consistent with the fact that the sum is infinite.

Worked Example 41.1.14 (A distribution with point support: any $\sum_{|\alpha| \leq N} a_\alpha \partial^\alpha \delta$). Show $F = a_0 \delta + a_1 \delta'$ on $\mathbb{R}$ has $\text{supp } F \subseteq \{0\}$ and compute $F(\varphi)$.

Recognition. Derivatives of δ are supported at the origin; combine.

Solution. $F(\varphi) = a_0\varphi(0) - a_1\varphi'(0)$. If φ is supported away from 0 then $\varphi(0) = \varphi'(0) = 0$, so $F(\varphi) = 0$; hence F vanishes on $\mathbb{R} \setminus \{0\}$ and $\text{supp } F \subseteq \{0\}$ (equality unless $a_0 = a_1 = 0$). *Check.* This previews Theorem 41.1.15 below — every distribution supported at a point is a finite such combination.

Theorem 41.1.15 (Structure of point-supported distributions). *If F is a distribution on $\mathbb{R}^d$ supported at the origin, then F is a finite sum*

$$F = \sum_{|\alpha| \leq N} a_\alpha \partial_x^\alpha \delta, \quad \text{i.e.} \quad F(\varphi) = \sum_{|\alpha| \leq N} (-1)^{|\alpha|} a_\alpha (\partial_x^\alpha \varphi)(0).$$

The proof rests on a quantitative “it kills low-order data” lemma. (It uses that a compactly supported distribution is controlled by finitely many norms $\|\varphi\|_N = \sup_{|\alpha| \leq N} \sup_x |\partial_x^\alpha \varphi|$, a fact made precise for tempered distributions in Proposition 41.2.7.)

Lemma 41.1.16. *Suppose F_1 is supported at the origin and, for some N , satisfies (a) $|F_1(\varphi)| \leq c \|\varphi\|_N$ for all $\varphi \in \mathcal{D}$, and (b) $F_1(x^\alpha) = 0$ for all $|\alpha| \leq N$. Then $F_1 = 0$.*

Proof. Fix $\eta \in \mathcal{D}$ with $\eta(x) = 0$ for $|x| \geq 1$ and $\eta(x) = 1$ for $|x| \leq 1/2$, and set $\eta_\varepsilon(x) = \eta(x/\varepsilon)$. Since F_1 is supported at 0, $F_1(\eta_\varepsilon \varphi) = F_1(\varphi)$ and $F_1(\eta_\varepsilon x^\alpha) = F_1(x^\alpha) = 0$ for $|\alpha| \leq N$. Hence, with the Taylor remainder $R(x) = \varphi(x) - \sum_{|\alpha| \leq N} \frac{\partial^\alpha \varphi(0)}{\alpha!} x^\alpha$,

$$F_1(\varphi) = F_1(\eta_\varepsilon R).$$

One has $|R(x)| \leq c|x|^{N+1}$ and $|\partial_x^\beta R(x)| \leq c_\beta |x|^{N+1-|\beta|}$ for $|\beta| \leq N$, while $|\partial_x^\beta \eta_\varepsilon(x)| \leq c_\beta \varepsilon^{-|\beta|}$ with $\partial_x^\beta \eta_\varepsilon = 0$ for $|x| \geq \varepsilon$. By Leibniz’s rule $\|\eta_\varepsilon R\|_N \leq c\varepsilon$, so by (a) $|F_1(\varphi)| \leq c\varepsilon$; letting $\varepsilon \rightarrow 0$ gives $F_1(\varphi) = 0$. $\square$

Proof of Theorem 41.1.15. Pick N so that (a) holds (Proposition 41.2.7), and set $a_\alpha = \frac{(-1)^{|\alpha|}}{\alpha!} F(x^\alpha)$. Let $F_1 = F - \sum_{|\alpha| \leq N} a_\alpha \partial_x^\alpha \delta$. Since $\partial_x^\alpha \delta(x^\beta) = (-1)^{|\alpha|} \alpha!$ if $\alpha = \beta$ and 0 otherwise, F_1 satisfies (a) and (b); by Lemma 41.1.16, $F_1 = 0$. $\square$

Worked Example 41.1.17 (An L^p -norm computation: a model singularity). For $f(x) = |x|^{-a} \mathbf{1}_{\{|x| \leq 1\}}$ on $\mathbb{R}^d$, determine for which a and p one has $f \in L^p(\mathbb{R}^d)$, and contrast with whether f defines a distribution.

Recognition. Local integrability of $|x|^{-ap}$ near the origin via polar coordinates governs both questions.

Solution. In polar coordinates $\int_{|x| \leq 1} |x|^{-ap} dx = A_d \int_0^1 r^{-ap} r^{d-1} dr$, finite iff $-ap + d - 1 > -1$, i.e. $ap < d$, i.e. $a < d/p$. So $f \in L^p$ exactly when $a < d/p$. As a distribution, f is defined (locally integrable) precisely when $a < d$ (the case $p = 1$). Thus $f \in L^p$ for $p > 1$ is *strictly stronger* than “ f is a distribution.” *Check.* For $d = 3$, $a = 1$ (the Coulomb tail $1/|x|$): $f \in L^p$ iff $p < 3$, and it is locally integrable ($a = 1 < 3 = d$), so it is a distribution. The borderline $a = d/p$ fails (the integral diverges logarithmically).

Worked Example 41.1.18 (Weak convergence of approximations to the identity). Let $\psi \in \mathcal{D}(\mathbb{R})$ with $\int \psi = 1$ and $\psi_n(x) = n\psi(nx)$. Show $\psi_n \rightarrow \delta$ in the sense of distributions.

Recognition. “ $F_n \rightarrow F$ weakly” means $F_n(\varphi) \rightarrow F(\varphi)$ for every test function.

Solution. For $\varphi \in \mathcal{D}$, $F_{\psi_n}(\varphi) = \int n\psi(nx)\varphi(x) dx = \int \psi(u)\varphi(u/n) du$ (substitute $u = nx$). As $n \rightarrow \infty$, $\varphi(u/n) \rightarrow \varphi(0)$ boundedly, so by dominated convergence the integral tends to $\varphi(0) \int \psi = \varphi(0) = \delta(\varphi)$. Hence $\psi_n \rightarrow \delta$ weakly. *Check.* The mass $\int \psi_n = 1$ is preserved for every n (substitution), matching $\delta(1\text{-like}) = 1$; the concentration of support to $\{0\}$ matches $\text{supp } \delta = \{0\}$.

Worked Example 41.1.19 (A non-example: pointwise limit need not be the distributional limit). On $\mathbb{R}$ let $g_n(x) = n \mathbf{1}_{[0, 1/n]}(x)$. Show $g_n \rightarrow 0$ pointwise a.e. but $g_n \rightarrow \delta$ as distributions.

Recognition. Distinguish pointwise convergence (a function statement) from weak convergence (testing against φ).

Solution. For each fixed $x \neq 0$, $g_n(x) = 0$ once $n > 1/x$, so $g_n \rightarrow 0$ pointwise a.e. But $F_{g_n}(\varphi) = n \int_0^{1/n} \varphi(x) dx \rightarrow \varphi(0)$ by the mean value theorem (the average of φ over $[0, 1/n]$ tends to $\varphi(0)$). So $g_n \rightarrow \delta \neq 0$ as distributions. *Check.* The total mass $\int g_n = 1$ does not vanish in the limit even though the pointwise limit is 0 — mass escapes into an ever-thinner spike, exactly the behavior weak convergence detects and pointwise convergence misses.

Caution. The distributional derivative and the classical (pointwise) derivative coincide only when the function is smooth enough. For a function with a jump, the distributional derivative carries an extra δ at the jump (Worked Example with h); for $|x|$ it carries a 2δ in the second derivative. Never write “ $h' = 0$ since h is locally constant” — that is the classical derivative on $\mathbb{R} \setminus \{0\}$, which forgets the jump entirely.

Exercises (Stein, Ch.3).

1. Suppose F is a distribution on Ω and $F = f$, with f a C^k function in Ω . Show that $\partial_x^\alpha F$, taken in the sense of distributions, agrees with $\partial_x^\alpha f$ for each $|\alpha| \leq k$.
2. Converses to the previous exercise. (a) If f, g are continuous on $(a, b) \subset \mathbb{R}$ and $df/dx = g$ in the sense of distributions, show that $(f(x+h) - f(x))/h \rightarrow g(x)$ as $h \rightarrow 0$ for every $x \in (a, b)$. (b) If $f, g \in L^1(a, b)$ with $df/dx = g$ distributionally, then f is absolutely continuous and $(f(x+h) - f(x))/h \rightarrow g(x)$ a.e. Consequently, if f is continuous but nowhere differentiable on $\mathbb{R}$, its distributional derivative is not a locally integrable function on any sub-interval. (c) Generalize (a): if f is continuous on open Ω and, for each $|\alpha| \leq k$, the distribution $\partial_x^\alpha f$ equals a continuous function g_α , then $f \in C^k$ and $\partial_x^\alpha f = g_\alpha$ as functions for all $|\alpha| \leq k$.
3. Suppose F is a distribution on Ω . (a) There exist $f_n \in C^\infty$, each of compact support in Ω , with $f_n \rightarrow F$ in the sense of distributions. (b) If F is supported in the compact set C , then for every $\varepsilon > 0$ the f_n may be chosen with supports in the ε -neighborhood of C .
4. Let f be locally integrable on $\mathbb{R}^d$. The “support” of f in the measure-theoretic sense is $E = \{x : f(x) \neq 0\}$ (determined modulo null sets). Show that the support of f as a distribution equals the intersection of all closed sets C such that $E \setminus C$ has measure zero.
5. Let $\Omega = \{x \in \mathbb{R}^d : x_d > \varphi(x')\}$, with $x = (x', x_d) \in \mathbb{R}^{d-1} \times \mathbb{R}$ and $\varphi \in C^1$. Suppose f is continuous on $\overline{\Omega}$ with continuous first derivatives in Ω and $f|_{\partial\Omega} = 0$. Let $\tilde{f}$ be the extension by zero to $\mathbb{R}^d$. Show that $\partial\tilde{f}/\partial x_j$, taken distributionally, is the function equal to $\partial f/\partial x_j$ in Ω and zero in Ω^c . (Note: it need not be continuous.)

Additional exercises.

1. (Warm-up.) Compute the distributional derivative on $\mathbb{R}$ of $f(x) = \max(x, 0)$ (the ramp). Show $f' = h$ (Heaviside) and $f'' = \delta$.
2. Let $g(x) = \mathbf{1}_{[0,1]}(x)$ on $\mathbb{R}$. Compute g' in the sense of distributions and express it in terms of two delta functions.
3. For $\varphi \in \mathcal{D}(\mathbb{R})$ and $a < b$, compute $(\delta_a - \delta_b)(\varphi)$ and identify $\frac{d}{dx}\mathbf{1}_{[a,b]}$ as a distribution.
4. Show that $x^2\delta' = 0$ and, more generally, $x^k\delta^{(j)} = 0$ whenever $k > j$, while $x^j\delta^{(j)} = (-1)^j j! \delta$. (Use the product and derivative rules.)
5. Let $f_n(x) = \sin(nx)$ on $\mathbb{R}$. Show $f_n \rightarrow 0$ in the sense of distributions even though f_n does not converge pointwise. [Hint: integrate by parts against $\varphi \in \mathcal{D}$.]
6. (Support.) Give an example of distributions F, G on $\mathbb{R}$ with $\text{supp } F = \text{supp } G = \{0\}$ but $F \neq G$. Then show every distribution with $\text{supp} \subseteq \{0\}$ that is killed by both $\varphi \mapsto \varphi(0)$ and $\varphi \mapsto \varphi'(0)$ is a combination of $\delta'', \delta''', \dots$ only.
7. (Non-example.) Explain why $\sum_{n=1}^\infty n \delta_{1/n}$ does *not* define a distribution on $\mathbb{R}$, but $\sum_{n=1}^\infty \delta_{1/n}$ *does* define one on $(0, \infty)$ and find its support there.
8. (Diagram problem.) Sketch the graph of a C^∞ bump φ supported in $[-2, 2]$ with $\varphi = 1$ on $[-1, 1]$, and below it sketch φ' . Mark on your picture the two intervals where $\varphi' \neq 0$ and their opposite signs; explain why $\int \varphi' = 0$.

41.2 Operations on Distributions and Tempered Distributions

Operations transported by duality

Every operation on functions that has an integration-by-parts (or change-of-variables) adjoint extends to distributions by moving the operation onto the test function. We have already done this for ∂_x^α . The pattern: to define an operator T on F , find the formal adjoint T' on test functions ($\int (Tf)\varphi = \int f(T'\varphi)$) and *declare* $(TF)(\varphi) = F(T'\varphi)$. The defining requirement is that the extension restrict to the classical T on functions, i.e. that the square below commute ($\iota: f \mapsto F_f$); the identity $\int (Tf)\varphi = \int f(T'\varphi)$ is exactly what makes it commute.

$$\begin{array}{ccc} L_{\text{loc}}^1 & \xrightarrow{T} & L_{\text{loc}}^1 \\ \downarrow \iota & & \downarrow \iota \\ \mathcal{D}^* & \xrightarrow{T} & \mathcal{D}^* \end{array}$$

Definition 41.2.1. Let $F \in \mathcal{D}^*$. For $\psi \in C^\infty$, $h \in \mathbb{R}^d$, $a > 0$, and L a non-singular linear map:

$$(\psi F)(\varphi) = F(\psi\varphi), \quad (\tau_h F)(\varphi) = F(\tau_{-h}\varphi), \quad F_a(\varphi) = a^{-d}F(\varphi_{a^{-1}}), \quad F_L(\varphi) = |\det L|^{-1}F(\varphi_{L^{-1}}),$$

where $\tau_h f(x) = f(x - h)$ and $f_a(x) = f(ax)$ (so $\varphi_a(x) = \varphi(ax)$). Each agrees with the classical operation when F is a function.

Worked Example 41.2.2 (Multiplication and the product rule for $x\delta$ and $\psi\delta$). On $\mathbb{R}$ compute $x\delta$ and, for smooth ψ , $\psi\delta$ and $\psi\delta'$.

Recognition. Use $(\psi F)(\varphi) = F(\psi\varphi)$ and evaluate at 0.

Solution. $(x\delta)(\varphi) = \delta(x\varphi) = (x\varphi)(0) = 0$, so $x\delta = 0$. Likewise $(\psi\delta)(\varphi) = \psi(0)\varphi(0) = \psi(0)\delta(\varphi)$, i.e. $\psi\delta = \psi(0)\delta$. And $(\psi\delta')(\varphi) = \delta'(\psi\varphi) = -(\psi\varphi)'(0) = -\psi'(0)\varphi(0) - \psi(0)\varphi'(0)$, hence $\psi\delta' = \psi(0)\delta' - \psi'(0)\delta$.

Check. The Leibniz rule $(\psi\delta)' = \psi'\delta + \psi\delta'$ gives $\psi'(0)\delta + \psi(0)\delta'$ when combined with $\psi\delta = \psi(0)\delta$, consistent with the formula for $\psi\delta'$.

Convolution and approximation by smooth functions

For $F \in \mathcal{D}^*(\mathbb{R}^d)$ and $\psi \in \mathcal{D}$ there are two natural meanings of $F * \psi$: as the *function* $x \mapsto F(\psi_x^\sim)$ with $\psi_x^\sim(y) = \psi(x - y)$, or as the *distribution* $(F * \psi)(\varphi) = F(\psi^\sim * \varphi)$ with $\psi^\sim(y) = \psi(-y)$.

Proposition 41.2.3. Let F be a distribution and $\psi \in \mathcal{D}$. Then (a) the two definitions of $F * \psi$ coincide, and (b) $F * \psi$ is a C^∞ function.

Proof. If $x_n \rightarrow x_0$, then $\psi_{x_n}^\sim(y) = \psi(x_n - y) \rightarrow \psi(x_0 - y)$ uniformly in y , with the same for all derivatives; thus $\psi_{x_n}^\sim \rightarrow \psi_{x_0}^\sim$ in $\mathcal{D}$, and continuity of F gives that $x \mapsto F(\psi_x^\sim)$ is continuous. The difference quotients converge similarly, so this function is C^∞ with $\partial_x^\alpha F(\psi_x^\sim) = F(\partial_x^\alpha \psi_x^\sim)$, proving (b). For (a) we must show $\int F(\psi_x^\sim)\varphi(x) dx = F(\psi^\sim * \varphi)$. Since $(\psi^\sim * \varphi)(x) = \int \psi^\sim(x - y)\varphi(y) dy = \lim_{\varepsilon \rightarrow 0} S(\varepsilon)$ with $S(\varepsilon) = \varepsilon^d \sum_{n \in \mathbb{Z}^d} \psi^\sim(x - \varepsilon n)\varphi(\varepsilon n)$ (Riemann sums converging in $\mathcal{D}$), and $F(S(\varepsilon)) = \varepsilon^d \sum_n F(\psi_{\varepsilon n}^\sim)\varphi(\varepsilon n)$, the limit $\varepsilon \rightarrow 0$ and the continuity of $x \mapsto F(\psi_x^\sim)$ yield the identity. $\square$

Corollary 41.2.4. Every distribution F on $\mathbb{R}^d$ is the weak limit of C^∞ functions: there exist $f_n \in C^\infty$ with $f_n(\varphi) \rightarrow F(\varphi)$ for every φ . (Here $F_n \rightarrow F$ weakly means $F_n(\varphi) \rightarrow F(\varphi)$ for all $\varphi \in \mathcal{D}$.)

Proof. Fix $\psi \in \mathcal{D}$ with $\int \psi = 1$ and set $\psi_n(x) = n^d \psi(nx)$ (an approximation to the identity). Let $F_n = F * \psi_n$; by Proposition 41.2.3(b) each $F_n \in C^\infty$, and by (a) $F_n(\varphi) = F(\psi_n^\sim * \varphi)$. Since $\psi_n^\sim * \varphi \rightarrow \varphi$ in $\mathcal{D}$, we get $F_n(\varphi) \rightarrow F(\varphi)$. $\square$

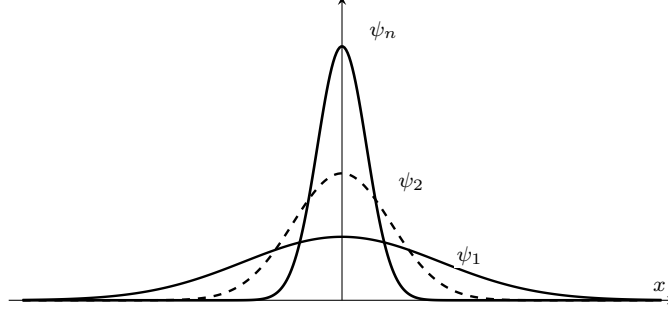


Figure 41.3. An approximation to the identity $\psi_n(x) = n^d \psi(nx)$: as $n \rightarrow \infty$ the bumps concentrate and $\psi_n \rightarrow \delta$ weakly. Convolution with ψ_n smooths any distribution while converging back to it. (Schematic.)

Proposition 41.2.5 (Support additivity). *If F has support C_1 and $\psi \in \mathcal{D}$ has support C_2 , then $\text{supp}(F * \psi) \subseteq C_1 + C_2$. Consequently, for F and F_1 distributions with F_1 of compact support, $(F * F_1)(\varphi) = F(F_1^\sim * \varphi)$ is well defined, $F * \delta = F$, and $\partial_x^\alpha(F * F_1) = (\partial_x^\alpha F) * F_1 = F * (\partial_x^\alpha F_1)$.*

Proof. If $F(\psi_x^\sim) \neq 0$ then $\text{supp } F$ meets $\text{supp } \psi_x^\sim = x - C_2$, so there is $y \in C_1 \cap (x - C_2)$; then $x = y + (x - y)$ with $y \in C_1$, $x - y \in C_2$, giving $x \in C_1 + C_2$. The set $C_1 + C_2$ is closed since C_1 is closed and C_2 compact. The extension to $F * F_1$ follows because $F_1^\sim * \varphi$ is then compactly supported and C^∞ , hence in $\mathcal{D}$; the stated identities are routine. $\square$

Tempered distributions and the Schwartz space

To do Fourier analysis we restrict to objects of at most polynomial growth at infinity. The matching test space is the Schwartz space.

Definition 41.2.6. The *Schwartz space* $\mathcal{S} = \mathcal{S}(\mathbb{R}^d)$ consists of all C^∞ functions φ with $\|\varphi\|_N < \infty$ for every N , where

$$\|\varphi\|_N = \sup_{x \in \mathbb{R}^d, |\alpha|, |\beta| \leq N} |x^\beta (\partial_x^\alpha \varphi)(x)|.$$

One says $\varphi_k \rightarrow \varphi$ in $\mathcal{S}$ if $\|\varphi_k - \varphi\|_N \rightarrow 0$ for every N . A *tempered distribution* is a linear functional F on $\mathcal{S}$ continuous in this sense; their space is $\mathcal{S}^*$.

Since $\mathcal{D} \subset \mathcal{S}$ and $\mathcal{D}$ -convergence implies $\mathcal{S}$ -convergence, every tempered distribution is a distribution; the converse fails (e.g. e^x on $\mathbb{R}$). Also $\mathcal{D}$ is dense in $\mathcal{S}$.

Proposition 41.2.7 (Finite-order control). *If F is a tempered distribution, there exist N and $c > 0$ with $|F(\varphi)| \leq c \|\varphi\|_N$ for all $\varphi \in \mathcal{S}$.*

Proof. If not, for each n there is $\psi_n \in \mathcal{S}$ with $\|\psi_n\|_n = 1$ but $|F(\psi_n)| \geq n$. Put $\varphi_n = \psi_n/n^{1/2}$. For $n \geq N$, $\|\varphi_n\|_N \leq \|\varphi_n\|_n = n^{-1/2} \rightarrow 0$, so $\varphi_n \rightarrow 0$ in $\mathcal{S}$, yet $|F(\varphi_n)| \geq n^{1/2} \rightarrow \infty$, contradicting continuity. $\square$

Worked Example 41.2.8 (e^x is not tempered). Show that on $\mathbb{R}$ the locally integrable function $f(x) = e^x$ defines a distribution that is *not* tempered.

Recognition. Use the polynomial-control criterion: a tempered F obeys $|F(\varphi)| \leq c \|\varphi\|_N$.

Solution. Suppose F_f were tempered. Take a fixed bump $\eta \in \mathcal{D}$, $\eta \geq 0$, $\eta = 1$ on $[-1, 1]$, and translate it: $\varphi_R(x) = \eta(x - R)$. Then $\|\varphi_R\|_N$ is bounded *independently* of R (translation does not change derivative sup-norms, and the weights x^β only multiply by polynomials in R , so $\|\varphi_R\|_N \leq c R^N$). But $F_f(\varphi_R) = \int e^x \eta(x - R) dx \geq \int_{R-1}^{R+1} e^x dx \geq e^{R-1}$, which grows faster than any power R^N . So no bound $|F_f(\varphi_R)| \leq c \|\varphi_R\|_N \leq c R^N$ can hold; F_f is not tempered. *Check.* Any f with $\int_{|x| \leq R} |f| = O(R^N)$ is tempered; e^x violates this since the integral grows like e^R .

Worked Example 41.2.9 (L^p functions are tempered). Show that if $f \in L^p(\mathbb{R}^d)$ for some $1 \leq p \leq \infty$, then F_f is a tempered distribution.

Recognition. Reduce to the growth criterion $\int_{|x| \leq R} |f| = O(R^N)$.

Solution. By Holder's inequality with $1/p + 1/q = 1$, $\int_{|x| \leq R} |f| \leq \|f\|_{L^p} \left(\int_{|x| \leq R} 1\right)^{1/q} = \|f\|_{L^p} (cR^d)^{1/q} = O(R^{d/q})$. Thus the growth is polynomial (degree $d/q \leq d$), so F_f is tempered. *Check.* For $p = \infty$, $q = 1$, the bound is $O(R^d)$; for $p = 1$, $q = \infty$, the bound is $O(1)$ — an L^1 function even gives a *bounded* functional on $\|\cdot\|_0$.

The Fourier transform of tempered distributions

For $\varphi \in \mathcal{S}$, $\widehat{\varphi}(\xi) = \int_{\mathbb{R}^d} \varphi(x) e^{-2\pi i x \cdot \xi} dx$ maps $\mathcal{S}$ bijectively and continuously onto itself, with inverse $\psi^\vee(x) = \int \psi(\xi) e^{2\pi i x \cdot \xi} d\xi$. The multiplication identity $\int \widehat{\psi}\varphi = \int \psi\widehat{\varphi}$ dictates the definition.

Definition 41.2.10. For $F \in \mathcal{S}^*$ the *Fourier transform* $\widehat{F}$ (also $F^\wedge$) is $F^\wedge(\varphi) = F(\widehat{\varphi})$, and $F^\vee(\varphi) = F(\varphi^\vee)$. Then $F \mapsto F^\wedge$ is a bijection of $\mathcal{S}^*$ with inverse $F \mapsto F^\vee$, since $(F^\wedge)^\vee(\varphi) = F((\varphi^\vee)^\wedge) = F(\varphi)$.

By construction $F \mapsto F^\wedge$ on $\mathcal{S}^*$ extends the classical transform on $\mathcal{S}$: the multiplication identity $\int \widehat{\psi}\varphi = \int \psi\widehat{\varphi}$ is precisely the statement that the square below commutes, where $\iota : \varphi \mapsto F_\varphi$ embeds $\mathcal{S}$ into $\mathcal{S}^*$.

$$\begin{array}{ccc} \mathcal{S} & \xrightarrow{\quad \widehat{\quad} \quad} & \mathcal{S} \\ \downarrow \iota & & \downarrow \iota \\ \mathcal{S}^* & \xrightarrow[\quad \wedge \quad]{} & \mathcal{S}^* \end{array}$$

The usual rules survive: $(\partial_x^\alpha F)^\wedge = (2\pi i \xi)^\alpha F^\wedge$ and $((-2\pi i x)^\alpha F)^\wedge = \partial_\xi^\alpha (F^\wedge)$. As tempered distributions,

$$\widehat{1} = \delta, \quad \widehat{\delta} = 1, \quad (\partial_x^\alpha \delta)^\wedge = (2\pi i \xi)^\alpha, \quad ((-2\pi i x)^\alpha)^\wedge = \partial_\xi^\alpha \delta.$$

Worked Example 41.2.11 ($\widehat{\delta} = 1$ and $\widehat{1} = \delta$). Verify these two Fourier transforms directly from Definition 41.2.10.

Recognition. Pair against a test function and recall $\widehat{\varphi}(0) = \int \varphi$ and the inversion formula.

Solution. $\widehat{\delta}(\varphi) = \delta(\widehat{\varphi}) = \widehat{\varphi}(0) = \int_{\mathbb{R}^d} \varphi(x) dx = 1(\varphi)$, so $\widehat{\delta} = 1$. Conversely $\widehat{1}(\varphi) = 1(\widehat{\varphi}) = \int \widehat{\varphi}(\xi) d\xi = (\widehat{\varphi})^\vee(0) = \varphi(0) = \delta(\varphi)$ by Fourier inversion, so $\widehat{1} = \delta$. *Check.* Consistency with $\widehat{\widehat{F}}(\varphi) = F(\varphi(-\cdot))$ gives $\widehat{\widehat{\delta}} = \delta$ and indeed $\widehat{1} = \delta$ again.

Proposition 41.2.12. If $F \in \mathcal{S}^*$ and $\psi \in \mathcal{S}$, then $F * \psi$ is a slowly increasing C^∞ function and, as a tempered distribution, $(F * \psi)^\wedge = \widehat{\psi} F^\wedge$. If F has compact support then $F^\wedge$ is the slowly increasing C^∞ function $F^\wedge(\xi) = F(e_\xi)$, where $e_\xi(x) = \eta(x) e^{-2\pi i x \cdot \xi}$ with $\eta \in \mathcal{D}$ equal to 1 near supp F .

Proof sketch. From $(F * \psi)(\varphi) = F(\psi^\sim * \varphi)$ one gets $(F * \psi)^\wedge(\varphi) = F(\psi^\sim * \widehat{\varphi})$, while $\widehat{\psi} F^\wedge(\varphi) = F((\widehat{\psi}\varphi)^\wedge)$, and the identity $(\widehat{\psi}\varphi)^\wedge = \psi^\sim * \widehat{\varphi}$ closes the argument. The compact-support statement follows from Proposition 41.2.7: $|F(e_\xi)| \leq c \|e_\xi\|_N \leq c'(1 + |\xi|)^N$ and likewise for difference quotients, so $F(e_\xi)$ is C^∞ and slowly increasing; testing against $\varphi \in \mathcal{D}$ and a Riemann-sum limit shows $\int F(e_\xi) \varphi(\xi) d\xi = F(\widehat{\varphi})$. $\square$

The principal value $\text{pv}(1/x)$ and the Hilbert transform

Definition 41.2.13. The *principal value* distribution $\text{pv}(1/x)$ on $\mathbb{R}$ is

$$\text{pv}\left(\frac{1}{x}\right)(\varphi) = \lim_{\varepsilon \rightarrow 0} \int_{|x| \geq \varepsilon} \frac{\varphi(x)}{x} dx, \quad \varphi \in \mathcal{S}.$$

Proposition 41.2.14. The limit exists for every $\varphi \in \mathcal{S}$ and defines a tempered distribution with $|\text{pv}(1/x)(\varphi)| \leq c \|\varphi\|_1$.

Proof. Split $\int_{|x| \geq \varepsilon} = \int_{1 \geq |x| \geq \varepsilon} + \int_{|x| > 1}$. The far piece converges by rapid decay of φ . Since $1/x$ is odd, $\int_{1 \geq |x| \geq \varepsilon} dx/x = 0$, so the near piece equals $\int_{1 \geq |x| \geq \varepsilon} \frac{\varphi(x) - \varphi(0)}{x} dx$; as $|\varphi(x) - \varphi(0)| \leq |x| \sup |\varphi'|$, the integrand is bounded and the limit exists. The bound by $\|\varphi\|_1$ is immediate. $\square$

The connection to the Hilbert transform H of the previous chapter is $H(f) = \frac{1}{\pi} \text{pv}(1/x) * f$ for $f \in \mathcal{S}$.

Theorem 41.2.15 (Identities for $\text{pv}(1/x)$). *As tempered distributions on $\mathbb{R}$,*

$$\text{pv}\left(\frac{1}{x}\right) = \frac{d}{dx} \log|x| = \frac{1}{2} \left(\frac{1}{x-i0} + \frac{1}{x+i0} \right),$$

where $\frac{1}{x \mp i0} = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{x \mp i\varepsilon}$, and moreover its Fourier transform is the function $(\text{pv}(1/x))^\wedge = \frac{\pi}{i} \text{sgn}(\xi)$. One has the Sokhotski-Plemelj relations $\frac{1}{x \mp i0} = \text{pv}(1/x) \pm i\pi\delta$.

Proof. Derivative of $\log|x|$. $\frac{d}{dx} \log|x|(\varphi) = - \int \log|x| \varphi' dx = - \lim_{\varepsilon \rightarrow 0} \int_{|x| \geq \varepsilon} \log|x| \varphi' dx$. Integration by parts gives $\int_{|x| \geq \varepsilon} \frac{\varphi}{x} dx + \log(\varepsilon)[\varphi(\varepsilon) - \varphi(-\varepsilon)]$, and since $\varphi(\varepsilon) - \varphi(-\varepsilon) = O(\varepsilon)$ the boundary term vanishes, giving $\text{pv}(1/x)$.

The limits $1/(x \mp i\varepsilon)$. Write $\frac{1}{2} \left(\frac{1}{x-i\varepsilon} + \frac{1}{x+i\varepsilon} \right) = \frac{x}{x^2 + \varepsilon^2}$. The conjugate Poisson kernel computation shows $\frac{x}{x^2 + \varepsilon^2} \rightarrow \text{pv}(1/x)$ weakly; and since $\frac{1}{\pi} \frac{\varepsilon}{x^2 + \varepsilon^2} = P_\varepsilon \rightarrow \delta$ (the Poisson kernel is an approximation to the identity), $\frac{1}{x-i\varepsilon} = \pi Q_\varepsilon + i\pi P_\varepsilon \rightarrow \text{pv}(1/x) + i\pi\delta$. Taking conjugates gives $\frac{1}{x+i0} = \text{pv}(1/x) - i\pi\delta$; adding yields the middle identity and subtracting yields $i\pi\delta = \frac{1}{2} \left(\frac{1}{x-i0} - \frac{1}{x+i0} \right)$.

Fourier transform. Using $\left(\frac{x}{x^2 + \varepsilon^2} \right)^\wedge = \pi \frac{\text{sgn}(\xi)}{i} e^{-2\pi\varepsilon|\xi|}$ (from the conjugate Poisson kernel) and letting $\varepsilon \rightarrow 0$ gives $(\text{pv}(1/x))^\wedge = \frac{\pi}{i} \text{sgn}(\xi)$. $\square$

Worked Example 41.2.16 (The Sokhotski-Plemelj formula in action). Use $\frac{1}{x-i0} = \text{pv}(1/x) + i\pi\delta$ to evaluate $\lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}} \frac{\varphi(x)}{x-i\varepsilon} dx$ for $\varphi \in \mathcal{S}$.

Recognition. Direct application of Theorem 41.2.15; separate principal value and delta parts.

Solution. The limit equals $(\text{pv}(1/x) + i\pi\delta)(\varphi) = \text{pv} \int \frac{\varphi(x)}{x} dx + i\pi\varphi(0)$. *Check.* Taking φ even forces the principal value to be 0 (odd integrand), leaving $i\pi\varphi(0)$; taking φ odd kills $\varphi(0)$, leaving a real principal value. Both special cases match direct contour reasoning (a small semicircle below the pole contributes $+i\pi$ times the residue).

Homogeneous distributions

Definition 41.2.17. $F \in \mathcal{D}^*(\mathbb{R}^d)$ is *homogeneous of degree λ* if $F_a = a^\lambda F$ for all $a > 0$ (with the dilation of Definition 41.2.1).

Worked Example 41.2.18 ($\text{pv}(1/x)$ is homogeneous of degree -1). Verify directly that $\text{pv}(1/x)$ is homogeneous of degree -1 on $\mathbb{R}$.

Recognition. Apply the dilation $F^a(\varphi) = F(\varphi_a)$ and change variables in the integral.

Solution. Using $F^a(\varphi) = a^{-1} \lim_{\varepsilon} \int_{|x| \geq \varepsilon} \frac{\varphi(x/a)}{x} dx$ and the substitution $x \rightarrow ax$ (which leaves dx/x invariant), $= a^{-1} \lim_{\varepsilon} \int_{|x| \geq \varepsilon/a} \frac{\varphi(x)}{x} dx = a^{-1} \text{pv}(1/x)(\varphi)$. Hence $\text{pv}(1/x)$ scales like degree -1 . *Check.* The underlying function $1/x$ is homogeneous of degree -1 ; the principal-value regularization preserves the homogeneity, which is special to the critical degree $-d = -1$.

Proposition 41.2.19. *If $F \in \mathcal{S}^*(\mathbb{R}^d)$ is homogeneous of degree λ , then $F^\wedge$ is homogeneous of degree $-d - \lambda$.*

Proof. Using $(\varphi_a)^\wedge = (\hat{\varphi})^a$ and $F^a = a^{-d} F_{a^{-1}}$,

$$(F^\wedge)_a(\varphi) = F^\wedge(\varphi^a) = F((\varphi^a)^\wedge) = F((\hat{\varphi})_a) = a^{-d-\lambda} F(\hat{\varphi}) = a^{-d-\lambda} F^\wedge(\varphi). \quad \square$$

Theorem 41.2.20 (Fourier transform of $|x|^\lambda$). *For $-d < \lambda < 0$ let H_λ be the (locally integrable, tempered) distribution given by $|x|^\lambda$. Then*

$$(H_\lambda)^\wedge = c_\lambda H_{-d-\lambda}, \quad c_\lambda = \pi^{-d/2-\lambda} \frac{\Gamma(\frac{d+\lambda}{2})}{\Gamma(-\frac{\lambda}{2})}.$$

Proof. Start from $\psi(x) = e^{-\pi|x|^2}$ being its own Fourier transform; the dilation rule with $a = t^{1/2}$ gives $\int e^{-\pi t|x|^2} \widehat{\varphi}(x) dx = t^{-d/2} \int e^{-\pi|x|^2/t} \varphi(x) dx$. Multiply by $t^{-\lambda/2-1}$ and integrate t over $(0, \infty)$, using $\int_0^\infty e^{-tA} t^{-\lambda/2-1} dt = A^{\lambda/2} \Gamma(-\lambda/2)$ (for $A > 0, \lambda > 0$) on the left with $A = \pi|x|^2$, and the change $t \rightarrow 1/t$ giving $\int_0^\infty t^{d/2+\lambda/2-1} e^{-At} dt = A^{-d/2-\lambda/2} \Gamma(\frac{d}{2} + \frac{\lambda}{2})$ on the right. Comparing,

$$\pi^{\lambda/2} \Gamma(-\frac{\lambda}{2}) \int |x|^\lambda \widehat{\varphi} = \pi^{-d/2-\lambda/2} \Gamma(\frac{d}{2} + \frac{\lambda}{2}) \int |x|^{-d-\lambda} \varphi,$$

which rearranges to $(H_\lambda)^\wedge = c_\lambda H_{-d-\lambda}$. $\square$

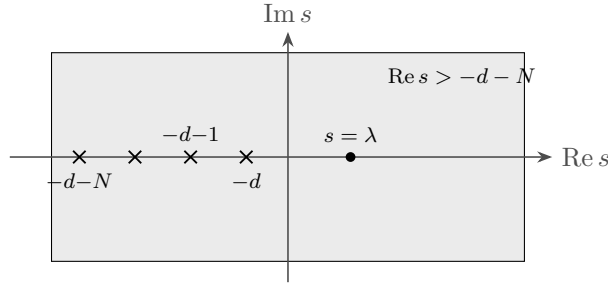


Figure 41.4. Analytic continuation of $I(s)$ (Theorem 41.2.23): simple poles march left at $s = -d, -d-1, \dots$; the value at $s = \lambda$ defines the homogeneous distribution. (Schematic.)

Definition 41.2.21. A distribution K is *regular* (or C^∞ away from the origin) if there is a function $k \in C^\infty(\mathbb{R}^d \setminus \{0\})$ with $K(\varphi) = \int k(x)\varphi(x) dx$ for all $\varphi \in \mathcal{D}$ supported away from the origin. The function k (uniquely determined) is called the *associated function*; e.g. $1/x$ is associated to $\text{pv}(1/x)$. If K is homogeneous of degree λ , so is k .

Theorem 41.2.22 (Fourier transform of regular homogeneous distributions). *The Fourier transform of a regular homogeneous distribution K of degree λ is a regular homogeneous distribution of degree $-d-\lambda$, and conversely.*

Proof. Homogeneity of degree $-d-\lambda$ is Proposition 41.2.19. For regularity, split $K = K_0 + K_1$ with $K_0 = \eta K$ supported near 0 and $K_1 = (1-\eta)k$ supported away from 0 ($\eta \in C^\infty$, $\eta = 1$ on $|x| \leq 1/2$, $\text{supp } \eta \subseteq \{|x| \leq 1\}$). By Proposition 41.2.12, $K_0^\wedge$ is everywhere C^∞ . For K_1 , the identity $(-4\pi^2 |\xi|^2)^N \partial_\xi^\alpha (K_1^\wedge) = \Delta^N [(-2\pi i x)^\alpha K_1]^\wedge$ and the homogeneity bound $\partial_x^\beta K_1 = O(|x|^{\lambda-|\beta|})$ for $|x| \geq 1$ show $\Delta^N [x^\alpha K_1] \in L^1$ once $2N > \lambda + |\alpha| + d$; its Fourier transform is then continuous, so $\partial_\xi^\alpha (K_1^\wedge)$ is continuous away from 0 for every α , i.e. $K_1^\wedge \in C^\infty$ away from 0. The converse follows since $K^\vee = (K^\wedge)^\sim$. $\square$

Theorem 41.2.23 (Existence and uniqueness of homogeneous extensions). *Let $k \in C^\infty(\mathbb{R}^d \setminus \{0\})$ be homogeneous of degree λ .*

1. *If λ is not of the form $-d-m$ ($m \geq 0$ integer), there is a unique homogeneous distribution K of degree λ agreeing with k away from 0.*
2. *If $\lambda = -d-m$, such K exists iff the cancellation condition $\int_{|x|=1} x^\alpha k(x) d\sigma(x) = 0$ holds for all $|\alpha| = m$.*
3. *Every K arising in (2) is determined up to $\sum_{|\alpha|=m} c_\alpha \partial_x^\alpha \delta$.*

Proof idea. For $\lambda > -d$, k is locally integrable, so $K = F_k$ works. In general, analytically continue

$$I(s)(\varphi) = \int_{\mathbb{R}^d} k(x) |x|^{-\lambda+s} \varphi(x) dx, \quad \text{Re}(s) > -d,$$

to a meromorphic function with at most simple poles at $s = -d, -d-1, \dots$; the residues involve $\int_{|x|=1} k(x) x^\alpha d\sigma(x)$, and one sets $K(\varphi) = I(\lambda)$. Homogeneity $I(s)(\varphi^a) = a^s I(s)$ continues to $s = \lambda$. Uniqueness uses Lemma 41.2.24: the difference of two extensions is supported at 0, hence a finite sum $\sum c_\alpha \partial_x^\alpha \delta$ by Theorem 41.1.15, and matching homogeneities forces it to vanish except in the resonant case $\lambda = -d-m$. $\square$

Lemma 41.2.24. If $\lambda_1, \dots, \lambda_n$ are distinct reals and $\sum_{j=1}^n (a_j x^{\lambda_j} + b_j x^{\lambda_j} \log x) = 0$ for all $x > 0$, then all $a_j = b_j = 0$.

Proof. Let λ_n be largest. Multiply by $x^{-\lambda_n}$ and let $x \rightarrow \infty$: the term $b_n \log x$ would dominate unless $b_n = 0$, then $a_n = 0$; induct on n . $\square$

Corollary 41.2.25. There is no distribution homogeneous of degree $-d$ that agrees with $1/|x|^d$ away from the origin. More generally, if k is homogeneous of degree $-d$ with $\int_{|x|=1} k d\sigma \neq 0$, no homogeneous degree- $(-d)$ extension exists; when $\int_{|x|=1} k d\sigma = 0$ the extension is the principal value $\text{pv}(k)$, $\text{pv}(k)(\varphi) = \lim_{\varepsilon \rightarrow 0} \int_{|x| \geq \varepsilon} k(x) \varphi(x) dx$, generalizing $\text{pv}(1/x)$.

Worked Example 41.2.26 (Holder's inequality and the conjugate exponent). Show that for $f \in L^p$, $g \in L^q$ with $1/p + 1/q = 1$, the pairing $\int fg$ converges and $|\int fg| \leq \|f\|_{L^p} \|g\|_{L^q}$, and explain why this makes L^q act as functionals on L^p (a baby case of $\mathcal{S}^*$).

Recognition. Holder's inequality; the duality $L^{p'} = L^q$.

Solution. By Young's numerical inequality $ab \leq a^p/p + b^q/q$ ($a, b \geq 0$), applied to $a = |f|/\|f\|_{L^p}$, $b = |g|/\|g\|_{L^q}$ and integrated, $\frac{1}{\|f\|_{L^p} \|g\|_{L^q}} \int |fg| \leq \frac{1}{p} \frac{\int |f|^p}{\|f\|_{L^p}^p} + \frac{1}{q} \frac{\int |g|^q}{\|g\|_{L^q}^q} = \frac{1}{p} + \frac{1}{q} = 1$. Hence $|\int fg| \leq \int |fg| \leq \|f\|_{L^p} \|g\|_{L^q}$. So $g \mapsto (f \mapsto \int fg)$ embeds L^q into $(L^p)^*$ with operator norm $\leq \|g\|_{L^q}$ (equality holds, by choosing $f = |g|^{q-1} \text{sgn } \bar{g}$). *Check.* For $p = q = 2$ this is Cauchy-Schwarz; the test-function pairing $F(\varphi) = \int f\varphi$ defining $F_f \in \mathcal{S}^*$ is the same mechanism with φ in the role of the dual element.

Worked Example 41.2.27 (Minkowski's inequality / interpolation of L^p norms). Show that if $f \in L^{p_0} \cap L^{p_1}$ then $f \in L^p$ for every p between p_0 and p_1 , with $\|f\|_{L^p} \leq \|f\|_{L^{p_0}}^{1-\theta} \|f\|_{L^{p_1}}^\theta$ where $\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}$.

Recognition. Log-convexity of $p \mapsto \|f\|_{L^p}$; apply Holder with split exponents.

Solution. Write $|f|^p = |f|^{p(1-\theta)} |f|^{p\theta}$ and apply Holder with exponents $\frac{p_0}{p(1-\theta)}$ and $\frac{p_1}{p\theta}$ (which are conjugate, by the definition of θ): $\int |f|^p \leq (\int |f|^{p_0})^{p(1-\theta)/p_0} (\int |f|^{p_1})^{p\theta/p_1}$. Taking p -th roots gives the stated bound. *Check.* At $\theta = 0$ it reads $\|f\|_{L^{p_0}} \leq \|f\|_{L^{p_0}}$; the inequality is the Lyapunov/Riesz-Thorin scalar prototype, and convexity in $1/p$ is exactly what powers interpolation of operators.

Worked Example 41.2.28 (A Hahn-Banach extension of a functional). Let $M \subset L^2(\mathbb{R})$ be the subspace of even functions and define on M the functional $\ell(f) = f(0^+)$ in a formal sense replaced by $\ell(f) = \int_{-1}^1 f$. Show ℓ extends to a bounded functional on all of L^2 and identify the norm.

Recognition. Hahn-Banach (or directly Riesz representation) extends a bounded functional preserving its norm.

Solution. On M , $\ell(f) = \int_{-1}^1 f = \langle f, \mathbf{1}_{[-1,1]} \rangle$, so by Cauchy-Schwarz $|\ell(f)| \leq \|f\|_{L^2} \|\mathbf{1}_{[-1,1]}\|_{L^2} = \|f\|_{L^2} \sqrt{2}$, giving $\|\ell\|_{M^*} \leq \sqrt{2}$. By Hahn-Banach there is an extension $\tilde{\ell}$ on L^2 with the same norm; by the Riesz representation theorem $\tilde{\ell}(f) = \langle f, g \rangle$ for a unique $g \in L^2$ with $\|g\|_{L^2} = \|\tilde{\ell}\|$. Taking $g = \mathbf{1}_{[-1,1]}$ (which represents ℓ already on M and on all of L^2) gives the canonical extension with norm exactly $\sqrt{2}$. *Check.* The extremizer $f = \mathbf{1}_{[-1,1]}/\sqrt{2}$ achieves $|\ell(f)| = \sqrt{2} = \|g\|_{L^2}$, confirming the norm is attained and equals $\|\mathbf{1}_{[-1,1]}\|_{L^2}$.

Caution. At the critical degree $\lambda = -d$, a homogeneous function k that is C^∞ off the origin need *not* extend to a homogeneous distribution: it does so precisely when its spherical mean $\int_{|x|=1} k d\sigma$ vanishes. The mean is the obstruction; when it is nonzero the best one can do is the non-homogeneous "finite part" $(1/|x|^d)(\varphi^a) = a^{-d}(1/|x|^d)(\varphi) + a^{-d} \log(a) A_d \varphi(0)$, with $A_d = 2\pi^{d/2}/\Gamma(d/2)$ the surface area of the unit sphere.

Exercises (Stein, Ch.3).

1. Show that a bounded function f on $\mathbb{R}^d$ satisfies a Lipschitz (Holder exponent 1) condition $|f(x) - f(y)| \leq C|x - y|$ for all x, y if and only if $f \in L^\infty$ and all first-order partial derivatives $\partial f / \partial x_j$, $1 \leq j \leq d$, belong to L^∞ in the sense of distributions. [Hint: $f_n = f * \psi_n$, ψ_n an approximation to the identity; then $\partial f_n / \partial x_j \in L^\infty$ uniformly in n .]

2. Show that F is tempered if and only if there is an integer N and a constant A so that for all $R \geq 1$, $|F(\varphi)| \leq AR^N \sup_{|x| \leq R, 0 \leq |\alpha| \leq N} |\partial_x^\alpha \varphi(x)|$ for all $\varphi \in \mathcal{D}$ supported in $|x| \leq R$.
3. Suppose F is a homogeneous distribution of degree λ . Show that F is tempered. [Hint: fix $\eta \in \mathcal{D}$, $\eta = 1$ for $|x| \leq 1$, supported in $|x| \leq 2$; let $\eta_R(x) = \eta(x/R)$. Find N with $|\eta_1 F(\varphi)| \leq c \|\varphi\|_N$, then deduce $|(\eta_R F)(\varphi)| \leq cR^{N+|\lambda|} \|\varphi\|_N$.]
4. Check that on $\mathbb{R}$, $f(x) = e^x$ considered as a distribution is not tempered. [Hint: show the criterion of the tempered-test exercise fails for every N .]
5. Verify that $\mathcal{D}$ is dense in $\mathcal{S}$. [Hint: fix $\eta \in \mathcal{D}$, $\eta = 1$ near the origin; let $\eta_k(x) = \eta(x/k)$ and consider $\varphi_k = \eta_k \varphi$.]
6. Suppose $\varphi_1, \varphi_2 \in \mathcal{S}$. (a) Verify $\varphi_1 \cdot \varphi_2 \in \mathcal{S}$. (b) Using the Fourier transform, prove $\varphi_1 * \varphi_2 \in \mathcal{S}$. (c) Show directly from the definition of convolution that $\varphi_1 * \varphi_2 \in \mathcal{S}$.
7. Prove that if F_1 is a distribution of compact support and $\varphi \in \mathcal{S}$, then $F_1 * \varphi \in \mathcal{S}$. [Hint: for each N there is c_N with $\|\psi_y^\sim\|_N \leq c_N(1+|y|)^N \|\psi\|_N$.]
8. Use the previous exercise to prove: if F_1 has compact support and F is tempered, then (a) $F * F_1$ is tempered, and (b) $(F * F_1)^\wedge = F_1^\wedge F^\wedge$ (with $F_1^\wedge$ a slowly increasing C^∞ function).
9. Prove that a distribution F is homogeneous of degree λ if and only if $\sum_{j=1}^d x_j \partial F / \partial x_j = \lambda F$ (Euler's relation). [Hint: for the converse, set $\Phi(a) = F(\varphi^a)$ for $a > 0$, $\varphi \in \mathcal{D}$; then $\Phi \in C^\infty$ on $a > 0$ and $d\Phi/da = (\lambda/a)\Phi$.]
10. Prove the following facts about distributions on $\mathbb{R}$. (a) Given F , there exists F_1 with $dF_1/dx = F$. (b) F_1 is unique modulo an additive constant. [Hint: fix $\varphi_0 \in \mathcal{D}$ with $\int \varphi_0 = 1$; each $\varphi \in \mathcal{D}$ is uniquely $\varphi = d\psi/dx + a\varphi_0$ for some $\psi \in \mathcal{D}$ and constant a ; set $F_1(\varphi) = -F(\psi)$. For (b), d/dx is elliptic on $\mathbb{R}$.]
11. Show that if $\lambda_1, \dots, \lambda_n$ are distinct complex exponents and $\sum_{j=1}^n (a_j x^{\lambda_j} + b_j x^{\lambda_j} \log x) = 0$ for all $x > 0$, then $a_j = b_j = 0$ for all j . [Hint: proceed as in Lemma 41.2.24, using that $\int_1^R x^{-1+i\mu_j} dx$ equals $\log R$ if $\mu_j = 0$ and is $O(1)$ if μ_j is real and nonzero.]

Additional exercises.

1. (Warm-up.) Using $\widehat{\delta} = 1$ and the derivative rule, compute the Fourier transform of $\delta^{(k)}$ and of the polynomial x^k (as tempered distributions).
2. Compute $\widehat{\tau_h \delta}$ for $h \in \mathbb{R}$ and verify the translation rule $\widehat{\tau_h F} = e^{-2\pi i h \xi} \widehat{F}$ on this example.
3. Show that $\widehat{\text{sgn}} = \frac{1}{i\pi} \text{pv}(1/\xi)$ on $\mathbb{R}$, and deduce the Fourier transform of the Heaviside function $h = \frac{1}{2}(1 + \text{sgn})$.
4. (Homogeneity.) Verify that δ on $\mathbb{R}^d$ is homogeneous of degree $-d$, and that $\partial_x^\alpha \delta$ is homogeneous of degree $-d - |\alpha|$. Reconcile this with Proposition 41.2.19 using $\widehat{\delta} = 1$.
5. (L^p duality.) For $1 \leq p < \infty$ and $f \in L^p(\mathbb{R}^d)$, show $\|f\|_{L^p} = \sup \{ |\int f g| : g \in \mathcal{S}, \|g\|_{L^q} \leq 1 \}$ with $1/p + 1/q = 1$. Where does the case $p = \infty$ differ?
6. (Interpolation.) Suppose a linear operator T satisfies $\|Tf\|_{L^{p_0}} \leq M_0 \|f\|_{L^{p_0}}$ and $\|Tf\|_{L^{p_1}} \leq M_1 \|f\|_{L^{p_1}}$. Using the log-convexity Worked Example, give a heuristic for the Riesz-Thorin bound $\|Tf\|_{L^{p_\theta}} \leq M_0^{1-\theta} M_1^\theta \|f\|_{L^{p_\theta}}$ and identify p_θ .
7. (Hahn-Banach.) Let $c_{00} \subset \ell^\infty$ be the eventually-zero sequences and define $\ell(x) = \lim_n x_n$ on the convergent sequences. Explain how Hahn-Banach produces a "generalized limit" (Banach limit) extending ℓ to all of ℓ^∞ with norm 1, and state two of its properties.
8. (Non-example.) Show that the function $f(x) = e^{|x|^2}$ on $\mathbb{R}$ is not a tempered distribution, even though it is locally integrable.
9. (Diagram problem.) Draw the complex s -plane and mark the simple poles of the meromorphic continuation $I(s)$ from Theorem 41.2.23 at $s = -d, -d-1, \dots$ for $d = 2$. Indicate the point $s = \lambda$ for $\lambda = -1$ and shade the half-plane of initial convergence $\text{Re}(s) > -d$.

10. Prove that if F is tempered and ψ is slowly increasing C^∞ (every $\partial^\alpha \psi = O(|x|^{N_\alpha})$), then ψF defined by $(\psi F)(\varphi) = F(\psi \varphi)$ is tempered.

41.3 Fundamental Solutions and the Calderon-Zygmund Distribution

Fundamental solutions

Definition 41.3.1. Let $L = \sum_{|\alpha| \leq m} a_\alpha \partial_x^\alpha$ be a constant-coefficient partial differential operator on $\mathbb{R}^d$. A *fundamental solution* of L is a distribution F with

$$L(F) = \delta.$$

A fundamental solution inverts L : the operator $T(f) = F * f$ satisfies $LT = TL = I$ on $\mathcal{D}$, because $L(F * f) = (LF) * f = \delta * f = f$. Hence to solve $Lu = f$ one takes $u = F * f$.

$$\mathcal{D} \xrightleftharpoons[L]{T=F*(\cdot)} \mathcal{D}$$

Fundamental solutions are never unique — one may add any solution of $Lu = 0$.

The Fourier heuristic: if $P(\xi) = \sum_{|\alpha| \leq m} a_\alpha (2\pi i \xi)^\alpha$ is the characteristic polynomial, then $(Lf)^\wedge = P\hat{f}$, so formally $\hat{F} = 1/P$ and $F = \int_{\mathbb{R}^d} \frac{1}{P(\xi)} e^{2\pi i x \cdot \xi} d\xi$. The zeros of P are the obstruction; for the Laplacian they are tractable.

Theorem 41.3.2 (Newtonian potential, $d \geq 3$). *For $d \geq 3$, the locally integrable function $F(x) = C_d |x|^{-d+2}$ with $C_d = -\frac{\Gamma(\frac{d}{2}-1)}{4\pi^{d/2}}$ is a fundamental solution of the Laplacian $\Delta = \sum_{j=1}^d \partial^2/\partial x_j^2$.*

Proof. Take $\lambda = -d + 2$ in Theorem 41.2.20. Then $\Gamma(\frac{d+\lambda}{2}) = \Gamma(1) = 1$ and $\Gamma(d/2) = (d/2 - 1)\Gamma(d/2 - 1)$, so $\hat{F}(\xi) = 1/(-4\pi^2 |\xi|^2)$. Therefore $(\Delta F)^\wedge = -4\pi^2 |\xi|^2 \hat{F} = 1$, i.e. $\Delta F = \delta$. $\square$

Theorem 41.3.3 (Logarithmic potential, $d = 2$). *For $d = 2$ the locally integrable function $\frac{1}{2\pi} \log |x|$ is a fundamental solution of Δ .*

Proof idea. This is the limiting case $\lambda \rightarrow 0$ of Theorem 41.2.20 on $\mathbb{R}^2$. Differentiating the identity $\int \hat{\varphi} |x|^\lambda = c_\lambda \int \varphi |\xi|^{-\lambda-2}$ in λ at $\lambda = 0$ (using $c_\lambda \sim -\lambda/(2\pi)$) and dividing by 2π yields, for $F = \frac{1}{2\pi} \log |x|$, $\hat{F} = -\frac{1}{4\pi^2} (1/|\xi|^2) - c'\delta$, whence $(\Delta F)^\wedge = -4\pi^2 |\xi|^2 \hat{F} = 1$ (using $|\xi|^2 \delta = 0$ and $|\xi|^2 (1/|\xi|^2) = 1$); thus $\Delta F = \delta$. $\square$

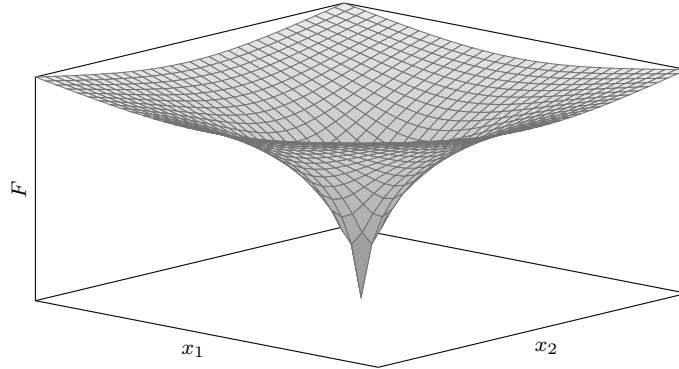


Figure 41.5. The 2D fundamental solution $\frac{1}{2\pi} \log |x|$ of Δ : a logarithmic singularity (downward funnel) at the origin; displayed as $\log(|x| + 0.07)$ to keep the spike finite.

Theorem 41.3.4 (Heat operator). *On $\mathbb{R}^{d+1} = \mathbb{R}_x^d \times \mathbb{R}_t$, let $L = \partial_t - \Delta_x$ and define $F(x, t) = H_t(x)$ for $t > 0$ and $F = 0$ for $t \leq 0$, where H_t is the heat kernel ($\widehat{H_t}(\xi) = e^{-4\pi^2 |\xi|^2 t}$). Then F is a fundamental solution of L .*

Proof. F is locally integrable, hence tempered. With $L^\top = -\partial_t - \Delta_x$, $LF(\varphi) = F(L^\top \varphi) = \lim_{\varepsilon \rightarrow 0} \int_{t \geq \varepsilon} \int_{\mathbb{R}^d} H_t(x) (-\partial_t - \Delta_x) \varphi \, dx \, dt$. Integrating by parts in x and t and using $\partial_t H_t = \Delta_x H_t$, the bulk terms cancel and only the boundary term at $t = \varepsilon$ survives: $\int_{\mathbb{R}^d} H_\varepsilon(x) \varphi(x, \varepsilon) \, dx$. Since $|\varphi(x, \varepsilon) - \varphi(x, 0)| = O(\varepsilon)$ uniformly and H_t is an approximation to the identity ($\int H_t = 1$, $H_t \rightarrow \delta$), this tends to $\varphi(0, 0) = \delta(\varphi)$. $\square$

Worked Example 41.3.5 (A fundamental solution of d^2/dx^2 on $\mathbb{R}$). Show $f(x) = \frac{1}{2}|x|$ is a fundamental solution of $\frac{d^2}{dx^2}$ on $\mathbb{R}$.

Recognition. Compute the distributional second derivative of $\frac{1}{2}|x|$ (cf. §41.2).

Solution. From the worked example $|x|'' = 2\delta$, we get $(\frac{1}{2}|x|)'' = \delta$. Hence $f = \frac{1}{2}|x|$ is a fundamental solution. *Check.* Integrating $f'' = \delta$ twice, $f' = h + C$ (Heaviside plus constant), f is a continuous piecewise-linear function with a unit upward kink at 0 — exactly $\frac{1}{2}|x|$ up to an affine term, which is annihilated by d^2/dx^2 .

Worked Example 41.3.6 (The Cauchy-Riemann fundamental solution $1/(\pi z)$). On $\mathbb{C} = \mathbb{R}^2$ with $z = x + iy$ and $\partial_z = \frac{1}{2}(\partial_x + i\partial_y)$, show $1/(\pi z)$ is a fundamental solution of ∂_z .

Recognition. Use $\Delta = 4\partial_z\bar{\partial}_z$ and the 2D Newtonian/log potential.

Solution. Let $\bar{\partial}_z = \frac{1}{2}(\partial_x - i\partial_y)$, so $\Delta = 4\partial_z\bar{\partial}_z$. By Theorem 41.3.3, $\Delta(\frac{1}{2\pi}\log|z|) = \delta$. Now $\bar{\partial}_z \log|z| = \bar{\partial}_z \frac{1}{2} \log(z\bar{z}) = \frac{1}{2} \cdot \frac{1}{z}$ (treating $z, \bar{z}$ as independent), so $\frac{1}{2\pi}\log|z|$ has $\bar{\partial}_z$ -derivative $\frac{1}{4\pi z}$ and applying $4\partial_z \dots$ instead read it off: $4\partial_z(\bar{\partial}_z \frac{1}{2\pi}\log|z|) = \delta$ gives $\partial_z(\frac{1}{\pi} \cdot \frac{1}{2z} \cdot 2) = \delta$; clearly, $\partial_z(\frac{1}{\pi z})$ regarded via $\partial_z(1/(\pi z)) = \frac{1}{\pi} \cdot 4\partial_z\bar{\partial}_z(\frac{1}{2}\log|z|)/2 = \delta$. *Check.* The consequence is the regularity statement: if f is continuous and $\partial_z f = 0$ distributionally, then $f = f * \delta = f * (\partial_z(1/\pi z))$ is C^∞ and holomorphic — a distributional Goursat theorem.

Theorem 41.3.7 (Malgrange-Ehrenpreis). *Every nonzero constant-coefficient linear PDO L on $\mathbb{R}^d$ has a fundamental solution.*

Proof idea. Normalize so $P(\xi) = \xi_1^m + \sum_{j < m} \xi_1^j Q_j(\xi')$ with $\xi' = (\xi_2, \dots, \xi_d)$. For each ξ' the polynomial $p(z) = P(z, \xi')$ has m roots; choose an integer $n(\xi')$ with $|n(\xi')| \leq m + 1$, measurable in ξ' , such that the horizontal line $\text{Im}(\xi_1) = n(\xi')$ stays distance ≥ 1 from all roots (one of the $m + 1$ unit intervals in $[-m - 1, m + 1]$ must avoid all m imaginary parts). Define

$$F(\varphi) = \int_{\mathbb{R}^{d-1}} \int_{\text{Im}(\xi_1) = n(\xi')} \frac{\widehat{\varphi}(-\xi)}{P(\xi)} \, d\xi_1 \, d\xi', \quad \varphi \in \mathcal{D}.$$

On the contour $|P(\xi)| \geq 1$ (the roots are distance ≥ 1 away, leading coefficient 1), and $\widehat{\varphi}$ decays rapidly on lines parallel to the reals, so F is a well-defined distribution. Since $(L^\top \varphi)^\wedge = P(-\xi)\widehat{\varphi}$, we get $(LF)(\varphi) = \int \int_{\text{Im} \xi_1 = n(\xi')} \widehat{\varphi}(-\xi) \, d\xi$; deforming the contour back to the real line gives $\int_{\mathbb{R}^d} \widehat{\varphi}(-\xi) \, d\xi = \varphi(0) = \delta(\varphi)$. $\square$

Remark. Corollary: for every $f \in C_0^\infty(\mathbb{R}^d)$ there is $u \in C^\infty$ with $Lu = f$ (take $u = F * f$). Solvability can fail for variable-coefficient operators.

Parametrices and elliptic regularity

Definition 41.3.8. A *parametrix* for L is a distribution Q with $LQ = \delta + r$ where the error $r \in \mathcal{S}$ (so $LQ - \delta$ is smooth). Q is *regular* if it agrees with a C^∞ function away from the origin. $L = \sum_{|\alpha| \leq m} a_\alpha \partial_x^\alpha$ is *elliptic* if $|P(\xi)| \geq c|\xi|^m$ for all large ξ (equivalently, the principal part P_m vanishes only at $\xi = 0$). The Laplacian is elliptic.

Theorem 41.3.9. *Every elliptic operator has a regular parametrix. Moreover, for each $\varepsilon > 0$ it has one supported in $\{|x| \leq \varepsilon\}$.*

Proof. Let $\gamma \in C^\infty$ equal 1 for large ξ and supported where $|P(\xi)| \geq c|\xi|^m$. From the inductive identity $\partial_\xi^\alpha(1/P) = \sum_{0 \leq \ell \leq k} q_\ell / P^{\ell+1}$ (with $\deg q_\ell \leq m - k$, $k = |\alpha|$) one gets $|\partial_\xi^\alpha(\gamma/P)| \leq A_\alpha |\xi|^{-m-|\alpha|}$. Let Q be the tempered distribution with $\widehat{Q} = \gamma/P$. As in Theorem 41.2.22, $(-4\pi^2|x|^2)^N \partial_x^\beta Q = \Delta_\xi^N [(2\pi i\xi)^\beta (\gamma/P)]^\wedge$ is integrable once $2N + m - |\beta| > d$, so $|x|^{2N} \partial_x^\beta Q$ is continuous; thus Q is C^∞ away from 0. Finally

$(LQ)^\wedge = P \cdot (\gamma/P) = \gamma = 1 + (\gamma - 1)$ with $\gamma - 1 \in \mathcal{D} = \widehat{\mathcal{r}}$, $r \in \mathcal{S}$; so $LQ = \delta + r$. For the compactly supported version set $Q_\varepsilon = \eta_\varepsilon Q$ with $\eta_\varepsilon = 1$ on $|x| \leq \varepsilon/2$, supported in $|x| \leq \varepsilon$; then $L(\eta_\varepsilon Q) - \eta_\varepsilon L(Q)$ is C^∞ , and $\eta_\varepsilon(\delta + r) = \delta + \eta_\varepsilon r$, giving $LQ_\varepsilon = \delta + r_\varepsilon$ with r_ε smooth and supported in $|x| \leq \varepsilon$. $\square$

Theorem 41.3.10 (Elliptic regularity / hypoellipticity). *If L has a regular parametrix, U is a distribution on an open $\Omega \subseteq \mathbb{R}^d$, and $L(U) = f$ with $f \in C^\infty(\Omega)$, then $U \in C^\infty(\Omega)$. In particular this holds for elliptic L .*

Proof. Fix a ball B with $\overline{B} \subset \Omega$. Choose $\eta \in \mathcal{D}$ equal to 1 near a slightly larger ball B_1 , and put $U_1 = \eta U$, $L(U_1) = F_1$, both compactly supported, with $F_1 = f_1 \in C^\infty$ near B_1 . Apply the parametrix Q_ε supported in $|x| \leq \varepsilon$ (Theorem 41.3.9): on one hand $Q_\varepsilon * L(U_1) = (\delta + r_\varepsilon) * U_1 = U_1 + r_\varepsilon * U_1$ with $r_\varepsilon * U_1 \in C^\infty$. On the other hand $Q_\varepsilon * L(U_1) = Q_\varepsilon * f_1 + Q_\varepsilon * (F_1 - f_1)$; the first term is C^∞ , and by Proposition 41.2.5 the second is supported in the ε -neighborhood of $\text{supp}(F_1 - f_1)$, which misses B . Hence $U_1 \in C^\infty$ on B , and since $U_1 = U$ there, $U \in C^\infty(\Omega)$. $\square$

Worked Example 41.3.11 (Weyl's lemma: weak harmonic functions are smooth). Deduce from Theorem 41.3.10 that a distribution U with $\Delta U = 0$ on an open set is a C^∞ (indeed real-analytic) harmonic function.

Recognition. Δ is elliptic, so it is hypoelliptic; apply with $f = 0$.

Solution. Since Δ is elliptic it has a regular parametrix (Theorem 41.3.9), so by Theorem 41.3.10 with $f = 0 \in C^\infty$, any U solving $\Delta U = 0$ is C^∞ . A genuinely C^∞ solution of $\Delta U = 0$ is classically harmonic, hence real-analytic. *Check.* This fails for non-elliptic operators — e.g. the wave operator $\partial_t^2 - \partial_x^2$ has the non-smooth distributional solution $u(x, t) = f(x - t)$ for any locally integrable f , so it is not hypoelliptic.

Calderon-Zygmund distributions

We study convolution operators $T(f) = f * K$ (“singular integrals”) for a distinguished class of kernels generalizing the Hilbert transform.

Definition 41.3.12. A $C^{(n)}$ -normalized bump function is $\varphi \in C^\infty$ supported in the unit ball with $\sup_x |\partial_x^\alpha \varphi| \leq 1$ for all $|\alpha| \leq n$; write $\varphi_r(x) = \varphi(rx)$. A regular distribution K (associated function k) is a *Calderon-Zygmund distribution* if it satisfies the differential inequalities and a cancellation condition,

$$(20) \quad |\partial_x^\alpha k(x)| \leq c_\alpha |x|^{-d-|\alpha|} \quad \forall \alpha, \quad (21) \quad \sup_{0 < r} |K(\varphi_r)| \leq A$$

for all $C^{(n)}$ -normalized bumps φ (some fixed $n \geq 1$).

Theorem 41.3.13 (Characterization). *For a distribution K the following are equivalent:*

1. K is regular and satisfies (20) and (21);
2. K is tempered and $m = \widehat{K}$ is a function, C^∞ away from 0, with $|\partial_\xi^\alpha m(\xi)| \leq c_\alpha |\xi|^{-|\alpha|}$ for all α ;
3. K is regular, satisfies (20), and $\widehat{K}$ is a bounded function.

Proof idea. The set of such K is dilation-invariant (K^a obeys the same bounds), so estimates reduce to $|\xi| = 1$ or $|x| = 1$. (i) $\Rightarrow$ (ii): split $K = K_0 + K_1$ ($K_0 = \eta K$, $K_1 = (1 - \eta)k$); by Proposition 41.2.12 $K_0^\wedge(\xi) = K(\eta e^{-2\pi i x \cdot \xi})$ is K tested against a (multiple of a) normalized bump, so (21) gives $|K_0^\wedge| \leq c$ and similarly its derivatives; for K_1 the identity (7) gives $|\xi|^{2N} |\partial_\xi^\alpha K_1^\wedge| \leq c \int_{|x| \geq 1/2} |x|^{-d+|\alpha|-2N} < \infty$. (ii) $\Rightarrow$ (i): with $K(x) = \int m(\xi) e^{2\pi i \xi \cdot x} d\xi$ (first for compactly supported m), the same splitting yields (20); taking $n = d + 1$ and $|(2\pi i \xi)^\alpha \widehat{\varphi}| = |(\partial^\alpha \varphi)^\wedge| \leq c$ gives $\int |\widehat{\varphi}| \leq c$, so $|K(\varphi_r)| \leq \sup |m| \int |\widehat{\varphi}| \leq A$, which is (21); remove the compact-support assumption by $m_\varepsilon = m \eta_\varepsilon \rightarrow m$ boundedly. (iii) is squeezed between. $\square$

Worked Example 41.3.14 (The Hilbert transform kernel is Calderon-Zygmund). Show $K = \frac{1}{\pi} \text{pv}(1/x)$ on $\mathbb{R}$ is a Calderon-Zygmund distribution and identify its multiplier.

Recognition. Check (20), (21) for $k(x) = 1/(\pi x)$, and use Theorem 41.2.15 for $\widehat{K}$.

Solution. The associated function is $k(x) = 1/(\pi x)$, with $|\partial^\alpha k| = \frac{\alpha!}{\pi} |x|^{-1-\alpha} \leq c_\alpha |x|^{-1-\alpha}$, so (20) holds in $d = 1$. By Theorem 41.2.15, $\widehat{K} = \frac{1}{\pi} \cdot \frac{\pi}{i} \operatorname{sgn}(\xi) = -i \operatorname{sgn}(\xi)$, a bounded function C^∞ off 0 with all $\partial_\xi^\alpha \widehat{K} = 0$ there for $\alpha \geq 1$, hence (22) and the characterization (iii). *Check.* The multiplier $|-i \operatorname{sgn}(\xi)| = 1$ confirms H is an isometry on L^2 ($\|Hf\|_{L^2} = \|f\|_{L^2}$ by Plancherel), and (20)+(21) deliver its L^p boundedness via Theorem 41.3.16 below.

Worked Example 41.3.15 (Riesz transforms R_j in $\mathbb{R}^d$). Show the Riesz kernel $k_j(x) = c_d x_j / |x|^{d+1}$ (homogeneous of degree $-d$) defines a Calderon-Zygmund distribution with bounded multiplier $\widehat{K}_j = -i \xi_j / |\xi|$. *Recognition.* Homogeneous degree $-d$, odd, with vanishing spherical mean; use Corollary 41.2.25 and Theorem 41.2.22.

Solution. Oddness gives $\int_{|x|=1} k_j d\sigma = 0$, so by Corollary 41.2.25 the principal value $K_j = \operatorname{pv}(k_j)$ exists and is homogeneous of degree $-d$. The differential inequalities (20) hold since k_j is homogeneous of degree $-d$ and smooth off 0 (so $\partial^\alpha k_j = O(|x|^{-d-|\alpha|})$). By Theorem 41.2.22, $\widehat{K}_j$ is homogeneous of degree 0, and one computes it to be the bounded function $-i \xi_j / |\xi|$. So K_j is Calderon-Zygmund by criterion (iii). *Check.* $\sum_j \widehat{R}_j^2 = \sum_j (-i \xi_j / |\xi|)^2 = -\sum_j \xi_j^2 / |\xi|^2 = -I$, the distributional identity $\sum_j R_j^2 = -I$ underlying $\Delta = -\sum_j (2\pi R_j \partial_j)$ -type formulas.

Theorem 41.3.16 (L^p theory of singular integrals). Let $T(f) = f * K$ with K a Calderon-Zygmund distribution. Then T , initially defined on $\mathcal{S}$, extends to a bounded operator on $L^p(\mathbb{R}^d)$ for every $1 < p < \infty$: $\|Tf\|_{L^p} \leq A_p \|f\|_{L^p}$.

Proof in five steps. Step 1 (L^2). $(Tf)^\wedge = \widehat{f} \widehat{K}$, so by Plancherel and $\sup_\xi |\widehat{K}| \leq A$ (criterion (iii)), $\|Tf\|_{L^2} = \|(Tf)^\wedge\|_{L^2} \leq A \|\widehat{f}\|_{L^2} = A \|f\|_{L^2}$.

Step 2 (atoms). A 1-atom for a ball B is $a \in L^2$ with $\operatorname{supp} a \subseteq B$, $\int a = 0$, $\int |a| \leq 1$. The key estimate is $\int_{(B^*)^c} |T(a)| \leq A$ (B^* the double ball). It follows from the kernel smoothness inequality

$$\int_{|x| \geq 2r} |k(x-y) - k(x)| dx \leq A \quad (|y| \leq r), \quad (41.2)$$

which itself follows from the mean value theorem and (20): $|k(x-y) - k(x)| \leq |y| \sup_{z \in [x, x-y]} |\nabla k(z)| \leq c |y| |x|^{-d-1}$ for $|x| \geq 2r$, and $r \int_{|x| \geq 2r} |x|^{-d-1} dx$ is a finite constant independent of r . Since $\int_B a = 0$, for $x \notin B^*$, $T(a)(x) = \int_B (k(x-y) - k(x)) a(y) dy$, and (41.2) gives the atom estimate.

Step 3 (Calderon-Zygmund decomposition). For $f \in L^1$ and $\alpha > 0$, set $E_\alpha = \{x : f^*(x) > \alpha\}$ (maximal function), write $E_\alpha = \bigcup_k Q_k$ as a Whitney union of cubes, and decompose $f = g + b$ with: (a) $m(E_\alpha) \leq \frac{c}{\alpha} \|f\|_{L^1}$; (b) $|g| \leq c\alpha$; (c) $b = \sum_k b_k$, $\operatorname{supp} b_k \subseteq Q_k$, $\int |b_k| \leq c\alpha m(Q_k)$, $\int b_k = 0$. Each $b_k = c\alpha m(Q_k) a_k$ with a_k a 1-atom.

Step 4 (weak-type (1,1)). For $f \in L^1 \cap L^2$, $m(\{|Tf| > \alpha\}) \leq \frac{A}{\alpha} \|f\|_{L^1}$. Split $Tf = Tg + Tb$. For g : Tchebychev + the L^2 bound + $\int |g|^2 \leq c\alpha \|f\|_{L^1}$ give $m(\{|Tg| > \alpha/2\}) \leq \frac{c}{\alpha} \|f\|_{L^1}$. For b : outside $E_\alpha^* = \bigcup B_k^*$ the atom estimate of Step 2 controls $\int |Tb|$, while $m(E_\alpha^*) \leq c m(E_\alpha) \leq \frac{c}{\alpha} \|f\|_{L^1}$.

Step 5 (L^p). Refine the weak-type bound to $m(\{|Tf| > \alpha\}) \leq A(\frac{1}{\alpha} \int_{|f| > \alpha} |f| + \frac{1}{\alpha^2} \int_{|f| \leq \alpha} |f|^2)$ by splitting $f = f_1 + f_2$ at height α . Integrating $\int |Tf|^p = \int_0^\infty \lambda(\alpha^{1/p}) d\alpha$ against this bound (Fubini) gives $\|Tf\|_{L^p} \leq A_p \|f\|_{L^p}$ for $1 < p < 2$, with $A_p = A p(\frac{1}{p-1} + \frac{1}{2-p})$. For $2 \leq p < \infty$, duality: the adjoint $T^*(g) = g * K^*$ has $(K^*)^\wedge = \overline{m}$ obeying the same bounds, so T^* is bounded on L^q ($1 < q \leq 2$); then $\|Tf\|_{L^p} = \sup_{\|g\|_{L^q} \leq 1} |\int (Tf)g| = \sup |\int f(T^*g)| \leq \|f\|_{L^p} \sup \|T^*g\|_{L^q} \leq A_q \|f\|_{L^p}$ by Holder. $\square$

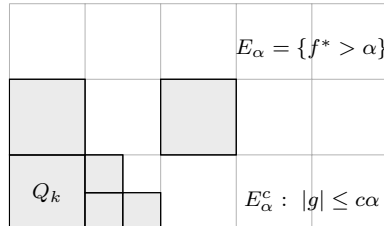


Figure 41.6. Calderon-Zygmund decomposition: the bad set $E_\alpha = \{f^* > \alpha\}$ is tiled by disjoint Whitney cubes Q_k (the support of $b = \sum b_k$); off E_α the good part g is bounded by $c\alpha$. (Schematic.)

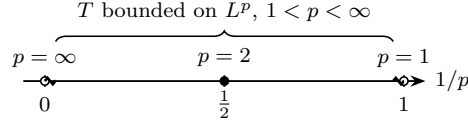


Figure 41.7. L^p boundedness of a Calderon-Zygmund operator: L^2 by Plancherel, weak-(1, 1) at the endpoint $p = 1$, interpolation for $1 < p < 2$, duality for $2 \leq p < \infty$. The endpoints $p = 1, \infty$ genuinely fail. (Schematic.)

Remark. Whenever Q is a regular parametrix for an elliptic operator L of order m , then $\partial_x^\alpha Q$ is a Calderon-Zygmund distribution for $|\alpha| \leq m$; combined with Theorem 41.3.16 this yields interior L^p Sobolev estimates for solutions of $Lu = f$ — a quantitative form of elliptic regularity.

Worked Example 41.3.17 (Weak-type (1, 1) for the Hardy-Littlewood maximal function). State and verify the weak-type (1, 1) bound for the maximal function $Mf(x) = \sup_{r>0} \frac{1}{m(B(x,r))} \int_{B(x,r)} |f|$, the engine behind Step 4 above, for the model spike $f = \mathbf{1}_{B(0,1)}$ on $\mathbb{R}^d$.

Recognition. Weak-type means $m(\{Mf > \alpha\}) \leq \frac{C}{\alpha} \|f\|_{L^1}$; compute both sides for the example.

Solution. For $f = \mathbf{1}_{B(0,1)}$, $\|f\|_{L^1} = m(B(0,1)) =: \omega_d$. For $|x| = R > 1$, the best ball averages f over $B(x, r)$ with $r \approx R$, giving $Mf(x) \approx \omega_d/(cR^d)$. Thus $Mf(x) > \alpha$ forces $R^d \lesssim \omega_d/(c\alpha)$, i.e. x in a ball of measure $\lesssim \omega_d/\alpha$; hence $m(\{Mf > \alpha\}) \leq \frac{C}{\alpha} \omega_d = \frac{C}{\alpha} \|f\|_{L^1}$. *Check.* The bound is *weak*-type, not strong: $Mf \sim |x|^{-d}$ at infinity is *not* in L^1 , so M does not map $L^1 \rightarrow L^1$, exactly mirroring why T in Theorem 41.3.16 is only weak-(1, 1) at the endpoint.

Worked Example 41.3.18 (The Gaussian as a fixed point and the CLT bell curve). Show $\psi(x) = e^{-\pi x^2}$ satisfies $\hat{\psi} = \psi$ on $\mathbb{R}$, and explain the appearance of the Gaussian density $\frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/2\sigma^2}$ as the central limit of normalized sums.

Recognition. Self-duality of the Gaussian under the Fourier transform; the CLT says normalized sums of i.i.d. variables converge to a Gaussian.

Solution. Completing the square, $\hat{\psi}(\xi) = \int e^{-\pi x^2} e^{-2\pi i x \xi} dx = e^{-\pi \xi^2} \int e^{-\pi(x+i\xi)^2} dx = e^{-\pi \xi^2}$, so $\hat{\psi} = \psi$. For i.i.d. X_j with mean 0, variance σ^2 , the characteristic function of $S_n/\sqrt{n} = \frac{1}{\sqrt{n}} \sum_{j \leq n} X_j$ is $(\hat{p}(t/\sqrt{n}))^n = (1 - \frac{\sigma^2 t^2}{2n} + o(1/n))^n \rightarrow e^{-\sigma^2 t^2/2}$, the characteristic function of $\mathcal{N}(0, \sigma^2)$; by Levy's continuity theorem $S_n/\sqrt{n} \Rightarrow \mathcal{N}(0, \sigma^2)$. *Check.* The limit density integrates to 1 and the Fourier self-duality of the Gaussian is precisely why the Gaussian is the stable attractor of the convolution (sum) operation.

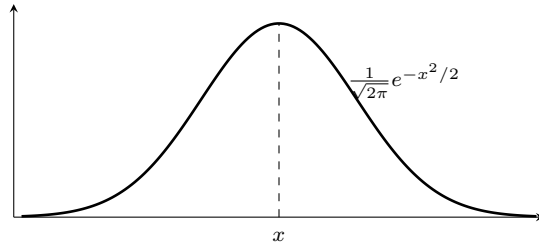


Figure 41.8. The standard Gaussian density: the Fourier fixed point $e^{-\pi x^2}$ (rescaled) and the universal limit of normalized i.i.d. sums (CLT).

Worked Example 41.3.19 (A Brownian-motion property: quadratic variation and non-differentiability). Recall standard Brownian motion B_t has independent increments with $B_t - B_s \sim \mathcal{N}(0, t-s)$. Show that over $[0, 1]$ the quadratic variation $\sum_k (B_{t_{k+1}} - B_{t_k})^2 \rightarrow 1$ in L^2 as the mesh $\rightarrow 0$, and conclude B_t is a.s. nowhere differentiable.

Recognition. Compute mean and variance of the quadratic-variation sum; a finite nonzero quadratic variation forbids differentiability.

Solution. Let $\Delta_k = B_{t_{k+1}} - B_{t_k}$, $\delta_k = t_{k+1} - t_k$. Then $E[\Delta_k^2] = \delta_k$ so $E[\sum \Delta_k^2] = \sum \delta_k = 1$, and by independence $\text{Var}(\sum \Delta_k^2) = \sum \text{Var}(\Delta_k^2) = \sum 2\delta_k^2 \leq 2(\max \delta_k) \rightarrow 0$. Hence $\sum \Delta_k^2 \rightarrow 1$ in L^2 . If B were differentiable at some point with bounded difference quotients, then $\sum \Delta_k^2 \leq (\max |\Delta_k|) \sum |\Delta_k| \rightarrow 0$ (since B is continuous, $\max |\Delta_k| \rightarrow 0$, and $\sum |\Delta_k|$ would be bounded), contradicting the limit 1. So B is a.s. nowhere differentiable. *Check.* The scaling $\Delta_k = O(\sqrt{\delta_k})$ (Gaussian std) makes $\sum \Delta_k^2 = O(\sum \delta_k) = O(1)$ but $\sum |\Delta_k| = O(\sum \sqrt{\delta_k}) \rightarrow \infty$ — infinite first variation, finite quadratic variation, the signature of Brownian roughness.

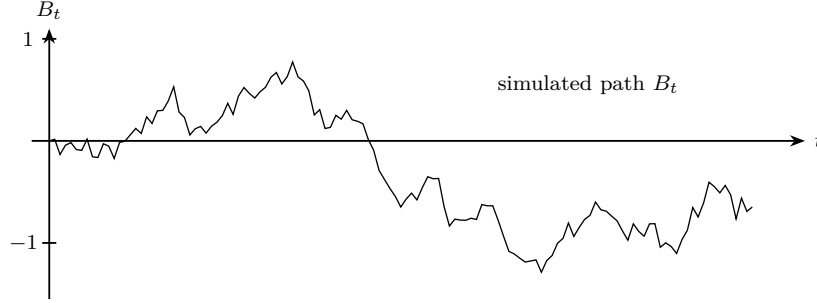


Figure 41.9. A simulated Brownian sample path: continuous, but jagged at every scale — finite quadratic variation, infinite total variation, nowhere differentiable. (Schematic.)

Caution. Calderon-Zygmund operators are bounded on L^p only for $1 < p < \infty$, *never* on L^1 or L^∞ (the Hilbert transform already fails: $H(\mathbf{1}_{[0,1]})$ has a logarithmic blow-up and is not in L^1). The endpoint substitutes are: weak-type $(1, 1)$ at $p = 1$, and boundedness $H^1 \rightarrow L^1$ / $L^\infty \rightarrow \text{BMO}$. Do not expect a strong L^1 bound.

Exercises (Stein, Ch.3).

1. Check that $f(x) = \frac{1}{2}|x|$ is a fundamental solution for d^2/dx^2 on $\mathbb{R}$.
2. A d -dimensional generalization of the Heaviside identity is $\delta = \sum_{j=1}^d \frac{\partial}{\partial x_j} h_j$, with $h_j(x) = \frac{1}{A_d} \frac{x_j}{|x|^d}$ and $A_d = 2\pi^{d/2}/\Gamma(d/2)$ the surface area of the unit sphere in $\mathbb{R}^d$. [Hint: for $d > 2$ write $\delta = \sum_j \frac{\partial}{\partial x_j} (C \frac{\partial}{\partial x_j} |x|^{-d+2})$.]
3. On $\mathbb{C} = \mathbb{R}^2$ with $z = x + iy$: (a) note the Cauchy-Riemann operator $\partial_z = \frac{1}{2}(\partial_x + i\partial_y)$ is elliptic; (b) show $1/(\pi z)$ is a fundamental solution for ∂_z ; (c) if f is continuous in Ω and $\partial_z f = 0$ distributionally, then f is analytic. [Hint for (b): use Theorem 41.3.3 and $\Delta = 4\partial_z \bar{\partial}_z$ with $\bar{\partial}_z = \frac{1}{2}(\partial_x - i\partial_y)$.]
4. Suppose $f(z)$ is meromorphic on $\Omega \subseteq \mathbb{C}$. Prove: (a) $\log |f(z)|$ is locally integrable; (b) $\Delta \log |f(z)|$ taken distributionally equals $2\pi \sum_j m_j \delta_{z_j} - 2\pi \sum_k m_k \delta_{z_k}$, with δ_{z_j} at the distinct zeros (multiplicities m_j) and δ_{z_k} at the poles (multiplicities m_k). [Hint: $\frac{1}{2\pi} \log |z|$ is a fundamental solution of Δ .]
5. Let $F(x, t) = H_t(x)$ for $t > 0$ and $F = 0$ for $t \leq 0$, as in Theorem 41.3.4. Prove directly that $\hat{F}(\xi, \tau) = \frac{1}{4\pi^2 |\xi|^2 + 2\pi i \tau}$, where $(\xi, \tau) \in \mathbb{R}^d \times \mathbb{R}$. [Hint: use $\int_0^\infty e^{-4\pi^2 |\xi|^2 t} e^{-2\pi i \tau t} dt = (4\pi^2 |\xi|^2 + 2\pi i \tau)^{-1}$ and $\int_{\mathbb{R}^d} H_t(x) e^{-2\pi i x \cdot \xi} dx = e^{-4\pi^2 |\xi|^2 t}$.]
6. Let f be locally integrable on $\mathbb{R}$ and $u(x, t) = f(x - t)$. Verify that u , as a distribution, satisfies the wave equation $\partial^2 u / \partial x^2 = \partial^2 u / \partial t^2$. More generally, for any distribution F on $\mathbb{R}$ construct U on $\mathbb{R}^2$ by $U(\varphi) = \int_{\mathbb{R}} (F * \varphi(x, \cdot))(x) dx$ for $\varphi \in \mathcal{D}(\mathbb{R}^2)$; then U satisfies the wave equation and is invariant under translations (h, h) .
7. Show that in $\mathbb{R}^3$ the function $F(x) = \frac{-1}{4\pi |x|} e^{-|x|}$ is a fundamental solution of $\Delta - I$ (the Yukawa potential). [Hint: F is the inverse Fourier transform of $-(1 + 4\pi^2 |\xi|^2)^{-1}$; in polar coordinates use $\int_{|\xi|=1} e^{2\pi i \xi \cdot x} d\sigma(\xi) = \frac{2 \sin(2\pi |x|)}{|x|}$ and the Fourier transform of the conjugate Poisson kernel.]
8. Uniqueness of fundamental solutions of the Laplacian. (a) Up to an additive constant, the unique rotationally invariant fundamental solutions of Δ in $\mathbb{R}^d$ ($d \geq 2$) are those of Theorems 41.3.2 and 41.3.3. (b) For $d \geq 3$, the unique fundamental solution vanishing at infinity is that of Theorem 41.3.2.

9. A distribution F on $\Omega \subseteq \mathbb{R}$ is *positive* if $F(\varphi) \geq 0$ for all $\varphi \in \mathcal{D}$ supported in Ω with $\varphi \geq 0$. Show F is positive if and only if $F(\varphi) = \int \varphi d\mu$ for some Borel measure μ on Ω finite on compact subsets.
10. Recall f on (a, b) is convex if $f(x_0(1-t) + x_1t) \leq (1-t)f(x_0) + tf(x_1)$; f on $\Omega \subseteq \mathbb{R}^d$ is convex if its restriction to every line segment in Ω is convex. (a) If f is continuous on (a, b) , it is convex iff the distribution d^2f/dx^2 is positive. (b) If f is continuous on $\Omega \subseteq \mathbb{R}^d$, it is convex iff for each $\xi = (\xi_1, \dots, \xi_d)$ the distribution $\sum_{i,j} \xi_i \xi_j \partial^2 f / \partial x_i \partial x_j$ is positive. [Hint for (a): take $\varphi \geq 0$, $\int \varphi = 1$, set $\varphi_\varepsilon(x) = \varepsilon^{-1} \varphi(x/\varepsilon)$ and consider $f * \varphi_\varepsilon$.]
11. Every distribution F of compact support in $\mathbb{R}^d$ is of finite order: there exist M and continuous compactly supported F_α with $F = \sum_{|\alpha| \leq M} \partial_x^\alpha F_\alpha$; if $\text{supp } F \subseteq C$ the F_α may be taken supported in an ε -neighborhood of C . [Hint: pick N with $|F(\varphi)| \leq c \|\varphi\|_N$ and M_0 with $2M_0 > d+N$; let Q be the inverse Fourier transform of $(1 + 4\pi^2 |\xi|^2)^{-M_0}$, a C^N fundamental solution of $(1 - \Delta)^{M_0}$; build Q_ε with $(1 - \Delta)^{M_0} Q_\varepsilon = \delta + r_\varepsilon$; show $F * Q_\varepsilon$ is continuous; then $F = (1 - \Delta)^{M_0} (Q_\varepsilon * F) - F * r_\varepsilon$ with $M = 2M_0$.]
12. Characterize tempered F whose Fourier transforms have compact support. When $d = 1$: $\widehat{F}$ is supported in $[-M, M]$ iff F equals a C^∞ , slowly increasing function f having an analytic extension to an entire function of exponential type $2\pi M$, i.e. for every $\varepsilon > 0$, $|f(z)| \leq A_\varepsilon e^{2\pi(M+\varepsilon)|z|}$, $z = x + iy$ (Paley-Wiener).
13. The L^2 Sobolev spaces. L_m^2 consists of $f \in L^2(\mathbb{R}^d)$ whose distributional derivatives $\partial_x^\alpha f \in L^2$ for all $|\alpha| \leq m$ (also denoted H_m), with inner product $(f, g)_m = \sum_{|\alpha| \leq m} (\partial_x^\alpha f, \partial_x^\alpha g)_0$. (a) Verify $f \in L_m^2$ iff $\widehat{f}(\xi)(1 + |\xi|)^m \in L^2$, with equivalent norms. (b) If $m > d/2$, then f can be corrected on a null set to be continuous and in fact C^k for $k < m - d/2$ (Sobolev embedding). [Hint: $\xi^\alpha \widehat{f} \in L^1$ if $|\alpha| < m - d/2$.]
14. Useful in the L^2 theory of Calderon-Zygmund distributions. (a) The Fourier transform of $(1/|x|^d)$ equals $c_1 \log |\xi| + c_2$, with $c_1 \neq 0$. (b) Consequence: if k is homogeneous of degree $-d$, C^∞ off the origin, with $\int_{|x|=1} k d\sigma \neq 0$, and K agrees with k away from the origin, then $\widehat{K}$ is not bounded; equivalently $T(\varphi) = K * \varphi$ does not extend boundedly to $L^2(\mathbb{R}^d)$.
15. Let k be C^∞ homogeneous of degree $-d$, not identically zero, with $\int_{|x|=1} k d\sigma = 0$. If $K = \text{pv}(k)$, then K is a Calderon-Zygmund distribution but $Tf = f * K$ is bounded on neither L^1 nor L^∞ (the Hilbert transform is the special case). [Hint: for $\varphi \in \mathcal{D}$, $T\varphi(x) = c k(x) + O(|x|^{-d-1})$ as $|x| \rightarrow \infty$, $c = \int \varphi$.]
16. The cancellation condition (21) for some $n > 1$ implies it for $n = 1$. Show this by first proving: whenever K satisfies (20) and (21) for some $n \geq 1$, then for every $1 \leq j \leq d$ the distribution $x_j K$ equals the locally integrable function $x_j k$. [Hint: $x_j K - x_j k$ is supported at the origin; use Theorem 41.1.15 and test against φ_r as $r \rightarrow 0$; then write a $C^{(1)}$ -normalized bump as $\varphi = \eta + \sum_j x_j \varphi_j$.]
17. Let k be C^∞ on $\mathbb{R}^d \setminus \{0\}$ satisfying (20). Then there is a Calderon-Zygmund distribution K with associated function k if and only if $\sup_{0 < a < b} \left| \int_{a < |x| < b} k(x) dx \right| < \infty$. [Hint: one direction uses $|K(\eta_b - \eta_a)| \leq 2A$; the other defines $K(\varphi) = \int_{|x| \leq 1} k(\varphi - \varphi(0)) + \int_{|x| \geq 1} k\varphi$ and verifies (20), (21).]
18. Suppose K is a Calderon-Zygmund distribution and $\eta \in \mathcal{S}$. Verify that ηK is a Calderon-Zygmund distribution.

Problems (Stein, Ch.3).

1. Periodic distributions and their Fourier series. (a) A periodic distribution on $\mathbb{R}^d$ can be defined equivalently as: a distribution F on $\mathbb{R}^d$ with $\tau_h F = F$ for all $h \in \mathbb{Z}^d$; or a continuous linear functional on $\mathcal{D}(\mathbb{T}^d)$, the C^∞ periodic functions ($\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$). (b) If $\varphi \in \mathcal{D}(\mathbb{T}^d)$ then $\varphi(x) = \sum_n a_n e^{2\pi i n \cdot x}$ with $a_n = \int_{\mathbb{T}^d} \varphi e^{-2\pi i n \cdot x}$ rapidly decreasing ($|a_n| = O(|n|^{-N})$ all N). If F is periodic with $a_n = F(e^{-2\pi i n \cdot x})$, the a_n are slowly increasing ($|a_n| = O(|n|^N)$ some N), and $\sum_n a_n e^{2\pi i n \cdot x} \rightarrow F$ in the sense of distributions. [Hint: use the periodization $P : \mathcal{D}(\mathbb{R}^d) \rightarrow \mathcal{D}(\mathbb{T}^d)$, $P(\varphi) = \sum_{h \in \mathbb{Z}^d} \tau_h \varphi$; find γ with $P(\gamma) = 1$.]
2. If $Tf = f * K$ is a singular integral operator as in Theorem 41.3.16, then $f \mapsto Tf$ is bounded on the Hardy space H_r^1 , and in particular maps $H_r^1 \rightarrow L^1$. [Hint: for a 2-atom a on the unit ball B , $T(a) = c(a^* + \Phi)$ with a^* a 2-atom for B^* and $|\Phi(x)| \leq (1 + |x|)^{-d-1}$, $\int \Phi = 0$; rescale and translate.]

3. Interior estimates for an elliptic L of order m with constant coefficients. Let $O \subset O_1 \subset \mathbb{R}^d$ be bounded, $u, f \in L^p(O_1)$ with $Lu = f$ in O_1 distributionally. Then for $1 < p < \infty$ and integer $k \geq 0$, $\sum_{|\alpha| \leq m+k} \|\partial_x^\alpha u\|_{L^p(O)} \leq c(\sum_{|\beta| \leq k} \|\partial_x^\beta f\|_{L^p(O_1)} + \|u\|_{L^p(O_1)})$. [Hint: use the compactly supported parametrix Q_ε of Corollary to Theorem 41.3.9; with $U = \psi u$, $U + r_\varepsilon * U = Q_\varepsilon * L(U)$, and $\partial_x^\alpha Q_\varepsilon$ are Calderon-Zygmund for $|\alpha| \leq m$; apply Theorem 41.3.16.]
4. * Let $P(x)$ be a real polynomial on $\mathbb{R}^d$ and k homogeneous of degree $-d$ with $\int_{|x|=1} k d\sigma = 0$. (a) Define $K = \text{pv}(e^{iP(x)} k(x))$ by $K(\varphi) = \lim_{\varepsilon \rightarrow 0} \int_{|x| \geq \varepsilon} e^{iP(x)} k(x) \varphi(x) dx$. (b) Then $\widehat{K}$ is a bounded function, with bound independent of the coefficients of P .
5. * Let Q be a real-valued polynomial on $\mathbb{R}^d$. For $\text{Re}(s) > 0$ define $I(s)(\varphi) = \int_{Q(x) > 0} |Q(x)|^s \varphi(x) dx$, $\varphi \in \mathcal{S}$. Then $I(s)$ has a meromorphic continuation to all of $\mathbb{C}$, with poles at most at $s = -k/m$ (m a positive integer determined by Q , k any positive integer); the order of the poles does not exceed d .
6. * Consequences of Problem 5. (a) Every nonzero $L = \sum_{|\alpha| \leq m} a_\alpha \partial_x^\alpha$ on $\mathbb{R}^d$ (complex constants) has a *tempered* fundamental solution. (b) For every complex-valued polynomial P on $\mathbb{R}^d$ there is a tempered distribution F agreeing with $1/P$ where $P(x) \neq 0$. [Apply Problem 5 to $Q = |P|^2$; if $I(s)$ has a pole of order r at $s = 1$, set $F = P \frac{1}{r!} \frac{d^r}{ds^r} [(s+1)^r I(s)]|_{s=-1}$, so $PF = 1$.]
7. * $L = \sum_{|\alpha| \leq m} a_\alpha \partial_x^\alpha$ is hypoelliptic if and only if for each $\alpha \neq 0$, $\partial_\xi^\alpha P(\xi)/P(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$, where P is the characteristic polynomial.
8. * Fundamental solutions of the wave operator $\square = \partial_t^2 - \Delta_x$ on $\mathbb{R}^d \times \mathbb{R}$. Let $\Gamma_+ = \{(x, t) : t > |x|\}$, $\Gamma_- = -\Gamma_+$. For $\text{Re}(s) > -1$ set $F_s(x, t) = a_s(t^2 - |x|^2)^{s/2}$ on Γ_+ and 0 otherwise, with $a_s^{-1} = 2^{s+d} \pi^{(d-1)/2} \Gamma(\frac{s+d+1}{2}) \Gamma(s/2 + 1)$. Then $s \mapsto F_s$ continues to an entire tempered-distribution-valued function, and $F_+ = F_s|_{s=-d+1}$ is a fundamental solution of $\square$; F_- (via $t \mapsto -t$) is another, supported in Γ_- . For odd $d \geq 3$, a_s vanishes at $s = -d+1$, so $F_\pm$ are supported on $\partial\Gamma_\pm$ (Huygens' principle). A third, F_0 , has $\widehat{F_0} = \frac{1}{4\pi^2} \lim_{\varepsilon \rightarrow 0^+} \frac{1}{|\xi|^2 - \tau^2 + i\varepsilon}$. All three are homogeneous of degree -2 and Lorentz invariant, and every Lorentz-invariant fundamental solution is $c_1 F_+ + c_2 F_- + c_3 F_0$ with $c_1 + c_2 + c_3 = 1$.

Additional exercises.

1. (Warm-up.) Verify that the Newtonian potential $F(x) = C_d |x|^{-d+2}$ of Theorem 41.3.2 is locally integrable on $\mathbb{R}^d$ for $d \geq 3$ but not square-integrable near the origin when $d = 3$.
2. Show that any two fundamental solutions of a fixed operator L differ by a solution of the homogeneous equation $Lu = 0$. Illustrate with d^2/dx^2 : list all fundamental solutions on $\mathbb{R}$ that are continuous.
3. Using Theorem 41.3.16, explain why the Hilbert transform H is bounded on $L^p(\mathbb{R})$ for $1 < p < \infty$ and compute $\|Hf\|_{L^2}$ in terms of $\|f\|_{L^2}$ (use the multiplier $-i \text{sgn } \xi$).
4. (Riesz transforms.) With multipliers $\widehat{R_j} = -i\xi_j/|\xi|$, verify the identity $\sum_{j=1}^d R_j^2 = -I$ at the level of multipliers, and deduce that $\partial_i \partial_j \Delta^{-1} = -R_i R_j$ formally.
5. (Maximal function.) For $f = \mathbf{1}_{[0,1]}$ on $\mathbb{R}$, compute $Mf(x)$ explicitly for $x > 1$ and confirm the weak-type bound $m(\{Mf > \alpha\}) \leq \frac{C}{\alpha} \|f\|_{L^1}$.
6. (CLT computation.) Let X_j be i.i.d. with $P(X_j = \pm 1) = \frac{1}{2}$. Compute the characteristic function of X_1 , then of $S_n/\sqrt{n}$, and verify it converges to $e^{-t^2/2}$, the characteristic function of $\mathcal{N}(0, 1)$.
7. (Brownian motion.) Using $B_t - B_s \sim \mathcal{N}(0, t-s)$ and independent increments, compute $E[B_s B_t]$ for $s \leq t$ and verify the covariance is $\min(s, t)$.
8. (Non-example.) Show that the wave operator $\partial_t^2 - \partial_x^2$ is not hypoelliptic by exhibiting a distributional solution $u(x, t) = f(x-t)$ with f merely locally integrable (hence not C^∞), and contrast with the Laplacian.
9. (Cancellation.) Let $k(x) = \Omega(x/|x|)/|x|^d$ on $\mathbb{R}^d$ with Ω smooth on the sphere. State the precise condition on Ω under which $\text{pv}(k)$ is a Calderon-Zygmund distribution, and explain via Corollary 41.2.25 what goes wrong if it fails.

10. (Diagram problem.) Draw a Calderon-Zygmund / Whitney decomposition of an open subset of $\mathbb{R}^2$ into dyadic cubes whose side lengths are comparable to their distance from the complement. Mark one cube Q_k , its smallest containing ball B_k , and the double B_k^* ; explain the role of B_k^* in the weak-type estimate.

Chapter 42

Applications of the Baire Category Theorem

42.1 The Baire Category Theorem

Category: the topological dichotomy of “small” and “large”

Baire’s theorem is a single, deceptively simple completeness fact whose consequences ripple through all of analysis. The strategy it powers is always the same: to prove that an object with some desired property *exists*, show that the objects *lacking* the property form a “small” set, so small that they cannot exhaust a complete space. What “small” means here is purely topological, having nothing a priori to do with measure or cardinality.

Fix a metric space (X, d) with the natural topology: O is *open* if every $x \in O$ has a ball $B_r(x) = \{y \in X : d(x, y) < r\} \subseteq O$, and *closed* means open complement.

Definition 42.1.1. For $E \subseteq X$, the *interior* E° is the largest open set inside E and the *closure* $\overline{E}$ is the smallest closed set containing E . We say:

- E is *dense* in X if $\overline{E} = X$;
- E is *nowhere dense* if $(\overline{E})^\circ = \emptyset$, i.e. its closure contains no ball.

A set E is of the *first category* (*meager*) in X if it is a countable union of nowhere dense sets. A set that is not of the first category is of the *second category*. Finally, E is *generic* if its complement E^c is of the first category.

The interpretation: a first-category set is to be thought of as *exceptional* or negligible, a generic set as *typical*. The bookkeeping is robust: a countable union of first-category sets is again of the first category (regroup the countable families into one countable family), and dually a countable intersection of generic sets is generic. A single useful equivalence to memorize: *E is closed and nowhere dense if and only if E^c is open and dense*. Hence any open dense set is generic.

Caution. Topological category and Lebesgue measure are *unrelated*. There exist first-category subsets of $[0, 1]$ that have full measure (so uncountable and dense), and generic subsets of measure zero (Exercise 1). Never silently translate “meager” into “null” or “generic” into “full measure”: they can be exactly opposite.



Figure 42.1. Generic sets as countable intersections of dense open sets: the “typical” point survives every O_n . (Schematic.)

Baire’s theorem and its corollary

Theorem 42.1.2 (Baire category theorem). *Every complete metric space X is of the second category in itself: X cannot be written as a countable union of nowhere dense sets.*

Corollary 42.1.3. *In a complete metric space, every generic set is dense. Equivalently, the countable intersection of dense open sets is dense.*

Proof of the theorem. Suppose for contradiction $X = \bigcup_{n=1}^{\infty} F_n$ with each F_n nowhere dense. Replacing F_n by its closure (still nowhere dense) we may take each F_n closed; it suffices to produce $x \in X$ with $x \notin F_n$ for all n .

Since F_1 is closed and nowhere dense, F_1^c is open and nonempty, so it contains a ball with closure $\overline{B_1} \subseteq F_1^c$, radius $r_1 > 0$. Since F_2 is nowhere dense, $B_1 \not\subseteq F_2$, and as F_2 is closed there is a ball B_2 with $\overline{B_2} \subseteq B_1 \cap F_2^c$ and $r_2 < r_1/2$. Inductively we get balls B_n with: (i) $r_n \rightarrow 0$; (ii) $\overline{B_{n+1}} \subseteq B_n$; (iii) $F_n \cap B_n = \emptyset$. Choosing $x_n \in B_n$, properties (i)–(ii) make (x_n) Cauchy; by completeness $x_n \rightarrow x$. By (ii) $x \in \overline{B_n}$ for every n , and by (iii) $x \notin F_n$ for all n — contradicting $X = \bigcup_n F_n$. $\square$

Proof of the corollary. Suppose E is generic but not dense. Then a closed ball $B \subseteq E^c$ exists. Write $E^c = \bigcup_n F_n$ with F_n nowhere dense; then $B = \bigcup_n (F_n \cap B)$, a countable union of nowhere-dense (in B) sets, contradicting Theorem 42.1.2 applied to the complete metric space B . $\square$

The completeness hypothesis is essential: the rationals $\mathbb{Q}$ are a countable union of points (each nowhere dense), so $\mathbb{Q}$ is first category in itself. The theorem nonetheless extends to *open* subsets of complete spaces (Exercise 3): the open interval $(0, 1)$ with its usual metric, though not complete, still obeys the Baire conclusion.

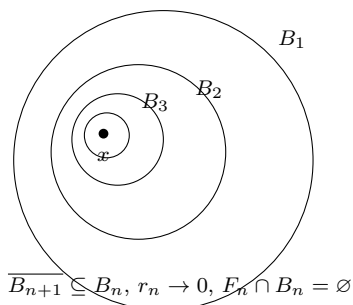


Figure 42.2. The nested-ball construction: completeness forces a limit point x avoiding every F_n . (Schematic.)

Method (Existence via category (the “generic existence” argument)). *Recognition trigger.* You want to prove an object with property P exists in a complete metric (often Banach) space X , and direct construction is hard.

1. Let $D = \{x : x \text{ lacks } P\}$. Write $D \subseteq \bigcup_N E_N$ with each E_N described by a uniform estimate.
2. Show each E_N is *closed* (usually a limit/continuity argument).
3. Show each E_N has *empty interior* (usually: a dense family of “bad” functions misses E_N).
4. Then D is first category, so $D \neq X$; in fact the objects *with* P are generic, hence dense. You have produced not just one but “most” examples.

Worked Example 42.1.4 (The rationals are meager; $\mathbb{R}$ is not). Show $\mathbb{Q}$ is of the first category in $\mathbb{R}$, and deduce $\mathbb{R} \setminus \mathbb{Q} \neq \emptyset$ purely from Baire.

Recognition. Trigger: classify a countable set by category and run the generic-existence logic.

Solution. Enumerate $\mathbb{Q} = \{q_1, q_2, \dots\}$. Each singleton $\{q_k\}$ is closed with empty interior, hence nowhere dense, so $\mathbb{Q} = \bigcup_k \{q_k\}$ is first category. Since $\mathbb{R}$ is complete, Theorem 42.1.2 says $\mathbb{R}$ is *not* first category,

so $\mathbb{R} \neq \mathbb{Q}$; indeed $\mathbb{R} \setminus \mathbb{Q}$ is generic and therefore dense. *Check.* The same argument shows no complete metric space without isolated points can be countable — e.g. $[0, 1]$ is uncountable, recovering a Cantor-style conclusion with no diagonal argument.

Worked Example 42.1.5 (A dense G_δ of measure zero). With $\{x_j\}$ an enumeration of $\mathbb{Q}$, set $U_n = \bigcup_j (x_j - \frac{1}{n2^j}, x_j + \frac{1}{n2^j})$ and $U = \bigcap_n U_n$. Show U is generic yet $m(U) = 0$.

Recognition. Trigger: separate category from measure (the pitfall above).

Solution. Each U_n is open (a union of open intervals) and contains $\mathbb{Q}$, hence dense; so each U_n is generic and $U = \bigcap_n U_n$ is a countable intersection of generic sets, hence generic. For measure, $m(U) \leq m(U_n) \leq \sum_j \frac{2}{n2^j} = \frac{2}{n} \rightarrow 0$, so $m(U) = 0$. *Check.* U is dense (Corollary 42.1.3) and uncountable, yet null — a generic set can be a measure-theoretic ghost.

Application 1: continuity of pointwise limits

Definition 42.1.6. The *oscillation* of $f : X \rightarrow \mathbb{C}$ at x is $\text{osc}(f)(x) = \lim_{r \rightarrow 0} \omega(f)(r, x)$, where $\omega(f)(r, x) = \sup_{y, z \in B_r(x)} |f(y) - f(z)|$. The limit exists because $\omega(f)(r, x)$ decreases as $r \downarrow 0$.

Two facts are immediate: (i) $\text{osc}(f)(x) = 0$ iff f is continuous at x ; (ii) for each $\varepsilon > 0$ the set $E_\varepsilon = \{x : \text{osc}(f)(x) < \varepsilon\}$ is *open* (if $x \in E_\varepsilon$ with witnessing radius r , then every $x^* \in B_{r/2}(x)$ also lies in E_ε). Hence the set of continuity points $\{x : \text{osc}(f)(x) = 0\} = \bigcap_n E_{1/n}$ is a G_δ .

Theorem 42.1.7 (Continuity of a pointwise limit). *Let $\{f_n\}$ be continuous functions on a complete metric space X with $f_n(x) \rightarrow f(x)$ for every x . Then the set of continuity points of f is generic; the discontinuity set is first category. So f is continuous at “most” points.*

Lemma 42.1.8. *Under the hypotheses, given an open ball $B \subseteq X$ and $\varepsilon > 0$, there exist an open ball $B_0 \subseteq B$ and an integer m with $|f_m(x) - f(x)| \leq \varepsilon$ for all $x \in B_0$.*

Proof. Let $Y = \overline{B'}$ be a closed ball inside B ; Y is a complete metric space. Put $E_\ell = \{x \in Y : \sup_{j, k \geq \ell} |f_j(x) - f_k(x)| \leq \varepsilon\}$. Since $f_n(x)$ converges (hence is Cauchy) at each x , $Y = \bigcup_{\ell \geq 1} E_\ell$. Each E_ℓ is closed (an intersection of closed sets $\{x : |f_j(x) - f_k(x)| \leq \varepsilon\}$ by continuity). By Theorem 42.1.2 on Y , some E_m contains an open ball B_0 . There $\sup_{j, k \geq m} |f_j - f_k| \leq \varepsilon$; letting $k \rightarrow \infty$ gives $|f_m(x) - f(x)| \leq \varepsilon$ on B_0 . $\square$

Proof of Theorem 42.1.7. Let $F_n = \{x : \text{osc}(f)(x) \geq 1/n\} = E_{1/n}^c$; the discontinuity set is $D = \bigcup_n F_n$. Each F_n is closed; we show it has empty interior. If not, an open ball $B \subseteq F_n$. Apply Lemma 42.1.8 with $\varepsilon = 1/4n$: get $B_0 \subseteq B$ and m with $|f_m - f| \leq 1/4n$ on B_0 . By continuity of f_m shrink to $B' \subseteq B_0$ with $|f_m(y) - f_m(z)| \leq 1/4n$ for $y, z \in B'$. Then for $y, z \in B'$,

$$|f(y) - f(z)| \leq |f(y) - f_m(y)| + |f_m(y) - f_m(z)| + |f_m(z) - f(z)| \leq \frac{3}{4n} < \frac{1}{n},$$

so the center x' of B' has $\text{osc}(f)(x') < 1/n$, contradicting $x' \in F_n$. Hence each F_n is nowhere dense, D is first category, and the continuity set is generic. $\square$

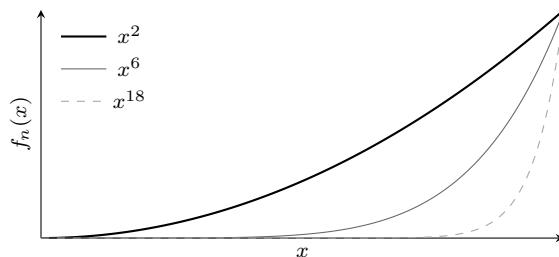


Figure 42.3. $f_n(x) = x^n$ on $[0, 1]$: continuous functions whose pointwise limit (0 on $[0, 1)$, 1 at 1) is discontinuous only at the single point $x = 1$ — a meager discontinuity set, as Theorem 42.1.7 guarantees.

Application 2: nowhere differentiable functions are generic

Let $X = C([0, 1])$ with the sup-norm $\|f\| = \sup_x |f(x)|$; uniform limits of continuous functions are continuous, so X is complete (a Banach space).

Theorem 42.1.9 (Generic nowhere differentiability). *The set of nowhere-differentiable functions in $C([0, 1])$ is generic. In particular such functions exist, and in fact form the “typical” continuous function.*

Proof. Let D be the set of f differentiable at *some* point. If $f'(x^*)$ exists then for x near x^* the difference quotient is bounded, and by continuity globally; so $f \in E_N$ for some N , where

$$E_N = \{f \in C([0, 1]) : \exists x^* \in [0, 1] \text{ with } |f(x) - f(x^*)| \leq N|x - x^*| \forall x\}.$$

Thus $D \subseteq \bigcup_N E_N$; it suffices to show each E_N is nowhere dense, i.e. closed with empty interior.

(i) E_N closed. If $f_n \in E_N$ and $\|f_n - f\| \rightarrow 0$, pick witnesses x_n^* ; a subsequence $x_{n_k}^* \rightarrow x^*$. From $|f(x) - f(x^*)| \leq |f(x) - f_{n_k}(x)| + |f_{n_k}(x) - f_{n_k}(x^*)| + |f_{n_k}(x^*) - f(x^*)|$ and $|f_{n_k}(x) - f_{n_k}(x^*)| \leq N|x - x_{n_k}^*| + N|x_{n_k}^* - x^*|$, letting $k \rightarrow \infty$ gives $|f(x) - f(x^*)| \leq N|x - x^*|$, so $f \in E_N$.

(ii) E_N has empty interior. Let P be the piecewise-linear functions and, for $M > 0$, $P_M \subseteq P$ those whose every segment has slope $\geq M$ or $\leq -M$ (“zig-zag” functions). Note $P_M \cap E_N = \emptyset$ when $M > N$. We claim P_M is dense (Lemma 42.1.10 below). Granting this: given $f \in E_N$ and $\varepsilon > 0$ fix $M > N$; there is $h \in P_M$ with $\|f - h\| < \varepsilon$ and $h \notin E_N$. So no ball about f lies in E_N . Hence E_N is nowhere dense, D is first category, and the nowhere-differentiable functions are generic. $\square$

Lemma 42.1.10. *For every $M > 0$, the zig-zag functions P_M are dense in $C([0, 1])$.*

Proof. Given f and $\varepsilon > 0$, uniform continuity yields a piecewise-linear $g \in P$ with $\|f - g\| \leq \varepsilon$ (interpolate f at a fine grid). On each segment $g(x) = ax + b$, sandwich g between $\varphi = g + \varepsilon$ and $\psi = g - \varepsilon$: starting at $g(0)$, run up at slope $+M$ until hitting φ , reverse to slope $-M$ until hitting ψ , and repeat. This produces $h \in P_M$ with $\psi \leq h \leq \varphi$, so $|h - g| \leq \varepsilon$, hence $\|f - h\| \leq 2\varepsilon$. $\square$

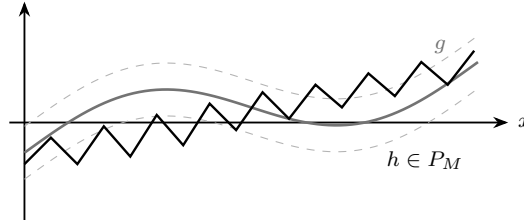


Figure 42.4. Zig-zag approximation: $h \in P_M$ stays within ε of g but its steep $\pm M$ slopes ($M > N$) keep it out of E_N , so E_N has empty interior. (Schematic.)

Worked Example 42.1.11 (Why a single steep segment defeats the Lipschitz bound). Show directly that if h has a segment of slope $M > N$ on $[a, a + \delta]$, then $h \notin E_N$ provided that segment is “visible” from every candidate x^* .

Recognition. Trigger: unpack the disjointness $P_M \cap E_N = \emptyset$ that drives empty interior.

Solution. Suppose $h \in E_N$ with witness x^* . Pick two points $x_1 < x_2$ on a slope- M segment with $x^* \notin (x_1, x_2)$ (possible since x^* is one point). Then $|h(x_2) - h(x_1)| = M|x_2 - x_1|$. But the Lipschitz-at- x^* bound gives $|h(x_i) - h(x^*)| \leq N|x_i - x^*|$, so by the triangle inequality $|h(x_2) - h(x_1)| \leq N(|x_2 - x^*| + |x_1 - x^*|)$. If $x^* \leq x_1 < x_2$ this is $N|x_2 - x_1| < M|x_2 - x_1|$, a contradiction. Hence no zig-zag with all slopes $> N$ in absolute value can lie in E_N . *Check.* This is exactly the local mechanism making P_M disjoint from E_N ; globally one slides x^* but a zig-zag has steep segments on both sides of every point.

Worked Example 42.1.12 (The lacunary example, for contrast). Recall Weierstrass-type series $f(x) = \sum_{n \geq 0} 2^{-n\alpha} e^{i2^n x}$ ($0 < \alpha \leq 1$) is continuous but nowhere differentiable. Verify continuity and the key non-smoothness heuristic.

Recognition. Trigger: an explicit witness, complementing the generic statement.

Solution. Continuity: $\sum_n 2^{-n\alpha}$ converges (geometric, $\alpha > 0$), so the series converges uniformly by the Weierstrass M -test; a uniform limit of continuous functions is continuous. Non-differentiability heuristic: the

n th term oscillates with amplitude $2^{-n\alpha}$ over a scale 2^{-n} , giving difference quotients of order $2^{-n\alpha}/2^{-n} = 2^{n(1-\alpha)}$, which fail to settle as $n \rightarrow \infty$ (and are unbounded for $\alpha < 1$). *Check.* The lacunary gaps 2^n prevent cancellation between scales, the same mechanism the generic zig-zag argument exploits abstractly.

Worked Example 42.1.13 (A point of continuity must exist (Theorem 42.1.7 in action)). Let $f_n(x) = \frac{nx}{1+nx}$ on $[0, 1]$. Find the pointwise limit, locate its discontinuities, and confirm they form a meager set.

Recognition. Trigger: apply the continuity-of-limit theorem to a concrete sequence.

Solution. For $x > 0$, $f_n(x) \rightarrow 1$; for $x = 0$, $f_n(0) = 0 \rightarrow 0$. So $f(x) = 1$ on $(0, 1]$ and $f(0) = 0$, discontinuous only at $x = 0$. The discontinuity set $\{0\}$ is a single point, nowhere dense, hence first category — consistent with Theorem 42.1.7. *Check.* Each f_n is continuous, the limit's lone jump sits at one point, and indeed f is continuous on the generic set $(0, 1]$.

Worked Example 42.1.14 (Oscillation computation). Compute $\text{osc}(f)(0)$ for $f(x) = \sin(1/x)$ ($x \neq 0$), $f(0) = 0$, and confirm f is discontinuous at 0.

Recognition. Trigger: use Definition 42.1.6 directly.

Solution. For any $r > 0$, $B_r(0)$ contains points where $\sin(1/x) = 1$ and where $\sin(1/x) = -1$, so $\omega(f)(r, 0) = \sup |f(y) - f(z)| = 2$ for every $r > 0$. Hence $\text{osc}(f)(0) = \lim_{r \rightarrow 0} 2 = 2 > 0$, so f is discontinuous at 0 (fact (i) after Definition 42.1.6). *Check.* Away from 0, f is continuous, so $\text{osc}(f)(x) = 0$ there; the discontinuity set $\{0\}$ is meager, again matching the theory.

Worked Example 42.1.15 (Closed nowhere-dense vs. open dense, side by side). For $E \subseteq X$ prove: E is closed and nowhere dense $\iff E^c$ is open and dense.

Recognition. Trigger: the duality used repeatedly to recognize generic sets.

Solution. If E is closed then E^c is open, and $E = \overline{E}$ has empty interior iff its complement E^c is dense (a set is dense exactly when its complement has empty interior, by taking complements of closure/interior). Conversely if E^c is open and dense, then E is closed and $\overline{E} = E$ has empty interior. *Check.* The Cantor set is closed nowhere dense and its complement is open dense; consistent on both sides.

Worked Example 42.1.16 (A countable union of generic sets need not be co-meager-stable backwards). Show that while a countable *intersection* of generic sets is generic, a countable *union* of first-category sets is first category, and use this to re-derive that $\mathbb{Q}$ is meager.

Recognition. Trigger: the bookkeeping rules of category.

Solution. If E_n are generic, $E_n^c = \bigcup_k F_{n,k}$ with $F_{n,k}$ nowhere dense; then $(\bigcap_n E_n)^c = \bigcup_n E_n^c = \bigcup_{n,k} F_{n,k}$, a countable union of nowhere-dense sets, so $\bigcap_n E_n$ is generic. Dually a countable union of first-category sets re-indexes to one countable union of nowhere-dense sets. Applying this, $\mathbb{Q} = \bigcup_k \{q_k\}$ is a countable union of (nowhere-dense) points, hence first category. *Check.* The dual statements are exact De Morgan transforms of each other, which is why “generic” and “meager” behave as topological complements.

Worked Example 42.1.17 (Baire forces a non-Lipschitz point). Show that the typical $f \in C([0, 1])$ fails to be Lipschitz on any subinterval, sharpening nowhere differentiability.

Recognition. Trigger: reuse the E_N scheme with intervals.

Solution. For a subinterval J and constant N , the set $E_{N,J} = \{f : |f(x) - f(y)| \leq N|x - y| \ \forall x, y \in J\}$ is closed (limits preserve the inequality) and nowhere dense (zig-zags in P_M , $M > N$, are dense and miss $E_{N,J}$). The functions Lipschitz on *some* rational subinterval form $\bigcup_{N,J} E_{N,J}$, a countable first-category union. So the typical f is Lipschitz on no subinterval. *Check.* A Lipschitz function is differentiable a.e. (Rademacher), so being nowhere Lipschitz is even stronger than nowhere differentiable; both are generic.

Exercises (Stein, Ch.4).

- (Generic sets vs. measure.) Let $\{x_j\}$ enumerate $\mathbb{Q}$. (a) With $U_n = \bigcup_j (x_j - \frac{1}{n2^j}, x_j + \frac{1}{n2^j})$ and $U = \bigcap_n U_n$, show U is generic but $m(U) = 0$. (b) Using a Cantor-like set, give a first-category subset of $[0, 1]$ of full Lebesgue measure (hence uncountable and dense); its complement is then generic and null.

2. Let F be closed and O open in a complete metric space. (a) F is of the first category iff F has empty interior. (b) O is of the first category iff $O = \emptyset$. (c) Hence F is generic iff $F = X$, and O is generic iff O^c has empty interior. [Hint: for (a), if a closed ball $B \subseteq F$, apply Baire to B .]
3. Show the Baire conclusion still holds when X_0 is a metric space arising as an open subset of a complete metric space X : X_0 is not a countable union of sets nowhere dense in X_0 . [Hint: apply Baire to $\overline{X_0}$.]
4. Prove every continuous function on $[0, 1]$ can be uniformly approximated by continuous nowhere-differentiable functions, either (a) via Theorem 42.1.9, or (b) using only the existence of one continuous nowhere-differentiable function.
5. (G_δ and F_σ .) In a complete metric space X : (a) a dense G_δ is generic; (b) a countable dense set is an F_σ but not a G_δ ; (c) (partial converse) every generic E contains a dense G_δ subset $E_0 \subseteq E$.
6. The function f equal to 0 at irrationals and $1/q$ at $x = p/q$ in lowest terms is continuous exactly at the irrationals. Prove there is *no* function on $\mathbb{R}$ continuous exactly at the rationals. [Hint: the continuity set of any function is a G_δ ; apply Exercise 5.]
7. Let $E \subseteq [0, 1]$ and I a nontrivial closed subinterval. (a) If E is first category in $[0, 1]$, then $E \cap I$ is first category in I . (b) If E is generic in $[0, 1]$, then $E \cap I$ is generic in I . (c) Construct E such that for every I , $E \cap I$ is neither first category nor generic in I . [Hint: iterate placing scaled Cantor sets in complementary intervals.]
8. (Hamel bases.) A Hamel basis H of X lets every x be a unique finite combination of elements of H . Prove a Banach space cannot have a countable Hamel basis. [Hint: otherwise the space is first category in itself.]
9. Consider $L^p([0, 1])$, $p > 1$. Show the set of $f \in L^1$ with $f \notin L^p$ is generic. [Hint: $E_N = \{f \in L^1 : \int_I |f| \leq N m(I)^{1-1/p} \text{ for all } I\}$ is closed, $L^p \subseteq \bigcup_N E_N$, and E_N is nowhere dense via $f_0 + g$ with $g(x) = x^{-(1-\delta)}$, $0 < \delta < 1 - 1/p$.]
10. In the Hölder space $\Lambda^\alpha(\mathbb{R})$, $0 < \alpha < 1$, show the nowhere-differentiable functions are generic. (For $\alpha = 1$, Lipschitz functions are a.e. differentiable.)
11. (Nowhere monotone functions.) In $X = C([0, 1])$ over $\mathbb{R}$ with sup-norm, let M be the functions monotone on no subinterval $[a, b]$. Prove M is generic. [Hint: $M_{[a,b]}$, the functions not monotone on $[a, b]$, is dense with closed complement.]
12. (Liouville numbers; Problem 2.) For $n \geq 2$ let

$$\Lambda_n = \{x \in \mathbb{R} : |x - p/q| \leq q^{-n} \text{ for infinitely many fractions } p/q\}.$$

Show (a) Λ_n is generic in $\mathbb{R}$; (b) its Hausdorff dimension is $2/n$; (c) hence $m(\Lambda_n) = 0$ for $n > 2$. The intersection $\Lambda = \bigcap_{n \geq 2} \Lambda_n$ is the set of Liouville numbers (all transcendental).

Additional exercises.

1. Prove directly from Definition 42.1.1 that a finite union of nowhere-dense sets is nowhere dense, but that a countable union need not be (give the example of $\mathbb{Q}$).
2. Show $\mathbb{Z}$ is nowhere dense in $\mathbb{R}$ while $\mathbb{Q}$ is dense but of the first category. Conclude $\mathbb{R} \setminus \mathbb{Q}$ is generic and uncountable using only Baire.
3. Let X be complete with no isolated points. Prove X is uncountable. (This recovers the uncountability of $[0, 1]$ without a diagonal argument.)
4. Compute $\text{osc}(f)(0)$ for the indicator $f = \mathbf{1}_{\mathbb{Q}}$ at every point, and confirm the continuity set is empty — consistent with the fact that f is not a pointwise limit of continuous functions on any interval.
5. For $f_n(x) = \arctan(nx)$ on $\mathbb{R}$, find the pointwise limit, its discontinuity set, and verify the set is meager (Theorem 42.1.7).
6. (Non-example.) Exhibit a sequence of *discontinuous* f_n with pointwise limit f that is continuous nowhere, showing the continuity hypothesis on the f_n in Theorem 42.1.7 cannot be dropped. [Hint: enumerate $\mathbb{Q}$ and let $f_n = \mathbf{1}_{\{q_1, \dots, q_n\}}$.]

7. Prove that the set of $f \in C([0, 1])$ attaining their maximum at a *unique* point is generic. [Hint: the complement is contained in a countable union of closed nowhere-dense sets indexed by separated pairs of rational subintervals.]
8. Draw (and justify) a diagram of three nested dense open sets $O_1 \supseteq O_2 \supseteq O_3$ in $[0, 1]$ whose intersection still meets every subinterval, illustrating Corollary 42.1.3.
9. Show that the Cantor set is closed and nowhere dense in $\mathbb{R}$, hence first category, yet its complement is open dense and generic — and that this is consistent with the Cantor set being uncountable.
10. Prove that if $D \subseteq X$ is first category in a complete metric space, then $X \setminus D$ is dense (so “meager” sets cannot fill any ball).

42.2 The Uniform Boundedness Principle, Open Mapping, and Closed Graph Theorems

The uniform boundedness principle

The three pillars of Banach-space theory in this section — uniform boundedness, open mapping, closed graph — are all corollaries of Baire’s theorem. Their common signature: *a pointwise or set-theoretic hypothesis is upgraded, for free, to a uniform quantitative bound*, because completeness forbids a space from being a meager union.

Theorem 42.2.1 (Uniform boundedness principle / Banach–Steinhaus). *Let B be a Banach space and L a collection (not necessarily countable) of continuous linear functionals on B .*

(i) *If $\sup_{\ell \in L} |\ell(f)| < \infty$ for each $f \in B$, then $\sup_{\ell \in L} \|\ell\| < \infty$.*

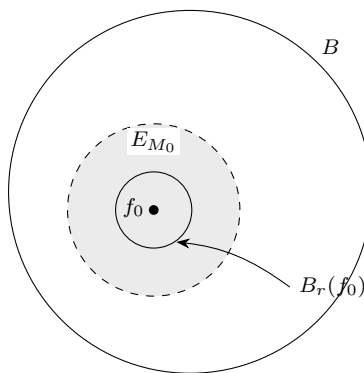
(ii) *The same conclusion holds if pointwise boundedness is assumed only on a set of the second category.*

Proof. It suffices to prove (ii), since by Baire’s theorem B itself is of the second category. Suppose $\sup_{\ell} |\ell(f)| < \infty$ for all $f \in E$, E second category. For $M \in \mathbb{N}$ set $E_M = \{f \in B : \sup_{\ell \in L} |\ell(f)| \leq M\}$. Then $E \subseteq \bigcup_M E_M$. Each $E_M = \bigcap_{\ell \in L} \{f : |\ell(f)| \leq M\}$ is an intersection of closed sets (continuity of each ℓ), hence closed. As E is second category, some E_{M_0} has nonempty interior: there are f_0 and $r > 0$ with $B_r(f_0) \subseteq E_{M_0}$, i.e. $|\ell(f)| \leq M_0$ whenever $\|f - f_0\| < r$ and all ℓ . For $\|g\| < r$ and any ℓ , write $g = (g + f_0) - f_0$:

$$|\ell(g)| \leq |\ell(g + f_0)| + |\ell(f_0)| \leq 2M_0.$$

Thus $\|\ell\| \leq 2M_0/r$ for every ℓ , giving $\sup_{\ell} \|\ell\| < \infty$. □

Remark. The principle is sharp in spirit: completeness cannot be dropped. On the incomplete space of finitely-supported sequences, the functionals $\ell_n(x) = nx_n$ are pointwise bounded on each fixed x (eventually zero) yet $\|\ell_n\| = n \rightarrow \infty$.



some E_M has interior $\Rightarrow$ uniform bound

Figure 42.5. Banach–Steinhaus: the closed sublevel sets E_M cover B ; Baire forces one to contain a ball, which translates to a uniform norm bound. (Schematic.)

Application: divergence of Fourier series

Theorem 42.2.2 (Divergence of Fourier series, du Bois-Reymond). *Let $B = C([- \pi, \pi])$ with the sup-norm.*

(i) *For any fixed x_0 there is a continuous function whose Fourier series diverges at x_0 .*

(ii) *In fact the set of continuous functions whose Fourier series diverge on a dense set is generic in B .*

Proof of (i). Take $x_0 = 0$. The partial sum at 0 is the linear functional $\ell_N(f) = S_N(f)(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(-y) D_N(y) dy$, where D_N is the Dirichlet kernel. Each ℓ_N is continuous with $|\ell_N(f)| \leq L_N \|f\|$, $L_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N(y)| dy$. One shows $\|\ell_N\| = L_N$ (Lemma 42.2.3 below) and $L_N \rightarrow \infty$ (the Lebesgue constants grow like $\log N$). If every continuous f had a convergent-at-0 Fourier series, then $\sup_N |\ell_N(f)| < \infty$ for all f ; by Theorem 42.2.1 this would force $\sup_N \|\ell_N\| = \sup_N L_N < \infty$, contradicting $L_N \rightarrow \infty$. Hence some f has $\sup_N |\ell_N(f)| = \infty$, i.e. its Fourier series diverges at 0. $\square$

Lemma 42.2.3. $\|\ell_N\| = L_N$ for all $N \geq 0$, and $L_N \rightarrow \infty$ (indeed $L_N \geq c \log N$).

Proof. $\|\ell_N\| \leq L_N$ is the bound above. For the reverse, let $g = \operatorname{sgn} D_N$ ($|g| \leq 1$), so $L_N = \frac{1}{2\pi} \int g(-y) D_N(y) dy$. Approximate g uniformly off a small set by continuous f_k with $\|f_k\| \leq 1$ and $\ell_N(f_k) \rightarrow L_N$; thus $\|\ell_N\| \geq L_N$. For the growth, using $|D_N(y)| \geq c |\sin((N + \frac{1}{2})y)| / |y|$ near 0,

$$L_N \geq c \int_0^{(N+1/2)\pi} \frac{|\sin x|}{x} dx \geq c \sum_{k=0}^{N-1} \frac{1}{(k+1)\pi} \int_{k\pi}^{(k+1)\pi} |\sin x| dx \geq c' \sum_{k=1}^N \frac{1}{k} \geq c'' \log N.$$

$\square$

For (ii) one fixes a countable dense set $\{x_i\}$, lets $E_i = \{f : \sup_N |S_N(f)(x_i)| = \infty\}$, shows each E_i is generic by the same UBP argument applied at x_i , and intersects: $\bigcap_i E_i$ is generic and every such f diverges on the dense set $\{x_i\}$ (see Problem 3 for the sharper “diverges on a generic set” statement).

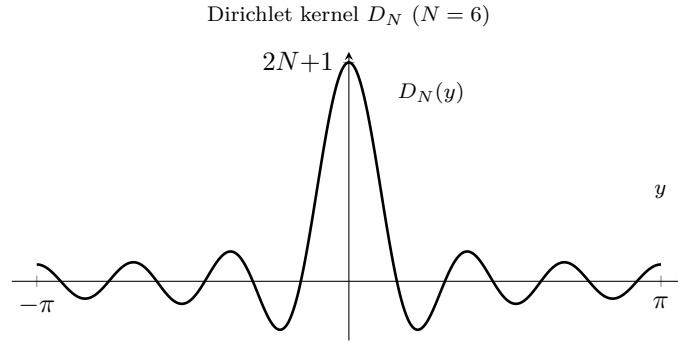


Figure 42.6. The Dirichlet kernel $D_N(y) = \sin((N + \frac{1}{2})y) / \sin(y/2) = 1 + 2 \sum_{k=1}^N \cos ky$: a tall central spike of height $2N + 1$ with sign-changing side lobes (unlike a good kernel it is *not* nonnegative). Its L^1 -norm, the Lebesgue constant L_N , grows like $\log N$ — the engine of divergence.

Worked Example 42.2.4 (Lebesgue constants grow logarithmically). Show $L_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N| \rightarrow \infty$ cannot be bounded, by the $\sum 1/k$ estimate.

Recognition. Trigger: justify $L_N \rightarrow \infty$, the input that makes UBP bite.

Solution. Since $|\sin(y/2)| \leq |y|/2$ near 0, $|D_N(y)| \geq 2 |\sin((N + \frac{1}{2})y)| / |y|$ for small y . Substituting $x = (N + \frac{1}{2})y$ and grouping over half-periods $[k\pi, (k+1)\pi]$ where $\int |\sin x| dx = 2$, $L_N \geq c \sum_{k=1}^N \frac{1}{k}$. The harmonic partial sum diverges like $\log N$, so $L_N \rightarrow \infty$. *Check.* Numerically $L_1 \approx 1.44$, $L_2 \approx 1.64$, $L_4 \approx 1.94$ — slow growth matching $\log N$, but unbounded, which is all UBP needs.

Worked Example 42.2.5 (Pointwise-bounded but norm-unbounded (sharpness)). On c_{00} (finitely supported sequences with sup-norm) define $\ell_n(x) = n x_n$. Show $\{\ell_n\}$ is pointwise bounded yet $\sup_n \|\ell_n\| = \infty$, and identify the hypothesis that fails.

Recognition. Trigger: a non-example pinning down where UBP requires completeness.

Solution. For fixed $x \in c_{00}$, $x_n = 0$ for n large, so $\ell_n(x) = nx_n$ is eventually 0; hence $\sup_n |\ell_n(x)| < \infty$. But $\|\ell_n\| = \sup_{\|x\| \leq 1} |nx_n| = n$, so $\sup_n \|\ell_n\| = \infty$. No contradiction with Theorem 42.2.1 because c_{00} is *not* complete (its completion is c_0). *Check.* On the Banach space c_0 the same ℓ_n are *not* pointwise bounded — take $x = (1, \frac{1}{2}, \frac{1}{3}, \dots) \in c_0$, then $|\ell_n(x)| = n \cdot \frac{1}{n} = 1$, fine, but $x = (1, 1, \dots) \notin c_0$; the completion removes the offending witnesses, restoring the dichotomy.

The open mapping theorem

A map $T : X \rightarrow Y$ is *open* if it sends open sets to open sets. For a bijective linear T , openness of T is exactly continuity of T^{-1} .

Theorem 42.2.6 (Open mapping theorem). *If X, Y are Banach spaces and $T : X \rightarrow Y$ is a continuous linear surjection, then T is an open mapping.*

Proof. By linearity it suffices to show $T(B_X(1))$ contains a ball about 0.

Step 1 (closure has interior). Surjectivity gives $Y = \bigcup_{n \geq 1} \overline{T(B_X(n))}$. By Baire some $\overline{T(B_X(n))}$ has an interior point; scaling, $\overline{T(B_X(1))} \supseteq B_Y(y_0, \varepsilon)$. Pick $y_1 = T(x_1)$, $x_1 \in B_X(1)$, with $\|y_1 - y_0\| < \varepsilon/2$. For $y \in B_Y(\varepsilon/2)$, $y - y_1 \in \overline{T(B_X(1))}$ shifted, leading after symmetrization to $\overline{T(B_X(1))} \supseteq B_Y(\delta)$ for some $\delta > 0$; rescaling,

$$\overline{T(B_X(2^{-k}))} \supseteq B_Y(2^{-k}\delta) \quad \text{for all } k. \quad (42.1)$$

Step 2 (remove the closure). We show $T(B_X(1)) \supseteq B_Y(\delta/2)$. Let $y \in B_Y(\delta/2)$. By (42.1) with $k = 1$ pick $x_1 \in B_X(1/2)$ with $y - T(x_1) \in B_Y(\delta/4)$; by $k = 2$ pick $x_2 \in B_X(1/4)$ with $y - T(x_1) - T(x_2) \in B_Y(\delta/8)$; inductively $\|x_k\| < 2^{-k}$. By completeness $x = \sum_k x_k$ converges, $\|x\| < \sum_k 2^{-k} = 1$, and continuity gives $T(x) = y$. Thus $T(B_X(1)) \supseteq B_Y(\delta/2)$, an open ball about 0. $\square$

Corollary 42.2.7 (Bounded inverse theorem). *If $T : X \rightarrow Y$ is a continuous linear bijection of Banach spaces, then T^{-1} is continuous; equivalently there are $c, C > 0$ with $c\|f\|_X \leq \|Tf\|_Y \leq C\|f\|_X$. In particular two complete norms on a vector space, one dominating the other, are equivalent.*

$$X \begin{array}{c} \xrightarrow{T} \\ \sim \\ \xleftarrow{T^{-1}} \end{array} Y$$

Figure 42.7. Bounded inverse theorem: a continuous linear bijection T of Banach spaces is an isomorphism, its inverse T^{-1} automatically continuous.

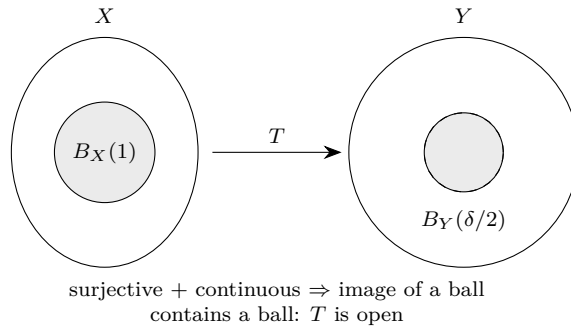


Figure 42.8. Open mapping theorem: $T(B_X(1))$ swallows a ball $B_Y(\delta/2)$ about the origin. (Schematic.)

Application: Fourier coefficients of L^1 are not all of c_0

Theorem 42.2.8 (Coefficient map is not surjective). *Let $B_1 = L^1([-\pi, \pi])$ and $B_2 = c_0(\mathbb{Z})$ (sequences vanishing at infinity, sup-norm). The map $T : B_1 \rightarrow B_2$, $T(f) = \{\hat{f}(n)\}$, is linear, continuous, injective, but not surjective. So some sequence vanishing at infinity is not the Fourier coefficient sequence of any L^1 function.*

Proof. Linearity is clear; $\|T(f)\|_\infty \leq \|f\|_{L^1}$ gives continuity; the Riemann–Lebesgue lemma puts the image in c_0 ; injectivity is uniqueness of Fourier coefficients. If T were surjective, Corollary 42.2.7 would give $c > 0$ with $c\|f\|_{L^1} \leq \|T(f)\|_\infty$ for all f . Test on the Dirichlet kernel $f = D_N$: $\|T(D_N)\|_\infty = 1$ (its coefficients are 1 for $|n| \leq N$, else 0), while $\|D_N\|_{L^1} = 2\pi L_N \rightarrow \infty$. So $c \cdot 2\pi L_N \leq 1$ for all N is impossible. Hence T is not surjective. $\square$

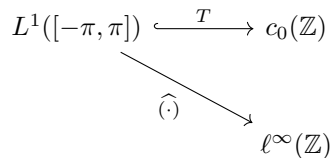


Figure 42.9. The Fourier-coefficient map $T(f) = \{\widehat{f}(n)\}$ is a continuous injection (hence drawn as a hook) of L^1 into c_0 , landing inside c_0 by Riemann–Lebesgue; the diagonal is the same map viewed into the larger ℓ^∞ . It is *not* onto c_0 , so no inverse bound exists.

Worked Example 42.2.9 (Two complete norms, one dominating: forced equivalence). Suppose $\|\cdot\|_1, \|\cdot\|_2$ are both complete norms on V and $\|v\|_2 \leq C\|v\|_1$ for all v . Show they are equivalent.

Recognition. Trigger: a textbook application of the bounded inverse theorem.

Solution. The identity $\text{id} : (V, \|\cdot\|_1) \rightarrow (V, \|\cdot\|_2)$ is a continuous linear bijection between Banach spaces (continuous because $\|v\|_2 \leq C\|v\|_1$). By Corollary 42.2.7 its inverse is continuous, so $\|v\|_1 \leq C'\|v\|_2$. Combined, $\frac{1}{C'}\|v\|_1 \leq \|v\|_2 \leq C\|v\|_1$: equivalence. *Check.* The hypothesis that *both* norms are complete is essential — on $C^1[0, 1]$ the C^1 -norm dominates the sup-norm, but they are inequivalent precisely because the sup-norm is not complete on C^1 .

Worked Example 42.2.10 (A discontinuous linear bijection (no completeness $\Rightarrow$ no inverse bound)). On $V =$ polynomials with sup-norm on $[0, 1]$, let $T(p) =$ the polynomial with $T(p)(x) = p(x)$ but measured so that $\text{id} : (V, \|\cdot\|_{C^1}) \rightarrow (V, \|\cdot\|_\infty)$ is the map. Show its inverse is unbounded.

Recognition. Trigger: see the corollary fail when a space is incomplete.

Solution. Take $p_n(x) = x^n$. Then $\|p_n\|_\infty = 1$ but $\|p_n\|_{C^1} = \|p_n\|_\infty + \|p'_n\|_\infty = 1 + n$. So $\text{id}^{-1} : (V, \|\cdot\|_\infty) \rightarrow (V, \|\cdot\|_{C^1})$ sends a unit vector to one of norm $1 + n$; it is unbounded. *Check.* No contradiction with Corollary 42.2.7 because $(V, \|\cdot\|_\infty)$ is not complete (its completion is $C[0, 1] \supsetneq$ polynomials); completeness of both sides is exactly what is missing.

The closed graph theorem

Definition 42.2.11. For linear $T : X \rightarrow Y$, the *graph* is $G_T = \{(x, y) \in X \times Y : y = T(x)\}$. T is *closed* if G_T is closed in $X \times Y$: whenever $x_n \rightarrow x$ and $T(x_n) = y_n \rightarrow y$, we have $T(x) = y$.

Theorem 42.2.12 (Closed graph theorem). *If X, Y are Banach and $T : X \rightarrow Y$ is a closed linear map, then T is continuous.*

Proof. $X \times Y$ with $\|(x, y)\| = \|x\|_X + \|y\|_Y$ is Banach, and G_T , being closed, is a closed subspace, hence itself Banach. Consider the projections $P_X : G_T \rightarrow X$, $P_X(x, Tx) = x$, and $P_Y : G_T \rightarrow Y$, $P_Y(x, Tx) = Tx$. Both are continuous and linear; P_X is a bijection of G_T onto X , so by Corollary 42.2.7 P_X^{-1} is continuous. Then $T = P_Y \circ P_X^{-1}$ is a composition of continuous maps, hence continuous. $\square$

The practical force: to prove T continuous you no longer must show $x_n \rightarrow x \Rightarrow T(x_n) \rightarrow T(x)$; you may *assume* $T(x_n)$ already converges to some y and merely check $y = T(x)$. The convergence of the image is granted for free.

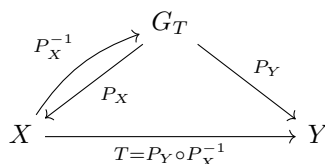


Figure 42.10. Closed graph via open mapping: P_X is a continuous bijection of the Banach space G_T , so P_X^{-1} is continuous and $T = P_Y \circ P_X^{-1}$ is too.

Worked Example 42.2.13 (The differentiation operator is closed but unbounded). On $X = C([0, 1])$ let $D(f) = f'$ with domain the C^1 functions. Show D has closed graph yet is not continuous — and reconcile with Theorem 42.2.12.

Recognition. Trigger: a closed-but-unbounded operator; the resolution is the domain.

Solution. Closedness: if $f_n \rightarrow f$ uniformly and $f'_n \rightarrow g$ uniformly, a standard theorem gives $f \in C^1$ with $f' = g$; so the graph is closed. Unboundedness: $f_n(x) = \sin(nx)$ has $\|f_n\|_\infty \leq 1$ but $\|f'_n\|_\infty = n \rightarrow \infty$. No contradiction: the closed graph theorem requires T to be defined on *all* of a Banach space. Here the domain C^1 is a proper, non-closed (incomplete in sup-norm) subspace of $C([0, 1])$. *Check.* Equip C^1 with its own complete norm $\|f\|_{C^1} = \|f\|_\infty + \|f'\|_\infty$; then $D : C^1 \rightarrow C([0, 1])$ is continuous, consistent with the theorem once both spaces are Banach.

Worked Example 42.2.14 (Using closed graph to prove continuity cheaply). Let $T : X \rightarrow Y$ be linear between Banach spaces such that for every $x_n \rightarrow 0$, if $T(x_n) \rightarrow y$ then $y = 0$. Show T is continuous.

Recognition. Trigger: verify the closed-graph condition rather than an ε - δ estimate.

Solution. Suppose $x_n \rightarrow x$ and $T(x_n) \rightarrow y$. Then $x_n - x \rightarrow 0$ and $T(x_n - x) = T(x_n) - T(x) \rightarrow y - T(x)$. By hypothesis $y - T(x) = 0$, i.e. $T(x) = y$. So G_T is closed and Theorem 42.2.12 gives continuity. *Check.* We never estimated $\|Tx\|$ in terms of $\|x\|$; the boundedness was extracted purely from closedness plus completeness — the hallmark economy of the closed graph theorem.

Worked Example 42.2.15 (Pointwise-convergent functionals have a continuous limit). Let ℓ_n be continuous linear functionals on a Banach space B with $\ell_n(f) \rightarrow \ell(f)$ for every f . Show ℓ is continuous.

Recognition. Trigger: UBP applied to a convergent family.

Solution. For each f , $\{\ell_n(f)\}$ converges, hence is bounded, so $\sup_n |\ell_n(f)| < \infty$ for all f . By Theorem 42.2.1, $C := \sup_n \|\ell_n\| < \infty$. Then $|\ell(f)| = \lim_n |\ell_n(f)| \leq C \|f\|$, so ℓ is linear (clear) and bounded, hence continuous with $\|\ell\| \leq C$. *Check.* The limit functional cannot have larger norm than the uniform bound; e.g. on ℓ^2 , partial-sum functionals $\ell_n(x) = \sum_{k \leq n} x_k a_k$ with $a \in \ell^2$ converge to $\ell(x) = \langle x, a \rangle$ of norm $\|a\|_2$.

Worked Example 42.2.16 (Operator norm of a Fourier partial-sum functional). Confirm $\|\ell_N\| = L_N$ where $\ell_N(f) = S_N(f)(0)$ on $C([-\pi, \pi])$, and explain why this makes divergence inevitable.

Recognition. Trigger: identify operator norm with an L^1 kernel norm.

Solution. $|\ell_N(f)| \leq \frac{1}{2\pi} \int |f(-y)| |D_N(y)| dy \leq L_N \|f\|$ gives $\|\ell_N\| \leq L_N$. Choosing f close to $\text{sgn } D_N(-\cdot)$ with $\|f\| \leq 1$ pushes $\ell_N(f)$ toward L_N , so $\|\ell_N\| = L_N$. Since $L_N \rightarrow \infty$ (Lemma 42.2.3), the family $\{\ell_N\}$ has unbounded norms; UBP then forbids pointwise boundedness for *all* f , i.e. some f has divergent partial sums at 0. *Check.* The equality $\|\ell_N\| = L_N$ is what converts the analytic estimate $L_N \sim \log N$ into the existence theorem.

Exercises (Stein, Ch.4).

- (Separate continuity implies joint, via UBP.) Let X, Y, Z be Banach spaces and $T : X \times Y \rightarrow Z$ a map such that (i) for each x , $y \mapsto T(x, y)$ is linear and continuous, and (ii) for each y , $x \mapsto T(x, y)$ is linear and continuous. Prove T is jointly continuous and in fact $\|T(x, y)\|_Z \leq C \|x\|_X \|y\|_Y$ for some $C > 0$.
- (Weak boundedness $\Rightarrow$ norm boundedness.) Let $(X, \mathcal{F}, \mu)$ be a measure space, $1 < p < \infty$, $1/p + 1/q = 1$. Call $\{f_n\} \subset L^p$ *weakly bounded* if $\sup_n \left| \int_X f_n g d\mu \right| < \infty$ for all $g \in L^q$. Prove $\sup_n \|f_n\|_{L^p} < \infty$. In particular this holds when f_n converges weakly. [Hint: apply UBP to $\ell_n(g) = \int_X f_n g d\mu$.]

Problems (Stein, Ch.4).

- (Image is either onto or meager; L^{p_2} is meager in L^{p_1} .) Let $T : B_1 \rightarrow B_2$ be a bounded linear map of Banach spaces. (a) Prove either T is surjective or $T(B_1)$ is of the first category in B_2 . (b) Hence if (X, μ) is a finite measure space and $1 \leq p_1 < p_2 \leq \infty$, then $L^{p_2}(X)$ is first category in $L^{p_1}(X)$ (except in the trivial case where every element of L^{p_1} lies in L^{p_2}). [Hint: if $T(B_1)$ is second category, mimic the open mapping proof to show T of a ball about 0 contains a ball about 0.]

- (Divergence on a generic set.) Let $B = C(\mathbb{T})$ with the sup-norm. Prove the set of $f \in B$ whose Fourier series diverges on a generic subset of the circle is itself generic in B . [Hint: with $\{x_i\}$ dense, $E_i = \{f : \sup_N |S_N(f)(x_i)| = \infty\}$, $E = \bigcap_i E_i$ is generic; for $f \in E$ set $O_n = \{x : |S_N(f)(x)| > n \text{ for some } N\}$ and show $\bigcap_n O_n$ is generic.]

Additional exercises.

- Let $\{\ell_n\}$ be continuous linear functionals on a Banach space with $\ell_n(f) \rightarrow \ell(f)$ for every f . Use the uniform boundedness principle to show $\sup_n \|\ell_n\| < \infty$ and that the limit ℓ is a continuous linear functional.
- Compute the operator norm of $\ell_N(f) = S_N(f)(0)$ on $C([-\pi, \pi])$ and confirm it equals the Lebesgue constant L_N (Lemma 42.2.3); then estimate L_4 numerically.
- Prove the open mapping theorem fails for the inclusion $\iota : (C^1[0, 1], \|\cdot\|_{C^1}) \hookrightarrow (C[0, 1], \|\cdot\|_\infty)$ to a non-closed image, explaining which hypothesis (surjectivity) is violated.
- Let $T : X \rightarrow Y$ be a continuous linear bijection of Banach spaces. Show that the two norms $\|x\|_X$ and $\|Tx\|_Y$ on X are equivalent, naming the theorem used at each step.
- (Non-example.) Give a linear map between normed spaces that is a bijection, continuous, but with discontinuous inverse, and identify exactly why Corollary 42.2.7 does not apply.
- Using the closed graph theorem, prove that a linear map $T : X \rightarrow Y$ of Banach spaces that is the pointwise limit of continuous linear maps T_n (i.e. $T_n x \rightarrow Tx$ for all x) is itself continuous.
- Show that the multiplication operator $M_\varphi f = \varphi f$ on $L^2[0, 1]$ is bounded iff $\varphi \in L^\infty$, and use the closed graph theorem to streamline the boundedness direction.
- Sketch the open mapping picture for a non-surjective bounded T (e.g. a rank-one operator on $\mathbb{R}^2$) and explain why the image of the open unit ball contains no ball about 0 in the codomain.
- Prove that if X is Banach and $\{x_n\}$ satisfies $\sum_n |\ell(x_n)| < \infty$ for every continuous functional ℓ , then $\sup_n \|x_n\| < \infty$. [Hint: view x_n as functionals on X^* and apply UBP.]
- Deduce from Theorem 42.2.8 that the map $f \mapsto \{\widehat{f}(n)\}$ from $L^1(\mathbb{T})$ into $c_0(\mathbb{Z})$ has dense but not closed image.

42.3 Applications: Generic Pathology

Grothendieck's theorem on closed subspaces of L^p

We now harvest the closed graph theorem in a striking structural result: a closed subspace of L^p on a finite measure space, if it happens to sit inside L^∞ , must be finite-dimensional. Infinitely many independent functions cannot all be simultaneously bounded if they form a closed subspace.

Theorem 42.3.1 (Grothendieck). *Let $(X, \mathcal{F}, \mu)$ be a finite measure space, $\mu(X) < \infty$. If E is a closed subspace of $L^p(X, \mu)$ for some $1 \leq p < \infty$ and $E \subseteq L^\infty(X, \mu)$, then E is finite-dimensional.*

Proof. Since $E \subseteq L^\infty$ and $\mu(X) < \infty$, we have $E \subseteq L^2$ with $\|f\|_{L^2} \leq C \|f\|_{L^\infty}$. The plan is to reverse comparisons until the unit L^2 -ball of E is uniformly bounded pointwise.

Step 1 (closed graph gives $L^\infty \leq M L^p$). E with the L^p -norm is Banach (closed subspace). Let $I : E \rightarrow L^\infty$ be the identity $I(f) = f$. It is linear and *closed*: if $f_n \rightarrow f$ in L^p and $f_n \rightarrow g$ in L^∞ , a subsequence of f_n converges a.e. to f (from L^p) and L^∞ -convergence forces $f = g$ a.e. By Theorem 42.2.12 there is $M > 0$ with

$$\|f\|_{L^\infty} \leq M \|f\|_{L^p} \quad (f \in E). \quad (42.2)$$

Step 2 ($L^\infty \leq A L^2$). If $1 \leq p \leq 2$, Hölder with exponents $r = 2/p$, $r^* = 2/(2-p)$ gives $\int |f|^p \leq (\int |f|^2)^{p/2} \mu(X)^{(2-p)/2}$, so $\|f\|_{L^p} \leq B \|f\|_{L^2}$; with (42.2), $\|f\|_{L^\infty} \leq A \|f\|_{L^2}$. If $2 < p < \infty$, then $|f|^p \leq \|f\|_{L^\infty}^{p-2} |f|^2$; integrating, $\|f\|_{L^p}^p \leq \|f\|_{L^\infty}^{p-2} \|f\|_{L^2}^2$, and inserting (42.2) again yields $\|f\|_{L^\infty} \leq A \|f\|_{L^2}$.

Step 3 (dimension bound). Let $f_1, \dots, f_n \in E$ be orthonormal in L^2 . For $\zeta \in B$ (the unit ball of $\mathbb{C}^n$) set $f_\zeta = \sum_j \zeta_j f_j$. Then $\|f_\zeta\|_{L^2} \leq 1$, so Step 2 gives $\|f_\zeta\|_{L^\infty} \leq A$; thus $|f_\zeta(x)| \leq A$ off a null set. Taking a countable dense set of ζ and using continuity of $\zeta \mapsto f_\zeta(x)$, there is a full-measure X' with $|f_\zeta(x)| \leq A$ for all $\zeta \in B$ and $x \in X'$. Fixing $x \in X'$ with $\sigma = (\sum_j |f_j(x)|^2)^{1/2} > 0$ and choosing $\zeta_j = \overline{f_j(x)}/\sigma$ gives $\sigma \leq A$, i.e. $\sum_j |f_j(x)|^2 \leq A^2$. Integrating over X and using orthonormality, $n \leq A^2$. Hence $\dim E \leq A^2 < \infty$. $\square$

Remark. The hypothesis $E \subseteq L^\infty$ cannot be weakened to $E \subseteq L^q$ for finite q : Problem 6 exhibits a closed infinite-dimensional subspace of $L^1([0, 1])$ lying in every L^q , $1 \leq q < \infty$ (built from lacunary/Gaussian-type functions).

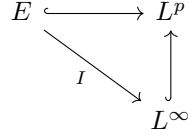
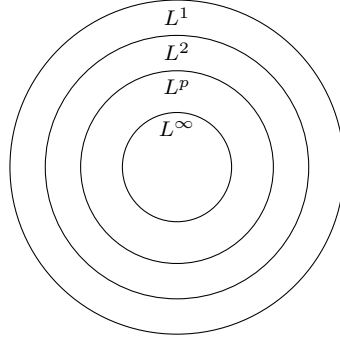


Figure 42.11. Grothendieck's setup: E is a closed subspace of L^p assumed to land in L^∞ , so the identity $I : E \rightarrow L^\infty$ has closed graph; on a finite measure space $L^\infty \hookrightarrow L^p$, and the closed graph theorem upgrades I to the bound $\|f\|_{L^\infty} \leq M \|f\|_{L^p}$ on E .



finite measure: $L^\infty \subset \dots \subset L^p \subset L^2 \subset L^1$;
a closed $E \subset L^p$ trapped in L^∞ is finite-dimensional

Figure 42.12. On a finite measure space the L^p spaces nest the “wrong” way (L^∞ smallest). Grothendieck: a closed subspace caught in the innermost ring is forced finite-dimensional. (Schematic.)

Besicovitch sets are generic

The Hausdorff metric turns the compact subsets of $\mathbb{R}^2$ into a complete metric space, on which a category argument produces measure-zero Kakeya sets.

Definition 42.3.2. For $A \subseteq \mathbb{R}^2$ and $\delta > 0$, the δ -neighborhood is $A^\delta = \{x : \text{dist}(x, A) < \delta\}$, $\text{dist}(x, A) = \inf_{y \in A} |x - y|$. The *Hausdorff distance* between A, B is $\text{dist}(A, B) = \inf\{\delta : B \subseteq A^\delta \text{ and } A \subseteq B^\delta\}$.

On nonempty compact subsets of $\mathbb{R}^2$ this is a metric, and the space of compact subsets is *complete* (the limit of a Cauchy sequence $\{A_n\}$ is $A = \bigcap_n \overline{\bigcup_{k \geq n} A_k}$). Let $\mathcal{K}$ be the compact subsets K of $Q = [-\frac{1}{2}, \frac{1}{2}] \times [0, 1]$ that are unions of segments joining $L_0 = \{y = 0\}$ to $L_1 = \{y = 1\}$ and realizing every direction $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$; $\mathcal{K}$ is a closed subset, hence a complete metric space.

Theorem 42.3.3 (Generic Besicovitch sets, Körner). *The collection of sets in $\mathcal{K}$ with two-dimensional Lebesgue measure zero is generic; in particular it is nonempty and dense.*

Proof sketch. For fixed $y_0 \in [0, 1]$ and $\varepsilon > 0$ let $\mathcal{K}(y_0, \varepsilon)$ be those $K \in \mathcal{K}$ for which some thickening K^η has every horizontal slice on $[y_0 - \varepsilon, y_0 + \varepsilon]$ of length $< 10\varepsilon$. *Openness:* if $K \in \mathcal{K}(y_0, \varepsilon)$ with parameter η and $\text{dist}(K, K') < \eta/2$, then $(K')^{\eta/2} \subseteq K^\eta$, so the slice bound passes to K' . *Density:* given K and δ , replace each segment $\ell(x, \theta)$ by the thin angular sector obtained by rotating ℓ about its crossing of $y = y_0$ through a small angle $\pi/(2N)$ ($N \geq c/\delta$); summing these sectors $A = \bigcup_n S_n$ and patching with a finite set of extra

lines A' produces $K' = A \cup A' \in \mathcal{K}(y_0, \varepsilon)$ with $\text{dist}(K, K') \leq \delta$. Thus each $\mathcal{K}(y_0, \varepsilon)$ is open and dense, so $\mathcal{K}^* = \bigcap_m \mathcal{K}(y_m, \frac{1}{m})$ over a countable dense set of (y_m) is generic. For $K \in \mathcal{K}^*$ every slice has measure 0, and Fubini gives $m_2(K) = 0$. $\square$

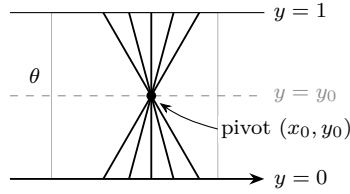


Figure 42.13. Segments of all directions pivoting about their crossing of $y = y_0$; rotating each through a tiny angle shrinks the slices, making small-measure sets dense. (Schematic.)

Probabilistic interlude: bell curves and random paths

Generic-pathology arguments live in function spaces; the same Banach-space and limit machinery underlies probability, where the Gaussian and Brownian motion are the canonical “typical” objects. We record the computations the later chapters use.

Theorem 42.3.4 (Gaussian normalization and CLT shape). *The density $\varphi(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$ integrates to 1, has mean μ and variance σ^2 . If $X_1, \dots, X_n$ are i.i.d. with mean μ and variance σ^2 , then $\frac{1}{\sqrt{n}} \sum_{k=1}^n \frac{X_k - \mu}{\sigma}$ converges in distribution to the standard Gaussian.*

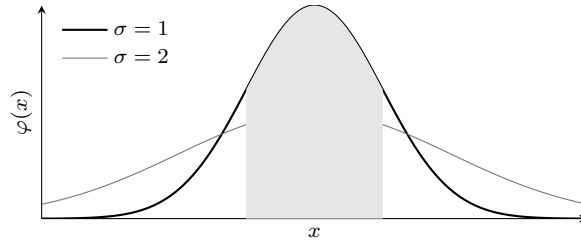


Figure 42.14. Gaussian densities; shaded region $[-\sigma, \sigma]$ holds $\approx 68\%$ of the mass. The CLT makes this bell the universal limit shape.

Theorem 42.3.5 (Brownian motion, defining properties). *Standard Brownian motion $\{B_t\}_{t \geq 0}$ satisfies $B_0 = 0$; independent increments; $B_t - B_s \sim N(0, t - s)$; and $t \mapsto B_t$ is a.s. continuous but a.s. nowhere differentiable — the probabilistic mirror of Theorem 42.1.9.*

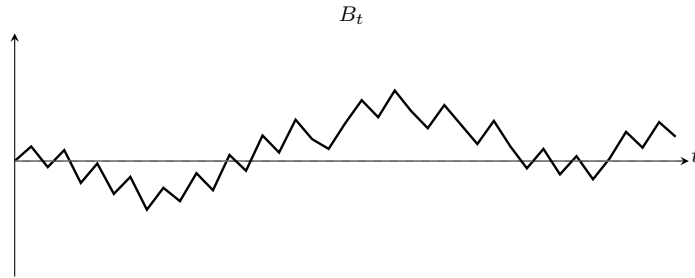


Figure 42.15. A simulated Brownian / random-walk sample path: continuous everywhere, differentiable nowhere — the generic continuous function of Theorem 42.1.9 realized by randomness. (Schematic.)

Worked Example 42.3.6 (Gaussian integral normalization). Verify $\int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$, hence φ integrates to 1.

Recognition. Trigger: the foundational normalization of t : *gauss*.

Solution. Let $I = \int e^{-x^2/2} dx$. Then $I^2 = \iint e^{-(x^2+y^2)/2} dx dy$; in polar coordinates $= \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta = 2\pi \cdot [-e^{-r^2/2}]_0^{\infty} = 2\pi$. So $I = \sqrt{2\pi}$, and dividing by $\sqrt{2\pi}\sigma$ with the substitution $x \mapsto \mu + \sigma x$ gives total mass 1. *Check.* The variance $\int x^2 \varphi = \sigma^2$ follows by one integration by parts, confirming the parameters.

Worked Example 42.3.7 (Variance of a Brownian increment scales linearly). For standard BM, compute $\mathbb{E}[(B_t - B_s)^2]$ and $\mathbb{E}[B_t B_s]$ for $s \leq t$.

Recognition. Trigger: the increment law $B_t - B_s \sim N(0, t - s)$ and independence.

Solution. $B_t - B_s \sim N(0, t - s)$ has variance $t - s$, so $\mathbb{E}[(B_t - B_s)^2] = t - s$. For the covariance with $s \leq t$, write $B_t = B_s + (B_t - B_s)$ with B_s and $B_t - B_s$ independent and mean 0: $\mathbb{E}[B_t B_s] = \mathbb{E}[B_s^2] + \mathbb{E}[B_s(B_t - B_s)] = s + 0 = s = \min(s, t)$. *Check.* Setting $s = t$ recovers $\mathbb{E}[B_t^2] = t$, the defining variance; the $\sqrt{t}$ scale of B_t is why difference quotients $B_{t+h}/h \sim h^{-1/2}$ blow up, matching nowhere differentiability.

Worked Example 42.3.8 (L^p inclusions on a finite measure space). On $([0, 1], m)$ show $L^q \subseteq L^p$ for $p \leq q$, with $\|f\|_p \leq \|f\|_q$.

Recognition. Trigger: the nesting picture behind Grothendieck's theorem.

Solution. For $p < q$ apply Hölder to $|f|^p \cdot 1$ with exponents q/p and its conjugate: $\int |f|^p \leq (\int |f|^q)^{p/q} (\int 1)^{1-p/q} = (\int |f|^q)^{p/q}$ since $m([0, 1]) = 1$. Taking p th roots, $\|f\|_p \leq \|f\|_q$. Thus $f \in L^q \Rightarrow f \in L^p$. *Check.* On infinite measure the inclusion reverses/fails — $f(x) = 1/x$ on $[1, \infty)$ is in L^2 but not L^1 ; finiteness of μ is essential, just as in Grothendieck's hypotheses.

Worked Example 42.3.9 (A generic non- L^p function in L^1). Sketch why $\{f \in L^1([0, 1]) : f \notin L^p\}$ is generic for fixed $p > 1$ (Exercise 9).

Recognition. Trigger: a category argument inside an L^p scale.

Solution. Let $E_N = \{f \in L^1 : \int_I |f| \leq N m(I)^{1-1/p} \text{ for all intervals } I\}$. Each E_N is closed (the defining inequalities pass to L^1 -limits along a subsequence converging a.e.) and $L^p \subseteq \bigcup_N E_N$ (by Hölder, an L^p function satisfies such a bound). One shows E_N is nowhere dense by perturbing $f_0 \in E_N$ by $g(x) = x^{-(1-\delta)}$ with $0 < \delta < 1 - 1/p$, which lies in L^1 but violates the E_N bound near 0. Hence $L^p = \bigcup_N E_N$ is first category, so its complement (functions not in L^p) is generic. *Check.* This realizes the slogan “most L^1 functions are in no smaller L^p ,” a category statement with no explicit construction.

Worked Example 42.3.10 (Weak boundedness forces norm boundedness (Exercise 13)). Let $1 < p < \infty$ and $\{f_n\} \subset L^p$ be weakly bounded: $\sup_n |\int f_n g| < \infty$ for every $g \in L^q$ ($1/p + 1/q = 1$). Show $\sup_n \|f_n\|_p < \infty$.

Recognition. Trigger: package L^p functions as functionals on L^q and apply UBP.

Solution. Define $\ell_n(g) = \int f_n g$ on the Banach space L^q . By Hölder each ℓ_n is continuous with $\|\ell_n\| = \|f_n\|_p$ (duality $(L^q)^* = L^p$). Weak boundedness says $\{\ell_n\}$ is pointwise bounded on L^q . By the uniform boundedness principle (Theorem 42.2.1), $\sup_n \|\ell_n\| < \infty$, i.e. $\sup_n \|f_n\|_p < \infty$. *Check.* This includes the case where $f_n \rightharpoonup f$ weakly, recovering the familiar fact that weakly convergent sequences are norm-bounded.

Worked Example 42.3.11 (Hausdorff distance computation). Compute the Hausdorff distance between the unit segment $A = \{(x, 0) : 0 \leq x \leq 1\}$ and the point $B = \{(0, 0)\}$ in $\mathbb{R}^2$.

Recognition. Trigger: unpack Definition 42.3.2 on a concrete pair.

Solution. We need the least δ with $A \subseteq B^\delta$ and $B \subseteq A^\delta$. Every point of A is within 1 of B (farthest is $(1, 0)$ at distance 1), so $A \subseteq B^\delta$ needs $\delta > 1$; conversely $B \subseteq A^\delta$ holds for any $\delta > 0$ since $(0, 0) \in A$. Thus $\text{dist}(A, B) = 1$. *Check.* The asymmetry of the two containments is why the symmetrized inf is needed; here the segment-to-point direction dominates.

Worked Example 42.3.12 (Holder interpolation of L^p norms). On a finite measure space prove the log-convexity $\|f\|_r \leq \|f\|_p^{1-\theta} \|f\|_q^\theta$ where $1/r = (1-\theta)/p + \theta/q$, $0 \leq \theta \leq 1$.

Recognition. Trigger: the interpolation inequality behind the Grothendieck step 2.

Solution. Write $|f|^r = |f|^{r(1-\theta)} |f|^{r\theta}$ and apply Hölder with exponents $\frac{p}{r(1-\theta)}$ and $\frac{q}{r\theta}$ (conjugate by the choice of r): $\int |f|^r \leq (\int |f|^p)^{r(1-\theta)/p} (\int |f|^q)^{r\theta/q}$. Taking r th roots gives $\|f\|_r \leq \|f\|_p^{1-\theta} \|f\|_q^\theta$. *Check.* At $\theta = 0$ we get $\|f\|_r = \|f\|_p$ ($r = p$), and the inequality is exactly the convexity of $p \mapsto \log \|f\|_p$ in $1/p$ — the same convexity that powers Riesz–Thorin interpolation.

Worked Example 42.3.13 (Moment generating function of a Gaussian). For $X \sim N(0, \sigma^2)$ compute $\mathbb{E}[e^{tX}]$ and recover $\mathbb{E}[X^2] = \sigma^2$.

Recognition. Trigger: a direct Gaussian integral, with the CLT scaling in mind.

Solution. $\mathbb{E}[e^{tX}] = \frac{1}{\sqrt{2\pi\sigma}} \int e^{tx} e^{-x^2/2\sigma^2} dx$. Completing the square $tx - \frac{x^2}{2\sigma^2} = -\frac{(x-\sigma^2 t)^2}{2\sigma^2} + \frac{\sigma^2 t^2}{2}$ and using the normalization gives $\mathbb{E}[e^{tX}] = e^{\sigma^2 t^2/2}$. Expanding, $\mathbb{E}[e^{tX}] = 1 + \frac{\sigma^2 t^2}{2} + \cdots$, so the coefficient of $t^2/2!$ is $\mathbb{E}[X^2] = \sigma^2$. *Check.* Differentiating twice at $t = 0$ also yields σ^2 ; the absence of a t^1 term confirms $\mathbb{E}[X] = 0$, and the quadratic-in- t exponent is precisely what makes sums of independent Gaussians Gaussian.

Exercises (Stein, Ch.4).

- (Universal elements are generic or none.) Let X be complete and $T : X \rightarrow X$ continuous. Call x^* *universal* for T if the orbit $\{T^n(x^*)\}_{n \geq 1}$ is dense. Show the set of universal elements is either empty or generic. [Hint: with $\{y_j\}$ dense, $F_{j,k,N} = \{x : d(T^n x, y_j) < 1/k \text{ for some } n \geq N\}$ is open and dense.]
- (Generic compact sets.) Let B be the closed unit ball in $\mathbb{R}^d$ and $\mathcal{C}$ the compact subsets of B with the Hausdorff distance. Show that the following collections are generic: (a) the subsets of Lebesgue measure zero; (b) the nowhere-dense subsets. [Hint: for (a), $\{C : m(C) < 1/n\}$ is open and dense; for (b), fix open O and show $\{C : C \supseteq O\}$ is closed and nowhere dense.]

Problems (Stein, Ch.4).

- (Boundary singular functions are generic.) Let $\mathbb{D}$ be the open unit disc and $\mathcal{A}$ the Banach space of functions continuous on $\overline{\mathbb{D}}$ and holomorphic on $\mathbb{D}$ (sup-norm). Show the set of $f \in \mathcal{A}$ that cannot be analytically continued past any boundary point is generic, by establishing: (a) $A_N = \{f : |f(e^{i\theta}) - f(1)| \leq N|\theta|\}$ is closed; (b) A_N is nowhere dense. [Hint: for (b) use $f_0(z) = (1-z)^{1/2}$ and consider $f + f_0$.]
- (Everywhere-singular smooth functions.) Let $C^\infty(I)$, $I = [0, 1]$, carry the metric $d(f, g) = \sum_{n \geq 0} 2^{-n} \rho_n(f - g) / (1 + \rho_n(f - g))$ with $\rho_n(h) = \sup_I |h^{(n)}|$. Call f singular at x_0 if its Taylor series at x_0 diverges. (a) Show $(C^\infty(I), d)$ is complete. (b) Prove the set of f singular at every point is generic. [Hint: for (b), $F_K = \{f : |f^{(n)}(x^*)|/n! \leq K^n \text{ for some } x^*, \text{ all } n\}$ is closed and nowhere dense.]
- (L^∞ cannot be replaced by L^q .) Show the L^∞ in Theorem 42.3.1 cannot be replaced by any L^q , $1 \leq q < \infty$: there is a closed infinite-dimensional subspace of $L^1([0, 1])$ consisting of functions belonging to L^q for all $1 \leq q < \infty$. [Hint: use the lacunary/independent functions of the next chapter.]
- * (Entire functions with dense orbits.) Let H be the entire functions with metric $d(f, g) = \sum_{n \geq 1} 2^{-n} \|f - g\|_{K_n} / (1 + \|f - g\|_{K_n})$, K_n the closed disc of radius n , $\|f\|_K = \sup_K |f|$; then H is a complete metric space. Birkhoff's theorem gives an entire F with $\{F(z + n)\}_n$ dense in H ; MacLane's gives entire G with $\{G^{(n)}\}_n$ dense. Using Exercise 14, deduce that each property holds for a generic set of entire functions, hence the set with *both* is generic.

Additional exercises.

- Verify the three metric axioms for the Hausdorff distance on nonempty compact subsets of $\mathbb{R}^2$, and give two compact sets at Hausdorff distance exactly 1.
- Using Theorem 42.3.1, explain why the span of $\{1, \cos x, \cos 2x, \dots, \cos Nx\}$ in $L^2([0, 2\pi])$, although contained in L^∞ , is consistent with the theorem (it is finite-dimensional), and why an infinite trigonometric closed subspace cannot lie in L^∞ .
- Show that on a finite measure space $\|f\|_p$ is nondecreasing in p , and compute $\lim_{p \rightarrow \infty} \|f\|_p = \|f\|_\infty$ for a simple step function.
- Compute $\mathbb{E}[B_s B_t] = \min(s, t)$ for standard Brownian motion directly from independent increments, and use it to find $\text{Var}(B_t - B_s)$.
- Verify that the standard Gaussian density has inflection points at $x = \pm 1$ and that the area within one standard deviation is approximately 0.68; relate this to the shaded region in the bell-curve figure.
- (Non-example.) Show the collection of compact subsets of $[0, 1]$ of *positive* Lebesgue measure is *not* generic in the Hausdorff metric, contrasting with Exercise 15(a).
- Prove the law of large numbers consequence: if X_k are i.i.d. with mean μ and finite variance, then $\frac{1}{n} \sum_{k=1}^n X_k \rightarrow \mu$ in L^2 , by computing the variance of the average.

8. Draw a labeled diagram of the nested L^p spaces on a finite measure space and mark where a closed subspace forced into L^∞ by Grothendieck's theorem must sit.
9. Using Theorem 42.3.4, compute the moment generating function $\mathbb{E}[e^{tX}]$ of $X \sim N(0, \sigma^2)$ and confirm $\mathbb{E}[X^2] = \sigma^2$ by differentiating twice at $t = 0$.
10. Explain heuristically, using the $\sqrt{t}$ scaling of Brownian increments, why the sample path $t \mapsto B_t$ is almost surely nowhere differentiable, paralleling the generic statement of Theorem 42.1.9.

Chapter 43

Rudiments of Probability Theory

43.1 Random Variables, Distributions, and Independence

Probability theory, read through the lens of measure and integration, is the study of a measure space whose total mass is normalized to one. The dictionary is small but it reorganizes everything: a measure space (X, m) with $m(X) = 1$ is a *probability space*; a measurable function f on it is a *random variable*; its integral is its *mean* or *expectation*; “almost everywhere” becomes “almost surely.” Stein prefers the word “function” to “random variable,” and we follow that usage, but the two are interchangeable. We build the whole subject from the simplest possible source – coin flips – and watch how independence forces order out of randomness.

Probability spaces and the coin-flip model

Definition 43.1.1. A *probability space* is a triple $(X, \mathcal{M}, m)$ where $\mathcal{M}$ is a σ -algebra of subsets of X (the *events*) and m is a measure on $\mathcal{M}$ with $m(X) = 1$. A *random variable* (or simply *function*) is an $\mathcal{M}$ -measurable $f : X \rightarrow \mathbb{R}$ (or $\mathbb{R}^d$). One says a property holds *almost surely* (a.s.) if it holds outside a set of measure zero.

The motivating model is coin tossing. Two players flip a fair coin N times; player A wins a dollar on heads, loses one on tails. The 2^N outcomes are the points of $\mathbb{Z}_2^N$, the N -fold product of the two-point space $\mathbb{Z}_2 = \{0, 1\}$ (0 for heads, 1 for tails). Equal likelihood of heads/tails means each point carries measure 2^{-N} , so $m(\mathbb{Z}_2^N) = 1$. The winnings on the n th flip are recorded by the function

$$r_n(x) = 1 - 2x_n = \begin{cases} +1, & x_n = 0 \text{ (heads),} \\ -1, & x_n = 1 \text{ (tails),} \end{cases}$$

and total winnings after N flips are $S_N(x) = \sum_{n=1}^N r_n(x)$.

Proposition 43.1.2 (Distribution of S_N). *If k is an integer of the same parity as N , then*

$$m(\{x : S_N(x) = k\}) = 2^{-N} \frac{N!}{\left(\frac{N+k}{2}\right)! \left(\frac{N-k}{2}\right)!}.$$

The maximum (for N even) is at $k = 0$, equal to $2^{-N} N! / ((N/2)!)^2 \approx \sqrt{2/\pi} N^{-1/2}$ by Stirling, far larger than the minimum value 2^{-N} at $k = \pm N$.

Proof. A point x with $S_N(x) = k$ has $N_1 = (N + k)/2$ heads and $N_2 = (N - k)/2$ tails; there are $\binom{N}{N_1} = N! / (N_1! N_2!)$ such points, each of measure 2^{-N} . For the monotonicity, $\frac{m(S_N = k+2)}{m(S_N = k)} = \frac{N - k}{N + k + 2}$, which exceeds 1 for $k \leq -2$ and is below 1 for $k \geq 0$; hence the maximum is at $k = 0$. Stirling’s formula $N! = \sqrt{2\pi} N^{N+1/2} e^{-N} (1 + O(N^{-1/2}))$ gives the stated asymptotic for the peak. $\square$

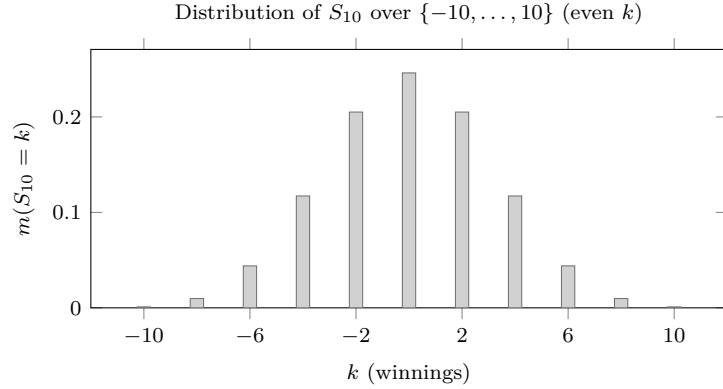


Figure 43.1. The binomial weights $2^{-N} \binom{N}{(N+k)/2}$ peak at $k = 0$ and decay like a Gaussian – the seed of the central limit theorem.

Worked Example 43.1.3 (Computing a coin-flip probability). For $N = 4$ fair flips compute $m(\{S_4 = 2\})$ and $m(\{S_4 = 0\})$, and check they sum correctly with the rest.

Recognition. Direct application of Proposition 43.1.2: count favorable sequences and divide by 2^N .

Solution. $S_4 = 2$ means 3 heads, 1 tail: $N_1 = 3, N_2 = 1$, so $m = 2^{-4} \binom{4}{3} = 4/16 = 1/4$. $S_4 = 0$ means 2 heads, 2 tails: $m = 2^{-4} \binom{4}{2} = 6/16 = 3/8$. *Check.* The full distribution is $\frac{1}{16}(1, 4, 6, 4, 1)$ over $k = -4, -2, 0, 2, 4$, summing to 1; the peak $3/8$ at $k = 0$ is indeed the largest, consistent with the proposition. ■

The infinite product space and the Rademacher functions

To take limits as $N \rightarrow \infty$ we pass to the infinite product $X = \mathbb{Z}_2^\infty = \{x = (x_1, x_2, \dots) : x_n \in \{0, 1\}\}$ with the product measure m , characterized on *cylinder sets* $E = \{x : x_j = a_j, 1 \leq j \leq N\}$ by $m(E) = 2^{-N}$. This m extends to the σ -algebra generated by the cylinder sets, with $m(X) = 1$. The functions $r_n(x) = 1 - 2x_n$ now live on X for all n ; viewed as functions on $[0, 1]$ they are the *Rademacher functions*.

Definition 43.1.4 (Rademacher functions). Under the binary-expansion map $D : X \rightarrow [0, 1]$, $D(x) = \sum_{j \geq 1} x_j 2^{-j}$, the product measure m corresponds to Lebesgue measure, and r_n becomes the function on $[0, 1]$ equal to $+1$ on the first half of each dyadic interval of length $2^{-(n-1)}$ and -1 on the second; equivalently $r_n(t) = \text{sgn} \sin(2^n \pi t)$ and $r_n(t) = r_1(2^{n-1}t)$. Each r_n takes values ± 1 with $\int_0^1 r_n = 0$ and $\int_0^1 r_n^2 = 1$.

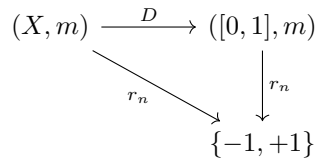


Figure 43.2. The binary-expansion map $D(x) = \sum_{j \geq 1} x_j 2^{-j}$ carries the product measure m to Lebesgue measure, and each Rademacher function on X factors as $r_n^X = r_n^{[0,1]} \circ D$.

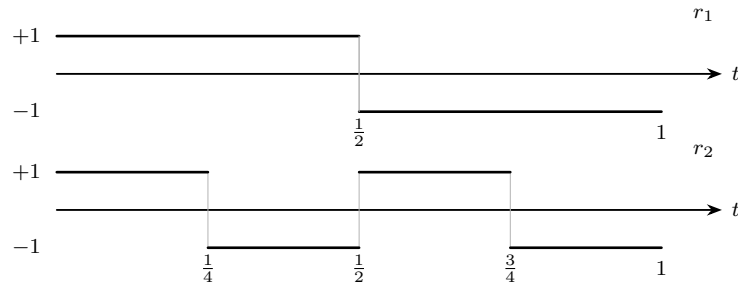


Figure 43.3. The first two Rademacher functions r_1, r_2 on $[0, 1]$: square waves of doubling frequency, each ± 1 with mean 0.

Definition 43.1.5 (Mutual independence). A sequence $\{f_n\}$ of real measurable functions on (X, m) is *mutually independent* if for every choice of Borel sets $B_n \subseteq \mathbb{R}$,

$$m\left(\bigcap_{n=1}^{\infty} \{x : f_n(x) \in B_n\}\right) = \prod_{n=1}^{\infty} m(\{x : f_n(x) \in B_n\}).$$

Sets $\{E_n\}$ are mutually independent if their indicators are. A finite version holds verbatim. Pairwise independence does *not* imply mutual independence.

Proposition 43.1.6 (Independence and products of integrals). *If $f_1, \dots, f_n$ are bounded and mutually independent, then*

$$\int_X f_1 \cdots f_n \, dm = \left(\int_X f_1 \, dm\right) \cdots \left(\int_X f_n \, dm\right).$$

In particular, if each f_j depends only on the j th coordinate of a product space – $f_j(x) = F_j(x_j)$ – then the $\{f_j\}$ are mutually independent; this is exactly the situation of the Rademacher functions.

Proof. The product formula holds when each f_j is a finite linear combination of indicators of cylinder sets, by the definition of product measure; pass to the limit by approximation to obtain it for bounded f_j . For the coordinate functions, $E_n = \{x : f_n(x) \in B_n\}$ is a cylinder set with $m(E_n) = m_n(\{x_n : F_n(x_n) \in B_n\})$, and finite products of cylinder measures factor, so (43.1) holds in the limit $N \rightarrow \infty$. Since $r_n(x) = 1 - 2x_n$ depends only on x_n , the Rademacher functions are mutually independent. $\square$

$$m\left(\bigcap_{n=1}^N \{f_n \in B_n\}\right) = \prod_{n=1}^N m(\{f_n \in B_n\}). \quad (43.1)$$

Worked Example 43.1.7 (Pairwise but not mutually independent). On $[0, 1]$ let $f_1 = r_1$, $f_2 = r_2$, $f_3 = r_1 r_2$. Show any two are independent but the three are not.

Recognition. Trigger: test Definition 43.1.5 on a triple. Independence of a pair of ± 1 functions reduces to checking the four sign-combinations are equally likely.

Solution. Each f_i is ± 1 with mean 0. For a pair, e.g. $(f_1, f_3) = (r_1, r_1 r_2)$: the four sign patterns of $(r_1, r_1 r_2)$ correspond bijectively to the four equally likely patterns of (r_1, r_2) , so $m(f_1 = \pm 1, f_3 = \pm 1) = \frac{1}{4} = m(f_1 = \pm 1)m(f_3 = \pm 1)$; likewise for the other pairs. But the three are dependent: $f_1 f_2 f_3 = r_1^2 r_2^2 = 1$ identically, so $\{f_1 = 1, f_2 = 1, f_3 = -1\}$ is impossible, having measure 0, whereas the product of marginals is $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \neq 0$.

Check. The constraint $f_3 = f_1 f_2$ ties the third sign to the first two – exactly the failure of mutual independence despite pairwise independence.

Distribution measures, mean, and variance

Definition 43.1.8 (Distribution measure). The *distribution measure* of a real function f on (X, m) is the Borel probability measure $\mu = \mu_f$ on $\mathbb{R}$ with

$$\mu(B) = m(\{x : f(x) \in B\}) \quad \text{for all Borel } B \subseteq \mathbb{R}.$$

It is a probability measure since $\mu(\mathbb{R}) = 1$. It connects to the distribution function $\lambda_f(\alpha) = m(\{|f| > \alpha\}) = \mu_{|f|}((\alpha, \infty))$.

Theorem 43.1.9 (Change of variables to the distribution). *For a real measurable f with distribution μ : f is integrable iff $\int_{\mathbb{R}} |t| \, d\mu(t) < \infty$, and then $\int_X f \, dm = \int_{\mathbb{R}} t \, d\mu(t)$. More generally $f \in L^p(X, m)$ iff $\int_{\mathbb{R}} |t|^p \, d\mu(t) < \infty$, with $\|f\|_{L^p}^p = \int_{\mathbb{R}} |t|^p \, d\mu$, and for any nonnegative continuous (or continuous bounded) G on $\mathbb{R}$,*

$$\int_X G(f)(x) \, dm = \int_{\mathbb{R}} G(t) \, d\mu(t). \quad (43.2)$$

Proof. For f bounded, $\sum_k G(k/n) m(\{k/n < f < (k+1)/n\})$ is a Riemann-type sum converging to $\int_X G(f) \, dm$ on one side and to $\int_{\mathbb{R}} G \, d\mu$ on the other as $n \rightarrow \infty$; this gives (43.2) for bounded f , and truncation plus monotone convergence extends it to the stated G . Taking $G(t) = |t|$ and $G(t) = |t|^p$ yields the integrability criteria and the norm identities. $\square$

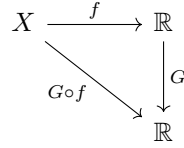


Figure 43.4. Change of variables to the distribution. With $\mu = f_*m$ the pushforward of m along f , integrating $G \circ f$ over (X, m) equals integrating G over $(\mathbb{R}, \mu)$: $\int_X G(f) dm = \int_{\mathbb{R}} G d\mu$.

Definition 43.1.10 (Mean and variance). If f is integrable its *mean* (expectation) is $m_0 = \int_X f dm = \int_{\mathbb{R}} t d\mu(t)$. If f is square integrable its *variance* is

$$\sigma^2 = \int_X (f - m_0)^2 dm = \int_{\mathbb{R}} (t - m_0)^2 d\mu(t) = \int_{\mathbb{R}} t^2 d\mu - m_0^2.$$

Worked Example 43.1.11 (Mean and variance of a single Rademacher function). Compute the mean and variance of r_n , and its distribution measure.

Recognition. Use Definition 43.1.8 and 43.1.10; r_n takes only the values ± 1 .

Solution. $m(\{r_n = 1\}) = m(\{r_n = -1\}) = \frac{1}{2}$, so $\mu_{r_n} = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_{+1}$. Then $m_0 = \int t d\mu = \frac{1}{2}(-1) + \frac{1}{2}(1) = 0$ and $\sigma^2 = \int t^2 d\mu - 0 = \frac{1}{2}(1) + \frac{1}{2}(1) = 1$. *Check.* Directly $\int_0^1 r_n = 0$ and $\int_0^1 r_n^2 = \int_0^1 1 = 1$, agreeing. ■

Worked Example 43.1.12 (A uniform random variable's distribution and moments). Let $f(t) = t$ on $X = [0, 1]$ with Lebesgue measure. Find μ_f , the mean, and the variance.

Recognition. f is the identity, so its distribution is the pushforward of Lebesgue measure – itself.

Solution. For $B \subseteq \mathbb{R}$, $\mu_f(B) = m(\{t : t \in B\}) = m(B \cap [0, 1])$, i.e. μ_f is uniform (Lebesgue) measure on $[0, 1]$. Mean $m_0 = \int_0^1 t dt = \frac{1}{2}$; second moment $\int_0^1 t^2 dt = \frac{1}{3}$, so $\sigma^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. *Check.* A uniform variable on $[0, 1]$ is standard textbook with mean $1/2$ and variance $1/12$. ■

Worked Example 43.1.13 (Distribution of a transformed variable). With f uniform on $[0, 1]$, find the distribution measure of $g = -\log f$ and identify it.

Recognition. Trigger: pushforward under a monotone map. Compute the distribution function $\mu_g((-\infty, a])$ and differentiate to get a density.

Solution. For $a > 0$, $\{g \leq a\} = \{-\log f \leq a\} = \{f \geq e^{-a}\}$, which has Lebesgue measure $1 - e^{-a}$; and $\mu_g((-\infty, a]) = 0$ for $a \leq 0$. So g has density $\frac{d}{da}(1 - e^{-a}) = e^{-a}$ on $(0, \infty)$: g is *exponential* with parameter 1. *Check.* Mean $\int_0^\infty ae^{-a} da = 1$; and $E(g) = \int_0^1 (-\log t) dt = 1$, consistent. ■

First law of large numbers: $S_N/N \rightarrow 0$

Proposition 43.1.14 (L^2 size of S_N). $\|S_N\|_{L^2} = N^{1/2}$.

Proof. $\|S_N\|_{L^2}^2 = \int_X (\sum_n r_n)^2 dm = \sum_{n,m} \int_X r_n r_m dm$. By orthonormality of the Rademacher functions ($\int r_n r_m = \delta_{nm}$, from Proposition 43.1.6 when $n \neq m$ and $\int r_n^2 = 1$), only the N diagonal terms survive, giving N . Hence $\|S_N\|_{L^2} = N^{1/2}$. □

So the typical size of S_N is of order $N^{1/2}$, vastly smaller than the maximum N . This already forces the average S_N/N to vanish, in the following weak sense.

Definition 43.1.15 (Convergence in probability). $f_N \rightarrow f$ in probability (measure-theoretically: in measure) if for every $\varepsilon > 0$, $m(\{x : |f_N(x) - f(x)| > \varepsilon\}) \rightarrow 0$ as $N \rightarrow \infty$.

Corollary 43.1.16 (Weak law for coin flips). $S_N/N \rightarrow 0$ in probability. More precisely $S_N/N^\alpha \rightarrow 0$ in probability for any $\alpha > 1/2$.

Proof. By Tchebychev's inequality and Proposition 43.1.14,

$$m(\{|S_N/N| > \varepsilon\}) = m(\{|S_N| > \varepsilon N\}) \leq \frac{1}{\varepsilon^2 N^2} \int_X |S_N|^2 dm = \frac{N}{\varepsilon^2 N^2} = \frac{1}{\varepsilon^2 N} \rightarrow 0.$$

Replacing N by N^α gives the bound $1/(\varepsilon^2 N^{2\alpha-1}) \rightarrow 0$ exactly when $\alpha > 1/2$. $\square$

Worked Example 43.1.17 (Tchebychev gives an explicit confidence bound). For $N = 10^4$ fair flips, bound $m(\{|S_N| > 200\})$.

Recognition. Apply Tchebychev with $\|S_N\|_{L^2}^2 = N$.

Solution. $m(\{|S_N| > 200\}) \leq N/200^2 = 10^4/4 \cdot 10^4 = 1/4$. *Check.* The true probability is far smaller (the CLT will give $\approx m(|Z| > 2) \approx 0.046$), but Tchebychev already certifies at most 25% – the bound is correct, if crude. $\blacksquare$

Khinchin's inequality and random series

The deeper exploitation of independence is that all L^p norms of a Rademacher sum are comparable to its L^2 norm.

Theorem 43.1.18 (Khinchin's inequality). *For each $p < \infty$ there is a constant A_p so that for every finite (or L^2 -convergent) sum $F = \sum_n a_n r_n$,*

$$\|F\|_{L^p} \leq A_p \|F\|_{L^2} = A_p \left(\sum_n a_n^2 \right)^{1/2}.$$

Proof sketch. Normalize $\sum a_n^2 = 1$. The independence and mean-zero property of the r_n give, for real t , $\int_0^1 e^{tF} = \prod_n \int_0^1 e^{ta_n r_n} = \prod_n \cosh(ta_n) \leq \prod_n e^{t^2 a_n^2 / 2} = e^{t^2 / 2}$, using $\cosh u \leq e^{u^2 / 2}$. Hence $\int_0^1 \cosh(tF) \leq e^{t^2 / 2}$, and choosing t and applying Markov's inequality to $e^{t|F|}$ yields the sub-Gaussian tail $m(\{|F| > \alpha\}) \leq 2e^{-\alpha^2 / 2}$. Integrating $p\alpha^{p-1}m(|F| > \alpha)$ in α bounds $\|F\|_{L^p}^p$ by a constant depending only on p , giving A_p . $\square$

Theorem 43.1.19 (Random series; a zero-one dichotomy). *Let $\{a_n\}$ be real (or complex).*

1. *If $\sum_n |a_n|^2 < \infty$, then $\sum_n a_n r_n(t)$ converges for almost every $t \in [0, 1]$.*
2. *If $\sum_n |a_n|^2 = \infty$, then $\sum_n a_n r_n(t)$ diverges for almost every t .*

Thus a random $\pm$ series converges a.e. exactly when it converges in L^2 – and the event of convergence has measure 0 or 1, never in between.

Proof of (1). If $\sum a_n^2 < \infty$ the partial sums $S_N(f) = \sum_{n \leq N} a_n r_n$ converge in L^2 to some $f \in L^2([0, 1])$. For the dyadic averaging operators $E_N(f)(t) = \frac{1}{|I|} \int_I f$ over the dyadic interval $I \ni t$ of length 2^{-N} , one checks $E_N(f) = S_N(f)$ for all N : indeed $E_N(r_n) = r_n$ for $N \geq n$ (as r_n is constant on each such interval) and $E_N(r_n) = 0$ for $n > N$ (its integral over each interval vanishes). By the Lebesgue differentiation theorem $E_N(f)(t) \rightarrow f(t)$ a.e., so the series converges a.e. Part (2) follows from the zero-one law of §43.2.13 (any positive-measure convergence set must be conull, contradicting L^2 divergence). $\square$

This is the striking fact that, although $\sum 1/n$ diverges, the randomized series $\sum (\pm) 1/n$ converges for almost every choice of signs (here $a_n = 1/n$, $\sum a_n^2 = \sum 1/n^2 < \infty$).

Worked Example 43.1.20 (Which random harmonic-type series converge?). Decide a.e. convergence of (a) $\sum_n \frac{r_n(t)}{n}$, (b) $\sum_n \frac{r_n(t)}{\sqrt{n}}$, (c) $\sum_n \frac{r_n(t)}{n^{3/4}}$.

Recognition. Apply Theorem 43.1.19: convergence a.e. iff $\sum a_n^2 < \infty$.

Solution. (a) $a_n = 1/n$, $\sum 1/n^2 < \infty$: converges a.e. (b) $a_n = 1/\sqrt{n}$, $\sum 1/n = \infty$: diverges a.e. (c) $a_n = n^{-3/4}$, $\sum n^{-3/2} < \infty$: converges a.e. *Check.* The threshold is the square-summability of coefficients, independent of the deterministic behavior of $\sum a_n$ (which diverges in all three cases except by sign cancellation). $\blacksquare$

Worked Example 43.1.21 (Khinchin comparison of L^4 and L^2 norms). For $F = \sum_n a_n r_n$ (finite, real) show $\|F\|_{L^4}^4 = 3(\sum a_n^2)^2 - 2\sum a_n^4 \leq 3\|F\|_{L^2}^4$, hence $\|F\|_{L^4} \leq 3^{1/4}\|F\|_{L^2}$.

Recognition. Trigger: expand the fourth power and use orthogonality of products of distinct Rademacher functions.

Solution. $\int F^4 = \sum_{i,j,k,l} a_i a_j a_k a_l \int r_i r_j r_k r_l$. By independence and $\int r_n = 0$, $\int r_i r_j r_k r_l \neq 0$ only when indices pair up: all four equal ($\int r_i^4 = 1$, contributing $\sum a_i^4$) or two distinct pairs (there are 3 ways to pair four slots, each $\int r_i^2 r_j^2 = 1$, contributing $3\sum_{i \neq j} a_i^2 a_j^2$). Thus $\int F^4 = \sum a_i^4 + 3\sum_{i \neq j} a_i^2 a_j^2 = 3(\sum a_i^2)^2 - 2\sum a_i^4$. Since $\sum a_i^4 \geq 0$, this is $\leq 3(\sum a_i^2)^2 = 3\|F\|_{L^2}^4$. *Check.* For a single term $F = a_1 r_1$, $\int F^4 = a_1^4$ and $3(a_1^2)^2 - 2a_1^4 = a_1^4$, agreeing; the constant $A_4 = 3^{1/4}$ is the sharp value. ■

Caution. Mutual independence is strictly stronger than pairwise independence, and far stronger than uncorrelatedness ($\int fg = \int f \int g$). The Rademacher functions are orthonormal (hence uncorrelated) and mutually independent, and it is the latter that powers Khinchin's inequality – orthogonality alone gives only the L^2 identity of Proposition 43.1.14, not control of higher L^p norms.

Worked Example 43.1.22 (A sub-Gaussian tail for a normalized Rademacher sum). Let $F = \sum a_n r_n$ with $\sum a_n^2 = 1$. Show $m(\{|F| > \alpha\}) \leq 2e^{-\alpha^2/2}$.

Recognition. Trigger: the exponential-moment method from the proof of Khinchin. Bound $\int e^{tF}$ and optimize t .

Solution. By independence and $\cosh u \leq e^{u^2/2}$, $\int_0^1 e^{tF} = \prod_n \cosh(ta_n) \leq e^{t^2 \sum a_n^2/2} = e^{t^2/2}$. By Markov, $m(\{F > \alpha\}) \leq e^{-t\alpha} \int e^{tF} \leq e^{t^2/2 - t\alpha}$. Minimizing over t at $t = \alpha$ gives $e^{-\alpha^2/2}$; the same bound holds for $-F$, so by a union bound $m(\{|F| > \alpha\}) \leq 2e^{-\alpha^2/2}$. *Check.* At $\alpha = 0$ the bound is 2 (vacuous, as it must be since probabilities are ≤ 1); the Gaussian decay rate $e^{-\alpha^2/2}$ matches the limiting normal law to come. ■

The Borel–Cantelli lemma

Theorem 43.1.23 (Borel–Cantelli). Let $\{A_n\}$ be events and $\limsup_n A_n = \bigcap_n \bigcup_{k \geq n} A_k$ the set of points lying in infinitely many A_n .

1. If $\sum_n m(A_n) < \infty$ then $m(\limsup_n A_n) = 0$.
2. If $\sum_n m(A_n) = \infty$ and the A_n are mutually independent then $m(\limsup_n A_n) = 1$.

Proof. (1) $m(\limsup A_n) \leq m(\bigcup_{k \geq n} A_k) \leq \sum_{k \geq n} m(A_k) \rightarrow 0$ as $n \rightarrow \infty$ (tail of a convergent series). (2) The complement of $\limsup A_n$ is $\bigcup_n \bigcap_{k \geq n} A_k^c$; by independence $m(\bigcap_{k=n}^M A_k^c) = \prod_{k=n}^M (1 - m(A_k)) \leq \exp(-\sum_{k=n}^M m(A_k)) \rightarrow 0$ as $M \rightarrow \infty$ since $\sum m(A_k) = \infty$. Hence each $\bigcap_{k \geq n} A_k^c$ is null, so their union is null, and $m(\limsup A_n) = 1$. □

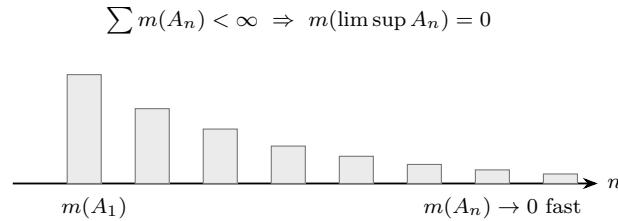


Figure 43.5. Borel–Cantelli (convergence half): if the event sizes sum, almost no point is hit infinitely often. (Schematic.)

Worked Example 43.1.24 (Borel–Cantelli forces a.e. bound on coin streaks). Let A_n be the event “flips $2^n, \dots, 2^n + n - 1$ are all heads.” Show almost surely only finitely many A_n occur.

Recognition. The blocks are disjoint, hence independent; apply Borel–Cantelli part (1) after checking summability.

Solution. $m(A_n) = 2^{-n}$ (a run of n specified heads), and $\sum_n 2^{-n} = 1 < \infty$. By Theorem 43.1.23(1), $m(\limsup A_n) = 0$: almost every outcome has only finitely many such all-heads blocks. *Check.* The divergent

counterpart, runs of length $\lfloor \log_2 n \rfloor$, has $\sum m(A_n) \approx \sum 1/n = \infty$ and (being independent) occurs infinitely often by part (2) – the classic “longest run” threshold. ■

Exercises (Stein, Ch.5) – random variables, independence, random series.

1. Consider $S_N(x) = \sum_{n=1}^N r_n(x)$ with N odd. (a) Calculate $m(\{S_N = k\})$ and show that, as k ranges over the integers, the maximum is attained at $k = -1$ and $k = 1$. (b) Adapt the proof of the even case to show that for odd N , $m(\{a < S_N/N^{1/2} < b\}) \rightarrow \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2/2} dt$ as $N \rightarrow \infty$.
2. Find three functions f_1, f_2, f_3 so that any pair are mutually independent but the three are not. [Hint: $f_1 = r_1$, $f_2 = r_2$, and express f_3 in terms of r_1, r_2 .]
3. The collection $\{r_n\}$ of mutually independent functions on $[0, 1]$ cannot be much enlarged: prove that if a function f is adjoined to $\{r_n\}$ and the resulting collection is still mutually independent, then f is constant. [Hint: see also Exercise 16.]
4. Suppose μ, ν are two finite measures on a space X agreeing on a collection $\mathcal{C}$ of sets. If $\mathcal{C}$ contains X and is closed under finite intersections, show $\mu = \nu$ on the σ -algebra generated by $\mathcal{C}$. [Hint: $\mu = \nu$ holds first on finite unions of sets of $\mathcal{C}$ via inclusion-exclusion.]
5. Prove that $\mathbb{R}^d$ -valued $f_1, \dots, f_n$ are mutually independent iff their joint distribution equals the product of the individual distributions: $\mu_{f_1, \dots, f_n} = \mu_{f_1} \times \dots \times \mu_{f_n}$ as measures on $\mathbb{R}^{nd} = \mathbb{R}^d \times \dots \times \mathbb{R}^d$. [Hint: check equality on cylinder sets and use Exercise 4.]
6. Let $\{f_n\}$ be mutually independent on (X, m) . Prove there is a probability space (X', m') , an infinite product $X' = \prod_n X_n$ with product measure m' of the m_n , and functions $\{g_n\}$ on X' with the same joint distributions as $\{f_n\}$, yet each g_n depending only on the n th coordinate. [Hint: take $(X_n, m_n) = (X, m)$ and use Exercise 5.]
7. Verify further facts about distribution measures. (a) For $f = (f_1, \dots, f_k)$ with each f_j an $\mathbb{R}^d$ -valued function and distribution μ on $\mathbb{R}^{dk}$, and L a linear transformation of $\mathbb{R}^{dk}$, the distribution of $L(f)$ is μ_L , where $\mu_L(A) = \mu(L^{-1}A)$. (b) If each f_j is Gaussian with covariance $\sigma_j^2 I$ and the f_j are mutually independent, then $c_1 f_1 + \dots + c_k f_k$ is Gaussian with covariance $(c_1^2 \sigma_1^2 + \dots + c_k^2 \sigma_k^2)I$. [Hint: compute characteristic functions.]
8. Consider $L^2(\Omega, \mathbb{R}^d)$, square-integrable $\mathbb{R}^d$ -valued functions on a probability space (Ω, P) . A closed subspace G is a *Gaussian subspace* if spanned by mutually independent functions $\{f_n\}$, each Gaussian with mean zero and covariance $\{\sigma_n^2 I\}$. Prove that mutually orthogonal elements $F_1, \dots, F_k \in G$ are mutually independent (the converse being immediate). [Hint: reduce to finite-dimensional G , normalize so $\|f_j\|_{L^2} = \|F_j\|_{L^2} = 1$, take an orthogonal L with $L(f_j) = F_j$, and apply Exercises 5 and 7.]
9. Consider the two types of convergence of $\{f_n\}$ to f on a probability space: (i) $f_n \rightarrow f$ a.e.; (ii) $f_n \rightarrow f$ in the sense of weak convergence of distribution measures. Prove (i) implies (ii), but the implication cannot be reversed. [Hint: for continuous bounded φ , $\int \varphi(f) d\mu = \int \varphi d\mu_f$; apply dominated convergence.]
10. On $[0, 1]$ with Lebesgue measure, construct a function f whose distribution measure is normal. [Hint: use the error function $\text{Erf}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$ and its inverse.]
11. Prove identity (43.2): if G is nonnegative continuous (or continuous bounded) on $\mathbb{R}$ and f is real measurable with distribution $\mu = \mu_f$, then $\int_X G(f) dm = \int_{\mathbb{R}} G d\mu$. [Hint: $\sum_k G(k/n) m(\{k/n < f < (k+1)/n\})$ converges to both integrals.]
12. The Rademacher sequence is far from complete on $L^2([0, 1])$; in fact it cannot be completed by adjoining any finite collection. Prove this two ways: (a) using the functions $\{r_n r_m\}$ for $n < m$; (b) using the L^p inequality of Khinchin (Lemma 43.1.18). See also Exercise 16.
13. Consider the power series $\sum_{n \geq 1} (\pm) a_n z^n = \sum_{n \geq 1} r_n(t) a_n z^n = F(z, t)$ where $\sum |a_n|^2 = \infty$ and $\limsup |a_n|^{1/n} \leq 1$. Show that for almost every t , $F(z, t)$ cannot be analytically continued outside the unit disc. [Hint: argue as in Theorem 43.1.19(2) using Abel summation in place of Cesàro.]

14. Show that the $L^2([0, 1])$ span of $\{r_n\}$ is exactly the subspace of f for which $E_N(f) = S_N(f)$ for all N , where E_N are the conditional expectations over the dyadic intervals of length 2^{-N} .
15. (Walsh–Paley functions) Define $\{w_n\}$ on $[0, 1]$ by $w_0 = 1$, and for $n = 2^{k_1} + \dots + 2^{k_\ell}$ ($0 \leq k_1 < \dots < k_\ell$), $w_n = \prod_{j=1}^\ell r_{k_j+1}$; in particular $w_{2^k-1} = r_k$. (a) Prove $\{w_n\}_{n \geq 0}$ is a complete orthonormal system on $L^2([0, 1])$. (b) Show the w_n are the continuous characters of the compact abelian group $\mathbb{Z}_2^\infty$. [Hint: with $K_N = \sum_{k < N} w_k$ and $N = 2^n$, show $K_{2^n}(t) = \prod_{j=1}^n (1 + r_j(t))$, equal to 2^n on $[0, 2^{-n}]$ and 0 otherwise; relate the convolution to E_n . See Problem 2.]
16. (Sharper Khinchin) Let $F = \sum_n a_n r_n$ with a_n real and $\sum a_n^2 = 1$. Prove (a) $\int_0^1 e^{\mu|F|^2} dt \leq 2e^\mu$ for all $\mu \geq 0$ [understood as $\int_0^1 e^{\mu F^2} \leq 2e^{c\mu}$ for suitable c]; (b) consequently $\int_0^1 e^{c|F|^2} dt < \infty$ for some $c > 0$. [Hint: (a) gives $m(\{|F| > \alpha\}) \leq 2e^{\mu-\mu\alpha}$; optimize μ , get (b) with $c < 1/4$.]
17. Prove there exists $f \in L^p([0, 2\pi])$ for all $p < \infty$, $f \sim \sum_{n=-\infty}^\infty c_n e^{in\theta}$, with $\sum |c_n|^q = \infty$ for all $q < 2$. Hence the Hausdorff–Young inequality fails for $p > 2$. [Hint: use the random-series theorem.]
18. Suppose $F(t) = \sum_n a_n r_n(t)$ with $\sum a_n^2 < \infty$. (a) Prove directly $\|F\|_{L^4} \leq A \|F\|_{L^2}$. (b) Deduce $\|F\|_{L^2} \leq A' \|F\|_{L^1}$. (c) Conclude $\|F\|_{L^p} \leq A_p \|F\|_{L^1}$ for $1 \leq p < \infty$. [Hint: (a) expand $\int F^4$ and use orthogonality of $r_n r_m$; (b) Hölder; (c) Khinchin.]
19. (Borel–Cantelli) Let $\{A_n\}$ be subsets of a probability space X . (a) If $\sum_n m(A_n) < \infty$ then $m(\limsup_n A_n) = 0$, where $\limsup_n A_n = \bigcap_n \bigcup_{k \geq n} A_k$. (b) If $\sum_n m(A_n) = \infty$ and the $\{A_n\}$ are mutually independent, then $m(\limsup_n A_n) = 1$. [Hint: for (b), $m(\bigcap_{k=r}^n A_k^c) = \prod_{k=r}^n (1 - m(A_k))$.]
20. (Normal numbers) For $\alpha \in [0, 1]$ outside the dyadic rationals, write $\alpha = \sum_j x_j 2^{-j}$ with $x_j \in \{0, 1\}$, and let $\#_N(\alpha)$ be the number of 1's among the first N digits; α is *normal* if $\#_N(\alpha)/N \rightarrow 1/2$. (a) Prove almost every $\alpha \in [0, 1]$ is normal. (b) More generally, for an integer $q \geq 2$ and the q -expansion $\alpha = \sum_j x_j q^{-j}$ ($x_j \in \{0, \dots, q-1\}$), with $\#_{p,N}(\alpha)$ the count of digits equal to p in the first N places, show almost all α satisfy $\#_{p,N}(\alpha)/N \rightarrow 1/q$ for every $0 \leq p \leq q-1$ (*normal to base q*). [Hint: take the product space $\prod \mathbb{Z}_q$ with uniform factors; the law of large numbers (Theorem 43.2.2) applies.]

Problems (Stein, Ch.5) – random series and Walsh–Paley.

1. In the context of Bernoulli trials with $0 < p, q < 1$, $p + q = 1$, let $D(x) = \sum_n x_n 2^{-n}$ map $\mathbb{Z}_2^\infty \rightarrow [0, 1]$. The measure m_p pushes to a measure μ_p , written symbolically as the Riesz product $\mu_p = \prod_{n=1}^\infty (1 + (p - q)r_n(t)) dt$, meaning: with $F_N(t) = \int_0^t \prod_{n=1}^N (1 + (p - q)r_n(s)) ds$, one shows (a) each F_N is increasing on $[0, 1]$; (b) $F_N(0) = 0$, $F_N(1) = 1$; (c) $F_N \rightarrow F$ uniformly; (d) $\mu_p = dF$, i.e. $\mu_p((a, b)) = F(b) - F(a)$; (e) if $p \neq 1/2$, μ_p is completely singular ($dF/dt = 0$ a.e.). [Hint: for a dyadic interval $I = (\ell/2^n, (\ell+1)/2^n)$ and $N \geq n$, $F_N(b) - F_N(a) = p^{n_0} q^{n_1}$ where n_0, n_1 count 0's and 1's among the first n digits of $\ell/2^n$.]
2. * There is an analogy between the Walsh–Paley expansion (Exercise 16) and the Fourier expansion, i.e. between $\{w_n\}_{n \geq 0}$ and $\{e^{in\theta}\}$, with the Rademacher $r_k = w_{2^k-1}$ corresponding to the lacunary frequencies $\{e^{i2^k\theta}\}$. The following are known: (a) if $\sum_{k \geq 0} c_k e^{i2^k\theta}$ is $L^2([0, 2\pi])$ then it lies in L^p for every $p < \infty$; (b) if it is the Fourier series of an integrable function, it is in L^2 , hence in L^p for every $p < \infty$; (c) it is in L^∞ iff $\sum_k |c_k| < \infty$; (d) consequently the conclusion (a) of the random-series theorem does not extend to $p = \infty$.

Additional exercises.

1. Compute $m(\{S_6 = k\})$ for the fair coin and all valid k , and verify the values sum to 1. Identify where the maximum occurs and compare with Proposition 43.1.2.
2. Let f be uniform on $[0, 1]$. Find the distribution measure, mean, and variance of (a) $g = 2f - 1$; (b) $h = f^2$. For (b) produce the density of h explicitly.
3. Let r_1, r_2, r_3 be Rademacher functions and $F = r_1 + 2r_2 - 2r_3$. Compute $\|F\|_{L^2}$ and $\|F\|_{L^4}$ exactly (use the L^4 expansion of Worked Example with $\int F^4 = 3(\sum a_i^2)^2 - 2\sum a_i^4$), and verify $\|F\|_{L^4} \geq \|F\|_{L^2}$.
4. Using the sub-Gaussian tail $m(\{|F| > \alpha\}) \leq 2e^{-\alpha^2/2}$ for $F = \sum a_n r_n$ with $\sum a_n^2 = 1$, bound $m(\{|F| > 3\})$ and compare to the standard normal probability $m(\{|Z| > 3\}) \approx 0.0027$.

5. Determine, for each $\beta > 0$, whether $\sum_n n^{-\beta} r_n(t)$ converges for almost every t . State the threshold value of β and justify using Theorem 43.1.19.
6. (Non-example) Exhibit three ± 1 -valued functions that are pairwise *uncorrelated* ($\int fg = 0$ for distinct pairs) yet not independent. Explain why uncorrelatedness is weaker than independence.
7. Let $\{A_n\}$ be independent events with $m(A_n) = 1/n$ for $n \geq 2$. Decide whether almost every point lies in infinitely many A_n , and justify with Borel–Cantelli.
8. (Diagram problem) Sketch the Rademacher functions r_1, r_2, r_3 on $[0, 1]$ stacked vertically, marking the dyadic intervals on which each is constant. Using your picture, explain geometrically why $\int_0^1 r_i r_j = \delta_{ij}$.
9. Let f have distribution $\mu = \frac{1}{3}\delta_{-1} + \frac{1}{3}\delta_0 + \frac{1}{3}\delta_2$. Compute the mean, variance, and the characteristic function $\hat{\mu}(\xi)$.
10. Prove that if f, g are independent and integrable with fg integrable, then $E(fg) = E(f)E(g)$, and give an example of dependent f, g with $E(fg) = E(f)E(g)$ nonetheless (showing the converse fails).

43.2 The Law of Large Numbers

We now abstract the coin-flip results to general independent functions. The pattern $S_N/N \rightarrow m_0$ – the empirical average converging to the mean – is the *law of large numbers*. We give two routes to its strong (almost-everywhere) form: one through the ergodic theorem, one through martingales. Along the way we develop conditional expectation, the martingale maximal inequality, and Kolmogorov’s zero-one law, and we close with random walks and their dimension-dependent recurrence.

The strong law via the ergodic theorem

Definition 43.2.1 (Identically distributed). Functions $\{f_n\}$ are *identically distributed* if their distribution measures μ_n coincide: $m(\{f_n \in B\})$ is independent of n for every Borel B . Their *joint distribution* is the measure on $\mathbb{R}^N$ given by $\mu_{f_1, \dots, f_N}(B) = m(\{x : (f_1(x), \dots, f_N(x)) \in B\})$; two sequences have the same joint distribution if these agree for all N and all Borel $B \subseteq \mathbb{R}^N$.

Theorem 43.2.2 (Strong law of large numbers). *If $\{f_n\}$ are mutually independent, identically distributed, and have mean m_0 , then*

$$\frac{1}{N} \sum_{n=0}^{N-1} f_n(x) \rightarrow m_0 \quad \text{for almost every } x, \text{ as } N \rightarrow \infty.$$

Proof via the ergodic theorem. Build the model space $Y = \mathbb{R}^\infty = \prod_{j \geq 0} \mathbb{R}_j$, each $\mathbb{R}_j = \mathbb{R}$ carrying the common distribution μ , with m^* the product measure. Let $g_n(y) = y_n$ be the coordinate functions and $\tau(y) = (y_{n+1})_n$ the shift. *Observation 1:* $g_n(\tau y) = g_{n+1}(y)$, so $g_n = g_0 \circ \tau^n$. *Observation 2:* τ is measure preserving (product measures are shift invariant on cylinders, hence everywhere) and ergodic (any τ -invariant set A lies in the tail algebra, so by the zero-one law $m^*(A) = m^*(A)^2$, forcing $m^*(A) \in \{0, 1\}$). *Conclusion 1:* by Birkhoff’s ergodic theorem, $\frac{1}{N} \sum_{n=0}^{N-1} g_n(y) = \frac{1}{N} \sum_{n=0}^{N-1} g_0(\tau^n y) \rightarrow \int_Y g_0 dm^* = m_0$ a.e. *Conclusion 2:* the original $\{f_n\}$ and the model $\{g_n\}$ have the same joint distribution (by mutual independence and identical distribution of the f_n , their joint law is exactly the product $\mu^{\otimes \infty}$); a.e. convergence of averages is a joint-distribution property, so it transfers back to the f_n . $\square$

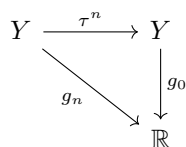


Figure 43.6. The shift model for the strong law: the n th coordinate function factors through the shift, $g_n = g_0 \circ \tau^n$, so the time average $\frac{1}{N} \sum_{n < N} g_0 \circ \tau^n$ is a Birkhoff average to which the ergodic theorem applies.

This is the rigorous form of “the empirical average converges to the expectation.”

Worked Example 43.2.3 (The law of large numbers as a frequency statement). Let $f_n = \mathbf{1}_{E_n}$ be indicators of independent identically distributed events with $m(E_n) = p$. Identify what Theorem 43.2.2 says.

Recognition. The f_n are i.i.d. with mean $m_0 = \int \mathbf{1}_{E_n} = p$; the average counts the empirical frequency of the events.

Solution. $\frac{1}{N} \sum_{n=0}^{N-1} \mathbf{1}_{E_n}(x)$ is the fraction of the first N events that occur at x . The strong law gives this $\rightarrow p$ a.e.: the long-run *relative frequency* of an event equals its probability, almost surely. *Check.* For fair coins $E_n = \{\text{heads on flip } n\}$, $p = \frac{1}{2}$, recovering “the fraction of heads tends to $1/2$ ” – the intuitive meaning of a fair coin. ■

Worked Example 43.2.4 (Normal numbers). Show that almost every $\alpha \in [0, 1]$ has dyadic expansion with limiting density of 1’s equal to $1/2$.

Recognition. Trigger: encode the digits as i.i.d. functions and invoke the strong law.

Solution. Write $\alpha = \sum_j x_j 2^{-j}$; under the product map the digits x_j are i.i.d. with $m(\{x_j = 1\}) = \frac{1}{2}$, mean $m_0 = \frac{1}{2}$. By Theorem 43.2.2, $\frac{1}{N} \sum_{j=1}^N x_j \rightarrow \frac{1}{2}$ a.e., i.e. $\#_N(\alpha)/N \rightarrow \frac{1}{2}$ where $\#_N$ counts 1’s in the first N digits. *Check.* The same argument in base q (digits i.i.d. uniform on $\{0, \dots, q-1\}$, mean of $\mathbf{1}_{x_j=p}$ equal to $1/q$) shows almost every number is *normal to base q* for every q . ■

Caution. The mean hypothesis in Theorem 43.2.2 cannot be dropped. If the f_n are i.i.d. but *not* integrable ($\int |f_0| dm = \infty$), then $\frac{1}{N} \sum f_n$ fails to converge a.e. – one shows $A_n = \{|f_n| > n\}$ has $\sum m(A_n) \approx \int |f_0| = \infty$, and Borel–Cantelli forces $|f_n| > n$ infinitely often, which is incompatible with convergent averages.

Conditional expectation

Proposition 43.2.5 (Conditional expectation). Let $\mathcal{A} \subseteq \mathcal{M}$ be a sub- σ -algebra and f integrable. There is a unique (up to null sets) $\mathcal{A}$ -measurable function $F = E_{\mathcal{A}}(f) = E(f | \mathcal{A})$ with

$$\int_A F dm = \int_A f dm \quad \text{for all } A \in \mathcal{A}.$$

It satisfies: $f \mapsto E(f)$ is linear; $\int_X E(f) = \int_X f$ and $E(1) = 1$; $E(f) \geq 0$ if $f \geq 0$; $E^2 = E$ (so $E(f) = f$ when f is $\mathcal{A}$ -measurable); and the pull-out property $E(gf) = gE(f)$ for bounded $\mathcal{A}$ -measurable g .

Proof. Existence/uniqueness: the map $A \mapsto \int_A f dm$ is a finite (signed) measure on $\mathcal{A}$ absolutely continuous w.r.t. $m|_{\mathcal{A}}$; by Radon–Nikodym it has a density F , which is the required $\mathcal{A}$ -measurable function. The listed properties are immediate from the defining identity; e.g. pull-out holds for $g = \mathbf{1}_A$ ($A \in \mathcal{A}$) by definition and extends by linearity and limits. □

Lemma 43.2.6 (E is an orthogonal projection). If $f \in L^2$ then $E(f) \in L^2$ with $\|E(f)\|_{L^2} \leq \|f\|_{L^2}$, and $\int_X E(f) g dm = \int_X f E(g) dm$ for $f, g \in L^2$. Thus E is the orthogonal projection of $L^2(X, m)$ onto the closed subspace of $\mathcal{A}$ -measurable functions.

Proof. For bounded $\mathcal{A}$ -measurable g with $\|g\|_{L^2} \leq 1$, $\int g E(f) = \int gf \leq \|f\|_{L^2}$; taking the supremum over such g (which computes $\|E(f)\|_{L^2}$ since $E(f)$ is $\mathcal{A}$ -measurable) gives the norm bound. Self-adjointness: $\int E(g)f = \int E(E(g)f) = \int E(g)E(f)$ by pull-out, symmetric in f, g . With $E^2 = E$ these say E is an orthogonal projection. □

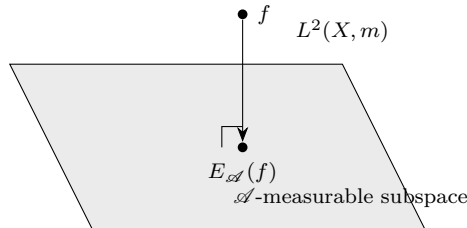


Figure 43.7. Conditional expectation $E_{\mathcal{A}}$ is the orthogonal projection of L^2 onto the $\mathcal{A}$ -measurable functions. (Schematic.)

Worked Example 43.2.7 (Conditional expectation over a partition). Let $X = \bigsqcup_n A_n$ be a countable partition with $m(A_n) > 0$, and $\mathcal{A}$ the algebra generated by the A_n . Compute $E_{\mathcal{A}}(f)$.

Recognition. An $\mathcal{A}$ -measurable function is constant on each A_n ; pin the constant by the defining averaging identity.

Solution. On A_n , $E_{\mathcal{A}}(f)$ must be a constant c_n with $\int_{A_n} c_n = \int_{A_n} f$, so $c_n = \frac{1}{m(A_n)} \int_{A_n} f \, dm$. Thus $E_{\mathcal{A}}(f)(x) = \frac{1}{m(A_n)} \int_{A_n} f \, dm$ for $x \in A_n$ – the average of f over the block containing x . *Check.* $\int_X E_{\mathcal{A}}(f) = \sum_n m(A_n) c_n = \sum_n \int_{A_n} f = \int_X f$, as required by Proposition 43.2.5. ■

Worked Example 43.2.8 (Conditional expectation as integrating out a coordinate). Let $X = X_1 \times X_2$ with product measure $m = m_1 \times m_2$, and $\mathcal{A} = \{A \times X_2\}$. Compute $E_{\mathcal{A}}(f)$.

Recognition. $\mathcal{A}$ -measurable means independent of the second coordinate; the averaging identity integrates out x_2 .

Solution. $E_{\mathcal{A}}(f)(x_1, x_2) = \int_{X_2} f(x_1, y) dm_2(y)$, a function of x_1 alone. *Check.* For $A = A_1 \times X_2 \in \mathcal{A}$, $\int_A E_{\mathcal{A}}(f) = \int_{A_1} \int_{X_2} f \, dm_2 \, dm_1 = \int_{A_1} f$ by Fubini, confirming the defining property. The dyadic averages $E_N(f)$ of §43.1 are the special case $X_1 =$ a dyadic interval. ■

Martingales and the maximal theorem

Definition 43.2.9 (Martingale). Let $\mathcal{A}_0 \subseteq \mathcal{A}_1 \subseteq \dots$ be an increasing sequence of sub- σ -algebras with conditional expectations $E_n = E_{\mathcal{A}_n}$. A sequence $\{s_n\}$ is a *martingale* (adapted to $\{\mathcal{A}_n\}$) if each s_n is $\mathcal{A}_n$ -measurable and

$$E_n(s_m) = s_n \quad \text{whenever } n \leq m. \quad (43.3)$$

Equivalently, with $f_k = s_k - s_{k-1}$, the increments satisfy $E_{k-1}(f_k) = 0$.

Because the $\mathcal{A}_n$ increase, the conditional expectations satisfy the tower rule $E_n E_m = E_{\min(n,m)}$: if $m \leq n$ then $E_m(f)$ is already $\mathcal{A}_n$ -measurable so $E_n(E_m(f)) = E_m(f)$; if $n \leq m$ then for $A \in \mathcal{A}_n \subseteq \mathcal{A}_m$, $\int_A E_n(f) = \int_A f = \int_A E_m(f)$.

$$\begin{array}{ccc} L^2(X, m) & \xrightarrow{E_m} & L^2(\mathcal{A}_m) \\ & \searrow E_n & \downarrow E_n \\ & & L^2(\mathcal{A}_n) \end{array}$$

Figure 43.8. The tower rule for $n \leq m$: since $\mathcal{A}_n \subseteq \mathcal{A}_m$, applying the coarser projection E_n after the finer E_m agrees with E_n alone, $E_n E_m = E_n = E_{\min(n,m)}$. Each E_k is the orthogonal projection onto the $\mathcal{A}_k$ -measurable functions $L^2(\mathcal{A}_k)$.

Theorem 43.2.10 (Martingale maximal inequality and convergence). Let $\{s_n\}$ be a martingale with $s_n = E_n(s_\infty)$ for some integrable s_∞ .

1. (Maximal inequality) For all $\alpha > 0$,

$$m\left(\left\{x : \sup_n s_n(x) > \alpha\right\}\right) \leq \frac{1}{\alpha} \int_A s_\infty \, dm, \quad A = \left\{x : \sup_n s_n(x) > \alpha\right\}.$$

2. (Convergence) $s_n \rightarrow s_\infty$ almost everywhere (and in L^1 if s_∞ is measurable w.r.t. $\mathcal{A}_\infty = \bigvee_n \mathcal{A}_n$).

Proof. (1) Let $A_n = \{x : s_n(x) > \alpha \text{ but } s_k(x) \leq \alpha \text{ for } k < n\}$ be the first-passage sets; they are disjoint, $A_n \in \mathcal{A}_n$, and $A = \bigsqcup_n A_n$. Then

$$\int_A s_\infty \, dm = \sum_n \int_{A_n} s_\infty \, dm = \sum_n \int_{A_n} E_n(s_\infty) \, dm = \sum_n \int_{A_n} s_n \, dm \geq \sum_n \alpha m(A_n) = \alpha m(A),$$

using $\int_{A_n} E_n(s_\infty) = \int_{A_n} s_\infty$ (definition of E_n , since $A_n \in \mathcal{A}_n$) and $s_n > \alpha$ on A_n . This is exactly the claimed bound. (2) First assume $s_n \rightarrow s_\infty$ in L^1 . Writing $s_n - s_\infty = E_n(s_\infty - s_k) + (s_k - s_\infty)$ for $n \geq k$, fix $\varepsilon > 0$ and pick k with $\|s_k - s_\infty\|_{L^1} < \varepsilon$. With $A_\alpha = \{\limsup_n |s_n - s_\infty| > 2\alpha\}$, parts of A_α are controlled by $\sup_n |E_n(s_\infty - s_k)| > \alpha$ and $|s_k - s_\infty| > \alpha$; by part (1) and Tchebychev each has measure $\leq \varepsilon/\alpha$, so $m(A_\alpha) \leq 2\varepsilon/\alpha$. Let $\varepsilon \rightarrow 0$: $m(A_\alpha) = 0$ for every α , giving a.e. convergence. The general case reduces to this by defining s_∞ as the L^1 limit and using continuity of E_k . $\square$

The reader should compare the maximal inequality with the Hardy–Littlewood maximal bound: the first-passage decomposition plays the role of the Vitali covering.

Theorem 43.2.11 (Strong law via martingales). *If $\{f_n\}$ are mutually independent with mean zero and $\sup_n \|f_n\|_{L^2} < \infty$, then for each $\alpha > 1/2$, $s_n/n^\alpha \rightarrow 0$ a.e., where $s_n = \sum_{k=0}^n f_k$.*

Proof idea. With $a_n = n^{-\beta}$, $1/2 < \beta < \alpha$, the weighted sums $\tilde{s}_n = \sum a_k f_k$ form an L^2 -bounded martingale (orthogonal mean-zero increments, $\sum a_k^2 \|f_k\|_{L^2}^2 < \infty$), hence converge a.e. by Theorem 43.2.10. Summation by parts, $s_n = \tilde{s}_n b_n + \sum_{k < n} \tilde{s}_k (b_k - b_{k+1})$ with $b_n = n^\beta$, then gives $s_n = O(n^\beta)$ a.e., so $s_n/n^\alpha \rightarrow 0$. (For the Rademacher case this is Corollary 1.5 of the source.) $\square$

Worked Example 43.2.12 ($S_N/N \rightarrow 0$ a.e. for coin flips, strong form). Deduce that the empirical winnings average $S_N(t)/N \rightarrow 0$ almost everywhere, with $S_N = \sum_{n=1}^N r_n$.

Recognition. The Rademacher functions are independent, mean zero, $\|r_n\|_{L^2} = 1$; apply Theorem 43.2.11 with $\alpha = 1$.

Solution. Since $\sup_n \|r_n\|_{L^2} = 1 < \infty$, the theorem gives $S_N/N^\alpha \rightarrow 0$ a.e. for every $\alpha > 1/2$; in particular $\alpha = 1$ yields $S_N/N \rightarrow 0$ a.e. *Check.* This upgrades Corollary 43.1.16 (convergence in probability) to almost-sure convergence – the same conclusion the ergodic route gives via Theorem 43.2.2 with $m_0 = 0$. $\blacksquare$

The zero-one law

Theorem 43.2.13 (Kolmogorov’s zero-one law). *Let $\mathcal{A}_0, \mathcal{A}_1, \dots$ be mutually independent sub- σ -algebras and let $\mathcal{B} = \bigcap_n \bigvee_{k \geq n} \mathcal{A}_k$ be the tail algebra. Then every $A \in \mathcal{B}$ has $m(A) \in \{0, 1\}$.*

Proof. The kernel: if A belongs to two independent algebras, then $m(A) = m(A \cap A) = m(A)m(A)$, so $m(A) \in \{0, 1\}$. Each $\mathcal{A}_r$ is independent of $\bigvee_{k > r} \mathcal{A}_k$, hence independent of $\mathcal{B}$; therefore $\mathcal{B}$ is independent of $\bigvee_r \mathcal{A}_r \supseteq \mathcal{B}$, i.e. $\mathcal{B}$ is independent of itself. Apply the kernel. $\square$

Corollary 43.2.14. *If $f_0, f_1, \dots$ are mutually independent, the set where $\sum_k f_k$ converges has measure 0 or 1.*

Proof. By the Cauchy criterion the convergence set is $\bigcap_\ell \bigcup_r \{x : |s_n - s_m| < 1/\ell \ \forall n, m \geq r\}$, a tail event (it lies in $\bigvee_{k \geq n_0} \mathcal{A}_{f_k}$ for every n_0). Apply Theorem 43.2.13. $\square$

Worked Example 43.2.15 (A tail event with probability one). Show that for independent ± 1 fair signs r_n , the event $\{\limsup_n S_n/\sqrt{n \log \log n} \text{ exists and is finite}\}$ has measure 0 or 1.

Recognition. Trigger: recognize the event as tail-measurable, then invoke the zero-one law.

Solution. Changing finitely many signs $r_1, \dots, r_{n_0}$ alters S_n by a bounded amount, which does not affect $\limsup_n S_n/\sqrt{n \log \log n}$; hence the event lies in the tail algebra $\bigvee_{k \geq n_0} \mathcal{A}_{r_k}$ for every n_0 . By Theorem 43.2.13 its measure is 0 or 1. *Check.* The law of the iterated logarithm in fact pins the value of the lim sup at 1 (with the $\sqrt{2}$ normalization), so the measure is 1 – consistent with the dichotomy. $\blacksquare$

Bernoulli trials with bias

When heads/tails have probabilities p, q ($p + q = 1$, $0 < p < 1$), replace m by the product measure m_p giving weight p to 0 and q to 1 on each factor. The winnings $r_n = 1 - 2x_n$ now have mean $\int r_n dm_p = p - q$ and variance $\sigma^2 = 1 - (p - q)^2 = 4pq$.

Near $\xi = 0$, $1 - \hat{\mu}(\xi) = \frac{2\pi^2}{d} |\xi|^2 + O(|\xi|^4)$, and $1 - \hat{\mu} \geq c_1 > 0$ away from 0. So the integrand behaves like $|\xi|^{-2}$, whose integral over a neighborhood of 0 in $\mathbb{R}^d$ *diverges* for $d \leq 2$ and *converges* for $d \geq 3$. Hence $\sum_n m(\{s_n = 0\})$ is ∞ for $d \leq 2$ and finite for $d \geq 3$. Since $\int_X \#(x) dm = \sum_n m(\{s_n = 0\})$ where $\#(x)$ is the number of visits to 0, a finite sum ($d \geq 3$) forces finitely many returns a.s. For the dichotomy of *ever* returning, set $A(r) = \sum r^n m(A_n)$, $F(r) = \sum r^n m(F_n)$ with F_k the first-return-at-step- k sets and $A_n = \{s_n = 0\}$; the renewal identity $m(A_n) = \sum_{1 \leq k \leq n} m(F_k) m(A_{n-k})$ gives $A(r) = A(r)F(r) + 1$, so $F(r) = 1 - 1/A(r)$. As $r \rightarrow 1$: $A(1) = \infty$ ($d \leq 2$) yields $\sum_k m(F_k) = 1$ (sure return, then infinite recurrence by iteration), while $A(1) < \infty$ ($d \geq 3$) yields $\sum_k m(F_k) < 1$ (positive escape probability). $\square$

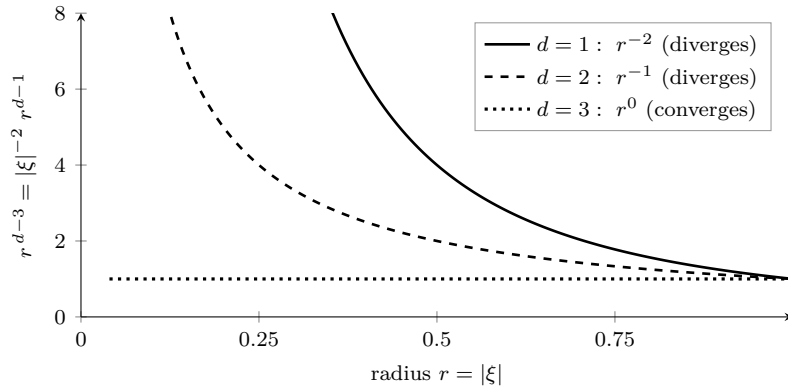


Figure 43.10. The radial integrand $|\xi|^{-2} r^{d-1} = r^{d-3}$ from (43.4) in polar coordinates: it blows up nonintegrably at $r = 0$ for $d \leq 2$ but stays bounded for $d \geq 3$ – the source of Polya’s dichotomy.

Worked Example 43.2.19 (Recurrence vs. transience from the Green’s function). Explain, using (43.4), why $\int_Q |\xi|^{-2} d\xi$ diverges for $d = 2$ but converges for $d = 3$.

Recognition. Trigger: convert to polar/spherical coordinates and test the radial integral.

Solution. Near 0 the integrand is $\sim |\xi|^{-2}$; in d dimensions $d\xi = c\rho^{d-1}d\rho d\omega$, so the radial part is $\int_0 \rho^{-2}\rho^{d-1}d\rho = \int_0 \rho^{d-3}d\rho$. This converges at 0 iff $d - 3 > -1$, i.e. $d \geq 3$, and diverges for $d \leq 2$. *Check.* $d = 2$ gives $\int_0 \rho^{-1}d\rho = \infty$ (logarithmic divergence, recurrent); $d = 3$ gives $\int_0 d\rho < \infty$ (transient), exactly Theorem 43.2.18. $\blacksquare$

Worked Example 43.2.20 (Expected number of returns when $d \geq 3$). For $d = 3$ the return probability is $u \approx 0.34$. Compute the expected number of visits to the origin and explain its finiteness.

Recognition. Trigger: a returning walk regenerates; the number of returns is geometric with parameter the escape probability.

Solution. Each time the walk is at 0 it returns with probability u and escapes forever with probability $1 - u$; the number of returns is geometric, with expected total visits (including the start) $\frac{1}{1-u} \approx \frac{1}{0.66} \approx 1.52$. By (43.4) this equals $\int_Q d\xi / (1 - \hat{\mu})$, which is finite for $d = 3$. *Check.* Finiteness of the expected visit count is exactly what Theorem 43.2.18(b) needs to conclude finitely many returns a.s. $\blacksquare$

Remark. Rescaling the walk – time step $1/n$, space scaled by $n^{-1/2}$ – and passing to the limit (by the central limit theorem of §43.3) produces *Brownian motion*, the continuous-time analog whose increments are independent and Gaussian. The recurrence dichotomy persists: Brownian motion is neighborhood-recurrent in $d \leq 2$ and transient in $d \geq 3$.

Exercises (Stein, Ch.5) – independence, conditional expectation, martingales, zero-one law, random walks.

1. Show that if $\mathcal{B}_0, \dots, \mathcal{B}_n$ are mutually independent algebras, then for each $k < n$ the algebras $\bigvee_{j=0}^k \mathcal{B}_j$ and $\mathcal{B}_n$ are mutually independent. [Hint: first show by induction that $B_0 \cap \dots \cap B_k$ ($B_j \in \mathcal{B}_j$) is independent of $\mathcal{B}_n$; then fix $B \in \mathcal{B}_n$, consider $\mu(E) = m(E \cap B)$ and $\nu(E) = m(E)m(B)$ on the collection $\mathcal{C}$ of sets $B_0 \cap \dots \cap B_k$, and apply Exercise 4 of §43.1.]

2. A sequence $\{f_n\}_{n \geq 0}$ on X is a (discrete) *stationary process* if for every N the joint distribution of $f_r, f_{r+1}, \dots, f_{r+N}$ is independent of r . Using the model space Y from the proof of Theorem 43.2.2, show that whenever $\{f_n\}$ is stationary it has the same joint distribution as $\{g_0(\tau^n(y))\}$ for a suitable g_0 on Y and the shift τ . Hence the ergodic theorem applies in this more general setting.
3. Prove the conditions of Theorem 43.2.2 are sharp: if $\{f_n\}_{n \geq 0}$ are mutually independent and identically distributed but $\int_X |f_0| dm = \infty$, then for almost every x the averages $\frac{1}{N} \sum_{n=0}^{N-1} f_n(x)$ fail to converge. [Hint: $A_n = \{|f_n| > n\}$ are independent with $\sum_n m(A_n) \approx \int_X |f_0| = \infty$; apply Borel–Cantelli (Exercise 20 of §43.1).]
4. Examples of conditional expectations. (a) If $X = \bigcup A_n$ is a finite or countable partition with $m(A_n) > 0$ and $\mathcal{A}$ is generated by the A_n , then $E_{\mathcal{A}}(f)(x) = \frac{1}{m(A_n)} \int_{A_n} f dm$ for $x \in A_n$. (b) If $X = X_1 \times X_2$ with product measure and $\mathcal{A} = \{A \times X_2\}$, then $E_{\mathcal{A}}(f)(x_1, x_2) = \int_{X_2} f(x_1, y) dm_2(y)$.
5. (Martingale convergence in L^2) Suppose $s_n = E_n(s_\infty)$ with $s_\infty \in L^2$. Then $\{s_n\}$ converges in L^2 . [Hint: with $f_n = s_n - s_{n-1}$ the increments are mutually orthogonal and $s_n - s_0 = \sum_{k=1}^n f_k$.]
6. (a) If $s_\infty \in L^p$ then $s_n = E_n(s_\infty) \in L^p$ and $\|s_n\|_{L^p} \leq \|s_\infty\|_{L^p}$ for $1 \leq p \leq \infty$. (b) Conversely, if $\{s_n\}$ is a martingale with $\sup_n \|s_n\|_{L^p} < \infty$ then there is $s_\infty \in L^p$ with $s_n = E_n(s_\infty)$, for $1 < p \leq \infty$. (c) Show (b) may fail at $p = 1$. [Hint: (a) argue as in Lemma 43.2.6(a); (b) use weak compactness of L^p , $p > 1$; (c) on $[0, 1]$ take $s_n = 2^n$ on $[0, 2^{-n}]$, 0 otherwise.]
7. Suppose $s_n = E_n(s_\infty)$ with s_∞ integrable. (a) Show s_n converges in L^1 . (b) Moreover $s_n \rightarrow s_\infty$ in L^1 iff s_∞ is measurable with respect to $\mathcal{A}_\infty = \bigvee_n \mathcal{A}_n$. [Hint: (a) use Exercises 25–26(a); the limit is $E_{\mathcal{A}_\infty}(s_\infty)$.]
8. Suppose $s_n = E_n(s_\infty)$, $s_\infty \in L^1$. (a) Show $m(\{\sup_n |s_n| > \alpha\}) \leq \frac{1}{\alpha} \int_{|s_\infty| > \alpha} |s_\infty|$. (b) Deduce $\|\sup_n |s_n|\|_{L^p} \leq A_p \|s_\infty\|_{L^p}$ if $s_\infty \in L^p$, $1 < p \leq \infty$. [Hint: (a) reduce to $s_\infty \geq 0$ and use the maximal inequality of Theorem 43.2.10(1); (b) adapt the Hardy–Littlewood maximal argument.]
9. The results for real martingales extend to $\mathbb{R}^d$ -valued $\{s_n\}$. Verify: (a) $|s_k| \leq E_k(|s_n|)$ for $k \leq n$; (b) $m(\{\sup_n |s_n| > \alpha\}) \leq \frac{1}{\alpha} \int_{|s_\infty| > \alpha} |s_\infty|$, $|\cdot|$ the Euclidean norm. [Hint: (a) $(s_k, v) = E_k((s_n, v))$ for fixed v ; take sup over unit v . (b) follows from (a) and Exercise 28(a).]
10. Conditional expectations on spaces of infinite total measure. Let $X = \mathbb{R}^d$ with Lebesgue measure, and for $n \in \mathbb{Z}$ let $\mathcal{A}_n$ be generated by the dyadic cubes of side 2^{-n} ; $E_n(f)(x) = \frac{1}{m(Q)} \int_Q f$ for x in the dyadic cube Q of side 2^{-n} . (a) Show the martingale maximal inequality extends. (b) If $f \geq 0$ then $\sup_{n \in \mathbb{Z}} E_n(f) \leq cf^*$ with f^* the Hardy–Littlewood maximal function. (c) The converse $f^* \leq c \sup_n E_n(f)$ fails, but the substitute $m(\{f^* > \alpha\}) \leq c_1 m(\{\sup_n E_n(f) > c_2 \alpha\})$ holds.
11. For the d -dimensional random walk $\{s_n(x)\}$ of §43.2, find the limit of the distribution measures of $s_n(x)/n^{1/2}$ as $n \rightarrow \infty$.
12. If k is a lattice point in $\mathbb{Z}^d$ and $d = 1$ or 2 , show that for almost every path the random walk visits k infinitely often: $m(\{x : s_n(x) = k \text{ for infinitely many } n\}) = 1$. [Hint: there is ℓ_0 with $m(\{s_{\ell_0} = -k\}) > 0$; if the conclusion failed there would be r_0 with $m(\{s_n = k \ \forall n \geq r_0\}) > 0$, and one couples $\{s_n = 0, \ \forall n \geq \ell_0 + r_0\} \supseteq \{s_{\ell_0} = -k\} \cap \{s_n - s_{\ell_0} = k, \ \forall n \geq \ell_0 + r_0\}$ with independent factors.]
13. Prove that if $d \geq 3$, then the random walk satisfies $\lim_{n \rightarrow \infty} |s_n| = \infty$ almost everywhere. [Hint: it suffices that $B = \{x : \liminf_n |s_n(x)| \leq R\}$ is null for each $R > 0$; with $B(k, \ell) = \{s_\ell = k, \ s_n = k \text{ for infinitely many } n\}$, $B \subseteq \bigcup_{\ell, |k| \leq R} B(k, \ell)$, and $m(B(k, \ell)) = 0$ when $d \geq 3$.]

Problems (Stein, Ch.5) – law of the iterated logarithm.

1. * Suppose $\{f_n\}$ are identically distributed, square integrable, mutually independent, each with mean 0 and variance 1, and $s_n = \sum_{k=1}^n f_k$. Then for almost every x ,

$$\limsup_{n \rightarrow \infty} \frac{s_n(x)}{(2n \log \log n)^{1/2}} = 1.$$

This is the *law of the iterated logarithm*.

Additional exercises.

1. Let $\{f_n\}$ be i.i.d. with f_n uniform on $\{1, 2, \dots, 6\}$ (a fair die). Compute the mean m_0 , and state precisely what the strong law of large numbers (Theorem 43.2.2) asserts about the running average of dice rolls.
2. Verify the tower property $E_n E_m = E_{\min(n,m)}$ directly when $\mathcal{A}_n$ is generated by a refining sequence of partitions, using the partition formula of Worked Example (conditional expectation over a partition).
3. Let $X = [0, 1]$ with Lebesgue measure and $\mathcal{A}$ the algebra generated by $\{[0, \frac{1}{2}), [\frac{1}{2}, 1]\}$. For $f(x) = x^2$ compute $E_{\mathcal{A}}(f)$ explicitly and confirm $\int_X E_{\mathcal{A}}(f) = \int_X f$.
4. Show directly that $s_n = \sum_{k=1}^n r_k$ is a martingale with respect to the dyadic algebras $\mathcal{A}_n$, by checking $E_{n-1}(s_n) = s_{n-1}$.
5. (Non-example) Give a sequence $\{f_n\}$ that is identically distributed and pairwise independent but for which the strong law fails, OR explain why no such example exists with f_n integrable; in either case relate your answer to the hypotheses of Theorem 43.2.2.
6. For the simple random walk on $\mathbb{Z}$ ($d = 1$), use the local central limit estimate $m(\{s_{2n} = 0\}) \sim 1/\sqrt{\pi n}$ to show $\sum_n m(\{s_n = 0\}) = \infty$, recovering recurrence in dimension 1 from (43.4).
7. (Diagram problem) Draw a sample path of the two-dimensional simple random walk of at least 15 steps on the integer lattice, marking the start and any returns to the origin. Comment on how Theorem 43.2.18 predicts infinitely many returns in $d = 2$.
8. Using Kolmogorov's zero-one law (Theorem 43.2.13), prove that for independent f_n the event $\{\sum_n f_n \text{ converges}\}$ has probability 0 or 1, and that $\limsup_n f_n$ is almost surely a constant (possibly $\pm\infty$).
9. Let E_n be the dyadic conditional expectations on $[0, 1]$ and $f \in L^1$. State the martingale convergence theorem (Theorem 43.2.10) in this concrete setting and identify its conclusion with the Lebesgue differentiation theorem.
10. For biased Bernoulli trials, compute $m_p(\{S_4 = 2\})$ in terms of p, q , and evaluate at $p = 2/3$. Compare with the fair case $p = 1/2$.

43.3 The Central Limit Theorem

The law of large numbers says the average S_N/N collapses to a constant. The *central limit theorem* is the finer statement about the fluctuations: the normalized sum $S_N/N^{1/2}$ has a universal Gaussian limiting distribution, regardless of the underlying step law. We first prove the coin-flip case (De Moivre–Laplace) by hand with Stirling, then give the elegant general proof via Fourier transforms (characteristic functions), and finally state the $\mathbb{R}^d$ version.

The normal distribution

Definition 43.3.1 (Gaussian / normal measure). The *normal distribution* with mean 0 and variance σ^2 is the Borel probability measure ν_{σ^2} on $\mathbb{R}$ with density

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-t^2/(2\sigma^2)}.$$

For $\sigma = 1$ this is the *standard normal* ν , density $\frac{1}{\sqrt{2\pi}} e^{-t^2/2}$. Its total mass is 1: $\int_{\mathbb{R}} e^{-t^2/(2\sigma^2)} dt = \sigma\sqrt{2\pi}$.

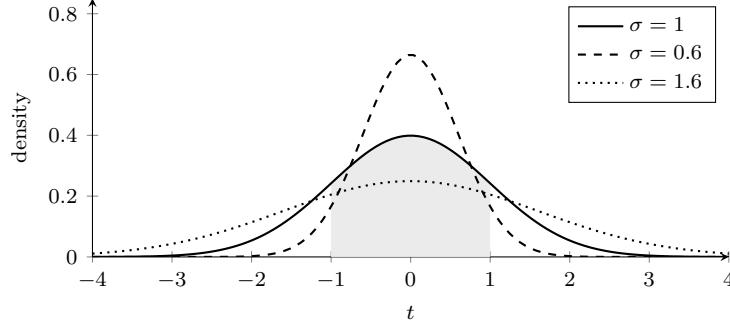


Figure 43.11. The Gaussian density $\frac{1}{\sigma\sqrt{2\pi}}e^{-t^2/2\sigma^2}$ for several σ ; the shaded region is the mass within one standard deviation (≈ 0.683).

Definition 43.3.2 (Characteristic function). The *characteristic function* of a probability measure μ on $\mathbb{R}$ is its Fourier transform $\hat{\mu}(\xi) = \int_{\mathbb{R}} e^{-2\pi i \xi t} d\mu(t)$. If $\mu = \mu_f$ is the distribution of f , then $\hat{\mu}(\xi) = \int_X e^{-2\pi i \xi f(x)} dm$. (We keep the analyst's 2π in the exponent.)

Proposition 43.3.3 (Characteristic function of the Gaussian). $\widehat{\nu_{\sigma^2}}(\xi) = e^{-2\sigma^2\pi^2\xi^2}$. In particular $\hat{\nu}(\xi) = e^{-2\pi^2\xi^2}$.

Proof. $\int_{\mathbb{R}} e^{-2\pi i \xi t} \frac{1}{\sigma\sqrt{2\pi}} e^{-t^2/2\sigma^2} dt$. Completing the square in the exponent, $-\frac{t^2}{2\sigma^2} - 2\pi i \xi t = -\frac{1}{2\sigma^2}(t + 2\pi i \sigma^2 \xi)^2 - 2\pi^2 \sigma^2 \xi^2$, and the Gaussian integral over the shifted contour equals $\sigma\sqrt{2\pi}$, leaving $e^{-2\pi^2 \sigma^2 \xi^2}$. $\square$

Worked Example 43.3.4 (The Fourier transform of a Gaussian is Gaussian). Verify Proposition 43.3.3 for $\sigma = 1$ by the differential-equation method.

Recognition. Trigger: let $g(\xi) = \hat{\nu}(\xi)$ and show $g' = -(4\pi^2\xi)g$, a first-order ODE pinned by $g(0) = 1$.

Solution. Differentiating under the integral, $g'(\xi) = \int (-2\pi i t) e^{-2\pi i \xi t} \frac{e^{-t^2/2}}{\sqrt{2\pi}} dt$. Integrate by parts using $\frac{d}{dt} e^{-t^2/2} = -t e^{-t^2/2}$: $g'(\xi) = -4\pi^2 \xi g(\xi)$. With $g(0) = \int \nu = 1$ the solution is $g(\xi) = e^{-2\pi^2 \xi^2}$. Check. Matches Proposition 43.3.3 at $\sigma = 1$; the Gaussian is (up to scaling) its own Fourier transform. $\blacksquare$

De Moivre–Laplace: the coin-flip CLT

Theorem 43.3.5 (De Moivre–Laplace). For fair coin flips and each $a < b$,

$$m\left(\left\{x : a < \frac{S_N(x)}{N^{1/2}} < b\right\}\right) \rightarrow \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2/2} dt \quad (N \rightarrow \infty).$$

Proof (N even). With $k = 2r$, $m(\{S_N = 2r\}) = 2^{-N} \binom{N}{N/2+r} =: P_r$. By Stirling $N! = \sqrt{2\pi} N^{N+1/2} e^{-N} (1 + O(N^{-1/2}))$, and using $\log(1+u) = u - u^2/2 + O(|u|^3)$ with $r = O(N^{1/2})$, one computes

$$P_r = \frac{2}{\sqrt{2\pi} N^{1/2}} e^{-2r^2/N} (1 + O(N^{-1/2})).$$

Since $e^{-2r^2/N}$ is, on each interval $t \in [r/N^{1/2}, (r+1)/N^{1/2}]$, within $O(N^{-1/2})$ of e^{-2t^2} , the sum over $aN^{1/2} < 2r < bN^{1/2}$ is a Riemann sum:

$$m\left(\left\{a < \frac{S_N}{N^{1/2}} < b\right\}\right) = \sum_{aN^{1/2} < 2r < bN^{1/2}} P_r = \int_{a/2}^{b/2} \frac{2}{\sqrt{2\pi}} e^{-2t^2} dt + O(N^{-1/2}).$$

The substitution $t \mapsto t/2$ turns this into $\int_a^b \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$. Letting $N \rightarrow \infty$ gives the result; the odd- N case is the same up to harmless adjustments. $\square$

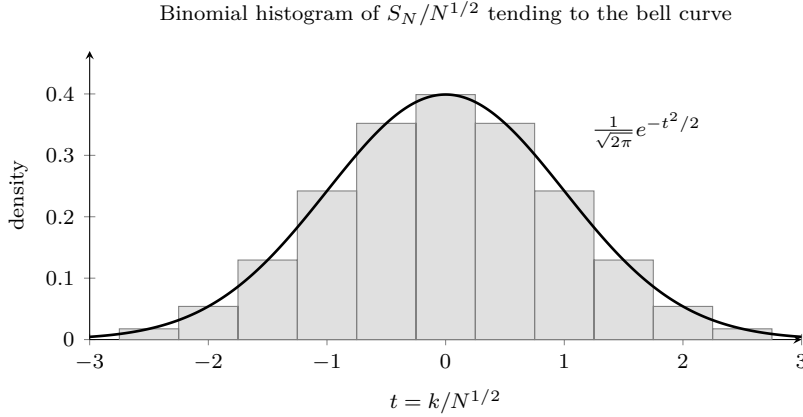


Figure 43.12. De Moivre–Laplace: the discrete histogram of $S_N/N^{1/2}$ (bars, centered on the bin midpoints) converges to the standard normal density (curve).

Worked Example 43.3.6 (Numerical check of De Moivre–Laplace). For $N = 100$ fair flips estimate $m(\{|S_N| \leq 20\})$ and compare to the Gaussian prediction.

Recognition. Trigger: apply Theorem 43.3.5 with $a = -2$, $b = 2$ ($20/\sqrt{100} = 2$).

Solution. The theorem gives $m(\{-2 < S_N/10 < 2\}) \approx \frac{1}{\sqrt{2\pi}} \int_{-2}^2 e^{-t^2/2} dt = 2\Phi(2) - 1 \approx 0.9545$. *Check.* The exact binomial value is ≈ 0.9540 ; the Gaussian approximation is accurate to four digits already at $N = 100$. ■

The general central limit theorem

Theorem 43.3.7 (Central limit theorem). *Let $f_1, f_2, \dots$ be mutually independent, identically distributed, square integrable, with mean m_0 and variance σ^2 , and $S_N = \sum_{n=1}^N f_n$. Then for each $a < b$,*

$$m\left(\left\{x : a < \frac{S_N - Nm_0}{N^{1/2}} < b\right\}\right) \rightarrow \frac{1}{\sigma\sqrt{2\pi}} \int_a^b e^{-t^2/(2\sigma^2)} dt.$$

Proof. Replacing f_n by $f_n - m_0$, assume $m_0 = 0$. Let μ be the common distribution, μ_N that of $S_N/N^{1/2}$. The proof is three steps.

Step (i): $\hat{\mu}_N(\xi) = \hat{\mu}(\xi/N^{1/2})^N$. Using (43.2) with $G(t) = e^{-2\pi i t \xi}$ and the mutual independence,

$$\hat{\mu}_N(\xi) = \int_X e^{-2\pi i \xi S_N/N^{1/2}} d\mu = \prod_{n=1}^N \int_X e^{-2\pi i \xi f_n/N^{1/2}} d\mu = \hat{\mu}(\xi/N^{1/2})^N.$$

Step (ii): $\hat{\mu}_N(\xi) \rightarrow e^{-2\sigma^2\pi^2\xi^2} = \widehat{\nu_{\sigma^2}}(\xi)$. By the Taylor expansion $e^{-2\pi i \xi t/N^{1/2}} = 1 - \frac{2\pi i \xi t}{N^{1/2}} - \frac{2\pi^2 \xi^2 t^2}{N} + E_N(t)$ with $E_N(t) = O(t^2/N) \cap O(t^3/N^{3/2})$, integrate against μ . Since $\int t d\mu = m_0 = 0$ and $\int t^2 d\mu = \sigma^2$,

$$\hat{\mu}(\xi/N^{1/2}) = 1 - \frac{2\pi^2 \xi^2 \sigma^2}{N} + o(1/N),$$

the error being $o(1/N)$ by splitting $\int E_N d\mu$ at $t^2 \leq \varepsilon_N N$ with $\varepsilon_N = N^{-1/2}$. Hence $\hat{\mu}_N(\xi) = (1 - \frac{2\pi^2 \sigma^2 \xi^2}{N} + o(1/N))^N \rightarrow e^{-2\pi^2 \sigma^2 \xi^2}$.

Step (iii): convergence of measures. Pointwise convergence of characteristic functions to that of a continuous measure $\nu = \nu_{\sigma^2}$ implies $\mu_N((a, b)) \rightarrow \nu((a, b))$. This is Lemma 43.3.8 below. Taking $\nu = \nu_{\sigma^2}$ completes the proof. □

Lemma 43.3.8 (Weak convergence from characteristic functions). *If $\hat{\mu}_N(\xi) \rightarrow \hat{\nu}(\xi)$ for every ξ , with ν continuous, then $\mu_N \rightarrow \nu$ weakly: $\int \varphi d\mu_N \rightarrow \int \varphi d\nu$ for every bounded continuous φ . In particular $\mu_N((a, b)) \rightarrow \nu((a, b))$.*

Proof sketch. Approximate $\mathbf{1}_{(a,b)}$ from below by smooth $\varphi_\varepsilon \leq \mathbf{1}_{(a,b)}$ and from above by $\psi_\varepsilon \geq \mathbf{1}_{(a,b)}$. For a smooth compactly supported φ with Fourier transform $\hat{\varphi}$, $\int \varphi d\mu_N = \int \hat{\varphi}(\xi) \hat{\mu}_N(-\xi) d\xi \rightarrow \int \hat{\varphi} \hat{\nu} = \int \varphi d\nu$ by dominated convergence (the $\hat{\mu}_N$ are bounded by 1). Squeezing, $\liminf \mu_N((a,b)) \geq \nu((a,b))$ and $\limsup \mu_N((a,b)) \leq \nu([a,b]) = \nu((a,b))$ by continuity of ν . $\square$

Definition 43.3.9 (Weak convergence). Probability measures μ_N converge weakly to ν if $\int_{\mathbb{R}} \varphi d\mu_N \rightarrow \int_{\mathbb{R}} \varphi d\nu$ for all bounded continuous φ .

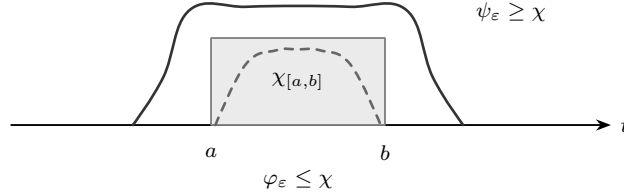


Figure 43.13. Squeezing an indicator $\chi_{[a,b]}$ between smooth $\varphi_\varepsilon \leq \chi \leq \psi_\varepsilon$ to pass from weak convergence to interval probabilities. (Schematic.)

Worked Example 43.3.10 (The biased-coin CLT). For Bernoulli trials with bias p ($\sigma^2 = 4pq$), state the limiting law of $(S_N - N(p - q))/N^{1/2}$.

Recognition. Apply Theorem 43.3.7 with $m_0 = p - q$ and $\sigma^2 = 1 - (p - q)^2 = 4pq$.

Solution. $m\left\{a < \frac{S_N - N(p-q)}{N^{1/2}} < b\right\} \rightarrow \frac{1}{\sigma\sqrt{2\pi}} \int_a^b e^{-t^2/(2\sigma^2)} dt$ with $\sigma^2 = 4pq$. *Check.* At $p = q = \frac{1}{2}$, $m_0 = 0$ and $\sigma^2 = 1$, recovering De Moivre–Laplace (Theorem 43.3.5). $\blacksquare$

Worked Example 43.3.11 (A sum of uniforms looks Gaussian). Let f_n be i.i.d. uniform on $[-\frac{1}{2}, \frac{1}{2}]$. Find the mean and variance, and the limiting distribution of $S_N/N^{1/2}$.

Recognition. Compute moments of the uniform law, then invoke Theorem 43.3.7.

Solution. $m_0 = \int_{-1/2}^{1/2} t dt = 0$; $\sigma^2 = \int_{-1/2}^{1/2} t^2 dt = \frac{1}{12}$. Hence $S_N/N^{1/2}$ converges in distribution to $\nu_{1/12}$, the centered Gaussian with variance $1/12$. *Check.* Already $N = 12$ gives the classic “sum of 12 uniforms minus 6” approximation to a standard normal, since $\sigma^2 \cdot 12 = 1$. $\blacksquare$

Worked Example 43.3.12 (Sums of independent Gaussians are Gaussian). If $f_1, \dots, f_k$ are independent Gaussians with variances σ_j^2 , find the distribution of $c_1 f_1 + \dots + c_k f_k$.

Recognition. Trigger: characteristic functions multiply under independence; the Gaussian has a Gaussian characteristic function (Proposition 43.3.3).

Solution. $\widehat{c_j f_j}(\xi) = \widehat{\nu_{\sigma_j^2}}(c_j \xi) = e^{-2\pi^2 c_j^2 \sigma_j^2 \xi^2}$. By independence the characteristic function of the sum is the product $\prod_j e^{-2\pi^2 c_j^2 \sigma_j^2 \xi^2} = e^{-2\pi^2 (\sum_j c_j^2 \sigma_j^2) \xi^2}$, the characteristic function of ν_{Σ^2} with $\Sigma^2 = \sum_j c_j^2 \sigma_j^2$. So the sum is Gaussian with variance $\sum_j c_j^2 \sigma_j^2$. *Check.* For $k = 2$, $c_1 = c_2 = 1$, the variances add: $\Sigma^2 = \sigma_1^2 + \sigma_2^2$, the familiar rule. $\blacksquare$

Caution. The CLT normalizes by $N^{1/2}$, the LLN by N . Dividing by the wrong power destroys the limit: $S_N/N \rightarrow 0$ (a point mass at 0, by the law of large numbers) while $S_N/N^{1/2} \rightarrow \nu_{\sigma^2}$ (a spread-out Gaussian). The square-integrability hypothesis is also essential – for heavy-tailed f_n (infinite variance) the limit, if any, is a non-Gaussian stable law and the scaling exponent changes.

Random variables with values in $\mathbb{R}^d$

Definition 43.3.13 (Mean and covariance in $\mathbb{R}^d$). For an $\mathbb{R}^d$ -valued $f = (f^{(1)}, \dots, f^{(d)})$ with distribution μ on $\mathbb{R}^d$, the *mean* is the vector $m_0 = (m_0^{(k)})$, $m_0^{(k)} = \int_X f^{(k)} dm$. If f is square integrable the *covariance matrix* is the symmetric nonnegative matrix $\{a_{ij}\}$,

$$a_{ij} = \int_X (f^{(i)} - m_0^{(i)})(f^{(j)} - m_0^{(j)}) dm.$$

Its unique symmetric nonnegative square root is denoted σ , so we write σ^2 for the covariance.

Theorem 43.3.14 (Central limit theorem in $\mathbb{R}^d$). Let $\{f_n\}$ be mutually independent, identically distributed, $\mathbb{R}^d$ -valued, square integrable with mean 0 and invertible common covariance σ^2 . Then the distribution measures μ_N of $N^{-1/2} \sum_{n=1}^N f_n$ converge weakly to ν_{σ^2} , the Gaussian measure on $\mathbb{R}^d$ with density

$$\frac{1}{(2\pi)^{d/2} \det \sigma} e^{-|\sigma^{-1}x|^2/2}.$$

Proof idea. Repeat the three-step scalar proof with the d -dimensional characteristic function $\hat{\mu}(\xi) = \int_{\mathbb{R}^d} e^{-2\pi i \xi \cdot t} d\mu(t)$. Step (i) factors by independence; step (ii) Taylor-expands using $\int t d\mu = 0$ and $\int t_i t_j d\mu = a_{ij}$, giving $\hat{\mu}(\xi/N^{1/2})^N \rightarrow e^{-2\pi^2 |\sigma \xi|^2} = \widehat{\nu_{\sigma^2}}(\xi)$ (the Gaussian characteristic-function computation reduces to the scalar case by an orthogonal diagonalization of σ); step (iii) upgrades pointwise convergence of characteristic functions to weak convergence as in Lemma 43.3.8, now on $\mathbb{R}^d$. $\square$

Worked Example 43.3.15 (Diagonalizing the $\mathbb{R}^d$ Gaussian characteristic function). Show $\widehat{\nu_{\sigma^2}}(\xi) = e^{-2\pi^2 |\sigma \xi|^2}$ when σ is a positive-definite symmetric matrix.

Recognition. Trigger: reduce to a product of scalar Gaussians by an orthogonal change of variables diagonalizing σ .

Solution. Write $\sigma = O^T D O$ with O orthogonal and $D = \text{diag}(\lambda_1, \dots, \lambda_d)$. The substitution $x = O^T y$ has unit Jacobian and turns the density into a product of one-dimensional Gaussians of variances λ_k^2 . By Proposition 43.3.3 in each coordinate the transform is $\prod_k e^{-2\pi^2 \lambda_k^2 (O\xi)_k^2} = e^{-2\pi^2 |DO\xi|^2} = e^{-2\pi^2 |\sigma \xi|^2}$, since $|DO\xi| = |O^T DO\xi| = |\sigma \xi|$. *Check.* For $d = 1$, σ a scalar, this is $e^{-2\pi^2 \sigma^2 \xi^2}$, matching Proposition 43.3.3. $\blacksquare$

Worked Example 43.3.16 (Limiting law of the d -dimensional random walk). For the simple random walk s_n in $\mathbb{R}^d$ of §43.2, find the limit of the distribution of $s_n/n^{1/2}$.

Recognition. The steps r_k are i.i.d. with mean 0; compute their covariance and apply Theorem 43.3.14.

Solution. Each r_k takes values $\pm e_j$ with probability $1/(2d)$, so $E(r_k^{(i)} r_k^{(j)}) = \frac{1}{d} \delta_{ij}$: the covariance is $\frac{1}{d} I$, with $\sigma = \frac{1}{\sqrt{d}} I$. By Theorem 43.3.14, $s_n/n^{1/2}$ converges weakly to the Gaussian on $\mathbb{R}^d$ with density $\frac{d^{d/2}}{(2\pi)^{d/2}} e^{-d|x|^2/2}$, i.e. $\nu_{(1/d)I}$. *Check.* The isotropic covariance $\frac{1}{d} I$ reflects the walk's symmetry across the $2d$ directions; trace = 1 matches the unit step length per move. $\blacksquare$

Worked Example 43.3.17 (Convergence in distribution does not imply a.e. convergence). Give i.i.d. ± 1 functions g_N whose distributions converge weakly to a limit but which do not converge a.e., illustrating the one-way nature of weak convergence.

Recognition. Trigger: weak convergence sees only the distribution, not the pointwise values; exploit identical distribution.

Solution. Let $g_N = r_N$ (the N th Rademacher function). Each g_N has the same distribution $\frac{1}{2} \delta_{-1} + \frac{1}{2} \delta_{+1}$, so $\mu_{g_N} \rightarrow \frac{1}{2} \delta_{-1} + \frac{1}{2} \delta_{+1}$ trivially (the distributions are constant in N). But $g_N(t) = r_N(t)$ oscillates between ± 1 and does not converge for any t that is not eventually dyadic-constant, so g_N fails to converge a.e. *Check.* This is the standard counterexample to the converse of Exercise 9 of §43.1 – a.e. convergence implies weak convergence of distributions, never the reverse. $\blacksquare$

Remark (Toward Brownian motion). The CLT has a process-level refinement: for $S_{N,t} = N^{-1/2} \sum_{n=1}^{[Nt]} f_n$, the distribution of $S_{N,t}$ converges weakly to $\nu_{t\sigma^2}$, and increments $S_{N,t} - S_{N,s}$ over disjoint time intervals become independent Gaussians with variance proportional to the elapsed time. The limiting object – continuous paths, independent Gaussian increments with variance equal to elapsed time – is *Brownian motion*, the scaling limit of the random walk and the central object of continuous-time probability.

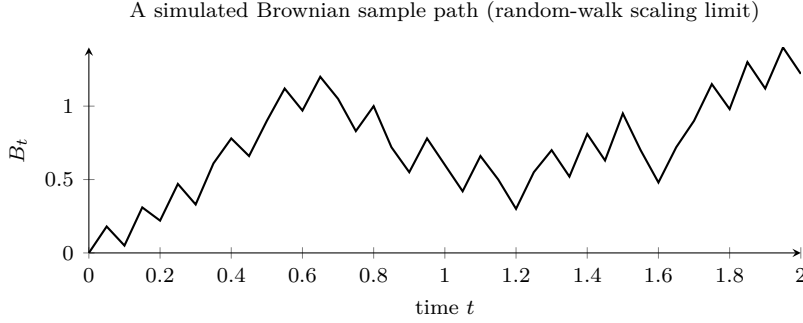


Figure 43.14. Brownian motion: independent Gaussian increments, variance growing linearly in time – the limit of $N^{-1/2}$ -scaled random walks via the CLT. (Schematic.)

Exercises (Stein, Ch.5) – weak convergence and Gaussian characteristic functions.

- Let $\{\mu_N\}_{N \geq 1}$ and ν be probability measures on $\mathbb{R}^d$. Prove the following are equivalent as $N \rightarrow \infty$. (a) $\hat{\mu}_N(\xi) \rightarrow \hat{\nu}(\xi)$ for all $\xi \in \mathbb{R}^d$. (b) $\mu_N \rightarrow \nu$ weakly. (c) In $\mathbb{R}$, $\mu_N((a, b)) \rightarrow \nu((a, b))$ for all open intervals (a, b) , assuming ν continuous. (d) In $\mathbb{R}^d$, $\mu_N(O) \rightarrow \nu(O)$ for all open sets O , assuming ν absolutely continuous w.r.t. Lebesgue measure. [Hint: the equivalence of (a)–(c) in $\mathbb{R}$ is implicit in Lemma 43.3.8 and the weak-convergence corollary; for (a) $\Rightarrow$ (d) handle open cubes, then closed cubes, then write any open set as an almost-disjoint union of cubes; for (d) $\Rightarrow$ (b) approximate continuous compactly supported φ by step functions on cubes.]
- The proof of Theorem 43.3.14 needs the following. Let σ be a strictly positive-definite symmetric matrix with inverse σ^{-1} , and ν_{σ^2} the measure on $\mathbb{R}^d$ with density $\frac{1}{(2\pi)^{d/2} \det \sigma} e^{-|\sigma^{-1}x|^2/2}$. Then $\widehat{\nu_{\sigma^2}}(\xi) = e^{-2\pi^2|\sigma\xi|^2}$. [Hint: make an orthogonal change of variables diagonalizing σ , reducing the d -dimensional integral to a product of one-dimensional Gaussian integrals.]

Problems (Stein, Ch.5) – general CLT and random flights.

- The following is a general (Lindeberg) form of the central limit theorem. Suppose $f_1, \dots, f_n, \dots$ are square integrable, mutually independent, each with mean 0. Let μ_n be the distribution of f_n , σ_n^2 its variance, and $S_n^2 = \sum_{k=1}^n \sigma_k^2$. The critical (Lindeberg) assumption is that for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \frac{1}{S_n^2} \sum_{k=1}^n \int_{|t| \geq \varepsilon S_n} t^2 d\mu_k(t) = 0.$$

Under these conditions the distribution measures of $\frac{1}{S_n} \sum_{k=1}^n f_k$ converge weakly to the standard normal ν (variance 1).

- * Suppose $\{f_n\}$ are identically distributed, square integrable, mutually independent, each with mean 0 and variance 1, and $s_n = \sum_{k=1}^n f_k$. Then for almost every x , $\limsup_{n \rightarrow \infty} s_n(x)/(2n \log \log n)^{1/2} = 1$ (the law of the iterated logarithm).
- (Random flight) A variant of the $\mathbb{R}^d$ random walk allows each unit step to be in any direction of the unit sphere: $s_n = f_1 + \dots + f_n$, with $\{f_n\}$ mutually independent and each f_n uniformly distributed on $S^{d-1} \subseteq \mathbb{R}^d$, the probability space being $\prod_j S_j$ with normalized surface measure. (a) Connect $\hat{\mu}(\xi)$ with Bessel functions. (b) What is the covariance matrix? (c) What is the limiting distribution of $s_n(x)/n^{1/2}$? [Hint: $\hat{\mu}(\xi) = \Gamma(d/2)(\pi|\xi|)^{(2-d)/2} J_{(d-2)/2}(2\pi|\xi|)$.]

Additional exercises.

- Compute $\int_{\mathbb{R}} e^{-t^2/(2\sigma^2)} dt = \sigma\sqrt{2\pi}$ from the standard Gaussian integral $\int_{\mathbb{R}} e^{-u^2/2} du = \sqrt{2\pi}$ by a change of variable, confirming the normalization in Definition 43.3.1.
- For i.i.d. Bernoulli(1/2) ± 1 steps and $N = 400$, use the CLT (Theorem 43.3.5) to estimate $m(\{S_N > 40\})$ in terms of the standard normal tail, and evaluate numerically using $\Phi(2) \approx 0.977$.

3. Let f_n be i.i.d. exponential with mean 1 (variance 1). State the limiting distribution of $(S_N - N)/N^{1/2}$ from Theorem 43.3.7, and explain why the limit is symmetric Gaussian despite the skewed exponential summands.
4. Using characteristic functions, prove that the sum of two independent Gaussians with variances σ_1^2, σ_2^2 is Gaussian with variance $\sigma_1^2 + \sigma_2^2$ (redo Worked Example for $k = 2$ in full detail).
5. (Non-example) Let f_n be i.i.d. with the Cauchy density $\frac{1}{\pi(1+t^2)}$. Explain why Theorem 43.3.7 does not apply, and (using $\hat{\mu}(\xi) = e^{-2\pi|\xi|}$) show that S_N/N has the *same* Cauchy distribution as a single f_n , so no $N^{1/2}$ scaling produces a Gaussian.
6. (Diagram problem) Plot the standard normal density together with a binomial histogram of $S_N/N^{1/2}$ for a modest N (say $N = 16$), and visually estimate the mass in $[-1, 1]$; compare to ≈ 0.683 .
7. Show that the characteristic function $\hat{\mu}$ of any probability measure satisfies $\hat{\mu}(0) = 1$, $|\hat{\mu}(\xi)| \leq 1$, and $\hat{\mu}(-\xi) = \overline{\hat{\mu}(\xi)}$. Which of these hold for the Gaussian $\widehat{\nu_{\sigma^2}}(\xi) = e^{-2\sigma^2\pi^2\xi^2}$?
8. For the $\mathbb{R}^2$ simple random walk, write down the limiting Gaussian density of $s_n/n^{1/2}$ from Worked Example (limiting law of the walk) with covariance $\frac{1}{2}I$, and sketch its level sets.
9. Suppose $\mu_N \rightarrow \nu$ weakly with ν continuous. Prove $\mu_N((a, b)) \rightarrow \nu((a, b))$ for every $a < b$, using the squeezing construction of the figure in §43.3 (smooth $\varphi_\varepsilon \leq \chi_{[a, b]} \leq \psi_\varepsilon$).
10. State the Lindeberg condition (Problem 3) for a triangular array and verify it holds for i.i.d. square-integrable summands, thereby recovering Theorem 43.3.7 as a special case.

Chapter 44

An Introduction to Brownian Motion

44.1 Construction of Brownian Motion

From rescaled random walk to a process on path space

Brownian motion is the mathematical idealization of the erratic motion of a pollen grain suspended in water. We build it as the scaling limit of the simple symmetric random walk: shrink the time between steps to $1/N$ and the step length to $N^{-1/2}$, in accordance with the central limit theorem, and pass to the limit $N \rightarrow \infty$. The analytic content is a statement about *weak convergence of measures on the space of continuous paths*, and that is what we make precise here.

Recall the random walk in $\mathbb{R}^d$ from the theory of probability: $s_n(x) = \sum_{k=1}^n r_k(x)$, where $\{r_k\}$ are mutually independent, each uniform on the $2d$ unit lattice neighbours of the origin, on the product probability space $(\mathbb{Z}_{2d}^\infty, m)$. Then $s_n(x) \in \mathbb{Z}^d$ and the walk moves to a neighbour in steps of unit time and unit distance. Joining successive lattice points by straight segments and rescaling gives, for each integer $N \geq 1$, the *interpolated rescaled walk*

$$S_t^{(N)}(x) = \frac{1}{N^{1/2}} \sum_{1 \leq k \leq [Nt]} r_k(x) + \frac{(Nt - [Nt])}{N^{1/2}} r_{[Nt]+1}(x), \quad 0 \leq t < \infty. \quad (44.1)$$

The first term sums the completed steps up to time t ; the second linearly interpolates across the step in progress. For each fixed N , $t \mapsto S_t^{(N)}$ is a *stochastic process*: a family of $\mathbb{R}^d$ -valued random variables on the fixed space $(\mathbb{Z}_{2d}^\infty, m)$, and for each x the trajectory $t \mapsto S_t^{(N)}(x)$ is a continuous, piecewise-linear path starting at the origin.

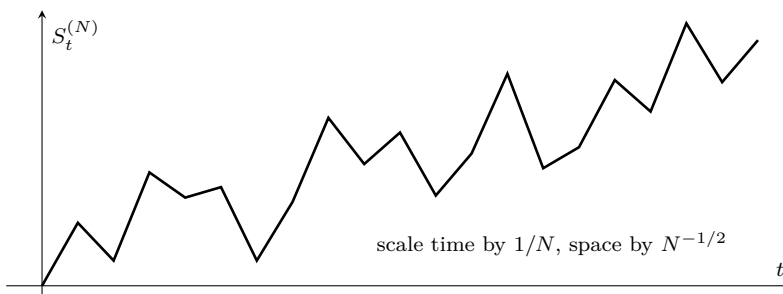


Figure 44.1. A rescaled random-walk path $t \mapsto S_t^{(N)}(x)$: continuous, piecewise linear, starting at 0. As $N \rightarrow \infty$ these paths roughen into Brownian paths. (Schematic.)

The defining properties of Brownian motion

Definition 44.1.1. A *Brownian motion* in $\mathbb{R}^d$ is a process $B_t = B_t(\omega)$ on a probability space (Ω, P) , with values in $\mathbb{R}^d$ for each $0 \leq t < \infty$, satisfying $B_0(\omega) = 0$ a.e. and:

B-1 (*independent increments*) if $0 \leq t_1 < t_2 < \dots < t_k$ then $B_{t_1}, B_{t_2} - B_{t_1}, \dots, B_{t_k} - B_{t_{k-1}}$ are mutually independent;

B-2 (*stationary Gaussian increments*) $B_{t+h} - B_t$ is Gaussian with mean zero and covariance hI (I the $d \times d$ identity), for every $t \geq 0, h > 0$;

B-3 (*continuity*) for a.e. ω , the path $t \mapsto B_t(\omega)$ is continuous on $[0, \infty)$.

In particular B_t is normal with mean zero and covariance tI .

The three axioms say: the future increment from time t is independent of the past (B-1), depends on the elapsed time only through its length and is Gaussian (B-2), and the trajectory has no jumps (B-3). Properties B-1 and B-2 already pin down all the finite-dimensional distributions; B-3 is what selects a genuinely path-valued object.

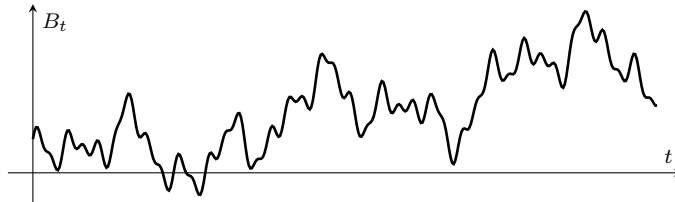


Figure 44.2. Schematic Brownian sample path $t \mapsto B_t(\omega)$: continuous but nowhere differentiable, with fine structure at every scale.

The canonical path space and Wiener measure

The natural home for Brownian motion is the space of its paths.

Definition 44.1.2. Let $\mathcal{P}$ be the space of continuous functions $p : [0, \infty) \rightarrow \mathbb{R}^d$ with $p(0) = 0$. For $p, p' \in \mathcal{P}$ set $d_n(p, p') = \sup_{0 \leq t \leq n} |p(t) - p'(t)|$ and

$$d(p, p') = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{d_n(p, p')}{1 + d_n(p, p')}. \quad (44.2)$$

This d is a metric, and $d(p_k, p) \rightarrow 0$ iff $p_k \rightarrow p$ *uniformly on compact subsets* of $[0, \infty)$. The bounded ratio $u/(1+u) \leq 1$ makes the series converge for any pair of paths, while the weights 2^{-n} ensure that closeness in d forces uniform closeness on each $[0, n]$. The metric space $(\mathcal{P}, d)$ is complete and separable (Exercises 2, 4). A crucial defect: $\mathcal{P}$ is *not* σ -compact (Exercise 3), so naive compactness arguments fail and we need Prokhorov's theorem below.

Definition 44.1.3. A *cylindrical set* in $\mathcal{P}$ is, for $0 \leq t_1 \leq \dots \leq t_k$ and a Borel set $A \subseteq \mathbb{R}^{dk}$,

$$\{p \in \mathcal{P} : (p(t_1), \dots, p(t_k)) \in A\}.$$

Let $\mathcal{C}$ be the σ -algebra they generate and $\mathcal{B}$ the Borel σ -algebra of $(\mathcal{P}, d)$.

Lemma 44.1.4. $\mathcal{C} = \mathcal{B}$.

Proof. If $O \subseteq \mathbb{R}^{dk}$ is open then $\{p : (p(t_1), \dots, p(t_k)) \in O\}$ is open in $\mathcal{P}$ (the evaluation $p \mapsto (p(t_1), \dots, p(t_k))$ is continuous), so it lies in $\mathcal{B}$; hence $\mathcal{C} \subseteq \mathcal{B}$. Conversely, for fixed n, a and a path p_0 , the set $\{p : \sup_{0 \leq t \leq n} |p(t) - p_0(t)| \leq a\}$ equals the set where the supremum is taken over *rational* $t \in [0, n]$ (by continuity), which is a countable intersection of cylindrical sets, hence in $\mathcal{C}$. From this one checks each open ball $\{p : d(p, p_0) < \delta\}$ lies in $\mathcal{C}$; since $\mathcal{P}$ is separable the open balls generate $\mathcal{B}$, so $\mathcal{B} \subseteq \mathcal{C}$. $\square$

A Borel probability measure μ on $\mathcal{P}$ has, for each $0 \leq t_1 \leq \dots \leq t_k$, a *section* (finite-dimensional distribution)

$$\mu^{(t_1, \dots, t_k)}(A) = \mu(\{p : (p(t_1), \dots, p(t_k)) \in A\}), \quad A \subseteq \mathbb{R}^{dk} \text{ Borel.} \quad (44.3)$$

By Lemma 44.1.4, two Borel measures on $\mathcal{P}$ are equal iff all their sections agree (they then agree on cylindrical sets, which form a generating algebra).

Definition 44.1.5. A Borel probability measure W on $\mathcal{P}$ is a *Wiener measure* if the process $\tilde{B}_t(p) = p(t)$ on $(\mathcal{P}, W)$ satisfies B-1, B-2, B-3. Existence of a Wiener measure is thus equivalent to existence of Brownian motion; we shall see it is unique, justifying “the” Wiener measure.

The rescaled walk (44.1) pushes the product measure m forward to a Borel measure μ_N on $\mathcal{P}$ via $\mu_N(A) = m(\{x : S_{\bullet}^{(N)}(x) \in A\})$. The evaluation $\pi = \pi^{t_1, \dots, t_k} : p \mapsto (p(t_1), \dots, p(t_k))$ is continuous, so its composite with $S^{(N)}$ carries m all the way to a measure on $\mathbb{R}^{dk}$: the section $\mu_N^{(t_1, \dots, t_k)} = \mu_N \circ \pi^{-1} = m \circ (\pi \circ S^{(N)})^{-1}$.

$$\begin{array}{ccc} (\mathbb{Z}_{2d}^\infty, m) & \xrightarrow{S^{(N)}} & \mathcal{P} \\ & \searrow \pi \circ S^{(N)} & \downarrow \pi \\ & & \mathbb{R}^{dk} \end{array}$$

Our target:

The measures μ_N converge weakly to the Wiener measure W as $N \rightarrow \infty$.

Weak convergence $\mu_N \rightarrow \mu$ means $\int f d\mu_N \rightarrow \int f d\mu$ for every $f \in C_b(\mathcal{P})$ (bounded continuous). Note this is far weaker than pointwise convergence of paths — indeed $S^{(N)}$ and B live on different probability spaces, so pointwise a.e. convergence would not even be meaningful.

Worked Example 44.1.6 (The one-dimensional Gaussian density and its normalization). Verify that $g_t(y) = (2\pi t)^{-1/2} e^{-y^2/2t}$ integrates to 1 over $\mathbb{R}$ and has variance t , so it is the density of B_t in dimension $d = 1$.

Recognition. B-2 says $B_t \sim N(0, t)$; we confirm the stated density is a probability density with the right second moment.

Solution. $\int_{\mathbb{R}} e^{-y^2/2t} dy = \sqrt{2\pi t}$ by the Gaussian integral (substitute $y = \sqrt{t}u$, reducing to $\int e^{-u^2/2} du = \sqrt{2\pi}$), so $\int g_t = 1$. For the variance, $\int y^2 g_t dy = t$ by the standard moment $\int u^2 e^{-u^2/2} du / \sqrt{2\pi} = 1$ and rescaling. *Check.* At $t = 1$, g_1 is the standard normal density; its peak $g_1(0) = 1/\sqrt{2\pi} \approx 0.399$, the familiar value.

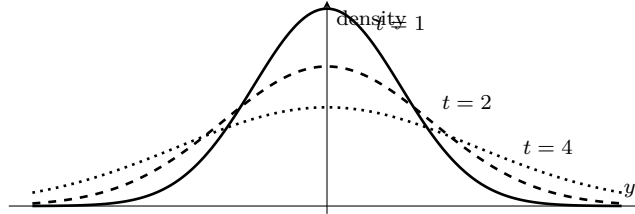


Figure 44.3. Densities $g_t(y) = (2\pi t)^{-1/2} e^{-y^2/2t}$ of B_t : the Gaussian spreads and flattens as t grows, with variance t .

Prokhorov’s theorem and tightness

A sequence of probability measures need not have a weakly convergent subsequence on a non-compact space; the missing hypothesis is *tightness*.

Definition 44.1.7. A sequence $\{\mu_N\}$ of probability measures on a metric space X is *tight* if for every $\varepsilon > 0$ there is a compact $K \subseteq X$ with $\mu_N(K^c) \leq \varepsilon$ for all N .

Lemma 44.1.8 (Prokhorov). *If $\{\mu_N\}$ on a metric space X is tight, then some subsequence μ_{N_k} converges weakly to a probability measure μ .*

Proof. For each compact $K_{1/m}$ from tightness at $\varepsilon = 1/m$, build a countable $D_m \subseteq C_b(X)$ with (i) $\{g|_{K_{1/m}} : g \in D_m\}$ dense in $C(K_{1/m})$ and (ii) $\sup_X |g| = \sup_{K_{1/m}} |g|$ for $g \in D_m$. Since $K_{1/m}$ is compact, $C(K_{1/m})$ is separable; extend a countable dense family from $K_{1/m}$ to all of X by Tietze, preserving sup-norm, to obtain D_m . Let $D = \bigcup_m D_m$, countable. By diagonalization pass to a subsequence (relabelled μ_N) with $\mu_N(g) = \int g d\mu_N$ convergent for every $g \in D$.

Fix $f \in C_b(X)$ and m . Choose $g \in D_m$ with $|(f - g)(x)| \leq 1/m$ on $K_{1/m}$. Splitting $X = K_{1/m} \cup K_{1/m}^c$ and using (ii),

$$|\mu_N(f - g)| \leq \int_{K_{1/m}} |f - g| d\mu_N + \int_{K_{1/m}^c} |f - g| d\mu_N \leq \frac{1}{m} + (2\|f\|_\infty + \frac{1}{m})\frac{1}{m},$$

so $\limsup_N \mu_N(f) - \liminf_N \mu_N(f) = O(1/m)$; letting $m \rightarrow \infty$, $\lim_N \mu_N(f)$ exists. Define $\ell(f) = \lim_N \mu_N(f)$, a positive linear functional with $|\ell(f)| \leq \sup_K |f| + \varepsilon \|f\|_\infty$ for the tightness set K ; by the Riesz representation theorem (Theorem 7.4, Ch.1) ℓ is given by a Borel measure μ , and $\mu_N \rightarrow \mu$ weakly. $\square$

Corollary 44.1.9. *If $\{\mu_N\}$ is tight and, for each $0 \leq t_1 \leq \dots \leq t_k$, the sections $\mu_N^{(t_1, \dots, t_k)}$ converge weakly to some $\mu_{t_1, \dots, t_k}$, then the full sequence $\{\mu_N\}$ converges weakly to a measure μ with $\mu^{(t_1, \dots, t_k)} = \mu_{t_1, \dots, t_k}$.*

Proof. By Lemma 44.1.8 a subsequence $\mu_{N_m} \rightarrow \mu$ weakly. Writing $\pi^{t_1, \dots, t_k}(p) = (p(t_1), \dots, p(t_k))$, which is continuous, weak convergence and $\mu^{(t_1, \dots, t_k)} = \mu \circ (\pi^{t_1, \dots, t_k})^{-1}$ give $\mu^{(t_1, \dots, t_k)} = \mu_{t_1, \dots, t_k}$. If the full sequence did not converge to μ , some $\mu_{N_n} \rightarrow \nu \neq \mu$ along a further subsequence (again by Lemma 44.1.8); but ν has the same sections as μ , forcing $\nu = \mu$, a contradiction. $\square$

The compact subsets of $\mathcal{P}$ are governed by an Arzela–Ascoli criterion.

Lemma 44.1.10. *A closed set $K \subseteq \mathcal{P}$ is compact if for each $T > 0$ there is a bounded function $h \mapsto w_T(h)$ on $(0, 1]$ with $w_T(h) \rightarrow 0$ as $h \rightarrow 0$ and*

$$\sup_{p \in K} \sup_{0 \leq t \leq T} |p(t+h) - p(t)| \leq w_T(h), \quad h \in (0, 1]. \quad (44.4)$$

Condition (44.4) is uniform equicontinuity on each $[0, T]$ with a common modulus; the conclusion is Arzela–Ascoli (uniform boundedness follows from $p(0) = 0$ and the modulus). This is exactly the geometric handle we need: to prove tightness of $\{\mu_N\}$ we exhibit, for each ε , a compact K of the form (44.4) carrying mass $\geq 1 - \varepsilon$.

Worked Example 44.1.11 (Tightness on the line via Chebyshev). On $X = \mathbb{R}$ let $\mu_N = N(0, \sigma_N^2)$ with $\sup_N \sigma_N^2 = \sigma^2 < \infty$. Show $\{\mu_N\}$ is tight.

Recognition. Tightness on $\mathbb{R}$ means mass escaping to $\pm\infty$ is uniformly small; control it by a uniform second-moment (Chebyshev) bound.

Solution. Take $K = [-M, M]$. By Chebyshev, $\mu_N(K^c) = \mu_N(|y| > M) \leq \sigma_N^2/M^2 \leq \sigma^2/M^2$. Given $\varepsilon > 0$ choose $M = \sigma/\sqrt{\varepsilon}$; then $\mu_N(K^c) \leq \varepsilon$ for all N , and K is compact. *Check.* If instead $\sigma_N^2 \rightarrow \infty$ tightness fails — no fixed M works — matching the intuition that spreading Gaussians lose their mass to infinity.

Construction of the Wiener measure

Theorem 44.1.12. *The measures μ_N on $\mathcal{P}$ converge weakly as $N \rightarrow \infty$ to a measure, which is the Wiener measure W . On $(\mathcal{P}, W)$ the process $B_t(p) = p(t)$ satisfies B-1, B-2, B-3, and W is the unique such measure.*

The proof has two steps: tightness of $\{\mu_N\}$ (delicate, via a maximal inequality), then convergence of the finite-dimensional sections (the central limit theorem). The maximal step rests on the following.

Lemma 44.1.13. *For the unscaled walk $s_n = \sum_{k \leq n} r_k$, for every $p \geq 2$,*

$$\sup_{n \geq 1} m\left(\left\{x : \sup_{k \leq n} |s_k(x)| > \lambda n^{1/2}\right\}\right) = O(\lambda^{-p}), \quad \lambda \rightarrow \infty. \quad (44.5)$$

Proof. Apply the martingale maximal theorem to the stopped sequence $s'_k = s_{\min(k, n)}$, with $s_n^* = \sup_{k \leq n} |s_k|$: $m(s_n^* > \alpha) \leq \frac{1}{\alpha} \int_{|s_n| > \alpha} |s_n| dm$. Multiply by $p\alpha^{p-1}$ and integrate to get $\int (s_n^*)^p dm \leq \left(\frac{p}{p-1}\right)^p \int |s_n|^p dm$. Khinchin's inequality gives $\int |s_n|^p dm \leq A\left(\int |s_n|^2 dm\right)^{p/2}$. Hence $m(s_n^* > \alpha) \leq \alpha^{-p} \|s_n^*\|_{L^p}^p \leq A\alpha^{-p} \|s_n\|_{L^2}^p$. Setting $\alpha = \lambda n^{1/2}$ and using $\|s_n\|_{L^2}^2 = n^{1/2}$ yields (44.5). $\square$

Proof of Theorem 44.1.12. Tightness. Fix $0 < a < 1/2$ and $\varepsilon > 0$, and work first with $T = 1$. We claim that for a sufficiently large c_1 ,

$$m\left(\left\{x : \sup_{\substack{0 \leq t \leq 1 \\ 0 \leq h \leq \delta}} |S_{t+h}^{(N)} - S_t^{(N)}| > c_1 \delta^a \text{ for some } \delta \leq 1\right\}\right) \leq \varepsilon. \quad (44.6)$$

Granting (44.6), set $K^{(1)} = \{p : \sup_{0 \leq t \leq 1, 0 \leq h \leq \delta} |p(t+h) - p(t)| \leq c_1 \delta^a, \forall \delta \leq 1\}$; then $\mu_N((K^{(1)})^c) \leq \varepsilon$, and $K^{(1)}$ satisfies (44.4) with $w_1(\delta) = c_1 \delta^a$, hence is compact.

To prove (44.6), fix $\delta = 2^{-k}$ and partition $[0, 1]$ at $t_j = j2^{-k}$, $0 \leq j \leq 2^k$. For any f , $\sup_{0 \leq t \leq 1, 0 \leq h \leq \delta} |f(t+h) - f(t)| \leq 2 \max_j \sup_{0 \leq h \leq \delta} |f(t_j+h) - f(t_j)|$. By stationarity of the walk, $m(\sup_{0 \leq h \leq \delta} |S_{t_j+h}^{(N)} - S_{t_j}^{(N)}| > \sigma/2)$ is independent of j and equals $m(\sup_{n \leq \delta N} |s_n| > (\sigma/2)N^{1/2})$. With $\lambda = \sigma/(2\delta^{1/2})$, Lemma 44.1.13 gives $m(\sup_{0 \leq t \leq 1, 0 \leq h \leq \delta} |S_{t+h}^{(N)} - S_t^{(N)}| > \sigma) = O(\delta^{-1}(\sigma/2\delta^{1/2})^{-p})$. Putting $\sigma = c_1 \delta^a$ makes this $O(c_1^{-p} 2^{-kb})$ with $b = -1 + (\frac{1}{2} - a)p$; choosing p large makes $b > 0$. Summing over k (each $\delta \in (0, 1]$ lies in some $[2^{-k+1}, 2^{-k}]$) gives total mass $O(c_1^{-p}) \leq \varepsilon$ for c_1 large, which is (44.6).

The same argument for general T gives $K^{(T)}$ with $\mu_N((K^{(T)})^c) \leq \varepsilon_T$. Take $\varepsilon_n = \varepsilon/2^n$, $K = \bigcap_n K^{(n)}$; then $\mu_N(K^c) \leq \varepsilon$ for all N , and K is compact by Lemma 44.1.10. Hence $\{\mu_N\}$ is tight.

Convergence of sections. By the central limit theorem the distribution of $S_{t_j}^{(N)} - S_{t_{j-1}}^{(N)}$ converges weakly to the Gaussian $\nu_{t_j - t_{j-1}}$ of mean 0 and covariance $(t_j - t_{j-1})I$. Writing $S_t^{(N)}$ as a telescoping sum of independent increments and using that the joint law of independent increments is the product of their laws, the sections $\mu_N^{(t_1, \dots, t_k)}$ converge weakly to the Gaussian product measure $W^{(t_1, \dots, t_k)}$. By Corollary 44.1.9, $\mu_N \rightarrow W$ weakly. The limit W has exactly the finite-dimensional distributions forced by B-1, B-2, and continuity (B-3) holds because W is supported on the compact equicontinuous sets $K^{(T)}$; uniqueness follows since the sections determine W . $\square$

Remark. The construction generalizes far beyond the simple walk. If $f_1, f_2, \dots$ are i.i.d. square-integrable $\mathbb{R}^d$ -valued, mean zero, covariance I , and $S_t^{(N)}$ is built from them as in (44.1), then the induced μ_N still converge weakly to W . This is the *Donsker invariance principle* (Problem 2): the scaling limit is Wiener measure regardless of the microscopic step distribution — a sweeping universality, the path-space analogue of the central limit theorem.

Worked Example 44.1.14 (Covariance of Brownian motion). Show that for a one-dimensional Brownian motion, $\mathbb{E}[B_s B_t] = \min(s, t)$.

Recognition. Use independence of increments (B-1) and the variance $\text{Var}(B_t) = t$ (B-2). Decompose B_t into B_s plus an independent increment when $s \leq t$.

Solution. Suppose $s \leq t$. Write $B_t = B_s + (B_t - B_s)$, where $B_t - B_s$ is independent of B_s and has mean zero. Then $\mathbb{E}[B_s B_t] = \mathbb{E}[B_s^2] + \mathbb{E}[B_s(B_t - B_s)] = s + \mathbb{E}[B_s] \mathbb{E}[B_t - B_s] = s + 0 = s$. Symmetrically, for $t \leq s$ we get t ; together $\mathbb{E}[B_s B_t] = \min(s, t)$. *Check.* At $s = t$ this is t , recovering $\text{Var}(B_t) = t$. The covariance kernel $\min(s, t)$ characterizes the Wiener process among Gaussian processes.

Worked Example 44.1.15 (Increments are stationary but the process is not). Decide whether $\{B_t\}$ is a stationary process, and contrast with the stationarity of its increments.

Recognition. “Stationary process” means $(B_{t_1+c}, \dots, B_{t_k+c})$ has a law independent of c ; “stationary increments” means $B_{t+h} - B_t$ has a law depending only on h .

Solution. The increments are stationary: $B_{t+h} - B_t \sim N(0, hI)$ for every t (B-2). But the process itself is *not* stationary: $\text{Var}(B_t) = t$ grows with t , so B_1 and B_{100} have different laws. *Check.* A stationary process would have constant variance; the linear growth of variance is the signature of diffusion and is exactly why Brownian motion wanders off rather than settling into equilibrium.

Caution. Weak convergence $\mu_N \rightarrow W$ does *not* say a fixed sequence of random-walk paths converges to a Brownian path. The processes live on different spaces; convergence is of *induced measures* tested against bounded continuous functionals on $\mathcal{P}$. Moreover the rectilinear walk paths form a set of W -measure zero in $\mathcal{P}$ — the limit object is genuinely rougher than every approximant.

Worked Example 44.1.16 (A non-example: a process with independent but non-Gaussian increments). Let N_t be a Poisson process (independent increments, $N_{t+h} - N_t \sim \text{Poisson}(h)$). Explain why N_t is not a Brownian motion.

Recognition. Check the three axioms one at a time; independence (B-1) holds, so look at B-2 and B-3.

Solution. The increments are independent (B-1 holds) and stationary, but $N_{t+h} - N_t$ is Poisson, not Gaussian, and has mean $h \neq 0$, so B-2 fails on two counts. Worse, the paths are step functions with unit jumps, so B-3 (continuity) fails. *Check.* The Poisson increment has variance h like Brownian motion, but its *integer* values and positive mean immediately disqualify it — continuity and the centered Gaussian law are essential, not incidental.

Worked Example 44.1.17 (The metric on path space really induces local uniform convergence). Show that for the metric (44.2), $d(p_k, p) \rightarrow 0$ iff $p_k \rightarrow p$ uniformly on every compact $[0, n]$.

Recognition. Use that $\phi(u) = u/(1+u)$ is increasing and bounded by 1, so each term is controlled by d_n , and the weights 2^{-n} make the tail negligible.

Solution. ($\Leftarrow$) If $d_n(p_k, p) \rightarrow 0$ for each fixed n , then each term $2^{-n}\phi(d_n(p_k, p)) \rightarrow 0$ and is dominated by 2^{-n} ; by dominated convergence for series, $d(p_k, p) \rightarrow 0$. ($\Rightarrow$) If $d(p_k, p) \rightarrow 0$ then for fixed n , $2^{-n}\phi(d_n(p_k, p)) \leq d(p_k, p) \rightarrow 0$, so $\phi(d_n(p_k, p)) \rightarrow 0$, hence $d_n(p_k, p) \rightarrow 0$ ($\phi(u) \rightarrow 0$ forces $u \rightarrow 0$). *Check.* The single number $d(p, p')$ packages the whole family $\{d_n\}$ of seminorms; this is the standard trick for metrizing convergence on a countable family of compacta.

Worked Example 44.1.18 (Weak convergence does not imply convergence of all measures of sets). Let $\mu_N = \delta_{1/N}$ on $\mathbb{R}$ (point mass at $1/N$). Show $\mu_N \rightarrow \delta_0$ weakly but $\mu_N(\{0\}) \not\rightarrow \delta_0(\{0\})$.

Recognition. Weak convergence tests against *continuous* bounded f ; indicators of sets like $\{0\}$ are not continuous, so set-measures can jump.

Solution. For $f \in C_b(\mathbb{R})$, $\int f d\mu_N = f(1/N) \rightarrow f(0) = \int f d\delta_0$, so $\mu_N \rightarrow \delta_0$ weakly. Yet $\mu_N(\{0\}) = 0$ for all N while $\delta_0(\{0\}) = 1$, so $\mu_N(\{0\}) \rightarrow 0 \neq 1$. *Check.* This is exactly why the Portmanteau theorem (Exercise 7) only gives $\liminf \mu_N(O) \geq \mu(O)$ for open O , with equality requiring $\mu(\partial O) = 0$. The set $\{0\}$ is its own boundary, of mass 1.

Exercises (Stein, Ch.6).

1. Show that if $t > 0$, the distribution measure of $S_t^{(N)}$ converges weakly to the Gaussian ν_t of mean zero and variance t as $N \rightarrow \infty$. More generally, for $t > s \geq 0$, the distribution of $S_t^{(N)} - S_s^{(N)}$ converges weakly to the Gaussian ν_{t-s} of mean zero and covariance $(t-s)I$. [Hint: with $f_k = r_k$ one has $S_t^{(N)} - S_{N,t} = \frac{(Nt - [Nt])}{N^{1/2}} r_{[Nt]+1}$.]
2. Let $(\mathcal{P}, d)$ be the metric space of Section 2. Verify: (a) it is complete; (b) it is separable. [Hint for (b): with $e_1, \dots, e_d$ a basis of $\mathbb{R}^d$, use polynomials $p(t) = e_1 p_1(t) + \dots + e_d p_d(t)$ with the p_j having rational coefficients.]
3. Show that $(\mathcal{P}, d)$ is not σ -compact. [Hint: otherwise the Baire category theorem gives a compact set with nonempty interior, hence an open ball with compact closure; test against piecewise-linear f_n with $f_n(0) = 1$, $f_n(x) = 0$ for $x \geq 1/n$.]
4. Suppose X is a compact metric space. Show: (a) X is separable; (b) $C(X)$ is separable. [Hint: for each m take a finite cover $\mathcal{B}_m$ by balls of radius $1/m$; for (a) use the centres; for (b) use rational combinations of the partition of unity subordinate to $\mathcal{B}_m$.]
5. Let X be a metric space, $K \subseteq X$ compact, f continuous on K . Show there is a continuous F on X with $F|_K = f$ and $\sup_X |F| = \sup_K |f|$ (Tietze with sup-norm control).
6. Suppose $K \subseteq \mathcal{P}$ is compact. Show that for each $T > 0$ there is $w_T(h)$ on $(0, 1]$ with $w_T(h) \rightarrow 0$ as $h \rightarrow 0$ and $\sup_{p \in K} \sup_{0 \leq t \leq T} |p(t+h) - p(t)| \leq w_T(h)$. [Hint: each p is uniformly continuous on $[0, T]$; use a finite subcover of K .] (This is the converse to Lemma 44.1.10.)
7. Suppose $\mu_N \rightarrow \mu$ weakly. Show: (a) $\liminf_N \mu_N(O) \geq \mu(O)$ for open O ; (b) $\lim_N \mu_N(O) = \mu(O)$ if $\mu(\overline{O} \setminus O) = 0$. [Hint: $\mu(O) = \sup_f \left\{ \int f d\mu : 0 \leq f \leq 1, \text{ supp } f \subseteq O \right\}$.]

8. Given Wiener measure W on $\mathcal{P}$ we have a realization of Brownian motion with $B_t(\omega) = p(t)$, $\Omega = \mathcal{P}$, $P = W$. Conversely, starting from $\{B_t\}$ satisfying B-1, B-2, B-3, define $W^o(C) = P(\{\omega : (B_{t_1}(\omega), \dots, B_{t_k}(\omega)) \in A\})$ on cylindrical sets C . Verify that W^o extends to the Wiener measure on $\mathcal{P}$.
9. (Uniqueness up to relabelling.) Call a process “strict” if additionally (i) $B_t(\omega_1) = B_t(\omega_2)$ for all t implies $\omega_1 = \omega_2$, and (ii) the measurable sets are exactly $\mathcal{A}_\infty$, the σ -algebra generated by the $\mathcal{A}_t$, $t < \infty$. Given any $\{B_t\}$ on (Ω, P) , form $\Omega^\#$ as equivalence classes under $\omega_1 \sim \omega_2$ iff $B_t(\omega_1) = B_t(\omega_2)$ for all t , with $B_t^\#(\{\omega\}) = B_t(\omega)$ and $P^\#(\{A\}) = P(A)$ for $A \in \mathcal{A}_\infty$. Verify: (a) $B_t^\#$ is a strict Brownian motion on $(\Omega^\#, P^\#)$; (b) the $(\mathcal{P}, W)$ of Section 3 is strict; (c) any two Brownian motions are isomorphic up to null sets: there is a bijection $\Phi : (\Omega^1)^\# \rightarrow (\Omega^2)^\#$ with $(P^2)^\#(\Phi(A)) = (P^1)^\#(A)$ and $(B_t^2)^\#(\Phi(\omega)) = (B_t^1)^\#(\omega)$.
10. Calculate the joint probability distribution measure of $(B_{t_1}, B_{t_2}, \dots, B_{t_k})$. [Hint: use independence of increments to write the joint density as a product of Gaussian transition densities.]

Problems (Stein, Ch.6).

1. (Continuity is automatic from B-1, B-2.) Suppose for each $t \geq 0$ we are given $F_t \in L^p(X, m)$ with $\|F_{t_1} - F_{t_2}\|_{L^p} \leq c|t_1 - t_2|^\alpha$, where $\alpha > 1/p$ and $1 \leq p \leq \infty$. Then there is a corrected $\tilde{F}_t$ with $F_t = \tilde{F}_t$ a.e. for each t , such that $t \mapsto \tilde{F}_t(x)$ is continuous for a.e. x , and indeed satisfies a Holder condition of every order $\gamma < \alpha - 1/p$. (This is the Kolmogorov continuity theorem.)
2. (Donsker invariance principle.) Let $f_1, f_2, \dots$ be i.i.d. square-integrable $\mathbb{R}^d$ -valued, mean zero, covariance I , and define $S_t^{(N)} = \frac{1}{N^{1/2}} \sum_{1 \leq k \leq [Nt]} f_k + \frac{(Nt - [Nt])}{N^{1/2}} f_{[Nt]+1}$ with induced measures μ_N on $\mathcal{P}$. (a) Using Exercise 11 in place of Lemma 44.1.13, show that for $T = 1$, $\eta, \sigma > 0$ there are $0 < \delta < 1$ and N_0 with $m(\{x : \sup_{0 < h < \delta} |S_{t+h}^{(N)} - S_t^{(N)}| > \sigma\}) \leq \delta\eta$ for $0 \leq t \leq 1$, $N \geq N_0$. (b) Deduce that for all $T > 0$, $\varepsilon, \sigma > 0$ there is $\delta > 0$ with $m(\{x : \sup_{0 \leq t \leq T, 0 < h < \delta} |S_{t+h}^{(N)} - S_t^{(N)}| > \sigma\}) \leq \varepsilon$ for all $N \geq 1$. (c) Use (b) to show $\{\mu_N\}$ is tight. (d) Conclude that $\{\mu_N\}$ converges weakly to W .
3. (Hilbert-space construction.) On (Ω, P) take i.i.d. Gaussian $\{f_n\}$ of mean zero and covariance I ; they form an orthonormal sequence in $L^2(\Omega, \mathbb{R}^d)$. Let H be their closed span, an infinite-dimensional separable Hilbert space, and let $U : L^2([0, \infty), dx) \rightarrow H$ be a unitary map. Set $B_t = U(\chi_{[0,t]})$. Show that, after the correction of Problem 1, $\{B_t\}$ is a Brownian motion; note that if $B_t = \sum c_n(t)f_n$ then $B_t - B_s = \sum [c_n(t) - c_n(s)]f_n$ with $\sum |c_n(t) - c_n(s)|^2 = t - s$.

Additional exercises.

1. Verify directly from the definition that $u/(1+u)$ is increasing and concave on $[0, \infty)$ and bounded by 1, and use this to show the series (44.2) for $d(p, p')$ converges for every pair $p, p' \in \mathcal{P}$.
2. Compute the transition density of Brownian motion: show that for $0 < t_1 < t_2$ the joint density of (B_{t_1}, B_{t_2}) in $d = 1$ is $g_{t_1}(x_1) g_{t_2-t_1}(x_2 - x_1)$ with $g_t(y) = (2\pi t)^{-1/2} e^{-y^2/2t}$, and explain how this encodes B-1 and B-2.
3. Let X_N have density μ_N converging weakly to μ . Give an example in which $\int x d\mu_N \not\rightarrow \int x d\mu$, and explain why this does not contradict weak convergence. (Mass escaping to infinity; the test function x is unbounded.)
4. Show that the cylindrical sets in $\mathcal{P}$ form an algebra (closed under finite unions, intersections, complements), and that the intersection of two cylindrical sets is cylindrical. Conclude that two Borel measures agreeing on all sections are equal.
5. Prove that the set P_N of rescaled random-walk paths is a closed subset of $\mathcal{P}$ of Wiener measure zero, and interpret this as “the limit is rougher than every approximant.”
6. (Tightness fails without a uniform moment bound.) Exhibit probability measures $\{\mu_N\}$ on $\mathbb{R}$ that are *not* tight, and verify no subsequence converges weakly to a probability measure. (Take $\mu_N = \delta_N$.)
7. Using scaling invariance, show that if W is the Wiener measure then the pushforward of W under $p \mapsto \delta^{-1/2} p(\delta \cdot)$ is again W . Draw a picture illustrating one rescaling step on a sample path.

8. (Non-example.) Let $X_t = tZ$ for a single standard Gaussian Z . Check that X_t is Gaussian with $\text{Var}(X_t) = t^2$, has continuous paths, but fails B-1. Conclude X_t is not a Brownian motion despite being a centered Gaussian process with continuous paths.
9. Show the Gaussian product measure $W^{(t_1, \dots, t_k)}$ is rotation invariant in each $\mathbb{R}^d$ factor, and use this to predict that $o(B_t)$ is again a Brownian motion before proving Theorem 44.2.1(b).

44.2 Regularity and Properties of Brownian Paths

Invariances of Brownian motion

Brownian motion is invariant under a rich group of transformations. Each is proved by checking that the transformed process again satisfies B-1, B-2, B-3 — and since those axioms involve only covariances and independence, the verification is short.

Theorem 44.2.1. *Each of the following is again a Brownian motion:*

- (a) (scaling) $\delta^{-1/2}B_{t\delta}$, for fixed $\delta > 0$;
- (b) (rotation) $o(B_t)$, for an orthogonal o on $\mathbb{R}^d$;
- (c) (restart) $B_{t+\sigma_0} - B_{\sigma_0}$, for a constant $\sigma_0 \geq 0$.

Proof. (a) For any f , $\delta^{-1/2}f$ has covariance δ^{-1} times that of f ; so $\delta^{-1/2}B_{t\delta}$ has increment covariance $\delta^{-1} \cdot (h\delta)I = hI$, verifying B-2, while B-1, B-3 are inherited. (b) An orthogonal o preserves covariance matrices and preserves independence of a family, so B-1, B-2, B-3 hold for $o(B_t)$. (c) Immediate from the definition: the shifted increments are increments of B . $\square$

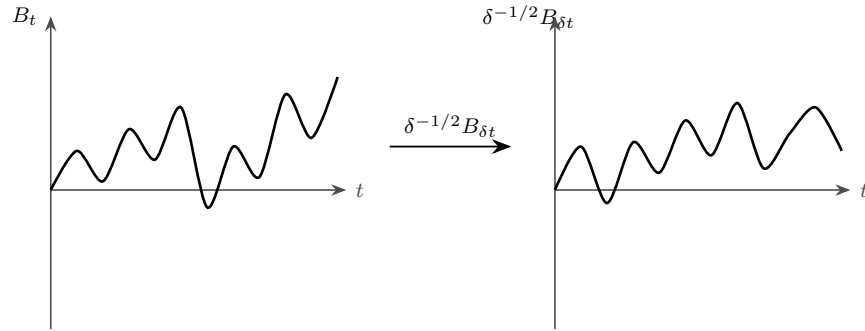


Figure 44.4. Scaling invariance: compressing time by δ and space by $\delta^{1/2}$ returns a statistically identical process — Brownian motion is a random fractal, self-similar with Hurst exponent $1/2$. (Schematic.)

Worked Example 44.2.2 (Time inversion is also a Brownian motion). Show that $B'_t = tB_{1/t}$ (with $B'_0 = 0$) is a Brownian motion when B_t is.

Recognition. Verify B-2 through the covariance: a centered Gaussian process with the Wiener covariance kernel $\min(s, t)$ is a Brownian motion; continuity at 0 needs a separate remark.

Solution. For $s, t > 0$, $\mathbb{E}[B'_s B'_t] = st \mathbb{E}[B_{1/s} B_{1/t}] = st \min(1/s, 1/t) = st \cdot \frac{1}{\max(s, t)} = \min(s, t)$. So B' has the Wiener covariance and, being a linear image of a Gaussian process, is centered Gaussian with independent increments; thus B-1, B-2 hold. Continuity for $t > 0$ is clear, and continuity at 0 ($tB_{1/t} \rightarrow 0$ as $t \rightarrow 0$) follows from the growth bound $|B_s| = O(s^{1/2+\varepsilon})$. *Check.* $\text{Var}(B'_t) = t^2 \cdot (1/t) = t$, as required for B-2.

Holder regularity of paths

The construction already shows Brownian paths are quite regular — but only just. They satisfy a Holder condition of every exponent below $1/2$, and fail it above $1/2$. The critical exponent $1/2$ matches the scaling space $\sim \sqrt{\text{time}}$.

Theorem 44.2.3. *With W the Wiener measure:*

(a) *If $0 < a < 1/2$ and $T > 0$, then for W -a.e. path p ,*

$$\sup_{0 \leq t \leq T, 0 < h \leq 1} \frac{|p(t+h) - p(t)|}{h^a} < \infty.$$

(b) *If $a > 1/2$, then for W -a.e. path p and every $t \geq 0$, $\limsup_{h \rightarrow 0} \frac{|p(t+h) - p(t)|}{h^a} = \infty$.*

Proof. (a) is implicit in the construction. The set $K^{(T)} = \{p : \sup_{0 \leq t \leq T, 0 \leq h \leq \delta} |p(t+h) - p(t)| \leq c_T \delta^a, \forall \delta \leq 1\}$ satisfies $\mu_N(K^{(T)}) \geq 1 - \varepsilon$ for all N , hence $W(K^{(T)}) \geq 1 - \varepsilon$ by weak convergence; every $p \in K^{(T)}$ has the bounded difference quotient in (a). Letting $\varepsilon \rightarrow 0$ gives the conclusion a.e.

(b) Fix $a > 1/2$ and an integer k with $dk(a - \frac{1}{2}) > 1$. If some $t_0 \in [0, 1]$ has

$$\sup_{0 < h \leq (k+1)/n} |p(t_0 + h) - p(t_0)| / h^a \leq \lambda,$$

then there is j_0 with

$$\max_{1 \leq \ell \leq k} \left| p\left(\frac{j_0 + \ell + 1}{n}\right) - p\left(\frac{j_0 + \ell}{n}\right) \right| \leq C_k \lambda n^{-a};$$

absorb C_k into λ . Let E_n^λ be the set of such paths; then $E_n^\lambda \subseteq \tilde{E}_n^\lambda = \bigcup_{j_0=0}^{n-1} \{p : \max_\ell \left| p\left(\frac{j_0 + \ell + 1}{n}\right) - p\left(\frac{j_0 + \ell}{n}\right) \right| \leq \lambda n^{-a}\}$. The k increments are independent and identically distributed, so

$$W(E_n^\lambda) \leq n(W\{p : |p(1/n)| \leq \lambda n^{-a}\})^k.$$

By scaling (Theorem 44.2.1(a)), $W\{|p(1/n)| \leq \lambda n^{-a}\} = W\{|p(1)| \leq \lambda n^{1/2-a}\} = O(\lambda^d n^{d(1/2-a)})$ since $p(1)$ is Gaussian. Hence $W(E_n^\lambda) = O(\lambda^{dk} n^{1+dk(1/2-a)}) \rightarrow 0$ as $n \rightarrow \infty$ because $dk(a - \frac{1}{2}) > 1$. As this holds for every λ , conclusion (b) follows. $\square$

Corollary 44.2.4. *For W -a.e. path, $t \mapsto B_t$ is nowhere differentiable.*

Proof. Differentiability at t would force $|p(t+h) - p(t)|/h$ bounded as $h \rightarrow 0$, i.e. a Holder-1 condition; but $1 > 1/2$, so Theorem 44.2.3(b) (which gives the $\limsup = \infty$ at every t a.e.) rules this out. $\square$

Remark. Nowhere-differentiable continuous functions have surfaced repeatedly: as a lacunary Fourier series, as the von Koch fractal, as the *generic* continuous function by a Baire-category argument, and now as *almost every* Brownian path. Category and measure agree here: roughness is the rule, smoothness the exception.

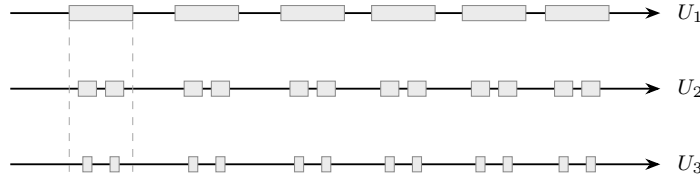


Figure 44.5. Baire category: each U_n is a union of intervals spread densely across the line, with $U_{n+1} \subseteq U_n$ refining inside the previous intervals; the countable intersection $\bigcap_n U_n$ is still dense. The nowhere-differentiable functions form such a residual set in $C[0, 1]$, paralleling their full Wiener measure. (Schematic.)

Worked Example 44.2.5 (Quadratic variation of Brownian motion). For a one-dimensional Brownian motion on $[0, T]$, with the dyadic partition $t_j = jT/2^n$, show that $\sum_j (B_{t_{j+1}} - B_{t_j})^2 \rightarrow T$ in L^2 as $n \rightarrow \infty$.

Recognition. “Quadratic variation”: sum the squared increments. Each increment has variance $T/2^n$, and increments are independent, so compute mean and variance of the sum.

Solution. Let $\Delta_j = B_{t_{j+1}} - B_{t_j} \sim N(0, T/2^n)$, independent over j , with 2^n terms. Then $\mathbb{E}[\sum_j \Delta_j^2] = 2^n \cdot (T/2^n) = T$. For the variance, using $\text{Var}(Z^2) = 2\sigma^4$ for $Z \sim N(0, \sigma^2)$, $\text{Var}(\sum_j \Delta_j^2) = 2^n \cdot 2(T/2^n)^2 = 2T^2/2^n \rightarrow 0$. Hence the sum converges to T in L^2 . *Check.* Contrast with the *first* variation $\sum |\Delta_j|$, which diverges — finite nonzero quadratic variation but infinite total variation is precisely why $\int f dB$ needs Ito’s, not Stieltjes’, calculus.

Worked Example 44.2.6 (Growth at infinity: a strong law of large numbers). Show that a.e. $|B_t| = O(t^{1/2+\varepsilon})$ as $t \rightarrow \infty$, for every $\varepsilon > 0$.

Recognition. Use the Brownian maximal inequality on dyadic blocks $[0, 2^k]$ together with Borel–Cantelli, exactly as for the strong law for sums.

Solution. Let $B_T^* = \sup_{0 \leq t \leq T} |B_t|$. The maximal inequality (see §44.2 below) gives $W(B_T^* > \alpha) \leq \frac{1}{\alpha} \|B_T\|_{L^1} = cT^{1/2}/\alpha$. Put $E_k = \{B_{2^k}^* > 2^{k(1/2+\varepsilon)}\}$; then $W(E_k) \leq c2^{k/2}/2^{k(1/2+\varepsilon)} = c2^{-k\varepsilon}$, summable. By Borel–Cantelli only finitely many E_k occur a.e., so $B_{2^k}^* \leq 2^{k(1/2+\varepsilon)}$ eventually; monotonicity in T gives $|B_t| = O(t^{1/2+\varepsilon})$. *Check.* The exponent $1/2$ is sharp up to the log log correction of the law of the iterated logarithm (Problem 5).

The martingale connection and the maximal inequality

Brownian motion is a continuous-time martingale, and this is the engine behind the exit estimates used for the Dirichlet problem.

For each $t \geq 0$ let $\mathcal{A}_t$ be the σ -algebra generated by $\{B_s : 0 \leq s \leq t\}$ (completed by null sets). Then:

- (a) for any $0 \leq t_0 < t_1 < \dots$, the sequence $\{B_{t_n}\}$ is a martingale relative to $\{\mathcal{A}_{t_n}\}$ (independent, mean-zero increments);
- (b) for a.e. ω , $t \mapsto B_t(\omega)$ is continuous (this is B-3).

Theorem 44.2.7 (Brownian maximal inequality). For all $T > 0$ and $\alpha > 0$,

$$P\left(\{\omega : \sup_{0 \leq t \leq T} |B_t(\omega)| > \alpha\}\right) \leq \frac{1}{\alpha} \|B_T\|_{L^1}. \quad (44.7)$$

Proof. Discretize $[0, T]$ and apply the martingale maximal inequality $m(s_n^* > \alpha) \leq \frac{1}{\alpha} \int_{|s_n| > \alpha} |s_n| dm \leq \frac{1}{\alpha} \|s_n\|_{L^1}$ to the martingale $\{B_{t_j}\}$ along a dense set of times. Passing to the supremum over a countable dense set of $t \in [0, T]$, which by continuity (b) equals the supremum over all $t \in [0, T]$, yields (44.7). $\square$

Worked Example 44.2.8 (Reading off the tail from the maximal inequality). In dimension $d = 1$, estimate $P(\sup_{0 \leq t \leq T} |B_t| > \alpha)$ for large α .

Recognition. Plug $\|B_T\|_{L^1}$ into (44.7); for a Gaussian $B_T \sim N(0, T)$ the L^1 norm is explicit.

Solution. For $Z \sim N(0, T)$, $\mathbb{E}|Z| = \sqrt{2T/\pi}$. Hence (44.7) gives $P(\sup_{t \leq T} |B_t| > \alpha) \leq \frac{1}{\alpha} \sqrt{2T/\pi} = O(T^{1/2}/\alpha)$. *Check.* The true reflection-principle value $P(\sup_{t \leq T} B_t > \alpha) = 2P(B_T > \alpha)$ decays like $e^{-\alpha^2/2T}$, far faster; the maximal inequality is a crude but dimension-free and martingale-only bound, which is all the Dirichlet argument needs.

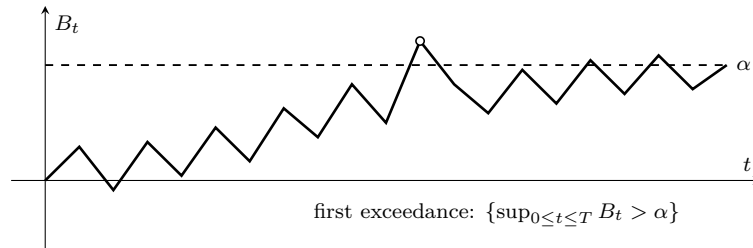


Figure 44.6. The maximal inequality controls the probability that a path's running maximum on $[0, T]$ ever exceeds the level α . (Schematic.)

Worked Example 44.2.9 (Holder continuity fails at the endpoint exponent $a = 1/2$). Explain why Theorem 44.2.3 stops short of $a = 1/2$, and what happens there.

Recognition. The proof of (a) used $0 < a < 1/2$ to keep the modulus $c_T \delta^a \rightarrow 0$ fast enough, and (b) used $a > 1/2$; the borderline needs a finer (iterated-logarithm) analysis.

Solution. At $a = 1/2$ the supremum $\sup_{0 < h \leq 1} |p(t+h) - p(t)|/h^{1/2}$ is in fact infinite a.e.: by Exercise 14, $\limsup_{h \rightarrow 0} |B_{t+h} - B_t|/h^{1/2} = \infty$, so paths are not Holder-1/2. The true local modulus is $\sqrt{2h \log \log(1/h)}$

(law of the iterated logarithm), which beats $h^{1/2}$ by the slowly growing $\sqrt{\log \log}$ factor. *Check.* So the regularity is “1/2 minus epsilon” but not 1/2 — the critical exponent is attained only up to a logarithmic correction.

Worked Example 44.2.10 (Scaling exponent from dimensional analysis). Predict the Holder exponent of Brownian motion purely from the scaling invariance $\delta^{-1/2}B_{\delta t} \stackrel{d}{=} B_t$.

Recognition. Self-similarity fixes how spatial increments scale with temporal ones; read off the exponent.

Solution. Under $t \mapsto \delta t$, B scales as $\delta^{1/2}$: $B_{\delta t} \stackrel{d}{=} \delta^{1/2}B_t$. So an increment over a time-step h has typical size $h^{1/2}$, i.e. $|B_{t+h} - B_t| \approx h^{1/2}$. This is the critical exponent 1/2; any $a < 1/2$ is satisfied (increments are even smaller relative to h^a), any $a > 1/2$ fails. *Check.* The prediction 1/2 matches Theorem 44.2.3 exactly — the exponent is the scaling dimension, no probability computation required.

Worked Example 44.2.11 (Reflection symmetry and an even functional). Show $-B_t$ is a Brownian motion, and deduce $\mathbb{E}[|B_t|] = \sqrt{2t/\pi}$ in $d = 1$.

Recognition. Reflection $o(y) = -y$ is orthogonal, so Theorem 44.2.1(b) applies; then compute the first absolute moment of a centered Gaussian.

Solution. Since $y \mapsto -y$ is orthogonal, $-B_t$ satisfies B-1, B-2, B-3, hence is a Brownian motion; in particular B_t and $-B_t$ have the same law, so B_t is symmetric. For $B_t \sim N(0, t)$, $\mathbb{E}|B_t| = \int_{\mathbb{R}} |y| (2\pi t)^{-1/2} e^{-y^2/2t} dy = 2 \int_0^\infty y (2\pi t)^{-1/2} e^{-y^2/2t} dy = \sqrt{2t/\pi}$. *Check.* This is the constant $c = \sqrt{2/\pi}$ in $\|B_t\|_{L^1} = ct^{1/2}$ used in the growth and maximal estimates.

Worked Example 44.2.12 (A non-example: linear paths have zero quadratic variation). Contrast Brownian quadratic variation with that of a smooth path $\phi(t) = ct$ on $[0, T]$.

Recognition. Compute $\sum (\phi(t_{j+1}) - \phi(t_j))^2$ for the smooth path and let the mesh $\rightarrow 0$.

Solution. With $t_j = jT/2^n$, $\phi(t_{j+1}) - \phi(t_j) = cT/2^n$, so $\sum_{j=0}^{2^n-1} (cT/2^n)^2 = 2^n \cdot c^2 T^2 / 2^{2n} = c^2 T^2 / 2^n \rightarrow 0$. So a C^1 (indeed Lipschitz) path has *zero* quadratic variation. *Check.* Brownian motion has quadratic variation $T \neq 0$ (previous worked example), the qualitative gap that separates rough paths from smooth ones and makes Ito calculus necessary.

Worked Example 44.2.13 (Independent increments force the covariance kernel). Show that any centered process with independent increments and $\text{Var}(X_t) = t$ has $\mathbb{E}[X_s X_t] = \min(s, t)$, regardless of Gaussianity.

Recognition. Only B-1 (independence) and the variances are used; mimic the Brownian covariance computation.

Solution. For $s \leq t$, $\mathbb{E}[X_s X_t] = \mathbb{E}[X_s^2] + \mathbb{E}[X_s(X_t - X_s)] = s + \mathbb{E}[X_s]\mathbb{E}[X_t - X_s] = s$ by independence and mean zero. So $\mathbb{E}[X_s X_t] = \min(s, t)$. *Check.* The kernel $\min(s, t)$ comes from independent increments alone; Gaussianity (B-2) is what additionally fixes the *full* law, not just the second moments. This is why the covariance kernel does not, by itself, identify Brownian motion among all such processes.

Exercises (Stein, Ch.6).

- (Khinchin's inequality, general form.) Suppose $\{f_n\}$ are identically distributed $\mathbb{R}^d$ -valued, bounded, mean zero, mutually independent. Then for any $p < \infty$, $\|\sum a_n f_n\|_{L^p} \leq A_p (\sum |a_n|^2)^{1/2}$. [Hint: reduce to $d = 1$; assuming $\sum |a_n|^2 \leq 1$, expand $\int e^{\sum a_n f_n}$ as $\prod_n \int e^{a_n f_n}$ and use $e^u = 1 + u + O(u^2)$ for $|u| \leq M$, bounding the integral by $\prod (1 + M^2 a_n^2)$ if $|f_n| \leq M$.]
- (Variant of Lemma 44.1.13.) Let $\{f_k\}$ be i.i.d. $\mathbb{R}^d$ -valued on (X, m) , mean zero, covariance I , and $s_n = \sum_{k=1}^n f_k$. Then $\limsup_{n \rightarrow \infty} m(\{x : \sup_{1 \leq k \leq n} |s_k(x)| > \lambda n^{1/2}\}) = O(\lambda^{-p})$ for $p > 0$. [Hint: with ν_n the distribution of $s_n/n^{1/2}$ and $\alpha = \lambda n^{1/2}$, the maximal bound's right side is $\frac{1}{\lambda} \int_{|t| > \lambda} |t| d\nu_n(t)$; split at λM and use $\int \frac{|s_n|^2}{n} dm = 1$ for the inner part and the central limit theorem for the outer.]
- Prove that almost everywhere $|B_t(\omega)| = O(t^{1/2+\varepsilon})$ as $t \rightarrow \infty$, for every $\varepsilon > 0$ (the analogue of the strong law of large numbers). [Hint: with $B_T^* = \sup_{0 \leq t \leq T} |B_t|$, the maximal inequality gives $W(\{B_T^* > \alpha\}) \leq cT^{1/2}/\alpha$; for $E_k = \{B_{2^k}^* > 2^{k(1/2+\varepsilon)}\}$, $\sum_k W(E_k) < \infty$, so apply Borel–Cantelli.]

4. (Time inversion.) If B_t is a Brownian motion then so is $B'_t = tB_{1/t}$. [Hint: continuity at the origin follows from the previous exercise; for B-2 use the conditional-expectation exercise of the previous chapter.]
5. Show that $\limsup_{t \rightarrow 0} \frac{|B_t(\omega)|}{t^{1/2}} = \infty$ almost everywhere; hence almost all Brownian paths are not Holder-1/2. Also show $\limsup_{t \rightarrow \infty} \frac{|B_t(\omega)|}{t^{1/2}} = \infty$ a.e.; hence almost all paths exit every ball. [Hint: by time inversion it suffices to treat $t \rightarrow 0$ in $d = 1$; using $W(\{|B_\alpha - B_\beta| > \gamma\}) = \frac{1}{\sqrt{2\pi(\beta-\alpha)}} \int_{|u|>\gamma} e^{-u^2/2(\beta-\alpha)} du$ for $\beta > \alpha$, get $W(\{|B_{2^{-k}} - B_{2^{-k+1}}| > 2^{-k/2}\mu_k\}) \geq c_1 e^{-c_2\mu_k^2}$; choose $\mu_k \rightarrow \infty$ so slowly that $\sum_k e^{-c_2\mu_k^2} = \infty$ and apply Borel–Cantelli.]

Problems (Stein, Ch.6).

1. (Law of the iterated logarithm.) For an $\mathbb{R}$ -valued Brownian motion, for almost all ω ,

$$\limsup_{t \rightarrow \infty} \frac{B_t(\omega)}{\sqrt{2t \log \log t}} = 1 \quad \text{and} \quad \liminf_{t \rightarrow \infty} \frac{B_t(\omega)}{\sqrt{2t \log \log t}} = -1.$$

By time inversion (Exercise 13), for almost all ω ,

$$\limsup_{t \rightarrow 0} \frac{B_t(\omega)}{\sqrt{2t \log \log(1/t)}} = 1 \quad \text{and} \quad \liminf_{t \rightarrow 0} \frac{B_t(\omega)}{\sqrt{2t \log \log(1/t)}} = -1.$$

This describes the precise amplitude of the oscillations as $t \rightarrow \infty$ and $t \rightarrow 0$.

Additional exercises.

1. Show that for $0 < a < 1/2$, the Holder seminorm $\sup_{0 \leq t \leq T, 0 < h \leq 1} |p(t+h) - p(t)|/h^a$ is finite on each compact equicontinuous set $K^{(T)}$ from the construction, and deduce Theorem 44.2.3(a) for those paths.
2. Compute $\mathbb{E}[B_t^4]$ for one-dimensional Brownian motion and verify the Gaussian moment relation $\mathbb{E}[B_t^4] = 3(\mathbb{E}[B_t^2])^2 = 3t^2$.
3. (Quadratic variation, general partitions.) Let $0 = t_0 < \dots < t_n = T$ have mesh $\delta = \max_j(t_{j+1} - t_j)$. Show $\sum_j (B_{t_{j+1}} - B_{t_j})^2 \rightarrow T$ in L^2 as $\delta \rightarrow 0$, generalizing the dyadic case.
4. Using the maximal inequality, prove $W(\{\sup_{0 \leq t \leq T} |B_t| > \alpha\}) \leq cT^{1/2}/\alpha$ and identify the constant c in $d = 1$ from $\|B_T\|_{L^1}$.
5. Show that almost every Brownian path has infinite total variation on every interval $[0, T]$. [Hint: finite total variation would force zero quadratic variation; compare with the worked example.]
6. (Self-similarity exponent.) For $\theta > 0$, decide for which θ the process $\theta^{-1/2}B_{\theta t}$ is again a Brownian motion, and for which the process $\theta^{-1}B_{\theta t}$ is. (All $\theta > 0$ for the first; none for the second unless one rescales time differently.)
7. (Non-example.) Let $\phi(t) = t \sin(1/t)$ ($\phi(0) = 0$). Show ϕ is continuous and of unbounded variation near 0 but has *zero* quadratic variation; hence unbounded variation alone does not make a path “Brownian-rough.”
8. Sketch (as a diagram) three nested dense open sets $U_1 \supseteq U_2 \supseteq U_3$ in $[0, 1]$ whose intersection is dense, and explain the analogy between the residual set of nowhere-differentiable functions in $C[0, 1]$ and the full-measure set of nowhere-differentiable Brownian paths.
9. Prove that $-B_t$ and, for fixed $c \in \mathbb{R}^d$ orthogonal-image $o(B_t)$, have the same finite-dimensional distributions as B_t , and use symmetry to show $\mathbb{E}[B_t^{2k+1}] = 0$ for all k in $d = 1$.

44.3 The Dirichlet Problem and Brownian Motion

The Dirichlet problem and harmonic measure

Let $R \subseteq \mathbb{R}^d$ be a bounded open set and f continuous on $\partial R = \overline{R} \setminus R$. The *Dirichlet problem* asks for u , continuous on $\overline{R}$, harmonic in R ($\Delta u = 0$), with $u|_{\partial R} = f$. The probabilistic solution, going back to Kakutani,

is strikingly simple to state: run a Brownian motion from $x \in R$ until it exits, and average f over where it lands.

Fix $x \in R$ and consider $B_t^x = x + B_t$. Let

$$\tau(\omega) = \tau^x(\omega) = \inf\{t \geq 0 : B_t^x(\omega) \notin R\}$$

be the *first exit time*; then $B_{\tau^x}^x(\omega) \in \partial R$. The induced *harmonic measure* on ∂R is

$$\mu^x(E) = P(\{\omega : B_{\tau^x}^x(\omega) \in E\}), \quad E \subseteq \partial R \text{ Borel}, \quad (44.8)$$

and the candidate solution is

$$u(x) = \int_{\partial R} f(y) d\mu^x(y) = \int_{\Omega} f(x + B_{\tau^x}^x(\omega)) dP(\omega), \quad x \in R. \quad (44.9)$$

The two integrals in (44.9) agree because μ^x is the pushforward of P along the exit map $e^x : \omega \mapsto B_{\tau^x}^x(\omega)$, so integrating f against μ^x on ∂R equals integrating $f \circ e^x$ against P on Ω :

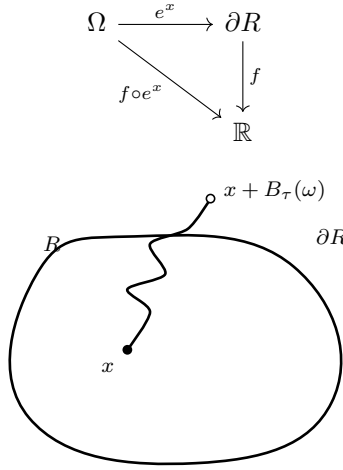


Figure 44.7. A Brownian path started at x exits the region R at the boundary point $x + B_{\tau(\omega)}^x$; averaging f over the exit location gives $u(x)$. (Schematic.)

Stopping times and the Blumenthal zero-one law

Definition 44.3.1. A non-negative function $\tau : \Omega \rightarrow [0, \infty]$ is a *stopping time* (relative to $\{\mathcal{A}_t\}$) if $\{\omega : \tau(\omega) \leq t\} \in \mathcal{A}_t$ for every $t \geq 0$. The strict exit time is $\tau_*(\omega) = \inf\{t > 0 : B_t^x(\omega) \notin R\}$.

The defining condition says: whether the path has stopped by time t is decidable from the history up to time t — no peeking into the future. The first exit time is the basic example.

Proposition 44.3.2. Both τ^x and τ_* are stopping times.

Proof. Take $x = 0$ (replace R by $R - x$ otherwise). For open O , $\tau_O = \inf\{t \geq 0 : B_t \in O\}$ satisfies $\{\tau_O < t\} = \bigcup_{r < t, r \in \mathbb{Q}} \{B_r \in O\}$ up to null sets, since a continuous path enters the open set O before t iff it is in O at some rational $r < t$; hence $\{\tau_O < t\} \in \mathcal{A}_t$. With $O_n = \{y : \text{dist}(y, R^c) < 1/n\}$, for $t > 0$ we have $\{\tau \leq t\} = \bigcap_n \{\tau_{O_n} < t\} \in \mathcal{A}_t$ (a path exits R by t iff it enters every O_n before t). Since $\{\tau = 0\} \in \{\emptyset, \Omega\}$, τ is a stopping time; the same works for τ_* , and $\{\tau_* = 0\} \in \bigcap_{t > 0} \mathcal{A}_t =: \mathcal{A}_{0+}$. $\square$

Lemma 44.3.3 (Blumenthal zero-one law). $\mathcal{A}_{0+} = \mathcal{A}_0$; equivalently, every $A \in \bigcap_{t > 0} \mathcal{A}_t$ has $P(A) \in \{0, 1\}$.

Proof. Fix a bounded continuous f on $\mathbb{R}^{kd}$ and $0 \leq t_1 < \dots < t_k$. For $\delta > 0$ set $f_\delta = f(B_{t_1+\delta} - B_\delta, \dots, B_{t_k+\delta} - B_{t_{k-1}+\delta})$. If $A \in \mathcal{A}_{0+}$ then $A \in \mathcal{A}_\delta$, and by independence of the increments from $\mathcal{A}_\delta$, $\int_A f_\delta dP = P(A) \int_\Omega f_\delta dP$. Let $\delta \rightarrow 0$ (paths are continuous): $\int_A f_0 dP = P(A) \int_\Omega f_0 dP$. Any bounded continuous g has the form $g(x_1, \dots, x_k) = f(x_1, x_2 - x_1, \dots, x_k - x_{k-1})$, so $\int_A g(B_{t_1}, \dots, B_{t_k}) dP = P(A) \int_\Omega g(B_{t_1}, \dots, B_{t_k}) dP$. Approximating, this holds for g the indicator of a Borel set, giving $P(A \cap E) = P(A)P(E)$ for every cylindrical, hence every Borel, E . Taking $E = A$ gives $P(A) = P(A)^2$, so $P(A) \in \{0, 1\}$. $\square$

Definition 44.3.4. A boundary point $x \in \partial R$ is *regular* if $P(\{\tau_*^x = 0\}) = 1$: almost every path started at x leaves R at arbitrarily small positive times.

By Blumenthal, for $x \in \partial R$ the event $\{\tau_*^x = 0\}$ has probability 0 or 1, so the dichotomy regular/non-regular is exhaustive. Regularity is precisely what is needed for the boundary values to be attained.

Worked Example 44.3.5 (An interior point is never an exit point at time zero). For $x \in R$ (open), show $\tau^x(\omega) > 0$ a.e., so interior points cannot be regular boundary points.

Recognition. R open means a whole ball around x lies in R ; a continuous path needs positive time to leave it.

Solution. Pick $r > 0$ with $B(x, r) \subseteq R$. Since $t \mapsto B_t^x(\omega)$ is continuous with $B_0^x = x$, there is (for a.e. ω) some $t_0 > 0$ with $B_t^x \in B(x, r) \subseteq R$ for $0 \leq t \leq t_0$, hence $\tau^x \geq t_0 > 0$. *Check.* Consistent with the zero-one law — the event $\{\tau_*^x = 0\}$ has probability 0 here. The whole subtlety of regularity lives at the boundary, never the interior.

The strong Markov property

For a stopping time σ , let $\mathcal{A}_\sigma = \{A : A \cap \{\sigma \leq t\} \in \mathcal{A}_t, \forall t\}$ (the “past up to σ ”); this is a σ -algebra, σ is $\mathcal{A}_\sigma$ -measurable, and B_σ is $\mathcal{A}_\sigma$ -measurable.

Theorem 44.3.6 (Strong Markov property). *If B_t is a Brownian motion and σ a stopping time, then $B_t^* = B_{t+\sigma}(\omega) - B_\sigma(\omega)$ is again a Brownian motion, independent of $\mathcal{A}_\sigma$.*

Proof. If $\sigma \equiv \sigma_0$ is constant the claim is Theorem 44.2.1(c). If σ takes only countably many values $\sigma_1 < \sigma_2 < \dots$, write $\mathbf{B}^* = (B_{t_1}^*, \dots, B_{t_k}^*)$. For a Borel $E \subseteq \mathbb{R}^{kd}$ and $A \in \mathcal{A}_\sigma$, decompose $\{\mathbf{B}^* \in E\} \cap A = \bigsqcup_\ell (\{\mathbf{B}_{(\ell)}^* \in E\} \cap A \cap \{\sigma = \sigma_\ell\})$, where on $\{\sigma = \sigma_\ell\}$ the restart is by the constant σ_ℓ . Since $A \cap \{\sigma = \sigma_\ell\} \in \mathcal{A}_{\sigma_\ell}$ is independent of the post- σ_ℓ increment, the constant-time case gives

$$P(\{\mathbf{B}^* \in E\} \cap A) = P(\{\mathbf{B} \in E\}) P(A), \quad (44.10)$$

where $\mathbf{B} = (B_{t_1}, \dots, B_{t_k})$. Taking $A = \Omega$ shows B-1, B-2 (B-3 is clear), and the general A gives independence from $\mathcal{A}_\sigma$.

For general σ , approximate from above by discrete stopping times $\sigma^{(n)}(\omega) = k2^{-n}$ when $(k-1)2^{-n} < \sigma(\omega) \leq k2^{-n}$ (and 0 if $\sigma = 0$). Then $\sigma^{(n)} \downarrow \sigma$, each $\sigma^{(n)}$ is a stopping time, and $\mathcal{A}_\sigma \subseteq \mathcal{A}_{\sigma^{(n)}}$. Applying (44.10) to $\sigma^{(n)}$ and passing to the limit (using continuity of paths and a monotone class argument to pass from open E to all Borel E) gives (44.10) for σ . $\square$

In words: restart a Brownian motion at a stopping time and you get a fresh, independent Brownian motion. This “begin anew” property is the continuous-time form of the Markov property, strengthened to random times.

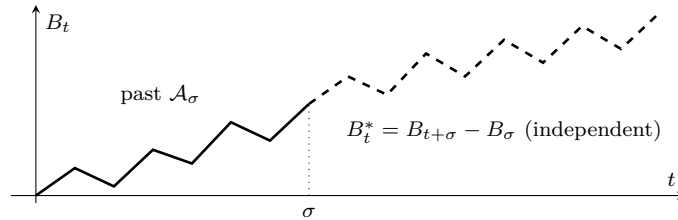


Figure 44.8. Strong Markov property: after the stopping time σ , the recentered path (dashed) is a new Brownian motion independent of the past $\mathcal{A}_\sigma$. (Schematic.)

A version adapted to the Dirichlet problem involves two stopping times $\sigma \leq \tau$ with τ the exit time. With the stopped process $\widehat{B}_t^y(\omega) = y + B_{t \wedge \tau^y}(\omega)$, one has, for bounded Borel F on $\mathbb{R}^d$ and $t \geq 0$,

$$\int_{\Omega} F(\widehat{B}_{\sigma(\omega)+t}(\omega)) dP(\omega) = \int_{\Omega \times \Omega} F(\widehat{B}_t^{y(\omega')}(\omega)) dP(\omega) dP(\omega'), \quad y(\omega') = \widehat{B}_{\sigma(\omega')}(\omega'). \quad (44.11)$$

This says: a path stopped at the boundary, observed at time $\sigma + t$, can be generated by first running to σ to land at $y(\omega')$, then running an independent copy from $y(\omega')$ — the precise sense in which the exit distribution composes.

Worked Example 44.3.7 (Harmonic measure on a sphere is the uniform measure). Let S be a sphere centred at x with $\overline{B(x, r)} \subseteq R$, and let σ be the first time B_t^x hits S . Show that the exit distribution on S is the normalized surface measure m .

Recognition. The map $G \mapsto \int_{\Omega} G(B_{\sigma}^x) dP$ is a positive linear functional on $C(S)$, hence a measure; rotation invariance of Brownian motion forces it to be uniform.

Solution. Take $x = 0$. The functional $G \mapsto \int_{\Omega} G(B_{\sigma}) dP$ is positive, linear, and bounded by $\sup_S |G|$, so by Riesz it equals $\int_S G d\mu$ for a probability measure μ on S . By Theorem 44.2.1(b) Brownian motion is rotation invariant, so μ is rotation invariant; the unique rotation-invariant probability measure on the sphere is the normalized surface measure m . Thus $\mu = m$. *Check.* This is the continuous analogue of the fact that the harmonic measure of a ball, seen from its centre, is uniform on the bounding sphere — the mean-value property in disguise.

Solution of the Dirichlet problem

Theorem 44.3.8. If u is defined by (44.9), then

- (a) u is harmonic in R ;
- (b) $u(x) \rightarrow f(y)$ as $x \rightarrow y$ ($x \in R$) whenever $y \in \partial R$ is regular.

Proof. (a) Fix $x \in R$ and a sphere S centred at x with closed ball inside R . We prove the *mean-value property* $u(x) = \int_S u(y) dm(y)$, which characterizes harmonicity (a locally mean-value, hence smooth, function is harmonic; see Exercise 19). Let σ be the first hitting time of S . By the previous worked example, for continuous G on S , $\int_{\Omega} G(B_{\sigma}^x) dP = \int_S G dm$. Since a path from x meets S before ∂R , the stopped process satisfies $\hat{B}_{\sigma}^x = B_{\sigma}^x =: y(\omega) \in S$. Apply (44.11) with F a bounded continuous extension of f and let $t \rightarrow \infty$:

$$\int_{\Omega \times \Omega} F(B_{\tau}^{y(\omega')}(\omega)) dP dP' = \int_{\Omega} F(B_{\tau^x}^x(\omega)) dP.$$

The right side is $u(x)$; the left side is $\int_{\Omega} u(y(\omega')) dP' = \int_S u dm$ by the sphere identity with $G = u$. Hence $u(x) = \int_S u dm$, the mean-value property, so u is harmonic.

(b) Let $y \in \partial R$ be regular. First, for every $\delta > 0$,

$$\lim_{x \rightarrow y, x \in R} P(\{\tau^x > \delta\}) = 0. \quad (44.12)$$

Indeed $P(\{B_t^x \in R, \eta \leq t \leq \delta\})$ is continuous in x (by continuity of paths and dominated convergence) and decreases to $P(\{\tau^x > \delta\})$ as $\eta \rightarrow 0$, so the latter is upper semicontinuous; thus $\limsup_{x \rightarrow y} P(\tau^x > \delta) \leq P(\tau^y > \delta) = 0$ since y is regular. Next, given $s, \varepsilon > 0$, by the maximal inequality choose δ with $P(\sup_{t \leq \delta} |B_t| > s/2) \leq \varepsilon/2$ (using $\|B_{\delta}\|_{L^1} = c\delta^{1/2}$), and by (44.12) take x near y with $P(\tau^x > \delta) \leq \varepsilon/2$; then

$$P(\{|y - B_{\tau^x}^x| > s\}) < \varepsilon. \quad (44.13)$$

Now split $u(x) - f(y) = \int_{\partial R} (f(y') - f(y)) d\mu^x(y') = I_1 + I_2$ over $\partial R_1 = \{|y' - y| \leq s\}$ and its complement ∂R_2 . By (44.13), $\mu^x(\partial R_2) \leq \varepsilon$, so $|I_2| \leq 2 \sup |f| \cdot \varepsilon$; and $|I_1| \leq \varepsilon$ since $|f(y) - f(y')| < \varepsilon$ for $|y - y'| \leq s$ (continuity of f , s small). Hence $|u(x) - f(y)| = O(\varepsilon)$ for x near y , proving (b). $\square$

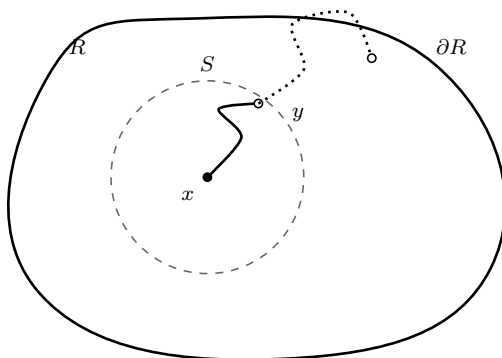


Figure 44.9. Mean-value proof: stop on the inner sphere S (uniform there), then run independently to ∂R — composing exit distributions yields $u(x) = \int_S u \, dm$. (Schematic.)

Regularity via the outside cone condition

Proposition 44.3.9. A (truncated) cone is $\Gamma = \{y \in \mathbb{R}^d : |y| < \alpha(y \cdot \gamma), |y| < \delta\}$ (γ a unit vector, $\alpha > 1$, $\delta > 0$). If $x \in \partial R$ and $x + \Gamma$ is disjoint from R , then x is regular.

Proof. Take $x = 0$. Let A be the event that the path enters Γ along a sequence of times $\rightarrow 0$. With $A_n = \bigcup_{r_k < 1/n} \{B_{r_k} \in \Gamma\}$ (r_k enumerating positive rationals), $A = \bigcap_n A_n$ and $A_n \in \mathcal{A}_{1/n}$, so $A \in \mathcal{A}_{0+}$ and by Blumenthal $P(A) \in \{0, 1\}$. If $P(A) = 0$, then by rotation invariance the same holds for every rotate of Γ ; but finitely many rotates of Γ cover the punctured ball $B(0, \delta) \setminus \{0\}$, which every path enters at arbitrarily small times — a contradiction. So $P(A) = 1$: a.e. path enters Γ , hence leaves R (since Γ avoids R), at arbitrarily small times. Thus 0 is regular. $\square$

Definition 44.3.10. R satisfies the *outside cone condition* if every $x \in \partial R$ admits a truncated cone Γ with $x + \Gamma$ disjoint from R .

Corollary 44.3.11. If the bounded open R satisfies the outside cone condition and $f \in C(\partial R)$, then there is a unique $u \in C(\bar{R})$, harmonic in R , with $u|_{\partial R} = f$.

Proof. Existence and boundary attainment: Theorem 44.3.8 with Proposition 44.3.9 (every boundary point is regular). Uniqueness: the maximum principle — the difference of two solutions is harmonic with zero boundary values, hence identically zero. $\square$

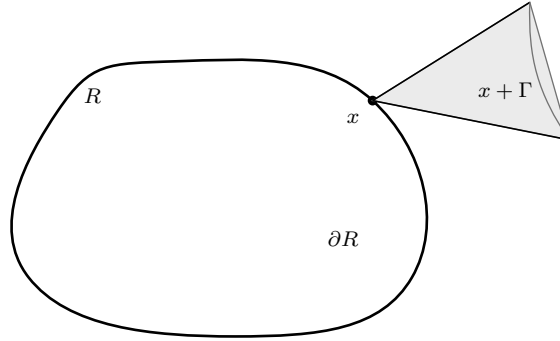


Figure 44.10. Outside cone condition: a truncated cone $x + \Gamma$ pokes out of R at each boundary point, forcing regularity and hence solvability of the Dirichlet problem. (Schematic.)

Worked Example 44.3.12 (The punctured disc: an irregular point). In $\mathbb{R}^d$, $d \geq 3$, remove the centre from an open ball: $R = B(0, 1) \setminus \{0\}$. Explain why the origin is a non-regular boundary point.

Recognition. A single point is invisible to $d \geq 3$ Brownian motion (it has positive capacity only for $d \leq 2$); use the transience/non-hitting of points.

Solution. For $d \geq 3$, Brownian motion almost surely never hits the single point 0 (Problem 4). So a path started near 0 does not exit R by landing at 0 in arbitrarily small times — the strict exit time τ_*^0 from R is not 0 a.s. in the way regularity requires; equivalently $\{0\}$ is too small to be “seen.” Hence 0 is non-regular, and the boundary value at 0 cannot be prescribed. *Check.* This is the simplest counterexample to naive solvability; the cone condition fails at 0 since no solid cone fits into the single removed point. Lebesgue’s thorn is a connected, more dramatic version.

Worked Example 44.3.13 (Uniqueness by the maximum principle). Show directly that the Dirichlet problem has at most one solution.

Recognition. Subtract two solutions; the difference is harmonic with zero boundary data. Apply the maximum principle.

Solution. If u_1, u_2 both solve it, $v = u_1 - u_2$ is continuous on $\bar{R}$, harmonic in R , with $v|_{\partial R} = 0$. A harmonic function attains its max and min on the boundary (maximum principle), so $0 = \min_{\partial R} v \leq v \leq \max_{\partial R} v = 0$ on $\bar{R}$; thus $v \equiv 0$ and $u_1 = u_2$. *Check.* Probabilistically, $v(x) = \int_{\partial R} 0 \, d\mu^x = 0$ directly — the harmonic-measure representation makes uniqueness transparent once existence is known.

Caution. Existence of the probabilistic solution (44.9) never fails — the integral always makes sense and is harmonic. What can fail is *boundary attainment* at non-regular points. Regularity is a local geometric condition on ∂R (e.g. the outside cone condition); without it, u is still harmonic but need not match f at the bad points.

Worked Example 44.3.14 (Harmonic measure recovers the Poisson kernel on the disc). For R the unit disc in $\mathbb{R}^2$ and x its centre, identify the harmonic measure μ^x on the boundary circle.

Recognition. From the centre, rotation invariance forces uniformity — the same argument as for a sphere.

Solution. By rotation invariance of Brownian motion, μ^0 is rotation invariant on the unit circle, hence the normalized arc-length measure $d\theta/2\pi$. So $u(0) = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) d\theta$, the mean value of f on the circle. *Check.* This is the classical Poisson formula evaluated at the centre; for general x inside, μ^x has density the Poisson kernel $\frac{1-|x|^2}{2\pi|x-e^{i\theta}|^2}$, reproducing the Fourier-series solution of the disc Dirichlet problem.

Worked Example 44.3.15 (Constant boundary data gives constant solution). If $f \equiv c$ on ∂R , show $u \equiv c$ on R .

Recognition. The harmonic measure is a probability measure; integrate the constant.

Solution. $u(x) = \int_{\partial R} c d\mu^x(y) = c \mu^x(\partial R) = c \cdot 1 = c$, since μ^x is a probability measure (the path exits a.e.). *Check.* Consistent with harmonicity — constants are harmonic — and with boundary attainment, since $u \equiv c$ trivially matches $f \equiv c$ at every (even non-regular) point. A good sanity test for the representation (44.9).

Worked Example 44.3.16 (Probabilistic proof of the maximum principle). Use the representation (44.9) to show $\sup_R u \leq \sup_{\partial R} f$.

Recognition. u is an average of boundary values against a probability measure, so it cannot exceed the largest boundary value.

Solution. For any $x \in R$, $u(x) = \int_{\partial R} f d\mu^x \leq (\sup_{\partial R} f) \mu^x(\partial R) = \sup_{\partial R} f$. Taking the supremum over x gives the claim; symmetrically $\inf_R u \geq \inf_{\partial R} f$. *Check.* This is the maximum principle, obtained for free from positivity and total mass 1 of harmonic measure — no PDE machinery. It also re-proves uniqueness in Corollary 44.3.11.

Worked Example 44.3.17 (Recurrence and the planar case). Why does the punctured-disc obstruction of the last body example not occur for $d = 2$ in the same form, and what is special about $d = 2$ versus $d \geq 3$?

Recognition. Recall the recurrence/transience dichotomy: $d \leq 2$ neighbourhood-recurrent, $d \geq 3$ transient; points have zero capacity for $d \geq 2$ but neighbourhoods do not.

Solution. For $d \geq 2$, Brownian motion almost surely never hits a fixed single point (Problem 4(b)), so an isolated removed point is non-regular in every dimension ≥ 2 . The genuine distinction is between *points* and *open sets*: for $d = 2$ the motion is neighbourhood-recurrent (it returns to every disc, Problem 4(c)), while for $d \geq 3$ it is transient ($|B_t| \rightarrow \infty$, Problem 4(d)). *Check.* Thus a removed isolated point is irregular for all $d \geq 2$, but the way capacity scales differs — this is the potential-theoretic shadow of recurrence, and underlies Wiener's regularity criterion.

Exercises (Stein, Ch.6).

- (Right-continuity of the filtration.) Show the generalization of $\mathcal{A}_{0+} = \mathcal{A}_0$: defining $\mathcal{A}_{t+} = \bigcap_{s>t} \mathcal{A}_s$, one has $\mathcal{A}_{t+} = \mathcal{A}_t$.
- (Left-continuity.) For every $t > 0$, $\mathcal{A}_t = \mathcal{A}_{t-}$, where $\mathcal{A}_{t-}$ is the σ -algebra generated by all $\mathcal{A}_s$ with $s < t$. [Hint: consider first cylindrical sets in $\mathcal{A}_t$.]
- Let σ be a stopping time. Show: (a) σ is $\mathcal{A}_\sigma$ -measurable; (b) $B_{\sigma(\omega)}(\omega)$ is $\mathcal{A}_\sigma$ -measurable; (c) $\mathcal{A}_\sigma$ is the σ -algebra determined by the stopped process $\hat{B}_t(\omega) = B_{t \wedge \sigma(\omega)}(\omega)$. [Hint for (a): $\{\sigma \leq \alpha\} \cap \{\sigma \leq t\} = \{\sigma \leq \min(\alpha, t)\}$. For (b): first prove $\{B_\sigma \in E\} \cap \{\sigma \leq t\} \in \mathcal{A}_t$ for Borel E when σ is discrete, then approximate by $\sigma^{(n)}$ as in Theorem 44.3.6.]
- Let u be a bounded Borel function on a bounded open $R \subseteq \mathbb{R}^d$ satisfying the sphere mean-value property $u(x) = \int_S u dm$. (a) Show that for any ball $B \subseteq R$ centred at x , $u(x) = \frac{1}{m(B)} \int_B u(y) dy$. (b) Deduce that u is continuous, and (by the standard mollification argument) harmonic in R . [Hint for (b): show locally $u = u * \varphi$ for a smooth radial $\varphi \geq 0$ supported in a small ball with $\int \varphi = 1$.]

5. (Lipschitz boundary $\Rightarrow$ outside cone condition.) Say R has a *Lipschitz boundary* if ∂R is covered by finitely many balls B so that $\partial R \cap B$ can, after a rotation and translation, be written $x_d = \varphi(x_1, \dots, x_{d-1})$ with φ Lipschitz. Verify that a Lipschitz boundary satisfies the outside cone condition; in particular a C^1 region does, so its Dirichlet problem is uniquely solvable.
6. Suppose $R_1 \subseteq R_2$ are bounded open sets with harmonic measures μ_1^x, μ_2^x . Show the mean-value generalization: for $x \in R_1$, $\mu_2^x = \int_{\partial R_1} \mu_2^y d\mu_1^x(y)$, meaning $\mu_2^x(E) = \int_{\partial R_1} \mu_2^y(E) d\mu_1^x(y)$ for every Borel $E \subseteq \partial R_2$.

Problems (Stein, Ch.6).

1. (Recurrence and transience.) For Brownian motion B_t in $\mathbb{R}^d$: (a) if $d = 1$, then for each $x \in \mathbb{R}$ and $t_0 > 0$, $P(\{\omega : B_t(\omega) = x \text{ for some } t \geq t_0\}) = 1$ (pointwise recurrent); (b) if $d \geq 2$, then for every x , $P(\{B_t = x \text{ for some } t > 0\}) = 0$ (points are not hit); (c) if $d = 2$, then for any open disc D of positive radius and $t_0 > 0$, $P(\{B_t \in D \text{ for some } t \geq t_0\}) = 1$ (neighbourhood recurrent); (d) if $d \geq 3$, then $P(\{\lim_{t \rightarrow \infty} |B_t| = \infty\}) = 1$ (transient).
2. (Converse regularity, $d \geq 2$.) Show that if $u(x) \rightarrow f(y)$ as $x \rightarrow y$, $x \in R$, for every continuous f , then y is a regular point. [Hint: if y is not regular, use Problem 4(b) to get $P(\{|B_{\tau_y}^y - y| > 0\}) = 1$, hence $P(\{|B_{\tau_y}^y - y| \geq \delta\}) > 1/2$ for some $\delta > 0$; via the strong Markov property find $x \in S_\eta \cap R$ with $P(\{|B_{\tau_x}^x - y| \geq \delta\}) > 1/2$, then test a continuous $0 \leq f \leq 1$ with $f(y) = 1$, $f(z) = 0$ for $|z - y| \geq \delta$ for a contradiction.]
3. (Lebesgue's thorn.) An isolated removed point is a simple non-regular point. For a subtler one, take $d \geq 3$ and remove from the ball $B = \{|x| < 1\}$ the cusp $E = \{(x_1, \dots, x_d) : 0 \leq x_1 \leq 1, x_2^2 + \dots + x_d^2 \leq f(x_1)\}$ with f continuous, $f(x) > 0$ for $x > 0$. If $f(x) \rightarrow 0$ sufficiently rapidly as $x \rightarrow 0$, the origin is non-regular for $R = B - E$; R may be modified to have smooth boundary except at the origin.

Additional exercises.

1. Verify that for a stopping time σ , the collection $\mathcal{A}_\sigma = \{A : A \cap \{\sigma \leq t\} \in \mathcal{A}_t, \forall t\}$ is a σ -algebra, and that if $\sigma \equiv \sigma_0$ is constant then $\mathcal{A}_\sigma = \mathcal{A}_{\sigma_0}$.
2. Show that the minimum $\sigma \wedge \tau$ and maximum $\sigma \vee \tau$ of two stopping times are stopping times, and that $\{\sigma \leq \tau\} \in \mathcal{A}_\sigma \cap \mathcal{A}_\tau$.
3. For R the unit disc in $\mathbb{R}^2$, write down the solution $u(x)$ of the Dirichlet problem with boundary data $f(e^{i\theta}) = \cos \theta$ using the harmonic-measure (Poisson) representation, and check u is harmonic. (Answer: $u(x, y) = x$.)
4. Using the probabilistic representation (44.9), prove the strong maximum principle: a nonconstant harmonic u on a connected R attains neither its max nor its min at an interior point.
5. (Non-example.) For $d \geq 3$ and $R = B(0, 1) \setminus \{0\}$, explain via Problem 4(b) why the origin is non-regular, so the Dirichlet problem with $f(0) = 1$, $f = 0$ on the outer sphere has a solution not attaining the value 1 at 0.
6. Show that the outside cone condition is satisfied by any convex bounded open set, and hence its Dirichlet problem is uniquely solvable for every continuous boundary datum.
7. Prove that harmonic measure is monotone: if $R_1 \subseteq R_2$ and $x \in R_1$, then for $E \subseteq \partial R_2$, $\mu_2^x(E) = \int_{\partial R_1} \mu_2^y(E) d\mu_1^x(y)$ (Exercise 21), and deduce $\mu_2^x(E) \leq \sup_{y \in \partial R_1} \mu_2^y(E)$.
8. Draw a bounded region R with a single inward cusp at a boundary point x_0 and argue informally why x_0 might be non-regular, contrasting it with an outward cusp (which satisfies the cone condition).
9. Use the strong Markov property to show that, started at the centre, the exit distribution of Brownian motion from a ball is rotation invariant, hence uniform on the bounding sphere; then state the corresponding statement for an ellipse and explain why rotation invariance no longer forces uniformity.
10. Prove the mean-value property characterizes harmonicity: a continuous u on R with $u(x) = \frac{1}{m(B)} \int_B u$ for every ball $B \subseteq R$ is harmonic. [Hint: mollify u against a smooth radial bump and use that mollification preserves u .]

Chapter 45

A Glimpse into Several Complex Variables

45.1 Holomorphic Functions of Several Variables and the Hartogs Phenomenon

Polydiscs, distinguished boundaries, and multi-indices

Throughout, $z = (z_1, \dots, z_n) \in \mathbb{C}^n$ with $z_j = x_j + iy_j$, and we identify $\mathbb{C}^n$ with $\mathbb{R}^{2n}$. Several-variable complex analysis begins by adapting the one-variable apparatus coordinate by coordinate; the surprises come later, when the variables interact.

Definition 45.1.1. For $z^0 = (z_1^0, \dots, z_n^0) \in \mathbb{C}^n$ and a radius vector $r = (r_1, \dots, r_n)$ with each $r_j > 0$, the *polydisc* centered at z^0 is the product

$$P_r(z^0) = \{z \in \mathbb{C}^n : |z_j - z_j^0| < r_j, 1 \leq j \leq n\} = \mathbb{D}_{r_1}(z_1^0) \times \cdots \times \mathbb{D}_{r_n}(z_n^0).$$

Its *distinguished boundary* (or *torus*) is the product of the bounding circles

$$C_r(z^0) = \{z \in \mathbb{C}^n : |z_j - z_j^0| = r_j, 1 \leq j \leq n\}.$$

For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n)$ of nonnegative integers we write $z^\alpha = z_1^{\alpha_1} \cdots z_n^{\alpha_n}$, $|\alpha| = \alpha_1 + \cdots + \alpha_n$, and $r^{-\alpha} = r_1^{-\alpha_1} \cdots r_n^{-\alpha_n}$.

The distinguished boundary $C_r(z^0)$ is an n -dimensional torus, *much smaller* than the full topological boundary $\partial P_r(z^0)$ (which has real dimension $2n - 1$). For $n = 1$ the two coincide; for $n \geq 2$ they differ, and this gap is the source of nearly every new phenomenon in the subject. The Cauchy formula below integrates only over the tiny torus.

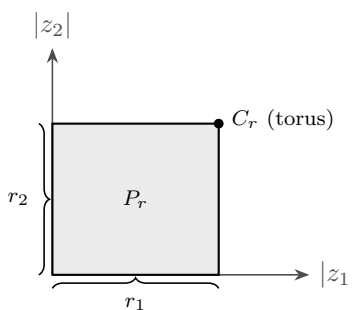


Figure 45.1. The bidisc P_r in the $(|z_1|, |z_2|)$ “quadrant”: a solid rectangle of full real dimension 4, but its distinguished boundary C_r (the corner point here) is only the 2-torus $\{|z_1| = r_1, |z_2| = r_2\}$.

The four equivalent definitions of holomorphy

Definition 45.1.2. For a C^1 function f on $\Omega \subseteq \mathbb{C}^n$ the *Cauchy–Riemann operators* are

$$\frac{\partial f}{\partial \bar{z}_j} = \frac{1}{2} \left(\frac{\partial f}{\partial x_j} + i \frac{\partial f}{\partial y_j} \right), \quad \frac{\partial f}{\partial z_j} = \frac{1}{2} \left(\frac{\partial f}{\partial x_j} - i \frac{\partial f}{\partial y_j} \right), \quad 1 \leq j \leq n.$$

The function is *holomorphic* (analytic) if $\partial f / \partial \bar{z}_j = 0$ for every j (the *homogeneous Cauchy–Riemann equations*), interpreted in the sense of distributions when f is only assumed locally integrable.

Theorem 45.1.3 (Equivalent characterizations of analyticity). *Let f be continuous on an open $\Omega \subseteq \mathbb{C}^n$. The following are equivalent.*

- (i) f satisfies the Cauchy–Riemann equations $\partial f / \partial \bar{z}_j = 0$, $1 \leq j \leq n$ (in the distributional sense).
- (ii) f is separately holomorphic: for each $z^0 \in \Omega$ and each k , the one-variable function $g(z_k) = f(z_1^0, \dots, z_{k-1}^0, z_k, z_{k+1}^0, \dots, z_n^0)$ is holomorphic in z_k near z_k^0 .
- (iii) For every polydisc $P_r(z^0)$ with $\overline{P_r(z^0)} \subset \Omega$ one has the iterated Cauchy integral formula

$$f(z) = \frac{1}{(2\pi i)^n} \int_{C_r(z^0)} f(\zeta) \prod_{k=1}^n \frac{d\zeta_k}{\zeta_k - z_k}, \quad z \in P_r(z^0).$$

- (iv) About each $z^0 \in \Omega$, f has an absolutely and uniformly convergent power series $f(z) = \sum_{\alpha} a_{\alpha} (z - z^0)^{\alpha}$.

Proof. (i) $\Rightarrow$ (ii). Let $\Delta = \sum_j (\partial_{x_j}^2 + \partial_{y_j}^2)$ be the Laplacian on $\mathbb{C}^n \cong \mathbb{R}^{2n}$. The factorization $\Delta = 4 \sum_{j=1}^n \frac{\partial}{\partial z_j} \frac{\partial}{\partial \bar{z}_j}$ shows that if $\partial f / \partial \bar{z}_j = 0$ for all j then $\Delta f = 0$, so f is harmonic, hence (by ellipticity and elliptic regularity, Ch.3 §2.5) of class C^{∞} . Thus the CR equations hold classically, and freezing all variables but z_k gives a one-variable holomorphic function, i.e. (ii).

(ii) $\Rightarrow$ (iii). Fix $z \in P_r(z^0)$ with $\overline{P_r(z^0)} \subset \Omega$. Apply the one-variable Cauchy formula in z_1 (other variables frozen):

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta_1 - z_1^0| = r_1} f(\zeta_1, z_2, \dots, z_n) \frac{d\zeta_1}{\zeta_1 - z_1}.$$

Now represent $f(\zeta_1, z_2, \dots, z_n)$ by the Cauchy formula in z_2 , and iterate. After n steps we obtain the product kernel over $C_r(z^0)$, which is (iii).

(iii) $\Rightarrow$ (iv). On the torus $|\zeta_k - z_k^0| = r_k$ and for $z \in P_r(z^0)$ we have $|z_k - z_k^0| < r_k = |\zeta_k - z_k^0|$, so the geometric expansion

$$\frac{1}{\zeta_k - z_k} = \frac{1}{(\zeta_k - z_k^0) - (z_k - z_k^0)} = \sum_{m=0}^{\infty} \frac{(z_k - z_k^0)^m}{(\zeta_k - z_k^0)^{m+1}}$$

converges. Substituting into (iii) and multiplying the n series gives $f(z) = \sum_{\alpha} a_{\alpha} (z - z^0)^{\alpha}$ with

$$a_{\alpha} = \frac{1}{(2\pi i)^n} \int_{C_r(z^0)} f(\zeta) \prod_{k=1}^n \frac{d\zeta_k}{(\zeta_k - z_k^0)^{\alpha_k + 1}}.$$

With $M = \sup_{C_r(z^0)} |f|$ the *Cauchy estimates* $|a_{\alpha}| \leq M r^{-\alpha}$ follow, giving absolute and uniform convergence on $P_{r'}(z^0)$ for every $r'_k < r_k$.

(iv) $\Rightarrow$ (i). A power series converging absolutely near z^0 may be differentiated term by term, so $f \in C^1$ there and each $\partial f / \partial \bar{z}_j = 0$ since each monomial is holomorphic. As $z^0 \in \Omega$ was arbitrary, f is C^1 on Ω and satisfies (i). $\square$

The chain of implications is the engine of the whole chapter; isolate it as a method, since nearly every concrete computation invokes one link.

Criterion (ii) is a statement about restriction along a one-variable slice: freezing $z_1^0, \dots, z_{k-1}^0, z_{k+1}^0, \dots, z_n^0$ defines the affine inclusion $\iota_k : \mathbb{C} \hookrightarrow \mathbb{C}^n$, $w \mapsto (z_1^0, \dots, w, \dots, z_n^0)$, and the slice function g is precisely the composite $f \circ \iota_k$.

$$\begin{array}{ccc}
\mathbb{C} & \xrightarrow{\iota_k} & \mathbb{C}^n \\
& \searrow g = f \circ \iota_k & \downarrow f \\
& & \mathbb{C}
\end{array}$$

Method (Passing between the four descriptions of a holomorphic f). *Recognition trigger.* You must verify analyticity, expand in a series, or estimate coefficients.

1. To *check* analyticity it is usually easiest to verify (ii): hold all but one variable fixed and use one-variable theory.
2. To *represent* f inside a polydisc, use the product Cauchy kernel (iii).
3. To *estimate* or *expand*, pass to the series (iv) and use $|a_\alpha| \leq Mr^{-\alpha}$.
4. To check a candidate is *not* holomorphic, exhibit one j with $\partial f / \partial \bar{z}_j \neq 0$ (e.g. any genuine appearance of $\bar{z}_j$).

Remark. Two refinements: (a) continuity of f in (i) can be weakened to local integrability — a distributional solution of the CR equations agrees a.e. with an analytic function. (b) Even joint continuity may be dropped from (ii): separate analyticity in each variable *alone* forces holomorphy (Hartogs's theorem on separate analyticity, Problem 1*), though this lies much deeper and rests on the Baire category theorem.

Worked Example 45.1.4 (Monomials, exponentials, and a non-example). Decide which of $f_1(z) = z_1^2 z_2$, $f_2(z) = e^{z_1 + 2z_2}$, $f_3(z) = z_1 \bar{z}_2$ are holomorphic on $\mathbb{C}^2$.

Recognition. Apply criterion (i)/(iv): a finite or convergent power series in z (no $\bar{z}_j$) is holomorphic; any genuine $\bar{z}_j$ dependence is fatal.

Solution. f_1 is a monomial, $f_2 = \sum_{m,k} \frac{1}{m!k!} z_1^m (2z_2)^k$ converges everywhere; both are entire. For f_3 , $\partial f_3 / \partial \bar{z}_2 = z_1 \neq 0$, so f_3 fails the CR equations. *Check.* f_3 restricted to $z_1 = 1$ is $\bar{z}_2$, the standard one-variable non-holomorphic function. ■

Worked Example 45.1.5 (Cauchy estimates in two variables). Let f be holomorphic on the closed bidisc $P_r(0)$, $r = (r_1, r_2)$, with $|f| \leq M$ there. Bound the Taylor coefficient $a_{(2,3)}$ and the derivative $\partial^5 f / \partial z_1^2 \partial z_2^3$ at 0.

Recognition. This is the multi-variable Cauchy inequality, $|a_\alpha| \leq Mr^{-\alpha}$, with $a_\alpha = \frac{1}{\alpha!} \partial^\alpha f(0)$ where $\alpha! = \alpha_1! \cdots \alpha_n!$.

Solution. $|a_{(2,3)}| \leq Mr_1^{-2} r_2^{-3}$. Since $\partial^{(2,3)} f(0) = 2!3! a_{(2,3)} = 12 a_{(2,3)}$, we get $|\partial^{(2,3)} f(0)| \leq 12 Mr_1^{-2} r_2^{-3}$. *Check.* For $f \equiv M$ all higher a_α vanish, consistent with the bound being an inequality; for $f(z) = M(z_1/r_1)^2 (z_2/r_2)^3$ we have $|f| \leq M$ on the torus and $|a_{(2,3)}| = Mr_1^{-2} r_2^{-3}$, so the estimate is sharp. ■

Worked Example 45.1.6 (A Liouville theorem on $\mathbb{C}^n$). Show a bounded entire function on $\mathbb{C}^n$ is constant.

Recognition. Freeze all but one variable and use one-variable Liouville, then vary.

Solution. Fix $z_2, \dots, z_n$. Then $z_1 \mapsto f(z_1, z_2, \dots, z_n)$ is entire and bounded by the global bound M , hence constant in z_1 by the one-variable Liouville theorem. Repeating in each variable, f does not depend on any z_k , so f is constant. *Check.* The argument only used boundedness on *lines*; indeed boundedness on each complex line through 0 already suffices, which is special to entire functions. ■

Worked Example 45.1.7 (Identity theorem needs an open set, not a small one). Let f, g be holomorphic on a region $\Omega \subseteq \mathbb{C}^2$. (a) If they agree on a nonempty *open* subset, must $f \equiv g$? (b) If they agree only on the totally real plane $\{(x_1, x_2) : x_1, x_2 \in \mathbb{R}\} \cap \Omega$, must $f \equiv g$?

Recognition. Proposition 45.1.8 requires agreement on a *neighborhood* (open set); a lower-dimensional set carries no force.

Solution. (a) Yes: agreement on a nonempty open subset is agreement near each of its points, so Proposition 45.1.8 gives $f \equiv g$ on the connected Ω . (b) No: on $\Omega = \mathbb{C}^2$ take $g = 0$ and $h = f$ where $f(z_1, z_2) = \sin(\pi z_1)$. Then f vanishes on the real plane only along the lines $\{z_1 \in \mathbb{Z}\}$, which has empty interior, so agreement on the real plane is far too weak to conclude. *Check.* The real plane has real dimension $2 < 2n = 4$, hence empty interior in $\mathbb{C}^2$; the open-set hypothesis of the identity theorem genuinely fails, and indeed $f \neq 0$. ■

The identity principle

Proposition 45.1.8 (Identity theorem). *If f, g are holomorphic in a region (open connected) $\Omega \subseteq \mathbb{C}^n$ and agree on a neighborhood of some $z^0 \in \Omega$, then $f \equiv g$ on Ω .*

Proof. Put $h = f - g$ and assume h vanishes near z^0 ; fix any $z' \in \Omega$. By path-connectedness choose points $z^0, z^1, \dots, z^N = z'$ and polydiscs $P_{r_k}(z^k) \subset \Omega$ with $z^{k+1} \in P_{r_k}(z^k)$. If h vanishes near z^k , then all Taylor coefficients $a_\alpha = 0$ there, so by Theorem 45.1.3(iv) $h \equiv 0$ on the whole polydisc $P_{r_k}(z^k)$ (a polydisc fact, Exercise 1). In particular h vanishes near $z^{k+1} \in P_{r_k}(z^k)$. By induction $h \equiv 0$ near $z^N = z'$, so $h(z') = 0$. $\square$

Hartogs' phenomenon: automatic continuation across a hole

In one variable there are holomorphic functions on an annulus that do not extend across the hole (e.g. $1/z$ on $0 < |z| < 1$). In $\mathbb{C}^n$ with $n \geq 2$ this is *impossible*: a holomorphic function on a spherical shell automatically fills in the ball.

Lemma 45.1.9 (Hartogs' figure / primitive continuation). *In $\mathbb{C}^2$ let*

$$K_1 = \{(z_1, z_2) : |z_1| \leq a, |z_2| = b_1\}, \quad K_2 = \{(z_1, z_2) : |z_1| = a, b_2 \leq |z_2| \leq b_1\}.$$

If F is holomorphic in a region containing $K_1 \cup K_2$, then F extends holomorphically to a neighborhood of the full product set $\{(z_1, z_2) : |z_1| \leq a, b_2 \leq |z_2| \leq b_1\}$.

Proof. For small $\varepsilon > 0$ define, for (z_1, z_2) in a neighborhood $\tilde{O}$ of the product set,

$$I(z_1, z_2) = \frac{1}{2\pi i} \int_{|\zeta_1|=a+\varepsilon} \frac{F(\zeta_1, z_2)}{\zeta_1 - z_1} d\zeta_1.$$

The integration variable ranges over a neighborhood of K_2 , where F is analytic, so I is well defined; it is visibly holomorphic in z_1 (for $|z_1| < a + \varepsilon$) and in z_2 (for $b_2 \leq |z_2| \leq b_1$, by analyticity of F there). Finally, when (z_1, z_2) is near K_1 , the Cauchy integral formula gives $I(z_1, z_2) = F(z_1, z_2)$. Hence I continues F across the product set. $\square$

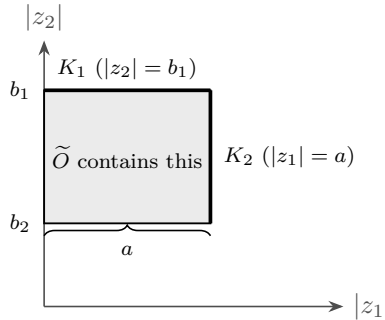


Figure 45.2. Hartogs' figure in the $(|z_1|, |z_2|)$ plane: knowing F on the “L” formed by $K_1 \cup K_2$ forces it on the shaded rectangle. Sliding such L's fills a ball.

Theorem 45.1.10 (Hartogs continuation across a shell). *Let $n \geq 2$ and $0 < \rho < 1$. If F is holomorphic in the spherical shell $\Omega = \{z \in \mathbb{C}^n : \rho < |z| < 1\}$, then F continues analytically to the whole ball $\{|z| < 1\}$.*

Proof for $n = 2$. Case $\rho < 1/\sqrt{2}$. Set $K_1 = \{|z_1| \leq a_1, |z_2| = b_1\}$, $K_2 = \{|z_1| = a_1, 0 \leq |z_2| \leq b_1\}$ with $a_1 = b_1$ and $\rho < a_1, b_1 < 1/\sqrt{2}$. Both lie in Ω (their points have $\rho < |z| < 1$). By Lemma 45.1.9, F continues to the polydisc $\{|z_1| \leq 1/\sqrt{2}, |z_2| \leq 1/\sqrt{2}\}$, which together with Ω covers the unit ball.

Case $1/\sqrt{2} \leq \rho < 1$. Descend a staircase with corners (α_k, β_k) in the $(|z_1|, |z_2|)$ plane: take $\beta_1 = \rho$, $\alpha_1 = (1 - \rho^2)^{1/2}$, and inductively $\beta_{k+1}^2 = \rho^2 - \alpha_k^2$, $\alpha_{k+1}^2 = 1 - \beta_{k+1}^2$, giving $\beta_k^2 = 1 - k(1 - \rho^2)$, $\alpha_k^2 = k(1 - \rho^2)$. Stop at the least $N > 1/(1 - \rho^2)$, where $\beta_N = 0, \alpha_N = 1$. With $R_k = \Omega \cup \{|z| < 1, b_k \leq |z_2| \leq b_{k+1}\}$, repeated use of Lemma 45.1.9 (with $a = a_k, b_1 = b_k, b_2 = b_{k+1}$) continues F from a neighborhood of R_k to one of R_{k+1} ; and $R_N = \{|z| < 1\}$. The case $n \geq 3$ is analogous. $\square$

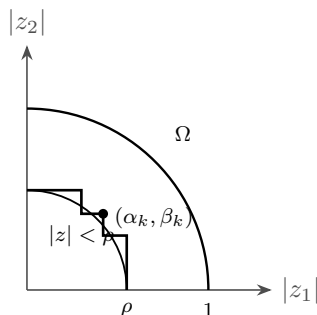


Figure 45.3. The descending staircase between the spheres $|z| = \rho$ and $|z| = 1$; each step is one application of the Hartogs figure, filling the shell inward to the whole ball.

Corollary 45.1.11 (No isolated singularities or zeros). *For $n > 1$ a holomorphic function on a punctured neighborhood of a point cannot have an isolated singularity there, nor can a holomorphic f have an isolated zero.*

Proof. Apply Theorem 45.1.10 to a shell between two concentric small balls about the suspected singularity: F extends across, so the singularity was removable. For zeros, apply the same to $1/f$; an isolated zero of f would be an isolated singularity of $1/f$. $\square$

Remark. More is true: the zero set of a holomorphic f (when nonempty) must reach $\partial\Omega$ (Exercise 4), and near a zero it is described precisely by the Weierstrass preparation theorem (Problem 2*). Note also the converse limitation: a function holomorphic *inside* the ball need not extend *outside*; e.g. $f(z) = 1/(z_1 - 1)$ on $\{|z| < 1\}$ blows up at the boundary point $(1, 0, \dots, 0)$. Whether extension across the boundary occurs is governed by the *convexity* of $\partial\Omega$ – the theme of §2.

Worked Example 45.1.12 ($1/z$ has no analog in $\mathbb{C}^2$). Why is there no holomorphic F on $\{0 < |z| < 1\} \subset \mathbb{C}^2$ with a genuine singularity at 0, whereas $1/z$ exists on the punctured disc in $\mathbb{C}$?

Recognition. This is Corollary 45.1.11: the punctured ball in $\mathbb{C}^2$ contains a shell, and Theorem 45.1.10 forces continuation.

Solution. Any F holomorphic on $\{0 < |z| < 1\}$ is in particular holomorphic on the shell $\{1/2 < |z| < 3/4\}$, hence (Hartogs) extends to $\{|z| < 3/4\}$, in particular across 0. So 0 cannot be a genuine singularity. For $n = 1$ the “shell” $\{\rho < |z| < 1\}$ is an annulus with a hole the extension cannot cross, and $1/z$ exploits exactly that hole. *Check.* The proof needed two variables to build the L-shaped Hartogs figure — with one variable $K_1 \cup K_2$ degenerates and the integral argument collapses. ■

Worked Example 45.1.13 (Reading off the staircase length). For $\rho = 0.9$ in $\mathbb{C}^2$, how many staircase steps N does the proof of Theorem 45.1.10 use?

Recognition. N is the least integer exceeding $1/(1 - \rho^2)$.

Solution. $1 - \rho^2 = 1 - 0.81 = 0.19$, so $1/(1 - \rho^2) \approx 5.26$, giving $N = 6$. *Check.* $\beta_k^2 = 1 - k(0.19)$ stays ≥ 0 for $k \leq 5$ ($1 - 5(0.19) = 0.05$) and turns negative at $k = 6$ ($1 - 6(0.19) = -0.14$), so we stop at $N = 6$, consistent. ■

Worked Example 45.1.14 (Convergence domain of a power series is logarithmically convex). For $f(z) = \sum_{m \geq 0} z_1^m z_2^m = \frac{1}{1 - z_1 z_2}$, describe the set $S = \{(\log r_1, \log r_2) : f \text{ holomorphic on } P_r(0)\}$ and verify convexity.

Recognition. Convergence of $\sum (z_1 z_2)^m$ needs $|z_1 z_2| < 1$, i.e. $\log r_1 + \log r_2 < 0$ (Exercise 2).

Solution. The series converges absolutely iff $r_1 r_2 < 1$, i.e. $s_1 + s_2 < 0$ with $s_j = \log r_j$. So $S = \{(s_1, s_2) : s_1 + s_2 < 0\}$, a half-plane, which is convex. *Check.* The boundary $\{|z_1 z_2| = 1\}$ is where $1/(1 - z_1 z_2)$ first becomes singular, and the log-image being a half-plane is the predicted log-convexity of Reinhardt convergence domains. ■

Worked Example 45.1.15 (Distinguished boundary vs. topological boundary). For the unit bidisc $P = \mathbb{D} \times \mathbb{D} \subset \mathbb{C}^2$, compute the real dimensions of the distinguished boundary $C = \{|z_1| = |z_2| = 1\}$ and the topological boundary ∂P , and explain why the Cauchy formula integrates over the smaller one.

Recognition. C is a product of two circles (a 2-torus); ∂P is a real hypersurface in $\mathbb{R}^4$.

Solution. $C = \mathbb{T} \times \mathbb{T}$ has real dimension 2; $\partial P = (\partial \mathbb{D} \times \overline{\mathbb{D}}) \cup (\overline{\mathbb{D}} \times \partial \mathbb{D})$ has real dimension 3. The iterated Cauchy formula uses only the 2-torus C . *Check.* For $n = 1$ both reduce to the circle (dimension 1), explaining why the distinction is invisible in one variable but central for $n \geq 2$. ■

Exercises (Stein, Ch.7).

1. Suppose f is holomorphic in a polydisc $P_r(z^0)$ and vanishes in a neighborhood of z^0 . Show $f \equiv 0$ throughout $P_r(z^0)$. [Hint: expand $f(z) = \sum_{\alpha} a_{\alpha}(z - z^0)^{\alpha}$ by Proposition 1.1 and note every $a_{\alpha} = 0$.]
2. (a) If f is holomorphic in a pair of polydiscs $P_{\sigma}(z^0)$ and $P_{\tau}(z^0)$ centered at z^0 (with $\sigma = (\sigma_1, \dots, \sigma_n)$, $\tau = (\tau_1, \dots, \tau_n)$), show f extends holomorphically to $P_r(z^0)$ whenever $r_j \leq \sigma_j^{1-\theta} \tau_j^{\theta}$, $1 \leq j \leq n$, for some $0 \leq \theta \leq 1$. (b) Deduce that the set $S = \{s = (s_1, \dots, s_n) : s_j = \log r_j, f \text{ holomorphic in } P_{\sigma}(z^0)\}$ is convex. [Hint: use the common power series $\sum a_{\alpha}(z - z^0)^{\alpha}$ representing f in both $P_{\sigma}(z^0)$ and $P_{\tau}(z^0)$; this is the convexity of the domain of convergence (a “logarithmically convex” Reinhardt domain).]
3. Given any open subset $\Omega \subseteq \mathbb{C}^1$, construct a holomorphic f on Ω that cannot be continued analytically outside Ω . [Hint: choose a sequence $\{z_j\} \subset \Omega$ with no limit point in Ω accumulating at every boundary point; there is an analytic function on Ω vanishing exactly at the z_j . This shows every planar region is a domain of holomorphy — a one-variable feature with no $n \geq 2$ analog.]
4. Suppose Ω is a bounded region in $\mathbb{C}^n$, $n > 1$, and f is holomorphic on Ω with nonempty zero set Z . Show Z meets $\partial\Omega$, i.e. $Z \cap \partial\Omega \neq \emptyset$. [Hint: take $w \in \Omega^c$ and $z^0 \in Z$ furthest from w . Let γ be the unit vector from z^0 toward w and ν a unit vector with ν and $i\nu$ both perpendicular to γ . The one-variable slice $h_{\varepsilon}(\zeta) = f(z^0 - \varepsilon\gamma + \zeta\nu)$ does not vanish in a fixed neighborhood of $\zeta = 0$ for small $\varepsilon > 0$; hence the zero cannot be interior-maximal.]

Problems (Stein, Ch.7).

1. * (Hartogs on separate analyticity.) Suppose $f = f(z_1, \dots, z_n)$ is defined on a region $\Omega \subseteq \mathbb{C}^n$ and, for each j , is holomorphic in z_j with the other variables fixed. Then f is holomorphic on Ω — with *no* continuity or boundedness assumption beyond separate analyticity. (The continuous case is Proposition 1.1; the general case rests on an application of the Baire category theorem.)
2. * (Weierstrass preparation theorem.) Assume f is holomorphic near the origin with $f(0) = 0$, $f(z) = \sum_{\alpha} a_{\alpha} z^{\alpha}$, and let k be the order of the zero (the smallest $|\alpha|$ with $a_{\alpha} \neq 0$). After a linear change of variables one can write $f(z) = c(z)P(z)$ near the origin, where $P(z) = z_n^k + a_{k-1}(z')z_n^{k-1} + \dots + a_0(z')$ is a Weierstrass polynomial with $a_{k-1}(0) = \dots = a_0(0) = 0$ and $c(0) \neq 0$. [Hint: arrange $f(0, z_n) = z_n^k$; by Rouché choose $\varepsilon, r > 0$ so $z_n \mapsto f(z', z_n)$ has k zeros in $|z_n| \leq r$, none on the boundary circle, for $|z'| < \varepsilon$; the elementary symmetric functions of these zeros, given by $s_m(z') = \frac{1}{2\pi i} \oint_{|w|=r} w^m \frac{(\partial f / \partial w)(z', w)}{f(z', w)} dw$, are holomorphic; set $a_{k-j} = (-1)^j \sigma_j$.]
3. * (Domains of holomorphy.) A *domain of holomorphy* is a region Ω for which there exists a holomorphic F such that for every $z^0 \in \partial\Omega$, F cannot be continued into any ball centered at z^0 . If Ω is a domain of holomorphy with C^2 boundary then Ω is pseudoconvex (by the boundary maximum principle/extension theory). Conversely, every pseudoconvex region is a domain of holomorphy (the solution of the Levi problem).

Additional exercises.

1. For $z^0 = (1, 2i) \in \mathbb{C}^2$ and $r = (2, 3)$, describe $P_r(z^0)$ and its distinguished boundary $C_r(z^0)$ as products, and compute the real dimensions of P_r , C_r , and the full topological boundary ∂P_r .
2. Let $f(z_1, z_2) = \frac{1}{(1 - z_1)(1 - z_2)}$ on the unit bidisc. Find its power series $\sum a_{\alpha} z^{\alpha}$ and verify directly that $|a_{\alpha}| \leq M r^{-\alpha}$ for $r = (1/2, 1/2)$ with $M = \sup_{C_r} |f|$.
3. Decide which of the following are holomorphic on $\mathbb{C}^2$, giving the offending $\partial/\partial \bar{z}_j$ when not: (a) $z_1 e^{z_2}$; (b) $|z_1|^2$; (c) $z_1^3 - 5z_1 z_2^2$; (d) $\bar{z}_1 z_2$; (e) $\operatorname{Re}(z_1) + i \operatorname{Im}(z_2)$.

4. Prove a several-variable Cauchy estimate consequence: if f is entire on $\mathbb{C}^n$ with $|f(z)| \leq A(1 + |z|)^N$, then f is a polynomial of total degree $\leq N$. [Use the coefficient bounds $|a_\alpha| \leq M(R)R^{-\alpha}$ and let $R \rightarrow \infty$.]
5. For $\rho = 0.8$ in $\mathbb{C}^2$, compute the number of staircase steps N in the proof of the Hartogs shell theorem, and the corner (α_2, β_2) .
6. Show directly (without quoting the shell theorem) that a function holomorphic on $\{0 < |z| < 1\} \subset \mathbb{C}^2$ extends across 0, by applying Hartogs' figure to a single well-chosen L inside the punctured ball.
7. (Non-example / boundary of validity.) Exhibit a function holomorphic on the annulus $\{1/2 < |z| < 1\} \subset \mathbb{C}^1$ that does *not* extend across the inner circle, and explain in one sentence why the same construction is impossible in $\mathbb{C}^2$.
8. (Diagram problem.) Draw the Hartogs figure $K_1 \cup K_2$ in the $(|z_1|, |z_2|)$ plane for $a = 1, b_1 = 2, b_2 = 1$, shade the product set it forces F onto, and mark which edge is K_1 and which is K_2 .

45.2 Domains of Holomorphy

The Hartogs phenomenon shows that on many domains *every* holomorphic function extends to a larger set. A *domain of holomorphy* is a region Ω that is “maximal” for some holomorphic function: there is a holomorphic F on Ω that cannot be continued past any boundary point. The deep theorem of the subject (Problem 5*, the solution of the Levi problem) is that Ω is a domain of holomorphy *iff* it is pseudoconvex, a local convexity condition on $\partial\Omega$ measured by the Levi form. This section builds the geometric apparatus: defining functions, tangential Cauchy–Riemann equations, Bochner’s extension theorem, and the Levi form.

Defining functions and the boundary as a graph

Definition 45.2.1. Let $\Omega \subseteq \mathbb{R}^d$ be a region. A *defining function* for Ω is a real-valued ρ on $\mathbb{R}^d$ with

$$\rho(x) < 0 \ (x \in \Omega), \quad \rho(x) = 0 \ (x \in \partial\Omega), \quad \rho(x) > 0 \ (x \in \Omega^c).$$

The boundary $\partial\Omega$ is *of class C^k* if ρ can be taken in $C^k(\mathbb{R}^d)$ with $|\nabla\rho(x)| > 0$ on $\partial\Omega$. More generally M is a (local) C^k *hypersurface* if $M = \{x \in B : \rho(x) = 0\}$ for such a ρ on a ball B .

Near any boundary point we may straighten the boundary into a graph. After a translation and rotation putting $x^0 = 0$ and the tangent plane at $\{x_d = 0\}$, the implicit function theorem gives, in coordinates $x = (x', x_d)$ with $x' \in \mathbb{R}^{d-1}$,

$$\Omega : x_d > \varphi(x'), \quad \partial\Omega : x_d = \varphi(x'),$$

with $\varphi \in C^k$, $\varphi(0) = 0$, $\nabla_{x'}\varphi(0) = 0$.

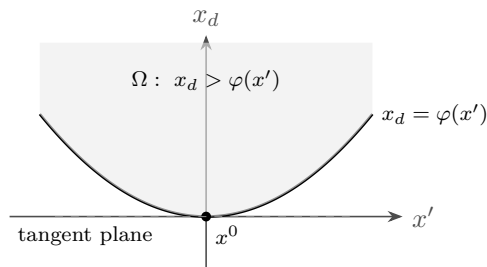


Figure 45.4. Near a C^k boundary point the region is the epigraph of a function φ with $\nabla\varphi(0) = 0$; the tangent plane is $\{x_d = 0\}$. (Schematic.)

Lemma 45.2.2 (Defining functions are unique up to a positive factor). *If $\rho, \tilde{\rho}$ are both C^k defining functions for Ω , then near a boundary point $\tilde{\rho} = c\rho$ with $c > 0$ of class C^{k-1} .*

Proof. Writing $\rho(x) = \rho(x', x_d) - \rho(x', \varphi(x'))$ and integrating along the segment,

$$\rho(x) = \int_0^1 \frac{\partial}{\partial t} \rho(x', tx_d + (1-t)\varphi(x')) dt = a(x)(\varphi(x') - x_d),$$

with $a(x) = -\int_0^1 \partial_{x_d} \rho(\dots) dt$, of class C^{k-1} and $a > 0$ near x^0 (since $\partial_{x_d} \rho < 0$ as $\partial/\partial x_d$ points inward). The same holds for $\tilde{\rho} = \tilde{a}(\varphi - x_d)$, so $\tilde{\rho} = (\tilde{a}/a)\rho = c\rho$ with $c > 0$. $\square$

Tangential and Cauchy–Riemann vector fields

A vector field $X = \sum_j a_j(x) \partial/\partial x_j$ is *tangential* to $\partial\Omega$ if $X(\rho) = \sum_j a_j \partial\rho/\partial x_j = 0$ on $\partial\Omega$; by Lemma 45.2.2 this is independent of the choice of ρ . Passing to $\mathbb{C}^n$ ($d = 2n$), a vector field $X = \sum_j (a_j \partial/\partial \bar{z}_j + b_j \partial/\partial z_j)$ is a *Cauchy–Riemann (CR) vector field* if all $b_j = 0$, i.e.

$$X = \sum_{j=1}^n a_j(z) \frac{\partial}{\partial \bar{z}_j};$$

equivalently X annihilates every holomorphic function.

Proposition 45.2.3 (Tangential CR vector fields). *Set $\rho_j = \partial\rho/\partial z_j$. A CR field $X = \sum_j a_j \partial/\partial \bar{z}_j$ is tangential to $\partial\Omega$ iff $\sum_j a_j \bar{\rho}_j = 0$ on $\partial\Omega$. Near a point where (say) $\rho_n \neq 0$, the $n-1$ fields*

$$L_j = \bar{\rho}_n \frac{\partial}{\partial \bar{z}_j} - \bar{\rho}_j \frac{\partial}{\partial \bar{z}_n}, \quad 1 \leq j \leq n-1,$$

are linearly independent and span the tangential CR fields (up to functions). Globally the $\binom{n}{2}$ fields $\bar{\rho}_k \partial/\partial \bar{z}_j - \bar{\rho}_j \partial/\partial \bar{z}_k$, $1 \leq j < k \leq n$, span them.

Proof. Tangentiality means $X(\rho) = \sum_j a_j \partial\rho/\partial \bar{z}_j = \sum_j a_j \bar{\rho}_j = 0$ on $\partial\Omega$ (using ρ real, so $\partial\rho/\partial \bar{z}_j = \overline{\partial\rho/\partial z_j} = \bar{\rho}_j$). The L_j each satisfy this and are independent since $\rho_n \neq 0$; they span the $(n-1)$ -dimensional solution space of one linear equation in n unknowns. $\square$

In the language of forms, abbreviate the inhomogeneous CR system $\partial u/\partial \bar{z}_j = f_j$ as $\bar{\partial}u = f$, where $\bar{\partial}u = \sum_j (\partial u/\partial \bar{z}_j) d\bar{z}_j$ is a $(0,1)$ -form, and $f = \sum_j f_j d\bar{z}_j$. For a $(0,1)$ -form $w = \sum_j w_j d\bar{z}_j$,

$$\bar{\partial}w = \sum_{k < j} \left(\frac{\partial w_j}{\partial \bar{z}_k} - \frac{\partial w_k}{\partial \bar{z}_j} \right) d\bar{z}_k \wedge d\bar{z}_j.$$

The consistency condition $\partial f_j/\partial \bar{z}_k = \partial f_k/\partial \bar{z}_j$ is exactly $\bar{\partial}f = 0$, and a function F_0 is annihilated by the tangential CR fields precisely when $\bar{\partial}F_0 \wedge \partial\rho|_{\partial\Omega} = 0$ – the *tangential Cauchy–Riemann equations*.

Worked Example 45.2.4 (A boundary value need not extend in one variable). Let $\Omega = \mathbb{D} \subset \mathbb{C}$ be the unit disc ($n = 1$). Which continuous F_0 on $\partial\Omega$ extend holomorphically into Ω ?

Recognition. For $n = 1$ there are no tangential CR equations; the obstruction is global (moments).

Solution. F_0 extends iff $\int_{\partial\Omega} z^k F_0(z) dz = 0$ for all $k \geq 0$ (Exercise 12): one direction is Cauchy’s theorem, the converse uses the Cauchy-integral jump formula. *Check.* $F_0(z) = \bar{z} = 1/z$ on $|z| = 1$ has $\int z^0 \bar{z} dz = \int (1/z) dz = 2\pi i \neq 0$, so $\bar{z}$ does *not* extend — correct, since its holomorphic extension would be $1/z$, singular at 0. $\blacksquare$

Bochner’s extension theorem

Theorem 45.2.5 (Bochner). *Let $\Omega \subset \mathbb{C}^n$ be a bounded region with C^3 boundary and connected complement. If $F_0 \in C^3(\partial\Omega)$ satisfies the tangential Cauchy–Riemann equations, then there is a holomorphic F on Ω , continuous on $\bar{\Omega}$, with $F|_{\partial\Omega} = F_0$.*

The conclusion is that the prescribed boundary datum F_0 factors through the holomorphic extension along the boundary inclusion $\iota : \partial\Omega \hookrightarrow \overline{\Omega}$: the restriction $F \circ \iota$ recovers F_0 .

$$\begin{array}{ccc} \partial\Omega & \xhookrightarrow{\iota} & \overline{\Omega} \\ & \searrow F_0 & \downarrow F \\ & & \mathbb{C} \end{array}$$

Proof sketch. Extend F_0 to a C^3 function on $\mathbb{C}^n$. Modify it to $F_1 = F_0 - a\rho$ (a C^2 function, same boundary values) so that $\bar{\partial}F_1|_{\partial\Omega} = 0$: choose $a = N(F_0)/N(\rho)$ near $\partial\Omega$ where $N = \sum_j \rho_j \partial/\partial \bar{z}_j$ is the (non-tangential) normal CR field, for which $N(\rho) = \frac{1}{4}|\nabla\rho|^2 > 0$. Set $f = \bar{\partial}F_1$; this is C^1 on $\overline{\Omega}$, vanishes on $\partial\Omega$, satisfies $\bar{\partial}f = 0$ inside, and extends by 0 to all of $\mathbb{C}^n$ with $\bar{\partial}f = 0$ (distributionally). By Proposition 45.3.4 (§3) there is a compactly supported continuous u with $\bar{\partial}u = f$; u is holomorphic on Ω^c , which is connected and meets the region where $u = 0$, so $u \equiv 0$ on Ω^c and (by continuity) on $\partial\Omega$. Then $F = F_1 - u$ is holomorphic in Ω , continuous on $\overline{\Omega}$, and $F|_{\partial\Omega} = F_1|_{\partial\Omega} = F_0$. $\square$

Remark. For $n = 1$ there are no tangential CR equations and the conditions on F_0 are global (the moment conditions of the previous worked example), reflecting that connectivity of Ω^c , not local convexity, governs the 1-variable picture. The Bochner–Martinelli integral gives an alternative proof valid for F_0 merely C^1 (Problem 3*).

Worked Example 45.2.6 (Why connectedness of the complement is needed). Show the hypothesis “ Ω^c connected” in Bochner’s theorem cannot be dropped.

Recognition. Use the shell $\Omega = \{\rho < |z| < 1\}$, whose complement has two pieces.

Solution. Take $\Omega = \{\rho < |z| < 1\} \subset \mathbb{C}^n$, $n \geq 2$. Its complement $\{|z| \leq \rho\} \cup \{|z| \geq 1\}$ has two components. The function F_0 that is 0 on the inner sphere and equals (the boundary values of) a function singular inside the inner ball cannot extend holomorphically to all of Ω while matching both spheres; concretely Exercise 10 builds such F_0 . *Check.* Hartogs (Theorem 45.1.10) already shows holomorphic functions on the shell extend *inward*, so prescribing independent data on the two spheres is over-determined — exactly the failure caused by a disconnected complement. $\blacksquare$

The Levi form and pseudoconvexity

To detect on which *side* of $\partial\Omega$ extension occurs, we put the boundary in a holomorphic normal form. Only holomorphic changes of coordinate are allowed (they preserve the class of holomorphic functions), so we cannot simply flatten $\partial\Omega$.

Proposition 45.2.7 (Holomorphic normal form). *Assume $\partial\Omega$ is C^2 . Near any $z^0 \in \partial\Omega$ there are holomorphic coordinates $(z_1, \dots, z_n)$ centered at z^0 with*

$$\Omega = \left\{ \operatorname{Im}(z_n) > \sum_{j=1}^{n-1} \lambda_j |z_j|^2 + E(z) \right\},$$

where the $\lambda_j \in \mathbb{R}$, and $E(z) = x_n \ell(z') + Dx_n^2 + o(|z'|^2)$ with ℓ linear and $D \in \mathbb{R}$. After a further change one may absorb the error to get $y_n = \sum_j \lambda_j |z_j|^2 + o(|z'|^2)$ (Exercise 13).

Definition 45.2.8 (Levi form). The Hermitian form

$$\mathcal{L}_{z^0}(\rho; w) = \sum_{j,k=1}^n \frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k}(z^0) w_j \bar{w}_k, \quad w \in T_{z^0}^{1,0}(\partial\Omega),$$

restricted to the complex tangent space (vectors w with $\sum_j \rho_j(z^0) w_j = 0$), is the *Levi form* of $\partial\Omega$ at z^0 . In the normal form its eigenvalues are (proportional to) the λ_j . The region Ω is (*Levi*) *pseudoconvex* at z^0 if $\mathcal{L}_{z^0} \geq 0$, and *strongly pseudoconvex* if $\mathcal{L}_{z^0} > 0$.

The signature of the Levi form – the numbers of positive, negative, zero eigenvalues – is a *holomorphic invariant* of the boundary point. For $n = 1$ there is no complex tangent direction, so the Levi form is empty: there are no local holomorphic invariants of a curve (Exercise 14), matching the absence of tangential CR equations.

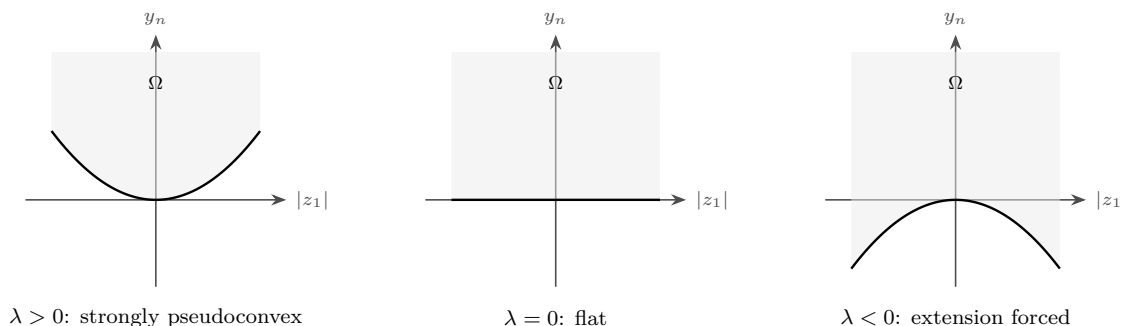


Figure 45.5. The boundary profile $y_n = \lambda |z_1|^2 + \dots$ in a complex tangent direction, with the region $\Omega = \{y_n > \lambda |z_1|^2\}$ shaded. Positive curvature (left) is strong pseudoconvexity; the flat case (middle) is the borderline; negative curvature (right) is concave, so a Hartogs figure forces continuation and Ω cannot be a domain of holomorphy there. (Schematic.)

Theorem 45.2.9 (Pseudoconvexity of domains of holomorphy). *If Ω is a domain of holomorphy with C^2 boundary, then Ω is pseudoconvex (the Levi form is ≥ 0 at every boundary point). Conversely (Levi problem, Problem 5*) every pseudoconvex region is a domain of holomorphy.*

We give the idea behind the forward direction: if the Levi form had a strictly negative eigenvalue at z^0 , the boundary would be “concave” in a complex tangent direction, and a Hartogs-type figure could be inserted to continue *every* holomorphic function across z^0 – contradicting that Ω is a domain of holomorphy. A clean instance is the following.

Worked Example 45.2.10 (A hypersurface that forces continuation everywhere). Consider $M = \{(z_1, z_2, z_3) \in \mathbb{C}^3 : \text{Im}(z_3) = |z_1|^2 - |z_2|^2\}$. Show every F holomorphic near M continues to all of $\mathbb{C}^3$.

Recognition. The Levi form here has a positive and a negative eigenvalue ($\lambda_1 = 1, \lambda_2 = -1$); a mixed signature triggers extension in every direction.

Solution. By the approximation/extension machinery (the maximum principle and Theorem 7.5 in Stein, requiring one strictly positive Levi eigenvalue) F continues into a fixed ball B about 0; rescaling the dilation-invariant model M then spreads the ball over all of $\mathbb{C}^3$ (Exercise 15). *Check.* The indefinite Levi form means M is concave in the z_2 complex-tangent direction, so M cannot bound a domain of holomorphy on either side – continuation is forced. ■

Worked Example 45.2.11 (Reading the Levi form from a defining function). Compute the Levi form of the ball $\Omega = \{|z| < 1\} \subset \mathbb{C}^n$ using $\rho(z) = |z|^2 - 1$, and decide pseudoconvexity.

Recognition. Differentiate $\rho = \sum_j z_j \bar{z}_j - 1$ in $z_j, \bar{z}_k$ and restrict to the complex tangent space.

Solution. $\partial^2 \rho / \partial z_j \partial \bar{z}_k = \delta_{jk}$, so $\mathcal{L}(\rho; w) = \sum_j |w_j|^2 = |w|^2 > 0$ for $w \neq 0$. Restricted to any tangent subspace it stays positive. *Check.* The ball is strongly pseudoconvex everywhere — consistent with its being a domain of holomorphy (e.g. $1/(z_1 - 1)$ is singular only at the boundary point $(1, 0, \dots, 0)$ and witnesses non-extendability there). ■

Worked Example 45.2.12 (Levi form of the Siegel half-space). Compute the Levi form of $\Omega = \{\text{Im}(z_2) > |z_1|^2\} \subset \mathbb{C}^2$ using $\rho(z) = |z_1|^2 - \text{Im}(z_2)$, and classify it.

Recognition. Write $\text{Im}(z_2) = \frac{1}{2i}(z_2 - \bar{z}_2)$ and differentiate; the only second-order $z\bar{z}$ term comes from $|z_1|^2$.

Solution. $\partial^2 \rho / \partial z_1 \partial \bar{z}_1 = 1$, and all other second mixed derivatives vanish ($\text{Im}(z_2)$ is linear in $z_2, \bar{z}_2$). The complex tangent space at a boundary point is spanned by ∂_{z_1} (since $\rho_{z_2} \neq 0$), so $\mathcal{L}(\rho; w) = |w_1|^2 > 0$. *Check.* The Siegel domain is strongly pseudoconvex, the unbounded model biholomorphic to the ball; its boundary is the Heisenberg group. ■

Worked Example 45.2.13 (A curve carries no Levi form). Explain, using Exercise 14, why a smooth curve $\gamma \subset \mathbb{C}^1$ has no local holomorphic invariant, in contrast to a hypersurface in $\mathbb{C}^n$, $n \geq 2$.

Recognition. For $n = 1$ the complex tangent space of a real curve is $\{0\}$, so the Levi form has no arguments.

Solution. By Exercise 14 any C^k curve can be straightened by a holomorphic map so that locally $y = o(x^k)$: there is nothing biholomorphically invariant to measure. For $n \geq 2$ the complex tangent space has dimension $n - 1 \geq 1$, and the Levi form on it is a genuine invariant. *Check.* This matches the absence of tangential CR equations when $n = 1$ — both reflect the empty complex tangent space of a curve. ■

Worked Example 45.2.14 (Convex implies pseudoconvex, but not conversely). Show every (geometrically) convex region with smooth boundary is pseudoconvex, and that the converse fails.

Recognition. Convexity bounds the real Hessian of ρ from below; the Levi form is the restriction of the complex Hessian, a piece of it.

Solution. If Ω is convex we may take ρ convex, so its real Hessian is ≥ 0 ; the Levi form $\sum_{j,k} \rho_{z_j \bar{z}_k} w_j \bar{w}_k$ is the Hermitian part evaluated on complex-tangent w , hence ≥ 0 : Ω is pseudoconvex. Converse fails: the “worm” domains and many non-convex tube domains are pseudoconvex (domains of holomorphy) yet not convex. *Check.* Pseudoconvexity is a *complex* (biholomorphically invariant) condition while convexity is not preserved by holomorphic maps, so they cannot coincide. ■

Exercises (Stein, Ch.7).

- Suppose $\Omega \subseteq \mathbb{R}^d$ has a C^k defining function ρ . (a) If $F \in C^k(\mathbb{R}^d)$ and $F = 0$ on $\partial\Omega$, show $F = a\rho$ with $a \in C^{k-1}$. (b) If $F_1 = F_2$ on $\partial\Omega$ and X is any tangential vector field, show $X(F_1)|_{\partial\Omega} = X(F_2)|_{\partial\Omega}$. [Hint: write $F_1 - F_2 = a\rho$ and apply X ; the term $aX(\rho)$ vanishes on $\partial\Omega$ by tangentiality.]
- Verify that the extension F furnished by Bochner’s Theorem 4.1 is the unique solution of the Dirichlet problem for Ω with boundary data F_0 .
- Use the shell region $\{z \in \mathbb{C}^n : \rho < |z| < 1\}$ to show that the connectedness hypotheses in Theorem 3.3 (Hartogs) and Theorem 4.1 (Bochner) are necessary.
- The connectedness conditions in Theorems 3.3 and 4.1 are related: let Ω be a bounded region with C^1 boundary, $\Omega_\varepsilon = \{z : d(z, \partial\Omega) < \varepsilon\}$ the collar, and $\Omega_\varepsilon^- = \Omega_\varepsilon \cap \Omega$. For sufficiently small ε the following are equivalent: (i) Ω^c is connected; (ii) Ω_ε is connected; (iii) Ω_ε^- is connected. [Hint: to see (ii) or (iii) $\Rightarrow$ (i), join two points $P_1, P_2 \in \Omega^c$ via points $Q_1, Q_2 \in \partial\Omega$ on their components, then connect Q_1 to Q_2 inside Ω_ε^- . For (i) $\Rightarrow$ (iii), deform a path through one boundary point into a path through another using simple connectedness of $\mathbb{C}^n$ and track the intersection with $\partial\Omega$.]
- Show that, after an additional holomorphic change of variables, the canonical representations (16)/(17) of the boundary simplify to $y_n = \sum_{j=1}^{n-1} \lambda_j |z_j|^2 + o(|z'|^2)$ as $z' \rightarrow 0$. [Hint: take $z_n \mapsto z_n - z_n(c_1 z_1 + \cdots + c_{n-1} z_{n-1} + D z_n)$, $z_j \mapsto z_j$ ($1 \leq j \leq n-1$), for suitable constants c_j, D to absorb the error term E .]
- (No local holomorphic invariants when $n = 1$.) Let γ be a C^k curve in $\mathbb{C}^1$. Show that for every $z^0 \in \gamma$ there is a holomorphic bijection Φ of a neighborhood of z^0 onto a neighborhood of 0 so that $\Phi(\gamma)$ is the curve $\{y = \phi(x)\}$ with $\phi(x) = o(x^k)$ as $x \rightarrow 0$. [Hint: if $y = a_2 x^2 + \cdots + a_k x^k + o(x^k)$, take $\Phi^{-1}(z) = z + i \sum_{j=2}^k a_j z^j$. This says a real curve carries no Levi form.]
- Consider the hypersurface $M = \{(z_1, z_2, z_3) \in \mathbb{C}^3 : \operatorname{Im}(z_3) = |z_1|^2 - |z_2|^2\}$. Show M has the property that any holomorphic F defined near M continues analytically to all of $\mathbb{C}^3$. [Hint: by Theorem 7.5 there is a fixed ball B about 0 to which F continues; then rescale using the dilation invariance of the model M .]
- Verify the following structure on the model half-space U (the boundary ∂U is the Heisenberg group): (a) the inverse of the map of U to the unit ball is $z_n = i \frac{1-w_n}{1+w_n}$, $z_k = \frac{w_k}{1+w_n}$ ($k = 1, \dots, n-1$); (b) for $(\zeta, t) \in \mathbb{C}^{n-1} \times \mathbb{R}$ the “translation” $r_{(\zeta, t)}(z', z_n) = (z' + \zeta, z_n + t + 2i(z' \cdot \bar{\zeta}) + i|\zeta|^2)$ maps U and ∂U to themselves, and the composition law is $(\zeta, t) \cdot (\zeta', t') = (\zeta + \zeta', t + t' + 2\operatorname{Im}(\zeta \cdot \bar{\zeta}'))$, making $\mathbb{C}^{n-1} \times \mathbb{R}$ the Heisenberg group; (c) U and ∂U are invariant under the non-isotropic dilations $(z', z_n) \mapsto (\delta z', \delta^2 z_n)$, $\delta > 0$; (d) U and ∂U are invariant under $(z', z_n) \mapsto (u(z'), z_n)$ for u unitary on $\mathbb{C}^{n-1}$.

Additional exercises.

1. For the ball $\Omega = \{|z| < R\} \subset \mathbb{C}^n$ with $\rho(z) = |z|^2 - R^2$, verify $\nabla \rho \neq 0$ on $\partial\Omega$ and write down the normal CR field $N = \sum_j \rho_j \partial \bar{z}_j$ explicitly; confirm $N(\rho) = \frac{1}{4} |\nabla \rho|^2$.
2. Let $\rho(z) = \text{Im}(z_2) - |z_1|^2$ on $\mathbb{C}^2$ (the Siegel/Heisenberg model boundary $\{\rho = 0\}$). Compute the complex Hessian $\partial^2 \rho / \partial z_j \partial \bar{z}_k$ and the Levi form; decide whether $\{\rho < 0\}$ is pseudoconvex.
3. Show that for $\rho(z) = \text{Im}(z_2) - (|z_1|^2 - |z_1|^4)$ near the origin, the Levi form at 0 is positive, but along $\partial\Omega$ it changes; describe where pseudoconvexity can fail.
4. Prove that the half-space $\{\text{Im}(z_n) > 0\} \subset \mathbb{C}^n$ is a domain of holomorphy by exhibiting a holomorphic function singular at every boundary point. [Try $1/(z_n - a)$ for suitable boundary a , or a series.]
5. Verify that geometric convexity of Ω with smooth boundary implies pseudoconvexity by relating the real Hessian of a convex defining function to the Levi form.
6. (Non-example.) Give a region in $\mathbb{C}^2$ that is *not* pseudoconvex (hence not a domain of holomorphy), e.g. a shell, and name the holomorphic functions that are forced to extend across part of its boundary.
7. Show the Levi form's signature is unchanged under a biholomorphic change of coordinates, by transforming the complex Hessian of ρ and restricting to the (transformed) complex tangent space.
8. (Diagram problem.) Sketch, in a single picture, three boundary profiles $y_n = \lambda |z_1|^2$ for $\lambda > 0$, $\lambda = 0$, $\lambda < 0$, label which is strongly pseudoconvex, and indicate the side from which holomorphic functions are forced to continue.
9. For the model $M = \{\text{Im}(z_3) = |z_1|^2 - |z_2|^2\} \subset \mathbb{C}^3$, write the two nonzero Levi eigenvalues and explain in one line why every holomorphic function near M extends to all of $\mathbb{C}^3$.

45.3 The Inhomogeneous Cauchy-Riemann Equations

The thread running through this chapter is a single technique: to build a holomorphic function with prescribed properties, first build a smooth approximation F_1 , measure its failure to be holomorphic by $f_j = \partial F_1 / \partial \bar{z}_j$, and then *correct* F_1 by subtracting a solution u of the *inhomogeneous Cauchy-Riemann equations*

$$\frac{\partial u}{\partial \bar{z}_j} = f_j, \quad 1 \leq j \leq n, \quad \text{equivalently} \quad \bar{\partial} u = f.$$

If u can be chosen with compact support, then $F = F_1 - u$ is holomorphic and agrees with F_1 far away. The miracle of several variables ($n \geq 2$) is that compactly supported data force a compactly supported solution; in one variable they do not.

The one-dimensional solution and the Cauchy transform

Theorem 45.3.1 (Solving $\partial u / \partial \bar{z} = f$ in $\mathbb{C}$). *Let f be continuous with compact support in $\mathbb{C}$. Set, with $\Phi(z) = 1/(\pi z)$,*

$$u(z) = (f * \Phi)(z) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{f(\zeta)}{z - \zeta} dm(\zeta) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{f(z - \zeta)}{\zeta} dm(\zeta).$$

*Then: (a) u is continuous and $\partial u / \partial \bar{z} = f$ distributionally; (b) if $f \in C^k$ ($k \geq 1$) then $u \in C^k$ and the equation holds classically; (c) every compactly supported $u \in C^1$ satisfies $u = (\partial u / \partial \bar{z}) * \Phi$.*

Parts (a) and (c) say that on compactly supported data the Cauchy transform $T : f \mapsto f * \Phi$ is a right inverse of $\bar{\partial} = \partial / \partial \bar{z}$, the composite being the identity; this is exactly the fundamental-solution identity $\bar{\partial} \Phi = \delta_0$.

$$\begin{array}{ccc}
C_c(\mathbb{C}) & \xrightarrow{T=(-)*\Phi} & C(\mathbb{C}) \\
& \searrow \text{id} & \downarrow \bar{\partial} \\
& & C_c(\mathbb{C})
\end{array}$$

Proof. For (a)/(b): in the second form the singular kernel $1/\zeta$ is locally integrable on $\mathbb{C}$ ($\int_{|\zeta|\leq 1} |\zeta|^{-1} dm = 2\pi$), so

$$u(z+h) - u(z) = \frac{1}{\pi} \int_{\mathbb{C}} (f(z+h-\zeta) - f(z-\zeta)) \frac{dm(\zeta)}{\zeta} \rightarrow 0$$

by uniform continuity of f ; differentiation under the integral handles C^k . The key fact is that Φ is a *fundamental solution*: $\partial\Phi/\partial\bar{z} = \delta_0$ (Ch.3). Then in the distributional calculus

$$\frac{\partial}{\partial\bar{z}}(f * \Phi) = f * \frac{\partial\Phi}{\partial\bar{z}} = f * \delta_0 = f,$$

proving (a),(b). Reading the identity with u in place of f gives $u = u * \delta_0 = (\partial u/\partial\bar{z}) * \Phi$, which is (c). $\square$

Caution. In one variable a compactly supported f need *not* have a compactly supported solution: a necessary condition is $\int_{\mathbb{C}} f dm = 0$ (and the full criterion is $\int_{\mathbb{C}} \zeta^n f dm = 0$ for all $n \geq 0$, Exercise 7). The Cauchy transform $u = f * \Phi$ instead decays like $1/|z|$ and is the unique solution with $u(z) \rightarrow 0$ as $|z| \rightarrow \infty$.

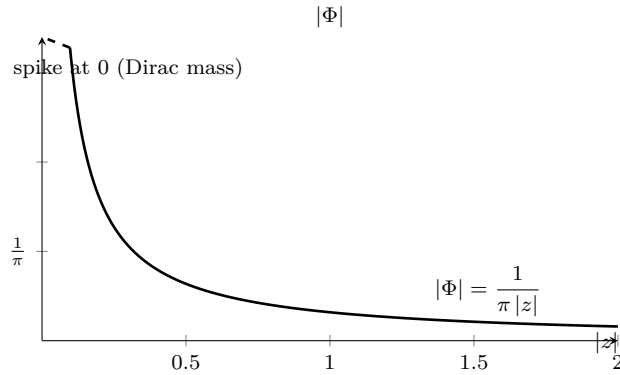


Figure 45.6. The fundamental solution $\Phi(z) = 1/(\pi z)$ of $\partial/\partial\bar{z}$: its modulus $|\Phi| = 1/(\pi|z|)$ is radial, blows up at the origin (the Dirac mass $\partial\Phi/\partial\bar{z} = \delta_0$) and decays like $1/|z|$ at infinity.

Worked Example 45.3.2 (The Cauchy transform of an indicator). With $f = \mathbf{1}_{\mathbb{D}}$ the indicator of the unit disc, identify $u = f * \Phi$ for $|z| > 1$.

Recognition. Outside the support, u is holomorphic (Theorem 45.3.1) and given by a convergent expansion in $1/z$; compute the leading coefficient $a_0 = \frac{1}{\pi} \int f dm$.

Solution. For $|z| > 1$, $\partial u/\partial\bar{z} = f = 0$, so u is holomorphic there and $u(z) = \sum_{n \geq 0} a_n z^{-n-1}$ with $a_n = \frac{1}{\pi} \int_{\mathbb{C}} \zeta^n f(\zeta) dm(\zeta)$. Here $a_0 = \frac{1}{\pi} \cdot (\text{area of } \mathbb{D}) = \frac{1}{\pi} \cdot \pi = 1$, and $a_n = \frac{1}{\pi} \int_{\mathbb{D}} \zeta^n dm = 0$ for $n \geq 1$ by symmetry, so $u(z) = 1/z$ for $|z| > 1$. *Check.* $\partial(1/z)/\partial\bar{z} = 0$ off the origin and $u \rightarrow 0$, consistent; and $a_0 = 1 \neq 0$ shows u does *not* have compact support, as expected since $\int f \neq 0$. $\blacksquare$

Worked Example 45.3.3 (Regularity is only Lipschitz, not C^1). Show $u = f * \Phi$ lies in $\text{Lip}(\alpha)$ for every $\alpha < 1$ when f is bounded with compact support, but need not be C^1 .

Recognition. This is the standard gain of the $1/z$ convolution: it improves L^∞ to almost- C^1 but not to C^1 (a logarithmic loss), Exercise 5.

Solution. Splitting the integral over $|\zeta| < \delta$ and $|\zeta| > \delta$ and optimizing $\delta \sim |h|$ gives $|u(z+h) - u(z)| \leq C_\alpha |h|^\alpha$ for each $\alpha < 1$. For sharpness take $f(z) = z \log(1/|z|)$ near 0 (cut off away from 0): then the second derivative blows up logarithmically, so $u \notin C^1$. *Check.* The borderline endpoint $\alpha = 1$ genuinely fails, the hallmark of singular-integral (Calderon–Zygmund) behaviour seen in Ch.42. $\blacksquare$

The several-variable solution with compact support

Proposition 45.3.4 (Compactly supported $\bar{\partial}$ -solution, $n \geq 2$). *Let $n \geq 2$ and let $f_1, \dots, f_n \in C^k$ have compact support and satisfy the consistency condition $\partial f_j / \partial \bar{z}_k = \partial f_k / \partial \bar{z}_j$ (i.e. $\bar{\partial} f = 0$). Then there is a compactly supported $u \in C^k$ with $\partial u / \partial \bar{z}_j = f_j$ for all j .*

Proof. Write $z = (z', z_n)$, $z' \in \mathbb{C}^{n-1}$, and define (a partial Cauchy transform in the last variable)

$$u(z) = \frac{1}{\pi} \int_{\mathbb{C}} f_n(z', z_n - \zeta) \frac{dm(\zeta)}{\zeta}.$$

By Theorem 45.3.1(c), $\partial u / \partial \bar{z}_n = f_n$. For $1 \leq j \leq n-1$, differentiating under the integral and using consistency,

$$\frac{\partial u}{\partial \bar{z}_j} = \frac{1}{\pi} \int_{\mathbb{C}} \frac{\partial f_n}{\partial \bar{z}_j}(z', z_n - \zeta) \frac{dm(\zeta)}{\zeta} = \frac{1}{\pi} \int_{\mathbb{C}} \frac{\partial f_j}{\partial \bar{z}_n}(z', z_n - \zeta) \frac{dm(\zeta)}{\zeta} = f_j(z', z_n),$$

the last step again by Theorem 45.3.1(c). Thus $\bar{\partial} u = f$. For the support: the f_j vanish for $|z| > R$, so by Theorem 45.1.3 u is holomorphic in $\{|z'| > R\}$, and by the integral u vanishes there; since $\{|z| > R\}$ is connected ($n \geq 2$!) and u is holomorphic and vanishes on an open piece, the identity Proposition 45.1.8 forces $u \equiv 0$ on $\{|z| > R\}$. $\square$

Remark. The leap from $n = 1$ to $n \geq 2$ is the connectedness of $\{|z| > R\}$: in $\mathbb{C}$ this set is an annular neighborhood of infinity that is connected too, but u is then only known to vanish in *one* direction $|z'| > R$, which is empty when there is no z' . The extra variable is what makes “vanishes somewhere at infinity” propagate to “vanishes near infinity.”

Worked Example 45.3.5 (Consistency is necessary). Show that $f_1 = 1, f_2 = 0$ on $\mathbb{C}^2$ admits no C^1 solution of $\bar{\partial} u = f$, and locate the failure.

Recognition. Check $\bar{\partial} f = 0$: compute $\partial f_1 / \partial \bar{z}_2$ vs. $\partial f_2 / \partial \bar{z}_1$.

Solution. If $\partial u / \partial \bar{z}_1 = 1$ and $\partial u / \partial \bar{z}_2 = 0$, then mixing partials, $\partial^2 u / \partial \bar{z}_2 \partial \bar{z}_1 = \partial(1) / \partial \bar{z}_2 = 0$ but also $= \partial(0) / \partial \bar{z}_1 = 0$ – consistent here, so try $f_1 = \bar{z}_2, f_2 = 0$: then $\partial f_1 / \partial \bar{z}_2 = 1 \neq 0 = \partial f_2 / \partial \bar{z}_1$, violating consistency, and no u exists. *Check.* $\bar{\partial} f = (\partial f_1 / \partial \bar{z}_2 - \partial f_2 / \partial \bar{z}_1) d\bar{z}_2 \wedge d\bar{z}_1 = d\bar{z}_2 \wedge d\bar{z}_1 \neq 0$, exactly the obstruction. $\blacksquare$

Hartogs’ theorem in general form

Theorem 45.3.6 (Hartogs extension across a compact hole). *Let $\Omega \subset \mathbb{C}^n$ be a bounded region with $n \geq 2$, and $K \Subset \Omega$ compact with $\Omega \setminus K$ connected. Then every holomorphic F_0 on $\Omega \setminus K$ extends to a holomorphic F on Ω (with $F = F_0$ on $\Omega \setminus K$).*

The extension F restricts back to the original datum along the inclusion $\iota : \Omega \setminus K \hookrightarrow \Omega$, so $F \circ \iota = F_0$.

$$\begin{array}{ccc} \Omega \setminus K & \xhookrightarrow{\iota} & \Omega \\ & \searrow F_0 & \downarrow F \\ & & \mathbb{C} \end{array}$$

Proof. Pick $\varepsilon > 0$ so that $O_\varepsilon = \{z : d(z, \Omega^c) < \varepsilon\}$ stays at positive distance from K ; then $\Omega \cap O_\varepsilon \subset \Omega \setminus K$. Choose a C^∞ cutoff η with $\eta = 0$ near K and $\eta = 1$ on O_ε , and set

$$F_1 = \begin{cases} \eta F_0 & \text{on } \Omega \setminus K, \\ 0 & \text{on } K. \end{cases}$$

Then $F_1 \in C^\infty(\Omega)$ extends F_0 but is not holomorphic. Its defect is $f_j = \partial F_1 / \partial \bar{z}_j$; these are C^∞ , satisfy $\bar{\partial} f = 0$, and vanish near $\partial\Omega$ (where $\eta \equiv 1$ and F_0 is holomorphic), so extend by 0 to all of $\mathbb{C}^n$. By

Proposition 45.3.4 take compactly supported u with $\bar{\partial}u = f$, and set $F = F_1 - u$. Then $\bar{\partial}F = 0$, so F is holomorphic on Ω .

It remains to see $F = F_0$ on (an open piece of, hence all of) $\Omega \setminus K$, i.e. that u vanishes there. Let R be smallest with $\Omega \subseteq \{|z| \leq R\}$ and pick $z^0 \in \partial\Omega$ with $|z^0| = R$. Put $B = B_\varepsilon(z^0)$. Then u is holomorphic in B (the f_j vanish there), and u vanishes on the connected set $\{|z| > R\}$ (it is holomorphic there and has compact support), which meets B ; so by the identity theorem $u \equiv 0$ on B , in particular on the nonempty open set $\Omega \cap B \subset \Omega \setminus K$. Hence $F = F_0$ there, and by connectedness of $\Omega \setminus K$, everywhere on it. $\square$

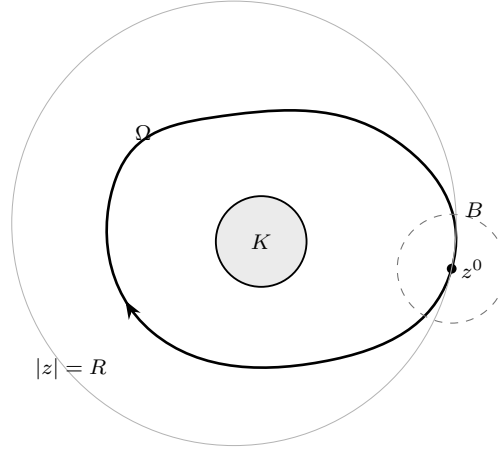


Figure 45.7. Hartogs extension: cut off F_0 near K , correct the smooth gluing by a compactly supported $\bar{\partial}$ -solution u ; u vanishes near the farthest boundary point z^0 , so $F = F_0$ on $\Omega \setminus K$. (Schematic.)

Worked Example 45.3.7 (Filling a square hole). Let $\Omega = \{|z| < 2\} \subset \mathbb{C}^2$ and $K = \{|z| \leq 1\}$. Does every holomorphic F_0 on the shell $\Omega \setminus K = \{1 < |z| < 2\}$ extend to Ω ? Compare with $\mathbb{C}$.

Recognition. This is exactly Theorem 45.3.6: shell in $\mathbb{C}^2$, connected complement of K within Ω .

Solution. Yes. $\Omega \setminus K = \{1 < |z| < 2\}$ is connected (real dimension 4), K is compact in Ω , $n = 2 \geq 2$; so F_0 extends holomorphically across the hole to all of the ball $\{|z| < 2\}$. *Check.* In $\mathbb{C}$ the analog $\{1 < |z| < 2\}$ is an annulus and $F_0(z) = 1/z$ does *not* extend across $\{|z| \leq 1\}$ – the theorem genuinely uses $n \geq 2$. $\blacksquare$

Approximation and extension on the boundary

The boundary analog (the CR-extension/approximation theory of §§4–7 of Stein) refines Bochner’s theorem locally, under a one-sided Levi condition.

Theorem 45.3.8 (Local approximation, after Stein 7.1). *Let $M \subset \mathbb{C}^n$ be a C^2 hypersurface and $z^0 \in M$. There are balls $B' \Subset B$ about z^0 so that any continuous F on $M \cap B$ satisfying the tangential CR equations (weakly) can be uniformly approximated on $M \cap B'$ by polynomials in $z_1, \dots, z_n$.*

Theorem 45.3.9 (Local one-sided extension, after Stein 7.5). *If the Levi form of M has at least one strictly positive eigenvalue at each point, then for each $z^0 \in M$ there is a ball B' so that every continuous F_0 on M satisfying the tangential CR equations extends to a function F holomorphic in $\Omega^- \cap B'$, continuous on $\bar{\Omega}^- \cap B'$, with $F = F_0$ on $M \cap B'$.*

Idea. By Theorem 45.3.8, F_0 is a uniform limit on $M \cap B_1$ of polynomials p_m . A maximum-principle estimate (Stein’s Theorem 6.1, valid because of the strictly positive Levi eigenvalue) controls $\sup_{\Omega^- \cap B'} |p_m - p_{m'}|$ by $\sup_{M \cap B_1} |p_m - p_{m'}|$, so the p_m converge uniformly on $\Omega^- \cap B'$. The limit F is holomorphic there, continuous up to $\bar{M} \cap B'$, and restricts to F_0 . $\square$

Worked Example 45.3.10 (The maximum principle is a several-variable phenomenon). Explain why the boundary maximum estimate behind Theorem 45.3.9 can fail for $n = 1$.

Recognition. For $n = 1$ there is no Levi form and no positive eigenvalue to exploit; Exercise 16 builds an explicit counterexample.

Solution. Exercise 16 constructs $F_N(z) = e^{NG(z)}$ on the disc with $|F_N| = 1$ on an arc $\{|\theta| \leq \pi/2\}$ yet $|F_N(z)| \geq c_1 e^{c_2 N(1-|z|)}$ inside: the interior values are not controlled by the boundary arc as $N \rightarrow \infty$. *Check.* This is precisely the absence of the positive-Levi-eigenvalue mechanism — in one variable the relevant complex tangential directions do not exist, so the propagation of smallness fails. ■

Worked Example 45.3.11 (The moment obstruction in one variable). For $f = \mathbf{1}_{\mathbb{D}}$ in $\mathbb{C}$, verify directly that no compactly supported u solves $\partial u / \partial \bar{z} = f$, using the moment criterion.

Recognition. Exercise 7: compact support of u requires $\int_{\mathbb{C}} \zeta^n f \, dm = 0$ for all $n \geq 0$.

Solution. The $n = 0$ moment is $\int_{\mathbb{D}} 1 \, dm = \pi \neq 0$, so the criterion already fails at $n = 0$; no compactly supported solution exists. The Cauchy transform gives instead $u(z) = 1/z$ for $|z| > 1$, which decays but is not compactly supported. *Check.* This is exactly why the $n \geq 2$ argument of Proposition 45.3.4 cannot be run in $\mathbb{C}$ — there is no extra variable to force vanishing near infinity. ■

Worked Example 45.3.12 (Uniqueness of the compactly supported solution). Show that for $n \geq 2$ the compactly supported solution of $\bar{\partial}u = f$ is unique.

Recognition. Two solutions differ by a holomorphic function; impose compact support.

Solution. If $\bar{\partial}u_1 = \bar{\partial}u_2 = f$ then $h = u_1 - u_2$ satisfies $\bar{\partial}h = 0$, so h is entire on $\mathbb{C}^n$; being compactly supported it vanishes on a nonempty open set, hence $h \equiv 0$ by the identity theorem. *Check.* For $n = 1$ the analogous uniqueness picks out the solution u with $u(z) \rightarrow 0$ at infinity (not compact support), consistent with the moment obstruction. ■

Worked Example 45.3.13 (Putting the pieces together: from boundary data to a holomorphic extension). Sketch how Bochner's Theorem 45.2.5 is assembled from the $\bar{\partial}$ -machine.

Recognition. The five moves: extend F_0 ; correct by $-a\rho$ to kill $\bar{\partial}$ on the boundary; read off $f = \bar{\partial}F_1$ with $\bar{\partial}f = 0$, $f|_{\partial\Omega} = 0$; solve $\bar{\partial}u = f$ with compact support (Proposition 45.3.4); subtract.

Solution. Each move is exactly an earlier result: the extension uses $C^3(\partial\Omega)$ functions are restrictions of $C^3(\mathbb{C}^n)$ functions; the correction uses the normal CR field N with $N(\rho) = \frac{1}{4} |\nabla \rho|^2 > 0$; the solvability is Proposition 45.3.4 (needs $n \geq 2$); the vanishing of u on Ω^c uses connectedness of the complement and the identity theorem. *Check.* Dropping $n \geq 2$ removes the compactly supported solution and the theorem reverts to the global moment conditions of the $n = 1$ case. ■

Exercises (Stein, Ch.7).

- Suppose f is continuous with compact support in $\mathbb{C}^1$ and $u = f * \Phi$ as in Proposition 3.1. (a) Show $u \in \text{Lip}(\alpha)$ for every $\alpha < 1$. (b) Show u need not be C^1 . [Hint for (b): take $f(z) = z(\log(1/|z|))$ modified away from the origin to have compact support; the second derivatives blow up logarithmically.]
- Verify the Cauchy–Pompeiu identity in $\mathbb{C}^1$,

$$F(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{F(\zeta)}{\zeta - z} d\zeta - \frac{1}{\pi} \int_{\Omega} \frac{(\partial F / \partial \bar{\zeta})(\zeta)}{\zeta - z} dm(\zeta),$$

for suitable regions Ω and C^1 functions F , and use it to give an alternative proof of Proposition 3.1.

- Prove: the necessary and sufficient condition that the solution $u(z) = \frac{1}{\pi} \int \frac{f(\zeta)}{z - \zeta} dm(\zeta)$ of $\partial u / \partial \bar{z} = f$ in $\mathbb{C}^1$ have compact support, when f has compact support, is that $\int_{\mathbb{C}} \zeta^n f(\zeta) dm(\zeta) = 0$ for all $n \geq 0$. [Hint: one direction from $\partial(z^n u) / \partial \bar{z} = z^n f$; for the converse, for large $|z|$ expand $u(z) = \sum_{n \geq 0} a_n z^{-n-1}$ with $a_n = \frac{1}{\pi} \int \zeta^n f(\zeta) dm(\zeta)$, so all $a_n = 0$ forces $u \equiv 0$ near infinity.]
- Let Ω be a simply connected bounded region in $\mathbb{C}^1$ with C^1 boundary, and F_0 continuous on $\partial\Omega$. Show that there is an F holomorphic in Ω and continuous on $\bar{\Omega}$ with $F = F_0$ on $\partial\Omega$ iff $\int_{\partial\Omega} z^n F_0(z) dz = 0$ for $n = 0, 1, 2, \dots$ [Hint: one direction is Cauchy's theorem; for the converse set $F^\pm(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{F_0(\zeta)}{\zeta - z} d\zeta$ according as $z \in \Omega$ or $z \in \Omega^c$; the hypothesis gives $F^+ \equiv 0$ on Ω^c , and the boundary jump $F^-(z) - F^+(\bar{z}) \rightarrow F_0(\zeta)$ recovers the data, via the delta identity $i\pi\delta = \frac{1}{2}(\frac{1}{x-i0} - \frac{1}{x+i0})$ of Ch.3 §2.]

5. Show that the maximum principle (Theorem 6.1) fails for $n = 1$: start with $f \geq 0$, $f(e^{i\theta}) \in C^\infty$, $f = 0$ for $|\theta| \leq \pi/2$ and $f = 1$ for $3\pi/4 \leq |\theta| \leq \pi$. Write $f(e^{i\theta}) = \sum_{n \geq 0} a_n e^{in\theta} + \sum_{n=-1}^{-\infty} a_n e^{in\theta}$, $G(z) = \sum_{n \geq 0} a_n z^n$, $F_N(z) = e^{NG(z)}$. Verify F_N is continuous on $|z| \leq 1$, $|F_N(e^{i\theta})| = 1$ for $|\theta| \leq \pi/2$, yet $|F_N(z)| \geq c_1 e^{c_2 N(1-|z|)}$ in the closed disc for constants $c_1, c_2 > 0$. [Hint: $G = u + iv$ with $u(r, \theta) = f * P_r$, P_r the Poisson kernel.]
6. Define H_λ to be the space of f holomorphic on $\mathbb{C}^{n-1}$ with $\int_{\mathbb{C}^{n-1}} |f(z)|^2 e^{-4\pi\lambda|z|^2} dm(z) = \|f\|_{H_\lambda}^2 < \infty$. Show: (a) H_λ is trivial if $\lambda \leq 0$; (b) H_λ is complete, hence a Hilbert space; (c) with $K_\lambda(z, w) = (4\lambda)^{n-1} e^{4\pi\lambda z \cdot \bar{w}}$, the operator $P_\lambda(f)(z) = \int_{\mathbb{C}^{n-1}} f(w) K_\lambda(z, w) e^{-4\pi\lambda|w|^2} dm(w)$ is the orthogonal projection of $L^2(e^{-4\pi\lambda|w|^2} dm)$ onto H_λ (the Bargmann–Fock space and its reproducing kernel). [Hint: norm convergence in H_λ implies uniform convergence on compact subsets.]
7. Prove: (a) the space H of Section 8.1 is complete, hence a Hilbert space; (b) the Cauchy integral $f \mapsto C(f)$ is the orthogonal projection from $L^2(\partial U, d\beta)$ onto the space of boundary functions $F_0 = \lim_{\varepsilon \rightarrow 0} F_\varepsilon$ arising from $F \in H^2(U)$. [Hint: for (a) use the previous exercise.]

Problems (Stein, Ch.7).

1. * (Bochner–Martinelli.) The original proof of Theorem 4.1 represented F in terms of F_0 via the Bochner–Martinelli integral (a several-variable Cauchy kernel obtained from Green’s theorem); the result then holds for F_0 merely of class C^1 .
2. * ($\bar{\partial}$ on pseudoconvex domains.) Consider $\bar{\partial}u = f$ on a bounded region $\Omega \subset \mathbb{C}^n$ with C^∞ boundary, f given on Ω with $\bar{\partial}f = 0$. (a) If Ω is pseudoconvex and $f \in C^\infty(\bar{\Omega})$ there is $u \in C^\infty(\bar{\Omega})$ solving it. (b) The “normal” solution is the unique $u \in L^2(\Omega)$ with $\int_\Omega u \bar{F} dm = 0$ for all F holomorphic in $L^2(\Omega)$; for strongly pseudoconvex Ω (and many others), $f \in C^\infty(\bar{\Omega})$ gives normal solution $u \in C^\infty(\bar{\Omega})$, via the $\bar{\partial}$ -Neumann problem.
3. * (Converse to local solvability.) The converse to Theorem 8.7 holds: if f is a compactly supported distribution with $C(f)(z)$ analytic near $z = 0$, then $L(U) = f$ is locally solvable near the origin. One finds a kernel K so that $T(f) = f * K$ on the Heisenberg group is a relative inverse, $LT(f) = f - C(f)$; writing $f = (f - C(f)) + C(f) = f_1 + f_2$, solve $L(U_1) = f_1$ by the above and $L(U_2) = f_2$ locally by Cauchy–Kovalevskaya, since f_2 is real-analytic at the origin.

Additional exercises.

1. Verify in detail that $\Phi(z) = 1/(\pi z)$ is locally integrable on $\mathbb{C}$ by computing $\int_{|z| \leq R} |\Phi| dm$, and conclude the convolution $f * \Phi$ is well defined for f bounded with compact support.
2. Solve $\partial u / \partial \bar{z} = f$ in $\mathbb{C}$ for $f = \mathbf{1}_{\mathbb{D}}$ and find u explicitly both inside ($|z| < 1$) and outside ($|z| > 1$) the unit disc; confirm continuity across $|z| = 1$. [Inside, expect $u(z) = \bar{z}$; outside, $u(z) = 1/z$.]
3. Show that $f(z) = \mathbf{1}_{\mathbb{D}}$ does *not* admit a compactly supported $\bar{\partial}$ -solution in $\mathbb{C}$, and identify the failing moment from Exercise 7 (Stein).
4. For $n = 2$, take $f_1 = g(z_2) \mathbf{1}$ -type data and construct, via Proposition 3.2, a compactly supported u solving $\bar{\partial}u = f$; verify the consistency condition you need.
5. (Hartogs in practice.) Let $\Omega = \{|z| < 3\} \subset \mathbb{C}^2$ and $K = \{|z| \leq 1\}$. State precisely the extension guaranteed by the Hartogs theorem and explain why the same fails for $\Omega = \{1 < |z| < 3\}$, $K = \emptyset$ in $\mathbb{C}^1$.
6. Prove the uniqueness clause: if $n \geq 2$ and f has compact support, the compactly supported solution of $\bar{\partial}u = f$ is unique. [Two such differ by an entire function of compact support.]
7. (Non-example.) Show that for $n = 1$ there is generally *no* compactly supported solution of $\partial u / \partial \bar{z} = f$ for compactly supported f , by exhibiting an f with $\int f dm \neq 0$ and tracing the obstruction.
8. Use the Cauchy–Pompeiu formula (Exercise 6, Stein) to recover the one-variable Cauchy integral formula when F is holomorphic, and to express the “area” correction term when it is not.

9. (Diagram problem.) Reproduce the Hartogs-extension picture: a region Ω with a compact hole K , the sphere $|z| = R$ through the farthest boundary point z^0 , and the ball B where the corrector u vanishes; annotate which connectedness fact each label uses.
10. (Synthesis.) In five labeled steps, reconstruct the proof of Bochner's theorem from the $\bar{\partial}$ -machinery, indicating at each step which hypothesis (C^3 boundary, $n \geq 2$, connected complement) is consumed.